



VyOS Documentation

Release rolling (current)

VyOS maintainers and contributors

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Get / Build VyOS Quickly *Build* your own Image or take a look at how to *download* a free or supported version.

Install VyOS Read about how to install VyOS on *Bare Metal* or in a *VM* and how to use an image with the usual *cloud* providers

Configuration and Operation Use the *Quickstart Guide*, to have a fast overview. Or go deeper and set up *advanced routing*, *VRFs*, or *VPNs* for example.

Automate Integrate VyOS in your automation Workflow with *Ansible*, have your own *local scripts*, or configure VyOS with the *HTTPS-API*.

Examples Get some inspiration from the *Blueprints* to build your infrastructure.

Contribute and Community There are many ways to contribute to the project. Add missing parts or improve the *Documentation*.

Discuss in *Slack* or the *Forum*.

Or you can pick up a *Task* and fix the *code*.

ABOUT

VyOS is an open-source network operating system that provides a single unified CLI and API to manage routing protocols, firewall and NAT, QoS, load balancing, DHCP and DNS servers, and many other features.

VyOS runs on a wide variety of commodity hardware, virtual machines, and multiple cloud environments.

We provide a dedicated user guide for each major VyOS release that receives long-term support (LTS). We maintain multiple user guide versions, all hosted at <https://docs.vyos.io>. To switch between versions, select the appropriate version in the bottom-right corner.

VyOS CLI syntax may vary between major and sometimes minor releases. Always refer to the documentation matching your current running installation. If a change in the CLI is required, VyOS provides a migration script to handle the syntax adjustments. No user action is required.

HISTORY

2.1 In the beginning...

There was a network operating system based on Debian GNU/Linux, called Vyatta.¹ Introduced in 2006, it served as a great free-software alternative to proprietary products. Vyatta came in two editions: Vyatta Core (formerly known as Vyatta Community Edition), which was free software, and Vyatta Subscription Edition, which included proprietary features and was available only to paying customers.

Brocade Communications Systems acquired Vyatta in 2012. Shortly after, Brocade renamed Vyatta Subscription Edition to Brocade vRouter, discontinued Vyatta Core, and shut down the community forum without notice. The bug tracker and Git repositories were closed the following year.

By the time Brocade acquired Vyatta, the development of Vyatta Core had already stagnated. The focus had shifted to Vyatta Subscription Edition, where core components were replaced with proprietary software. As a result, Vyatta Core received fewer new features, and some of those added faced issues.

In 2013, shortly after Vyatta Core was discontinued, the community forked its final version (6.6R1) to create the VyOS project. In 2014, the maintainers established a company to fund VyOS development through technical support, consulting services, and LTS release access subscriptions. The company was originally named Sentrion and was later reorganized under the VyOS brand.

2.2 Major releases

VyOS originally named its major versions after elements by atomic number. Beginning with version 1.2, this naming scheme was changed. It now uses the Latin names of constellations recognized by the International Astronomical Union (IAU), ordered by their solid angle area, beginning with the smallest.

2.2.1 Hydrogen (1.0)

Released just in time for the holidays on 22 December 2013, Hydrogen was the first major VyOS release. It fixed features that were broken in Vyatta Core 6.6, such as IPv4 BGP peer groups and DHCPv6 relay, and introduced command scripting, a task scheduler, and web proxy LDAP authentication.

¹ From the Sanskrit adjective “Vyāta” (), meaning opened.

2.2.2 Helium (1.1)

Helium, released on 9 October 2014, marked the first anniversary of the VyOS Project. The release introduced an event handler, L2TPv3 support, 802.1ad (QinQ), and IGMP proxy, as well as experimental support for VXLAN and DMVPN. Notably, DMVPN remained non-functional in Vyatta Core due to its reliance on a proprietary NHRP implementation.

2.2.3 Crux (1.2)

Crux (the Southern Cross) was released on 28 January 2019 and marked a departure from legacy Vyatta codebase and the start of the migration from Perl to Python as the primary language. The underlying base system was upgraded from Debian 6 (Squeeze) to Debian 8 (Jessie).

Crux introduced many new features, some of the most noteworthy are: an mDNS repeater, a broadcast relay, a high-performance PPPoE server, an HFSC scheduler, and support for Wireguard, unicast VRRP, RPKI for BGP, and fully 802.1ad-compliant QinQ ethernet. The telnet server and support for P2P filtering were removed.

Crux was the first VyOS release to feature a modular image build system. CLI definitions were written using an XML syntax automatically checked against a schema at build time. Python APIs were introduced for command scripting and configuration migration. New Perl code and old-style (non-XML) command definition were no longer accepted from that point.

Crux reached the end of support in 2023.

2.2.4 Equuleus (1.3)

Equuleus (the Little Horse) was a long-term support version released on 21 December 2021, just in time for the winter holidays.

Equuleus brought many long-awaited features, most notably an SSTP VPN server, an IPoE server, an OpenConnect VPN server, and a serial console server. It also introduced reworked support for WWAN interfaces, support for GEN-EVE and MACSec interfaces, VRF, IS-IS routing, and preliminary support for MPLS and LDP.

Equuleus reached the end of support in 2025.

2.2.5 Sagitta (1.4)

Sagitta (the Arrow), the current LTS release, became generally available on 4 June 2024. Its development began in late 2021 and focused on eliminating remaining legacy components and reworking core subsystems.

The transition to XML-defined command definitions and script refactoring with separate verify, update, and apply stages were completed. The firewall subsystem was rebuilt on nftables, introducing interface-independent rulesets and the reimplemented zone-based firewall model. The PKI subsystem was redesigned to manage cryptographic material directly within the configuration file.

Sagitta introduced rollback without reboot, support for Babel and PIM6 routing protocols, failover routes, segment routing, NAT64, an IKEv2 remote-access VPN server, Zabbix monitoring, HTTP load balancing, and configuration synchronization using the HTTP API.

The underlying base system was upgraded to Debian 12 (Bookworm).

2.2.6 Circinus (1.5)

Circinus (the Drawing Compass) became generally available as an LTS release on 31 March 2026. Its development began in 2024 and focused on major performance upgrades and modernizing core subsystems.

Circinus introduces several major architectural improvements, most notably an optional VPP-based accelerated dataplane. Using the DPDK driver, this dataplane can offer performance up to 15x faster than the Linux kernel dataplane and allows administrators to selectively enable hardware acceleration on a per-interface and per-feature basis.

Other significant additions and updates include:

- A high-performance kernel-mode NetFlow sensor based on ipt-netflow, replacing the older pmacct implementation.
- Unification of sFlow to exclusively use the much faster hsflowd implementation.
- Transition of the DHCP server to a Kea-based implementation, replacing the legacy, end-of-life ISC DHCPD.
- A completely rewritten WAN load balancing implementation to resolve long-standing stability issues and introduce support for firewall groups in load balancing rules.
- A new `execute` operational mode command family to separate action commands that do not depend on or modify system configuration.

The release also cleans up several legacy and underutilized components. FastNetMon was removed, OpenVPN support for Blowfish and Twofish ciphers was dropped for security reasons, and the Salt minion integration was deprecated.

Like Sagitta (1.4), the underlying base system for Circinus remains Debian 12 (Bookworm).

2.2.7 Scutum (1.6)

Scutum (the Shield) is the codename for the upcoming development branch. VyOS 1.6 Scutum has not been released yet.

2.3 A note on copyright

Unlike Vyatta, VyOS has never had closed-source code and never will. The only proprietary material in VyOS is non-code assets, such as graphics and the trademark “VyOS”.²

Note that we do not provide support for images distributed by a third party. See the [artwork license](#) and the end-user license agreement at `/usr/share/vyos/EULA` in any pre-built image for more information.

² This is similar to how Linus Torvalds owns the Linux trademark.

INSTALLATION AND IMAGE MANAGEMENT

Note: This information applies primarily to virtual installations:

When installing VyOS, ensure that the MAC address you select for your NICs is not a locally administered MAC address. Locally administered addresses are distinguished from universally administered addresses by setting the second-least-significant bit of the first octet to 1:

Example: `02:00:00:00:00:01`, where the second-least-significant bit (`02` in hexadecimal) is set to 1.

3.1 Installation

VyOS installation requires a VyOS .iso file. This file is a live installation image that you can use to boot a live VyOS system. From there, you can proceed with a permanent installation on a hard drive or other storage device.

Table 1: Comparison of VyOS image releases

Release Type	Description	Release Cycle	Intended Use	Access to Images	Access to Source
Nightly (Current)	Automatically built from the rolling branch. Always up to date with cutting edge development but guaranteed to contain bugs.	Every night	Developing VyOS, testing new features, experimenting.	Everyone	Everyone
Stream	VyOS Stream serves as a technology preview and a quality gate for the upcoming LTS release. Allows everyone to try new features and check if they work well or need improvements.	Every quarter	Non-critical production environments, preparing for the LTS release.	Everyone	Everyone
Release Candidate	Rather stable. All development focuses on testing and hunting down remaining bugs following the feature freeze.	Irregularly until EPA comes out	Labs, small offices and non-critical production systems backed by a high-availability setup.	Everyone	Everyone
Early Production Access	Highly stable with no known bugs. Needs to be tested repeatedly under different conditions before it can become the final release.	Irregularly until LTS comes out	Non-critical production environments, preparing for the LTS release.	Everyone	Everyone
Long-Term Support	Guaranteed to be stable and carefully maintained for several years after the release. No features are introduced but security updates are released in a timely manner.	Every major version	Large-scale enterprise networks, internet service providers, critical production environments that call for minimum downtime.	Subscribers, contributors, non-profits, emergency services, academic institutions	Subscribers, contributors, non-profits, emergency services, academic institutions

3.1.1 Hardware requirements

The minimum system requirements for VyOS are 4 GB RAM and 10 GB storage. Depending on your use case, you might need additional RAM and CPU resources.

3.1.2 Download

Registered Subscribers

Registered subscribers can log into <https://support.vyos.io/> to access a variety of different downloads via the “Downloads” link. These downloads include LTS (Long-Term Support), the associated hot-fix releases, early public access releases, pre-built VM images, as well as device specific installation ISOs. See this [article](#) for more information on downloads.

Note: The .qcow2 image provided for Proxmox deployment can also be used to deploy VyOS on KVM environments. This image includes cloud-init support.

Building from source

Subscribers can download the source code for the LTS release from the “Downloads” link. Non-subscribers can access the source code for the Rolling release. For instructions, see the *Build VyOS* section. The VyOS source code repository is available at <https://github.com/vyos/vyos-build>.

Rolling Release

Everyone can download bleeding-edge VyOS rolling images from: <https://downloads.vyos.io/>

Note: Rolling releases contain the latest enhancements and fixes. This means there may be new bugs. If you encounter a bug, follow the guide at *Bug Report/Issue*. We depend on your feedback to improve VyOS.

The following link contains the most recent VyOS builds for AMD64 systems from the rolling branch: <https://vyos.net/get/nightly-builds/>

Download Verification

LTS images are signed with the VyOS lead package maintainer’s private key. You can verify the authenticity of the package using the official public key and Minisign.

Minisign verification

VyOS uses *Minisign* for release signing. Minisign is a tool for signing files and verifying signatures.

OpenBSD introduced signify in 2015. Minisign is an alternative implementation of the same protocol, available for Windows, macOS, and most GNU/Linux distributions. Minisign is portable, lightweight, and uses the Ed25519 public-key signature system.

T2108 switched the validation system to prefer Minisign over GPG keys.

To verify a VyOS image starting with VyOS 1.3.0-rc6, run:

```
$ minisign -V -P RWSIhKR/dkM2DSaBRniv/bbbAf8hmDqdbOEmgXkf1RxRoxzodgKcDyGq -m vyos-1.5-
↳rolling-202409250007-generic-amd64.iso vyos-1.5-rolling-202409250007-generic-amd64.iso.
↳minisig

Signature and comment signature verified
Trusted comment: timestamp:1727223408 file:vyos-1.5-rolling-202409250007-generic-amd64.
↳iso      hashed
```

During an image upgrade, VyOS runs the following command:

```
$ minisign -V -p /usr/share/vyos/keys/vyos-release.minisign.pub -m vyos-1.3.0-rc6-amd64.
iso vyos-1.3.0-rc6-amd64.iso.minisig
Signature and comment signature verified
Trusted comment: timestamp:1629997936 file:vyos-1.3.0-rc6-amd64.iso
```

Note: Starting with version 1.4.3, VyOS uses Minisign exclusively. If you see an unexpected verification error, update your system to version 1.4.2 first. Support for GnuPG signatures has been removed ([T7301](#)).

3.1.3 Live installation

Note: To permanently install VyOS, you must first complete a live installation.

You can test VyOS without installing it on your hard drive. **Using your downloaded VyOS .iso file, you can create a bootable USB drive to boot into a fully functional VyOS system.** After testing it, you can start a *Permanent installation* on your hard drive or power off your system and remove the USB drive.

If you have a GNU/Linux system, you can create a bootable VyOS USB drive using the `dd` command:

1. Open your terminal emulator.
2. Find the device name of your USB drive (use the `lsblk` command).
3. Unmount the USB drive. Replace `X` with your device letter and keep the asterisk (*) to unmount all partitions.

```
$ umount /dev/sdX*
```

1. Write the image (your VyOS .iso file) to the USB drive. Use the device name (for example, `/dev/sdb`), not the partition name (for example, `/dev/sdb1`).

Warning: This will destroy all data on the USB drive!

```
# dd if=/path/to/vyos.iso of=/dev/sdX bs=8M; sync
```

1. Wait for the operation to complete (bytes copied). On some systems, this may take more than one minute.
2. Once `dd` has finished, pull the USB drive out and plug it into the powered-off computer where you want to install (or test) VyOS.
3. Power on the computer and ensure it boots from the USB drive (you may need to select the boot device or change boot settings).
4. When VyOS finishes loading, sign in using the default credentials (login: `vyos`, password: `vyos`).

If you encounter issues with this method, prefer a different operating system, or want a GUI program, you can use other tools to create a bootable USB drive, such as [balenaEtcher](#) (GNU/Linux, macOS, and Windows), [Rufus](#) (Windows), and [many others](#). Follow their instructions to create a bootable USB drive from an .iso file.

Hint: The default username and password for the live system is `vyos`.

3.1.4 Permanent installation

Note: Before a permanent installation, VyOS requires a [Live installation](#).

Unlike general-purpose Linux distributions, VyOS uses “image installation”, which mimics the user experience of traditional hardware routers and allows you to keep multiple VyOS versions installed simultaneously. This lets you switch to a previous version if something breaks or misbehaves after an image upgrade.

Each version is contained in its own squashfs image mounted in a union filesystem along with a directory for mutable data such as configurations, keys, and custom scripts.

In order to proceed with a permanent installation:

1. Sign in to the VyOS live system using the default credentials (login: vyos, password: vyos).
2. Run the `install image` command and follow the wizard:

```
vyos@vyos:~$ install image
Welcome to VyOS installation!
This command will install VyOS to your permanent storage.
Would you like to continue? [y/N] y
What would you like to name this image? (Default: 2025.09.17-0018-rolling)
Please enter a password for the "vyos" user:
Please confirm password for the "vyos" user:
What console should be used by default? (K: KVM, S: Serial)? (Default: S)
Probing disks
1 disk(s) found
The following disks were found:
Drive: /dev/vda (10.0 GB)
Which one should be used for installation? (Default: /dev/vda)
Installation will delete all data on the drive. Continue? [y/N] y
Searching for data from previous installations
No previous installation found
Would you like to use all the free space on the drive? [Y/n] Y
Creating partition table...
The following config files are available for boot:
    1: /opt/vyatta/etc/config/config.boot
    2: /opt/vyatta/etc/config.config.boot.default
Which file would you like as boot config? (Default: 1)
Creating temporary directories
Mounting new partitions
Creating a configuration file
Copying system image files
Installing GRUB configuration files
Installing GRUB to the drive
Cleaning up
Unmounting target filesystems
Removing temporary files
The image installed successfully; please reboot now.
```

3. After installation completes, remove the live USB drive or CD.
4. Reboot the system.

```
vyos@vyos:~$ reboot
Proceed with reboot? (Yes/No) [No] Yes
```

You will boot now into a permanent VyOS system.

3.1.5 PXE Boot

You can also install VyOS using PXE, a more complex installation method that allows you to deploy VyOS over the network.

Requirements

- A machine (client) with a PXE-enabled NIC.
- *DHCP Server*
- *TFTP server*
- Webserver (HTTP). Optional, but speeds up installation.
- VyOS ISO image (do not use images prior to VyOS 1.2.3).
- Files *pxelinux.0* and *ldlinux.c32* from the [Syslinux distribution](#).

Configuration

Step 1: DHCP

Configure a DHCP server to provide the client with:

- An IP address
- The TFTP server address (DHCP option 66), sometimes referred to as the *boot server*
- The *bootfile name* (DHCP option 67): *pxelinux.0*

In this example we configured an existent VyOS as the DHCP server:

```
vyos@vyos# show service dhcp-server
shared-network-name mydhcp {
  subnet 192.168.1.0/24 {
    option {
      bootfile-name pxelinux.0
      bootfile-server 192.168.1.50
      default-router 192.168.1.50
    }
    range 0 {
      start 192.168.1.70
      stop 192.168.1.100
    }
    subnet-id 1
  }
}
```

Step 2: TFTP

Configure a TFTP server to serve the following:

- The *pxelinux.0* file from the Syslinux distribution
- The *ldlinux.c32* file from the Syslinux distribution
- The VyOS kernel you want to deploy (*vmlinuz* file from the */live* directory in the extracted ISO file)
- The VyOS initial ramdisk (*initrd.img* file from the */live* directory in the extracted ISO file). Do not use an empty (0 bytes) *initrd.img* file; the correct file may have a longer name.
- A directory named *pxelinux.cfg* containing the configuration file. By default, the VyOS configuration file is named *default*.

In the example you configured your existent VyOS as the TFTP server too:

```
vyos@vyos# show service tftp-server
directory /config/tftpboot
listen-address 192.168.1.50
```

Example of the contents of the TFTP server:

```
vyos@vyos# ls -hal /config/tftpboot/
total 29M
drwxr-sr-x 3 tftp tftp      4.0K Oct 14 00:23 .
drwxrwsr-x 9 root vyattacfg 4.0K Oct 18 00:05 ..
-r--r--r-- 1 root vyattacfg 25M Oct 13 23:24 initrd.img-4.19.54-amd64-vyos
-rwxr-xr-x 1 root vyattacfg 120K Oct 13 23:44 ldlinux.c32
-rw-r--r-- 1 root vyattacfg 46K Oct 13 23:24 pxelinux.0
drwxr-xr-x 2 root vyattacfg 4.0K Oct 14 01:10 pxelinux.cfg
-r--r--r-- 1 root vyattacfg 3.7M Oct 13 23:24 vmlinuz

vyos@vyos# ls -hal /config/tftpboot/pxelinux.cfg
total 12K
drwxr-xr-x 2 root vyattacfg 4.0K Oct 14 01:10 .
drwxr-sr-x 3 tftp tftp      4.0K Oct 14 00:23 ..
-rw-r--r-- 1 root root      191 Oct 14 01:10 default
```

Example of simple (no menu) configuration file:

```
vyos@vyos# cat /config/tftpboot/pxelinux.cfg/default
DEFAULT VyOS123

LABEL VyOS123
  KERNEL vmlinuz
  APPEND initrd=initrd.img-4.19.54-amd64-vyos boot=live nopersistence noautologin_
  ↪nonetworking fetch=http://address:8000/filesystem.squashfs
```

Step 3: HTTP

You also need to provide the *filesystem.squashfs* file. Because this is a large file and TFTP is slow, you can send it through HTTP to speed up the transfer. In our example, we do this—see the configuration file above.

1. Start a web server. You can use one like [Python's SimpleHTTPServer](#) to serve the *filesystem.squashfs* file. The file is in the */live* directory of the extracted ISO file.
2. Edit the *Step 2: TFTP* configuration file to show the correct URL: `fetch=http://<address_of_your_HTTP_server>/filesystem.squashfs`.

Note: Do not rename the *filesystem.squashfs* file. If you're working with different versions, create different directories instead.

3. restart the TFTP service. If you're using VyOS as your TFTP server, restart the service with `sudo service tftpd-hpa restart`.

Note: Ensure the directories and files on both the TFTP and HTTP servers have the correct permissions for the booting clients to access them.

Client Boot

Finally, power on your PXE-enabled clients. They will automatically receive an IP address from the DHCP server and boot into VyOS live using files from the TFTP and HTTP servers.

Once finished you will be able to proceed with the `install image` command as in a regular VyOS installation.

3.1.6 Known Issues

This is a list of known issues that can arise during installation.

Black screen on install

GRUB redirects all output to a serial port to facilitate installation on headless hosts. On some hardware that lacks a serial port, this causes a hard lockup and displays a black screen after you select the `Live system` option from the installation image.

The workaround is to press `e` when the boot menu appears and edit the GRUB boot options. Specifically, remove the:

```
console=ttyS0,115200
```

option, and type `CTRL-X` to boot.

Installation can then continue as outlined above.

3.2 Virtual Environments

3.2.1 Run VyOS on Libvirt QEMU/KVM

Libvirt is an open-source API, daemon, and management tool for managing platform virtualization. You can deploy VyOS on libvirt KVM in several ways: using Virt-Manager or the native CLI. This example uses 4 gigabytes of memory, 2 CPU cores, and the default network `virbr0`.

CLI

Deploy from ISO

Create VM name `vyos_r1`. You must specify the path to the ISO image, the disk `qcow2` will be created automatically. The default network is the virtual network (type `Virtio`) created by the hypervisor with NAT.

```
$ virt-install -n vyos_r1 \
  --ram 4096 \
  --vcpus 2 \
  --cdrom /var/lib/libvirt/images/vyos.iso \
  --os-variant debian10 \
  --network network=default \
  --graphics vnc \
  --hvm \
  --virt-type kvm \
  --disk path=/var/lib/libvirt/images/vyos_r1.qcow2,bus=virtio,size=8 \
  --noautoconsole
```

Connect to the VM with the command `virsh console vyos_r1`

```
$ virsh console vyos_r1

Connected to domain vyos_r1
Escape character is ^]

vyos login: vyos
Password:

vyos@vyos:~$ install image
```

After installation, exit the console using the key combination `Ctrl + ]` and reboot the system.

Deploy from qcow2

The benefit of using KVM (Kernel-based Virtual Machine) images is that they don't require installation. Download the predefined VyOS `.qcow2` image.

```
curl --url link_to_vyos_kvm.qcow2 --output /var/lib/libvirt/images/vyos_kvm.qcow2
```

Create VM with `import qcow2` disk option.


```
$ virt-install -n vyos_r2 \
  --ram 4096 \
  --vcpus 2 \
  --os-variant debian10 \
  --network network=default \
  --graphics vnc \
  --hvm \
  --virt-type kvm \
  --disk path=/var/lib/libvirt/images/vyos_kvm.qcow2,bus=virtio \
  --import \
  --noautoconsole
```

Connect to the VM with the command `virsh console vyos_r2`

```
$ virsh console vyos_r2

Connected to domain vyos_r2
Escape character is ^]

vyos login: vyos
Password:

vyos@vyos:~$
```

If you cannot access the login screen, the KVM console may be set as the default boot option.

Open a secondary session and run this command to reboot the VM:

```
$ virsh reboot vyos_r2
```

Then go to the first session where you opened the console. Select VyOS 1.4.x for QEMU (Serial console) and press **Enter**.

The system is fully operational.

Virt-Manager

The Virt-Manager application is a desktop user interface for managing virtual machines through libvirt. On Linux, open the VMM (Virtual Machine Manager).

Deploy from ISO

1. Open VMM and create a new VM (Virtual Machine)
2. Choose Local install media (ISO)
3. Choose the path to the VyOS ISO image. Select any Debian-based operating system.
4. Choose Memory and CPU
5. Disk size

6. Name of VM and network selection
7. Then the system will be taken to the console.

Deploy from qcow2

Download the predefined VyOS .qcow2 image.

```
curl --url link_to_vyos_kvm.qcow2 --output /var/lib/libvirt/images/vyos_kvm.qcow2
```

1. Open VMM and create a new VM
2. Choose **Import existing disk image**
3. Choose the path to the vyos_kvm.qcow2 image that you downloaded. Select any Debian-based operating system.
4. Choose Memory and CPU
5. Name of VM and network selection
6. Then the system will be taken to the console.

3.2.2 Running on Proxmox

Proxmox is an open-source platform for virtualization.

Deploy VyOS from CLI with qcow2 image

1. Download the .qcow2 image from <https://support.vyos.io/>. Official images are available to users with a valid subscription.
2. Copy the .qcow2 image to a temporary directory on the Proxmox server.
3. The following commands assume that virtual machine (VM) ID 200 is unused and that the imported disk will be stored in a storage pool named local-lvm.

```
$ qm create 200 --name vyos --memory 4096 --net0 virtio,bridge=vbr0
$ qm importdisk 200 /var/lib/vz/images/vyos-<version>-proxmox-amd64.qcow2
↪ local-lvm
$ qm set 200 --virtio0 local-lvm:vm-200-disk-0
$ qm set 200 --boot order=virtio0
```

4. When using a qcow2 image on Proxmox, the system **does not include any preconfigured user accounts**. You must define a user account using **Cloud-Init** before the first boot. Otherwise, login access is not possible.

Attach a Cloud-Init data source to the VM. For example, using local-lvm storage:

```
$ qm set 200 --ide2 local-lvm:cloudinit
```

Alternatively, add a Cloud-Init drive using the Proxmox GUI:

1. Open the VM and navigate to **Hardware**

2. Click **Add** → **CloudInit Drive**
3. Select a storage (for example, `local-lvm`)
4. Click **Add**
5. Start the virtual machine using the Proxmox GUI or by running `qm start 200`.

Deploy VyOS from CLI with rolling release ISO

1. Download the rolling release ISO from <https://vyos.net/get/nightly-builds/>.
2. Prepare the VM for ISO installation. The commands below assume that the ISO image is available in the `local` storage, a VM ID `200` is unused, and a 15GB disk will be created on storage pool `local-lvm`.

```
qm create 200 --name vyos --memory 4096 \  
--net0 virtio,bridge=vbr0 \  
--scsihw virtio-scsi-pci \  
--scsi0 local-lvm:15 \  
--ide2 local:iso/vyos-<version>.iso,media=cdrom \  
--boot order=ide2
```

3. Start the VM using `qm start 200` or by clicking the **Start** button in the Proxmox GUI.
4. In the Proxmox GUI, open the virtual console for your new VM. The login username and password are `vyos/vyos`.
5. After booting into the live system, type `install image` and follow the prompts to install VyOS to the virtual drive.
6. After installation completes, remove the installation ISO using the GUI or run `qm set 200 --ide2 none`, then set the boot device with `qm set 200 --boot order=scsi0`.
7. Reboot the virtual machine using the GUI or run `qm reboot 200`.

For more information about downloading and installing Proxmox, visit <https://www.proxmox.com/en/>.

3.2.3 Running on VMware ESXi

ESXi 5.5 or later

.ova files are available for supporting users. You can also set up VyOS using a generic Linux instance by attaching the bootable ISO file and installing using the `install image` command.

Note: Previous issues have been documented with GRE/IPSEC tunneling using the E1000 adapter on VyOS guests. Use the VMXNET3 adapter instead.

Memory Contention Considerations

When the underlying ESXi host reaches approximately 92% memory utilization, it begins the balloon process to reclaim memory from guest operating systems. This creates artificial memory pressure through the `vmmemctl` driver. Because VyOS does not have a swap file by default, this pressure cannot move memory data to a paging file. Instead, it consumes memory and forces the guest into a low memory state with no recovery option. The balloon can expand to 65% of guest allocated memory, so a VyOS guest using more than 35% of memory can encounter an out-of-memory situation and trigger the kernel `oom_kill` process. The `oom_kill` process then terminates memory-hungry processes.

To prevent ballooning, configure VyOS routers in a resource group with adequate memory reservations.

References

<https://muralidba.blogspot.com/2018/03/how-does-linux-out-of-memory-oom-killer.html>

3.2.4 Run VyOS on GNS3

You may want to test VyOS in a lab environment. [GNS3](#) is a network emulation software that you can use for this purpose.

This guide will provide the necessary steps for installing and setting up VyOS on GNS3.

Requirements

The following items are required:

- A VyOS installation image (.iso file). You can find how to get it on the [Installation](#) page
- A working GNS3 installation. For further information see the [GNS3 documentation](#).

VM setup

First, a virtual machine (VM) for the VyOS installation must be created in GNS3.

Go to the GNS3 **File** menu, click **New template**, and select **Manually create a new Template**.

Select **Qemu VMs** and then click the **New** button.

Write a name for your VM, such as “VyOS”, and click **Next**.

Select **qemu-system-x86_64** as Qemu binary, then **512MB** of RAM and click **Next**.

Select **telnet** as your console type and click **Next**.

Select **New image** for the base disk image of your VM and click **Create**.

Use the defaults in the **Binary and format** window and click **Next**.

Use the defaults in the **Qcow2 options** window and click **Next**.

Set the disk size to 2000 MiB, and click **Finish** to end the **Qemu image creator**.

Click **Finish** to end the **New QEMU VM template** wizard.

Now you need to edit the VM settings.

In the **Preferences** window, with **Qemu VMs** selected and your new VM selected, click the **Edit** button.

In the **General settings** tab of your **QEMU VM template configuration**, do the following:

- Click on the **Browse . . .** button to choose the **Symbol** you want to have representing your VM.
- In **Category** select in which group you want to find your VM.
- Set the **Boot priority** to **CD/DVD-ROM**.

At the **HDD** tab, change the Disk interface to **sata** to speed up the boot process.

At the **CD/DVD** tab click on **Browse . . .** and locate the VyOS image you want to install.

Note: You probably will want to accept to copy the .iso file to your default image directory when you are asked.

In the **Network** tab, set the number of adapters to **0**, set the **Name format** to **eth{0}**, and set the **Type** to **Paravirtualized Network I/O (virtio-net-pci)**.

In the **Advanced** tab, unmark the checkbox **Use as a linked base VM** and click **OK**, which will save and close the **QEMU VM template configuration** window.

At the general **Preferences** window, click **OK** to save and close.

VyOS installation

- Create a new project.
- Drag the newly created VyOS VM into it.
- Start the VM.
- Open a console. The console displays the system booting. It prompts for login credentials. You're now at the VyOS live system.
- *Install VyOS* as normal (that is, using the `install imageOS` command).
- After successful installation, shut down the VM with the `poweroff` command.
- **Delete the VM** from the GNS3 project.

The *VyOS-hda.qcow2* file now contains a working VyOS image and can be used as a template. But it still needs some fixes before we can deploy VyOS in our labs.

VyOS VM configuration

To turn the template into a working VyOS machine, further steps are necessary as outlined below:

General settings tab: Set the boot priority to **HDD**

CD/DVD tab: Clear the **Image** entry field to unmount the installation image.

Set the number of required network adapters. For example, set it to **4**.

Advanced settings tab: Check the **Use as a linked base VM** checkbox and click OK to save the changes.

The VyOS VM is now ready to be deployed.

3.2.5 EVE-NG

Note: This page is a stub and needs expansion. Contributions welcome via the [VyOS documentation repository](#).

References

<https://www.eve-ng.net/>

3.2.6 Run VyOS in a Docker Container

Docker is an open-source project for deploying applications as standardized units called containers. Deploying VyOS in a container provides a simple and lightweight mechanism for both testing and packet routing for container workloads.

IPv6 support for Docker

VyOS requires an IPv6-enabled Docker network. Currently Linux distributions do not enable Docker IPv6 support by default. You can enable IPv6 support in two ways.

Method 1: Create a docker network with IPv6 support

Here's an example using the macvlan driver.

```
docker network create --ipv6 -d macvlan -o parent=eth0 --subnet 2001:db8::/64 --subnet_
↪ 192.0.2.0/24 mynet
```

Method 2: Add IPv6 support to the Docker daemon

Edit `/etc/docker/daemon.json` to set the `ipv6` key to `true` and specify the `fixed-cidr-v6` to your desired IPv6 subnet.

```
{
  "ipv6": true,
  "fixed-cidr-v6": "2001:db8::/64"
}
```

Reload the Docker configuration.

```
$ sudo systemctl reload docker
```

Deploy container from ISO

Download the ISO you want to base the container on. In this example, the ISO is `vyos-1.4-rolling-202308240020-amd64.iso`. If you created a custom IPv6-enabled network, include it as the `--net` parameter to `docker run`.

```
$ mkdir vyos && cd vyos
$ curl -o vyos-1.4-rolling-202308240020-amd64.iso https://github.com/vyos/vyos-rolling-
↳ nightly-builds/releases/download/1.4-rolling-202308240020/vyos-1.4-rolling-
↳ 202308240020-amd64.iso
$ mkdir rootfs
$ sudo mount -o loop vyos-1.4-rolling-202308240020-amd64.iso rootfs
$ sudo apt-get install -y squashfs-tools
$ mkdir unsquashfs
$ sudo unsquashfs -f -d unsquashfs/ rootfs/live/filesystem.squashfs
$ sudo tar -C unsquashfs -c . | docker import - vyos:1.4-rolling-202111281249
$ sudo umount rootfs
$ cd ..
$ sudo rm -rf vyos
$ docker run -d --rm --name vyos --privileged -v /lib/modules:/lib/modules \
> vyos:1.4-rolling-202111281249 /sbin/init
$ docker exec -ti vyos su - vyos
```

To stop the container, run `docker stop vyos`.

3.3 Cloud Environments

3.3.1 Amazon AWS

Deploy VM

Deploy VyOS on Amazon AWS (Amazon Web Services).

1. Click **Instances** and then click **Launch Instance**.

2. Search for “VyOS” in the Marketplace.
3. Choose the instance type. The recommended minimum is `m3.medium`.
4. Configure the instance for your requirements. Select the number of instances, network, and subnet.
5. Configure additional storage. You can remove the additional storage `/dev/sdb`. The root device will be `/dev/xvda`. You can skip this step.
6. Configure the security group. We recommend configuring SSH access only from specific sources, or you can permit any IP address (the default).
7. Select the SSH key pair and click **Launch Instances**.
8. Note your public IP address.
9. Connect to the instance using your SSH key.

```
ssh -i ~/.ssh/amazon.pem vyos@203.0.113.3
vyos@ip-192-0-2-10:~$
```

Amazon CloudWatch Agent Usage

To use the Amazon CloudWatch agent, configure it in the Amazon Systems Manager Parameter Store. For instructions on creating a configuration, see [CloudWatch SSM Configuration creation](#).

1. Create an IAM (Identity and Access Management) role for the EC2 (Elastic Compute Cloud) instance to access CloudWatch service, and name it `CloudWatchAgentServerRole`. The role should contain two default policies: `CloudWatchAgentServerPolicy` and `AmazonSSMManagedInstanceCore`.
2. Attach the created role to your VyOS EC2 instance.
3. Ensure the `amazon-cloudwatch-agent` package is installed.

```
$ sudo apt list --installed | grep amazon-cloudwatch-agent
```

Note: The `amazon-cloudwatch-agent` package is normally included in VyOS 1.3.3+ and 1.4+

4. Retrieve an existing CloudWatch Agent configuration from the SSM (Systems Manager) Parameter Store.

```
$ sudo /opt/aws/amazon-cloudwatch-agent/bin/amazon-cloudwatch-agent-ctl -a
↪fetch-config -m ec2 -s -c ssm:<your-configuration-name>
```

This step also enables `systemd` service and runs it.

Note: The VyOS platform-specific scripts feature is under development. Thus, this step should be repeated manually after changing system image ([Update VyOS](#))

CloudWatch SSM Configuration creation

Creating the Amazon CloudWatch Agent Configuration in Amazon SSM Parameter Store.

1. Create an IAM role for your EC2 instance to access the CloudWatch service. Name it `CloudWatchAgentAdminRole`. The role must contain at least two policies: `CloudWatchAgentAdminPolicy` and `AmazonSSMManagedInstanceCore`.

Note: `CloudWatchAgentServerRole` is too permissive and should be used only for creating and deploying a single configuration. After step 3, we recommend replacing the `CloudWatchAgentAdminRole` with the `CloudWatchAgentServerRole`.

2. Run the CloudWatch configuration wizard.

```
$ sudo /opt/aws/amazon-cloudwatch-agent/bin/amazon-cloudwatch-agent-config-
↪ wizard
```

3. When prompted, enter “yes” to the question “Do you want to store the config in the SSM parameter store?”.

AWS Gateway Load Balancer

VyOS supports the AWS Gateway Load Balancer (GWLB) tunnel handler (`gwlb-tun`), which enables VyOS to act as an inspection or processing target for GWLB. GWLB uses Geneve encapsulation with custom metadata to deliver traffic to VyOS for packet filtering, shaping, deep packet inspection, NAT, or other traffic manipulation functions. The tunnel handler automatically creates Linux tunnel interfaces (`gwi-*` for ingress and `gwo-*` for egress) per endpoint, allowing you to use standard Linux utilities like `iptables`, `tc`, and `netfilter` to implement your inspection or processing logic. This enables VyOS to serve as a centralized appliance for traffic inspection in your AWS infrastructure, supporting both single-endpoint (1-arm) and multi-endpoint (2-arm) deployment modes.

For more information about integrating with AWS Gateway Load Balancer, see the following article from AWS:

[How to integrate Linux instances with AWS Gateway Load Balancer.](#)

Configuration Example

Configure the AWS GWLB service with the following commands:

```
set service aws glb script on-create '/config/scripts/glb-create.sh'
set service aws glb script on-destroy '/config/scripts/glb-destroy.sh'
set service aws glb status format 'simple'
set service aws glb status port '8282'
set service aws glb threads tunnel '4'
set service aws glb threads tunnel-affinity '1-2'
set service aws glb threads udp '4'
set service aws glb threads udp-affinity '0-3'
```

References

- <https://console.aws.amazon.com/>
- <https://docs.aws.amazon.com/AmazonCloudWatch/latest/monitoring/create-iam-roles-for-cloudwatch-agent.html>
- <https://docs.aws.amazon.com/AmazonCloudWatch/latest/monitoring/install-CloudWatch-Agent-on-EC2-Instance-fleet.html>
- <https://aws.amazon.com/blogs/networking-and-content-delivery/how-to-integrate-linux-instances-with-aws-gateway-load-balancing/>

3.3.2 Azure

Deploy VM

Deploy VyOS on Azure.

1. Go to Azure services and click **Add new Virtual machine**.
2. Choose a VM name, resource group, and region, then click **Browse all public and private images**.
3. Search for “VyOS” in the marketplace and choose the appropriate subscription.
4. Generate new SSH key pair or use existing.
5. Configure the network, subnet, and public IP. Or use the defaults.
6. Click **Review + create**. Your deployment completes in a few seconds.
7. Select your new VM and note your public IP address.
8. Connect to the instance with your SSH key.

```
ssh -i ~/.ssh/vyos_azure vyos@203.0.113.3
vyos@vyos-doc-r1:~$
```

Add interface

If your instance was deployed with one **eth0** (WAN) interface and you want to add another, you must shut down the instance. To add a new interface, such as **eth1** (LAN), attach it in the Azure portal and then restart the instance.

Note: Azure doesn't allow you to attach an interface while the instance is running.

Absorbing Routes

If you're using the VM as a router, you can use a route table to absorb some or all traffic from your virtual network (VNET) with your LAN interface.

1. Create a route table and navigate to **Configuration**.
2. Add one or more routes for the networks you want to route through the VyOS VM. For **Next hop type**, select **Virtual Appliance** and set the **Next Hop Address** to the VyOS LAN interface.

Note: To create a default route for VMs on the subnet, use **Address Prefix 0.0.0.0/0**. For a typical edge device configuration, configure masquerade NAT on the WAN interface.

Serial Console

VyOS includes serial console support by default. However, if you replace the `config.boot` file and reboot, ensure this configuration is present:

```
set system console device ttyS0 speed '9600'
```

References

<https://azure.microsoft.com>

3.3.3 Google Cloud Platform

Deploy VM

To deploy VyOS on Google Cloud Platform (GCP):

1. Generate an SSH key pair of type **ssh-rsa** on the host that will connect to VyOS.

Example:

```
ssh-keygen -t rsa -f ~/.ssh/vyos_gcp -C "vyos@mypc"
```

Note: The SSH key comment must begin with `vyos@` because that's the default VyOS user. GCP uses this value to set the username on the instance.

2. Open the GCP Console and navigate to **Metadata**. Select **SSH Keys** and click **Edit**.
3. Search for "VyOS" in the Marketplace.
4. Configure the deployment name, zone, and machine type, then click **Deploy**.
5. After a few seconds, select your **instance**.
6. Note your external IP address.

7. Connect to the instance using the SSH key you generated in step 1.

```
ssh -i ~/.ssh/vyos_gcp vyos@203.0.113.3
vyos@vyos-r1-vm:~$
```

References

<https://console.cloud.google.com/>

3.3.4 Oracle

Note: This page is a stub and needs expansion. Contributions welcome via the [VyOS documentation repository](#).

References

<https://www.oracle.com/cloud/>

<https://docs.oracle.com/en/cloud/paas/developer-cloud-classic/csdc/develop-application.html>

3.4 Bare Metal Deployment

3.4.1 Supermicro A2SDi (Atom C3000)

I opted to get one of the new Intel Atom C3000 CPUs to spawn VyOS on it. Running VyOS on an UEFI only device is supported as of VyOS release 1.2.

Supermicro Shopping Cart

- 1x Supermicro CSE-505-203B (19" 1U chassis, inkl. 200W PSU)
- 1x Supermicro MCP-260-00085-0B (I/O Shield for A2SDi-2C-HLN4F)
- 1x Supermicro A2SDi-2C-HLN4F (Intel Atom C3338, 2C/2T, 4MB cache, Quad LAN with Intel C3000 SoC 1GbE)
- 1x Crucial CT4G4DFS824A (4GB DDR4 RAM 2400 MT/s, PC4-19200)
- 1x SanDisk Ultra Fit 32GB (USB-A 3.0 SDCZ43-032G-G46 mass storage for OS)
- 1x Supermicro MCP-320-81302-0B (optional FAN tray)

Optional (10GE)

If you want to get additional ethernet ports or even 10GE connectivity the following optional parts will be required:

- 1x Supermicro RSC-RR1U-E8 (Riser Card)
- 1x Supermicro MCP-120-00063-0N (Riser Card Bracket)

Latest VyOS rolling releases boot without any problem on this board. You also receive a nice IPMI interface realized with an ASPEED AST2400 BMC (no information about [OpenBMC](#) so far on this motherboard).

Pictures

3.4.2 PC Engines APU4

As this platform seems to be quite common in terms of noise, cost, power and performance it makes sense to write a small installation manual.

This guide was developed using an APU4C4 board with the following specs:

- AMD Embedded G series GX-412TC, 1 GHz quad Jaguar core with 64 bit and AES-NI support, 32K data + 32K instruction cache per core, shared 2MB L2 cache.
- 4 GB DDR3-1333 DRAM, with optional ECC support
- About 6 to 10W of 12V DC power depending on CPU load
- 2 miniPCI express (one with SIM socket for 3G modem).
- 4 Gigabit Ethernet channels using Intel i211AT NICs

The board can be powered via 12V from the front or via a 5V onboard connector.

APU4 Shopping Cart

- 1x apu4c4 = 4 i211AT LAN / AMD GX-412TC CPU / 4 GB DRAM / dual SIM
- 1x Kingston SUV500MS/120G
- 1x VARIA Group Item 326745 19" dual rack for APU4

The 19" enclosure can accommodate up to two APU4 boards - there is a single and dual front cover.

Extension Modules

WiFi

Refer to [Wireless LAN / Wi-Fi](#) for additional information, below listed modules have been tested successfully on this Hardware platform:

- Compex WLE900VX mini-PCIe WiFi module, only supported in mPCIe slot 1.
- Intel Corporation AX200 mini-PCIe WiFi module, only supported in mPCIe slot 1. (see [Intel AX200](#))

WWAN

Refer to [WWAN](#) for additional information, below listed modules have been tested successfully on this Hardware platform using VyOS 1.3 (equuleus):

- Sierra Wireless AirPrime MC7304 miniPCIe card (LTE)
- Sierra Wireless AirPrime MC7430 miniPCIe card (LTE)
- Sierra Wireless AirPrime MC7455 miniPCIe card (LTE)
- Sierra Wireless AirPrime MC7710 miniPCIe card (LTE)
- Huawei ME909u-521 miniPCIe card (LTE)

VyOS 1.4 (sagitta)

Depending on the VyOS versions you intend to install there is a difference in the serial port settings ([T1327](#)).

Create a bootable USB pendrive using e.g. [Rufus](#) on a Windows machine.

Connect serial port to a PC through null modem cable (RXD / TXD crossed over). Set terminal emulator to 115200 8N1.

```
PC Engines apu4
coreboot build 20171130
BIOS version v4.6.4
4080 MB ECC DRAM
SeaBIOS (version rel-1.11.0.1-0-g90da88d)

Press F10 key now for boot menu:

Select boot device:

1. ata0-0: KINGSTON SUV500MS120G ATA-11 Hard-Disk (111 GiBytes)
2. USB MSC Drive Generic Flash Disk 8.07
3. Payload [memtest]
4. Payload [setup]
```

Now boot from the USB MSC Drive Generic Flash Disk 8.07 media by pressing 2, the VyOS boot menu will appear, just wait 10 seconds or press **Enter** to continue.

```
lqqqqqqqqqqqqqqqqqqqqqqqqqqqqqqqqqqqqqqqqqqqqqqqqqqqqqqqqk
x                               VyOS - Boot Menu                      x
tqqqqqqqqqqqqqqqqqqqqqqqqqqqqqqqqqqqqqqqqqqqqqqqqqqqqqqqqqu
x Live system (amd64-vyos)                                           x
x Live system (amd64-vyos fail-safe mode)                           x
x Live system (amd64-vyos) - Serial console                         x
x                                                                       x
mqqqqqqPress ENAutomatic boot in 10 seconds...nu entryqqqqqqqj
```

The image will be loaded and the last lines you will get will be:

```
Loading /live/vmlinuz... ok
Loading /live/initrd.img...
...
Welcome to VyOS - vyos ttyS0

vyos login:
```

You can now proceed with a regular image installation as described in [Installation](#).

Pictures

Note: Both device types operate without any moving parts and emit zero noise.

Rack Mount

VyOS custom print

Desktop / Bench Top

3.4.3 Qotom Q355G4

The install on this Q355G4 box is pretty much plug and play. The port numbering the OS does might differ from the labels on the outside, but the UEFI firmware has a port blink test built in with MAC addresses so you can very quickly identify which is which. MAC labels are on the inside as well, and this test can be done from VyOS or plain Linux too. Default settings in the UEFI will make it boot, but depending on your installation wishes (i.e. storage type, boot type, console type) you might want to adjust them. This Qotom company seems to be the real OEM/ODM for many other relabelling companies like Protectli.

Hardware

There are a number of other options, but they all seem to be close to Intel reference designs, with added features like more serial ports, more network interfaces and the likes. Because they don't deviate too much from standard designs all the hardware is well-supported by mainline. It accepts one LPDDR3 SO-DIMM, but chances are that if you need more than that, you'll also want something even beefier than an i5. There are options for antenna holes, and SIM slots, so you could in theory add an LTE/Cell modem (not tested so far).

The chassis is a U-shaped alu extrusion with removable I/O plates and removable bottom plate. Cooling is completely passive with a heatsink on the SoC with internal and external fins, a flat interface surface, thermal pad on top of that, which then directly attaches to the chassis, which has fins as well. It comes with mounting hardware and rubber feet, so you could place it like a desktop model or mount it on a VESA mount, or even wall mount it with the provided mounting plate. The closing plate doubles as internal 2.5" mounting place for an HDD or SSD, and comes supplied with a small SATA cable and SATA power cable.

Power supply is a 12VDC barrel jack, and included switching power supply, which is why SATA power regulation is on-board. Internally it has a NUC-board-style on-board 12V input header as well, the molex locking style.

Fig. 1: Click **Add item**, paste your public SSH key, and click **Save**.

There are WDT options and auto-boot on power enable, which is great for remote setups. Firmware is reasonably secure (no backdoors found, BootGuard is enabled in enforcement mode, which is good but also means no coreboot option), yet has most options available to configure (so it's not locked out like most firmwares are).

An external RS232 serial port is available, internally a GPIO header as well. It does have Realtek based audio on board for some reason, but you can disable that. Booting works on both USB2 and USB3 ports. Switching between serial BIOS mode and HDMI BIOS mode depends on what is connected at startup; it goes into serial mode if you disconnect HDMI and plug in serial, in all other cases it's HDMI mode.

3.4.4 Partaker i5

I believe this is actually the same hardware as the Protectli. I purchased it in June 2018. It came pre-loaded with pfSense.

[Manufacturer product page](#).

Installation

- Write VyOS ISO to USB drive of some sort
- Plug in VGA, power, USB keyboard, and USB drive
- Press “SW” button on the front (this is the power button; I don't know what “SW” is supposed to mean).
- Begin rapidly pressing delete on the keyboard. The boot prompt is very quick, but with a few tries you should be able to get into the BIOS.
- Chipset > South Bridge > USB Configuration: set XHCI to Disabled and USB 2.0 (EHCI) to Enabled. Without doing this, the USB drive won't boot.
- Boot to the VyOS installer and install as usual.

Warning the interface labels on my device are backwards; the left-most “LAN4” port is eth0 and the right-most “LAN1” port is eth3.

3.4.5 Acrosser AND-J190N1

This microbox network appliance was build to create OpenVPN bridges. It can saturate a 100Mbps link. It is a small (serial console only) PC with 6 Gb LAN

You may have to add your own RAM and HDD/SSD. There is no VGA connector. But Acrosser provides a DB25 adapter for the VGA header on the motherboard (not used).

BIOS Settings:

First thing you want to do is getting a more user friendly console to configure BIOS. Default VT100 brings a lot of issues. Configure VT100+ instead.

For practical issues change speed from 115200 to 9600. 9600 is the default speed at which both linux kernel and VyOS will reconfigure the serial port when loading.

Connect to serial (115200bps). Power on the appliance and press Del in the console when requested to enter BIOS settings.

Advanced > Serial Port Console Redirection > Console Redirection Settings:

- Terminal Type : VT100+
- Bits per second : 9600

Save, reboot and change serial speed to 9600 on your client.

Some options have to be changed for VyOS to boot correctly. With XHCI enabled the installer can't access the USB key. Enable EHCI instead.

Reboot into BIOS, Chipset > South Bridge > USB Configuration:

- Disable XHCI
- Enable USB 2.0 (EHCI) Support

Perform Image installation using `install image` CLI command.

3.4.6 Gowin GW-FN-1UR1-10G

A platform utilizing an Intel Alder Lake-N100 CPU with 6M cache, TDP 6W. Onboard LPDDR5 16GB RAM and 128GB eMMC (can be used for image installation).

The appliance comes with 2 * 2.5GbE Intel I226-V and 3 * 1GbE Intel I210 where one supports IEEE802.3at PoE+ (Typical 30W).

In addition there is a Mellanox ConnectX-3 2* 10GbE SFP+ NIC available.

NOTE: This is the entry level platform. Other derivatives exist with i3-N305 CPU and 2x 25GbE!

Gowin Shopping Cart

- 1x Gowin GW-FN-1UR1-10G
- 2x 128GB M.2 NVMe SSDs

Optional (WiFi + WWAN)

- 1x MediaTek 7921E M.2 NGFF WIFI module (not tested as this currently leads to a Kernel crash)
- 1x HP LT4120 Snapdragon X5 LTE WWAN module

Pictures

Cooling

The device itself is passively cooled, whereas the power supply has an active fan. Even if the main processor is powered off, the power supply fan is operating and the entire chassis draws 7.5W. During operation the chassis drew around 38W.

BIOS Settings

No settings needed to be altered, everything worked out of the box!

Installation

The system provides a regular RS232 console port using 115200,8n1 setting which is sufficient to install VyOS from a USB pendrive.

First Boot

Please note that there is a weirdness on the network interface mapping. The interface <-> MAC mapping is going upwards but the NICs are placed somehow swapped on the mainboard/MACs programmed in a swapped order.

See interface description for more detailed mapping.

```
vyos@vyos:~$ show interfaces
Codes: S - State, L - Link, u - Up, D - Down, A - Admin Down
```

Interface	IP Address	MAC	VRF	MTU	S/L	Description
eth0	-	00:f0:cb:00:00:99	default	1500	u/D	Intel I226-V - u
↪Front eth2						
eth1	-	00:f0:cb:00:00:9a	default	1500	u/D	Intel I226-V - u
↪Front eth1						
eth2	-	00:f0:cb:00:00:9b	default	1500	u/D	Intel I210 - u
↪Front eth4						
eth3	-	00:f0:cb:00:00:9c	default	1500	u/D	Intel I210 - u
↪Front eth3						
eth4	-	00:f0:cb:00:00:9d	default	1500	u/D	Intel I210 - u
↪Front POE						
eth5	-	00:02:c9:00:00:30	default	1500	u/D	Mellanox ConnectX-
↪3 - SFP2						
eth6	-	00:02:c9:00:00:31	default	1500	u/D	Mellanox ConnectX-
↪3 - SFP1						

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```
lo          127.0.0.1/8      00:00:00:00:00:00 default 65536 u/u
           ::1/128
wan0        -              d2:39:76:8e:05:12 default 1500 A/D
```

VyOS 1.4 (sagitta)

Connect serial port to a PC through a USB <-> RJ45 console cable. Set terminal emulator to 115200 8N1. You can also perform the installation using VGA or HDMI ports.

In this example I choose to install VyOS as RAID-1 on both NVMe drives. However, a previous installation on the 128GB eMMC storage worked without any issues, too.

```
Welcome to VyOS - vynos ttyS0
vynos login:
```

Perform Image installation using `install image` CLI command. This installation uses two 128GB NVMe disks setup as RAID1.

```
Welcome to VyOS!
```

```
—
. VyOS 1.4.0
l — sagitta

* Support portal: https://support.vyos.io
* Documentation: https://docs.vyos.io/en/sagitta
* Project news:  https://blog.vyos.io
* Bug reports:   https://vyos.dev
```

You can change this banner using "set system login banner post-login" command.

VyOS is a free software distribution that includes multiple components, you can check individual component licenses under `/usr/share/doc/*/copyright`. Use of this pre-built image is governed by the EULA you can find in `/usr/share/vyos/EULA`

```
vynos@vynos:~$ install image
```

```
Welcome to VyOS installation!
This command will install VyOS to your permanent storage.
Would you like to continue? [y/N] y
```

```
What would you like to name this image? (Default: 1.4.0)
```

```
Please enter a password for the "vynos" user:
Please confirm password for the "vynos" user:
```

(continues on next page)

(continued from previous page)

What console should be used by default? (K: KVM, S: Serial)? (Default: S)

Probing disks

4 disk(s) found

Would you like to configure RAID-1 mirroring? [Y/n] y

The following disks were found:

/dev/sda (14.4 GB)

/dev/mmcblk0 (116.5 GB)

Would you like to configure RAID-1 mirroring on them? [Y/n] n

Would you like to choose two disks for RAID-1 mirroring? [Y/n] y

Disks available:

1: /dev/sda (14.4 GB)

2: /dev/mmcblk0 (116.5 GB)

3: /dev/nvme1n1 (119.2 GB)

4: /dev/nvme0n1 (119.2 GB)

Select first disk: 3

Remaining disks:

1: /dev/sda (14.4 GB)

2: /dev/mmcblk0 (116.5 GB)

3: /dev/nvme0n1 (119.2 GB)

Select second disk: 3

Installation will delete all data on both drives. Continue? [y/N] y

Searching for data from previous installations

No previous installation found

Creating partitions on /dev/nvme1n1

Creating partition table...

Creating partitions on /dev/nvme0n1

Creating partition table...

Creating RAID array

Updating initramfs

Creating filesystem on RAID array

The following config files are available for boot:

1: /opt/vyatta/etc/config/config.boot

2: /opt/vyatta/etc/config/boot.default

Which file would you like as boot config? (Default: 1)

Creating temporary directories

Mounting new partitions

Creating a configuration file

Copying system image files

Installing GRUB configuration files

Installing GRUB to the drives

Cleaning up

(continues on next page)

(continued from previous page)

```

Unmounting target filesystems
Removing temporary files
The image installed successfully; please reboot now.

```

Hardware

```

vyos@vyos:~$ lspci
00:00.0 Host bridge: Intel Corporation Device 461c
00:02.0 VGA compatible controller: Intel Corporation Alder Lake-N [UHD Graphics]
00:0a.0 Signal processing controller: Intel Corporation Platform Monitoring Technology
↳ (rev 01)
00:0d.0 USB controller: Intel Corporation Device 464e
00:14.0 USB controller: Intel Corporation Device 54ed
00:14.2 RAM memory: Intel Corporation Device 54ef
00:15.0 Serial bus controller: Intel Corporation Device 54e8
00:16.0 Communication controller: Intel Corporation Device 54e0
00:1a.0 SD Host controller: Intel Corporation Device 54c4
00:1c.0 PCI bridge: Intel Corporation Device 54b8
00:1c.2 PCI bridge: Intel Corporation Device 54ba
00:1c.3 PCI bridge: Intel Corporation Device 54bb
00:1c.6 PCI bridge: Intel Corporation Device 54be
00:1d.0 PCI bridge: Intel Corporation Device 54b0
00:1f.0 ISA bridge: Intel Corporation Device 5481
00:1f.4 SMBus: Intel Corporation Device 54a3
00:1f.5 Serial bus controller: Intel Corporation Device 54a4
01:00.0 PCI bridge: ASMedia Technology Inc. Device 1806 (rev 01)
02:00.0 PCI bridge: ASMedia Technology Inc. Device 1806 (rev 01)
02:02.0 PCI bridge: ASMedia Technology Inc. Device 1806 (rev 01)
02:06.0 PCI bridge: ASMedia Technology Inc. Device 1806 (rev 01)
02:0e.0 PCI bridge: ASMedia Technology Inc. Device 1806 (rev 01)
03:00.0 Ethernet controller: Intel Corporation Ethernet Controller I226-V (rev 04)
04:00.0 Ethernet controller: Intel Corporation Ethernet Controller I226-V (rev 04)
05:00.0 Network controller: MEDIATEK Corp. MT7922 802.11ax PCI Express Wireless Network
↳ Adapter
06:00.0 SATA controller: ASMedia Technology Inc. Device 0622 (rev 01)
07:00.0 PCI bridge: ASMedia Technology Inc. Device 1806 (rev 01)
08:00.0 PCI bridge: ASMedia Technology Inc. Device 1806 (rev 01)
08:02.0 PCI bridge: ASMedia Technology Inc. Device 1806 (rev 01)
08:06.0 PCI bridge: ASMedia Technology Inc. Device 1806 (rev 01)
08:0e.0 PCI bridge: ASMedia Technology Inc. Device 1806 (rev 01)
09:00.0 Ethernet controller: Intel Corporation I210 Gigabit Network Connection (rev 03)
0a:00.0 Ethernet controller: Intel Corporation I210 Gigabit Network Connection (rev 03)
0b:00.0 Ethernet controller: Intel Corporation I210 Gigabit Network Connection (rev 03)
0d:00.0 Non-Volatile memory controller: Device 1ed0:2283
0f:00.0 Non-Volatile memory controller: Device 1ed0:2283
11:00.0 Ethernet controller: Mellanox Technologies MT27500 Family [ConnectX-3]

```

```
vyos@vyos:~$ lsusb
Bus 004 Device 001: ID 1d6b:0003 Linux Foundation 3.0 root hub
Bus 003 Device 005: ID 0e8d:c616 MediaTek Inc. Wireless_Device
Bus 003 Device 003: ID 413c:2113 Dell Computer Corp. KB216 Wired Keyboard
Bus 003 Device 004: ID 03f0:9d1d HP, Inc HP 1t4120 Snapdragon X5 LTE
Bus 003 Device 002: ID 05e3:0610 Genesys Logic, Inc. Hub
Bus 003 Device 001: ID 1d6b:0002 Linux Foundation 2.0 root hub
Bus 002 Device 002: ID 05e3:0620 Genesys Logic, Inc. GL3523 Hub
Bus 002 Device 001: ID 1d6b:0003 Linux Foundation 3.0 root hub
Bus 001 Device 001: ID 1d6b:0002 Linux Foundation 2.0 root hub
```

WWAN

The LTE module can be enabled as simple as this config snippet:

```
interfaces {
    wwan wwan0 {
        address "dhcp"
        apn "YOUR-APN-GOES-HERE"
    }
}
```

For more information please refer to chapter: [WWAN](#)

3.5 Update VyOS

New system images can be added using the `add system image` command. This command extracts the image and prompts you to use the current system configuration and SSH security keys, allowing the new image to boot with your current configuration.

Note: Only LTS releases are PGP-signed.

add system image <url | path> | [latest] [vrf name] [username user [password pass]]

Use this command to install a new system image. You can retrieve the image from the web (<http://>, <https://> /) or from your local system. For example: `/tmp/vyos-1.2.3-amd64.iso`.

The `add system image` command also supports installing new VyOS versions through an optional VRF. If the URL requires authentication, you can specify an optional username and password on the command line, which will be passed as “Basic-Auth” to the server.

If there isn’t enough free disk space, the installation will be canceled. To delete images, use the `delete system image` command.

VyOS associates configuration with each image, and each image has its own unique configuration copy. This differs from traditional network routers where the configuration is shared across all images.

Note: If you have personal files such as scripts that you want to preserve during the upgrade, store them in `/config` since this directory is always copied to newly installed images.

You can access files from a previous installation and copy them to your current image if they were stored in the `/config` directory. Use the `copy` command to do this. For example, to copy `/config/config.boot` from the VyOS 1.2.1 image, run:

```
copy file 1.2.1://config/config.boot to /tmp/config.boot.1.2.1
```

3.5.1 Example

```
vyos@vyos:~$ add system image https://s3.amazonaws.com/s3-us.vyos.io/rolling/current/
↪vyos-1.4-rolling-202201120317-amd64.iso
Trying to fetch ISO file from https://s3.amazonaws.com/s3-us.vyos.io/rolling/current/
↪vyos-1.4-rolling-202201120317-amd64.iso
  % Total      % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left   Speed
100 338M 100 338M    0     0 3837k      0  0:01:30  0:01:30 --:--:-- 3929k
ISO download succeeded.
Checking for digital signature file...
  % Total      % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left   Speed
   0     0    0     0    0     0   0      0  --:--:-- --:--:-- --:--:--    0
curl: (22) The requested URL returned error: 404 Not Found

Unable to fetch digital signature file.
Do you want to continue without signature check? (yes/no) [yes]
Checking MD5 checksums of files on the ISO image...OK.
Done!

What would you like to name this image? [vyos-1.3-rolling-201912201452]:

OK. This image will be named: vyos-1.3-rolling-201912201452
```

You can use `latest` option. It loads the latest available Rolling release.

```
vyos@vyos:~$ add system image latest
```

Note: To use the `latest` option, “system update-check url” must be configured appropriately for your installed release.

For updates to the Rolling Release for AMD64, the following URL may be used:

<https://raw.githubusercontent.com/vyos/vyos-nightly-build/rolling/version.json>

Hint: You can access the latest Rolling Release for AMD64 from a web browser at:

<https://vyos.net/get/nightly-builds/>

After rebooting, verify the version you're running using the `show version` command.

3.6 Image Management

VyOS uses an image-based installation that creates a directory for each image on the storage device you select during installation.

The boot device has the following directory structure:

```
/
/boot
/boot/grub
/boot/2025.07.16-0020-rolling.squashfs
```

The image directory contains the system kernel, a compressed root filesystem image, and a directory for persistent storage (such as configuration). During boot, the system extracts the OS image into memory and mounts the appropriate live-rw subdirectories to provide persistent storage for system configuration.

This process ensures that the system always boots to a known working state, since the OS image is fixed and non-persistent. You can also install multiple VyOS releases on the same storage device. You can manually select the image at boot if needed, but the system boots the default image by default.

show system image

List all available system images which can be booted on the current system.

```
vyos@vyos:~$ show system image
Name                Default boot    Running
-----
2025.07.16-0020-rolling  Yes            Yes
1.4.1
1.4.0
```

delete system image [image-name]

Delete unused images from the system. You can specify an optional image name to delete. Use the `show system image` command to list available images.

```
vyos@vyos:~$ delete system image
The following images are installed:
    1: 2025.07.16-0020-rolling (running) (default boot)
    2: 1.4.1
    3: 1.4.0
Select an image to delete: 3
Do you really want to delete the image 1.4.0? [y/N] y
The image "1.4.0" was successfully deleted
```

show version

Show current system image version.

```
vyos@vyos:~$ show version
Version:      VyOS 2025.07.16-0020-rolling
Release train: current
Release flavor: generic

Built by:      autobuild@vyos.net
Built on:      Wed 16 Jul 2025 00:21 UTC
Build UUID:    20d432ee-6d55-4ebc-8462-46fe836246c9
Build Commit ID: f7ce0d8a692f2d

Architecture: x86_64
Boot via:      installed image
System type:   KVM guest
Secure Boot:   n/a (BIOS)

Hardware vendor: QEMU
Hardware model: Standard PC (i440FX + PIIX, 1996)
Hardware S/N:
Hardware UUID: b9831d42-c1fe-b2bd-7d3d-49db9418f5c9

Copyright:     VyOS maintainers and contributors
```

3.6.1 System rollback

To roll back to a previous image, first view the available images by using the `show system image` command, then select your image with the following command:

```
set system image default-boot [image-name]
```

Select the default boot image which will be started on the next boot of the system.

Then reboot the system.

Note: VyOS automatically associates the configuration with each image, so you don't need to manage this separately. Each image has its own unique configuration copy.

If you have console access, you can also select the boot image by restarting the system and using the GRUB menu at startup.

3.7 Secure Boot

Initial UEFI Secure Boot support is available (T861). VyOS uses shim from Debian 12 (Bookworm), which is properly signed by the UEFI Secure Boot key from Microsoft.

Note: There is yet no signed version of `shim` for VyOS, thus we provide no signed image for secure boot yet. If you are interested in secure boot you can build an image on your own.

To generate a custom ISO with your own secure boot keys, run the following commands prior to your ISO image build:

```
cd vyos-build
CA_DIR="data/certificates"
SHIM_CERT_NAME="vyos-dev-2025-shim"
VYOS_KERNEL_CERT_NAME="vyos-dev-2025-linux"

openssl req -new -x509 -newkey rsa:4096 -keyout ${CA_DIR}/${SHIM_CERT_NAME}.key -out $
↪ ${CA_DIR}/${SHIM_CERT_NAME}.der \
  -outform DER -days 36500 -subj "/CN=VyOS Networks Secure Boot CA/" -nodes
openssl x509 -inform der -in ${CA_DIR}/${SHIM_CERT_NAME}.der -out ${CA_DIR}/${SHIM_CERT_
↪ NAME}.pem

openssl req -newkey rsa:4096 -sha256 -nodes -keyout ${CA_DIR}/${VYOS_KERNEL_CERT_NAME}.
↪ key \
  -out ${CA_DIR}/${VYOS_KERNEL_CERT_NAME}.csr -outform PEM -days 3650 \
  -subj "/CN=VyOS Networks Secure Boot Signer 2025 - linux/"
openssl x509 -req -in ${CA_DIR}/${VYOS_KERNEL_CERT_NAME}.csr -CA ${CA_DIR}/${SHIM_CERT_
↪ NAME}.pem \
  -CAkey ${CA_DIR}/${SHIM_CERT_NAME}.key -CAcreateserial -out ${CA_DIR}/${VYOS_KERNEL_
↪ CERT_NAME}.pem -days 3650 -sha256
```

3.7.1 Installation

As our version of `shim` is not signed by Microsoft we need to enroll the previously generated MOK (Machine Owner Key) to the system.

First, disable UEFI Secure Boot for the installation.

Proceed with the standard VyOS *installation* on your system. Instead of the final `reboot` command, enroll the MOK.

```
vyos@vyos:~$ install mok
input password:
input password again:
```

You can set the `input password` to any value you choose. You'll need this password after reboot when MOK Manager launches to permanently install the keys.

With the next reboot, MOK Manager will automatically launch

Select Enroll MOK

You can now view the key to be installed and continue with key installation.

Now you need to enter the password you defined previously.

Now reboot and re-enable UEFI Secure Boot.

VyOS will now launch in UEFI Secure Boot mode. You can verify this by running one of the following commands:

```
vyos@vyos:~$ show secure-boot
SecureBoot enabled
```

```
vyos@vyos:~$ show log kernel | match Secure
Oct 08 19:15:41 kernel: Secure boot enabled
```

```
vyos@vyos:~$ show version
Version:      VyOS 1.5-secureboot
Release train: current
Release flavor: generic

Built by:     autobuild@vyos.net
Built on:     Tue 08 Oct 2024 18:00 UTC
Build UUID:   5702ca38-e6f4-470f-b89e-ffc29baee474
Build commit ID: 9eb61d3b6cf426

Architecture: x86_64
Boot via:     installed image
System type:  KVM guest
Secure Boot:  enabled <-- UEFI secure boot indicator

Hardware vendor: QEMU
Hardware model: Standard PC (i440FX + PIIX, 1996)
Hardware S/N:
Hardware UUID: 1f6e7f5c-fb52-4c33-96c9-782fbea36436

Copyright:    VyOS maintainers and contributors
```

3.7.2 Image Update

Note: Currently, there is no signed version of shim for VyOS. If you want Secure Boot support, you can build a custom image with your own keys.

During image installation, you install your MOK into the UEFI variables to add trust to this key. After you re-enable Secure Boot in UEFI, you can only boot into your signed image.

You can no longer boot into a CI-generated rolling release because those are not signed by a trusted party (T861 work in progress). This also means you must sign all successor builds with the same key; otherwise, you'll see this error:

```
error: bad shim signature
error: you need to load the kernel first
```

3.7.3 Linux Kernel

In addition to Secure Boot support, VyOS uses ephemeral key signing of Linux Kernel modules for an extra security layer in both Secure and non-Secure boot images.

<https://patchwork.kernel.org/project/linux-integrity/patch/20210218220011.67625-5-nayna@linux.ibm.com/>

When the CI system builds a Kernel package and required third-party modules, it generates a temporary (ephemeral) key pair for signing the modules. The public key is embedded in the Kernel binary to verify loaded modules.

After the Kernel CI build completes, the generated key is discarded, meaning we can no longer sign additional modules with that key. The Kernel configuration also includes the option `CONFIG_MODULE_SIG_FORCE=y`, which enforces signature verification for all modules. If you try to load an unsigned module, you'll get this error:

```
insmod: ERROR: could not insert module malicious.ko: Key was rejected by service
```

This prevents loading any malicious code after the image is assembled into the Kernel as a module. You can disable this behavior on custom builds if needed.

3.7.4 Troubleshoot

In most cases, if something goes wrong during system boot, you'll see this error message:

```
error: bad shim signature
error: you need to load the kernel first
```

This error means the Machine Owner Key used to sign the Kernel is not trusted by your UEFI. Install the MOK using the `install mok` command as described above.

QUICK START

This chapter will guide you on how to get up to speed quickly using your new VyOS system. It will show you a very basic configuration example that will provide a *NAT* gateway for a device with two network interfaces (*eth0* and *eth1*).

4.1 Configuration Mode

By default, VyOS is in operational mode, and the command prompt displays a `$`. To configure VyOS, you will need to enter configuration mode, resulting in the command prompt displaying a `#`, as demonstrated below:

```
vyos@vyos$ configure
vyos@vyos#
```

4.2 Commit and Save

After every configuration change, you need to apply the changes by using the following command:

```
commit
```

Once your configuration works as expected, you can save it permanently by using the following command:

```
save
```

4.3 Interface Configuration

- Your outside/WAN interface will be *eth0*. It will receive its interface address via DHCP.
- Your internal/LAN interface will be *eth1*. It will use a static IP address of `192.168.0.1/24`.

After switching to *Configuration Mode* issue the following commands:

```
set interfaces ethernet eth0 address dhcp
set interfaces ethernet eth0 description 'OUTSIDE'
set interfaces ethernet eth1 address '192.168.0.1/24'
set interfaces ethernet eth1 description 'LAN'
```

4.4 SSH Management

After switching to *Configuration Mode* issue the following commands, and your system will listen on every interface for incoming SSH connections. You might want to check the *SSH* chapter on how to listen on specific addresses only.

```
set service ssh port '22'
```

4.5 DHCP/DNS quick-start

The following settings will configure DHCP and DNS services on your internal/LAN network, where VyOS will act as the default gateway and DNS server.

- The default gateway and DNS recursor address will be 192.168.0.1/24
- The address range 192.168.0.2/24 - 192.168.0.8/24 will be reserved for static assignments
- DHCP clients will be assigned IP addresses within the range of 192.168.0.9 - 192.168.0.254 and have a domain name of internal-network
- DHCP leases will hold for one day (86400 seconds)
- VyOS will serve as a full DNS recursor, replacing the need to utilize Google, Cloudflare, or other public DNS servers (which is good for privacy)
- Only hosts from your internal/LAN network can use the DNS recursor

```
set service dhcp-server shared-network-name LAN subnet 192.168.0.0/24 option default-
↪router '192.168.0.1'
set service dhcp-server shared-network-name LAN subnet 192.168.0.0/24 option name-server
↪'192.168.0.1'
set service dhcp-server shared-network-name LAN subnet 192.168.0.0/24 option domain-name
↪'vyos.net'
set service dhcp-server shared-network-name LAN subnet 192.168.0.0/24 lease '86400'
set service dhcp-server shared-network-name LAN subnet 192.168.0.0/24 range 0 start '192.
↪168.0.9'
set service dhcp-server shared-network-name LAN subnet 192.168.0.0/24 range 0 stop '192.
↪168.0.254'
set service dhcp-server shared-network-name LAN subnet 192.168.0.0/24 subnet-id '1'

set service dns forwarding cache-size '0'
set service dns forwarding listen-address '192.168.0.1'
set service dns forwarding allow-from '192.168.0.0/24'
```

4.6 NAT

The following settings will configure *SNAT* rules for our internal/LAN network, allowing hosts to communicate through the outside/WAN network via IP masquerade.

```
set nat source rule 100 outbound-interface name 'eth0'
set nat source rule 100 source address '192.168.0.0/24'
set nat source rule 100 translation address masquerade
```

4.7 Firewall

A new firewall structure—which uses the `nftables` backend, rather than `iptables`—is available on all installations starting from VyOS 1.4-rolling-202308040557. The firewall supports creation of distinct, interlinked chains for each `Netfilter` hook and allows for more granular control over the packet filtering process.

The firewall begins with the base `filter` tables you define for each of the `forward`, `input`, and `output` `Netfilter` hooks. Each of these tables is populated with rules that are processed in order and can jump to other chains for more granular filtering.

4.7.1 Configure Firewall Groups

To make firewall configuration easier, we can create groups of interfaces, networks, addresses, ports, and domains that describe different parts of our network. We can then use them for filtering within our firewall rulesets, allowing for more concise and readable configuration.

In this case, we will create two interface groups — a `WAN` group for our interfaces connected to the public internet and a `LAN` group for the interfaces connected to our internal network. Additionally, we will create a network group, `NET-INSIDE-v4`, that contains our internal subnet.

```
set firewall group interface-group WAN interface eth0
set firewall group interface-group LAN interface eth1
set firewall group network-group NET-INSIDE-v4 network '192.168.0.0/24'
```

4.7.2 Configure Stateful Packet Filtering

With the new firewall structure, we have have a lot of flexibility in how we group and order our rules, as shown by the three alternative approaches below.

Option 1: Global State Policies

Using options defined in `set firewall global-options state-policy`, state policy rules that applies for both IPv4 and IPv6 are created. These global state policies also applies for all traffic that passes through the router (transit) and for traffic originated/destinated to/from the router itself, and will be evaluated before any other rule defined in the firewall.

Most installations would choose this option, and will contain:

```
set firewall global-options state-policy established action accept
set firewall global-options state-policy related action accept
set firewall global-options state-policy invalid action drop
```

Option 2: Common/Custom Chain

We can create a common chain for stateful connection filtering of multiple interfaces (or multiple netfilter hooks on one interface). Those individual chains can then jump to the common chain for stateful connection filtering, returning to the original chain for further rule processing if no action is taken on the packet.

The chain we will create is called `CONN_FILTER` and has three rules:

- A default action of `return`, which returns the packet back to the original chain if no action is taken.
- A rule to `accept` packets from established and related connections.
- A rule to `drop` packets from invalid connections.

```
set firewall ipv4 name CONN_FILTER default-action 'return'

set firewall ipv4 name CONN_FILTER rule 10 action 'accept'
set firewall ipv4 name CONN_FILTER rule 10 state established
set firewall ipv4 name CONN_FILTER rule 10 state related

set firewall ipv4 name CONN_FILTER rule 20 action 'drop'
set firewall ipv4 name CONN_FILTER rule 20 state invalid
```

Then, we can jump to the common chain from both the `forward` and `input` hooks as the first filtering rule in the respective chains:

```
set firewall ipv4 forward filter rule 10 action 'jump'
set firewall ipv4 forward filter rule 10 jump-target CONN_FILTER

set firewall ipv4 input filter rule 10 action 'jump'
set firewall ipv4 input filter rule 10 jump-target CONN_FILTER
```

Option 3: Per-Hook Chain

Alternatively, you can take the more traditional stateful connection filtering approach by creating rules on each base hook's chain:

```
set firewall ipv4 forward filter rule 5 action 'accept'
set firewall ipv4 forward filter rule 5 state established
set firewall ipv4 forward filter rule 5 state related
set firewall ipv4 forward filter rule 10 action 'drop'
set firewall ipv4 forward filter rule 10 state invalid

set firewall ipv4 input filter rule 5 action 'accept'
set firewall ipv4 input filter rule 5 state established
set firewall ipv4 input filter rule 5 state related
set firewall ipv4 input filter rule 10 action 'drop'
set firewall ipv4 input filter rule 10 state invalid
```


4.7.3 Block Incoming Traffic

Now that we have configured stateful connection filtering to allow traffic from established and related connections, we can block all other incoming traffic addressed to our local network.

Create a new chain (OUTSIDE-IN) which will drop all traffic that is not explicitly allowed at some point in the chain. Then, we can jump to that chain from the forward hook when traffic is coming from the WAN interface group and is addressed to our local network.

```
set firewall ipv4 name OUTSIDE-IN default-action 'drop'

set firewall ipv4 forward filter rule 100 action jump
set firewall ipv4 forward filter rule 100 jump-target OUTSIDE-IN
set firewall ipv4 forward filter rule 100 inbound-interface group WAN
set firewall ipv4 forward filter rule 100 destination group network-group NET-INSIDE-v4
```

We should also block all traffic destined to the router itself that isn't explicitly allowed at some point in the chain for the input hook. As we've already configured stateful packet filtering above, we only need to set the default action to drop:

```
set firewall ipv4 input filter default-action 'drop'
```

4.7.4 Allow Management Access

We can now configure access to the router itself, allowing SSH access from the inside/LAN network and rate limiting SSH access from the outside/WAN network.

First, create a new dedicated chain (VyOS_MANAGEMENT) for management access, which returns to the parent chain if no action is taken. Add a rule to accept traffic from the LAN interface group:

```
set firewall ipv4 name VyOS_MANAGEMENT default-action 'return'
```

Configure a rule on the input hook filter to jump to the VyOS_MANAGEMENT chain when new connections are addressed to port 22 (SSH) on the router itself:

```
set firewall ipv4 input filter rule 20 action jump
set firewall ipv4 input filter rule 20 jump-target VyOS_MANAGEMENT
set firewall ipv4 input filter rule 20 destination port 22
set firewall ipv4 input filter rule 20 protocol tcp
```

Finally, configure the VyOS_MANAGEMENT chain to accept connection from the LAN interface group while limiting requests coming from the WAN interface group to 4 per minute:

```
set firewall ipv4 name VyOS_MANAGEMENT rule 15 action 'accept'
set firewall ipv4 name VyOS_MANAGEMENT rule 15 inbound-interface group 'LAN'

set firewall ipv4 name VyOS_MANAGEMENT rule 20 action 'drop'
set firewall ipv4 name VyOS_MANAGEMENT rule 20 recent count 4
set firewall ipv4 name VyOS_MANAGEMENT rule 20 recent time minute
set firewall ipv4 name VyOS_MANAGEMENT rule 20 state new
set firewall ipv4 name VyOS_MANAGEMENT rule 20 inbound-interface group 'WAN'

set firewall ipv4 name VyOS_MANAGEMENT rule 21 action 'accept'
```

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```
set firewall ipv4 name VyOS_MANAGEMENT rule 21 state new
set firewall ipv4 name VyOS_MANAGEMENT rule 21 inbound-interface group 'WAN'
```

4.7.5 Allow Access to Services

Here we're allowing the router to respond to pings. Then, we can allow access to the DNS recursor we configured earlier, accepting traffic bound for port 53 from all hosts on the NET-INSIDE-v4 network:

```
set firewall ipv4 input filter rule 30 action 'accept'
set firewall ipv4 input filter rule 30 icmp type-name 'echo-request'
set firewall ipv4 input filter rule 30 protocol 'icmp'
set firewall ipv4 input filter rule 30 state new

set firewall ipv4 input filter rule 40 action 'accept'
set firewall ipv4 input filter rule 40 destination port '53'
set firewall ipv4 input filter rule 40 protocol 'tcp_udp'
set firewall ipv4 input filter rule 40 source group network-group NET-INSIDE-v4
```

Finally, we can now configure access to the services running on this router, allowing all connections coming from localhost:

```
set firewall ipv4 input filter rule 50 action 'accept'
set firewall ipv4 input filter rule 50 source address 127.0.0.0/8
```

Commit changes, save the configuration, and exit configuration mode:

```
vyos@vyos# commit
vyos@vyos# save
Saving configuration to '/config/config.boot'...
Done
vyos@vyos# exit
vyos@vyos$
```

4.8 Hardening

Especially if you are allowing SSH remote access from the outside/WAN interface, there are a few additional configuration steps that should be taken.

Replace the default vyos system user:

```
set system login user myvyosuser authentication plaintext-password mysecurepassword
```

Set up *Key-based authentication*:

```
set system login user myvyosuser authentication public-keys myusername@mydesktop type_
↪ssh-rsa
set system login user myvyosuser authentication public-keys myusername@mydesktop key_
↪contents_of_id_rsa.pub
```

Finally, try and SSH into the VyOS install as your new user. Once you have confirmed that your new user can access your router without a password, delete the original vyos user and completely disable password authentication for *SSH*:

```
delete system login user vyos
set service ssh disable-password-authentication
```

As above, commit your changes, save the configuration, and exit configuration mode:

```
vyos@vyos# commit
vyos@vyos# save
Saving configuration to '/config/config.boot'...
Done
vyos@vyos# exit
vyos@vyos$
```

You now should have a simple yet secure and functioning router to experiment with further. Enjoy!

COMMAND LINE INTERFACE

The VyOS CLI (Command-Line Interface) comprises an operational and a configuration mode.

5.1 Operational Mode

Operational mode allows for commands to perform operational system tasks and view system and service status, while configuration mode allows for the modification of system configuration.

The CLI provides a built-in help system. In the CLI the ? key may be used to display available commands. The TAB key can be used to auto-complete commands and will present the help system upon a conflict or unknown value.

For example typing `sh` followed by the TAB key will complete to `show`. Pressing TAB a second time will display the possible sub-commands of the `show` command.

```
vyos@vyos:~$ s[tab]
set      show
```

Example showing possible show commands:

```
vyos@vyos:~$ show [tab]
Possible completions:
  arp          Show Address Resolution Protocol (ARP) information
  bridge       Show bridging information
  cluster      Show clustering information
  configuration Show running configuration
  conntrack    Show conntrack entries in the conntrack table
  conntrack-sync
               Show connection syncing information
  date         Show system date and time
  dhcp         Show Dynamic Host Configuration Protocol (DHCP) information
  dhcpv6       Show status related to DHCPv6
  disk         Show status of disk device
  dns          Show Domain Name Server (DNS) information
  file         Show files for a particular image
  firewall     Show firewall information
  flow-accounting
               Show flow accounting statistics
  hardware     Show system hardware details
  history      show command history
  host         Show host information
```

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```
incoming      Show ethernet input-policy information
: q
```

You can scroll up with the keys [Shift]+[PageUp] and scroll down with [Shift]+[PageDown].

When the output of a command results in more lines than can be displayed on the terminal screen the output is paginated as indicated by a `:` prompt.

When viewing in page mode the following commands are available:

- `q` key can be used to cancel output
- `space` will scroll down one page
- `b` will scroll back one page
- `return` will scroll down one line
- `up-arrow` and `down-arrow` will scroll up or down one line at a time respectively
- `left-arrow` and `right-arrow` can be used to scroll left or right in the event that the output has lines which exceed the terminal size.

5.1.1 Operational mode command families

Many operational mode commands in VyOS are placed in families such as `show`, `clear`, or `reset`. Every such family has a specific meaning to allow the user to guess how the command is going to behave — in particular, whether it will be disruptive to the system or not.

Note that this convention was not always followed with perfect consistency and some commands may still be in wrong families, so you should always check the command help and documentation if you are not sure what exactly it does.

clear

“Clear” commands are completely non-disruptive to any system operations. Generally, they can be used freely without hesitation.

Most often their purpose is to remove or reset various debug and diagnostic information such as system logs and packet counters.

Examples:

- `clear console` — clears the screen.
- `clear interfaces ethernet eth0 counters` — zeroes packet counters on `eth0`.
- `clear log` — deletes all system log entries.

reset

“Reset” commands can be locally-disruptive. They may, for example, terminate a single user session or a session with a dynamic routing protocol peer.

They should be used with caution since they may have a significant impact on a particular users in the network.

- `reset pppoe-server username jsmith` — terminate all PPPoE sessions from user `jsmith`.
- `reset bgp 192.0.2.54` — terminates the BGP session with neighbor `192.0.2.54`.
- `reset vpn ipsec site-to-site peer vpn.example.com` — terminates IPsec tunnels to `vpn.example.com`.
- `reset session tty1` — terminates the TTY user session `tty1`

restart

“Restart” operations may disrupt an entire subsystem. Most often they initiate a restart of a server process, which causes it to be unavailable for a brief period and resets all the process state.

They should be used with extreme caution.

- `restart dhcp server` — restarts the IPv4 DHCP server process (DHCP requests are not served while it is restarting).
- `restart ipsec` — restarts the IPsec process (which forces all sessions and all IPsec process state to reset).

force

“Force” commands force the system to perform an action that it might perform by itself at a later point.

Examples:

- `force arp request interface eth1 address 10.3.0.2` — send a gratuitous ARP request.
- `force root-partition-auto-resize` — grow the root filesystem to the size of the system partition (this is also done on startup, but this command can do it without a reboot).

execute

“Execute” commands are for executing various diagnostic and auxiliary actions that the system would never perform by itself.

Examples:

- `execute wake-on-lan interface <intf> host <MAC>` — send a Wake-On-LAN packet to a host.

show

“Show” commands display various system information. They may occasionally use a pager for long outputs, that you can quit by pressing the Q button. Their output is always finite, however.

Examples:

- `show system login` — displays current system users.
- `show ip route` — displays the IPv4 routing table.

monitor

“Monitor” commands initiate various monitoring operations that may output information continuously, until terminated with Ctrl-C or disabled.

Examples:

- `monitor log` — continuously outputs latest system logs.

5.2 Configuration Mode

To enter configuration mode use the `configure` command:

```
vyos@vyos:~$ configure
[edit]
vyos@vyos:~#
```

Note: Prompt changes from \$ to #. To exit configuration mode, type `exit`.

```
vyos@vyos:~# exit
exit
vyos@vyos:~$
```

See the configuration section of this document for more information on configuration mode.

CONFIGURATION OVERVIEW

VyOS makes use of a unified configuration file for the entire system's configuration: `/config/config.boot`. This allows easy template creation, backup, and replication of system configuration. A system can thus also be easily cloned by simply copying the required configuration files.

6.1 Terminology

A VyOS system has three major types of configurations:

- **Active or running configuration** is the system configuration that is loaded and currently active (used by VyOS). Any change in the configuration will have to be committed to belong to the active/running configuration.
- **Working configuration** is the one that is currently being modified in configuration mode. Changes made to the working configuration do not go into effect until the changes are committed with the `commit` command. At which time the working configuration will become the active or running configuration.
- **Saved configuration** is the one saved to a file using the `save` command. It allows you to keep safe a configuration for future uses. There can be multiple configuration files. The default or “boot” configuration is saved and loaded from the file `/config/config.boot`.

6.2 Seeing and navigating the configuration

`show configuration`

View the current active configuration, also known as the running configuration, from the operational mode.

```
vyos@vyos:~$ show configuration
interfaces {
  ethernet eth0 {
    address dhcp
    hw-id 00:53:00:00:aa:01
  }
  loopback lo {
  }
}
service {
  ssh {
    port 22
  }
}
```

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```

system {
  config-management {
    commit-revisions 20
  }
  console {
    device ttyS0 {
      speed 9600
    }
  }
  login {
    user vyos {
      authentication {
        encrypted-password *****
      }
      level admin
    }
  }
  ntp {
    server 0.pool.ntp.org {
    }
    server 1.pool.ntp.org {
    }
    server 2.pool.ntp.org {
    }
  }
  syslog {
    global {
      facility all {
        level notice
      }
      facility protocols {
        level debug
      }
    }
  }
}

```

By default, the configuration is displayed in a hierarchy like the above example, this is only one of the possible ways to display the configuration. When the configuration is generated and the device is configured, changes are added through a collection of set and delete commands.

show configuration commands

Get a collection of all the set commands required which led to the running configuration.

```

vyos@vyos:~$ show configuration commands
set interfaces ethernet eth0 address 'dhcp'
set interfaces ethernet eth0 hw-id '00:53:dd:44:3b:0f'
set interfaces loopback 'lo'
set service ssh port '22'
set system config-management commit-revisions '20'
set system console device ttyS0 speed '9600'
set system login user vyos authentication encrypted-password '$6$Vt68...QzF0'

```

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```

set system login user vyos level 'admin'
set system ntp server '0.pool.ntp.org'
set system ntp server '1.pool.ntp.org'
set system ntp server '2.pool.ntp.org'
set system syslog global facility all level 'notice'
set system syslog global facility protocols level 'debug'

```

Both these show commands should be executed when in operational mode, they do not work directly in configuration mode. There is a special way on how to *Access opmode from config mode*.

Hint: Use the `show configuration commands | strip-private` command when you want to hide private data. You may want to do so if you want to share your configuration on the [forum](#).

show configuration json

View the current active configuration in JSON format.

```

{"interfaces": {"ethernet": {"eth0": {"address": ["192.0.2.11/24", "192.0.2.35/24"],
→ "hw-id": "52:54:00:48:a0:c6"}, "eth1": {"address": ["203.0.113.1/24"], "hw-id":
→ "52:54:00:fc:50:0b"}}, "loopback": {"lo": {}}, "protocols": {"static": {"route":
→ {"0.0.0.0/0": {"next-hop": {"192.0.2.254": {}}}}}, "service": {"ssh": {"disable-
→ host-validation": {}}, "system": {"config-management": {"commit-revisions": "100
→ "}, "console": {"device": {"ttyS0": {"speed": "115200"}}}, "host-name": "r11-vyos
→ ", "login": {"user": {"vyos": {"authentication": {"encrypted-password": "$6$Vt68..
→ F0", "plaintext-password": "", "public-keys": {"vyos@vyos": {"key": "AAAAxxx=",
→ "type": "ssh-rsa"}}}}}, "name-server": ["203.0.113.254"], "ntp": {"server": {
→ "time1.vyos.net": {}, "time2.vyos.net": {}, "time3.vyos.net": {}}, "syslog": {
→ "global": {"facility": {"all": {"level": "info"}, "protocols": {"level": "debug"}}
→ }, "time-zone": "America/New_York"}}}

```

show configuration json pretty

View the current active configuration in readable JSON format.

```

{
  "interfaces": {
    "ethernet": {
      "eth0": {
        "address": [
          "192.0.2.11/24",
          "192.0.2.35/24"
        ],
        "hw-id": "52:54:00:48:a0:c6"
      },
      "eth1": {
        "address": [
          "203.0.113.1/24"
        ],
        "hw-id": "52:54:00:fc:50:0b"
      }
    },
    "loopback": {

```

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```

        "lo": {}
    },
    "protocols": {
        "static": {
            "route": {
                "0.0.0.0/0": {
                    "next-hop": {
                        "192.0.2.254": {}
                    }
                }
            }
        }
    },
    "service": {
        "ssh": {
            "disable-host-validation": {}
        }
    },
    "system": {
        "config-management": {
            "commit-revisions": "100"
        },
        "console": {
            "device": {
                "ttyS0": {
                    "speed": "115200"
                }
            }
        },
        "host-name": "r11-vyos",
        "login": {
            "user": {
                "vyos": {
                    "authentication": {
                        "encrypted-password": "$6$Vt68...F0",
                        "plaintext-password": "",
                        "public-keys": {
                            "vyos@vyos": {
                                "key": "AAAAxxx=",
                                "type": "ssh-rsa"
                            }
                        }
                    }
                }
            }
        },
        "name-server": [
            "203.0.113.254"
        ],
        "ntp": {
            "server": {

```

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```

        "time1.vyos.net": {},
        "time2.vyos.net": {},
        "time3.vyos.net": {}
    },
    "syslog": {
        "global": {
            "facility": {
                "all": {
                    "level": "info"
                },
                "protocols": {
                    "level": "debug"
                }
            }
        }
    },
    "time-zone": "America/New_York"
}

```

6.2.1 The config mode

When entering the configuration mode you are navigating inside a tree structure, to enter configuration mode enter the command `configure` when in operational mode.

```

vyos@vyos$ configure
[edit]
vyos@vyos#

```

Note: When going into configuration mode, prompt changes from `$` to `#`.

All commands executed here are relative to the configuration level you have entered. You can do everything from the top level, but commands will be quite lengthy when manually typing them.

The current hierarchy level can be changed by the `edit` command.

```

[edit]
vyos@vyos# edit interfaces ethernet eth0

[edit interfaces ethernet eth0]
vyos@vyos#

```

You are now in a sublevel relative to `interfaces ethernet eth0`, all commands executed from this point on are relative to this sublevel. Use either the `top` or `exit` command to go back to the top of the hierarchy. You can also use the `up` command to move only one level up at a time.

show

The `show` command within configuration mode will show the working configuration indicating line changes with `+` for additions, `>` for replacements and `-` for deletions. **Example:**

```
vyos@vyos:~$ configure
[edit]
vyos@vyos# show interfaces
  ethernet eth0 {
    description MY_OLD_DESCRIPTION
    disable
    hw-id 00:53:dd:44:3b:03
  }
  loopback lo {
  }
[edit]
vyos@vyos# set interfaces ethernet eth0 address dhcp
[edit]
vyos@vyos# set interfaces ethernet eth0 description MY_NEW_DESCRIPTION
[edit]
vyos@vyos# delete interfaces ethernet eth0 disable
[edit]
vyos@vyos# show interfaces
  ethernet eth0 {
+   address dhcp
>   description MY_NEW_DESCRIPTION
-   disable
    hw-id 00:53:dd:44:3b:03
  }
  loopback lo {
  }
```

It is also possible to display all set commands within configuration mode using `show | commands`

```
vyos@vyos# show interfaces ethernet eth0 | commands
set address dhcp
set hw-id 00:53:ad:44:3b:03
```

These commands are also relative to the level you are inside and only relevant configuration blocks will be displayed when entering a sub-level.

```
[edit interfaces ethernet eth0]
vyos@vyos# show
  address dhcp
  hw-id 00:53:ad:44:3b:03
```

Exiting from the configuration mode is done via the `exit` command from the top level, executing `exit` from within a sub-level takes you back to the top level.

```
[edit interfaces ethernet eth0]
vyos@vyos# exit
[edit]
vyos@vyos# exit
Warning: configuration changes have not been saved.
```

6.3 Editing the configuration

The configuration can be edited by the use of `set` and `delete` commands from within configuration mode.

set

Use this command to set the value of a parameter or to create a new element.

Configuration commands are flattened from the tree into ‘one-liner’ commands shown in `show configuration` commands from operation mode. Commands are relative to the level where they are executed and all redundant information from the current level is removed from the command entered.

```
[edit]
vyos@vyos# set interfaces ethernet eth0 address 192.0.2.100/24
```

```
[edit interfaces ethernet eth0]
vyos@vyos# set address 203.0.113.6/24
```

These two commands above are essentially the same, just executed from different levels in the hierarchy.

delete

To delete a configuration entry use the `delete` command, this also deletes all sub-levels under the current level you’ve specified in the `delete` command. Deleting an entry will also result in the element reverting back to its default value if one exists.

```
[edit interfaces ethernet eth0]
vyos@vyos# delete address 192.0.2.100/24
```

commit

Any change you do on the configuration, will not take effect until committed using the `commit` command in configuration mode.

```
vyos@vyos# commit
[edit]
vyos@vyos# exit
Warning: configuration changes have not been saved.
vyos@vyos:~$
```

Hint: You can specify a commit message with `commit comment <message>`.

save

Use this command to preserve configuration changes upon reboot. By default it is stored at `/config/config.boot`. In the case you want to store the configuration file somewhere else, you can add a local path, a SCP address, a FTP address or a TFTP address.

```
vyos@vyos# save
Saving configuration to '/config/config.boot'...
Done
```

```
vyos@vyos# save [tab]
Possible completions:
<Enter>      Save to system config file
```

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```
<file>          Save to file on local machine
scp://<user>:<passwd>@<host>:<file> Save to file on remote machine
ftp://<user>:<passwd>@<host>:<file> Save to file on remote machine
tftp://<host>:<file>          Save to file on remote machine
vyos@vyos# save tftp://192.168.0.100/vyos-test.config.boot
Saving configuration to 'tftp://192.168.0.100/vyos-test.config.boot'...
Done
```

exit [discard]

Configuration mode can not be exited while uncommitted changes exist. To exit configuration mode without applying changes, the `exit discard` command must be used.

All changes in the working config will thus be lost.

```
vyos@vyos# exit
Cannot exit: configuration modified.

Use 'exit discard' to discard the changes and exit.
[edit]
vyos@vyos# exit discard
```

commit-confirm <minutes>

Use this command to temporarily commit your changes and set the number of minutes available for confirmation. `confirm` must be entered within those minutes, otherwise the system will revert into a previous configuration. The default value is 10 minutes.

The definition of ‘revert’ and ‘a previous configuration’ depends on the setting:

```
vyos@vyos# set system config-management commit-confirm action
Possible completions:
reload          Reload previous configuration if not confirmed
reboot          Reboot to saved configuration if not confirmed (default)
```

Note that ‘reload’ loads the most recent completed configuration and does not require a reboot.

What if you are doing something dangerous? Suppose you want to setup a firewall, and you are not sure there are no mistakes that will lock you out of your system. You can use confirmed commit. If you issue the `commit-confirm` command, your changes will be committed, and if you don’t issue the `confirm` command in 10 minutes, your system will reboot into previous config revision.

```
vyos@router# set firewall interface eth0 local name FromWorld
vyos@router# commit-confirm
commit confirm will be automatically reboot in 10 minutes unless confirmed
Proceed? [confirm]y
[edit]
vyos@router# confirm
[edit]
```

copy

Copy a configuration element.

You can copy and remove configuration subtrees. Suppose you set up a firewall ruleset `FromWorld` with one rule that allows traffic from specific subnet. Now you want to setup a similar rule, but for different subnet. Change

your edit level to firewall name FromWorld and use copy rule 10 to rule 20, then modify rule 20.

```
vyos@router# show firewall name FromWorld
default-action drop
rule 10 {
    action accept
    source {
        address 203.0.113.0/24
    }
}
[edit]
vyos@router# edit firewall name FromWorld
[edit firewall name FromWorld]
vyos@router# copy rule 10 to rule 20
[edit firewall name FromWorld]
vyos@router# set rule 20 source address 198.51.100.0/24
[edit firewall name FromWorld]
vyos@router# commit
[edit firewall name FromWorld]
```

rename

Rename a configuration element.

You can also rename config subtrees:

```
vyos@router# rename rule 10 to rule 5
[edit firewall name FromWorld]
vyos@router# commit
[edit firewall name FromWorld]
```

Note that show command respects your edit level and from this level you can view the modified firewall ruleset with just show with no parameters.

```
vyos@router# show
default-action drop
rule 5 {
    action accept
    source {
        address 203.0.113.0/24
    }
}
rule 20 {
    action accept
    source {
        address 198.51.100.0/24
    }
}
```

comment <config node> “comment text”

Add comment as an annotation to a configuration node.

The `comment` command allows you to insert a comment above the <config node> configuration section. When shown, comments are enclosed with `/*` and `*/` as open/close delimiters. Comments need to be committed, just like other config changes.

To remove an existing comment from your current configuration, specify an empty string enclosed in double quote marks ("") as the comment text.

Example:

```
vyos@vyos# comment firewall all-ping "Yes I know this VyOS is cool"
vyos@vyos# commit
vyos@vyos# show
firewall {
    /* Yes I know this VyOS is cool */
    all-ping enable
    broadcast-ping disable
    ...
}
```

Note: An important thing to note is that since the comment is added on top of the section, it will not appear if the `show <section>` command is used. With the above example, the `show firewall` command would return starting after the `firewall {` line, hiding the comment.

6.4 Access opmode from config mode

When inside configuration mode you are not directly able to execute operational commands.

run

Access to these commands are possible through the use of the `run [command]` command. From this command you will have access to everything accessible from operational mode.

Command completion and syntax help with `?` and `[tab]` will also work.

```
[edit]
vyos@vyos# run show interfaces
Codes: S - State, L - Link, u - Up, D - Down, A - Admin Down
Interface      IP Address      S/L  Description
-----
eth0           0.0.0.0/0       u/u
```

6.5 Managing configurations

VyOS comes with an integrated versioning system for the system configuration. It automatically maintains a backup of every previous configuration which has been committed to the system. The configurations are versioned locally for rollback but they can also be stored on a remote host for archiving/backup reasons.

6.5.1 Local Archive

Revisions are stored on disk. You can view, compare and rollback them to any previous revisions if something goes wrong.

show system commit

View all existing revisions on the local system.

```
vyos@vyos:~$ show system commit
0   2015-03-30 08:53:03 by vyos via cli
1   2015-03-30 08:52:20 by vyos via cli
2   2015-03-26 21:26:01 by root via boot-config-loader
3   2015-03-26 20:43:18 by root via boot-config-loader
4   2015-03-25 11:06:14 by root via boot-config-loader
5   2015-03-25 01:04:28 by root via boot-config-loader
6   2015-03-25 00:16:47 by vyos via cli
7   2015-03-24 23:43:45 by root via boot-config-loader
```

set system config-management commit-revisions <N>

You can specify the number of revisions stored on disk. N can be in the range of 0 - 65535. When the number of revisions exceeds the configured value, the oldest revision is removed. The default setting for this value is to store 100 revisions locally.

6.5.2 Compare configurations

VyOS lets you compare different configurations.

compare <saved | N> <M>

Use this command to spot what the differences are between different configurations.

```
vyos@vyos# compare [tab]
Possible completions:
<Enter> Compare working & active configurations
saved    Compare working & saved configurations
<N>      Compare working with revision N
<N> <M> Compare revision N with M
Revisions:
0   2013-12-17 20:01:37 root by boot-config-loader
1   2013-12-13 15:59:31 root by boot-config-loader
2   2013-12-12 21:56:22 vyos by cli
3   2013-12-12 21:55:11 vyos by cli
4   2013-12-12 21:27:54 vyos by cli
5   2013-12-12 21:23:29 vyos by cli
6   2013-12-12 21:13:59 root by boot-config-loader
7   2013-12-12 16:25:19 vyos by cli
8   2013-12-12 15:44:36 vyos by cli
9   2013-12-12 15:42:07 root by boot-config-loader
10  2013-12-12 15:42:06 root by init
```

The command `compare` allows you to compare different type of configurations. It also lets you compare different revisions through the `compare N M` command, where N and M are revision numbers. The output will describe how the configuration N is when compared to M indicating with a plus sign (+) the additional parts N has when compared to M, and indicating with a minus sign (-) the lacking parts N misses when compared to M.

```
vyos@vyos# compare 0 6
[edit interfaces]
+dummy dum1 {
+  address 10.189.0.1/31
+}
[edit interfaces ethernet eth0]
+vif 99 {
+  address 10.199.0.1/31
+}
-vif 900 {
-  address 192.0.2.4/24
-}
```

show system commit diff <number>

Show commit revision difference.

The command above also lets you see the difference between two commits. By default the difference with the running config is shown.

```
vyos@router# run show system commit diff 4
[edit system]
+ipv6 {
+  disable-forwarding
+}
```

This means four commits ago we did set `system ipv6 disable-forwarding`.

6.5.3 Rollback Changes

You can rollback configuration changes using the rollback command. This will apply the selected revision and trigger a system reboot.

rollback <N>

Rollback to revision N (currently requires reboot)

```
vyos@vyos# compare 1
[edit system]
>host-name vyos-1
[edit]
vyos@vyos# rollback 1
Proceed with reboot? [confirm][y]
Broadcast message from root@vyos-1 (pts/0) (Tue Dec 17 21:07:45 2013):

The system is going down for reboot NOW!
```

6.5.4 Remote Archive

VyOS can upload the configuration to a remote location after each call to `commit`. You will have to set the commit-archive location. TFTP, FTP, SCP and SFTP servers are supported. Every time a `commit` is successful the `config.boot` file will be copied to the defined destination(s). The filename used on the remote host will be `config.boot-hostname.YYYYMMDD_HHMMSS`.

set system config-management commit-archive location <URI>

Specify remote location of commit archive as any of the below URI (Uniform Resource Identifier)

- `http://<user>:<passwd>@<host>:/<dir>`
- `https://<user>:<passwd>@<host>:/<dir>`
- `ftp://<user>:<passwd>@<host>/<dir>`
- `sftp://<user>:<passwd>@<host>/<dir>`
- `scp://<user>:<passwd>@<host>:/<dir>`
- `tftp://<host>/<dir>`
- `git+https://<user>:<passwd>@<host>/<path>`

Since username and password are part of the URI, they need to be properly url encoded if containing special characters.

Note: The number of revisions don't affect the commit-archive.

When using Git as destination for the commit archive the `source-address` CLI option has no effect.

You may find VyOS not allowing the secure connection because it cannot verify the legitimacy of the remote server. You can use the workaround below to quickly add the remote host's SSH fingerprint to your `~/.ssh/known_hosts` file:

```
vyos@vyos# ssh-keyscan <host> >> ~/.ssh/known_hosts
```

set system config-management commit-archive vrf <name>

Specify name of the VRF (Virtual Routing and Forwarding) instance used to upload the configuration to the remote system.

6.5.5 Saving and loading manually

You can use the `save` and `load` commands if you want to manually manage specific configuration files.

When using the `save` command, you can add a specific location where to store your configuration file. And, when needed it, you will be able to load it with the `load` command:

load <URI>

Use this command to load a configuration which will replace the running configuration. Define the location of the configuration file to be loaded. You can use a path to a local file, an SCP address, an SFTP address, an FTP address, an HTTP address, an HTTPS address or a TFTP address.

```
vyos@vyos# load
Possible completions:
<Enter>                               Load from system config file
```

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```
<file>                                Load from file on local machine
scp://<user>:<passwd>@<host>:/<file> Load from file on remote machine
sftp://<user>:<passwd>@<host>/<file> Load from file on remote machine
ftp://<user>:<passwd>@<host>/<file>  Load from file on remote machine
http://<host>/<file>                  Load from file on remote machine
https://<host>/<file>                 Load from file on remote machine
tftp://<host>/<file>                 Load from file on remote machine
```

If you are remotely connected, you will lose your connection. You may want to copy first the config, edit it to ensure connectivity, and load the edited config.

6.5.6 Restore Default

In the case you want to completely delete your configuration and restore the default one, you can enter the following command in configuration mode:

```
load /opt/vyatta/etc/config.boot.default
```

You will be asked if you want to continue. If you accept, you will have to use `commit` if you want to make the changes active.

Then you may want to `save` in order to delete the saved configuration too.

Note: Prompt changes from `$` to `#`. To exit configuration mode, type `exit`.

CONFIGURATION GUIDE

The following structure represents the CLI structure.

7.1 Container

The VyOS container implementation is based on [Podman](#) as a daemonless container engine.

7.1.1 Configuration

set container name <name> image

Sets the image name in the hub registry

```
set container name mysql-server image mysql:8.0
```

If a registry is not specified, Docker.io will be used as the container registry unless an alternative registry is specified using **set container registry <name>** or the registry is included in the image name

```
set container name mysql-server image quay.io/mysql:8.0
```

set container name <name> entrypoint <entrypoint>

Override the default entrypoint from the image for a container.

set container name <name> command <command>

Override the default command from the image for a container.

set container name <name> arguments <arguments>

Set the command arguments for a container.

set container name <name> host-name <hostname>

Set the host name for a container.

set container name <name> allow-host-pid

The container and the host share the same process namespace. This means that processes running on the host are visible inside the container, and processes inside the container are visible on the host.

The command translates to “-pid host” when the container is created.

set container name <name> allow-host-networks

Allow host networking in a container. The network stack of the container is not isolated from the host and will use the host IP.

The command translates to “-net host” when the container is created.

Note: `allow-host-networks` cannot be used with `network`

set container name <name> network <networkname>

Attaches user-defined network to a container. Only one network must be specified and must already exist.

set container name <name> network <networkname> address <address>

Optionally set a specific static IPv4 or IPv6 address for the container. This address must be within the named network prefix.

Note: The first IP in the container network is reserved by the engine and cannot be used

set container name <name> name-server <address>

Optionally set a custom name server. If a container network is used with DNS enabled, this setting will not have any effect.

set container name <name> description <text>

Set a container description

set container name <name> environment <key> value <value>

Add custom environment variables. Multiple environment variables are allowed. The following commands translate to “-e key=value” when the container is created.

```
set container name mysql-server environment MYSQL_DATABASE value 'zabbix'
set container name mysql-server environment MYSQL_USER value 'zabbix'
set container name mysql-server environment MYSQL_PASSWORD value 'zabbix_pwd'
set container name mysql-server environment MYSQL_ROOT_PASSWORD value 'root_pwd'
```

set container name <name> port <portname> source <portnumber>

set container name <name> port <portname> destination <portnumber>

set container name <name> port <portname> protocol <tcp | udp>

Publish a port for the container.

```
set container name zabbix-web-nginx-mysql port http source 80
set container name zabbix-web-nginx-mysql port http destination 8080
set container name zabbix-web-nginx-mysql port http protocol tcp
```

Note: Port publishing cannot be used with `network`. For this purpose, a workaround using destination NAT and static IP assignment for the container is available.

set container name <name> volume <volumename> source <path>

set container name <name> volume <volumename> destination <path>

Mount a volume into the container

```
set container name coredns volume 'corefile' source /config/coredns/Corefile
set container name coredns volume 'corefile' destination /etc/Corefile
```

set container name <name> volume <volumename> mode <ro | rw>

Volume is either mounted as rw (read-write - default) or ro (read-only)

set container name <name> tmpfs <tmpfsname> destination <path>

Mount a tmpfs (*ramdisk*) filesystem to the given path within the container.

set container name <name> tmpfs <tmpfsname> size <MB>

Size in MB for tmpfs filesystem, maximum size is 64GB or 50% of the systems total available memory.

set container name <name> uid <number>

set container name <name> gid <number>

Set the User ID or Group ID of the container

set container name <name> restart [no | on-failure | always]

Set the restart behavior of the container.

- **no**: Do not restart containers on exit
- **on-failure**: Restart containers when they exit with a non-zero exit code, retrying indefinitely (default)
- **always**: Restart containers when they exit, regardless of status, retrying indefinitely

set container name <name> cpu-quota <num>

This specifies the number of CPU resources the container can use.

Default is 0 for unlimited. For example, 1.25 limits the container to use up to 1.25 cores worth of CPU time. This can be a decimal number with up to three decimal places.

The command translates to “-cpus=<num>” when the container is created.

set container name <name> memory <MB>

Constrain the memory available to the container.

Default is 512 MB. Use 0 MB for unlimited memory.

set container name <name> device <devicename> source <path>

set container name <name> device <devicename> destination <path>

Add a host device to the container.

set container name <name> capability <text>

Set container capabilities or permissions.

- **net-admin**: Network operations (interface, firewall, routing tables)
- **net-bind-service**: Bind a socket to privileged ports (port numbers less than 1024)
- **net-raw**: Permission to create raw network sockets
- **setpcap**: Capability sets (from bounded or inherited set)
- **sys-admin**: Administration operations (quotactl, mount, sethostname, setdomainname)
- **sys-time**: Permission to set system clock

set container name <name> sysctl parameter <parameter> value <value>

Set container sysctl values.

The subset of possible parameters are:

- Kernel Parameters: kernel.msgmax, kernel.msgmnb, kernel.msgmni, kernel.sem, kernel.shmall, kernel.shmmax, kernel.shmmni, kernel.shm_rmid_forced
- Parameters beginning with fs.mqueue.*
- Parameters beginning with net.* (only if user-defined network is used)

set container name <name> label <label> value <value>

Add metadata label for this container.

set container name <name> disable

Disable a container.

Container Health checks

By default, no health checks are run, even when defined by the image.

set container name <name> health-check

Default health check is run for the container if defined by the image.

set container name <name> health-check command <command>

Override the default health check command from the image for a container.

set container name <name> health-check interval <interval>

Override the default health-check interval. For example: 60

set container name <name> health-check timeout <timeout>

Override the default health-check timeout. For example: 10

set container name <name> health-check retries <retries>

Number of health check retries before container is considered unhealthy. For example: 1

Container Networks

set container network <name>

Creates a named container network

set container network <name> description

A brief description what this network is all about.

set container network <name> prefix <ipv4|ipv6>

Define IPv4 and/or IPv6 prefix for a given network name. Both IPv4 and IPv6 can be used in parallel.

set container network <name> mtu <number>

Configure MTU (Maximum Transmission Unit) for a given network. It is the size (in bytes) of the largest ethernet frame sent on this link.

set container network <name> no-name-server

Disable Domain Name System (DNS) plugin for this network.

```
set container network <name> vrf <name>
```

Bind container network to a given VRF instance.

Container Registry

```
set container registry <name>
```

Adds registry to list of unqualified-search-registries. By default, for any image that does not include the registry in the image name, VyOS will use docker.io and quay.io as the container registry.

```
set container registry <name> disable
```

Disable a given container registry

```
set container registry <name> authentication username
```

```
set container registry <name> authentication password
```

Some container registries require credentials to be used.

Credentials can be defined here and will only be used when adding a container image to the system.

```
set container registry <name> insecure
```

Allow registry access over unencrypted HTTP or TLS connections with untrusted certificates.

```
set container registry <name> mirror address <address>
```

```
set container registry <name> mirror host-name <host-name>
```

```
set container registry <name> mirror port <port>
```

```
set container registry <name> mirror path <path>
```

Registry mirror, use (host-name|address)[:port][/path].

If you have mirror http://192.168.1.1:8080 for docker.io, you can use docker.io/some/repo or run podman pull docker.io/some/repo

```
set container registry docker.io mirror address 192.168.1.1
set container registry docker.io mirror port 8080
set container registry docker.io insecure
```

If http://192.168.1.1:8080 is your own registry, you can use 192.168.1.1:8080/some/repo or run podman pull 192.168.1.1:8080/some/repo

```
set container registry 192.168.1.1:8080 insecure
```

Log Configuration

```
set container name <name> log-driver [k8s-file | journald | none]
```

Set the default log driver for containers.

- **k8s-file**: Log to a plain text file in Kubernetes-style format.
- **journald**: Log to the system journal
- **none**: Disable logging for the container

Current default is journald.

7.1.2 Operation Commands

add container image <containername>

Pull a new image for container

show container

Show the list of all active containers.

show container image

Show the local container images.

show container log <containername>

Show logs from a given container

show container network

Show a list available container networks

restart container <containername>

Restart a given container

update container image <containername>

Update container image

delete container image <image id|all> [force]

Delete a particular container image based on it's image ID. You can also delete all container images at once.

You can not delete a container image if it has more then one tag assigned, this is why there is a force option to pass down to the container image to also remove those images.

7.1.3 Example Configuration

For the sake of demonstration, [example #1 in the official documentation](#) to the declarative VyOS CLI syntax.

```
set container network zabbix prefix 172.20.0.0/16
set container network zabbix description 'Network for Zabbix component containers'

set container name mysql-server image mysql:8.0
set container name mysql-server network zabbix

set container name mysql-server environment 'MYSQL_DATABASE' value 'zabbix'
set container name mysql-server environment 'MYSQL_USER' value 'zabbix'
set container name mysql-server environment 'MYSQL_PASSWORD' value 'zabbix_pwd'
set container name mysql-server environment 'MYSQL_ROOT_PASSWORD' value 'root_pwd'

set container name zabbix-java-gateway image zabbix/zabbix-java-gateway:alpine-5.2-latest
set container name zabbix-java-gateway network zabbix

set container name zabbix-server-mysql image zabbix/zabbix-server-mysql:alpine-5.2-latest
set container name zabbix-server-mysql network zabbix
```

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```

set container name zabbix-server-mysql environment 'DB_SERVER_HOST' value 'mysql-server'
set container name zabbix-server-mysql environment 'MYSQL_DATABASE' value 'zabbix'
set container name zabbix-server-mysql environment 'MYSQL_USER' value 'zabbix'
set container name zabbix-server-mysql environment 'MYSQL_PASSWORD' value 'zabbix_pwd'
set container name zabbix-server-mysql environment 'MYSQL_ROOT_PASSWORD' value 'root_pwd'
set container name zabbix-server-mysql environment 'ZBX_JAVAGATEWAY' value 'zabbix-java-
↪gateway'

set container name zabbix-server-mysql port zabbix source 10051
set container name zabbix-server-mysql port zabbix destination 10051

set container name zabbix-web-nginx-mysql image zabbix/zabbix-web-nginx-mysql:alpine-5.2-
↪latest
set container name zabbix-web-nginx-mysql network zabbix

set container name zabbix-web-nginx-mysql environment 'MYSQL_DATABASE' value 'zabbix'
set container name zabbix-web-nginx-mysql environment 'ZBX_SERVER_HOST' value 'zabbix-
↪server-mysql'
set container name zabbix-web-nginx-mysql environment 'DB_SERVER_HOST' value 'mysql-
↪server'
set container name zabbix-web-nginx-mysql environment 'MYSQL_USER' value 'zabbix'
set container name zabbix-web-nginx-mysql environment 'MYSQL_PASSWORD' value 'zabbix_pwd'
set container name zabbix-web-nginx-mysql environment 'MYSQL_ROOT_PASSWORD' value 'root_
↪pwd'

set container name zabbix-web-nginx-mysql port http source 80
set container name zabbix-web-nginx-mysql port http destination 8080

```

7.2 Firewall

Warning: Due to a boot-time race condition, all interfaces initialize before the firewall. This temporarily leaves the system open to all traffic and poses a security risk.

VyOS uses Netfilter. The Netfilter project developed `iptables` and its successor `nftables` for the Linux kernel to process packet data flows directly. This extends the concept of zone-based security to let you manipulate data at multiple stages after the network interface and driver accept it, and before sending it to its destination (for example, a web server or another device).

The following is a simplified traffic flow diagram based on Netfilter packet flow. This diagram provides an overview of how packets are processed and the possible paths traffic can take.

The main points regarding packet flow and terminology in VyOS firewall are:

- **Bridge Port?:** Choose the appropriate path based on whether the interface where the packet was received is part of a bridge.

If the interface where the packet was received is not part of a bridge, the packet is processed at the **IP Layer**:

- **Prerouting:** The router processes all packets in this stage, regardless of the destination. You can perform several actions in this stage, and these actions are also defined in different parts of the VyOS configuration. Order is

important. The relevant configuration that applies in this stage includes:

- **Firewall prerouting:** Rules you define under `set firewall [ipv4 | ipv6] prerouting raw....`. The system processes all rules in this section before the connection tracking subsystem.
- **Conntrack Ignore:** Rules you define under `set system conntrack ignore [ipv4 | ipv6] ....`. You can configure this section with `firewall [ipv4 | ipv6] prerouting ....`. For compatibility reasons, this feature is supported, but will be deprecated in the future.
- **Policy Route:** Rules you define under `set policy [route | route6] ....`
- **Destination NAT:** Rules you define under `set [nat | nat66] destination....`
- **Destination is the router?:** Choose the appropriate path based on the destination IP address. Transit traffic continues to **forward**, while traffic destined for the router continues to **input**.
- **Input:** The stage where you filter and control traffic destined for the router itself. This is where you enforce all rules for securing the router. This includes IPv4 and IPv6 filtering rules, defined in:
 - `set firewall ipv4 input filter ....`
 - `set firewall ipv6 input filter ....`
- **Forward:** The stage where you filter and control transit traffic. This includes IPv4 and IPv6 filtering rules, defined in:
 - `set firewall ipv4 forward filter ....`
 - `set firewall ipv6 forward filter ....`
- **Output:** The stage where you filter and control traffic that the router originates. Note that this traffic comes from either a new connection that an internal process on the VyOS router (such as NTP) originates or a response to traffic the router receives externally through **input** (for example, a response to an SSH login attempt). This includes IPv4 and IPv6 rules, and two different sections apply:
 - **Output Prerouting:** `set firewall [ipv4 | ipv6] output raw ....`. As described in **Prerouting**, the system processes rules in this section before the connection tracking subsystem.
 - **Output Filter:** `set firewall [ipv4 | ipv6] output filter ....`
- **Postrouting:** As in **Prerouting**, you can perform several actions defined in different parts of VyOS configuration in this stage. This includes:
 - **Source NAT:** Rules you define under `set [nat | nat66] source....`

If the interface where the packet was received is part of a bridge, the packet is processed at the **Bridge Layer**:

- **Prerouting (Bridge):** The bridge processes all packets it receives in this stage, regardless of the destination. First, you can apply filters here, or you can configure rules that ignore the connection tracking system. The relevant configuration that applies:
 - `set firewall bridge prerouting filter ....`
- **Forward (Bridge):** The stage where you filter and control traffic that passes through the bridge:
 - `set firewall bridge forward filter ....`
- **Input (Bridge):** The stage where you filter and control traffic destined for the bridge itself:
 - `set firewall bridge input filter ....`
- **Output (Bridge):** The stage where you filter and control traffic that the bridge originates:
 - `set firewall bridge output filter ....`

The following is the overall structure of the VyOS firewall CLI:

```

- set firewall
  * bridge
    - forward
      + filter
    - input
      + filter
    - output
      + filter
    - prerouting
      + filter
    - name
      + custom_name
  * flowtable
    - custom_flow_table
      + ...
  * global-options
    + all-ping
    + broadcast-ping
    + ...
  * group
    - address-group
    - ipv6-address-group
    - network-group
    - ipv6-network-group
    - interface-group
    - mac-group
    - port-group
    - domain-group
  * ipv4
    - forward
      + filter
    - input
      + filter
    - output
      + filter
      + raw
    - prerouting
      + raw
    - name
      + custom_name
  * ipv6
    - forward
      + filter
    - input
      + filter
    - output
      + filter
      + raw
    - prerouting
      + raw
    - ipv6-name
      + custom_name
  * zone

```

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```
- custom_zone_name  
+ ...
```

Here is a list of VyOS firewall CLI subcommands and their corresponding pages in the documentation:

set firewall bridge ...

Configure bridge firewall rules for traffic at the bridge layer. See the Bridge Firewall Configuration page for detailed information.

set firewall flowtable ...

Configure firewall flowtables for stateful connection tracking and rules. See the Flowtables Firewall Configuration page for detailed information.

set firewall global-options ...

Configure global firewall options such as `all-ping`, `broadcast-ping`, `syn-cookies`, and other system-wide firewall settings. See the Global Firewall Options page for detailed information.

set firewall group ...

Organize firewall rules by creating reusable address, network, interface, MAC, port, and domain groups. Use groups in multiple rules to simplify configuration and maintenance. See the Firewall Groups page for detailed information.

set firewall ipv4 ...

Configure IPv4-specific firewall rules. See the IPv4 Firewall Configuration page for detailed information.

set firewall ipv6 ...

Configure IPv6-specific firewall rules. See the IPv6 Firewall Configuration page for detailed information.

set firewall zone ...

Configure zone-based firewall policies for controlling traffic between different network zones. See the Zone-Based Firewall Configuration page for detailed information.

For more information on firewall configuration, see the following pages:

7.2.1 Global Options Firewall Configuration

Overview

Some firewall settings are global and affect the entire system. This section provides information about these global options that you can configure using the VyOS CLI.

Configuration commands covered in this section:

set firewall global-options ...

Configuration

set firewall global-options all-ping [enable | disable]

By default, when VyOS receives an ICMP echo request packet destined for itself, it answers with an ICMP echo reply, unless your firewall prevents it.

You can set firewall rules to accept, drop, or reject ICMP in, out, or local traffic. You can also use the **firewall global-options all-ping** command. This command affects only LOCAL traffic (packets destined for your VyOS system), not IN or OUT traffic.

Note: **firewall global-options all-ping** affects only LOCAL traffic and always behaves in the most restrictive way

```
set firewall global-options all-ping enable
```

When you set this command, VyOS answers every ICMP echo request addressed to itself, but that response occurs only if no other rule drops or rejects local echo requests. In case of conflict, VyOS does not answer ICMP echo requests.

```
set firewall global-options all-ping disable
```

When you set this command, VyOS answers no ICMP echo requests addressed to itself, regardless of where they come from or what specific rules accept them.

set firewall global-options apply-to-bridged-traffic [ipv4 | ipv6]

Apply IPv4 or IPv6 firewall rules to bridged traffic.

set firewall global-options broadcast-ping [enable | disable]

Enable or disable the response to ICMP broadcast messages. The system alters the following parameter:

- `net.ipv4.icmp_echo_ignore_broadcasts`

set firewall global-options ip-src-route [enable | disable]

set firewall global-options ipv6-src-route [enable | disable]

Set whether VyOS accepts packets with a source route option. The following sysctl parameters will be changed:

- `net.ipv4.conf.all.accept_source_route`
- `net.ipv6.conf.all.accept_source_route`

set firewall global-options receive-redirects [enable | disable]

set firewall global-options ipv6-receive-redirects [enable | disable]

Allow VyOS to accept ICMPv4 and ICMPv6 redirect messages. The following sysctl parameters will be changed:

- `net.ipv4.conf.all.accept_redirects`
- `net.ipv6.conf.all.accept_redirects`

set firewall global-options send-redirects [enable | disable]

Allow VyOS to send ICMPv4 redirect messages. The following sysctl parameter will be changed:

- `net.ipv4.conf.all.send_redirects`

set firewall global-options log-martians [enable | disable]

Allow VyOS to log martian IPv4 packets. The following sysctl parameter will be changed:

- `net.ipv4.conf.all.log_martians`

```
set firewall global-options source-validation [strict | loose | disable]
```

Set the IPv4 source validation mode. The following sysctl parameter will be changed:

- `net.ipv4.conf.all.rp_filter`

```
set firewall global-options syn-cookies [enable | disable]
```

Allow VyOS to use IPv4 TCP SYN Cookies. The following sysctl parameter will be changed:

- `net.ipv4.tcp_syncookies`

```
set firewall global-options twa-hazards-protection [enable | disable]
```

Enable or disable VyOS [RFC 1337](#) conformance. The following sysctl parameter will be changed:

- `net.ipv4.tcp_rfc1337`

```
set firewall global-options state-policy established action [accept | drop | reject]
```

```
set firewall global-options state-policy established log
```

```
set firewall global-options state-policy established log-level [emerg | alert | crit |  
err | warn | notice | info | debug]
```

Set the global setting for an established connection.

```
set firewall global-options state-policy invalid action [accept | drop | reject]
```

```
set firewall global-options state-policy invalid log
```

```
set firewall global-options state-policy invalid log-level [emerg | alert | crit | err |  
warn | notice | info | debug]
```

Set the global setting for invalid packets.

```
set firewall global-options state-policy related action [accept | drop | reject]
```

```
set firewall global-options state-policy related log
```

```
set firewall global-options state-policy related log-level [emerg | alert | crit | err |  
warn | notice | info | debug]
```

Set the global setting for related connections.

VyOS supports setting timeouts for connections by connection type. You can set timeout values for generic connections, ICMP connections, UDP connections, or TCP connections in various states.

```
set firewall global-options timeout icmp <1-21474836>
```

```
set firewall global-options timeout other <1-21474836>
```

```
set firewall global-options timeout tcp close <1-21474836>
```

```
set firewall global-options timeout tcp close-wait <1-21474836>
```

```
set firewall global-options timeout tcp established <1-21474836>
```

```
set firewall global-options timeout tcp fin-wait <1-21474836>
```

```
set firewall global-options timeout tcp last-ack <1-21474836>
```

```
set firewall global-options timeout tcp syn-recv <1-21474836>
```

```
set firewall global-options timeout tcp syn-sent <1-21474836>
```

```
set firewall global-options timeout tcp time-wait <1-21474836>
set firewall global-options timeout udp other <1-21474836>
set firewall global-options timeout udp stream <1-21474836>
```

Set the timeout in seconds for a protocol or state.

7.2.2 Firewall groups

Configuration

Firewall groups represent collections of IP addresses, networks, ports, MAC addresses, domains, or interfaces. You can reference a group in firewall, NAT, and policy route rules as either a source or destination matcher, and/or as inbound or outbound in the case of interface groups.

Address Groups

An **address group** contains a single IP address or IP address range.

```
set firewall group address-group <name> address [address | address range]
```

```
set firewall group ipv6-address-group <name> address <address>
```

Define an IPv4 or IPv6 address group.

```
set firewall group address-group ADR-INSIDE-v4 address 192.168.0.1
set firewall group address-group ADR-INSIDE-v4 address 10.0.0.1-10.0.0.8
set firewall group ipv6-address-group ADR-INSIDE-v6 address 2001:db8::1
```

```
set firewall group address-group <name> description <text>
```

```
set firewall group ipv6-address-group <name> description <text>
```

Provide an IPv4 or IPv6 address group description.

Remote Groups

A **remote-group** uses a URL that hosts a newline-delimited list of IPv4 and/or IPv6 addresses, CIDRs, and ranges. VyOS pulls this list periodically according to the frequency you define in the firewall **resolver-interval** and loads matching entries into the group for use in rules. The list is cached in persistent storage, so rules continue to function if updates fail.

```
set firewall group remote-group <name> url <http(s) url>
```

Specify a remote list of IPv4 and/or IPv6 addresses, ranges, and CIDRs to fetch.

```
set firewall group remote-group <name> description <text>
```

Set a description for a remote group.

The remote list format is flexible. VyOS attempts to parse the first word of each line as an entry and skips lines it cannot match. Lines that begin with an alphanumeric character but do not match valid IPv4 or IPv6 addresses, ranges, or CIDRs are logged to the system log. The following examples show acceptable formats that VyOS parses correctly:

```
127.0.0.1
127.0.0.0/24
127.0.0.1-127.0.0.254
2001:db8::1
2001:db8:cafe::/48
2001:db8:cafe::1-2001:db8:cafe::ffff
```

Network Groups

Network groups accept IP networks in CIDR notation. You can add specific IP addresses as a 32-bit prefix. If you need to add a mix of addresses and networks, use a network group.

```
set firewall group network-group <name> network <CIDR>
```

```
set firewall group ipv6-network-group <name> network <CIDR>
```

Define an IPv4 or IPv6 network group.

```
set firewall group network-group NET-INSIDE-v4 network 192.168.0.0/24
set firewall group network-group NET-INSIDE-v4 network 192.168.1.0/24
set firewall group ipv6-network-group NET-INSIDE-v6 network 2001:db8::/64
```

```
set firewall group network-group <name> description <text>
```

```
set firewall group ipv6-network-group <name> description <text>
```

Provide an IPv4 or IPv6 network group description.

Interface Groups

An **interface group** represents a collection of interfaces.

```
set firewall group interface-group <name> interface <text>
```

Define an interface group. Wildcard * is supported. For example: eth3*. Prepend the character ! to invert the criteria. For example: !eth2.

```
set firewall group interface-group LAN interface bond1001
set firewall group interface-group LAN interface eth3*
```

```
set firewall group interface-group <name> description <text>
```

Provide an interface group description.

Port Groups

A **port group** represents only port numbers, not the protocol. You can reference port groups for either TCP or UDP. Create TCP and UDP groups separately to avoid accidentally filtering unnecessary ports. Specify port ranges by using -.

```
set firewall group port-group <name> port [portname | portnumber | startport-endport]
```

Define a port group. A port name can be any name defined in /etc/services. For example, http.

```
set firewall group port-group PORT-TCP-SERVER1 port http
set firewall group port-group PORT-TCP-SERVER1 port 443
set firewall group port-group PORT-TCP-SERVER1 port 5000-5010
```

set firewall group port-group <name> description <text>

Provide a port group description.

MAC Groups

A **mac group** represents a collection of mac addresses.

set firewall group mac-group <name> mac-address <mac-address>

Define a mac group.

```
set firewall group mac-group MAC-G01 mac-address 88:a4:c2:15:b6:4f
set firewall group mac-group MAC-G01 mac-address 4c:d5:77:c0:19:81
```

set firewall group mac-group <name> description <text>

Provide a MAC group description.

Domain Groups

A **domain group** represents a collection of domains.

set firewall group domain-group <name> address <domain>

Define a domain group.

```
set firewall group domain-group DOM address example.com
```

set firewall group domain-group <name> description <text>

Provide a domain group description.

Dynamic Groups

Firewall dynamic groups differ from other groups because you can use them as source/destination in firewall rules, and members are not defined statically in VyOS configuration. Instead, firewall rules dynamically add members to these groups.

Defining Dynamic Address Groups

Dynamic address groups support both IPv4 and IPv6 families. Use these commands to define dynamic IPv4 and IPv6 address groups:

set firewall group dynamic-group address-group <name>

set firewall group dynamic-group ipv6-address-group <name>

Add description to firewall groups:

set firewall group dynamic-group address-group <name> description <text>

```
set firewall group dynamic-group ipv6-address-group <name> description <text>
```

Adding elements to Dynamic Firewall Groups

After you define dynamic firewall groups, use them in firewall rules to dynamically add elements to them.

Commands used for this task are:

- Add destination IP address of the connection to a dynamic address group:

```
set firewall ipv4 [forward | input | output] filter rule <1-999999> add-address-to-group
destination-address address-group <name>
```

```
set firewall ipv4 name <name> rule <1-999999> add-address-to-group destination-address
address-group <name>
```

```
set firewall ipv6 [forward | input | output] filter rule <1-999999> add-address-to-group
destination-address address-group <name>
```

```
set firewall ipv6 name <name> rule <1-999999> add-address-to-group destination-address
address-group <name>
```

- Add source IP address of the connection to a dynamic address group:

```
set firewall ipv4 [forward | input | output] filter rule <1-999999> add-address-to-group
source-address address-group <name>
```

```
set firewall ipv4 name <name> rule <1-999999> add-address-to-group source-address
address-group <name>
```

```
set firewall ipv6 [forward | input | output] filter rule <1-999999> add-address-to-group
source-address address-group <name>
```

```
set firewall ipv6 name <name> rule <1-999999> add-address-to-group source-address
address-group <name>
```

You can define specific timeouts per rule. When a rule matches, the source or destination address is added to the group, and the element remains in the group until the timeout expires. If you do not define a timeout, the element remains in the group until the next reboot or until you commit firewall configuration changes.

```
set firewall ipv4 [forward | input | output] filter rule <1-999999> add-address-to-group
[destination-address | source-address] timeout <timeout>
```

```
set firewall ipv4 name <name> rule <1-999999> add-address-to-group [destination-address |
source-address] timeout <timeout>
```

```
set firewall ipv6 [forward | input | output] filter rule <1-999999> add-address-to-group
[destination-address | source-address] timeout <timeout>
```

```
set firewall ipv6 name <name> rule <1-999999> add-address-to-group [destination-address |
source-address] timeout <timeout>
```

Timeout can be defined using seconds, minutes, hours or days:

```
set firewall ipv6 name F00 rule 10 add-address-to-group source-address timeout
```

Possible completions:

<number>s	Timeout value in seconds
<number>m	Timeout value in minutes
<number>h	Timeout value in hours
<number>d	Timeout value in days

Using Dynamic Firewall Groups

Like other firewall groups, you can use dynamic firewall groups in firewall rules as matching options. For example:

```
set firewall ipv4 input filter rule 10 source group dynamic-address-group F00
set firewall ipv4 input filter rule 10 destination group dynamic-address-group BAR
```

Examples

General example

After you create firewall groups, you can reference them in firewall, NAT, NAT66, and/or policy-route rules. The following example creates multiple groups:

```
set firewall group address-group SERVERS address 198.51.100.101
set firewall group address-group SERVERS address 198.51.100.102
set firewall group network-group TRUSTEDv4 network 192.0.2.0/30
set firewall group network-group TRUSTEDv4 network 203.0.113.128/25
set firewall group ipv6-network-group TRUSTEDv6 network 2001:db8::/64
set firewall group interface-group LAN interface eth2.2001
set firewall group interface-group LAN interface bon0
set firewall group port-group PORT-SERVERS port http
set firewall group port-group PORT-SERVERS port 443
set firewall group port-group PORT-SERVERS port 5000-5010
```

And next, some configuration example where groups are used:

```
set firewall ipv4 output filter rule 10 action accept
set firewall ipv4 output filter rule 10 outbound-interface group !LAN
set firewall ipv4 forward filter rule 20 action accept
set firewall ipv4 forward filter rule 20 source group network-group TRUSTEDv4
set firewall ipv6 input filter rule 10 action accept
set firewall ipv6 input filter rule 10 source group network-group TRUSTEDv6
set nat destination rule 101 inbound-interface group LAN
set nat destination rule 101 destination group address-group SERVERS
set nat destination rule 101 protocol tcp
set nat destination rule 101 destination group port-group PORT-SERVERS
set nat destination rule 101 translation address 203.0.113.250
set policy route PBR rule 201 destination group port-group PORT-SERVERS
set policy route PBR rule 201 protocol tcp
set policy route PBR rule 201 set table 15
```

Port knocking example

You can use dynamic firewall groups with port knocking to secure access to the router or any other device. The following example shows a 4-step port knocking configuration:

```
set firewall global-options state-policy established action 'accept'
set firewall global-options state-policy invalid action 'drop'
set firewall global-options state-policy related action 'accept'
set firewall group dynamic-group address-group ALLOWED
set firewall group dynamic-group address-group PN_01
set firewall group dynamic-group address-group PN_02
set firewall ipv4 input filter default-action 'drop'
set firewall ipv4 input filter rule 5 action 'accept'
set firewall ipv4 input filter rule 5 protocol 'icmp'
set firewall ipv4 input filter rule 10 action 'drop'
set firewall ipv4 input filter rule 10 add-address-to-group source-address_
↪address-group 'PN_01'
set firewall ipv4 input filter rule 10 add-address-to-group source-address_
↪timeout '2m'
set firewall ipv4 input filter rule 10 description 'Port_nock 01'
set firewall ipv4 input filter rule 10 destination port '9990'
set firewall ipv4 input filter rule 10 protocol 'tcp'
set firewall ipv4 input filter rule 20 action 'drop'
set firewall ipv4 input filter rule 20 add-address-to-group source-address_
↪address-group 'PN_02'
set firewall ipv4 input filter rule 20 add-address-to-group source-address_
↪timeout '3m'
set firewall ipv4 input filter rule 20 description 'Port_nock 02'
set firewall ipv4 input filter rule 20 destination port '9991'
set firewall ipv4 input filter rule 20 protocol 'tcp'
set firewall ipv4 input filter rule 20 source group dynamic-address-group 'PN_
↪01'
set firewall ipv4 input filter rule 30 action 'drop'
set firewall ipv4 input filter rule 30 add-address-to-group source-address_
↪address-group 'ALLOWED'
set firewall ipv4 input filter rule 30 add-address-to-group source-address_
↪timeout '2h'
set firewall ipv4 input filter rule 30 description 'Port_nock 03'
set firewall ipv4 input filter rule 30 destination port '9992'
set firewall ipv4 input filter rule 30 protocol 'tcp'
set firewall ipv4 input filter rule 30 source group dynamic-address-group 'PN_
↪02'
set firewall ipv4 input filter rule 99 action 'accept'
set firewall ipv4 input filter rule 99 description 'Port_nock 04 - Allow ssh'
set firewall ipv4 input filter rule 99 destination port '22'
set firewall ipv4 input filter rule 99 protocol 'tcp'
set firewall ipv4 input filter rule 99 source group dynamic-address-group
↪'ALLOWED'
```

Before testing, we can check the members of firewall groups:

```
vyos@vyos# run show firewall group
Firewall Groups
```

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Name	Type	References	Members	Timeout	Expires
↩ -					
ALLOWED	address_group(dynamic)	ipv4-input-filter-30	N/D	N/D	N/D
PN_01	address_group(dynamic)	ipv4-input-filter-10	N/D	N/D	N/D
PN_02	address_group(dynamic)	ipv4-input-filter-20	N/D	N/D	N/D
[edit]					
vyos@vyos#					

With this configuration, to gain SSH access to the router, the user must:

1. Create a new TCP connection to destination port 9990. A new entry is added to dynamic firewall group PN_01.

```
vyos@vyos# run show firewall group
```

Firewall Groups

Name	Type	References	Members	Timeout	Expires
↩ Expires					
↩ -----					
↩ -----					
ALLOWED	address_group(dynamic)	ipv4-input-filter-30	N/D	N/D	N/D
PN_01	address_group(dynamic)	ipv4-input-filter-10	192.168.89.31	120	119
PN_02	address_group(dynamic)	ipv4-input-filter-20	N/D	N/D	N/D
[edit]					
vyos@vyos#					

2. Create a new TCP connection to destination port 9991. A new entry is added to dynamic firewall group PN_02.

```
vyos@vyos# run show firewall group
```

Firewall Groups

Name	Type	References	Members	Timeout	Expires
↩ Expires					
↩ -----					
↩ -----					
ALLOWED	address_group(dynamic)	ipv4-input-filter-30	N/D	N/D	N/D
PN_01	address_group(dynamic)	ipv4-input-filter-10	192.168.89.31	120	106
PN_02	address_group(dynamic)	ipv4-input-filter-20	192.168.89.31	180	179
[edit]					
vyos@vyos#					

3. Create a new TCP connection to destination port 9992. A new entry is added to dynamic firewall group ALLOWED.

```
vyos@vyos# run show firewall group
```

Firewall Groups

Name	Type	References	Members	Timeout	Expires
↩ Expires					
↩ -----					
↩ -----					
ALLOWED	address_group(dynamic)	ipv4-input-filter-30	192.168.89.31	7200	↩
↩ 7199					

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```
PN_01    address_group(dynamic)  ipv4-input-filter-10  192.168.89.31    120
↪      89
PN_02    address_group(dynamic)  ipv4-input-filter-20  192.168.89.31    180
↪      170
[edit]
vyos@vyos#
```

- Now you can connect via SSH to the router (assuming SSH is configured).

Operation-mode

show firewall group

show firewall group <name>

Display an overview of defined groups, including the firewall group name, type, references (where the group is used), members, timeout, and expiration (the last two only apply to dynamic firewall groups).

Here is an example of such command:

```
vyos@vyos:~$ show firewall group
Firewall Groups
```

Name	Type	References	Members	
↪ Timeout	Expires			
-----	-----	-----	-----	-----
↪ -----				
SERVERS	address_group	nat-destination-101	198.51.100.101 198.51.100.102	
ALLOWED	address_group(dynamic)	ipv4-input-filter-30	192.168.77.39	
↪ 7200	7174			
PN_01	address_group(dynamic)	ipv4-input-filter-10	192.168.0.245	
↪ 120	112		192.168.77.39	
↪ 120	85			
PN_02	address_group(dynamic)	ipv4-input-filter-20	192.168.77.39	
↪ 180	151			
LAN	interface_group	ipv4-output-filter-10 nat-destination-101	bon0 eth2.2001	
TRUSTEDv6	ipv6_network_group	ipv6-input-filter-10	2001:db8::/64	
TRUSTEDv4	network_group	ipv4-forward-filter-20	192.0.2.0/30 203.0.113.128/25	
PORT-SERVERS	port_group	route-PBR-201 route-PBR-201 nat-destination-101	443 5000-5010 http	

```
vyos@vyos:~$
```

7.2.3 Bridge Firewall Configuration

Overview

Learn more about bridge firewall configuration and related op-mode commands.

The following commands are covered in this section:

set firewall bridge <options>

From the main structure defined in *Firewall Overview* in this section you can find detailed information only for the next part of the general structure:

```
- set firewall
  * bridge
    - forward
      + filter
    - input
      + filter
    - output
      + filter
    - prerouting
      + filter
    - name
      + custom_name
```

Traffic that is received by the router on an interface that is a member of a bridge is processed on the **Bridge Layer**. Before the bridge decision is made, all packets are analyzed at **Prerouting**. First filters can be applied here, and also rules for ignoring connection tracking system can be configured. The relevant configuration that acts in **prerouting** is:

- set firewall bridge prerouting filter

For traffic that needs to be switched internally by the bridge, the base chain is **forward**, and its base command for filtering is **set firewall bridge forward filter ...**, which happens in stage 4, highlighted with red color.

For traffic destined to the router itself or that needs to be routed (assuming a layer3 bridge is configured), the base chain is **input**, and the base command is **set firewall bridge input filter ...** and the path is:

If it's not dropped, then the packet is sent to **IP Layer**, and will be processed by the **IP Layer** firewall: IPv4 or IPv6 ruleset. Check once again the *general packet flow diagram* if needed.

For traffic that originates from the bridge itself, the base chain is **output**, and the base command is **set firewall bridge output filter ...**, and the path is:

Custom bridge firewall chains can be created with the command **set firewall bridge name <name>** To use such a custom chain, a rule with action jump and the appropriate target must be defined in a base chain.

Bridge Rules

For firewall filtering, firewall rules need to be created. Each rule is numbered, has an action to apply if the rule is matched, and the ability to specify multiple matching criteria. Data packets go through the rules from 1 - 999999, so order is crucial. At the first match the action of the rule will be executed.

Actions

If a rule is defined, an action must also be defined for it. This tells the firewall what to do if all matching criteria in the rule are met.

In firewall bridge rules, the action can be:

- **accept**: accept the packet.
- **continue**: continue parsing next rule.
- **drop**: drop the packet.
- **jump**: jump to another custom chain.
- **return**: Return from the current chain and continue at the next rule of the last chain.
- **queue**: Enqueue packet to userspace.
- **notrack**: ignore connection tracking system. This action is only available in prerouting chain.

```
set firewall bridge forward filter rule <1-999999> action [accept | continue | drop |
jump | queue | return]
```

```
set firewall bridge input filter rule <1-999999> action [accept | continue | drop | jump
| queue | return]
```

```
set firewall bridge output filter rule <1-999999> action [accept | continue | drop | jump
| queue | return]
```

```
set firewall bridge prerouting filter rule <1-999999> action [accept | continue | drop |
jump | notrack | queue | return]
```

```
set firewall bridge name <name> rule <1-999999> action [accept | continue | drop | jump |
queue | return]
```

This required setting defines the action of the current rule. If action is set to jump, then jump-target is also needed.

```
set firewall bridge forward filter rule <1-999999> jump-target <text>
```

```
set firewall bridge input filter rule <1-999999> jump-target <text>
```

```
set firewall bridge output filter rule <1-999999> jump-target <text>
```

```
set firewall bridge prerouting filter rule <1-999999> jump-target <text>
```

```
set firewall bridge name <name> rule <1-999999> jump-target <text>
```

If action is set to queue, use next command to specify the queue target. Range is also supported:

```
set firewall bridge forward filter rule <1-999999> queue <0-65535>
```

```
set firewall bridge input filter rule <1-999999> queue <0-65535>
```

```
set firewall bridge output filter rule <1-999999> queue <0-65535>
```

```
set firewall bridge prerouting filter rule <1-999999> queue <0-65535>
```

```
set firewall bridge name <name> rule <1-999999> queue <0-65535>
```

Also, if action is set to queue, use next command to specify the queue options. Possible options are bypass and fanout:

```
set firewall bridge forward filter rule <1-999999> queue-options bypass
set firewall bridge input filter rule <1-999999> queue-options bypass
set firewall bridge output filter rule <1-999999> queue-options bypass
set firewall bridge prerouting filter rule <1-999999> queue-options bypass
set firewall bridge name <name> rule <1-999999> queue-options bypass
set firewall bridge forward filter rule <1-999999> queue-options fanout
set firewall bridge input filter rule <1-999999> queue-options fanout
set firewall bridge output filter rule <1-999999> queue-options fanout
set firewall bridge prerouting filter rule <1-999999> queue-options fanout
set firewall bridge name <name> rule <1-999999> queue-options fanout
```

Also, **default-action** is an action that takes place whenever a packet does not match any rule in its chain. For base chains, possible options for **default-action** are **accept** or **drop**.

```
set firewall bridge forward filter default-action [accept | drop]
set firewall bridge input filter default-action [accept | drop]
set firewall bridge output filter default-action [accept | drop]
set firewall bridge prerouting filter default-action [accept | drop]
set firewall bridge name <name> default-action [accept | continue | drop | jump | reject
| return]
```

This sets the default action of the rule-set if a packet does not match any of the rules in that chain. If default-action is set to **jump**, then **default-jump-target** is also needed. Note that for base chains, default action can only be set to **accept** or **drop**, while on custom chains more actions are available.

```
set firewall bridge name <name> default-jump-target <text>
```

To be used only when **default-action** is set to **jump**. Use this command to specify jump target for default rule.

Note: Important note about default-actions: If the default action for any base chain is not defined, then the default action is set to **accept** for that chain. For custom chains, if the default action is not defined, then the default-action is set to **drop**.

Firewall Logs

You can enable logging for every firewall rule. If enabled, other log options can be configured.

```
set firewall bridge forward filter rule <1-999999> log
set firewall bridge input filter rule <1-999999> log
set firewall bridge output filter rule <1-999999> log
set firewall bridge prerouting filter rule <1-999999> log
set firewall bridge name <name> rule <1-999999> log
```

Enable logging for the matched packet. If this configuration command is not present, then the log is not enabled.

```
set firewall bridge forward filter default-log
set firewall bridge input filter default-log
set firewall bridge output filter default-log
set firewall bridge prerouting filter default-log
set firewall bridge name <name> default-log
```

Use this command to enable the logging of the default action on the specified chain.

```
set firewall bridge forward filter rule <1-999999> log-options level [emerg | alert |
crit | err | warn | notice | info | debug]
set firewall bridge input filter rule <1-999999> log-options level [emerg | alert | crit
| err | warn | notice | info | debug]
set firewall bridge output filter rule <1-999999> log-options level [emerg | alert | crit
| err | warn | notice | info | debug]
set firewall bridge prerouting filter rule <1-999999> log-options level [emerg | alert |
crit | err | warn | notice | info | debug]
set firewall bridge name <name> rule <1-999999> log-options level [emerg | alert | crit |
err | warn | notice | info | debug]
```

Define log-level. Only applicable if rule log is enabled.

```
set firewall bridge forward filter rule <1-999999> log-options group <0-65535>
set firewall bridge input filter rule <1-999999> log-options group <0-65535>
set firewall bridge output filter rule <1-999999> log-options group <0-65535>
set firewall bridge prerouting filter rule <1-999999> log-options group <0-65535>
set firewall bridge name <name> rule <1-999999> log-options group <0-65535>
```

Define the log group to send messages to. Only applicable if rule log is enabled.

```
set firewall bridge forward filter rule <1-999999> log-options snapshot-length <0-9000>
set firewall bridge input filter rule <1-999999> log-options snapshot-length <0-9000>
set firewall bridge output filter rule <1-999999> log-options snapshot-length <0-9000>
set firewall bridge prerouting filter rule <1-999999> log-options snapshot-length
<0-9000>
set firewall bridge name <name> rule <1-999999> log-options snapshot-length <0-9000>
```

Define length of packet payload to include in netlink message. Only applicable if rule log is enabled and the log group is defined.

```
set firewall bridge forward filter rule <1-999999> log-options queue-threshold <0-65535>
set firewall bridge input filter rule <1-999999> log-options queue-threshold <0-65535>
set firewall bridge output filter rule <1-999999> log-options queue-threshold <0-65535>
set firewall bridge prerouting filter rule <1-999999> log-options queue-threshold
<0-65535>
set firewall bridge name <name> rule <1-999999> log-options queue-threshold <0-65535>
```

Define the number of packets to queue inside the kernel before sending them to userspace. Only applicable if rule log is enabled and the log group is defined.

Firewall Description

You can define a description for reference for every custom chain.

```
set firewall bridge name <name> description <text>
```

Provide a rule-set description to a custom firewall chain.

```
set firewall bridge forward filter rule <1-999999> description <text>
set firewall bridge input filter rule <1-999999> description <text>
set firewall bridge output filter rule <1-999999> description <text>
set firewall bridge prerouting filter rule <1-999999> description <text>
set firewall bridge name <name> rule <1-999999> description <text>
```

Provide a description for each rule.

Rule Status

By default, when you define a rule, it is enabled. In some cases, it is useful to disable the rule instead of removing it.

```
set firewall bridge forward filter rule <1-999999> disable
set firewall bridge input filter rule <1-999999> disable
set firewall bridge output filter rule <1-999999> disable
set firewall bridge prerouting filter rule <1-999999> disable
set firewall bridge name <name> rule <1-999999> disable
```

Command for disabling a rule but keep it in the configuration.

Matching criteria

There are many matching criteria against which a packet can be tested. Refer to *IPv4* and *IPv6* matching criteria for more details.

Since bridges operate at layer 2, both matchers for IPv4 and IPv6 are supported in bridge firewall configuration. Same applies to firewall groups.

Same specific matching criteria that can be used in bridge firewall are described in this section:

```
set firewall bridge forward filter rule <1-999999> ethernet-type [802.1q | 802.1ad | arp | ipv4 | ipv6]
```

```
set firewall bridge input filter rule <1-999999> ethernet-type [802.1q | 802.1ad | arp | ipv4 | ipv6]
```

```
set firewall bridge output filter rule <1-999999> ethernet-type [802.1q | 802.1ad | arp | ipv4 | ipv6]
```

```
set firewall bridge prerouting filter rule <1-999999> ethernet-type [802.1q | 802.1ad | arp | ipv4 | ipv6]
```

```
set firewall bridge name <name> rule <1-999999> ethernet-type [802.1q | 802.1ad | arp | ipv4 | ipv6]
```

Match based on the Ethernet type of the packet.

```
set firewall bridge forward filter rule <1-999999> vlan ethernet-type [802.1q | 802.1ad | arp | ipv4 | ipv6]
```

```
set firewall bridge input filter rule <1-999999> vlan ethernet-type [802.1q | 802.1ad | arp | ipv4 | ipv6]
```

```
set firewall bridge output filter rule <1-999999> vlan ethernet-type [802.1q | 802.1ad | arp | ipv4 | ipv6]
```

```
set firewall bridge prerouting filter rule <1-999999> vlan ethernet-type [802.1q | 802.1ad | arp | ipv4 | ipv6]
```

```
set firewall bridge name <name> rule <1-999999> vlan ethernet-type [802.1q | 802.1ad | arp | ipv4 | ipv6]
```

Match based on the Ethernet type of the packet when it is VLAN tagged.

```
set firewall bridge forward filter rule <1-999999> vlan id <0-4096>
```

```
set firewall bridge input filter rule <1-999999> vlan id <0-4096>
```

```
set firewall bridge output filter rule <1-999999> vlan id <0-4096>
```

```
set firewall bridge prerouting filter rule <1-999999> vlan id <0-4096>
```

```
set firewall bridge name <name> rule <1-999999> vlan id <0-4096>
```

Match based on VLAN identifier. Range is also supported.

```
set firewall bridge forward filter rule <1-999999> vlan priority <0-7>
```

```
set firewall bridge input filter rule <1-999999> vlan priority <0-7>
```

```
set firewall bridge output filter rule <1-999999> vlan priority <0-7>
```

```
set firewall bridge prerouting filter rule <1-999999> vlan priority <0-7>
```

```
set firewall bridge name <name> rule <1-999999> vlan priority <0-7>
```

Match based on VLAN priority (Priority Code Point - PCP). Range is also supported.

Packet Modifications

Starting from **VyOS-1.5-rolling-202410060007**, the firewall can modify packets before they are sent out. This feature provides more flexibility in packet handling.

```
set firewall bridge [prerouting | forward | output] filter rule <1-999999> set dscp  
<0-63>
```

Set a specific value of Differentiated Services Codepoint (DSCP).

```
set firewall bridge [prerouting | forward | output] filter rule <1-999999> set mark  
<1-2147483647>
```

Set a specific packet mark value.

```
set firewall bridge [prerouting | forward | output] filter rule <1-999999> set tcp-mss  
<500-1460>
```

Set the TCP-MSS (TCP maximum segment size) for the connection.

```
set firewall bridge [prerouting | forward | output] filter rule <1-999999> set ttl  
<0-255>
```

Set the TTL (Time to Live) value.

```
set firewall bridge [prerouting | forward | output] filter rule <1-999999> set hop-limit  
<0-255>
```

Set hop limit value.

```
set firewall bridge [forward | output] filter rule <1-999999> set connection-mark  
<0-2147483647>
```

Set connection mark value.

Use IP firewall

By default, for switched traffic, only the rules defined under `set firewall bridge` are applied. There are two global-options that can be configured in order to force deeper analysis of the packet on the IP layer. These options are:

```
set firewall global-options apply-to-bridged-traffic ipv4
```

This command enables the IPv4 firewall for bridged traffic. If this option is used, packets are also parsed by rules defined in `set firewall ipv4 ...`

```
set firewall global-options apply-to-bridged-traffic ipv6
```

This command enables the IPv6 firewall for bridged traffic. If this option is used, packets are also parsed by rules defined in `set firewall ipv6 ...`

Operation-mode Firewall

Rule-set overview

In this section you can find all useful firewall op-mode commands. General commands for firewall configuration, counter and statistics:

```
show firewall
```

```
show firewall summary
```



```
show firewall statistics
```

And, to print only bridge firewall information:

```
show firewall bridge
```

```
show firewall bridge forward filter
```

```
show firewall bridge forward filter rule <rule>
```

```
show firewall bridge name <name>
```

```
show firewall bridge name <name> rule <rule>
```

Show Firewall log

```
show log firewall
```

```
show log firewall bridge
```

```
show log firewall bridge forward
```

```
show log firewall bridge forward filter
```

```
show log firewall bridge name <name>
```

```
show log firewall bridge forward filter rule <rule>
```

```
show log firewall bridge name <name> rule <rule>
```

Show the logs of all firewall; show all bridge firewall logs; show all logs for forward hook; show all logs for forward hook and priority filter; show all logs for particular custom chain; show logs for specific Rule-Set.

Example

Configuration example:

```
set firewall bridge forward filter default-action 'drop'
set firewall bridge forward filter default-log
set firewall bridge forward filter rule 10 action 'continue'
set firewall bridge forward filter rule 10 inbound-interface name 'eth2'
set firewall bridge forward filter rule 10 vlan id '22'
set firewall bridge forward filter rule 20 action 'drop'
set firewall bridge forward filter rule 20 inbound-interface group 'TRUNK-RIGHT'
set firewall bridge forward filter rule 20 vlan id '60'
set firewall bridge forward filter rule 30 action 'jump'
set firewall bridge forward filter rule 30 jump-target 'TEST'
set firewall bridge forward filter rule 30 outbound-interface name '!eth1'
set firewall bridge forward filter rule 35 action 'accept'
set firewall bridge forward filter rule 35 vlan id '11'
set firewall bridge forward filter rule 40 action 'continue'
set firewall bridge forward filter rule 40 destination mac-address '66:55:44:33:22:11'
set firewall bridge forward filter rule 40 source mac-address '11:22:33:44:55:66'
set firewall bridge name TEST default-action 'accept'
set firewall bridge name TEST default-log
set firewall bridge name TEST rule 10 action 'continue'
set firewall bridge name TEST rule 10 log
set firewall bridge name TEST rule 10 vlan priority '0'
```

And op-mode commands:

```
vyos@BRI:~$ show firewall bridge
Rulesets bridge Information

-----
bridge Firewall "forward filter"

Rule      Action    Protocol    Packets    Bytes    Conditions
-----
10        continue  all         0           0        iifname "eth2" vlan id 22  continue
20        drop      all         0           0        iifname @I_TRUNK-RIGHT vlan id 60
30        jump      all         2130        170688   oifname != "eth1"  jump NAME_TEST
35        accept    all         2080        168616   vlan id 11  accept
40        continue  all         0           0        ether daddr 66:55:44:33:22:11 ether_
↳saddr 11:22:33:44:55:66  continue
default   drop      all         0           0

-----
bridge Firewall "name TEST"

Rule      Action    Protocol    Packets    Bytes    Conditions
-----
10        continue  all         2130        170688   vlan pcp 0  prefix "[bri-NAM-TEST-10-
↳C]" continue
default   accept    all         2130        170688

vyos@BRI:~$
vyos@BRI:~$ show firewall bridge name TEST
Ruleset Information

-----
bridge Firewall "name TEST"

Rule      Action    Protocol    Packets    Bytes    Conditions
-----
10        continue  all         2130        170688   vlan pcp 0  prefix "[bri-NAM-TEST-10-
↳C]" continue
default   accept    all         2130        170688

vyos@BRI:~$
```

Inspect logs:

```
vyos@BRI:~$ show log firewall bridge
Dec 05 14:37:47 kernel: [bri-NAM-TEST-10-C]IN=eth1 OUT=eth2 ARP HTYPE=1 PTYPE=0x0800_
↳OPCODE=1 MACSRC=50:00:00:04:00:00 IPSRC=10.11.11.101 MACDST=00:00:00:00:00:00 IPDST=10.
↳11.11.102
Dec 05 14:37:48 kernel: [bri-NAM-TEST-10-C]IN=eth1 OUT=eth2 ARP HTYPE=1 PTYPE=0x0800_
↳OPCODE=1 MACSRC=50:00:00:04:00:00 IPSRC=10.11.11.101 MACDST=00:00:00:00:00:00 IPDST=10.
↳11.11.102
```

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```
Dec 05 14:37:49 kernel: [bri-NAM-TEST-10-C]IN=eth1 OUT=eth2 ARP HTYPE=1 PTYPE=0x0800
↳ OPCODE=1 MACSRC=50:00:00:04:00:00 IPSRC=10.11.11.101 MACDST=00:00:00:00:00:00 IPDST=10.
↳ 11.11.102
...
vyos@BRI:~$ show log firewall bridge forward filter
Dec 05 14:42:22 kernel: [bri-FWD-filter-default-D]IN=eth2 OUT=eth1
↳ MAC=33:33:00:00:00:16:50:00:00:06:00:00:86:dd
↳ SRC=0000:0000:0000:0000:0000:0000:0000:0000
↳ DST=ff02:0000:0000:0000:0000:0000:0000:0016 LEN=96 TC=0 HOPLIMIT=1 FLOWLBL=0
↳ PROTO=ICMPv6 TYPE=143 CODE=0
Dec 05 14:42:22 kernel: [bri-FWD-filter-default-D]IN=eth2 OUT=eth1
↳ MAC=33:33:00:00:00:16:50:00:00:06:00:00:86:dd
↳ SRC=0000:0000:0000:0000:0000:0000:0000:0000
↳ DST=ff02:0000:0000:0000:0000:0000:0000:0016 LEN=96 TC=0 HOPLIMIT=1 FLOWLBL=0
↳ PROTO=ICMPv6 TYPE=143 CODE=0
```

7.2.4 IPv4 Firewall Configuration

Overview

This section provides information on IPv4 firewall configuration and appropriate operation-mode commands. This section covers the following configuration commands:

set firewall ipv4 ...

To learn about the general traffic flow in VyOS firewalls, see [Firewall](#).

```
- set firewall
  * ipv4
    - forward
      + filter
    - input
      + filter
    - output
      + filter
      + raw
    - prerouting
      + raw
    - name
      + custom_name
```

First, the router receives all traffic and processes it in the **prerouting** stage.

This stage includes:

- **Firewall Prerouting:** commands found under `set firewall ipv4 prerouting raw ...`
- *Conntrack Ignore:* `set system conntrack ignore ipv4...`
- *Policy Route:* commands found under `set policy route ...`
- *Destination NAT:* commands found under `set nat destination ...`

For transit traffic, which is received by the router and forwarded, the base chain is **forward**. The following is a simplified packet flow diagram for transit traffic:

The base firewall chain for configuring filtering rules for transit traffic is `set firewall ipv4 forward filter ...`, which occurs in stage 5, highlighted in red.

For traffic to the router itself, the base chain is **input**. For traffic the router originates, the base chain is **output**. A simplified packet flow diagram is shown next, which shows the path for traffic destined to the router itself and traffic the router generates (starting from circle number 6):

The base chain for traffic towards the router is `set firewall ipv4 input filter ...`

The base chain for traffic the router generates is `set firewall ipv4 output ...`, where two sub-chains are available: **filter** and **raw**:

- **Output Prerouting**: `set firewall ipv4 output raw ....` As described in **Prerouting**, the system processes rules in this section before the connection tracking subsystem.
- **Output Filter**: `set firewall ipv4 output filter ....` The system processes rules in this section after the connection tracking subsystem.

Note: Important note about default-actions: If you do not define a default action for a base chain, the system sets the default action to **accept** for that chain. For custom chains, if you do not define a default action, the system sets the default-action to **drop**.

You can create custom firewall chains using the following commands: `set firewall ipv4 name <name> ....` To use a custom chain, you must define a rule with the **action jump** and the appropriate **target** in a base chain.

Firewall - IPv4 Rules

Each firewall rule has a number, an action to apply if the rule matches, and the ability to specify multiple matching criteria. Packets traverse rules numbered 1-999999, so order is crucial. The system executes the rule action at the first match.

Actions

If you define a rule, you must define an action for it. The action tells the firewall what to do if all the criteria you define for that rule are met.

The action can be:

- **accept**: Accept the packet.
- **continue**: Continue parsing the next rule.
- **drop**: Drop the packet.
- **reject**: Reject the packet.
- **jump**: Jump to another custom chain.
- **return**: Return from the current chain and continue at the next rule of the last chain.
- **queue**: Enqueue packet to userspace.
- **synproxy**: Synproxy the packet.

```
set firewall ipv4 forward filter rule <1-999999> action [accept | continue | drop | jump  
| queue | reject | return | synproxy]
```

```
set firewall ipv4 input filter rule <1-999999> action [accept | continue | drop | jump |
queue | reject | return | synproxy]
set firewall ipv4 output filter rule <1-999999> action [accept | continue | drop | jump |
queue | reject | return]
set firewall ipv4 name <name> rule <1-999999> action [accept | continue | drop | jump |
queue | reject | return]
```

This required setting defines the action of the current rule. If you set the action to jump, you must also specify a jump-target.

```
set firewall ipv4 forward filter rule <1-999999> jump-target <text>
set firewall ipv4 input filter rule <1-999999> jump-target <text>
set firewall ipv4 output filter rule <1-999999> jump-target <text>
set firewall ipv4 name <name> rule <1-999999> jump-target <text>
```

Use this command only when the action is set to jump. Specify the jump target.

```
set firewall ipv4 forward filter rule <1-999999> queue <0-65535>
set firewall ipv4 input filter rule <1-999999> queue <0-65535>
set firewall ipv4 output filter rule <1-999999> queue <0-65535>
set firewall ipv4 name <name> rule <1-999999> queue <0-65535>
```

Use this command only when the action is set to queue. Specify the queue target to use. Queue range is also supported.

```
set firewall ipv4 forward filter rule <1-999999> queue-options bypass
set firewall ipv4 input filter rule <1-999999> queue-options bypass
set firewall ipv4 output filter rule <1-999999> queue-options bypass
set firewall ipv4 name <name> rule <1-999999> queue-options bypass
```

Use this command only when the action is set to queue. Allow the packet to pass through the firewall when no userspace software is connected to the queue.

```
set firewall ipv4 forward filter rule <1-999999> queue-options fanout
set firewall ipv4 input filter rule <1-999999> queue-options fanout
set firewall ipv4 output filter rule <1-999999> queue-options fanout
set firewall ipv4 name <name> rule <1-999999> queue-options fanout
```

Use this command only when the action is set to queue. Distribute packets between several queues.

Also, **default-action** is an action that applies when a packet does not match any rule in its chain. For base chains, possible options for **default-action** are **accept** or **drop**.

```
set firewall ipv4 forward filter default-action [accept | drop]
set firewall ipv4 input filter default-action [accept | drop]
set firewall ipv4 output filter default-action [accept | drop]
set firewall ipv4 name <name> default-action [accept | drop | jump | queue | reject |
return]
```

This command sets the default action of the rule-set if a packet does not match the criteria of any rule. If you set the default-action to `jump`, you must also specify `default-jump-target`. Note that for base chains, you can set the default action only to `accept` or `drop`, while on custom chains, more actions are available.

```
set firewall ipv4 name <name> default-jump-target <text>
```

Use this command only when you set `default-action` to `jump`. Specify the jump target for the default rule.

Note: Important note about default-actions: If you do not define a default action for a base chain, the system sets the default action to `accept` for that chain. For custom chains, if you do not define a default action, the system sets the default-action to `drop`.

Firewall Logs

You can enable logging for every single firewall rule. If you enable logging, you can define other log options.

```
set firewall ipv4 forward filter rule <1-999999> log
set firewall ipv4 input filter rule <1-999999> log
set firewall ipv4 output filter rule <1-999999> log
set firewall ipv4 name <name> rule <1-999999> log
```

Enable logging for the matched packet. If this command is not present, then logging is not enabled.

```
set firewall ipv4 forward filter default-log
set firewall ipv4 input filter default-log
set firewall ipv4 output filter default-log
set firewall ipv4 name <name> default-log
```

Use this command to enable logging of the default action on the specified chain.

```
set firewall ipv4 forward filter rule <1-999999> log-options level [emerg | alert | crit
| err | warn | notice | info | debug]
set firewall ipv4 input filter rule <1-999999> log-options level [emerg | alert | crit |
err | warn | notice | info | debug]
set firewall ipv4 output filter rule <1-999999> log-options level [emerg | alert | crit |
err | warn | notice | info | debug]
set firewall ipv4 name <name> rule <1-999999> log-options level [emerg | alert | crit |
err | warn | notice | info | debug]
```

Define the log level. Only applicable if you enable rule logging.

```
set firewall ipv4 forward filter rule <1-999999> log-options group <0-65535>
set firewall ipv4 input filter rule <1-999999> log-options group <0-65535>
set firewall ipv4 output filter rule <1-999999> log-options group <0-65535>
set firewall ipv4 name <name> rule <1-999999> log-options group <0-65535>
```

Define the log group to send messages to. Only applicable if you enable rule logging.

```
set firewall ipv4 forward filter rule <1-999999> log-options snapshot-length <0-9000>
set firewall ipv4 input filter rule <1-999999> log-options snapshot-length <0-9000>
```

```
set firewall ipv4 output filter rule <1-999999> log-options snapshot-length <0-9000>
```

```
set firewall ipv4 name <name> rule <1-999999> log-options snapshot-length <0-9000>
```

Define the length of packet payload to include in a netlink message. Only applicable if you enable rule logging and define the log group.

```
set firewall ipv4 forward filter rule <1-999999> log-options queue-threshold <0-65535>
```

```
set firewall ipv4 input filter rule <1-999999> log-options queue-threshold <0-65535>
```

```
set firewall ipv4 output filter rule <1-999999> log-options queue-threshold <0-65535>
```

```
set firewall ipv4 name <name> rule <1-999999> log-options queue-threshold <0-65535>
```

Define the number of packets to queue inside the kernel before sending them to userspace. Only applicable if you enable rule logging and define the log group.

Firewall Description

You can add a description for reference for every single rule and for every defined custom chain.

```
set firewall ipv4 name <name> description <text>
```

Provide a rule-set description for a custom firewall chain.

```
set firewall ipv4 forward filter rule <1-999999> description <text>
```

```
set firewall ipv4 input filter rule <1-999999> description <text>
```

```
set firewall ipv4 output filter rule <1-999999> description <text>
```

```
set firewall ipv4 name <name> rule <1-999999> description <text>
```

Provide a description for each rule.

Rule Status

When you define a rule, it is enabled by default. In some cases, it is useful to disable the rule rather than removing it.

```
set firewall ipv4 forward filter rule <1-999999> disable
```

```
set firewall ipv4 input filter rule <1-999999> disable
```

```
set firewall ipv4 output filter rule <1-999999> disable
```

```
set firewall ipv4 name <name> rule <1-999999> disable
```

Command for disabling a rule but keeping it in the configuration.

Matching criteria

There are a lot of matching criteria against which the packet can be tested.

```
set firewall ipv4 forward filter rule <1-999999> connection-status nat [destination | source]
```

```
set firewall ipv4 input filter rule <1-999999> connection-status nat [destination | source]
```

```
set firewall ipv4 output filter rule <1-999999> connection-status nat [destination | source]
```

```
set firewall ipv4 name <name> rule <1-999999> connection-status nat [destination |
source]
```

Match based on nat connection status.

```
set firewall ipv4 forward filter rule <1-999999> connection-mark <1-2147483647>
set firewall ipv4 input filter rule <1-999999> connection-mark <1-2147483647>
set firewall ipv4 output filter rule <1-999999> connection-mark <1-2147483647>
set firewall ipv4 name <name> rule <1-999999> connection-mark <1-2147483647>
```

Match based on connection mark.

```
set firewall ipv4 forward filter rule <1-999999> conntrack-helper <module>
set firewall ipv4 input filter rule <1-999999> conntrack-helper <module>
set firewall ipv4 output filter rule <1-999999> conntrack-helper <module>
set firewall ipv4 name <name> rule <1-999999> conntrack-helper <module>
```

Match based on connection tracking protocol helper module to secure use of that helper module. See below for possible completions <module>.

Possible completions:

ftp	Related traffic from FTP helper
h323	Related traffic from H.323 helper
pptp	Related traffic from PPTP helper
nfs	Related traffic from NFS helper
sip	Related traffic from SIP helper
tftp	Related traffic from TFTP helper
sqlnet	Related traffic from SQLNet helper

```
set firewall ipv4 forward filter rule <1-999999> source address [address | addressrange |
CIDR]
```

```
set firewall ipv4 input filter rule <1-999999> source address [address | addressrange |
CIDR]
```

```
set firewall ipv4 output filter rule <1-999999> source address [address | addressrange |
CIDR]
```

```
set firewall ipv4 name <name> rule <1-999999> source address [address | addressrange |
CIDR]
```

```
set firewall ipv4 forward filter rule <1-999999> destination address [address |
addressrange | CIDR]
```

```
set firewall ipv4 input filter rule <1-999999> destination address [address |
addressrange | CIDR]
```

```
set firewall ipv4 output filter rule <1-999999> destination address [address |
addressrange | CIDR]
```

```
set firewall ipv4 name <name> rule <1-999999> destination address [address | addressrange
| CIDR]
```

Match criteria based on source and/or destination address. This is similar to the network groups part, but here you are able to negate the matching addresses.


```
set firewall ipv4 name F00 rule 50 source address 192.0.2.10-192.0.2.11
# with a '!' the rule match everything except the specified subnet
set firewall ipv4 name F00 rule 51 source address !203.0.113.0/24
```

```
set firewall ipv4 forward filter rule <1-999999> source address-mask [address]
set firewall ipv4 input filter rule <1-999999> source address-mask [address]
set firewall ipv4 output filter rule <1-999999> source address-mask [address]
set firewall ipv4 name <name> rule <1-999999> source address-mask [address]
set firewall ipv4 forward filter rule <1-999999> destination address-mask [address]
set firewall ipv4 input filter rule <1-999999> destination address-mask [address]
set firewall ipv4 output filter rule <1-999999> destination address-mask [address]
set firewall ipv4 name <name> rule <1-999999> destination address-mask [address]
```

An arbitrary netmask can be applied to mask addresses to only match against a specific portion.

This functions for both individual addresses and address groups.

```
# Match any IPv4 address with `11` as the 2nd octet and `13` as the forth octet
set firewall ipv4 name F00 rule 100 destination address 0.11.0.13
set firewall ipv4 name F00 rule 100 destination address-mask 0.255.0.255
```

```
set firewall ipv4 forward filter rule <1-999999> source fqdn <fqdn>
set firewall ipv4 input filter rule <1-999999> source fqdn <fqdn>
set firewall ipv4 output filter rule <1-999999> source fqdn <fqdn>
set firewall ipv4 name <name> rule <1-999999> source fqdn <fqdn>
set firewall ipv4 forward filter rule <1-999999> destination fqdn <fqdn>
set firewall ipv4 input filter rule <1-999999> destination fqdn <fqdn>
set firewall ipv4 output filter rule <1-999999> destination fqdn <fqdn>
set firewall ipv4 name <name> rule <1-999999> destination fqdn <fqdn>
```

Specify a Fully Qualified Domain Name as source/destination to match. Ensure that the router is able to resolve this dns query.

```
set firewall ipv4 forward filter rule <1-999999> source geoip country-code <country>
set firewall ipv4 input filter rule <1-999999> source geoip country-code <country>
set firewall ipv4 output filter rule <1-999999> source geoip country-code <country>
set firewall ipv4 name <name> rule <1-999999> source geoip country-code <country>
set firewall ipv4 forward filter rule <1-999999> source geoip asn <1-4294967294>
set firewall ipv4 input filter rule <1-999999> source geoip asn <1-4294967294>
set firewall ipv4 output filter rule <1-999999> source geoip asn <1-4294967294>
set firewall ipv4 name <name> rule <1-999999> source geoip asn <1-4294967294>
set firewall ipv4 forward filter rule <1-999999> destination geoip country-code <country>
set firewall ipv4 input filter rule <1-999999> destination geoip country-code <country>
set firewall ipv4 output filter rule <1-999999> destination geoip country-code <country>
```

```

set firewall ipv4 name <name> rule <1-999999> destination geoip country-code <country>
set firewall ipv4 forward filter rule <1-999999> destination geoip asn <1-4294967294>
set firewall ipv4 input filter rule <1-999999> destination geoip asn <1-4294967294>
set firewall ipv4 output filter rule <1-999999> destination geoip asn <1-4294967294>
set firewall ipv4 name <name> rule <1-999999> destination geoip asn <1-4294967294>
set firewall ipv4 forward filter rule <1-999999> source geoip inverse-match
set firewall ipv4 input filter rule <1-999999> source geoip inverse-match
set firewall ipv4 output filter rule <1-999999> source geoip inverse-match
set firewall ipv4 name <name> rule <1-999999> source geoip inverse-match
set firewall ipv4 forward filter rule <1-999999> destination geoip inverse-match
set firewall ipv4 input filter rule <1-999999> destination geoip inverse-match
set firewall ipv4 output filter rule <1-999999> destination geoip inverse-match
set firewall ipv4 name <name> rule <1-999999> destination geoip inverse-match

```

Match IP addresses based on its geolocation. More info: [geoip matching](#). Use country-code or asn to match based on geography or origin network. Use inverse-match to match anything except the given country code or ASN.

Data is provided by DB-IP.com under CC-BY-4.0 license. Attribution required, permits redistribution so we can include a database in images (~3MB compressed). Includes cron script (manually callable by op-mode update geoip) to keep database and rules updated.

```

set firewall ipv4 forward filter rule <1-999999> source mac-address <mac-address>
set firewall ipv4 input filter rule <1-999999> source mac-address <mac-address>
set firewall ipv4 output filter rule <1-999999> source mac-address <mac-address>
set firewall ipv4 name <name> rule <1-999999> source mac-address <mac-address>

```

You can only specify a source mac-address to match.

```

set firewall ipv4 input filter rule 100 source mac-address 00:53:00:11:22:33
set firewall ipv4 input filter rule 101 source mac-address !00:53:00:aa:12:34

```

```

set firewall ipv4 forward filter rule <1-999999> source port [1-65535 | portname |
start-end]
set firewall ipv4 input filter rule <1-999999> source port [1-65535 | portname |
start-end]
set firewall ipv4 output filter rule <1-999999> source port [1-65535 | portname |
start-end]
set firewall ipv4 name <name> rule <1-999999> source port [1-65535 | portname |
start-end]
set firewall ipv4 forward filter rule <1-999999> destination port [1-65535 | portname |
start-end]
set firewall ipv4 input filter rule <1-999999> destination port [1-65535 | portname |
start-end]
set firewall ipv4 output filter rule <1-999999> destination port [1-65535 | portname |
start-end]

```

```
set firewall ipv4 name <name> rule <1-999999> destination port [1-65535 | portname |
start-end]
```

A port can be set by number or name as defined in `/etc/services`.

```
set firewall ipv4 forward filter rule 10 source port '22'
set firewall ipv4 forward filter rule 11 source port '!http'
set firewall ipv4 forward filter rule 12 source port 'https'
```

Multiple source ports can be specified as a comma-separated list. The whole list can also be “negated” using `!`. For example:

```
set firewall ipv4 forward filter rule <1-999999> source group address-group <name | !
name>
set firewall ipv4 input filter rule <1-999999> source group address-group <name | !name>
set firewall ipv4 output filter rule <1-999999> source group address-group <name | !name>
set firewall ipv4 name <name> rule <1-999999> source group address-group <name | !name>
set firewall ipv4 forward filter rule <1-999999> destination group address-group <name |
!name>
set firewall ipv4 input filter rule <1-999999> destination group address-group <name |
!name>
set firewall ipv4 output filter rule <1-999999> destination group address-group <name |
!name>
set firewall ipv4 name <name> rule <1-999999> destination group address-group <name |
!name>
```

Use a specific address-group. Prepending the character `!` to invert the criteria to match is also supported.

```
set firewall ipv4 forward filter rule <1-999999> source group dynamic-address-group <name
| !name>
set firewall ipv4 input filter rule <1-999999> source group dynamic-address-group <name |
!name>
set firewall ipv4 output filter rule <1-999999> source group dynamic-address-group <name
| !name>
set firewall ipv4 name <name> rule <1-999999> source group dynamic-address-group <name |
!name>
set firewall ipv4 forward filter rule <1-999999> destination group dynamic-address-group
<name | !name>
set firewall ipv4 input filter rule <1-999999> destination group dynamic-address-group
<name | !name>
set firewall ipv4 output filter rule <1-999999> destination group dynamic-address-group
<name | !name>
set firewall ipv4 name <name> rule <1-999999> destination group dynamic-address-group
<name | !name>
```

Use a specific dynamic-address-group. Prepending the character `!` to invert the criteria to match is also supported.

```
set firewall ipv4 forward filter rule <1-999999> source group network-group <name | !
name>
```

```

set firewall ipv4 input filter rule <1-999999> source group network-group <name | !name>
set firewall ipv4 output filter rule <1-999999> source group network-group <name | !name>
set firewall ipv4 name <name> rule <1-999999> source group network-group <name | !name>
set firewall ipv4 forward filter rule <1-999999> destination group network-group <name | !name>
set firewall ipv4 input filter rule <1-999999> destination group network-group <name | !name>
set firewall ipv4 output filter rule <1-999999> destination group network-group <name | !name>
set firewall ipv4 name <name> rule <1-999999> destination group network-group <name | !name>

```

Use a specific network-group. Prepending the character ! to invert the criteria to match is also supported.

```

set firewall ipv4 forward filter rule <1-999999> source group port-group <name | !name>
set firewall ipv4 input filter rule <1-999999> source group port-group <name | !name>
set firewall ipv4 output filter rule <1-999999> source group port-group <name | !name>
set firewall ipv4 name <name> rule <1-999999> source group port-group <name | !name>
set firewall ipv4 forward filter rule <1-999999> destination group port-group <name | !name>
set firewall ipv4 input filter rule <1-999999> destination group port-group <name | !name>
set firewall ipv4 output filter rule <1-999999> destination group port-group <name | !name>
set firewall ipv4 name <name> rule <1-999999> destination group port-group <name | !name>

```

Use a specific port-group. Prepending the character ! to invert the criteria to match is also supported.

```

set firewall ipv4 forward filter rule <1-999999> source group domain-group <name | !name>
set firewall ipv4 input filter rule <1-999999> source group domain-group <name | !name>
set firewall ipv4 output filter rule <1-999999> source group domain-group <name | !name>
set firewall ipv4 name <name> rule <1-999999> source group domain-group <name | !name>
set firewall ipv4 forward filter rule <1-999999> destination group domain-group <name | !name>
set firewall ipv4 input filter rule <1-999999> destination group domain-group <name | !name>
set firewall ipv4 output filter rule <1-999999> destination group domain-group <name | !name>
set firewall ipv4 name <name> rule <1-999999> destination group domain-group <name | !name>

```

Use a specific domain-group. Prepending the character ! to invert the criteria to match is also supported.

```

set firewall ipv4 forward filter rule <1-999999> source group mac-group <name | !name>
set firewall ipv4 input filter rule <1-999999> source group mac-group <name | !name>
set firewall ipv4 output filter rule <1-999999> source group mac-group <name | !name>

```

```

set firewall ipv4 name <name> rule <1-999999> source group mac-group <name | !name>
set firewall ipv4 forward filter rule <1-999999> destination group mac-group <name | !name>
set firewall ipv4 input filter rule <1-999999> destination group mac-group <name | !name>
set firewall ipv4 output filter rule <1-999999> destination group mac-group <name | !name>
set firewall ipv4 name <name> rule <1-999999> destination group mac-group <name | !name>

```

Use a specific mac-group. Prepending the character ! to invert the criteria to match is also supported.

```

set firewall ipv4 forward filter rule <1-999999> dscp [0-63 | start-end]
set firewall ipv4 input filter rule <1-999999> dscp [0-63 | start-end]
set firewall ipv4 output filter rule <1-999999> dscp [0-63 | start-end]
set firewall ipv4 name <name> rule <1-999999> dscp [0-63 | start-end]
set firewall ipv4 forward filter rule <1-999999> dscp-exclude [0-63 | start-end]
set firewall ipv4 input filter rule <1-999999> dscp-exclude [0-63 | start-end]
set firewall ipv4 output filter rule <1-999999> dscp-exclude [0-63 | start-end]
set firewall ipv4 name <name> rule <1-999999> dscp-exclude [0-63 | start-end]

```

Match based on dscp value.

```

set firewall ipv4 forward filter rule <1-999999> fragment [match-frag | match-non-frag]
set firewall ipv4 input filter rule <1-999999> fragment [match-frag | match-non-frag]
set firewall ipv4 output filter rule <1-999999> fragment [match-frag | match-non-frag]
set firewall ipv4 name <name> rule <1-999999> fragment [match-frag | match-non-frag]

```

Match based on fragmentation.

```

set firewall ipv4 forward filter rule <1-999999> icmp [code | type] <0-255>
set firewall ipv4 input filter rule <1-999999> icmp [code | type] <0-255>
set firewall ipv4 output filter rule <1-999999> icmp [code | type] <0-255>
set firewall ipv4 name <name> rule <1-999999> icmp [code | type] <0-255>

```

Match based on icmp code and type.

```

set firewall ipv4 forward filter rule <1-999999> icmp type-name <text>
set firewall ipv4 input filter rule <1-999999> icmp type-name <text>
set firewall ipv4 output filter rule <1-999999> icmp type-name <text>
set firewall ipv4 name <name> rule <1-999999> icmp type-name <text>

```

Match based on icmp type-name. Use tab for information about what **type-name** criteria are supported.

```

set firewall ipv4 forward filter rule <1-999999> inbound-interface name <iface>
set firewall ipv4 input filter rule <1-999999> inbound-interface name <iface>
set firewall ipv4 name <name> rule <1-999999> inbound-interface name <iface>

```

Match based on inbound interface. Wildcard * is supported. For example: eth2*. Prepend the character ! to invert the criteria. For example: !eth2

Note: If an interface is attached to a non-default vrf, when using **inbound-interface**, the vrf name must be used. For example `set firewall ipv4 forward filter rule 10 inbound-interface name MGMT`

```
set firewall ipv4 forward filter rule <1-999999> inbound-interface group <iface_group>
set firewall ipv4 input filter rule <1-999999> inbound-interface group <iface_group>
set firewall ipv4 name <name> rule <1-999999> inbound-interface group <iface_group>
```

Match based on the inbound interface group. Prepend the character `!` to invert the criteria. For example, `!IFACE_GROUP`

```
set firewall ipv4 forward filter rule <1-999999> outbound-interface name <iface>
set firewall ipv4 output filter rule <1-999999> outbound-interface name <iface>
set firewall ipv4 name <name> rule <1-999999> outbound-interface name <iface>
```

Match based on outbound interface. Wildcard `*` is supported. For example: `eth2*`. Prepend the character `!` to invert the criteria. For example: `!eth2`

Note: If an interface is attached to a non-default vrf, when using **outbound-interface**, the real interface name must be used. For example `set firewall ipv4 forward filter rule 10 outbound-interface name eth0`

```
set firewall ipv4 forward filter rule <1-999999> outbound-interface group <iface_group>
set firewall ipv4 output filter rule <1-999999> outbound-interface group <iface_group>
set firewall ipv4 name <name> rule <1-999999> outbound-interface group <iface_group>
```

Match based on outbound interface group. Prepend the character `!` to invert the criteria. For example: `!IFACE_GROUP`

```
set firewall ipv4 forward filter rule <1-999999> ipsec [match-ipsec-in | match-ipsec-out
| match-none-in | match-none-out]
set firewall ipv4 input filter rule <1-999999> ipsec [match-ipsec-in | match-none-in]
set firewall ipv4 output filter rule <1-999999> ipsec [match-ipsec-out | match-none-out]
set firewall ipv4 name <name> rule <1-999999> ipsec [match-ipsec-in | match-ipsec-out |
match-none-in | match-none-out]
```

Match based on ipsec.

```
set firewall ipv4 forward filter rule <1-999999> limit burst <0-4294967295>
set firewall ipv4 input filter rule <1-999999> limit burst <0-4294967295>
set firewall ipv4 output filter rule <1-999999> limit burst <0-4294967295>
set firewall ipv4 name <name> rule <1-999999> limit burst <0-4294967295>
```

Match based on the maximum number of packets to allow in excess of rate.

```
set firewall ipv4 forward filter rule <1-999999> limit rate <text>
set firewall ipv4 input filter rule <1-999999> limit rate <text>
set firewall ipv4 output filter rule <1-999999> limit rate <text>
set firewall ipv4 name <name> rule <1-999999> limit rate <text>
```

Specify the maximum average rate as **integer/unit**. For example: `5/minutes`

```

set firewall ipv4 forward filter rule <1-999999> packet-length <text>
set firewall ipv4 input filter rule <1-999999> packet-length <text>
set firewall ipv4 output filter rule <1-999999> packet-length <text>
set firewall ipv4 name <name> rule <1-999999> packet-length <text>
set firewall ipv4 forward filter rule <1-999999> packet-length-exclude <text>
set firewall ipv4 input filter rule <1-999999> packet-length-exclude <text>
set firewall ipv4 output filter rule <1-999999> packet-length-exclude <text>
set firewall ipv4 name <name> rule <1-999999> packet-length-exclude <text>

```

Match based on packet length. Specify multiple values from 1 to 65535 and ranges.

```

set firewall ipv4 forward filter rule <1-999999> packet-type [broadcast | host |
multicast | other]
set firewall ipv4 input filter rule <1-999999> packet-type [broadcast | host | multicast
| other]
set firewall ipv4 output filter rule <1-999999> packet-type [broadcast | host | multicast
| other]
set firewall ipv4 name <name> rule <1-999999> packet-type [broadcast | host | multicast |
other]

```

Match based on the packet type.

```

set firewall ipv4 forward filter rule <1-999999> protocol [<text> | <0-255> | all |
tcp_udp]
set firewall ipv4 input filter rule <1-999999> protocol [<text> | <0-255> | all |
tcp_udp]
set firewall ipv4 output filter rule <1-999999> protocol [<text> | <0-255> | all |
tcp_udp]
set firewall ipv4 name <name> rule <1-999999> protocol [<text> | <0-255> | all | tcp_udp]

```

Match based on protocol number or name as defined in /etc/protocols. Special names are all for all protocols and tcp_udp for TCP and UDP based packets. The ! character negates the selected protocol.

```

set firewall ipv4 forward filter rule 10 protocol tcp_udp
set firewall ipv4 forward filter rule 11 protocol !tcp_udp

```

```

set firewall ipv4 forward filter rule <1-999999> recent count <1-255>
set firewall ipv4 input filter rule <1-999999> recent count <1-255>
set firewall ipv4 output filter rule <1-999999> recent count <1-255>
set firewall ipv4 name <name> rule <1-999999> recent count <1-255>
set firewall ipv4 forward filter rule <1-999999> recent time [second | minute | hour]
set firewall ipv4 input filter rule <1-999999> recent time [second | minute | hour]
set firewall ipv4 output filter rule <1-999999> recent time [second | minute | hour]
set firewall ipv4 name <name> rule <1-999999> recent time [second | minute | hour]

```

Match based on recently seen sources.

```

set firewall ipv4 forward filter rule <1-999999> tcp flags [not] <text>

```

```
set firewall ipv4 input filter rule <1-999999> tcp flags [not] <text>
set firewall ipv4 output filter rule <1-999999> tcp flags [not] <text>
set firewall ipv4 name <name> rule <1-999999> tcp flags [not] <text>
```

Specify TCP flags. Allowed values are ack, cwr, ecn, fin, psh, rst, syn, and urg. Specify multiple values, and use not for inverted selection, as shown in the example.

```
set firewall ipv4 input filter rule 10 tcp flags 'ack'
set firewall ipv4 input filter rule 12 tcp flags 'syn'
set firewall ipv4 input filter rule 13 tcp flags not 'fin'
```

```
set firewall ipv4 forward filter rule <1-999999> state [established | invalid | new |
related]
set firewall ipv4 input filter rule <1-999999> state [established | invalid | new |
related]
set firewall ipv4 output filter rule <1-999999> state [established | invalid | new |
related]
set firewall ipv4 name <name> rule <1-999999> state [established | invalid | new |
related]
```

Match against the state of a packet.

```
set firewall ipv4 forward filter rule <1-999999> time startdate <text>
set firewall ipv4 input filter rule <1-999999> time startdate <text>
set firewall ipv4 output filter rule <1-999999> time startdate <text>
set firewall ipv4 name <name> rule <1-999999> time startdate <text>
set firewall ipv4 forward filter rule <1-999999> time starttime <text>
set firewall ipv4 input filter rule <1-999999> time starttime <text>
set firewall ipv4 output filter rule <1-999999> time starttime <text>
set firewall ipv4 name <name> rule <1-999999> time starttime <text>
set firewall ipv4 forward filter rule <1-999999> time stopdate <text>
set firewall ipv4 input filter rule <1-999999> time stopdate <text>
set firewall ipv4 output filter rule <1-999999> time stopdate <text>
set firewall ipv4 name <name> rule <1-999999> time stopdate <text>
set firewall ipv4 forward filter rule <1-999999> time stoptime <text>
set firewall ipv4 input filter rule <1-999999> time stoptime <text>
set firewall ipv4 output filter rule <1-999999> time stoptime <text>
set firewall ipv4 name <name> rule <1-999999> time stoptime <text>
set firewall ipv4 forward filter rule <1-999999> time weekdays <text>
set firewall ipv4 input filter rule <1-999999> time weekdays <text>
set firewall ipv4 output filter rule <1-999999> time weekdays <text>
set firewall ipv4 name <name> rule <1-999999> time weekdays <text>
```

Time to match the defined rule.


```
set firewall ipv4 forward filter rule <1-999999> ttl <eq | gt | lt> <0-255>
set firewall ipv4 input filter rule <1-999999> ttl <eq | gt | lt> <0-255>
set firewall ipv4 output filter rule <1-999999> ttl <eq | gt | lt> <0-255>
set firewall ipv4 name <name> rule <1-999999> ttl <eq | gt | lt> <0-255>
```

Match the time to live parameter, where ‘eq’ means ‘equal’, ‘gt’ means ‘greater than’, and ‘lt’ means ‘less than’.

Packet Modifications

Starting from **VyOS-1.5-rolling-202410060007**, the firewall can modify packets before sending them out. This feature provides more flexibility in packet handling.

```
set firewall ipv4 prerouting raw rule <1-999999> set dscp <0-63>
set firewall ipv4 forward filter rule <1-999999> set dscp <0-63>
set firewall ipv4 output [filter | raw] rule <1-999999> set dscp <0-63>
```

Set a specific value of Differentiated Services Codepoint (DSCP).

```
set firewall ipv4 prerouting raw rule <1-999999> set mark <1-2147483647>
set firewall ipv4 forward filter rule <1-999999> set mark <1-2147483647>
set firewall ipv4 output [filter | raw] rule <1-999999> set mark <1-2147483647>
```

Set a specific packet mark value.

```
set firewall ipv4 prerouting raw rule <1-999999> set tcp-mss <500-1460>
set firewall ipv4 forward filter rule <1-999999> set tcp-mss <500-1460>
set firewall ipv4 output [filter | raw] rule <1-999999> set tcp-mss <500-1460>
```

Set the TCP-MSS (TCP maximum segment size) for the connection.

```
set firewall ipv4 prerouting raw rule <1-999999> set ttl <0-255>
set firewall ipv4 forward filter rule <1-999999> set ttl <0-255>
set firewall ipv4 output [filter | raw] rule <1-999999> set ttl <0-255>
```

Set the TTL (Time to Live) value.

```
set firewall ipv4 forward filter rule <1-999999> set connection-mark <0-2147483647>
set firewall ipv4 output [filter | raw] rule <1-999999> set connection-mark
<0-2147483647>
```

Set connection mark value.

Synproxy

Synproxy connections

```
set firewall ipv4 [input | forward] filter rule <1-999999> action synproxy
set firewall ipv4 [input | forward] filter rule <1-999999> protocol tcp
set firewall ipv4 [input | forward] filter rule <1-999999> synproxy tcp mss <501-65535>
```

Set the TCP-MSS (maximum segment size) for the connection

```
set firewall ipv4 [input | forward] filter rule <1-999999> synproxy tcp window-scale <1-14>
```

Set the window scale factor for TCP window scaling

Example synproxy

Requirements to enable synproxy:

- Traffic must be symmetric.
- Synproxy relies on syncookies and TCP timestamps, ensure these are enabled.
- Disable conntrack loose track option.

```
set system sysctl parameter net.ipv4.tcp_timestamps value '1'
set system conntrack tcp loose disable
set system conntrack ignore ipv4 rule 10 destination port '8080'
set system conntrack ignore ipv4 rule 10 protocol 'tcp'
set system conntrack ignore ipv4 rule 10 tcp flags syn
set firewall global-options syn-cookies 'enable'
set firewall ipv4 input filter rule 10 action 'synproxy'
set firewall ipv4 input filter rule 10 destination port '8080'
set firewall ipv4 input filter rule 10 inbound-interface name 'eth1'
set firewall ipv4 input filter rule 10 protocol 'tcp'
set firewall ipv4 input filter rule 10 synproxy tcp mss '1460'
set firewall ipv4 input filter rule 10 synproxy tcp window-scale '7'
set firewall ipv4 input filter rule 1000 action 'drop'
set firewall ipv4 input filter rule 1000 state invalid
```

Operation-mode Firewall

Rule-set overview

show firewall

This will show you a basic firewall overview, for all rule-sets, not only for IPv4.

```
vyos@vyos:~$ show firewall
Rulesets Information

-----
ipv4 Firewall "forward filter"

Rule      Action    Protocol    Packets    Bytes    Conditions
-----
20        accept    all         0           0        ip saddr @N_TRUSTEDv4  accept
21        jump      all         0           0        jump NAME_AUX
default   accept    all         0           0

-----
ipv4 Firewall "input filter"

Rule      Action    Protocol    Packets    Bytes    Conditions
-----
10        accept    all         156        14377    iifname != @I_LAN  accept
default   accept    all         0           0

-----
ipv4 Firewall "name AUX"

Rule      Action    Protocol    Packets    Bytes    Conditions
-----
→ -----
10        accept    icmp         0           0        meta l4proto icmp  accept
20        accept    udp          0           0        meta l4proto udp ip saddr @A_SERVERS  →
→ accept
30        drop      all          0           0        ip saddr != @A_SERVERS iifname "eth2"

-----
ipv4 Firewall "output filter"

Rule      Action    Protocol    Packets    Bytes    Conditions
-----
→ -----
10        reject    all          0           0        oifname @I_LAN
20        accept    icmp         2           168      meta l4proto icmp oifname "eth0" →
→ accept
default   accept    all         72          9258

-----
ipv6 Firewall "input filter"

Rule      Action    Protocol    Packets    Bytes    Conditions
-----
10        accept    all         0           0        ip6 saddr @N6_TRUSTEDv6  accept
default   accept    all         2           112

vyos@vyos:~$
```

show firewall summary

This shows you a summary of rule-sets and groups.

```
vyos@vyos:~$ show firewall summary
Ruleset Summary

IPv6 Ruleset:

Ruleset Hook      Ruleset Priority      Description
-----
forward           filter
input             filter
ipv6_name         IPV6-VyOS_MANAGEMENT
ipv6_name         IPV6-WAN_IN          PUBLIC_INTERNET

IPv4 Ruleset:

Ruleset Hook      Ruleset Priority      Description
-----
forward           filter
input             filter
name              VyOS_MANAGEMENT
name              WAN_IN               PUBLIC_INTERNET

Firewall Groups

Name              Type              References          Members
-----
--
PBX               address_group      WAN_IN-100          198.51.100.77
SERVERS           address_group      WAN_IN-110          192.0.2.10
WAN_IN-111        192.0.2.11
WAN_IN-112        192.0.2.12
WAN_IN-120
WAN_IN-121
WAN_IN-122
SUPPORT           address_group      VyOS_MANAGEMENT-20  192.168.1.2
WAN_IN-20
PHONE_VPN_SERVERS address_group      WAN_IN-160          10.6.32.2
PINGABLE_ADDRESSES address_group      WAN_IN-170          192.168.5.2
WAN_IN-171
PBX               ipv6_address_group IPV6-WAN_IN-100      2001:db8::1
SERVERS           ipv6_address_group IPV6-WAN_IN-110      2001:db8::2
IPV6-WAN_IN-111   2001:db8::3
IPV6-WAN_IN-112   2001:db8::4
IPV6-WAN_IN-120
IPV6-WAN_IN-121
IPV6-WAN_IN-122
SUPPORT           ipv6_address_group IPV6-VyOS_MANAGEMENT-20 2001:db8::5
IPV6-WAN_IN-20
```

show firewall ipv4 [forward | input | output] filter

show firewall ipv4 name <name>

This command will give an overview of a single rule-set.

```
vyos@vyos:~$ show firewall ipv4 input filter
Ruleset Information
-----
IPv4 Firewall "input filter"
Rule      Action    Protocol    Packets    Bytes    Conditions
-----
5         jump      all         0          0        iifname "eth2" jump NAME_VyOS_
MANAGEMENT
default   accept    all
```

show firewall ipv4 [forward | input | output] filter rule <1-999999>

show firewall ipv4 name <name> rule <1-999999>

This command gives an overview of a rule in a single rule-set, plus information for default action.

```
vyos@vyos:~$ show firewall ipv4 output filter rule 20
Rule Information
-----
ipv4 Firewall "output filter"
Rule      Action    Protocol    Packets    Bytes    Conditions
-----
20        accept    icmp         2          168      meta l4proto icmp oifname "eth0"
accept
default   accept    all         286        47614
```

show firewall statistics

This will show you statistics of all rule-sets since the last boot.

Show Firewall log

show log firewall

show log firewall ipv4

show log firewall ipv4 [forward | input | output | name]

show log firewall ipv4 [forward | input | output] filter

show log firewall ipv4 name <name>

show log firewall ipv4 [forward | input | output] filter rule <rule>

show log firewall ipv4 name <name> rule <rule>

Show the logs of all firewall; show all IPv4 firewall logs; show all logs for particular hook; show all logs for particular hook and priority; show all logs for particular custom chain; show logs for specific rule-set.

Example Partial Config

```

firewall {
  group {
    network-group BAD-NETWORKS {
      network 198.51.100.0/24
      network 203.0.113.0/24
    }
    network-group GOOD-NETWORKS {
      network 192.0.2.0/24
    }
    port-group BAD-PORTS {
      port 65535
    }
  }
  ipv4 {
    forward {
      filter {
        default-action accept
        rule 5 {
          action accept
          source {
            group {
              network-group GOOD-NETWORKS
            }
          }
        }
        rule 10 {
          action drop
          description "Bad Networks"
          protocol all
          source {
            group {
              network-group BAD-NETWORKS
            }
          }
        }
      }
    }
  }
}

```

Update geoip database

update geoip

Command to update GeoIP database and firewall sets.

7.2.5 IPv6 Firewall Configuration

Overview

This section covers useful information about IPv6 firewall configuration and appropriate operation-mode commands.

This section describes the following configuration commands:

set firewall ipv6 ...

To learn about the general traffic flow in VyOS firewalls, see [Firewall](#).

```
- set firewall
  * ipv6
    - forward
      + filter
    - input
      + filter
    - output
      + filter
      + raw
    - prerouting
      + raw
    - name
      + custom_name
```

The router first receives all traffic and processes it in the **prerouting** section.

This stage includes:

- **Firewall Prerouting**: commands found under `set firewall ipv6 prerouting raw ...`
- *Conntrack Ignore*: `set system conntrack ignore ipv6...`
- *Policy Route*: commands found under `set policy route6 ...`
- *Destination NAT*: commands found under `set nat66 destination ...`

For transit traffic that the router receives and forwards, the base chain is **forward**. The following diagram shows a simplified packet flow for transit traffic:

Use `set firewall ipv6 forward filter ...` to configure filtering rules for transit traffic. This command corresponds to stage 5 and is highlighted in red in the diagram.

For traffic destined to the router, use the **input** chain. For traffic the router generates, use the **output** chain. The following diagram shows the packet flow for traffic destined to the router and traffic generated by the router (starting from circle number 6):

Use `set firewall ipv6 input filter ...` to configure traffic destined to the router.

Use `set firewall ipv6 output ...` to configure traffic the router generates. Two sub-chains are available: **filter** and **raw**:

- **Output Prerouting:** `set firewall ipv6 output raw ....` As described in **Prerouting**, the firewall processes rules in this section before the connection tracking subsystem.
- **Output Filter:** `set firewall ipv6 output filter ....` The firewall processes rules in this section after the connection tracking subsystem.

Note: Important note about default-actions: If you do not define a default action for a base chain, the system sets the default action to **accept** for that chain. For custom chains, if you do not define a default action, the system sets the default-action to **drop**

Create custom firewall chains using the commands `set firewall ipv6 name <name> ....` To use the custom chain, define a rule with **action jump** and the appropriate **target** in a base chain.

Firewall - IPv6 Rules

Create firewall rules for firewall filtering. Each rule is numbered and has an action to apply when the rule is matched. You can specify multiple matching criteria. Packets go through rules from 1 - 999999, so order is crucial. The firewall executes the action of the first matching rule.

Actions

If you define a rule, you must define an action for it. The action tells the firewall what to do when all criteria for that rule are met.

The action can be :

- **accept:** accept the packet.
- **continue:** continue parsing next rule.
- **drop:** drop the packet.
- **reject:** reject the packet.
- **jump:** jump to another custom chain.
- **return:** Return from the current chain and continue at the next rule of the last chain.
- **queue:** Enqueue packet to userspace.
- **synproxy:** synproxy the packet.

```
set firewall ipv6 forward filter rule <1-999999> action [accept | continue | drop | jump
| queue | reject | return | synproxy]
```

```
set firewall ipv6 input filter rule <1-999999> action [accept | continue | drop | jump |
queue | reject | return | synproxy]
```

```
set firewall ipv6 output filter rule <1-999999> action [accept | continue | drop | jump |
queue | reject | return]
```

```
set firewall ipv6 name <name> rule <1-999999> action [accept | continue | drop | jump |
queue | reject | return]
```

This required setting defines the action of the current rule. If you set the action to jump, you must also define a jump-target.


```
set firewall ipv6 forward filter rule <1-999999> jump-target <text>
set firewall ipv6 input filter rule <1-999999> jump-target <text>
set firewall ipv6 output filter rule <1-999999> jump-target <text>
set firewall ipv6 name <name> rule <1-999999> jump-target <text>
```

Use this command only when action is set to jump. Specify the jump target.

```
set firewall ipv6 forward filter rule <1-999999> queue <0-65535>
set firewall ipv6 input filter rule <1-999999> queue <0-65535>
set firewall ipv6 output filter rule <1-999999> queue <0-65535>
set firewall ipv6 name <name> rule <1-999999> queue <0-65535>
```

Use this command only when action is set to queue. Specify the queue target. Queue ranges are also supported.

```
set firewall ipv6 forward filter rule <1-999999> queue-options bypass
set firewall ipv6 input filter rule <1-999999> queue-options bypass
set firewall ipv6 output filter rule <1-999999> queue-options bypass
set firewall ipv6 name <name> rule <1-999999> queue-options bypass
```

Use this command only when action is set to queue. This command allows the packet to go through the firewall when no userspace software is connected to the queue.

```
set firewall ipv6 forward filter rule <1-999999> queue-options fanout
set firewall ipv6 input filter rule <1-999999> queue-options fanout
set firewall ipv6 output filter rule <1-999999> queue-options fanout
set firewall ipv6 name <name> rule <1-999999> queue-options fanout
```

Use this command only when action is set to queue. This command distributes packets among multiple queues.

Also, **default-action** is an action that takes place whenever a packet does not match any rule in its chain. For base chains, possible options for **default-action** are **accept** or **drop**.

```
set firewall ipv6 forward filter default-action [accept | drop]
set firewall ipv6 input filter default-action [accept | drop]
set firewall ipv6 output filter default-action [accept | drop]
set firewall ipv6 name <name> default-action [accept | drop | jump | queue | reject |
return]
```

Set the default action of the rule-set if a packet does not match any rule criteria. If you set default-action to jump, you must also define default-jump-target. For base chains, you can only set the default action to accept or drop. For custom chains, more actions are available.

```
set firewall ipv6 name <name> default-jump-target <text>
```

To be used only when default-action is set to jump. Use this command to specify the jump target for the default rule.

Note: Important note about default-actions: If you do not define the default action for a base chain, the system sets the default action to **accept** for that chain. For custom chains, if you do not define a default action, the system sets the default-action to **drop**.

Firewall Logs

You can enable logging for each firewall rule. When enabled, you can also define other log options.

```
set firewall ipv6 forward filter rule <1-999999> log
set firewall ipv6 input filter rule <1-999999> log
set firewall ipv6 output filter rule <1-999999> log
set firewall ipv6 name <name> rule <1-999999> log
```

Enable logging for matched packets. If this configuration command is not present, logging is disabled.

```
set firewall ipv6 forward filter default-log
set firewall ipv6 input filter default-log
set firewall ipv6 output filter default-log
set firewall ipv6 name <name> default-log
```

Use this command to enable the logging of the default action on the specified chain.

```
set firewall ipv6 forward filter rule <1-999999> log-options level [emerg | alert | crit
| err | warn | notice | info | debug]
set firewall ipv6 input filter rule <1-999999> log-options level [emerg | alert | crit |
err | warn | notice | info | debug]
set firewall ipv6 output filter rule <1-999999> log-options level [emerg | alert | crit |
err | warn | notice | info | debug]
set firewall ipv6 name <name> rule <1-999999> log-options level [emerg | alert | crit |
err | warn | notice | info | debug]
```

Define log-level. Only applicable if rule log is enabled.

```
set firewall ipv6 forward filter rule <1-999999> log-options group <0-65535>
set firewall ipv6 input filter rule <1-999999> log-options group <0-65535>
set firewall ipv6 output filter rule <1-999999> log-options group <0-65535>
set firewall ipv6 name <name> rule <1-999999> log-options group <0-65535>
```

Define the log group to send messages to. Only applicable if rule log is enabled.

```
set firewall ipv6 forward filter rule <1-999999> log-options snapshot-length <0-9000>
set firewall ipv6 input filter rule <1-999999> log-options snapshot-length <0-9000>
set firewall ipv6 output filter rule <1-999999> log-options snapshot-length <0-9000>
set firewall ipv6 name <name> rule <1-999999> log-options snapshot-length <0-9000>
```

Define the length of packet payload to include in a netlink message. Only applicable when rule logging is enabled and log group is defined.

```
set firewall ipv6 forward filter rule <1-999999> log-options queue-threshold <0-65535>
set firewall ipv6 input filter rule <1-999999> log-options queue-threshold <0-65535>
set firewall ipv6 output filter rule <1-999999> log-options queue-threshold <0-65535>
set firewall ipv6 name <name> rule <1-999999> log-options queue-threshold <0-65535>
```

Define the number of packets to queue inside the kernel before sending them to userspace. Only applicable when rule logging is enabled and log group is defined.

Firewall Description

For reference, you can define descriptions on every rule and custom chain.

```
set firewall ipv6 name <name> description <text>
```

Provide a rule-set description to a custom firewall chain.

```
set firewall ipv6 forward filter rule <1-999999> description <text>
```

```
set firewall ipv6 input filter rule <1-999999> description <text>
```

```
set firewall ipv6 output filter rule <1-999999> description <text>
```

```
set firewall ipv6 name <name> rule <1-999999> description <text>
```

Provide a description for each rule.

Rule Status

New rules are enabled by default. In some cases, you may want to disable a rule rather than remove it.

```
set firewall ipv6 forward filter rule <1-999999> disable
```

```
set firewall ipv6 input filter rule <1-999999> disable
```

```
set firewall ipv6 output filter rule <1-999999> disable
```

```
set firewall ipv6 name <name> rule <1-999999> disable
```

Command for disabling a rule but keep it in the configuration.

Matching criteria

There are a lot of matching criteria against which the packet can be tested.

```
set firewall ipv6 forward filter rule <1-999999> connection-status nat [destination | source]
```

```
set firewall ipv6 input filter rule <1-999999> connection-status nat [destination | source]
```

```
set firewall ipv6 output filter rule <1-999999> connection-status nat [destination | source]
```

```
set firewall ipv6 name <name> rule <1-999999> connection-status nat [destination | source]
```

Match packets based on NAT connection status.

```
set firewall ipv6 forward filter rule <1-999999> connection-mark <1-2147483647>
```

```
set firewall ipv6 input filter rule <1-999999> connection-mark <1-2147483647>
```

```
set firewall ipv6 output filter rule <1-999999> connection-mark <1-2147483647>
```

```
set firewall ipv6 name <name> rule <1-999999> connection-mark <1-2147483647>
```

Match packets based on connection mark.

```
set firewall ipv6 forward filter rule <1-999999> source address [address | addressrange | CIDR]
```

```

set firewall ipv6 input filter rule <1-999999> source address [address | addressrange |
CIDR]
set firewall ipv6 output filter rule <1-999999> source address [address | addressrange |
CIDR]
set firewall ipv6 name <name> rule <1-999999> source address [address | addressrange |
CIDR]
set firewall ipv6 forward filter rule <1-999999> destination address [address |
addressrange | CIDR]
set firewall ipv6 input filter rule <1-999999> destination address [address |
addressrange | CIDR]
set firewall ipv6 output filter rule <1-999999> destination address [address |
addressrange | CIDR]
set firewall ipv6 name <name> rule <1-999999> destination address [address | addressrange
| CIDR]

```

Match based on source or destination address. This is similar to network groups, but you can negate the matching addresses here.

```
set firewall ipv6 name F00 rule 100 source address 2001:db8::202
```

```

set firewall ipv6 forward filter rule <1-999999> source address-mask [address]
set firewall ipv6 input filter rule <1-999999> source address-mask [address]
set firewall ipv6 output filter rule <1-999999> source address-mask [address]
set firewall ipv6 name <name> rule <1-999999> source address-mask [address]
set firewall ipv6 forward filter rule <1-999999> destination address-mask [address]
set firewall ipv6 input filter rule <1-999999> destination address-mask [address]
set firewall ipv6 output filter rule <1-999999> destination address-mask [address]
set firewall ipv6 name <name> rule <1-999999> destination address-mask [address]

```

Apply an arbitrary netmask to mask addresses and match only a specific portion. This is useful for IPv6 because rules remain valid when the IPv6 prefix changes if the host portion of the system's IPv6 address is static. Examples include SLAAC and tokenised IPv6 addresses

This function works for both individual addresses and address groups.

```

# Match any IPv6 address with the suffix ::0000:0000:0000:beef
set firewall ipv6 forward filter rule 100 destination address ::beef
set firewall ipv6 forward filter rule 100 destination address-mask_
→ ::ffff:ffff:ffff:ffff
# Address groups
set firewall group ipv6-address-group WEBSERVERS address ::1000
set firewall group ipv6-address-group WEBSERVERS address ::2000
set firewall ipv6 forward filter rule 200 source group address-group WEBSERVERS
set firewall ipv6 forward filter rule 200 source address-mask ::ffff:ffff:ffff:ffff

```

```

set firewall ipv6 forward filter rule <1-999999> source fqdn <fqdn>
set firewall ipv6 input filter rule <1-999999> source fqdn <fqdn>
set firewall ipv6 output filter rule <1-999999> source fqdn <fqdn>

```

```

set firewall ipv6 name <name> rule <1-999999> source fqdn <fqdn>
set firewall ipv6 forward filter rule <1-999999> destination fqdn <fqdn>
set firewall ipv6 input filter rule <1-999999> destination fqdn <fqdn>
set firewall ipv6 output filter rule <1-999999> destination fqdn <fqdn>
set firewall ipv6 name <name> rule <1-999999> destination fqdn <fqdn>

```

Specify a Fully Qualified Domain Name as source or destination to match. Ensure that the router can resolve the DNS query.

```

set firewall ipv6 forward filter rule <1-999999> source geoip country-code <country>
set firewall ipv6 input filter rule <1-999999> source geoip country-code <country>
set firewall ipv6 output filter rule <1-999999> source geoip country-code <country>
set firewall ipv6 name <name> rule <1-999999> source geoip country-code <country>
set firewall ipv6 forward filter rule <1-999999> source geoip asn <1-4294967294>
set firewall ipv6 input filter rule <1-999999> source geoip asn <1-4294967294>
set firewall ipv6 output filter rule <1-999999> source geoip asn <1-4294967294>
set firewall ipv6 name <name> rule <1-999999> source geoip asn <1-4294967294>
set firewall ipv6 forward filter rule <1-999999> destination geoip country-code <country>
set firewall ipv6 input filter rule <1-999999> destination geoip country-code <country>
set firewall ipv6 output filter rule <1-999999> destination geoip country-code <country>
set firewall ipv6 name <name> rule <1-999999> destination geoip country-code <country>
set firewall ipv6 forward filter rule <1-999999> destination geoip asn <1-4294967294>
set firewall ipv6 input filter rule <1-999999> destination geoip asn <1-4294967294>
set firewall ipv6 output filter rule <1-999999> destination geoip asn <1-4294967294>
set firewall ipv6 name <name> rule <1-999999> destination geoip asn <1-4294967294>
set firewall ipv6 forward filter rule <1-999999> source geoip inverse-match
set firewall ipv6 input filter rule <1-999999> source geoip inverse-match
set firewall ipv6 output filter rule <1-999999> source geoip inverse-match
set firewall ipv6 name <name> rule <1-999999> source geoip inverse-match
set firewall ipv6 forward filter rule <1-999999> destination geoip inverse-match
set firewall ipv6 input filter rule <1-999999> destination geoip inverse-match
set firewall ipv6 output filter rule <1-999999> destination geoip inverse-match
set firewall ipv6 name <name> rule <1-999999> destination geoip inverse-match

```

Match IP addresses based on their geolocation. For more information, see [GeoIP matching](#). Use country-code or asn to match based on geography or origin network. Use inverse-match to match anything except the specified country code or ASN.

DB-IP.com provides data under CC-BY-4.0 license. Attribution is required and redistribution is permitted, allowing VyOS to include a database in images (approximately 3 MB compressed). The package includes a cron script that you can manually call through op-mode `update geoip` to keep the database and rules updated.

```

set firewall ipv6 forward filter rule <1-999999> source mac-address <mac-address>

```

```
set firewall ipv6 input filter rule <1-999999> source mac-address <mac-address>
set firewall ipv6 output filter rule <1-999999> source mac-address <mac-address>
set firewall ipv6 name <name> rule <1-999999> source mac-address <mac-address>
```

You can specify only a source MAC address to match.

```
set firewall ipv6 input filter rule 100 source mac-address 00:53:00:11:22:33
set firewall ipv6 input filter rule 101 source mac-address !00:53:00:aa:12:34
```

```
set firewall ipv6 forward filter rule <1-999999> source port [1-65535 | portname |
start-end]
set firewall ipv6 input filter rule <1-999999> source port [1-65535 | portname |
start-end]
set firewall ipv6 output filter rule <1-999999> source port [1-65535 | portname |
start-end]
set firewall ipv6 name <name> rule <1-999999> source port [1-65535 | portname |
start-end]
set firewall ipv6 forward filter rule <1-999999> destination port [1-65535 | portname |
start-end]
set firewall ipv6 input filter rule <1-999999> destination port [1-65535 | portname |
start-end]
set firewall ipv6 output filter rule <1-999999> destination port [1-65535 | portname |
start-end]
set firewall ipv6 name <name> rule <1-999999> destination port [1-65535 | portname |
start-end]
```

Specify a port by number or by name as defined in /etc/services.

```
set firewall ipv6 forward filter rule 10 source port '22'
set firewall ipv6 forward filter rule 11 source port '!http'
set firewall ipv6 forward filter rule 12 source port 'https'
```

Multiple source ports can be specified as a comma-separated list. The whole list can also be “negated” using !. For example:

```
set firewall ipv6 forward filter rule 10 source port '!22,https,3333-3338'
```

```
set firewall ipv6 forward filter rule <1-999999> source group address-group <name | !
name>
set firewall ipv6 input filter rule <1-999999> source group address-group <name | !name>
set firewall ipv6 output filter rule <1-999999> source group address-group <name | !name>
set firewall ipv6 name <name> rule <1-999999> source group address-group <name | !name>
set firewall ipv6 forward filter rule <1-999999> destination group address-group <name |
!name>
set firewall ipv6 input filter rule <1-999999> destination group address-group <name |
!name>
set firewall ipv6 output filter rule <1-999999> destination group address-group <name |
!name>
```

```
set firewall ipv6 name <name> rule <1-999999> destination group address-group <name | !name>
```

Specify an address group. You can prepend the character ! to invert the matching criteria.

```
set firewall ipv6 forward filter rule <1-999999> source group dynamic-address-group <name | !name>
```

```
set firewall ipv6 input filter rule <1-999999> source group dynamic-address-group <name | !name>
```

```
set firewall ipv6 output filter rule <1-999999> source group dynamic-address-group <name | !name>
```

```
set firewall ipv6 name <name> rule <1-999999> source group dynamic-address-group <name | !name>
```

```
set firewall ipv6 forward filter rule <1-999999> destination group dynamic-address-group <name | !name>
```

```
set firewall ipv6 input filter rule <1-999999> destination group dynamic-address-group <name | !name>
```

```
set firewall ipv6 output filter rule <1-999999> destination group dynamic-address-group <name | !name>
```

```
set firewall ipv6 name <name> rule <1-999999> destination group dynamic-address-group <name | !name>
```

Specify a dynamic address group. You can prepend the character ! to invert the matching criteria.

```
set firewall ipv6 forward filter rule <1-999999> source group network-group <name | !name>
```

```
set firewall ipv6 input filter rule <1-999999> source group network-group <name | !name>
```

```
set firewall ipv6 output filter rule <1-999999> source group network-group <name | !name>
```

```
set firewall ipv6 name <name> rule <1-999999> source group network-group <name | !name>
```

```
set firewall ipv6 forward filter rule <1-999999> destination group network-group <name | !name>
```

```
set firewall ipv6 input filter rule <1-999999> destination group network-group <name | !name>
```

```
set firewall ipv6 output filter rule <1-999999> destination group network-group <name | !name>
```

```
set firewall ipv6 name <name> rule <1-999999> destination group network-group <name | !name>
```

Specify a network group. You can prepend the character ! to invert the matching criteria.

```
set firewall ipv6 forward filter rule <1-999999> source group port-group <name | !name>
```

```
set firewall ipv6 input filter rule <1-999999> source group port-group <name | !name>
```

```
set firewall ipv6 output filter rule <1-999999> source group port-group <name | !name>
```

```
set firewall ipv6 name <name> rule <1-999999> source group port-group <name | !name>
```

```
set firewall ipv6 forward filter rule <1-999999> destination group port-group <name | !name>
```

```

set firewall ipv6 input filter rule <1-999999> destination group port-group <name | !
name>
set firewall ipv6 output filter rule <1-999999> destination group port-group <name | !
name>
set firewall ipv6 name <name> rule <1-999999> destination group port-group <name | !name>

```

Specify a port group. You can prepend the character ! to invert the matching criteria.

```

set firewall ipv6 forward filter rule <1-999999> source group domain-group <name | !name>
set firewall ipv6 input filter rule <1-999999> source group domain-group <name | !name>
set firewall ipv6 output filter rule <1-999999> source group domain-group <name | !name>
set firewall ipv6 name <name> rule <1-999999> source group domain-group <name | !name>
set firewall ipv6 forward filter rule <1-999999> destination group domain-group <name |
!name>
set firewall ipv6 input filter rule <1-999999> destination group domain-group <name |
!name>
set firewall ipv6 output filter rule <1-999999> destination group domain-group <name |
!name>
set firewall ipv6 name <name> rule <1-999999> destination group domain-group <name | !
name>

```

Specify a domain group. You can prepend the character ! to invert the matching criteria.

```

set firewall ipv6 forward filter rule <1-999999> source group mac-group <name | !name>
set firewall ipv6 input filter rule <1-999999> source group mac-group <name | !name>
set firewall ipv6 output filter rule <1-999999> source group mac-group <name | !name>
set firewall ipv6 name <name> rule <1-999999> source group mac-group <name | !name>
set firewall ipv6 forward filter rule <1-999999> destination group mac-group <name | !
name>
set firewall ipv6 input filter rule <1-999999> destination group mac-group <name | !name>
set firewall ipv6 output filter rule <1-999999> destination group mac-group <name | !
name>
set firewall ipv6 name <name> rule <1-999999> destination group mac-group <name | !name>

```

Specify a MAC group. You can prepend the character ! to invert the matching criteria.

```

set firewall ipv6 forward filter rule <1-999999> dscp [0-63 | start-end]
set firewall ipv6 input filter rule <1-999999> dscp [0-63 | start-end]
set firewall ipv6 output filter rule <1-999999> dscp [0-63 | start-end]
set firewall ipv6 name <name> rule <1-999999> dscp [0-63 | start-end]
set firewall ipv6 forward filter rule <1-999999> dscp-exclude [0-63 | start-end]
set firewall ipv6 input filter rule <1-999999> dscp-exclude [0-63 | start-end]
set firewall ipv6 output filter rule <1-999999> dscp-exclude [0-63 | start-end]
set firewall ipv6 name <name> rule <1-999999> dscp-exclude [0-63 | start-end]

```

Match based on dscp value.


```

set firewall ipv6 forward filter rule <1-999999> fragment [match-frag | match-non-frag]
set firewall ipv6 input filter rule <1-999999> fragment [match-frag | match-non-frag]
set firewall ipv6 output filter rule <1-999999> fragment [match-frag | match-non-frag]
set firewall ipv6 name <name> rule <1-999999> fragment [match-frag | match-non-frag]

```

Match packets based on fragmentation.

```

set firewall ipv6 forward filter rule <1-999999> icmpv6 [code | type] <0-255>
set firewall ipv6 input filter rule <1-999999> icmpv6 [code | type] <0-255>
set firewall ipv6 output filter rule <1-999999> icmpv6 [code | type] <0-255>
set firewall ipv6 name <name> rule <1-999999> icmpv6 [code | type] <0-255>

```

Match packets based on ICMP or ICMPv6 code and type.

```

set firewall ipv6 forward filter rule <1-999999> icmpv6 type-name <text>
set firewall ipv6 input filter rule <1-999999> icmpv6 type-name <text>
set firewall ipv6 output filter rule <1-999999> icmpv6 type-name <text>
set firewall ipv6 name <name> rule <1-999999> icmpv6 type-name <text>

```

Match based on ICMPv6 type-name. Press **Tab** for information about supported **type-name** criteria.

```

set firewall ipv6 forward filter rule <1-999999> inbound-interface name <iface>
set firewall ipv6 input filter rule <1-999999> inbound-interface name <iface>
set firewall ipv6 name <name> rule <1-999999> inbound-interface name <iface>

```

Match based on inbound interface. You can use the wildcard *. For example: eth2*. You can prepend the character ! to invert the matching criteria. For example !eth2

Note: If an interface is attached to a non-default VRF, when using **inbound-interface**, use the VRF name. For example: set firewall ipv6 forward filter rule 10 inbound-interface name MGMT

```

set firewall ipv6 forward filter rule <1-999999> inbound-interface group <iface_group>
set firewall ipv6 input filter rule <1-999999> inbound-interface group <iface_group>
set firewall ipv6 name <name> rule <1-999999> inbound-interface group <iface_group>

```

Match based on the inbound interface group. You can prepend the character ! to invert the matching criteria. For example !IFACE_GROUP

```

set firewall ipv6 forward filter rule <1-999999> outbound-interface name <iface>
set firewall ipv6 output filter rule <1-999999> outbound-interface name <iface>
set firewall ipv6 name <name> rule <1-999999> outbound-interface name <iface>

```

Match based on outbound interface. You can use the wildcard *. For example: eth2*. You can prepend the character ! to invert the matching criteria. For example !eth2

Note: If an interface is attached to a non-default VRF, when using **outbound-interface**, use the physical interface name. For example: set firewall ipv6 forward filter rule 10 outbound-interface name eth0

```

set firewall ipv6 forward filter rule <1-999999> outbound-interface group <iface_group>

```

```
set firewall ipv6 output filter rule <1-999999> outbound-interface group <iface_group>
```

```
set firewall ipv6 name <name> rule <1-999999> outbound-interface group <iface_group>
```

Match based on outbound interface group. You can prepend the character ! to invert the matching criteria. For example !IFACE_GROUP

```
set firewall ipv6 forward filter rule <1-999999> ipsec [match-ipsec-in | match-ipsec-out  
| match-none-in | match-none-out]
```

```
set firewall ipv6 input filter rule <1-999999> ipsec [match-ipsec-in | match-none-in]
```

```
set firewall ipv6 output filter rule <1-999999> ipsec [match-ipsec-out | match-none-out]
```

```
set firewall ipv6 name <name> rule <1-999999> ipsec [match-ipsec-in | match-ipsec-out |  
match-none-in | match-none-out]
```

Match packets based on IPsec.

```
set firewall ipv6 forward filter rule <1-999999> limit burst <0-4294967295>
```

```
set firewall ipv6 input filter rule <1-999999> limit burst <0-4294967295>
```

```
set firewall ipv6 output filter rule <1-999999> limit burst <0-4294967295>
```

```
set firewall ipv6 name <name> rule <1-999999> limit burst <0-4294967295>
```

Match based on the maximum number of packets allowed to exceed the rate limit.

```
set firewall ipv6 forward filter rule <1-999999> limit rate <text>
```

```
set firewall ipv6 input filter rule <1-999999> limit rate <text>
```

```
set firewall ipv6 output filter rule <1-999999> limit rate <text>
```

```
set firewall ipv6 name <name> rule <1-999999> limit rate <text>
```

Match based on the maximum average rate, specified as integer/unit. For example, specify 5/minutes.

```
set firewall ipv6 forward filter rule <1-999999> packet-length <text>
```

```
set firewall ipv6 input filter rule <1-999999> packet-length <text>
```

```
set firewall ipv6 output filter rule <1-999999> packet-length <text>
```

```
set firewall ipv6 name <name> rule <1-999999> packet-length <text>
```

```
set firewall ipv6 forward filter rule <1-999999> packet-length-exclude <text>
```

```
set firewall ipv6 input filter rule <1-999999> packet-length-exclude <text>
```

```
set firewall ipv6 output filter rule <1-999999> packet-length-exclude <text>
```

```
set firewall ipv6 name <name> rule <1-999999> packet-length-exclude <text>
```

Match based on packet length. You can specify multiple values from 1 to 65535 and ranges.

```
set firewall ipv6 forward filter rule <1-999999> packet-type [broadcast | host |  
multicast | other]
```

```
set firewall ipv6 input filter rule <1-999999> packet-type [broadcast | host | multicast  
| other]
```

```
set firewall ipv6 output filter rule <1-999999> packet-type [broadcast | host | multicast  
| other]
```

```
set firewall ipv6 name <name> rule <1-999999> packet-type [broadcast | host | multicast |  
other]
```

Match based on packet type.

```
set firewall ipv6 forward filter rule <1-999999> protocol [<text> | <0-255> | all |
tcp_udp]

set firewall ipv6 input filter rule <1-999999> protocol [<text> | <0-255> | all |
tcp_udp]

set firewall ipv6 output filter rule <1-999999> protocol [<text> | <0-255> | all |
tcp_udp]

set firewall ipv6 name <name> rule <1-999999> protocol [<text> | <0-255> | all | tcp_udp]
```

Match based on protocol number or name as defined in /etc/protocols. Specify all for all protocols and tcp_udp for TCP and UDP packets. Prepend ! to negate the protocol selection.

```
set firewall ipv6 input filter rule 10 protocol tcp
```

```
set firewall ipv6 forward filter rule <1-999999> recent count <1-255>
set firewall ipv6 input filter rule <1-999999> recent count <1-255>
set firewall ipv6 output filter rule <1-999999> recent count <1-255>
set firewall ipv6 name <name> rule <1-999999> recent count <1-255>
set firewall ipv6 forward filter rule <1-999999> recent time [second | minute | hour]
set firewall ipv6 input filter rule <1-999999> recent time [second | minute | hour]
set firewall ipv6 output filter rule <1-999999> recent time [second | minute | hour]
set firewall ipv6 name <name> rule <1-999999> recent time [second | minute | hour]
```

Match packets based on recently seen sources.

```
set firewall ipv6 forward filter rule <1-999999> tcp flags [not] <text>
set firewall ipv6 input filter rule <1-999999> tcp flags [not] <text>
set firewall ipv6 output filter rule <1-999999> tcp flags [not] <text>
set firewall ipv6 name <name> rule <1-999999> tcp flags [not] <text>
```

Allowed values for TCP flags: ack, cwr, ecn, fin, psh, rst, syn, and urg. You can specify multiple values. To invert the selection, use not, as shown in the following example.

```
set firewall ipv6 input filter rule 10 tcp flags 'ack'
set firewall ipv6 input filter rule 12 tcp flags 'syn'
set firewall ipv6 input filter rule 13 tcp flags not 'fin'
```

```
set firewall ipv6 forward filter rule <1-999999> state [established | invalid | new |
related]

set firewall ipv6 input filter rule <1-999999> state [established | invalid | new |
related]

set firewall ipv6 output filter rule <1-999999> state [established | invalid | new |
related]

set firewall ipv6 name <name> rule <1-999999> state [established | invalid | new |
related]
```

Match based on packet state.

```
set firewall ipv6 forward filter rule <1-999999> time startdate <text>
```

```

set firewall ipv6 input filter rule <1-999999> time startdate <text>
set firewall ipv6 output filter rule <1-999999> time startdate <text>
set firewall ipv6 name <name> rule <1-999999> time startdate <text>
set firewall ipv6 forward filter rule <1-999999> time starttime <text>
set firewall ipv6 input filter rule <1-999999> time starttime <text>
set firewall ipv6 output filter rule <1-999999> time starttime <text>
set firewall ipv6 name <name> rule <1-999999> time starttime <text>
set firewall ipv6 forward filter rule <1-999999> time stopdate <text>
set firewall ipv6 input filter rule <1-999999> time stopdate <text>
set firewall ipv6 output filter rule <1-999999> time stopdate <text>
set firewall ipv6 name <name> rule <1-999999> time stopdate <text>
set firewall ipv6 forward filter rule <1-999999> time stoptime <text>
set firewall ipv6 input filter rule <1-999999> time stoptime <text>
set firewall ipv6 output filter rule <1-999999> time stoptime <text>
set firewall ipv6 name <name> rule <1-999999> time stoptime <text>
set firewall ipv6 forward filter rule <1-999999> time weekdays <text>
set firewall ipv6 input filter rule <1-999999> time weekdays <text>
set firewall ipv6 output filter rule <1-999999> time weekdays <text>
set firewall ipv6 name <name> rule <1-999999> time weekdays <text>

```

Match packets based on time criteria.

```

set firewall ipv6 forward filter rule <1-999999> hop-limit <eq | gt | lt> <0-255>
set firewall ipv6 input filter rule <1-999999> hop-limit <eq | gt | lt> <0-255>
set firewall ipv6 output filter rule <1-999999> hop-limit <eq | gt | lt> <0-255>
set firewall ipv6 name <name> rule <1-999999> hop-limit <eq | gt | lt> <0-255>

```

Match the hop-limit parameter. Use eq for equal, gt for greater than, and lt for less than.

```

set firewall ipv6 forward filter rule <1-999999> recent count <1-255>
set firewall ipv6 input filter rule <1-999999> recent count <1-255>
set firewall ipv6 output filter rule <1-999999> recent count <1-255>
set firewall ipv6 name <name> rule <1-999999> recent count <1-255>
set firewall ipv6 forward filter rule <1-999999> recent time <second | minute | hour>
set firewall ipv6 input filter rule <1-999999> recent time <second | minute | hour>
set firewall ipv6 output filter rule <1-999999> recent time <second | minute | hour>
set firewall ipv6 name <name> rule <1-999999> recent time <second | minute | hour>

```

Match when the specified number of connections occur within the specified time period. Use these criteria to block brute-force attempts.

Packet Modifications

The firewall can modify packets before sending them. This feature provides more flexibility for packet handling.

```
set firewall ipv6 prerouting raw rule <1-999999> set dscp <0-63>
set firewall ipv6 forward filter rule <1-999999> set dscp <0-63>
set firewall ipv6 output [filter | raw] rule <1-999999> set dscp <0-63>
```

Set a specific value of Differentiated Services Codepoint (DSCP).

```
set firewall ipv6 prerouting raw rule <1-999999> set mark <1-2147483647>
set firewall ipv6 forward filter rule <1-999999> set mark <1-2147483647>
set firewall ipv6 output [filter | raw] rule <1-999999> set mark <1-2147483647>
```

Set a specific packet mark value.

```
set firewall ipv6 prerouting raw rule <1-999999> set tcp-mss <500-1460>
set firewall ipv6 forward filter rule <1-999999> set tcp-mss <500-1460>
set firewall ipv6 output [filter | raw] rule <1-999999> set tcp-mss <500-1460>
```

Set the TCP-MSS (TCP maximum segment size) for the connection.

```
set firewall ipv6 prerouting raw rule <1-999999> set hop-limit <0-255>
set firewall ipv6 forward filter rule <1-999999> set hop-limit <0-255>
set firewall ipv6 output [filter | raw] rule <1-999999> set hop-limit <0-255>
```

Set hop limit value.

```
set firewall ipv6 forward filter rule <1-999999> set connection-mark <0-2147483647>
set firewall ipv6 output [filter | raw] rule <1-999999> set connection-mark
<0-2147483647>
```

Set connection mark value.

Synproxy

Synproxy connections

```
set firewall ipv6 [input | forward] filter rule <1-999999> action synproxy
set firewall ipv6 [input | forward] filter rule <1-999999> protocol tcp
set firewall ipv6 [input | forward] filter rule <1-999999> synproxy tcp mss <501-65535>
```

Set the TCP MSS (maximum segment size) for the connection.

```
set firewall ipv6 [input | forward] filter rule <1-999999> synproxy tcp window-scale
<1-14>
```

Set the window scale factor for TCP window scaling.

Example synproxy

Requirements to enable synproxy:

- Traffic must be symmetric
- Synproxy relies on syncookies and TCP timestamps, ensure these are enabled
- Disable conntrack loose track option

```
set system sysctl parameter net.ipv4.tcp_timestamps value '1'

set system conntrack tcp loose disable

set system conntrack ignore ipv6 rule 10 destination port '8080'

set system conntrack ignore ipv6 rule 10 protocol 'tcp'

set system conntrack ignore ipv6 rule 10 tcp flags syn

set firewall global-options syn-cookies 'enable'

set firewall ipv6 input filter rule 10 action 'synproxy'

set firewall ipv6 input filter rule 10 destination port '8080'

set firewall ipv6 input filter rule 10 inbound-interface name 'eth1'

set firewall ipv6 input filter rule 10 protocol 'tcp'

set firewall ipv6 input filter rule 10 synproxy tcp mss '1460'

set firewall ipv6 input filter rule 10 synproxy tcp window-scale '7'

set firewall ipv6 input filter rule 1000 action 'drop'

set firewall ipv6 input filter rule 1000 state invalid
```

Operation-mode Firewall

Rule-set overview

show firewall

Show a basic firewall overview for all rule-sets, not only for IPv6:

```
vyos@vyos:~$ show firewall
Rulesets Information
-----
IPv4 Firewall "forward filter"
```

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Rule	Action	Protocol	Packets	Bytes	Conditions
5	jump	all	0	0	iifname "eth1" jump NAME_VyOS_
10	jump	all	0	0	oifname "eth1" jump NAME_WAN_IN
15	jump	all	0	0	iifname "eth3" jump NAME_WAN_IN
default	accept	all			
IPv4 Firewall "name VyOS_MANAGEMENT"					
Rule	Action	Protocol	Packets	Bytes	Conditions
5	accept	all	0	0	ct state established accept
10	drop	all	0	0	ct state invalid
20	accept	all	0	0	ip saddr @A_GOOD_GUYS accept
30	accept	all	0	0	ip saddr @N_ENTIRE_RANGE accept
40	accept	all	0	0	ip saddr @A_VyOS_SERVERS accept
50	accept	icmp	0	0	meta l4proto icmp accept
default	drop	all	0	0	
IPv6 Firewall "forward filter"					
Rule	Action	Protocol			
5	jump	all			
10	jump	all			
15	jump	all			
default	accept	all			
IPv6 Firewall "input filter"					
Rule	Action	Protocol			
5	jump	all			
default	accept	all			
IPv6 Firewall "ipv6_name IPV6-VyOS_MANAGEMENT"					
Rule	Action	Protocol			
5	accept	all			
10	drop	all			
20	accept	all			
30	accept	all			
40	accept	all			
50	accept	ipv6-icmp			

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```
default drop all
```

show firewall summary

This will show you a summary of rule-sets and groups

```
vyos@vyos:~$ show firewall summary
Ruleset Summary
```

IPv6 Ruleset:

Ruleset Hook	Ruleset Priority	Description
forward	filter	
input	filter	
ipv6_name	IPV6-VyOS_MANAGEMENT	
ipv6_name	IPV6-WAN_IN	PUBLIC_INTERNET

IPv4 Ruleset:

Ruleset Hook	Ruleset Priority	Description
forward	filter	
input	filter	
name	VyOS_MANAGEMENT	
name	WAN_IN	PUBLIC_INTERNET

Firewall Groups

Name	Type	References	Members
↔--			
PBX	address_group	WAN_IN-100	198.51.100.77
SERVERS	address_group	WAN_IN-110	192.0.2.10
WAN_IN-111		192.0.2.11	
WAN_IN-112		192.0.2.12	
WAN_IN-120			
WAN_IN-121			
WAN_IN-122			
SUPPORT	address_group	VyOS_MANAGEMENT-20	192.168.1.2
WAN_IN-20			
PHONE_VPN_SERVERS	address_group	WAN_IN-160	10.6.32.2
PINGABLE_ADDRESSES	address_group	WAN_IN-170	192.168.5.2
WAN_IN-171			
PBX	ipv6_address_group	IPV6-WAN_IN-100	2001:db8::1
SERVERS	ipv6_address_group	IPV6-WAN_IN-110	2001:db8::2
IPV6-WAN_IN-111		2001:db8::3	
IPV6-WAN_IN-112		2001:db8::4	
IPV6-WAN_IN-120			
IPV6-WAN_IN-121			
IPV6-WAN_IN-122			
SUPPORT	ipv6_address_group	IPV6-VyOS_MANAGEMENT-20	2001:db8::5
IPV6-WAN_IN-20			

show firewall ipv6 [forward | input | output] filter

show firewall ipv6 ipv6-name <name>

This command will give an overview of a single rule-set.

```
vyos@vyos:~$ show firewall ipv6 input filter
Ruleset Information

-----
ipv6 Firewall "input filter"

Rule      Action    Protocol    Packets    Bytes    Conditions
-----
10        jump      all         13         1456     iifname "eth1"  jump NAME6_INP-
→ETH1
20        accept    ipv6-icmp   10         1112     meta l4proto ipv6-icmp iifname
→"eth0" prefix "[ipv6-INP-filter-20-A]" accept
default   accept    all         14         1584
```

show firewall ipv6 [forward | input | output] filter rule <1-999999>

show firewall ipv6 name <name> rule <1-999999>

show firewall ipv6 ipv6-name <name> rule <1-999999>

This command will give an overview of a rule in a single rule-set

show firewall group <name>

Show an overview of defined groups, including the type, members, and where the group is used.

```
vyos@vyos:~$ show firewall group LAN
Firewall Groups

Name      Type              References          Members
-----
LAN       ipv6_network_group  IPV6-VyOS_MANAGEMENT-30  2001:db8::0/64
IPV6-WAN_IN-30
LAN       network_group      VyOS_MANAGEMENT-30      192.168.200.0/24
WAN_IN-30
```

show firewall statistics

Show statistics of all rule-sets since the last boot.

Show Firewall log

```
show log firewall
```

```
show log firewall ipv6
```

```
show log firewall ipv6 [forward | input | output | name]
```

```
show log firewall ipv6 [forward | input | output] filter
```

```
show log firewall ipv6 name <name>
```

```
show log firewall ipv6 [forward | input | output] filter rule <rule>
```

```
show log firewall ipv6 name <name> rule <rule>
```

Show firewall logs for all firewalls, all IPv6 firewalls, specific hooks, specific priorities, specific custom chains, or specific rule-sets.

Example Partial Config

```

firewall {
    ipv6 {
        input {
            filter {
                rule 10 {
                    action jump
                    inbound-interface {
                        name eth1
                    }
                    jump-target INP-ETH1
                }
                rule 20 {
                    action accept
                    inbound-interface {
                        name eth0
                    }
                    log
                    protocol ipv6-icmp
                }
            }
        }
        name INP-ETH1 {
            default-action drop
            default-log
            rule 10 {
                action accept
                protocol tcp_udp
            }
        }
    }
}

```

Update geoip database

update geoip

Command used to update GeoIP database and firewall sets.

7.2.6 Flowtables Firewall Configuration

Overview

This section provides information on firewall configuration for flowtables.

set firewall flowtable ...

To learn about the general traffic flow in VyOS firewalls, see *Firewall*.

```
- set firewall
  * flowtable
    - custom_flow_table
    + ...
```

Flowtables let you define a fastpath through the flowtable datapath. Flowtables support layer 3 (IPv4 and IPv6) and layer 4 (TCP and UDP) protocols.

After the first packet successfully traverses the IP forwarding path (black circles path), you can offload subsequent packets to the flowtable through your ruleset. You specify when to add a flow to the flowtable during forward filtering (red circle number 6).

When a packet finds a matching entry in the flowtable (flowtable hit), the system transmits it to the output netdevice. This means packets bypass the classic IP forwarding path and use the **Fast Path** (orange circles path). As a result, you do not see these packets from any Netfilter hooks after ingress. If no matching entry exists in the flowtable (flowtable miss), the packet traverses the classic IP forwarding path.

Note: Flowtable Reference: https://docs.kernel.org/networking/nf_flowtable.html

Flowtable Configuration

To use flowtables, you need to configure the following:

- Create a flowtable that includes the interfaces that are going to be used by the flowtable.
- Create a firewall rule. Set the action to offload and use your desired flowtable for offload-target.

Creating a flow table:

```
set firewall flowtable <flow_table_name> interface <iface>
```

Specify interfaces to use in the flowtable.

```
set firewall flowtable <flow_table_name> description <text>
```

Provide a description for the flow table.

```
set firewall flowtable <flow_table_name> offload <hardware | software>
```

Specify the offload type the flowtable uses: hardware or software. The default is software offload.

Note: Hardware offload: Make sure your network interface controller (NIC) supports hardware offloading and that you have the necessary drivers

installed before enabling this option.

Creating rules for using flow tables:

```
set firewall [ipv4 | ipv6] forward filter rule <1-999999> action offload
```

Create a firewall rule in the forward chain with the action set to offload.

```
set firewall [ipv4 | ipv6] forward filter rule <1-999999> offload-target <flowtable>
```

Create a firewall rule in the forward chain and specify which flowtable to use. Only applicable if the action is offload.

Configuration Example

Consider the following in this setup:

- This example uses two interfaces in the flowtables: `eth0` and `eth1`.
- The example provides a minimal firewall ruleset with filtering rules and rules for using flowtable offload capabilities.

The first packet is evaluated by the firewall path, so a desired connection should be explicitly accepted. The same should occur for traffic in reverse order. In most cases, state policies are used to accept a connection in the reverse path.

In the following example only traffic coming from interface `eth0`, TCP protocol, and destination port 1122 is accepted. All other traffic to the router is dropped.

Commands

```
set firewall flowtable FT01 interface 'eth0'
set firewall flowtable FT01 interface 'eth1'
set firewall ipv4 forward filter default-action 'drop'
set firewall ipv4 forward filter rule 10 action 'offload'
set firewall ipv4 forward filter rule 10 offload-target 'FT01'
set firewall ipv4 forward filter rule 10 state 'established'
set firewall ipv4 forward filter rule 10 state 'related'
set firewall ipv4 forward filter rule 20 action 'accept'
set firewall ipv4 forward filter rule 20 state 'established'
set firewall ipv4 forward filter rule 20 state 'related'
set firewall ipv4 forward filter rule 110 action 'accept'
set firewall ipv4 forward filter rule 110 destination address '192.0.2.100'
set firewall ipv4 forward filter rule 110 destination port '1122'
set firewall ipv4 forward filter rule 110 inbound-interface name 'eth0'
set firewall ipv4 forward filter rule 110 protocol 'tcp'
```

Explanation

Here's what happens for a desired connection:

1. A packet arrives on **eth0** with destination address **192.0.2.100**, TCP protocol, and destination port **1122**. Assume this address is reachable through interface **eth1**.
2. For this first packet, the connection state is **new**. Neither rule 10 nor rule 20 applies.
3. Rule 110 matches, so the connection is accepted.
4. When the server **192.0.2.100** replies, the connection state becomes **established**, and rule 20 accepts the reply.
5. The router receives the second packet for this connection. Because the connection state is **established**, rule 10 matches and adds a new entry in the flowtable FT01 for this connection.
6. Subsequent packets skip the traditional path and use the **Fast Path** for offloading.

Checks

Check the conntrack table to verify that the system accepted and properly offloaded connections.

```
vyos@FlowTables:~$ show firewall ipv4 forward filter
Ruleset Information

-----
ipv4 Firewall "forward filter"

Rule      Action    Protocol  Packets  Bytes  Conditions
-----
10         offload   all       8        468    ct state { established, related }
→flow add @VYOS_FLOWTABLE_FT01
20         accept    all       8        468    ct state { established, related }
→accept
110        accept    tcp       2        120    ip daddr 192.0.2.100 tcp dport 1122
→iifname "eth0" accept
default    drop      all       7        420

vyos@FlowTables:~$ sudo conntrack -L | grep tcp
conntrack v1.4.6 (conntrack-tools): 5 flow entries have been shown.
tcp        6 src=198.51.100.100 dst=192.0.2.100 sport=41676 dport=1122 src=192.0.2.100
→dst=198.51.100.100 sport=1122 dport=41676 [OFFLOAD] mark=0 use=2
vyos@FlowTables:~$
```

Note: For more information on Netfilter hooks and Linux networking packet flows, see the [Netfilter-Hooks](#) documentation.

7.2.7 Zone-Based firewall

Zone-Based Firewall

Overview

Note: All VyOS versions built after 2023-10-22 (VyOS 1.4 and 1.5) support this feature.

This section provides information on firewall configuration for the zone-based firewall. This section covers the following configuration commands:

set firewall zone ...

To learn about the general traffic flow in VyOS firewalls, see [Firewall](#).

```
- set firewall
  * zone
    - custom_zone_name
    + ...
```

In zone-based policy, you assign interfaces to zones and apply inspection policy to traffic moving between zones. The firewall acts on traffic according to rules. A zone is a group of interfaces that have similar functions or features. It establishes the security borders of a network. A zone defines a boundary where the system subjects traffic to policy restrictions as it crosses to another region of a network.

Key Points:

- A zone must be configured before you assign an interface to it, and you can assign an interface to only a single zone.
- All traffic to and from an interface within a zone flows freely.
- Existing policies affect all traffic between zones.
- Traffic cannot flow between a zone member interface and any interface that is not a zone member.
- You must define 2 separate firewalls to define traffic: one for each direction.

Note: In T2199 the syntax of the zone configuration was changed. The zone configuration moved from `zone-policy zone <name>` to `firewall zone <name>`.

Configuration

As an alternative to applying policy to an interface directly, you can create a zone-based firewall to simplify configuration when multiple interfaces belong to the same security zone. Instead of applying rule-sets to interfaces, you apply them to source-destination zone pairs.

You can find a basic introduction to zone-based firewalls in the [VyOS Knowledge Base](#), and an example at [Zone-Policy example](#).

The following steps are required to create a zone-based firewall:

1. Define both the source and destination zones
2. Define the rule-set

3. Apply the rule-set to the zones

Define a Zone

To define a zone, set up either one with interfaces or as the local zone.

set firewall zone <name> interface <interface>

Assign interfaces as a member of a zone.

Note:

- An interface can only be a member of one zone.
 - You can have multiple interfaces in a zone. Traffic between interfaces in the same zone follows the intra-zone-filtering policy (allowed by default).
-

set firewall zone <name> local-zone

Define the zone as the local zone for traffic that originates from or is destined to the router itself.

Note:

- A local zone cannot have any member interfaces
 - You cannot have multiple local zones
-

set firewall zone <name> default-action [drop | reject]

Modify the zone default-action, which applies to traffic destined to this zone that does not match any of the source zone rulesets applied.

set firewall zone <name> default-log

Enable logging of packets that match this zone's default-action (disabled by default).

set firewall zone <name> description

Add a meaningful description.

Defining a Rule-Set

Zone-based firewall rule-sets define traffic from a *Source Zone* to a *Destination Zone*.

You create rule-sets as a custom firewall chain using the commands below (refer to the firewall IPv4/IPv6 sections for the full syntax):

- For *IPv4*: **set firewall ipv4 name <name> ...**
- For *IPv6*: **set firewall ipv6 name <name> ...**

It is helpful to name the rule-sets in the format <Source Zone>-<Destination Zone>-<v4 | v6> to make them easily identifiable.

Applying a Rule-Set to a Zone

After you define a rule-set, apply it to the source and destination zones. The configuration syntax anchors to the destination zone, with each of the source zone rule-sets listed against the destination.

```
set firewall zone <Destination Zone> from <Source Zone> firewall name
<ipv4-rule-set-name>
```

```
set firewall zone <Destination Zone> from <Source Zone> firewall ipv6-name
<ipv6-rule-set-name>
```

You should create two rule-sets for each source-destination zone pair.

```
set firewall zone DMZ from LAN firewall name LAN-DMZ-v4
set firewall zone LAN from DMZ firewall name DMZ-LAN-v4
```

Applying a Default Rule-Set to a Zone

When a destination zone shares a common rule-set for multiple source zones, or when you require a complex set of default policies, you can apply an optional default rule-set. The default rule-set applies to all zones that do not have a rule-set configured as defined in *IPv4*

```
set firewall zone <Destination Zone> default-firewall name <ipv4-rule-set-name>
```

```
set firewall zone <Destination Zone> default-firewall ipv6-name <ipv6-rule-set-name>
```

Operation-mode

show firewall zone-policy

Display a basic summary of the zone configuration.

```
vyos@vyos:~$ show firewall zone-policy
Zone    Interfaces    From Zone    Firewall IPv4    Firewall IPv6
-----
LAN     eth1           WAN          WAN-LAN-v4
eth2
LOCAL   LOCAL         LAN          LAN-LOCAL-v4
WAN     WAN-LOCAL-v4   WAN-LOCAL-v6
WAN     eth3           LAN          LAN-WAN-v4
eth0    LOCAL         LOCAL-WAN-v4
```

show firewall zone-policy zone <zone>

Display a basic summary of a particular zone.

```
vyos@vyos:~$ show firewall zone-policy zone WAN
Zone    Interfaces    From Zone    Firewall IPv4    Firewall IPv6
-----
WAN     eth3           LAN          LAN-WAN-v4
eth0    LOCAL         LOCAL-WAN-v4
vyos@vyos:~$ show firewall zone-policy zone LOCAL
Zone    Interfaces    From Zone    Firewall IPv4    Firewall IPv6
-----
```

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LOCAL	LOCAL	LAN	LAN-LOCAL-v4
WAN	WAN-LOCAL-v4		WAN-LOCAL-v6

With zone-based firewalls, a new concept applies. In addition to the standard in and out traffic flows, a local flow enables traffic originating from and destined to the router itself. This means you must configure additional rules to secure the firewall from the network, in addition to the existing inbound and outbound rules.

To configure VyOS with zone-based firewall, see [Zone-Based Firewall Configuration](#).

As the following example image shows, you must configure rules to allow or block traffic to or from the services running on the device that have open connections on that interface.

7.3 High availability

VRRP (Virtual Router Redundancy Protocol) provides active/backup redundancy for routers. Every VRRP router has a physical IP/IPv6 address, and a virtual address. On startup, routers elect the master, and the router with the highest priority becomes the master and assigns the virtual address to its interface. All routers with lower priorities become backup routers. The master then starts sending keepalive packets to notify other routers that it's available. If the master fails and stops sending keepalive packets, the router with the next highest priority becomes the new master and takes over the virtual address.

VRRP keepalive packets use multicast, and VRRP setups are limited to a single datalink layer segment. You can setup multiple VRRP groups (also called virtual routers). Virtual routers are identified by a VRID (Virtual Router Identifier). If you setup multiple groups on the same interface, their VRIDs must be unique if they use the same address family, but it's possible (even if not recommended for readability reasons) to use duplicate VRIDs on different interfaces.

7.3.1 Basic setup

VRRP groups are created with the `set high-availability vrrp group $GROUP_NAME` commands. The required parameters are interface, vrid, and address.

minimal config

```
set high-availability vrrp group Foo vrid 10
set high-availability vrrp group Foo interface eth0
set high-availability vrrp group Foo address 192.0.2.1/24
```

You can verify your VRRP group status with the operational mode `run show vrrp` command:

```
vyos@vyos# run show vrrp
Name      Interface  VRID  State  Last Transition
-----
Foo       eth1       10    MASTER 2s
```

7.3.2 IPv6 support

The address parameter can be either an IPv4 or IPv6 address, but you can not mix IPv4 and IPv6 in the same group, and will need to create groups with different VRIDs specially for IPv4 and IPv6. If you want to use IPv4 + IPv6 address you can use option `excluded-address`

7.3.3 Address

The address can be configured either on the VRRP interface or on not VRRP interface.

```
set high-availability vrrp group Foo address 192.0.2.1/24
set high-availability vrrp group Foo address 203.0.113.22/24 interface eth2
set high-availability vrrp group Foo address 198.51.100.33/24 interface eth3
```

7.3.4 Disabling a VRRP group

You can disable a VRRP group with `disable` option:

```
set high-availability vrrp group Foo disable
```

A disabled group will be removed from the VRRP process and your router will not participate in VRRP for that VRID. It will disappear from operational mode commands output, rather than enter the backup state.

7.3.5 Exclude address

Exclude IP addresses from VRRP packets. This option `excluded-address` is used when you want to set IPv4 + IPv6 addresses on the same virtual interface or when used more than 20 IP addresses.

```
set high-availability vrrp group Foo excluded-address '203.0.113.254/24'
set high-availability vrrp group Foo excluded-address '2001:db8:aa::1/64'
set high-availability vrrp group Foo excluded-address '2001:db8:22::1/64'
```

7.3.6 Setting VRRP group priority

VRRP priority can be set with `priority` option:

```
set high-availability vrrp group Foo priority 200
```

The priority must be an integer number from 1 to 255. Higher priority value increases router's precedence in the master elections.

7.3.7 Sync groups

A sync group allows VRRP groups to transition together.

```
edit high-availability vrrp
set sync-group MAIN member VLAN9
set sync-group MAIN member VLAN20
```

In the following example, when VLAN9 transitions, VLAN20 will also transition:

```
vrrp {
  group VLAN9 {
    interface eth0.9
    address 10.9.1.1/24
    priority 200
    vrid 9
  }
  group VLAN20 {
    interface eth0.20
    priority 200
    address 10.20.20.1/24
    vrid 20
  }
  sync-group MAIN {
    member VLAN20
    member VLAN9
  }
}
```

Warning: All items in a sync group should be similarly configured. If one VRRP group is set to a different preemption delay or priority, it would result in an endless transition loop.

7.3.8 Preemption

VRRP can use two modes: preemptive and non-preemptive. In the preemptive mode, if a router with a higher priority fails and then comes back, routers with lower priority will give up their master status. In non-preemptive mode, the newly elected master will keep the master status and the virtual address indefinitely.

By default VRRP uses preemption. You can disable it with the “no-preempt” option:

```
set high-availability vrrp group Foo no-preempt
```

You can also configure the time interval for preemption with the “preempt-delay” option. For example, to set the higher priority router to take over in 180 seconds, use:

```
set high-availability vrrp group Foo preempt-delay 180
```

7.3.9 Track

Track option to track non VRRP interface states. VRRP changes status to FAULT if one of the track interfaces in state down.

```
set high-availability vrrp group Foo track interface eth0
set high-availability vrrp group Foo track interface eth1
```

Ignore VRRP main interface faults

```
set high-availability vrrp group Foo track exclude-vrrp-interface
```

7.3.10 Unicast VRRP

By default VRRP uses multicast packets. If your network does not support multicast for whatever reason, you can make VRRP use unicast communication instead.

```
set high-availability vrrp group Foo peer-address 192.0.2.10
set high-availability vrrp group Foo hello-source-address 192.0.2.15
```

7.3.11 rfc3768-compatibility

RFC 3768 defines a virtual MAC address to each VRRP virtual router. This virtual router MAC address will be used as the source in all periodic VRRP messages sent by the active node. When the rfc3768-compatibility option is set, a new VRRP interface is created, to which the MAC address and the virtual IP address is automatically assigned.

```
set high-availability vrrp group Foo rfc3768-compatibility
```

Verification

```
$show interfaces ethernet eth0v10
eth0v10@eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue
state UP group default qlen 1000
link/ether 00:00:5e:00:01:0a brd ff:ff:ff:ff:ff:ff
inet 172.25.0.247/16 scope global eth0v10
valid_lft forever preferred_lft forever
```

Warning: RFC 3768 creates a virtual interface. If you want to apply the destination NAT rule to the traffic sent to the virtual MAC, set the created virtual interface as `inbound-interface`.

7.3.12 Global options

On most scenarios, there's no need to change specific parameters, and using default configuration is enough. But there are cases where extra configuration is needed.

```
set high-availability vrrp global-parameters startup_delay <1-600>
```

This option specifies a delay in seconds before vrrp instances start up after keepalived starts.

7.3.13 Gratuitous ARP

These configuration is not mandatory and in most cases there's no need to configure it. But if necessary, Gratuitous ARP can be configured in `global-parameters` and/or in `group` section.

```
set high-availability vrrp global-parameters garp interval <0.000-1000>
```

```
set high-availability vrrp group <name> garp interval <0.000-1000>
```

Set delay between gratuitous ARP messages sent on an interface.

0 if not defined.

```
set high-availability vrrp global-parameters garp master-delay <1-255>
```

```
set high-availability vrrp group <name> garp master-delay <1-255>
```

Set delay for second set of gratuitous ARPs after transition to MASTER.

5 if not defined.

```
set high-availability vrrp global-parameters garp master-refresh <1-600>
```

```
set high-availability vrrp group <name> garp master-refresh <1-600>
```

Set minimum time interval for refreshing gratuitous ARPs while MASTER.

0 if not defined, which means no refreshing.

```
set high-availability vrrp global-parameters garp master-refresh-repeat <1-600>
```

```
set high-availability vrrp group <name> garp master-refresh-repeat <1-600>
```

Set number of gratuitous ARP messages to send at a time while MASTER.

1 if not defined.

```
set high-availability vrrp global-parameters garp master-repeat <1-600>
```

```
set high-availability vrrp group <name> garp master-repeat <1-600>
```

Set number of gratuitous ARP messages to send at a time after transition to MASTER.

5 if not defined.

7.3.14 Version

```
set high-availability vrrp global-parameters version 2|3
```

Set the default VRRP version to use. This defaults to 2, but IPv6 instances will always use version 3.

7.3.15 Scripting

VRRP functionality can be extended with scripts. VyOS supports two kinds of scripts: health check scripts and transition scripts. Health check scripts execute custom checks in addition to the master router reachability. Transition scripts are executed when VRRP state changes from master to backup or fault and vice versa and can be used to enable or disable certain services, for example.

Note: Simply placing script files in `/config/scripts/` does not mean the system can execute them. To make custom scripts executable, grant them **execute permissions**. Use the following command:

```
chmod +x /config/scripts/script-name.sh
```

Warning: It is not recommended to change VRRP configuration inside health-check and transition scripts.

Health check scripts

There is the ability to run an arbitrary script at regular intervals according to health-check parameters. If a script returns 0, it indicates success. If a script returns anything else, it will indicate that the VRRP instance should enter the FAULT state.

This setup will make the VRRP process execute the `/config/scripts/vrrp-check.sh` script every 60 seconds, and transition the group to the fault state if it fails (i.e. exits with non-zero status) three times:

```
set high-availability vrrp group Foo health-check script /config/scripts/vrrp-check.sh
set high-availability vrrp group Foo health-check interval 60
set high-availability vrrp group Foo health-check failure-count 3
```

An optional `timeout` can be set to define the maximum number of seconds the script is allowed to run. This is useful for scripts that may hang or take longer than expected — setting the timeout higher than the interval allows longer-running scripts to complete before being considered failed.

```
set high-availability vrrp group Foo health-check script /config/scripts/vrrp-check.sh
set high-availability vrrp group Foo health-check interval 20
set high-availability vrrp group Foo health-check failure-count 3
set high-availability vrrp group Foo health-check timeout 40
```

When the vrrp group is a member of the sync group will use only the sync group health check script. This example shows how to configure it for the sync group:

```
set high-availability vrrp sync-group Bar health-check script /config/scripts/vrrp-check.
↪ sh
set high-availability vrrp sync-group Bar health-check interval 60
set high-availability vrrp sync-group Bar health-check failure-count 3
```

Transition scripts

Transition scripts can help you implement various fixups, such as starting and stopping services, or even modifying the VyOS config on VRRP transition. This setup will make the VRRP process execute the `/config/scripts/vrrp-fail.sh` with argument `Foo` when VRRP fails, and the `/config/scripts/vrrp-master.sh` when the router becomes the master:

```
set high-availability vrrp group Foo transition-script backup "/config/scripts/vrrp-fail.
↪ sh Foo"
set high-availability vrrp group Foo transition-script fault "/config/scripts/vrrp-fail.
↪ sh Foo"
set high-availability vrrp group Foo transition-script master "/config/scripts/vrrp-
↪ master.sh Foo"
```

To know more about scripting, check the [Command scripting](#) section.

7.3.16 Virtual-server

Virtual Server allows to Load-balance traffic destination virtual-address:port between several real servers.

Algorithm

Load-balancing schedule algorithm:

- round-robin
- weighted-round-robin
- least-connection
- weighted-least-connection
- source-hashing
- destination-hashing
- locality-based-least-connection

```
set high-availability virtual-server 203.0.113.1 algorithm 'least-connection'
```

Forward method

- NAT
- direct
- tunnel

```
set high-availability virtual-server 203.0.113.1 forward-method 'nat'
```

Persistence timeout

Timeout in seconds for persistent connections. IPVS will forward all connections from the same source IP to the same real server for the duration of the timeout. Default value is 300 seconds.

Set to 0 to disable persistence and allow the load-balancing algorithm to distribute every connection independently.

```
set high-availability virtual-server 203.0.113.1 persistence-timeout '180'
```

Health-check

Custom health-check script allows checking real-server availability

```
set high-availability virtual-server 203.0.113.1 real-server 192.0.2.11 health-check ↵  
↵script <path-to-script>
```

fwmark

Firewall mark. It possible to loadbalancing traffic based on fwmark value

```
set high-availability virtual-server 203.0.113.1 fwmark '111'
```

Real server

Real server IP address and port

```
set high-availability virtual-server 203.0.113.1 real-server 192.0.2.11 port '80'
```

Example

Virtual-server can be configured with VRRP virtual address or without VRRP.

In the next example all traffic destined to 203.0.113.1 and port 8280 protocol TCP is balanced between 2 real servers 192.0.2.11 and 192.0.2.12 to port 80

Real server is auto-excluded if port check with this server fail.

```
set interfaces ethernet eth0 address '203.0.113.1/24'
set interfaces ethernet eth1 address '192.0.2.1/24'
set high-availability vrrp group F00 interface 'eth0'
set high-availability vrrp group F00 no-preempt
set high-availability vrrp group F00 priority '150'
set high-availability vrrp group F00 address '203.0.113.1/24'
set high-availability vrrp group F00 vrid '10'

set high-availability virtual-server 203.0.113.1 algorithm 'source-hashing'
set high-availability virtual-server 203.0.113.1 delay-loop '10'
set high-availability virtual-server 203.0.113.1 forward-method 'nat'
set high-availability virtual-server 203.0.113.1 persistence-timeout '180'
set high-availability virtual-server 203.0.113.1 port '8280'
set high-availability virtual-server 203.0.113.1 protocol 'tcp'
set high-availability virtual-server 203.0.113.1 real-server 192.0.2.11 port '80'
set high-availability virtual-server 203.0.113.1 real-server 192.0.2.12 port '80'
```

A firewall mark fwmark allows using multiple ports for high-availability virtual-server. It uses fwmark value.

In this example all traffic destined to ports “80, 2222, 8888” protocol TCP marks to fwmark “111” and balanced between 2 real servers. Port “0” is required if multiple ports are used.

```
set interfaces ethernet eth0 address 'dhcp'
set interfaces ethernet eth0 description 'WAN'
set interfaces ethernet eth1 address '192.0.2.1/24'
set interfaces ethernet eth1 description 'LAN'

set policy route PR interface 'eth0'
set policy route PR rule 10 destination port '80,2222,8888'
set policy route PR rule 10 protocol 'tcp'
set policy route PR rule 10 set mark '111'
```

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```

set high-availability virtual-server vyos fwmark '111'
set high-availability virtual-server vyos protocol 'tcp'
set high-availability virtual-server vyos real-server 192.0.2.11 health-check script '/
↪config/scripts/check-real-server-first.sh'
set high-availability virtual-server vyos real-server 192.0.2.11 port '0'
set high-availability virtual-server vyos real-server 192.0.2.12 health-check script '/
↪config/scripts/check-real-server-second.sh'
set high-availability virtual-server vyos real-server 192.0.2.12 port '0'

set nat source rule 100 outbound-interface name 'eth0'
set nat source rule 100 source address '192.0.2.0/24'
set nat source rule 100 translation address 'masquerade'

```

Op-mode check virtual-server status

```

vyos@r14:~$ run show virtual-server
IP Virtual Server version 1.2.1 (size=4096)
Prot LocalAddress:Port Scheduler Flags
  -> RemoteAddress:Port      Forward Weight ActiveConn InActConn
FWM  111 lc persistent 300
  -> 192.0.2.11:0             Masq    1      0          0
  -> 192.0.2.12:0             Masq    1      1          0

```

7.4 Interfaces

7.4.1 Bond / link aggregation

A **bonding interface** aggregates multiple network interfaces into a single logical interface (referred to as a bond, LAG (Link Aggregation Group), EtherChannel, or port-channel).

The behavior of a bonding interface depends on the selected mode. Modes provide either fault tolerance or a combination of load balancing and fault tolerance. Additionally, the bonding interface can be configured for link integrity monitoring.

Configuration

Common interface configuration

```
set interfaces bonding <interface> address <address | dhcp | dhcpv6>
```

Configure the interface with one or more IP addresses.

The following options are available:

- **address:** Assign one or more IPv4 or IPv6 addresses to the interface. For example, use 192.0.2.1/24 for IPv4 or 2001:db8::1/64 for IPv6.
- **dhcp:** The interface obtains an IPv4 address from a DHCP server on the same network segment.
- **dhcpv6:** The interface obtains an IPv6 address from a DHCPv6 server on the same network segment.

Note: If the interface obtains an IPv4 address via DHCP, and specific adjustments are needed before/after the IP address is obtained, use the provided hook scripts:

- /config/scripts/dhcp-client/pre-hooks.d/
- /config/scripts/dhcp-client/post-hooks.d/

Example:

```
set interfaces bonding bond0 address 192.0.2.1/24
set interfaces bonding bond0 address 2001:db8::1/64
set interfaces bonding bond0 address dhcp
set interfaces bonding bond0 address dhcpv6
```

set interfaces bonding <interface> description <description>

Configure a clear, descriptive alias for the interface.

This alias appears in the `show interfaces` command and SNMP-based monitoring tools.

Example:

```
set interfaces bonding bond0 description 'This is an interface running on VyOS.'
```

set interfaces bonding <interface> disable

Disable the interface.

The interface will be set to the administratively down (A/D) state.

Example:

```
set interfaces bonding bond0 disable
```

set interfaces bonding <interface> disable-flow-control

Disable Ethernet flow control (IEEE 802.3x pause frames) on the interface.

Ethernet flow control, defined by the IEEE 802.3x standard, temporarily stops data transmission to prevent packet loss during network congestion. For example, when a sender transmits data faster than the receiver can process it.

Disabling Ethernet flow control means the interface will not signal the connected device to pause transmission and will drop packets if overwhelmed.

Example:

```
set interfaces bonding bond0 disable-flow-control
```

set interfaces bonding <interface> disable-link-detect

Disable physical link-state change detection on the interface, such as when a cable is unplugged.

By default, the interface detects physical link-state changes.

Example:

```
set interfaces bonding bond0 disable-link-detect
```

set interfaces bonding <interface> mac <xx:xx:xx:xx:xx:xx>

Configure a custom MAC address on the interface.

Example:

```
set interfaces bonding bond0 mac '00:53:01:02:03:04'
```

set interfaces bonding <interface> mtu <mtu>**Configure the MTU on the interface.**

This value defines the largest packet size, in bytes, that the interface transmits without fragmentation.

Example:

```
set interfaces bonding bond0 mtu 1600
```

set interfaces bonding <interface> ip adjust-mss <mss | clamp-mss-to-pmtu>**Configure the MSS (Maximum Segment Size) advertised in outgoing TCP SYN packets on the specified interface.**

By clamping the MSS value in TCP SYN packets, you explicitly inform the remote side not to send packets larger than that size. This prevents connection issues that occur when Path MTU Discovery (PMTUD) fails.

The following options are available:

- **mss:** Sets the MSS to a specific value, in bytes. Use this option if you need to enforce a specific MSS, for example, to troubleshoot connectivity issues or accommodate specific network requirements.
- **clamp-mss-to-pmtu:** The router automatically calculates the MSS to be the interface's MTU minus 40 bytes for IPv4 traffic (20 bytes for the IPv4 header and 20 bytes for the TCP header). This option is recommended to automatically set the proper value.

set interfaces bonding <interface> ip arp-cache-timeout**Configure how long an ARP entry remains valid after learning an IP-to-MAC address mapping on this interface.**

The default duration is 30 seconds.

An ARP entry remains valid if it receives positive feedback from higher-level protocols.

Example:

```
set interfaces bonding bond0 ip arp-cache-timeout 180
```

set interfaces bonding <interface> ip disable-arp-filter**Configure ARP filtering on this interface.**

Default behavior: The kernel responds to ARP requests on this interface only if the traffic would be routed back to the ARP sender through that specific interface.

If configured: The kernel responds to ARP requests on this interface for any IP address configured on the local host, regardless of which specific interface that IP address is assigned to, and regardless of the routing table. This reflects the Linux concept that IP addresses belong to the host, not individual interfaces.

Example:

```
set interfaces bonding bond0 ip disable-arp-filter
```

set interfaces bonding <interface> ip disable-forwarding

Configure the interface for host or router behavior.

If configured, the interface switches to host mode, and IPv4 forwarding is disabled on it.

Example:

```
set interfaces bonding bond0 ip disable-forwarding
```

set interfaces bonding <interface> ip enable-directed-broadcast

Configure whether to forward IP-directed broadcast packets received on this interface.

Default behavior: IP-directed broadcast packets are dropped.

If configured: IP-directed broadcast packets are forwarded to all hosts on the destination subnet, as defined in [RFC 1812](#) and [RFC 2644](#).

Example:

```
set interfaces bonding bond0 ip enable-directed-broadcast
```

set interfaces bonding <interface> ip enable-arp-accept

Configure how to process gratuitous ARPs on this interface.

If configured, an IP-to-MAC address mapping is added to the ARP table based on gratuitous ARP requests or replies.

Note: If the ARP table already contains the IP address from a gratuitous ARP, its entry is updated regardless of whether this setting is configured.

Example:

```
set interfaces bonding bond0 ip enable-arp-accept
```

set interfaces bonding <interface> ip enable-arp-announce

Configure the source IP selection for ARP requests on this interface.

Default behavior: The kernel can use any IP address the host owns as the source IP address in ARP requests on this interface.

If configured: The kernel first attempts to select a source IP address configured on the interface that shares a common subnet with the target IP address. If there is no such subnet, the kernel selects the IP address it would normally use (based on the routing table to reach the target destination).

Example:

```
set interfaces bonding bond0 ip enable-arp-announce
```

set interfaces bonding <interface> ip enable-arp-ignore

Configure which ARP requests will be ignored on this interface.

Default behavior: The kernel responds to ARP requests for any local IP addresses, regardless of which interface they are assigned to.

If configured: The kernel responds to ARP requests only if the target IP address is assigned to this specific interface.

Example:

```
set interfaces bonding bond0 ip enable-arp-ignore
```

set interfaces bonding <interface> ip enable-proxy-arp

Configure proxy ARP on this interface.

If configured, the router (kernel) intercepts ARP requests for non-local IP addresses and replies with the MAC (Media Access Control) address of the interface that received the request. Subsequent packets destined to these IP addresses are forwarded to their actual destinations on remote subnets.

Example:

```
set interfaces bonding bond0 ip enable-proxy-arp
```

set interfaces bonding <interface> ip proxy-arp-pvlan

Configure local proxy ARP on the interface.

If configured, the router (kernel) responds to ARP requests on this VLAN interface even if the target IP address resides on the same subnet and interface.

This is used to support network isolation requirements (RFC 3069) for private VLANs (PVLANS). In PVLAN configurations, hosts on isolated ports are NOT allowed to communicate directly with each other at Layer 2, but they can communicate with the upstream router.

By replying to inter-host ARP requests with its own MAC address, the router (kernel) directs inter-host traffic through itself instead of directly between hosts.

Note: This command works independently and does not require enabling the standard proxy ARP on the interface.

Local proxy ARP is also known as:

- VLAN aggregation ([RFC 3069](#)).
- Private VLAN (Cisco, Allied Telesyn).
- Source-port filtering or port isolation (Hewlett-Packard).
- MAC-Forced Forwarding (Ericsson).

set interfaces bonding <interface> ip source-validation <strict | loose | disable>

Configure source IP address validation using RPF (Reverse Path Forwarding) on this interface, as specified in [RFC 3704](#).

The following options are available:

- **strict:** Each incoming packet's source IP address is checked against the Forwarding Information Base (FIB). If the interface is not the best route back to that source, validation fails, and the packet is dropped.
- **loose:** Each incoming packet's source IP address is checked against the FIB. If the source IP address is unreachable through any interface, validation fails.
- **disable:** No source IP address validation is performed. All incoming packets are accepted.

RFC 3704 recommends enabling strict mode to prevent IP spoofing, such as DDoS attacks. For asymmetric or other complex routing scenarios, use loose mode.

set interfaces bonding <interface> ipv6 address autoconf

Configure the interface to automatically obtain an IPv6 address using SLAAC (Stateless Address Autoconfiguration), as specified in [RFC 4862](#).

IPv6 hosts can configure themselves automatically when connected to an IPv6 network using the Neighbor Discovery Protocol via ICMPv6 router discovery messages. When first connected to a network, a host sends a link-local router solicitation multicast request for its configuration parameters. The router responds with a router advertisement packet containing Internet Layer configuration parameters.

Note: This method automatically disables IPv6 traffic forwarding on the interface.

Example:

```
set interfaces bonding bond0 ipv6 address autoconf
```

set interfaces bonding <interface> ipv6 address eui64 <prefix>

Configure the interface to assign itself an IPv6 address using the EUI-64 (64-Bit Extended Unique Identifier) method, as specified in [RFC 4291](#).

Example:

```
set interfaces bonding bond0 ipv6 address eui64 2001:db8:beef::/64
```

set interfaces bonding <interface> ipv6 address no-default-link-local

Disable the automatic assignment of a link-local IPv6 address to this interface.

Example:

```
set interfaces bonding bond0 ipv6 address no-default-link-local
```

set interfaces bonding <interface> ipv6 disable-forwarding

Configure the interface for host or router behavior.

If configured, the interface switches to host mode, and IPv6 forwarding is disabled on it.

Example:

```
set interfaces bonding bond0 ipv6 disable-forwarding
```

set interfaces bonding <interface> ipv6 adjust-mss <mss | clamp-mss-to-pmtu>

Configure the MSS advertised in outgoing TCP SYN packets on the specified interface.

By clamping the MSS value in TCP SYN packets, you explicitly inform the remote side not to send packets larger than that size. This prevents connection issues when Path MTU Discovery (PMTUD) fails.

The following options are available:

- **mss:** Set the MSS to a specific value, in bytes. Use this option to enforce a specific MSS, for example, to troubleshoot connectivity issues or accommodate specific network requirements.
- **clamp-mss-to-pmtu:** The router calculates the MSS to be the interface's MTU minus 60 bytes for IPv6 traffic (40 bytes for the IPv6 header and 20 bytes for the TCP header). This option is recommended to automatically set the proper value.

set interfaces bonding <interface> ipv6 accept-dad <0 | 1 | 2>

Configure IPv6 DAD (Duplicate Address Detection) on the interface.

The following options are available:

- **0:** Disables DAD. No duplicate address detection is performed.
- **1:** Enables DAD (default). Duplicate addresses are detected. The interface's IPv6 operation continues for valid IPv6 addresses.
- **2:** Enables DAD and, if a MAC-based duplicate link-local address is found, disables IPv6 operation on this interface.

Example:

```
set interfaces bonding bond0 ipv6 accept-dad 2
```

set interfaces bonding <interface> ipv6 dup-addr-detect-transmits <n>

Configure the number of DAD messages that the router (kernel) sends during IPv6 address assignment on this interface.

The default value is 1.

Example:

```
set interfaces bonding bond0 ipv6 dup-addr-detect-transmits 5
```

set interfaces bonding <interface> vrf <vrf>

Assign the interface to a specific VRF instance.

See also:

For information on configuring a VRF, refer to the [VRF](#) section.

Example:

```
set interfaces bonding bond0 vrf red
```

DHCP(v6)

set interfaces bonding <interface> dhcp-options client-id <description>

Configure a DHCP client identifier for the interface, as specified in [RFC 2131](#).

The `client-id` is an identifier that the DHCP client sends to the DHCP server to uniquely identify itself for IP address assignment. By default, the client uses its MAC address. The `<description>` is a user-defined string that will be sent to the DHCP server as the DHCP client identifier.

Example:

```
set interfaces bonding bond0 dhcp-options client-id 'foo-bar'
```

set interfaces bonding <interface> dhcp-options host-name <hostname>

Configure a specific hostname for the interface.

Instead of the real hostname, the DHCP client will send the specific hostname to the DHCP server when requesting an IP address.

Example:

```
set interfaces bonding bond0 dhcp-options host-name 'VyOS'
```

set interfaces bonding <interface> dhcp-options vendor-class-id <vendor-id>

Configure the DHCP client to include a vendor-class identifier in its DHCP requests on this interface.

The vendor-class identifier is a vendor-specific byte string that enables the DHCP server to identify the device and, in some cases, provide configuration options.

Example:

```
set interfaces bonding bond0 dhcp-options vendor-class-id 'VyOS'
```

set interfaces bonding <interface> dhcp-options no-default-route

Configure the DHCP client to obtain an IP address, but ignore any default gateway provided by the DHCP server on this interface.

Example:

```
set interfaces bonding bond0 dhcp-options no-default-route
```

set interfaces bonding <interface> dhcp-options default-route-distance <distance>

Configure the distance for the default route obtained from the DHCP server on this interface.

Example:

```
set interfaces bonding bond0 dhcp-options default-route-distance 220
```

set interfaces bonding <interface> dhcp-options reject <address>

Configure the DHCP client to reject the specific IP address or IP address range from the DHCP server on this interface.

This is useful when a modem assigns a local IP address upon start. To reject multiple addresses, run this command multiple times with different values. You can reject individual addresses (192.168.100.1) or entire subnets (192.168.100.0/24).

Example:

```
set interfaces bonding bond0 dhcp-options reject 192.168.100.0/24
```

set interfaces bonding <interface> dhcp-options user-class <string>

Configure the DHCP client to send a specific user-class identifier in its DHCP requests on this interface.

The DHCP server can interpret this identifier and provide specific configuration options based on it (for example, default routes). The user-class value typically groups DHCP clients with similar configuration needs (for example, employees, guests, or printers).

Example:

```
set interfaces bonding bond0 dhcp-options user-class VyOS
```

set interfaces bonding <interface> dhcpv6-options duid <duid>

Configure a specific DUID (DHCP Unique Identifier) for the DHCPv6 client on this interface.

The DUID is an identifier used by a DHCPv6 client to get an IPv6 address from a DHCPv6 server. It consists of a 2-byte type field, followed by a variable-length identifier field up to 128 bytes. The format of the identifier part depends on the DUID type:

- **DUID-LLT:** The most common type, which includes a hardware type, a timestamp, and a MAC address.
- **DUID-EN:** Is based on a vendor's enterprise number and a unique identifier assigned by the vendor.

- **DUID-LL:** Includes only a MAC address.

The DHCP server matches the DUID against its database and provides configuration data (such as address, lease times, DNS servers, etc.) to the DHCP client.

Example:

```
set interfaces bonding bond0 duid '0e:00:00:01:00:01:27:71:db:f0:00:50:56:bf:c5:6d'
```

set interfaces bonding <interface> dhcpv6-options no-release

Configure the DHCP client not to send a release message when it stops running on this interface.

This helps retain the assigned address or prefix.

Example:

```
set interfaces bonding bond0 dhcpv6-options no-release
```

set interfaces bonding <interface> dhcpv6-options parameters-only

Enable a stateless DHCPv6 client mode on this interface.

In stateless mode, the DHCPv6 client requests only stateless configuration parameters from the DHCP server (for example, DNS server addresses). It doesn't request a stateful configuration, such as IPv6 addresses or prefixes.

Example:

```
set interfaces bonding bond0 dhcpv6-options parameters-only
```

set interfaces bonding <interface> dhcpv6-options rapid-commit

Enable DHCPv6 rapid commit on this interface.

When enabled, the DHCP client and server skip the negotiation steps (Advertise and Request), completing the DHCPv6 configuration process in just two messages (Solicit and final Reply).

Example:

```
set interfaces bonding bond0 dhcpv6-options rapid-commit
```

set interfaces bonding <interface> dhcpv6-options temporary

Configure the DHCPv6 client to request a temporary IPv6 address on this interface.

When configured, the DHCP client doesn't form an Identity Association for Non-temporary Addresses (IA_NA) partnership. Consequently, it only obtains a temporary IPv6 address and doesn't obtain a permanent one.

Example:

```
set interfaces bonding bond0 dhcpv6-options temporary
```

DHCPv6 Prefix Delegation (PD)

VyOS supports DHCPv6 Prefix Delegation (DHCPv6-PD) as described in [RFC 3633](#). DHCPv6-PD is supported by most ISPs that provide native IPv6 for consumers on fixed networks.

set interfaces bonding <interface> dhcpv6-options pd <id> length <length>

Configure a specific prefix length for DHCPv6-PD requests on this interface.

Some ISPs provide only a /64 prefix by default. Use this command to request a different prefix length for a specific DHCPv6-PD request, ranging from /32 (if allowed by your ISP) down to /64. <id> is a unique identifier for the DHCPv6-PD request.

The default value is 64.

To request a /56 prefix from your ISP, use:

```
set interfaces bonding bond0 dhcpv6-options pd 0 length 56
```

set interfaces bonding <interface> dhcpv6-options pd <id> interface <delegatee> address <address>

Configure the IPv6 interface identifier (host portion) for the delegatee interface.

The value must be a decimal integer. It is appended to the delegated prefix and the configured SLA ID (Site-Level Aggregation ID) to form the final IPv6 address.

By default, the host portion is generated based on the parent interface's MAC address (EUI-64 format).

Example:

If a /64 prefix is delegated to interface eth8 and you configure the host portion as 65535, the resulting IPv6 address will end with ::ffff, as 65535 corresponds to ffff in hexadecimal notation.

```
set interfaces bonding bond0 dhcpv6-options pd 0 interface eth8 address 65534
```

set interfaces bonding <interface> dhcpv6-options pd <id> interface <delegatee> sla-id <id>

Configure the SLA ID for the delegatee interface.

The value must be a decimal integer greater than 0 and fit in the length of SLA IDs. It is converted to hexadecimal and appended to the delegated prefix to form the specific subnet prefix for the delegatee interface.

Example:

If SLA ID is 1 and the delegated prefix is 2001:db8:ffff::/48, the resulting subnet prefix for the delegatee interface will be 2001:db8:ffff:1::/64.

```
set interfaces bonding bond0 dhcpv6-options pd 0 interface eth8 sla-id 1
```

Member interfaces

set interfaces bonding <interface> member interface <member>

Add an interface to the bonding group.

Example:

To configure eth0 and eth1 as members of the bonding interface bond0, execute the following commands:

```
set interfaces bonding bond0 member interface eth0
set interfaces bonding bond0 member interface eth1
```

Bond modes

```
set interfaces bonding <interface> mode <802.3ad | active-backup | broadcast |
round-robin | transmit-load-balance | adaptive-load-balance | xor-hash>
```

Configure the bonding mode on the interface. The default mode is 802.3ad.

The available modes are:

- 802.3ad

Description:	IEEE 802.3ad Dynamic Link Aggregation. Groups only member interfaces with the same speed (e.g., 1 Gbps) and duplex settings. Member interfaces with different speed and duplex settings are not included in the active bond. Provides load balancing and fault tolerance. Uses the LACP (Link Aggregation Control Protocol) to negotiate the bond with the switch.
Traffic distribution:	Traffic is distributed according to the transmit hash policy (default: XOR). The bonding driver applies an XOR operation to specific packet header fields, generating a hash value that maps to a particular member interface. This ensures the same network flow is consistently transmitted over the same member interface. The transmit hash policy is configured via the <code>hash-policy</code> option.
Failover:	If a member interface fails, the hash is recalculated to distribute traffic among the remaining active member interfaces.

Note: Not all transmit hash policies comply with 802.3ad, particularly section 43.2.4. Using a non-compliant policy may result in out-of-order packet delivery.

- active-backup

Description:	Provides fault tolerance. Only one member interface is active at a time. Other member interfaces remain in a standby mode.
Traffic distribution:	All traffic (incoming and outgoing) is routed via one active member interface.
Failover:	If the designated member interface fails, all traffic is routed to another member interface. The bonding driver sends a Gratuitous ARP to update the peer's MAC address table, linking the bond's MAC address to another physical port.

- broadcast

Description:	Provides maximum fault tolerance by duplicating traffic.
Traffic distribution:	Every packet is duplicated and transmitted on all member interfaces.
Failover:	Traffic flow is not interrupted as long as at least one member interface remains active.

- round-robin

Description:	Provides load balancing and fault tolerance.
Traffic distribution:	Packets are transmitted in sequential order across the member interfaces (e.g., packet 1 > interface A, packet 2 > interface B, etc.).
Failover:	If a member interface fails, the sequence skips the failed interface and continues with the remaining active members.

- `transmit-load-balance`

Description:	Provides adaptive transmit load balancing and fault tolerance.
Traffic distribution:	Outgoing: Distributed across all active member interfaces based on the current load. Incoming: Received by a designated member interface (active receiver).
Failover:	If the active receiver fails, another member interface takes over as the new active receiver.

- `adaptive-load-balance`

Description:	Provides adaptive transmit load balancing identical to <code>transmit-load-balance</code> , receive load balancing for IPv4 traffic, and fault tolerance for both incoming and outgoing traffic.
Traffic distribution:	Outgoing: Identical to <code>transmit-load-balance</code> . Incoming: Distributed based on ARP manipulation. For both local and remote connections, the bonding driver intercepts ARP traffic and changes the source MAC address to the MAC address of the least loaded member interface. All traffic from that peer is then routed to the chosen member interface.
Failover:	If a member interface's state changes (fails, recovers, is added, or excluded), the traffic is redistributed among all active member interfaces.

- `xor-hash`: Provides load balancing and fault tolerance based on a hash formula. Distributes traffic and handles failover identically to `802.3ad`, but operates without the LACP.

set interfaces bonding <interface> min-links <0-16>

Configure how many member interfaces must be active (in the link-up state) to mark the bonding interface UP (carrier asserted).

This command applies only when the bonding interface is configured in `802.3ad` mode and functions like the Cisco EtherChannel `min-links` feature. It ensures that a bonding interface is marked UP (carrier asserted) only when a specified number of member interfaces are active (in the link-up state). This helps guarantee a minimum level of bandwidth for higher-level services (such as clustering) relying on the bonding interface.

The default value is 0. This marks the bonding interface UP (carrier asserted) whenever an active LACP aggregator exists, regardless of the number of member interfaces in that aggregator.

Note: In `802.3ad` mode, a bond cannot be active without at least one active member interface. Therefore, setting `min-links` to 0 or 1 has the same result: the bonding interface is marked UP (carrier asserted).

set interfaces bonding <interface> lacp-rate <slow|fast>

Configure the rate at which the bonding interface requests its link partner to send LACPDU's (Link Aggregation Control Protocol Data Units) in `802.3ad` mode.

This command applies only when the bonding interface is configured in `802.3ad` mode.

The following options are available:

- **slow (default):** Requests the link partner to transmit LACPDU's every 30 seconds.
- **fast:** Requests the link partner to transmit LACPDU's every 1 second.

set interfaces bonding <interface> system-mac <mac address>

Configure a specific MAC address for the bonding interface.

This sets the 802.3ad system MAC address, which is used for LACPDU (Link Aggregation Control Protocol Data Unit) exchanges with the link partner. You can assign a fixed MAC address or generate a random one for these LACPDU exchanges.

set interfaces bonding <interface> hash-policy <policy>

Configure which transmit hash policy to use for distributing traffic across member interfaces.

The following policies are available:

- layer2

Description: Routes all traffic destined for a specific network peer through the same member interface. The policy is 802.3ad-compliant.

Hash inputs: Source MAC address, destination MAC address, and Ethernet packet type ID.

Formula:

```
hash = source MAC address XOR destination MAC address XOR packet type ID
member interface number = hash modulo member interface count
```

- layer2+3

Description: Similar to layer2, routes all traffic destined for a specific network peer through the same member interface and is IEEE 802.3ad-compliant. Uses both Layer 2 and Layer 3 information to provide a more balanced traffic distribution.

Hash inputs:

- Source MAC address, destination MAC address, and Ethernet packet type ID.
- Source IP address, destination IP address. IPv6 addresses are first hashed using IPv6_addr_hash.

Formula:

```
hash = source MAC address XOR destination MAC address XOR packet type ID
hash = hash XOR source IP address XOR destination IP address
hash = hash XOR (hash RSHIFT 16)
hash = hash XOR (hash RSHIFT 8)
member interface number = hash modulo member interface count
```

For non-IP traffic, the formula is the same as for layer2.

- layer3+4

Description: Routes different connections (flows) destined for a specific network peer through multiple member interfaces, but ensures each individual flow is routed through only one member interface.

Note: This policy is not fully 802.3ad-compliant. When a single TCP or UDP flow contains both fragmented and unfragmented packets, the algorithm may distribute them across different member interfaces. This may result in out-of-order packet delivery, violating the 802.3ad standard.

Hash inputs:

- Source port, destination port (if available).
- Source IP address, destination IP address. IPv6 addresses are first hashed using IPv6_addr_hash.

Formula:

```

hash = source port, destination port (as in the header)
hash = hash XOR source IP address XOR destination IP address
hash = hash XOR (hash RSHIFT 16)
hash = hash XOR (hash RSHIFT 8)
member interface number = hash modulo member interface count

```

For fragmented TCP or UDP packets and all other IPv4 and IPv6 traffic, the source and destination port information is omitted.

For non-IP traffic, the formula is the same as for layer2.

set interfaces bonding <interface> primary <interface>

Configure the primary member interface in the bond.

The primary member interface remains active as long as it is operational; alternative member interfaces are used only if it fails.

Use this configuration when a specific member interface is preferred, such as one with higher throughput.

This command applies only to active-backup, transmit-load-balance, and adaptive-load-balance modes.

set interfaces bonding <interface> arp-monitor interval <time>

Configure the ARP monitoring interval, in seconds, for the bonding interface.

ARP monitoring periodically assesses the health of each member interface by checking whether it has recently sent or received traffic (this criterion varies depending on the bonding mode and the member interface's state). ARP probes are sent to the IP addresses specified with the arp-monitor target option.

When ARP monitoring is used with EtherChannel-compatible modes (such as round-robin or xor-hash), the switch should be configured to distribute traffic across all member interfaces. If the switch distributes traffic using an XOR-based policy, all ARP replies will be received on one member interface, causing other member interfaces to be incorrectly marked as failed.

Setting this value to 0 disables ARP monitoring.

The default value is 0.

set interfaces bonding <interface> arp-monitor target <address>

Configure the IP addresses for ARP monitoring requests.

The bonding driver sends ARP requests to these IP addresses to check the state of member interfaces.

To enable ARP monitoring, configure at least one IP address (up to 16 per bonding interface).

By default, no IP addresses are configured.

VLAN (Virtual Local Area Network)

IEEE 802.1q, often referred to as Dot1q, is the industry standard for implementing VLANs on Ethernet networks. It defines VLAN tagging for Ethernet frames and outlines procedures for bridges and switches. The standard also includes quality-of-service prioritization (IEEE 802.1p) and defines the Generic Attribute Registration Protocol.

VLAN-aware network segments (i.e., IEEE 802.1q conformant) use VLAN tags. When a frame enters such a segment, a tag is added to indicate VLAN membership. Each frame can belong to only one VLAN. If a frame arrives without a tag, it is assumed to be part of the native VLAN.

IEEE 802.1, a working group of the IEEE 802 standards committee, has developed the standard and continues to revise it. One notable revision is 802.1Q-2014, which incorporated IEEE 802.1aq (Shortest Path Bridging) and much of the IEEE 802.1d standard.

In VyOS, 802.1q VLAN interfaces are represented as virtual subinterfaces, referred to as `vif`.

set interfaces bonding <interface> vif <vlan-id>

Configure a VLAN interface with a unique VLAN ID.

VLAN ID identifies a specific VLAN and ranges from 0 to 4094.

You can configure multiple VLAN interfaces on a single physical interface.

Note: Only 802.1Q-tagged packets are accepted on Ethernet vifs.

set interfaces bonding <interface> vif <vlan-id> address <address | dhcp | dhcpv6>

Configure the interface with one or more IP addresses.

The following options are available:

- **address:** Assign one or more IPv4 or IPv6 addresses to the interface. For example, use 192.0.2.1/24 for IPv4 or 2001:db8::1/64 for IPv6.
- **dhcp:** The interface obtains an IPv4 address from a DHCP server on the same network segment.
- **dhcpv6:** The interface obtains an IPv6 address from a DHCPv6 server on the same network segment.

Note: If the interface obtains an IPv4 address via DHCP, and specific adjustments are needed before/after the IP address is obtained, use the provided hook scripts:

- /config/scripts/dhcp-client/pre-hooks.d/
 - /config/scripts/dhcp-client/post-hooks.d/
-

Example:

```
set interfaces bonding bond0 vif 10 address 192.0.2.1/24
set interfaces bonding bond0 vif 10 address 2001:db8::1/64
set interfaces bonding bond0 vif 10 address dhcp
set interfaces bonding bond0 vif 10 address dhcpv6
```

set interfaces bonding <interface> vif <vlan-id> description <description>

Configure a clear, descriptive alias for the interface.

This alias appears in the `show interfaces` command and SNMP-based monitoring tools.

Example:

```
set interfaces bonding bond0 vif 10 description 'This is an interface running on
↳ VyOS.'
```

set interfaces bonding <interface> vif <vlan-id> disable

Disable the interface.

The interface will be set to the administratively down (A/D) state.

Example:

```
set interfaces bonding bond0 vif 10 disable
```

set interfaces bonding <interface> vif <vlan-id> disable-link-detect

Disable physical link-state change detection on the interface, such as when a cable is unplugged.

By default, the interface detects physical link-state changes.

Example:

```
set interfaces bonding bond0 vif 10 disable-link-detect
```

set interfaces bonding <interface> vif <vlan-id> mac <xx:xx:xx:xx:xx:xx>

Configure a custom MAC address on the interface.

Example:

```
set interfaces bonding bond0 vif 10 mac '00:53:01:02:03:04'
```

set interfaces bonding <interface> vif <vlan-id> mtu <mtu>

Configure the MTU on the interface.

This value defines the largest packet size, in bytes, that the interface transmits without fragmentation.

Example:

```
set interfaces bonding bond0 vif 10 mtu 1600
```

set interfaces bonding <interface> vif <vlan-id> ip adjust-mss <mss | clamp-mss-to-pmtu>

Configure the MSS advertised in outgoing TCP SYN packets on the specified interface.

By clamping the MSS value in TCP SYN packets, you explicitly inform the remote side not to send packets larger than that size. This prevents connection issues that occur when Path MTU Discovery (PMTUD) fails.

The following options are available:

- **mss:** Sets the MSS to a specific value, in bytes. Use this option if you need to enforce a specific MSS, for example, to troubleshoot connectivity issues or accommodate specific network requirements.
- **clamp-mss-to-pmtu:** The router automatically calculates the MSS to be the interface's MTU minus 40 bytes for IPv4 traffic (20 bytes for the IPv4 header and 20 bytes for the TCP header). This option is recommended to automatically set the proper value.

set interfaces bonding <interface> vif <vlan-id> ip arp-cache-timeout

Configure how long an ARP entry remains valid after learning an IP-to-MAC address mapping on this interface.

The default duration is 30 seconds.

An ARP entry remains valid if it receives positive feedback from higher-level protocols.

Example:

```
set interfaces bonding bond0 vif 10 ip arp-cache-timeout 180
```

set interfaces bonding <interface> vif <vlan-id> ip disable-arp-filter

Configure ARP filtering on this interface.

Default behavior: The kernel responds to ARP requests on this interface only if the traffic would be routed back to the ARP sender through that specific interface.

If configured: The kernel responds to ARP requests on this interface for any IP address configured on the local host, regardless of which specific interface that IP address is assigned to, and regardless of the routing table. This reflects the Linux concept that IP addresses belong to the host, not individual interfaces.

Example:

```
set interfaces bonding bond0 vif 10 ip disable-arp-filter
```

set interfaces bonding <interface> vif <vlan-id> ip disable-forwarding

Configure the interface for host or router behavior.

If configured, the interface switches to host mode, and IPv4 forwarding is disabled on it.

Example:

```
set interfaces bonding bond0 vif 10 ip disable-forwarding
```

set interfaces bonding <interface> vif <vlan-id> ip enable-directed-broadcast

Configure whether to forward IP-directed broadcast packets received on this interface.

Default behavior: IP-directed broadcast packets are dropped.

If configured: IP-directed broadcast packets are forwarded to all hosts on the destination subnet, as defined in [RFC 1812](#) and [RFC 2644](#).

Example:

```
set interfaces bonding bond0 vif 10 ip enable-directed-broadcast
```

set interfaces bonding <interface> vif <vlan-id> ip enable-arp-accept

Configure how to process gratuitous ARPs on this interface.

If configured, an IP-to-MAC address mapping is added to the ARP table based on gratuitous ARP requests or replies.

Note: If the ARP table already contains the IP address from a gratuitous ARP, its entry is updated regardless of whether this setting is configured.

Example:

```
set interfaces bonding bond0 vif 10 ip enable-arp-accept
```

set interfaces bonding <interface> vif <vlan-id> ip enable-arp-announce

Configure the source IP selection for ARP requests on this interface.

Default behavior: The kernel can use any IP address the host owns as the source IP address in ARP requests on this interface.

If configured: The kernel first attempts to select a source IP address configured on the interface that shares a common subnet with the target IP address. If there is no such subnet, the kernel selects the IP address it would normally use (based on the routing table to reach the target destination).

Example:

```
set interfaces bonding bond0 vif 10 ip enable-arp-announce
```

set interfaces bonding <interface> vif <vlan-id> ip enable-arp-ignore

Configure which ARP requests will be ignored on this interface.

Default behavior: The kernel responds to ARP requests for any local IP addresses, regardless of which interface they are assigned to.

If configured: The kernel responds to ARP requests only if the target IP address is assigned to this specific interface.

Example:

```
set interfaces bonding bond0 vif 10 ip enable-arp-ignore
```

set interfaces bonding <interface> vif <vlan-id> ip enable-proxy-arp

Configure proxy ARP on this interface.

If configured, the router (kernel) intercepts ARP requests for non-local IP addresses and replies with the MAC address of the interface that received the request. Subsequent packets destined to these IP addresses are forwarded to their actual destinations on remote subnets.

Example:

```
set interfaces bonding bond0 vif 10 ip enable-proxy-arp
```

set interfaces bonding <interface> vif <vlan-id> ip proxy-arp-pvlan

Configure local proxy ARP on the interface.

If configured, the router (kernel) responds to ARP requests on this VLAN interface even if the target IP address resides on the same subnet and interface.

This is used to support network isolation requirements (RFC 3069) for private VLANs (PVLANS). In PVLAN configurations, hosts on isolated ports are NOT allowed to communicate directly with each other at Layer 2, but they can communicate with the upstream router.

By replying to inter-host ARP requests with its own MAC address, the router (kernel) directs inter-host traffic through itself instead of directly between hosts.

Note: This command works independently and does not require enabling the standard proxy ARP on the interface.

Local proxy ARP is also known as:

- VLAN aggregation ([RFC 3069](#)).
- Private VLAN (Cisco, Allied Telesyn).
- Source-port filtering or port isolation (Hewlett-Packard).
- MAC-Forced Forwarding (Ericsson).

set interfaces bonding <interface> vif <vlan-id> ip source-validation <strict | loose | disable>

Configure source IP address validation using RPF on this interface, as specified in [RFC 3704](#).

The following options are available:

- **strict:** Each incoming packet's source IP address is checked against the Forwarding Information Base (FIB). If the interface is not the best route back to that source, validation fails, and the packet is dropped.
- **loose:** Each incoming packet's source IP address is checked against the FIB. If the source IP address is unreachable through any interface, validation fails.
- **disable:** No source IP address validation is performed. All incoming packets are accepted.

RFC 3704 recommends enabling strict mode to prevent IP spoofing, such as DDoS attacks. For asymmetric or other complex routing scenarios, use loose mode.

set interfaces bonding <interface> vif <vlan-id> ipv6 address autoconf

Configure the interface to automatically obtain an IPv6 address using SLAAC, as specified in [RFC 4862](#).

IPv6 hosts can configure themselves automatically when connected to an IPv6 network using the Neighbor Discovery Protocol via ICMPv6 router discovery messages. When first connected to a network, a host sends a link-local router solicitation multicast request for its configuration parameters. The router responds with a router advertisement packet containing Internet Layer configuration parameters.

Note: This method automatically disables IPv6 traffic forwarding on the interface.

Example:

```
set interfaces bonding bond0 vif 10 ipv6 address autoconf
```

set interfaces bonding <interface> vif <vlan-id> ipv6 address eui64 <prefix>

Configure the interface to assign itself an IPv6 address using the EUI-64 method, as specified in [RFC 4291](#).

Example:

```
set interfaces bonding bond0 vif 10 ipv6 address eui64 2001:db8:beef::/64
```

set interfaces bonding <interface> vif <vlan-id> ipv6 address no-default-link-local

Disable the automatic assignment of a link-local IPv6 address to this interface.

Example:

```
set interfaces bonding bond0 vif 10 ipv6 address no-default-link-local
```

set interfaces bonding <interface> vif <vlan-id> ipv6 disable-forwarding

Configure the interface for host or router behavior.

If configured, the interface switches to host mode, and IPv6 forwarding is disabled on it.

Example:

```
set interfaces bonding bond0 vif 10 ipv6 disable-forwarding
```

set interfaces bonding <interface> vif <vlan-id> ipv6 adjust-mss <mss | clamp-mss-to-pmtu>

Configure the MSS advertised in outgoing TCP SYN packets on the specified interface.

By clamping the MSS value in TCP SYN packets, you explicitly inform the remote side not to send packets larger than that size. This prevents connection issues when Path MTU Discovery (PMTUD) fails.

The following options are available:

- **mss**: Set the MSS to a specific value, in bytes. Use this option to enforce a specific MSS, for example, to troubleshoot connectivity issues or accommodate specific network requirements.
- **clamp-mss-to-pmtu**: The router calculates the MSS to be the interface's MTU minus 60 bytes for IPv6 traffic (40 bytes for the IPv6 header and 20 bytes for the TCP header). This option is recommended to automatically set the proper value.

```
set interfaces bonding <interface> vif <vlan-id> ipv6 accept-dad <0 | 1 | 2>
```

Configure IPv6 DAD on the interface.

The following options are available:

- **0**: Disables DAD. No duplicate address detection is performed.
- **1**: Enables DAD (default). Duplicate addresses are detected. The interface's IPv6 operation continues for valid IPv6 addresses.
- **2**: Enables DAD and, if a MAC-based duplicate link-local address is found, disables IPv6 operation on this interface.

Example:

```
set interfaces bonding bond0 vif 10 ipv6 accept-dad 2
```

```
set interfaces bonding <interface> vif <vlan-id> ipv6 dup-addr-detect-transmits <n>
```

Configure the number of DAD messages that the router (kernel) sends during IPv6 address assignment on this interface.

The default value is 1.

Example:

```
set interfaces bonding bond0 vif 10 ipv6 dup-addr-detect-transmits 5
```

```
set interfaces bonding <interface> vif <vlan-id> vrf <vrf>
```

Assign the interface to a specific VRF instance.

See also:

For information on configuring a VRF, refer to the [VRF](#) section.

Example:

```
set interfaces bonding bond0 vif 10 vrf red
```

DHCP(v6)

```
set interfaces bonding <interface> vif <vlan-id> dhcp-options client-id <description>
```

Configure a DHCP client identifier for the interface, as specified in [RFC 2131](#).

The `client-id` is an identifier that the DHCP client sends to the DHCP server to uniquely identify itself for IP address assignment. By default, the client uses its MAC address. The `<description>` is a user-defined string that will be sent to the DHCP server as the DHCP client identifier.

Example:

```
set interfaces bonding bond0 vif 10 dhcp-options client-id 'foo-bar'
```

```
set interfaces bonding <interface> vif <vlan-id> dhcp-options host-name <hostname>
```

Configure a specific hostname for the interface.

Instead of the real hostname, the DHCP client will send the specific hostname to the DHCP server when requesting an IP address.

Example:

```
set interfaces bonding bond0 vif 10 dhcp-options host-name 'VyOS'
```

set interfaces bonding <interface> vif <vlan-id> dhcp-options vendor-class-id <vendor-id>

Configure the DHCP client to include a vendor-class identifier in its DHCP requests on this interface.

The vendor-class identifier is a vendor-specific byte string that enables the DHCP server to identify the device and, in some cases, provide configuration options.

Example:

```
set interfaces bonding bond0 vif 10 dhcp-options vendor-class-id 'VyOS'
```

set interfaces bonding <interface> vif <vlan-id> dhcp-options no-default-route

Configure the DHCP client to obtain an IP address, but ignore any default gateway provided by the DHCP server on this interface.

Example:

```
set interfaces bonding bond0 vif 10 dhcp-options no-default-route
```

set interfaces bonding <interface> vif <vlan-id> dhcp-options default-route-distance <distance>

Configure the distance for the default route obtained from the DHCP server on this interface.

Example:

```
set interfaces bonding bond0 vif 10 dhcp-options default-route-distance 220
```

set interfaces bonding <interface> vif <vlan-id> dhcp-options reject <address>

Configure the DHCP client to reject the specific IP address or IP address range from the DHCP server on this interface.

This is useful when a modem assigns a local IP address upon start. To reject multiple addresses, run this command multiple times with different values. You can reject individual addresses (192.168.100.1) or entire subnets (192.168.100.0/24).

Example:

```
set interfaces bonding bond0 vif 10 dhcp-options reject 192.168.100.0/24
```

set interfaces bonding <interface> vif <vlan-id> dhcp-options user-class <string>

Configure the DHCP client to send a specific user-class identifier in its DHCP requests on this interface.

The DHCP server can interpret this identifier and provide specific configuration options based on it (for example, default routes). The user-class value typically groups DHCP clients with similar configuration needs (for example, employees, guests, or printers).

Example:

```
set interfaces bonding bond0 vif 10 dhcp-options user-class VyOS
```

```
set interfaces bonding <interface> vif <vlan-id> dhcpv6-options duid <duid>
```

Configure a specific DUID for the DHCPv6 client on this interface.

The DUID is an identifier used by a DHCPv6 client to get an IPv6 address from a DHCPv6 server. It consists of a 2-byte type field, followed by a variable-length identifier field up to 128 bytes. The format of the identifier part depends on the DUID type:

- **DUID-LLT:** The most common type, which includes a hardware type, a timestamp, and a MAC address.
- **DUID-EN:** Is based on a vendor's enterprise number and a unique identifier assigned by the vendor.
- **DUID-LL:** Includes only a MAC address.

The DHCP server matches the DUID against its database and provides configuration data (such as address, lease times, DNS servers, etc.) to the DHCP client.

Example:

```
set interfaces bonding bond0 vif 10 duid
→ '0e:00:00:01:00:01:27:71:db:f0:00:50:56:bf:c5:6d'
```

```
set interfaces bonding <interface> vif <vlan-id> dhcpv6-options no-release
```

Configure the DHCP client not to send a release message when it stops running on this interface.

This helps retain the assigned address or prefix.

Example:

```
set interfaces bonding bond0 vif 10 dhcpv6-options no-release
```

```
set interfaces bonding <interface> vif <vlan-id> dhcpv6-options parameters-only
```

Enable a stateless DHCPv6 client mode on this interface.

In stateless mode, the DHCPv6 client requests only stateless configuration parameters from the DHCP server (for example, DNS server addresses). It doesn't request a stateful configuration, such as IPv6 addresses or prefixes.

Example:

```
set interfaces bonding bond0 vif 10 dhcpv6-options parameters-only
```

```
set interfaces bonding <interface> vif <vlan-id> dhcpv6-options rapid-commit
```

Enable DHCPv6 rapid commit on this interface.

When enabled, the DHCP client and server skip the negotiation steps (Advertise and Request), completing the DHCPv6 configuration process in just two messages (Solicit and final Reply).

Example:

```
set interfaces bonding bond0 vif 10 dhcpv6-options rapid-commit
```

```
set interfaces bonding <interface> vif <vlan-id> dhcpv6-options temporary
```

Configure the DHCPv6 client to request a temporary IPv6 address on this interface.

When configured, the DHCP client doesn't form an Identity Association for Non-temporary Addresses (IA_NA) partnership. Consequently, it only obtains a temporary IPv6 address and doesn't obtain a permanent one.

Example:

```
set interfaces bonding bond0 vif 10 dhcpv6-options temporary
```

DHCPv6 Prefix Delegation (PD)

VyOS supports DHCPv6 Prefix Delegation (DHCPv6-PD) as described in [RFC 3633](#). DHCPv6-PD is supported by most ISPs that provide native IPv6 for consumers on fixed networks.

```
set interfaces bonding <interface> vif <vlan-id> dhcpv6-options pd <id> length <length>
```

Configure a specific prefix length for DHCPv6-PD requests on this interface.

Some ISPs provide only a /64 prefix by default. Use this command to request a different prefix length for a specific DHCPv6-PD request, ranging from /32 (if allowed by your ISP) down to /64. <id> is a unique identifier for the DHCPv6-PD request.

The default value is 64.

To request a /56 prefix from your ISP, use:

```
set interfaces bonding bond0 vif 10 dhcpv6-options pd 0 length 56
```

```
set interfaces bonding <interface> vif <vlan-id> dhcpv6-options pd <id> interface <delegatee> address <address>
```

Configure the IPv6 interface identifier (host portion) for the delegatee interface.

The value must be a decimal integer. It is appended to the delegated prefix and the configured SLA ID to form the final IPv6 address.

By default, the host portion is generated based on the parent interface's MAC address (EUI-64 format).

Example:

If a /64 prefix is delegated to interface eth8 and you configure the host portion as 65535, the resulting IPv6 address will end with ::ffff, as 65535 corresponds to ffff in hexadecimal notation.

```
set interfaces bonding bond0 vif 10 dhcpv6-options pd 0 interface eth8 address 65534
```

```
set interfaces bonding <interface> vif <vlan-id> dhcpv6-options pd <id> interface <delegatee> sla-id <id>
```

Configure the SLA ID for the delegatee interface.

The value must be a decimal integer greater than 0 and fit in the length of SLA IDs. It is converted to hexadecimal and appended to the delegated prefix to form the specific subnet prefix for the delegatee interface.

Example:

If SLA ID is 1 and the delegated prefix is 2001:db8:ffff::/48, the resulting subnet prefix for the delegatee interface will be 2001:db8:ffff:1::/64.

```
set interfaces bonding bond0 vif 10 dhcpv6-options pd 0 interface eth8 sla-id 1
```

SPAN port mirroring

SPAN (Switched Port Analyser) port mirroring copies inbound and outbound traffic from one interface to another specified interface.

The destination interface is usually connected to specialized equipment, such as a behavior control system, an intrusion detection system, or a traffic collector, and copies all related traffic from this port. The benefit of mirroring traffic is that the application is isolated from the source traffic, so application processing does not affect the traffic or system performance.

To configure SPAN port mirroring, VyOS uses the `mirror` parameter. You can mirror ingress traffic (traffic entering the router) and egress traffic (traffic leaving the router) separately. Both directions can be mirrored to the same destination interface or split to different ones.

set interfaces bonding <interface> mirror ingress <monitor-interface>

Mirror ingress traffic from a bonding interface to the specified monitoring interface.

Example: Mirror the ingress traffic from *bond1* to *eth3*.

```
set interfaces bonding bond1 mirror ingress eth3
```

set interfaces bonding <interface> mirror egress <monitor-interface>

Mirror egress traffic from a bonding interface to the specified monitoring interface.

Example: Mirror the egress traffic from *bond1* to *eth3*.

```
set interfaces bonding bond1 mirror egress eth3
```

EVPN multihoming

EVPN multihoming (EVPN-MH) is a standards-based solution (RFC 7432, RFC 8365) that enables Customer Edge (CE) devices, such as servers, to connect to two or more Provider Edge (PE) devices for redundancy and load balancing.

EVPN-MH is often used as a modern, standards-based alternative to MLAG (Multi-Chassis Link Aggregation) and VTEPs (Virtual Tunnel Endpoints).

Ethernet Segment (ES) and Ethernet Segment Identifier (ESI)

Physical links that connect a CE device to PE devices are bundled using link aggregation. This logical bundle is called an Ethernet Segment (ES) and is uniquely identified by an Ethernet Segment Identifier (ESI) within the EVPN domain.

To enable EVPN-MH, configure the same ESI on the bonding interfaces of all PE devices connected to a single CE device.

An ESI is configured by specifying either a system MAC address and a local discriminator, or an Ethernet Segment Identifier Name (ESINAME).

The following two commands generate a 10-byte Type-3 ESI by combining the system MAC and local discriminator:

set interfaces bonding <interface> evpn es-id <1-16777215|10-byte ID>

set interfaces bonding <interface> evpn es-sys-mac <xx:xx:xx:xx:xx:xx>

Alternatively, assign an ESINAME directly as a 10-byte Type-0 ESI using the following format:
00:AA:BB:CC:DD:EE:FF:GG:HH:II.

BGP-EVPN route usage

EVPN-MH uses BGP-EVPN route types 1 and 2 for ES discovery and MAC-IP synchronization:

- **Type 1 (EAD-per-ES and EAD-per-EVI)** routes advertise the locally attached ESs and discover remote ESs in the network.
- **Type 2 (MAC-IP advertisement)** routes are advertised with a destination ESI, enabling MAC-IP synchronization between ES peers.

set interfaces bonding <interface> evpn es-df-pref <1-65535>

Configure the DF (Designated Forwarder) preference (1-65535) for the interface. A higher value indicates a higher preference to become the DF. The DF preference is configured per-ES.

The DF election process determines which interface in a specific ES forwards BUM (Broadcast, Unknown Unicast, and Multicast) traffic from the EVPN overlay to the connected CE device. EVPN Type-4 (Ethernet Segment) routes are used to elect the DF, implementing the preference-based election method defined in RFC 9785.

Interfaces not elected as the DF drop any BUM traffic from the EVPN overlay using non-DF filters. Similarly, traffic received from ES peers via the EVPN overlay is blocked from forwarding to the CE device to maintain split-horizon filtering with local bias.

set interfaces bonding <interface> evpn uplink

Configure this interface as an EVPN-MH uplink interface.

If all uplink interfaces on a PE device go down, this PE device loses access to the VXLAN overlay. To prevent traffic blackholing, the PE device forces a protocol shutdown (protodown) of its downstream EVPN-MH interfaces.

The following example configures bond0 as an EVPN-MH uplink interface:

```
set interfaces bonding bond0 evpn uplink
```

Example

The following configuration example applies to all listed third-party vendors. It creates a bonding interface with two member interfaces, defines VLANs 10 and 100 on the bonding interface, and assigns an IPv4 address to each VLAN subinterface.

```
# Create the bonding interface bond0 with 802.3ad LACP
set interfaces bonding bond0 hash-policy 'layer2'
set interfaces bonding bond0 mode '802.3ad'

# Add the required VLANs and IPv4 addresses on them
set interfaces bonding bond0 vif 10 address 192.168.0.1/24
set interfaces bonding bond0 vif 100 address 10.10.10.1/24

# Add the member interfaces to the bonding interface
set interfaces bonding bond0 member interface eth1
set interfaces bonding bond0 member interface eth2
```

Note: If you are running this configuration in a virtual environment like EVE-NG, ensure the e1000 driver is chosen for your VyOS NIC. The default drivers, such as virtio-net-pci or vmxnet3, are incompatible with this configuration. Specifically, ICMP messages will not be processed correctly.

To check your NIC driver, use the following command: `show interfaces ethernet eth0 physical | grep -i driver`

Cisco Catalyst configuration

Configure a Cisco Catalyst switch to integrate with a two-member VyOS bonding interface.

Assign member interfaces to PortChannel:

```
interface GigabitEthernet1/0/23
description VyOS eth1
channel-group 1 mode active
!
interface GigabitEthernet1/0/24
description VyOS eth2
channel-group 1 mode active
!
```

A new interface, Port-channel1, becomes available; all configuration, such as allowed VLAN interfaces and STP, is applied here.

```
interface Port-channel1
description LACP Channel for VyOS
switchport trunk encapsulation dot1q
switchport trunk allowed vlan 10,100
switchport mode trunk
spanning-tree portfast trunk
!
```

Juniper EX Switch configuration

Configure a Juniper EX Series switch to integrate with a two-member VyOS bonding interface.

```
# Create aggregated ethernet device with 802.3ad LACP and port speeds of 10gbit/s
set interfaces ae0 aggregated-ether-options link-speed 10g
set interfaces ae0 aggregated-ether-options lacp active

# Create layer 2 on the aggregated ethernet device with trunking for our VLANs
set interfaces ae0 unit 0 family ethernet-switching port-mode trunk

# Add the required vlans to the device
set interfaces ae0 unit 0 family ethernet-switching vlan members 10
set interfaces ae0 unit 0 family ethernet-switching vlan members 100

# Add the two interfaces to the aggregated ethernet device, in this setup both
# ports are on the same switch (switch 0, module 1, port 0 and 1)
set interfaces xe-0/1/0 ether-options 802.3ad ae0
set interfaces xe-0/1/1 ether-options 802.3ad ae0

# But this can also be done with multiple switches in a stack, a virtual
# chassis on Juniper (switch 0 and switch 1, module 1, port 0 on both switches)
```

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```
set interfaces xe-0/1/0 ether-options 802.3ad ae0
set interfaces xe-1/1/0 ether-options 802.3ad ae0
```

Aruba/HP configuration

Configure an Aruba/HP 2510G switch to integrate with a two-member VyOS bonding interface.

```
# Create trunk with 2 member interfaces (interface 1 and 2) and LACP
trunk 1-2 Trk1 LACP

# Add the required VLANs to the trunk
vlan 10 tagged Trk1
vlan 100 tagged Trk1
```

Arista EOS configuration

When deploying VyOS in environments with Arista switches, use the following blueprint as an initial setup to configure an operational LACP port-channel between the two devices.

Let's assume the following topology:

R1

```
interfaces {
    bonding bond10 {
        hash-policy layer3+4
        member {
            interface eth1
            interface eth2
        }
        mode 802.3ad
        vif 100 {
            address 192.0.2.1/30
            address 2001:db8::1/64
        }
    }
}
```

R2

```
interfaces {
    bonding bond10 {
        hash-policy layer3+4
        member {
            interface eth1
            interface eth2
        }
        mode 802.3ad
        vif 100 {
            address 192.0.2.2/30
        }
    }
}
```

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```

        address 2001:db8::2/64
    }
}

```

SW1

```

!
vlan 100
    name F00
!
interface Port-Channel10
    switchport trunk allowed vlan 100
    switchport mode trunk
    spanning-tree portfast
!
interface Port-Channel20
    switchport mode trunk
    no spanning-tree portfast auto
    spanning-tree portfast network
!
interface Ethernet1
    channel-group 10 mode active
!
interface Ethernet2
    channel-group 10 mode active
!
interface Ethernet3
    channel-group 20 mode active
!
interface Ethernet4
    channel-group 20 mode active
!

```

SW2

```

!
vlan 100
    name F00
!
interface Port-Channel10
    switchport trunk allowed vlan 100
    switchport mode trunk
    spanning-tree portfast
!
interface Port-Channel20
    switchport mode trunk
    no spanning-tree portfast auto
    spanning-tree portfast network
!
interface Ethernet1
    channel-group 10 mode active
!

```

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```
interface Ethernet2
    channel-group 10 mode active
!
interface Ethernet3
    channel-group 20 mode active
!
interface Ethernet4
    channel-group 20 mode active
!
```

Note: When testing this environment in EVE-NG, ensure the e1000 driver is chosen for your VyOS network interfaces. If the default virtio driver is used, VyOS will not transmit LACP PDUs, preventing the port-channel from ever becoming active.

Operation

show interfaces bonding

Show brief interface information.

```
vyos@vyos:~$ show interfaces bonding
Codes: S - State, L - Link, u - Up, D - Down, A - Admin Down
Interface      IP Address      S/L  Description
-----
bond0          -               u/u  my-sw1 int 23 and 24
bond0.10       192.168.0.1/24  u/u  office-net
bond0.100      10.10.10.1/24   u/u  management-net
```

show interfaces bonding <interface>

Show detailed interface information.

```
vyos@vyos:~$ show interfaces bonding bond5
bond5: <NO-CARRIER,BROADCAST,MULTICAST,MASTER,UP> mtu 1500 qdisc noqueue state DOWN
↳ group default qlen 1000
    link/ether 00:50:56:bf:ef:aa brd ff:ff:ff:ff:ff:ff
    inet6 fe80::e862:26ff:fe72:2dac/64 scope link tentative
        valid_lft forever preferred_lft forever

    RX:  bytes  packets  errors  dropped  overrun        mcast
         0         0         0         0         0             0
    TX:  bytes  packets  errors  dropped  carrier  collisions
         0         0         0         0         0             0
```

show interfaces bonding <interface> detail

Show detailed information about the underlying physical links on the given bonding interface.

```
vyos@vyos:~$ show interfaces bonding bond5 detail
Ethernet Channel Bonding Driver: v3.7.1 (April 27, 2011)
Bonding Mode: IEEE 802.3ad Dynamic link aggregation
```

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```

Transmit Hash Policy: layer2 (0)
MII Status: down
MII Polling Interval (ms): 100
Up Delay (ms): 0
Down Delay (ms): 0
802.3ad info
LACP rate: slow
Min links: 0
Aggregator selection policy (ad_select): stable
Slave Interface: eth1
MII Status: down
Speed: Unknown
Duplex: Unknown
Link Failure Count: 0
Permanent HW addr: 00:50:56:bf:ef:aa
Slave queue ID: 0
Aggregator ID: 1
Actor Churn State: churned
Partner Churn State: churned
Actor Churned Count: 1
Partner Churned Count: 1
Slave Interface: eth2
MII Status: down
Speed: Unknown
Duplex: Unknown
Link Failure Count: 0
Permanent HW addr: 00:50:56:bf:19:26
Slave queue ID: 0
Aggregator ID: 2
Actor Churn State: churned
Partner Churn State: churned
Actor Churned Count: 1
Partner Churned Count: 1

```

7.4.2 Bridge

VyOS bridges connect Ethernet segments by grouping multiple interfaces into a single bridge interface, which acts as a virtual software switch. Unlike routers, which forward traffic based on Layer 3 IP addresses, bridges operate at Layer 2 and forward traffic based on MAC addresses. Operating at Layer 2, bridges are protocol-agnostic and transparently forward all Ethernet- encapsulated traffic, whether it is IPv4, IPv6, or specialized industrial protocols.

This implementation utilizes the Linux bridge subsystem to support a subset of the ANSI/IEEE 802.1d standard for transparent bridging and MAC address learning.

Note: STP (Spanning Tree Protocol) is disabled by default in VyOS and must be explicitly enabled if required. See [STP configuration](#) for details.

Configuration

Common interface configuration

set interfaces bridge <interface> address <address | dhcp | dhcpv6>

Configure the interface with one or more IP addresses.

The following options are available:

- **address:** Assign one or more IPv4 or IPv6 addresses to the interface. For example, use 192.0.2.1/24 for IPv4 or 2001:db8::1/64 for IPv6.
- **dhcp:** The interface obtains an IPv4 address from a DHCP server on the same network segment.
- **dhcpv6:** The interface obtains an IPv6 address from a DHCPv6 server on the same network segment.

Note: If the interface obtains an IPv4 address via DHCP, and specific adjustments are needed before/after the IP address is obtained, use the provided hook scripts:

- /config/scripts/dhcp-client/pre-hooks.d/
 - /config/scripts/dhcp-client/post-hooks.d/
-

Example:

```
set interfaces bridge br0 address 192.0.2.1/24
set interfaces bridge br0 address 2001:db8::1/64
set interfaces bridge br0 address dhcp
set interfaces bridge br0 address dhcpv6
```

set interfaces bridge <interface> description <description>

Configure a clear, descriptive alias for the interface.

This alias appears in the `show interfaces` command and SNMP-based monitoring tools.

Example:

```
set interfaces bridge br0 description 'This is an interface running on VyOS.'
```

set interfaces bridge <interface> disable

Disable the interface.

The interface will be set to the administratively down (A/D) state.

Example:

```
set interfaces bridge br0 disable
```

set interfaces bridge <interface> disable-flow-control

Disable Ethernet flow control (IEEE 802.3x pause frames) on the interface.

Ethernet flow control, defined by the IEEE 802.3x standard, temporarily stops data transmission to prevent packet loss during network congestion. For example, when a sender transmits data faster than the receiver can process it.

Disabling Ethernet flow control means the interface will not signal the connected device to pause transmission and will drop packets if overwhelmed.

Example:

```
set interfaces bridge br0 disable-flow-control
```

set interfaces bridge <interface> disable-link-detect

Disable physical link-state change detection on the interface, such as when a cable is unplugged.

By default, the interface detects physical link-state changes.

Example:

```
set interfaces bridge br0 disable-link-detect
```

set interfaces bridge <interface> mac <xx:xx:xx:xx:xx:xx>

Configure a custom MAC address on the interface.

Example:

```
set interfaces bridge br0 mac '00:53:01:02:03:04'
```

set interfaces bridge <interface> mtu <mtu>

Configure the MTU on the interface.

This value defines the largest packet size, in bytes, that the interface transmits without fragmentation.

Example:

```
set interfaces bridge br0 mtu 1600
```

set interfaces bridge <interface> ip adjust-mss <mss | clamp-mss-to-pmtu>

Configure the MSS advertised in outgoing TCP SYN packets on the specified interface.

By clamping the MSS value in TCP SYN packets, you explicitly inform the remote side not to send packets larger than that size. This prevents connection issues that occur when Path MTU Discovery (PMTUD) fails.

The following options are available:

- **mss:** Sets the MSS to a specific value, in bytes. Use this option if you need to enforce a specific MSS, for example, to troubleshoot connectivity issues or accommodate specific network requirements.
- **clamp-mss-to-pmtu:** The router automatically calculates the MSS to be the interface's MTU minus 40 bytes for IPv4 traffic (20 bytes for the IPv4 header and 20 bytes for the TCP header). This option is recommended to automatically set the proper value.

set interfaces bridge <interface> ip arp-cache-timeout

Configure how long an ARP entry remains valid after learning an IP-to-MAC address mapping on this interface.

The default duration is 30 seconds.

An ARP entry remains valid if it receives positive feedback from higher-level protocols.

Example:

```
set interfaces bridge br0 ip arp-cache-timeout 180
```

set interfaces bridge <interface> ip disable-arp-filter

Configure ARP filtering on this interface.

Default behavior: The kernel responds to ARP requests on this interface only if the traffic would be routed back to the ARP sender through that specific interface.

If configured: The kernel responds to ARP requests on this interface for any IP address configured on the local host, regardless of which specific interface that IP address is assigned to, and regardless of the routing table. This reflects the Linux concept that IP addresses belong to the host, not individual interfaces.

Example:

```
set interfaces bridge br0 ip disable-arp-filter
```

set interfaces bridge <interface> ip disable-forwarding**Configure the interface for host or router behavior.**

If configured, the interface switches to host mode, and IPv4 forwarding is disabled on it.

Example:

```
set interfaces bridge br0 ip disable-forwarding
```

set interfaces bridge <interface> ip enable-directed-broadcast**Configure whether to forward IP-directed broadcast packets received on this interface.**

Default behavior: IP-directed broadcast packets are dropped.

If configured: IP-directed broadcast packets are forwarded to all hosts on the destination subnet, as defined in [RFC 1812](#) and [RFC 2644](#).

Example:

```
set interfaces bridge br0 ip enable-directed-broadcast
```

set interfaces bridge <interface> ip enable-arp-accept**Configure how to process gratuitous ARPs on this interface.**

If configured, an IP-to-MAC address mapping is added to the ARP table based on gratuitous ARP requests or replies.

Note: If the ARP table already contains the IP address from a gratuitous ARP, its entry is updated regardless of whether this setting is configured.

Example:

```
set interfaces bridge br0 ip enable-arp-accept
```

set interfaces bridge <interface> ip enable-arp-announce**Configure the source IP selection for ARP requests on this interface.**

Default behavior: The kernel can use any IP address the host owns as the source IP address in ARP requests on this interface.

If configured: The kernel first attempts to select a source IP address configured on the interface that shares a common subnet with the target IP address. If there is no such subnet, the kernel selects the IP address it would normally use (based on the routing table to reach the target destination).

Example:

```
set interfaces bridge br0 ip enable-arp-announce
```

set interfaces bridge <interface> ip enable-arp-ignore

Configure which ARP requests will be ignored on this interface.

Default behavior: The kernel responds to ARP requests for any local IP addresses, regardless of which interface they are assigned to.

If configured: The kernel responds to ARP requests only if the target IP address is assigned to this specific interface.

Example:

```
set interfaces bridge br0 ip enable-arp-ignore
```

set interfaces bridge <interface> ip enable-proxy-arp

Configure proxy ARP on this interface.

If configured, the router (kernel) intercepts ARP requests for non-local IP addresses and replies with the MAC address of the interface that received the request. Subsequent packets destined to these IP addresses are forwarded to their actual destinations on remote subnets.

Example:

```
set interfaces bridge br0 ip enable-proxy-arp
```

set interfaces bridge <interface> ip proxy-arp-pvlan

Configure local proxy ARP on the interface.

If configured, the router (kernel) responds to ARP requests on this VLAN interface even if the target IP address resides on the same subnet and interface.

This is used to support network isolation requirements (RFC 3069) for private VLANs (PVLANS). In PVLAN configurations, hosts on isolated ports are NOT allowed to communicate directly with each other at Layer 2, but they can communicate with the upstream router.

By replying to inter-host ARP requests with its own MAC address, the router (kernel) directs inter-host traffic through itself instead of directly between hosts.

Note: This command works independently and does not require enabling the standard proxy ARP on the interface.

Local proxy ARP is also known as:

- VLAN aggregation ([RFC 3069](#)).
- Private VLAN (Cisco, Allied Telesyn).
- Source-port filtering or port isolation (Hewlett-Packard).
- MAC-Forced Forwarding (Ericsson).

set interfaces bridge <interface> ip source-validation <strict | loose | disable>

Configure source IP address validation using RPF on this interface, as specified in [RFC 3704](#).

The following options are available:

- **strict:** Each incoming packet's source IP address is checked against the Forwarding Information Base (FIB). If the interface is not the best route back to that source, validation fails, and the packet is dropped.
- **loose:** Each incoming packet's source IP address is checked against the FIB. If the source IP address is unreachable through any interface, validation fails.
- **disable:** No source IP address validation is performed. All incoming packets are accepted.

RFC 3704 recommends enabling strict mode to prevent IP spoofing, such as DDoS attacks. For asymmetric or other complex routing scenarios, use loose mode.

set interfaces bridge <interface> ipv6 address autoconf

Configure the interface to automatically obtain an IPv6 address using SLAAC, as specified in [RFC 4862](#).

IPv6 hosts can configure themselves automatically when connected to an IPv6 network using the Neighbor Discovery Protocol via ICMPv6 router discovery messages. When first connected to a network, a host sends a link-local router solicitation multicast request for its configuration parameters. The router responds with a router advertisement packet containing Internet Layer configuration parameters.

Note: This method automatically disables IPv6 traffic forwarding on the interface.

Example:

```
set interfaces bridge br0 ipv6 address autoconf
```

set interfaces bridge <interface> ipv6 address eui64 <prefix>

Configure the interface to assign itself an IPv6 address using the EUI-64 method, as specified in [RFC 4291](#).

Example:

```
set interfaces bridge br0 ipv6 address eui64 2001:db8:beef::/64
```

set interfaces bridge <interface> ipv6 address no-default-link-local

Disable the automatic assignment of a link-local IPv6 address to this interface.

Example:

```
set interfaces bridge br0 ipv6 address no-default-link-local
```

set interfaces bridge <interface> ipv6 disable-forwarding

Configure the interface for host or router behavior.

If configured, the interface switches to host mode, and IPv6 forwarding is disabled on it.

Example:

```
set interfaces bridge br0 ipv6 disable-forwarding
```

set interfaces bridge <interface> ipv6 adjust-mss <mss | clamp-mss-to-pmtu>

Configure the MSS advertised in outgoing TCP SYN packets on the specified interface.

By clamping the MSS value in TCP SYN packets, you explicitly inform the remote side not to send packets larger than that size. This prevents connection issues when Path MTU Discovery (PMTUD) fails.

The following options are available:

- **mss:** Set the MSS to a specific value, in bytes. Use this option to enforce a specific MSS, for example, to troubleshoot connectivity issues or accommodate specific network requirements.
- **clamp-mss-to-pmtu:** The router calculates the MSS to be the interface's MTU minus 60 bytes for IPv6 traffic (40 bytes for the IPv6 header and 20 bytes for the TCP header). This option is recommended to automatically set the proper value.

set interfaces bridge <interface> ipv6 accept-dad <0 | 1 | 2>

Configure IPv6 DAD on the interface.

The following options are available:

- **0:** Disables DAD. No duplicate address detection is performed.
- **1:** Enables DAD (default). Duplicate addresses are detected. The interface's IPv6 operation continues for valid IPv6 addresses.
- **2:** Enables DAD and, if a MAC-based duplicate link-local address is found, disables IPv6 operation on this interface.

Example:

```
set interfaces bridge br0 ipv6 accept-dad 2
```

set interfaces bridge <interface> ipv6 dup-addr-detect-transmits <n>

Configure the number of DAD messages that the router (kernel) sends during IPv6 address assignment on this interface.

The default value is 1.

Example:

```
set interfaces bridge br0 ipv6 dup-addr-detect-transmits 5
```

set interfaces bridge <interface> vrf <vrf>

Assign the interface to a specific VRF instance.

See also:

For information on configuring a VRF, refer to the [VRF](#) section.

Example:

```
set interfaces bridge br0 vrf red
```

DHCP(v6)

set interfaces bridge <interface> dhcp-options client-id <description>

Configure a DHCP client identifier for the interface, as specified in [RFC 2131](#).

The `client-id` is an identifier that the DHCP client sends to the DHCP server to uniquely identify itself for IP address assignment. By default, the client uses its MAC address. The `<description>` is a user-defined string that will be sent to the DHCP server as the DHCP client identifier.

Example:

```
set interfaces bridge br0 dhcp-options client-id 'foo-bar'
```

set interfaces bridge <interface> dhcp-options host-name <hostname>

Configure a specific hostname for the interface.

Instead of the real hostname, the DHCP client will send the specific hostname to the DHCP server when requesting an IP address.

Example:

```
set interfaces bridge br0 dhcp-options host-name 'VyOS'
```

set interfaces bridge <interface> dhcp-options vendor-class-id <vendor-id>

Configure the DHCP client to include a vendor-class identifier in its DHCP requests on this interface.

The vendor-class identifier is a vendor-specific byte string that enables the DHCP server to identify the device and, in some cases, provide configuration options.

Example:

```
set interfaces bridge br0 dhcp-options vendor-class-id 'VyOS'
```

set interfaces bridge <interface> dhcp-options no-default-route

Configure the DHCP client to obtain an IP address, but ignore any default gateway provided by the DHCP server on this interface.

Example:

```
set interfaces bridge br0 dhcp-options no-default-route
```

set interfaces bridge <interface> dhcp-options default-route-distance <distance>

Configure the distance for the default route obtained from the DHCP server on this interface.

Example:

```
set interfaces bridge br0 dhcp-options default-route-distance 220
```

set interfaces bridge <interface> dhcp-options reject <address>

Configure the DHCP client to reject the specific IP address or IP address range from the DHCP server on this interface.

This is useful when a modem assigns a local IP address upon start. To reject multiple addresses, run this command multiple times with different values. You can reject individual addresses (192.168.100.1) or entire subnets (192.168.100.0/24).

Example:

```
set interfaces bridge br0 dhcp-options reject 192.168.100.0/24
```

set interfaces bridge <interface> dhcp-options user-class <string>

Configure the DHCP client to send a specific user-class identifier in its DHCP requests on this interface.

The DHCP server can interpret this identifier and provide specific configuration options based on it (for example, default routes). The user-class value typically groups DHCP clients with similar configuration needs (for example, employees, guests, or printers).

Example:

```
set interfaces bridge br0 dhcp-options user-class VyOS
```

set interfaces bridge <interface> dhcpv6-options duid <duid>

Configure a specific DUID for the DHCPv6 client on this interface.

The DUID is an identifier used by a DHCPv6 client to get an IPv6 address from a DHCPv6 server. It consists of a 2-byte type field, followed by a variable-length identifier field up to 128 bytes. The format of the identifier part depends on the DUID type:

- **DUID-LLT:** The most common type, which includes a hardware type, a timestamp, and a MAC address.
- **DUID-EN:** Is based on a vendor's enterprise number and a unique identifier assigned by the vendor.
- **DUID-LL:** Includes only a MAC address.

The DHCP server matches the DUID against its database and provides configuration data (such as address, lease times, DNS servers, etc.) to the DHCP client.

Example:

```
set interfaces bridge br0 duid '0e:00:00:01:00:01:27:71:db:f0:00:50:56:bf:c5:6d'
```

set interfaces bridge <interface> dhcpv6-options no-release

Configure the DHCP client not to send a release message when it stops running on this interface.

This helps retain the assigned address or prefix.

Example:

```
set interfaces bridge br0 dhcpv6-options no-release
```

set interfaces bridge <interface> dhcpv6-options parameters-only

Enable a stateless DHCPv6 client mode on this interface.

In stateless mode, the DHCPv6 client requests only stateless configuration parameters from the DHCP server (for example, DNS server addresses). It doesn't request a stateful configuration, such as IPv6 addresses or prefixes.

Example:

```
set interfaces bridge br0 dhcpv6-options parameters-only
```

set interfaces bridge <interface> dhcpv6-options rapid-commit

Enable DHCPv6 rapid commit on this interface.

When enabled, the DHCP client and server skip the negotiation steps (Advertise and Request), completing the DHCPv6 configuration process in just two messages (Solicit and final Reply).

Example:

```
set interfaces bridge br0 dhcpv6-options rapid-commit
```

set interfaces bridge <interface> dhcpv6-options temporary

Configure the DHCPv6 client to request a temporary IPv6 address on this interface.

When configured, the DHCP client doesn't form an Identity Association for Non-temporary Addresses (IA_NA) partnership. Consequently, it only obtains a temporary IPv6 address and doesn't obtain a permanent one.

Example:

```
set interfaces bridge br0 dhcpv6-options temporary
```

DHCPv6 Prefix Delegation (PD)

VyOS supports DHCPv6 Prefix Delegation (DHCPv6-PD) as described in [RFC 3633](#). DHCPv6-PD is supported by most ISPs that provide native IPv6 for consumers on fixed networks.

set interfaces bridge <interface> dhcpv6-options pd <id> length <length>

Configure a specific prefix length for DHCPv6-PD requests on this interface.

Some ISPs provide only a /64 prefix by default. Use this command to request a different prefix length for a specific DHCPv6-PD request, ranging from /32 (if allowed by your ISP) down to /64. <id> is a unique identifier for the DHCPv6-PD request.

The default value is 64.

To request a /56 prefix from your ISP, use:

```
set interfaces bridge br0 dhcpv6-options pd 0 length 56
```

set interfaces bridge <interface> dhcpv6-options pd <id> interface <delegatee> address <address>

Configure the IPv6 interface identifier (host portion) for the delegatee interface.

The value must be a decimal integer. It is appended to the delegated prefix and the configured SLA ID to form the final IPv6 address.

By default, the host portion is generated based on the parent interface's MAC address (EUI-64 format).

Example:

If a /64 prefix is delegated to interface eth8 and you configure the host portion as 65535, the resulting IPv6 address will end with ::ffff, as 65535 corresponds to ffff in hexadecimal notation.

```
set interfaces bridge br0 dhcpv6-options pd 0 interface eth8 address 65534
```

set interfaces bridge <interface> dhcpv6-options pd <id> interface <delegatee> sla-id <id>

Configure the SLA ID for the delegatee interface.

The value must be a decimal integer greater than 0 and fit in the length of SLA IDs. It is converted to hexadecimal and appended to the delegated prefix to form the specific subnet prefix for the delegatee interface.

Example:

If SLA ID is 1 and the delegated prefix is 2001:db8:ffff::/48, the resulting subnet prefix for the delegatee interface will be 2001:db8:ffff:1::/64.

```
set interfaces bridge br0 dhcpv6-options pd 0 interface eth8 sla-id 1
```

Member interfaces

```
set interfaces bridge <interface> member interface <member>
```

Configure an interface as a bridge member.

Valid interface types are: *Ethernet*, *Bond/link aggregation*, *L2TPv3*, *OpenVPN*, *VXLAN*, *Wireless LAN/Wi-Fi*, *Tunnel*, and *Geneve*.

Use tab completion to list interfaces that can be bridged.

```
set interfaces bridge <interface> member interface <member> priority <priority>
```

Configure the STP port priority for a specific member interface within a bridge.

Within the STP topology, each member interface in a bridge operates as a port with an assigned **priority** and **path cost**. STP uses these values to determine the **lowest-cost path** to the root bridge, maintaining a loop-free topology. Traffic flows through the path with the lowest path cost, while alternate paths remain in standby.

A **lower** priority value means **higher** precedence in path selection.

STP considers the port priority only if multiple member interfaces have the same path costs.

```
set interfaces bridge <interface> member interface <member> cost <cost>
```

Configure the STP path cost for a specific member interface within the bridge.

Path cost is the primary metric STP uses to determine the path to the root bridge. This value is based on interface bandwidth; faster interfaces receive lower costs.

By assigning a lower cost, you give the interface higher precedence during path selection.

```
set interfaces bridge <interface> member interface <member> disable-learning
```

Disable MAC address learning for a specific member interface within a bridge.

When learning is disabled, the bridge will not add source MAC addresses observed on this port to its forwarding database (FDB). Frames destined to MACs not present in the FDB are then flooded to all bridge ports rather than unicast-forwarded.

Bridge options

Configure how bridge interfaces maintain their FDB (Forwarding Database), react to topology changes, and optimize multicast data streams.

```
set interfaces bridge <interface> aging <time>
```

Configure the MAC address aging time for the bridge.

The duration in seconds that a MAC address remains in the bridge's FDB before removal if no traffic is received from that address.

The default value is 300 seconds.

```
set interfaces bridge <interface> max-age <time>
```

Configure the STP max age timer for the bridge.

The duration in seconds that the bridge waits for a BPDU (Bridge Protocol Data Unit) from the root bridge.

If the bridge does not receive a BPDU within this period, it recalculates the path to the root bridge or initiates a new root bridge election.

```
set interfaces bridge <interface> igmp querier
```


Configure the bridge interface to act as the IGMP (Internet Group Management Protocol)/MLD (Multicast Listener Discovery) Querier.

When configured: The bridge interface sends IGMP (IPv4) and MLD (IPv6) general queries to all connected hosts to identify active multicast listeners.

```
set interfaces bridge <interface> igmp snooping
```

Configure the bridge interface to perform IGMP/MLD snooping.

When configured: The bridge interface monitors IGMP (IPv4) and MLD (IPv6) join requests and restricts multicast traffic forwarding to only active listeners. This prevents network flooding.

STP configuration

STP is a Layer 2 protocol that prevents loops in Ethernet networks by ensuring only one logical path exists between any two bridges. This creates a loop-free topology and prevents broadcast storms that can crash the network.

By default, STP is disabled on bridge interfaces. To activate loop prevention, you must explicitly enable the protocol and configure its parameters.

```
set interfaces bridge <interface> stp
```

Enable STP on the bridge interface.

```
set interfaces bridge <interface> forwarding-delay <delay>
```

Configure the STP delay, in seconds, for the bridge interface.

This parameter defines how long the bridge interface remains in the listening and learning states before forwarding traffic. The delay ensures that the bridge has sufficient time to detect loops (in the listening state) and learn the MAC addresses of connected devices (in the learning state).

The default value is 15 seconds. The total time before forwarding begins is twice this value.

```
set interfaces bridge <interface> hello-time <interval>
```

Configure the STP Hello advertisement interval, in seconds.

This parameter sets the frequency at which the bridge interface transmits Hello packets (BPDUs (Bridge Protocol Data Units)). These packets originate from the root bridge and are propagated by designated bridges. If neighbors stop receiving Hello packets, they assume a connection failure and trigger a topology recalculation.

The default value is 2 seconds.

VLAN

VLAN-aware bridges

```
set interfaces bridge <interface> enable-vlan
```

Enable VLAN filtering (also known as VLAN awareness) on the bridge interface.

When enabled, the bridge strictly segregates traffic among VLANs configured on its member interfaces.

Note: Do not configure **vif 1** on a VLAN-aware bridge. The main bridge interface acts as VLAN 1 (the default native VLAN) and automatically handles all untagged traffic.

```
set interfaces bridge <interface> protocol <802.1ad | 802.1q>
```

Configure the VLAN protocol (EtherType) for the bridge interface.

The following options are available:

- **802.1q** (default): Sets the EtherType to **0x8100**. Used for standard enterprise VLANs.
- **802.1ad**: Sets the EtherType to **0x88a8**. Used for QinQ (provider bridging).

VLAN configuration

IEEE 802.1q, often referred to as Dot1q, is the industry standard for implementing VLANs on Ethernet networks. It defines VLAN tagging for Ethernet frames and outlines procedures for bridges and switches. The standard also includes quality-of-service prioritization (IEEE 802.1p) and defines the Generic Attribute Registration Protocol.

VLAN-aware network segments (i.e., IEEE 802.1q conformant) use VLAN tags. When a frame enters such a segment, a tag is added to indicate VLAN membership. Each frame can belong to only one VLAN. If a frame arrives without a tag, it is assumed to be part of the native VLAN.

IEEE 802.1, a working group of the IEEE 802 standards committee, has developed the standard and continues to revise it. One notable revision is 802.1Q-2014, which incorporated IEEE 802.1aq (Shortest Path Bridging) and much of the IEEE 802.1d standard.

In VyOS, 802.1q VLAN interfaces are represented as virtual subinterfaces, referred to as **vif**.

set interfaces bridge <interface> vif <vlan-id>

Configure a VLAN interface with a unique VLAN ID.

VLAN ID identifies a specific VLAN and ranges from 0 to 4094.

You can configure multiple VLAN interfaces on a single physical interface.

Note: Only 802.1Q-tagged packets are accepted on Ethernet vifs.

set interfaces bridge <interface> vif <vlan-id> address <address | dhcp | dhcpv6>

Configure the interface with one or more IP addresses.

The following options are available:

- **address:** Assign one or more IPv4 or IPv6 addresses to the interface. For example, use 192.0.2.1/24 for IPv4 or 2001:db8::1/64 for IPv6.
- **dhcp:** The interface obtains an IPv4 address from a DHCP server on the same network segment.
- **dhcpv6:** The interface obtains an IPv6 address from a DHCPv6 server on the same network segment.

Note: If the interface obtains an IPv4 address via DHCP, and specific adjustments are needed before/after the IP address is obtained, use the provided hook scripts:

- /config/scripts/dhcp-client/pre-hooks.d/
 - /config/scripts/dhcp-client/post-hooks.d/
-

Example:

```
set interfaces bridge br0 vif 10 address 192.0.2.1/24
set interfaces bridge br0 vif 10 address 2001:db8::1/64
```

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```
set interfaces bridge br0 vif 10 address dhcp
set interfaces bridge br0 vif 10 address dhcpv6
```

set interfaces bridge <interface> vif <vlan-id> description <description>

Configure a clear, descriptive alias for the interface.

This alias appears in the `show interfaces` command and SNMP-based monitoring tools.

Example:

```
set interfaces bridge br0 vif 10 description 'This is an interface running on VyOS.'
```

set interfaces bridge <interface> vif <vlan-id> disable

Disable the interface.

The interface will be set to the administratively down (A/D) state.

Example:

```
set interfaces bridge br0 vif 10 disable
```

set interfaces bridge <interface> vif <vlan-id> disable-link-detect

Disable physical link-state change detection on the interface, such as when a cable is unplugged.

By default, the interface detects physical link-state changes.

Example:

```
set interfaces bridge br0 vif 10 disable-link-detect
```

set interfaces bridge <interface> vif <vlan-id> mac <xx:xx:xx:xx:xx:xx>

Configure a custom MAC address on the interface.

Example:

```
set interfaces bridge br0 vif 10 mac '00:53:01:02:03:04'
```

set interfaces bridge <interface> vif <vlan-id> mtu <mtu>

Configure the MTU on the interface.

This value defines the largest packet size, in bytes, that the interface transmits without fragmentation.

Example:

```
set interfaces bridge br0 vif 10 mtu 1600
```

set interfaces bridge <interface> vif <vlan-id> ip adjust-mss <mss | clamp-mss-to-pmtu>

Configure the MSS advertised in outgoing TCP SYN packets on the specified interface.

By clamping the MSS value in TCP SYN packets, you explicitly inform the remote side not to send packets larger than that size. This prevents connection issues that occur when Path MTU Discovery (PMTUD) fails.

The following options are available:

- **mss:** Sets the MSS to a specific value, in bytes. Use this option if you need to enforce a specific MSS, for example, to troubleshoot connectivity issues or accommodate specific network requirements.

- **clamp-mss-to-pmtu:** The router automatically calculates the MSS to be the interface's MTU minus 40 bytes for IPv4 traffic (20 bytes for the IPv4 header and 20 bytes for the TCP header). This option is recommended to automatically set the proper value.

set interfaces bridge <interface> vif <vlan-id> ip arp-cache-timeout

Configure how long an ARP entry remains valid after learning an IP-to-MAC address mapping on this interface.

The default duration is 30 seconds.

An ARP entry remains valid if it receives positive feedback from higher-level protocols.

Example:

```
set interfaces bridge br0 vif 10 ip arp-cache-timeout 180
```

set interfaces bridge <interface> vif <vlan-id> ip disable-arp-filter

Configure ARP filtering on this interface.

Default behavior: The kernel responds to ARP requests on this interface only if the traffic would be routed back to the ARP sender through that specific interface.

If configured: The kernel responds to ARP requests on this interface for any IP address configured on the local host, regardless of which specific interface that IP address is assigned to, and regardless of the routing table. This reflects the Linux concept that IP addresses belong to the host, not individual interfaces.

Example:

```
set interfaces bridge br0 vif 10 ip disable-arp-filter
```

set interfaces bridge <interface> vif <vlan-id> ip disable-forwarding

Configure the interface for host or router behavior.

If configured, the interface switches to host mode, and IPv4 forwarding is disabled on it.

Example:

```
set interfaces bridge br0 vif 10 ip disable-forwarding
```

set interfaces bridge <interface> vif <vlan-id> ip enable-directed-broadcast

Configure whether to forward IP-directed broadcast packets received on this interface.

Default behavior: IP-directed broadcast packets are dropped.

If configured: IP-directed broadcast packets are forwarded to all hosts on the destination subnet, as defined in [RFC 1812](#) and [RFC 2644](#).

Example:

```
set interfaces bridge br0 vif 10 ip enable-directed-broadcast
```

set interfaces bridge <interface> vif <vlan-id> ip enable-arp-accept

Configure how to process gratuitous ARPs on this interface.

If configured, an IP-to-MAC address mapping is added to the ARP table based on gratuitous ARP requests or replies.

Note: If the ARP table already contains the IP address from a gratuitous ARP, its entry is updated regardless of whether this setting is configured.

Example:

```
set interfaces bridge br0 vif 10 ip enable-arp-accept
```

set interfaces bridge <interface> vif <vlan-id> ip enable-arp-announce

Configure the source IP selection for ARP requests on this interface.

Default behavior: The kernel can use any IP address the host owns as the source IP address in ARP requests on this interface.

If configured: The kernel first attempts to select a source IP address configured on the interface that shares a common subnet with the target IP address. If there is no such subnet, the kernel selects the IP address it would normally use (based on the routing table to reach the target destination).

Example:

```
set interfaces bridge br0 vif 10 ip enable-arp-announce
```

set interfaces bridge <interface> vif <vlan-id> ip enable-arp-ignore

Configure which ARP requests will be ignored on this interface.

Default behavior: The kernel responds to ARP requests for any local IP addresses, regardless of which interface they are assigned to.

If configured: The kernel responds to ARP requests only if the target IP address is assigned to this specific interface.

Example:

```
set interfaces bridge br0 vif 10 ip enable-arp-ignore
```

set interfaces bridge <interface> vif <vlan-id> ip enable-proxy-arp

Configure proxy ARP on this interface.

If configured, the router (kernel) intercepts ARP requests for non-local IP addresses and replies with the MAC address of the interface that received the request. Subsequent packets destined to these IP addresses are forwarded to their actual destinations on remote subnets.

Example:

```
set interfaces bridge br0 vif 10 ip enable-proxy-arp
```

set interfaces bridge <interface> vif <vlan-id> ip proxy-arp-pvlan

Configure local proxy ARP on the interface.

If configured, the router (kernel) responds to ARP requests on this VLAN interface even if the target IP address resides on the same subnet and interface.

This is used to support network isolation requirements (RFC 3069) for private VLANs (PVLANS). In PVLAN configurations, hosts on isolated ports are NOT allowed to communicate directly with each other at Layer 2, but they can communicate with the upstream router.

By replying to inter-host ARP requests with its own MAC address, the router (kernel) directs inter-host traffic through itself instead of directly between hosts.

Note: This command works independently and does not require enabling the standard proxy ARP on the interface.

Local proxy ARP is also known as:

- VLAN aggregation ([RFC 3069](#)).
- Private VLAN (Cisco, Allied Telesyn).
- Source-port filtering or port isolation (Hewlett-Packard).
- MAC-Forced Forwarding (Ericsson).

set interfaces bridge <interface> vif <vlan-id> ip source-validation <strict | loose | disable>

Configure source IP address validation using RPF on this interface, as specified in [RFC 3704](#).

The following options are available:

- **strict:** Each incoming packet's source IP address is checked against the Forwarding Information Base (FIB). If the interface is not the best route back to that source, validation fails, and the packet is dropped.
- **loose:** Each incoming packet's source IP address is checked against the FIB. If the source IP address is unreachable through any interface, validation fails.
- **disable:** No source IP address validation is performed. All incoming packets are accepted.

RFC 3704 recommends enabling strict mode to prevent IP spoofing, such as DDoS attacks. For asymmetric or other complex routing scenarios, use loose mode.

set interfaces bridge <interface> vif <vlan-id> ipv6 address autoconf

Configure the interface to automatically obtain an IPv6 address using SLAAC, as specified in [RFC 4862](#).

IPv6 hosts can configure themselves automatically when connected to an IPv6 network using the Neighbor Discovery Protocol via ICMPv6 router discovery messages. When first connected to a network, a host sends a link-local router solicitation multicast request for its configuration parameters. The router responds with a router advertisement packet containing Internet Layer configuration parameters.

Note: This method automatically disables IPv6 traffic forwarding on the interface.

Example:

```
set interfaces bridge br0 vif 10 ipv6 address autoconf
```

set interfaces bridge <interface> vif <vlan-id> ipv6 address eui64 <prefix>

Configure the interface to assign itself an IPv6 address using the EUI-64 method, as specified in [RFC 4291](#).

Example:

```
set interfaces bridge br0 vif 10 ipv6 address eui64 2001:db8:beef::/64
```

set interfaces bridge <interface> vif <vlan-id> ipv6 address no-default-link-local

Disable the automatic assignment of a link-local IPv6 address to this interface.

Example:

```
set interfaces bridge br0 vif 10 ipv6 address no-default-link-local
```

set interfaces bridge <interface> vif <vlan-id> ipv6 disable-forwarding

Configure the interface for host or router behavior.

If configured, the interface switches to host mode, and IPv6 forwarding is disabled on it.

Example:

```
set interfaces bridge br0 vif 10 ipv6 disable-forwarding
```

set interfaces bridge <interface> vif <vlan-id> ipv6 adjust-mss <mss | clamp-mss-to-pmtu>

Configure the MSS advertised in outgoing TCP SYN packets on the specified interface.

By clamping the MSS value in TCP SYN packets, you explicitly inform the remote side not to send packets larger than that size. This prevents connection issues when Path MTU Discovery (PMTUD) fails.

The following options are available:

- **mss:** Set the MSS to a specific value, in bytes. Use this option to enforce a specific MSS, for example, to troubleshoot connectivity issues or accommodate specific network requirements.
- **clamp-mss-to-pmtu:** The router calculates the MSS to be the interface's MTU minus 60 bytes for IPv6 traffic (40 bytes for the IPv6 header and 20 bytes for the TCP header). This option is recommended to automatically set the proper value.

set interfaces bridge <interface> vif <vlan-id> ipv6 accept-dad <0 | 1 | 2>

Configure IPv6 DAD on the interface.

The following options are available:

- **0:** Disables DAD. No duplicate address detection is performed.
- **1:** Enables DAD (default). Duplicate addresses are detected. The interface's IPv6 operation continues for valid IPv6 addresses.
- **2:** Enables DAD and, if a MAC-based duplicate link-local address is found, disables IPv6 operation on this interface.

Example:

```
set interfaces bridge br0 vif 10 ipv6 accept-dad 2
```

set interfaces bridge <interface> vif <vlan-id> ipv6 dup-addr-detect-transmits <n>

Configure the number of DAD messages that the router (kernel) sends during IPv6 address assignment on this interface.

The default value is 1.

Example:

```
set interfaces bridge br0 vif 10 ipv6 dup-addr-detect-transmits 5
```

set interfaces bridge <interface> vif <vlan-id> vrf <vrf>

Assign the interface to a specific VRF instance.

See also:

For information on configuring a VRF, refer to the [VRF](#) section.

Example:

```
set interfaces bridge br0 vif 10 vrf red
```

DHCP(v6)

set interfaces bridge <interface> vif <vlan-id> dhcp-options client-id <description>

Configure a DHCP client identifier for the interface, as specified in [RFC 2131](#).

The `client-id` is an identifier that the DHCP client sends to the DHCP server to uniquely identify itself for IP address assignment. By default, the client uses its MAC address. The `<description>` is a user-defined string that will be sent to the DHCP server as the DHCP client identifier.

Example:

```
set interfaces bridge br0 vif 10 dhcp-options client-id 'foo-bar'
```

set interfaces bridge <interface> vif <vlan-id> dhcp-options host-name <hostname>

Configure a specific hostname for the interface.

Instead of the real hostname, the DHCP client will send the specific hostname to the DHCP server when requesting an IP address.

Example:

```
set interfaces bridge br0 vif 10 dhcp-options host-name 'VyOS'
```

set interfaces bridge <interface> vif <vlan-id> dhcp-options vendor-class-id <vendor-id>

Configure the DHCP client to include a vendor-class identifier in its DHCP requests on this interface.

The vendor-class identifier is a vendor-specific byte string that enables the DHCP server to identify the device and, in some cases, provide configuration options.

Example:

```
set interfaces bridge br0 vif 10 dhcp-options vendor-class-id 'VyOS'
```

set interfaces bridge <interface> vif <vlan-id> dhcp-options no-default-route

Configure the DHCP client to obtain an IP address, but ignore any default gateway provided by the DHCP server on this interface.

Example:

```
set interfaces bridge br0 vif 10 dhcp-options no-default-route
```

set interfaces bridge <interface> vif <vlan-id> dhcp-options default-route-distance <distance>

Configure the distance for the default route obtained from the DHCP server on this interface.

Example:

```
set interfaces bridge br0 vif 10 dhcp-options default-route-distance 220
```

set interfaces bridge <interface> vif <vlan-id> dhcp-options reject <address>

Configure the DHCP client to reject the specific IP address or IP address range from the DHCP server on this interface.

This is useful when a modem assigns a local IP address upon start. To reject multiple addresses, run this command multiple times with different values. You can reject individual addresses (192.168.100.1) or entire subnets (192.168.100.0/24).

Example:

```
set interfaces bridge br0 vif 10 dhcp-options reject 192.168.100.0/24
```

set interfaces bridge <interface> vif <vlan-id> dhcp-options user-class <string>

Configure the DHCP client to send a specific user-class identifier in its DHCP requests on this interface.

The DHCP server can interpret this identifier and provide specific configuration options based on it (for example, default routes). The user-class value typically groups DHCP clients with similar configuration needs (for example, employees, guests, or printers).

Example:

```
set interfaces bridge br0 vif 10 dhcp-options user-class VyOS
```

set interfaces bridge <interface> vif <vlan-id> dhcpv6-options duid <duid>

Configure a specific DUID for the DHCPv6 client on this interface.

The DUID is an identifier used by a DHCPv6 client to get an IPv6 address from a DHCPv6 server. It consists of a 2-byte type field, followed by a variable-length identifier field up to 128 bytes. The format of the identifier part depends on the DUID type:

- **DUID-LLT:** The most common type, which includes a hardware type, a timestamp, and a MAC address.
- **DUID-EN:** Is based on a vendor's enterprise number and a unique identifier assigned by the vendor.
- **DUID-LL:** Includes only a MAC address.

The DHCP server matches the DUID against its database and provides configuration data (such as address, lease times, DNS servers, etc.) to the DHCP client.

Example:

```
set interfaces bridge br0 vif 10 duid
→ '0e:00:00:01:00:01:27:71:db:f0:00:50:56:bf:c5:6d'
```

set interfaces bridge <interface> vif <vlan-id> dhcpv6-options no-release

Configure the DHCP client not to send a release message when it stops running on this interface.

This helps retain the assigned address or prefix.

Example:

```
set interfaces bridge br0 vif 10 dhcpv6-options no-release
```

set interfaces bridge <interface> vif <vlan-id> dhcpv6-options parameters-only

Enable a stateless DHCPv6 client mode on this interface.

In stateless mode, the DHCPv6 client requests only stateless configuration parameters from the DHCP server (for example, DNS server addresses). It doesn't request a stateful configuration, such as IPv6 addresses or prefixes.

Example:

```
set interfaces bridge br0 vif 10 dhcpv6-options parameters-only
```

```
set interfaces bridge <interface> vif <vlan-id> dhcpv6-options rapid-commit
```

Enable DHCPv6 rapid commit on this interface.

When enabled, the DHCP client and server skip the negotiation steps (Advertise and Request), completing the DHCPv6 configuration process in just two messages (Solicit and final Reply).

Example:

```
set interfaces bridge br0 vif 10 dhcpv6-options rapid-commit
```

```
set interfaces bridge <interface> vif <vlan-id> dhcpv6-options temporary
```

Configure the DHCPv6 client to request a temporary IPv6 address on this interface.

When configured, the DHCP client doesn't form an Identity Association for Non-temporary Addresses (IA_NA) partnership. Consequently, it only obtains a temporary IPv6 address and doesn't obtain a permanent one.

Example:

```
set interfaces bridge br0 vif 10 dhcpv6-options temporary
```

DHCPv6 Prefix Delegation (PD)

VyOS supports DHCPv6 Prefix Delegation (DHCPv6-PD) as described in [RFC 3633](#). DHCPv6-PD is supported by most ISPs that provide native IPv6 for consumers on fixed networks.

```
set interfaces bridge <interface> vif <vlan-id> dhcpv6-options pd <id> length <length>
```

Configure a specific prefix length for DHCPv6-PD requests on this interface.

Some ISPs provide only a /64 prefix by default. Use this command to request a different prefix length for a specific DHCPv6-PD request, ranging from /32 (if allowed by your ISP) down to /64. <id> is a unique identifier for the DHCPv6-PD request.

The default value is 64.

To request a /56 prefix from your ISP, use:

```
set interfaces bridge br0 vif 10 dhcpv6-options pd 0 length 56
```

```
set interfaces bridge <interface> vif <vlan-id> dhcpv6-options pd <id> interface
<delegatee> address <address>
```

Configure the IPv6 interface identifier (host portion) for the delegatee interface.

The value must be a decimal integer. It is appended to the delegated prefix and the configured SLA ID to form the final IPv6 address.

By default, the host portion is generated based on the parent interface's MAC address (EUI-64 format).

Example:

If a /64 prefix is delegated to interface eth8 and you configure the host portion as 65535, the resulting IPv6 address will end with ::ffff, as 65535 corresponds to ffff in hexadecimal notation.

```
set interfaces bridge br0 vif 10 dhcpv6-options pd 0 interface eth8 address 65534
```

```
set interfaces bridge <interface> vif <vlan-id> dhcpv6-options pd <id> interface
<delegatee> sla-id <id>
```

Configure the SLA ID for the delegatee interface.

The value must be a decimal integer greater than 0 and fit in the length of SLA IDs. It is converted to hexadecimal and appended to the delegated prefix to form the specific subnet prefix for the delegatee interface.

Example:

If SLA ID is 1 and the delegated prefix is 2001:db8:ffff::/48, the resulting subnet prefix for the delegatee interface will be 2001:db8:ffff:1::/64.

```
set interfaces bridge br0 vif 10 dhcpv6-options pd 0 interface eth8 sla-id 1
```

```
set interfaces bridge <interface> member interface <member> native-vlan <vlan-id>
```

Configure the native VLAN ID for a specific member interface within a VLAN-aware bridge.

This assigns the specified <vlan-id> to untagged traffic entering the member interface. The bridge strips the VLAN tag from outgoing traffic matching this ID.

Example:

Set the native VLAN ID to 2 for the member interface eth0:

```
set interfaces bridge br1 member interface eth0 native-vlan 2
```

```
set interfaces bridge <interface> member interface <member> allowed-vlan <vlan-id>
```

Configure allowed VLAN IDs for a specific member interface within a VLAN-aware bridge.

Enter a single VLAN ID or a range of VLAN IDs separated by a hyphen.

Example:

To allow VLAN ID 4 on member interface eth0:

```
set interfaces bridge br1 member interface eth0 allowed-vlan 4
```

Example:

To allow VLAN IDs 6 through 8 on member interface eth0:

```
set interfaces bridge br1 member interface eth0 allowed-vlan 6-8
```

SPAN port mirroring

SPAN port mirroring copies inbound and outbound traffic from one interface to another specified interface.

The destination interface is usually connected to specialized equipment, such as a behavior control system, an intrusion detection system, or a traffic collector, and copies all related traffic from this port. The benefit of mirroring traffic is that the application is isolated from the source traffic, so application processing does not affect the traffic or system performance.

To configure SPAN port mirroring, VyOS uses the `mirror` parameter. You can mirror ingress traffic (traffic entering the router) and egress traffic (traffic leaving the router) separately. Both directions can be mirrored to the same destination interface or split to different ones.

```
set interfaces bridge <interface> mirror ingress <monitor-interface>
```

Mirror ingress traffic from a bonding interface to the specified monitoring interface.

Example: Mirror the ingress traffic from *br1* to *eth3*.

```
set interfaces bridge br1 mirror ingress eth3
```

set interfaces bridge <interface> mirror egress <monitor-interface>

Mirror egress traffic from a bonding interface to the specified monitoring interface.

Example: Mirror the egress traffic from *br1* to *eth3*.

```
set interfaces bridge br1 mirror egress eth3
```

Examples

Configure a standard bridge

The following example creates a bridge named *br100* with STP enabled.

Configuration requirements:

- **Bridge name:** *br100*
- **Member interfaces:** Physical interface *eth1* and VLAN interface *eth2.10*.
- **STP:** Enabled.
- **Bridge IP addresses:** *192.0.2.1/24* (IPv4) and *2001:db8::ffff/64* (IPv6).

```
set interfaces bridge br100 address 192.0.2.1/24
set interfaces bridge br100 address 2001:db8::ffff/64
set interfaces bridge br100 member interface eth1
set interfaces bridge br100 member interface eth2.10
set interfaces bridge br100 stp
```

Verify the configuration:

```
vyos@vyos# show interfaces bridge br100
address 192.0.2.1/24
address 2001:db8::ffff/64
member {
    interface eth1 {
    }
    interface eth2.10 {
    }
}
stp
```

Configure a VLAN-aware bridge

The following example creates a VLAN-aware bridge named br100. In this setup, one member interface is configured as a trunk port, and the other as an access port. The VLAN interface is configured with IP addresses.

Configuration requirements:

- **Bridge name:** br100.
- **Trunk port** (eth1): Handles **tagged** traffic for VLAN 10.
- **Access port** (eth2): Handles **untagged** traffic (assigned to native VLAN 10).
- **STP:** Enabled.
- **VLAN IP addresses** (vif 10): 192.0.2.1/24 (IPv4) and 2001:db8::ffff/64 (IPv6).

```
set interfaces bridge br100 enable-vlan
set interfaces bridge br100 member interface eth1 allowed-vlan 10
set interfaces bridge br100 member interface eth2 native-vlan 10
set interfaces bridge br100 vif 10 address 192.0.2.1/24
set interfaces bridge br100 vif 10 address 2001:db8::ffff/64
set interfaces bridge br100 stp
```

Verify the configuration:

```
vyos@vyos# show interfaces bridge br100
enable-vlan
member {
    interface eth1 {
        allowed-vlan 10
    }
    interface eth2 {
        native-vlan 10
    }
}
stp
vif 10 {
    address 192.0.2.1/24
    address 2001:db8::ffff/64
}
```

Operation

show bridge

Show the status of member interfaces for all configured bridges.

```
vyos@vyos:~$ show bridge
3: eth1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 master br0 state forwarding
priority 32 cost 100
4: eth2: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 master br0 state forwarding
priority 32 cost 100
```

show bridge <name> fdb

Show the FDB for the specified bridge.

```
vyos@vyos:~$ show bridge br0 fdb
50:00:00:08:00:01 dev eth1 vlan 20 master br0 permanent
50:00:00:08:00:01 dev eth1 vlan 10 master br0 permanent
50:00:00:08:00:01 dev eth1 master br0 permanent
33:33:00:00:00:01 dev eth1 self permanent
33:33:00:00:00:02 dev eth1 self permanent
01:00:5e:00:00:01 dev eth1 self permanent
50:00:00:08:00:02 dev eth2 vlan 20 master br0 permanent
50:00:00:08:00:02 dev eth2 vlan 10 master br0 permanent
50:00:00:08:00:02 dev eth2 master br0 permanent
33:33:00:00:00:01 dev eth2 self permanent
33:33:00:00:00:02 dev eth2 self permanent
01:00:5e:00:00:01 dev eth2 self permanent
33:33:00:00:00:01 dev br0 self permanent
33:33:00:00:00:02 dev br0 self permanent
33:33:ff:08:00:01 dev br0 self permanent
01:00:5e:00:00:6a dev br0 self permanent
33:33:00:00:00:6a dev br0 self permanent
01:00:5e:00:00:01 dev br0 self permanent
33:33:ff:00:00:00 dev br0 self permanent
```

show bridge <name> mdb

Show the MDB (Multicast group Database) for the specified bridge.

The MDB is populated by IGMP/MLD snooping and lists the multicast groups currently active on the bridge.

```
vyos@vyos:~$ show bridge br0 mdb
dev br0 port br0 grp ff02::1:ff00:0 temp vid 1
dev br0 port br0 grp ff02::2 temp vid 1
dev br0 port br0 grp ff02::1:ff08:1 temp vid 1
dev br0 port br0 grp ff02::6a temp vid 1
```

show bridge <name> macs

Show the learned MAC address table for the specified bridge.

```
vyos@vyos:~$ show bridge br100 macs
port no mac addr is local? ageing timer
1 00:53:29:44:3b:19 yes 0.00
```

7.4.3 Dummy

A dummy interface is a virtual network interface that operates like the loopback interface, accepting traffic and routing it back to the local host. Unlike the loopback interface, which is limited to one per system and reserved for internal system use, multiple dummy interfaces can be created, removed, and managed without impacting core operations.

As a software-based interface, the dummy interface does not depend on physical link state and remains active as long as the operating system is running.

Dummy interfaces are commonly used in environments with multiple redundant uplinks (e.g., a server connected to two different switches), where assigning a management IP address to a specific physical interface is risky. If that interface fails, the management IP address becomes unreachable.

Assigning the management IP address to a dummy interface and advertising it over all available physical links ensures the address remains reachable as long as at least one physical path is active.

Dummy interfaces are also used for testing and simulation purposes.

Configuration

Common interface configuration

set interfaces dummy <interface> address <address>

Configure one or more interface IP addresses.

<address> supports IPv4 and IPv6 in CIDR notation. For example, use 192.0.2.1/24 for IPv4 or 2001:db8::1/64 for IPv6.

Example:

```
set interfaces dummy dum0 address 192.0.2.1/24
set interfaces dummy dum0 address 2001:db8::1/64
```

set interfaces dummy <interface> description <description>

Configure a clear, descriptive alias for the interface.

This alias appears in the `show interfaces` command and SNMP-based monitoring tools.

Example:

```
set interfaces dummy dum0 description 'This is an interface running on VyOS.'
```

set interfaces dummy <interface> disable

Disable the interface.

The interface will be set to the administratively down (A/D) state.

Example:

```
set interfaces dummy dum0 disable
```

set interfaces dummy <interface> vrf <vrf>

Assign the interface to a specific VRF instance.

See also:

For information on configuring a VRF, refer to the [VRF](#) section.

Example:

```
set interfaces dummy dum0 vrf red
```

Operation

show interfaces dummy

Show brief interface information.

```
vyos@vyos:~$ show interfaces dummy
Codes: S - State, L - Link, u - Up, D - Down, A - Admin Down
Interface      IP Address      S/L  Description
-----
dum0           172.18.254.201/32  u/u
```

show interfaces dummy <interface>

Show detailed interface information.

```
vyos@vyos:~$ show interfaces dummy dum0
dum0: <BROADCAST,NOARP,UP,LOWER_UP> mtu 1500 qdisc noqueue state UNKNOWN group_
↳ default qlen 1000
    link/ether 26:7c:8e:bc:fc:f5 brd ff:ff:ff:ff:ff:ff
    inet 172.18.254.201/32 scope global dum0
        valid_lft forever preferred_lft forever
    inet6 fe80::247c:8eff:febc:fcf5/64 scope link
        valid_lft forever preferred_lft forever

    RX:  bytes    packets   errors   dropped   overrun    mcast
         0         0         0         0         0         0
    TX:  bytes    packets   errors   dropped   carrier collisions
    1369707      4267         0         0         0         0
```

7.4.4 Ethernet

Ethernet interfaces (e.g., `eth0`, `eth1`) represent the host's physical or virtual network ports.

They are the most common interface type, serving as the base layer upon which IP addresses, VLANs, and tunnels are configured to carry traffic across both LANs and WANs.

Configuration

Common interface configuration

set interfaces ethernet <interface> address <address | dhcp | dhcpv6>

Configure the interface with one or more IP addresses.

The following options are available:

- **address:** Assign one or more IPv4 or IPv6 addresses to the interface. For example, use `192.0.2.1/24` for IPv4 or `2001:db8::1/64` for IPv6.
- **dhcp:** The interface obtains an IPv4 address from a DHCP server on the same network segment.
- **dhcpv6:** The interface obtains an IPv6 address from a DHCPv6 server on the same network segment.

Note: If the interface obtains an IPv4 address via DHCP, and specific adjustments are needed before/after the IP address is obtained, use the provided hook scripts:

- /config/scripts/dhcp-client/pre-hooks.d/
- /config/scripts/dhcp-client/post-hooks.d/

Example:

```
set interfaces ethernet eth0 address 192.0.2.1/24
set interfaces ethernet eth0 address 2001:db8::1/64
set interfaces ethernet eth0 address dhcp
set interfaces ethernet eth0 address dhcpv6
```

set interfaces ethernet <interface> description <description>

Configure a clear, descriptive alias for the interface.

This alias appears in the `show interfaces` command and SNMP-based monitoring tools.

Example:

```
set interfaces ethernet eth0 description 'This is an interface running on VyOS.'
```

set interfaces ethernet <interface> disable

Disable the interface.

The interface will be set to the administratively down (A/D) state.

Example:

```
set interfaces ethernet eth0 disable
```

set interfaces ethernet <interface> disable-flow-control

Disable Ethernet flow control (IEEE 802.3x pause frames) on the interface.

Ethernet flow control, defined by the IEEE 802.3x standard, temporarily stops data transmission to prevent packet loss during network congestion. For example, when a sender transmits data faster than the receiver can process it.

Disabling Ethernet flow control means the interface will not signal the connected device to pause transmission and will drop packets if overwhelmed.

Example:

```
set interfaces ethernet eth0 disable-flow-control
```

set interfaces ethernet <interface> disable-link-detect

Disable physical link-state change detection on the interface, such as when a cable is unplugged.

By default, the interface detects physical link-state changes.

Example:

```
set interfaces ethernet eth0 disable-link-detect
```

set interfaces ethernet <interface> mac <xx:xx:xx:xx:xx:xx>

Configure a custom MAC address on the interface.

Example:

```
set interfaces ethernet eth0 mac '00:53:01:02:03:04'
```

set interfaces ethernet <interface> mtu <mtu>

Configure the MTU on the interface.

This value defines the largest packet size, in bytes, that the interface transmits without fragmentation.

Example:

```
set interfaces ethernet eth0 mtu 1600
```

set interfaces ethernet <interface> ip adjust-mss <mss | clamp-mss-to-pmtu>

Configure the MSS advertised in outgoing TCP SYN packets on the specified interface.

By clamping the MSS value in TCP SYN packets, you explicitly inform the remote side not to send packets larger than that size. This prevents connection issues that occur when Path MTU Discovery (PMTUD) fails.

The following options are available:

- **mss:** Sets the MSS to a specific value, in bytes. Use this option if you need to enforce a specific MSS, for example, to troubleshoot connectivity issues or accommodate specific network requirements.
- **clamp-mss-to-pmtu:** The router automatically calculates the MSS to be the interface's MTU minus 40 bytes for IPv4 traffic (20 bytes for the IPv4 header and 20 bytes for the TCP header). This option is recommended to automatically set the proper value.

set interfaces ethernet <interface> ip arp-cache-timeout

Configure how long an ARP entry remains valid after learning an IP-to-MAC address mapping on this interface.

The default duration is 30 seconds.

An ARP entry remains valid if it receives positive feedback from higher-level protocols.

Example:

```
set interfaces ethernet eth0 ip arp-cache-timeout 180
```

set interfaces ethernet <interface> ip disable-arp-filter

Configure ARP filtering on this interface.

Default behavior: The kernel responds to ARP requests on this interface only if the traffic would be routed back to the ARP sender through that specific interface.

If configured: The kernel responds to ARP requests on this interface for any IP address configured on the local host, regardless of which specific interface that IP address is assigned to, and regardless of the routing table. This reflects the Linux concept that IP addresses belong to the host, not individual interfaces.

Example:

```
set interfaces ethernet eth0 ip disable-arp-filter
```

set interfaces ethernet <interface> ip disable-forwarding

Configure the interface for host or router behavior.

If configured, the interface switches to host mode, and IPv4 forwarding is disabled on it.

Example:

```
set interfaces ethernet eth0 ip disable-forwarding
```

set interfaces ethernet <interface> ip enable-directed-broadcast

Configure whether to forward IP-directed broadcast packets received on this interface.

Default behavior: IP-directed broadcast packets are dropped.

If configured: IP-directed broadcast packets are forwarded to all hosts on the destination subnet, as defined in [RFC 1812](#) and [RFC 2644](#).

Example:

```
set interfaces ethernet eth0 ip enable-directed-broadcast
```

set interfaces ethernet <interface> ip enable-arp-accept

Configure how to process gratuitous ARPs on this interface.

If configured, an IP-to-MAC address mapping is added to the ARP table based on gratuitous ARP requests or replies.

Note: If the ARP table already contains the IP address from a gratuitous ARP, its entry is updated regardless of whether this setting is configured.

Example:

```
set interfaces ethernet eth0 ip enable-arp-accept
```

set interfaces ethernet <interface> ip enable-arp-announce

Configure the source IP selection for ARP requests on this interface.

Default behavior: The kernel can use any IP address the host owns as the source IP address in ARP requests on this interface.

If configured: The kernel first attempts to select a source IP address configured on the interface that shares a common subnet with the target IP address. If there is no such subnet, the kernel selects the IP address it would normally use (based on the routing table to reach the target destination).

Example:

```
set interfaces ethernet eth0 ip enable-arp-announce
```

set interfaces ethernet <interface> ip enable-arp-ignore

Configure which ARP requests will be ignored on this interface.

Default behavior: The kernel responds to ARP requests for any local IP addresses, regardless of which interface they are assigned to.

If configured: The kernel responds to ARP requests only if the target IP address is assigned to this specific interface.

Example:

```
set interfaces ethernet eth0 ip enable-arp-ignore
```

set interfaces ethernet <interface> ip enable-proxy-arp

Configure proxy ARP on this interface.

If configured, the router (kernel) intercepts ARP requests for non-local IP addresses and replies with the MAC address of the interface that received the request. Subsequent packets destined to these IP addresses are forwarded to their actual destinations on remote subnets.

Example:

```
set interfaces ethernet eth0 ip enable-proxy-arp
```

set interfaces ethernet <interface> ip proxy-arp-pvlan**Configure local proxy ARP on the interface.**

If configured, the router (kernel) responds to ARP requests on this VLAN interface even if the target IP address resides on the same subnet and interface.

This is used to support network isolation requirements (RFC 3069) for private VLANs (PVLANS). In PVLAN configurations, hosts on isolated ports are NOT allowed to communicate directly with each other at Layer 2, but they can communicate with the upstream router.

By replying to inter-host ARP requests with its own MAC address, the router (kernel) directs inter-host traffic through itself instead of directly between hosts.

Note: This command works independently and does not require enabling the standard proxy ARP on the interface.

Local proxy ARP is also known as:

- VLAN aggregation ([RFC 3069](#)).
- Private VLAN (Cisco, Allied Telesyn).
- Source-port filtering or port isolation (Hewlett-Packard).
- MAC-Forced Forwarding (Ericsson).

set interfaces ethernet <interface> ip source-validation <strict | loose | disable>**Configure source IP address validation using RPF on this interface, as specified in [RFC 3704](#).**

The following options are available:

- **strict:** Each incoming packet's source IP address is checked against the Forwarding Information Base (FIB). If the interface is not the best route back to that source, validation fails, and the packet is dropped.
- **loose:** Each incoming packet's source IP address is checked against the FIB. If the source IP address is unreachable through any interface, validation fails.
- **disable:** No source IP address validation is performed. All incoming packets are accepted.

RFC 3704 recommends enabling strict mode to prevent IP spoofing, such as DDoS attacks. For asymmetric or other complex routing scenarios, use loose mode.

set interfaces ethernet <interface> ipv6 address autoconf**Configure the interface to automatically obtain an IPv6 address using SLAAC, as specified in [RFC 4862](#).**

IPv6 hosts can configure themselves automatically when connected to an IPv6 network using the Neighbor Discovery Protocol via ICMPv6 router discovery messages. When first connected to a network, a host sends a link-local router solicitation multicast request for its configuration parameters. The router responds with a router advertisement packet containing Internet Layer configuration parameters.

Note: This method automatically disables IPv6 traffic forwarding on the interface.

Example:

```
set interfaces ethernet eth0 ipv6 address autoconf
```

set interfaces ethernet <interface> ipv6 address eui64 <prefix>

Configure the interface to assign itself an IPv6 address using the EUI-64 method, as specified in [RFC 4291](#).

Example:

```
set interfaces ethernet eth0 ipv6 address eui64 2001:db8:beef::/64
```

set interfaces ethernet <interface> ipv6 address no-default-link-local

Disable the automatic assignment of a link-local IPv6 address to this interface.

Example:

```
set interfaces ethernet eth0 ipv6 address no-default-link-local
```

set interfaces ethernet <interface> ipv6 disable-forwarding

Configure the interface for host or router behavior.

If configured, the interface switches to host mode, and IPv6 forwarding is disabled on it.

Example:

```
set interfaces ethernet eth0 ipv6 disable-forwarding
```

set interfaces ethernet <interface> ipv6 adjust-mss <mss | clamp-mss-to-pmtu>

Configure the MSS advertised in outgoing TCP SYN packets on the specified interface.

By clamping the MSS value in TCP SYN packets, you explicitly inform the remote side not to send packets larger than that size. This prevents connection issues when Path MTU Discovery (PMTUD) fails.

The following options are available:

- **mss:** Set the MSS to a specific value, in bytes. Use this option to enforce a specific MSS, for example, to troubleshoot connectivity issues or accommodate specific network requirements.
- **clamp-mss-to-pmtu:** The router calculates the MSS to be the interface's MTU minus 60 bytes for IPv6 traffic (40 bytes for the IPv6 header and 20 bytes for the TCP header). This option is recommended to automatically set the proper value.

set interfaces ethernet <interface> ipv6 accept-dad <0 | 1 | 2>

Configure IPv6 DAD on the interface.

The following options are available:

- **0:** Disables DAD. No duplicate address detection is performed.
- **1:** Enables DAD (default). Duplicate addresses are detected. The interface's IPv6 operation continues for valid IPv6 addresses.
- **2:** Enables DAD and, if a MAC-based duplicate link-local address is found, disables IPv6 operation on this interface.

Example:

```
set interfaces ethernet eth0 ipv6 accept-dad 2
```

set interfaces ethernet <interface> ipv6 dup-addr-detect-transmits <n>

Configure the number of DAD messages that the router (kernel) sends during IPv6 address assignment on this interface.

The default value is 1.

Example:

```
set interfaces ethernet eth0 ipv6 dup-addr-detect-transmits 5
```

set interfaces ethernet <interface> vrf <vrf>

Assign the interface to a specific VRF instance.

See also:

For information on configuring a VRF, refer to the [VRF](#) section.

Example:

```
set interfaces ethernet eth0 vrf red
```

DHCP(v6)

set interfaces ethernet <interface> dhcp-options client-id <description>

Configure a DHCP client identifier for the interface, as specified in [RFC 2131](#).

The `client-id` is an identifier that the DHCP client sends to the DHCP server to uniquely identify itself for IP address assignment. By default, the client uses its MAC address. The `<description>` is a user-defined string that will be sent to the DHCP server as the DHCP client identifier.

Example:

```
set interfaces ethernet eth0 dhcp-options client-id 'foo-bar'
```

set interfaces ethernet <interface> dhcp-options host-name <hostname>

Configure a specific hostname for the interface.

Instead of the real hostname, the DHCP client will send the specific hostname to the DHCP server when requesting an IP address.

Example:

```
set interfaces ethernet eth0 dhcp-options host-name 'VyOS'
```

set interfaces ethernet <interface> dhcp-options vendor-class-id <vendor-id>

Configure the DHCP client to include a vendor-class identifier in its DHCP requests on this interface.

The vendor-class identifier is a vendor-specific byte string that enables the DHCP server to identify the device and, in some cases, provide configuration options.

Example:

```
set interfaces ethernet eth0 dhcp-options vendor-class-id 'VyOS'
```

set interfaces ethernet <interface> dhcp-options no-default-route

Configure the DHCP client to obtain an IP address, but ignore any default gateway provided by the DHCP server on this interface.

Example:

```
set interfaces ethernet eth0 dhcp-options no-default-route
```

set interfaces ethernet <interface> dhcp-options default-route-distance <distance>

Configure the distance for the default route obtained from the DHCP server on this interface.

Example:

```
set interfaces ethernet eth0 dhcp-options default-route-distance 220
```

set interfaces ethernet <interface> dhcp-options reject <address>

Configure the DHCP client to reject the specific IP address or IP address range from the DHCP server on this interface.

This is useful when a modem assigns a local IP address upon start. To reject multiple addresses, run this command multiple times with different values. You can reject individual addresses (192.168.100.1) or entire subnets (192.168.100.0/24).

Example:

```
set interfaces ethernet eth0 dhcp-options reject 192.168.100.0/24
```

set interfaces ethernet <interface> dhcp-options user-class <string>

Configure the DHCP client to send a specific user-class identifier in its DHCP requests on this interface.

The DHCP server can interpret this identifier and provide specific configuration options based on it (for example, default routes). The user-class value typically groups DHCP clients with similar configuration needs (for example, employees, guests, or printers).

Example:

```
set interfaces ethernet eth0 dhcp-options user-class VyOS
```

set interfaces ethernet <interface> dhcpv6-options duid <duid>

Configure a specific DUID for the DHCPv6 client on this interface.

The DUID is an identifier used by a DHCPv6 client to get an IPv6 address from a DHCPv6 server. It consists of a 2-byte type field, followed by a variable-length identifier field up to 128 bytes. The format of the identifier part depends on the DUID type:

- **DUID-LLT:** The most common type, which includes a hardware type, a timestamp, and a MAC address.
- **DUID-EN:** Is based on a vendor's enterprise number and a unique identifier assigned by the vendor.
- **DUID-LL:** Includes only a MAC address.

The DHCP server matches the DUID against its database and provides configuration data (such as address, lease times, DNS servers, etc.) to the DHCP client.

Example:

```
set interfaces ethernet eth0 duid '0e:00:00:01:00:01:27:71:db:f0:00:50:56:bf:c5:6d'
```

set interfaces ethernet <interface> dhcpv6-options no-release

Configure the DHCP client not to send a release message when it stops running on this interface.

This helps retain the assigned address or prefix.

Example:

```
set interfaces ethernet eth0 dhcpv6-options no-release
```

set interfaces ethernet <interface> dhcpv6-options parameters-only

Enable a stateless DHCPv6 client mode on this interface.

In stateless mode, the DHCPv6 client requests only stateless configuration parameters from the DHCP server (for example, DNS server addresses). It doesn't request a stateful configuration, such as IPv6 addresses or prefixes.

Example:

```
set interfaces ethernet eth0 dhcpv6-options parameters-only
```

set interfaces ethernet <interface> dhcpv6-options rapid-commit

Enable DHCPv6 rapid commit on this interface.

When enabled, the DHCP client and server skip the negotiation steps (Advertise and Request), completing the DHCPv6 configuration process in just two messages (Solicit and final Reply).

Example:

```
set interfaces ethernet eth0 dhcpv6-options rapid-commit
```

set interfaces ethernet <interface> dhcpv6-options temporary

Configure the DHCPv6 client to request a temporary IPv6 address on this interface.

When configured, the DHCP client doesn't form an Identity Association for Non-temporary Addresses (IA_NA) partnership. Consequently, it only obtains a temporary IPv6 address and doesn't obtain a permanent one.

Example:

```
set interfaces ethernet eth0 dhcpv6-options temporary
```

DHCPv6 Prefix Delegation (PD)

VyOS supports DHCPv6 Prefix Delegation (DHCPv6-PD) as described in [RFC 3633](#). DHCPv6-PD is supported by most ISPs that provide native IPv6 for consumers on fixed networks.

set interfaces ethernet <interface> dhcpv6-options pd <id> length <length>

Configure a specific prefix length for DHCPv6-PD requests on this interface.

Some ISPs provide only a /64 prefix by default. Use this command to request a different prefix length for a specific DHCPv6-PD request, ranging from /32 (if allowed by your ISP) down to /64. <id> is a unique identifier for the DHCPv6-PD request.

The default value is 64.

To request a /56 prefix from your ISP, use:

```
set interfaces ethernet eth0 dhcpv6-options pd 0 length 56
```

set interfaces ethernet <interface> dhcpv6-options pd <id> interface <delegatee> address <address>

Configure the IPv6 interface identifier (host portion) for the delegatee interface.

The value must be a decimal integer. It is appended to the delegated prefix and the configured SLA ID to form the final IPv6 address.

By default, the host portion is generated based on the parent interface's MAC address (EUI-64 format).

Example:

If a /64 prefix is delegated to interface eth8 and you configure the host portion as 65535, the resulting IPv6 address will end with ::ffff, as 65535 corresponds to ffff in hexadecimal notation.

```
set interfaces ethernet eth0 dhcpv6-options pd 0 interface eth8 address 65534
```

```
set interfaces ethernet <interface> dhcpv6-options pd <id> interface <delegatee> sla-id <id>
```

Configure the SLA ID for the delegatee interface.

The value must be a decimal integer greater than 0 and fit in the length of SLA IDs. It is converted to hexadecimal and appended to the delegated prefix to form the specific subnet prefix for the delegatee interface.

Example:

If SLA ID is 1 and the delegated prefix is 2001:db8:ffff::/48, the resulting subnet prefix for the delegatee interface will be 2001:db8:ffff:1::/64.

```
set interfaces ethernet eth0 dhcpv6-options pd 0 interface eth8 sla-id 1
```

```
set interfaces ethernet <interface> switchdev
```

Enable switchdev mode for the interface.

In switchdev mode, the interface offloads traffic switching between ports to the hardware, bypassing the host CPU. This increases the interface's traffic-handling capacity and reduces its forwarding delay.

Note: switchdev mode is available only on certain physical network interfaces and requires a switchdev-compatible driver.

Ethernet options

```
set interfaces ethernet <interface> duplex <auto | full | half>
```

Configure duplex mode for the interface.

The following duplex modes are available:

- **auto:** The interface negotiates the duplex mode with the connected device.
- **full:** The interface sends and receives data simultaneously. The connected device must also be set to full-duplex to avoid a duplex mismatch.
- **half:** The interface either sends or receives data, but not both at the same time.

The default duplex mode is auto.

```
set interfaces ethernet <interface> speed <auto | 10 | 100 | 1000 | 2500 | 5000 | 10000 | 25000 | 40000 | 50000 | 100000>
```

Configure the interface's speed, in Mbit/s.

The following options are available:

- **auto**: The interface negotiates the speed with the connected device.
- **10, 100, 1000 ...**: The interface operates at the selected speed. The connected device must be set to the same speed to establish a connection.

The default option is **auto**.

```
set interfaces ethernet <interface> ring-buffer rx <value>
```

Configure the receive (RX) ring buffer size for the interface.

The RX ring buffer size defines the number of incoming packets the interface can queue in hardware before the CPU processes them.

Higher values reduce the risk of drops when the NIC receives network traffic faster than the CPU can process it, though latency may increase. Lower values reduce latency but increase the risk of packet drops during incoming traffic bursts.

To view supported values for a specific interface, use:

```
ethtool -g <interface>
```

```
set interfaces ethernet <interface> ring-buffer tx <value>
```

Configure the transmit (TX) ring buffer size.

The TX ring buffer size defines the number of outgoing packets the interface can queue in hardware before they are transmitted onto the network.

Higher values reduce the risk of drops when the CPU generates traffic faster than the NIC can handle, though latency may increase. Lower values reduce latency but increase the risk of packet drops during outgoing traffic bursts.

To view supported values for a specific interface, use:

```
ethtool -g <interface>
```

Interrupt Coalescing

Interrupt coalescing is a mechanism that reduces CPU interrupt load by bundling multiple packets into a single interrupt event instead of interrupting the CPU for every packet arrival or transmission.

Note: Not all network drivers or virtual interfaces support all coalescing parameters. Use **ethtool --show-coalesce <interface>** to verify which settings are supported by your hardware and driver.

Basic adaptive coalescing

```
set interfaces ethernet <interface> interrupt-coalescing adaptive-rx
```

```
set interfaces ethernet <interface> interrupt-coalescing adaptive-tx
```

Enable adaptive interrupt coalescing. The NIC automatically tunes RX/TX interrupt pacing based on traffic patterns to reduce CPU utilization during high throughput while preserving latency at low packet rates.

Basic interrupt delay

```
set interfaces ethernet <interface> interrupt-coalescing rx-usecs <0-16384>
```

```
set interfaces ethernet <interface> interrupt-coalescing tx-usecs <0-16384>
```

Set the delay in microseconds before generating an RX/TX interrupt after receiving or transmitting a packet. Lower values reduce latency; higher values reduce CPU load.

Interrupt frame thresholds

```
set interfaces ethernet <interface> interrupt-coalescing rx-frames <number>
```

```
set interfaces ethernet <interface> interrupt-coalescing tx-frames <number>
```

Generate an RX/TX interrupt only after the specified number of packets have been received or transmitted.

IRQ-specific coalescing

```
set interfaces ethernet <interface> interrupt-coalescing rx-usecs-irq <number>
```

```
set interfaces ethernet <interface> interrupt-coalescing rx-frames-irq <number>
```

```
set interfaces ethernet <interface> interrupt-coalescing tx-usecs-irq <number>
```

```
set interfaces ethernet <interface> interrupt-coalescing tx-frames-irq <number>
```

Control interrupt coalescing parameters while the driver is already servicing an interrupt (IRQ context). These settings allow finer tuning of interrupt behavior under sustained load.

Adaptive rate thresholds

```
set interfaces ethernet <interface> interrupt-coalescing pkt-rate-low <number>
```

```
set interfaces ethernet <interface> interrupt-coalescing pkt-rate-high <number>
```

Define packet-rate thresholds (packets per second) used by adaptive coalescing to switch between low-rate and high-rate interrupt coalescing profiles.

Low-rate adaptive parameters

```
set interfaces ethernet <interface> interrupt-coalescing rx-usecs-low <number>
```

```
set interfaces ethernet <interface> interrupt-coalescing rx-frame-low <number>
```

```
set interfaces ethernet <interface> interrupt-coalescing tx-usecs-low <number>
```

```
set interfaces ethernet <interface> interrupt-coalescing tx-frame-low <number>
```

Interrupt coalescing parameters applied when the packet rate is below pkt-rate-low. Typically optimized for lower latency.

High-rate adaptive parameters

```
set interfaces ethernet <interface> interrupt-coalescing rx-usecs-high <number>
```

```
set interfaces ethernet <interface> interrupt-coalescing rx-frame-high <number>
```

```
set interfaces ethernet <interface> interrupt-coalescing tx-usecs-high <number>
```

```
set interfaces ethernet <interface> interrupt-coalescing tx-frame-high <number>
```

Interrupt coalescing parameters applied when the packet rate exceeds pkt-rate-high. Typically optimized for maximum throughput and reduced CPU utilization.

Statistics and sampling

```
set interfaces ethernet <interface> interrupt-coalescing stats-block-usecs <number>
```

```
set interfaces ethernet <interface> interrupt-coalescing sample-interval <number>
```

Control how frequently coalescing statistics are updated and how often the NIC samples traffic rates for adaptive coalescing decisions.

Completion queue (CQE) mode

```
set interfaces ethernet <interface> interrupt-coalescing cqe-mode-rx
```

```
set interfaces ethernet <interface> interrupt-coalescing cqe-mode-tx
```

Enable RX/TX Completion Queue Entry (CQE) mode, if supported by the driver. CQE mode can improve performance on high-speed NICs by optimizing completion handling.

Transmit aggregation

```
set interfaces ethernet <interface> interrupt-coalescing tx-aggr-max-bytes <number>
```

```
set interfaces ethernet <interface> interrupt-coalescing tx-aggr-max-frames <number>
```

```
set interfaces ethernet <interface> interrupt-coalescing tx-aggr-time-usecs <number>
```

Control transmit packet aggregation. Packets may be buffered and sent together until one of the configured limits (bytes, frames, or time) is reached, reducing interrupt and DMA overhead.

Offloading

```
set interfaces ethernet <interface> offload <lro | tso | gso | gro | rps | sg>
```

Configure the offloading features for the interface.

The interface offloading features define whether specific packet-processing tasks are performed by hardware (the NIC) or by software (the kernel). You can enable multiple offloading features for a single interface.

- **lro (Large Receive Offload):** Instructs the NIC to merge multiple incoming packets into one larger packet before sending it to the CPU.

Note: LRO (Large Receive Offload) hardware support is often limited to TCP/IPv4 packets. For details on LRO limitations, see <https://lwn.net/Articles/358910/>

Warning: LRO irreversibly alters packet headers during merging. This prevents the merged packet from being correctly split back into the original packets, causing packet drops and forwarding failures on routers and bridges. Use LRO only for end-hosts that do not forward traffic.

- **tso (TCP Segmentation Offload):** Instructs the NIC to split large TCP packets into smaller ones before transmitting them to the network.

Important: SG (Scatter-Gather/Scatter-Gather DMA) must be enabled for TSO (TCP Segmentation Offload) to work. Additionally, GSO (Generic Segmentation Offload) should be enabled as a safety fallback; it ensures that if traffic is rerouted to hardware without TSO support, the kernel can still segment the packets, preventing transmission failures.

- **gso (Generic Segmentation Offload):** Instructs the kernel to split large packets into smaller ones before sending them to the NIC.

GSO serves as a software fallback for hardware that does not support TSO or for protocols (like UDP) that hardware cannot offload.

Important: SG must be enabled for GSO to work.

- **gro (Generic Receive Offload):** Instructs the kernel to merge multiple incoming packets into one larger packet before passing it to upper protocol layers.

Unlike LRO, GRO preserves the necessary packet metadata so the merged packet can be correctly split back into the original packets. This makes GRO safe for use on routers and bridges.

Note: The exception is for IPv4 IDs. If the “Don’t Fragment” (DF) bit is set and IDs are not sequential, GSO alters them to maintain a consistent sequence for GSO compatibility.

- **rps (Receive Packet Steering):** Instructs the kernel to distribute the processing of incoming packets across multiple CPU cores.

The kernel calculates a hash from packet headers (IP addresses and ports) to ensure packets from the same flow are processed by the same CPU core.

Note: RPS (Receive Packet Steering) is a software version of RSS (Receive Side Scaling) and is useful for NICs without hardware multi-queue support.

- **sg (Scatter-Gather/Scatter-Gather DMA):** Instructs the NIC to fetch data fragments from various RAM locations and transmit them as a single packet to the network, eliminating the need for the kernel to copy them into a contiguous block first.

802.1X (EAPOL) authentication

Overview

IEEE 802.1X is a security standard that enforces access control at the data link layer. It blocks all traffic on a port until the connecting device proves its identity. The EAPOL (Extensible Authentication Protocol over LAN) protocol transports credentials between the client (supplicant) and the network switch (authenticator). The switch forwards these credentials to a backend authentication server, typically RADIUS, which verifies them and authorizes the connection.

The VyOS router acts as the supplicant, authenticating with upstream network equipment such as ISP gateways or enterprise switches. Authentication uses X.509 certificates to validate the identities of both the router and the authentication server.

The EAPOL protocol requires the supplicant (the router) to provide an identity string to the authentication server during the initial handshake. If no identity is configured, VyOS uses the Ethernet interface’s MAC address as the identity string.

Configuration

Prerequisites: Before configuring 802.1X (EAPOL) authentication, upload the required CA (Certificate Authority) certificate, client certificate, and private key to the router and import them into the PKI system.

Note: The client certificate and private key must share the **same** PKI name.

See also:

For more information about managing certificates and keys, see the [PKI](#) section.

set interfaces ethernet <interface> eapol ca-certificate <name>

Configure the trusted CA certificate for the interface.

The router uses this certificate to validate the authentication server’s identity.

<name> is the CA certificate name as defined in the PKI system.

Note: If you specify an intermediate CA certificate, ensure the full certificate chain, including the root and all higher-level intermediate CA certificates, is available to the system.

Example:

```
set pki ca eapol-server-intermediate-ca <server intermediate CA contents>
set pki ca eapol-server-root-ca <server root CA contents>
set interfaces ethernet eth0 eapol ca-certificate eapol-server-intermediate-ca
```

set interfaces ethernet <interface> eapol certificate <name>

Configure the client certificate for the interface.

The router uses this certificate to prove its identity to the authentication server.

<name> is the client certificate name as defined in the PKI system.

During authentication, all parent CA certificates of the client certificate, such as intermediate and root CA certificates, are automatically sent as part of the EAP-TLS handshake.

Example:

```
set pki ca eapol-client-intermediate-ca <client intermediate CA contents>
set pki ca eapol-client-root-ca <client root CA contents>
set pki certificate eapol-client certificate <client certificate contents>
set pki certificate eapol-client private key <client private key contents>
set interfaces ethernet eth0 eapol certificate eapol-client
```

EVPN Multihoming

Uplink/core tracking.

set interfaces ethernet <interface> evpn uplink

Configure this interface as an EVPN-MH uplink interface.

If all uplink interfaces on a PE device go down, this PE device loses access to the VXLAN overlay. To prevent traffic blackholing, the PE device forces a protocol shutdown (protodown) of its downstream EVPN-MH interfaces.

The following example configures eth0 as an EVPN-MH uplink interface:

```
set interfaces ethernet eth0 evpn uplink
```

VLAN

Regular VLANs (802.1q)

IEEE 802.1q, often referred to as Dot1q, is the industry standard for implementing VLANs on Ethernet networks. It defines VLAN tagging for Ethernet frames and outlines procedures for bridges and switches. The standard also includes quality-of-service prioritization (IEEE 802.1p) and defines the Generic Attribute Registration Protocol.

VLAN-aware network segments (i.e., IEEE 802.1q conformant) use VLAN tags. When a frame enters such a segment, a tag is added to indicate VLAN membership. Each frame can belong to only one VLAN. If a frame arrives without a tag, it is assumed to be part of the native VLAN.

IEEE 802.1, a working group of the IEEE 802 standards committee, has developed the standard and continues to revise it. One notable revision is 802.1Q-2014, which incorporated IEEE 802.1aq (Shortest Path Bridging) and much of the IEEE 802.1d standard.

In VyOS, 802.1q VLAN interfaces are represented as virtual subinterfaces, referred to as `vif`.

set interfaces ethernet <interface> vif <vlan-id>

Configure a VLAN interface with a unique VLAN ID.

VLAN ID identifies a specific VLAN and ranges from 0 to 4094.

You can configure multiple VLAN interfaces on a single physical interface.

Note: Only 802.1Q-tagged packets are accepted on Ethernet vifs.

set interfaces ethernet <interface> vif <vlan-id> address <address | dhcp | dhcpv6>

Configure the interface with one or more IP addresses.

The following options are available:

- **address:** Assign one or more IPv4 or IPv6 addresses to the interface. For example, use 192.0.2.1/24 for IPv4 or 2001:db8::1/64 for IPv6.
- **dhcp:** The interface obtains an IPv4 address from a DHCP server on the same network segment.
- **dhcpv6:** The interface obtains an IPv6 address from a DHCPv6 server on the same network segment.

Note: If the interface obtains an IPv4 address via DHCP, and specific adjustments are needed before/after the IP address is obtained, use the provided hook scripts:

- /config/scripts/dhcp-client/pre-hooks.d/
 - /config/scripts/dhcp-client/post-hooks.d/
-

Example:

```
set interfaces ethernet eth0 vif 10 address 192.0.2.1/24
set interfaces ethernet eth0 vif 10 address 2001:db8::1/64
set interfaces ethernet eth0 vif 10 address dhcp
set interfaces ethernet eth0 vif 10 address dhcpv6
```

set interfaces ethernet <interface> vif <vlan-id> description <description>

Configure a clear, descriptive alias for the interface.

This alias appears in the `show interfaces` command and SNMP-based monitoring tools.

Example:

```
set interfaces ethernet eth0 vif 10 description 'This is an interface running on
↳ VyOS.'
```

set interfaces ethernet <interface> vif <vlan-id> disable

Disable the interface.

The interface will be set to the administratively down (A/D) state.

Example:

```
set interfaces ethernet eth0 vif 10 disable
```

set interfaces ethernet <interface> vif <vlan-id> disable-link-detect

Disable physical link-state change detection on the interface, such as when a cable is unplugged.

By default, the interface detects physical link-state changes.

Example:

```
set interfaces ethernet eth0 vif 10 disable-link-detect
```

set interfaces ethernet <interface> vif <vlan-id> mac <xx:xx:xx:xx:xx:xx>

Configure a custom MAC address on the interface.

Example:

```
set interfaces ethernet eth0 vif 10 mac '00:53:01:02:03:04'
```

set interfaces ethernet <interface> vif <vlan-id> mtu <mtu>

Configure the MTU on the interface.

This value defines the largest packet size, in bytes, that the interface transmits without fragmentation.

Example:

```
set interfaces ethernet eth0 vif 10 mtu 1600
```

set interfaces ethernet <interface> vif <vlan-id> ip adjust-mss <mss | clamp-mss-to-pmtu>

Configure the MSS advertised in outgoing TCP SYN packets on the specified interface.

By clamping the MSS value in TCP SYN packets, you explicitly inform the remote side not to send packets larger than that size. This prevents connection issues that occur when Path MTU Discovery (PMTUD) fails.

The following options are available:

- **mss:** Sets the MSS to a specific value, in bytes. Use this option if you need to enforce a specific MSS, for example, to troubleshoot connectivity issues or accommodate specific network requirements.
- **clamp-mss-to-pmtu:** The router automatically calculates the MSS to be the interface's MTU minus 40 bytes for IPv4 traffic (20 bytes for the IPv4 header and 20 bytes for the TCP header). This option is recommended to automatically set the proper value.

set interfaces ethernet <interface> vif <vlan-id> ip arp-cache-timeout

Configure how long an ARP entry remains valid after learning an IP-to-MAC address mapping on this interface.

The default duration is 30 seconds.

An ARP entry remains valid if it receives positive feedback from higher-level protocols.

Example:

```
set interfaces ethernet eth0 vif 10 ip arp-cache-timeout 180
```

set interfaces ethernet <interface> vif <vlan-id> ip disable-arp-filter

Configure ARP filtering on this interface.

Default behavior: The kernel responds to ARP requests on this interface only if the traffic would be routed back to the ARP sender through that specific interface.

If configured: The kernel responds to ARP requests on this interface for any IP address configured on the local host, regardless of which specific interface that IP address is assigned to, and regardless of the routing table. This reflects the Linux concept that IP addresses belong to the host, not individual interfaces.

Example:

```
set interfaces ethernet eth0 vif 10 ip disable-arp-filter
```

set interfaces ethernet <interface> vif <vlan-id> ip disable-forwarding

Configure the interface for host or router behavior.

If configured, the interface switches to host mode, and IPv4 forwarding is disabled on it.

Example:

```
set interfaces ethernet eth0 vif 10 ip disable-forwarding
```

set interfaces ethernet <interface> vif <vlan-id> ip enable-directed-broadcast

Configure whether to forward IP-directed broadcast packets received on this interface.

Default behavior: IP-directed broadcast packets are dropped.

If configured: IP-directed broadcast packets are forwarded to all hosts on the destination subnet, as defined in [RFC 1812](#) and [RFC 2644](#).

Example:

```
set interfaces ethernet eth0 vif 10 ip enable-directed-broadcast
```

set interfaces ethernet <interface> vif <vlan-id> ip enable-arp-accept

Configure how to process gratuitous ARPs on this interface.

If configured, an IP-to-MAC address mapping is added to the ARP table based on gratuitous ARP requests or replies.

Note: If the ARP table already contains the IP address from a gratuitous ARP, its entry is updated regardless of whether this setting is configured.

Example:

```
set interfaces ethernet eth0 vif 10 ip enable-arp-accept
```

set interfaces ethernet <interface> vif <vlan-id> ip enable-arp-announce

Configure the source IP selection for ARP requests on this interface.

Default behavior: The kernel can use any IP address the host owns as the source IP address in ARP requests on this interface.

If configured: The kernel first attempts to select a source IP address configured on the interface that shares a common subnet with the target IP address. If there is no such subnet, the kernel selects the IP address it would normally use (based on the routing table to reach the target destination).

Example:

```
set interfaces ethernet eth0 vif 10 ip enable-arp-announce
```

set interfaces ethernet <interface> vif <vlan-id> ip enable-arp-ignore

Configure which ARP requests will be ignored on this interface.

Default behavior: The kernel responds to ARP requests for any local IP addresses, regardless of which interface they are assigned to.

If configured: The kernel responds to ARP requests only if the target IP address is assigned to this specific interface.

Example:

```
set interfaces ethernet eth0 vif 10 ip enable-arp-ignore
```

set interfaces ethernet <interface> vif <vlan-id> ip enable-proxy-arp

Configure proxy ARP on this interface.

If configured, the router (kernel) intercepts ARP requests for non-local IP addresses and replies with the MAC address of the interface that received the request. Subsequent packets destined to these IP addresses are forwarded to their actual destinations on remote subnets.

Example:

```
set interfaces ethernet eth0 vif 10 ip enable-proxy-arp
```

set interfaces ethernet <interface> vif <vlan-id> ip proxy-arp-pvlan

Configure local proxy ARP on the interface.

If configured, the router (kernel) responds to ARP requests on this VLAN interface even if the target IP address resides on the same subnet and interface.

This is used to support network isolation requirements (RFC 3069) for private VLANs (PVLANS). In PVLAN configurations, hosts on isolated ports are NOT allowed to communicate directly with each other at Layer 2, but they can communicate with the upstream router.

By replying to inter-host ARP requests with its own MAC address, the router (kernel) directs inter-host traffic through itself instead of directly between hosts.

Note: This command works independently and does not require enabling the standard proxy ARP on the interface.

Local proxy ARP is also known as:

- VLAN aggregation ([RFC 3069](#)).
- Private VLAN (Cisco, Allied Telesyn).
- Source-port filtering or port isolation (Hewlett-Packard).
- MAC-Forced Forwarding (Ericsson).

set interfaces ethernet <interface> vif <vlan-id> ip source-validation <strict | loose | disable>

Configure source IP address validation using RPF on this interface, as specified in [RFC 3704](#).

The following options are available:

- **strict:** Each incoming packet's source IP address is checked against the Forwarding Information Base (FIB). If the interface is not the best route back to that source, validation fails, and the packet is dropped.
- **loose:** Each incoming packet's source IP address is checked against the FIB. If the source IP address is unreachable through any interface, validation fails.
- **disable:** No source IP address validation is performed. All incoming packets are accepted.

RFC 3704 recommends enabling strict mode to prevent IP spoofing, such as DDoS attacks. For asymmetric or other complex routing scenarios, use loose mode.

set interfaces ethernet <interface> vif <vlan-id> ipv6 address autoconf

Configure the interface to automatically obtain an IPv6 address using SLAAC, as specified in [RFC 4862](#).

IPv6 hosts can configure themselves automatically when connected to an IPv6 network using the Neighbor Discovery Protocol via ICMPv6 router discovery messages. When first connected to a network, a host sends a link-local router solicitation multicast request for its configuration parameters. The router responds with a router advertisement packet containing Internet Layer configuration parameters.

Note: This method automatically disables IPv6 traffic forwarding on the interface.

Example:

```
set interfaces ethernet eth0 vif 10 ipv6 address autoconf
```

set interfaces ethernet <interface> vif <vlan-id> ipv6 address eui64 <prefix>

Configure the interface to assign itself an IPv6 address using the EUI-64 method, as specified in [RFC 4291](#).

Example:

```
set interfaces ethernet eth0 vif 10 ipv6 address eui64 2001:db8:beef::/64
```

set interfaces ethernet <interface> vif <vlan-id> ipv6 address no-default-link-local

Disable the automatic assignment of a link-local IPv6 address to this interface.

Example:

```
set interfaces ethernet eth0 vif 10 ipv6 address no-default-link-local
```

set interfaces ethernet <interface> vif <vlan-id> ipv6 disable-forwarding

Configure the interface for host or router behavior.

If configured, the interface switches to host mode, and IPv6 forwarding is disabled on it.

Example:

```
set interfaces ethernet eth0 vif 10 ipv6 disable-forwarding
```

set interfaces ethernet <interface> vif <vlan-id> ipv6 adjust-mss <mss | clamp-mss-to-pmtu>

Configure the MSS advertised in outgoing TCP SYN packets on the specified interface.

By clamping the MSS value in TCP SYN packets, you explicitly inform the remote side not to send packets larger than that size. This prevents connection issues when Path MTU Discovery (PMTUD) fails.

The following options are available:

- **mss**: Set the MSS to a specific value, in bytes. Use this option to enforce a specific MSS, for example, to troubleshoot connectivity issues or accommodate specific network requirements.
- **clamp-mss-to-pmtu**: The router calculates the MSS to be the interface's MTU minus 60 bytes for IPv6 traffic (40 bytes for the IPv6 header and 20 bytes for the TCP header). This option is recommended to automatically set the proper value.

```
set interfaces ethernet <interface> vif <vlan-id> ipv6 accept-dad <0 | 1 | 2>
```

Configure IPv6 DAD on the interface.

The following options are available:

- **0**: Disables DAD. No duplicate address detection is performed.
- **1**: Enables DAD (default). Duplicate addresses are detected. The interface's IPv6 operation continues for valid IPv6 addresses.
- **2**: Enables DAD and, if a MAC-based duplicate link-local address is found, disables IPv6 operation on this interface.

Example:

```
set interfaces ethernet eth0 vif 10 ipv6 accept-dad 2
```

```
set interfaces ethernet <interface> vif <vlan-id> ipv6 dup-addr-detect-transmits <n>
```

Configure the number of DAD messages that the router (kernel) sends during IPv6 address assignment on this interface.

The default value is 1.

Example:

```
set interfaces ethernet eth0 vif 10 ipv6 dup-addr-detect-transmits 5
```

```
set interfaces ethernet <interface> vif <vlan-id> vrf <vrf>
```

Assign the interface to a specific VRF instance.

See also:

For information on configuring a VRF, refer to the [VRF](#) section.

Example:

```
set interfaces ethernet eth0 vif 10 vrf red
```

DHCP(v6)

```
set interfaces ethernet <interface> vif <vlan-id> dhcp-options client-id <description>
```

Configure a DHCP client identifier for the interface, as specified in [RFC 2131](#).

The `client-id` is an identifier that the DHCP client sends to the DHCP server to uniquely identify itself for IP address assignment. By default, the client uses its MAC address. The `<description>` is a user-defined string that will be sent to the DHCP server as the DHCP client identifier.

Example:

```
set interfaces ethernet eth0 vif 10 dhcp-options client-id 'foo-bar'
```

```
set interfaces ethernet <interface> vif <vlan-id> dhcp-options host-name <hostname>
```

Configure a specific hostname for the interface.

Instead of the real hostname, the DHCP client will send the specific hostname to the DHCP server when requesting an IP address.

Example:

```
set interfaces ethernet eth0 vif 10 dhcp-options host-name 'VyOS'
```

```
set interfaces ethernet <interface> vif <vlan-id> dhcp-options vendor-class-id <vendor-id>
```

Configure the DHCP client to include a vendor-class identifier in its DHCP requests on this interface.

The vendor-class identifier is a vendor-specific byte string that enables the DHCP server to identify the device and, in some cases, provide configuration options.

Example:

```
set interfaces ethernet eth0 vif 10 dhcp-options vendor-class-id 'VyOS'
```

```
set interfaces ethernet <interface> vif <vlan-id> dhcp-options no-default-route
```

Configure the DHCP client to obtain an IP address, but ignore any default gateway provided by the DHCP server on this interface.

Example:

```
set interfaces ethernet eth0 vif 10 dhcp-options no-default-route
```

```
set interfaces ethernet <interface> vif <vlan-id> dhcp-options default-route-distance <distance>
```

Configure the distance for the default route obtained from the DHCP server on this interface.

Example:

```
set interfaces ethernet eth0 vif 10 dhcp-options default-route-distance 220
```

```
set interfaces ethernet <interface> vif <vlan-id> dhcp-options reject <address>
```

Configure the DHCP client to reject the specific IP address or IP address range from the DHCP server on this interface.

This is useful when a modem assigns a local IP address upon start. To reject multiple addresses, run this command multiple times with different values. You can reject individual addresses (192.168.100.1) or entire subnets (192.168.100.0/24).

Example:

```
set interfaces ethernet eth0 vif 10 dhcp-options reject 192.168.100.0/24
```

```
set interfaces ethernet <interface> vif <vlan-id> dhcp-options user-class <string>
```

Configure the DHCP client to send a specific user-class identifier in its DHCP requests on this interface.

The DHCP server can interpret this identifier and provide specific configuration options based on it (for example, default routes). The user-class value typically groups DHCP clients with similar configuration needs (for example, employees, guests, or printers).

Example:

```
set interfaces ethernet eth0 vif 10 dhcp-options user-class VyOS
```

```
set interfaces ethernet <interface> vif <vlan-id> dhcpv6-options duid <duid>
```

Configure a specific DUID for the DHCPv6 client on this interface.

The DUID is an identifier used by a DHCPv6 client to get an IPv6 address from a DHCPv6 server. It consists of a 2-byte type field, followed by a variable-length identifier field up to 128 bytes. The format of the identifier part depends on the DUID type:

- **DUID-LLT:** The most common type, which includes a hardware type, a timestamp, and a MAC address.
- **DUID-EN:** Is based on a vendor's enterprise number and a unique identifier assigned by the vendor.
- **DUID-LL:** Includes only a MAC address.

The DHCP server matches the DUID against its database and provides configuration data (such as address, lease times, DNS servers, etc.) to the DHCP client.

Example:

```
set interfaces ethernet eth0 vif 10 duid
→ '0e:00:00:01:00:01:27:71:db:f0:00:50:56:bf:c5:6d'
```

```
set interfaces ethernet <interface> vif <vlan-id> dhcpv6-options no-release
```

Configure the DHCP client not to send a release message when it stops running on this interface.

This helps retain the assigned address or prefix.

Example:

```
set interfaces ethernet eth0 vif 10 dhcpv6-options no-release
```

```
set interfaces ethernet <interface> vif <vlan-id> dhcpv6-options parameters-only
```

Enable a stateless DHCPv6 client mode on this interface.

In stateless mode, the DHCPv6 client requests only stateless configuration parameters from the DHCP server (for example, DNS server addresses). It doesn't request a stateful configuration, such as IPv6 addresses or prefixes.

Example:

```
set interfaces ethernet eth0 vif 10 dhcpv6-options parameters-only
```

```
set interfaces ethernet <interface> vif <vlan-id> dhcpv6-options rapid-commit
```

Enable DHCPv6 rapid commit on this interface.

When enabled, the DHCP client and server skip the negotiation steps (Advertise and Request), completing the DHCPv6 configuration process in just two messages (Solicit and final Reply).

Example:

```
set interfaces ethernet eth0 vif 10 dhcpv6-options rapid-commit
```

```
set interfaces ethernet <interface> vif <vlan-id> dhcpv6-options temporary
```

Configure the DHCPv6 client to request a temporary IPv6 address on this interface.

When configured, the DHCP client doesn't form an Identity Association for Non-temporary Addresses (IA_NA) partnership. Consequently, it only obtains a temporary IPv6 address and doesn't obtain a permanent one.

Example:

```
set interfaces ethernet eth0 vif 10 dhcpv6-options temporary
```

DHCPv6 Prefix Delegation (PD)

VyOS supports DHCPv6 Prefix Delegation (DHCPv6-PD) as described in [RFC 3633](#). DHCPv6-PD is supported by most ISPs that provide native IPv6 for consumers on fixed networks.

set interfaces ethernet <interface> vif <vlan-id> dhcpv6-options pd <id> length <length>

Configure a specific prefix length for DHCPv6-PD requests on this interface.

Some ISPs provide only a /64 prefix by default. Use this command to request a different prefix length for a specific DHCPv6-PD request, ranging from /32 (if allowed by your ISP) down to /64. <id> is a unique identifier for the DHCPv6-PD request.

The default value is 64.

To request a /56 prefix from your ISP, use:

```
set interfaces ethernet eth0 vif 10 dhcpv6-options pd 0 length 56
```

set interfaces ethernet <interface> vif <vlan-id> dhcpv6-options pd <id> interface <delegatee> address <address>

Configure the IPv6 interface identifier (host portion) for the delegatee interface.

The value must be a decimal integer. It is appended to the delegated prefix and the configured SLA ID to form the final IPv6 address.

By default, the host portion is generated based on the parent interface's MAC address (EUI-64 format).

Example:

If a /64 prefix is delegated to interface eth8 and you configure the host portion as 65535, the resulting IPv6 address will end with ::ffff, as 65535 corresponds to ffff in hexadecimal notation.

```
set interfaces ethernet eth0 vif 10 dhcpv6-options pd 0 interface eth8 address 65534
```

set interfaces ethernet <interface> vif <vlan-id> dhcpv6-options pd <id> interface <delegatee> sla-id <id>

Configure the SLA ID for the delegatee interface.

The value must be a decimal integer greater than 0 and fit in the length of SLA IDs. It is converted to hexadecimal and appended to the delegated prefix to form the specific subnet prefix for the delegatee interface.

Example:

If SLA ID is 1 and the delegated prefix is 2001:db8:ffff::/48, the resulting subnet prefix for the delegatee interface will be 2001:db8:ffff:1::/64.

```
set interfaces ethernet eth0 vif 10 dhcpv6-options pd 0 interface eth8 sla-id 1
```

802.1ad (QinQ)

Overview

IEEE *802.1ad*, commonly known as QinQ, is an Ethernet standard first published as an amendment to *802.1q* in 2005, then officially merged into the base standard in 2011.

Unlike the original *802.1q*, which allows a single VLAN header per Ethernet frame, QinQ allows two VLAN headers per Ethernet frame, for the inner and the outer VLAN tags. Most often the inner VLAN tag comes from a customer while the outer tag is used by the service provider to differentiate between traffic of different customers.

Frame structure and ethertypes

The IEEE *802.1ad* (QinQ) frame includes two VLAN tags:

- **The outer service tag (S-TAG):** The S-TAG is typically added by the provider.

It uses the Ethertype 0x88a8 by default.

- **The inner customer tag (C-TAG):** The C-TAG is generated by the customer's equipment and remains unchanged during transit. It uses the Ethertype 0x8100.

Implementation in VyOS

In VyOS, these tag types are associated with the following CLI options:

- **vif-s:** Corresponds to the S-TAG (Ethertype 0x88a8).
- **vif-c:** Corresponds to the C-TAG (Ethertype 0x8100).

set interfaces ethernet <interface> vif-s <vlan-id> vif-c <vlan-id> address <address | dhcp | dhcpv6>

Configure the interface with one or more IP addresses.

The following options are available:

- **address:** Assign one or more IPv4 or IPv6 addresses to the interface. For example, use 192.0.2.1/24 for IPv4 or 2001:db8::1/64 for IPv6.
- **dhcp:** The interface obtains an IPv4 address from a DHCP server on the same network segment.
- **dhcpv6:** The interface obtains an IPv6 address from a DHCPv6 server on the same network segment.

Note: If the interface obtains an IPv4 address via DHCP, and specific adjustments are needed before/after the IP address is obtained, use the provided hook scripts:

- /config/scripts/dhcp-client/pre-hooks.d/
- /config/scripts/dhcp-client/post-hooks.d/

Example:

```
set interfaces ethernet eth0 vif-s 1000 vif-c 20 address 192.0.2.1/24
set interfaces ethernet eth0 vif-s 1000 vif-c 20 address 2001:db8::1/64
set interfaces ethernet eth0 vif-s 1000 vif-c 20 address dhcp
set interfaces ethernet eth0 vif-s 1000 vif-c 20 address dhcpv6
```

set interfaces ethernet <interface> vif-s <vlan-id> vif-c <vlan-id> description <description>

Configure a clear, descriptive alias for the interface.

This alias appears in the `show interfaces` command and SNMP-based monitoring tools.

Example:

```
set interfaces ethernet eth0 vif-s 1000 vif-c 20 description 'This is an interface.  
↪running on VyOS.'
```

set interfaces ethernet <interface> vif-s <vlan-id> vif-c <vlan-id> disable

Disable the interface.

The interface will be set to the administratively down (A/D) state.

Example:

```
set interfaces ethernet eth0 vif-s 1000 vif-c 20 disable
```

set interfaces ethernet <interface> vif-s <vlan-id> vif-c <vlan-id> disable-link-detect

Disable physical link-state change detection on the interface, such as when a cable is unplugged.

By default, the interface detects physical link-state changes.

Example:

```
set interfaces ethernet eth0 vif-s 1000 vif-c 20 disable-link-detect
```

set interfaces ethernet <interface> vif-s <vlan-id> vif-c <vlan-id> mac

<xx:xx:xx:xx:xx:xx>

Configure a custom MAC address on the interface.

Example:

```
set interfaces ethernet eth0 vif-s 1000 vif-c 20 mac '00:53:01:02:03:04'
```

set interfaces ethernet <interface> vif-s <vlan-id> vif-c <vlan-id> mtu <mtu>

Configure the MTU on the interface.

This value defines the largest packet size, in bytes, that the interface transmits without fragmentation.

Example:

```
set interfaces ethernet eth0 vif-s 1000 vif-c 20 mtu 1600
```

set interfaces ethernet <interface> vif-s <vlan-id> vif-c <vlan-id> ip adjust-mss <mss | clamp-mss-to-pmtu>

Configure the MSS advertised in outgoing TCP SYN packets on the specified interface.

By clamping the MSS value in TCP SYN packets, you explicitly inform the remote side not to send packets larger than that size. This prevents connection issues that occur when Path MTU Discovery (PMTUD) fails.

The following options are available:

- **mss:** Sets the MSS to a specific value, in bytes. Use this option if you need to enforce a specific MSS, for example, to troubleshoot connectivity issues or accommodate specific network requirements.
- **clamp-mss-to-pmtu:** The router automatically calculates the MSS to be the interface's MTU minus 40 bytes for IPv4 traffic (20 bytes for the IPv4 header and 20 bytes for the TCP header). This option is recommended to automatically set the proper value.

set interfaces ethernet <interface> vif-s <vlan-id> vif-c <vlan-id> ip arp-cache-timeout

Configure how long an ARP entry remains valid after learning an IP-to-MAC address mapping on this interface.

The default duration is 30 seconds.

An ARP entry remains valid if it receives positive feedback from higher-level protocols.

Example:

```
set interfaces ethernet eth0 vif-s 1000 vif-c 20 ip arp-cache-timeout 180
```

set interfaces ethernet <interface> vif-s <vlan-id> vif-c <vlan-id> ip disable-arp-filter

Configure ARP filtering on this interface.

Default behavior: The kernel responds to ARP requests on this interface only if the traffic would be routed back to the ARP sender through that specific interface.

If configured: The kernel responds to ARP requests on this interface for any IP address configured on the local host, regardless of which specific interface that IP address is assigned to, and regardless of the routing table. This reflects the Linux concept that IP addresses belong to the host, not individual interfaces.

Example:

```
set interfaces ethernet eth0 vif-s 1000 vif-c 20 ip disable-arp-filter
```

set interfaces ethernet <interface> vif-s <vlan-id> vif-c <vlan-id> ip disable-forwarding

Configure the interface for host or router behavior.

If configured, the interface switches to host mode, and IPv4 forwarding is disabled on it.

Example:

```
set interfaces ethernet eth0 vif-s 1000 vif-c 20 ip disable-forwarding
```

set interfaces ethernet <interface> vif-s <vlan-id> vif-c <vlan-id> ip enable-directed-broadcast

Configure whether to forward IP-directed broadcast packets received on this interface.

Default behavior: IP-directed broadcast packets are dropped.

If configured: IP-directed broadcast packets are forwarded to all hosts on the destination subnet, as defined in [RFC 1812](#) and [RFC 2644](#).

Example:

```
set interfaces ethernet eth0 vif-s 1000 vif-c 20 ip enable-directed-broadcast
```

set interfaces ethernet <interface> vif-s <vlan-id> vif-c <vlan-id> ip enable-arp-accept

Configure how to process gratuitous ARPs on this interface.

If configured, an IP-to-MAC address mapping is added to the ARP table based on gratuitous ARP requests or replies.

Note: If the ARP table already contains the IP address from a gratuitous ARP, its entry is updated regardless of whether this setting is configured.

Example:

```
set interfaces ethernet eth0 vif-s 1000 vif-c 20 ip enable-arp-accept
```

set interfaces ethernet <interface> vif-s <vlan-id> vif-c <vlan-id> ip enable-arp-announce

Configure the source IP selection for ARP requests on this interface.

Default behavior: The kernel can use any IP address the host owns as the source IP address in ARP requests on this interface.

If configured: The kernel first attempts to select a source IP address configured on the interface that shares a common subnet with the target IP address. If there is no such subnet, the kernel selects the IP address it would normally use (based on the routing table to reach the target destination).

Example:

```
set interfaces ethernet eth0 vif-s 1000 vif-c 20 ip enable-arp-announce
```

set interfaces ethernet <interface> vif-s <vlan-id> vif-c <vlan-id> ip enable-arp-ignore

Configure which ARP requests will be ignored on this interface.

Default behavior: The kernel responds to ARP requests for any local IP addresses, regardless of which interface they are assigned to.

If configured: The kernel responds to ARP requests only if the target IP address is assigned to this specific interface.

Example:

```
set interfaces ethernet eth0 vif-s 1000 vif-c 20 ip enable-arp-ignore
```

set interfaces ethernet <interface> vif-s <vlan-id> vif-c <vlan-id> ip enable-proxy-arp

Configure proxy ARP on this interface.

If configured, the router (kernel) intercepts ARP requests for non-local IP addresses and replies with the MAC address of the interface that received the request. Subsequent packets destined to these IP addresses are forwarded to their actual destinations on remote subnets.

Example:

```
set interfaces ethernet eth0 vif-s 1000 vif-c 20 ip enable-proxy-arp
```

set interfaces ethernet <interface> vif-s <vlan-id> vif-c <vlan-id> ip proxy-arp-pvlan

Configure local proxy ARP on the interface.

If configured, the router (kernel) responds to ARP requests on this VLAN interface even if the target IP address resides on the same subnet and interface.

This is used to support network isolation requirements (RFC 3069) for private VLANs (PVLANS). In PVLAN configurations, hosts on isolated ports are NOT allowed to communicate directly with each other at Layer 2, but they can communicate with the upstream router.

By replying to inter-host ARP requests with its own MAC address, the router (kernel) directs inter-host traffic through itself instead of directly between hosts.

Note: This command works independently and does not require enabling the standard proxy ARP on the interface.

Local proxy ARP is also known as:

- VLAN aggregation ([RFC 3069](#)).
- Private VLAN (Cisco, Allied Telesyn).
- Source-port filtering or port isolation (Hewlett-Packard).
- MAC-Forced Forwarding (Ericsson).

set interfaces ethernet <interface> vif-s <vlan-id> vif-c <vlan-id> ip source-validation <strict | loose | disable>

Configure source IP address validation using RPF on this interface, as specified in [RFC 3704](#).

The following options are available:

- **strict:** Each incoming packet's source IP address is checked against the Forwarding Information Base (FIB). If the interface is not the best route back to that source, validation fails, and the packet is dropped.
- **loose:** Each incoming packet's source IP address is checked against the FIB. If the source IP address is unreachable through any interface, validation fails.
- **disable:** No source IP address validation is performed. All incoming packets are accepted.

RFC 3704 recommends enabling strict mode to prevent IP spoofing, such as DDoS attacks. For asymmetric or other complex routing scenarios, use loose mode.

set interfaces ethernet <interface> vif-s <vlan-id> vif-c <vlan-id> ipv6 address autoconf

Configure the interface to automatically obtain an IPv6 address using SLAAC, as specified in [RFC 4862](#).

IPv6 hosts can configure themselves automatically when connected to an IPv6 network using the Neighbor Discovery Protocol via ICMPv6 router discovery messages. When first connected to a network, a host sends a link-local router solicitation multicast request for its configuration parameters. The router responds with a router advertisement packet containing Internet Layer configuration parameters.

Note: This method automatically disables IPv6 traffic forwarding on the interface.

Example:

```
set interfaces ethernet eth0 vif-s 1000 vif-c 20 ipv6 address autoconf
```

set interfaces ethernet <interface> vif-s <vlan-id> vif-c <vlan-id> ipv6 address eui64 <prefix>

Configure the interface to assign itself an IPv6 address using the EUI-64 method, as specified in [RFC 4291](#).

Example:

```
set interfaces ethernet eth0 vif-s 1000 vif-c 20 ipv6 address eui64 2001:db8:beef::/
→ 64
```

set interfaces ethernet <interface> vif-s <vlan-id> vif-c <vlan-id> ipv6 address no-default-link-local

Disable the automatic assignment of a link-local IPv6 address to this interface.

Example:

```
set interfaces ethernet eth0 vif-s 1000 vif-c 20 ipv6 address no-default-link-local
```

set interfaces ethernet <interface> vif-s <vlan-id> vif-c <vlan-id> ipv6 disable-forwarding

Configure the interface for host or router behavior.

If configured, the interface switches to host mode, and IPv6 forwarding is disabled on it.

Example:

```
set interfaces ethernet eth0 vif-s 1000 vif-c 20 ipv6 disable-forwarding
```

set interfaces ethernet <interface> vif-s <vlan-id> vif-c <vlan-id> ipv6 adjust-mss <mss> | clamp-mss-to-pmtu>

Configure the MSS advertised in outgoing TCP SYN packets on the specified interface.

By clamping the MSS value in TCP SYN packets, you explicitly inform the remote side not to send packets larger than that size. This prevents connection issues when Path MTU Discovery (PMTUD) fails.

The following options are available:

- **mss:** Set the MSS to a specific value, in bytes. Use this option to enforce a specific MSS, for example, to troubleshoot connectivity issues or accommodate specific network requirements.
- **clamp-mss-to-pmtu:** The router calculates the MSS to be the interface's MTU minus 60 bytes for IPv6 traffic (40 bytes for the IPv6 header and 20 bytes for the TCP header). This option is recommended to automatically set the proper value.

set interfaces ethernet <interface> vif-s <vlan-id> vif-c <vlan-id> ipv6 accept-dad <0 | 1 | 2>

Configure IPv6 DAD on the interface.

The following options are available:

- **0:** Disables DAD. No duplicate address detection is performed.
- **1:** Enables DAD (default). Duplicate addresses are detected. The interface's IPv6 operation continues for valid IPv6 addresses.
- **2:** Enables DAD and, if a MAC-based duplicate link-local address is found, disables IPv6 operation on this interface.

Example:

```
set interfaces ethernet eth0 vif-s 1000 vif-c 20 ipv6 accept-dad 2
```

set interfaces ethernet <interface> vif-s <vlan-id> vif-c <vlan-id> ipv6 dup-addr-detect-transmits <n>

Configure the number of DAD messages that the router (kernel) sends during IPv6 address assignment on this interface.

The default value is 1.

Example:

```
set interfaces ethernet eth0 vif-s 1000 vif-c 20 ipv6 dup-addr-detect-transmits 5
```

```
set interfaces ethernet <interface> vif-s <vlan-id> vif-c <vlan-id> vrf <vrf>
```

Assign the interface to a specific VRF instance.

See also:

For information on configuring a VRF, refer to the [VRF](#) section.

Example:

```
set interfaces ethernet eth0 vif-s 1000 vif-c 20 vrf red
```

DHCP(v6)

```
set interfaces ethernet <interface> vif-s <vlan-id> vif-c <vlan-id> dhcp-options
client-id <description>
```

Configure a DHCP client identifier for the interface, as specified in [RFC 2131](#).

The `client-id` is an identifier that the DHCP client sends to the DHCP server to uniquely identify itself for IP address assignment. By default, the client uses its MAC address. The `<description>` is a user-defined string that will be sent to the DHCP server as the DHCP client identifier.

Example:

```
set interfaces ethernet eth0 vif-s 1000 vif-c 20 dhcp-options client-id 'foo-bar'
```

```
set interfaces ethernet <interface> vif-s <vlan-id> vif-c <vlan-id> dhcp-options
host-name <hostname>
```

Configure a specific hostname for the interface.

Instead of the real hostname, the DHCP client will send the specific hostname to the DHCP server when requesting an IP address.

Example:

```
set interfaces ethernet eth0 vif-s 1000 vif-c 20 dhcp-options host-name 'VyOS'
```

```
set interfaces ethernet <interface> vif-s <vlan-id> vif-c <vlan-id> dhcp-options
vendor-class-id <vendor-id>
```

Configure the DHCP client to include a vendor-class identifier in its DHCP requests on this interface.

The vendor-class identifier is a vendor-specific byte string that enables the DHCP server to identify the device and, in some cases, provide configuration options.

Example:

```
set interfaces ethernet eth0 vif-s 1000 vif-c 20 dhcp-options vendor-class-id 'VyOS'
```

```
set interfaces ethernet <interface> vif-s <vlan-id> vif-c <vlan-id> dhcp-options
no-default-route
```

Configure the DHCP client to obtain an IP address, but ignore any default gateway provided by the DHCP server on this interface.

Example:

```
set interfaces ethernet eth0 vif-s 1000 vif-c 20 dhcp-options no-default-route
```

set interfaces ethernet <interface> vif-s <vlan-id> vif-c <vlan-id> dhcp-options default-route-distance <distance>

Configure the distance for the default route obtained from the DHCP server on this interface.

Example:

```
set interfaces ethernet eth0 vif-s 1000 vif-c 20 dhcp-options default-route-
→distance 220
```

set interfaces ethernet <interface> vif-s <vlan-id> vif-c <vlan-id> dhcp-options reject <address>

Configure the DHCP client to reject the specific IP address or IP address range from the DHCP server on this interface.

This is useful when a modem assigns a local IP address upon start. To reject multiple addresses, run this command multiple times with different values. You can reject individual addresses (192.168.100.1) or entire subnets (192.168.100.0/24).

Example:

```
set interfaces ethernet eth0 vif-s 1000 vif-c 20 dhcp-options reject 192.168.100.0/
→24
```

set interfaces ethernet <interface> vif-s <vlan-id> vif-c <vlan-id> dhcp-options user-class <string>

Configure the DHCP client to send a specific user-class identifier in its DHCP requests on this interface.

The DHCP server can interpret this identifier and provide specific configuration options based on it (for example, default routes). The user-class value typically groups DHCP clients with similar configuration needs (for example, employees, guests, or printers).

Example:

```
set interfaces ethernet eth0 vif-s 1000 vif-c 20 dhcp-options user-class VyOS
```

set interfaces ethernet <interface> vif-s <vlan-id> vif-c <vlan-id> dhcpv6-options duid <duid>

Configure a specific DUID for the DHCPv6 client on this interface.

The DUID is an identifier used by a DHCPv6 client to get an IPv6 address from a DHCPv6 server. It consists of a 2-byte type field, followed by a variable-length identifier field up to 128 bytes. The format of the identifier part depends on the DUID type:

- **DUID-LLT:** The most common type, which includes a hardware type, a timestamp, and a MAC address.
- **DUID-EN:** Is based on a vendor's enterprise number and a unique identifier assigned by the vendor.
- **DUID-LL:** Includes only a MAC address.

The DHCP server matches the DUID against its database and provides configuration data (such as address, lease times, DNS servers, etc.) to the DHCP client.

Example:

```
set interfaces ethernet eth0 vif-s 1000 vif-c 20 duid
→ '0e:00:00:01:00:01:27:71:db:f0:00:50:56:bf:c5:6d'
```

set interfaces ethernet <interface> vif-s <vlan-id> vif-c <vlan-id> dhcpv6-options no-release

Configure the DHCP client not to send a release message when it stops running on this interface.

This helps retain the assigned address or prefix.

Example:

```
set interfaces ethernet eth0 vif-s 1000 vif-c 20 dhcpv6-options no-release
```

set interfaces ethernet <interface> vif-s <vlan-id> vif-c <vlan-id> dhcpv6-options parameters-only

Enable a stateless DHCPv6 client mode on this interface.

In stateless mode, the DHCPv6 client requests only stateless configuration parameters from the DHCP server (for example, DNS server addresses). It doesn't request a stateful configuration, such as IPv6 addresses or prefixes.

Example:

```
set interfaces ethernet eth0 vif-s 1000 vif-c 20 dhcpv6-options parameters-only
```

set interfaces ethernet <interface> vif-s <vlan-id> vif-c <vlan-id> dhcpv6-options rapid-commit

Enable DHCPv6 rapid commit on this interface.

When enabled, the DHCP client and server skip the negotiation steps (Advertise and Request), completing the DHCPv6 configuration process in just two messages (Solicit and final Reply).

Example:

```
set interfaces ethernet eth0 vif-s 1000 vif-c 20 dhcpv6-options rapid-commit
```

set interfaces ethernet <interface> vif-s <vlan-id> vif-c <vlan-id> dhcpv6-options temporary

Configure the DHCPv6 client to request a temporary IPv6 address on this interface.

When configured, the DHCP client doesn't form an Identity Association for Non-temporary Addresses (IA_NA) partnership. Consequently, it only obtains a temporary IPv6 address and doesn't obtain a permanent one.

Example:

```
set interfaces ethernet eth0 vif-s 1000 vif-c 20 dhcpv6-options temporary
```

DHCPv6 Prefix Delegation (PD)

VyOS supports DHCPv6 Prefix Delegation (DHCPv6-PD) as described in [RFC 3633](#). DHCPv6-PD is supported by most ISPs that provide native IPv6 for consumers on fixed networks.

set interfaces ethernet <interface> vif-s <vlan-id> vif-c <vlan-id> dhcpv6-options pd <id> length <length>

Configure a specific prefix length for DHCPv6-PD requests on this interface.

Some ISPs provide only a /64 prefix by default. Use this command to request a different prefix length for a specific DHCPv6-PD request, ranging from /32 (if allowed by your ISP) down to /64. <id> is a unique identifier for the DHCPv6-PD request.

The default value is 64.

To request a /56 prefix from your ISP, use:

```
set interfaces ethernet eth0 vif-s 1000 vif-c 20 dhcpv6-options pd 0 length 56
```

```
set interfaces ethernet <interface> vif-s <vlan-id> vif-c <vlan-id> dhcpv6-options pd
<id> interface <delegatee> address <address>
```

Configure the IPv6 interface identifier (host portion) for the delegatee interface.

The value must be a decimal integer. It is appended to the delegated prefix and the configured SLA ID to form the final IPv6 address.

By default, the host portion is generated based on the parent interface's MAC address (EUI-64 format).

Example:

If a /64 prefix is delegated to interface eth8 and you configure the host portion as 65535, the resulting IPv6 address will end with ::fff, as 65535 corresponds to fff in hexadecimal notation.

```
set interfaces ethernet eth0 vif-s 1000 vif-c 20 dhcpv6-options pd 0 interface eth8
↪address 65534
```

```
set interfaces ethernet <interface> vif-s <vlan-id> vif-c <vlan-id> dhcpv6-options pd
<id> interface <delegatee> sla-id <id>
```

Configure the SLA ID for the delegatee interface.

The value must be a decimal integer greater than 0 and fit in the length of SLA IDs. It is converted to hexadecimal and appended to the delegated prefix to form the specific subnet prefix for the delegatee interface.

Example:

If SLA ID is 1 and the delegated prefix is 2001:db8:ffff::/48, the resulting subnet prefix for the delegatee interface will be 2001:db8:ffff:1::/64.

```
set interfaces ethernet eth0 vif-s 1000 vif-c 20 dhcpv6-options pd 0 interface eth8
↪sla-id 1
```

SPAN port mirroring

SPAN port mirroring copies inbound and outbound traffic from one interface to another specified interface.

The destination interface is usually connected to specialized equipment, such as a behavior control system, an intrusion detection system, or a traffic collector, and copies all related traffic from this port. The benefit of mirroring traffic is that the application is isolated from the source traffic, so application processing does not affect the traffic or system performance.

To configure SPAN port mirroring, VyOS uses the `mirror` parameter. You can mirror ingress traffic (traffic entering the router) and egress traffic (traffic leaving the router) separately. Both directions can be mirrored to the same destination interface or split to different ones.

```
set interfaces ethernet <interface> mirror ingress <monitor-interface>
```

Mirror ingress traffic from a bonding interface to the specified monitoring interface.

Example: Mirror the ingress traffic from *eth1* to *eth3*.

```
set interfaces ethernet eth1 mirror ingress eth3
```

set interfaces ethernet <interface> mirror egress <monitor-interface>

Mirror egress traffic from a bonding interface to the specified monitoring interface.

Example: Mirror the egress traffic from *eth1* to *eth3*.

```
set interfaces ethernet eth1 mirror egress eth3
```

Operation

show interfaces ethernet

Show brief interface information.

```
vyos@vyos:~$ show interfaces ethernet
Codes: S - State, L - Link, u - Up, D - Down, A - Admin Down
Interface      IP Address      S/L Description
-----
eth0            172.18.201.10/24 u/u LAN
eth1            172.18.202.11/24 u/u WAN
eth2            -               u/D
```

show interfaces ethernet <interface>

Show detailed interface information.

```
vyos@vyos:~$ show interfaces ethernet eth0
eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group_
->default qlen 1000
    link/ether 00:50:44:00:f5:c9 brd ff:ff:ff:ff:ff:ff
    inet6 fe80::250:44ff:fe00:f5c9/64 scope link
        valid_lft forever preferred_lft forever

    RX:  bytes    packets   errors   dropped   overrun    mcast
         56735451    179841        0         0         0    142380
    TX:  bytes    packets   errors   dropped   carrier collisions
         5601460     62595        0         0         0         0
```

show interfaces ethernet <interface> physical

Show interface hardware-level and driver details.

```
vyos@vyos:~$ show interfaces ethernet eth0 physical
Settings for eth0:
    Supported ports: [ TP ]
    Supported link modes:   1000baseT/Full
                           10000baseT/Full
    Supported pause frame use: No
    Supports auto-negotiation: No
    Supported FEC modes: Not reported
```

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```

    Advertised link modes:  Not reported
    Advertised pause frame use: No
    Advertised auto-negotiation: No
    Advertised FEC modes: Not reported
    Speed: 10000Mb/s
    Duplex: Full
    Port: Twisted Pair
    PHYAD: 0
    Transceiver: internal
    Auto-negotiation: off
    MDI-X: Unknown
    Supports Wake-on: uag
    Wake-on: d
    Link detected: yes
driver: vmxnet3
version: 1.4.16.0-k-NAPI
firmware-version:
expansion-rom-version:
bus-info: 0000:0b:00.0
supports-statistics: yes
supports-test: no
supports-eeprom-access: no
supports-register-dump: yes
supports-priv-flags: no

```

show interfaces ethernet <interface> physical offload

Show the status of the interface offloading features.

```

vyos@vyos:~$ show interfaces ethernet eth0 physical offload
rx-checksumming                on
tx-checksumming                on
tx-checksum-ip-generic         on
scatter-gather                 off
tx-scatter-gather              off
tcp-segmentation-offload       off
tx-tcp-segmentation            off
tx-tcp-mangleid-segmentation   off
tx-tcp6-segmentation           off
udp-fragmentation-offload      off
generic-segmentation-offload   off
generic-receive-offload        off
large-receive-offload          off
rx-vlan-offload                on
tx-vlan-offload                on
ntuple-filters                 off
receive-hashing                on
tx-gre-segmentation            on
tx-gre-csum-segmentation       on
tx-udp_tnl-segmentation        on
tx-udp_tnl-csum-segmentation   on
tx-gso-partial                 on
tx-nocache-copy                off

```

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rx-all	off
--------	-----

show interfaces ethernet <interface> transceiver

Show information about the transceiver module plugged into the interface (e.g., SFP+, QSFP).

```
vyos@vyos:~$ show interfaces ethernet eth5 transceiver
Identifier           : 0x03 (SFP)
Extended identifier   : 0x04 (GBIC/SFP defined by 2-wire interface ID)
Connector            : 0x07 (LC)
Transceiver codes     : 0x00 0x00 0x00 0x01 0x00 0x00 0x00 0x00
Transceiver type      : Ethernet: 1000BASE-SX
Encoding              : 0x01 (8B/10B)
BR, Nominal           : 1300MBd
Rate identifier       : 0x00 (unspecified)
Length (SMF,km)       : 0km
Length (SMF)          : 0m
Length (50um)         : 550m
Length (62.5um)       : 270m
Length (Copper)       : 0m
Length (OM3)          : 0m
Laser wavelength     : 850nm
Vendor name           : CISCO-FINISAR
Vendor OUI            : 00:90:65
Vendor PN             : FTRJ-8519-7D-CS4
Vendor rev            : A
Option values         : 0x00 0x1a
Option                : RX_LOS implemented
Option                : TX_FAULT implemented
Option                : TX_DISABLE implemented
BR margin, max        : 0%
BR margin, min        : 0%
Vendor SN             : FNS092xxxxx
Date code             : 0506xx
```

7.4.5 Geneve

GENEVE (Generic Network Virtualization Encapsulation) interfaces operate as virtual network ports. Administrators can apply standard network configurations on them, such as IP addressing, bridging, or firewall rules, just as they would on physical Ethernet ports.

The Geneve protocol encapsulates Layer 2 Ethernet frames originating from endpoints such as virtual machines, containers, or physical servers inside UDP packets. It unifies the features of earlier encapsulation protocols, including VXLAN, NVGRE, and STT, and addresses their limitations, such as fixed header structures and a lack of metadata support. Because of its extensibility, Geneve may eventually replace those older protocols.

Geneve tunnels are used to connect virtual switches residing within hypervisors, physical switches, middleboxes, and other network appliances.

Geneve tunnels operate over any standard IP network. In larger deployments, the underlying network (underlay) is often built using a **Clos** topology, also known as a *leaf-and-spine* or *fat-tree* topology.

Geneve header:

Ver	Opt Len	O C	Rsvd.	Protocol Type
Virtual Network Identifier (VNI)				
Variable Length Options				Reserved

Configuration

Common interface configuration

set interfaces geneve <interface> address <address>

Configure one or more interface IP addresses.

<address> supports IPv4 and IPv6 in CIDR notation. For example, use 192.0.2.1/24 for IPv4 or 2001:db8::1/64 for IPv6.

Example:

```
set interfaces geneve gnv0 address 192.0.2.1/24
set interfaces geneve gnv0 address 2001:db8::1/64
```

set interfaces geneve <interface> description <description>

Configure a clear, descriptive alias for the interface.

This alias appears in the show interfaces command and SNMP-based monitoring tools.

Example:

```
set interfaces geneve gnv0 description 'This is an interface running on VyOS.'
```

set interfaces geneve <interface> disable

Disable the interface.

The interface will be set to the administratively down (A/D) state.

Example:

```
set interfaces geneve gnv0 disable
```

set interfaces geneve <interface> mac <xx:xx:xx:xx:xx:xx>

Configure a custom MAC address on the interface.

Example:

```
set interfaces geneve gnv0 mac '00:53:01:02:03:04'
```

set interfaces geneve <interface> mtu <mtu>

Configure the MTU on the interface.

This value defines the largest packet size, in bytes, that the interface transmits without fragmentation.

Example:

```
set interfaces geneve gnv0 mtu 1600
```

set interfaces geneve <interface> ip adjust-mss <mss | clamp-mss-to-pmtu>

Configure the MSS advertised in outgoing TCP SYN packets on the specified interface.

By clamping the MSS value in TCP SYN packets, you explicitly inform the remote side not to send packets larger than that size. This prevents connection issues that occur when Path MTU Discovery (PMTUD) fails.

The following options are available:

- **mss:** Sets the MSS to a specific value, in bytes. Use this option if you need to enforce a specific MSS, for example, to troubleshoot connectivity issues or accommodate specific network requirements.
- **clamp-mss-to-pmtu:** The router automatically calculates the MSS to be the interface's MTU minus 40 bytes for IPv4 traffic (20 bytes for the IPv4 header and 20 bytes for the TCP header). This option is recommended to automatically set the proper value.

set interfaces geneve <interface> ip arp-cache-timeout

Configure how long an ARP entry remains valid after learning an IP-to-MAC address mapping on this interface.

The default duration is 30 seconds.

An ARP entry remains valid if it receives positive feedback from higher-level protocols.

Example:

```
set interfaces geneve gnv0 ip arp-cache-timeout 180
```

set interfaces geneve <interface> ip disable-arp-filter

Configure ARP filtering on this interface.

Default behavior: The kernel responds to ARP requests on this interface only if the traffic would be routed back to the ARP sender through that specific interface.

If configured: The kernel responds to ARP requests on this interface for any IP address configured on the local host, regardless of which specific interface that IP address is assigned to, and regardless of the routing table. This reflects the Linux concept that IP addresses belong to the host, not individual interfaces.

Example:

```
set interfaces geneve gnv0 ip disable-arp-filter
```

set interfaces geneve <interface> ip disable-forwarding

Configure the interface for host or router behavior.

If configured, the interface switches to host mode, and IPv4 forwarding is disabled on it.

Example:

```
set interfaces geneve gnv0 ip disable-forwarding
```

set interfaces geneve <interface> ip enable-directed-broadcast

Configure whether to forward IP-directed broadcast packets received on this interface.

Default behavior: IP-directed broadcast packets are dropped.

If configured: IP-directed broadcast packets are forwarded to all hosts on the destination subnet, as defined in [RFC 1812](#) and [RFC 2644](#).

Example:

```
set interfaces geneve gnv0 ip enable-directed-broadcast
```

set interfaces geneve <interface> ip enable-arp-accept

Configure how to process gratuitous ARPs on this interface.

If configured, an IP-to-MAC address mapping is added to the ARP table based on gratuitous ARP requests or replies.

Note: If the ARP table already contains the IP address from a gratuitous ARP, its entry is updated regardless of whether this setting is configured.

Example:

```
set interfaces geneve gnv0 ip enable-arp-accept
```

set interfaces geneve <interface> ip enable-arp-announce

Configure the source IP selection for ARP requests on this interface.

Default behavior: The kernel can use any IP address the host owns as the source IP address in ARP requests on this interface.

If configured: The kernel first attempts to select a source IP address configured on the interface that shares a common subnet with the target IP address. If there is no such subnet, the kernel selects the IP address it would normally use (based on the routing table to reach the target destination).

Example:

```
set interfaces geneve gnv0 ip enable-arp-announce
```

set interfaces geneve <interface> ip enable-arp-ignore

Configure which ARP requests will be ignored on this interface.

Default behavior: The kernel responds to ARP requests for any local IP addresses, regardless of which interface they are assigned to.

If configured: The kernel responds to ARP requests only if the target IP address is assigned to this specific interface.

Example:

```
set interfaces geneve gnv0 ip enable-arp-ignore
```

set interfaces geneve <interface> ip enable-proxy-arp

Configure proxy ARP on this interface.

If configured, the router (kernel) intercepts ARP requests for non-local IP addresses and replies with the MAC address of the interface that received the request. Subsequent packets destined to these IP addresses are forwarded to their actual destinations on remote subnets.

Example:

```
set interfaces geneve gnv0 ip enable-proxy-arp
```

set interfaces geneve <interface> ip proxy-arp-pvlan

Configure local proxy ARP on the interface.

If configured, the router (kernel) responds to ARP requests on this VLAN interface even if the target IP address resides on the same subnet and interface.

This is used to support network isolation requirements (RFC 3069) for private VLANs (PVLANS). In PVLAN configurations, hosts on isolated ports are NOT allowed to communicate directly with each other at Layer 2, but they can communicate with the upstream router.

By replying to inter-host ARP requests with its own MAC address, the router (kernel) directs inter-host traffic through itself instead of directly between hosts.

Note: This command works independently and does not require enabling the standard proxy ARP on the interface.

Local proxy ARP is also known as:

- VLAN aggregation ([RFC 3069](#)).
- Private VLAN (Cisco, Allied Telesyn).
- Source-port filtering or port isolation (Hewlett-Packard).
- MAC-Forced Forwarding (Ericsson).

set interfaces geneve <interface> ip source-validation <strict | loose | disable>

Configure source IP address validation using RPF on this interface, as specified in [RFC 3704](#).

The following options are available:

- **strict:** Each incoming packet's source IP address is checked against the Forwarding Information Base (FIB). If the interface is not the best route back to that source, validation fails, and the packet is dropped.
- **loose:** Each incoming packet's source IP address is checked against the FIB. If the source IP address is unreachable through any interface, validation fails.
- **disable:** No source IP address validation is performed. All incoming packets are accepted.

RFC 3704 recommends enabling strict mode to prevent IP spoofing, such as DDoS attacks. For asymmetric or other complex routing scenarios, use loose mode.

set interfaces geneve <interface> ipv6 address autoconf

Configure the interface to automatically obtain an IPv6 address using SLAAC, as specified in [RFC 4862](#).

IPv6 hosts can configure themselves automatically when connected to an IPv6 network using the Neighbor Discovery Protocol via ICMPv6 router discovery messages. When first connected to a network, a host sends a link-local router solicitation multicast request for its configuration parameters. The router responds with a router advertisement packet containing Internet Layer configuration parameters.

Note: This method automatically disables IPv6 traffic forwarding on the interface.

Example:

```
set interfaces geneve gnv0 ipv6 address autoconf
```

set interfaces geneve <interface> ipv6 address eui64 <prefix>

Configure the interface to assign itself an IPv6 address using the EUI-64 method, as specified in [RFC 4291](#).

Example:

```
set interfaces geneve gnv0 ipv6 address eui64 2001:db8:beef::/64
```

set interfaces geneve <interface> ipv6 address no-default-link-local

Disable the automatic assignment of a link-local IPv6 address to this interface.

Example:

```
set interfaces geneve gnv0 ipv6 address no-default-link-local
```

set interfaces geneve <interface> ipv6 disable-forwarding

Configure the interface for host or router behavior.

If configured, the interface switches to host mode, and IPv6 forwarding is disabled on it.

Example:

```
set interfaces geneve gnv0 ipv6 disable-forwarding
```

set interfaces geneve <interface> ipv6 adjust-mss <mss | clamp-mss-to-pmtu>

Configure the MSS advertised in outgoing TCP SYN packets on the specified interface.

By clamping the MSS value in TCP SYN packets, you explicitly inform the remote side not to send packets larger than that size. This prevents connection issues when Path MTU Discovery (PMTUD) fails.

The following options are available:

- **mss:** Set the MSS to a specific value, in bytes. Use this option to enforce a specific MSS, for example, to troubleshoot connectivity issues or accommodate specific network requirements.
- **clamp-mss-to-pmtu:** The router calculates the MSS to be the interface's MTU minus 60 bytes for IPv6 traffic (40 bytes for the IPv6 header and 20 bytes for the TCP header). This option is recommended to automatically set the proper value.

set interfaces geneve <interface> ipv6 accept-dad <0 | 1 | 2>

Configure IPv6 DAD on the interface.

The following options are available:

- **0:** Disables DAD. No duplicate address detection is performed.
- **1:** Enables DAD (default). Duplicate addresses are detected. The interface's IPv6 operation continues for valid IPv6 addresses.
- **2:** Enables DAD and, if a MAC-based duplicate link-local address is found, disables IPv6 operation on this interface.

Example:

```
set interfaces geneve gnv0 ipv6 accept-dad 2
```

set interfaces geneve <interface> ipv6 dup-addr-detect-transmits <n>

Configure the number of DAD messages that the router (kernel) sends during IPv6 address assignment on this interface.

The default value is 1.

Example:

```
set interfaces geneve gnv0 ipv6 dup-addr-detect-transmits 5
```

Geneve options

set interfaces geneve gnv0 remote <address>

Configure the remote endpoint IP address for the Geneve tunnel.

set interfaces geneve gnv0 vni <vni>

Configure VNI (Virtual Network Identifier) for the Geneve interface.

The VNI is a virtual network identifier. It allows multiple virtual networks to share the same physical infrastructure and remain isolated.

The VNI is also used to distribute traffic after it leaves the tunnel, for example, to map packets with overlapping IP addresses to specific routing tables.

set interfaces gnv0 <interface> port <port>

Configure the destination UDP port for the remote Geneve tunnel endpoint. Ensure the remote peer is configured to listen on this specific port.

7.4.6 L2TPv3

L2TPv3 (Layer 2 Tunneling Protocol version 3) interfaces let you establish L2TPv3 tunnels to transport Layer 2 traffic over IP networks.

The L2TPv3 protocol (defined in RFC 3931) wraps Layer 2 frames (e.g., Ethernet, Frame Relay, HDLC) within IP packets, allowing them to traverse the underlying IP infrastructure.

Unlike L2TPv2, which strictly requires UDP encapsulation, the L2TPv3 protocol is more flexible and supports two encapsulation types:

- **Direct IP:** Tunnel data is encapsulated directly inside IP packets (Protocol 115) for lower overhead.
- **UDP:** Tunnel data is encapsulated inside a UDP datagram. This allows the tunnel to traverse NAT more easily.

L2TPv3 tunnels connect geographically separated sites, serving as a simpler alternative to *MPLS* by operating over basic IP connectivity rather than requiring a full MPLS infrastructure.

L2TPv3 tunnels can be established over both IPv4 and IPv6 underlying networks.

Configuration

Common interface configuration

set interfaces l2tpv3 <interface> address <address>

Configure one or more interface IP addresses.

<address> supports IPv4 and IPv6 in CIDR notation. For example, use 192.0.2.1/24 for IPv4 or 2001:db8::1/64 for IPv6.

Example:

```
set interfaces l2tpv3 l2tpeth0 address 192.0.2.1/24
set interfaces l2tpv3 l2tpeth0 address 2001:db8::1/64
```

```
set interfaces l2tpv3 <interface> description <description>
```

Configure a clear, descriptive alias for the interface.

This alias appears in the `show interfaces` command and SNMP-based monitoring tools.

Example:

```
set interfaces l2tpv3 l2tpeth0 description 'This is an interface running on VyOS.'
```

```
set interfaces l2tpv3 <interface> disable
```

Disable the interface.

The interface will be set to the administratively down (A/D) state.

Example:

```
set interfaces l2tpv3 l2tpeth0 disable
```

```
set interfaces l2tpv3 <interface> disable-flow-control
```

Disable Ethernet flow control (IEEE 802.3x pause frames) on the interface.

Ethernet flow control, defined by the IEEE 802.3x standard, temporarily stops data transmission to prevent packet loss during network congestion. For example, when a sender transmits data faster than the receiver can process it.

Disabling Ethernet flow control means the interface will not signal the connected device to pause transmission and will drop packets if overwhelmed.

Example:

```
set interfaces l2tpv3 l2tpeth0 disable-flow-control
```

```
set interfaces l2tpv3 <interface> disable-link-detect
```

Disable physical link-state change detection on the interface, such as when a cable is unplugged.

By default, the interface detects physical link-state changes.

Example:

```
set interfaces l2tpv3 l2tpeth0 disable-link-detect
```

```
set interfaces l2tpv3 <interface> mac <xx:xx:xx:xx:xx:xx>
```

Configure a custom MAC address on the interface.

Example:

```
set interfaces l2tpv3 l2tpeth0 mac '00:53:01:02:03:04'
```

```
set interfaces l2tpv3 <interface> mtu <mtu>
```

Configure the MTU on the interface.

This value defines the largest packet size, in bytes, that the interface transmits without fragmentation.

Example:

```
set interfaces l2tpv3 l2tpeth0 mtu 1600
```

```
set interfaces l2tpv3 <interface> ip adjust-mss <mss | clamp-mss-to-pmtu>
```

Configure the MSS advertised in outgoing TCP SYN packets on the specified interface.

By clamping the MSS value in TCP SYN packets, you explicitly inform the remote side not to send packets larger than that size. This prevents connection issues that occur when Path MTU Discovery (PMTUD) fails.

The following options are available:

- **mss:** Sets the MSS to a specific value, in bytes. Use this option if you need to enforce a specific MSS, for example, to troubleshoot connectivity issues or accommodate specific network requirements.
- **clamp-mss-to-pmtu:** The router automatically calculates the MSS to be the interface's MTU minus 40 bytes for IPv4 traffic (20 bytes for the IPv4 header and 20 bytes for the TCP header). This option is recommended to automatically set the proper value.

set interfaces l2tpv3 <interface> ip arp-cache-timeout

Configure how long an ARP entry remains valid after learning an IP-to-MAC address mapping on this interface.

The default duration is 30 seconds.

An ARP entry remains valid if it receives positive feedback from higher-level protocols.

Example:

```
set interfaces l2tpv3 l2tpeth0 ip arp-cache-timeout 180
```

set interfaces l2tpv3 <interface> ip disable-arp-filter

Configure ARP filtering on this interface.

Default behavior: The kernel responds to ARP requests on this interface only if the traffic would be routed back to the ARP sender through that specific interface.

If configured: The kernel responds to ARP requests on this interface for any IP address configured on the local host, regardless of which specific interface that IP address is assigned to, and regardless of the routing table. This reflects the Linux concept that IP addresses belong to the host, not individual interfaces.

Example:

```
set interfaces l2tpv3 l2tpeth0 ip disable-arp-filter
```

set interfaces l2tpv3 <interface> ip disable-forwarding

Configure the interface for host or router behavior.

If configured, the interface switches to host mode, and IPv4 forwarding is disabled on it.

Example:

```
set interfaces l2tpv3 l2tpeth0 ip disable-forwarding
```

set interfaces l2tpv3 <interface> ip enable-directed-broadcast

Configure whether to forward IP-directed broadcast packets received on this interface.

Default behavior: IP-directed broadcast packets are dropped.

If configured: IP-directed broadcast packets are forwarded to all hosts on the destination subnet, as defined in [RFC 1812](#) and [RFC 2644](#).

Example:

```
set interfaces l2tpv3 l2tpeth0 ip enable-directed-broadcast
```

set interfaces l2tpv3 <interface> ip enable-arp-accept

Configure how to process gratuitous ARPs on this interface.

If configured, an IP-to-MAC address mapping is added to the ARP table based on gratuitous ARP requests or replies.

Note: If the ARP table already contains the IP address from a gratuitous ARP, its entry is updated regardless of whether this setting is configured.

Example:

```
set interfaces l2tpv3 l2tpeth0 ip enable-arp-accept
```

set interfaces l2tpv3 <interface> ip enable-arp-announce

Configure the source IP selection for ARP requests on this interface.

Default behavior: The kernel can use any IP address the host owns as the source IP address in ARP requests on this interface.

If configured: The kernel first attempts to select a source IP address configured on the interface that shares a common subnet with the target IP address. If there is no such subnet, the kernel selects the IP address it would normally use (based on the routing table to reach the target destination).

Example:

```
set interfaces l2tpv3 l2tpeth0 ip enable-arp-announce
```

set interfaces l2tpv3 <interface> ip enable-arp-ignore

Configure which ARP requests will be ignored on this interface.

Default behavior: The kernel responds to ARP requests for any local IP addresses, regardless of which interface they are assigned to.

If configured: The kernel responds to ARP requests only if the target IP address is assigned to this specific interface.

Example:

```
set interfaces l2tpv3 l2tpeth0 ip enable-arp-ignore
```

set interfaces l2tpv3 <interface> ip enable-proxy-arp

Configure proxy ARP on this interface.

If configured, the router (kernel) intercepts ARP requests for non-local IP addresses and replies with the MAC address of the interface that received the request. Subsequent packets destined to these IP addresses are forwarded to their actual destinations on remote subnets.

Example:

```
set interfaces l2tpv3 l2tpeth0 ip enable-proxy-arp
```

set interfaces l2tpv3 <interface> ip proxy-arp-pvlan

Configure local proxy ARP on the interface.

If configured, the router (kernel) responds to ARP requests on this VLAN interface even if the target IP address resides on the same subnet and interface.

This is used to support network isolation requirements (RFC 3069) for private VLANs (PVLANS). In PVLAN configurations, hosts on isolated ports are NOT allowed to communicate directly with each other at Layer 2, but they can communicate with the upstream router.

By replying to inter-host ARP requests with its own MAC address, the router (kernel) directs inter-host traffic through itself instead of directly between hosts.

Note: This command works independently and does not require enabling the standard proxy ARP on the interface.

Local proxy ARP is also known as:

- VLAN aggregation ([RFC 3069](#)).
- Private VLAN (Cisco, Allied Telesyn).
- Source-port filtering or port isolation (Hewlett-Packard).
- MAC-Forced Forwarding (Ericsson).

set interfaces l2tpv3 <interface> ip source-validation <strict | loose | disable>

Configure source IP address validation using RPF on this interface, as specified in [RFC 3704](#).

The following options are available:

- **strict:** Each incoming packet's source IP address is checked against the Forwarding Information Base (FIB). If the interface is not the best route back to that source, validation fails, and the packet is dropped.
- **loose:** Each incoming packet's source IP address is checked against the FIB. If the source IP address is unreachable through any interface, validation fails.
- **disable:** No source IP address validation is performed. All incoming packets are accepted.

RFC 3704 recommends enabling strict mode to prevent IP spoofing, such as DDoS attacks. For asymmetric or other complex routing scenarios, use loose mode.

set interfaces l2tpv3 <interface> ipv6 address autoconf

Configure the interface to automatically obtain an IPv6 address using SLAAC, as specified in [RFC 4862](#).

IPv6 hosts can configure themselves automatically when connected to an IPv6 network using the Neighbor Discovery Protocol via ICMPv6 router discovery messages. When first connected to a network, a host sends a link-local router solicitation multicast request for its configuration parameters. The router responds with a router advertisement packet containing Internet Layer configuration parameters.

Note: This method automatically disables IPv6 traffic forwarding on the interface.

Example:

```
set interfaces l2tpv3 l2tpeth0 ipv6 address autoconf
```

set interfaces l2tpv3 <interface> ipv6 address eui64 <prefix>

Configure the interface to assign itself an IPv6 address using the EUI-64 method, as specified in [RFC 4291](#).

Example:

```
set interfaces l2tpv3 l2tpeth0 ipv6 address eui64 2001:db8:beef::/64
```

```
set interfaces l2tpv3 <interface> ipv6 address no-default-link-local
```

Disable the automatic assignment of a link-local IPv6 address to this interface.

Example:

```
set interfaces l2tpv3 l2tpeth0 ipv6 address no-default-link-local
```

```
set interfaces l2tpv3 <interface> ipv6 disable-forwarding
```

Configure the interface for host or router behavior.

If configured, the interface switches to host mode, and IPv6 forwarding is disabled on it.

Example:

```
set interfaces l2tpv3 l2tpeth0 ipv6 disable-forwarding
```

```
set interfaces l2tpv3 <interface> ipv6 adjust-mss <mss | clamp-mss-to-pmtu>
```

Configure the MSS advertised in outgoing TCP SYN packets on the specified interface.

By clamping the MSS value in TCP SYN packets, you explicitly inform the remote side not to send packets larger than that size. This prevents connection issues when Path MTU Discovery (PMTUD) fails.

The following options are available:

- **mss:** Set the MSS to a specific value, in bytes. Use this option to enforce a specific MSS, for example, to troubleshoot connectivity issues or accommodate specific network requirements.
- **clamp-mss-to-pmtu:** The router calculates the MSS to be the interface's MTU minus 60 bytes for IPv6 traffic (40 bytes for the IPv6 header and 20 bytes for the TCP header). This option is recommended to automatically set the proper value.

```
set interfaces l2tpv3 <interface> ipv6 accept-dad <0 | 1 | 2>
```

Configure IPv6 DAD on the interface.

The following options are available:

- **0:** Disables DAD. No duplicate address detection is performed.
- **1:** Enables DAD (default). Duplicate addresses are detected. The interface's IPv6 operation continues for valid IPv6 addresses.
- **2:** Enables DAD and, if a MAC-based duplicate link-local address is found, disables IPv6 operation on this interface.

Example:

```
set interfaces l2tpv3 l2tpeth0 ipv6 accept-dad 2
```

```
set interfaces l2tpv3 <interface> ipv6 dup-addr-detect-transmits <n>
```

Configure the number of DAD messages that the router (kernel) sends during IPv6 address assignment on this interface.

The default value is 1.

Example:

```
set interfaces l2tpv3 l2tpeth0 ipv6 dup-addr-detect-transmits 5
```

```
set interfaces l2tpv3 <interface> vrf <vrf>
```

Assign the interface to a specific VRF instance.

See also:

For information on configuring a VRF, refer to the [VRF](#) section.

Example:

```
set interfaces l2tpv3 l2tpeth0 vrf red
```

L2TPv3 options

Use the following commands to configure the L2TPv3 tunnel's specific parameters.

set interfaces l2tpv3 <interface> encapsulation <udp | ip>

Configure the encapsulation type for the L2TPv3 tunnel.

Valid values are udp and ip.

The default encapsulation type is udp.

Note: The encapsulation type must match on both the local and remote peers for the tunnel to establish.

set interfaces l2tpv3 <interface> source-address <address>

Configure the L2TPv3 tunnel source IP address.

The specified address must be a local interface IP address and can be either IPv4 or IPv6.

set interfaces l2tpv3 <interface> remote <address>

Configure the L2TPv3 tunnel destination IP address.

The specified address must be a remote peer's interface IP address and can be either IPv4 or IPv6.

set interfaces l2tpv3 <interface> session-id <id>

Configure the local session ID within the L2TPv3 tunnel.

The session-id is a 32-bit value that identifies an incoming tunnel session on the local peer.

The peer-session-id that identifies this session on the remote peer must be set to the same value.

set interfaces l2tpv3 <interface> peer-session-id <id>

Configure the peer session ID within the L2TPv3 tunnel.

The peer-session-id is a 32-bit value that identifies an outgoing tunnel session from the local peer.

The peer-session-id must match the session-id configured for this session on the remote peer.

set interfaces l2tpv3 <interface> tunnel-id <id>

Configure the local identifier for the L2TPv3 tunnel.

The tunnel-id is a 32-bit value that identifies the L2TPv3 tunnel on the local peer.

The peer-tunnel-id that identifies this tunnel on the remote peer must be set to the same value.

set interfaces l2tpv3 <interface> peer-tunnel-id <id>

Configure the peer identifier for the L2TPv3 tunnel.

The `peer-tunnel-id` is a 32-bit value that identifies the L2TPv3 tunnel on the remote peer and must correspond to the `tunnel-id` configured for that tunnel on that peer.

The `peer-tunnel-id` must match the `tunnel-id` that identifies this tunnel on the remote peer.

Example**L2TPv3 tunnel with IP encapsulation**

The following example shows the configuration of an L2TPv3 tunnel using direct IP encapsulation:

```
# show interfaces l2tpv3
l2tpv3 l2tpeth10 {
    address 192.168.37.1/27
    encapsulation ip
    source-address 192.0.2.1
    peer-session-id 100
    peer-tunnel-id 200
    remote 203.0.113.24
    session-id 100
    tunnel-id 200
}
```

The inverse configuration must be applied to the remote peer.

L2TPv3 tunnel with UDP encapsulation

The following example shows the configuration of an L2TPv3 tunnel using UDP encapsulation.

This setup is recommended when the tunnel traverses NAT devices.

Configuration notes:

- Use a local LAN IP address as the `source-address`.
- Configure a forwarding rule to allow tunnel traffic on the specified UDP port on the upstream NAT device.
- Use a distinct UDP port for each individual tunnel.

```
# show interfaces l2tpv3
l2tpv3 l2tpeth10 {
    address 192.168.37.1/27
    destination-port 9001
    encapsulation udp
    source-address 192.0.2.1
    peer-session-id 100
    peer-tunnel-id 200
    remote 203.0.113.24
    session-id 100
    source-port 9000
    tunnel-id 200
}
```

7.4.7 Loopback

The loopback interface is a virtual, software-based network interface. All traffic sent to it loops back and only targets services on the local host.

Note: Only one loopback `lo` interface is allowed per operating system. If you require multiple virtual interfaces, use the *Dummy* interface type.

Configuration

Common interface configuration

set interfaces loopback <interface> address <address>

Configure one or more interface IP addresses.

<address> supports IPv4 and IPv6 in CIDR notation. For example, use `192.0.2.1/24` for IPv4 or `2001:db8::1/64` for IPv6.

Example:

```
set interfaces loopback lo address 192.0.2.1/24
set interfaces loopback lo address 2001:db8::1/64
```

set interfaces loopback <interface> description <description>

Configure a clear, descriptive alias for the interface.

This alias appears in the `show interfaces` command and SNMP-based monitoring tools.

Example:

```
set interfaces loopback lo description 'This is an interface running on VyOS.'
```

Operation

show interfaces loopback

Show brief interface information.

```
vyos@vyos:~$ show interfaces loopback
Codes: S - State, L - Link, u - Up, D - Down, A - Admin Down
Interface      IP Address      S/L  Description
-----
lo             127.0.0.1/8     u/u
               ::1/128
```

show interfaces loopback lo

Show detailed interface information.

```
vyos@vyos:~$ show interfaces loopback lo
lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen.
↪ 1000
```

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```

link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
inet 127.0.0.1/8 scope host lo
    valid_lft forever preferred_lft forever
inet6 ::1/128 scope host
    valid_lft forever preferred_lft forever

RX:  bytes    packets    errors    dropped    overrun    mcast
     300         6         0         0         0         0
TX:  bytes    packets    errors    dropped    carrier    collisions
     300         6         0         0         0         0

```

7.4.8 MACsec

MACsec is an IEEE standard (IEEE 802.1AE) for MAC security, introduced in 2006. It enables protocol-independent connectivity between two hosts, providing data confidentiality, authenticity, and integrity using GCM-AES ciphers. MACsec operates at the Ethernet layer as a Layer 2 protocol and secures traffic within Layer 2 networks, including DHCP and ARP requests. It does not compete with other security solutions, such as IPsec (Layer 3) or TLS (Layer 4), as each addresses distinct use cases.

Configuration

Common interface configuration

set interfaces macsec <interface> address <address | dhcp | dhcpv6>

Configure the interface with one or more IP addresses.

The following options are available:

- **address:** Assign one or more IPv4 or IPv6 addresses to the interface. For example, use 192.0.2.1/24 for IPv4 or 2001:db8::1/64 for IPv6.
- **dhcp:** The interface obtains an IPv4 address from a DHCP server on the same network segment.
- **dhcpv6:** The interface obtains an IPv6 address from a DHCPv6 server on the same network segment.

Note: If the interface obtains an IPv4 address via DHCP, and specific adjustments are needed before/after the IP address is obtained, use the provided hook scripts:

- /config/scripts/dhcp-client/pre-hooks.d/
- /config/scripts/dhcp-client/post-hooks.d/

Example:

```

set interfaces macsec macsec0 address 192.0.2.1/24
set interfaces macsec macsec0 address 2001:db8::1/64
set interfaces macsec macsec0 address dhcp
set interfaces macsec macsec0 address dhcpv6

```

set interfaces macsec <interface> description <description>

Configure a clear, descriptive alias for the interface.

This alias appears in the `show interfaces` command and SNMP-based monitoring tools.

Example:

```
set interfaces macsec macsec0 description 'This is an interface running on VyOS.'
```

set interfaces macsec <interface> disable**Disable the interface.**

The interface will be set to the administratively down (A/D) state.

Example:

```
set interfaces macsec macsec0 disable
```

set interfaces macsec <interface> disable-flow-control**Disable Ethernet flow control (IEEE 802.3x pause frames) on the interface.**

Ethernet flow control, defined by the IEEE 802.3x standard, temporarily stops data transmission to prevent packet loss during network congestion. For example, when a sender transmits data faster than the receiver can process it.

Disabling Ethernet flow control means the interface will not signal the connected device to pause transmission and will drop packets if overwhelmed.

Example:

```
set interfaces macsec macsec0 disable-flow-control
```

set interfaces macsec <interface> disable-link-detect**Disable physical link-state change detection on the interface, such as when a cable is unplugged.**

By default, the interface detects physical link-state changes.

Example:

```
set interfaces macsec macsec0 disable-link-detect
```

set interfaces macsec <interface> mac <xx:xx:xx:xx:xx:xx>**Configure a custom MAC address on the interface.**

Example:

```
set interfaces macsec macsec0 mac '00:53:01:02:03:04'
```

set interfaces macsec <interface> mtu <mtu>**Configure the MTU on the interface.**

This value defines the largest packet size, in bytes, that the interface transmits without fragmentation.

Example:

```
set interfaces macsec macsec0 mtu 1600
```

set interfaces macsec <interface> ip adjust-mss <mss | clamp-mss-to-pmtu>

Configure the MSS advertised in outgoing TCP SYN packets on the specified interface.

By clamping the MSS value in TCP SYN packets, you explicitly inform the remote side not to send packets larger than that size. This prevents connection issues that occur when Path MTU Discovery (PMTUD) fails.

The following options are available:

- **mss:** Sets the MSS to a specific value, in bytes. Use this option if you need to enforce a specific MSS, for example, to troubleshoot connectivity issues or accommodate specific network requirements.
- **clamp-mss-to-pmtu:** The router automatically calculates the MSS to be the interface's MTU minus 40 bytes for IPv4 traffic (20 bytes for the IPv4 header and 20 bytes for the TCP header). This option is recommended to automatically set the proper value.

set interfaces macsec <interface> ip arp-cache-timeout

Configure how long an ARP entry remains valid after learning an IP-to-MAC address mapping on this interface.

The default duration is 30 seconds.

An ARP entry remains valid if it receives positive feedback from higher-level protocols.

Example:

```
set interfaces macsec macsec0 ip arp-cache-timeout 180
```

set interfaces macsec <interface> ip disable-arp-filter

Configure ARP filtering on this interface.

Default behavior: The kernel responds to ARP requests on this interface only if the traffic would be routed back to the ARP sender through that specific interface.

If configured: The kernel responds to ARP requests on this interface for any IP address configured on the local host, regardless of which specific interface that IP address is assigned to, and regardless of the routing table. This reflects the Linux concept that IP addresses belong to the host, not individual interfaces.

Example:

```
set interfaces macsec macsec0 ip disable-arp-filter
```

set interfaces macsec <interface> ip disable-forwarding

Configure the interface for host or router behavior.

If configured, the interface switches to host mode, and IPv4 forwarding is disabled on it.

Example:

```
set interfaces macsec macsec0 ip disable-forwarding
```

set interfaces macsec <interface> ip enable-directed-broadcast

Configure whether to forward IP-directed broadcast packets received on this interface.

Default behavior: IP-directed broadcast packets are dropped.

If configured: IP-directed broadcast packets are forwarded to all hosts on the destination subnet, as defined in [RFC 1812](#) and [RFC 2644](#).

Example:

```
set interfaces macsec macsec0 ip enable-directed-broadcast
```

set interfaces macsec <interface> ip enable-arp-accept

Configure how to process gratuitous ARPs on this interface.

If configured, an IP-to-MAC address mapping is added to the ARP table based on gratuitous ARP requests or replies.

Note: If the ARP table already contains the IP address from a gratuitous ARP, its entry is updated regardless of whether this setting is configured.

Example:

```
set interfaces macsec macsec0 ip enable-arp-accept
```

set interfaces macsec <interface> ip enable-arp-announce

Configure the source IP selection for ARP requests on this interface.

Default behavior: The kernel can use any IP address the host owns as the source IP address in ARP requests on this interface.

If configured: The kernel first attempts to select a source IP address configured on the interface that shares a common subnet with the target IP address. If there is no such subnet, the kernel selects the IP address it would normally use (based on the routing table to reach the target destination).

Example:

```
set interfaces macsec macsec0 ip enable-arp-announce
```

set interfaces macsec <interface> ip enable-arp-ignore

Configure which ARP requests will be ignored on this interface.

Default behavior: The kernel responds to ARP requests for any local IP addresses, regardless of which interface they are assigned to.

If configured: The kernel responds to ARP requests only if the target IP address is assigned to this specific interface.

Example:

```
set interfaces macsec macsec0 ip enable-arp-ignore
```

set interfaces macsec <interface> ip enable-proxy-arp

Configure proxy ARP on this interface.

If configured, the router (kernel) intercepts ARP requests for non-local IP addresses and replies with the MAC address of the interface that received the request. Subsequent packets destined to these IP addresses are forwarded to their actual destinations on remote subnets.

Example:

```
set interfaces macsec macsec0 ip enable-proxy-arp
```

set interfaces macsec <interface> ip proxy-arp-pvlan

Configure local proxy ARP on the interface.

If configured, the router (kernel) responds to ARP requests on this VLAN interface even if the target IP address resides on the same subnet and interface.

This is used to support network isolation requirements (RFC 3069) for private VLANs (PVLANS). In PVLAN configurations, hosts on isolated ports are NOT allowed to communicate directly with each other at Layer 2, but they can communicate with the upstream router.

By replying to inter-host ARP requests with its own MAC address, the router (kernel) directs inter-host traffic through itself instead of directly between hosts.

Note: This command works independently and does not require enabling the standard proxy ARP on the interface.

Local proxy ARP is also known as:

- VLAN aggregation ([RFC 3069](#)).
- Private VLAN (Cisco, Allied Telesyn).
- Source-port filtering or port isolation (Hewlett-Packard).
- MAC-Forced Forwarding (Ericsson).

set interfaces macsec <interface> ip source-validation <strict | loose | disable>

Configure source IP address validation using RPF on this interface, as specified in [RFC 3704](#).

The following options are available:

- **strict:** Each incoming packet's source IP address is checked against the Forwarding Information Base (FIB). If the interface is not the best route back to that source, validation fails, and the packet is dropped.
- **loose:** Each incoming packet's source IP address is checked against the FIB. If the source IP address is unreachable through any interface, validation fails.
- **disable:** No source IP address validation is performed. All incoming packets are accepted.

RFC 3704 recommends enabling strict mode to prevent IP spoofing, such as DDoS attacks. For asymmetric or other complex routing scenarios, use loose mode.

set interfaces macsec <interface> ipv6 address autoconf

Configure the interface to automatically obtain an IPv6 address using SLAAC, as specified in [RFC 4862](#).

IPv6 hosts can configure themselves automatically when connected to an IPv6 network using the Neighbor Discovery Protocol via ICMPv6 router discovery messages. When first connected to a network, a host sends a link-local router solicitation multicast request for its configuration parameters. The router responds with a router advertisement packet containing Internet Layer configuration parameters.

Note: This method automatically disables IPv6 traffic forwarding on the interface.

Example:

```
set interfaces macsec macsec0 ipv6 address autoconf
```

set interfaces macsec <interface> ipv6 address eui64 <prefix>

Configure the interface to assign itself an IPv6 address using the EUI-64 method, as specified in [RFC 4291](#).

Example:

```
set interfaces macsec macsec0 ipv6 address eui64 2001:db8:beef::/64
```

```
set interfaces macsec <interface> ipv6 address no-default-link-local
```

Disable the automatic assignment of a link-local IPv6 address to this interface.

Example:

```
set interfaces macsec macsec0 ipv6 address no-default-link-local
```

```
set interfaces macsec <interface> ipv6 disable-forwarding
```

Configure the interface for host or router behavior.

If configured, the interface switches to host mode, and IPv6 forwarding is disabled on it.

Example:

```
set interfaces macsec macsec0 ipv6 disable-forwarding
```

```
set interfaces macsec <interface> ipv6 adjust-mss <mss | clamp-mss-to-pmtu>
```

Configure the MSS advertised in outgoing TCP SYN packets on the specified interface.

By clamping the MSS value in TCP SYN packets, you explicitly inform the remote side not to send packets larger than that size. This prevents connection issues when Path MTU Discovery (PMTUD) fails.

The following options are available:

- **mss:** Set the MSS to a specific value, in bytes. Use this option to enforce a specific MSS, for example, to troubleshoot connectivity issues or accommodate specific network requirements.
- **clamp-mss-to-pmtu:** The router calculates the MSS to be the interface's MTU minus 60 bytes for IPv6 traffic (40 bytes for the IPv6 header and 20 bytes for the TCP header). This option is recommended to automatically set the proper value.

```
set interfaces macsec <interface> ipv6 accept-dad <0 | 1 | 2>
```

Configure IPv6 DAD on the interface.

The following options are available:

- **0:** Disables DAD. No duplicate address detection is performed.
- **1:** Enables DAD (default). Duplicate addresses are detected. The interface's IPv6 operation continues for valid IPv6 addresses.
- **2:** Enables DAD and, if a MAC-based duplicate link-local address is found, disables IPv6 operation on this interface.

Example:

```
set interfaces macsec macsec0 ipv6 accept-dad 2
```

```
set interfaces macsec <interface> ipv6 dup-addr-detect-transmits <n>
```

Configure the number of DAD messages that the router (kernel) sends during IPv6 address assignment on this interface.

The default value is 1.

Example:

```
set interfaces macsec macsec0 ipv6 dup-addr-detect-transmits 5
```

```
set interfaces macsec <interface> vrf <vrf>
```


Assign the interface to a specific VRF instance.

See also:

For information on configuring a VRF, refer to the [VRF](#) section.

Example:

```
set interfaces macsec macsec0 vrf red
```

DHCP(v6)

set interfaces macsec <interface> dhcp-options client-id <description>

Configure a DHCP client identifier for the interface, as specified in [RFC 2131](#).

The `client-id` is an identifier that the DHCP client sends to the DHCP server to uniquely identify itself for IP address assignment. By default, the client uses its MAC address. The `<description>` is a user-defined string that will be sent to the DHCP server as the DHCP client identifier.

Example:

```
set interfaces macsec macsec0 dhcp-options client-id 'foo-bar'
```

set interfaces macsec <interface> dhcp-options host-name <hostname>

Configure a specific hostname for the interface.

Instead of the real hostname, the DHCP client will send the specific hostname to the DHCP server when requesting an IP address.

Example:

```
set interfaces macsec macsec0 dhcp-options host-name 'VyOS'
```

set interfaces macsec <interface> dhcp-options vendor-class-id <vendor-id>

Configure the DHCP client to include a vendor-class identifier in its DHCP requests on this interface.

The vendor-class identifier is a vendor-specific byte string that enables the DHCP server to identify the device and, in some cases, provide configuration options.

Example:

```
set interfaces macsec macsec0 dhcp-options vendor-class-id 'VyOS'
```

set interfaces macsec <interface> dhcp-options no-default-route

Configure the DHCP client to obtain an IP address, but ignore any default gateway provided by the DHCP server on this interface.

Example:

```
set interfaces macsec macsec0 dhcp-options no-default-route
```

set interfaces macsec <interface> dhcp-options default-route-distance <distance>

Configure the distance for the default route obtained from the DHCP server on this interface.

Example:

```
set interfaces macsec macsec0 dhcp-options default-route-distance 220
```

set interfaces macsec <interface> dhcp-options reject <address>

Configure the DHCP client to reject the specific IP address or IP address range from the DHCP server on this interface.

This is useful when a modem assigns a local IP address upon start. To reject multiple addresses, run this command multiple times with different values. You can reject individual addresses (192.168.100.1) or entire subnets (192.168.100.0/24).

Example:

```
set interfaces macsec macsec0 dhcp-options reject 192.168.100.0/24
```

set interfaces macsec <interface> dhcp-options user-class <string>**Configure the DHCP client to send a specific user-class identifier in its DHCP requests on this interface.**

The DHCP server can interpret this identifier and provide specific configuration options based on it (for example, default routes). The user-class value typically groups DHCP clients with similar configuration needs (for example, employees, guests, or printers).

Example:

```
set interfaces macsec macsec0 dhcp-options user-class VyOS
```

set interfaces macsec <interface> dhcpv6-options duid <duid>**Configure a specific DUID for the DHCPv6 client on this interface.**

The DUID is an identifier used by a DHCPv6 client to get an IPv6 address from a DHCPv6 server. It consists of a 2-byte type field, followed by a variable-length identifier field up to 128 bytes. The format of the identifier part depends on the DUID type:

- **DUID-LLT:** The most common type, which includes a hardware type, a timestamp, and a MAC address.
- **DUID-EN:** Is based on a vendor's enterprise number and a unique identifier assigned by the vendor.
- **DUID-LL:** Includes only a MAC address.

The DHCP server matches the DUID against its database and provides configuration data (such as address, lease times, DNS servers, etc.) to the DHCP client.

Example:

```
set interfaces macsec macsec0 duid '0e:00:00:01:00:01:27:71:db:f0:00:50:56:bf:c5:6d'
```

set interfaces macsec <interface> dhcpv6-options no-release**Configure the DHCP client not to send a release message when it stops running on this interface.**

This helps retain the assigned address or prefix.

Example:

```
set interfaces macsec macsec0 dhcpv6-options no-release
```

set interfaces macsec <interface> dhcpv6-options parameters-only**Enable a stateless DHCPv6 client mode on this interface.**

In stateless mode, the DHCPv6 client requests only stateless configuration parameters from the DHCP server (for example, DNS server addresses). It doesn't request a stateful configuration, such as IPv6 addresses or prefixes.

Example:

```
set interfaces macsec macsec0 dhcpv6-options parameters-only
```

set interfaces macsec <interface> dhcpv6-options rapid-commit

Enable DHCPv6 rapid commit on this interface.

When enabled, the DHCP client and server skip the negotiation steps (Advertise and Request), completing the DHCPv6 configuration process in just two messages (Solicit and final Reply).

Example:

```
set interfaces macsec macsec0 dhcpv6-options rapid-commit
```

set interfaces macsec <interface> dhcpv6-options temporary

Configure the DHCPv6 client to request a temporary IPv6 address on this interface.

When configured, the DHCP client doesn't form an Identity Association for Non-temporary Addresses (IA_NA) partnership. Consequently, it only obtains a temporary IPv6 address and doesn't obtain a permanent one.

Example:

```
set interfaces macsec macsec0 dhcpv6-options temporary
```

DHCPv6 Prefix Delegation (PD)

VyOS supports DHCPv6 Prefix Delegation (DHCPv6-PD) as described in [RFC 3633](#). DHCPv6-PD is supported by most ISPs that provide native IPv6 for consumers on fixed networks.

set interfaces macsec <interface> dhcpv6-options pd <id> length <length>

Configure a specific prefix length for DHCPv6-PD requests on this interface.

Some ISPs provide only a /64 prefix by default. Use this command to request a different prefix length for a specific DHCPv6-PD request, ranging from /32 (if allowed by your ISP) down to /64. <id> is a unique identifier for the DHCPv6-PD request.

The default value is 64.

To request a /56 prefix from your ISP, use:

```
set interfaces macsec macsec0 dhcpv6-options pd 0 length 56
```

set interfaces macsec <interface> dhcpv6-options pd <id> interface <delegatee> address <address>

Configure the IPv6 interface identifier (host portion) for the delegatee interface.

The value must be a decimal integer. It is appended to the delegated prefix and the configured SLA ID to form the final IPv6 address.

By default, the host portion is generated based on the parent interface's MAC address (EUI-64 format).

Example:

If a /64 prefix is delegated to interface eth8 and you configure the host portion as 65535, the resulting IPv6 address will end with ::ffff, as 65535 corresponds to ffff in hexadecimal notation.

```
set interfaces macsec macsec0 dhcpv6-options pd 0 interface eth8 address 65534
```

```
set interfaces macsec <interface> dhcpv6-options pd <id> interface <delegatee> sla-id <id>
```

Configure the SLA ID for the delegatee interface.

The value must be a decimal integer greater than 0 and fit in the length of SLA IDs. It is converted to hexadecimal and appended to the delegated prefix to form the specific subnet prefix for the delegatee interface.

Example:

If SLA ID is 1 and the delegated prefix is 2001:db8:ffff::/48, the resulting subnet prefix for the delegatee interface will be 2001:db8:ffff:1::/64.

```
set interfaces macsec macsec0 dhcpv6-options pd 0 interface eth8 sla-id 1
```

MACsec options

```
set interfaces macsec <interface> security cipher <gcm-aes-128|gcm-aes-256>
```

Configure the cipher suite for the MACsec interface.

This configuration parameter is mandatory.

```
set interfaces macsec <interface> security encrypt
```

Enable encryption on the MACsec interface.

By default, MACsec interfaces only provide authentication; encryption is optional. When enabled, outgoing packets are encrypted using the configured cipher suite.

```
set interfaces macsec <interface> source-interface <physical-source>
```

Configure a physical source interface for the MACsec interface.

Traffic transmitted through this interface is authenticated and, if configured, encrypted.

MACsec key management

Static SAK (Secure Authentication Key) mode

In static SAK mode, administrators must manually configure and update SAKs on each MACsec peer. MKA (MACsec Key Agreement protocol) cannot be used in this mode.

```
set interfaces macsec <interface> security static key <key>
```

Configure the Transmit (TX) SAK for the MACsec interface.

The key must be a 16-byte (GCM-AES-128) or 64-byte (GCM-AES-256) hexadecimal string.

```
set interfaces macsec <interface> security static peer <peer> mac <mac address>
```

Configure the MAC address associated with the MACsec peer.

```
set interfaces macsec <interface> security static peer <peer> key <key>
```

Configure the RX SAK for traffic from the MACsec peer.

The key must be a 16-byte (GCM-AES-128) or 64-byte (GCM-AES-256) hexadecimal string.

```
set interfaces macsec <interface> security static peer <peer> disable
```

Dynamic MKA mode

In this mode, the MKA protocol is used to generate, distribute, and update CAKs (MACsec Connectivity Association Keys), and to authenticate MACsec peers.

set interfaces macsec <interface> security mka cak <key>

Configure the CAK (MACsec Connectivity Association Key) for the MACsec interface.

The CAK and its CKN (MACsec Connectivity Association Key Name) form the pre-shared master key pair used to authenticate MACsec peers.

set interfaces macsec <interface> security mka ckn <key>

Configure the CKN for the MACsec interface.

set interfaces macsec <interface> security mka priority <priority>

Configure the MKA key server priority for the MACsec interface. The peer with the lowest priority is elected as the key server.

Replay protection

set interfaces macsec <interface> security replay-window <window>

The replay protection window defines how many out-of-order frames can be received before they are dropped as a potential replay attack. The following values are valid:

- 0: Any out-of-order frame is immediately dropped.
- 1-4294967295: Allows the specified number of out-of-order frames.

Operation

run generate macsec mka cak <gcm-aes-128|gcm-aes-256>

Generate a 128-bit (GCM-AES-128) or 256-bit (GCM-AES-256) MKA CAK.

```
vyos@vyos:~$ generate macsec mka cak gcm-aes-128
20693b6e08bfa482703a563898c9e3ad
```

run generate macsec mka ckn

Generate an MKA CAK.

```
vyos@vyos:~$ generate macsec mka ckn
88737efef314ee319b2cbf30210a5f164957d884672c143aefdc0f5f6bc49eb2
```

show interfaces macsec

Show all MACsec interfaces.

```
vyos@vyos:~$ show interfaces macsec
17: macsec1: protect on validate strict sc off sa off encrypt on send_sci on end_
↪station off scb off replay off
cipher suite: GCM-AES-128, using ICV length 16
TXSC: 005056bfe0001 on SA 0
20: macsec0: protect on validate strict sc off sa off encrypt off send_sci on end_
↪station off scb off replay off
```

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```
cipher suite: GCM-AES-128, using ICV length 16
TXSC: 005056bfefaa0001 on SA 0
```

show interfaces macsec <interface>

Show information for a specific MACsec interface.

```
vyos@vyos:~$ show interfaces macsec macsec1
17: macsec1: protect on validate strict sc off sa off encrypt on send_sci on end_
↳ station off scb off replay off
cipher suite: GCM-AES-128, using ICV length 16
TXSC: 005056bfefaa0001 on SA 0
```

Examples**Site-to-site MACsec with dynamic MKA over an untrusted network**

In the following example, two routers (R1 and R2) are connected via an untrusted switch, using their eth1 interfaces as the underlay. The MACsec interface (macsec1) with dynamic MKA encrypts traffic between them.

Topology details:

- R1 IP addresses: 192.0.2.1/24 and 2001:db8::1/64.
- R2 IP addresses: 192.0.2.2/24 and 2001:db8::2/64.

R1

```
set interfaces macsec macsec1 address '192.0.2.1/24'
set interfaces macsec macsec1 address '2001:db8::1/64'
set interfaces macsec macsec1 security cipher 'gcm-aes-128'
set interfaces macsec macsec1 security encrypt
set interfaces macsec macsec1 security mka cak '232e44b7fda6f8e2d88a07bf78a7aff4'
set interfaces macsec macsec1 security mka ckn
↳ '40916f4b23e3d548ad27eedd2d10c6f98c2d21684699647d63d41b500dfe8836'
set interfaces macsec macsec1 source-interface 'eth1'
```

R2

```
set interfaces macsec macsec1 address '192.0.2.2/24'
set interfaces macsec macsec1 address '2001:db8::2/64'
set interfaces macsec macsec1 security cipher 'gcm-aes-128'
set interfaces macsec macsec1 security encrypt
set interfaces macsec macsec1 security mka cak '232e44b7fda6f8e2d88a07bf78a7aff4'
set interfaces macsec macsec1 security mka ckn
↳ '40916f4b23e3d548ad27eedd2d10c6f98c2d21684699647d63d41b500dfe8836'
set interfaces macsec macsec1 source-interface 'eth1'
```

Pinging (IPv6) the other host and intercepting traffic on eth1 confirm that the content is encrypted.

```
17:35:44.586668 00:50:56:bf:ef:aa > 00:50:56:b3:ad:d6, ethertype Unknown (0x88e5),
↳ length 150:
    0x0000: 2c00 0000 000a 0050 56bf efaa 0001 d9fb ,.....PV.....
    0x0010: 920a 8b8d 68ed 9609 29dd e767 25a4 4466 ....h...)..g%.Df
    0x0020: 5293 487b 9990 8517 3b15 22c7 ea5c ac83 R.H{....;."..\..
```

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```

0x0030: 4c6e 13cf 0743 f917 2c4e 694e 87d1 0f09 Ln...C...,NiN....
0x0040: 0f77 5d53 ed75 cfe1 54df 0e5a c766 93cb .w]S.u..T..Z.f..
0x0050: c4f2 6e23 f200 6dfe 3216 c858 dcaa a73b ..n#..m.2..X...;
0x0060: 4dd1 9358 d9e4 ed0e 072f 1acc 31c4 f669 M..X...../..1..i
0x0070: e93a 9f38 8a62 17c6 2857 6ac5 ec11 8b0e ..:8.b..(Wj.....
0x0080: 6b30 92a5 7ccc 720b k0..|.r.

```

Disabling encryption on the MACsec interface by removing the security encrypt option shows the unencrypted but authenticated content.

```

17:37:00.746155 00:50:56:bf:ef:aa > 00:50:56:b3:ad:d6, ethertype Unknown (0x88e5),
↳ length 150:
0x0000: 2000 0000 0009 0050 56bf efaa 0001 86dd .....PV.....
0x0010: 6009 86f3 0040 3a40 2001 0db8 0000 0000 `....@:@.....
0x0020: 0000 0000 0000 0001 2001 0db8 0000 0000 .....
0x0030: 0000 0000 0000 0002 8100 d977 0f30 0003 .....w.0..
0x0040: 1ca0 c65e 0000 0000 8d93 0b00 0000 0000 ...^.....
0x0050: 1011 1213 1415 1617 1819 1a1b 1c1d 1e1f .....
0x0060: 2021 2223 2425 2627 2829 2a2b 2c2d 2e2f .!"#$%&'()*+,-./
0x0070: 3031 3233 3435 3637 87d5 eed3 3a39 d52b 01234567....:9.+
0x0080: a282 c842 5254 ef28 ...BRT.(

```

Site-to-site MACsec with static SAK over an untrusted network

This example uses the same topology as above, but applies static SAK mode to the MACsec interface configuration.

R1

```

set interfaces macsec macsec1 address '192.0.2.1/24'
set interfaces macsec macsec1 address '2001:db8::1/64'
set interfaces macsec macsec1 security cipher 'gcm-aes-128'
set interfaces macsec macsec1 security encrypt
set interfaces macsec macsec1 security static key 'ddd6f4a7be4d8bbaf88b26f10e1c05f7'
set interfaces macsec macsec1 security static peer R2 mac 00:11:22:33:44:02
set interfaces macsec macsec1 security static peer R2 key
↳ 'eadcc0aa9cf203f3ce651b332bd6e6c7'
set interfaces macsec macsec1 source-interface 'eth1'

```

R2

```

set interfaces macsec macsec1 address '192.0.2.2/24'
set interfaces macsec macsec1 address '2001:db8::2/64'
set interfaces macsec macsec1 security cipher 'gcm-aes-128'
set interfaces macsec macsec1 security encrypt
set interfaces macsec macsec1 security static key 'eadcc0aa9cf203f3ce651b332bd6e6c7'
set interfaces macsec macsec1 security static peer R1 mac 00:11:22:33:44:01
set interfaces macsec macsec1 security static peer R1 key
↳ 'ddd6f4a7be4d8bbaf88b26f10e1c05f7'
set interfaces macsec macsec1 source-interface 'eth1'

```

MACsec over WAN

MACsec offers an alternative to traditional tunneling solutions by securing Layer 2 with integrity, origin authentication, and optional encryption.

While typically deployed between hosts and access switches, MACsec can also secure traffic over a WAN. In the following example, we combine VXLAN (for transport) and MACsec (for security) to create a secure tunnel between two sites.

R1 MACsec01

```
set interfaces macsec macsec1 address '192.0.2.1/24'
set interfaces macsec macsec1 address '2001:db8::1/64'
set interfaces macsec macsec1 security cipher 'gcm-aes-128'
set interfaces macsec macsec1 security encrypt
set interfaces macsec macsec1 security static key 'ddd6f4a7be4d8bbaf88b26f10e1c05f7'
set interfaces macsec macsec1 security static peer SEC02 key
↪ 'eadcc0aa9cf203f3ce651b332bd6e6c7'
set interfaces macsec macsec1 security static peer SEC02 mac '00:11:22:33:44:02'
set interfaces macsec macsec1 source-interface 'vxlan1'
set interfaces vxlan vxlan1 mac '00:11:22:33:44:01'
set interfaces vxlan vxlan1 remote '10.1.3.3'
set interfaces vxlan vxlan1 source-address '172.16.100.1'
set interfaces vxlan vxlan1 vni '10'
set protocols static route 10.1.3.3/32 next-hop 172.16.100.2
```

R2 MACsec02

```
set interfaces macsec macsec1 address '192.0.2.2/24'
set interfaces macsec macsec1 address '2001:db8::2/64'
set interfaces macsec macsec1 security cipher 'gcm-aes-128'
set interfaces macsec macsec1 security encrypt
set interfaces macsec macsec1 security static key 'eadcc0aa9cf203f3ce651b332bd6e6c7'
set interfaces macsec macsec1 security static peer SEC01 key
↪ 'ddd6f4a7be4d8bbaf88b26f10e1c05f7'
set interfaces macsec macsec1 security static peer SEC01 mac '00:11:22:33:44:01'
set interfaces macsec macsec1 source-interface 'vxlan1'
set interfaces vxlan vxlan1 mac '00:11:22:33:44:02'
set interfaces vxlan vxlan1 remote '10.1.2.2'
set interfaces vxlan vxlan1 source-address '172.16.100.2'
set interfaces vxlan vxlan1 vni '10'
set protocols static route 10.1.2.2/32 next-hop 172.16.100.1
```

7.4.9 OpenVPN

Traditionally, hardware routers use IPsec exclusively because it is easy to implement in hardware, and their CPUs lack sufficient power for software-based encryption. This limitation is less relevant for VyOS, as it is a software router.

OpenVPN has been widely used on UNIX platforms for a long time and is a popular choice for remote-access VPNs. It also supports site-to-site connections.

OpenVPN offers the following advantages:

- It uses a single TCP or UDP connection and does not rely on packet source addresses, so it works even through double NAT. This makes it well-suited for public hotspots.

- It is easy to set up and offers very flexible split tunneling.
- A variety of client GUI frontends are available for any platform.

Disadvantages include:

- It is slower than IPsec due to higher protocol overhead and because it runs in user mode, while IPsec on Linux runs in kernel mode.
- No operating system includes OpenVPN client software by default.

In the VyOS CLI, OpenVPN is configured as a network interface using `set interfaces openvpn` rather than `set vpn`, which is often overlooked.

Configuration

set interfaces openvpn <interface> authentication password <text>

Configure the password for the auth-user-pass authentication method.

This option applies only to OpenVPN clients.

set interfaces openvpn <interface> authentication username <text>

Configure the username for the auth-user-pass authentication method.

This option applies only to OpenVPN clients.

set interfaces openvpn <interface> description <description>

Configure the description for the OpenVPN interface.

set interfaces openvpn <interface> device-type <tap | tun>

Configure the virtual network device type for the OpenVPN interface:

- **tun (default):** Operates at Layer 3, encapsulating IPv4 or IPv6 packets.
- **tap:** Operates at Layer 2, encapsulating Ethernet 802.3 frames.

set interfaces openvpn <interface> disable

Disable the specific OpenVPN interface.

set interfaces openvpn <interface> encryption cipher < 3des | aes128 | aes128gcm | aes192 | aes192gcm | aes256 | aes256gcm | none >

Configure the static encryption cipher for the OpenVPN tunnel.

The `cipher` option maps to OpenVPN's `--cipher` directive and specifies the symmetric encryption algorithm for both control and data channels.

This was previously the default encryption method in all OpenVPN modes. In newer OpenVPN versions, the `--cipher` directive is considered **legacy** and should be used only in compatibility scenarios.

set interfaces openvpn <interface> encryption data-ciphers < 3des | aes128 | aes128gcm | aes192 | aes192gcm | aes256 | aes256gcm | none >

Configure a prioritized list of negotiated ciphers for OpenVPN in client or server mode.

The `data-ciphers` option represents a list of supported encryption algorithms. It corresponds to OpenVPN's `--data-ciphers` directive and enables cipher negotiation, where both peers automatically agree on a mutually supported cipher during session startup.

Note: This option is not compatible with `site-to-site` mode.

```
set interfaces openvpn <interface> encryption data-ciphers-fallback < 3des | aes128 |
aes128gcm | aes192 | aes192gcm | aes256 | aes256gcm | none >
```

Configure the fallback cipher for site-to-site mode.

The `data-ciphers-fallback` option maps to OpenVPN's `--data-ciphers-fallback` directive. It defines the cipher to use if negotiation is **not supported**.

Note: This option ensures consistent encryption between two static peers without cipher negotiation capability.

```
set interfaces openvpn <interface> hash <md5 | sha1 | sha256 | ...>
```

Configure the hashing algorithm for the OpenVPN interface.

```
set interfaces openvpn <interface> ip adjust-mss <mss | clamp-mss-to-pmtu>
```

Configure the MSS advertised in outgoing TCP SYN packets on the specified interface.

By clamping the MSS value in TCP SYN packets, you explicitly inform the remote side not to send packets larger than that size. This prevents connection issues that occur when Path MTU Discovery (PMTUD) fails.

The following options are available:

- **mss:** Sets the MSS to a specific value, in bytes. Use this option if you need to enforce a specific MSS, for example, to troubleshoot connectivity issues or accommodate specific network requirements.
- **clamp-mss-to-pmtu:** The router automatically calculates the MSS to be the interface's MTU minus 40 bytes for IPv4 traffic (20 bytes for the IPv4 header and 20 bytes for the TCP header). This option is recommended to automatically set the proper value.

```
set interfaces openvpn <interface> ip arp-cache-timeout
```

Configure how long an ARP entry remains valid after learning an IP-to-MAC address mapping on this interface.

The default duration is 30 seconds.

An ARP entry remains valid if it receives positive feedback from higher-level protocols.

Example:

```
set interfaces openvpn vtun0 ip arp-cache-timeout 180
```

```
set interfaces openvpn <interface> ip disable-arp-filter
```

Configure ARP filtering on this interface.

Default behavior: The kernel responds to ARP requests on this interface only if the traffic would be routed back to the ARP sender through that specific interface.

If configured: The kernel responds to ARP requests on this interface for any IP address configured on the local host, regardless of which specific interface that IP address is assigned to, and regardless of the routing table. This reflects the Linux concept that IP addresses belong to the host, not individual interfaces.

Example:

```
set interfaces openvpn vtun0 ip disable-arp-filter
```

```
set interfaces openvpn <interface> ip disable-forwarding
```

Configure the interface for host or router behavior.

If configured, the interface switches to host mode, and IPv4 forwarding is disabled on it.

Example:

```
set interfaces openvpn vtun0 ip disable-forwarding
```

set interfaces openvpn <interface> ip enable-directed-broadcast

Configure whether to forward IP-directed broadcast packets received on this interface.

Default behavior: IP-directed broadcast packets are dropped.

If configured: IP-directed broadcast packets are forwarded to all hosts on the destination subnet, as defined in [RFC 1812](#) and [RFC 2644](#).

Example:

```
set interfaces openvpn vtun0 ip enable-directed-broadcast
```

set interfaces openvpn <interface> ip enable-arp-accept

Configure how to process gratuitous ARPs on this interface.

If configured, an IP-to-MAC address mapping is added to the ARP table based on gratuitous ARP requests or replies.

Note: If the ARP table already contains the IP address from a gratuitous ARP, its entry is updated regardless of whether this setting is configured.

Example:

```
set interfaces openvpn vtun0 ip enable-arp-accept
```

set interfaces openvpn <interface> ip enable-arp-announce

Configure the source IP selection for ARP requests on this interface.

Default behavior: The kernel can use any IP address the host owns as the source IP address in ARP requests on this interface.

If configured: The kernel first attempts to select a source IP address configured on the interface that shares a common subnet with the target IP address. If there is no such subnet, the kernel selects the IP address it would normally use (based on the routing table to reach the target destination).

Example:

```
set interfaces openvpn vtun0 ip enable-arp-announce
```

set interfaces openvpn <interface> ip enable-arp-ignore

Configure which ARP requests will be ignored on this interface.

Default behavior: The kernel responds to ARP requests for any local IP addresses, regardless of which interface they are assigned to.

If configured: The kernel responds to ARP requests only if the target IP address is assigned to this specific interface.

Example:

```
set interfaces openvpn vtun0 ip enable-arp-ignore
```

set interfaces openvpn <interface> ip enable-proxy-arp

Configure proxy ARP on this interface.

If configured, the router (kernel) intercepts ARP requests for non-local IP addresses and replies with the MAC address of the interface that received the request. Subsequent packets destined to these IP addresses are forwarded to their actual destinations on remote subnets.

Example:

```
set interfaces openvpn vtun0 ip enable-proxy-arp
```

set interfaces openvpn <interface> ip proxy-arp-pvlan

Configure local proxy ARP on the interface.

If configured, the router (kernel) responds to ARP requests on this VLAN interface even if the target IP address resides on the same subnet and interface.

This is used to support network isolation requirements (RFC 3069) for private VLANs (PVLANS). In PVLAN configurations, hosts on isolated ports are NOT allowed to communicate directly with each other at Layer 2, but they can communicate with the upstream router.

By replying to inter-host ARP requests with its own MAC address, the router (kernel) directs inter-host traffic through itself instead of directly between hosts.

Note: This command works independently and does not require enabling the standard proxy ARP on the interface.

Local proxy ARP is also known as:

- VLAN aggregation ([RFC 3069](#)).
- Private VLAN (Cisco, Allied Telesyn).
- Source-port filtering or port isolation (Hewlett-Packard).
- MAC-Forced Forwarding (Ericsson).

set interfaces openvpn <interface> ip source-validation <strict | loose | disable>

Configure source IP address validation using RPF on this interface, as specified in [RFC 3704](#).

The following options are available:

- **strict:** Each incoming packet's source IP address is checked against the Forwarding Information Base (FIB). If the interface is not the best route back to that source, validation fails, and the packet is dropped.
- **loose:** Each incoming packet's source IP address is checked against the FIB. If the source IP address is unreachable through any interface, validation fails.
- **disable:** No source IP address validation is performed. All incoming packets are accepted.

RFC 3704 recommends enabling strict mode to prevent IP spoofing, such as DDoS attacks. For asymmetric or other complex routing scenarios, use loose mode.

set interfaces openvpn <interface> ipv6 address autoconf

Configure the interface to automatically obtain an IPv6 address using SLAAC, as specified in [RFC 4862](#).

IPv6 hosts can configure themselves automatically when connected to an IPv6 network using the Neighbor Discovery Protocol via ICMPv6 router discovery messages. When first connected to a network, a host sends a link-local router solicitation multicast request for its configuration parameters. The router responds with a router advertisement packet containing Internet Layer configuration parameters.

Note: This method automatically disables IPv6 traffic forwarding on the interface.

Example:

```
set interfaces openvpn vtun0 ipv6 address autoconf
```

set interfaces openvpn <interface> ipv6 address eui64 <prefix>

Configure the interface to assign itself an IPv6 address using the EUI-64 method, as specified in [RFC 4291](#).

Example:

```
set interfaces openvpn vtun0 ipv6 address eui64 2001:db8:beef::/64
```

set interfaces openvpn <interface> ipv6 address no-default-link-local

Disable the automatic assignment of a link-local IPv6 address to this interface.

Example:

```
set interfaces openvpn vtun0 ipv6 address no-default-link-local
```

set interfaces openvpn <interface> ipv6 disable-forwarding

Configure the interface for host or router behavior.

If configured, the interface switches to host mode, and IPv6 forwarding is disabled on it.

Example:

```
set interfaces openvpn vtun0 ipv6 disable-forwarding
```

set interfaces openvpn <interface> ipv6 adjust-mss <mss | clamp-mss-to-pmtu>

Configure the MSS advertised in outgoing TCP SYN packets on the specified interface.

By clamping the MSS value in TCP SYN packets, you explicitly inform the remote side not to send packets larger than that size. This prevents connection issues when Path MTU Discovery (PMTUD) fails.

The following options are available:

- **mss:** Set the MSS to a specific value, in bytes. Use this option to enforce a specific MSS, for example, to troubleshoot connectivity issues or accommodate specific network requirements.
- **clamp-mss-to-pmtu:** The router calculates the MSS to be the interface's MTU minus 60 bytes for IPv6 traffic (40 bytes for the IPv6 header and 20 bytes for the TCP header). This option is recommended to automatically set the proper value.

set interfaces openvpn <interface> ipv6 accept-dad <0 | 1 | 2>

Configure IPv6 DAD on the interface.

The following options are available:

- **0:** Disables DAD. No duplicate address detection is performed.

- **1:** Enables DAD (default). Duplicate addresses are detected. The interface's IPv6 operation continues for valid IPv6 addresses.
- **2:** Enables DAD and, if a MAC-based duplicate link-local address is found, disables IPv6 operation on this interface.

Example:

```
set interfaces openvpn vtun0 ipv6 accept-dad 2
```

set interfaces openvpn <interface> ipv6 dup-addr-detect-transmits <n>

Configure the number of DAD messages that the router (kernel) sends during IPv6 address assignment on this interface.

The default value is 1.

Example:

```
set interfaces openvpn vtun0 ipv6 dup-addr-detect-transmits 5
```

set interfaces openvpn <interface> keep-alive failure-count <value>

Configure the number of tolerated keepalive packet failures.

Default: 60 consecutive failures.

set interfaces openvpn <interface> keep-alive interval <value>

Configure the frequency, in seconds, at which keepalive packets are sent.

Default: 10 seconds.

set interfaces openvpn <interface> local-address <address>

Configure the local tunnel IP address for site-to-site mode.

set interfaces openvpn <interface> local-host <address>

Configure the local IP address to accept connections.

If configured, OpenVPN binds to this IP address only.

By default, OpenVPN binds to all interfaces.

set interfaces openvpn <interface> local-port <port>

Configure the local port to accept connections.

set interfaces openvpn <interface> mirror egress <monitor-interface>

Configure mirroring of outgoing traffic from this OpenVPN interface to the designated monitor interface.

set interfaces openvpn <interface> mirror ingress <monitor-interface>

Configure mirroring of incoming traffic from this OpenVPN interface to the designated monitor interface.

set interfaces openvpn <interface> mode <site-to-site | server | client>

Configure OpenVPN operation mode:

- **site-to-site:** Establishes a site-to-site VPN connection.
- **client:** Operates as a client in server-client mode.
- **server:** Operates as a server in server-client mode.

OpenVPN Data Channel Offload (DCO)

OpenVPN DCO (Data Channel Offload) improves the performance of encrypted OpenVPN data processing by keeping most data handling in the kernel and avoiding frequent context switches between the kernel and user space.

As a result, packet processing becomes more efficient and may utilize hardware encryption offload support available in the kernel.

Note:

- DCO is an **experimental**, not fully supported OpenVPN feature. Some OpenVPN features and deployment scenarios are **not compatible** with DCO.

For a complete list of supported features, visit: <https://community.openvpn.net/openvpn/wiki/DataChannelOffload/Features>

- DCO is configured per tunnel and disabled by default. Existing tunnels operate without DCO unless it is explicitly enabled.
 - Enabling DCO resets the interface.
-

Best practice: Create a new tunnel with DCO enabled to avoid compatibility issues with existing clients.

set interfaces openvpn <interface> offload dco

Enable DCO for the specified OpenVPN interface.

Example:

```
set interfaces openvpn vtun0 offload dco
```

This command enables DCO and loads the required kernel module.

set interfaces openvpn <interface> openvpn-option <text>

Add raw OpenVPN configuration options to the openvpn.conf file.

OpenVPN provides many configuration options, but not all are available in the VyOS CLI.

If a required option is missing, you may submit a feature request at Phabricator so all users can benefit from it (see Contributing/Issues and Features).

Alternatively, use `openvpn-option` to pass raw OpenVPN configuration options to the `openvpn.conf` file.

Warning: Use this option only as a last resort. Invalid options or syntax may prevent OpenVPN from starting. Check system logs for errors after applying changes.

Example:

```
set interfaces openvpn vtun0 openvpn-option 'persist-key'
```

This command adds `persist-key` to the configuration file. This solves the problem by persisting keys across resets, so they do not need to be re-read.

```
set interfaces openvpn vtun0 openvpn-option 'route-up &quot;/config/auth/tun_up.sh_
↪ arg1&quot;;'
```

This command adds `route-up "/config/auth/tun_up.sh arg1"` to the configuration file. This option is executed after connection authentication, either immediately or after a short delay, as defined.

Ensure the path and arguments are enclosed in single or double quotes.

Note: Some raw configuration options require quotes. To include them, use the “ statement.

set interfaces openvpn <interface> persistent-tunnel

Enable always-active mode for the TUN/TAP device.

When enabled, the TUN/TAP device remains active upon connection resets or daemon reloads.

set interfaces openvpn <interface> protocol <udp | tcp-passive | tcp-active >

Configure the protocol for OpenVPN communication with a remote host:

- **udp (default):** Uses the UDP protocol.
- **tcp-passive:** Uses the TCP protocol and accepts connections passively.
- **tcp-active:** Uses the TCP protocol and initiates connections actively.

set interfaces openvpn <interface> redirect <interface>

Enable redirection of incoming packets to the specified interface.

set interfaces openvpn <interface> remote-address <address>

Configure the remote tunnel IP address for site-to-site mode.

set interfaces openvpn <interface> remote-host <address | host>

Configure the IPv4/IPv6 address or hostname for a server device if OpenVPN runs in client mode.

This setting is not used in server mode.

set interfaces openvpn <interface> remote-port <port>

Configure the remote port to connect to the server.

set interfaces openvpn <interface> replace-default-route

Configure the OpenVPN tunnel as the default route.

set interfaces openvpn <interface> server bridge disable

Disable the given instance.

set interfaces openvpn <interface> server bridge gateway <ipv4 address>

Configure the gateway IP address.

set interfaces openvpn <interface> server bridge start <ipv4 address>

Configure the first IP address in the pool to allocate to connecting clients.

set interfaces openvpn <interface> server bridge stop <ipv4 address>

Configure the last IP address in the pool to allocate to connecting clients.

set interfaces openvpn <interface> server bridge subnet-mask <ipv4 subnet mask>

Configure the subnet mask pushed to dynamic clients.

set interfaces openvpn <interface> server client <name>

Configure the Common Name (CN) specified in the client certificate.


```
set interfaces openvpn <interface> server client <name> disable
```

Disable the client connection.

```
set interfaces openvpn <interface> server client <name> ip <address>
```

Configure the IPv4/IPv6 address for the client.

```
set interfaces openvpn <interface> server client <name> push-route <subnet>
```

Configure a route to be pushed to the specific client.

```
set interfaces openvpn <interface> server client <name> subnet <subnet>
```

Configure a fixed subnet to be routed from the server to the specified client.

Used as OpenVPN's `iroute` directive.

```
set interfaces openvpn <interface> server client-ip-pool start <address>
```

Configure the first IP address in the subnet's IPv4 pool to be dynamically allocated to connecting clients.

```
set interfaces openvpn <interface> server client-ip-pool stop <address>
```

Configure the last IP address in the subnet's IPv4 pool to be dynamically allocated to connecting clients.

```
set interfaces openvpn <interface> server client-ip-pool subnet <netmask>
```

Configure the subnet mask pushed to dynamic clients.

Use this command only for the TAP device type. Do not use it for bridged interfaces.

```
set interfaces openvpn <interface> server client-ipv6-pool base <ipv6addr/bits>
```

Configure the IPv6 address pool for dynamic assignment to clients.

```
set interfaces openvpn <interface> server domain-name <name>
```

Configure the DNS suffix to be pushed to all clients.

```
set interfaces openvpn <interface> server max-connections <1-4096>
```

Configure the maximum number of client connections.

```
set interfaces openvpn <interface> server mfa totp challenge <enable | disable>
```

If enabled, `openvpn-otp` expects a password as a result of the challenge/ response protocol.

```
set interfaces openvpn <interface> server mfa totp digits <1-65535>
```

Configure the number of digits to use for the TOTP (Time-based One-Time Password) hash.

Default: 6.

```
set interfaces openvpn <interface> server mfa totp drift <1-65535>
```

Configure the time drift in seconds.

Default: 0.

```
set interfaces openvpn <interface> server mfa totp slop <1-65535>
```

Configure the allowed clock slop in seconds.

Default: 180.

```
set interfaces openvpn <interface> server mfa totp step <1-65535>
```

Configure the step value for TOTP in seconds.

Default: 30.

```
set interfaces openvpn <interface> server name-server <address>
```

Define the client DNS configuration to be used with the connection.

```
set interfaces openvpn <interface> server push-route <subnet>
```

Configure the route to be pushed to all clients.

```
set interfaces openvpn <interface> server reject-unconfigured-client
```

Reject connections from clients that are not explicitly configured.

```
set interfaces openvpn <interface> server subnet <subnet>
```

Configure the IPv4 or IPv6 network.

This parameter is mandatory when operating in server mode.

```
set interfaces openvpn <interface> server topology < net30 | point-to-point | subnet>
```

Configure the virtual addressing topology for tun mode.

This command does not affect tap mode, which always uses the subnet topology.

- **subnet (default):** Allocates a single IP address to each connecting client. This is the recommended topology.
- **net30:** Allocates a /30 subnet to each connecting client. This is a legacy topology used to support Windows clients. It is now effectively deprecated.
- **point-to-point:** Creates a point-to-point topology where the remote endpoint of the client's tun interface always points to the local endpoint of the server's tun interface.

Like subnet, this topology allocates a single IP address per client. Use it only if no clients run Windows operating systems.

```
set interfaces openvpn <interface> shared-secret-key <key>
```

Configure the static secret key for a site-to-site OpenVPN connection.

```
set interfaces openvpn <interface> tls auth-key <key>
```

Configure the TLS secret key for tls-auth.

This adds an HMAC signature to all SSL/TLS handshake packets to verify integrity.

Use `run generate pki openvpn shared-secret install <name>` to generate the key.

```
set interfaces openvpn <interface> tls ca-certificate <name>
```

Configure the Certificate Authority chain in the PKI configuration.

```
set interfaces openvpn <interface> tls certificate <name>
```

Configure the certificate name in the PKI configuration.

```
set interfaces openvpn <interface> tls crypt-key
```

Configure a shared secret key to provide an additional level of security, a variant similar to tls-auth.

```
set interfaces openvpn <interface> tls dh-params
```

Configure Diffie-Hellman parameters for server mode.

```
set interfaces openvpn <interface> tls peer-fingerprint <text>
```

Configure the peer certificate SHA256 fingerprint for site-to-site mode.

```
set interfaces openvpn <interface> tls role <active | passive>
```

Configure the TLS negotiation role, preferably used in site-to-site mode:

- **active:** Initiates TLS negotiation actively.
- **passive:** Waits for incoming TLS connections.

set interfaces openvpn <interface> tls tls-version-min <1.0 | 1.1 | 1.2 | 1.3 >

Configure the minimum TLS version to be accepted from the peer.

set interfaces openvpn <interface> use-lzo-compression

Configure fast LZO compression on this TUN/TAP interface.

set interfaces openvpn <interface> vrf <name>

Assign the interface to a specific VRF instance.

Operation mode

show openvpn site-to-site

Show tunnel status for OpenVPN site-to-site interfaces.

show openvpn server

Show tunnel status for OpenVPN server interfaces.

show openvpn client

Show tunnel status for OpenVPN client interfaces.

show log openvpn

Show logs for all OpenVPN interfaces.

show log openvpn interface <interface>

Show logs for the specific OpenVPN interface.

reset openvpn client <text>

Reset the specified OpenVPN client.

reset openvpn interface <interface>

Reset the OpenVPN process on the specified interface.

generate openvpn client-config interface <interface> ca <name> certificate <name>

Generate an OpenVPN client configuration file in the .ovpn format for client machines.

Examples

This section covers examples of OpenVPN configurations for various deployments.

Site-to-site

Todo: Convert raw command blocks in this file to `cfgcmd/opcmd` directives for command coverage tracking.

OpenVPN is popular for client-server setups, but its site-to-site mode is less common and often not supported by router appliances. Despite limited support, it is effective for quickly establishing tunnels between routers.

As of VyOS 1.4, OpenVPN site-to-site mode can use either pre-shared keys or x.509 certificates.

Pre-shared key mode is now deprecated and will be removed from future OpenVPN versions. VyOS will also discontinue support for this option because pre-shared keys are significantly less secure than TLS.

We will configure OpenVPN with self-signed certificates, and then discuss the legacy pre-shared key mode.

In both cases, we will use the following settings:

- The public IP address of the local VPN endpoint is 198.51.100.10.
- The public IP address of the remote VPN endpoint is 203.0.113.11.
- The tunnel uses 10.255.1.1 for the local IP address and 10.255.1.2 for the remote IP address.
- The local site has a subnet of 10.0.0.0/16.
- The remote site has a subnet of 10.1.0.0/16.
- The official OpenVPN port 1194 is reserved for client VPN. For site-to-site VPN, port 1195 is used.
- The `persistent-tunnel` directive allows us to configure tunnel-related attributes, such as firewall policy, as we would on any standard network interface.
- If known, the remote router's IP address can be configured using the `remote-host` directive. If unknown, it can be omitted. We assume the remote router has a dynamic IP address.

Set up site-to-site certificates

Deploying a complete Public Key Infrastructure (PKI) with a Certificate Authority (CA) would overcomplicate site-to-site OpenVPN setups, which are primarily designed for simplicity. To keep their configuration simple without compromising security, VyOS 1.4 and later lets you verify self-signed certificates using certificate fingerprints.

Generate a self-signed certificate on each router, preferably using the Elliptic Curve (EC) type. In configuration mode, run the following command: `run generate pki certificate self-signed install <name>`. This adds the certificate to the configuration session's `pki` subtree. Review and commit the changes.

```
vyos@vyos# run generate pki certificate self-signed install openvpn-local
Enter private key type: [rsa, dsa, ec] (Default: rsa) ec
Enter private key bits: (Default: 256)
Enter country code: (Default: GB)
Enter state: (Default: Some-State)
Enter locality: (Default: Some-City)
Enter organization name: (Default: VyOS)
Enter common name: (Default: vyos.io)
Do you want to configure Subject Alternative Names? [y/N]
Enter how many days certificate will be valid: (Default: 365)
Enter certificate type: (client, server) (Default: server)
```

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Note: If you plan to use the generated key on this router, do not encrypt the private_

key.

Do you want to encrypt the private key with a passphrase? [y/N]

2 value(s) installed. Use "compare" to see the pending changes, and "commit" to apply.
[edit]

```
vyos@vyos# compare
```

```
[pki]
```

```
+ certificate openvpn-local {
```

```
+   certificate "MIICJTCCAcugAwIBAgIUMXLfRNJ5i0jk/
```

```
→ uAZqUe4phW8MdgwCgYIKoZIzj0EAwIwVzELMAkGA1UEBhMCR0IxExARBgNVBAGMC1NvbWUtU3RhdGUxEjAQBgNVBACMCVnbWUtQ2
```

```
→ lw9Eq9Q89r247AJR6ec/GT26AicVA1bsongV1YaWvRwzTPC/yi5pkzV/PcT/
```

```
→ WU7JQIyMWo3UwczAMBgNVHRMBAf8EAjAAMA4GA1UdDwEB/
```

```
→ wQEAWIHgDATBgNVHSUEDDAKBggrBgEFBQcDATAdbGgNVHQ4EFgQUBrAxRdFppdG/
```

```
→ UBRdo7qNyHutaTQwHwYDVR0jBBgwFoAUBrAxRdFppdG/
```

```
→ UBRdo7qNyHutaTQwCgYIKoZIzj0EAwIDSAAwRQIhAI2+8C92z9wTcTWkQ/
```

```
→ goRxs10EBC+h780+vgo9k97z5iAiBSeqfaVr5taQTS31+McGTAK3cYWNTg0Dl0BI8aK02oRg=="
```

```
+   private {
```

```
+       key "MIGHAgEAMBMGBYqGSM49AgEGCCqGSM49AwEHBG0wawIBAQQgtOeEb0dMb5P/
```

```
→ 2Exi09WWvk6Cvz0oBoDuP68ZimS2LShRANCAASp7D0vE3SKSAWAzr/lw9Eq9Q89r247AJR6ec/
```

```
→ GT26AicVA1bsongV1YaWvRwzTPC/yi5pkzV/PcT/WU7JQIyMW"
```

```
+   }
```

```
+ }
```

```
[edit]
```

```
vyos@vyos# commit
```

You do **not** need to copy the certificate to the other router. Instead, retrieve its SHA-256 fingerprint. Since OpenVPN currently supports only SHA-256 fingerprints, use the following command:

```
vyos@vyos# run show pki certificate openvpn-local fingerprint sha256
```

```
5C:B8:09:64:8B:59:51:DC:F4:DF:2C:12:5C:B7:03:D1:68:94:D7:5B:62:C2:E1:83:79:F1:F0:68:B2:81:26:79
```

Note: Certificate names are arbitrary. While `openvpn-local` and `openvpn-remote` are used here, you may choose any names.

Repeat the procedure on the other router.

Set up site-to-site OpenVPN

Local configuration:

Configure the tunnel:

```
set interfaces openvpn vtun1 mode site-to-site
```

```
set interfaces openvpn vtun1 protocol udp
```

```
set interfaces openvpn vtun1 persistent-tunnel
```

```
set interfaces openvpn vtun1 remote-host '203.0.113.11' # Public_
```

```
→ IP of the other side
```

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```
set interfaces openvpn vtun1 local-port '1195'
set interfaces openvpn vtun1 remote-port '1195'
set interfaces openvpn vtun1 local-address '10.255.1.1' # Local
↳ IP of vtun interface
set interfaces openvpn vtun1 remote-address '10.255.1.2' # Remote
↳ IP of vtun interface
set interfaces openvpn vtun1 tls certificate 'openvpn-local' # The
↳ self-signed certificate
set interfaces openvpn vtun1 tls peer-fingerprint <remote cert fingerprint> # The
↳ output of 'run show pki certificate <name> fingerprint sha256' on the remote router
set interfaces openvpn vtun1 tls role active
```

Remote configuration:

```
set interfaces openvpn vtun1 mode site-to-site
set interfaces openvpn vtun1 protocol udp
set interfaces openvpn vtun1 persistent-tunnel
set interfaces openvpn vtun1 remote-host '198.51.100.10' # Pub
↳ IP of other site
set interfaces openvpn vtun1 local-port '1195'
set interfaces openvpn vtun1 remote-port '1195'
set interfaces openvpn vtun1 local-address '10.255.1.2' # Local
↳ IP of vtun interface
set interfaces openvpn vtun1 remote-address '10.255.1.1' #
↳ Remote IP of vtun interface
set interfaces openvpn vtun1 tls certificate 'openvpn-remote' # The
↳ self-signed certificate
set interfaces openvpn vtun1 tls peer-fingerprint <local cert fingerprint> # The
↳ output of 'run show pki certificate <name> fingerprint sha256' on the local router
set interfaces openvpn vtun1 tls role passive
```

Set up pre-shared keys

Before VyOS 1.4, site-to-site OpenVPN without PKI required pre-shared keys. This option is still available but is deprecated and will be removed in future releases. If you need to set up a tunnel to an older VyOS version or a system with older OpenVPN, you still need to use pre-shared keys.

First, generate a key by running `run generate pki openvpn shared-secret install <name>` in configuration mode. You can use any name; in this example, we use `s2s`.

```
vyos@local# run generate pki openvpn shared-secret install s2s
2 value(s) installed. Use "compare" to see the pending changes, and "commit" to apply.
[edit]
vyos@local# compare
[pki openvpn shared-secret]
+ s2s {
+   key
↳ "7c73046a9da91e874d31c7ad894a32688cda054bde157c64270f28eceeabc0bb2f44dbb70335fad45148b0456aaa78cb34a34"
↳ "
+   version "1"
+ }
```

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```
[edit]
vyos@local# commit
[edit]
```

Next, install the key on the remote router:

```
vyos@remote# set pki openvpn shared-secret s2s key <generated key string>
```

Finally, configure the key in your OpenVPN interface settings:

```
set interfaces openvpn vtun1 shared-secret-key s2s
```

Set up firewall exceptions

To allow OpenVPN traffic to pass through the WAN interface, create a firewall exception:

```
set firewall ipv4 name OUTSIDE_LOCAL rule 10 action 'accept'
set firewall ipv4 name OUTSIDE_LOCAL rule 10 description 'Allow established/related'
set firewall ipv4 name OUTSIDE_LOCAL rule 10 state 'established'
set firewall ipv4 name OUTSIDE_LOCAL rule 10 state 'related'
set firewall ipv4 name OUTSIDE_LOCAL rule 20 action 'accept'
set firewall ipv4 name OUTSIDE_LOCAL rule 20 description 'OpenVPN_IN'
set firewall ipv4 name OUTSIDE_LOCAL rule 20 destination port '1195'
set firewall ipv4 name OUTSIDE_LOCAL rule 20 log
set firewall ipv4 name OUTSIDE_LOCAL rule 20 protocol 'udp'
```

Apply the OUTSIDE_LOCAL firewall group to the WAN interface and to the input filter for traffic destined for the router itself:

```
set firewall ipv4 input filter rule 10 action 'jump'
set firewall ipv4 input filter rule 10 inbound-interface name eth0
set firewall ipv4 input filter rule 10 jump-target OUTSIDE_LOCAL
```

Static routing:

Configure static routes by referencing the tunnel interface. For example, if the local router's network is 10.0.0.0/16 and the remote router's network is 10.1.0.0/16, define the routes as follows:

Local configuration:

```
set protocols static route 10.1.0.0/16 interface vtun1
```

Remote configuration:

```
set protocols static route 10.0.0.0/16 interface vtun1
```

As with standard Ethernet interfaces, you can apply firewall policies to the tunnel interface for input, output, and forward directions.

If you use multiple tunnels, OpenVPN must distinguish between them beyond just the pre-shared key. To achieve this, assign either unique IP addresses or unique ports to each tunnel.

Verify OpenVPN status using the show openvpn operational commands.

In OpenVPN's server-client mode, the server acts as a central hub, allowing multiple clients to connect and securely route their traffic or access a private network. Multi-client server is the most popular OpenVPN mode for routers.

Server-client mode always uses x.509 authentication and therefore requires a PKI setup. The PKI utility now simplifies the creation of Certificate Authorities (CAs), server and client certificates, and Diffie-Hellman keys directly in VyOS using configuration or operational mode commands.

Certificate Authority (CA):

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```

→BAQDagGGB0GA1UdJQQWMBQGCCsGAQUFBwMCBggrBgEFBQcDATAdbGnVHQ4EFgQUAG9lvr7AzJ/y4vY/
→XlWxXru+6m0wDQYJKoZIhvcNAQELBQADggEBAKsu4eZa8Fha9aKfuKqLGQHPpEFfVdaVJmebw0uMw+b5Y8EpBxzZrqbqbk3Mty8pB
→+xMTqkukkpLbioTeIDg6FCU2HYPY5emDF5DDZAZWxtTqi0zdT3Y6FqiTvs5VuWwXCcp+HM+Lwe1/
→VVJhwi4CHTq0CKWnQIH5blyjmyxzRBlrlZm4ntWlL5Mtepa1A3DJirY4kw/SqMAAh/
→Q9lh41JzBc8epf+OdnOzK55YmtmctGO2o+NBCFi0="
+   private {
+       key
→"MIIEvgIBADANBgkqhkiG9w0BAQEFAASCBAKgwggSkAgEAAoIBAQDi+v6i241T9ABxq1ngjWxDQITkqjV0nq2Jb3HSSuQpXRCu7DWd
→WTL0RNgkhS4iV/
→WYhE+bLihwiZ0GTeQnUd1QJJSkusFROX46w6kKXYUR5IQtcBC+vdky8PESynPd+DXsJn5X9JTWqDeviUAQz/
→ZjDzWk+71MBCqa+Zps1zpIjK0ywn7pR/HnDrxJOQXlBMNgvbw8U3IAZ2jJp0jTB8TnuDtWSA+XZeJmM/EN/
→AWUQyliX60JFSCiHBL2BZ9lmVms4/
→HkRpb50k3vvCoz+laOEE6VsH0fEdLC3lZ+CtXZ7kjp2wdWWUs5ggIJYZZkixsCisbtEmbAgMBAACggEAAZdykF6wV8Z4n8NsoG4
→ZJbWEhWj03x1y3JNtJw735LhmmysMSsreToXtxGfgYRTgNwt5l7oHoqmGHCsLx01NBb5A7JB1lIkIwUYqn31syOJofg0NsJpuwZ2
→WiCNjsEotNx8Kk70Vwt43DbkN/
→rbF5lxquuLedaSspOHuhIAOfZB5ZySfqohQalSAaguVD66rGPMrerZ2Vc7B1iJ6Mn/
→KZrSaQeHwyWrwDDHdzVwG9waydevtGTVO0dvH4etWnRypDx8p1FPJJKD4XVcsl3rR6oQKBgQD497Ep2RjcbNnKVj+kNXGriSGmyRSj
→otMHMAxIOcDMX6vRn2MB2pqVqwQUBQy6JfLrSemdpIjMN9xlX6Dw3BWP39SdewZ896/
→Eo0Q1ythMj1ORp+u3Pq0lSa14Cy9aPwDWmNy2deD68YDnsQKBgQDpZE/
→T84BMQ0FzL6NRijZrZr6Dc775hRfmmSTYI0KqpG0gXNSc5UgrWSLN5H7fnx36mT01P7UkgXCInV0AlJOfkt4a8QTqM1Fh/
→rZbLLWpQE55S6Fs28GDiFYl2kvZT/TtxhA/E0P0f/YXl/
→8KITS7ZVAZxE8rxBelhVUfDbnlCwKBgQDeWUguGaGeTDCMNk8MnNbYPdaCAB+mRp3G6ls51sF4qi5Lltva2jKn3H/
→AooHzaP3vGzUm0WLACdsAct2QQXtnCsN9FBtJK2+qzKEN0dPR7X/
→s3IGdRse6BX+b6BFgSnfGmumxI7L86L1JoHXCTnTQ0x0FOjNjdI3ZnplZRIPdYQKBgFJacASU9l9yl+SiGZnLEDG7FBpEPE3lVbKr
→FZwdAisMhtU/61TdY65BTAoGBAM4jKeImiXta5lzlRgNiW/TPhft3UbOLj3TbzchNCNAamqCv4Tmh9YKB2d/
→mz2hNxbnAGe2cYn4iRYKcjJLMZ0UfBL2WxlrgQYQPPGzSD0fh1pLIXPohpBZpsGqNR3Nc8Jd+Uw3IiIJ2oxPCOPT0JsklNB0Xf1Al
→"
+   }
+ }

[edit]
vyos@vyos# commit

```

Server certificate:

```

vyos@vyos# run generate pki certificate sign ca-1 install srv-1
Do you already have a certificate request? [y/N] N
Enter private key type: [rsa, dsa, ec] (Default: rsa)
Enter private key bits: (Default: 2048)
Enter country code: (Default: GB)
Enter state: (Default: Some-State)
Enter locality: (Default: Some-City)
Enter organization name: (Default: VyOS)
Enter common name: (Default: vyos.io) srv-1
Do you want to configure Subject Alternative Names? [y/N]
Enter how many days certificate will be valid: (Default: 365)
Enter certificate type: (client, server) (Default: server) server
Note: If you plan to use the generated key on this router, do not encrypt the private
→key.
Do you want to encrypt the private key with a passphrase? [y/N]
2 value(s) installed. Use "compare" to see the pending changes, and "commit" to apply.
[edit]

```

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```

vyos@vyos# compare
[pki certificate]
+ srv-1 {
+   certificate
  ↪ "MIIDrDCCApSgAwIBAgIULpu+qZj fG01kUI58XNmQXbQC3qQwDQYJKoZIhvcNAQELBQAwVDELMAkGA1UEBhMCROIx EzARBgNVBAGM
  ↪ 8mhYGjNY4OnZ7NYz3FU/
  ↪ iglo8qFtaZ26mWtPWpv2xW1F8JAEK715BAg42cBucxiIZFeRm+jkE6VN1bcNU0utnn3sbCwZMyH6pS9k08G1qrrFLA7ZFhv5AmgJc
  ↪ 603e06ICnVjvIfHLb+9DKKUEffHzFV8RrkqVCGmgisz9fF+j1Rvg9s+y1Nc2lZJTbb1XnzixvSRro4t9I3uIWdpJ0iOu09YiTXGQg
  ↪ wQEAWIHgDATBgNVHSUEDDAKBggrBgEFBQCdATAcBgNVHQ4EFgQUZuH0h4vBxma89HF9rUQ+DW052c5swHwYDVR0jBBgwFoAUAG9lvr
  ↪ y4vY/XlWxXru+6m0wDQYJKoZIhvcNAQELBQADggEBAI/Cyd0y7AJ7wY3yRssCud2iJA19/
  ↪ ZjgXzXOUo6ibawYIYOnSf9tS3eD4CIH4BgppDXoJZ/
  ↪ qEA4WvIsLx3yvny0xiqyk3TQmKIZ27VJH+yQkgzPeiKrHn1pCXBKEb1/jlT80zu4Lmn/
  ↪ oFwDH6nk8toxI8DM+qsTxqUFlTA3ea9yaRtxeNPMWJdaxZSUyGVSZL0wVKw5ZuQ1Gn7vGVApWlYDKYbMozCuZUG1q8wMzFBRa7x0a
  ↪ CvmmQ7k3uPg95GYSNddFqEMbQHoyd8hdCo="
+   private {
+     key "MIIEvQIBADANBgkqhkiG9w0BAQEFAASCBKcwggSjAgEAAoIBAQDKx0sx9UfrdpWJ0lPtsh/
  ↪ 7MvCQuX7Tucay/hqWIYRu5+vbflEEedXgmfwttD6cH/yaFgaM1jg6dns1jPcVT+KCWjyoW1pnbqZa09am/
  ↪ bFbUXwkaQRuXkECDjZwG5zGIhkV5Gb60QTPU3Vtw1TS62efexsLBkzIfqlL2TTwbWqusUsDtkWG/
  ↪ kCaAlw40Y7yyKBIC7utFL/
  ↪ o7d47ogKdW08h8ctv70MopQR98fMVXxGuSpUIaaCKzP18X6PVG+D2z7KU1zaVklNtvVef0LG9JGuji30je4hZ2knSI67T1iJNcZCA
  ↪ 4wkJXriELWayD7xXc2vD1AgMBAAECggEACsUk3PVzSX11+ekTDigM7NHK11UpEQPoGu/GR70mBKIK9BCyI/
  ↪ N9W0YAPE09kn4p9KNrINGXzKV3sVLBnXEyTmzyRl5Fs9YxLBF0X7eIcSVPHBVvU2CVHKez5uX2ypKfNax7A6FRUNq1FbwtXdNfLoU
  ↪ TTSGQkaP/z/cwIsXOLfG9m6iFkw98ShUzalrUWN0/
  ↪ 4fJKl01+DvXVYE9sv9rjD8J7DtAbr5KyKQ5n0AALZTCWQ7jwMybSnjjY9ypZUms0l17raJrfhrdbWayc6xMDvtrmNIDebkF+J7CHU
  ↪ 7yjjZgUSM2ANcHMDzQKBgQDmTi5tUeaj1JUS19lAP/
  ↪ XUzcElw2tcU1B8qpX69J4ofjTNgj5okLWQZVIy1UyAfLOI3LJbHTBUtSvedhH0VaMulq99NXs5qnbPGG3//
  ↪ RBAC0wKhJknB5Qv0D3FxmI14kM06jzPly+aIGek4dTtHvZuHbbVHbKSZ5MMouLYt+SS7wKBgQDhZETARZ0MazeWRaPjwdkjlfnCqq
  ↪ mPcv3tF07uUmeYvRC5fGYrghq29NADmUefOopAc06Izd4A3iqWwKBgQC1uPrpR7a1jwgRo7/
  ↪ I8q8H01MseQY903+u3ut5GYuyZ+NCRYL4/
  ↪ zZEua4ibivvNnZzh7E0M9PvAwWag4+n0+uG11+hbJH07rLQtnYVh5lLQa6+neI66cAD+kzDwH1+BwriufFB3Amzk9kTQR7B+6x3Nv
  ↪ HYi8A0iv/i4mzf00KXzF4JU7bIJYrUVlk/w8x77gzDRlphsPqpMBJkTI0jisQHZKWNEe7IbmM/
  ↪ dWW2S4jvCkrhB7F5Szf72Q+j/lPbfx2g/8="
+   }
+ }

[edit]
vyos@vyos# commit

```

Diffie-Hellman key:

```

vyos@vyos# run generate pki dh install dh-1
Enter DH parameters key size: (Default: 2048)
Generating parameters...
1 value(s) installed. Use "compare" to see the pending changes, and "commit" to apply.
[edit]
vyos@vyos# compare
[pki]
+ dh dh-1 {
+   parameters "MIIBCAKCAQEAp25kxwZeLZ7wcbRii5E5RD4uWCUOBxarzKEE0msa84omh5nZ9dv/
  ↪ 4bfJw4gIXlA2+sGc2lLV/
  ↪ jajZminMryiSwJdisyVuUd0B7sJWZwrzHBAY0qFbNyaRMVJBar2xVm+XcKd3A2eNTEgn10G7rPPvf6CJ5isUKFaKT8ymUv+mI0upL
  ↪ yS3sAojzeulCf49fa5SiaGccZZkdOI3Nbylu/

```

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```

→ ZG4okqJ2wE2c2hRVLS1k5qrrono00F4Dh0B91ihnywRfp1xPYeqpiln+OPh+PPgTuBxkz4VxwRDoQ+NhVr/
→ LOCb3vbhnyFisxI0w4r3109cA3QiDmo1L14aK11wIBAg=="
+ }

```

```

[edit]
vyos@vyos# commit

```

Client certificate:

```

vyos@vyos:~$ generate pki certificate sign ca-1 install client1
Do you already have a certificate request? [y/N] N
Enter private key type: [rsa, dsa, ec] (Default: rsa)
Enter private key bits: (Default: 2048)
Enter country code: (Default: GB)
Enter state: (Default: Some-State)
Enter locality: (Default: Some-City)
Enter organization name: (Default: VyOS)
Enter common name: (Default: vyos.io) client1
Do you want to configure Subject Alternative Names? [y/N]
Enter how many days certificate will be valid: (Default: 365)
Enter certificate type: (client, server) (Default: server) client
Note: If you plan to use the generated key on this router, do not encrypt the private_
→ key.
Do you want to encrypt the private key with a passphrase? [y/N]
You are not in configure mode, commands to install manually from configure mode:
set pki certificate client1 certificate
→ 'MIIDrjCCApagAwIBAgIUPvtffeYTdoOiHxu++wdrjHwwVX4wDQYJKoZIhvcNAQELBQAwVDELMakGA1UEBhMCRC0IXEzARBgNVBAGM
→ WoA9xlg89/
→ bnScLmfztVDnluyNn8epE6zAi2GWBhtj4ixLllIwLdkJ7L2mF3yUZtA1Q0oYbGIqTbnaZ37JydCygVGn1LT7UX9zfRfS3KebCivIt
→ 9Vyn4UBAR8QQNbezRB3XPMk9gzULnuLhmEDP6GVcPq7RzGXoXUMqsCxfE0JBjeJ0y4ANKH07HGVVrfVRiY+z1GkM4TFjVuZKnEA0B
→ DOECIifBoYeJ9DITu8VxyyZmgaINHEN2gGb0LRH07rvQapc+XZ2z9DAgMBAAAGjdTBzMAwGA1UdEWEB/
→ wQCMAAwDgYDVROPAQH/
→ BAQDAgeAMBGA1UdJQMMaAGCCsGAQUFBwMCMCB0GA1UdDgQWBBQnUyqEzG+AqZzsdSud5MDqs0xiXTafBgNVHSMEGDAWgBQAb2W+v
→ Li9j9eVbFue77qbTANBgkqhkiG9w0BAQsFAAOCAQEAp1ItvZpoX/
→ joG3QREu9tHVkwdTmXB2lwUM5G8iKPgd6D6oOILZMe2KuvWt12dcdEzUCGfJwJJ8M8R2WD00mcLdFqvM/
→ 8UM1hYzUP2BCnFCLtElVD+b4wM1QNpdHqNbdckw8J4MLQlhUgu9rZAZ0XjWCpr+U50bX++vYRw7Un3Ds6ETEjflm5WAPb2e0V1h
→ QwA17Iy8x86RpOgqmesbt0U7e9Rmo81aVgiy/
→ V40CV7u6bPX03fmZNS8UwwJuRUlXkj0+epHNYB2cn0cjSkUxaIJ9Hv3tMWHQEtVZSNYSOZozw=='
set pki certificate client1 private key
→ 'MIIEvAIBADANBgkqhkiG9w0BAQEFAASCBKYwggSiAgEAAoIBAQC9H6E6gm0PfX01n/WoA9xlg89/
→ bnScLmfztVDnluyNn8epE6zAi2GWBhtj4ixLllIwLdkJ7L2mF3yUZtA1Q0oYbGIqTbnaZ37JydCygVGn1LT7UX9zfRfS3KebCivIt
→ 9Vyn4UBAR8QQNbezRB3XPMk9gzULnuLhmEDP6GVcPq7RzGXoXUMqsCxfE0JBjeJ0y4ANKH07HGVVrfVRiY+z1GkM4TFjVuZKnEA0B
→ DOECIifBoYeJ9DITu8VxyyZmgaINHEN2gGb0LRH07rvQapc+XZ2z9DAgMBAAECggEAPS/
→ Fhtt5k2BgFivZW3FcVc+OS0keGwV8hkFsGoXTZIKelZSFWIn/
→ mX0CUY90C0Rn9MRwiqB4Pwss0ASHY6QQjdRK8irRbVK812ZeydHC7DfVudXtKR0YnxTaePML3nTV/
→ TqPF14Rx6EINTHrkLeBbu2DhGsKfhoHioTVbvUiKLHa2TkGJ0khvjsyMSPKzUXA1AzLmu+UBIhRYpEPHj0SQUUJjNkgIb7mTR2fhJ
→ MGqcuuHncTxWBAwYgKhf26a/
→ tqFZRnUrJ6GowxDiAcQEc1mWnmDngRa+dvvCwNbXvGVqfVEQKBgQDgKi447TqD76QfvrPn/
→ fRSjM+QE1duk+XorEJ0HHIha5HV9kCrZdV/olGRjDLwPJ06JW7iE2FUS9SsIrcCFE/
→ 9P2ZUqfYP2wL5vN05kAmoLLU10gwqg1WnBTPJfXeKReTj2uGmOdEuMPXpL/49hDuPViiE2Q4MGe2Z+oEYN/
→ EwKBgHfQMuTEl2e9qaDn8OM6Sr1uC5V4fjunh6dLnfgwaCx1fk17021onQuJWzsim19o4rao06PP4AGqzphz2PsKSJ2ya1NnIJRDF
→ tHeDcX7ZNVxSKLFhElh+i02S+DkqHb2GRT47I2hkFAaqBtBMPiKgZ/ftaNDP46nLEuRYHQdXu4zhfHTV+a/

```

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```

↪CHtqAWGLuddyjaYJNM96SQ6eqjzxcMcCgYAzdx0F2e27hIgo2ttjsROMGqW0/0r/
↪HsKGKPNao7xHQNCAswTnBT+QGugPCe0NXjuxbySP7V1GeWMWF+WV5khtteWerT1/ELAC48NSDpaMxVa4GP8Q/
↪0w6+ZyJty3UGbCYQzZZue81dU+42LUIaVJ4NAc2tYj3jD780udasawS6w== '

```

Manually copy the CA, client certificate, and Diffie-Hellman key to the client device, then commit them before configuring the OpenVPN interface.

For more options, refer to [PKI](#).

Set up server-client OpenVPN

The following example demonstrates the most complicated scenario: each client acts as a router with its own subnet (e.g., an HQ and multiple branch offices). Simpler setups are subsets of it.

In this scenario, the 10.23.1.0/24 network is used for client tunnel endpoints, and all client subnets belong to 10.23.0.0/20. Each client needs access to the 192.168.0.0/16 network.

Server configuration:

```

set interfaces openvpn vtun10 encryption data-ciphers 'aes256'
set interfaces openvpn vtun10 hash 'sha512'
set interfaces openvpn vtun10 local-host '172.18.201.10'
set interfaces openvpn vtun10 local-port '1194'
set interfaces openvpn vtun10 mode 'server'
set interfaces openvpn vtun10 persistent-tunnel
set interfaces openvpn vtun10 protocol 'udp'
set interfaces openvpn vtun10 server client client1 ip '10.23.1.10'
set interfaces openvpn vtun10 server client client1 subnet '10.23.2.0/25'
set interfaces openvpn vtun10 server domain-name 'vyos.net'
set interfaces openvpn vtun10 server max-connections '250'
set interfaces openvpn vtun10 server name-server '172.16.254.30'
set interfaces openvpn vtun10 server subnet '10.23.1.0/24'
set interfaces openvpn vtun10 server topology 'subnet'
set interfaces openvpn vtun10 tls ca-certificate ca-1
set interfaces openvpn vtun10 tls certificate srv-1
set interfaces openvpn vtun10 tls dh-params dh-1

```

The configuration above uses the default 1194/UDP port, 256-bit AES encryption, SHA-512 for HMAC authentication, and the persistent-tunnel option. Persistent-tunnel is recommended as it keeps the TUN/TAP device active during connection resets or daemon reloads. Clients are identified by the CN attribute in their SSL certificates.

To grant clients access to a specific network behind the router, use the push-route option to automatically install the appropriate route on each client.

```

set interfaces openvpn vtun10 server push-route 192.168.0.0/16

```

OpenVPN does not automatically create kernel routes for client subnets when clients connect; it only uses client-subnet association internally. Therefore, you must manually create a route to the 10.23.0.0/20 network:

```

set protocols static route 10.23.0.0/20 interface vtun10

```

Set up OpenVPN client

VyOS can operate not only as an OpenVPN site-to-site peer or a server for multiple clients, but also as an OpenVPN client. Any VyOS OpenVPN interface can be configured to connect to another VyOS or third-party OpenVPN server.

Client configuration:

```
set interfaces openvpn vtun10 encryption data-ciphers 'aes256'
set interfaces openvpn vtun10 hash 'sha512'
set interfaces openvpn vtun10 mode 'client'
set interfaces openvpn vtun10 persistent-tunnel
set interfaces openvpn vtun10 protocol 'udp'
set interfaces openvpn vtun10 remote-host '172.18.201.10'
set interfaces openvpn vtun10 remote-port '1194'
set interfaces openvpn vtun10 tls ca-certificate ca-1
set interfaces openvpn vtun10 tls certificate client1
```

Verification

Check the tunnel status:

```
vyos@vyos:~$ show openvpn server
```

OpenVPN status on vtun10

Client CN	Remote Host	Tunnel IP	Local Host	TX bytes	RX bytes	
↪Connected Since						
-----	-----	-----	-----	-----	-----	---
↪-----						
client1	172.16.12.54:33166	10.23.1.10	172.18.201.10:1194	3.4 KB	3.4 KB	↪
↪ 2024-06-11 12:07:25						

Server bridge

In Ethernet bridging configurations, an OpenVPN interface operating in server mode with the device type set to TAP can be added to a bridge. By encapsulating entire Ethernet frames (up to 1514 bytes) rather than just IP packets (up to 1500 bytes), this setup enables clients to transmit Layer 2 frames through the OpenVPN tunnel.

The following is a basic configuration example:

Server side:

```
set interfaces bridge br10 member interface eth1.10
set interfaces bridge br10 member interface vtun10
set interfaces openvpn vtun10 device-type 'tap'
set interfaces openvpn vtun10 encryption data-ciphers 'aes192'
set interfaces openvpn vtun10 hash 'sha256'
set interfaces openvpn vtun10 local-host '172.18.201.10'
set interfaces openvpn vtun10 local-port '1194'
set interfaces openvpn vtun10 mode 'server'
set interfaces openvpn vtun10 server bridge gateway '10.10.0.1'
set interfaces openvpn vtun10 server bridge start '10.10.0.100'
```

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```
set interfaces openvpn vtun10 server bridge stop '10.10.0.200'
set interfaces openvpn vtun10 server bridge subnet-mask '255.255.255.0'
set interfaces openvpn vtun10 server topology 'subnet'
set interfaces openvpn vtun10 tls ca-certificate 'ca-1'
set interfaces openvpn vtun10 tls certificate 'srv-1'
set interfaces openvpn vtun10 tls dh-params 'dh-1'
```

Client side:

```
set interfaces openvpn vtun10 device-type 'tap'
set interfaces openvpn vtun10 encryption data-ciphers 'aes192'
set interfaces openvpn vtun10 hash 'sha256'
set interfaces openvpn vtun10 mode 'client'
set interfaces openvpn vtun10 protocol 'udp'
set interfaces openvpn vtun10 remote-host '172.18.201.10'
set interfaces openvpn vtun10 remote-port '1194'
set interfaces openvpn vtun10 tls ca-certificate 'ca-1'
set interfaces openvpn vtun10 tls certificate 'client-1'
```

Server LDAP authentication

LDAP

Enterprise installations usually include a directory service to centralize employee password management. VyOS and OpenVPN support using LDAP and Active Directory as a single user backend.

Authentication is performed by the `openvpn-auth-ldap.so` plugin, included with every VyOS installation. To use it, you must create a dedicated configuration file. **Best practice:** Store the configuration file in the `/config` directory to ensure it is preserved after image updates.

```
set interfaces openvpn vtun0 openvpn-option "--plugin /usr/lib/openvpn/openvpn-auth-ldap.
↪so /config/auth/ldap-auth.config"
```

A sample configuration file is shown below:

```
<LDAP>
# LDAP server URL
URL ldap://ldap.example.com
# Bind DN (If your LDAP server doesn't support anonymous binds)
BindDN cn=LDAPUser,dc=example,dc=com
# Bind Password password
Password S3cr3t
# Network timeout (in seconds)
Timeout 15
</LDAP>

<Authorization>
# Base DN
BaseDN "ou=people,dc=example,dc=com"
# User Search Filter
SearchFilter "(&(uid=%u)(objectClass=shadowAccount))"
```

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```
# Require Group Membership - allow all users
RequireGroup    false
</Authorization>
```

Active Directory

A sample configuration file is shown below:

```
<LDAP>
# LDAP server URL
URL ldap://dc01.example.com
# Bind DN (If your LDAP server doesn't support anonymous binds)
BindDN CN=LDAPUser,DC=example,DC=com
# Bind Password
Password mysecretpassword
# Network timeout (in seconds)
Timeout 15
# Enable Start TLS
TLSEnable no
# Follow LDAP Referrals (anonymously)
FollowReferrals no
</LDAP>

<Authorization>
# Base DN
BaseDN "DC=example,DC=com"
# User Search Filter, user must be a member of the VPN AD group
SearchFilter "(&(sAMAccountName=%u)(memberOf=CN=VPN,OU=Groups,DC=example,DC=com))"
# Require Group Membership
RequireGroup    false # already handled by SearchFilter
<Group>
  BaseDN "OU=Groups,DC=example,DC=com"
  SearchFilter "(|(cn=VPN))"
  MemberAttribute memberOf
</Group>
</Authorization>
```

If you only want to check that the user account is enabled and can authenticate (against the primary group), the following snippet is sufficient:

```
<LDAP>
URL ldap://dc01.example.com
BindDN CN=SA_OPENVPN,OU=ServiceAccounts,DC=example,DC=com
Password ThisIsTopSecret
Timeout 15
TLSEnable no
FollowReferrals no
</LDAP>

<Authorization>
BaseDN "DC=example,DC=com"
```

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```

SearchFilter      "sAMAccountName=%u"
RequireGroup      false
</Authorization>

```

A complete example of an LDAP authentication configuration for OpenVPN is shown below:

```

vyos@vyos# show interfaces openvpn
openvpn vtun0 {
    mode server
    openvpn-option "--tun-mtu 1500 --fragment 1300 --mssfix"
    openvpn-option "--plugin /usr/lib/openvpn/openvpn-auth-ldap.so /config/auth/ldap-
    ↪auth.config"
    openvpn-option "--push redirect-gateway"
    openvpn-option --duplicate-cn
    openvpn-option "--verify-client-cert none"
    openvpn-option --comp-lzo
    openvpn-option --persist-key
    openvpn-option --persist-tun
    server {
        domain-name example.com
        max-connections 5
        name-server 203.0.113.0.10
        name-server 198.51.100.3
        subnet 172.18.100.128/29
    }
    tls {
        ca-certificate ca.crt
        certificate server.crt
        dh-params dh1024.pem
    }
}

```

For a detailed example, refer to [OpenVPN with LDAP](#).

Multi-factor authentication

VyOS supports multi-factor authentication (MFA) or two-factor authentication using Time-based One-Time Passwords (TOTP). It is compatible with Google Authenticator and other software tokens.

Server side

```

set interfaces openvpn vtun20 encryption cipher 'aes256'
set interfaces openvpn vtun20 hash 'sha512'
set interfaces openvpn vtun20 mode 'server'
set interfaces openvpn vtun20 persistent-tunnel
set interfaces openvpn vtun20 server client user1
set interfaces openvpn vtun20 server mfa totp challenge 'disable'
set interfaces openvpn vtun20 server subnet '10.10.2.0/24'
set interfaces openvpn vtun20 server topology 'subnet'
set interfaces openvpn vtun20 tls ca-certificate 'openvpn_vtun20'

```

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```
set interfaces openvpn vtun20 tls certificate 'openvpn_vtun20'  
set interfaces openvpn vtun20 tls dh-params 'dh-pem'
```

A TOTP secret is created for each client in the OpenVPN server configuration. To display authentication information, use the following command: `show interfaces openvpn vtun20 user user1 mfa qrcode`.

Example:

```
vyos@vyos:~$ sh interfaces openvpn vtun20 user user1 mfa qrcode
```

Scan the QR code to add the user account to Google Authenticator. On the client side, use the generated OTP as the password.

Authentication with username/password

An OpenVPN server can securely obtain a username and password from a connecting client and use this information for authentication.

First, configure the server to use an authentication plugin or script. The server calls this plugin every time a client tries to connect, passing it the client's credentials.

In the following example, the `--auth-user-pass-verify` directive is used with the `via-env` method and a specified script path to validate the client's username and password.

Server configuration

```
set interfaces openvpn vtun10 local-port '1194'
set interfaces openvpn vtun10 mode 'server'
set interfaces openvpn vtun10 openvpn-option '--auth-user-pass-verify /config/auth/check_
↪user.sh via-env'
set interfaces openvpn vtun10 openvpn-option '--script-security 3'
set interfaces openvpn vtun10 persistent-tunnel
set interfaces openvpn vtun10 protocol 'udp'
set interfaces openvpn vtun10 server client client-1 ip '10.10.10.55'
set interfaces openvpn vtun10 server push-route 192.0.2.0/24
set interfaces openvpn vtun10 server subnet '10.10.10.0/24'
set interfaces openvpn vtun10 server topology 'subnet'
set interfaces openvpn vtun10 tls ca-certificate 'ca-1'
set interfaces openvpn vtun10 tls certificate 'srv-1'
set interfaces openvpn vtun10 tls dh-params 'dh-1'
```

The /config/auth/check_user.sh example includes two test users:

```
#!/bin/bash
USERNAME="$username"
PASSWORD="$password"

# Replace this with real user checking logic or use getent
if [[ "$USERNAME" == "client1" && "$PASSWORD" == "pass123" ]]; then
    exit 0
elif [[ "$USERNAME" == "peter" && "$PASSWORD" == "qwerty" ]]; then
    exit 0
else
    exit 1
fi
```

Client configuration

Storing the client certificate locally lets you generate the OpenVPN client configuration file. Use the following command:

```
vyos@vyos:~$ generate openvpn client-config interface vtun10 ca ca-1 certificate client1
```

Copy the output and save it as a .ovpn file. Add the auth-user-pass directive to the file. This instructs the OpenVPN client to prompt the user for a username and password, which are then sent to the server over the TLS channel. You can now import this file into any OpenVPN client application.

```
client
dev tun
proto udp
remote 192.168.77.10 1194

remote-cert-tls server
proto udp
dev tun
dev-type tun
```

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```

8IauRfghMHJf3t0C+y0YIoXshF3lPq4nYso5Jc/HGCHlsboCODMCnY3CZsH7/01n
/MI710KpzZTCLnv4Qtx9JpZxR7FTddl3600uYUXU3GcnsG==
-----END CERTIFICATE-----

</cert>

<key>
-----BEGIN PRIVATE KEY-----
MIIEvQIBADANBgkqhkiG9w0BAQEFAASCBCwggSjAgEAAoIBAQCd0Wq8vd08CznG
N83uAXCuN4PcdTJaRFEDJIEfqHjlcG0MZQuPIAlDbOU+IWmuQBmeCj7SlbYtVYo1
uQOMUaIrAvxLIQUaL1Y60oLVTF5eAPrGV+NSTQR5uMApCH9/RcZcW530pu/QpYin
KtBgEd54so6YRVPmYbIOPNUMbnZbccpinYi5t2dqubBb585A7L40043VtsVVbPj
Qq5V0HDursvqlaMqMRcfhR8H4B4ByU/EPRK4yTKmlhi19v3UtRHiq74CfGtJzYt
plgrLJBON7TsbIi/fEux4q1yhbKA0S66L6e5DZldRxNZ0XG6QjEL0RkYloMgkbv/
2HLCu09hAgMBAAECggEAOR3xRVU09Sr816JRSQwz486eNDpNSxazgwtOb3JUTUH9
E7onq1y/kM0gOmSIEHoP9GaTcQxbbPe86IxomhLT/50ri52YzWzx/heY2SVPyQXB
FMo79putKw0vnj5UyydNiyLrbMQyrhFc5iFmWVdz5/c4cWHwjIThPp7V4znXYWHZ
OB/Xn1NNHDNy872oQn5wZWzuA4ml00qjU5D+Ne9srODl3r40To3lb1N3JuH3a0SA
cACl1JnN/KElN8IotIdweeUFAdn2jsGjZnCPGaJvZQ+2iMn6doJXHGFIF5+GMF7o
a0atglElIuqgPtB/4nvnegSL0DSnB36ojqv2PAh24wKBgQDPBt4S4muqo8SqP2e0
8X78MyK3tz1VmgPKn3068Vdi1V7FPz0RHRGsw/kdgxXsJlfZTWgzczq2NNFu0yPBj
A/h7qo16mv8GW7cJCd2exjb+/oq4r5iWeqLdSsMUXN87x02LRaMNd9wz1mls1Z73
oQ5hJ7zTtlyYXnvKPQo8XlImjwKBgQDCaptQxZ/a3tcUQqlXAFMAScviODZd0LCL
30ZalwpNs6nVVIPoZHD3tlzWN5Es74gndfkC7/Gm2cnsOW9QQAu56q+5LeNXItW8
rc6yXq3vNqerqJxHNUmKWwLCQtSyLRjFqpGTl/PyX2bGXQ7/zjTL3W8VMD5otf4Y
SJJb+sKjDwKBgHSVX3WvAAamFtfwwMwKuwH3IfPnQqj0BHKUfK2nvxgvJCFbzV3X
yt5Jtf3ClhPYO9xpV0a0C7va4lHaXkYf8Exj7SxAIKFKALccUStaYBoU6bW7X0hQ
w2pu8ZCEBEo7oBVv77Rj7SNb+R6K5ex5TAm2QQXQSjCb9IYc/ail3TNNAoGBALu6
GPMrgKnlfyV1j0E1DPBwUbDEuqpoArFtDRAYXFiFLVTS4PQbWIG403f9++659Gy2
G5ZcfqiWd6xL4VJLSPF1zewvhr/0gRJJEhb+GVGrkRa0HykbKUGxk75kreDGBu8f
PqaXyXS17hWiCh1Lzes0jDiXdwvA//Q0zztqmVq9AoGAVMbmF04+QtzckLoLAP4q
Uwr5svfy14A7V3IGkwlSHZdm37L26lfxW0kp00E7g7D6gdinuALo6oopP7RN/IDq
PLaaHaGrIoLAEVFfa0bRLGsrU2q87ytwfSgdra4jmsTn+xEabdI4IgmqWgwSRvGVf
KN18e19Ssw5x7Wq0Rsw/3VM=
-----END PRIVATE KEY-----

</key>

```

When prompted, log in with the username and password.

7.4.10 WireGuard

WireGuard is an extremely simple, fast, and modern VPN that utilizes state-of-the-art cryptography. See <https://www.wireguard.com> for more information.

Site-to-site VPN

The following diagram illustrates a site-to-site VPN setup.

Keypairs

WireGuard requires a keypair, which includes a **private** key to decrypt incoming traffic, and a **public** key for peer(s) to encrypt outgoing traffic.

Generate keypair

generate pki wireguard key-pair

Generate a keypair: a public and a private key.

Note: This command only outputs the keys to your console. It neither stores them in the system nor applies them to the system configuration.

```
vyos@vyos:~$ generate pki wireguard key-pair
Private key: iJJyEARGK52Ls1GYRCcFvPuTj7WyWYDo//BknoDU0XY=
Public key: EKY0dxRrSD98QHjfhOK13mZ5PJ7hnddRZt5woB3szyw=
```

generate pki wireguard key-pair install interface <interface>

Generate a keypair and output the private key assignment command for the specified interface.

```
vyos@vyos:~$ generate pki wireguard key-pair install interface wg10
"generate" CLI command executed from operational level.
Generated private key is not automatically added to the VyOS configuration, use the
→ following configuration mode commands to install key:

set interfaces wireguard wg10 private-key
→ '4KrkV8h6NkAYMMaBWI957yYDJDMvj9URTHstdl0cDU0='

Corresponding public-key to use on peer system is:
→ 'UxDsYT6EnpTI0KUzvMlw2p0sNOKQvFxEdSVrnNrX1Ro='
```

Note: If you invoke this command from configuration mode with the run prefix, the generated private key is automatically assigned to the specified interface.

```
vyos@vyos# run generate pki wireguard key-pair install interface wg10
"generate" CLI command executed from config session.
Generated private-key was imported to CLI!

Use the following command to verify: show interfaces wireguard wg10
Corresponding public-key to use on peer system is: '7d9KwabjLhHpJiEJeIGd0CBla0/
→eTwFOh6xyCovTfG8='

vyos@vyos# compare
[edit interfaces]
+wireguard wg10 {
+  private-key CJweb8FC6BU3Loj4PC2pn5V82cDjIPs7G1saW0ZfLWc=
+}
```

show interfaces wireguard <interface> public-key

Show the public key assigned to the interface.

```
vyos@vyos:~$ show interfaces wireguard wg01 public-key
EKY0dxRrSD98QHjFHOK13mZ5PJ7hnddRZt5woB3szyw=
```

Optional

generate pki wireguard preshared-key

Generate a pre-shared key.

The pre-shared key is optional. It adds an additional layer of symmetric-key cryptography on top of the asymmetric cryptography.

```
vyos@vyos:~$ generate pki wireguard preshared-key
Pre-shared key: OHH2EwZfMNK+1L6BXbYw3bKCtMrfjpR4mCAEeBlFnRs=
```

generate pki wireguard preshared-key install interface <interface> peer <peer>

Generate a pre-shared key and output the key assignment command for the specified peer.

```
vyos@vyos:~$ generate pki wireguard preshared-key install interface wg10 peer foo
"generate" CLI command executed from operational level.
Generated preshared-key is not stored to CLI, use configure mode commands to
→install key:

set interfaces wireguard wg10 peer foo preshared-key
→'32vQ1w1yFKTna8n7Gu7EimubSe2Y63m8bafz55EG3Ro='

Pre-shared key: +LuaZ8W6DjsDFJFX3jJzoNqrsXHhvfq08JztM9z8LHCs=
```

Note: If you invoke this command from configuration mode with the run prefix, the generated key is automatically assigned to the specified peer.

Interface configuration

The next step is to configure your local WireGuard interface and define the networks you want to tunnel (allowed-ips).

If your system only initiates connections, specifying the listen port is optional. If your system accepts incoming connections, you must define a port for peers to connect to. Otherwise, WireGuard selects a random port at each reboot, and that may break your peers' ability to connect if that port is not enabled in your firewall rules.

To configure a WireGuard tunnel, you also need your peer's public key.

Note: The public key specified in the peer configuration block is always the **remote** peer's public key, never your local one.

Local side configuration

The local side is configured with the following parameters:

- Local WireGuard interface IP: 10.1.0.1/30
- Local listen port: 51820
- Remote peer name: to-wg02
- Remote peer endpoint: 192.0.2.1 on port 51820
- Remote peer public key: XMrlPykaxhdAAiSjhtPlvi30NVkvLQliQuKP7AI7CyI=
- Allowed networks: 192.168.2.0/24

```
set interfaces wireguard wg01 address '10.1.0.1/30'
set interfaces wireguard wg01 description 'VPN-to-wg02'
set interfaces wireguard wg01 peer to-wg02 allowed-ips '192.168.2.0/24'
set interfaces wireguard wg01 peer to-wg02 address '192.0.2.1'
set interfaces wireguard wg01 peer to-wg02 port '51820'
set interfaces wireguard wg01 peer to-wg02 public-key
↪ 'XMrlPykaxhdAAiSjhtPlvi30NVkvLQliQuKP7AI7CyI='
set interfaces wireguard wg01 port '51820'

set protocols static route 192.168.2.0/24 interface wg01
```

To send traffic destined for 192.168.2.0/24 through the WireGuard interface (wg01), configure a static route. Multiple IP addresses or networks can be defined and routed. The final check is performed against allowed-ips, which either permits or drops the traffic.

Warning: You cannot assign the same allowed-ips to multiple WireGuard peers. This is a strict design restriction. For more information, check the [WireGuard mailing list](#).

set interfaces wireguard <interface> private-key <private-key>

Assign a private key to the specified WireGuard interface.

Example:

```
set interfaces wireguard wg01 private-key 'iJlyEARGK52Ls1GYRCcFvPuTj7WyWYDo//
↪ BknoDU0XY='
```

To generate a private key, use the following command: `generate pki wireguard key-pair`.

To view the public key assigned to the interface so you can share it with a peer, use the following command:
`show interfaces wireguard wg01 public-key.`

set interfaces wireguard <interface> per-client-thread

Enable threaded mode for all NAPI instances of the specified network device.

Enabling threaded mode doesn't affect device operation and can be done while the device is up.

This mode is disabled by default.

Example:

```
set interfaces wireguard wg01 per-client-thread
```

Remote side configuration

```
set interfaces wireguard wg01 address '10.1.0.2/30'
set interfaces wireguard wg01 description 'VPN-to-wg01'
set interfaces wireguard wg01 peer to-wg01 allowed-ips '192.168.1.0/24'
set interfaces wireguard wg01 peer to-wg01 address '192.0.2.2'
set interfaces wireguard wg01 peer to-wg01 port '51820'
set interfaces wireguard wg01 peer to-wg01 public-key
↳ 'EKY0dxRrSD98QHjfhOK13mZ5PJ7hnddRZt5woB3szyw='
set interfaces wireguard wg01 port '51820'
set interfaces wireguard wg01 private-key 'OLTQY3HuK5qWDgVs6fJR093SwPgOmCKkDI1+vJLGoFU='

set protocols static route 192.168.1.0/24 interface wg01
```

Firewall exceptions

To allow WireGuard traffic through the WAN interface, create a firewall exception:

```
set firewall ipv4 name OUTSIDE_LOCAL rule 10 action accept
set firewall ipv4 name OUTSIDE_LOCAL rule 10 description 'Allow established/related'
set firewall ipv4 name OUTSIDE_LOCAL rule 10 state established enable
set firewall ipv4 name OUTSIDE_LOCAL rule 10 state related enable
set firewall ipv4 name OUTSIDE_LOCAL rule 20 action accept
set firewall ipv4 name OUTSIDE_LOCAL rule 20 description WireGuard_IN
set firewall ipv4 name OUTSIDE_LOCAL rule 20 destination port 51820
set firewall ipv4 name OUTSIDE_LOCAL rule 20 log enable
set firewall ipv4 name OUTSIDE_LOCAL rule 20 protocol udp
```

Ensure that the OUTSIDE_LOCAL firewall group is applied to the WAN interface and in an input (local) direction.

```
set firewall ipv4 input filter rule 10 action jump
set firewall ipv4 input filter rule 10 jump-target 'OUTSIDE_LOCAL'
set firewall ipv4 input filter rule 10 inbound-interface name 'eth0'
```

Verify that your firewall rules permit traffic. If so, your WireGuard VPN should be operational.

```
wg01# ping 192.168.1.1
PING 192.168.1.1 (192.168.1.1) 56(84) bytes of data.
64 bytes from 192.168.1.1: icmp_seq=1 ttl=64 time=1.16 ms
64 bytes from 192.168.1.1: icmp_seq=2 ttl=64 time=1.77 ms
```

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```
wg02# ping 192.168.2.1
PING 192.168.2.1 (192.168.2.1) 56(84) bytes of data.
64 bytes from 192.168.2.1: icmp_seq=1 ttl=64 time=4.40 ms
64 bytes from 192.168.2.1: icmp_seq=2 ttl=64 time=1.02 ms
```

An additional layer of symmetric-key cryptography can be used on top of the asymmetric cryptography. This is optional.

```
vyos@vyos:~$ generate pki wireguard preshared-key
Pre-shared key: rvVD0oc2IYEnV+k5p7TNAmHBMEGTHbPU8Qqg8c/sUqc=
```

Copy the key, as it is not stored locally. Since it is a symmetric key, only you and your peer should know its contents. Distribute the key securely.

```
wg01# set interfaces wireguard wg01 peer to-wg02 preshared-key
↪ 'rvVD0oc2IYEnV+k5p7TNAmHBMEGTHbPU8Qqg8c/sUqc='
wg02# set interfaces wireguard wg01 peer to-wg01 preshared-key
↪ 'rvVD0oc2IYEnV+k5p7TNAmHBMEGTHbPU8Qqg8c/sUqc='
```

Remote access (road warrior)

With WireGuard, a road warrior VPN configuration is similar to a site-to-site VPN. It just omits the address and port statements.

In the following example, the IP addresses for remote clients are defined within each peer configuration. This allows peers to communicate with each other.

Additionally, this setup uses a `persistent-keepalive` flag set to 15 seconds to keep the connection alive. This setting is mainly relevant if a peer is behind NAT and cannot be reached if the connection is lost. For effectiveness, the value should be lower than the UDP timeout.

```
wireguard wg01 {
  address 10.172.24.1/24
  address 2001:db8:470:22::1/64
  description RoadWarrior
  peer MacBook {
    allowed-ips 10.172.24.30/32
    allowed-ips 2001:db8:470:22::30/128
    persistent-keepalive 15
    pubkey F5MbW7ye7Dsoxd0aixjdrudshjjxN5UdNV+pGFHqehc=
  }
  peer iPhone {
    allowed-ips 10.172.24.20/32
    allowed-ips 2001:db8:470:22::20/128
    persistent-keepalive 15
    pubkey BknHcLFo8nOo8Dwq2CjaC/TedchKQ0ebxC7GYn7Al00=
  }
  port 2224
  private-key OLTQY3HuK5qWDgVs6fJR093SwPgOmCKkDI1+vJLGoFU=
}
```

Below is the configuration for the iPhone peer. The `AllowedIPs` wildcard setting directs all IPv4 and IPv6 traffic through the VPN connection.

```
[Interface]
PrivateKey = ARAKLSDJsadlkfjasdfiowqeruriowgeuasdf=
Address = 10.172.24.20/24, 2001:db8:470:22::20/64
DNS = 10.0.0.53, 10.0.0.54

[Peer]
PublicKey = RIbtUTCfgzNjnLNPQ/ulkGnnB2vMWHm7l2H/xUfbyjc=
AllowedIPs = 0.0.0.0/0, ::/0
Endpoint = 192.0.2.1:2224
PersistentKeepalive = 15
```

To enable split tunneling, specify the remote subnets. This ensures that only traffic destined for the remote site is sent through the tunnel, while all other traffic remains unaffected.

```
[Interface]
PrivateKey = 8Iasdfweirousd1EVGUk5XsT+wYFZ9mhPnQhmjzaJE6Go=
Address = 10.172.24.30/24, 2001:db8:470:22::30/64

[Peer]
PublicKey = RIbtUTCfgzNjnLNPQ/ulkGnnB2vMWHm7l2H/xUfbyjc=
AllowedIPs = 10.172.24.30/24, 2001:db8:470:22::/64
Endpoint = 192.0.2.1:2224
PersistentKeepalive = 15
```

Operational commands

Status

show interfaces wireguard wg01 summary

Show information about the WireGuard service, including the latest handshake.

```
vyos@vyos:~$ show interfaces wireguard wg01 summary
interface: wg01
public key:
private key: (hidden)
listening port: 51820

peer: <peer pubkey>
endpoint: <peer public IP>
allowed ips: 10.69.69.2/32
latest handshake: 23 hours, 45 minutes, 26 seconds ago
transfer: 1.26 MiB received, 6.47 MiB sent
```

show interfaces wireguard

Show a list of all WireGuard interfaces.

```
Codes: S - State, L - Link, u - Up, D - Down, A - Admin Down
Interface      IP Address      S/L  Description
-----
wg01           10.0.0.1/24     u/u
```

show interfaces wireguard <interface>

Show general information about a specific WireGuard interface.

```
vyos@vyos:~$ show interfaces wireguard wg01
interface: wg01
address: 10.0.0.1/24
public key: h1HkYlSuHdJN6Qv4Hz4bBzjGg5WUty+U1L7DJsZy1iE=
private key: (hidden)
listening port: 41751
RX:  bytes  packets  errors  dropped  overrun      mcast
    0         0         0         0         0           0
TX:  bytes  packets  errors  dropped  carrier  collisions
    0         0         0         0         0           0
```

Remote access (road warrior) clients

Some users connect mobile devices to their VyOS router using WireGuard. To simplify deployment, generate a per-mobile configuration from the VyOS CLI.

Warning: From a security perspective, it is not recommended to let a third party create and share the private key for a secure connection. You should create the private portion yourself and hand out only the public key.

generate wireguard client-config <name> interface <interface> server <ip|fqdn> address <client-ip>

Generate a client configuration file that establishes a connection to the specified interface.

The public key from the specified interface is automatically included in the configuration file.

The command also generates a configuration snippet that can be copied into the VyOS CLI. The <name> you provide will be used as the peer name in the snippet.

You must also specify the IP address or FQDN of the server the client connects to. The address parameter can be used twice to assign both an IPv4 (/32) and an IPv6 (/128) address to the client.

7.4.11 PPPoE

PPPoE (Point-to-Point Protocol over Ethernet) is a network protocol that encapsulates PPP frames within Ethernet frames. It's often used for connecting ISP clients to a broadband access server.

Configuration

Common interface configuration

set interfaces pppoe <interface> description <description>

Configure a clear, descriptive alias for the interface.

This alias appears in the `show interfaces` command and SNMP-based monitoring tools.

Example:

```
set interfaces pppoe pppoe0 description 'This is an interface running on VyOS.'
```

set interfaces pppoe <interface> disable

Disable the interface.

The interface will be set to the administratively down (A/D) state.

Example:

```
set interfaces pppoe pppoe0 disable
```

set interfaces pppoe <interface> mtu <mtu>

Configure the MTU on the interface.

This value defines the largest packet size, in bytes, that the interface transmits without fragmentation.

Example:

```
set interfaces pppoe pppoe0 mtu 1600
```

set interfaces pppoe <interface> vrf <vrf>

Assign the interface to a specific VRF instance.

See also:

For information on configuring a VRF, refer to the [VRF](#) section.

Example:

```
set interfaces pppoe pppoe0 vrf red
```

PPPoE options

set interfaces pppoe <interface> access-concentrator <name>

Configure the name of the target access concentrator for the PPPoE session.

During the PPPoE discovery process, the client sends a PPPoE initiation packet. Multiple access concentrators may respond with offer packets, and the client selects one of them.

This setting restricts the client to establishing sessions only with the specified access concentrator.

set interfaces pppoe <interface> authentication username <username>

Configure the username for PPPoE session authentication.

Although authentication is optional in the interface configuration, most ISPs require it to establish a connection.

set interfaces pppoe <interface> authentication password <password>

Configure the password for PPPoE session authentication.

Although authentication is optional in the interface configuration, most ISPs require it to establish a connection.

set interfaces pppoe <interface> connect-on-demand

Enable dial-on-demand on the PPPoE interface.

When enabled, the system establishes a PPPoE connection only when traffic passes through the interface. If the connection fails, it is reestablished when traffic resumes.

For on-demand connections, you must also configure an `idle-timeout` period to disconnect the session after inactivity.

Note: Setting the idle timeout to zero, or leaving it unconfigured, keeps the connection active continuously once established.

By default, the PPPoE connection is established at boot and remains active continuously; if the connection fails, it is reestablished immediately.

set interfaces pppoe <interface> no-default-route

Request an IP address from the PPPoE server without installing a default route.

Example:

```
set interfaces pppoe pppoe0 no-default-route
```

Note: Introduced in VyOS 1.4, this command inverts the logic of the former `default-route` CLI option.

set interfaces pppoe <interface> default-route-distance <distance>

Configure the distance for the default gateway provided by the PPPoE server.

Example:

```
set interfaces pppoe pppoe0 default-route-distance 220
```

set interfaces pppoe <interface> mru <mru>

Configure the MRU (Maximum Receive Unit) for the PPPoE interface.

This setting instructs the `pppd` daemon to restrict the remote peer from sending packets larger than the configured MRU. Allowed MRU values range from 128 to 16384 bytes.

An MRU of 296 is suitable for very slow links (40 bytes for the TCP/IP header and 256 bytes for data).

The default MRU is 1492 bytes.

Note: When using the IPv6 protocol, the MRU must be at least 1280 bytes.

set interfaces pppoe <interface> idle-timeout <time>

Configure the idle timeout for on-demand PPPoE sessions.

This setting defines how long the connection remains active without any traffic before being disconnected.

Note: Setting the idle timeout to zero, or leaving it unconfigured, keeps the connection active continuously once established.

set interfaces pppoe <interface> holdoff <time>

Configure the redial delay for persistent PPPoE sessions.

If a persistent session (with `connect-on-demand` disabled) is terminated by the remote peer or drops unexpectedly, the router waits the specified interval before attempting to reconnect.

The default redial delay is 30 seconds.

```
set interfaces pppoe <interface> local-address <address>
```

Configure the local endpoint IP address for PPPoE sessions.

By default, this IP address is negotiated.

```
set interfaces pppoe <interface> no-peer-dns
```

Disable the installation of advertised DNS nameservers on the local system.

```
set interfaces pppoe <interface> remote-address <address>
```

Configure the remote endpoint IP address for PPPoE sessions.

By default, this IP address is negotiated.

```
set interfaces pppoe <interface> service-name <name>
```

Configure the service name of the target access concentrator for the PPPoE session.

By default, the PPPoE interface connects to any available access concentrator.

```
set interfaces pppoe <interface> source-interface <source-interface>
```

Configure the underlying interface for the PPPoE connection.

Each PPPoE connection is established over an underlying interface, which can be an Ethernet interface, a VIF, or a bonding interface.

```
set interfaces pppoe <interface> ip adjust-mss <mss | clamp-mss-to-pmtu>
```

Configure the MSS advertised in outgoing TCP SYN packets on the specified interface.

By clamping the MSS value in TCP SYN packets, you instruct the remote side not to send packets larger than the specified size. This helps prevent connection issues if PMTUD (Path MTU Discovery) fails.

The following options are available:

- `mss`: Sets the MSS to a specific value in bytes.
- `clamp-mss-to-pmtu`: Sets the MSS to the interface's MTU minus 40 bytes for IPv4 traffic (20 bytes for the IPv4 header and 20 bytes for the TCP header). This option is recommended to automatically set the proper value.

Note: Introduced in VyOS 1.4, this command replaces the older `set firewall options interface <name> adjust-mss <value>` syntax.

```
set interfaces pppoe <interface> ip disable-forwarding
```

Configure the interface for host or router behavior.

If configured, the interface switches to host mode, and IPv4 forwarding is disabled on it.

```
set interfaces pppoe <interface> ip source-validation <strict | loose | disable>
```

Configure source IP address validation using RPF on this interface, as specified in [RFC 3704](#).

The following options are available:

- `strict`: Each incoming packet's source IP address is checked against the FIB (Forwarding Information Base). If the interface is not the best route back to that source, validation fails, and the packet is dropped.
- `loose`: Each incoming packet's source IP address is checked against the FIB. If the source IP address is unreachable through any interface, validation fails.
- `disable`: No source IP address validation is performed. All incoming packets are accepted.

RFC 3704 recommends enabling strict mode to prevent IP spoofing, such as DDoS attacks. For asymmetric or other complex routing scenarios, use loose mode.

IPv6

set interfaces pppoe <interface> ipv6 address autoconf

Enable IPv6 address assignment via SLAAC on this interface.

set interfaces pppoe <interface> ipv6 adjust-mss <mss | clamp-mss-to-pmtu>

Configure the MSS advertised in outgoing TCP SYN packets on the specified interface.

By clamping the MSS value in TCP SYN packets, you instruct the remote side not to send packets larger than the specified size. This helps prevent connection issues if PMTUD fails.

The following options are available:

- **mss**: Sets the MSS to a specific value in bytes.
- **clamp-mss-to-pmtu**: Sets the MSS to the interface's MTU minus 60 bytes for IPv6 traffic (40 bytes for the IPv6 header and 20 bytes for the TCP header). This option is recommended to automatically set the proper value.

Note: Introduced in VyOS 1.4, this command replaces the older `set firewall options interface <name> adjust-mss <value>` syntax.

set interfaces pppoe <interface> ipv6 disable-forwarding

Configure the interface for host or router behavior.

If configured, the interface switches to host mode, and IPv6 forwarding is disabled on it.

DHCPv6 Prefix Delegation (PD)

VyOS supports DHCPv6 Prefix Delegation (DHCPv6-PD) as described in **RFC 3633**. DHCPv6-PD is supported by most ISPs that provide native IPv6 for consumers on fixed networks.

set interfaces pppoe <interface> dhcpv6-options pd <id> length <length>

Configure a specific prefix length for DHCPv6-PD requests on this interface.

Some ISPs provide only a /64 prefix by default. Use this command to request a different prefix length for a specific DHCPv6-PD request, ranging from /32 (if allowed by your ISP) down to /64. <id> is a unique identifier for the DHCPv6-PD request.

The default value is 64.

To request a /56 prefix from your ISP, use:

```
set interfaces pppoe pppoe0 dhcpv6-options pd 0 length 56
```

set interfaces pppoe <interface> dhcpv6-options pd <id> interface <delegatee> address <address>

Configure the IPv6 interface identifier (host portion) for the delegatee interface.

The value must be a decimal integer. It is appended to the delegated prefix and the configured SLA ID to form the final IPv6 address.

By default, the host portion is generated based on the parent interface's MAC address (EUI-64 format).

Example:

If a /64 prefix is delegated to interface eth8 and you configure the host portion as 65535, the resulting IPv6 address will end with ::ffff, as 65535 corresponds to ffff in hexadecimal notation.

```
set interfaces pppoe pppoe0 dhcpv6-options pd 0 interface eth8 address 65534
```

set interfaces pppoe <interface> dhcpv6-options pd <id> interface <delegatee> sla-id <id>

Configure the SLA ID for the delegatee interface.

The value must be a decimal integer greater than 0 and fit in the length of SLA IDs. It is converted to hexadecimal and appended to the delegated prefix to form the specific subnet prefix for the delegatee interface.

Example:

If SLA ID is 1 and the delegated prefix is 2001:db8:ffff::/48, the resulting subnet prefix for the delegatee interface will be 2001:db8:ffff:1::/64.

```
set interfaces pppoe pppoe0 dhcpv6-options pd 0 interface eth8 sla-id 1
```

Operation

show interfaces pppoe <interface>

Show detailed information about a specific PPPoE interface.

```
vyos@vyos:~$ show interfaces pppoe pppoe0
pppoe0: <POINTOPOINT,MULTICAST,NOARP,UP,LOWER_UP> mtu 1492 qdisc pfifo_fast state_
↪UNKNOWN group default qlen 3
  link/ppp
  inet 192.0.2.1 peer 192.0.2.255/32 scope global pppoe0
    valid_lft forever preferred_lft forever

RX:  bytes    packets    errors    dropped    overrun    mcast
    7002658233  5064967      0         0         0         0
TX:  bytes    packets    errors    dropped    carrier    collisions
    533822843  1620173      0         0         0         0
```

show interfaces pppoe <interface> queue

Show queue information for a specific PPPoE interface.

```
vyos@vyos:~$ show interfaces pppoe pppoe0 queue
qdisc pfifo_fast 0: root refcnt 2 bands 3 priomap  1 2 2 2 1 2 0 0 1 1 1 1 1 1 1 1
Sent 534625359 bytes 1626761 pkt (dropped 62, overlimits 0 requeues 0)
backlog 0b 0p requeues 0
```


Connect/disconnect

disconnect interface <interface>

Disconnect the specified interface.

connect interface <interface>

Initiate a session on the specified interface.

Example

PPPoE over DSL

Configuration scenario:

- Your ISP's DSL modem is connected to the `eth0` interface on your VyOS router.
- Your ISP does not require VLAN tagging.
- PPPoE credentials are provided by your ISP. The typical username format is `name@host.net`, though this may vary.

Configuration notes:

- The maximum MTU size for DSL is 1492 because of PPPoE overhead. If you are switching from a DHCP-based ISP (e.g., a standard cable connection), ensure VPN links have MTU sizes adjusted accordingly.
- To ignore ISP-provided nameservers and use only your statically configured ones, set the `name-server` option to `none`.
- A default route is automatically installed once the interface is up. To change this behavior, use the `no-default-route` CLI option.

Note: The PPPoE configuration syntax changed after VyOS 1.2 (Crux) and is automatically migrated during an upgrade.

```
set interfaces pppoe pppoe0 authentication username 'userid'
set interfaces pppoe pppoe0 authentication password 'secret'
set interfaces pppoe pppoe0 source-interface 'eth0'
```

Secure your setup by creating rules matching the `pppoe0` interface in the firewall chains:

```
set firewall ipv4 input filter rule 10 inbound-interface name 'pppoe0'
set firewall ipv4 forward filter rule 10 inbound-interface name 'pppoe0'
```

PPPoE over VLAN

Some ISPs require PPPoE connections to be established over a VLAN interface. This specific topology is fully supported by VyOS.

The following configuration establishes the PPPoE connection through VLAN 7, which is the default VLAN for Deutsche Telekom:

```
set interfaces pppoe pppoe0 authentication username 'userid'
set interfaces pppoe pppoe0 authentication password 'secret'
set interfaces pppoe pppoe0 source-interface 'eth0.7'
```

IPv6 DHCPv6 prefix delegation

Configuration scenario:

The following configuration establishes a PPPoE session on the eth1 interface, requests a /56 IPv6 prefix delegation from the ISP, and assigns a /64 subnet from that delegation to the eth0 interface.

Configuration notes:

- The IPv6 address assigned to eth0 is <prefix>::1/64.
- If you do not know your delegated prefix size, begin with sla-len 0.
- To advertise the prefix on the eth0 link, configure IPv6 Router Advertisement.

```
set interfaces pppoe pppoe0 authentication username vyos
set interfaces pppoe pppoe0 authentication password vyos
set interfaces pppoe pppoe0 dhcpv6-options pd 0 interface eth0 address '1'
set interfaces pppoe pppoe0 dhcpv6-options pd 0 interface eth0 sla-id '0'
set interfaces pppoe pppoe0 dhcpv6-options pd 0 length '56'
set interfaces pppoe pppoe0 ipv6 address autoconf
set interfaces pppoe pppoe0 source-interface eth1

set service router-advert interface eth0 prefix ::/64
```

7.4.12 MACVLAN (pseudo-Ethernet)

MACVLAN, or pseudo-Ethernet interfaces, operate as logical subinterfaces of standard Ethernet interfaces. Each subinterface has a unique MAC address but shares a single physical Ethernet port. That allows the user to send packets from different source IPv4 or IPv6 addresses using a different MAC address.

Pseudo-Ethernet interfaces behave like physical Ethernet interfaces. They support IPv4 and IPv6 addressing, can obtain IP addresses through DHCP or DHCPv6, and are mapped to a physical Ethernet port. They inherit characteristics such as speed and duplex from their parent interface and can be referenced like standard Ethernet interfaces once created.

Pseudo-Ethernet interfaces may not work in environments that require a

NIC (Network Interface Card) to have only one MAC address. This includes:

- VMware machines with default settings.
- Network switches that permit only a single MAC address.
- xDSL modems that learn the NIC's MAC address.

Configuration

Common interface configuration

set interfaces pseudo-ethernet <interface> address <address | dhcp | dhcpv6>

Configure the interface with one or more IP addresses.

The following options are available:

- **address:** Assign one or more IPv4 or IPv6 addresses to the interface. For example, use 192.0.2.1/24 for IPv4 or 2001:db8::1/64 for IPv6.
- **dhcp:** The interface obtains an IPv4 address from a DHCP server on the same network segment.
- **dhcpv6:** The interface obtains an IPv6 address from a DHCPv6 server on the same network segment.

Note: If the interface obtains an IPv4 address via DHCP, and specific adjustments are needed before/after the IP address is obtained, use the provided hook scripts:

- /config/scripts/dhcp-client/pre-hooks.d/
- /config/scripts/dhcp-client/post-hooks.d/

Example:

```
set interfaces pseudo-ethernet peth0 address 192.0.2.1/24
set interfaces pseudo-ethernet peth0 address 2001:db8::1/64
set interfaces pseudo-ethernet peth0 address dhcp
set interfaces pseudo-ethernet peth0 address dhcpv6
```

set interfaces pseudo-ethernet <interface> description <description>

Configure a clear, descriptive alias for the interface.

This alias appears in the `show interfaces` command and SNMP-based monitoring tools.

Example:

```
set interfaces pseudo-ethernet peth0 description 'This is an interface running on
↳ VyOS.'
```

set interfaces pseudo-ethernet <interface> disable

Disable the interface.

The interface will be set to the administratively down (A/D) state.

Example:

```
set interfaces pseudo-ethernet peth0 disable
```

set interfaces pseudo-ethernet <interface> disable-flow-control

Disable Ethernet flow control (IEEE 802.3x pause frames) on the interface.

Ethernet flow control, defined by the IEEE 802.3x standard, temporarily stops data transmission to prevent packet loss during network congestion. For example, when a sender transmits data faster than the receiver can process it.

Disabling Ethernet flow control means the interface will not signal the connected device to pause transmission and will drop packets if overwhelmed.

Example:

```
set interfaces pseudo-ethernet peth0 disable-flow-control
```

set interfaces pseudo-ethernet <interface> disable-link-detect

Disable physical link-state change detection on the interface, such as when a cable is unplugged.

By default, the interface detects physical link-state changes.

Example:

```
set interfaces pseudo-ethernet peth0 disable-link-detect
```

set interfaces pseudo-ethernet <interface> mac <xx:xx:xx:xx:xx:xx>

Configure a custom MAC address on the interface.

Example:

```
set interfaces pseudo-ethernet peth0 mac '00:53:01:02:03:04'
```

set interfaces pseudo-ethernet <interface> mtu <mtu>

Configure the MTU on the interface.

This value defines the largest packet size, in bytes, that the interface transmits without fragmentation.

Example:

```
set interfaces pseudo-ethernet peth0 mtu 1600
```

set interfaces pseudo-ethernet <interface> ip adjust-mss <mss | clamp-mss-to-pmtu>

Configure the MSS advertised in outgoing TCP SYN packets on the specified interface.

By clamping the MSS value in TCP SYN packets, you explicitly inform the remote side not to send packets larger than that size. This prevents connection issues that occur when Path MTU Discovery (PMTUD) fails.

The following options are available:

- **mss:** Sets the MSS to a specific value, in bytes. Use this option if you need to enforce a specific MSS, for example, to troubleshoot connectivity issues or accommodate specific network requirements.
- **clamp-mss-to-pmtu:** The router automatically calculates the MSS to be the interface's MTU minus 40 bytes for IPv4 traffic (20 bytes for the IPv4 header and 20 bytes for the TCP header). This option is recommended to automatically set the proper value.

set interfaces pseudo-ethernet <interface> ip arp-cache-timeout

Configure how long an ARP entry remains valid after learning an IP-to-MAC address mapping on this interface.

The default duration is 30 seconds.

An ARP entry remains valid if it receives positive feedback from higher-level protocols.

Example:

```
set interfaces pseudo-ethernet peth0 ip arp-cache-timeout 180
```

set interfaces pseudo-ethernet <interface> ip disable-arp-filter

Configure ARP filtering on this interface.

Default behavior: The kernel responds to ARP requests on this interface only if the traffic would be routed back to the ARP sender through that specific interface.

If configured: The kernel responds to ARP requests on this interface for any IP address configured on the local host, regardless of which specific interface that IP address is assigned to, and regardless of the routing table. This reflects the Linux concept that IP addresses belong to the host, not individual interfaces.

Example:

```
set interfaces pseudo-ethernet peth0 ip disable-arp-filter
```

set interfaces pseudo-ethernet <interface> ip disable-forwarding

Configure the interface for host or router behavior.

If configured, the interface switches to host mode, and IPv4 forwarding is disabled on it.

Example:

```
set interfaces pseudo-ethernet peth0 ip disable-forwarding
```

set interfaces pseudo-ethernet <interface> ip enable-directed-broadcast

Configure whether to forward IP-directed broadcast packets received on this interface.

Default behavior: IP-directed broadcast packets are dropped.

If configured: IP-directed broadcast packets are forwarded to all hosts on the destination subnet, as defined in [RFC 1812](#) and [RFC 2644](#).

Example:

```
set interfaces pseudo-ethernet peth0 ip enable-directed-broadcast
```

set interfaces pseudo-ethernet <interface> ip enable-arp-accept

Configure how to process gratuitous ARPs on this interface.

If configured, an IP-to-MAC address mapping is added to the ARP table based on gratuitous ARP requests or replies.

Note: If the ARP table already contains the IP address from a gratuitous ARP, its entry is updated regardless of whether this setting is configured.

Example:

```
set interfaces pseudo-ethernet peth0 ip enable-arp-accept
```

set interfaces pseudo-ethernet <interface> ip enable-arp-announce

Configure the source IP selection for ARP requests on this interface.

Default behavior: The kernel can use any IP address the host owns as the source IP address in ARP requests on this interface.

If configured: The kernel first attempts to select a source IP address configured on the interface that shares a common subnet with the target IP address. If there is no such subnet, the kernel selects the IP address it would normally use (based on the routing table to reach the target destination).

Example:

```
set interfaces pseudo-ethernet peth0 ip enable-arp-announce
```

set interfaces pseudo-ethernet <interface> ip enable-arp-ignore

Configure which ARP requests will be ignored on this interface.

Default behavior: The kernel responds to ARP requests for any local IP addresses, regardless of which interface they are assigned to.

If configured: The kernel responds to ARP requests only if the target IP address is assigned to this specific interface.

Example:

```
set interfaces pseudo-ethernet peth0 ip enable-arp-ignore
```

set interfaces pseudo-ethernet <interface> ip enable-proxy-arp

Configure proxy ARP on this interface.

If configured, the router (kernel) intercepts ARP requests for non-local IP addresses and replies with the MAC address of the interface that received the request. Subsequent packets destined to these IP addresses are forwarded to their actual destinations on remote subnets.

Example:

```
set interfaces pseudo-ethernet peth0 ip enable-proxy-arp
```

set interfaces pseudo-ethernet <interface> ip proxy-arp-pvlan

Configure local proxy ARP on the interface.

If configured, the router (kernel) responds to ARP requests on this VLAN interface even if the target IP address resides on the same subnet and interface.

This is used to support network isolation requirements (RFC 3069) for private VLANs (PVLANS). In PVLAN configurations, hosts on isolated ports are NOT allowed to communicate directly with each other at Layer 2, but they can communicate with the upstream router.

By replying to inter-host ARP requests with its own MAC address, the router (kernel) directs inter-host traffic through itself instead of directly between hosts.

Note: This command works independently and does not require enabling the standard proxy ARP on the interface.

Local proxy ARP is also known as:

- VLAN aggregation ([RFC 3069](#)).
- Private VLAN (Cisco, Allied Telesyn).
- Source-port filtering or port isolation (Hewlett-Packard).
- MAC-Forced Forwarding (Ericsson).

set interfaces pseudo-ethernet <interface> ip source-validation <strict | loose | disable>

Configure source IP address validation using RPF on this interface, as specified in [RFC 3704](#).

The following options are available:

- **strict:** Each incoming packet's source IP address is checked against the Forwarding Information Base (FIB). If the interface is not the best route back to that source, validation fails, and the packet is dropped.
- **loose:** Each incoming packet's source IP address is checked against the FIB. If the source IP address is unreachable through any interface, validation fails.
- **disable:** No source IP address validation is performed. All incoming packets are accepted.

RFC 3704 recommends enabling strict mode to prevent IP spoofing, such as DDoS attacks. For asymmetric or other complex routing scenarios, use loose mode.

set interfaces pseudo-ethernet <interface> ipv6 address autoconf

Configure the interface to automatically obtain an IPv6 address using SLAAC, as specified in [RFC 4862](#).

IPv6 hosts can configure themselves automatically when connected to an IPv6 network using the Neighbor Discovery Protocol via ICMPv6 router discovery messages. When first connected to a network, a host sends a link-local router solicitation multicast request for its configuration parameters. The router responds with a router advertisement packet containing Internet Layer configuration parameters.

Note: This method automatically disables IPv6 traffic forwarding on the interface.

Example:

```
set interfaces pseudo-ethernet peth0 ipv6 address autoconf
```

set interfaces pseudo-ethernet <interface> ipv6 address eui64 <prefix>

Configure the interface to assign itself an IPv6 address using the EUI-64 method, as specified in [RFC 4291](#).

Example:

```
set interfaces pseudo-ethernet peth0 ipv6 address eui64 2001:db8:beef::/64
```

set interfaces pseudo-ethernet <interface> ipv6 address no-default-link-local

Disable the automatic assignment of a link-local IPv6 address to this interface.

Example:

```
set interfaces pseudo-ethernet peth0 ipv6 address no-default-link-local
```

set interfaces pseudo-ethernet <interface> ipv6 disable-forwarding

Configure the interface for host or router behavior.

If configured, the interface switches to host mode, and IPv6 forwarding is disabled on it.

Example:

```
set interfaces pseudo-ethernet peth0 ipv6 disable-forwarding
```

set interfaces pseudo-ethernet <interface> ipv6 adjust-mss <mss | clamp-mss-to-pmtu>

Configure the MSS advertised in outgoing TCP SYN packets on the specified interface.

By clamping the MSS value in TCP SYN packets, you explicitly inform the remote side not to send packets larger than that size. This prevents connection issues when Path MTU Discovery (PMTUD) fails.

The following options are available:

- **mss:** Set the MSS to a specific value, in bytes. Use this option to enforce a specific MSS, for example, to troubleshoot connectivity issues or accommodate specific network requirements.
- **clamp-mss-to-pmtu:** The router calculates the MSS to be the interface's MTU minus 60 bytes for IPv6 traffic (40 bytes for the IPv6 header and 20 bytes for the TCP header). This option is recommended to automatically set the proper value.

```
set interfaces pseudo-ethernet <interface> ipv6 accept-dad <0 | 1 | 2>
```

Configure IPv6 DAD on the interface.

The following options are available:

- **0:** Disables DAD. No duplicate address detection is performed.
- **1:** Enables DAD (default). Duplicate addresses are detected. The interface's IPv6 operation continues for valid IPv6 addresses.
- **2:** Enables DAD and, if a MAC-based duplicate link-local address is found, disables IPv6 operation on this interface.

Example:

```
set interfaces pseudo-ethernet peth0 ipv6 accept-dad 2
```

```
set interfaces pseudo-ethernet <interface> ipv6 dup-addr-detect-transmits <n>
```

Configure the number of DAD messages that the router (kernel) sends during IPv6 address assignment on this interface.

The default value is 1.

Example:

```
set interfaces pseudo-ethernet peth0 ipv6 dup-addr-detect-transmits 5
```

```
set interfaces pseudo-ethernet <interface> vrf <vrf>
```

Assign the interface to a specific VRF instance.

See also:

For information on configuring a VRF, refer to the [VRF](#) section.

Example:

```
set interfaces pseudo-ethernet peth0 vrf red
```

DHCP(v6)

```
set interfaces pseudo-ethernet <interface> dhcp-options client-id <description>
```

Configure a DHCP client identifier for the interface, as specified in [RFC 2131](#).

The `client-id` is an identifier that the DHCP client sends to the DHCP server to uniquely identify itself for IP address assignment. By default, the client uses its MAC address. The `<description>` is a user-defined string that will be sent to the DHCP server as the DHCP client identifier.

Example:

```
set interfaces pseudo-ethernet peth0 dhcp-options client-id 'foo-bar'
```

```
set interfaces pseudo-ethernet <interface> dhcp-options host-name <hostname>
```


Configure a specific hostname for the interface.

Instead of the real hostname, the DHCP client will send the specific hostname to the DHCP server when requesting an IP address.

Example:

```
set interfaces pseudo-ethernet peth0 dhcp-options host-name 'VyOS'
```

set interfaces pseudo-ethernet <interface> dhcp-options vendor-class-id <vendor-id>

Configure the DHCP client to include a vendor-class identifier in its DHCP requests on this interface.

The vendor-class identifier is a vendor-specific byte string that enables the DHCP server to identify the device and, in some cases, provide configuration options.

Example:

```
set interfaces pseudo-ethernet peth0 dhcp-options vendor-class-id 'VyOS'
```

set interfaces pseudo-ethernet <interface> dhcp-options no-default-route

Configure the DHCP client to obtain an IP address, but ignore any default gateway provided by the DHCP server on this interface.

Example:

```
set interfaces pseudo-ethernet peth0 dhcp-options no-default-route
```

set interfaces pseudo-ethernet <interface> dhcp-options default-route-distance <distance>

Configure the distance for the default route obtained from the DHCP server on this interface.

Example:

```
set interfaces pseudo-ethernet peth0 dhcp-options default-route-distance 220
```

set interfaces pseudo-ethernet <interface> dhcp-options reject <address>

Configure the DHCP client to reject the specific IP address or IP address range from the DHCP server on this interface.

This is useful when a modem assigns a local IP address upon start. To reject multiple addresses, run this command multiple times with different values. You can reject individual addresses (192.168.100.1) or entire subnets (192.168.100.0/24).

Example:

```
set interfaces pseudo-ethernet peth0 dhcp-options reject 192.168.100.0/24
```

set interfaces pseudo-ethernet <interface> dhcp-options user-class <string>

Configure the DHCP client to send a specific user-class identifier in its DHCP requests on this interface.

The DHCP server can interpret this identifier and provide specific configuration options based on it (for example, default routes). The user-class value typically groups DHCP clients with similar configuration needs (for example, employees, guests, or printers).

Example:

```
set interfaces pseudo-ethernet peth0 dhcp-options user-class VyOS
```

set interfaces pseudo-ethernet <interface> dhcpv6-options duid <duid>

Configure a specific DUID for the DHCPv6 client on this interface.

The DUID is an identifier used by a DHCPv6 client to get an IPv6 address from a DHCPv6 server. It consists of a 2-byte type field, followed by a variable-length identifier field up to 128 bytes. The format of the identifier part depends on the DUID type:

- **DUID-LLT:** The most common type, which includes a hardware type, a timestamp, and a MAC address.
- **DUID-EN:** Is based on a vendor's enterprise number and a unique identifier assigned by the vendor.
- **DUID-LL:** Includes only a MAC address.

The DHCP server matches the DUID against its database and provides configuration data (such as address, lease times, DNS servers, etc.) to the DHCP client.

Example:

```
set interfaces pseudo-ethernet peth0 duid
→ '0e:00:00:01:00:01:27:71:db:f0:00:50:56:bf:c5:6d'
```

set interfaces pseudo-ethernet <interface> dhcpv6-options no-release

Configure the DHCP client not to send a release message when it stops running on this interface.

This helps retain the assigned address or prefix.

Example:

```
set interfaces pseudo-ethernet peth0 dhcpv6-options no-release
```

set interfaces pseudo-ethernet <interface> dhcpv6-options parameters-only

Enable a stateless DHCPv6 client mode on this interface.

In stateless mode, the DHCPv6 client requests only stateless configuration parameters from the DHCP server (for example, DNS server addresses). It doesn't request a stateful configuration, such as IPv6 addresses or prefixes.

Example:

```
set interfaces pseudo-ethernet peth0 dhcpv6-options parameters-only
```

set interfaces pseudo-ethernet <interface> dhcpv6-options rapid-commit

Enable DHCPv6 rapid commit on this interface.

When enabled, the DHCP client and server skip the negotiation steps (Advertise and Request), completing the DHCPv6 configuration process in just two messages (Solicit and final Reply).

Example:

```
set interfaces pseudo-ethernet peth0 dhcpv6-options rapid-commit
```

set interfaces pseudo-ethernet <interface> dhcpv6-options temporary

Configure the DHCPv6 client to request a temporary IPv6 address on this interface.

When configured, the DHCP client doesn't form an Identity Association for Non-temporary Addresses (IA_NA) partnership. Consequently, it only obtains a temporary IPv6 address and doesn't obtain a permanent one.

Example:

```
set interfaces pseudo-ethernet peth0 dhcpv6-options temporary
```

DHCPv6 Prefix Delegation (PD)

VyOS supports DHCPv6 Prefix Delegation (DHCPv6-PD) as described in [RFC 3633](#). DHCPv6-PD is supported by most ISPs that provide native IPv6 for consumers on fixed networks.

set interfaces pseudo-ethernet <interface> dhcpv6-options pd <id> length <length>

Configure a specific prefix length for DHCPv6-PD requests on this interface.

Some ISPs provide only a /64 prefix by default. Use this command to request a different prefix length for a specific DHCPv6-PD request, ranging from /32 (if allowed by your ISP) down to /64. <id> is a unique identifier for the DHCPv6-PD request.

The default value is 64.

To request a /56 prefix from your ISP, use:

```
set interfaces pseudo-ethernet peth0 dhcpv6-options pd 0 length 56
```

set interfaces pseudo-ethernet <interface> dhcpv6-options pd <id> interface <delegatee> address <address>

Configure the IPv6 interface identifier (host portion) for the delegatee interface.

The value must be a decimal integer. It is appended to the delegated prefix and the configured SLA ID to form the final IPv6 address.

By default, the host portion is generated based on the parent interface's MAC address (EUI-64 format).

Example:

If a /64 prefix is delegated to interface eth8 and you configure the host portion as 65535, the resulting IPv6 address will end with ::ffff, as 65535 corresponds to ffff in hexadecimal notation.

```
set interfaces pseudo-ethernet peth0 dhcpv6-options pd 0 interface eth8 address ↵  
↵ 65534
```

set interfaces pseudo-ethernet <interface> dhcpv6-options pd <id> interface <delegatee> sla-id <id>

Configure the SLA ID for the delegatee interface.

The value must be a decimal integer greater than 0 and fit in the length of SLA IDs. It is converted to hexadecimal and appended to the delegated prefix to form the specific subnet prefix for the delegatee interface.

Example:

If SLA ID is 1 and the delegated prefix is 2001:db8:ffff::/48, the resulting subnet prefix for the delegatee interface will be 2001:db8:ffff:1::/64.

```
set interfaces pseudo-ethernet peth0 dhcpv6-options pd 0 interface eth8 sla-id 1
```

MACVLAN (pseudo-Ethernet) options

set interfaces pseudo-ethernet <interface> source-interface <ethX>

Assign a physical Ethernet interface to the specified pseudo-Ethernet interface.

set interfaces pseudo-ethernet <interface> anycast-gateway

Enable EVPN anycast gateway mode on the specified pseudo-Ethernet interface.

In this mode, VyOS installs a local Forwarding Database (FDB) entry on the pseudo-Ethernet interface's parent bridge. This ensures traffic destined for the pseudo-Ethernet's anycast MAC address is terminated locally rather than forwarded across the VXLAN overlay.

VyOS automatically removes the FDB entry when you disable EVPN anycast gateway mode or delete the pseudo-Ethernet interface.

Note: The following requirements must be met to enable EVPN anycast gateway mode:

- The MAC address must be explicitly configured on the pseudo-Ethernet interface.
 - The pseudo-Ethernet interface's **source-interface** must be a bridge or bridge sub-interface (e.g., **br0** or **br0.100**). VyOS always installs the FDB entry on the base bridge (e.g., **br0**), even if the **source-interface** is a sub-interface like **br0.100**.
-

Example:

```
set interfaces pseudo-ethernet peth0 anycast-gateway
```

VLAN

IEEE 802.1q, often referred to as Dot1q, is the industry standard for implementing VLANs on Ethernet networks. It defines VLAN tagging for Ethernet frames and outlines procedures for bridges and switches. The standard also includes quality-of-service prioritization (IEEE 802.1p) and defines the Generic Attribute Registration Protocol.

VLAN-aware network segments (i.e., IEEE 802.1q conformant) use VLAN tags. When a frame enters such a segment, a tag is added to indicate VLAN membership. Each frame can belong to only one VLAN. If a frame arrives without a tag, it is assumed to be part of the native VLAN.

IEEE 802.1, a working group of the IEEE 802 standards committee, has developed the standard and continues to revise it. One notable revision is 802.1Q-2014, which incorporated IEEE 802.1aq (Shortest Path Bridging) and much of the IEEE 802.1d standard.

In VyOS, 802.1q VLAN interfaces are represented as virtual subinterfaces, referred to as **vif**.

set interfaces pseudo-ethernet <interface> vif <vlan-id>

Configure a VLAN interface with a unique VLAN ID.

VLAN ID identifies a specific VLAN and ranges from 0 to 4094.

You can configure multiple VLAN interfaces on a single physical interface.

Note: Only 802.1Q-tagged packets are accepted on Ethernet vifs.

set interfaces pseudo-ethernet <interface> vif <vlan-id> address <address | dhcp | dhcpv6>

Configure the interface with one or more IP addresses.

The following options are available:

- **address:** Assign one or more IPv4 or IPv6 addresses to the interface. For example, use 192.0.2.1/24 for IPv4 or 2001:db8::1/64 for IPv6.
- **dhcp:** The interface obtains an IPv4 address from a DHCP server on the same network segment.
- **dhcpv6:** The interface obtains an IPv6 address from a DHCPv6 server on the same network segment.

Note: If the interface obtains an IPv4 address via DHCP, and specific adjustments are needed before/after the IP address is obtained, use the provided hook scripts:

- /config/scripts/dhcp-client/pre-hooks.d/
 - /config/scripts/dhcp-client/post-hooks.d/
-

Example:

```
set interfaces pseudo-ethernet peth0 vif 10 address 192.0.2.1/24
set interfaces pseudo-ethernet peth0 vif 10 address 2001:db8::1/64
set interfaces pseudo-ethernet peth0 vif 10 address dhcp
set interfaces pseudo-ethernet peth0 vif 10 address dhcpv6
```

set interfaces pseudo-ethernet <interface> vif <vlan-id> description <description>

Configure a clear, descriptive alias for the interface.

This alias appears in the `show interfaces` command and SNMP-based monitoring tools.

Example:

```
set interfaces pseudo-ethernet peth0 vif 10 description 'This is an interface_
↳running on VyOS.'
```

set interfaces pseudo-ethernet <interface> vif <vlan-id> disable

Disable the interface.

The interface will be set to the administratively down (A/D) state.

Example:

```
set interfaces pseudo-ethernet peth0 vif 10 disable
```

set interfaces pseudo-ethernet <interface> vif <vlan-id> disable-link-detect

Disable physical link-state change detection on the interface, such as when a cable is unplugged.

By default, the interface detects physical link-state changes.

Example:

```
set interfaces pseudo-ethernet peth0 vif 10 disable-link-detect
```

set interfaces pseudo-ethernet <interface> vif <vlan-id> mac <xx:xx:xx:xx:xx:xx>

Configure a custom MAC address on the interface.

Example:

```
set interfaces pseudo-ethernet peth0 vif 10 mac '00:53:01:02:03:04'
```

set interfaces pseudo-ethernet <interface> vif <vlan-id> mtu <mtu>

Configure the MTU on the interface.

This value defines the largest packet size, in bytes, that the interface transmits without fragmentation.

Example:

```
set interfaces pseudo-ethernet peth0 vif 10 mtu 1600
```

set interfaces pseudo-ethernet <interface> vif <vlan-id> ip adjust-mss <mss | clamp-mss-to-pmtu>

Configure the MSS advertised in outgoing TCP SYN packets on the specified interface.

By clamping the MSS value in TCP SYN packets, you explicitly inform the remote side not to send packets larger than that size. This prevents connection issues that occur when Path MTU Discovery (PMTUD) fails.

The following options are available:

- **mss:** Sets the MSS to a specific value, in bytes. Use this option if you need to enforce a specific MSS, for example, to troubleshoot connectivity issues or accommodate specific network requirements.
- **clamp-mss-to-pmtu:** The router automatically calculates the MSS to be the interface's MTU minus 40 bytes for IPv4 traffic (20 bytes for the IPv4 header and 20 bytes for the TCP header). This option is recommended to automatically set the proper value.

set interfaces pseudo-ethernet <interface> vif <vlan-id> ip arp-cache-timeout

Configure how long an ARP entry remains valid after learning an IP-to-MAC address mapping on this interface.

The default duration is 30 seconds.

An ARP entry remains valid if it receives positive feedback from higher-level protocols.

Example:

```
set interfaces pseudo-ethernet peth0 vif 10 ip arp-cache-timeout 180
```

set interfaces pseudo-ethernet <interface> vif <vlan-id> ip disable-arp-filter

Configure ARP filtering on this interface.

Default behavior: The kernel responds to ARP requests on this interface only if the traffic would be routed back to the ARP sender through that specific interface.

If configured: The kernel responds to ARP requests on this interface for any IP address configured on the local host, regardless of which specific interface that IP address is assigned to, and regardless of the routing table. This reflects the Linux concept that IP addresses belong to the host, not individual interfaces.

Example:

```
set interfaces pseudo-ethernet peth0 vif 10 ip disable-arp-filter
```

set interfaces pseudo-ethernet <interface> vif <vlan-id> ip disable-forwarding

Configure the interface for host or router behavior.

If configured, the interface switches to host mode, and IPv4 forwarding is disabled on it.

Example:

```
set interfaces pseudo-ethernet peth0 vif 10 ip disable-forwarding
```

set interfaces pseudo-ethernet <interface> vif <vlan-id> ip enable-directed-broadcast

Configure whether to forward IP-directed broadcast packets received on this interface.

Default behavior: IP-directed broadcast packets are dropped.

If configured: IP-directed broadcast packets are forwarded to all hosts on the destination subnet, as defined in [RFC 1812](#) and [RFC 2644](#).

Example:

```
set interfaces pseudo-ethernet peth0 vif 10 ip enable-directed-broadcast
```

set interfaces pseudo-ethernet <interface> vif <vlan-id> ip enable-arp-accept

Configure how to process gratuitous ARPs on this interface.

If configured, an IP-to-MAC address mapping is added to the ARP table based on gratuitous ARP requests or replies.

Note: If the ARP table already contains the IP address from a gratuitous ARP, its entry is updated regardless of whether this setting is configured.

Example:

```
set interfaces pseudo-ethernet peth0 vif 10 ip enable-arp-accept
```

set interfaces pseudo-ethernet <interface> vif <vlan-id> ip enable-arp-announce

Configure the source IP selection for ARP requests on this interface.

Default behavior: The kernel can use any IP address the host owns as the source IP address in ARP requests on this interface.

If configured: The kernel first attempts to select a source IP address configured on the interface that shares a common subnet with the target IP address. If there is no such subnet, the kernel selects the IP address it would normally use (based on the routing table to reach the target destination).

Example:

```
set interfaces pseudo-ethernet peth0 vif 10 ip enable-arp-announce
```

set interfaces pseudo-ethernet <interface> vif <vlan-id> ip enable-arp-ignore

Configure which ARP requests will be ignored on this interface.

Default behavior: The kernel responds to ARP requests for any local IP addresses, regardless of which interface they are assigned to.

If configured: The kernel responds to ARP requests only if the target IP address is assigned to this specific interface.

Example:

```
set interfaces pseudo-ethernet peth0 vif 10 ip enable-arp-ignore
```

set interfaces pseudo-ethernet <interface> vif <vlan-id> ip enable-proxy-arp

Configure proxy ARP on this interface.

If configured, the router (kernel) intercepts ARP requests for non-local IP addresses and replies with the MAC address of the interface that received the request. Subsequent packets destined to these IP addresses are forwarded to their actual destinations on remote subnets.

Example:

```
set interfaces pseudo-ethernet peth0 vif 10 ip enable-proxy-arp
```

```
set interfaces pseudo-ethernet <interface> vif <vlan-id> ip proxy-arp-pvlan
```

Configure local proxy ARP on the interface.

If configured, the router (kernel) responds to ARP requests on this VLAN interface even if the target IP address resides on the same subnet and interface.

This is used to support network isolation requirements (RFC 3069) for private VLANs (PVLANS). In PVLAN configurations, hosts on isolated ports are NOT allowed to communicate directly with each other at Layer 2, but they can communicate with the upstream router.

By replying to inter-host ARP requests with its own MAC address, the router (kernel) directs inter-host traffic through itself instead of directly between hosts.

Note: This command works independently and does not require enabling the standard proxy ARP on the interface.

Local proxy ARP is also known as:

- VLAN aggregation ([RFC 3069](#)).
- Private VLAN (Cisco, Allied Telesyn).
- Source-port filtering or port isolation (Hewlett-Packard).
- MAC-Forced Forwarding (Ericsson).

```
set interfaces pseudo-ethernet <interface> vif <vlan-id> ip source-validation <strict | loose | disable>
```

Configure source IP address validation using RPF on this interface, as specified in [RFC 3704](#).

The following options are available:

- **strict:** Each incoming packet's source IP address is checked against the Forwarding Information Base (FIB). If the interface is not the best route back to that source, validation fails, and the packet is dropped.
- **loose:** Each incoming packet's source IP address is checked against the FIB. If the source IP address is unreachable through any interface, validation fails.
- **disable:** No source IP address validation is performed. All incoming packets are accepted.

RFC 3704 recommends enabling strict mode to prevent IP spoofing, such as DDoS attacks. For asymmetric or other complex routing scenarios, use loose mode.

```
set interfaces pseudo-ethernet <interface> vif <vlan-id> ipv6 address autoconf
```

Configure the interface to automatically obtain an IPv6 address using SLAAC, as specified in [RFC 4862](#).

IPv6 hosts can configure themselves automatically when connected to an IPv6 network using the Neighbor Discovery Protocol via ICMPv6 router discovery messages. When first connected to a network, a host sends a link-local router solicitation multicast request for its configuration parameters. The router responds with a router advertisement packet containing Internet Layer configuration parameters.

Note: This method automatically disables IPv6 traffic forwarding on the interface.

Example:

```
set interfaces pseudo-ethernet peth0 vif 10 ipv6 address autoconf
```

set interfaces pseudo-ethernet <interface> vif <vlan-id> ipv6 address eui64 <prefix>

Configure the interface to assign itself an IPv6 address using the EUI-64 method, as specified in [RFC 4291](#).

Example:

```
set interfaces pseudo-ethernet peth0 vif 10 ipv6 address eui64 2001:db8:beef::/64
```

set interfaces pseudo-ethernet <interface> vif <vlan-id> ipv6 address no-default-link-local

Disable the automatic assignment of a link-local IPv6 address to this interface.

Example:

```
set interfaces pseudo-ethernet peth0 vif 10 ipv6 address no-default-link-local
```

set interfaces pseudo-ethernet <interface> vif <vlan-id> ipv6 disable-forwarding

Configure the interface for host or router behavior.

If configured, the interface switches to host mode, and IPv6 forwarding is disabled on it.

Example:

```
set interfaces pseudo-ethernet peth0 vif 10 ipv6 disable-forwarding
```

set interfaces pseudo-ethernet <interface> vif <vlan-id> ipv6 adjust-mss <mss | clamp-mss-to-pmtu>

Configure the MSS advertised in outgoing TCP SYN packets on the specified interface.

By clamping the MSS value in TCP SYN packets, you explicitly inform the remote side not to send packets larger than that size. This prevents connection issues when Path MTU Discovery (PMTUD) fails.

The following options are available:

- **mss:** Set the MSS to a specific value, in bytes. Use this option to enforce a specific MSS, for example, to troubleshoot connectivity issues or accommodate specific network requirements.
- **clamp-mss-to-pmtu:** The router calculates the MSS to be the interface's MTU minus 60 bytes for IPv6 traffic (40 bytes for the IPv6 header and 20 bytes for the TCP header). This option is recommended to automatically set the proper value.

set interfaces pseudo-ethernet <interface> vif <vlan-id> ipv6 accept-dad <0 | 1 | 2>

Configure IPv6 DAD on the interface.

The following options are available:

- **0:** Disables DAD. No duplicate address detection is performed.
- **1:** Enables DAD (default). Duplicate addresses are detected. The interface's IPv6 operation continues for valid IPv6 addresses.

- **2:** Enables DAD and, if a MAC-based duplicate link-local address is found, disables IPv6 operation on this interface.

Example:

```
set interfaces pseudo-ethernet peth0 vif 10 ipv6 accept-dad 2
```

```
set interfaces pseudo-ethernet <interface> vif <vlan-id> ipv6 dup-addr-detect-transmits <n>
```

Configure the number of DAD messages that the router (kernel) sends during IPv6 address assignment on this interface.

The default value is 1.

Example:

```
set interfaces pseudo-ethernet peth0 vif 10 ipv6 dup-addr-detect-transmits 5
```

```
set interfaces pseudo-ethernet <interface> vif <vlan-id> vrf <vrf>
```

Assign the interface to a specific VRF instance.

See also:

For information on configuring a VRF, refer to the [VRF](#) section.

Example:

```
set interfaces pseudo-ethernet peth0 vif 10 vrf red
```

DHCP(v6)

```
set interfaces pseudo-ethernet <interface> vif <vlan-id> dhcp-options client-id <description>
```

Configure a DHCP client identifier for the interface, as specified in [RFC 2131](#).

The `client-id` is an identifier that the DHCP client sends to the DHCP server to uniquely identify itself for IP address assignment. By default, the client uses its MAC address. The `<description>` is a user-defined string that will be sent to the DHCP server as the DHCP client identifier.

Example:

```
set interfaces pseudo-ethernet peth0 vif 10 dhcp-options client-id 'foo-bar'
```

```
set interfaces pseudo-ethernet <interface> vif <vlan-id> dhcp-options host-name <hostname>
```

Configure a specific hostname for the interface.

Instead of the real hostname, the DHCP client will send the specific hostname to the DHCP server when requesting an IP address.

Example:

```
set interfaces pseudo-ethernet peth0 vif 10 dhcp-options host-name 'VyOS'
```

```
set interfaces pseudo-ethernet <interface> vif <vlan-id> dhcp-options vendor-class-id <vendor-id>
```

Configure the DHCP client to include a vendor-class identifier in its DHCP requests on this interface.

The vendor-class identifier is a vendor-specific byte string that enables the DHCP server to identify the device and, in some cases, provide configuration options.

Example:

```
set interfaces pseudo-ethernet peth0 vif 10 dhcp-options vendor-class-id 'VyOS'
```

set interfaces pseudo-ethernet <interface> vif <vlan-id> dhcp-options no-default-route

Configure the DHCP client to obtain an IP address, but ignore any default gateway provided by the DHCP server on this interface.

Example:

```
set interfaces pseudo-ethernet peth0 vif 10 dhcp-options no-default-route
```

set interfaces pseudo-ethernet <interface> vif <vlan-id> dhcp-options default-route-distance <distance>

Configure the distance for the default route obtained from the DHCP server on this interface.

Example:

```
set interfaces pseudo-ethernet peth0 vif 10 dhcp-options default-route-distance 220
```

set interfaces pseudo-ethernet <interface> vif <vlan-id> dhcp-options reject <address>

Configure the DHCP client to reject the specific IP address or IP address range from the DHCP server on this interface.

This is useful when a modem assigns a local IP address upon start. To reject multiple addresses, run this command multiple times with different values. You can reject individual addresses (192.168.100.1) or entire subnets (192.168.100.0/24).

Example:

```
set interfaces pseudo-ethernet peth0 vif 10 dhcp-options reject 192.168.100.0/24
```

set interfaces pseudo-ethernet <interface> vif <vlan-id> dhcp-options user-class <string>

Configure the DHCP client to send a specific user-class identifier in its DHCP requests on this interface.

The DHCP server can interpret this identifier and provide specific configuration options based on it (for example, default routes). The user-class value typically groups DHCP clients with similar configuration needs (for example, employees, guests, or printers).

Example:

```
set interfaces pseudo-ethernet peth0 vif 10 dhcp-options user-class VyOS
```

set interfaces pseudo-ethernet <interface> vif <vlan-id> dhcpv6-options duid <duid>

Configure a specific DUID for the DHCPv6 client on this interface.

The DUID is an identifier used by a DHCPv6 client to get an IPv6 address from a DHCPv6 server. It consists of a 2-byte type field, followed by a variable-length identifier field up to 128 bytes. The format of the identifier part depends on the DUID type:

- **DUID-LLT:** The most common type, which includes a hardware type, a timestamp, and a MAC address.
- **DUID-EN:** Is based on a vendor's enterprise number and a unique identifier assigned by the vendor.

- **DUID-LL:** Includes only a MAC address.

The DHCP server matches the DUID against its database and provides configuration data (such as address, lease times, DNS servers, etc.) to the DHCP client.

Example:

```
set interfaces pseudo-ethernet peth0 vif 10 duid
↪ '0e:00:00:01:00:01:27:71:db:f0:00:50:56:bf:c5:6d'
```

set interfaces pseudo-ethernet <interface> vif <vlan-id> dhcpv6-options no-release

Configure the DHCP client not to send a release message when it stops running on this interface.

This helps retain the assigned address or prefix.

Example:

```
set interfaces pseudo-ethernet peth0 vif 10 dhcpv6-options no-release
```

set interfaces pseudo-ethernet <interface> vif <vlan-id> dhcpv6-options parameters-only

Enable a stateless DHCPv6 client mode on this interface.

In stateless mode, the DHCPv6 client requests only stateless configuration parameters from the DHCP server (for example, DNS server addresses). It doesn't request a stateful configuration, such as IPv6 addresses or prefixes.

Example:

```
set interfaces pseudo-ethernet peth0 vif 10 dhcpv6-options parameters-only
```

set interfaces pseudo-ethernet <interface> vif <vlan-id> dhcpv6-options rapid-commit

Enable DHCPv6 rapid commit on this interface.

When enabled, the DHCP client and server skip the negotiation steps (Advertise and Request), completing the DHCPv6 configuration process in just two messages (Solicit and final Reply).

Example:

```
set interfaces pseudo-ethernet peth0 vif 10 dhcpv6-options rapid-commit
```

set interfaces pseudo-ethernet <interface> vif <vlan-id> dhcpv6-options temporary

Configure the DHCPv6 client to request a temporary IPv6 address on this interface.

When configured, the DHCP client doesn't form an Identity Association for Non-temporary Addresses (IA_NA) partnership. Consequently, it only obtains a temporary IPv6 address and doesn't obtain a permanent one.

Example:

```
set interfaces pseudo-ethernet peth0 vif 10 dhcpv6-options temporary
```

DHCPv6 Prefix Delegation (PD)

VyOS supports DHCPv6 Prefix Delegation (DHCPv6-PD) as described in [RFC 3633](#). DHCPv6-PD is supported by most ISPs that provide native IPv6 for consumers on fixed networks.

set interfaces pseudo-ethernet <interface> vif <vlan-id> dhcpv6-options pd <id> length <length>

Configure a specific prefix length for DHCPv6-PD requests on this interface.

Some ISPs provide only a /64 prefix by default. Use this command to request a different prefix length for a specific DHCPv6-PD request, ranging from /32 (if allowed by your ISP) down to /64. <id> is a unique identifier for the DHCPv6-PD request.

The default value is 64.

To request a /56 prefix from your ISP, use:

```
set interfaces pseudo-ethernet peth0 vif 10 dhcpv6-options pd 0 length 56
```

```
set interfaces pseudo-ethernet <interface> vif <vlan-id> dhcpv6-options pd <id> interface
<delegatee> address <address>
```

Configure the IPv6 interface identifier (host portion) for the delegatee interface.

The value must be a decimal integer. It is appended to the delegated prefix and the configured SLA ID to form the final IPv6 address.

By default, the host portion is generated based on the parent interface's MAC address (EUI-64 format).

Example:

If a /64 prefix is delegated to interface eth8 and you configure the host portion as 65535, the resulting IPv6 address will end with ::ffff, as 65535 corresponds to ffff in hexadecimal notation.

```
set interfaces pseudo-ethernet peth0 vif 10 dhcpv6-options pd 0 interface eth8
↪address 65534
```

```
set interfaces pseudo-ethernet <interface> vif <vlan-id> dhcpv6-options pd <id> interface
<delegatee> sla-id <id>
```

Configure the SLA ID for the delegatee interface.

The value must be a decimal integer greater than 0 and fit in the length of SLA IDs. It is converted to hexadecimal and appended to the delegated prefix to form the specific subnet prefix for the delegatee interface.

Example:

If SLA ID is 1 and the delegated prefix is 2001:db8:ffff::/48, the resulting subnet prefix for the delegatee interface will be 2001:db8:ffff:1::/64.

```
set interfaces pseudo-ethernet peth0 vif 10 dhcpv6-options pd 0 interface eth8 sla-
↪id 1
```

7.4.13 SSTP client

SSTP (Secure Socket Tunneling Protocol) transports PPP traffic over an SSL/TLS channel, providing transport-level security through key negotiation, encryption, and traffic integrity checking. The use of SSL/TLS over TCP port 443 (by default, the port can be changed) allows SSTP to pass through virtually all firewalls and proxy servers, except for authenticated web proxies.

Note: VyOS includes a built-in SSTP server. For more information, see [SSTP Server](#).

Configuration

Common interface configuration

set interfaces sstpc <interface> description <description>

Configure a clear, descriptive alias for the interface.

This alias appears in the `show interfaces` command and SNMP-based monitoring tools.

Example:

```
set interfaces sstpc sstpc0 description 'This is an interface running on VyOS.'
```

set interfaces sstpc <interface> disable

Disable the interface.

The interface will be set to the administratively down (A/D) state.

Example:

```
set interfaces sstpc sstpc0 disable
```

set interfaces sstpc <interface> mtu <mtu>

Configure the MTU on the interface.

This value defines the largest packet size, in bytes, that the interface transmits without fragmentation.

Example:

```
set interfaces sstpc sstpc0 mtu 1600
```

set interfaces sstpc <interface> vrf <vrf>

Assign the interface to a specific VRF instance.

See also:

For information on configuring a VRF, refer to the [VRF](#) section.

Example:

```
set interfaces sstpc sstpc0 vrf red
```

SSTP client options

set interfaces sstpc <interface> no-default-route

Request an IP address from the SSTP server without installing a default route.

Example:

```
set interfaces sstpc sstpc0 no-default-route
```

Note: Introduced in VyOS 1.4, this command inverts the logic of the former `default-route` CLI option.

set interfaces sstpc <interface> default-route-distance <distance>

Configure the distance for the default gateway provided by the SSTP server.

Example:

```
set interfaces sstpc sstpc0 default-route-distance 220
```

set interfaces sstpc <interface> no-peer-dns

Disable the installation of advertised DNS nameservers on the local system.

set interfaces sstpc <interface> server <address>

Configure the remote SSTP server address for the client connection.

The address can be either an IP address or a FQDN (Fully Qualified Domain Name).

set interfaces sstpc <interface> ip adjust-mss <mss | clamp-mss-to-pmtu>

Configure the MSS advertised in outgoing TCP SYN packets on the specified interface.

By clamping the MSS value in TCP SYN packets, you instruct the remote side not to send packets larger than the specified size. This helps prevent connection issues if PMTUD fails.

The following options are available:

- **mss**: Sets the MSS to a specific value in bytes.
- **clamp-mss-to-pmtu**: Sets the MSS to the interface's MTU minus 40 bytes for IPv4 traffic (20 bytes for the IPv4 header and 20 bytes for the TCP header). This option is recommended to automatically set the proper value.

Note: Introduced in VyOS 1.4, this command replaces the older `set firewall options interface <name> adjust-mss <value>` syntax.

set interfaces sstpc <interface> ip disable-forwarding

Configure the interface for host or router behavior.

If configured, the interface switches to host mode, and IPv4 forwarding is disabled on it.

set interfaces sstpc <interface> ip source-validation <strict | loose | disable>

Configure source IP address validation using RPF on this interface, as specified in [RFC 3704](#).

The following options are available:

- **strict**: Each incoming packet's source IP address is checked against the FIB. If the interface is not the best route back to that source, validation fails, and the packet is dropped.
- **loose**: Each incoming packet's source IP address is checked against the FIB. If the source IP address is unreachable through any interface, validation fails.
- **disable**: No source IP address validation is performed. All incoming packets are accepted.

[RFC 3704](#) recommends enabling **strict** mode to prevent IP spoofing, such as DDoS attacks. For asymmetric or other complex routing scenarios, use **loose** mode.

Operation

show interfaces sstpc <interface>

Show detailed information about the specified interface.

```
vyos@vyos:~$ show interfaces sstpc sstpc10
sstpc10: <POINTOPOINT,MULTICAST,NOARP,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state_
->UNKNOWN group default qlen 3
    link/ppp
    inet 192.0.2.5 peer 192.0.2.254/32 scope global sstpc10
        valid_lft forever preferred_lft forever
    inet6 fe80::fd53:c7ff:fe8b:144f/64 scope link
        valid_lft forever preferred_lft forever

RX:  bytes  packets  errors  dropped  overrun      mcast
    215         9        0        0         0          0
TX:  bytes  packets  errors  dropped  carrier  collisions
    539        14        0        0         0          0
```

Connect/disconnect

disconnect interface <interface>

Disconnect the specified interface.

connect interface <interface>

Initiate a session on the specified interface.

7.4.14 Tunnel

Tunnel interfaces are virtual links that transmit encapsulated traffic between private networks or hosts across public infrastructure, such as the Internet. They operate using encapsulation protocols to wrap original traffic for transport. The supported protocols include GRE (Generic Routing Encapsulation), IPIP, IPIP6, IP6IP6, and 6in4 (SIT).

While GRE is often the preferred one-size-fits-all solution due to its versatility, other encapsulation protocols may be better suited for specific use cases.

VyOS uses a single tunnel interface type for all of these protocols. There are no separate GRE, IPIP, or IP6IP6 interface types; instead, the desired encapsulation protocol is selected within the `set interfaces tunnel` configuration.

Configuration options for each protocol are described below.

Warning: Do not change the encapsulation type for already configured tunnel interfaces, as this may break their dependent configurations.

Common interface configuration

set interfaces tunnel <interface> address <address>

Configure one or more interface IP addresses.

<address> supports IPv4 and IPv6 in CIDR notation. For example, use 192.0.2.1/24 for IPv4 or 2001:db8::1/64 for IPv6.

Example:

```
set interfaces tunnel tun0 address 192.0.2.1/24
set interfaces tunnel tun0 address 2001:db8::1/64
```

set interfaces tunnel <interface> description <description>

Configure a clear, descriptive alias for the interface.

This alias appears in the show interfaces command and SNMP-based monitoring tools.

Example:

```
set interfaces tunnel tun0 description 'This is an interface running on VyOS.'
```

set interfaces tunnel <interface> disable

Disable the interface.

The interface will be set to the administratively down (A/D) state.

Example:

```
set interfaces tunnel tun0 disable
```

set interfaces tunnel <interface> disable-flow-control

Disable Ethernet flow control (IEEE 802.3x pause frames) on the interface.

Ethernet flow control, defined by the IEEE 802.3x standard, temporarily stops data transmission to prevent packet loss during network congestion. For example, when a sender transmits data faster than the receiver can process it.

Disabling Ethernet flow control means the interface will not signal the connected device to pause transmission and will drop packets if overwhelmed.

Example:

```
set interfaces tunnel tun0 disable-flow-control
```

set interfaces tunnel <interface> disable-link-detect

Disable physical link-state change detection on the interface, such as when a cable is unplugged.

By default, the interface detects physical link-state changes.

Example:

```
set interfaces tunnel tun0 disable-link-detect
```

set interfaces tunnel <interface> mtu <mtu>

Configure the MTU on the interface.

This value defines the largest packet size, in bytes, that the interface transmits without fragmentation.

Example:

```
set interfaces tunnel tun0 mtu 1600
```

```
set interfaces tunnel <interface> ip adjust-mss <mss | clamp-mss-to-pmtu>
```

Configure the MSS advertised in outgoing TCP SYN packets on the specified interface.

By clamping the MSS value in TCP SYN packets, you explicitly inform the remote side not to send packets larger than that size. This prevents connection issues that occur when Path MTU Discovery (PMTUD) fails.

The following options are available:

- **mss:** Sets the MSS to a specific value, in bytes. Use this option if you need to enforce a specific MSS, for example, to troubleshoot connectivity issues or accommodate specific network requirements.
- **clamp-mss-to-pmtu:** The router automatically calculates the MSS to be the interface's MTU minus 40 bytes for IPv4 traffic (20 bytes for the IPv4 header and 20 bytes for the TCP header). This option is recommended to automatically set the proper value.

```
set interfaces tunnel <interface> ip arp-cache-timeout
```

Configure how long an ARP entry remains valid after learning an IP-to-MAC address mapping on this interface.

The default duration is 30 seconds.

An ARP entry remains valid if it receives positive feedback from higher-level protocols.

Example:

```
set interfaces tunnel tun0 ip arp-cache-timeout 180
```

```
set interfaces tunnel <interface> ip disable-arp-filter
```

Configure ARP filtering on this interface.

Default behavior: The kernel responds to ARP requests on this interface only if the traffic would be routed back to the ARP sender through that specific interface.

If configured: The kernel responds to ARP requests on this interface for any IP address configured on the local host, regardless of which specific interface that IP address is assigned to, and regardless of the routing table. This reflects the Linux concept that IP addresses belong to the host, not individual interfaces.

Example:

```
set interfaces tunnel tun0 ip disable-arp-filter
```

```
set interfaces tunnel <interface> ip disable-forwarding
```

Configure the interface for host or router behavior.

If configured, the interface switches to host mode, and IPv4 forwarding is disabled on it.

Example:

```
set interfaces tunnel tun0 ip disable-forwarding
```

```
set interfaces tunnel <interface> ip enable-directed-broadcast
```

Configure whether to forward IP-directed broadcast packets received on this interface.

Default behavior: IP-directed broadcast packets are dropped.

If configured: IP-directed broadcast packets are forwarded to all hosts on the destination subnet, as defined in [RFC 1812](#) and [RFC 2644](#).

Example:

```
set interfaces tunnel tun0 ip enable-directed-broadcast
```

set interfaces tunnel <interface> ip enable-arp-accept

Configure how to process gratuitous ARPs on this interface.

If configured, an IP-to-MAC address mapping is added to the ARP table based on gratuitous ARP requests or replies.

Note: If the ARP table already contains the IP address from a gratuitous ARP, its entry is updated regardless of whether this setting is configured.

Example:

```
set interfaces tunnel tun0 ip enable-arp-accept
```

set interfaces tunnel <interface> ip enable-arp-announce

Configure the source IP selection for ARP requests on this interface.

Default behavior: The kernel can use any IP address the host owns as the source IP address in ARP requests on this interface.

If configured: The kernel first attempts to select a source IP address configured on the interface that shares a common subnet with the target IP address. If there is no such subnet, the kernel selects the IP address it would normally use (based on the routing table to reach the target destination).

Example:

```
set interfaces tunnel tun0 ip enable-arp-announce
```

set interfaces tunnel <interface> ip enable-arp-ignore

Configure which ARP requests will be ignored on this interface.

Default behavior: The kernel responds to ARP requests for any local IP addresses, regardless of which interface they are assigned to.

If configured: The kernel responds to ARP requests only if the target IP address is assigned to this specific interface.

Example:

```
set interfaces tunnel tun0 ip enable-arp-ignore
```

set interfaces tunnel <interface> ip enable-proxy-arp

Configure proxy ARP on this interface.

If configured, the router (kernel) intercepts ARP requests for non-local IP addresses and replies with the MAC address of the interface that received the request. Subsequent packets destined to these IP addresses are forwarded to their actual destinations on remote subnets.

Example:

```
set interfaces tunnel tun0 ip enable-proxy-arp
```

set interfaces tunnel <interface> ip proxy-arp-pvlan

Configure local proxy ARP on the interface.

If configured, the router (kernel) responds to ARP requests on this VLAN interface even if the target IP address resides on the same subnet and interface.

This is used to support network isolation requirements (RFC 3069) for private VLANs (PVLANS). In PVLAN configurations, hosts on isolated ports are NOT allowed to communicate directly with each other at Layer 2, but they can communicate with the upstream router.

By replying to inter-host ARP requests with its own MAC address, the router (kernel) directs inter-host traffic through itself instead of directly between hosts.

Note: This command works independently and does not require enabling the standard proxy ARP on the interface.

Local proxy ARP is also known as:

- VLAN aggregation ([RFC 3069](#)).
- Private VLAN (Cisco, Allied Telesyn).
- Source-port filtering or port isolation (Hewlett-Packard).
- MAC-Forced Forwarding (Ericsson).

set interfaces tunnel <interface> ip source-validation <strict | loose | disable>

Configure source IP address validation using RPF on this interface, as specified in [RFC 3704](#).

The following options are available:

- **strict:** Each incoming packet's source IP address is checked against the Forwarding Information Base (FIB). If the interface is not the best route back to that source, validation fails, and the packet is dropped.
- **loose:** Each incoming packet's source IP address is checked against the FIB. If the source IP address is unreachable through any interface, validation fails.
- **disable:** No source IP address validation is performed. All incoming packets are accepted.

RFC 3704 recommends enabling strict mode to prevent IP spoofing, such as DDoS attacks. For asymmetric or other complex routing scenarios, use loose mode.

set interfaces tunnel <interface> ipv6 address autoconf

Configure the interface to automatically obtain an IPv6 address using SLAAC, as specified in [RFC 4862](#).

IPv6 hosts can configure themselves automatically when connected to an IPv6 network using the Neighbor Discovery Protocol via ICMPv6 router discovery messages. When first connected to a network, a host sends a link-local router solicitation multicast request for its configuration parameters. The router responds with a router advertisement packet containing Internet Layer configuration parameters.

Note: This method automatically disables IPv6 traffic forwarding on the interface.

Example:

```
set interfaces tunnel tun0 ipv6 address autoconf
```

set interfaces tunnel <interface> ipv6 address eui64 <prefix>

Configure the interface to assign itself an IPv6 address using the EUI-64 method, as specified in [RFC 4291](#).

Example:

```
set interfaces tunnel tun0 ipv6 address eui64 2001:db8:beef::/64
```

set interfaces tunnel <interface> ipv6 address no-default-link-local

Disable the automatic assignment of a link-local IPv6 address to this interface.

Example:

```
set interfaces tunnel tun0 ipv6 address no-default-link-local
```

set interfaces tunnel <interface> ipv6 disable-forwarding

Configure the interface for host or router behavior.

If configured, the interface switches to host mode, and IPv6 forwarding is disabled on it.

Example:

```
set interfaces tunnel tun0 ipv6 disable-forwarding
```

set interfaces tunnel <interface> ipv6 adjust-mss <mss | clamp-mss-to-pmtu>

Configure the MSS advertised in outgoing TCP SYN packets on the specified interface.

By clamping the MSS value in TCP SYN packets, you explicitly inform the remote side not to send packets larger than that size. This prevents connection issues when Path MTU Discovery (PMTUD) fails.

The following options are available:

- **mss:** Set the MSS to a specific value, in bytes. Use this option to enforce a specific MSS, for example, to troubleshoot connectivity issues or accommodate specific network requirements.
- **clamp-mss-to-pmtu:** The router calculates the MSS to be the interface's MTU minus 60 bytes for IPv6 traffic (40 bytes for the IPv6 header and 20 bytes for the TCP header). This option is recommended to automatically set the proper value.

set interfaces tunnel <interface> ipv6 accept-dad <0 | 1 | 2>

Configure IPv6 DAD on the interface.

The following options are available:

- **0:** Disables DAD. No duplicate address detection is performed.
- **1:** Enables DAD (default). Duplicate addresses are detected. The interface's IPv6 operation continues for valid IPv6 addresses.
- **2:** Enables DAD and, if a MAC-based duplicate link-local address is found, disables IPv6 operation on this interface.

Example:

```
set interfaces tunnel tun0 ipv6 accept-dad 2
```

set interfaces tunnel <interface> ipv6 dup-addr-detect-transmits <n>

Configure the number of DAD messages that the router (kernel) sends during IPv6 address assignment on this interface.

The default value is 1.

Example:

```
set interfaces tunnel tun0 ipv6 dup-addr-detect-transmits 5
```

set interfaces tunnel <interface> vrf <vrf>

Assign the interface to a specific VRF instance.

See also:

For information on configuring a VRF, refer to the [VRF](#) section.

Example:

```
set interfaces tunnel tun0 vrf red
```

IPIP

IPIP is a straightforward encapsulation protocol defined in RFC 2003. It encapsulates one IPv4 packet inside another IPv4 packet.

Tunnels with IPIP encapsulation do not have protocol-specific configuration options except for explicitly defining the encapsulation type as IPIP (see the example below).

Example:

```
set interfaces tunnel tun0 encapsulation ipip
set interfaces tunnel tun0 source-address 192.0.2.10
set interfaces tunnel tun0 remote 203.0.113.20
set interfaces tunnel tun0 address 192.168.100.200/24
```

IP6IP6

IP6IP6 is the IPv6 counterpart to IPIP. It encapsulates one IPv6 packet inside another IPv6 packet.

Similar to their IPIP counterparts, tunnels with IP6IP6 encapsulation do not have protocol-specific configuration options except for explicitly defining the encapsulation type as IP6IP6.

Example:

```
set interfaces tunnel tun0 encapsulation ip6ip6
set interfaces tunnel tun0 source-address 2001:db8:aa::1
set interfaces tunnel tun0 remote 2001:db8:aa::2
set interfaces tunnel tun0 address 2001:db8:bb::1/64
```

IPIP6

IPIP6 is an encapsulation protocol that wraps IPv4 packets inside IPv6 packets.

Similar to IPIP and IP6IP6, protocol-specific configuration for tunnels with IPIP6 encapsulation only requires defining the encapsulation type as IP6IP6.

Example:

```
set interfaces tunnel tun0 encapsulation ipip6
set interfaces tunnel tun0 source-address 2001:db8:aa::1
set interfaces tunnel tun0 remote 2001:db8:aa::2
set interfaces tunnel tun0 address 192.168.70.80/24
```

6in4 (SIT)

6in4, also known as SIT (Simple Internet Transition), is an encapsulation protocol defined in [RFC 4213](#) that wraps IPv6 packets inside IPv4 packets. The encapsulating IPv4 headers use IP protocol number 41, which is reserved exclusively for IPv6 encapsulation.

The encapsulation process adds a 20-byte IPv4 header to each IPv6 packet. Consequently, 6in4 tunnel interfaces can transmit IPv6 packets up to 1480 bytes over an underlying network with a standard MTU of 1500 bytes without fragmentation.

6in4 tunnel interfaces are frequently used by IPv6 tunnel brokers (such as [Hurricane Electric](#)) to connect isolated IPv6 networks or individual hosts to the IPv6 internet.

Example:

```
set interfaces tunnel tun0 encapsulation sit
set interfaces tunnel tun0 source-address 192.0.2.10
set interfaces tunnel tun0 remote 192.0.2.20
set interfaces tunnel tun0 address 2001:db8:bb::1/64
```

See also:

For a practical configuration example, see the [Tunnelbroker.net \(IPv6\)](#) section.

Generic Routing Encapsulation (GRE)

GRE is a versatile encapsulation protocol defined in RFC 2784. Unlike simpler protocols such as IPIP, it allows both IPv4 and IPv6 to be transported through the same tunnel.

GRE encapsulates original data packets by adding a GRE header, followed by an IP header (the delivery header). The delivery header uses IP protocol number 47 to identify GRE-encapsulated traffic.

In VyOS, GRE tunnels can be established over both IPv4 (encapsulation gre) and IPv6 (encapsulation ip6gre) transport networks.

Configuration

To configure a GRE tunnel, you need to define a tunnel source IP address, a tunnel destination IP address, an encapsulation type (GRE), and a tunnel interface IP address.

Example:

The following example shows how to configure an IPv4/IPv6-over-IPv6 GRE tunnel between a VyOS router and a Linux host running `systemd-networkd`.

VyOS router:

```
set interfaces tunnel tun101 address '2001:db8:feed:beef::1/126'
set interfaces tunnel tun101 address '192.168.5.1/30'
set interfaces tunnel tun101 encapsulation 'ip6gre'
set interfaces tunnel tun101 source-address '2001:db8:babe:face::3afe:3'
set interfaces tunnel tun101 remote '2001:db8:9bb:3ce::5'
```

Linux systemd-networkd:

The `systemd-networkd` setup requires two configuration files: `xxx.netdev` to create the GRE tunnel interface, and `xxx.network` to assign IP addresses to it.

```
# cat /etc/systemd/network/gre-example.netdev
[NetDev]
Name=gre-example
Kind=ip6gre
MTUBytes=14180

[Tunnel]
Remote=2001:db8:babe:face::3afe:3

# cat /etc/systemd/network/gre-example.network
[Match]
Name=gre-example

[Network]
Address=2001:db8:feed:beef::2/126

[Address]
Address=192.168.5.2/30
```

GRE keys

A GRE key is an optional 32-bit field in the GRE header that allows multiple GRE tunnels to operate between the same source and destination endpoints. When a packet arrives, the receiver checks the GRE key to determine which tunnel interface should process it.

Although it may sound security-related, the GRE key is only an identifier and provides no encryption or data protection.

Example:

```
set interfaces tunnel tun0 source-address 192.0.2.10
set interfaces tunnel tun0 remote 192.0.2.20
```

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```
set interfaces tunnel tun0 address 10.40.50.60/24
set interfaces tunnel tun0 parameters ip key 10
```

```
set interfaces tunnel tun1 source-address 192.0.2.10
set interfaces tunnel tun1 remote 192.0.2.20
set interfaces tunnel tun1 address 172.16.17.18/24
set interfaces tunnel tun1 parameters ip key 20
```

GRETAP

Unlike GRE, which encapsulates only Layer 3 (IP) traffic, GRETAP encapsulates Layer 2 (Ethernet) frames.

That means that GRETAP tunnel interfaces can be members of a bridge interface. This allows two geographically distant sites to connect as if they were on the same LAN.

GRETAP tunnels can be established over both IPv4 and IPv6 transport networks.

Example:

```
set interfaces bridge br0 member interface eth0
set interfaces bridge br0 member interface tun0
set interfaces tunnel tun0 encapsulation gretap
set interfaces tunnel tun0 source-address 198.51.100.2
set interfaces tunnel tun0 remote 203.0.113.10
```

Troubleshooting

GRE is a standardized tunneling protocol used in many network environments.

Although the GRE tunnel setup is straightforward, connectivity failures frequently occur because ACLs or firewall rules block IP protocol 47 or prevent direct communication between the tunnel endpoints.

If your GRE tunnel fails to establish, perform these diagnostic steps:

1. Verify that the remote peer is reachable from the configured `source-address`.

This ensures that the underlying physical path between the two endpoints is functional.

```
vyos@vyos:~$ ping 203.0.113.10 interface 198.51.100.2 count 4
PING 203.0.113.10 (203.0.113.10) from 198.51.100.2 : 56(84) bytes of data.
64 bytes from 203.0.113.10: icmp_seq=1 ttl=254 time=0.807 ms
64 bytes from 203.0.113.10: icmp_seq=2 ttl=254 time=1.50 ms
64 bytes from 203.0.113.10: icmp_seq=3 ttl=254 time=0.624 ms
64 bytes from 203.0.113.10: icmp_seq=4 ttl=254 time=1.41 ms

--- 203.0.113.10 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3007ms
rtt min/avg/max/mdev = 0.624/1.087/1.509/0.381 ms
```

2. Verify that the tunnel interface is correctly configured (with the link type set to GRE) and is actively processing traffic.

```
vyos@vyos:~$ show interfaces tunnel tun100
tun100@NONE: <POINTOPOINT,NOARP,UP,LOWER_UP> mtu 1476 qdisc noqueue state UNKNOWN group
↪ default qlen 1000
  link/gre 198.51.100.2 peer 203.0.113.10
  inet 10.0.0.1/30 brd 10.0.0.3 scope global tun100
    valid_lft forever preferred_lft forever
  inet6 fe80::5efe:c612:2/64 scope link
    valid_lft forever preferred_lft forever

RX:  bytes    packets    errors    dropped    overrun    mcast
    2183         27         0         0         0         0
TX:  bytes    packets    errors    dropped    carrier    collisions
    836          9         0         0         0         0
```

3. Test the connection through the tunnel using the private IP addresses assigned to each tunnel endpoint.

```
vyos@vyos:~$ ping 10.0.0.2 interface 10.0.0.1 count 4
PING 10.0.0.2 (10.0.0.2) from 10.0.0.1 : 56(84) bytes of data.
64 bytes from 10.0.0.2: icmp_seq=1 ttl=255 time=1.05 ms
64 bytes from 10.0.0.2: icmp_seq=2 ttl=255 time=1.88 ms
64 bytes from 10.0.0.2: icmp_seq=3 ttl=255 time=1.98 ms
64 bytes from 10.0.0.2: icmp_seq=4 ttl=255 time=1.98 ms

--- 10.0.0.2 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3008ms
rtt min/avg/max/mdev = 1.055/1.729/1.989/0.395 ms
```

7.4.15 Virtual Ethernet

Virtual Ethernet (veth) interfaces are software-based interfaces that operate in pairs, creating a tunnel between each other. Traffic transmitted into one interface of the pair (e.g., `veth0`) is delivered directly to its peer interface (e.g., `veth1`).

Veth interfaces are commonly used to connect network namespaces or VRFs, but they can also function as standalone virtual network interfaces.

Note: Veth interfaces must be created in pairs, where each interface acts as the peer of the other.

Configuration

Common interface configuration

```
set interfaces virtual-ethernet <interface> address <address | dhcp | dhcpv6>
```

Configure the interface with one or more IP addresses.

The following options are available:

- **address:** Assign one or more IPv4 or IPv6 addresses to the interface. For example, use `192.0.2.1/24` for IPv4 or `2001:db8::1/64` for IPv6.
- **dhcp:** The interface obtains an IPv4 address from a DHCP server on the same network segment.

- **dhcpv6:** The interface obtains an IPv6 address from a DHCPv6 server on the same network segment.

Note: If the interface obtains an IPv4 address via DHCP, and specific adjustments are needed before/after the IP address is obtained, use the provided hook scripts:

- /config/scripts/dhcp-client/pre-hooks.d/
 - /config/scripts/dhcp-client/post-hooks.d/
-

Example:

```
set interfaces virtual-ethernet veth0 address 192.0.2.1/24
set interfaces virtual-ethernet veth0 address 2001:db8::1/64
set interfaces virtual-ethernet veth0 address dhcp
set interfaces virtual-ethernet veth0 address dhcpv6
```

set interfaces virtual-ethernet <interface> description <description>

Configure a clear, descriptive alias for the interface.

This alias appears in the `show interfaces` command and SNMP-based monitoring tools.

Example:

```
set interfaces virtual-ethernet veth0 description 'This is an interface running on
↳ VyOS.'
```

VLAN

Regular VLANs (802.1q)

IEEE 802.1q, often referred to as Dot1q, is the industry standard for implementing VLANs on Ethernet networks. It defines VLAN tagging for Ethernet frames and outlines procedures for bridges and switches. The standard also includes quality-of-service prioritization (IEEE 802.1p) and defines the Generic Attribute Registration Protocol.

VLAN-aware network segments (i.e., IEEE 802.1q conformant) use VLAN tags. When a frame enters such a segment, a tag is added to indicate VLAN membership. Each frame can belong to only one VLAN. If a frame arrives without a tag, it is assumed to be part of the native VLAN.

IEEE 802.1, a working group of the IEEE 802 standards committee, has developed the standard and continues to revise it. One notable revision is 802.1Q-2014, which incorporated IEEE 802.1aq (Shortest Path Bridging) and much of the IEEE 802.1d standard.

In VyOS, 802.1q VLAN interfaces are represented as virtual subinterfaces, referred to as `vif`.

set interfaces virtual-ethernet <interface> vif <vlan-id>

Configure a VLAN interface with a unique VLAN ID.

VLAN ID identifies a specific VLAN and ranges from 0 to 4094.

You can configure multiple VLAN interfaces on a single physical interface.

Note: Only 802.1Q-tagged packets are accepted on Ethernet vifs.

set interfaces virtual-ethernet <interface> vif <vlan-id> address <address | dhcp | dhcpv6>

Configure the interface with one or more IP addresses.

The following options are available:

- **address:** Assign one or more IPv4 or IPv6 addresses to the interface. For example, use 192.0.2.1/24 for IPv4 or 2001:db8::1/64 for IPv6.
- **dhcp:** The interface obtains an IPv4 address from a DHCP server on the same network segment.
- **dhcpv6:** The interface obtains an IPv6 address from a DHCPv6 server on the same network segment.

Note: If the interface obtains an IPv4 address via DHCP, and specific adjustments are needed before/after the IP address is obtained, use the provided hook scripts:

- /config/scripts/dhcp-client/pre-hooks.d/
 - /config/scripts/dhcp-client/post-hooks.d/
-

Example:

```
set interfaces virtual-ethernet veth0 vif 10 address 192.0.2.1/24
set interfaces virtual-ethernet veth0 vif 10 address 2001:db8::1/64
set interfaces virtual-ethernet veth0 vif 10 address dhcp
set interfaces virtual-ethernet veth0 vif 10 address dhcpv6
```

set interfaces virtual-ethernet <interface> vif <vlan-id> description <description>

Configure a clear, descriptive alias for the interface.

This alias appears in the `show interfaces` command and SNMP-based monitoring tools.

Example:

```
set interfaces virtual-ethernet veth0 vif 10 description 'This is an interface_
↳running on VyOS.'
```

set interfaces virtual-ethernet <interface> vif <vlan-id> disable

Disable the interface.

The interface will be set to the administratively down (A/D) state.

Example:

```
set interfaces virtual-ethernet veth0 vif 10 disable
```

set interfaces virtual-ethernet <interface> vif <vlan-id> disable-link-detect

Disable physical link-state change detection on the interface, such as when a cable is unplugged.

By default, the interface detects physical link-state changes.

Example:

```
set interfaces virtual-ethernet veth0 vif 10 disable-link-detect
```

set interfaces virtual-ethernet <interface> vif <vlan-id> mac <xx:xx:xx:xx:xx:xx>

Configure a custom MAC address on the interface.

Example:

```
set interfaces virtual-ethernet veth0 vif 10 mac '00:53:01:02:03:04'
```

```
set interfaces virtual-ethernet <interface> vif <vlan-id> mtu <mtu>
```

Configure the MTU on the interface.

This value defines the largest packet size, in bytes, that the interface transmits without fragmentation.

Example:

```
set interfaces virtual-ethernet veth0 vif 10 mtu 1600
```

```
set interfaces virtual-ethernet <interface> vif <vlan-id> ip adjust-mss <mss | clamp-mss-to-pmtu>
```

Configure the MSS advertised in outgoing TCP SYN packets on the specified interface.

By clamping the MSS value in TCP SYN packets, you explicitly inform the remote side not to send packets larger than that size. This prevents connection issues that occur when Path MTU Discovery (PMTUD) fails.

The following options are available:

- **mss:** Sets the MSS to a specific value, in bytes. Use this option if you need to enforce a specific MSS, for example, to troubleshoot connectivity issues or accommodate specific network requirements.
- **clamp-mss-to-pmtu:** The router automatically calculates the MSS to be the interface's MTU minus 40 bytes for IPv4 traffic (20 bytes for the IPv4 header and 20 bytes for the TCP header). This option is recommended to automatically set the proper value.

```
set interfaces virtual-ethernet <interface> vif <vlan-id> ip arp-cache-timeout
```

Configure how long an ARP entry remains valid after learning an IP-to-MAC address mapping on this interface.

The default duration is 30 seconds.

An ARP entry remains valid if it receives positive feedback from higher-level protocols.

Example:

```
set interfaces virtual-ethernet veth0 vif 10 ip arp-cache-timeout 180
```

```
set interfaces virtual-ethernet <interface> vif <vlan-id> ip disable-arp-filter
```

Configure ARP filtering on this interface.

Default behavior: The kernel responds to ARP requests on this interface only if the traffic would be routed back to the ARP sender through that specific interface.

If configured: The kernel responds to ARP requests on this interface for any IP address configured on the local host, regardless of which specific interface that IP address is assigned to, and regardless of the routing table. This reflects the Linux concept that IP addresses belong to the host, not individual interfaces.

Example:

```
set interfaces virtual-ethernet veth0 vif 10 ip disable-arp-filter
```

```
set interfaces virtual-ethernet <interface> vif <vlan-id> ip disable-forwarding
```

Configure the interface for host or router behavior.

If configured, the interface switches to host mode, and IPv4 forwarding is disabled on it.

Example:

```
set interfaces virtual-ethernet veth0 vif 10 ip disable-forwarding
```

set interfaces virtual-ethernet <interface> vif <vlan-id> ip enable-directed-broadcast

Configure whether to forward IP-directed broadcast packets received on this interface.

Default behavior: IP-directed broadcast packets are dropped.

If configured: IP-directed broadcast packets are forwarded to all hosts on the destination subnet, as defined in [RFC 1812](#) and [RFC 2644](#).

Example:

```
set interfaces virtual-ethernet veth0 vif 10 ip enable-directed-broadcast
```

set interfaces virtual-ethernet <interface> vif <vlan-id> ip enable-arp-accept

Configure how to process gratuitous ARPs on this interface.

If configured, an IP-to-MAC address mapping is added to the ARP table based on gratuitous ARP requests or replies.

Note: If the ARP table already contains the IP address from a gratuitous ARP, its entry is updated regardless of whether this setting is configured.

Example:

```
set interfaces virtual-ethernet veth0 vif 10 ip enable-arp-accept
```

set interfaces virtual-ethernet <interface> vif <vlan-id> ip enable-arp-announce

Configure the source IP selection for ARP requests on this interface.

Default behavior: The kernel can use any IP address the host owns as the source IP address in ARP requests on this interface.

If configured: The kernel first attempts to select a source IP address configured on the interface that shares a common subnet with the target IP address. If there is no such subnet, the kernel selects the IP address it would normally use (based on the routing table to reach the target destination).

Example:

```
set interfaces virtual-ethernet veth0 vif 10 ip enable-arp-announce
```

set interfaces virtual-ethernet <interface> vif <vlan-id> ip enable-arp-ignore

Configure which ARP requests will be ignored on this interface.

Default behavior: The kernel responds to ARP requests for any local IP addresses, regardless of which interface they are assigned to.

If configured: The kernel responds to ARP requests only if the target IP address is assigned to this specific interface.

Example:

```
set interfaces virtual-ethernet veth0 vif 10 ip enable-arp-ignore
```

set interfaces virtual-ethernet <interface> vif <vlan-id> ip enable-proxy-arp

Configure proxy ARP on this interface.

If configured, the router (kernel) intercepts ARP requests for non-local IP addresses and replies with the MAC address of the interface that received the request. Subsequent packets destined to these IP addresses are forwarded to their actual destinations on remote subnets.

Example:

```
set interfaces virtual-ethernet veth0 vif 10 ip enable-proxy-arp
```

set interfaces virtual-ethernet <interface> vif <vlan-id> ip proxy-arp-pvlan

Configure local proxy ARP on the interface.

If configured, the router (kernel) responds to ARP requests on this VLAN interface even if the target IP address resides on the same subnet and interface.

This is used to support network isolation requirements (RFC 3069) for private VLANs (PVLANS). In PVLAN configurations, hosts on isolated ports are NOT allowed to communicate directly with each other at Layer 2, but they can communicate with the upstream router.

By replying to inter-host ARP requests with its own MAC address, the router (kernel) directs inter-host traffic through itself instead of directly between hosts.

Note: This command works independently and does not require enabling the standard proxy ARP on the interface.

Local proxy ARP is also known as:

- VLAN aggregation ([RFC 3069](#)).
- Private VLAN (Cisco, Allied Telesyn).
- Source-port filtering or port isolation (Hewlett-Packard).
- MAC-Forced Forwarding (Ericsson).

set interfaces virtual-ethernet <interface> vif <vlan-id> ip source-validation <strict | loose | disable>

Configure source IP address validation using RPF on this interface, as specified in [RFC 3704](#).

The following options are available:

- **strict:** Each incoming packet's source IP address is checked against the Forwarding Information Base (FIB). If the interface is not the best route back to that source, validation fails, and the packet is dropped.
- **loose:** Each incoming packet's source IP address is checked against the FIB. If the source IP address is unreachable through any interface, validation fails.
- **disable:** No source IP address validation is performed. All incoming packets are accepted.

RFC 3704 recommends enabling strict mode to prevent IP spoofing, such as DDoS attacks. For asymmetric or other complex routing scenarios, use loose mode.

set interfaces virtual-ethernet <interface> vif <vlan-id> ipv6 address autoconf

Configure the interface to automatically obtain an IPv6 address using SLAAC, as specified in [RFC 4862](#).

IPv6 hosts can configure themselves automatically when connected to an IPv6 network using the Neighbor Discovery Protocol via ICMPv6 router discovery messages. When first connected to a network, a host sends a link-local router solicitation multicast request for its configuration parameters. The router responds with a router advertisement packet containing Internet Layer configuration parameters.

Note: This method automatically disables IPv6 traffic forwarding on the interface.

Example:

```
set interfaces virtual-ethernet veth0 vif 10 ipv6 address autoconf
```

set interfaces virtual-ethernet <interface> vif <vlan-id> ipv6 address eui64 <prefix>

Configure the interface to assign itself an IPv6 address using the EUI-64 method, as specified in [RFC 4291](#).

Example:

```
set interfaces virtual-ethernet veth0 vif 10 ipv6 address eui64 2001:db8:beef::/64
```

set interfaces virtual-ethernet <interface> vif <vlan-id> ipv6 address no-default-link-local

Disable the automatic assignment of a link-local IPv6 address to this interface.

Example:

```
set interfaces virtual-ethernet veth0 vif 10 ipv6 address no-default-link-local
```

set interfaces virtual-ethernet <interface> vif <vlan-id> ipv6 disable-forwarding

Configure the interface for host or router behavior.

If configured, the interface switches to host mode, and IPv6 forwarding is disabled on it.

Example:

```
set interfaces virtual-ethernet veth0 vif 10 ipv6 disable-forwarding
```

set interfaces virtual-ethernet <interface> vif <vlan-id> ipv6 adjust-mss <mss | clamp-mss-to-pmtu>

Configure the MSS advertised in outgoing TCP SYN packets on the specified interface.

By clamping the MSS value in TCP SYN packets, you explicitly inform the remote side not to send packets larger than that size. This prevents connection issues when Path MTU Discovery (PMTUD) fails.

The following options are available:

- **mss:** Set the MSS to a specific value, in bytes. Use this option to enforce a specific MSS, for example, to troubleshoot connectivity issues or accommodate specific network requirements.
- **clamp-mss-to-pmtu:** The router calculates the MSS to be the interface's MTU minus 60 bytes for IPv6 traffic (40 bytes for the IPv6 header and 20 bytes for the TCP header). This option is recommended to automatically set the proper value.

set interfaces virtual-ethernet <interface> vif <vlan-id> ipv6 accept-dad <0 | 1 | 2>

Configure IPv6 DAD on the interface.

The following options are available:

- **0:** Disables DAD. No duplicate address detection is performed.
- **1:** Enables DAD (default). Duplicate addresses are detected. The interface's IPv6 operation continues for valid IPv6 addresses.

- **2:** Enables DAD and, if a MAC-based duplicate link-local address is found, disables IPv6 operation on this interface.

Example:

```
set interfaces virtual-ethernet veth0 vif 10 ipv6 accept-dad 2
```

```
set interfaces virtual-ethernet <interface> vif <vlan-id> ipv6 dup-addr-detect-transmits <n>
```

Configure the number of DAD messages that the router (kernel) sends during IPv6 address assignment on this interface.

The default value is 1.

Example:

```
set interfaces virtual-ethernet veth0 vif 10 ipv6 dup-addr-detect-transmits 5
```

```
set interfaces virtual-ethernet <interface> vif <vlan-id> vrf <vrf>
```

Assign the interface to a specific VRF instance.

See also:

For information on configuring a VRF, refer to the [VRF](#) section.

Example:

```
set interfaces virtual-ethernet veth0 vif 10 vrf red
```

DHCP(v6)

```
set interfaces virtual-ethernet <interface> vif <vlan-id> dhcp-options client-id <description>
```

Configure a DHCP client identifier for the interface, as specified in [RFC 2131](#).

The `client-id` is an identifier that the DHCP client sends to the DHCP server to uniquely identify itself for IP address assignment. By default, the client uses its MAC address. The `<description>` is a user-defined string that will be sent to the DHCP server as the DHCP client identifier.

Example:

```
set interfaces virtual-ethernet veth0 vif 10 dhcp-options client-id 'foo-bar'
```

```
set interfaces virtual-ethernet <interface> vif <vlan-id> dhcp-options host-name <hostname>
```

Configure a specific hostname for the interface.

Instead of the real hostname, the DHCP client will send the specific hostname to the DHCP server when requesting an IP address.

Example:

```
set interfaces virtual-ethernet veth0 vif 10 dhcp-options host-name 'VyOS'
```

```
set interfaces virtual-ethernet <interface> vif <vlan-id> dhcp-options vendor-class-id <vendor-id>
```

Configure the DHCP client to include a vendor-class identifier in its DHCP requests on this interface.

The vendor-class identifier is a vendor-specific byte string that enables the DHCP server to identify the device and, in some cases, provide configuration options.

Example:

```
set interfaces virtual-ethernet veth0 vif 10 dhcp-options vendor-class-id 'VyOS'
```

set interfaces virtual-ethernet <interface> vif <vlan-id> dhcp-options no-default-route

Configure the DHCP client to obtain an IP address, but ignore any default gateway provided by the DHCP server on this interface.

Example:

```
set interfaces virtual-ethernet veth0 vif 10 dhcp-options no-default-route
```

set interfaces virtual-ethernet <interface> vif <vlan-id> dhcp-options default-route-distance <distance>

Configure the distance for the default route obtained from the DHCP server on this interface.

Example:

```
set interfaces virtual-ethernet veth0 vif 10 dhcp-options default-route-distance 220
```

set interfaces virtual-ethernet <interface> vif <vlan-id> dhcp-options reject <address>

Configure the DHCP client to reject the specific IP address or IP address range from the DHCP server on this interface.

This is useful when a modem assigns a local IP address upon start. To reject multiple addresses, run this command multiple times with different values. You can reject individual addresses (192.168.100.1) or entire subnets (192.168.100.0/24).

Example:

```
set interfaces virtual-ethernet veth0 vif 10 dhcp-options reject 192.168.100.0/24
```

set interfaces virtual-ethernet <interface> vif <vlan-id> dhcp-options user-class <string>

Configure the DHCP client to send a specific user-class identifier in its DHCP requests on this interface.

The DHCP server can interpret this identifier and provide specific configuration options based on it (for example, default routes). The user-class value typically groups DHCP clients with similar configuration needs (for example, employees, guests, or printers).

Example:

```
set interfaces virtual-ethernet veth0 vif 10 dhcp-options user-class VyOS
```

set interfaces virtual-ethernet <interface> vif <vlan-id> dhcpv6-options duid <duid>

Configure a specific DUID for the DHCPv6 client on this interface.

The DUID is an identifier used by a DHCPv6 client to get an IPv6 address from a DHCPv6 server. It consists of a 2-byte type field, followed by a variable-length identifier field up to 128 bytes. The format of the identifier part depends on the DUID type:

- **DUID-LLT:** The most common type, which includes a hardware type, a timestamp, and a MAC address.

- **DUID-EN:** Is based on a vendor's enterprise number and a unique identifier assigned by the vendor.
- **DUID-LL:** Includes only a MAC address.

The DHCP server matches the DUID against its database and provides configuration data (such as address, lease times, DNS servers, etc.) to the DHCP client.

Example:

```
set interfaces virtual-ethernet veth0 vif 10 duid
→ '0e:00:00:01:00:01:27:71:db:f0:00:50:56:bf:c5:6d'
```

set interfaces virtual-ethernet <interface> vif <vlan-id> dhcpv6-options no-release

Configure the DHCP client not to send a release message when it stops running on this interface.

This helps retain the assigned address or prefix.

Example:

```
set interfaces virtual-ethernet veth0 vif 10 dhcpv6-options no-release
```

set interfaces virtual-ethernet <interface> vif <vlan-id> dhcpv6-options parameters-only

Enable a stateless DHCPv6 client mode on this interface.

In stateless mode, the DHCPv6 client requests only stateless configuration parameters from the DHCP server (for example, DNS server addresses). It doesn't request a stateful configuration, such as IPv6 addresses or prefixes.

Example:

```
set interfaces virtual-ethernet veth0 vif 10 dhcpv6-options parameters-only
```

set interfaces virtual-ethernet <interface> vif <vlan-id> dhcpv6-options rapid-commit

Enable DHCPv6 rapid commit on this interface.

When enabled, the DHCP client and server skip the negotiation steps (Advertise and Request), completing the DHCPv6 configuration process in just two messages (Solicit and final Reply).

Example:

```
set interfaces virtual-ethernet veth0 vif 10 dhcpv6-options rapid-commit
```

set interfaces virtual-ethernet <interface> vif <vlan-id> dhcpv6-options temporary

Configure the DHCPv6 client to request a temporary IPv6 address on this interface.

When configured, the DHCP client doesn't form an Identity Association for Non-temporary Addresses (IA_NA) partnership. Consequently, it only obtains a temporary IPv6 address and doesn't obtain a permanent one.

Example:

```
set interfaces virtual-ethernet veth0 vif 10 dhcpv6-options temporary
```

DHCPv6 Prefix Delegation (PD)

VyOS supports DHCPv6 Prefix Delegation (DHCPv6-PD) as described in [RFC 3633](#). DHCPv6-PD is supported by most ISPs that provide native IPv6 for consumers on fixed networks.

set interfaces virtual-ethernet <interface> vif <vlan-id> dhcpv6-options pd <id> length <length>

Configure a specific prefix length for DHCPv6-PD requests on this interface.

Some ISPs provide only a /64 prefix by default. Use this command to request a different prefix length for a specific DHCPv6-PD request, ranging from /32 (if allowed by your ISP) down to /64. <id> is a unique identifier for the DHCPv6-PD request.

The default value is 64.

To request a /56 prefix from your ISP, use:

```
set interfaces virtual-ethernet veth0 vif 10 dhcpv6-options pd 0 length 56
```

```
set interfaces virtual-ethernet <interface> vif <vlan-id> dhcpv6-options pd <id>
interface <delegatee> address <address>
```

Configure the IPv6 interface identifier (host portion) for the delegatee interface.

The value must be a decimal integer. It is appended to the delegated prefix and the configured SLA ID to form the final IPv6 address.

By default, the host portion is generated based on the parent interface's MAC address (EUI-64 format).

Example:

If a /64 prefix is delegated to interface eth8 and you configure the host portion as 65535, the resulting IPv6 address will end with ::ffff, as 65535 corresponds to ffff in hexadecimal notation.

```
set interfaces virtual-ethernet veth0 vif 10 dhcpv6-options pd 0 interface eth8
→address 65534
```

```
set interfaces virtual-ethernet <interface> vif <vlan-id> dhcpv6-options pd <id>
interface <delegatee> sla-id <id>
```

Configure the SLA ID for the delegatee interface.

The value must be a decimal integer greater than 0 and fit in the length of SLA IDs. It is converted to hexadecimal and appended to the delegated prefix to form the specific subnet prefix for the delegatee interface.

Example:

If SLA ID is 1 and the delegated prefix is 2001:db8:ffff::/48, the resulting subnet prefix for the delegatee interface will be 2001:db8:ffff:1::/64.

```
set interfaces virtual-ethernet veth0 vif 10 dhcpv6-options pd 0 interface eth8 sla-
→id 1
```

802.1ad (QinQ)**Overview**

IEEE 802.1ad, commonly known as QinQ, is an Ethernet standard first published as an amendment to 802.1q in 2005, then officially merged into the base standard in 2011.

Unlike the original 802.1q, which allows a single VLAN header per Ethernet frame, QinQ allows two VLAN headers per Ethernet frame, for the inner and the outer VLAN tags. Most often the inner VLAN tag comes from a customer while the outer tag is used by the service provider to differentiate between traffic of different customers.

Frame structure and ethertypes

The IEEE 802.1ad (QinQ) frame includes two VLAN tags:

- **The outer service tag (S-TAG):** The S-TAG is typically added by the provider.

It uses the Ethertype 0x88a8 by default.

- **The inner customer tag (C-TAG):** The C-TAG is generated by the customer's equipment and remains unchanged during transit. It uses the Ethertype 0x8100.

Implementation in VyOS

In VyOS, these tag types are associated with the following CLI options:

- **vif-s:** Corresponds to the S-TAG (Ethertype 0x88a8).
- **vif-c:** Corresponds to the C-TAG (Ethertype 0x8100).

```
set interfaces virtual-ethernet <interface> vif-s <vlan-id> vif-c <vlan-id> address
<address | dhcp | dhcpv6>
```

Configure the interface with one or more IP addresses.

The following options are available:

- **address:** Assign one or more IPv4 or IPv6 addresses to the interface. For example, use 192.0.2.1/24 for IPv4 or 2001:db8::1/64 for IPv6.
- **dhcp:** The interface obtains an IPv4 address from a DHCP server on the same network segment.
- **dhcpv6:** The interface obtains an IPv6 address from a DHCPv6 server on the same network segment.

Note: If the interface obtains an IPv4 address via DHCP, and specific adjustments are needed before/after the IP address is obtained, use the provided hook scripts:

- /config/scripts/dhcp-client/pre-hooks.d/
 - /config/scripts/dhcp-client/post-hooks.d/
-

Example:

```
set interfaces virtual-ethernet veth0 vif-s 1000 vif-c 20 address 192.0.2.1/24
set interfaces virtual-ethernet veth0 vif-s 1000 vif-c 20 address 2001:db8::1/64
set interfaces virtual-ethernet veth0 vif-s 1000 vif-c 20 address dhcp
set interfaces virtual-ethernet veth0 vif-s 1000 vif-c 20 address dhcpv6
```

```
set interfaces virtual-ethernet <interface> vif-s <vlan-id> vif-c <vlan-id> description
<description>
```

Configure a clear, descriptive alias for the interface.

This alias appears in the `show interfaces` command and SNMP-based monitoring tools.

Example:

```
set interfaces virtual-ethernet veth0 vif-s 1000 vif-c 20 description 'This is an
↪ interface running on VyOS.'
```

```
set interfaces virtual-ethernet <interface> vif-s <vlan-id> vif-c <vlan-id> disable
```

Disable the interface.

The interface will be set to the administratively down (A/D) state.

Example:

```
set interfaces virtual-ethernet veth0 vif-s 1000 vif-c 20 disable
```

```
set interfaces virtual-ethernet <interface> vif-s <vlan-id> vif-c <vlan-id>
disable-link-detect
```

Disable physical link-state change detection on the interface, such as when a cable is unplugged.

By default, the interface detects physical link-state changes.

Example:

```
set interfaces virtual-ethernet veth0 vif-s 1000 vif-c 20 disable-link-detect
```

```
set interfaces virtual-ethernet <interface> vif-s <vlan-id> vif-c <vlan-id> mac
<xx:xx:xx:xx:xx:xx>
```

Configure a custom MAC address on the interface.

Example:

```
set interfaces virtual-ethernet veth0 vif-s 1000 vif-c 20 mac '00:53:01:02:03:04'
```

```
set interfaces virtual-ethernet <interface> vif-s <vlan-id> vif-c <vlan-id> mtu <mtu>
```

Configure the MTU on the interface.

This value defines the largest packet size, in bytes, that the interface transmits without fragmentation.

Example:

```
set interfaces virtual-ethernet veth0 vif-s 1000 vif-c 20 mtu 1600
```

```
set interfaces virtual-ethernet <interface> vif-s <vlan-id> vif-c <vlan-id> ip adjust-mss
<mss | clamp-mss-to-pmtu>
```

Configure the MSS advertised in outgoing TCP SYN packets on the specified interface.

By clamping the MSS value in TCP SYN packets, you explicitly inform the remote side not to send packets larger than that size. This prevents connection issues that occur when Path MTU Discovery (PMTUD) fails.

The following options are available:

- **mss:** Sets the MSS to a specific value, in bytes. Use this option if you need to enforce a specific MSS, for example, to troubleshoot connectivity issues or accommodate specific network requirements.
- **clamp-mss-to-pmtu:** The router automatically calculates the MSS to be the interface's MTU minus 40 bytes for IPv4 traffic (20 bytes for the IPv4 header and 20 bytes for the TCP header). This option is recommended to automatically set the proper value.

```
set interfaces virtual-ethernet <interface> vif-s <vlan-id> vif-c <vlan-id> ip
arp-cache-timeout
```

Configure how long an ARP entry remains valid after learning an IP-to-MAC address mapping on this interface.

The default duration is 30 seconds.

An ARP entry remains valid if it receives positive feedback from higher-level protocols.

Example:

```
set interfaces virtual-ethernet veth0 vif-s 1000 vif-c 20 ip arp-cache-timeout 180
```

```
set interfaces virtual-ethernet <interface> vif-s <vlan-id> vif-c <vlan-id> ip
disable-arp-filter
```

Configure ARP filtering on this interface.

Default behavior: The kernel responds to ARP requests on this interface only if the traffic would be routed back to the ARP sender through that specific interface.

If configured: The kernel responds to ARP requests on this interface for any IP address configured on the local host, regardless of which specific interface that IP address is assigned to, and regardless of the routing table. This reflects the Linux concept that IP addresses belong to the host, not individual interfaces.

Example:

```
set interfaces virtual-ethernet veth0 vif-s 1000 vif-c 20 ip disable-arp-filter
```

```
set interfaces virtual-ethernet <interface> vif-s <vlan-id> vif-c <vlan-id> ip
disable-forwarding
```

Configure the interface for host or router behavior.

If configured, the interface switches to host mode, and IPv4 forwarding is disabled on it.

Example:

```
set interfaces virtual-ethernet veth0 vif-s 1000 vif-c 20 ip disable-forwarding
```

```
set interfaces virtual-ethernet <interface> vif-s <vlan-id> vif-c <vlan-id> ip
enable-directed-broadcast
```

Configure whether to forward IP-directed broadcast packets received on this interface.

Default behavior: IP-directed broadcast packets are dropped.

If configured: IP-directed broadcast packets are forwarded to all hosts on the destination subnet, as defined in [RFC 1812](#) and [RFC 2644](#).

Example:

```
set interfaces virtual-ethernet veth0 vif-s 1000 vif-c 20 ip enable-directed-
→broadcast
```

```
set interfaces virtual-ethernet <interface> vif-s <vlan-id> vif-c <vlan-id> ip
enable-arp-accept
```

Configure how to process gratuitous ARPs on this interface.

If configured, an IP-to-MAC address mapping is added to the ARP table based on gratuitous ARP requests or replies.

Note: If the ARP table already contains the IP address from a gratuitous ARP, its entry is updated regardless of whether this setting is configured.

Example:

```
set interfaces virtual-ethernet veth0 vif-s 1000 vif-c 20 ip enable-arp-accept
```

```
set interfaces virtual-ethernet <interface> vif-s <vlan-id> vif-c <vlan-id> ip
enable-arp-announce
```

Configure the source IP selection for ARP requests on this interface.

Default behavior: The kernel can use any IP address the host owns as the source IP address in ARP requests on this interface.

If configured: The kernel first attempts to select a source IP address configured on the interface that shares a common subnet with the target IP address. If there is no such subnet, the kernel selects the IP address it would normally use (based on the routing table to reach the target destination).

Example:

```
set interfaces virtual-ethernet veth0 vif-s 1000 vif-c 20 ip enable-arp-announce
```

```
set interfaces virtual-ethernet <interface> vif-s <vlan-id> vif-c <vlan-id> ip
enable-arp-ignore
```

Configure which ARP requests will be ignored on this interface.

Default behavior: The kernel responds to ARP requests for any local IP addresses, regardless of which interface they are assigned to.

If configured: The kernel responds to ARP requests only if the target IP address is assigned to this specific interface.

Example:

```
set interfaces virtual-ethernet veth0 vif-s 1000 vif-c 20 ip enable-arp-ignore
```

```
set interfaces virtual-ethernet <interface> vif-s <vlan-id> vif-c <vlan-id> ip
enable-proxy-arp
```

Configure proxy ARP on this interface.

If configured, the router (kernel) intercepts ARP requests for non-local IP addresses and replies with the MAC address of the interface that received the request. Subsequent packets destined to these IP addresses are forwarded to their actual destinations on remote subnets.

Example:

```
set interfaces virtual-ethernet veth0 vif-s 1000 vif-c 20 ip enable-proxy-arp
```

```
set interfaces virtual-ethernet <interface> vif-s <vlan-id> vif-c <vlan-id> ip
proxy-arp-pvlan
```

Configure local proxy ARP on the interface.

If configured, the router (kernel) responds to ARP requests on this VLAN interface even if the target IP address resides on the same subnet and interface.

This is used to support network isolation requirements (RFC 3069) for private VLANs (PVLANS). In PVLAN configurations, hosts on isolated ports are NOT allowed to communicate directly with each other at Layer 2, but they can communicate with the upstream router.

By replying to inter-host ARP requests with its own MAC address, the router (kernel) directs inter-host traffic through itself instead of directly between hosts.

Note: This command works independently and does not require enabling the standard proxy ARP on the interface.

Local proxy ARP is also known as:

- VLAN aggregation ([RFC 3069](#)).
- Private VLAN (Cisco, Allied Telesyn).
- Source-port filtering or port isolation (Hewlett-Packard).
- MAC-Forced Forwarding (Ericsson).

```
set interfaces virtual-ethernet <interface> vif-s <vlan-id> vif-c <vlan-id> ip
source-validation <strict | loose | disable>
```

Configure source IP address validation using RPF on this interface, as specified in [RFC 3704](#).

The following options are available:

- **strict:** Each incoming packet's source IP address is checked against the Forwarding Information Base (FIB). If the interface is not the best route back to that source, validation fails, and the packet is dropped.
- **loose:** Each incoming packet's source IP address is checked against the FIB. If the source IP address is unreachable through any interface, validation fails.
- **disable:** No source IP address validation is performed. All incoming packets are accepted.

RFC 3704 recommends enabling strict mode to prevent IP spoofing, such as DDoS attacks. For asymmetric or other complex routing scenarios, use loose mode.

```
set interfaces virtual-ethernet <interface> vif-s <vlan-id> vif-c <vlan-id> ipv6 address
autoconf
```

Configure the interface to automatically obtain an IPv6 address using SLAAC, as specified in [RFC 4862](#).

IPv6 hosts can configure themselves automatically when connected to an IPv6 network using the Neighbor Discovery Protocol via ICMPv6 router discovery messages. When first connected to a network, a host sends a link-local router solicitation multicast request for its configuration parameters. The router responds with a router advertisement packet containing Internet Layer configuration parameters.

Note: This method automatically disables IPv6 traffic forwarding on the interface.

Example:

```
set interfaces virtual-ethernet veth0 vif-s 1000 vif-c 20 ipv6 address autoconf
```

```
set interfaces virtual-ethernet <interface> vif-s <vlan-id> vif-c <vlan-id> ipv6 address
eui64 <prefix>
```

Configure the interface to assign itself an IPv6 address using the EUI-64 method, as specified in [RFC 4291](#).

Example:

```
set interfaces virtual-ethernet veth0 vif-s 1000 vif-c 20 ipv6 address eui64
↪2001:db8:beef::/64
```

```
set interfaces virtual-ethernet <interface> vif-s <vlan-id> vif-c <vlan-id> ipv6 address
no-default-link-local
```

Disable the automatic assignment of a link-local IPv6 address to this interface.

Example:

```
set interfaces virtual-ethernet veth0 vif-s 1000 vif-c 20 ipv6 address no-default-
↪link-local
```

```
set interfaces virtual-ethernet <interface> vif-s <vlan-id> vif-c <vlan-id> ipv6
disable-forwarding
```

Configure the interface for host or router behavior.

If configured, the interface switches to host mode, and IPv6 forwarding is disabled on it.

Example:

```
set interfaces virtual-ethernet veth0 vif-s 1000 vif-c 20 ipv6 disable-forwarding
```

```
set interfaces virtual-ethernet <interface> vif-s <vlan-id> vif-c <vlan-id> ipv6
adjust-mss <mss | clamp-mss-to-pmtu>
```

Configure the MSS advertised in outgoing TCP SYN packets on the specified interface.

By clamping the MSS value in TCP SYN packets, you explicitly inform the remote side not to send packets larger than that size. This prevents connection issues when Path MTU Discovery (PMTUD) fails.

The following options are available:

- **mss:** Set the MSS to a specific value, in bytes. Use this option to enforce a specific MSS, for example, to troubleshoot connectivity issues or accommodate specific network requirements.
- **clamp-mss-to-pmtu:** The router calculates the MSS to be the interface's MTU minus 60 bytes for IPv6 traffic (40 bytes for the IPv6 header and 20 bytes for the TCP header). This option is recommended to automatically set the proper value.

```
set interfaces virtual-ethernet <interface> vif-s <vlan-id> vif-c <vlan-id> ipv6
accept-dad <0 | 1 | 2>
```

Configure IPv6 DAD on the interface.

The following options are available:

- **0:** Disables DAD. No duplicate address detection is performed.
- **1:** Enables DAD (default). Duplicate addresses are detected. The interface's IPv6 operation continues for valid IPv6 addresses.
- **2:** Enables DAD and, if a MAC-based duplicate link-local address is found, disables IPv6 operation on this interface.

Example:

```
set interfaces virtual-ethernet veth0 vif-s 1000 vif-c 20 ipv6 accept-dad 2
```

```
set interfaces virtual-ethernet <interface> vif-s <vlan-id> vif-c <vlan-id> ipv6
dup-addr-detect-transmits <n>
```

Configure the number of DAD messages that the router (kernel) sends during IPv6 address assignment on this interface.

The default value is 1.

Example:

```
set interfaces virtual-ethernet veth0 vif-s 1000 vif-c 20 ipv6 dup-addr-detect-
↪transmits 5
```

```
set interfaces virtual-ethernet <interface> vif-s <vlan-id> vif-c <vlan-id> vrf <vrf>
```

Assign the interface to a specific VRF instance.

See also:

For information on configuring a VRF, refer to the [VRF](#) section.

Example:

```
set interfaces virtual-ethernet veth0 vif-s 1000 vif-c 20 vrf red
```

DHCP(v6)

set interfaces virtual-ethernet <interface> vif-s <vlan-id> vif-c <vlan-id> dhcp-options client-id <description>

Configure a DHCP client identifier for the interface, as specified in [RFC 2131](#).

The `client-id` is an identifier that the DHCP client sends to the DHCP server to uniquely identify itself for IP address assignment. By default, the client uses its MAC address. The `<description>` is a user-defined string that will be sent to the DHCP server as the DHCP client identifier.

Example:

```
set interfaces virtual-ethernet veth0 vif-s 1000 vif-c 20 dhcp-options client-id
  ↪ 'foo-bar'
```

set interfaces virtual-ethernet <interface> vif-s <vlan-id> vif-c <vlan-id> dhcp-options host-name <hostname>

Configure a specific hostname for the interface.

Instead of the real hostname, the DHCP client will send the specific hostname to the DHCP server when requesting an IP address.

Example:

```
set interfaces virtual-ethernet veth0 vif-s 1000 vif-c 20 dhcp-options host-name
  ↪ 'VyOS'
```

set interfaces virtual-ethernet <interface> vif-s <vlan-id> vif-c <vlan-id> dhcp-options vendor-class-id <vendor-id>

Configure the DHCP client to include a vendor-class identifier in its DHCP requests on this interface.

The vendor-class identifier is a vendor-specific byte string that enables the DHCP server to identify the device and, in some cases, provide configuration options.

Example:

```
set interfaces virtual-ethernet veth0 vif-s 1000 vif-c 20 dhcp-options vendor-class-
  ↪ id 'VyOS'
```

set interfaces virtual-ethernet <interface> vif-s <vlan-id> vif-c <vlan-id> dhcp-options no-default-route

Configure the DHCP client to obtain an IP address, but ignore any default gateway provided by the DHCP server on this interface.

Example:

```
set interfaces virtual-ethernet veth0 vif-s 1000 vif-c 20 dhcp-options no-default-
  ↪ route
```

```
set interfaces virtual-ethernet <interface> vif-s <vlan-id> vif-c <vlan-id> dhcp-options
default-route-distance <distance>
```

Configure the distance for the default route obtained from the DHCP server on this interface.

Example:

```
set interfaces virtual-ethernet veth0 vif-s 1000 vif-c 20 dhcp-options default-
↳ route-distance 220
```

```
set interfaces virtual-ethernet <interface> vif-s <vlan-id> vif-c <vlan-id> dhcp-options
reject <address>
```

Configure the DHCP client to reject the specific IP address or IP address range from the DHCP server on this interface.

This is useful when a modem assigns a local IP address upon start. To reject multiple addresses, run this command multiple times with different values. You can reject individual addresses (192.168.100.1) or entire subnets (192.168.100.0/24).

Example:

```
set interfaces virtual-ethernet veth0 vif-s 1000 vif-c 20 dhcp-options reject 192.
↳ 168.100.0/24
```

```
set interfaces virtual-ethernet <interface> vif-s <vlan-id> vif-c <vlan-id> dhcp-options
user-class <string>
```

Configure the DHCP client to send a specific user-class identifier in its DHCP requests on this interface.

The DHCP server can interpret this identifier and provide specific configuration options based on it (for example, default routes). The user-class value typically groups DHCP clients with similar configuration needs (for example, employees, guests, or printers).

Example:

```
set interfaces virtual-ethernet veth0 vif-s 1000 vif-c 20 dhcp-options user-class
↳ VyOS
```

```
set interfaces virtual-ethernet <interface> vif-s <vlan-id> vif-c <vlan-id>
dhcpv6-options duid <duid>
```

Configure a specific DUID for the DHCPv6 client on this interface.

The DUID is an identifier used by a DHCPv6 client to get an IPv6 address from a DHCPv6 server. It consists of a 2-byte type field, followed by a variable-length identifier field up to 128 bytes. The format of the identifier part depends on the DUID type:

- **DUID-LLT:** The most common type, which includes a hardware type, a timestamp, and a MAC address.
- **DUID-EN:** Is based on a vendor's enterprise number and a unique identifier assigned by the vendor.
- **DUID-LL:** Includes only a MAC address.

The DHCP server matches the DUID against its database and provides configuration data (such as address, lease times, DNS servers, etc.) to the DHCP client.

Example:

```
set interfaces virtual-ethernet veth0 vif-s 1000 vif-c 20 duid
↳ '0e:00:00:01:00:01:27:71:db:f0:00:50:56:bf:c5:6d'
```

```
set interfaces virtual-ethernet <interface> vif-s <vlan-id> vif-c <vlan-id>
dhcpv6-options no-release
```

Configure the DHCP client not to send a release message when it stops running on this interface.

This helps retain the assigned address or prefix.

Example:

```
set interfaces virtual-ethernet veth0 vif-s 1000 vif-c 20 dhcpv6-options no-release
```

```
set interfaces virtual-ethernet <interface> vif-s <vlan-id> vif-c <vlan-id>
dhcpv6-options parameters-only
```

Enable a stateless DHCPv6 client mode on this interface.

In stateless mode, the DHCPv6 client requests only stateless configuration parameters from the DHCP server (for example, DNS server addresses). It doesn't request a stateful configuration, such as IPv6 addresses or prefixes.

Example:

```
set interfaces virtual-ethernet veth0 vif-s 1000 vif-c 20 dhcpv6-options parameters-
→only
```

```
set interfaces virtual-ethernet <interface> vif-s <vlan-id> vif-c <vlan-id>
dhcpv6-options rapid-commit
```

Enable DHCPv6 rapid commit on this interface.

When enabled, the DHCP client and server skip the negotiation steps (Advertise and Request), completing the DHCPv6 configuration process in just two messages (Solicit and final Reply).

Example:

```
set interfaces virtual-ethernet veth0 vif-s 1000 vif-c 20 dhcpv6-options rapid-
→commit
```

```
set interfaces virtual-ethernet <interface> vif-s <vlan-id> vif-c <vlan-id>
dhcpv6-options temporary
```

Configure the DHCPv6 client to request a temporary IPv6 address on this interface.

When configured, the DHCP client doesn't form an Identity Association for Non-temporary Addresses (IA_NA) partnership. Consequently, it only obtains a temporary IPv6 address and doesn't obtain a permanent one.

Example:

```
set interfaces virtual-ethernet veth0 vif-s 1000 vif-c 20 dhcpv6-options temporary
```

DHCPv6 Prefix Delegation (PD)

VyOS supports DHCPv6 Prefix Delegation (DHCPv6-PD) as described in [RFC 3633](#). DHCPv6-PD is supported by most ISPs that provide native IPv6 for consumers on fixed networks.

```
set interfaces virtual-ethernet <interface> vif-s <vlan-id> vif-c <vlan-id>
dhcpv6-options pd <id> length <length>
```

Configure a specific prefix length for DHCPv6-PD requests on this interface.

Some ISPs provide only a /64 prefix by default. Use this command to request a different prefix length for a specific DHCPv6-PD request, ranging from /32 (if allowed by your ISP) down to /64. <id> is a unique identifier for the DHCPv6-PD request.

The default value is 64.

To request a /56 prefix from your ISP, use:

```
set interfaces virtual-ethernet veth0 vif-s 1000 vif-c 20 dhcpv6-options pd 0
↪length 56
```

**set interfaces virtual-ethernet <interface> vif-s <vlan-id> vif-c <vlan-id>
dhcpv6-options pd <id> interface <delegatee> address <address>**

Configure the IPv6 interface identifier (host portion) for the delegatee interface.

The value must be a decimal integer. It is appended to the delegated prefix and the configured SLA ID to form the final IPv6 address.

By default, the host portion is generated based on the parent interface's MAC address (EUI-64 format).

Example:

If a /64 prefix is delegated to interface eth8 and you configure the host portion as 65535, the resulting IPv6 address will end with ::ffff, as 65535 corresponds to ffff in hexadecimal notation.

```
set interfaces virtual-ethernet veth0 vif-s 1000 vif-c 20 dhcpv6-options pd 0
↪interface eth8 address 65534
```

**set interfaces virtual-ethernet <interface> vif-s <vlan-id> vif-c <vlan-id>
dhcpv6-options pd <id> interface <delegatee> sla-id <id>**

Configure the SLA ID for the delegatee interface.

The value must be a decimal integer greater than 0 and fit in the length of SLA IDs. It is converted to hexadecimal and appended to the delegated prefix to form the specific subnet prefix for the delegatee interface.

Example:

If SLA ID is 1 and the delegated prefix is 2001:db8:ffff::/48, the resulting subnet prefix for the delegatee interface will be 2001:db8:ffff:1::/64.

```
set interfaces virtual-ethernet veth0 vif-s 1000 vif-c 20 dhcpv6-options pd 0
↪interface eth8 sla-id 1
```

set interfaces virtual-ethernet <interface> disable

Disable the interface.

The interface will be set to the administratively down (A/D) state.

Example:

```
set interfaces virtual-ethernet veth0 disable
```

set interfaces virtual-ethernet <interface> vrf <vrf>

Assign the interface to a specific VRF instance.

See also:

For information on configuring a VRF, refer to the [VRF](#) section.

Example:

```
set interfaces virtual-ethernet veth0 vrf red
```

Operation

show interfaces virtual-ethernet

Show brief interface information.

```
vyos@vyos:~$ show interfaces virtual-ethernet
Codes: S - State, L - Link, u - Up, D - Down, A - Admin Down
Interface      IP Address      S/L  Description
-----
veth10         100.64.0.0/31   u/u
veth11         100.64.0.1/31   u/u
```

show interfaces virtual-ethernet <interface>

Show detailed interface information.

```
vyos@vyos:~$ show interfaces virtual-ethernet veth11
10: veth11@veth10: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue master_
→red state UP group default qlen 1000
link/ether b2:7b:df:e9:11:11 brd ff:ff:ff:ff:ff:ff
inet 100.64.0.1/31 scope global veth11
valid_lft forever preferred_lft forever
inet6 fe80::b07b:dfff:fe47:e911/64 scope link
valid_lft forever preferred_lft forever

RX:  bytes    packets    errors    dropped    overrun    mcast
0           0           0           0           0
TX:  bytes    packets    errors    dropped    carrier    collisions
1369707     4267        0           0           0           0
```

Example

The following example shows how to connect the global VRF to VRF 'red' using the veth10 and veth11 veth pair.

```
set interfaces virtual-ethernet veth10 address '100.64.0.0/31'
set interfaces virtual-ethernet veth10 peer-name 'veth11'
set interfaces virtual-ethernet veth11 address '100.64.0.1/31'
set interfaces virtual-ethernet veth11 peer-name 'veth10'
set interfaces virtual-ethernet veth11 vrf 'red'
set vrf name red table '1000'
```

```
vyos@vyos:~$ ping 100.64.0.1
PING 100.64.0.1 (100.64.0.1) 56(84) bytes of data.
64 bytes from 100.64.0.1: icmp_seq=1 ttl=64 time=0.080 ms
64 bytes from 100.64.0.1: icmp_seq=2 ttl=64 time=0.119 ms
```

7.4.16 VTI (virtual tunnel interface)

VTIs (virtual tunnel interfaces) let you create secure, encrypted tunnels between private networks or hosts across public infrastructure, such as the Internet. They operate alongside an underlying IPsec tunnel, which handles encapsulation and encryption, while VTIs function exclusively as routing interfaces.

Configuration

Common interface configuration

set interfaces vti <interface> address <address>

Configure one or more interface IP addresses.

<address> supports IPv4 and IPv6 in CIDR notation. For example, use 192.0.2.1/24 for IPv4 or 2001:db8::1/64 for IPv6.

Example:

```
set interfaces vti vti0 address 192.0.2.1/24
set interfaces vti vti0 address 2001:db8::1/64
```

set interfaces vti <interface> description <description>

Configure a clear, descriptive alias for the interface.

This alias appears in the `show interfaces` command and SNMP-based monitoring tools.

Example:

```
set interfaces vti vti0 description 'This is an interface running on VyOS.'
```

set interfaces vti <interface> disable

Disable the interface.

The interface will be set to the administratively down (A/D) state.

Example:

```
set interfaces vti vti0 disable
```

set interfaces vti <interface> ip adjust-mss <mss | clamp-mss-to-pmtu>

Configure the MSS advertised in outgoing TCP SYN packets on the specified interface.

By clamping the MSS value in TCP SYN packets, you explicitly inform the remote side not to send packets larger than that size. This prevents connection issues that occur when Path MTU Discovery (PMTUD) fails.

The following options are available:

- **mss:** Sets the MSS to a specific value, in bytes. Use this option if you need to enforce a specific MSS, for example, to troubleshoot connectivity issues or accommodate specific network requirements.
- **clamp-mss-to-pmtu:** The router automatically calculates the MSS to be the interface's MTU minus 40 bytes for IPv4 traffic (20 bytes for the IPv4 header and 20 bytes for the TCP header). This option is recommended to automatically set the proper value.

set interfaces vti <interface> ip arp-cache-timeout

Configure how long an ARP entry remains valid after learning an IP-to-MAC address mapping on this interface.

The default duration is 30 seconds.

An ARP entry remains valid if it receives positive feedback from higher-level protocols.

Example:

```
set interfaces vti vti0 ip arp-cache-timeout 180
```

set interfaces vti <interface> ip disable-arp-filter

Configure ARP filtering on this interface.

Default behavior: The kernel responds to ARP requests on this interface only if the traffic would be routed back to the ARP sender through that specific interface.

If configured: The kernel responds to ARP requests on this interface for any IP address configured on the local host, regardless of which specific interface that IP address is assigned to, and regardless of the routing table. This reflects the Linux concept that IP addresses belong to the host, not individual interfaces.

Example:

```
set interfaces vti vti0 ip disable-arp-filter
```

set interfaces vti <interface> ip disable-forwarding

Configure the interface for host or router behavior.

If configured, the interface switches to host mode, and IPv4 forwarding is disabled on it.

Example:

```
set interfaces vti vti0 ip disable-forwarding
```

set interfaces vti <interface> ip enable-directed-broadcast

Configure whether to forward IP-directed broadcast packets received on this interface.

Default behavior: IP-directed broadcast packets are dropped.

If configured: IP-directed broadcast packets are forwarded to all hosts on the destination subnet, as defined in [RFC 1812](#) and [RFC 2644](#).

Example:

```
set interfaces vti vti0 ip enable-directed-broadcast
```

set interfaces vti <interface> ip enable-arp-accept

Configure how to process gratuitous ARPs on this interface.

If configured, an IP-to-MAC address mapping is added to the ARP table based on gratuitous ARP requests or replies.

Note: If the ARP table already contains the IP address from a gratuitous ARP, its entry is updated regardless of whether this setting is configured.

Example:

```
set interfaces vti vti0 ip enable-arp-accept
```

set interfaces vti <interface> ip enable-arp-announce

Configure the source IP selection for ARP requests on this interface.

Default behavior: The kernel can use any IP address the host owns as the source IP address in ARP requests on this interface.

If configured: The kernel first attempts to select a source IP address configured on the interface that shares a common subnet with the target IP address. If there is no such subnet, the kernel selects the IP address it would normally use (based on the routing table to reach the target destination).

Example:

```
set interfaces vti vti0 ip enable-arp-announce
```

set interfaces vti <interface> ip enable-arp-ignore

Configure which ARP requests will be ignored on this interface.

Default behavior: The kernel responds to ARP requests for any local IP addresses, regardless of which interface they are assigned to.

If configured: The kernel responds to ARP requests only if the target IP address is assigned to this specific interface.

Example:

```
set interfaces vti vti0 ip enable-arp-ignore
```

set interfaces vti <interface> ip enable-proxy-arp

Configure proxy ARP on this interface.

If configured, the router (kernel) intercepts ARP requests for non-local IP addresses and replies with the MAC address of the interface that received the request. Subsequent packets destined to these IP addresses are forwarded to their actual destinations on remote subnets.

Example:

```
set interfaces vti vti0 ip enable-proxy-arp
```

set interfaces vti <interface> ip proxy-arp-pvlan

Configure local proxy ARP on the interface.

If configured, the router (kernel) responds to ARP requests on this VLAN interface even if the target IP address resides on the same subnet and interface.

This is used to support network isolation requirements (RFC 3069) for private VLANs (PVLANS). In PVLAN configurations, hosts on isolated ports are NOT allowed to communicate directly with each other at Layer 2, but they can communicate with the upstream router.

By replying to inter-host ARP requests with its own MAC address, the router (kernel) directs inter-host traffic through itself instead of directly between hosts.

Note: This command works independently and does not require enabling the standard proxy ARP on the interface.

Local proxy ARP is also known as:

- VLAN aggregation ([RFC 3069](#)).
- Private VLAN (Cisco, Allied Telesyn).
- Source-port filtering or port isolation (Hewlett-Packard).
- MAC-Forced Forwarding (Ericsson).

set interfaces vti <interface> ip source-validation <strict | loose | disable>

Configure source IP address validation using RPF on this interface, as specified in [RFC 3704](#).

The following options are available:

- **strict:** Each incoming packet's source IP address is checked against the Forwarding Information Base (FIB). If the interface is not the best route back to that source, validation fails, and the packet is dropped.
- **loose:** Each incoming packet's source IP address is checked against the FIB. If the source IP address is unreachable through any interface, validation fails.
- **disable:** No source IP address validation is performed. All incoming packets are accepted.

RFC 3704 recommends enabling strict mode to prevent IP spoofing, such as DDoS attacks. For asymmetric or other complex routing scenarios, use loose mode.

set interfaces vti <interface> ipv6 address autoconf

Configure the interface to automatically obtain an IPv6 address using SLAAC, as specified in [RFC 4862](#).

IPv6 hosts can configure themselves automatically when connected to an IPv6 network using the Neighbor Discovery Protocol via ICMPv6 router discovery messages. When first connected to a network, a host sends a link-local router solicitation multicast request for its configuration parameters. The router responds with a router advertisement packet containing Internet Layer configuration parameters.

Note: This method automatically disables IPv6 traffic forwarding on the interface.

Example:

```
set interfaces vti vti0 ipv6 address autoconf
```

set interfaces vti <interface> ipv6 address eui64 <prefix>

Configure the interface to assign itself an IPv6 address using the EUI-64 method, as specified in [RFC 4291](#).

Example:

```
set interfaces vti vti0 ipv6 address eui64 2001:db8:beef::/64
```

set interfaces vti <interface> ipv6 address no-default-link-local

Disable the automatic assignment of a link-local IPv6 address to this interface.

Example:

```
set interfaces vti vti0 ipv6 address no-default-link-local
```

set interfaces vti <interface> ipv6 disable-forwarding

Configure the interface for host or router behavior.

If configured, the interface switches to host mode, and IPv6 forwarding is disabled on it.

Example:

```
set interfaces vti vti0 ipv6 disable-forwarding
```

set interfaces vti <interface> ipv6 adjust-mss <mss | clamp-mss-to-pmtu>

Configure the MSS advertised in outgoing TCP SYN packets on the specified interface.

By clamping the MSS value in TCP SYN packets, you explicitly inform the remote side not to send packets larger than that size. This prevents connection issues when Path MTU Discovery (PMTUD) fails.

The following options are available:

- **mss:** Set the MSS to a specific value, in bytes. Use this option to enforce a specific MSS, for example, to troubleshoot connectivity issues or accommodate specific network requirements.
- **clamp-mss-to-pmtu:** The router calculates the MSS to be the interface's MTU minus 60 bytes for IPv6 traffic (40 bytes for the IPv6 header and 20 bytes for the TCP header). This option is recommended to automatically set the proper value.

set interfaces vti <interface> ipv6 accept-dad <0 | 1 | 2>

Configure IPv6 DAD on the interface.

The following options are available:

- **0:** Disables DAD. No duplicate address detection is performed.
- **1:** Enables DAD (default). Duplicate addresses are detected. The interface's IPv6 operation continues for valid IPv6 addresses.
- **2:** Enables DAD and, if a MAC-based duplicate link-local address is found, disables IPv6 operation on this interface.

Example:

```
set interfaces vti vti0 ipv6 accept-dad 2
```

set interfaces vti <interface> ipv6 dup-addr-detect-transmits <n>

Configure the number of DAD messages that the router (kernel) sends during IPv6 address assignment on this interface.

The default value is 1.

Example:

```
set interfaces vti vti0 ipv6 dup-addr-detect-transmits 5
```

set interfaces vti <interface> mtu <mtu>

Configure the MTU on the interface.

This value defines the largest packet size, in bytes, that the interface transmits without fragmentation.

Example:

```
set interfaces vti vti0 mtu 1600
```

set interfaces vti <interface> mirror egress <monitor-interface>

Configure mirroring of outgoing traffic from the specified VTI to the designated monitor interface.

set interfaces vti <interface> mirror ingress <monitor-interface>

Configure mirroring of incoming traffic from the specified VTI to the designated monitor interface.

```
set interfaces vti <interface> redirect <interface>
```

Enable redirection of incoming packets to the specified interface.

```
set interfaces vti <interface> vrf <vrf>
```

Assign the interface to a specific VRF instance.

See also:

For information on configuring a VRF, refer to the [VRF](#) section.

Example:

```
set interfaces vti vti0 vrf red
```

Operation

```
show interfaces vti <vtiX>
```

Show the operational status and traffic statistics for the specified VTI.

```
show interfaces vti <vtiX> brief
```

Show a brief operational status summary for the specified VTI.

Example

Configure a VTI

Assign IPv4 and IPv6 addresses to the VTI, along with a brief description:

```
set interfaces vti vti0 address 192.168.2.249/30
set interfaces vti vti0 address 2001:db8:2::249/64
set interfaces vti vti0 description "Description"
```

Resulting configuration:

```
vyos@vyos# show interfaces vti
vti vti0 {
    address 192.168.2.249/30
    address 2001:db8:2::249/64
    description "Description"
}
```

Warning: When configuring site-to-site IPsec with VTIs, ensure that route autoinstall is disabled.

```
set vpn ipsec options disable-route-autoinstall
```

For more information about the IPsec and VTI issue, as well as the `disable-route-autoinstall` option, see: <https://blog.vyos.io/vyos-1-dot-2-0-development-news-in-july>.

The root cause of the problem is that VTI tunnels require their traffic selectors to be set to `0.0.0.0/0` for traffic to match the tunnel, even though routing decisions are based on netfilter marks. Unless route insertion is explicitly disabled, strongSWAN incorrectly inserts a default route through the VTI peer address, causing all traffic to be misrouted.

7.4.17 VXLAN

VXLAN (Virtual Extensible LAN) is a network virtualization technology that addresses scalability challenges in large cloud computing environments. It encapsulates Ethernet frames (Layer 2) within UDP datagrams (Layer 4), which are then transmitted via UDP port 4789, as assigned by IANA. VXLAN endpoints, called VTEPs, terminate VXLAN tunnels and can be either virtual or physical switch ports.

VXLAN supports up to 16 million logical networks and enables Layer 2 adjacency across Layer 3 IP networks. It uses multicast or unicast with head-end replication (HER) to flood broadcast, unknown unicast, and multicast (BUM) traffic.

The VXLAN specification was initially developed by VMware, Arista Networks, and Cisco. Other supporters include Huawei, Broadcom, Citrix, Pica8, Big Switch Networks, Cumulus Networks, Dell EMC, Ericsson, Mellanox, FreeBSD, OpenBSD, Red Hat, Joyent, and Juniper Networks.

VXLAN is officially documented by the IETF in [RFC 7348](#).

When configuring VXLAN in a VyOS virtual machine, ensure that MAC spoofing (Hyper-V) or Forged Transmits (ESX) are permitted. Otherwise, the hypervisor may block forwarded frames.

Note: Although the IANA-assigned VXLAN port is **4789**, VyOS uses the Linux default UDP port **8472** for VXLAN interfaces. To ensure compatibility with other vendors, set the port to the IANA standard **4789**.

Configuration

Common interface configuration

set interfaces vxlan <interface> address <address>

Configure one or more interface IP addresses.

<address> supports IPv4 and IPv6 in CIDR notation. For example, use 192.0.2.1/24 for IPv4 or 2001:db8::1/64 for IPv6.

Example:

```
set interfaces vxlan vxlan0 address 192.0.2.1/24
set interfaces vxlan vxlan0 address 2001:db8::1/64
```

set interfaces vxlan <interface> description <description>

Configure a clear, descriptive alias for the interface.

This alias appears in the `show interfaces` command and SNMP-based monitoring tools.

Example:

```
set interfaces vxlan vxlan0 description 'This is an interface running on VyOS.'
```

set interfaces vxlan <interface> disable

Disable the interface.

The interface will be set to the administratively down (A/D) state.

Example:

```
set interfaces vxlan vxlan0 disable
```

set interfaces vxlan <interface> disable-flow-control

Disable Ethernet flow control (IEEE 802.3x pause frames) on the interface.

Ethernet flow control, defined by the IEEE 802.3x standard, temporarily stops data transmission to prevent packet loss during network congestion. For example, when a sender transmits data faster than the receiver can process it.

Disabling Ethernet flow control means the interface will not signal the connected device to pause transmission and will drop packets if overwhelmed.

Example:

```
set interfaces vxlan vxlan0 disable-flow-control
```

set interfaces vxlan <interface> disable-link-detect

Disable physical link-state change detection on the interface, such as when a cable is unplugged.

By default, the interface detects physical link-state changes.

Example:

```
set interfaces vxlan vxlan0 disable-link-detect
```

set interfaces vxlan <interface> mac <xx:xx:xx:xx:xx:xx>

Configure a custom MAC address on the interface.

Example:

```
set interfaces vxlan vxlan0 mac '00:53:01:02:03:04'
```

set interfaces vxlan <interface> mtu <mtu>

Configure the MTU on the interface.

This value defines the largest packet size, in bytes, that the interface transmits without fragmentation.

Example:

```
set interfaces vxlan vxlan0 mtu 1600
```

set interfaces vxlan <interface> ip adjust-mss <mss | clamp-mss-to-pmtu>

Configure the MSS advertised in outgoing TCP SYN packets on the specified interface.

By clamping the MSS value in TCP SYN packets, you explicitly inform the remote side not to send packets larger than that size. This prevents connection issues that occur when Path MTU Discovery (PMTUD) fails.

The following options are available:

- **mss:** Sets the MSS to a specific value, in bytes. Use this option if you need to enforce a specific MSS, for example, to troubleshoot connectivity issues or accommodate specific network requirements.
- **clamp-mss-to-pmtu:** The router automatically calculates the MSS to be the interface's MTU minus 40 bytes for IPv4 traffic (20 bytes for the IPv4 header and 20 bytes for the TCP header). This option is recommended to automatically set the proper value.

set interfaces vxlan <interface> ip arp-cache-timeout

Configure how long an ARP entry remains valid after learning an IP-to-MAC address mapping on this interface.

The default duration is 30 seconds.

An ARP entry remains valid if it receives positive feedback from higher-level protocols.

Example:

```
set interfaces vxlan vxlan0 ip arp-cache-timeout 180
```

set interfaces vxlan <interface> ip disable-arp-filter

Configure ARP filtering on this interface.

Default behavior: The kernel responds to ARP requests on this interface only if the traffic would be routed back to the ARP sender through that specific interface.

If configured: The kernel responds to ARP requests on this interface for any IP address configured on the local host, regardless of which specific interface that IP address is assigned to, and regardless of the routing table. This reflects the Linux concept that IP addresses belong to the host, not individual interfaces.

Example:

```
set interfaces vxlan vxlan0 ip disable-arp-filter
```

set interfaces vxlan <interface> ip disable-forwarding

Configure the interface for host or router behavior.

If configured, the interface switches to host mode, and IPv4 forwarding is disabled on it.

Example:

```
set interfaces vxlan vxlan0 ip disable-forwarding
```

set interfaces vxlan <interface> ip enable-directed-broadcast

Configure whether to forward IP-directed broadcast packets received on this interface.

Default behavior: IP-directed broadcast packets are dropped.

If configured: IP-directed broadcast packets are forwarded to all hosts on the destination subnet, as defined in [RFC 1812](#) and [RFC 2644](#).

Example:

```
set interfaces vxlan vxlan0 ip enable-directed-broadcast
```

set interfaces vxlan <interface> ip enable-arp-accept

Configure how to process gratuitous ARPs on this interface.

If configured, an IP-to-MAC address mapping is added to the ARP table based on gratuitous ARP requests or replies.

Note: If the ARP table already contains the IP address from a gratuitous ARP, its entry is updated regardless of whether this setting is configured.

Example:

```
set interfaces vxlan vxlan0 ip enable-arp-accept
```

set interfaces vxlan <interface> ip enable-arp-announce

Configure the source IP selection for ARP requests on this interface.

Default behavior: The kernel can use any IP address the host owns as the source IP address in ARP requests on this interface.

If configured: The kernel first attempts to select a source IP address configured on the interface that shares a common subnet with the target IP address. If there is no such subnet, the kernel selects the IP address it would normally use (based on the routing table to reach the target destination).

Example:

```
set interfaces vxlan vxlan0 ip enable-arp-announce
```

set interfaces vxlan <interface> ip enable-arp-ignore**Configure which ARP requests will be ignored on this interface.**

Default behavior: The kernel responds to ARP requests for any local IP addresses, regardless of which interface they are assigned to.

If configured: The kernel responds to ARP requests only if the target IP address is assigned to this specific interface.

Example:

```
set interfaces vxlan vxlan0 ip enable-arp-ignore
```

set interfaces vxlan <interface> ip enable-proxy-arp**Configure proxy ARP on this interface.**

If configured, the router (kernel) intercepts ARP requests for non-local IP addresses and replies with the MAC address of the interface that received the request. Subsequent packets destined to these IP addresses are forwarded to their actual destinations on remote subnets.

Example:

```
set interfaces vxlan vxlan0 ip enable-proxy-arp
```

set interfaces vxlan <interface> ip proxy-arp-pvlan**Configure local proxy ARP on the interface.**

If configured, the router (kernel) responds to ARP requests on this VLAN interface even if the target IP address resides on the same subnet and interface.

This is used to support network isolation requirements (RFC 3069) for private VLANs (PVLANS). In PVLAN configurations, hosts on isolated ports are NOT allowed to communicate directly with each other at Layer 2, but they can communicate with the upstream router.

By replying to inter-host ARP requests with its own MAC address, the router (kernel) directs inter-host traffic through itself instead of directly between hosts.

Note: This command works independently and does not require enabling the standard proxy ARP on the interface.

Local proxy ARP is also known as:

- VLAN aggregation ([RFC 3069](#)).
- Private VLAN (Cisco, Allied Telesyn).

- Source-port filtering or port isolation (Hewlett-Packard).
- MAC-Forced Forwarding (Ericsson).

set interfaces vxlan <interface> ip source-validation <strict | loose | disable>

Configure source IP address validation using RPF on this interface, as specified in [RFC 3704](#).

The following options are available:

- **strict:** Each incoming packet's source IP address is checked against the Forwarding Information Base (FIB). If the interface is not the best route back to that source, validation fails, and the packet is dropped.
- **loose:** Each incoming packet's source IP address is checked against the FIB. If the source IP address is unreachable through any interface, validation fails.
- **disable:** No source IP address validation is performed. All incoming packets are accepted.

RFC 3704 recommends enabling strict mode to prevent IP spoofing, such as DDoS attacks. For asymmetric or other complex routing scenarios, use loose mode.

set interfaces vxlan <interface> ipv6 address autoconf

Configure the interface to automatically obtain an IPv6 address using SLAAC, as specified in [RFC 4862](#).

IPv6 hosts can configure themselves automatically when connected to an IPv6 network using the Neighbor Discovery Protocol via ICMPv6 router discovery messages. When first connected to a network, a host sends a link-local router solicitation multicast request for its configuration parameters. The router responds with a router advertisement packet containing Internet Layer configuration parameters.

Note: This method automatically disables IPv6 traffic forwarding on the interface.

Example:

```
set interfaces vxlan vxlan0 ipv6 address autoconf
```

set interfaces vxlan <interface> ipv6 address eui64 <prefix>

Configure the interface to assign itself an IPv6 address using the EUI-64 method, as specified in [RFC 4291](#).

Example:

```
set interfaces vxlan vxlan0 ipv6 address eui64 2001:db8:beef::/64
```

set interfaces vxlan <interface> ipv6 address no-default-link-local

Disable the automatic assignment of a link-local IPv6 address to this interface.

Example:

```
set interfaces vxlan vxlan0 ipv6 address no-default-link-local
```

set interfaces vxlan <interface> ipv6 disable-forwarding

Configure the interface for host or router behavior.

If configured, the interface switches to host mode, and IPv6 forwarding is disabled on it.

Example:

```
set interfaces vxlan vxlan0 ipv6 disable-forwarding
```

```
set interfaces vxlan <interface> ipv6 adjust-mss <mss | clamp-mss-to-pmtu>
```

Configure the MSS advertised in outgoing TCP SYN packets on the specified interface.

By clamping the MSS value in TCP SYN packets, you explicitly inform the remote side not to send packets larger than that size. This prevents connection issues when Path MTU Discovery (PMTUD) fails.

The following options are available:

- **mss:** Set the MSS to a specific value, in bytes. Use this option to enforce a specific MSS, for example, to troubleshoot connectivity issues or accommodate specific network requirements.
- **clamp-mss-to-pmtu:** The router calculates the MSS to be the interface's MTU minus 60 bytes for IPv6 traffic (40 bytes for the IPv6 header and 20 bytes for the TCP header). This option is recommended to automatically set the proper value.

```
set interfaces vxlan <interface> ipv6 accept-dad <0 | 1 | 2>
```

Configure IPv6 DAD on the interface.

The following options are available:

- **0:** Disables DAD. No duplicate address detection is performed.
- **1:** Enables DAD (default). Duplicate addresses are detected. The interface's IPv6 operation continues for valid IPv6 addresses.
- **2:** Enables DAD and, if a MAC-based duplicate link-local address is found, disables IPv6 operation on this interface.

Example:

```
set interfaces vxlan vxlan0 ipv6 accept-dad 2
```

```
set interfaces vxlan <interface> ipv6 dup-addr-detect-transmits <n>
```

Configure the number of DAD messages that the router (kernel) sends during IPv6 address assignment on this interface.

The default value is 1.

Example:

```
set interfaces vxlan vxlan0 ipv6 dup-addr-detect-transmits 5
```

```
set interfaces vxlan <interface> vrf <vrf>
```

Assign the interface to a specific VRF instance.

See also:

For information on configuring a VRF, refer to the [VRF](#) section.

Example:

```
set interfaces vxlan vxlan0 vrf red
```

VXLAN-specific options

set interfaces vxlan <interface> vni <number>

Configure a VNI for the VXLAN interface.

Each VXLAN segment is identified by this 24-bit VNI, allowing up to 16 million segments to coexist within the same administrative domain.

set interfaces vxlan <interface> port <port>

Configure the UDP port of the remote VXLAN endpoint.

Note: Although the IANA-assigned VXLAN port is **4789**, VyOS uses the Linux default UDP port **8472** for VXLAN interfaces.

set interfaces vxlan <interface> source-address <address>

Configure the source IP address for the VXLAN underlay.

Warning: This setting is mandatory when deploying VXLAN via L2VPN/EVPN.

set interfaces vxlan <interface> gpe

Enable the GPE (Generic Protocol Extension) for the VXLAN interface.

To use this feature, you must configure the interface with the `external` parameter.

set interfaces vxlan <interface> parameters external

Configure the VXLAN interface to use an external control plane, such as BGP L2VPN/EVPN, for remote endpoint discovery.

If not configured, the internal FDB is used.

set interfaces vxlan <interface> parameters neighbor-suppress

Enable ARP and ND suppression on the VXLAN interface.

This reduces ARP and ND message flooding across the VXLAN network. As defined in [RFC 7432#section-10](#), participating VTEPs use known MAC-to-IP bindings to reply to local requests on behalf of remote hosts.

set interfaces vxlan <interface> parameters nolearning

Disable SLLA (Source Link-Layer Address) and IP address learning on the VXLAN interface.

set interfaces vxlan <interface> parameters vni-filter

Enable VNI filtering on the VXLAN interface.

When enabled, the interface only receives packets with VNIs configured in its VNI filtering table.

Note: VNI filtering works only if the interface is configured with the `external` parameter.

Unicast

set interfaces vxlan <interface> remote <address>

Configure the IPv4 or IPv6 address of the remote VTEP.

Unlike multicast setups, this command allows you to directly configure the remote IPv4 or IPv6 address.

Multicast

set interfaces vxlan <interface> source-interface <interface>

Configure the source interface for the VXLAN underlay.

All VXLAN traffic is sent and received through the specified interface. This setting is mandatory when deploying VXLAN over a multicast network.

set interfaces vxlan <interface> group <address>

Configure the IPv4 or IPv6 multicast group address for the VXLAN interface.

VXLAN tunnels can be built using either multicast group or unicast IP addresses.

Multicast VXLAN

Topology: PC4 - Leaf2 - Spine1 - Leaf3 - PC5

PC4 uses the IP address 10.0.0.4/24, and PC5 uses the IP address 10.0.0.5/24. Both devices assume they reside within the same broadcast domain.

Assume PC4 on Leaf2 pings PC5 on Leaf3. Rather than manually specifying Leaf3 as the remote endpoint, Leaf2 encapsulates the packet into a UDP datagram and sends it to the designated multicast address via Spine1. Spine1 forwards the packet to all leaves in the same multicast group, including Leaf3. Upon receiving the datagram, Leaf3 forwards it to PC5 and learns that PC4 is reachable through Leaf2 by inspecting the source IP in the encapsulated datagram.

PC5 receives the ping and responds with an echo reply. Leaf3, now aware of PC4's location, forwards the reply directly to Leaf2's unicast address. Upon receiving the echo reply, Leaf2 learns that PC5 is reachable through Leaf3.

After this discovery, subsequent traffic between PC4 and PC5 will not use the multicast address between the leaves, as both leaves have learned the PCs' locations. This reduces multicast traffic and network load, improving scalability as more leaves are added.

Single VXLAN device (SVD)

In VyOS, you can configure multiple **VLAN-to-VNI mappings** for EVPN-VXLAN on a single container interface, known as a single VXLAN device (SVD). This enables significant VNI scaling because a separate VXLAN interface is not required for each VNI.

set interfaces vxlan <interface> vlan-to-vni <vlan> vni <vni>

Map a VLAN ID to a VNI on the specified VXLAN interface.

The VXLAN interface can be added to a bridge.

The following example shows an SVD configuration with multiple VLAN-to-VNI mappings.

```

set interfaces bridge br0 member interface vxlan0
set interfaces vxlan vxlan0 parameters external
set interfaces vxlan vxlan0 source-interface 'dum0'
set interfaces vxlan vxlan0 vlan-to-vni 10 vni '10010'
set interfaces vxlan vxlan0 vlan-to-vni 11 vni '10011'
set interfaces vxlan vxlan0 vlan-to-vni 30 vni '10030'
set interfaces vxlan vxlan0 vlan-to-vni 31 vni '10031'

```

Example

The following example demonstrates a multicast VXLAN deployment.

The setup includes three routers: Spine1, a Cisco IOS router, and Leaf2 and Leaf3, which are VyOS routers.

Topology: Leaf2 - Spine1 - Leaf3.

The topology is built using GNS3.

```

Spine1:
fa0/2 towards Leaf2, IP-address: 10.1.2.1/24
fa0/3 towards Leaf3, IP-address: 10.1.3.1/24

Leaf2:
Eth0 towards Spine1, IP-address: 10.1.2.2/24
Eth1 towards a VLAN-aware switch

Leaf3:
Eth0 towards Spine1, IP-address 10.1.3.3/24
Eth1 towards a VLAN-aware switch

```

Spine1 configuration:

```

conf t
ip multicast-routing
!
interface fastethernet0/2
ip address 10.1.2.1 255.255.255.0
ip pim sparse-dense-mode
!
interface fastethernet0/3
ip address 10.1.3.1 255.255.255.0
ip pim sparse-dense-mode
!
router ospf 1
network 10.0.0.0 0.255.255.255 area 0

```

Multicast routing is required for scalable traffic forwarding between leaves. PIM (Protocol Independent Multicast) must be enabled towards the leaves so the spine can learn from which multicast groups each leaf expects traffic.

Leaf2 configuration:

```

set interfaces ethernet eth0 address '10.1.2.2/24'
set protocols ospf area 0 network '10.0.0.0/8'

```

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```

! First VXLAN interface
set interfaces bridge br241 address '172.16.241.1/24'
set interfaces bridge br241 member interface 'eth1.241'
set interfaces bridge br241 member interface 'vxlan241'

set interfaces vxlan vxlan241 group '239.0.0.241'
set interfaces vxlan vxlan241 source-interface 'eth0'
set interfaces vxlan vxlan241 vni '241'

! Second VXLAN interface
set interfaces bridge br242 address '172.16.242.1/24'
set interfaces bridge br242 member interface 'eth1.242'
set interfaces bridge br242 member interface 'vxlan242'

set interfaces vxlan vxlan242 group '239.0.0.242'
set interfaces vxlan vxlan242 source-interface 'eth0'
set interfaces vxlan vxlan242 vni '242'

```

Leaf3 configuration:

```

set interfaces ethernet eth0 address '10.1.3.3/24'
set protocols ospf area 0 network '10.0.0.0/8'

! First VXLAN interface
set interfaces bridge br241 address '172.16.241.1/24'
set interfaces bridge br241 member interface 'eth1.241'
set interfaces bridge br241 member interface 'vxlan241'

set interfaces vxlan vxlan241 group '239.0.0.241'
set interfaces vxlan vxlan241 source-interface 'eth0'
set interfaces vxlan vxlan241 vni '241'

! Second VXLAN interface
set interfaces bridge br242 address '172.16.242.1/24'
set interfaces bridge br242 member interface 'eth1.242'
set interfaces bridge br242 member interface 'vxlan242'

set interfaces vxlan vxlan242 group '239.0.0.242'
set interfaces vxlan vxlan242 source-interface 'eth0'
set interfaces vxlan vxlan242 vni '242'

```

The configurations for Leaf2 and Leaf3 are nearly identical. Detailed explanations for each command are provided below.

```
set interfaces bridge br241 address '172.16.241.1/24'
```

This command creates a bridge to bind traffic on eth1 VLAN 241 with the vxlan241 interface. The IP address is optional. If configured, it can serve as the default gateway for each leaf, allowing devices on the VLAN to reach other subnets. Subnets must be redistributed by OSPF (Open Shortest Path First) so the spine can learn how to reach them. To advertise 172.16/12 networks, change the OSPF network from 10.0.0.0/8 to 0.0.0.0/0.

```
set interfaces bridge br241 member interface 'eth1.241'
set interfaces bridge br241 member interface 'vxlan241'
```

These commands bind `eth1.241` and `vlan241` as member interfaces of the same bridge.

```
set interfaces vxlan vxlan241 group '239.0.0.241'
```

This command configures the multicast group used by all leaves for this VLAN extension. It must be the same on all leaves that have this interface.

```
set interfaces vxlan vxlan241 source-interface 'eth0'
```

This command configures the interface that listens for multicast packets. It can also be a loopback interface.

```
set interfaces vxlan vxlan241 vni '241'
```

This command configures the unique ID for the VXLAN interface.

```
set interfaces vxlan vxlan241 port 12345
```

VyOS uses the Linux default UDP port **8472** for VXLAN interfaces. This command allows you to configure a different UDP port.

Unicast VXLAN

As an alternative to multicast, you can configure the VXLAN tunnel by specifying the remote IPv4 address directly. The following updates the previous multicast example:

```
# leaf2 and leaf3
delete interfaces vxlan vxlan241 group '239.0.0.241'
delete interfaces vxlan vxlan241 source-interface 'eth0'

# leaf2
set interfaces vxlan vxlan241 remote 10.1.3.3

# leaf3
set interfaces vxlan vxlan241 remote 10.1.2.2
```

The default UDP port is 8472. To configure a different port, use `set interfaces vxlan <vxlanN> port <port>`.

7.4.18 Wireless LAN / Wi-Fi

WLAN (Wireless LAN) interfaces provide 802.11 (a/b/g/n/ac) wireless connectivity, referred to as Wi-Fi, and operate in one of the following modes:

- WAP (Wireless Access-Point) mode provides network access to connecting stations if the physical hardware supports acting as a WAP
- Station mode acts as a Wi-Fi client accessing the network through an available WAP
- Monitor mode lets the system passively monitor wireless traffic

If the system detects an unconfigured wireless device, it will be automatically added to the configuration tree, specifying any detected settings (for example, its MAC address) and configured to run in monitor mode.

Configuration

Common interface configuration

set interfaces wireless <interface> address <address | dhcp | dhcpv6>

Configure the interface with one or more IP addresses.

The following options are available:

- **address:** Assign one or more IPv4 or IPv6 addresses to the interface. For example, use 192.0.2.1/24 for IPv4 or 2001:db8::1/64 for IPv6.
- **dhcp:** The interface obtains an IPv4 address from a DHCP server on the same network segment.
- **dhcpv6:** The interface obtains an IPv6 address from a DHCPv6 server on the same network segment.

Note: If the interface obtains an IPv4 address via DHCP, and specific adjustments are needed before/after the IP address is obtained, use the provided hook scripts:

- /config/scripts/dhcp-client/pre-hooks.d/
 - /config/scripts/dhcp-client/post-hooks.d/
-

Example:

```
set interfaces wireless wlan0 address 192.0.2.1/24
set interfaces wireless wlan0 address 2001:db8::1/64
set interfaces wireless wlan0 address dhcp
set interfaces wireless wlan0 address dhcpv6
```

set interfaces wireless <interface> description <description>

Configure a clear, descriptive alias for the interface.

This alias appears in the `show interfaces` command and SNMP-based monitoring tools.

Example:

```
set interfaces wireless wlan0 description 'This is an interface running on VyOS.'
```

set interfaces wireless <interface> disable

Disable the interface.

The interface will be set to the administratively down (A/D) state.

Example:

```
set interfaces wireless wlan0 disable
```

set interfaces wireless <interface> disable-flow-control

Disable Ethernet flow control (IEEE 802.3x pause frames) on the interface.

Ethernet flow control, defined by the IEEE 802.3x standard, temporarily stops data transmission to prevent packet loss during network congestion. For example, when a sender transmits data faster than the receiver can process it.

Disabling Ethernet flow control means the interface will not signal the connected device to pause transmission and will drop packets if overwhelmed.

Example:

```
set interfaces wireless wlan0 disable-flow-control
```

set interfaces wireless <interface> disable-link-detect

Disable physical link-state change detection on the interface, such as when a cable is unplugged.

By default, the interface detects physical link-state changes.

Example:

```
set interfaces wireless wlan0 disable-link-detect
```

set interfaces wireless <interface> mac <xx:xx:xx:xx:xx:xx>

Configure a custom MAC address on the interface.

Example:

```
set interfaces wireless wlan0 mac '00:53:01:02:03:04'
```

set interfaces wireless <interface> mtu <mtu>

Configure the MTU on the interface.

This value defines the largest packet size, in bytes, that the interface transmits without fragmentation.

Example:

```
set interfaces wireless wlan0 mtu 1600
```

set interfaces wireless <interface> ip adjust-mss <mss | clamp-mss-to-pmtu>

Configure the MSS advertised in outgoing TCP SYN packets on the specified interface.

By clamping the MSS value in TCP SYN packets, you explicitly inform the remote side not to send packets larger than that size. This prevents connection issues that occur when Path MTU Discovery (PMTUD) fails.

The following options are available:

- **mss:** Sets the MSS to a specific value, in bytes. Use this option if you need to enforce a specific MSS, for example, to troubleshoot connectivity issues or accommodate specific network requirements.
- **clamp-mss-to-pmtu:** The router automatically calculates the MSS to be the interface's MTU minus 40 bytes for IPv4 traffic (20 bytes for the IPv4 header and 20 bytes for the TCP header). This option is recommended to automatically set the proper value.

set interfaces wireless <interface> ip arp-cache-timeout

Configure how long an ARP entry remains valid after learning an IP-to-MAC address mapping on this interface.

The default duration is 30 seconds.

An ARP entry remains valid if it receives positive feedback from higher-level protocols.

Example:

```
set interfaces wireless wlan0 ip arp-cache-timeout 180
```

set interfaces wireless <interface> ip disable-arp-filter

Configure ARP filtering on this interface.

Default behavior: The kernel responds to ARP requests on this interface only if the traffic would be routed back to the ARP sender through that specific interface.

If configured: The kernel responds to ARP requests on this interface for any IP address configured on the local host, regardless of which specific interface that IP address is assigned to, and regardless of the routing table. This reflects the Linux concept that IP addresses belong to the host, not individual interfaces.

Example:

```
set interfaces wireless wlan0 ip disable-arp-filter
```

set interfaces wireless <interface> ip disable-forwarding**Configure the interface for host or router behavior.**

If configured, the interface switches to host mode, and IPv4 forwarding is disabled on it.

Example:

```
set interfaces wireless wlan0 ip disable-forwarding
```

set interfaces wireless <interface> ip enable-directed-broadcast**Configure whether to forward IP-directed broadcast packets received on this interface.**

Default behavior: IP-directed broadcast packets are dropped.

If configured: IP-directed broadcast packets are forwarded to all hosts on the destination subnet, as defined in [RFC 1812](#) and [RFC 2644](#).

Example:

```
set interfaces wireless wlan0 ip enable-directed-broadcast
```

set interfaces wireless <interface> ip enable-arp-accept**Configure how to process gratuitous ARPs on this interface.**

If configured, an IP-to-MAC address mapping is added to the ARP table based on gratuitous ARP requests or replies.

Note: If the ARP table already contains the IP address from a gratuitous ARP, its entry is updated regardless of whether this setting is configured.

Example:

```
set interfaces wireless wlan0 ip enable-arp-accept
```

set interfaces wireless <interface> ip enable-arp-announce**Configure the source IP selection for ARP requests on this interface.**

Default behavior: The kernel can use any IP address the host owns as the source IP address in ARP requests on this interface.

If configured: The kernel first attempts to select a source IP address configured on the interface that shares a common subnet with the target IP address. If there is no such subnet, the kernel selects the IP address it would normally use (based on the routing table to reach the target destination).

Example:

```
set interfaces wireless wlan0 ip enable-arp-announce
```

set interfaces wireless <interface> ip enable-arp-ignore

Configure which ARP requests will be ignored on this interface.

Default behavior: The kernel responds to ARP requests for any local IP addresses, regardless of which interface they are assigned to.

If configured: The kernel responds to ARP requests only if the target IP address is assigned to this specific interface.

Example:

```
set interfaces wireless wlan0 ip enable-arp-ignore
```

set interfaces wireless <interface> ip enable-proxy-arp

Configure proxy ARP on this interface.

If configured, the router (kernel) intercepts ARP requests for non-local IP addresses and replies with the MAC address of the interface that received the request. Subsequent packets destined to these IP addresses are forwarded to their actual destinations on remote subnets.

Example:

```
set interfaces wireless wlan0 ip enable-proxy-arp
```

set interfaces wireless <interface> ip proxy-arp-pvlan

Configure local proxy ARP on the interface.

If configured, the router (kernel) responds to ARP requests on this VLAN interface even if the target IP address resides on the same subnet and interface.

This is used to support network isolation requirements (RFC 3069) for private VLANs (PVLANS). In PVLAN configurations, hosts on isolated ports are NOT allowed to communicate directly with each other at Layer 2, but they can communicate with the upstream router.

By replying to inter-host ARP requests with its own MAC address, the router (kernel) directs inter-host traffic through itself instead of directly between hosts.

Note: This command works independently and does not require enabling the standard proxy ARP on the interface.

Local proxy ARP is also known as:

- VLAN aggregation ([RFC 3069](#)).
- Private VLAN (Cisco, Allied Telesyn).
- Source-port filtering or port isolation (Hewlett-Packard).
- MAC-Forced Forwarding (Ericsson).

set interfaces wireless <interface> ip source-validation <strict | loose | disable>

Configure source IP address validation using RPF on this interface, as specified in [RFC 3704](#).

The following options are available:

- **strict:** Each incoming packet's source IP address is checked against the Forwarding Information Base (FIB). If the interface is not the best route back to that source, validation fails, and the packet is dropped.
- **loose:** Each incoming packet's source IP address is checked against the FIB. If the source IP address is unreachable through any interface, validation fails.
- **disable:** No source IP address validation is performed. All incoming packets are accepted.

RFC 3704 recommends enabling strict mode to prevent IP spoofing, such as DDoS attacks. For asymmetric or other complex routing scenarios, use loose mode.

set interfaces wireless <interface> ipv6 address autoconf

Configure the interface to automatically obtain an IPv6 address using SLAAC, as specified in [RFC 4862](#).

IPv6 hosts can configure themselves automatically when connected to an IPv6 network using the Neighbor Discovery Protocol via ICMPv6 router discovery messages. When first connected to a network, a host sends a link-local router solicitation multicast request for its configuration parameters. The router responds with a router advertisement packet containing Internet Layer configuration parameters.

Note: This method automatically disables IPv6 traffic forwarding on the interface.

Example:

```
set interfaces wireless wlan0 ipv6 address autoconf
```

set interfaces wireless <interface> ipv6 address eui64 <prefix>

Configure the interface to assign itself an IPv6 address using the EUI-64 method, as specified in [RFC 4291](#).

Example:

```
set interfaces wireless wlan0 ipv6 address eui64 2001:db8:beef::/64
```

set interfaces wireless <interface> ipv6 address no-default-link-local

Disable the automatic assignment of a link-local IPv6 address to this interface.

Example:

```
set interfaces wireless wlan0 ipv6 address no-default-link-local
```

set interfaces wireless <interface> ipv6 disable-forwarding

Configure the interface for host or router behavior.

If configured, the interface switches to host mode, and IPv6 forwarding is disabled on it.

Example:

```
set interfaces wireless wlan0 ipv6 disable-forwarding
```

set interfaces wireless <interface> ipv6 adjust-mss <mss | clamp-mss-to-pmtu>

Configure the MSS advertised in outgoing TCP SYN packets on the specified interface.

By clamping the MSS value in TCP SYN packets, you explicitly inform the remote side not to send packets larger than that size. This prevents connection issues when Path MTU Discovery (PMTUD) fails.

The following options are available:

- **mss:** Set the MSS to a specific value, in bytes. Use this option to enforce a specific MSS, for example, to troubleshoot connectivity issues or accommodate specific network requirements.
- **clamp-mss-to-pmtu:** The router calculates the MSS to be the interface's MTU minus 60 bytes for IPv6 traffic (40 bytes for the IPv6 header and 20 bytes for the TCP header). This option is recommended to automatically set the proper value.

set interfaces wireless <interface> ipv6 accept-dad <0 | 1 | 2>

Configure IPv6 DAD on the interface.

The following options are available:

- **0:** Disables DAD. No duplicate address detection is performed.
- **1:** Enables DAD (default). Duplicate addresses are detected. The interface's IPv6 operation continues for valid IPv6 addresses.
- **2:** Enables DAD and, if a MAC-based duplicate link-local address is found, disables IPv6 operation on this interface.

Example:

```
set interfaces wireless wlan0 ipv6 accept-dad 2
```

set interfaces wireless <interface> ipv6 dup-addr-detect-transmits <n>

Configure the number of DAD messages that the router (kernel) sends during IPv6 address assignment on this interface.

The default value is 1.

Example:

```
set interfaces wireless wlan0 ipv6 dup-addr-detect-transmits 5
```

set interfaces wireless <interface> vrf <vrf>

Assign the interface to a specific VRF instance.

See also:

For information on configuring a VRF, refer to the [VRF](#) section.

Example:

```
set interfaces wireless wlan0 vrf red
```

DHCP(v6)

set interfaces wireless <interface> dhcp-options client-id <description>

Configure a DHCP client identifier for the interface, as specified in [RFC 2131](#).

The `client-id` is an identifier that the DHCP client sends to the DHCP server to uniquely identify itself for IP address assignment. By default, the client uses its MAC address. The `<description>` is a user-defined string that will be sent to the DHCP server as the DHCP client identifier.

Example:

```
set interfaces wireless wlan0 dhcp-options client-id 'foo-bar'
```

set interfaces wireless <interface> dhcp-options host-name <hostname>

Configure a specific hostname for the interface.

Instead of the real hostname, the DHCP client will send the specific hostname to the DHCP server when requesting an IP address.

Example:

```
set interfaces wireless wlan0 dhcp-options host-name 'VyOS'
```

set interfaces wireless <interface> dhcp-options vendor-class-id <vendor-id>

Configure the DHCP client to include a vendor-class identifier in its DHCP requests on this interface.

The vendor-class identifier is a vendor-specific byte string that enables the DHCP server to identify the device and, in some cases, provide configuration options.

Example:

```
set interfaces wireless wlan0 dhcp-options vendor-class-id 'VyOS'
```

set interfaces wireless <interface> dhcp-options no-default-route

Configure the DHCP client to obtain an IP address, but ignore any default gateway provided by the DHCP server on this interface.

Example:

```
set interfaces wireless wlan0 dhcp-options no-default-route
```

set interfaces wireless <interface> dhcp-options default-route-distance <distance>

Configure the distance for the default route obtained from the DHCP server on this interface.

Example:

```
set interfaces wireless wlan0 dhcp-options default-route-distance 220
```

set interfaces wireless <interface> dhcp-options reject <address>

Configure the DHCP client to reject the specific IP address or IP address range from the DHCP server on this interface.

This is useful when a modem assigns a local IP address upon start. To reject multiple addresses, run this command multiple times with different values. You can reject individual addresses (192.168.100.1) or entire subnets (192.168.100.0/24).

Example:

```
set interfaces wireless wlan0 dhcp-options reject 192.168.100.0/24
```

set interfaces wireless <interface> dhcp-options user-class <string>

Configure the DHCP client to send a specific user-class identifier in its DHCP requests on this interface.

The DHCP server can interpret this identifier and provide specific configuration options based on it (for example, default routes). The user-class value typically groups DHCP clients with similar configuration needs (for example, employees, guests, or printers).

Example:

```
set interfaces wireless wlan0 dhcp-options user-class VyOS
```

set interfaces wireless <interface> dhcpv6-options duid <duid>

Configure a specific DUID for the DHCPv6 client on this interface.

The DUID is an identifier used by a DHCPv6 client to get an IPv6 address from a DHCPv6 server. It consists of a 2-byte type field, followed by a variable-length identifier field up to 128 bytes. The format of the identifier part depends on the DUID type:

- **DUID-LLT:** The most common type, which includes a hardware type, a timestamp, and a MAC address.
- **DUID-EN:** Is based on a vendor's enterprise number and a unique identifier assigned by the vendor.
- **DUID-LL:** Includes only a MAC address.

The DHCP server matches the DUID against its database and provides configuration data (such as address, lease times, DNS servers, etc.) to the DHCP client.

Example:

```
set interfaces wireless wlan0 duid '0e:00:00:01:00:01:27:71:db:f0:00:50:56:bf:c5:6d'
```

set interfaces wireless <interface> dhcpv6-options no-release

Configure the DHCP client not to send a release message when it stops running on this interface.

This helps retain the assigned address or prefix.

Example:

```
set interfaces wireless wlan0 dhcpv6-options no-release
```

set interfaces wireless <interface> dhcpv6-options parameters-only

Enable a stateless DHCPv6 client mode on this interface.

In stateless mode, the DHCPv6 client requests only stateless configuration parameters from the DHCP server (for example, DNS server addresses). It doesn't request a stateful configuration, such as IPv6 addresses or prefixes.

Example:

```
set interfaces wireless wlan0 dhcpv6-options parameters-only
```

set interfaces wireless <interface> dhcpv6-options rapid-commit

Enable DHCPv6 rapid commit on this interface.

When enabled, the DHCP client and server skip the negotiation steps (Advertise and Request), completing the DHCPv6 configuration process in just two messages (Solicit and final Reply).

Example:

```
set interfaces wireless wlan0 dhcpv6-options rapid-commit
```

set interfaces wireless <interface> dhcpv6-options temporary

Configure the DHCPv6 client to request a temporary IPv6 address on this interface.

When configured, the DHCP client doesn't form an Identity Association for Non-temporary Addresses (IA_NA) partnership. Consequently, it only obtains a temporary IPv6 address and doesn't obtain a permanent one.

Example:

```
set interfaces wireless wlan0 dhcpv6-options temporary
```


DHCPv6 Prefix Delegation (PD)

VyOS supports DHCPv6 Prefix Delegation (DHCPv6-PD) as described in [RFC 3633](#). DHCPv6-PD is supported by most ISPs that provide native IPv6 for consumers on fixed networks.

set interfaces wireless <interface> dhcpv6-options pd <id> length <length>

Configure a specific prefix length for DHCPv6-PD requests on this interface.

Some ISPs provide only a /64 prefix by default. Use this command to request a different prefix length for a specific DHCPv6-PD request, ranging from /32 (if allowed by your ISP) down to /64. <id> is a unique identifier for the DHCPv6-PD request.

The default value is 64.

To request a /56 prefix from your ISP, use:

```
set interfaces wireless wlan0 dhcpv6-options pd 0 length 56
```

set interfaces wireless <interface> dhcpv6-options pd <id> interface <delegatee> address <address>

Configure the IPv6 interface identifier (host portion) for the delegatee interface.

The value must be a decimal integer. It is appended to the delegated prefix and the configured SLA ID to form the final IPv6 address.

By default, the host portion is generated based on the parent interface's MAC address (EUI-64 format).

Example:

If a /64 prefix is delegated to interface eth8 and you configure the host portion as 65535, the resulting IPv6 address will end with ::ffff, as 65535 corresponds to ffff in hexadecimal notation.

```
set interfaces wireless wlan0 dhcpv6-options pd 0 interface eth8 address 65534
```

set interfaces wireless <interface> dhcpv6-options pd <id> interface <delegatee> sla-id <id>

Configure the SLA ID for the delegatee interface.

The value must be a decimal integer greater than 0 and fit in the length of SLA IDs. It is converted to hexadecimal and appended to the delegated prefix to form the specific subnet prefix for the delegatee interface.

Example:

If SLA ID is 1 and the delegated prefix is 2001:db8:ffff::/48, the resulting subnet prefix for the delegatee interface will be 2001:db8:ffff:1::/64.

```
set interfaces wireless wlan0 dhcpv6-options pd 0 interface eth8 sla-id 1
```

System-wide configuration

set system wireless country-code <cc>

Country code (ISO/IEC 3166-1). Used to set regulatory domain. Set as needed to indicate country in which device is operating. This can limit available channels and transmit power.

Note: This option is mandatory in access-point mode.

Wireless options

set interfaces wireless <interface> channel <number>

Configure the IEEE 802.11 wireless radio channel for the interface. Channel allocation depends on the frequency band:

- **2.4 GHz** (802.11b/g/n/ax): Channels range from 1 to 14.
- **5 GHz** (802.11a/h/j/n/ac/ax): Channels range from 34 to 177.
- **6 GHz** (802.11ax): Channels range from 1 to 233.
- **Automatic channel selection:** 0.

set interfaces wireless <interface> disable-broadcast-ssid

Send empty SSID in beacons and ignore probe request frames that do not specify full SSID, i.e., require stations to know the SSID.

set interfaces wireless <interface> expunge-failing-stations

Disassociate stations based on excessive transmission failures or other indications of connection loss.

This depends on the driver capabilities and may not be available with all drivers.

set interfaces wireless <interface> isolate-stations

Client isolation can be used to prevent low-level bridging of frames between associated stations in the BSS.

By default, this bridging is allowed.

set interfaces wireless <interface> max-stations <count>

Maximum number of stations allowed in station table. New stations will be rejected after the station table is full. IEEE 802.11 has a limit of 2007 different association IDs, so this number should not be larger than that.

This defaults to 2007.

set interfaces wireless <interface> mgmt-frame-protection

Management Frame Protection (MFP) according to IEEE 802.11w

Note: MFP (Management Frame Protection) is required for WPA3.

set interfaces wireless <interface> enable-bf-protection

Beacon Protection: management frame protection for Beacon frames.

Note: This option requires MFP to be enabled.

set interfaces wireless <interface> mode <a | b | g | n | ac | ax>

Operation mode of wireless radio.

- a - 802.11a - 54 Mbits/sec
- b - 802.11b - 11 Mbits/sec
- g - 802.11g - 54 Mbits/sec (default)
- n - 802.11n - 600 Mbits/sec
- ac - 802.11ac - 1300 Mbits/sec
- ax - 802.11ax - exceeds 1Gbit/sec

Note: In VyOS, 802.11ax is only implemented for 2.4GHz and 6GHz.

set interfaces wireless <interface> physical-device <device>

Wireless hardware device used as underlay radio.

This defaults to phy0.

set interfaces wireless <interface> reduce-transmit-power <number>

Adds the Power Constraint information element to Beacon and Probe Response frames.

This option adds the Power Constraint information element when applicable and the Country information element is configured. The Power Constraint element is required by Transmit Power Control.

Valid values are 0..255.

set interfaces wireless <interface> ssid <ssid>

SSID to be used in IEEE 802.11 management frames

set interfaces wireless <interface> type <access-point | station | monitor>

Wireless device type for this interface

- access-point: Forwards packets between other nodes.
- station: Connects to another AP (Access Point).
- monitor: Passively monitors all packets on the frequency/channel.

set interfaces wireless <interface> per-client-thread

Enable threaded mode for all NAPI instances of the specified network device.

Enabling threaded mode doesn't affect device operation and can be done while the device is up.

This mode is disabled by default.

Example:

```
set interfaces wireless wlan0 per-client-thread
```

PPDU

```
set interfaces wireless <interface> capabilities require-ht
```

```
set interfaces wireless <interface> capabilities require-vht
```

```
set interfaces wireless <interface> capabilities require-he
```

HT (High Throughput) capabilities (802.11n)

Configuring HT mode options is required when using 802.11n or 802.11ax at 2.4GHz.

```
set interfaces wireless <interface> capabilities ht 40mhz-incapable
```

Device is incapable of 40 MHz, do not advertise. This sets [40-INTOLERANT]

```
set interfaces wireless <interface> capabilities ht auto-powersave
```

WMM-PS Unscheduled Automatic Power Save Delivery [U-APSD]

```
set interfaces wireless <interface> capabilities ht channel-set-width <ht20 | ht40+ | ht40->
```

Supported channel width set.

- ht20 - 20 MHz channel width
- ht40- - Both 20 MHz and 40 MHz with secondary channel below the primary channel
- ht40+ - Both 20 MHz and 40 MHz with secondary channel above the primary channel

Note: Channel availability for HT40- and HT40+ is limited. The following table lists channels permitted for HT40- and HT40+ according to IEEE 802.11n Annex J. Channel availability may vary by location.

freq	HT40-	HT40+
2.4 GHz	5-13	1-7 (1-9 in Europe/Japan)
5 GHz	40, 48, 56, 64	36, 44, 52, 60

Note: 40 MHz channels may switch their primary and secondary channels if needed or creation of 40 MHz channel may be rejected based on overlapping BSSes. These changes are done automatically when hostapd is setting up the 40 MHz channel.

```
set interfaces wireless <interface> capabilities ht delayed-block-ack
```

Enable HT-delayed Block Ack [DELAYED-BA]

```
set interfaces wireless <interface> capabilities ht dsss-cck-40
```

DSSS/CCK Mode in 40 MHz, this sets [DSSS_CCK-40]

```
set interfaces wireless <interface> capabilities ht greenfield
```

This enables the greenfield option which sets the [GF] option

```
set interfaces wireless <interface> capabilities ht ldpc
```

Enable LDPC coding capability

```
set interfaces wireless <interface> capabilities ht lsig-protection
```

Enable L-SIG TXOP protection capability

```
set interfaces wireless <interface> capabilities ht max-amsdu <3839 | 7935>
```

Maximum A-MSDU length 3839 (default) or 7935 octets

```
set interfaces wireless <interface> capabilities ht short-gi <20 | 40>
```

Short GI capabilities for 20 and 40 MHz

```
set interfaces wireless <interface> capabilities ht smps <static | dynamic>
```

Spatial Multiplexing Power Save (SMPS) settings

```
set interfaces wireless <interface> capabilities ht stbc rx <num>
```

Enable receiving PPDU using STBC (Space Time Block Coding)

```
set interfaces wireless <interface> capabilities ht stbc tx
```

Enable sending PPDU using STBC (Space Time Block Coding)

VHT (Very High Throughput) capabilities (802.11ac)

```
set interfaces wireless <interface> capabilities vht antenna-count <count>
```

```
set interfaces wireless <interface> capabilities vht antenna-pattern-fixed
```

Set if antenna pattern does not change during the lifetime of an association

```
set interfaces wireless <interface> capabilities vht beamform <single-user-beamformer |  
single-user-beamformee | multi-user-beamformer | multi-user-beamformee>
```

Beamforming capabilities:

- single-user-beamformer - Support for operation as single user beamformer
- single-user-beamformee - Support for operation as single user beamformee
- multi-user-beamformer - Support for operation as multi user beamformer
- multi-user-beamformee - Support for operation as multi user beamformee

```
set interfaces wireless <interface> capabilities vht center-channel-freq <freq-1 |  
freq-2> <number>
```

VHT operating channel center frequency - center freq 1 (for use with 80, 80+80 and 160 modes)

VHT operating channel center frequency - center freq 2 (for use with the 80+80 mode)

<number> must be from 34 - 173. For 80 MHz channels it should be channel + 6.

```
set interfaces wireless <interface> capabilities vht channel-set-width <0 | 1 | 2 | 3>
```

- 0 - 20 or 40 MHz channel width (default)
- 1 - 80 MHz channel width
- 2 - 160 MHz channel width
- 3 - 80+80 MHz channel width

```
set interfaces wireless <interface> capabilities vht ldpc
```

Enable LDPC (Low Density Parity Check) coding capability

```
set interfaces wireless <interface> capabilities vht link-adaptation
```

VHT link adaptation capabilities

```
set interfaces wireless <interface> capabilities vht max-mpdu <value>
```

Increase Maximum MPDU length to 7991 or 11454 octets (default 3895 octets)

```
set interfaces wireless <interface> capabilities vht max-mpdu-exp <value>
```

Set the maximum length of A-MPDU pre-EOF padding that the station can receive

```
set interfaces wireless <interface> capabilities vht short-gi <80 | 160>
```

Short GI capabilities

```
set interfaces wireless <interface> capabilities vht stbc rx <num>
```

Enable receiving PPDU using STBC (Space Time Block Coding)

```
set interfaces wireless <interface> capabilities vht stbc tx
```

Enable sending PPDU using STBC (Space Time Block Coding)

```
set interfaces wireless <interface> capabilities vht tx-powersave
```

Enable VHT TXOP Power Save Mode

```
set interfaces wireless <interface> capabilities vht vht-cf
```

Station supports receiving VHT variant HT Control field

HE (High Efficiency) capabilities (802.11ax)

```
set interfaces wireless <interface> capabilities he antenna-pattern-fixed
```

Tell the AP that antenna positions are fixed and will not change during the lifetime of an association.

```
set interfaces wireless <interface> capabilities he beamform <single-user-beamformer |  
single-user-beamformee | multi-user-beamformer>
```

Beamforming capabilities:

- single-user-beamformer - Support for operation as single user beamformer
- single-user-beamformee - Support for operation as single user beamformee
- multi-user-beamformer - Support for operation as multi user beamformer

```
set interfaces wireless <interface> capabilities he bss-color <number>
```

BSS coloring helps to prevent channel jamming when multiple APs use the same channels.

Valid values are 1..63

```
set interfaces wireless <interface> capabilities he center-channel-freq <freq-1 | freq-2>  
<number>
```

HE operating channel center frequency - center freq 1 (for use with 80, 80+80 and 160 modes)

HE operating channel center frequency - center freq 2 (for use with the 80+80 mode)

<number> must be within 1..233. For 80 MHz channels it should be channel + 6 and for 160 MHz channels, it should be channel + 14.

```
set interfaces wireless <interface> capabilities he channel-set-width <number>
```

<number> must be one of:

- 81 - 20 MHz channel width (2.4GHz)
- 83 - 40 MHz channel width, secondary 20MHz channel above primary channel (2.4GHz)
- 84 - 40 MHz channel width, secondary 20MHz channel below primary channel (2.4GHz)
- 131 - 20 MHz channel width (6GHz)
- 132 - 40 MHz channel width (6GHz)
- 133 - 80 MHz channel width (6GHz)
- 134 - 160 MHz channel width (6GHz)
- 135 - 80+80 MHz channel width (6GHz)

set interfaces wireless <interface> capabilities he coding-scheme <number>

This setting configures Spatial Stream and Modulation Coding Scheme settings for HE mode (HE-MCS). It is usually not needed to set this explicitly, but it might help with some WiFi adapters.

<number> must be one of:

- 0 - HE-MCS 0-7
- 1 - HE-MCS 0-9
- 2 - HE-MCS 0-11
- 3 - HE-MCS is not supported

Wireless options (Station/Client)

The example creates a wireless station (commonly referred to as Wi-Fi client) that accesses the network through the WAP defined in the above example. The default physical device (phy0) is used.

```
set system wireless country-code de
set interfaces wireless wlan0 type station
set interfaces wireless wlan0 address dhcp
set interfaces wireless wlan0 ssid 'TEST'
set interfaces wireless wlan0 security wpa passphrase '12345678'
```

Resulting configuration:

```
system {
  wireless {
    country-code de
  }
}
interfaces {
  wireless wlan0 {
    address dhcp
    security {
      wpa {
        passphrase "12345678"
      }
    }
    ssid TEST
    type station
  }
}
```

Security

WPA (Wi-Fi Protected Access), WPA2 Enterprise and WPA3 Enterprise in combination with 802.1X based authentication can be used to authenticate users or computers in a domain.

The wireless client (supplicant) authenticates against the RADIUS server (authentication server) using an EAP (Extensible Authentication Protocol) method configured on the RADIUS server. The WAP (also referred to as authenticator) role is to send all authentication messages between the supplicant and the configured authentication server, thus the RADIUS server is responsible for authenticating the users.

The WAP in this example has the following characteristics:

- IP address 192.168.2.1/24
- Network ID (SSID) Enterprise-TEST
- WPA passphrase 12345678
- Use 802.11n protocol
- Wireless channel 1
- RADIUS server at 192.168.3.10 with shared-secret VyOSPassword

```
set system wireless country-code de
set interfaces wireless wlan0 address '192.168.2.1/24'
set interfaces wireless wlan0 type access-point
set interfaces wireless wlan0 channel 1
set interfaces wireless wlan0 mode n
set interfaces wireless wlan0 ssid 'Enterprise-TEST'
set interfaces wireless wlan0 security wpa mode wpa2
set interfaces wireless wlan0 security wpa cipher CCMP
set interfaces wireless wlan0 security wpa radius server 192.168.3.10 key 'VyOSPassword'
set interfaces wireless wlan0 security wpa radius server 192.168.3.10 port 1812
```

Resulting configuration:

```
system {
  wireless {
    country-code de
  }
}
interfaces {
  [...]
  wireless wlan0 {
    address 192.168.2.1/24
    channel 1
    mode n
    security {
      wpa {
        cipher CCMP
        mode wpa2
        radius {
          server 192.168.3.10 {
            key 'VyOSPassword'
            port 1812
          }
        }
      }
    }
  }
}
```

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```

    }
  }
}
ssid "Enterprise-TEST"
type access-point
}

```

VLAN

Regular VLANs (802.1q)

IEEE 802.1q, often referred to as Dot1q, is the industry standard for implementing VLANs on Ethernet networks. It defines VLAN tagging for Ethernet frames and outlines procedures for bridges and switches. The standard also includes quality-of-service prioritization (IEEE 802.1p) and defines the Generic Attribute Registration Protocol.

VLAN-aware network segments (i.e., IEEE 802.1q conformant) use VLAN tags. When a frame enters such a segment, a tag is added to indicate VLAN membership. Each frame can belong to only one VLAN. If a frame arrives without a tag, it is assumed to be part of the native VLAN.

IEEE 802.1, a working group of the IEEE 802 standards committee, has developed the standard and continues to revise it. One notable revision is 802.1Q-2014, which incorporated IEEE 802.1aq (Shortest Path Bridging) and much of the IEEE 802.1d standard.

In VyOS, 802.1q VLAN interfaces are represented as virtual subinterfaces, referred to as `vif`.

set interfaces wireless <interface> vif <vlan-id>

Configure a VLAN interface with a unique VLAN ID.

VLAN ID identifies a specific VLAN and ranges from 0 to 4094.

You can configure multiple VLAN interfaces on a single physical interface.

Note: Only 802.1Q-tagged packets are accepted on Ethernet vifs.

set interfaces wireless <interface> vif <vlan-id> address <address | dhcp | dhcpv6>

Configure the interface with one or more IP addresses.

The following options are available:

- **address:** Assign one or more IPv4 or IPv6 addresses to the interface. For example, use 192.0.2.1/24 for IPv4 or 2001:db8::1/64 for IPv6.
- **dhcp:** The interface obtains an IPv4 address from a DHCP server on the same network segment.
- **dhcpv6:** The interface obtains an IPv6 address from a DHCPv6 server on the same network segment.

Note: If the interface obtains an IPv4 address via DHCP, and specific adjustments are needed before/after the IP address is obtained, use the provided hook scripts:

- `/config/scripts/dhcp-client/pre-hooks.d/`
- `/config/scripts/dhcp-client/post-hooks.d/`

Example:

```
set interfaces wireless wlan0 vif 10 address 192.0.2.1/24
set interfaces wireless wlan0 vif 10 address 2001:db8::1/64
set interfaces wireless wlan0 vif 10 address dhcp
set interfaces wireless wlan0 vif 10 address dhcpv6
```

set interfaces wireless <interface> vif <vlan-id> description <description>

Configure a clear, descriptive alias for the interface.

This alias appears in the `show interfaces` command and SNMP-based monitoring tools.

Example:

```
set interfaces wireless wlan0 vif 10 description 'This is an interface running on
↳ VyOS.'
```

set interfaces wireless <interface> vif <vlan-id> disable

Disable the interface.

The interface will be set to the administratively down (A/D) state.

Example:

```
set interfaces wireless wlan0 vif 10 disable
```

set interfaces wireless <interface> vif <vlan-id> disable-link-detect

Disable physical link-state change detection on the interface, such as when a cable is unplugged.

By default, the interface detects physical link-state changes.

Example:

```
set interfaces wireless wlan0 vif 10 disable-link-detect
```

set interfaces wireless <interface> vif <vlan-id> mac <xx:xx:xx:xx:xx:xx>

Configure a custom MAC address on the interface.

Example:

```
set interfaces wireless wlan0 vif 10 mac '00:53:01:02:03:04'
```

set interfaces wireless <interface> vif <vlan-id> mtu <mtu>

Configure the MTU on the interface.

This value defines the largest packet size, in bytes, that the interface transmits without fragmentation.

Example:

```
set interfaces wireless wlan0 vif 10 mtu 1600
```

set interfaces wireless <interface> vif <vlan-id> ip adjust-mss <mss | clamp-mss-to-pmtu>

Configure the MSS advertised in outgoing TCP SYN packets on the specified interface.

By clamping the MSS value in TCP SYN packets, you explicitly inform the remote side not to send packets larger than that size. This prevents connection issues that occur when Path MTU Discovery (PMTUD) fails.

The following options are available:

- **mss:** Sets the MSS to a specific value, in bytes. Use this option if you need to enforce a specific MSS, for example, to troubleshoot connectivity issues or accommodate specific network requirements.
- **clamp-mss-to-pmtu:** The router automatically calculates the MSS to be the interface's MTU minus 40 bytes for IPv4 traffic (20 bytes for the IPv4 header and 20 bytes for the TCP header). This option is recommended to automatically set the proper value.

set interfaces wireless <interface> vif <vlan-id> ip arp-cache-timeout

Configure how long an ARP entry remains valid after learning an IP-to-MAC address mapping on this interface.

The default duration is 30 seconds.

An ARP entry remains valid if it receives positive feedback from higher-level protocols.

Example:

```
set interfaces wireless wlan0 vif 10 ip arp-cache-timeout 180
```

set interfaces wireless <interface> vif <vlan-id> ip disable-arp-filter

Configure ARP filtering on this interface.

Default behavior: The kernel responds to ARP requests on this interface only if the traffic would be routed back to the ARP sender through that specific interface.

If configured: The kernel responds to ARP requests on this interface for any IP address configured on the local host, regardless of which specific interface that IP address is assigned to, and regardless of the routing table. This reflects the Linux concept that IP addresses belong to the host, not individual interfaces.

Example:

```
set interfaces wireless wlan0 vif 10 ip disable-arp-filter
```

set interfaces wireless <interface> vif <vlan-id> ip disable-forwarding

Configure the interface for host or router behavior.

If configured, the interface switches to host mode, and IPv4 forwarding is disabled on it.

Example:

```
set interfaces wireless wlan0 vif 10 ip disable-forwarding
```

set interfaces wireless <interface> vif <vlan-id> ip enable-directed-broadcast

Configure whether to forward IP-directed broadcast packets received on this interface.

Default behavior: IP-directed broadcast packets are dropped.

If configured: IP-directed broadcast packets are forwarded to all hosts on the destination subnet, as defined in [RFC 1812](#) and [RFC 2644](#).

Example:

```
set interfaces wireless wlan0 vif 10 ip enable-directed-broadcast
```

set interfaces wireless <interface> vif <vlan-id> ip enable-arp-accept

Configure how to process gratuitous ARPs on this interface.

If configured, an IP-to-MAC address mapping is added to the ARP table based on gratuitous ARP requests or replies.

Note: If the ARP table already contains the IP address from a gratuitous ARP, its entry is updated regardless of whether this setting is configured.

Example:

```
set interfaces wireless wlan0 vif 10 ip enable-arp-accept
```

set interfaces wireless <interface> vif <vlan-id> ip enable-arp-announce**Configure the source IP selection for ARP requests on this interface.**

Default behavior: The kernel can use any IP address the host owns as the source IP address in ARP requests on this interface.

If configured: The kernel first attempts to select a source IP address configured on the interface that shares a common subnet with the target IP address. If there is no such subnet, the kernel selects the IP address it would normally use (based on the routing table to reach the target destination).

Example:

```
set interfaces wireless wlan0 vif 10 ip enable-arp-announce
```

set interfaces wireless <interface> vif <vlan-id> ip enable-arp-ignore**Configure which ARP requests will be ignored on this interface.**

Default behavior: The kernel responds to ARP requests for any local IP addresses, regardless of which interface they are assigned to.

If configured: The kernel responds to ARP requests only if the target IP address is assigned to this specific interface.

Example:

```
set interfaces wireless wlan0 vif 10 ip enable-arp-ignore
```

set interfaces wireless <interface> vif <vlan-id> ip enable-proxy-arp**Configure proxy ARP on this interface.**

If configured, the router (kernel) intercepts ARP requests for non-local IP addresses and replies with the MAC address of the interface that received the request. Subsequent packets destined to these IP addresses are forwarded to their actual destinations on remote subnets.

Example:

```
set interfaces wireless wlan0 vif 10 ip enable-proxy-arp
```

set interfaces wireless <interface> vif <vlan-id> ip proxy-arp-pvlan**Configure local proxy ARP on the interface.**

If configured, the router (kernel) responds to ARP requests on this VLAN interface even if the target IP address resides on the same subnet and interface.

This is used to support network isolation requirements (RFC 3069) for private VLANs (PVLANS). In PVLAN configurations, hosts on isolated ports are NOT allowed to communicate directly with each other at Layer 2, but they can communicate with the upstream router.

By replying to inter-host ARP requests with its own MAC address, the router (kernel) directs inter-host traffic through itself instead of directly between hosts.

Note: This command works independently and does not require enabling the standard proxy ARP on the interface.

Local proxy ARP is also known as:

- VLAN aggregation ([RFC 3069](#)).
- Private VLAN (Cisco, Allied Telesyn).
- Source-port filtering or port isolation (Hewlett-Packard).
- MAC-Forced Forwarding (Ericsson).

set interfaces wireless <interface> vif <vlan-id> ip source-validation <strict | loose | disable>

Configure source IP address validation using RPF on this interface, as specified in [RFC 3704](#).

The following options are available:

- **strict:** Each incoming packet's source IP address is checked against the Forwarding Information Base (FIB). If the interface is not the best route back to that source, validation fails, and the packet is dropped.
- **loose:** Each incoming packet's source IP address is checked against the FIB. If the source IP address is unreachable through any interface, validation fails.
- **disable:** No source IP address validation is performed. All incoming packets are accepted.

RFC 3704 recommends enabling strict mode to prevent IP spoofing, such as DDoS attacks. For asymmetric or other complex routing scenarios, use loose mode.

set interfaces wireless <interface> vif <vlan-id> ipv6 address autoconf

Configure the interface to automatically obtain an IPv6 address using SLAAC, as specified in [RFC 4862](#).

IPv6 hosts can configure themselves automatically when connected to an IPv6 network using the Neighbor Discovery Protocol via ICMPv6 router discovery messages. When first connected to a network, a host sends a link-local router solicitation multicast request for its configuration parameters. The router responds with a router advertisement packet containing Internet Layer configuration parameters.

Note: This method automatically disables IPv6 traffic forwarding on the interface.

Example:

```
set interfaces wireless wlan0 vif 10 ipv6 address autoconf
```

set interfaces wireless <interface> vif <vlan-id> ipv6 address eui64 <prefix>

Configure the interface to assign itself an IPv6 address using the EUI-64 method, as specified in [RFC 4291](#).

Example:

```
set interfaces wireless wlan0 vif 10 ipv6 address eui64 2001:db8:beef::/64
```

```
set interfaces wireless <interface> vif <vlan-id> ipv6 address no-default-link-local
```

Disable the automatic assignment of a link-local IPv6 address to this interface.

Example:

```
set interfaces wireless wlan0 vif 10 ipv6 address no-default-link-local
```

```
set interfaces wireless <interface> vif <vlan-id> ipv6 disable-forwarding
```

Configure the interface for host or router behavior.

If configured, the interface switches to host mode, and IPv6 forwarding is disabled on it.

Example:

```
set interfaces wireless wlan0 vif 10 ipv6 disable-forwarding
```

```
set interfaces wireless <interface> vif <vlan-id> ipv6 adjust-mss <mss | clamp-mss-to-pmtu>
```

Configure the MSS advertised in outgoing TCP SYN packets on the specified interface.

By clamping the MSS value in TCP SYN packets, you explicitly inform the remote side not to send packets larger than that size. This prevents connection issues when Path MTU Discovery (PMTUD) fails.

The following options are available:

- **mss:** Set the MSS to a specific value, in bytes. Use this option to enforce a specific MSS, for example, to troubleshoot connectivity issues or accommodate specific network requirements.
- **clamp-mss-to-pmtu:** The router calculates the MSS to be the interface's MTU minus 60 bytes for IPv6 traffic (40 bytes for the IPv6 header and 20 bytes for the TCP header). This option is recommended to automatically set the proper value.

```
set interfaces wireless <interface> vif <vlan-id> ipv6 accept-dad <0 | 1 | 2>
```

Configure IPv6 DAD on the interface.

The following options are available:

- **0:** Disables DAD. No duplicate address detection is performed.
- **1:** Enables DAD (default). Duplicate addresses are detected. The interface's IPv6 operation continues for valid IPv6 addresses.
- **2:** Enables DAD and, if a MAC-based duplicate link-local address is found, disables IPv6 operation on this interface.

Example:

```
set interfaces wireless wlan0 vif 10 ipv6 accept-dad 2
```

```
set interfaces wireless <interface> vif <vlan-id> ipv6 dup-addr-detect-transmits <n>
```

Configure the number of DAD messages that the router (kernel) sends during IPv6 address assignment on this interface.

The default value is 1.

Example:

```
set interfaces wireless wlan0 vif 10 ipv6 dup-addr-detect-transmits 5
```

set interfaces wireless <interface> vif <vlan-id> vrf <vrf>

Assign the interface to a specific VRF instance.

See also:

For information on configuring a VRF, refer to the [VRF](#) section.

Example:

```
set interfaces wireless wlan0 vif 10 vrf red
```

DHCP(v6)

set interfaces wireless <interface> vif <vlan-id> dhcp-options client-id <description>

Configure a DHCP client identifier for the interface, as specified in [RFC 2131](#).

The `client-id` is an identifier that the DHCP client sends to the DHCP server to uniquely identify itself for IP address assignment. By default, the client uses its MAC address. The `<description>` is a user-defined string that will be sent to the DHCP server as the DHCP client identifier.

Example:

```
set interfaces wireless wlan0 vif 10 dhcp-options client-id 'foo-bar'
```

set interfaces wireless <interface> vif <vlan-id> dhcp-options host-name <hostname>

Configure a specific hostname for the interface.

Instead of the real hostname, the DHCP client will send the specific hostname to the DHCP server when requesting an IP address.

Example:

```
set interfaces wireless wlan0 vif 10 dhcp-options host-name 'VyOS'
```

set interfaces wireless <interface> vif <vlan-id> dhcp-options vendor-class-id <vendor-id>

Configure the DHCP client to include a vendor-class identifier in its DHCP requests on this interface.

The vendor-class identifier is a vendor-specific byte string that enables the DHCP server to identify the device and, in some cases, provide configuration options.

Example:

```
set interfaces wireless wlan0 vif 10 dhcp-options vendor-class-id 'VyOS'
```

set interfaces wireless <interface> vif <vlan-id> dhcp-options no-default-route

Configure the DHCP client to obtain an IP address, but ignore any default gateway provided by the DHCP server on this interface.

Example:

```
set interfaces wireless wlan0 vif 10 dhcp-options no-default-route
```

set interfaces wireless <interface> vif <vlan-id> dhcp-options default-route-distance <distance>

Configure the distance for the default route obtained from the DHCP server on this interface.

Example:

```
set interfaces wireless wlan0 vif 10 dhcp-options default-route-distance 220
```

set interfaces wireless <interface> vif <vlan-id> dhcp-options reject <address>

Configure the DHCP client to reject the specific IP address or IP address range from the DHCP server on this interface.

This is useful when a modem assigns a local IP address upon start. To reject multiple addresses, run this command multiple times with different values. You can reject individual addresses (192.168.100.1) or entire subnets (192.168.100.0/24).

Example:

```
set interfaces wireless wlan0 vif 10 dhcp-options reject 192.168.100.0/24
```

set interfaces wireless <interface> vif <vlan-id> dhcp-options user-class <string>

Configure the DHCP client to send a specific user-class identifier in its DHCP requests on this interface.

The DHCP server can interpret this identifier and provide specific configuration options based on it (for example, default routes). The user-class value typically groups DHCP clients with similar configuration needs (for example, employees, guests, or printers).

Example:

```
set interfaces wireless wlan0 vif 10 dhcp-options user-class VyOS
```

set interfaces wireless <interface> vif <vlan-id> dhcpv6-options duid <duid>

Configure a specific DUID for the DHCPv6 client on this interface.

The DUID is an identifier used by a DHCPv6 client to get an IPv6 address from a DHCPv6 server. It consists of a 2-byte type field, followed by a variable-length identifier field up to 128 bytes. The format of the identifier part depends on the DUID type:

- **DUID-LLT:** The most common type, which includes a hardware type, a timestamp, and a MAC address.
- **DUID-EN:** Is based on a vendor's enterprise number and a unique identifier assigned by the vendor.
- **DUID-LL:** Includes only a MAC address.

The DHCP server matches the DUID against its database and provides configuration data (such as address, lease times, DNS servers, etc.) to the DHCP client.

Example:

```
set interfaces wireless wlan0 vif 10 duid
→ '0e:00:00:01:00:01:27:71:db:f0:00:50:56:bf:c5:6d'
```

set interfaces wireless <interface> vif <vlan-id> dhcpv6-options no-release

Configure the DHCP client not to send a release message when it stops running on this interface.

This helps retain the assigned address or prefix.

Example:

```
set interfaces wireless wlan0 vif 10 dhcpv6-options no-release
```



```
set interfaces wireless <interface> vif <vlan-id> dhcpv6-options parameters-only
```

Enable a stateless DHCPv6 client mode on this interface.

In stateless mode, the DHCPv6 client requests only stateless configuration parameters from the DHCP server (for example, DNS server addresses). It doesn't request a stateful configuration, such as IPv6 addresses or prefixes.

Example:

```
set interfaces wireless wlan0 vif 10 dhcpv6-options parameters-only
```

```
set interfaces wireless <interface> vif <vlan-id> dhcpv6-options rapid-commit
```

Enable DHCPv6 rapid commit on this interface.

When enabled, the DHCP client and server skip the negotiation steps (Advertise and Request), completing the DHCPv6 configuration process in just two messages (Solicit and final Reply).

Example:

```
set interfaces wireless wlan0 vif 10 dhcpv6-options rapid-commit
```

```
set interfaces wireless <interface> vif <vlan-id> dhcpv6-options temporary
```

Configure the DHCPv6 client to request a temporary IPv6 address on this interface.

When configured, the DHCP client doesn't form an Identity Association for Non-temporary Addresses (IA_NA) partnership. Consequently, it only obtains a temporary IPv6 address and doesn't obtain a permanent one.

Example:

```
set interfaces wireless wlan0 vif 10 dhcpv6-options temporary
```

DHCPv6 Prefix Delegation (PD)

VyOS supports DHCPv6 Prefix Delegation (DHCPv6-PD) as described in [RFC 3633](#). DHCPv6-PD is supported by most ISPs that provide native IPv6 for consumers on fixed networks.

```
set interfaces wireless <interface> vif <vlan-id> dhcpv6-options pd <id> length <length>
```

Configure a specific prefix length for DHCPv6-PD requests on this interface.

Some ISPs provide only a /64 prefix by default. Use this command to request a different prefix length for a specific DHCPv6-PD request, ranging from /32 (if allowed by your ISP) down to /64. <id> is a unique identifier for the DHCPv6-PD request.

The default value is 64.

To request a /56 prefix from your ISP, use:

```
set interfaces wireless wlan0 vif 10 dhcpv6-options pd 0 length 56
```

```
set interfaces wireless <interface> vif <vlan-id> dhcpv6-options pd <id> interface <delegatee> address <address>
```

Configure the IPv6 interface identifier (host portion) for the delegatee interface.

The value must be a decimal integer. It is appended to the delegated prefix and the configured SLA ID to form the final IPv6 address.

By default, the host portion is generated based on the parent interface's MAC address (EUI-64 format).

Example:

If a /64 prefix is delegated to interface eth8 and you configure the host portion as 65535, the resulting IPv6 address will end with ::ffff, as 65535 corresponds to ffff in hexadecimal notation.

```
set interfaces wireless wlan0 vif 10 dhcpv6-options pd 0 interface eth8 address_
↪ 65534
```

set interfaces wireless <interface> vif <vlan-id> dhcpv6-options pd <id> interface <delegatee> sla-id <id>

Configure the SLA ID for the delegatee interface.

The value must be a decimal integer greater than 0 and fit in the length of SLA IDs. It is converted to hexadecimal and appended to the delegated prefix to form the specific subnet prefix for the delegatee interface.

Example:

If SLA ID is 1 and the delegated prefix is 2001:db8:ffff::/48, the resulting subnet prefix for the delegatee interface will be 2001:db8:ffff:1::/64.

```
set interfaces wireless wlan0 vif 10 dhcpv6-options pd 0 interface eth8 sla-id 1
```

QinQ (802.1ad)**Overview**

IEEE 802.1ad, commonly known as QinQ, is an Ethernet standard first published as an amendment to 802.1q in 2005, then officially merged into the base standard in 2011.

Unlike the original 802.1q, which allows a single VLAN header per Ethernet frame, QinQ allows two VLAN headers per Ethernet frame, for the inner and the outer VLAN tags. Most often the inner VLAN tag comes from a customer while the outer tag is used by the service provider to differentiate between traffic of different customers.

Frame structure and ethertypes

The IEEE 802.1ad (QinQ) frame includes two VLAN tags:

- **The outer service tag (S-TAG):** The S-TAG is typically added by the provider.

It uses the Ethertype 0x88a8 by default.

- **The inner customer tag (C-TAG):** The C-TAG is generated by the customer's equipment and remains unchanged during transit. It uses the Ethertype 0x8100.

Implementation in VyOS

In VyOS, these tag types are associated with the following CLI options:

- **vif-s:** Corresponds to the S-TAG (Ethertype 0x88a8).
- **vif-c:** Corresponds to the C-TAG (Ethertype 0x8100).

set interfaces wireless <interface> vif-s <vlan-id> vif-c <vlan-id> address <address | dhcp | dhcpv6>

Configure the interface with one or more IP addresses.

The following options are available:

- **address:** Assign one or more IPv4 or IPv6 addresses to the interface. For example, use 192.0.2.1/24 for IPv4 or 2001:db8::1/64 for IPv6.

- **dhcp:** The interface obtains an IPv4 address from a DHCP server on the same network segment.
- **dhcpv6:** The interface obtains an IPv6 address from a DHCPv6 server on the same network segment.

Note: If the interface obtains an IPv4 address via DHCP, and specific adjustments are needed before/after the IP address is obtained, use the provided hook scripts:

- /config/scripts/dhcp-client/pre-hooks.d/
- /config/scripts/dhcp-client/post-hooks.d/

Example:

```
set interfaces wireless wlan0 vif-s 1000 vif-c 20 address 192.0.2.1/24
set interfaces wireless wlan0 vif-s 1000 vif-c 20 address 2001:db8::1/64
set interfaces wireless wlan0 vif-s 1000 vif-c 20 address dhcp
set interfaces wireless wlan0 vif-s 1000 vif-c 20 address dhcpv6
```

set interfaces wireless <interface> vif-s <vlan-id> vif-c <vlan-id> description <description>

Configure a clear, descriptive alias for the interface.

This alias appears in the `show interfaces` command and SNMP-based monitoring tools.

Example:

```
set interfaces wireless wlan0 vif-s 1000 vif-c 20 description 'This is an interface
↳ running on VyOS.'
```

set interfaces wireless <interface> vif-s <vlan-id> vif-c <vlan-id> disable

Disable the interface.

The interface will be set to the administratively down (A/D) state.

Example:

```
set interfaces wireless wlan0 vif-s 1000 vif-c 20 disable
```

set interfaces wireless <interface> vif-s <vlan-id> vif-c <vlan-id> disable-link-detect

Disable physical link-state change detection on the interface, such as when a cable is unplugged.

By default, the interface detects physical link-state changes.

Example:

```
set interfaces wireless wlan0 vif-s 1000 vif-c 20 disable-link-detect
```

set interfaces wireless <interface> vif-s <vlan-id> vif-c <vlan-id> mac <xx:xx:xx:xx:xx:xx>

Configure a custom MAC address on the interface.

Example:

```
set interfaces wireless wlan0 vif-s 1000 vif-c 20 mac '00:53:01:02:03:04'
```

set interfaces wireless <interface> vif-s <vlan-id> vif-c <vlan-id> mtu <mtu>

Configure the MTU on the interface.

This value defines the largest packet size, in bytes, that the interface transmits without fragmentation.

Example:

```
set interfaces wireless wlan0 vif-s 1000 vif-c 20 mtu 1600
```

```
set interfaces wireless <interface> vif-s <vlan-id> vif-c <vlan-id> ip adjust-mss <mss | clamp-mss-to-pmtu>
```

Configure the MSS advertised in outgoing TCP SYN packets on the specified interface.

By clamping the MSS value in TCP SYN packets, you explicitly inform the remote side not to send packets larger than that size. This prevents connection issues that occur when Path MTU Discovery (PMTUD) fails.

The following options are available:

- **mss:** Sets the MSS to a specific value, in bytes. Use this option if you need to enforce a specific MSS, for example, to troubleshoot connectivity issues or accommodate specific network requirements.
- **clamp-mss-to-pmtu:** The router automatically calculates the MSS to be the interface's MTU minus 40 bytes for IPv4 traffic (20 bytes for the IPv4 header and 20 bytes for the TCP header). This option is recommended to automatically set the proper value.

```
set interfaces wireless <interface> vif-s <vlan-id> vif-c <vlan-id> ip arp-cache-timeout
```

Configure how long an ARP entry remains valid after learning an IP-to-MAC address mapping on this interface.

The default duration is 30 seconds.

An ARP entry remains valid if it receives positive feedback from higher-level protocols.

Example:

```
set interfaces wireless wlan0 vif-s 1000 vif-c 20 ip arp-cache-timeout 180
```

```
set interfaces wireless <interface> vif-s <vlan-id> vif-c <vlan-id> ip disable-arp-filter
```

Configure ARP filtering on this interface.

Default behavior: The kernel responds to ARP requests on this interface only if the traffic would be routed back to the ARP sender through that specific interface.

If configured: The kernel responds to ARP requests on this interface for any IP address configured on the local host, regardless of which specific interface that IP address is assigned to, and regardless of the routing table. This reflects the Linux concept that IP addresses belong to the host, not individual interfaces.

Example:

```
set interfaces wireless wlan0 vif-s 1000 vif-c 20 ip disable-arp-filter
```

```
set interfaces wireless <interface> vif-s <vlan-id> vif-c <vlan-id> ip disable-forwarding
```

Configure the interface for host or router behavior.

If configured, the interface switches to host mode, and IPv4 forwarding is disabled on it.

Example:

```
set interfaces wireless wlan0 vif-s 1000 vif-c 20 ip disable-forwarding
```

```
set interfaces wireless <interface> vif-s <vlan-id> vif-c <vlan-id> ip
enable-directed-broadcast
```

Configure whether to forward IP-directed broadcast packets received on this interface.

Default behavior: IP-directed broadcast packets are dropped.

If configured: IP-directed broadcast packets are forwarded to all hosts on the destination subnet, as defined in [RFC 1812](#) and [RFC 2644](#).

Example:

```
set interfaces wireless wlan0 vif-s 1000 vif-c 20 ip enable-directed-broadcast
```

```
set interfaces wireless <interface> vif-s <vlan-id> vif-c <vlan-id> ip enable-arp-accept
```

Configure how to process gratuitous ARPs on this interface.

If configured, an IP-to-MAC address mapping is added to the ARP table based on gratuitous ARP requests or replies.

Note: If the ARP table already contains the IP address from a gratuitous ARP, its entry is updated regardless of whether this setting is configured.

Example:

```
set interfaces wireless wlan0 vif-s 1000 vif-c 20 ip enable-arp-accept
```

```
set interfaces wireless <interface> vif-s <vlan-id> vif-c <vlan-id> ip
enable-arp-announce
```

Configure the source IP selection for ARP requests on this interface.

Default behavior: The kernel can use any IP address the host owns as the source IP address in ARP requests on this interface.

If configured: The kernel first attempts to select a source IP address configured on the interface that shares a common subnet with the target IP address. If there is no such subnet, the kernel selects the IP address it would normally use (based on the routing table to reach the target destination).

Example:

```
set interfaces wireless wlan0 vif-s 1000 vif-c 20 ip enable-arp-announce
```

```
set interfaces wireless <interface> vif-s <vlan-id> vif-c <vlan-id> ip enable-arp-ignore
```

Configure which ARP requests will be ignored on this interface.

Default behavior: The kernel responds to ARP requests for any local IP addresses, regardless of which interface they are assigned to.

If configured: The kernel responds to ARP requests only if the target IP address is assigned to this specific interface.

Example:

```
set interfaces wireless wlan0 vif-s 1000 vif-c 20 ip enable-arp-ignore
```

```
set interfaces wireless <interface> vif-s <vlan-id> vif-c <vlan-id> ip enable-proxy-arp
```

Configure proxy ARP on this interface.

If configured, the router (kernel) intercepts ARP requests for non-local IP addresses and replies with the MAC address of the interface that received the request. Subsequent packets destined to these IP addresses are forwarded to their actual destinations on remote subnets.

Example:

```
set interfaces wireless wlan0 vif-s 1000 vif-c 20 ip enable-proxy-arp
```

```
set interfaces wireless <interface> vif-s <vlan-id> vif-c <vlan-id> ip proxy-arp-pvlan
```

Configure local proxy ARP on the interface.

If configured, the router (kernel) responds to ARP requests on this VLAN interface even if the target IP address resides on the same subnet and interface.

This is used to support network isolation requirements (RFC 3069) for private VLANs (PVLANS). In PVLAN configurations, hosts on isolated ports are NOT allowed to communicate directly with each other at Layer 2, but they can communicate with the upstream router.

By replying to inter-host ARP requests with its own MAC address, the router (kernel) directs inter-host traffic through itself instead of directly between hosts.

Note: This command works independently and does not require enabling the standard proxy ARP on the interface.

Local proxy ARP is also known as:

- VLAN aggregation ([RFC 3069](#)).
- Private VLAN (Cisco, Allied Telesyn).
- Source-port filtering or port isolation (Hewlett-Packard).
- MAC-Forced Forwarding (Ericsson).

```
set interfaces wireless <interface> vif-s <vlan-id> vif-c <vlan-id> ip source-validation
<strict | loose | disable>
```

Configure source IP address validation using RPF on this interface, as specified in [RFC 3704](#).

The following options are available:

- **strict:** Each incoming packet's source IP address is checked against the Forwarding Information Base (FIB). If the interface is not the best route back to that source, validation fails, and the packet is dropped.
- **loose:** Each incoming packet's source IP address is checked against the FIB. If the source IP address is unreachable through any interface, validation fails.
- **disable:** No source IP address validation is performed. All incoming packets are accepted.

RFC 3704 recommends enabling strict mode to prevent IP spoofing, such as DDoS attacks. For asymmetric or other complex routing scenarios, use loose mode.

```
set interfaces wireless <interface> vif-s <vlan-id> vif-c <vlan-id> ipv6 address autoconf
```

Configure the interface to automatically obtain an IPv6 address using SLAAC, as specified in [RFC 4862](#).

IPv6 hosts can configure themselves automatically when connected to an IPv6 network using the Neighbor Discovery Protocol via ICMPv6 router discovery messages. When first connected to a network, a host sends a link-local router solicitation multicast request for its configuration parameters. The router responds with a router advertisement packet containing Internet Layer configuration parameters.

Note: This method automatically disables IPv6 traffic forwarding on the interface.

Example:

```
set interfaces wireless wlan0 vif-s 1000 vif-c 20 ipv6 address autoconf
```

set interfaces wireless <interface> vif-s <vlan-id> vif-c <vlan-id> ipv6 address eui64 <prefix>

Configure the interface to assign itself an IPv6 address using the EUI-64 method, as specified in [RFC 4291](#).

Example:

```
set interfaces wireless wlan0 vif-s 1000 vif-c 20 ipv6 address eui64
↪ 2001:db8:beef::/64
```

set interfaces wireless <interface> vif-s <vlan-id> vif-c <vlan-id> ipv6 address no-default-link-local

Disable the automatic assignment of a link-local IPv6 address to this interface.

Example:

```
set interfaces wireless wlan0 vif-s 1000 vif-c 20 ipv6 address no-default-link-local
```

set interfaces wireless <interface> vif-s <vlan-id> vif-c <vlan-id> ipv6 disable-forwarding

Configure the interface for host or router behavior.

If configured, the interface switches to host mode, and IPv6 forwarding is disabled on it.

Example:

```
set interfaces wireless wlan0 vif-s 1000 vif-c 20 ipv6 disable-forwarding
```

set interfaces wireless <interface> vif-s <vlan-id> vif-c <vlan-id> ipv6 adjust-mss <mss> | clamp-mss-to-pmtu>

Configure the MSS advertised in outgoing TCP SYN packets on the specified interface.

By clamping the MSS value in TCP SYN packets, you explicitly inform the remote side not to send packets larger than that size. This prevents connection issues when Path MTU Discovery (PMTUD) fails.

The following options are available:

- **mss:** Set the MSS to a specific value, in bytes. Use this option to enforce a specific MSS, for example, to troubleshoot connectivity issues or accommodate specific network requirements.
- **clamp-mss-to-pmtu:** The router calculates the MSS to be the interface's MTU minus 60 bytes for IPv6 traffic (40 bytes for the IPv6 header and 20 bytes for the TCP header). This option is recommended to automatically set the proper value.

set interfaces wireless <interface> vif-s <vlan-id> vif-c <vlan-id> ipv6 accept-dad <0 | 1 | 2>

Configure IPv6 DAD on the interface.

The following options are available:

- **0:** Disables DAD. No duplicate address detection is performed.
- **1:** Enables DAD (default). Duplicate addresses are detected. The interface's IPv6 operation continues for valid IPv6 addresses.
- **2:** Enables DAD and, if a MAC-based duplicate link-local address is found, disables IPv6 operation on this interface.

Example:

```
set interfaces wireless wlan0 vif-s 1000 vif-c 20 ipv6 accept-dad 2
```

**set interfaces wireless <interface> vif-s <vlan-id> vif-c <vlan-id> ipv6
dup-addr-detect-transmits <n>**

Configure the number of DAD messages that the router (kernel) sends during IPv6 address assignment on this interface.

The default value is 1.

Example:

```
set interfaces wireless wlan0 vif-s 1000 vif-c 20 ipv6 dup-addr-detect-transmits 5
```

set interfaces wireless <interface> vif-s <vlan-id> vif-c <vlan-id> vrf <vrf>

Assign the interface to a specific VRF instance.

See also:

For information on configuring a VRF, refer to the [VRF](#) section.

Example:

```
set interfaces wireless wlan0 vif-s 1000 vif-c 20 vrf red
```

DHCP(v6)

**set interfaces wireless <interface> vif-s <vlan-id> vif-c <vlan-id> dhcp-options
client-id <description>**

Configure a DHCP client identifier for the interface, as specified in [RFC 2131](#).

The `client-id` is an identifier that the DHCP client sends to the DHCP server to uniquely identify itself for IP address assignment. By default, the client uses its MAC address. The `<description>` is a user-defined string that will be sent to the DHCP server as the DHCP client identifier.

Example:

```
set interfaces wireless wlan0 vif-s 1000 vif-c 20 dhcp-options client-id 'foo-bar'
```

**set interfaces wireless <interface> vif-s <vlan-id> vif-c <vlan-id> dhcp-options
host-name <hostname>**

Configure a specific hostname for the interface.

Instead of the real hostname, the DHCP client will send the specific hostname to the DHCP server when requesting an IP address.

Example:

```
set interfaces wireless wlan0 vif-s 1000 vif-c 20 dhcp-options host-name 'VyOS'
```



```
set interfaces wireless <interface> vif-s <vlan-id> vif-c <vlan-id> dhcp-options
vendor-class-id <vendor-id>
```

Configure the DHCP client to include a vendor-class identifier in its DHCP requests on this interface.

The vendor-class identifier is a vendor-specific byte string that enables the DHCP server to identify the device and, in some cases, provide configuration options.

Example:

```
set interfaces wireless wlan0 vif-s 1000 vif-c 20 dhcp-options vendor-class-id 'VyOS
→ '
```

```
set interfaces wireless <interface> vif-s <vlan-id> vif-c <vlan-id> dhcp-options
no-default-route
```

Configure the DHCP client to obtain an IP address, but ignore any default gateway provided by the DHCP server on this interface.

Example:

```
set interfaces wireless wlan0 vif-s 1000 vif-c 20 dhcp-options no-default-route
```

```
set interfaces wireless <interface> vif-s <vlan-id> vif-c <vlan-id> dhcp-options
default-route-distance <distance>
```

Configure the distance for the default route obtained from the DHCP server on this interface.

Example:

```
set interfaces wireless wlan0 vif-s 1000 vif-c 20 dhcp-options default-route-
→ distance 220
```

```
set interfaces wireless <interface> vif-s <vlan-id> vif-c <vlan-id> dhcp-options reject
<address>
```

Configure the DHCP client to reject the specific IP address or IP address range from the DHCP server on this interface.

This is useful when a modem assigns a local IP address upon start. To reject multiple addresses, run this command multiple times with different values. You can reject individual addresses (192.168.100.1) or entire subnets (192.168.100.0/24).

Example:

```
set interfaces wireless wlan0 vif-s 1000 vif-c 20 dhcp-options reject 192.168.100.0/
→ 24
```

```
set interfaces wireless <interface> vif-s <vlan-id> vif-c <vlan-id> dhcp-options
user-class <string>
```

Configure the DHCP client to send a specific user-class identifier in its DHCP requests on this interface.

The DHCP server can interpret this identifier and provide specific configuration options based on it (for example, default routes). The user-class value typically groups DHCP clients with similar configuration needs (for example, employees, guests, or printers).

Example:

```
set interfaces wireless wlan0 vif-s 1000 vif-c 20 dhcp-options user-class VyOS
```

```
set interfaces wireless <interface> vif-s <vlan-id> vif-c <vlan-id> dhcpv6-options duid
<duid>
```

Configure a specific DUID for the DHCPv6 client on this interface.

The DUID is an identifier used by a DHCPv6 client to get an IPv6 address from a DHCPv6 server. It consists of a 2-byte type field, followed by a variable-length identifier field up to 128 bytes. The format of the identifier part depends on the DUID type:

- **DUID-LLT:** The most common type, which includes a hardware type, a timestamp, and a MAC address.
- **DUID-EN:** Is based on a vendor's enterprise number and a unique identifier assigned by the vendor.
- **DUID-LL:** Includes only a MAC address.

The DHCP server matches the DUID against its database and provides configuration data (such as address, lease times, DNS servers, etc.) to the DHCP client.

Example:

```
set interfaces wireless wlan0 vif-s 1000 vif-c 20 duid
→ '0e:00:00:01:00:01:27:71:db:f0:00:50:56:bf:c5:6d'
```

```
set interfaces wireless <interface> vif-s <vlan-id> vif-c <vlan-id> dhcpv6-options
no-release
```

Configure the DHCP client not to send a release message when it stops running on this interface.

This helps retain the assigned address or prefix.

Example:

```
set interfaces wireless wlan0 vif-s 1000 vif-c 20 dhcpv6-options no-release
```

```
set interfaces wireless <interface> vif-s <vlan-id> vif-c <vlan-id> dhcpv6-options
parameters-only
```

Enable a stateless DHCPv6 client mode on this interface.

In stateless mode, the DHCPv6 client requests only stateless configuration parameters from the DHCP server (for example, DNS server addresses). It doesn't request a stateful configuration, such as IPv6 addresses or prefixes.

Example:

```
set interfaces wireless wlan0 vif-s 1000 vif-c 20 dhcpv6-options parameters-only
```

```
set interfaces wireless <interface> vif-s <vlan-id> vif-c <vlan-id> dhcpv6-options
rapid-commit
```

Enable DHCPv6 rapid commit on this interface.

When enabled, the DHCP client and server skip the negotiation steps (Advertise and Request), completing the DHCPv6 configuration process in just two messages (Solicit and final Reply).

Example:

```
set interfaces wireless wlan0 vif-s 1000 vif-c 20 dhcpv6-options rapid-commit
```

```
set interfaces wireless <interface> vif-s <vlan-id> vif-c <vlan-id> dhcpv6-options
temporary
```

Configure the DHCPv6 client to request a temporary IPv6 address on this interface.

When configured, the DHCP client doesn't form an Identity Association for Non-temporary Addresses (IA_NA) partnership. Consequently, it only obtains a temporary IPv6 address and doesn't obtain a permanent one.

Example:

```
set interfaces wireless wlan0 vif-s 1000 vif-c 20 dhcpv6-options temporary
```

DHCPv6 Prefix Delegation (PD)

VyOS supports DHCPv6 Prefix Delegation (DHCPv6-PD) as described in [RFC 3633](#). DHCPv6-PD is supported by most ISPs that provide native IPv6 for consumers on fixed networks.

```
set interfaces wireless <interface> vif-s <vlan-id> vif-c <vlan-id> dhcpv6-options pd
<id> length <length>
```

Configure a specific prefix length for DHCPv6-PD requests on this interface.

Some ISPs provide only a /64 prefix by default. Use this command to request a different prefix length for a specific DHCPv6-PD request, ranging from /32 (if allowed by your ISP) down to /64. <id> is a unique identifier for the DHCPv6-PD request.

The default value is 64.

To request a /56 prefix from your ISP, use:

```
set interfaces wireless wlan0 vif-s 1000 vif-c 20 dhcpv6-options pd 0 length 56
```

```
set interfaces wireless <interface> vif-s <vlan-id> vif-c <vlan-id> dhcpv6-options pd
<id> interface <delegatee> address <address>
```

Configure the IPv6 interface identifier (host portion) for the delegatee interface.

The value must be a decimal integer. It is appended to the delegated prefix and the configured SLA ID to form the final IPv6 address.

By default, the host portion is generated based on the parent interface's MAC address (EUI-64 format).

Example:

If a /64 prefix is delegated to interface eth8 and you configure the host portion as 65535, the resulting IPv6 address will end with ::ffff, as 65535 corresponds to ffff in hexadecimal notation.

```
set interfaces wireless wlan0 vif-s 1000 vif-c 20 dhcpv6-options pd 0 interface_
→eth8 address 65534
```

```
set interfaces wireless <interface> vif-s <vlan-id> vif-c <vlan-id> dhcpv6-options pd
<id> interface <delegatee> sla-id <id>
```

Configure the SLA ID for the delegatee interface.

The value must be a decimal integer greater than 0 and fit in the length of SLA IDs. It is converted to hexadecimal and appended to the delegated prefix to form the specific subnet prefix for the delegatee interface.

Example:

If SLA ID is 1 and the delegated prefix is 2001:db8:ffff::/48, the resulting subnet prefix for the delegatee interface will be 2001:db8:ffff:1::/64.

```
set interfaces wireless wlan0 vif-s 1000 vif-c 20 dhcpv6-options pd 0 interface_
↳ eth8 sla-id 1
```

Operation

show interfaces wireless info

Use this command to view operational status and wireless-specific information about all wireless interfaces.

```
vyos@vyos:~$ show interfaces wireless info
Interface Type      SSID                      Channel
wlan0      access-point  VyOS-TEST-0              1
```

show interfaces wireless detail

Show the operational status and detailed wireless-specific information about all wireless interfaces.

```
vyos@vyos:~$ show interfaces wireless detail
wlan0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP group default_
↳ qlen 1000
    link/ether XX:XX:XX:XX:XX:c3 brd XX:XX:XX:XX:XX:ff
    inet xxx.xxx.99.254/24 scope global wlan0
        valid_lft forever preferred_lft forever
    inet6 fe80::xxxx:xxxx:fe54:2fc3/64 scope link
        valid_lft forever preferred_lft forever

    RX:  bytes    packets    errors    dropped    overrun    mcast
         66072         282         0         0         0         0
    TX:  bytes    packets    errors    dropped    carrier    collisions
         83413         430         0         0         0         0

wlan1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP group default_
↳ qlen 1000
    link/ether XX:XX:XX:XX:XX:c3 brd XX:XX:XX:XX:XX:ff
    inet xxx.xxx.100.254/24 scope global wlan0
        valid_lft forever preferred_lft forever
    inet6 fe80::xxxx:xxxx:ffff:2ed3/64 scope link
        valid_lft forever preferred_lft forever

    RX:  bytes    packets    errors    dropped    overrun    mcast
        166072        5282         0         0         0         0
    TX:  bytes    packets    errors    dropped    carrier    collisions
        183413        5430         0         0         0         0
```

show interfaces wireless <wlanX>

This command shows both status and statistics on the specified wireless interface. The wireless interface identifier can range from wlan0 to wlan999.

```
vyos@vyos:~$ show interfaces wireless wlan0
wlan0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP group default_
↳ qlen 1000
    link/ether XX:XX:XX:XX:XX:c3 brd XX:XX:XX:XX:XX:ff
    inet xxx.xxx.99.254/24 scope global wlan0
```

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```
valid_lft forever preferred_lft forever
inet6 fe80::xxxx:xxxx:fe54:2fc3/64 scope link
valid_lft forever preferred_lft forever
```

RX:	bytes	packets	errors	dropped	overrun	mcast
	66072	282	0	0	0	0
TX:	bytes	packets	errors	dropped	carrier	collisions
	83413	430	0	0	0	0

show interfaces wireless <wlanX> brief

This command gives a brief status overview of a specified wireless interface. The wireless interface identifier can range from wlan0 to wlan999.

```
vyos@vyos:~$ show interfaces wireless wlan0 brief
Codes: S - State, L - Link, u - Up, D - Down, A - Admin Down
Interface      IP Address      S/L Description
-----
wlan0          192.168.2.254/24      u/u
```

show interfaces wireless <wlanX> queue

Use this command to view wireless interface queue information. The wireless interface identifier can range from wlan0 to wlan999.

```
vyos@vyos:~$ show interfaces wireless wlan0 queue
qdisc pfifo_fast 0: root bands 3 priomap 1 2 2 2 1 2 0 0 1 1 1 1 1 1 1
Sent 810323 bytes 6016 pkt (dropped 0, overlimits 0 requeues 0)
rate 0bit 0pps backlog 0b 0p requeues 0
```

show interfaces wireless <wlanX> scan

This command is used to retrieve information about WAP within the range of your wireless interface. This command is useful on wireless interfaces configured in station mode.

Note: Scanning is not supported on all wireless drivers and wireless hardware. Refer to your driver and wireless hardware documentation for further details.

```
vyos@vyos:~$ show interfaces wireless wlan0 scan
Address      SSID      Channel  Signal (dbm)
00:53:3b:88:6e:d8 WLAN-576405      1 -64.00
00:53:3b:88:6e:da Telekom_FON      1 -64.00
00:53:00:f2:c2:a4 BabyView_F2C2A4  6 -60.00
00:53:3b:88:6e:d6 Telekom_FON      100 -72.00
00:53:3b:88:6e:d4 WLAN-576405      100 -71.00
00:53:44:a4:96:ec KabelBox-4DC8    56 -81.00
00:53:d9:7a:67:c2 WLAN-741980      1 -75.00
00:53:7c:99:ce:76 Vodafone Homespot 1 -86.00
00:53:44:a4:97:21 KabelBox-4DC8    1 -78.00
00:53:44:a4:97:21 Vodafone Hotspot 1 -79.00
00:53:44:a4:97:21 Vodafone Homespot 1 -79.00
00:53:86:40:30:da Telekom_FON      1 -86.00
```

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00:53:7c:99:ce:76	Vodafone Hotspot	1	-86.00
00:53:44:46:d2:0b	Vodafone Hotspot	1	-87.00

Examples

The following example creates a WAP. When configuring multiple WAP interfaces, you must specify unique IP addresses, channels, Network IDs commonly referred to as SSID (Service Set Identifier), and MAC addresses.

The WAP in this example has the following characteristics:

- IP address 192.168.2.1/24
- Network ID (SSID) TEST
- WPA passphrase 12345678
- Use 802.11n protocol
- Wireless channel 1

```
set system wireless country-code de
set interfaces wireless wlan0 address '192.168.2.1/24'
set interfaces wireless wlan0 type access-point
set interfaces wireless wlan0 channel 1
set interfaces wireless wlan0 mode n
set interfaces wireless wlan0 ssid 'TEST'
set interfaces wireless wlan0 security wpa mode wpa2
set interfaces wireless wlan0 security wpa cipher CCMP
set interfaces wireless wlan0 security wpa passphrase '12345678'
```

Resulting configuration:

```
system {
  wireless {
    country-code de
  }
}
interfaces {
  [...]
  wireless wlan0 {
    address 192.168.2.1/24
    channel 1
    mode n
    security {
      wpa {
        cipher CCMP
        mode wpa2
        passphrase "12345678"
      }
    }
    ssid "TEST"
    type access-point
  }
}
```

To enable access point functionality, configure a DHCP server for this interface's network, or add the interface to an existing local bridge (see [Bridge](#) for details).

Wi-Fi 6/6E (802.11ax)

The following examples configure Wi-Fi 6 (2.4 GHz) and Wi-Fi 6E (6 GHz) APs (Access Points) with the following parameters:

- Network ID (SSID): `test.ax`
- WPA passphrase: `super-doooper-secure-passphrase`
- Protocol: 802.11ax
- Wireless channel for 2.4 GHz: 11
- Wireless channel for 6 GHz: 5

Example configuration: Wi-Fi 6 at 2.4 GHz

You may expect real throughput around 10 MB/s or higher in crowded areas.

```
set system wireless country-code de
set interfaces wireless wlan0 capabilities he antenna-pattern-fixed
set interfaces wireless wlan0 capabilities he beamform multi-user-beamformer
set interfaces wireless wlan0 capabilities he beamform single-user-beamformee
set interfaces wireless wlan0 capabilities he beamform single-user-beamformer
set interfaces wireless wlan0 capabilities he bss-color 13
set interfaces wireless wlan0 capabilities he channel-set-width 81
set interfaces wireless wlan0 capabilities ht 40mhz-incapable
set interfaces wireless wlan0 capabilities ht channel-set-width ht20
set interfaces wireless wlan0 capabilities ht channel-set-width ht40+
set interfaces wireless wlan0 capabilities ht channel-set-width ht40-
set interfaces wireless wlan0 capabilities ht short-gi 20
set interfaces wireless wlan0 capabilities ht short-gi 40
set interfaces wireless wlan0 capabilities ht stbc rx 2
set interfaces wireless wlan0 capabilities ht stbc tx
set interfaces wireless wlan0 channel 11
set interfaces wireless wlan0 description "802.11ax 2.4GHz"
set interfaces wireless wlan0 mode ax
set interfaces wireless wlan0 security wpa cipher CCMP
set interfaces wireless wlan0 security wpa cipher CCMP-256
set interfaces wireless wlan0 security wpa cipher GCMP-256
set interfaces wireless wlan0 security wpa cipher GCMP
set interfaces wireless wlan0 security wpa mode wpa2
set interfaces wireless wlan0 security wpa passphrase super-doooper-secure-passphrase
set interfaces wireless wlan0 ssid test.ax
set interfaces wireless wlan0 type access-point
commit
```

Resulting configuration:

```
system {
  wireless {
```

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```

    country-code de
  }
}
interfaces {
  [...]
  wireless wlan0 {
    capabilities {
      he {
        antenna-pattern-fixed
        beamform {
          multi-user-beamformer
          single-user-beamformee
          single-user-beamformer
        }
        bss-color 13
        channel-set-width 81
      }
      ht {
        40mhz-incapable
        channel-set-width ht20
        channel-set-width ht40+
        channel-set-width ht40-
        short-gi 20
        short-gi 40
        stbc {
          rx 2
          tx
        }
      }
    }
    channel 11
    description "802.11ax 2.4GHz"
    hw-id [...]
    mode ax
    physical-device phy0
    security {
      wpa {
        cipher CCMP
        cipher CCMP-256
        cipher GCMP-256
        cipher GCMP
        mode wpa2
        passphrase super-doooper-secure-passphrase
      }
    }
    ssid test.ax
    type access-point
  }
}

```


Example configuration: Wi-Fi 6E at 6 GHz

You may expect real throughput between 50 MB/s and 150 MB/s, depending on obstructions from walls, water, metal, or other materials with high electromagnetic damping at 6 GHz. Best results are achieved with the AP being in the same room and in line-of-sight.

```

set system wireless country-code de
set interfaces wireless wlan0 capabilities he antenna-pattern-fixed
set interfaces wireless wlan0 capabilities he beamform multi-user-beamformer
set interfaces wireless wlan0 capabilities he beamform single-user-beamformee
set interfaces wireless wlan0 capabilities he beamform single-user-beamformer
set interfaces wireless wlan0 capabilities he bss-color 13
set interfaces wireless wlan0 capabilities he channel-set-width 134
set interfaces wireless wlan0 capabilities he center-channel-freq freq-1 15
set interfaces wireless wlan0 channel 5
set interfaces wireless wlan0 description "802.11ax 6GHz"
set interfaces wireless wlan0 mode ax
set interfaces wireless wlan0 security wpa cipher CCMP
set interfaces wireless wlan0 security wpa cipher CCMP-256
set interfaces wireless wlan0 security wpa cipher GCMP-256
set interfaces wireless wlan0 security wpa cipher GCMP
set interfaces wireless wlan0 security wpa mode wpa3
set interfaces wireless wlan0 security wpa passphrase super-doooper-secure-passphrase
set interfaces wireless wlan0 mgmt-frame-protection required
set interfaces wireless wlan0 enable-bf-protection
set interfaces wireless wlan0 ssid test.ax
set interfaces wireless wlan0 type access-point
set interfaces wireless wlan0 stationary-ap
commit

```

Resulting configuration:

```

system {
  wireless {
    country-code de
  }
}
interfaces {
  [...]
  wireless wlan0 {
    capabilities {
      he {
        antenna-pattern-fixed
        beamform {
          multi-user-beamformer
          single-user-beamformee
          single-user-beamformer
        }
        bss-color 13
        center-channel-freq {
          freq-1 15
        }
        channel-set-width 134

```

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```

    }
  }
  channel 5
  description "802.11ax 6GHz"
  enable-bf-protection
  hw-id [...]
  mgmt-frame-protection required
  mode ax
  physical-device phy0
  security {
    wpa {
      cipher CCMP
      cipher CCMP-256
      cipher GCMP-256
      cipher GCMP
      mode wpa3
      passphrase super-doooper-secure-passphrase
    }
  }
  ssid test.ax
  stationary-ap
  type access-point
}

```

Intel AX200

The Intel AX200 card does not work out of the box in AP mode. You can still put this card into AP mode using the following configuration:

```

set system wireless country-code 'us'
set interfaces wireless wlan0 channel '1'
set interfaces wireless wlan0 mode 'n'
set interfaces wireless wlan0 physical-device 'phy0'
set interfaces wireless wlan0 ssid 'VyOS'
set interfaces wireless wlan0 type 'access-point'

```

7.4.19 WWAN

WWAN (Wireless Wide Area Network) interfaces provide access to cellular networks via a cellular modem or card. Configure these interfaces under the `interfaces wwan` node.

Configuration

Common interface configuration

set interfaces wwan <interface> address <address | dhcp | dhcpv6>

Configure the interface with one or more IP addresses.

The following options are available:

- **address:** Assign one or more IPv4 or IPv6 addresses to the interface. For example, use 192.0.2.1/24 for IPv4 or 2001:db8::1/64 for IPv6.
- **dhcp:** The interface obtains an IPv4 address from a DHCP server on the same network segment.
- **dhcpv6:** The interface obtains an IPv6 address from a DHCPv6 server on the same network segment.

Note: If the interface obtains an IPv4 address via DHCP, and specific adjustments are needed before/after the IP address is obtained, use the provided hook scripts:

- /config/scripts/dhcp-client/pre-hooks.d/
- /config/scripts/dhcp-client/post-hooks.d/

Example:

```
set interfaces wwan wwan0 address 192.0.2.1/24
set interfaces wwan wwan0 address 2001:db8::1/64
set interfaces wwan wwan0 address dhcp
set interfaces wwan wwan0 address dhcpv6
```

set interfaces wwan <interface> description <description>

Configure a clear, descriptive alias for the interface.

This alias appears in the `show interfaces` command and SNMP-based monitoring tools.

Example:

```
set interfaces wwan wwan0 description 'This is an interface running on VyOS.'
```

set interfaces wwan <interface> disable

Disable the interface.

The interface will be set to the administratively down (A/D) state.

Example:

```
set interfaces wwan wwan0 disable
```

set interfaces wwan <interface> disable-link-detect

Disable physical link-state change detection on the interface, such as when a cable is unplugged.

By default, the interface detects physical link-state changes.

Example:

```
set interfaces wwan wwan0 disable-link-detect
```

set interfaces wwan <interface> mtu <mtu>

Configure the MTU on the interface.

This value defines the largest packet size, in bytes, that the interface transmits without fragmentation.

Example:

```
set interfaces wwan wwan0 mtu 1600
```

set interfaces wwan <interface> ip adjust-mss <mss | clamp-mss-to-pmtu>

Configure the MSS advertised in outgoing TCP SYN packets on the specified interface.

By clamping the MSS value in TCP SYN packets, you explicitly inform the remote side not to send packets larger than that size. This prevents connection issues that occur when Path MTU Discovery (PMTUD) fails.

The following options are available:

- **mss:** Sets the MSS to a specific value, in bytes. Use this option if you need to enforce a specific MSS, for example, to troubleshoot connectivity issues or accommodate specific network requirements.
- **clamp-mss-to-pmtu:** The router automatically calculates the MSS to be the interface's MTU minus 40 bytes for IPv4 traffic (20 bytes for the IPv4 header and 20 bytes for the TCP header). This option is recommended to automatically set the proper value.

set interfaces wwan <interface> ip arp-cache-timeout

Configure how long an ARP entry remains valid after learning an IP-to-MAC address mapping on this interface.

The default duration is 30 seconds.

An ARP entry remains valid if it receives positive feedback from higher-level protocols.

Example:

```
set interfaces wwan wwan0 ip arp-cache-timeout 180
```

set interfaces wwan <interface> ip disable-arp-filter

Configure ARP filtering on this interface.

Default behavior: The kernel responds to ARP requests on this interface only if the traffic would be routed back to the ARP sender through that specific interface.

If configured: The kernel responds to ARP requests on this interface for any IP address configured on the local host, regardless of which specific interface that IP address is assigned to, and regardless of the routing table. This reflects the Linux concept that IP addresses belong to the host, not individual interfaces.

Example:

```
set interfaces wwan wwan0 ip disable-arp-filter
```

set interfaces wwan <interface> ip disable-forwarding

Configure the interface for host or router behavior.

If configured, the interface switches to host mode, and IPv4 forwarding is disabled on it.

Example:

```
set interfaces wwan wwan0 ip disable-forwarding
```

set interfaces wwan <interface> ip enable-directed-broadcast

Configure whether to forward IP-directed broadcast packets received on this interface.

Default behavior: IP-directed broadcast packets are dropped.

If configured: IP-directed broadcast packets are forwarded to all hosts on the destination subnet, as defined in [RFC 1812](#) and [RFC 2644](#).

Example:

```
set interfaces wwan wwan0 ip enable-directed-broadcast
```

set interfaces wwan <interface> ip enable-arp-accept

Configure how to process gratuitous ARPs on this interface.

If configured, an IP-to-MAC address mapping is added to the ARP table based on gratuitous ARP requests or replies.

Note: If the ARP table already contains the IP address from a gratuitous ARP, its entry is updated regardless of whether this setting is configured.

Example:

```
set interfaces wwan wwan0 ip enable-arp-accept
```

set interfaces wwan <interface> ip enable-arp-announce

Configure the source IP selection for ARP requests on this interface.

Default behavior: The kernel can use any IP address the host owns as the source IP address in ARP requests on this interface.

If configured: The kernel first attempts to select a source IP address configured on the interface that shares a common subnet with the target IP address. If there is no such subnet, the kernel selects the IP address it would normally use (based on the routing table to reach the target destination).

Example:

```
set interfaces wwan wwan0 ip enable-arp-announce
```

set interfaces wwan <interface> ip enable-arp-ignore

Configure which ARP requests will be ignored on this interface.

Default behavior: The kernel responds to ARP requests for any local IP addresses, regardless of which interface they are assigned to.

If configured: The kernel responds to ARP requests only if the target IP address is assigned to this specific interface.

Example:

```
set interfaces wwan wwan0 ip enable-arp-ignore
```

set interfaces wwan <interface> ip enable-proxy-arp

Configure proxy ARP on this interface.

If configured, the router (kernel) intercepts ARP requests for non-local IP addresses and replies with the MAC address of the interface that received the request. Subsequent packets destined to these IP addresses are forwarded to their actual destinations on remote subnets.

Example:

```
set interfaces wwan wwan0 ip enable-proxy-arp
```

set interfaces wwan <interface> ip proxy-arp-pvlan

Configure local proxy ARP on the interface.

If configured, the router (kernel) responds to ARP requests on this VLAN interface even if the target IP address resides on the same subnet and interface.

This is used to support network isolation requirements (RFC 3069) for private VLANs (PVLANS). In PVLAN configurations, hosts on isolated ports are NOT allowed to communicate directly with each other at Layer 2, but they can communicate with the upstream router.

By replying to inter-host ARP requests with its own MAC address, the router (kernel) directs inter-host traffic through itself instead of directly between hosts.

Note: This command works independently and does not require enabling the standard proxy ARP on the interface.

Local proxy ARP is also known as:

- VLAN aggregation ([RFC 3069](#)).
- Private VLAN (Cisco, Allied Telesyn).
- Source-port filtering or port isolation (Hewlett-Packard).
- MAC-Forced Forwarding (Ericsson).

set interfaces wwan <interface> ip source-validation <strict | loose | disable>

Configure source IP address validation using RPF on this interface, as specified in [RFC 3704](#).

The following options are available:

- **strict:** Each incoming packet's source IP address is checked against the Forwarding Information Base (FIB). If the interface is not the best route back to that source, validation fails, and the packet is dropped.
- **loose:** Each incoming packet's source IP address is checked against the FIB. If the source IP address is unreachable through any interface, validation fails.
- **disable:** No source IP address validation is performed. All incoming packets are accepted.

RFC 3704 recommends enabling strict mode to prevent IP spoofing, such as DDoS attacks. For asymmetric or other complex routing scenarios, use loose mode.

set interfaces wwan <interface> ipv6 address autoconf

Configure the interface to automatically obtain an IPv6 address using SLAAC, as specified in [RFC 4862](#).

IPv6 hosts can configure themselves automatically when connected to an IPv6 network using the Neighbor Discovery Protocol via ICMPv6 router discovery messages. When first connected to a network, a host sends a link-local router solicitation multicast request for its configuration parameters. The router responds with a router advertisement packet containing Internet Layer configuration parameters.

Note: This method automatically disables IPv6 traffic forwarding on the interface.

Example:

```
set interfaces wwan wwan0 ipv6 address autoconf
```

set interfaces wwan <interface> ipv6 address eui64 <prefix>

Configure the interface to assign itself an IPv6 address using the EUI-64 method, as specified in [RFC 4291](#).

Example:

```
set interfaces wwan wwan0 ipv6 address eui64 2001:db8:beef::/64
```

set interfaces wwan <interface> ipv6 address no-default-link-local

Disable the automatic assignment of a link-local IPv6 address to this interface.

Example:

```
set interfaces wwan wwan0 ipv6 address no-default-link-local
```

set interfaces wwan <interface> ipv6 disable-forwarding

Configure the interface for host or router behavior.

If configured, the interface switches to host mode, and IPv6 forwarding is disabled on it.

Example:

```
set interfaces wwan wwan0 ipv6 disable-forwarding
```

set interfaces wwan <interface> ipv6 adjust-mss <mss | clamp-mss-to-pmtu>

Configure the MSS advertised in outgoing TCP SYN packets on the specified interface.

By clamping the MSS value in TCP SYN packets, you explicitly inform the remote side not to send packets larger than that size. This prevents connection issues when Path MTU Discovery (PMTUD) fails.

The following options are available:

- **mss:** Set the MSS to a specific value, in bytes. Use this option to enforce a specific MSS, for example, to troubleshoot connectivity issues or accommodate specific network requirements.
- **clamp-mss-to-pmtu:** The router calculates the MSS to be the interface's MTU minus 60 bytes for IPv6 traffic (40 bytes for the IPv6 header and 20 bytes for the TCP header). This option is recommended to automatically set the proper value.

set interfaces wwan <interface> ipv6 accept-dad <0 | 1 | 2>

Configure IPv6 DAD on the interface.

The following options are available:

- **0:** Disables DAD. No duplicate address detection is performed.

- **1:** Enables DAD (default). Duplicate addresses are detected. The interface's IPv6 operation continues for valid IPv6 addresses.
- **2:** Enables DAD and, if a MAC-based duplicate link-local address is found, disables IPv6 operation on this interface.

Example:

```
set interfaces wwan wwan0 ipv6 accept-dad 2
```

set interfaces wwan <interface> ipv6 dup-addr-detect-transmits <n>

Configure the number of DAD messages that the router (kernel) sends during IPv6 address assignment on this interface.

The default value is 1.

Example:

```
set interfaces wwan wwan0 ipv6 dup-addr-detect-transmits 5
```

set interfaces wwan <interface> vrf <vrf>

Assign the interface to a specific VRF instance.

See also:

For information on configuring a VRF, refer to the [VRF](#) section.

Example:

```
set interfaces wwan wwan0 vrf red
```

DHCP(v6)

set interfaces wwan <interface> dhcp-options client-id <description>

Configure a DHCP client identifier for the interface, as specified in [RFC 2131](#).

The `client-id` is an identifier that the DHCP client sends to the DHCP server to uniquely identify itself for IP address assignment. By default, the client uses its MAC address. The `<description>` is a user-defined string that will be sent to the DHCP server as the DHCP client identifier.

Example:

```
set interfaces wwan wwan0 dhcp-options client-id 'foo-bar'
```

set interfaces wwan <interface> dhcp-options host-name <hostname>

Configure a specific hostname for the interface.

Instead of the real hostname, the DHCP client will send the specific hostname to the DHCP server when requesting an IP address.

Example:

```
set interfaces wwan wwan0 dhcp-options host-name 'VyOS'
```

set interfaces wwan <interface> dhcp-options vendor-class-id <vendor-id>

Configure the DHCP client to include a vendor-class identifier in its DHCP requests on this interface.

The vendor-class identifier is a vendor-specific byte string that enables the DHCP server to identify the device and, in some cases, provide configuration options.

Example:

```
set interfaces wwan wwan0 dhcp-options vendor-class-id 'VyOS'
```

set interfaces wwan <interface> dhcp-options no-default-route

Configure the DHCP client to obtain an IP address, but ignore any default gateway provided by the DHCP server on this interface.

Example:

```
set interfaces wwan wwan0 dhcp-options no-default-route
```

set interfaces wwan <interface> dhcp-options default-route-distance <distance>

Configure the distance for the default route obtained from the DHCP server on this interface.

Example:

```
set interfaces wwan wwan0 dhcp-options default-route-distance 220
```

set interfaces wwan <interface> dhcp-options reject <address>

Configure the DHCP client to reject the specific IP address or IP address range from the DHCP server on this interface.

This is useful when a modem assigns a local IP address upon start. To reject multiple addresses, run this command multiple times with different values. You can reject individual addresses (192.168.100.1) or entire subnets (192.168.100.0/24).

Example:

```
set interfaces wwan wwan0 dhcp-options reject 192.168.100.0/24
```

set interfaces wwan <interface> dhcp-options user-class <string>

Configure the DHCP client to send a specific user-class identifier in its DHCP requests on this interface.

The DHCP server can interpret this identifier and provide specific configuration options based on it (for example, default routes). The user-class value typically groups DHCP clients with similar configuration needs (for example, employees, guests, or printers).

Example:

```
set interfaces wwan wwan0 dhcp-options user-class VyOS
```

set interfaces wwan <interface> dhcpv6-options duid <duid>

Configure a specific DUID for the DHCPv6 client on this interface.

The DUID is an identifier used by a DHCPv6 client to get an IPv6 address from a DHCPv6 server. It consists of a 2-byte type field, followed by a variable-length identifier field up to 128 bytes. The format of the identifier part depends on the DUID type:

- **DUID-LLT:** The most common type, which includes a hardware type, a timestamp, and a MAC address.
- **DUID-EN:** Is based on a vendor's enterprise number and a unique identifier assigned by the vendor.
- **DUID-LL:** Includes only a MAC address.

The DHCP server matches the DUID against its database and provides configuration data (such as address, lease times, DNS servers, etc.) to the DHCP client.

Example:

```
set interfaces wwan wwan0 duid '0e:00:00:01:00:01:27:71:db:f0:00:50:56:bf:c5:6d'
```

set interfaces wwan <interface> dhcpv6-options no-release

Configure the DHCP client not to send a release message when it stops running on this interface.

This helps retain the assigned address or prefix.

Example:

```
set interfaces wwan wwan0 dhcpv6-options no-release
```

set interfaces wwan <interface> dhcpv6-options parameters-only

Enable a stateless DHCPv6 client mode on this interface.

In stateless mode, the DHCPv6 client requests only stateless configuration parameters from the DHCP server (for example, DNS server addresses). It doesn't request a stateful configuration, such as IPv6 addresses or prefixes.

Example:

```
set interfaces wwan wwan0 dhcpv6-options parameters-only
```

set interfaces wwan <interface> dhcpv6-options rapid-commit

Enable DHCPv6 rapid commit on this interface.

When enabled, the DHCP client and server skip the negotiation steps (Advertise and Request), completing the DHCPv6 configuration process in just two messages (Solicit and final Reply).

Example:

```
set interfaces wwan wwan0 dhcpv6-options rapid-commit
```

set interfaces wwan <interface> dhcpv6-options temporary

Configure the DHCPv6 client to request a temporary IPv6 address on this interface.

When configured, the DHCP client doesn't form an Identity Association for Non-temporary Addresses (IA_NA) partnership. Consequently, it only obtains a temporary IPv6 address and doesn't obtain a permanent one.

Example:

```
set interfaces wwan wwan0 dhcpv6-options temporary
```

DHCPv6 Prefix Delegation (PD)

VyOS supports DHCPv6 Prefix Delegation (DHCPv6-PD) as described in [RFC 3633](#). DHCPv6-PD is supported by most ISPs that provide native IPv6 for consumers on fixed networks.

set interfaces wwan <interface> dhcpv6-options pd <id> length <length>

Configure a specific prefix length for DHCPv6-PD requests on this interface.

Some ISPs provide only a /64 prefix by default. Use this command to request a different prefix length for a specific DHCPv6-PD request, ranging from /32 (if allowed by your ISP) down to /64. <id> is a unique identifier for the DHCPv6-PD request.

The default value is 64.

To request a /56 prefix from your ISP, use:

```
set interfaces wwan wwan0 dhcpv6-options pd 0 length 56
```

set interfaces wwan <interface> dhcpv6-options pd <id> interface <delegatee> address <address>

Configure the IPv6 interface identifier (host portion) for the delegatee interface.

The value must be a decimal integer. It is appended to the delegated prefix and the configured SLA ID to form the final IPv6 address.

By default, the host portion is generated based on the parent interface's MAC address (EUI-64 format).

Example:

If a /64 prefix is delegated to interface eth8 and you configure the host portion as 65535, the resulting IPv6 address will end with ::ffff, as 65535 corresponds to ffff in hexadecimal notation.

```
set interfaces wwan wwan0 dhcpv6-options pd 0 interface eth8 address 65534
```

set interfaces wwan <interface> dhcpv6-options pd <id> interface <delegatee> sla-id <id>

Configure the SLA ID for the delegatee interface.

The value must be a decimal integer greater than 0 and fit in the length of SLA IDs. It is converted to hexadecimal and appended to the delegated prefix to form the specific subnet prefix for the delegatee interface.

Example:

If SLA ID is 1 and the delegated prefix is 2001:db8:ffff::/48, the resulting subnet prefix for the delegatee interface will be 2001:db8:ffff:1::/64.

```
set interfaces wwan wwan0 dhcpv6-options pd 0 interface eth8 sla-id 1
```

WWAN options

set interfaces wwan <interface> apn <apn>

Configure the APN (Access Point Name) for the WWAN connection.

Every WWAN connection requires an APN to connect to the cellular network.

This parameter is mandatory. Contact your service provider for the correct APN.

Operation

show interfaces wwan <interface>

Show the operational status and traffic statistics for the specified WWAN interface.

```
vyos@vyos:~$ show interfaces wwan wwan0
wwan0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UNKNOWN
↪group default qlen 1000
link/ether 02:c2:f3:00:01:02 brd ff:ff:ff:ff:ff:ff
inet 10.155.144.12/30 brd 10.155.144.15 scope global dynamic wwan0
valid_lft 7012sec preferred_lft 7012sec
inet6 fe80::c2:f3ff:fe00:0102/64 scope link
```

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```

valid_lft forever preferred_lft forever

RX:  bytes  packets  errors  dropped  overrun     mcast
640           2         0         0         0           0
TX:  bytes  packets  errors  dropped  carrier  collisions
3229         16         0         0         0           0

```

show interfaces wwan <interface> summary

Show WWAN module hardware characteristics and connection information.

```

vyos@vyos:~$ show interfaces wwan wwan0 summary
-----
General |          dbus path: /org/freedesktop/ModemManager1/Modem/0
|          device id: 79f4e9cc2e9fc8d4a3b8c8f6327c2e363170194d
-----
Hardware |          manufacturer: Sierra Wireless, Incorporated
|          model: MC7710
|          revision: SWI9200X_03.05.29.03ap r6485 CNSHZ-ED-XP0031 2014/12/02
→17:53:15
|          h/w revision: 1.0
|          supported: gsm-umts, lte
|          current: gsm-umts, lte
|          equipment id: 358xxxxxxxxxxx
-----
System   |          device: /sys/devices/pci0000:00/0000:00:13.0/usb3/3-1/3-1.3
|          drivers: qcserial, qmi_wwan
|          plugin: Generic
|          primary port: cdc-wdm0
|          ports: ttyUSB0 (qcdm), ttyUSB2 (at), cdc-wdm0 (qmi), wwan0 (net)
-----
Numbers  |          own: 4917xxxxxxxx
-----
Status   |          lock: sim-pin2
|          unlock retries: sim-pin (3), sim-pin2 (3), sim-puk (10), sim-puk2 (10)
|          state: connected
|          power state: on
|          access tech: lte
|          signal quality: 63% (recent)
-----
Modes    |          supported: allowed: 2g; preferred: none
|          allowed: 3g; preferred: none
|          allowed: 4g; preferred: none
|          allowed: 2g, 3g; preferred: 3g
|          allowed: 2g, 3g; preferred: 2g
|          allowed: 2g, 4g; preferred: 4g
|          allowed: 2g, 4g; preferred: 2g
|          allowed: 3g, 4g; preferred: 3g
|          allowed: 3g, 4g; preferred: 4g
|          allowed: 2g, 3g, 4g; preferred: 4g
|          allowed: 2g, 3g, 4g; preferred: 3g
|          allowed: 2g, 3g, 4g; preferred: 2g
|          current: allowed: 2g, 3g, 4g; preferred: 2g

```

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```
-----
Bands      | supported: egsm, dcs, pcs, utran-1, utran-8, eutran-1, eutran-
->3,
|           | eutran-7, eutran-8, eutran-20
|           | current: egsm, dcs, pcs, utran-1, utran-8, eutran-1, eutran-3,
|           | eutran-7, eutran-8, eutran-20
-----
IP          | supported: ipv4, ipv6, ipv4v6
-----
3GPP        | imei: 358xxxxxxxxxxxxx
|           | operator id: 26201
|           | operator name: Telekom.de
|           | registration: home
-----
3GPP EPS    | ue mode of operation: ps-1
-----
SIM          | dbus path: /org/freedesktop/ModemManager1/SIM/0
-----
Bearer       | dbus path: /org/freedesktop/ModemManager1/Bearer/0
-----
```

show interfaces wwan <interface> capabilities

Show WWAN module radio capabilities.

```
vyos@vyos:~$ show interfaces wwan wwan0 capabilities
Max TX channel rate: '500000000'
Max RX channel rate: '1000000000'
Data Service: 'simultaneous-cs-ps'
SIM: 'supported'
Networks: 'gsm, umts, lte'
Bands: 'gsm-dcs-1800, gsm-900-extended, gsm-900-primary, gsm-pcs-1900, wcdma-2100,
->wcdma-900'
LTE bands: '1, 3, 7, 8, 20'
```

show interfaces wwan <interface> firmware

Show WWAN module firmware information.

```
vyos@vyos:~$ show interfaces wwan wwan0 firmware
Model: MC7710
Boot version: SWI9200X_03.05.29.03bt r6485 CNSHZ-ED-XP0031 2014/12/02 17:33:08
AMSS version: SWI9200X_03.05.29.03ap r6485 CNSHZ-ED-XP0031 2014/12/02 17:53:15
SKU ID: unknown
Package ID: unknown
Carrier ID: 0
Config version: unknown
```

show interfaces wwan <interface> imei

Show WWAN module IMEI.

```
vyos@vyos:~$ show interfaces wwan wwan0 imei
ESN: '0'
IMEI: '358xxxxxxxxxxxxx'
```

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```
MEID: 'unknown'
```

show interfaces wwan <interface> imsi

Show the IMSI of the associated SIM card.

```
vyos@vyos:~$ show interfaces wwan wwan0 imsi
IMSI: '262xxxxxxxxxxx'
```

show interfaces wwan <interface> model

Show WWAN module model.

```
vyos@vyos:~$ show interfaces wwan wwan0 model
Model: 'MC7710'
```

show interfaces wwan <interface> msisdn

Show the MSISDN of the associated SIM card.

```
vyos@vyos:~$ show interfaces wwan wwan0 msisdn
MSISDN: '4917xxxxxxxx'
```

show interfaces wwan <interface> revision

Show WWAN module hardware revision.

```
vyos@vyos:~$ show interfaces wwan wwan0 revision
Revision: 'SWI9200X_03.05.29.03ap r6485 CNSHZ-ED-XP0031 2014/12/02 17:53:15'
```

show interfaces wwan <interface> signal

Show signal information for the cellular connection.

```
vyos@vyos:~$ show interfaces wwan wwan0 signal
LTE:
RSSI: '-74 dBm'
RSRQ: '-7 dB'
RSRP: '-100 dBm'
SNR: '13.0 dB'
Radio Interface: 'lte'
Active Band Class: 'eutran-3'
Active Channel: '1300'
```

show interfaces wwan <interface> sim

Show WWAN module SIM card information.

```
vyos@vyos:~$ show interfaces wwan wwan0 sim
Provisioning applications:
Primary GW: slot '1', application '1'
Primary 1X: session doesn't exist
Secondary GW: session doesn't exist
Secondary 1X: session doesn't exist
Slot [1]:
Card state: 'present'
```

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```

UPIN state: 'not-initialized'
UPIN retries: '0'
UPUK retries: '0'
Application [1]:
Application type: 'usim (2)'
Application state: 'ready'
Application ID:
A0:00:00:00:87:10:02:FF:49:94:20:89:03:10:00:00
Personalization state: 'ready'
UPIN replaces PIN1: 'no'
PIN1 state: 'disabled'
PIN1 retries: '3'
PUK1 retries: '10'
PIN2 state: 'enabled-not-verified'
PIN2 retries: '3'
PUK2 retries: '10'

```

Example

The following example shows how to configure a cellular connection using a Sierra Wireless MC7710 miniPCIe card that operates over USB despite its form factor. The card is installed in a *PC Engines APU4*.

```

set interfaces wwan wwan0 apn 'internet.telekom'
set interfaces wwan wwan0 address 'dhcp'

```

Supported hardware

The following WWAN modules have been successfully tested with a *PC Engines APU4* board:

- Sierra Wireless AirPrime MC7304 miniPCIe card (LTE)
- Sierra Wireless AirPrime MC7430 miniPCIe card (LTE)
- Sierra Wireless AirPrime MC7455 miniPCIe card (LTE)
- Sierra Wireless AirPrime MC7710 miniPCIe card (LTE)
- Huawei ME909u-521 miniPCIe card (LTE)
- Huawei ME909s-120 miniPCIe card (LTE)
- HP LT4120 Snapdragon X5 LTE

Firmware update

WWAN modules include reprogrammable firmware, and most vendors regularly provide updates for it.

Since VyOS communicates with these devices via the QMI interface, you can update firmware directly within the system using the `qmi-firmware-update` utility.

The following example shows how to update the firmware for a Sierra Wireless MC7710 module using the provided .cwe file.

```
$ sudo qmi-firmware-update --update -d 1199:68a2 \
  9999999_9999999_9200_03.05.14.00_00_generic_000.000_001_SPKG_MC.cwe
```

7.5 Load-balancing

7.5.1 WAN load balancing

Todo: Convert raw command blocks in this file to cfgcmd/opcmd directives for command coverage tracking.

The load balancer distributes outbound traffic across two or more interfaces. If a path fails, the load balancer balances traffic across the remaining healthy paths. When a path recovers, it is automatically added back to the routing table. The load balancer adds routes for each path and distributes traffic based on interface health and weight.

In a minimal configuration, the following must be provided:

- An interface with a `nexthop`.
- One rule with a LAN (inbound-interface) and the WAN (interface).

The following examples uses two DHCP WAN interfaces and one LAN (`eth2`):

```
set load-balancing wan interface-health eth0 nexthop 'dhcp'
set load-balancing wan interface-health eth1 nexthop 'dhcp'
set load-balancing wan rule 1 inbound-interface 'eth2'
set load-balancing wan rule 1 interface eth0
set load-balancing wan rule 1 interface eth1
```

Note: Do not use WAN load balancing with dynamic routing protocols. This feature creates customized routing tables and firewall rules that are incompatible with routing protocols.

Load balancing rules

You define interfaces, their weight, and the traffic type to balance in numbered rule sets. The load balancer executes rules in numerical order against outgoing packets. When a packet matches a rule, it is sent through the specified interface. Packets that do not match any rule use the system routing table. You cannot change rule numbers.

Create a load balancing rule, it can be a number between 1 and 9999:

```
vyos@vyos# set load-balancing wan rule 1
Possible completions:
description          Description for this rule
> destination        Destination
exclude              Exclude packets matching this rule from wan load balance
failover             Enable failover for packets matching this rule from wan load_
↪ balance
inbound-interface    Inbound interface name (e.g., "eth0") [REQUIRED]
+> interface         Interface name [REQUIRED]
> limit              Enable packet limit for this rule
per-packet-balancing Option to match traffic per-packet instead of the default, per-
```

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```

↪ flow
protocol          Protocol to match
> source          Source information

```

Interface weight

By default, the load balancer distributes outbound traffic randomly across available interfaces. You can assign weights to interfaces to influence the distribution. If `eth0` has more bandwidth than `eth1`, you can assign a higher weight to `eth0` to send more traffic through it:

```

set load-balancing wan rule 1 interface eth0 weight 2
set load-balancing wan rule 1 interface eth1 weight 1

```

In this example, `eth0` receives 66% of traffic, and `eth1` receives 33% of traffic.

Rate limit

Set a packet rate limit for a rule to apply it to traffic above or below a specified threshold. To configure rate limiting, use:

```

set load-balancing wan rule <rule> limit <parameter>

```

- **burst:** Number of packets allowed to overshoot the limit within period. Default 5.
- **period:** Time window for rate calculation. Possible values: `second` (one second), `minute` (one minute), `hour` (one hour). Default is `second`.
- **rate:** Number of packets. Default: 5.
- **threshold:** below or above the specified rate limit.

Flow and packet-based balancing

The load balancer balances outgoing traffic by flow. A connection tracking table tracks flows by source address, destination address, and port. Each flow is assigned to an interface based on the balancing rules, and subsequent packets use the same interface. This ensures packets arrive in order when links have different speeds.

Packet-based balancing can improve balance across interfaces when packet order is not critical. Enable per-packet balancing for a rule with:

```

set load-balancing wan rule <rule> per-packet-balancing

```

Exclude traffic

To exclude traffic from load balancing, traffic matching an exclude rule bypasses load balancing and uses the system routing table instead:

```
set load-balancing wan rule <rule> exclude
```

Health checks

The load balancer periodically checks the health of interfaces and paths by sending ICMP packets (ping) to remote destinations, performing TTL tests, or executing a user-defined script. If an interface fails the health check, the load balancer removes it from its interface pool. To enable health checking for an interface:

```
vyos@vyos# set load-balancing wan interface-health <interface>
Possible completions:
failure-count      Failure count
nexthop            Outbound interface nexthop address. Can be 'dhcp or ip address'
→ [REQUIRED]
success-count      Success count
+> test           Rule number
```

Specify the nexthop on the path to the destination. You can set `ipv4-address` to `dhcp`.

```
set load-balancing wan interface-health <interface> nexthop <ipv4-address>
```

Set the number of health check failures before the load balancer marks an interface as unavailable (range 1-10, default 1). Or set the number of successful health checks before adding an interface back to the pool (range 1-10, default 1).

```
set load-balancing wan interface-health <interface> failure-count <number>
set load-balancing wan interface-health <interface> success-count <number>
```

Configure each health check in its own test. Tests are numbered and processed in numeric order. You can define multiple tests for multi-target health checking:

```
vyos@vyos# set load-balancing wan interface-health eth1 test 0
Possible completions:
resp-time          Ping response time (seconds)
target             Health target address
test-script        Path to user defined script
ttl-limit          Ttl limit (hop count)
type              WLB test type
```

- **resp-time:** The maximum response time for ping in seconds. Range 1-30, default 5.
- **target:** The target to receive ICMP packets. The address can be an IPv4 address or hostname.
- **test-script:** A user-defined script must return 0 to succeed and non-zero to fail. Scripts reside in `/config/scripts`. For other locations, provide the full path.
- **ttl-limit:** For the UDP TTL limit test, specify the hop count limit. The limit must be shorter than the path length. The test succeeds when an ICMP time-expired message is returned. Default 1.
- **type:** Specify the test type: `ping`, `ttl`, or a user-defined script.

Source NAT rules

By default, interfaces in a load balancing pool replace the source IP of each outgoing packet with their own address to ensure replies arrive on the same interface. The load balancer handles this through automatically generated Source NAT (SNAT) rules applied only to balanced traffic. To disable the automatic generation of SNAT rules when this behavior is not desired, use:

```
set load-balancing wan disable-source-nat
```

Sticky connections

Inbound connections to a WAN interface can be improperly handled when replies are sent back to the client.

When responding to an incoming packet, you may want to ensure the response leaves from the same interface as the incoming packet. Enable sticky connections in the load balancer to do this:

```
set load-balancing wan sticky-connections inbound
```

Failover

In failover mode, one interface is primary and other interfaces are secondary or spare. The load balancer uses only the primary interface. If it fails, a secondary interface from the available pool takes over. The load balancer selects the primary interface based on its weight and health. Other interfaces become secondary. Secondary interfaces are chosen based on their weight and health. You can also select interface roles based on rule order by including interfaces in balancing rules and ordering those rules accordingly. To enable failover mode, create a failover rule:

```
set load-balancing wan rule <number> failover
```

Existing sessions do not automatically fail over to a new path. Flush the session table on each connection state change to enable failover:

```
set load-balancing wan flush-connections
```

Warning: Flushing the session table causes other connections to revert from flow-based to packet-based balancing until each flow is reestablished.

Script execution

Run a script when an interface state changes. Scripts run from the `/config/scripts` directory. To use a script in another location, specify the full path:

```
set load-balancing wan hook script-name
```

Two environment variables are available:

- `WLB_INTERFACE_NAME=[interfacename]`: Interface to be monitored
- `WLB_INTERFACE_STATE=[ACTIVE|FAILED]`: Interface state

Warning: Blocking call with no timeout: VyOS becomes unresponsive if the script does not return.

Handling and monitoring

The following command shows WAN load balancer information including test types and targets. The character at the start of each line indicates the test state:

- + successful.
- - failed.
- A blank indicates that no test has been carried out.

```
vyos@vyos:~$ show wan-load-balance
Interface: eth0
Status: failed
Last Status Change: Tue Jun 11 20:12:19 2019
-Test: ping Target:
    Last Interface Success: 55s
    Last Interface Failure: 0s
    # Interface Failure(s): 5

Interface: eth1
Status: active
Last Status Change: Tue Jun 11 20:06:42 2019
+Test: ping Target:
    Last Interface Success: 0s
    Last Interface Failure: 6m26s
    # Interface Failure(s): 0
```

Show connection data of load balanced traffic:

```
vyos@vyos:~$ show wan-load-balance connection
conntrack v1.4.2 (conntrack-tools): 3 flow entries have been shown.
Type      State      Src          Dst          Packets Bytes
tcp       TIME_WAIT  10.1.1.13:38040 203.0.113.2:80 203.0.113.2 192.
↪168.188.71
udp       10.1.1.13:41891 198.51.100.3:53 198.51.100.3 192.
↪168.188.71
udp       10.1.1.13:55437 198.51.100.3:53 198.51.100.3 192.
↪168.188.71
```

Restart

```
restart wan-load-balance
```

7.5.2 HAproxy

HAProxy is a load balancer and proxy server that provides high-availability, load balancing, and proxying for TCP (level 4) and HTTP-based (level 7) applications.

Configuration

Service configuration specifies the port to bind to. Backend configuration defines the load balancing method and specifies the backend servers.

Service

set load-balancing haproxy service <name> listen-address <address>

Set the IP address for the service to bind to. By default, the service listens on all IPv4 and IPv6 addresses.

set load-balancing haproxy service <name> port <port>

Create service to listen on <port>

set load-balancing haproxy service <name> mode <tcp|http>

Configure service mode TCP or HTTP

set load-balancing haproxy service <name> backend <name>

Configure service to use the backend <name>

set load-balancing haproxy service <name> ssl certificate <name>

Set the SSL certificate <name> for service <name>. You can define multiple certificates.

set load-balancing haproxy service <name> http-response-headers <header-name> value <header-value>

Set custom HTTP headers to include in all responses.

set load-balancing haproxy service <name> logging facility <facility> level <level>

Specify facility and level for logging. For an explanation on *Syslog facilities* and *Severity levels*, see tables in the syslog configuration section.

set load-balancing haproxy service <name> timeout client <seconds>

Set the maximum inactivity time on the client side for this service. Value range 1-3600 seconds.

set load-balancing haproxy service <name> http-compression algorithm <gzip | deflate | identity | raw-deflate>

Set the compression algorithm to be used when compressing HTTP responses.

set load-balancing haproxy service <name> http-compression mime-type <mime-type>

Set the list of HTTP response MIME types which haproxy will attempt to compress, if received uncompressed from backend server.

Rules

Rules control and route incoming traffic to specific backends based on predefined conditions. Rules define matching criteria and specify actions to perform.

set load-balancing haproxy service <name> rule <rule> domain-name <name>

Match domain name

set load-balancing haproxy service <name> rule <rule> ssl <sni>

SSL match Server Name Indication (SNI) option:

- req-ssl-sni SSL Server Name Indication (SNI) request match
- ssl-fc-sni SSL frontend connection Server Name Indication match
- ssl-fc-sni-end SSL frontend match end of connection Server Name Indication

set load-balancing haproxy service <name> rule <rule> url-path <match> <url>

Define URL path matching rules for a specific service. Use this command to specify how to match the URL path against incoming requests.

The available options for <match> are:

- begin Matches the beginning of the URL path
- end Matches the end of the URL path.
- exact Matches the URL path exactly.

set load-balancing haproxy service <name> rule <rule> set backend <name>

Assign a specific backend to a rule

set load-balancing haproxy service <name> rule <rule> redirect-location <url>

Redirect URL to a new location.

Backend

set load-balancing haproxy backend <name> balance <balance>

Specify the load balancing algorithm for distributing requests among available servers.

Balance algorithms:

- source-address Distributes requests based on the source IP address of the client.
- round-robin Distributes requests in a circular manner, sequentially sending each request to the next server in line.
- least-connection Distributes requests to the server with the fewest active connections.

set load-balancing haproxy backend <name> mode <mode>

Configure backend mode TCP or HTTP.

set load-balancing haproxy backend <name> server <name> address <x.x.x.x>

Set the address of the backend server that receives incoming traffic.

set load-balancing haproxy backend <name> server <name> port <port>

Set the address of the backend port.

```
set load-balancing haproxy backend <name> server <name> check
```

Active health check backend server.

```
set load-balancing haproxy backend <name> server <name> check port <port>
```

Set an alternative port number for health checks. Overrides the default server port used for TCP/HTTP checks.

```
set load-balancing haproxy backend <name> server <name> send-proxy
```

Send a Proxy Protocol version 1 header (text format).

```
set load-balancing haproxy backend <name> server <name> send-proxy-v2
```

Send a Proxy Protocol version 2 header (binary format).

```
set load-balancing haproxy backend <name> ssl ca-certificate <ca-certificate>
```

Use SSL encryption for backend requests and authenticate the backend against <ca-certificate>.

```
set load-balancing haproxy backend <name> ssl no-verify
```

Use SSL encryption for backend requests without validating the server certificate.

```
set load-balancing haproxy backend <name> http-response-headers <header-name> value <header-value>
```

Set custom HTTP headers to include in all responses from the backend.

```
set load-balancing haproxy backend <name> logging facility <facility> level <level>
```

Specify facility and level for logging. For an explanation on *Syslog facilities* and *Severity levels*, see tables in the syslog configuration section.

```
set load-balancing haproxy backend <name> timeout check <seconds>
```

Set the timeout in seconds for established connections. Value range 1-3600 seconds.

```
set load-balancing haproxy backend <name> timeout connect <seconds>
```

Set the maximum time to wait for a connection attempt to a server to succeed. Value range 1-3600 seconds.

```
set load-balancing haproxy backend <name> timeout server <seconds>
```

Set the maximum inactivity time on the server side. Value range 1-3600 seconds.

Global

Global configuration parameters:

```
set load-balancing haproxy global-parameters max-connections <num>
```

Limit maximum number of connections

```
set load-balancing haproxy global-parameters ssl-bind-ciphers <ciphers>
```

Limit the cipher algorithms allowed during SSL/TLS handshake.

```
set load-balancing haproxy global-parameters tls-version-min <version>
```

Specify the minimum required TLS version 1.2 or 1.3

```
set load-balancing haproxy global-parameters logging facility <facility> level <level>
```

Specify facility and level for logging. For an explanation on *Syslog facilities* and *Severity levels*, see tables in the syslog configuration section.

set load-balancing haproxy timeout check <seconds>

Set the timeout in seconds for established connections. Value range 1-3600 seconds. Default is 5 seconds.

set load-balancing haproxy timeout client <seconds>

Set the maximum inactivity time on the client side. Value range 1-3600 seconds. Default is 50 seconds.

set load-balancing haproxy timeout connect <seconds>

Set the maximum time to wait for a connection attempt to a server to succeed. Value range 1-3600 seconds. Default is 10 seconds.

set load-balancing haproxy timeout server <seconds>

Set the maximum inactivity time on the server side. Value range 1-3600 seconds. Default is 50 seconds.

Health checks

HTTP checks

Use HTTP health checks to monitor web applications that provide health status information and determine their availability.

set load-balancing haproxy backend <name> http-check

Enables HTTP health checks using OPTION HTTP requests against '/' and expecting a successful response code in the 200-399 range.

set load-balancing haproxy backend <name> http-check method <method>

Set the HTTP method: OPTION, GET, POST, or PUT.

set load-balancing haproxy backend <name> http-check uri <path>

Set the endpoint to use for health checks.

set load-balancing haproxy backend <name> http-check expect <condition>

Set the expected result condition for a server to be considered healthy.

Some possible examples are:

- **status 200** Expecting a 200 response code
- **status 200-399** Expecting a non-failure response code
- **string success** Expecting the string success in the response body

TCP checks

Configure health checks for TCP mode backends. You can configure protocol-aware checks for a range of Layer 7 protocols:

set load-balancing haproxy backend <name> health-check <protocol>

Available health check protocols:

- **ldap** LDAP protocol check.
- **redis** Redis protocol check.
- **mysql** MySQL protocol check.

- pgsql PostgreSQL protocol check.
- smtp SMTP protocol check.

Note: If you specify a server to check but do not configure a protocol, HAProxy performs a basic TCP health check. A server is online if it responds to a connection attempt with a valid SYN/ACK packet.

Redirect HTTP to HTTPS

Configure a HAProxy service for HTTP that listens on port 80 and redirects incoming requests to HTTPS:

```
set load-balancing haproxy service http port '80'
set load-balancing haproxy service http redirect-http-to-https
```

You can use a different service name; in this example, `http` is just for convenience.

Examples

Level 4 balancing

This configuration enables the TCP reverse proxy for the `my-tcp-api` service. Incoming TCP connections on port 8888 are load balanced across the backend servers (`srv01` and `srv02`) using the round-robin load balancing algorithm.

```
set load-balancing haproxy service my-tcp-api backend 'bk-01'
set load-balancing haproxy service my-tcp-api mode 'tcp'
set load-balancing haproxy service my-tcp-api port '8888'

set load-balancing haproxy backend bk-01 balance 'round-robin'
set load-balancing haproxy backend bk-01 mode 'tcp'

set load-balancing haproxy backend bk-01 server srv01 address '192.0.2.11'
set load-balancing haproxy backend bk-01 server srv01 port '8881'
set load-balancing haproxy backend bk-01 server srv02 address '192.0.2.12'
set load-balancing haproxy backend bk-01 server srv02 port '8882'
```

Balancing based on domain name

The following configuration demonstrates how to use VyOS to achieve load balancing based on the domain name:

The HTTP service listens on TCP port 80.

Rule 10 matches requests with the domain name `node1.example.com` and forwards them to the backend `bk-api-01`.

Rule 20 matches requests with the domain name `node2.example.com` and forwards them to the backend `bk-api-02`.

```
set load-balancing haproxy service http description 'bind app listen on 443 port'
set load-balancing haproxy service http mode 'tcp'
set load-balancing haproxy service http port '80'

set load-balancing haproxy service http rule 10 domain-name 'node1.example.com'
set load-balancing haproxy service http rule 10 set backend 'bk-api-01'
```

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```

set load-balancing haproxy service http rule 20 domain-name 'node2.example.com'
set load-balancing haproxy service http rule 20 set backend 'bk-api-02'

set load-balancing haproxy backend bk-api-01 description 'My API-1'
set load-balancing haproxy backend bk-api-01 mode 'tcp'
set load-balancing haproxy backend bk-api-01 server api01 address '127.0.0.1'
set load-balancing haproxy backend bk-api-01 server api01 port '4431'
set load-balancing haproxy backend bk-api-02 description 'My API-2'
set load-balancing haproxy backend bk-api-02 mode 'tcp'
set load-balancing haproxy backend bk-api-02 server api01 address '127.0.0.2'
set load-balancing haproxy backend bk-api-02 server api01 port '4432'

```

Terminate SSL

The following configuration terminates SSL on the router.

The `http` service listens on port 80 and redirects HTTP requests to HTTPS.

The `https` service listens on port 443 with the `bk-default` backend and handles HTTPS traffic using the `cert` certificate for SSL termination. The HSTS header is set with a 1-year expiry to tell browsers to always use SSL for the site.

Rule 10 matches requests with the exact URL path `/.well-known/xxx` and redirects them to `/certs/`.

Rule 20 matches requests with URL paths ending in `/mail` or the exact path `/email/bar` and redirects them to `/postfix/`.

Global parameters include a maximum connection limit of 4000 and a minimum TLS version of 1.3.

```

set load-balancing haproxy service http description 'Force redirect to HTTPS'
set load-balancing haproxy service http port '80'
set load-balancing haproxy service http redirect-http-to-https

set load-balancing haproxy service https backend 'bk-default'
set load-balancing haproxy service https description 'listen on 443 port'
set load-balancing haproxy service https mode 'http'
set load-balancing haproxy service https port '443'
set load-balancing haproxy service https ssl certificate 'cert'
set load-balancing haproxy service https http-response-headers Strict-Transport-Security_
↪value 'max-age=31536000'

set load-balancing haproxy service https rule 10 url-path exact '/.well-known/xxx'
set load-balancing haproxy service https rule 10 set redirect-location '/certs/'
set load-balancing haproxy service https rule 20 url-path end '/mail'
set load-balancing haproxy service https rule 20 url-path exact '/email/bar'
set load-balancing haproxy service https rule 20 set redirect-location '/postfix/'

set load-balancing haproxy backend bk-default description 'Default backend'
set load-balancing haproxy backend bk-default mode 'http'
set load-balancing haproxy backend bk-default server sr01 address '192.0.2.23'
set load-balancing haproxy backend bk-default server sr01 port '80'

```

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```
set load-balancing haproxy global-parameters max-connections '4000'
set load-balancing haproxy global-parameters tls-version-min '1.3'
```

SSL Bridging

The following configuration terminates incoming HTTPS traffic on the router, then re-encrypts the traffic and sends it to the backend server via HTTPS. Use this when encryption is required for both paths but you do not want to install publicly trusted certificates on each backend server.

Backend service certificates are checked against the certificate authority specified in the configuration, which could be an internal CA.

The https service listens on port 443 with backend bk-bridge-ssl to handle HTTPS traffic. It uses certificate named cert for SSL termination.

The bk-bridge-ssl backend connects to sr01 server on port 443 via HTTPS and checks backend server has a valid certificate trusted by CA cacert

```
set load-balancing haproxy service https backend 'bk-bridge-ssl'
set load-balancing haproxy service https description 'listen on 443 port'
set load-balancing haproxy service https mode 'http'
set load-balancing haproxy service https port '443'
set load-balancing haproxy service https ssl certificate 'cert'

set load-balancing haproxy backend bk-bridge-ssl description 'SSL backend'
set load-balancing haproxy backend bk-bridge-ssl mode 'http'
set load-balancing haproxy backend bk-bridge-ssl ssl ca-certificate 'cacert'
set load-balancing haproxy backend bk-bridge-ssl server sr01 address '192.0.2.23'
set load-balancing haproxy backend bk-bridge-ssl server sr01 port '443'
```

Balancing with HTTP health checks

This configuration enables HTTP health checks for backend servers.

```
set load-balancing haproxy service my-tcp-api backend 'bk-01'
set load-balancing haproxy service my-tcp-api mode 'tcp'
set load-balancing haproxy service my-tcp-api port '8888'

set load-balancing haproxy backend bk-01 balance 'round-robin'
set load-balancing haproxy backend bk-01 mode 'tcp'

set load-balancing haproxy backend bk-01 http-check method 'get'
set load-balancing haproxy backend bk-01 http-check uri '/health'
set load-balancing haproxy backend bk-01 http-check expect 'status 200'

set load-balancing haproxy backend bk-01 server srv01 address '192.0.2.11'
set load-balancing haproxy backend bk-01 server srv01 port '8881'
set load-balancing haproxy backend bk-01 server srv01 check
set load-balancing haproxy backend bk-01 server srv02 address '192.0.2.12'
set load-balancing haproxy backend bk-01 server srv02 port '8882'
set load-balancing haproxy backend bk-01 server srv02 check port '8892'
```

7.6 NAT

7.6.1 NAT44

NAT (Network Address Translation) is a common method of remapping one IP address space into another by modifying network address information in the IP header of packets while they are in transit across a traffic routing device. The technique was originally used as a shortcut to avoid the need to readdress every host when a network was moved. It has become a popular and essential tool in conserving global address space in the face of IPv4 address exhaustion. One Internet-routable IP address of a NAT gateway can be used for an entire private network.

IP masquerading is a technique that hides an entire IP address space, usually consisting of private IP addresses, behind a single IP address in another, usually public address space. The hidden addresses are changed into a single (public) IP address as the source address of the outgoing IP packets so they appear as originating not from the hidden host but from the routing device itself. Because of the popularity of this technique to conserve IPv4 address space, the term NAT has become virtually synonymous with IP masquerading.

As network address translation modifies the IP address information in packets, NAT implementations may vary in their specific behavior in various addressing cases and their effect on network traffic. The specifics of NAT behavior are not commonly documented by vendors of equipment containing NAT implementations.

The computers on an internal network can use any of the addresses set aside by the IANA (Internet Assigned Numbers Authority) for private addressing (see [RFC 1918](#)). These reserved IP addresses are not in use on the Internet, so an external machine will not directly route to them. The following addresses are reserved for private use:

- 10.0.0.0 to 10.255.255.255 (CIDR: 10.0.0.0/8)
- 172.16.0.0 to 172.31.255.255 (CIDR: 172.16.0.0/12)
- 192.168.0.0 to 192.168.255.255 (CIDR: 192.168.0.0/16)

If an ISP deploys a CGN (Carrier-grade NAT), and uses [RFC 1918](#) address space to number customer gateways, the risk of address collision, and therefore routing failures, arises when the customer network already uses an [RFC 1918](#) address space.

This prompted some ISPs to develop a policy within the ARIN (American Registry for Internet Numbers) to allocate new private address space for CGNs, but ARIN deferred to the IETF before implementing the policy indicating that the matter was not a typical allocation issue but a reservation of addresses for technical purposes (per [RFC 2860](#)).

IETF published [RFC 6598](#), detailing a shared address space for use in ISP CGN deployments that can handle the same network prefixes occurring both on inbound and outbound interfaces. ARIN returned address space to the IANA for this allocation.

The allocated address block is 100.64.0.0/10.

Devices evaluating whether an IPv4 address is public must be updated to recognize the new address space. Allocating more private IPv4 address space for NAT devices might prolong the transition to IPv6.

Overview

Different NAT Types

SNAT

SNAT (Source Network Address Translation) is the most common form of NAT and is typically referred to simply as NAT. To be more correct, what most people refer to as NAT is actually the process of PAT (Port Address Translation), or NAT overload. SNAT is typically used by internal users/private hosts to access the Internet - the source address is translated and thus kept private.

DNAT

DNAT (Destination Network Address Translation) changes the destination address of packets passing through the router, while *SNAT* changes the source address of packets. DNAT is typically used when an external (public) host needs to initiate a session with an internal (private) host. A customer needs to access a private service behind the routers public IP. A connection is established with the routers public IP address on a well known port and thus all traffic for this port is rewritten to address the internal (private) host.

Bidirectional NAT

This is a common scenario where both *SNAT* and *DNAT* are configured at the same time. It's commonly used when internal (private) hosts need to establish a connection with external resources and external systems need to access internal (private) resources.

NAT, Routing, Firewall Interaction

There is a very nice picture/explanation in the Vyatta documentation which should be rewritten here.

NAT Ruleset

NAT is configured entirely on a series of so called *rules*. Rules are numbered and evaluated by the underlying OS in numerical order! The rule numbers can be changes by utilizing the `rename` and `copy` commands.

Note: Changes to the NAT system only affect newly established connections. Already established connections are not affected.

Hint: When designing your NAT ruleset leave some space between consecutive rules for later extension. Your ruleset could start with numbers 10, 20, 30. You thus can later extend the ruleset and place new rules between existing ones.

Rules will be created for both *SNAT* and *DNAT*.

For *Bidirectional NAT* a rule for both *SNAT* and *DNAT* needs to be created.

Traffic Filters

Traffic Filters are used to control which packets will have the defined NAT rules applied. Five different filters can be applied within a NAT rule.

- **outbound-interface** - applicable only to *SNAT*. It configures the interface which is used for the outside traffic that this translation rule applies to. Interface groups, inverted selection and wildcard, are also supported.

Examples:

```
set nat source rule 20 outbound-interface name eth0
set nat source rule 30 outbound-interface name bond1*
set nat source rule 20 outbound-interface name !vtun2
set nat source rule 20 outbound-interface group GROUP1
set nat source rule 20 outbound-interface group !GROUP2
```

- **inbound-interface** - applicable only to *DNAT*. It configures the interface which is used for the inside traffic the translation rule applies to. Interface groups, inverted selection and wildcard, are also supported.

Example:

```
set nat destination rule 20 inbound-interface name eth0
set nat destination rule 30 inbound-interface name bond1*
set nat destination rule 20 inbound-interface name !vtun2
set nat destination rule 20 inbound-interface group GROUP1
set nat destination rule 20 inbound-interface group !GROUP2
```

- **protocol** - specify which types of protocols this translation rule applies to. Only packets matching the specified protocol are NATed. By default this applies to *all* protocols.

Example:

- Set SNAT rule 20 to only NAT TCP and UDP packets
- Set DNAT rule 20 to only NAT UDP packets

```
set nat source rule 20 protocol tcp_udp
set nat destination rule 20 protocol udp
```

- **source** - specifies which packets the NAT translation rule applies to based on the packets source IP address and/or source port. Only matching packets are considered for NAT.

Example:

- Set SNAT rule 20 to only NAT packets arriving from the 192.0.2.0/24 network
- Set SNAT rule 30 to only NAT packets arriving from the 203.0.113.0/24 network with a source port of 80 and 443

```
set nat source rule 20 source address 192.0.2.0/24
set nat source rule 30 source address 203.0.113.0/24
set nat source rule 30 source port 80,443
```

- **destination** - specify which packets the translation will be applied to, only based on the destination address and/or port number configured.

Note: If no destination is specified the rule will match on any destination address and port.

Example:

- Configure SNAT rule (40) to only NAT packets with a destination address of 192.0.2.1.

```
set nat source rule 40 destination address 192.0.2.1
```

Address Conversion

Every NAT rule has a translation command defined. The address defined for the translation is the address used when the address information in a packet is replaced.

Source Address

For *SNAT* rules the packets source address will be replaced with the address specified in the translation command. A port translation can also be specified and is part of the translation address.

Note: The translation address must be set to one of the available addresses on the configured *outbound-interface* or it must be set to *masquerade* which will use the primary IP address of the *outbound-interface* as its translation address.

Note: When using NAT for a large number of host systems it recommended that a minimum of 1 IP address is used to NAT every 256 private host systems. This is due to the limit of 65,000 port numbers available for unique translations and a reserving an average of 200-300 sessions per host system.

Example:

- Define a discrete source IP address of 100.64.0.1 for SNAT rule 20
- Use address *masquerade* (the interfaces primary address) on rule 30
- For a large amount of private machines behind the NAT your address pool might to be bigger. Use any address in the range 100.64.0.10 - 100.64.0.20 on SNAT rule 40 when doing the translation

```
set nat source rule 20 translation address 100.64.0.1
set nat source rule 30 translation address 'masquerade'
set nat source rule 40 translation address 100.64.0.10-100.64.0.20
```

Destination Address

For *DNAT* rules the packets destination address will be replaced by the specified address in the *translation address* command.

Example:

- DNAT rule 10 replaces the destination address of an inbound packet with 192.0.2.10

```
set nat destination rule 10 translation address 192.0.2.10
```

Also, in *DNAT*, redirection to localhost is supported. The redirect statement is a special form of dn timer which always translates the destination address to the local host's one.

Example of redirection:

```
set nat destination rule 10 translation redirect port 22
```

NAT Load Balance

Advanced configuration can be used in order to apply source or destination NAT, and within a single rule, be able to define multiple translated addresses, so NAT balances the translations among them.

NAT Load Balance uses an algorithm that generates a hash and based on it, then it applies corresponding translation. This hash can be generated randomly, or can use data from the ip header: source-address, destination-address, source-port and/or destination-port. By default, it will generate the hash randomly.

When defining the translated address, called **backends**, a **weight** must be configured. This lets the user define load balance distribution according to their needs. The sum of all the weights defined for the backends should be equal to 100. In other words, the weight defined for the backend is the percentage of the connections that will receive such backend.

```
set nat [source | destination] rule <rule> load-balance hash [source-address |
destination-address | source-port | destination-port | random]
set nat [source | destination] rule <rule> load-balance backend <x.x.x.x> weight <1-100>
```

Configuration Examples

To setup SNAT, we need to know:

- The internal IP addresses we want to translate
- The outgoing interface to perform the translation on
- The external IP address to translate to

In the example used for the Quick Start configuration above, we demonstrate the following configuration:

```
set nat source rule 100 outbound-interface name 'eth0'
set nat source rule 100 source address '192.168.0.0/24'
set nat source rule 100 translation address 'masquerade'
```

Which generates the following configuration:

```
rule 100 {
    outbound-interface {
        name eth0
    }
    source {
        address 192.168.0.0/24
    }
    translation {
        address masquerade
    }
}
```

In this example, we use **masquerade** as the translation address instead of an IP address. The **masquerade** target is effectively an alias to say “use whatever IP address is on the outgoing interface”, rather than a statically configured IP address. This is useful if you use DHCP for your outgoing interface and do not know what the external address will be.

When using NAT for a large number of host systems it recommended that a minimum of 1 IP address is used to NAT every 256 host systems. This is due to the limit of 65,000 port numbers available for unique translations and a reserving an average of 200-300 sessions per host system.

Example: For an ~8,000 host network a source NAT pool of 32 IP addresses is recommended.

A pool of addresses can be defined by using a hyphen between two IP addresses:

```
set nat source rule 100 translation address '203.0.113.32-203.0.113.63'
```

Avoiding “leaky” NAT

Linux netfilter will not NAT traffic marked as INVALID. This often confuses people into thinking that Linux (or specifically VyOS) has a broken NAT implementation because non-NATed traffic is seen leaving an external interface. This is actually working as intended, and a packet capture of the “leaky” traffic should reveal that the traffic is either an additional TCP “RST”, “FIN,ACK”, or “RST,ACK” sent by client systems after Linux netfilter considers the connection closed. The most common is the additional TCP RST some host implementations send after terminating a connection (which is implementation-specific).

In other words, connection tracking has already observed the connection be closed and has transition the flow to INVALID to prevent attacks from attempting to reuse the connection.

You can avoid the “leaky” behavior by using a firewall policy that drops “invalid” state packets.

Having control over the matching of INVALID state traffic, e.g. the ability to selectively log, is an important troubleshooting tool for observing broken protocol behavior. For this reason, VyOS does not globally drop invalid state traffic, instead allowing the operator to make the determination on how the traffic is handled.

Hairpin NAT/NAT Reflection

A typical problem with using NAT and hosting public servers is the ability for internal systems to reach an internal server using its external IP address. The solution to this is usually the use of split-DNS to correctly point host systems to the internal address when requests are made internally. Because many smaller networks lack DNS infrastructure, a work-around is commonly deployed to facilitate the traffic by NATing the request from internal hosts to the source address of the internal interface on the firewall.

This technique is commonly referred to as NAT Reflection or Hairpin NAT.

Example:

- Redirect Microsoft RDP traffic from the outside (WAN, external) world via *DNAT* in rule 100 to the internal, private host 192.0.2.40.
- Redirect Microsoft RDP traffic from the internal (LAN, private) network via *DNAT* in rule 110 to the internal, private host 192.0.2.40. We also need a *SNAT* rule 110 for the reverse path of the traffic. The internal network 192.0.2.0/24 is reachable via interface *eth0.10*.

```
set nat destination rule 100 description 'Regular destination NAT from external'
set nat destination rule 100 destination port '3389'
set nat destination rule 100 inbound-interface name 'pppoe0'
set nat destination rule 100 protocol 'tcp'
set nat destination rule 100 translation address '192.0.2.40'

set nat destination rule 110 description 'NAT Reflection: INSIDE'
set nat destination rule 110 destination port '3389'
set nat destination rule 110 inbound-interface name 'eth0.10'
set nat destination rule 110 protocol 'tcp'
set nat destination rule 110 translation address '192.0.2.40'

set nat source rule 110 description 'NAT Reflection: INSIDE'
set nat source rule 110 destination address '192.0.2.0/24'
```

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```
set nat source rule 110 outbound-interface name 'eth0.10'
set nat source rule 110 protocol 'tcp'
set nat source rule 110 source address '192.0.2.0/24'
set nat source rule 110 translation address 'masquerade'
```

Which results in a configuration of:

```
vyos@vyos# show nat
destination {
  rule 100 {
    description "Regular destination NAT from external"
    destination {
      port 3389
    }
    inbound-interface {
      name pppoe0
    }
    protocol tcp
    translation {
      address 192.0.2.40
    }
  }
  rule 110 {
    description "NAT Reflection: INSIDE"
    destination {
      port 3389
    }
    inbound-interface {
      name eth0.10
    }
    protocol tcp
    translation {
      address 192.0.2.40
    }
  }
}
source {
  rule 110 {
    description "NAT Reflection: INSIDE"
    destination {
      address 192.0.2.0/24
    }
    outbound-interface {
      name eth0.10
    }
    protocol tcp
    source {
      address 192.0.2.0/24
    }
    translation {
      address masquerade
    }
  }
}
```

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```
}
}
```

Destination NAT

DNAT is typically referred to as a **Port Forward**. When using VyOS as a NAT router and firewall, a common configuration task is to redirect incoming traffic to a system behind the firewall.

In this example, we will be using the example Quick Start configuration above as a starting point.

To setup a destination NAT rule we need to gather:

- The interface traffic will be coming in on;
- The protocol and port we wish to forward;
- The IP address of the internal system we wish to forward traffic to.

In our example, we will be forwarding web server traffic to an internal web server on 192.168.0.100. HTTP traffic makes use of the TCP protocol on port 80. For other common port numbers, see: https://en.wikipedia.org/wiki/List_of_TCP_and_UDP_port_numbers

Our configuration commands would be:

```
set nat destination rule 10 description 'Port Forward: HTTP to 192.168.0.100'
set nat destination rule 10 destination port '80'
set nat destination rule 10 inbound-interface name 'eth0'
set nat destination rule 10 protocol 'tcp'
set nat destination rule 10 translation address '192.168.0.100'
```

Which would generate the following NAT destination configuration:

```
nat {
    destination {
        rule 10 {
            description "Port Forward: HTTP to 192.168.0.100"
            destination {
                port 80
            }
            inbound-interface {
                name eth0
            }
            protocol tcp
            translation {
                address 192.168.0.100
            }
        }
    }
}
```

Note: If forwarding traffic to a different port than it is arriving on, you may also configure the translation port using *set nat destination rule [n] translation port*.

This establishes our Port Forward rule, but if we created a firewall policy it will likely block the traffic.

Firewall rules for Destination NAT

It is important to note that when creating firewall rules, the DNAT translation occurs **before** traffic traverses the firewall. In other words, the destination address has already been translated to 192.168.0.100.

So in our firewall ruleset, we want to allow traffic which previously matched a destination nat rule. In order to avoid creating many rules, one for each destination nat rule, we can accept all **'dnat'** connections with one simple rule, using **connection-status** matcher:

```
set firewall ipv4 forward filter rule 10 action accept
set firewall ipv4 forward filter rule 10 connection-status nat destination
set firewall ipv4 forward filter rule 10 state new
```

This would generate the following configuration:

```
ipv4 {
  forward {
    filter {
      rule 10 {
        action accept
        connection-status {
          nat destination
        }
        state new
      }
    }
  }
}
```

1-to-1 NAT

Another term often used for DNAT is **1-to-1 NAT**. For a 1-to-1 NAT configuration, both DNAT and SNAT are used to NAT all traffic from an external IP address to an internal IP address and vice-versa.

Typically, a 1-to-1 NAT rule omits the destination port (all ports) and replaces the protocol with either **all** or **ip**.

Then a corresponding SNAT rule is created to NAT outgoing traffic for the internal IP to a reserved external IP. This dedicates an external IP address to an internal IP address and is useful for protocols which don't have the notion of ports, such as GRE.

Here's an extract of a simple 1-to-1 NAT configuration with one internal and one external interface:

```
set interfaces ethernet eth0 address '192.168.1.1/24'
set interfaces ethernet eth0 description 'Inside interface'
set interfaces ethernet eth1 address '192.0.2.30/24'
set interfaces ethernet eth1 description 'Outside interface'
set nat destination rule 2000 description '1-to-1 NAT example'
set nat destination rule 2000 destination address '192.0.2.30'
set nat destination rule 2000 inbound-interface name 'eth1'
set nat destination rule 2000 translation address '192.168.1.10'
set nat source rule 2000 description '1-to-1 NAT example'
set nat source rule 2000 outbound-interface name 'eth1'
set nat source rule 2000 source address '192.168.1.10'
set nat source rule 2000 translation address '192.0.2.30'
```

Firewall rules are written as normal, using the internal IP address as the source of outbound rules and the destination of inbound rules.

NAT before VPN

Some application service providers (ASPs) operate a VPN gateway to provide access to their internal resources, and require that a connecting organisation translate all traffic to the service provider network to a source address provided by the ASP.

Load Balance

Here we provide two examples on how to apply NAT Load Balance.

First scenario: apply destination NAT for all HTTP traffic coming through interface eth0, and user 4 backends. First backend should received 30% of the request, second backend should get 20%, third 15% and the fourth 35%. We will use source and destination address for hash generation.

```
set nat destination rule 10 inbound-interface name eth0
set nat destination rule 10 protocol tcp
set nat destination rule 10 destination port 80
set nat destination rule 10 load-balance hash source-address
set nat destination rule 10 load-balance hash destination-address
set nat destination rule 10 load-balance backend 198.51.100.101 weight 30
set nat destination rule 10 load-balance backend 198.51.100.102 weight 20
set nat destination rule 10 load-balance backend 198.51.100.103 weight 15
set nat destination rule 10 load-balance backend 198.51.100.104 weight 35
```

Second scenario: apply source NAT for all outgoing connections from LAN 10.0.0.0/8, using 3 public addresses and equal distribution. We will generate the hash randomly.

```
set nat source rule 10 outbound-interface name eth0
set nat source rule 10 source address 10.0.0.0/8
set nat source rule 10 load-balance hash random
set nat source rule 10 load-balance backend 192.0.2.251 weight 33
set nat source rule 10 load-balance backend 192.0.2.252 weight 33
set nat source rule 10 load-balance backend 192.0.2.253 weight 34
```

Example Network

Here's one example of a network environment for an ASP. The ASP requests that all connections from this company should come from 172.29.41.89 - an address that is assigned by the ASP and not in use at the customer site.

Fig. 1: NAT before VPN Topology

Configuration

The required configuration can be broken down into 4 major pieces:

- A dummy interface for the provider-assigned IP;
- NAT (specifically, Source NAT);
- IPSec IKE and ESP Groups;
- IPSec VPN tunnels.

Dummy interface

The dummy interface allows us to have an equivalent of the Cisco IOS Loopback interface - a router-internal interface we can use for IP addresses the router must know about, but which are not actually assigned to a real network.

We only need a single step for this interface:

```
set interfaces dummy dum0 address '172.29.41.89/32'
```

NAT Configuration

```
set nat source rule 110 description 'Internal to ASP'
set nat source rule 110 destination address '172.27.1.0/24'
set nat source rule 110 source address '192.168.43.0/24'
set nat source rule 110 translation address '172.29.41.89'
set nat source rule 120 description 'Internal to ASP'
set nat source rule 120 destination address '10.125.0.0/16'
set nat source rule 120 source address '192.168.43.0/24'
set nat source rule 120 translation address '172.29.41.89'
```

IPSec IKE and ESP

The ASP has documented their IPSec requirements:

- IKE Phase:
 - aes256 Encryption
 - sha256 Hashes
- ESP Phase:
 - aes256 Encryption
 - sha256 Hashes
 - DH Group 14

Additionally, we want to use VPNs only on our eth1 interface (the external interface in the image above)

```
set vpn ipsec ike-group my-ike key-exchange 'ikev1'
set vpn ipsec ike-group my-ike lifetime '7800'
set vpn ipsec ike-group my-ike proposal 1 dh-group '14'
```

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```

set vpn ipsec ike-group my-ike proposal 1 encryption 'aes256'
set vpn ipsec ike-group my-ike proposal 1 hash 'sha256'

set vpn ipsec esp-group my-esp lifetime '3600'
set vpn ipsec esp-group my-esp mode 'tunnel'
set vpn ipsec esp-group my-esp pfs 'disable'
set vpn ipsec esp-group my-esp proposal 1 encryption 'aes256'
set vpn ipsec esp-group my-esp proposal 1 hash 'sha256'

set vpn ipsec interface 'eth1'

```

IPSec VPN Tunnels

We'll use the IKE and ESP groups created above for this VPN. Because we need access to 2 different subnets on the far side, we will need two different tunnels. If you changed the names of the ESP group and IKE group in the previous step, make sure you use the correct names here too.

```

set vpn ipsec authentication psk vyos id '203.0.113.46'
set vpn ipsec authentication psk vyos id '198.51.100.243'
set vpn ipsec authentication psk vyos secret 'MYSECRETPASSWORD'
set vpn ipsec site-to-site peer branch authentication local-id '203.0.113.46'
set vpn ipsec site-to-site peer branch authentication mode 'pre-shared-secret'
set vpn ipsec site-to-site peer branch authentication remote-id '198.51.100.243'
set vpn ipsec site-to-site peer branch connection-type 'initiate'
set vpn ipsec site-to-site peer branch default-esp-group 'my-esp'
set vpn ipsec site-to-site peer branch ike-group 'my-ike'
set vpn ipsec site-to-site peer branch ikev2-reauth 'inherit'
set vpn ipsec site-to-site peer branch local-address '203.0.113.46'
set vpn ipsec site-to-site peer branch remote-address '198.51.100.243'
set vpn ipsec site-to-site peer branch tunnel 0 local prefix '172.29.41.89/32'
set vpn ipsec site-to-site peer branch tunnel 0 remote prefix '172.27.1.0/24'
set vpn ipsec site-to-site peer branch tunnel 1 local prefix '172.29.41.89/32'
set vpn ipsec site-to-site peer branch tunnel 1 remote prefix '10.125.0.0/16'

```

Testing and Validation

If you've completed all the above steps you no doubt want to see if it's all working.

Start by checking for IPSec SAs (Security Associations) with:

```
$ show vpn ipsec sa
```

Peer ID / IP	Local ID / IP								
-----	-----								
198.51.100.243	203.0.113.46								
Tunnel	State	Bytes Out/In	Encrypt	Hash	NAT-T	A-Time	L-Time	Proto	
-----	-----	-----	-----	-----	-----	-----	-----	-----	
0	up	0.0/0.0	aes256	sha256	no	1647	3600	all	
1	up	0.0/0.0	aes256	sha256	no	865	3600	all	

That looks good - we defined 2 tunnels and they're both up and running.

7.6.2 NAT64

NAT64 (IPv6-to-IPv4 Prefix Translation) is a critical component in modern networking, facilitating communication between IPv6 and IPv4 networks. This documentation outlines the setup, configuration, and usage of the NAT64 feature in your project. Whether you are transitioning to IPv6 or need to seamlessly connect IPv4 and IPv6 devices. NAT64 is a stateful translation mechanism that translates IPv6 addresses to IPv4 addresses and IPv4 addresses to IPv6 addresses. NAT64 is used to enable IPv6-only clients to contact IPv4 servers using unicast UDP, TCP, or ICMP.

Overview

Different NAT Types

SNAT64

SNAT64 (IPv6-to-IPv4 Source Address Translation) is a stateful translation mechanism that translates IPv6 addresses to IPv4 addresses.

64:ff9b::/96 is the well-known prefix for IPv4-embedded IPv6 addresses. The prefix is used to represent IPv4 addresses in an IPv6 address format. The IPv4 address is encoded in the low-order 32 bits of the IPv6 address. The high-order 32 bits are set to the well-known prefix 64:ff9b::/96.

Configuration Examples

The following examples show how to configure NAT64 on a VyOS router. The 192.0.2.10 address is used as the IPv4 address for the translation pool.

NAT64 server configuration:

```
set interfaces ethernet eth0 address '192.0.2.1/24'
set interfaces ethernet eth0 address '192.0.2.10/24'
set interfaces ethernet eth0 description 'WAN'
set interfaces ethernet eth1 address '2001:db8::1/64'
set interfaces ethernet eth1 description 'LAN'

set service dns forwarding allow-from '2001:db8::/64'
set service dns forwarding dns64-prefix '64:ff9b::/96'
set service dns forwarding listen-address '2001:db8::1'

set nat64 source rule 100 source prefix '64:ff9b::/96'
set nat64 source rule 100 translation pool 10 address '192.0.2.10'
set nat64 source rule 100 translation pool 10 port '1-65535'
```

NAT64 client configuration:

```
set interfaces ethernet eth1 address '2001:db8::2/64'
set protocols static route6 64:ff9b::/96 next-hop 2001:db8::1
set system name-server '2001:db8::1'
```

Test from the IPv6 only client:


```
vyos@r1:~$ ping 64:ff9b::192.0.2.1 count 2
PING 64:ff9b::192.0.2.1(64:ff9b::c000:201) 56 data bytes
64 bytes from 64:ff9b::c000:201: icmp_seq=1 ttl=63 time=0.351 ms
64 bytes from 64:ff9b::c000:201: icmp_seq=2 ttl=63 time=0.373 ms

--- 64:ff9b::192.0.2.1 ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1023ms
rtt min/avg/max/mdev = 0.351/0.362/0.373/0.011 ms
```

7.6.3 NAT66(NPTv6)

Todo: Convert raw command blocks in this file to cfgcmd/opcmd directives for command coverage tracking.

NPTv6 (IPv6-to-IPv6 Network Prefix Translation) is an address translation technology based on IPv6 networks, used to convert an IPv6 address prefix in an IPv6 message into another IPv6 address prefix. We call this address translation method NAT66. Devices that support the NAT66 function are called NAT66 devices, which can provide NAT66 source and destination address translation functions.

Overview

Different NAT Types

SNAT66

SNPTv6 (Source IPv6-to-IPv6 Network Prefix Translation) The conversion function is mainly used in the following scenarios:

- A single internal network and external network. Use the NAT66 device to connect a single internal network and public network, and the hosts in the internal network use IPv6 address prefixes that only support routing within the local range. When a host in the internal network accesses the external network, the source IPv6 address prefix in the message will be converted into a global unicast IPv6 address prefix by the NAT66 device.
- Redundancy and load sharing. There are multiple NAT66 devices at the edge of an IPv6 network to another IPv6 network. The path through the NAT66 device to another IPv6 network forms an equivalent route, and traffic can be load-shared on these NAT66 devices. In this case, you can configure the same source address translation rules on these NAT66 devices, so that any NAT66 device can handle IPv6 traffic between different sites.
- Multi-homed. In a multi-homed network environment, the NAT66 device connects to an internal network and simultaneously connects to different external networks. Address translation can be configured on each external network side interface of the NAT66 device to convert the same internal network address into different external network addresses, and realize the mapping of the same internal address to multiple external addresses.

DNAT66

The DNPTv6 (Destination IPv6-to-IPv6 Network Prefix Translation) destination address translation function is used in scenarios where the server in the internal network provides services to the external network, such as providing Web services or FTP services to the external network. By configuring the mapping relationship between the internal server address and the external network address on the external network side interface of the NAT66 device, external network users can access the internal network server through the designated external network address.

Prefix Conversion

Source Prefix

Every SNAT66 rule has a translation command defined. The prefix defined for the translation is the prefix used when the address information in a packet is replaced.

The *SNAT66* rule replaces the source address of the packet and calculates the converted address using the prefix specified in the rule.

Example:

- Convert the address prefix of a single `fc01::/64` network to `fc00::/64`
- Output from `eth0` network interface

```
set nat66 source rule 1 outbound-interface name 'eth0'
set nat66 source rule 1 source prefix 'fc01::/64'
set nat66 source rule 1 translation address 'fc00::/64'
```

Destination Prefix

For the *DNAT66* rule, the destination address of the packet is replaced by the address calculated from the specified address or prefix in the `translation address` command

Example:

- Convert the address prefix of a single `fc00::/64` network to `fc01::/64`
- Input from `eth0` network interface

```
set nat66 destination rule 1 inbound-interface name 'eth0'
set nat66 destination rule 1 destination address 'fc00::/64'
set nat66 destination rule 1 translation address 'fc01::/64'
```

For the destination, groups can also be used instead of an address.

Example:

```
set firewall group ipv6-address-group ADR-INSIDE-v6 address fc00::1

set nat66 destination rule 1 inbound-interface name 'eth0'
set nat66 destination rule 1 destination group address-group ADR-INSIDE-v6
set nat66 destination rule 1 translation address 'fc01::/64'
```

Configuration Examples

Use the following topology to build a nat66 based isolated network between internal and external networks (dynamic prefix is not supported):

R1:

```
set interfaces ethernet eth0 ipv6 address autoconf
set interfaces ethernet eth1 address 'fc01::1/64'
set nat66 destination rule 1 destination address 'fc00:470:f1cd:101::/64'
set nat66 destination rule 1 inbound-interface name 'eth0'
set nat66 destination rule 1 translation address 'fc01::/64'
set nat66 source rule 1 outbound-interface name 'eth0'
set nat66 source rule 1 source prefix 'fc01::/64'
set nat66 source rule 1 translation address 'fc00:470:f1cd:101::/64'
```

R2:

```
set interfaces bridge br1 address 'fc01::2/64'
set interfaces bridge br1 member interface eth0
set interfaces bridge br1 member interface eth1
set protocols static route6 ::/0 next-hop fc01::1
set service router-advert interface br1 prefix ::/0
```

Use the following topology to translate internal user local addresses (fc::/7) to DHCPv6-PD provided prefixes from an ISP connected to a VyOS HA pair.

Configure both routers (a and b) for DHCPv6-PD via dummy interface:

```
set interfaces dummy dum1 description 'DHCPv6-PD NPT dummy'
set interfaces bonding bond0 vif 20 dhcpv6-options pd 0 interface dum1 address '0'
set interfaces bonding bond0 vif 20 dhcpv6-options pd 1 interface dum1 address '0'
set interfaces bonding bond0 vif 20 dhcpv6-options pd 2 interface dum1 address '0'
set interfaces bonding bond0 vif 20 dhcpv6-options pd 3 interface dum1 address '0'
set interfaces bonding bond0 vif 20 dhcpv6-options rapid-commit
commit
```

Get the DHCPv6-PD prefixes from both routers:

```
trae@cr01a-vyos# run show interfaces dummy dum1 br
Codes: S - State, L - Link, u - Up, D - Down, A - Admin Down
Interface      IP Address      S/L  Description
-----
dum1           2001:db8:123:b008::/64    u/u  DHCPv6-PD NPT dummy
              2001:db8:123:b00a::/64
              2001:db8:123:b00b::/64
              2001:db8:123:b009::/64
```

```
trae@cr01b-vyos# run show int dummy dum1 brief
Codes: S - State, L - Link, u - Up, D - Down, A - Admin Down
Interface      IP Address      S/L  Description
-----
              -----
              -----
```

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```
dum1          2001:db8:123:b00d::/64          u/u  DHCPv6-PD NPT dummy
              2001:db8:123:b00c::/64
              2001:db8:123:b00e::/64
              2001:db8:123:b00f::/64
```

Configure the A-side router for NPTv6 using the prefixes above:

```
set nat66 source rule 10 description 'NPT to VLAN 10'
set nat66 source rule 10 outbound-interface name 'bond0.20'
set nat66 source rule 10 source prefix 'fd52:d62e:8011:a::/64'
set nat66 source rule 10 translation address '2001:db8:123:b008::/64'
set nat66 source rule 20 description 'NPT to VLAN 70'
set nat66 source rule 20 outbound-interface name 'bond0.20'
set nat66 source rule 20 source prefix 'fd52:d62e:8011:46::/64'
set nat66 source rule 20 translation address '2001:db8:123:b009::/64'
set nat66 source rule 30 description 'NPT to VLAN 200'
set nat66 source rule 30 outbound-interface name 'bond0.20'
set nat66 source rule 30 source prefix 'fd52:d62e:8011:c8::/64'
set nat66 source rule 30 translation address '2001:db8:123:b00a::/64'
set nat66 source rule 40 description 'NPT to VLAN 240'
set nat66 source rule 40 outbound-interface name 'bond0.20'
set nat66 source rule 40 source prefix 'fd52:d62e:8011:f0::/64'
set nat66 source rule 40 translation address '2001:db8:123:b00b::/64'
commit
```

Configure the B-side router for NPTv6 using the prefixes above:

```
set nat66 source rule 10 description 'NPT to VLAN 10'
set nat66 source rule 10 outbound-interface name 'bond0.20'
set nat66 source rule 10 source prefix 'fd52:d62e:8011:a::/64'
set nat66 source rule 10 translation address '2001:db8:123:b00c::/64'
set nat66 source rule 20 description 'NPT to VLAN 70'
set nat66 source rule 20 outbound-interface name 'bond0.20'
set nat66 source rule 20 source prefix 'fd52:d62e:8011:46::/64'
set nat66 source rule 20 translation address '2001:db8:123:b00d::/64'
set nat66 source rule 30 description 'NPT to VLAN 200'
set nat66 source rule 30 outbound-interface name 'bond0.20'
set nat66 source rule 30 source prefix 'fd52:d62e:8011:c8::/64'
set nat66 source rule 30 translation address '2001:db8:123:b00e::/64'
set nat66 source rule 40 description 'NPT to VLAN 240'
set nat66 source rule 40 outbound-interface name 'bond0.20'
set nat66 source rule 40 source prefix 'fd52:d62e:8011:f0::/64'
set nat66 source rule 40 translation address '2001:db8:123:b00f::/64'
commit
```

Verify that connections are hitting the rule on both sides:

```
trae@cr01a-vyos# run show nat66 source statistics
Rule   Packets   Bytes   Interface
-----
10      1         104     bond0.20
20      1         104     bond0.20
30     8093     669445  bond0.20
```

(continues on next page)

(continued from previous page)

40 2446 216912 bond0.20

7.6.4 CGNAT

CGNAT (Carrier-Grade Network Address Translation), also known as Large-Scale NAT (LSN), is a type of network address translation used by Internet Service Providers (ISPs) to enable multiple private IP addresses to share a single public IP address. This technique helps to conserve the limited IPv4 address space. The 100.64.0.0/10 address block is reserved for use in carrier-grade NAT.

Overview

CGNAT works by placing a NAT device within the ISP's network. This device translates private IP addresses from customer networks to a limited pool of public IP addresses assigned to the ISP. This allows many customers to share a smaller number of public IP addresses.

Not all [RFC 6888](#) requirements are implemented in CGNAT.

Implemented the following [RFC 6888](#) requirements:

- **REQ 2:** A CGN must have a default “IP address pooling” behavior of “Paired”. CGN must use the same external IP address mapping for all sessions associated with the same internal IP address, be they TCP, UDP, ICMP, something else, or a mix of different protocols.
- **REQ 3:** The CGN function should not have any limitations on the size or the contiguity of the external address pool.
- **REQ 4:** A CGN must support limiting the number of external ports (or, equivalently, “identifiers” for ICMP) that are assigned per subscriber.

Advantages of CGNAT

- **IPv4 Address Conservation:** CGNAT helps mitigate the exhaustion of IPv4 addresses by allowing multiple customers to share a single public IP address.
- **Scalability:** ISPs can support more customers without needing a proportional increase in public IP addresses.
- **Cost-Effective:** Reduces the cost associated with acquiring additional public IPv4 addresses.

Considerations

- **Traceability Issues:** Since multiple users share the same public IP address, tracking individual users for security and legal purposes can be challenging.
- **Performance Overheads:** The translation process can introduce latency and potential performance bottlenecks, especially under high load.
- **Application Compatibility:** Some applications and protocols may not work well with CGNAT due to their reliance on unique public IP addresses.
- **Port Allocation Limits:** Each public IP address has a limited number of ports, which can be exhausted, affecting the ability to establish new connections.
- **Port Control Protocol:** PCP is not implemented.

Port calculation

When implementing CGNAT, ensuring that there are enough ports allocated per subscriber is critical. Below is a summary based on RFC 6888.

1. Total Ports Available:

- Total Ports: 65536 (0 to 65535)
- Reserved Ports: Assume 1024 ports are reserved for well-known services and administrative purposes.
- Usable Ports: $65536 - 1024 = 64512$

2. Estimate Ports Needed per Subscriber:

- Example: A household might need 1000 ports to ensure smooth operation for multiple devices and applications.

3. Calculate the Number of Subscribers per Public IP:

- Usable Ports / Ports per Subscriber
- $64512 / 1000 = 64$ subscribers per public IP

Configuration

```
set nat cgnat pool external <pool-name> external-port-range <port-range>
```

Set an external port-range for the external pool, the default range is 1024-65535. Multiple entries can be added to the same pool.

```
set nat cgnat pool external <pool-name> per-user-limit port <num>
```

Set external source port limits that will be allocated to each subscriber individually. The default value is 2000.

```
set nat cgnat pool external <pool-name> range [address | address range | network] [seq]
```

Set the range of external IP addresses for the CGNAT pool. The sequence is optional; if set, a lower value means higher priority.

```
set nat cgnat pool internal <pool-name> range [address range | network]
```

Set the range of internal IP addresses for the CGNAT pool.

```
set nat cgnat rule <num> source pool <internal-pool-name>
```

Set the rule for the source pool.

```
set nat cgnat rule <num> translation pool <external-pool-name>
```

Set the rule for the translation pool.

```
set nat cgnat log-allocation
```

Enable logging of IP address and ports allocations.

Configuration Examples

Single external address

Example of setting up a basic CGNAT configuration: In the following example, we define an external pool named `ext-1` with one external IP address.

Each subscriber will be allocated a maximum of 2000 ports from the external pool.

```
set nat cgnat pool external ext1 external-port-range '1024-65535'
set nat cgnat pool external ext1 per-user-limit port '2000'
set nat cgnat pool external ext1 range '192.0.2.222/32'
set nat cgnat pool internal int1 range '100.64.0.0/28'
set nat cgnat rule 10 source pool 'int1'
set nat cgnat rule 10 translation pool 'ext1'
```

Multiple external addresses

```
set nat cgnat pool external ext1 external-port-range '1024-65535'
set nat cgnat pool external ext1 per-user-limit port '8000'
set nat cgnat pool external ext1 range '192.0.2.1-192.0.2.2'
set nat cgnat pool external ext1 range '203.0.113.253-203.0.113.254'
set nat cgnat pool internal int1 range '100.64.0.1-100.64.0.32'
set nat cgnat rule 10 source pool 'int1'
set nat cgnat rule 10 translation pool 'ext1'
```

External address sequences

```
set nat cgnat pool external ext-01 per-user-limit port '16000'
set nat cgnat pool external ext-01 range 203.0.113.1/32 seq '10'
set nat cgnat pool external ext-01 range 192.0.2.1/32 seq '20'
set nat cgnat pool internal int-01 range '100.64.0.0/29'
set nat cgnat rule 10 source pool 'int-01'
set nat cgnat rule 10 translation pool 'ext-01'
```

Operation commands

show nat cgnat allocation

Show address and port allocations

show nat cgnat allocation external-address <address>

Show all allocations for an external IP address

show nat cgnat allocation internal-address <address>

Show all allocations for an internal IP address

Show CGNAT allocations

```
vyos@vyos:~$ show nat cgnat allocation
Internal IP      External IP      Port range
-----
100.64.0.0       203.0.113.1     1024-17023
100.64.0.1       203.0.113.1     17024-33023
100.64.0.2       203.0.113.1     33024-49023
100.64.0.3       203.0.113.1     49024-65023
100.64.0.4       192.0.2.1       1024-17023
100.64.0.5       192.0.2.1       17024-33023
100.64.0.6       192.0.2.1       33024-49023
100.64.0.7       192.0.2.1       49024-65023

vyos@vyos:~$ show nat cgnat allocation internal-address 100.64.0.4
Internal IP      External IP      Port range
-----
100.64.0.4       192.0.2.1       1024-17023
```

Further Reading

- [RFC 6598](#) - IANA-Reserved IPv4 Prefix for Shared Address Space
- [RFC 6888](#) - Requirements for CGNAT

7.7 Policy

Policies are used for filtering and traffic management. With policies, network administrators could filter and treat traffic according to their needs.

There could be a wide range of routing policies. Some examples are listed below:

- Filter traffic based on source/destination address.
- Set some metric to routes learned from a particular neighbor.
- Set some attributes (like AS PATH or Community value) to advertised routes to neighbors.
- Prefer a specific routing protocol routes over another routing protocol running on the same router.

Policies, in VyOS, are implemented using FRR filtering and route maps. Detailed information of FRR could be found in <http://docs.frrouting.org/>

7.7.1 Policy Sections

Access List Policy

Filtering is used for both input and output of the routing information. Once filtering is defined, it can be applied in any direction. VyOS makes filtering possible using acls and prefix lists.

Basic filtering can be done using access-list and access-list6.

Configuration

Access Lists

set policy access-list <acl_number>

This command creates the new access list policy, where <acl_number> must be a number from 1 to 2699.

set policy access-list <acl_number> description <text>

Set description for the access list.

set policy access-list <acl_number> rule <1-65535> action <permit|deny>

This command creates a new rule in the access list and defines an action.

**set policy access-list <acl_number> rule <1-65535> <destination|source>
<any|host|inverse-mask|network>**

This command defines matching parameters for access list rule. Matching criteria could be applied to destination or source parameters:

- any: any IP address to match.
- host: single host IP address to match.
- inverse-match: network/netmask to match (requires network be defined).
- network: network/netmask to match (requires inverse-match be defined).

IPv6 Access List

Basic filtering could also be applied to IPv6 traffic.

set policy access-list6 <text>

This command creates the new IPv6 access list, identified by <text>

set policy access-list6 <text> description <text>

Set description for the IPv6 access list.

set policy access-list6 <text> rule <1-65535> action <permit|deny>

This command creates a new rule in the IPv6 access list and defines an action.

set policy access-list6 <text> rule <1-65535> source <any|exact-match|network>

This command defines matching parameters for IPv6 access list rule. Matching criteria could be applied to source parameters:

- any: any IPv6 address to match.
- exact-match: exact match of the network prefixes.
- network: network/netmask to match (requires inverse-match be defined) BUG, NO invert-match option in access-list6

Prefix List Policy

Prefix lists provides the most powerful prefix based filtering mechanism. In addition to access-list functionality, ip prefix-list has prefix length range specification.

If no ip prefix list is specified, it acts as permit. If ip prefix list is defined, and no match is found, default deny is applied.

Prefix filtering can be done using prefix-list and prefix-list6.

Configuration

IPv4 Prefix Lists (prefix-list)

set policy prefix-list <text>

This command creates the new prefix-list policy, identified by <text>.

set policy prefix-list <text> description <text>

Set description for the prefix-list policy.

set policy prefix-list <text> rule <1-65535> action <permit|deny>

This command creates a new rule in the prefix-list and defines an action.

set policy prefix-list <text> rule <1-65535> description <text>

Set description for rule in the prefix-list.

set policy prefix-list <text> rule <1-65535> prefix <x.x.x.x/x>

Prefix to match against.

set policy prefix-list <text> rule <1-65535> ge <0-32>

Netmask greater than length.

set policy prefix-list <text> rule <1-65535> le <0-32>

Netmask less than length

Example: IPv4 Prefix Lists (prefix-list)

This example creates an IPv4 prefix-list named PL4-EXAMPLE-NAME, defines 3 rules each with 1 prefix, and matches le (less than/equal to) /32.

set policy prefix-list PL4-EXAMPLE-NAME rule 10 action 'permit'

set policy prefix-list PL4-EXAMPLE-NAME rule 10 le '32'

set policy prefix-list PL4-EXAMPLE-NAME rule 10 prefix '192.0.2.0/24'

set policy prefix-list PL4-EXAMPLE-NAME rule 20 action 'permit'

set policy prefix-list PL4-EXAMPLE-NAME rule 20 le '32'

set policy prefix-list PL4-EXAMPLE-NAME rule 20 prefix '198.51.100.0/24'

set policy prefix-list PL4-EXAMPLE-NAME rule 30 action 'permit'

set policy prefix-list PL4-EXAMPLE-NAME rule 30 le '32'

```
set policy prefix-list PL4-EXAMPLE-NAME rule 30 prefix '203.0.113.0/24'
```

IPv6 Prefix Lists (prefix-list6)

```
set policy prefix-list6 <text>
```

This command creates the new IPv6 prefix-list policy, identified by <text>.

```
set policy prefix-list6 <text> description <text>
```

Set description for the IPv6 prefix-list policy.

```
set policy prefix-list6 <text> rule <1-65535> action <permit|deny>
```

This command creates a new rule in the IPv6 prefix-list and defines an action.

```
set policy prefix-list6 <text> rule <1-65535> description <text>
```

Set description for rule in IPv6 prefix-list.

```
set policy prefix-list6 <text> rule <1-65535> prefix <h:h:h:h:h:h:h/x>
```

IPv6 prefix.

```
set policy prefix-list6 <text> rule <1-65535> ge <0-128>
```

Netmask greater than length.

```
set policy prefix-list6 <text> rule <1-65535> le <0-128>
```

Netmask less than length

Example: IPv6 Prefix Lists (prefix-list6)

This example creates an IPv6 prefix-list6 named PL6-EXAMPLE-NAME, defines 3 rules each with 1 prefix, and matches le (less than/equal to) /128.

```
set policy prefix-list6 PL6-EXAMPLE-NAME rule 10 action 'permit'
set policy prefix-list6 PL6-EXAMPLE-NAME rule 10 le '128'
set policy prefix-list6 PL6-EXAMPLE-NAME rule 10 prefix '2001:db8:0:0::/64'
set policy prefix-list6 PL6-EXAMPLE-NAME rule 20 action 'permit'
set policy prefix-list6 PL6-EXAMPLE-NAME rule 20 le '128'
set policy prefix-list6 PL6-EXAMPLE-NAME rule 20 prefix '2001:db8:0:1::/64'
set policy prefix-list6 PL6-EXAMPLE-NAME rule 30 action 'permit'
set policy prefix-list6 PL6-EXAMPLE-NAME rule 30 le '128'
set policy prefix-list6 PL6-EXAMPLE-NAME rule 30 prefix '2001:db8:0:2::/64'
```

Route and Route6 Policy

IPv4 route and IPv6 route policies are defined in this section. These route policies can then be associated to interfaces.

Rule-Sets

A rule-set is a named collection of rules that can be applied to an interface. Each rule is numbered, has an action to apply if the rule is matched, and the ability to specify the criteria to match. Data packets go through the rules from 1 - 999999, at the first match the action of the rule will be executed.

```
set policy route <name> description <text>
```

```
set policy route6 <name> description <text>
```

Provide a rule-set description.

```
set policy route <name> default-log
```

```
set policy route6 <name> default-log
```

Option to log packets hitting default-action.

```
set policy route <name> interface <interface>
```

```
set policy route6 <name> interface <interface>
```

Apply routing policy to interface

```
set policy route <name> rule <n> description <text>
```

```
set policy route6 <name> rule <n> description <text>
```

Provide a description for each rule.

```
set policy route <name> rule <n> log <enable|disable>
```

```
set policy route6 <name> rule <n> log <enable|disable>
```

Option to enable or disable log matching rule.

Matching criteria

There are a lot of matching criteria options available, both for policy route and policy route6. These options are listed in this section.

```
set policy route <name> rule <n> connection-mark <1-2147483647>
```

```
set policy route6 <name> rule <n> connection-mark <1-2147483647>
```

Set match criteria based on connection mark.

```
set policy route <name> rule <n> mark <match_criteria>
```

```
set policy route6 <name> rule <n> mark <match_criteria>
```

Match based on the firewall mark (fwmark), where <match_criteria> can be:

- <0-2147483647> a single fwmark
- !<0-2147483647> everything except a single fwmark
- <start-end> a range of marks

- `!<start-end>` everything except the range of marks

Note: When using the `set table` or `set vrf` commands the mark settings are ignored and overwritten with a table-specific mark that is set to `0x7FFFFFFF` - the id of the table/VRF.

```
set policy route <name> rule <n> source address <match_criteria>
set policy route <name> rule <n> destination address <match_criteria>
set policy route6 <name> rule <n> source address <match_criteria>
set policy route6 <name> rule <n> destination address <match_criteria>
```

Set match criteria based on source or destination ipv4|ipv6 address, where `<match_criteria>` could be:

For ipv4:

- `<x.x.x.x>`: IP address to match.
- `<x.x.x.x/x>`: Subnet to match.
- `<x.x.x.x>-<x.x.x.x>`: IP range to match.
- `!<x.x.x.x>`: Match everything except the specified address.
- `!<x.x.x.x/x>`: Match everything except the specified subnet.
- `!<x.x.x.x>-<x.x.x.x>`: Match everything except the specified range.

And for ipv6:

- `<h:h:h:h:h:h:h>`: IPv6 address to match.
- `<h:h:h:h:h:h:h/x>`: IPv6 prefix to match.
- `<h:h:h:h:h:h:h>-<h:h:h:h:h:h:h>`: IPv6 range to match.
- `!<h:h:h:h:h:h:h>`: Match everything except the specified address.
- `!<h:h:h:h:h:h:h/x>`: Match everything except the specified prefix.
- `!<h:h:h:h:h:h:h>-<h:h:h:h:h:h:h>`: Match everything except the specified range.

```
set policy route <name> rule <n> source group <address-group|domain-group|mac-group|network-group|port-
<text>
set policy route <name> rule <n> destination group <address-group|domain-group|mac-group|network-group|
<text>
set policy route6 <name> rule <n> source group <address-group|domain-group|mac-group|network-group|port-
<text>
set policy route6 <name> rule <n> destination group <address-group|domain-group|mac-group|network-group
<text>
```

Set match criteria based on source or destination groups, where `<text>` would be the group name/identifier. Prepend character `!` for inverted matching criteria.

```
set policy route <name> rule <n> destination port <match_criteria>
set policy route6 <name> rule <n> destination port <match_criteria>
```

Set match criteria based on destination port, where `<match_criteria>` could be:

- `<port name>`: Named port (any name in `/etc/services`, e.g., `http`).
- `<1-65535>`: Numbered port.

- <start>-<end>: Numbered port range (e.g., 1001-1005).

Multiple destination ports can be specified as a comma-separated list. The whole list can also be “negated” using ‘!’. For example: ‘!22,telnet,http,123,1001-1005’

set policy route <name> rule <n> disable

set policy route6 <name> rule <n> disable

Option to disable rule.

set policy route <name> rule <n> dscp <text>

set policy route6 <name> rule <n> dscp <text>

set policy route <name> rule <n> dscp-exclude <text>

set policy route6 <name> rule <n> dscp-exclude <text>

Match based on dscp value criteria. Multiple values from 0 to 63 and ranges are supported.

set policy route <name> rule <n> fragment <match-grag|match-non-frag>

set policy route6 <name> rule <n> fragment <match-grag|match-non-frag>

Set IP fragment match, where:

- match-frag: Second and further fragments of fragmented packets.
- match-non-frag: Head fragments or unfragmented packets.

set policy route <name> rule <n> icmp <code | type>

set policy route6 <name> rule <n> icmpv6 <code | type>

Match based on icmp|icmpv6 code and type.

set policy route <name> rule <n> icmp type-name <text>

set policy route6 <name> rule <n> icmpv6 type-name <text>

Match based on icmp|icmpv6 type-name criteria. Use tab for information about what type-name criteria are supported.

set policy route <name> rule <n> ipsec <match-ipsec|match-none>

set policy route6 <name> rule <n> ipsec <match-ipsec|match-none>

Set IPSec inbound match criteria, where:

- match-ipsec: match inbound IPsec packets.
- match-none: match inbound non-IPsec packets.

set policy route <name> rule <n> limit burst <0-4294967295>

set policy route6 <name> rule <n> limit burst <0-4294967295>

Set maximum number of packets to allow in excess of rate.

set policy route <name> rule <n> limit rate <text>

set policy route6 <name> rule <n> limit rate <text>

Set maximum average matching rate. Format for rate: integer/time_unit, where time_unit could be any one of second, minute, hour or day. For example 1/second implies rule to be matched at an average of once per second.

set policy route <name> rule <n> protocol <text | 0-255 | tcp_udp | all >

set policy route6 <name> rule <n> protocol <text | 0-255 | tcp_udp | all >

Match a protocol criteria. A protocol number or a name which is defined in: /etc/protocols. Special names are all for all protocols and tcp_udp for tcp and udp based packets. The ! negates the selected protocol.

```
set policy route <name> rule <n> packet-length <text>
set policy route6 <name> rule <n> packet-length <text>
set policy route <name> rule <n> packet-length-exclude <text>
set policy route6 <name> rule <n> packet-length-exclude <text>
```

Match based on packet length criteria. Multiple values from 1 to 65535 and ranges are supported.

```
set policy route <name> rule <n> packet-type [broadcast | host | multicast | other]
set policy route6 <name> rule <n> packet-type [broadcast | host | multicast | other]
```

Match based on packet type criteria.

```
set policy route <name> rule <n> recent count <1-255>
set policy route6 <name> rule <n> recent count <1-255>
set policy route <name> rule <n> recent time <1-4294967295>
set policy route6 <name> rule <n> recent time <1-4294967295>
```

Set parameters for matching recently seen sources. This match could be used by setting count (source address seen more than <1-255> times) and/or time (source address seen in the last <0-4294967295> seconds).

```
set policy route <name> rule <n> state <established | invalid | new | related>
set policy route6 <name> rule <n> state <established | invalid | new | related>
```

Set match criteria based on session state.

```
set policy route <name> rule <n> tcp flags <text>
set policy route6 <name> rule <n> tcp flags <text>
```

Set match criteria based on tcp flags. Allowed values for TCP flags: SYN ACK FIN RST URG PSH ALL. When specifying more than one flag, flags should be comma-separated. For example : value of 'SYN,!ACK,!FIN,!RST' will only match packets with the SYN flag set, and the ACK, FIN and RST flags unset.

```
set policy route <name> rule <n> time monthdays <text>
set policy route6 <name> rule <n> time monthdays <text>
set policy route <name> rule <n> time startdate <text>
set policy route6 <name> rule <n> time startdate <text>
set policy route <name> rule <n> time starttime <text>
set policy route6 <name> rule <n> time starttime <text>
set policy route <name> rule <n> time stopdate <text>
set policy route6 <name> rule <n> time stopdate <text>
set policy route <name> rule <n> time stoptime <text>
set policy route6 <name> rule <n> time stoptime <text>
set policy route <name> rule <n> time weekdays <text>
set policy route6 <name> rule <n> time weekdays <text>
set policy route <name> rule <n> time utc
```

```
set policy route6 <name> rule <n> time utc
```

Time to match the defined rule.

```
set policy route rule <n> ttl <eq | gt | lt> <0-255>
```

Match time to live parameter, where 'eq' stands for 'equal'; 'gt' stands for 'greater than', and 'lt' stands for 'less than'.

```
set policy route6 rule <n> hop-limit <eq | gt | lt> <0-255>
```

Match hop-limit parameter, where 'eq' stands for 'equal'; 'gt' stands for 'greater than', and 'lt' stands for 'less than'.

Actions

When matching all patterns defined in a rule, then different actions can be made. This includes dropping the packet, modifying certain data, or setting a different routing table.

```
set policy route <name> rule <n> action drop
```

```
set policy route6 <name> rule <n> action drop
```

Set rule action to drop.

```
set policy route <name> rule <n> set connection-mark <1-2147483647>
```

```
set policy route6 <name> rule <n> set connection-mark <1-2147483647>
```

Set a specific connection mark.

```
set policy route <name> rule <n> set dscp <0-63>
```

```
set policy route6 <name> rule <n> set dscp <0-63>
```

Set packet modifications: Packet Differentiated Services Codepoint (DSCP)

```
set policy route <name> rule <n> set mark <1-2147483647>
```

```
set policy route6 <name> rule <n> set mark <1-2147483647>
```

Set a specific packet mark.

```
set policy route <name> rule <n> set table <main | 1-200>
```

```
set policy route6 <name> rule <n> set table <main | 1-200>
```

Set the routing table to forward packet with.

Note: When using the `set table` or `set vrf` commands matching against the mark is not possible, because it gets overwritten with a table-specific mark that is 0x7FFFFFFF - the id of the table/VRF.

```
set policy route <name> rule <n> set tcp-mss <500-1460>
```

```
set policy route6 <name> rule <n> set tcp-mss <500-1460>
```

Set packet modifications: Explicitly set TCP Maximum segment size value.

```
set policy route <name> rule <n> set vrf <default | text >
```

```
set policy route6 <name> rule <n> set vrf <default | text >
```


Set the VRF to forward packet with.

Note: When using the `set table` or `set vrf` commands matching against the mark is not possible, because it gets overwritten with a table-specific mark that is 0x7FFFFFFF - the id of the table/VRF.

Route Map Policy

Route map is a powerful command, that gives network administrators a very useful and flexible tool for traffic manipulation.

Configuration

Route Map

set policy route-map <text>

This command creates a new route-map policy, identified by <text>.

set policy route-map <text> description <text>

Set description for the route-map policy.

set policy route-map <text> rule <1-65535> action <permit|deny>

Set action for the route-map policy.

set policy route-map <text> rule <1-65535> call <text>

Call another route-map policy on match.

set policy route-map <text> rule <1-65535> continue <1-65535>

Jump to a different rule in this route-map on a match.

set policy route-map <text> rule <1-65535> description <text>

Set description for the rule in the route-map policy.

set policy route-map <text> rule <1-65535> match as-path <text>

BGP as-path list to match.

set policy route-map <text> rule <1-65535> match community community-list <text>

BGP community-list to match.

set policy route-map <text> rule <1-65535> match community exact-match

Set BGP community-list to exactly match.

set policy route-map <text> rule <1-65535> match extcommunity <text>

BGP extended community to match.

set policy route-map <text> rule <1-65535> match interface <text>

First hop interface of a route to match.

set policy route-map <text> rule <1-65535> match ip address access-list <1-2699>

IP address of route to match, based on access-list.

```
set policy route-map <text> rule <1-65535> match ip address prefix-list <text>
```

IP address of route to match, based on prefix-list.

```
set policy route-map <text> rule <1-65535> match ip address prefix-len <0-32>
```

IP address of route to match, based on specified prefix-length. Note that this can be used for kernel routes only. Do not apply to the routes of dynamic routing protocols (e.g. BGP, RIP, OSPF), as this can lead to unexpected results..

```
set policy route-map <text> rule <1-65535> match ip nexthop access-list <1-2699>
```

IP next-hop of route to match, based on access-list.

```
set policy route-map <text> rule <1-65535> match ip nexthop address <x.x.x.x>
```

IP next-hop of route to match, based on ip address.

```
set policy route-map <text> rule <1-65535> match ip nexthop prefix-len <0-32>
```

IP next-hop of route to match, based on prefix length.

```
set policy route-map <text> rule <1-65535> match ip nexthop prefix-list <text>
```

IP next-hop of route to match, based on prefix-list.

```
set policy route-map <text> rule <1-65535> match ip nexthop type <blackhole>
```

IP next-hop of route to match, based on type.

```
set policy route-map <text> rule <1-65535> match ip route-source access-list <1-2699>
```

IP route source of route to match, based on access-list.

```
set policy route-map <text> rule <1-65535> match ip route-source prefix-list <text>
```

IP route source of route to match, based on prefix-list.

```
set policy route-map <text> rule <1-65535> match ipv6 address access-list <text>
```

IPv6 address of route to match, based on IPv6 access-list.

```
set policy route-map <text> rule <1-65535> match ipv6 address prefix-list <text>
```

IPv6 address of route to match, based on IPv6 prefix-list.

```
set policy route-map <text> rule <1-65535> match ipv6 address prefix-len <0-128>
```

IPv6 address of route to match, based on specified prefix-length. Note that this can be used for kernel routes only. Do not apply to the routes of dynamic routing protocols (e.g. BGP, RIP, OSPF), as this can lead to unexpected results..

```
set policy route-map <text> rule <1-65535> match ipv6 nexthop <h:h:h:h:h:h:h>
```

Nexthop IPv6 address to match.

```
set policy route-map <text> rule <1-65535> match large-community large-community-list <text>
```

Match BGP large communities.

```
set policy route-map <text> rule <1-65535> match local-preference <0-4294967295>
```

Match local preference.

```
set policy route-map <text> rule <1-65535> match metric <1-65535>
```

Match route metric.

```
set policy route-map <text> rule <1-65535> match origin <egp|igp|incomplete>
```

Border Gateway Protocol (BGP) origin code to match.

```
set policy route-map <text> rule <1-65535> match peer <x.x.x.x>
```

Peer IP address to match.

```
set policy route-map <text> rule <1-65535> match protocol <protocol>
```

Source protocol to match.

- `babel` - Babel routing protocol (Babel)
- `bgp` - Border Gateway Protocol (BGP)
- `connected` - Connected routes (directly attached subnet or host)
- `isis` - Intermediate System to Intermediate System (IS-IS)
- `kernel` - Kernel routes
- `ospf` - Open Shortest Path First (OSPFv2)
- `ospfv3` - Open Shortest Path First (IPv6) (OSPFv3)
- `rip` - Routing Information Protocol (RIP)
- `ripng` - Routing Information Protocol next-generation (IPv6) (RIPng)
- `static` - Statically configured routes
- `table` - Non-main Kernel Routing Table
- `vnc` - Virtual Network Control (VNC)

```
set policy route-map <text> rule <1-65535> match rpki <invalid|notfound|valid>
```

Match RPKI validation result.

```
set policy route-map <text> rule <1-65535> match source-vrf <text>
```

Source VRF to match.

```
set policy route-map <text> rule <1-65535> match tag <1-65535>
```

Route tag to match.

```
set policy route-map <text> rule <1-65535> on-match goto <1-65535>
```

Exit policy on match: go to rule <1-65535>

```
set policy route-map <text> rule <1-65535> on-match next
```

Exit policy on match: go to next sequence number.

```
set policy route-map <text> rule <1-65535> set aggregator <as|ip> <1-4294967295|x.x.x.x>
```

BGP aggregator attribute: AS number or IP address of an aggregation.

```
set policy route-map <text> rule <1-65535> set as-path exclude <1-4294967295 | all>
```

Drop AS-NUMBER from the BGP AS path.

If `all` is specified, remove all AS numbers from the `AS_PATH` of the BGP path's NLRI.

```
set policy route-map <text> rule <1-65535> set as-path prepend <1-4294967295>
```

Prepend the given string of AS numbers to the `AS_PATH` of the BGP path's NLRI.

```
set policy route-map <text> rule <1-65535> set as-path prepend-last-as <n>
```

Prepend the existing last AS number (the leftmost ASN) to the `AS_PATH`.

```
set policy route-map <text> rule <1-65535> set atomic-aggregate
```

BGP atomic aggregate attribute.

```
set policy route-map <text> rule <1-65535> set community <add|replace> <community>
```

Add or replace BGP community attribute in format <0-65535:0-65535> or from well-known community list

```
set policy route-map <text> rule <1-65535> set community none
```

Delete all BGP communities

```
set policy route-map <text> rule <1-65535> set community delete <text>
```

Delete BGP communities matching the community-list.

```
set policy route-map <text> rule <1-65535> set large-community <add|replace>  
<GA:LDP1:LDP2>
```

Add or replace BGP large-community attribute in format <0-4294967295:0-4294967295:0-4294967295>

```
set policy route-map <text> rule <1-65535> set large-community none
```

Delete all BGP large-communities

```
set policy route-map <text> rule <1-65535> set large-community delete <text>
```

Delete BGP communities matching the large-community-list.

```
set policy route-map <text> rule <1-65535> set extcommunity bandwidth  
<1-25600|cumulative|num-multipaths>
```

Set extcommunity bandwidth

```
set policy route-map <text> rule <1-65535> set extcommunity bandwidth-non-transitive
```

The link bandwidth extended community is encoded as non-transitive

```
set policy route-map <text> rule <1-65535> set extcommunity rt <text>
```

Set route target value in format <0-65535:0-4294967295> or <IP:0-65535>.

```
set policy route-map <text> rule <1-65535> set extcommunity soo <text>
```

Set site of origin value in format <0-65535:0-4294967295> or <IP:0-65535>.

```
set policy route-map <text> rule <1-65535> set extcommunity none
```

Clear all BGP extcommunities.

```
set policy route-map <text> rule <1-65535> set distance <0-255>
```

Locally significant administrative distance.

```
set policy route-map <text> rule <1-65535> set ip-next-hop <x.x.x.x>
```

Nexthop IP address.

```
set policy route-map <text> rule <1-65535> set ip-next-hop unchanged
```

Set the next-hop as unchanged. Pass through the route-map without changing its value

```
set policy route-map <text> rule <1-65535> set ip-next-hop peer-address
```

Set the BGP nexthop address to the address of the peer. For an incoming route-map this means the ip address of our peer is used. For an outgoing route-map this means the ip address of our self is used to establish the peering with our neighbor.

```
set policy route-map <text> rule <1-65535> set ipv6-next-hop <global|local>  
<h:h:h:h:h:h:h>
```

Nexthop IPv6 address.

set policy route-map <text> rule <1-65535> set ipv6-next-hop peer-address

Set the BGP nexthop address to the address of the peer. For an incoming route-map this means the ip address of our peer is used. For an outgoing route-map this means the ip address of our self is used to establish the peering with our neighbor.

set policy route-map <text> rule <1-65535> set ipv6-next-hop prefer-global

For Incoming and Import Route-maps if we receive a v6 global and v6 LL address for the route, then prefer to use the global address as the nexthop.

set policy route-map <text> rule <1-65535> set local-preference <0-4294967295>

Set BGP local preference attribute.

set policy route-map <text> rule <1-65535> set metric <+/-metric|0-4294967295|rtt|+rtt|-rtt>

Set the route metric. When used with BGP, set the BGP attribute MED to a specific value. Use +/- to add or subtract the specified value to/from the existing/MED. Use rtt to set the MED to the round trip time or +rtt/-rtt to add/subtract the round trip time to/from the MED.

set policy route-map <text> rule <1-65535> set metric-type <type-1|type-2>

Set OSPF external metric-type.

set policy route-map <text> rule <1-65535> set origin <igp|egp|incomplete>

Set BGP origin code.

set policy route-map <text> rule <1-65535> set originator-id <x.x.x.x>

Set BGP originator ID attribute.

set policy route-map <text> rule <1-65535> set src <x.x.x.x|h:h:h:h:h:h:h>

Set source IP/IPv6 address for route.

set policy route-map <text> rule <1-65535> set table <1-200>

Set prefixes to table.

set policy route-map <text> rule <1-65535> set tag <1-65535>

Set tag value for routing protocol.

set policy route-map <text> rule <1-65535> set weight <0-4294967295>

Set BGP weight attribute

List of well-known communities

- local-as - Well-known communities value NO_EXPORT_SUBCONFED 0xFFFFFFFF03
- no-advertise - Well-known communities value NO_ADVERTISE 0xFFFFFFFF02
- no-export - Well-known communities value NO_EXPORT 0xFFFFFFFF01
- graceful-shutdown - Well-known communities value GRACEFUL_SHUTDOWN 0xFFFF0000
- accept-own - Well-known communities value ACCEPT_OWN 0xFFFF0001
- route-filter-translated-v4 - Well-known communities value ROUTE_FILTER_TRANSLATED_v4 0xFFFF0002
- route-filter-v4 - Well-known communities value ROUTE_FILTER_v4 0xFFFF0003

- `route-filter-translated-v6` - Well-known communities value `ROUTE_FILTER_TRANSLATED_v6 0xFFFF0004`
- `route-filter-v6` - Well-known communities value `ROUTE_FILTER_v6 0xFFFF0005`
- `llgr-stale` - Well-known communities value `LLGR_STALE 0xFFFF0006`
- `no-llgr` - Well-known communities value `NO_LLGR 0xFFFF0007`
- `accept-own-nexthop` - Well-known communities value `accept-own-nexthop 0xFFFF0008`
- `blackhole` - Well-known communities value `BLACKHOLE 0xFFFF029A`
- `no-peer` - Well-known communities value `NOPEER 0xFFFF0004`

Local Route Policy

Policies for local traffic are defined in this section.

Configuration

Local Route IPv4

```
set policy local-route rule <1-32765> set table <1-200|main>
```

Set the routing table to use for forwarding matching packets.

```
set policy local-route rule <1-32765> set vrf <vrf|default>
```

Set the VRF to use for forwarding matching packets.

```
set policy local-route rule <1-32765> protocol <protocol>
```

Match specified protocol (name or number).

```
set policy local-route rule <1-32765> fwmark <1-2147483647>
```

Match specified firewall mark (fwmark).

```
set policy local-route rule <1-32765> source address <x.x.x.x|x.x.x.x/x>
```

Match specified source address or prefix.

```
set policy local-route rule <1-32765> source port <1-65535>
```

Match specified source port.

```
set policy local-route rule <1-32765> destination address <x.x.x.x|x.x.x.x/x>
```

Match specified destination address or prefix.

```
set policy local-route rule <1-32765> destination port <1-65535>
```

Match specified destination port.

```
set policy local-route rule <1-32765> inbound-interface <interface>
```

Match specified inbound interface.

Local Route IPv6

set policy local-route6 rule <1-32765> set table <1-200|main>

Set the routing table to use for forwarding matching packets.

set policy local-route6 rule <1-32765> set vrf <vrf|default>

Set the VRF to use for forwarding matching packets.

set policy local-route6 rule <1-32765> protocol <protocol>

Match specified protocol (name or number).

set policy local-route6 rule <1-32765> fwmark <1-2147483647>

Match specified firewall mark (fwmark).

set policy local-route6 rule <1-32765> source address <h:h:h:h:h:h:h|h:h:h:h:h:h:h/x>

Match specified source address or prefix.

set policy local-route6 rule <1-32765> source port <1-65535>

Match specified source port.

set policy local-route6 rule <1-32765> destination address <h:h:h:h:h:h:h|h:h:h:h:h:h:h/x>

Match specified destination address or prefix.

set policy local-route6 rule <1-32765> destination port <1-65535>

Match specified destination port.

set policy local-route6 rule <1-32765> inbound-interface <interface>

Match specified inbound interface.

BGP - AS Path Policy

VyOS provides policies commands exclusively for BGP traffic filtering and manipulation: **as-path-list** is one of them.

Configuration

policy as-path-list

set policy as-path-list <text>

Create as-path-policy identified by name <text>.

set policy as-path-list <text> description <text>

Set description for as-path-list policy.

set policy as-path-list <text> rule <1-65535> action <permit|deny>

Set action to take on entries matching this rule.

set policy as-path-list <text> rule <1-65535> description <text>

Set description for rule.

set policy as-path-list <text> rule <1-65535> regex <text>

Regular expression to match against an AS path. For example “64501 64502”.

BGP - Community List

VyOS provides policies commands exclusively for BGP traffic filtering and manipulation: **community-list** is one of them.

Configuration

policy community-list

```
set policy community-list <text>
```

Creat community-list policy identified by name <text>.

```
set policy community-list <text> description <text>
```

Set description for community-list policy.

```
set policy community-list <text> rule <1-65535> action <permit|deny>
```

Set action to take on entries matching this rule.

```
set policy community-list <text> rule <1-65535> description <text>
```

Set description for rule.

```
set policy community-list <text> rule <1-65535> regex <aa:nn|local-AS|no-advertise|no-export|additive>
```

Regular expression to match against a community-list.

BGP - Extended Community List

VyOS provides policies commands exclusively for BGP traffic filtering and manipulation: **extcommunity-list** is one of them.

Configuration

policy extcommunity-list

```
set policy extcommunity-list <text>
```

Creat extcommunity-list policy identified by name <text>.

```
set policy extcommunity-list <text> description <text>
```

Set description for extcommunity-list policy.

```
set policy extcommunity-list <text> rule <1-65535> action <permit|deny>
```

Set action to take on entries matching this rule.

```
set policy extcommunity-list <text> rule <1-65535> description <text>
```

Set description for rule.

```
set policy extcommunity-list <text> rule <1-65535> regex <text>
```

Regular expression to match against an extended community list, where text could be:

- <aa:nn:nn>: Extended community list regular expression.
- <rt aa:nn:nn>: Route Target regular expression.
- <soo aa:nn:nn>: Site of Origin regular expression.

BGP - Large Community List

VyOS provides policies commands exclusively for BGP traffic filtering and manipulation: **large-community-list** is one of them.

Configuration

policy large-community-list

set policy large-community-list <text>

Create large-community-list policy identified by name <text>.

set policy large-community-list <text> description <text>

Set description for large-community-list policy.

set policy large-community-list <text> rule <1-65535> action <permit|deny>

Set action to take on entries matching this rule.

set policy large-community-list <text> rule <1-65535> description <text>

Set description for rule.

set policy large-community-list <text> rule <1-65535> regex <aa:nn:nn>

Regular expression to match against a large community list.

7.7.2 Examples

Examples of policies usage:

BGP Example

Policy definition:

```
# Create policy
set policy route-map setmet rule 2 action 'permit'
set policy route-map setmet rule 2 set as-path prepend '2 2 2'

# Apply policy to BGP
set protocols bgp system-as 1
set protocols bgp neighbor 203.0.113.2 address-family ipv4-unicast route-map import
↪ 'setmet'
set protocols bgp neighbor 203.0.113.2 address-family ipv4-unicast soft-reconfiguration
↪ 'inbound'
```

Using 'soft-reconfiguration' we get the policy update without bouncing the neighbor.

Routes learned before routing policy applied:

```
vyos@vos1:~$ show ip bgp
BGP table version is 0, local router ID is 192.168.56.101
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
               r RIB-failure, S Stale, R Removed
Origin codes: i - IGP, e - EGP, ? - incomplete

   Network          Next Hop          Metric LocPrf Weight Path
*> 198.51.100.3/32   203.0.113.2          1             0 2 i < Path

Total number of prefixes 1
```

Routes learned after routing policy applied:

```
vyos@vos1:~$ show ip bgp
BGP table version is 0, local router ID is 192.168.56.101
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
               r RIB-failure, S Stale, R Removed
Origin codes: i - IGP, e - EGP, ? - incomplete

   Network          Next Hop          Metric LocPrf Weight Path
*> 198.51.100.3/32   203.0.113.2          1             0 2 2 2 2 i

Total number of prefixes 1
vyos@vos1:~$
```

You now see the longer AS path.

Transparent Proxy

The following example will show how VyOS can be used to redirect web traffic to an external transparent proxy:

```
set policy route FILTER-WEB rule 1000 destination port 80
set policy route FILTER-WEB rule 1000 protocol tcp
set policy route FILTER-WEB rule 1000 set table 100
```

This creates a route policy called FILTER-WEB with one rule to set the routing table for matching traffic (TCP port 80) to table ID 100 instead of the default routing table.

To create routing table 100 and add a new default gateway to be used by traffic matching our route policy:

```
set protocols static table 100 route 0.0.0.0/0 next-hop 10.255.0.2
```

This can be confirmed using the `show ip route table 100` operational command.

Finally, to apply the policy route to ingress traffic on our LAN interface, we use:

```
set policy route FILTER-WEB interface eth1
```

Multiple Uplinks

VyOS Policy-Based Routing (PBR) works by matching source IP address ranges and forwarding the traffic using different routing tables.

Routing tables that will be used in this example are:

- **table 10** Routing table used for VLAN 10 (192.168.188.0/24)
- **table 11** Routing table used for VLAN 11 (192.168.189.0/24)
- **main** Routing table used by VyOS and other interfaces not participating in PBR

Fig. 2: Policy-Based Routing with multiple ISP uplinks (source [./draw.io/pbr_example_1.drawio](https://draw.io/pbr_example_1.drawio))

Add default routes for routing table 10 and table 11

```
set protocols static table 10 route 0.0.0.0/0 next-hop 192.0.2.1
set protocols static table 11 route 0.0.0.0/0 next-hop 192.0.2.2
```

Add policy route matching VLAN source addresses

```
set policy route PBR rule 20 set table '10'
set policy route PBR rule 20 description 'Route VLAN10 traffic to table 10'
set policy route PBR rule 20 source address '192.168.188.0/24'

set policy route PBR rule 30 set table '11'
set policy route PBR rule 30 description 'Route VLAN11 traffic to table 11'
set policy route PBR rule 30 source address '192.168.189.0/24'
```

Apply routing policy to **inbound** direction of out VLAN interfaces

```
set policy route 'PBR' interface eth0.10
set policy route 'PBR' interface eth0.11
```

OPTIONAL: Exclude Inter-VLAN traffic (between VLAN10 and VLAN11) from PBR

```
set firewall group network-group VLANS-GR description 'VLANs networks'
set firewall group network-group VLANS-GR network '192.168.188.0/24'
set firewall group network-group VLANS-GR network '192.168.189.0/24'

set policy route PBR rule 10 description 'VLAN10 <-> VLAN11 shortcut'
set policy route PBR rule 10 destination group network-group 'VLANS-GR'
set policy route PBR rule 10 set table 'main'
```

These commands allow the VLAN10 and VLAN11 hosts to communicate with each other using the main routing table.

Local route

The following example allows VyOS to use PBR (Policy-Based Routing) for traffic, which originated from the router itself. That solution for multiple ISP's and VyOS router will respond from the same interface that the packet was received. Also, it used, if we want that one VPN tunnel to be through one provider, and the second through another.

- 203.0.113.254 IP address on VyOS eth1 from ISP1
- 192.168.2.254 IP address on VyOS eth2 from ISP2
- table 10 Routing table used for ISP1
- table 11 Routing table used for ISP2

```
set policy local-route rule 101 set table '10'
set policy local-route rule 101 source address '203.0.113.254'
set policy local-route rule 102 set table '11'
set policy local-route rule 102 source address '192.0.2.254'
set protocols static table 10 route 0.0.0.0/0 next-hop '203.0.113.1'
set protocols static table 11 route 0.0.0.0/0 next-hop '192.0.2.2'
```

Add multiple source IP in one rule with same priority

```
set policy local-route rule 101 set table '10'
set policy local-route rule 101 source address '203.0.113.254'
set policy local-route rule 101 source address '203.0.113.253'
set policy local-route rule 101 source address '198.51.100.0/24'
```

Clamp MSS for a specific IP

This example shows how to target an MSS clamp (in our example to 1360 bytes) to a specific destination IP.

```
set policy route IP-MSS-CLAMP rule 10 description 'Clamp TCP session MSS to 1360 for 198.
↪51.100.30'
set policy route IP-MSS-CLAMP rule 10 destination address '198.51.100.30/32'
set policy route IP-MSS-CLAMP rule 10 protocol 'tcp'
set policy route IP-MSS-CLAMP rule 10 set tcp-mss '1360'
set policy route IP-MSS-CLAMP rule 10 tcp flags 'SYN'
```

To apply this policy to the correct interface, configure it on the interface the inbound local host will send through to reach our destined target host (in our example eth1).

```
set policy route IP-MSS-CLAMP interface eth1
```

You can view that the policy is being correctly (or incorrectly) utilised with the following command:

```
show policy route statistics
```

7.8 PKI

VyOS 1.4 changed the way in how encryption keys or certificates are stored on the system. In the pre VyOS 1.4 era, certificates got stored under /config and every service referenced a file. That made copying a running configuration from system A to system B a bit harder, as you had to copy the files and their permissions by hand.

T3642 describes a new CLI subsystem that serves as a “certstore” to all services requiring any kind of encryption key(s). In short, public and private certificates are now stored in PKCS#8 format in the regular VyOS CLI. Keys can now be added, edited, and deleted using the regular set/edit/delete CLI commands.

VyOS not only can now manage certificates issued by 3rd party Certificate Authorities, it can also act as a CA on its own. You can create your own root CA and sign keys with it by making use of some simple op-mode commands.

Don't be afraid that you need to re-do your configuration. Key transformation is handled, as always, by our migration scripts, so this will be a smooth transition for you!

7.8.1 Key Generation

Certificate Authority (CA)

VyOS now also has the ability to create CAs, keys, Diffie-Hellman and other keypairs from an easy to access operational level command.

generate pki ca

Create a new CA and output the CAs public and private key on the console.

generate pki ca install <name>

Create a new CA and output the CAs public and private key on the console.

Note: In addition to the command above, the output is in a format which can be used to directly import the key into the VyOS CLI by simply copy-pasting the output from op-mode into configuration mode.

name is used for the VyOS CLI command to identify this key. This key name is then used in the CLI configuration to reference the key instance.

generate pki ca sign <ca-name>

Create a new subordinate CA and sign it using the private key referenced by ca-name.

generate pki ca sign <ca-name> install <name>

Create a new subordinate CA and sign it using the private key referenced by name.

Note: In addition to the command above, the output is in a format which can be used to directly import the key into the VyOS CLI by simply copy-pasting the output from op-mode into configuration mode.

name is used for the VyOS CLI command to identify this key. This key name is then used in the CLI configuration to reference the key instance.

Certificates

generate pki certificate

Create a new public/private keypair and output the certificate on the console.

generate pki certificate install <name>

Create a new public/private keypair and output the certificate on the console.

Note: In addition to the command above, the output is in a format which can be used to directly import the key into the VyOS CLI by simply copy-pasting the output from op-mode into configuration mode.

name is used for the VyOS CLI command to identify this key. This key name is then used in the CLI configuration to reference the key instance.

generate pki certificate self-signed

Create a new self-signed certificate. The public/private is then shown on the console.

generate pki certificate self-signed install <name>

Create a new self-signed certificate. The public/private is then shown on the console.

Note: In addition to the command above, the output is in a format which can be used to directly import the key into the VyOS CLI by simply copy-pasting the output from op-mode into configuration mode.

name is used for the VyOS CLI command to identify this key. This key name is then used in the CLI configuration to reference the key instance.

generate pki certificate sign <ca-name>

Create a new public/private keypair which is signed by the CA referenced by ca-name. The signed certificate is then output to the console.

generate pki certificate sign <ca-name> install <name>

Create a new public/private keypair which is signed by the CA referenced by ca-name. The signed certificate is then output to the console.

Note: In addition to the command above, the output is in a format which can be used to directly import the key into the VyOS CLI by simply copy-pasting the output from op-mode into configuration mode.

name is used for the VyOS CLI command to identify this key. This key name is then used in the CLI configuration to reference the key instance.

Diffie-Hellman parameters

generate pki dh

Generate a new set of DH (Diffie-Hellman) parameters. The key size is requested by the CLI and defaults to 2048 bit.

The generated parameters are then output to the console.

generate pki dh install <name>

Generate a new set of DH parameters. The key size is requested by the CLI and defaults to 2048 bit.

Note: In addition to the command above, the output is in a format which can be used to directly import the key into the VyOS CLI by simply copy-pasting the output from op-mode into configuration mode.

name is used for the VyOS CLI command to identify this key. This key name is then used in the CLI configuration to reference the key instance.

OpenVPN

generate pki openvpn shared-secret

Generate a new OpenVPN shared secret. The generated secret is the output to the console.

generate pki openvpn shared-secret install <name>

Generate a new OpenVPN shared secret. The generated secret is the output to the console.

Note: In addition to the command above, the output is in a format which can be used to directly import the key into the VyOS CLI by simply copy-pasting the output from op-mode into configuration mode.

name is used for the VyOS CLI command to identify this key. This key name is then used in the CLI configuration to reference the key instance.

WireGuard

generate pki wireguard key-pair

Generate a new WireGuard public/private key portion and output the result to the console.

generate pki wireguard key-pair install <interface>

Generate a new WireGuard public/private key portion and output the result to the console.

Note: In addition to the command above, the output is in a format which can be used to directly import the key into the VyOS CLI by simply copy-pasting the output from op-mode into configuration mode.

interface is used for the VyOS CLI command to identify the WireGuard interface where this private key is to be used.

generate pki wireguard preshared-key

Generate a WireGuard pre-shared secret used for peers to communicate.

generate pki wireguard preshared-key install <peer>

Generate a WireGuard pre-shared secret used for peers to communicate.

Note: In addition to the command above, the output is in a format which can be used to directly import the key into the VyOS CLI by simply copy-pasting the output from op-mode into configuration mode.

peer is used for the VyOS CLI command to identify the WireGuard peer where this secret is to be used.

7.8.2 Key usage (CLI)

CA (Certificate Authority)

set pki ca <name> certificate

Add the public CA certificate for the CA named name to the VyOS CLI.

Note: When loading the certificate you need to manually strip the -----BEGIN CERTIFICATE----- and -----END CERTIFICATE----- tags. Also, the certificate/key needs to be presented in a single line without line breaks (\n), this can be done using the following shell command:

```
$ tail -n +2 ca.pem | head -n -1 | tr -d '\n'
```

set pki ca <name> crl

Certificate revocation list in PEM format.

set pki ca <name> description

A human readable description what this CA is about.

set pki ca <name> private key

Add the CAs private key to the VyOS CLI. This should never leave the system, and is only required if you use VyOS as your certificate generator as mentioned above.

Note: When loading the certificate you need to manually strip the -----BEGIN KEY----- and -----END KEY----- tags. Also, the certificate/key needs to be presented in a single line without line breaks (\n), this can be done using the following shell command:

```
$ tail -n +2 ca.key | head -n -1 | tr -d '\n'
```

set pki ca <name> private password-protected

Mark the CAs private key as password protected. User is asked for the password when the key is referenced.

Server Certificate

After we have imported the CA certificate(s) we can now import and add certificates used by services on this router.

set pki certificate <name> certificate

Add public key portion for the certificate named `name` to the VyOS CLI.

Note: When loading the certificate you need to manually strip the -----BEGIN CERTIFICATE----- and -----END CERTIFICATE----- tags. Also, the certificate/key needs to be presented in a single line without line breaks (\n), this can be done using the following shell command:

```
$ tail -n +2 cert.pem | head -n -1 | tr -d '\n'
```

set pki certificate <name> description

A human readable description what this certificate is about.

set pki certificate <name> private key

Add the private key portion of this certificate to the CLI. This should never leave the system as it is used to decrypt the data.

Note: When loading the certificate you need to manually strip the -----BEGIN KEY----- and -----END KEY----- tags. Also, the certificate/key needs to be presented in a single line without line breaks (\n), this can be done using the following shell command:

```
$ tail -n +2 cert.key | head -n -1 | tr -d '\n'
```

set pki certificate <name> private password-protected

Mark the private key as password protected. User is asked for the password when the key is referenced.

set pki certificate <name> revoke

If CA is present, this certificate will be included in generated CRLs

Import files to PKI format

VyOS provides this utility to import existing certificates/key files directly into PKI from op-mode. Previous to VyOS 1.4, certificates were stored under the /config folder permanently and will be retained post upgrade.

import pki ca <name> file <Path to CA certificate file>

Import the public CA certificate from the defined file to VyOS CLI.

import pki ca <name> key-file <Path to private key file>

Import the CAs private key portion to the CLI. This should never leave the system as it is used to decrypt the data. The key is required if you use VyOS as your certificate generator.

import pki certificate <name> file <path to certificate>

Import the certificate from the file to VyOS CLI.

import pki certificate <name> key-file <path to private key>

Import the private key of the certificate to the VyOS CLI. This should never leave the system as it is used to decrypt the data.

import pki openvpn shared-secret <name> file <path to OpenVPN secret key>

Import the OpenVPN shared secret stored in file to the VyOS CLI.

ACME

The VyOS PKI subsystem can also be used to automatically retrieve Certificates using the ACME (Automatic Certificate Management Environment) protocol.

set pki certificate <name> acme domain-name <name>

Domain names to apply, multiple domain-names can be specified.

This is a mandatory option

set pki certificate <name> acme email <address>

Email used for registration and recovery contact.

This is a mandatory option

set pki certificate <name> acme listen-address <address>

The address the server listens to during http-01 challenge

set pki certificate <name> acme rsa-key-size <2048 | 3072 | 4096>

Size of the RSA key.

This options defaults to 2048

set pki certificate <name> acme url <url>

ACME Directory Resource URI.

This defaults to <https://acme-v02.api.letsencrypt.org/directory>

Note: During initial deployment we recommend using the staging API of LetsEncrypt to prevent and black-listing of your system. The API endpoint is <https://acme-staging-v02.api.letsencrypt.org/directory>

7.8.3 Operation

VyOS operational mode commands are not only available for generating keys but also to display them.

show pki ca

Show a list of installed CA certificates.

```
vyos@vyos:~$ show pki ca
Certificate Authorities:
Name          Subject          Issuer CN
-----
→          Issued          Expiry          Private Key    Parent
-----
→-----
DST_Root_CA_X3 CN=ISRG Root X1,0=Internet Security Research Group,C=US CN=DST
→Root CA X3    2021-01-20 19:14:03 2024-09-30 18:14:03 No N/A
R3              CN=R3,0=Let's Encrypt,C=US CN=ISRG
→Root X1       2020-09-04 00:00:00 2025-09-15 16:00:00 No DST_Root_CA_X3
vyos_rw        CN=VyOS RW CA,0=VyOS,L=Some-City,ST=Some-State,C=GB CN=VyOS RW
→CA           2021-07-05 13:46:03 2026-07-04 13:46:03 Yes N/A
```

show pki ca <name>

Show only information for specified Certificate Authority.

show pki certificate

Show a list of installed certificates

```
vyos@vyos:~$ show pki certificate
Certificates:
Name          Type      Subject CN          Issuer CN          Issued
→Expiry      Revoked   Private Key        CA Present
-----
→-----
ac2           Server   CN=ac2.vyos.net    CN=R3              2021-07-05 07:29:59  2021-
→10-03 07:29:58 No        Yes                Yes (R3)
rw_server     Server   CN=VyOS RW         CN=VyOS RW CA      2021-07-05 13:48:02  2022-
→07-05 13:48:02 No        Yes                Yes (vyos_rw)
```

show pki certificate <name>

Show only information for specified certificate.

show pki crl

Show a list of installed CRLs (Certificate Revocation List).

renew certbot

Manually trigger certificate renewal. This will be done twice a day.

7.8.4 Examples

Create a CA chain and leaf certificates

This configuration generates & installs into the VyOS PKI system a root certificate authority, alongside two intermediary certificate authorities for client & server certificates. These CAs are then used to generate a server certificate for the router, and a client certificate for a user.

- vyos_root_ca is the root certificate authority.
- vyos_client_ca and vyos_server_ca are intermediary certificate authorities, which are signed by the root CA.
- vyos_cert is a leaf server certificate used to identify the VyOS router, signed by the server intermediary CA.
- vyos_example_user is a leaf client certificate used to identify a user, signed by client intermediary CA.

First, we create the root certificate authority.

```
[edit]
vyos@vyos# run generate pki ca install vyos_root_ca
Enter private key type: [rsa, dsa, ec] (Default: rsa) rsa
Enter private key bits: (Default: 2048) 2048
Enter country code: (Default: GB) GB
Enter state: (Default: Some-State) Some-State
Enter locality: (Default: Some-City) Some-City
Enter organization name: (Default: VyOS) VyOS
Enter common name: (Default: vyos.io) VyOS Root CA
```

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```

Enter how many days certificate will be valid: (Default: 1825) 1825
Note: If you plan to use the generated key on this router, do not encrypt the private_
key.
Do you want to encrypt the private key with a passphrase? [y/N] n
2 value(s) installed. Use "compare" to see the pending changes, and "commit" to apply.

```

Secondly, we create the intermediary certificate authorities, which are used to sign the leaf certificates.

```

[edit]
vyos@vyos# run generate pki ca sign vyos_root_ca install vyos_server_ca
Do you already have a certificate request? [y/N] n
Enter private key type: [rsa, dsa, ec] (Default: rsa) rsa
Enter private key bits: (Default: 2048) 2048
Enter country code: (Default: GB) GB
Enter state: (Default: Some-State) Some-State
Enter locality: (Default: Some-City) Some-City
Enter organization name: (Default: VyOS) VyOS
Enter common name: (Default: vyos.io) VyOS Intermediary Server CA
Enter how many days certificate will be valid: (Default: 1825) 1095
Note: If you plan to use the generated key on this router, do not encrypt the private_
key.
Do you want to encrypt the private key with a passphrase? [y/N] n
2 value(s) installed. Use "compare" to see the pending changes, and "commit" to apply.

```

```

[edit]
vyos@vyos# run generate pki ca sign vyos_root_ca install vyos_client_ca
Do you already have a certificate request? [y/N] n
Enter private key type: [rsa, dsa, ec] (Default: rsa) rsa
Enter private key bits: (Default: 2048) 2048
Enter country code: (Default: GB) GB
Enter state: (Default: Some-State) Some-State
Enter locality: (Default: Some-City) Some-City
Enter organization name: (Default: VyOS) VyOS
Enter common name: (Default: vyos.io) VyOS Intermediary Client CA
Enter how many days certificate will be valid: (Default: 1825) 1095
Note: If you plan to use the generated key on this router, do not encrypt the private_
key.
Do you want to encrypt the private key with a passphrase? [y/N] n
2 value(s) installed. Use "compare" to see the pending changes, and "commit" to apply.

```

Lastly, we can create the leaf certificates that devices and users will utilise.

```

[edit]
vyos@vyos# run generate pki certificate sign vyos_server_ca install vyos_cert
Do you already have a certificate request? [y/N] n
Enter private key type: [rsa, dsa, ec] (Default: rsa) rsa
Enter private key bits: (Default: 2048) 2048
Enter country code: (Default: GB) GB
Enter state: (Default: Some-State) Some-State
Enter locality: (Default: Some-City) Some-City
Enter organization name: (Default: VyOS) VyOS

```

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```

Enter common name: (Default: vyos.io) vyos.net
Do you want to configure Subject Alternative Names? [y/N] y
Enter alternative names in a comma separate list, example: ipv4:1.1.1.1,ipv6:fe80::1,
↪ dns:vyos.net
Enter Subject Alternative Names: dns:vyos.net,dns:www.vyos.net
Enter how many days certificate will be valid: (Default: 365) 365
Enter certificate type: (client, server) (Default: server) server
Note: If you plan to use the generated key on this router, do not encrypt the private_
↪ key.
Do you want to encrypt the private key with a passphrase? [y/N] n
2 value(s) installed. Use "compare" to see the pending changes, and "commit" to apply.

```

```

[edit]
vyos@vyos# run generate pki certificate sign vyos_client_ca install vyos_example_user
Do you already have a certificate request? [y/N] n
Enter private key type: [rsa, dsa, ec] (Default: rsa) rsa
Enter private key bits: (Default: 2048) 2048
Enter country code: (Default: GB) GB
Enter state: (Default: Some-State) Some-State
Enter locality: (Default: Some-City) Some-City
Enter organization name: (Default: VyOS) VyOS
Enter common name: (Default: vyos.io) Example User
Do you want to configure Subject Alternative Names? [y/N] y
Enter alternative names in a comma separate list, example: ipv4:1.1.1.1,ipv6:fe80::1,
↪ dns:vyos.net,rfc822:user@vyos.net
Enter Subject Alternative Names: rfc822:example.user@vyos.net
Enter how many days certificate will be valid: (Default: 365) 365
Enter certificate type: (client, server) (Default: server) client
Note: If you plan to use the generated key on this router, do not encrypt the private_
↪ key.
Do you want to encrypt the private key with a passphrase? [y/N] n
2 value(s) installed. Use "compare" to see the pending changes, and "commit" to apply.

```

7.9 Protocols

7.9.1 ARP

The ARP (Address Resolution Protocol) resolves IPv4 network layer addresses to link layer MAC addresses. addresses. This mapping is essential for communication within the Internet Protocol suite. ARP was standardized in 1982 by [RFC 826](#) (STD 37).

Note: In Internet Protocol version 6 (IPv6) networks, address resolution is performed by the Neighbor Discovery Protocol (NDP).

Use the following commands to configure or view ARP table entries.

Configuration

set protocols static arp interface <interface> address <host> mac <mac>

Configure a static ARP entry on the specified interface.

This creates a permanent mapping between an IP address and a MAC address on the specified interface.

Example:

```
set protocols static arp interface eth0 address 192.0.2.1 mac 01:23:45:67:89:01
```

Operation

show protocols static arp

Show all ARP table entries across all interfaces.

```
vyos@vyos:~$ show protocols static arp
```

Address	HWtype	HWaddress	Flags Mask	Iface
10.1.1.1	ether	00:53:00:de:23:2e	C	eth1
10.1.1.100	ether	00:53:00:de:23:aa	CM	eth1

show protocols static arp interface <interface>

Show all ARP table entries for the specific interface.

Example for eth1:

```
vyos@vyos:~$ show protocols static arp interface eth1
```

Address	HWtype	HWaddress	Flags Mask	Iface
10.1.1.1	ether	00:53:00:de:23:2e	C	eth1
10.1.1.100	ether	00:53:00:de:23:aa	CM	eth1

7.9.2 Babel

The Babel protocol provides robust and efficient routing for both wired and wireless mesh networks. By default, Babel uses hop-count metrics on wired links and a variant of Expected Transmission Count (ETX) on wireless links. Administrators can configure Babel to account for radio diversity, automatically compute link latency, and include that latency in the routing metric. [RFC 8966](#) defines the Babel protocol.

Babel is a dual-stack protocol. A single Babel instance routes both IPv4 and IPv6 traffic simultaneously.

General configuration

VyOS does not require a specific command to start the Babel process. The system automatically starts the routing process when you configure the first Babel-enabled interface.

set protocols babel interface <interface>

Enable Babel routing on the specified interface.

The system immediately begins sending and receiving Babel packets on this interface.

Optional configuration

set protocols babel parameters diversity

Enable radio-frequency diversity routing for the Babel process.

Enabling this feature is highly recommended for networks with many wireless nodes.

Note: When you enable diversity routing, you should also configure the `diversity-factor` and `channel` parameters.

set protocols babel parameters diversity-factor <1-256>

Configure the multiplicative factor for diversity routing, in units of 1/256.

Lower multiplicative factors give greater weight to diversity in route selection. The default value is 256, which disables diversity routing. On nodes with multiple independent radios, configure a value of 128 or less.

set protocols babel parameters resend-delay <20-655340>

Configure the delay in milliseconds before the system resends an important request or update.

The default value is 2000 ms.

set protocols babel parameters smoothing-half-life <0-65534>

Configure the time constant, in seconds, for the smoothing algorithm used to implement hysteresis.

Higher values reduce route oscillation but slightly increase convergence time. A value of 0 disables hysteresis and is suitable for wired networks. The default is 4 seconds.

Interfaces configuration

set protocols babel interface <interface> type <auto|wired|wireless>

Configure the network type for the Babel-enabled interface.

Choose from the following:

- `auto`: Babel automatically detects if an interface is wired or wireless.
- `wired`: Babel enables optimizations for wired interfaces.
- `wireless`: Babel disables optimizations suitable only for wired interfaces. Specifying wireless is always correct, but may cause slower convergence and increased routing traffic.

set protocols babel interface <interface> split-horizon <default|disable|enable>

Configure the split-horizon routing behavior for the specified interface.

Use one of the following options:

- `default`: Babel automatically enables split-horizon on wired interfaces and disables it on wireless interfaces.
- `enable`: Babel enables split-horizon on the interface. This optimization should be used only on symmetric, transitive (wired) networks.
- `disable`: Babel disables split-horizon on the interface. Disabling split-horizon is always safe and correct.

set protocols babel interface <interface> hello-interval <20-655340>

Configure the interval, in milliseconds, between scheduled hello messages on the specified interface.

On wired links, Babel detects link failures within two hello intervals. On wireless links, link quality is reestimated at each interval. The default is 4000 ms.

```
set protocols babel interface <interface> update-interval <20-655340>
```

Configure the interval, in milliseconds, between scheduled routing updates on the specified interface.

Because Babel uses triggered updates extensively, you can increase this value on reliable links with minimal packet loss. The default is 20000 ms.

```
set protocols babel interface <interface> rxcost <1-65534>
```

Configure the base receive cost for the specified interface.

Babel applies this value based on the configured network type:

- **wired:** The value is the routing cost advertised to neighboring routers.
- **wireless:** The value is a multiplier used to compute the ETX (Expected Transmission Count) reception cost.

The default value is 256.

```
set protocols babel interface <interface> rtt-decay <1-256>
```

Configure the decay factor for the exponential moving average of RTT samples, in units of 1/256.

Higher values discard older samples faster. The default value is 42.

```
set protocols babel interface <interface> rtt-min <1-65535>
```

Configure the minimum RTT, in milliseconds, at which the cost to a neighbor begins to increase.

The additional cost is linear in (rtt - rtt-min). The default value is 10 ms.

```
set protocols babel interface <interface> rtt-max <1-65535>
```

Configure the maximum RTT, in milliseconds, above which the cost to a neighbor stops increasing.

The default value is 120 ms.

```
set protocols babel interface <interface> max-rtt-penalty <0-65535>
```

Configure the maximum cost added to a neighbor when RTT meets or exceeds rtt-max.

Setting this value to 0 disables RTT-based costs. The default value is 150.

```
set protocols babel interface <interface> enable-timestamps
```

Configure adding timestamps to each Hello and IHU message to calculate RTT values.

Enabling timestamps is recommended for tunnel interfaces.

```
set protocols babel interface <interface> channel <1-254|interfering|noninterfering>
```

Configure the channel identifier that diversity routing uses for the specified interface.

Interfaces interfere with each other based on the assigned channel identifier:

- **1-254:** The interface interferes with interfaces sharing the same channel number and with interfaces configured as **interfering**.
- **interfering:** The interface interferes with all others except those configured as **noninterfering**.
- **noninterfering:** The interface interferes only with itself.

Redistribution configuration

set protocols babel redistribute <ipv4|ipv6> <route source>

Configure the redistribution of routing information from the specified route source into the Babel process.

The following route sources are available:

- **ipv4:** bgp, connected, eigrp, isis, kernel, nhrp, ospf, rip, static
- **ipv6:** bgp, connected, eigrp, isis, kernel, nhrp, ospfv3, ripng, static

set protocols babel distribute-list <ipv4|ipv6> access-list <in|out> <number>

Configure global Babel route filtering using an access list.

Specify the direction in which the access list is applied:

- **in:** Filters incoming routes.
- **out:** Filters outgoing routes.

set protocols babel distribute-list <ipv4|ipv6> interface <interface> access-list <in|out> <number>

Configure Babel route filtering on the specified interface using an access list.

Specify the direction in which the access list is applied:

- **in:** Filters incoming routes.
- **out:** Filters outgoing routes.

set protocols babel distribute-list <ipv4|ipv6> prefix-list <in|out> <name>

Configure global Babel route filtering using a prefix list.

Specify the direction in which the prefix list is applied:

- **in:** Filters incoming routes.
- **out:** Filters outgoing routes.

set protocols babel distribute-list <ipv4|ipv6> interface <interface> prefix-list <in|out> <name>

Configure Babel route filtering on the specified interface using a prefix list.

Specify the direction in which the prefix list is applied:

- **in:** Filters incoming routes.
- **out:** Filters outgoing routes.

Configuration example

Basic two-node babel network

Goal: The following example connects two routers (Node 1 and Node 2) via their eth0 interfaces and uses the Babel routing protocol to advertise (redistribute) each router's locally configured networks (represented by loopback addresses) to one another.

Node 1:

```
# Configure the loopback (local networks) and physical (eth0) addresses
set interfaces loopback lo address 10.1.1.1/32
set interfaces loopback lo address fd12:3456:dead:beef::1/128
set interfaces ethernet eth0 address 192.168.1.1/24

# Enable Babel on the physical link
set protocols babel interface eth0 type wired

# Instruct Babel to advertise (redistribute) the locally configured networks
set protocols babel redistribute ipv4 connected
set protocols babel redistribute ipv6 connected
```

Node 2:

```
# Configure the loopback (local networks) and physical (eth0) addresses
set interfaces loopback lo address 10.2.2.2/32
set interfaces loopback lo address fd12:3456:beef:dead::2/128
set interfaces ethernet eth0 address 192.168.1.2/24

# Enable Babel on the physical link
set protocols babel interface eth0 type wired

# Tell Babel to advertise (redistribute) the locally configured networks
set protocols babel redistribute ipv4 connected
set protocols babel redistribute ipv6 connected
```

7.9.3 BFD

BFD (Bidirectional Forwarding Detection) is a network protocol that detects communication failures between two connected routing devices in mere milliseconds — far faster than traditional routing protocols. BFD does not route traffic itself. Instead, it provides rapid failure detection for other protocols, such as BGP (Border Gateway Protocol), OSPF, and static routes.

BFD is described and extended by the following RFCs: [RFC 5880](#), [RFC 5881](#) and [RFC 5883](#).

How BFD works

BFD establishes a session between two peers and exchanges rapid UDP (User Datagram Protocol) control packets at a negotiated interval (e.g., 300 ms between consecutive packets). If a peer doesn't receive these packets within the detection time (the negotiated interval multiplied by a detection multiplier), BFD assumes a connection failure and notifies the routing protocols.

Configuration

```
set protocols bfd peer <address>
```

Configure the IPv4 or IPv6 address of the BFD peer.

You can configure multiple BFD peers by specifying a different IP address for each neighbor you want to monitor.

Note: For IPv6 peers, you must configure a source IP address or interface.

Example:

```
set protocols bfd peer 198.51.100.33
```

set protocols bfd peer <address> echo-mode

Enable BFD echo transmission mode for the peer at the specified IP address.

Example:

```
set protocols bfd peer 198.51.100.33 echo-mode
```

set protocols bfd peer <address> multihop

Configure the peer at the specified IP address as a multi-hop neighbor.

Note: For BFD multi-hop peers, you must configure a local source IP address.

Example:

```
set protocols bfd peer 198.51.100.33 multihop
set protocols bfd peer 198.51.100.33 source address 198.51.100.1
```

set protocols bfd peer <address> source address <address>

Configure the source IP address for the BFD session with the peer at the specified IP address.

Note: Configuring a source IP address or interface is mandatory for IPv6 BFD peers.

Example:

```
set protocols bfd peer 198.51.100.33 source address 198.51.100.1
```

set protocols bfd peer <address> source interface <interface>

Configure the source interface for the BFD session with the peer at the specified IP address.

Note: Configuring a source IP address or interface is mandatory for IPv6 BFD peers.

Example:

```
set protocols bfd peer 2001:db8::2 source interface eth0
```

set protocols bfd peer <address> interval echo-interval <10-60000>

Configure the minimum interval, in milliseconds, at which the local system can receive echo packets from the peer at the specified IP address.

Example:

```
set protocols bfd peer 198.51.100.33 interval echo-interval 50
```

set protocols bfd peer <address> interval multiplier <2-255>

Configure the detection multiplier for the peer at the specified IP address.

BFD multiplies the negotiated interval by this value to calculate the detection time — the period after which it declares the session down if no control packets arrive.

Example:

```
set protocols bfd peer 192.0.2.1 interval multiplier 3
```

```
set protocols bfd peer <address> interval receive <10-60000>
```

Configure the interval, in milliseconds, at which the local system can receive BFD control packets from the peer at the specified IP address.

Example:

```
set protocols bfd peer 192.0.2.1 interval receive 300
```

```
set protocols bfd peer <address> interval transmit <10-60000>
```

Configure the interval, in milliseconds, at which you want the local system to transmit BFD control packets to the peer at the specified IP address.

Example:

```
set protocols bfd peer 192.0.2.1 interval transmit 300
```

```
set protocols bfd peer <address> shutdown
```

Disable the BFD session with the peer at the specified IP address.

Example:

```
set protocols bfd peer 198.51.100.33 shutdown
```

```
set protocols bfd peer <address> minimum-ttl <1-254>
```

Configure the minimum TTL (Time to Live) value that incoming BFD control packets must have to be accepted from the peer at the specified IP address.

BFD discards any packet whose TTL is below this threshold.

Note: This setting applies only to multi-hop sessions.

Example:

```
set protocols bfd peer 192.0.2.1 minimum-ttl 250
```

```
set protocols bfd peer <address> passive
```

Configure the BFD session with the peer at the specified IP address to operate in passive mode.

In passive mode, the local system does not initiate the BFD session. Instead, it waits to receive the first BFD control packet from the remote peer.

Note: At least one of the two peers must remain in active mode for the session to establish. If both peers are passive, neither will initiate, and the session never establishes.

Example:

```
set protocols bfd peer 198.51.100.33 passive
```

```
set protocols bfd peer <address> profile <profile>
```

Apply settings from a designated BFD profile to the peer at the specified IP address.

Example:

```
set protocols bfd peer 192.0.2.1 profile datacenter-fast
```

```
set protocols bfd peer <address> vrf <vrf>
```

Bind the BFD session with the peer at the specified IP address to a specific VRF instance.

Example:

```
set protocols bfd peer 192.0.2.1 vrf customer-a
```

Enable BFD in BGP

```
set protocols bgp neighbor <neighbor> bfd
```

Enable BFD failure detection for the BGP session with the neighbor at the specified IP address.

Example:

```
set protocols bgp neighbor 192.0.2.1 bfd
```

```
set protocols bgp peer-group <neighbor> bfd
```

Enable BFD failure detection for all neighbors in the specified BGP peer group.

Example:

```
set protocols bgp peer-group INTERNAL-PEERS bfd
```

Enable BFD in OSPF

```
set protocols ospf interface <interface> bfd
```

Enable BFD failure detection for OSPF neighbors discovered on the specified interface.

When OSPF forms an adjacency with a neighbor on this interface, it requests BFD to establish a session with that neighbor.

Example:

```
set protocols ospf interface eth0 bfd
```

```
set protocols ospfv3 interface <interface> bfd
```

Enable BFD failure detection for OSPFv3 neighbors discovered on the specified interface.

When OSPFv3 forms an adjacency with a neighbor on this interface, it requests BFD to establish a session with that neighbor.

Example:

```
set protocols ospfv3 interface eth0 bfd
```

Enable BFD in IS-IS

set protocols isis interface <interface> bfd

Enable BFD failure detection for IS-IS neighbors discovered on the specified interface.

When IS-IS forms an adjacency with a neighbor on this interface, it requests BFD to establish a session with that neighbor.

Example:

```
set protocols isis interface eth0 bfd
```

Operation

show bfd peers

Show all BFD peers.

```
vyos@vyos:~$ show bfd peers
BFD Peers:
  peer 198.51.100.33 vrf default
    ID: 4182341893
    Remote ID: 3267892964
    Active mode
    Status: up
    Uptime: 1 month(s), 16 hour(s), 29 minute(s), 38 second(s)
    Diagnostics: ok
    Remote diagnostics: ok
    Peer Type: configured
    RTT min/avg/max: 245/267/289 usec
    Local timers:
      Detect-multiplier: 3
      Receive interval: 300ms
      Transmission interval: 300ms
      Echo receive interval: 50ms
      Echo transmission interval: disabled
    Remote timers:
      Detect-multiplier: 3
      Receive interval: 300ms
      Transmission interval: 300ms
      Echo receive interval: disabled

  peer 198.51.100.55 multihop local-address 198.51.100.1 vrf default
    ID: 3618932327
    Remote ID: 3312345688
    Active mode
    Minimum TTL: 254
    Status: up
    Uptime: 20 hour(s), 16 minute(s), 19 second(s)
    Diagnostics: ok
    Remote diagnostics: ok
    Peer Type: configured
    RTT min/avg/max: 1240/1289/1356 usec
```

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```

Local timers:
    Detect-multiplier: 3
    Receive interval: 300ms
    Transmission interval: 300ms
    Echo receive interval: 50ms
    Echo transmission interval: disabled
Remote timers:
    Detect-multiplier: 3
    Receive interval: 1000ms
    Transmission interval: 1000ms
    Echo receive interval: disabled

```

BFD static route monitoring

A monitored static route is present in the RIB (Routing Information Base) only when the BFD session is up. If the session goes down, the route is removed.

set protocols static route <subnet> next-hop <address> bfd profile <profile>

Configure BFD failure detection for a single-hop static route.

BFD uses the next-hop IP address as the BFD peer IP address and applies settings from the specified BFD profile.

Example:

```
set protocols static route 10.0.0.0/24 next-hop 192.0.2.1 bfd profile datacenter-fast
```

set protocols static route <subnet> next-hop <address> bfd multi-hop source-address <address>

Configure BFD failure detection for a multi-hop static route.

BFD uses the next-hop IP address as the BFD peer IP address and the multi-hop source IP address as the local source IP for outgoing BFD packets.

Example:

```
set protocols static route 10.0.0.0/24 next-hop 198.51.100.33 bfd multi-hop source-
↪address 198.51.100.1
```

set protocols static route6 <subnet> next-hop <address> bfd profile <profile>

Configure BFD failure detection for a single-hop IPv6 static route.

BFD uses the next-hop IPv6 address as the BFD peer IPv6 address and applies settings from the specified BFD profile.

Example:

```
set protocols static route6 2001:db8:1::/64 next-hop 2001:db8::1 bfd profile datacenter-
↪fast
```

set protocols static route6 <subnet> next-hop <address> bfd multi-hop source-address <address>

Configure BFD failure detection for a multi-hop IPv6 static route.

BFD uses the next-hop IPv6 address as the BFD peer IPv6 address and the multi-hop source IPv6 address as the local source IPv6 address for outgoing BFD packets.

Example:

```
set protocols static route6 2001:db8:1::/64 next-hop 2001:db8:100::33 bfd multi-hop
↪ source-address 2001:db8::1
```

7.9.4 BGP

BGP is one of the Exterior Gateway Protocols and the de facto standard interdomain routing protocol. The latest BGP version is 4. BGP-4 is described in [RFC 1771](#) and updated by [RFC 4271](#). [RFC 2858](#) adds multiprotocol support to BGP.

VyOS makes use of FRR (Free Range Routing) and we would like to thank them for their effort!

Basic Concepts

Autonomous Systems

From [RFC 1930](#):

An AS is a connected group of one or more IP prefixes run by one or more network operators which has a SINGLE and CLEARLY DEFINED routing policy.

Each AS (Autonomous System) has an identifying number associated with it called an ASN (Autonomous System Number). This is a two octet value ranging in value from 1 to 65535. The AS numbers 64512 through 65535 are defined as private AS numbers. Private AS numbers must not be advertised on the global Internet. The 2-byte AS number range has been exhausted. 4-byte AS numbers are specified in [RFC 6793](#), and provide a pool of 4294967296 AS numbers.

The ASN is one of the essential elements of BGP. BGP is a distance vector routing protocol, and the AS-Path framework provides distance vector metric and loop detection to BGP.

```
set protocols bgp system-as <asn>
```

Set local ASN that this router represents. This is a mandatory option!

Address Families

Multiprotocol extensions enable BGP to carry routing information for multiple network layer protocols. BGP supports an Address Family Identifier (AFI) for IPv4 and IPv6.

Route Selection

The route selection process used by FRR's BGP implementation uses the following decision criterion, starting at the top of the list and going towards the bottom until one of the factors can be used.

1. **Weight check**

Prefer higher local weight routes to lower routes.

2. **Local preference check**

Prefer higher local preference routes to lower.

3. **Local route check**

Prefer local routes (statics, aggregates, redistributed) to received routes.

4. **AS path length check**

Prefer shortest hop-count AS_PATHs.

5. **Origin check**

Prefer the lowest origin type route. That is, prefer IGP origin routes to EGP, to Incomplete routes.

6. **MED check**

Where routes with a MED were received from the same AS, prefer the route with the lowest MED.

7. **External check**

Prefer the route received from an external, eBGP peer over routes received from other types of peers.

8. **IGP cost check**

Prefer the route with the lower IGP cost.

9. **Multi-path check**

If multi-pathing is enabled, then check whether the routes not yet distinguished in preference may be considered equal. If `bgp bestpath as-path multipath-relax` is set, all such routes are considered equal, otherwise routes received via iBGP with identical AS_PATHs or routes received from eBGP neighbours in the same AS are considered equal.

10. **Already-selected external check**

Where both routes were received from eBGP peers, then prefer the route which is already selected. Note that this check is not applied if `bgp bestpath compare-routerid` is configured. This check can prevent some cases of oscillation.

11. **Router-ID check**

Prefer the route with the lowest router-ID. If the route has an `ORIGINATOR_ID` attribute, through iBGP reflection, then that router ID is used, otherwise the router-ID of the peer the route was received from is used.

12. **Cluster-List length check**

The route with the shortest cluster-list length is used. The cluster-list reflects the iBGP reflection path the route has taken.

13. **Peer address**

Prefer the route received from the peer with the higher transport layer address, as a last-resort tie-breaker.

Capability Negotiation

When adding IPv6 routing information exchange feature to BGP. There were some proposals. IETF (Internet Engineering Task Force) IDR (Inter Domain Routing) adopted a proposal called Multiprotocol Extension for BGP. The specification is described in [RFC 2283](#). The protocol does not define new protocols. It defines new attributes to existing BGP. When it is used exchanging IPv6 routing information it is called BGP-4+. When it is used for exchanging multicast routing information it is called MBGP.

bgpd supports Multiprotocol Extension for BGP. So if a remote peer supports the protocol, *bgpd* can exchange IPv6 and/or multicast routing information.

Traditional BGP did not have the feature to detect a remote peer's capabilities, e.g. whether it can handle prefix types other than IPv4 unicast routes. This was a big problem using Multiprotocol Extension for BGP in an operational network. [RFC 2842](#) adopted a feature called Capability Negotiation. *bgpd* use this Capability Negotiation to detect the remote peer's capabilities. If a peer is only configured as an IPv4 unicast neighbor, *bgpd* does not send these Capability Negotiation packets (at least not unless other optional BGP features require capability negotiation).

By default, FRR will bring up peering with minimal common capability for the both sides. For example, if the local router has unicast and multicast capabilities and the remote router only has unicast capability the local router will establish the connection with unicast only capability. When there are no common capabilities, FRR sends Unsupported Capability error and then resets the connection.

Configuration

BGP Router Configuration

First of all you must configure BGP router with the ASN. The AS number is an identifier for the autonomous system. The BGP protocol uses the AS number for detecting whether the BGP connection is internal or external. VyOS does not have a special command to start the BGP process. The BGP process starts when the first neighbor is configured.

```
set protocols bgp system-as <asn>
```

Set local autonomous system number that this router represents. This is a mandatory option!

Peers Configuration

Defining Peers

```
set protocols bgp neighbor <address|interface> remote-as <asn>
```

This command creates a new neighbor whose remote-as is <asn>. The neighbor address can be an IPv4 address or an IPv6 address or an interface to use for the connection. The command is applicable for peer and peer group.

```
set protocols bgp neighbor <address|interface> remote-as internal
```

Create a peer as you would when you specify an ASN, except that if the peers ASN is different than mine as specified under the `protocols bgp <asn>` command the connection will be denied.

```
set protocols bgp neighbor <address|interface> remote-as external
```

Create a peer as you would when you specify an ASN, except that if the peers ASN is the same as mine as specified under the `protocols bgp <asn>` command the connection will be denied.

```
set protocols bgp neighbor <address|interface> remote-as auto
```

Create a peer as you would when you specify an ASN, except that the peers remote ASN is detected automatically from the OPEN message.

set protocols bgp neighbor <address|interface> local-role <role> [strict]

BGP roles are defined in RFC [RFC 9234](#) and provide an easy way to add route leak prevention, detection and mitigation. The local Role value is negotiated with the new BGP Role capability which has a built-in check of the corresponding value. In case of a mismatch the new OPEN Roles Mismatch Notification <2, 11> would be sent. The correct Role pairs are:

Provider - Customer

Peer - Peer

RS-Server - RS-Client

If **strict** is set the BGP session won't become established until the BGP neighbor sets local Role on its side. This configuration parameter is defined in RFC [RFC 9234](#) and is used to enforce the corresponding configuration at your counter-parts side.

Routes that are sent from provider, rs-server, or the peer local-role (or if received by customer, rs-client, or the peer local-role) will be marked with a new Only to Customer (OTC) attribute.

Routes with this attribute can only be sent to your neighbor if your local-role is provider or rs-server. Routes with this attribute can be received only if your local-role is customer or rs-client.

In case of peer-peer relationship routes can be received only if OTC value is equal to your neighbor AS number.

All these rules with OTC will help to detect and mitigate route leaks and happen automatically if local-role is set.

set protocols bgp neighbor <address|interface> shutdown

This command disables the peer or peer group. To re-enable the peer, use the delete form of this command.

set protocols bgp neighbor <address|interface> description <text>

Set description of the peer or peer group.

set protocols bgp neighbor <address|interface> update-source <address|interface>

Specify the IPv4 source address to use for the BGP session to this neighbor, may be specified as either an IPv4 address directly or as an interface name.

Capability Negotiation

set protocols bgp neighbor <address|interface> capability dynamic

This command would allow the dynamic update of capabilities over an established BGP session.

set protocols bgp neighbor <address|interface> capability extended-nexthop

Allow bgp to negotiate the extended-nexthop capability with it's peer. If you are peering over a IPv6 Link-Local address then this capability is turned on automatically. If you are peering over a IPv6 Global Address then turning on this command will allow BGP to install IPv4 routes with IPv6 nexthops if you do not have IPv4 configured on interfaces.

set protocols bgp neighbor <address|interface> disable-capability-negotiation

Suppress sending Capability Negotiation as OPEN message optional parameter to the peer. This command only affects the peer is configured other than IPv4 unicast configuration.

When remote peer does not have capability negotiation feature, remote peer will not send any capabilities at all. In that case, bgp configures the peer with configured capabilities.

You may prefer locally configured capabilities more than the negotiated capabilities even though remote peer sends capabilities. If the peer is configured by `override-capability`, VyOS ignores received capabilities then override negotiated capabilities with configured values.

Additionally you should keep in mind that this feature fundamentally disables the ability to use widely deployed BGP features. BGP unnumbered, hostname support, AS4, Addpath, Route Refresh, ORF, Dynamic Capabilities, and graceful restart.

set protocols bgp neighbor <address|interface> override-capability

This command allow override the result of Capability Negotiation with local configuration. Ignore remote peer's capability value.

set protocols bgp neighbor <address|interface> strict-capability-match

This command forces strictly compare remote capabilities and local capabilities. If capabilities are different, send Unsupported Capability error then reset connection.

You may want to disable sending Capability Negotiation OPEN message optional parameter to the peer when remote peer does not implement Capability Negotiation. Please use `disable-capability-negotiation` command to disable the feature.

Peer Parameters

set protocols bgp neighbor <address|interface> address-family <ipv4-unicast|ipv6-unicast> allowas-in number <number>

This command accept incoming routes with AS path containing AS number with the same value as the current system AS. This is used when you want to use the same AS number in your sites, but you can't connect them directly.

The number parameter (1-10) configures the amount of accepted occurrences of the system AS number in AS path.

This command is only allowed for eBGP peers. It is not applicable for peer groups.

set protocols bgp neighbor <address|interface> address-family <ipv4-unicast|ipv6-unicast> as-override

This command override AS number of the originating router with the local AS number.

Usually this configuration is used in PEs (Provider Edge) to replace the incoming customer AS number so the connected CE (Customer Edge) can use the same AS number as the other customer sites. This allows customers of the provider network to use the same AS number across their sites.

This command is only allowed for eBGP peers.

set protocols bgp neighbor <address|interface> address-family <ipv4-unicast|ipv6-unicast> attribute-unchanged <as-path|med|next-hop>

This command specifies attributes to be left unchanged for advertisements sent to a peer or peer group.

set protocols bgp neighbor <address|interface> address-family <ipv4-unicast|ipv6-unicast> maximum-prefix <number>

This command specifies a maximum number of prefixes we can receive from a given peer. If this number is exceeded, the BGP session will be destroyed. The number range is 1 to 4294967295.

set protocols bgp neighbor <address|interface> address-family <ipv4-unicast|ipv6-unicast> nexthop-self

This command forces the BGP speaker to report itself as the next hop for an advertised route it advertised to a neighbor.

**set protocols bgp neighbor <address|interface> address-family <ipv4-unicast|ipv6-unicast>
remove-private-as**

This command removes the private ASN of routes that are advertised to the configured peer. It removes only private ASNs on routes advertised to EBGP peers.

If the AS-Path for the route has only private ASNs, the private ASNs are removed.

If the AS-Path for the route has a private ASN between public ASNs, it is assumed that this is a design choice, and the private ASN is not removed.

**set protocols bgp neighbor <address|interface> address-family <ipv4-unicast|ipv6-unicast>
soft-reconfiguration inbound**

Changes in BGP policies require the BGP session to be cleared. Clearing has a large negative impact on network operations. Soft reconfiguration enables you to generate inbound updates from a neighbor, change and activate BGP policies without clearing the BGP session.

This command specifies that route updates received from this neighbor will be stored unmodified, regardless of the inbound policy. When inbound soft reconfiguration is enabled, the stored updates are processed by the new policy configuration to create new inbound updates.

Note: Storage of route updates uses memory. If you enable soft reconfiguration inbound for multiple neighbors, the amount of memory used can become significant.

**set protocols bgp neighbor <address|interface> address-family <ipv4-unicast|ipv6-unicast>
weight <number>**

This command specifies a default weight value for the neighbor's routes. The number range is 1 to 65535.

set protocols bgp neighbor <address|interface> advertisement-interval <seconds>

This command specifies the minimum route advertisement interval for the peer. The interval value is 0 to 600 seconds, with the default advertisement interval being 0.

set protocols bgp neighbor <address|interface> disable-connected-check

This command allows peerings between directly connected eBGP peers using loopback addresses without adjusting the default TTL of 1.

set protocols bgp neighbor <address|interface> disable-send-community <extended|standard>

This command specifies that the community attribute should not be sent in route updates to a peer. By default community attribute is sent.

set protocols bgp neighbor <address|interface> ebgp-multihop <number>

This command allows sessions to be established with eBGP neighbors when they are multiple hops away. When the neighbor is not directly connected and this knob is not enabled, the session will not establish. The number of hops range is 1 to 255. This command is mutually exclusive with `ttl-security hops`.

set protocols bgp neighbor <address|interface> local-as <asn> [no-prepend] [replace-as]

Specify an alternate AS for this BGP process when interacting with the specified peer or peer group. With no modifiers, the specified local-as is prepended to the received AS_PATH when receiving routing updates from the peer, and prepended to the outgoing AS_PATH (after the process local AS) when transmitting local routes to the peer.

If the no-prepend attribute is specified, then the supplied local-as is not prepended to the received AS_PATH.

If the `replace-as` attribute is specified, then only the supplied local-as is prepended to the `AS_PATH` when transmitting local-route updates to this peer.

Note: This command is only allowed for eBGP peers.

set protocols bgp neighbor <address|interface> passive

Configures the BGP speaker so that it only accepts inbound connections from, but does not initiate outbound connections to the peer or peer group.

set protocols bgp neighbor <address|interface> password <text>

This command specifies a MD5 password to be used with the tcp socket that is being used to connect to the remote peer.

set protocols bgp neighbor <address|interface> ttl-security hops <number>

This command enforces Generalized TTL Security Mechanism (GTSM), as specified in [RFC 5082](#). With this command, only neighbors that are specified number of hops away will be allowed to become neighbors. The number of hops range is 1 to 254. This command is mutually exclusive with `ebgp-multihop`.

Peer Groups

Peer groups are used to help improve scaling by generating the same update information to all members of a peer group. Note that this means that the routes generated by a member of a peer group will be sent back to that originating peer with the originator identifier attribute set to indicated the originating peer. All peers not associated with a specific peer group are treated as belonging to a default peer group, and will share updates.

set protocols bgp peer-group <name>

This command defines a new peer group. You can specify to the group the same parameters that you can specify for specific neighbors.

Note: If you apply a parameter to an individual neighbor IP address, you override the action defined for a peer group that includes that IP address.

set protocols bgp neighbor <address|interface> peer-group <name>

This command bind specific peer to peer group with a given name.

Network Advertisement Configuration

set protocols bgp address-family <ipv4-unicast|ipv6-unicast> network <prefix>

This command is used for advertising IPv4 or IPv6 networks.

Note: By default, the BGP prefix is advertised even if it's not present in the routing table. This behaviour differs from the implementation of some vendors.

set protocols bgp parameters network-import-check

This configuration modifies the behavior of the network statement. If you have this configured the underlying network must exist in the routing table.

```
set protocols bgp neighbor <address|interface> address-family <ipv4-unicast|ipv6-unicast>
default-originate [route-map <name>]
```

By default, VyOS does not advertise a default route (0.0.0.0/0) even if it is in routing table. When you want to announce default routes to the peer, use this command. Using optional argument `route-map` you can inject the default route to given neighbor only if the conditions in the route map are met.

Route Aggregation Configuration

```
set protocols bgp address-family <ipv4-unicast|ipv6-unicast> aggregate-address <prefix>
```

This command specifies an aggregate address. The router will also announce longer-prefixes inside of the aggregate address.

```
set protocols bgp address-family <ipv4-unicast|ipv6-unicast> aggregate-address <prefix>
as-set
```

This command specifies an aggregate address with a mathematical set of autonomous systems. This command summarizes the `AS_PATH` attributes of all the individual routes.

```
set protocols bgp address-family <ipv4-unicast|ipv6-unicast> aggregate-address <prefix>
summary-only
```

This command specifies an aggregate address and provides that longer-prefixes inside of the aggregate address are suppressed before sending BGP updates out to peers.

```
set protocols bgp neighbor <address|interface> address-family <ipv4-unicast|ipv6-unicast>
unsuppress-map <name>
```

This command applies route-map to selectively unsuppress prefixes suppressed by summarisation.

Redistribution Configuration

```
set protocols bgp address-family <ipv4-unicast|ipv6-unicast> redistribute
```

This command redistributes routing information from the given route source to the BGP process. There are six modes available for route source: `connected`, `kernel`, `ospf`, `rip`, `static`, `table`.

```
set protocols bgp address-family <ipv4-unicast|ipv6-unicast> redistribute metric <number>
```

This command specifies metric (MED) for redistributed routes. The metric range is 0 to 4294967295. There are six modes available for route source: `connected`, `kernel`, `ospf`, `rip`, `static`, `table`.

```
set protocols bgp address-family <ipv4-unicast|ipv6-unicast> redistribute route-map
<name>
```

This command allows to use route map to filter redistributed routes. There are six modes available for route source: `connected`, `kernel`, `ospf`, `rip`, `static`, `table`.

General Configuration

Common parameters

set protocols bgp parameters allow-martian-nexthop

When a peer receives a martian nexthop as part of the NLRI for a route permit the nexthop to be used as such, instead of rejecting and resetting the connection.

set protocols bgp parameters router-id <id>

This command specifies the router-ID. If router ID is not specified it will use the highest interface IP address.

set protocols bgp address-family <ipv4-unicast|ipv6-unicast> maximum-paths <ebgp|ibgp> <number>

This command defines the maximum number of parallel routes that the BGP can support. In order for BGP to use the second path, the following attributes have to match: Weight, Local Preference, AS Path (both AS number and AS path length), Origin code, MED, IGP metric. Also, the next hop address for each path must be different.

set protocols bgp parameters no-hard-administrative-reset

Do not send Hard Reset CEASE Notification for “Administrative Reset” events. When set and Graceful Restart Notification capability is exchanged between the peers, Graceful Restart procedures apply, and routes will be retained.

set protocols bgp parameters log-neighbor-changes

This command enable logging neighbor up/down changes and reset reason.

set protocols bgp parameters no-client-to-client-reflection

This command disables route reflection between route reflector clients. By default, the clients of a route reflector are not required to be fully meshed and the routes from a client are reflected to other clients. However, if the clients are fully meshed, route reflection is not required. In this case, use the no-client-to-client-reflection command to disable client-to-client reflection.

set protocols bgp parameters no-fast-external-failover

Disable immediate session reset if peer’s connected link goes down.

set protocols bgp parameters no-ipv6-auto-ra

By default, FRR sends router advertisement packets when Extended Next Hop is on or when a connection is established directly using the device name (Unnumbered BGP). Setting this option prevents FRR from sending router advertisement packets, but could break Unnumbered BGP.

set protocols bgp listen range <prefix> peer-group <name>

This command is useful if one desires to loosen the requirement for BGP to have strictly defined neighbors. Specifically what is allowed is for the local router to listen to a range of IPv4 or IPv6 addresses defined by a prefix and to accept BGP open messages. When a TCP connection (and subsequently a BGP open message) from within this range tries to connect the local router then the local router will respond and connect with the parameters that are defined within the peer group. One must define a peer-group for each range that is listed. If no peer-group is defined then an error will keep you from committing the configuration.

set protocols bgp listen limit <number>

This command goes hand in hand with the listen range command to limit the amount of BGP neighbors that are allowed to connect to the local router. The limit range is 1 to 5000.

set protocols bgp parameters ebgp-requires-policy

This command changes the eBGP behavior of FRR. By default FRR enables [RFC 8212](#) functionality which affects how eBGP routes are advertised, namely no routes are advertised across eBGP sessions without some sort of egress route-map/policy in place. In VyOS however we have this RFC functionality disabled by default so that we can preserve backwards compatibility with older versions of VyOS. With this option one can enable [RFC 8212](#) functionality to operate.

```
set protocols bgp parameters labeled-unicast <explicit-null | ipv4-explicit-null |  
ipv6-explicit-null>
```

By default, locally advertised prefixes use the implicit-null label to encode in the outgoing NLRI.

The following command uses the explicit-null label value for all the BGP instances.

Administrative Distance

```
set protocols bgp parameters distance global <external|internal|local> <distance>
```

This command change distance value of BGP. The arguments are the distance values for external routes, internal routes and local routes respectively. The distance range is 1 to 255.

```
set protocols bgp parameters distance prefix <subnet> distance <distance>
```

This command sets the administrative distance for a particular route. The distance range is 1 to 255.

Note: Routes with a distance of 255 are effectively disabled and not installed into the kernel.

BGP message queue limits

```
set protocols bgp parameters input-queue-limit <messages>
```

Set the BGP Input Queue limit for all peers during message parsing. Increase this only if you have the memory to handle large queues of messages at once.

The default is 10000.

```
set protocols bgp parameters output-queue-limit <messages>
```

Set the BGP Output Queue limit for all peers during message parsing. Increase this only if you have the memory to handle large queues of messages at once.

The default is 10000.

Timers

```
set protocols bgp timers holdtime <seconds>
```

This command specifies hold-time in seconds. The timer range is 4 to 65535. The default value is 180 second. If you set value to 0 VyOS will not hold routes.

```
set protocols bgp timers keepalive <seconds>
```

This command specifies keep-alive time in seconds. The timer can range from 4 to 65535. The default value is 60 second.

BGP read-only mode

set protocols bgp parameters update-delay max-delay <seconds>

Enable BGP read-only mode after a BGP process restart or clearing all BGP sessions, and configure the mode's maximum delay.

In read-only mode, BGP suspends best-path selection and does not send routing updates to peers. The mode ends when the router receives all expected End-of-RIB (EOR) messages from BGP peers or when the maximum delay is reached, whichever occurs first.

The router expects EOR messages from all configured BGP peers except those that are administratively shut down. If the `establish-wait` parameter is configured, the router waits for EOR messages only from BGP peers that reach the Established state within the `establish-wait` window.

The allowed range is 0 to 3600 seconds. By default, this feature is disabled.

Example:

```
set protocols bgp parameters update-delay max-delay 120
```

set protocols bgp parameters update-delay establish-wait <seconds>

Configure how long the router waits for BGP peers to reach the Established state after read-only mode begins.

The router waits for EOR messages only from BGP peers that reach the Established state within the `establish-wait` window. The `establish-wait` value must not exceed the `max-delay` value.

The allowed range is 1 to 3600 seconds. This parameter requires `max-delay` to be configured.

Example:

```
set protocols bgp parameters update-delay establish-wait 90
```

Route Dampening

When a route fails, a routing update is sent to withdraw the route from the network's routing tables. When the route is re-enabled, the change in availability is also advertised. A route that continually fails and returns requires a great deal of network traffic to update the network about the route's status.

Route dampening, described in [RFC 2439](#), enables you to identify routes that repeatedly fail and return. If route dampening is enabled, an unstable route accumulates penalties each time the route fails and returns. If the accumulated penalties exceed a threshold, the route is no longer advertised. This is route suppression. Routes that have been suppressed are re-entered into the routing table only when the amount of their penalty falls below a threshold.

A penalty of 1000 is assessed each time the route fails. When the penalties reach a predefined threshold (`suppress-value`), the router stops advertising the route.

Once a route is assessed a penalty, the penalty is decreased by half each time a predefined amount of time elapses (`half-life-time`). When the accumulated penalties fall below a predefined threshold (`reuse-value`), the route is unsuppressed and added back into the BGP routing table.

No route is suppressed indefinitely. `Maximum-suppress-time` defines the maximum time a route can be suppressed before it is re-advertised.

set protocols bgp parameters dampening half-life <minutes>

This command defines the amount of time in minutes after which a penalty is reduced by half. The timer range is 10 to 45 minutes.

set protocols bgp parameters dampening re-use <seconds>

This command defines the accumulated penalty amount at which the route is re-advertised. The penalty range is 1 to 20000.

set protocols bgp parameters dampening start-suppress-time <seconds>

This command defines the accumulated penalty amount at which the route is suppressed. The penalty range is 1 to 20000.

set protocols bgp parameters dampening max-suppress-time <seconds>

This command defines the maximum time in minutes that a route is suppressed. The timer range is 1 to 255 minutes.

Route Selection Configuration

set protocols bgp parameters always-compare-med

This command provides to compare the MED on routes, even when they were received from different neighbouring ASes. Setting this option makes the order of preference of routes more defined, and should eliminate MED induced oscillations.

set protocols bgp parameters bestpath as-path confed

This command specifies that the length of confederation path sets and sequences should be taken into account during the BGP best path decision process.

set protocols bgp parameters bestpath as-path multipath-relax

This command specifies that BGP decision process should consider paths of equal AS_PATH length candidates for multipath computation. Without the knob, the entire AS_PATH must match for multipath computation.

set protocols bgp parameters bestpath as-path ignore

Ignore AS_PATH length when selecting a route

set protocols bgp parameters bestpath compare-routerid

Ensure that when comparing routes where both are equal on most metrics, including local-pref, AS_PATH length, IGP cost, MED, that the tie is broken based on router-ID.

If this option is enabled, then the already-selected check, where already selected eBGP routes are preferred, is skipped.

If a route has an ORIGINATOR_ID attribute because it has been reflected, that ORIGINATOR_ID will be used. Otherwise, the router-ID of the peer the route was received from will be used.

The advantage of this is that the route-selection (at this point) will be more deterministic. The disadvantage is that a few or even one lowest-ID router may attract all traffic to otherwise-equal paths because of this check. It may increase the possibility of MED or IGP oscillation, unless other measures were taken to avoid these. The exact behaviour will be sensitive to the iBGP and reflection topology.

set protocols bgp parameters bestpath med confed

This command specifies that BGP considers the MED when comparing routes originated from different sub-ASs within the confederation to which this BGP speaker belongs. The default state, where the MED attribute is not considered.

set protocols bgp parameters bestpath med missing-as-worst

This command specifies that a route with a MED is always considered to be better than a route without a MED by causing the missing MED attribute to have a value of infinity. The default state, where the missing MED attribute is considered to have a value of zero.

set protocols bgp parameters default local-pref

This command specifies the default local preference value. The local preference range is 0 to 4294967295.

set protocols bgp parameters deterministic-med

This command provides to compare different MED values that advertised by neighbours in the same AS for routes selection. When this command is enabled, routes from the same autonomous system are grouped together, and the best entries of each group are compared.

set protocols bgp address-family ipv4-unicast network <prefix> backdoor

This command allows the router to prefer route to specified prefix learned via IGP through backdoor link instead of a route to the same prefix learned via EBGp.

Route Filtering Configuration

In order to control and modify routing information that is exchanged between peers you can use route-map, filter-list, prefix-list, distribute-list.

For inbound updates the order of preference is:

- route-map
- filter-list
- prefix-list, distribute-list

For outbound updates the order of preference is:

- prefix-list, distribute-list
- filter-list
- route-map

Note: The attributes `prefix-list` and `distribute-list` are mutually exclusive, and only one command (`distribute-list` or `prefix-list`) can be applied to each inbound or outbound direction for a particular neighbor.

set protocols bgp neighbor <address|interface> address-family <ipv4-unicast|ipv6-unicast> distribute-list <export|import> <number>

This command applies the access list filters named in <number> to the specified BGP neighbor to restrict the routing information that BGP learns and/or advertises. The arguments `export` and `import` specify the direction in which the access list are applied.

set protocols bgp neighbor <address|interface> address-family <ipv4-unicast|ipv6-unicast> prefix-list <export|import> <name>

This command applies the prefix list filters named in <name> to the specified BGP neighbor to restrict the routing information that BGP learns and/or advertises. The arguments `export` and `import` specify the direction in which the prefix list are applied.

set protocols bgp neighbor <address|interface> address-family <ipv4-unicast|ipv6-unicast> route-map <export|import> <name>

This command applies the route map named in <name> to the specified BGP neighbor to control and modify routing information that is exchanged between peers. The arguments `export` and `import` specify the direction in which the route map are applied.

```
set protocols bgp neighbor <address|interface> address-family <ipv4-unicast|ipv6-unicast>  
filter-list <export|import> <name>
```

This command applies the AS path access list filters named in <name> to the specified BGP neighbor to restrict the routing information that BGP learns and/or advertises. The arguments `export` and `import` specify the direction in which the AS path access list are applied.

```
set protocols bgp neighbor <address|interface> address-family <ipv4-unicast|ipv6-unicast>  
capability orf <receive|send>
```

This command enables the ORF capability (described in [RFC 5291](#)) on the local router, and enables ORF capability advertisement to the specified BGP peer. The `receive` keyword configures a router to advertise ORF receive capabilities. The `send` keyword configures a router to advertise ORF send capabilities. To advertise a filter from a sender, you must create an IP prefix list for the specified BGP peer applied in inbound direction.

```
set protocols bgp neighbor <address|interface> solo
```

This command prevents from sending back prefixes learned from the neighbor.

BGP Scaling Configuration

BGP routers connected inside the same AS through BGP belong to an internal BGP session, or IBGP. In order to prevent routing table loops, IBGP speaker does not advertise IBGP-learned routes to other IBGP speaker (Split Horizon mechanism). As such, IBGP requires a full mesh of all peers. For large networks, this quickly becomes unscalable.

There are two ways that help us to mitigate the BGPs full-mesh requirement in a network:

- Using BGP route-reflectors
- Using BGP confederation

Route Reflector Configuration

Introducing route reflectors removes the need for the full-mesh. When you configure a route reflector you have to tell the router whether the other IBGP router is a client or non-client. A client is an IBGP router that the route reflector will “reflect” routes to, the non-client is just a regular IBGP neighbor. Route reflectors mechanism is described in [RFC 4456](#) and updated by [RFC 7606](#).

```
set protocols bgp neighbor <address> address-family <ipv4-unicast|ipv6-unicast>  
route-reflector-client
```

This command specifies the given neighbor as route reflector client.

```
set protocols bgp parameters cluster-id <id>
```

This command specifies cluster ID which identifies a collection of route reflectors and their clients, and is used by route reflectors to avoid looping. By default cluster ID is set to the BGP router id value, but can be set to an arbitrary 32-bit value.

Confederation Configuration

A BGP confederation divides our AS into sub-ASes to reduce the number of required IBGP peerings. Within a sub-AS we still require full-mesh IBGP but between these sub-ASes we use something that looks like EBGP but behaves like IBGP (called confederation BGP). Confederation mechanism is described in [RFC 5065](#)

set protocols bgp parameters confederation identifier <asn>

This command specifies a BGP confederation identifier. <asn> is the number of the autonomous system that internally includes multiple sub-autonomous systems (a confederation).

set protocols bgp parameters confederation peers <subasn>

This command sets other confederations <subasn> as members of autonomous system specified by confederation identifier <asn>.

Operational Mode Commands

Show

show bgp <ipv4|ipv6>

This command displays all entries in BGP routing table.

```
BGP table version is 10, local router ID is 10.0.35.3, vrf id 0
Default local pref 100, local AS 65000
Status codes: s suppressed, d damped, h history, * valid, > best, = multipath,
               i internal, r RIB-failure, S Stale, R Removed
Nexthop codes: @NNN nexthop's vrf id, < announce-nh-self
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

   Network          Next Hop           Metric LocPrf Weight Path
*> 198.51.100.0/24  10.0.34.4             0             0 65004 i
*> 203.0.113.0/24  10.0.35.5             0             0 65005 i

Displayed 2 routes and 2 total paths
```

show bgp <ipv4|ipv6> <address|prefix>

This command displays information about the particular entry in the BGP routing table.

```
BGP routing table entry for 198.51.100.0/24
Paths: (1 available, best #1, table default)
  Advertised to non peer-group peers:
    10.0.13.1 10.0.23.2 10.0.34.4 10.0.35.5
  65004
    10.0.34.4 from 10.0.34.4 (10.0.34.4)
      Origin IGP, metric 0, valid, external, best (First path received)
      Last update: Wed Jan  6 12:18:53 2021
```

show bgp cidr-only

This command displays routes with classless interdomain routing (CIDR).

show bgp <ipv4|ipv6> community <value>

This command displays routes that belong to specified BGP communities. Valid value is a community number in the range from 1 to 4294967200, or AA:NN (autonomous system-community number/2-byte number), no-export, local-as, or no-advertise.

show bgp <ipv4|ipv6> community-list <name>

This command displays routes that are permitted by the BGP community list.

show bgp <ipv4|ipv6> dampening dampened-paths

This command displays BGP dampened routes.

show bgp <ipv4|ipv6> dampening flap-statistics

This command displays information about flapping BGP routes.

show bgp <ipv4|ipv6> filter-list <name>

This command displays BGP routes allowed by the specified AS Path access list.

show bgp <ipv4|ipv6> neighbors <address> advertised-routes

This command displays BGP routes advertised to a neighbor.

show bgp <ipv4|ipv6> neighbors <address> received-routes

This command displays BGP routes originating from the specified BGP neighbor before inbound policy is applied. To use this command inbound soft reconfiguration must be enabled.

show bgp <ipv4|ipv6> neighbors <address> routes

This command displays BGP received-routes that are accepted after filtering.

show bgp <ipv4|ipv6> neighbors <address> dampened-routes

This command displays dampened routes received from BGP neighbor.

show bgp <ipv4|ipv6> regexp <text>

This command displays information about BGP routes whose AS path matches the specified regular expression.

show bgp <ipv4|ipv6> summary

This command displays the status of all BGP connections.

```
IPv4 Unicast Summary:
BGP router identifier 10.0.35.3, local AS number 65000 vrf-id 0
BGP table version 11
RIB entries 5, using 920 bytes of memory
Peers 4, using 82 KiB of memory

Neighbor      V      AS MsgRcvd MsgSent   TblVer  InQ OutQ  Up/Down State/PfxRcd
10.0.13.1      4      65000    148     159       0    0    0 02:16:01         0
10.0.23.2      4      65000    136     143       0    0    0 02:13:21         0
10.0.34.4      4      65004    161     163       0    0    0 02:16:01         1
10.0.35.5      4      65005    162     166       0    0    0 02:16:01         1

Total number of neighbors 4
```

Reset

reset bgp <ipv4|ipv6> <address> [soft [in|out]]

This command resets BGP connections to the specified neighbor IP address. With argument `soft` this command initiates a soft reset. If you do not specify the `in` or `out` options, both inbound and outbound soft reconfiguration are triggered.

reset bgp all

This command resets all BGP connections of given router.

reset bgp <ipv4|ipv6> external

This command resets all external BGP peers of given router.

reset bgp <ipv4|ipv6> peer-group <name> [soft [in|out]]

This command resets BGP connections to the specified peer group. With argument `soft` this command initiates a soft reset. If you do not specify the `in` or `out` options, both inbound and outbound soft reconfiguration are triggered.

Examples

IPv4 peering

A simple eBGP configuration:

Node 1:

```
set protocols bgp system-as 65534
set protocols bgp neighbor 192.168.0.2 ebgp-multihop '2'
set protocols bgp neighbor 192.168.0.2 remote-as '65535'
set protocols bgp neighbor 192.168.0.2 update-source '192.168.0.1'
set protocols bgp neighbor 192.168.0.2 address-family ipv4-unicast
set protocols bgp address-family ipv4-unicast network '172.16.0.0/16'
set protocols bgp parameters router-id '192.168.0.1'
```

Node 2:

```
set protocols bgp system-as 65535
set protocols bgp neighbor 192.168.0.1 ebgp-multihop '2'
set protocols bgp neighbor 192.168.0.1 remote-as '65534'
set protocols bgp neighbor 192.168.0.1 update-source '192.168.0.2'
set protocols bgp neighbor 192.168.0.1 address-family ipv4-unicast
set protocols bgp address-family ipv4-unicast network '172.17.0.0/16'
set protocols bgp parameters router-id '192.168.0.2'
```

Don't forget, the CIDR declared in the network statement **MUST exist in your routing table (dynamic or static)**, the best way to make sure that is true is creating a static route:

Node 1:

```
set protocols static route 172.16.0.0/16 blackhole distance '254'
```

Node 2:


```
set protocols static route 172.17.0.0/16 blackhole distance '254'
```

IPv6 peering

A simple BGP configuration via IPv6.

Node 1:

```
set protocols bgp system-as 65534
set protocols bgp neighbor 2001:db8::2 ebgp-multihop '2'
set protocols bgp neighbor 2001:db8::2 remote-as '65535'
set protocols bgp neighbor 2001:db8::2 update-source '2001:db8::1'
set protocols bgp neighbor 2001:db8::2 address-family ipv6-unicast
set protocols bgp address-family ipv6-unicast network '2001:db8:1::/48'
set protocols bgp parameters router-id '10.1.1.1'
```

Node 2:

```
set protocols bgp system-as 65535
set protocols bgp neighbor 2001:db8::1 ebgp-multihop '2'
set protocols bgp neighbor 2001:db8::1 remote-as '65534'
set protocols bgp neighbor 2001:db8::1 update-source '2001:db8::2'
set protocols bgp neighbor 2001:db8::1 address-family ipv6-unicast
set protocols bgp address-family ipv6-unicast network '2001:db8:2::/48'
set protocols bgp parameters router-id '10.1.1.2'
```

Don't forget, the CIDR declared in the network statement **MUST** exist in your routing table (dynamic or static), the **best way to make sure that is true is creating a static route:**

Node 1:

```
set protocols static route6 2001:db8:1::/48 blackhole distance '254'
```

Node 2:

```
set protocols static route6 2001:db8:2::/48 blackhole distance '254'
```

Route Filtering

Route filter can be applied using a route-map:

Node1:

```
set policy prefix-list AS65535-IN rule 10 action 'permit'
set policy prefix-list AS65535-IN rule 10 prefix '172.16.0.0/16'
set policy prefix-list AS65535-OUT rule 10 action 'deny'
set policy prefix-list AS65535-OUT rule 10 prefix '172.16.0.0/16'
set policy prefix-list6 AS65535-IN rule 10 action 'permit'
set policy prefix-list6 AS65535-IN rule 10 prefix '2001:db8:2::/48'
set policy prefix-list6 AS65535-OUT rule 10 action 'deny'
set policy prefix-list6 AS65535-OUT rule 10 prefix '2001:db8:2::/48'
```

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```

set policy route-map AS65535-IN rule 10 action 'permit'
set policy route-map AS65535-IN rule 10 match ip address prefix-list 'AS65535-IN'
set policy route-map AS65535-IN rule 10 match ipv6 address prefix-list 'AS65535-IN'
set policy route-map AS65535-IN rule 20 action 'deny'
set policy route-map AS65535-OUT rule 10 action 'deny'
set policy route-map AS65535-OUT rule 10 match ip address prefix-list 'AS65535-OUT'
set policy route-map AS65535-OUT rule 10 match ipv6 address prefix-list 'AS65535-OUT'
set policy route-map AS65535-OUT rule 20 action 'permit'

set protocols bgp system-as 65534
set protocols bgp neighbor 2001:db8::2 address-family ipv4-unicast route-map export
↪ 'AS65535-OUT'
set protocols bgp neighbor 2001:db8::2 address-family ipv4-unicast route-map import
↪ 'AS65535-IN'
set protocols bgp neighbor 2001:db8::2 address-family ipv6-unicast route-map export
↪ 'AS65535-OUT'
set protocols bgp neighbor 2001:db8::2 address-family ipv6-unicast route-map import
↪ 'AS65535-IN'

```

Node2:

```

set policy prefix-list AS65534-IN rule 10 action 'permit'
set policy prefix-list AS65534-IN rule 10 prefix '172.17.0.0/16'
set policy prefix-list AS65534-OUT rule 10 action 'deny'
set policy prefix-list AS65534-OUT rule 10 prefix '172.17.0.0/16'
set policy prefix-list6 AS65534-IN rule 10 action 'permit'
set policy prefix-list6 AS65534-IN rule 10 prefix '2001:db8:1::/48'
set policy prefix-list6 AS65534-OUT rule 10 action 'deny'
set policy prefix-list6 AS65534-OUT rule 10 prefix '2001:db8:1::/48'

set policy route-map AS65534-IN rule 10 action 'permit'
set policy route-map AS65534-IN rule 10 match ip address prefix-list 'AS65534-IN'
set policy route-map AS65534-IN rule 10 match ipv6 address prefix-list 'AS65534-IN'
set policy route-map AS65534-IN rule 20 action 'deny'
set policy route-map AS65534-OUT rule 10 action 'deny'
set policy route-map AS65534-OUT rule 10 match ip address prefix-list 'AS65534-OUT'
set policy route-map AS65534-OUT rule 10 match ipv6 address prefix-list 'AS65534-OUT'
set policy route-map AS65534-OUT rule 20 action 'permit'

set protocols bgp system-as 65535
set protocols bgp neighbor 2001:db8::1 address-family ipv4-unicast route-map export
↪ 'AS65534-OUT'
set protocols bgp neighbor 2001:db8::1 address-family ipv4-unicast route-map import
↪ 'AS65534-IN'
set protocols bgp neighbor 2001:db8::1 address-family ipv6-unicast route-map export
↪ 'AS65534-OUT'
set protocols bgp neighbor 2001:db8::1 address-family ipv6-unicast route-map import
↪ 'AS65534-IN'

```

We could expand on this and also deny link local and multicast in the rule 20 action deny.

7.9.5 Failover

Failover routes are manually configured network paths used only while their health-check targets are reachable. If the target stops responding, VyOS removes the route from the routing table and reinstalls it once the target recovers.

Configuration

Use the following commands to configure failover routes for a specific remote <subnet> reachable via next-hop <address>.

set protocols failover route <subnet> next-hop <address> check target <target-address>

Configure the health check target IP address.

This is typically a highly available host, either within the destination subnet or on the public internet.

Example:

```
set protocols failover route 203.0.113.1/32 next-hop 10.217.37.254 check target 8.8.8.8
```

set protocols failover route <subnet> next-hop <address> check timeout <timeout>

Configure the timeout interval, in seconds, between target health checks.

The valid range is 1 to 300 seconds. The default is 10 seconds.

Example:

```
set protocols failover route 203.0.113.1/32 next-hop 10.217.37.254 check timeout 2
```

set protocols failover route <subnet> next-hop <address> check type <protocol>

Configure the protocol to use for health checks.

The following protocols are available:

- **icmp**: VyOS sends two ICMP echo request packets with a 1-second response timeout. The health check is successful if at least one response is received.
- **arp**: VyOS sends two ARP requests with a 1-second response timeout. The health check is successful if at least one response is received.
- **tcp**: VyOS verifies whether the destination TCP port is open. The health check is successful if a TCP connection is successfully established with the target port.

The default protocol is **icmp**.

Note: When the check type is set to **tcp**, you must also define the target TCP port.

Example:

```
set protocols failover route 203.0.113.1/32 next-hop 10.217.37.254 check type tcp
```

set protocols failover route <subnet> next-hop <address> check port <port>

Configure the destination TCP port on the health check target.

This parameter applies only when the check type is configured as **tcp**.

The valid port range is 1 to 65535.

Example:

```
set protocols failover route 203.0.113.1/32 next-hop 10.217.37.254 check port 443
```

set protocols failover route <subnet> next-hop <address> check policy <policy>

Configure the health check success policy for multiple targets.

The following policies are available:

- **any-available:** The health check succeeds if at least one of the configured targets responds successfully.
- **all-available:** The health check succeeds only if every configured target responds successfully.

The default policy is any-available.

Example:

```
set protocols failover route 203.0.113.1/32 next-hop 10.217.37.254 check policy all-
↪available
```

set protocols failover route <subnet> next-hop <address> interface <interface>

Configure the local interface used to reach the next-hop address.

This parameter is mandatory.

Example:

```
set protocols failover route 203.0.113.1/32 next-hop 10.217.37.254 interface eth0
```

set protocols failover route <subnet> next-hop <address> metric <1-255>

Configure the metric (cost) for the failover route.

The metric defines the route priority. A lower metric value indicates a more preferred route.

The default value is 1.

Example:

```
set protocols failover route 203.0.113.1/32 next-hop 10.217.37.254 metric 50
```

set protocols failover route <subnet> next-hop <address> onlink

Configure the next-hop to be reachable via the assigned interface, even when <address> is outside any subnet configured on that interface.

Example:

```
set protocols failover route 203.0.113.1/32 next-hop 10.217.37.254 onlink
```

Examples

Failover route with a single next-hop and ICMP health check

The following example configures a failover route to 203.0.113.1/32 through next-hop 192.0.2.1 on eth0. The next-hop is monitored with ICMP probes to 192.0.2.1 every 5 seconds, and the route is installed with a metric of 10.

```
set protocols failover route 203.0.113.1/32 next-hop 192.0.2.1 check target '192.0.2.1'
set protocols failover route 203.0.113.1/32 next-hop 192.0.2.1 check timeout '5'
set protocols failover route 203.0.113.1/32 next-hop 192.0.2.1 check type 'icmp'
set protocols failover route 203.0.113.1/32 next-hop 192.0.2.1 interface 'eth0'
set protocols failover route 203.0.113.1/32 next-hop 192.0.2.1 metric '10'
```

Verify the route:

```
vyos@vyos:~$ show ip route 203.0.113.1
Routing entry for 203.0.113.1/32
  Known via "kernel", distance 0, metric 10, best
  Last update 00:00:39 ago
  Flags: Selected
  Status: Installed
  * 192.0.2.1, via eth0, weight 1
```

Two failover routes with different metrics

The following example configures two failover routes to 203.0.113.1/32, each through a different next-hop. The primary next-hop 192.0.2.1 is reached on eth0 with metric 10, and the backup next-hop 198.51.100.1 is reached on eth2 with metric 20. Both next-hops are monitored with ICMP probes every 5 seconds.

While both health checks succeed, the lower-metric route through eth0 is preferred. If the primary target stops responding, its route is removed from the routing table, and traffic falls over to 198.51.100.1 via eth2.

```
set protocols failover route 203.0.113.1/32 next-hop 192.0.2.1 check target '192.0.2.1'
set protocols failover route 203.0.113.1/32 next-hop 192.0.2.1 check timeout '5'
set protocols failover route 203.0.113.1/32 next-hop 192.0.2.1 check type 'icmp'
set protocols failover route 203.0.113.1/32 next-hop 192.0.2.1 interface 'eth0'
set protocols failover route 203.0.113.1/32 next-hop 192.0.2.1 metric '10'

set protocols failover route 203.0.113.1/32 next-hop 198.51.100.1 check target '198.51.
↪100.99'
set protocols failover route 203.0.113.1/32 next-hop 198.51.100.1 check timeout '5'
set protocols failover route 203.0.113.1/32 next-hop 198.51.100.1 check type 'icmp'
set protocols failover route 203.0.113.1/32 next-hop 198.51.100.1 interface 'eth2'
set protocols failover route 203.0.113.1/32 next-hop 198.51.100.1 metric '20'
```

Verify routes:

```
vyos@vyos:~$ show ip route 203.0.113.1
Routing entry for 203.0.113.1/32
  Known via "kernel", distance 0, metric 10, best
  Last update 00:08:06 ago
  Flags: Selected
  Status: Installed
```

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```
* 192.0.2.1, via eth0, weight 1

Routing entry for 203.0.113.1/32
  Known via "kernel", distance 0, metric 20
  Last update 00:08:14 ago
  Flags: None
  Status: Installed
  * 198.51.100.1, via eth2, weight 1
```

7.9.6 IGMP proxy

The IGMP proxy enables routers to forward multicast traffic, such as IPTV or video streams, between different networks, such as home networks and ISPs. The IGMP proxy relies on an upstream interface that faces upstream multicast sources, and one or more downstream interfaces that face local networks or hosts and forward traffic to requesting clients.

IGMP proxy configuration requires exactly one upstream interface and at least one downstream interface.

Configuration

set protocols igmp-proxy interface <interface> role <upstream | downstream>

Configure the operational role for the specified IGMP proxy interface.

- **upstream:** Communicates with upstream multicast sources to retrieve the multicast traffic. A valid configuration requires exactly one upstream interface.
- **downstream:** Communicates with downstream local subnets or hosts to distribute the requested traffic. A valid configuration requires at least one downstream interface.

Example:

```
set protocols igmp-proxy interface eth0 role upstream
set protocols igmp-proxy interface eth1 role downstream
```

set protocols igmp-proxy interface <interface> alt-subnet <network>

Configure an allowed remote subnet for incoming multicast traffic on the specified IGMP proxy interface

By default, the IGMP proxy accepts multicast traffic only from directly connected subnets. If a multicast source resides on a remote network, you must explicitly define the allowed remote subnet to permit the traffic.

You can configure multiple remote subnets for an **upstream** IGMP proxy interface.

Upstream interfaces frequently require this configuration because multicast sources typically reside on external subnets.

Example:

```
set protocols igmp-proxy interface eth0 alt-subnet 10.0.0.0/8
```

set protocols igmp-proxy interface <interface> whitelist <network>

Configure a permitted destination network for multicast traffic requests on the specified IGMP proxy interface.

By default, the IGMP proxy accepts requests for all multicast destination networks. When you define a whitelist, the IGMP proxy forwards requests only for the specified multicast networks.

You can configure multiple whitelist entries per **downstream** IGMP proxy interface.

Example:

```
set protocols igmp-proxy interface eth1 whitelist 239.0.0.0/8
```

set protocols igmp-proxy interface <interface> threshold <1-255>

Configure the Time-to-Live (TTL) threshold for the specified IGMP proxy interface.

The IGMP proxy drops any multicast packet with a TTL value lower than the configured threshold.

Example:

```
set protocols igmp-proxy interface eth0 threshold 5
```

set protocols igmp-proxy disable-quickleave

Disable quickleave mode for the IGMP proxy.

If disabled, the IGMP proxy does not send an upstream Leave message upon receiving a downstream Leave message, preventing the immediate termination of upstream multicast traffic. The IGMP proxy also stops querying downstream interfaces for membership reports. If a downstream client submits a new report, the IGMP proxy discards the message and does not resume the delivery of requested traffic.

Note: Disabling quickleave mode forces the IGMP proxy to act exactly like a standard multicast client on the upstream interface.

Note: Disabling quickleave mode increases the risk of network bandwidth saturation.

Example:

```
set protocols igmp-proxy disable-quickleave
```

set protocols igmp-proxy disable

Disable the IGMP proxy on the router.

Example:

```
set protocols igmp-proxy disable
```

Operation

restart igmp-proxy

Restart the IGMP proxy process.

Example

In this example, the local LAN on interface eth1 operates behind NAT. To allow local clients to receive multicast traffic originating from the 198.51.100.0/24 source network on the WAN interface (eth0), configure the IGMP proxy as follows:

```
set protocols igmp-proxy interface eth0 role upstream
set protocols igmp-proxy interface eth0 alt-subnet 198.51.100.0/24
set protocols igmp-proxy interface eth1 role downstream
```

7.9.7 IS-IS

Intermediate System-to-Intermediate System (IS-IS) is a link-state IGP (Interior Gateway Protocol) described in ISO/IEC 10589, [RFC 1195](#), and [RFC 5308](#). It uses Dijkstra's SPF (Shortest Path First) algorithm to build a network topology database (LSDB (Link State Database)) and compute the shortest path to each destination.

In an IS-IS network, routers are called Intermediate Systems (ISs). They exchange topology information with directly connected neighbors via the IS-IS protocol, whose PDUs are carried directly in Layer 2 frames rather than over IP. IS-IS routers are identified by a NET (Network Entity Title), which ranges from 8 to 20 bytes (typically 10 bytes).

IS-IS vs OSPF

IS-IS builds an LSDB from link-state information flooded by other routers and computes routing paths using Dijkstra's algorithm, just like OSPF. Because both protocols share the same link-state architecture, they make nearly identical path-selection decisions. Given the similarities between IS-IS and OSPF, comparing their routing behaviors is an effective way to understand how a network will respond with either IGP.

Configuration

Mandatory settings

Each IS-IS router must be configured with a unique NET, the CLNS (Connectionless Network Service) equivalent of a Router ID. The 6-byte system identifier portion of the NET must be unique across the IS-IS routing domain.

set protocols isis net <network-entity-title>

Configure the Network Entity Title (NET) for the router.

A typical NET looks like 49.0001.1921.6800.1002.00.

The NET consists of the following parts:

- AFI (Authority and Format Identifier) (49): IS-IS conventionally uses AFI value 49 for private addressing.
- Area identifier (0001): The area number within the IS-IS routing domain (in this example, Area 1).
- System identifier (1921.6800.1002): Uniquely identifies the router within the IS-IS routing domain. We recommend deriving this value from the router IP address or MAC address. To construct the system identifier from an IPv4 address (for example, 192.168.1.2):
 - Pad each octet with leading zeros: 192.168.1.2 → 192.168.001.002.
 - Regroup the digits into three 4-digit blocks: 192.168.001.002 → 1921.6800.1002.
- NET selector (00): Must always be 00 to indicate the local system.

Example:


```
set protocols isis net 49.0001.1921.6800.1002.00
```

set protocols isis interface <interface>

Enable IS-IS on the specified interface.

This enables the router to form adjacencies with directly connected peers.

Example:

```
set protocols isis interface eth0
```

IS-IS global configuration

set protocols isis dynamic-hostname

Enable Dynamic Hostname TLV (Type-Length-Value) support on the router (RFC 5301).

This enables the router to include its human-readable system name alongside its System ID in IS-IS advertisements for easier peer identification.

Example:

```
set protocols isis dynamic-hostname
```

set protocols isis level <level-1 | level-1-2 | level-2>

Configure the IS-IS level at which the router operates:

- level-1: Participates exclusively in Level-1 (intra-area) routing.
- level-1-2: Participates in both Level-1 (intra-area) and Level-2 (inter-area) routing.
- level-2: Participates exclusively in Level-2 (inter-area) routing.

Example:

```
set protocols isis level level-1
```

set protocols isis lsp-mtu <128-4352>

Configure the MTU size, in bytes, for originating or receiving IS-IS LSPs (Link State PDUs).

Example:

```
set protocols isis lsp-mtu 1400
```

set protocols isis metric-style <narrow | transition | wide>

Configure the TLV format (metric style) the router uses when originating or processing IS-IS LSPs:

- narrow: Originates and processes only the original ISO/IEC 10589 TLVs.
- transition: Originates and processes both narrow and wide TLV formats.
- wide: Originates and processes only extended TLVs.

Example:

```
set protocols isis metric-style wide
```

set protocols isis purge-originator

Enable POI (Purge Originator Identification) (RFC 6232) for IS-IS purges triggered by the router.

When enabled, the router includes the POI TLV with its System ID in each purge it triggers. This identifies which IS (Intermediate System) triggered the purge.

Example:

```
set protocols isis purge-originator
```

set protocols isis set-attached-bit

Configure the L1/L2 IS-IS router to set the ATT (Attached) bit in the Level-1 LSPs it originates (RFC 3787).

This signals to pure Level-1 routers that this L1/L2 router has Level-2 reachability.

Example:

```
set protocols isis set-attached-bit
```

set protocols isis set-overload-bit

Configure the router to set the OL (LSP Database Overload) bit in LSPs it originates (RFC 3787).

When configured, other IS-IS routers stop sending transit traffic through this router but can still reach its directly connected networks.

Example:

```
set protocols isis set-overload-bit
```

set protocols isis advertise-high-metrics

Configure the router to advertise a high metric value on all of its IS-IS interfaces, regardless of the metric configured on each interface.

The advertised value depends on the metric style: 63 for narrow, 16777215 for wide, and 62 for transition.

Example:

```
set protocols isis advertise-high-metrics
```

set protocols isis advertise-passive-only

Configure the router to advertise in its LSPs only the IP prefixes of passive interfaces.

Prefixes of non-passive interfaces are not advertised, but those interfaces are still used to form adjacencies and participate in SPF.

Example:

```
set protocols isis advertise-passive-only
```

set protocols isis log-adjacency-changes

Configure the router to log IS-IS adjacency state changes to syslog.

Example:

```
set protocols isis log-adjacency-changes
```

```
set protocols isis topology <ipv4-multicast | ipv4-mgmt | ipv6-unicast | ipv6-multicast |  
ipv6-mgmt | ipv6-dstsrc>
```

Enable an additional IS-IS Multi-Topology for the router:

- `ipv4-multicast`: IPv4 multicast topology (MT 3).
- `ipv4-mgmt`: IPv4 management topology (MT 1).
- `ipv6-unicast`: IPv6 unicast topology (MT 2).
- `ipv6-multicast`: IPv6 multicast topology (MT 4).
- `ipv6-mgmt`: IPv6 management topology (MT 5).
- `ipv6-dstsrc`: IPv6 destination/source routing topology.

The default IPv4-unicast topology (MT 0) is always present and does not need to be explicitly configured.

Example:

```
set protocols isis topology ipv6-unicast
```

Authentication

IS-IS supports two protocol-wide passwords:

- **area-password**: applied to Level-1 LSPs/SNPs (within the area).
- **domain-password**: applied to Level-2 LSPs/SNPs (across the Level-2 backbone).

These are distinct from the per-interface password command (see [Interface configuration](#)). That command authenticates only the Hello (IIH) PDUs exchanged with the neighbor on the interface where it is configured.

set protocols isis area-password plaintext-password <text>

Configure a plaintext password used to authenticate Level-1 LSPs and SNPs originated and received by this router.

All Level-1 routers in the area must be configured with the same password.

Example:

```
set protocols isis area-password plaintext-password mysharedsecret
```

set protocols isis area-password md5 <text>

Configure an HMAC-MD5 key used to authenticate Level-1 LSPs and SNPs originated and received by this router.

All Level-1 routers in the area must be configured with the same key.

Note: For each of `area-password` and `domain-password`, configure either `plaintext-password` or `md5`, but not both.

Example:

```
set protocols isis area-password md5 mysharedsecret
```

set protocols isis domain-password plaintext-password <text>

Configure a plaintext password used to authenticate Level-2 LSPs and SNPs originated and received by this router.

All Level-2 routers in the routing domain must be configured with the same password.

Example:

```
set protocols isis domain-password plaintext-password mysharedsecret
```

```
set protocols isis domain-password md5 <text>
```

Configure an HMAC-MD5 key used to authenticate Level-2 LSPs and SNPs originated and received by this router.

All Level-2 routers in the routing domain must be configured with the same key.

Example:

```
set protocols isis domain-password md5 mysharedsecret
```

Default route advertisement

Level-1 IPv4

```
set protocols isis default-information originate ipv4 level-1
```

Configure the router to originate the IPv4 default route (0.0.0.0/0) and advertise it in locally generated LSPs throughout the Level-1 area.

Note: This command requires the router's IS-IS level to be set to level-1 or level-1-2.

Example:

```
set protocols isis default-information originate ipv4 level-1
```

```
set protocols isis default-information originate ipv4 level-1 always
```

Configure the router to unconditionally originate the IPv4 default route and advertise it throughout the Level-1 area, even if the router lacks a default route in its RIB.

Note: This command requires the router's IS-IS level to be set to level-1 or level-1-2.

Example:

```
set protocols isis default-information originate ipv4 level-1 always
```

```
set protocols isis default-information originate ipv4 level-1 metric <0-16777215>
```

Configure the IS-IS metric for the IPv4 default route advertised at Level 1.

Note: This command requires the router's IS-IS level to be set to level-1 or level-1-2.

Example:

```
set protocols isis default-information originate ipv4 level-1 metric 100
```

```
set protocols isis default-information originate ipv4 level-1 route-map <name>
```

Apply a route-map to the IPv4 default route advertised at Level 1.

The route-map can permit or deny the advertisement and, if permitted, modify the route's metric.

Note: This command requires the router's IS-IS level to be set to level-1 or level-1-2.

Example:

```
set protocols isis default-information originate ipv4 level-1 route-map ISIS-DEFAULT
```

Level-1 IPv6**set protocols isis default-information originate ipv6 level-1**

Configure the router to originate the IPv6 default route (::/0) and advertise it in locally generated LSPs throughout the Level-1 area.

Note: This command requires the router's IS-IS level to be set to level-1 or level-1-2.

Example:

```
set protocols isis default-information originate ipv6 level-1
```

set protocols isis default-information originate ipv6 level-1 always

Configure the router to unconditionally originate the IPv6 default route and advertise it throughout the Level-1 area, even if the router lacks a default route in its RIB.

Note: This command requires the router's IS-IS level to be set to level-1 or level-1-2.

Example:

```
set protocols isis default-information originate ipv6 level-1 always
```

set protocols isis default-information originate ipv6 level-1 metric <0-16777215>

Configure the IS-IS metric for the IPv6 default route advertised at Level 1.

Note: This command requires the router's IS-IS level to be set to level-1 or level-1-2.

Example:

```
set protocols isis default-information originate ipv6 level-1 metric 100
```

set protocols isis default-information originate ipv6 level-1 route-map <name>

Apply a route-map to the IPv6 default route advertised at Level 1.

The route-map can permit or deny the advertisement and, if permitted, modify the route's metric.

Note: This command requires the router's IS-IS level to be set to level-1 or level-1-2.

Example:

```
set protocols isis default-information originate ipv6 level-1 route-map ISIS-DEFAULT
```

Level-2 IPv4

```
set protocols isis default-information originate ipv4 level-2
```

Configure the router to originate the IPv4 default route (0.0.0.0/0) and advertise it in locally generated LSPs throughout the Level-2 backbone.

Note: This command requires the router's IS-IS level to be set to level-2 or level-1-2.

Example:

```
set protocols isis default-information originate ipv4 level-2
```

```
set protocols isis default-information originate ipv4 level-2 always
```

Configure the router to unconditionally originate the IPv4 default route and advertise it throughout the Level-2 backbone, even if the router lacks a default route in its RIB.

Note: This command requires the router's IS-IS level to be set to level-2 or level-1-2.

Example:

```
set protocols isis default-information originate ipv4 level-2 always
```

```
set protocols isis default-information originate ipv4 level-2 metric <0-16777215>
```

Configure the IS-IS metric for the IPv4 default route advertised into the Level-2 backbone.

Note: This command requires the router's IS-IS level to be set to level-2 or level-1-2.

Example:

```
set protocols isis default-information originate ipv4 level-2 metric 100
```

```
set protocols isis default-information originate ipv4 level-2 route-map <name>
```

Apply a route-map to the IPv4 default route advertised into the Level-2 backbone.

The route-map can permit or deny the advertisement and, if permitted, modify the route's metric.

Note: This command requires the router's IS-IS level to be set to level-2 or level-1-2.

Example:

```
set protocols isis default-information originate ipv4 level-2 route-map ISIS-DEFAULT
```

Level-2 IPv6

```
set protocols isis default-information originate ipv6 level-2
```

Configure the router to originate the IPv6 default route (::/0) and advertise it in locally generated LSPs throughout the Level-2 backbone.

Note: This command requires the router's IS-IS level to be set to level-2 or level-1-2.

Example:

```
set protocols isis default-information originate ipv6 level-2
```

```
set protocols isis default-information originate ipv6 level-2 always
```

Configure the router to unconditionally originate the IPv6 default route and advertise it throughout the Level-2 backbone, even if the router lacks a default route in its RIB.

Note: This command requires the router's IS-IS level to be set to level-2 or level-1-2.

Example:

```
set protocols isis default-information originate ipv6 level-2 always
```

```
set protocols isis default-information originate ipv6 level-2 metric <0-16777215>
```

Configure the IS-IS metric for the IPv6 default route advertised into the Level-2 backbone.

Note: This command requires the router's IS-IS level to be set to level-2 or level-1-2.

Example:

```
set protocols isis default-information originate ipv6 level-2 metric 100
```

```
set protocols isis default-information originate ipv6 level-2 route-map <name>
```

Apply a route-map to the IPv6 default route advertised into the Level-2 backbone.

The route-map can permit or deny the advertisement and, if permitted, modify the route's metric.

Note: This command requires the router's IS-IS level to be set to level-2 or level-1-2.

Example:

```
set protocols isis default-information originate ipv6 level-2 route-map ISIS-DEFAULT
```

LDP synchronization

set protocols isis ldp-sync

Enable LDP IGP synchronization ([RFC 5443](#)) for the IS-IS routing process.

When enabled, all operational IS-IS interfaces automatically participate in synchronization, except for loopback interfaces.

Note: LDP must be configured and functional on the router for synchronization to operate.

Example:

```
set protocols isis ldp-sync
```

set protocols isis ldp-sync holddown <0-10000>

Configure the time, in seconds, that IS-IS keeps any of its interfaces at max-metric while waiting for LDP-IGP synchronization to complete.

When this time expires on an interface, IS-IS restores the configured interface metric even if LDP-IGP synchronization has not completed.

The default value is 0, which causes IS-IS to wait indefinitely.

Example:

```
set protocols isis ldp-sync holddown 60
```

Interface configuration

set protocols isis interface <interface> circuit-type <level-1 | level-1-2 | level-2-only>

Configure the IS-IS level at which the router can form adjacencies on the specified interface:

- **level-1:** Permits only Level-1 (intra-area) adjacencies.
- **level-2-only:** Permits only Level-2 (inter-area) adjacencies.
- **level-1-2:** Permits both Level-1 (intra-area) and Level-2 (inter-area) adjacencies.

Note: The selected level must be supported by the router's IS-IS process.

Example:

```
set protocols isis interface eth0 circuit-type level-1
```

set protocols isis interface <interface> hello-interval <1-600>

Configure the interval, in seconds, between successive IS-IS Hello PDUs (IIH) sent on the specified interface.

Example:

```
set protocols isis interface eth1 hello-interval 5
```



```
set protocols isis interface <interface> hello-multiplier <2-100>
```

Configure the multiplier applied to the Hello interval to derive the Holding Time advertised in IS-IS Hello PDUs (IIH) sent on the specified interface.

The receiving neighbor uses the advertised Holding Time as its adjacency timeout. If no IIH arrives within that period, the adjacency is declared down.

Example:

```
set protocols isis interface eth1 hello-multiplier 5
```

```
set protocols isis interface <interface> hello-padding
```

Enable padding of IS-IS Hello PDUs (IIH) sent on the specified interface to that interface's full MTU (RFC 3719).

This ensures that neighbors with asymmetric MTUs cannot establish an adjacency. Without padding, asymmetric MTUs bypass detection during adjacency setup.

Example:

```
set protocols isis interface eth0 hello-padding
```

```
set protocols isis interface <interface> metric <0-16777215>
```

Configure the IS-IS metric (cost) advertised for the specified interface.

The SPF algorithm uses this value to calculate the optimal routing path to destinations.

The valid range depends on the configured metric style: narrow limits the metric to 0-63, while wide extends it to 0-16777215.

The default value is 10.

Example:

```
set protocols isis interface eth1 metric 100
```

```
set protocols isis interface <interface> network point-to-point
```

Configure the IS-IS network type for the specified interface as point-to-point.

The default network type is broadcast.

Note: The neighboring interface must be configured with the same network type; otherwise, the adjacency does not form.

Example:

```
set protocols isis interface eth1 network point-to-point
```

```
set protocols isis interface <interface> passive
```

Enable passive mode for the specified interface.

On a passive interface, the router neither sends nor processes IS-IS Hello PDUs (IIH), so no adjacency forms. The interface's IP prefix is still advertised in this router's LSP.

Example:

```
set protocols isis interface lo passive
```

set protocols isis interface <interface> password plaintext-password <text>

Configure the plaintext authentication password for the specified interface.

This password is included in IS-IS Hello (IIH) PDUs sent on the interface and validated on IIH PDUs received from neighbors.

A mismatch prevents adjacency formation and tears down an established adjacency.

Example:

```
set protocols isis interface eth1 password plaintext-password mysharedsecret
```

set protocols isis interface <interface> password md5 <text>

Configure the MD5 authentication key for the specified interface.

This key is used to generate a cryptographic hash that is included in IS-IS Hello (IIH) PDUs sent on the interface and validated in IIH PDUs received from neighbors.

A mismatch prevents adjacency formation and tears down an established adjacency.

Example:

```
set protocols isis interface eth1 password md5 mysharedsecret
```

set protocols isis interface <interface> priority <0-127>

Configure the IS-IS DIS (Designated Intermediate System) election priority for the specified interface.

The priority is used in DIS election on the broadcast (LAN) segment. The router whose interface advertises the highest priority wins.

The default priority is 64.

Note: The configured value applies only to broadcast interfaces and has no effect on point-to-point interfaces.

Example:

```
set protocols isis interface eth1 priority 100
```

set protocols isis interface <interface> psnp-interval <0-127>

Configure the interval, in seconds, between successive IS-IS PSNP (Partial Sequence Number PDU) transmissions on the specified interface.

Example:

```
set protocols isis interface eth1 psnp-interval 2
```

set protocols isis interface <interface> no-three-way-handshake

Disable the Three-Way Handshake for Point-to-Point (P2P) adjacencies ([RFC 5303](#)).

The three-way handshake is enabled by default.

Note: This command applies only to point-to-point interfaces and has no effect on broadcast interfaces.

Example:

```
set protocols isis interface eth1 no-three-way-handshake
```

set protocols isis interface <interface> ldp-sync disable

Disable LDP-IGP synchronization on the specified IS-IS interface.

With LDP-IGP synchronization enabled, IS-IS advertises a maximum metric on the interface while LDP is not yet synchronized on it.

Disabling it on this interface causes IS-IS to advertise the configured metric regardless of LDP synchronization state.

Note: This command applies only if LDP-IGP synchronization is enabled for the IS-IS routing process.

Example:

```
set protocols isis interface eth1 ldp-sync disable
```

set protocols isis interface <interface> ldp-sync holddown <0-10000>

Configure the time, in seconds, that IS-IS keeps the specified interface at max-metric while waiting for LDP-IGP synchronization to complete.

When this time expires, IS-IS restores the configured interface metric even if LDP-IGP synchronization has not completed.

The default value is 0, which causes IS-IS to wait indefinitely.

Note: This setting overrides the LDP-IGP synchronization hold-down time configured for the IS-IS routing process.

Example:

```
set protocols isis interface eth1 ldp-sync holddown 60
```

set protocols isis interface <interface> bfd profile <profile-name>

Attach a BFD profile to the IS-IS adjacency on the specified interface.

When configured, IS-IS uses the BFD session described by the named profile to detect adjacency failure. If BFD reports the session down, the IS-IS adjacency is torn down immediately rather than waiting for the Hello holding time to expire.

The profile itself is configured under `set protocols bfd profile <profile-name>`.

Example:

```
set protocols isis interface eth1 bfd profile ISIS-FAST
```

Level-1 fast-reroute

set protocols isis interface <interface> fast-reroute lfa level-1 enable

Enable Level-1 LFA (Loop-Free Alternate) computation on the specified interface.

When enabled, IS-IS precomputes a backup next-hop for each Level-1 destination reached through this interface. If the primary next-hop fails, the router uses the precomputed backup instead.

Example:

```
set protocols isis interface eth1 fast-reroute lfa level-1 enable
```

set protocols isis interface <interface-1> fast-reroute lfa level-1 exclude interface <interface-2>

Exclude an interface from being selected as a Level-1 LFA backup next-hop on the specified interface.

In the command syntax, <interface-1> identifies the protected interface, and <interface-2> identifies the interface that must not be used as an LFA backup next-hop.

Example:

```
set protocols isis interface eth1 fast-reroute lfa level-1 exclude interface eth2
```

set protocols isis interface <interface> fast-reroute remote-lfa level-1 tunnel mpls-ldp

Enable Level-1 remote LFA computation using MPLS-LDP tunnels on the specified interface.

Remote LFA provides backup paths when local LFA cannot find a loop-free alternate neighbor.

If the primary next-hop fails, the router sends traffic into an MPLS-LDP tunnel to a precomputed remote LFA node. From there, normal IS-IS forwarding delivers it to the destination.

Note: Remote LFA requires the corresponding local LFA to be configured on this interface.

Note: Remote LFA requires all potential remote LFA endpoints in the IS-IS routing instance to accept targeted LDP Hello messages.

Example:

```
set protocols isis interface eth1 fast-reroute remote-lfa level-1 tunnel mpls-ldp
```

set protocols isis interface <interface> fast-reroute remote-lfa level-1 maximum-metric <1-16777215>

Configure the maximum metric for Level-1 remote LFA node selection on the specified interface.

Remote LFA nodes with a metric exceeding this value are excluded from selection.

Note: The maximum metric applies only when the corresponding remote LFA is enabled on this interface.

Example:

```
set protocols isis interface eth1 fast-reroute remote-lfa level-1 maximum-metric 100
```

```
set protocols isis interface <interface> fast-reroute ti-lfa level-1
```

Enable Level-1 TI-LFA (Topology Independent Loop-Free Alternate) computation on the specified interface.

By default, TI-LFA operates in link protection mode, computing backup paths that protect against the failure of this interface's link.

Note: LFA and TI-LFA cannot be configured at the same level on the same interface.

Example:

```
set protocols isis interface eth1 fast-reroute ti-lfa level-1
```

```
set protocols isis interface <interface> fast-reroute ti-lfa level-1 node-protection
```

Enable node protection mode for Level-1 TI-LFA on the specified interface.

In node protection mode, TI-LFA computes backup paths that protect against the failure of the next-hop node.

Note: LFA and TI-LFA cannot be configured at the same level on the same interface.

Example:

```
set protocols isis interface eth0 fast-reroute ti-lfa level-1 node-protection
```

```
set protocols isis interface <interface> fast-reroute ti-lfa level-1 node-protection link-fallback
```

Enable link-protection fallback for Level-1 TI-LFA on the specified interface.

When node protection cannot compute a backup path, the computation falls back to link protection.

Example:

```
set protocols isis interface eth1 fast-reroute ti-lfa level-1 node-protection link-
↪ fallback
```

Level-2 fast-reroute

```
set protocols isis interface <interface> fast-reroute lfa level-2 enable
```

Enable Level-2 LFA computation on the specified interface.

When enabled, IS-IS precomputes a backup next-hop for each Level-2 destination reached through this interface. If the primary next-hop fails, the router uses the precomputed backup instead.

Example:

```
set protocols isis interface eth1 fast-reroute lfa level-2 enable
```

```
set protocols isis interface <interface-1> fast-reroute lfa level-2 exclude interface
<interface-2>
```

Exclude an interface from being selected as a Level-2 LFA backup next-hop on the specified interface.

In the command syntax, <interface-1> identifies the protected interface, and <interface-2> identifies the interface that must not be used as an LFA backup next-hop.

Example:

```
set protocols isis interface eth1 fast-reroute lfa level-2 exclude interface eth2
```

```
set protocols isis interface <interface> fast-reroute remote-lfa level-2 tunnel mpls-ldp
```

Enable Level-2 remote LFA computation using MPLS-LDP tunnels on the specified interface.

Remote LFA provides backup paths when local LFA cannot find a loop-free alternate neighbor.

If the primary next-hop fails, the router sends traffic into an MPLS-LDP tunnel to a precomputed remote LFA node. From there, normal IS-IS forwarding delivers it to the destination.

Note: Remote LFA requires the corresponding local LFA to be configured on this interface.

Note: Remote LFA requires all potential remote LFA endpoints in the IS-IS routing instance to accept targeted LDP Hello messages.

Example:

```
set protocols isis interface eth1 fast-reroute remote-lfa level-2 tunnel mpls-ldp
```

```
set protocols isis interface <interface> fast-reroute remote-lfa level-2 maximum-metric <1-16777215>
```

Configure the maximum metric for Level-2 remote LFA node selection on the specified interface.

Remote LFA nodes with a metric exceeding this value are excluded from selection.

Note: The maximum metric applies only when the corresponding remote LFA is enabled on this interface.

Example:

```
set protocols isis interface eth1 fast-reroute remote-lfa level-2 maximum-metric 100
```

```
set protocols isis interface <interface> fast-reroute ti-lfa level-2
```

Enable Level-2 TI-LFA computation on the specified interface.

By default, TI-LFA operates in link protection mode, computing backup paths that protect against the failure of this interface's link.

Note: LFA and TI-LFA cannot be configured at the same level on the same interface.

Example:

```
set protocols isis interface eth1 fast-reroute ti-lfa level-2
```

```
set protocols isis interface <interface> fast-reroute ti-lfa level-2 node-protection
```

Enable node protection mode for Level-2 TI-LFA on the specified interface.

In node protection mode, TI-LFA computes backup paths that protect against the failure of the next-hop node.

Note: LFA and TI-LFA cannot be configured at the same level on the same interface.

Example:

```
set protocols isis interface eth0 fast-reroute ti-lfa level-2 node-protection
```

set protocols isis interface <interface> fast-reroute ti-lfa level-2 node-protection link-fallback

Enable link-protection fallback for Level-2 TI-LFA on the specified interface.

When node protection cannot compute a backup path, the computation falls back to link protection.

Note: LFA and TI-LFA cannot be configured at the same level on the same interface.

Example:

```
set protocols isis interface eth1 fast-reroute ti-lfa level-2 node-protection link-
↪ fallback
```

Route redistribution

Level-1 IPv4

set protocols isis redistribute ipv4 <bgp | connected | kernel | nhrp | ospf | rip | babel | static> level-1

Configure the redistribution of IPv4 routing information from the specified route source into IS-IS Level-1.

Note: This command requires the router's IS-IS level to be set to level-1 or level-1-2.

Example:

```
set protocols isis redistribute ipv4 bgp level-1
```

set protocols isis redistribute ipv4 <bgp | connected | kernel | nhrp | ospf | rip | babel | static> level-1 metric <0-16777215>

Configure the IS-IS metric for IPv4 routes redistributed from the specified route source into IS-IS Level-1.

Note: This command requires the router's IS-IS level to be set to level-1 or level-1-2.

Example:

```
set protocols isis redistribute ipv4 bgp level-1 metric 50
```

```
set protocols isis redistribute ipv4 <bgp | connected | kernel | nhrp | ospf | rip |
babel | static> level-1 route-map <name>
```

Apply a route-map to IPv4 routes redistributed from the specified route source into IS-IS Level-1.

Only routes permitted by the route-map are redistributed. The route-map can modify the metric and tag of permitted routes.

Note: This command requires the router's IS-IS level to be set to level-1 or level-1-2.

Example:

```
set protocols isis redistribute ipv4 bgp level-1 route-map BGP-TO-ISIS
```

Level-1 IPv6

```
set protocols isis redistribute ipv6 <bgp | connected | kernel | ospf6 | ripng | babel |
static> level-1
```

Configure the redistribution of IPv6 routing information from the specified route source into IS-IS Level-1.

Note: This command requires the router's IS-IS level to be set to level-1 or level-1-2.

Example:

```
set protocols isis redistribute ipv6 bgp level-1
```

```
set protocols isis redistribute ipv6 <bgp | connected | kernel | ospf6 | ripng | babel |
static> level-1 metric <0-16777215>
```

Configure the IS-IS metric for IPv6 routes redistributed from the specified route source into IS-IS Level-1.

Note: This command requires the router's IS-IS level to be set to level-1 or level-1-2.

Example:

```
set protocols isis redistribute ipv6 bgp level-1 metric 50
```

```
set protocols isis redistribute ipv6 <bgp | connected | kernel | ospf6 | ripng | babel |
static> level-1 route-map <name>
```

Apply a route-map to IPv6 routes redistributed from the specified route source into IS-IS Level-1.

Only routes permitted by the route-map are redistributed. The route-map can modify the metric and tag of permitted routes.

Note: This command requires the router's IS-IS level to be set to level-1 or level-1-2.

Example:


```
set protocols isis redistribute ipv6 bgp level-1 route-map BGP-T0-ISIS
```

Level-2 IPv4

```
set protocols isis redistribute ipv4 <bgp | connected | kernel | nhrp | ospf | rip |  
babel | static> level-2
```

Configure the redistribution of IPv4 routing information from the specified route source into IS-IS Level-2.

Note: This command requires the router's IS-IS level to be set to level-2 or level-1-2.

Example:

```
set protocols isis redistribute ipv4 bgp level-2
```

```
set protocols isis redistribute ipv4 <bgp | connected | kernel | nhrp | ospf | rip |  
babel | static> level-2 metric <0-16777215>
```

Configure the IS-IS metric for IPv4 routes redistributed from the specified route source into IS-IS Level-2.

Note: This command requires the router's IS-IS level to be set to level-2 or level-1-2.

Example:

```
set protocols isis redistribute ipv4 bgp level-2 metric 50
```

```
set protocols isis redistribute ipv4 <bgp | connected | kernel | nhrp | ospf | rip |  
babel | static> level-2 route-map <name>
```

Apply a route-map to IPv4 routes redistributed from the specified route source into IS-IS Level-2.

Only routes permitted by the route-map are redistributed. The route-map can modify the metric and tag of permitted routes.

Note: This command requires the router's IS-IS level to be set to level-2 or level-1-2.

Example:

```
set protocols isis redistribute ipv4 bgp level-2 route-map BGP-T0-ISIS
```

Level-2 IPv6

```
set protocols isis redistribute ipv6 <bgp | connected | kernel | ospf6 | ripng | babel | static> level-2
```

Configure the redistribution of IPv6 routing information from the specified route source into IS-IS Level-2.

Note: This command requires the router's IS-IS level to be set to level-2 or level-1-2.

Example:

```
set protocols isis redistribute ipv6 bgp level-2
```

```
set protocols isis redistribute ipv6 <bgp | connected | kernel | ospf6 | ripng | babel | static> level-2 metric <0-16777215>
```

Configure the IS-IS metric for IPv6 routes redistributed from the specified route source into IS-IS Level-2.

Note: This command requires the router's IS-IS level to be set to level-2 or level-1-2.

Example:

```
set protocols isis redistribute ipv6 bgp level-2 metric 50
```

```
set protocols isis redistribute ipv6 <bgp | connected | kernel | ospf6 | ripng | babel | static> level-2 route-map <name>
```

Apply a route-map to IPv6 routes redistributed from the specified route source into IS-IS Level-2.

Only routes permitted by the route-map are redistributed. The route-map can modify the metric and tag of permitted routes.

Note: This command requires the router's IS-IS level to be set to level-2 or level-1-2.

Example:

```
set protocols isis redistribute ipv6 bgp level-2 route-map BGP-TO-ISIS
```

Timers

```
set protocols isis lsp-gen-interval <1-120>
```

Configure the minimum interval, in seconds, between consecutive regenerations of this router's own LSP.

Regenerations triggered by events, such as network topology changes, are postponed until the specified interval has elapsed since the previous regeneration.

The default interval is 30 seconds.

Example:

```
set protocols isis lsp-gen-interval 5
```

```
set protocols isis lsp-refresh-interval <2-65235>
```

Configure the maximum interval, in seconds, between consecutive regenerations of this router's own LSP.

When this interval elapses, the router regenerates its LSP even if no event has triggered a regeneration.

Note: The value must be less than the configured `max-lsp-lifetime` so that the LSP is refreshed before its remaining lifetime expires.

Example:

```
set protocols isis lsp-refresh-interval 600
```

```
set protocols isis max-lsp-lifetime <350-65535>
```

Configure the lifetime, in seconds, of LSPs originated by this router.

The default is 1200 seconds.

Note: This value must be greater than `lsp-refresh-interval` so that LSPs are refreshed before their lifetime expires.

Example:

```
set protocols isis max-lsp-lifetime 1500
```

```
set protocols isis spf-interval <1-120>
```

Configure the interval, in seconds, between consecutive SPF calculations on this router.

SPF calculations triggered by events, such as network topology changes, are postponed until the specified interval has elapsed since the previous calculation.

Example:

```
set protocols isis spf-interval 5
```

The following commands implement the IETF SPF delay algorithm for IS-IS described in [RFC 8405](#). They control how quickly the router runs the SPF calculation after topology changes are detected.

Note: All five `spf-delay-ietf` parameters must be configured; setting only some of them causes a commit failure.

```
set protocols isis spf-delay-ietf holddown <0-60000>
```

Configure the interval, in milliseconds, that must elapse without topology changes before the router considers the network stable.

Once this time passes, the router returns to its initial response state and applies `init-delay` to the next SPF calculation.

Example:

```
set protocols isis spf-delay-ietf holddown 10000
```

```
set protocols isis spf-delay-ietf init-delay <0-60000>
```

Configure the interval, in milliseconds, that the router waits before running the first SPF calculation when a new topology change arrives after the network is considered stable.

Example:

```
set protocols isis spf-delay-ietf init-delay 500
```

```
set protocols isis spf-delay-ietf time-to-learn <0-60000>
```

Configure the learning interval, in milliseconds, that begins when the first topology change is received.

Within this window, the router applies the short-delay before performing SPF recalculations triggered by additional topology changes.

If topology changes continue to arrive after this window expires, the router considers the network unstable and delays subsequent SPF recalculations by the configured long-delay.

Example:

```
set protocols isis spf-delay-ietf time-to-learn 5000
```

```
set protocols isis spf-delay-ietf long-delay <0-60000>
```

Configure the interval, in milliseconds, the router applies before running an SPF calculation when topology changes arrive after the time-to-learn window expires.

Example:

```
set protocols isis spf-delay-ietf long-delay 10000
```

```
set protocols isis spf-delay-ietf short-delay <0-60000>
```

Configure the interval, in milliseconds, the router applies before running subsequent SPF calculations triggered by additional topology changes within the time-to-learn window.

The first SPF in the window is delayed by init-delay.

Example:

```
set protocols isis spf-delay-ietf short-delay 1000
```

Loop-Free Alternate (LFA)

Level-1

```
set protocols isis fast-reroute lfa remote prefix-list <name> level-1
```

Filter which PQ nodes the router evaluates as Remote LFA tunnel endpoints for IS-IS Level-1 using the specified prefix-list.

Only PQ nodes permitted by the prefix-list are evaluated as potential backup routers.

Example:

```
set protocols isis fast-reroute lfa remote prefix-list CRITICAL-ROUTES level-1
```

```
set protocols isis fast-reroute lfa local load-sharing disable level-1
```

Disable load sharing across multiple local LFAs for IS-IS Level-1.

When multiple LFAs are available to back up a given destination, the router uses only one LFA instead of distributing rerouted traffic across all of them.

Example:

```
set protocols isis fast-reroute lfa local load-sharing disable level-1
```

```
set protocols isis fast-reroute lfa local tiebreaker <downstream | lowest-backup-metric | node-protecting> index <1-255> level-1
```

Configure a tiebreaker rule for selecting a single LFA when multiple local LFAs exist for an IS-IS Level-1 prefix.

Tiebreaker rules with lower index numbers are evaluated first. You can choose one of the following tiebreaker behaviors:

- **downstream:** Prefers an alternate that is closer to the destination than this router.
- **lowest-backup-metric:** Prefers the alternate with the shortest path to the destination.
- **node-protecting:** Prefers an alternate that bypasses the primary next-hop router, protecting against a complete node failure rather than just link failure.

Example:

```
set protocols isis fast-reroute lfa local tiebreaker node-protecting index 10 level-1
```

```
set protocols isis fast-reroute lfa local priority-limit <critical | high | medium> level-1
```

Configure the priority limit for calculating local LFAs in IS-IS Level-1.

The router calculates LFAs only for prefixes with a priority greater than or equal to the specified limit. For example, setting the limit to high protects both high and critical prefixes.

Note: Prefixes carry one of the following priorities: low, medium, high, or critical. By default, the router assigns medium priority to loopbacks and low priority to all other prefixes. The limit itself accepts only medium, high, or critical.

Example:

```
set protocols isis fast-reroute lfa local priority-limit critical level-1
```

Level-2

```
set protocols isis fast-reroute lfa remote prefix-list <name> level-2
```

Filter which PQ nodes the router evaluates as Remote LFA tunnel endpoints for IS-IS Level-2 using the specified prefix-list.

Only PQ nodes permitted by the prefix-list are evaluated as potential backup routers.

Example:

```
set protocols isis fast-reroute lfa remote prefix-list CRITICAL-ROUTES level-2
```

```
set protocols isis fast-reroute lfa local load-sharing disable level-2
```

Disable load sharing across multiple local LFAs for IS-IS Level-2.

When multiple LFAs are available to back up a given destination, the router uses only one LFA instead of distributing rerouted traffic across all of them.

Example:

```
set protocols isis fast-reroute lfa local load-sharing disable level-2
```

```
set protocols isis fast-reroute lfa local tiebreaker <downstream | lowest-backup-metric | node-protecting> index <1-255> level-2
```

Configure a tiebreaker rule for selecting a single LFA when multiple local LFAs exist for an IS-IS Level-2 prefix.

Tiebreaker rules with lower index numbers are evaluated first. You can choose one of the following tiebreaker behaviors:

- **downstream:** Prefers an alternate that is closer to the destination than this router.
- **lowest-backup-metric:** Prefers the alternate with the shortest path to the destination.
- **node-protecting:** Prefers an alternate that bypasses the primary next-hop router, protecting against a complete node failure rather than just link failure.

Example:

```
set protocols isis fast-reroute lfa local tiebreaker node-protecting index 10 level-2
```

```
set protocols isis fast-reroute lfa local priority-limit <critical | high | medium> level-2
```

Configure the priority limit for calculating local LFAs in IS-IS Level-2.

Note: Prefixes carry one of the following priorities: low, medium, high, or critical. By default, the router assigns medium priority to loopbacks and low priority to all other prefixes. The limit itself accepts only medium, high, or critical.

Example:

```
set protocols isis fast-reroute lfa local priority-limit critical level-2
```

Segment Routing over IPv6 (SRv6)

```
set protocols isis segment-routing srv6 interface <interface>
```

Enable IS-IS Segment Routing over IPv6 (SRv6) on the specified interface.

Once enabled, IS-IS allocates and advertises an adjacency SID for each IS-IS neighbor reached through this interface, so other routers can steer SRv6 traffic over it.

Example:

```
set protocols isis segment-routing srv6 interface eth1
```

```
set protocols isis segment-routing srv6 locator <locator>
```

Configure IS-IS to use a globally defined SRv6 locator.

The locator itself must be configured separately using `set protocols segment-routing srv6 locator <name> prefix <ipv6-prefix>`.

Once you attach the locator to IS-IS, the routing process automatically:

- Allocates a node SID for the router and an adjacency SID for each of its IS-IS neighbors.
- Advertises the locator and these SIDs to the network so other routers can route SRv6 traffic through this node.

Example:

```
set protocols isis segment-routing srv6 locator MAIN-LOCATOR
```

```
set protocols isis segment-routing srv6 node-msd max-end-d <0-255>
```

Configure the Maximum End D MSD value advertised by the router (RFC 9352).

This value indicates the maximum number of SIDs in the SRH that this router can handle when performing a decapsulation behavior (e.g., End.DX6, End.DT4, End.DT46, End with USD, End.X with USD) defined in RFC 8986.

If this value is set to 0 or left unconfigured, the router advertises that it cannot decapsulate and forward packets when an SRH is present.

Example:

```
set protocols isis segment-routing srv6 node-msd max-end-d 8
```

```
set protocols isis segment-routing srv6 node-msd max-end-pop <0-255>
```

Configure the Maximum End Pop MSD value advertised by the router (RFC 9352).

This value indicates the maximum number of SIDs in the received SRH to which this router can apply the PSP (Penultimate Segment Pop) or USP (Ultimate Segment Pop) flavors defined in RFC 8986, §4.16.

If this value is set to 0 or left unconfigured, the router advertises that it cannot apply the PSP or USP flavors.

Example:

```
set protocols isis segment-routing srv6 node-msd max-end-pop 16
```

```
set protocols isis segment-routing srv6 node-msd max-h-encaps <0-255>
```

Configure the Maximum H.Encaps MSD value advertised by the router (RFC 9352).

This value indicates the maximum number of SIDs that this router can insert into a new SRH (Segment Routing Header) when encapsulating traffic (the H.Encaps behavior), as defined in RFC 8986.

If set to 0 or left unconfigured, the router advertises that it can only apply an SR Policy containing a single segment, without inserting an SRH.

Example:

```
set protocols isis segment-routing srv6 node-msd max-h-encaps 8
```

```
set protocols isis segment-routing srv6 node-msd max-segs-left <0-255>
```

Configure the Maximum Segments Left MSD value advertised by the router (RFC 9352).

This value indicates the maximum Segments Left value (RFC 8754) in the SRH of a received packet that this router can process before applying the Endpoint behavior associated with a SID.

If set to 0 or left unconfigured, the router advertises that it can only be the last segment of an SRv6 path. Set a value greater than 0 to also allow the router to be used as a segment in the middle of a path.

Example:

```
set protocols isis segment-routing srv6 node-msd max-segs-left 8
```

Examples

Enable IS-IS

The following example demonstrates a basic IS-IS routing protocol setup between two VyOS routers.

Node 1:

```
set interfaces loopback lo address '198.51.100.1/32'
set interfaces ethernet eth1 address '192.0.2.1/24'

set protocols isis interface eth1
set protocols isis interface lo
set protocols isis net '49.0001.1980.5110.0001.00'
```

Node 2:

```
set interfaces loopback lo address '198.51.100.2/32'
set interfaces ethernet eth1 address '192.0.2.2/24'

set protocols isis interface eth1
set protocols isis interface lo
set protocols isis net '49.0001.1980.5110.0002.00'
```

This gives us the following neighborships, Level 1 and Level 2:

```
Node-1@vyos:~$ show isis neighbor
Area VyOS:
System Id      Interface  L  State      Holdtime  SNPA
vyos           eth1      1  Up         28         0c87.6c09.0001
vyos           eth1      2  Up         28         0c87.6c09.0001

Node-2@vyos:~$ show isis neighbor
Area VyOS:
System Id      Interface  L  State      Holdtime  SNPA
vyos           eth1      1  Up         29         0c33.0280.0001
vyos           eth1      2  Up         28         0c33.0280.0001
```

Here's the IP routes that are populated. Just the loopback:

```
Node-1@vyos:~$ show ip route isis
Codes: K - kernel route, C - connected, S - static, R - RIP,
       O - OSPF, I - IS-IS, B - BGP, E - EIGRP, N - NHRP,
       T - Table, v - VNC, V - VNC-Direct, A - Babel, F - PBR,
       f - OpenFabric,
       > - selected route, * - FIB route, q - queued, r - rejected, b - backup
       t - trapped, o - offload failure
```

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```
I 192.0.2.0/24 [115/20] via 192.0.2.2, eth1 inactive, weight 1, 00:02:22
I>* 198.51.100.2/32 [115/20] via 192.0.2.2, eth1, weight 1, 00:02:22

Node-2@vyos:~$ show ip route isis
Codes: K - kernel route, C - connected, S - static, R - RIP,
        O - OSPF, I - IS-IS, B - BGP, E - EIGRP, N - NHRP,
        T - Table, v - VNC, V - VNC-Direct, A - Babel, F - PBR,
        f - OpenFabric,
        > - selected route, * - FIB route, q - queued, r - rejected, b - backup
        t - trapped, o - offload failure

I 192.0.2.0/24 [115/20] via 192.0.2.1, eth1 inactive, weight 1, 00:02:21
I>* 198.51.100.1/32 [115/20] via 192.0.2.1, eth1, weight 1, 00:02:21
```

Enable IS-IS and redistribute routes not natively in IS-IS

The following example demonstrates a basic IS-IS setup between two VyOS routers, with Node 1 also redistributing a connected network that isn't running IS-IS.

Node 1:

```
set interfaces dummy dum0 address '203.0.113.1/24'
set interfaces ethernet eth1 address '192.0.2.1/24'

set policy prefix-list EXPORT-ISIS rule 10 action 'permit'
set policy prefix-list EXPORT-ISIS rule 10 prefix '203.0.113.0/24'
set policy route-map EXPORT-ISIS rule 10 action 'permit'
set policy route-map EXPORT-ISIS rule 10 match ip address prefix-list 'EXPORT-ISIS'

set protocols isis interface eth1
set protocols isis net '49.0001.2030.0011.3001.00'
set protocols isis redistribute ipv4 connected level-2 route-map 'EXPORT-ISIS'
```

Node 2:

```
set interfaces ethernet eth1 address '192.0.2.2/24'

set protocols isis interface eth1
set protocols isis net '49.0001.1920.0000.2002.00'
```

Routes on Node 2:

```
Node-2@r2:~$ show ip route isis
Codes: K - kernel route, C - connected, S - static, R - RIP,
        O - OSPF, I - IS-IS, B - BGP, E - EIGRP, N - NHRP,
        T - Table, v - VNC, V - VNC-Direct, A - Babel, D - SHARP,
        F - PBR, f - OpenFabric,
        > - selected route, * - FIB route, q - queued route, r - rejected route

I 203.0.113.0/24 [115/10] via 192.0.2.1, eth1, 00:03:42
```

Enable IS-IS and IGP-LDP synchronization

The following example demonstrates an IS-IS setup with IGP-LDP synchronization enabled on Node 1.

Node 1:

```
set interfaces loopback lo address 198.51.100.1/32
set interfaces ethernet eth0 address 192.0.2.1/24

set protocols isis interface eth0
set protocols isis interface lo passive
set protocols isis ldp-sync
set protocols isis net 49.0001.1980.5110.0001.00

set protocols mpls interface eth0
set protocols mpls ldp discovery transport-ipv4-address 198.51.100.1
set protocols mpls ldp interface lo
set protocols mpls ldp interface eth0
set protocols mpls ldp parameters transport-prefer-ipv4
set protocols mpls ldp router-id 198.51.100.1
```

This gives us IGP-LDP synchronization for all non-loopback interfaces with a holddown timer of zero seconds:

```
Node-1@vyos:~$ show isis mpls ldp-sync
eth0
  LDP-IGP Synchronization enabled: yes
  holddown timer in seconds: 0
  State: Sync achieved
```

Enable IS-IS with Segment Routing (experimental)

The following example demonstrates IS-IS with segment routing between two VyOS routers.

Node 1:

```
set interfaces loopback lo address '198.51.100.1/32'
set interfaces ethernet eth1 address '192.0.2.1/24'

set protocols isis interface eth1
set protocols isis interface lo
set protocols isis net '49.0001.1980.5110.0001.00'
set protocols isis segment-routing global-block high-label-value '599'
set protocols isis segment-routing global-block low-label-value '550'
set protocols isis segment-routing prefix 198.51.100.1/32 index value '1'
set protocols isis segment-routing prefix 198.51.100.1/32 index explicit-null
set protocols mpls interface 'eth1'
```

Node 2:

```
set interfaces loopback lo address '198.51.100.2/32'
set interfaces ethernet eth1 address '192.0.2.2/24'

set protocols isis interface eth1
```

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```
set protocols isis interface lo
set protocols isis net '49.0001.1980.5110.0002.00'
set protocols isis segment-routing global-block high-label-value '599'
set protocols isis segment-routing global-block low-label-value '550'
set protocols isis segment-routing prefix 198.51.100.2/32 index value '2'
set protocols isis segment-routing prefix 198.51.100.2/32 index explicit-null
set protocols mpls interface 'eth1'
```

This gives us MPLS segment routing enabled and labels for far end loopbacks:

```
Node-1@vyos:~$ show mpls table
Inbound Label  Type      Nexthop      Outbound Label
-----
552            SR (IS-IS)  192.0.2.2    IPv4 Explicit Null <-- Node-2
↳ loopback learned on Node-1
15000          SR (IS-IS)  192.0.2.2    implicit-null
15001          SR (IS-IS)  fe80::e87:6cff:fe09:1  implicit-null
15002          SR (IS-IS)  192.0.2.2    implicit-null
15003          SR (IS-IS)  fe80::e87:6cff:fe09:1  implicit-null

Node-2@vyos:~$ show mpls table
Inbound Label  Type      Nexthop      Outbound Label
-----
551            SR (IS-IS)  192.0.2.1    IPv4 Explicit Null <-- Node-1 loopback
↳ learned on Node-2
15000          SR (IS-IS)  192.0.2.1    implicit-null
15001          SR (IS-IS)  fe80::e33:2ff:fe80:1  implicit-null
15002          SR (IS-IS)  192.0.2.1    implicit-null
15003          SR (IS-IS)  fe80::e33:2ff:fe80:1  implicit-null
```

Here is the routing tables showing the MPLS segment routing label operations:

```
Node-1@vyos:~$ show ip route isis
Codes: K - kernel route, C - connected, S - static, R - RIP,
       O - OSPF, I - IS-IS, B - BGP, E - EIGRP, N - NHRP,
       T - Table, v - VNC, V - VNC-Direct, A - Babel, F - PBR,
       f - OpenFabric,
       > - selected route, * - FIB route, q - queued, r - rejected, b - backup
       t - trapped, o - offload failure

I 192.0.2.0/24 [115/20] via 192.0.2.2, eth1 inactive, weight 1, 00:07:48
I>* 198.51.100.2/32 [115/20] via 192.0.2.2, eth1, label IPv4 Explicit Null, weight 1,
↳ 00:03:39

Node-2@vyos:~$ show ip route isis
Codes: K - kernel route, C - connected, S - static, R - RIP,
       O - OSPF, I - IS-IS, B - BGP, E - EIGRP, N - NHRP,
       T - Table, v - VNC, V - VNC-Direct, A - Babel, F - PBR,
       f - OpenFabric,
       > - selected route, * - FIB route, q - queued, r - rejected, b - backup
       t - trapped, o - offload failure
```

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```
I 192.0.2.0/24 [115/20] via 192.0.2.1, eth1 inactive, weight 1, 00:07:46
I>* 198.51.100.1/32 [115/20] via 192.0.2.1, eth1, label IPv4 Explicit Null, weight 1,
↪ 00:03:43
```

Enable IS-IS with Segment Routing over IPv6 (experimental)

The following example demonstrates IS-IS with Segment Routing over IPv6 between two VyOS routers.

Node 1:

```
set interfaces dummy dum6 description "SRv6 IS-IS"
set interfaces ethernet eth1 address '192.0.2.1/24'
set interfaces loopback lo address '198.51.100.1/32'

set protocols segment-routing srv6 locator MAIN prefix 2001:db8:1::/64
set protocols segment-routing interface eth1

set protocols isis interface eth1
set protocols isis interface lo
set protocols isis net '49.0001.1980.5110.0001.00'
set protocols isis segment-routing srv6 locator MAIN
set protocols isis segment-routing srv6 interface dum6
```

Node 2:

```
set interfaces dummy dum6 description "SRv6 IS-IS"
set interfaces ethernet eth1 address '192.0.2.2/24'
set interfaces loopback lo address '198.51.100.2/32'

set protocols segment-routing srv6 locator MAIN prefix 2001:db8:2::/64
set protocols segment-routing interface eth1

set protocols isis interface eth1
set protocols isis interface lo
set protocols isis net '49.0001.1980.5110.0002.00'
set protocols isis segment-routing srv6 locator MAIN
set protocols isis segment-routing srv6 interface dum6
```

Enable IS-IS with Segment Routing over IPv6 (uSID) (experimental)

The following example demonstrates IS-IS with SRv6 uSID (micro-SID) between two VyOS routers.

Node 1:

```
set interfaces dummy dum6 description "SRv6 IS-IS"
set interfaces ethernet eth1 address '192.0.2.1/24'
set interfaces loopback lo address '198.51.100.1/32'

set protocols segment-routing interface eth1
set protocols segment-routing srv6 locator MAIN prefix 2001:db8:1::/48
set protocols segment-routing srv6 locator MAIN behavior-usid
```

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```

set protocols segment-routing srv6 locator MAIN block-len 32
set protocols segment-routing srv6 locator MAIN format usid-f3216
set protocols segment-routing srv6 locator MAIN func-bits 16
set protocols segment-routing srv6 locator MAIN node-len 16

set protocols isis interface eth1
set protocols isis interface lo
set protocols isis net '49.0001.1980.5110.0001.00'
set protocols isis segment-routing srv6 interface dum6
set protocols isis segment-routing srv6 locator MAIN

```

Node 2:

```

set interfaces dummy dum6 description "SRv6 IS-IS"
set interfaces ethernet eth1 address '192.0.2.2/24'
set interfaces loopback lo address '198.51.100.2/32'

set protocols segment-routing interface eth1
set protocols segment-routing srv6 locator MAIN prefix 2001:db8:2::/48
set protocols segment-routing srv6 locator MAIN behavior-usid
set protocols segment-routing srv6 locator MAIN block-len 32
set protocols segment-routing srv6 locator MAIN format usid-f3216
set protocols segment-routing srv6 locator MAIN func-bits 16
set protocols segment-routing srv6 locator MAIN node-len 16

set protocols isis interface eth1
set protocols isis interface lo
set protocols isis net '49.0001.1980.5110.0002.00'
set protocols isis segment-routing srv6 interface dum6
set protocols isis segment-routing srv6 locator MAIN

```

7.9.8 MPLS

MPLS (Multiprotocol Label Switching) is a packet-forwarding method that differs from traditional IP forwarding. Instead of performing IP address lookups, MPLS routers forward traffic using exact-match lookups on MPLS labels inserted between the Layer 2 (Ethernet) and Layer 3 (IP) headers.

MPLS labels can be assigned statically or dynamically. This section focuses on dynamic label allocation using label distribution protocols such as the LDP (Label Distribution Protocol) and RSVP-TE (Resource Reservation Protocol for Traffic Engineering), or via SR (Segment Routing) extensions to OSPF or IS-IS.

These protocols establish a unidirectional path called a LSP (Label-Switched Path) across the network. When a packet enters the MPLS network, the ingress router pushes an MPLS label onto it based on the destination and forwards the packet along the LSP. Each intermediate router uses the incoming label to look up the outgoing interface and next label value, swaps the label, and forwards the packet on that interface. The egress router pops the label and forwards the original packet using normal IP routing.

MPLS support in VyOS

MPLS support in VyOS is under development, so its feature set is currently limited. Basic MPLS forwarding is functional, as the underlying FRR routing stack supports [RFC 3031](#).

Supported features:

- **LDP:** Implemented according to [RFC 5036](#).

Note: Other related LDP standards include RFCs [6720](#), [6667](#), [5919](#), [5561](#), and [7552](#).

Current limitations:

- **MPLS VPNs:** No support yet for MPLS-enabled VPN services, such as L2VPNs and mVPNs.
- **RSVP-TE:** Not supported, as the underlying FRR stack does not currently implement it.

Label Distribution Protocol (LDP)

MPLS supports various protocols for path creation. VyOS implements LDP via the underlying FRR routing stack, per [RFC 5036](#).

LDP is a TCP-based MPLS signaling protocol that dynamically distributes labels to create label-switched paths. It is not a routing protocol and depends on existing routing protocols for forwarding decisions and communication with other LDP-enabled routers.

To exchange label advertisements, LDP establishes TCP sessions with automatically discovered or statically configured neighbors. Each session is formed using the peer's transport address, which must be present in the routing table and continuously reachable.

Note: It is recommended to use the same address for both the LDP router ID and the discovery transport address. For MPLS LDP to function in VyOS, both parameters must be explicitly configured.

Note: LDP has no mechanism to refresh an existing session. To apply a configuration change or a change to negotiated session capabilities, the LDP neighbor must be reset on the local router.

Configuration

set protocols mpls interface <interface>

Enable MPLS on the specified interface.

Example:

```
set protocols mpls interface eth1
```

set protocols mpls parameters no-propagate-ttl

Disable propagation of the IP TTL into the MPLS header at label imposition.

Example:

```
set protocols mpls parameters no-propagate-ttl
```

```
set protocols mpls parameters maximum-ttl <1-255>
```

Configure the maximum TTL value for MPLS packets.

This value is used when the TTL is not propagated from the IP header. The default is 255.

Example:

```
set protocols mpls parameters maximum-ttl 15
```

```
set protocols mpls ldp interface <interface>
```

Enable LDP on the specified interface.

Example:

```
set protocols mpls ldp interface eth1
```

```
set protocols mpls ldp interface <interface> disable-establish-hello
```

Disable the triggered Hello that the router sends on the specified interface in response to receiving an LDP Hello from a peer.

Example:

```
set protocols mpls ldp interface eth0 disable-establish-hello
```

```
set protocols mpls ldp router-id <address>
```

Configure the IP address used as the LDP router ID for the local device.

Example:

```
set protocols mpls ldp router-id 192.0.2.1
```

```
set protocols mpls ldp discovery transport-ipv4-address <address>
```

Configure the IPv4 transport address for IPv4 LDP connections.

Example:

```
set protocols mpls ldp discovery transport-ipv4-address 192.0.2.1
```

```
set protocols mpls ldp discovery transport-ipv6-address <address>
```

Configure the IPv6 transport address for IPv6 LDP connections.

Example:

```
set protocols mpls ldp discovery transport-ipv6-address 2001:db8::1
```

```
set protocols mpls ldp neighbor <ipv4-address> password <password>
```

Configure authentication for the LDP session with the specified neighbor.

Note: For the session to establish successfully, both peers must be configured with the same password.

Example:

```
set protocols mpls ldp neighbor 192.0.2.2 password mysharedsecret
```

```
set protocols mpls ldp neighbor <ipv4-address> session-holdtime <15-65535>
```

Configure the LDP session hold time, in seconds, advertised to the specified neighbor.

The neighbor must be reset for this change to take effect.

Example:

```
set protocols mpls ldp neighbor 192.0.2.2 session-holdtime 180
```

```
set protocols mpls ldp neighbor <ipv4-address> ttl-security <disable | 1-254>
```

Configure TTL security (GTSM (Generalized TTL Security Mechanism)) for the LDP session with the specified neighbor.

Use `disable` to turn off TTL security, or set a hop-count value (1-254) to restrict which incoming packets are accepted based on their TTL.

Example:

```
set protocols mpls ldp neighbor 192.0.2.2 ttl-security 5
```

```
set protocols mpls ldp discovery hello-ipv4-interval <1-65535>
```

Configure the interval, in seconds, at which the router sends IPv4 LDP Hello messages.

Example:

```
set protocols mpls ldp discovery hello-ipv4-interval 5
```

```
set protocols mpls ldp discovery hello-ipv4-holdtime <1-65535>
```

Configure the hold time, in seconds, advertised in IPv4 LDP Hello messages.

Example:

```
set protocols mpls ldp discovery hello-ipv4-holdtime 15
```

```
set protocols mpls ldp discovery hello-ipv6-interval <1-65535>
```

Configure the interval, in seconds, at which the router sends IPv6 LDP Hello messages.

Example:

```
set protocols mpls ldp discovery hello-ipv6-interval 5
```

```
set protocols mpls ldp discovery hello-ipv6-holdtime <1-65535>
```

Configure the hold time, in seconds, advertised in IPv6 LDP Hello messages.

Example:

```
set protocols mpls ldp discovery hello-ipv6-holdtime 15
```

```
set protocols mpls ldp discovery session-ipv4-holdtime <15-65535>
```

Configure the LDP session hold time, in seconds, advertised in IPv4 LDP Initialization messages.

Example:

```
set protocols mpls ldp discovery session-ipv4-holdtime 180
```

```
set protocols mpls ldp discovery session-ipv6-holdtime <15-65535>
```


Configure the LDP session hold time, in seconds, advertised in IPv6 LDP Initialization messages.

Example:

```
set protocols mpls ldp discovery session-ipv6-holdtime 180
```

```
set protocols mpls ldp import ipv4 import-filter filter-access-list <1-2699>
```

Filter the IPv4 label bindings (associations between an IPv4 prefix and an MPLS label) from LDP peers using the specified access list.

Only label bindings whose prefix is permitted by the access list are accepted.

Example:

```
set protocols mpls ldp import ipv4 import-filter filter-access-list 10
```

```
set protocols mpls ldp import ipv4 import-filter neighbor-access-list <1-2699>
```

Filter the IPv4 label bindings (associations between an IPv4 prefix and an MPLS label) from LDP peers using the specified access list.

Only label bindings received from a peer whose IPv4 address is permitted by the access list are accepted.

Example:

```
set protocols mpls ldp import ipv4 import-filter neighbor-access-list 20
```

```
set protocols mpls ldp import ipv6 import-filter filter-access-list6 <1-2699>
```

Filter the IPv6 label bindings (associations between an IPv6 prefix and an MPLS label) from LDP peers using the specified access list.

Only label bindings whose prefix is permitted by the access list are accepted.

Example:

```
set protocols mpls ldp import ipv6 import-filter filter-access-list6 10
```

```
set protocols mpls ldp import ipv6 import-filter neighbor-access-list6 <1-2699>
```

Filter the IPv6 label bindings (associations between an IPv6 prefix and an MPLS label) from LDP peers using the specified access list.

Only label bindings received from a peer whose IPv6 address is permitted by the access list are accepted.

Example:

```
set protocols mpls ldp import ipv6 import-filter neighbor-access-list6 20
```

```
set protocols mpls ldp export ipv4 export-filter filter-access-list <1-2699>
```

Filter the IPv4 label bindings (associations between an IPv4 prefix and an MPLS label) advertised to LDP peers using the specified access list.

Only label bindings whose prefix is permitted by the access list are advertised.

Example:

```
set protocols mpls ldp export ipv4 export-filter filter-access-list 10
```

```
set protocols mpls ldp export ipv4 export-filter neighbor-access-list <1-2699>
```

Filter the IPv4 label bindings (associations between an IPv4 prefix and an MPLS label) advertised to LDP peers using the specified access list.

Label bindings are advertised only to peers whose IPv4 address is permitted by the access list.

Example:

```
set protocols mpls ldp export ipv4 export-filter neighbor-access-list 20
```

```
set protocols mpls ldp export ipv6 export-filter filter-access-list6 <1-2699>
```

Filter the IPv6 label bindings (associations between an IPv6 prefix and an MPLS label) advertised to LDP peers using the specified access list.

Only label bindings whose prefix is permitted by the access list are advertised.

Example:

```
set protocols mpls ldp export ipv6 export-filter filter-access-list6 10
```

```
set protocols mpls ldp export ipv6 export-filter neighbor-access-list6 <1-2699>
```

Filter the IPv6 label bindings (associations between an IPv6 prefix and an MPLS label) advertised to LDP peers using the specified access list.

Label bindings are advertised only to peers whose IPv6 address is permitted by the access list.

Example:

```
set protocols mpls ldp export ipv6 export-filter neighbor-access-list6 20
```

```
set protocols mpls ldp export ipv4 explicit-null
```

Configure the router to advertise the Explicit Null label for its directly connected IPv4 prefixes to LDP neighbors.

This instructs the penultimate-hop router to forward labeled packets unchanged without removing the MPLS label.

Example:

```
set protocols mpls ldp export ipv4 explicit-null
```

```
set protocols mpls ldp export ipv6 explicit-null
```

Configure the router to advertise the Explicit Null label for its directly connected IPv6 prefixes to LDP neighbors.

This instructs the penultimate-hop router to forward labeled packets unchanged without removing the MPLS label.

Example:

```
set protocols mpls ldp export ipv6 explicit-null
```

```
set protocols mpls ldp allocation ipv4 access-list <1-2699>
```

Restrict MPLS label allocation to IPv4 prefixes permitted by the specified access list.

By default, LDP allocates a label for every IPv4 prefix in the routing table. A common best practice is to limit allocation to loopback addresses.

Example:

```
set protocols mpls ldp allocation ipv4 access-list 10
```

```
set protocols mpls ldp allocation ipv6 access-list6 <1-2699>
```

Restrict MPLS label allocation to IPv6 prefixes permitted by the specified access list.

By default, LDP allocates a label for every IPv6 prefix in the routing table. A common best practice is to limit allocation to loopback addresses.

Example:

```
set protocols mpls ldp allocation ipv6 access-list6 10
```

```
set protocols mpls ldp parameters cisco-interop-tlv
```

Configure the router to negotiate the Dual-Stack capability TLV ([RFC 7552](#)) using a Cisco non-compliant format for dual-stack LDP sessions.

Example:

```
set protocols mpls ldp parameters cisco-interop-tlv
```

```
set protocols mpls ldp parameters ordered-control
```

Enable LDP Ordered Label Distribution Control mode ([RFC 5036](#)) for the router.

By default, the router operates in Independent Label Distribution Control mode.

Example:

```
set protocols mpls ldp parameters ordered-control
```

```
set protocols mpls ldp parameters transport-prefer-ipv4
```

Configure the router to prefer an IPv4 TCP transport connection for LDP peering when LDP is configured for dual-stack operation (both IPv4 and IPv6 address families are enabled).

Example:

```
set protocols mpls ldp parameters transport-prefer-ipv4
```

```
set protocols mpls ldp targeted-neighbor ipv4 enable
```

Enable the router to accept targeted IPv4 LDP sessions.

This allows the router to establish LDP adjacencies with non-directly connected peers by responding to inbound targeted (unicast) LDP Hello messages.

Example:

```
set protocols mpls ldp targeted-neighbor ipv4 enable
```

```
set protocols mpls ldp targeted-neighbor ipv6 enable
```

Enable the router to accept targeted IPv6 LDP sessions.

This allows the router to establish LDP adjacencies with non-directly connected peers by responding to inbound targeted (unicast) LDP Hello messages.

Example:

```
set protocols mpls ldp targeted-neighbor ipv6 enable
```

```
set protocols mpls ldp targeted-neighbor ipv4 address <address>
```

Configure the router to initiate a targeted IPv4 LDP session with the specified remote LSR (Label Switching Router).

Example:

```
set protocols mpls ldp targeted-neighbor ipv4 address 192.0.2.5
```

```
set protocols mpls ldp targeted-neighbor ipv6 address <address>
```

Configure the router to initiate a targeted IPv6 LDP session with the specified remote LSR.

Example:

```
set protocols mpls ldp targeted-neighbor ipv6 address 2001:db8::5
```

```
set protocols mpls ldp targeted-neighbor ipv4 hello-holdtime <1-65535>
```

Configure the hold time, in seconds, advertised in targeted IPv4 LDP Hello messages.

Example:

```
set protocols mpls ldp targeted-neighbor ipv4 hello-holdtime 45
```

```
set protocols mpls ldp targeted-neighbor ipv4 hello-interval <1-65535>
```

Configure the interval, in seconds, between targeted IPv4 LDP Hello messages sent to remote LSRs.

Example:

```
set protocols mpls ldp targeted-neighbor ipv4 hello-interval 15
```

```
set protocols mpls ldp targeted-neighbor ipv6 hello-holdtime <1-65535>
```

Configure the hold time, in seconds, advertised in targeted IPv6 LDP Hello messages.

Example:

```
set protocols mpls ldp targeted-neighbor ipv6 hello-holdtime 45
```

```
set protocols mpls ldp targeted-neighbor ipv6 hello-interval <1-65535>
```

Configure the interval, in seconds, between targeted IPv6 LDP Hello messages sent to remote LSRs.

Example:

```
set protocols mpls ldp targeted-neighbor ipv6 hello-interval 15
```

Operation

When LDP is established, you can view label assignments directly within the `show ip route` or `show ipv6 route` outputs. To inspect LDP operational states, use the following commands.

Show

show mpls ldp binding

Show the LIB (Label Information Base).

show mpls ldp binding <prefix>

Show LDP label bindings for the specified prefix.

The <prefix> parameter accepts an IPv4 prefix (x.x.x.x/x) or IPv6 prefix (h:h:h:h:h:h:h/h).

show mpls ldp binding detail

Show detailed LDP Label Information Base (LIB) entries.

show mpls ldp binding neighbor <address>

Show LDP label bindings learned from the specified peer.

The <address> parameter accepts an IPv4 or IPv6 address.

show mpls ldp binding local-label <label>

Show LDP label bindings associated with the specified locally assigned label value.

show mpls ldp binding remote-label <label>

Show LDP label bindings associated with the specified remotely assigned label value.

show mpls ldp discovery

Show LDP discovery Hello information.

show mpls ldp discovery detail

Show detailed LDP discovery Hello information.

show mpls ldp interface

Show LDP interface information.

show mpls ldp neighbor

Show basic LDP neighbor information.

show mpls ldp neighbor detail

Show detailed LDP neighbor information.

show mpls ldp neighbor capabilities

Show the LDP capabilities negotiated with each peer during session initialization.

show mpls ldp neighbor <address>

Show LDP neighbor information for the specified peer.

The <address> parameter accepts an IPv4 or IPv6 address.

show mpls ldp neighbor <address> detail

Show detailed LDP session information for the specified peer.

show mpls ldp neighbor <address> capabilities

Show the LDP capabilities negotiated with the specified peer.

show mpls table

Show the MPLS table.

show mpls pseudowire

Show MPLS pseudowire interfaces.

Reset**reset mpls ldp neighbor <address>**

Reset the established LDP session with the specified neighbor.

The <address> parameter accepts an IPv4 or IPv6 address.

Example

The following configuration sets up a basic MPLS LDP-enabled router. OSPF is used as the underlying IGP to provide reachability between LDP peers' loopback addresses, which are then used as the LDP transport addresses and router IDs. MPLS forwarding and LDP are enabled on the interface connecting to the network.

```
set protocols ospf area 0 network '192.0.2.1/32'
set protocols ospf area 0 network '192.0.2.4/31'
set protocols ospf parameters router-id '192.0.2.1'
set protocols mpls interface 'eth1'
set protocols mpls ldp discovery transport-ipv4-address '192.0.2.1'
set protocols mpls ldp interface 'eth1'
set protocols mpls ldp interface 'lo'
set protocols mpls ldp router-id '192.0.2.1'
set interfaces ethernet eth1 address '192.0.2.5/31'
set interfaces loopback lo address '192.0.2.1/32'
```

Note: This example assumes a transit (P-router) role: MPLS forwarding is enabled on the transit interface only, not on the loopback. If the router needs to terminate MPLS-encapsulated traffic on its loopback (PE-router role), also configure `set protocols mpls interface 'lo'`.

Apply the same configuration on the LDP peer (using its own loopback and transit interface addresses) to establish a basic LDP session.

7.9.9 Multicast

Static multicast routes (`mroutes`) explicitly define the expected upstream path (either a next-hop IP address or an incoming interface) for RPF checks on arriving multicast traffic, overriding the default RPF behavior.

When a multicast packet arrives, the router performs the RPF check against the matching `mroute` in the dedicated **multicast** routing table (MRIB (Multicast Routing Information Base)), rather than looking up the standard **unicast** routing table (RIB).

If the packet arrives via the expected upstream path defined in the MRIB, the RPF check passes, and the packet is forwarded toward the multicast receivers. Otherwise, the packet is dropped.

Administrators build the MRIB by manually configuring `mroutes`. The ZEBRA routing daemon uses these entries exclusively for multicast RPF lookups. ZEBRA neither evaluates `mroutes` in the unicast RIB nor installs them into the kernel FIB.

Configuration

Use the following commands to configure `mroutes` for a specific multicast source subnet.

Note: Because `mroutes` bypass standard forwarding logic, they may cause unpredictable network behavior. Use `mroutes` with caution; in most cases, overriding the default RPF check is unnecessary.

Next-hop `mroutes`

`set protocols static mroute <subnet> next-hop <address>`

Configure the next-hop IP address for the specified source subnet.

When multicast traffic originates from this subnet, the router expects it to arrive from the upstream neighbor at the specified IP address.

You can configure multiple next-hop IP addresses for the same source subnet.

Example:

```
set protocols static mroute 192.0.2.0/24 next-hop 203.0.113.1
```

`set protocols static mroute <subnet> next-hop <address> distance <1-255>`

Configure the administrative distance for the `mroute` via the specified next-hop IP address.

`Mroutes` with a lower administrative distance are prioritized over `mroutes` with a higher distance.

The allowed range is 1 to 255.

Example:

```
set protocols static mroute 192.0.2.0/24 next-hop 203.0.113.1 distance 50
```

`set protocols static mroute <subnet> next-hop <address> disable`

Disable the `mroute` via the specified next-hop IP address.

This command temporarily deactivates the `mroute` while preserving the configured subnet, next-hop, and distance parameters.

Example:

```
set protocols static mroute 192.0.2.0/24 next-hop 203.0.113.1 disable
```

Interface mroutes

set protocols static mroute <subnet> interface <interface>

Configure the valid interface for the specified source subnet.

When multicast traffic originates from this subnet, the router expects it to arrive on the specified interface.

You can configure multiple interfaces for the same source subnet.

Example:

```
set protocols static mroute 192.0.2.0/24 interface eth0
```

set protocols static mroute <subnet> interface <interface> distance <1-255>

Configure the administrative distance for the mroute via the specified interface.

Mroutes with a lower administrative distance are prioritized over mroutes with a higher distance.

The allowed range is 1 to 255.

Example:

```
set protocols static mroute 192.0.2.0/24 interface eth0 distance 30
```

set protocols static mroute <subnet> interface <interface> disable

Disable the mroute via the specified interface.

This command temporarily deactivates the mroute while preserving the configured subnet, interface, and distance parameters.

Example:

```
set protocols static mroute 192.0.2.0/24 interface eth0 disable
```

7.9.10 Segment Routing

SR is a network architecture similar to source routing. The ingress router attaches a list of segments, known as SIDs (Segment Identifiers), to each packet as it enters the network.

The SID list explicitly defines the path the packet will follow. At each node, the router reads the first SID, executes its forwarding instruction, and typically removes it so the next node can process the subsequent SID.

SR relies on IGP (Interior Gateway Protocols) such as IS-IS or OSPF to advertise SIDs across the network.

Note: SR defines a control plane architecture and can be applied to existing MPLS-based data planes. In MPLS networks, segments are encoded as MPLS labels and applied at the ingress router. These labels are then distributed across the routing domain by IGPs such as IS-IS (as described in [RFC 8667](#)) or OSPF.

SR for the MPLS data plane supports IPv4, IPv6, and ECMP (Equal-Cost Multi-Path), and has been tested with Cisco and Juniper routers. However, this deployment is still experimental for FRR.

IS-IS SR configuration

Use the following commands to enable SR on IS-IS.

Known limitations:

- No support for level redistribution (L1 to L2 or L2 to L1).
- No support for BSID (Binding SID).
- Only a single SRGB (Segment Routing Global Block) and the default SPF algorithm are supported.

set protocols isis segment-routing global-block high-label-value <16-1048575>

Configure the upper bound of the Segment Routing Global Block (SRGB).

The SRGB defines the range of MPLS labels reserved for mapping global segments, such as Prefix-SIDs, to FIB entries.

The range cannot exceed 65535 labels.

Example:

```
set protocols isis segment-routing global-block high-label-value 23999
```

set protocols isis segment-routing global-block low-label-value <16-1048575>

Configure the lower bound of the Segment Routing Global Block (SRGB).

The SRGB defines the range of MPLS labels reserved for mapping global segments, such as Prefix-SIDs, to FIB entries.

The range cannot exceed 65535 labels.

Example:

```
set protocols isis segment-routing global-block low-label-value 16000
```

set protocols isis segment-routing local-block high-label-value <16-1048575>

Configure the upper bound of the Segment Routing Local Block (SRLB).

The SRLB defines the range of MPLS labels that a router reserves for its local segments, such as Adjacency-SIDs.

The range cannot exceed 65535 labels.

Example:

```
set protocols isis segment-routing local-block high-label-value 15999
```

set protocols isis segment-routing local-block low-label-value <16-1048575>

Configure the lower bound of the Segment Routing Local Block (SRLB).

The SRLB defines the range of MPLS labels that a router reserves for its local segments, such as Adjacency-SIDs.

The range cannot exceed 65535 labels.

Example:

```
set protocols isis segment-routing local-block low-label-value 15000
```

Note: SR label blocks have the following configuration constraints:

- Both bounds (high-label-value and low-label-value) must be set. Removing either bound also removes the other.
- local-block requires global-block to be configured.
- The SRGB and SRLB ranges must not overlap.

Violating any of these causes the commit to fail.

set protocols isis segment-routing maximum-label-depth <1-16>

Configure the MSD (Maximum SID Depth) supported by the router.

The value depends on the MPLS data plane.

Example:

```
set protocols isis segment-routing maximum-label-depth 10
```

set protocols isis segment-routing prefix <address> index value <0-65535>

Configure a Prefix-SID with an index value for the specified IP prefix.

A Prefix-SID is a Segment Identifier associated with an IP prefix and distributed by the IGP (IS-IS).

Example:

```
set protocols isis segment-routing prefix 192.0.2.1/32 index value 1
```

set protocols isis segment-routing prefix <address> index <no-php-flag | explicit-null>

Configure a label processing flag for the indexed Prefix-SID associated with the specified IP prefix:

- no-php-flag: Requests the upstream neighbor not to pop the Prefix-SID label before forwarding the packet.
- explicit-null: Requests that the upstream neighbor forwards the packet with the Explicit-Null label.

Example:

```
set protocols isis segment-routing prefix 192.0.2.1/32 index no-php-flag
```

set protocols isis segment-routing prefix <address> absolute value <16-1048575>

Configure a Prefix-SID with an absolute value for the specified IP prefix

A Prefix-SID is a Segment Identifier associated with an IP prefix and distributed by the IGP (IS-IS).

Example:

```
set protocols isis segment-routing prefix 192.0.2.1/32 absolute value 16001
```

set protocols isis segment-routing prefix <address> absolute <no-php-flag | explicit-null>

Configure a label processing flag for the absolute Prefix-SID associated with the specified IP prefix:

- no-php-flag: Requests the upstream neighbor not to pop the Prefix-SID label before forwarding the packet.
- explicit-null: Requests that the upstream neighbor forwards the packet with the Explicit-Null label.

Example:

```
set protocols isis segment-routing prefix 192.0.2.1/32 absolute no-php-flag
```

Operational commands

show isis segment-routing node

Show all learned SR nodes.

show isis route prefix-sid

Show all learned Prefix-SIDs and their MPLS labels.

Note: For more information, see [IS-IS](#).

OSPF SR configuration

Use the following commands to enable SR on OSPF.

set protocols ospf parameters opaque-lsa

Enable Opaque LSA (Link State Advertisement) ([RFC 2370](#)) in OSPF.

Opaque LSAs are required to transport MPLS labels via OSPF.

Example:

```
set protocols ospf parameters opaque-lsa
```

set protocols ospf segment-routing global-block high-label-value <16-1048575>

Configure the upper bound of the Segment Routing Global Block (SRGB).

The SRGB defines the range of MPLS labels reserved for mapping Prefix-SIDs to FIB entries.

The range cannot exceed 65535 labels.

Example:

```
set protocols ospf segment-routing global-block high-label-value 23999
```

set protocols ospf segment-routing global-block low-label-value <16-1048575>

Configure the lower bound of the Segment Routing Global Block (SRGB).

The SRGB defines the range of MPLS labels reserved for mapping Prefix-SIDs to FIB entries.

The range cannot exceed 65535 labels.

Example:

```
set protocols ospf segment-routing global-block low-label-value 16000
```

set protocols ospf segment-routing local-block high-label-value <16-1048575>

Configure the upper bound of the Segment Routing Local Block (SRLB).

The SRLB defines the range of MPLS labels that a router reserves for its local segments, such as Adjacency-SIDs.

The range cannot exceed 65535 labels.

Example:

```
set protocols ospf segment-routing local-block high-label-value 15999
```

```
set protocols ospf segment-routing local-block low-label-value <16-1048575>
```

Configure the lower bound of the Segment Routing Local Block (SRLB).

The SRLB defines the range of MPLS labels that a router reserves for its local segments, such as Adjacency-SIDs.

The range cannot exceed 65535 labels.

Example:

```
set protocols ospf segment-routing local-block low-label-value 15000
```

Note: SR label blocks have the following configuration constraints:

- Both bounds (high-label-value and low-label-value) must be set. Removing either bound also removes the other.
- local-block requires global-block to be configured.
- The SRGB and SRLB ranges must not overlap.

Violating any of these causes the commit to fail.

```
set protocols ospf segment-routing maximum-label-depth <1-16>
```

Configure the MSD supported by the router.

The value depends on the MPLS data plane.

Example:

```
set protocols ospf segment-routing maximum-label-depth 10
```

```
set protocols ospf segment-routing prefix <address> index value <0-65535>
```

Configure a Prefix-SID for the specified IP prefix.

A Prefix-SID is a Segment Identifier associated with an IP prefix and distributed by the IGP (OSPF).

Prefix-SIDs are unique within an SR domain.

Example:

```
set protocols ospf segment-routing prefix 192.0.2.1/32 index value 1
```

```
set protocols ospf segment-routing prefix <address> index <no-php-flag | explicit-null>
```

Configure a label processing flag for the Prefix-SID associated with the specified IP prefix:

- no-php-flag: Requests the upstream neighbor not to pop the Prefix-SID label before forwarding the packet.

- **explicit-null**: Requests that the upstream neighbor forwards the packet with the Explicit-Null label.

Example:

```
set protocols ospf segment-routing prefix 192.0.2.1/32 index no-php-flag
```

Note: For more information, see ospf.

Examples

Enable SR on IS-IS (experimental)

The following example demonstrates a basic two-node SR topology using IS-IS.

Node 1:

```
set interfaces loopback lo address '198.51.100.1/32'
set interfaces ethernet eth1 address '192.0.2.1/24'

set protocols isis interface eth1
set protocols isis interface lo
set protocols isis net '49.0001.1980.5110.0001.00'
set protocols isis segment-routing global-block high-label-value '599'
set protocols isis segment-routing global-block low-label-value '550'
set protocols isis segment-routing prefix 198.51.100.1/32 index value '1'
set protocols isis segment-routing prefix 198.51.100.1/32 index explicit-null
set protocols mpls interface 'eth1'
```

Node 2:

```
set interfaces loopback lo address '198.51.100.2/32'
set interfaces ethernet eth1 address '192.0.2.2/24'

set protocols isis interface eth1
set protocols isis interface lo
set protocols isis net '49.0001.1980.5110.0002.00'
set protocols isis segment-routing global-block high-label-value '599'
set protocols isis segment-routing global-block low-label-value '550'
set protocols isis segment-routing prefix 198.51.100.2/32 index value '2'
set protocols isis segment-routing prefix 198.51.100.2/32 index explicit-null
set protocols mpls interface 'eth1'
```

The following outputs show that MPLS Segment Routing is enabled and that each router has learned MPLS labels for the other router's loopback:

Node-1@vyos:~\$ show mpls table

Inbound Label	Type	Nexthop	Outbound Label	
552	SR (IS-IS)	192.0.2.2	IPv4 Explicit Null	<-- Node-2
↳ loopback learned on Node-1				
15000	SR (IS-IS)	192.0.2.2	implicit-null	
15001	SR (IS-IS)	fe80::e87:6cff:fe09:1	implicit-null	

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15002	SR (IS-IS)	192.0.2.2	implicit-null
15003	SR (IS-IS)	fe80::e87:6cff:fe09:1	implicit-null

```
Node-2@vyos:~$ show mpls table
Inbound Label  Type           Nexthop           Outbound Label
-----
551            SR (IS-IS)  192.0.2.1         IPv4 Explicit Null  <-- Node-1
↳ loopback learned on Node-2
15000          SR (IS-IS)  192.0.2.1         implicit-null
15001          SR (IS-IS)  fe80::e33:2ff:fe80:1  implicit-null
15002          SR (IS-IS)  192.0.2.1         implicit-null
15003          SR (IS-IS)  fe80::e33:2ff:fe80:1  implicit-null
```

The following outputs show MPLS Segment Routing label assignments for IP routes:

```
Node-1@vyos:~$ show ip route isis
Codes: K - kernel route, C - connected, S - static, R - RIP,
       O - OSPF, I - IS-IS, B - BGP, E - EIGRP, N - NHRP,
       T - Table, v - VNC, V - VNC-Direct, A - Babel, F - PBR,
       f - OpenFabric,
       > - selected route, * - FIB route, q - queued, r - rejected, b - backup
       t - trapped, o - offload failure

I  192.0.2.0/24 [115/20] via 192.0.2.2, eth1 inactive, weight 1, 00:07:48
I>* 198.51.100.2/32 [115/20] via 192.0.2.2, eth1, label IPv4 Explicit Null, weight 1,
↳ 00:03:39
```

```
Node-2@vyos:~$ show ip route isis
Codes: K - kernel route, C - connected, S - static, R - RIP,
       O - OSPF, I - IS-IS, B - BGP, E - EIGRP, N - NHRP,
       T - Table, v - VNC, V - VNC-Direct, A - Babel, F - PBR,
       f - OpenFabric,
       > - selected route, * - FIB route, q - queued, r - rejected, b - backup
       t - trapped, o - offload failure

I  192.0.2.0/24 [115/20] via 192.0.2.1, eth1 inactive, weight 1, 00:07:46
I>* 198.51.100.1/32 [115/20] via 192.0.2.1, eth1, label IPv4 Explicit Null, weight 1,
↳ 00:03:43
```

Enable SR on OSPF (experimental)

The following example demonstrates a basic two-node SR topology using OSPF.

Node 1:

```
set interfaces loopback lo address 198.51.100.1/32
set interfaces ethernet eth0 address 192.0.2.1/24
set protocols ospf area 0 network '192.0.2.0/24'
set protocols ospf area 0 network '198.51.100.1/32'
set protocols ospf parameters opaque-lsa
set protocols ospf parameters router-id '198.51.100.1'
```

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```
set protocols ospf segment-routing global-block high-label-value '1100'
set protocols ospf segment-routing global-block low-label-value '1000'
set protocols ospf segment-routing prefix 198.51.100.1/32 index explicit-null
set protocols ospf segment-routing prefix 198.51.100.1/32 index value '1'
```

Node 2:

```
set interfaces loopback lo address 198.51.100.2/32
set interfaces ethernet eth0 address 192.0.2.2/24
set protocols ospf area 0 network '192.0.2.0/24'
set protocols ospf area 0 network '198.51.100.2/32'
set protocols ospf parameters opaque-lsa
set protocols ospf parameters router-id '198.51.100.2'
set protocols ospf segment-routing global-block high-label-value '1100'
set protocols ospf segment-routing global-block low-label-value '1000'
set protocols ospf segment-routing prefix 198.51.100.2/32 index explicit-null
set protocols ospf segment-routing prefix 198.51.100.2/32 index value '2'
```

The following outputs show MPLS Segment Routing label assignments:

Node-1@vyos:~\$ show mpls table

Inbound Label	Type	Nexthop	Outbound Label
1002	SR (OSPF)	192.0.2.2	IPv4 Explicit Null <-- Node-2 loopback learned
on Node-1			
15000	SR (OSPF)	192.0.2.2	implicit-null
15001	SR (OSPF)	192.0.2.2	implicit-null

Node-2@vyos:~\$ show mpls table

Inbound Label	Type	Nexthop	Outbound Label
1001	SR (OSPF)	192.0.2.1	IPv4 Explicit Null <-- Node-1 loopback learned
on Node-2			
15000	SR (OSPF)	192.0.2.1	implicit-null
15001	SR (OSPF)	192.0.2.1	implicit-null

The following outputs show MPLS Segment Routing label assignments for IP routes:

Node-1@vyos:~\$ show ip route ospf

Codes: K - kernel route, C - connected, S - static, R - RIP,
 O - OSPF, I - IS-IS, B - BGP, E - EIGRP, N - NHRP,
 T - Table, v - VNC, V - VNC-Direct, A - Babel, F - PBR,
 f - OpenFabric,
 > - selected route, * - FIB route, q - queued, r - rejected, b - backup
 t - trapped, o - offload failure

```
O 198.51.100.1/32 [110/0] is directly connected, lo, weight 1, 00:03:43
O>* 198.51.100.2/32 [110/1] via 192.0.2.2, eth0, label IPv4 Explicit Null, weight 1, 00:03:32
O 192.0.2.0/24 [110/1] is directly connected, eth0, weight 1, 00:03:43
```

```

Node-2@vyos:~$ show ip route ospf
Codes: K - kernel route, C - connected, S - static, R - RIP,
       O - OSPF, I - IS-IS, B - BGP, E - EIGRP, N - NHRP,
       T - Table, v - VNC, V - VNC-Direct, A - Babel, F - PBR,
       f - OpenFabric,
       > - selected route, * - FIB route, q - queued, r - rejected, b - backup
       t - trapped, o - offload failure

O>* 198.51.100.1/32 [110/1] via 192.0.2.1, eth0, label IPv4 Explicit Null, weight 1, 00:03:36
O  198.51.100.2/32 [110/0] is directly connected, lo, weight 1, 00:03:51
O  192.0.2.0/24 [110/1] is directly connected, eth0, weight 1, 00:03:51

```

7.9.11 Traffic Engineering

TE (Traffic Engineering) is a set of techniques for steering traffic along paths different from the shortest path computed by the IGP.

To enforce these paths, TE typically uses a form of source routing, such as MPLS labels or SRv6 (Segment Routing over IPv6) extension headers, combined with path constraints, such as administrative groups or bandwidth requirements, and a method to map traffic to TE paths.

Use the commands below to configure link-level TE attributes (administrative groups and bandwidth) and enable TE in the IGP (specifically, IS-IS) so this information is distributed across the network.

Common link parameters

The following commands define link-level TE attributes that can be advertised by both IS-IS and OSPF (although OSPF advertisement is not supported yet).

set protocols traffic-engineering admin-group <admin-group-name> bit-position <0-31>

Configure an administrative group and associate it with a bit position.

These groups can then be referenced by the per-interface commands below.

Note: Two administrative groups cannot share the same bit-position value.

Example:

```

set protocols traffic-engineering admin-group primary bit-position 0
set protocols traffic-engineering admin-group backup bit-position 1

```

set protocols traffic-engineering interface <ifname> admin-group <admin-group-name>

Configure an administrative group on the specified interface.

You can configure multiple administrative groups on the same interface.

Example:

```

set protocols traffic-engineering interface eth0 admin-group primary

```

```

set protocols traffic-engineering interface <ifname> metric <1-4294967295>

```


Configure the TE metric for the specified interface (distinct from the OSPF/ISIS metric).

Example:

```
set protocols traffic-engineering interface eth0 metric 100
```

```
set protocols traffic-engineering interface <ifname> max-bandwidth <1-4294967295>
```

Configure the maximum bandwidth, in Mbps, for the specified interface.

Example:

```
set protocols traffic-engineering interface eth0 max-bandwidth 1000
```

```
set protocols traffic-engineering interface <ifname> max-reservable-bandwidth  
<1-4294967295>
```

Configure the maximum reservable bandwidth, in Mbps, for the specified interface.

Example:

```
set protocols traffic-engineering interface eth0 max-reservable-bandwidth 800
```

IS-IS TE configuration

The following commands enable TE for IS-IS.

```
set protocols isis traffic-engineering enable
```

Enable Traffic Engineering for IS-IS.

Example:

```
set protocols isis traffic-engineering enable
```

```
set protocols isis traffic-engineering export
```

Export the local TED (Traffic Engineering Database) to other FRR daemons.

Example:

```
set protocols isis traffic-engineering export
```

```
set protocols isis traffic-engineering address <ipv4-address>
```

Configure the IPv4 address used as the TE Router ID for MPLS-TE.

Example:

```
set protocols isis traffic-engineering address 198.51.100.1
```

7.9.12 OpenFabric

OpenFabric, specified in [draft-white-openfabric-06.txt](#),

is a routing protocol derived from IS-IS. It provides link-state routing with optimized flooding, making it well-suited for spine-leaf topologies.

OpenFabric is a dual-stack protocol. A single instance can simultaneously route both IPv4 and IPv6 traffic.

Configuration

Mandatory settings

Each OpenFabric router must be configured with a unique NET, the CLNS equivalent of a Router ID. The 6-byte system identifier portion of the NET must be unique across the fabric.

set protocols openfabric net <network-entity-title>

Configure the Network Entity Title (NET) for the router.

A typical NET looks like 49.0001.1921.6800.1002.00.

The NET consists of the following parts:

- **AFI (49)**: OpenFabric conventionally uses AFI value 49 for private addressing.
- **Area identifier (0001)**: The area number within the fabric (in this case, Area 1).
- **System identifier (1921.6800.1002)**: Uniquely identifies the router within the fabric. We recommend deriving this value from the router IP address or MAC address. To construct the system identifier from an IPv4 address (for example, 192.168.1.2):
 - Pad each octet with leading zeros: 192.168.1.2 → 192.168.001.002.
 - Regroup the digits into three 4-digit blocks: 192.168.001.002 → 1921.6800.1002.
- **NET selector (00)**: Must always be 00 to indicate the local system.

Example:

```
set protocols openfabric net 49.0001.1921.6800.1002.00
```

set protocols openfabric domain <name> interface <interface> address-family <ipv4|ipv6>

Configure the named OpenFabric domain on a specific interface for the given address family (IPv4 or IPv6).

For dual-stack operation, run the command twice: once with ipv4 and once with ipv6.

Example:

```
set protocols openfabric domain fabric1 interface eth1 address-family ipv4
set protocols openfabric domain fabric1 interface eth1 address-family ipv6
```

OpenFabric global configuration

```
set protocols openfabric domain <name> domain-password <plaintext-password|md5>  
<password>
```

Configure the authentication password for the specified OpenFabric domain.

The password can be specified as either plain text or an MD5 (Message-Digest Algorithm 5) hash.

Example:

```
set protocols openfabric domain fabric1 domain-password md5 'shared-secret'
```

```
set protocols openfabric domain <name> purge-originator
```

Enable POI for the specified OpenFabric domain.

POI is defined in RFC 6232.

Example:

```
set protocols openfabric domain fabric1 purge-originator
```

```
set protocols openfabric domain <name> set-overload-bit
```

Configure the overload bit for the specified OpenFabric domain.

This instructs other routers in the domain not to use this router for transit traffic.

Example:

```
set protocols openfabric domain fabric1 set-overload-bit
```

```
set protocols openfabric domain <name> log-adjacency-changes
```

Enable logging of adjacency state changes in the specified OpenFabric domain.

Example:

```
set protocols openfabric domain fabric1 log-adjacency-changes
```

```
set protocols openfabric domain <name> fabric-tier <0-14>
```

Configure a static tier number for the specified OpenFabric domain.

The router advertises the tier value to indicate its location in the OpenFabric domain.

Example:

```
set protocols openfabric domain fabric1 fabric-tier 1
```

Interface configuration

set protocols openfabric domain <name> interface <interface> hello-interval <1-600>

Configure the interval, in seconds, at which the router transmits Hello packets on this interface.

Hello packets are exchanged with neighbors to establish and maintain adjacencies.

Example:

```
set protocols openfabric domain fabric1 interface eth0 hello-interval 5
```

set protocols openfabric domain <name> interface <interface> hello-multiplier <2-100>

Configure the multiplier used to compute the Hello hold time on the specified interface.

Example:

```
set protocols openfabric domain fabric1 interface eth0 hello-multiplier 3
```

set protocols openfabric domain <name> interface <interface> metric <0-16777215>

Configure the routing metric for the specified interface.

This metric is used in path selection to determine the most efficient route. Lower metrics indicate preferred paths.

Example:

```
set protocols openfabric domain fabric1 interface eth0 metric 100
```

set protocols openfabric domain <name> interface <interface> passive

Enable passive mode for the specified interface.

In passive mode, the router does not send Hello packets on the interface and does not form adjacencies.

Example:

```
set protocols openfabric domain fabric1 interface eth0 passive
```

**set protocols openfabric domain <name> interface <interface> password
<plaintext-password|md5> <password>**

Configure the authentication password for the specified interface.

The password can be specified as either plain text or an MD5 hash.

Example:

```
set protocols openfabric domain fabric1 interface eth0 password plaintext-password link-
↪secret
```

set protocols openfabric domain <name> interface <interface> csnp-interval <1-600>

Configure the interval, in seconds, at which CSNPs (Complete Sequence Number PDUs) are sent on the specified interface.

Example:

```
set protocols openfabric domain fabric1 interface eth0 csnp-interval 10
```

set protocols openfabric domain <name> interface <interface> psnp-interval <0-120>

Configure the interval, in seconds, at which PSNPs (Partial Sequence Number PDUs) are sent on the specified interface.

Example:

```
set protocols openfabric domain fabric1 interface eth0 psnp-interval 2
```

Timers

```
set protocols openfabric domain <name> lsp-gen-interval <1-120>
```

Configure the minimum interval, in seconds, between successive generations of the same LSP in the OpenFabric domain.

Example:

```
set protocols openfabric domain fabric1 lsp-gen-interval 5
```

```
set protocols openfabric domain <name> lsp-refresh-interval <1-65235>
```

Configure the LSP refresh interval, in seconds, for the OpenFabric domain.

Note: The value must be lower than `max-lsp-lifetime`. Otherwise, LSPs will time out before they can be refreshed.

Example:

```
set protocols openfabric domain fabric1 lsp-refresh-interval 900
```

```
set protocols openfabric domain <name> max-lsp-lifetime <360-65535>
```

Configure the maximum lifetime, in seconds, for LSPs in the OpenFabric domain.

By default, LSPs remain in the link-state database for 1200 seconds and are deleted if they are not refreshed.

Example:

```
set protocols openfabric domain fabric1 max-lsp-lifetime 1200
```

```
set protocols openfabric domain <name> spf-interval <1-120>
```

Configure the minimum interval, in seconds, between consecutive SPF calculations in the OpenFabric domain.

Example:

```
set protocols openfabric domain fabric1 spf-interval 5
```

Example

The following example demonstrates a basic OpenFabric configuration between two routers.

Node 1:

```
set interfaces loopback lo address '198.51.100.1/32'
set interfaces ethernet eth1 address '192.0.2.1/24'

set protocols openfabric domain VyOS interface eth1 address-family ipv4
set protocols openfabric domain VyOS interface lo address-family ipv4
set protocols openfabric domain VyOS interface lo passive
set protocols openfabric net '49.0001.1980.5110.0001.00'
```

Node 2:

```
set interfaces loopback lo address '198.51.100.2/32'
set interfaces ethernet eth1 address '192.0.2.2/24'

set protocols openfabric domain VyOS interface eth1 address-family ipv4
set protocols openfabric domain VyOS interface lo address-family ipv4
set protocols openfabric domain VyOS interface lo passive
set protocols openfabric net '49.0001.1980.5110.0002.00'
```

After committing the configuration, verify the neighbor adjacencies on both nodes:

```
vyos@node-1:~$ show openfabric neighbor
show openfabric neighbor
Area VyOS:
  System Id      Interface  L  State      Holdtime  SNPA
  node-2         eth1      2  Up         27        2020.2020.2020

vyos@node-2:~$ show openfabric neighbor
show openfabric neighbor
Area VyOS:
  System Id      Interface  L  State      Holdtime  SNPA
  node-1         eth1      2  Up         30        2020.2020.2020
```

Verify that the OpenFabric routes have successfully populated:

```
vyos@node-1:~$ show ip route openfabric
show ip route openfabric
Codes: K - kernel route, C - connected, S - static, R - RIP,
       O - OSPF, I - IS-IS, B - BGP, E - EIGRP, N - NHRP,
       T - Table, v - VNC, V - VNC-Direct, A - Babel, F - PBR,
       f - OpenFabric,
       > - selected route, * - FIB route, q - queued, r - rejected, b - backup
       t - trapped, o - offload failure

f  192.0.2.0/24 [115/20] via 192.0.2.2, eth1 onlink, weight 1, 00:00:10
f>* 198.51.100.2/32 [115/20] via 192.0.2.2, eth1 onlink, weight 1, 00:00:10

vyos@node-2:~$ show ip route openfabric
```

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```
show ip route openfabric
Codes: K - kernel route, C - connected, S - static, R - RIP,
       O - OSPF, I - IS-IS, B - BGP, E - EIGRP, N - NHRP,
       T - Table, v - VNC, V - VNC-Direct, A - Babel, F - PBR,
       f - OpenFabric,
       > - selected route, * - FIB route, q - queued, r - rejected, b - backup
       t - trapped, o - offload failure

f 192.0.2.0/24 [115/20] via 192.0.2.1, eth1 onlink, weight 1, 00:00:48
f>* 198.51.100.1/32 [115/20] via 192.0.2.1, eth1 onlink, weight 1, 00:00:48
```

7.9.13 OSPF

OSPF is a routing protocol for Internet Protocol (IP) networks. It uses a link state routing (LSR) algorithm and falls into the group of interior gateway protocols (IGPs), operating within a single autonomous system (AS). It is defined as OSPF Version 2 in [RFC 2328](#) (1998) for IPv4. Updates for IPv6 are specified as OSPF Version 3 in [RFC 5340](#) (2008). OSPF supports the CIDR (Classless Inter-Domain Routing) addressing model.

OSPF is a widely used IGP in large enterprise networks.

OSPFv2 (IPv4)

Configuration

General

VyOS does not have a special command to start the OSPF process. The OSPF process starts when the first ospf enabled interface is configured.

set protocols ospf area <number> network <A.B.C.D/M>

This command specifies the OSPF enabled interface(s). If the interface has an address from defined range then the command enables OSPF on this interface so router can provide network information to the other ospf routers via this interface.

This command is also used to enable the OSPF process. The area number can be specified in decimal notation in the range from 0 to 4294967295. Or it can be specified in dotted decimal notation similar to ip address.

Prefix length in interface must be equal or bigger (i.e. smaller network) than prefix length in network statement. For example statement above doesn't enable ospf on interface with address 192.168.1.1/23, but it does on interface with address 192.168.1.129/25.

In some cases it may be more convenient to enable OSPF on a per interface/subnet basis **set protocols ospf interface <interface> area <x.x.x.x | x>**

set protocols ospf auto-cost reference-bandwidth <number>

This command sets the reference bandwidth for cost calculations, where bandwidth can be in range from 1 to 4294967, specified in Mbits/s. The default is 100Mbit/s (i.e. a link of bandwidth 100Mbit/s or higher will have a cost of 1. Cost of lower bandwidth links will be scaled with reference to this cost).

set protocols ospf parameters router-id <rid>

This command sets the router-ID of the OSPF process. The router-ID may be an IP address of the router, but need not be – it can be any arbitrary 32bit number. However it **MUST** be unique within the entire OSPF domain to the OSPF speaker – bad things will happen if multiple OSPF speakers are configured with the same router-ID!

Optional

set protocols ospf default-information originate [always] [metric <number>] [metric-type <1|2>] [route-map <name>]

Originate an AS-External (type-5) LSA describing a default route into all external-routing capable areas, of the specified metric and metric type. If the `always` keyword is given then the default is always advertised, even when there is no default present in the routing table. The argument `route-map` specifies to advertise the default route if the route map is satisfied.

set protocols ospf distance global <distance>

This command change distance value of OSPF globally. The distance range is 1 to 255.

set protocols ospf distance ospf <external|inter-area|intra-area> <distance>

This command change distance value of OSPF. The arguments are the distance values for external routes, inter-area routes and intra-area routes respectively. The distance range is 1 to 255.

Note: Routes with a distance of 255 are effectively disabled and not installed into the kernel.

set protocols ospf log-adjacency-changes [detail]

This command allows to log changes in adjacency. With the optional `detail` argument, all changes in adjacency status are shown. Without `detail`, only changes to full or regressions are shown.

set protocols ospf max-metric router-lsa <administrative|on-shutdown <seconds>|on-startup <seconds>>

This enables [RFC 3137](#) support, where the OSPF process describes its transit links in its router-LSA as having infinite distance so that other routers will avoid calculating transit paths through the router while still being able to reach networks through the router.

This support may be enabled administratively (and indefinitely) with the `administrative` command. It may also be enabled conditionally. Conditional enabling of max-metric router-lsas can be for a period of seconds after startup with the `on-startup <seconds>` command and/or for a period of seconds prior to shutdown with the `on-shutdown <seconds>` command. The time range is 5 to 86400.

set protocols ospf parameters abr-type <cisco|ibm|shortcut|standard>

This command selects ABR model. OSPF router supports four ABR models:

cisco – a router will be considered as ABR if it has several configured links to the networks in different areas one of which is a backbone area. Moreover, the link to the backbone area should be active (working). **ibm** – identical to “cisco” model but in this case a backbone area link may not be active. **standard** – router has several active links to different areas. **shortcut** – identical to “standard” but in this model a router is allowed to use a connected areas topology without involving a backbone area for inter-area connections.

Detailed information about “cisco” and “ibm” models differences can be found in [RFC 3509](#). A “shortcut” model allows ABR to create routes between areas based on the topology of the areas connected to this router but not using a backbone area in case if non-backbone route will be cheaper. For more information about “shortcut” model, see `ospf-shortcut-abr-02.txt`

set protocols ospf parameters rfc1583-compatibility

RFC 2328, the successor to **RFC 1583**, suggests according to section G.2 (changes) in section 16.4.1 a change to the path preference algorithm that prevents possible routing loops that were possible in the old version of OSPFv2. More specifically it demands that inter-area paths and intra-area backbone path are now of equal preference but still both preferred to external paths.

This command should NOT be set normally.

set protocols ospf interface <interface> passive [disable]

This command specifies interface as passive. Passive interface advertises its address, but does not run the OSPF protocol (adjacencies are not formed and hello packets are not generated).

The optional disable option allows to exclude interface from passive state. This command is used if the command `passive-interface default` was configured.

set protocols ospf passive-interface default

This command specifies all interfaces as passive by default. Because this command changes the configuration logic to a default passive; therefore, interfaces where router adjacencies are expected need to be configured with the `passive-interface-exclude` command.

set protocols ospf maximum-paths <1-64>

Use this command to control the maximum number of equal cost paths to reach a specific destination. The upper limit may differ if you change the value of `MULTIPATH_NUM` during compilation. The default is `MULTIPATH_NUM` (64).

set protocols ospf refresh timers <seconds>

The router automatically updates link-state information with its neighbors. Only an obsolete information is updated which age has exceeded a specific threshold. This parameter changes a threshold value, which by default is 1800 seconds (half an hour). The value is applied to the whole OSPF router. The timer range is 10 to 1800.

set protocols ospf timers throttle spf <delay|initial-holdtime|max-holdtime> <seconds>

This command sets the initial delay, the initial-holdtime and the maximum-holdtime between when SPF is calculated and the event which triggered the calculation. The times are specified in milliseconds and must be in the range of 0 to 600000 milliseconds. `delay` sets the initial SPF schedule delay in milliseconds. The default value is 200 ms. `initial-holdtime` sets the minimum hold time between two consecutive SPF calculations. The default value is 1000 ms. `max-holdtime` sets the maximum wait time between two consecutive SPF calculations. The default value is 10000 ms.

set protocols ospf ldp-sync

This command will enable IGP-LDP synchronization globally for OSPF. This requires for LDP to be functional. This is described in **RFC 5443**. By default all interfaces operational in OSPF are enabled for synchronization. Loopbacks are exempt.

set protocols ospf ldp-sync holddown <seconds>

This command will change the hold down value globally for IGP-LDP synchronization during convergence/interface flap events.

set protocols ospf capability opaque

ospfd supports Opaque LSA **RFC 2370** as partial support for MPLS Traffic Engineering LSAs. The opaque-lsa capability must be enabled in the configuration.

An alternate command could be “mpls-te on” (Traffic Engineering)

Note: FRR offers only partial support for some of the routing protocol extensions that are used with MPLS-TE; it does not support a complete RSVP-TE solution.

Area Configuration

set protocols ospf area <number> area-type stub

This command specifies the area to be a Stub Area. That is, an area where no router originates routes external to OSPF and hence an area where all external routes are via the ABR(s). Hence, ABRs for such an area do not need to pass AS-External LSAs (type-5) or ASBR-Summary LSAs (type-4) into the area. They need only pass Network-Summary (type-3) LSAs into such an area, along with a default-route summary.

set protocols ospf area <number> area-type stub no-summary

This command specifies the area to be a Totally Stub Area. In addition to stub area limitations this area type prevents an ABR from injecting Network-Summary (type-3) LSAs into the specified stub area. Only default summary route is allowed.

set protocols ospf area <number> area-type stub default-cost <number>

This command sets the cost of default-summary LSAs announced to stubby areas. The cost range is 0 to 16777215.

set protocols ospf area <number> area-type nssa

This command specifies the area to be a Not So Stubby Area. External routing information is imported into an NSSA in Type-7 LSAs. Type-7 LSAs are similar to Type-5 AS-external LSAs, except that they can only be flooded into the NSSA. In order to further propagate the NSSA external information, the Type-7 LSA must be translated to a Type-5 AS-external-LSA by the NSSA ABR.

set protocols ospf area <number> area-type nssa no-summary

This command specifies the area to be a NSSA Totally Stub Area. ABRs for such an area do not need to pass Network-Summary (type-3) LSAs (except the default summary route), ASBR-Summary LSAs (type-4) and AS-External LSAs (type-5) into the area. But Type-7 LSAs that convert to Type-5 at the NSSA ABR are allowed.

set protocols ospf area <number> area-type nssa default-cost <number>

This command sets the default cost of LSAs announced to NSSA areas. The cost range is 0 to 16777215.

set protocols ospf area <number> area-type nssa translate <always|candidate|never>

Specifies whether this NSSA border router will unconditionally translate Type-7 LSAs into Type-5 LSAs. When role is Always, Type-7 LSAs are translated into Type-5 LSAs regardless of the translator state of other NSSA border routers. When role is Candidate, this router participates in the translator election to determine if it will perform the translations duties. When role is Never, this router will never translate Type-7 LSAs into Type-5 LSAs.

set protocols ospf area <number> authentication plaintext-password

This command specifies that simple password authentication should be used for the given area. The password must also be configured on a per-interface basis.

set protocols ospf area <number> authentication md5

This command specify that OSPF packets must be authenticated with MD5 HMACs within the given area. Keying material must also be configured on a per-interface basis.

set protocols ospf area <number> range <A.B.C.D/M> [cost <number>]

This command summarizes intra area paths from specified area into one summary-LSA (Type-3) announced to other areas. This command can be used only in ABR and ONLY router-LSAs (Type-1) and network-LSAs (Type-2) (i.e. LSAs with scope area) can be summarized. AS-external-LSAs (Type-5) can't be summarized - their scope is AS. The optional argument `cost` specifies the aggregated link metric. The metric range is 0 to 16777215.

set protocols ospf area <number> range <A.B.C.D/M> not-advertise

This command instead of summarizing intra area paths filter them - i.e. intra area paths from this range are not advertised into other areas. This command makes sense in ABR only.

set protocols ospf area <number> export-list <acl_number>

Filter Type-3 summary-LSAs announced to other areas originated from intra- area paths from specified area. This command makes sense in ABR only.

set protocols ospf area <number> import-list <acl_number>

Same as export-list, but it applies to paths announced into specified area as Type-3 summary-LSAs. This command makes sense in ABR only.

set protocols ospf area <number> range <A.B.C.D/M> substitute <E.F.G.H/M>

One Type-3 summary-LSA with routing info <E.F.G.H/M> is announced into backbone area if defined area contains at least one intra-area network (i.e. described with router-LSA or network-LSA) from range <A.B.C.D/M>. This command makes sense in ABR only.

set protocols ospf area <number> shortcut <default|disable|enable>

This parameter allows to “shortcut” routes (non-backbone) for inter-area routes. There are three modes available for routes shortcutting:

default – this area will be used for shortcutting only if ABR does not have a link to the backbone area or this link was lost. **enable** – the area will be used for shortcutting every time the route that goes through it is cheaper. **disable** – this area is never used by ABR for routes shortcutting.

set protocols ospf area <number> virtual-link <A.B.C.D>

Provides a backbone area coherence by virtual link establishment.

In general, OSPF protocol requires a backbone area (area 0) to be coherent and fully connected. I.e. any backbone area router must have a route to any other backbone area router. Moreover, every ABR must have a link to backbone area. However, it is not always possible to have a physical link to a backbone area. In this case between two ABR (one of them has a link to the backbone area) in the area (not stub area) a virtual link is organized.

<number> – area identifier through which a virtual link goes. <A.B.C.D> – ABR router-id with which a virtual link is established. Virtual link must be configured on both routers.

Formally, a virtual link looks like a point-to-point network connecting two ABR from one area one of which physically connected to a backbone area. This pseudo-network is considered to belong to a backbone area.

Interface Configuration

set protocols ospf interface <interface> area <x.x.x.x | x>

Enable ospf on an interface and set associated area.

If you have a lot of interfaces, and/or a lot of subnets, then enabling OSPF via this command may result in a slight performance improvement.

set protocols ospf interface <interface> authentication plaintext-password <text>

This command sets OSPF authentication key to a simple password. After setting, all OSPF packets are authenticated. Key has length up to 8 chars.

Simple text password authentication is insecure and deprecated in favour of MD5 HMAC authentication.

set protocols ospf interface <interface> authentication md5 key-id <id> md5-key <text>

This command specifies that MD5 HMAC authentication must be used on this interface. It sets OSPF authentication key to a cryptographic password. Key-id identifies secret key used to create the message digest. This ID is part of the protocol and must be consistent across routers on a link. The key can be long up to 16 chars (larger strings will be truncated), and is associated with the given key-id.

set protocols ospf interface <interface> bandwidth <number>

This command sets the interface bandwidth for cost calculations, where bandwidth can be in range from 1 to 100000, specified in Mbits/s.

set protocols ospf interface <interface> cost <number>

This command sets link cost for the specified interface. The cost value is set to router-LSA's metric field and used for SPF calculation. The cost range is 1 to 65535.

set protocols ospf interface <interface> dead-interval <number>

Set number of seconds for router Dead Interval timer value used for Wait Timer and Inactivity Timer. This value must be the same for all routers attached to a common network. The default value is 40 seconds. The interval range is 1 to 65535.

set protocols ospf interface <interface> hello-multiplier <number>

The hello-multiplier specifies how many Hellos to send per second, from 1 (every second) to 10 (every 100ms). Thus one can have 1s convergence time for OSPF. If this form is specified, then the hello-interval advertised in Hello packets is set to 0 and the hello-interval on received Hello packets is not checked, thus the hello-multiplier need NOT be the same across multiple routers on a common link.

set protocols ospf interface <interface> hello-interval <number>

Set number of seconds for Hello Interval timer value. Setting this value, Hello packet will be sent every timer value seconds on the specified interface. This value must be the same for all routers attached to a common network. The default value is 10 seconds. The interval range is 1 to 65535.

set protocols ospf interface <interface> bfd

This command enables BFD on this OSPF link interface.

set protocols ospf interface <interface> mtu-ignore

This command disables check of the MTU value in the OSPF DBD packets. Thus, use of this command allows the OSPF adjacency to reach the FULL state even though there is an interface MTU mismatch between two OSPF routers.

set protocols ospf interface <interface> network <type>

This command allows to specify the distribution type for the network connected to this interface:

broadcast – broadcast IP addresses distribution. **non-broadcast** – address distribution in NBMA networks topology. **point-to-multipoint** – address distribution in point-to-multipoint networks. **point-to-point** – address distribution in point-to-point networks.

set protocols ospf interface <interface> priority <number>

This command sets Router Priority integer value. The router with the highest priority will be more eligible to become Designated Router. Setting the value to 0, makes the router ineligible to become Designated Router. The default value is 1. The interval range is 0 to 255.

set protocols ospf interface <interface> retransmit-interval <number>

This command sets number of seconds for RxmtInterval timer value. This value is used when retransmitting Database Description and Link State Request packets if acknowledge was not received. The default value is 5 seconds. The interval range is 3 to 65535.

set protocols ospf interface <interface> transmit-delay <number>

This command sets number of seconds for InfTransDelay value. It allows to set and adjust for each interface the delay interval before starting the synchronizing process of the router's database with all neighbors. The default value is 1 seconds. The interval range is 3 to 65535.

set protocols ospf interface <interface> ldp-sync disable

This command disables IGP-LDP sync for this specific interface.

set protocols ospf interface <interface> ldp-sync holddown <seconds>

This command will change the hold down value for IGP-LDP synchronization during convergence/interface flap events, but for this interface only.

External Route Summarisation

This feature summarises originated external LSAs (Type-5 and Type-7). Summary Route will be originated on-behalf of all matched external LSAs.

set protocols ospf aggregation timer <seconds>

Configure aggregation delay timer interval.

Summarisation starts only after this delay timer expiry.

set protocols ospf summary-address x.x.x.x/y [tag (1-4294967295)]

This command enable/disables summarisation for the configured address range.

Tag is the optional parameter. If tag configured Summary route will be originated with the configured tag.

set protocols ospf summary-address x.x.x.x/y no-advertise

This command to ensure not advertise the summary lsa for the matched external LSAs.

Graceful Restart

set protocols ospf graceful-restart [grace-period (1-1800)]

Configure Graceful Restart [RFC 3623](#) restarting support. When enabled, the default grace period is 120 seconds.

To perform a graceful shutdown, the FRR graceful-restart prepare ip ospf EXEC-level command needs to be issued before restarting the ospfd daemon.

set protocols ospf graceful-restart helper enable [router-id A.B.C.D]

Configure Graceful Restart [RFC 3623](#) helper support. By default, helper support is disabled for all neighbours. This config enables/disables helper support on this router for all neighbours.

To enable/disable helper support for a specific neighbour, the router-id (A.B.C.D) has to be specified.

set protocols ospf graceful-restart helper no-strict-lsa-checking

By default strict-lsa-checking is configured then the helper will abort the Graceful Restart when a LSA change occurs which affects the restarting router.

This command disables it.

set protocols ospf graceful-restart helper supported-grace-time

Supports as HELPER for configured grace period.

set protocols ospf graceful-restart helper planned-only

It helps to support as HELPER only for planned restarts.

By default, it supports both planned and unplanned outages.

Manual Neighbor Configuration

OSPF routing devices normally discover their neighbors dynamically by listening to the broadcast or multicast hello packets on the network. Because an NBMA network does not support broadcast (or multicast), the device cannot discover its neighbors dynamically, so you must configure all the neighbors statically.

set protocols ospf neighbor <A.B.C.D>

This command specifies the IP address of the neighboring device.

set protocols ospf neighbor <A.B.C.D> poll-interval <seconds>

This command specifies the length of time, in seconds, before the routing device sends hello packets out of the interface before it establishes adjacency with a neighbor. The range is 1 to 65535 seconds. The default value is 60 seconds.

set protocols ospf neighbor <A.B.C.D> priority <number>

This command specifies the router priority value of the nonbroadcast neighbor associated with the IP address specified. The default is 0. This keyword does not apply to point-to-multipoint interfaces.

Redistribution Configuration

set protocols ospf redistribute <route source>

This command redistributes routing information from the given route source to the OSPF process. There are five modes available for route source: bgp, connected, kernel, rip, static.

set protocols ospf default-metric <number>

This command specifies the default metric value of redistributed routes. The metric range is 0 to 16777214.

set protocols ospf redistribute <route source> metric <number>

This command specifies metric for redistributed routes from the given route source. There are five modes available for route source: bgp, connected, kernel, rip, static. The metric range is 1 to 16777214.

set protocols ospf redistribute <route source> metric-type <1|2>

This command specifies metric type for redistributed routes. Difference between two metric types that metric type 1 is a metric which is “commensurable” with inner OSPF links. When calculating a metric to the external destination, the full path metric is calculated as a metric sum path of a router which had advertised this link plus the link metric. Thus, a route with the least summary metric will be selected. If external link is advertised with metric type 2 the path is selected which lies through the router which advertised this link with the least metric despite of the fact that internal path to this router is longer (with more cost). However, if two routers advertised an external link and with metric type 2 the preference is given to the path which lies through the router with a shorter internal path. If two different routers advertised two links to the same external destination but with different metric type, metric type 1 is preferred. If type of a metric left undefined the router will consider these external links to have a default metric type 2.

set protocols ospf redistribute <route source> route-map <name>

This command allows to use route map to filter redistributed routes from the given route source. There are five modes available for route source: bgp, connected, kernel, rip, static.

Operational Mode Commands

show ip ospf neighbor

This command displays the neighbors status.

Neighbor ID	Pri	State	Dead Time	Address	Interface
↪ RXmtL RqstL DBsmL					
10.0.13.1	1	Full/DR	38.365s	10.0.13.1	eth0:10.0.13.3
↪ 0	0	0			
10.0.23.2	1	Full/Backup	39.175s	10.0.23.2	eth1:10.0.23.3
↪ 0	0	0			

show ip ospf neighbor detail

This command displays the neighbors information in a detailed form, not just a summary table.

```
Neighbor 10.0.13.1, interface address 10.0.13.1
```

```
In the area 0.0.0.0 via interface eth0
```

```
Neighbor priority is 1, State is Full, 5 state changes
```

```
Most recent state change statistics:
```

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```

    Progressive change 11m55s ago

    DR is 10.0.13.1, BDR is 10.0.13.3

    Options 2 *|---|E|
    Dead timer due in 34.854s

    Database Summary List 0

    Link State Request List 0

    Link State Retransmission List 0

    Thread Inactivity Timer on

    Thread Database Description Retransmission off

    Thread Link State Request Retransmission on

    Thread Link State Update Retransmission on

Neighbor 10.0.23.2, interface address 10.0.23.2

    In the area 0.0.0.1 via interface eth1

    Neighbor priority is 1, State is Full, 4 state changes

    Most recent state change statistics:

        Progressive change 41.193s ago

        DR is 10.0.23.3, BDR is 10.0.23.2

        Options 2 *|---|E|
        Dead timer due in 35.661s

        Database Summary List 0

        Link State Request List 0

        Link State Retransmission List 0

        Thread Inactivity Timer on

        Thread Database Description Retransmission off

        Thread Link State Request Retransmission on

```

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Thread Link State Update Retransmission on

show ip ospf neighbor <A.B.C.D>

This command displays the neighbors information in a detailed form for a neighbor whose IP address is specified.

show ip ospf neighbor <interface>

This command displays the neighbors status for a neighbor on the specified interface.

show ip ospf interface [<interface>]

This command displays state and configuration of OSPF for the specified interface, or all interfaces if no interface is given.

```
eth0 is up
  ifindex 2, MTU 1500 bytes, BW 4294967295 Mbit <UP,BROADCAST,RUNNING,MULTICAST>
  Internet Address 10.0.13.3/24, Broadcast 10.0.13.255, Area 0.0.0.0
  MTU mismatch detection: enabled
  Router ID 10.0.23.3, Network Type BROADCAST, Cost: 1
  Transmit Delay is 1 sec, State Backup, Priority 1
  Backup Designated Router (ID) 10.0.23.3, Interface Address 10.0.13.3
  Multicast group memberships: OSPFAllRouters OSPFDesignatedRouters
  Timer intervals configured, Hello 10s, Dead 40s, Wait 40s, Retransmit 5
    Hello due in 4.470s
  Neighbor Count is 1, Adjacent neighbor count is 1
eth1 is up
  ifindex 3, MTU 1500 bytes, BW 4294967295 Mbit <UP,BROADCAST,RUNNING,MULTICAST>
  Internet Address 10.0.23.3/24, Broadcast 10.0.23.255, Area 0.0.0.1
  MTU mismatch detection: enabled
  Router ID 10.0.23.3, Network Type BROADCAST, Cost: 1
  Transmit Delay is 1 sec, State DR, Priority 1
  Backup Designated Router (ID) 10.0.23.2, Interface Address 10.0.23.2
  Saved Network-LSA sequence number 0x80000002
  Multicast group memberships: OSPFAllRouters OSPFDesignatedRouters
  Timer intervals configured, Hello 10s, Dead 40s, Wait 40s, Retransmit 5
    Hello due in 4.563s
  Neighbor Count is 1, Adjacent neighbor count is 1
```

show ip ospf route [detail]

This command displays the OSPF routing table, as determined by the most recent SPF calculation. With the optional detail argument, each route item's advertiser router and network attribute will be shown.

```
===== OSPF network routing table =====
N IA 10.0.12.0/24      [3] area: 0.0.0.0
                        via 10.0.13.3, eth0
N   10.0.13.0/24      [1] area: 0.0.0.0
                        directly attached to eth0
N IA 10.0.23.0/24      [2] area: 0.0.0.0
                        via 10.0.13.3, eth0
N   10.0.34.0/24      [2] area: 0.0.0.0
                        via 10.0.13.3, eth0

===== OSPF router routing table =====
```

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```

R    10.0.23.3          [1] area: 0.0.0.0, ABR
                        via 10.0.13.3, eth0
R    10.0.34.4          [2] area: 0.0.0.0, ASBR
                        via 10.0.13.3, eth0

===== OSPF external routing table =====
N E2 172.16.0.0/24      [2/20] tag: 0
                        via 10.0.13.3, eth0

```

The table consists of following data:

OSPF network routing table – includes a list of acquired routes for all accessible networks (or aggregated area ranges) of OSPF system. “IA” flag means that route destination is in the area to which the router is not connected, i.e. it’s an inter-area path. In square brackets a summary metric for all links through which a path lies to this network is specified. “via” prefix defines a router-gateway, i.e. the first router on the way to the destination (next hop). **OSPF router routing table** – includes a list of acquired routes to all accessible ABRs and ASBRs. **OSPF external routing table** – includes a list of acquired routes that are external to the OSPF process. “E” flag points to the external link metric type (E1 – metric type 1, E2 – metric type 2). External link metric is printed in the “<metric of the router which advertised the link>/<link metric>” format.

show ip ospf border-routers

This command displays a table of paths to area boundary and autonomous system boundary routers.

show ip ospf database

This command displays a summary table with a database contents (LSA).

```

OSPF Router with ID (10.0.13.1)

      Router Link States (Area 0.0.0.0)

Link ID        ADV Router    Age  Seq#           CkSum  Link count
10.0.13.1      10.0.13.1      984  0x800000005   0xd915  1
10.0.23.3      10.0.23.3      1186 0x800000008   0xfe62  2
10.0.34.4      10.0.34.4      1063 0x800000004   0x4e3f  1

      Net Link States (Area 0.0.0.0)

Link ID        ADV Router    Age  Seq#           CkSum
10.0.13.1      10.0.13.1      994  0x800000003   0x30bb
10.0.34.4      10.0.34.4      1188 0x800000001   0x9411

      Summary Link States (Area 0.0.0.0)

Link ID        ADV Router    Age  Seq#           CkSum  Route
10.0.12.0      10.0.23.3      1608 0x800000001   0x6ab6  10.0.12.0/24
10.0.23.0      10.0.23.3      981  0x800000003   0xe232  10.0.23.0/24

      AS External Link States

Link ID        ADV Router    Age  Seq#           CkSum  Route
172.16.0.0     10.0.34.4      1063 0x800000001   0xc40d  E2 172.16.0.0/24 [0x0]

```

show ip ospf database <type> [A.B.C.D] [adv-router <A.B.C.D>|self-originate]

This command displays a database contents for a specific link advertisement type.

The type can be the following: asbr-summary, external, network, nssa-external, opaque-area, opaque-as, opaque-link, router, summary.

[A.B.C.D] – link-state-id. With this specified the command displays portion of the network environment that is being described by the advertisement. The value entered depends on the advertisement's LS type. It must be entered in the form of an IP address.

adv-router <A.B.C.D> – router id, which link advertisements need to be reviewed.

self-originate displays only self-originated LSAs from the local router.

```

      OSPF Router with ID (10.0.13.1)

      Router Link States (Area 0.0.0.0)

LS age: 1213
Options: 0x2 : *|---|E|
LS Flags: 0x3
Flags: 0x0
LS Type: router-LSA
Link State ID: 10.0.13.1
Advertising Router: 10.0.13.1
LS Seq Number: 80000009
Checksum: 0xd119
Length: 36

Number of Links: 1

Link connected to: a Transit Network
(Link ID) Designated Router address: 10.0.13.1
(Link Data) Router Interface address: 10.0.13.1
Number of TOS metrics: 0
TOS 0 Metric: 1
    
```

show ip ospf database max-age

This command displays LSAs in MaxAge list.

Examples

Enable OSPF

Node 1

```

set interfaces loopback lo address 10.1.1.1/32
set interfaces ethernet eth0 address 192.168.0.1/24
set protocols ospf area 0 network 192.168.0.0/24
set protocols ospf area 0 network 10.1.1.1/32
set protocols ospf parameters router-id 10.1.1.1
    
```

Node 2

```
set interfaces loopback lo address 10.1.1.2/32
set interfaces ethernet eth0 address 192.168.0.2/24
set protocols ospf area 0 network 192.168.0.0/24
set protocols ospf area 0 network 10.1.1.2/32
set protocols ospf parameters router-id 10.1.1.2
```

Here's the neighbors up:

```
Node-1@vyos:~$ show ip ospf neighbor
```

Neighbor ID	Pri	State	Up Time	Dead Time	Address	Interface
↪ 10.1.1.2	1	Full/DR	3m43s	36.094s	192.168.0.2	eth0:192.
↪ 168.0.1			0	0		

```
Node-2@vyos:~$ show ip ospf neighbor
```

Neighbor ID	Pri	State	Up Time	Dead Time	Address	Interface
↪ 10.1.1.1	1	Full/Backup	3m47s	31.736s	192.168.0.1	eth0:192.
↪ 168.0.2			0	0		

Here's the routes:

```
Node-1@vyos:~$ show ip route ospf
```

```
Codes: K - kernel route, C - connected, S - static, R - RIP,
       O - OSPF, I - IS-IS, B - BGP, E - EIGRP, N - NHRP,
       T - Table, v - VNC, V - VNC-Direct, A - Babel, F - PBR,
       f - OpenFabric,
       > - selected route, * - FIB route, q - queued, r - rejected, b - backup
       t - trapped, o - offload failure
```

```
O 10.1.1.1/32 [110/0] is directly connected, lo, weight 1, 00:00:14
O>* 10.1.1.2/32 [110/1] via 192.168.0.2, eth0, weight 1, 00:00:07
O 192.168.0.0/24 [110/1] is directly connected, eth0, weight 1, 00:03:32
```

```
Node-2@vyos:~$ show ip route ospf
```

```
Codes: K - kernel route, C - connected, S - static, R - RIP,
       O - OSPF, I - IS-IS, B - BGP, E - EIGRP, N - NHRP,
       T - Table, v - VNC, V - VNC-Direct, A - Babel, F - PBR,
       f - OpenFabric,
       > - selected route, * - FIB route, q - queued, r - rejected, b - backup
       t - trapped, o - offload failure
```

```
O>* 10.1.1.1/32 [110/1] via 192.168.0.1, eth0, weight 1, 00:00:11
O 10.1.1.2/32 [110/0] is directly connected, lo, weight 1, 00:00:04
O 192.168.0.0/24 [110/1] is directly connected, eth0, weight 1, 00:03:18
```

Enable OSPF with route redistribution of the loopback and default originate:**Node 1**

```

set interfaces loopback lo address 10.1.1.1/32
set protocols ospf area 0 network 192.168.0.0/24
set protocols ospf default-information originate always
set protocols ospf default-information originate metric 10
set protocols ospf default-information originate metric-type 2
set protocols ospf log-adjacency-changes
set protocols ospf parameters router-id 10.1.1.1
set protocols ospf redistribute connected metric-type 2
set protocols ospf redistribute connected route-map CONNECT

set policy route-map CONNECT rule 10 action permit
set policy route-map CONNECT rule 10 match interface lo

```

Node 2

```

set interfaces loopback lo address 10.2.2.2/32
set protocols ospf area 0 network 192.168.0.0/24
set protocols ospf log-adjacency-changes
set protocols ospf parameters router-id 10.2.2.2
set protocols ospf redistribute connected metric-type 2
set protocols ospf redistribute connected route-map CONNECT

set policy route-map CONNECT rule 10 action permit
set policy route-map CONNECT rule 10 match interface lo

```

Enable OSPF and IGP-LDP synchronization:**Node 1:**

```

set interfaces loopback lo address 10.1.1.1/32
set interfaces ethernet eth0 address 192.168.0.1/24

set protocols ospf area 0 network '192.168.0.0/24'
set protocols ospf area 0 network '10.1.1.1/32'
set protocols ospf parameters router-id '10.1.1.1'
set protocols ospf ldp-sync

set protocols mpls interface eth0
set protocols mpls ldp discovery transport-ipv4-address 10.1.1.1
set protocols mpls ldp interface lo
set protocols mpls ldp interface eth0
set protocols mpls ldp parameters transport-prefer-ipv4
set protocols mpls ldp router-id 10.1.1.1

```

This gives us IGP-LDP synchronization for all non-loopback interfaces with a holddown timer of zero seconds:

```

Node-1@vyos:~$ show ip ospf mpls ldp-sync
eth0

```

(continues on next page)

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```
LDP-IGP Synchronization enabled: yes
Holddown timer in seconds: 0
State: Sync achieved
```

Enable OSPF with Segment Routing (Experimental):

Node 1

```
set interfaces loopback lo address 10.1.1.1/32
set interfaces ethernet eth0 address 192.168.0.1/24

set protocols ospf area 0 network '192.168.0.0/24'
set protocols ospf area 0 network '10.1.1.1/32'
set protocols ospf parameters opaque-lsa
set protocols ospf parameters router-id '10.1.1.1'
set protocols ospf segment-routing global-block high-label-value '1100'
set protocols ospf segment-routing global-block low-label-value '1000'
set protocols ospf segment-routing prefix 10.1.1.1/32 index explicit-null
set protocols ospf segment-routing prefix 10.1.1.1/32 index value '1'
```

Node 2

```
set interfaces loopback lo address 10.1.1.2/32
set interfaces ethernet eth0 address 192.168.0.2/24

set protocols ospf area 0 network '192.168.0.0/24'
set protocols ospf area 0 network '10.1.1.2/32'
set protocols ospf parameters opaque-lsa
set protocols ospf parameters router-id '10.1.1.2'
set protocols ospf segment-routing global-block high-label-value '1100'
set protocols ospf segment-routing global-block low-label-value '1000'
set protocols ospf segment-routing prefix 10.1.1.2/32 index explicit-null
set protocols ospf segment-routing prefix 10.1.1.2/32 index value '2'
```

This gives us MPLS segment routing enabled and labels for far end loopbacks:

```
Node-1@vyos:~$ show mpls table
Inbound Label  Type      Nexthop      Outbound Label
-----
1002           SR (OSPF)  192.168.0.2  IPv4 Explicit Null  <-- Node-2 loopback learned
↳ on Node-1
15000          SR (OSPF)  192.168.0.2  implicit-null
15001          SR (OSPF)  192.168.0.2  implicit-null

Node-2@vyos:~$ show mpls table
Inbound Label  Type      Nexthop      Outbound Label
-----
1001           SR (OSPF)  192.168.0.1  IPv4 Explicit Null  <-- Node-1 loopback learned
↳ on Node-2
15000          SR (OSPF)  192.168.0.1  implicit-null
15001          SR (OSPF)  192.168.0.1  implicit-null
```

Here is the routing tables showing the MPLS segment routing label operations:

```
Node-1@vyos:~$ show ip route ospf
Codes: K - kernel route, C - connected, S - static, R - RIP,
       O - OSPF, I - IS-IS, B - BGP, E - EIGRP, N - NHRP,
       T - Table, v - VNC, V - VNC-Direct, A - Babel, F - PBR,
       f - OpenFabric,
       > - selected route, * - FIB route, q - queued, r - rejected, b - backup
       t - trapped, o - offload failure

O   10.1.1.1/32 [110/0] is directly connected, lo, weight 1, 00:03:43
O>* 10.1.1.2/32 [110/1] via 192.168.0.2, eth0, label IPv4 Explicit Null, weight 1, 00:03:32
O   192.168.0.0/24 [110/1] is directly connected, eth0, weight 1, 00:03:43

Node-2@vyos:~$ show ip route ospf
Codes: K - kernel route, C - connected, S - static, R - RIP,
       O - OSPF, I - IS-IS, B - BGP, E - EIGRP, N - NHRP,
       T - Table, v - VNC, V - VNC-Direct, A - Babel, F - PBR,
       f - OpenFabric,
       > - selected route, * - FIB route, q - queued, r - rejected, b - backup
       t - trapped, o - offload failure

O>* 10.1.1.1/32 [110/1] via 192.168.0.1, eth0, label IPv4 Explicit Null, weight 1, 00:03:36
O   10.1.1.2/32 [110/0] is directly connected, lo, weight 1, 00:03:51
O   192.168.0.0/24 [110/1] is directly connected, eth0, weight 1, 00:03:51
```

OSPFv3 (IPv6)

Configuration

General

VyOS does not have a special command to start the OSPFv3 process. The OSPFv3 process starts when the first ospf enabled interface is configured.

set protocols ospfv3 interface <interface> area <number>

This command specifies the OSPFv3 enabled interface. This command is also used to enable the OSPF process. The area number can be specified in decimal notation in the range from 0 to 4294967295. Or it can be specified in dotted decimal notation similar to ip address.

set protocols ospfv3 parameters router-id <rid>

This command sets the router-ID of the OSPFv3 process. The router-ID may be an IP address of the router, but need not be – it can be any arbitrary 32bit number. However it MUST be unique within the entire OSPFv3 domain to the OSPFv3 speaker – bad things will happen if multiple OSPFv3 speakers are configured with the same router-ID!

Optional

set protocols ospfv3 distance global <distance>

This command change distance value of OSPFv3 globally. The distance range is 1 to 255.

set protocols ospfv3 distance ospfv3 <external|inter-area|intra-area> <distance>

This command change distance value of OSPFv3. The arguments are the distance values for external routes, inter-area routes and intra-area routes respectively. The distance range is 1 to 255.

Area Configuration

set protocols ospfv3 area <number> range <prefix>

This command summarizes intra area paths from specified area into one Type-3 Inter-Area Prefix LSA announced to other areas. This command can be used only in ABR.

set protocols ospfv3 area <number> range <prefix> not-advertise

This command instead of summarizing intra area paths filter them - i.e. intra area paths from this range are not advertised into other areas. This command makes sense in ABR only.

Interface Configuration

set protocols ospfv3 interface <interface> ipv6 cost <number>

This command sets link cost for the specified interface. The cost value is set to router-LSA's metric field and used for SPF calculation. The cost range is 1 to 65535.

set protocols ospfv3 interface <interface> dead-interval <number>

Set number of seconds for router Dead Interval timer value used for Wait Timer and Inactivity Timer. This value must be the same for all routers attached to a common network. The default value is 40 seconds. The interval range is 1 to 65535.

set protocols ospfv3 interface <interface> hello-interval <number>

Set number of seconds for Hello Interval timer value. Setting this value, Hello packet will be sent every timer value seconds on the specified interface. This value must be the same for all routers attached to a common network. The default value is 10 seconds. The interval range is 1 to 65535.

set protocols ospfv3 interface <interface> mtu-ignore

This command disables check of the MTU value in the OSPF DBD packets. Thus, use of this command allows the OSPF adjacency to reach the FULL state even though there is an interface MTU mismatch between two OSPF routers.

set protocols ospfv3 interface <interface> network <type>

This command allows to specify the distribution type for the network connected to this interface:

broadcast – broadcast IP addresses distribution. **point-to-point** – address distribution in point-to-point networks.

set protocols ospfv3 interface <interface> priority <number>

This command sets Router Priority integer value. The router with the highest priority will be more eligible to become Designated Router. Setting the value to 0, makes the router ineligible to become Designated Router. The default value is 1. The interval range is 0 to 255.

set protocols ospfv3 interface <interface> passive

This command specifies interface as passive. Passive interface advertises its address, but does not run the OSPF protocol (adjacencies are not formed and hello packets are not generated).

set protocols ospfv3 interface <interface> retransmit-interval <number>

This command sets number of seconds for RxmtInterval timer value. This value is used when retransmitting Database Description and Link State Request packets if acknowledge was not received. The default value is 5 seconds. The interval range is 3 to 65535.

set protocols ospfv3 interface <interface> transmit-delay <number>

This command sets number of seconds for InfTransDelay value. It allows to set and adjust for each interface the delay interval before starting the synchronizing process of the router's database with all neighbors. The default value is 1 seconds. The interval range is 3 to 65535.

Graceful Restart**set protocols ospfv3 graceful-restart [grace-period (1-1800)]**

Configure Graceful Restart [RFC 3623](#) restarting support. When enabled, the default grace period is 120 seconds.

To perform a graceful shutdown, the FRR graceful-restart prepare ip ospf EXEC-level command needs to be issued before restarting the ospfd daemon.

set protocols ospfv3 graceful-restart helper enable [router-id A.B.C.D]

Configure Graceful Restart [RFC 3623](#) helper support. By default, helper support is disabled for all neighbours. This config enables/disables helper support on this router for all neighbours.

To enable/disable helper support for a specific neighbour, the router-id (A.B.C.D) has to be specified.

set protocols ospfv3 graceful-restart helper lsa-check-disable

By default strict-lsa-checking is configured then the helper will abort the Graceful Restart when a LSA change occurs which affects the restarting router.

This command disables it.

set protocols ospfv3 graceful-restart helper supported-grace-time

Supports as HELPER for configured grace period.

set protocols ospfv3 graceful-restart helper planned-only

It helps to support as HELPER only for planned restarts. By default, it supports both planned and unplanned outages.

Redistribution Configuration**set protocols ospfv3 redistribute <route source>**

This command redistributes routing information from the given route source to the OSPFv3 process. There are five modes available for route source: bgp, connected, kernel, ripng, static.

set protocols ospf redistribute <route source> route-map <name>

This command allows to use route map to filter redistributed routes from given route source. There are five modes available for route source: bgp, connected, kernel, ripng, static.

Operational Mode Commands

show ipv6 ospfv3 neighbor

This command displays the neighbors status.

show ipv6 ospfv3 neighbor detail

This command displays the neighbors information in a detailed form, not just a summary table.

show ipv6 ospfv3 neighbor drchoice

This command displays the neighbor DR choice information.

show ipv6 ospfv3 interface [prefix] [[<interface> [prefix]]

This command displays state and configuration of OSPF for the specified interface, or all interfaces if no interface is given. With the argument **prefix** this command shows connected prefixes to advertise.

show ipv6 ospfv3 route

This command displays the OSPF routing table, as determined by the most recent SPF calculation.

show ipv6 ospfv3 border-routers

This command displays a table of paths to area boundary and autonomous system boundary routers.

show ipv6 ospfv3 database

This command displays a summary table with a database contents (LSA).

show ipv6 ospfv3 database <type> [A.B.C.D] [adv-router <A.B.C.D>|self-originate]

This command displays a database contents for a specific link advertisement type.

show ipv6 ospfv3 redistribute

This command displays external information redistributed into OSPFv3

Configuration Example

A typical configuration using 2 nodes.

Node 1:

```
set protocols ospfv3 interface eth1 area 0.0.0.0
set protocols ospfv3 area 0.0.0.0 range 2001:db8:1::/64
set protocols ospfv3 parameters router-id 192.168.1.1
set protocols ospfv3 redistribute connected
```

Node 2:

```
set protocols ospfv3 interface eth1 area 0.0.0.0
set protocols ospfv3 area 0.0.0.0 range 2001:db8:2::/64
set protocols ospfv3 parameters router-id 192.168.2.1
set protocols ospfv3 redistribute connected
```

To see the redistributed routes:

```
show ipv6 ospfv3 redistribute
```

Cost calculation for WireGuard interfaces is unreliable as ospfv3 uses the link speed to calculate the link cost. You might therefore want to set the link cost to a fixed value on WireGuard tunnels.

Example configuration for WireGuard interfaces:

Node 1

```
set interfaces wireguard wg01 peer ospf02 allowed-ips '::/0'
set interfaces wireguard wg01 peer ospf02 endpoint '10.1.1.101:12345'
set interfaces wireguard wg01 peer ospf02 pubkey 'ie3...='
set interfaces wireguard wg01 port '12345'
set protocols ospfv3 parameters router-id 192.168.1.1
set protocols ospfv3 interface 'wg01' area 0.0.0.0
set protocols ospfv3 interface 'wg01' cost 10
set protocols ospfv3 interface 'lo' area 0.0.0.0
```

Node 2

```
set interfaces wireguard wg01 peer ospf01 allowed-ips '::/0'
set interfaces wireguard wg01 peer ospf01 endpoint '10.1.1.100:12345'
set interfaces wireguard wg01 peer ospf01 pubkey 'NHI...='
set interfaces wireguard wg01 port '12345'
set protocols ospfv3 parameters router-id 192.168.1.2
set protocols ospfv3 interface 'wg01' area 0.0.0.0
set protocols ospfv3 interface 'wg01' cost 10
set protocols ospfv3 interface 'lo' area 0.0.0.0
```

Status

```
vyos@ospf01:~$ sh ipv6 ospfv3 neighbor
Neighbor ID    Pri    DeadTime    State/IfState    Duration I/F[State]
192.168.0.2      1      00:00:37    Full/PointToPoint    00:18:03 wg01[PointToPoint]

vyos@ospf02# run sh ipv6 ospfv3 neighbor
Neighbor ID    Pri    DeadTime    State/IfState    Duration I/F[State]
192.168.0.1      1      00:00:39    Full/PointToPoint    00:19:44 wg01[PointToPoint]
```

7.9.14 PIM

PIM routes multicast traffic between routers using the existing unicast routing table for path decisions. VyOS supports PIM Sparse Mode (PIM-SM) and IGMP versions 2 and 3.

In PIM-SM, traffic from multicast sources is forwarded toward the RP (Rendezvous Point). Multicast receivers use IGMP to signal their local router which groups they want to receive, and the router uses PIM to pull that traffic from the RP via a shared tree.

A working PIM-SM deployment requires:

- PIM enabled on every interface along the multicast path.
- The location of the RP declared identically on every router in the PIM domain.
- IGMP enabled on interfaces facing multicast receivers.

Configuration

PIM-SM

set protocols pim ecmp

Enable ECMP routing for multicast traffic.

When a specific RPF lookup returns multiple equal-cost next-hops, enabling this feature distributes (S,G) multicast flows across all of them.

If ECMP is not enabled, PIM defaults to using only the first next-hop returned by the RPF lookup.

Example:

```
set protocols pim ecmp
```

set protocols pim ecmp rebalance

Enable automatic rebalancing of all PIM (S,G) multicast flows when an ECMP next-hop interface fails.

When ECMP is enabled, and one of the interfaces used by PIM goes down, this feature forces PIM to redistribute all existing (S,G) flows across the remaining equal-cost interfaces.

If rebalancing is not configured, PIM only reroutes the specific flows that were actively using the failed interface.

Example:

```
set protocols pim ecmp rebalance
```

set protocols pim join-prune-interval <1-65535>

Configure the interval, in seconds, at which PIM sends Join/Prune messages to upstream neighbors.

The default interval is 60 seconds. Setting a value below 60 seconds can negatively impact convergence at scale.

Example:

```
set protocols pim join-prune-interval 120
```

set protocols pim keep-alive-timer <1-65535>

Configure the keep-alive timeout interval, in seconds, for (S,G) multicast flow state.

Note: Some hardware cannot detect traffic flowing in intervals shorter than 30 seconds. For such hardware, ensure the value is set to 31 seconds or higher.

Example:

```
set protocols pim keep-alive-timer 60
```

set protocols pim packets <1-255>

Configure the number of incoming PIM packets processed consecutively from a neighbor before the system moves to the next task.

The default value is 3 packets. This command is only useful in deployments with a high number of PIM control packets.

Example:

```
set protocols pim packets 10
```

set protocols pim register-accept-list prefix-list <prefix-list>

Configure a prefix-list to filter incoming PIM Register messages.

When PIM receives a Register packet, the multicast source IP address is evaluated against the specified prefix-list. If the source IP address is permitted, normal PIM processing continues. If the source IP address is denied, a Register-Stop message is sent to the sender.

Example:

```
set protocols pim register-accept-list prefix-list MULTICAST-SOURCES
```

set protocols pim register-suppress-time <1-65535>

Configure the duration, in seconds, during which a FHR (First-Hop Router) suppresses sending PIM Register messages.

Example:

```
set protocols pim register-suppress-time 60
```

set protocols pim rp address <ip-address> group <prefix>

Configure a Rendezvous Point (RP) for the specified multicast group range.

To use PIM, you must configure an RP to receive Join messages. VyOS currently supports only static RP configuration.

The <address> parameter defines the RP's IP address, and the <group> parameter defines the prefix of the multicast group range served by this RP.

A single RP can serve multiple group ranges. Repeat the command with different <group> prefixes to map several ranges to the same RP.

Example:

```
set protocols pim rp address 192.0.2.1 group 239.0.0.0/8
```

set protocols pim rp keep-alive-timer <1-65535>

Configure the keep-alive timeout interval, in seconds, for (S,G) multicast flow state at the Rendezvous Point (RP).

This value must be at least equal to the register-suppression-time; otherwise, the RP may time out the (S,G) state before the next Null-Register message arrives from the FHR.

When a Register-Stop is sent, the KAT(S,G) at the RP is set to the maximum of the global Keepalive_Period and the RP_Keepalive_Period.

The default value is 210 seconds.

See [RFC 7761, Section 4.1](#) for details.

Note: As some hardware platforms cannot detect data flowing in intervals shorter than 30 seconds, ensure the value is set to 31 seconds or higher.

Example:

```
set protocols pim rp keep-alive-timer 300
```

set protocols pim no-v6-secondary

Disable sending IPv6 secondary addresses in PIM Hello messages.

By default, PIM Hello messages include all IPv6 secondary addresses configured on the interface, allowing PIM neighbors to use them as IPv6 next-hops for RPF lookups.

Example:

```
set protocols pim no-v6-secondary
```

set protocols pim spt-switchover infinity-and-beyond prefix-list <list>

On the LHR (Last-Hop Router), suppress the switchover from the shared (*,G) tree to the source-specific (S,G) SPT (Shortest-Path Tree).

The optional *prefix-list* controls which groups are affected: groups matched by a *permit* entry stay on the shared tree, groups matched by a *deny* entry switch over normally. Without a *prefix-list*, the suppression applies to all groups.

Example:

```
set protocols pim spt-switchover infinity-and-beyond
set protocols pim spt-switchover infinity-and-beyond prefix-list SHARED-TREE-GROUPS
```

set protocols pim ssm prefix-list <list>

Configure a prefix-list with the range of multicast group IP addresses that operate in SSM mode.

Example:

```
set protocols pim ssm prefix-list SSM-RANGE
```

Interface-specific commands

set protocols pim interface <interface> bfd

Enable BFD for each PIM neighbor discovered on the specified interface.

When BFD detects a link failure, the PIM neighbor relationship is torn down, and all associated multicast (S,G) entries are removed.

Example:

```
set protocols pim interface eth0 bfd
```

set protocols pim interface <interface> bfd profile <name>

Apply the specified BFD profile to BFD sessions on this PIM interface.

Example:

```
set protocols pim interface eth0 bfd profile fast-failover
```

set protocols pim interface <interface> dr-priority <1-4294967295>

Configure the DR (Designated Router) priority for the specified interface.

The router with the highest interface priority is elected as the DR for a LAN segment.

Example:

```
set protocols pim interface eth0 dr-priority 100
```

```
set protocols pim interface <interface> hello <1-180>
```

Configure the PIM Hello interval, in seconds, for the specified interface.

PIM Hello messages are sent periodically on the interface to discover and maintain PIM neighbor relationships. The hold time (the duration after which a neighbor is considered dead if no Hello is received) is calculated based on the Hello interval.

Example:

```
set protocols pim interface eth0 hello 10
```

```
set protocols pim interface <interface> no-bsm
```

Disable processing of PIM BSMs (Bootstrap Messages) on the specified interface.

Example:

```
set protocols pim interface eth0 no-bsm
```

```
set protocols pim interface <interface> no-unicast-bsm
```

Disable processing of unicast PIM BSMs on this interface.

Example:

```
set protocols pim interface eth0 no-unicast-bsm
```

```
set protocols pim interface <interface> passive
```

Disable sending and receiving PIM control packets on this interface.

Example:

```
set protocols pim interface eth0 passive
```

```
set protocols pim interface <interface> source-address <ip-address>
```

Configure the source IP address PIM uses for control messages sent on the specified interface.

This command is useful when multiple IP addresses are configured on a single interface.

Example:

```
set protocols pim interface eth0 source-address 192.0.2.10
```

IGMP

```
set protocols pim igmp watermark-warning <1-65535>
```

Configure watermark warning generation based on the number of joined IGMP groups.

When the total number of joined IGMP groups reaches this specified limit, the router generates a warning.

Example:

```
set protocols pim igmp watermark-warning 500
```

Interface-specific commands

```
set protocols pim interface <interface> igmp
```

Enable IGMP on the specified interface.

Example:

```
set protocols pim interface eth0 igmp
```

```
set protocols pim interface <interface> igmp disable
```

Administratively disable IGMP on the specified interface without removing the existing IGMP configuration.

Example:

```
set protocols pim interface eth0 igmp disable
```

```
set protocols pim interface <interface> igmp join <multicast-address> source-address <IP-address>
```

Configure the interface to join a specific (S,G) channel.

Specify both the multicast IP address (G) and the source IP address (S). Each command joins one (S,G) channel. To join multiple channels sharing the same group, repeat the command with different source-address values.

Example:

```
set protocols pim interface eth0 igmp join 232.1.1.100 source-address 192.0.2.50
set protocols pim interface eth0 igmp join 232.1.1.100 source-address 192.0.2.51
```

```
set protocols pim interface <interface> igmp query-interval <1-1800>
```

Configure the IGMP Host Query Interval, in seconds, on the specified interface.

This setting determines the interval between IGMP queries sent by the router to discover which multicast groups have active listeners on the attached network.

Example:

```
set protocols pim interface eth1 igmp query-interval 125
```

```
set protocols pim interface <interface> igmp query-max-response-time <10-250>
```

Configure the IGMP maximum query response time, in deciseconds, on the specified interface.

This value is inserted in IGMP queries and specifies the maximum time hosts have to respond with a membership report. If no report is received within the specified window, PIM assumes there are no longer any active local receivers and may time out the associated (*,G) or (S,G) state (see [RFC 7761, Section 4.1](#)).

Example:

```
set protocols pim interface eth0 igmp query-max-response-time 50
```

```
set protocols pim interface <interface> igmp version <2-3>
```


Configure the IGMP version on the specified interface.

The default version is 3.

Example:

```
set protocols pim interface eth0 igmp version 2
```

Example

The following example demonstrates a basic PIM Sparse Mode (PIM-SM) multicast deployment across three routers, showing how to configure each router's role within a multicast domain. Together, the three routers form a complete multicast distribution path from sources to receivers, with one router acting as the Rendezvous Point (RP).

Topology and roles

- Router 1 is the LHR.
- Router 3 is the RP.
- Router 2 is the FHR.

Router 1:

```
set interfaces ethernet eth2 address '198.51.100.1/30'
set interfaces ethernet eth1 address '192.0.2.1/24'
set protocols ospf area 0 network '198.51.100.0/30'
set protocols ospf area 0 network '192.0.2.0/24'
set protocols pim interface eth1
set protocols pim interface eth1 igmp
set protocols pim interface eth2
set protocols pim rp address 203.0.113.1 group '224.0.0.0/4'
```

Router 3:

```
set interfaces dummy dum0 address '203.0.113.1/32'
set interfaces ethernet eth0 address '198.51.100.2/30'
set interfaces ethernet eth1 address '198.51.100.5/30'
set protocols ospf area 0 network '198.51.100.0/30'
set protocols ospf area 0 network '203.0.113.1/32'
set protocols ospf area 0 network '198.51.100.4/30'
set protocols pim interface dum0
set protocols pim interface eth0
set protocols pim interface eth1
set protocols pim rp address 203.0.113.1 group '224.0.0.0/4'
```

Router 2:

```
set interfaces ethernet eth1 address '203.0.113.129/25'
set interfaces ethernet eth2 address '198.51.100.6/30'
set protocols ospf area 0 network '203.0.113.128/25'
set protocols ospf area 0 network '198.51.100.4/30'
set protocols pim interface eth1
set protocols pim interface eth2
set protocols pim rp address 203.0.113.1 group '224.0.0.0/4'
```

7.9.15 PIMv6

VyOS supports IPv6 multicast routing through PIMv6 (Protocol Independent Multicast for IPv6) and MLD versions 1 and 2.

PIMv6 operates similarly to its IPv4 counterpart: it uses the existing unicast IPv6 routing table for path decisions, and traffic from multicast sources is forwarded toward a RP. IPv6 multicast receivers use MLD to signal their local router which groups they want to receive, and the router uses PIMv6 to pull that traffic from the RP via a shared distribution tree.

A working PIMv6 deployment requires:

- PIMv6 enabled on every interface participating in IPv6 multicast forwarding.
- The location of the Rendezvous Point (RP) declared identically on every router in the PIMv6 domain.
- MLD enabled on interfaces facing multicast receivers.

Basic commands

Use the following commands for a basic PIMv6 setup.

set protocols pim6 interface <interface-name>

Enable PIMv6 on the specified PIMv6 interface.

This command also enables MLD, allowing the router to process MLD queries and reports from IPv6 multicast receivers.

Example:

```
set protocols pim6 interface eth0
```

set protocols pim6 interface <interface-name> mld disable

Disable MLD on the specified PIMv6 interface.

This command turns off MLD processing on the interface while keeping PIMv6 enabled.

Example:

```
set protocols pim6 interface eth0 mld disable
```

Tuning commands

You can also tune multicast with the following commands.

set protocols pim6 interface <interface-name> mld interval <1-65535>

Configure the MLD Query Interval, in seconds, on the specified interface.

This setting determines how often the router sends MLD queries to discover which IPv6 multicast groups have active listeners on the attached network.

The default value is 125 seconds.

Example:

```
set protocols pim6 interface eth0 mld interval 100
```

set protocols pim6 interface <interface-name> mld join <multicast-address>

Configure the interface to join a specific IPv6 multicast group.

Example:

```
set protocols pim6 interface eth0 mld join ff0e::db8:0:1234
```

```
set protocols pim6 interface <interface-name> mld join <multicast-address> source <source-address>
```

Configure the interface to join a specific (S,G) channel.

A channel is identified by the combination of a source IPv6 address (S) and a multicast IPv6 address (G).

Example:

```
set protocols pim6 interface eth0 mld join ff3e::1234 source 2001:db8::1
```

```
set protocols pim6 interface <interface-name> mld last-member-query-count <1-255>
```

Configure the number of MLD Last Listener Queries the router sends on the specified interface after a listener leaves a multicast group.

The default value is 2.

Example:

```
set protocols pim6 interface eth0 mld last-member-query-count 3
```

```
set protocols pim6 interface <interface-name> mld last-member-query-interval <100-6553500>
```

Configure the MLD Last Listener Query Interval, in milliseconds, on the specified interface.

This setting determines how long the router waits between Last Listener Queries.

The default value is 1000 milliseconds.

Example:

```
set protocols pim6 interface eth0 mld last-member-query-interval 500
```

```
set protocols pim6 interface <interface-name> mld max-response-time <100-6553500>
```

Configure the MLD maximum response delay, in milliseconds, on the specified interface.

This value is inserted in MLD queries and specifies the maximum time hosts have to respond with a membership report. If no report is received within the specified window, PIMv6 assumes there are no longer any active local receivers and may time out the associated (*,G) or (S,G) state.

Example:

```
set protocols pim6 interface eth0 mld max-response-time 5000
```

```
set protocols pim6 interface <interface-name> mld version <1-2>
```

Configure the MLD version on the specified interface.

The default version is 2.

Example:

```
set protocols pim6 interface eth0 mld version 1
```

Configuration example

To enable PIMv6 on interfaces eth0 and eth1 (which also enables MLD automatically, allowing the router to process MLD queries and reports from connected IPv6 multicast listeners):

```
set protocols pim6 interface eth0
set protocols pim6 interface eth1
```

The following configuration explicitly joins the IPv6 multicast group ff0e::db8:0:1234 on interface eth0, and the (S,G) channel (2001:db8::1, ff3e::5678) on interface eth1:

```
set protocols pim6 interface eth0 mld join ff0e::db8:0:1234
set protocols pim6 interface eth1 mld join ff3e::5678 source 2001:db8::1
```

7.9.16 RIP

Developed in the 1970s at Xerox Labs, RIP is a distance-vector routing protocol based on the Bellman-Ford algorithm.

RIP determines the best route based on hop count (number of routers traversed). RIP-enabled routers periodically advertise their entire routing table to neighbors, along with the metric (hop count) for each destination.

Supported versions:

- RIPv1: As described in [RFC 1058](#)
- RIPv2: As described in [RFC 2453](#)

General configuration

set protocols rip network <A.B.C.D/M>

Enable RIP on all interfaces whose IP addresses fall within the specified network.

Example:

```
set protocols rip network 192.0.2.0/24
```

set protocols rip interface <interface>

Enable RIP on the specified interface.

Once applied, the interface sends and receives RIP packets.

Example:

```
set protocols rip interface eth0
```

set protocols rip neighbor <A.B.C.D>

Configure a RIP neighbor using its IP address.

Use this command when a neighbor cannot process multicast RIP updates. Once configured, the local router sends updates directly to the specified neighbor as unicast packets.

Example:

```
set protocols rip neighbor 192.0.2.2
```

set protocols rip passive-interface <interface>

Enable passive mode for the specified interface.

This prevents the router from sending outgoing RIP updates on the interface, except to explicitly configured neighbors. Incoming RIP updates are still accepted and processed.

Example:

```
set protocols rip passive-interface eth0
```

set protocols rip passive-interface default**Enable passive mode for all interfaces.**

This prevents the router from sending outgoing RIP updates on all interfaces, except to explicitly configured neighbors. The router still receives and processes incoming RIP updates normally.

Example:

```
set protocols rip passive-interface default
```

Optional configuration**set protocols rip default-distance <1-255>****Configure the administrative distance for all RIP-learned routes.**

The administrative distance determines how RIP-learned routes are prioritized against routes learned from other sources (such as BGP, OSPF, or static routes) when multiple paths to the same destination exist.

Note: Routes assigned a distance of 255 are disabled and will not be installed in the kernel forwarding table.

Example:

```
set protocols rip default-distance 100
```

set protocols rip network-distance <A.B.C.D/M> distance <1-255>**Configure an administrative distance for RIP routes whose source IP address falls within the specified network.**

Example:

```
set protocols rip network-distance 192.0.2.0/24 distance 80
```

set protocols rip network-distance <A.B.C.D/M> access-list <name>**Restrict administrative distance assignments to RIP routes whose source IP address matches the specified network and the route itself matches the specified access-list.**

Example:

```
set protocols rip network-distance 192.0.2.0/24 distance 80
set protocols rip network-distance 192.0.2.0/24 access-list 10
```

set protocols rip default-information originate**Generate and advertise a default route (0.0.0.0/0) to all RIP neighbors.**

Example:

```
set protocols rip default-information originate
```

```
set protocols rip distribute-list access-list <in|out> <0-4294967295>
```

Apply an access-list to filter RIP routes in the inbound or outbound direction.

- in: Filters routes received from RIP neighbors.
- out: Filters routes advertised to RIP neighbors.

Example:

```
set protocols rip distribute-list access-list in 20
```

```
set protocols rip distribute-list interface <interface> access-list <in|out> <0-4294967295>
```

Apply an access-list to filter RIP routes in the inbound or outbound direction on the specified interface.

- in: Filters routes received from RIP neighbors on this interface.
- out: Filters routes advertised to RIP neighbors on this interface.

Example:

```
set protocols rip distribute-list interface eth0 access-list in 20
```

```
set protocols rip distribute-list prefix-list <in|out> <name>
```

Apply a prefix-list to filter RIP routes in the inbound or outbound direction.

- in: Filters routes received from RIP neighbors.
- out: Filters routes advertised to RIP neighbors.

Example:

```
set protocols rip distribute-list prefix-list in INBOUND-FILTER
```

```
set protocols rip distribute-list interface <interface> prefix-list <in|out> <name>
```

Apply a prefix-list to filter RIP routes on the specified interface in the inbound or outbound direction.

- in: Filters routes received from RIP neighbors on this interface.
- out: Filters routes advertised to RIP neighbors on this interface.

Example:

```
set protocols rip distribute-list interface eth0 prefix-list in RIP-IN
```

```
set protocols rip route <A.B.C.D/M>
```

Add a static route to the RIP routing process.

The route is created only inside RIP.

This command is specific to FRR and VyOS and should only be used by advanced users with a strong understanding of the RIP protocol.

For most deployments, the recommended approach is to create a standard static route and redistribute it into RIP using `set protocols rip redistribute static`.

Example:

```
set protocols rip route 192.0.2.0/24
```

set protocols rip timers update <5-2147483647>

Configure the RIP update timer, in seconds.

The update timer defines how often the router sends unsolicited Response messages to its RIP neighbors. Each Response contains the complete RIP routing table.

The default value is 30 seconds.

Example:

```
set protocols rip timers update 60
```

set protocols rip timers timeout <5-2147483647>

Configure the RIP route timeout, in seconds.

If a route is not refreshed by an incoming update within this interval, it becomes invalid. The router temporarily retains the invalid route in the RIP routing table so that neighbors learn the route has been dropped.

The default value is 180 seconds.

Example:

```
set protocols rip timers timeout 300
```

set protocols rip timers garbage-collection <5-2147483647>

Configure the RIP garbage-collection timer, in seconds.

This timer starts when a route becomes invalid. When the timer expires, the route is removed from the routing table.

The default value is 120 seconds.

Example:

```
set protocols rip timers garbage-collection 60
```

Redistribution configuration

set protocols rip redistribute <babel|bgp|connected|isis|kernel|nhrrp|ospf|static>

Redistribute routes from the specified source into RIP.

Routes learned from the chosen source are imported into the RIP routing table and advertised to RIP neighbors as if they had been learned through RIP itself.

The supported sources are:

- **babel:** Routes learned via Babel.
- **bgp:** Routes learned via BGP.
- **connected:** Directly connected routes.
- **isis:** Routes learned via IS-IS.
- **kernel:** Routes installed in the kernel routing table.
- **nhrrp:** Routes learned via NHRP.

- `ospf`: Routes learned via OSPF.
- `static`: Routes configured statically.

Example:

```
set protocols rip redistribute static
```

```
set protocols rip redistribute <babel|bgp|connected|isis|kernel|nhrp|ospf|static> metric <1-16>
```

Configure the metric for routes redistributed from the specified source.

Example:

```
set protocols rip redistribute static metric 5
```

```
set protocols rip redistribute <babel|bgp|connected|isis|kernel|nhrp|ospf|static> route-map <name>
```

Apply a route-map to filter routes redistributed from the specified source.

Example:

```
set protocols rip redistribute static route-map RIP-REDISTRIBUTE
```

```
set protocols rip default-metric <1-16>
```

Configure the default metric (hop count) applied to redistributed routes.

The default value is 1.

Note: This command does not affect connected routes redistributed via `set protocols rip redistribute connected`. To configure the metric for connected routes, use `set protocols rip redistribute connected metric` explicitly.

Example:

```
set protocols rip default-metric 5
```

Interfaces configuration

```
set protocols rip interface <interface> authentication plaintext-password <text>
```

Enable simple password authentication for RIPv2 on the specified interface and set a password.

The password must not exceed 16 characters.

Example:

```
set protocols rip interface eth0 authentication plaintext-password vyos-secret
```

```
set protocols rip interface <interface> authentication md5 <id> password <text>
```

Enable MD5 authentication for RIPv2 on the specified interface and set the MD5 key.

The MD5 key must not exceed 16 characters.

Example:


```
set protocols rip interface eth0 authentication md5 1 password vyos-secret
```

set protocols rip interface <interface> split-horizon disable

Disable split-horizon on the specified interface.

By default, VyOS does not advertise RIP routes back through the interface on which they were learned (split horizon). This command turns off that default.

Example:

```
set protocols rip interface eth0 split-horizon disable
```

set protocols rip interface <interface> split-horizon poison-reverse

Enable split horizon with poison reverse on the specified interface.

If enabled, the router advertises the learned routes as unreachable on the interface where they were learned.

Example:

```
set protocols rip interface eth0 split-horizon poison-reverse
```

Operational mode commands

show ip rip

Show all RIP routes.

```
vyos@vyos-1:~$ show ip rip
Codes: K - kernel route, C - connected, L - local, S - static,
       R - RIP, O - OSPF, I - IS-IS, B - BGP, E - EIGRP, N - NHRP,
       T - Table, v - VNC, V - VNC-Direct, A - Babel, F - PBR,
       f - OpenFabric, t - Table-Direct
Sub-codes:
       (n) - normal, (s) - static, (d) - default, (r) - redistribute,
       (i) - interface

Network        Next Hop      Metric From      Tag Time
C(i) 192.0.2.0/24  0.0.0.0          1 self           0
C(i) 198.51.100.0/24  0.0.0.0          1 self           0
R(n) 203.0.113.0/24  192.0.2.2        2 192.0.2.2      0 02:53
```

show ip rip status

Show the current RIP status.

```
Routing Protocol is "rip"
  Sending updates every 30 seconds with +/-50%, next due in 11 seconds
  Timeout after 180 seconds, garbage collect after 120 seconds
  Outgoing update filter list for all interface is not set
  Incoming update filter list for all interface is not set
  Default redistribution metric is 1
  Redistributing:
  Default version control: send version 2, receive any version
    Interface      Send Recv  Key-chain
```

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```

eth0          2      1 2
eth2          2      1 2
Routing for Networks:
  192.0.2.0/24
  eth0
Routing Information Sources:
  Gateway      BadPackets  BadRoutes   Distance  Last Update
  192.0.2.2      0           0           120      00:00:11
Distance: (default is 120)

```

Example

The following example demonstrates a basic RIP configuration between two routers, where directly connected networks are redistributed into RIP.

Node 1:

```

set interfaces ethernet eth0 address '198.51.100.1/24'
set interfaces loopback lo address '192.0.2.1/32'
set protocols rip network 198.51.100.0/24
set protocols rip redistribute connected

```

Node 2:

```

set interfaces ethernet eth0 address '198.51.100.2/24'
set interfaces loopback lo address '192.0.2.2/32'
set protocols rip network 198.51.100.0/24
set protocols rip redistribute connected

```

7.9.17 RPKI

RPKI (Resource Public Key Infrastructure) is an architecture designed to secure the internet routing infrastructure. It cryptographically associates IP address prefixes with the correct originating ASNs (Autonomous System Numbers). BGP routers can use this data to check the origin of each route against the corresponding ROA (Route Origin Authorization) for validity. RPKI is described in [RFC 6480](#).

BGP routers can retrieve ROA information from RPKI Relying Party software, also known as an RPKI server or validator, using the RTR protocol. Open-source implementations include NLnet Labs' [Routinator](#) (Rust), OpenBSD's [rpki-client](#) (C), and [StayRTR](#) (Go). The RTR protocol is described in [RFC 8210](#).

Tip: If you are new to routing security, the [RPKI Documentation](#) maintained by NLnet Labs will quickly get you up to speed. It covers everything from fundamental concepts to production deployment.

Getting started

First, you need to deploy an RPKI validator for your routers. The [RPKI Documentation](#) maintained by NLnet Labs provides a comprehensive list of open-source software packages you can compare and choose from. Once your chosen server is running, your routers can start validating announcements.

After validation, imported routes may have one of the following states:

- **valid:** The prefix and originating ASN match a signed ROA. These route announcements are considered trustworthy.
- **invalid:** The prefix, prefix length, or originating ASN does not match any existing ROA. This may indicate a prefix hijack or misconfiguration and should be treated as untrustworthy.
- **not found:** No ROA covers the prefix. As of early 2024, this applies to approximately 40% to 50% of prefixes announced to the DFZ (Default-Free Zone).

Note: If you manage global IP addresses for your network, ensure their prefixes have associated ROAs to prevent them from being marked as **not found** by RPKI.

For most network operators, this involves publishing ROAs through their respective RIR (Regional Internet Registry), such as ARIN, RIPE NCC, APNIC, LACNIC, or AFRINIC. It is highly recommended to create ROAs for any IP prefixes you plan to announce into the DFZ.

Particularly large networks may choose to run their own RPKI certificate authority and publication server instead of publishing ROAs via their RIR. This subject is beyond the scope of VyOS documentation. For more information, consider reviewing resources about [Krill](#).

Current implementation

The current implementation includes the following features:

- The BGP router connects to one or more RPKI cache servers to receive validated prefix-to-origin AS mappings. Advanced failover can be implemented by configuring multiple cache servers with different preference values.
- If no connection to an RPKI cache server is established after a predefined timeout, the router processes routes without prefix origin validation. It will continue attempting to connect to an RPKI cache server in the background.
- By default, enabling RPKI does not affect best path selection. Invalid prefixes are still considered during best path selection, but you can configure the router to ignore them.
- You can configure route maps to match a specific RPKI validation state. This enables the creation of local policies that handle BGP routes based on Prefix Origin Validation results.
- Updates from RPKI cache servers are applied directly, and path selection is updated accordingly. Note: BGP soft reconfiguration must be enabled for this feature to work.

Configuration

set protocols rpki polling-period <1-86400>

Configure the interval, in seconds, at which the router polls the RPKI cache server for updates.

The default value is 300 seconds.

Example:

```
set protocols rpki polling-period 600
```

set protocols rpki expire-interval <600-172800>

Configure the interval, in seconds, after which the router expires its cached RPKI data.

The default value is 7200 seconds.

Example:

```
set protocols rpki expire-interval 14400
```

set protocols rpki retry-interval <1-7200>

Configure the interval, in seconds, before the router retries a failed connection to the RPKI cache server.

The default value is 600 seconds.

Example:

```
set protocols rpki retry-interval 60
```

set protocols rpki cache <address> port <1-65535>

Configure the address and port of an RPKI cache server.

The address can be specified as an IPv4 or IPv6 address, or as a FQDN.

Note: You need to configure both port and preference for each cache server IP address.

Example:

```
set protocols rpki cache 192.0.2.1 port 3323
```

set protocols rpki cache <address> preference <1-255>

Configure the preference value for an RPKI cache server.

When multiple cache servers are configured, the router prioritizes connecting to the server with the lowest preference value.

Note: You need to configure both port and preference for each cache server IP address.

Note: preference values must be unique across cache servers.

Example:

```
set protocols rpki cache 192.0.2.1 preference 1
```

SSH

You can connect to the RPKI cache server using the RTR protocol over plain TCP or over SSH. SSH provides transport integrity and confidentiality. Use SSH if your validation software supports this option.

To use SSH authentication, complete the following steps:

Step 1. Generate an SSH client keypair:

```
generate ssh client-key /config/auth/id_rsa_rpki
```

Step 2. Extract the base64 portion from both keys, plus the public key's algorithm identifier:

```
grep -v "PRIVATE KEY" /config/auth/id_rsa_rpki | tr -d '\n'
awk '{print $2}' /config/auth/id_rsa_rpki.pub
awk '{print $1}' /config/auth/id_rsa_rpki.pub
```

Note: The PKI subsystem accepts only the base64 portion of each key, without any PEM/OpenSSH headers, footer, or line breaks. If you paste the file contents directly, the commit fails.

Step 3. Import the keypair into the PKI subsystem:

```
set pki openssh <name> private key <base64-encoded-private-key>
set pki openssh <name> public key <base64-encoded-public-key>
set pki openssh <name> public type <type>
```

Once the keypair is imported, set up the SSH connection using the commands below.

```
set protocols rpki cache <address> ssh username <user>
```

Configure the SSH username for authenticating to the RPKI cache server.

Example:

```
set protocols rpki cache 192.0.2.1 ssh username rpki-user
```

```
set protocols rpki cache <address> ssh key <name>
```

Configure the name of the OpenSSH keypair used to authenticate to the RPKI cache server.

The <name> must match a keypair name previously defined under the `set pki openssh` commands.

Example:

```
set protocols rpki cache 192.0.2.1 ssh key rpki
```

Example

The following example demonstrates how to connect to an RPKI cache server (such as Routinator) and build an import route-map to filter prefixes based on their validation state.

First, configure the connection to the RPKI cache server at 192.0.2.1:

```
set protocols rpki cache 192.0.2.1 port '3323'
set protocols rpki cache 192.0.2.1 preference '1'
```

Next, create a route-map to apply to your incoming BGP updates. This policy dictates the following behavior:

- Permits valid prefixes and assigns a high local-preference (300).
- Permits notfound prefixes but assigns a lower local-preference (125).
- Rejects prefixes with an invalid state.

```
set policy route-map ROUTES-IN rule 10 action 'permit'
set policy route-map ROUTES-IN rule 10 match rpki 'valid'
set policy route-map ROUTES-IN rule 10 set local-preference '300'

set policy route-map ROUTES-IN rule 20 action 'permit'
set policy route-map ROUTES-IN rule 20 match rpki 'notfound'
set policy route-map ROUTES-IN rule 20 set local-preference '125'

set policy route-map ROUTES-IN rule 30 action 'deny'
set policy route-map ROUTES-IN rule 30 match rpki 'invalid'
```

After configuring your routers to reject RPKI-invalid prefixes, verify the setup using [Cloudflare's test website](#).

Note: Ensure there are no default routes or other paths that would direct traffic to RPKI-invalid destinations. Otherwise, the test may report the router as unsafe.

7.9.18 Static

Static routes are manually configured entries in the router's RIB that define how traffic should reach specific destinations. Unlike dynamic routing protocols, static routes are not automatically updated when the network topology changes.

Static routes are best suited for simple network topologies or to override dynamic routing protocol behavior for a limited number of routes.

The RIB stores all routing information, including manually configured static routes and any dynamically learned routes. From the RIB, the router derives unicast routes to build the FIB.

IPv4 unicast routes

IPv4 unicast routes direct traffic destined for a single IPv4 host or subnet toward a specific next hop. Use the following commands to configure IPv4 unicast routes for a specific remote subnet.

set protocols static route <subnet> next-hop <address>

Configure the next-hop IP address for the specified subnet.

You can configure multiple next-hop IP addresses for the same destination subnet.

Example:

```
set protocols static route 192.0.2.0/24 next-hop 10.0.0.254
```

set protocols static route <subnet> next-hop <address> disable

Disable the route via the specified next-hop IP address.

This command temporarily deactivates the route while preserving the configured subnet, next-hop, and distance parameters.

Example:

```
set protocols static route 192.0.2.0/24 next-hop 10.0.0.254 disable
```

set protocols static route <subnet> next-hop <address> distance <distance>

Configure the administrative distance for the route via the specified next-hop IP address.

Routes with a lower administrative distance are prioritized over routes with a higher distance.

The allowed range is 1 to 255. The default distance is 1.

Note: Routes with a distance of 255 are disabled and not used for packet forwarding.

Example:

```
set protocols static route 192.0.2.0/24 next-hop 10.0.0.254 distance 10
```

IPv4 interface routes

IPv4 interface routes direct traffic destined for a single IPv4 host or subnet out of a specific local interface. Use the following commands to configure IPv4 interface routes for a specific remote subnet.

set protocols static route <subnet> interface <interface>

Configure the next-hop interface for the specified subnet.

You can configure multiple next-hop interfaces for the same destination subnet.

Example:

```
set protocols static route 192.0.2.0/24 interface eth0
```

set protocols static route <subnet> interface <interface> disable

Disable the route via the specified next-hop interface.

This command temporarily deactivates the route while preserving the configured subnet, interface, and distance parameters.

Example:

```
set protocols static route 192.0.2.0/24 interface eth0 disable
```

```
set protocols static route <subnet> interface <interface> distance <distance>
```

Configure the administrative distance for the route via the specified next-hop interface.

Routes with a lower administrative distance are prioritized over routes with a higher distance.

The allowed range is 1 to 255. The default distance is 1.

Example:

```
set protocols static route 192.0.2.0/24 interface eth0 distance 10
```

IPv4 BFD

BFD monitors the reachability of a static route's next-hop IP address. Use the following commands to configure BFD parameters for your IPv4 static routes.

```
set protocols static route <subnet> next-hop <address> bfd
```

Enable BFD monitoring for the route via the specified next-hop IP address.

The system uses the next-hop IP address as the BFD peer destination.

Example:

```
set protocols static route 192.0.2.0/24 next-hop 10.0.0.254 bfd
```

```
set protocols static route <subnet> next-hop <address> bfd profile <profile>
```

Apply the settings from a designated BFD profile to the BFD session that monitors the specified next-hop IP address.

Example:

```
set protocols static route 192.0.2.0/24 next-hop 10.0.0.254 bfd profile custom-profile
```

```
set protocols static route <subnet> next-hop <address> bfd multi-hop source-address <source-address>
```

Enable a multi-hop BFD session to monitor the reachability of the specified next-hop IP address.

The system initiates the BFD session from the specified local source IPv4 address.

Example:

```
set protocols static route 192.0.2.0/24 next-hop 10.0.0.254 bfd multi-hop source-address 172.16.1.1
```


DHCP interface routes

DHCP interface routes direct traffic destined for a single IPv4 host or subnet toward a next-hop gateway obtained from a DHCP-enabled interface. Use the following command to configure DHCP interface routes.

set protocols static route <subnet> dhcp-interface <interface>

Configure a static route that derives its next-hop IP address from the specified DHCP-enabled interface.

Example:

```
set protocols static route 192.0.2.0/24 dhcp-interface eth0
```

IPv4 reject routes

IPv4 reject routes explicitly drop traffic destined for a single IPv4 host or subnet and return an ICMP unreachable message to the sending device. Use the following commands to configure IPv4 reject routes.

set protocols static route <subnet> reject

Configure a reject route for the specified subnet.

The system discards traffic destined for this subnet and returns an ICMP destination unreachable message to the sending device.

Example:

```
set protocols static route 192.0.2.0/24 reject
```

set protocols static route <subnet> reject distance <distance>

Configure the administrative distance for the reject route.

Routes with a lower administrative distance are prioritized over routes with a higher distance.

The allowed range is 1 to 255.

Example:

```
set protocols static route 192.0.2.0/24 reject distance 200
```

set protocols static route <subnet> reject tag <tag>

Configure a numeric tag for the reject route.

The allowed range is 1 to 4294967295.

Example:

```
set protocols static route 192.0.2.0/24 reject tag 100
```

IPv4 blackhole routes

Blackhole routes silently discard traffic destined for a specific subnet. Unlike a reject route, which notifies the sender with an ICMP destination unreachable message, a blackhole route drops the packets without generating any response.

set protocols static route <subnet> blackhole

Configure a blackhole route for the specified subnet.

The system silently discards matching packets without sending an ICMP response to the source.

Example:

```
set protocols static route 10.0.0.0/8 blackhole
```

set protocols static route <subnet> blackhole distance <distance>

Configure the administrative distance for the blackhole route.

Routes with a lower administrative distance are prioritized over routes with a higher distance.

The allowed range is 1 to 255.

Example:

```
set protocols static route 10.0.0.0/8 blackhole distance 200
```

set protocols static route <subnet> blackhole tag <tag>

Configure a numeric tag for the blackhole route.

The allowed range is 1 to 4294967295.

Example:

```
set protocols static route 10.0.0.0/8 blackhole tag 100
```

IPv6 unicast routes

IPv6 unicast routes direct traffic destined for a single IPv6 host or subnet toward a specific next hop. Use the following commands to configure IPv6 unicast routes for a specific remote subnet.

set protocols static route6 <subnet> next-hop <address>

Configure the next-hop IPv6 address for the specified subnet.

You can configure multiple next-hop IPv6 addresses for the same destination subnet.

Example:

```
set protocols static route6 2001:db8:1::/64 next-hop 2001:db8:100::1
```

set protocols static route6 <subnet> next-hop <address> disable

Disable the route via the specified next-hop IPv6 address.

This command temporarily deactivates the route while preserving the configured subnet, next-hop, and distance parameters.

Example:

```
set protocols static route6 2001:db8:1::/64 next-hop 2001:db8:100::1 disable
```

set protocols static route6 <subnet> next-hop <address> distance <distance>

Configure the administrative distance for the route via the specified next-hop IPv6 address.

Routes with a lower administrative distance are prioritized over routes with a higher distance.

The allowed range is 1 to 255. The default distance is 1.

Note: Routes with a distance of 255 are disabled and not used for packet forwarding.

Example:

```
set protocols static route6 2001:db8:1::/64 next-hop 2001:db8:100::1 distance 10
```

set protocols static route6 <subnet> next-hop <address> segments <segments>

Configure an IPv6 unicast route that utilizes Segment Routing over IPv6 (SRv6) to reach the destination subnet.

Use a forward slash (/) to separate multiple IPv6 segment identifiers (SIDs) in your sequence.

Example:

```
set protocols static route6 2001:db8:1000::/36 next-hop 2001:db8:201::ffff segments
↳ '2001:db8:aaaa::7/2002::4/2002::3/2002::2'
```

```
vyos@vyos:~$ show ipv6 route
Codes: K - kernel route, C - connected, S - static, R - RIPng,
       O - OSPFv3, I - IS-IS, B - BGP, N - NHRP, T - Table,
       v - VNC, V - VNC-Direct, A - Babel, F - PBR,
       f - OpenFabric,
       > - selected route, * - FIB route, q - queued, r - rejected, b - backup
       t - trapped, o - offload failure
C>* 2001:db8:201::/64 is directly connected, eth0.201, 00:00:46
S>* 2001:db8:1000::/36 [1/0] via 2001:db8:201::ffff, eth0.201, seg6 2001:db8:aaaa::7,
↳ 2002::4,2002::3,2002::2, weight 1, 00:00:08
```

IPv6 interface routes

IPv6 interface routes direct traffic destined for a single IPv6 host or subnet out of a specific local interface. Use the following commands to configure IPv6 interface routes for a specific remote subnet.

set protocols static route6 <subnet> interface <interface>

Configure the next-hop interface for the specified subnet.

You can configure multiple next-hop interfaces for the same destination subnet.

Example:

```
set protocols static route6 2001:db8:1::/64 interface eth0
```

set protocols static route6 <subnet> interface <interface> disable

Disable the route via the specified next-hop interface.

This command temporarily deactivates the route while preserving the configured subnet, interface, and distance parameters.

Example:

```
set protocols static route6 2001:db8:1::/64 interface eth0 disable
```

```
set protocols static route6 <subnet> interface <interface> distance <distance>
```

Configure the administrative distance for the route via the specified next-hop interface.

Routes with a lower administrative distance are prioritized over routes with a higher distance.

The allowed range is 1 to 255. The default distance is 1.

Example:

```
set protocols static route6 2001:db8:1::/64 interface eth0 distance 10
```

```
set protocols static route6 <subnet> interface <interface> segments <segments>
```

Configure an IPv6 interface route that utilizes Segment Routing over IPv6 (SRv6) to reach a destination subnet.

Use a forward slash (/) to separate multiple IPv6 segment identifiers (SIDs) in your sequence.

Example:

```
set protocols static route6 2001:db8:1000::/36 interface eth0 segments '2001:db8:aaaa::7/
↪2002::4/2002::3/2002::2'
```

IPv6 BFD

BFD monitors the reachability of a static route's next-hop IPv6 address. Use the following commands to configure BFD parameters for your IPv6 static routes.

```
set protocols static route6 <subnet> next-hop <address> bfd
```

Enable BFD monitoring for the route via the specified next-hop IPv6 address.

The system uses the next-hop IPv6 address as the BFD peer destination.

Example:

```
set protocols static route6 2001:db8:1::/64 next-hop 2001:db8:100::1 bfd
```

```
set protocols static route6 <subnet> next-hop <address> bfd profile <profile>
```

Apply the settings from a designated BFD profile to the BFD session that monitors the specified next-hop IPv6 address.

Example:

```
set protocols static route6 2001:db8:1::/64 next-hop 2001:db8:100::1 bfd profile custom-
↪profile
```

```
set protocols static route6 <subnet> next-hop <address> bfd multi-hop source-address
<source-address>
```

Enable a multi-hop BFD session to monitor the reachability of the specified next-hop IPv6 address.

The system initiates the BFD session from the specified local source IPv6 address.

Example:

```
set protocols static route6 2001:db8:1::/64 next-hop 2001:db8:100::1 bfd multi-hop_
↪source-address 2001:db8:200::1
```

IPv6 reject routes

IPv6 reject routes explicitly drop traffic destined for a single IPv6 host or subnet and return an ICMPv6 unreachable message to the sending device. Use the following commands to configure IPv6 reject routes.

set protocols static route6 <subnet> reject

Configure a reject route for the specified subnet.

The system discards traffic destined for this subnet and returns an ICMPv6 destination unreachable message to the sending device.

Example:

```
set protocols static route6 2001:db8:1::/64 reject
```

set protocols static route6 <subnet> reject distance <distance>

Configure the administrative distance for the reject route.

Routes with a lower administrative distance are prioritized over routes with a higher distance.

The allowed range is 1 to 255.

Example:

```
set protocols static route6 2001:db8:1::/64 reject distance 200
```

set protocols static route6 <subnet> reject tag <tag>

Configure a numeric tag for the reject route.

The allowed range is 1 to 4294967295.

Example:

```
set protocols static route6 2001:db8:1::/64 reject tag 100
```

IPv6 blackhole routes

Blackhole routes silently discard traffic destined for a specific subnet. Unlike a reject route, which notifies the sender with an ICMPv6 destination unreachable message, a blackhole route drops the packets without generating any response.

set protocols static route6 <subnet> blackhole

Configure a blackhole route for the specified subnet.

The system silently discards matching packets without sending an ICMPv6 response to the source.

Example:

```
set protocols static route6 2001:db8:1::/64 blackhole
```

set protocols static route6 <subnet> blackhole distance <distance>

Configure the administrative distance for the blackhole route.

Routes with a lower administrative distance are prioritized over routes with a higher distance.

The allowed range is 1 to 255.

Example:

```
set protocols static route6 2001:db8:1::/64 blackhole distance 200
```

set protocols static route6 <subnet> blackhole tag <tag>

Configure a numeric tag for the blackhole route.

The allowed range is 1 to 4294967295.

Example:

```
set protocols static route6 2001:db8:1::/64 blackhole tag 100
```

Alternate routing tables

Alternate routing tables are utilized to implement both Policy-Based Routing (PBR) and Virtual Routing and Forwarding (*VRF*).

7.10 Service

7.10.1 UDP broadcast relay

Certain applications use UDP broadcasts to discover peers or services within a local subnet. Since IP routers do not forward broadcast packets between subnets, applications in different subnets cannot use broadcast discovery.

The UDP broadcast relay service bridges this gap. When an application sends a UDP broadcast packet to discover peers or services, the relay listens for that message on any of its configured interfaces and rebroadcasts it on the others, reaching the subnets they connect to.

The unit of configuration is a UDP relay instance. Each instance is defined by the following settings:

- a numeric ID in the range 1–99 that identifies the instance
- a single UDP port number that the instance relays
- two or more interfaces across which the instance operates

Every interface in the instance must be assigned an IPv4 address.

Note: Multiple routers may run UDP broadcast relay on a shared subnet, provided that any routers relaying the same UDP port use the same instance ID. Multiple routers relaying the same UDP port with different instance IDs will produce a relay loop and a packet storm.

Configuration

set service broadcast-relay id <1-99> description <description>

Configure a description for the specified relay instance.

Example:

```
set service broadcast-relay id 1 description 'SONOS discovery'
```

set service broadcast-relay id <1-99> interface <interface>

Add an interface to the specified relay instance.

Repeat the command to add additional interfaces.

Note: At least two interfaces must be configured per instance, and each interface must be assigned an IPv4 address.

Example:

```
set service broadcast-relay id 1 interface eth1
set service broadcast-relay id 1 interface eth2
```

set service broadcast-relay id <1-99> address <ipv4-address>

Configure the source IPv4 address used in forwarded packets.

If unset, the original sender's source IP address is preserved.

Example:

```
set service broadcast-relay id 1 address 192.0.2.1
```

set service broadcast-relay id <1-99> port <1-65535>

Configure the UDP destination port for the specified relay instance.

The relay listens for broadcasts on this port on any of the instance's interfaces and forwards them to the same port on the others.

Note: This setting is mandatory for each relay instance. Otherwise, the commit is rejected.

Example:

```
set service broadcast-relay id 1 port 1900
```

set service broadcast-relay id <1-99> disable

Administratively disable the specified relay instance while preserving its configuration.

Example:

```
set service broadcast-relay id 1 disable
```

set service broadcast-relay disable

Administratively disable the UDP broadcast relay service on the router without removing its configured instances.

Example:

```
set service broadcast-relay disable
```

Example

The following example shows how to configure the UDP broadcast relay to forward broadcasts on UDP port 1900 between eth1, eth2, and eth3. A broadcast received on any of these interfaces is forwarded to the other two. Each interface must already have an IPv4 address configured.

```
set service broadcast-relay id 1 description 'SONOS discovery'
set service broadcast-relay id 1 port '1900'
set service broadcast-relay id 1 interface 'eth1'
set service broadcast-relay id 1 interface 'eth2'
set service broadcast-relay id 1 interface 'eth3'
```

7.10.2 Config sync

Configuration synchronization (config-sync) is a VyOS service that replicates selected sections of one router's configuration to another router. It eliminates the need to manually mirror configuration changes from a primary router to a secondary.

Synchronization is performed by a post-commit hook on the primary router. The hook watches the sections selected for synchronization. When any of them is modified by a commit, the hook calls the HTTPS API on the secondary router and sends the current state of all selected sections in one of the following modes:

- **load:** Replaces the selected sections on the secondary with the primary's version. Anything that exists only on the secondary inside those sections is removed.
- **set:** Merges the primary's version of the selected sections into the secondary. Conflicting values are overwritten by the primary. Entries that exist only on the secondary are kept.

Synchronization is strictly unidirectional from primary to secondary; changes made on the secondary are not sent back.

Both routers must be online during a commit on the primary that modifies a configured section. If the secondary is unreachable, the primary's commit still succeeds, but the configuration is not synchronized, and the failure is recorded in the system log.

Configuration

Secondary router parameters

```
set service config-sync secondary address <address>
```

Configure the address of the secondary router.

The primary router pushes the synchronized configuration to this address over the secondary's HTTPS API. The value can be an IPv4 address, an IPv6 address, or an FQDN. This value is mandatory.

Note: The address must be set whenever `service config-sync` is configured. Otherwise, the commit is rejected.

Example:


```
set service config-sync secondary address 192.0.2.112
```

set service config-sync secondary port <1-65535>

Configure the TCP port on which the secondary router's HTTPS API listens.

The value must match the port configured under `service https port` on the secondary router. If they diverge, the primary's commit succeeds, but the post-commit sync fails. The secondary is not updated, and the primary logs the error to the system log.

The default is 443, matching the default of `service https port`.

Example:

```
set service config-sync secondary port 8443
```

set service config-sync secondary key <key>

Configure the HTTPS API key the primary router uses to authenticate to the secondary router's HTTPS API.

The value must match a key configured under `service https api keys id <ID> key` on the secondary. If no matching key exists on the secondary, the primary's commit still succeeds, but the post-commit sync fails. The secondary is not updated, and the primary logs the error to the system log.

Note: This value must be set whenever `service config-sync` is configured. Otherwise, the commit is rejected.

Example:

```
set service config-sync secondary key 'foo'
```

set service config-sync secondary timeout <1-3600>

Configure how long, in seconds, the primary router waits for an HTTPS response from the secondary after pushing configuration.

The default is 60 seconds.

Example:

```
set service config-sync secondary timeout 120
```

Synchronization mode

set service config-sync mode <load | set>

Configure how the secondary router applies the configuration pushed by the primary:

- **load:** Replaces the selected sections on the secondary with the primary's version. Anything that exists only on the secondary inside those sections is removed.
- **set:** Merges the primary's version of the selected sections into the secondary. Conflicting values are overwritten by the primary. Entries that exist only on the secondary are kept.

Note: This value must be set whenever `service config-sync` is configured. Otherwise, the commit is rejected.

Example:

```
set service config-sync mode load
```

Sections to synchronize

set service config-sync section <section>

Configure one or more configuration sections to be synchronized from the primary to the secondary router.

Repeat the command to add additional sections. If any configured section was modified by a commit on the primary, the hook pushes the current state of all configured sections to the secondary. If none of the configured sections were modified, the hook does nothing.

Supported sections:

- firewall
- interfaces <bonding | bridge | dummy | ethernet | geneve | input | l2tpv3 | loopback | macsec | openvpn | ppoe | pseudo-ethernet | sstp | tunnel | virtual-ethernet | vti | vxlan | wireguard | wireless | wwan>
- nat
- nat66
- pki
- policy
- protocols <babel | bfd | bgp | failover | igmp-proxy | isis | mpls | nhrp | ospf | ospfv3 | pim | pim6 | rip | ripng | rpki | segment-routing | static>
- qos <interface | policy>
- service <console-server | dhcp-relay | dhcp-server | dhcpv6-relay | dhcpv6-server | dns | lldp | mdns | monitoring | ndp-proxy | ntp | snmp | tftp-server | webproxy>
- system <conntrack | flow-accounting | login | option | sflow | static-host-mapping | sysctl | time-zone>
- vpn
- vrf

Note: At least one section must be set whenever `service config-sync` is configured. Otherwise, the commit is rejected.

Example:

```
set service config-sync section protocols ospf
set service config-sync section system time-zone
```

Operation

**show configuration secondary sync [commands] [running | candidate | saved
[<config-node-path>]]**

Show the difference between the local configuration on the primary and the running configuration on the secondary.

The command fetches the secondary's running configuration live over the HTTPS API, so the secondary must be reachable.

The local configuration defaults to **running** and can be set to **candidate** (uncommitted changes) or **saved** (last saved configuration). The remote side is always the secondary's running configuration.

By default, the difference is shown in the curly-brace configuration format with **-** prefixing lines that exist only on the secondary and **+** prefixing lines that exist only on the primary. The optional **commands** keyword renders the same difference as a sequence of **set** and **delete** configuration commands, where **set** lists paths present only on the primary, and **delete** lists paths present only on the secondary.

The optional **<config-node-path>** argument limits the difference to a single section and must be one of the sections already configured under **set service config-sync** section. If omitted, the difference covers all configured sections.

Examples:

```
# Compare the full running configuration against the secondary
show configuration secondary sync

# Compare only the OSPF protocol configuration against the secondary
show configuration secondary sync running protocols ospf

# Compare the candidate configuration against the secondary,
# rendered as commands
show configuration secondary sync commands candidate
```

Example

The following example synchronizes the **system time-zone** and **protocols ospf** sections from Router A (primary, 192.0.2.1) to Router B (secondary, 192.0.2.112). Router B's HTTPS API listens on TCP port 8443.

Enable and configure the HTTPS API service on Router B:

```
set service https listen-address '192.0.2.112'
set service https port '8443'
set service https api keys id primary key 'foo'
set service https api rest
```

Configure the config-sync service on Router A:

```
set service config-sync mode 'load'
set service config-sync secondary address '192.0.2.112'
set service config-sync secondary port '8443'
set service config-sync secondary key 'foo'
set service config-sync section protocols 'ospf'
set service config-sync section system 'time-zone'
```

Make changes on Router A that fall under one of the synchronized sections. The post-commit hook fires automatically on commit:

```
vyos@vyos-A# set system time-zone 'America/Los_Angeles'
vyos@vyos-A# commit
INFO:vyos_config_sync:Config synchronization: Mode=load, Secondary=192.0.2.112
vyos@vyos-A# save

vyos@vyos-A# set protocols ospf area 0 network '198.51.100.0/30'
vyos@vyos-A# commit
INFO:vyos_config_sync:Config synchronization: Mode=load, Secondary=192.0.2.112
vyos@vyos-A# save
```

Each commit on Router A modifies one section, but because the hook pushes all configured sections whenever any of them is modified, both commits replace the full state of `system` and `protocols ospf` on Router B.

Verify that the changes have been applied on Router B:

```
vyos@vyos-B:~$ show configuration commands | match time-zone
set system time-zone 'America/Los_Angeles'

vyos@vyos-B:~$ show configuration commands | match ospf
set protocols ospf area 0 network '198.51.100.0/30'
```

7.10.3 Conntrack Sync

One of the important features built on top of the Netfilter framework is connection tracking. Connection tracking allows the kernel to keep track of all logical network connections or sessions, and thereby relate all of the packets which may make up that connection. NAT relies on this information to translate all related packets in the same way, and iptables can use this information to act as a stateful firewall.

The connection state however is completely independent of any upper-level state, such as TCP's or SCTP's state. Part of the reason for this is that when merely forwarding packets, i.e. no local delivery, the TCP engine may not necessarily be invoked at all. Even connectionless-mode transmissions such as UDP, IPsec (AH/ESP), GRE and other tunneling protocols have, at least, a pseudo connection state. The heuristic for such protocols is often based upon a preset timeout value for inactivity, after whose expiration a Netfilter connection is dropped.

Each Netfilter connection is uniquely identified by a (layer-3 protocol, source address, destination address, layer-4 protocol, layer-4 key) tuple. The layer-4 key depends on the transport protocol; for TCP/UDP it is the port numbers, for tunnels it can be their tunnel ID, but otherwise is just zero, as if it were not part of the tuple. To be able to inspect the TCP port in all cases, packets will be mandatorily defragmented.

It is possible to use either Multicast or Unicast to sync conntrack traffic. Most examples below show Multicast, but unicast can be specified by using the “peer” keyword after the specified interface, as in the following example:

```
set service conntrack-sync interface eth0 peer 192.168.0.250
```

Configuration

set service conntrack-sync accept-protocol

Accept only certain protocols: You may want to replicate the state of flows depending on their layer 4 protocol.

Protocols are: tcp, sctp, dccp, udp, icmp and ipv6-icmp.

set service conntrack-sync event-listen-queue-size <size>

The daemon doubles the size of the netlink event socket buffer size if it detects netlink event message dropping. This clause sets the maximum buffer size growth that can be reached.

Queue size for listening to local conntrack events in MB.

set service conntrack-sync expect-sync <all|ftp|h323|nfs|sip|sqlnet>

Protocol for which expect entries need to be synchronized.

set service conntrack-sync failover-mechanism vrrp sync-group <group>

Failover mechanism to use for conntrack-sync.

Only VRRP is supported. Required option.

set service conntrack-sync ignore-address <x.x.x.x>

IP addresses or networks for which local conntrack entries will not be synced

set service conntrack-sync interface <name>

Interface to use for syncing conntrack entries.

set service conntrack-sync interface <name> port <port>

Port number used by connection.

set service conntrack-sync listen-address <ipv4address>

Local IPv4 addresses for service to listen on.

set service conntrack-sync mcast-group <x.x.x.x>

Multicast group to use for syncing conntrack entries.

Defaults to 225.0.0.50.

set service conntrack-sync interface <name> peer <address>

Peer to send unicast UDP conntrack sync entries to, if not using Multicast configuration from above.

set service conntrack-sync sync-queue-size <size>

Queue size for syncing conntrack entries in MB.

set service conntrack-sync disable-external-cache

This disables the external cache and directly injects the flow-states into the in-kernel Connection Tracking System of the backup firewall.

set service conntrack-sync purge-timeout <timeout>

Timeout (in seconds) for purging synchronized entries on handover events.

On handover, `conntrackd -t` is invoked, which schedules a conntrack table flush after `<timeout>` seconds to purge stale (“zombie”) entries and reduce clashes when multiple handovers occur in a short period. The default is 60 seconds.

Note: In VRRP stateful firewall deployments, align VRRP timing with this behavior: because synchronized conntrack state is purged after the purge timeout, set **VRRP preempt-delay** to **purge-timeout** so mastership can be restored before conntrack state is purged.

set service conntrack-sync disable-syslog

Disable connection logging via Syslog.

set service conntrack-sync startup-resync

Order conntrackd to request a complete conntrack table resync against the other node at startup.

Operation

show conntrack table <ipv4|ipv6>

Make sure conntrack is enabled by running and show connection tracking table.

```
vyos@vyos:~$ show conntrack table ipv4
```

Id	Original src	Original dst	Original packets	Original bytes	Protocol	State
→ Reply src	→ Reply dst	→ Reply packets	→ Reply bytes			
→ Timeout	→ Mark	→ Zone				
-----	-----	-----	-----	-----	-----	-----
→ -	→ -	→ -	→ -	→ -	→ -	→ -
282920088	10.0.2.2:41724	10.0.2.15:22	79	8741		
→ 10.0.2.15:22	→ 10.0.2.2:41724	→ 52	→ 8495	→ tcp	→ TIME_	→
→ WAIT 3	→ 0					
102466872	192.168.50.1	192.168.50.4	31953	2684052		
→ 192.168.50.4	→ 192.168.50.1	→ 31953	→ 2684052	→ icmp	→	→
→ 29	→ 0	→ 110				
1445684978	10.0.2.2:37762	10.0.2.15:22	111	9969		
→ 10.0.2.15:22	→ 10.0.2.2:37762	→ 86	→ 13323	→ tcp	→	→
→ ESTABLISHED	→ 431999	→ 0				
3302843234	10.0.2.2:37758	10.0.2.15:22	2612	3845685		
→ 10.0.2.15:22	→ 10.0.2.2:37758	→ 254	→ 17447	→ tcp	→ TIME_	→
→ WAIT 11	→ 0					

Note: If the table is empty and you have a warning message, it means conntrack is not enabled. To enable conntrack, just create a NAT or a firewall rule. **set firewall global-options state-policy established action accept**

show conntrack table <ipv4|ipv6> vrf <vrf-name>

Show connection tracking table filtered by VRF name. Only entries originating from the specified VRF are displayed.

show conntrack-sync cache external

Show connection syncing external cache entries

show conntrack-sync cache internal

Show connection syncing internal cache entries

show conntrack-sync statistics

Retrieve current statistics of connection tracking subsystem.

```
vyos@vyos:~$ show conntrack-sync statistics
Main Table Statistics:

cache internal:
current active connections:          19606
connections created:                 6298470    failed:          0
connections updated:                 3786793    failed:          0
connections destroyed:               6278864    failed:          0

cache external:
current active connections:          15771
connections created:                 1660193    failed:          0
connections updated:                 77204     failed:          0
connections destroyed:               1644422    failed:          0

traffic processed:
0 Bytes                               0 Pckts

multicast traffic (active device=eth0.5):
976826240 Bytes sent                 212898000 Bytes recv
8302333 Pckts sent                   2009929 Pckts recv
0 Error send                          0 Error recv

message tracking:
0 Malformed msgs                     263 Lost msgs
```

show conntrack-sync status

Retrieve current status of connection tracking subsystem.

```
vyos@vyos:~$ show conntrack-sync status
sync-interface      : eth0.5
failover-mechanism  : vrrp [sync-group GEFOEKOM]
last state transition : no transition yet!
ExpectationSync     : disabled
```

Example

The next example is a simple configuration of conntrack-sync.

Now configure conntrack-sync service on router1 and router2

```
set high-availability vrrp group internal virtual-address ... etc ...
set high-availability vrrp sync-group syncgrp member 'internal'
set service conntrack-sync accept-protocol 'tcp'
set service conntrack-sync accept-protocol 'udp'
set service conntrack-sync accept-protocol 'icmp'
set service conntrack-sync failover-mechanism vrrp sync-group 'syncgrp'
```

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```
set service conntrack-sync interface 'eth0'
set service conntrack-sync mcast-group '225.0.0.50'
```

On the active router, you should have information in the internal-cache of conntrack-sync. The same current active connections number should be shown in the external-cache of the standby router

On active router run:

```
$ show conntrack-sync statistics
```

Main Table Statistics:

cache internal:

current active connections:	10		
connections created:	8517	failed:	0
connections updated:	127	failed:	0
connections destroyed:	8507	failed:	0

cache external:

current active connections:	0		
connections created:	0	failed:	0
connections updated:	0	failed:	0
connections destroyed:	0	failed:	0

traffic processed:

0 Bytes

0 Pkts

multicast traffic (active device=eth0):

868780 Bytes sent

224136 Bytes recv

20595 Pkts sent

14034 Pkts recv

0 Error send

0 Error recv

message tracking:

0 Malformed msgs

0 Lost msgs

On standby router run:

```
$ show conntrack-sync statistics
```

Main Table Statistics:

cache internal:

current active connections:	0		
connections created:	0	failed:	0
connections updated:	0	failed:	0
connections destroyed:	0	failed:	0

cache external:

current active connections:	10		
connections created:	888	failed:	0
connections updated:	134	failed:	0
connections destroyed:	878	failed:	0

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```

traffic processed:
                0 Bytes                0 Pkts

multicast traffic (active device=eth0):
    234184 Bytes sent          907504 Bytes recv
    14663 Pkts sent           21495 Pkts recv
        0 Error send          0 Error recv

message tracking:
        0 Malformed msgs          0 Lost msgs

```

7.10.4 Console server

VyOS can serve as an OOB (Out-of-Band) management device, providing SSH-based remote access to the serial consoles of directly attached devices.

The following serial interfaces are supported:

- On-board UARTs (Universal Asynchronous Receiver-Transmitters), exposed by the Linux kernel as `/dev/ttyS<N>`.
- USB-to-serial adapters supported by the Linux USB serial driver, including Prolific PL2303 and FTDI FT232/FT4232 based chips.

You can view available devices in the Tab completion of `set service console-server device`.

See [USB](#) for more details on the naming scheme.

Configuration

A console server only works when its serial port is configured to match the framing the attached device uses. The framing consists of three parameters that the device's documentation will specify: data bits per character (typically 8, sometimes 7), parity (none, even, or odd), and stop bits (1 or 2), conventionally written together as a line, such as 8N1 (eight data bits, no parity, one stop bit).

8N1 is the default framing for most network equipment, so VyOS also defaults to it. For an 8N1 device, only the line rate (speed) must be configured. The `data-bits`, `parity`, and `stop-bits` commands below are needed only for devices that use a different framing.

```

set service console-server device <device> speed <300 | 1200 | 2400 | 4800 | 9600 | 19200
| 38400 | 57600 | 115200>

```

Configure the line rate, in baud, for the specified serial device.

Note: This setting is mandatory for each serial device. Otherwise, the commit is rejected.

Note: A serial device already in use by VyOS's system console (configured via `set system console device <device>`) cannot be configured as a console server device. Configuring the same device under both `system console` and `console-server` causes a commit failure.

Example:

```
set service console-server device usb0b2.4p1.0 speed 9600
```

```
set service console-server device <device> data-bits <7 | 8>
```

Configure the number of data bits per character for the specified serial device.

The default is 8.

Example:

```
set service console-server device usb0b2.4p1.0 data-bits 8
```

```
set service console-server device <device> stop-bits <1 | 2>
```

Configure the number of stop bits that mark the end of each character for the specified serial device.

The default is 1.

Example:

```
set service console-server device usb0b2.4p1.0 stop-bits 1
```

```
set service console-server device <device> parity <even | odd | none>
```

Configure the parity mode for the specified serial device.

The default is none.

Example:

```
set service console-server device usb0b2.4p1.0 parity none
```

```
set service console-server device <device> description <description>
```

Configure a description for the specified serial device.

Limited to 255 characters.

Example:

```
set service console-server device usb0b2.4p1.0 description 'core-sw-1 console'
```

```
set service console-server device <device> alias <alias>
```

Configure an alias for the specified serial device.

The alias can be used in place of the device name in the `connect console` operational command. Aliases must be unique across all configured serial devices and are restricted to the character class `[-_a-zA-Z0-9.]`, with a maximum length of 128 characters.

Example:

```
set service console-server device usb0b2.4p1.0 alias core-sw-1
```

Remote access

Each configured console device can be exposed for direct SSH access on a dedicated TCP port. This SSH access is served by a separate SSH instance, independent of the main `service ssh` daemon.

set service console-server device <device> ssh port <1-65535>

Configure a dedicated TCP port on which SSH access to the specified serial device is exposed.

After successful authentication, the user is connected directly to the device's serial console.

Note: Multiple users can SSH to the same serial device simultaneously, but only one can type at a time. The others see the same output in read-only mode.

Example:

```
set service console-server device usb0b2.4p1.0 ssh port 2201
```

Operation

show console-server ports

Show each configured console device together with its line rate.

```
vyos@vyos:~$ show console-server ports
usb0b2.4p1.0          on /dev/serial/by-bus/usb0b2.4p1.0@ at   9600n
```

show console-server user

Show each configured console device, its up/down state, and the user currently typing in the console, if any.

```
vyos@vyos:~$ show console-server user
usb0b2.4p1.0          up   vyos@localhost
```

connect console <device | alias>

Connect to the specified serial device's console from the VyOS CLI.

If an alias is configured for the device, it can be used in place of the device name.

Note: Multiple users can connect to the same serial device simultaneously, but only one can type at a time. The others see the same output in read-only mode.

Note: Press the keys in sequence: `Ctrl+E c ?` to list in-session commands, `Ctrl+E c .` to disconnect from the session. Both escape sequences are interpreted by the console-server client locally and are not sent to the attached device.

```
vyos@vyos-r1:~$ connect console usb0b2.4p1.0
[Enter '^Ec?' for help]
[-- MOTD -- VyOS Console Server]

vyos-r2 login:
```

show log console-server

Show the console server log since the most recent boot, in live mode.

Use Ctrl+C to exit.

7.10.5 DHCP Relay

If you want your router to forward DHCP requests to an external DHCP server you can configure the system to act as a DHCP relay agent. The DHCP relay agent works with IPv4 and IPv6 addresses.

All interfaces used for the DHCP relay must be configured. This includes the uplink to the DHCP server.

IPv4 relay**Configuration****set service dhcp-relay interface <interface>**

Interfaces that participate in the DHCP relay process. If this command is used, at least two entries of it are required: one for the interface that captures the dhcp-requests, and one for the interface to forward such requests. A warning message will be shown if this command is used, since new implementations should use `listen-interface` and `upstream-interface`.

set service dhcp-relay listen-interface <interface>

Interface for DHCP Relay Agent to listen for requests.

set service dhcp-relay upstream-interface <interface>

Interface for DHCP Relay Agent to forward requests out.

set service dhcp-relay server <server>

Configure IP address of the DHCP <server> which will handle the relayed packets.

set service dhcp-relay relay-options relay-agents-packets discard

The router should discard DHCP packages already containing relay agent information to ensure that only requests from DHCP clients are forwarded.

set service dhcp-relay disable

Disable dhcp-relay service.

Options**set service dhcp-relay relay-options hop-count <count>**

Set the maximum hop <count> before packets are discarded. Range 0...255, default 10.

set service dhcp-relay relay-options max-size <size>

Set maximum <size> of DHCP packets including relay agent information. If a DHCP packet size surpasses this value it will be forwarded without appending relay agent information. Range 64...1400, default 576.

set service dhcp-relay relay-options relay-agents-packets <append | discard | forward | replace>

Four policies for reforwarding DHCP packets exist:

- **append:** The relay agent is allowed to append its own relay information to a received DHCP packet, disregarding relay information already present in the packet.
- **discard:** Received packets which already contain relay information will be discarded.
- **forward:** All packets are forwarded, relay information already present will be ignored.
- **replace:** Relay information already present in a packet is stripped and replaced with the router's own relay information set.

Example

- Listen for DHCP requests on interface `eth1`.
- DHCP server is located at IPv4 address `10.0.1.4` on `eth2`.
- Router receives DHCP client requests on `eth1` and relays them to the server at `10.0.1.4` on `eth2`.

Fig. 3: DHCP relay example

The generated configuration will look like:

```
show service dhcp-relay
  listen-interface eth1
  upstream-interface eth2
  server 10.0.1.4
  relay-options {
    relay-agents-packets discard
  }
```

Also, for backwards compatibility this configuration, which uses generic interface definition, is still valid:

```
show service dhcp-relay
  interface eth1
  interface eth2
  server 10.0.1.4
  relay-options {
    relay-agents-packets discard
  }
```

Operation

restart dhcp relay-agent

Restart DHCP relay service

IPv6 relay

Configuration

set service dhcpv6-relay listen-interface <interface>

Set eth1 to be the listening interface for the DHCPv6 relay.

Multiple interfaces may be specified.

set service dhcpv6-relay upstream-interface <interface> address <server>

Specifies an upstream network <interface> from which replies from <server> and other relay agents will be accepted.

set service dhcpv6-relay disable

Disable dhcpv6-relay service.

Options

set service dhcpv6-relay max-hop-count <count>

Set maximum hop count before packets are discarded, default: 10

set service dhcpv6-relay use-interface-id-option

If this is set the relay agent will insert the interface ID. This option is set automatically if more than one listening interfaces are in use.

Example

- DHCPv6 requests are received by the router on listening interface eth1
- Requests are forwarded through eth2 as the upstream interface
- External DHCPv6 server is at 2001:db8::4

Fig. 4: DHCPv6 relay example

The generated configuration will look like:

```
commit
show service dhcpv6-relay
  listen-interface eth1 {
  }
  upstream-interface eth2 {
    address 2001:db8::4
  }
```

Operation

restart dhcpv6 relay-agent

Restart DHCPv6 relay agent immediately.

7.10.6 DHCP Server

VyOS uses Kea DHCP server for both IPv4 and IPv6 address assignment.

IPv4 server

The network topology is declared by shared-network-name and the subnet declarations. The DHCP service can serve multiple shared networks, with each shared network having 1 or more subnets. Each subnet must be present on an interface. A range can be declared inside a subnet to define a pool of dynamic addresses. Multiple ranges can be defined and can contain holes. Static mappings can be set to assign “static” addresses to clients based on their MAC address.

Configuration

set service dhcp-server hostfile-update

Create DNS record per client lease, by adding clients to /etc/hosts file. Entry will have format:
<shared-network-name>_<hostname>.<domain-name>

Shared network options

The following DHCP options apply to an entire shared network. All subnets inherit these values unless the same option is set locally.

set service dhcp-server shared-network-name <name> option bootfile-name <filename>

Bootstrap file name (DHCP Option 67).

set service dhcp-server shared-network-name <name> option bootfile-server <address>

Server from which the initial boot file is to be loaded (DHCP siaddr field).

set service dhcp-server shared-network-name <name> option bootfile-size <size>

Bootstrap file size in 512-octet blocks (DHCP Option 13).

set service dhcp-server shared-network-name <name> option captive-portal <url>

Captive portal API endpoint (DHCP Option 114).

set service dhcp-server shared-network-name <name> option capwap-controller <address>

IP address of CAPWAP access controller (DHCP Option 138).

set service dhcp-server shared-network-name <name> option client-prefix-length <prefix-length>

Specifies the client subnet mask as per RFC 950. If unset, the subnet declaration is used (DHCP Option 1).

set service dhcp-server shared-network-name <name> option default-router <address>

IP address of default router (DHCP Option 3).

```
set service dhcp-server shared-network-name <name> option domain-name <domain-name>
```

The domain-name parameter should be the domain name that will be appended to the client's hostname to form a fully-qualified domain-name (FQDN) (DHCP Option 15).

This is the configuration parameter for the entire shared network definition. All subnets will inherit this configuration item if not specified locally.

```
set service dhcp-server shared-network-name <name> option domain-search <domain-name>
```

The domain-name parameter should be the domain name used when completing DNS request where no full FQDN is passed. This option can be given multiple times if you need multiple search domains (DHCP Option 119).

This is the configuration parameter for the entire shared network definition. All subnets will inherit this configuration item if not specified locally.

```
set service dhcp-server shared-network-name <name> option interface-mtu <mtu>
```

Client interface MTU (DHCP Option 26).

```
set service dhcp-server shared-network-name <name> option ip-forwarding
```

Enable IP forwarding on the client (DHCP Option 19).

```
set service dhcp-server shared-network-name <name> option ipv6-only-preferred <seconds>
```

Disable IPv4 on IPv6-only hosts for the specified number of seconds (RFC 8925, DHCP Option 108).

```
set service dhcp-server shared-network-name <name> option name-server <address>
```

Inform client that the DNS server can be found at <address> (DHCP Option 6).

This is the configuration parameter for the entire shared network definition. All subnets will inherit this configuration item if not specified locally. Multiple DNS servers can be defined.

```
set service dhcp-server shared-network-name <name> option ntp-server <address>
```

IP address of NTP server (DHCP Option 42). This option can be specified multiple times.

```
set service dhcp-server shared-network-name <name> option pop-server <address>
```

IP address of POP3 server (DHCP Option 70). This option can be specified multiple times.

```
set service dhcp-server shared-network-name <name> option server-identifier <address>
```

Address for DHCP server identifier (DHCP Option 54).

```
set service dhcp-server shared-network-name <name> option smtp-server <address>
```

IP address of SMTP server (DHCP Option 69). This option can be specified multiple times.

```
set service dhcp-server shared-network-name <name> option static-route <subnet> next-hop <address>
```

Classless static route destination subnet (DHCP Options 121 and 249). This option can be specified multiple times.

```
set service dhcp-server shared-network-name <name> option tftp-server-name <server-name>
```

TFTP server name (DHCP Option 66).

```
set service dhcp-server shared-network-name <name> option time-offset <seconds>
```

Client subnet offset in seconds from Coordinated Universal Time (UTC) (DHCP Option 2).

```
set service dhcp-server shared-network-name <name> option time-server <address>
```

IP address of time server (DHCP Option 4). This option can be specified multiple times.


```
set service dhcp-server shared-network-name <name> option time-zone <timezone>
```

Time zone to send to clients (DHCP Options 100 and 101, pcode and tcode, RFC 4833).

```
set service dhcp-server shared-network-name <name> option vendor-option <vendor>  
<option-name> <value>
```

This configuration parameter lets you specify a vendor-option for the entire shared network definition. All subnets will inherit this configuration item if not specified locally (DHCP Option 43). An example for Ubiquiti is shown below:

Example:

Pass address of Unifi controller at 172.16.100.1 to all clients of NET1

```
set service dhcp-server shared-network-name 'NET1' option vendor-option  
ubiquiti unifi-controller '172.16.100.1'
```

```
set service dhcp-server shared-network-name <name> option wins-server <address>
```

IP address for Windows Internet Name Service (WINS) server (DHCP Option 44). This option can be specified multiple times.

```
set service dhcp-server shared-network-name <name> option wpad-url <url>
```

Web Proxy Autodiscovery (WPAD) URL (DHCP Option 252).

```
set service dhcp-server listen-address <address>
```

This configuration parameter lets the DHCP server to listen for DHCP requests sent to the specified address, it is only realistically useful for a server whose only clients are reached via unicasts, such as via DHCP relay agents.

```
set service dhcp-server log-level <fatal | error | warn | info | debug>
```

Set the logging verbosity of the Kea DHCP server. The default level is info.

Individual Client Subnet

```
set service dhcp-server shared-network-name <name> authoritative
```

This says that this device is the only DHCP server for this network. If other devices are trying to offer DHCP leases, this machine will send 'DHCPNAK' to any device trying to request an IP address that is not valid for this network.

```
set service dhcp-server shared-network-name <name> subnet <subnet> subnet-id <id>
```

This configuration parameter is required and must be unique to each subnet. It is required to map subnets to lease file entries.

```
set service dhcp-server shared-network-name <name> subnet <subnet> option default-router  
<address>
```

This is a configuration parameter for the <subnet>, saying that as part of the response, tell the client that the default gateway can be reached at <address> (DHCP Option 3).

```
set service dhcp-server shared-network-name <name> subnet <subnet> option name-server  
<address>
```

This is a configuration parameter for the subnet, saying that as part of the response, tell the client that the DNS server can be found at <address> (DHCP Option 6).

Multiple DNS servers can be defined.

```
set service dhcp-server shared-network-name <name> subnet <subnet> lease <time>
```

Assign the IP address to this machine for <time> seconds.

The default value is 86400 seconds which corresponds to one day.

```
set service dhcp-server shared-network-name <name> subnet <subnet> range <n> start  
<address>
```

Create DHCP address range with a range id of <n>. DHCP leases are taken from this pool. The pool starts at address <address>.

```
set service dhcp-server shared-network-name <name> subnet <subnet> range <n> stop  
<address>
```

Create DHCP address range with a range id of <n>. DHCP leases are taken from this pool. The pool stops with address <address>.

```
set service dhcp-server shared-network-name <name> subnet <subnet> exclude <address>
```

Always exclude this address from any defined range. This address will never be assigned by the DHCP server.

This option can be specified multiple times.

```
set service dhcp-server shared-network-name <name> subnet <subnet> option domain-name  
<domain-name>
```

The domain-name parameter should be the domain name that will be appended to the client's hostname to form a fully-qualified domain-name (FQDN) (DHCP Option 15).

```
set service dhcp-server shared-network-name <name> subnet <subnet> option domain-search  
<domain-name>
```

The domain-name parameter should be the domain name used when completing DNS request where no full FQDN is passed. This option can be given multiple times if you need multiple search domains (DHCP Option 119).

```
set service dhcp-server shared-network-name <name> subnet <subnet> option vendor-option  
<vendor> <option-name> <value>
```

This configuration parameter lets you specify a vendor-option for the subnet specified within the shared network definition (DHCP Option 43). An example for Ubiquiti is shown below:

Example:

Create 172.18.201.0/24 as a subnet within NET1 and pass address of Unifi controller at 172.16.100.1 to clients of that subnet.

```
set service dhcp-server shared-network-name 'NET1' subnet  
'172.18.201.0/24' option vendor-option ubiquiti unifi-controller '172.16.100.1'
```

Dynamic DNS Update (RFC 2136)

VyOS DHCP service supports RFC-2136 DDNS protocol. Based on DHCP lease change events, DHCP server generates DDNS update requests (defines as NameChangeRequests or NCRs) and posts them to a compliant DNS server, that will update its name database accordingly.

VyOS built-in DNS Forwarder does not support DDNS, you will need an external DNS server with RFC-2136 DDNS support.

```
set service dhcp-server dynamic-dns-update
```

Enables DDNS globally.

Behavioral settings

These settings can be configured on the global level and overridden on the scope level, i.e. for individual shared networks or subnets. See examples below.

set service dhcp-server dynamic-dns-update send-updates <enable | disable>

If set to **enable** on global level, updates for all scopes will be enabled, except if explicitly set to **disable** on the scope level. If set to **disable**, updates will only be sent for scopes, where **send-updates** is explicitly set to **enable**.

This model is followed for a few behavioral settings below: if the option is not set, the setting is inherited from the parent scope. You can override the parent scope setting by setting the option explicitly.

set service dhcp-server dynamic-dns-update override-no-update <enable | disable>

VyOS will ignore client request not to update DNS records and send DDNS update requests regardless.

set service dhcp-server dynamic-dns-update override-client-update <enable | disable>

VyOS will override client DDNS request settings and always update both forward and reverse DNS records.

set service dhcp-server dynamic-dns-update update-on-renew <enable | disable>

Issue DDNS update requests on DHCP lease renew. In busy networks this may generate a lot of traffic.

set service dhcp-server dynamic-dns-update conflict-resolution <enable | disable>

Use RFC-4703 conflict resolution. This algorithm helps in situation when multiple clients reserve same IP addresses or advertise identical hostnames. Should be used in most situations.

set service dhcp-server dynamic-dns-update replace-client-name [never | always | when-present | when-not-present]

- **never**: use the name sent by the client. If the client didn't provide any, do not generate one. This is the default behavior
- **always**: always generate a name for the client
- **when-present**: replace the name the client sent with a generated one, if the client didn't send any, do not generate one
- **when-not-present**: use the name sent by the client. If the client didn't send any, generate one for the client

The names are generated using **generated-prefix**, **qualifying-suffix** and the client's IP address string.

set service dhcp-server dynamic-dns-update generated-prefix <prefix>

Prefix used in client name generation.

set service dhcp-server dynamic-dns-update qualifying-suffix <suffix>

DNS suffix used in client name generation.

set service dhcp-server dynamic-dns-update ttl-percent <0-100>

TTL of the DNS record as a percentage of the DHCP lease time.

set service dhcp-server dynamic-dns-update hostname-char-set <character string>

Characters, that are considered invalid in the client name. They will be replaced with **hostname-char-replacement** string.

set service dhcp-server dynamic-dns-update hostname-char-replacement <character string>

Replacement string for the invalid characters defined by **hostname-char-set**.

TSIG keys definition

This is the global list of TSIG keys for DDNS updates. They need to be specified by the name in the DNS domain definitions.

```
set service dhcp-server dynamic-dns-update tsig-key <key-name> algorithm <algorithm>
```

Sets the algorithm for the TSIG key. Supported algorithms are hmac-md5, hmac-sha1, hmac-sha224, hmac-sha256, hmac-sha384, hmac-sha512

```
set service dhcp-server dynamic-dns-update tsig-key <key-name> secret <key-secret>
```

base64-encoded TSIG key secret value

DNS domains definition

This is global configuration of DNS servers for the updatable forward and reverse DNS domains. For every domain multiple DNS servers can be specified.

```
set service dhcp-server dynamic-dns-update [forward|reverse]-domain <domain-name>  
key-name <tsig-key-name>
```

TSIG key used for the domain.

```
set service dhcp-server dynamic-dns-update [forward|reverse]-domain <domain-name>  
dns-server <number> address <ip-address>
```

IP address of the DNS server.

```
set service dhcp-server dynamic-dns-update [forward|reverse]-domain <domain-name>  
dns-server <number> port <port>
```

UDP port of the DNS server. 53 is the default.

Example:

Global configuration you will most likely want:

```
set service dhcp-server dynamic-dns-update send-updates enable  
set service dhcp-server dynamic-dns-update conflict-resolution enable
```

Override the above configuration for a shared network NET1:

```
set service dhcp-server shared-network-name 'NET1' dynamic-dns-update replace-client-  
↪ name when-not-present  
set service dhcp-server shared-network-name 'NET1' dynamic-dns-update generated-prefix ip  
set service dhcp-server shared-network-name 'NET1' dynamic-dns-update qualifying-suffix,↪  
↪ mybigdomain.net
```

And in a subnet within the same shared network:

```
set service dhcp-server shared-network-name 'NET1' subnet '172.18.201.0/24' dynamic-dns-  
↪ update qualifying-suffix mydomain.net
```

Configure TSIG keys:

```
set service dhcp-server dynamic-dns-update tsig-key mydomain-net algorithm hmac-sha256  
set service dhcp-server dynamic-dns-update tsig-key mydomain-net secret,↪  
↪ eWF5YW15bGl0dGxla2V5IQ==  
set service dhcp-server dynamic-dns-update tsig-key reverse-172-18-201 algorithm hmac-  
↪ sha256
```

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```
set service dhcp-server dynamic-dns-update tsig-key reverse-172-18-201 secret_
↳ eWF5YW15YW5vdGhlcmxpdHRsZWtleSE=
```

Configure DDNS domains:

```
set service dhcp-server dynamic-dns-update forward-domain mydomain.net key-name mydomain-
↳ net
set service dhcp-server dynamic-dns-update forward-domain mydomain.net dns-server 1_
↳ address '172.18.0.254'
set service dhcp-server dynamic-dns-update forward-domain mydomain.net dns-server 1 port_
↳ 1053
set service dhcp-server dynamic-dns-update forward-domain mydomain.net dns-server 2_
↳ address '192.168.124.254'
set service dhcp-server dynamic-dns-update forward-domain mydomain.net dns-server 2 port_
↳ 53
set service dhcp-server dynamic-dns-update forward-domain 201.18.172.in-addr.arpa key-
↳ name reverse-172-18-201
set service dhcp-server dynamic-dns-update reverse-domain 201.18.172.in-addr.arpa dns-
↳ server 1 address '172.18.0.254'
set service dhcp-server dynamic-dns-update reverse-domain 201.18.172.in-addr.arpa dns-
↳ server 1 port 1053
set service dhcp-server dynamic-dns-update reverse-domain 201.18.172.in-addr.arpa dns-
↳ server 2 address '192.168.124.254'
set service dhcp-server dynamic-dns-update reverse-domain 201.18.172.in-addr.arpa dns-
↳ server 2 port 53
```

High Availability

VyOS provides High Availability support for DHCP server. DHCP High Availability can act in two different modes:

- **Active-active:** both DHCP servers will respond to DHCP requests. If mode is not defined, this is the default behavior.
- **Active-passive:** only primary server will respond to DHCP requests. If this server goes offline, then secondary server will take place.

DHCP High Availability must be configured explicitly by the following statements on both servers:

```
set service dhcp-server high-availability mode [active-active | active-passive]
```

Define operation mode of High Availability feature. Default value if command is not specified is active-active

```
set service dhcp-server high-availability source-address <address>
```

Local IP <address> used when communicating to the HA peer.

```
set service dhcp-server high-availability remote <address>
```

Remote peer IP <address> of the second DHCP server in this HA cluster.

```
set service dhcp-server high-availability name <name>
```

Define the name of the peer server to establish and identify the HA (High Availability) connection.

Note: Make sure the specified value does not conflict with the system host-name.

set service dhcp-server high-availability status <primary | secondary>

The primary and secondary statements determines whether the server is primary or secondary.

Note: In order for the primary and the secondary DHCP server to keep their lease tables in sync, they must be able to reach each other on TCP port 647. If you have firewall rules in effect, adjust them accordingly.

Hint: The dialogue between HA partners is neither encrypted nor authenticated. Since most DHCP servers exist within an organisation's own secure Intranet, this would be an unnecessary overhead. However, if you have DHCP HA peers whose communications traverse insecure networks, then we recommend that you consider the use of VPN tunneling between them to ensure that the HA partnership is immune to disruption (accidental or otherwise) via third parties.

Static mappings

You can specify a static DHCP assignment on a per-host basis (DHCP Reservations). You will need the MAC address of the station and your desired IP address. The address must be inside the subnet definition but can be outside the range statement.

set service dhcp-server shared-network-name <name> subnet <subnet> static-mapping <hostname> mac <address>

Create a new DHCP static mapping named <hostname> which is valid for the host identified by its MAC <address>.

set service dhcp-server shared-network-name <name> subnet <subnet> static-mapping <hostname> duid <identifier>

Create a new DHCP static mapping named <hostname> which is valid for the host identified by its DHCP unique identifier (DUID) <identifier>.

set service dhcp-server shared-network-name <name> subnet <subnet> static-mapping <hostname> ip-address <address>

Static DHCP IP address assigned to the host identified by <hostname>. IP address must be inside the <subnet> which is defined but can be outside the dynamic range created with {cfgcmd} **set service dhcp-server shared-network-name <name> subnet <subnet> range <n>**. If no ip-address is specified, an IP address from the dynamic pool is used.

This is useful, for example, in combination with hostfile update.

Hint: This is the equivalent of the host block in dhcpd.conf of isc-dhcpd.

Example:

- IP address 192.168.1.100 shall be statically mapped to client named client1

```
set service dhcp-server shared-network-name 'NET1' subnet 192.168.1.0/24 subnet-id 1
set service dhcp-server shared-network-name 'NET1' subnet 192.168.1.0/24 static-mapping_
```

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```

↪client1 ip-address 192.168.1.100
set service dhcp-server shared-network-name 'NET1' subnet 192.168.1.0/24 static-mapping_
↪client1 mac aa:bb:11:22:33:00

```

The configuration will look as follows:

```

show service dhcp-server shared-network-name NET1
  subnet 192.168.1.0/24 {
    static-mapping client1 {
      ip-address 192.168.1.100
      mac aa:bb:11:22:33:00
    }
    subnet-id 1
  }

```

Relay agent information (Option 82)

Some DHCP relays support the injection of information into a DHCP request, depending on where the request originated from. This is commonly used to determine the behaviour of the DHCP server, based on the port/switch combination where the request was first detected. I.e. the device plugged into a particular port (or set of ports) always gets the same IP address (or range of IP addresses). This information is usually included in the request using Option 82, hence this is what we call this part of the configuration.

This behaviour is controlled in two parts. First, “client classes” are defined which determine which inputs match. Once a positive match has been found the request is “tagged” with this client class. Second, when the DHCP server processes the request it checks to see if the configuration has a client class defined. If it does then that part of the configuration will override the others

Client classes can be applied at either the subnet or range level, depending on how you want the server to behave.

Client Class definition

set service dhcp-server client-class <name> relay-agent-information circuit-id <value>

Create a new client class (if not already defined) and set it to match on the “Circuit ID” part of the Option 82 field in the DHCP request. This is sub option “1” as specified by RFC 3046. The value specified here is either interpreted as a raw hex value, if it starts with the prefix 0x, or ASCII text otherwise. e.g. e1-5 and 0x65312d35 are the same

set service dhcp-server client-class <name> relay-agent-information remote-id <value>

Create a new client class (if not already defined) and set it to match on the “Remote ID” part of the Option 82 field in the DHCP request. This is sub option “2” as specified by RFC 3046. The value specified here is either interpreted as a raw hex value, if it starts with the prefix 0x, or ASCII text otherwise. e.g. 10.100.0.41 and 0x31302e3130302e302e3431 are the same

Client Class application

set service dhcp-server shared-network-name <subnet-name> subnet <CIDR> client-class <class-name>

Applies the Client Class with the name <class-name> to the subnet <subnet-name>. This means that whenever the client class matches a request it is always routed to this subnet definition first.

set service dhcp-server shared-network-name <subnet-name> subnet <CIDR> range <range-name> client-class <class-name>

Applies the Client Class with the name <class-name> to the range <range-name> which belongs to subnet <subnet-name>. This means that whenever the client class matches a request it is always routed to this range definition first.

NB: Kea (the DHCP server used by VyOS) is programmed to offer as many alternatives as it can to repeated DHCP Discover requests. Some operating systems (Notably Microsoft Windows) make multiple DHCP Discover requests before settling on an address. This particularly seems to happen when the DHCP server isn't set to authoritative. This may explain why the address you expect isn't being chosen. Wireshark is helpful in these situations.

Example:

The following configuration example will classify requests coming in on port e1-5 from DHCP Relay 192.0.2.1 and make sure that they are allocated the address 192.0.2.4. Any requests which do not match the circuit and remote ID will, instead, be allocated from the range otherRange in the usual manner.

NB: Both the Circuit ID and Remote ID fields are arbitrary free text. *Most* switches set the Remote ID to the IP address of the management interface but that should not be relied upon. Check the documentation of your DHCP Relay for more detail or, as a measure of last resort, inspect the DHCP requests in Wireshark.

```
service {
  dhcp-server {
    client-class className {
      relay-agent-information {
        circuit-id e1-5
        remote-id 192.0.2.1
      }
    }
    shared-network-name test {
      subnet 192.0.2.0/24 {
        range classNameRange {
          client-class className
          start 192.0.2.4
          stop 192.0.2.4
        }
        range otherRange {
          start 192.0.2.5
          stop 192.0.2.100
        }
      }
      subnet-id 1
    }
  }
}
```

Options

The following DHCP options can be set under `set service dhcp-server shared-network-name <name> option ...` or `set service dhcp-server shared-network-name <name> subnet <subnet> option ...`.

Setting name	Option number	DHCP option name	Option description	Multi
bootfile-name	67	boot-file-name	Bootstrap file name	N
bootfile-server	siaddr	next-server	Server from which the initial boot file is to be loaded	N
bootfile-size	13	boot-size	Bootstrap file size	N
captive-portal	114	v4-captive-portal	Captive portal API endpoint	N
capwap-controller	138	capwap-ac-v4	IPv4 address of CAPWAP access controller	N
client-prefix-length	1	subnet-mask	Specifies the clients subnet mask as per RFC 950. If unset, subnet declaration is used.	N
default-router	3	routers	IPv4 address of router on the client's subnet	N
domain-name	15	domain-name	Client domain name	N
domain-search	119	domain-search	Client domain name search list	Y
interface-mtu	26	interface-mtu	Client interface MTU	N
ip-forwarding	19	ip-forwarding	Enable IP forwarding on client	N
ipv6-only-preferred	108	v6-only-preferred	Disable IPv4 on IPv6-only hosts (RFC 8925)	N
name-server	6	domain-name-servers	Domain Name Servers (DNS) addresses	Y
ntp-server	42	ntp-servers	IPv4 address of NTP server	Y
pop-server	70	pop-server	IPv4 address of POP3 server	Y
server-identifier	54	dhcp-server-identifier	IPv4 address for DHCP server identifier	N
smtp-server	69	smtp-server	IPv4 address of SMTP server	Y
static-route	121, 249	classless-static-route, ms-classless-static-route	Classless static route destination subnet	Y
tftp-server-name	66	tftp-server-name	TFTP server name	N
time-offset	2	time-offset	Client subnet offset in seconds from Coordinated Universal Time (UTC)	N
time-server	4	time-servers	IPv4 address of time server	Y
time-zone	100, 101	pcode, tcode	Time zone to send to clients (RFC 4833)	N
vendor-option	43	vendor-encapsulated-options	Vendor-specific options	Y
wins-server	44	netbios-name-servers	IPv4 address for Windows Internet Name Service (WINS) server	Y
wpad-url	252	wpad-url	Web Proxy Autodiscovery (WPAD) URL	N

Multi: can be specified multiple times.

Example

Please see the *DHCP/DNS quick-start* configuration.

High Availability

Configuration of a DHCP HA pair:

- Setup DHCP HA for network 192.0.2.0/24
- Use active-active HA mode.
- Default gateway and DNS server is at 192.0.2.254
- The primary DHCP server named dhcp-primary uses address 192.168.189.252
- The secondary DHCP server with named dhcp-secondary uses address 192.168.189.253
- DHCP range spans from 192.168.189.10 - 192.168.189.250

Common configuration, valid for both primary and secondary node.

```
set service dhcp-server shared-network-name NET-VYOS subnet 192.0.2.0/24 option default-
↪router '192.0.2.254'
set service dhcp-server shared-network-name NET-VYOS subnet 192.0.2.0/24 option name-
↪server '192.0.2.254'
set service dhcp-server shared-network-name NET-VYOS subnet 192.0.2.0/24 option domain-
↪name 'vyos.net'
set service dhcp-server shared-network-name NET-VYOS subnet 192.0.2.0/24 range 0 start
↪'192.0.2.10'
set service dhcp-server shared-network-name NET-VYOS subnet 192.0.2.0/24 range 0 stop
↪'192.0.2.250'
set service dhcp-server shared-network-name NET-VYOS subnet 192.0.2.0/24 subnet-id '1'
```

Primary

```
set service dhcp-server high-availability mode 'active-active'
set service dhcp-server high-availability source-address '192.168.189.252'
set service dhcp-server high-availability name 'dhcp-secondary'
set service dhcp-server high-availability remote '192.168.189.253'
set service dhcp-server high-availability status 'primary'
```

Secondary

```
set service dhcp-server high-availability mode 'active-active'
set service dhcp-server high-availability source-address '192.168.189.253'
set service dhcp-server high-availability name 'dhcp-primary'
set service dhcp-server high-availability remote '192.168.189.252'
set service dhcp-server high-availability status 'secondary'
```

Operation Mode

show log dhcp server

Show DHCP server daemon log file

show log dhcp client

Show logs from all DHCP client processes.

show log dhcp client interface <interface>

Show logs from specific interface DHCP client process.

restart dhcp server

Restart the DHCP server

show dhcp server statistics

Show the DHCP server statistics:

```
vyos@vyos:~$ show dhcp server statistics
Pool           Size    Leases    Available    Usage
-----
dhcpexample    99      2         97          2%
```

show dhcp server statistics pool <pool>

Show the DHCP server statistics for the specified pool.

show dhcp server leases

Show statuses of all active leases:

```
vyos@vyos:~$ show dhcp server leases
IP Address      MAC address      State    Lease start      Lease expiration
↳ Remaining    Pool            Hostname    Origin
-----
↳
192.168.11.134  00:50:79:66:68:09 active    2023/11/29 09:51:05 2023/11/29 10:21:05 ↳
↳ 0:24:10      LAN            VPCS1      local
192.168.11.133  50:00:00:06:00:00 active    2023/11/29 09:51:38 2023/11/29 10:21:38 ↳
↳ 0:24:43      LAN            VYOS-6     local
10.11.11.108    50:00:00:05:00:00 active    2023/11/29 09:51:43 2023/11/29 10:21:43 ↳
↳ 0:24:48      VIF-1001      VYOS5      local
192.168.11.135  00:50:79:66:68:07 active    2023/11/29 09:55:16 2023/11/29 09:59:16 ↳
↳ 0:02:21                      remote
vyos@vyos:~$
```

Hint: Static mappings aren't shown. To show all states, use `show dhcp server leases state all`.

show dhcp server leases origin [local | remote]

Show statuses of all active leases granted by local (this server) or remote (failover server):

```
vyos@vyos:~$ show dhcp server leases origin remote
IP Address      MAC address      State    Lease start      Lease expiration ↳
```

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↪ Remaining	Pool	Hostname	Origin			
↪ -----	-----	-----	-----	-----	-----	-----
192.168.11.135	00:50:79:66:68:07	active	2023/11/29 09:55:16	2023/11/29 09:59:16	↪	
↪ 0:02:21		remote				

vyos@vyos:~\$

show dhcp server leases pool <pool>

Show only leases in the specified pool.

```
vyos@vyos:~$ show dhcp server leases pool LAN
```

IP Address	MAC address	State	Lease start	Lease expiration	↪
↪ Remaining	Pool	Hostname	Origin		
↪ -----	-----	-----	-----	-----	-----
192.168.11.134	00:50:79:66:68:09	active	2023/11/29 09:51:05	2023/11/29 10:21:05	↪
↪ 0:23:55	LAN	VPCS1	local		
192.168.11.133	50:00:00:06:00:00	active	2023/11/29 09:51:38	2023/11/29 10:21:38	↪
↪ 0:24:28	LAN	VYOS-6	local		

vyos@vyos:~\$

show dhcp server leases sort <key>

Sort the output by the specified key. Possible keys: ip, hardware_address, state, start, end, remaining, pool, hostname (default = ip)

show dhcp server leases state <state>

Show only leases with the specified state. Possible states: all, active, free, expired, released, abandoned, reset, backup (default = active)

IPv6 server

The network topology is declared by shared-network-name and subnet declarations. The DHCPv6 service can serve multiple shared networks, with each shared network having one or more subnets. Each subnet must be present on an interface. A range can be declared inside a subnet to define a pool of dynamic addresses. Prefix delegation and static mappings can assign prefixes or fixed addresses to clients based on their DUID.

Configuration

set service dhcpv6-server preference <preference value>

Clients receiving advertise messages from multiple servers choose the server with the highest preference value. The range for this value is 0...255.

set service dhcpv6-server log-level <fatal | error | warn | info | debug>

Set the logging verbosity of the Kea DHCPv6 server. The default level is info.

Shared network options

The following DHCPv6 options apply to an entire shared network. All subnets inherit these values unless the same option is set locally.

set service dhcpv6-server shared-network-name <name> option captive-portal <url>

Captive portal API endpoint (DHCPv6 Option 103).

set service dhcpv6-server shared-network-name <name> option capwap-controller <address>

IP address of CAPWAP access controller (DHCPv6 Option 52).

set service dhcpv6-server shared-network-name <name> option domain-search <domain-name>

Domain name used when completing DNS requests where no full FQDN is passed. This option can be given multiple times if you need multiple search domains (DHCPv6 Option 24).

set service dhcpv6-server shared-network-name <name> option info-refresh-time <seconds>

Time in seconds that stateless clients should wait between refreshing the information they were given (DHCPv6 Option 32).

set service dhcpv6-server shared-network-name <name> option name-server <address>

Inform the client that the DNS server can be found at <address> (DHCPv6 Option 23). Multiple DNS servers can be defined.

set service dhcpv6-server shared-network-name <name> option nis-domain <domain-name>

A NIS (Network Information Service) domain name for clients to use (DHCPv6 Option 29).

set service dhcpv6-server shared-network-name <name> option nis-server <address>

IPv6 address of a NIS server (DHCPv6 Option 27). This option can be specified multiple times.

set service dhcpv6-server shared-network-name <name> option nisplus-domain <domain-name>

A NIS+ (Network Information Service Plus) domain name for clients to use (DHCPv6 Option 30).

set service dhcpv6-server shared-network-name <name> option nisplus-server <address>

IPv6 address of a NIS+ server (DHCPv6 Option 28). This option can be specified multiple times.

set service dhcpv6-server shared-network-name <name> option sip-server <address>

IPv6 address of SIP (Session Initiation Protocol) server (DHCPv6 Option 21 and 22, sip-server-dns and sip-server-addr). This option can be specified multiple times.

set service dhcpv6-server shared-network-name <name> option sntp-server <address>

IPv6 address of an SNTP server for clients to use (DHCPv6 Option 31). This option can be specified multiple times.

set service dhcpv6-server shared-network-name <name> option time-zone <timezone>

Time zone to send to clients (DHCPv6 Options 41 and 42, new-posix-timezone and new-tzdb-timezone, RFC 4833).

**set service dhcpv6-server shared-network-name <name> option vendor-option <vendor>
<option-name> <value>**

Specify a vendor-specific DHCPv6 option for the shared network (DHCPv6 Option 17). This option can be specified multiple times.

Individual Client Subnet

```
set service dhcpv6-server shared-network-name <name> interface <interface>
```

Bind shared network <name> to <interface>.

```
set service dhcpv6-server shared-network-name <name> subnet <prefix> subnet-id <id>
```

This configuration parameter is required and must be unique to each subnet. It is required to map subnets to lease file entries.

```
set service dhcpv6-server shared-network-name <name> subnet <prefix> lease-time {default  
| maximum | minimum} <seconds>
```

The default lease time for DHCPv6 leases is 24 hours. This can be changed by supplying a default, maximum and minimum. All values need to be supplied in seconds.

```
set service dhcpv6-server shared-network-name <name> subnet <prefix> range <n> start  
<address>
```

Create DHCPv6 address range with a range id of <n>. DHCPv6 leases are taken from this pool. The pool starts at address <address>.

```
set service dhcpv6-server shared-network-name <name> subnet <prefix> range <n> stop  
<address>
```

Create DHCPv6 address range with a range id of <n>. DHCPv6 leases are taken from this pool. The pool stops at address <address>.

Prefix Delegation

To hand out individual prefixes to your clients the following configuration is used:

```
set service dhcpv6-server shared-network-name <name> subnet <prefix> prefix-delegation  
prefix <pd-prefix> prefix-length <length>
```

Delegate prefixes from <pd-prefix> to clients in subnet <prefix>. Range is defined by <length> in bits, 32 to 64.

```
set service dhcpv6-server shared-network-name <name> subnet <prefix> prefix-delegation  
prefix <pd-prefix> delegated-length <length>
```

Hand out prefixes of size <length> in bits from <pd-prefix> to clients in subnet <prefix> when the request for prefix delegation.

```
set service dhcpv6-server shared-network-name <name> subnet <prefix> prefix-delegation  
prefix <pd-prefix> excluded-prefix <exclude-prefix>
```

Exclude <exclude-prefix> from <pd-prefix>.

```
set service dhcpv6-server shared-network-name <name> subnet <prefix> prefix-delegation  
prefix <pd-prefix> excluded-prefix-length <length>
```

Define length of exclude prefix in <pd-prefix>.

Example:

- A shared network named PD-NET serves subnet 2001:db8::/64.
- It is connected to eth1.
- Address pool shall be 2001:db8::100 through 2001:db8::199.
- It hands out prefixes 2001:db8:0:10::/64 through 2001:db8:0:1f::/64.

```

set service dhcpv6-server shared-network-name 'PD-NET' interface 'eth1'
set service dhcpv6-server shared-network-name 'PD-NET' subnet 2001:db8::/64 range 1
↪start 2001:db8::100
set service dhcpv6-server shared-network-name 'PD-NET' subnet 2001:db8::/64 range 1 stop
↪2001:db8::199
set service dhcpv6-server shared-network-name 'PD-NET' subnet 2001:db8::/64 prefix-
↪delegation prefix 2001:db8:0:10:: delegated-length '64'
set service dhcpv6-server shared-network-name 'PD-NET' subnet 2001:db8::/64 prefix-
↪delegation prefix 2001:db8:0:10:: prefix-length '60'
set service dhcpv6-server shared-network-name 'PD-NET' subnet 2001:db8::/64 subnet-id 1

```

Address pools

DHCPv6 address pools must be configured for the system to act as a DHCPv6 server. The following example describes a common scenario.

Example:

- A shared network named NET1 serves subnet 2001:db8::/64
- It is connected to eth1
- DNS server is located at 2001:db8::ffff
- Address pool shall be 2001:db8::100 through 2001:db8::199.
- Lease time will be left at the default value which is 24 hours

```

set service dhcpv6-server shared-network-name 'NET' interface 'eth1'
set service dhcpv6-server shared-network-name 'NET1' subnet 2001:db8::/64 range 1 start
↪2001:db8::100
set service dhcpv6-server shared-network-name 'NET1' subnet 2001:db8::/64 range 1 stop
↪2001:db8::199
set service dhcpv6-server shared-network-name 'NET1' subnet 2001:db8::/64 option name-
↪server 2001:db8::ffff
set service dhcpv6-server shared-network-name 'NET1' subnet 2001:db8::/64 subnet-id 1

```

The configuration will look as follows:

```

show service dhcpv6-server
  shared-network-name NET1 {
    subnet 2001:db8::/64 {
      range 1 {
        start 2001:db8::100
        stop 2001:db8::199
      }
      option {
        name-server 2001:db8::ffff
      }
      subnet-id 1
    }
  }

```

Static mappings

In order to map specific IPv6 addresses to specific hosts static mappings can be created (DHCPv6 Reservations). The following example explains the process.

```
set service dhcpv6-server shared-network-name <name> subnet <prefix> static-mapping
<hostname> duid <identifier>
```

Create a new DHCPv6 static mapping named <hostname> which is valid for the host identified by its DHCP unique identifier (DUID) <identifier>.

```
set service dhcpv6-server shared-network-name <name> subnet <prefix> static-mapping
<hostname> ipv6-address <address>
```

Static IPv6 address assigned to the host identified by <hostname>. This option can be specified multiple times.

```
set service dhcpv6-server shared-network-name <name> subnet <prefix> static-mapping
<hostname> ipv6-prefix <delegated-prefix>
```

Static IPv6 prefix assigned to the host identified by <hostname>.

Example:

- IPv6 address 2001:db8::101 shall be statically mapped
- IPv6 prefix 2001:db8:0:101::/64 shall be statically mapped
- Host specific mapping shall be named client1

Hint: The identifier is the device's DUID: colon-separated hex list (as used by isc-dhcp option dhcpv6.client-id). If the device already has a dynamic lease from the DHCPv6 server, its DUID can be found with `show service dhcpv6 server leases`. The DUID begins at the 5th octet (after the 4th colon) of IAID_DUID.

```
set service dhcpv6-server shared-network-name 'NET1' subnet 2001:db8::/64 static-mapping_
↪client1 ipv6-address 2001:db8::101
set service dhcpv6-server shared-network-name 'NET1' subnet 2001:db8::/64 static-mapping_
↪client1 ipv6-prefix 2001:db8:0:101::/64
set service dhcpv6-server shared-network-name 'NET1' subnet 2001:db8::/64 static-mapping_
↪client1 duid 00:01:00:01:12:34:56:78:aa:bb:cc:dd:ee:ff
```

The configuration will look as follows:

```
show service dhcpv6-server shared-network-name NET1
  subnet 2001:db8::/64 {
    static-mapping client1 {
      duid 00:01:00:01:12:34:56:78:aa:bb:cc:dd:ee:ff
      ipv6-address 2001:db8::101
      ipv6-prefix 2001:db8:0:101::/64
    }
  }
```


Options

The following DHCPv6 options can be set under `set service dhcpv6-server shared-network-name <name> option ...` or under `subnet <prefix> option ...` within the same shared network.

Setting name	Option number	DHCP option name	Option description	Multi
captive-portal	103	v6-captive-portal	Captive portal API endpoint	N
capwap-controller	52	capwap-ac-v6	IPv6 address of CAPWAP access controller	N
domain-search	24	domain-search	Client domain name search list	Y
info-refresh-time	32	information-refresh-time	Time in seconds that stateless clients should wait between refreshing information	N
name-server	23	dns-servers	Domain Name Servers (DNS) addresses	Y
nis-domain	29	nis-domain-name	NIS domain name for client to use	N
nis-server	27	nis-servers	IPv6 address of a NIS server	Y
nisplus-domain	30	nisp-domain-name	NIS+ domain name for client to use	N
nisplus-server	28	nisp-servers	IPv6 address of a NIS+ server	Y
sip-server	21, 22	sip-server-dns, sip-server-addr	IPv6 address of SIP server	Y
sntp-server	31	sntp-servers	IPv6 address of an SNTP server for client to use	Y
time-zone	41, 42	new-posix-timezone, new-tzdb-timezone	Time zone to send to clients (RFC 4833)	N
vendor-option	17	vendor-opts	Vendor-specific options	Y

Multi: can be specified multiple times.

Example

DHCPv6 address pools must be configured for the system to act as a DHCPv6 server. The following example describes a common scenario.

- A shared network named NET1 serves subnet `2001:db8::/64`
- It is connected to `eth1`
- DNS server is located at `2001:db8::ffff`
- Address pool shall be `2001:db8::100` through `2001:db8::199`.
- Lease time will be left at the default value which is 24 hours

```

set service dhcpv6-server shared-network-name 'NET1' interface 'eth1'
set service dhcpv6-server shared-network-name 'NET1' subnet 2001:db8::/64 range 1 start_
↪ 2001:db8::100
set service dhcpv6-server shared-network-name 'NET1' subnet 2001:db8::/64 range 1 stop_
↪ 2001:db8::199
set service dhcpv6-server shared-network-name 'NET1' subnet 2001:db8::/64 option name-
↪ server 2001:db8::ffff
set service dhcpv6-server shared-network-name 'NET1' subnet 2001:db8::/64 subnet-id 1

```

The configuration will look as follows:

```

show service dhcpv6-server
  shared-network-name NET1 {
    subnet 2001:db8::/64 {
      range 1 {
        start 2001:db8::100
        stop 2001:db8::199
      }
      option {
        name-server 2001:db8::ffff
      }
      subnet-id 1
    }
  }

```

Operation Mode

show log dhcpv6 server

Show DHCPv6 server daemon log file

show log dhcpv6 client

Show logs from all DHCPv6 client processes.

show log dhcpv6 client interface <interface>

Show logs from specific interface DHCPv6 client process.

restart dhcpv6 server

To restart the DHCPv6 server

show dhcpv6 server leases

Shows status of all assigned leases:

```

vyos@vyos:~$ show dhcpv6 server leases
IPv6 address      State      Last communication    Lease expiration      Remaining      Type
↪ Pool          DUID
-----
↪ -----
2001:db8::101     active    2019/12/05 19:40:10    2019/12/06 07:40:10    11:45:21      IA_NA
↪ NET1          98:76:54:32:00:01:00:01:12:34:56:78:aa:bb:cc:dd:ee:ff
2001:db8::102     active    2019/12/05 14:01:23    2019/12/06 02:01:23    6:06:34       IA_NA
↪ NET1          87:65:43:21:00:01:00:01:11:22:33:44:fa:fb:fc:fd:fe:ff

```

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```
2001:db8:10::/64 active 2019/12/05 23:20:10 2019/12/06 11:40:10 11:45:21 IA_PD_
↪ PD-NET1 98:76:54:32:00:01:00:01:12:34:56:78:aa:bb:cc:dd:ee:ff
```

Hint: Static mappings aren't shown. To show all states, use `show dhcp server leases state all`.

show dhcpv6 server leases pool <pool>

Show only leases in the specified pool.

show dhcpv6 server leases sort <key>

Sort the output by the specified key. Possible keys: expires, ia_id_duid, ip, last_comm, pool, remaining, state, type (default = ip)

show dhcpv6 server leases state <state>

Show only leases with the specified state. Possible states: abandoned, active, all, backup, expired, free, released, reset (default = active)

7.10.7 DNS Forwarding

Configuration

VyOS provides DNS infrastructure for small networks. It is designed to be lightweight and have a small footprint, suitable for resource constrained routers and firewalls. For this we utilize PowerDNS recursor.

The VyOS DNS forwarder does not require an upstream DNS server. It can serve as a full recursive DNS server - but it can also forward queries to configurable upstream DNS servers. By not configuring any upstream DNS servers you also avoid being tracked by the provider of your upstream DNS server.

set service dns forwarding system

Forward incoming DNS queries to the DNS servers configured under the `system name-server` nodes.

set service dns forwarding dhcp <interface>

Interfaces whose DHCP client nameservers to forward requests to.

set service dns forwarding name-server <address> port <port>

Send all DNS queries to the IPv4/IPv6 DNS server specified under <address> on optional port specified under <port>. The port defaults to 53. You can configure multiple nameservers here.

set service dns forwarding domain <domain-name> name-server <address>

Forward received queries for a particular domain (specified via `domain-name`) to a given nameserver. Multiple nameservers can be specified. You can use this feature for a DNS split-horizon configuration.

Note: This also works for reverse-lookup zones (`18.172.in-addr.arpa`).

set service dns forwarding domain <domain-name> addnta

Add NTA (negative trust anchor) for this domain. This must be set if the domain does not support DNSSEC.

set service dns forwarding domain <domain-name> recursion-desired

Set the "recursion desired" bit in requests to the upstream nameserver.

set service dns forwarding allow-from <network>

Given the fact that open DNS recursors could be used on DDoS amplification attacks, you must configure the networks which are allowed to use this recursor. A network of 0.0.0.0/0 or ::/0 would allow all IPv4 and IPv6 networks to query this server. This is generally a bad idea.

set service dns forwarding dnssec <off | process-no-validate | process | log-fail | validate>

The PowerDNS recursor has 5 different levels of DNSSEC processing, which can be set with the dnssec setting. In order from least to most processing, these are:

- **off** In this mode, no DNSSEC processing takes place. The recursor will not set the DNSSEC OK (DO) bit in the outgoing queries and will ignore the DO and AD bits in queries.
- **process-no-validate** In this mode the recursor acts as a “security aware, non-validating” nameserver, meaning it will set the DO-bit on outgoing queries and will provide DNSSEC related RRsets (NSEC, RRSIG) to clients that ask for them (by means of a DO-bit in the query), except for zones provided through the auth-zones setting. It will not do any validation in this mode, not even when requested by the client.
- **process** When dnssec is set to process the behavior is similar to process-no-validate. However, the recursor will try to validate the data if at least one of the DO or AD bits is set in the query; in that case, it will set the AD-bit in the response when the data is validated successfully, or send SERVFAIL when the validation comes up bogus.
- **log-fail** In this mode, the recursor will attempt to validate all data it retrieves from authoritative servers, regardless of the client’s DNSSEC desires, and will log the validation result. This mode can be used to determine the extra load and amount of possibly bogus answers before turning on full-blown validation. Responses to client queries are the same as with process.
- **validate** The highest mode of DNSSEC processing. In this mode, all queries will be validated and will be answered with a SERVFAIL in case of bogus data, regardless of the client’s request.

Note: The popular Unix/Linux dig tool sets the AD-bit in the query. This might lead to unexpected query results when testing. Set +noad on the dig command line when this is the case.

Note: The CD-bit is honored correctly for process and validate. For log-fail, failures will be logged too.

set service dns forwarding ignore-hosts-file

Do not use the local /etc/hosts file in name resolution. VyOS DHCP server will use this file to add resolvers to assigned addresses.

set service dns forwarding cache-size <0-2147483647>

Maximum number of DNS cache entries. 1 million per CPU core will generally suffice for most installations.

This defaults to 10000.

set service dns forwarding negative-ttl <0-7200>

A query for which there is authoritatively no answer is cached to quickly deny a record’s existence later on, without putting a heavy load on the remote server. In practice, caches can become saturated with hundreds of thousands of hosts which are tried only once.

This setting, which defaults to 3600 seconds, puts a maximum on the amount of time negative entries are cached.

set service dns forwarding timeout <10-60000>

The number of milliseconds to wait for a remote authoritative server to respond before timing out and responding with SERVFAIL.

This setting defaults to 1500 and is valid between 10 and 60000.

set service dns forwarding listen-address <address>

The local IPv4 or IPv6 addresses to bind the DNS forwarder to. The forwarder will listen on this address for incoming connections.

set service dns forwarding source-address <address>

The local IPv4 or IPv6 addresses to use as a source address for sending queries. The forwarder will send forwarded outbound DNS requests from this address.

set service dns forwarding no-serve-rfc1918

This makes the server authoritatively not aware of: 10.in-addr.arpa, 168.192.in-addr.arpa, 16-31.172.in-addr.arpa, which enabling upstream DNS server(s) to be used for reverse lookups of these zones.

Authoritative zones

The VyOS DNS forwarder can also be configured to host authoritative records for a domain.

set service dns forwarding authoritative-domain <domain-name> disable

Disable hosting authoritative zone for <domain-name> without deleting from configuration.

set service dns forwarding authoritative-domain <domain-name> records <type> <name> disable

Disable specific record without deleting it from configuration.

set service dns forwarding authoritative-domain <domain-name> records <type> <name> ttl <seconds>

Set the TTL for the record in seconds. Default is 300 seconds.

Record types

Below are a list of record types available to be configured within VyOS. Some records support special <name> keywords:

- @ Use @ as record name to set the record for the root domain.
- any Use any as record name to configure the record as a wildcard.

set service dns forwarding authoritative-domain <domain-name> records a <name> address <x.x.x.x>

Set an A (Address) record. Supports @ and any keywords.

set service dns forwarding authoritative-domain <domain-name> records aaaa <name> address <h:h:h:h:h:h:h:h>

Set an AAAA (IPv6 Address) record. Supports @ and any keywords.

set service dns forwarding authoritative-domain <domain-name> records cname <name> target <target-domain-name>

Set an CNAME (Canonical name) record. Supports @ keyword.

```
set service dns forwarding authoritative-domain <domain-name> records naptr <name> rule
<rule-number> <option> <value>
```

Set an NAPTR (Naming authority pointer) record. Supports @ keyword. NAPTR records support the following options:

- **lookup-a** A Flag.
- **lookup-srv** S flag.
- **order** Rule order. Requires <value>.
- **preference** Rule preference. Requires <value>. Defaults to 0 if not set.
- **protocol-specific** P flag.
- **regexp** Regular expression. Requires <value>.
- **replacement** Replacement DNS name.
- **resolve-uri** U flag.
- **service** Service type. Requires <value>.

```
set service dns forwarding authoritative-domain <domain-name> records ns <name> target
<target-name>
```

Set an NS (Nameserver) record.

```
set service dns forwarding authoritative-domain <domain-name> records ptr <name> target
<target-name>
```

Set an PTR (Pointer record) record. Supports @ keyword.

```
set service dns forwarding authoritative-domain <domain-name> records spf <name> value
<value>
```

Set an SPF record. Supports @ keyword.

```
set service dns forwarding authoritative-domain <domain-name> records srv <name> entry
<entry-number> [hostname | port | priority | weight] <value>
```

Set an SRV (Service) record. Supports @ keyword.

```
set service dns forwarding authoritative-domain <domain-name> records txt <name> value
<value>
```

Set an TXT (Text) record. Supports @ keyword.

Example

A VyOS router with two interfaces - eth0 (WAN) and eth1 (LAN) - is required to implement a split-horizon DNS configuration for example.com.

In this scenario:

- All DNS requests for example.com must be forwarded to a DNS server at 192.0.2.254 and 2001:db8:cafe::1
- All other DNS requests will be forwarded to a different set of DNS servers at 192.0.2.1, 192.0.2.2, 2001:db8::1:ffff and 2001:db8::2:ffff
- The VyOS DNS forwarder will only listen for requests on the eth1 (LAN) interface addresses - 192.168.1.254 for IPv4 and 2001:db8::ffff for IPv6
- The VyOS DNS forwarder will only accept lookup requests from the LAN subnets - 192.168.1.0/24 and 2001:db8::/64

- The VyOS DNS forwarder will pass reverse lookups for 10.in-addr.arpa, 168.192.in-addr.arpa, 16-31.172.in-addr.arpa zones to upstream server.

```
set service dns forwarding domain example.com name-server 192.0.2.254
set service dns forwarding domain example.com name-server 2001:db8:cafe::1
set service dns forwarding name-server 192.0.2.1
set service dns forwarding name-server 192.0.2.2
set service dns forwarding name-server 192.0.2.3 port 853
set service dns forwarding name-server 2001:db8::1:ffff
set service dns forwarding name-server 2001:db8::2:ffff
set service dns forwarding name-server 2001:db8::3:ffff port 8053
set service dns forwarding listen-address 192.168.1.254
set service dns forwarding listen-address 2001:db8::ffff
set service dns forwarding allow-from 192.168.1.0/24
set service dns forwarding allow-from 2001:db8::/64
set service dns forwarding no-serve-rfc1918
```

Operation

reset dns forwarding <all | domain>

Resets the local DNS forwarding cache database. You can reset the cache for all entries or only for entries to a specific domain.

restart dns forwarding

Restarts the DNS recursor process. This also invalidates the local DNS forwarding cache.

7.10.8 Dynamic DNS

VyOS is able to update a remote DNS record when an interface gets a new IP address. In order to do so, VyOS includes `ddclient`, a Perl script written for this only one purpose.

`ddclient` uses two methods to update a DNS record. The first one will send updates directly to the DNS daemon, in compliance with [RFC 2136](#). The second one involves a third party service, like DynDNS.com or any other such service provider. This method uses HTTP requests to transmit the new IP address. You can configure both in VyOS.

Configuration

RFC 2136 Based

set service dns dynamic name <service-name> address interface <interface>

Create new dynamic DNS update configuration which will update the IP address assigned to <interface> on the service you configured under <service-name>.

set service dns dynamic name <service-name> description <text>

Set description <text> for dynamic DNS service being configured.

set service dns dynamic name <service-name> key <filename>

File identified by <filename> containing the TSIG authentication key for RFC2136 nsupdate on remote DNS server.

set service dns dynamic name <service-name> server <server>

Configure the DNS <server> IP/FQDN used when updating this dynamic assignment.

```
set service dns dynamic name <service-name> zone <zone>
```

Configure DNS <zone> to be updated.

```
set service dns dynamic name <service-name> host-name <record>
```

Configure DNS <record> which should be updated. This can be set multiple times.

```
set service dns dynamic name <service-name> ttl <ttl>
```

Configure optional TTL value on the given resource record. This defaults to 600 seconds.

```
set service dns dynamic interval <60-3600>
```

Specify interval in seconds to wait between Dynamic DNS updates. The default is 300 seconds.

Example

- Register DNS record example.vyos.io on DNS server ns1.vyos.io
- Use auth key file at /config/auth/my.key
- Set TTL to 300 seconds

```
# Configuration commands entered:
#
set service dns dynamic name 'VyOS-DNS' address interface 'eth0'
set service dns dynamic name 'VyOS-DNS' description 'RFC 2136 dynamic dns service'
set service dns dynamic name 'VyOS-DNS' key '/config/auth/my.key'
set service dns dynamic name 'VyOS-DNS' server 'ns1.vyos.io'
set service dns dynamic name 'VyOS-DNS' zone 'vyos.io'
set service dns dynamic name 'VyOS-DNS' host-name 'example.vyos.io'
set service dns dynamic name 'VyOS-DNS' protocol 'nsupdate'
set service dns dynamic name 'VyOS-DNS' ttl '300'

# Resulting config:
#
vyos@vyos# show service dns dynamic
name VyOS-DNS {
    address {
        interface eth0
    }
    description "RFC 2136 dynamic dns service"
    host-name example.vyos.io
    key /config/auth/my.key
    protocol nsupdate
    server ns1.vyos.io
    ttl 300
    zone vyos.io
}
```

This will render the following `ddclient` configuration entry:

```
# ddclient configuration for interface "eth0":
#
```

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```
# Web service dynamic DNS configuration for VyOS-DNS: [nsupdate, example.vyos.io]
use=if, \
if=eth0, \
protocol=nsupdate, \
server=ns1.vyos.io, \
zone=vyos.io, \
password='/config/auth/my.key', \
ttl=300 \
example.vyos.io
```

Note: You can also keep different DNS zone updated. Just create a new config node: `set service dns dynamic interface <interface> rfc2136 <other-service-name>`

HTTP based services

VyOS is also able to use any service relying on protocols supported by ddclient.

To use such a service, one must define a login, password, one or multiple hostnames, protocol and server.

set service dns dynamic name <service-name> address interface <interface>

Create new dynamic DNS update configuration which will update the IP address assigned to <interface> on the service you configured under <service-name>.

set service dns dynamic name <service-name> description <text>

Set description <text> for dynamic DNS service being configured.

set service dns dynamic name <service-name> host-name <hostname>

Setup the dynamic DNS hostname <hostname> associated with the DynDNS provider identified by <service-name>.

set service dns dynamic name <service-name> username <username>

Configure <username> used when authenticating the update request for DynDNS service identified by <service-name>.

set service dns dynamic name <service-name> password <password>

Configure <password> used when authenticating the update request for DynDNS service identified by <service-name>.

set service dns dynamic name <service-name> protocol <protocol>

When a custom DynDNS provider is used, the protocol used for communicating to the provider must be specified under <protocol>. See the embedded completion helper when entering above command for available protocols.

set service dns dynamic name <service-name> server <server>

When a custom DynDNS provider is used the <server> where update requests are being sent to must be specified.

set service dns dynamic name <service-name> ip-version 'ipv6'

Allow explicit IPv6 address for the interface.

Example:

Use deSEC (dedyn.io) as your preferred provider:

```
set service dns dynamic name dedyn description 'deSEC dynamic dns service'
set service dns dynamic name dedyn username 'myusername'
set service dns dynamic name dedyn password 'mypassword'
set service dns dynamic name dedyn host-name 'myhostname.dedyn.io'
set service dns dynamic name dedyn protocol 'dyndns2'
set service dns dynamic name dedyn server 'update.dedyn.io'
set service dns dynamic name dedyn address interface 'eth0'
```

Note: Multiple services can be used per interface. Just specify as many services per interface as you like!

Example IPv6 only:

```
set service dns dynamic name dedyn description 'deSEC ipv6 dynamic dns service'
set service dns dynamic name dedyn username 'myusername'
set service dns dynamic name dedyn password 'mypassword'
set service dns dynamic name dedyn host-name 'myhostname.dedyn.io'
set service dns dynamic name dedyn protocol 'dyndns2'
set service dns dynamic name dedyn ip-version 'ipv6'
set service dns dynamic name dedyn server 'update6.dedyn.io'
set service dns dynamic name dedyn address interface 'eth0'
```

Running Behind NAT

By default, `ddclient` will update a dynamic dns record using the IP address directly attached to the interface. If your VyOS instance is behind NAT, your record will be updated to point to your internal IP.

`ddclient` has another way to determine the WAN IP address. This is controlled by:

set service dns dynamic name <service-name> address web <url>

Use configured <url> to determine your IP address. `ddclient` will load <url> and tries to extract your IP address from the response.

set service dns dynamic name <service-name> address web skip <pattern>

`ddclient` will skip any address located before the string set in <pattern>.

7.10.9 Event Handler**Event Handler Technology Overview**

Event handler allows you to execute scripts when a string that matches a regex or a regex with a service name appears in journald logs. You can pass variables, arguments, and a full matching string to the script.

How to configure Event Handler

1. *Create an event handler*
2. *Add regex to the script*
3. *Add a full path to the script*
4. *Add optional parameters*

Event Handler Configuration Steps

1. Create an event handler

```
set service event-handler event <event-handler name>
```

This is an optional command because the event handler will be automatically created after any of the next commands.

2. Add regex to the script

```
set service event-handler event <event-handler name> filter pattern <regex>
```

This is a mandatory command. Sets regular expression to match against log string message.

Note: The regular expression matches if and only if the entire string matches the pattern.

3. Add a full path to the script

```
set service event-handler event <event-handler name> script path <path to script>
```

This is a mandatory command. Sets the full path to the script. The script file must be executable.

4. Add optional parameters

```
set service event-handler event <event-handler name> filter syslog-identifier <syslogid name>
```

This is an optional command. Filters log messages by syslog-identifier.

```
set service event-handler event <event-handler name> script environment <env name> value <env value>
```

This is an optional command. Adds environment and its value to the script. Use separate commands for each environment.

One implicit environment exists.

- **message:** Full message that has triggered the script.

```
set service event-handler event <event-handler name> script arguments <arguments>
```

This is an optional command. Adds arguments to the script. Arguments must be separated by spaces.

Note: We don't recommend using arguments. Using environment variables is preferable.

Example

Event handler that monitors the state of interface eth0.

```
set service event-handler event INTERFACE_STATE_DOWN filter pattern '.*eth0.*,RUNNING,.*-
↳>.*'
set service event-handler event INTERFACE_STATE_DOWN filter syslog-identifier 'netplugd'
set service event-handler event INTERFACE_STATE_DOWN script environment interface_action_
↳value 'down'
set service event-handler event INTERFACE_STATE_DOWN script environment interface_name_
↳value 'eth0'
set service event-handler event INTERFACE_STATE_DOWN script path '/config/scripts/
↳eventhandler.py'
```

Event handler script

```
#!/usr/bin/env python3
#
# VyOS event-handler script example
from os import environ
import subprocess
from sys import exit

# Perform actions according to requirements
def process_event() -> None:
    # Get variables
    message_text = environ.get('message')
    interface_name = environ.get('interface_name')
    interface_action = environ.get('interface_action')
    # Print the message that triggered this script
    print(f'Logged message: {message_text}')
    # Prepare a command to run
    command = f'sudo ip link set {interface_name} {interface_action}'.split()
    # Execute a command
    subprocess.run(command)

if __name__ == '__main__':
    try:
        # Run script actions and exit
        process_event()
        exit(0)
    except Exception as err:
        # Exit properly in case if something in the script goes wrong
        print(f'Error running script: {err}')
        exit(1)
```

7.10.10 HTTP API

VyOS provide an HTTP API. You can use it to execute op-mode commands, update VyOS, set or delete config. Please take a look at the [VyOS API](#) page for an detailed how-to.

Configuration

set service https allow-client address <address>

Only allow certain IP addresses or prefixes to access the https webserver.

set service https certificates ca-certificate <name>

Use CA certificate from PKI subsystem

set service https certificates certificate <name>

Use certificate from PKI subsystem

set service https certificates dh-params <name>

Use DH parameters from PKI subsystem. Must be at least 2048 bits in length.

set service https listen-address <address>

Webserver should only listen on specified IP address

set service https port <number>

Webserver should listen on specified port.

Default: 443

set service https enable-http-redirect

Enable automatic redirect from http to https.

set service https tls-version <1.2 | 1.3>

Select TLS version used.

This defaults to both 1.2 and 1.3.

set service https vrf <name>

Start Webserver in given VRF.

set service https request-body-size-limit <size>

Set the maximum request body size in megabytes. Default is 1MB.

API

set service https api keys id <name> key <apikey>

Set a named api key. Every key has the same, full permissions on the system.

REST

```
set service https api rest
```

Enable REST API

```
set service https api rest debug
```

To enable debug messages. Available via `show log` or `monitor log`

```
set service https api rest strict
```

Enforce strict path checking.

GraphQL

```
set service https api graphql introspection
```

Enable GraphQL Schema introspection.

Note: Do not leave introspection enabled in production, it is a security risk.

```
set service https api graphql authentication type <key | token>
```

Set the authentication type for GraphQL, default option is key. Available options are:

- key use API keys configured in `service https api keys`
- token use JWT tokens.

```
set service https api graphql authentication expiration
```

Set the lifetime for JWT tokens in seconds. Default is 3600 seconds.

```
set service https api graphql authentication secret-length
```

Set the byte length of the JWT secret. Default is 32.

```
set service https api graphql cors allow-origin <origin>
```

Allow cross-origin requests from <origin>.

Example Configuration

Setting REST API and an API-KEY is the minimal configuration to get a working API Endpoint.

```
set service https api keys id MY-HTTPS-API-ID key MY-HTTPS-API-PLAINTEXT-KEY
set service https api rest
```

7.10.11 IPoE Server

VyOS utilizes `accel-ppp` to provide IPoE (Internet Protocol over Ethernet) server functionality. It can be used with local authentication (mac-address) or a connected RADIUS server.

IPoE is a method of delivering an IP payload over an Ethernet-based access network or an access network using bridged Ethernet over Asynchronous Transfer Mode (ATM) without using PPPoE. It directly encapsulates the IP datagrams in Ethernet frames, using the standard [RFC 894](#) encapsulation.

The use of IPoE addresses the disadvantage that PPP is unsuited for multicast delivery to multiple users. Typically, IPoE uses Dynamic Host Configuration Protocol and Extensible Authentication Protocol to provide the same functionality as PPPoE, but in a less robust manner.

Note: Please be aware, due to an upstream bug, config changes/commits will restart the ppp daemon and will reset existing IPoE sessions, in order to become effective.

Configuring IPoE Server

IPoE can be configured on different interfaces, it will depend on each specific situation which interface will provide IPoE to clients. The client's mac address and the incoming interface is being used as control parameter, to authenticate a client.

The example configuration below will assign an IP to the client on the incoming interface eth1 with the client mac address 00:50:79:66:68:00. Other DHCP discovery requests will be ignored, unless the client mac has been enabled in the configuration.

```
set interfaces ethernet eth1 address '192.168.0.1/24'
set service ipoe-server authentication interface eth1.100 mac 00:50:79:66:68:00
set service ipoe-server authentication interface eth1.101 mac 00:50:79:66:68:01
set service ipoe-server authentication mode 'local'
set service ipoe-server client-ip-pool IPOE-POOL range '192.168.0.2-192.168.0.254'
set service ipoe-server default-pool 'IPOE-POOL'
set service ipoe-server gateway-address '192.168.0.1/24'
set service ipoe-server interface eth1 mode 'l2'
set service ipoe-server interface eth1 network 'vlan'
set service ipoe-server interface eth1 vlan '100-200'
```

set service ipoe-server authentication interface <interface> mac <MAC>

Creates local IPoE user with username=****<interface>**** and password=****<MAC>**** (mac-address)

set service ipoe-server authentication mode <local | radius>

Set authentication backend. The configured authentication backend is used for all queries.

- **radius:** All authentication queries are handled by a configured RADIUS server.
- **local:** All authentication queries are handled locally.
- **noauth:** Authentication disabled

set service ipoe-server client-ip-pool <POOL-NAME> range <x.x.x.x-x.x.x.x | x.x.x.x/x>

Use this command to define the first IP address of a pool of addresses to be given to IPoE clients. If notation **x.x.x.x-x.x.x.x**, it must be within a /24 subnet. If notation **x.x.x.x/x** is used there is possibility to set host/netmask.

set service ipoe-server default-pool <POOL-NAME>

Use this command to define default address pool name.

```
set service ipoe-server gateway-address <x.x.x.x/x>
```

Specifies address to be used as server ip address if radius can assign only client address. In such case if client address is matched network and mask then specified address and mask will be used. You can specify multiple such options.

```
set service ipoe-server interface <interface> mode <12 | 13>
```

Specifies the client connectivity mode.

- **12:** It means that clients are on same network where interface is. ****(default)****
- **13:** It means that client are behind some router.

```
set service ipoe-server interface <interface> network <shared | vlan>
```

Specify where interface is shared by multiple users or it is vlan-per-user.

- **shared:** Multiple clients share the same network. **(default)**
- **vlan:** One VLAN per client.

```
vyos@vyos:~$ show ipoe-server sessions
```

ifname	username	calling-sid	ip	rate-limit	type	comp	state
uptime							
ipoe0	eth1.100	00:50:79:66:68:00	192.168.0.2		ipoe		active
00:04:55							
ipoe1	eth1.101	00:50:79:66:68:01	192.168.0.3		ipoe		active
00:04:44							

Configuring RADIUS authentication

To enable RADIUS based authentication, the authentication mode needs to be changed within the configuration. Previous settings like the local users, still exists within the configuration, however they are not used if the mode has been changed from local to radius. Once changed back to local, it will use all local accounts again.

```
set service ipoe-server authentication mode radius
```

```
set service ipoe-server authentication radius server <server> key <secret>
```

Configure RADIUS <server> and its required shared <secret> for communicating with the RADIUS server.

Since the RADIUS server would be a single point of failure, multiple RADIUS servers can be setup and will be used subsequently. For example:

```
set service ipoe-server authentication radius server 10.0.0.1 key 'foo'  
set service ipoe-server authentication radius server 10.0.0.2 key 'foo'
```

Note: Some RADIUS servers use an access control list which allows or denies queries, make sure to add your VyOS router to the allowed client list.

RADIUS source address

If you are using OSPF as IGP, always the closest interface connected to the RADIUS server is used. With VyOS 1.2 you can bind all outgoing RADIUS requests to a single source IP e.g. the loopback interface.

```
set service ipoe-server authentication radius source-address <address>
```

Source IPv4 address used in all RADIUS server queries.

Note: The source-address must be configured on one of VyOS interface. Best practice would be a loopback or dummy interface.

RADIUS advanced options

```
set service ipoe-server authentication radius server <server> port <port>
```

Configure RADIUS <server> and its required port for authentication requests.

```
set service ipoe-server authentication radius server <server> fail-time <time>
```

Mark RADIUS server as offline for this given <time> in seconds.

```
set service ipoe-server authentication radius server <server> disable
```

Temporary disable this RADIUS server.

```
set service ipoe-server authentication radius acct-timeout <timeout>
```

Timeout to wait reply for Interim-Update packets. (default 3 seconds)

```
set service ipoe-server authentication radius dynamic-author server <address>
```

Specifies IP address for Dynamic Authorization Extension server (DM/CoA). This IP must exist on any VyOS interface or it can be 0.0.0.0.

```
set service ipoe-server authentication radius dynamic-author port <port>
```

UDP port for Dynamic Authorization Extension server (DM/CoA)

```
set service ipoe-server authentication radius dynamic-author key <secret>
```

Secret for Dynamic Authorization Extension server (DM/CoA)

```
set service ipoe-server authentication radius max-try <number>
```

Maximum number of tries to send Access-Request/Accounting-Request queries

```
set service ipoe-server authentication radius timeout <timeout>
```

Timeout to wait response from server (seconds)

```
set service ipoe-server authentication radius nas-identifier <identifier>
```

Value to send to RADIUS server in NAS-Identifier attribute and to be matched in DM/CoA requests.

```
set service ipoe-server authentication radius nas-ip-address <address>
```

Value to send to RADIUS server in NAS-IP-Address attribute and to be matched in DM/CoA requests. Also DM/CoA server will bind to that address.

```
set service ipoe-server authentication radius source-address <address>
```

Source IPv4 address used in all RADIUS server queries.

```
set service ipoe-server authentication radius rate-limit attribute <attribute>
```

Specifies which RADIUS server attribute contains the rate limit information. The default attribute is *Filter-Id*.

Note: If you set a custom RADIUS attribute you must define it on both dictionaries at RADIUS server and client.

```
set service ipoe-server authentication radius rate-limit enable
```

Enables bandwidth shaping via RADIUS.

```
set service ipoe-server authentication radius rate-limit vendor
```

Specifies the vendor dictionary, dictionary needs to be in /usr/share/accel-ppp/radius.

Received RADIUS attributes have a higher priority than parameters defined within the CLI configuration, refer to the explanation below.

Allocation clients ip addresses by RADIUS

If the RADIUS server sends the attribute Framed-IP-Address then this IP address will be allocated to the client and the option default-pool within the CLI config is being ignored.

If the RADIUS server sends the attribute Framed-Pool, IP address will be allocated from a predefined IP pool whose name equals the attribute value.

If the RADIUS server sends the attribute Stateful-IPv6-Address-Pool, IPv6 address will be allocated from a predefined IPv6 pool prefix whose name equals the attribute value.

If the RADIUS server sends the attribute Delegated-IPv6-Prefix-Pool, IPv6 delegation prefix will be allocated from a predefined IPv6 pool delegate whose name equals the attribute value.

Note: Stateful-IPv6-Address-Pool and Delegated-IPv6-Prefix-Pool are defined in RFC6911. If they are not defined in your RADIUS server, add new [dictionary](#).

User interface can be put to VRF context via RADIUS Access-Accept packet, or change it via RADIUS CoA. Accel-VRF-Name is used from these purposes. It is custom [ACCEL-PPP attribute](#). Define it in your RADIUS server.

IPv6

```
set service ipoe-server client-ipv6-pool <IPv6-POOL-NAME> prefix <address> mask  
<number-of-bits>
```

Use this command to set the IPv6 address pool from which an IPoE client will get an IPv6 prefix of your defined length (mask) to terminate the IPoE endpoint at their side. The mask length can be set from 48 to 128 bit long, the default value is 64.

```
set service ipoe-server client-ipv6-pool <IPv6-POOL-NAME> delegate <address>  
delegation-prefix <number-of-bits>
```

Use this command to configure DHCPv6 Prefix Delegation (RFC3633) on IPoE. You will have to set your IPv6 pool and the length of the delegation prefix. From the defined IPv6 pool you will be handing out networks of the defined length (delegation-prefix). The length of the delegation prefix can be set from 32 to 64 bit long.

```
set service ipoe-server default-ipv6-pool <IPv6-POOL-NAME>
```

Use this command to define default IPv6 address pool name.

```
set service ipoe-server client-ipv6-pool IPv6-POOL delegate '2001:db8:8003::/48'
↪delegation-prefix '56'
set service ipoe-server client-ipv6-pool IPv6-POOL prefix '2001:db8:8002::/48' mask '64'
set service ipoe-server default-ipv6-pool IPv6-POOL
```

Scripting

set service ipoe-server extended-scripts on-change <path_to_script>

Script to run when session interface changed by RADIUS CoA handling

set service ipoe-server extended-scripts on-down <path_to_script>

Script to run when session interface going to terminate

set service ipoe-server extended-scripts on-pre-up <path_to_script>

Script to run before session interface comes up

set service ipoe-server extended-scripts on-up <path_to_script>

Script to run when session interface is completely configured and started

Advanced Options

Authentication Advanced Options

set service ipoe-server authentication interface <interface> mac <MAC> vlan <vlan-id>

VLAN monitor for automatic creation of VLAN interfaces for specific user on specific <interface>

set service ipoe-server authentication interface <interface> mac <MAC> rate-limit download <bandwidth>

Download bandwidth limit in kbit/s for user on interface <interface>.

set service ipoe-server authentication interface <interface> mac <MAC> rate-limit upload <bandwidth>

Upload bandwidth limit in kbit/s for for user on interface <interface>.

Client IP Pool Advanced Options

set service ipoe-server client-ip-pool <POOL-NAME> next-pool <NEXT-POOL-NAME>

Use this command to define the next address pool name.

Advanced Interface Options

set service ipoe-server interface <interface> client-subnet <x.x.x.x/x>

Specify local range of ip address to give to dhcp clients. First IP in range is router IP. If you need more customization use *client-ip-pool*

set service ipoe-server interface <interface> external-dhcp dhcp-relay <x.x.x.x>

Specify DHCPv4 relay IP address to pass requests to. If specified giaddr is also needed.

set service ipoe-server interface <interface> external-dhcp giaddr <x.x.x.x>

Specifies relay agent IP address

Global Advanced options

set service ipoe-server description <description>

Set description.

set service ipoe-server limits burst <value>

Burst count

set service ipoe-server limits connection-limit <value>

Acceptable rate of connections (e.g. 1/min, 60/sec)

set service ipoe-server limits timeout <value>

Timeout in seconds

set service ipoe-server max-concurrent-sessions

Maximum number of concurrent session start attempts

set service ipoe-server name-server <address>

Connected client should use <address> as their DNS server. This command accepts both IPv4 and IPv6 addresses. Up to two nameservers can be configured for IPv4, up to three for IPv6.

set service ipoe-server shaper fwmark <1-2147483647>

Match firewall mark value

set service ipoe-server snmp master-agent

Enable SNMP

Monitoring

show ipoe-server sessions

Use this command to locally check the active sessions in the IPoE server.

```
vyos@vyos:~$ show ipoe-server sessions
ifname | username |   calling-sid   |   ip   | rate-limit | type | comp | state |
--|-----|-----|-----|-----|-----|-----|-----|
--|-----|-----|-----|-----|-----|-----|-----|
```

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```
eth1.100 | eth1.100 | 0c:98:bd:b8:00:01 | 192.168.0.3 | | ipoe | |
↪active | 03:03:58
```

```
vyos@vyos:~$ show ipoe-server statistics
```

```
uptime: 0.03:31:36
```

```
cpu: 0%
```

```
mem(rss/virt): 6044/101360 kB
```

```
core:
```

```
  mempool_allocated: 148628
```

```
  mempool_available: 144748
```

```
  thread_count: 1
```

```
  thread_active: 1
```

```
  context_count: 10
```

```
  context_sleeping: 0
```

```
  context_pending: 0
```

```
  md_handler_count: 6
```

```
  md_handler_pending: 0
```

```
  timer_count: 1
```

```
  timer_pending: 0
```

```
sessions:
```

```
  starting: 0
```

```
  active: 1
```

```
  finishing: 0
```

```
ipoe:
```

```
  starting: 0
```

```
  active: 1
```

```
  delayed: 0
```

Troubleshooting

```
vyos@vyos:~$ show log ipoe-server
```

```
Feb 27 14:29:27 vyos accel-ipoe[2262]: eth1.100:: recv [DHCPv4 Discover xid=55df9228,
↪chaddr=0c:98:bd:b8:00:01 <Message-Type Discover> <Request-IP 192.168.0.3> <Host-Name,
↪vyos> <Request-List Subnet,Broadcast,Router,DNS,Classless-Route,Domain-Name,MTU>]
```

```
Feb 27 14:29:27 vyos accel-ipoe[2262]: eth1.100:eth1.100: eth1.100: authentication,
↪succeeded
```

```
Feb 27 14:29:27 vyos accel-ipoe[2262]: eth1.100:eth1.100: send [DHCPv4 Offer,
↪xid=55df9228 yiaddr=192.168.0.4 chaddr=0c:98:bd:b8:00:01 <Message-Type Offer> <Server-
↪ID 192.168.0.1> <Lease-Time 600> <T1 300> <T2 525> <Router 192.168.0.1> <Subnet 255.
↪255.255.0>]
```

```
Feb 27 14:29:27 vyos accel-ipoe[2262]: eth1.100:eth1.100: recv [DHCPv4 Request,
↪xid=55df9228 chaddr=0c:98:bd:b8:00:01 <Message-Type Request> <Server-ID 192.168.0.1>
↪<Request-IP 192.168.0.4> <Host-Name vyos> <Request-List Subnet,Broadcast,Router,DNS,
↪Classless-Route,Domain-Name,MTU>]
```

```
Feb 27 14:29:27 vyos accel-ipoe[2262]: eth1.100:eth1.100: ipoe: activate session
```

```
Feb 27 14:29:27 vyos accel-ipoe[2262]: eth1.100:eth1.100: ipoe: no free IPv6 address
```

```
Feb 27 14:29:27 vyos accel-ipoe[2262]: eth1.100:eth1.100: ipoe: session started
```

```
Feb 27 14:29:27 vyos accel-ipoe[2262]: eth1.100:eth1.100: send [DHCPv4 Ack xid=55df9228,
↪yiaddr=192.168.0.4 chaddr=0c:98:bd:b8:00:01 <Message-Type Ack> <Server-ID 192.168.0.1>
↪<Lease-Time 600> <T1 300> <T2 525> <Router 192.168.0.1> <Subnet 255.255.255.0>]
```

7.10.12 LLDP

LLDP (Link Layer Discovery Protocol) is a vendor-neutral link layer protocol in the Internet Protocol Suite used by network devices for advertising their identity, capabilities, and neighbors on an IEEE 802 local area network, principally wired Ethernet. The protocol is formally referred to by the IEEE as Station and Media Access Control Connectivity Discovery specified in IEEE 802.1AB and IEEE 802.3-2012 section 6 clause 79.

LLDP performs functions similar to several proprietary protocols, such as CDP (Cisco Discovery Protocol), FDP (Foundry Discovery Protocol), NDP (Nortel Discovery Protocol) and LLTD (Link Layer Topology Discovery).

Information gathered with LLDP is stored in the device as a MIB (Management Information Database) and can be queried with SNMP (Simple Network Management Protocol) as specified in [RFC 2922](#). The topology of an LLDP-enabled network can be discovered by crawling the hosts and querying this database. Information that may be retrieved include:

- System Name and Description
- Port name and description
- VLAN name
- IP management address
- System capabilities (switching, routing, etc.)
- MAC/PHY information
- MDI power
- Link aggregation

Configuration

set service lldp

Enable LLDP service

set service lldp management-address <address>

Define IPv4/IPv6 management address transmitted via LLDP. Multiple addresses can be defined. Only addresses connected to the system will be transmitted.

set service lldp interface <interface>

Enable transmission of LLDP information on given <interface>. You can also say `all` here so LLDP is turned on on every interface.

set service lldp interface <interface> mode [disable|rx-tx|rx|tx]

Configure the administrative status of the given port.

By default, all ports are configured to be in rx-tx mode. This means they can receive and transmit LLDP frames.

In rx mode, they won't emit any frames. In tx mode, they won't receive any frames. In disabled mode, no frame will be sent and any incoming frame will be discarded.

set service lldp snmp

Enable SNMP queries of the LLDP database

set service lldp legacy-protocols <cdp|edp|fdp|sonmp>

Enable given legacy protocol on this LLDP instance. Legacy protocols include:

- `cdp` - Listen for CDP for Cisco routers/switches

- edp - Listen for EDP for Extreme routers/switches
- fdp - Listen for FDP for Foundry routers/switches
- sonmp - Listen for SONMP for Nortel routers/switches

Operation

show lldp neighbors

Displays information about all neighbors discovered via LLDP.

```
vyos@vyos:~$ show lldp neighbors
Capability Codes: R - Router, B - Bridge, W - Wlan r - Repeater, S - Station
                  D - Docsis, T - Telephone, O - Other

Device ID          Local    Proto  Cap  Platform          Port ID
-----
BR2.vyos.net       eth0    LLDP   R    VyOS 1.2.4        eth1
BR3.vyos.net       eth0    LLDP   RB   VyOS 1.2.4        eth2
SW1.vyos.net       eth0    LLDP   B    Cisco IOS Software
↳GigabitEthernet0/6
```

show lldp neighbors detail

Get detailed information about LLDP neighbors.

```
vyos@vyos:~$ show lldp neighbors detail
-----
LLDP neighbors:
-----
Interface:      eth0, via: LLDP, RID: 28, Time: 0 day, 00:24:33
Chassis:
  ChassisID:    mac 00:53:00:01:02:c9
  SysName:      BR2.vyos.net
  SysDescr:     VyOS 1.3-rolling-201912230217
  MgmtIP:       192.0.2.1
  MgmtIP:       2001:db8::ffff
  Capability:   Bridge, on
  Capability:   Router, on
  Capability:   Wlan, off
  Capability:   Station, off
Port:
  PortID:       mac 00:53:00:01:02:c9
  PortDescr:    eth0
  TTL:          120
  PMD autoneg:  supported: no, enabled: no
  MAU oper type: 10GigBaseCX4 - X copper over 8 pair 100-Ohm balanced cable
VLAN:          201 eth0.201
VLAN:          205 eth0.205
LLDP-MED:
  Device Type:  Network Connectivity Device
  Capability:   Capabilities, yes
  Capability:   Policy, yes
  Capability:   Location, yes
```

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```

Capability: MDI/PSE, yes
Capability: MDI/PD, yes
Capability: Inventory, yes
Inventory:
  Hardware Revision: None
  Software Revision: 4.19.89-amd64-vyos
  Firmware Revision: 6.00
  Serial Number: VMware-42 1d 83 b9 fe c1 bd b2-7
  Manufacturer: VMware, Inc.
  Model: VMware Virtual Platform
  Asset ID: No Asset Tag
-----

```

show lldp neighbors interface <interface>

Show LLDP neighbors connected via interface <interface>.

show log lldp

Used for troubleshooting.

7.10.13 mDNS Repeater

Starting with VyOS 1.2 a mDNS (Multicast DNS) repeater functionality is provided. Additional information can be obtained from https://en.wikipedia.org/wiki/Multicast_DNS.

Multicast DNS uses the reserved address 224.0.0.251, which is “administratively scoped” and does not leave the subnet. mDNS repeater retransmits mDNS packets from one interface to other interfaces. This enables support for devices using mDNS discovery (like network printers, Apple Airplay, Chromecast, various IP based home-automation devices etc) across multiple VLANs.

Since the mDNS protocol sends the AA (Authoritative Answer) records in the packet itself, the repeater does not need to forge the source address. Instead, the source address is of the interface that repeats the packet.

Note: You can not run this in a VRRP setup, if multiple mDNS repeaters are launched in a subnet you will experience the mDNS packet storm death!

Configuration**set service mdns repeater interface <interface>**

To enable mDNS repeater you need to configure at least two interfaces so that all incoming mDNS packets from one interface configured here can be re-broadcasted to any other interface(s) configured under this section.

set service mdns repeater disable

mDNS repeater can be temporarily disabled without deleting the service using

set service mdns repeater ip-version <ipv4 | ipv6 | both>

mDNS repeater can be enabled either on IPv4 socket or on IPv6 socket or both to re-broadcast. By default, mDNS repeater will listen on both IPv4 and IPv6.

set service mdns repeater allow-service <service>

mDNS repeater can be configured to re-broadcast only specific services. By default, all services are re-broadcasted.

set service mdns repeater browse-domain <domain>

Allow listing additional custom domains to be browsed (in addition to the default `local`) so that they can be reflected.

set service mdns repeater cache-entries <entries>

Specify how many resource records are cached per interface. Bigger values allow mDNS work correctly in large LANs but also increase memory consumption.

Defaults to: 4096

Firewall recommendations

Unlike typical routed traffic, mDNS packets relayed between interfaces do not traverse the FORWARD hook chain in the firewall. Instead, they are processed through the following hooks:

- **INPUT:** For packets received by the local system
- **OUTPUT:** For packets sent from the local system

To control or allow mDNS packet forwarding via the relay, you must define appropriate rules in the INPUT and OUTPUT directions. Rules in the FORWARD direction will have no effect on mDNS relay traffic.

```
set firewall ipv4 input filter rule 10 action 'accept'
set firewall ipv4 input filter rule 10 destination address '224.0.0.251'
set firewall ipv4 input filter rule 10 destination port '5353'
set firewall ipv4 input filter rule 10 protocol 'udp'
set firewall ipv4 output filter rule 10 action 'accept'
set firewall ipv4 output filter rule 10 destination address '224.0.0.251'
set firewall ipv4 output filter rule 10 destination port '5353'
set firewall ipv4 output filter rule 10 protocol 'udp'
```

Example

To listen on both `eth0` and `eth1` mDNS packets and also repeat packets received on `eth0` to `eth1` (and vice-versa) use the following commands:

```
set service mdns repeater interface 'eth0'
set service mdns repeater interface 'eth1'
```

To allow only specific services, for example `_airplay._tcp` or `_ipp._tcp`, (instead of all services) to be re-broadcasted, use the following command:

```
set service mdns repeater allow-service '_airplay._tcp'
set service mdns repeater allow-service '_ipp._tcp'
```

To allow listing additional custom domain, for example `openthread.thread.home.arpa`, so that it can be reflected in addition to the default `local`, use the following command:

```
set service mdns repeater browse-domain 'openthread.thread.home.arpa'
```

Operation

restart mdns repeater

Restart mDNS repeater service.

show log mdns repeater

Show logs for mDNS repeater service.

monitor log mdns repeater

Follow the logs for mDNS repeater service.

7.10.14 Monitoring

VyOS supports monitoring through Telegraf as well as through Prometheus exporters.

Telegraf

Telegraf is the open source server agent to help you collect metrics, events and logs from your routers.

The following Telegraf plugins are configurable to export metrics and logs:

- Azure Data Explorer
- Prometheus Client
- Splunk
- InfluxDB
- Loki

Azure data explorer

Telegraf output plugin [azure-data-explorer](#).

```
set service monitoring telegraf azure-data-explorer authentication client-id <client-id>
```

Authentication application client-id.

```
set service monitoring telegraf azure-data-explorer authentication client-secret  
<client-secret>
```

Authentication application client-secret.

```
set service monitoring telegraf azure-data-explorer authentication tenant-id <tenant-id>
```

Authentication application tenant-id

```
set service monitoring telegraf azure-data-explorer database <name>
```

Remote database name.

```
set service monitoring telegraf azure-data-explorer group-metrics <single-table |  
table-per-metric>
```

Type of metrics grouping when push to Azure Data Explorer. The default is table-per-metric.

```
set service monitoring telegraf azure-data-explorer table <name>
```

Name of the single table Only if set group-metrics single-table.

```
set service monitoring telegraf azure-data-explorer url <url>
```

Remote URL.

Prometheus client

Telegraf output plugin [prometheus-client](#) This plugin allows export of Telegraf metrics to Prometheus, for Prometheus native metrics through exporters see section below.

```
set service monitoring telegraf prometheus-client
```

Output plugin Prometheus client

```
set service monitoring telegraf prometheus-client allow-from <prefix>
```

Networks allowed to query this server

```
set service monitoring telegraf prometheus-client authentication username <username>
```

HTTP basic authentication username

```
set service monitoring telegraf prometheus-client authentication password <password>
```

HTTP basic authentication password

```
set service monitoring telegraf prometheus-client listen-address <address>
```

Local IP addresses to listen on

```
set service monitoring telegraf prometheus-client metric-version <1 | 2>
```

Metrics version, the default is 2

```
set service monitoring telegraf prometheus-client port <port>
```

Port number used by connection, default is 9273

Example:

```
set service monitoring telegraf prometheus-client
```

```
vyos@r14:~$ curl --silent localhost:9273/metrics | egrep -v "#" | grep cpu_usage_system
cpu_usage_system{cpu="cpu-total",host="r14"} 0.20040080160320556
cpu_usage_system{cpu="cpu0",host="r14"} 0.17182130584191915
cpu_usage_system{cpu="cpu1",host="r14"} 0.22896393817971655
```

Splunk

Telegraf output plugin [splunk](#) HTTP Event Collector.

```
set service monitoring telegraf splunk authentication insecure
```

Use TLS but skip host validation

```
set service monitoring telegraf splunk authentication token <token>
```

Authorization token

```
set service monitoring telegraf splunk authentication url <url>
```

Remote URL to Splunk collector

Example:

```
set service monitoring telegraf splunk authentication insecure
set service monitoring telegraf splunk authentication token 'xxxxf5b8-xxxx-452a-xxxx-
↪43828911xxxx'
set service monitoring telegraf splunk url 'https://192.0.2.10:8088/services/collector'
```

InfluxDB

Telegraf output plugin `influxdb` to write metrics to InfluxDB via HTTP.

```
set service monitoring telegraf influxdb authentication organization <organization>
```

Authentication organization name

```
set service monitoring telegraf influxdb authentication token <token>
```

Authentication token

```
set service monitoring telegraf bucket <bucket>
```

Remote InfluxDB bucket name

```
set service monitoring telegraf influxdb port <port>
```

Remote port

```
set service monitoring telegraf influxdb url <url>
```

Remote URL

Example:

```
set service monitoring telegraf influxdb authentication organization 'vyos'
set service monitoring telegraf influxdb authentication token 'ZAm19Uy5wrhA...=='
set service monitoring telegraf influxdb bucket 'bucket_vyos'
set service monitoring telegraf influxdb port '8086'
set service monitoring telegraf influxdb url 'http://r1.influxdb2.local'
```

Loki

Telegraf can be used to send logs to `loki` using tags as labels.

```
set service monitoring telegraf loki port <port>
```

Remote Loki port

Default is 3100

```
set service monitoring telegraf loki url <url>
```

Remote Loki url

```
set service monitoring telegraf loki authentication username <username>
```

```
set service monitoring telegraf loki authentication password <password>
```

HTTP basic authentication.

If either is set both must be set.

```
set service monitoring telegraf loki metric-name-label <label>
```

Label to use for the metric name when sending metrics.

If set to an empty string, the label will not be added. This is NOT recommended, as it makes it impossible to differentiate between multiple metrics.

Prometheus

The following Prometheus exporters are configurable to export metrics:

- Node Exporter
- FRR Exporter

Node Exporter

Prometheus `node_exporter` which provides a wide range of hardware and OS metrics.

```
set service monitoring prometheus node-exporter listen-address <address>
```

Configure the address `node_exporter` is listening on.

```
set service monitoring prometheus node-exporter port <port>
```

Configure the port number `node_exporter` is listening on.

```
set service monitoring prometheus node-exporter vrf <name>
```

Configure name of the VRF instance.

```
set service monitoring prometheus node-exporter collectors textfile
```

Configure textfile collector to export custom metrics read from `/run/node_exporter/collector`

FRR Exporter

Prometheus `frr_exporter` which provides free range routing metrics.

```
set service monitoring prometheus frr-exporter listen-address <address>
```

Configure the address `frr_exporter` is listening on.

```
set service monitoring prometheus frr-exporter port <port>
```

Configure the port number `frr_exporter` is listening on.

```
set service monitoring prometheus frr-exporter vrf <name>
```

Configure name of the VRF instance.

Blackbox Exporter

Prometheus `blackbox_exporter` which allows probing of endpoints over HTTP, HTTPS, DNS, TCP, ICMP and gRPC .

```
set service monitoring prometheus blackbox-exporter listen-address <address>
```

Configure the address `blackbox_exporter` is listening on.

```
set service monitoring prometheus blackbox-exporter port <port>
```

Configure the port number `blackbox_exporter` is listening on.

set service monitoring prometheus blackbox-exporter vrf <name>

Configure name of the VRF instance.

Configuring modules

Blackbox exporter can be configured with different modules for probing DNS or ICMP.

DNS module example:

```
set service monitoring prometheus blackbox-exporter modules dns name dns4 preferred-ip-
↪protocol ipv4
set service monitoring prometheus blackbox-exporter modules dns name dns4 query-name_
↪vyos.io
set service monitoring prometheus blackbox-exporter modules dns name dns4 query-type A
```

ICMP module example:

```
set service monitoring prometheus blackbox-exporter modules icmp name ping6 preferred-ip-
↪protocol ipv6
set service monitoring prometheus blackbox-exporter modules icmp name ping6 ip-protocol-
↪fallback
set service monitoring prometheus blackbox-exporter modules icmp name ping6 timeout 3
```

7.10.15 NTP

NTP (NETWORK TIME PROTOCOL) is a networking protocol for clock synchronization between computer systems over packet-switched, variable-latency data networks. In operation since before 1985, NTP is one of the oldest Internet protocols in current use.

NTP is intended to synchronize all participating computers to within a few milliseconds of UTC (Coordinated Universal Time). It uses the intersection algorithm, a modified version of Marzullo's algorithm, to select accurate time servers and is designed to mitigate the effects of variable network latency. NTP can usually maintain time to within tens of milliseconds over the public Internet, and can achieve better than one millisecond accuracy in local area networks under ideal conditions. Asymmetric routes and network congestion can cause errors of 100 ms or more.

The protocol is usually described in terms of a client-server model, but can as easily be used in peer-to-peer relationships where both peers consider the other to be a potential time source. Implementations send and receive timestamps using UDP on port number 123.

NTP supplies a warning of any impending leap second adjustment, but no information about local time zones or daylight saving time is transmitted.

The current protocol is version 4 (NTPv4), which is a proposed standard as documented in [RFC 5905](#). It is backward compatible with version 3, specified in [RFC 1305](#).

Note: VyOS 1.4 uses chrony instead of ntpd (see [T3008](#)) which will no longer accept anonymous NTP requests as in VyOS 1.3. All configurations will be migrated to keep the anonymous functionality. For new setups if you have clients using your VyOS installation as NTP server, you must specify the `allow-client` directive.

Configuration

set service ntp server <address>

Configure one or more servers for synchronisation. Server name can be either an IP address or FQDN.

There are 3 default NTP server set. You are able to change them.

- time1.vyos.net
- time2.vyos.net
- time3.vyos.net

set service ntp server <address> <noselect | nts | pool | prefer | ptp | interleave>

Configure one or more attributes to the given NTP server.

- **noselect** marks the server as unused, except for display purposes. The server is discarded by the selection algorithm.
- **nts** enables Network Time Security (NTS) for the server as specified in [RFC 8915](#)
- **pool** mobilizes persistent client mode association with a number of remote servers.
- **prefer** marks the server as preferred. All other things being equal, this host will be chosen for synchronization among a set of correctly operating hosts.
- **ptp** enables the PTP transport for this server (see *PTP Transport of NTP Packets*).
- **interleave** enables NTP interleaved mode (see [draft-ntp-interleaved-modes](#)), which can improve synchronization accuracy and stability when supported by both parties.

set service ntp listen-address <address>

NTP process will only listen on the specified IP address. You must specify the <address> and optionally the permitted clients. Multiple listen addresses for same IP family is no longer supported. Only one IPv4 and one IPv6 address can be configured, using separate commands for each.

set service ntp allow-client address <address>

List of networks or client addresses permitted to contact this NTP server.

Multiple networks/client IP addresses can be configured.

set service ntp vrf <name>

Specify name of the VRF instance.

set service ntp leap-second [ignore|smear|system|timezone]

Define how to handle leap-seconds.

- **ignore**: No correction is applied to the clock for the leap second. The clock will be corrected later in normal operation when new measurements are made and the estimated offset includes the one second error.
- **smear**: When smearing a leap second, the leap status is suppressed on the server and the served time is corrected slowly by slewing instead of stepping. The clients do not need any special configuration as they do not know there is any leap second and they follow the server time which eventually brings them back to UTC. Care must be taken to ensure they use only NTP servers which smear the leap second in exactly the same way for synchronisation.
- **system**: When inserting a leap second, the kernel steps the system clock backwards by one second when the clock gets to 00:00:00 UTC. When deleting a leap second, it steps forward by one second when the clock gets to 23:59:59 UTC.

- **timezone:** This directive specifies a timezone in the system timezone database which chronyd can use to determine when will the next leap second occur and what is the current offset between TAI and UTC. It will periodically check if 23:59:59 and 23:59:60 are valid times in the timezone. This normally works with the right/UTC timezone which is the default

Hardware Timestamping of NTP Packets

The chrony daemon on VyOS can leverage NIC hardware capabilities to record the exact time packets are received on the interface, as well as when packets were actually transmitted. This provides improved accuracy and stability when the system is under load, as queuing and OS context switching can introduce a variable delay between when the packet is received on the network and when it is actually processed by the NTP daemon.

Hardware timestamping depends on NIC support. Some NICs can be configured to apply timestamps to any incoming packet, while others only support applying timestamps to specific protocols (e.g. PTP).

When timestamping is enabled on an interface, chrony's default behavior is to try to configure the interface to only timestamp NTP packets. If this mode is not supported, chrony will attempt to set it to timestamp all packets. If neither option is supported (e.g. the NIC can only timestamp received PTP packets), chrony will leverage timestamping on transmitted packets only, which still provides some benefit.

set service ntp timestamp interface <interface>

Configures hardware timestamping on the interface <interface>. The special value `all` can also be specified to enable timestamping on all interfaces that support it.

Configure the timestamping behavior with the following option:

- **receive-filter** [`all|ntp|ptp|none`] selects the receive filter mode, which controls which inbound packets the NIC applies timestamps to. The selected mode must be supported by the NIC, or timestamping will be disabled for the interface.

The following **receive-filter** modes can be selected:

- **all:** All received packets will be timestamped.
- **ntp:** Only received NTP protocol packets will be timestamped.
- **ptp:** Only received PTP protocol packets will be timestamped. Combined with the PTP transport for NTP packets, this can be leveraged to take advantage of hardware timestamping on NICs that only support the ptp filter mode.
- **none:** No received packets will be timestamped. Hardware timestamping of transmitted packets will still be leveraged, if supported by the NIC.

PTP Transport of NTP Packets

The Precision Time Protocol (IEEE 1588) is a local network time synchronization protocol that provides high precision time synchronization by leveraging hardware clocks in NICs and other network elements. VyOS does not currently support standards-based PTP, which can be deployed independently of NTP.

For networks consisting of VyOS and other Linux systems running relatively recent versions of the chrony daemon, NTP packets can be “tunneled” over PTP. NTP over PTP provides the best of both worlds, leveraging hardware support for timestamping PTP packets while retaining the configuration flexibility and fault tolerance of NTP.

set service ntp ptp

Enables the NTP daemon PTP transport. The NTP daemon will listen on the configured PTP port. Note that one or more servers must be individually enabled for PTP before the daemon will synchronize over the transport.

set service ntp ptp port <port>

Configures the PTP port. By default, the standard port 319 is used.

7.10.16 PPPoE Server

VyOS utilizes `accel-ppp` to provide PPPoE server functionality. It can be used with local authentication or a connected RADIUS server.

Note: Please be aware, due to an upstream bug, config changes/commits will restart the ppp daemon and will reset existing PPPoE connections from connected users, in order to become effective.

Configuring PPPoE Server

```
set service pppoe-server access-concentrator PPPoE-Server
set service pppoe-server authentication mode local
set service pppoe-server authentication local-users username test password 'test'
set service pppoe-server client-ip-pool PPPoE-POOL range 192.168.255.2-192.168.255.254
set service pppoe-server default-pool 'PPPoE-POOL'
set service pppoe-server gateway-address 192.168.255.1
set service pppoe-server interface eth0
```

set service pppoe-server access-concentrator <name>

Use this command to set a name for this PPPoE-server access concentrator.

set service pppoe-server authentication mode <local | radius>

Set authentication backend. The configured authentication backend is used for all queries.

- **radius:** All authentication queries are handled by a configured RADIUS server.
- **local:** All authentication queries are handled locally.
- **noauth:** Authentication disabled.

set service pppoe-server authentication local-users username <name> password <password>

Create <user> for local authentication on this system. The users password will be set to <pass>.

set service pppoe-server client-ip-pool <POOL-NAME> range <x.x.x.x-x.x.x.x | x.x.x.x/x>

Use this command to define the first IP address of a pool of addresses to be given to pppoe clients. If notation `x.x.x.x-x.x.x.x`, it must be within a /24 subnet. If notation `x.x.x.x/x` is used there is possibility to set host/netmask.

set service pppoe-server default-pool <POOL-NAME>

Use this command to define default address pool name.

set service pppoe-server interface <interface>

Use this command to define the interface the PPPoE server will use to listen for PPPoE clients.

set service pppoe-server gateway-address <address>

Specifies single <gateway> IP address to be used as local address of PPP interfaces.

Configuring RADIUS authentication

To enable RADIUS based authentication, the authentication mode needs to be changed within the configuration. Previous settings like the local users, still exists within the configuration, however they are not used if the mode has been changed from local to radius. Once changed back to local, it will use all local accounts again.

```
set service pppoe-server authentication mode radius
```

```
set service pppoe-server authentication radius server <server> key <secret>
```

Configure RADIUS <server> and its required shared <secret> for communicating with the RADIUS server.

Since the RADIUS server would be a single point of failure, multiple RADIUS servers can be setup and will be used subsequently. For example:

```
set service pppoe-server authentication radius server 10.0.0.1 key 'foo'
set service pppoe-server authentication radius server 10.0.0.2 key 'foo'
```

Note: Some RADIUS servers use an access control list which allows or denies queries, make sure to add your VyOS router to the allowed client list.

RADIUS source address

If you are using OSPF as IGP, always the closest interface connected to the RADIUS server is used. With VyOS 1.2 you can bind all outgoing RADIUS requests to a single source IP e.g. the loopback interface.

```
set service pppoe-server authentication radius source-address <address>
```

Source IPv4 address used in all RADIUS server queries.

Note: The source-address must be configured on one of VyOS interface. Best practice would be a loopback or dummy interface.

RADIUS advanced options

```
set service pppoe-server authentication radius server <server> port <port>
```

Configure RADIUS <server> and its required port for authentication requests.

```
set service pppoe-server authentication radius server <server> fail-time <time>
```

Mark RADIUS server as offline for this given <time> in seconds.

```
set service pppoe-server authentication radius server <server> disable
```

Temporary disable this RADIUS server.

```
set service pppoe-server authentication radius acct-timeout <timeout>
```

Timeout to wait reply for Interim-Update packets. (default 3 seconds)

```
set service pppoe-server authentication radius dynamic-author server <address>
```

Specifies IP address for Dynamic Authorization Extension server (DM/CoA). This IP must exist on any VyOS interface or it can be 0.0.0.0.

```
set service pppoe-server authentication radius dynamic-author port <port>
```

UDP port for Dynamic Authorization Extension server (DM/CoA)

```
set service pppoe-server authentication radius dynamic-author key <secret>
```

Secret for Dynamic Authorization Extension server (DM/CoA)

```
set service pppoe-server authentication radius max-try <number>
```

Maximum number of tries to send Access-Request/Accounting-Request queries

```
set service pppoe-server authentication radius timeout <timeout>
```

Timeout to wait response from server (seconds)

```
set service pppoe-server authentication radius nas-identifier <identifier>
```

Value to send to RADIUS server in NAS-Identifier attribute and to be matched in DM/CoA requests.

```
set service pppoe-server authentication radius nas-ip-address <address>
```

Value to send to RADIUS server in NAS-IP-Address attribute and to be matched in DM/CoA requests. Also DM/CoA server will bind to that address.

```
set service pppoe-server authentication radius source-address <address>
```

Source IPv4 address used in all RADIUS server queries.

```
set service pppoe-server authentication radius rate-limit attribute <attribute>
```

Specifies which RADIUS server attribute contains the rate limit information. The default attribute is Filter-Id.

Note: If you set a custom RADIUS attribute you must define it on both dictionaries at RADIUS server and client.

```
set service pppoe-server authentication radius rate-limit enable
```

Enables bandwidth shaping via RADIUS.

```
set service pppoe-server authentication radius rate-limit vendor
```

Specifies the vendor dictionary, dictionary needs to be in /usr/share/accel-ppp/radius.

Received RADIUS attributes have a higher priority than parameters defined within the CLI configuration, refer to the explanation below.

Allocation clients ip addresses by RADIUS

If the RADIUS server sends the attribute Framed-IP-Address then this IP address will be allocated to the client and the option default-pool within the CLI config is being ignored.

If the RADIUS server sends the attribute Framed-Pool, IP address will be allocated from a predefined IP pool whose name equals the attribute value.

If the RADIUS server sends the attribute Stateful-IPv6-Address-Pool, IPv6 address will be allocated from a predefined IPv6 pool prefix whose name equals the attribute value.

If the RADIUS server sends the attribute Delegated-IPv6-Prefix-Pool, IPv6 delegation prefix will be allocated from a predefined IPv6 pool delegate whose name equals the attribute value.

Note: Stateful-IPv6-Address-Pool and Delegated-IPv6-Prefix-Pool are defined in RFC6911. If they are not defined in your RADIUS server, add new [dictionary](#).

User interface can be put to VRF context via RADIUS Access-Accept packet, or change it via RADIUS CoA. Accel-VRF-Name is used from these purposes. It is custom [ACCEL-PPP attribute](#). Define it in your RADIUS server.

Renaming clients interfaces by RADIUS

If the RADIUS server uses the attribute NAS-Port-Id, ppp tunnels will be renamed.

Note: The value of the attribute NAS-Port-Id must be less than 16 characters, otherwise the interface won't be renamed.

Automatic VLAN Creation

set service pppoe-server interface <interface> vlan <id | range>

VLAN's can be created by Accel-ppp on the fly via the use of a Kernel module named `vlan_mon`, which is monitoring incoming vlans and creates the necessary VLAN if required and allowed. VyOS supports the use of either VLAN ID's or entire ranges, both values can be defined at the same time for an interface.

When configured, PPPoE will create the necessary VLANs when required. Once the user session has been cancelled and the VLAN is not needed anymore, VyOS will remove it again.

```
set service pppoe-server interface eth3 vlan 100
set service pppoe-server interface eth3 vlan 200
set service pppoe-server interface eth3 vlan 500-1000
set service pppoe-server interface eth3 vlan 2000-3000
```

Bandwidth Shaping

Bandwidth rate limits can be set for local users or RADIUS based attributes.

For Local Users

set service pppoe-server authentication local-users username <user> rate-limit download <bandwidth>

Download bandwidth limit in kbit/s for <user>.

set service pppoe-server authentication local-users username <user> rate-limit upload <bandwidth>

Upload bandwidth limit in kbit/s for <user>.

```
set service pppoe-server access-concentrator 'ACN'
set service pppoe-server authentication local-users username foo password 'bar'
set service pppoe-server authentication local-users username foo rate-limit download
→ '20480'
```

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```

set service pppoe-server authentication local-users username foo rate-limit upload '10240
↪ '
set service pppoe-server authentication mode 'local'
set service pppoe-server client-ip-pool IP-POOL range '10.1.1.100/24'
set service pppoe-server default-pool 'IP-POOL'
set service pppoe-server name-server '10.100.100.1'
set service pppoe-server name-server '10.100.200.1'
set service pppoe-server interface 'eth1'
set service pppoe-server gateway-address '10.1.1.2'

```

Once the user is connected, the user session is using the set limits and can be displayed via `show pppoe-server sessions`.

```

show pppoe-server sessions
ifname | username | ip | calling-sid | rate-limit | state | uptime |
↪ rx-bytes | tx-bytes
-----+-----+-----+-----+-----+-----+-----+-----+
↪ -----+-----+
ppp0 | foo | 10.1.1.100 | 00:53:00:ba:db:15 | 20480/10240 | active | 00:00:11 |
↪ 214 B | 76 B

```

For RADIUS users

The current attribute `Filter-Id` is being used as default and can be setup within RADIUS:

`Filter-Id=2000/3000` (means 2000Kbit down-stream rate and 3000Kbit up-stream rate)

The command below enables it, assuming the RADIUS connection has been setup and is working.

set service pppoe-server authentication radius rate-limit enable

Use this command to enable bandwidth shaping via RADIUS.

Other attributes can be used, but they have to be in one of the dictionaries in `/usr/share/accel-ppp/radius`.

Load Balancing

set service pppoe-server pado-delay <number-of-ms> sessions <number-of-sessions>

Use this command to enable the delay of PADO (PPPoE Active Discovery Offer) packets, which can be used as a session balancing mechanism with other PPPoE servers.

```

set service pppoe-server pado-delay 50 sessions '500'
set service pppoe-server pado-delay 100 sessions '1000'
set service pppoe-server pado-delay 300 sessions '3000'

```

In the example above, the first 499 sessions connect without delay. PADO packets will be delayed 50 ms for connection from 500 to 999, this trick allows other PPPoE servers send PADO faster and clients will connect to other servers. Last command says that this PPPoE server can serve only 3000 clients.

IPv6

```
set service pppoe-server ppp-options ipv6 <require | prefer | allow | deny>
```

Specifies IPv6 negotiation preference.

- **require** - Require IPv6 negotiation
- **prefer** - Ask client for IPv6 negotiation, do not fail if it rejects
- **allow** - Negotiate IPv6 only if client requests
- **deny** - Do not negotiate IPv6 (default value)

```
set service pppoe-server client-ipv6-pool <IPv6-POOL-NAME> prefix <address> mask <number-of-bits>
```

Use this command to set the IPv6 address pool from which a PPPoE client will get an IPv6 prefix of your defined length (mask) to terminate the PPPoE endpoint at their side. The mask length can be set from 48 to 128 bit long, the default value is 64.

```
set service pppoe-server client-ipv6-pool <IPv6-POOL-NAME> delegate <address> delegation-prefix <number-of-bits>
```

Use this command to configure DHCPv6 Prefix Delegation (RFC3633) on PPPoE. You will have to set your IPv6 pool and the length of the delegation prefix. From the defined IPv6 pool you will be handing out networks of the defined length (delegation-prefix). The length of the delegation prefix can be set from 32 to 64 bit long.

```
set service pppoe-server default-ipv6-pool <IPv6-POOL-NAME>
```

Use this command to define default IPv6 address pool name.

```
set service pppoe-server ppp-options ipv6 allow
set service pppoe-server client-ipv6-pool IPv6-POOL delegate '2001:db8:8003::/48'
↪delegation-prefix '56'
set service pppoe-server client-ipv6-pool IPv6-POOL prefix '2001:db8:8002::/48' mask '64'
set service pppoe-server default-ipv6-pool IPv6-POOL
```

IPv6 Advanced Options

```
set service pppoe-server ppp-options ipv6-accept-peer-interface-id
```

Accept peer interface identifier. By default is not defined.

```
set service pppoe-server ppp-options ipv6-interface-id <random | x:x:x:x>
```

Specifies fixed or random interface identifier for IPv6. By default is fixed.

- **random** - Random interface identifier for IPv6
- **x:x:x:x** - Specify interface identifier for IPv6

```
set service pppoe-server ppp-options ipv6-interface-id <random | x:x:x:x>
```

Specifies peer interface identifier for IPv6. By default is fixed.

- **random** - Random interface identifier for IPv6
- **x:x:x:x** - Specify interface identifier for IPv6
- **ipv4-addr** - Calculate interface identifier from IPv4 address.
- **calling-sid** - Calculate interface identifier from calling-station-id.

Scripting

set service pppoe-server extended-scripts on-change <path_to_script>

Script to run when session interface changed by RADIUS CoA handling

set service pppoe-server extended-scripts on-down <path_to_script>

Script to run when session interface going to terminate

set service pppoe-server extended-scripts on-pre-up <path_to_script>

Script to run before session interface comes up

set service pppoe-server extended-scripts on-up <path_to_script>

Script to run when session interface is completely configured and started

Advanced Options

Authentication Advanced Options

set service pppoe-server authentication local-users username <user> disable

Disable <user> account.

set service pppoe-server authentication local-users username <user> static-ip <address>

Assign static IP address to <user> account.

set service pppoe-server authentication protocols <pap | chap | mschap | mschap-v2>

Require the peer to authenticate itself using one of the following protocols: pap, chap, mschap, mschap-v2.

Client IP Pool Advanced Options

set service pppoe-server client-ip-pool <POOL-NAME> next-pool <NEXT-POOL-NAME>

Use this command to define the next address pool name.

PPP Advanced Options

set service pppoe-server ppp-options disable-ccp

Disable Compression Control Protocol (CCP). CCP is enabled by default.

set service pppoe-server ppp-options interface-cache <number>

Specifies number of interfaces to keep in cache. It means that don't destroy interface after corresponding session is destroyed, instead place it to cache and use it later for new sessions repeatedly. This should reduce kernel-level interface creation/deletion rate lack. Default value is 0.

set service pppoe-server ppp-options ipv4 <require | prefer | allow | deny>

Specifies IPv4 negotiation preference.

- **require** - Require IPv4 negotiation
- **prefer** - Ask client for IPv4 negotiation, do not fail if it rejects
- **allow** - Negotiate IPv4 only if client requests (Default value)

- **deny** - Do not negotiate IPv4

set service pppoe-server ppp-options lcp-echo-failure <number>

Defines the maximum <number> of unanswered echo requests. Upon reaching the value <number>, the session will be reset. Default value is **3**.

set service pppoe-server ppp-options lcp-echo-interval <interval>

If this option is specified and is greater than 0, then the PPP module will send LCP pings of the echo request every <interval> seconds. Default value is **30**.

set service pppoe-server ppp-options lcp-echo-timeout

Specifies timeout in seconds to wait for any peer activity. If this option specified it turns on adaptive lcp echo functionality and “lcp-echo-failure” is not used. Default value is **0**.

set service pppoe-server ppp-options min-mtu <number>

Defines minimum acceptable MTU. If client will try to negotiate less then specified MTU then it will be NAKed or disconnected if rejects greater MTU. Default value is **100**.

set service pppoe-server ppp-options mppe <require | prefer | deny>

Specifies MPPE (Microsoft Point-to-Point Encryption) negotiation preference.

- **require** - ask client for mppe, if it rejects drop connection
- **prefer** - ask client for mppe, if it rejects don't fail. (Default value)
- **deny** - deny mppe

Default behavior - don't ask client for mppe, but allow it if client wants. Please note that RADIUS may override this option by MS-MPPE-Encryption-Policy attribute.

set service pppoe-server ppp-options mru <number>

Defines preferred MRU. By default is not defined.

Global Advanced options

set service pppoe-server description <description>

Set description.

set service pppoe-server limits burst <value>

Burst count

set service pppoe-server limits connection-limit <value>

Acceptable rate of connections (e.g. 1/min, 60/sec)

set service pppoe-server limits timeout <value>

Timeout in seconds

set service pppoe-server mtu

Maximum Transmission Unit (MTU) (default: **1492**)

set service pppoe-server max-concurrent-sessions

Maximum number of concurrent session start attempts

set service pppoe-server name-server <address>

Connected client should use <address> as their DNS server. This command accepts both IPv4 and IPv6 addresses. Up to two nameservers can be configured for IPv4, up to three for IPv6.

set service pppoe-server service-name <names>

Specifies Service-Name to respond. If absent any Service-Name is acceptable and client's Service-Name will be sent back. Also possible set multiple service-names: sn1, sn2, sn3

Per default the user session is being replaced if a second authentication request succeeds. Such session requests can be either denied or allowed entirely, which would allow multiple sessions for a user in the latter case. If it is denied, the second session is being rejected even if the authentication succeeds, the user has to terminate its first session and can then authentication again.

set service pppoe-server session-control

- **disable**: Disables session control.
- **deny**: Deny second session authorization.
- **replace**: Terminate first session when second is authorized (**default**)

set service pppoe-server shaper fwmark <1-2147483647>

Match firewall mark value

set service pppoe-server snmp master-agent

Enable SNMP

set service pppoe-server wins-server <address>

Windows Internet Name Service (WINS) servers propagated to client

Monitoring

show pppoe-server sessions

Use this command to locally check the active sessions in the PPPoE server.

```
show pppoe-server sessions
ifname | username | ip | calling-sid | rate-limit | state | uptime |
↳ rx-bytes | tx-bytes
-----+-----+-----+-----+-----+-----+-----+
↳ -----+-----+-----+-----+-----+-----+-----+
ppp0 | foo | 10.1.1.100 | 00:53:00:ba:db:15 | 20480/10240 | active | 00:00:11 |
↳ 214 B | 76 B
```

Examples

IPv4

The example below uses ACN as access-concentrator name, assigns an address from the pool 10.1.1.100-111, terminates at the local endpoint 10.1.1.1 and serves requests only on eth1.

```
set service pppoe-server access-concentrator 'ACN'
set service pppoe-server authentication local-users username foo password 'bar'
set service pppoe-server authentication mode 'local'
set service pppoe-server client-ip-pool IP-POOL range '10.1.1.100-10.1.1.111'
```

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```
set service pppoe-server default-pool 'IP-POOL'
set service pppoe-server interface eth1
set service pppoe-server gateway-address '10.1.1.2'
set service pppoe-server name-server '10.100.100.1'
set service pppoe-server name-server '10.100.200.1'
```

Dual-Stack IPv4/IPv6 provisioning with Prefix Delegation

The example below covers a dual-stack configuration.

```
set service pppoe-server authentication local-users username test password 'test'
set service pppoe-server authentication mode 'local'
set service pppoe-server client-ip-pool IP-POOL range '192.168.0.1/24'
set service pppoe-server default-pool 'IP-POOL'
set service pppoe-server client-ipv6-pool IPv6-POOL delegate '2001:db8:8003::/48'
↳ delegation-prefix '56'
set service pppoe-server client-ipv6-pool IPv6-POOL prefix '2001:db8:8002::/48' mask '64'
set service pppoe-server default-ipv6-pool IPv6-POOL
set service pppoe-server ppp-options ipv6 allow
set service pppoe-server name-server '10.1.1.1'
set service pppoe-server name-server '2001:db8:4860::8888'
set service pppoe-server interface 'eth2'
set service pppoe-server gateway-address '10.100.100.1'
```

The client, once successfully authenticated, will receive an IPv4 and an IPv6 /64 address to terminate the PPPoE endpoint on the client side and a /56 subnet for the clients internal use.

```
vyos@pppoe-server:~$ sh pppoe-server sessions
ifname | username | ip | ip6 | ip6-dp |
↳ calling-sid | rate-limit | state | uptime | rx-bytes | tx-bytes
-----+-----+-----+-----+-----+-----
↳ -----+-----+-----+-----+-----+-----
ppp0 | test | 192.168.0.1 | 2001:db8:8002:0:200::/64 | 2001:db8:8003::1/56 |
↳ 00:53:00:12:42:eb | | active | 00:00:49 | 875 B | 2.1 KiB
```

7.10.17 Router Advertisements

RAs (Router advertisements) are described in [RFC 4861#section-4.6.2](#). They are part of what is known as SLAAC.

Supported interface types:

- bonding
- bridge
- ethernet
- geneve
- l2tpv3
- openvpn
- pseudo-ethernet

- tunnel
- vxlan
- wireguard
- wireless
- wwan

Configuration

set service router-advert interface <interface> ...

Field	VyOS Option	Description
Cur Hop Limit	hop-limit	Hop count field of the outgoing RA packets
“Managed address configuration” flag	managed-flag	Tell hosts to use the administered stateful protocol (i.e. DHCP) for autoconfiguration
“Other configuration” flag	other-config-flag	Tell hosts to use the administered (stateful) protocol (i.e. DHCP) for autoconfiguration of other (non-address) information
MTU	link-mtu	Link MTU value placed in RAs, excluded in RAs if unset
Router Lifetime	default-lifetime	Lifetime associated with the default router in units of seconds
Reachable Time	reachable-time	Time, in milliseconds, that a node assumes a neighbor is reachable after having received a reachability confirmation
Retransmit Timer	retrans-timer	Time in milliseconds between retransmitted Neighbor Solicitation messages
Default Router Preference	default-preference	Preference associated with the default router
Interval	interval	Min and max intervals between unsolicited multicast RAs
DNSSL	dnssl	DNS search list to advertise
Name Server	name-server	Advertise DNS server per https://tools.ietf.org/html/rfc6106
Auto Ignore Prefix	auto-ignore	Exclude a prefix from being advertised when the wildcard <code>::/64</code> prefix is used
Captive Portal	captive-portal	Advertise a URL pointing to an RFC 8908-compliant API to tell hosts that they are behind a captive portal

Advertising a Prefix

set service router-advert interface <interface> prefix <prefix/mask>

Note: You can also opt for using `::/64` as prefix for your RAs. This is a special wildcard prefix that will emit RAs for every prefix assigned to the interface. This comes in handy when using dynamically obtained prefixes from DHCPv6-PD.

VyOS Field	Description
decrement-lifetime	Lifetime is decremented by the number of seconds since the last RA - use in conjunction with a DHCPv6-PD prefix
deprecate-prefix	Upon shutdown, this option will deprecate the prefix by announcing it in the shutdown RA
no-autonomous-flag	Prefix can not be used for stateless address auto-configuration
no-on-link-flag	Prefix can not be used for on-link determination
preferred-lifetime	Time in seconds that the prefix will remain preferred (default 4 hours)
valid-lifetime	Time in seconds that the prefix will remain valid (default: 30 days)

Advertising a NAT64 Prefix

set service router-advert interface <interface> nat64prefix <prefix/mask>

Enable PREF64 option as outlined in [RFC 8781](#).

NAT64 prefix mask must be one of: /32, /40, /48, /56, /64 or 96.

Note: The well known NAT64 prefix is 64:ff9b::/96

VyOS Field	Description
valid-lifetime	Time in seconds that the prefix will remain valid (default: 65528 seconds)

Disabling Advertisements

To disable advertisements without deleting the configuration:

set service router-advert interface <interface> no-send-advert

If set, the router will no longer send periodic router advertisements and will not respond to router solicitations.

set service router-advert interface <interface> no-send-interval

Advertisement Interval Option (specified by Mobile IPv6) is always included in Router Advertisements unless this option is set.

Example

Your LAN connected on eth0 uses prefix 2001:db8:beef:2::/64 with the router being 2001:db8:beef:2::1

```
set interfaces ethernet eth0 address 2001:db8:beef:2::1/64

set service router-advert interface eth0 default-preference 'high'
set service router-advert interface eth0 name-server '2001:db8::1'
set service router-advert interface eth0 name-server '2001:db8::2'
set service router-advert interface eth0 other-config-flag
set service router-advert interface eth0 prefix 2001:db8:beef:2::/64
```

7.10.18 SNMP

SNMP is an Internet Standard protocol for collecting and organizing information about managed devices on IP networks and for modifying that information to change device behavior. Devices that typically support SNMP include cable modems, routers, switches, servers, workstations, printers, and more.

SNMP is widely used in network management for network monitoring. SNMP exposes management data in the form of variables on the managed systems organized in a management information base (MIB) which describe the system status and configuration. These variables can then be remotely queried (and, in some circumstances, manipulated) by managing applications.

Three significant versions of SNMP have been developed and deployed. SNMPv1 is the original version of the protocol. More recent versions, SNMPv2c and SNMPv3, feature improvements in performance, flexibility and security.

SNMP is a component of the Internet Protocol Suite as defined by the Internet Engineering Task Force (IETF). It consists of a set of standards for network management, including an application layer protocol, a database schema, and a set of data objects.

Overview and basic concepts

In typical uses of SNMP, one or more administrative computers called managers have the task of monitoring or managing a group of hosts or devices on a computer network. Each managed system executes a software component called an agent which reports information via SNMP to the manager.

An SNMP-managed network consists of three key components:

- Managed devices
- Agent - software which runs on managed devices
- Network management station (NMS) - software which runs on the manager

A managed device is a network node that implements an SNMP interface that allows unidirectional (read-only) or bidirectional (read and write) access to node-specific information. Managed devices exchange node-specific information with the NMSs. Sometimes called network elements, the managed devices can be any type of device, including, but not limited to, routers, access servers, switches, cable modems, bridges, hubs, IP telephones, IP video cameras, computer hosts, and printers.

An agent is a network-management software module that resides on a managed device. An agent has local knowledge of management information and translates that information to or from an SNMP-specific form.

A network management station executes applications that monitor and control managed devices. NMSs provide the bulk of the processing and memory resources required for network management. One or more NMSs may exist on any managed network.

Fig. 5: Image thankfully borrowed from https://en.wikipedia.org/wiki/File:SNMP_communication_principles_diagram.PNG which is under the GNU Free Documentation License

Note: VyOS SNMP supports both IPv4 and IPv6.

SNMP Protocol Versions

VyOS itself supports [SNMPv2](#) (version 2) and [SNMPv3](#) (version 3) where the later is recommended because of improved security (optional authentication and encryption).

SNMPv2

SNMPv2 is the original and most commonly used version. For authorizing clients, SNMP uses the concept of communities. Communities may have authorization set to read only (this is most common) or to read and write (this option is not actively used in VyOS).

SNMP can work synchronously or asynchronously. In synchronous communication, the monitoring system queries the router periodically. In asynchronous, the router sends notification to the “trap” (the monitoring host).

SNMPv2 does not support any authentication mechanisms, other than client source address, so you should specify addresses of clients allowed to monitor the router. Note that SNMPv2 also supports no encryption and always sends data in plain text.

Example

```
# Define a community
set service snmp community routers authorization ro

# Allow monitoring access from the entire network
set service snmp community routers network 192.0.2.0/24
set service snmp community routers network 2001::db8:ffff:eeee::/64

# Allow monitoring access from specific addresses
set service snmp community routers client 203.0.113.10
set service snmp community routers client 203.0.113.20

# Define optional router information
set service snmp location "UK, London"
set service snmp contact "admin@example.com"

# Trap target if you want asynchronous communication
set service snmp trap-target 203.0.113.10

# Listen only on specific IP addresses (port defaults to 161)
set service snmp listen-address 172.16.254.36 port 161
set service snmp listen-address 2001::db8::f00::1
```

SNMPv3

SNMPv3 (version 3 of the SNMP protocol) introduced a whole slew of new security related features that have been missing from the previous versions. Security was one of the biggest weakness of SNMP until v3. Authentication in SNMP Versions 1 and 2 amounts to nothing more than a password (community string) sent in clear text between a manager and agent. Each SNMPv3 message contains security parameters which are encoded as an octet string. The meaning of these security parameters depends on the security model being used.

The security approach in SNMPv3 targets:

- Confidentiality – Encryption of packets to prevent snooping by an unauthorized source.
- Integrity – Message integrity to ensure that a packet has not been tampered while in transit including an optional packet replay protection mechanism.
- Authentication – to verify that the message is from a valid source.

Example

- Let SNMP daemon listen only on IP address 192.0.2.1
- Configure new SNMP user named “vyos” with password “vyos12345678”
- New user will use SHA/AES for authentication and privacy

```
set service snmp listen-address 192.0.2.1
set service snmp location 'VyOS Datacenter'
set service snmp v3 engineid '000000000000000000000002'
set service snmp v3 group default mode 'ro'
set service snmp v3 group default view 'default'
set service snmp v3 user vyos auth plaintext-password 'vyos12345678'
set service snmp v3 user vyos auth type 'sha'
set service snmp v3 user vyos group 'default'
set service snmp v3 user vyos privacy plaintext-password 'vyos12345678'
set service snmp v3 user vyos privacy type 'aes'
set service snmp v3 view default oid 1
```

After commit the plaintext passwords will be hashed and stored in your configuration. The resulting CLI config will look like:

```
vyos@vyos# show service snmp
listen-address 192.0.2.1 {
}
location "VyOS Datacenter"
v3 {
  engineid 000000000000000000000002
  group default {
    mode ro
    view default
  }
  user vyos {
    auth {
      encrypted-password 4e52fe55fd011c9c51ae2c65f4b78ca93dcafdfe
      type sha
    }
  }
}
```

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```

group default
privacy {
    encrypted-password 4e52fe55fd011c9c51ae2c65f4b78ca93dcafdfe
    type aes
}
}
view default {
    oid 1 {
    }
}
}
}

```

You can test the SNMPv3 functionality from any linux based system, just run the following command: `snmpwalk -v 3 -u vyos -a SHA -A vyos12345678 -x AES -X vyos12345678 -l authPriv 192.0.2.1 .1`

VyOS MIBs

All SNMP MIBs are located in each image of VyOS here: `/usr/share/snmp/mibs/`

You are be able to download the files using SCP, once the SSH service has been activated like so

```
scp -r vyos@your_router:/usr/share/snmp/mibs /your_folder/mibs
```

SNMP Extensions

To extend SNMP agent functionality, custom scripts can be executed every time the agent is being called. This can be achieved by using arbitrary `extensioncommands`. The first step is to create a functional script of course, then upload it to your VyOS instance via the command `scp your_script.sh vyos@your_router:/config/user-data`. Once the script is uploaded, it needs to be configured via the command below.

```
set service snmp script-extensions extension-name my-extension script your_script.sh
commit
```

The OID `.1.3.6.1.4.1.8072.1.3.2.3.1.1.4.116.101.115.116`, once called, will contain the output of the extension.

```

root@vyos:/home/vyos# snmpwalk -v2c -c public 127.0.0.1 nsExtendOutput1
NET-SNMP-EXTEND-MIB::nsExtendOutput1Line."my-extension" = STRING: hello
NET-SNMP-EXTEND-MIB::nsExtendOutputFull."my-extension" = STRING: hello
NET-SNMP-EXTEND-MIB::nsExtendOutNumLines."my-extension" = INTEGER: 1
NET-SNMP-EXTEND-MIB::nsExtendResult."my-extension" = INTEGER: 0

```


SolarWinds

If you happen to use SolarWinds Orion as NMS you can also use the Device Templates Management. A template for VyOS can be easily imported.

Create a file named VyOS-1.3.6.1.4.1.44641.ConfigMgmt-Commands using the following content:

```
<Configuration-Management Device="VyOS" SystemOID="1.3.6.1.4.1.44641">
  <Commands>
    <Command Name="Reset" Value="set terminal width 0${CRLF}set terminal length 0"/>
    <Command Name="Reboot" Value="reboot${CRLF}Yes"/>
    <Command Name="EnterConfigMode" Value="configure"/>
    <Command Name="ExitConfigMode" Value="commit${CRLF}exit"/>
    <Command Name="DownloadConfig" Value="show configuration commands"/>
    <Command Name="SaveConfig" Value="commit${CRLF}save"/>
    <Command Name="Version" Value="show version"/>
    <Command Name="MenuBased" Value="False"/>
    <Command Name="VirtualPrompt" Value=":~"/>
  </Commands>
</Configuration-Management>
```

7.10.19 SSH

SSH (Secure Shell) is a cryptographic network protocol for operating network services securely over an unsecured network. The standard TCP port for SSH is 22. The best known example application is for remote login to computer systems by users.

SSH provides a secure channel over an unsecured network in a client-server architecture, connecting an SSH client application with an SSH server. Common applications include remote command-line login and remote command execution, but any network service can be secured with SSH. The protocol specification distinguishes between two major versions, referred to as SSH-1 and SSH-2.

The most visible application of the protocol is for access to shell accounts on Unix-like operating systems, but it sees some limited use on Windows as well. In 2015, Microsoft announced that they would include native support for SSH in a future release.

SSH was designed as a replacement for Telnet and for unsecured remote shell protocols such as the Berkeley rlogin, rsh, and rexec protocols. Those protocols send information, notably passwords, in plaintext, rendering them susceptible to interception and disclosure using packet analysis. The encryption used by SSH is intended to provide confidentiality and integrity of data over an unsecured network, such as the Internet.

Note: VyOS 1.1 supported login as user root. This has been removed due to tighter security in VyOS 1.2.

See also:

SSH *Key-based authentication*

Configuration

set service ssh port <port>

Enabling SSH only requires you to specify the port <port> you want SSH to listen on. By default, SSH runs on port 22.

set service ssh listen-address <address>

Specify IPv4/IPv6 listen address of SSH server. Multiple addresses can be defined.

set service ssh cipher <cipher>

Define allowed ciphers used for the SSH connection. A number of allowed ciphers can be specified, use multiple occurrences to allow multiple ciphers.

List of supported ciphers: 3des-cbc, aes128-cbc, aes192-cbc, aes256-cbc, aes128-ctr, aes192-ctr, aes256-ctr, aes128-gcm@openssh.com, aes256-gcm@openssh.com, chacha20-poly1305@openssh.com

set service ssh disable-password-authentication

Disable password based authentication. Login via SSH keys only. This hardens security!

set service ssh fido pin-required

Require FIDO2 keys to attest that a user has been verified (e.g. via a PIN).

set service ssh fido touch-required

Require FIDO2 keys to attest that a user is physically present.

VyOS supports SSH authentication using FIDO2-backed keys generated by OpenSSH. Two FIDO2 key types are supported by OpenSSH: ed25519-sk, ecdsa-sk

Generic FIDO2-backed SSH key generation example:

```
ssh-keygen -t ecdsa-sk -O verify-required -C "fido2-ssh-key"
```

During key generation, OpenSSH will:

- Request user presence (for example, a physical touch or confirmation)
- Optionally request user verification (PIN), if supported by the authenticator
- Create a local key handle file and a corresponding public key (.pub)

The private key material never leaves the authenticator device.

VyOS configuration example:

```
# Generate a FIDO2 SSH key on the client system
# Copy the public key to the VyOS instance
set system login user vyos authentication public-keys fido key '<public-key>'
set system login user vyos authentication public-keys fido type 'sk-ecdsa-sha2-
↪nistp256@openssh.com'
set service ssh fido touch-required
```

You can now log into the system using: `ssh -i ~/.ssh/id_fido_key vyos@192.0.2.1`

set service ssh disable-host-validation

Disable the host validation through reverse DNS lookups - can speedup login time when reverse lookup is not possible.

set service ssh mac <mac>

Specifies the available MAC algorithms. The MAC algorithm is used in protocol version 2 for data integrity protection. Multiple algorithms can be provided by using multiple commands, defining one algorithm per command.

List of supported MACs: hmac-md5, hmac-md5-96, hmac-ripemd160, hmac-sha1, hmac-sha1-96, hmac-sha2-256, hmac-sha2-512, umac-64@openssh.com, umac-128@openssh.com, hmac-md5-etm@openssh.com, hmac-md5-96-etm@openssh.com, hmac-ripemd160-etm@openssh.com, hmac-sha1-etm@openssh.com, hmac-sha1-96-etm@openssh.com, hmac-sha2-256-etm@openssh.com, hmac-sha2-512-etm@openssh.com, umac-64-etm@openssh.com, umac-128-etm@openssh.com

set service ssh access-control <allow | deny> <group | user> <name>

Add access-control directive to allow or deny users and groups. Directives are processed in the following order of precedence: deny-users, allow-users, deny-groups and allow-groups.

set service ssh client-keepalive-interval <interval>

Specify timeout interval for keepalive message in seconds.

set service ssh key-exchange <kex>

Specify allowed KEX (Key Exchange) algorithms.

List of supported algorithms: diffie-hellman-group1-sha1, diffie-hellman-group14-sha1, diffie-hellman-group14-sha256, diffie-hellman-group16-sha512, diffie-hellman-group18-sha512, diffie-hellman-group-exchange-sha1, diffie-hellman-group-exchange-sha256, ecdh-sha2-nistp256, ecdh-sha2-nistp384, ecdh-sha2-nistp521, curve25519-sha256 and curve25519-sha256@libssh.org.

set service ssh loglevel <quiet | fatal | error | info | verbose>

Set the sshd log level. The default is info.

set service ssh vrf <name>

Specify name of the VRF instance.

set service ssh pubkey-accepted-algorithm <name>

Specifies the signature algorithms that will be accepted for public key authentication

List of supported algorithms: ssh-ed25519, ssh-ed25519-cert-v01@openssh.com, sk-ssh-ed25519@openssh.com, sk-ssh-ed25519-cert-v01@openssh.com, ecdsa-sha2-nistp256, ecdsa-sha2-nistp256-cert-v01@openssh.com, ecdsa-sha2-nistp384, ecdsa-sha2-nistp384-cert-v01@openssh.com, ecdsa-sha2-nistp521, ecdsa-sha2-nistp521-cert-v01@openssh.com, sk-ecdsa-sha2-nistp256@openssh.com, sk-ecdsa-sha2-nistp256-cert-v01@openssh.com, webauthn-sk-ecdsa-sha2-nistp256@openssh.com, ssh-dss, ssh-dss-cert-v01@openssh.com, ssh-rsa, ssh-rsa-cert-v01@openssh.com, rsa-sha2-256, rsa-sha2-256-cert-v01@openssh.com, rsa-sha2-512, rsa-sha2-512-cert-v01@openssh.com

set service ssh trusted-user-ca <name>

Specify the name of the OpenSSH key-pair that acts as certificate authority and will be used to verify user certificates.

You can use it by adding the OpenSSH key-pair under the PKI subsystem.

Example:

```
# Generate key-pair acting as CA
$ ssh-keygen -f vyos-ssh-ca.key

# Generate key for user: vyos_testca
$ ssh-keygen -f vyos_testca -C "vyos_testca@vyos.net"

# Sign public key from user vyos_testca and insert principal names: vyos, vyos_
→testca
# with a key lifetime of two weeks - after which the key is unusable
$ ssh-keygen -s vyos-ssh-ca.key -I vyos_testca@vyos.net -n vyos,vyos_testca -V +2w_
→vyos_testca.pub

$ set system login user vyos_testca
$ set pki openssh test_ca public key AAAAB3N....
$ set pki openssh test_ca public type ssh-rsa
$ set service ssh trusted-user-ca test_ca
```

You can now log into the system using: `ssh -i vyos_testca vyos_testca@vyos.test.com`

Dynamic-protection

Protects host from brute-force attacks against SSH. Log messages are parsed, line-by-line, for recognized patterns. If an attack, such as several login failures within a few seconds, is detected, the offending IP is blocked. Offenders are unblocked after a set interval.

set service ssh dynamic-protection

Allow ssh dynamic-protection.

set service ssh dynamic-protection allow-from <address | prefix>

Whitelist of addresses and networks. Always allow inbound connections from these systems.

set service ssh dynamic-protection block-time <sec>

Block source IP in seconds. Subsequent blocks increase by a factor of 1.5 The default is 120.

set service ssh dynamic-protection detect-time <sec>

Remember source IP in seconds before reset their score. The default is 1800.

set service ssh dynamic-protection threshold <sec>

Block source IP when their cumulative attack score exceeds threshold. The default is 30.

Operation

restart ssh

Restart the SSH daemon process, the current session is not affected, only the background daemon is restarted.

generate ssh server-key

Re-generated the public/private keyportion which SSH uses to secure connections.

Note: Already learned known_hosts files of clients need an update as the public key will change.

generate ssh client-key /path/to/private_key

Re-generated a known pub/private keyfile which can be used to connect to other services (e.g. RPKI cache).

Example:

```
vyos@vyos:~$ generate ssh client-key /config/auth/id_rsa_rpki
Generating public/private rsa key pair.
Your identification has been saved in /config/auth/id_rsa_rpki.
Your public key has been saved in /config/auth/id_rsa_rpki.pub.
The key fingerprint is:
SHA256:XGv2PpdOzVCzpmEzJZga8hTRq7B/ZYL3fXaioLFLS5Q vyos@vyos
The key's randomart image is:
+---[RSA 2048]-----+
|           oo       |
|          ..O       |
|         . o.o.. o. |
|        o+ooo  o.o |
|         Eo*  =.o |
|        o = +.o*+ |
|         = o *.o.o |
|        o * +.o+.+ |
|         =.. o=.oo |
+-----[SHA256]-----+
```

Two new files /config/auth/id_rsa_rpki and /config/auth/id_rsa_rpki.pub will be created.

generate public-key-command user <username> path <location>

Generate the configuration mode commands to add a public key for *Key-based authentication*.
<location> can be a local path or a URL pointing at a remote file.

Supported remote protocols are FTP, FTPS, HTTP, HTTPS, SCP/SFTP and TFTP.

Example:

```
alyssa@vyos:~$ generate public-key-command user alyssa path sftp://example.net/home/alyssa/.ssh/id_rsa.pub
# To add this key as an embedded key, run the following commands:
configure
set system login user alyssa authentication public-keys alyssa@example.net key AAA..
.
set system login user alyssa authentication public-keys alyssa@example.net type ssh-
rsa
commit
save
exit

ben@vyos:~$ generate public-key-command user ben path ~/.ssh/id_rsa.pub
# To add this key as an embedded key, run the following commands:
configure
set system login user ben authentication public-keys ben@vyos key AAA...
set system login user ben authentication public-keys ben@vyos type ssh-dss
commit
save
exit
```

show log ssh

Show SSH server log.

monitor log ssh

Follow the SSH server log.

show log ssh dynamic-protection

Show SSH dynamic-protection log.

monitor log ssh dynamic-protection

Follow the SSH dynamic-protection log.

show ssh dynamic-protection

Show list of IPs currently blocked by SSH dynamic-protection.

show ssh fingerprints

Show SSH server public key fingerprints.

show ssh fingerprints ascii

Show SSH server public key fingerprints, including a visual ASCII art representation.

7.10.20 TFTP server

TFTP (Trivial File Transfer Protocol) is a simple, UDP-based file-transfer protocol defined in [RFC 1350](#) that lets clients upload files to and download files from a remote host. One of its main uses is during the early stages of network boot, where devices fetch a boot image or initial configuration from a TFTP server before they are fully operational.

VyOS ships a TFTP server that lets clients download files from a configured directory. Uploads are disabled by default but can be enabled. The TFTP server is configured under `service tftp-server`.

Configuration

set service tftp-server directory <directory>

Configure the directory from which clients download files and, when uploads are enabled, upload files to.

The directory is created if it does not exist, and its owner and group are set to the `tftp` system user. This value is mandatory.

Note: This setting is mandatory. The commit fails if no directory is configured.

Note: Choose the location carefully, or the contents will be lost on image upgrade. Any path under `/config` is preserved and migrated across image upgrades.

Example:

```
set service tftp-server directory /config/tftpboot
```

```
set service tftp-server listen-address <address>
```

Configure a local IPv4 or IPv6 address on which the TFTP server accepts incoming requests.

Repeat the command to listen on multiple IP addresses. The IP address must be assigned to a local interface. Otherwise, the commit succeeds with a warning, and the TFTP server does not accept connections on that IP address.

Note: At least one listen address must be set whenever `service tftp-server` is configured. Otherwise, the commit is rejected.

Example:

```
set service tftp-server listen-address 192.0.2.1
set service tftp-server listen-address 2001:db8::1
```

set service tftp-server listen-address <address> vrf <name>

Expose the TFTP server on the specified IP address within a non-default VRF.

The VRF must already be configured under `vrf name <name>`. Otherwise, the commit is rejected.

Example:

```
set service tftp-server listen-address 192.0.2.1 vrf MGMT
```

set service tftp-server port <1-65535>

Configure the UDP port on which the TFTP server listens.

The default is 69, the port assigned to TFTP by the IANA.

Example:

```
set service tftp-server port 6969
```

set service tftp-server allow-upload

Enable client uploads (TFTP write requests) to the server.

By default, the server operates in read-only mode and rejects write requests.

Example:

```
set service tftp-server allow-upload
```

Example

The following example configures a TFTP server that uses `/config/tftpboot` as its file directory, listens on both IPv4 and IPv6 addresses, and accepts client uploads.

```
set service tftp-server directory '/config/tftpboot'
set service tftp-server listen-address '192.0.2.1'
set service tftp-server listen-address '2001:db8::1'
set service tftp-server allow-upload
```

Verification

Upload a file from a TFTP client to the server (write request):

```
vyos@client:~$ tftp -p -l /config/config.boot -r backup 192.0.2.1
backup          100% |*****|          723  0:00:00 ETA
```

On the server, the uploaded file appears in the configured directory, owned by the `tftp` user:

```
vyos@vyos:~$ ls -ltr /config/tftpboot/
total 1
-rw-rw-rw- 1 tftp tftp 723 May 19 16:02 backup
```

7.10.21 Webproxy

The proxy service in VyOS is based on [Squid](#) and some related modules.

[Squid](#) is a caching and forwarding HTTP web proxy. It has a wide variety of uses, including speeding up a web server by caching repeated requests, caching web, DNS and other computer network lookups for a group of people sharing network resources, and aiding security by filtering traffic. Although primarily used for HTTP and FTP, Squid includes limited support for several other protocols including Internet Gopher, SSL,[6] TLS and HTTPS. Squid does not support the SOCKS protocol.

URL Filtering is provided by [SquidGuard](#).

Configuration

set service webproxy append-domain <domain>

Use this command to specify a domain name to be appended to domain-names within URLs that do not include a dot . the domain is appended.

Example: to be appended is set to `vyos.net` and the URL received is `www/foo.html`, the system will use the generated, final URL of `www.vyos.net/foo.html`.

```
set service webproxy append-domain vyos.net
```

set service webproxy cache-size <size>

The size of the on-disk Proxy cache is user configurable. The Proxies default cache-size is configured to 100 MB.

Unit of this command is MB.

```
set service webproxy cache-size 1024
```

set service webproxy default-port <port>

Specify the port used on which the proxy service is listening for requests. This port is the default port used for the specified listen-address.

Default port is 3128.

```
set service webproxy default-port 8080
```

set service webproxy domain-block <domain>

Used to block specific domains by the Proxy. Specifying “vyos.net” will block all access to vyos.net, and specifying “.xxx” will block all access to URLs having an URL ending on .xxx.

```
set service webproxy domain-block vyos.net
```

set service webproxy domain-noncache <domain>

Allow access to sites in a domain without retrieving them from the Proxy cache. Specifying “vyos.net” will allow access to vyos.net but the pages accessed will not be cached. It useful for working around problems with “If-Modified-Since” checking at certain sites.

```
set service webproxy domain-noncache vyos.net
```

set service webproxy listen-address <address>

Specifies proxy service listening address. The listen address is the IP address on which the web proxy service listens for client requests.

For security, the listen address should only be used on internal/trusted networks!

```
set service webproxy listen-address 192.0.2.1
```

set service webproxy listen-address <address> disable-transparent

Disables web proxy transparent mode at a listening address.

In transparent proxy mode, all traffic arriving on port 80 and destined for the Internet is automatically forwarded through the proxy. This allows immediate proxy forwarding without configuring client browsers.

Non-transparent proxying requires that the client browsers be configured with the proxy settings before requests are redirected. The advantage of this is that the client web browser can detect that a proxy is in use and can behave accordingly. In addition, web-transmitted malware can sometimes be blocked by a non-transparent web proxy, since they are not aware of the proxy settings.

```
set service webproxy listen-address 192.0.2.1 disable-transparent
```

set service webproxy listen-address <address> port <port>

Sets the listening port for a listening address. This overrides the default port of 3128 on the specific listen address.

```
set service webproxy listen-address 192.0.2.1 port 8080
```

set service webproxy reply-block-mime <mime>

Used to block a specific mime-type.

```
# block all PDFs
set service webproxy reply-block-mime application/pdf
```

set service webproxy reply-body-max-size <size>

Specifies the maximum size of a reply body in KB, used to limit the reply size.

All reply sizes are accepted by default.

```
set service webproxy reply-body-max-size 2048
```

set service webproxy safe-ports <port>

Add new port to Safe-ports acl. Ports included by default in Safe-ports acl: 21, 70, 80, 210, 280, 443, 488, 591, 777, 873, 1025-65535

set service webproxy ssl-safe-ports <port>

Add new port to SSL-ports acl. Ports included by default in SSL-ports acl: 443

Authentication

The embedded Squid proxy can use LDAP to authenticate users against a company wide directory. The following configuration is an example of how to use Active Directory as authentication backend. Queries are done via LDAP.

set service webproxy authentication children <number>

Maximum number of authenticator processes to spawn. If you start too few Squid will have to wait for them to process a backlog of credential verifications, slowing it down. When password verifications are done via a (slow) network you are likely to need lots of authenticator processes.

This defaults to 5.

```
set service webproxy authentication children 10
```

set service webproxy authentication credentials-ttl <time>

Specifies how long squid assumes an externally validated username:password pair is valid for - in other words how often the helper program is called for that user. Set this low to force revalidation with short lived passwords.

Time is in minutes and defaults to 60.

```
set service webproxy authentication credentials-ttl 120
```

set service webproxy authentication method <ldap>

Proxy authentication method, currently only LDAP is supported.

```
set service webproxy authentication method ldap
```

set service webproxy authentication realm

Specifies the protection scope (aka realm name) which is to be reported to the client for the authentication scheme. It is commonly part of the text the user will see when prompted for their username and password.

```
set service webproxy authentication realm "VyOS proxy auth"
```

LDAP

set service webproxy authentication ldap base-dn <base-dn>

Specifies the base DN under which the users are located.

```
set service webproxy authentication ldap base-dn DC=vyos,DC=net
```

set service webproxy authentication ldap bind-dn <bind-dn>

The DN and password to bind as while performing searches.

```
set service webproxy authentication ldap bind-dn CN=proxyuser,CN=Users,DC=vyos,
↪DC=net
```

set service webproxy authentication ldap filter-expression <expr>

LDAP search filter to locate the user DN. Required if the users are in a hierarchy below the base DN, or if the login name is not what builds the user specific part of the users DN.

The search filter can contain up to 15 occurrences of %s which will be replaced by the username, as in “uid=%s” for [RFC 2037](#) directories. For a detailed description of LDAP search filter syntax see [RFC 2254](#).

```
set service webproxy authentication ldap filter-expression (cn=%s)
```

set service webproxy authentication ldap password <password>

The DN and password to bind as while performing searches. As the password needs to be printed in plain text in your Squid configuration it is strongly recommended to use an account with minimal associated privileges. This to limit the damage in case someone could get hold of a copy of your Squid configuration file.

```
set service webproxy authentication ldap password vyos
```

set service webproxy authentication ldap persistent-connection

Use a persistent LDAP connection. Normally the LDAP connection is only open while validating a username to preserve resources at the LDAP server. This option causes the LDAP connection to be kept open, allowing it to be reused for further user validations.

Recommended for larger installations.

```
set service webproxy authentication ldap persistent-connection
```

set service webproxy authentication ldap port <port>

Specify an alternate TCP port where the ldap server is listening if other than the default LDAP port 389.

```
set service webproxy authentication ldap port 389
```

set service webproxy authentication ldap server <server>

Specify the LDAP server to connect to.

```
set service webproxy authentication ldap server ldap.vyos.net
```

set service webproxy authentication ldap use-ssl

Use TLS encryption.

```
set service webproxy authentication ldap use-ssl
```

set service webproxy authentication ldap username-attribute <attr>

Specifies the name of the DN attribute that contains the username/login. Combined with the base DN to construct the users DN when no search filter is specified (filter-expression).

Defaults to ‘uid’

Note: This can only be done if all your users are located directly under the same position in the LDAP tree and the login name is used for naming each user object. If your LDAP tree does not match these criteria or if you want to filter who are valid users then you need to use a search filter to search for your users DN (filter-expression).

```
set service webproxy authentication ldap username-attribute uid
```

```
set service webproxy authentication ldap version <2 | 3>
```

LDAP protocol version. Defaults to 3 if not specified.

```
set service webproxy authentication ldap version 2
```

URL filtering

```
set service webproxy url-filtering disable
```

Disables web filtering without discarding configuration.

```
set service webproxy url-filtering disable
```

Operation

Filtering

Update

If you want to use existing blacklists you have to create/download a database first. Otherwise you will not be able to commit the config changes.

```
update webproxy blacklists
```

Download/Update complete blacklist

```
vyos@vyos:~$ update webproxy blacklists
Warning: No url-filtering blacklist installed
Would you like to download a default blacklist? [confirm][y]
Connecting to ftp.example.com (192.0.2.249:21)
blacklists.gz          100%
→ | *****
→ 17.0M  0:00:00 ETA
Uncompressing blacklist...
Checking permissions...
Skip link for  [ads] -> [publicite]
Building DB for [adult/domains] - 2467177 entries
Building DB for [adult/urls] - 67798 entries
Skip link for  [aggressive] -> [agressif]
Building DB for [agressif/domains] - 348 entries
Building DB for [agressif/urls] - 36 entries
Building DB for [arjel/domains] - 69 entries
...
Building DB for [webmail/domains] - 374 entries
Building DB for [webmail/urls] - 9 entries
The webproxy daemon must be restarted
Would you like to restart it now? [confirm][y]
[ ok ] Restarting squid (via systemctl): squid.service.
vyos@vyos:~$
```

```
update webproxy blacklists category <category>
```

Download/Update partial blacklist.

Use tab completion to get a list of categories.

- To auto update the blacklist files


```
set service webproxy url-filtering squidguard auto-update update-hour 23
```
- To configure blocking add the following to the configuration


```
set service webproxy url-filtering squidguard block-category ads
set service webproxy url-filtering squidguard block-category malware
```

Bypassing the webproxy

Some services don't work correctly when being handled via a web proxy. So sometimes it is useful to bypass a transparent proxy:

- To bypass the proxy for every request that is directed to a specific destination:


```
set service webproxy whitelist destination-address 198.51.100.33
set service webproxy whitelist destination-address 192.0.2.0/24
```
- To bypass the proxy for every request that is coming from a specific source:


```
set service webproxy whitelist source-address 192.168.1.2
set service webproxy whitelist source-address 192.168.2.0/24
```

(This can be useful when a called service has many and/or often changing destination addresses - e.g. Netflix.)

Examples

```
vyos@vyos# show service webproxy
authentication {
  children 5
  credentials-ttl 60
  ldap {
    base-dn DC=example,DC=local
    bind-dn CN=proxyuser,CN=Users,DC=example,DC=local
    filter-expression (cn=%s)
    password Qwert1234
    server ldap.example.local
    username-attribute cn
  }
  method ldap
  realm "VyOS Webproxy"
}
cache-size 100
default-port 3128
listen-address 192.168.188.103 {
  disable-transparent
}
```

7.10.22 Suricata

Suricata is an open-source network threat detection engine that performs intrusion detection (IDS (Intrusion Detection System)), inline intrusion prevention (IPS (Intrusion Prevention System)), and network security monitoring (NSM (Network Security Monitoring)) by matching live traffic against a set of rules, also called signatures. Each event is recorded in the EVE (Extensible Event Format) event log in JSON (JavaScript Object Notation) format.

VyOS integrates Suricata as a managed service, configurable under `service suricata`. When the configuration is committed, VyOS generates Suricata configuration from these settings and starts the service.

Under `service suricata`, you can configure the following:

- **Monitored interfaces:** One or more interfaces on which Suricata captures traffic.
- **Address groups:** Named lists of IPv4/IPv6 addresses or prefixes, used in place of literal IP addresses in Suricata rules.
- **Port groups:** Named lists of ports or port ranges, used in place of literal ports in Suricata rules.
- **EVE log:** Destination (regular file or syslog), file name or path, and the event types to log.

Rules are fetched separately by running `update suricata` from operational mode. Until this command is run, the service has no rules to match against.

Note: A Suricata configuration is committable only when at least one interface, one address group, and one port group are defined. Otherwise, the commit fails.

Configuration

Monitored interfaces

Use the following command to configure the interfaces Suricata monitors.

```
set service suricata interface <interface>
```

Configure an interface on which Suricata captures and inspects traffic.

Repeat the command to monitor several interfaces. At least one interface must be configured for a successful commit.

Example:

```
set service suricata interface eth1
```

Address groups

Use the following commands to configure address groups.

Note: Group names must be lowercase letters, digits, or hyphens.

Note: Suricata rules reference a conventional set of group names, such as `home-net`, `external-net`, `http-servers`, and others. Defining a group with a non-standard name is allowed, but no shipped rules will reference it. Typical address groups include:

- `home-net`: the networks Suricata should treat as internal (for example, `192.0.2.0/24` and `2001:db8::/32`).
- `external-net`: everything not in `home-net`.
- `http-servers`, `sql-servers`, `dns-servers`: hosts running the corresponding services.

```
set service suricata address-group <name> address <address>
```

Configure an IPv4 or IPv6 address or prefix as a member of the specified address group.

`<address>` accepts an IPv4 or IPv6 address, an IPv4 or IPv6 prefix, or any of these prefixed with `!` to exclude it from matches.

Repeat the command to add more addresses or prefixes to the same group.

Example:

```
set service suricata address-group home-net address 192.0.2.0/24
set service suricata address-group home-net address 2001:db8::/32
set service suricata address-group home-net address !192.0.2.5
```

```
set service suricata address-group <name> group <name>
```

Configure another address group as a member of the specified address group.

Prefix the referenced name with `!` to exclude that group's members. The referenced group must be defined at commit time, and cyclic references between groups are rejected.

Repeat the command to add more groups to the same parent group.

Example:

```
set service suricata address-group home-net address 192.0.2.0/24
set service suricata address-group external-net group !home-net
```

Port groups

Use the following commands to configure port groups.

Note: Group names must be lowercase letters, digits, or hyphens.

Note: Suricata rules reference a conventional set of port-group names, such as `http-ports`, `ssh-ports`, `oracle-ports`, `shellcode-ports`, and others. Defining a port group with a non-standard name is allowed, but no shipped rules will reference it. Typical port groups include:

- `http-ports`: ports where HTTP services run.
 - `ssh-ports`: ports where SSH services listen.
 - `oracle-ports`: ports used by Oracle database services.
 - `shellcode-ports`: ports inspected by shellcode-detection rules.
-

```
set service suricata port-group <name> port <port>
```

Configure a port or port range as a member of the specified port group.

<port> accepts a single port (1–65535), a numeric range in start–end form (e.g., 1001–1005), or either form prefixed with ! to exclude it from matches.

Repeat the command to add more ports or port ranges to the same group.

Example:

```
set service suricata port-group http-ports port 80
set service suricata port-group http-ports port 443
set service suricata port-group http-ports port 8000-8999
set service suricata port-group http-ports port !8080
```

set service suricata port-group <name> group <name>

Configure another port group as a member of the specified port group.

Prefix the referenced name with ! to exclude that group’s members from matches. The referenced group must be defined at commit time, and cyclic references between groups are rejected.

Repeat the command to add multiple groups to the same parent group.

Example:

```
set service suricata port-group http-ports port 80
set service suricata port-group shellcode-ports group !http-ports
```

EVE log

Use the following commands to configure the EVE log.

set service suricata log eve filename <filename>

Configure the filename or path for the Suricata EVE log.

A bare filename (e.g., eve.json) places the log in Suricata’s default log directory, /var/log/suricata/. An absolute path is used as-is.

The default is eve.json.

Example:

```
set service suricata log eve filename eve.json
set service suricata log eve filename /var/log/custom/suricata-eve.json
```

set service suricata log eve filetype <regular | syslog>

Configure the destination for the Suricata EVE log:

- regular: Writes the EVE log to the file specified by set service suricata log eve filename.
- syslog: Sends the EVE log to syslog.

The default is regular.

Example:

```
set service suricata log eve filetype syslog
```

set service suricata log eve type <type>

Configure which EVE event types are logged.

Accepted values:

- General event types: alert, anomaly, drop, files, flow, netflow.
- Per-protocol event types: http, http2, dns, tls, smtp, ftp, smb, ssh, dhcp, tftp, nfs, rdp, sip, snmp, ikev2, krb5, dcerpc, dnp3, rfb, mqtt.

Repeat the command to log multiple event types.

Example:

```
set service suricata log eve type alert
set service suricata log eve type flow
set service suricata log eve type http
```

Operation

update suricata

Fetch the current rule set with `suricata-update` and restart the Suricata service to load it.

If service `suricata` is not configured, the command outputs an error message and exits.

Example:

```
update suricata
```

restart suricata

Restart the Suricata service.

If the service is not configured (no `service suricata` configuration exists), or a configuration commit is in progress, the command prints an error message and exits.

Example:

```
restart suricata
```

7.11 System

7.11.1 Acceleration

In this command tree, all hardware acceleration options will be handled. At the moment only [Intel® QAT](#) is supported

Intel® QAT

show system acceleration qat

use this command to check if there is an Intel® QAT supported Processor in your system.

```
vyos@vyos:~$ show system acceleration qat
01:00.0 Co-processor [0b40]: Intel Corporation Atom Processor C3000 Series
↳ QuickAssist Technology [8086:19e2] (rev 11)
```

if there is non device the command will show `No QAT device found`

set system acceleration qat

if there is a supported device, enable Intel® QAT

show system acceleration qat status

Check if the Intel® QAT device is up and ready to do the job.

```
vyos@vyos:~$ show system acceleration qat status
Checking status of all devices.
There is 1 QAT acceleration device(s) in the system:
qat_dev0 - type: c3xxx, inst_id: 0, node_id: 0, bsf: 0000:01:00.0, #accel: 3
↪#engines: 6 state: up
```

Operation Mode

show system acceleration qat device <device> config

Show the full config uploaded to the QAT device.

show system acceleration qat device <device> flows

Get an overview over the encryption counters.

show system acceleration qat interrupts

Show binded qat device interrupts to certain core.

Example

Let's build a simple VPN between 2 Intel® QAT ready devices.

Side A:

```
set interfaces vti vti1 address '192.168.1.2/24'
set vpn ipsec authentication psk right id '10.10.10.2'
set vpn ipsec authentication psk right id '10.10.10.1'
set vpn ipsec authentication psk right secret 'Qwerty123'
set vpn ipsec esp-group MyESPGroup proposal 1 encryption 'aes256'
set vpn ipsec esp-group MyESPGroup proposal 1 hash 'sha256'
set vpn ipsec ike-group MyIKEGroup proposal 1 dh-group '14'
set vpn ipsec ike-group MyIKEGroup proposal 1 encryption 'aes256'
set vpn ipsec ike-group MyIKEGroup proposal 1 hash 'sha256'
set vpn ipsec interface 'eth0'
set vpn ipsec site-to-site peer right authentication local-id '10.10.10.2'
set vpn ipsec site-to-site peer right authentication mode 'pre-shared-secret'
set vpn ipsec site-to-site peer right authentication remote-id '10.10.10.1'
set vpn ipsec site-to-site peer right connection-type 'initiate'
set vpn ipsec site-to-site peer right default-esp-group 'MyESPGroup'
set vpn ipsec site-to-site peer right ike-group 'MyIKEGroup'
set vpn ipsec site-to-site peer right local-address '10.10.10.2'
set vpn ipsec site-to-site peer right remote-address '10.10.10.1'
set vpn ipsec site-to-site peer right vti bind 'vti1'
```

Side B:

```

set interfaces vti vti1 address '192.168.1.1/24'
set vpn ipsec authentication psk left id '10.10.10.2'
set vpn ipsec authentication psk left id '10.10.10.1'
set vpn ipsec authentication psk left secret 'Qwerty123'
set vpn ipsec esp-group MyESPGroup proposal 1 encryption 'aes256'
set vpn ipsec esp-group MyESPGroup proposal 1 hash 'sha256'
set vpn ipsec ike-group MyIKEGroup proposal 1 dh-group '14'
set vpn ipsec ike-group MyIKEGroup proposal 1 encryption 'aes256'
set vpn ipsec ike-group MyIKEGroup proposal 1 hash 'sha256'
set vpn ipsec interface 'eth0'
set vpn ipsec site-to-site peer left authentication local-id '10.10.10.1'
set vpn ipsec site-to-site peer left authentication mode 'pre-shared-secret'
set vpn ipsec site-to-site peer left authentication remote-id '10.10.10.2'
set vpn ipsec site-to-site peer left connection-type 'initiate'
set vpn ipsec site-to-site peer left default-esp-group 'MyESPGroup'
set vpn ipsec site-to-site peer left ike-group 'MyIKEGroup'
set vpn ipsec site-to-site peer left local-address '10.10.10.1'
set vpn ipsec site-to-site peer left remote-address '10.10.10.2'
set vpn ipsec site-to-site peer left vti bind 'vti1'

```

a bandwidth test over the VPN got these results:

```

Connecting to host 192.168.1.2, port 5201
[ 9] local 192.168.1.1 port 51344 connected to 192.168.1.2 port 5201
[ ID] Interval          Transfer    Bitrate      Retr  Cwnd
[ 9]  0.00-1.01    sec  32.3 MBytes  268 Mbits/sec  0    196 KBytes
[ 9]  1.01-2.03    sec  32.5 MBytes  268 Mbits/sec  0    208 KBytes
[ 9]  2.03-3.03    sec  32.5 MBytes  271 Mbits/sec  0    208 KBytes
[ 9]  3.03-4.04    sec  32.5 MBytes  272 Mbits/sec  0    208 KBytes
[ 9]  4.04-5.00    sec  31.2 MBytes  272 Mbits/sec  0    208 KBytes
[ 9]  5.00-6.01    sec  32.5 MBytes  272 Mbits/sec  0    234 KBytes
[ 9]  6.01-7.04    sec  32.5 MBytes  265 Mbits/sec  0    234 KBytes
[ 9]  7.04-8.04    sec  32.5 MBytes  272 Mbits/sec  0    234 KBytes
[ 9]  8.04-9.04    sec  32.5 MBytes  273 Mbits/sec  0    336 KBytes
[ 9]  9.04-10.00   sec  31.2 MBytes  272 Mbits/sec  0    336 KBytes
- - - - -
[ ID] Interval          Transfer    Bitrate      Retr
[ 9]  0.00-10.00   sec  322 MBytes  270 Mbits/sec  0
[ 9]  0.00-10.00   sec  322 MBytes  270 Mbits/sec

```

sender
receiver

with set system acceleration qat on both systems the bandwidth increases.

```

Connecting to host 192.168.1.2, port 5201
[ 9] local 192.168.1.1 port 51340 connected to 192.168.1.2 port 5201
[ ID] Interval          Transfer    Bitrate      Retr  Cwnd
[ 9]  0.00-1.00    sec  97.3 MBytes  817 Mbits/sec  0    1000 KBytes
[ 9]  1.00-2.00    sec  92.5 MBytes  776 Mbits/sec  0    1.07 MBytes
[ 9]  2.00-3.00    sec  92.5 MBytes  776 Mbits/sec  0    820 KBytes
[ 9]  3.00-4.00    sec  92.5 MBytes  776 Mbits/sec  0    899 KBytes
[ 9]  4.00-5.00    sec  91.2 MBytes  765 Mbits/sec  0    972 KBytes
[ 9]  5.00-6.00    sec  92.5 MBytes  776 Mbits/sec  0    1.02 MBytes
[ 9]  6.00-7.00    sec  92.5 MBytes  776 Mbits/sec  0    1.08 MBytes
[ 9]  7.00-8.00    sec  92.5 MBytes  776 Mbits/sec  0    1.14 MBytes

```

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[9]	8.00-9.00	sec	91.2 MBytes	765 Mbits/sec	0	915 KBytes
[9]	9.00-10.00	sec	92.5 MBytes	776 Mbits/sec	0	1000 KBytes

[ID]	Interval		Transfer	Bitrate	Retr	
[9]	0.00-10.00	sec	927 MBytes	778 Mbits/sec	0	sender
[9]	0.00-10.01	sec	925 MBytes	775 Mbits/sec		receiver

7.11.2 Conntrack

VyOS can be configured to track connections using the connection tracking subsystem. Connection tracking becomes operational once either stateful firewall or NAT is configured.

Configure

set system conntrack table-size <1-50000000>

The connection tracking table contains one entry for each connection being tracked by the system.

set system conntrack expect-table-size <1-50000000>

The connection tracking expect table contains one entry for each expected connection related to an existing connection. These are generally used by “connection tracking helper” modules such as FTP. The default size of the expect table is 2048 entries.

set system conntrack hash-size <1-50000000>

Set the size of the hash table. The connection tracking hash table makes searching the connection tracking table faster. The hash table uses “buckets” to record entries in the connection tracking table.

set system conntrack modules ftp

set system conntrack modules h323

set system conntrack modules nfs

set system conntrack modules pptp

set system conntrack modules sip

set system conntrack modules sqlnet

set system conntrack modules tftp

Configure the connection tracking protocol helper modules. All modules are enabled by default.

Use *delete system conntrack modules* to deactivate all modules.

Or, for example ftp, *delete system conntrack modules ftp*.

set system conntrack tcp half-open-connections <1-21474836>

Set the maximum number of TCP half-open connections.

set system conntrack tcp loose <enable | disable>

Policy to track previously established connections.

set system conntrack tcp max-retrans <1-2147483647>

Set the number of TCP maximum retransmit attempts.

Conntrack Timeouts

You can define custom timeout values to apply to a specific subset of connections, based on a packet and flow selector. To do this, you need to create a rule defining the packet and flow selector.

```
set system conntrack timeout custom [ipv4 | ipv6] rule <1-999999> description <test>
```

Set a rule description.

```
set system conntrack timeout custom [ipv4 | ipv6] rule <1-999999> destination address <ip-address>
```

```
set system conntrack timeout custom [ipv4 | ipv6] rule <1-999999> source address <ip-address>
```

Set a destination and/or source address. Accepted input for ipv4:

```
set system conntrack timeout custom ipv4 rule <1-999999> [source | ↵
↵destination] address
Possible completions:
<x.x.x.x>           IPv4 address to match
<x.x.x.x/x>         IPv4 prefix to match
<x.x.x.x>-<x.x.x.x> IPv4 address range to match
!<x.x.x.x>          Match everything except the specified address
!<x.x.x.x/x>        Match everything except the specified prefix
!<x.x.x.x>-<x.x.x.x> Match everything except the specified range

set system conntrack timeout custom ipv6 rule <1-999999> [source | ↵
↵destination] address
Possible completions:
<h:h:h:h:h:h:h>    IP address to match
<h:h:h:h:h:h:h/x>  Subnet to match
<h:h:h:h:h:h:h>-<h:h:h:h:h:h:h>
                    IP range to match
!<h:h:h:h:h:h:h>    Match everything except the specified address
!<h:h:h:h:h:h:h/x> Match everything except the specified prefix
!<h:h:h:h:h:h:h>-<h:h:h:h:h:h:h>
                    Match everything except the specified range
```

```
set system conntrack timeout custom [ipv4 | ipv6] rule <1-999999> destination port <value>
```

```
set system conntrack timeout custom [ipv4 | ipv6] rule <1-999999> source port <value>
```

Set a destination and/or source port. Accepted input:

```
<port name>    Named port (any name in /etc/services, e.g., http)
<1-65535>      Numbered port
<start>-<end>  Numbered port range (e.g., 1001-1005)
```

Multiple destination ports can be specified as a comma-separated list. The whole list can also be “negated” using ‘!’. For example: `!22,telnet,http,123,1001-1005`

```
set system conntrack timeout custom [ipv4 | ipv6] rule <1-999999> protocol tcp close <1-21474836>
```

```

set system conntrack timeout custom [ipv4 | ipv6] rule <1-999999> protocol tcp close-wait
<1-21474836>

set system conntrack timeout custom [ipv4 | ipv6] rule <1-999999> protocol tcp
established <1-21474836>

set system conntrack timeout custom [ipv4 | ipv6] rule <1-999999> protocol tcp fin-wait
<1-21474836>

set system conntrack timeout custom [ipv4 | ipv6] rule <1-999999> protocol tcp last-ack
<1-21474836>

set system conntrack timeout custom [ipv4 | ipv6] rule <1-999999> protocol tcp syn-recv
<1-21474836>

set system conntrack timeout custom [ipv4 | ipv6] rule <1-999999> protocol tcp syn-sent
<1-21474836>

set system conntrack timeout custom [ipv4 | ipv6] rule <1-999999> protocol tcp time-wait
<1-21474836>

set system conntrack timeout custom [ipv4 | ipv6] rule <1-999999> protocol udp replied
<1-21474836>

set system conntrack timeout custom [ipv4 | ipv6] rule <1-999999> protocol udp unreplied
<1-21474836>

```

Set the timeout in seconds for a protocol or state in a custom rule.

Conntrack ignore rules

Note: Important note about conntrack ignore rules: Starting from vyos-1.5-rolling-202406120020, ignore rules can be defined in `set firewall [ipv4 | ipv6] prerouting raw ...`. It's expected that in the future the conntrack ignore rules will be removed.

Customized ignore rules, based on a packet and flow selector.

```

set system conntrack ignore [ipv4 | ipv6] rule <1-999999> description <text>

set system conntrack ignore [ipv4 | ipv6] rule <1-999999> destination address
<ip-address>

set system conntrack ignore [ipv4 | ipv6] rule <1-999999> destination port <port>

set system conntrack ignore [ipv4 | ipv6] rule <1-999999> inbound-interface <interface>

set system conntrack ignore [ipv4 | ipv6] rule <1-999999> protocol <protocol>

set system conntrack ignore [ipv4 | ipv6] rule <1-999999> source address <ip-address>

set system conntrack ignore [ipv4 | ipv6] rule <1-999999> source port <port>

set system conntrack ignore [ipv4 | ipv6] rule <1-999999> tcp flags [not] <text>

```

Allowed values for TCP flags: ack, cwr, ecn, fin, psh, rst, syn and urg. Multiple values are supported, and for inverted selection use not, as shown in the example.

Conntrack log

```
set system conntrack log event destroy
```

```
set system conntrack log event new
```

```
set system conntrack log event update
```

Log the connection tracking events per type.

```
set system conntrack log event destroy icmp
```

```
set system conntrack log event destroy other
```

```
set system conntrack log event destroy tcp
```

```
set system conntrack log event destroy udp
```

```
set system conntrack log event new icmp
```

```
set system conntrack log event new other
```

```
set system conntrack log event new tcp
```

```
set system conntrack log event new udp
```

```
set system conntrack log event update icmp
```

```
set system conntrack log event update other
```

```
set system conntrack log event update tcp
```

```
set system conntrack log event update udp
```

Log the connection tracking events per protocol.

```
set system conntrack log timestamp
```

Turn on flow-based timestamp extension.

```
set system conntrack log queue-size <100-999999>
```

Manage internal queue size, default size is 4096 events.

```
set system conntrack log log-level <info | debug>
```

Manage log level

7.11.3 Serial Console

For the average user a serial console has no advantage over a console offered by a directly attached keyboard and screen. Serial consoles are much slower, taking up to a second to fill a 80 column by 24 line screen. Serial consoles generally only support non-proportional ASCII text, with limited support for languages other than English.

There are some scenarios where serial consoles are useful. System administration of remote computers is usually done using *SSH*, but there are times when access to the console is the only way to diagnose and correct software failures. Major upgrades to the installed distribution may also require console access.

```
set system console device <device>
```

Defines the specified device as a system console. Available console devices can be (see completion helper):

- `ttySN` - Serial device name
- `ttyAMAN` - Serial device name for some arm64 systems
- `ttyUSBX` - USB Serial device name

- `hvc0` - Xen console

set system console device <device> kernel

When set, the selected serial console is used as the kernel boot console. When removed, the kernel boot console falls back to `tty0`.

Note: Only one serial console can carry the `kernel` option. When VyOS is installed via serial console, this option is set automatically for the serial interface used during installation; usually `ttyS0` or `ttyAMA0`.

set system console device <device> speed <speed>

The speed (baudrate) of the console device. Supported values are:

- `1200` - 1200 bps
- `2400` - 2400 bps
- `4800` - 4800 bps
- `9600` - 9600 bps
- `19200` - 19,200 bps
- `38400` - 38,400 bps (default for Xen console)
- `57600` - 57,600 bps
- `115200` - 115,200 bps (default for serial console)

Note: If you use USB to serial converters for connecting to your VyOS appliance please note that most of them use software emulation without flow control. This means you should start with a common baud rate (most likely 9600 baud) as otherwise you probably can not connect to the device using high speed baud rates as your serial converter simply can not process this data rate.

7.11.4 Flow Accounting

VyOS supports flow-accounting for both IPv4 and IPv6 traffic. The system acts as a flow exporter, and you are free to use it with any compatible collector.

Flows can be exported via protocol NetFlow (versions 5, 9 and 10/IPFIX). Additionally, you may save flows to an in-memory table internally in a router.

Warning: You need to disable the in-memory table in production environments! Using IMT (In-Memory Table) may lead to heavy CPU overloading and unstable flow-accounting behavior.

NetFlow / IPFIX

NetFlow is a feature that was introduced on Cisco routers around 1996 that provides the ability to collect IP network traffic as it enters or exits an interface. By analyzing the data provided by NetFlow, a network administrator can determine things such as the source and destination of traffic, class of service, and the causes of congestion. A typical flow monitoring setup (using NetFlow) consists of three main components:

- **exporter:** aggregates packets into flows and exports flow records towards one or more flow collectors
- **collector:** responsible for reception, storage and pre-processing of flow data received from a flow exporter
- **application:** analyzes received flow data in the context of intrusion detection or traffic profiling, for example

For connectionless protocols as like ICMP and UDP, a flow is considered complete once no more packets for this flow appear after configurable timeout.

NetFlow is usually enabled on a per-interface basis to limit load on the router components involved in NetFlow, or to limit the amount of NetFlow records exported.

Configuration

Warning: Using NetFlow on routers with high traffic levels may lead to high CPU usage and may affect the router's performance. In such cases, consider using sFlow instead.

In order for flow accounting information to be collected and displayed for an interface, the interface must be configured for flow accounting.

set system flow-accounting interface <interface>

Configure and enable collection of flow information for the interface identified by <interface>.

You can configure multiple interfaces which would participate in flow accounting.

Note: Will be recorded only packets/flows on **incoming** direction in configured interfaces by default.

By default, recorded flows will be saved internally and can be listed with the CLI command. You may disable using the local in-memory table with the command:

set system flow-accounting disable-imt

If you need to sample also egress traffic, you may want to configure egress flow-accounting:

set system flow-accounting enable-egress

Internally, in flow-accounting processes exist a buffer for data exchanging between core process and plugins (each export target is a separated plugin). If you have high traffic levels or noted some problems with missed records or stopping exporting, you may try to increase a default buffer size (10 MiB) with the next command:

set system flow-accounting buffer-size <buffer size>

In case, if you need to catch some logs from flow-accounting daemon, you may configure logging facility:

set system flow-accounting syslog-facility <facility>

Set the syslog facility for flow-accounting log messages. Supported values include `daemon`, `local0` through `local7`, and other standard syslog facilities.

Flow Export

In addition to displaying flow accounting information locally, one can also exported them to a collection server.

NetFlow

set system flow-accounting netflow version <version>

There are multiple versions available for the NetFlow data. The <version> used in the exported flow data can be configured here. The following versions are supported:

- **5** - Most common version, but restricted to IPv4 flows only
- **9** - NetFlow version 9 (default)
- **10** - IPFIX (IP Flow Information Export) as per [RFC 3917](#)

set system flow-accounting netflow server <address>

Configure address of NetFlow collector. NetFlow server at <address> can be both listening on an IPv4 or IPv6 address.

set system flow-accounting netflow source-ip <address>

IPv4 or IPv6 source address of NetFlow packets

set system flow-accounting netflow engine-id <id>

NetFlow engine-id which will appear in NetFlow data. The range is 0 to 255.

set system flow-accounting netflow sampling-rate <rate>

Use this command to configure the sampling rate for flow accounting. The system samples one in every <rate> packets, where <rate> is the value configured for the sampling-rate option. The advantage of sampling every n packets, where n > 1, allows you to decrease the amount of processing resources required for flow accounting. The disadvantage of not sampling every packet is that the statistics produced are estimates of actual data flows.

Per default every packet is sampled (that is, the sampling rate is 1).

set system flow-accounting netflow timeout expiry-interval <interval>

Specifies the interval at which Netflow data will be sent to a collector. As per default, Netflow data will be sent every 60 seconds.

You may also additionally configure timeouts for different types of connections.

set system flow-accounting netflow max-flows <n>

If you want to change the maximum number of flows, which are tracking simultaneously, you may do this with this command (default 8192).

Example:

NetFlow v5 example:

```
set system flow-accounting netflow engine-id 100
set system flow-accounting netflow version 5
set system flow-accounting netflow server 192.168.2.10 port 2055
```

Operation

Once flow accounting is configured on an interfaces it provides the ability to display captured network traffic information for all configured interfaces.

show flow-accounting interface <interface>

Show flow accounting information for given <interface>.

```
vyos@vyos:~$ show flow-accounting interface eth0
```

IN_IFACE	SRC_MAC	DST_MAC	SRC_IP	DST_IP				
→	SRC_PORT	DST_PORT	PROTOCOL	TOS	PACKETS	FLows	BYTES	
-----	-----	-----	-----	-----	-----	-----	-----	-----
eth0	00:53:01:a8:28:ac	ff:ff:ff:ff:ff:ff	192.0.2.2				255.255.	
→255.255	5678	5678	udp	0	1	1	178	
eth0	00:53:01:b2:2f:34	33:33:ff:00:00:00	fe80::253:01ff:feb2:2f34					
→ff02::1:ff00:0	0	0	ipv6-icmp	0	0	2	1	
→144								
eth0	00:53:01:1a:b4:53	33:33:ff:00:00:00	fe80::253:01ff:fe1a:b453					
→ff02::1:ff00:0	0	0	ipv6-icmp	0	0	1	1	
→72								
eth0	00:53:01:b2:22:48	00:53:02:58:a2:92	192.0.2.100				192.0.2.	
→14	40152	22	tcp	16	39	1	2064	
eth0	00:53:01:c8:33:af	ff:ff:ff:ff:ff:ff	192.0.2.3				255.255.	
→255.255	5678	5678	udp	0	1	1	154	
eth0	00:53:01:b2:22:48	00:53:02:58:a2:92	192.0.2.100				192.0.2.	
→14	40006	22	tcp	16	146	1	9444	
eth0	00:53:01:b2:22:48	00:53:02:58:a2:92	192.0.2.100				192.0.2.	
→14	0	0	icmp	192	27	1	4455	

show flow-accounting interface <interface> host <address>

Show flow accounting information for given <interface> for a specific host only.

```
vyos@vyos:~$ show flow-accounting interface eth0 host 192.0.2.14
```

IN_IFACE	SRC_MAC	DST_MAC	SRC_IP	DST_IP	SRC_			
→PORT	DST_PORT	PROTOCOL	TOS	PACKETS	FLows	BYTES		
-----	-----	-----	-----	-----	-----	-----	-----	-----
eth0	00:53:01:b2:22:48	00:53:02:58:a2:92	192.0.2.100	192.0.2.14				
→40006	22	tcp	16	197	2	12940		
eth0	00:53:01:b2:22:48	00:53:02:58:a2:92	192.0.2.100	192.0.2.14				
→40152	22	tcp	16	94	1	4924		
eth0	00:53:01:b2:22:48	00:53:02:58:a2:92	192.0.2.100	192.0.2.14				
→0	0	icmp	192	36	1	5877		

7.11.5 FRR

VyOS uses [FRRouting](#) as the control plane for dynamic and static routing. The routing daemon behavior can be adjusted during runtime, but requires either a restart of the routing daemon, or a reboot of the system.

set system frr bmp

Enable BMP (BGP Monitoring Protocol) support.

set system frr descriptors <number>

This allows the operator to control the number of open file descriptors each daemon is allowed to start with. If the operator plans to run bgp with several thousands of peers then this is where we would modify FRR to allow this to happen.

set system frr irdp

Enable ICMP Router Discovery Protocol support.

set system frr profile <traditional | datacenter>

Select an FRR profile to adapt its default settings. If unset, the traditional profile is applied.

set system frr snmp <daemon>

Enable SNMP support for an individual routing daemon.

Supported daemons:

- bgpd
- isisd
- ldpd
- ospf6d
- ospfd
- ripd
- zebra

7.11.6 Host Information

This section describes the system's host information and how to configure them, it covers the following topics:

- Host name
- Domain
- IP address
- Aliases

Hostname

A hostname is the label (name) assigned to a network device (a host) on a network and is used to distinguish one device from another on specific networks or over the internet. On the other hand this will be the name which appears on the command line prompt.

set system host-name <hostname>

The hostname can be up to 63 characters. A hostname must start and end with a letter or digit, and have as interior characters only letters, digits, or a hyphen.

The default hostname used is vyos.

Domain Name

A domain name is the label (name) assigned to a computer network and is thus unique. VyOS appends the domain name as a suffix to any unqualified name. For example, if you set the domain name `example.com`, and you would ping the unqualified name of `crux`, then VyOS qualifies the name to `crux.example.com`.

set system domain-name <domain>

Configure system domain name. A domain name must start and end with a letter or digit, and have as interior characters only letters, digits, or a hyphen.

Static Hostname Mapping

How an IP address is assigned to an interface in *Ethernet*. This section shows how to statically map an IP address to a hostname for local (meaning on this VyOS instance) name resolution. This is the VyOS equivalent to `/etc/hosts` file entries.

Note: Do *not* manually edit `/etc/hosts`. This file will automatically be regenerated on boot based on the settings in this section, which means you'll lose all your manual edits. Instead, configure static host mappings as follows.

set system static-host-mapping host-name <hostname> inet <address>

Create a static hostname mapping which will always resolve the name `<hostname>` to IP address `<address>`.

set system static-host-mapping host-name <hostname> alias <alias>

Create named `<alias>` for the configured static mapping for `<hostname>`. Thus the address configured as `set system static-host-mapping host-name <hostname> inet <address>` can be reached via multiple names.

Multiple aliases can be specified per host-name.

7.11.7 IP

System configuration commands

set system ip disable-forwarding

Use this command to disable IPv4 forwarding on all interfaces.

set system ip disable-directed-broadcast

Use this command to disable IPv4 directed broadcast forwarding on all interfaces.

If set, IPv4 directed broadcast forwarding will be completely disabled regardless of whether per-interface directed broadcast forwarding is enabled or not.

set system ip arp table-size <number>

Use this command to define the maximum number of entries to keep in the ARP cache (1024, 2048, 4096, 8192, 16384, 32768).

set system ip multipath layer4-hashing

Use this command to use Layer 4 information for IPv4 ECMP hashing.

set system ip import-table <table-id>

Use this command to import the table, by given table id, into the main RIB.

set system ip import-table <table-id> distance <distance>

Use this command to override the default distance when importing routers from the alternate table.

set system ip import-table <table-id> route-map <route-map>

Use this command to filter routes that are imported into the main table from alternate table using route-map.

Zebra/Kernel route filtering

Zebra supports prefix-lists and Route Maps to match routes received from other FRR components. The permit/deny facilities provided by these commands can be used to filter which routes zebra will install in the kernel.

set system ip protocol <protocol> route-map <route-map>

Apply a route-map filter to routes for the specified protocol. The following protocols can be used: any, babel, bgp, eigrp, isis, ospf, rip, static

Note: If you choose any as the option that will cause all protocols that are sending routes to zebra.

NextHop Tracking

NextHop tracking resolve nexthops via the default route by default. This is enabled by default for a traditional profile of FRR which we use. It and can be disabled if you do not want to e.g. allow BGP to peer across the default route.

set system ip nht no-resolve-via-default

Do not allow IPv4 nexthop tracking to resolve via the default route. This parameter is configured per-VRF, so the command is also available in the VRF subnode.

Operational commands

show commands

See below the different parameters available for the IPv4 **show** command:

```
vyos@vyos:~$ show ip
Possible completions:
  access-list      Show all IP access-lists
  as-path-access-list
                  Show all as-path-access-lists
  bgp              Show Border Gateway Protocol (BGP) information
  community-list   Show IP community-lists
  extcommunity-list
                  Show extended IP community-lists
  forwarding       Show IP forwarding status
  groups           Show IP multicast group membership
  igmp            Show IGMP (Internet Group Management Protocol) information
  large-community-list
                  Show IP large-community-lists
  multicast        Show IP multicast
  ospf            Show IPv4 Open Shortest Path First (OSPF) routing information
  pim             Show PIM (Protocol Independent Multicast) information
  ports           Show IP ports in use by various system services
  prefix-list      Show all IP prefix-lists
  protocol         Show IP route-maps per protocol
  rip             Show Routing Information Protocol (RIP) information
  route           Show IP routes
```

reset commands

And the different IPv4 **reset** commands available:

```
vyos@vyos:~$ reset ip
Possible completions:
  arp             Reset Address Resolution Protocol (ARP) cache
  bgp            Clear Border Gateway Protocol (BGP) statistics or status
  igmp           IGMP clear commands
  multicast      IP multicast routing table
  route         Reset IP route
```

7.11.8 IPv6

System configuration commands

set system ipv6 disable-forwarding

Use this command to disable IPv6 forwarding on all interfaces.

set system ipv6 neighbor table-size <number>

Use this command to define the maximum number of entries to keep in the Neighbor cache (1024, 2048, 4096, 8192, 16384, 32768).

set system ipv6 strict-dad

Use this command to disable IPv6 operation on interface when Duplicate Address Detection fails on Link-Local address.

set system ipv6 multipath layer4-hashing

Use this command to user Layer 4 information for ECMP hashing.

Zebra/Kernel route filtering

Zebra supports prefix-lists and Route Maps to match routes received from other FRR components. The permit/deny facilities provided by these commands can be used to filter which routes zebra will install in the kernel.

set system ipv6 protocol <protocol> route-map <route-map>

Apply a route-map filter to routes for the specified protocol. The following protocols can be used: any, babel, bgp, isis, ospfv3, ripng, static

Note: If you choose any as the option that will cause all protocols that are sending routes to zebra.

Nexthop Tracking

Nexthop tracking resolve nexthops via the default route by default. This is enabled by default for a traditional profile of FRR which we use. It can be disabled if you do not want to e.g. allow BGP to peer across the default route.

set system ipv6 nht no-resolve-via-default

Do not allow IPv6 nexthop tracking to resolve via the default route. This parameter is configured per-VRF, so the command is also available in the VRF subnode.

Operational commands

Show commands

show ipv6 neighbors

Use this command to show IPv6 Neighbor Discovery Protocol information.

show ipv6 groups

Use this command to show IPv6 multicast group membership.

show ipv6 forwarding

Use this command to show IPv6 forwarding status.

show ipv6 route

Use this command to show IPv6 routes.

Check the many parameters available for the show ipv6 route command:

```
vyos@vyos:~$ show ipv6 route
Possible completions:
<Enter>          Execute the current command
<X:X::X:X>       Show IPv6 routes of given address or prefix
<X:X::X:X/M>
bgp              Show IPv6 BGP routes
cache           Show kernel IPv6 route cache
connected       Show IPv6 connected routes
forward         Show kernel IPv6 route table
isis            Show IPv6 ISIS routes
kernel          Show IPv6 kernel routes
ospfv3          Show IPv6 OSPF6 routes
ripng           Show IPv6 RIPNG routes
static          Show IPv6 static routes
summary         Show IPv6 routes summary
table           Show IP routes in policy table
tag             Show only routes with tag
vrf             Show IPv6 routes in VRF
```

show ipv6 prefix-list

Use this command to show all IPv6 prefix lists

There are different parameters for getting prefix-list information:

```
vyos@vyos:~$ show ipv6 prefix-list
Possible completions:
<Enter>          Execute the current command
<WORD>           Show specified IPv6 prefix-list
detail           Show detail of IPv6 prefix-lists
summary          Show summary of IPv6 prefix-lists
```

show ipv6 access-list

Use this command to show all IPv6 access lists

You can also specify which IPv6 access-list should be shown:

```
vyos@vyos:~$ show ipv6 access-list
Possible completions:
<Enter>          Execute the current command
<text>           Show specified IPv6 access-list
```

show ipv6 ospfv3

Use this command to get information about OSPFv3.

You can get more specific OSPFv3 information by using the parameters shown below:

```
vyos@vyos:~$ show ipv6 ospfv3
Possible completions:
<Enter>      Execute the current command
area         Show OSPFv3 spf-tree information
border-routers
             Show OSPFv3 border-router (ABR and ASBR) information
database     Show OSPFv3 Link state database information
interface    Show OSPFv3 interface information
linkstate    Show OSPFv3 linkstate routing information
neighbor     Show OSPFv3 neighbor information
redistribute Show OSPFv3 redistribute External information
route        Show OSPFv3 routing table information
```

show ipv6 ripng

Use this command to get information about the RIPNG protocol

show ipv6 ripng status

Use this command to show the status of the RIPNG protocol

Reset commands**reset bgp ipv6 <address>**

Use this command to clear Border Gateway Protocol statistics or status.

reset ipv6 neighbors <address | interface>

Use this command to reset IPv6 Neighbor Discovery Protocol cache for an address or interface.

reset ipv6 route cache

Use this command to flush the kernel IPv6 route cache. An address can be added to flush it only for that route.

7.11.9 System Display (LCD)

The system LCD LCD (Liquid-crystal display) option is for users running VyOS on hardware that features an LCD display. This is typically a small display built in an 19 inch rack-mountable appliance. Those displays are used to show runtime data.

To configure your LCD display you must first identify the used hardware, and connectivity of the display to your system. This can be any serial port (`ttysxx`) or serial via USB or even old parallel port interfaces.

Configuration**set system lcd device <device>**

This is the name of the physical interface used to connect to your LCD display. Tab completion is supported and it will list you all available serial interface.

For serial via USB port information please refer to the USB hardware section.

set system lcd model <model>

This is the LCD model used in your system.

At the time of this writing the following displays are supported:

- Crystalfontz CFA-533
- Crystalfontz CFA-631
- Crystalfontz CFA-633
- Crystalfontz CFA-635

Note: We can't support all displays from the beginning. If your display type is missing, please create a feature request via Phabricator.

7.11.10 Login/user management

The default VyOS user account (vyos), as well as newly created user accounts, possess full system configuration privileges. These accounts are granted sudo privileges, allowing them to execute commands as the root user.

VyOS supports both local authentication and remote authentication via RADIUS (Remote Authentication Dial-In User Service)/ TACACS+ (Terminal Access Controller Access-Control System).

Local authentication

set system login user <name> full-name "<string>"

Configure the real name or description for a system user.

If the description includes spaces, enclose <string> in double quotes.

If the user <name> already exists, the command updates the current description. If not, it creates a new user with the specified description.

set system login user <name> authentication plaintext-password <password>

Configure a password for a system user.

Enter the password in plaintext. Upon `commit`, VyOS hashes the password for secure storage and removes the plaintext value.

If the user <name> already exists, the command updates the current password. If not, it creates a new user with the specified plaintext password.

set system login user <name> authentication encrypted-password <password>

Configure a pre-encrypted password for a system user.

Enter the password in its hashed format. Upon `commit`, VyOS stores this value directly without modification.

If the user <name> already exists, the command updates the current password. If not, it creates a new user with the specified pre-encrypted password.

set system login user <name> authentication principal <principal>

Configure an SSH certificate principal for a system user.

Enter the principal (a string included in the user's signed SSH certificate). Upon `commit`, VyOS stores this mapping, allowing the user to log in if the certificate they present contains this principal.

If the user <name> already exists, the command updates the principal. If not, it creates a new user linked to the specified principal.

If not configured, the principal defaults to <name>.

```
set system login user <name> disable
```

Disable a system user account.

VyOS locks the account, preventing the user from logging in.

Key-based authentication

Key-based authentication is the recommended method for securing SSH access in VyOS. It uses a **public/private key pair** to verify user identity without requiring a password. To authorize access, you assign **SSH public keys** to user accounts on the router, while SSH private keys remain on local devices. VyOS allows assigning multiple SSH public keys to a single user account, which is useful for accessing a router from different devices.

Generate the key pair

Generate an SSH key pair on your **local machine** using the `ssh-keygen` command. This creates two files:

- **Private key** (e.g., `id_rsa`): Remains on your local machine and must never be shared.
- **Public key** (e.g., `id_rsa.pub`): Is used to configure the VyOS user account. By default, it is saved to `~/.ssh/id_rsa.pub`.

Each SSH public key consists of three parts, separated by spaces:

- **Encryption algorithm type:** `ssh-rsa`, `ssh-ed25519`, etc.
- **Key:** The actual data (a long string beginning with `AAAA...`).
- **Comment:** An identifier for your reference (e.g., `user@host`).

Only the encryption algorithm type and key parts are required to configure the authorization entry in VyOS. The comment part is optional.

See also:

SSH operation

Warning: SSH key strings are long. When copying and pasting, ensure your terminal does not insert line breaks. The key must be entered as a **single line** to function correctly.

Configure the router

To configure SSH public key authentication for a user account, run the following two commands using the same `<identifier>`:

```
set system login user <username> authentication public-keys <identifier> key <key>
```

Configure the SSH public key for the user account.

- `<identifier>`: A unique label that identifies this specific key entry.
- `<key>`: The actual string of characters from your public key.

```
set system login user <username> authentication public-keys <identifier> type <type>
```

Configure the SSH key's encryption type.

The following encryption algorithm types are available:

- ecdsa-sha2-nistp256
- ecdsa-sha2-nistp384
- ecdsa-sha2-nistp521
- ssh-dss
- ssh-ed25519
- ssh-rsa

Note: To assign multiple SSH public keys to a user account, repeat the commands above with a unique identifier for each key.

set system login user <username> authentication public-keys <identifier> options <options>

Configure specific restrictions or behaviors for an SSH public key.

<options>: A string of comma-separated values that define permissions or restrictions for this key.

The command accepts standard OpenSSH options listed in the router's ~/.ssh/authorized_keys file.

To include a " character in the options string, use ".

For example, to restrict allowed source IP addresses for an SSH public key, use: from="10.0.0.0/24".

OTP-based MFA

VyOS lets you enhance user access security by enabling OTP (One-time password)-based MFA (Multi-factor Authentication) for individual users. Users with OTP-based MFA must enter a valid OTP along with their password at login. Users without OTP-based MFA use standard authentication.

set system login user <username> authentication otp key <key>

Configure OTP-based MFA for a user.

<key>: A Base32-encoded secret key. This key must be added to the user's authenticator app to generate valid OTPs (One-time passwords).

When configured, the user is required to enter their password followed by a valid OTP for all subsequent logins.

OTP settings

set system login user <username> authentication otp rate-limit <limit>

Configure the number of OTP authentication attempts allowed within a specified time period.

If this limit is exceeded, the user is temporarily blocked.

The default value is 3 attempts. The valid range is 1 to 10 attempts.

set system login user <username> authentication otp rate-time <seconds>

Configure the time period, in seconds, for tracking OTP authentication attempts.

The default value is 30 seconds. The valid range is 1 to 600 seconds.

set system login user <username> authentication otp window-size <size>

Configure the OTP window size for a user.

The OTP window size defines the number of concurrently valid OTPs that the authentication server accepts. This setting assumes a new token is generated every 30 seconds.

The default value is 3. This permits 3 concurrent codes: the code for the current 30-second interval, the preceding code, and the following code. This allows up to 30 seconds of time skew between the authentication server and client.

If the window size is increased to 17, the system permits 17 concurrent codes (the current code, the 8 preceding codes, and the 8 following codes). This allows for a time skew of up to 4 minutes.

The valid range is 1 to 21.

Generate an OTP-key

Use the following command to generate an OTP key:

```
generate system login username <username> otp-key hotp-time rate-limit <1-10> rate-time <15-600> window-size <1-21>
```

Key generation example:

```
vyos@vyos:~$ generate system login username opttester otp-key hotp-time rate-limit 2
↳rate-time 20 window-size 5
# You can share it with the user, he just needs to scan the QR in his OTP app
# username:  opttester
# OTP KEY:   J5A64ERPMGJOZXY6FMHHLKXXKANNI6TCY
# OTP URL:  otpauth://totp/opttester@vyos?secret=J5A64ERPMGJOZXY6FMHHLKXXKANNI6TCY&
↳digits=6&period=30
```

```
# To add this OTP key to configuration, run the following commands:
set system login user opttester authentication otp key 'J5A64ERPMGJOZXY6FMHHLKXXKANNI6TCY'
```

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```
set system login user otptester authentication otp rate-limit '2'  
set system login user otptester authentication otp rate-time '20'  
set system login user otptester authentication otp window-size '5'
```

Display the OTP key for a user

Use the following command to display the OTP key for a user:

```
sh system login authentication user <username> otp <full | key-b32 | qrcode | uri>
```

Example:

```
vyos@vyos:~$ sh system login authentication user otptester otp full  
# You can share the OTP key with the user. They just need to scan the QR in their OTP_  
→app.  
# username: otptester  
# OTP KEY: J5A64ERPMGJOZXY6FMHHLKXXKANNI6TCY  
# OTP URL: otpauth://totp/otptester@vyos?secret=J5A64ERPMGJOZXY6FMHHLKXXKANNI6TCY&  
→digits=6&period=30
```

```
# To add this OTP key to configuration, run the following commands:  
set system login user otptester authentication otp key 'J5A64ERPMGJOZXY6FMHHLKXXKANNI6TCY'  
set system login user otptester authentication otp rate-limit '2'  
set system login user otptester authentication otp rate-time '20'  
set system login user otptester authentication otp window-size '5'
```

Once OTP-based MFA is configured for a user account, this user must enter their standard password followed by the current 6-digit OTP code at login. For example, if the user's password is vyosrocks and the OTP is 817454, they should enter vyosrocks817454.

RADIUS authentication

For large-scale deployments, managing individual user accounts across multiple VyOS instances is inefficient. VyOS supports centralized authentication via RADIUS, consolidating user account management on a single backend server.

Configuration

set system login radius server <address> key <secret>

Configure the RADIUS server's IP address and shared secret.

The shared secret is used to verify the router's identity and to encrypt user passwords during authentication.

You can configure multiple RADIUS servers.

set system login radius server <address> port <port>

Configure the UDP port for communication with the RADIUS server.

The default port is 1812.

set system login radius server <address> disable

Disable a RADIUS server from the authentication process.

Disabling a specific RADIUS server doesn't remove its configuration settings (the server's IP address and shared secret).

set system login radius server <address> timeout <timeout>

Configure the duration, in seconds, that the VyOS router waits for a response from the RADIUS server after sending an authentication request.

If the server does not respond within this timeframe, the VyOS router tries to connect to another configured server or falls back to local authentication.

set system login radius source-address <address>

Configure the source IP address the router uses for RADIUS authentication requests.

A consistent source IP address is recommended as RADIUS servers typically accept requests only from known, trusted IP addresses.

If not explicitly defined, the router uses the current egress interface address, which may change (e.g., due to a link outage), causing authentication failures.

set system login radius vrf <name>

Configure the router to send all RADIUS authentication requests via a specific VRF.

By default, RADIUS authentication requests are sent via the global routing table.

Configuration example

```
set system login radius server 192.168.0.2 key 'test-vyos'
set system login radius server 192.168.0.2 port '1812'
set system login radius server 192.168.0.2 timeout '5'
set system login radius source-address '192.168.0.1'
```

If communication with the RADIUS server fails, the router falls back to local user authentication. During this process, users may experience a login delay while the system waits for the RADIUS request to time out. This delay depends on the configured timeout value.

Hint: To grant administrative privileges to RADIUS-authenticated users, the server must return the Cisco-AV-Pair attribute set to `shell:priv-lvl=15`. Otherwise, users receive standard privileges and cannot perform configuration tasks.

TACACS+ authentication

In addition to RADIUS, VyOS supports TACACS+, which is commonly used in large enterprise environments.

Unlike RADIUS, TACACS+ separates Authentication, Authorization, and Accounting (AAA) into independent processes and encrypts the entire packet body for enhanced security.

TACACS+ is defined in [RFC 8907](#).

Configuration

set system login tacacs server <address> key <secret>

Configure the TACACS+ server IP address and shared secret.

Unlike RADIUS, which encrypts only passwords, TACACS+ encrypts the entire packet body for enhanced security.

You can configure multiple TACACS+ servers.

set system login tacacs server <address> port <port>

Configure the TCP port for communication with the TACACS+ server.

The default port is 49.

set system login tacacs server <address> disable

Disable a TACACS+ server from the authentication process.

Disabling a specific TACACS+ server doesn't remove its configuration settings (the server's IP address and shared secret).

set system login tacacs server <address> timeout <timeout>

Configure the duration, in seconds, that the VyOS router waits for a response from the TACACS+ server after sending an authentication request.

If the server does not respond within this timeframe, the VyOS router tries to connect to another configured server or falls back to local authentication.

set system login tacacs source-address <address>

Configure the source IP address the router uses for TACACS+ authentication requests.

A consistent source IP address is recommended as TACACS+ servers typically accept requests only from known, trusted IP addresses.

If not explicitly defined, the router uses the current egress interface address, which may change (e.g., due to a link outage), causing authentication failures.

set system login tacacs vrf <name>

Configure the router to send all TACACS+ authentication requests via a specific VRF.

By default, TACACS+ authentication requests are sent via the global routing table.

Configuration example

```
set system login tacacs server 192.168.0.2 key 'test-vyos'
set system login tacacs server 192.168.0.2 port '49'
set system login tacacs source-address '192.168.0.1'
```

If communication with the TACACS+ server fails, the router falls back to local user authentication.

Login banners

VyOS allows you to configure **pre-login** and **post-login** banners. Pre-login banners are typically used for system identification, legal disclaimers, or security warnings displayed before authentication, while post-login banners provide system information or operational notices to users after login.

set system login banner pre-login <message>

Configure a message to be shown to users before the username and password prompts appear.

set system login banner post-login <message>

Configure a message to be shown to users after successful authentication.

Note: Use `\\n` to insert line breaks in multi-line banner messages.

Login session limits

set system login max-login-session <number>

Configure the maximum number of concurrent login sessions.

Note: If you limit concurrent login sessions, you must also configure a session `<timeout>`. This clears inactive sessions and prevents blocking new login attempts.

set system login timeout <timeout>

Configure the login session timeout, in seconds.

Idle login sessions are terminated after this period.

Configuration examples

Example 1: Multi-key SSH with MFA and source restrictions

In this configuration, User1 and User2 both use the vyos user account, each with a unique SSH key. User1 is restricted to authentication from a single IP address.

For both users, password-based logins require OTP -based MFA.

```
set system login user vyos authentication public-keys 'User1' key "AAAAB3Nz...KwEW"
set system login user vyos authentication public-keys 'User1' type ssh-rsa
set system login user vyos authentication public-keys 'User1' options "from="192.
↪168.0.100";"

set system login user vyos authentication public-keys 'User2' key "AAAAQ39x...fbV3"
set system login user vyos authentication public-keys 'User2' type ssh-rsa

set system login user vyos authentication otp key OHZ30J7U2N25BK4G7S0FFJTZDTCFUUE2
set system login user vyos authentication plaintext-password vyos
```

Example 2: Containerized TACACS+ deployment with redundancy.

In this configuration, the VyOS router hosts its own authentication infrastructure using two containerized TACACS+ servers (tacacs1 and tacacs2) on a private network for redundancy.

System logins are authenticated against credentials stored within these internal containers rather than the router's local user database.

First, download the image in operational mode:

```
add container image lfkeitel/tacacs_plus:latest
```

Next, configure the containers in configuration mode:

```
set container network tac-test prefix '100.64.0.0/24'

set container name tacacs1 image 'lfkeitel/tacacs_plus:latest'
set container name tacacs1 network tac-test address '100.64.0.11'

set container name tacacs2 image 'lfkeitel/tacacs_plus:latest'
set container name tacacs2 network tac-test address '100.64.0.12'

set system login tacacs server 100.64.0.11 key 'tac_plus_key'
set system login tacacs server 100.64.0.12 key 'tac_plus_key'

commit
```

You can now log in via SSH or console using admin/admin credentials supplied by the container image.

7.11.11 System DNS

Warning: If you are configuring a VRF for management purposes, there is currently no way to force system DNS traffic via a specific VRF.

This section describes configuring DNS on the system, namely:

- DNS name servers
- Domain search order

DNS name servers

set system name-server <address>

Use this command to specify a DNS server for the system to be used for DNS lookups. More than one DNS server can be added, configuring one at a time. Both IPv4 and IPv6 addresses are supported.

Example

In this example, some *OpenNIC* servers are used, two IPv4 addresses and two IPv6 addresses:

```
set system name-server 176.9.37.132
set system name-server 195.10.195.195
set system name-server 2a01:4f8:161:3441::1
set system name-server 2a00:f826:8:2::195
```

Domain search order

In order for the system to use and complete unqualified host names, a list can be defined which will be used for domain searches.

set system domain-search <domain>

Use this command to define domains, one at a time, so that the system uses them to complete unqualified host names. Maximum: 6 entries.

Note: Domain names can include letters, numbers, hyphens and periods with a maximum length of 253 characters.

Example

The system is configured to attempt domain completion in the following order: vyos.io (first), vyos.net (second) and vyos.network (last):

```
set system domain-search vyos.io
set system domain-search vyos.net
set system domain-search vyos.network
```

7.11.12 Option

This chapter describe the possibilities of advanced system behavior.

General

set system option ctrl-alt-delete <ignore | reboot | poweroff>

Action which will be run once the ctrl-alt-del keystroke is received.

set system option reboot-on-panic

Automatically reboot system on kernel panic after 60 seconds.

set system option reboot-on-upgrade-failure <timeout>

Automatically reboot after `timeout` minutes into the previous running image, that was used to perform the image upgrade.

Reboot `timeout` is configurable in minutes. This gives the user the change to log into the system and perform some analysis before automatic rebooting.

Automatic reboot can be cancelled after login using: `reboot cancel`

set system option startup-beep

Play an audible beep to the system speaker when system is ready.

set system option root-partition-auto-resize

Enables the root partition auto-extension and resizes to the maximum available space on system boot.

Kernel

set system option kernel disable-mitigations

Disable all optional CPU mitigations. This improves system performance, but it may also expose users to several CPU vulnerabilities.

This will add the following option to the Kernel commandline:

- `mitigations=off`

Note: Setting will only become active with the next reboot!

set system option kernel disable-power-saving

This will add the following two options to the Kernel commandline:

- `intel_idle.max_cstate=0` Disable `intel_idle` and fall back on `acpi_idle`
- `processor.max_cstate=1` Limit processor to maximum C-state 1

Note: Setting will only become active with the next reboot!

set system option kernel amd-pstate-driver <mode>

Enables and configures p-state driver for modern AMD Ryzen and Epyc CPUs.

The available modes are:

- **active** This is the low-level firmware control mode based on the profile set and the system governor has no effect.
- **passive** The driver allows the system governor to manage CPU frequency while providing available performance states.
- **guided** The driver allows to set desired performance levels and the firmware selects a performance level in this range and fitting to the current workload.

This will add the following two options to the Kernel commandline:

- `initcall_blacklist=acpi_cpufreq_init` Disable default ACPI CPU frequency scale
- `amd_pstate={mode}` Sets the p-state mode

Note: Setting will only become active with the next reboot!

See also:

<https://docs.kernel.org/admin-guide/pm/amd-pstate.html>

set system option kernel quiet

Suppress most kernel messages during boot. This is useful for systems with embedded serial console interfaces to speed up the boot process.

HTTP client

set system option http-client source-address <address>

Several commands utilize cURL to initiate transfers. Configure the local source IPv4/IPv6 address used for all cURL operations.

set system option http-client source-interface <interface>

Several commands utilize curl to initiate transfers. Configure the local source interface used for all CURL operations.

Note: source-address and source-interface can not be used at the same time.

SSH client

set system option ssh-client source-address <address>

Use the specified address on the local machine as the source address of the connection. Only useful on systems with more than one address.

set system option ssh-client source-interface <interface>

Use the address of the specified interface on the local machine as the source address of the connection.

Keyboard Layout

When starting a VyOS live system (the installation CD) the configured keyboard layout defaults to US. As this might not suite everyone's use case you can adjust the used keyboard layout on the system console.

```
set system option keyboard-layout <us | fr | de | fi | no | dk>
```

Change system keyboard layout to given language.

Defaults to us.

Note: Changing the keymap only has an effect on the system console, using SSH or Serial remote access to the device is not affected as the keyboard layout here corresponds to your access system.

Performance

As more and more routers run on Hypervisors, especially with a NOS (Network Operating System) as VyOS, it makes fewer and fewer sense to use static resource bindings like `smp-affinity` as present in VyOS 1.2 and earlier to pin certain interrupt handlers to specific CPUs.

We now utilize `tuned` for dynamic resource balancing based on profiles.

See also:

<https://access.redhat.com/sites/default/files/attachments/201501-perf-brief-low-latency-tuning-rhel7-v2.1.pdf>

```
set system option performance < throughput | latency >
```

Configure one of the predefined system performance profiles.

- **throughput:** A server profile focused on improving network throughput. This profile favors performance over power savings by setting `intel_pstate` and `max_perf_pct=100` and increasing kernel network buffer sizes.

It enables transparent huge pages, and uses `cpupower` to set the performance `cpufreq` governor. It also sets `kernel.sched_min_granularity_ns` to 10 us, `kernel.sched_wakeup_granularity_ns` to 15 uss, and `vm.dirty_ratio` to 40%.

- **latency:** A server profile focused on lowering network latency. This profile favors performance over power savings by setting `intel_pstate` and `min_perf_pct=100`.

It disables transparent huge pages, and automatic NUMA balancing. It also uses `cpupower` to set the performance `cpufreq` governor, and requests a `cpu_dma_latency` value of 1. It also sets `busy_read` and `busy_poll` times to 50 us, and `tcp_fastopen` to 3.

7.11.13 System Proxy

Some IT environments require the use of a proxy to connect to the Internet. Without this configuration VyOS updates could not be installed directly by using the `add system image` command (*Update VyOS*).

```
set system proxy url <url>
```

Set proxy for all connections initiated by VyOS, including HTTP, HTTPS, and FTP (anonymous ftp).

```
set system proxy port <port>
```

Configure proxy port if it does not listen to the default port 80.

```
set system proxy username <username>
```

Some proxies require/support the “basic” HTTP authentication scheme as per [RFC 7617](#), thus a username can be configured.

set system proxy password <password>

Some proxies require/support the “basic” HTTP authentication scheme as per [RFC 7617](#), thus a password can be configured.

7.11.14 sFlow

VyOS supports sFlow accounting for both IPv4 and IPv6 traffic. The system acts as a flow exporter, and you are free to use it with any compatible collector.

sFlow is a technology that enables monitoring of network traffic by sending sampled packets to a collector device.

The sFlow accounting based on hsfowd <https://sflow.net/>

Configuration

set system sflow agent-address <address>

Configure sFlow agent IPv4 or IPv6 address

set system sflow agent-interface <interface>

Configure agent IP address associated with this interface.

set system sflow drop-monitor-limit <limit>

Dropped packets reported on DROPMON Netlink channel by Linux kernel are exported via the standard sFlow v5 extension for reporting dropped packets

set system sflow interface <interface>

Configure and enable collection of flow information for the interface identified by <interface>.

You can configure multiple interfaces which would participate in sflow accounting.

set system sflow polling <sec>

Configure schedule counter-polling in seconds (default: 30)

set system sflow sampling-rate <rate>

Use this command to configure the sampling rate for sFlow accounting (default: 1000)

set system sflow server <address> port <port>

Configure address of sFlow collector. sFlow server at <address> can be both listening on an IPv4 or IPv6 address.

set system sflow enable-egress

Use this command to if you need to sample also egress traffic

Example

```

set system sflow agent-address '192.0.2.14'
set system sflow agent-interface 'eth0'
set system sflow drop-monitor-limit '50'
set system sflow interface 'eth0'
set system sflow interface 'eth1'
set system sflow polling '30'
set system sflow sampling-rate '1000'
set system sflow server 192.0.2.1 port '6343'
set system sflow server 203.0.113.23 port '6343'

```

7.11.15 Syslog

Overview

By default, VyOS provides a minimal logging configuration with local storage and log rotation. All errors, including local7 messages, are saved to a local file. Emergency alerts are sent to the console.

To change these settings, enter configuration mode.

Syslog configuration

Syslog supports logging to multiple destinations: a local file, a console, or a remote syslog server over UDP or TCP.

The syslog configuration is organized into the following categories:

- Global settings
- Local logging
- Console logging
- Remote logging
- TLS-encrypted remote logging

Global settings

Configure the general behavior of the syslog service.

set system syslog marker interval <number>

Configure the interval, in seconds, for sending syslog mark messages.

Syslog mark messages confirm the logging service is operational.

Default: 1200 seconds.

set system syslog marker disable

Disable sending syslog mark messages.

set system syslog preserve-fqdn

Configure how the logging device's hostname appears in log messages sent to a remote syslog server.

If configured, the device includes its FQDN in log messages, even if the syslog server is in the same domain.

Local logging

Configure which log messages to save to a local log file.

```
set system syslog local <filename> facility <keyword> level <keyword>
```

Configure syslog to save log messages for a specific facility and severity level to ``/var/log/messages``.

Refer to the tables below for valid facility and severity options.

Console logging

Configure which log messages to send to /dev/console.

```
set system syslog console facility <keyword> level <keyword>
```

Configure syslog to send log messages for a specific facility and severity level to the device's console.

Refer to the tables below for valid facility and severity options.

Remote logging

Configure **remote logging** to send log messages to a remote syslog server.

Remote logging does not affect either **local** or **console logging** and runs in parallel with them. Remote logging supports sending log messages to multiple hosts.

```
set system syslog remote <address> facility <keyword> level <keyword>
```

Configure log transmission to the remote syslog server for a specific facility and severity level.

The server's address can be specified using either a FQDN or an IP address.

Refer to the tables below for valid facility and severity options.

```
set system syslog remote <address> protocol <udp | tcp>
```

Configure the protocol for log transmission.

The protocol can be either UDP or TCP. By default, log messages are sent over UDP.

```
set system syslog remote <address> port <port>
```

Configure the port for log transmission.

By default, the standard port 514 is used.

```
set system syslog remote <address> format include-timezone
```

Configure log transmission in the RFC 5424 format.

The RFC 5424 format includes the timezone in the timestamp. For example:

```
<34>1 2003-10-11T22:14:15.003-07:00 mymachine.example.com su - ID47 - BOM'su root'
↳ failed for lonvick on /dev/pts/8.
```

By default, log messages are sent in the RFC 3164 format. For example:

```
<34>Oct 11 22:14:15 mymachine su: 'su root' failed for lonvick on /dev/pts/8
```

```
set system syslog remote <address> format octet-counted
```

Enable octet-counted framing for log transmission.

When enabled, multi-line log messages are sent without splitting. Ensure the remote server supports octet-counted framing to avoid parsing errors.

Octet-counted framing is not available for the UDP protocol.

```
set system syslog remote <address> vrf <name>
```

Configure the VRF instance for log transmission.

```
set system syslog remote <address> source-address <address>
```

Configure the source IP address (IPv4 or IPv6) for log transmission.

TLS (Transport Layer Security)-encrypted remote logging

VyOS supports TLS-encrypted remote logging over TCP to ensure secure transmission of syslog data to remote syslog servers.

Prerequisites: Before configuring TLS-encrypted remote logging, ensure you have:

- A valid remote syslog server address.
- Valid CA and client certificates uploaded to the local PKI (Public Key Infrastructure) storage.
- The **remote syslog transport protocol** is set to **TCP**:

```
set system syslog remote <address> protocol tcp
```

Note: TLS-encrypted remote logging is **not supported** over **UDP**.

```
set system syslog remote <address> tls
```

Enable TLS-encrypted remote logging.

```
set system syslog remote <address> tls ca-certificate <ca_name>
```

Configure the CA certificate.

The syslog client uses the CA certificate to verify the identity of the remote syslog server.

The CA certificate is required for **all** authentication modes except anon.

```
set system syslog remote <address> tls certificate <cert_name>
```

Configure the client certificate.

The remote syslog server uses the client certificate to verify the identity of the syslog client.

The client certificate is required if the remote syslog server enforces client certificate verification.

```
set system syslog remote <address> tls auth-mode <anon | fingerprint | certvalid | name>
```

Configure the authentication mode.

The authentication mode defines how the syslog client verifies the syslog server's identity.

The following authentication modes are available:

- **anon (default):** Allows encrypted connections without verifying the syslog server's identity. This mode is **not recommended**, as it is vulnerable to MITM (Man-in-the-Middle) attacks.
- **fingerprint:** Verifies the server's certificate fingerprint against the value preconfigured with:

```
set system syslog remote <address> tls permitted-peer <peer>
```

- **certvalid**: Verifies the server certificate is signed by a trusted CA, skipping CN (Common Name) check.
- **name**: Verifies that:
 - The server's certificate is signed by a trusted CA.
 - The CN in the certificate matches the value preconfigured with:

```
set system syslog remote <address> tls permitted-peer <peer>
```

This is a **recommended** secure mode for production environments.

set system syslog remote <address> tls permitted-peer <peer>

Configure the peer certificate identifiers.

The certificate identifier format depends on the authentication mode:

- **fingerprint**: Enter the expected certificate fingerprints (SHA-1 or SHA-256).
- **name**: Enter the expected certificate CNs (Common Names).

For anon and certvalid authentication modes, certificate identifiers are not required.

Examples:

```
# Example of 'anon' authentication mode
set system syslog remote 10.10.2.3 facility all level debug
set system syslog remote 10.10.2.3 port 6514
set system syslog remote 10.10.2.3 protocol tcp
set system syslog remote 10.10.2.3 tls auth-mode anon
# or just use 'set system syslog remote 10.10.2.3 tls'

# Example of 'certvalid' authentication mode
set system syslog remote elk.example.com facility all level debug
set system syslog remote elk.example.com port 6514
set system syslog remote elk.example.com protocol tcp
set system syslog remote elk.example.com tls ca-certificate my-ca
set system syslog remote elk.example.com tls auth-mode certvalid

# Example of 'fingerprint' authentication mode
set system syslog remote syslog.example.com facility all level debug
set system syslog remote syslog.example.com port 6514
set system syslog remote syslog.example.com protocol tcp
set system syslog remote syslog.example.com tls ca-certificate my-ca
set system syslog remote syslog.example.com tls auth-mode fingerprint
set system syslog remote syslog.example.com tls permitted-peers 'SHA1:10:C4:26:...,
↪SHA256:7B:4B:10:...'

# Example of 'name' authentication mode
set system syslog remote graylog.example.com facility all level debug
set system syslog remote graylog.example.com port 6514
set system syslog remote graylog.example.com protocol tcp
set system syslog remote graylog.example.com tls ca-certificate my-ca
```

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```
set system syslog remote graylog.example.com tls certificate syslog-client
set system syslog remote graylog.example.com tls auth-mode name
set system syslog remote graylog.example.com tls permitted-peers 'graylog.example.com'
```

Security recommendations

- For secure deployments, always use the `name` authentication mode. It ensures that the server is validated by a trusted CA and that the hostname matches the certificate.
- Use the `anon` authentication mode only in testing environments, as it doesn't provide server authentication.
- Ensure private keys are generated, stored, and maintained exclusively within the *PKI system*.

Syslog facilities

This section lists facilities used by syslog. Most facility names are self-explanatory. The `local0`–`local7` facilities are used for custom purposes, such as logging from network nodes and equipment. Facility assignment is flexible and should be tailored to your company's needs. Consider facilities as categorization tools, rather than strict directives.

Facility code	Keyword	Description
	<code>all</code>	All facilities
0	<code>kern</code>	Kernel messages
1	<code>user</code>	User-level messages
2	<code>mail</code>	Mail system
3	<code>daemon</code>	System daemons
4	<code>auth</code>	Security/authentication messages
5	<code>syslog</code>	Messages generated internally by syslog
6	<code>lpr</code>	Line printer subsystem
7	<code>news</code>	Network news subsystem
8	<code>uucp</code>	UUCP subsystem
9	<code>cron</code>	Clock daemon
10	<code>security</code>	Security/authentication messages
11	<code>ftp</code>	FTP daemon
12	<code>ntp</code>	NTP subsystem
13	<code>logaudit</code>	Log audit
14	<code>logalert</code>	Log alert
15	<code>clock</code>	clock daemon (note 2)
16	<code>local0</code>	local use 0 (local0)
17	<code>local1</code>	local use 1 (local1)
18	<code>local2</code>	local use 2 (local2)
19	<code>local3</code>	local use 3 (local3)
20	<code>local4</code>	local use 4 (local4)
21	<code>local5</code>	local use 5 (local5)
22	<code>local6</code>	local use 6 (local6)
23	<code>local7</code>	local use 7 (local7)

Severity levels

Value	Severity	Key-word	Description
0	Emergency	all emerg	Log everything. System is unusable - a panic condition.
1	Alert	alert	Action must be taken immediately - A condition that should be corrected immediately, such as a corrupted system database.
2	Critical	crit	Critical conditions - e.g., hard drive errors.
3	Error	err	Error conditions.
4	Warning	warning	Warning conditions.
5	Notice	notice	Normal but significant conditions - conditions that are not error conditions, but that may require special handling.
6	Informational	info	Informational messages.
7	Debug	debug	Debug-level messages - Messages that contain information normally of use only when debugging a program.

Display logs

show log [all | authorization | cluster | conntrack-sync | ...]

Display logs for a specific category on the console.

Use tab completion to view a list of available categories.

If no category is specified, all logs are shown.

show log image <name> [all | authorization | directory | file <file name> | tail <lines>]

Display logs for a specific image on the console.

Available log categories:

all	Displays the contents of system log files of the specified image.
authorization	Displays authorization attempts of the specified image.
directory	Displays user-defined log files of the specified image.
file <file name>	Displays the contents of a specified user-defined log file of the specified image.
tail	Displays last lines of the system log of the specified image.
<lines>	Number of lines to be displayed, default 10.

If no category is specified, the contents of the main syslog file are displayed.

Hint: Use `show log | strip-private` to hide private data when displaying your logs.

7.11.16 Sysctl

Note: This page is a stub and needs expansion. Contributions welcome via the [VyOS documentation repository](#).

This chapter describes how to configure kernel parameters at runtime.

sysctl is used to modify kernel parameters at runtime. The parameters available are those listed under `/proc/sys/`.

set system sysctl parameter <parameter> value <value>

7.11.17 Task Scheduler

The task scheduler allows you to execute tasks on a given schedule. It makes use of UNIX [cron](#).

Note: All scripts executed this way are executed as root user - this may be dangerous. Together with *Command scripting* this can be used for automating (re-)configuration.

set system task-scheduler task <task> interval <interval>

Specify the time interval when <task> should be executed. The interval is specified as number with one of the following suffixes:

- none - Execution interval in minutes
- m - Execution interval in minutes
- h - Execution interval in hours
- d - Execution interval in days

Note: If suffix is omitted, minutes are implied.

set system task-scheduler task <task> crontab-spec <spec>

Set execution time in common cron time format. A cron <spec> of `30 */6 * * *` would execute the <task> at minute 30 past every 6th hour.

set system task-scheduler task <task> executable path <path>

Specify absolute <path> to script which will be run when <task> is executed.

set system task-scheduler task <task> executable arguments <args>

Arguments which will be passed to the executable.

7.11.18 Time Zone

Time Zone setting is very important as e.g all your logfile entries will be based on the configured zone. Without proper time zone configuration it will be very difficult to compare logfiles from different systems.

set system time-zone <timezone>

Specify the systems <timezone> as the Region/Location that best defines your location. For example, specifying US/Pacific sets the time zone to US Pacific time.

Command completion can be used to list available time zones. The adjustment for daylight time will take place automatically based on the time of year.

7.11.19 Updates

VyOS supports online checking for updates

Configuration

set system update-check auto-check

Configure auto-checking for new images

set system update-check url <url>

Configure a URL that contains information about images.

Example

```
set system update-check auto-check
set system update-check url 'https://raw.githubusercontent.com/vyos/vyos-nightly-build/
↪rolling/version.json'
```

Check:

```
vyos@r4:~$ show system updates
Current version: 1.5-rolling-202312220023

Update available: 1.5-rolling-202312250024
Update URL: https://github.com/vyos/vyos-nightly-build/releases/download/1.5-rolling-
↪202312250024/1.5-rolling-202312250024-amd64.iso
vyos@r4:~$

vyos@r4:~$ add system image latest
```


7.11.20 Watchdog

VyOS supports hardware watchdog timers to automatically reboot the system if it becomes unresponsive. This is particularly useful for remote or embedded systems where physical access is limited.

A watchdog timer is a hardware or software mechanism that automatically resets the system if the operating system stops responding within a configured timeout period. The system will periodically notify the watchdog that it is still running. If the watchdog is not notified within the timeout period, the watchdog will reset the system.

Configuration

The watchdog feature is configured under the `system watchdog` configuration tree. The presence of the `system watchdog` node enables the watchdog feature.

set system watchdog

Enable watchdog support.

The watchdog is enabled only when a watchdog device is available as `/dev/watchdog0`.

Note: If multiple watchdog devices are present, only the first watchdog device is supported (VyOS uses `/dev/watchdog0` only).

If `/dev/watchdog0` does not exist and no module is configured, commit will fail. If a module is configured but `/dev/watchdog0` still cannot be created, VyOS will emit a warning and will not enable the systemd watchdog.

set system watchdog module <module-name>

Specify the kernel watchdog driver module to load for `/dev/watchdog0`.

The configured module must be a watchdog driver module, not an arbitrary kernel module.

In most cases, this option is not required as the kernel will automatically load the appropriate watchdog driver for your system. Use this option if the kernel fails to load the required driver, or when you want to use the software watchdog (`softdog`).

Common modules include:

- `softdog` - Software watchdog timer (available on all systems)
- `iTCO_wdt` - Intel TCO watchdog timer
- `sp5100_tco` - AMD SP5100 TCO watchdog timer
- `i6300esb` - Intel 6300ESB watchdog timer
- `ipmi_watchdog` - IPMI watchdog timer

Warning: `softdog` is not a hardware watchdog. It is implemented using kernel timers and therefore depends on the Linux kernel continuing to run. In some fault conditions (for example, a kernel hang), `softdog` may not be able to trigger a reset.

Prefer a hardware watchdog driver whenever possible, as hardware watchdogs can operate independently of the operating system.

If no module is specified, VyOS will use an existing `/dev/watchdog0` device if available.

Note: If a module is specified but a different driver is actually bound to watchdog0, VyOS will emit a warning during commit.

Example:

```
set system watchdog module softdog
```

set system watchdog timeout <seconds>

Set the watchdog timeout for normal runtime operation in seconds.

Valid range: 1-65535 seconds

Note: Some watchdog drivers expose minimum and maximum supported runtime timeouts via sysfs. When available, VyOS validates timeout against those driver limits during commit.

This is the interval during which the system must respond to the watchdog. If the system does not respond within this time, the watchdog will trigger a reboot.

Example:

```
set system watchdog timeout 30
```

set system watchdog shutdown-timeout <seconds>

Set the watchdog timeout during system shutdown in seconds.

Valid range: 60-65535 seconds

This extended timeout allows the system to complete a graceful shutdown without triggering the watchdog.

Warning: Setting this value too low (below 120 seconds) may cause unclean shutdowns, as the system may not have enough time to properly stop all services and flush disk buffers. The recommended minimum value is 120 seconds.

Example:

```
set system watchdog shutdown-timeout 180
```

set system watchdog reboot-timeout <seconds>

Set the watchdog timeout during system reboot in seconds.

Valid range: 60-65535 seconds

This extended timeout allows the system to complete the reboot process without triggering the watchdog during the transition.

Warning: Setting this value too low (below 120 seconds) may cause unclean reboots, as the system may not have enough time to properly stop all services before restarting. The recommended minimum value is 120 seconds.

Example:

```
set system watchdog reboot-timeout 180
```

Examples

Basic Configuration with Software Watchdog

This example configures a basic software watchdog with default timeouts:

```
set system watchdog module softdog
```

This will:

- Enable the watchdog feature
- Load the `softdog` kernel module
- Use a 10-second runtime timeout (default)
- Use 120-second shutdown and reboot timeouts (default)

Advanced Configuration

This example shows a more customized configuration suitable for a production system:

```
set system watchdog module iTCO_wdt
set system watchdog timeout 30
set system watchdog shutdown-timeout 300
set system watchdog reboot-timeout 300
```

This configuration:

- Enables the watchdog feature
- Loads the Intel TCO hardware watchdog module
- Sets a 30-second runtime timeout
- Allows 5 minutes for shutdown and reboot operations

Best Practices

- **Start with conservative timeouts:** Use longer timeouts initially and reduce them as you gain confidence in system stability.
- **Test before deployment:** Verify the watchdog works as expected in a non-production environment before deploying to production systems.
- **Choose appropriate modules:** Use hardware watchdog modules (like `iTCO_wdt`) when available, as they are more reliable than software watchdogs.
- **Consider shutdown time:** Set `shutdown-timeout` and `reboot-timeout` values high enough to allow for normal shutdown procedures, especially on systems with many services or slow storage.
- **Monitor watchdog events:** Check system logs after any unexpected reboots to determine if the watchdog triggered the reboot.

- **Remote systems:** For systems without physical console access, use conservative timeout values to avoid false-positive reboots during high load conditions.

Note: The watchdog configuration takes effect immediately after commit, but systemd must be reloaded. This happens automatically during commit.

Warning: Incorrect watchdog configuration on remote systems can result in unexpected reboots. Always test watchdog settings in a controlled environment before deploying to production systems.

7.11.21 Default Gateway/Route

In the past (VyOS 1.1) used a gateway-address configured under the system tree (`set system gateway-address <address>`), this is no longer supported and existing configurations are migrated to the new CLI command.

Configuration

set protocols static route 0.0.0.0/0 next-hop <address>

Specify static route into the routing table sending all non local traffic to the nexthop address <address>.

delete protocols static route 0.0.0.0/0

Delete default route from the system.

Operation

show ip route 0.0.0.0

Show routing table entry for the default route.

```
vyos@vyos:~$ show ip route 0.0.0.0
Routing entry for 0.0.0.0/0
Known via "static", distance 10, metric 0, best
Last update 09:46:30 ago
* 172.18.201.254, via eth0.201
```

See also:

Configuration of *Static*

7.12 Traffic Policy

7.12.1 QoS

The generic name of Quality of Service or Traffic Control involves things like shaping traffic, scheduling or dropping packets, which are the kind of things you may want to play with when you have, for instance, a bandwidth bottleneck in a link and you want to somehow prioritize some type of traffic over another.

`tc` is a powerful tool for Traffic Control found at the Linux kernel. However, its configuration is often considered a cumbersome task. Fortunately, VyOS eases the job through its CLI, while using `tc` as backend.

How to make it work

In order to have VyOS Traffic Control working you need to follow 2 steps:

1. **Create a traffic policy.**
2. **Apply the traffic policy to an interface ingress or egress.**

But before learning to configure your policy, we will warn you about the different units you can use and also show you what *classes* are and how they work, as some policies may require you to configure them.

Units

When configuring your traffic policy, you will have to set data rate values, watch out the units you are managing, it is easy to get confused with the different prefixes and suffixes you can use. VyOS will always show you the different units you can use.

Prefixes

They can be **decimal** prefixes.

kbit	(10 ³)	kilobit per second
mbit	(10 ⁶)	megabit per second
gbit	(10 ⁹)	gigabit per second
tbit	(10 ¹²)	terabit per second
kbps	(8*10 ³)	kilobyte per second
mbps	(8*10 ⁶)	megabyte per second
gbps	(8*10 ⁹)	gigabyte per second
tbps	(8*10 ¹²)	terabyte per second

Or **binary** prefixes.

kibit	(2 ¹⁰ = 1024)	kibibit per second
mibit	(2 ²⁰ = 1024 ²)	mebibit per second
gibit	(2 ³⁰ = 1024 ³)	gibibit per second
tbit	(2 ⁴⁰ = 1024 ⁴)	tebibit per second
kibps	(1024*8)	kibibyte (KiB) per second
mibps	(1024 ² *8)	mebibyte (MiB) per second
gibps	(1024 ³ *8)	gibibyte (GiB) per second
tibps	(1024 ⁴ *8)	tebibyte (TiB) per second

Suffixes

A *bit* is written as **bit**,

kbit	(kilobits per second)
mbit	(megabits per second)
gbit	(gigabits per second)
tbit	(terabits per second)

while a *byte* is written as a single **b**.

```

kbps (kilobytes per second)
mbps (megabytes per second)
gbps (gigabytes per second)

```

Classes

In the *Creating a traffic policy* section you will see that some of the policies use *classes*. Those policies let you distribute traffic into different classes according to different parameters you can choose. So, a class is just a specific type of traffic you select.

The ultimate goal of classifying traffic is to give each class a different treatment.

Matching traffic

In order to define which traffic goes into which class, you define filters (that is, the matching criteria). Packets go through these matching rules (as in the rules of a firewall) and, if a packet matches the filter, it is assigned to that class.

In VyOS, a class is identified by a number you can choose when configuring it.

Note: The meaning of the Class ID is not the same for every type of policy. Normally policies just need a meaningless number to identify a class (Class ID), but that does not apply to every policy. The number of a class in a Priority Queue it does not only identify it, it also defines its priority.

```
set qos policy <policy> <policy-name> class <class-ID> match <class-matching-rule-name>
```

In the command above, we set the type of policy we are going to work with and the name we choose for it; a class (so that we can differentiate some traffic) and an identifiable number for that class; then we configure a matching rule (or filter) and a name for it.

A class can have multiple match filters:

```
set qos policy shaper MY-SHAPER class 30 match HTTP
set qos policy shaper MY-SHAPER class 30 match HTTPs
```

A match filter can contain multiple criteria and will match traffic if all those criteria are true.

For example:

```
set qos policy shaper MY-SHAPER class 30 match HTTP ip protocol tcp
set qos policy shaper MY-SHAPER class 30 match HTTP ip source port 80
```

This will match TCP traffic with source port 80.

There are many parameters you will be able to use in order to match the traffic you want for a class:

- Ethernet (protocol, destination address or source address)
- Interface name
- IPv4 (DSCP value, maximum packet length, protocol, source address, destination address, source port, destination port or TCP flags)
- IPv6 (DSCP value, maximum payload length, protocol, source address, destination address, source port, destination port or TCP flags)

- Firewall mark
- VLAN ID

When configuring your filter, you can use the Tab key to see the many different parameters you can configure.

```
vyos@vyos# set qos policy shaper MY-SHAPER class 30 match MY-FIRST-FILTER
Possible completions:
  description  Description
> ether       Ethernet header match
  interface    Interface to use
> ip          Match IP protocol header
> ipv6        Match IPV6 protocol header
  mark         Match on mark applied by firewall
  vif          Virtual Local Area Network (VLAN) ID for this match
```

As shown in the example above, one of the possibilities to match packets is based on marks done by the firewall, [that can give you a great deal of flexibility](#).

You can also write a description for a filter:

```
set qos policy shaper MY-SHAPER class 30 match MY-FIRST-FILTER description "My filter.
↪description"
```

Note: An IPv4 TCP filter will only match packets with an IPv4 header length of 20 bytes (which is the majority of IPv4 packets anyway).

Note: IPv6 TCP filters will only match IPv6 packets with no header extension, see https://en.wikipedia.org/wiki/IPv6_packet#Extension_headers

Traffic Match Group

In some case where we need to have an organization of our matching selection, in order to be more flexible and organize with our filter definition. We can apply traffic match groups, allowing us to create distinct filter groups within our policy and define various parameters for each group:

```
set qos traffic-match-group <group_name> match <match_name>
Possible completions:
  description  Description
> ip          Match IP protocol header
> ipv6        Match IPv6 protocol header
  mark         Match on mark applied by firewall
  vif          Virtual Local Area Network (VLAN) ID for this match
```

inherit matches from another group

```
set qos traffic-match-group <group_name> match-group <match_group_name>
```

A match group can contain multiple criteria and inherit them in the same policy.

For example:

```
set qos traffic-match-group Mission-Critical match AF31 ip dscp 'AF31'
set qos traffic-match-group Mission-Critical match AF32 ip dscp 'AF42'
set qos traffic-match-group Mission-Critical match CS3 ip dscp 'CS3'
set qos traffic-match-group Streaming-Video match AF11 ip dscp 'AF11'
set qos traffic-match-group Streaming-Video match AF41 ip dscp 'AF41'
set qos traffic-match-group Streaming-Video match AF43 ip dscp 'AF43'
set qos policy shaper VyOS-HTB class 10 bandwidth '30%'
set qos policy shaper VyOS-HTB class 10 description 'Multimedia'
set qos policy shaper VyOS-HTB class 10 match CS4 ip dscp 'CS4'
set qos policy shaper VyOS-HTB class 10 match-group 'Streaming-Video'
set qos policy shaper VyOS-HTB class 10 priority '1'
set qos policy shaper VyOS-HTB class 10 queue-type 'fair-queue'
set qos policy shaper VyOS-HTB class 20 description 'MC'
set qos policy shaper VyOS-HTB class 20 match-group 'Mission-Critical'
set qos policy shaper VyOS-HTB class 20 priority '2'
set qos policy shaper VyOS-HTB class 20 queue-type 'fair-queue'
set qos policy shaper VyOS-HTB default bandwidth '20%'
set qos policy shaper VyOS-HTB default queue-type 'fq-codel'
```

In this example, we can observe that different DSCP criteria are defined based on our QoS configuration within the same policy group.

Default

Often you will also have to configure your *default* traffic in the same way you do with a class. *Default* can be considered a class as it behaves like that. It contains any traffic that did not match any of the defined classes, so it is like an open class, a class without matching filters.

Class treatment

Once a class has a filter configured, you will also have to define what you want to do with the traffic of that class, what specific Traffic-Control treatment you want to give it. You will have different possibilities depending on the Traffic Policy you are configuring.

```
vyos@vyos# set qos policy shaper MY-SHAPER class 30
Possible completions:
  bandwidth    Available bandwidth for this policy (default: auto)
  burst        Burst size for this class (default: 15k)
  ceiling      Bandwidth limit for this class
  codel-quantum
                Deficit in the fair queuing algorithm (default 1514)
  description  Description
  flows        Number of flows into which the incoming packets are classified(default.
→ 1024)
  interval     Interval used to measure the delay (default 100)
+> match       Class matching rule name
  priority     Priority for rule evaluation
  queue-limit  Maximum queue size
  queue-type   Queue type for default traffic (default: fq-codel)
  set-dscp     Change the Differentiated Services (DiffServ) field in the IP header
  target       Acceptable minimum standing/persistent queue delay (default: 5)
```


For instance, with `set qos policy shaper MY-SHAPER class 30 set-dscp EF` you would be modifying the DSCP field value of packets in that class to Expedite Forwarding.

DSCP values as per [RFC 2474](#) and [RFC 4595](#):

Binary value	Configured value	Drop rate	Description
101110	46	-	Expedited forwarding (EF)
000000	0	-	Best effort traffic, default
001010	10	Low	Assured Forwarding(AF) 11
001100	12	Medium	Assured Forwarding(AF) 12
001110	14	High	Assured Forwarding(AF) 13
010010	18	Low	Assured Forwarding(AF) 21
010100	20	Medium	Assured Forwarding(AF) 22
010110	22	High	Assured Forwarding(AF) 23
011010	26	Low	Assured Forwarding(AF) 31
011100	28	Medium	Assured Forwarding(AF) 32
011110	30	High	Assured Forwarding(AF) 33
100010	34	Low	Assured Forwarding(AF) 41
100100	36	Medium	Assured Forwarding(AF) 42
100110	38	High	Assured Forwarding(AF) 43

Embedding one policy into another one

Often we need to embed one policy into another one. It is possible to do so on classful policies, by attaching a new policy into a class. For instance, you might want to apply different policies to the different classes of a Round-Robin policy you have configured.

A common example is the case of some policies which, in order to be effective, they need to be applied to an interface that is directly connected where the bottleneck is. If your router is not directly connected to the bottleneck, but some hop before it, you can emulate the bottleneck by embedding your non-shaping policy into a classful shaping one so that it takes effect.

You can configure a policy into a class through the `queue-type` setting.

```
set qos policy shaper FQ-SHAPER bandwidth 4gbit
set qos policy shaper FQ-SHAPER default bandwidth 100%
set qos policy shaper FQ-SHAPER default queue-type fq-codec
```

As shown in the last command of the example above, the `queue-type` setting allows these combinations. You will be able to use it in many policies.

Note: Some policies already include other embedded policies inside. That is the case of Shaper: each of its classes use fair-queue unless you change it.

Creating a traffic policy

VyOS lets you control traffic in many different ways, here we will cover every possibility. You can configure as many policies as you want, but you will only be able to apply one policy per interface and direction (inbound or outbound).

Some policies can be combined, you will be able to embed a different policy that will be applied to a class of the main policy.

Hint: If you are looking for a policy for your outbound traffic but you don't know which one you need and you don't want to go through every possible policy shown here, **our bet is that highly likely you are looking for a Shaper policy and you want to *set its queues* as FQ-CoDel.**

Drop Tail

Queueing discipline: PFIFO (Packet First In First Out).

Applies to: Outbound traffic.

This the simplest queue possible you can apply to your traffic. Traffic must go through a finite queue before it is actually sent. You must define how many packets that queue can contain.

When a packet is to be sent, it will have to go through that queue, so the packet will be placed at the tail of it. When the packet completely goes through it, it will be dequeued emptying its place in the queue and being eventually handed to the NIC to be actually sent out.

Despite the Drop-Tail policy does not slow down packets, if many packets are to be sent, they could get dropped when trying to get enqueued at the tail. This can happen if the queue has still not been able to release enough packets from its head.

This is the policy that requires the lowest resources for the same amount of traffic. But **very likely you do not need it as you cannot get much from it. Sometimes it is used just to enable logging.**

set qos policy drop-tail <policy-name> queue-limit <number-of-packets>

Use this command to configure a drop-tail policy (PFIFO). Choose a unique name for this policy and the size of the queue by setting the number of packets it can contain (maximum 4294967295).

Fair Queue

Queueing discipline: SFQ (Stochastic Fairness Queuing).

Applies to: Outbound traffic.

Fair Queue is a work-conserving scheduler which schedules the transmission of packets based on flows, that is, it balances traffic distributing it through different sub-queues in order to ensure fairness so that each flow is able to send data in turn, preventing any single one from drowning out the rest.

set qos policy fair-queue <policy-name>

Use this command to create a Fair-Queue policy and give it a name. It is based on the Stochastic Fairness Queueing and can be applied to outbound traffic.

In order to separate traffic, Fair Queue uses a classifier based on source address, destination address and source port. The algorithm enqueues packets to hash buckets based on those tree parameters. Each of these buckets should represent

a unique flow. Because multiple flows may get hashed to the same bucket, the hashing algorithm is perturbed at configurable intervals so that the unfairness lasts only for a short while. Perturbation may however cause some inadvertent packet reordering to occur. An advisable value could be 10 seconds.

One of the uses of Fair Queue might be the mitigation of Denial of Service attacks.

set qos policy fair-queue <policy-name> hash-interval <seconds>

Use this command to define a Fair-Queue policy, based on the Stochastic Fairness Queueing, and set the number of seconds at which a new queue algorithm perturbation will occur (maximum 4294967295).

When dequeuing, each hash-bucket with data is queried in a round robin fashion. You can configure the length of the queue.

set qos policy fair-queue <policy-name> queue-limit <limit>

Use this command to define a Fair-Queue policy, based on the Stochastic Fairness Queueing, and set the number of maximum packets allowed to wait in the queue. Any other packet will be dropped.

Note: Fair Queue is a non-shaping (work-conserving) policy, so it will only be useful if your outgoing interface is really full. If it is not, VyOS will not own the queue and Fair Queue will have no effect. If there is bandwidth available on the physical link, you can embed Fair-Queue into a classful shaping policy to make sure it owns the queue.

FQ-CoDel

Queueing discipline: Fair/Flow Queue CoDel.

Applies to: Outbound Traffic.

The FQ-CoDel policy distributes the traffic into 1024 FIFO queues and tries to provide good service between all of them. It also tries to keep the length of all the queues short.

FQ-CoDel fights bufferbloat and reduces latency without the need of complex configurations. It has become the new default Queueing Discipline for the interfaces of some GNU/Linux distributions.

It uses a stochastic model to classify incoming packets into different flows and is used to provide a fair share of the bandwidth to all the flows using the queue. Each flow is managed by the CoDel queueing discipline. Reordering within a flow is avoided since CoDel internally uses a FIFO queue.

FQ-CoDel is based on a modified Deficit Round Robin (DRR) queue scheduler with the CoDel Active Queue Management (AQM) algorithm operating on each queue.

Note: FQ-Codel is a non-shaping (work-conserving) policy, so it will only be useful if your outgoing interface is really full. If it is not, VyOS will not own the queue and FQ-Codel will have no effect. If there is bandwidth available on the physical link, you can embed FQ-Codel into a classful shaping policy to make sure it owns the queue. If you are not sure if you need to embed your FQ-CoDel policy into a Shaper, do it.

FQ-CoDel is tuned to run ok with its default parameters at 10Gbit speeds. It might work ok too at other speeds without configuring anything, but here we will explain some cases when you might want to tune its parameters.

When running it at 1Gbit and lower, you may want to reduce the `queue-limit` to 1000 packets or less. In rates like 10Mbit, you may want to set it to 600 packets.

If you are using FQ-CoDel embedded into Shaper and you have large rates (100Mbit and above), you may consider increasing quantum to 8000 or higher so that the scheduler saves CPU.

On low rates (below 40Mbit) you may want to tune quantum down to something like 300 bytes.

At very low rates (below 3Mbit), besides tuning quantum (300 keeps being ok) you may also want to increase target to something like 15ms and increase interval to something around 150 ms.

set qos policy fq-codel <policy name> codel-quantum <bytes>

Use this command to configure an fq-codel policy, set its name and the maximum number of bytes (default: 1514) to be dequeued from a queue at once.

set qos policy fq-codel <policy name> flows <number-of-flows>

Use this command to configure an fq-codel policy, set its name and the number of sub-queues (default: 1024) into which packets are classified.

set qos policy fq-codel <policy name> interval <milliseconds>

Use this command to configure an fq-codel policy, set its name and the time period used by the control loop of CoDel to detect when a persistent queue is developing, ensuring that the measured minimum delay does not become too stale (default: 100ms).

set qos policy fq-codel <policy-name> queue-limit <number-of-packets>

Use this command to configure an fq-codel policy, set its name, and define a hard limit on the real queue size. When this limit is reached, new packets are dropped (default: 10240 packets).

set qos policy fq-codel <policy-name> target <milliseconds>

Use this command to configure an fq-codel policy, set its name, and define the acceptable minimum standing/persistent queue delay. This minimum delay is identified by tracking the local minimum queue delay that packets experience (default: 5ms).

Example

A simple example of an FQ-CoDel policy working inside a Shaper one.

```
set qos policy shaper FQ-CODEL-SHAPER bandwidth 2gbit
set qos policy shaper FQ-CODEL-SHAPER default bandwidth 100%
set qos policy shaper FQ-CODEL-SHAPER default queue-type fq-codel
```

Limiter

Queueing discipline: Ingress policer.

Applies to: Inbound traffic.

Limiter is one of those policies that uses classes (Ingress qdisc is actually a classless policy but filters do work in it).

The limiter performs basic ingress policing of traffic flows. Multiple classes of traffic can be defined and traffic limits can be applied to each class. Although the policer uses a token bucket mechanism internally, it does not have the capability to delay a packet as a shaping mechanism does. Traffic exceeding the defined bandwidth limits is directly dropped. A maximum allowed burst can be configured too.

You can configure classes (up to 4090) with different settings and a default policy which will be applied to any traffic not matching any of the configured classes.

Note: In the case you want to apply some kind of **shaping** to your **inbound** traffic, check the ingress-shaping section.

```
set qos policy limiter <policy-name> class <class ID> match <match-name> description <description>
```

Use this command to configure an Ingress Policer, defining its name, a class identifier (1-4090), a class matching rule name and its description.

Once the matching rules are set for a class, you can start configuring how you want matching traffic to behave.

```
set qos policy limiter <policy-name> class <class-ID> bandwidth <rate>
```

Use this command to configure an Ingress Policer, defining its name, a class identifier (1-4090) and the maximum allowed bandwidth for this class.

```
set qos policy limiter <policy-name> class <class-ID> burst <burst-size>
```

Use this command to configure an Ingress Policer, defining its name, a class identifier (1-4090) and the burst size in bytes for this class (default: 15).

```
set qos policy limiter <policy-name> default bandwidth <rate>
```

Use this command to configure an Ingress Policer, defining its name and the maximum allowed bandwidth for its default policy.

```
set qos policy limiter <policy-name> default burst <burst-size>
```

Use this command to configure an Ingress Policer, defining its name and the burst size in bytes (default: 15) for its default policy.

```
set qos policy limiter <policy-name> class <class ID> priority <value>
```

Use this command to configure an Ingress Policer, defining its name, a class identifier (1-4090), and the priority (0-20, default 20) in which the rule is evaluated (the lower the number, the higher the priority).

Network Emulator

Queueing discipline: netem (Network Emulator) + TBF (Token Bucket Filter).

Applies to: Outbound traffic.

VyOS Network Emulator policy emulates the conditions you can suffer in a real network. You will be able to configure things like rate, burst, delay, packet loss, packet corruption or packet reordering.

This could be helpful if you want to test how an application behaves under certain network conditions.

```
set qos policy network-emulator <policy-name> bandwidth <rate>
```

Use this command to configure the maximum rate at which traffic will be shaped in a Network Emulator policy. Define the name of the policy and the rate.

```
set qos policy network-emulator <policy-name> burst <burst-size>
```

Use this command to configure the burst size of the traffic in a Network Emulator policy. Define the name of the Network Emulator policy and its traffic burst size (it will be configured through the Token Bucket Filter qdisc). Default: 15kb. It will only take effect if you have configured its bandwidth too.

```
set qos policy network-emulator <policy-name> delay <delay>
```

Use this command to configure a Network Emulator policy defining its name and the fixed amount of time you want to add to all packet going out of the interface. The latency will be added through the Token Bucket Filter qdisc. It will only take effect if you have configured its bandwidth too. You can use secs, ms and us. Default: 50ms.

```
set qos policy network-emulator <policy-name> corruption <percent>
```

Use this command to emulate noise in a Network Emulator policy. Set the policy name and the percentage of corrupted packets you want. A random error will be introduced in a random position for the chosen percent of packets.

```
set qos policy network-emulator <policy-name> loss <percent>
```

Use this command to emulate packet-loss conditions in a Network Emulator policy. Set the policy name and the percentage of loss packets your traffic will suffer.

```
set traffic-policy network-emulator <policy-name> reordering <percent>
```

Use this command to emulate packet-reordering conditions in a Network Emulator policy. Set the policy name and the percentage of reordered packets your traffic will suffer.

```
set traffic-policy network-emulator <policy-name> queue-limit <limit>
```

Use this command to define the length of the queue of your Network Emulator policy. Set the policy name and the maximum number of packets (1-4294967295) the queue may hold queued at a time.

Priority Queue

Queueing discipline: PRIO.

Applies to: Outbound traffic.

The Priority Queue is a classful scheduling policy. It does not delay packets (Priority Queue is not a shaping policy), it simply dequeues packets according to their priority.

Note: Priority Queue, as other non-shaping policies, is only useful if your outgoing interface is really full. If it is not, VyOS will not own the queue and Priority Queue will have no effect. If there is bandwidth available on the physical link, you can embed Priority Queue into a classful shaping policy to make sure it owns the queue. In that case packets can be prioritized based on DSCP.

Up to seven queues -defined as classes with different priorities- can be configured. Packets are placed into queues based on associated match criteria. Packets are transmitted from the queues in priority order. If classes with a higher priority are being filled with packets continuously, packets from lower priority classes will only be transmitted after traffic volume from higher priority classes decreases.

Note: In Priority Queue we do not define classes with a meaningless class ID number but with a class priority number (1-7). The lower the number, the higher the priority.

As with other policies, you can define different type of matching rules for your classes:

```
vyos@vyos# set qos policy priority-queue MY-PRIO class 3 match MY-MATCH-RULE
```

Possible completions:

description	Description
> ether	Ethernet header match
interface	Interface to use
> ip	Match IP protocol header
> ipv6	Match IPV6 protocol header
mark	Match on mark applied by firewall
vif	Virtual Local Area Network (VLAN) ID for this match

As with other policies, you can embed other policies into the classes (and default) of your Priority Queue policy through the `queue-type` setting:

```
vyos@vyos# set qos policy priority-queue MY-PRIO class 3 queue-type
Possible completions:
  drop-tail      First-In-First-Out (FIFO) (default)
  fq-codel       Fair Queue Codel
  fair-queue     Stochastic Fair Queue (SFQ)
  priority       Priority queueing
  random-detect  Random Early Detection (RED)
```

set qos policy priority-queue <policy-name> class <class-ID> queue-limit <limit>

Use this command to configure a Priority Queue policy, set its name, set a class with a priority from 1 to 7 and define a hard limit on the real queue size. When this limit is reached, new packets are dropped.

Random-Detect

Queueing discipline: Generalized Random Early Drop.

Applies to: Outbound traffic.

A simple Random Early Detection (RED) policy would start randomly dropping packets from a queue before it reaches its queue limit thus avoiding congestion. That is good for TCP connections as the gradual dropping of packets acts as a signal for the sender to decrease its transmission rate.

In contrast to simple RED, VyOS' Random-Detect uses a Generalized Random Early Detect policy that provides different virtual queues based on the IP Precedence value so that some virtual queues can drop more packets than others.

This is achieved by using the first three bits of the ToS (Type of Service) field to categorize data streams and, in accordance with the defined precedence parameters, a decision is made.

IP precedence as defined in [RFC 791](#):

Precedence	Priority
7	Network Control
6	Internetwork Control
5	CRITIC/ECP
4	Flash Override
3	Flash
2	Immediate
1	Priority
0	Routine

Random-Detect could be useful for heavy traffic. One use of this algorithm might be to prevent a backbone overload. But only for TCP (because dropped packets could be retransmitted), not for UDP.

set qos policy random-detect <policy-name> bandwidth <bandwidth>

Use this command to configure a Random-Detect policy, set its name and set the available bandwidth for this policy. It is used for calculating the average queue size after some idle time. It should be set to the bandwidth of your interface. Random Detect is not a shaping policy, this command will not shape.

```
set qos policy random-detect <policy-name> precedence <IP-precedence-value>
average-packet <bytes>
```

Use this command to configure a Random-Detect policy and set its name, then state the IP Precedence for the virtual queue you are configuring and what the size of its average-packet should be (in bytes, default: 1024).

Note: When configuring a Random-Detect policy: **the higher the precedence number, the higher the priority.**

```
set qos policy random-detect <policy-name> precedence <IP-precedence-value>
mark-probability <value>
```

Use this command to configure a Random-Detect policy and set its name, then state the IP Precedence for the virtual queue you are configuring and what its mark (drop) probability will be. Set the probability by giving the N value of the fraction 1/N (default: 10).

```
set qos policy random-detect <policy-name> precedence <IP-precedence-value>
maximum-threshold <packets>
```

Use this command to configure a Random-Detect policy and set its name, then state the IP Precedence for the virtual queue you are configuring and what its maximum threshold for random detection will be (from 0 to 4096 packets, default: 18). At this size, the marking (drop) probability is maximal.

```
set qos policy random-detect <policy-name> precedence <IP-precedence-value>
minimum-threshold <packets>
```

Use this command to configure a Random-Detect policy and set its name, then state the IP Precedence for the virtual queue you are configuring and what its minimum threshold for random detection will be (from 0 to 4096 packets). If this value is exceeded, packets start being eligible for being dropped.

The default values for the minimum-threshold depend on IP precedence:

Precedence	default min-threshold
7	16
6	15
5	14
4	13
3	12
2	11
1	10
0	9

```
set qos policy random-detect <policy-name> precedence <IP-precedence-value> queue-limit
<packets>
```

Use this command to configure a Random-Detect policy and set its name, then name the IP Precedence for the virtual queue you are configuring and what the maximum size of its queue will be (from 1 to 1-4294967295 packets). Packets are dropped when the current queue length reaches this value.

If the average queue size is lower than the **min-threshold**, an arriving packet will be placed in the queue.

In the case the average queue size is between **min-threshold** and **max-threshold**, then an arriving packet would be either dropped or placed in the queue, it will depend on the defined **mark-probability**.

If the current queue size is larger than **queue-limit**, then packets will be dropped. The average queue size depends on its former average size and its current one.

If **max-threshold** is set but **min-threshold** is not, then ****min-threshold** is scaled to 50% of **max-threshold**.

In principle, values must be **min-threshold < max-threshold < queue-limit**.

Rate Control

Queueing discipline: Token Bucket Filter.

Applies to: Outbound traffic.

Rate-Control is a classless policy that limits the packet flow to a set rate. It is a pure shaper, it does not schedule traffic. Traffic is filtered based on the expenditure of tokens. Tokens roughly correspond to bytes.

Short bursts can be allowed to exceed the limit. On creation, the Rate-Control traffic is stocked with tokens which correspond to the amount of traffic that can be burst in one go. Tokens arrive at a steady rate, until the bucket is full.

set qos policy rate-control <policy-name> bandwidth <rate>

Use this command to configure a Rate-Control policy, set its name and the rate limit you want to have.

set qos policy rate-control <policy-name> burst <burst-size>

Use this command to configure a Rate-Control policy, set its name and the size of the bucket in bytes which will be available for burst.

As a reference: for 10mbit/s on Intel, you might need at least 10kbyte buffer if you want to reach your configured rate.

A very small buffer will soon start dropping packets.

set qos policy rate-control <policy-name> latency

Use this command to configure a Rate-Control policy, set its name and the maximum amount of time a packet can be queued (default: 50 ms).

Rate-Control is a CPU-friendly policy. You might consider using it when you just simply want to slow traffic down.

Round Robin

Queueing discipline: Deficit Round Robin. **Applies to:** Outbound traffic.

The round-robin policy is a classful scheduler that divides traffic in different classes you can configure (up to 4096). You can embed a new policy into each of those classes (default included).

Each class is assigned a deficit counter (the number of bytes that a flow is allowed to transmit when it is its turn) initialized to quantum. Quantum is a parameter you configure which acts like a credit of fix bytes the counter receives on each round. Then the Round-Robin policy starts moving its Round Robin pointer through the queues. If the deficit counter is greater than the packet's size at the head of the queue, this packet will be sent and the value of the counter will be decremented by the packet size. Then, the size of the next packet will be compared to the counter value again, repeating the process. Once the queue is empty or the value of the counter is insufficient, the Round-Robin pointer will move to the next queue. If the queue is empty, the value of the deficit counter is reset to 0.

At every round, the deficit counter adds the quantum so that even large packets will have their opportunity to be de-queued.

set qos policy round-robin <policy name> class <class-ID> quantum <packets>

Use this command to configure a Round-Robin policy, set its name, set a class ID, and the quantum for that class. The deficit counter will add that value each round.

set qos policy round-robin <policy name> class queue-limit <packets>

Use this command to configure a Round-Robin policy, set its name, set a class ID, and the queue size in packets.

As with other policies, Round-Robin can embed another policy into a class through the `queue-type` setting.

```
vyos@vyos# set qos policy round-robin DRR class 10 queue-type
Possible completions:
  drop-tail      First-In-First-Out (FIFO) (default)
  fq-codel       Fair Queue Codel
  fair-queue     Stochastic Fair Queue (SFQ)
  priority       Priority queueing based
  random-detect  Random Early Detection (RED)
```

Shaper

Queueing discipline: Hierarchical Token Bucket.

Applies to: Outbound traffic.

The Shaper policy does not guarantee a low delay, but it does guarantee bandwidth to different traffic classes and also lets you decide how to allocate more traffic once the guarantees are met.

Each class can have a guaranteed part of the total bandwidth defined for the whole policy, so all those shares together should not be higher than the policy's whole bandwidth.

If guaranteed traffic for a class is met and there is room for more traffic, the ceiling parameter can be used to set how much more bandwidth could be used. If guaranteed traffic is met and there are several classes willing to use their ceilings, the priority parameter will establish the order in which that additional traffic will be allocated. Priority can be any number from 0 to 7. The lower the number, the higher the priority.

set qos policy shaper <policy-name> bandwidth <rate>

Use this command to configure a Shaper policy, set its name and the maximum bandwidth for all combined traffic.

set qos policy shaper <policy-name> class <class-ID> bandwidth <rate>

Use this command to configure a Shaper policy, set its name, define a class and set the guaranteed traffic you want to allocate to that class.

set qos policy shaper <policy-name> class <class-ID> burst <bytes>

Use this command to configure a Shaper policy, set its name, define a class and set the size of the token bucket in bytes, which will be available to be sent at ceiling speed (default: 15Kb).

set qos policy shaper <policy-name> class <class-ID> ceiling <bandwidth>

Use this command to configure a Shaper policy, set its name, define a class and set the maximum speed possible for this class. The default ceiling value is the bandwidth value.

set qos policy shaper <policy-name> class <class-ID> priority <0-7>

Use this command to configure a Shaper policy, set its name, define a class and set the priority for usage of available bandwidth once guarantees have been met. The lower the priority number, the higher the priority. The default priority value is 0, the highest priority.

As with other policies, Shaper can embed other policies into its classes through the `queue-type` setting and then configure their parameters.

```
vyos@vyos# set qos policy shaper HTB class 10 queue-type
Possible completions:
  fq-codel      Fair Queue Codel (default)
```

(continues on next page)

(continued from previous page)

fair-queue	Stochastic Fair Queue (SFQ)
drop-tail	First-In-First-Out (FIFO)
priority	Priority queueing
random-detect	Random Early Detection (RED)

```
vyos@vyos# set qos policy shaper HTB class 10
```

Possible completions:

bandwidth	Available bandwidth for this policy (default: auto)
burst	Burst size for this class (default: 15k)
ceiling	Bandwidth limit for this class
codel-quantum	Deficit in the fair queuing algorithm (default 1514)
description	Description
flows	Number of flows into which the incoming packets are classified (default 1024)
interval	Interval used to measure the delay (default 100)
match	Class matching rule name
priority	Priority for rule evaluation
queue-limit	Maximum queue size (packets)
queue-type	Queue type for default traffic (default: fq-codel)
set-dscp	Change the Differentiated Services (DiffServ) field in the IP header
target	Acceptable minimum standing/persistent queue delay (default: 5)

Note: If you configure a class for **VoIP traffic**, don't give it any *ceiling*, otherwise new VoIP calls could start when the link is available and get suddenly dropped when other classes start using their assigned *bandwidth* share.

Example

A simple example of Shaper using priorities.

```
set qos policy shaper MY-HTB bandwidth '50mbit'
set qos policy shaper MY-HTB class 10 bandwidth '20%'
set qos policy shaper MY-HTB class 10 match DSCP ip dscp 'EF'
set qos policy shaper MY-HTB class 10 queue-type 'fq-codel'
set qos policy shaper MY-HTB class 20 bandwidth '10%'
set qos policy shaper MY-HTB class 20 ceiling '50%'
set qos policy shaper MY-HTB class 20 match PORT666 ip destination port '666'
set qos policy shaper MY-HTB class 20 priority '3'
set qos policy shaper MY-HTB class 20 queue-type 'fair-queue'
set qos policy shaper MY-HTB class 30 bandwidth '10%'
set qos policy shaper MY-HTB class 30 ceiling '50%'
set qos policy shaper MY-HTB class 30 match ADDRESS30 ip source address '192.168.30.0/24'
set qos policy shaper MY-HTB class 30 priority '5'
set qos policy shaper MY-HTB class 30 queue-type 'fair-queue'
set qos policy shaper MY-HTB default bandwidth '10%'
set qos policy shaper MY-HTB default ceiling '100%'
set qos policy shaper MY-HTB default priority '7'
set qos policy shaper MY-HTB default queue-type 'fair-queue'
```

CAKE

Queueing discipline: Deficit mode.

Applies to: Outbound traffic.

Common Applications Kept Enhanced (CAKE) is a comprehensive queue management system, implemented as a queue discipline (qdisc) for the Linux kernel. It is designed to replace and improve upon the complex hierarchy of simple qdiscs presently required to effectively tackle the bufferbloat problem at the network edge.

set qos policy cake <text> bandwidth <value>

Set the shaper bandwidth, either as an explicit bitrate or a percentage of the interface bandwidth.

set qos policy cake <text> description

Set a description for the shaper.

set qos policy cake <text> flow-isolation blind

Disables flow isolation, all traffic passes through a single queue.

set qos policy cake <text> flow-isolation dst-host

Flows are defined only by destination address.

set qos policy cake <text> flow-isolation dual-dst-host

Flows are defined by the 5-tuple. Fairness is applied first over destination addresses, then over individual flows.

set qos policy cake <text> flow-isolation dual-src-host

Flows are defined by the 5-tuple. Fairness is applied first over source addresses, then over individual flows.

set qos policy cake <text> flow-isolation flow

Flows are defined by the entire 5-tuple (source IP address, source port, destination IP address, destination port, transport protocol).

set qos policy cake <text> flow-isolation host

Flows are defined by source-destination host pairs.

set qos policy cake <text> flow-isolation nat

Perform NAT lookup before applying flow-isolation rules.

set qos policy cake <text> flow-isolation src-host

Flows are defined only by source address.

set qos policy cake <text> flow-isolation triple-isolate

(Default) Flows are defined by the 5-tuple, fairness is applied over source and destination addresses and also over individual flows.

set qos policy cake <text> rtt

Defines the round-trip time used for active queue management (AQM) in milliseconds. The default value is 100.

Applying a traffic policy

Once a traffic-policy is created, you can apply it to an interface:

```
set qos interface eth0 egress WAN-OUT
```

You can only apply one policy per interface and direction, but you could reuse a policy on different interfaces and directions:

```
set qos interface eth0 ingress WAN-IN
set qos interface eth0 egress WAN-OUT
set qos interface eth1 ingress LAN-IN
set qos interface eth1 egress LAN-OUT
set qos interface eth2 ingress LAN-IN
set qos interface eth2 egress LAN-OUT
set qos interface eth3 ingress TWO-WAY-POLICY
set qos interface eth3 egress TWO-WAY-POLICY
set qos interface eth4 ingress TWO-WAY-POLICY
set qos interface eth4 egress TWO-WAY-POLICY
```

The case of ingress shaping

Applies to: Inbound traffic.

For the ingress traffic of an interface, there is only one policy you can directly apply, a **Limiter** policy. You cannot apply a shaping policy directly to the ingress traffic of any interface because shaping only works for outbound traffic.

This workaround lets you apply a shaping policy to the ingress traffic by first redirecting it to an in-between virtual interface ([Intermediate Functional Block](#)). There, in that virtual interface, you will be able to apply any of the policies that work for outbound traffic, for instance, a shaping one.

That is how it is possible to do the so-called “ingress shaping”.

```
set qos policy shaper MY-INGRESS-SHAPING bandwidth 1000kbit
set qos policy shaper MY-INGRESS-SHAPING default bandwidth 1000kbit
set qos policy shaper MY-INGRESS-SHAPING default queue-type fair-queue

set qos interface ifb0 egress MY-INGRESS-SHAPING
set interfaces ethernet eth0 redirect ifb0

set interfaces input ifb0
```

Warning: Do not configure IFB as the first step. First create everything else of your traffic-policy, and then you can configure IFB. Otherwise you might get the `RTNETLINK answer: File exists` error, which can be solved with `sudo ip link delete ifb0`.

7.13 VPN

7.13.1 IPsec

IPsec General Information

Information about IPsec

IPsec is the framework used to secure data. IPsec accomplishes these goals by providing authentication, encryption of IP network packets, key exchange, and key management. VyOS uses Strongswan package to implement IPsec.

Authentication Header (AH) is defined in [RFC 4302](#). It creates a hash using the IP header and data payload, and prepends it to the packet. This hash is used to validate that the data has not been changed during transfer over the network.

Encapsulating Security Payload (ESP) is defined in [RFC 4303](#). It provides encryption and authentication of the data.

There are two IPsec modes:

IPsec Transport Mode:

In transport mode, an IPsec header (AH or ESP) is inserted between the IP header and the upper layer protocol header.

IPsec Tunnel Mode:

In tunnel mode, the original IP packet is encapsulated in another IP datagram, and an IPsec header (AH or ESP) is inserted between the outer and inner headers.

IKE (Internet Key Exchange)

The default IPsec method for secure key negotiation is the Internet Key Exchange (IKE) protocol. IKE is designed to provide mutual authentication of systems, as well as to establish a shared secret key to create IPsec security associations. A security association (SA) includes all relevant attributes of the connection, including the cryptographic algorithm used, the IPsec mode, the encryption key, and other parameters related to the transmission of data over the VPN connection.

IKEv1

IKEv1 is the older version and is still used today. Nowadays, most manufacturers recommend using IKEv2 protocol.

IKEv1 is described in the next RFCs: [RFC 2409](#) (IKE), [RFC 3407](#) (IPsec DOI), [RFC 3947](#) (NAT-T), [RFC 3948](#) (UDP Encapsulation of ESP Packets), [RFC 3706](#) (DPD)

IKEv1 operates in two phases to establish these IKE and IPsec SAs:

- **Phase 1** provides mutual authentication of the IKE peers and establishment of the session key. This phase creates an IKE SA (a security association for IKE) using a DH exchange, cookies, and an ID exchange. Once an IKE SA is established, all IKE communication between the initiator and responder is protected with encryption and an integrity check that is authenticated. The purpose of IKE phase 1 is to facilitate a secure channel between the peers so that phase 2 negotiations can occur securely. IKE phase 1 offers two modes: Main and Aggressive.
 - **Main Mode** is used for site-to-site VPN connections.
 - **Aggressive Mode** is used for remote access VPN connections.

- **Phase 2** provides for the negotiation and establishment of the IPsec SAs using ESP or AH to protect IP data traffic.

IKEv2

IKEv2 is described in [RFC 7296](#). The biggest difference between IKEv1 and IKEv2 is that IKEv2 is much simpler and more reliable than IKEv1 because fewer messages are exchanged during the establishment of the VPN and additional security capabilities are available.

IKE Authentication

VyOS supports 3 authentication methods.

- **Pre-shared keys:** In this method, both peers of the IPsec tunnel must have the same preshared keys.
- **Digital certificates:** PKI is used in this method.
- **RSA-keys:** If the RSA-keys method is used in your IKE policy, you need to make sure each peer has the other peer's public keys.

DPD (Dead Peer Detection)

This is a mechanism used to detect when a VPN peer is no longer active. This mechanism has different algorithms in IKEv1 and IKEv2 in VyOS. DPD Requests are sent as ISAKMP R-U-THERE messages and DPD Responses are sent as ISAKMP R-U-THERE-ACK messages. In IKEv1, DPD sends messages every configured interval. The remote peer is considered unreachable if no response to these packets is received within the DPD timeout. In IKEv2, DPD sends messages every configured interval. If one request is not responded, Strongswan execute its retransmission algorithm with its timers. *[IKEv2 Retransmission](#)*

Post-Quantum Preshared Keys (PPK)

Post-Quantum Preshared Keys help provide some quantum resistance to IPsec tunnels when a post-quantum key exchange algorithm such as ML-KEM is not available. The use of PPKs in IKEv2 is described in [RFC 8784](#).

PPKs can be configured within VyOS under the `vpn ipsec authentication ppk` config.

PPKs need an id and a secret value. The ID and the secret must match if PPKs are required for a successful IPsec connection. The secret can be plain text, a hex value, or a Base64 value. The default is plain text. If using another type of value, you must define the secret type.

To use a PPK within a site-to-site or remote access connection, define the PPK id under the connection.

Optionally, you can require the use of PPK to have a successful connection.

You can view the PPK column for information on if PPK is configured, and if it is in use. The output is in the format of `<configured> / <in use>`. The options for configured are none if not configured, opt if configured but optional, and req is configured and required. The in use will show yes Possible values of the `configured` field are none if not configured, opt if configured but optional, and req is configured and required. The in use will show yes

Configuration IKE

IKE (Internet Key Exchange) Attributes

VyOS IKE group has the next options:

set vpn ipsec ike-group <name> close-action <action>

Defines the action to take if the remote peer unexpectedly closes a CHILD_SA:

- **none** - Set action to none (default),
- **trap** - Installs a trap policy (IPsec policy without Security Association) for the CHILD_SA and traffic matching these policies will trigger acquire events that cause the daemon to establish the required IKE/IPsec SAs.
- **start** - Tries to immediately re-create the CHILD_SA.

set vpn ipsec ike-group <name> ikev2-reauth

Whether rekeying of an IKE_SA should also reauthenticate the peer. In IKEv1, reauthentication is always done. Setting this parameter enables remote host re-authentication during an IKE rekey.

set vpn ipsec ike-group <name> key-exchange

Which protocol should be used to initialize the connection. If not set both protocols are handled and connections will use IKEv2 when initiating, but accept any protocol version when responding:

- **ikev1** - Use IKEv1 for Key Exchange.
- **ikev2** - Use IKEv2 for Key Exchange.

set vpn ipsec ike-group <name> lifetime

IKE lifetime in seconds <0-86400> (default 28800).

set vpn ipsec ike-group <name> mode

IKEv1 Phase 1 Mode Selection:

- **main** - Use Main mode for Key Exchanges in the IKEv1 Protocol (Recommended Default).
- **aggressive** - Use Aggressive mode for Key Exchanges in the IKEv1 protocol. Aggressive mode is much more insecure compared to Main mode.

set vpn ipsec ike-group <name> proposal <number> dh-group <dh-group number>

Dh-group. Default value is 2.

set vpn ipsec ike-group <name> proposal <number> encryption <encryption>

Encryption algorithm. Default value is aes128.

set vpn ipsec ike-group <name> proposal <number> hash <hash>

Hash algorithm. Default value is sha1.

set vpn ipsec ike-group <name> proposal <number> prf <prf>

Pseudo-random function.

DPD (Dead Peer Detection) Configuration

set vpn ipsec ike-group <name> dead-peer-detection action <action>

Action to perform for this CHILD_SA on DPD timeout.

- **trap** - Installs a trap policy (IPsec policy without Security Association), which will catch matching traffic and tries to re-negotiate the tunnel on-demand.
- **clear** - Closes the CHILD_SA and does not take further action (default).
- **restart** - Immediately tries to re-negotiate the CHILD_SA under a fresh IKE_SA.

set vpn ipsec ike-group <name> dead-peer-detection interval <interval>

Keep-alive interval in seconds <2-86400> (default 30).

set vpn ipsec ike-group <name> dead-peer-detection timeout <timeout>

Keep-alive timeout in seconds <2-86400> (default 120) **IKEv1 only**

ESP (Encapsulating Security Payload) Attributes

In VyOS, ESP attributes are specified through ESP groups. Multiple proposals can be specified in a single group.

VyOS ESP group has the next options:

set vpn ipsec esp-group <name> compression

Enables the IPComp(IP Payload Compression) protocol which allows compressing the content of IP packets.

set vpn ipsec esp-group <name> disable-rekey

Do not locally initiate a re-key of the SA, remote peer must re-key before expiration.

set vpn ipsec esp-group <name> life-bytes <bytes>

ESP life in bytes <1024-26843545600000>. Number of bytes transmitted over an IPsec SA before it expires.

set vpn ipsec esp-group <name> life-packets <packets>

ESP life in packets <1000-26843545600000>. Number of packets transmitted over an IPsec SA before it expires.

set vpn ipsec esp-group <name> lifetime <timeout>

ESP lifetime in seconds <30-86400> (default 3600). How long a particular instance of a connection (a set of encryption/authentication keys for user packets) should last, from successful negotiation to expiry.

set vpn ipsec esp-group <name> mode <mode>

The type of the connection:

- **tunnel** - Tunnel mode (default).
- **transport** - Transport mode.

set vpn ipsec esp-group <name> pfs < dh-group>

Whether Perfect Forward Secrecy of keys is desired on the connection's keying channel and defines a Diffie-Hellman group for PFS:

- **enable** - Inherit Diffie-Hellman group from IKE group (default).
- **disable** - Disable PFS.
- **<dh-group>** - Defines a Diffie-Hellman group for PFS.

```
set vpn ipsec esp-group <name> proposal <number> encryption <encryption>
```

Encryption algorithm. Default value is **aes128**.

```
set vpn ipsec esp-group <name> proposal <number> hash <hash>
```

Hash algorithm. Default value is **sha1**.

Global IPsec Settings

```
set vpn ipsec interface <name>
```

Interface name to restrict outbound IPsec policies. There is a possibility to specify multiple interfaces. If an interfaces are not specified, IPsec policies apply to all interfaces.

```
set vpn ipsec log level <number>
```

Level of logging. Default value is **0**.

```
set vpn ipsec log subsystem <name>
```

Subsystem of the daemon.

Options

```
set vpn ipsec options disable-route-autoinstall
```

Do not automatically install routes to remote networks.

```
set vpn ipsec options flexvpn
```

Allows FlexVPN vendor ID payload (IKEv2 only). Send the Cisco FlexVPN vendor ID payload (IKEv2 only), which is required in order to make Cisco brand devices allow negotiating a local traffic selector (from strongSwan's point of view) that is not the assigned virtual IP address if such an address is requested by strongSwan. Sending the Cisco FlexVPN vendor ID prevents the peer from narrowing the initiator's local traffic selector and allows it to e.g. negotiate a TS of 0.0.0.0/0 == 0.0.0.0/0 instead. This has been tested with a "tunnel mode ipsec ipv4" Cisco template but should also work for GRE encapsulation.

```
set vpn ipsec options interface <name>
```

Interface Name to use. The name of the interface on which virtual IP addresses should be installed. If not specified the addresses will be installed on the outbound interface.

```
set vpn ipsec options virtual-ip
```

Allows the installation of virtual-ip addresses.

IKEv2 Retransmission

If the peer does not respond on DPD packet, the router starts retransmission procedure.

The following formula is used to calculate the timeout:

$$\text{relative timeout} = \text{timeout} * \text{base} ^ (\text{attempts}-1)$$

```
set vpn ipsec options retransmission attempts
```

Number of attempts before the peer is considered to be in the down state. Default value is **5**.

```
set vpn ipsec options retransmission base
```

Base number of exponential backoff. Default value is **1.8**.

set vpn ipsec options retransmission timeout

Timeout in seconds before the first retransmission. Default value is **4**.

Using the default values, packets are retransmitted as follows:

Attempts	Formula	Relative timeout	Absolute timeout
1	$4 * 1.8 ^ 0$	4s	4s
2	$4 * 1.8 ^ 1$	7s	11s
3	$4 * 1.8 ^ 2$	13s	24s
4	$4 * 1.8 ^ 3$	23s	47s
5	$4 * 1.8 ^ 4$	42s	89s
peer down	$4 * 1.8 ^ 5$	76s	165s

IPsec Site-to-Site VPN

IPsec Site-to-Site VPN Types

VyOS supports two types of IPsec VPN: Policy-based IPsec VPN and Route-based IPsec VPN.

Policy-based VPN

Policy-based VPN is based on static configured policies. Each policy creates individual IPsec SA. Traffic matches these SAs encrypted and directed to the remote peer.

Route-Based VPN

Route-based VPN is based on secure traffic passing over Virtual Tunnel Interfaces (VTIs). This type of IPsec VPNs allows using routing protocols.

Configuration Site-to-Site VPN

Requirements and Prerequisites for Site-to-Site VPN

Negotiated parameters that need to match

Phase 1

- IKE version
- Authentication
- Encryption
- Hashing
- PRF
- Lifetime

Note: Strongswan recommends to use the same lifetime value on both peers

Phase 2

- Encryption
- Hashing
- PFS
- Mode (tunnel or transport)
- Lifetime

Note: Strongswan recommends to use the same lifetime value on both peers

- Remote and Local networks in SA must be compatible on both peers

Configuration Steps for Site-to-Site VPN

The next example shows the configuration one of the router participating in IPsec VPN.

Tunnel information:

- **Phase 1:**
 - encryption: AES256
 - hash: SHA256
 - PRF: SHA256
 - DH: 14
 - lifetime: 28800
- **Phase 2:**
 - IPsec mode: tunnel
 - encryption: AES256
 - hash: SHA256
 - PFS: inherited from DH Phase 1
 - lifetime: 3600
- **If Policy based VPN is used**
 - Remote network is 192.168.50.0/24. Local network is 192.168.10.0/24
- **If Route based VPN is used**
 - IP of the VTI interface is 10.0.0.1/30

Note: We do not recommend using policy-based vpn and route-based vpn configurations to the same peer.

1. Configure ike-group (IKE Phase 1)

```
set vpn ipsec ike-group IKE close-action 'start'
set vpn ipsec ike-group IKE key-exchange 'ikev1'
set vpn ipsec ike-group IKE lifetime '28800'
set vpn ipsec ike-group IKE proposal 10 dh-group '14'
set vpn ipsec ike-group IKE proposal 10 encryption 'aes256'
set vpn ipsec ike-group IKE proposal 10 hash 'sha256'
set vpn ipsec ike-group IKE proposal 10 prf 'prfsha256'
```

2. Configure ESP-group (IKE Phase 2)

```
set vpn ipsec esp-group ESP lifetime '3600'
set vpn ipsec esp-group ESP mode 'tunnel'
set vpn ipsec esp-group ESP pfs 'enable'
set vpn ipsec esp-group ESP proposal 10 encryption 'aes256'
set vpn ipsec esp-group ESP proposal 10 hash 'sha256'
```

3. Specify interface facing to the protected destination.

```
set vpn ipsec interface eth0
```

4. Configure PSK keys and authentication ids for this key if authentication type is PSK

```
set vpn ipsec authentication psk PSK-KEY id '192.168.0.2'
set vpn ipsec authentication psk PSK-KEY id '192.168.5.2'
set vpn ipsec authentication psk PSK-KEY secret 'vyos'
```

To set base64 secret encode plaintext password to base64 and set secret-type

```
echo -n "vyos" | base64
dnlvcw==
```

```
set vpn ipsec authentication psk PSK-KEY secret 'dnlvcw=='
set vpn ipsec authentication psk PSK-KEY secret-type base64
```

5. Configure peer and apply IKE-group and esp-group to peer.

```
set vpn ipsec site-to-site peer PEER1 authentication local-id '192.168.0.2'
set vpn ipsec site-to-site peer PEER1 authentication mode 'pre-shared-secret'
set vpn ipsec site-to-site peer PEER1 authentication remote-id '192.168.5.2'
set vpn ipsec site-to-site peer PEER1 connection-type 'initiate'
set vpn ipsec site-to-site peer PEER1 default-esp-group 'ESP'
set vpn ipsec site-to-site peer PEER1 ike-group 'IKE'
set vpn ipsec site-to-site peer PEER1 local-address '192.168.0.2'
set vpn ipsec site-to-site peer PEER1 remote-address '192.168.5.2'
```

Peer selects the key from step 4 according to local-id/remote-id pair.

6. Depends to vpn type (route-based vpn or policy-based vpn).

6.1 For Policy-based VPN configure SAs using tunnel command specifying remote and local networks.

```
set vpn ipsec site-to-site peer PEER1 tunnel 1 local prefix '192.168.
↪10.0/24'
```

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```
set vpn ipsec site-to-site peer PEER1 tunnel 1 remote prefix '192.168.
↪50.0/24'
```

6.2 For Route-based VPN create VTI interface, set IP address to this interface and bind this interface to the vpn peer.

```
set interfaces vti vti1 address 10.0.0.1/30
set vpn ipsec site-to-site peer PEER1 vti bind vti1
set vpn ipsec options disable-route-autoinstall
```

Create routing between local networks via VTI interface using dynamic or static routing.

```
set protocol static route 192.168.50.0/24 next-hop 10.0.0.2
```

Initiator and Responder Connection Types

In Site-to-Site IPsec VPN it is recommended that one peer should be an initiator and the other - the responder. The initiator actively establishes the VPN tunnel. The responder passively waits for the remote peer to establish the VPN tunnel. Depends on selected role it is recommended select proper values for close-action and DPD action.

The result of wrong value selection can be unstable work of the VPN.

- Duplicate CHILD SA creation.
- None of the VPN sides initiates the tunnel establishment.

Below flow-chart could be a quick reference for the close-action combination depending on how the peer is configured.

Similar combinations are applicable for the dead-peer-detection.

Detailed Configuration Commands

PSK Key Authentication

set vpn ipsec authentication psk <name> dhcp-interface

ID for authentication generated from DHCP address dynamically.

set vpn ipsec authentication psk id <id>

static ID's for authentication. In general local and remote address <x.x.x.x>, <h:h:h:h:h:h:h:h> or %any.

set vpn ipsec authentication psk secret <secret>

A predefined shared secret used in configured mode pre-shared-secret. Base64-encoded secrets are allowed if secret-type base64 is configured.

set vpn ipsec authentication psk secret-type <type>

Specifies the secret type:

- **plaintext** - Plain text type (default value).
- **base64** - Base64 type.

Peer Configuration

Peer Authentication Commands

set vpn ipsec site-to-site peer <name> authentication mode <mode>

Mode for authentication between VyOS and remote peer:

- **pre-shared-secret** - Use predefined shared secret phrase.
- **rsa** - Use simple shared RSA key.
- **x509** - Use certificates infrastructure for authentication.

set vpn ipsec site-to-site peer <name> authentication local-id <id>

ID for the local VyOS router. If defined, during the authentication it will be send to remote peer.

set vpn ipsec site-to-site peer <name> authentication remote-id <id>

ID for remote peer, instead of using peer name or address. Useful in case if the remote peer is behind NAT or if mode **x509** is used.

set vpn ipsec site-to-site peer <name> authentication rsa local-key <key>

Name of PKI key-pair with local private key.

set vpn ipsec site-to-site peer <name> authentication rsa remote-key <key>

Name of PKI key-pair with remote public key.

set vpn ipsec site-to-site peer <name> authentication rsa passphrase <passphrase>

Local private key passphrase.

set vpn ipsec site-to-site peer <name> authentication use-x509-id <id>

Use local ID from x509 certificate. Cannot be used when id is defined.

set vpn ipsec site-to-site peer <name> authentication x509 ca-certificate <name>

Name of CA certificate in PKI configuration. Using for authenticating remote peer in x509 mode.

set vpn ipsec site-to-site peer <name> authentication x509 certificate <name>

Name of certificate in PKI configuration, which will be used for authenticating local router on remote peer.

set vpn ipsec authentication x509 passphrase <passphrase>

Private key passphrase, if needed.

Global Peer Configuration Commands

set vpn ipsec site-to-site peer <name> connection-type <type>

Operational mode defines how to handle this connection process.

- **initiate** - does initial connection to remote peer immediately after configuring and after boot. In this mode the connection will not be restarted in case of disconnection, therefore should be used only together with DPD or another session tracking methods.
- **trap** - does not try to initiate a connection to a remote peer immediately. Instead, it installs a trap policy that will trigger IKE negotiation and establish the IPsec session when matching traffic is sent from the local side. This can be useful when there is no direct connectivity to the peer due to firewall or NAT in the middle of the local and remote side.

Warning: The `trap` mode is not needed in most environments and can lead to connection confusion or unintended tunnel uptime behavior if used incorrectly. Using this mode requires careful coordination with parameters such as `close-action` and `DPD`. For most deployments, use `initiate` and `none` as described below.

- **none** - loads the connection only, which then can be manually initiated or used as a responder configuration.

Note: For most site-to-site VPNs, configure one peer with `connection-type initiate` (active side) and the other peer with `connection-type none` (passive side) to ensure stable and predictable tunnel behavior. When using `connection-type initiate`, you must also configure `DPD` or another session tracking method (such as `close-action`) to automatically re-establish the tunnel after a disconnection. Otherwise, the tunnel will not reconnect automatically if it goes down.

set vpn ipsec site-to-site peer <name> default-esp-group <name>

Name of ESP group to use by default for traffic encryption. Might be overwritten by individual settings for tunnel or VTI interface binding.

set vpn ipsec site-to-site peer <name> description <description>

Description for this peer.

set vpn ipsec site-to-site peer <name> dhcp-interface <interface>

Specify the interface which IP address, received from DHCP for IPsec connection with this peer, will be used as `local-address`.

set vpn ipsec site-to-site peer <name> force-udp-encapsulation

Force encapsulation of ESP into UDP datagrams. Useful in case if between local and remote side is firewall or NAT, which not allows passing plain ESP packets between them.

set vpn ipsec site-to-site peer <name> ike-group <name>

Name of IKE group to use for key exchanges.

set vpn ipsec site-to-site peer <name> ikev2-reauth <yes|no|inherit>

Configure IKEv2 reauthentication behavior for the specified IPsec site-to-site peer.

The following options are available:

- **yes:** The active IKE SA channel between the two peers is terminated and re-established at each rekeying interval, explicitly re-verifying each peer's identity.
- **no (default):** Reauthentication (identity re-verification) is disabled for this peer. The active IKE SA channel is seamlessly renewed at each rekeying interval without termination.
- **inherit:** The peer inherits reauthentication behavior from the assigned IKE group. Reauthentication is applied only if it is explicitly enabled in that group's configuration.

set vpn ipsec site-to-site peer <name> local-address <address>

Local IP address for IPsec connection with this peer. If defined any, then an IP address which configured on interface with default route will be used.

set vpn ipsec site-to-site peer <name> remote-address <address>

Remote IP address or hostname for IPsec connection. IPv4 or IPv6 address is used when a peer has a public static IP address. Hostname is a DNS name which could be used when a peer has a public IP address and DNS name, but an IP address could be changed from time to time.


```
set vpn ipsec site-to-site peer <name> replay-window <size>
```

IPsec replay window to configure for CHILD_SAs (default: 32), a value of 0 disables IPsec replay protection.

```
set vpn ipsec site-to-site peer <name> virtual-address <address>
```

Defines a virtual IP address which is requested by the initiator and one or several IPv4 and/or IPv6 addresses are assigned from multiple pools by the responder. The wildcard addresses 0.0.0.0 and :: request an arbitrary address, specific addresses may be defined.

CHILD SAs Configuration Commands

Policy-Based CHILD SAs Configuration Commands

Every configured tunnel under peer configuration is a new CHILD SA.

```
set vpn ipsec site-to-site peer <name> tunnel <number> disable
```

Disable this tunnel.

```
set vpn ipsec site-to-site peer <name> tunnel <number> esp-group <name>
```

Specify ESP group for this CHILD SA.

```
set vpn ipsec site-to-site peer <name> tunnel <number> priority <number>
```

Priority for policy-based IPsec VPN tunnels (lowest value more preferable).

```
set vpn ipsec site-to-site peer <name> tunnel <number> protocol <name>
```

Define the protocol for match traffic, which should be encrypted and send to this peer.

```
set vpn ipsec site-to-site peer <name> tunnel <number> local prefix <network>
```

IP network at the local side.

```
set vpn ipsec site-to-site peer <name> tunnel <number> local port <number>
```

Local port number. Have effect only when used together with prefix.

```
set vpn ipsec site-to-site peer <name> tunnel <number> remote prefix <network>
```

IP network at the remote side.

```
set vpn ipsec site-to-site peer <name> tunnel <number> remote port <number>
```

Remote port number. Have effect only when used together with prefix.

Route-Based CHILD SAs Configuration Commands

To configure route-based VPN it is enough to create vti interface and bind it to the peer. Any traffic, which will be send to VTI interface will be encrypted and send to this peer. Using VTI makes IPsec configuration much flexible and easier in complex situation, and allows to dynamically add/delete remote networks, reachable via a peer, as in this mode router don't need to create additional SA/policy for each remote network.

Warning: When using site-to-site IPsec with VTI interfaces, be sure to disable route autoinstall.

```
set vpn ipsec options disable-route-autoinstall
```

```
set vpn ipsec site-to-site peer <name> vti bind <interface>
```

VTI interface to bind to this peer.

```
set vpn ipsec site-to-site peer <name> vti esp-group <name>
```

ESP group for encrypt traffic, passed this VTI interface.

Traffic-selectors parameters for traffic that should pass via vti interface.

```
set vpn ipsec site-to-site peer <name> vti traffic-selector local prefix <network>
```

Local prefix for interesting traffic.

```
set vpn ipsec site-to-site peer <name> vti traffic-selector remote prefix <network>
```

Remote prefix for interesting traffic.

IPsec Op-mode Commands

```
show vpn ike sa
```

Shows active IKE SAs information.

```
show vpn ike secrets
```

Shows configured authentication keys.

```
show vpn ike status
```

Shows Strongswan daemon status.

```
show vpn ipsec connections
```

Shows summary status of all configured IKE and IPsec SAs.

```
show vpn ipsec sa [detail]
```

Shows active IPsec SAs information.

```
show vpn ipsec status
```

Shows status of IPsec process.

```
show vpn ipsec policy
```

Shows the in-kernel crypto policies.

```
show vpn ipsec state
```

Shows the in-kernel crypto state.

```
show log ipsec
```

Shows IPsec logs.

```
reset vpn ipsec site-to-site all
```

Clear all ipsec connection and reinitiate them if VyOS is configured as initiator.

```
reset vpn ipsec site-to-site peer <name>
```

Clear all peer IKE SAs with IPsec SAs and reinitiate them if VyOS is configured as initiator.

```
reset vpn ipsec site-to-site peer <name> tunnel <number>
```

Clear specific IPsec SA and reinitiate it if VyOS is configured as initiator.

```
reset vpn ipsec site-to-site peer <name> vti <number>
```

Clear IPsec SA which is map to vti interface of this peer and reinitiate it if VyOS is configured as initiator.

restart ipsec

Restart Strongswan daemon.

Examples:

Policy-Based VPN Example

PEER1:

- WAN interface on eth0
- eth0 interface IP: 10.0.1.2/30
- dum0 interface IP: 192.168.0.1/24 (for testing purposes)
- Initiator

PEER2:

- WAN interface on eth0
- eth0 interface IP: 10.0.2.2/30
- dum0 interface IP: 192.168.1.0/24 (for testing purposes)
- Responder

```
# PEER1
set interfaces dummy dum0 address '192.168.0.1/32'
set interfaces ethernet eth0 address '10.0.1.2/30'
set protocols static route 0.0.0.0/0 next-hop 10.0.1.1
set vpn ipsec authentication psk AUTH-PSK id '10.0.1.2'
set vpn ipsec authentication psk AUTH-PSK id '10.0.2.2'
set vpn ipsec authentication psk AUTH-PSK secret 'test'
set vpn ipsec esp-group ESP-GRPOUP lifetime '3600'
set vpn ipsec esp-group ESP-GRPOUP proposal 10 encryption 'aes256'
set vpn ipsec esp-group ESP-GRPOUP proposal 10 hash 'sha1'
set vpn ipsec ike-group IKE-GROUP close-action 'start'
set vpn ipsec ike-group IKE-GROUP dead-peer-detection action 'restart'
set vpn ipsec ike-group IKE-GROUP dead-peer-detection interval '30'
set vpn ipsec ike-group IKE-GROUP dead-peer-detection timeout '120'
set vpn ipsec ike-group IKE-GROUP key-exchange 'ikev1'
set vpn ipsec ike-group IKE-GROUP lifetime '28800'
set vpn ipsec ike-group IKE-GROUP proposal 10 dh-group '14'
set vpn ipsec ike-group IKE-GROUP proposal 10 encryption 'aes256'
set vpn ipsec ike-group IKE-GROUP proposal 10 hash 'sha1'
set vpn ipsec interface 'eth0'
set vpn ipsec site-to-site peer PEER2 authentication local-id '10.0.1.2'
set vpn ipsec site-to-site peer PEER2 authentication mode 'pre-shared-secret'
set vpn ipsec site-to-site peer PEER2 authentication remote-id '10.0.2.2'
set vpn ipsec site-to-site peer PEER2 connection-type 'initiate'
set vpn ipsec site-to-site peer PEER2 default-esp-group 'ESP-GRPOUP'
set vpn ipsec site-to-site peer PEER2 ike-group 'IKE-GROUP'
set vpn ipsec site-to-site peer PEER2 local-address '10.0.1.2'
set vpn ipsec site-to-site peer PEER2 remote-address '10.0.2.2'
```

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```

set vpn ipsec site-to-site peer PEER2 tunnel 0 local prefix '192.168.0.0/24'
set vpn ipsec site-to-site peer PEER2 tunnel 0 remote prefix '192.168.1.0/24'

# PEER2
set interfaces dummy dum0 address '192.168.1.1/32'
set interfaces ethernet eth0 address '10.0.2.2/30'
set protocols static route 0.0.0.0/0 next-hop 10.0.2.1
set vpn ipsec authentication psk AUTH-PSK id '10.0.1.2'
set vpn ipsec authentication psk AUTH-PSK id '10.0.2.2'
set vpn ipsec authentication psk AUTH-PSK secret 'test'
set vpn ipsec esp-group ESP-GRPOUP lifetime '3600'
set vpn ipsec esp-group ESP-GRPOUP proposal 10 encryption 'aes256'
set vpn ipsec esp-group ESP-GRPOUP proposal 10 hash 'sha1'
set vpn ipsec ike-group IKE-GROUP close-action 'none'
set vpn ipsec ike-group IKE-GROUP dead-peer-detection action 'clear'
set vpn ipsec ike-group IKE-GROUP dead-peer-detection interval '30'
set vpn ipsec ike-group IKE-GROUP dead-peer-detection timeout '120'
set vpn ipsec ike-group IKE-GROUP key-exchange 'ikev1'
set vpn ipsec ike-group IKE-GROUP lifetime '28800'
set vpn ipsec ike-group IKE-GROUP proposal 10 dh-group '14'
set vpn ipsec ike-group IKE-GROUP proposal 10 encryption 'aes256'
set vpn ipsec ike-group IKE-GROUP proposal 10 hash 'sha1'
set vpn ipsec interface 'eth0'
set vpn ipsec site-to-site peer PEER1 authentication local-id '10.0.2.2'
set vpn ipsec site-to-site peer PEER1 authentication mode 'pre-shared-secret'
set vpn ipsec site-to-site peer PEER1 authentication remote-id '10.0.1.2'
set vpn ipsec site-to-site peer PEER1 connection-type 'none'
set vpn ipsec site-to-site peer PEER1 default-esp-group 'ESP-GRPOUP'
set vpn ipsec site-to-site peer PEER1 ike-group 'IKE-GROUP'
set vpn ipsec site-to-site peer PEER1 local-address '10.0.2.2'
set vpn ipsec site-to-site peer PEER1 remote-address '10.0.1.2'
set vpn ipsec site-to-site peer PEER1 tunnel 0 local prefix '192.168.1.0/24'
set vpn ipsec site-to-site peer PEER1 tunnel 0 remote prefix '192.168.0.0/24'

```

Show status of policy-based IPsec VPN setup:

```

vyos@PEER2:~$ show vpn ike sa
Peer ID / IP                               Local ID / IP
-----
10.0.1.2 10.0.1.2                           10.0.2.2 10.0.2.2

    State  IKEVer  Encrypt      Hash           D-H Group      NAT-T  A-Time  L-Time
    ----  -
    up     IKEv1   AES_CBC_256  HMAC_SHA1_96   MODP_2048      no     1254   25633

vyos@srv-gw0:~$ show vpn ipsec sa
Connection      State  Uptime  Bytes In/Out  Packets In/Out  Remote address
↳ Remote ID    Proposal
-----
↳ -----

```

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```

PEER1-tunnel-0 up      20m42s    0B/0B      0/0        10.0.1.2    ↵
↪10.0.1.2      AES_CBC_256/HMAC_SHA1_96/MODP_2048

vyos@PEER2:~$ show vpn ipsec connections
Connection      State    Type      Remote address  Local TS      Remote TS      Local
↪id             Remote id Proposal
-----
↪-----
PEER1           up      IKEv1     10.0.1.2        -             -             10.0.
↪2.2           10.0.1.2 AES_CBC/256/HMAC_SHA1_96/MODP_2048
PEER1-tunnel-0 up      IPsec     10.0.1.2        192.168.1.0/24 192.168.0.0/24 10.0.
↪2.2           10.0.1.2 AES_CBC/256/HMAC_SHA1_96/MODP_2048

```

If there is SNAT rules on eth0, need to add exclude rule

```

# PEER1 side
set nat source rule 10 destination address '192.168.1.0/24'
set nat source rule 10 'exclude'
set nat source rule 10 outbound-interface name 'eth0'
set nat source rule 10 source address '192.168.0.0/24'

# PEER2 side
set nat source rule 10 destination address '192.168.0.0/24'
set nat source rule 10 'exclude'
set nat source rule 10 outbound-interface name 'eth0'
set nat source rule 10 source address '192.168.1.0/24'

```

Route-Based VPN Example

PEER1:

- WAN interface on eth0
- eth0 interface IP: 10.0.1.2/30
- 'vti0' interface IP: 10.100.100.1/30
- dum0 interface IP: 192.168.0.1/24 (for testing purposes)
- Role: Initiator

PEER2:

- WAN interface on eth0
- eth0 interface IP: 10.0.2.2/30
- 'vti0' interface IP: 10.100.100.2/30
- dum0 interface IP: 192.168.1.0/24 (for testing purposes)
- Role: Responder

```

# PEER1
set interfaces dummy dum0 address '192.168.0.1/32'
set interfaces ethernet eth0 address '10.0.1.2/30'

```

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```

set interfaces vti vti0 address '10.100.100.1/30'
set protocols static route 0.0.0.0/0 next-hop 10.0.1.1
set protocols static route 192.168.1.0/24 next-hop 10.100.100.2
set vpn ipsec authentication psk AUTH-PSK id '10.0.1.2'
set vpn ipsec authentication psk AUTH-PSK id '10.0.2.2'
set vpn ipsec authentication psk AUTH-PSK secret 'test'
set vpn ipsec esp-group ESP-GRPOUP lifetime '3600'
set vpn ipsec esp-group ESP-GRPOUP proposal 10 encryption 'aes256'
set vpn ipsec esp-group ESP-GRPOUP proposal 10 hash 'sha1'
set vpn ipsec ike-group IKE-GROUP close-action 'start'
set vpn ipsec ike-group IKE-GROUP dead-peer-detection action 'restart'
set vpn ipsec ike-group IKE-GROUP dead-peer-detection interval '30'
set vpn ipsec ike-group IKE-GROUP key-exchange 'ikev2'
set vpn ipsec ike-group IKE-GROUP lifetime '28800'
set vpn ipsec ike-group IKE-GROUP proposal 10 dh-group '14'
set vpn ipsec ike-group IKE-GROUP proposal 10 encryption 'aes256'
set vpn ipsec ike-group IKE-GROUP proposal 10 hash 'sha1'
set vpn ipsec interface 'eth0'
set vpn ipsec options disable-route-autoinstall
set vpn ipsec site-to-site peer PEER2 authentication local-id '10.0.1.2'
set vpn ipsec site-to-site peer PEER2 authentication mode 'pre-shared-secret'
set vpn ipsec site-to-site peer PEER2 authentication remote-id '10.0.2.2'
set vpn ipsec site-to-site peer PEER2 connection-type 'initiate'
set vpn ipsec site-to-site peer PEER2 default-esp-group 'ESP-GRPOUP'
set vpn ipsec site-to-site peer PEER2 ike-group 'IKE-GROUP'
set vpn ipsec site-to-site peer PEER2 local-address '10.0.1.2'
set vpn ipsec site-to-site peer PEER2 remote-address '10.0.2.2'
set vpn ipsec site-to-site peer PEER2 vti bind 'vti0'

# PEER2
set interfaces dummy dum0 address '192.168.1.1/32'
set interfaces ethernet eth0 address '10.0.2.2/30'
set interfaces vti vti0 address '10.100.100.2/30'
set protocols static route 0.0.0.0/0 next-hop 10.0.2.1
set protocols static route 192.168.0.0/24 next-hop 10.100.100.1
set vpn ipsec authentication psk AUTH-PSK id '10.0.1.2'
set vpn ipsec authentication psk AUTH-PSK id '10.0.2.2'
set vpn ipsec authentication psk AUTH-PSK secret 'test'
set vpn ipsec esp-group ESP-GRPOUP lifetime '3600'
set vpn ipsec esp-group ESP-GRPOUP proposal 10 encryption 'aes256'
set vpn ipsec esp-group ESP-GRPOUP proposal 10 hash 'sha1'
set vpn ipsec ike-group IKE-GROUP close-action 'none'
set vpn ipsec ike-group IKE-GROUP dead-peer-detection action 'clear'
set vpn ipsec ike-group IKE-GROUP dead-peer-detection interval '30'
set vpn ipsec ike-group IKE-GROUP key-exchange 'ikev2'
set vpn ipsec ike-group IKE-GROUP lifetime '28800'
set vpn ipsec ike-group IKE-GROUP proposal 10 dh-group '14'
set vpn ipsec ike-group IKE-GROUP proposal 10 encryption 'aes256'
set vpn ipsec ike-group IKE-GROUP proposal 10 hash 'sha1'
set vpn ipsec interface 'eth0'
set vpn ipsec options disable-route-autoinstall

```

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```

set vpn ipsec site-to-site peer PEER1 authentication local-id '10.0.2.2'
set vpn ipsec site-to-site peer PEER1 authentication mode 'pre-shared-secret'
set vpn ipsec site-to-site peer PEER1 authentication remote-id '10.0.1.2'
set vpn ipsec site-to-site peer PEER1 connection-type 'none'
set vpn ipsec site-to-site peer PEER1 default-esp-group 'ESP-GRPOUP'
set vpn ipsec site-to-site peer PEER1 ike-group 'IKE-GROUP'
set vpn ipsec site-to-site peer PEER1 local-address '10.0.2.2'
set vpn ipsec site-to-site peer PEER1 remote-address '10.0.1.2'
set vpn ipsec site-to-site peer PEER1 vti bind 'vti0'

```

Show status of route-based IPsec VPN setup:

```

vyos@PEER2:~$ show vpn ike sa
Peer ID / IP                               Local ID / IP
-----
10.0.1.2 10.0.1.2                         10.0.2.2 10.0.2.2

      State  IKEVer  Encrypt      Hash          D-H Group      NAT-T  A-Time  L-Time
      ----  -
      up     IKEv2   AES_CBC_256  HMAC_SHA1_96  MODP_2048      no     404     27650

vyos@PEER2:~$ show vpn ipsec sa
Connection      State  Uptime    Bytes In/Out    Packets In/Out    Remote address
↳ Remote ID    Proposal
-----
↳ -----
PEER1-vti      up     3m28s     0B/0B           0/0               10.0.1.2          10.
↳ 0.1.2        AES_CBC_256/HMAC_SHA1_96/MODP_2048

vyos@PEER2:~$ show vpn ipsec connections
Connection      State  Type      Remote address    Local TS    Remote TS    Local id
↳ Remote id    Proposal
-----
↳ -----
PEER1          up     IKEv2     10.0.1.2         -           -           10.0.2.2      10.
↳ 0.1.2        AES_CBC/256/HMAC_SHA1_96/MODP_2048
PEER1-vti      up     IPsec     10.0.1.2         0.0.0.0/0   0.0.0.0/0   10.0.2.2      10.
↳ 0.1.2        AES_CBC/256/HMAC_SHA1_96/MODP_2048
::/0           ::/0

```

IPSec IKEv2 Remote Access VPN

Todo: Convert raw command blocks in this file to cfgcmd/opcmd directives for command coverage tracking.

Internet Key Exchange version 2 (IKEv2) is a tunneling protocol, based on IPsec, that establishes a secure VPN communication between VPN devices, and defines negotiation and authentication processes for IPsec security associations (SAs). It is often known as IKEv2/IPSec or IPsec IKEv2 remote-access — or road-warriors as others call it.

Key exchange and payload encryption is done using IKE and ESP proposals as known from IKEv1 but the connections are faster to establish, more reliable, and also support roaming from IP to IP (called MOBIKE which makes sure your

connection does not drop when changing networks from e.g. WIFI to LTE and back). Authentication can be achieved with X.509 certificates.

Setting up certificates:

First of all, we need to create a CA root certificate and server certificate on the server side.

```
vyos@vpn.vyos.net# run generate pki ca install ca_root
Enter private key type: [rsa, dsa, ec] (Default: rsa)
Enter private key bits: (Default: 2048)
Enter country code: (Default: GB)
Enter state: (Default: Some-State)
Enter locality: (Default: Some-City)
Enter organization name: (Default: VyOS)
Enter common name: (Default: vyos.io)
Enter how many days certificate will be valid: (Default: 1825)
Note: If you plan to use the generated key on this router, do not encrypt the private_
↵key.
Do you want to encrypt the private key with a passphrase? [y/N] N
2 value(s) installed. Use "compare" to see the pending changes, and "commit" to apply.
[edit]
```

```
vyos@vpn.vyos.net# comp
[pki ca]
+ ca_root {
+   certificate "MIIDnTCCAoWgAwI...."
+   private {
+       key "MIIEvAIBADANBgkqhkiG9...."
```

```
vyos@vpn.vyos.net# run generate pki certificate sign ca_root install server_cert
Do you already have a certificate request? [y/N] N
Enter private key type: [rsa, dsa, ec] (Default: rsa)
Enter private key bits: (Default: 2048)
Enter country code: (Default: GB)
Enter state: (Default: Some-State)
Enter locality: (Default: Some-City)
Enter organization name: (Default: VyOS)
Enter common name: (Default: vyos.io) vpn.vyos.net
Do you want to configure Subject Alternative Names? [y/N] N
Enter how many days certificate will be valid: (Default: 365)
Enter certificate type: (client, server) (Default: server)
Note: If you plan to use the generated key on this router, do not encrypt the private_
↵key.
Do you want to encrypt the private key with a passphrase? [y/N] N
2 value(s) installed. Use "compare" to see the pending changes, and "commit" to apply.
```

```
vyos@vpn.vyos.net# comp
[pki certificate]
+ server_cert {
+   certificate "MIIDuzCCAqOgAwIBAgIUaSrCPWx....."
+   private {
```

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```
+         key "MIIEvQIBADANBgkqhkiG9w0BAQEFAASCBK...."
+     }
+ }
```

Once the command is completed, it will add the certificate to the configuration session, to the pki subtree. You can then review the proposed changes and commit them.

Setting up IPsec:

After the PKI certs are all set up we can start configuring our IPsec/IKE proposals used for key-exchange end data encryption. The used encryption ciphers and integrity algorithms vary from operating system to operating system. The ones used in this example are validated to work on Windows 10.

```
set vpn ipsec esp-group ESP-RW lifetime '3600'
set vpn ipsec esp-group ESP-RW pfs 'disable'
set vpn ipsec esp-group ESP-RW proposal 10 encryption 'aes128gcm128'
set vpn ipsec esp-group ESP-RW proposal 10 hash 'sha256'

set vpn ipsec ike-group IKE-RW key-exchange 'ikev2'
set vpn ipsec ike-group IKE-RW lifetime '7200'
set vpn ipsec ike-group IKE-RW proposal 10 dh-group '14'
set vpn ipsec ike-group IKE-RW proposal 10 encryption 'aes128gcm128'
set vpn ipsec ike-group IKE-RW proposal 10 hash 'sha256'
```

Every connection/remote-access pool we configure also needs a pool where we can draw our client IP addresses from. We provide one IPv4 and IPv6 pool. Authorized clients will receive an IPv4 address from the configured IPv4 prefix and an IPv6 address from the IPv6 prefix. We can also send some DNS nameservers down to our clients used on their connection.

```
set vpn ipsec remote-access pool ra-rw-ipv4 name-server '192.0.2.1'
set vpn ipsec remote-access pool ra-rw-ipv4 prefix '192.0.2.128/25'

set vpn ipsec remote-access pool ra-rw-ipv6 name-server '2001:db8:1000::1'
set vpn ipsec remote-access pool ra-rw-ipv6 prefix '2001:db8:2000::/64'
```

Setting up tunnel:

```
set vpn ipsec remote-access connection rw authentication local-id '192.0.2.1'
set vpn ipsec remote-access connection rw authentication server-mode 'x509'
set vpn ipsec remote-access connection rw authentication x509 ca-certificate 'ca_root'
set vpn ipsec remote-access connection rw authentication x509 certificate 'server_cert'
set vpn ipsec remote-access connection rw esp-group 'ESP-RW'
set vpn ipsec remote-access connection rw ike-group 'IKE-RW'
set vpn ipsec remote-access connection rw local-address '192.0.2.1'
set vpn ipsec remote-access connection rw pool 'ra-rw-ipv4'
set vpn ipsec remote-access connection rw pool 'ra-rw-ipv6'
```

VyOS also supports two different modes of authentication, local and RADIUS. To create a new local user named “vyos” with a password of “vyos” use the following commands.

```
set vpn ipsec remote-access connection rw authentication client-mode 'eap-mschapv2'
set vpn ipsec remote-access connection rw authentication local-users username vyos
password 'vyos'
```

Some client operating systems like to see the servers certificate. The following option causes the server to voluntarily send its certificate, even if it wasn't requested.

```
set vpn ipsec remote-access connection rw authentication always-send-cert
```

Client Configuration

Most operating systems include native client support for IPsec IKEv2 VPN connections, and others typically have an app or add-on package which adds the capability. This section covers IPsec IKEv2 client configuration for Windows 10.

VyOS provides a command to generate a connection profile used by Windows clients that will connect to the “rw” connection on our VyOS server.

Note: Windows expects the server name to be also used in the server's certificate common name, so it's best to use this DNS name for your VPN connection.

```
vyos@vpn.vyos.net:~$ generate ipsec profile windows-remote-access rw remote vpn.vyos.net

==== <snip> ====
Add-VpnConnection -Name "VyOS IKEv2 VPN" -ServerAddress "vpn.vyos.net" -TunnelType "Ikev2"

Set-VpnConnectionIPsecConfiguration -ConnectionName "VyOS IKEv2 VPN" -
AuthenticationTransformConstants GCMAES128 -CipherTransformConstants
GCMAES128 -EncryptionMethod GCMAES128 -IntegrityCheckMethod SHA256128 -PfsGroup None -
DHGroup "Group14" -PassThru -Force
==== </snip> =====
```

Add the commands from Snippet in the Windows side via PowerShell. Also import the root CA cert to the Windows “Trusted Root Certification Authorities” and establish the connection.

Verification:

```
vyos@vpn.vyos.net:~$ show vpn ipsec remote-access summary
```

Connection ID	Username	Protocol	State	Uptime	Tunnel IP	Remote Host
Remote ID	IKE Proposal				IPSec Proposal	
5	vyos	IKEv2	UP	37s	192.0.2.129	10.0.0.2
10.0.0.2	AES_GCM_16-128/PRF_HMAC_SHA2_256/MODP_2048				ESP:AES_GCM_16-128	

Troubleshooting Site-to-Site VPN IPsec

Todo: Convert raw command blocks in this file to cfgcmd/opcmd directives for command coverage tracking.

Introduction

This document describes the methodology to monitor and troubleshoot Site-to-Site VPN IPsec.

Steps for troubleshooting problems with Site-to-Site VPN IPsec:

1. Ping the remote site through the tunnel using the source and destination IPs included in the policy.
2. Check connectivity between the routers using the ping command (if ICMP traffic is allowed).
3. Check the IKE SAs' statuses.
4. Check the IPsec SAs' statuses.
5. Check logs to view debug messages.

Checking IKE SA Status

The next command shows IKE SAs' statuses.

```
vyos@vyos:~$ show vpn ike sa
```

Peer ID / IP	Local ID / IP							
-----	-----							
192.168.1.2 192.168.1.2	192.168.0.1 192.168.0.1							
State	IKEVer	Encrypt	Hash	D-H Group	NAT-T	A-Time	L-Time	
----	-----	-----	----	-----	-----	-----	-----	
up	IKEv2	AES_CBC_128	HMAC_SHA1_96	MODP_2048	no	162	27023	

This command shows the next information:

- IKE SA status.
- Selected IKE version.
- Selected Encryption, Hash and Diffie-Hellman Group.
- NAT-T.
- ID and IP of both peers.
- A-Time: established time, L-Time: time for next rekeying.

IPsec SA (CHILD SA) Status

The next commands show IPsec SAs' statuses.

```
vyos@vyos:~$ show vpn ipsec sa
Connection      State    Uptime    Bytes In/Out    Packets In/Out    Remote address
↳ Remote ID     Proposal
-----
↳ -----
PEER-tunnel-1  up      16m30s    168B/168B      2/2              192.168.1.2
↳ 192.168.1.2  AES_CBC_128/HMAC_SHA1_96/MODP_2048
```

```
vyos@vyos:~$ show vpn ipsec sa detail
PEER: #1, ESTABLISHED, IKEv2, 101275ac719d5a1b_i* 68ea4ec3bed3bf0c_r
  local '192.168.0.1' @ 192.168.0.1[4500]
  remote '192.168.1.2' @ 192.168.1.2[4500]
  AES_CBC-128/HMAC_SHA1_96/PRF_HMAC_SHA1/MODP_2048
  established 4054s ago, rekeying in 23131s
PEER-tunnel-1: #2, reqid 1, INSTALLED, TUNNEL, ESP:AES_CBC-128/HMAC_SHA1_96/MODP_2048
  installed 1065s ago, rekeying in 1998s, expires in 2535s
  in c5821882, 168 bytes, 2 packets, 81s ago
  out c433406a, 168 bytes, 2 packets, 81s ago
  local 10.0.0.0/24
  remote 10.0.1.0/24
```

These commands show the next information:

- IPsec SA status.
- Uptime and time for the next rekeying.
- Amount of transferred data.
- Remote and local ID and IP.
- Selected Encryption, Hash and Diffie-Hellman Group.
- Mode (tunnel or transport).
- Remote and local prefixes which are use for policy.

There is a possibility to view the summarized information of SAs' status

```
vyos@vyos:~$ show vpn ipsec connections
Connection      State    Type    Remote address    Local TS    Remote TS    Local id
↳ Remote id     Proposal
-----
↳ -----
PEER            up      IKEv2    192.168.1.2      -          -          192.168.0.1
↳ 192.168.1.2  AES_CBC/128/HMAC_SHA1_96/MODP_2048
PEER-tunnel-1  up      IPsec    192.168.1.2      10.0.0.0/24 10.0.1.0/24 192.168.0.1
↳ 192.168.1.2  AES_CBC/128/HMAC_SHA1_96/MODP_2048
```

Viewing Logs for Debugging

If IKE SAs or IPsec SAs are down, need to debug IPsec connectivity using `logs show log ipsec`

The next example of the successful IPsec connection initialization.

```
vyos@vyos:~$ show log ipsec
Jun 20 14:29:47 charon[2428]: 02[NET] <PEER|1> received packet: from 192.168.1.2[500] to 192.168.0.1[500] (472 bytes)
Jun 20 14:29:47 charon[2428]: 02[ENC] <PEER|1> parsed IKE_SA_INIT response 0 [ SA KE No N(NATD_S_IP) N(NATD_D_IP) N(FRAG_SUP) N(HASH_ALG) N(CHKDLESS_SUP) N(MULT_AUTH) ]
Jun 20 14:29:47 charon-systemd[2428]: received packet: from 192.168.1.2[500] to 192.168.0.1[500] (472 bytes)
Jun 20 14:29:47 charon[2428]: 02[CFG] <PEER|1> selected proposal: IKE:AES_CBC_128/HMAC_SHA1_96/PRF_HMAC_SHA1/MODP_2048
Jun 20 14:29:47 charon-systemd[2428]: parsed IKE_SA_INIT response 0 [ SA KE No N(NATD_S_IP) N(NATD_D_IP) N(FRAG_SUP) N(HASH_ALG) N(CHKDLESS_SUP) N(MULT_AUTH) ]
Jun 20 14:29:47 charon-systemd[2428]: selected proposal: IKE:AES_CBC_128/HMAC_SHA1_96/PRF_HMAC_SHA1/MODP_2048
Jun 20 14:29:47 charon[2428]: 02[IKE] <PEER|1> authentication of '192.168.0.1' (myself) with pre-shared key
Jun 20 14:29:47 charon-systemd[2428]: authentication of '192.168.0.1' (myself) with pre-shared key
Jun 20 14:29:47 charon[2428]: 02[IKE] <PEER|1> establishing CHILD_SA PEER-tunnel-1{1}
Jun 20 14:29:47 charon-systemd[2428]: establishing CHILD_SA PEER-tunnel-1{1}
Jun 20 14:29:47 charon[2428]: 02[ENC] <PEER|1> generating IKE_AUTH request 1 [ IDi N(INIT_CONTACT) IDr AUTH SA TSi TSr N(MOBIKE_SUP) N(NO_ADD_ADDR) N(MULT_AUTH) N(EAP_ONLY) N(MSG_ID_SYN_SUP) ]
Jun 20 14:29:47 charon-systemd[2428]: generating IKE_AUTH request 1 [ IDi N(INIT_CONTACT) IDr AUTH SA TSi TSr N(MOBIKE_SUP) N(NO_ADD_ADDR) N(MULT_AUTH) N(EAP_ONLY) N(MSG_ID_SYN_SUP) ]
Jun 20 14:29:47 charon[2428]: 02[NET] <PEER|1> sending packet: from 192.168.0.1[4500] to 192.168.1.2[4500] (268 bytes)
Jun 20 14:29:47 charon-systemd[2428]: sending packet: from 192.168.0.1[4500] to 192.168.1.2[4500] (268 bytes)
Jun 20 14:29:47 charon[2428]: 13[NET] <PEER|1> received packet: from 192.168.1.2[4500] to 192.168.0.1[4500] (220 bytes)
Jun 20 14:29:47 charon[2428]: 13[ENC] <PEER|1> parsed IKE_AUTH response 1 [ IDr AUTH SA TSi TSr N(MOBIKE_SUP) N(NO_ADD_ADDR) ]
Jun 20 14:29:47 charon-systemd[2428]: received packet: from 192.168.1.2[4500] to 192.168.0.1[4500] (220 bytes)
Jun 20 14:29:47 charon[2428]: 13[IKE] <PEER|1> authentication of '192.168.1.2' with pre-shared key successful
Jun 20 14:29:47 charon-systemd[2428]: parsed IKE_AUTH response 1 [ IDr AUTH SA TSi TSr N(MOBIKE_SUP) N(NO_ADD_ADDR) ]
Jun 20 14:29:47 charon[2428]: 13[IKE] <PEER|1> peer supports MOBIKE
Jun 20 14:29:47 charon-systemd[2428]: authentication of '192.168.1.2' with pre-shared key successful
Jun 20 14:29:47 charon[2428]: 13[IKE] <PEER|1> IKE_SA PEER[1] established between 192.168.0.1[192.168.0.1]...192.168.1.2[192.168.1.2]
Jun 20 14:29:47 charon-systemd[2428]: peer supports MOBIKE
Jun 20 14:29:47 charon[2428]: 13[IKE] <PEER|1> scheduling rekeying in 27703s
Jun 20 14:29:47 charon-systemd[2428]: IKE_SA PEER[1] established between 192.168.0.1[192.168.0.1]...192.168.1.2[192.168.1.2]
```

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```

Jun 20 14:29:47 charon[2428]: 13[IKE] <PEER|1> maximum IKE_SA lifetime 30583s
Jun 20 14:29:47 charon-systemd[2428]: scheduling rekeying in 27703s
Jun 20 14:29:47 charon[2428]: 13[CFG] <PEER|1> selected proposal: ESP:AES_CBC_128/HMAC_
↳ SHA1_96/NO_EXT_SEQ
Jun 20 14:29:47 charon-systemd[2428]: maximum IKE_SA lifetime 30583s
Jun 20 14:29:47 charon-systemd[2428]: selected proposal: ESP:AES_CBC_128/HMAC_SHA1_96/NO_
↳ EXT_SEQ
Jun 20 14:29:47 charon[2428]: 13[IKE] <PEER|1> CHILD_SA PEER-tunnel-1{1} established_
↳ with SPIs cb94fb3f_i ca99c8a9_o and TS 10.0.0.0/24 == 10.0.1.0/24
Jun 20 14:29:47 charon-systemd[2428]: CHILD_SA PEER-tunnel-1{1} established with SPIs_
↳ cb94fb3f_i ca99c8a9_o and TS 10.0.0.0/24 == 10.0.1.0/24

```

Troubleshooting Examples

IKE PROPOSAL are Different

In this situation, IKE SAs can be down or not active.

```
vyos@vyos:~$ show vpn ike sa
```

The problem is in IKE phase (Phase 1). The next step is checking debug logs.

Responder Side:

```

Jun 23 07:36:33 charon[2440]: 01[CFG] <1> received proposals: IKE:AES_CBC_256/HMAC_SHA1_
↳ 96/PRF_HMAC_SHA1/MODP_2048
Jun 23 07:36:33 charon-systemd[2440]: received proposals: IKE:AES_CBC_256/HMAC_SHA1_96/
↳ PRF_HMAC_SHA1/MODP_2048
Jun 23 07:36:33 charon[2440]: 01[CFG] <1> configured proposals: IKE:AES_CBC_128/HMAC_
↳ SHA1_96/PRF_HMAC_SHA1/MODP_2048
Jun 23 07:36:33 charon-systemd[2440]: configured proposals: IKE:AES_CBC_128/HMAC_SHA1_96/
↳ PRF_HMAC_SHA1/MODP_2048
Jun 23 07:36:33 charon[2440]: 01[IKE] <1> received proposals unacceptable
Jun 23 07:36:33 charon-systemd[2440]: received proposals unacceptable
Jun 23 07:36:33 charon[2440]: 01[ENC] <1> generating IKE_SA_INIT response 0 [ N(NO_PROPOSAL)_
↳ ]

```

Initiator side:

```

Jun 23 07:36:32 charon-systemd[2444]: parsed IKE_SA_INIT response 0 [ N(NO_PROPOSAL) ]
Jun 23 07:36:32 charon[2444]: 14[IKE] <PEER|1> received NO_PROPOSAL_CHOSEN notify error
Jun 23 07:36:32 charon-systemd[2444]: received NO_PROPOSAL_CHOSEN notify error

```

The notification **NO_PROPOSAL_CHOSEN** means that the proposal mismatch. On the Responder side there is concrete information where is mismatch. Encryption **AES_CBC_128** is configured in IKE policy on the responder but **AES_CBC_256** is configured on the initiator side.

PSK Secret Mismatch

In this situation, IKE SAs can be down or not active.

```
vyos@vyos:~$ show vpn ike sa
```

The problem is in IKE phase (Phase 1). The next step is checking debug logs.

Responder:

```
Jun 23 08:07:26 charon-systemd[2440]: tried 1 shared key for '192.168.1.2' - '192.168.0.1
↳ ', but MAC mismatched
Jun 23 08:07:26 charon[2440]: 13[ENC] <PEER|3> generating IKE_AUTH response 1 [ N(AUTH_
↳ FAILED) ]
```

Initiator side:

```
Jun 23 08:07:24 charon[2436]: 12[ENC] <PEER|1> parsed IKE_AUTH response 1 [ N(AUTH_
↳ FAILED) ]
Jun 23 08:07:24 charon-systemd[2436]: parsed IKE_AUTH response 1 [ N(AUTH_FAILED) ]
Jun 23 08:07:24 charon[2436]: 12[IKE] <PEER|1> received AUTHENTICATION_FAILED notify_
↳ error
Jun 23 08:07:24 charon-systemd[2436]: received AUTHENTICATION_FAILED notify error
```

The notification **AUTHENTICATION_FAILED** means that the authentication is failed. There is a reason to check PSK on both side.

ESP Proposal Mismatch

The output of **show** commands shows us that IKE SA is established but IPSec SA is not.

```
vyos@vyos:~$ show vpn ike sa
```

Peer ID / IP	Local ID / IP
192.168.1.2 192.168.1.2	192.168.0.1 192.168.0.1

State	IKEVer	Encrypt	Hash	D-H Group	NAT-T	A-Time	L-Time
up	IKEv2	AES_CBC_128	HMAC_SHA1_96	MODP_2048	no	158	26817

```
vyos@vyos:~$ show vpn ipsec sa
```

Connection	State	Uptime	Bytes In/Out	Packets In/Out	Remote address
↳ Remote ID	Proposal				

The next step is checking debug logs.

Initiator side:

```
Jun 23 08:16:10 charon[3789]: 13[NET] <PEER|1> received packet: from 192.168.1.2[500] to_
↳ 192.168.0.1[500] (472 bytes)
Jun 23 08:16:10 charon[3789]: 13[ENC] <PEER|1> parsed IKE_SA_INIT response 0 [ SA KE No_
↳ N(NATD_S_IP) N(NATD_D_IP) N(FRAG_SUP) N(HASH_ALG) N(CHDLESS_SUP) N(MULT_AUTH) ]
```

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```

Jun 23 08:16:10 charon-systemd[3789]: received packet: from 192.168.1.2[500] to 192.168.
↳0.1[500] (472 bytes)
Jun 23 08:16:10 charon[3789]: 13[CFG] <PEER|1> selected proposal: IKE:AES_CBC_128/HMAC_
↳SHA1_96/PRF_HMAC_SHA1/MODP_2048
Jun 23 08:16:10 charon-systemd[3789]: parsed IKE_SA_INIT response 0 [ SA KE No N(NATD_S_
↳IP) N(NATD_D_IP) N(FRAG_SUP) N(HASH_ALG) N(CHDLESS_SUP) N(MULT_AUTH) ]
Jun 23 08:16:10 charon-systemd[3789]: selected proposal: IKE:AES_CBC_128/HMAC_SHA1_96/
↳PRF_HMAC_SHA1/MODP_2048
Jun 23 08:16:10 charon[3789]: 13[IKE] <PEER|1> authentication of '192.168.0.1' (myself)
↳with pre-shared key
Jun 23 08:16:10 charon-systemd[3789]: authentication of '192.168.0.1' (myself) with pre-
↳shared key
Jun 23 08:16:10 charon[3789]: 13[IKE] <PEER|1> establishing CHILD_SA PEER-tunnel-1{1}
Jun 23 08:16:10 charon-systemd[3789]: establishing CHILD_SA PEER-tunnel-1{1}
Jun 23 08:16:10 charon[3789]: 13[ENC] <PEER|1> generating IKE_AUTH request 1 [ IDi
↳N(INIT_CONTACT) IDr AUTH SA TSi TSr N(MOBIKE_SUP) N(NO_ADD_ADDR) N(MULT_AUTH) N(EAP_
↳ONLY) N(MSG_ID_SYN_SUP) ]
Jun 23 08:16:10 charon-systemd[3789]: generating IKE_AUTH request 1 [ IDi N(INIT_
↳CONTACT) IDr AUTH SA TSi TSr N(MOBIKE_SUP) N(NO_ADD_ADDR) N(MULT_AUTH) N(EAP_ONLY)
↳N(MSG_ID_SYN_SUP) ]
Jun 23 08:16:10 charon[3789]: 13[NET] <PEER|1> sending packet: from 192.168.0.1[4500] to
↳192.168.1.2[4500] (268 bytes)
Jun 23 08:16:10 charon-systemd[3789]: sending packet: from 192.168.0.1[4500] to 192.168.
↳1.2[4500] (268 bytes)
Jun 23 08:16:10 charon[3789]: 09[NET] <PEER|1> received packet: from 192.168.1.2[4500]
↳to 192.168.0.1[4500] (140 bytes)
Jun 23 08:16:10 charon-systemd[3789]: received packet: from 192.168.1.2[4500] to 192.168.
↳0.1[4500] (140 bytes)
Jun 23 08:16:10 charon[3789]: 09[ENC] <PEER|1> parsed IKE_AUTH response 1 [ IDr AUTH
↳N(MOBIKE_SUP) N(NO_ADD_ADDR) N(NO_PROP) ]
Jun 23 08:16:10 charon-systemd[3789]: parsed IKE_AUTH response 1 [ IDr AUTH N(MOBIKE_
↳SUP) N(NO_ADD_ADDR) N(NO_PROP) ]
Jun 23 08:16:10 charon[3789]: 09[IKE] <PEER|1> authentication of '192.168.1.2' with pre-
↳shared key successful
Jun 23 08:16:10 charon-systemd[3789]: authentication of '192.168.1.2' with pre-shared
↳key successful
Jun 23 08:16:10 charon[3789]: 09[IKE] <PEER|1> peer supports MOBIKE
Jun 23 08:16:10 charon-systemd[3789]: peer supports MOBIKE
Jun 23 08:16:10 charon[3789]: 09[IKE] <PEER|1> IKE_SA PEER[1] established between 192.
↳168.0.1[192.168.0.1]...192.168.1.2[192.168.1.2]
Jun 23 08:16:10 charon-systemd[3789]: IKE_SA PEER[1] established between 192.168.0.1[192.
↳168.0.1]...192.168.1.2[192.168.1.2]
Jun 23 08:16:10 charon[3789]: 09[IKE] <PEER|1> scheduling rekeying in 26975s
Jun 23 08:16:10 charon-systemd[3789]: scheduling rekeying in 26975s
Jun 23 08:16:10 charon[3789]: 09[IKE] <PEER|1> maximum IKE_SA lifetime 29855s
Jun 23 08:16:10 charon-systemd[3789]: maximum IKE_SA lifetime 29855s
Jun 23 08:16:10 charon[3789]: 09[IKE] <PEER|1> received NO_PROPOSAL_CHOSEN notify, no
↳CHILD_SA built
Jun 23 08:16:10 charon-systemd[3789]: received NO_PROPOSAL_CHOSEN notify, no CHILD_SA
↳built
Jun 23 08:16:10 charon[3789]: 09[IKE] <PEER|1> failed to establish CHILD_SA, keeping IKE_
↳SA

```

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```
Jun 23 08:16:10 charon-systemd[3789]: failed to establish CHILD_SA, keeping IKE_SA
```

There are messages: **NO_PROPOSAL_CHOSEN** and **failed to establish CHILD_SA** which refers that the problem is in the IPsec(ESP) proposal mismatch.

The reason of this problem is showed on the responder side.

```
Jun 23 08:16:12 charon[2440]: 01[CFG] <PEER|5> received proposals: ESP:AES_CBC_256/HMAC_
↳ SHA1_96/NO_EXT_SEQ
Jun 23 08:16:12 charon-systemd[2440]: received proposals: ESP:AES_CBC_256/HMAC_SHA1_96/
↳ NO_EXT_SEQ
Jun 23 08:16:12 charon[2440]: 01[CFG] <PEER|5> configured proposals: ESP:AES_CBC_128/
↳ HMAC_SHA1_96/MODP_2048/NO_EXT_SEQ
Jun 23 08:16:12 charon-systemd[2440]: configured proposals: ESP:AES_CBC_128/HMAC_SHA1_96/
↳ MODP_2048/NO_EXT_SEQ
Jun 23 08:16:12 charon[2440]: 01[IKE] <PEER|5> no acceptable proposal found
Jun 23 08:16:12 charon-systemd[2440]: no acceptable proposal found
Jun 23 08:16:12 charon[2440]: 01[IKE] <PEER|5> failed to establish CHILD_SA, keeping IKE_
↳ SA
```

Encryption **AES_CBC_128** is configured in IKE policy on the responder but **AES_CBC_256** is configured on the initiator side.

Prefixes in Policies Mismatch

As in previous situation, IKE SA is in up state but IPsec SA is not up. According to logs we can see **TS_UNACCEPTABLE** notification. It means that prefixes (traffic selectors) mismatch on both sides

Initiator:

```
Jun 23 14:13:17 charon[4996]: 11[IKE] <PEER|1> received TS_UNACCEPTABLE notify, no CHILD_
↳ SA built
Jun 23 14:13:17 charon-systemd[4996]: maximum IKE_SA lifetime 29437s
Jun 23 14:13:17 charon[4996]: 11[IKE] <PEER|1> failed to establish CHILD_SA, keeping IKE_
↳ SA
Jun 23 14:13:17 charon-systemd[4996]: received TS_UNACCEPTABLE notify, no CHILD_SA built
Jun 23 14:13:17 charon-systemd[4996]: failed to establish CHILD_SA, keeping IKE_SA
```

The reason of this problem is showed on the responder side.

```
Jun 23 14:13:19 charon[2440]: 01[IKE] <PEER|7> traffic selectors 10.0.2.0/24 === 10.0.0.
↳ 0/24 unacceptable
Jun 23 14:13:19 charon-systemd[2440]: traffic selectors 10.0.2.0/24 === 10.0.0.0/24
↳ unacceptable
Jun 23 14:13:19 charon[2440]: 01[IKE] <PEER|7> failed to establish CHILD_SA, keeping IKE_
↳ SA
Jun 23 14:13:19 charon-systemd[2440]: failed to establish CHILD_SA, keeping IKE_SA
Jun 23 14:13:19 charon[2440]: 01[ENC] <PEER|7> generating IKE_AUTH response 1 [ IDr AUTH
↳ N(MOBIKE_SUP) N(NO_ADD_ADDR) N(TS_UNACCEPT) ]
Jun 23 14:13:19 charon-systemd[2440]: generating IKE_AUTH response 1 [ IDr AUTH N(MOBIKE_
↳ SUP) N(NO_ADD_ADDR) N(TS_UNACCEPT) ]
```

Traffic selectors **10.0.2.0/24 === 10.0.0.0/24** are unacceptable on the responder side.

7.13.2 L2TP

VyOS utilizes `accel-ppp` to provide L2TP server functionality. It can be used with local authentication or a connected RADIUS server.

Configuring L2TP Server

```
set vpn l2tp remote-access authentication mode local
set vpn l2tp remote-access authentication local-users username test password 'test'
set vpn l2tp remote-access client-ip-pool L2TP-POOL range 192.168.255.2-192.168.255.254
set vpn l2tp remote-access default-pool 'L2TP-POOL'
set vpn l2tp remote-access outside-address 192.0.2.2
set vpn l2tp remote-access gateway-address 192.168.255.1
```

set vpn l2tp remote-access authentication mode <local | radius>

Set authentication backend. The configured authentication backend is used for all queries.

- **radius:** All authentication queries are handled by a configured RADIUS server.
- **local:** All authentication queries are handled locally.

set vpn l2tp remote-access authentication local-users username <user> password <pass>

Create <user> for local authentication on this system. The users password will be set to <pass>.

set vpn l2tp remote-access client-ip-pool <POOL-NAME> range <x.x.x.x-x.x.x.x | x.x.x.x/x>

Use this command to define the first IP address of a pool of addresses to be given to l2tp clients. If notation `x.x.x.x-x.x.x.x`, it must be within a /24 subnet. If notation `x.x.x.x/x` is used there is possibility to set host/netmask.

set vpn l2tp remote-access default-pool <POOL-NAME>

Use this command to define default address pool name.

set vpn l2tp remote-access gateway-address <gateway>

Specifies single <gateway> IP address to be used as local address of PPP interfaces.

Configuring IPsec

```
set vpn ipsec interface eth0
set vpn l2tp remote-access ipsec-settings authentication mode pre-shared-secret
set vpn l2tp remote-access ipsec-settings authentication pre-shared-secret secret
```

set vpn ipsec interface <INTERFACE>

Use this command to define IPsec interface.

set vpn l2tp remote-access ipsec-settings authentication mode <pre-shared-secret | x509>

Set mode for IPsec authentication between VyOS and L2TP clients.

set vpn l2tp remote-access ipsec-settings authentication pre-shared-secret <secret>

Set predefined shared secret phrase.

If a local firewall policy is in place on your external interface you will need to allow the ports below:

- UDP port 500 (IKE)

- IP protocol number 50 (ESP)
- UDP port 1701 for IPsec

As well as the below to allow NAT-traversal (when NAT is detected by the VPN client, ESP is encapsulated in UDP for NAT-traversal):

- UDP port 4500 (NAT-T)

Example:

```
set firewall ipv4 name OUTSIDE-LOCAL rule 40 action 'accept'
set firewall ipv4 name OUTSIDE-LOCAL rule 40 protocol 'esp'
set firewall ipv4 name OUTSIDE-LOCAL rule 41 action 'accept'
set firewall ipv4 name OUTSIDE-LOCAL rule 41 destination port '500'
set firewall ipv4 name OUTSIDE-LOCAL rule 41 protocol 'udp'
set firewall ipv4 name OUTSIDE-LOCAL rule 42 action 'accept'
set firewall ipv4 name OUTSIDE-LOCAL rule 42 destination port '4500'
set firewall ipv4 name OUTSIDE-LOCAL rule 42 protocol 'udp'
set firewall ipv4 name OUTSIDE-LOCAL rule 43 action 'accept'
set firewall ipv4 name OUTSIDE-LOCAL rule 43 destination port '1701'
set firewall ipv4 name OUTSIDE-LOCAL rule 43 ipsec 'match-ipsec'
set firewall ipv4 name OUTSIDE-LOCAL rule 43 protocol 'udp'
```

To allow VPN-clients access via your external address, a NAT rule is required:

```
set nat source rule 110 outbound-interface name 'eth0'
set nat source rule 110 source address '192.168.255.0/24'
set nat source rule 110 translation address masquerade
```

Configuring RADIUS authentication

To enable RADIUS based authentication, the authentication mode needs to be changed within the configuration. Previous settings like the local users, still exists within the configuration, however they are not used if the mode has been changed from local to radius. Once changed back to local, it will use all local accounts again.

```
set vpn l2tp remote-access authentication mode radius
```

```
set vpn l2tp remote-access authentication radius server <server> key <secret>
```

Configure RADIUS <server> and its required shared <secret> for communicating with the RADIUS server.

Since the RADIUS server would be a single point of failure, multiple RADIUS servers can be setup and will be used subsequently. For example:

```
set vpn l2tp remote-access authentication radius server 10.0.0.1 key 'foo'
set vpn l2tp remote-access authentication radius server 10.0.0.2 key 'foo'
```

Note: Some RADIUS servers use an access control list which allows or denies queries, make sure to add your VyOS router to the allowed client list.

RADIUS source address

If you are using OSPF as your IGP, use the interface connected closest to the RADIUS server. You can bind all outgoing RADIUS requests to a single source IP e.g. the loopback interface.

```
set vpn l2tp remote-access authentication radius source-address <address>
```

Source IPv4 address used in all RADIUS server queries.

Note: The source-address must be configured to that of an interface. Best practice would be a loopback or dummy interface.

RADIUS advanced options

```
set vpn l2tp remote-access authentication radius server <server> port <port>
```

Configure RADIUS <server> and its required port for authentication requests.

```
set vpn l2tp remote-access authentication radius server <server> fail-time <time>
```

Mark RADIUS server as offline for this given <time> in seconds.

```
set vpn l2tp remote-access authentication radius server <server> disable
```

Temporary disable this RADIUS server.

```
set vpn l2tp remote-access authentication radius acct-timeout <timeout>
```

Timeout to wait reply for Interim-Update packets. (default 3 seconds)

```
set vpn l2tp remote-access authentication radius dynamic-author server <address>
```

Specifies IP address for Dynamic Authorization Extension server (DM/CoA). This IP must exist on any VyOS interface or it can be 0.0.0.0.

```
set vpn l2tp remote-access authentication radius dynamic-author port <port>
```

UDP port for Dynamic Authorization Extension server (DM/CoA)

```
set vpn l2tp remote-access authentication radius dynamic-author key <secret>
```

Secret for Dynamic Authorization Extension server (DM/CoA)

```
set vpn l2tp remote-access authentication radius max-try <number>
```

Maximum number of tries to send Access-Request/Accounting-Request queries

```
set vpn l2tp remote-access authentication radius timeout <timeout>
```

Timeout to wait response from server (seconds)

```
set vpn l2tp remote-access authentication radius nas-identifier <identifier>
```

Value to send to RADIUS server in NAS-Identifier attribute and to be matched in DM/CoA requests.

```
set vpn l2tp remote-access authentication radius nas-ip-address <address>
```

Value to send to RADIUS server in NAS-IP-Address attribute and to be matched in DM/CoA requests. Also DM/CoA server will bind to that address.

```
set vpn l2tp remote-access authentication radius source-address <address>
```

Source IPv4 address used in all RADIUS server queries.

set vpn l2tp remote-access authentication radius rate-limit attribute <attribute>

Specifies which RADIUS server attribute contains the rate limit information. The default attribute is `Filter-Id`.

Note: If you set a custom RADIUS attribute you must define it on both dictionaries on the RADIUS server and client.

set vpn l2tp remote-access authentication radius rate-limit enable

Enables bandwidth shaping via RADIUS.

set vpn l2tp remote-access authentication radius rate-limit vendor

Specifies the vendor dictionary. This dictionary needs to be present in `/usr/share/accel-ppp/radius`.

Received RADIUS attributes have a higher priority than parameters defined within the CLI configuration, refer to the explanation below.

Allocation clients ip addresses by RADIUS

If the RADIUS server sends the attribute `Framed-IP-Address` then this IP address will be allocated to the client and the option `default-pool` within the CLI config will be ignored.

If the RADIUS server sends the attribute `Framed-Pool`, then the IP address will be allocated from a predefined IP pool whose name equals the attribute value.

If the RADIUS server sends the attribute `Stateful-IPv6-Address-Pool`, the IPv6 address will be allocated from a predefined IPv6 pool `prefix` whose name equals the attribute value.

If the RADIUS server sends the attribute `Delegated-IPv6-Prefix-Pool`, an IPv6 delegation prefix will be allocated from a predefined IPv6 pool `delegate` whose name equals the attribute value.

Note: `Stateful-IPv6-Address-Pool` and `Delegated-IPv6-Prefix-Pool` are defined in RFC6911. If they are not defined in your RADIUS server, add new [dictionary](#).

The client's interface can be put into a VRF context via a RADIUS Access-Accept packet, or changed via RADIUS CoA. `Acce1-VRF-Name` is used for these purposes. This is a custom [ACCEL-PPP attribute](#). Define it in your RADIUS server.

Renaming clients interfaces by RADIUS

If the RADIUS server uses the attribute `NAS-Port-Id`, ppp tunnels will be renamed.

Note: The value of the attribute `NAS-Port-Id` must be less than 16 characters, otherwise the interface won't be renamed.

Configuring LNS (L2TP Network Server)

LNS are often used to connect to a LAC (L2TP Access Concentrator).

```
set vpn l2tp remote-access lns host-name <hostname>
```

Sent to the client (LAC) in the Host-Name attribute

```
set vpn l2tp remote-access lns shared-secret <secret>
```

Tunnel password used to authenticate the client (LAC)

To explain the usage of LNS follow our blueprint [PPPoE over L2TP](#).

IPv6

```
set vpn l2tp remote-access ppp-options ipv6 <require | prefer | allow | deny>
```

Specifies IPv6 negotiation preference.

- **require** - Require IPv6 negotiation
- **prefer** - Ask client for IPv6 negotiation, do not fail if it rejects
- **allow** - Negotiate IPv6 only if client requests
- **deny** - Do not negotiate IPv6 (default value)

```
set vpn l2tp remote-access client-ipv6-pool <IPv6-POOL-NAME> prefix <address> mask <number-of-bits>
```

Use this command to set the IPv6 address pool from which an l2tp client will get an IPv6 prefix of your defined length (mask) to terminate the l2tp endpoint at their side. The mask length can be set between 48 and 128 bits long, the default value is 64.

```
set vpn l2tp remote-access client-ipv6-pool <IPv6-POOL-NAME> delegate <address> delegation-prefix <number-of-bits>
```

Use this command to configure DHCPv6 Prefix Delegation (RFC3633) on l2tp. You will have to set your IPv6 pool and the length of the delegation prefix. From the defined IPv6 pool you will be handing out networks of the defined length (delegation-prefix). The length of the delegation prefix can be between 32 and 64 bits long.

```
set vpn l2tp remote-access default-ipv6-pool <IPv6-POOL-NAME>
```

Use this command to define default IPv6 address pool name.

```
set vpn l2tp remote-access ppp-options ipv6 allow
set vpn l2tp remote-access client-ipv6-pool IPv6-POOL delegate '2001:db8:8003::/48'
↪delegation-prefix '56'
set vpn l2tp remote-access client-ipv6-pool IPv6-POOL prefix '2001:db8:8002::/48' mask
↪'64'
set vpn l2tp remote-access default-ipv6-pool IPv6-POOL
```

IPv6 Advanced Options

set vpn l2tp remote-access ppp-options ipv6-accept-peer-interface-id

Accept peer interface identifier. By default this is not defined.

set vpn l2tp remote-access ppp-options ipv6-interface-id <random | x:x:x:x>

Specifies if a fixed or random interface identifier is used for IPv6. The default is fixed.

- **random** - Random interface identifier for IPv6
- **x:x:x:x** - Specify interface identifier for IPv6

set vpn l2tp remote-access ppp-options ipv6-interface-id <random | x:x:x:x>

Specifies the peer interface identifier for IPv6. The default is fixed.

- **random** - Random interface identifier for IPv6
- **x:x:x:x** - Specify interface identifier for IPv6
- **ipv4-addr** - Calculate interface identifier from IPv4 address.
- **calling-sid** - Calculate interface identifier from calling-station-id.

Scripting

set vpn l2tp remote-access extended-scripts on-change <path_to_script>

Script to run when the session interface is changed by RADIUS CoA handling

set vpn l2tp remote-access extended-scripts on-down <path_to_script>

Script to run when the session interface is about to terminate

set vpn l2tp remote-access extended-scripts on-pre-up <path_to_script>

Script to run before the session interface comes up

set vpn l2tp remote-access extended-scripts on-up <path_to_script>

Script to run when the session interface is completely configured and started

Advanced Options

Authentication Advanced Options

set vpn l2tp remote-access authentication local-users username <user> disable

Disable <user> account.

set vpn l2tp remote-access authentication local-users username <user> static-ip <address>

Assign a static IP address to <user> account.

set vpn l2tp remote-access authentication local-users username <user> rate-limit download <bandwidth>

Rate limit the download bandwidth for <user> to <bandwidth> kbit/s.

set vpn l2tp remote-access authentication local-users username <user> rate-limit upload <bandwidth>

Rate limit the upload bandwidth for <user> to <bandwidth> kbit/s

```
set vpn l2tp remote-access authentication protocols <pap | chap | mschap | mschap-v2>
```

Require the peer to authenticate itself using one of the following protocols: pap, chap, mschap, mschap-v2.

Client IP Pool Advanced Options

```
set vpn l2tp remote-access client-ip-pool <POOL-NAME> next-pool <NEXT-POOL-NAME>
```

Use this command to define the next address pool name.

PPP Advanced Options

```
set vpn l2tp remote-access ppp-options disable-ccp
```

Disable Compression Control Protocol (CCP). CCP is enabled by default.

```
set vpn l2tp remote-access ppp-options interface-cache <number>
```

Specifies number of interfaces to cache. This prevents interfaces from being removed once the corresponding session is destroyed. Instead, interfaces are cached for later use in new sessions. This should reduce the kernel-level interface creation/deletion rate. Default value is **0**.

```
set vpn l2tp remote-access ppp-options ipv4 <require | prefer | allow | deny>
```

Specifies IPv4 negotiation preference.

- **require** - Require IPv4 negotiation
- **prefer** - Ask client for IPv4 negotiation, do not fail if it rejects
- **allow** - Negotiate IPv4 only if client requests (Default value)
- **deny** - Do not negotiate IPv4

```
set vpn l2tp remote-access ppp-options lcp-echo-failure <number>
```

Defines the maximum <number> of unanswered echo requests. Upon reaching the value <number>, the session will be reset. Default value is **3**.

```
set vpn l2tp remote-access ppp-options lcp-echo-interval <interval>
```

If this option is specified and is greater than 0, then the PPP module will send LCP echo requests every <interval> seconds. Default value is **30**.

```
set vpn l2tp remote-access ppp-options lcp-echo-timeout
```

Specifies timeout in seconds to wait for any peer activity. If this option is specified it turns on adaptive lcp echo functionality and “lcp-echo-failure” is not used. Default value is **0**.

```
set vpn l2tp remote-access ppp-options min-mtu <number>
```

Defines the minimum acceptable MTU. If a client tries to negotiate an MTU lower than this it will be NAKed, and disconnected if it rejects a greater MTU. Default value is **100**.

```
set vpn l2tp remote-access ppp-options mppe <require | prefer | deny>
```

Specifies MPPE negotiation preference.

- **require** - ask client for mppe, if it rejects drop connection
- **prefer** - ask client for mppe, if it rejects don't fail. (Default value)

- **deny** - deny mppe

Default behavior - don't ask the client for mppe, but allow it if the client wants. Please note that RADIUS may override this option with the MS-MPPE-Encryption-Policy attribute.

set vpn l2tp remote-access ppp-options mru <number>

Defines preferred MRU. By default is not defined.

Global Advanced options

set vpn l2tp remote-access description <description>

Set description.

set vpn l2tp remote-access limits burst <value>

Burst count

set vpn l2tp remote-access limits connection-limit <value>

Maximum accepted connection rate (e.g. 1/min, 60/sec)

set vpn l2tp remote-access limits timeout <value>

Timeout in seconds

set vpn l2tp remote-access mtu

Maximum Transmission Unit (MTU) (default: **1436**)

set vpn l2tp remote-access max-concurrent-sessions

Maximum number of concurrent session start attempts

set vpn l2tp remote-access name-server <address>

Connected clients should use <address> as their DNS server. This command accepts both IPv4 and IPv6 addresses. Up to two nameservers can be configured for IPv4, up to three for IPv6.

set vpn l2tp remote-access shaper fwmark <1-2147483647>

Match firewall mark value

set vpn l2tp remote-access snmp master-agent

Enable SNMP

set vpn l2tp remote-access wins-server <address>

Windows Internet Name Service (WINS) servers propagated to client

Monitoring

```
vyos@vyos:~$ show l2tp-server sessions
ifname | username |      ip      | ip6 | ip6-dp | calling-sid | rate-limit | state | 
--uptime | rx-bytes | tx-bytes
-----+-----+-----+-----+-----+-----+-----+-----+
l2tp0 | test      | 192.168.255.3 |     |         | 192.168.0.36 |           | active | 
02:01:47 | 7.7 KiB  | 1.2 KiB
```

```
vyos@vyos:~$ show l2tp-server statistics
uptime: 0.02:49:49
cpu: 0%
mem(rss/virt): 5920/100892 kB
core:
  mempool_allocated: 133202
  mempool_available: 131770
  thread_count: 1
  thread_active: 1
  context_count: 5
  context_sleeping: 0
  context_pending: 0
  md_handler_count: 3
  md_handler_pending: 0
  timer_count: 0
  timer_pending: 0
sessions:
  starting: 0
  active: 0
  finishing: 0
l2tp:
  tunnels:
    starting: 0
    active: 0
    finishing: 0
  sessions (control channels):
    starting: 0
    active: 0
    finishing: 0
  sessions (data channels):
    starting: 0
    active: 0
    finishing: 0
```

7.13.3 OpenConnect

Todo: Convert raw command blocks in this file to cfgcmd/opcmd directives for command coverage tracking.

OpenConnect-compatible server feature has been available since Equuleus (1.3). Openconnect VPN supports SSL connection and offers full network access. SSL VPN network extension connects the end-user system to the corporate network with access controls based only on network layer information, such as destination IP address and port number. So, it provides safe communication for all types of device traffic across public networks and private networks, also encrypts the traffic with SSL protocol.

The remote user will use the openconnect client to connect to the router and will receive an IP address from a VPN pool, allowing full access to the network.

Configuration

SSL Certificates

We need to generate the certificate which authenticates users who attempt to access the network resource through the SSL VPN tunnels. The following commands will create a self signed certificates and will be stored in configuration:

```
run generate pki ca install <CA name>
run generate pki certificate sign <CA name> install <Server name>
```

We can also create the certificates using Certbot which is an easy-to-use client that fetches a certificate from Let's Encrypt an open certificate authority launched by the EFF, Mozilla, and others and deploys it to a web server.

```
sudo certbot certonly --standalone --preferred-challenges http -d <domain name>
```

Server Configuration

```
set vpn openconnect authentication local-users username <user> password <pass>
set vpn openconnect authentication mode <local password|radius|certificate>
set vpn openconnect network-settings client-ip-settings subnet <subnet>
set vpn openconnect network-settings name-server <address>
set vpn openconnect network-settings name-server <address>
set vpn openconnect ssl ca-certificate <pki-ca-name>
set vpn openconnect ssl certificate <pki-cert-name>
set vpn openconnect ssl passphrase <pki-password>
```

2FA OTP support

Instead of password only authentication, 2FA password authentication + OTP key can be used. Alternatively, OTP authentication only, without a password, can be used. To do this, an OTP configuration must be added to the configuration above:

```
set vpn openconnect authentication mode local <password-otp|otp>
set vpn openconnect authentication local-users username <user> otp <key>
set vpn openconnect authentication local-users username <user> interval <interval>
↳(optional)>
set vpn openconnect authentication local-users username <user> otp-length <otp-length>
↳(optional)>
set vpn openconnect authentication local-users username <user> token-type <token-type>
↳(optional)>
```

For generating an OTP key in VyOS, you can use the CLI command (operational mode):

```
generate openconnect username <user> otp-key hotp-time
```

User Certificate Authentication

You can configure users to be authenticated by certificate by setting the authentication mode to `certificate`, and defining what field (by OID) in the certificate will be used to identify the username. Two pre-defined

shortcuts for Common Name (OID 2.5.4.3) and User ID (OID 0.9.2342.19200300.100.1.1) have been provided as `cn` or `uid`.

Otherwise a specific OID value must be provided.

The user's certificate must be signed by the certificate authority defined in the configuration for it to be validated for authentication.

```
set vpn openconnect authentication mode certificate
set vpn openconnect authentication mode certificate user-identifier-field cn
set vpn openconnect ssl ca-certificate <cert>
```

Verification

```
vyos@vyos:~$ sh openconnect-server sessions
interface      username      ip            remote IP     RX            TX            state         ↵
↵uptime
-----
↵---
sslvpn0        tst          172.20.20.198 192.168.6.1  0 bytes      152 bytes    connected    3s
```

Note: It is compatible with Cisco (R) AnyConnect (R) clients.

Example

SSL Certificates generation

Follow the instructions to generate CA cert (in configuration mode):

```
vyos@vyos# run generate pki ca install ca-ocserv
Enter private key type: [rsa, dsa, ec] (Default: rsa)
Enter private key bits: (Default: 2048)
Enter country code: (Default: GB) US
Enter state: (Default: Some-State) Delaware
Enter locality: (Default: Some-City) Mycity
Enter organization name: (Default: VyOS) MyORG
Enter common name: (Default: vyos.io) oc-ca
Enter how many days certificate will be valid: (Default: 1825) 3650
Note: If you plan to use the generated key on this router, do not encrypt the private.
key.
Do you want to encrypt the private key with a passphrase? [y/N] N
2 value(s) installed. Use "compare" to see the pending changes, and "commit" to apply.
[edit]
```

Follow the instructions to generate server cert (in configuration mode):

```
vyos@vyos# run generate pki certificate sign ca-ocserv install srv-ocserv
Do you already have a certificate request? [y/N] N
Enter private key type: [rsa, dsa, ec] (Default: rsa)
Enter private key bits: (Default: 2048)
Enter country code: (Default: GB) US
Enter state: (Default: Some-State) Delaware
Enter locality: (Default: Some-City) Mycity
Enter organization name: (Default: VyOS) MyORG
Enter common name: (Default: vyos.io) oc-srv
Do you want to configure Subject Alternative Names? [y/N] N
Enter how many days certificate will be valid: (Default: 365) 1830
Enter certificate type: (client, server) (Default: server)
Note: If you plan to use the generated key on this router, do not encrypt the private_
key.
Do you want to encrypt the private key with a passphrase? [y/N] N
2 value(s) installed. Use "compare" to see the pending changes, and "commit" to apply.
[edit]
```

Each of the install commands should be applied to the configuration and committed before using under the openconnect configuration:

```
vyos@vyos# commit
[edit]
vyos@vyos# save
Saving configuration to '/config/config.boot'...
Done
[edit]
```

Openconnect Configuration

Simple setup with one user added and password authentication:

```
set vpn openconnect authentication local-users username tst password 'OC_bad_Secret'
set vpn openconnect authentication mode local password
set vpn openconnect network-settings client-ip-settings subnet '172.20.20.0/24'
set vpn openconnect network-settings name-server '10.1.1.1'
set vpn openconnect network-settings name-server '10.1.1.2'
set vpn openconnect ssl ca-certificate 'ca-ocserv'
set vpn openconnect ssl certificate 'srv-ocserv'
```

To enable the HTTP security headers in the configuration file, use the command:

```
set vpn openconnect http-security-headers
```

Adding a 2FA with an OTP-key

First the OTP keys must be generated and sent to the user and to the configuration:

```
vyos@vyos:~$ generate openconnect username tst otp-key hotp-time
# You can share it with the user, he just needs to scan the QR in his OTP app
# username:  tst
# OTP KEY:  5PA4SGYTQSGOB03H3EQSSNCUNZAYAPH2
# OTP URL:  otpauth://totp/tst@vyos?secret=5PA4SGYTQSGOB03H3EQSSNCUNZAYAPH2&digits=6&
↳ period=30
```

```
# To add this OTP key to configuration, run the following commands:
set vpn openconnect authentication local-users username tst otp key
↳ 'ebc1c91b13848ce0bb67d9212934546e41803cfa'
```

Next it is necessary to configure 2FA for OpenConnect:

```
set vpn openconnect authentication mode local password-otp
set vpn openconnect authentication local-users username tst otp key
↳ 'ebc1c91b13848ce0bb67d9212934546e41803cfa'
```

Now when connecting the user will first be asked for the password and then the OTP key.

Warning: When using Time-based one-time password (TOTP) (OTP HOTP-time), be sure that the time on the server and the OTP token generator are synchronized by NTP

To display the configured OTP user settings, use the command:

```
show openconnect-server user <username> otp <full|key-b32|key-hex|qrcode|uri>
```

Identity Based Configuration

OpenConnect supports a subset of its configuration options to be applied on a per user/group basis, for configuration purposes we refer to this functionality as “Identity based config”. The following [OpenConnect Server Manual](#) outlines the set of configuration options that are allowed. This can be leveraged to apply different sets of configs to different users or groups of users.

```
sudo mkdir -p /config/auth/ocserv/config-per-user
sudo touch /config/auth/ocserv/default-user.conf

set vpn openconnect authentication identity-based-config mode user
set vpn openconnect authentication identity-based-config directory /config/auth/ocserv/
↪config-per-user
set vpn openconnect authentication identity-based-config default-config /config/auth/
↪ocserv/default-user.conf
```

Warning: The above directory and default-config must be a child directory of /config/auth, since files outside this directory are not persisted after an image upgrade.

Once you commit the above changes you can create a config file in the /config/auth/ocserv/config-per-user directory that matches a username of a user you have created e.g. “tst”. Now when logging in with the “tst” user the config options you set in this file will be loaded.

Be sure to set a sane default config in the default config file, this will be loaded in the case that a user is authenticated and no file is found in the configured directory matching the users username/group.

```
sudo nano /config/auth/ocserv/config-per-user/tst
```

The same configuration options apply when Identity based config is configured in group mode except that group mode can only be used with RADIUS authentication.

Warning: OpenConnect server matches the filename in a case sensitive manner, make sure the username/group name you configure matches the filename exactly.

Configuring RADIUS accounting

OpenConnect can be configured to send accounting information to a RADIUS server to capture user session data such as time of connect/disconnect, data transferred, and so on.

Configure an accounting server and enable accounting with:

```
set vpn openconnect accounting mode radius
set vpn openconnect accounting radius server 172.20.20.10
set vpn openconnect accounting radius server 172.20.20.10 port 1813
set vpn openconnect accounting radius server 172.20.20.10 key your_radius_secret
```

Warning: The RADIUS accounting feature must be used with the OpenConnect authentication mode RADIUS. It cannot be used with local authentication. You must configure the OpenConnect authentication mode to “radius”.

An example of the data captured by a FREERADIUS server with sql accounting:

```
mysql> SELECT username, nasipaddress, acctstarttime, acctstoptime, acctinputoctets,
↪acctoutputoctets, callingstationid, framedipaddress, connectinfo_start FROM radacct;
+-----+-----+-----+-----+-----+-----+-----+
↪+-----+
↪-----+
| username | nasipaddress | acctstarttime | acctstoptime | acctinputoctets|
↪| acctoutputoctets | callingstationid | framedipaddress | connectinfo_start |
↪|
+-----+-----+-----+-----+-----+-----+-----+
↪+-----+
↪-----+
| test | 198.51.100.15 | 2023-01-13 00:59:15 | 2023-01-13 00:59:21 | 10606|
↪| 152 | 192.168.6.1 | 172.20.20.198 | Open AnyConnect VPN Agent|
↪v8.05-1 |
+-----+-----+-----+-----+-----+-----+-----+
↪+-----+
↪-----+
```

7.13.4 PPTP-Server

The Point-to-Point Tunneling Protocol (PPTP) has been implemented in VyOS only for backwards compatibility. PPTP has many well known security issues and you should use one of the many other new VPN implementations.

Configuring PPTP Server

```
set vpn pptp remote-access authentication mode local
set vpn pptp remote-access authentication local-users username test password 'test'
set vpn pptp remote-access client-ip-pool PPTP-POOL range 192.168.255.2-192.168.255.254
set vpn pptp remote-access default-pool 'PPTP-POOL'
set vpn pptp remote-access outside-address 192.0.2.2
set vpn pptp remote-access gateway-address 192.168.255.1
```

set vpn pptp remote-access authentication mode <local | radius>

Set authentication backend. The configured authentication backend is used for all queries.

- **radius:** All authentication queries are handled by a configured RADIUS server.
- **local:** All authentication queries are handled locally.
- **noauth:** Authentication disabled.

set vpn pptp remote-access authentication local-users username <user> password <pass>

Create <user> for local authentication on this system. The users password will be set to <pass>.

set vpn pptp remote-access client-ip-pool <POOL-NAME> range <x.x.x.x-x.x.x.x | x.x.x.x/x>

Use this command to define the first IP address of a pool of addresses to be given to PPTP clients. If notation x.x.x.x-x.x.x.x, it must be within a /24 subnet. If notation x.x.x.x/x is used there is possibility to set host/netmask.

set vpn pptp remote-access default-pool <POOL-NAME>

Use this command to define default address pool name.


```
set vpn pptp remote-access gateway-address <gateway>
```

Specifies single <gateway> IP address to be used as local address of PPP interfaces.

Configuring RADIUS authentication

To enable RADIUS based authentication, the authentication mode needs to be changed within the configuration. Previous settings like the local users, still exists within the configuration, however they are not used if the mode has been changed from local to radius. Once changed back to local, it will use all local accounts again.

```
set vpn pptp remote-access authentication mode radius
```

```
set vpn pptp remote-access authentication radius server <server> key <secret>
```

Configure RADIUS <server> and its required shared <secret> for communicating with the RADIUS server.

Since the RADIUS server would be a single point of failure, multiple RADIUS servers can be setup and will be used subsequently. For example:

```
set vpn pptp remote-access authentication radius server 10.0.0.1 key 'foo'
set vpn pptp remote-access authentication radius server 10.0.0.2 key 'foo'
```

Note: Some RADIUS servers use an access control list which allows or denies queries, make sure to add your VyOS router to the allowed client list.

RADIUS source address

If you are using OSPF as IGP, always the closest interface connected to the RADIUS server is used. You can bind all outgoing RADIUS requests to a single source IP e.g. the loopback interface.

```
set vpn pptp remote-access authentication radius source-address <address>
```

Source IPv4 address used in all RADIUS server queries.

Note: Some RADIUS servers use an access control list which allows or denies queries, make sure to add your VyOS router to the allowed client list.

RADIUS advanced options

```
set vpn pptp remote-access authentication radius server <server> port <port>
```

Configure RADIUS <server> and its required port for authentication requests.

```
set vpn pptp remote-access authentication radius server <server> fail-time <time>
```

Mark RADIUS server as offline for this given <time> in seconds.

```
set vpn pptp remote-access authentication radius server <server> disable
```

Temporary disable this RADIUS server.

```
set vpn pptp remote-access authentication radius acct-timeout <timeout>
```

Timeout to wait reply for Interim-Update packets. (default 3 seconds)

set vpn pptp remote-access authentication radius dynamic-author server <address>

Specifies IP address for Dynamic Authorization Extension server (DM/CoA). This IP must exist on any VyOS interface or it can be 0.0.0.0.

set vpn pptp remote-access authentication radius dynamic-author port <port>

UDP port for Dynamic Authorization Extension server (DM/CoA)

set vpn pptp remote-access authentication radius dynamic-author key <secret>

Secret for Dynamic Authorization Extension server (DM/CoA)

set vpn pptp remote-access authentication radius max-try <number>

Maximum number of tries to send Access-Request/Accounting-Request queries

set vpn pptp remote-access authentication radius timeout <timeout>

Timeout to wait response from server (seconds)

set vpn pptp remote-access authentication radius nas-identifier <identifier>

Value to send to RADIUS server in NAS-Identifier attribute and to be matched in DM/CoA requests.

set vpn pptp remote-access authentication radius nas-ip-address <address>

Value to send to RADIUS server in NAS-IP-Address attribute and to be matched in DM/CoA requests. Also DM/CoA server will bind to that address.

set vpn pptp remote-access authentication radius source-address <address>

Source IPv4 address used in all RADIUS server queries.

set vpn pptp remote-access authentication radius rate-limit attribute <attribute>

Specifies which RADIUS server attribute contains the rate limit information. The default attribute is Filter-Id.

Note: If you set a custom RADIUS attribute you must define it on both dictionaries at RADIUS server and client.

set vpn pptp remote-access authentication radius rate-limit enable

Enables bandwidth shaping via RADIUS.

set vpn pptp remote-access authentication radius rate-limit vendor

Specifies the vendor dictionary, dictionary needs to be in /usr/share/accel-ppp/radius.

Received RADIUS attributes have a higher priority than parameters defined within the CLI configuration, refer to the explanation below.

Allocation clients ip addresses by RADIUS

If the RADIUS server sends the attribute Framed-IP-Address then this IP address will be allocated to the client and the option default-pool within the CLI config is being ignored.

If the RADIUS server sends the attribute Framed-Pool, IP address will be allocated from a predefined IP pool whose name equals the attribute value.

If the RADIUS server sends the attribute Stateful-IPv6-Address-Pool, IPv6 address will be allocated from a predefined IPv6 pool prefix whose name equals the attribute value.

If the RADIUS server sends the attribute `Delegated-IPv6-Prefix-Pool`, IPv6 delegation prefix will be allocated from a predefined IPv6 pool `delegate` whose name equals the attribute value.

Note: `Stateful-IPv6-Address-Pool` and `Delegated-IPv6-Prefix-Pool` are defined in RFC6911. If they are not defined in your RADIUS server, add new [dictionary](#).

User interface can be put to VRF context via RADIUS Access-Accept packet, or change it via RADIUS CoA. `Accel-VRF-Name` is used from these purposes. It is custom [ACCEL-PPP attribute](#). Define it in your RADIUS server.

Renaming clients interfaces by RADIUS

If the RADIUS server uses the attribute `NAS-Port-Id`, ppp tunnels will be renamed.

Note: The value of the attribute `NAS-Port-Id` must be less than 16 characters, otherwise the interface won't be renamed.

IPv6

```
set vpn pptp remote-access ppp-options ipv6 <require | prefer | allow | deny>
```

Specifies IPv6 negotiation preference.

- **require** - Require IPv6 negotiation
- **prefer** - Ask client for IPv6 negotiation, do not fail if it rejects
- **allow** - Negotiate IPv6 only if client requests
- **deny** - Do not negotiate IPv6 (default value)

```
set vpn pptp remote-access client-ipv6-pool <IPv6-POOL-NAME> prefix <address> mask <number-of-bits>
```

Use this command to set the IPv6 address pool from which a PPTP client will get an IPv6 prefix of your defined length (mask) to terminate the PPTP endpoint at their side. The mask length can be set from 48 to 128 bit long, the default value is 64.

```
set vpn pptp remote-access client-ipv6-pool <IPv6-POOL-NAME> delegate <address> delegation-prefix <number-of-bits>
```

Use this command to configure DHCPv6 Prefix Delegation (RFC3633) on PPTP. You will have to set your IPv6 pool and the length of the delegation prefix. From the defined IPv6 pool you will be handing out networks of the defined length (delegation-prefix). The length of the delegation prefix can be set from 32 to 64 bit long.

```
set vpn pptp remote-access default-ipv6-pool <IPv6-POOL-NAME>
```

Use this command to define default IPv6 address pool name.

```
set vpn pptp remote-access ppp-options ipv6 allow
set vpn pptp remote-access client-ipv6-pool IPv6-POOL delegate '2001:db8:8003::/48'
↪ delegation-prefix '56'
set vpn pptp remote-access client-ipv6-pool IPv6-POOL prefix '2001:db8:8002::/48' mask
↪ '64'
set vpn pptp remote-access default-ipv6-pool IPv6-POOL
```

IPv6 Advanced Options

set vpn pptp remote-access ppp-options ipv6-accept-peer-interface-id

Accept peer interface identifier. By default is not defined.

set vpn pptp remote-access ppp-options ipv6-interface-id <random | x:x:x:x>

Specifies fixed or random interface identifier for IPv6. By default is fixed.

- **random** - Random interface identifier for IPv6
- **x:x:x:x** - Specify interface identifier for IPv6

set vpn pptp remote-access ppp-options ipv6-interface-id <random | x:x:x:x>

Specifies peer interface identifier for IPv6. By default is fixed.

- **random** - Random interface identifier for IPv6
- **x:x:x:x** - Specify interface identifier for IPv6
- **ipv4-addr** - Calculate interface identifier from IPv4 address.
- **calling-sid** - Calculate interface identifier from calling-station-id.

Scripting

set vpn pptp remote-access extended-scripts on-change <path_to_script>

Script to run when session interface changed by RADIUS CoA handling

set vpn pptp remote-access extended-scripts on-down <path_to_script>

Script to run when session interface going to terminate

set vpn pptp remote-access extended-scripts on-pre-up <path_to_script>

Script to run before session interface comes up

set vpn pptp remote-access extended-scripts on-up <path_to_script>

Script to run when session interface is completely configured and started

Advanced Options

Authentication Advanced Options

set vpn pptp remote-access authentication local-users username <user> disable

Disable <user> account.

set vpn pptp remote-access authentication local-users username <user> static-ip <address>

Assign static IP address to <user> account.

set vpn pptp remote-access authentication local-users username <user> rate-limit download <bandwidth>

Download bandwidth limit in kbit/s for <user>.

set vpn pptp remote-access authentication local-users username <user> rate-limit upload <bandwidth>

Upload bandwidth limit in kbit/s for <user>.

```
set vpn pptp remote-access authentication protocols <pap | chap | mschap | mschap-v2>
```

Require the peer to authenticate itself using one of the following protocols: pap, chap, mschap, mschap-v2.

Client IP Pool Advanced Options

```
set vpn pptp remote-access client-ip-pool <POOL-NAME> next-pool <NEXT-POOL-NAME>
```

Use this command to define the next address pool name.

PPP Advanced Options

```
set vpn pptp remote-access ppp-options disable-ccp
```

Disable Compression Control Protocol (CCP). CCP is enabled by default.

```
set vpn pptp remote-access ppp-options interface-cache <number>
```

Specifies number of interfaces to keep in cache. It means that don't destroy interface after corresponding session is destroyed, instead place it to cache and use it later for new sessions repeatedly. This should reduce kernel-level interface creation/deletion rate lack. Default value is 0.

```
set vpn pptp remote-access ppp-options ipv4 <require | prefer | allow | deny>
```

Specifies IPv4 negotiation preference.

- **require** - Require IPv4 negotiation
- **prefer** - Ask client for IPv4 negotiation, do not fail if it rejects
- **allow** - Negotiate IPv4 only if client requests (Default value)
- **deny** - Do not negotiate IPv4

```
set vpn pptp remote-access ppp-options lcp-echo-failure <number>
```

Defines the maximum <number> of unanswered echo requests. Upon reaching the value <number>, the session will be reset. Default value is 3.

```
set vpn pptp remote-access ppp-options lcp-echo-interval <interval>
```

If this option is specified and is greater than 0, then the PPP module will send LCP pings of the echo request every <interval> seconds. Default value is 30.

```
set vpn pptp remote-access ppp-options lcp-echo-timeout
```

Specifies timeout in seconds to wait for any peer activity. If this option specified it turns on adaptive lcp echo functionality and "lcp-echo-failure" is not used. Default value is 0.

```
set vpn pptp remote-access ppp-options min-mtu <number>
```

Defines minimum acceptable MTU. If client will try to negotiate less then specified MTU then it will be NAKed or disconnected if rejects greater MTU. Default value is 100.

```
set vpn pptp remote-access ppp-options mppe <require | prefer | deny>
```

Specifies MPPE negotiation preference.

- **require** - ask client for mppe, if it rejects drop connection
- **prefer** - ask client for mppe, if it rejects don't fail. (Default value)

- **deny** - deny mppe

Default behavior - don't ask client for mppe, but allow it if client wants. Please note that RADIUS may override this option by MS-MPPE-Encryption-Policy attribute.

set vpn pptp remote-access ppp-options mru <number>

Defines preferred MRU. By default is not defined.

Global Advanced options

set vpn pptp remote-access description <description>

Set description.

set vpn pptp remote-access limits burst <value>

Burst count

set vpn pptp remote-access limits connection-limit <value>

Acceptable rate of connections (e.g. 1/min, 60/sec)

set vpn pptp remote-access limits timeout <value>

Timeout in seconds

set vpn pptp remote-access mtu

Maximum Transmission Unit (MTU) (default: **1436**)

set vpn pptp remote-access max-concurrent-sessions

Maximum number of concurrent session start attempts

set vpn pptp remote-access name-server <address>

Connected client should use <address> as their DNS server. This command accepts both IPv4 and IPv6 addresses. Up to two nameservers can be configured for IPv4, up to three for IPv6.

set vpn pptp remote-access shaper fwmark <1-2147483647>

Match firewall mark value

set vpn pptp remote-access snmp master-agent

Enable SNMP

set vpn pptp remote-access wins-server <address>

Windows Internet Name Service (WINS) servers propagated to client

Monitoring

show pptp-server sessions

Use this command to locally check the active sessions in the PPTP server.

```
vyos@vyos:~$ show pptp-server sessions
ifname | username |   ip   | ip6 | ip6-dp |   calling-sid   | rate-limit | state | 
↪uptime | rx-bytes | tx-bytes
-----+-----+-----+-----+-----+-----+-----+-----+-----
↪-----+-----+-----
```

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```
pptp0 | test | 10.0.0.2 | | | 192.168.10.100 | | active |
↪ 00:01:26 | 6.9 KiB | 220 B
```

```
vyos@vyos:~$ show pptp-server statistics
```

```
uptime: 0.00:04:52
cpu: 0%
mem(rss/virt): 5504/100176 kB
core:
  mempool_allocated: 152007
  mempool_available: 149007
  thread_count: 1
  thread_active: 1
  context_count: 6
  context_sleeping: 0
  context_pending: 0
  md_handler_count: 6
  md_handler_pending: 0
  timer_count: 2
  timer_pending: 0
sessions:
  starting: 0
  active: 1
  finishing: 0
pptp:
  starting: 0
  active: 1
```

Troubleshooting

```
vyos@vyos:~$ sudo journalctl -u accel-ppp@pptp -b 0
```

```
Feb 29 14:58:57 vyos accel-pptp[4629]: pptp: new connection from 192.168.10.100
Feb 29 14:58:57 vyos accel-pptp[4629]: :: recv [PPTP Start-Ctrl-Conn-Request <Version 1>
↪ <Framing 1> <Bearer 1> <Max-Chan 0>]
Feb 29 14:58:57 vyos accel-pptp[4629]: :: send [PPTP Start-Ctrl-Conn-Reply <Version 1>
↪ <Result 1> <Error 0> <Framing 3> <Bearer 3> <Max-Chan 1>]
Feb 29 14:58:57 vyos accel-pptp[4629]: :: recv [PPTP Outgoing-Call-Request <Call-ID 2961>
↪ <Call-Serial 2> <Min-BPS 300> <Max-BPS 1000000000> <Bearer 3> <Framing 3> <Window-Size 64>
↪ <Delay 0>]
Feb 29 14:58:57 vyos accel-pptp[4629]: :: send [PPTP Outgoing-Call-Reply <Call-ID 2>
↪ <Peer-Call-ID 2961> <Result 1> <Error 0> <Cause 0> <Speed 1000000000> <Window-Size 64>
↪ <Delay 0> <Channel 0>]
Feb 29 14:58:57 vyos accel-pptp[4629]: :: lcp_layer_init
Feb 29 14:58:57 vyos accel-pptp[4629]: :: auth_layer_init
Feb 29 14:58:57 vyos accel-pptp[4629]: :: ccp_layer_init
Feb 29 14:58:57 vyos accel-pptp[4629]: :: ipcp_layer_init
Feb 29 14:58:57 vyos accel-pptp[4629]: :: ipv6cp_layer_init
Feb 29 14:58:57 vyos accel-pptp[4629]: :: ppp establishing
Feb 29 14:58:57 vyos accel-pptp[4629]: :: lcp_layer_start
Feb 29 14:58:57 vyos accel-pptp[4629]: :: send [LCP ConfReq id=75 <auth PAP> <mru 1436>
```

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```

↪<magic 483920bd>]
Feb 29 14:58:57 vyos accel-pptp[4629]: :: recv [PPTP Set-Link-Info]
Feb 29 14:58:57 vyos accel-pptp[4629]: :: recv [LCP ConfReq id=0 <mru 1400> <magic
↪0142785a> <pcomp> <accomp> < d 3 6 >]
Feb 29 14:58:57 vyos accel-pptp[4629]: :: send [LCP ConfRej id=0 <pcomp> <accomp> < d 3
↪6 >]
Feb 29 14:58:57 vyos accel-pptp[4629]: :: recv [LCP ConfReq id=1 <mru 1400> <magic
↪0142785a>]
Feb 29 14:58:57 vyos accel-pptp[4629]: :: send [LCP ConfAck id=1]
Feb 29 14:59:00 vyos accel-pptp[4629]: :: fsm timeout 9
Feb 29 14:59:00 vyos accel-pptp[4629]: :: send [LCP ConfReq id=75 <auth PAP> <mru 1436>
↪<magic 483920bd>]
Feb 29 14:59:00 vyos accel-pptp[4629]: :: recv [LCP ConfNak id=75 <auth MSCHAP-v2>]
Feb 29 14:59:00 vyos accel-pptp[4629]: :: send [LCP ConfReq id=76 <auth CHAP-md5> <mru
↪1436> <magic 483920bd>]
Feb 29 14:59:00 vyos accel-pptp[4629]: :: recv [LCP ConfNak id=76 <auth MSCHAP-v2>]
Feb 29 14:59:00 vyos accel-pptp[4629]: :: send [LCP ConfReq id=77 <auth MSCHAP-v1> <mru
↪1436> <magic 483920bd>]
Feb 29 14:59:00 vyos accel-pptp[4629]: :: recv [LCP ConfNak id=77 <auth MSCHAP-v2>]
Feb 29 14:59:00 vyos accel-pptp[4629]: :: send [LCP ConfReq id=78 <auth MSCHAP-v2> <mru
↪1436> <magic 483920bd>]
Feb 29 14:59:00 vyos accel-pptp[4629]: :: recv [LCP ConfAck id=78 <auth MSCHAP-v2> <mru
↪1436> <magic 483920bd>]
Feb 29 14:59:00 vyos accel-pptp[4629]: :: lcp_layer_started
Feb 29 14:59:00 vyos accel-pptp[4629]: :: auth_layer_start
Feb 29 14:59:00 vyos accel-pptp[4629]: :: send [MSCHAP-v2 Challenge id=1
↪<8aa758781676e6a8e85c11963ee010>]
Feb 29 14:59:00 vyos accel-pptp[4629]: :: recv [LCP Ident id=2 <MSRASV5.20>]
Feb 29 14:59:00 vyos accel-pptp[4629]: :: recv [LCP Ident id=3 <MSRAS-0-MSEdgeWIN10>]
Feb 29 14:59:00 vyos accel-pptp[4629]: [43B blob data]
Feb 29 14:59:00 vyos accel-pptp[4629]: :: recv [PPTP Set-Link-Info]
Feb 29 14:59:00 vyos accel-pptp[4629]: :: recv [MSCHAP-v2 Response id=1
↪<90c21af1091f745e8bf22388b058>, <e695ae5aae274c88a3fa1ee3dc9057aece4d53c87b9fea>, F=0,
↪name="test"]
Feb 29 14:59:00 vyos accel-pptp[4629]: ppp0:test: connect: ppp0 <--> pptp(192.168.10.100)
Feb 29 14:59:00 vyos accel-pptp[4629]: ppp0:test: ppp connected
Feb 29 14:59:00 vyos accel-pptp[4629]: ppp0:test: send [MSCHAP-v2 Success id=1
↪"S=347F417CF04BEBBC7F75CFA7F43474C36FB218F9 M=Authentication succeeded"]
Feb 29 14:59:00 vyos accel-pptp[4629]: ppp0:test: test: authentication succeeded
Feb 29 14:59:00 vyos accel-pptp[4629]: ppp0:test: auth_layer_started
Feb 29 14:59:00 vyos accel-pptp[4629]: ppp0:test: ccp_layer_start
Feb 29 14:59:00 vyos accel-pptp[4629]: ppp0:test: send [CCP ConfReq id=b9 <mppe +H -M +S
↪-L -D -C>]
Feb 29 14:59:00 vyos accel-pptp[4629]: ppp0:test: ipcp_layer_start
Feb 29 14:59:00 vyos accel-pptp[4629]: ppp0:test: ipv6cp_layer_start
Feb 29 14:59:00 vyos accel-pptp[4629]: ppp0:test: IPV6CP: discarding packet
Feb 29 14:59:00 vyos accel-pptp[4629]: ppp0:test: send [LCP ProtoRej id=122 <8057>]
Feb 29 14:59:00 vyos accel-pptp[4629]: ppp0:test: recv [IPCP ConfReq id=6 <addr 0.0.0.0>
↪<dns1 0.0.0.0> <wins1 0.0.0.0> <dns2 0.0.0.0> <wins2 0.0.0.0>]
Feb 29 14:59:00 vyos accel-pptp[4629]: ppp0:test: send [IPCP ConfReq id=3b <addr 10.0.0.
↪1>]
Feb 29 14:59:00 vyos accel-pptp[4629]: ppp0:test: send [IPCP ConfRej id=6 <dns1 0.0.0.0>

```

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```

↪<wins1 0.0.0.0> <dns2 0.0.0.0> <wins2 0.0.0.0>]
Feb 29 14:59:00 vyos accel-pptp[4629]: ppp0:test: recv [LCP ProtoRej id=7 <80fd>]
Feb 29 14:59:00 vyos accel-pptp[4629]: ppp0:test: ccp_layer_finished
Feb 29 14:59:00 vyos accel-pptp[4629]: ppp0:test: recv [IPCP ConfAck id=3b <addr 10.0.0.
↪1>]
Feb 29 14:59:00 vyos accel-pptp[4629]: ppp0:test: recv [IPCP ConfReq id=8 <addr 0.0.0.0>]
Feb 29 14:59:00 vyos accel-pptp[4629]: ppp0:test: send [IPCP ConfNak id=8 <addr 10.0.0.2>
↪]
Feb 29 14:59:00 vyos accel-pptp[4629]: ppp0:test: recv [IPCP ConfReq id=9 <addr 10.0.0.2>
↪]
Feb 29 14:59:00 vyos accel-pptp[4629]: ppp0:test: send [IPCP ConfAck id=9]
Feb 29 14:59:00 vyos accel-pptp[4629]: ppp0:test: ipcp_layer_started
Feb 29 14:59:00 vyos accel-pptp[4629]: ppp0:test: rename interface to 'pptp0'
Feb 29 14:59:00 vyos accel-pptp[4629]: pptp0:test: pptp: ppp started

```

7.13.5 RSA-Keys

Todo: Convert raw command blocks in this file to cfgcmd/opcmd directives for command coverage tracking.

RSA can be used for services such as key exchanges and for encryption purposes. To make IPsec work with dynamic address on one/both sides, we will have to use RSA keys for authentication. They are very fast and easy to setup.

First, on both routers run the operational command “generate pki key-pair install <key-pair name>”. You may choose different length than 2048 of course.

```

vyos@left# run generate pki key-pair install ipsec-LEFT
Enter private key type: [rsa, dsa, ec] (Default: rsa)
Enter private key bits: (Default: 2048)
Note: If you plan to use the generated key on this router, do not encrypt the private_
↪key.
Do you want to encrypt the private key with a passphrase? [y/N] N
Configure mode commands to install key pair:
Do you want to install the public key? [Y/n] Y
set pki key-pair ipsec-LEFT public key 'MIIBIjANBgkqh...'
Do you want to install the private key? [Y/n] Y
set pki key-pair ipsec-LEFT private key 'MIIEvgIBADAN...'
[edit]

```

Configuration commands will display. Note the command with the public key (set pki key-pair ipsec-LEFT public key 'MIIBIjANBgkqh...'). Then do the same on the opposite router:

```
vyos@left# run generate pki key-pair install ipsec-RIGHT
```

Note the command with the public key (set pki key-pair ipsec-RIGHT public key 'FAAOCAQ8AMII...').

The noted public keys should be entered on the opposite routers.

On the LEFT:

```
set pki key-pair ipsec-RIGHT public key 'FAAOCAQ8AMII...'
```

On the RIGHT:

```
set pki key-pair ipsec-LEFT public key 'MIIBIjANBgkqh...'

```

Now you are ready to setup IPsec. The key points:

1. Since both routers do not know their effective public addresses, we set the local-address of the peer to “any”.
2. On the initiator, we set the peer address to its public address, but on the responder we only set the id.
3. On the initiator, we need to set the remote-id option so that it can identify IKE traffic from the responder correctly.
4. On the responder, we need to set the local id so that initiator can know who’s talking to it for the point #3 to work.

On the LEFT (static address):

```
set vpn ipsec interface eth0

set vpn ipsec esp-group MyESPGroup proposal 1 encryption aes128
set vpn ipsec esp-group MyESPGroup proposal 1 hash sha1

set vpn ipsec ike-group MyIKEGroup proposal 1 dh-group 2
set vpn ipsec ike-group MyIKEGroup proposal 1 encryption aes128
set vpn ipsec ike-group MyIKEGroup proposal 1 hash sha1

set vpn ipsec site-to-site peer @RIGHT authentication id LEFT
set vpn ipsec site-to-site peer @RIGHT authentication mode rsa
set vpn ipsec site-to-site peer @RIGHT authentication rsa local-key ipsec-LEFT
set vpn ipsec site-to-site peer @RIGHT authentication rsa remote-key ipsec-RIGHT
set vpn ipsec site-to-site peer @RIGHT authentication remote-id RIGHT
set vpn ipsec site-to-site peer @RIGHT default-esp-group MyESPGroup
set vpn ipsec site-to-site peer @RIGHT ike-group MyIKEGroup
set vpn ipsec site-to-site peer @RIGHT local-address 192.0.2.10
set vpn ipsec site-to-site peer @RIGHT connection-type none
set vpn ipsec site-to-site peer @RIGHT tunnel 1 local prefix 192.168.99.1/32 #_
↪Additional loopback address on the local
set vpn ipsec site-to-site peer @RIGHT tunnel 1 remote prefix 192.168.99.2/32 #_
↪Additional loopback address on the remote

```

On the RIGHT (dynamic address):

```
set vpn ipsec interface eth0

set vpn ipsec esp-group MyESPGroup proposal 1 encryption aes128
set vpn ipsec esp-group MyESPGroup proposal 1 hash sha1

set vpn ipsec ike-group MyIKEGroup proposal 1 dh-group 2
set vpn ipsec ike-group MyIKEGroup proposal 1 encryption aes128
set vpn ipsec ike-group MyIKEGroup proposal 1 hash sha1

set vpn ipsec site-to-site peer 192.0.2.10 authentication id RIGHT
set vpn ipsec site-to-site peer 192.0.2.10 authentication mode rsa
set vpn ipsec site-to-site peer 192.0.2.10 authentication rsa local-key ipsec-RIGHT
set vpn ipsec site-to-site peer 192.0.2.10 authentication rsa remote-key ipsec-LEFT
set vpn ipsec site-to-site peer 192.0.2.10 authentication remote-id LEFT
set vpn ipsec site-to-site peer 192.0.2.10 connection-type initiate
set vpn ipsec site-to-site peer 192.0.2.10 default-esp-group MyESPGroup

```

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```

set vpn ipsec site-to-site peer 192.0.2.10 ike-group MyIKEGroup
set vpn ipsec site-to-site peer 192.0.2.10 local-address any
set vpn ipsec site-to-site peer 192.0.2.10 tunnel 1 local prefix 192.168.99.2/32 #_
↪ Additional loopback address on the local
set vpn ipsec site-to-site peer 192.0.2.10 tunnel 1 remote prefix 192.168.99.1/32 #_
↪ Additional loopback address on the remote

```

7.13.6 SSTP Server

SSTP is a form of VPN (Virtual Private Network) tunnel that provides a mechanism to transport PPP traffic through an SSL/TLS channel. SSL/TLS provides transport-level security with key negotiation, encryption and traffic integrity checking. The use of SSL/TLS over TCP port 443 allows SSTP to pass through virtually all firewalls and proxy servers except for authenticated web proxies.

SSTP is available for Linux, BSD, and Windows.

VyOS utilizes [accel-ppp](#) to provide SSTP server functionality. We support both local and RADIUS authentication.

As SSTP provides PPP via a SSL/TLS channel the use of either publicly signed certificates or private PKI is required.

Configuring SSTP Server

Certificates

Using our documentation chapter - [PKI](#) generate and install CA and Server certificate

```
vyos@vyos:~$ generate pki ca install CA
```

```
vyos@vyos:~$ generate pki certificate sign CA install Server
```

Configuration

```

set vpn sstp authentication local-users username test password 'test'
set vpn sstp authentication mode 'local'
set vpn sstp client-ip-pool SSTP-POOL range '10.0.0.2-10.0.0.100'
set vpn sstp default-pool 'SSTP-POOL'
set vpn sstp gateway-address '10.0.0.1'
set vpn sstp ssl ca-certificate 'CA1'
set vpn sstp ssl certificate 'Server'

```

set vpn sstp authentication mode <local | radius>

Set authentication backend. The configured authentication backend is used for all queries.

- **radius:** All authentication queries are handled by a configured RADIUS server.
- **local:** All authentication queries are handled locally.

set vpn sstp authentication local-users username <user> password <pass>

Create <user> for local authentication on this system. The users password will be set to <pass>.

set vpn sstp client-ip-pool <POOL-NAME> range <x.x.x.x-x.x.x.x | x.x.x.x/x>

Use this command to define the first IP address of a pool of addresses to be given to SSTP clients. If notation `x.x.x.x-x.x.x.x`, it must be within a /24 subnet. If notation `x.x.x.x/x` is used there is possibility to set host/netmask.

set vpn sstp default-pool <POOL-NAME>

Use this command to define default address pool name.

set vpn sstp gateway-address <gateway>

Specifies single <gateway> IP address to be used as local address of PPP interfaces.

set vpn sstp ssl ca-certificate <file>

Name of installed certificate authority certificate.

set vpn sstp ssl certificate <file>

Name of installed server certificate.

Configuring RADIUS authentication

To enable RADIUS based authentication, the authentication mode needs to be changed within the configuration. Previous settings like the local users still exist within the configuration, however they are not used if the mode has been changed from local to radius. Once changed back to local, it will use all local accounts again.

```
set vpn sstp authentication mode radius
```

set vpn sstp authentication radius server <server> key <secret>

Configure RADIUS <server> and its required shared <secret> for communicating with the RADIUS server.

Since the RADIUS server would be a single point of failure, multiple RADIUS servers can be setup and will be used subsequently. For example:

```
set vpn sstp authentication radius server 10.0.0.1 key 'foo'
set vpn sstp authentication radius server 10.0.0.2 key 'foo'
```

Note: Some RADIUS servers use an access control list which allows or denies queries, make sure to add your VyOS router to the allowed client list.

RADIUS source address

If you are using OSPF as your IGP, use the interface connected closest to the RADIUS server. You can bind all outgoing RADIUS requests to a single source IP e.g. the loopback interface.

set vpn sstp authentication radius source-address <address>

Source IPv4 address used in all RADIUS server queries.

Note: The source-address must be configured to that of an interface. Best practice would be a loopback or dummy interface.

RADIUS advanced options

set vpn sstp authentication radius server <server> port <port>

Configure RADIUS <server> and its required port for authentication requests.

set vpn sstp authentication radius server <server> fail-time <time>

Mark RADIUS server as offline for this given <time> in seconds.

set vpn sstp authentication radius server <server> disable

Temporary disable this RADIUS server.

set vpn sstp authentication radius acct-timeout <timeout>

Timeout to wait reply for Interim-Update packets. (default 3 seconds)

set vpn sstp authentication radius dynamic-author server <address>

Specifies IP address for Dynamic Authorization Extension server (DM/CoA). This IP must exist on any VyOS interface or it can be 0.0.0.0.

set vpn sstp authentication radius dynamic-author port <port>

UDP port for Dynamic Authorization Extension server (DM/CoA)

set vpn sstp authentication radius dynamic-author key <secret>

Secret for Dynamic Authorization Extension server (DM/CoA)

set vpn sstp authentication radius max-try <number>

Maximum number of tries to send Access-Request/Accounting-Request queries

set vpn sstp authentication radius timeout <timeout>

Timeout to wait response from server (seconds)

set vpn sstp authentication radius nas-identifier <identifier>

Value to send to RADIUS server in NAS-Identifier attribute and to be matched in DM/CoA requests.

set vpn sstp authentication radius nas-ip-address <address>

Value to send to RADIUS server in NAS-IP-Address attribute and to be matched in DM/CoA requests. Also DM/CoA server will bind to that address.

set vpn sstp authentication radius source-address <address>

Source IPv4 address used in all RADIUS server queries.

set vpn sstp authentication radius rate-limit attribute <attribute>

Specifies which RADIUS server attribute contains the rate limit information. The default attribute is Filter-Id.

Note: If you set a custom RADIUS attribute you must define it on both dictionaries on the RADIUS server and client.

set vpn sstp authentication radius rate-limit enable

Enables bandwidth shaping via RADIUS.

set vpn sstp authentication radius rate-limit vendor

Specifies the vendor dictionary, This dictionary needs to be present in /usr/share/accel-ppp/radius.

Received RADIUS attributes have a higher priority than parameters defined within the CLI configuration, refer to the explanation below.

Allocation clients ip addresses by RADIUS

If the RADIUS server sends the attribute `Framed-IP-Address` then this IP address will be allocated to the client and the option `default-pool` within the CLI config will be ignored.

If the RADIUS server sends the attribute `Framed-Pool`, then the IP address will be allocated from a predefined IP pool whose name equals the attribute value.

If the RADIUS server sends the attribute `Stateful-IPv6-Address-Pool`, the IPv6 address will be allocated from a predefined IPv6 pool `prefix` whose name equals the attribute value.

If the RADIUS server sends the attribute `Delegated-IPv6-Prefix-Pool`, an IPv6 delegation prefix will be allocated from a predefined IPv6 pool `delegate` whose name equals the attribute value.

Note: `Stateful-IPv6-Address-Pool` and `Delegated-IPv6-Prefix-Pool` are defined in RFC6911. If they are not defined in your RADIUS server, add new [dictionary](#).

The client's interface can be put into a VRF context via a RADIUS Access-Accept packet, or changed via RADIUS CoA. `Acce1-VRF-Name` is used for these purposes. This is a custom [ACCEL-PPP attribute](#). Define it in your RADIUS server.

Renaming clients interfaces by RADIUS

If the RADIUS server uses the attribute `NAS-Port-Id`, ppp tunnels will be renamed.

Note: The value of the attribute `NAS-Port-Id` must be less than 16 characters, otherwise the interface won't be renamed.

IPv6

set vpn sstp ppp-options ipv6 <require | prefer | allow | deny>

Specifies IPv6 negotiation preference.

- **require** - Require IPv6 negotiation
- **prefer** - Ask client for IPv6 negotiation, do not fail if it rejects
- **allow** - Negotiate IPv6 only if client requests
- **deny** - Do not negotiate IPv6 (default value)

set vpn sstp client-ipv6-pool <IPv6-POOL-NAME> prefix <address> mask <number-of-bits>

Use this command to set the IPv6 address pool from which an SSTP client will get an IPv6 prefix of your defined length (mask) to terminate the SSTP endpoint at their side. The mask length can be set between 48 and 128 bits long, the default value is 64.

set vpn sstp client-ipv6-pool <IPv6-POOL-NAME> delegate <address> delegation-prefix <number-of-bits>

Use this command to configure DHCPv6 Prefix Delegation (RFC3633) on SSTP. You will have to set your IPv6 pool and the length of the delegation prefix. From the defined IPv6 pool you will be handing out networks of the defined length (delegation-prefix). The length of the delegation prefix can be set between 32 and 64 bits long.

set vpn sstp default-ipv6-pool <IPv6-POOL-NAME>

Use this command to define default IPv6 address pool name.

```
set vpn sstp ppp-options ipv6 allow
set vpn sstp client-ipv6-pool IPv6-POOL delegate '2001:db8:8003::/48' delegation-prefix
↪ '56'
set vpn sstp client-ipv6-pool IPv6-POOL prefix '2001:db8:8002::/48' mask '64'
set vpn sstp default-ipv6-pool IPv6-POOL
```

IPv6 Advanced Options

set vpn sstp ppp-options ipv6-accept-peer-interface-id

Accept peer interface identifier. By default this is not defined.

set vpn sstp ppp-options ipv6-interface-id <random | x:x:x:x>

Specifies if a fixed or random interface identifier is used for IPv6. The default is fixed.

- **random** - Random interface identifier for IPv6
- **x:x:x:x** - Specify interface identifier for IPv6

set vpn sstp ppp-options ipv6-interface-id <random | x:x:x:x>

Specifies the peer interface identifier for IPv6. The default is fixed.

- **random** - Random interface identifier for IPv6
- **x:x:x:x** - Specify interface identifier for IPv6
- **ipv4-addr** - Calculate interface identifier from IPv4 address.
- **calling-sid** - Calculate interface identifier from calling-station-id.

Scripting

set vpn sstp extended-scripts on-change <path_to_script>

Script to run when the session interface is changed by RADIUS CoA handling

set vpn sstp extended-scripts on-down <path_to_script>

Script to run when the session interface about to terminate

set vpn sstp extended-scripts on-pre-up <path_to_script>

Script to run before the session interface comes up

set vpn sstp extended-scripts on-up <path_to_script>

Script to run when the session interface is completely configured and started

Advanced Options

Authentication Advanced Options

set vpn sstp authentication local-users username <user> disable

Disable <user> account.

set vpn sstp authentication local-users username <user> static-ip <address>

Assign a static IP address to <user> account.

set vpn sstp authentication local-users username <user> rate-limit download <bandwidth>

Rate limit the download bandwidth for <user> to <bandwidth> kbit/s.

set vpn sstp authentication local-users username <user> rate-limit upload <bandwidth>

Rate limit the upload bandwidth for <user> to <bandwidth> kbit/s.

set vpn sstp authentication protocols <pap | chap | mschap | mschap-v2>

Require the peer to authenticate itself using one of the following protocols: pap, chap, mschap, mschap-v2.

Client IP Pool Advanced Options

set vpn sstp client-ip-pool <POOL-NAME> next-pool <NEXT-POOL-NAME>

Use this command to define the next address pool name.

PPP Advanced Options

set vpn sstp ppp-options disable-ccp

Disable Compression Control Protocol (CCP). CCP is enabled by default.

set vpn sstp ppp-options interface-cache <number>

Specifies number of interfaces to cache. This prevents interfaces from being removed once the corresponding session is destroyed. Instead, interfaces are cached for later use in new sessions. This should reduce the kernel-level interface creation/deletion rate. Default value is 0.

set vpn sstp ppp-options ipv4 <require | prefer | allow | deny>

Specifies IPv4 negotiation preference.

- **require** - Require IPv4 negotiation
- **prefer** - Ask client for IPv4 negotiation, do not fail if it rejects
- **allow** - Negotiate IPv4 only if client requests (Default value)
- **deny** - Do not negotiate IPv4

set vpn sstp ppp-options lcp-echo-failure <number>

Defines the maximum <number> of unanswered echo requests. Upon reaching the value <number>, the session will be reset. Default value is 3.

set vpn sstp ppp-options lcp-echo-interval <interval>

If this option is specified and is greater than 0, then the PPP module will send LCP echo requests every <interval> seconds. Default value is 30.

set vpn sstp ppp-options lcp-echo-timeout

Specifies timeout in seconds to wait for any peer activity. If this option is specified it turns on adaptive lcp echo functionality and “lcp-echo-failure” is not used. Default value is **0**.

set vpn sstp ppp-options min-mtu <number>

Defines the minimum acceptable MTU. If a client tries to negotiate an MTU lower than this it will be NAKed, and disconnected if it rejects a greater MTU. Default value is **100**.

set vpn sstp ppp-options mppe <require | prefer | deny>

Specifies MPPE negotiation preference.

- **require** - ask client for mppe, if it rejects drop connection
- **prefer** - ask client for mppe, if it rejects don't fail. (Default value)
- **deny** - deny mppe

Default behavior - don't ask the client for mppe, but allow it if the client wants. Please note that RADIUS may override this option by MS-MPPE-Encryption-Policy attribute.

set vpn sstp ppp-options mru <number>

Defines preferred MRU. By default is not defined.

Global Advanced options**set vpn sstp description <description>**

Set description.

set vpn sstp limits burst <value>

Burst count

set vpn sstp limits connection-limit <value>

Maximum accepted connection rate (e.g. 1/min, 60/sec)

set vpn sstp limits timeout <value>

Timeout in seconds

set vpn sstp mtu

Maximum Transmission Unit (MTU) (default: **1500**)

set vpn sstp max-concurrent-sessions

Maximum number of concurrent session start attempts

set vpn sstp name-server <address>

Connected clients should use <address> as their DNS server. This command accepts both IPv4 and IPv6 addresses. Up to two nameservers can be configured for IPv4, up to three for IPv6.

set vpn sstp shaper fwmark <1-2147483647>

Match firewall mark value

set vpn sstp snmp master-agent

Enable SNMP

set vpn sstp wins-server <address>

Windows Internet Name Service (WINS) servers propagated to client

```
set vpn sstp host-name <hostname>
```

If this option is given, only SSTP connections to the specified host and with the same TLS SNI will be allowed.

Configuring SSTP client

Once you have setup your SSTP server there comes the time to do some basic testing. The Linux client used for testing is called `sstpc`. `sstpc` requires a PPP configuration/peer file.

If you use a self-signed certificate, do not forget to install CA on the client side.

The following PPP configuration tests MSCHAP-v2:

```
$ cat /etc/ppp/peers/vyos
usepeerdns
#require-mppe
#require-pap
require-mschap-v2
noauth
lock
refuse-pap
refuse-eap
refuse-chap
refuse-mschap
#refuse-mschap-v2
nobsdcomp
nodeflate
debug
```

You can now “dial” the peer with the following command: `sstpc --log-level 4 --log-stderr --user vyos --password vyos vpn.example.com -- call vyos`.

A connection attempt will be shown as:

```
$ sstpc --log-level 4 --log-stderr --user vyos --password vyos vpn.example.com -- call_
↪vyos

Mar 22 13:29:12 sstpc[12344]: Resolved vpn.example.com to 192.0.2.1
Mar 22 13:29:12 sstpc[12344]: Connected to vpn.example.com
Mar 22 13:29:12 sstpc[12344]: Sending Connect-Request Message
Mar 22 13:29:12 sstpc[12344]: SEND SSTP CTRL PKT(14)
Mar 22 13:29:12 sstpc[12344]:   TYPE(1): CONNECT REQUEST, ATTR(1):
Mar 22 13:29:12 sstpc[12344]:     ENCAP PROTO(1): 6
Mar 22 13:29:12 sstpc[12344]: RECV SSTP CTRL PKT(48)
Mar 22 13:29:12 sstpc[12344]:   TYPE(2): CONNECT ACK, ATTR(1):
Mar 22 13:29:12 sstpc[12344]:     CRYPTO BIND REQ(4): 40
Mar 22 13:29:12 sstpc[12344]: Started PPP Link Negotiation
Mar 22 13:29:15 sstpc[12344]: Sending Connected Message
Mar 22 13:29:15 sstpc[12344]: SEND SSTP CTRL PKT(112)
Mar 22 13:29:15 sstpc[12344]:   TYPE(4): CONNECTED, ATTR(1):
Mar 22 13:29:15 sstpc[12344]:     CRYPTO BIND(3): 104
Mar 22 13:29:15 sstpc[12344]: Connection Established
```

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```
$ ip addr show ppp0
164: ppp0: <POINTOPOINT,MULTICAST,NOARP,UP,LOWER_UP> mtu 1452 qdisc fq_codel state_
↳ UNKNOWN group default qlen 3
    link/ppp promiscuity 0
    inet 100.64.2.2 peer 100.64.1.1/32 scope global ppp0
        valid_lft forever preferred_lft forever
```

Monitoring

show sstp-server sessions

Use this command to locally check the active sessions in the SSTP server.

```
vyos@vyos:~$ show sstp-server sessions
ifname | username | ip | ip6 | ip6-dp | calling-sid | rate-limit | state |
↳ uptime | rx-bytes | tx-bytes
-----+-----+-----+-----+-----+-----+-----+-----+
↳ -----+-----+-----+
sstp0 | test | 10.0.0.2 | | | 192.168.10.100 | | active |
↳ 00:15:46 | 16.3 KiB | 210 B
```

```
vyos@vyos:~$ show sstp-server statistics
uptime: 0.01:21:54
cpu: 0%
mem(rss/virt): 6688/100464 kB
core:
  mempool_allocated: 149420
  mempool_available: 146092
  thread_count: 1
  thread_active: 1
  context_count: 6
  context_sleeping: 0
  context_pending: 0
  md_handler_count: 7
  md_handler_pending: 0
  timer_count: 2
  timer_pending: 0
sessions:
  starting: 0
  active: 1
  finishing: 0
sstp:
  starting: 0
  active: 1
```

Troubleshooting

```
vyos@vyos:~$sudo journalctl -u accel-ppp@sstp -b 0

Feb 28 17:03:04 vyos accel-sstp[2492]: sstp: new connection from 192.168.10.100:49852
Feb 28 17:03:04 vyos accel-sstp[2492]: sstp: starting
Feb 28 17:03:04 vyos accel-sstp[2492]: sstp: started
Feb 28 17:03:04 vyos accel-sstp[2492]: :: recv [HTTP <SSTP_DUPLEX_POST /sra_{BA195980-
↳CD49-458b-9E23-C84EE0ADCD75}/ HTTP/1.1>]
Feb 28 17:03:04 vyos accel-sstp[2492]: :: recv [HTTP <SSTPCORRELATIONID: {48B82435-099A-
↳4158-A987-052E7570CFAA}>]
Feb 28 17:03:04 vyos accel-sstp[2492]: :: recv [HTTP <Content-Length:
↳18446744073709551615>]
Feb 28 17:03:04 vyos accel-sstp[2492]: :: recv [HTTP <Host: vyos.io>]
Feb 28 17:03:04 vyos accel-sstp[2492]: :: send [HTTP <HTTP/1.1 200 OK>]
Feb 28 17:03:04 vyos accel-sstp[2492]: :: send [HTTP <Date: Wed, 28 Feb 2024 17:03:04
↳GMT>]
Feb 28 17:03:04 vyos accel-sstp[2492]: :: send [HTTP <Content-Length:
↳18446744073709551615>]
Feb 28 17:03:04 vyos accel-sstp[2492]: :: recv [SSTP SSTP_MSG_CALL_CONNECT_REQUEST]
Feb 28 17:03:04 vyos accel-sstp[2492]: :: send [SSTP SSTP_MSG_CALL_CONNECT_ACK]
Feb 28 17:03:04 vyos accel-sstp[2492]: :: lcp_layer_init
Feb 28 17:03:04 vyos accel-sstp[2492]: :: auth_layer_init
Feb 28 17:03:04 vyos accel-sstp[2492]: :: ccp_layer_init
Feb 28 17:03:04 vyos accel-sstp[2492]: :: ipcp_layer_init
Feb 28 17:03:04 vyos accel-sstp[2492]: :: ipv6cp_layer_init
Feb 28 17:03:04 vyos accel-sstp[2492]: :: ppp establishing
Feb 28 17:03:04 vyos accel-sstp[2492]: :: lcp_layer_start
Feb 28 17:03:04 vyos accel-sstp[2492]: :: send [LCP ConfReq id=56 <auth PAP> <mru 1452>
↳<magic 1cd9ad05>]
Feb 28 17:03:04 vyos accel-sstp[2492]: :: recv [LCP ConfReq id=0 <mru 4091> <magic
↳345f64ca> <pcomp> <accomp> < d 3 6 >]
Feb 28 17:03:04 vyos accel-sstp[2492]: :: send [LCP ConfRej id=0 <pcomp> <accomp> < d 3
↳6 >]
Feb 28 17:03:04 vyos accel-sstp[2492]: :: recv [LCP ConfReq id=1 <mru 4091> <magic
↳345f64ca>]
Feb 28 17:03:04 vyos accel-sstp[2492]: :: send [LCP ConfNak id=1 <mru 1452>]
Feb 28 17:03:04 vyos accel-sstp[2492]: :: recv [LCP ConfReq id=2 <mru 1452> <magic
↳345f64ca>]
Feb 28 17:03:04 vyos accel-sstp[2492]: :: send [LCP ConfAck id=2]
Feb 28 17:03:07 vyos accel-sstp[2492]: :: fsm timeout 9
Feb 28 17:03:07 vyos accel-sstp[2492]: :: send [LCP ConfReq id=56 <auth PAP> <mru 1452>
↳<magic 1cd9ad05>]
Feb 28 17:03:07 vyos accel-sstp[2492]: :: recv [LCP ConfAck id=56 <auth PAP> <mru 1452>
↳<magic 1cd9ad05>]
Feb 28 17:03:07 vyos accel-sstp[2492]: :: lcp_layer_started
Feb 28 17:03:07 vyos accel-sstp[2492]: :: auth_layer_start
Feb 28 17:03:07 vyos accel-sstp[2492]: :: recv [LCP Ident id=3 <MSRASV5.20>]
Feb 28 17:03:07 vyos accel-sstp[2492]: :: recv [LCP Ident id=4 <MSRAS-0-MSEDGEWIN10>]
Feb 28 17:03:07 vyos accel-sstp[2492]: [50B blob data]
Feb 28 17:03:07 vyos accel-sstp[2492]: :: recv [PAP AuthReq id=3]
Feb 28 17:03:07 vyos accel-sstp[2492]: ppp0:test: connect: ppp0 <--> sstp(192.168.10.
↳100:49852)
```

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```

Feb 28 17:03:07 vyos accel-sstp[2492]: ppp0:test: ppp connected
Feb 28 17:03:07 vyos accel-sstp[2492]: ppp0:test: send [PAP AuthAck id=3 "Authentication_
↪succeeded"]
Feb 28 17:03:07 vyos accel-sstp[2492]: ppp0:test: test: authentication succeeded
Feb 28 17:03:07 vyos accel-sstp[2492]: ppp0:test: auth_layer_started
Feb 28 17:03:07 vyos accel-sstp[2492]: ppp0:test: ccp_layer_start
Feb 28 17:03:07 vyos accel-sstp[2492]: ppp0:test: ipcp_layer_start
Feb 28 17:03:07 vyos accel-sstp[2492]: ppp0:test: ipv6cp_layer_start
Feb 28 17:03:07 vyos accel-sstp[2492]: ppp0:test: recv [SSTP SSTP_MSG_CALL_CONNECTED]
Feb 28 17:03:07 vyos accel-sstp[2492]: ppp0:test: IPV6CP: discarding packet
Feb 28 17:03:07 vyos accel-sstp[2492]: ppp0:test: send [LCP ProtoRej id=88 <8057>]
Feb 28 17:03:07 vyos accel-sstp[2492]: ppp0:test: recv [IPCP ConfReq id=7 <addr 0.0.0.0>
↪<dns1 0.0.0.0> <wins1 0.0.0.0> <dns2 0.0.0.0> <wins2 0.0.0.0>]
Feb 28 17:03:07 vyos accel-sstp[2492]: ppp0:test: send [IPCP ConfReq id=25 <addr 10.0.0.
↪1>]
Feb 28 17:03:07 vyos accel-sstp[2492]: ppp0:test: send [IPCP ConfRej id=7 <dns1 0.0.0.0>
↪<wins1 0.0.0.0> <dns2 0.0.0.0> <wins2 0.0.0.0>]
Feb 28 17:03:07 vyos accel-sstp[2492]: ppp0:test: recv [IPCP ConfAck id=25 <addr 10.0.0.
↪1>]
Feb 28 17:03:07 vyos accel-sstp[2492]: ppp0:test: recv [IPCP ConfReq id=8 <addr 0.0.0.0>]
Feb 28 17:03:07 vyos accel-sstp[2492]: ppp0:test: send [IPCP ConfNak id=8 <addr 10.0.0.5>
↪]
Feb 28 17:03:07 vyos accel-sstp[2492]: ppp0:test: recv [IPCP ConfReq id=9 <addr 10.0.0.5>
↪]
Feb 28 17:03:07 vyos accel-sstp[2492]: ppp0:test: send [IPCP ConfAck id=9]
Feb 28 17:03:07 vyos accel-sstp[2492]: ppp0:test: ipcp_layer_started
Feb 28 17:03:07 vyos accel-sstp[2492]: ppp0:test: rename interface to 'sstp0'
Feb 28 17:03:07 vyos accel-sstp[2492]: sstp0:test: sstp: ppp: started

```

7.13.7 DMVPN

DMVPN (Dynamic Multipoint Virtual Private Network) is a dynamic VPN technology originally developed by Cisco. While their implementation was somewhat proprietary, the underlying technologies are actually standards based. The three technologies are:

- NHRP (Next Hop Resolution Protocol) [RFC 2332](#)
- mGRE (Multipoint Generic Routing Encapsulation) [RFC 1702](#)
- IPSEC (IP Security) - too many RFCs to list, but start with [RFC 4301](#)

NHRP provides the dynamic tunnel endpoint discovery mechanism (endpoint registration, and endpoint discovery/lookup), mGRE provides the tunnel encapsulation itself, and the IPSEC protocols handle the key exchange, and crypto mechanism.

In short, DMVPN provides the capability for creating a dynamic-mesh VPN network without having to pre-configure (static) all possible tunnel end-point peers.

Note: DMVPN only automates the tunnel endpoint discovery and setup. A complete solution also incorporates the use of a routing protocol. BGP is particularly well suited for use with DMVPN.

Fig. 6: Baseline DMVPN topology

Configuration

Tunnel interface configuration

NHRP never handles routing of prefixes itself. You need to run some real routing protocol (e.g. BGP) to advertise routes over the tunnels. What nhrpd does it establishes ‘shortcut routes’ that optimizes the routing protocol to avoid going through extra nodes in NBMA GRE mesh.

NHRP does route NHRP domain addresses individually using per-host prefixes. This is similar to Cisco FlexVPN, but in contrast to opennhrp which uses a generic subnet route.

To create NBMA GRE tunnel you might use the following:

```
set interfaces tunnel tun100 address '10.0.0.1/32'
set interfaces tunnel tun100 enable-multicast
set interfaces tunnel tun100 encapsulation 'gre'
set interfaces tunnel tun100 ip adjust-mss '1360'
set interfaces tunnel tun100 mtu '1400'
set interfaces tunnel tun100 parameters ip key '42'
set interfaces tunnel tun100 source-interface 'eth0'
```

- Please refer to the [Tunnel](#) documentation for the individual tunnel related options.

Note: The IP-address is assigned as host prefix to tunnel interface. NHRP will automatically create additional host routes pointing to tunnel interface when a connection with these hosts is established.

The tunnel interface subnet prefix should be announced by routing protocol from the hub nodes (e.g. BGP ‘network’ announce). This allows the routing protocol to decide which is the closest hub and determine the relay hub on prefix basis when direct tunnel is not established.

NHRP protocol configuration

set protocols nhrp tunnel <tunnel> authentication <secret>

Enables Cisco style authentication on NHRP packets. This embeds the plaintext password to the outgoing NHRP packets. Maximum length of the password is 8 characters.

set protocols nhrp tunnel <tunnel> holdtime <timeout>

Holdtime is the number of seconds that have to pass before stopping to advertise an NHRP NBMA address as valid. It also controls how often NHRP registration requests are sent. By default registrations are sent every one third of the holdtime

set protocols nhrp tunnel <tunnel> map tunnel-ip <tunnel-ip> nbma <nbma-ip>

- **tunnel-ip** - Tunnel ip address in format **x.x.x.x**.
- **nbma-ip** - NBMA ip address in format **x.x.x.x** or **local**

Map an IP address of a station to the station’s NBMA address.

set protocols nhrp tunnel <tunnel> mtu <mtu>

Configure NHRP advertised MTU.

set protocols nhrp tunnel <tunnel> multicast <nbma-ip>

- **nbma-ip** - NBMA ip address in format **x.x.x.x** or **dynamic**

Sends multicast packets to the specified NBMA address. If dynamic is specified then destination NBMA address (or addresses) are learnt dynamically.

set protocols nhrp tunnel <tunnel> network-id <network-id>

- **network-id** - NHRP network id <1-4294967295>

Enable NHRP on this interface and set the interface's network ID. The network ID is used to allow creating multiple nhrp domains on a router when multiple interfaces are configured on the router. Interfaces configured with the same ID are part of the same logical NBMA network. The ID is a local only parameter and is not sent to other NHRP nodes and so IDs on different nodes do not need to match. When NHRP packets are received on an interface they are assigned to the local NHRP domain for that interface.

set protocols nhrp tunnel <tunnel> nhs tunnel-ip <tunnel-ip> nbma <nbma-ip>

- **tunnel-ip** - Tunnel ip address in format **x.x.x.x** or **dynamic**
- **nbma-ip** - NBMA ip address in format **x.x.x.x**

Configure the Next Hop Server address and its NBMA address. If dynamic is specified then Next Hop Server can have dynamic address which maps to its NBMA address.

set protocols nhrp tunnel <tunnel> redirect

This enable redirect replies on the NHS similar to ICMP redirects except this is managed by the nhrp protocol. This setting allows spokes to communicate with each others directly.

set protocols nhrp tunnel <tunnel> registration-no-unique

Allow the client to not set the unique flag in the NHRP packets. This is useful when a station has a dynamic IP address that could change over time.

set protocols nhrp tunnel <tunnel> shortcut

Enable shortcut (spoke-to-spoke) tunnels to allow NHC to talk to each others directly after establishing a connection without going through the hub.

IPSEC configuration

- Please refer to the [IPsec General Information](#) documentation for the individual IPsec related options.

Note: NHRP daemon based on FRR nhrpd. It controls IPSEC. That's why 'close-action' parameter in IKE configuration always is set to 'close' and 'dead-peer-detection action' always is set to 'clear'.

set vpn ipsec profile <profile-name> authentication mode pre-shared-secret

Set preshared secret mode authentication

set vpn ipsec profile <profile-name> authentication pre-shared-secret <secret>

Set preshared secret

set vpn ipsec profile <profile-name> bind tunnel <tunnel name>

Bind IPSEC profile to the specific tunnel interface.

set vpn ipsec profile <profile-name> esp-group 'ESP-HUB'

Map ESP group to IPSEC profile

```
set vpn ipsec profile <profile-name> ike-group 'IKE-HUB'
```

Map IKE group to IPSEC profile

Monitoring

```
show ip nhrp cache
```

Forwarding cache information.

```
show ip nhrp nhs
```

Next hop server information.

```
show ip nhrp shortcut
```

Shortcut information.

Example

This blueprint uses VyOS as the DMVPN Hub and Cisco IOSv 15.5(3)M and VyOS as multiple spoke sites.

Fig. 7: DMVPN Network Topology Diagram

Each node (Hub and Spoke) uses an IP address from the network 10.0.0.0/24.

The below referenced IP address 192.168.0.2 is used as example address representing a global unicast address under which the HUB can be contacted by each and every individual spoke.

Configuration

Hub

VyOS-HUB-1

```
set interfaces ethernet eth0 address '192.168.0.2/30'

set interfaces tunnel tun100 address '10.0.0.100/32'
set interfaces tunnel tun100 enable-multicast
set interfaces tunnel tun100 encapsulation 'gre'
set interfaces tunnel tun100 parameters ip key '42'
set interfaces tunnel tun100 source-interface 'eth0'

set protocols nhrp tunnel tun100 authentication 'test123'
set protocols nhrp tunnel tun100 holdtime '300'
set protocols nhrp tunnel tun100 multicast 'dynamic'
set protocols nhrp tunnel tun100 network-id '1'
set protocols nhrp tunnel tun100 redirect
set protocols nhrp tunnel tun100 registration-no-unique

set protocols static route 0.0.0.0/0 next-hop 192.168.0.1

set vpn ipsec esp-group ESP-HUB lifetime '1800'
```

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```

set vpn ipsec esp-group ESP-HUB mode 'transport'
set vpn ipsec esp-group ESP-HUB pfs 'dh-group2'
set vpn ipsec esp-group ESP-HUB proposal 1 encryption 'aes256'
set vpn ipsec esp-group ESP-HUB proposal 1 hash 'sha1'
set vpn ipsec ike-group IKE-HUB key-exchange 'ikev1'
set vpn ipsec ike-group IKE-HUB lifetime '3600'
set vpn ipsec ike-group IKE-HUB proposal 1 dh-group '2'
set vpn ipsec ike-group IKE-HUB proposal 1 encryption 'aes256'
set vpn ipsec ike-group IKE-HUB proposal 1 hash 'sha1'
set vpn ipsec interface 'eth0'
set vpn ipsec profile NHRPVPN authentication mode 'pre-shared-secret'
set vpn ipsec profile NHRPVPN authentication pre-shared-secret 'secret'
set vpn ipsec profile NHRPVPN bind tunnel 'tun100'
set vpn ipsec profile NHRPVPN esp-group 'ESP-HUB'
set vpn ipsec profile NHRPVPN ike-group 'IKE-HUB'

```

Note: Setting this up on AWS will require a “Custom Protocol Rule” for protocol number “47” (GRE) Allow Rule in TWO places. Firstly on the VPC Network ACL, and secondly on the security group network ACL attached to the EC2 instance. This has been tested as working for the official AMI image on the AWS Marketplace. (Locate the correct VPC and security group by navigating through the details pane below your EC2 instance in the AWS console).

Spokes

The individual spoke configurations only differ in interface IP addresses.

VyOS-Spoke-1 and VyOS-Spoke-2

```

set interfaces ethernet eth0 address '192.168.1.2/30'

set interfaces tunnel tun100 address '10.0.0.1/32'
set interfaces tunnel tun100 enable-multicast
set interfaces tunnel tun100 encapsulation 'gre'
set interfaces tunnel tun100 parameters ip key '42'
set interfaces tunnel tun100 source-interface 'eth0'

set protocols nhrp tunnel tun100 authentication 'test123'
set protocols nhrp tunnel tun100 holdtime '300'
set protocols nhrp tunnel tun100 multicast 'dynamic'
set protocols nhrp tunnel tun100 network-id '1'
set protocols nhrp tunnel tun100 nhs tunnel-ip dynamic nbma '192.168.0.2'
set protocols nhrp tunnel tun100 registration-no-unique
set protocols nhrp tunnel tun100 shortcut

set protocols static route 0.0.0.0/0 next-hop 192.168.1.1
set protocols static route 10.0.0.0/24 next-hop 10.0.0.100

set vpn ipsec esp-group ESP-HUB lifetime '1800'
set vpn ipsec esp-group ESP-HUB mode 'transport'

```

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```

set vpn ipsec esp-group ESP-HUB pfs 'dh-group2'
set vpn ipsec esp-group ESP-HUB proposal 1 encryption 'aes256'
set vpn ipsec esp-group ESP-HUB proposal 1 hash 'sha1'
set vpn ipsec ike-group IKE-HUB key-exchange 'ikev1'
set vpn ipsec ike-group IKE-HUB lifetime '3600'
set vpn ipsec ike-group IKE-HUB proposal 1 dh-group '2'
set vpn ipsec ike-group IKE-HUB proposal 1 encryption 'aes256'
set vpn ipsec ike-group IKE-HUB proposal 1 hash 'sha1'
set vpn ipsec interface 'eth0'
set vpn ipsec profile NHRPVPN authentication mode 'pre-shared-secret'
set vpn ipsec profile NHRPVPN authentication pre-shared-secret 'secret'
set vpn ipsec profile NHRPVPN bind tunnel 'tun100'
set vpn ipsec profile NHRPVPN esp-group 'ESP-HUB'
set vpn ipsec profile NHRPVPN ike-group 'IKE-HUB'

```

Cisco-Spoke-3

```

crypto isakmp policy 10
  encr aes 256
  authentication pre-share
  group 2
  lifetime 3600
crypto isakmp key secret address 0.0.0.0
!
!
crypto ipsec transform-set DMVPNESP esp-aes 256 esp-sha-hmac
  mode transport
!
crypto ipsec profile DMVPNPROFILE
  set security-association lifetime seconds 1800
  set transform-set DMVPNESP
  set pfs group2
!
!
!
!
!
!
!
interface Tunnel100
  ip address 10.0.0.3 255.255.255.0
  no ip redirects
  ip nhrp authentication test123
  ip nhrp map multicast dynamic
  ip nhrp network-id 1
  ip nhrp holdtime 300
  ip nhrp nhs 10.0.0.100 nbma 192.168.0.2
  ip nhrp registration no-unique
  ip nhrp redirect
  tunnel source GigabitEthernet0/0

```

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```

tunnel mode gre multipoint
tunnel key 42
tunnel protection ipsec profile DMVPNPROFILE
!
interface GigabitEthernet0/0
ip address 192.168.3.2 255.255.255.252
duplex auto
speed auto
media-type rj45
!
ip route 0.0.0.0 0.0.0.0 192.168.3.1

```

Monitoring DMVPN Network

Let send ICMP packets from VyOS-SPOKE-1 to Cisco-SPOKE-3

```

vyos@vyos:~$ ping 10.0.0.3
PING 10.0.0.3 (10.0.0.3) 56(84) bytes of data.
64 bytes from 10.0.0.3: icmp_seq=1 ttl=255 time=3.44 ms
64 bytes from 10.0.0.3: icmp_seq=2 ttl=255 time=3.07 ms
^C
--- 10.0.0.3 ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1002ms
rtt min/avg/max/mdev = 3.072/3.257/3.442/0.185 ms

```

Monitoring on HUB

```

vyos@vyos:~$ show ip nhrp cache

```

Iface	Type	Protocol	NBMA	Claimed NBMA
tun100	dynamic	10.0.0.1	192.168.1.2	192.168.1.2
tun100	dynamic	10.0.0.3	192.168.3.2	192.168.3.2
tun100	dynamic	10.0.0.2	192.168.2.2	192.168.2.2
tun100	local	10.0.0.100	192.168.0.2	192.168.0.2

```

vyos@vyos:~$ show vpn ipsec sa

```

Connection	Remote ID	State	Uptime	Bytes In/Out	Packets In/Out	Remote
dmvpn-NHRPVPN-tun100-child 1.2	192.168.1.2	up	3m46s	230B/270B	2/2	192.168.
dmvpn-NHRPVPN-tun100-child 2.2	192.168.2.2	up	5m48s	460B/540B	4/4	192.168.

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```
dmvpn-NHRPVPN-tun100-child up      16m26s    1K/1K      13/12      192.168.
↪ 3.2      192.168.3.2 AES_CBC_256/HMAC_SHA1_96/MODP_1024
```

Monitoring on Spokes

```
vyos@vyos:~$ show ip nhrp cache
Iface   Type   Protocol      NBMA      Claimed NBMA
↪      Flags Identity
tun100  local  10.0.0.1      192.168.1.2  192.168.1.2
↪      -
tun100  dynamic 10.0.0.3      192.168.3.2  192.168.3.2
↪      T    192.168.3.2
tun100  nhs     10.0.0.100    192.168.0.2  192.168.0.2
↪      T    192.168.0.2

vyos@vyos:~$ show ip nhrp nhs
Iface   FQDN      NBMA      Protocol
tun100  192.168.0.2  192.168.0.2  10.0.0.100

vyos@vyos:~$ show ip nhrp shortcut
Type    Prefix      Via      Identity
dynamic 10.0.0.3/32  10.0.0.3  192.168.3.2

vyos@vyos:~$ show vpn ipsec sa
Connection      State   Uptime   Bytes In/Out   Packets In/Out   Remote
↪address      Remote ID   Proposal
-----
↪-----
dmvpn-NHRPVPN-tun100-child up      6m43s    898B/695B      7/6      192.168.
↪0.2      192.168.0.2 AES_CBC_256/HMAC_SHA1_96/MODP_1024
dmvpn-NHRPVPN-tun100-child up      49s      215B/187B      2/2      192.168.
↪3.2      192.168.3.2 AES_CBC_256/HMAC_SHA1_96/MODP_1024
```

7.14 VRF

VRF devices combined with ip rules provides the ability to create virtual routing and forwarding domains (aka VRFs, VRF-lite to be specific) in the Linux network stack. One use case is the multi-tenancy problem where each tenant has their own unique routing tables and in the very least need different default gateways.

7.14.1 Configuration

A VRF device is created with an associated route table. Network interfaces are then enslaved to a VRF device.

set vrf name <name> table <id>

Create a new VRF instance with <name> and <id>. The name is used when placing individual interfaces into the VRF.

Note: A routing table ID can not be modified once it is assigned. It can only be changed by deleting and re-adding the VRF instance.

set vrf bind-to-all

By default the scope of the port bindings for unbound sockets is limited to the default VRF. That is, it will not be matched by packets arriving on interfaces enslaved to a VRF and processes may bind to the same port if they bind to a VRF.

TCP & UDP services running in the default VRF context (ie., not bound to any VRF device) can work across all VRF domains by enabling this option.

Zebra/Kernel route filtering

Zebra supports prefix-lists and Route Maps to match routes received from other FRR components. The permit/deny facilities provided by these commands can be used to filter which routes zebra will install in the kernel.

set vrf <name> ip protocol <protocol> route-map <route-map>

Apply a route-map filter to routes for the specified protocol.

The following protocols can be used: any, babel, bgp, eigrp, isis, ospf, rip, static

Note: If you choose any as the option that will cause all protocols that are sending routes to zebra.

set vrf <name> ipv6 protocol <protocol> route-map <route-map>

Apply a route-map filter to routes for the specified protocol.

The following protocols can be used: any, babel, bgp, isis, ospfv3, ripng, static

Note: If you choose any as the option that will cause all protocols that are sending routes to zebra.

Nexthop Tracking

Nexthop tracking resolve nexthops via the default route by default. This is enabled by default for a traditional profile of FRR which we use. It can be disabled if you do not want to e.g. allow BGP to peer across the default route.

set vrf name <name> ip nht no-resolve-via-default

Do not allow IPv4 nexthop tracking to resolve via the default route. This parameter is configured per-VRF, so the command is also available in the VRF subnode.

set vrf name <name> ipv6 nht no-resolve-via-default

Do not allow IPv6 nexthop tracking to resolve via the default route. This parameter is configured per-VRF, so the command is also available in the VRF subnode.

Interfaces

When VRFs are used it is not only mandatory to create a VRF but also the VRF itself needs to be assigned to an interface.

set interfaces <dummy | ethernet | bonding | bridge | pppoe> <interface> vrf <name>

Assign interface identified by <interface> to VRF named <name>.

Routing

Note: VyOS 1.4 (sagitta) introduced dynamic routing support for VRFs.

Currently dynamic routing is supported for the following protocols:

- *BGP*
- routing-isis
- *OSPF*
- *OSPFv3 (IPv6)*
- *Static*

The CLI configuration is same as mentioned in above articles. The only difference is, that each routing protocol used, must be prefixed with the `vrf name <name>` command.

Example

The following commands would be required to set options for a given dynamic routing protocol inside a given vrf:

- *BGP*: `set vrf name <name> protocols bgp ...`
- routing-isis: `set vrf name <name> protocols isis ...`
- *OSPF*: `set vrf name <name> protocols ospf ...`
- *OSPFv3 (IPv6)*: `set vrf name <name> protocols ospfv3 ...`
- *Static*: `set vrf name <name> protocols static ...`

Services

Currently the following services can be created isolated in VRFs

- *DHCP Server*

The CLI configuration is same as mentioned in above articles. The only difference is, that each service used, must be prefixed with the `vrf name <name>` command.

Example

The following commands would be required to set options for a given service inside a given vrf:

- *DHCP Server*: `set vrf name <name> service dhcp-server ...`
- *DHCP Server*: `set vrf name <name> service dhcpv6-server ...`

7.14.2 Operation

It is not sufficient to only configure a VRF but VRFs must be maintained, too. For VRF maintenance the following operational commands are in place.

show vrf

Lists VRFs that have been created

```
vyos@vyos:~$ show vrf
VRF name      state      mac address      flags      interfaces
-----
blue          up        00:53:12:d8:74:24  noarp, master, up, lower_up  dum200,
→eth0.302
red          up        00:53:de:02:df:aa  noarp, master, up, lower_up  dum100,
→eth0.300, bond0.100, peth0
```

Note: Command should probably be extended to list also the real interfaces assigned to this one VRF to get a better overview.

show vrf <name>

```
vyos@vyos:~$ show vrf name blue
VRF name      state      mac address      flags      interfaces
-----
blue          up        00:53:12:d8:74:24  noarp, master, up, lower_up  dum200,
→eth0.302
```

show ip route vrf <name>

Display IPv4 routing table for VRF identified by <name>.

```
vyos@vyos:~$ show ip route vrf blue
Codes: K - kernel route, C - connected, S - static, R - RIP,
       O - OSPF, I - IS-IS, B - BGP, E - EIGRP, N - NHRP,
       T - Table, v - VNC, V - VNC-Direct, A - Babel, D - SHARP,
       F - PBR, f - OpenFabric,
       > - selected route, * - FIB route, q - queued route, r - rejected route

VRF blue:
K  0.0.0.0/0 [255/8192] unreachable (ICMP unreachable), 00:00:50
S>* 172.16.0.0/16 [1/0] via 192.0.2.1, dum1, 00:00:02
C>* 192.0.2.0/24 is directly connected, dum1, 00:00:06
```

show ipv6 route vrf <name>

Display IPv6 routing table for VRF identified by <name>.

```
vyos@vyos:~$ show ipv6 route vrf red
Codes: K - kernel route, C - connected, S - static, R - RIPng,
        O - OSPFv3, I - IS-IS, B - BGP, N - NHRP, T - Table,
        v - VNC, V - VNC-Direct, A - Babel, D - SHARP, F - PBR,
        f - OpenFabric,
        > - selected route, * - FIB route, q - queued route, r - rejected route

VRF red:
K  ::/0 [255/8192] unreachable (ICMP unreachable), 00:43:20
C>* 2001:db8::/64 is directly connected, dum1, 00:02:19
C>* fe80::/64 is directly connected, dum1, 00:43:19
K>* ff00::/8 [0/256] is directly connected, dum1, 00:43:19
```

ping <host> vrf <name>

The ping command is used to test whether a network host is reachable or not.

Ping uses ICMP protocol's mandatory ECHO_REQUEST datagram to elicit an ICMP ECHO_RESPONSE from a host or gateway. ECHO_REQUEST datagrams (pings) will have an IP and ICMP header, followed by "struct timeval" and an arbitrary number of pad bytes used to fill out the packet.

When doing fault isolation with ping, you should first run it on the local host, to verify that the local network interface is up and running. Then, continue with hosts and gateways further down the road towards your destination. Round-trip time and packet loss statistics are computed.

Duplicate packets are not included in the packet loss calculation, although the round-trip time of these packets is used in calculating the minimum/ average/maximum round-trip time numbers.

Note: Ping command can be interrupted at any given time using <Ctrl>+c. A brief statistic is shown afterwards.

```
vyos@vyos:~$ ping 192.0.2.1 vrf red
PING 192.0.2.1 (192.0.2.1) 56(84) bytes of data.
64 bytes from 192.0.2.1: icmp_seq=1 ttl=64 time=0.070 ms
64 bytes from 192.0.2.1: icmp_seq=2 ttl=64 time=0.078 ms
^C
--- 192.0.2.1 ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 4ms
rtt min/avg/max/mdev = 0.070/0.074/0.078/0.004 ms
```

traceroute vrf <name> [ipv4 | ipv6] <host>

Displays the route packets taken to a network host utilizing VRF instance identified by <name>. When using the IPv4 or IPv6 option, displays the route packets taken to the given hosts IP address family. This option is useful when the host is specified as a hostname rather than an IP address.

force vrf <name>

Join a given VRF. This will open a new subshell within the specified VRF.

The prompt is adjusted to reflect this change in both config and op-mode.

```
vyos@vyos:~$ force vrf blue
vyos@vyos(vrf:blue):~$
```


7.14.3 Example

VRF route leaking

The following example topology was built using EVE-NG.

Fig. 8: VRF route leaking

- PC1 is in the default VRF and acting as e.g. a “fileserver”
- PC2 is in VRF blue which is the development department
- PC3 and PC4 are connected to a bridge device on router R1 which is in VRF red. Say this is the HR department.
- R1 is managed through an out-of-band network that resides in VRF mgmt

Configuration

```
set interfaces bridge br10 address '10.30.0.254/24'
set interfaces bridge br10 member interface eth3
set interfaces bridge br10 member interface eth4
set interfaces bridge br10 vrf 'red'

set interfaces ethernet eth0 address 'dhcp'
set interfaces ethernet eth0 vrf 'mgmt'
set interfaces ethernet eth1 address '10.0.0.254/24'
set interfaces ethernet eth2 address '10.20.0.254/24'
set interfaces ethernet eth2 vrf 'blue'

set protocols static route 10.20.0.0/24 interface eth2 vrf 'blue'
set protocols static route 10.30.0.0/24 interface br10 vrf 'red'

set service ssh disable-host-validation
set service ssh vrf 'mgmt'

set system name-server 'eth0'

set vrf name blue protocols static route 10.0.0.0/24 interface eth1 vrf 'default'
set vrf name blue table '3000'
set vrf name mgmt table '1000'
set vrf name red protocols static route 10.0.0.0/24 interface eth1 vrf 'default'
set vrf name red table '2000'
```

VRF and NAT

Configuration

```
set interfaces ethernet eth0 address '172.16.50.12/24'
set interfaces ethernet eth0 vrf 'red'

set interfaces ethernet eth1 address '192.168.130.100/24'
set interfaces ethernet eth1 vrf 'blue'

set nat destination rule 110 description 'NAT ssh- INSIDE'
set nat destination rule 110 destination port '2022'
set nat destination rule 110 inbound-interface name 'eth0'
set nat destination rule 110 protocol 'tcp'
set nat destination rule 110 translation address '192.168.130.40'

set nat source rule 100 outbound-interface name 'eth0'
set nat source rule 100 protocol 'all'
set nat source rule 100 source address '192.168.130.0/24'
set nat source rule 100 translation address 'masquerade'

set service ssh vrf 'red'

set vrf bind-to-all
set vrf name blue protocols static route 0.0.0.0/0 next-hop 172.16.50.1 vrf 'red'
set vrf name blue protocols static route 172.16.50.0/24 interface eth0 vrf 'red'
set vrf name blue table '1010'

set vrf name red protocols static route 0.0.0.0/0 next-hop 172.16.50.1
set vrf name red protocols static route 192.168.130.0/24 interface eth1 vrf 'blue'
set vrf name red table '2020'
```

Operation

After committing the configuration we can verify all leaked routes are installed, and try to ICMP ping PC1 from PC3.

```
PCS> ping 10.0.0.1

84 bytes from 10.0.0.1 icmp_seq=1 ttl=63 time=1.943 ms
84 bytes from 10.0.0.1 icmp_seq=2 ttl=63 time=1.618 ms
84 bytes from 10.0.0.1 icmp_seq=3 ttl=63 time=1.745 ms
```

```
VPCS> show ip
NAME       : VPCS[1]
IP/MASK    : 10.30.0.1/24
GATEWAY    : 10.30.0.254
DNS        :
MAC        : 00:50:79:66:68:0f
```

VRF default routing table

```
vyos@R1:~$ show ip route
Codes: K - kernel route, C - connected, S - static, R - RIP,
       O - OSPF, I - IS-IS, B - BGP, E - EIGRP, N - NHRP,
       T - Table, v - VNC, V - VNC-Direct, A - Babel, D - SHARP,
       F - PBR, f - OpenFabric,
       > - selected route, * - FIB route, q - queued, r - rejected, b - backup

C>* 10.0.0.0/24 is directly connected, eth1, 00:07:44
S>* 10.20.0.0/24 [1/0] is directly connected, eth2 (vrf blue), weight 1, 00:07:38
S>* 10.30.0.0/24 [1/0] is directly connected, br10 (vrf red), weight 1, 00:07:38
```

VRF red routing table

```
vyos@R1:~$ show ip route vrf red
Codes: K - kernel route, C - connected, S - static, R - RIP,
       O - OSPF, I - IS-IS, B - BGP, E - EIGRP, N - NHRP,
       T - Table, v - VNC, V - VNC-Direct, A - Babel, D - SHARP,
       F - PBR, f - OpenFabric,
       > - selected route, * - FIB route, q - queued, r - rejected, b - backup

VRF red:
K>* 0.0.0.0/0 [255/8192] unreachable (ICMP unreachable), 00:07:57
S>* 10.0.0.0/24 [1/0] is directly connected, eth1 (vrf default), weight 1, 00:07:40
C>* 10.30.0.0/24 is directly connected, br10, 00:07:54
```

VRF blue routing table

```
vyos@R1:~$ show ip route vrf blue
Codes: K - kernel route, C - connected, S - static, R - RIP,
       O - OSPF, I - IS-IS, B - BGP, E - EIGRP, N - NHRP,
       T - Table, v - VNC, V - VNC-Direct, A - Babel, D - SHARP,
       F - PBR, f - OpenFabric,
       > - selected route, * - FIB route, q - queued, r - rejected, b - backup

VRF blue:
K>* 0.0.0.0/0 [255/8192] unreachable (ICMP unreachable), 00:08:00
S>* 10.0.0.0/24 [1/0] is directly connected, eth1 (vrf default), weight 1, 00:07:44
C>* 10.20.0.0/24 is directly connected, eth2, 00:07:53
```

7.15 L3VPN VRFs

L3VPN VRFs (Layer 3 Virtual Private Networks) bgpd supports for IPv4 RFC 4364 and IPv6 RFC 4659. L3VPN routes, and their associated VRF MPLS labels, can be distributed to VPN SAFI neighbors in the default, i.e., non VRF, BGP instance. VRF MPLS labels are reached using core MPLS labels which are distributed using LDP or BGP labeled unicast. bgpd also supports inter-VRF route leaking.

7.15.1 VRF Route Leaking

BGP routes may be leaked (i.e. copied) between a unicast VRF RIB and the VPN SAFI RIB of the default VRF for use in MPLS-based L3VPNs. Unicast routes may also be leaked between any VRFs (including the unicast RIB of the default BGP instance). A shortcut syntax is also available for specifying leaking from one VRF to another VRF using the default instance's VPN RIB as the intermediary. A common application of the VRF-VRF feature is to connect a customer's private routing domain to a provider's VPN service. Leaking is configured from the point of view of an individual VRF: import refers to routes leaked from VPN to a unicast VRF, whereas export refers to routes leaked from a unicast VRF to VPN.

Note: Routes exported from a unicast VRF to the VPN RIB must be augmented by two parameters:

an RD / RTLIST

Configuration for these exported routes must, at a minimum, specify these two parameters.

7.15.2 Configuration

Configuration of route leaking between a unicast VRF RIB and the VPN SAFI RIB of the default VRF is accomplished via commands in the context of a VRF address-family.

```
set vrf name <name> protocols bgp address-family <ipv4-unicast|ipv6-unicast> rd vpn  
export <asn:nn|address:nn>
```

Specifies the route distinguisher to be added to a route exported from the current unicast VRF to VPN.

```
set vrf name <name> protocols bgp address-family <ipv4-unicast|ipv6-unicast> route-target  
vpn <import|export|both> [RTLIST]
```

Specifies the route-target list to be attached to a route (export) or the route-target list to match against (import) when exporting/importing between the current unicast VRF and VPN. The RTLIST is a space-separated list of route-targets, which are BGP extended community values as described in Extended Communities Attribute.

```
set vrf name <name> protocols bgp address-family <ipv4-unicast|ipv6-unicast> label vpn  
export <0-1048575|auto>
```

Enables an MPLS label to be attached to a route exported from the current unicast VRF to VPN. If the value specified is auto, the label value is automatically assigned from a pool maintained.

```
set vrf name <name> protocols bgp address-family <ipv4-unicast|ipv6-unicast> label vpn  
allocation-mode per-nexthop
```

Select how labels are allocated in the given VRF. By default, the per-vrf mode is selected, and one label is used for all prefixes from the VRF. The per-nexthop will use a unique label for all prefixes that are reachable via the same nexthop.

```
set vrf name <name> protocols bgp address-family <ipv4-unicast|ipv6-unicast> route-map  
vpn <import|export> [route-map <name>]
```

Specifies an optional route-map to be applied to routes imported or exported between the current unicast VRF and VPN.

```
set vrf name <name> protocols bgp address-family <ipv4-unicast|ipv6-unicast>
<import|export> vpn
```

Enables import or export of routes between the current unicast VRF and VPN.

```
set vrf name <name> protocols bgp address-family <ipv4-unicast|ipv6-unicast> import vrf
<name>
```

Shortcut syntax for specifying automatic leaking from vrf VRFNAME to the current VRF using the VPN RIB as intermediary. The RD and RT are auto derived and should not be specified explicitly for either the source or destination VRF's.

```
set vrf name <name> protocols bgp address-family <ipv4-unicast|ipv6-unicast> route-map
vrf import [route-map <name>]
```

Specifies an optional route-map to be applied to routes imported from VRFs.

```
set vrf name <name> protocols bgp interface <interface> mpls forwarding
```

It is possible to permit BGP install VPN prefixes without transport labels. This configuration will install VPN prefixes originated from an e-bgp session, and with the next-hop directly connected.

7.15.3 Operation

It is not sufficient to only configure a L3VPN VRFs but L3VPN VRFs must be maintained, too. For L3VPN VRF maintenance the following operational commands are in place.

```
show bgp <ipv4|ipv6> vpn
```

Print active IPV4 or IPV6 routes advertised via the VPN SAFI.

```
BGP table version is 2, local router ID is 10.0.1.1, vrf id 0
Default local pref 100, local AS 65001
Status codes: s suppressed, d damped, h history, * valid, > best, = multipath,
i internal, r RIB-failure, S Stale, R Removed
Nexthop codes: @NNN nexthop's vrf id, < announce-nh-self
Origin codes: i - IGP, e - EGP, ? - incomplete

Network          Next Hop          Metric LocPrf Weight Path
Route Distinguisher: 10.50.50.1:1011
*>i10.50.50.0/24   10.0.0.7           0      100      0 i
UN=10.0.0.7 EC{65035:1011} label=80 type=bgp, subtype=0
Route Distinguisher: 10.60.60.1:1011
*>i10.60.60.0/24   10.0.0.10          0      100      0 i
UN=10.0.0.10 EC{65035:1011} label=80 type=bgp, subtype=0
```

```
show bgp <ipv4|ipv6> vpn summary
```

Print a summary of neighbor connections for the specified AFI/SAFI combination.

```
BGP router identifier 10.0.1.1, local AS number 65001 vrf-id 0
BGP table version 0
RIB entries 9, using 1728 bytes of memory
Peers 4, using 85 KiB of memory
Peer groups 1, using 64 bytes of memory
```

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Neighbor	V	AS	MsgRcvd	MsgSent	TblVer	InQ	OutQ	Up/Down	State/
↔ PfxRcd	PfxSnt								
10.0.0.7	4	65001	2860	2870	0	0	0	1d23h34m	↔
↔ 2	10								

OPERATION MODE

8.1 System Information

VyOS features a rich set of operational level commands to retrieve arbitrary information about your running system. For more information on the VyOS command line interface (CLI), see *Command Line Interface*.

8.2 Hardware

8.2.1 USB

In the past, serial interfaces were defined as `ttySx` and `ttyUSBx` where `x` was the instance number. However, the mapping of USB-based serial interfaces can change from one system boot to another, depending on which driver the operating system loads first. This inconsistency can be problematic when you use multiple serial interfaces. For example, both console-server connections and a serial-backed *WWAN*.

To address this issue, and because many low-cost USB-to-serial converters do not have a programmed serial number, VyOS now identifies USB-to-serial interfaces by the USB root bridge and the bus they connect to. This approach is similar to the network interface naming conventions used in recent Linux distributions.

show hardware usb

Retrieve a tree-like representation of all connected USB devices.

Note: If a device is unplugged and plugged in again, it is assigned a new `Port`, `Dev`, and `If`.

show hardware usb serial

Retrieve a list and description of all connected USB serial devices. The device name displayed, (for example `usb0b2.4p1.0`), can be used directly when accessing the serial console as console-server device.

8.3 Version

show version

Return the currently running VyOS version and build information. This includes the name of the release train, e.g., *sagitta* on VyOS 1.4, and *circinus* on VyOS 1.5.

```
vyos@vyos:~$ show version

Version:          VyOS 1.4-rolling-202106270801
Release Train:    sagitta

Built by:         autobuild@vyos.net
Built on:         Sun 27 Jun 2021 09:50 UTC
Build UUID:       ab43e735-edcb-405a-9f51-f16a1b104e52
Build Commit ID:  f544d75eab758f

Architecture:    x86_64
Boot via:         installed image
System type:      KVM guest

Hardware vendor:  QEMU
Hardware model:   Standard PC (i440FX + PIIX, 1996)
Hardware S/N:     Unknown
Hardware UUID:    Unknown

Copyright:        VyOS maintainers and contributors
```

show version kernel

Return the version number of the currently running Linux kernel.

```
vyos@vyos:~$ show version kernel
5.10.46-amd64-vyos
```

show version frr

Return the version number of FRR (Free Range Routing - <https://frrouting.org/>) used in this release. This is the routing control plane and a successor to GNU Zebra and Quagga.

```
vyos@vyos:~$ show version frr
FRRouting 7.5.1-20210625-00-gf07d935a2 (vyos).
Copyright 1996-2005 Kunihiro Ishiguro, et al.
```

8.4 Boot Options

Warning: This function can disrupt services. Run it only when necessary, and verify all input values before proceeding.

VyOS provides several kernel command-line options to modify the normal boot process. To add an option, select the desired image in the GRUB menu at load time. Type **e** to edit the first line, then type **Ctrl+X** to boot.

8.4.1 Specify custom config file

You can use a configuration file instead of the default `/config/config.boot` file. If the specified file doesn't exist or isn't readable, the system uses the default configuration file. No additional verification is performed, so specify a valid configuration file.

```
vyos-config=/path/to/file
```

To load the *factory default* configuration, use:

```
vyos-config=/opt/vyatta/etc/config.boot.default
```

8.4.2 Disable specific boot process steps

These options disable certain steps in the boot process. Understand the *boot process* before using them.

no-vyos-migrate

Do not perform config migration.

no-vyos-firewall

Do not initialize default firewall chains, renders any firewall configuration unusable.

8.5 Recovery after Failed Upgrades

Use **VyOS upgrade recovery** to restore the system to the last working version after a failed upgrade.

- *Configuration*: How to enable upgrade recovery
- *How it works*: Overview of the recovery process
- *Cancelling recovery*: Overview of the recovery process

8.5.1 Configuration

Warning: Upgrade recovery is disabled by default. To use it, **enable it first**.

To enable upgrade recovery, run the following command:

```
set system option reboot-on-upgrade-failure [timeout <min>]
```

- `timeout <min>`: The time in minutes (5 - 30) to cancel upgrade recovery before VyOS reboots. See *Cancelling Recovery*.

8.5.2 How it works

After a VyOS upgrade, the system monitors the boot process. Upon detecting a boot failure, VyOS initiates a revert to the last working version and displays the following warning:

```
Booting failed, reverting to previous image
Automatic reboot in xx minutes
Use "reboot cancel" to cancel
```

If no action is taken, the reboot happens automatically after the configured timeout. Upon successful recovery and reboot, the following message appears:

```
WARNING: Image update to "VyOS 1.5.xxxx" failed
Please check the logs:
/usr/lib/live/mount/persistence/boot/NAME/rw/var/log
Message is cleared on next reboot!
```

8.5.3 Cancelling recovery

Upon detecting a boot failure, you have the predefined timeout to cancel upgrade recovery. This is useful if you want to troubleshoot the faulty VyOS version on your own.

To cancel upgrade recovery, run the following command:

```
reboot cancel
```

8.6 Password Recovery

Restart VyOS from the console. The GRUB menu appears. Select **Boot options**.

Next, select **Select boot mode**.

Select **Password reset**.

Boot the desired VyOS version.

The standalone user password recovery tool runs and prompts you to reset the local system user password. VyOS automatically reboots after you reset your password.

```
Do you wish to reset the admin password? (y or n)
y
Which admin account do you want to reset?[vyos]
my_username
Enter my_username password:
Retype my_username password:
System will reboot in 10 seconds...
```

8.7 RAID 1

A Redundant Array of Independent Disks (RAID) uses two or more hard disk drives to improve disk speed, store more data, and/or provide fault tolerance. There are several storage schemes possible in a RAID array, each offering a different combination of storage, reliability, and performance. VyOS supports **RAID 1** deployments. RAID 1 uses two or more disks that mirror one another to provide system fault tolerance. In a RAID 1 configuration, every sector on one disk is duplicated on every sector of all disks in the array. Provided even one disk in the RAID 1 set is operational, the system continues to run, even through disk replacement (provided that the hardware supports in-service replacement of drives). RAID 1 can be implemented using special hardware or it can be implemented in software. VyOS supports software RAID 1 on two disks. The VyOS implementation of RAID 1 features the following:

- Detection and reporting of disk failure.
- Maintain system operation with one failed disk.
- Boot the system with one failed disk.
- Replace a failed disk and initiate re-mirroring.
- Monitor the status of re-mirroring.

8.7.1 Installation implications

The VyOS installation utility provides several options for installing to a RAID 1 set. You can:

- Use the install system to create the RAID 1 set.
- Use the built-in Linux commands to create a RAID 1 set before running the install system command.
- Use a previously-created RAID 1 set.

Note: Before a permanent installation, VyOS runs a live installation.

8.7.2 Configuration

Standard installation on a single disk

VyOS automatically detects the presence of two or more disks that are not currently part of a RAID array when installed. The VyOS installation utility automatically offers you the option to configure RAID 1 mirroring for eligible drives with the following prompt:

```
Would you like to configure RAID 1 mirroring on them?
```

- If you do not want to configure RAID 1 mirroring, enter **No** at the prompt.

Empty 2+ disk

If VyOS detects two identical disks that are not currently part of a RAID 1 set, the VyOS installation utility automatically offers the option to configure RAID 1 mirroring for the drives with the following prompt:

```
Would you like to configure RAID 1 mirroring on them?
```

1. To create a new RAID 1 array, enter **Yes** at the prompt. If VyOS detects a filesystem on the partitions being used for RAID 1, it will prompt you to indicate whether you want to continue creating the RAID 1 array.

```
Continue creating array?
```

2. To overwrite the old filesystem, enter **Yes**.

3. The system informs you that all data on both drives will be erased. Confirm you want to continue.

```
Are you sure you want to do this?
```

4. Enter **Yes** at the prompt to retain the current VyOS configuration. Enter **No** to delete the current VyOS configuration.

```
Would you like me to save the data on it before I delete it?
```

5. Enter **Yes** at the prompt to retain the current VyOS configuration. Enter **No** to delete the current VyOS configuration.

6. Continue installing VyOS.

Preexisting RAID 1 configuration

When VyOS detects a previously configured RAID 1 set, the installation utility displays the following prompt:

```
Would you like to use this one?
```

1. To break up the current RAID 1 set, enter **No** at the prompt. The installation utility detects that there are two identical disks and offers you the option of configuring RAID 1 mirroring with the following prompt:

```
Would you like to configure RAID 1 mirroring on them?
```

2. To decline to set up a new RAID 1 configuration on the disks, enter **No** at the prompt. VyOS prompts you to indicate which partition you would like the system installed on.

```
Which partition should I install the root on? [sda1]:
```

3. Enter the partition where you would like the system installed. The system then prompts you to indicate whether you want to save the old configuration data. This represents the current VyOS configuration.

```
Would you like me to save the data on it before I delete it?
```

4. Enter **Yes** at the prompt to retain the current VyOS configuration once installation is complete. Enter **No** to delete the current VyOS configuration.

5. Continue installing VyOS.

Detecting and replacing a failed RAID 1 disk

VyOS system detects disk failures within a RAID 1 set and reports them to the system console. You can verify the failure by running the `show raid` command.

To replace a bad disk within a RAID 1 set:

1. Remove the failed disk from the RAID 1 set:

```
delete raid <RAID-1-device> member <disk-partition>
```

where `RAID-1-device` is the name of the RAID 1 device. For example, `md0` and `disk-partition` is the name of the failed disk partition. For example, `sdb2`.

2. Physically remove the failed disk from the system. If the drives are not hot-swappable, then you must shut down the system before removing the disk.
3. Replace the failed drive with a drive of the same size or larger.
4. Format the new disk for RAID 1 by running the following command:

```
format disk <disk-device1> like <disk-device2>
```

where `disk-device1` is the replacement disk. For example, `sdb` and `disk-device2` is the existing healthy disk. For example, `sda`.

5. Add the replacement disk to the RAID 1 set by running the following command:

```
add raid <RAID-1-device> member <disk-partition>
```

where `RAID-1-device` is the name of the RAID 1 device. For example, `md0` and `disk-partition` is the name of the replacement disk partition. For example, `sdb2`.

8.7.3 Operation

Learn how to add a disk partition to a RAID 1 set, initiate mirror synchronization, and check and display information.

```
add raid <RAID-1-device> member <disk-partition>
```

Use this command to add a member disk partition to the RAID 1 set. Adding a disk partition to a RAID 1 set initiates mirror synchronization, where all data on the existing member partition is copied to the new partition.

```
format disk <disk-device1> like <disk-device2>
```

This command is typically used to prepare a disk to be added to a preexisting RAID 1 set (of which `disk-device2` is already a member).

```
show raid <RAID-1-device>
```

shows output for `show raid md0` as `sdb1` is being added to the RAID 1 set and is in the process of being resynchronized.

```
vyos@vyos:~$ show raid md0
/dev/md0:
  Version : 00.90
Creation Time : Wed Oct 29 09:19:09 2008
  Raid Level : raid1
  Array Size : 1044800 (1020.48 MiB 1069.88 MB)
Used Dev Size : 1044800 (1020.48 MiB 1069.88 MB)
  Raid Devices : 2
Total Devices : 2
```

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```

Preferred Minor : 0
Persistence : Superblock is persistent
Update Time : Wed Oct 29 19:34:23 2008
      State : active, degraded, recovering
Active Devices : 1
Working Devices : 2
Failed Devices : 0
Spare Devices : 1
Rebuild Status : 17% complete
      UUID : 981abd77:9f8c8dd8:fdbf4de4:3436c70f
      Events : 0.103
Number   Major   Minor   RaidDevice State
    0         8         1         0    active sync   /dev/sda1
    2         8        17         1    spare rebuilding /dev/sdb1

```

show disk sda format

Use this command to display the formatting of a hard disk.

```

vyos@vyos:~$ show disk sda format
Disk /dev/sda: 1073 MB, 1073741824 bytes
85 heads, 9 sectors/track, 2741 cylinders
Units = cylinders of 765 * 512 = 391680 bytes
Disk identifier: 0x000b7179
Device Boot      Start         End      Blocks   Id  System
/dev/sda1                6         2737     1044922+  fd  Linux raid autodetect

```

VYOS AUTOMATION

9.1 VyOS API

For instructions on configuring and enabling the API, see [HTTP API](#).

9.1.1 Authentication

All endpoints, except one, accept HTTP POST requests. The API key must be provided as the `key` field in the form data. The only public endpoint accepts HTTP GET requests and supports optional query parameters.

Below are examples of API requests in cURL and Python. All other code examples in this documentation use cURL.

```
curl --location --request POST 'https://vyos/retrieve' \
--form data='{ "op": "showConfig", "path": []}' \
--form key='MY-HTTPS-API-PLAINTEXT-KEY'
```

```
import requests
url = "https://vyos/retrieve"
payload={'data': '{"op": "showConfig", "path": []}',
        'key': 'MY-HTTPS-API-PLAINTEXT-KEY'
        }
headers = {}
response = requests.request("POST", url, headers=headers, data=payload)
print(response.text)
```

9.1.2 API endpoints

/info

This is the only API endpoint that does not require authentication and can be accessed by anonymous users. The info endpoint returns general system information, including the VyOS version, system hostname, and a welcome banner.

This endpoint accepts **only** HTTP GET requests.

```
curl --location --request GET 'https://vyos/info'

response
{
  "success": true,
```

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```
"data": {
  "version": "1.5-rolling",
  "hostname": "vyos",
  "banner": "Welcome to VyOS"
},
"error": null
}
```

Query parameters

This endpoint accepts two optional query parameters, version and hostname. Each parameter accepts values convertible to Boolean (e.g., yes/no, 1/0, or true/false) to control the inclusion of related fields in the response.

If no query parameters are provided, both parameters default to true.

```
curl --location --request GET 'https://vyos/info?version=1&hostname=1'

response {
  "success": true,
  "data": {
    "version": "1.5-rolling",
    "hostname": "vyos",
    "banner": "Welcome to VyOS"
  },
  "error": null
}
```

If either parameter is set to a value corresponding to false, its field is returned as an empty string in the response:

```
curl --location --request GET 'https://vyos/info?version=0&hostname=1'

response {
  "success": true,
  "data": {
    "version": "",
    "hostname": "vyos",
    "banner": "Welcome to VyOS"
  },
  "error": null
}
```

You do not need to specify both parameters if you want to hide only one. Any missing query parameter defaults to true.

```
curl --location --request GET 'https://vyos/info?hostname=no'

response {
  "success": true,
  "data": {
    "version": "1.5-rolling",
    "hostname": "",
    "banner": "Welcome to VyOS"
  },
  "error": null
}
```


Note that while you can disable output for both `hostname` and `version`, the banner is always included in the response.

Important: The endpoint accepts **ONLY** `hostname` and `version` query parameters. Including any other parameters results in an HTTP 400 Bad Request.

```
curl --location --request GET \
  'https://192.168.56.119/info?hostname=1&url=https://evilsite.com'

response {
  "success": false,
  "error": "{ 'type': 'extra_forbidden', 'loc': ('query', 'url'), 'msg': 'Extra inputs_
  ↪are not permitted', 'input': 'https://evilsite.com'}",
  "data": null
}
```

Values passed to the query string are validated to ensure they are strictly Boolean. Other data types are not accepted.

```
curl --location --request GET 'https://vyos/info?hostname=1; eval"sudo rm -rf /"'

response
{
  "success": false,
  "error": "{ 'type': 'bool_parsing', 'loc': ('query', 'hostname'), 'msg': 'Input_
  ↪should be a valid boolean, unable to interpret input', 'input': '1; eval \"sudo rm -rf_
  ↪/\",'',
  "data": null
}
```

/retrieve

The `/retrieve` endpoint returns either specific parts or the entire configuration.

To retrieve the entire configuration, pass an empty list to the `path` field.

```
curl --location --request POST 'https://vyos/retrieve' \
--form data='{ "op": "showConfig", "path": [] }' \
--form key='MY-HTTPS-API-PLAINTEXT-KEY'

response (shortened)
{
  "success": true,
  "data": {
    "interfaces": {
      "ethernet": {
        "eth0": {
          "address": "dhcp",
          "duplex": "auto",
          "hw-id": "50:00:00:01:00:00",
          "speed": "auto"
        },
        "eth1": {
```

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```

        "duplex": "auto",
        "hw-id": "50:00:00:01:00:01",
        "speed": "auto"
    },
    "error": null
}

```

To retrieve a specific configuration part, such as `system syslog`, specify the desired path.

```

curl -k --location --request POST 'https://vyos/retrieve' \
--form data='{ "op": "showConfig", "path": ["system", "syslog"] }' \
--form key='MY-HTTPS-API-PLAINTEXT-KEY'

```

```

response:
{
  "success": true,
  "data": {
    "global": {
      "facility": {
        "all": {
          "level": "info"
        },
        "protocols": {
          "level": "debug"
        }
      }
    }
  },
  "error": null
}

```

If you only need the value of a multi-valued node, use the `returnValues` operation.

For example, to get the addresses of a `dum0` interface:

```

curl -k --location --request POST 'https://vyos/retrieve' \
--form data='{ "op": "returnValues", "path": ["interfaces", "dummy", "dum0", "address"] }' \
--form key='MY-HTTPS-API-PLAINTEXT-KEY'

```

```

response:
{
  "success": true,
  "data": [
    "10.10.10.10/24",
    "10.10.10.11/24",
    "10.10.10.12/24"
  ],
  "error": null
}

```

To check whether a configuration path exists, use the `exists` operation. It returns `true` for an existing path:

```
curl -k --location --request POST 'https://vyos/retrieve' \
--form data='{ "op": "exists", "path": ["service", "https", "api"] }' \
--form key='MY-HTTPS-API-PLAINTEXT-KEY'
```

```
response:
{
  "success": true,
  "data": true,
  "error": null
}
```

It returns false for a non-existing path:

```
curl -k --location --request POST 'https://vyos/retrieve' \
--form data='{ "op": "exists", "path": ["service", "non", "existent", "path"] }' \
--form key='MY-HTTPS-API-PLAINTEXT-KEY'
```

```
response:
{
  "success": true,
  "data": false,
  "error": null
}
```

/reset

The /reset endpoint runs the reset command.

```
curl --location --request POST 'https://vyos/reset' \
--form data='{ "op": "reset", "path": ["ip", "bgp", "192.0.2.11"] }' \
--form key='MY-HTTPS-API-PLAINTEXT-KEY'
```

```
response:
{
  "success": true,
  "data": "",
  "error": null
}
```

/reboot

To initiate a reboot, use the /reboot endpoint.

```
curl --location --request POST 'https://vyos/reboot' \
--form data='{ "op": "reboot", "path": ["now"] }' \
--form key='MY-HTTPS-API-PLAINTEXT-KEY'
```

```
response:
{
  "success": true,
  "data": "",

```

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```
"error": null
}
```

/poweroff

To power off the system, use the /poweroff endpoint.

```
curl --location --request POST 'https://vyos/poweroff' \
--form data='{ "op": "poweroff", "path": ["now"] }' \
--form key='MY-HTTPS-API-PLAINTEXT-KEY'

response:
{
  "success": true,
  "data": "",
  "error": null
}
```

/image

To add or delete an image, use the /image endpoint.

To add an image:

```
curl -k --location --request POST 'https://vyos/image' \
--form data='{ "op": "add", "url": "https://downloads.vyos.io/rolling/current/amd64/vyos-
rolling-latest.iso" }' \
--form key='MY-HTTPS-API-PLAINTEXT-KEY'

response (shortened):
{
  "success": true,
  "data": "Trying to fetch ISO file from https://downloads.vyos.io/rolling-latest.iso\n
  ...
  Setting up grub configuration...\nDone.\n",
  "error": null
}
```

To delete an image, for example 1.3-rolling-202006070117:

```
curl -k --location --request POST 'https://vyos/image' \
--form data='{ "op": "delete", "name": "1.3-rolling-202006070117" }' \
--form key='MY-HTTPS-API-PLAINTEXT-KEY'

response:
{
  "success": true,
  "data": "Deleting the \"1.3-rolling-202006070117\" image...\nDone\n",
  "error": null
}
```

/show

The /show endpoint runs operational mode commands and returns the resulting output.

For example, to show the installed images:

```
curl -k --location --request POST 'https://vyos/show' \
--form data='{ "op": "show", "path": ["system", "image"] }' \
--form key='MY-HTTPS-API-PLAINTEXT-KEY'
```

response:

```
{
  "success": true,
  "data": "The system currently has the following image(s) installed:\n\n
          1: 1.4-rolling-202102280559 (default boot)\n
          2: 1.4-rolling-202102230218\n
          3: 1.3-beta-202102210443\n\n",
  "error": null
}
```

/generate

The /generate endpoint runs a generate command.

```
curl -k --location --request POST 'https://vyos/generate' \
--form data='{ "op": "generate", "path": ["pki", "wireguard", "key-pair"] }' \
--form key='MY-HTTPS-API-PLAINTEXT-KEY'
```

response:

```
{
  "success": true,
  "data": "Private key: CFZR2eyhoVZwk4n3JFPMJx3E145f1EYgDM+ubytXYVY=\n
          Public key: jjtpPT8ycI1Q0bNtrWuxAk04k88Xwzg5VHV9xGZ58lU=\n\n",
  "error": null
}
```

/configure

The /configure endpoint accepts set, delete, and comment commands.

To apply a set command:

```
curl -k --location --request POST 'https://vyos/configure' \
--form data='{ "op": "set", "path": ["interfaces", "dummy", "dum1", "address", "10.11.0.1/\n
→ 32"] }' \
--form key='MY-HTTPS-API-PLAINTEXT-KEY'
```

response:

```
{
  "success": true,
  "data": null,
  "error": null
}
```

To apply a delete command:

```
curl -k --location --request POST 'https://vyos/configure' \
--form data='{ "op": "delete", "path": ["interfaces", "dummy", "dum1", "address", "10.11.
↪0.1/32"] }' \
--form key='MY-HTTPS-API-PLAINTEXT-KEY'

response:
{
  "success": true,
  "data": null,
  "error": null
}
```

The API processes each request in a session and commits it. For components such as DHCP and PPPoE servers, IPsec, VXLAN, and other tunnels, VyOS requires the entire configuration block for a commit.

The endpoint can process multiple commands if you pass them as a list to the data field.

```
curl -k --location --request POST 'https://vyos/configure' \
--form data='[{ "op": "set", "path": ["interfaces", "vxlan", "vxlan1", "remote", "203.0.113.99
↪"] }, { "op": "set", "path": ["interfaces", "vxlan", "vxlan1", "vni", "1"] } ]' \
--form key='MY-HTTPS-API-PLAINTEXT-KEY'

response:
{
  "success": true,
  "data": null,
  "error": null
}
```

/config-file

The /config-file endpoint allows you to save, load, or merge a configuration.

If you do not specify a file during the save operation, the configuration is automatically saved to /config/config.boot.

```
curl -k --location --request POST 'https://vyos/config-file' \
--form data='{ "op": "save" }' \
--form key='MY-HTTPS-API-PLAINTEXT-KEY'

response:
{
  "success": true,
  "data": "Saving configuration to '/config/config.boot'...\nDone\n",
  "error": null
}
```

To save a running configuration to a file:

```
curl -k --location --request POST 'https://vyos/config-file' \
--form data='{ "op": "save", "file": "/config/test.config" }' \
--form key='MY-HTTPS-API-PLAINTEXT-KEY'
```

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```
response:
{
  "success": true,
  "data": "Saving configuration to '/config/test.config'...\nDone\n",
  "error": null
}
```

To load a configuration file:

```
curl -k --location --request POST 'https://vyos/config-file' \
--form data='{ "op": "load", "file": "/config/test.config" }' \
--form key='MY-HTTPS-API-PLAINTEXT-KEY'

response:
{
  "success": true,
  "data": null,
  "error": null
}
```

To merge a configuration file:

```
curl -k --location --request POST 'https://vyos/config-file' \
--form data='{ "op": "merge", "file": "/config/test.config" }' \
--form key='MY-HTTPS-API-PLAINTEXT-KEY'

response:
{
  "success": true,
  "data": null,
  "error": null
}
```

For both load and merge operations, you can pass a string in the request body. For example:

```
curl -k --location --request POST 'https://vyos/config-file' \
--form data='{ "op": "merge", "string": "interfaces {\nethernet eth1 {\naddress \"192.168.\n2.137/24\"\ndescription \"test\"\n}\n}\n"}' \
--form key='MY-HTTPS-API-PLAINTEXT-KEY'

response:
{
  "success": true,
  "data": null,
  "error": null
}
```

9.1.3 Commit-confirm

For the previous two endpoints, a `commit` command is executed automatically after a successful request operation (`set`, `delete`, `load`, `merge`, or a list of `set` and `delete` operations).

Alternatively, you can initiate a `commit-confirm`. Include the `confirm_time` field in your request and set it to an integer greater than 0.

The following example uses the JSON format for brevity, though the standard form data format is equally valid:

```
curl -k -X POST -d '{"key": "MY-HTTPS-API-PLAINTEXT-KEY", "op": "merge", "string":
↪ "interfaces {\nethernet eth1 {\naddress \"192.168.137.1/24\"\ndescription \"internal\"\n
↪n}\n}\n", "confirm_time": 1}' https://vyos/config-file
```

response:

```
{
  "success": true,
  "data": "Initialized commit-confirm; 1 minutes to confirm before reload\n",
  "error": null
}
```

If not confirmed within the specified time, the committed changes will be reverted. To confirm and keep the changes:

```
curl -k -X POST -d '{"key": "MY-HTTPS-API-PLAINTEXT-KEY", "op": "confirm"}' https://vyos/
↪config-file
```

response:

```
{
  "success": true,
  "data": "Reload timer stopped\n",
  "error": null
}
```

If the commit is not confirmed, the revert behavior is controlled by:

```
vyos@vyos# set system config-management commit-confirm action
```

Possible completions:

reload	Reload previous configuration if not confirmed
reboot	Reboot to saved configuration if not confirmed (default)

9.2 Ansible

VyOS can be configured using Ansible. To use it, install the `ansible` package and the `python3-paramiko` module.

9.2.1 Directory structure

Arrange your Ansible project directory as follows:

```
.
├── ansible.cfg
├── files
│   └── id_rsa_docker.pub
├── hosts
└── main.yml
```

9.2.2 File contents

- `ansible.cfg`

```
[defaults]
host_key_checking = no
retry_files_enabled = False
ANSIBLE_INVENTORY_UNPARSED_FAILED = true
```

- `id_rsa_docker.pub`

Contains only the SSH public key.

```
AAAAB3NzaC1yc2EAAAADAQABAAQCoDgfhQJuJRFWJijHn7ZinZ3NWp4hWVrt7HFcvn0kgtP/5PeCtMt
```

- `hosts`

Defines the target VyOS devices and the connection parameters required to reach them.

```
[vyos_hosts]
r11 ansible_ssh_host=192.0.2.11

[vyos_hosts:vars]
ansible_python_interpreter=/usr/bin/python3
ansible_user=vyos
ansible_ssh_pass=vyos
ansible_network_os=vyos
ansible_connection=network_cli
```

- `main.yml`

Defines the configuration tasks to be applied to the target VyOS devices.

```
---

- hosts: r11

  connection: network_cli
```

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```
gather_facts: 'no'

tasks:
  - name: Configure remote r11
    vyos_config:
      lines:
        - set system host-name r11
        - set system name-server 203.0.113.254
        - set service ssh disable-host-validation
        - set system login user vyos authentication public-keys docker@work type ssh-
↪rsa
        - set system login user vyos authentication public-keys docker@work key "{{
↪lookup('file', 'id_rsa_docker.pub') }}"
        - set system time-zone America/Los_Angeles
        - set interfaces ethernet eth0 description WAN
```

9.2.3 Run Ansible

To apply the configuration, use the following command:

```
$ ansible-playbook -i hosts main.yml

PLAY [r11]
↪*****

TASK [Configure remote r11]
↪*****

PLAY RECAP
↪*****

r11                                : ok=1    changed=1    unreachable=0    failed=0
↪skipped=0    rescued=0    ignored=0
```

9.3 VyOS Terraform

VyOS supports development infrastructure via Terraform and provisioning via Ansible. Terraform allows you to automate the deployment of instances on a number of cloud and virtual platforms. This section shows how to deploy VyOS on multiple platforms: AWS, Microsoft Azure, Google Cloud Platform (GCP), and VMware vSphere. For more information, see the official documentation for [Terraform](#) and [Ansible](#).

9.3.1 Terraform for VyOS

VyOS supports development infrastructure via Terraform and provisioning via Ansible. Terraform allows you to automate the process of deploying instances on many cloud and virtual platforms. In this article, we will look at using Terraform to deploy VyOS on platforms - AWS, Azure, and vSphere. For more details about Terraform, see the [Terraform introduction](#).

You will need to [install Terraform](#) before proceeding.

Structure of files in the standard Terraform project:

```
.
├── main.tf           # The main script
├── version.tf        # File for the changing version of Terraform.
├── variables.tf      # The file of all variables in "main.tf"
└── terraform.tfvars  # The value of all variables (passwords, login, IP addresses and
↳so on)
```

General commands that we will use for running Terraform scripts

```
cd /<your folder>    # go to the Terraform project
terraform init        # install all add-ons and providers (AWS, Azure, and so on)
terraform plan        # show what is changing
terraform apply       # run script
yes                  # apply running
```

9.3.2 Deploy VyOS on AWS with Terraform and Ansible

You can use Terraform to quickly deploy VyOS-based infrastructure on AWS and remove infrastructure when it's no longer needed. Additionally, you can use Ansible for provisioning.

On this page you'll learn how to:

- Create the necessary files for Terraform and Ansible.
- Use Terraform to create a single instance on AWS and use Ansible for provisioning.

Prepare to deploy VyOS with Terraform on AWS

To create a single instance and install your configuration using Terraform, Ansible, and AWS, follow these steps:

AWS

1. Create an account with AWS and get your `access_key` and `secret_key`.
2. Create a key [pair](#) and download your `.pem` key.
3. Create a security [group](#) for the new VyOS instance and open all traffic.

Terraform

1. Create an UNIX or Windows instance.
2. Download and install [Terraform](#).
3. Create a folder, for example `/root/awsterraform`:

```
mkdir /root/awsterraform
```

4. Copy all files into your Terraform project (`vyos.tf`, `var.tf`, `terraform.tfvars`, `version.tf`). See [Structure of files in Terraform for AWS](#) for more details.
5. Run the following commands:

```
cd /<your folder>
terraform init
```

Ansible

1. Create a UNIX instance whenever you need.
2. Download and install Ansible
3. Create a folder, for example `/root/aws/`.
4. Copy all files into your Ansible project (`ansible.cfg`, `instance.yml`, `mykey.pem`, and `all`). See [Structure of files in Ansible for AWS](#) for more details. You can obtain `mykey.pem` by creating a key pair in AWS and downloading your `.pem` key.

Deploy with Terraform

Run the following commands on your Terraform instance:

```
cd /<your folder>
terraform plan
terraform apply
yes
```

Create an AWS instance and check its configuration

```
root@localhost:~/awsterraform# terraform apply

Terraform used the selected providers to generate the following execution plan.
Resource actions are indicated with the following symbols:
  + create

Terraform will perform the following actions:

# aws_instance.myVy0Sec2 will be created
+ resource "aws_instance" "myVy0Sec2" {
    + ami                                = "ami-*****62c2d"
```

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```

+ arn = (known after apply)
+ associate_public_ip_address = (known after apply)
+ availability_zone = (known after apply)
+ cpu_core_count = (known after apply)
+ cpu_threads_per_core = (known after apply)
+ disable_api_stop = (known after apply)
+ disable_api_termination = (known after apply)
+ ebs_optimized = (known after apply)
+ get_password_data = false
+ host_id = (known after apply)
+ host_resource_group_arn = (known after apply)
+ iam_instance_profile = (known after apply)
+ id = (known after apply)
+ instance_initiated_shutdown_behavior = (known after apply)
+ instance_lifecycle = (known after apply)
+ instance_state = (known after apply)
+ instance_type = "t2.micro"
+ ipv6_address_count = (known after apply)
+ ipv6_addresses = (known after apply)
+ key_name = "awsterraform"
+ monitoring = (known after apply)
+ outpost_arn = (known after apply)
+ password_data = (known after apply)
+ placement_group = (known after apply)
+ placement_partition_number = (known after apply)
+ primary_network_interface_id = (known after apply)
+ private_dns = (known after apply)
+ private_ip = (known after apply)
+ public_dns = (known after apply)
+ public_ip = (known after apply)
+ secondary_private_ips = (known after apply)
+ security_groups = [
  + "awsterraformsg",
]
+ source_dest_check = true
+ spot_instance_request_id = (known after apply)
+ subnet_id = (known after apply)
+ tags = {
  + "name" = "VyOS System"
}
+ tags_all = {
  + "name" = "VyOS System"
}
+ tenancy = (known after apply)
+ user_data = (known after apply)
+ user_data_base64 = (known after apply)
+ user_data_replace_on_change = false
+ vpc_security_group_ids = (known after apply)
}

# local_file.ip will be created
+ resource "local_file" "ip" {

```

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```

+ content           = (known after apply)
+ content_base64sha256 = (known after apply)
+ content_base64sha512 = (known after apply)
+ content_md5        = (known after apply)
+ content_sha1        = (known after apply)
+ content_sha256      = (known after apply)
+ content_sha512      = (known after apply)
+ directory_permission = "0777"
+ file_permission     = "0777"
+ filename            = "ip.txt"
+ id                  = (known after apply)
}

```

```

# null_resource.SSHconnection1 will be created
+ resource "null_resource" "SSHconnection1" {
  + id = (known after apply)
}

```

```

# null_resource.SSHconnection2 will be created
+ resource "null_resource" "SSHconnection2" {
  + id = (known after apply)
}

```

Plan: 4 to add, 0 to change, 0 to destroy.

Changes to Outputs:

```

+ my_IP = (known after apply)

```

Do you want to perform these actions?

Terraform will perform the actions described above.

Only 'yes' will be accepted to approve.

Enter a value: yes

```

aws_instance.myVy0Sec2: Creating...
aws_instance.myVy0Sec2: Still creating... [10s elapsed]
aws_instance.myVy0Sec2: Still creating... [20s elapsed]
aws_instance.myVy0Sec2: Still creating... [30s elapsed]
aws_instance.myVy0Sec2: Still creating... [40s elapsed]
aws_instance.myVy0Sec2: Creation complete after 44s [id=i-09edfca15aac2fe0a]
null_resource.SSHconnection1: Creating...
null_resource.SSHconnection2: Creating...
null_resource.SSHconnection1: Provisioning with 'file'...
null_resource.SSHconnection2: Provisioning with 'remote-exec'...
null_resource.SSHconnection2 (remote-exec): Connecting to remote host via SSH...
null_resource.SSHconnection2 (remote-exec):   Host: 10.217.80.104
null_resource.SSHconnection2 (remote-exec):   User: root
null_resource.SSHconnection2 (remote-exec):   Password: true
null_resource.SSHconnection2 (remote-exec):   Private key: false
null_resource.SSHconnection2 (remote-exec):   Certificate: false
null_resource.SSHconnection2 (remote-exec):   SSH Agent: false
null_resource.SSHconnection2 (remote-exec):   Checking Host Key: false

```

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```

null_resource.SSHconnection2 (remote-exec): Target Platform: unix
local_file.ip: Creating...
local_file.ip: Creation complete after 0s [id=e8e91f2e24579cd28b92e2d152c0c24c3bf4b52c]
null_resource.SSHconnection2 (remote-exec): Connected!
null_resource.SSHconnection1: Creation complete after 0s [id=7070868940858935600]

null_resource.SSHconnection2 (remote-exec): PLAY [integration of terraform and ansible]
↳ *****

null_resource.SSHconnection2 (remote-exec): TASK [Wait 300 seconds, but only start
↳ checking after 60 seconds] *****
null_resource.SSHconnection2: Still creating... [10s elapsed]
null_resource.SSHconnection2: Still creating... [20s elapsed]
null_resource.SSHconnection2: Still creating... [30s elapsed]
null_resource.SSHconnection2: Still creating... [40s elapsed]
null_resource.SSHconnection2: Still creating... [50s elapsed]
null_resource.SSHconnection2: Still creating... [1m0s elapsed]
null_resource.SSHconnection2 (remote-exec): ok: [54.xxx.xxx.xxx]

null_resource.SSHconnection2 (remote-exec): TASK [Configure general settings for the
↳ vyos hosts group] *****
null_resource.SSHconnection2: Still creating... [1m10s elapsed]
null_resource.SSHconnection2 (remote-exec): changed: [54.xxx.xxx.xxx]

null_resource.SSHconnection2 (remote-exec): PLAY RECAP
↳ *****
null_resource.SSHconnection2 (remote-exec): 54.xxx.xxx.xxx : ok=2
↳ changed=1 unreachable=0 failed=0 skipped=0 rescued=0 ignored=0

null_resource.SSHconnection2: Creation complete after 1m16s [id=4902256962410024771]

Apply complete! Resources: 4 added, 0 changed, 0 destroyed.

Outputs:

my_IP = "54.xxx.xxx.xxx"

```

After running all the commands, your VyOS instance is deployed on AWS with your specified configuration. To delete the instance, type the following command:

```
terraform destroy
```

Troubleshooting

1. If Ansible doesn't connect via SSH to your AWS instance, verify that your SSH key is in the path `/root/aws/`. You might need to increase the timeout in `instance.yml` from 300 seconds to 500 seconds or more, depending on your location. Make sure that the security group allows access to the instance.
2. If Terraform doesn't connect via SSH to your Ansible instance, verify the correct login and password in the `VyOS.tf` file.

```
connection {
  type      = "ssh"
  user      = "root"                # open root access using login and password on_
  ↪your Ansible
  password  = var.password          # check password in the file terraform.tfvars_
  ↪isn't empty
    host    = var.host              # check the correct IP address of your Ansible_
  ↪host
}
```

Make sure Ansible can ping from Terraform.

Structure of files in Terraform for AWS

```
.
├── vyos.tf                # The main script
├── var.tf                 # The file of all variables in "vyos.tf"
├── versions.tf            # File for the changing version of Terraform.
└── terraform.tfvars      # The value of all variables (passwords, login, ip_
  ↪addresses and so on)
```

File contents of Terraform for AWS

`vyos.tf`

```
#####
# Build a VyOS VM from the Marketplace.
# Find the necessary AMI image_ in AWS.
#
# The vyos.tf script uses default values (you can change them as
# needed)
# AWS Region = "us-east-1"
# AMI         = "standard AMI of VyOS from AWS Marketplace"
# Size of VM = "t2.micro"
# AWS Region = "us-east-1"
# After deploying the AWS instance and getting an IP address, the IP address is copied_
  ↪into the file
#"ip.txt" and copied to the Ansible node for provisioning.
#####

provider "aws" {
  access_key = var.access
  secret_key = var.secret
```

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```

    region = var.region
}

variable "region" {
    default = "us-east-1"
    description = "AWS Region"
}

variable "ami" {
    default = "ami-*****3b3" # ami image please enter your
    details
    description = "Amazon Machine Image ID for VyOS"
}

variable "type" {
    default = "t2.micro"
    description = "Size of VM"
}

# my resource for VyOS

resource "aws_instance" "myVyOSec2" {
    ami = var.ami
    key_name = "awsterraform" # Please enter your
    details from 1.2 of Preparation steps for deploying VyOS on AWS
    security_groups = ["awsterraformsg"] # Please enter your
    details from 1.3 of Preparation steps for deploying VyOS on AWS
    instance_type = var.type
    tags = {
        name = "VyOS System"
    }
}

#####
# Specific variable (to getting type "terraform plan"):
# aws_instance.myVyOSec2.public_ip - the information about public IP address
# of our instance, needs for provisioning and SSH connection from Ansible
#####

output "my_IP"{
    value = aws_instance.myVyOSec2.public_ip
}

#####
# The IP address of the AWS instance is copied to the ip.txt file
# on the local Terraform system. The ip.txt file contains the public
# IP address in the format: xxx.xxx.xxx.xxx
#####

resource "local_file" "ip" {
    content = aws_instance.myVyOSec2.public_ip
    filename = "ip.txt"
}

```

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```

}

#connecting to the Ansible control node using SSH connection

#####
# The "SSHconnection1" and "SSHconnection2" steps retrieve ip.txt
# from the Terraform node and run the Ansible playbook remotely.
#####

resource "null_resource" "SSHconnection1" {
  depends_on = [aws_instance.myVyOSec2]
  connection {
    type      = "ssh"
    user      = "root"
    password  = var.password
    host      = var.host
  }

  # Copy the ip.txt file to the Ansible control node from the local
  # system
  provisioner "file" {
    source      = "ip.txt"
    destination = "/root/aws/ip.txt"                # The folder of your
    ↪ Ansible project
  }
}

resource "null_resource" "SSHconnection2" {
  depends_on = [aws_instance.myVyOSec2]
  connection {
    type      = "ssh"
    user      = "root"
    password  = var.password
    host      = var.host
  }

  # Run Ansible playbook on remote Linux OS
  provisioner "remote-exec" {
    inline = [
      "cd /root/aws/",
      "ansible-playbook instance.yml"                # more detailed in
    ↪ "File contents of Ansible for AWS"
    ]
  }
}

```

var.tf

```

variable "password" {
  description = "pass for Ansible"
  type        = string
  sensitive   = true
}

```

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```

variable "host"{
    description = "The IP of my Ansible"
    type = string
}
variable "access" {
    description = "my access_key for AWS"
    type = string
    sensitive = true
}
variable "secret" {
    description = "my secret_key for AWS"
    type = string
    sensitive = true
}

```

versions.tf

```

terraform {
  required_providers {
    aws = {
      source  = "hashicorp/aws"
      version = "~> 5.0"
    }
  }
}

```

terraform.tfvars

```

password = ""    # password for Ansible SSH
host     = ""    # IP of my Ansible
access   = ""    # access_key for AWS
secret   = ""    # secret_key for AWS

```

Structure of files in Ansible for AWS

```

.
├── group_vars
│   └── all
├── ansible.cfg
├── mykey.pem
└── instance.yml

```

File contents of Ansible for AWS

ansible.cfg

```
[defaults]
inventory = /root/aws/ip.txt
host_key_checking= False
private_key_file = /root/aws/awsterraform.pem          # check the name
remote_user=vyos
```

mykey.pem

Copy your key.pem from AWS

instance.yml

```
#####
# About tasks:
# "Wait 300 seconds, but only start checking after 60 seconds" -
# attempts SSH connection every 60 seconds until 300 seconds
# "Configure general settings for the VyOS hosts group" -
# provisions the AWS VyOS node
# Add all necessary VyOS commands under the "lines:" block
#####

- name: integration of terraform and ansible
  hosts: all
  gather_facts: 'no'

  tasks:

    - name: "Wait 300 seconds, but only start checking after 60 seconds"
      wait_for_connection:
        delay: 60
        timeout: 300

    - name: "Configure general settings for the VyOS hosts group"
      vyos_config:
        lines:
          - set system name-server xxx.xxx.xxx.xxx
        save:
          true
```

group_vars/all

```
ansible_connection: ansible.netcommon.network_cli
ansible_network_os: vyos.vyos.vyos
ansible_user: vyos
```

Source files on GitHub

All files related to deploying VyOS on AWS with Terraform and Ansible can be found in the [vyos-automation](#) repository.

9.3.3 Deploy VyOS on Microsoft Azure with Terraform and Ansible

You can use Terraform to quickly deploy VyOS-based infrastructure on Microsoft Azure (hereafter referred to as *Azure*) and remove infrastructure when it's no longer needed. Additionally, you can use Ansible for provisioning.

On this page you'll learn how to:

- Create the necessary files for Terraform and Ansible.
- Use Terraform to create a single instance on Azure and use Ansible for provisioning.

Prepare to deploy VyOS with Terraform on Azure

To create a single instance and install your configuration using Terraform, Ansible, and Azure, follow these steps:

Azure

- Create an [Azure account](#).

Terraform

1. Create an UNIX or Windows instance.
2. Download and install [Terraform](#).
3. Create the folder for example `/root/azvyos/`.

```
mkdir /root/azvyos
```

4. Copy all files into your Terraform project `"/root/azvyos"` (`vyos.tf`, `var.tf`, `terraform.tfvars`). For more details, see *Structure of files in Terraform for Azure*.
5. Log in to Azure using the command:

```
az login
```

6. Run the following commands to initialize Terraform:

```
cd /<your folder>
terraform init
```

Ansible

1. Create an UNIX instance either locally or in the cloud.
2. Download and install Ansible
3. Create a folder, for example /root/az/.
4. Copy all files into your Ansible project /root/az/ (ansible.cfg, instance.yml, all). For more details, see *Structure of files in Ansible for Azure*

Deploy with Terraform

Run the following commands on your Terraform instance:

```
cd /<your folder>
terraform plan
terraform apply
yes
```

After executing all the commands, your VyOS instance is deployed to Azure with your configuration. If you need to delete the instance, run the following command:

```
terraform destroy
```

Structure of files in Terraform for Azure

```
.
├── vyos.tf                # The main script
├── var.tf                 # File for the Terraform version.
└── terraform.tfvars      # Values for all variables (passwords,
                          # login, IP addresses, etc.)
```

File contents of Terraform for Azure

vyos.tf

```
#####
# HashiCorp Guide to Using Terraform on Azure
# This Terraform configuration will create the following:
# Resource group with a virtual network and subnet
# A VyOS server without SSH key (only login+password)
#####

# Choose a provider

provider "azurerm" {
  features {}
}

# Create a resource group. In Azure, every resource belongs to a
```

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```

# resource group.

resource "azurerm_resource_group" "azure_vyos" {
  name      = "${var.resource_group}"
  location = "${var.location}"
}

# The next resource is a Virtual Network.

resource "azurerm_virtual_network" "vnet" {
  name                = "${var.virtual_network_name}"
  location            = "${var.location}"
  address_space       = ["${var.address_space}"]
  resource_group_name = "${var.resource_group}"
}

# Build a subnet to run your VMs.

resource "azurerm_subnet" "subnet" {
  name                = "${var.prefix}subnet"
  virtual_network_name = "${azurerm_virtual_network.vnet.name}"
  resource_group_name = "${var.resource_group}"
  address_prefixes     = ["${var.subnet_prefix}"]
}

#####
# Build a VyOS VM from the Marketplace.
# To find the necessary image, use the command:
#
# az vm image list --offer vyos --all
#
# Now that you have a network, you can deploy a VyOS server.
# An Azure Virtual Machine has several components. In this example,
# you build a security group, a network interface, a public IP
# address, a storage account, and finally the VM itself. Terraform
# handles all the dependencies automatically, and each resource is
# named with user-defined variables.
#####

# Security group to allow inbound access on port 22 (SSH)

resource "azurerm_network_security_group" "vyos-sg" {
  name      = "${var.prefix}-sg"
  location  = "${var.location}"
  resource_group_name = "${var.resource_group}"

  security_rule {
    name        = "SSH"
    priority    = 100
    direction   = "Inbound"
    access      = "Allow"
  }
}

```

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```

    protocol                = "Tcp"
    source_port_range        = "*"
    destination_port_range   = "22"
    source_address_prefix    = "${var.source_network}"
    destination_address_prefix = "*"
  }
}

# A network interface.

resource "azurerm_network_interface" "vyos-nic" {
  name                = "${var.prefix}vyos-nic"
  location            = "${var.location}"
  resource_group_name = "${var.resource_group}"

  ip_configuration {
    name                = "${var.prefix}ipconfig"
    subnet_id          = "${azurerm_subnet.subnet.id}"
    private_ip_address_allocation = "Dynamic"
    public_ip_address_id = "${azurerm_public_ip.vyos-pip.id}"
  }
}

# Add a public IP address.

resource "azurerm_public_ip" "vyos-pip" {
  name                = "${var.prefix}-ip"
  location            = "${var.location}"
  resource_group_name = "${var.resource_group}"
  allocation_method   = "Dynamic"
}

# Build a virtual machine. This is a standard VyOS instance from
# Marketplace.

resource "azurerm_virtual_machine" "vyos" {
  name                = "${var.hostname}-vyos"
  location            = "${var.location}"
  resource_group_name = "${var.resource_group}"
  vm_size             = "${var.vm_size}"

  network_interface_ids = ["${azurerm_network_interface.vyos-nic.id}"]
  delete_os_disk_on_termination = "true"

  # To find information about the plan, use the command:
  # az vm image list --offer vyos --all

  plan {
    publisher = "sentrionmsl"
    name      = "vyos-1-3"
    product   = "vyos-1-2-lts-on-azure"
  }
}

```

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```

storage_image_reference {
  publisher = "${var.image_publisher}"
  offer     = "${var.image_offer}"
  sku       = "${var.image_sku}"
  version   = "${var.image_version}"
}

storage_os_disk {
  name          = "${var.hostname}-osdisk"
  managed_disk_type = "Standard_LRS"
  caching       = "ReadWrite"
  create_option  = "FromImage"
}

os_profile {
  computer_name  = "${var.hostname}"
  admin_username = "${var.admin_username}"
  admin_password = "${var.admin_password}"
}

os_profile_linux_config {
  disable_password_authentication = false
}
}

data "azurerm_public_ip" "example" {
  depends_on = ["azurerm_virtual_machine.vyos"]
  name       = "vyos-ip"
  resource_group_name = "${var.resource_group}"
}

output "public_ip_address" {
  value = data.azurerm_public_ip.example.ip_address
}

# IP of AZ instance copied to a file ip.txt in the local system.

resource "local_file" "ip" {
  content  = data.azurerm_public_ip.example.ip_address
  filename = "ip.txt"
}

# Connect to the Ansible control node via SSH

resource "null_resource" "nullremotel" {
  depends_on = ["azurerm_virtual_machine.vyos"]
  connection {
    type      = "ssh"
    user      = "root"
    password  = var.password
    host      = var.host
  }
}

```

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```
# Copy the ip.txt file to the Ansible control node from the local
# system

provisioner "file" {
    source      = "ip.txt"
    destination = "/root/az/ip.txt"
}

resource "null_resource" "nullremote2" {
    depends_on = ["azurerm_virtual_machine.vyos"]
    connection {
        type      = "ssh"
        user      = "root"
        password = var.password
        host      = var.host
    }

    # Run the Ansible playbook on the remote Linux OS

    provisioner "remote-exec" {

        inline = [
            "cd /root/az/",
            "ansible-playbook instance.yml"
        ]
    }
}
```

var.tf

```
#####
# Variables File
#
# Default values for all variables used in Terraform code.
#####

variable "resource_group" {
    description = "The name of your Azure Resource Group."
    default     = "my_resource_group"
}

variable "prefix" {
    description = "This prefix will be included in the name of some resources."
    default     = "vyos"
}

variable "hostname" {
    description = "Virtual machine hostname. Used for local hostname, DNS, and storage-
    related names."
    default     = "vyos_terraform"
```

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```

}

variable "location" {
  description = "The region where the virtual network is created."
  default     = "centralus"
}

variable "virtual_network_name" {
  description = "The name for your virtual network."
  default     = "vnet"
}

variable "address_space" {
  description = "The address space that is used by the virtual network. You can supply
↳ more than one address space. Changing this forces a new resource to be created."
  default     = "10.0.0.0/16"
}

variable "subnet_prefix" {
  description = "The address prefix to use for the subnet."
  default     = "10.0.10.0/24"
}

variable "storage_account_tier" {
  description = "Defines the storage tier. Valid options are Standard and Premium."
  default     = "Standard"
}

variable "storage_replication_type" {
  description = "Defines the replication type to use for this storage account. Valid
↳ options include LRS, GRS etc."
  default     = "LRS"
}

# The most cost-effective size

variable "vm_size" {
  description = "Specifies the size of the virtual machine."
  default     = "Standard_B1s"
}

variable "image_publisher" {
  description = "Name of the publisher of the image (az vm image list)"
  default     = "sentrionmsl"
}

variable "image_offer" {
  description = "Name of the offer (az vm image list)"
  default     = "vyos-1-2-lts-on-azure"
}

variable "image_sku" {

```

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```

    description = "Image SKU to apply (az vm image list)"
    default     = "vyos-1-3"
}

variable "image_version" {
    description = "Version of the image to apply (az vm image list)"
    default     = "1.3.3"
}

variable "admin_username" {
    description = "Administrator user name"
    default     = "vyos"
}

variable "admin_password" {
    description = "Administrator password"
    type        = string
    sensitive    = true
}

variable "source_network" {
    description = "Allow access from this network prefix. Defaults to '*'."
    default     = "*"
}

variable "password" {
    description = "pass for Ansible"
    type        = string
    sensitive    = true
}
variable "host"{
    description = "IP of my Ansible"
}

```

terraform.tfvars

```

password = "" # password for Ansible SSH
host     = "" # IP of my Ansible

```

Structure of files in Ansible for Azure

```

.
├── group_vars
│   └── all
├── ansible.cfg
└── instance.yml

```

File contents of Ansible for Azure

ansible.cfg

```
[defaults]
inventory = /root/az/ip.txt
host_key_checking= False
remote_user=vyos
```

instance.yml

```
#####
# About tasks:
# "Wait 300 seconds, but only start checking after 60 seconds" - Tries
# to make SSH connection every 60 seconds until 300 seconds.
# "Configure general settings for the VyOS hosts group" - Provision
# the Azure VyOS node.
# Add all necessary commands for VyOS under the block "lines:"
#####

- name: integration of terraform and ansible
  hosts: all
  gather_facts: 'no'

  tasks:

    - name: "Wait 300 seconds, but only start checking after 60 seconds"
      wait_for_connection:
        delay: 60
        timeout: 300

    - name: "Configure general settings for the VyOS hosts group"
      vyos_config:
        lines:
          - set system name-server xxx.xxx.xxx.xxx
        save:
          true
```

group_vars/all

```
ansible_connection: ansible.netcommon.network_cli
ansible_network_os: vyos.vyos.vyos

# user and password gets from terraform variables "admin_username" and "admin_password"
↳ in the file /root/azvyos/var.tf
ansible_user: vyos
ansible_ssh_pass: "{{ vault_vyos_ssh_pass }}"
```

Source files on GitHub

All files related to deploying VyOS on Azure with Terraform and Ansible can be found in the [vyos-automation](#) repository.

9.3.4 Deploy VyOS on Google Cloud with Terraform and Ansible

Using Terraform, you can quickly deploy VyOS-based infrastructure on Google Cloud Platform (GCP) and remove the infrastructure when it's no longer needed. Additionally, you can use Ansible for provisioning.

On this page you'll learn how to:

- Create the necessary files for Terraform and Ansible.
- Use Terraform to create a single instance on GCP and use Ansible for provisioning.

Prepare to deploy VyOS with Terraform on GCP

To create a single instance and install your configuration using Terraform, Ansible, and GCP, follow these steps:

GCP

1. Create an account with GCP and a new project.
2. Create a service account and download your key (a JSON file).

The .JSON file downloads automatically after you create it and looks like the following:

Terraform

1. Create an UNIX or Windows instance.
2. Download and install [Terraform](#).
3. Create the folder. For example, /root/google.

```
mkdir /root/google
```

4. Copy all files into your Terraform project /root/google (vyos.tf, var.tf, terraform.tfvars, mykey.json). For more details, see *Structure of files Terraform for Google Cloud*
5. Run the following commands:

```
cd /<your folder>
terraform init
```

Ansible

1. Create an UNIX instance either locally or in the cloud.
2. Download and install Ansible
3. Create the folder for example /root/google/
4. Copy all files into your Ansible project /root/google/ (ansible.cfg, instance.yml, mykey.json, and all). For more details, see [Structure of files in Ansible for Google Cloud](#)

You obtain mykey.json when you create a service account in GCP and download the key (a JSON file).

Deploy with Terraform

Run the following commands on your Terraform instance:

```
cd /<your folder>
terraform plan
terraform apply
yes
```

Create a GCP instance and check its configuration

```
# terraform apply
```

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:

```
+ create
```

Terraform will perform the following actions:

```
# google_compute_firewall.tcp_22[0] will be created
+ resource "google_compute_firewall" "tcp_22" {
  + creation_timestamp = (known after apply)
  + destination_ranges = (known after apply)
  + direction          = (known after apply)
  + enable_logging      = (known after apply)
  + id                 = (known after apply)
  + name               = "vyos-tcp-22"
  + network             = "default"
  + priority            = 1000
  + project             = "vyosproject"
  + self_link           = (known after apply)
  + source_ranges       = [
    + "0.0.0.0/0",
  ]
  + target_tags         = [
    + "vyos-deployment",
  ]

  + allow {
    + ports = [
```

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```

        + "22",
      ]
      + protocol = "tcp"
    }
  }

# google_compute_firewall.udp_500_4500[0] will be created
+ resource "google_compute_firewall" "udp_500_4500" {
  + creation_timestamp = (known after apply)
  + destination_ranges = (known after apply)
  + direction          = (known after apply)
  + enable_logging      = (known after apply)
  + id                 = (known after apply)
+ name                 = "vyos-udp-500-4500"
+ network              = "default"
+ priority             = 1000
+ project              = "vyosproject"
+ self_link            = (known after apply)
+ source_ranges        = [
  + "0.0.0.0/0",
  ]
+ target_tags          = [
  + "vyos-deployment",
  ]

+ allow {
  + ports = [
    + "500",
    + "4500",
  ]
  + protocol = "udp"
}
}

# google_compute_instance.default will be created
+ resource "google_compute_instance" "default" {
  + can_ip_forward      = true
  + cpu_platform        = (known after apply)
  + current_status      = (known after apply)
  + deletion_protection = false
  + effective_labels    = (known after apply)
  + guest_accelerator   = (known after apply)
  + id                  = (known after apply)
  + instance_id         = (known after apply)
  + label_fingerprint   = (known after apply)
  + machine_type         = "n2-highcpu-4"
  + metadata             = {
    + "enable-oslogin"    = "FALSE"
    + "serial-port-enable" = "TRUE"
    + "user-data"         = ""
  }
+ metadata_fingerprint = (known after apply)

```

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```

+ min_cpu_platform    = (known after apply)
+ name                = "vyos"
+ project              = "vyosproject"
+ self_link            = (known after apply)
+ tags_fingerprint    = (known after apply)
+ terraform_labels    = (known after apply)
+ zone                = "us-west1-a"

+ boot_disk {
  + auto_delete        = true
  + device_name         = (known after apply)
  + disk_encryption_key_sha256 = (known after apply)
  + kms_key_self_link   = (known after apply)
  + mode               = "READ_WRITE"
  + source              = (known after apply)

  + initialize_params {
    + image              = "projects/sentrium-public/global/images/vyos-1-
↪ 3-5-20231222143039"
    + labels             = (known after apply)
    + provisioned_iops   = (known after apply)
    + provisioned_throughput = (known after apply)
    + size               = (known after apply)
    + type               = (known after apply)
  }
}

+ network_interface {
  + internal_ipv6_prefix_length = (known after apply)
  + ipv6_access_type           = (known after apply)
  + ipv6_address               = (known after apply)
  + name                       = (known after apply)
  + network                    = "default"
  + network_ip                 = (known after apply)
  + nic_type                   = "GVNIC"
  + stack_type                 = (known after apply)
  + subnetwork                 = "default"
  + subnetwork_project         = (known after apply)

  + access_config {
    + nat_ip      = (known after apply)
    + network_tier = (known after apply)
  }
}

# local_file.ip will be created
+ resource "local_file" "ip" {
  + content          = (known after apply)
  + content_base64sha256 = (known after apply)
  + content_base64sha512 = (known after apply)
  + content_md5       = (known after apply)

```

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```

+ content_sha1          = (known after apply)
+ content_sha256        = (known after apply)
+ content_sha512        = (known after apply)
+ directory_permission  = "0777"
+ file_permission       = "0777"
+ filename              = "ip.txt"
+ id                    = (known after apply)
}

# null_resource.SSHconnection1 will be created
+ resource "null_resource" "SSHconnection1" {
  + id = (known after apply)
}

# null_resource.SSHconnection2 will be created
+ resource "null_resource" "SSHconnection2" {
  + id = (known after apply)
}

```

Plan: 6 to add, 0 to change, 0 to destroy.

Changes to Outputs:

```
+ public_ip_address = (known after apply)
```

Warning: Quoted references are deprecated

```

on vyos.tf line 126, in resource "null_resource" "SSHconnection1":
126:   depends_on = ["google_compute_instance.default"]

```

In this context, references are expected literally rather than in quotes. Terraform 0.11 and earlier required quotes, but quoted references are now deprecated and will be removed in a future version of Terraform. Remove the quotes surrounding this reference to silence this warning.

(and one more similar warning elsewhere)

Do you want to perform these actions?

Terraform will perform the actions described above.

Only 'yes' will be accepted to approve.

Enter a value: yes

```

google_compute_firewall.udp_500_4500[0]: Creating...
google_compute_firewall.tcp_22[0]: Creating...
google_compute_instance.default: Creating...
google_compute_firewall.udp_500_4500[0]: Still creating... [10s elapsed]
google_compute_firewall.tcp_22[0]: Still creating... [10s elapsed]
google_compute_instance.default: Still creating... [10s elapsed]
google_compute_firewall.tcp_22[0]: Creation complete after 16s [id=projects/vyosproject/
↳ global/firewalls/vyos-tcp-22]

```

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```

google_compute_firewall.udp_500_4500[0]: Creation complete after 16s [id=projects/
↳vyosproject/global/firewalls/vyos-udp-500-4500]
google_compute_instance.default: Creation complete after 20s [id=projects/vyosproject/
↳zones/us-west1-a/instances/vyos]
null_resource.SSHconnection1: Creating...
null_resource.SSHconnection2: Creating...
null_resource.SSHconnection1: Provisioning with 'file'...
null_resource.SSHconnection2: Provisioning with 'remote-exec'...
null_resource.SSHconnection2 (remote-exec): Connecting to remote host via SSH...
null_resource.SSHconnection2 (remote-exec):   Host: 10.***.***.104
null_resource.SSHconnection2 (remote-exec):   User: root
null_resource.SSHconnection2 (remote-exec):   Password: true
null_resource.SSHconnection2 (remote-exec):   Private key: false
null_resource.SSHconnection2 (remote-exec):   Certificate: false
null_resource.SSHconnection2 (remote-exec):   SSH Agent: false
null_resource.SSHconnection2 (remote-exec):   Checking Host Key: false
null_resource.SSHconnection2 (remote-exec):   Target Platform: unix
local_file.ip: Creating...
local_file.ip: Creation complete after 0s [id=7d568c3b994a018c942a3cdb952ccbf3c729d0ca]
null_resource.SSHconnection2 (remote-exec): Connected!
null_resource.SSHconnection1: Creation complete after 4s [id=5175298735911137161]

null_resource.SSHconnection2 (remote-exec): PLAY [integration of terraform and ansible]_
↳*****

null_resource.SSHconnection2 (remote-exec): TASK [Wait 300 seconds, but only start_
↳checking after 60 seconds] *****

null_resource.SSHconnection2: Still creating... [10s elapsed]
null_resource.SSHconnection2: Still creating... [20s elapsed]
null_resource.SSHconnection2: Still creating... [30s elapsed]
null_resource.SSHconnection2: Still creating... [40s elapsed]
null_resource.SSHconnection2: Still creating... [50s elapsed]
null_resource.SSHconnection2: Still creating... [1m0s elapsed]
null_resource.SSHconnection2: Still creating... [1m10s elapsed]
null_resource.SSHconnection2 (remote-exec): ok: [104.***.***.158]

null_resource.SSHconnection2 (remote-exec): TASK [Configure general settings for the_
↳vyos hosts group] *****

null_resource.SSHconnection2: Still creating... [1m20s elapsed]
null_resource.SSHconnection2 (remote-exec): changed: [104.***.***.158]

null_resource.SSHconnection2 (remote-exec): PLAY RECAP_
↳*****

null_resource.SSHconnection2 (remote-exec): 104.***.***.158           : ok=2   _
↳changed=1    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0

null_resource.SSHconnection2: Creation complete after 1m22s [id=3355727070503709742]

Apply complete! Resources: 6 added, 0 changed, 0 destroyed.

Outputs:

```

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```
public_ip_address = "104.***.***.158"
```

After running all the commands, your VyOS instance is deployed on GCP with your specified configuration. To delete the instance, type the following command:

```
terraform destroy
```

Troubleshooting

- Increase the timeout value in `instance.yml` from 300 seconds to 500 seconds or more (depends on your location). Ensure that the security group allows access to the instance.
- If Terraform doesn't connect via SSH to your Ansible instance: Check the correct login and password in the `VyOS.tf` file.

```
connection {
  type      = "ssh"
  user      = "root"                # open root access using login and password on your
  ↪ Ansible
  password  = var.password          # check password in the file terraform.tfvars isn't empty
  host      = var.host              # check the correct IP address of your Ansible host
}
```

Verify that Ansible can ping from Terraform.

Structure of files in Terraform for Google Cloud

```
.
├── vyos.tf                # The main script
├── ***.JSON               # The credential file from GCP
├── var.tf                 # The file of all variables in "vyos.tf"
├── terraform.tfvars       # The value of all variables (passwords, login, IP
↪ addresses and so on)
```

File contents of Terraform for Google Cloud

`vyos.tf`

```
#####
# Build a VyOS VM from the Marketplace
#
# After deploying the GCP instance and getting an IP address, the IP address is copied
↪ into the file
#"ip.txt" and copied to the Ansible node for provisioning.
#####

terraform {
  required_providers {
    google = {
      source = "hashicorp/google"
```

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```

    }
  }
}

provider "google" {
  project          = var.project_id
  request_timeout = "60s"
  credentials = file(var.gcp_auth_file)
}

locals {
  network_interfaces = [for i, n in var.networks : {
    network      = n,
    subnetwork   = length(var.sub_networks) > i ? element(var.sub_networks, i) : null
    external_ip  = length(var.external_ips) > i ? element(var.external_ips, i) : "NONE"
  }
]
}

resource "google_compute_instance" "default" {
  name          = var.goog_cm_deployment_name
  machine_type  = var.machine_type
  zone          = var.zone

  metadata = {
    enable-oslogin      = "FALSE"
    serial-port-enable = "TRUE"
    user-data            = var.vyos_user_data
  }
  boot_disk {
    initialize_params {
      image = var.image
    }
  }
}

can_ip_forward = true

dynamic "network_interface" {
  for_each = local.network_interfaces
  content {
    network      = network_interface.value.network
    subnetwork   = network_interface.value.subnetwork
    nic_type     = "GVNIC"
    dynamic "access_config" {
      for_each = network_interface.value.external_ip == "NONE" ? [] : [1]
      content {
        nat_ip = network_interface.value.external_ip == "EPHEMERAL" ? null : network_
↪ interface.value.external_ip
      }
    }
  }
}

```

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```

}

resource "google_compute_firewall" "tcp_22" {
  count = var.enable_tcp_22 ? 1 : 0

  name      = "${var.goog_cm_deployment_name}-tcp-22"
  network   = element(var.networks, 0)

  allow {
    ports    = ["22"]
    protocol = "tcp"
  }

  source_ranges = ["0.0.0.0/0"]

  target_tags = ["${var.goog_cm_deployment_name}-deployment"]
}

resource "google_compute_firewall" "udp_500_4500" {
  count = var.enable_udp_500_4500 ? 1 : 0

  name      = "${var.goog_cm_deployment_name}-udp-500-4500"
  network   = element(var.networks, 0)

  allow {
    ports    = ["500", "4500"]
    protocol = "udp"
  }

  source_ranges = ["0.0.0.0/0"]

  target_tags = ["${var.goog_cm_deployment_name}-deployment"]
}

output "public_ip_address" {
  value = google_compute_instance.default.network_interface[0].access_config[0].nat_ip
}

#####
#
# IP of google instance copied to a file ip.txt in local system Terraform
# ip.txt looks like:
# cat ./ip.txt
# ...
#####

resource "local_file" "ip" {
  content = google_compute_instance.default.network_interface[0].access_config[0].nat_ip
  ip
  filename = "ip.txt"
}

```

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```
#connecting to the Ansible control node using SSH connection

#####
# Steps "SSHconnection1" and "SSHconnection2" need to get file ip.txt from the terraform_
↪node and start remotely the playbook of Ansible.
#####

resource "null_resource" "SSHconnection1" {
  depends_on = ["google_compute_instance.default"]
  connection {
    type      = "ssh"
    user      = "root"
    password  = var.password
    host      = var.host
  }
}

#copying the ip.txt file to the Ansible control node from local system

  provisioner "file" {
    source      = "ip.txt"
    destination = "/root/google/ip.txt"           # The folder of your_
↪Ansible project
  }
}

resource "null_resource" "SSHconnection2" {
  depends_on = ["google_compute_instance.default"]
  connection {
    type      = "ssh"
    user      = "root"
    password  = var.password
    host      = var.host
  }
}

#command to run Ansible playbook on remote Linux OS

provisioner "remote-exec" {
  inline = [
    "cd /root/google/",
    "ansible-playbook instance.yml"               # more detailed in
↪"File contents of Ansible for Google Cloud"
  ]
}
}
```

var.tf

```
variable "image" {
  type      = string
  default   = "projects/sentrium-public/global/images/vyos-1-3-5-20231222143039"
}
```

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```

variable "project_id" {
    type = string
}

variable "zone" {
    type = string
}

#####
# You can choose a lower cost machine type than n2-highcpu-4
#####

variable "machine_type" {
    type      = string
    default   = "n2-highcpu-4"
}

variable "networks" {
    description = "The network name to attach the VM instance."
    type        = list(string)
    default     = ["default"]
}

variable "sub_networks" {
    description = "The sub network name to attach the VM instance."
    type        = list(string)
    default     = ["default"]
}

variable "external_ips" {
    description = "The external IPs assigned to the VM for public access."
    type        = list(string)
    default     = ["EPHEMERAL"]
}

variable "enable_tcp_22" {
    description = "Allow SSH traffic from the Internet"
    type        = bool
    default     = true
}

variable "enable_udp_500_4500" {
    description = "Allow IKE/IPSec traffic from the Internet"
    type        = bool
    default     = true
}

variable "vyos_user_data" {
    type      = string
    default   = ""
}

```

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```
// Marketplace requires this variable name to be declared
variable "goog_cm_deployment_name" {
  description = "VyOS Universal Router Deployment"
  type        = string
  default     = "vyos"
}

# GCP authentication file
variable "gcp_auth_file" {
  type        = string
  description = "GCP authentication file"
}

variable "password" {
  description = "pass for Ansible"
  type        = string
  sensitive   = true
}

variable "host" {
  description = "The IP of my Ansible"
  type        = string
}
```

terraform.tfvars

```
#####
# Must be filled in
#####

zone = "us-west1-a"
gcp_auth_file = "/root/**/**.json" # path of your .json file
project_id    = ""                 # the google project
password      = ""                 # password for Ansible SSH
host          = ""                 # IP of my Ansible
```

Structure of files in Ansible for Google Cloud

```
.
├── group_vars
│   └── all
├── ansible.cfg
└── instance.yml
```

File contents of Ansible for Google Cloud

ansible.cfg

```
[defaults]
inventory = /root/google/ip.txt
host_key_checking= False
remote_user=vyos
```

instance.yml

```
#####
# About tasks:
# "Wait 300 seconds, but only start checking after 60 seconds" - try to make ssh_
↳ connection every 60 seconds until 300 seconds
# "Configure general settings for the VyOS hosts group" - make provisioning into Google_
↳ Cloud VyOS node
# Add all necessary VyOS commands under the "lines:" block
#####

- name: integration of terraform and ansible
  hosts: all
  gather_facts: 'no'

  tasks:

    - name: "Wait 300 seconds, but only start checking after 60 seconds"
      wait_for_connection:
        delay: 60
        timeout: 300

    - name: "Configure general settings for the VyOS hosts group"
      vyos_config:
        lines:
          - set system name-server xxx.xxx.xxx.xxx
        save:
          true
```

group_vars/all

```
ansible_connection: ansible.netcommon.network_cli
ansible_network_os: vyos.vyos.vyos
ansible_user: vyos
ansible_ssh_pass: vyos
```

Source files on GitHub

All files related to deploying VyOS on Google Cloud Platform with Terraform and Ansible can be found in the [vyos-automation](#) repository.

9.3.5 Deploy VyOS on VMware vSphere with Terraform and Ansible

You can use Terraform to quickly deploy VyOS-based infrastructure on VMware vSphere (hereafter referred to as *vSphere*) and remove infrastructure when it's no longer needed. Additionally, you can use Ansible for provisioning.

On this page you'll learn how to:

- Create the necessary files for Terraform and Ansible.
- Use Terraform to create a single instance on vSphere and use Ansible for provisioning.

Prepare to deploy VyOS with Terraform on vSphere

To create a single instance and install your configuration using Terraform, Ansible, and vSphere, follow these steps:

vSphere

- Add all necessary data to the `terraform.tfvars` [file](#) and create resources.

Terraform

- Create an UNIX or Windows instance.
- Download and install [Terraform](#).
- Create the folder for example `/root/vsphereterraform`.

```
mkdir /root/vsphereterraform
```

- Copy all files into your Terraform project `/root/vsphereterraform` (`vyos.tf`, `var.tf`, `terraform.tfvars`, `version.tf`). For more details, see [Structure of files in Terraform for vSphere](#)
- Run the following commands:

```
cd /<your folder>
terraform init
```

Ansible

- Create an UNIX instance either locally or in the cloud.
- Download and install Ansible.
- Create the folder. For example, `/root/vsphereterraform/`.
- Copy all files into your Ansible project `/root/vsphereterraform/` (`ansible.cfg`, `instance.yml`, `all`). For more details, see [Structure of files in Ansible for vSphere](#)

Deploy with Terraform

Run the following commands on your Terraform instance:

```
cd /<your folder>
terraform plan
terraform apply
yes
```

After executing these commands, your VyOS instance is deployed to vSphere with your configuration. If you need to delete the instance, run the following command:

```
terraform destroy
```

Structure of files in Terraform for vSphere

```
.
├── vyos.tf                # The main script.
├── versions.tf            # File for Terraform version.
├── var.tf                 # File for Terraform version.
└── terraform.tfvars      # Values for all variables (passwords,
                           # login, IP addresses, etc.).
```

File contents of Terraform for vSphere

vyos.tf

```
provider "vsphere" {
  user          = var.vsphere_user
  password      = var.vsphere_password
  vsphere_server = var.vsphere_server
  allow_unverified_ssl = true
}

data "vsphere_datacenter" "datacenter" {
  name = var.datacenter
}

data "vsphere_datastore" "datastore" {
  name          = var.datastore
  datacenter_id = data.vsphere_datacenter.datacenter.id
}

data "vsphere_compute_cluster" "cluster" {
  name          = var.cluster
  datacenter_id = data.vsphere_datacenter.datacenter.id
}

data "vsphere_resource_pool" "default" {
  name          = format("%s%s", data.vsphere_compute_cluster.cluster.name, "/Resources/
↳ terraform") # set as you need
```

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```

    datacenter_id = data.vsphere_datacenter.datacenter.id
  }

data "vsphere_host" "host" {
  name          = var.host
  datacenter_id = data.vsphere_datacenter.datacenter.id
}

data "vsphere_network" "network" {
  name          = var.network_name
  datacenter_id = data.vsphere_datacenter.datacenter.id
}

# Deployment of VM from Remote OVF
resource "vsphere_virtual_machine" "vmFromRemoteOvf" {
  name                = var.remotename
  datacenter_id       = data.vsphere_datacenter.datacenter.id
  datastore_id        = data.vsphere_datastore.datastore.id
  host_system_id      = data.vsphere_host.host.id
  resource_pool_id    = data.vsphere_resource_pool.default.id
  network_interface {
    network_id = data.vsphere_network.network.id
  }
  wait_for_guest_net_timeout = 2
  wait_for_guest_ip_timeout  = 2

  ovf_deploy {
    allow_unverified_ssl_cert = true
    remote_ovf_url            = var.url_ova
    disk_provisioning         = "thin"
    ip_protocol                = "IPv4"
    ip_allocation_policy      = "dhcpPolicy"
    ovf_network_map = {
      "Network 1" = data.vsphere_network.network.id
      "Network 2" = data.vsphere_network.network.id
    }
  }
}

vapp {
  properties = {
    "password"      = "12345678",
    "local-hostname" = "terraform_vyos"
  }
}

output "ip" {
  description = "default ip address of the deployed VM"
  value       = vsphere_virtual_machine.vmFromRemoteOvf.default_ip_address
}

# IP of vSphere instance copied to a file ip.txt in local system

```

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```

resource "local_file" "ip" {
  content = vsphere_virtual_machine.vmFromRemoteOvf.default_ip_address
  filename = "ip.txt"
}

#Connecting to the Ansible control node using SSH connection

resource "null_resource" "nullremote1" {
  depends_on = ["vsphere_virtual_machine.vmFromRemoteOvf"]
  connection {
    type      = "ssh"
    user      = "root"
    password  = var.ansiblepassword
    host      = var.ansiblehost
  }
}

# Copying the ip.txt file to the Ansible control node from local system

provisioner "file" {
  source      = "ip.txt"
  destination = "/root/vsphere/ip.txt"
}

resource "null_resource" "nullremote2" {
  depends_on = ["vsphere_virtual_machine.vmFromRemoteOvf"]
  connection {
    type      = "ssh"
    user      = "root"
    password  = var.ansiblepassword
    host      = var.ansiblehost
  }
}

# Command to run ansible playbook on remote Linux OS

provisioner "remote-exec" {

  inline = [
    "cd /root/vsphere/",
    "ansible-playbook instance.yml"
  ]
}

```

versions.tf

```

# Copyright (c) HashiCorp, Inc.
# SPDX-License-Identifier: MPL-2.0

terraform {
  required_providers {

```

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```

    vsphere = {
        source = "hashicorp/vsphere"
        version = "2.4.0"
    }
}
}

```

var.tf

```

# Copyright (c) HashiCorp, Inc.
# SPDX-License-Identifier: MPL-2.0

variable "vsphere_server" {
    description = "vSphere server"
    type        = string
}

variable "vsphere_user" {
    description = "vSphere username"
    type        = string
}

variable "vsphere_password" {
    description = "vSphere password"
    type        = string
    sensitive   = true
}

variable "datacenter" {
    description = "vSphere data center"
    type        = string
}

variable "cluster" {
    description = "vSphere cluster"
    type        = string
}

variable "datastore" {
    description = "vSphere datastore"
    type        = string
}

variable "network_name" {
    description = "vSphere network name"
    type        = string
}

variable "host" {
    description = "Name of your host"
    type        = string
}

```

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```

variable "remotename" {
    description = "The name of your VM"
    type        = string
}

variable "url_ova" {
    description = "The URL to the .OVA file or cloud storage"
    type        = string
}

variable "ansiblepassword" {
    description = "Ansible password"
    type        = string
}

variable "ansiblehost" {
    description = "Ansible host name or IP"
    type        = string
}

```

terraform.tfvars

```

vsphere_user      = ""
vsphere_password  = ""
vsphere_server    = ""
datacenter        = ""
datastore         = ""
cluster           = ""
network_name      = ""
host              = ""
url_ova           = ""
ansiblepassword   = ""
ansiblehost       = ""
remotename        = ""

```

Structure of files in Ansible for vSphere

```

.
├── group_vars
│   └── all
├── ansible.cfg
└── instance.yml

```


File contents of Ansible for vSphere

ansible.cfg

```
[defaults]
inventory = /root/vsphere/ip.txt
host_key_checking= False
remote_user=vyos
```

instance.yml

```
#####
# About tasks:
# "Wait 300 seconds, but only start checking after 60 seconds" - try to make ssh_
↳ connection every 60 seconds until 300 seconds
# "Configure general settings for the VyOS hosts group" - make provisioning into vSphere_
↳ VyOS node
# You have to add all necessary commands of VyOS under the block "lines:"
#####

- name: integration of terraform and ansible
  hosts: all
  gather_facts: 'no'

  tasks:

    - name: "Wait 300 seconds, but only start checking after 60 seconds"
      wait_for_connection:
        delay: 60
        timeout: 300
    - name: "Configure general settings for the VyOS hosts group"
      vyos_config:
        lines:
          - set system name-server 192.0.2.1
        save:
          true
```

group_vars/all

```
ansible_connection: ansible.netcommon.network_cli
ansible_network_os: vyos.vyos.vyos

# user and password gets from terraform variables "admin_username" and "admin_password"
ansible_user: vyos
# get from vyos.tf "vapp"
ansible_ssh_pass: 12345678
```

Source files on GitHub

All files related to deploying VyOS on vSphere with Terraform and Ansible can be found in the [vyos-automation](#) repository.

9.4 NAPALM VyOS driver

VyOS can be configured using the [NAPALM VyOS driver](#), which enables you to retrieve device data and apply configurations via SSH.

Note: The `napalm-vyos` module is currently in testing.

To use the NAPALM VyOS driver, install the following packages:

```
apt install python3-pip
pip3 install napalm
pip3 install napalm-vyos
```

9.4.1 Retrieve device data

The following script connects to a VyOS device, retrieves device facts and the ARP table, and prints the output in JSON format.

```
#!/usr/bin/env python3

import json
from napalm import get_network_driver

driver = get_network_driver('vyos')

vyos_router = driver(
    hostname="192.0.2.1",
    username="vyos",
    password="vyospass",
    optional_args={"port": 22},
)

vyos_router.open()
output = vyos_router.get_facts()
print(json.dumps(output, indent=4))

output = vyos_router.get_arp_table()
print(json.dumps(output, indent=4))

vyos_router.close()
```

Output:

```
$ ./vyos-napalm.py
{
```

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```

    "uptime": 7185,
    "vendor": "VyOS",
    "os_version": "1.3.0-rc5",
    "serial_number": "",
    "model": "Standard PC (Q35 + ICH9, 2009)",
    "hostname": "r4-1.3",
    "fqdn": "vyos.local",
    "interface_list": [
        "eth0",
        "eth1",
        "eth2",
        "lo",
        "vtun10"
    ]
}
[
    {
        "interface": "eth1",
        "mac": "52:54:00:b2:38:2c",
        "ip": "192.0.2.2",
        "age": 0.0
    },
    {
        "interface": "eth0",
        "mac": "52:54:00:a2:b9:5b",
        "ip": "203.0.113.11",
        "age": 0.0
    }
]

```

9.4.2 Apply a configuration

To apply a configuration using NAPALM VyOS driver, you will need a file with configuration commands (`commands.conf`) and a script that executes and commits them (`vyos-napalm.py`).

- `commands.conf`

```

set service ssh disable-host-validation
set service ssh port '2222'
set system name-server '192.0.2.8'
set system name-server '203.0.113.8'
set interfaces ethernet eth1 description 'F00'

```

- `vyos-napalm.py`

```

#!/usr/bin/env python3

from napalm import get_network_driver

driver = get_network_driver('vyos')

vyos_router = driver(

```

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```

    hostname="192.0.2.1",
    username="vyos",
    password="vyospass",
    optional_args={"port": 22},
)

vyos_router.open()
vyos_router.load_merge_candidate(filename='commands.conf')
diffs = vyos_router.compare_config()

if bool(diffs) == True:
    print(diffs)
    vyos_router.commit_config()
else:
    print('No configuration changes to commit')
    vyos_router.discard_config()

vyos_router.close()

```

Output:

```

$./vyos-napalm.py
[edit interfaces ethernet eth1]
+description F00
[edit service ssh]
+disable-host-validation
+port 2222
[edit system]
+name-server 192.0.2.8
+name-server 203.0.113.8
[edit]

```

9.5 Netmiko

VyOS can be configured using [Netmiko](#). To use Netmiko, install the `python3-netmiko` module.

9.5.1 Example

The following script connects to a VyOS device, applies configuration changes, commits them, and runs an operational mode command to verify the updated configuration.

```

#!/usr/bin/env python3

from netmiko import ConnectHandler

vyos_router = {
    "device_type": "vyos",
    "host": "192.0.2.1",
    "username": "vyos",

```

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```

    "password": "vyospass",
    "port": 22,
}

net_connect = ConnectHandler(**vyos_router)

config_commands = [
    'set interfaces ethernet eth0 description WAN',
    'set interfaces ethernet eth1 description LAN',
]

# set configuration
output = net_connect.send_config_set(config_commands, exit_config_mode=False)
print(output)

# commit configuration
output = net_connect.commit()
print(output)

# operational mode commands
output = net_connect.send_command("run show interfaces")
print(output)

```

Output

```

$ ./vyos-netmiko.py
configure
set interfaces ethernet eth0 description WAN
[edit]
vyos@r4-1.5# set interfaces ethernet eth1 description LAN
[edit]
vyos@r4-1.5#
commit
[edit]
vyos@r4-1.5#
Codes: S - State, L - Link, u - Up, D - Down, A - Admin Down
Interface      IP Address      S/L  Description
-----
eth0           203.0.113.1/24  u/u   WAN
eth1           192.0.2.1/30    u/u   LAN
eth2           -               u/u
lo             127.0.0.1/8     u/u
              ::1/128
vtun10         10.10.0.1/24    u/u
[edit]

```

9.6 Salt

VyOS supports op-mode and configuration via [salt](#).

Without proxy it requires VyOS minion configuration and supports op-mode data:

```
set service salt-minion id 'r14'
set service salt-minion master '192.0.2.250'
```

Check salt-keys on the salt master

```
/ # salt-key --list-all
Accepted Keys:
r11
Denied Keys:
Unaccepted Keys:
r14
Rejected Keys:
```

Accept minion key

```
/ # salt-key --accept r14
The following keys are going to be accepted:
Unaccepted Keys:
r14
Proceed? [n/Y] y
Key for minion r14 accepted.
```

Check that salt master can communicate with minions

```
/ # salt '*' test.ping
r14:
    True
r11:
    True
```

At this step we can get some op-mode information from VyOS nodes:

```
/ # salt '*' network.interface eth0
r11:
  |_
  -----
  address:
    192.0.2.11
  broadcast:
    192.0.2.255
  label:
    eth0
  netmask:
    255.255.255.0
r14:
  |_
  -----
  address:
```

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```

        192.0.2.14
    broadcast:
        192.0.2.255
    label:
        eth0
    netmask:
        255.255.255.0

/ # salt r14 network.arp
r14:
-----
aa:bb:cc:dd:f3:db:
    192.0.2.1
aa:bb:cc:dd:2e:80:
    203.0.113.1

```

9.6.1 Netmiko-proxy

It is possible to configure VyOS via [netmiko](#) proxy module. It requires a minion with installed `packet python3-netmiko` module who has a connection to VyOS nodes. Salt-minion have to communicate with salt master

Configuration

Salt master configuration:

```

/ # cat /etc/salt/master
file_roots:
    base:
        - /srv/salt/states

pillar_roots:
    base:
        - /srv/salt/pillars

```

Structure of `/srv/salt`:

```

/ # tree /srv/salt/
/srv/salt/
|___ pillars
|   |__ r11-proxy.sls
|   |__ top.sls
|___ states
|   |__ commands.txt

```

`top.sls`

```

/ # cat /srv/salt/pillars/top.sls
base:
    r11-proxy:
        - r11-proxy

```

r11-proxy.sls Includes parameters for connecting to salt-proxy minion

```
/ # cat /srv/salt/pillars/r11-proxy.sls
proxy:
  proxytype: netmiko # how to connect to proxy minion, change it
  device_type: vyos #
  host: 192.0.2.250
  username: user
  password: secret_passwd
```

commands.txt

```
/ # cat /srv/salt/states/commands.txt
set interfaces ethernet eth0 description 'WAN'
set interfaces ethernet eth1 description 'LAN'
```

Check that proxy minion is alive:

```
/ # salt r11-proxy test.ping
r11-proxy:
  True
/ #
```

Examples

Example of op-mode:

```
/ # salt r11-proxy netmiko.send_command 'show interfaces ethernet eth0 brief' host=192.0.
↳ 2.14 device_type=vyos username=vyos password=vyos
r11-proxy:
  Codes: S - State, L - Link, u - Up, D - Down, A - Admin Down
  Interface      IP Address      S/L  Description
  -----
  eth0           192.0.2.14/24   u/u  Upstream
/ #
```

Example of configuration:

```
/ # salt r11-proxy netmiko.send_config config_commands=['set interfaces ethernet eth0
↳ description Link_to_WAN'] commit=True host=192.0.2.14 device_type=vyos username=vyos
↳ password=vyos
r11-proxy:
  configure
  set interfaces ethernet eth0 description Link_to_WAN
  [edit]
  vyos@r14# commit
  [edit]
  vyos@r14#
/ #
```

Example of configuration commands from the file “/srv/salt/states/commands.txt”


```
/ # salt r11-proxy netmiko.send_config config_file=salt://commands.txt commit=True_
↪host=192.0.2.11 device_type=vyos username=vyos password=vyos
r11-proxy:
  configure
  set interfaces ethernet eth0 description 'WAN'
  [edit]
vyos@r1# set interfaces ethernet eth1 description 'LAN'
  [edit]
vyos@r1# commit
  [edit]
vyos@r1#
/ #
```

9.7 Command scripting

VyOS supports executing configuration and operational commands non-interactively from shell scripts.

To include VyOS-specific functions and aliases, source the `/opt/vyatta/etc/functions/script-template` file at the beginning of your script.

```
#!/bin/vbash
source /opt/vyatta/etc/functions/script-template
exit
```

9.7.1 Script execute permissions

Simply placing script files in `/config/scripts/` does not mean the system can execute them.

To make your scripts executable, grant them **execute permissions**. Use the following command:

```
chmod +x /config/scripts/script-name.sh
```

9.7.2 Run configuration commands

In scripts, present configuration commands as in a standard configuration session.

For example, to disable a BGP peer during a VRRP transition to the backup state, use the following syntax:

```
#!/bin/vbash
source /opt/vyatta/etc/functions/script-template
configure
set protocols bgp system-as 65536
set protocols bgp neighbor 192.168.2.1 shutdown
commit
exit
```

9.7.3 Run operational commands

In scripts, **always** prefix operational commands with `run`.

```
#!/bin/vbash
source /opt/vyatta/etc/functions/script-template
run show interfaces
exit
```

9.7.4 Run commands remotely

You can execute multiple **operational commands** on a remote VyOS system by passing a script block over SSH.

```
ssh 192.0.2.1 'vbash -s' <<EOF
source /opt/vyatta/etc/functions/script-template
run show interfaces
exit
EOF
```

Example output:

```
Welcome to VyOS
Codes: S - State, L - Link, u - Up, D - Down, A - Admin Down
Interface      IP Address      S/L  Description
-----
eth0           192.0.2.1/24    u/u
lo             127.0.0.1/8     u/u
:::1/128
```

9.7.5 Other script languages

If you use a scripting language other than bash, configure your script to output the relevant commands, and then source that output into a bash script.

The following example demonstrates this two-step process:

```
#!/usr/bin/env python3
print("delete firewall group address-group somehosts")
print("set firewall group address-group somehosts address '192.0.2.3'")
print("set firewall group address-group somehosts address '203.0.113.55'")
```

```
#!/bin/vbash
source /opt/vyatta/etc/functions/script-template
configure
source <(/config/scripts/setfirewallgroup.py)
commit
```

9.7.6 Execute configuration scripts

In Linux, it is common practice to prefix system commands with `sudo`.

In VyOS, if you prefix a script that modifies the configuration with `sudo` (see the code snippet below), subsequent manual configuration changes fail with the `Set failed` error. Recovery requires a system reboot.

```
sudo ./myscript.sh # Modifies config
configure
set ... # Any configuration parameter
```

To avoid this issue, run scripts under the `vyattacfg` group using the `sg` command:

```
sg vyattacfg -c ./myscript.sh
```

To ensure the script is executed under the `vyattacfg` group, safeguard it as follows:

```
if [ "$(id -g -n)" != 'vyattacfg' ] ; then
    exec sg vyattacfg -c "/bin/vbash $(readlink -f $0) $@"
fi
```

9.7.7 Executing pre-hooks/post-hooks scripts

VyOS allows you to run custom scripts **before** and **after** each commit.

Place your custom scripts in the following default directories:

```
/config/scripts/commit/pre-hooks.d - Directory with scripts that run before
                                     each commit.

/config/scripts/commit/post-hooks.d - Directory with scripts that run after
                                     each commit.
```

Scripts run in alphabetical order. Filenames must consist only of ASCII letters (upper and lowercase), digits (0-9), underscores (`_`), and hyphens (`-`). No other characters are allowed.

Note: Custom scripts are executed **without** root privileges. Prefix specific commands with `sudo` in your script when required.

The following example shows the output after executing a post-hook script that runs the `show interfaces` command:

```
vyos@vyos# set interfaces ethernet eth1 address 192.0.2.3/24
vyos@vyos# commit
Codes: S - State, L - Link, u - Up, D - Down, A - Admin Down
Interface      IP Address      S/L  Description
-----
eth0           198.51.100.10/24 u/u
eth1           192.0.2.3/24    u/u
eth2           -               u/u
eth3           -               u/u
lo             203.0.113.5/24  u/u
```

9.7.8 Preconfig script on boot

VyOS runs `/config/scripts/vyos-preconfig-bootup.script` at boot, **before** the system configuration is applied.

Use this script to apply **pre-configuration** workarounds for unresolved bugs or enhancements not yet available in VyOS.

The default script contains the following:

```
#!/bin/sh
# This script is executed at boot time before VyOS configuration is applied.
# Any modifications required to work around unfixed bugs or use
# services not available through the VyOS CLI system can be placed here.
```

9.7.9 Postconfig script on boot

VyOS runs `/config/scripts/vyos-postconfig-bootup.script` at boot, **after** the system configuration is applied.

Use this script to apply **post-configuration** workarounds for unresolved bugs or enhancements not yet available in VyOS.

The default script contains the following:

```
#!/bin/sh
# This script is executed at boot time after VyOS configuration is fully
# applied. Any modifications required to work around unfixed bugs or use
# services not available through the VyOS CLI system can be placed here.
```

Warning: For configuration or upgrade management issues, modify this script only as a last resort. Always try CLI-based solutions first.

9.8 PyVyOS

PyVyOS is a Python library for configuring and managing VyOS devices through their API.

Key resources:

- [Documentation](#): Provides installation, configuration, and usage instructions.
- [GitHub repository](#): Hosts the source code.
- [PyPI](#): Hosts distribution packages for installation via the Python package installer (`pip`).

9.8.1 Installation

To install PyVyOS via pip, run:

```
pip install pyvyos
```

9.8.2 Getting started

Import and disable warnings for `verify=false`

```
import urllib3
urllib3.disable_warnings()
```

Use API response class

```
@dataclass
class ApiResponse:
    status: int
    request: dict
    result: dict
    error: str
```

Initialize a `VyDevice` object

```
from dotenv import load_dotenv
load_dotenv()

hostname = os.getenv('VYDEVICE_HOSTNAME')
apikey = os.getenv('VYDEVICE_APIKEY')
port = os.getenv('VYDEVICE_PORT')
protocol = os.getenv('VYDEVICE_PROTOCOL')
verify_ssl = os.getenv('VYDEVICE_VERIFY_SSL')

verify = verify_ssl.lower() == "true" if verify_ssl else True

device = VyDevice(hostname=hostname, apikey=apikey, port=port, protocol=protocol,
    ↪ verify=verify)
```

9.8.3 Use PyVyOS

Configure, then set

```
response = device.configure_set(path=["interfaces", "ethernet", "eth0", "address", "192.
    ↪ 168.1.1/24"])
if not response.error:
    print(response.result)
```

Configure, then show a single object value

```
response = device.retrieve_return_values(path=["interfaces", "dummy", "dum1", "address"])
print(response.result)
```

Configure, then show object

```
response = device.retrieve_show_config(path=[])
if not response.error:
    print(response.result)
```

Configure, then delete object

```
response = device.configure_delete(path=["interfaces", "dummy", "dum1"])
```

Configure, then save

```
response = device.config_file_save()
```

Configure, then save file

```
response = device.config_file_save(file="/config/test300.config")
```

Show object

```
response = device.show(path=["system", "image"])
print(response.result)
```

Generate object

```
randstring = ''.join(random.choice(string.ascii_letters + string.digits) for _ in
↳ range(20))
keyrand = f'/tmp/key_{randstring}'
response = device.generate(path=["ssh", "client-key", keyrand])
```

Reset object

```
response = device.reset(path=["conntrack-sync", "internal-cache"])
if not response.error:
    print(response.result)
```

Configure, then load file

```
response = device.config_file_load(file="/config/test300.config")
```

9.9 Go-VyOS

Go-VyOS is a Go library for configuring and managing VyOS devices through their API.

- [GitHub repository](#): Hosts the source code.
- [Documentation](#): Provides the complete API reference, including available types, functions, and methods.

9.9.1 Installation

To install Go-VyOS, run:

```
go install "github.com/ganawaj/go-vyos/vyos"
```

9.9.2 Getting started

Import and disable TLS verification

```
import "github.com/ganawaj/go-vyos/vyos"
client := vyos.NewClient(nil).WithToken("AUTH_KEY").WithURL("https://192.168.0.1").
↳ Insecure()
```

Initialize a VyDevice object

```
import (
    "github.com/ganawaj/go-vyos/vyos"
    "os"
)

hostname := os.Getenv("VYDEVICE_HOSTNAME")
port := os.Getenv("VYDEVICE_PORT")
url := fmt.Sprintf("https://%s:%s", hostname, port)

apikey := os.Getenv("VYDEVICE_APIKEY")
verify_ssl := os.Getenv("VYDEVICE_VERIFY_SSL")
```

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```

client := vyos.NewClient(nil).WithToken(apikey).WithURL(url)

if verify_ssl == "false" {
    client = client.Insecure()
}

```

9.9.3 Use Go-VyOS

Configure, then set

```

out, resp, err := c.Conf.Set(ctx, "interfaces ethernet eth0 address 192.168.1.1/24")
if err != nil {
    panic(fmt.Sprintf("Error: %v", err))
}

fmt.Println(out.Success)

```

Show a single object value

```

out, resp, err := c.Show.Do(ctx, "interfaces dummy dum1 address")
if err != nil {
    panic(fmt.Sprintf("Error: %v", err))
}

fmt.Println(out.Success)
fmt.Printf("Data: %v\n", out.Data)

```

Configure, then show object

```

out, resp, err := c.Conf.Get(ctx, "interfaces dummy dum1", nil)
if err != nil {
    panic(fmt.Sprintf("Error: %v", err))
}

fmt.Println(out.Success)
fmt.Printf("Data: %v\n", out.Data)

```

Configure, then show multivalue object

```

options := RetrieveOptions{
    Multivalue: true,
}

out, resp, err := c.Conf.Get(ctx, "interfaces dummy dum1", options)
if err != nil {
    panic(fmt.Sprintf("Error: %v", err))
}

```

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```
}  
  
fmt.Println(out.Success)
```

Configure, then delete object

```
out, resp, err := c.Conf.Delete(ctx, "interfaces dummy dum1")  
if err != nil {  
    panic(fmt.Sprintf("Error: %v", err))  
}  
  
fmt.Println(out.Success)
```

Configure, then save

```
out, resp, err := c.Conf.Save(ctx, "")  
  
if err != nil {  
    panic(fmt.Sprintf("Error: %v", err))  
}  
  
fmt.Println(out.Success)
```

Configure, then save file

```
out, resp, err := c.Conf.Save(ctx, "/config/test300.config")  
  
if err != nil {  
    panic(fmt.Sprintf("Error: %v", err))  
}  
  
fmt.Println(out.Success)
```

Show object

```
out, resp, err := c.Show.Do(ctx, "system image")  
if err != nil {  
    panic(fmt.Sprintf("Error: %v", err))  
}  
  
fmt.Println(out.Success)  
fmt.Printf("Data: %v\n", out.Data)
```

Generate object

```
out, resp, err := c.Generate.Do(ctx, "pki wireguard key-pair")
if err != nil {
    panic(fmt.Sprintf("Error: %v", err))
}

fmt.Println(out.Success)
fmt.Printf("Data: %v\n", out.Data)
```

Reset object

```
out, resp, err := c.Reset.Do(ctx, "ip bgp 192.0.2.11")
if err != nil {
    panic(fmt.Sprintf("Error: %v", err))
}

fmt.Println(out.Success)
fmt.Printf("Data: %v\n", out.Data)
```

Configure, then load file

```
out, resp, err := c.ConfigFile.Load(ctx, "/config/test300.config")
```

TROUBLESHOOTING

Sometimes things break or don't work as expected. This section describes several troubleshooting tools provided by VyOS that can help when something goes wrong.

10.1 Connectivity Tests

10.1.1 Basic Connectivity Tests

Verifying connectivity can be done with the familiar ping and traceroute commands. The options for each are shown (the options for each command were displayed using the built-in help as described in the *Command Line Interface* section and are omitted from the output here):

ping <destination>

Send ICMP echo requests to destination host. There are multiple options to ping, including VRF support.

```
vyos@vyos:~$ ping 10.1.1.1
Possible completions:
<Enter>          Execute the current command
adaptive         Ping options
allow-broadcast
audible
bypass-route
count
deadline
do-not-fragment
flood
interface
interval
mark
no-loopback
numeric
pattern
quiet
record-route
size
timestamp
tos
ttl
verbose
vrf
```

ping tcp <host> port <port> [count <n>] [interface <ifname>] [vrf <name>] [source-address <address>]

Test TCP connectivity to a destination host and port. This is useful when ICMP is blocked or when you need to verify that a specific TCP service is reachable.

The **count** option sets the number of TCP connection attempts. Use **interface** to send probes through a specific interface, **vrf** to run the test in a VRF, and **source-address** to select the local source address.

Example:

```
vyos@vyos:~$ ping tcp 10.1.1.1 port 443 count 3
vyos@vyos:~$ ping tcp example.com port 80 interface eth0
vyos@vyos:~$ ping tcp 192.0.2.10 port 179 vrf CUSTOMER source-address 192.0.2.1
```

traceroute <destination>

Trace path to target.

```
vyos@vyos:~$ traceroute
Possible completions:
<hostname>      Track network path to specified node
<x.x.x.x>
<h:h:h:h:h:h:h>
ipv4             Track network path to <hostname|IPv4 address>
ipv6             Track network path to <hostname|IPv6 address>
```

10.1.2 Advanced Connectivity Tests

monitor traceroute <destination>

However, another helper is available which combines ping and traceroute into a single tool. An example of its output is shown:

```
vyos@vyos:~$ mtr 10.62.212.12

                My traceroute  [v0.85]
vyos (0.0.0.0)
Keys:  Help    Display mode    Restart statistics   Order of fields    quit
                Packets
Host      Loss%  Snt   Last   Avg    Best  Wrst  StDev
1. 10.11.110.4      0.0%   34    0.5    0.5    0.4    0.8    0.1
2. 10.62.255.184    0.0%   34    1.1    1.0    0.9    1.4    0.1
3. 10.62.255.71     0.0%   34    1.4    1.4    1.3    2.0    0.1
4. 10.62.212.12     0.0%   34    1.6    1.6    1.6    1.7    0.0
```

Note: The output consumes the screen and will replace your command prompt.

Several options are available for changing the display output. Press **h** to invoke the built in help system. To quit, just press **q** and you'll be returned to the VyOS command prompt.

10.1.3 IPv6 Topology Discovery

IPv6 uses different techniques to discover its Neighbors/topology.

Router Discovery

force ipv6-rd interface <interface> [address <ipv6-address>]

Discover routers via eth0.

Example:

```
vyos@vyos:~$ force ipv6-rd interface eth0
Soliciting ff02::2 (ff02::2) on eth0...

Hop limit           :           60 (      0x3c)
Stateful address conf. :           No
Stateful other conf.  :           No
Mobile home agent    :           No
Router preference     :           high
Neighbor discovery proxy :           No
Router lifetime       :       1800 (0x00000708) seconds
Reachable time        : unspecified (0x00000000)
Retransmit time       : unspecified (0x00000000)
Prefix               : 240e:fe:8ca7:ea01::/64
On-link              :           Yes
Autonomous address conf.:           Yes
Valid time           :       2592000 (0x00278d00) seconds
Pref. time           :       14400 (0x00003840) seconds
Prefix              : fc00:470:f1cd:101::/64
On-link              :           Yes
Autonomous address conf.:           Yes
Valid time           :       2592000 (0x00278d00) seconds
Pref. time           :       14400 (0x00003840) seconds
Recursive DNS server  : fc00:470:f1cd::ff00
DNS server lifetime   :           600 (0x00000258) seconds
Source link-layer address: 00:98:2B:F8:3F:11
from fe80::298:2bff:fe8:3f11
```

Neighbor Discovery

force ipv6-nd interface <interface> address <ipv6-address>

Example:

```
vyos@vyos:~$ force ipv6-nd interface eth0 address fc00:470:f1cd:101::1

Soliciting fc00:470:f1cd:101::1 (fc00:470:f1cd:101::1) on eth0...
Target link-layer address: 00:98:2B:F8:3F:11 from fc00:470:f1cd:101::1
```

10.2 Interface Names

If you find the names of your interfaces have changed, this could be because your MAC addresses have changed.

- For example, you have a VyOS VM with 4 Ethernet interfaces named eth0, eth1, eth2 and eth3. Then, you migrate your VyOS VM to a different host and find your interfaces now are eth4, eth5, eth6 and eth7.

One way to fix this issue **taking control of the MAC addresses** is:

Log into VyOS and run this command to display your interface settings.

```
show interfaces detail
```

Take note of MAC addresses.

Now, in order to update a MAC address in the configuration, run this command specifying the interface name and MAC address you want.

```
set interfaces ethernet eth0 hw-id 00:0c:29:da:a4:fe
```

If it is a VM, go into the settings of the host and set the MAC address to the settings found in the config.boot file. You can also set the MAC to static if the host allows so.

- Another example could be when cloning VyOS VMs in GNS3 and you get into the same issue: interface names have changed.

And **a more generic way to fix it** is just deleting every MAC address at the configuration file of the cloned machine. They will be correctly regenerated automatically.

10.3 Monitoring

VyOS features several monitoring tools.

```
vyos@vyos:~$ monitor
Possible completions:
bandwidth      Monitor interface bandwidth in real time
bandwidth-test  Initiate or wait for bandwidth test
cluster        Monitor clustering service
command        Monitor an operational mode command (refreshes every 2 seconds)
conntrack-sync Monitor conntrack-sync
content-inspection
                Monitor Content-Inspection
dhcp          Monitor Dynamic Host Control Protocol (DHCP)
dns           Monitor a Domain Name Service (DNS) daemon
firewall      Monitor Firewall
https         Monitor the Secure Hypertext Transfer Protocol (HTTPS) service
lldp         Monitor Link Layer Discovery Protocol (LLDP) daemon
log           Monitor last lines of messages file
nat           Monitor network address translation (NAT)
ndp           Monitor the NDP information received by the router through the device
openvpn       Monitor OpenVPN
protocol      Monitor routing protocols
```

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snmp	Monitor Simple Network Management Protocol (SNMP) daemon
stop-all	Stop all current background monitoring processes
traceroute	Monitor the path to a destination in realtime
traffic	Monitor traffic dumps
vpn	Monitor VPN
vrrp	Monitor Virtual Router Redundancy Protocol (VRRP)
webproxy	Monitor Webproxy service

10.3.1 Traffic Dumps

To monitor interface traffic, issue the `monitor traffic interface <name>` command, replacing `<name>` with your chosen interface.

```
vyos@vyos:~$ monitor traffic interface eth0
tcpdump: verbose output suppressed, use -v or -vv for full protocol decode
listening on eth0, link-type EN10MB (Ethernet), capture size 262144 bytes
15:54:28.581601 IP 192.168.0.1 > vyos: ICMP echo request, id 1870, seq 3848, length 64
15:54:28.581660 IP vyos > 192.168.0.1: ICMP echo reply, id 1870, seq 3848, length 64
15:54:29.583399 IP 192.168.0.1 > vyos: ICMP echo request, id 1870, seq 3849, length 64
15:54:29.583454 IP vyos > 192.168.0.1: ICMP echo reply, id 1870, seq 3849, length 64
^C
4 packets captured
4 packets received by filter
0 packets dropped by kernel
vyos@vyos:~$
```

To quit monitoring, press Ctrl-C and you'll be returned to the VyOS command prompt.

Traffic can be filtered and saved.

```
vyos@vyos:~$ monitor traffic interface eth0
Possible completions:
<Enter>      Execute the current command
filter       Monitor traffic matching filter conditions
save         Save traffic dump from an interface to a file
```

10.3.2 Interface Bandwidth Usage

To quickly view the bandwidth usage of an interface, use the `monitor bandwidth` command:

```
vyos@vyos:~$ monitor bandwidth interface eth0
```

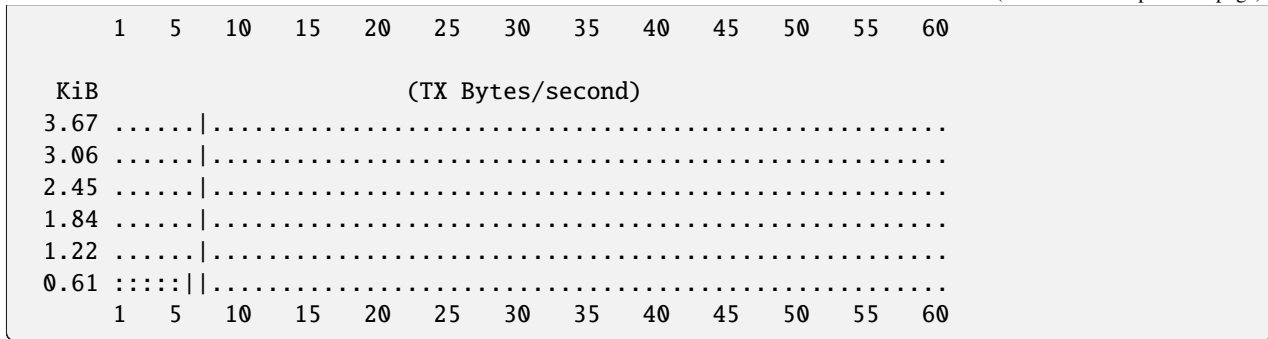
This shows the following:

```

      B                      (RX Bytes/second)
198.00 .|....|.....
165.00 .|....|.....
132.00 ||...|.....
 99.00 ||...|.....
 66.00 |||||.....
 33.00 |||||.....
```

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10.3.3 Interface Performance

To take a look on the network bandwidth between two nodes, the `monitor bandwidth-test` command is used to run iperf.

```
vyos@vyos:~$ monitor bandwidth-test
Possible completions:
  accept      Wait for bandwidth test connections (port TCP/5001)
  initiate    Initiate a bandwidth test
```

- The `accept` command opens a listening iperf server on TCP Port 5001
- The `initiate` command connects to that server to perform the test.

```
vyos@vyos:~$ monitor bandwidth-test initiate
Possible completions:
<hostname>   Initiate a bandwidth test to specified host (port TCP/5001)
<x.x.x.x>
<h:h:h:h:h:h:h>
```

10.3.4 Monitor command

The `monitor command` command allows you to repeatedly run a command to view a continuously refreshed output. The command is run and output every 2 seconds, allowing you to monitor the output continuously without having to re-run the command. This can be useful to follow routing adjacency formation.

```
vyos@router:~$ monitor command "show interfaces"
```

Will clear the screen and show you the output of `show interfaces` every 2 seconds.

```
Every 2.0s: /opt/vyatta/bin/vyatta-op-cmd-wrapper    Sun Mar 26 02:49:46 2019

Codes: S - State, L - Link, u - Up, D - Down, A - Admin Down
Interface      IP Address      S/L  Description
-----
eth0           192.168.1.1/24   u/u
eth0.5         198.51.100.4/24  u/u  WAN
lo             127.0.0.1/8     u/u
::1/128
```

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vti0	172.25.254.2/30	u/u
vti1	172.25.254.9/30	u/u

10.4 Terminal/Console

Sometimes you need to clear counters or statistics to troubleshoot better.

To do this use the `clear` command in Operational mode.

to clear the console output

```
vyos@vyos:~$ clear console
```

to clear interface counters

```
# clear all interfaces
vyos@vyos:~$ clear interface ethernet counters
# clear specific interface
vyos@vyos:~$ clear interface ethernet eth0 counters
```

The command follows the same logic as the `set` command in configuration mode.

```
# clear all counters of an interface type
vyos@vyos:~$ clear interface <interface_type> counters
# clear counter of an interface in interface_type
vyos@vyos:~$ clear interface <interface_type> <interface_name> counters
```

to clear counters on firewall rulesets or single rules

```
vyos@vyos:~$ clear firewall name <ipv4 ruleset name> counters
vyos@vyos:~$ clear firewall name <ipv4 ruleset name> rule <rule#> counters

vyos@vyos:~$ clear firewall ipv6-name <ipv6 ruleset name> counters
vyos@vyos:~$ clear firewall ipv6-name <ipv6 ruleset name> rule <rule#> counters
```

10.5 System Information

10.5.1 Boot Steps

VyOS 1.2 uses [Debian Jessie](#) as the base Linux operating system. Jessie was the first version of Debian that uses [systemd](#) as the default init system.

These are the boot steps for VyOS 1.2

1. The BIOS loads Grub (or isolinux for the Live CD)
2. Grub then starts the Linux boot and loads the Linux Kernel `/boot/vmlinuz`
3. Kernel Launches Systemd `/lib/systemd/systemd`
4. Systemd loads the VyOS service file `/lib/systemd/system/vyos-router.service`

5. The service file launches the VyOS router init script `/usr/libexec/vyos/init/vyos-router` - this is part of the `vyatta-cfg` Debian package
 1. Starts `FRR` - successor to `GNU Zebra` and `Quagga`
 2. Initialises the boot configuration file - copies over `config.boot.default` if there is no configuration
 3. Runs the configuration migration, if the configuration is for an older version of VyOS
 4. Runs The pre-config script, if there is one `/config/scripts/vyos-preconfig-bootup.script`
 5. If the config file was upgraded, runs any post upgrade scripts `/config/scripts/post-upgrade.d`
 6. Starts `rl-system` and `firewall`
 7. Mounts the `/boot` partition
 8. The boot configuration file is then applied by `/opt/vyatta/sbin/vyatta-boot-config-loader/opt/vyatta/etc/config/config.boot`
 1. The config loader script writes log entries to `/var/log/vyatta-config-loader.log`
 9. Runs `telinit q` to tell the init system to reload `/etc/inittab`
 10. Finally it runs the post-config script `/config/scripts/vyos-postconfig-bootup.script`

CONFIGURATION BLUEPRINTS

This chapter contains various configuration examples:

11.1 Firewall Examples

This section contains examples of firewall configurations for various deployments.

11.1.1 VRF and firewall example

Scenario and requirements

This example shows how to configure a VyOS router with VRFs and firewall rules.

Diagram used in this example:

As exposed in the diagram, there are four VRFs. These VRFs are **MGMT**, **WAN**, **LAN** and **PROD**, and their requirements are:

- **VRF MGMT:**
 - Allow connections to LAN and PROD.
 - Deny connections to internet(WAN).
 - Allow connections to the router.
- **VRF LAN:**
 - Allow connections to PROD.
 - Allow connections to internet(WAN).
- **VRF PROD:**
 - Only accepts connections.
- **VRF WAN:**
 - Allow connection to PROD.

Configuration

First, we need to configure the interfaces and VRFs:

```
set interfaces ethernet eth1 address '10.100.100.1/24'
set interfaces ethernet eth1 vrf 'MGMT'
set interfaces ethernet eth2 vif 150 address '10.150.150.1/24'
set interfaces ethernet eth2 vif 150 vrf 'LAN'
set interfaces ethernet eth2 vif 160 address '10.160.160.1/24'
set interfaces ethernet eth2 vif 160 vrf 'LAN'
set interfaces ethernet eth2 vif 3500 address '172.16.20.1/24'
set interfaces ethernet eth2 vif 3500 vrf 'PROD'
set interfaces loopback lo
set interfaces pppoe pppoe0 authentication password 'p4ssw0rd'
set interfaces pppoe pppoe0 authentication username 'vyos'
set interfaces pppoe pppoe0 source-interface 'eth0'
set interfaces pppoe pppoe0 vrf 'WAN'
set vrf bind-to-all
set vrf name LAN protocols static route 0.0.0.0/0 interface pppoe0 vrf 'WAN'
set vrf name LAN protocols static route 10.100.100.0/24 interface eth1 vrf 'MGMT'
set vrf name LAN protocols static route 172.16.20.0/24 interface eth2.3500 vrf 'PROD'
set vrf name LAN table '103'
set vrf name MGMT protocols static route 10.150.150.0/24 interface eth2.150 vrf 'LAN'
set vrf name MGMT protocols static route 10.160.160.0/24 interface eth2.160 vrf 'LAN'
set vrf name MGMT protocols static route 172.16.20.0/24 interface eth2.3500 vrf 'PROD'
set vrf name MGMT table '102'
set vrf name PROD protocols static route 0.0.0.0/0 interface pppoe0 vrf 'WAN'
set vrf name PROD protocols static route 10.100.100.0/24 interface eth1 vrf 'MGMT'
set vrf name PROD protocols static route 10.150.150.0/24 interface eth2.150 vrf 'LAN'
set vrf name PROD protocols static route 10.160.160.0/24 interface eth2.160 vrf 'LAN'
set vrf name PROD table '104'
set vrf name WAN protocols static route 10.150.150.0/24 interface eth2.150 vrf 'LAN'
set vrf name WAN protocols static route 10.160.160.0/24 interface eth2.160 vrf 'LAN'
set vrf name WAN protocols static route 172.16.20.0/24 interface eth2.3500 vrf 'PROD'
set vrf name WAN table '101'
```

And before firewall rules are shown, we need to pay attention how to configure and match interfaces and VRFs. In case where an interface is assigned to a non-default VRF, if we want to use inbound-interface or outbound-interface in firewall rules, we need to:

- For **inbound-interface**: use the interface name with the VRF name, like MGMT or LAN.
- For **outbound-interface**: use the interface name, like eth0, vtun0, eth2* or similar.

Next, we need to configure the firewall rules. First we will define all rules for transit traffic between VRFs.

```
set firewall ipv4 forward filter default-action 'drop'
set firewall ipv4 forward filter default-log
set firewall ipv4 forward filter rule 10 action 'accept'
set firewall ipv4 forward filter rule 10 description 'MGMT - Allow to LAN and PROD'
set firewall ipv4 forward filter rule 10 inbound-interface name 'MGMT'
set firewall ipv4 forward filter rule 10 outbound-interface name 'eth2*'
set firewall ipv4 forward filter rule 99 action 'drop'
set firewall ipv4 forward filter rule 99 description 'MGMT - Drop all going to mgmt'
set firewall ipv4 forward filter rule 99 outbound-interface name 'eth1'
```

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```

set firewall ipv4 forward filter rule 120 action 'accept'
set firewall ipv4 forward filter rule 120 description 'LAN - Allow to PROD'
set firewall ipv4 forward filter rule 120 inbound-interface name 'LAN'
set firewall ipv4 forward filter rule 120 outbound-interface name 'eth2.3500'
set firewall ipv4 forward filter rule 130 action 'accept'
set firewall ipv4 forward filter rule 130 description 'LAN - Allow internet'
set firewall ipv4 forward filter rule 130 inbound-interface name 'LAN'
set firewall ipv4 forward filter rule 130 outbound-interface name 'pppoe0'

```

Also, we are adding global state policies, in order to allow established and related traffic, in order not to drop valid responses:

```

set firewall global-options state-policy established action 'accept'
set firewall global-options state-policy invalid action 'drop'
set firewall global-options state-policy related action 'accept'

```

And finally, we need to allow input connections to the router itself only from vrf MGMT:

```

set firewall ipv4 input filter default-action 'drop'
set firewall ipv4 input filter default-log
set firewall ipv4 input filter rule 10 action 'accept'
set firewall ipv4 input filter rule 10 description 'MGMT - Allow input'
set firewall ipv4 input filter rule 10 inbound-interface name 'MGMT'

```

11.1.2 Bridge and firewall example

Scenario and requirements

This example shows how to configure a VyOS router with bridge interfaces and firewall rules.

Three non VLAN-aware bridges are going to be configured, and each one has its own requirements.

- **Bridge br0:**
 - Isolated layer 2 bridge.
 - Accept only IPv6 communication within the bridge.
- **Bridge br1:**
 - Drop all DHCP discover packets.
 - Accept all ARP packets.
 - Within the bridge, accept only new IPv4 connections from host 10.1.1.102
 - Drop all other IPv4 connections.
 - Drop all IPv6 connections.
 - Accept access to router itself.
 - Allow connections to internet
 - Drop connections to other LANs.
- **Bridge br2:**
 - Accept all DHCP discover packets.

- Accept only DHCP offers from valid server and/or trusted bridge port.
- Accept all ARP packets.
- Accept all IPv4 connections.
- Drop all IPv6 connections.
- Deny access to the router.
- Allow connections to internet.
- Allow connections to bridge br1.

Configuration

Bridges and interfaces configuration

First, we need to configure the interfaces and bridges:

```
# Bridge br0
set interfaces bridge br0 description 'Isolated L2 bridge'
set interfaces bridge br0 member interface eth1
set interfaces bridge br0 member interface eth2
set interfaces ethernet eth1 description 'br0'
set interfaces ethernet eth2 description 'br0'

# Bridge br1:
set interfaces bridge br1 address '10.1.1.1/24'
set interfaces bridge br1 description 'L3 bridge br1'
set interfaces bridge br1 member interface eth3
set interfaces bridge br1 member interface eth4
set interfaces ethernet eth3 description 'br1'
set interfaces ethernet eth4 description 'br1'

# Bridge br2:
set interfaces bridge br2 address '10.2.2.1/24'
set interfaces bridge br2 description 'L3 bridge br2'
set interfaces bridge br2 member interface eth5
set interfaces bridge br2 member interface eth6
set interfaces bridge br2 member interface eth7
set interfaces ethernet eth5 description 'br2 - Host'
set interfaces ethernet eth6 description 'br2 - Trusted DHCP Server'
set interfaces ethernet eth7 description 'br2'
```

Bridge firewall configuration

In this section, we are going to configure the firewall rules that will be used in bridge firewall, and will control the traffic within each bridge.

We are going to use custom firewall rulesets, one for each bridge that will be used in prerouting, and one for each bridge that will be used in the forward chain.

Also, we are going to use firewall interface groups in order to simplify the firewall configuration.

So first, let's create the required firewall interface groups:

```
# Bridge br0 interface-group:
set firewall group interface-group br0-ifaces interface 'br0'
set firewall group interface-group br0-ifaces interface 'eth1'
set firewall group interface-group br0-ifaces interface 'eth2'

# Bridge br1 interface-group:
set firewall group interface-group br1-ifaces interface 'br1'
set firewall group interface-group br1-ifaces interface 'eth3'
set firewall group interface-group br1-ifaces interface 'eth4'

# Bridge br2 interface-group:
set firewall group interface-group br2-ifaces interface 'br2'
set firewall group interface-group br2-ifaces interface 'eth5'
set firewall group interface-group br2-ifaces interface 'eth6'
set firewall group interface-group br2-ifaces interface 'eth7'
```

As said before, we are going to create custom firewall rulesets for each bridge, that will be used in the prerouting chain, in order to drop as much unwanted traffic as early as possible. So, custom rulesets used in prerouting chain are going to be br0-pre, br1-pre, and br2-pre:

```
# Prerouting - Catch all traffic for br0
set firewall bridge prerouting filter rule 10 action 'jump'
set firewall bridge prerouting filter rule 10 description 'br0 traffic'
set firewall bridge prerouting filter rule 10 inbound-interface group 'br0-ifaces'
set firewall bridge prerouting filter rule 10 jump-target 'br0-pre'

# Prerouting - Catch all traffic for br1
set firewall bridge prerouting filter rule 20 action 'jump'
set firewall bridge prerouting filter rule 20 description 'br1 traffic'
set firewall bridge prerouting filter rule 20 inbound-interface group 'br1-ifaces'
set firewall bridge prerouting filter rule 20 jump-target 'br1-pre'

# Prerouting - Catch all traffic for br2
set firewall bridge prerouting filter rule 30 action 'jump'
set firewall bridge prerouting filter rule 30 description 'br2 traffic'
set firewall bridge prerouting filter rule 30 inbound-interface group 'br2-ifaces'
set firewall bridge prerouting filter rule 30 jump-target 'br2-pre'
```

And then create the custom rulesets:

```
### br0 - br0-pre
# Requirements: accept only IPv6 communication within the bridge
set firewall bridge name br0-pre rule 10 description 'Accept IPv6 traffic'
set firewall bridge name br0-pre rule 10 action 'accept'
set firewall bridge name br0-pre rule 10 ethernet-type 'ipv6'
# And drop everything else
set firewall bridge name br0-pre default-action 'drop'

### br1 - br1-pre
# Requirements: drop all DHCP discover packets
set firewall bridge name br1-pre rule 10 description 'Drop DHCP discover'
set firewall bridge name br1-pre rule 10 action 'drop'
set firewall bridge name br1-pre rule 10 protocol 'udp'
```

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```

set firewall bridge name br1-pre rule 10 source port '68'
set firewall bridge name br1-pre rule 10 destination port '67'
set firewall bridge name br1-pre rule 10 destination mac-address 'ff:ff:ff:ff:ff:ff'
set firewall bridge name br1-pre rule 10 log
  # Requirement: drop all IPv6 connections
set firewall bridge name br1-pre rule 20 description 'Drop IPv6 traffic'
set firewall bridge name br1-pre rule 20 action 'drop'
set firewall bridge name br1-pre rule 20 ethernet-type 'ipv6'
  # Accept everything else so it can be parsed later
set firewall bridge name br1-pre default-action 'accept'

### br2 - br2-pre
  # Requirements: drop all IPv6 connections
set firewall bridge name br2-pre rule 10 description 'Drop IPv6 traffic'
set firewall bridge name br2-pre rule 10 action 'drop'
set firewall bridge name br2-pre rule 10 ethernet-type 'ipv6'
  # Accept everything else so it can be parsed later
set firewall bridge name br2-pre default-action 'accept'

```

Now, in the forward chain, we are going to define state policies, and custom rulesets for each bridge that would be used in the forward chain. These rulesets are br0-fwd, br1-fwd, and br2-fwd:

```

# Forward - State policies if not defined globally
set firewall bridge forward filter rule 5 action 'accept'
set firewall bridge forward filter rule 5 state 'established'
set firewall bridge forward filter rule 5 state 'related'
set firewall bridge forward filter rule 10 action 'drop'
set firewall bridge forward filter rule 10 state 'invalid'

# Forward - Catch all traffic for br0
set firewall bridge forward filter rule 110 description 'br0 traffic'
set firewall bridge forward filter rule 110 action 'jump'
set firewall bridge forward filter rule 110 inbound-interface group 'br0-ifaces'
set firewall bridge forward filter rule 110 jump-target 'br0-fwd'

# Forward - Catch all traffic for br1
set firewall bridge forward filter rule 120 description 'br1 traffic'
set firewall bridge forward filter rule 120 action 'jump'
set firewall bridge forward filter rule 120 inbound-interface group 'br1-ifaces'
set firewall bridge forward filter rule 120 jump-target 'br1-fwd'

# Forward - Catch all traffic for br2
set firewall bridge forward filter rule 130 description 'br2 traffic'
set firewall bridge forward filter rule 130 action 'jump'
set firewall bridge forward filter rule 130 inbound-interface group 'br2-ifaces'
set firewall bridge forward filter rule 130 jump-target 'br2-fwd'

# Forward - Default action drop:
set firewall bridge forward filter default-action 'drop'

```

And the content of the custom rulesets:


```

### br0 - br0-fwd
# Accept everything that wasn't dropped in prerouting
set firewall bridge name br0-fwd default-action 'accept'

### br1 - br1-fwd
# Requirement: Accept all ARP packets
set firewall bridge name br1-fwd rule 10 description 'Accept ARP'
set firewall bridge name br1-fwd rule 10 action 'accept'
set firewall bridge name br1-fwd rule 10 ethernet-type 'arp'
# Requirement: Accept only new IPv4 connections from host 10.1.1.102
set firewall bridge name br1-fwd rule 20 description 'Accept ipv4 from host'
set firewall bridge name br1-fwd rule 20 action 'accept'
set firewall bridge name br1-fwd rule 20 source address '10.1.1.102'
set firewall bridge name br1-fwd rule 20 state 'new'
# Drop everything else within the bridge:
set firewall bridge name br1-fwd default-action 'drop'

### br2 - br2-fwd
# Requirement: Accept all DHCP discover packets
set firewall bridge name br2-fwd rule 10 description 'Accept DHCP discover'
set firewall bridge name br2-fwd rule 10 action 'accept'
set firewall bridge name br2-fwd rule 10 protocol 'udp'
set firewall bridge name br2-fwd rule 10 source port '68'
set firewall bridge name br2-fwd rule 10 destination port '67'
set firewall bridge name br2-fwd rule 10 destination mac-address 'ff:ff:ff:ff:ff:ff'
# Requirement: Accept only DHCP offers from valid server on port eth6
set firewall bridge name br2-fwd rule 20 description 'Accept DHCP offers from trusted
↳ interface'
set firewall bridge name br2-fwd rule 20 action 'accept'
set firewall bridge name br2-fwd rule 20 protocol 'udp'
set firewall bridge name br2-fwd rule 20 source port '67'
set firewall bridge name br2-fwd rule 20 destination port '68'
set firewall bridge name br2-fwd rule 20 inbound-interface name 'eth6'
set firewall bridge name br2-fwd rule 22 description 'Drop all other DHCP offers'
set firewall bridge name br2-fwd rule 22 action 'drop'
set firewall bridge name br2-fwd rule 22 protocol 'udp'
set firewall bridge name br2-fwd rule 22 source port '67'
set firewall bridge name br2-fwd rule 22 destination port '68'
set firewall bridge name br2-fwd rule 22 log

# Accept all ARP packets
set firewall bridge name br2-fwd rule 30 description 'Accept ARP'
set firewall bridge name br2-fwd rule 30 action 'accept'
set firewall bridge name br2-fwd rule 30 ethernet-type 'arp'
# Accept all IPv4 connections
set firewall bridge name br2-fwd rule 40 description 'Accept ipv4'
set firewall bridge name br2-fwd rule 40 action 'accept'
set firewall bridge name br2-fwd rule 40 ethernet-type 'ipv4'
# Drop everything else
set firewall bridge name br2-fwd default-action 'drop'

```

IP firewall configuration

Since some of the requirements listed above exceed the capabilities of the bridge firewall, we need to use the IP firewall to implement them. For bridge br1 and br2, we need to control the traffic that is going to the router itself, to other local networks, and to the Internet.

As a reminder, here's a link to the [firewall documentation](#), where you can find more information about the packet flow for traffic that comes from bridge layer and should be analyzed by the IP firewall.

Access to the router itself is controlled by the base chain input, and rules to accomplish all the requirements are:

```
# First of all, if not using global state policies, we need to define them:
set firewall ipv4 input filter rule 10 state 'established'
set firewall ipv4 input filter rule 10 state 'related'
set firewall ipv4 input filter rule 10 action 'accept'
set firewall ipv4 input filter rule 20 state 'invalid'
set firewall ipv4 input filter rule 20 action 'drop'

# Input - br1 - Accept access to router itself
set firewall ipv4 input filter rule 110 description "Accept access from br1"
set firewall ipv4 input filter rule 110 action 'accept'
set firewall ipv4 input filter rule 110 inbound-interface group 'br1-ifaces'

# Input - br2 - Deny access to the router
set firewall ipv4 input filter rule 120 description "Deny access from br2"
set firewall ipv4 input filter rule 120 action 'drop'
set firewall ipv4 input filter rule 120 inbound-interface group 'br2-ifaces'
```

And for traffic that is going to other local networks, and to the Internet, we need to use the base chain forward. As in the bridge firewall, we are going to use custom rulesets for each bridge, that would be used in the forward chain. Those rulesets are ip-br1-fwd and ip-br2-fwd:

```
# First of all, if not using global state policies, we need to define them:
set firewall ipv4 forward filter rule 5 action 'accept'
set firewall ipv4 forward filter rule 5 state 'established'
set firewall ipv4 forward filter rule 5 state 'related'
set firewall ipv4 forward filter rule 10 action 'drop'
set firewall ipv4 forward filter rule 10 state 'invalid'

# Forward - Catch all traffic for br1
set firewall ipv4 forward filter rule 110 description 'br1 traffic'
set firewall ipv4 forward filter rule 110 action 'jump'
set firewall ipv4 forward filter rule 110 inbound-interface group 'br1-ifaces'
set firewall ipv4 forward filter rule 110 jump-target 'ip-br1-fwd'

# Forward - Catch all traffic for br2
set firewall ipv4 forward filter rule 120 description 'br2 traffic'
set firewall ipv4 forward filter rule 120 action 'jump'
set firewall ipv4 forward filter rule 120 inbound-interface group 'br2-ifaces'
set firewall ipv4 forward filter rule 120 jump-target 'ip-br2-fwd'

# Forward - Default action drop:
set firewall ipv4 forward filter default-action 'drop'
```

And the content of the custom rulesets:

```

### br1 - ip-br1-fwd
# Requirement: Allow connections to internet
set firewall ipv4 name ip-br1-fwd rule 10 description 'br1 - allow internet access'
set firewall ipv4 name ip-br1-fwd rule 10 action 'accept'
set firewall ipv4 name ip-br1-fwd rule 10 outbound-interface name 'eth0'
# Requirement: Drop all other connections
set firewall ipv4 name ip-br1-fwd default-action 'drop'

### br2 - ip-br2-fwd
# Requirement: Allow connections to internet
set firewall ipv4 name ip-br2-fwd rule 10 description 'br2 - allow internet access'
set firewall ipv4 name ip-br2-fwd rule 10 action 'accept'
set firewall ipv4 name ip-br2-fwd rule 10 outbound-interface name 'eth0'
# Requirement: Allow connections to br1
set firewall ipv4 name ip-br2-fwd rule 20 description 'br2 - allow access to br1'
set firewall ipv4 name ip-br2-fwd rule 20 action 'accept'
set firewall ipv4 name ip-br2-fwd rule 20 outbound-interface group 'br1-ifaces'
# Requirement: Drop all other connections
set firewall ipv4 name ip-br2-fwd default-action 'drop'

```

Validation

While testing the configuration, we can check logs in order to ensure that we are accepting and/or blocking the correct traffic.

For example, while a host tries to get an IP address from a DHCP server in br1 all DHCP discover are dropped, and in br2, we can see that DHCP offers from untrusted servers are dropped:

```

vyos@bridge:~$ show log firewall bridge
Sep 17 14:22:35 kernel: [bri-NAM-br2-fwd-22-D]IN=eth7 OUT=eth5
↪MAC=50:00:00:09:00:00:50:00:00:04:00:00:08:00 SRC=10.2.2.199 DST=10.2.2.92 LEN=322
↪TOS=0x10 PREC=0x00 TTL=128 ID=0 DF PROTO=UDP SPT=67 DPT=68 LEN=302
Sep 17 14:28:18 kernel: [bri-NAM-br1-pre-10-D]IN=eth3 OUT=
↪MAC=ff:ff:ff:ff:ff:ff:00:50:79:66:68:0c:08:00 SRC=0.0.0.0 DST=255.255.255.255 LEN=392
↪TOS=0x10 PREC=0x00 TTL=16 ID=0 PROTO=UDP SPT=68 DPT=67 LEN=372
Sep 17 14:28:19 kernel: [bri-NAM-br1-pre-10-D]IN=eth3 OUT=
↪MAC=ff:ff:ff:ff:ff:ff:00:50:79:66:68:0c:08:00 SRC=0.0.0.0 DST=255.255.255.255 LEN=392
↪TOS=0x10 PREC=0x00 TTL=16 ID=0 PROTO=UDP SPT=68 DPT=67 LEN=372

```

And with operational mode commands, we can check rules matchers, actions, and counters.

Bridge firewall ruleset:

```

vyos@bri:~$ show firewall bridge
Rulesets bridge Information

-----
bridge Firewall "forward filter"

Rule      Action    Protocol  Packets  Bytes  Conditions
-----
↪-----
5         accept    all       19       1916   ct state { established, related }

```

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```

↪accept
10      drop      all          0          0  ct state invalid
110     jump      all          2          208 iifname @I_br0-ifaces  jump NAME_br0-
↪fwd
120     jump      all          10         670 iifname @I_br1-ifaces  jump NAME_br1-
↪fwd
130     jump      all          12        3086 iifname @I_br2-ifaces  jump NAME_br2-
↪fwd
default drop      all          0          0

```

```

-----
bridge Firewall "name br0-fwd"

```

Rule	Action	Protocol	Packets	Bytes
default	accept	all	2	208

```

-----
bridge Firewall "name br0-pre"

```

Rule	Action	Protocol	Packets	Bytes	Conditions
10	accept	all	18	1872	ether type ip6 accept
default	drop	all	9	1476	

```

-----
bridge Firewall "name br1-fwd"

```

Rule	Action	Protocol	Packets	Bytes	Conditions
↪--					
10	accept	all	5	250	ether type arp accept
20	accept	all	3	252	ct state new ip saddr 10.1.1.102 ↪
↪accept					
default	drop	all	2	168	

```

-----
bridge Firewall "name br1-pre"

```

Rule	Action	Protocol	Packets	Bytes	Conditions
↪--					
10	drop	udp	3	1176	ether daddr ff:ff:ff:ff:ff:ff udp ↪
↪sport 68	udp	dport 67			prefix "[bri-NAM-br1-pre-10-D]"
20	drop	all	0	0	ether type ip6
default	accept	all	58	4430	

```

-----
bridge Firewall "name br2-fwd"

```

Rule	Action	Protocol	Packets	Bytes	Conditions
------	--------	----------	---------	-------	------------

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```

↪ -----
10      accept    udp                4      1312  ether daddr ff:ff:ff:ff:ff:ff udp_
↪sport 68 udp dport 67  accept
20      accept    udp                2       656  udp sport 67 udp dport 68 iifname
↪"eth6"  accept
22      drop      udp                1       322  udp sport 67 udp dport 68  prefix
↪"[bri-NAM-br2-fwd-22-D]"
30      accept    all                2        92  ether type arp  accept
40      accept    all                3       704  ether type ip   accept
default drop      all                0         0

-----
bridge Firewall "name br2-pre"

Rule      Action    Protocol    Packets    Bytes    Conditions
-----
10        drop      all         7         728  ether type ip6
default   accept    all        77        7548

-----
bridge Firewall "prerouting filter"

Rule      Action    Protocol    Packets    Bytes    Conditions
-----
↪ --
10        jump      all         27        3348  iifname @I_br0-ifaces  jump NAME_br0-
↪pre
20        jump      all         61        5606  iifname @I_br1-ifaces  jump NAME_br1-
↪pre
30        jump      all         84        8276  iifname @I_br2-ifaces  jump NAME_br2-
↪pre
default   drop      all         0         0

vyos@bridge:~$

```

IPv4 firewall ruleset:

```

vyos@bridge:~$ show firewall ipv4
Rulesets ipv4 Information

-----
ipv4 Firewall "forward filter"

Rule      Action    Protocol    Packets    Bytes    Conditions
-----
↪ -----
5         accept    all         76        6384  ct state { established, related } _
↪accept
10        drop      all         0         0      ct state invalid
110       jump      all         13        1092  iifname @I_br1-ifaces  jump NAME_ip-
↪br1-fwd
120       jump      all         3         252   iifname @I_br2-ifaces  jump NAME_ip-

```

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```

↪br2-fw
default drop all 0 0

-----
ipv4 Firewall "input filter"

Rule Action Protocol Packets Bytes Conditions
-----
↪---
10 accept all 0 0 ct state { established, related } ↪
↪accept
20 drop all 0 0 ct state invalid
110 accept all 10 720 iifname @I_br1-ifaces accept
120 drop all 26 2672 iifname @I_br2-ifaces
default accept all 3037 991621

-----
ipv4 Firewall "name ip-br1-fw"

Rule Action Protocol Packets Bytes Conditions
-----
10 accept all 5 420 oifname "eth0" accept
default drop all 8 672

-----
ipv4 Firewall "name ip-br2-fw"

Rule Action Protocol Packets Bytes Conditions
-----
10 accept all 1 84 oifname "eth0" accept
20 accept all 2 168 oifname @I_br1-ifaces accept
default drop all 0 0

vyos@bridge:~$

```

11.1.3 Zone-Policy example

Note: In T2199 the syntax of the zone configuration was changed. The zone configuration moved from zone-policy zone <name> to firewall zone <name>.

Native IPv4 and IPv6

We have three networks.

```
WAN - 172.16.10.0/24, 2001:0DB8:0:9999::0/64
LAN - 192.168.100.0/24, 2001:0DB8:0:AAAA::0/64
DMZ - 192.168.200.0/24, 2001:0DB8:0:BBBB::0/64
```

This specific example is for a router on a stick, but is very easily adapted for however many NICs you have:

- Internet - 192.168.200.100 - TCP/80
- Internet - 192.168.200.100 - TCP/443
- Internet - 192.168.200.100 - TCP/25
- Internet - 192.168.200.100 - TCP/53
- VyOS acts as DHCP, DNS forwarder, NAT, router and firewall.
- 192.168.200.200/2001:0DB8:0:BBBB::200 is an internal/external DNS, web and mail (SMTP/IMAP) server.
- 192.168.100.10/2001:0DB8:0:AAAA::10 is the administrator's console. It can SSH to VyOS.
- LAN and DMZ hosts have basic outbound access: Web, FTP, SSH.
- LAN can access DMZ resources.
- DMZ cannot access LAN resources.
- Inbound WAN connect to DMZ host.

The VyOS interface is assigned the .1/:1 address of their respective networks. WAN is on VLAN 10, LAN on VLAN 20, and DMZ on VLAN 30.

It will look something like this:

```
interfaces {
    ethernet eth0 {
        duplex auto
        hw-id 00:53:ed:6e:2a:92
        smp_affinity auto
        speed auto
        vif 10 {
            address 172.16.10.1/24
            address 2001:db8:0:9999::1/64
        }
        vif 20 {
            address 192.168.100.1/24
            address 2001:db8:0:AAAA::1/64
        }
        vif 30 {
            address 192.168.200.1/24
            address 2001:db8:0:BBBB::1/64
        }
    }
    loopback lo {
    }
}
```

Zones Basics

Each interface is assigned to a zone. The interface can be physical or virtual such as tunnels (VPN, PPTP, GRE, etc) and are treated exactly the same.

Traffic flows from zone A to zone B. That flow is what I refer to as a zone-pair-direction. eg. A->B and B->A are two zone-pair-destinations.

Rulesets are created per zone-pair-direction.

I name rule sets to indicate which zone-pair-direction they represent. eg. ZoneA-ZoneB or ZoneB-ZoneA. LAN-DMZ, DMZ-LAN.

In VyOS, you have to have unique Ruleset names. In the event of overlap, I add a “-6” to the end of v6 rulesets. eg. LAN-DMZ, LAN-DMZ-6. This allows for each auto-completion and uniqueness.

In this example we have 4 zones. LAN, WAN, DMZ, Local. The local zone is the firewall itself.

If your computer is on the LAN and you need to SSH into your VyOS box, you would need a rule to allow it in the LAN-Local ruleset. If you want to access a webpage from your VyOS box, you need a rule to allow it in the Local-LAN ruleset.

In rules, it is good to keep them named consistently. As the number of rules you have grows, the more consistency you have, the easier your life will be.

```
Rule 1 - State Established, Related
Rule 2 - State Invalid
Rule 100 - ICMP
Rule 200 - Web
Rule 300 - FTP
Rule 400 - NTP
Rule 500 - SMTP
Rule 600 - DNS
Rule 700 - DHCP
Rule 800 - SSH
Rule 900 - IMAPS
```

The first two rules are to deal with the idiosyncrasies of VyOS and iptables.

Zones and Rulesets both have a default action statement. When using Zone-Policies, the default action is set by the zone-policy statement and is represented by rule 10000.

It is good practice to log both accepted and denied traffic. It can save you significant headaches when trying to troubleshoot a connectivity issue.

To add logging to the default rule, do:

```
set firewall name <ruleSet> default-log
```

By default, iptables does not allow traffic for established sessions to return, so you must explicitly allow this. I do this by adding two rules to every ruleset. 1 allows established and related state packets through and rule 2 drops and logs invalid state packets. We place the established/related rule at the top because the vast majority of traffic on a network is established and the invalid rule to prevent invalid state packets from mistakenly being matched against other rules. Having the most matched rule listed first reduces CPU load in high volume environments. Note: I have filed a bug to have this added as a default action as well.

“It is important to note, that you do not want to add logging to the established state rule as you will be logging both the inbound and outbound packets for each session instead of just the initiation of the session. Your logs will be massive in a very short period of time.”

In VyOS you must have the interfaces created before you can apply it to the zone and the rulesets must be created prior to applying it to a zone-policy.

I create/configure the interfaces first. Build out the rulesets for each zone-pair-direction which includes at least the three state rules. Then I setup the zone-policies.

Zones do not allow for a default action of accept; either drop or reject. It is important to remember this because if you apply an interface to a zone and commit, any active connections will be dropped. Specifically, if you are SSH'd into VyOS and add local or the interface you are connecting through to a zone and do not have rulesets in place to allow SSH and established sessions, you will not be able to connect.

The following are the rules that were created for this example (may not be complete), both in IPv4 and IPv6. If there is no IP specified, then the source/destination address is not explicit.

```
WAN - DMZ:192.168.200.200 - tcp/80
WAN - DMZ:192.168.200.200 - tcp/443
WAN - DMZ:192.168.200.200 - tcp/25
WAN - DMZ:192.168.200.200 - tcp/53
WAN - DMZ:2001:0DB8:0:BBBB::200 - tcp/80
WAN - DMZ:2001:0DB8:0:BBBB::200 - tcp/443
WAN - DMZ:2001:0DB8:0:BBBB::200 - tcp/25
WAN - DMZ:2001:0DB8:0:BBBB::200 - tcp/53

DMZ - Local - tcp/53
DMZ - Local - tcp/123
DMZ - Local - tcp/67,68

LAN - Local - tcp/53
LAN - Local - tcp/123
LAN - Local - tcp/67,68
LAN:192.168.100.10 - Local - tcp/22
LAN:2001:0DB8:0:AAAA::10 - Local - tcp/22

LAN - WAN - tcp/80
LAN - WAN - tcp/443
LAN - WAN - tcp/22
LAN - WAN - tcp/20,21

DMZ - WAN - tcp/80
DMZ - WAN - tcp/443
DMZ - WAN - tcp/22
DMZ - WAN - tcp/20,21
DMZ - WAN - tcp/53
DMZ - WAN - udp/53

Local - WAN - tcp/80
Local - WAN - tcp/443
Local - WAN - tcp/20,21

Local - DMZ - tcp/25
Local - DMZ - tcp/67,68
Local - DMZ - tcp/53
Local - DMZ - udp/53

Local - LAN - tcp/67,68
```

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```

LAN - DMZ - tcp/80
LAN - DMZ - tcp/443
LAN - DMZ - tcp/993
LAN:2001:0DB8:0:AAAA::10 - DMZ:2001:0DB8:0:BBBB::200 - tcp/22
LAN:192.168.100.10 - DMZ:192.168.200.200 - tcp/22

```

Since we have 4 zones, we need to setup the following rulesets.

```

Lan-wan
Lan-local
Lan-dmz
Wan-lan
Wan-local
Wan-dmz
Local-lan
Local-wan
Local-dmz
Dmz-lan
Dmz-wan
Dmz-local

```

Even if the two zones will never communicate, it is a good idea to create the zone-pair-direction rulesets and set default-log. This will allow you to log attempts to access the networks. Without it, you will never see the connection attempts.

This is an example of the three base rules.

```

name wan-lan {
    default-action drop
    default-log
    rule 1 {
        action accept
        state {
            established enable
            related enable
        }
    }
    rule 2 {
        action drop
        log enable
        state {
            invalid enable
        }
    }
}

```

Here is an example of an IPv6 DMZ-WAN ruleset.

```

ipv6-name dmz-wan-6 {
    default-action drop
    default-log
    rule 1 {
        action accept

```

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```
    state {
        established enable
        related enable
    }
}
rule 2 {
    action drop
    log enable
    state {
        invalid enable
    }
}
rule 100 {
    action accept
    log enable
    protocol ipv6-icmp
}
rule 200 {
    action accept
    destination {
        port 80,443
    }
    log enable
    protocol tcp
}
rule 300 {
    action accept
    destination {
        port 20,21
    }
    log enable
    protocol tcp
}
rule 500 {
    action accept
    destination {
        port 25
    }
    log enable
    protocol tcp
    source {
        address 2001:db8:0:BBBB::200
    }
}
rule 600 {
    action accept
    destination {
        port 53
    }
    log enable
    protocol tcp_udp
    source {
```

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```

        address 2001:db8:0:BBBB::200
    }
}
rule 800 {
    action accept
    destination {
        port 22
    }
    log enable
    protocol tcp
}
}

```

Once you have all of your rulesets built, then you need to create your zone-policy.

Start by setting the interface and default action for each zone.

```

set firewall zone dmz default-action drop
set firewall zone dmz interface eth0.30

```

In this case, we are setting the v6 ruleset that represents traffic sourced from the LAN, destined for the DMZ. Because the zone-policy firewall syntax is a little awkward, I keep it straight by thinking of it backwards.

```

set firewall zone dmz from lan firewall ipv6-name lan-dmz-6

```

DMZ-LAN policy is LAN-DMZ. You can get a rhythm to it when you build out a bunch at one time.

In the end, you will end up with something like this config. I took out everything but the Firewall, Interfaces, and zone-policy sections. It is long enough as is.

IPv6 Tunnel

If you are using a IPv6 tunnel from HE.net or someone else, the basis is the same except you have two WAN interfaces. One for v4 and one for v6.

You would have 5 zones instead of just 4 and you would configure your v6 ruleset between your tunnel interface and your LAN/DMZ zones instead of to the WAN.

LAN, WAN, DMZ, local and TUN (tunnel)

v6 pairs would be:

```

lan-tun
lan-local
lan-dmz
tun-lan
tun-local
tun-dmz
local-lan
local-tun
local-dmz
dmz-lan
dmz-tun
dmz-local

```

Notice, none go to WAN since WAN wouldn't have a v6 address on it.

You would have to add a couple of rules on your wan-local ruleset to allow protocol 41 in.

Something like:

```
rule 400 {
    action accept
    destination {
        address 172.16.10.1
    }
    log enable
    protocol 41
    source {
        address ip.of.tunnel.broker
    }
}
```

11.2 BGP IPv6 unnumbered with extended nexthop

General information can be found in the *BGP* chapter.

11.2.1 Configuration

- Router A:

```
set protocols bgp system-as 64496
set protocols bgp address-family ipv4-unicast redistribute connected
set protocols bgp address-family ipv6-unicast redistribute connected
set protocols bgp neighbor eth1 interface v6only
set protocols bgp neighbor eth1 interface v6only peer-group 'fabric'
set protocols bgp neighbor eth2 interface v6only
set protocols bgp neighbor eth2 interface v6only peer-group 'fabric'
set protocols bgp parameters bestpath as-path multipath-relax
set protocols bgp parameters bestpath compare-routerid
set protocols bgp parameters default no-ipv4-unicast
set protocols bgp parameters router-id '192.168.0.1'
set protocols bgp peer-group fabric address-family ipv4-unicast
set protocols bgp peer-group fabric address-family ipv6-unicast
set protocols bgp peer-group fabric capability extended-nexthop
set protocols bgp peer-group fabric remote-as 'external'
```

- Router B:

```
set protocols bgp system-as 64499
set protocols bgp address-family ipv4-unicast redistribute connected
set protocols bgp address-family ipv6-unicast redistribute connected
set protocols bgp neighbor eth1 interface v6only
set protocols bgp neighbor eth1 interface v6only peer-group 'fabric'
set protocols bgp neighbor eth2 interface v6only
set protocols bgp neighbor eth2 interface v6only peer-group 'fabric'
set protocols bgp parameters bestpath as-path multipath-relax
```

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```

set protocols bgp parameters bestpath compare-routerid
set protocols bgp parameters default no-ipv4-unicast
set protocols bgp parameters router-id '192.168.0.2'
set protocols bgp peer-group fabric address-family ipv4-unicast
set protocols bgp peer-group fabric address-family ipv6-unicast
set protocols bgp peer-group fabric capability extended-nexthop
set protocols bgp peer-group fabric remote-as 'external'

```

11.2.2 Results

- Router A:

```

vyos@vyos:~$ show interfaces
Codes: S - State, L - Link, u - Up, D - Down, A - Admin Down
Interface          IP Address          S/L  Description
-----
eth0               198.51.100.34/24    u/u
eth1               -                   u/u
eth2               -                   u/u
lo                 127.0.0.1/8         u/u
                  192.168.0.1/32
                  ::1/128

```

```

vyos@vyos:~$ show ip route
Codes: K - kernel route, C - connected, S - static, R - RIP,
       O - OSPF, I - IS-IS, B - BGP, E - EIGRP, N - NHRP,
       T - Table, v - VNC, V - VNC-Direct, A - Babel, D - SHARP,
       F - PBR, f - OpenFabric,
       > - selected route, * - FIB route

S>* 0.0.0.0/0 [210/0] via 198.51.100.34, eth0, 03:21:53
C>* 198.51.100.0/24 is directly connected, eth0, 03:21:53
C>* 192.168.0.1/32 is directly connected, lo, 03:21:56
B>* 192.168.0.2/32 [20/0] via fe80::a00:27ff:fe3b:7ed2, eth2, 00:05:07
   *                  via fe80::a00:27ff:fe7b:4000, eth1, 00:05:07

```

```

vyos@vyos:~$ ping 192.168.0.2
PING 192.168.0.2 (192.168.0.2) 56(84) bytes of data.
64 bytes from 192.168.0.2: icmp_seq=1 ttl=64 time=0.575 ms
64 bytes from 192.168.0.2: icmp_seq=2 ttl=64 time=0.628 ms
64 bytes from 192.168.0.2: icmp_seq=3 ttl=64 time=0.581 ms
64 bytes from 192.168.0.2: icmp_seq=4 ttl=64 time=0.682 ms
64 bytes from 192.168.0.2: icmp_seq=5 ttl=64 time=0.597 ms

--- 192.168.0.2 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4086ms
rtt min/avg/max/mdev = 0.575/0.612/0.682/0.047 ms

```

```
vyos@vyos:~$ show ip bgp summary
```

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```
IPv4 Unicast Summary:
BGP router identifier 192.168.0.1, local AS number 64496 vrf-id 0
BGP table version 4
RIB entries 5, using 800 bytes of memory
Peers 2, using 41 KiB of memory
Peer groups 1, using 64 bytes of memory
```

Neighbor	V	AS	MsgRcvd	MsgSent	TblVer	InQ	OutQ	Up/Down	State/PfxRcd
eth1	4	64499	13	13	0	0	0	00:05:33	2
eth2	4	64499	13	14	0	0	0	00:05:29	2

```
Total number of neighbors 2
```

- Router B:

```
vyos@vyos:~$ show interfaces
Codes: S - State, L - Link, u - Up, D - Down, A - Admin Down
Interface      IP Address      S/L  Description
-----
eth0           198.51.100.33/24  u/u
eth1           -                u/u
eth2           -                u/u
lo             127.0.0.1/8      u/u
              192.168.0.2/32
              ::1/128
```

```
vyos@vyos:~$ show ip route
Codes: K - kernel route, C - connected, S - static, R - RIP,
       O - OSPF, I - IS-IS, B - BGP, E - EIGRP, N - NHRP,
       T - Table, v - VNC, V - VNC-Direct, A - Babel, D - SHARP,
       F - PBR, f - OpenFabric,
       > - selected route, * - FIB route

S>* 0.0.0.0/0 [210/0] via 198.51.100.33, eth0, 00:44:08
C>* 198.51.100.0/24 is directly connected, eth0, 00:44:09
B>* 192.168.0.1/32 [20/0] via fe80::a00:27ff:fe2d:205d, eth1, 00:06:18
   *                  via fe80::a00:27ff:fe93:e142, eth2, 00:06:18
C>* 192.168.0.2/32 is directly connected, lo, 00:44:11
```

```
vyos@vyos:~$ ping 192.168.0.1
PING 192.168.0.1 (192.168.0.1) 56(84) bytes of data.
64 bytes from 192.168.0.1: icmp_seq=1 ttl=64 time=0.427 ms
64 bytes from 192.168.0.1: icmp_seq=2 ttl=64 time=0.471 ms
64 bytes from 192.168.0.1: icmp_seq=3 ttl=64 time=0.782 ms
64 bytes from 192.168.0.1: icmp_seq=4 ttl=64 time=0.715 ms

--- 192.168.0.1 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3051ms
rtt min/avg/max/mdev = 0.427/0.598/0.782/0.155 ms
```

```
vyos@vyos:~$ show ip bgp summary
```

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IPv4 Unicast Summary:

BGP router identifier 192.168.0.2, local AS number 64499 vrf-id 0

BGP table version 4

RIB entries 5, using 800 bytes of memory

Peers 2, using 41 KiB of memory

Peer groups 1, using 64 bytes of memory

Neighbor	V	AS	MsgRcvd	MsgSent	TblVer	InQ	OutQ	Up/Down	State/PfxRcd
eth1	4	64496	14	14	0	0	0	00:06:40	2
eth2	4	64496	14	14	0	0	0	00:06:37	2

Total number of neighbors 2

11.3 OSPF unnumbered with ECMP

General information can be found in the *OSPF* chapter.

11.3.1 Configuration

- Router A:

```
set interfaces ethernet eth0 address '10.0.0.1/24'
set interfaces ethernet eth1 address '192.168.0.1/32'
set interfaces ethernet eth1 ip ospf authentication md5 key-id 1 md5-key 'yourpassword'
set interfaces ethernet eth1 ip ospf network 'point-to-point'
set interfaces ethernet eth2 address '192.168.0.1/32'
set interfaces ethernet eth2 ip ospf authentication md5 key-id 1 md5-key 'yourpassword'
set interfaces ethernet eth2 ip ospf network 'point-to-point'
set interfaces loopback lo address '192.168.0.1/32'
set protocols ospf area 0.0.0.0 authentication 'md5'
set protocols ospf area 0.0.0.0 network '192.168.0.1/32'
set protocols ospf parameters router-id '192.168.0.1'
set protocols ospf redistribute connected
```

- Router B:

```
set interfaces ethernet eth0 address '10.0.0.2/24'
set interfaces ethernet eth1 address '192.168.0.2/32'
set interfaces ethernet eth1 ip ospf authentication md5 key-id 1 md5-key 'yourpassword'
set interfaces ethernet eth1 ip ospf network 'point-to-point'
set interfaces ethernet eth2 address '192.168.0.2/32'
set interfaces ethernet eth2 ip ospf authentication md5 key-id 1 md5-key 'yourpassword'
set interfaces ethernet eth2 ip ospf network 'point-to-point'
set interfaces loopback lo address '192.168.0.2/32'
set protocols ospf area 0.0.0.0 authentication 'md5'
set protocols ospf area 0.0.0.0 network '192.168.0.2/32'
set protocols ospf parameters router-id '192.168.0.2'
set protocols ospf redistribute connected
```


11.3.2 Results

- Router A:

```
vyos@vyos:~$ show interfaces
Codes: S - State, L - Link, u - Up, D - Down, A - Admin Down
Interface      IP Address      S/L  Description
-----
eth0           10.0.0.1/24     u/u
eth1           192.168.0.1/32  u/u
eth2           192.168.0.1/32  u/u
lo             127.0.0.1/8     u/u
               192.168.0.1/32
               ::1/128
```

```
vyos@vyos:~$ show ip route
Codes: K - kernel route, C - connected, S - static, R - RIP,
       O - OSPF, I - IS-IS, B - BGP, E - EIGRP, N - NHRP,
       T - Table, v - VNC, V - VNC-Direct, A - Babel, D - SHARP,
       F - PBR, f - OpenFabric,
       > - selected route, * - FIB route, q - queued route, r - rejected route

S>* 0.0.0.0/0 [210/0] via 10.0.0.254, eth0, 00:57:34
O  10.0.0.0/24 [110/20] via 192.168.0.2, eth1 onlink, 00:13:21
                        via 192.168.0.2, eth2 onlink, 00:13:21
C>* 10.0.0.0/24 is directly connected, eth0, 00:57:35
O  192.168.0.1/32 [110/0] is directly connected, lo, 00:48:53
C * 192.168.0.1/32 is directly connected, eth2, 00:56:31
C * 192.168.0.1/32 is directly connected, eth1, 00:56:31
C>* 192.168.0.1/32 is directly connected, lo, 00:57:36
O>* 192.168.0.2/32 [110/1] via 192.168.0.2, eth1 onlink, 00:29:03
*                                via 192.168.0.2, eth2 onlink, 00:29:03
```

- Router B:

```
vyos@vyos:~$ show interfaces
Codes: S - State, L - Link, u - Up, D - Down, A - Admin Down
Interface      IP Address      S/L  Description
-----
eth0           10.0.0.2/24     u/u
eth1           192.168.0.2/32  u/u
eth2           192.168.0.2/32  u/u
lo             127.0.0.1/8     u/u
               192.168.0.2/32
               ::1/128
```

```
vyos@vyos:~$ show ip route
Codes: K - kernel route, C - connected, S - static, R - RIP,
       O - OSPF, I - IS-IS, B - BGP, E - EIGRP, N - NHRP,
       T - Table, v - VNC, V - VNC-Direct, A - Babel, D - SHARP,
       F - PBR, f - OpenFabric,
       > - selected route, * - FIB route, q - queued route, r - rejected route
```

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```

S>* 0.0.0.0/0 [210/0] via 10.0.0.254, eth0, 00:57:34
O 10.0.0.0/24 [110/20] via 192.168.0.1, eth1 onlink, 00:13:21
    via 192.168.0.1, eth2 onlink, 00:13:21
C>* 10.0.0.0/24 is directly connected, eth0, 00:57:35
O 192.168.0.2/32 [110/0] is directly connected, lo, 00:48:53
C * 192.168.0.2/32 is directly connected, eth2, 00:56:31
C * 192.168.0.2/32 is directly connected, eth1, 00:56:31
C>* 192.168.0.2/32 is directly connected, lo, 00:57:36
O>* 192.168.0.1/32 [110/1] via 192.168.0.1, eth1 onlink, 00:29:03
    * via 192.168.0.1, eth2 onlink, 00:29:03

```

11.4 Route-Based Site-to-Site VPN to Azure (BGP over IKEv2/IPsec)

This guide shows an example of a route-based IKEv2 site-to-site VPN to Azure using VTI and BGP for dynamic routing updates.

For redundant / active-active configurations see [Route-Based Redundant Site-to-Site VPN to Azure \(BGP over IKEv2/IPsec\)](#)

11.4.1 Prerequisites

- A pair of Azure VNet Gateways deployed in active-passive configuration with BGP enabled.
- A local network gateway deployed in Azure representing the Vyos device, matching the below Vyos settings except for address space, which only requires the Vyos private IP, in this example 10.10.0.5/32
- A connection resource deployed in Azure linking the Azure VNet gateway and the local network gateway representing the Vyos device.

11.4.2 Example

WAN Interface	eth0
On-premises address space	10.10.0.0/16
Azure address space	10.0.0.0/16
Vyos public IP	198.51.100.3
Vyos private IP	10.10.0.5
Azure VNet Gateway public IP	203.0.113.2
Azure VNet Gateway BGP IP	10.0.0.4
Pre-shared key	ch00s3-4-s3cur3-psk
Vyos ASN	64499
Azure ASN	65540

11.4.3 Vyos configuration

- Configure the IKE and ESP settings to match a subset of those supported by Azure:

```
set vpn ipsec esp-group AZURE lifetime '3600'
set vpn ipsec esp-group AZURE mode 'tunnel'
set vpn ipsec esp-group AZURE pfs 'dh-group2'
set vpn ipsec esp-group AZURE proposal 1 encryption 'aes256'
set vpn ipsec esp-group AZURE proposal 1 hash 'sha1'

set vpn ipsec ike-group AZURE dead-peer-detection action 'restart'
set vpn ipsec ike-group AZURE dead-peer-detection interval '15'
set vpn ipsec ike-group AZURE dead-peer-detection timeout '30'
set vpn ipsec ike-group AZURE ikev2-reauth
set vpn ipsec ike-group AZURE key-exchange 'ikev2'
set vpn ipsec ike-group AZURE lifetime '28800'
set vpn ipsec ike-group AZURE proposal 1 dh-group '2'
set vpn ipsec ike-group AZURE proposal 1 encryption 'aes256'
set vpn ipsec ike-group AZURE proposal 1 hash 'sha1'
```

- Enable IPsec on eth0

```
set vpn ipsec interface 'eth0'
```

- Configure a VTI with a dummy IP address

```
set interfaces vti vti1 address '10.10.1.5/32'
set interfaces vti vti1 description 'Azure Tunnel'
```

- Clamp the VTI's MSS to 1350 to avoid PMTU blackholes.

```
set interfaces vti vti1 ip adjust-mss 1350
```

- Configure the VPN tunnel

```
set vpn ipsec authentication psk azure id '198.51.100.3'
set vpn ipsec authentication psk azure id '203.0.113.2'
set vpn ipsec authentication psk azure secret 'ch00s3-4-s3cur3-psk'
set vpn ipsec site-to-site peer 203.0.113.2 authentication local-id '198.51.100.3'
set vpn ipsec site-to-site peer 203.0.113.2 authentication mode 'pre-shared-secret'
set vpn ipsec site-to-site peer 203.0.113.2 authentication remote-id '203.0.113.2'
set vpn ipsec site-to-site peer 203.0.113.2 connection-type 'initiate'
set vpn ipsec site-to-site peer 203.0.113.2 description 'AZURE PRIMARY TUNNEL'
set vpn ipsec site-to-site peer 203.0.113.2 ike-group 'AZURE'
set vpn ipsec site-to-site peer 203.0.113.2 ikev2-reauth 'inherit'
set vpn ipsec site-to-site peer 203.0.113.2 local-address '10.10.0.5'
set vpn ipsec site-to-site peer 203.0.113.2 remote-address '203.0.113.2'
set vpn ipsec site-to-site peer 203.0.113.2 vti bind 'vti1'
set vpn ipsec site-to-site peer 203.0.113.2 vti esp-group 'AZURE'
```

- **Important:** Add an interface route to reach Azure's BGP listener

```
set protocols static route 10.0.0.4/32 interface vti1
```

- Configure your BGP settings

```
set protocols bgp system-as 64499
set protocols bgp neighbor 10.0.0.4 remote-as '65540'
set protocols bgp neighbor 10.0.0.4 address-family ipv4-unicast soft-reconfiguration
↪ 'inbound'
set protocols bgp neighbor 10.0.0.4 timers holdtime '30'
set protocols bgp neighbor 10.0.0.4 timers keepalive '10'
```

- **Important:** Disable connected check

```
set protocols bgp neighbor 10.0.0.4 disable-connected-check
```

11.5 Route-Based Redundant Site-to-Site VPN to Azure (BGP over IKEv2/IPsec)

This guide shows an example of a redundant (active-active) route-based IKEv2 site-to-site VPN to Azure using VTI and BGP for dynamic routing updates.

11.5.1 Prerequisites

- A pair of Azure VNet Gateways deployed in active-active configuration with BGP enabled.
- A local network gateway deployed in Azure representing the Vyos device, matching the below Vyos settings except for address space, which only requires the Vyos private IP, in this example 10.10.0.5/32
- A connection resource deployed in Azure linking the Azure VNet gateway and the local network gateway representing the Vyos device.

11.5.2 Example

WAN Interface	eth0
On-premises address space	10.10.0.0/16
Azure address space	10.0.0.0/16
Vyos public IP	198.51.100.3
Vyos private IP	10.10.0.5
Azure VNet Gateway 1 public IP	203.0.113.2
Azure VNet Gateway 2 public IP	203.0.113.3
Azure VNet Gateway BGP IP	10.0.0.4,10.0.0.5
Pre-shared key	ch00s3-4-s3cur3-psk
Vyos ASN	64499
Azure ASN	65540

11.5.3 Vyos configuration

- Configure the IKE and ESP settings to match a subset of those supported by Azure:

```
set vpn ipsec esp-group AZURE lifetime '3600'
set vpn ipsec esp-group AZURE mode 'tunnel'
set vpn ipsec esp-group AZURE pfs 'dh-group2'
set vpn ipsec esp-group AZURE proposal 1 encryption 'aes256'
set vpn ipsec esp-group AZURE proposal 1 hash 'sha1'

set vpn ipsec ike-group AZURE dead-peer-detection action 'restart'
set vpn ipsec ike-group AZURE dead-peer-detection interval '15'
set vpn ipsec ike-group AZURE dead-peer-detection timeout '30'
set vpn ipsec ike-group AZURE ikev2-reauth
set vpn ipsec ike-group AZURE key-exchange 'ikev2'
set vpn ipsec ike-group AZURE lifetime '28800'
set vpn ipsec ike-group AZURE proposal 1 dh-group '2'
set vpn ipsec ike-group AZURE proposal 1 encryption 'aes256'
set vpn ipsec ike-group AZURE proposal 1 hash 'sha1'
```

- Enable IPsec on eth0

```
set vpn ipsec interface 'eth0'
```

- Configure two VTIs with a dummy IP address each

```
set interfaces vti vti1 address '10.10.1.5/32'
set interfaces vti vti1 description 'Azure Primary Tunnel'

set interfaces vti vti2 address '10.10.1.6/32'
set interfaces vti vti2 description 'Azure Secondary Tunnel'
```

- Clamp the VTI's MSS to 1350 to avoid PMTU blackholes.

```
set interfaces vti vti1 ip adjust-mss 1350
set interfaces vti vti2 ip adjust-mss 1350
```

- Configure the VPN tunnels

```
set vpn ipsec authentication psk azure id '198.51.100.3'
set vpn ipsec authentication psk azure id '203.0.113.2'
set vpn ipsec authentication psk azure id '203.0.113.3'
set vpn ipsec authentication psk azure secret 'ch00s3-4-s3cur3-psk'

set vpn ipsec site-to-site peer azure-primary authentication local-id '198.51.100.3'
set vpn ipsec site-to-site peer azure-primary authentication mode 'pre-shared-secret'
set vpn ipsec site-to-site peer azure-primary authentication remote-id '203.0.113.2'
set vpn ipsec site-to-site peer azure-primary connection-type 'initiate'
set vpn ipsec site-to-site peer azure-primary description 'AZURE PRIMARY TUNNEL'
set vpn ipsec site-to-site peer azure-primary ike-group 'AZURE'
set vpn ipsec site-to-site peer azure-primary ikev2-reauth 'inherit'
set vpn ipsec site-to-site peer azure-primary local-address '10.10.0.5'
set vpn ipsec site-to-site peer azure-primary remote-address '203.0.113.2'
set vpn ipsec site-to-site peer azure-primary vti bind 'vti1'
```

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```

set vpn ipsec site-to-site peer azure-primary vti esp-group 'AZURE'

set vpn ipsec site-to-site peer azure-secondary authentication local-id '198.51.100.3'
set vpn ipsec site-to-site peer azure-secondary authentication mode 'pre-shared-secret'
set vpn ipsec site-to-site peer azure-secondary authentication remote-id '203.0.113.3'
set vpn ipsec site-to-site peer azure-secondary connection-type 'initiate'
set vpn ipsec site-to-site peer azure-secondary description 'AZURE secondary TUNNEL'
set vpn ipsec site-to-site peer azure-secondary ike-group 'AZURE'
set vpn ipsec site-to-site peer azure-secondary ikev2-reauth 'inherit'
set vpn ipsec site-to-site peer azure-secondary local-address '10.10.0.5'
set vpn ipsec site-to-site peer azure-secondary remote-address '203.0.113.3'
set vpn ipsec site-to-site peer azure-secondary vti bind 'vti2'
set vpn ipsec site-to-site peer azure-secondary vti esp-group 'AZURE'

```

- **Important:** Add an interface route to reach both Azure's BGP listeners

```

set protocols static route 10.0.0.4/32 interface vti1
set protocols static route 10.0.0.5/32 interface vti2

```

- Configure your BGP settings

```

set protocols bgp system-as 64499
set protocols bgp neighbor 10.0.0.4 remote-as '65540'
set protocols bgp neighbor 10.0.0.4 address-family ipv4-unicast soft-reconfiguration
  ↪ 'inbound'
set protocols bgp neighbor 10.0.0.4 timers holdtime '30'
set protocols bgp neighbor 10.0.0.4 timers keepalive '10'

set protocols bgp neighbor 10.0.0.5 remote-as '65540'
set protocols bgp neighbor 10.0.0.5 address-family ipv4-unicast soft-reconfiguration
  ↪ 'inbound'
set protocols bgp neighbor 10.0.0.5 timers holdtime '30'
set protocols bgp neighbor 10.0.0.5 timers keepalive '10'

```

- **Important:** Disable connected check, otherwise the routes learned from Azure will not be imported into the routing table.

```

set protocols bgp neighbor 10.0.0.4 disable-connected-check
set protocols bgp neighbor 10.0.0.5 disable-connected-check

```

11.6 High Availability Walkthrough

This document walks you through a complete HA setup of two VyOS machines. This design is based on a VM as the primary router and a physical machine as a backup, using VRRP, BGP, OSPF, and conntrack sharing.

This document aims to walk you through setting everything up, so at a point where you can reboot any machine and not lose more than a few seconds worth of connectivity.

11.6.1 Design

This is based on a real-life production design. One of the complex issues is ensuring you have redundant data INTO your network. We do this with a pair of Cisco Nexus switches and using Virtual PortChannels that are spanned across them. As a bonus, this also allows for complete switch failure without an outage. How you achieve this yourself is left as an exercise to the reader. But our setup is documented here.

Walkthrough suggestion

The `commit` command is implied after every section. If you make an error, `commit` will warn you and you can fix it before getting too far into things. Please ensure you commit early and commit often.

If you are following through this document, it is strongly suggested you complete the entire document, **ONLY** doing the virtual router1 steps, and then come back and walk through it **AGAIN** on the backup hardware router.

This ensures you don't go too fast or miss a step. However, it will make your life easier to configure the fixed IP address and default route now on the hardware router.

Example Network

In this document, we have been allocated 203.0.113.0/24 by our upstream provider, which we are publishing on VLAN100.

They want us to establish a BGP session to their routers on 192.0.2.11 and 192.0.2.12 from our routers 192.0.2.21 and 192.0.2.22. They are AS 65550 and we are AS 65551.

Our routers are going to have a floating IP address of 203.0.113.1, and use .2 and .3 as their fixed IPs.

We are going to use 10.200.201.0/24 for an 'internal' network on VLAN201.

When traffic is originated from the 10.200.201.0/24 network, it will be masqueraded to 203.0.113.1

For connection between sites, we are running a WireGuard link to two REMOTE routers and using OSPF over those links to distribute routes. That remote site is expected to send traffic from anything in 10.201.0.0/16

VLANs

These are the vlans we will be using:

- 50: Upstream, using the 192.0.2.0/24 network allocated by them.
- 100: 'Public' network, using our 203.0.113.0/24 network.
- 201: 'Internal' network, using 10.200.201.0/24

Hardware

- switch1 (Nexus 10gb Switch)
- switch2 (Nexus 10gb Switch)
- compute1 (VMware ESXi 6.5)
- compute2 (VMware ESXi 6.5)
- compute3 (VMware ESXi 6.5)
- router2 (Random 1RU machine with 4 NICs)

Note that router1 is a VM that runs on one of the compute nodes.

Network Cabling

- From Datacenter - This connects into port 1 on both switches, and is tagged as VLAN 50
- Cisco VPC Crossconnect - Ports 39 and 40 bonded between each switch
- Hardware Router - Port 8 of each switch
- compute1 - Port 9 of each switch
- compute2 - Port 10 of each switch
- compute3 - Port 11 of each switch

This is ignoring the extra Out-of-band management networking, which should be on totally different switches, and a different feed into the rack, and is out of scope of this.

Note: Our implementation uses VMware's Distributed Port Groups, which allows VMware to use LACP. This is a part of the ENTERPRISE licence, and is not available on a free licence. If you are implementing this and do not have access to DPGs, you should not use VMware, and use some other virtualization platform instead.

11.6.2 Basic Setup (via console)

Create your router1 VM. So it can withstand a VM Host failing or a network link failing. Using VMware, this is achieved by enabling vSphere DRS, vSphere Availability, and creating a Distributed Port Group that uses LACP.

Many other Hypervisors do this, and I'm hoping that this document will be expanded to document how to do this for others.

Create an 'All VLANs' network group, that passes all trunked traffic through to the VM. Attach this network group to router1 as eth0.

Note: VMware: You must DISABLE SECURITY on this Port group. Make sure that Promiscuous Mode, MAC address changes and Forged transmits are enabled. All of these will be done as part of failover.

Bonding on Hardware Router

Create a LACP bond on the hardware router. We are assuming that eth0 and eth1 are connected to port 8 on both switches, and that those ports are configured as a Port-Channel.

```
set interfaces bonding bond0 description 'Switch Port-Channel'
set interfaces bonding bond0 hash-policy 'layer2'
set interfaces bonding bond0 member interface 'eth0'
set interfaces bonding bond0 member interface 'eth1'
set interfaces bonding bond0 mode '802.3ad'
```


Assign external IP addresses

VLAN 100 and 201 will have floating IP addresses, but VLAN50 does not, as this is talking directly to upstream. Create our IP address on vlan50.

For the hardware router, replace `eth0` with `bond0`. As (almost) every command is identical, this will not be specified unless different things need to be performed on different hosts.

```
set interfaces ethernet eth0 vif 50 address '192.0.2.21/24'
```

In this case, the hardware router has a different IP, so it would be

```
set interfaces ethernet bond0 vif 50 address '192.0.2.22/24'
```

Add (temporary) default route

It is assumed that the routers provided by upstream are capable of acting as a default router, add that as a static route.

```
set protocols static route 0.0.0.0/0 next-hop 192.0.2.11
commit
save
```

Enable SSH

Enable SSH so you can now SSH into the routers, rather than using the console.

```
set service ssh
commit
save
```

At this point, you should be able to SSH into both of them, and will no longer need access to the console (unless you break something!)

11.6.3 VRRP Configuration

We are setting up VRRP so that it does NOT fail back when a machine returns into service, and it prioritizes router1 over router2.

Internal Network

This has a floating IP address of 10.200.201.1/24, using virtual router ID 201. The difference between them is the interface name, hello-source-address, and peer-address.

router1

```
set interfaces ethernet eth0 vif 201 address 10.200.201.2/24
set high-availability vrrp group int hello-source-address '10.200.201.2'
set high-availability vrrp group int interface 'eth0.201'
set high-availability vrrp group int peer-address '10.200.201.3'
set high-availability vrrp group int no-preempt
set high-availability vrrp group int priority '200'
```

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```
set high-availability vrrp group int address '10.200.201.1/24'
set high-availability vrrp group int vrid '201'
```

router2

```
set interfaces ethernet bond0 vif 201 address 10.200.201.3/24
set high-availability vrrp group int hello-source-address '10.200.201.3'
set high-availability vrrp group int interface 'bond0.201'
set high-availability vrrp group int peer-address '10.200.201.2'
set high-availability vrrp group int no-preempt
set high-availability vrrp group int priority '100'
set high-availability vrrp group int address '10.200.201.1/24'
set high-availability vrrp group int vrid '201'
```

Public Network

This has a floating IP address of 203.0.113.1/24, using virtual router ID 113. The virtual router ID is just a random number between 1 and 254, and can be set to whatever you want. Best practices suggest you try to keep them unique enterprise-wide.

router1

```
set interfaces ethernet eth0 vif 100 address 203.0.113.2/24
set high-availability vrrp group public hello-source-address '203.0.113.2'
set high-availability vrrp group public interface 'eth0.100'
set high-availability vrrp group public peer-address '203.0.113.3'
set high-availability vrrp group public no-preempt
set high-availability vrrp group public priority '200'
set high-availability vrrp group public address '203.0.113.1/24'
set high-availability vrrp group public vrid '113'
```

router2

```
set interfaces ethernet bond0 vif 100 address 203.0.113.3/24
set high-availability vrrp group public hello-source-address '203.0.113.3'
set high-availability vrrp group public interface 'bond0.100'
set high-availability vrrp group public peer-address '203.0.113.2'
set high-availability vrrp group public no-preempt
set high-availability vrrp group public priority '100'
set high-availability vrrp group public address '203.0.113.1/24'
set high-availability vrrp group public vrid '113'
```

Create VRRP sync-group

The sync group is used to replicate connection tracking. It needs to be assigned to a random VRRP group, and we are creating a sync group called sync using the vrrp group int.

```
set high-availability vrrp sync-group sync member 'int'
```

Testing

At this point, you should be able to see both IP addresses when you run `show interfaces`, and `show vrrp` should show both interfaces in MASTER state (and SLAVE state on router2).

```
vyos@router1:~$ show vrrp
Name      Interface  VRID  State  Last Transition
-----
int       eth0.201    201   MASTER 100s
public    eth0.100    113   MASTER 200s
vyos@router1:~$
```

You should be able to ping to and from all the IPs you have allocated.

11.6.4 NAT and conntrack-sync

Masquerade Traffic originating from 10.200.201.0/24 that is heading out the public interface.

Note: We explicitly exclude the primary upstream network so that BGP or OSPF traffic doesn't accidentally get NAT'ed.

```
set nat source rule 10 destination address '!192.0.2.0/24'
set nat source rule 10 outbound-interface name 'eth0.50'
set nat source rule 10 source address '10.200.201.0/24'
set nat source rule 10 translation address '203.0.113.1'
```

Configure conntrack-sync and enable helpers

Conntrack helper modules are enabled by default, but they tend to cause more problems than they're worth in complex networks. You can disable all of them at one go.

```
delete system conntrack modules
```

Now enable replication between nodes. Replace eth0.201 with bond0.201 on the hardware router.

```
set service conntrack-sync accept-protocol 'tcp,udp,icmp'
set service conntrack-sync event-listen-queue-size '8'
set service conntrack-sync failover-mechanism vrrp sync-group 'sync'
set service conntrack-sync interface eth0.201
set service conntrack-sync mcast-group '224.0.0.50'
set service conntrack-sync sync-queue-size '8'
```

Testing

The simplest way to test is to look at the connection tracking stats on the standby hardware router with the command `show conntrack-sync statistics`. The numbers should be very close to the numbers on the primary router.

When you have both routers up, you should be able to establish a connection from a NAT'ed machine out to the internet, reboot the active machine, and that connection should be preserved, and will not drop out.

11.6.5 OSPF Over WireGuard

Wireguard doesn't have the concept of an up or down link, due to its design. This complicates AND simplifies using it for network transport, as for reliable state detection you need to use SOMETHING to detect when the link is down.

If you use a routing protocol itself, you solve two problems at once. This is only a basic example, and is provided as a starting point.

Configure Wireguard

There is plenty of instructions and documentation on setting up Wireguard. The only important thing you need to remember is to only use one WireGuard interface per OSPF connection.

We use small /30's from 10.254.60/24 for the point-to-point links.

router1

Replace the 203.0.113.3 with whatever the other router's IP address is.

```
set interfaces wireguard wg01 address '10.254.60.1/30'
set interfaces wireguard wg01 description 'router1-to-offsite1'
set interfaces wireguard wg01 peer OFFSITE1 allowed-ips '0.0.0.0/0'
set interfaces wireguard wg01 peer OFFSITE1 endpoint '203.0.113.3:50001'
set interfaces wireguard wg01 peer OFFSITE1 persistent-keepalive '15'
set interfaces wireguard wg01 peer OFFSITE1 pubkey 'GEFMOWzAyau42/
↪HwdwfXnrfHdIISQF8YHj35r0gSZ0o='
set interfaces wireguard wg01 port '50001'
set protocols ospf interface wg01 authentication md5 key-id 1 md5-key 'i360KoCwUGZvPq7e'
set protocols ospf interface wg01 cost '11'
set protocols ospf interface wg01 dead-interval '5'
set protocols ospf interface wg01 hello-interval '1'
set protocols ospf interface wg01 network 'point-to-point'
set protocols ospf interface wg01 priority '1'
set protocols ospf interface wg01 retransmit-interval '5'
set protocols ospf interface wg01 transmit-delay '1'
```

offsite1

This is connecting back to the STATIC IP of router1, not the floating.

```
set interfaces wireguard wg01 address '10.254.60.2/30'
set interfaces wireguard wg01 description 'offsite1-to-router1'
set interfaces wireguard wg01 peer ROUTER1 allowed-ips '0.0.0.0/0'
set interfaces wireguard wg01 peer ROUTER1 endpoint '192.0.2.21:50001'
set interfaces wireguard wg01 peer ROUTER1 persistent-keepalive '15'
set interfaces wireguard wg01 peer ROUTER1 pubkey 'CKwMV3ZaLntMule2Kd3G7UyVBR7zE8/
↪qoZgLb82EE2Q='
```

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```

set interfaces wireguard wg01 port '50001'
set protocols ospf interface wg01 authentication md5 key-id 1 md5-key 'i360KoCwUGZvPq7e'
set protocols ospf interface wg01 cost '11'
set protocols ospf interface wg01 dead-interval '5'
set protocols ospf interface wg01 hello-interval '1'
set protocols ospf interface wg01 network 'point-to-point'
set protocols ospf interface wg01 priority '1'
set protocols ospf interface wg01 retransmit-interval '5'
set protocols ospf interface wg01 transmit-delay '1'

```

Test WireGuard

Make sure you can ping 10.254.60.1 and .2 from both routers.

Create Export Filter

We only want to export the networks we know. Always do a whitelist on your route filters, both importing and exporting. A good rule of thumb is **‘If you are not the default router for a network, don’t advertise it’**. This means we explicitly do not want to advertise the 192.0.2.0/24 network (but do want to advertise 10.200.201.0 and 203.0.113.0, which we ARE the default route for). This filter is applied to `redistribute connected`. If we WERE to advertise it, the remote machines would see 192.0.2.21 available via their default route, establish the connection, and then OSPF would say ‘192.0.2.0/24 is available via this tunnel’, at which point the tunnel would break, OSPF would drop the routes, and then 192.0.2.0/24 would be reachable via default again. This is called ‘flapping’.

```

set policy access-list 150 description 'Outbound OSPF Redistribution'
set policy access-list 150 rule 10 action 'permit'
set policy access-list 150 rule 10 destination any
set policy access-list 150 rule 10 source inverse-mask '0.0.0.255'
set policy access-list 150 rule 10 source network '10.200.201.0'
set policy access-list 150 rule 20 action 'permit'
set policy access-list 150 rule 20 destination any
set policy access-list 150 rule 20 source inverse-mask '0.0.0.255'
set policy access-list 150 rule 20 source network '203.0.113.0'
set policy access-list 150 rule 100 action 'deny'
set policy access-list 150 rule 100 destination any
set policy access-list 150 rule 100 source any

```

Create Import Filter

We only want to import networks we know. Our OSPF peer should only be advertising networks in the 10.201.0.0/16 range. Note that this is an INVERSE MATCH. You deny in access-list 100 to accept the route.

```

set policy access-list 100 description 'Inbound OSPF Routes from Peers'
set policy access-list 100 rule 10 action 'deny'
set policy access-list 100 rule 10 destination any
set policy access-list 100 rule 10 source inverse-mask '0.0.255.255'
set policy access-list 100 rule 10 source network '10.201.0.0'
set policy access-list 100 rule 100 action 'permit'
set policy access-list 100 rule 100 destination any

```

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```
set policy access-list 100 rule 100 source any
set policy route-map PUBOSPF rule 100 action 'deny'
set policy route-map PUBOSPF rule 100 match ip address access-list '100'
set policy route-map PUBOSPF rule 500 action 'permit'
```

Enable OSPF

Every router **must** have a unique router-id. The 'reference-bandwidth' is used because when OSPF was originally designed, the idea of a link faster than 1gbit was unheard of, and it does not scale correctly.

```
set protocols ospf area 0.0.0.0 authentication 'md5'
set protocols ospf area 0.0.0.0 network '10.254.60.0/24'
set protocols ospf auto-cost reference-bandwidth '10000'
set protocols ospf log-adjacency-changes
set protocols ospf parameters abr-type 'cisco'
set protocols ospf parameters router-id '10.254.60.2'
set system ip protocol ospf route-map PUBOSPF
```

Test OSPF

When you have enabled OSPF on both routers, you should be able to see each other with the command `show ip ospf neighbour`. The state must be 'Full' or '2-Way'. If it is not, then there is a network connectivity issue between the hosts. This is often caused by NAT or MTU issues. You should not see any new routes (unless this is the second pass) in the output of `show ip route`

11.6.6 Advertise connected routes

As a reminder, only advertise routes that you are the default router for. This is why we are NOT announcing the 192.0.2.0/24 network, because if that was announced into OSPF, the other routers would try to connect to that network over a tunnel that connects to that network!

```
set protocols ospf access-list 150 export 'connected'
set protocols ospf redistribute connected
```

You should now be able to see the advertised network on the other host.

Duplicate configuration

At this point, you now need to create the X link between all four routers. Use a different /30 for each link.

Priorities

Set the cost on the secondary links to be 200. This means that they will not be used unless the primary links are down.

```
set protocols ospf interface wg01 cost '10'
set protocols ospf interface wg01 cost '200'
```

This will be visible in 'show ip route'.

11.6.7 BGP

BGP is an extremely complex network protocol. An example is provided here.

Note: Router id's must be unique.

router1

The `redistribute ospf` command is there purely as an example of how this can be expanded. In this walkthrough, it will be filtered by BGPOUT rule 10000, as it is not 203.0.113.0/24.

```
set policy prefix-list BGPOUT description 'BGP Export List'
set policy prefix-list BGPOUT rule 10 action 'deny'
set policy prefix-list BGPOUT rule 10 description 'Do not advertise short masks'
set policy prefix-list BGPOUT rule 10 ge '25'
set policy prefix-list BGPOUT rule 10 prefix '0.0.0.0/0'
set policy prefix-list BGPOUT rule 100 action 'permit'
set policy prefix-list BGPOUT rule 100 description 'Our network'
set policy prefix-list BGPOUT rule 100 prefix '203.0.113.0/24'
set policy prefix-list BGPOUT rule 10000 action 'deny'
set policy prefix-list BGPOUT rule 10000 prefix '0.0.0.0/0'

set policy route-map BGPOUT description 'BGP Export Filter'
set policy route-map BGPOUT rule 10 action 'permit'
set policy route-map BGPOUT rule 10 match ip address prefix-list 'BGPOUT'
set policy route-map BGPOUT rule 10000 action 'deny'
set policy route-map BGPPREPENDOUT description 'BGP Export Filter'
set policy route-map BGPPREPENDOUT rule 10 action 'permit'
set policy route-map BGPPREPENDOUT rule 10 set as-path prepend '65551 65551 65551'
set policy route-map BGPPREPENDOUT rule 10 match ip address prefix-list 'BGPOUT'
set policy route-map BGPPREPENDOUT rule 10000 action 'deny'

set protocols bgp system-as 65551
set protocols bgp address-family ipv4-unicast network 192.0.2.0/24
set protocols bgp address-family ipv4-unicast redistribute connected metric '50'
set protocols bgp address-family ipv4-unicast redistribute ospf metric '50'
set protocols bgp neighbor 192.0.2.11 address-family ipv4-unicast route-map export
↪ 'BGPOUT'
set protocols bgp neighbor 192.0.2.11 address-family ipv4-unicast soft-reconfiguration_
↪ inbound
set protocols bgp neighbor 192.0.2.11 remote-as '65550'
set protocols bgp neighbor 192.0.2.11 update-source '192.0.2.21'
set protocols bgp parameters router-id '192.0.2.21'
```

router2

This is identical, but you use the BGPPEPENDOUT route-map to advertise the route with a longer path.

11.7 WAN Load Balancer examples

11.7.1 Example 1: Distributing load evenly

The setup used in this example is shown in the following diagram:

Overview

- All traffic coming in through eth2 is balanced between eth0 and eth1 on the router.
- Pings will be sent to four targets for health testing (33.44.55.66, 44.55.66.77, 55.66.77.88 and 66.77.88.99).
- All outgoing packets are assigned the source address of the assigned interface (SNAT).
- eth0 is set to be removed from the load balancer's interface pool after 5 ping failures, eth1 will be removed after 4 ping failures.

Create static routes to ping targets

Create static routes through the two ISPs towards the ping targets and commit the changes:

```
set protocols static route 33.44.55.66/32 next-hop 11.22.33.1
set protocols static route 44.55.66.77/32 next-hop 11.22.33.1
set protocols static route 55.66.77.88/32 next-hop 22.33.44.1
set protocols static route 66.77.88.99/32 next-hop 22.33.44.1
```

Configure the load balancer

Configure the WAN load balancer with the parameters described above:

```
set load-balancing wan interface-health eth0 failure-count 5
set load-balancing wan interface-health eth0 nexthop 11.22.33.1
set load-balancing wan interface-health eth0 test 10 type ping
set load-balancing wan interface-health eth0 test 10 target 33.44.55.66
set load-balancing wan interface-health eth0 test 20 type ping
set load-balancing wan interface-health eth0 test 20 target 44.55.66.77
set load-balancing wan interface-health eth1 failure-count 4
set load-balancing wan interface-health eth1 nexthop 22.33.44.1
set load-balancing wan interface-health eth1 test 10 type ping
set load-balancing wan interface-health eth1 test 10 target 55.66.77.88
set load-balancing wan interface-health eth1 test 20 type ping
set load-balancing wan interface-health eth1 test 20 target 66.77.88.99
set load-balancing wan rule 10 inbound-interface eth2
set load-balancing wan rule 10 interface eth0
set load-balancing wan rule 10 interface eth1
```


11.7.2 Example 2: Failover based on interface weights

This example uses the failover mode.

Overview

In this example, eth0 is the primary interface and eth1 is the secondary interface. To provide simple failover functionality. If eth0 fails, eth1 takes over.

Create interface weight based configuration

The configuration steps are the same as in the previous example, except rule 10. So we keep the configuration, remove rule 10 and add a new rule for the failover mode:

```
delete load-balancing wan rule 10
set load-balancing wan rule 10 failover
set load-balancing wan rule 10 inbound-interface eth2
set load-balancing wan rule 10 interface eth0 weight 10
set load-balancing wan rule 10 interface eth1 weight 1
```

11.7.3 Example 3: Failover based on rule order

The previous example used the failover command to send traffic through eth1 if eth0 fails. In this example, failover functionality is provided by rule order.

Overview

Two rules will be created, the first rule directs traffic coming in from eth2 to eth0 and the second rule directs the traffic to eth1. If eth0 fails the first rule is bypassed and the second rule matches, directing traffic to eth1.

Create rule order based configuration

We keep the configuration from the previous example, delete rule 10 and create the two new rules as described:

```
delete load-balancing wan rule 10
set load-balancing wan rule 10 inbound-interface eth2
set load-balancing wan rule 10 interface eth0
set load-balancing wan rule 20 inbound-interface eth2
set load-balancing wan rule 20 interface eth1
```

11.7.4 Example 4: Failover based on rule order - priority traffic

A rule order for prioritizing traffic is useful in scenarios where the secondary link has a lower speed and should only carry high priority traffic. It is assumed for this example that eth1 is connected to a slower connection than eth0 and should prioritize VoIP traffic.

Overview

A rule order for prioritizing traffic is useful in scenarios where the secondary link has a lower speed and should only carry high priority traffic. It is assumed for this example that eth1 is connected to a slower connection than eth0 and should prioritize VoIP traffic.

Create rule order based configuration with low speed secondary link

We keep the configuration from the previous example, delete rule 20 and create a new rule as described:

```
delete load-balancing wan rule 20
set load-balancing wan rule 20 inbound-interface eth2
set load-balancing wan rule 20 interface eth1
set load-balancing wan rule 20 destination port sip
set load-balancing wan rule 20 protocol tcp
set protocols static route 0.0.0.0/0 next-hop 11.22.33.1
```

11.7.5 Example 5: Exclude traffic from load balancing

In this example two LAN interfaces exist in different subnets instead of one like in the previous examples:

Adding a rule for the second interface

Based on the previous example, another rule for traffic from the second interface eth3 can be added to the load balancer. However, traffic meant to flow between the LAN subnets will be sent to eth0 and eth1 as well. To prevent this, another rule is required. This rule excludes traffic between the local subnets from the load balancer. It also excludes locally-sources packets (required for web caching with load balancing). eth+ is used as an alias that refers to all ethernet interfaces:

```
set load-balancing wan rule 5 exclude
set load-balancing wan rule 5 inbound-interface eth+
set load-balancing wan rule 5 destination address 10.0.0.0/8
```

11.8 PPPoE IPv6 Basic Setup for Home Network

This document is to describe a basic setup using PPPoE with DHCPv6-PD + SLAAC to construct a typical home network. The user can follow the steps described here to quickly setup a working network and use this as a starting point to further configure or fine-tune other settings.

To achieve this, your ISP is required to support DHCPv6-PD. If you're not sure, please contact your ISP for more information.

11.8.1 Network Topology

11.8.2 Configurations

PPPoE Setup

```
set interfaces pppoe pppoe0 authentication password <YOUR PASSWORD>
set interfaces pppoe pppoe0 authentication username <YOUR USERNAME>
set interfaces pppoe pppoe0 service-name <YOUR SERVICENAME>
set interfaces pppoe pppoe0 source-interface 'eth0'
```

- Fill password and user with the credential provided by your ISP.
- service-name can be an arbitrary string.

DHCPv6-PD Setup

During address configuration, in addition to assigning an address to the WAN interface, ISP also provides a prefix to allow the router to configure addresses of LAN interface and other nodes connecting to LAN, which is called prefix delegation (PD).

```
set interfaces pppoe pppoe0 ipv6 address autoconf
set interfaces pppoe pppoe0 dhcpv6-options pd 0 interface eth1 address '100'
```

- Here we use the prefix to configure the address of eth1 (LAN) to form <prefix>::64, where 64 is hexadecimal of address 100.
- For home network users, most of time ISP only provides /64 prefix, hence there is no need to set SLA ID and prefix length. See [PPPoE](#) for more information.

Router Advertisement

We need to enable router advertisement for LAN network so that PC can receive the prefix and use SLAAC to configure the address automatically.

```
set service router-advert interface eth1 link-mtu '1492'
set service router-advert interface eth1 name-server <NAME SERVER>
set service router-advert interface eth1 prefix ::/64 valid-lifetime '172800'
```

- Set MTU in advertisement to 1492 because of PPPoE header overhead.

- Set DNS server address in the advertisement so that clients can obtain it by using RDNSS option. Most operating systems (Windows, Linux, Mac) should already support it.
- Here we set the prefix to `::/64` to indicate advertising any /64 prefix the LAN interface is assigned.
- Since some ISPs disconnects continuous connection for every 2~3 days, we set `valid-lifetime` to 2 days to allow PC for phasing out old address.

Basic Firewall

To have basic protection while keeping IPv6 network functional, we need to:

- Allow all established and related traffic for router and LAN
- Allow all icmpv6 packets for router and LAN
- Allow DHCPv6 packets for router

```
set firewall ipv6 name WAN_IN default-action 'drop'
set firewall ipv6 name WAN_IN rule 10 action 'accept'
set firewall ipv6 name WAN_IN rule 10 state established 'enable'
set firewall ipv6 name WAN_IN rule 10 state related 'enable'
set firewall ipv6 name WAN_IN rule 20 action 'accept'
set firewall ipv6 name WAN_IN rule 20 protocol 'icmpv6'
set firewall ipv6 name WAN_LOCAL default-action 'drop'
set firewall ipv6 name WAN_LOCAL rule 10 action 'accept'
set firewall ipv6 name WAN_LOCAL rule 10 state established 'enable'
set firewall ipv6 name WAN_LOCAL rule 10 state related 'enable'
set firewall ipv6 name WAN_LOCAL rule 20 action 'accept'
set firewall ipv6 name WAN_LOCAL rule 20 protocol 'icmpv6'
set firewall ipv6 name WAN_LOCAL rule 30 action 'accept'
set firewall ipv6 name WAN_LOCAL rule 30 destination port '546'
set firewall ipv6 name WAN_LOCAL rule 30 protocol 'udp'
set firewall ipv6 name WAN_LOCAL rule 30 source port '547'
set firewall ipv6 forward filter rule 10 action jump
set firewall ipv6 forward filter rule 10 jump-target 'WAN_IN'
set firewall ipv6 forward filter rule 10 inbound-interface name 'pppoe0'
set firewall ipv6 input filter rule 10 action jump
set firewall ipv6 input filter rule 10 jump-target 'WAN_LOCAL'
set firewall ipv6 input filter rule 10 inbound-interface name 'pppoe0'
```

Note to allow the router to receive DHCPv6 response from ISP. We need to allow packets with source port 547 (server) and destination port 546 (client).

11.9 L3VPN for Hub-and-Spoke connectivity with VyOS

IP/MPLS technology is widely used by various service providers and large enterprises in order to achieve better network scalability, manageability and flexibility. It also provides the possibility to deliver different services for the customers in a seamless manner. Layer 3 VPN (L3VPN) is a type of VPN mode that is built and delivered through OSI layer 3 networking technologies. Often the border gateway protocol (BGP) is used to send and receive VPN-related data that is responsible for the control plane. L3VPN utilizes virtual routing and forwarding (VRF) techniques to receive and deliver user data as well as separate data planes of the end-users. It is built using a combination of IP- and MPLS-based information. Generally, L3VPNs are used to send data on back-end VPN infrastructures, such as for VPN connections between data centres, HQs and branches.

An L3VPN consists of multiple access links, multiple VPN routing and forwarding (VRF) tables, and multiple MPLS paths or multiple P2MP LSPs. An L3VPN can be configured to connect two or more customer sites. In hub-and-spoke MPLS L3VPN environments, the spoke routers need to have unique Route Distinguishers (RDs). In order to use the hub site as a transit point for connectivity in such an environment, the spoke sites export their routes to the hub. Spokes can talk to hubs, but never have direct paths to other spokes. All traffic between spokes is controlled and delivered over the hub site.

To deploy a Layer3 VPN with MPLS on VyOS, we should meet a couple requirements in order to properly implement the solution. We'll use the following nodes in our LAB environment:

- 2 x Route reflectors (VyOS-RRx)
- 4 x Provider routers (VyOS-Px)
- 3 x Provider Edge (VyOS-PEx)
- 3 x Customer Edge (VyOS-CEx)

The following software was used in the creation of this document:

- Operating system: VyOS
- Version: 1.4-rolling-202110310317
- Image name: vyos-1.4-rolling-202110310317-amd64.iso

NOTE: VyOS Router (tested with VyOS 1.4-rolling-202110310317) – The configurations below are specifically for VyOS 1.4.x.

General information can be found in the [L3VPN VRFs](#) chapter.

11.9.1 Topology

11.9.2 How does it work?

As we know the main assumption of L3VPN “Hub and Spoke” is, that the traffic between spokes have to pass via hub, in our scenario VyOS-PE2 is the Hub PE and the VyOS-CE1-HUB is the central customer office device that is responsible for controlling access between all spokes and announcing its network prefixes (10.0.0.100/32). VyOS-PE2 has the main VRF (its name is BLUE_HUB), its own Route-Distinguisher(RD) and route-target import/export lists. Multiprotocol-BGP(MP-BGP) delivers L3VPN related control-plane information to the nodes across network where PEs Spokes import the route-target 60535:1030 (this is export route-target of vrf BLUE_HUB) and export its own route-target 60535:1011(this is vrf BLUE_SPOKE export route-target). Therefore, the Customer edge nodes can only learn the network prefixes of the HUB site [10.0.0.100/32]. For this example VyOS-CE1 has network prefixes [10.0.0.80/32] / VyOS-CE2 has network prefixes [10.0.0.90/32]. Route-Reflector devices VyOS-RR1 and VyOS-RR2 are used to simplify network routes exchange and minimize iBGP peerings between devices.

L3VPN configuration parameters table:

Node	Role	VRF	RD	RT import	RT export
VyOS-PE2	Hub	BLUE_HUB	10.80.80.1:1011	65035:1011 65035:1030	65035:1030
VyOS-PE1	Spoke	BLUE_SPOKE	10.50.50.1:1011	65035:1030	65035:1011
VyOS-PE3	Spoke	BLUE_SPOKE	10.60.60.1:1011	65035:1030	65035:1011

11.9.3 Configuration

Step-1: Configuring IGP and enabling MPLS LDP

At the first step we need to configure the IP/MPLS backbone network using OSPF as IGP protocol and LDP as label-switching protocol for the base connectivity between **P** (rovider), **P** (rovider) **E** (dge) and **R** (oute) **R** (efflector) nodes:

- VyOS-P1:

```
# interfaces
set interfaces dummy dum10 address '10.0.0.3/32'
set interfaces ethernet eth0 address '172.16.30.1/24'
set interfaces ethernet eth1 address '172.16.40.1/24'
set interfaces ethernet eth2 address '172.16.90.1/24'
set interfaces ethernet eth3 address '172.16.10.1/24'
set interfaces ethernet eth5 address '172.16.100.1/24'

# protocols ospf+ldp
set protocols mpls interface 'eth1'
set protocols mpls interface 'eth2'
set protocols mpls interface 'eth3'
set protocols mpls interface 'eth5'
set protocols mpls interface 'eth0'
set protocols mpls ldp discovery transport-ipv4-address '10.0.0.3'
set protocols mpls ldp interface 'eth0'
set protocols mpls ldp interface 'eth1'
set protocols mpls ldp interface 'eth2'
set protocols mpls ldp interface 'eth3'
set protocols mpls ldp interface 'eth5'
set protocols mpls ldp router-id '10.0.0.3'
set protocols ospf area 0 network '0.0.0.0/0'
set protocols ospf parameters abr-type 'cisco'
set protocols ospf parameters router-id '10.0.0.3'
```

- VyOS-P2:

```
# interfaces
set interfaces dummy dum10 address '10.0.0.4/32'
set interfaces ethernet eth0 address '172.16.30.2/24'
set interfaces ethernet eth1 address '172.16.20.1/24'
set interfaces ethernet eth2 address '172.16.120.1/24'
set interfaces ethernet eth3 address '172.16.60.1/24'

# protocols ospf+ldp
set protocols mpls interface 'eth1'
set protocols mpls interface 'eth2'
set protocols mpls interface 'eth3'
set protocols mpls interface 'eth0'
set protocols mpls ldp discovery transport-ipv4-address '10.0.0.4'
set protocols mpls ldp interface 'eth0'
set protocols mpls ldp interface 'eth1'
set protocols mpls ldp interface 'eth2'
set protocols mpls ldp interface 'eth3'
set protocols mpls ldp router-id '10.0.0.4'
```

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```
set protocols ospf area 0 network '0.0.0.0/0'
set protocols ospf parameters abr-type 'cisco'
set protocols ospf parameters router-id '10.0.0.4'
```

- VyOS-P3:

```
# interfaces
set interfaces dummy dum10 address '10.0.0.5/32'
set interfaces ethernet eth0 address '172.16.110.1/24'
set interfaces ethernet eth1 address '172.16.40.2/24'
set interfaces ethernet eth2 address '172.16.50.1/24'
set interfaces ethernet eth3 address '172.16.70.1/24'

# protocols ospf + ldp
set protocols mpls interface 'eth1'
set protocols mpls interface 'eth2'
set protocols mpls interface 'eth3'
set protocols mpls interface 'eth0'
set protocols mpls ldp discovery transport-ipv4-address '10.0.0.5'
set protocols mpls ldp interface 'eth0'
set protocols mpls ldp interface 'eth1'
set protocols mpls ldp interface 'eth2'
set protocols mpls ldp interface 'eth3'
set protocols mpls ldp router-id '10.0.0.5'
set protocols ospf area 0 network '0.0.0.0/0'
set protocols ospf parameters abr-type 'cisco'
set protocols ospf parameters router-id '10.0.0.5'
```

- VyOS-P4:

```
# interfaces
set interfaces dummy dum10 address '10.0.0.6/32'
set interfaces ethernet eth0 address '172.16.80.2/24'
set interfaces ethernet eth1 address '172.16.130.1/24'
set interfaces ethernet eth2 address '172.16.50.2/24'
set interfaces ethernet eth3 address '172.16.60.2/24'
set interfaces ethernet eth5 address '172.16.140.1/24'

# protocols ospf + ldp
set protocols mpls interface 'eth1'
set protocols mpls interface 'eth2'
set protocols mpls interface 'eth3'
set protocols mpls interface 'eth0'
set protocols mpls interface 'eth5'
set protocols mpls ldp discovery transport-ipv4-address '10.0.0.6'
set protocols mpls ldp interface 'eth0'
set protocols mpls ldp interface 'eth1'
set protocols mpls ldp interface 'eth2'
set protocols mpls ldp interface 'eth3'
set protocols mpls ldp interface 'eth5'
set protocols mpls ldp router-id '10.0.0.6'
```

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```
set protocols ospf area 0 network '0.0.0.0/0'
set protocols ospf parameters abr-type 'cisco'
set protocols ospf parameters router-id '10.0.0.6'
```

- VyOS-PE1:

```
# interfaces
set interfaces dummy dum10 address '10.0.0.7/32'
set interfaces ethernet eth0 address '172.16.90.2/24'

# protocols ospf + ldp
set protocols mpls interface 'eth0'
set protocols mpls ldp discovery transport-ipv4-address '10.0.0.7'
set protocols mpls ldp interface 'eth0'
set protocols mpls ldp router-id '10.0.0.7'
set protocols ospf area 0 network '0.0.0.0/0'
set protocols ospf parameters abr-type 'cisco'
set protocols ospf parameters router-id '10.0.0.7'
```

- VyOS-PE2:

```
# interfaces
set interfaces dummy dum10 address '10.0.0.8/32'
set interfaces ethernet eth0 address '172.16.110.2/24'
set interfaces ethernet eth1 address '172.16.100.2/24'
set interfaces ethernet eth2 address '172.16.80.1/24'

# protocols ospf + ldp
set protocols mpls interface 'eth0'
set protocols mpls interface 'eth1'
set protocols mpls ldp discovery transport-ipv4-address '10.0.0.8'
set protocols mpls ldp interface 'eth0'
set protocols mpls ldp interface 'eth1'
set protocols mpls ldp router-id '10.0.0.8'
set protocols ospf area 0 network '0.0.0.0/0'
set protocols ospf parameters abr-type 'cisco'
set protocols ospf parameters router-id '10.0.0.8'
```

- VyOS-PE3:

```
# interfaces
set interfaces dummy dum10 address '10.0.0.10/32'
set interfaces ethernet eth0 address '172.16.140.2/24'

# protocols ospf + ldp
set protocols mpls interface 'eth0'
set protocols mpls ldp discovery transport-ipv4-address '10.0.0.10'
set protocols mpls ldp interface 'eth0'
set protocols mpls ldp router-id '10.0.0.10'
set protocols ospf area 0 network '0.0.0.0/0'
set protocols ospf parameters abr-type 'cisco'
set protocols ospf parameters router-id '10.0.0.10'
```


- VyOS-RR1:

```
# interfaces
set interfaces ethernet eth1 address '172.16.20.2/24'
set interfaces ethernet eth2 address '172.16.10.2/24'
set interfaces dummy dum10 address '10.0.0.1/32'

# protocols ospf + ldp
set protocols mpls interface 'eth1'
set protocols mpls interface 'eth2'
set protocols mpls ldp discovery transport-ipv4-address '10.0.0.1'
set protocols mpls ldp interface 'eth1'
set protocols mpls ldp interface 'eth2'
set protocols mpls ldp router-id '10.0.0.1'
set protocols ospf area 0 network '0.0.0.0/0'
set protocols ospf parameters abr-type 'cisco'
set protocols ospf parameters router-id '10.0.0.1'
```

- VyOS-RR2:

```
# interfaces
set interfaces ethernet eth0 address '172.16.80.1/24'
set interfaces ethernet eth1 address '172.16.70.2/24'
set interfaces dummy dum10 address '10.0.0.2/32'

# protocols ospf + ldp
set protocols mpls interface 'eth0'
set protocols mpls interface 'eth1'
set protocols mpls ldp discovery transport-ipv4-address '10.0.0.2'
set protocols mpls ldp interface 'eth1'
set protocols mpls ldp interface 'eth0'
set protocols mpls ldp router-id '10.0.0.2'
set protocols ospf area 0 network '0.0.0.0/0'
set protocols ospf parameters abr-type 'cisco'
set protocols ospf parameters router-id '10.0.0.2'
```

Step-2: Configuring iBGP for L3VPN control-plane

At this step we are going to enable iBGP protocol on MPLS nodes and Route Reflectors (two routers for redundancy) that will deliver IPv4 VPN (L3VPN) routes between them:

- VyOS-RR1:

```
set protocols bgp system-as '65001'
set protocols bgp neighbor 10.0.0.7 address-family ipv4-vpn route-reflector-client
set protocols bgp neighbor 10.0.0.7 peer-group 'RR_VPNv4'
set protocols bgp neighbor 10.0.0.8 address-family ipv4-vpn route-reflector-client
set protocols bgp neighbor 10.0.0.8 peer-group 'RR_VPNv4'
set protocols bgp neighbor 10.0.0.10 address-family ipv4-vpn route-reflector-client
set protocols bgp neighbor 10.0.0.10 peer-group 'RR_VPNv4'
set protocols bgp parameters cluster-id '10.0.0.1'
set protocols bgp parameters log-neighbor-changes
set protocols bgp parameters router-id '10.0.0.1'
```

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```
set protocols bgp peer-group RR_VPNv4 remote-as '65001'
set protocols bgp peer-group RR_VPNv4 update-source 'dum10'
```

- VyOS-RR2:

```
set protocols bgp system-as '65001'
set protocols bgp neighbor 10.0.0.7 address-family ipv4-vpn route-reflector-client
set protocols bgp neighbor 10.0.0.7 peer-group 'RR_VPNv4'
set protocols bgp neighbor 10.0.0.8 address-family ipv4-vpn route-reflector-client
set protocols bgp neighbor 10.0.0.8 peer-group 'RR_VPNv4'
set protocols bgp neighbor 10.0.0.10 address-family ipv4-vpn route-reflector-client
set protocols bgp neighbor 10.0.0.10 peer-group 'RR_VPNv4'
set protocols bgp parameters cluster-id '10.0.0.1'
set protocols bgp parameters log-neighbor-changes
set protocols bgp parameters router-id '10.0.0.2'
set protocols bgp peer-group RR_VPNv4 remote-as '65001'
set protocols bgp peer-group RR_VPNv4 update-source 'dum10'
```

- VyOS-PE1:

```
set protocols bgp system-as '65001'
set protocols bgp neighbor 10.0.0.1 address-family ipv4-vpn nexthop-self
set protocols bgp neighbor 10.0.0.1 peer-group 'RR_VPNv4'
set protocols bgp neighbor 10.0.0.2 address-family ipv4-vpn nexthop-self
set protocols bgp neighbor 10.0.0.2 peer-group 'RR_VPNv4'
set protocols bgp parameters log-neighbor-changes
set protocols bgp parameters router-id '10.0.0.7'
set protocols bgp peer-group RR_VPNv4 remote-as '65001'
set protocols bgp peer-group RR_VPNv4 update-source 'dum10'
```

- VyOS-PE2:

```
set protocols bgp system-as '65001'
set protocols bgp neighbor 10.0.0.1 address-family ipv4-vpn nexthop-self
set protocols bgp neighbor 10.0.0.1 peer-group 'RR_VPNv4'
set protocols bgp neighbor 10.0.0.2 address-family ipv4-vpn nexthop-self
set protocols bgp neighbor 10.0.0.2 peer-group 'RR_VPNv4'
set protocols bgp parameters log-neighbor-changes
set protocols bgp parameters router-id '10.0.0.8'
set protocols bgp peer-group RR_VPNv4 remote-as '65001'
set protocols bgp peer-group RR_VPNv4 update-source 'dum10'
```

- VyOS-PE3:

```
set protocols bgp system-as '65001'
set protocols bgp neighbor 10.0.0.1 address-family ipv4-vpn nexthop-self
set protocols bgp neighbor 10.0.0.1 peer-group 'RR_VPNv4'
set protocols bgp neighbor 10.0.0.2 address-family ipv4-vpn nexthop-self
set protocols bgp neighbor 10.0.0.2 peer-group 'RR_VPNv4'
set protocols bgp parameters log-neighbor-changes
set protocols bgp parameters router-id '10.0.0.10'
set protocols bgp peer-group RR_VPNv4 remote-as '65001'
set protocols bgp peer-group RR_VPNv4 update-source 'dum10'
```

Step-3: Configuring L3VPN VRFs on PE nodes

This section provides configuration steps for setting up VRFs on our PE nodes including CE facing interfaces, BGP, rd and route-target import/export based on the pre-defined parameters.

- VyOS-PE1:

```
# VRF settings
set vrf name BLUE_SPOKE table '200'
set vrf name BLUE_SPOKE protocols bgp address-family ipv4-unicast export vpn
set vrf name BLUE_SPOKE protocols bgp address-family ipv4-unicast import vpn
set vrf name BLUE_SPOKE protocols bgp address-family ipv4-unicast label vpn export 'auto'
set vrf name BLUE_SPOKE protocols bgp address-family ipv4-unicast network 10.50.50.0/24
set vrf name BLUE_SPOKE protocols bgp address-family ipv4-unicast rd vpn export '10.50.
↪50.1:1011'
set vrf name BLUE_SPOKE protocols bgp address-family ipv4-unicast redistribute connected
set vrf name BLUE_SPOKE protocols bgp address-family ipv4-unicast route-target vpn↪
↪export '65035:1011'
set vrf name BLUE_SPOKE protocols bgp address-family ipv4-unicast route-target vpn↪
↪import '65035:1030'
set vrf name BLUE_SPOKE protocols bgp neighbor 10.50.50.2 address-family ipv4-unicast as-
↪override
set vrf name BLUE_SPOKE protocols bgp neighbor 10.50.50.2 remote-as '65035'

# interfaces
set interfaces ethernet eth3 address '10.50.50.1/24'
set interfaces ethernet eth3 vrf 'BLUE_SPOKE'
```

- VyOS-PE2:

```
# VRF settings
set vrf name BLUE_HUB table '400'
set vrf name BLUE_HUB protocols bgp address-family ipv4-unicast export vpn
set vrf name BLUE_HUB protocols bgp address-family ipv4-unicast import vpn
set vrf name BLUE_HUB protocols bgp address-family ipv4-unicast label vpn export 'auto'
set vrf name BLUE_HUB protocols bgp address-family ipv4-unicast network 10.80.80.0/24
set vrf name BLUE_HUB protocols bgp address-family ipv4-unicast rd vpn export '10.80.80.
↪1:1011'
set vrf name BLUE_HUB protocols bgp address-family ipv4-unicast redistribute connected
set vrf name BLUE_HUB protocols bgp address-family ipv4-unicast route-target vpn export
↪'65035:1030'
set vrf name BLUE_HUB protocols bgp address-family ipv4-unicast route-target vpn import
↪'65035:1011 65050:2011 65035:1030'
set vrf name BLUE_HUB protocols bgp neighbor 10.80.80.2 address-family ipv4-unicast as-
↪override
set vrf name BLUE_HUB protocols bgp neighbor 10.80.80.2 remote-as '65035'

# interfaces
set interfaces ethernet eth3 address '10.80.80.1/24'
set interfaces ethernet eth3 vrf 'BLUE_HUB'
```

- VyOS-PE3:

```
# VRF settings
```

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```
set vrf name BLUE_SPOKE table '200'
set vrf name BLUE_SPOKE protocols bgp address-family ipv4-unicast export vpn
set vrf name BLUE_SPOKE protocols bgp address-family ipv4-unicast import vpn
set vrf name BLUE_SPOKE protocols bgp address-family ipv4-unicast label vpn export 'auto'
set vrf name BLUE_SPOKE protocols bgp address-family ipv4-unicast network 10.60.60.0/24
set vrf name BLUE_SPOKE protocols bgp address-family ipv4-unicast rd vpn export '10.60.
↪60.1:1011'
set vrf name BLUE_SPOKE protocols bgp address-family ipv4-unicast redistribute connected
set vrf name BLUE_SPOKE protocols bgp address-family ipv4-unicast route-target vpn↪
↪export '65035:1011'
set vrf name BLUE_SPOKE protocols bgp address-family ipv4-unicast route-target vpn↪
↪import '65035:1030'
set vrf name BLUE_SPOKE protocols bgp neighbor 10.60.60.2 address-family ipv4-unicast as-
↪override
set vrf name BLUE_SPOKE protocols bgp neighbor 10.60.60.2 remote-as '65035'

# interfaces
set interfaces ethernet eth3 address '10.60.60.1/24'
set interfaces ethernet eth3 vrf 'BLUE_SPOKE'
```

Step-4: Configuring CE nodes

Dynamic routing used between CE and PE nodes and eBGP peering established for the route exchanging between them. All routes received by PEs are then exported to L3VPN and delivered from Spoke sites to Hub and vice-versa based on previously configured L3VPN parameters.

- VyOS-CE1-SPOKE:

```
# interfaces
set interfaces dummy dum20 address '10.0.0.80/32'
set interfaces ethernet eth0 address '10.50.50.2/24'

# BGP for peering with PE
set protocols bgp system-as 65035
set protocols bgp address-family ipv4-unicast network 10.0.0.80/32
set protocols bgp neighbor 10.50.50.1 ebgp-multihop '2'
set protocols bgp neighbor 10.50.50.1 remote-as '65001'
set protocols bgp neighbor 10.50.50.1 update-source 'eth0'
set protocols bgp parameters log-neighbor-changes
set protocols bgp parameters router-id '10.50.50.2'
```

- VyOS-CE1-HUB:

```
# interfaces
set interfaces dummy dum20 address '10.0.0.100/32'
set interfaces ethernet eth0 address '10.80.80.2/24'

# BGP for peering with PE
set protocols bgp system-as 65035
set protocols bgp address-family ipv4-unicast network 10.0.0.100/32
set protocols bgp address-family ipv4-unicast redistribute connected
set protocols bgp neighbor 10.80.80.1 ebgp-multihop '2'
```

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```
set protocols bgp neighbor 10.80.80.1 remote-as '65001'
set protocols bgp neighbor 10.80.80.1 update-source 'eth0'
set protocols bgp parameters log-neighbor-changes
set protocols bgp parameters router-id '10.80.80.2'
```

- VyOS-CE2-SPOKE:

```
# interfaces
set interfaces dummy dum20 address '10.0.0.90/32'
set interfaces ethernet eth0 address '10.60.60.2/24'

# BGP for peering with PE
set protocols bgp system-as 65035
set protocols bgp address-family ipv4-unicast network 10.0.0.90/32
set protocols bgp neighbor 10.60.60.1 ebgp-multihop '2'
set protocols bgp neighbor 10.60.60.1 remote-as '65001'
set protocols bgp neighbor 10.60.60.1 update-source 'eth0'
set protocols bgp parameters log-neighbor-changes
set protocols bgp parameters router-id '10.60.60.2'
```

Step-5: Verification

This section describes verification commands for MPLS/BGP/LDP protocols and L3VPN related routes as well as diagnosis and reachability checks between CE nodes.

Let's check IPv4 routing and MPLS information on provider nodes (same procedure for all P nodes):

- “show ip ospf neighbor” for checking ospf relationship

```
vyos@VyOS-P1:~$ show ip ospf neighbor
```

Neighbor ID	Pri	State	Dead Time	Address	Interface	RXmtL
10.0.0.4	1	Full/Backup	34.718s	172.16.30.2	eth0:172.16.30.1	0
10.0.0.5	1	Full/Backup	35.132s	172.16.40.2	eth1:172.16.40.1	0
10.0.0.7	1	Full/Backup	34.764s	172.16.90.2	eth2:172.16.90.1	0
10.0.0.1	1	Full/Backup	35.642s	172.16.10.2	eth3:172.16.10.1	0
10.0.0.8	1	Full/Backup	35.484s	172.16.100.2	eth5:172.16.100.1	0

- “show mpls ldp neighbor “ for checking ldp neighbors

```
vyos@VyOS-P1:~$ show mpls ldp neighbor
```

AF	ID	State	Remote Address	Uptime
ipv4	10.0.0.1	OPERATIONAL	10.0.0.1	07w5d06h
ipv4	10.0.0.4	OPERATIONAL	10.0.0.4	09w3d00h
ipv4	10.0.0.5	OPERATIONAL	10.0.0.5	09w2d23h
ipv4	10.0.0.7	OPERATIONAL	10.0.0.7	03w0d01h
ipv4	10.0.0.8	OPERATIONAL	10.0.0.8	01w3d02h

- “show mpls ldp binding” for checking mpls label assignment

```
vyos@VyOS-P1:~$ show mpls ldp discovery
```

AF	Destination	Nexthop	Local	Label	Remote Label	In Use
ipv4	10.0.0.1/32	10.0.0.1	23		imp-null	yes
ipv4	10.0.0.1/32	10.0.0.4	23		20	no
ipv4	10.0.0.1/32	10.0.0.5	23		17	no
ipv4	10.0.0.1/32	10.0.0.7	23		16	no
ipv4	10.0.0.1/32	10.0.0.8	23		16	no
ipv4	10.0.0.2/32	10.0.0.1	20		16	no
ipv4	10.0.0.2/32	10.0.0.4	20		22	no
ipv4	10.0.0.2/32	10.0.0.5	20		24	yes
ipv4	10.0.0.2/32	10.0.0.7	20		17	no
ipv4	10.0.0.2/32	10.0.0.8	20		17	no
ipv4	10.0.0.3/32	10.0.0.1	imp-null		17	no
ipv4	10.0.0.3/32	10.0.0.4	imp-null		16	no
ipv4	10.0.0.3/32	10.0.0.5	imp-null		18	no
ipv4	10.0.0.3/32	10.0.0.7	imp-null		18	no
ipv4	10.0.0.3/32	10.0.0.8	imp-null		18	no
ipv4	10.0.0.4/32	10.0.0.1	16		18	no
ipv4	10.0.0.4/32	10.0.0.4	16		imp-null	yes
ipv4	10.0.0.4/32	10.0.0.5	16		19	no
ipv4	10.0.0.4/32	10.0.0.7	16		19	no
ipv4	10.0.0.4/32	10.0.0.8	16		19	no
ipv4	10.0.0.5/32	10.0.0.1	21		19	no
ipv4	10.0.0.5/32	10.0.0.4	21		17	no
ipv4	10.0.0.5/32	10.0.0.5	21		imp-null	yes
ipv4	10.0.0.5/32	10.0.0.7	21		20	no
ipv4	10.0.0.5/32	10.0.0.8	21		20	no
ipv4	10.0.0.6/32	10.0.0.1	17		20	no
ipv4	10.0.0.6/32	10.0.0.4	17		23	yes
ipv4	10.0.0.6/32	10.0.0.5	17		21	yes
ipv4	10.0.0.6/32	10.0.0.7	17		21	no
ipv4	10.0.0.6/32	10.0.0.8	17		21	no
ipv4	10.0.0.7/32	10.0.0.1	22		21	no
ipv4	10.0.0.7/32	10.0.0.4	22		18	no
ipv4	10.0.0.7/32	10.0.0.5	22		20	no
ipv4	10.0.0.7/32	10.0.0.7	22		imp-null	yes
ipv4	10.0.0.7/32	10.0.0.8	22		22	no
ipv4	10.0.0.8/32	10.0.0.1	24		22	no
ipv4	10.0.0.8/32	10.0.0.4	24		19	no
ipv4	10.0.0.8/32	10.0.0.5	24		16	no
ipv4	10.0.0.8/32	10.0.0.7	24		22	no
ipv4	10.0.0.8/32	10.0.0.8	24		imp-null	yes
ipv4	10.0.0.9/32	10.0.0.1	18		23	no
ipv4	10.0.0.9/32	10.0.0.4	18		21	yes
ipv4	10.0.0.9/32	10.0.0.5	18		22	no
ipv4	10.0.0.9/32	10.0.0.7	18		23	no
ipv4	10.0.0.9/32	10.0.0.8	18		23	no
ipv4	10.0.0.10/32	10.0.0.1	19		24	no
ipv4	10.0.0.10/32	10.0.0.4	19		24	yes
ipv4	10.0.0.10/32	10.0.0.5	19		23	yes
ipv4	10.0.0.10/32	10.0.0.7	19		24	no
ipv4	10.0.0.10/32	10.0.0.8	19		24	no

Now we're checking iBGP status and routes from route-reflector nodes to other devices:

- “show bgp ipv4 vpn summary” for checking BGP VPNv4 neighbors:

```
vyos@VyOS-RR1:~$ show bgp ipv4 vpn summary
BGP router identifier 10.0.0.1, local AS number 65001 vrf-id 0
BGP table version 0
RIB entries 9, using 1728 bytes of memory
Peers 4, using 85 KiB of memory
Peer groups 1, using 64 bytes of memory
```

Neighbor	V	AS	MsgRcvd	MsgSent	TblVer	InQ	OutQ	Up/Down	State/PfxRcd
↪ PfxSnt									
10.0.0.7		4	65001	7719	7733	0	0	0	5d07h56m
↪ 2	10								
10.0.0.8		4	65001	7715	7724	0	0	0	5d08h28m
↪ 4	10								
10.0.0.9		4	65001	7713	7724	0	0	0	5d08h28m
↪ 2	10								
10.0.0.10	4	65001	7713	7724		0	0	0	5d08h28m 2
↪ 10									

```
Total number of neighbors 4
```

- “show bgp ipv4 vpn” for checking all VPNv4 prefixes information:

```
vyos@VyOS-RR1:~$ show bgp ipv4 vpn
BGP table version is 2, local router ID is 10.0.0.1, vrf id 0
Default local pref 100, local AS 65001
Status codes: s suppressed, d damped, h history, * valid, > best, = multipath,
               i internal, r RIB-failure, S Stale, R Removed
Nexthop codes: @NNN nexthop's vrf id, < announce-nh-self
Origin codes: i - IGP, e - EGP, ? - incomplete
```

Network	Next Hop	Metric	LocPrf	Weight	Path
Route Distinguisher: 10.50.50.1:1011					
*>i10.50.50.0/24	10.0.0.7	0	100	0	i
UN=10.0.0.7 EC{65035:1011} label=80 type=bgp, subtype=0					
*>i80.80.80.80/32	10.0.0.7	0	100	0	65035 i
UN=10.0.0.7 EC{65035:1011} label=80 type=bgp, subtype=0					
Route Distinguisher: 10.60.60.1:1011					
*>i10.60.60.0/24	10.0.0.10	0	100	0	i
UN=10.0.0.10 EC{65035:1011} label=80 type=bgp, subtype=0					
*>i90.90.90.90/32	10.0.0.10	0	100	0	65035 i
UN=10.0.0.10 EC{65035:1011} label=80 type=bgp, subtype=0					
Route Distinguisher: 10.80.80.1:1011					
*>i10.80.80.0/24	10.0.0.8	0	100	0	i
UN=10.0.0.8 EC{65035:1030} label=80 type=bgp, subtype=0					
*>i100.100.100.100/32	10.0.0.8	0	100	0	65035 i
UN=10.0.0.8 EC{65035:1030} label=80 type=bgp, subtype=0					
Route Distinguisher: 172.16.80.1:2011					
*>i10.110.110.0/24	10.0.0.8	0	100	0	65050 i
UN=10.0.0.8 EC{65050:2011} label=81 type=bgp, subtype=0					

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```
*>i172.16.80.0/24 10.0.0.8 0 100 0 i
UN=10.0.0.8 EC{65050:2011} label=81 type=bgp, subtype=0
Route Distinguisher: 172.16.100.1:2011
*>i10.210.210.0/24 10.0.0.9 0 100 0 65050 i
UN=10.0.0.9 EC{65050:2011} label=80 type=bgp, subtype=0
*>i172.16.100.0/24 10.0.0.9 0 100 0 i
UN=10.0.0.9 EC{65050:2011} label=80 type=bgp, subtype=0
```

- “show bgp ipv4 vpn x.x.x.x/x” for checking best path selected for specific VPNv4 destination

```
vyos@VyOS-RR1:~$ show bgp ipv4 vpn 10.0.0.100/32
BGP routing table entry for 10.80.80.1:1011:10.0.0.100/32
not allocated
Paths: (1 available, best #1)
  Advertised to non peer-group peers:
    10.0.0.7 10.0.0.8 10.0.0.9 10.0.0.10
    65035, (Received from a RR-client)
    10.0.0.8 from 10.0.0.8 (10.0.0.8)
      Origin incomplete, metric 0, localpref 100, valid, internal, best (First path_
      received)
      Extended Community: RT:65035:1030
      Remote label: 80
      Last update: Tue Oct 19 13:45:32 202
```

Also we can verify how PE devices receives VPNv4 networks from the RRs and installing them to the specific customer VRFs:

- “show bgp ipv4 vpn summary” for checking iBGP neighbors against route-reflector devices:

```
vyos@VyOS-PE1:~$ show bgp ipv4 vpn summary
BGP router identifier 10.0.0.7, local AS number 65001 vrf-id 0
BGP table version 0
RIB entries 9, using 1728 bytes of memory
Peers 2, using 43 KiB of memory
Peer groups 1, using 64 bytes of memory
```

Neighbor	V	AS	MsgRcvd	MsgSent	TblVer	InQ	OutQ	Up/Down	State/PfxRcd
→PfxSnt									
10.0.0.1	4	65001	8812	8794	0	0	0	01:18:42	
→8	2								
10.0.0.2	4	65001	8800	8792	0	0	0	6d02h27m	
→8	2								

- “show bgp vrf all” for checking all the prefix learning on BGP within VRFs:

```
vyos@VyOS-PE1:~$ show bgp vrf all

Instance default:
No BGP prefixes displayed, 0 exist

Instance BLUE_SPOKE:
BGP table version is 8, local router ID is 10.50.50.1, vrf id 6
```

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```
Default local pref 100, local AS 65001
Status codes: s suppressed, d damped, h history, * valid, > best, = multipath,
               i internal, r RIB-failure, S Stale, R Removed
Nexthop codes: @NNN nexthop's vrf id, < announce-nh-self
Origin codes: i - IGP, e - EGP, ? - incomplete
```

	Network	Next Hop	Metric	LocPrf	Weight	Path
*	10.50.50.0/24	0.0.0.0	0		32768	?
*>		0.0.0.0	0		32768	i
*>	10.80.80.0/24	10.0.0.8@0<	0	100	0	i
*		10.0.0.8@0<	0	100	0	i
*>	10.0.0.80/32	10.50.50.2	0		0	65035 i
*>	10.0.0.100/32					
		10.0.0.8@0<	0	100	0	65035 ?
*		10.0.0.8@0<	0	100	0	65035 ?

- “show bgp vrf BLUE_SPOKE summary” for checking EBGp neighbor information between PE and CE:

```
vyos@VyOS-PE1:~$ show bgp vrf BLUE_SPOKE summary
```

```
IPv4 Unicast Summary:
BGP router identifier 10.50.50.1, local AS number 65001 vrf-id 6
BGP table version 8
RIB entries 7, using 1344 bytes of memory
Peers 1, using 21 KiB of memory
```

Neighbor	V	AS	MsgRcvd	MsgSent	TblVer	InQ	OutQ	Up/Down	State/PfxRcd
↪ PfxSnt									
10.50.50.2	4	65035	9019	9023	0	0	0	6d06h12m	1
↪ 4									

```
Total number of neighbors 1
```

- “show ip route vrf BLUE_SPOKE” for viewing the RIB in our Spoke PE.

Using this command we are also able to check the transport and customer label (inner/outer) for Hub network prefix (10.0.0.100/32):

```
vyos@VyOS-PE1:~$ show ip route vrf BLUE_SPOKE
```

```
Codes: K - kernel route, C - connected, S - static, R - RIP,
       O - OSPF, I - IS-IS, B - BGP, E - EIGRP, N - NHRP,
       T - Table, v - VNC, V - VNC-Direct, A - Babel, D - SHARP,
       F - PBR, f - OpenFabric,
       > - selected route, * - FIB route, q - queued, r - rejected, b - backup
```

```
VRF BLUE_SPOKE:
```

```
K>* 0.0.0.0/0 [255/8192] unreachable (ICMP unreachable), 03w0d23h
C>* 10.50.50.0/24 is directly connected, eth3, 03w0d23h
B> 10.80.80.0/24 [200/0] via 10.0.0.8 (vrf default) (recursive), label 80, weight 1,
↪ 04:22:00
* via 172.16.90.1, eth0 (vrf default), label 24/80, weight 1,
```

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```

↪04:22:00
B>* 10.0.0.80/32 [20/0] via 10.50.50.2, eth3, weight 1, 6d05h30m
B> 10.0.0.100/32 [200/0] via 10.0.0.8 (vrf default) (recursive), label 80, weight 1,↪
↪04:22:00
*
via 172.16.90.1, eth0 (vrf default), label 24/80,↪
↪weight 1, 04:22:00

```

- **“show bgp ipv4 vpn x.x.x.x/32” for checking the best-path to the**
specific VPNv4 destination including extended community and remotelabel information. This procedure is the same on all Spoke nodes:

```

vyos@VyOS-PE1:~$ show bgp ipv4 vpn 10.0.0.100/32
BGP routing table entry for 10.80.80.1:1011:10.0.0.100/32
not allocated
Paths: (2 available, best #1)
  Not advertised to any peer
  65035
  10.0.0.8 from 10.0.0.1 (10.0.0.8)
    Origin incomplete, metric 0, localpref 100, valid, internal, best (Neighbor IP)
    Extended Community: RT:65035:1030
    Originator: 10.0.0.8, Cluster list: 10.0.0.1
    Remote label: 80
    Last update: Tue Oct 19 13:45:26 2021
  65035
  10.0.0.8 from 10.0.0.2 (10.0.0.8)
    Origin incomplete, metric 0, localpref 100, valid, internal
    Extended Community: RT:65035:1030
    Originator: 10.0.0.8, Cluster list: 10.0.0.1
    Remote label: 80
    Last update: Wed Oct 13 12:39:34 202

```

Now, let’s check routing information on out Hub PE:

- **“show bgp ipv4 vpn summary” for checking iBGP neighbors again**
VyOS-RR1/RR2

```

vyos@VyOS-PE2:~$ show bgp ipv4 vpn summary
BGP router identifier 10.0.0.8, local AS number 65001 vrf-id 0
BGP table version 0
RIB entries 9, using 1728 bytes of memory
Peers 2, using 43 KiB of memory
Peer groups 1, using 64 bytes of memory

Neighbor      V      AS  MsgRcvd  MsgSent  TblVer  InQ  OutQ  Up/Down  State/PfxRcd  ↪
↪PfxSnt
10.0.0.1      4    65001   15982   15949      0    0    0 05:41:28      6      4
10.0.0.2      4    65001    9060    9054      0    0    0 6d06h47m      6      4

Total number of neighbors

```

- **“show bgp vrf all” for checking all the prefixes learning on BGP**

```
vyos@VyOS-PE2:~$ show bgp vrf all
```

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```
Instance default:
No BGP prefixes displayed, 0 exist

Instance BLUE_HUB:
BGP table version is 50, local router ID is 10.80.80.1, vrf id 8
Default local pref 100, local AS 65001
Status codes: s suppressed, d damped, h history, * valid, > best, = multipath,
               i internal, r RIB-failure, S Stale, R Removed
Nexthop codes: @NNN nexthop's vrf id, < announce-nh-self
Origin codes: i - IGP, e - EGP, ? - incomplete
```

Network	Next Hop	Metric	LocPrf	Weight	Path
*> 10.50.50.0/24	10.0.0.7@0<	0	100	0	i
*	10.0.0.7@0<	0	100	0	i
*> 10.60.60.0/24	10.0.0.10@0<	0	100	0	i
*	10.0.0.10@0<	0	100	0	i
* 10.80.80.0/24	10.80.80.2	0		0	65035 ?
*	0.0.0.0	0		32768	i
*>	0.0.0.0	0		32768	?
*> 10.110.110.0/24	172.16.80.2@9<		0		0 65050 i
*> 10.210.210.0/24	10.0.0.9@0<		0	100	0 65050 i
*	10.0.0.9@0<	0	100	0	65050 i
*> 10.0.0.80/32	10.0.0.7@0<	0	100	0	65035 i
*	10.0.0.7@0<	0	100	0	65035 i
*> 10.0.0.90/32	10.0.0.10@0<	0	100	0	65035 i
*	10.0.0.10@0<	0	100	0	65035 i
*> 10.0.0.100/32					
	10.80.80.2	0		0	65035 ?
*> 172.16.80.0/24	0.0.0.0@9<		0		32768 ?
	0.0.0.0@9<	0		32768	i
*> 172.16.100.0/24	10.0.0.9@0<		0	100	0 i
*	10.0.0.9@0<	0	100	0	i

- “show bgp vrf BLUE_HUB summary” for checking EBGp neighbor
CE Hub device

```
vyos@VyOS-PE2:~$ show bgp vrf BLUE_HUB summary
```

```
IPv4 Unicast Summary:
BGP router identifier 10.80.80.1, local AS number 65001 vrf-id 8
BGP table version 50
RIB entries 19, using 3648 bytes of memory
Peers 1, using 21 KiB of memory
```

Neighbor	V	AS	MsgRcvd	MsgSent	TblVer	InQ	OutQ	Up/Down	State/PfxRcd
↪PfxSnt									
10.80.80.2	4	65035	15954	15972	0	0	0	01w4d01h	2
↪10									

- “show ip route vrf BLUE_HUB” to view the RIB in our Hub PE.
With this command we are able to check the transport and customer label (inner/outer) for network spokes prefixes 10.0.0.80/32 - 10.0.0.90/32

```
vyos@VyOS-PE2:~$ show ip route vrf BLUE_HUB
```

```
Codes: K - kernel route, C - connected, S - static, R - RIP,
       O - OSPF, I - IS-IS, B - BGP, E - EIGRP, N - NHRP,
       T - Table, v - VNC, V - VNC-Direct, A - Babel, D - SHARP,
       F - PBR, f - OpenFabric,
       > - selected route, * - FIB route, q - queued, r - rejected, b - backup
```

```
VRF BLUE_HUB:
```

```
K>* 0.0.0.0/0 [255/8192] unreachable (ICMP unreachable), 01w4d01h
B> 10.50.50.0/24 [200/0] via 10.0.0.7 (vrf default) (recursive), label 144, weight 1,
  05:53:15
  *
    via 172.16.100.1, eth1 (vrf default), label 22/144, weight 1,
  05:53:15
B> 10.60.60.0/24 [200/0] via 10.0.0.10 (vrf default) (recursive), label 144, weight 1,
  05:53:15
  *
    via 172.16.110.1, eth0 (vrf default), label 23/144, weight 1,
  05:53:15
C>* 10.80.80.0/24 is directly connected, eth3, 01w4d01h
B>* 10.110.110.0/24 [200/0] via 172.16.80.2, eth2 (vrf GREEN), weight 1, 01w4d01h
B> 10.210.210.0/24 [200/0] via 10.0.0.9 (vrf default) (recursive), label 144, weight 1,
  05:53:15
  *
    via 172.16.100.1, eth1 (vrf default), label 18/144, weight
  1, 05:53:15
  *
    via 172.16.110.1, eth0 (vrf default), label 22/144, weight
  1, 05:53:15
B> 10.0.0.80/32 [200/0] via 10.0.0.7 (vrf default) (recursive), label 144, weight 1,
  05:53:15
  *
    via 172.16.100.1, eth1 (vrf default), label 22/144, weight
  1, 05:53:15
B> 10.0.0.90/32 [200/0] via 10.0.0.10 (vrf default) (recursive), label 144, weight 1,
  05:53:15
  *
    via 172.16.110.1, eth0 (vrf default), label 23/144, weight
  1, 05:53:15
B>* 10.0.0.100/32 [20/0] via 10.80.80.2, eth3, weight 1, 01w4d01h
B>* 172.16.80.0/24 [200/0] is directly connected, eth2 (vrf GREEN), weight 1, 01w4d01h
B> 172.16.100.0/24 [200/0] via 10.0.0.9 (vrf default) (recursive), label 144, weight 1,
  05:53:15
  *
    via 172.16.100.1, eth1 (vrf default), label 18/144, weight
  1, 05:53:15
  *
    via 172.16.110.1, eth0 (vrf default), label 22/144, weight
  1, 05:53:15
```

- “show bgp ipv4 vpn x.x.x.x/32” for checking best-path, extended community and remote label of specific destination

```
vyos@VyOS-PE2:~$ show bgp ipv4 vpn 10.0.0.80/32
BGP routing table entry for 10.50.50.1:1011:10.0.0.80/32
not allocated
Paths: (2 available, best #1)
  Not advertised to any peer
  65035
  10.0.0.7 from 10.0.0.1 (10.0.0.7)
```

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```

Origin IGP, metric 0, localpref 100, valid, internal, best (Neighbor IP)
Extended Community: RT:65035:1011
Originator: 10.0.0.7, Cluster list: 10.0.0.1
Remote label: 144
Last update: Tue Oct 19 13:45:30 2021
65035
10.0.0.7 from 10.0.0.2 (10.0.0.7)
Origin IGP, metric 0, localpref 100, valid, internal
Extended Community: RT:65035:1011
Originator: 10.0.0.7, Cluster list: 10.0.0.1
Remote label: 144
Last update: Wed Oct 13 12:39:37 2021

vyos@VyOS-PE2:~$ show bgp ipv4 vpn 10.0.0.90/32
BGP routing table entry for 10.60.60.1:1011:10.0.0.90/32
not allocated
Paths: (2 available, best #1)
  Not advertised to any peer
  65035
  10.0.0.10 from 10.0.0.1 (10.0.0.10)
    Origin IGP, metric 0, localpref 100, valid, internal, best (Neighbor IP)
    Extended Community: RT:65035:1011
    Originator: 10.0.0.10, Cluster list: 10.0.0.1
    Remote label: 144
    Last update: Tue Oct 19 13:45:30 2021
  65035
  10.0.0.10 from 10.0.0.2 (10.0.0.10)
    Origin IGP, metric 0, localpref 100, valid, internal
    Extended Community: RT:65035:1011
    Originator: 10.0.0.10, Cluster list: 10.0.0.1
    Remote label: 144
    Last update: Wed Oct 13 12:45:44 2021

```

Finally, let's check the reachability between CEs:

- VyOS-CE1-SPOKE —> VyOS-CE-HUB

```

# check rib
vyos@VyOS-CE1-SPOKE:~$ show ip route
Codes: K - kernel route, C - connected, S - static, R - RIP,
       O - OSPF, I - IS-IS, B - BGP, E - EIGRP, N - NHRP,
       T - Table, v - VNC, V - VNC-Direct, A - Babel, D - SHARP,
       F - PBR, f - OpenFabric,
       > - selected route, * - FIB route, q - queued, r - rejected, b - backup

B   10.50.50.0/24 [20/0] via 10.50.50.1 inactive, weight 1, 6d07h53m
C>* 10.50.50.0/24 is directly connected, eth0, 09w0d00h
B>* 10.80.80.0/24 [20/0] via 10.50.50.1, eth0, weight 1, 6d07h53m
C>* 10.0.0.80/32 is directly connected, dum20, 09w0d00h
B>* 10.0.0.100/32 [20/0] via 10.50.50.1, eth0, weight 1, 6d07h53m

# check icmp
vyos@VyOS-CE1-SPOKE:~$ ping 10.0.0.100 interface 10.0.0.80

```

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```

PING 10.0.0.100 (10.0.0.100) from 10.0.0.80 : 56(84) bytes of data.
64 bytes from 10.0.0.100: icmp_seq=1 ttl=62 time=6.52 ms
64 bytes from 10.0.0.100: icmp_seq=2 ttl=62 time=4.13 ms
64 bytes from 10.0.0.100: icmp_seq=3 ttl=62 time=4.04 ms
64 bytes from 10.0.0.100: icmp_seq=4 ttl=62 time=4.03 ms
^C
--- 10.0.0.100 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 8ms
rtt min/avg/max/mdev = 4.030/4.680/6.518/1.064 ms

# check network path
vyos@VyOS-CE1-SPOKE:~$ traceroute 10.0.0.100
traceroute to 10.0.0.100 (10.0.0.100), 30 hops max, 60 byte packets
 1  10.50.50.1 (10.50.50.1)  1.041 ms  1.252 ms  1.835 ms
 2  * * *
 3  10.0.0.100 (10.0.0.100)  9.225 ms  9.159 ms  9.121 m

```

- VyOS-CE-HUB —> VyOS-CE1-SPOKE
- VyOS-CE-HUB —> VyOS-CE2-SPOKE

```

# check rib
vyos@VyOS-CE-HUB:~$ show ip route
Codes: K - kernel route, C - connected, S - static, R - RIP,
       O - OSPF, I - IS-IS, B - BGP, E - EIGRP, N - NHRP,
       T - Table, v - VNC, V - VNC-Direct, A - Babel, D - SHARP,
       F - PBR, f - OpenFabric,
       > - selected route, * - FIB route, q - queued, r - rejected, b - backup

B>* 10.50.50.0/24 [20/0] via 10.80.80.1, eth0, weight 1, 6d08h04m
B>* 10.60.60.0/24 [20/0] via 10.80.80.1, eth0, weight 1, 6d08h35m
C>* 10.80.80.0/24 is directly connected, eth0, 01w6d07h
B>* 10.110.110.0/24 [20/0] via 10.80.80.1, eth0, weight 1, 01w4d02h
B>* 10.210.210.0/24 [20/0] via 10.80.80.1, eth0, weight 1, 6d08h35m
B>* 10.0.0.80/32 [20/0] via 10.80.80.1, eth0, weight 1, 6d08h04m
B>* 10.0.0.90/32 [20/0] via 10.80.80.1, eth0, weight 1, 6d08h35m
C>* 10.0.0.100/32 is directly connected, dum20, 01w6d07h
B>* 172.16.80.0/24 [20/0] via 10.80.80.1, eth0, weight 1, 01w4d02h
B>* 172.16.100.0/24 [20/0] via 10.80.80.1, eth0, weight 1, 6d08h35m

# check icmp
vyos@VyOS-CE-HUB:~$ ping 10.0.0.80 interface 10.0.0.100 c 4
PING 10.0.0.80 (10.0.0.80) from 10.0.0.100 : 56(84) bytes of data.
64 bytes from 10.0.0.80: icmp_seq=1 ttl=62 time=3.31 ms
64 bytes from 10.0.0.80: icmp_seq=2 ttl=62 time=4.23 ms
64 bytes from 10.0.0.80: icmp_seq=3 ttl=62 time=3.89 ms
64 bytes from 10.0.0.80: icmp_seq=4 ttl=62 time=3.22 ms

--- 10.0.0.80 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 9ms
rtt min/avg/max/mdev = 3.218/3.661/4.226/0.421 ms

vyos@VyOS-CE-HUB:~$ ping 10.0.0.90 interface 10.0.0.100 c 4

```

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```

PING 10.0.0.90 (10.0.0.90) from 10.0.0.100 : 56(84) bytes of data.
64 bytes from 10.0.0.90: icmp_seq=1 ttl=62 time=7.46 ms
64 bytes from 10.0.0.90: icmp_seq=2 ttl=62 time=4.43 ms
64 bytes from 10.0.0.90: icmp_seq=3 ttl=62 time=4.60 ms
^C
--- 10.0.0.90 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 6ms
rtt min/avg/max/mdev = 4.430/5.498/7.463/1.391 ms

# check network path
vyos@VyOS-CE-HUB:~$ traceroute 10.0.0.80
traceroute to 10.0.0.80 (10.0.0.80), 30 hops max, 60 byte packets
 1  10.80.80.1 (10.80.80.1)  1.563 ms  1.341 ms  1.075 ms
 2  * * *
 3  10.0.0.80 (10.0.0.80)  8.125 ms  8.019 ms  7.781 ms

vyos@VyOS-CE-HUB:~$ traceroute 10.0.0.90
traceroute to 10.0.0.90 (10.0.0.90), 30 hops max, 60 byte packets
 1  10.80.80.1 (10.80.80.1)  1.305 ms  1.137 ms  1.097 ms
 2  * * *
 3  * * *
 4  10.0.0.90 (10.0.0.90)  9.358 ms  9.325 ms  9.292 ms

```

• VyOS-CE2-SPOKE —> VyOS-CE-HUB

```

# check rib
vyos@rt-ce2-SPOKE:~$ show ip route
Codes: K - kernel route, C - connected, S - static, R - RIP,
       O - OSPF, I - IS-IS, B - BGP, E - EIGRP, N - NHRP,
       T - Table, v - VNC, V - VNC-Direct, A - Babel, D - SHARP,
       F - PBR, f - OpenFabric,
       > - selected route, * - FIB route, q - queued, r - rejected, b - backup

B   10.60.60.0/24 [20/0] via 10.60.60.1 inactive, weight 1, 02w6d00h
C>* 10.60.60.0/24 is directly connected, eth0, 02w6d00h
B>* 10.80.80.0/24 [20/0] via 10.60.60.1, eth0, weight 1, 6d08h46m
C>* 10.0.0.90/32 is directly connected, dum20, 02w6d00h
B>* 10.0.0.100/32 [20/0] via 10.60.60.1, eth0, weight 1, 6d08h46m

# check icmp
vyos@rt-ce2-SPOKE:~$ ping 10.0.0.100 interface 10.0.0.90 c 4
PING 10.0.0.100 (10.0.0.100) from 10.0.0.90 : 56(84) bytes of data.
64 bytes from 10.0.0.100: icmp_seq=1 ttl=62 time=4.97 ms
64 bytes from 10.0.0.100: icmp_seq=2 ttl=62 time=4.45 ms
64 bytes from 10.0.0.100: icmp_seq=3 ttl=62 time=4.20 ms
64 bytes from 10.0.0.100: icmp_seq=4 ttl=62 time=4.29 ms

--- 10.0.0.100 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 9ms
rtt min/avg/max/mdev = 4.201/4.476/4.971/0.309 ms

# check network path

```

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```
vyos@rt-ce2-SPOKE:~$ traceroute 10.0.0.100
traceroute to 10.0.0.100 (10.0.0.100), 30 hops max, 60 byte packets
 1  10.60.60.1 (10.60.60.1)  1.343 ms  1.190 ms  1.152 ms
 2  * * *
 3  * * *
 4  10.0.0.100 (10.0.0.100)  7.504 ms  7.480 ms  7.488 ms
```

Note: At the moment, trace mpls doesn't show labels/paths. So we'll see * * * for the transit routers of the mpls backbone.

11.10 PPPoE over L2TP

This document is to describe a basic setup using PPPoE over L2TP. LAC and LNS are components of the broadband topology. LAC - L2TP access concentrator LNS - L2TP Network Server LAC and LNS forms L2TP tunnel. LAC receives packets from PPPoE clients and forward them to LNS. LNS is the termination point that comes from PPP packets from the remote client.

In this example we use VyOS 1.5 as LNS and Cisco IOS as LAC. All users with domain **vyos.io** will be tunneled to LNS via L2TP.

11.10.1 Network Topology

11.10.2 Configurations

LAC

```
aaa new-model
!
aaa authentication ppp default local
!
vpdn enable
vpdn aaa attribute nas-ip-address vpdn-nas
!
vpdn-group LAC
 request-dialin
 protocol l2tp
 domain vyos.io
 initiate-to ip 192.168.139.100
 source-ip 192.168.139.101
 local name LAC
 l2tp tunnel password 0 test123
!
bba-group pppoe MAIN-BBA
 virtual-template 1
!
interface GigabitEthernet0/0
 description To LNS
```

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```
ip address 192.168.139.101 255.255.255.0
duplex auto
speed auto
media-type rj45
!
interface GigabitEthernet0/1
description To PPPoE clients
no ip address
duplex auto
speed auto
media-type rj45
pppoe enable group MAIN-BBA
!
interface Virtual-Template1
description pppoe MAIN-BBA
no ip address
no peer default ip address
ppp mtu adaptive
ppp authentication chap
!
```

LNS

```
set interfaces ethernet eth0 address '192.168.139.100/24'
set nat source rule 100 outbound-interface name 'eth0'
set nat source rule 100 source address '10.0.0.0/24'
set nat source rule 100 translation address 'masquerade'
set protocols static route 0.0.0.0/0 next-hop 192.168.139.2
set vpn l2tp remote-access authentication mode 'radius'
set vpn l2tp remote-access authentication radius server 192.168.139.110 key 'radiustest'
set vpn l2tp remote-access client-ip-pool TEST-POOL range '10.0.0.2-10.0.0.100'
set vpn l2tp remote-access default-pool 'TEST-POOL'
set vpn l2tp remote-access gateway-address '10.0.0.1'
set vpn l2tp remote-access lns host-name 'LAC'
set vpn l2tp remote-access lns shared-secret 'test123'
set vpn l2tp remote-access name-server '8.8.8.8'
set vpn l2tp remote-access ppp-options disable-ccp
```

Note: This setup requires the Compression Control Protocol (CCP) being disabled, the command `set vpn l2tp remote-access ppp-options disable-ccp` accomplishes that.

Client

In this lab we use Windows PPPoE client.

Monitoring

Monitoring on LNS side

```
vyos@vyos:~$ show l2tp-server sessions
ifname | username | ip | ip6 | ip6-dp | calling-sid | rate-limit | state
--|-----|-----|-----|-----|-----|-----|-----
l2tp0 | test@vyos.io | 10.0.0.2 | | | 192.168.139.101 | | active
00:00:35 | 188.4 KiB | 9.3 MiB
```

Monitoring on LAC side

```
Router#show pppoe session
1 session in FORWARDED (FWDED) State
1 session total
Uniq ID  PPPoE  RemMAC      Port      VT  VA      State
      SID  LocMAC
      1      1  000c.290b.20a6  Gi0/1     1  N/A     FWDED
      0c58.88ac.0001

Router#show l2tp
L2TP Tunnel and Session Information Total tunnels 1 sessions 1

LocTunID  RemTunID  Remote Name  State  Remote Address  Sessn L2TP Class/
Count VPDN Group
23238     2640     LAC          est    192.168.139.100 1     LAC

LocID     RemID     TunID        Username, Intf/  State  Last Chg Uniq ID
Vcid, Circuit
25641     25822     23238        test@vyos.io, Gi0/1  est    00:05:36 1
```

Monitoring on RADIUS Server side

```
root@Radius:~# cat /var/log/freeradius/radacct/192.168.139.100/detail-20240221
Wed Feb 21 13:37:17 2024
  User-Name = "test@vyos.io"
  NAS-Port = 0
  NAS-Port-Id = "l2tp0"
  NAS-Port-Type = Virtual
  Service-Type = Framed-User
  Framed-Protocol = PPP
  Calling-Station-Id = "192.168.139.101"
  Called-Station-Id = "192.168.139.100"
  Acct-Status-Type = Start
  Acct-Authentic = RADIUS
  Acct-Session-Id = "45c731e169d9a4f1"
```

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```

Acct-Session-Time = 0
Acct-Input-Octets = 0
Acct-Output-Octets = 0
Acct-Input-Packets = 0
Acct-Output-Packets = 0
Acct-Input-Gigawords = 0
Acct-Output-Gigawords = 0
Framed-IP-Address = 10.0.0.2
NAS-IP-Address = 192.168.139.100
Event-Timestamp = "Feb 21 2024 13:37:17 UTC"
Tmp-String-9 = "ai:"
Acct-Unique-Session-Id = "ea6a1089816f19c0d0f1819bc61c3318"
Timestamp = 1708522637

```

11.11 Inter-VRF Routing over VRF Lite

Virtual Routing and Forwarding is a technology that allow multiple instance of a routing table to exist within a single device. One of the key aspect of **VRFs** is that do not share the same routes or interfaces, therefore packets are forwarded between interfaces that belong to the same VRF only.

Any information related to a VRF is not exchanged between devices -or in the same device- by default, this is a technique called **VRF-Lite**.

Keep networks isolated is -in general- a good principle, but there are cases where you might need that some network can access other in a different VRF.

The scope of this document is to cover such cases in a dynamic way without the use of MPLS-LDP.

General information about L3VPNs can be found in the [L3VPN VRFs](#) chapter.

11.11.1 Overview

Let's say we have a requirement to have multiple networks.

- LAN 1
- LAN 2
- Management
- Internet

Both LANs have to be able to route between each other, both will have managed devices through a dedicated management network and both will need Internet access yet the LAN2 will need access to some set of outside networks, not all. The management network will need access to both LANs but cannot have access to/from the outside.

This scenario could be a nightmare applying regular routing and might need filtering in multiple interfaces.

A simple solution could be using different routing tables, or VRFs for all the networks so we can keep the routing restrictions. But for us to route between the different VRFs we would need a cable or a logical connection between each other:

- One cable/logical connection between LAN1 and LAN2
- One cable/logical connection between LAN1 and Internet
- One cable/logical connection between LAN2 and Internet

- One cable/logical connection between LAN1 and Management
- One cable/logical connection between LAN2 and Management

As we can see this is impractical.

To address this scenario we will use to our advantage an extension of the BGP routing protocol that will help us in the “Export” between VRFs without the need for MPLS.

MP-BGP or MultiProtocol BGP introduces two main concepts to solve this limitation: - Route Distinguisher (RD): Is used to distinguish between different VRFs –called VPNs- inside the BGP Process. The RD is appended to each IPv4 Network that is advertised into BGP for that VPN making it a unique VPNv4 route. - Route Target (RT): This is an extended BGP community append to the VPNv4 route in the Import/Export process. When a route passes from the VRF routing table into the BGP process it will add the configured export extended community(ies) for that VPN. When that route needs to go from BGP into the VRF routing table will only pass if that given VPN import policy matches any of the appended community(ies) into that prefix.

11.11.2 Topology

IP Schema

Device-A	Device-B	IPv4 Network	IPv6 Network
Core	LAN1	10.1.1.0/30	2001:db8::/127
Core	LAN2	172.16.2.0/30	2001:db8::2/127
Core	Management	192.168.3.0/30	2001:db8::4/127
Core	ISP	10.2.2.0/30	2001:db8::6/127

RD & RT Schema

VRF	RD	RT
LAN1	64496:1	64496:1
LAN2	64496:2	64496:2
Management	64496:50	64496:50
Internet	64496:100	64496:100

11.11.3 Configurations

Note: We use a static route configuration in between the Core and each LAN and Management router, and BGP between the Core router and the ISP router but any dynamic routing protocol can be used.

Remote Networks

The following template configuration can be used in each remote router based in our topology.

```
# Interface Configuration
set interface eth eth<N> address <IP ADDRESS/CIDR>

# Static default route back to Core
set protocols static route 0.0.0.0/0 next-hop <CORE IP ADDRESS>
```

Core Router

Step 1: VRF and Configurations to remote networks

- Configuration

Set the VRF name and Table ID, set interface address and bind it to the VRF. Last add the static route to the remote network.

```
# VRF name and table ID (MANDATORY)
set vrf name <VRF> table <ID>

# Interface Configuration
set interface eth eth<N> address <IP ADDRESS/CIDR>

# Assign interface to VRF
set interface eth eth<N> vrf <VRF>

# Static route to remote Network
set vrf name <VRF> protocols static route <NETWORK/CIDR> next-hop <REMOTE IP ADDRESS>
```

- Verification

Checking the routing table of the VRF should reveal both static and connected entries active. A PING test between the Core and remote router is a way to validate connectivity within the VRF.

```
# show ip route vrf <VRF>
# show ipv6 route vrf <VRF>

vyos@Core:~$ show ip route vrf LAN1
Codes: K - kernel route, C - connected, S - static, R - RIP,
       O - OSPF, I - IS-IS, B - BGP, E - EIGRP, N - NHRP,
       T - Table, v - VNC, V - VNC-Direct, A - Babel, F - PBR,
       f - OpenFabric,
       > - selected route, * - FIB route, q - queued, r - rejected, b - backup
       t - trapped, o - offload failure

VRF LAN1:
S>* 10.0.0.0/24 [1/0] via 10.1.1.2, eth0, weight 1, 00:05:41
C>* 10.1.1.0/30 is directly connected, eth0, 00:05:44

vyos@Core:~$ show ipv6 route vrf LAN1
Codes: K - kernel route, C - connected, S - static, R - RIPng,
```

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```

O - OSPFv3, I - IS-IS, B - BGP, N - NHRP, T - Table,
v - VNC, V - VNC-Direct, A - Babel, F - PBR,
f - OpenFabric,
> - selected route, * - FIB route, q - queued, r - rejected, b - backup
t - trapped, o - offload failure

VRF LAN1:
C>* 2001:db8::/127 is directly connected, eth0, 00:18:43
S>* 2001:db8:0:1::/64 [1/0] via 2001:db8::1, eth0, weight 1, 00:16:03
C>* fe80::/64 is directly connected, eth0, 00:18:43

# ping <DESTINATION> vrf <VRF>

vyos@Core:~$ ping 10.1.1.2 vrf LAN1
PING 10.1.1.2 (10.1.1.2) 56(84) bytes of data.
64 bytes from 10.1.1.2: icmp_seq=1 ttl=64 time=1.52 ms
64 bytes from 10.1.1.2: icmp_seq=2 ttl=64 time=0.830 ms
^C
--- 10.1.1.2 ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1002ms
rtt min/avg/max/mdev = 0.830/1.174/1.518/0.344 ms
vyos@Core:~$ ping 10.0.0.1 vrf LAN1
PING 10.0.0.1 (10.0.0.1) 56(84) bytes of data.
64 bytes from 10.0.0.1: icmp_seq=1 ttl=64 time=0.785 ms
64 bytes from 10.0.0.1: icmp_seq=2 ttl=64 time=0.948 ms
^C
--- 10.0.0.1 ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1002ms
rtt min/avg/max/mdev = 0.785/0.866/0.948/0.081 ms

vyos@Core:~$ ping 2001:db8:0:1::1 vrf LAN1
PING 2001:db8:0:1::1(2001:db8:0:1::1) 56 data bytes
64 bytes from 2001:db8:0:1::1: icmp_seq=1 ttl=64 time=3.04 ms
64 bytes from 2001:db8:0:1::1: icmp_seq=2 ttl=64 time=1.04 ms
64 bytes from 2001:db8:0:1::1: icmp_seq=3 ttl=64 time=0.925 ms
^C
--- 2001:db8:0:1::1 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2004ms
rtt min/avg/max/mdev = 0.925/1.665/3.035/0.969 ms

```

Step 2: BGP Configuration for VRF-Lite

- Configuration

Setting BGP global local-as as well inside the VRF. Redistribute static routes to inject configured networks into the BGP process but still inside the VRF.

```

# set BGP global local-as
set protocols bgp system-as <ASN>

# set BGP VRF local-as and redistribution

```

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```
set vrf name <VRF> protocols bgp address-family <AF IPv4/IPv6> redistribute static
```

- Verification

Check the BGP VRF table and verify if the static routes are injected showing the correct next-hop information.

```
# show ip bgp vrf <VRF>
# show bgp vrf <VRF> ipv6

vyos@Core:~$ show ip bgp vrf LAN1
BGP table version is 3, local router ID is 10.1.1.1, vrf id 8
Default local pref 100, local AS 64496
Status codes: s suppressed, d damped, h history, * valid, > best, = multipath,
               i internal, r RIB-failure, S Stale, R Removed
Nexthop codes: @NNN nexthop's vrf id, < announce-nh-self
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

   Network          Next Hop           Metric LocPrf Weight Path
*> 10.0.0.0/24      10.1.1.2             0           32768 ?

vyos@Core# run show bgp vrf LAN1 ipv6
BGP table version is 13, local router ID is 10.1.1.1, vrf id 8
Default local pref 100, local AS 64496
Status codes: s suppressed, d damped, h history, * valid, > best, = multipath,
               i internal, r RIB-failure, S Stale, R Removed
Nexthop codes: @NNN nexthop's vrf id, < announce-nh-self
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

   Network          Next Hop           Metric LocPrf Weight Path
*> 2001:db8:0:1::/64
                        2001:db8::1             0           32768 ?
```

Step 3: VPN Configuration

- Configuration

Within the VRF we set the Route-Distinguisher (RD) and Route-Targets (RT), then we enable the export/import VPN.

```
# set Route-distinguisher
set vrf name <VRF> protocols bgp address-family <AF IPv4/IPv6> rd vpn export '<RD>'

# set route-target for import/export
# Note: RT are a list that can be more than one community between apostrophe
#       and separated by blank space. Ex: '<RT:1> <RT:2> <RT:3>'
set vrf name <VRF> protocols bgp address-family <AF IPv4/IPv6> route-target vpn export '
↪<RT:Export>'
set vrf name <VRF> protocols bgp address-family <AF IPv4/IPv6> route-target vpn import '
↪<RT:Import>'

# Enable VPN export/import under this VRF
```

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```
set vrf name <VRF> protocols bgp address-family <AF IPv4/IPv6> export vpn
set vrf name <VRF> protocols bgp address-family <AF IPv4/IPv6> import vpn
```

A key point to understand is that if we need two VRFs to communicate between each other EXPORT rt from VRF1 has to be in the IMPORT rt list from VRF2. But this is only in ONE direction, to complete the communication the EXPORT rt from VRF2 has to be in the IMPORT rt list from VRF1.

There are some cases where this is not needed -for example, in some DDoS appliance- but most inter-vrf routing designs use the above configurations.

- Verification

After configured all the VRFs involved in this topology we take a deeper look at both BGP and Routing table for the VRF LAN1

```
# show ip bgp vrf <VRF>
# show bgp vrf <VRF> ipv6
```

```
vyos@Core# run show ip bgp vrf LAN1
BGP table version is 53, local router ID is 10.1.1.1, vrf id 8
Default local pref 100, local AS 64496
Status codes: s suppressed, d damped, h history, * valid, > best, = multipath,
               i internal, r RIB-failure, S Stale, R Removed
Nexthop codes: @NNN nexthop's vrf id, < announce-nh-self
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found
```

Network	Next Hop	Metric	LocPrf	Weight	Path
*> 0.0.0.0/0	10.2.2.2@7<			0	64497 i
*> 10.0.0.0/24	10.1.1.2	0		32768	?
*> 10.2.2.0/30	10.2.2.2@7<	0		0	64497 ?
*> 192.0.2.0/24	10.2.2.2@7<	0		0	64497 ?
*> 192.168.0.0/24	192.168.3.2@11<	0		32768	?
*> 198.51.100.0/24	10.2.2.2@7<	0		0	64497 ?
*> 203.0.113.0/24	10.2.2.2@7<	0		0	64497 ?

```
vyos@Core# run show bgp vrf LAN1 ipv6
BGP table version is 13, local router ID is 10.1.1.1, vrf id 8
Default local pref 100, local AS 64496
Status codes: s suppressed, d damped, h history, * valid, > best, = multipath,
               i internal, r RIB-failure, S Stale, R Removed
Nexthop codes: @NNN nexthop's vrf id, < announce-nh-self
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found
```

Network	Next Hop	Metric	LocPrf	Weight	Path
*> ::/0	fe80::5200:ff:fe02:3@7<			0	64497 i
*> 2001:db8::6/127	fe80::5200:ff:fe02:3@7<	0		0	64497 ?
*> 2001:db8:0:1::/64	2001:db8::1	0		32768	?
*> 2001:db8:0:3::/64	2001:db8::5@11<	0		32768	?

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```

*> 2001:db8:1::/48 fe80::5200:ff:fe02:3@7<
                                0          0 64497 ?
*> 2001:db8:2::/48 fe80::5200:ff:fe02:3@7<
                                0          0 64497 ?
*> 2001:db8:3::/48 fe80::5200:ff:fe02:3@7<
                                0          0 64497 ?

# show ip route vrf <VRF>
# show ipv6 route vrf <VRF>

vyos@Core:~$ show ip route vrf LAN1
Codes: K - kernel route, C - connected, S - static, R - RIP,
       O - OSPF, I - IS-IS, B - BGP, E - EIGRP, N - NHRP,
       T - Table, v - VNC, V - VNC-Direct, A - Babel, F - PBR,
       f - OpenFabric,
       > - selected route, * - FIB route, q - queued, r - rejected, b - backup
       t - trapped, o - offload failure

VRF LAN1:
B>* 0.0.0.0/0 [20/0] via 10.2.2.2, eth3 (vrf Internet), weight 1, 00:00:38
S>* 10.0.0.0/24 [1/0] via 10.1.1.2, eth0, weight 1, 00:29:57
C>* 10.1.1.0/30 is directly connected, eth0, 00:29:59
B 10.2.2.0/30 [20/0] via 10.2.2.2 (vrf Internet) inactive, weight 1, 00:00:38
B>* 172.16.0.0/24 [20/0] via 172.16.2.2, eth1 (vrf LAN2), weight 1, 00:00:38
B>* 192.0.2.0/24 [20/0] via 10.2.2.2, eth3 (vrf Internet), weight 1, 00:00:38
B>* 198.51.100.0/24 [20/0] via 10.2.2.2, eth3 (vrf Internet), weight 1, 00:00:38
B>* 203.0.113.0/24 [20/0] via 10.2.2.2, eth3 (vrf Internet), weight 1, 00:00:38

vyos@Core# run show ipv6 route vrf LAN1
Codes: K - kernel route, C - connected, S - static, R - RIPng,
       O - OSPFv3, I - IS-IS, B - BGP, N - NHRP, T - Table,
       v - VNC, V - VNC-Direct, A - Babel, F - PBR,
       f - OpenFabric,
       > - selected route, * - FIB route, q - queued, r - rejected, b - backup
       t - trapped, o - offload failure

VRF LAN1:
B>* ::/0 [20/0] via fe80::5200:ff:fe02:3, eth3 (vrf Internet), weight 1, 00:07:50
C>* 2001:db8::/127 is directly connected, eth0, 05:33:43
B>* 2001:db8::6/127 [20/0] via fe80::5200:ff:fe02:3, eth3 (vrf Internet), weight 1,
↳ 00:07:50
S>* 2001:db8:0:1::/64 [1/0] via 2001:db8::1, eth0, weight 1, 05:31:03
B>* 2001:db8:0:3::/64 [20/0] via 2001:db8::5, eth2 (vrf Management), weight 1, 00:07:50
B>* 2001:db8:1::/48 [20/0] via fe80::5200:ff:fe02:3, eth3 (vrf Internet), weight 1,
↳ 00:07:50
B>* 2001:db8:2::/48 [20/0] via fe80::5200:ff:fe02:3, eth3 (vrf Internet), weight 1,
↳ 00:07:50
B>* 2001:db8:3::/48 [20/0] via fe80::5200:ff:fe02:3, eth3 (vrf Internet), weight 1,
↳ 00:07:50
C>* fe80::/64 is directly connected, eth0, 05:33:43

```

As we can see in the BGP table any imported route has been injected with a “@” followed by the VPN id; In the routing table of the VRF, if the route was installed, we can see -between round brackets- the exported VRF table.

Step 4: End to End verification

Now we perform some end-to-end testing

- From Management to LAN1/LAN2

```
vyos@Management:~$ ping 10.0.0.1 source-address 192.168.0.1
PING 10.0.0.1 (10.0.0.1) from 192.168.0.1 : 56(84) bytes of data.
64 bytes from 10.0.0.1: icmp_seq=1 ttl=63 time=1.93 ms
64 bytes from 10.0.0.1: icmp_seq=2 ttl=63 time=2.12 ms
64 bytes from 10.0.0.1: icmp_seq=3 ttl=63 time=2.12 ms
^C
--- 10.0.0.1 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2005ms
rtt min/avg/max/mdev = 1.931/2.056/2.123/0.088 ms
vyos@Management:~$ ping 172.16.0.1 source-address 192.168.0.1
PING 172.16.0.1 (172.16.0.1) from 192.168.0.1 : 56(84) bytes of data.
64 bytes from 172.16.0.1: icmp_seq=1 ttl=63 time=1.62 ms
64 bytes from 172.16.0.1: icmp_seq=2 ttl=63 time=1.75 ms
^C
--- 172.16.0.1 ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1001ms
rtt min/avg/max/mdev = 1.621/1.686/1.752/0.065 ms
vyos@Management:~$ ping 2001:db8:0:1::1 source-address 2001:db8:0:3::1
PING 2001:db8:0:1::1(2001:db8:0:1::1) from 2001:db8:0:3::1 : 56 data bytes
64 bytes from 2001:db8:0:1::1: icmp_seq=1 ttl=63 time=2.44 ms
64 bytes from 2001:db8:0:1::1: icmp_seq=2 ttl=63 time=2.40 ms
64 bytes from 2001:db8:0:1::1: icmp_seq=3 ttl=63 time=2.41 ms
^C
--- 2001:db8:0:1::1 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2003ms
rtt min/avg/max/mdev = 2.399/2.418/2.442/0.017 ms
vyos@Management:~$ ping 2001:db8:0:2::1 source-address 2001:db8:0:3::1
PING 2001:db8:0:2::1(2001:db8:0:2::1) from 2001:db8:0:3::1 : 56 data bytes
64 bytes from 2001:db8:0:2::1: icmp_seq=1 ttl=63 time=1.66 ms
64 bytes from 2001:db8:0:2::1: icmp_seq=2 ttl=63 time=1.99 ms
64 bytes from 2001:db8:0:2::1: icmp_seq=3 ttl=63 time=1.88 ms
64 bytes from 2001:db8:0:2::1: icmp_seq=4 ttl=63 time=2.32 ms
^C
--- 2001:db8:0:2::1 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3005ms
rtt min/avg/max/mdev = 1.660/1.960/2.315/0.236 ms
```

- From Management to Outside (fails as intended)

```
vyos@Management:~$ show ip route
Codes: K - kernel route, C - connected, S - static, R - RIP,
       O - OSPF, I - IS-IS, B - BGP, E - EIGRP, N - NHRP,
       T - Table, v - VNC, V - VNC-Direct, A - Babel, F - PBR,
       f - OpenFabric,
       > - selected route, * - FIB route, q - queued, r - rejected, b - backup
       t - trapped, o - offload failure

S>* 0.0.0.0/0 [1/0] via 192.168.3.1, eth2, weight 1, 00:01:58
```

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```

C>* 192.168.0.0/24 is directly connected, dum0, 00:02:05
C>* 192.168.3.0/30 is directly connected, eth2, 00:02:03
vyos@Management:~$ ping 192.0.2.1
PING 192.0.2.1 (192.0.2.1) 56(84) bytes of data.
From 192.168.3.1 icmp_seq=1 Destination Net Unreachable
From 192.168.3.1 icmp_seq=2 Destination Net Unreachable
^C
--- 192.0.2.1 ping statistics ---
2 packets transmitted, 0 received, +2 errors, 100% packet loss, time 1002ms

vyos@Management:~$ ping 195.51.100.1
PING 195.51.100.1 (195.51.100.1) 56(84) bytes of data.
From 192.168.3.1 icmp_seq=1 Destination Net Unreachable
From 192.168.3.1 icmp_seq=2 Destination Net Unreachable
From 192.168.3.1 icmp_seq=3 Destination Net Unreachable
^C
--- 195.51.100.1 ping statistics ---
3 packets transmitted, 0 received, +3 errors, 100% packet loss, time 2003ms

vyos@Management:~$ ping 2001:db8:1::1
PING 2001:db8:1::1(2001:db8:1::1) 56 data bytes
From 2001:db8::4 icmp_seq=1 Destination unreachable: No route
From 2001:db8::4 icmp_seq=2 Destination unreachable: No route
^C
--- 2001:db8:1::1 ping statistics ---
2 packets transmitted, 0 received, +2 errors, 100% packet loss, time 1002ms

vyos@Management:~$ ping 2001:db8:2::1
PING 2001:db8:2::1(2001:db8:2::1) 56 data bytes
From 2001:db8::4 icmp_seq=1 Destination unreachable: No route
From 2001:db8::4 icmp_seq=2 Destination unreachable: No route
^C
--- 2001:db8:2::1 ping statistics ---
2 packets transmitted, 0 received, +2 errors, 100% packet loss, time 1002ms

```

- LAN1 to Outside

```

vyos@LAN1:~$ ping 192.0.2.1 source-address 10.0.0.1
PING 192.0.2.1 (192.0.2.1) from 10.0.0.1 : 56(84) bytes of data.
64 bytes from 192.0.2.1: icmp_seq=1 ttl=63 time=1.47 ms
64 bytes from 192.0.2.1: icmp_seq=2 ttl=63 time=1.41 ms
64 bytes from 192.0.2.1: icmp_seq=3 ttl=63 time=1.80 ms
^C
--- 192.0.2.1 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2004ms
rtt min/avg/max/mdev = 1.414/1.563/1.803/0.171 ms
vyos@LAN1:~$ ping 198.51.100.1 source-address 10.0.0.1
PING 198.51.100.1 (198.51.100.1) from 10.0.0.1 : 56(84) bytes of data.
64 bytes from 198.51.100.1: icmp_seq=1 ttl=63 time=1.71 ms
64 bytes from 198.51.100.1: icmp_seq=2 ttl=63 time=1.83 ms
^C
--- 198.51.100.1 ping statistics ---

```

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```

2 packets transmitted, 2 received, 0% packet loss, time 1002ms
rtt min/avg/max/mdev = 1.705/1.766/1.828/0.061 ms
vyos@LAN1:~$ ping 203.0.113.1 source-address 10.0.0.1
PING 203.0.113.1 (203.0.113.1) from 10.0.0.1 : 56(84) bytes of data.
64 bytes from 203.0.113.1: icmp_seq=1 ttl=63 time=1.25 ms
64 bytes from 203.0.113.1: icmp_seq=2 ttl=63 time=1.88 ms
^C
--- 203.0.113.1 ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1003ms
rtt min/avg/max/mdev = 1.249/1.566/1.884/0.317 ms
vyos@LAN1:~$ ping 2001:db8:1::1 source-address 2001:db8:0:1::1
PING 2001:db8:1::1(2001:db8:1::1) from 2001:db8:0:1::1 : 56 data bytes
64 bytes from 2001:db8:1::1: icmp_seq=1 ttl=63 time=2.35 ms
64 bytes from 2001:db8:1::1: icmp_seq=2 ttl=63 time=2.29 ms
64 bytes from 2001:db8:1::1: icmp_seq=3 ttl=63 time=2.22 ms
^C
--- 2001:db8:1::1 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2004ms
rtt min/avg/max/mdev = 2.215/2.285/2.352/0.055 ms
vyos@LAN1:~$ ping 2001:db8:2::1 source-address 2001:db8:0:1::1
PING 2001:db8:2::1(2001:db8:2::1) from 2001:db8:0:1::1 : 56 data bytes
64 bytes from 2001:db8:2::1: icmp_seq=1 ttl=63 time=1.37 ms
64 bytes from 2001:db8:2::1: icmp_seq=2 ttl=63 time=2.68 ms
64 bytes from 2001:db8:2::1: icmp_seq=3 ttl=63 time=2.00 ms
^C
--- 2001:db8:2::1 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2003ms
rtt min/avg/max/mdev = 1.367/2.015/2.679/0.535 ms

```

Note: we are using “source-address” option cause we are not redistributing connected interfaces into BGP on the Core router hence there is no comeback route and ping will fail.

- LAN1 to LAN2

```

vyos@LAN1:~$ ping 172.16.0.1 source-address 10.0.0.1
PING 172.16.0.1 (172.16.0.1) from 10.0.0.1 : 56(84) bytes of data.
64 bytes from 172.16.0.1: icmp_seq=1 ttl=63 time=3.00 ms
64 bytes from 172.16.0.1: icmp_seq=2 ttl=63 time=2.20 ms
^C
--- 172.16.0.1 ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1002ms
rtt min/avg/max/mdev = 2.199/2.600/3.001/0.401 ms
vyos@LAN1:~$ ping 2001:db8:0:2::1 source-address 2001:db8:0:1::1
PING 2001:db8:0:2::1(2001:db8:0:2::1) from 2001:db8:0:1::1 : 56 data bytes
64 bytes from 2001:db8:0:2::1: icmp_seq=1 ttl=63 time=4.82 ms
64 bytes from 2001:db8:0:2::1: icmp_seq=2 ttl=63 time=1.95 ms
64 bytes from 2001:db8:0:2::1: icmp_seq=3 ttl=63 time=1.98 ms
^C
--- 2001:db8:0:2::1 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2003ms
rtt min/avg/max/mdev = 1.949/2.915/4.815/1.343 ms

```

11.11.4 Conclusions

Inter-VRF routing is a well-known solution to address complex routing scenarios that enable -in a dynamic way- to leak routes between VRFs. Is recommended to take special consideration while designing route-targets and its application as it can minimize future interventions while creating a new VRF will automatically take the desired effect in its propagation.

11.11.5 Appendix-A

Full configuration from all devices

- Core

```
set interfaces ethernet eth0 address '10.1.1.1/30'
set interfaces ethernet eth0 address '2001:db8::/127'
set interfaces ethernet eth0 vrf 'LAN1'
set interfaces ethernet eth1 address '172.16.2.1/30'
set interfaces ethernet eth1 address '2001:db8::2/127'
set interfaces ethernet eth1 vrf 'LAN2'
set interfaces ethernet eth2 address '192.168.3.1/30'
set interfaces ethernet eth2 address '2001:db8::4/127'
set interfaces ethernet eth2 vrf 'Management'
set interfaces ethernet eth3 address '10.2.2.1/30'
set interfaces ethernet eth3 address '2001:db8::6/127'
set interfaces ethernet eth3 vrf 'Internet'
set protocols bgp address-family ipv4-unicast
set protocols bgp system-as '64496'
set vrf name Internet protocols bgp address-family ipv4-unicast export vpn
set vrf name Internet protocols bgp address-family ipv4-unicast import vpn
set vrf name Internet protocols bgp address-family ipv4-unicast rd vpn export '64496:100'
set vrf name Internet protocols bgp address-family ipv4-unicast route-target vpn export
↪ '64496:100'
set vrf name Internet protocols bgp address-family ipv4-unicast route-target vpn import
↪ '64496:1 64496:2'
set vrf name Internet protocols bgp address-family ipv6-unicast export vpn
set vrf name Internet protocols bgp address-family ipv6-unicast import vpn
set vrf name Internet protocols bgp address-family ipv6-unicast rd vpn export '64496:100'
set vrf name Internet protocols bgp address-family ipv6-unicast route-target vpn export
↪ '64496:100'
set vrf name Internet protocols bgp address-family ipv6-unicast route-target vpn import
↪ '64496:1 64496:2'
set vrf name Internet protocols bgp neighbor 10.2.2.2 address-family ipv4-unicast
set vrf name Internet protocols bgp neighbor 10.2.2.2 remote-as '64497'
set vrf name Internet protocols bgp neighbor 2001:db8::7 address-family ipv6-unicast
set vrf name Internet protocols bgp neighbor 2001:db8::7 remote-as '64497'
set vrf name Internet table '104'
set vrf name LAN1 protocols bgp address-family ipv4-unicast export vpn
set vrf name LAN1 protocols bgp address-family ipv4-unicast import vpn
set vrf name LAN1 protocols bgp address-family ipv4-unicast rd vpn export '64496:1'
set vrf name LAN1 protocols bgp address-family ipv4-unicast redistribute static
set vrf name LAN1 protocols bgp address-family ipv4-unicast route-target vpn export
↪ '64496:1'
```

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```

set vrf name LAN1 protocols bgp address-family ipv4-unicast route-target vpn import
↳ '64496:100 64496:50 64496:2'
set vrf name LAN1 protocols bgp address-family ipv6-unicast export vpn
set vrf name LAN1 protocols bgp address-family ipv6-unicast import vpn
set vrf name LAN1 protocols bgp address-family ipv6-unicast rd vpn export '64496:1'
set vrf name LAN1 protocols bgp address-family ipv6-unicast redistribute static
set vrf name LAN1 protocols bgp address-family ipv6-unicast route-target vpn export
↳ '64496:1'
set vrf name LAN1 protocols bgp address-family ipv6-unicast route-target vpn import
↳ '64496:100 64496:50 64496:2'
set vrf name LAN1 protocols static route 10.0.0.0/24 next-hop 10.1.1.2
set vrf name LAN1 protocols static route6 2001:db8:0:1::/64 next-hop 2001:db8::1
set vrf name LAN1 table '101'
set vrf name LAN2 protocols bgp address-family ipv4-unicast export vpn
set vrf name LAN2 protocols bgp address-family ipv4-unicast import vpn
set vrf name LAN2 protocols bgp address-family ipv4-unicast rd vpn export '64496:2'
set vrf name LAN2 protocols bgp address-family ipv4-unicast redistribute static
set vrf name LAN2 protocols bgp address-family ipv4-unicast route-target vpn export
↳ '64496:2'
set vrf name LAN2 protocols bgp address-family ipv4-unicast route-target vpn import
↳ '64496:100 64496:50 64496:1'
set vrf name LAN2 protocols bgp address-family ipv6-unicast export vpn
set vrf name LAN2 protocols bgp address-family ipv6-unicast import vpn
set vrf name LAN2 protocols bgp address-family ipv6-unicast rd vpn export '64496:2'
set vrf name LAN2 protocols bgp address-family ipv6-unicast redistribute static
set vrf name LAN2 protocols bgp address-family ipv6-unicast route-target vpn export
↳ '64496:2'
set vrf name LAN2 protocols bgp address-family ipv6-unicast route-target vpn import
↳ '64496:100 64496:50 64496:1'
set vrf name LAN2 protocols static route 172.16.0.0/24 next-hop 172.16.2.2
set vrf name LAN2 protocols static route6 2001:db8:0:2::/64 next-hop 2001:db8::3
set vrf name LAN2 table '102'
set vrf name Management protocols bgp address-family ipv4-unicast export vpn
set vrf name Management protocols bgp address-family ipv4-unicast import vpn
set vrf name Management protocols bgp address-family ipv4-unicast rd vpn export '64496:50
↳ '
set vrf name Management protocols bgp address-family ipv4-unicast redistribute static
set vrf name Management protocols bgp address-family ipv4-unicast route-target vpn
↳ export '64496:50'
set vrf name Management protocols bgp address-family ipv4-unicast route-target vpn
↳ import '64496:1 64496:2'
set vrf name Management protocols bgp address-family ipv6-unicast export vpn
set vrf name Management protocols bgp address-family ipv6-unicast import vpn
set vrf name Management protocols bgp address-family ipv6-unicast rd vpn export '64496:50
↳ '
set vrf name Management protocols bgp address-family ipv6-unicast redistribute static
set vrf name Management protocols bgp address-family ipv6-unicast route-target vpn
↳ export '64496:50'
set vrf name Management protocols bgp address-family ipv6-unicast route-target vpn
↳ import '64496:1 64496:2'
set vrf name Management protocols static route 192.168.0.0/24 next-hop 192.168.3.2
set vrf name Management protocols static route6 2001:db8:0:3::/64 next-hop 2001:db8::5

```

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```
set vrf name Management table '103'
```

- LAN1

```
set interfaces dummy dum0 address '10.0.0.1/24'
set interfaces dummy dum0 address '2001:db8:0:1::1/64'
set interfaces ethernet eth0 address '10.1.1.2/30'
set interfaces ethernet eth0 address '2001:db8::1/127'
set protocols static route 0.0.0.0/0 next-hop 10.1.1.1
set protocols static route6 ::/0 next-hop 2001:db8::1
```

- LAN2

```
set interfaces dummy dum0 address '172.16.0.1/24'
set interfaces dummy dum0 address '2001:db8:0:2::1/64'
set interfaces ethernet eth0 hw-id '50:00:00:03:00:00'
set interfaces ethernet eth1 address '172.16.2.2/30'
set interfaces ethernet eth1 address '2001:db8::3/127'
set protocols static route 0.0.0.0/0 next-hop 172.16.2.1
set protocols static route6 ::/0 next-hop 2001:db8::2
```

- Management

```
set interfaces dummy dum0 address '192.168.0.1/24'
set interfaces dummy dum0 address '2001:db8:0:3::1/64'
set interfaces ethernet eth2 address '192.168.3.2/30'
set interfaces ethernet eth2 address '2001:db8::5/127'
set protocols static route 0.0.0.0/0 next-hop 192.168.3.1
set protocols static route6 ::/0 next-hop 2001:db8::4
```

- ISP

```
set interfaces dummy dum0 address '192.0.2.1/24'
set interfaces dummy dum0 address '2001:db8:1::1/48'
set interfaces dummy dum1 address '198.51.100.1/24'
set interfaces dummy dum1 address '2001:db8:2::1/48'
set interfaces dummy dum2 address '203.0.113.1/24'
set interfaces dummy dum2 address '2001:db8:3::1/48'
set interfaces ethernet eth3 address '10.2.2.2/30'
set interfaces ethernet eth3 address '2001:db8::7/127'
set protocols bgp address-family ipv4-unicast redistribute connected
set protocols bgp address-family ipv6-unicast redistribute connected
set protocols bgp system-as '64497'
set protocols bgp neighbor 10.2.2.1 address-family ipv4-unicast default-originate
set protocols bgp neighbor 10.2.2.1 remote-as '64496'
set protocols bgp neighbor 2001:db8::6 address-family ipv6-unicast default-originate
set protocols bgp neighbor 2001:db8::6 remote-as '64496'
set protocols static route 0.0.0.0/0 next-hop 10.2.2.1
set protocols static route6 ::/0 next-hop 2001:db8::6
```

11.11.6 Appendix-B

Route-Filtering

When importing routes using MP-BGP it is possible to filter a subset of them before are injected in the BGP table. One of the most common case is to use a route-map with an prefix-list.

- Configuration

We create a prefix-list first and add all the routes we need to.

```
# set both ipv4 and ipv6 policies

set policy prefix-list LAN2-Internet rule 1 action 'permit'
set policy prefix-list LAN2-Internet rule 1 le '24'
set policy prefix-list LAN2-Internet rule 1 prefix '198.51.0.0/16'
set policy prefix-list LAN2-Internet rule 2 action 'permit'
set policy prefix-list LAN2-Internet rule 2 prefix '192.0.2.0/24'
set policy prefix-list LAN2-Internet rule 3 action 'permit'
set policy prefix-list LAN2-Internet rule 3 prefix '192.168.0.0/24'
set policy prefix-list LAN2-Internet rule 4 action 'permit'
set policy prefix-list LAN2-Internet rule 4 prefix '10.0.0.0/24'

set policy prefix-list6 LAN2-Internet-v6 rule 1 action 'permit'
set policy prefix-list6 LAN2-Internet-v6 rule 1 prefix '2001:db8:1::/48'
set policy prefix-list6 LAN2-Internet-v6 rule 2 action 'permit'
set policy prefix-list6 LAN2-Internet-v6 rule 2 prefix '2001:db8:2::/48'
set policy prefix-list6 LAN2-Internet-v6 rule 3 action 'permit'
set policy prefix-list6 LAN2-Internet-v6 rule 3 prefix '2001:db8:0:3::/64'
set policy prefix-list6 LAN2-Internet-v6 rule 4 action 'permit'
set policy prefix-list6 LAN2-Internet-v6 rule 4 prefix '2001:db8:0:1::/64'
```

Then add a route-map and reference to above prefix. Consider that the actions taken inside the prefix will MATCH the routes that will be affected by the actions inside the rules of the route-map.

```
set policy route-map LAN2-Internet rule 1 action 'permit'
set policy route-map LAN2-Internet rule 1 match ip address prefix-list 'LAN2-Internet'

set policy route-map LAN2-Internet-v6 rule 1 action 'permit'
set policy route-map LAN2-Internet-v6 rule 1 match ipv6 address prefix-list 'LAN2-
↳Internet-v6'
```

We are using a “white list” approach by allowing only what is necessary. In case that need to implement a “black list” approach then you will need to change the action in the route-map for a deny BUT you need to add a rule that permits the rest due to the implicit deny in the route-map.

Then we need to attach the policy to the BGP process. This needs to be under the import statement in the vrf we need to filter.

```
set vrf name LAN2 protocols bgp address-family ipv4-unicast route-map vpn import 'LAN2-
↳Internet'
set vrf name LAN2 protocols bgp address-family ipv6-unicast route-map vpn import 'LAN2-
↳Internet-v6'
```

- Verification


```
# show ip route vrf LAN2

B>* 10.0.0.0/24 [20/0] via 10.1.1.2, eth0 (vrf LAN1), weight 1, 00:45:28
S>* 172.16.0.0/24 [1/0] via 172.16.2.2, eth1, weight 1, 00:45:32
C>* 172.16.2.0/30 is directly connected, eth1, 00:45:39
B>* 192.0.2.0/24 [20/0] via 10.2.2.2, eth3 (vrf Internet), weight 1, 00:45:24
B>* 192.168.0.0/24 [20/0] via 192.168.3.2, eth2 (vrf Management), weight 1, 00:45:27
B>* 198.51.100.0/24 [20/0] via 10.2.2.2, eth3 (vrf Internet), weight 1, 00:45:24

# show ipv6 route vrf LAN2

C>* 2001:db8::2/127 is directly connected, eth1, 00:46:26
B>* 2001:db8:0:1::/64 [20/0] via 2001:db8::1, eth0 (vrf LAN1), weight 1, 00:46:17
S>* 2001:db8:0:2::/64 [1/0] via 2001:db8::3, eth1, weight 1, 00:46:21
B>* 2001:db8:0:3::/64 [20/0] via 2001:db8::5, eth2 (vrf Management), weight 1, 00:46:16
B>* 2001:db8:1::/48 [20/0] via fe80::5200:ff:fe02:3, eth3 (vrf Internet), weight 1, ↵
↵ 00:46:13
B>* 2001:db8:2::/48 [20/0] via fe80::5200:ff:fe02:3, eth3 (vrf Internet), weight 1, ↵
↵ 00:46:13
C>* fe80::/64 is directly connected, eth1, 00:46:27
```

As we can see even if both VRF LAN1 and LAN2 has the same import RTs we are able to select which routes are effectively imported and installed.

11.12 DMVPN Dual HUB Dual Cloud

This document is to describe a basic setup to build DMVPN network with two Hubs and two clouds using DMVPN Phase3. OSPF is used as routing protocol inside DMVPN.

In this example we use VyOS 1.5 as HUBs and Spokes (HUB-1, HUB-2, SPOKE-2, SPOKE-3) and Cisco IOSv 15.5(3)M (SPOKE-1) as a Spoke.

11.12.1 Network Topology

11.12.2 Configurations

Underlay configuration

Networks 192.168.X.0/24 are used as LANs for every spoke.

HUB-1

```
set interfaces ethernet eth0 address '10.0.0.2/30'
set protocols static route 0.0.0.0/0 next-hop 10.0.0.1
```

HUB-2

```
set interfaces ethernet eth0 address '10.0.1.2/30'
set protocols static route 0.0.0.0/0 next-hop 10.0.1.1
```

Spoke-1

```
interface GigabitEthernet0/0
ip address 10.0.11.2 255.255.255.252
duplex auto
speed auto
media-type rj45
!
interface GigabitEthernet0/1
ip address 192.168.11.1 255.255.255.0
ip ospf 1 area 0
duplex auto
speed auto
media-type rj45
!
ip route 0.0.0.0 0.0.0.0 10.0.11.1
```

Spoke-2

```
set interfaces ethernet eth0 address '10.0.12.2/30'
set interfaces ethernet eth1 address '192.168.12.1/24'
set protocols static route 0.0.0.0/0 next-hop 10.0.12.1
```

Spoke-3

```
set interfaces ethernet eth0 address '10.0.13.2/30'
set interfaces ethernet eth1 address '192.168.13.1/24'
set protocols static route 0.0.0.0/0 next-hop 10.0.13.1
```

NHRP configuration

The next step is to configure the NHRP protocol. In a Dual cloud network, every HUB has to be configured with one GRE multipoint tunnel interface and every spoke has to be configured with two tunnel interfaces, one tunnel to each hub. In this example tunnel networks are 10.100.100.0/24 for the first cloud and 10.100.101.0/24 for the second cloud. But VyOS uses FRR for NHRP, that is why the tunnel address mask must be /32.

HUB-1

```
set interfaces tunnel tun100 address '10.100.100.1/32'
set interfaces tunnel tun100 enable-multicast
set interfaces tunnel tun100 encapsulation 'gre'
set interfaces tunnel tun100 ip adjust-mss '1360'
set interfaces tunnel tun100 mtu '1436'
set interfaces tunnel tun100 parameters ip key '42'
set interfaces tunnel tun100 source-interface 'eth0'
set protocols nhrp tunnel tun100 authentication 'vyos'
set protocols nhrp tunnel tun100 holdtime '300'
set protocols nhrp tunnel tun100 multicast 'dynamic'
set protocols nhrp tunnel tun100 network-id '1'
set protocols nhrp tunnel tun100 redirect
set protocols nhrp tunnel tun100 registration-no-unique
```

HUB-2

```

set interfaces tunnel tun101 address '10.100.101.1/32'
set interfaces tunnel tun101 enable-multicast
set interfaces tunnel tun101 encapsulation 'gre'
set interfaces tunnel tun101 ip adjust-mss '1360'
set interfaces tunnel tun101 mtu '1436'
set interfaces tunnel tun101 parameters ip key '43'
set interfaces tunnel tun101 source-interface 'eth0'
set protocols nhrp tunnel tun101 authentication 'vyos'
set protocols nhrp tunnel tun101 holdtime '300'
set protocols nhrp tunnel tun101 multicast 'dynamic'
set protocols nhrp tunnel tun101 network-id '2'
set protocols nhrp tunnel tun101 redirect
set protocols nhrp tunnel tun101 registration-no-unique

```

Spoke-1

```

interface Tunnel100
 ip address 10.100.100.11 255.255.255.0
 no ip redirects
 ip mtu 1436
 ip nhrp authentication vyos
 ip nhrp map multicast 10.0.0.2
 ip nhrp network-id 1
 ip nhrp holdtime 300
 ip nhrp nhs 10.100.100.1 nbma 10.0.0.2
 ip nhrp shortcut
 ip tcp adjust-mss 1360
 tunnel source GigabitEthernet0/0
 tunnel mode gre multipoint
 tunnel key 42
!
interface Tunnel101
 ip address 10.100.101.11 255.255.255.0
 no ip redirects
 ip mtu 1436
 ip nhrp authentication vyos
 ip nhrp map multicast 10.0.1.2
 ip nhrp network-id 2
 ip nhrp holdtime 300
 ip nhrp nhs 10.100.101.1 nbma 10.0.1.2
 ip nhrp shortcut
 ip tcp adjust-mss 1360
 tunnel source GigabitEthernet0/0
 tunnel mode gre multipoint
 tunnel key 43

```

Spoke-2

```

set interfaces tunnel tun100 address '10.100.100.12/32'
set interfaces tunnel tun100 enable-multicast
set interfaces tunnel tun100 encapsulation 'gre'
set interfaces tunnel tun100 ip adjust-mss '1360'
set interfaces tunnel tun100 mtu '1436'

```

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```

set interfaces tunnel tun100 parameters ip key '42'
set interfaces tunnel tun100 source-interface 'eth0'
set interfaces tunnel tun101 address '10.100.101.12/32'
set interfaces tunnel tun101 enable-multicast
set interfaces tunnel tun101 encapsulation 'gre'
set interfaces tunnel tun101 ip adjust-mss '1360'
set interfaces tunnel tun101 mtu '1436'
set interfaces tunnel tun101 parameters ip key '43'
set interfaces tunnel tun101 source-interface 'eth0'
set protocols nhrp tunnel tun100 authentication 'vyos'
set protocols nhrp tunnel tun100 holdtime '300'
set protocols nhrp tunnel tun100 multicast '10.0.0.2'
set protocols nhrp tunnel tun100 network-id '1'
set protocols nhrp tunnel tun100 nhs tunnel-ip dynamic nbma '10.0.0.2'
set protocols nhrp tunnel tun100 registration-no-unique
set protocols nhrp tunnel tun100 shortcut
set protocols nhrp tunnel tun101 authentication 'vyos'
set protocols nhrp tunnel tun101 holdtime '300'
set protocols nhrp tunnel tun101 multicast '10.0.1.2'
set protocols nhrp tunnel tun101 network-id '2'
set protocols nhrp tunnel tun101 nhs tunnel-ip dynamic nbma '10.0.1.2'
set protocols nhrp tunnel tun101 registration-no-unique
set protocols nhrp tunnel tun101 shortcut

```

Spoke-3

```

set protocols nhrp tunnel tun100 authentication 'vyos'
set protocols nhrp tunnel tun100 holdtime '300'
set protocols nhrp tunnel tun100 multicast '10.0.0.2'
set protocols nhrp tunnel tun100 network-id '1'
set protocols nhrp tunnel tun100 nhs tunnel-ip dynamic nbma '10.0.0.2'
set protocols nhrp tunnel tun100 registration-no-unique
set protocols nhrp tunnel tun100 shortcut
set protocols nhrp tunnel tun101 authentication 'vyos'
set protocols nhrp tunnel tun101 holdtime '300'
set protocols nhrp tunnel tun101 multicast '10.0.1.2'
set protocols nhrp tunnel tun101 network-id '2'
set protocols nhrp tunnel tun101 nhs tunnel-ip dynamic nbma '10.0.1.2'
set protocols nhrp tunnel tun101 registration-no-unique
set protocols nhrp tunnel tun101 shortcut

```

Overlay configuration

The last step is to configure the routing protocol. In this scenario, OSPF was chosen as the dynamic routing protocol. But you can use iBGP or eBGP. To form fast convergence it is possible to use BFD protocol.

HUB-1

```

set protocols ospf interface tun100 area '0'
set protocols ospf interface tun100 network 'point-to-multipoint'
set protocols ospf interface tun100 passive disable
set protocols ospf passive-interface 'default'

```

HUB-2

```
set protocols ospf interface tun101 area '0'
set protocols ospf interface tun101 network 'point-to-multipoint'
set protocols ospf interface tun101 passive disable
set protocols ospf passive-interface 'default'
```

Spoke-1

```
interface Tunnel100
 ip ospf network point-to-multipoint
 ip ospf dead-interval 40
 ip ospf hello-interval 10
 ip ospf 1 area 0
!
interface Tunnel101
 ip ospf network point-to-multipoint
 ip ospf dead-interval 40
 ip ospf hello-interval 10
 ip ospf 1 area 0
!
router ospf 1
 passive-interface default
 no passive-interface Tunnel100
 no passive-interface Tunnel101
```

Spoke-2

```
set protocols ospf interface eth1 area '0'
set protocols ospf interface tun100 area '0'
set protocols ospf interface tun100 network 'point-to-multipoint'
set protocols ospf interface tun100 passive disable
set protocols ospf interface tun101 area '0'
set protocols ospf interface tun101 network 'point-to-multipoint'
set protocols ospf interface tun101 passive disable
set protocols ospf passive-interface 'default'
```

Spoke-3

```
set protocols ospf interface eth1 area '0'
set protocols ospf interface tun100 area '0'
set protocols ospf interface tun100 network 'point-to-multipoint'
set protocols ospf interface tun100 passive disable
set protocols ospf interface tun101 area '0'
set protocols ospf interface tun101 network 'point-to-multipoint'
set protocols ospf interface tun101 passive disable
set protocols ospf passive-interface 'default'
```

Security configuration

Tunnels can be encrypted by IPSEC for security.

HUB-1

```
set vpn ipsec esp-group ESP-HUB lifetime '1800'
set vpn ipsec esp-group ESP-HUB mode 'transport'
set vpn ipsec esp-group ESP-HUB pfs 'disable'
set vpn ipsec esp-group ESP-HUB proposal 1 encryption 'aes256'
set vpn ipsec esp-group ESP-HUB proposal 1 hash 'sha1'
set vpn ipsec ike-group IKE-HUB key-exchange 'ikev1'
set vpn ipsec ike-group IKE-HUB lifetime '3600'
set vpn ipsec ike-group IKE-HUB proposal 1 dh-group '2'
set vpn ipsec ike-group IKE-HUB proposal 1 encryption 'aes256'
set vpn ipsec ike-group IKE-HUB proposal 1 hash 'sha1'
set vpn ipsec interface 'eth0'
set vpn ipsec profile NHRPVPN authentication mode 'pre-shared-secret'
set vpn ipsec profile NHRPVPN authentication pre-shared-secret 'secret'
set vpn ipsec profile NHRPVPN bind tunnel 'tun100'
set vpn ipsec profile NHRPVPN esp-group 'ESP-HUB'
set vpn ipsec profile NHRPVPN ike-group 'IKE-HUB'
```

HUB-2

```
set vpn ipsec esp-group ESP-HUB lifetime '1800'
set vpn ipsec esp-group ESP-HUB mode 'transport'
set vpn ipsec esp-group ESP-HUB pfs 'disable'
set vpn ipsec esp-group ESP-HUB proposal 1 encryption 'aes256'
set vpn ipsec esp-group ESP-HUB proposal 1 hash 'sha1'
set vpn ipsec ike-group IKE-HUB key-exchange 'ikev1'
set vpn ipsec ike-group IKE-HUB lifetime '3600'
set vpn ipsec ike-group IKE-HUB proposal 1 dh-group '2'
set vpn ipsec ike-group IKE-HUB proposal 1 encryption 'aes256'
set vpn ipsec ike-group IKE-HUB proposal 1 hash 'sha1'
set vpn ipsec interface 'eth0'
set vpn ipsec profile NHRPVPN authentication mode 'pre-shared-secret'
set vpn ipsec profile NHRPVPN authentication pre-shared-secret 'secret'
set vpn ipsec profile NHRPVPN bind tunnel 'tun101'
set vpn ipsec profile NHRPVPN esp-group 'ESP-HUB'
set vpn ipsec profile NHRPVPN ike-group 'IKE-HUB'
```

VyOS Spokes have the same configuration

```
set vpn ipsec esp-group ESP-HUB lifetime '1800'
set vpn ipsec esp-group ESP-HUB mode 'transport'
set vpn ipsec esp-group ESP-HUB pfs 'disable'
set vpn ipsec esp-group ESP-HUB proposal 1 encryption 'aes256'
set vpn ipsec esp-group ESP-HUB proposal 1 hash 'sha1'
set vpn ipsec ike-group IKE-HUB key-exchange 'ikev1'
set vpn ipsec ike-group IKE-HUB lifetime '3600'
set vpn ipsec ike-group IKE-HUB proposal 1 dh-group '2'
set vpn ipsec ike-group IKE-HUB proposal 1 encryption 'aes256'
set vpn ipsec ike-group IKE-HUB proposal 1 hash 'sha1'
```

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```

set vpn ipsec interface 'eth0'
set vpn ipsec profile NHRPVPN authentication mode 'pre-shared-secret'
set vpn ipsec profile NHRPVPN authentication pre-shared-secret 'secret'
set vpn ipsec profile NHRPVPN bind tunnel 'tun100'
set vpn ipsec profile NHRPVPN bind tunnel 'tun101'
set vpn ipsec profile NHRPVPN esp-group 'ESP-HUB'
set vpn ipsec profile NHRPVPN ike-group 'IKE-HUB'

```

SPOKE-1

```

crypto isakmp policy 1
  encr aes 256
  authentication pre-share
  group 2
  lifetime 3600
crypto isakmp key secret address 0.0.0.0
!
!
crypto ipsec transform-set ESP_TRANSFORMSET esp-aes 256 esp-sha-hmac
mode transport
!
!
crypto ipsec profile gre_protection
  set security-association lifetime seconds 1800
  set transform-set ESP_TRANSFORMSET
!
interface Tunnel100
  tunnel protection ipsec profile gre_protection shared
!
interface Tunnel101
  tunnel protection ipsec profile gre_protection shared

```

11.12.3 Monitoring

All spokes created IPSec tunnels to Hubs, are registered on Hubs using NHRP protocol and formed adjacency in OSPF.

```

vyos@HUB-1:~$ show vpn ipsec sa

```

Connection	State	Uptime	Bytes In/Out	Packets In/Out	Remote
↪ address	Remote ID	Proposal			
dmvpn-NHRPVPN-tun100-child	up	6m1s	4K/5K	51/56	10.0.13.
↪ 2	10.0.13.2	AES_CBC_256/HMAC_SHA1_96			
dmvpn-NHRPVPN-tun100-child	up	6m36s	4K/6K	56/65	10.0.12.
↪ 2	10.0.12.2	AES_CBC_256/HMAC_SHA1_96			
dmvpn-NHRPVPN-tun100-child	up	8m49s	6K/6K	73/77	10.0.11.
↪ 2	10.0.11.2	AES_CBC_256/HMAC_SHA1_96			

```

vyos@HUB-1:~$ show ip nhrp cache

```

Iface	Type	Protocol	NBMA	Claimed NBMA
↪ Flags	Identity			

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```
tun100    dynamic  10.100.100.12          10.0.12.2          10.0.12.2
↪      T      10.0.12.2
tun100    dynamic  10.100.100.13          10.0.13.2          10.0.13.2
↪      T      10.0.13.2
tun100    dynamic  10.100.100.11          10.0.11.2          10.0.11.2
↪      T      10.0.11.2
tun100    local    10.100.100.1            10.0.0.2            10.0.0.2
↪      -
```

vyos@HUB-1:~\$ show ip ospf neighbor

Neighbor ID	Pri	State	Up Time	Dead Time	Address	Interface
			RXmtL RqstL DBsmL			
192.168.11.1	1	Full/DROther	17m01s	36.201s	10.100.100.11	tun100:10.
↪ 100.100.1		0	0 0			
192.168.12.1	1	Full/DROther	9m42s	37.443s	10.100.100.12	tun100:10.
↪ 100.100.1		0	0 0			
192.168.13.1	1	Full/DROther	9m15s	35.053s	10.100.100.13	tun100:10.
↪ 100.100.1		0	0 0			

First, we see that LANs are accessible through hubs using OSPF routes.

SPOKE-1#show ip route

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
a - application route
+ - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is 10.0.11.1 to network 0.0.0.0

```
.....
192.168.11.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.11.0/24 is directly connected, GigabitEthernet0/1
L    192.168.11.1/32 is directly connected, GigabitEthernet0/1
O    192.168.12.0/24 [110/1002] via 10.100.101.1, 00:14:36, Tunnel101
[110/1002] via 10.100.100.1, 00:16:13, Tunnel100
O    192.168.13.0/24 [110/1002] via 10.100.101.1, 00:14:36, Tunnel101
[110/1002] via 10.100.100.1, 00:15:45, Tunnel100
```

vyos@SPOKE-2:~\$ show ip route

Codes: K - kernel route, C - connected, L - local, S - static,
R - RIP, O - OSPF, I - IS-IS, B - BGP, E - EIGRP, N - NHRP,
T - Table, v - VNC, V - VNC-Direct, A - Babel, F - PBR,
f - OpenFabric, t - Table-Direct,
> - selected route, * - FIB route, q - queued, r - rejected, b - backup
t - trapped, o - offload failure

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```
.....
O>* 192.168.11.0/24 [110/3] via 10.100.100.1, tun100 onlink, weight 1, 00:12:36
*                               via 10.100.101.1, tun101 onlink, weight 1, 00:12:36
O 192.168.12.0/24 [110/1] is directly connected, eth1, weight 1, 01:24:40
C>* 192.168.12.0/24 is directly connected, eth1, weight 1, 01:24:43
L>* 192.168.12.1/32 is directly connected, eth1, weight 1, 01:24:43
O>* 192.168.13.0/24 [110/3] via 10.100.100.1, tun100 onlink, weight 1, 00:12:36
*                               via 10.100.101.1, tun101 onlink, weight 1, 00:12:36
```

After initiating traffic between SPOKES sites, Phase 3 of DMVPN will work. For instance, traceroute was generated from PC-SPOKE-2 to PC-SPOKE-1

```
PC-SPOKE-2 : 192.168.12.2 255.255.255.0 gateway 192.168.12.1
```

```
PC-SPOKE-2> trace 192.168.11.2
trace to 192.168.11.2, 8 hops max, press Ctrl+C to stop
 1 192.168.12.1 0.558 ms 0.378 ms 0.561 ms
 2 10.100.101.1 1.768 ms 1.158 ms 1.744 ms
 3 10.100.101.11 7.196 ms 4.971 ms 4.793 ms
 4 *192.168.11.2 7.747 ms (ICMP type:3, code:3, Destination port unreachable)
```

```
PC-SPOKE-2> trace 192.168.11.2
trace to 192.168.11.2, 8 hops max, press Ctrl+C to stop
 1 192.168.12.1 0.562 ms 0.396 ms 0.364 ms
 2 10.100.100.11 4.401 ms 4.399 ms 4.174 ms
 3 *192.168.11.2 3.241 ms (ICMP type:3, code:3, Destination port unreachable)
```

First trace goes via HUB but the second goes directly from SPOKE-1 to SPOKE-2. Now routing tables are changed. LAN networks 192.168.12.0/24 and 192.168.11.0/24 available directly via SPOKES.

```
vyos@SPOKE-2:~$ show ip route
Codes: K - kernel route, C - connected, L - local, S - static,
       R - RIP, O - OSPF, I - IS-IS, B - BGP, E - EIGRP, N - NHRP,
       T - Table, v - VNC, V - VNC-Direct, A - Babel, F - PBR,
       f - OpenFabric, t - Table-Direct,
       > - selected route, * - FIB route, q - queued, r - rejected, b - backup
       t - trapped, o - offload failure

N>* 192.168.11.0/24 [10/0] via 10.100.100.11, tun100 onlink, weight 1, 00:00:14
O 192.168.11.0/24 [110/3] via 10.100.100.1, tun100 onlink, weight 1, 00:00:54
                        via 10.100.101.1, tun101 onlink, weight 1, 00:00:54
```

```
SPOKE-1# show ip route next-hop-override
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
```

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+ - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is 10.0.11.1 to network 0.0.0.0

```
0  % 192.168.12.0/24 [110/1002] via 10.100.101.1, 00:24:09, Tunnel101
    [110/1002] via 10.100.100.1, 00:25:46, Tunnel100
    [NHO][110/1] via 10.100.100.12, 00:00:03, Tunnel100
```

NHRP shows shortcuts on Spokes

```
vyos@SPOKE-2:~$ show ip nhrp shortcut
```

Type	Prefix	Via	Identity
dynamic	192.168.11.0/24	10.100.100.11	10.0.11.2

```
SPOKE-1# show ip nhrp shortcut
```

```
10.100.100.12/32 via 10.100.100.12
  Tunnel100 created 00:09:59, expire 00:02:21
  Type: dynamic, Flags: router nhop rib nho
  NBMA address: 10.0.12.2
192.168.12.0/24 via 10.100.100.12
  Tunnel100 created 00:02:38, expire 00:02:21
  Type: dynamic, Flags: router rib nho
  NBMA address: 10.0.12.2
```

A new Spoke to Spoke IPSec tunnel is created

```
SPOKE-1#show crypto isakmp sa
```

```
IPv4 Crypto ISAKMP SA
```

dst	src	state	conn-id	status
10.0.0.2	10.0.11.2	QM_IDLE	1002	ACTIVE
10.0.12.2	10.0.11.2	QM_IDLE	1004	ACTIVE
10.0.1.2	10.0.11.2	QM_IDLE	1003	ACTIVE

```
vyos@SPOKE-2:~$ show vpn ipsec sa
```

Connection	State	Uptime	Bytes In/Out	Packets In/Out	Remote
↪address Remote ID Proposal	Proposal				
dmvpn-NHRPVPN-tun100-child	up	7m26s	4K/4K	57/53	10.0.0.
↪2 10.0.0.2	AES_CBC_256/HMAC_SHA1_96				
dmvpn-NHRPVPN-tun100-child	up	11m48s	316B/1K	3/15	10.0.11.
↪2 10.0.11.2	AES_CBC_256/HMAC_SHA1_96				
dmvpn-NHRPVPN-tun101-child	up	5m58s	5K/4K	62/51	10.0.1.
↪2 10.0.1.2	AES_CBC_256/HMAC_SHA1_96				

11.12.4 Summary

If one of the Hubs loses connectivity to the Internet, the other Hub will be available and take the main role. This is a simple example where only one internet connection is used. But in the real world, there can be two connections to the Internet. In this case, there is a recommendation to build each tunnel via each Internet connection, choose the main cloud, and manipulate traffic via a routing protocol. It allows the creation failover on link-level connections too.

11.13 QoS example

11.13.1 Configuration ‘dscp’ and shaper using QoS

In this case, we’ll try to make a simple lab using QoS and the general ability of the VyOS system. We recommend you to go through the main article about [QoS](#) first.

Using the general schema for example:

We have four hosts on the local network 172.17.1.0/24. All hosts are labeled CS0 by default. We need to replace labels on all hosts except vpc8. We will replace the labels on the nearest router “VyOS3” using the IP addresses of the sources.

- 172.17.1.2 CS0 -> CS4
- 172.17.1.3 CS0 -> CS5
- 172.17.1.4 CS0 -> CS6
- 172.17.1.40 CS0 by default

Next, we will replace only all CS4 labels on the “VyOS2” router.

- CS4 -> CS5

In the end, we will configure the traffic shaper using QoS mechanisms on the “VYOS2” router.

11.13.2 Configuration:

Set IP addresses on all VPCs and a default gateway 172.17.1.1. We’ll use in this case only static routes. On the VyOS3 router, we need to change the ‘dscp’ labels for the VPCs. To do this, we use this configuration.

```
set interfaces ethernet eth0 address '10.1.1.100/24'
set interfaces ethernet eth1 address '172.17.1.1/24'
set protocols static route 0.0.0.0/0 next-hop 10.1.1.1
set qos policy shaper vyos3 class 10 match ADDRESS10 ip source address '172.17.1.2/32'
set qos policy shaper vyos3 class 10 set-dscp 'CS4'
set qos policy shaper vyos3 class 20 match ADDRESS20 ip source address '172.17.1.3/32'
set qos policy shaper vyos3 class 20 set-dscp 'CS5'
set qos policy shaper vyos3 class 30 match ADDRESS30 ip source address '172.17.1.4/32'
set qos policy shaper vyos3 class 30 set-dscp 'CS6'
set qos policy shaper vyos3 default bandwidth '10%'
set qos policy shaper vyos3 default ceiling '100%'
set qos policy shaper vyos3 default priority '7'
set qos policy shaper vyos3 default queue-type 'fair-queue'
set qos interface eth0 egress 'vyos3'
```

Main rules:

- ADDRESS10 change CS0 -> CS4 source 172.17.1.2/32
- ADDRESS20 change CS0 -> CS5 source 172.17.1.3/32
- ADDRESS30 change CS0 -> CS6 source 172.17.1.4/32

Check the result

Before the interface eth0 on router VyOS3

After the interface eth0 on router VyOS3

On the router, VyOS4 set all traffic as CS4. We have to configure the default class and class for changing all labels from CS0 to CS4

```
set interfaces ethernet eth0 address '10.2.1.100/24'
set protocols static route 0.0.0.0/0 next-hop 10.2.1.1
set qos policy shaper vyos4 class 10 bandwidth '100%'
set qos policy shaper vyos4 class 10 burst '15k'
set qos policy shaper vyos4 class 10 match ALL ether protocol 'all'
set qos policy shaper vyos4 class 10 queue-type 'fair-queue'
set qos policy shaper vyos4 class 10 set-dscp 'CS4'
set qos policy shaper vyos4 default bandwidth '10%'
set qos policy shaper vyos4 default burst '15k'
set qos policy shaper vyos4 default ceiling '100%'
set qos policy shaper vyos4 default priority '7'
set qos policy shaper vyos4 default queue-type 'fair-queue'
set qos interface eth0 egress 'vyos4'
```

Next on the router VyOS2 we will change labels on all incoming traffic only from CS4-> CS6

```
set interfaces ethernet eth0 address '10.1.1.1/24'
set interfaces ethernet eth1 address '10.2.1.1/24'
set interfaces ethernet eth2 address '10.9.9.1/24'
set protocols static route 172.17.1.0/24 next-hop 10.1.1.100
set qos policy shaper vyos2 class 10 bandwidth '100%'
set qos policy shaper vyos2 class 10 burst '15k'
set qos policy shaper vyos2 class 10 match VYOS2 ip dscp 'CS4'
set qos policy shaper vyos2 class 10 queue-type 'fair-queue'
set qos policy shaper vyos2 class 10 set-dscp 'CS5'
set qos policy shaper vyos2 default bandwidth '100%'
set qos policy shaper vyos2 default burst '15k'
set qos policy shaper vyos2 default ceiling '100%'
set qos policy shaper vyos2 default priority '7'
set qos policy shaper vyos2 default queue-type 'fair-queue'
set qos interface eth2 egress 'vyos2'
```

- 172.17.1.2/24 CS0

- 172.17.1.2/24 CS0 - > CS4
- 172.17.1.2/24 CS4 - > CS5

In the end, on the router “VyOS2” we will set outgoing bandwidth limits between the “VyOS3” and “VyOS1” routers. Let’s set a limit for IP 10.1.1.100 = 5 Mbps(Tx). We will check the result of the work with the help of the “iPerf” utility.

Set up bandwidth limits on the eth2 interface of the router “VyOS2”.

```
vyos@vyos2# show qos policy shaper vyos2 class 20
bandwidth 5mbit
description "for VyOS3 eth0"
match VyOS3 {
    ip {
        source {
            address 10.1.1.100/32
        }
    }
}
```

Check the result.

As we see shaper is working and the traffic will not work over 5 Mbit/s.

11.14 Segment-routing IS-IS example

When utilizing VyOS in an environment with Cisco IOS-XR gear you can use this blue print as an initial setup to get MPLS ISIS-SR working between those two devices. The lab was build using EVE-NG (Emulated Virtual Environment NG).

Fig. 1: ISIS-SR example network

The below configuration is used as example where we keep focus on VyOS-P1/VyOS-P2/XRv-P3 which we share the settings.

11.14.1 Configuration

- VyOS-P1:

```
set interfaces dummy dum0 address '192.0.2.1/32'
set interfaces ethernet eth1 address '192.0.2.5/30'
set interfaces ethernet eth1 mtu '8000'
set interfaces ethernet eth3 address '192.0.2.21/30'
set interfaces ethernet eth3 mtu '8000'
set protocols isis interface dum0 passive
set protocols isis interface eth1 network point-to-point
set protocols isis interface eth3 network point-to-point
set protocols isis level 'level-2'
```

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```
set protocols isis log-adjacency-changes
set protocols isis metric-style 'wide'
set protocols isis net '49.0000.0000.0000.0001.00'
set protocols isis segment-routing maximum-label-depth '8'
set protocols isis segment-routing prefix 192.0.2.1/32 index value '1'
set protocols mpls interface 'eth1'
set protocols mpls interface 'eth3'
set system host-name 'P1-VyOS'
```

- XRv-P3:

```
hostname P3-VyOS
interface Loopback0
  ipv4 address 192.0.2.3 255.255.255.255
!
interface GigabitEthernet0/0/0/1
  mtu 8014
  ipv4 address 192.0.2.6 255.255.255.252
!
interface GigabitEthernet0/0/0/2
  mtu 8014
  ipv4 address 192.0.2.18 255.255.255.252
!
router isis VyOS
  is-type level-2-only
  net 49.0000.0000.0000.0003.00
  log adjacency changes
  address-family ipv4 unicast
    metric-style wide
    segment-routing mpls
!
interface Loopback0
  passive
  address-family ipv4 unicast
    prefix-sid index 3
!
!
interface GigabitEthernet0/0/0/1
  point-to-point
  address-family ipv4 unicast
!
!
interface GigabitEthernet0/0/0/2
  point-to-point
  address-family ipv4 unicast
!
!
!
```

- VyOS-P2:

```
set interfaces dummy dum0 address '192.0.2.2/32'
```

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```

set interfaces ethernet eth2 address '192.0.2.17/30'
set interfaces ethernet eth2 mtu '8000'
set interfaces ethernet eth3 address '192.0.2.26/30'
set interfaces ethernet eth3 mtu '8000'
set protocols isis interface dum0 passive
set protocols isis interface eth2 network point-to-point
set protocols isis interface eth3 network point-to-point
set protocols isis level 'level-2'
set protocols isis log-adjacency-changes
set protocols isis metric-style 'wide'
set protocols isis net '49.0000.0000.0000.0002.00'
set protocols isis segment-routing maximum-label-depth '8'
set protocols isis segment-routing prefix 192.0.2.2/32 index value '2'
set protocols mpls interface 'eth2'
set protocols mpls interface 'eth3'
set system host-name 'P2-VyOS'

```

This gives us MPLS segment routing enabled and labels forwarding :

```

vyos@P1-VyOS:~$ show mpls table
Inbound Label  Type           Nexthop           Outbound Label
-----
15000          SR (IS-IS)      192.0.2.6         implicit-null
15001          SR (IS-IS)      192.0.2.22        implicit-null
15002          SR (IS-IS)      fe80::5200:ff:fe04:3 implicit-null
16002          SR (IS-IS)      192.0.2.6         16002
16003          SR (IS-IS)      192.0.2.6         implicit-null
16011          SR (IS-IS)      192.0.2.22        implicit-null

```

```

vyos@P2-VyOS:~$ show mpls table
Inbound Label  Type           Nexthop           Outbound Label
-----
15000          SR (IS-IS)      192.0.2.18        implicit-null
16001          SR (IS-IS)      192.0.2.18        16001
16003          SR (IS-IS)      192.0.2.18        implicit-null
16011          SR (IS-IS)      192.0.2.18        16011

```

```
RP/0/0/CPU0:P3-VyOS#show mpls forwarding
```

```
Tue Mar 28 17:47:18.928 UTC
```

Local Label	Outgoing Label	Prefix or ID	Outgoing Interface	Next Hop	Bytes Switched
16001	Pop	SR Pfx (idx 1)	Gi0/0/0/1	192.0.2.5	0
16002	Pop	SR Pfx (idx 2)	Gi0/0/0/2	192.0.2.17	0
16011	16011	SR Pfx (idx 11)	Gi0/0/0/1	192.0.2.5	0
24000	Pop	SR Adj (idx 1)	Gi0/0/0/1	192.0.2.5	0
24001	Pop	SR Adj (idx 3)	Gi0/0/0/1	192.0.2.5	0
24002	Pop	SR Adj (idx 1)	Gi0/0/0/2	192.0.2.17	0
24003	Pop	SR Adj (idx 3)	Gi0/0/0/2	192.0.2.17	0

VyOS is able to check MSD per devices:

```
vyos@P1-VyOS:~$ show isis segment-routing node
```

```
Area VyOS:
```

```
IS-IS L1 SR-Nodes:
```

```
IS-IS L2 SR-Nodes:
```

System ID	SRGB	SRLB	Algorithm	MSD
0000.0000.0001	16000 - 23999	15000 - 15999	SPF	8
0000.0000.0002	16000 - 23999	15000 - 15999	SPF	8
0000.0000.0003	16000 - 23999	0 - 4294967295	SPF	10
0000.0000.0011	16000 - 23999	15000 - 15999	SPF	8

```
vyos@P2-VyOS:~$ show isis segment-routing node
```

```
Area VyOS:
```

```
IS-IS L1 SR-Nodes:
```

```
IS-IS L2 SR-Nodes:
```

System ID	SRGB	SRLB	Algorithm	MSD
0000.0000.0001	16000 - 23999	15000 - 15999	SPF	8
0000.0000.0002	16000 - 23999	15000 - 15999	SPF	8
0000.0000.0003	16000 - 23999	0 - 4294967295	SPF	10
0000.0000.0011	16000 - 23999	15000 - 15999	SPF	8

Here is the routing tables showing the MPLS segment routing label operations:

```
vyos@P1-VyOS:~$ show ip route isis
```

```
Codes: K - kernel route, C - connected, S - static, R - RIP,
```

```
O - OSPF, I - IS-IS, B - BGP, E - EIGRP, N - NHRP,
```

```
T - Table, v - VNC, V - VNC-Direct, A - Babel, F - PBR,
```

```
f - OpenFabric,
```

```
> - selected route, * - FIB route, q - queued, r - rejected, b - backup
```

```
t - trapped, o - offload failure
```

```
I>* 192.0.2.2/32 [115/30] via 192.0.2.6, eth1, label 16002, weight 1, 1d03h18m
```

```
I>* 192.0.2.3/32 [115/10] via 192.0.2.6, eth1, label implicit-null, weight 1, 1d03h18m
```

```
I 192.0.2.4/30 [115/20] via 192.0.2.6, eth1 inactive, weight 1, 1d03h18m
```

```
I>* 192.0.2.11/32 [115/20] via 192.0.2.22, eth3, label implicit-null, weight 1, 1d02h47m
```

```
I>* 192.0.2.16/30 [115/20] via 192.0.2.6, eth1, weight 1, 1d03h18m
```

```
I 192.0.2.20/30 [115/20] via 192.0.2.22, eth3 inactive, weight 1, 1d02h48m
```

```
I>* 192.0.2.24/30 [115/30] via 192.0.2.6, eth1, weight 1, 1d03h18m
```

```
vyos@P2-VyOS:~$ show ip route isis
```

```
Codes: K - kernel route, C - connected, S - static, R - RIP,
```

```
O - OSPF, I - IS-IS, B - BGP, E - EIGRP, N - NHRP,
```

```
T - Table, v - VNC, V - VNC-Direct, A - Babel, F - PBR,
```

```
f - OpenFabric,
```

```
> - selected route, * - FIB route, q - queued, r - rejected, b - backup
```

```
t - trapped, o - offload failure
```

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```

I>* 192.0.2.1/32 [115/30] via 192.0.2.18, eth2, label 16001, weight 1, 1d03h17m
I>* 192.0.2.3/32 [115/10] via 192.0.2.18, eth2, label implicit-null, weight 1, 1d03h17m
I>* 192.0.2.4/30 [115/20] via 192.0.2.18, eth2, weight 1, 1d03h17m
I>* 192.0.2.11/32 [115/40] via 192.0.2.18, eth2, label 16011, weight 1, 1d02h47m
I 192.0.2.16/30 [115/20] via 192.0.2.18, eth2 inactive, weight 1, 1d03h17m
I>* 192.0.2.20/30 [115/30] via 192.0.2.18, eth2, weight 1, 1d03h17m

```

```

RP/0/0/CPU0:P3-VyOS#show route isis
Tue Mar 28 18:19:16.417 UTC

```

```

i L2 192.0.2.1/32 [115/20] via 192.0.2.5, 1d03h, GigabitEthernet0/0/0/1
i L2 192.0.2.2/32 [115/20] via 192.0.2.17, 1d03h, GigabitEthernet0/0/0/2
i L2 192.0.2.11/32 [115/30] via 192.0.2.5, 1d02h, GigabitEthernet0/0/0/1
i L2 192.0.2.20/30 [115/20] via 192.0.2.5, 1d03h, GigabitEthernet0/0/0/1
i L2 192.0.2.24/30 [115/20] via 192.0.2.17, 1d03h, GigabitEthernet0/0/0/2

```

Information about prefix-sid and label-operation from VyOS

```

vyos@P1-VyOS:~$ show isis route prefix-sid
Area VyOS:
IS-IS L2 IPv4 routing table:

```

Prefix	Metric	Interface	Nexthop	SID	Label Op.
192.0.2.1/32	0	-	-	-	-
192.0.2.2/32	30	eth1	192.0.2.6	2	Swap(16002, 16002)
192.0.2.3/32	10	eth1	192.0.2.6	3	Pop(16003)
192.0.2.4/30	20	eth1	192.0.2.6	-	-
192.0.2.16/30	20	eth1	192.0.2.6	-	-
192.0.2.20/30	0	-	-	-	-
192.0.2.24/30	30	eth1	192.0.2.6	-	-

```

vyos@P2-VyOS:~$ show isis route prefix-sid
Area VyOS:
IS-IS L2 IPv4 routing table:

```

Prefix	Metric	Interface	Nexthop	SID	Label Op.
192.0.2.1/32	30	eth2	192.0.2.18	1	Swap(16001, 16001)
192.0.2.2/32	0	-	-	-	-
192.0.2.3/32	10	eth2	192.0.2.18	3	Pop(16003)
192.0.2.4/30	20	eth2	192.0.2.18	-	-
192.0.2.16/30	20	eth2	192.0.2.18	-	-
192.0.2.20/30	30	eth2	192.0.2.18	-	-
192.0.2.24/30	0	-	-	-	-

Ping between VyOS-P1 / VyOS-P2 to confirm reachability:

```

vyos@P1-VyOS:~$ ping 192.0.2.2 source-address 192.0.2.1
PING 192.0.2.2 (192.0.2.2) from 192.0.2.1 : 56(84) bytes of data.
64 bytes from 192.0.2.2: icmp_seq=1 ttl=63 time=3.47 ms
64 bytes from 192.0.2.2: icmp_seq=2 ttl=63 time=2.06 ms

```

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```

64 bytes from 192.0.2.2: icmp_seq=3 ttl=63 time=3.90 ms
64 bytes from 192.0.2.2: icmp_seq=4 ttl=63 time=3.87 ms
^C
--- 192.0.2.2 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3004ms
rtt min/avg/max/mdev = 2.064/3.326/3.903/0.748 ms

vyos@P2-VyOS:~$ ping 192.0.2.1 source-address 192.0.2.2
PING 192.0.2.1 (192.0.2.1) from 192.0.2.2 : 56(84) bytes of data.
64 bytes from 192.0.2.1: icmp_seq=1 ttl=63 time=2.91 ms
64 bytes from 192.0.2.1: icmp_seq=2 ttl=63 time=3.23 ms
64 bytes from 192.0.2.1: icmp_seq=3 ttl=63 time=2.91 ms
64 bytes from 192.0.2.1: icmp_seq=4 ttl=63 time=2.85 ms
^C
--- 192.0.2.1 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3005ms
rtt min/avg/max/mdev = 2.846/2.972/3.231/0.151 ms

```

11.15 NMP example

Consider how to quickly set up NMP and VyOS for monitoring. NMP is multi-vendor network monitoring from ‘SolarWinds’ built to scale and expand with the needs of your network.

11.15.1 Configuration ‘VyOS’

First prepare our VyOS router for connection to NMP. We have to set up the SNMP protocol and connectivity between the router and NMP.

```

set interfaces ethernet eth0 address 'dhcp'
set system name-server '8.8.8.8'
set service snmp community router authorization 'test'
set service snmp community router network '0.0.0.0/0'

```

11.15.2 Configuration ‘NMP’

Next, you should just follow the pictures:

In the end, you’ll get a powerful instrument for monitoring the VyOS systems.

11.16 Ansible example

11.16.1 Setting up Ansible on a server running the Debian operating system.

In this example, we will set up a simple use of Ansible to configure multiple VyOS routers. We have four pre-configured routers with this configuration:

Using the general schema for example:

We have four pre-configured routers with this configuration:

```
set interfaces ethernet eth0 address dhcp
set service ssh
commit
save
```

- vyos7 - 192.0.2.105
- vyos8 - 192.0.2.106
- vyos9 - 192.0.2.107
- vyos10 - 192.0.2.108

11.16.2 Install Ansible:

```
# apt-get install ansible
Do you want to continue? [Y/n] y
```

11.16.3 Install Paramiko:

```
#apt-get install -y python3-paramiko
```

11.16.4 Check the version:

```
# ansible --version
ansible 2.10.8
config file = None
configured module search path = ['/root/.ansible/plugins/modules', '/usr/share/ansible/
↳ plugins/modules']
ansible python module location = /usr/lib/python3/dist-packages/ansible
executable location = /usr/bin/ansible
python version = 3.9.2 (default, Feb 28 2021, 17:03:44) [GCC 10.2.1 20210110]
```

11.16.5 Basic configuration of ansible.cfg:

```
# nano /root/ansible.cfg
[defaults]
host_key_checking = no
```

11.16.6 Add all the VyOS hosts:

```
# nano /root/hosts
[vyos_hosts]
vyos7 ansible_ssh_host=192.0.2.105
vyos8 ansible_ssh_host=192.0.2.106
vyos9 ansible_ssh_host=192.0.2.107
vyos10 ansible_ssh_host=192.0.2.108
```

11.16.7 Add general variables:

```
# mkdir /root/group_vars/
# nano /root/group_vars/vyos_hosts
ansible_python_interpreter: /usr/bin/python3
ansible_network_os: vyos
ansible_connection: network_cli
ansible_user: vyos
ansible_ssh_pass: vyos
```

11.16.8 Add a simple playbook with the tasks for each router:

```
# nano /root/main.yml
---
- hosts: vyos_hosts
  gather_facts: 'no'
  tasks:
    - name: Configure general settings for the vyos hosts group
      vyos_config:
        lines:
          - set system name-server 192.0.2.1
          - set interfaces ethernet eth0 description '#WAN#'
          - set interfaces ethernet eth1 description '#LAN#'
          - set interfaces ethernet eth2 disable
          - set interfaces ethernet eth3 disable
          - set system host-name {{ inventory_hostname }}
        save: true
```

11.16.9 Start the playbook:

```

ansible-playbook -i hosts main.yml
PLAY [vyos_hosts] *****

TASK [Configure general settings for the vyos hosts group] *****
ok: [vyos9]
ok: [vyos10]
ok: [vyos7]
ok: [vyos8]

PLAY RECAP *****
vyos10                : ok=2    changed=0    unreachable=0    failed=0    skipped=0
↳ rescued=0    ignored=0
vyos7                  : ok=2    changed=0    unreachable=0    failed=0    skipped=0
↳ rescued=0    ignored=0
vyos8                  : ok=2    changed=0    unreachable=0    failed=0    skipped=0
↳ rescued=0    ignored=0
vyos9                  : ok=2    changed=0    unreachable=0    failed=0    skipped=0
↳ rescued=0    ignored=0

```

11.16.10 Check the result on the vyos10 router:

```

vyos@vyos10:~$ show interfaces
Codes: S - State, L - Link, u - Up, D - Down, A - Admin Down
Interface      IP Address      S/L  Description
-----
eth0            192.0.2.108/24  u/u  WAN
eth1            -               u/u  LAN
eth2            -               A/D
eth3            -               A/D
lo              127.0.0.1/8    u/u
::1/128

vyos@vyos10:~$ sh configuration commands | grep 192.0.2.1
set system name-server '192.0.2.1'

```

11.16.11 The simple way without configuration of the hostname (one task for all routers):

```

# nano /root/hosts_v2
[vyos_hosts_group]
vyos7 ansible_ssh_host=192.0.2.105
vyos8 ansible_ssh_host=192.0.2.106
vyos9 ansible_ssh_host=192.0.2.107
vyos10 ansible_ssh_host=192.0.2.108
[vyos_hosts_group:vars]
ansible_python_interpreter=/usr/bin/python3
ansible_user=vyos

```

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```

ansible_ssh_pass=vyos
ansible_network_os=vyos
ansible_connection=network_cli

```

```
# nano /root/main_v2.yml
```

```
---
```

```

- hosts: vyos_hosts_group
  connection: network_cli
  gather_facts: 'no'
  tasks:
    - name: Configure remote vyos_hosts_group
      vyos_config:
        lines:
          - set system name-server 192.0.2.1
          - set interfaces ethernet eth0 description WAN
          - set interfaces ethernet eth1 description LAN
          - set interfaces ethernet eth2 disable
          - set interfaces ethernet eth3 disable
        save: true

```

```
# ansible-playbook -i hosts_v2 main_v2.yml
```

```
PLAY [vyos_hosts_group] *****
```

```
TASK [Configure remote vyos_hosts_group] *****
```

```
ok: [vyos8]
```

```
ok: [vyos7]
```

```
ok: [vyos9]
```

```
ok: [vyos10]
```

```
PLAY RECAP *****
```

```

vyos10                                : ok=1    changed=0    unreachable=0    failed=0    skipped=0_
↪   rescued=0    ignored=0
vyos7                                : ok=1    changed=0    unreachable=0    failed=0    skipped=0_
↪   rescued=0    ignored=0
vyos8                                : ok=1    changed=0    unreachable=0    failed=0    skipped=0_
↪   rescued=0    ignored=0
vyos9                                : ok=1    changed=0    unreachable=0    failed=0    skipped=0_
↪   rescued=0    ignored=0

```

In the next chapter of the example, we'll use Ansible with jinja2 templates and variables.

11.17 Policy-based Site-to-Site VPN IPsec between VyOS and Cisco

This document is to describe a basic setup using policy-based site-to-site VPN IPsec. In this example we use VyOS 1.5 and Cisco IOS. Cisco initiates IPsec connection only if interesting traffic present. For stable work we recommend configuring an initiator role on VyOS side.

11.17.1 Network Topology

11.17.2 Prerequisites

VyOS:

WAN IP	10.0.1.2/30
LAN1 IP	192.168.0.1/24
LAN2 IP	192.168.1.1/24

Cisco:

WAN IP	10.0.2.2/30
LAN1 IP	192.168.10.1/24
LAN2 IP	192.168.11.1/24

IKE parameters:

Encryption	AES-256
HASH	SHA-1
Diff-Helman Group	14
Life-Time	28800
IKE Version	2

IPsec parameters:

Encryption	AES-256
HASH	SHA-256
Life-Time	3600
PFS	disable

Traffic Selectors

192.168.0.0/24 <==> 192.168.10.0/24

192.168.1.0/24 <==> 192.168.11.0/24

Hosts configuration

PC1 IP	192.168.0.2
PC2 IP	192.168.1.2
PC3 IP	192.168.10.2
PC4 IP	192.168.11.2

11.17.3 Configuration

Note: Pfs is disabled in Cisco by default.

VyOS

```

set interfaces ethernet eth0 address '10.0.1.2/30'
set interfaces ethernet eth1 address '192.168.0.1/24'
set interfaces ethernet eth2 address '192.168.1.1/24'
set protocols static route 0.0.0.0/0 next-hop 10.0.1.1
set vpn ipsec authentication psk AUTH-PSK id '10.0.1.2'
set vpn ipsec authentication psk AUTH-PSK id '10.0.2.2'
set vpn ipsec authentication psk AUTH-PSK secret 'dGVzdA=='
set vpn ipsec authentication psk AUTH-PSK secret-type 'base64'
set vpn ipsec esp-group ESP-GROUP lifetime '3600'
set vpn ipsec esp-group ESP-GROUP pfs 'disable'
set vpn ipsec esp-group ESP-GROUP proposal 10 encryption 'aes256'
set vpn ipsec esp-group ESP-GROUP proposal 10 hash 'sha256'
set vpn ipsec ike-group IKE-GROUP close-action 'start'
set vpn ipsec ike-group IKE-GROUP dead-peer-detection action 'restart'
set vpn ipsec ike-group IKE-GROUP dead-peer-detection interval '10'
set vpn ipsec ike-group IKE-GROUP key-exchange 'ikev2'
set vpn ipsec ike-group IKE-GROUP lifetime '28800'
set vpn ipsec ike-group IKE-GROUP proposal 10 dh-group '14'
set vpn ipsec ike-group IKE-GROUP proposal 10 encryption 'aes256'
set vpn ipsec ike-group IKE-GROUP proposal 10 hash 'sha1'
set vpn ipsec site-to-site peer CISCO authentication local-id '10.0.1.2'
set vpn ipsec site-to-site peer CISCO authentication mode 'pre-shared-secret'
set vpn ipsec site-to-site peer CISCO authentication remote-id '10.0.2.2'
set vpn ipsec site-to-site peer CISCO connection-type 'initiate'
set vpn ipsec site-to-site peer CISCO default-esp-group 'ESP-GROUP'
set vpn ipsec site-to-site peer CISCO ike-group 'IKE-GROUP'
set vpn ipsec site-to-site peer CISCO local-address '10.0.1.2'
set vpn ipsec site-to-site peer CISCO remote-address '10.0.2.2'
set vpn ipsec site-to-site peer CISCO tunnel 1 local prefix '192.168.0.0/24'
set vpn ipsec site-to-site peer CISCO tunnel 1 remote prefix '192.168.10.0/24'
set vpn ipsec site-to-site peer CISCO tunnel 2 local prefix '192.168.1.0/24'
set vpn ipsec site-to-site peer CISCO tunnel 2 remote prefix '192.168.11.0/24'

```


Cisco

```

crypto ikev2 proposal aes-cbc-256-proposal
  encryption aes-cbc-256
  integrity sha1
  group 14
!
crypto ikev2 policy policy1
  match address local 10.0.2.2
  proposal aes-cbc-256-proposal
!
crypto ikev2 keyring keys
  peer VyOS
    address 10.0.1.2
    pre-shared-key local test
    pre-shared-key remote test
!
crypto ikev2 profile IKEv2-profile
  match identity remote address 10.0.1.2 255.255.255.255
  authentication remote pre-share
  authentication local pre-share
  keyring local keys
  lifetime 28800
!
crypto ipsec transform-set TS esp-aes 256 esp-sha256-hmac
  mode tunnel
!
crypto map IPSEC-map 10 ipsec-isakmp
  set peer 10.0.1.2
  set security-association lifetime seconds 3600
  set transform-set TS
  set ikev2-profile IKEv2-profile
  match address cryptoacl
!
interface GigabitEthernet0/0
  ip address 10.0.2.2 255.255.255.252
  crypto map IPSEC-map
!
interface GigabitEthernet0/1
  ip address 192.168.10.1 255.255.255.0
!
interface GigabitEthernet0/2
  ip address 192.168.11.1 255.255.255.0
!
ip route 0.0.0.0 0.0.0.0 10.0.2.1
!
ip access-list extended cryptoacl
  permit ip 192.168.10.0 0.0.0.255 192.168.0.0 0.0.0.255
  permit ip 192.168.11.0 0.0.0.255 192.168.1.0 0.0.0.255

```

11.17.4 Monitoring

Monitoring on VyOS side

IKE SAs:

```
vyos@vyos:~$ show vpn ike sa
```

Peer ID / IP				Local ID / IP				
-----				-----				
10.0.2.2 10.0.2.2				10.0.1.2 10.0.1.2				
State	IKESVer	Encrypt	Hash	D-H Group	NAT-T	A-Time	L-Time	
----	-----	-----	----	-----	-----	-----	-----	-----
up	IKEv2	AES_CBC_256	HMAC_SHA1_96	MODP_2048	no	304	26528	

IPsec SAs:

```
vyos@vyos:~$ show vpn ipsec sa
```

Connection	State	Uptime	Bytes In/Out	Packets In/Out	Remote address	
↪ Remote ID	Proposal					

↪ -----						
CISCO-tunnel-1	up	6m6s	0B/0B	0/0	10.0.2.2	↪
↪ 10.0.2.2	AES_CBC_256/HMAC_SHA2_256_128					
CISCO-tunnel-2	up	6m6s	0B/0B	0/0	10.0.2.2	↪
↪ 10.0.2.2	AES_CBC_256/HMAC_SHA2_256_128					

Monitoring on Cisco side

IKE SAs:

```
Cisco#show crypto ikev2 sa
```

IPv4 Crypto IKEv2 SA			
Tunnel-id	Local	Remote	fvr/f/ivr/f
1	10.0.2.2/4500	10.0.1.2/4500	none/none
Encr: AES-CBC, keysize: 256, PRF: SHA1, Hash: SHA96, DH Grp:14, Auth sign: PSK,↪			
↪ Auth verify: PSK			
Life/Active Time: 28800/471 sec			
IPv6 Crypto IKEv2 SA			

IPsec SAs:

```
Cisco#show crypto ipsec sa
```

```
interface: GigabitEthernet0/0
```

```
  Crypto map tag: IPSEC-map, local addr 10.0.2.2
```

```
  protected vrf: (none)
```

```
  local ident (addr/mask/prot/port): (192.168.11.0/255.255.255.0/0/0)
```

```
  remote ident (addr/mask/prot/port): (192.168.1.0/255.255.255.0/0/0)
```

```
  current_peer 10.0.1.2 port 4500
```

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```

    PERMIT, flags={origin_is_acl,}
#pkts encaps: 0, #pkts encrypt: 0, #pkts digest: 0
#pkts decaps: 0, #pkts decrypt: 0, #pkts verify: 0
#pkts compressed: 0, #pkts decompressed: 0
#pkts not compressed: 0, #pkts compr. failed: 0
#pkts not decompressed: 0, #pkts decompress failed: 0
#send errors 0, #recv errors 0

local crypto endpt.: 10.0.2.2, remote crypto endpt.: 10.0.1.2
plaintext mtu 1438, path mtu 1500, ip mtu 1500, ip mtu idb GigabitEthernet0/0
current outbound spi: 0xC81F83DA(3357508570)
PFS (Y/N): N, DH group: none

inbound esp sas:
spi: 0x8C63C51E(2355348766)
  transform: esp-256-aes esp-sha256-hmac ,
  in use settings ={Tunnel, }
  conn id: 23, flow_id: SW:23, sibling_flags 80000040, crypto map: IPSEC-map
  sa timing: remaining key lifetime (k/sec): (4231729/3585)
  IV size: 16 bytes
  replay detection support: Y
  Status: ACTIVE(ACTIVE)

inbound ah sas:

inbound pcg sas:

outbound esp sas:
spi: 0xC81F83DA(3357508570)
  transform: esp-256-aes esp-sha256-hmac ,
  in use settings ={Tunnel, }
  conn id: 24, flow_id: SW:24, sibling_flags 80000040, crypto map: IPSEC-map
  sa timing: remaining key lifetime (k/sec): (4231729/3585)
  IV size: 16 bytes
  replay detection support: Y
  Status: ACTIVE(ACTIVE)

outbound ah sas:

outbound pcg sas:

protected vrf: (none)
local ident (addr/mask/prot/port): (192.168.10.0/255.255.255.0/0/0)
remote ident (addr/mask/prot/port): (192.168.0.0/255.255.255.0/0/0)
current_peer 10.0.1.2 port 4500
  PERMIT, flags={origin_is_acl,}
#pkts encaps: 0, #pkts encrypt: 0, #pkts digest: 0
#pkts decaps: 0, #pkts decrypt: 0, #pkts verify: 0
#pkts compressed: 0, #pkts decompressed: 0
#pkts not compressed: 0, #pkts compr. failed: 0
#pkts not decompressed: 0, #pkts decompress failed: 0
#send errors 0, #recv errors 0

```

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```

local crypto endpt.: 10.0.2.2, remote crypto endpt.: 10.0.1.2
plaintext mtu 1438, path mtu 1500, ip mtu 1500, ip mtu idb GigabitEthernet0/0
current outbound spi: 0xC40C7A20(3289152032)
PFS (Y/N): N, DH group: none

inbound esp sas:
spi: 0x2948B6CB(692631243)
  transform: esp-256-aes esp-sha256-hmac ,
  in use settings ={Tunnel, }
  conn id: 21, flow_id: SW:21, sibling_flags 80000040, crypto map: IPSEC-map
  sa timing: remaining key lifetime (k/sec): (4194891/3581)
  IV size: 16 bytes
  replay detection support: Y
  Status: ACTIVE(ACTIVE)

inbound ah sas:

inbound pcsp sas:

outbound esp sas:
spi: 0xC40C7A20(3289152032)
  transform: esp-256-aes esp-sha256-hmac ,
  in use settings ={Tunnel, }
  conn id: 22, flow_id: SW:22, sibling_flags 80000040, crypto map: IPSEC-map
  sa timing: remaining key lifetime (k/sec): (4194891/3581)
  IV size: 16 bytes
  replay detection support: Y
  Status: ACTIVE(ACTIVE)

outbound ah sas:

outbound pcsp sas:

```

Checking Connectivity

ICMP packets from PC1 to PC3.

```

PC1> ping 192.168.10.2

84 bytes from 192.168.10.2 icmp_seq=1 ttl=62 time=8.479 ms
84 bytes from 192.168.10.2 icmp_seq=2 ttl=62 time=3.344 ms
84 bytes from 192.168.10.2 icmp_seq=3 ttl=62 time=3.139 ms
84 bytes from 192.168.10.2 icmp_seq=4 ttl=62 time=3.176 ms
84 bytes from 192.168.10.2 icmp_seq=5 ttl=62 time=3.978 ms

```

ICMP packets from PC2 to PC4.

```

PC2> ping 192.168.11.2

84 bytes from 192.168.11.2 icmp_seq=1 ttl=62 time=9.687 ms

```

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```

84 bytes from 192.168.11.2 icmp_seq=2 ttl=62 time=3.286 ms
84 bytes from 192.168.11.2 icmp_seq=3 ttl=62 time=2.972 ms

```

11.18 Route-based Site-to-Site VPN IPsec between VyOS and Cisco

This document is to describe a basic setup using route-based site-to-site VPN IPsec. In this example we use VyOS 1.5 and Cisco IOS. Cisco initiates IPsec connection only if interesting traffic present. For stable work we recommend configuring an initiator role on VyOS side. OSPF is selected as routing protocol inside the tunnel.

11.18.1 Network Topology

11.18.2 Prerequisites

VyOS:

WAN IP	10.0.1.2/30
LAN1 IP	192.168.0.1/24
LAN2 IP	192.168.1.1/24

Cisco:

WAN IP	10.0.2.2/30
LAN1 IP	192.168.10.1/24
LAN2 IP	192.168.11.1/24

IKE parameters:

Encryption	AES-128
HASH	SHA-1
Diff-Helman Group	14
Life-Time	28800
IKE Version	1

IPsec parameters:

Encryption	AES-256
HASH	SHA-256
Life-Time	3600
PFS	disable

Hosts configuration

PC1 IP	192.168.0.2
PC2 IP	192.168.1.2
PC3 IP	192.168.10.2
PC4 IP	192.168.11.2

11.18.3 Configuration

Note: Pfs is disabled in Cisco by default.

VyOS

```

set interfaces ethernet eth0 address '10.0.1.2/30'
set interfaces ethernet eth1 address '192.168.0.1/24'
set interfaces ethernet eth2 address '192.168.1.1/24'
set interfaces vti vti1 address '10.100.100.1/30'
set interfaces vti vti1 mtu '1438'
set protocols ospf area 0 network '10.100.100.0/30'
set protocols ospf area 0 network '192.168.0.0/24'
set protocols ospf area 0 network '192.168.1.0/24'
set protocols ospf interface eth1 passive
set protocols ospf interface eth2 passive
set protocols ospf interface vti1 network 'point-to-point'
set protocols ospf parameters router-id '2.2.2.2'
set protocols static route 0.0.0.0/0 next-hop 10.0.1.1
set vpn ipsec authentication psk AUTH-PSK id '10.0.1.2'
set vpn ipsec authentication psk AUTH-PSK id '10.0.2.2'
set vpn ipsec authentication psk AUTH-PSK secret 'dGVzdA=='
set vpn ipsec authentication psk AUTH-PSK secret-type 'base64'
set vpn ipsec esp-group ESP-GROUP lifetime '3600'
set vpn ipsec esp-group ESP-GROUP pfs 'disable'
set vpn ipsec esp-group ESP-GROUP proposal 10 encryption 'aes256'
set vpn ipsec esp-group ESP-GROUP proposal 10 hash 'sha256'
set vpn ipsec ike-group IKE-GROUP close-action 'start'
set vpn ipsec ike-group IKE-GROUP dead-peer-detection action 'restart'
set vpn ipsec ike-group IKE-GROUP dead-peer-detection interval '10'
set vpn ipsec ike-group IKE-GROUP dead-peer-detection timeout '30'
set vpn ipsec ike-group IKE-GROUP key-exchange 'ikev1'
set vpn ipsec ike-group IKE-GROUP lifetime '28800'
set vpn ipsec ike-group IKE-GROUP proposal 10 dh-group '14'
set vpn ipsec ike-group IKE-GROUP proposal 10 encryption 'aes128'
set vpn ipsec ike-group IKE-GROUP proposal 10 hash 'sha1'
set vpn ipsec options disable-route-autoinstall
set vpn ipsec site-to-site peer CISCO authentication local-id '10.0.1.2'
set vpn ipsec site-to-site peer CISCO authentication mode 'pre-shared-secret'
set vpn ipsec site-to-site peer CISCO authentication remote-id '10.0.2.2'
set vpn ipsec site-to-site peer CISCO connection-type 'initiate'
set vpn ipsec site-to-site peer CISCO default-esp-group 'ESP-GROUP'
set vpn ipsec site-to-site peer CISCO ike-group 'IKE-GROUP'

```

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```
set vpn ipsec site-to-site peer CISCO local-address '10.0.1.2'
set vpn ipsec site-to-site peer CISCO remote-address '10.0.2.2'
set vpn ipsec site-to-site peer CISCO vti bind 'vti1'
```

Cisco

```
crypto isakmp policy 10
  encr aes
  authentication pre-share
  group 14
  lifetime 28800
crypto isakmp key test address 10.0.1.2
!
!
crypto ipsec transform-set TS esp-aes 256 esp-sha256-hmac
  mode transport
!
crypto ipsec profile IPsec-profile
  set transform-set TS
!
!
!
!
!
!
!
interface Loopback0
  ip address 1.1.1.1 255.255.255.255
!
interface Tunnel10
  ip address 10.100.100.2 255.255.255.252
  ip ospf network point-to-point
  tunnel source GigabitEthernet0/0
  tunnel mode ipsec ipv4
  tunnel destination 10.0.1.2
  tunnel protection ipsec profile IPsec-profile
!
interface GigabitEthernet0/0
  ip address 10.0.2.2 255.255.255.252
  duplex auto
  speed auto
  media-type rj45
!
interface GigabitEthernet0/1
  ip address 192.168.10.1 255.255.255.0
  duplex auto
  speed auto
  media-type rj45
!
interface GigabitEthernet0/2
```

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```

ip address 192.168.11.1 255.255.255.0
duplex auto
speed auto
media-type rj45
!
router ospf 1
router-id 1.1.1.1
passive-interface GigabitEthernet0/1
passive-interface GigabitEthernet0/2
network 10.100.100.0 0.0.0.3 area 0
network 192.168.10.0 0.0.0.255 area 0
network 192.168.11.0 0.0.0.255 area 0
!
ip route 0.0.0.0 0.0.0.0 10.0.2.1

```

11.18.4 Monitoring

Monitoring on VyOS side

IKE SAs:

```

vyos@vyos:~$ show vpn ike sa
Peer ID / IP                               Local ID / IP
-----
10.0.2.2 10.0.2.2                          10.0.1.2 10.0.1.2

  State  IKEVer  Encrypt      Hash          D-H Group    NAT-T  A-Time  L-Time
  ----  -
  up     IKEv1   AES_CBC_128  HMAC_SHA1_96  MODP_2048    no     8175   18439

```

IPsec SAs:

```

vyos@vyos:~$ show vpn ipsec sa
Connection  State  Uptime  Bytes In/Out  Packets In/Out  Remote address
↪ Remote ID  Proposal
-----
↪ -----
CISCO-vti   up     34m59s  17K/14K      224/213         10.0.2.2       10.
↪ 0.2.2     AES_CBC_256/HMAC_SHA2_256_128

```

OSPF Neighbor Status:

```

vyos@vyos:~$ show ip ospf neighbor
Neighbor ID  Pri State          Up Time          Dead Time Address      Interface
↪
↪          RXmtL RqstL DBsmL
1.1.1.1      1 Full/-         1h29m37s        39.317s 10.100.100.2  vti1:10.
↪ 100.100.1      0      0      0

```

Routing Table:


```
vyos@vyos:~$ show ip route
Codes: K - kernel route, C - connected, L - local, S - static,
       R - RIP, O - OSPF, I - IS-IS, B - BGP, E - EIGRP, N - NHRP,
       T - Table, v - VNC, V - VNC-Direct, A - Babel, F - PBR,
       f - OpenFabric, t - Table-Direct,
       > - selected route, * - FIB route, q - queued, r - rejected, b - backup
       t - trapped, o - offload failure
S>* 0.0.0.0/0 [1/0] via 10.0.1.1, eth0, weight 1, 00:07:54
C>* 10.0.1.0/30 is directly connected, eth0, weight 1, 00:07:59
L>* 10.0.1.2/32 is directly connected, eth0, weight 1, 00:07:59
O 10.100.100.0/30 [110/1] is directly connected, vti1, weight 1, 00:07:50
C>* 10.100.100.0/30 is directly connected, vti1, weight 1, 00:07:50
L>* 10.100.100.1/32 is directly connected, vti1, weight 1, 00:07:50
O 192.168.0.0/24 [110/1] is directly connected, eth1, weight 1, 00:07:54
C>* 192.168.0.0/24 is directly connected, eth1, weight 1, 00:07:59
L>* 192.168.0.1/32 is directly connected, eth1, weight 1, 00:07:59
O 192.168.1.0/24 [110/1] is directly connected, eth2, weight 1, 00:07:54
C>* 192.168.1.0/24 is directly connected, eth2, weight 1, 00:07:59
L>* 192.168.1.1/32 is directly connected, eth2, weight 1, 00:07:59
O>* 192.168.10.0/24 [110/2] via 10.100.100.2, vti1, weight 1, 00:07:34
O>* 192.168.11.0/24 [110/2] via 10.100.100.2, vti1, weight 1, 00:07:34
```

Monitoring on Cisco side

IKE SAs:

```
Cisco#show crypto isakmp sa
IPv4 Crypto ISAKMP SA
dst          src          state          conn-id status
10.0.1.2     10.0.2.2     QM_IDLE        1002 ACTIVE

IPv6 Crypto ISAKMP SA
```

IPsec SAs:

```
Cisco#show crypto ipsec sa

interface: Tunnel10
  Crypto map tag: Tunnel10-head-0, local addr 10.0.2.2

  protected vrf: (none)
  local ident (addr/mask/prot/port): (0.0.0.0/0.0.0.0/0/0)
  remote ident (addr/mask/prot/port): (0.0.0.0/0.0.0.0/0/0)
  current_peer 10.0.1.2 port 500
    PERMIT, flags={origin_is_acl,}
    #pkts encaps: 1295, #pkts encrypt: 1295, #pkts digest: 1295
    #pkts decaps: 1238, #pkts decrypt: 1238, #pkts verify: 1238
    #pkts compressed: 0, #pkts decompressed: 0
    #pkts not compressed: 0, #pkts compr. failed: 0
    #pkts not decompressed: 0, #pkts decompress failed: 0
    #send errors 0, #recv errors 0
```

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```

local crypto endpt.: 10.0.2.2, remote crypto endpt.: 10.0.1.2
plaintext mtu 1438, path mtu 1500, ip mtu 1500, ip mtu idb GigabitEthernet0/0
current outbound spi: 0xC3E9B307(3286872839)
PFS (Y/N): N, DH group: none

inbound esp sas:
spi: 0x2740C328(658555688)
  transform: esp-256-aes esp-sha256-hmac ,
  in use settings ={Tunnel, }
  conn id: 7, flow_id: SW:7, sibling_flags 80000040, crypto map: Tunnel10-head-0
  sa timing: remaining key lifetime (k/sec): (4173824/1401)
  IV size: 16 bytes
  replay detection support: Y
  Status: ACTIVE(ACTIVE)

inbound ah sas:

inbound pcip sas:

outbound esp sas:
spi: 0xC3E9B307(3286872839)
  transform: esp-256-aes esp-sha256-hmac ,
  in use settings ={Tunnel, }
  conn id: 8, flow_id: SW:8, sibling_flags 80000040, crypto map: Tunnel10-head-0
  sa timing: remaining key lifetime (k/sec): (4173819/1401)
  IV size: 16 bytes
  replay detection support: Y
  Status: ACTIVE(ACTIVE)

outbound ah sas:

outbound pcip sas:

```

OSPF Neighbor Status:

Cisco# show ip ospf neighbor

Neighbor ID	Pri	State	Dead Time	Address	Interface
2.2.2.2	0	FULL/ -	00:00:35	10.100.100.1	Tunnel10

Routing Table:

Cisco#show ip route

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
 D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
 N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
 E1 - OSPF external type 1, E2 - OSPF external type 2
 i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
 ia - IS-IS inter area, * - candidate default, U - per-user static route
 o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
 a - application route
 + - replicated route, % - next hop override, p - overrides from PfR

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```
Gateway of last resort is 10.0.2.1 to network 0.0.0.0

S*   0.0.0.0/0 [1/0] via 10.0.2.1
      1.0.0.0/32 is subnetted, 1 subnets
C     1.1.1.1 is directly connected, Loopback0
      10.0.0.0/8 is variably subnetted, 4 subnets, 2 masks
C     10.0.2.0/30 is directly connected, GigabitEthernet0/0
L     10.0.2.2/32 is directly connected, GigabitEthernet0/0
C     10.100.100.0/30 is directly connected, Tunnel10
L     10.100.100.2/32 is directly connected, Tunnel10
O     192.168.0.0/24 [110/1001] via 10.100.100.1, 00:09:36, Tunnel10
O     192.168.1.0/24 [110/1001] via 10.100.100.1, 00:09:36, Tunnel10
      192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks
C     192.168.10.0/24 is directly connected, GigabitEthernet0/1
L     192.168.10.1/32 is directly connected, GigabitEthernet0/1
      192.168.11.0/24 is variably subnetted, 2 subnets, 2 masks
C     192.168.11.0/24 is directly connected, GigabitEthernet0/2
L     192.168.11.1/32 is directly connected, GigabitEthernet0/2
```

Checking Connectivity

ICMP packets from PC1 to PC3.

```
PC1> ping 192.168.10.2

84 bytes from 192.168.10.2 icmp_seq=1 ttl=62 time=8.479 ms
84 bytes from 192.168.10.2 icmp_seq=2 ttl=62 time=3.344 ms
84 bytes from 192.168.10.2 icmp_seq=3 ttl=62 time=3.139 ms
84 bytes from 192.168.10.2 icmp_seq=4 ttl=62 time=3.176 ms
84 bytes from 192.168.10.2 icmp_seq=5 ttl=62 time=3.978 ms
```

ICMP packets from PC2 to PC4.

```
PC2> ping 192.168.11.2

84 bytes from 192.168.11.2 icmp_seq=1 ttl=62 time=9.687 ms
84 bytes from 192.168.11.2 icmp_seq=2 ttl=62 time=3.286 ms
84 bytes from 192.168.11.2 icmp_seq=3 ttl=62 time=2.972 ms
```

11.19 Route-based Site-to-Site VPN IPsec between VyOS and Palo Alto

This document is to describe a basic setup using route-based site-to-site VPN IPsec. In this example we use VyOS 1.5 and PA 11.0.0. OSPF is selected as routing protocol inside the tunnel.

Since this example focuses on IPsec configuration it does not include firewall configuration.

11.19.1 Network Topology

11.19.2 Prerequisites

VyOS:

WAN IP	10.0.1.2/30
LAN1 IP	192.168.0.1/24
LAN2 IP	192.168.1.1/24

Palo Alto:

WAN IP	10.0.2.2/30
LAN1 IP	192.168.10.1/24
LAN2 IP	192.168.11.1/24

IKE parameters:

Encryption	AES-128
HASH	SHA-1
Diff-Helman Group	14
Life-Time	28800
IKE Version	1

IPsec parameters:

Encryption	AES-256
HASH	SHA-256
Life-Time	3600
PFS	disable

Hosts configuration

PC1 IP	192.168.0.2
PC2 IP	192.168.1.2
PC3 IP	192.168.10.2
PC4 IP	192.168.11.2

11.19.3 Configuration

VyOS

```

set interfaces ethernet eth0 address '10.0.1.2/30'
set interfaces ethernet eth1 address '192.168.0.1/24'
set interfaces ethernet eth2 address '192.168.1.1/24'
set interfaces vti vti1 address '10.100.100.1/30'
set interfaces vti vti1 mtu '1438'
set protocols ospf area 0 network '10.100.100.0/30'
set protocols ospf area 0 network '192.168.0.0/24'
set protocols ospf area 0 network '192.168.1.0/24'
set protocols ospf interface eth1 passive
set protocols ospf interface eth2 passive
set protocols ospf interface vti1 network 'point-to-point'
set protocols ospf parameters router-id '2.2.2.2'
set protocols static route 0.0.0.0/0 next-hop 10.0.1.1
set vpn ipsec authentication psk AUTH-PSK id '10.0.1.2'
set vpn ipsec authentication psk AUTH-PSK id '10.0.2.2'
set vpn ipsec authentication psk AUTH-PSK secret 'dGVzdA=='
set vpn ipsec authentication psk AUTH-PSK secret-type 'base64'
set vpn ipsec esp-group ESP-GROUP lifetime '3600'
set vpn ipsec esp-group ESP-GROUP pfs 'disable'
set vpn ipsec esp-group ESP-GROUP proposal 10 encryption 'aes256'
set vpn ipsec esp-group ESP-GROUP proposal 10 hash 'sha256'
set vpn ipsec ike-group IKE-GROUP close-action 'start'
set vpn ipsec ike-group IKE-GROUP dead-peer-detection action 'restart'
set vpn ipsec ike-group IKE-GROUP dead-peer-detection interval '10'
set vpn ipsec ike-group IKE-GROUP dead-peer-detection timeout '30'
set vpn ipsec ike-group IKE-GROUP key-exchange 'ikev1'
set vpn ipsec ike-group IKE-GROUP lifetime '28800'
set vpn ipsec ike-group IKE-GROUP proposal 10 dh-group '14'
set vpn ipsec ike-group IKE-GROUP proposal 10 encryption 'aes128'
set vpn ipsec ike-group IKE-GROUP proposal 10 hash 'sha1'
set vpn ipsec options disable-route-autoinstall
set vpn ipsec site-to-site peer PA authentication local-id '10.0.1.2'
set vpn ipsec site-to-site peer PA authentication mode 'pre-shared-secret'
set vpn ipsec site-to-site peer PA authentication remote-id '10.0.2.2'
set vpn ipsec site-to-site peer PA connection-type 'initiate'
set vpn ipsec site-to-site peer PA default-esp-group 'ESP-GROUP'
set vpn ipsec site-to-site peer PA ike-group 'IKE-GROUP'
set vpn ipsec site-to-site peer PA local-address '10.0.1.2'
set vpn ipsec site-to-site peer PA remote-address '10.0.2.2'
set vpn ipsec site-to-site peer PA vti bind 'vti1'

```

Palo Alto

GUI Configuration:

Network -> Network Profiles -> IKE Crypto

Network -> Network Profiles -> IKE Gateways

Network -> Network Profiles -> IPsec Crypto

Network -> Interfaces

Network -> IPsec Tunnels

CLI configuration with OSPF:

```
set network interface ethernet ethernet1/1 layer3 ip 10.0.2.2/30
set network interface ethernet ethernet1/1 layer3 interface-management-profile Allow
set network interface ethernet ethernet1/2 layer3 ip 192.168.10.1/24
set network interface ethernet ethernet1/1 layer3 interface-management-profile Allow
set network interface ethernet ethernet1/3 layer3 ip 192.168.11.1/24
set network interface ethernet ethernet1/1 layer3 interface-management-profile Allow
set network interface tunnel units tunnel.1 ip 10.100.100.2/30
set network interface tunnel units tunnel.1 interface-management-profile Allow
set network interface tunnel units tunnel.1 mtu 1438
set network profiles interface-management-profile Allow ping yes
set network ike crypto-profiles ike-crypto-profiles IKE-GROUP hash sha1
set network ike crypto-profiles ike-crypto-profiles IKE-GROUP dh-group group14
set network ike crypto-profiles ike-crypto-profiles IKE-GROUP encryption aes-128-cbc
set network ike crypto-profiles ike-crypto-profiles IKE-GROUP lifetime seconds 28800
set network ike crypto-profiles ipsec-crypto-profiles ESP-GROUP esp authentication sha256
set network ike crypto-profiles ipsec-crypto-profiles ESP-GROUP esp encryption aes-256-
↪ cbc
set network ike crypto-profiles ipsec-crypto-profiles ESP-GROUP lifetime seconds 3600
set network ike crypto-profiles ipsec-crypto-profiles ESP-GROUP dh-group no-pfs
set network ike gateway VyOS authentication pre-shared-key key test
set network ike gateway VyOS protocol ikev1 dpd enable yes
set network ike gateway VyOS protocol ikev1 exchange-mode main
set network ike gateway VyOS protocol ikev1 ike-crypto-profile IKE-GROUP
set network ike gateway VyOS protocol ikev2 dpd enable yes
set network ike gateway VyOS protocol version ikev1
set network ike gateway VyOS protocol-common nat-traversal enable yes
set network ike gateway VyOS protocol-common fragmentation enable no
set network ike gateway VyOS protocol-common passive-mode yes
set network ike gateway VyOS local-address interface ethernet1/1
```

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```

set network ike gateway VyOS peer-address ip 10.0.1.2
set network ike gateway VyOS local-id id 10.0.2.2
set network ike gateway VyOS local-id type ipaddr
set network ike gateway VyOS peer-id id 10.0.1.2
set network ike gateway VyOS peer-id type ipaddr
set network tunnel ipsec VyOS-tunnel auto-key ike-gateway VyOS
set network tunnel ipsec VyOS-tunnel auto-key ipsec-crypto-profile ESP-GROUP
set network tunnel ipsec VyOS-tunnel tunnel-monitor enable no
set network tunnel ipsec VyOS-tunnel tunnel-interface tunnel.1
set network tunnel ipsec VyOS-tunnel anti-replay no
set network virtual-router default protocol ospf enable yes
set network virtual-router default protocol ospf area 0.0.0.0 type normal
set network virtual-router default protocol ospf area 0.0.0.0 interface tunnel.1 enable_
↪yes
set network virtual-router default protocol ospf area 0.0.0.0 interface tunnel.1 passive_
↪no
set network virtual-router default protocol ospf area 0.0.0.0 interface tunnel.1 link-
↪type p2p
set network virtual-router default protocol ospf area 0.0.0.0 interface ethernet1/2_
↪enable yes
set network virtual-router default protocol ospf area 0.0.0.0 interface ethernet1/2_
↪passive yes
set network virtual-router default protocol ospf area 0.0.0.0 interface ethernet1/2 link-
↪type broadcast
set network virtual-router default protocol ospf area 0.0.0.0 interface ethernet1/3_
↪enable yes
set network virtual-router default protocol ospf area 0.0.0.0 interface ethernet1/3_
↪passive yes
set network virtual-router default protocol ospf area 0.0.0.0 interface ethernet1/3 link-
↪type broadcast
set network virtual-router default protocol ospf router-id 1.1.1.1
set network virtual-router default interface [ ethernet1/1 ethernet1/2 ethernet1/3_
↪tunnel.1 ]

```

11.19.4 Monitoring

Monitoring on VyOS side

IKE SAs:

```

vyos@vyos:~$ show vpn ike sa
Peer ID / IP                               Local ID / IP
-----
10.0.2.2 10.0.2.2                           10.0.1.2 10.0.1.2

  State  IKEVer  Encrypt      Hash           D-H Group      NAT-T  A-Time  L-Time
  ----  -
  up     IKEv1    AES_CBC_128  HMAC_SHA1_96  MODP_2048      no     1372    25802

```

IPsec SAs:

```
vyos@vyos:~$ show vpn ipsec sa
```

Connection	State	Uptime	Bytes In/Out	Packets In/Out	Remote address	
↪ Remote ID	Proposal					
PA-vti	up	23m27s	9K/10K	149/151	10.0.2.2	10.
↪ 0.2.2	AES_CBC_256/HMAC_SHA2_256_128					

OSPF Neighbor Status:

```
vyos@vyos:~$ show ip ospf neighbor
```

Neighbor ID	Pri	State	Up Time	Dead Time	Address	Interface
↪		RXmtL RqstL DBsmL				
1.1.1.1	1	Full/-	23m56s	37.948s	10.100.100.2	vti1:10.
↪ 100.100.1		0	0	0		

Routing Table:

```
vyos@vyos:~$ show ip route
```

Codes: K - kernel route, C - connected, L - local, S - static,
R - RIP, O - OSPF, I - IS-IS, B - BGP, E - EIGRP, N - NHRP,
T - Table, v - VNC, V - VNC-Direct, A - Babel, F - PBR,
f - OpenFabric, t - Table-Direct,
> - selected route, * - FIB route, q - queued, r - rejected, b - backup
t - trapped, o - offload failure

```
S>* 0.0.0.0/0 [1/0] via 10.0.1.1, eth0, weight 1, 00:27:30
C>* 10.0.1.0/30 is directly connected, eth0, weight 1, 00:27:34
L>* 10.0.1.2/32 is directly connected, eth0, weight 1, 00:27:34
O 10.100.100.0/30 [110/1] is directly connected, vti1, weight 1, 00:24:34
C>* 10.100.100.0/30 is directly connected, vti1, weight 1, 00:24:34
L>* 10.100.100.1/32 is directly connected, vti1, weight 1, 00:24:34
O 192.168.0.0/24 [110/1] is directly connected, eth1, weight 1, 00:27:29
C>* 192.168.0.0/24 is directly connected, eth1, weight 1, 00:27:34
L>* 192.168.0.1/32 is directly connected, eth1, weight 1, 00:27:34
O 192.168.1.0/24 [110/1] is directly connected, eth2, weight 1, 00:27:29
C>* 192.168.1.0/24 is directly connected, eth2, weight 1, 00:27:34
L>* 192.168.1.1/32 is directly connected, eth2, weight 1, 00:27:34
O>* 192.168.10.0/24 [110/11] via 10.100.100.2, vti1, weight 1, 00:24:19
O>* 192.168.11.0/24 [110/11] via 10.100.100.2, vti1, weight 1, 00:24:19
```

Monitoring on Palo Alto side

IKE SAs:

```
admin@PA-VM> show vpn ike-sa
```

IKEv1 phase-1 SAs

GwID/client IP	Peer-Address	Gateway Name					
↪	Role Mode Algorithm	Established	Expiration	V	ST	Xt	
↪ Phase2							

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```

-----
--
--
--
1          10.0.1.2          VyOS
--          Resp Main PSK/DH14/A128/SHA1    Jul.31 01:35:00 Jul.31 09:35:00 v1 13 1
--1

Show IKEv1 IKE SA: Total 1 gateways found. 1 ike sa found.

IKEv1 phase-2 SAs
Gateway Name
--
GwID/IP          Role Algorithm          SPI(in)  SPI(out) MsgID      ST Xt
-----
--
VyOS
--          1          Resp ESP/    /tun1/SHA2 8827A3D9 C204F4FA BD202829 9 1

Show IKEv1 phase2 SA: Total 1 gateways found. 1 ike sa found.

There is no IKEv2 SA found.

```

IPsec SAs:

```

admin@PA-VM> show vpn ipsec-sa

GwID/client IP  TnID  Peer-Address          Tunnel(Gateway)
--
Algorithm          SPI(in)  SPI(out) life(Sec/KB)          remain-time(Sec)
-----
--
1          1          10.0.1.2          VyOS-tunnel(VyOS)
--
ESP/A256/SHA256    8827A3D9 C204F4FA 3600/Unlimited          2733

Show IPsec SA: Total 1 tunnels found. 1 ipsec sa found.

```

OSPF Neighbor Status:

```

admin@PA-VM> show routing protocol ospf neighbor

Options: 0x80:reserved, 0:Opaq-LSA capability, DC:demand circuits, EA:Ext-Attr LSA
capability,
N/P:NSSA option, MC:multicast, E:AS external LSA capability, T:TOS capability
=====
virtual router:          default
neighbor address:        10.100.100.1
local address binding:   0.0.0.0
type:                    dynamic
status:                  full

```

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```
neighbor router ID:      2.2.2.2
area id:                 0.0.0.0
neighbor priority:       1
lifetime remain:         32
messages pending:        0
LSA request pending:     0
options:                 0x02: E
hello suppressed:        no
restart helper status:    not helping
restart helper time remaining: 0
restart helper exit reason: none
```

Routing Table:

```
admin@PA-VM> show routing route
```

```
flags: A:active, ?:loose, C:connect, H:host, S:static, ~:internal, R:rip, O:ospf, B:bgp,
      Oi:ospf intra-area, Oo:ospf inter-area, O1:ospf ext-type-1, O2:ospf ext-type-2,
      ↪E:ecmp, M:multicast
```

VIRTUAL ROUTER: default (id 1)

```
=====
destination                nexthop
↪metric flags      age  interface      next-AS
0.0.0.0/0                10.0.2.1      10
↪  A S              ethernet1/1
10.0.2.0/30              10.0.2.2      0
↪  A C              ethernet1/1
10.0.2.2/32              0.0.0.0      0
↪  A H
10.100.100.0/30          0.0.0.0      10
↪    Oi      1273  tunnel.1
10.100.100.0/30          10.100.100.2  0
↪  A C              tunnel.1
10.100.100.2/32          0.0.0.0      0
↪  A H
192.168.0.0/24           10.100.100.1  11
↪  A Oi      1253  tunnel.1
192.168.1.0/24           10.100.100.1  11
↪  A Oi      1253  tunnel.1
192.168.10.0/24          0.0.0.0      10
↪    Oi      1273  ethernet1/2
192.168.10.0/24          192.168.10.1  0
↪  A C              ethernet1/2
192.168.10.1/32          0.0.0.0      0
↪  A H
192.168.11.0/24          0.0.0.0      10
↪    Oi      1273  ethernet1/3
192.168.11.0/24          192.168.11.1  0
↪  A C              ethernet1/3
192.168.11.1/32          0.0.0.0      0
```

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```

→ A H
total routes shown: 14

```

Checking Connectivity

ICMP packets from PC1 to PC3.

```

PC1> ping 192.168.10.2

84 bytes from 192.168.10.2 icmp_seq=1 ttl=62 time=8.479 ms
84 bytes from 192.168.10.2 icmp_seq=2 ttl=62 time=3.344 ms
84 bytes from 192.168.10.2 icmp_seq=3 ttl=62 time=3.139 ms
84 bytes from 192.168.10.2 icmp_seq=4 ttl=62 time=3.176 ms
84 bytes from 192.168.10.2 icmp_seq=5 ttl=62 time=3.978 ms

```

ICMP packets from PC2 to PC4.

```

PC2> ping 192.168.11.2

84 bytes from 192.168.11.2 icmp_seq=1 ttl=62 time=9.687 ms
84 bytes from 192.168.11.2 icmp_seq=2 ttl=62 time=3.286 ms
84 bytes from 192.168.11.2 icmp_seq=3 ttl=62 time=2.972 ms

```

11.20 Policy-Based Site-to-Site VPN and Firewall Configuration

This guide shows an example policy-based IKEv2 site-to-site VPN between two VyOS routers, and firewall configuration.

For simplicity, configuration and tests are done only using IPv4, and firewall configuration is done only on one router.

11.20.1 Network Topology and requirements

This configuration example and the requirements consists of:

- Two VyOS routers with public IP address.
- 2 private subnets on each site.
- Local subnets should be able to reach internet using source NAT.
- Communication between private subnets should be done through IPSec tunnel without NAT.
- Configuration of basic firewall in one site, in order to:
 - Protect the router on ‘WAN’ interface, allowing only IPSec connections and SSH access from trusted IPs.
 - Allow access to the router only from trusted networks.
 - Allow DNS requests only for local networks.
 - Allow ICMP on all interfaces.
 - Allow all new connections from local subnets.
 - Allow connections from LANs to LANs through the tunnel.

11.20.2 Configuration

Interface and routing configuration:

```
# LEFT router:
set interfaces ethernet eth0 address '198.51.100.14/30'
set interfaces ethernet eth1 vif 111 address '10.1.11.1/24'
set interfaces ethernet eth2 vif 112 address '10.1.12.1/24'
set protocols static route 0.0.0.0/0 next-hop 198.51.100.13

# RIGHT router:
set interfaces ethernet eth0 address '192.0.2.130/30'
set interfaces ethernet eth1 vif 221 address '10.2.21.1/24'
set interfaces ethernet eth2 vif 222 address '10.2.22.1/24'
```

IPSec configuration:

```
# LEFT router:
set vpn ipsec authentication psk RIGHT id '198.51.100.14'
set vpn ipsec authentication psk RIGHT id '192.0.2.130'
set vpn ipsec authentication psk RIGHT secret 'p4ssw0rd'
set vpn ipsec esp-group ESP-GROUP mode 'tunnel'
set vpn ipsec esp-group ESP-GROUP proposal 1 encryption 'aes256'
set vpn ipsec esp-group ESP-GROUP proposal 1 hash 'sha256'
set vpn ipsec ike-group IKE-GROUP key-exchange 'ikev2'
set vpn ipsec ike-group IKE-GROUP proposal 1 dh-group '14'
set vpn ipsec ike-group IKE-GROUP proposal 1 encryption 'aes256'
set vpn ipsec ike-group IKE-GROUP proposal 1 hash 'sha256'
set vpn ipsec interface 'eth0'
set vpn ipsec site-to-site peer RIGHT authentication mode 'pre-shared-secret'
set vpn ipsec site-to-site peer RIGHT connection-type 'initiate'
set vpn ipsec site-to-site peer RIGHT default-esp-group 'ESP-GROUP'
set vpn ipsec site-to-site peer RIGHT ike-group 'IKE-GROUP'
set vpn ipsec site-to-site peer RIGHT local-address '198.51.100.14'
set vpn ipsec site-to-site peer RIGHT remote-address '192.0.2.130'
set vpn ipsec site-to-site peer RIGHT tunnel 0 local prefix '10.1.11.0/24'
set vpn ipsec site-to-site peer RIGHT tunnel 0 remote prefix '10.2.21.0/24'
set vpn ipsec site-to-site peer RIGHT tunnel 1 local prefix '10.1.11.0/24'
set vpn ipsec site-to-site peer RIGHT tunnel 1 remote prefix '10.2.22.0/24'
set vpn ipsec site-to-site peer RIGHT tunnel 2 local prefix '10.1.12.0/24'
set vpn ipsec site-to-site peer RIGHT tunnel 2 remote prefix '10.2.21.0/24'
set vpn ipsec site-to-site peer RIGHT tunnel 3 local prefix '10.1.12.0/24'
set vpn ipsec site-to-site peer RIGHT tunnel 3 remote prefix '10.2.22.0/24'

# RIGHT router:
set vpn ipsec authentication psk LEFT id '192.0.2.130'
set vpn ipsec authentication psk LEFT id '198.51.100.14'
set vpn ipsec authentication psk LEFT secret 'p4ssw0rd'
set vpn ipsec esp-group ESP-GROUP mode 'tunnel'
set vpn ipsec esp-group ESP-GROUP proposal 1 encryption 'aes256'
set vpn ipsec esp-group ESP-GROUP proposal 1 hash 'sha256'
set vpn ipsec ike-group IKE-GROUP key-exchange 'ikev2'
set vpn ipsec ike-group IKE-GROUP proposal 1 dh-group '14'
set vpn ipsec ike-group IKE-GROUP proposal 1 encryption 'aes256'
```

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```

set vpn ipsec ike-group IKE-GROUP proposal 1 hash 'sha256'
set vpn ipsec interface 'eth0'
set vpn ipsec site-to-site peer LEFT authentication mode 'pre-shared-secret'
set vpn ipsec site-to-site peer LEFT connection-type 'none'
set vpn ipsec site-to-site peer LEFT default-esp-group 'ESP-GROUP'
set vpn ipsec site-to-site peer LEFT ike-group 'IKE-GROUP'
set vpn ipsec site-to-site peer LEFT local-address '192.0.2.130'
set vpn ipsec site-to-site peer LEFT remote-address '198.51.100.14'
set vpn ipsec site-to-site peer LEFT tunnel 0 local prefix '10.2.21.0/24'
set vpn ipsec site-to-site peer LEFT tunnel 0 remote prefix '10.1.11.0/24'
set vpn ipsec site-to-site peer LEFT tunnel 1 local prefix '10.2.22.0/24'
set vpn ipsec site-to-site peer LEFT tunnel 1 remote prefix '10.1.11.0/24'
set vpn ipsec site-to-site peer LEFT tunnel 2 local prefix '10.2.21.0/24'
set vpn ipsec site-to-site peer LEFT tunnel 2 remote prefix '10.1.12.0/24'
set vpn ipsec site-to-site peer LEFT tunnel 3 local prefix '10.2.22.0/24'
set vpn ipsec site-to-site peer LEFT tunnel 3 remote prefix '10.1.12.0/24'

```

Firewall Configuration:

```

# Firewall Groups:
set firewall group network-group LOCAL-NETS network '10.1.11.0/24'
set firewall group network-group LOCAL-NETS network '10.1.12.0/24'
set firewall group network-group REMOTE-NETS network '10.2.21.0/24'
set firewall group network-group REMOTE-NETS network '10.2.22.0/24'
set firewall group network-group TRUSTED network '198.51.100.125/32'
set firewall group network-group TRUSTED network '203.0.113.0/24'
set firewall group network-group TRUSTED network '10.1.11.0/24'
set firewall group network-group TRUSTED network '192.168.70.0/24'

# Forward traffic: default drop and only allow what is needed
set firewall ipv4 forward filter default-action 'drop'

# Forward traffic: global state policies
set firewall ipv4 forward filter rule 1 action 'accept'
set firewall ipv4 forward filter rule 1 state established 'enable'
set firewall ipv4 forward filter rule 1 state related 'enable'
set firewall ipv4 forward filter rule 2 action 'drop'
set firewall ipv4 forward filter rule 2 state invalid 'enable'

# Forward traffic: Accept all connections from local networks
set firewall ipv4 forward filter rule 10 action 'accept'
set firewall ipv4 forward filter rule 10 source group network-group 'LOCAL-NETS'

# Forward traffic: accept connections from remote LANs to local LANs
set firewall ipv4 forward filter rule 20 action 'accept'
set firewall ipv4 forward filter rule 20 destination group network-group 'LOCAL-NETS'
set firewall ipv4 forward filter rule 20 source group network-group 'REMOTE-NETS'

# Input traffic: default drop and only allow what is needed
set firewall ipv4 input filter default-action 'drop'

# Input traffic: global state policies

```

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```

set firewall ipv4 input filter rule 1 action 'accept'
set firewall ipv4 input filter rule 1 state established 'enable'
set firewall ipv4 input filter rule 1 state related 'enable'
set firewall ipv4 input filter rule 2 action 'drop'
set firewall ipv4 input filter rule 2 state invalid 'enable'

# Input traffic: add rules needed for ipsec connection
set firewall ipv4 input filter rule 10 action 'accept'
set firewall ipv4 input filter rule 10 destination port '500,4500'
set firewall ipv4 input filter rule 10 inbound-interface name 'eth0'
set firewall ipv4 input filter rule 10 protocol 'udp'
set firewall ipv4 input filter rule 15 action 'accept'
set firewall ipv4 input filter rule 15 inbound-interface name 'eth0'
set firewall ipv4 input filter rule 15 protocol 'esp'

# Input traffic: accept ssh connection from trusted ips
set firewall ipv4 input filter rule 20 action 'accept'
set firewall ipv4 input filter rule 20 destination port '22'
set firewall ipv4 input filter rule 20 protocol 'tcp'
set firewall ipv4 input filter rule 20 source group network-group 'TRUSTED'

# Input traffic: accept dns requests only from local networks.
set firewall ipv4 input filter rule 25 action 'accept'
set firewall ipv4 input filter rule 25 destination port '53'
set firewall ipv4 input filter rule 25 protocol 'udp'
set firewall ipv4 input filter rule 25 source group network-group 'LOCAL-NETS'

# Input traffic: allow icmp
set firewall ipv4 input filter rule 30 action 'accept'
set firewall ipv4 input filter rule 30 protocol 'icmp'

```

And NAT Configuration:

```

set nat source rule 10 destination group network-group 'REMOTE-NETS'
set nat source rule 10 exclude
set nat source rule 10 outbound-interface name 'eth0'
set nat source rule 10 source group network-group 'LOCAL-NETS'
set nat source rule 20 outbound-interface name 'eth0'
set nat source rule 20 source group network-group 'LOCAL-NETS'
set nat source rule 20 translation address 'masquerade'

```

11.20.3 Checking through op-mode commands

After some testing, we can check IPsec status, and counter on every tunnel:

```

vyos@LEFT:~$ show vpn ipsec sa

```

Connection	State	Uptime	Bytes In/Out	Packets In/Out	Remote address
↪ Remote ID	Proposal				
-----	-----	-----	-----	-----	-----
↪					
RIGHT-tunnel-0	up	36m24s	840B/840B	10/10	192.0.2.130

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```

↪192.0.2.130 AES_CBC_256/HMAC_SHA2_256_128/MODP_2048
RIGHT-tunnel-1 up 36m33s 588B/588B 7/7 192.0.2.130 ↪
↪192.0.2.130 AES_CBC_256/HMAC_SHA2_256_128/MODP_2048
RIGHT-tunnel-2 up 35m50s 1K/1K 15/15 192.0.2.130 ↪
↪192.0.2.130 AES_CBC_256/HMAC_SHA2_256_128/MODP_2048
RIGHT-tunnel-3 up 36m54s 2K/2K 32/32 192.0.2.130 ↪
↪192.0.2.130 AES_CBC_256/HMAC_SHA2_256_128/MODP_2048
vyos@LEFT:~$

```

Also, we can check firewall counters:

```

vyos@LEFT:~$ show firewall
Rulesets Information

-----
IPv4 Firewall "forward filter"

Rule      Action    Protocol    Packets    Bytes    Conditions
-----
↪-----
1         accept    all         681        96545    ct state { established, related } ↪
↪accept
2         drop      all         0          0        ct state invalid
10        accept    all         360        27205    ip saddr @N_LOCAL-NETS accept
20        accept    all         8          648      ip daddr @N_LOCAL-NETS ip saddr @N_
↪REMOTE-NETS accept
default   drop      all
-----

IPv4 Firewall "input filter"

Rule      Action    Protocol    Packets    Bytes    Conditions
-----
↪-----
1         accept    all         901        123709   ct state { established, related } ↪
↪accept
2         drop      all         0          0        ct state invalid
10        accept    udp         0          0        udp dport { 500, 4500 } iifname "eth0
↪" accept
15        accept    esp         0          0        meta l4proto esp iifname "eth0" ↪
↪accept
20        accept    tcp         1          60       tcp dport 22 ip saddr @N_TRUSTED ↪
↪accept
25        accept    udp         0          0        udp dport 53 ip saddr @N_LOCAL-NETS ↪
↪accept
30        accept    icmp        0          0        meta l4proto icmp accept
default   drop      all

vyos@LEFT:~$
vyos@LEFT:~$ show firewall statistics
Rulesets Statistics

```

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IPv4 Firewall "forward filter"							
Rule	Packets	Bytes	Action	Source	Destination	Inbound-Interface	
↪ Outbound-interface							

↪ -----							
1	681	96545	accept	any	any	any	
↪ any							
2	0	0	drop	any	any	any	
↪ any							
10	360	27205	accept	LOCAL-NETS	any	any	
↪ any							
20	8	648	accept	REMOTE-NETS	LOCAL-NETS	any	
↪ any							
default	N/A	N/A	drop	any	any	any	
↪ any							

IPv4 Firewall "input filter"							
Rule	Packets	Bytes	Action	Source	Destination	Inbound-Interface	
↪ Outbound-interface							

↪ -----							
1	905	124213	accept	any	any	any	
↪ any							
2	0	0	drop	any	any	any	
↪ any							
10	0	0	accept	any	any	eth0	
↪ any							
15	0	0	accept	any	any	eth0	
↪ any							
20	1	60	accept	TRUSTED	any	any	
↪ any							
25	0	0	accept	LOCAL-NETS	any	any	
↪ any							
30	0	0	accept	any	any	any	
↪ any							
default	N/A	N/A	drop	any	any	any	
↪ any							
vyos@LEFT:~\$							

11.21 Site-to-Site IPsec VPN to Cisco using FlexVPN

This guide shows a sample configuration for FlexVPN site-to-site Internet Protocol Security (IPsec)/Generic Routing Encapsulation (GRE) tunnel.

FlexVPN is a newer “solution” for deployment of VPNs and it utilizes IKEv2 as the key exchange protocol. The result is a flexible and scalable VPN solution that can be easily adapted to fit various network needs. It can also support a variety of encryption methods, including AES and 3DES.

The lab was built using EVE-NG.

11.21.1 Configuration

VyOS

- GRE:

```
set interfaces tunnel tun1 encapsulation 'gre'
set interfaces tunnel tun1 ip adjust-mss '1336'
set interfaces tunnel tun1 mtu '1376'
set interfaces tunnel tun1 remote '10.1.1.6'
set interfaces tunnel tun1 source-address '198.51.100.1'
```

- IPsec:

```
set vpn ipsec authentication psk vyos_cisco_1 id 'vyos.net'
set vpn ipsec authentication psk vyos_cisco_1 id 'cisco.hub.net'
set vpn ipsec authentication psk vyos_cisco_1 secret 'secret'
set vpn ipsec esp-group e1 lifetime '3600'
set vpn ipsec esp-group e1 mode 'tunnel'
set vpn ipsec esp-group e1 pfs 'disable'
set vpn ipsec esp-group e1 proposal 1 encryption 'aes128'
set vpn ipsec esp-group e1 proposal 1 hash 'sha256'
set vpn ipsec ike-group i1 key-exchange 'ikev2'
set vpn ipsec ike-group i1 lifetime '28800'
set vpn ipsec ike-group i1 proposal 1 dh-group '5'
set vpn ipsec ike-group i1 proposal 1 encryption 'aes256'
set vpn ipsec ike-group i1 proposal 1 hash 'sha256'
set vpn ipsec interface 'eth2'
set vpn ipsec options disable-route-autoinstall
set vpn ipsec options flexvpn
set vpn ipsec options interface 'tun1'
set vpn ipsec options virtual-ip
set vpn ipsec site-to-site peer cisco_hub authentication local-id 'vyos.net'
set vpn ipsec site-to-site peer cisco_hub authentication mode 'pre-shared-secret'
set vpn ipsec site-to-site peer cisco_hub authentication remote-id 'cisco.hub.net'
set vpn ipsec site-to-site peer cisco_hub connection-type 'initiate'
set vpn ipsec site-to-site peer cisco_hub default-esp-group 'e1'
set vpn ipsec site-to-site peer cisco_hub ike-group 'i1'
set vpn ipsec site-to-site peer cisco_hub local-address '198.51.100.1'
set vpn ipsec site-to-site peer cisco_hub remote-address '10.1.1.6'
set vpn ipsec site-to-site peer cisco_hub tunnel 1 local prefix '198.51.100.1/32'
set vpn ipsec site-to-site peer cisco_hub tunnel 1 protocol 'gre'
```

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```
set vpn ipsec site-to-site peer cisco_hub tunnel 1 remote prefix '10.1.1.6/32'
set vpn ipsec site-to-site peer cisco_hub virtual-address '0.0.0.0'
```

Cisco

```
aaa new-model
!
!
aaa authorization network default local
!
crypto ikev2 name-mangler GET_DOMAIN
  fqdn all
  email all
!
!
crypto ikev2 authorization policy vyos
  pool mypool
  aaa attribute list mylist
  route set interface
  route accept any tag 100 distance 5
!
crypto ikev2 keyring mykeys
  peer peer1
    identity fqdn vyos.net
    pre-shared-key local secret
    pre-shared-key remote secret
crypto ikev2 profile my_profile
  match identity remote fqdn vyos.net
  identity local fqdn cisco.hub.net
  authentication remote pre-share
  authentication local pre-share
  keyring local mykeys
  dpd 10 3 periodic
  aaa authorization group psk list local name-mangler GET_DOMAIN
  aaa authorization user psk cached
  virtual-template 1
!
!
!
crypto ipsec transform-set TSET esp-aes esp-sha256-hmac
  mode tunnel
!
!
crypto ipsec profile my-ipsec-profile
  set transform-set TSET
  set ikev2-profile my_profile
!
interface Virtual-Template1 type tunnel
  no ip address
  ip mtu 1376
```

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```

ip nhrp network-id 1
ip nhrp shortcut virtual-template 1
ip tcp adjust-mss 1336
tunnel path-mtu-discovery
tunnel protection ipsec profile my-ipsec-profile
!
ip local pool my_pool 172.16.122.1 172.16.122.254

```

Since the tunnel is a point-to-point GRE tunnel, it behaves like any other point-to-point interface (for example: serial, dialer), and it is possible to run any Interior Gateway Protocol (IGP)/Exterior Gateway Protocol (EGP) over the link in order to exchange routing information

11.21.2 Verification

```
vyos@vyos$ show interfaces
```

Codes: S - State, L - Link, u - Up, D - Down, A - Admin Down

Interface	IP Address	S/L	Description
eth0	-	u/u	
eth1	-	u/u	
eth2	198.51.100.1/24	u/u	
eth3	172.16.1.2/24	u/u	
lo	127.0.0.1/8	u/u	
tun1	::1/128		
	172.16.122.2/32	u/u	

```
vyos@vyos:~$ show vpn ipsec sa
```

Connection	State	Uptime	Bytes In/Out	Packets In/Out	Remote address
Remote ID	Proposal				
cisco_hub-tunnel-1	up	44m17s	35K/31K	382/367	10.1.1.6
cisco.hub.net	AES_CBC_128/HMAC_SHA2_256_128				

```
Hub#sh crypto ikev2 sa detailed
```

IPv4 Crypto IKEv2 SA

Tunnel-id	Local	Remote	fvr/f/ivrf	Status
5	10.1.1.6/4500	198.51.100.1/4500	none/none	READY
Encr: AES-CBC, keysize: 256, PRF: SHA256, Hash: SHA256, DH Grp:5, Auth sign: PSK,				
Auth verify: PSK				
Life/Active Time: 86400/2694 sec				
CE id: 0, Session-id: 2				
Status Description: Negotiation done				
Local spi: C94EE2DC92A60C47		Remote spi: 9AF0EF151BECF14C		
Local id: cisco.hub.net				
Remote id: vyos.net				
Local req msg id: 269		Remote req msg id: 0		
Local next msg id: 269		Remote next msg id: 0		
Local req queued: 269		Remote req queued: 0		

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```
Local window:      5           Remote window:      1
DPD configured for 10 seconds, retry 3
Fragmentation not configured.
Extended Authentication not configured.
NAT-T is not detected
Cisco Trust Security SGT is disabled
Assigned host addr: 172.16.122.2
```

11.22 Configuration Blueprints (autotest)

The next pages contain fully automated configuration examples.

Each lab will build and test from an external script. The page content is generated, so changes will not take effect.

A host `vyos-oobm` will be used as an SSH proxy. This host is just necessary for the lab tests.

The process will do the following steps:

1. create the lab on a eve-ng server
2. configure each host in the lab
3. do some defined tests
4. optional do an upgrade to a higher version and do step 3 again.
5. generate the documentation and include files
6. shutdown and destroy the lab, if there is no error

11.22.1 DHCP Relay through GRE-Bridge

Testdate: 2023-05-11

Version: 1.4-rolling-202305100734

This simple structure shows how to configure a DHCP Relay over a GRE Bridge interface.

Topology

The topology has 3 VyOS routers and one client. Between the DHCP Server and the DHCP Relay is a GRE tunnel. The transport VyOS represent a large Network.

Configuration

First, we configure the transport network and the Tunnel interface.

Transport:

```
set interfaces ethernet eth1 address '10.0.10.254/24'
set interfaces ethernet eth2 address '10.0.20.254/24'
```

DHCP-Server

```
set interfaces ethernet eth1 address '10.0.20.10/24'
set interfaces tunnel tun100 address '172.16.0.1/30'
set interfaces tunnel tun100 encapsulation 'gretap'
set interfaces tunnel tun100 remote '10.0.10.10'
set interfaces tunnel tun100 source-address '10.0.20.10'
set protocols static route 0.0.0.0/0 next-hop 172.16.0.2
set protocols static route 10.0.10.0/24 next-hop 10.0.20.254
set protocols static route 192.168.0.0/24 next-hop 127.16.0.2
```

DHCP-Relay

```
set interfaces ethernet eth1 address '10.0.10.10/24'
set interfaces ethernet eth2 address '192.168.0.254/24'
set interfaces tunnel tun100 address '172.16.0.2/30'
set interfaces tunnel tun100 encapsulation 'gretap'
set interfaces tunnel tun100 remote '10.0.20.10'
set interfaces tunnel tun100 source-address '10.0.10.10'
set protocols static route 0.0.0.0/0 next-hop 172.16.0.1
set protocols static route 10.0.20.0/24 next-hop 10.0.10.254
```

After this, we need the DHCP-Server and Relay configuration. To get a testable result, we just have one IP in the DHCP range. Expand it as you need it.

DHCP-Server

```
set service dhcp-server listen-address '172.16.0.1'
set service dhcp-server shared-network-name DHCP_Tun100 authoritative
set service dhcp-server shared-network-name DHCP_Tun100 subnet 192.168.0.0/24 option_
↳ default-router '192.168.0.254'
set service dhcp-server shared-network-name DHCP_Tun100 subnet 192.168.0.0/24 range 0_
↳ start '192.168.0.30'
set service dhcp-server shared-network-name DHCP_Tun100 subnet 192.168.0.0/24 range 0_
↳ stop '192.168.0.30'
```

DHCP-Relay

```
set service dhcp-relay interface 'eth2'
set service dhcp-relay interface 'tun100'
```

Test the result

Ping the Client from the DHCP Server.

```
vyos@dhcp-server:~$ ping 192.168.0.30 count 4
PING 192.168.0.30 (192.168.0.30) 56(84) bytes of data.
64 bytes from 192.168.0.30: icmp_seq=1 ttl=63 time=1.02 ms
64 bytes from 192.168.0.30: icmp_seq=2 ttl=63 time=1.06 ms
64 bytes from 192.168.0.30: icmp_seq=3 ttl=63 time=1.21 ms
64 bytes from 192.168.0.30: icmp_seq=4 ttl=63 time=1.16 ms

--- 192.168.0.30 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3004ms
rtt min/avg/max/mdev = 1.016/1.112/1.214/0.077 ms
```

And show all DHCP Leases

```
vyos@dhcp-server:~$ show dhcp server leases
IP Address      MAC address      State      Lease start      Lease expiration
↳ Remaining    Pool             Hostname
-----
↳ -----
192.168.0.30    00:50:79:66:68:05 active      2023/05/11 13:08:50 2023/05/12 13:08:50
↳ 23:59:16      DHCP Tun100      VPCS
```

11.22.2 Tunnelbroker.net (IPv6)

Testdate: 2024-01-13

Version: 1.5-rolling-202401121239

This guide walks through the setup of <https://www.tunnelbroker.net/> for an IPv6 Tunnel.

Prerequisites

- A public, routable IPv4 address. This does not necessarily need to be static, but you will need to update the tunnel endpoint when/if your IP address changes, which can be done with a script and a scheduled task.
- Account at <https://www.tunnelbroker.net/>
- Requested a “Regular Tunnel”. You want to choose a location that is closest to your physical location for the best response time.

Topology

The example topology has 2 VyOS routers. One as the WAN router and one as a client, to test a single LAN setup

Configuration

First, we configure the vyos-wan interface to get a DHCP address.

```
set interfaces ethernet eth1 address dhcp
```

Now we are able to setup the tunnel interface.

```
set interfaces tunnel tun0 address '2001:470:6c:779::2/64' #Tunnelbroker Client IPv6
↪address
set interfaces tunnel tun0 description 'HE.NET IPv6 Tunnel'
set interfaces tunnel tun0 encapsulation 'sit'
set interfaces tunnel tun0 remote '216.66.86.114' #Tunnelbroker Server IPv4 address
set interfaces tunnel tun0 source-address '172.29.129.60' # Tunnelbroker Client IPv4
↪address. See note below
```

Note: The source-address is the Tunnelbroker client IPv4 address or if there is NAT the current WAN interface address.

If source-address is dynamic, the tunnel will cease working once the address changes. To avoid having to manually update source-address each time the dynamic IP changes, an address of '0.0.0.0' can be specified.

Setup the IPv6 default route to the tunnel interface

```
set protocols static route6 ::/0 interface tun0
```

Now you should be able to ping a public IPv6 Address

```
vyos@vyos-wan:~$ ping 2001:470:20::2 count 4
PING 2001:470:20::2(2001:470:20::2) 56 data bytes
64 bytes from 2001:470:20::2: icmp_seq=1 ttl=64 time=33.8 ms
64 bytes from 2001:470:20::2: icmp_seq=2 ttl=64 time=43.9 ms
64 bytes from 2001:470:20::2: icmp_seq=3 ttl=64 time=43.4 ms
64 bytes from 2001:470:20::2: icmp_seq=4 ttl=64 time=42.5 ms

--- 2001:470:20::2 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 2999ms
rtt min/avg/max/mdev = 33.802/40.920/43.924/4.139 ms
```

Assuming the pings are successful, you need to add some DNS servers. Some options:

```
set system name-server 2001:470:20::2 #Tunnelbroker DNS Server
```

You should now be able to ping something by IPv6 DNS name:

```
vyos@vyos-wan:~$ ping tunnelbroker.net count 4
PING tunnelbroker.net(tunnelbroker.net (2001:470:0:63::2)) 56 data bytes
64 bytes from tunnelbroker.net (2001:470:0:63::2): icmp_seq=1 ttl=48 time=285 ms
```

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```
64 bytes from tunnelbroker.net (2001:470:0:63::2): icmp_seq=2 ttl=48 time=186 ms
64 bytes from tunnelbroker.net (2001:470:0:63::2): icmp_seq=3 ttl=48 time=178 ms
64 bytes from tunnelbroker.net (2001:470:0:63::2): icmp_seq=4 ttl=48 time=177 ms

--- tunnelbroker.net ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3002ms
rtt min/avg/max/mdev = 176.707/206.638/285.128/45.457 ms
```

LAN Configuration

At this point, your VyOS install should have full IPv6, but now your LAN devices need access.

With Tunnelbroker.net, you have two options:

- Routed /64. This is the default assignment. In IPv6-land, it's good for a single "LAN", and is somewhat equivalent to a /24.
- Routed /48. This is something you can request by clicking the "Assign /48" link in the Tunnelbroker.net tunnel config. It allows you to have up to 65k

Unlike IPv4, IPv6 is really not designed to be broken up smaller than /64. So if you ever want to have multiple LANs, VLANs, DMZ, etc, you'll want to ignore the assigned /64, and request the /48 and use that.

Single LAN Setup

Single LAN setup where eth2 is your LAN interface. Use the Tunnelbroker Routed /64 prefix:

```
set interface ethernet eth2 address '2001:470:6d:778::1/64' # Tunnelbroker Routed /64
↪ prefix
set service router-advert interface eth2 name-server '2001:470:20::2'
set service router-advert interface eth2 prefix 2001:470:6d:778::/64 # Tunnelbroker
↪ Routed /64 prefix
```

Please note, 'autonomous-flag' and 'on-link-flag' are enabled by default, 'valid-lifetime' and 'preferred-lifetime' are set to default values of 30 days and 4 hours respectively.

And the client to receive an IPv6 address with stateless autoconfig.

```
set interfaces ethernet eth1 ipv6 address autoconf
```

This accomplishes a few things:

- Sets your LAN interface's IP address
- Enables router advertisements. This is an IPv6 alternative for DHCP (though DHCPv6 can still be used). With RAs, Your devices will automatically find the information they need for routing and DNS.

Now the Client is able to ping a public IPv6 address

```
vyos@client:~$ ping 2001:470:20::2 count 4
PING 2001:470:20::2(2001:470:20::2) 56 data bytes
64 bytes from 2001:470:20::2: icmp_seq=1 ttl=63 time=32.1 ms
64 bytes from 2001:470:20::2: icmp_seq=2 ttl=63 time=41.8 ms
64 bytes from 2001:470:20::2: icmp_seq=3 ttl=63 time=41.7 ms
```

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```
64 bytes from 2001:470:20::2: icmp_seq=4 ttl=63 time=47.1 ms

--- 2001:470:20::2 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3005ms
rtt min/avg/max/mdev = 32.128/40.688/47.107/5.403 ms
```

Multiple LAN/DMZ Setup

That's how you can expand the example above. Use the Routed /48 information. This allows you to assign a different /64 to every interface, LAN, or even device. Or you could break your network into smaller chunks like /56 or /60.

The format of these addresses:

- 2001:470:xxxx::/48: The whole subnet. xxxx should come from Tunnelbroker.
- 2001:470:xxxx:1::/64: A subnet suitable for a LAN
- 2001:470:xxxx:2::/64: Another subnet
- 2001:470:xxxx:ffff::/64: The last usable /64 subnet.

In the above examples, 1,2,ffff are all chosen by you. You can use 1-ffff (1-65535).

So, when your LAN is eth1, your DMZ is eth2, your cameras are on eth3, etc:

```
set interfaces ethernet eth1 address '2001:470:xxxx:1::1/64'
set service router-advert interface eth1 name-server '2001:470:20::2'
set service router-advert interface eth1 prefix 2001:470:xxxx:1::/64

set interfaces ethernet eth2 address '2001:470:xxxx:2::1/64'
set service router-advert interface eth2 name-server '2001:470:20::2'
set service router-advert interface eth2 prefix 2001:470:xxxx:2::/64

set interfaces ethernet eth3 address '2001:470:xxxx:3::1/64'
set service router-advert interface eth3 name-server '2001:470:20::2'
set service router-advert interface eth3 prefix 2001:470:xxxx:3::/64
```

Please note, 'autonomous-flag' and 'on-link-flag' are enabled by default, 'valid-lifetime' and 'preferred-lifetime' are set to default values of 30 days and 4 hours respectively.

Firewall

Finally, don't forget the [Firewall](#). The usage is identical, except instead of `set firewall ipv4 name NAME`, you would use `set firewall ipv6 name NAME`.

Similarly, to attach the firewall, you would use `set firewall ipv6 name NAME rule N inbound-interface name eth0` or `set firewall zone LOCAL from WAN firewall ipv6-name`.

11.22.3 L3VPN EVPN with VyOS

Testdate: 2023-05-11

Version: 1.4-rolling-202305100734

I spun up a new lab in EVE-NG, which represents this as the “Foo Bar - Service Provider Inc.” that has 3 points of presence (PoP) in random datacenters/sites named PE1, PE2, and PE3. Each PoP aggregates at least two customers.

I named the customers blue, red and green which is common practice in VRF (Virtual Routing and Forwarding) documentation scenarios.

- PE1 is located in an industrial area that holds multiple office buildings. All customers have a site in this area.
- PE2 is located in a smaller area where by coincidence two customers (blue and red) share an office building.
- PE3 is located in a smaller area where by coincidence two customers (blue and green) are located.

Management VRF

A brief excursion into VRFs: This has been one of the longest-standing feature requests of VyOS (dating back to 2016) which can be described as “a VLAN for layer 2 is what a VRF is for layer 3”. With VRFs, a router/system can hold multiple, isolated routing tables on the same system. If you wonder what’s the difference between multiple tables that people used for policy-based routing since forever, it’s that a VRF also isolates connected routes rather than just static and dynamically learned routes, so it allows NICs in different VRFs to use conflicting network ranges without issues.

VyOS 1.3 added initial support for VRFs (including IPv4/IPv6 static routing) and VyOS 1.4 now enables full dynamic routing protocol support for OSPF, IS-IS, and BGP for individual VRFs.

The lab I built is using a VRF (called **mgmt**) to provide out-of-band SSH access to the PE (Provider Edge) routers.

```
set system host-name PE1
set interfaces ethernet eth0 address 'dhcp'
set interfaces ethernet eth0 description 'out-of-band management'
set interfaces ethernet eth0 vrf 'mgmt'
set service ssh vrf 'mgmt'
set vrf name mgmt table '1000'
```

Topology

We use the following network topology in this example:

Core network

I chose to run OSPF as the IGP (Interior Gateway Protocol). All required BGP sessions are established via a dummy interfaces (similar to the loopback, but in Linux you can have only one loopback, while there can be many dummy interfaces) on the PE routers. In case of a link failure, traffic is diverted in the other direction in this triangle setup and BGP sessions will not go down. One could even enable BFD (Bidirectional Forwarding Detection) on the links for a faster failover and resilience in the network.

Regular VyOS users will notice that the BGP syntax has changed in VyOS 1.4 from even the prior post about this subject. This is due to T1711, where it was finally decided to get rid of the redundant BGP ASN (Autonomous System Number) specification on the CLI and move it to a single leaf node (set protocols bgp local-as).

It's important to note that all your existing configurations will be migrated automatically on image upgrade. Nothing to do on your side.

PE1

```
set interfaces dummy dum0 address '172.29.255.1/32'

set interfaces ethernet eth1 address '172.29.0.2/31'
set interfaces ethernet eth1 description 'link to pe2'
set interfaces ethernet eth1 mtu '1600'
set interfaces ethernet eth3 address '172.29.0.6/31'
set interfaces ethernet eth3 description 'link to pe3'
set interfaces ethernet eth3 mtu '1600'

set protocols ospf area 0 network '172.29.0.2/31'
set protocols ospf area 0 network '172.29.0.6/31'
set protocols ospf interface eth1 network 'point-to-point'
set protocols ospf interface eth3 network 'point-to-point'
set protocols ospf interface eth1 passive disable
set protocols ospf interface eth3 passive disable
set protocols ospf log-adjacency-changes detail
set protocols ospf parameters abr-type 'cisco'
set protocols ospf parameters router-id '172.29.255.1'
set protocols ospf passive-interface 'default'
set protocols ospf redistribute connected

set protocols bgp address-family l2vpn-evpn advertise ipv4 unicast
set protocols bgp address-family l2vpn-evpn advertise-all-vni
set protocols bgp system-as '100'
set protocols bgp neighbor 172.29.255.2 peer-group 'ibgp'
set protocols bgp neighbor 172.29.255.3 peer-group 'ibgp'
set protocols bgp parameters log-neighbor-changes
set protocols bgp parameters router-id '172.29.255.1'
set protocols bgp peer-group ibgp address-family l2vpn-evpn
set protocols bgp peer-group ibgp remote-as '100'
set protocols bgp peer-group ibgp update-source 'dum0'
```

PE2

```
set interfaces dummy dum0 address '172.29.255.2/32'

set interfaces ethernet eth1 address '172.29.0.3/31'
set interfaces ethernet eth1 description 'link to pe1'
set interfaces ethernet eth1 mtu '1600'
set interfaces ethernet eth2 address '172.29.0.4/31'
set interfaces ethernet eth2 description 'link to pe3'
set interfaces ethernet eth2 mtu '1600'

set protocols ospf area 0 network '172.29.0.2/31'
set protocols ospf area 0 network '172.29.0.4/31'
set protocols ospf interface eth1 network 'point-to-point'
set protocols ospf interface eth2 network 'point-to-point'
set protocols ospf interface eth1 passive disable
set protocols ospf interface eth2 passive disable
```

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```

set protocols ospf log-adjacency-changes detail
set protocols ospf parameters abr-type 'cisco'
set protocols ospf parameters router-id '172.29.255.2'
set protocols ospf passive-interface 'default'
set protocols ospf redistribute connected

set protocols bgp address-family l2vpn-evpn advertise ipv4 unicast
set protocols bgp address-family l2vpn-evpn advertise-all-vni
set protocols bgp system-as '100'
set protocols bgp neighbor 172.29.255.1 peer-group 'ibgp'
set protocols bgp neighbor 172.29.255.3 peer-group 'ibgp'
set protocols bgp parameters log-neighbor-changes
set protocols bgp parameters router-id '172.29.255.2'
set protocols bgp peer-group ibgp address-family l2vpn-evpn
set protocols bgp peer-group ibgp remote-as '100'
set protocols bgp peer-group ibgp update-source 'dum0'

```

PE3

```

set interfaces dummy dum0 address '172.29.255.3/32'

set interfaces ethernet eth2 address '172.29.0.5/31'
set interfaces ethernet eth2 description 'link to pe2'
set interfaces ethernet eth2 mtu '1600'
set interfaces ethernet eth3 address '172.29.0.7/31'
set interfaces ethernet eth3 description 'link to pe1'
set interfaces ethernet eth3 mtu '1600'

set protocols ospf area 0 network '172.29.0.4/31'
set protocols ospf area 0 network '172.29.0.6/31'
set protocols ospf interface eth2 network 'point-to-point'
set protocols ospf interface eth3 network 'point-to-point'
set protocols ospf interface eth2 passive disable
set protocols ospf interface eth3 passive disable
set protocols ospf log-adjacency-changes detail
set protocols ospf parameters abr-type 'cisco'
set protocols ospf parameters router-id '172.29.255.3'
set protocols ospf passive-interface 'default'
set protocols ospf redistribute connected

set protocols bgp address-family l2vpn-evpn advertise ipv4 unicast
set protocols bgp address-family l2vpn-evpn advertise-all-vni
set protocols bgp system-as '100'
set protocols bgp neighbor 172.29.255.1 peer-group 'ibgp'
set protocols bgp neighbor 172.29.255.2 peer-group 'ibgp'
set protocols bgp parameters log-neighbor-changes
set protocols bgp parameters router-id '172.29.255.3'
set protocols bgp peer-group ibgp address-family l2vpn-evpn
set protocols bgp peer-group ibgp remote-as '100'
set protocols bgp peer-group ibgp update-source 'dum0'

```

Tenant networks (VRFs)

Once all routers can be safely remotely managed and the core network is operational, we can now setup the tenant networks.

Every tenant is assigned an individual VRF that would support overlapping address ranges for customers blue, red and green. In our example, we do not use overlapping ranges to make it easier when showing debug commands.

Thus you can easily match it to one of the devices/networks below.

Every router that provides access to a customer network needs to have the customer network (VRF + VNI) configured. To make our own lives easier, we utilize the same VRF table id (local routing table number) and VNI (Virtual Network Identifier) per tenant on all our routers.

- blue uses local routing table id and VNI 2000
- red uses local routing table id and VNI 3000
- green uses local routing table id and VNI 4000

PE1

```
set interfaces vxlan vxlan2000 mtu '1500'
set interfaces vxlan vxlan2000 parameters nolearning
set interfaces vxlan vxlan2000 port '4789'
set interfaces vxlan vxlan2000 source-address '172.29.255.1'
set interfaces vxlan vxlan2000 vni '2000'

set vrf name blue protocols bgp address-family ipv4-unicast redistribute connected
set vrf name blue protocols bgp address-family l2vpn-evpn advertise ipv4 unicast
set vrf name blue table '2000'
set vrf name blue vni '2000'

set interfaces vxlan vxlan3000 mtu '1500'
set interfaces vxlan vxlan3000 parameters nolearning
set interfaces vxlan vxlan3000 port '4789'
set interfaces vxlan vxlan3000 source-address '172.29.255.1'
set interfaces vxlan vxlan3000 vni '3000'

set vrf name red protocols bgp address-family ipv4-unicast redistribute connected
set vrf name red protocols bgp address-family l2vpn-evpn advertise ipv4 unicast
set vrf name red table '3000'
set vrf name red vni '3000'

set interfaces vxlan vxlan4000 mtu '1500'
set interfaces vxlan vxlan4000 parameters nolearning
set interfaces vxlan vxlan4000 port '4789'
set interfaces vxlan vxlan4000 source-address '172.29.255.1'
set interfaces vxlan vxlan4000 vni '4000'

set vrf name green protocols bgp address-family ipv4-unicast redistribute connected
set vrf name green protocols bgp address-family l2vpn-evpn advertise ipv4 unicast
set vrf name green table '4000'
set vrf name green vni '4000'

set interfaces bridge br2000 address '10.1.1.1/24'
set interfaces bridge br2000 description 'customer blue'
```

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```

set interfaces bridge br2000 member interface eth4
set interfaces bridge br2000 member interface vxlan2000
set interfaces bridge br2000 vrf 'blue'

set interfaces bridge br3000 address '10.2.1.1/24'
set interfaces bridge br3000 description 'customer red'
set interfaces bridge br3000 member interface eth5
set interfaces bridge br3000 member interface vxlan3000
set interfaces bridge br3000 vrf 'red'

set interfaces bridge br4000 address '10.3.1.1/24'
set interfaces bridge br4000 description 'customer green'
set interfaces bridge br4000 member interface eth6
set interfaces bridge br4000 member interface vxlan4000
set interfaces bridge br4000 vrf 'green'

set interfaces ethernet eth4 description 'customer blue'
set interfaces ethernet eth5 description 'customer red'
set interfaces ethernet eth6 description 'customer green'

```

PE2

```

set interfaces vxlan vxlan2000 mtu '1500'
set interfaces vxlan vxlan2000 parameters nolearning
set interfaces vxlan vxlan2000 port '4789'
set interfaces vxlan vxlan2000 source-address '172.29.255.2'
set interfaces vxlan vxlan2000 vni '2000'

set vrf name blue protocols bgp address-family ipv4-unicast redistribute connected
set vrf name blue protocols bgp address-family l2vpn-evpn advertise ipv4 unicast
set vrf name blue table '2000'
set vrf name blue vni '2000'

set interfaces vxlan vxlan3000 mtu '1500'
set interfaces vxlan vxlan3000 parameters nolearning
set interfaces vxlan vxlan3000 port '4789'
set interfaces vxlan vxlan3000 source-address '172.29.255.2'
set interfaces vxlan vxlan3000 vni '3000'

set vrf name red protocols bgp address-family ipv4-unicast redistribute connected
set vrf name red protocols bgp address-family l2vpn-evpn advertise ipv4 unicast
set vrf name red table '3000'
set vrf name red vni '3000'

set interfaces vxlan vxlan4000 mtu '1500'
set interfaces vxlan vxlan4000 parameters nolearning
set interfaces vxlan vxlan4000 port '4789'
set interfaces vxlan vxlan4000 source-address '172.29.255.2'
set interfaces vxlan vxlan4000 vni '4000'

set vrf name green protocols bgp address-family ipv4-unicast redistribute connected
set vrf name green protocols bgp address-family l2vpn-evpn advertise ipv4 unicast

```

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```

set vrf name green table '4000'
set vrf name green vni '4000'

set interfaces bridge br2000 address '10.1.2.1/24'
set interfaces bridge br2000 description 'customer blue'
set interfaces bridge br2000 member interface eth4
set interfaces bridge br2000 member interface vxlan2000
set interfaces bridge br2000 vrf 'blue'

set interfaces bridge br3000 address '10.2.2.1/24'
set interfaces bridge br3000 description 'customer red'
set interfaces bridge br3000 member interface eth5
set interfaces bridge br3000 member interface vxlan3000
set interfaces bridge br3000 vrf 'red'

set interfaces ethernet eth4 description 'customer blue'
set interfaces ethernet eth5 description 'customer red'

```

PE3

```

set interfaces vxlan vxlan2000 mtu '1500'
set interfaces vxlan vxlan2000 parameters nolearning
set interfaces vxlan vxlan2000 port '4789'
set interfaces vxlan vxlan2000 source-address '172.29.255.3'
set interfaces vxlan vxlan2000 vni '2000'

set vrf name blue protocols bgp address-family ipv4-unicast redistribute connected
set vrf name blue protocols bgp address-family l2vpn-evpn advertise ipv4 unicast
set vrf name blue table '2000'
set vrf name blue vni '2000'

set interfaces vxlan vxlan3000 mtu '1500'
set interfaces vxlan vxlan3000 parameters nolearning
set interfaces vxlan vxlan3000 port '4789'
set interfaces vxlan vxlan3000 source-address '172.29.255.3'
set interfaces vxlan vxlan3000 vni '3000'

set vrf name red protocols bgp address-family ipv4-unicast redistribute connected
set vrf name red protocols bgp address-family l2vpn-evpn advertise ipv4 unicast
set vrf name red table '3000'
set vrf name red vni '3000'

set interfaces vxlan vxlan4000 mtu '1500'
set interfaces vxlan vxlan4000 parameters nolearning
set interfaces vxlan vxlan4000 port '4789'
set interfaces vxlan vxlan4000 source-address '172.29.255.3'
set interfaces vxlan vxlan4000 vni '4000'

set vrf name green protocols bgp address-family ipv4-unicast redistribute connected
set vrf name green protocols bgp address-family l2vpn-evpn advertise ipv4 unicast
set vrf name green table '4000'
set vrf name green vni '4000'

```

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```

set interfaces bridge br2000 address '10.1.3.1/24'
set interfaces bridge br2000 description 'customer blue'
set interfaces bridge br2000 member interface eth4
set interfaces bridge br2000 member interface vxlan2000
set interfaces bridge br2000 vrf 'blue'

set interfaces bridge br4000 address '10.3.3.1/24'
set interfaces bridge br4000 description 'customer green'
set interfaces bridge br4000 member interface eth6
set interfaces bridge br4000 member interface vxlan4000
set interfaces bridge br4000 vrf 'green'

set interfaces ethernet eth4 description 'customer blue'
set interfaces ethernet eth6 description 'customer green'

```

Testing and debugging

You managed to come this far, now we want to see the network and routing tables in action.

Show routes for all VRFs

```

vyos@PE1:~$ show ip route vrf all
Codes: K - kernel route, C - connected, S - static, R - RIP,
       O - OSPF, I - IS-IS, B - BGP, E - EIGRP, N - NHRP,
       T - Table, v - VNC, V - VNC-Direct, A - Babel, F - PBR,
       f - OpenFabric,
       > - selected route, * - FIB route, q - queued, r - rejected, b - backup
       t - trapped, o - offload failure

VRF blue:
C>* 10.1.1.0/24 is directly connected, br2000, 00:01:13
B>* 10.1.2.0/24 [200/0] via 172.29.255.2, br2000 onlink, weight 1, 00:00:49
B>* 10.1.3.0/24 [200/0] via 172.29.255.3, br2000 onlink, weight 1, 00:00:49

VRF default:
O 172.29.0.2/31 [110/1] is directly connected, eth1, weight 1, 00:01:09
C>* 172.29.0.2/31 is directly connected, eth1, 00:01:12
O>* 172.29.0.4/31 [110/2] via 172.29.0.3, eth1, weight 1, 00:00:46
   *                               via 172.29.0.7, eth3, weight 1, 00:00:46
O 172.29.0.6/31 [110/1] is directly connected, eth3, weight 1, 00:01:09
C>* 172.29.0.6/31 is directly connected, eth3, 00:01:12
C>* 172.29.255.1/32 is directly connected, dum0, 00:01:14
O>* 172.29.255.2/32 [110/20] via 172.29.0.3, eth1, weight 1, 00:00:50
O>* 172.29.255.3/32 [110/20] via 172.29.0.7, eth3, weight 1, 00:00:45

VRF green:
C>* 10.3.1.0/24 is directly connected, br4000, 00:01:13
B>* 10.3.3.0/24 [200/0] via 172.29.255.3, br4000 onlink, weight 1, 00:00:49

VRF mgmt:

```

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```

S>* 0.0.0.0/0 [210/0] via 10.100.0.1, eth0, weight 1, 00:01:45
C>* 10.100.0.0/24 is directly connected, eth0, 00:01:45

VRF red:
C>* 10.2.1.0/24 is directly connected, br3000, 00:01:13
B>* 10.2.2.0/24 [200/0] via 172.29.255.2, br3000 onlink, weight 1, 00:00:49

```

Information about Ethernet Virtual Private Networks

```

vyos@PE1:~$ show bgp l2vpn evpn
BGP table version is 1, local router ID is 172.29.255.1
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal
Origin codes: i - IGP, e - EGP, ? - incomplete
EVPN type-1 prefix: [1]:[EthTag]:[ESI]:[IPlen]:[VTEP-IP]:[Frag-id]
EVPN type-2 prefix: [2]:[EthTag]:[MAClen]:[MAC]:[IPlen]:[IP]
EVPN type-3 prefix: [3]:[EthTag]:[IPlen]:[OrigIP]
EVPN type-4 prefix: [4]:[ESI]:[IPlen]:[OrigIP]
EVPN type-5 prefix: [5]:[EthTag]:[IPlen]:[IP]

  Network          Next Hop          Metric LocPrf Weight Path
Route Distinguisher: 10.1.1.1:5
*> [5]:[0]:[24]:[10.1.1.0]
      172.29.255.1          0          32768 ?
      ET:8 RT:100:2000 Rmac:4e:bb:3c:ba:bd:a6
Route Distinguisher: 10.1.2.1:4
*>i[5]:[0]:[24]:[10.1.2.0]
      172.29.255.2          0      100      0 ?
      RT:100:2000 ET:8 Rmac:26:07:da:eb:fc:ea
Route Distinguisher: 10.1.3.1:4
*>i[5]:[0]:[24]:[10.1.3.0]
      172.29.255.3          0      100      0 ?
      RT:100:2000 ET:8 Rmac:26:98:28:24:6e:54
Route Distinguisher: 10.2.1.1:6
*> [5]:[0]:[24]:[10.2.1.0]
      172.29.255.1          0          32768 ?
      ET:8 RT:100:3000 Rmac:50:00:00:01:00:05
Route Distinguisher: 10.2.2.1:5
*>i[5]:[0]:[24]:[10.2.2.0]
      172.29.255.2          0      100      0 ?
      RT:100:3000 ET:8 Rmac:50:00:00:02:00:05
Route Distinguisher: 10.3.1.1:7
*> [5]:[0]:[24]:[10.3.1.0]
      172.29.255.1          0          32768 ?
      ET:8 RT:100:4000 Rmac:50:00:00:01:00:06
Route Distinguisher: 10.3.3.1:6
*>i[5]:[0]:[24]:[10.3.3.0]
      172.29.255.3          0      100      0 ?
      RT:100:4000 ET:8 Rmac:06:32:9d:22:55:8a

Displayed 7 out of 7 total prefixes

```

If we need to retrieve information about a specific host/network inside the EVPN network we need to run

```
vyos@PE2:~$ show bgp l2vpn evpn 10.3.1.10
BGP routing table entry for 10.3.1.1:7:[5]:[0]:[24]:[10.3.1.0]
Paths: (1 available, best #1)
  Not advertised to any peer
  Route [5]:[0]:[24]:[10.3.1.0] VNI 4000
  Local
    172.29.255.1 (metric 20) from 172.29.255.1 (172.29.255.1)
      Origin incomplete, metric 0, localpref 100, valid, internal, best (First path
      ←received)
      Extended Community: RT:100:4000 ET:8 Rmac:50:00:00:01:00:06
      Last update: Thu May 11 13:31:13 2023
```

11.22.4 Wireguard

Testdate: 2024-01-13

Version: 1.5-rolling-202401121239

This simple structure show how to connect two offices. One remote branch and the central office.

Topology

The topology have a central and a branch VyOS router and one client, to test, in each site.

Configuration

Set the local subnet on eth2 and the public ip address eth1 on each site.

Central

```
set interface ethernet eth2 address 10.0.1.254/24
set interface ethernet eth1 address 198.51.100.1/24
```

Branch

```
set interface ethernet eth2 address 10.0.2.254/24
set interface ethernet eth1 address 198.51.100.2/24
```

Next thing to do, is to create a wireguard keypair on each side. After this, the public key can be displayed, to save for later.

```
vyos@central:~$ generate pki wireguard
Private key: wHQS+ib3eMIp2DxRiAeXfFVaSCMMP1YHBaKfSR1xfV8=
Public key: RCMY6BAER0uEcPvspUb3K38MHyHJpK5kiV5IOX943HI=
```

After you have each public key. The wireguard interfaces can be setup.

Central

```

set interfaces wireguard wg01 private-key 'wHQS+ib3eMIp2DxRiAeXfFVaSCMMP1YHBaKfSR1xfV8='
set interfaces wireguard wg01 address 192.168.0.1/24
set interfaces wireguard wg01 description 'VPN-to-Branch'
set interfaces wireguard wg01 peer branch allowed-ips 10.0.2.0/24
set interfaces wireguard wg01 peer branch allowed-ips 192.168.0.0/24
set interfaces wireguard wg01 peer branch address 198.51.100.2
set interfaces wireguard wg01 peer branch port 51820
set interfaces wireguard wg01 peer branch public-key 'nWhMTjGQbQiJwaNqHpZ/
↪p8+iAH29HaJDNsdfsRdW9As='
set interfaces wireguard wg01 port 51820

```

Branch

```

set interfaces wireguard wg01 private-key 'QM3ZtmaxstxIDoz00AVLE/F/UVjmdcrOvfKYW/TVw18='
set interfaces wireguard wg01 address 192.168.0.2/24
set interfaces wireguard wg01 description 'VPN-to-central'
set interfaces wireguard wg01 peer central allowed-ips 10.0.1.0/24
set interfaces wireguard wg01 peer central allowed-ips 192.168.0.0/24
set interfaces wireguard wg01 peer central address 198.51.100.1
set interfaces wireguard wg01 peer central port 51820
set interfaces wireguard wg01 peer central public-key
↪'RCMy6BAER0uEcPvspUb3K38MHyHJpK5kiV5IOX943HI='
set interfaces wireguard wg01 port 51820

```

To reach the network, a route must be set on each VyOS host. In this structure, a static interface route will fit the requirements.

Central

```
set protocols static route 10.0.2.0/24 interface wg01
```

Branch

```
set protocols static route 10.0.1.0/24 interface wg01
```

Testing and debugging

After all is done and commit, let's take a look if the Wireguard interface is up and running.

```

vyos@central:~$ show interfaces wireguard
Codes: S - State, L - Link, u - Up, D - Down, A - Admin Down
Interface      IP Address      S/L  Description
-----
wg01           192.168.0.1/24  u/u  VPN-to-Branch

```

And ping the Branch PC from your central router to check the response.

```

vyos@central:~$ ping 10.0.2.100 count 4
PING 10.0.2.100 (10.0.2.100) 56(84) bytes of data.
64 bytes from 10.0.2.100: icmp_seq=1 ttl=63 time=0.894 ms
64 bytes from 10.0.2.100: icmp_seq=2 ttl=63 time=0.869 ms
64 bytes from 10.0.2.100: icmp_seq=3 ttl=63 time=0.966 ms
64 bytes from 10.0.2.100: icmp_seq=4 ttl=63 time=0.998 ms

```

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```

--- 10.0.2.100 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3004ms
rtt min/avg/max/mdev = 0.869/0.931/0.998/0.052 ms

```

11.22.5 OpenVPN with LDAP

Testdate: 2023-05-11

Version: 1.4-rolling-202305100734

This LAB shows how to use OpenVPN with a Active Directory authentication method.

Topology consists of:

- Windows Server 2019 with a running Active Directory
- VyOS as a OpenVPN Server
- VyOS as Client

Active Directory on Windows server

The lab assumes a full running Active Directory on the Windows Server. Here are some PowerShell commands to quickly add a Test Active Directory.

```

# install the Active Directory Server role
Install-WindowsFeature AD-Domain-Services -IncludeManagementTools

# install the Active Directory Server role
Install-ADDSForest -DomainName "vyos.local" -DomainNetBiosName "VYOS" -InstallDns:$true -
↳ NoRebootCompletion:$true

# create test user01 and binduser
New-ADUser binduser -AccountPassword(Read-Host -AsSecureString "Input Password") -
↳ Enabled $true
New-ADUser user01 -AccountPassword(Read-Host -AsSecureString "Input Password") -Enabled
↳ $true

```

Configure VyOS as OpenVPN Server

In this example OpenVPN will be setup with a client certificate and username / password authentication.

First a CA, a signed server and client certificate and a Diffie-Hellman parameter must be generated and installed. Please look [here](#) for more information.

Add the LDAP plugin configuration file `/config/auth/ldap-auth.config`

Check all possible settings [here](#).

```
<LDAP>
URL ldap://192.168.1.10
BindDN bind_user@vyos.local
Password P4ssw0rd123
Timeout 15
TLSEnable no
FollowReferrals no
</LDAP>
<Authorization>
BaseDN "DC=vyos,DC=local"
SearchFilter "sAMAccountName=%u"
RequireGroup false
</Authorization>
```

Now generate all required certificates on the ovpn-server:

First the CA

```
vyos@ovpn-server# run generate pki ca install OVPN-CA
```

after this create a signed server and a client certificate

```
vyos@ovpn-server# run generate pki certificate sign OVPN-CA install SRV
vyos@ovpn-server# run generate pki certificate sign OVPN-CA install CLIENT
```

and last the DH Key

```
vyos@ovpn-server# run generate pki dh install DH
```

after all these steps the config look like this:

```
set pki ca OVPN-CA certificate
  ↪ 'MIIFnTCCA4WgAwIBAgIUIPFIXvCxYdavCnSPFNjr6lUtlsswDQYJKoZIhvcNAQELBQAwVzELMAkGA1UEBhMCROIxZzARBgNVBAgM
  ↪ az1C22Sbp3wPJLfgOmy0K3TA5qVsx/c/8gatsatMkCsekGnK5BPzCDd5eCCLo//
  ↪ B25HF06fBYRNVhVvYCUx7QEXw4FHFNG88zCIizx114AGtVwZfGGG9xCc53xjLPUph6iqTXme41cCFFQlqXwZ7fuySieSdoV8SAsJT
  ↪ EyauQMY/
  ↪ LC4tLc6moPaNlTWa9HJv8s6xUqpzNptDoUHK0qKuw2JRFnno5SCQ788KkKNGVWBy2o3BGoewfHFhAdr61CXeLpmuneuhi96GcM031
  ↪ RchtHRC6rtFavHJjB2cUcCkhhQofUE6IR2dYJZ1cw0Wy5CI3bXHf43BpvDGmuxILNGirTq8wf5RCWzDJJgmKQpYhUYe8x4faF4gTo
  ↪ QD+EIV9xXgOk14+BbnHKWbZ70u5emewFuE/
  ↪ bjl79oNjklpXdc4soRkCPCTEGK3zDBdmUtCYk1DwIDAQABo2EwXzAPBgNVHRMBAf8EBTADAQH/MA4GA1UdDwEB/
  ↪ wQEAWIBhjAdBgNVHSUEFjAUBggrBgEFBQcDAgYIKwYBBQUHAWEwHQYDVR00BBYEFP5NDac/
  ↪ yC+mQmaTpZDUv9GZMGMBMA0GCSqSgSIb3DQEBChUA4ICAQDEqpF2ibwYFxsF1XDIPS5/
  ↪ Gs0sZTZBuByNm5d2+jTy07d5a1ZUdbvobbwhxZ0hWasmFNyPLr4TYmZm5zF+efFsi0xjyRuEoVU+Fe8rZmpRIF/
  ↪ +6+nYX5r9vMI4QxGjeeyP200HJ85Kvz182CTsITrM15Vw/kVVjAVzFI5Gm/
  ↪ Qo1alAoFQza9rAL4kDqaUszjHjPhysvDpGF+NLPjiYDHXcty/
  ↪ BC48bnuzAeEM60SGZ7EXvf8l0X8Ys07z39w6780A/3rbZvFhCYMKp/
  ↪ +p5xBRDjnX91dM6DJw73RwYQ1KHbHk9wWUwnL1giL71jzp/
  ↪ y40j6SSK2PQv+On080J6Zg06WIQx9xYcyr108Xh9FotUrlG7GYPI3Udf95t6SjuydDhULAVD01MBxlDe9DHW1k1q1p0XaHZg926tY
  ↪ lda6dcuwJjA2Dx5JI6L0u9ureQmQAtxvnoTCtf+hRliX/IkskZCKs34SjNiCnBuw/
  ↪ DNfdOpfaABm7y+twiXBwnu5l/
  ↪ K8poXcQYQByyZj6YmmpgsbVPr5KNsLW0gRA81M6IPof8qxvnFrkazhiQWh1YHSjnaHtA3z5/
  ↪ BdgwHVICuFyrIOlbkKyJOjKcKBSdMwIV0tsnpnyli2xEPZKu1tAQFAavXrK/RGYyh0Z3e0aRSV8h1P8i/
  ↪ mf7p0I45cJiBCqPg=='
set pki ca OVPN-CA private key '<REDACTED>'
```

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[illegible]

Once all the required certificates and keys are installed, the remaining OpenVPN Server configuration can be carried out.

```

set interface ethernet eth1 address '192.168.1.1/24'
set interface ethernet eth2 address '198.51.100.254/24'
set interfaces openvpn vtun10 local-host '198.51.100.254'
set interfaces openvpn vtun10 local-port '1194'
set interfaces openvpn vtun10 mode 'server'
set interfaces openvpn vtun10 openvpn-option '--plugin /usr/lib/openvpn/openvpn-auth-
↳ ldap.so /config/auth/ldap-auth.config'
set interfaces openvpn vtun10 persistent-tunnel
set interfaces openvpn vtun10 protocol 'udp'
set interfaces openvpn vtun10 server push-route '192.168.1.0/24'
set interfaces openvpn vtun10 server subnet '10.23.1.0/24'

set interfaces openvpn vtun10 tls ca-certificate OVPN-CA
set interfaces openvpn vtun10 tls certificate SRV
set interfaces openvpn vtun10 tls dh-params DH
set protocols static route 10.1.1.0/24 interface vtun10

```

Client configuration

One advantage of having the client certificate stored is the ability to create the client configuration.

```

vyos@ovpn-server:~$ generate openvpn client-config interface vtun10 ca OVPN-CA
↳ certificate CLIENT

```

save the output to a file and import it in nearly all openvpn clients.

```

client
nobind
remote 198.51.100.254 1194
remote-cert-tls server
proto udp
dev tun
dev-type tun
persist-key
persist-tun
verb 3

# Encryption options

keysize 256
comp-lzo no

<ca>
-----BEGIN CERTIFICATE-----
MIIFnTCCA4WgAwIBAgIUIPFIXvCxYdavCnSPFNjr6lUtlsswDQYJKoZIhvcNAQEL
BQAwVzELMAkGA1UEBhMCR0IxExARBgNVBAGMC1NvbWUtU3RhdGUxEjAQBGNVBACM
CVNvbWUtQ2l0eTENMA5GA1UECgwEVnlpUzEQMA4GA1UEAwwHdnlvcy5pbzAeFw0y
MzA1MTExMjM4MjJaFw0zMzA1MDgxMjM4MjJaMFcxZCZAJBgNVBAYTAkdCMRMwEQYD
VQQUIDApTb21lLVN0YXRlMRItEAYDVQQHDA1Tb21lLUNpdHkxDALBgNVBAoMBFZ5
T1MxEDA0BgNVBAMMB3Z5b3MuaW8wggiMA0GCSqGSIb3DQEBAQUAA4ICDwAwggIK
AoICAQDg45vAzS6xNqU+Pa7wk1Imt1/az1C22Sbp3wPJLfgoMy0K3TA5qVsx/c/8
gatsatMkCsekGnK5BPzCDd5eCCLo//B25HF06fBYRNvHvVyCUx7QEXw4FHFNG88z

```

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```

CIizx114AGtVwZfGGG9xCc53xjLPUpH6iqTXme41cCFFQlqXwZ7fuySieSdoV8SA
sJTT0sGCEUEcDEnNPn6tX3KWTzNuyFPECy8WcmNgWNyG2nmH+U7WRTX0ehZ5dZyU
5au7TxpRN4a+JtE0gNqCwJ+nh1A543q2pcRoQpPAzHFclgj8wG/EyauQMY/LC4tL
c6moPaNlTwA9HJv8s6xUqpzNptDoUHK0qKuw2JRFnno5SCQ788KkKNgVWBy2o3BG
oewfHFhAdR61CXeLpmuneuhi96GcM031gW8ptXbd4DkCF7H6KRtqeIvwiYg79ttC
8kZf01Sn1fM5ftjGxaE38dAk/RchtHRC6rtFavHJjB2cUcCkhhQofUE6IR2dYJZ1
cw0Wy5CI3bXHf43BpvDgmuxlLNGirTq8wf5RCWzDJJgmKQpYhUYe8x4faF4gTo00
uH4ZvAYjQu3JNZGkb50p4kM9Mu5rQAiZJUeMAz/QD+EIV9xXgOk14+BbnHKWbZ70
u5emewFuE/bjl79oNjklpXdc4soRkCPCTEGK3zDBdmUtCYk1DwIDAQABo2EwXzAP
BgNVHRMBAf8EBTADAQH/MA4GA1UdDwEB/wQEAWIBhjAdBgNVHSUEFjAUBggrBgEF
BQcDAGYIKwYBBQUHAWewHQYDVR0OBYYEFP5NDac/yC+mQmaTpZDUv9GZMGBMA0G
CSqGSIb3DQEBChUA4ICAQDEqpf2ibwYFxsF1XDIPS5/Gs0sZTZBuByNm5d2+jTy
07d5alZUdbvobbwhxZohWasmfNyPLr4TYmZm5zF+efFsi0xjyRuEoVU+Fe8rZmpR
IF/+6+nYX5r9vMI4QxGjeeyP200HJ85Kvz182CTsITrM15Vw/kVVjAVzFI5Gm/Qo
lalAoFQza9rAL4kDqaUszjHjPhysvDpGF+NLPjiYDHXcty/BC48bnuzAeEM60SGZ
7EXvf8l0X8Ys07z39w6780A/3rbZvFhCYMKp/+p5xBRDjnX91dM6DJw73RwYQ1KH
bHk9wWUwnL1giL71jzp/y40j6SSK2PQv+On080J6Zg06WlQx9xYcXr108Xh9FotU
rlG7GYPI3Udf95t6SjuydDhULAVD01MBxlDe9DHW1k1q1p0XaHZg926tY66xx/ld
a6dcuWjA2Dx5JI6L0u9ureQmQAtxvnoTCtf+hR1iX/TkskZCKs34SjNiCnBuW/D
NfdOpfaABm7y+tWiXBwnu5l/K8poXcQYQByyZj6YMmpgsbVPr5KNsLW0gRA81M6I
Pof8qxvnrFrkazhiQWh1YHSjnaHtA3z5/BdgwHVICuFyrIOlbbKyJOjKcKBSddMwI
V0tsnpnyli2xEPZKu1tAQFAavXrK/RGYYhOZ3e0aRSV8hlP8i/mf7p0I45cJiBCq
Pg==

```

```
-----END CERTIFICATE-----
```

```
</ca>
```

```
<cert>
```

```
-----BEGIN CERTIFICATE-----
```

```

MIIFsDCCA5igAwIBAgIU5SgZgwzGsFjFecGxCwLXVsGCLMkAwDQYJKoZIhvcNAQEL
BQAwVzELMAkGA1UEBhMCRC0xExARBgNVBAGMC1NvbWUuU3RhdGUxEjAQBgNVBAcM
CVNvbWUuU3RhdGUxEjAQBgNVBAcMCMVnbWUuU3RhdGUxEjAQBgNVBAcMCMVnbWUuU3RhdGUxEjAQBgNVBAcM
MzA1MTExMjM0MzlaFw0zMzA1MDgxMjM0MzlaMFYxXzA1BgNVBAYTAkdCMRMEQYD
VQIDApTb211LVN0YXRlMRlweAYDVQQLDA1Tb211LUNpdHkxDTALBgNVBAoMBFZ5
T1MxMzANBgNVBAMMBmNsaWVudDCCAIwDQYJKoZIhvcNAQEBBQADggIPADCCAgoc
ggIBANHNJOSwCDBRqziLlgXyNHlq7P7vEUFvS8d/XLYJ1xIpcYTRXTut2CTGRar7
fZZicu7x0yoK4TzrHvGVf1o4NC4NSGV5RX6kwRdrfWBmvpIkjSLGtCREFYhb+PHD
pnsIS7cfN9udC0vocqVlx/xM/sfcP6Vja/uFp+9TQcneJIxYw34zkF+TtOVbE3pP
5VxU7ZAJ8F5/q1ONhTmdzG4014/0nBqZfdYA3LVDeSSNIJNF5jlaKXXFHZ1EJRem
TYDx+f5bfcVK2Qs8fU9jCFBlATjMu905rgk6nMLRwEnJZuZ1gj2tWQvz4e9yo5y
Uqf1PUhOrn3c81MRliUNHkr+CkxgQJal6P3Ar3q4iftJih3K+/j4o194mQ/Dt/Et
+/Qn/DUFk2FB0rTmcQwJLTEAzxtTdmBjeJpipIPDR0u7UMZLNh/raQ8s3FsbY4uY
ORt2f5YQ1CVHbth4dRa9xa+oRbm7eomNACIbWfkLh5Bzud1+qIfdBMZKaZbnf0HE
euH0J5LBJeova8EPxWbYmJPrRHzu5gowkIKl+uIxcy8IiNTA9YEOJVonCjmlr8NE
tYShrIVbicdMNSI3pOQR60MFhkHwBjSU21/z+4wwLxtzq/c2xKw9yrOZ46ZVLwGD
Fq8rPwp7/P9r6mDKsbn6jIvG0eH71dMZvoc41CaClw+hKIzLAgMBAAGjdTBzMAwG
A1UdEwEB/wQCMAAwDgYDVR0PAQH/BAQDAgeAMBGA1UdJQMMMAoGCCsGAQUFBwMC
MB0GA1UdDgQWBBS6j30FmL6kZW7rDH8QjRMoWoA/njAfBgNVHSMEGDAWgBT+TQ2n
P8gvpkJmk6WQ1L/RmTBjATANBgkqhkiG9w0BAQsFAAOCAgEANW2Y4bgaB9oexEjj
6rkGvePtQmXRkF/adVQREY9iZDGT72ePybVzrfMkZHjse3o7JvXWRIVVztWSzEp
v5noIOX71AioGG3wsFTHotTFR0zrYJHXHBCV2Neq4Kx2Ta/TZwD8QnZHAAXEQ1pY
b4fxwN/A60VELAZoz9zYsbrJyVrfuHDL9queQxPFzqis+7W1BiVICv4rn0DMQ560

```

(continues on next page)

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```

jTGh4t4rIm0Su5gUsUrQaih85XDd0BPxViSNwfVdZJIgbvamudpfEaKsIun/uCjc
xpNnzIp0rhyYmDeqVat4GnTV7Sy48e/Uvcq71ZWbBYJF4+yW4pylIU2Sh/Uy2sAz
4C2M71FlFB7qsmcnPRsFFHf+r1NyD1lkVI9k2371fTG/Kub9V0rOz4pvKz4Em5b4
MUPdDbZ0qJ8hQ+atGE3ovFJIovA3NFb00tnyC4l+kG7dfjqFudOrnmDa+Qsya+2Y0
xBZBIRfuhlXhb6Y6Smsk9R6x0jBmcQTPS5ZmvKaTxQCFc53xMdQNAswjiI2L9rw4
BcqQfVmf/vpoN+VusD/XEv2V0Ixm10YybA7BI/tixh9vwj3fdQXVly3jSYjVBd5W
OFPizbQZeD10ElvlLqZzyWrP/Wre7Nmi/gEOnhBXXmo034fFF/vXf0JRpQsd2oDs
24+4XwZYb8mbM31j7Nx8YvhR+64=
-----END CERTIFICATE-----

</cert>

<key>
-----BEGIN PRIVATE KEY-----
...REDACTED...
-----END PRIVATE KEY-----

</key>

```

Configure VyOS as client

```

set interfaces openvpn vtun10 authentication username 'user01'
set interfaces openvpn vtun10 authentication password '$secret'
set interfaces openvpn vtun10 encryption cipher 'aes256'
set interfaces openvpn vtun10 hash 'sha512'
set interfaces openvpn vtun10 mode 'client'
set interfaces openvpn vtun10 persistent-tunnel
set interfaces openvpn vtun10 protocol 'udp'
set interfaces openvpn vtun10 remote-host '198.51.100.254'
set interfaces openvpn vtun10 remote-port '1194'
set interfaces openvpn vtun10 tls ca-certificate 'OVPN-CA'
set interfaces openvpn vtun10 tls certificate 'CLIENT'

```

Monitoring

If the client is connected successfully you can check the status

```

vyos@ovpn-server:~$ show openvpn server
OpenVPN status on vtun10

```

Client CN	Remote Host	Tunnel IP	Local Host	TX bytes	RX bytes
Connected Since					
client	198.51.100.1:55150	10.23.1.6	198.51.100.254:1194	4.7 KB	4.7 KB
2023-05-11 12:47:11					

VPP DATAPLANE

VPP (Vector Packet Processing) is a high performance packet processing stack that runs in user space. VyOS can use VPP as an alternative dataplane to the Linux kernel networking stack.

12.1 VPP Dataplane Description

12.1.1 What is VPP in VyOS?

VyOS supports two packet forwarding dataplanes:

- **Linux kernel dataplane** (traditional)
- **Vector Packet Processor (VPP) dataplane** (optional)

VPP is a high-performance user space packet processor that improves throughput for demanding network workloads.

12.1.2 Key Benefits

Performance Improvement

VPP uses vector-based packet processing instead of one-by-one handling, delivering:

- **Higher throughput** compared to kernel forwarding.
- **Lower and more consistent latency** for time-sensitive applications.
- **Linear scaling** with additional CPU cores.

VyOS Hybrid Integration

VyOS supports both dataplanes simultaneously, providing:

- **Cross-dataplane forwarding:** Traffic can flow between the VPP dataplane and kernel interfaces seamlessly.
- **Transparent configuration:** Same CLI commands and most services work regardless of dataplane.
- **Gradual migration:** Enable VPP on high-traffic interfaces while keeping others on kernel.

12.1.3 When to Use VPP

Consider VPP if you have:

- High-throughput requirements
- Latency-sensitive applications requiring consistent performance

Stay with kernel dataplane if you have:

- Low to moderate traffic volumes
- No latency-sensitive workloads
- Applications requiring specific features not supported by VPP Dataplane

12.1.4 Packet Processing Integration

VPP Dataplane integration minimizes configuration changes. Features in the kernel dataplane continue to operate there. VPP Dataplane only handles packet forwarding for interfaces explicitly assigned to it.

Traffic flow examples between VPP and kernel dataplane interfaces:

Green path

Traffic between two VPP interfaces stays within VPP for maximum performance and can use only VPP dataplane features.

Blue path

Traffic between a VPP interface and a kernel interface is processed by both dataplanes and can use features from both.

Note: This path has slower performance than pure VPP or pure kernel forwarding because packets traverse both dataplanes.

Red path

Traffic between two kernel interfaces stays within the kernel dataplane without VPP acceleration. This is the traditional VyOS dataplane operation.

12.1.5 CLI Integration

VyOS CLI commands work with both dataplanes. Use the same commands to configure interfaces, routing, and other features regardless of the dataplane.

12.2 VPP Dataplane Requirements

VPP Dataplane requires specific hardware. Ensure your system meets these prerequisites before enabling VPP:

- **Deployment Platform**

VPP Dataplane is available on both bare-metal, on-premise virtualized, and cloud deployment platforms.

- **CPU Requirements**

Regardless of the platform, VPP Dataplane requires a CPU with:

- SSE4.2 support (available on most modern Intel and AMD CPUs).
- At least 4 physical CPU cores for a minimum configuration (more cores recommended for higher throughput).

Important: Physical Cores vs Logical Cores

VPP Dataplane requires 4 *physical* CPU cores, not logical cores. Systems with Simultaneous Multithreading (SMT) or Hyper-Threading (HT) present each physical core as 2 logical cores.

Cloud providers often display logical cores as “cores” or “vCPUs”. For example, a cloud instance showing “4 cores” may have only 2 physical cores with SMT/HT enabled. Always verify the actual physical core count in your cloud provider’s documentation.

For virtualized environments, ensure CPU features are passed through to the VM and that sufficient physical cores are allocated.

- **Memory Requirements**

Memory significantly affects VPP stability. Insufficient RAM can cause initialization failures or prevent the dataplane from starting.

- Minimum: 8 GB RAM. VyOS will not start the VPP Dataplane if less than 8 GB is available.
- Recommended: 16 GB or more (especially for high throughput, many interfaces, or large routing tables).

- **Network Interface Cards (NICs)**

Warning: VyOS supports only specific NICs for the VPP dataplane. Using unsupported hardware may cause activation failures, initialization errors, crashes, or degraded performance.

When enabling VPP, VyOS checks detected network interfaces against a list of validated NICs. Validation is based on the **PCI ID** of the device or the **kernel driver** used by the interface.

Supported NICs:

Filter Type	Filter Value	NIC Name/Description	Platform Where NIC Can Be Found
PCI ID	15b3:1019	Mellanox Technologies MT28800 Family [ConnectX-5 Ex]	Bare-metal
PCI ID	15b3:101d	Mellanox Technologies MT2892 Family [ConnectX-6 Dx]	Bare-metal
PCI ID	15b3:101e	Mellanox Technologies ConnectX Family mlx5Gen Virtual Function	Oracle Cloud
PCI ID	8086:1592	Intel Corporation Ethernet Controller E810-C for QSFP	Bare-metal
PCI ID	1ae0:0042	Google, Inc. Compute Engine Virtual Ethernet [gVNIC]	Google Cloud
PCI ID	1af4:1000	Red Hat, Inc. Virtio network device	KVM-based hypervisors, including with Open vSwitch; Google Cloud
PCI ID	1d0f:ec20	Amazon.com, Inc. Elastic Network Adapter (ENA)	AWS
Kernel Driver	hv_netvsc	Microsoft Hyper-V network interface card	Microsoft Azure

If no supported NIC is detected, VPP activation will be rejected.

In testing or advanced deployments, unsupported hardware can be explicitly allowed in the configuration:

```
set vpp settings allow-unsupported-nics
```

Note: This option bypasses the hardware validation checks for the specified devices. Stability and performance are not guaranteed when using unsupported NICs or drivers.

12.3 VPP Dataplane Limitations

VPP Dataplane provides significant performance advantages, but has some limitations you should consider.

- **Feature Parity**

VPP does not support all features available in the Linux kernel dataplane. Some networking features, specific protocols, or services may not be available.

While VPP supports various interface types similar to the kernel, their capabilities may differ.

- **NIC and Driver Compatibility**

VyOS currently supports only DPDK drivers for network interfaces. Not all network interface cards are compatible with DPDK drivers.

- **Data Path Limitations**

If a feature exists only in the kernel dataplane, traffic that uses that feature cannot traverse VPP interfaces. Examples include:

- Firewall
- QoS

When traffic uses the pure VPP path, it does not reach the kernel, where such features are implemented. Plan how traffic flows through your VyOS instance to ensure it reaches the necessary features.

VPP provides native alternatives for some features. For example, VPP native ACLs provide basic firewall functionality.

12.4 VPP Configuration

VPP settings consist of several main sections.

Main Dataplane settings and internal VPP interfaces:

12.4.1 VPP Dataplane Core Configuration

This section covers the core configuration options for the VPP dataplane in VyOS. It includes settings for memory management, CPU allocation, hugepages, and other essential parameters that influence the performance and behavior of the VPP dataplane.

Please review the general system configuration, before starting to configure VPP. Without proper VyOS preconditions, VPP will not start or its efficiency will be significantly degraded.

VyOS Configuration for VPP

Hugepages

VPP uses hugepages for efficient memory management. Hugepages are larger memory pages that reduce the overhead of page management and improve performance for applications that require large amounts of memory.

Hugepages can be configured in VyOS using the following commands:

Warning: Changes to hugepage settings require a system reboot to take effect.

Hugepages must be enabled before VPP configuration is applied.

To enable hugepages:

```
set system option kernel memory hugepage-size <size> hugepage-count '<count>'
```

Enables hugepages with the specified size and count. The size can be either 2MB or 1GB, and the count specifies the number of hugepages to allocate.

If your system has multiple NUMA nodes, the total amount of hugepages will be divided equally among them.

Resources Limits

Note: By default, system will calculate and set the recommended values for resource limits. Avoid tuning these values if you are not sure what you are doing.

During operations VPP utilizes a significant amount of system resources, especially memory. There are two main settings that may need to be adjusted to ensure VPP runs smoothly:

Maximum number of memory map areas a process may have:

```
set system option resource-limits max-map-count <value>
```

Maximum shared memory segment size:

```
set system option resource-limits shmmax <value>
```

Both settings are automatically calculated based on configured hugepages.

Kernel Tuning

VPP performance greatly benefits from proper kernel tuning, especially CPU isolation and disabling unnecessary kernel features. These optimizations ensure dedicated CPU cores are available exclusively for VPP dataplane processing without interference from the kernel scheduler or other system processes.

Warning: Kernel tuning changes require a system reboot to take effect.

Improper CPU isolation can lead to system instability if essential system processes are starved of CPU resources.

CPU Isolation and Optimization

CPU isolation is crucial for VPP performance as it dedicates specific CPU cores exclusively to VPP dataplane processing. The isolated cores are removed from the kernel scheduler and will not run regular system processes.

Disable NMI Watchdog

The NMI (Non-Maskable Interrupt) watchdog can interfere with VPP performance by generating interrupts on isolated cores and is not compatible with nohz-full mode:

```
set system option kernel cpu disable-nmi-watchdog
```

Disables the NMI watchdog for detecting hard CPU lockups. This prevents unnecessary interrupts on VPP worker cores.

CPU Core Isolation

```
set system option kernel cpu isolate-cpus <cpu-range>
```

Isolates specified CPUs from the kernel scheduler. Isolated cores will not run regular system processes and are dedicated to applications like VPP.

The <cpu-range> can be:

- Single core: 2
- Range: 2-5
- Mixed: 1,3-5,7

Important: Always reserve at least 2 cores for the operating system to ensure system stability. For example, on a 4-core system, isolate cores 2-3 for VPP and leave cores 0-1 for the OS.

Assign the first isolated core as the VPP main core and the remaining isolated cores as VPP worker cores. Ensure that VPP CPU assignments match the isolated CPU range.

Adaptive-Tick Mode

set system option kernel cpu nohz-full <cpu-range>

Enables adaptive-tick mode (NO_HZ_FULL) for specified CPUs. This causes the kernel to avoid sending scheduling-clock interrupts to CPUs that have only one runnable task, significantly reducing interrupt overhead for dedicated workloads like VPP.

Use the same CPU range as configured for `isolate-cpus`.

RCU Callback Offloading

set system option kernel cpu rcu-no-cbs <cpu-range>

Offloads Read-Copy-Update (RCU) callback processing from specified CPUs. This ensures that RCU callbacks do not prevent the specified CPUs from entering dyntick-idle or adaptive-tick mode, which is essential for nohz-full functionality.

Use the same CPU range as configured for `isolate-cpus`.

System Optimization

Additional kernel optimizations can further improve VPP performance by disabling unnecessary features and reducing system overhead.

Disable High Precision Event Timer

set system option kernel disable-hpet

Disables the High Precision Event Timer (HPET). HPET can cause additional interrupts and overhead that may impact VPP performance.

Disable Machine Check Exceptions

set system option kernel disable-mce

Disables Machine Check Exception (MCE) reporting and handling. While MCE provides hardware error detection, it can introduce latency in high-performance scenarios.

Disable CPU Power Saving

set system option kernel disable-power-saving

Disables CPU power saving mechanisms (C-states). This keeps CPU cores at maximum performance levels, eliminating latency from power state transitions.

Disable Soft Lockup Detection

set system option kernel disable-softlockup

Disables the soft lockup detector for kernel threads. This prevents false positives when VPP worker threads are busy processing packets.

Disable CPU Mitigations

set system option kernel disable-mitigations

Disables all optional CPU mitigations for security vulnerabilities (for example, Spectre, Meltdown). This may improve performance on some platforms.

Optimal Configuration Example

For a system with 4 CPU cores (0-3) where cores 2-3 are dedicated to VPP:

```
# Kernel CPU optimizations
set system option kernel cpu disable-nmi-watchdog
set system option kernel cpu isolate-cpus '2-3'
set system option kernel cpu nohz-full '2-3'
set system option kernel cpu rcu-no-cbs '2-3'

# System optimizations
set system option kernel disable-hpet
set system option kernel disable-mce
set system option kernel disable-power-saving
set system option kernel disable-softlockup

# VPP CPU assignment
set vpp settings resource-allocation cpu-cores '2'
```

VPP Dataplane Buffers Configuration

Buffers are essential for handling network packets efficiently. Proper configuration enhances performance and reliability, and is mandatory for VPP to work. Buffers temporarily store packets during processing. Therefore, their configuration must be in sync with NIC configuration, CPU threads, and overall system resources.

Important: VPP buffers are allocated from the physical memory pool (`physmem`). The total amount of memory available for buffer allocation is controlled by the `physmem-max-size` setting, while the buffer configuration parameters below control how that memory is used for buffer allocation.

See [VPP Physical Memory Configuration](#) for details on configuring `physmem`.

Buffer Configuration Parameters

The following parameters can be configured for VPP buffers:

buffers-per-numa

Number of buffers allocated per NUMA node. This setting optimizes memory access patterns for multi-CPU systems.

Typically, you need to tune this value if:

- The system has many interfaces
- NICs have many queues
- NICs have large descriptor sizes

Set this value carefully to balance memory usage and performance.

```
set vpp settings resource-allocation buffers buffers-per-numa <value>
```

The common approach for the calculation is to use the formula:

$$\text{buffers-per-numa} = (\text{num-rx-queues} * \text{num-rx-desc}) + (\text{num-tx-queues} * \text{num-tx-desc})$$

Calculate this formula for each NIC and sum the results. Multiply the total by 2.5 to get the minimum recommended value for `buffers-per-numa`.

Avoid setting this value too low to prevent packet drops.

data-size

This value sets how much payload data can be stored in a single buffer allocated by VPP. Larger values reduce buffer chains for large packets, while smaller values conserve memory for environments handling mostly small packets.

```
set vpp settings resource-allocation buffers data-size <value>
```

Optimal size depends on the typical packet size in your network. If unsure, use the largest MTU in your network plus overhead (for example, 128 bytes).

page-size

A memory pages type used for buffer allocation. Common values are 4K, 2M, or 1G.

Use page sizes configured in your system settings.

```
set vpp settings resource-allocation buffers page-size <value>
```

Potential Issues and Troubleshooting

Improper buffer configuration can lead to issues such as:

- Increased latency and packet loss
- Inefficient CPU utilization
- Interface initialization failures

Indicators of such issues are:

- Errors during interfaces initialization in VPP logs
- Packet drops observed in VPP statistics

To troubleshoot buffer-related issues, consider the following steps:

- Review VPP logs for errors related to buffer allocation. Look for error -5 messages.
- Tune available buffers by adjusting the `buffers-per-numa` and `data-size` parameters.

VPP Dataplane CPU Configuration

VPP can utilize multiple CPU cores for better packet processing performance. Proper CPU configuration is essential for optimal throughput and low latency.

VPP CPU assignment is handled automatically. You specify how many CPU cores VPP may use, and the system distributes them between the main thread and worker threads.

Important: Review the system configuration settings page before changing CPU settings: [VyOS Configuration for VPP](#).

If you don't configure CPU settings, VPP uses a single core for the main thread and doesn't create worker threads.

CPU Configuration Parameters

`cpu-cores`

This parameter defines the total number of CPU cores allocated to VPP.

set vpp settings resource-allocation cpu-cores <core-number>

The system automatically assigns cores using the following rules:

- The first two CPU cores are always reserved for the operating system and other services.
- The main VPP thread is assigned to the first available core after the reserved ones.
- The remaining allocated cores are used for worker threads.

For example:

- If `cpu-cores` is set to 1, VPP runs only a main thread.
- If `cpu-cores` is set to 4, VPP uses:
 - 1 core for the main thread
 - 3 cores for worker threads

Choose a value based on available hardware resources and expected traffic load. Too few cores may limit performance, while too many can negatively impact other system services.

Potential Issues and Troubleshooting

Improper CPU configuration can lead to issues such as:

- VPP underperformance when not enough cores are assigned, or kernel underperformance when too many cores are assigned to VPP.
- Resource conflicts with other processes and services.

Indicators of such issues are:

- VPP or kernel forwarding performance is lower than expected
- Degraded performance of system components or services, such as DNS, DHCP, and dynamic routing

VPP Dataplane Interfaces Configuration

Only Ethernet interfaces (physical or virtual) can be connected to the VPP dataplane. Interfaces configured here act as a bridge between VPP and the outside world, allowing VPP to send and receive network packets.

Interface Configuration Parameters

Interfaces connected to the VPP dataplane use the DPDK driver by default, providing high performance and low latency.

set vpp settings interface <interface-name>

Some network interface cards (NICs) may not be compatible with the DPDK driver.

DPDK interface options

This section shows how to configure DPDK-specific settings for an interface.

set vpp settings interface <interface-name> num-rx-queues <value>

Specifies the number of receive queues for the interface. More queues improve performance on multi-core systems by allowing parallel processing of incoming packets. Each queue is assigned to a separate CPU core.

set vpp settings interface <interface-name> num-tx-queues <value>

Specifies the number of transmit queues for the interface. Similar to receive queues, more transmit queues improve performance by enabling parallel processing of outgoing packets. By default, the VPP Dataplane has one TX queue per enabled CPU worker, or a single queue if no workers are configured.

See also:

VPP Dataplane CPU Configuration

set vpp settings interface <interface-name> num-rx-desc <value>

Defines the size of each receive queue. Larger queue sizes accommodate bursts of incoming traffic and reduce the likelihood of packet drops during high traffic periods.

set vpp settings interface <interface-name> num-tx-desc <value>

Defines the size of each transmit queue. Larger sizes help manage bursts of outgoing traffic more effectively.

Global Interface Parameters

interface-rx-mode

The `interface-rx-mode` parameter defines how VPP handles incoming packets on interfaces. There are several modes available, each with its own advantages and use cases:

- **interrupt:** In this mode, VPP relies on hardware interrupts to notify it of incoming packets. This mode suits low to moderate traffic loads and reduces CPU usage during idle periods. It is not recommended for low-latency processing. Some NICs may not support this mode.
- **polling:** In polling mode, VPP continuously checks the interface for incoming packets. This mode is ideal for high-throughput scenarios where low latency is critical, as it minimizes packet waiting time. However, it can increase CPU usage, especially during low traffic periods, as the polling process is always active.
- **adaptive:** Adaptive mode combines the benefits of interrupt and polling modes. VPP starts in interrupt mode and switches to polling mode when traffic load increases.

```
set vpp settings interface-rx-mode <mode>
```

Choose an rx-mode based on expected traffic patterns and performance requirements of your network.

Potential Issues and Troubleshooting

Improper interface configuration can lead to issues such as:

- Failure to initialize the interface
- Poor performance due to suboptimal driver selection or settings

Indicators of such issues are:

- Failed commits after adding or modifying an interface settings
- Low throughput or high latency on the interface

VPP IPsec Configuration

VPP supports IPsec (Internet Protocol Security) offloading from the kernel, which speeds up cryptographic operations by leveraging VPP's high-performance packet processing capabilities.

IPsec does not require any specific configuration on VPP side. If both sources and destinations of the IPsec traffic are reachable via VPP interfaces, VPP will automatically offload the IPsec processing from the kernel. IPsec tunnels are configured in the VPN configuration section, see [IPsec General Information](#).

IPsec Configuration Parameters

enable IPsec acceleration

When VPP is used for offloading IPsec, it creates a virtual interface to connect to peers. The interface type is always 'ipsec', which is used for IPsec tunnels.

```
set vpp settings ipsec-acceleration
```

Enabling this option allows VPP to handle IPsec traffic more efficiently by offloading processing from the kernel.

netlink

VPP uses netlink to receive IPsec event messages from the kernel. Proper settings of the following parameters are crucial for ensuring that VPP can process all such messages:

```
set vpp settings lcp netlink batch-delay-ms <milliseconds>
```

This parameter specifies the delay in milliseconds between processing batch netlink messages.

```
set vpp settings lcp netlink batch-size <number>
```

This parameter specifies the maximum number of netlink messages to process in a single batch.

```
set vpp settings lcp netlink rx-buffer-size <number>
```

This parameter specifies the size of the receive buffer for netlink socket. If you expect to offload many IPsec tunnels or get frequent and intensive rekeying, you may need to increase this value.

Note: IPsec uses the same netlink parameters as LCP, so tuning them affects both LCP and IPsec processing.

Potential Issues and Troubleshooting

Improper IPsec configuration can lead to various issues, including:

- Failure to offload IPsec tunnels to VPP
- Lost IPsec event messages due to insufficient netlink buffer size or batch settings
- IPsec states or SAs are not synchronized between kernel and VPP

VPP IPv6 Configuration

VPP lets you configure resources allocated for IPv6 traffic processing independently from IPv4. This helps ensure that in networks without IPv6 traffic, resources are not wasted. If IPv6 traffic is present, especially with large routing tables, you must allocate additional resources for IPv6 processing to keep the dataplane stable.

You can configure two main resources for IPv6 traffic processing:

set vpp settings resource-allocation ipv6 hash-buckets <value>

This parameter configures the number of hash buckets used for IPv6 routing. If you have a large IPv6 routing table, you may need to increase this value to ensure efficient routing table performance and fast lookups.

set vpp settings resource-allocation ipv6 heap-size <value>

This parameter configures the heap size used for IPv6 forwarding. If you have a large IPv6 routing table, you may need to increase this value to ensure the routing table can accommodate all routes.

Potential Issues and Troubleshooting

Improper IPv6 configuration can lead to various issues, including:

- Inefficient, slow routing table lookups and traffic processing due to insufficient hash buckets
- Dataplane crashes or instability due to insufficient heap size when handling a large number of IPv6 routes
- Overall dataplane instability when handling IPv6 traffic

Consider increasing configuration values if you experience issues with IPv6 traffic processing or if you have a large IPv6 routing table.

VPP L2LEARN Configuration

When VPP dataplane connects to an L2 domain, it learns MAC addresses of devices in the domain. By default, the number of MAC addresses it can learn is limited.

You can configure the limit using the following command:

set vpp settings resource-allocation mac-limit <value>

This parameter sets the maximum number of MAC addresses that can be learned in the L2 domain. If you have many devices, you may need to increase this limit to ensure VPP learns all MAC addresses.

Potential Issues and Troubleshooting

Improper L2LEARN configuration can lead to various issues, including:

- MAC address learning failure in the L2 domain if the limit is set too low
- Increased packet loss or latency for devices that aren't learned
- Overall dataplane instability when handling L2 traffic

Consider increasing the L2LEARN limit if you experience issues with MAC address learning or if you have many devices in the L2 domain.

VPP LCP Configuration

Linux Control Plane (LCP) is a core component of VPP that lets you offload various control plane functions to the Linux kernel. LCP provides seamless integration with other VyOS components, letting you use system components like DHCP clients and routing daemons together with the VPP dataplane.

VPP integration in VyOS relies heavily on LCP. Almost all control plane functions are handled by other daemons and services, while VPP handles high-performance packet forwarding exclusively. This approach also reduces VPP management processing load, improving overall dataplane performance and stability.

VyOS integrates the kernel and VPP routing tables uniquely. By default, all routes, even those not directly connected to VPP interfaces, are imported from the kernel routing table to the VPP routing table, pointing to the kernel. This lets you forward traffic to any destination known to the kernel, even if VPP doesn't have a route to that destination.

However, in some scenarios this behavior may not be desired. For example, if you have many routes in the kernel routing table not directly connected to VPP interfaces, and you don't need forwarding between those destinations and destinations reachable via VPP, you can disable this behavior using the following command:

set vpp settings ignore-kernel-routes

Pay attention that disabling this option leads to loss of connectivity to destinations if there are no direct routes in VPP routing table.

Potential Issues and Troubleshooting

Disabling kernel route import can result in:

- Loss of connectivity to certain destinations if kernel routes are ignored
- Incomplete route synchronization between the kernel and VPP

VPP Logging Configuration

VPP logging is an important part of monitoring and troubleshooting the performance and behavior of the VPP dataplane.

VPP stores logs in two places:

- `/var/log/vpp.log` — This file contains logs related to daemon startup and logs of commands executed directly via VPP CLI. Pay attention: VyOS does not use VPP CLI for configuration, so this log will not contain any configuration changes made via VyOS CLI and will not be informative in most cases.
- System journal — contains logs related to the VPP daemon work, including errors, warnings, and informational messages. It is the main destination of logs generated by VPP.

Logging detail level can be configured via the next command:

```
set vpp settings logging default-level <level>
```

Where <level> can be one of the following:

- **emerg** (Emergency) - System is unusable.
- **alert** (Alert) - Immediate action required.
- **crit** (Critical) - Critical conditions.
- **error** (Error) - Error conditions.
- **warn** (Warning) - Warning conditions.
- **notice** (Notice) - Normal but significant.
- **info** (Informational) - Routine informational messages.
- **debug** (Debug) - Detailed debugging messages.
- **disabled** (Disabled) - Logging disabled.

It is recommended to set logging level to **debug** only for troubleshooting purposes, as it can generate a large volume of log data. For regular operation, a level of **info** or **warn** is usually sufficient.

Troubleshooting

Improper logging configuration can lead to various issues, including:

- Excessive log file sizes if the logging level is set too high (for example, **debug**).
- Missing critical information if the logging level is set too low (for example, **alert**).
- Performance degradation due to excessive logging overhead

Consider adjusting the logging level if you experience issues mentioned above.

VPP Memory Configuration

VPP heavily relies on hugepages for its memory management. Hugepages are larger memory pages that reduce the overhead of page management and improve performance for applications that require large amounts of memory, such as VPP.

VPP supports both 2MB and 1GB hugepages, but the default and most commonly used size is 2MB. The choice of hugepage size can impact performance, with larger pages generally providing better performance for memory-intensive applications.

Before configuring memory in VPP dataplane settings, you need to ensure that hugepages are enabled and properly configured on your system.

See also:

[Hugepages in VyOS Configuration for VPP](#)

To configure memory settings for VPP, you can use the following commands in the VPP CLI:

VPP uses a main heap as a central memory pool for FIB data structures entry allocations.

Efficient memory management is crucial for VPP's performance, and the main heap plays a significant role in this.

It can be configured using the following command:

```
set vpp settings resource-allocation memory main-heap-page-size <size>
```

Sets the main heap page size for VPP.


```
set vpp settings resource-allocation memory main-heap-size <size>
```

Sets the main heap size for VPP.

Physical Memory Configuration

VPP uses physical memory for packet buffers and interface operations. The `physmem` setting controls how much memory VPP can allocate for these operations.

```
set vpp settings resource-allocation memory physmem-max-size <size>
```

Sets the maximum amount of physical memory VPP can use for packet processing and interface buffers.

Default: 16GB (usually sufficient for most deployments)

You may need to modify the value for high-throughput environments with many interfaces, large packet buffers, very high packet rates, or memory-constrained systems where you need to limit VPP's memory usage.

Physmem independent of main heap size — `physmem` is for packet buffers, main heap is for routing tables.

See also:

- [Hugepages in VyOS Configuration for VPP](#)
- [VPP Buffer Configuration](#) - for controlling buffer allocation within `physmem`

Common configurations

```
# Reduce for memory-constrained systems
set vpp settings physmem max-size 4G

# Increase for high-throughput environments
set vpp settings physmem max-size 32G
```

Stats Memory Configuration

VPP uses a dedicated statistics memory segment to store runtime counters and telemetry data. This segment is used by the VPP CLI and monitoring tools to access performance and status information.

The statistics segment is allocated from hugepage memory and can be configured independently from the main heap and `physmem` settings.

You can configure statistics memory using the following commands:

```
set vpp settings resource-allocation memory stats page-size <size>
```

Sets the hugepage page size used for the statistics memory segment.

```
set vpp settings resource-allocation memory stats size <size>
```

Sets the total size of the statistics memory segment.

Increasing this value may be required in large deployments with many interfaces or enabled features that generate a high number of counters.

Statistics memory is used only for telemetry and monitoring. It does not affect packet buffer allocation or routing table memory.

Troubleshooting

Improper configuration of main heap size can lead to performance degradation or even system instability. If VPP runs out of memory in the main heap, it may crash or exhibit erratic behavior. Symptoms you may observe include:

- Increased latency or packet loss
- Crashes or restarts of VPP processes, especially during routing table population (for example, BGP session establishment)
- Error messages related to memory allocation failures

You need to tune the main heap size based on expected FIB entries. Pay attention: the same amount of routes with a single next-hop and with multiple next-hops will consume different amounts of memory.

For physmem, insufficient allocation can lead to packet drops, interface initialization failures, and overall degraded performance. Symptoms include:

- Packet drops or failures to allocate buffers
- Increased latency or jitter in packet processing
- Crashes or restarts of VPP processes under heavy load

You need to tune the physmem settings based on expected traffic patterns and interface usage. Monitor memory usage closely and adjust the configuration as needed to ensure optimal performance.

VPP Unix Dataplane Configuration

The UNIX configuration section is used to control VPP's interaction with the underlying operating system, including operations scheduling.

VPP relies on the polling mechanism to efficiently manage I/O operations and system events. By default VPP continuously polls for events, which leads to permanent 100% CPU usage by all cores assigned to VPP dataplane. This is optimal for performance, but may not be desirable in all environments, especially where power consumption is a concern or where VPP is running inside a hypervisor, especially if the VM has burstable thresholds and CPU usage limits.

To mitigate this, VPP provides a configurable polling delay that allows reducing CPU usage by introducing a delay between polling cycles. This introduces a trade-off between CPU usage and latency, as longer delays can lead to increased latency in processing events.

You can configure the polling delay using the following command in the VyOS CLI:

```
set vpp settings poll-sleep-usec <delay>
```

Sets the polling delay in microseconds. A value of 0 means no delay (default), while higher values introduce a delay between polling cycles.

Troubleshooting

Setting the polling delay too high can lead to increased latency and reduced performance, as VPP may not respond to events as quickly. Conversely, setting it too low may result in high CPU usage, which can be problematic in resource-constrained environments.

Symptoms of improper configuration may include:

- Increased latency in packet processing
- Higher CPU usage than expected

- Packets lost due to buffer overruns

If you do not need to reduce CPU usage, it is recommended to leave the polling delay at its default value of 0 for optimal performance.

If you need to reduce CPU usage, you may also consider using `interrupt` or adaptive *DPDK driver modes*, which can provide a balance between performance and resource utilization without affecting polling behavior.

12.4.2 VPP Interfaces Configuration

VPP Bonding Configuration

VPP bonding interfaces provide link aggregation capabilities by combining multiple physical interfaces into a single logical interface for increased bandwidth and redundancy. VPP bonding offers high-performance packet processing compared to traditional Linux bonding.

Basic Configuration

Creating a Bonding Interface

To create a VPP bonding interface:

```
set interfaces vpp bonding <vppbondN>
```

Create a bonding interface where <vppbondN> follows the naming convention `vppbond0`, `vppbond1`, and so on. A kernel pair interface is automatically created for the VPP bonding interface. This allows standard Linux networking tools and services to interact with the VPP bond.

Example:

```
set interfaces vpp bonding vppbond0
```

Interface Description

```
set interfaces vpp bonding <vppbondN> description <description>
```

Set a descriptive name for the bonding interface.

Example:

```
set interfaces vpp bonding vppbond0 description "Primary uplink bond"
```

Administrative Control

```
set interfaces vpp bonding <vppbondN> disable
```

Administratively disable the bonding interface. By default, interfaces are enabled.

Member Interface Configuration

Adding Member Interfaces

```
set interfaces vpp bonding <vppbondN> member interface <interface-name>
```

Add physical interfaces as members of the bond. You can add multiple interfaces to the same bond.

Example:

```
set interfaces vpp bonding vppbond0 member interface eth0
set interfaces vpp bonding vppbond0 member interface eth1
```

Note: Member interfaces must have the same speed and duplex for optimal performance. They must already be attached to VPP.

Bonding Modes

```
set interfaces vpp bonding <vppbondN> mode <mode>
```

Configure the bonding mode. Available modes:

- **802.3ad:** IEEE 802.3ad Dynamic Link Aggregation (LACP) - Default
- **active-backup:** Fault tolerant, only one slave interface active
- **broadcast:** Transmits everything on all slave interfaces
- **round-robin:** Load balance by transmitting packets in sequential order
- **xor-hash:** Distribute based on hash policy

Examples:

```
# Use LACP (recommended for switch environments)
set interfaces vpp bonding vppbond0 mode 802.3ad

# Use active-backup for simple failover
set interfaces vpp bonding vppbond0 mode active-backup
```

Hash Policies

For load balancing modes, configure how the system distributes traffic across member interfaces:

```
set interfaces vpp bonding <vppbondN> hash-policy <policy>
```

Set the transmit hash policy:

- **layer2:** Use MAC addresses to generate hash (default)
- **layer2+3:** Combine MAC addresses and IP addresses
- **layer3+4:** Combine IP addresses and port numbers

Examples:

```
# Layer 2 hashing (default)
set interfaces vpp bonding vppbond0 hash-policy layer2

# Layer 3+4 for better distribution with multiple flows
set interfaces vpp bonding vppbond0 hash-policy layer3+4
```

MAC Address Configuration

set interfaces vpp bonding <vppbondN> mac <mac-address>

Set a specific MAC address for the bonding interface.

Example:

```
set interfaces vpp bonding vppbond0 mac 00:11:22:33:44:55
```

IP Address Configuration

set interfaces vpp bonding <vppbondN> address <ip-address/prefix>

Configure IPv4 or IPv6 addresses on the kernel interface. You can assign multiple addresses.

Examples:

```
# IPv4 address
set interfaces vpp bonding vppbond0 address 192.168.1.10/24

# IPv6 address
set interfaces vpp bonding vppbond0 address 2001:db8::10/64

# Multiple addresses
set interfaces vpp bonding vppbond0 address 192.168.1.10/24
set interfaces vpp bonding vppbond0 address 10.0.0.10/8
```

MTU Configuration

set interfaces vpp bonding <vppbondN> mtu <size>

Set the Maximum Transmission Unit (MTU) for the kernel interface. The MTU must be compatible with the connected VPP interface.

Example:

```
set interfaces vpp bonding vppbond0 mtu 9000
```

Note: The MTU setting must match or be smaller than the MTU supported by the associated VPP interface.

VLAN Configuration

VPP kernel interfaces support VLAN (Virtual LAN) sub-interfaces for network segmentation.

Creating VLAN Sub-interfaces

```
set interfaces vpp bonding <vppbondN> vif <vlan-id>
```

Create a VLAN sub-interface with the specified VLAN ID (0-4094).

Example:

```
set interfaces vpp bonding vppbond0 vif 100
```

VLAN Sub-interface Configuration

VLAN sub-interfaces support the same configuration options as the parent interface:

```
set interfaces vpp bonding <vppbondN> vif <vlan-id> address <ip-address/prefix>
```

```
set interfaces vpp bonding <vppbondN> vif <vlan-id> description <description>
```

```
set interfaces vpp bonding <vppbondN> vif <vlan-id> disable
```

```
set interfaces vpp bonding <vppbondN> vif <vlan-id> mtu <size>
```

Examples:

```
# Configure VLAN 100
set interfaces vpp bonding vppbond0 vif 100 address 192.168.100.1/24
set interfaces vpp bonding vppbond0 vif 100 description "Management VLAN"
set interfaces vpp bonding vppbond0 vif 100 mtu 1500

# Configure VLAN 200
set interfaces vpp bonding vppbond0 vif 200 address 192.168.200.1/24
set interfaces vpp bonding vppbond0 vif 200 description "Guest VLAN"
```

Complete Configuration Example

Here's a complete example configuring a bonding interface with LACP:

```
# Create bonding interface
set interfaces vpp bonding vppbond0
set interfaces vpp bonding vppbond0 description "Server uplink bond"

# Configure bonding parameters
set interfaces vpp bonding vppbond0 mode 802.3ad
set interfaces vpp bonding vppbond0 hash-policy layer3+4

# Add member interfaces
set interfaces vpp bonding vppbond0 member interface eth0
set interfaces vpp bonding vppbond0 member interface eth1
```

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```
# Configure IP on kernel interface
set interfaces vpp bonding vppbond0 address 192.168.1.10/24
```

Best Practices

- Use **802.3ad mode** with LACP-capable switches for best performance and standards compliance.
- Configure **layer3+4 hash policy** for environments with multiple traffic flows.
- Ensure member interfaces have identical settings (speed, duplex, MTU).

VPP Bridge Configuration

VPP bridge interfaces provide Layer 2 switching functionality, allowing multiple interfaces to be connected at the data link layer.

VPP bridges operate as learning bridges, automatically discovering MAC addresses and building forwarding tables to efficiently switch traffic between member interfaces. This provides transparent connectivity between different network segments while maintaining the performance benefits of VPP's optimized data plane.

Supported Member Interface Types:

VPP bridges support various interface types as members:

- Physical Ethernet interfaces (managed through linux-cp)
- *VPP Bonding Configuration* - VPP bonding interfaces
- *VPP GRE Configuration* - GRE tunnel interfaces
- *VPP Loopback Interface Configuration* - Loopback interfaces (required for BVI)
- *VPP VXLAN Configuration* - VXLAN tunnel interfaces

This flexibility allows you to create complex Layer 2 topologies combining different networking technologies.

Basic Configuration

Creating a Bridge Interface

```
set interfaces vpp bridge <vppbrN>
```

Create a bridge interface where <vppbrN> follows the naming convention vppbr1, vppbr2, etc.

Note: Bridge domain vppbr0 is reserved by VPP and cannot be configured through VyOS. Start with vppbr1 for your bridge configurations.

Example:

```
set interfaces vpp bridge vppbr1
```

Interface Description

```
set interfaces vpp bridge <vppbrN> description <description>
```

Set a descriptive name for the bridge interface.

Example:

```
set interfaces vpp bridge vppbr1 description "Main campus bridge"
```

Member Interface Configuration

Adding Member Interfaces

```
set interfaces vpp bridge <vppbrN> member interface <interface-name>
```

Add an interface as a member of the bridge.

Examples:

```
# Add physical interfaces
set interfaces vpp bridge vppbr1 member interface eth0
set interfaces vpp bridge vppbr1 member interface eth1

# Add other VPP interfaces
set interfaces vpp bridge vppbr1 member interface vppbond0
set interfaces vpp bridge vppbr1 member interface vppgre1
```

Important: Bridge members can include various interface types such as:

- Physical Ethernet interfaces (eth0, eth1, etc.)
 - *VPP Bonding Configuration* - VPP bonding interfaces (vppbond0, vppbond1, etc.)
 - *VPP GRE Configuration* - GRE tunnel interfaces
 - *VPP Loopback Interface Configuration* - Loopback interfaces
 - *VPP VXLAN Configuration* - VXLAN tunnel interfaces
-

Bridge Virtual Interface (BVI)

A Bridge Virtual Interface (BVI) provides Layer 3 connectivity to a bridge domain, allowing the bridge to have an IP address and participate in routing.

Configuring BVI

```
set interfaces vpp bridge <vppbrN> member interface <loopback-interface> bvi
```

Designate a loopback interface as the Bridge Virtual Interface for the bridge domain.

Example:

```
# Create a loopback interface first
set interfaces vpp loopback vpplo1

# Add it to the bridge as BVI
set interfaces vpp bridge vppbr1 member interface vpplo1 bvi
```

Important: BVI Restrictions:

- Only loopback interfaces can be configured as BVI
- Each bridge domain can have only one BVI interface

Configuration Examples

Basic Bridge Setup

```
# Create bridge interface
set interfaces vpp bridge vppbr1
set interfaces vpp bridge vppbr1 description "Office network bridge"

# Add member interfaces
set interfaces vpp bridge vppbr1 member interface eth0
set interfaces vpp bridge vppbr1 member interface eth1
set interfaces vpp bridge vppbr1 member interface eth2
```

Bridge with BVI

```
# Create bridge and loopback for BVI
set interfaces vpp bridge vppbr2
set interfaces vpp bridge vppbr2 description "Server segment with gateway"
set interfaces vpp loopback vpplo1

# Configure bridge members
set interfaces vpp bridge vppbr2 member interface eth3
set interfaces vpp bridge vppbr2 member interface eth4
set interfaces vpp bridge vppbr2 member interface vpplo1 bvi
```

Multi-Technology Bridge

```
# Create bridge combining different interface types
set interfaces vpp bridge vppbr3
set interfaces vpp bridge vppbr3 description "Hybrid network bridge"

# Add various interface types
set interfaces vpp bridge vppbr3 member interface vppbond1
set interfaces vpp bridge vppbr3 member interface vppgre1
set interfaces vpp bridge vppbr3 member interface vppvxlan1
set interfaces vpp bridge vppbr3 member interface vpplo2 bvi
```

Integration with Kernel Interfaces

Bridge interfaces can be integrated with kernel interfaces for management and compatibility with standard Linux networking services. This is accomplished by binding a kernel interface to the Bridge Virtual Interface (BVI).

Example Integration:

```
# Create VPP bridge with member interfaces
set interfaces vpp bridge vppbr1
set interfaces vpp bridge vppbr1 member interface eth1
set interfaces vpp bridge vppbr1 member interface eth2

# Create loopback interface and configure as BVI
set interfaces vpp loopback vpplo1
set interfaces vpp bridge vppbr1 member interface vpplo1 bvi

# Bind LCP kernel interface to the BVI loopback
set interfaces vpp loopback vpplo1 address '192.0.2.1/24'
```

This configuration creates a kernel interface bound to the BVI, allowing standard Linux applications and routing daemons to interact with the VPP bridge. The kernel interface provides Layer 3 access to the bridge domain.

VPP GRE Configuration

VPP GRE interfaces provide Generic Routing Encapsulation tunneling with high-performance packet processing. GRE tunnels encapsulate various protocols within IP packets, enabling connectivity across Layer 3 networks while maintaining the performance benefits of VPP's optimized data plane.

Basic Configuration

Creating a GRE Interface

```
set interfaces vpp gre <vppgreN>
```

Create a GRE interface where <vppgreN> follows the naming convention vppgre1, vppgre2, etc.

```
set interfaces vpp gre <vppgreN> remote <address>
```

Set the tunnel remote endpoint address. Supports both IPv4 and IPv6 addresses.

```
set interfaces vpp gre <vppgreN> source-address <address>
```

Set the tunnel source address. Must match an address configured on the local system.

Basic Example:

```
set interfaces vpp gre vppgre1
set interfaces vpp gre vppgre1 remote 203.0.113.2
set interfaces vpp gre vppgre1 source-address 192.168.1.1
```

Interface Configuration

Description and Administrative Control

set interfaces vpp gre <vppgreN> description <description>

Set a descriptive name for the GRE interface.

set interfaces vpp gre <vppgreN> disable

Administratively disable the GRE interface.

Tunnel Type

set interfaces vpp gre <vppgreN> tunnel-type <type>

Set the GRE tunnel encapsulation type:

- 13 - Generic Routing Encapsulation for network layer traffic (default).
- teb - Transparent Ethernet Bridge for Layer 2 frame transport.
- erspan - Encapsulated Remote Switched Port Analyzer for traffic mirroring.

Kernel Interface Integration

LCP kernel pair interface bound to the VPP GRE interface is created automatically. This allows standard Linux networking tools and services to interact with the VPP GRE.

IP Address Configuration

set interfaces vpp gre <vppgreN> address <ip-address/prefix>

Configure IPv4 or IPv6 addresses on the kernel interface. Multiple addresses can be assigned.

Examples:

```
# IPv4 address
set interfaces vpp gre vppgre0 address 192.168.1.10/24

# IPv6 address
set interfaces vpp gre vppgre0 address 2001:db8::10/64
```

MTU Configuration

```
set interfaces vpp gre <vppgreN> mtu <size>
```

Set the Maximum Transmission Unit (MTU) for the kernel interface. The MTU must be compatible with the connected VPP interface.

Example:

```
set interfaces vpp gre vppgre0 mtu 9000
```

Note: The MTU size must not exceed the MTU size supported by the associated VPP interface.

Configuration Examples

Layer 3 GRE Tunnel

```
# IPv4 GRE tunnel
set interfaces vpp gre vppgre1
set interfaces vpp gre vppgre1 description "Site-to-site tunnel"
set interfaces vpp gre vppgre1 remote 203.0.113.10
set interfaces vpp gre vppgre1 source-address 192.168.1.1
set interfaces vpp gre vppgre1 tunnel-type l3
```

Layer 2 GRE Tunnel (TEB)

```
# Transparent Ethernet Bridge
set interfaces vpp gre vppgre2
set interfaces vpp gre vppgre2 description "L2 extension tunnel"
set interfaces vpp gre vppgre2 remote 203.0.113.20
set interfaces vpp gre vppgre2 source-address 192.168.1.1
set interfaces vpp gre vppgre2 tunnel-type teb
```

IPv6 GRE Tunnel

```
# IPv6 endpoints
set interfaces vpp gre vppgre3
set interfaces vpp gre vppgre3 remote 2001:db8::2
set interfaces vpp gre vppgre3 source-address 2001:db8::1
```

GRE with Kernel Interface

```
# GRE tunnel with management interface
set interfaces vpp gre vppgre4
set interfaces vpp gre vppgre4 remote 203.0.113.30
set interfaces vpp gre vppgre4 source-address 192.168.1.1
set interfaces vpp gre vppgre4 address 10.0.1.1/30
```

Bridge Integration

GRE interfaces can be added as members to VPP bridges for Layer 2 switching. See [VPP Bridge Configuration](#) for detailed bridge configuration.

```
# Add TEB GRE tunnel to bridge
set interfaces vpp bridge vppbr1
set interfaces vpp bridge vppbr1 member interface vppgre2
set interfaces vpp bridge vppbr1 member interface eth1
```

VPP IPIP Configuration

VPP IPIP interfaces provide IP-in-IP tunneling with high-performance packet processing. IPIP tunnels encapsulate IP packets within IP packets, creating point-to-point connections across Layer 3 networks.

Basic Configuration

Creating an IPIP Interface

```
set interfaces vpp ipip <vppipipN>
```

Create an IPIP interface where <vppipipN> follows the naming convention vppipip1, vppipip2, etc.

```
set interfaces vpp ipip <vppipipN> remote <address>
```

Set the tunnel remote endpoint address. Supports both IPv4 and IPv6 addresses.

```
set interfaces vpp ipip <vppipipN> source-address <address>
```

Set the tunnel source address. The source address must match an address configured on the local system.

Basic Example:

```
set interfaces vpp ipip vppipip1
set interfaces vpp ipip vppipip1 remote 203.0.113.2
set interfaces vpp ipip vppipip1 source-address 192.168.1.1
```

Interface Configuration

Description and Administrative Control

set interfaces vpp ipip <vppipipN> description <description>

Set a descriptive name for the IPIP interface.

set interfaces vpp ipip <vppipipN> disable

Administratively disable the IPIP interface.

Kernel Interface Integration

Kernel interface is bound to the VPP IPIP interface for management and application compatibility.

IP Address Configuration

set interfaces vpp ipip <vppipipN> address <ip-address/prefix>

Configure IPv4 or IPv6 addresses on the kernel interface. Multiple addresses can be assigned.

Examples:

```
# IPv4 address
set interfaces vpp ipip vppipip0 address 192.168.1.10/24

# IPv6 address
set interfaces vpp ipip vppipip0 address 2001:db8::10/64
```

MTU Configuration

set interfaces vpp ipip <vppipipN> mtu <size>

Set the Maximum Transmission Unit (MTU) for the kernel interface. The MTU must be compatible with the connected VPP interface.

Configuration Examples

IPv4 IPIP Tunnel

```
# Basic IPv4 IPIP tunnel
set interfaces vpp ipip vppipip1
set interfaces vpp ipip vppipip1 description "Site-to-site IPIP tunnel"
set interfaces vpp ipip vppipip1 remote 203.0.113.10
set interfaces vpp ipip vppipip1 source-address 192.168.1.1
```

IPv6 IPIP Tunnel

```
# IPv6 endpoints
set interfaces vpp ipip vppipip2
set interfaces vpp ipip vppipip2 remote 2001:db8::2
set interfaces vpp ipip vppipip2 source-address 2001:db8::1
```

IPIP with Kernel Interface

```
# IPIP tunnel with management interface
set interfaces vpp ipip vppipip3
set interfaces vpp ipip vppipip3 remote 203.0.113.30
set interfaces vpp ipip vppipip3 source-address 192.168.1.1
set interfaces vpp ipip vppipip3 address 10.0.2.1/30
```

VPP Loopback Interface Configuration

VPP loopback interfaces provide virtual interfaces that remain administratively up and are commonly used for stable addressing, routing protocols, and as Bridge Virtual Interfaces (BVI). Loopback interfaces in VPP offer high-performance virtual connectivity with optimized packet processing.

Basic Configuration

Creating a Loopback Interface

```
set interfaces vpp loopback <vpploN>
```

Create a loopback interface where <vpploN> follows the naming convention vpplo1, vpplo2, etc.

Basic Example:

```
set interfaces vpp loopback vpplo1
```

Interface Configuration

Description and Administrative Control

```
set interfaces vpp loopback <vpploN> description <description>
```

Set a descriptive name for the loopback interface.

```
set interfaces vpp loopback <vpploN> disable
```

Administratively disable the loopback interface.

Kernel Interface Integration

Kernel interface is bound to the VPP loopback interface for management and application compatibility.

IP Address Configuration

set interfaces vpp loopback <vpplon> address <ip-address/prefix>

Configure IPv4 or IPv6 addresses on the kernel interface. Multiple addresses can be assigned.

Examples:

```
# IPv4 address
set interfaces vpp loopback vppl01 address 192.168.1.10/24

# IPv6 address
set interfaces vpp loopback vppl01 address 2001:db8::10/64
```

MTU Configuration

set interfaces vpp loopback <vpplon> mtu <size>

Set the Maximum Transmission Unit (MTU) for the kernel interface. The MTU must be compatible with the connected VPP interface.

VLAN Configuration

VPP kernel interfaces support VLAN (Virtual LAN) sub-interfaces for network segmentation.

Creating VLAN Sub-interfaces

set interfaces vpp loopback <vpplon> vif <vlan-id>

Create a VLAN sub-interface with the specified VLAN ID (0-4094).

VLAN Sub-interface Configuration

VLAN sub-interfaces support the same configuration options as the parent interface:

set interfaces vpp loopback <vpplon> vif <vlan-id> address <ip-address/prefix>

set interfaces vpp loopback <vpplon> vif <vlan-id> description <description>

set interfaces vpp loopback <vpplon> vif <vlan-id> disable

set interfaces vpp loopback <vpplon> vif <vlan-id> mtu <size>

Examples:


```
# Configure VLAN 100
set interfaces vpp loopback vpplo1 vif 100 address 192.168.100.1/24
set interfaces vpp loopback vpplo1 vif 100 description "Management VLAN"
set interfaces vpp loopback vpplo1 vif 100 mtu 1500

# Configure VLAN 200
set interfaces vpp loopback vpplo1 vif 200 address 192.168.200.1/24
set interfaces vpp loopback vpplo1 vif 200 description "Guest VLAN"
```

Configuration Examples

Basic Loopback Interface

```
# Create simple loopback
set interfaces vpp loopback vpplo1
set interfaces vpp loopback vpplo1 description "Router ID interface"
```

Loopback with Kernel Interface

```
# Loopback with management access
set interfaces vpp loopback vpplo2
set interfaces vpp loopback vpplo2 description "Management loopback"
set interfaces vpp loopback vpplo2 address 10.255.255.1/32
```

Bridge Virtual Interface (BVI)

```
# Loopback as BVI for bridge
set interfaces vpp loopback vpplo3
set interfaces vpp loopback vpplo3 description "Bridge gateway interface"
set interfaces vpp bridge vppbr1
set interfaces vpp bridge vppbr1 member interface vpplo3 bvi
set interfaces vpp loopback vpplo3 address 192.168.100.1/24
```

VPP VXLAN Configuration

VPP VXLAN interfaces provide virtual extensible local area network (VXLAN) tunneling with high-performance packet processing. VXLAN extends Layer 2 domains across Layer 3 networks using UDP encapsulation, enabling scalable multi-tenant networking while leveraging VPP's optimized data plane.

Basic Configuration

Creating a VXLAN Interface

set interfaces vpp vxlan <vppvxlanN>

Create a VXLAN interface where <vppvxlanN> follows the naming convention vppvxlan1, vppvxlan2, etc.

set interfaces vpp vxlan <vppvxlanN> vni <vni>

Set the Virtual Network Identifier (VNI) for the VXLAN tunnel. Valid range is 0-16777214.

set interfaces vpp vxlan <vppvxlanN> remote <address>

Set the tunnel remote endpoint address. Supports both IPv4 and IPv6 addresses.

set interfaces vpp vxlan <vppvxlanN> source-address <address>

Set the tunnel source address. Must match an address configured on the local system.

Basic Example:

```
set interfaces vpp vxlan vppvxlan1
set interfaces vpp vxlan vppvxlan1 vni 100
set interfaces vpp vxlan vppvxlan1 remote 203.0.113.2
set interfaces vpp vxlan vppvxlan1 source-address 192.168.1.1
```

Interface Configuration

Description and Administrative Control

set interfaces vpp vxlan <vppvxlanN> description <description>

Set a descriptive name for the VXLAN interface.

set interfaces vpp vxlan <vppvxlanN> disable

Administratively disable the VXLAN interface.

Kernel Interface Integration

The kernel interface is bound to the VXLAN tunnel for management and application compatibility.

IP Address Configuration

set interfaces vpp vxlan <vppvxlanN> address <ip-address/prefix>

Configure IPv4 or IPv6 addresses on the kernel interface. Multiple addresses can be assigned.

Examples:

```
set interfaces vpp vxlan vppvxlan1 address 192.168.1.10/24
set interfaces vpp vxlan vppvxlan1 address 2001:db8::10/64
```

MTU Configuration

```
set interfaces vpp vxlan <vppvxlanN> mtu <size>
```

Set the Maximum Transmission Unit (MTU) for the kernel interface. The MTU must be compatible with the connected VPP interface.

Configuration Examples

Basic VXLAN Tunnel

```
# IPv4 VXLAN tunnel
set interfaces vpp vxlan vppvxlan1
set interfaces vpp vxlan vppvxlan1 description "Tenant A network extension"
set interfaces vpp vxlan vppvxlan1 vni 1000
set interfaces vpp vxlan vppvxlan1 remote 203.0.113.10
set interfaces vpp vxlan vppvxlan1 source-address 192.168.1.1
```

IPv6 VXLAN Tunnel

```
# IPv6 endpoints
set interfaces vpp vxlan vppvxlan2
set interfaces vpp vxlan vppvxlan2 vni 2000
set interfaces vpp vxlan vppvxlan2 remote 2001:db8::2
set interfaces vpp vxlan vppvxlan2 source-address 2001:db8::1
```

VXLAN with Kernel Interface

```
# VXLAN tunnel with management interface
set interfaces vpp vxlan vppvxlan3
set interfaces vpp vxlan vppvxlan3 vni 3000
set interfaces vpp vxlan vppvxlan3 remote 203.0.113.30
set interfaces vpp vxlan vppvxlan3 source-address 192.168.1.1
set interfaces vpp vxlan vppvxlan3 address 10.0.3.1/24
```

Bridge Integration

VXLAN interfaces are commonly used as members in VPP bridges for Layer 2 extension. See [VPP Bridge Configuration](#) for more information.

```
# Add VXLAN tunnel to bridge
set interfaces vpp bridge vppbr1
set interfaces vpp bridge vppbr1 member interface vppvxlan1
set interfaces vpp bridge vppbr1 member interface eth1
set interfaces vpp bridge vppbr1 member interface vpplo1 bvi
```

Multi-Tenant Configuration

```
# Multiple VNIs for tenant separation
set interfaces vpp vxlan vppvxlan10
set interfaces vpp vxlan vppvxlan10 description "Tenant A - Production"
set interfaces vpp vxlan vppvxlan10 vni 1001
set interfaces vpp vxlan vppvxlan10 remote 203.0.113.20
set interfaces vpp vxlan vppvxlan10 source-address 192.168.1.1

set interfaces vpp vxlan vppvxlan11
set interfaces vpp vxlan vppvxlan11 description "Tenant A - Development"
set interfaces vpp vxlan vppvxlan11 vni 1002
set interfaces vpp vxlan vppvxlan11 remote 203.0.113.21
set interfaces vpp vxlan vppvxlan11 source-address 192.168.1.1
```

VPP XConnect Configuration

VPP XConnect provides direct Layer 2 packet forwarding between two interfaces with maximum transparency and minimal overhead. XConnect creates a simple point-to-point bridge that forwards all Layer 2 packets bidirectionally without MAC learning or flooding, making it ideal for transparent connectivity scenarios.

XConnect operates as a super-transparent bridge, forwarding all frames between the connected interfaces without any packet inspection or modification. This provides the simplest possible Layer 2 forwarding with VPP's high-performance packet processing.

Comparison with Bridges

- **XConnect:** Point-to-point only, no MAC learning, maximum transparency, minimal overhead
- **Bridge:** Multi-port, MAC learning, broadcast handling, more features but higher overhead

Choose XConnect when you need simple point-to-point Layer 2 forwarding with maximum performance and transparency. Use bridges when you need multi-port switching with MAC learning and broadcast handling.

Basic Configuration

Creating an XConnect Interface

```
set interfaces vpp xconnect <vppxconN>
```

Create an XConnect interface where <vppxconN> follows the naming convention vppxcon1, vppxcon2, etc.

```
set interfaces vpp xconnect <vppxconN> member interface <interface-name>
```

Add an interface as a member of the XConnect. Exactly two member interfaces must be configured to create bidirectional forwarding.

Basic Example:

```
set interfaces vpp xconnect vppxcon1
set interfaces vpp xconnect vppxcon1 member interface eth0
set interfaces vpp xconnect vppxcon1 member interface eth1
```

This configuration creates transparent forwarding between `eth0` and `eth1`, where any packet received on either interface is immediately forwarded to the other without any processing.

Interface Configuration

```
set interfaces vpp xconnect <vppxconN> description <description>
```

Set a descriptive name for the XConnect interface.

Configuration Examples

Physical Interface XConnect

```
# Connect two physical interfaces
set interfaces vpp xconnect vppxcon1
set interfaces vpp xconnect vppxcon1 description "Transparent wire between ports"
set interfaces vpp xconnect vppxcon1 member interface eth0
set interfaces vpp xconnect vppxcon1 member interface eth1
```

This creates a transparent wire between two physical ports, effectively making them function as a single cable.

Tunnel to Physical XConnect

```
# Connect tunnel to physical interface
set interfaces vpp xconnect vppxcon2
set interfaces vpp xconnect vppxcon2 description "GRE tunnel to physical bridge"
set interfaces vpp xconnect vppxcon2 member interface vppgre1
set interfaces vpp xconnect vppxcon2 member interface eth2
```

This forwards all traffic from a GRE tunnel directly to a physical interface and vice versa.

Mixed Interface Types

```
# Connect different interface types
set interfaces vpp xconnect vppxcon3
set interfaces vpp xconnect vppxcon3 description "VXLAN to bonding bridge"
set interfaces vpp xconnect vppxcon3 member interface vppvxlan1
set interfaces vpp xconnect vppxcon3 member interface vppbond0
```

This demonstrates XConnect's flexibility in connecting various VPP interface types.

VyOS utilizes VPP (Vector Packet Processor) to provide high-performance data plane processing. While physical interfaces are typically managed through the Linux kernel using `linux-cp` (Linux Control Plane) integration, VyOS also supports creating dedicated VPP interfaces for enhanced flexibility and performance.

Why VPP Interfaces?

VPP interfaces offer several advantages:

- **Total Isolation:** VPP interfaces operate entirely within the VPP data plane, providing isolation from the Linux kernel when needed.
- **Advanced Features:** Access to VPP-specific functionality not available in standard Linux interfaces.
- **Flexible Deployment:** Some interface types are only available as VPP interfaces or may not be supported by the kernel.
- **Specific scenarios:** Not all use cases require integration with the Linux Kernel.

Integration with Kernel

VyOS provides seamless integration between VPP and kernel networking. This allows you to leverage the strengths of both approaches: create interfaces inside VPP, and access them from the Linux kernel and other services.

Features that can be enabled on VPP Dataplane:

12.4.3 VPP ACL Configuration

VPP ACLs (Access Control Lists) provide a way to filter traffic passing through VPP interfaces. They offer a high-performance packet filtering solution that can be used as a fast firewall alternative.

VyOS VPP ACL implementation supports two main types of access control lists:

- **IP ACLs** - Layer 3 filtering based on IPv4/IPv6 addresses, ports, and protocols (can be applied to both input and output directions)
- **MAC ACLs** - Layer 2 filtering based on MAC addresses and IP prefixes (can only be applied to input direction)

Structure and Components

Tags

ACL tags are named rule sets that contain one or more access control entries (ACEs). Tags provide a way to group related rules and apply them consistently across different interfaces.

- Tag names are user-defined text strings
- Each tag can contain multiple numbered rules
- Tags can be applied to interfaces in input or output direction
- Multiple tags can be applied to a single interface

Interface Application

ACL tags are applied to interfaces to control traffic flow:

- **Input direction:** Filters traffic entering the interface
- **Output direction:** Filters traffic leaving the interface

Note: Important Limitation: MAC ACLs can only be applied to the input direction of interfaces. They cannot filter outbound traffic. Use IP ACLs if you need to filter traffic in both directions.

Rule Processing

Rules within an ACL are processed in numerical order (lowest to highest). The first matching rule determines the action taken on the packet.

Available actions:

- `permit` - Allow the packet to continue
- `deny` - Drop the packet
- `permit-reflect` - Allow traffic and automatically permit return traffic

L3/IP ACLs

IP ACLs provide Layer 3 filtering capabilities based on IPv4 and IPv6 addresses, port numbers, and protocols. They support both stateless and stateful (reflexive) filtering.

Creating IP ACL Tags

IP ACL tags are created under the `vpp acl ip` configuration node:

```
set vpp acl ip tag-name <tag-name>
set vpp acl ip tag-name <tag-name> description '<description>'
```

Example:

```
set vpp acl ip tag-name 'WEB-FILTER'
set vpp acl ip tag-name 'WEB-FILTER' description 'Web server access control'
```

Adding Rules to IP ACL Tags

Rules are added to IP ACL tags with specific rule numbers:

```
set vpp acl ip tag-name <tag-name> rule <rule-number>
```

Basic IP ACL Rule Configuration

Each rule requires an action and matching criteria:

```
set vpp acl ip tag-name <tag-name> rule <rule-number> action <permit|deny|permit-reflect>
set vpp acl ip tag-name <tag-name> rule <rule-number> description '<description>'
set vpp acl ip tag-name <tag-name> rule <rule-number> protocol <protocol>
```

Actions:

- permit - Allow matching traffic
- deny - Block matching traffic
- permit-reflect - Allow outbound traffic and automatically permit return traffic

Protocols:

- all - Match all IP protocols (default)
- Or specific protocol by name, e.g. tcp, udp, icmp

Source and Destination Matching

Configure source and destination parameters:

```
# Source configuration
set vpp acl ip tag-name <tag-name> rule <rule-number> source prefix <ip-prefix>
set vpp acl ip tag-name <tag-name> rule <rule-number> source port <port-spec>

# Destination configuration
set vpp acl ip tag-name <tag-name> rule <rule-number> destination prefix <ip-prefix>
set vpp acl ip tag-name <tag-name> rule <rule-number> destination port <port-spec>
```

Prefix Specification:

- <x.x.x.x/x> - IPv4 prefix in CIDR notation
- <h:h:h:h:h:h:h/x> - IPv6 prefix in CIDR notation

Port Specification:

- <1-65535> - Single port number
- <start>-<end> - Port range (e.g., 1001-1005)

TCP Flags Matching

For TCP protocol rules, you can match specific TCP flags:

```
# Match packets with specific flags set
set vpp acl ip tag-name <tag-name> rule <rule-number> tcp-flags is-set
  ↳<ack|cwr|ecn|fin|psh|rst|syn|urg>

# Match packets without specific flags set
set vpp acl ip tag-name <tag-name> rule <rule-number> tcp-flags is-not-set
  ↳<ack|cwr|ecn|fin|psh|rst|syn|urg>
```


IP ACL Configuration Examples

Example 1: Basic Web Server ACL

```
# Create ACL for web server access
set vpp acl ip tag-name 'WEB-SERVER'
set vpp acl ip tag-name 'WEB-SERVER' description 'Web server access control'

# Allow HTTP traffic
set vpp acl ip tag-name 'WEB-SERVER' rule 10 action permit
set vpp acl ip tag-name 'WEB-SERVER' rule 10 protocol tcp
set vpp acl ip tag-name 'WEB-SERVER' rule 10 destination port 80

# Allow HTTPS traffic
set vpp acl ip tag-name 'WEB-SERVER' rule 20 action permit
set vpp acl ip tag-name 'WEB-SERVER' rule 20 protocol tcp
set vpp acl ip tag-name 'WEB-SERVER' rule 20 destination port 443

# Deny all other traffic
set vpp acl ip tag-name 'WEB-SERVER' rule 999 action deny
set vpp acl ip tag-name 'WEB-SERVER' rule 999 protocol all
```

Example 2: Network Segmentation ACL

```
# Create ACL for network segmentation
set vpp acl ip tag-name 'DMZ-FILTER'
set vpp acl ip tag-name 'DMZ-FILTER' description 'DMZ to internal network filter'

# Allow specific internal subnet access
set vpp acl ip tag-name 'DMZ-FILTER' rule 10 action permit
set vpp acl ip tag-name 'DMZ-FILTER' rule 10 destination prefix '192.168.100.0/24'
set vpp acl ip tag-name 'DMZ-FILTER' rule 10 protocol tcp
set vpp acl ip tag-name 'DMZ-FILTER' rule 10 destination port 443

# Allow DNS queries
set vpp acl ip tag-name 'DMZ-FILTER' rule 20 action permit
set vpp acl ip tag-name 'DMZ-FILTER' rule 20 destination prefix '192.168.1.10/32'
set vpp acl ip tag-name 'DMZ-FILTER' rule 20 protocol udp
set vpp acl ip tag-name 'DMZ-FILTER' rule 20 destination port 53

# Block everything else to internal networks
set vpp acl ip tag-name 'DMZ-FILTER' rule 100 action deny
set vpp acl ip tag-name 'DMZ-FILTER' rule 100 destination prefix '192.168.0.0/16'
```

Example 3: Reflexive ACL

```
# Create reflexive ACL for outbound connections
set vpp acl ip tag-name 'OUTBOUND-REFLECT'
set vpp acl ip tag-name 'OUTBOUND-REFLECT' description 'Allow outbound with return
↳traffic'

# Allow outbound HTTP/HTTPS with return traffic
set vpp acl ip tag-name 'OUTBOUND-REFLECT' rule 10 action permit-reflect
set vpp acl ip tag-name 'OUTBOUND-REFLECT' rule 10 protocol tcp
set vpp acl ip tag-name 'OUTBOUND-REFLECT' rule 10 destination port 80

set vpp acl ip tag-name 'OUTBOUND-REFLECT' rule 20 action permit-reflect
set vpp acl ip tag-name 'OUTBOUND-REFLECT' rule 20 protocol tcp
set vpp acl ip tag-name 'OUTBOUND-REFLECT' rule 20 destination port 443
```

Applying IP ACL Tags to Interfaces

IP ACL tags are applied to interfaces using the interface configuration:

```
# Apply to input direction
set vpp acl ip interface <interface> input acl-tag <number> tag-name <tag-name>

# Apply to output direction
set vpp acl ip interface <interface> output acl-tag <number> tag-name <tag-name>
```

Where:

- <interface> - Interface name (e.g., eth0, eth1)
- <number> - ACL rule number (0-4294967295) for ordering multiple ACL tags
- <tag-name> - Name of the ACL tag to apply

Multiple tags can be applied to the same interface and direction by using different ACL rule numbers.

Example:

```
# Apply web server ACL to input direction
set vpp acl ip interface eth0 input acl-tag 10 tag-name 'WEB-SERVER'

# Apply outbound reflexive ACL to output direction
set vpp acl ip interface eth1 output acl-tag 10 tag-name 'OUTBOUND-REFLECT'

# Apply multiple ACLs to the same interface and direction
set vpp acl ip interface eth0 input acl-tag 20 tag-name 'FIREWALL'
```

L2/MAC ACLs

MAC ACLs provide Layer 2 filtering capabilities based on MAC addresses and IP prefixes. They are particularly useful for controlling access at the data link layer.

Important: Direction Limitation: MAC ACLs can **only** be applied to the **input direction** of interfaces. They cannot filter outbound/output traffic. If you need bidirectional filtering, use IP ACLs instead.

Creating MAC ACL Tags

MAC ACL tags are created under the `vpp acl mac` configuration node:

```
set vpp acl mac tag-name <tag-name>
set vpp acl mac tag-name <tag-name> description '<description>'
```

Example:

```
set vpp acl mac tag-name 'MAC-FILTER'
set vpp acl mac tag-name 'MAC-FILTER' description 'Layer 2 MAC address filtering'
```

Adding Rules to MAC ACL Tags

Rules are added to MAC ACL tags with specific rule numbers:

```
set vpp acl mac tag-name <tag-name> rule <rule-number>
```

Basic MAC ACL Rule Configuration

Each rule requires an action and matching criteria:

```
set vpp acl mac tag-name <tag-name> rule <rule-number> action <permit|deny>
set vpp acl mac tag-name <tag-name> rule <rule-number> description '<description>'
```

Actions:

- `permit` - Allow matching traffic
- `deny` - Block matching traffic

Note: MAC ACLs do not support the `permit-reflect` action available in IP ACLs.

MAC Address Matching

Configure MAC address matching criteria:

```
set vpp acl mac tag-name <tag-name> rule <rule-number> mac-address <mac-address>
set vpp acl mac tag-name <tag-name> rule <rule-number> mac-mask <mac-mask>
```

MAC Address Specification:

- mac-address - Source MAC address to match (format: xx:xx:xx:xx:xx:xx)
- mac-mask - MAC address mask (default: ff:ff:ff:ff:ff:ff for exact match)

The MAC mask allows for partial MAC address matching. For example: - ff:ff:ff:00:00:00 matches the first 3 octets (OUI) - ff:ff:ff:ff:ff:ff matches the complete MAC address (default)

IP Prefix Matching

Configure IP prefix matching for the source:

```
set vpp acl mac tag-name <tag-name> rule <rule-number> prefix <ip-prefix>
```

Prefix Specification:

- Supports both IPv4 and IPv6 prefixes in CIDR notation
- Examples: 192.168.1.0/24, 10.0.0.0/8, 2001:db8::/32

MAC ACL Configuration Examples

Example 1: Device Whitelist

```
# Create MAC ACL for device whitelisting
set vpp acl mac tag-name 'DEVICE-WHITELIST'
set vpp acl mac tag-name 'DEVICE-WHITELIST' description 'Allow only approved devices'

# Allow specific workstation
set vpp acl mac tag-name 'DEVICE-WHITELIST' rule 10 action permit
set vpp acl mac tag-name 'DEVICE-WHITELIST' rule 10 mac-address '00:1b:21:12:34:56'
set vpp acl mac tag-name 'DEVICE-WHITELIST' rule 10 prefix '192.168.1.100/32'
set vpp acl mac tag-name 'DEVICE-WHITELIST' rule 10 description 'Admin workstation'

# Allow specific server
set vpp acl mac tag-name 'DEVICE-WHITELIST' rule 20 action permit
set vpp acl mac tag-name 'DEVICE-WHITELIST' rule 20 mac-address '00:1b:21:78:90:ab'
set vpp acl mac tag-name 'DEVICE-WHITELIST' rule 20 prefix '192.168.1.10/32'
set vpp acl mac tag-name 'DEVICE-WHITELIST' rule 20 description 'Web server'

# Deny everything else
set vpp acl mac tag-name 'DEVICE-WHITELIST' rule 999 action deny
set vpp acl mac tag-name 'DEVICE-WHITELIST' rule 999 mac-address '00:00:00:00:00:00'
set vpp acl mac tag-name 'DEVICE-WHITELIST' rule 999 mac-mask '00:00:00:00:00:00'
```

Example 2: Vendor-Based Filtering

```
# Create MAC ACL for vendor-based filtering
set vpp acl mac tag-name 'VENDOR-FILTER'
set vpp acl mac tag-name 'VENDOR-FILTER' description 'Filter by MAC vendor OUI'

# Deny Realtek devices (OUI: 00:e0:4c)
set vpp acl mac tag-name 'VENDOR-FILTER' rule 10 action deny
set vpp acl mac tag-name 'VENDOR-FILTER' rule 10 mac-address '00:e0:4c:00:00:00'
set vpp acl mac tag-name 'VENDOR-FILTER' rule 10 mac-mask 'ff:ff:ff:00:00:00'
set vpp acl mac tag-name 'VENDOR-FILTER' rule 10 description 'Block Realtek devices'

# Allow all other devices
set vpp acl mac tag-name 'VENDOR-FILTER' rule 100 action permit
set vpp acl mac tag-name 'VENDOR-FILTER' rule 100 mac-address '00:00:00:00:00:00'
set vpp acl mac tag-name 'VENDOR-FILTER' rule 100 mac-mask '00:00:00:00:00:00'
set vpp acl mac tag-name 'VENDOR-FILTER' rule 100 description 'Allow all other vendors'
```

Example 3: Network Segmentation by MAC

```
# Create MAC ACL for network segmentation
set vpp acl mac tag-name 'SEGMENT-FILTER'
set vpp acl mac tag-name 'SEGMENT-FILTER' description 'Segment networks by MAC/IP binding
→ '

# Allow management VLAN devices
set vpp acl mac tag-name 'SEGMENT-FILTER' rule 10 action permit
set vpp acl mac tag-name 'SEGMENT-FILTER' rule 10 mac-address '02:01:00:00:00:00'
set vpp acl mac tag-name 'SEGMENT-FILTER' rule 10 mac-mask 'ff:ff:00:00:00:00'
set vpp acl mac tag-name 'SEGMENT-FILTER' rule 10 prefix '10.1.0.0/16'
set vpp acl mac tag-name 'SEGMENT-FILTER' rule 10 description 'Management VLAN'

# Allow user VLAN devices
set vpp acl mac tag-name 'SEGMENT-FILTER' rule 20 action permit
set vpp acl mac tag-name 'SEGMENT-FILTER' rule 20 mac-address '02:02:00:00:00:00'
set vpp acl mac tag-name 'SEGMENT-FILTER' rule 20 mac-mask 'ff:ff:00:00:00:00'
set vpp acl mac tag-name 'SEGMENT-FILTER' rule 20 prefix '10.2.0.0/16'
set vpp acl mac tag-name 'SEGMENT-FILTER' rule 20 description 'User VLAN'
```

Applying MAC ACL Tags to Interfaces

MAC ACL tags can only be applied to the input direction of interfaces:

```
set vpp acl mac interface <interface> tag-name <tag-name>
```

Note: Syntax Difference: Unlike IP ACLs, MAC ACL interface application does not use the `acl-tag <number>` structure since only single MAC ACLs can be applied.

Warning: Unlike IP ACLs, MAC ACLs do **not** support output direction filtering. There is no output option available for MAC ACL interface application.

Example:

```
# Apply MAC filtering to interface input
set vpp acl mac interface eth0 tag-name 'MAC-FILTER'
set vpp acl mac interface eth1 tag-name 'DEVICE-WHITELIST'
```

Configuration Best Practices

Rule Ordering

- **Number rules strategically:** Use gaps between rule numbers (10, 20, 30) to allow for future insertions
- **Place specific rules first:** More specific matches should have lower rule numbers
- **End with catch-all:** Always include a final rule that matches all traffic with explicit action
- **Document rules:** Use descriptions for complex rules to aid troubleshooting

Performance Considerations

- **Minimize rule count:** Fewer rules generally mean better performance
- **Use appropriate ACL type:** Use MAC ACLs for Layer 2/3 filtering, IP ACLs for Layer 3/4 filtering
- **Consider direction limitations:** Remember that MAC ACLs only work on input traffic; use IP ACLs for filtering in both directions
- **Combine related rules:** Group similar filtering requirements into single ACL tags
- **Apply strategically:** Apply ACLs at ingress points where possible to minimize processing

Troubleshooting

Common Issues

- **ACL not taking effect:**
 - Verify ACL is applied to correct interface and direction
 - Check rule numbering and order
 - Ensure interface is properly configured in VPP
- **Performance degradation:**
 - Review ACL complexity and rule count
 - Consider consolidating rules
 - Check for unnecessary broad matches
- **Traffic blocked unexpectedly:**
 - Review rule order (first match wins)

- Check for overly restrictive rules
- Verify protocol and port specifications

Verification Commands

Use these commands to verify ACL configuration and operation:

```
# Show VPP ACL configuration
show configuration commands | grep "vpp acl"

# Show VPP interface configuration
show configuration commands | grep "vpp acl.*interface"

# View commit history for ACL changes
show configuration commit-revisions | grep -A5 -B5 "vpp acl"
```

Operational Commands

VyOS provides several operational commands to monitor and troubleshoot VPP ACL configurations and their status.

Viewing All ACLs

Display all configured ACLs (both IP and MAC):

show vpp acl

This command shows a summary of all configured ACL tags with their rules, displaying both IP ACLs and MAC ACLs in a tabular format. Example output:

```
-----
IP ACL "tag-name WEB-SERVER" acl_index 0

Rule  Action    Src prefix    Src port    Dst prefix    Dst port    Proto  TCP flags
↪set      TCP flags not set
-----
↪-----
    10  permit    0.0.0.0/0    0-65535    0.0.0.0/0    80          6
    20  permit    0.0.0.0/0    0-65535    0.0.0.0/0    443         6
    999 deny      0.0.0.0/0    0-65535    0.0.0.0/0    0-65535     0

-----
MACIP ACL "tag-name VENDOR-FILTER" acl_index 0

Rule  Action    IP prefix    MAC address    MAC mask
-----
    10  deny      0.0.0.0/0    00:e0:4c:00:00:00  ff:ff:ff:00:00:00
   100  permit    0.0.0.0/0    00:00:00:00:00:00  00:00:00:00:00:00
```

IP ACL Commands

View all IP ACLs:

```
show vpp acl ip
```

View IP ACL interface assignments:

```
show vpp acl ip interface
```

Example output:

Interface	Input ACLs	Output ACLs
eth1	WEB-SERVER	

View specific IP ACL by tag name:

```
show vpp acl ip tag-name <tag-name>
```

Example:

```
vyos@vyos:~$ show vpp acl ip tag-name WEB-SERVER

-----
IP ACL "tag-name WEB-SERVER" acl_index 0

  Rule  Action  Src prefix  Src port  Dst prefix  Dst port  Proto  TCP flags
  ↪set   TCP flags not set

-----
  ↪-----
    10  permit  0.0.0.0/0  0-65535  0.0.0.0/0  80        6
    20  permit  0.0.0.0/0  0-65535  0.0.0.0/0  443       6
    999 deny    0.0.0.0/0  0-65535  0.0.0.0/0  0-65535   0
```

MAC ACL Commands

View all MAC ACLs:

```
show vpp acl mac
```

View MAC ACL interface assignments:

```
show vpp acl mac interface
```

Example output:

Interface	ACL
eth0	VENDOR-FILTER

View specific MAC ACL by tag name:

```
show vpp acl mac tag-name <tag-name>
```

Example:


```
vyos@vyos:~$ show vpp acl mac tag-name VENDOR-FILTER
```

```
-----
MACIP ACL "tag-name VENDOR-FILTER" acl_index 0
```

Rule	Action	IP prefix	MAC address	MAC mask
10	deny	0.0.0.0/0	00:e0:4c:00:00:00	ff:ff:ff:00:00:00
100	permit	0.0.0.0/0	00:00:00:00:00:00	00:00:00:00:00:00

Understanding Command Output

IP ACL Output Fields:

- **Rule:** Rule number within the ACL
- **Action:** permit, deny, or permit-reflect
- **Src prefix:** Source IP prefix (0.0.0.0/0 = any source)
- **Src port:** Source port range (0-65535 = any port)
- **Dst prefix:** Destination IP prefix
- **Dst port:** Destination port or port range
- **Proto:** IP protocol number (6=TCP, 17=UDP, 1=ICMP, 0=any)
- **TCP flags set:** Required TCP flags (for TCP protocol)
- **TCP flags not set:** Prohibited TCP flags (for TCP protocol)

MAC ACL Output Fields:

- **Rule:** Rule number within the ACL
- **Action:** permit or deny
- **IP prefix:** Source IP prefix constraint
- **MAC address:** Source MAC address to match
- **MAC mask:** MAC address mask for partial matching

Interface Assignment Output:

- Shows which interfaces have ACLs applied
- **Input ACLs:** ACL tags applied to incoming traffic
- **Output ACLs:** ACL tags applied to outgoing traffic (IP ACLs only)
- **ACL:** MAC ACL tag applied to interface (input only)

12.4.4 VPP IPFIX Configuration

VPP IPFIX in VyOS allows monitoring and exporting network traffic flows for analytics, security, and accounting. IPFIX works with the VPP (Vector Packet Processing) backend to provide high-performance flow tracking.

Overview

VyOS integrates VPP for high-performance packet processing. IPFIX configuration controls how flows are monitored, exported, and which interfaces are included.

Key IPFIX Concepts

- **Active timeout:** Maximum time a flow is kept active before export.
- **Inactive timeout:** Maximum time an idle flow is kept before export.
- **Collector:** The remote host and port to which flow records are sent.
- **Flow layers:** Determines which layer information is included (12, 13, 14).
- **Interfaces:** Physical or virtual interfaces to monitor.
- **Direction:** Which traffic to monitor (rx, tx, both).
- **Flow variant:** Optional filter for IPv4 or IPv6 flows.

Configuration Options

- **active-timeout:** Duration (in seconds) after which active flows are exported.
- **inactive-timeout:** Duration (in seconds) after which idle flows are exported.
- **collector <ip> port <port>:** IP and UDP port of the IPFIX collector.
- **collector <ip> source-address <ip>:** Source address for flow export.
- **flowprobe-record <l2|l3|l4>:** Layers to include in flow records.
- **interface <interface> [direction <rx|tx|both>] [flow-variant <ipv4|ipv6>]:** Interfaces to monitor, direction of traffic, and optional flow variant filter.

Example Configuration

```
set vpp ipfix active-timeout '15'
set vpp ipfix inactive-timeout '120'
set vpp ipfix collector 192.0.2.2 port '4739'
set vpp ipfix collector 192.0.2.2 source-address '192.0.2.1'
set vpp ipfix flowprobe-record '12'
set vpp ipfix flowprobe-record '13'
set vpp ipfix flowprobe-record '14'
set vpp ipfix interface eth0
set vpp ipfix interface eth1 direction 'both'
set vpp ipfix interface eth1 flow-variant 'ipv4'
```

12.4.5 VPP IPsec Configuration

VPP Dataplane in VyOS can offload IPsec processing from the kernel. This can significantly speed up IPsec traffic handling when the necessary conditions are met.

Note: VPP IPsec implementation is not as feature-rich as Linux kernel IPsec. It supports only a subset of algorithms and modes.

Requirements

To make IPsec offloading work, following requirements must be met:

- VPP dataplane must be configured.
- VPP *IPsec settings* should be configured as needed.
- IPsec should be configured in the VPN configuration section, see *IPsec*.
- Both source and destination of the IPsec traffic must be reachable via VPP interfaces, so it can perform both encryption and decryption of the traffic.

Integration Details

VPP Dataplane offloads IPsec processing from kernel, but does not handle IPsec configuration itself. IPsec configuration management and control-plane operation, like IKE negotiation, is still done by the kernel and other daemons.

After an IPsec tunnel is configured in the kernel, VPP receives the necessary information via netlink messages and creates corresponding SAs and policies to be able to offload the traffic.

When VPP is used for offloading IPsec, it creates a virtual interface of a specific type to connect to a peer. The type of the interface can be configured using the `interface-type` parameter in the dataplane settings.

Supported IPsec Modes

VPP supports offloading IPsec connections in the following IPsec modes:

- Tunnel mode
- Transport mode

Supported Encryption and Integrity Algorithms

Warning: Since VPP dataplane is used only to offload IPsec traffic processing, algorithms mentioned below are applicable to ESP profiles in the IPsec configuration. IKE profiles are not affected by these limitations and can use any algorithms supported by the kernel.

VPP **supports** only the following **encryption algorithms**:

- AES-CBC
- AES-GCM with ICV

VPP **does not** support the following **encryption algorithms**:

- Null encryption
- AES-CTR
- AES-CCM with ICV
- Null encryption with AES-GMAC
- 3DES-EDE-CBC
- Blowfish-CBC
- Camellia-CBC
- Camellia-COUNTER
- Camellia-CCM with ICV
- Serpent-CBC
- Twofish-CBC
- CAST-CBC
- ChaCha20/Poly1305 with ICV

VPP **supports** the following **integrity algorithms**:

- MD5 HMAC
- SHA1 HMAC
- SHA2_256_128 HMAC
- SHA2_384_192 HMAC
- SHA2_512_256 HMAC

VPP **does not** support the following **integrity algorithms**:

- MD5_128 HMAC
- SHA1_160 HMAC
- SHA2_256_96 HMAC
- AES XCBC
- AES CMAC
- AES-GMAC

If you have configured ESP profiles with algorithms not supported by VPP and the traffic for such peers flows through VPP interfaces, such traffic will be dropped.

Configuration Examples

ACL for VPP IPsec Traffic

When using VPP for offloading IPsec traffic, you may need to adjust your firewall rules to allow the necessary protocols and ports. Below is an example of how to configure ACLs for VPP IPsec traffic:

```
set vpp acl ip interface <interface-name> input acl-tag 10 tag-name 'IPSEC'
set vpp acl ip tag-name IPSEC description 'Allow IPsec traffic'
set vpp acl ip tag-name IPSEC rule 10 action 'permit'
set vpp acl ip tag-name IPSEC rule 10 destination port '500'
```

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```

set vpp acl ip tag-name IPSEC rule 10 protocol 'tcp'
set vpp acl ip tag-name IPSEC rule 20 action 'permit'
set vpp acl ip tag-name IPSEC rule 20 destination port '500'
set vpp acl ip tag-name IPSEC rule 20 protocol 'udp'
set vpp acl ip tag-name IPSEC rule 30 action 'permit'
set vpp acl ip tag-name IPSEC rule 30 destination port '4500'
set vpp acl ip tag-name IPSEC rule 30 protocol 'tcp'
set vpp acl ip tag-name IPSEC rule 40 action 'permit'
set vpp acl ip tag-name IPSEC rule 40 destination port '4500'
set vpp acl ip tag-name IPSEC rule 40 protocol 'udp'
set vpp acl ip tag-name IPSEC rule 50 action 'permit'
set vpp acl ip tag-name IPSEC rule 50 protocol 'esp'

```

Pay attention to the order of the rules, as they are processed sequentially. Make sure to place IPsec-related rules before any other rules that might deny traffic to ensure that IPsec traffic is allowed.

Simple VTI-based IPsec Tunnel

On the VPP host:

```

set interfaces ethernet eth1 address '192.168.1.1/24'
set interfaces ethernet eth2 address '192.168.100.1/24'

set interfaces vti vti1
set protocols static route 192.168.200.0/24 interface vti1

set vpn ipsec authentication psk psk1 id 'peerA'
set vpn ipsec authentication psk psk1 id 'peerB'
set vpn ipsec authentication psk psk1 secret 'AB'

set vpn ipsec esp-group esp1 mode 'tunnel'
set vpn ipsec esp-group esp1 pfs 'disable'
set vpn ipsec esp-group esp1 proposal 10 encryption 'aes256'
set vpn ipsec esp-group esp1 proposal 10 hash 'sha256'
set vpn ipsec ike-group ike1 close-action 'none'
set vpn ipsec ike-group ike1 dead-peer-detection action 'clear'
set vpn ipsec ike-group ike1 proposal 10 encryption 'aes256'
set vpn ipsec ike-group ike1 proposal 10 hash 'sha256'

set vpn ipsec interface 'eth1'

set vpn ipsec site-to-site peer peerB authentication local-id 'peerA'
set vpn ipsec site-to-site peer peerB authentication mode 'pre-shared-secret'
set vpn ipsec site-to-site peer peerB authentication remote-id 'peerB'
set vpn ipsec site-to-site peer peerB connection-type 'none'
set vpn ipsec site-to-site peer peerB default-esp-group 'esp1'
set vpn ipsec site-to-site peer peerB ike-group 'ike1'
set vpn ipsec site-to-site peer peerB local-address '192.168.1.1'
set vpn ipsec site-to-site peer peerB remote-address '192.168.1.3'
set vpn ipsec site-to-site peer peerB vti bind 'vti1'
set vpn ipsec site-to-site peer peerB vti traffic-selector remote prefix '192.168.200.0/
↪24'

```

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```
set vpp settings interface eth1
set vpp settings interface eth2
set vpp settings ipsec netlink rx-buffer-size '32000'
set vpp settings lcp ignore-kernel-routes
```

Where:

- **eth1** is the interface connected to the IPsec peer.
- **eth2** is the interface connected to the local subnet, where unencrypted traffic is expected.
- **192.168.100.0/24** is the local subnet that will be accessible through the IPsec tunnel.
- **192.168.200.0/24** is the remote subnet that will be accessible through the IPsec tunnel.
- **vti1** is the VTI interface created by VPP for the IPsec tunnel.
- **peerA** and **peerB** are the identifiers for the local side and remote peer, respectively.

Note: What is important in this configuration

VPP uses only remote traffic-selector to determine what traffic should be offloaded to the IPsec tunnel.

Adding additional routes via VTI interface does not affect actual VPP IPsec operation.

Potential Issues and Troubleshooting

Improper IPsec configuration can lead to various issues, including:

- **Unidirectional traffic flow or acceleration**

If kernel has a conflicting route for the remote subnet, such route may take precedence over the policy route created for the IPsec tunnel in VPP. This may lead to unidirectional traffic flow or acceleration only in one direction. This has no security impact because traffic will still be encrypted by the kernel, but it may lead to performance degradation. To avoid this, ensure that no conflicting routes exist in the kernel routing table.

- **Conflicting with kernel routes**

If the kernel routes synchronization option is enabled, VPP will install all the routes from kernel. If you have these routes configured via VTI interfaces to the IPsec peer, they will conflict with the policy routes created for the IPsec tunnel in VPP. Consider using policy-based IPsec configuration to avoid this or *disable the kernel routes synchronization*.

- **Unsupported algorithms**

If you have configured ESP profiles with algorithms not supported by VPP and the traffic for such peers flows through VPP interfaces, such traffic will be dropped. You can check system logs for messages from VPP with `linux-cp/ipsec: Invalid/Unsupported crypto algo` or `linux-cp/ipsec: Invalid/Unsupported integ algo` line to identify such cases.

- **Connection is established but no traffic flows**

Even if you use compatible algorithms, there can be other reasons why traffic is not flowing. One of most frequent is blocking traffic between peers - that is especially common in public clouds. Make sure that TCP/UDP ports 500 and 4500 and ESP protocol are allowed between the peers. Alternatively, consider enforcing UDP encapsulation on both sides of the tunnel:

```
set vpn ipsec site-to-site peer <peer-name> force-udp-encapsulation
```

12.4.6 VPP NAT Configuration

VPP CGNAT Configuration

Carrier-grade NAT (CGNAT) is a NAT type designed for Internet Service Providers (ISPs) to manage limited pools of public IP addresses. It solves two main problems:

- Enables fair sharing of a limited number of public IP addresses among multiple customers, ensuring all have internet access without interfering with each other.
- Enables tracking and logging of public IP address usage by different customers, which is often a regulatory requirement.

CGNAT configuration is straightforward. Define the inside and outside interfaces, then create rules to manage the translation of private IP addresses to public IP addresses.

Warning: Enabling CGNAT on an interface (both inside and outside) **disables normal routing** on these interfaces and **blocks management access** to the VyOS router itself.

Ensure you have an alternative management path to the router before applying your CGNAT configuration.

Interface Configuration

Define the inside and outside interfaces. The inside interface connects to the private network, while the outside interface connects to the public network.

```
set vpp nat cgnat interface inside <inside-interface>
```

```
set vpp nat cgnat interface outside <outside-interface>
```

This is a mandatory step, as the CGNAT needs to know on which interfaces it needs to apply rules and operate.

NAT Rules Configuration

Next, you need to create the NAT rules.

```
set vpp nat cgnat rule <rule-number> description <description>
```

Add a description to the rule for easier identification.

```
set vpp nat cgnat rule <rule-number> inside-prefix <inside-prefix>
```

Specify the inside prefix (private IP range) to translate.

```
set vpp nat cgnat rule <rule-number> outside-prefix <outside-prefix>
```

Specify the outside prefix (public IP range) to use for translation.

Exclude Rules Configuration

CGNAT exclude rules are implemented as DET44 identity mappings. Matching traffic is excluded from CGNAT translation and keeps its original address/port tuple.

set vpp nat cgnat exclude rule <rule-number> description <description>

Adds a description (stored as VPP identity-mapping tag) for easier identification.

set vpp nat cgnat exclude rule <rule-number> local-address <local-address>

Sets the local IPv4 address that should be excluded from translation. This option is mandatory for each exclude rule.

set vpp nat cgnat exclude rule <rule-number> protocol <tcp|udp|icmp|all>

Matches a specific protocol. Default is all.

set vpp nat cgnat exclude rule <rule-number> local-port <1-65535>

Matches a specific local port (or ICMP identifier in case of ICMP protocol).

Important: Exclude-rule validation rules:

- local-address must be specified.
 - **protocol and local-port must either both be specified or both be omitted.**
 - **Duplicate identity mappings are not allowed (same local-address, protocol, local-port tuple).**
-

Note: A common use case for exclude rules is preserving management-plane access to the router itself (for example SSH) and local-originated services (for example DNS queries) when CGNAT is enabled.

Important: Memory Requirements

CGNAT memory usage scales with the number of internal customers.

Each 256 customers (equivalent to a /24 subnet) requires approximately **4 MB of main heap memory**. This memory maintains customer-to-port mappings and session state information.

Configure your VPP main heap size appropriately based on your expected customer count. See [VPP Memory Configuration](#) for details on adjusting main heap size.

Session Limitations

CGNAT has built-in session limitations to ensure fair resource allocation:

Each customer (internal IP address) is limited to a maximum of 1000 simultaneous sessions, even if more than 1000 ports are allocated to that customer. This limitation applies to all session types (TCP, UDP, ICMP).

Timeouts Configuration

You can adjust NAT session timers to optimize address space usage by controlling how long sessions remain active and how long they occupy IP address and port combinations.

Adjust these settings for different protocols individually:

```
set vpp nat cgnat timeout icmp <timeout-value>
set vpp nat cgnat timeout tcp-established <timeout-value>
set vpp nat cgnat timeout tcp-transitory <timeout-value>
set vpp nat cgnat timeout udp <timeout-value>
```

Example Configuration

Here is an example CGNAT configuration with these assumptions:

- Inside interface: eth2
- Outside interface: eth1
- Inside prefix: 100.64.0.0/16
- Outside prefix: 203.0.113.0/24

```
set vpp nat cgnat interface inside eth2
set vpp nat cgnat interface outside eth1
set vpp nat cgnat rule 1 description "CGNAT Rule 1"
set vpp nat cgnat rule 1 inside-prefix 100.64.0.0/16
set vpp nat cgnat rule 1 outside-prefix 203.0.113.0/24
set vpp nat cgnat exclude rule 10 description "Bypass management host"
set vpp nat cgnat exclude rule 10 local-address 100.64.0.10
set vpp nat cgnat exclude rule 20 description "Bypass subscriber DNS"
set vpp nat cgnat exclude rule 20 local-address 100.64.0.20
set vpp nat cgnat exclude rule 20 protocol udp
set vpp nat cgnat exclude rule 20 local-port 53
```

Operational Commands

Once the CGNAT is configured, you can use the following commands to monitor its status and operation:

show vpp nat cgnat interfaces

Displays the configured inside and outside interfaces.

```
vyos@vyos:~$ show vpp nat cgnat interfaces
CGNAT interfaces:
  eth2 in
  eth1 out
```

show vpp nat cgnat sessions

Display active NAT sessions. This command may produce extensive output if many sessions are active.

show vpp nat cgnat mappings

Display current NAT mappings, including inside and outside address prefixes.

```
vyos@vyos:~$ show vpp nat cgnat mappings
```

Inside	Outside	Sharing ratio	Ports per host	Sessions
100.64.0.0/16	203.0.113.0/24	256	252	0

show vpp nat cgnat exclude-rules

Displays configured CGNAT exclude rules (identity mappings).

```
vyos@vyos:~$ show vpp nat cgnat exclude-rules
```

Address	Protocol	Port	VRF	Description
100.64.0.10	all	any	0	Bypass management host
100.64.0.20	udp	53	0	Bypass subscriber DNS

Potential Issues and Troubleshooting

Configuration fails to apply with an error similar to:

```
vpp_papi.vpp_papi.VPPIOError: [Errno 2] VPP API client: read failed
```

CGNAT utilizes main heap memory and if you are trying to configure big prefixes or a large number of NAT sessions, you may run into memory allocation issues. Try to increase the main heap size in VPP configuration.

SSH/DNS Reachability After Enabling CGNAT

If SSH access to the router (or local-originated DNS queries) stops working after enabling CGNAT, traffic may be dropped by DET44 when it does not match a translation mapping.

In this case, add an exclude rule for the router local address that must bypass CGNAT translation.

```
set vpp nat cgnat exclude rule 100 local-address <router-ip>
```

Then verify:

```
show vpp nat cgnat exclude-rules
```

VPP NAT44 Configuration

NAT44 has two main use cases:

- **Source NAT (SNAT):** Enabling internet access for hosts in private networks using dynamic or static address translation.
- **Destination NAT (DNAT):** Providing external access to internal services through static port forwarding rules.

VyOS supports both dynamic translation using address pools and static mappings for predictable address translation requirements.

Configuring NAT44 involves a few steps:

1. Define the inside and outside interfaces.
2. Create NAT rules for SNAT or DNAT.

Dynamic and Static Operations

NAT44 configuration can be done in one of two ways or in both ways simultaneously:

1. Dynamically performing NAT using a pool of public IP addresses.
2. Statically mapping private IP addresses to public IP addresses.

To configure dynamic NAT, you need to define a pool of public IP addresses that will be used for translation. This offers an easy way to provide internet access to internal users.

Static rules are suitable for scenarios where you need consistent and predictable mappings between private and public IP addresses. They are also the only way to configure DNAT.

NAT Rule Processing and Traffic Flow

This section explains how different combinations of NAT rules affect traffic handling on a router. There are three possible combinations of NAT rule configurations:

1. **Dynamic NAT Only**
 - **All** traffic received on the “in” interface is processed by dynamic NAT rules without exceptions.
2. **Dynamic + Static NAT**
 - **All** traffic received on the “in” interface is first matched against static NAT rules.
 - If no match is found, it is then processed against dynamic NAT rules.
3. **Static NAT Only**
 - **All** traffic on the “in” interface is checked against static NAT rules.
 - If no match is found, the traffic is routed **without NAT**.

Important:

- If **dynamic NAT rules** are present, **all** traffic received on “in” interfaces is subject to NAT processing.
 - If **only static NAT rules** are configured, traffic that does not match any static rule is routed unchanged.
-

Interfaces Configuration

The first step in configuring NAT44 is defining which interfaces handle inside (private) and outside (public) traffic. VyOS uses these interface designations to determine the direction of translation.

Inside Interfaces

Inside interfaces connect to private networks where hosts need source NAT to access external networks.

set vpp nat nat44 interface inside <inside-interface>

Traffic flowing **from** inside interfaces gets source NAT applied, translating private source addresses to public addresses from the translation pool.

Outside Interfaces

Outside interfaces connect to public networks where external hosts may need to access internal services.

set vpp nat nat44 interface outside <outside-interface>

Traffic flowing **to** outside interfaces can trigger destination NAT based on static rules, allowing external access to internal services.

Interface Roles and Traffic Flow

Note: While VyOS uses “inside” and “outside” as established conventions, the technical definitions are:

- **Inside interface:** Interface where traffic originates that needs source NAT (SNAT)
- **Outside interface:** Interface where traffic originates that needs destination NAT (DNAT)

In complex network topologies, the same physical interface can be configured as both inside and outside to handle bidirectional NAT scenarios.

Traffic Processing:

1. **Inside → Outside (SNAT):** Private hosts accessing external networks
2. **Outside → Inside (DNAT):** External hosts accessing internal services via static rules
3. **Dynamic NAT:** Created automatically for inside→outside traffic
4. **Static NAT:** Requires explicit configuration for outside→inside traffic

Multiple Interface Support

You can configure multiple interfaces as inside or outside to support complex network topologies:

```
# Multiple inside interfaces (different private networks)
set vpp nat nat44 interface inside eth0
set vpp nat nat44 interface inside eth2

# Multiple outside interfaces (redundancy or load balancing)
set vpp nat nat44 interface outside eth1
set vpp nat nat44 interface outside eth3
```

Address Pool Configuration

Address pools define ranges of IP addresses that can be used for NAT translations. VyOS NAT44 supports two types of address pools, each serving different purposes.

Translation Pools

Translation pools are used for dynamic source NAT (SNAT). They provide a range of public IP addresses that can be dynamically assigned to private hosts when they access external networks.

```
set vpp nat nat44 address-pool translation address <ip-address | ip-address-range>
```

```
set vpp nat nat44 address-pool translation interface <interface-name>
```

Examples:

```
# Single address pool
set vpp nat nat44 address-pool translation address 203.0.113.10

# Address range pool
set vpp nat nat44 address-pool translation address 203.0.113.10-203.0.113.20

# Interface-based pool (use a first IP assigned to the interface)
set vpp nat nat44 address-pool translation interface eth1
```

Twice-NAT Pools

Twice-NAT pools are used when performing both source and destination NAT on the same traffic flow. This is particularly useful in scenarios where you need to:

- Translate both source and destination addresses
- Provide access between networks with overlapping IP ranges
- Implement advanced NAT scenarios like self-twice-nat

```
set vpp nat nat44 address-pool twice-nat address <ip-address | ip-address-range>
```

```
set vpp nat nat44 address-pool twice-nat interface <interface-name>
```

Examples:

```
# Twice-NAT pool for advanced scenarios
set vpp nat nat44 address-pool twice-nat address 192.168.100.1-192.168.100.10

# Interface-based twice-nat pool
set vpp nat nat44 address-pool twice-nat interface eth2
```

Pool Requirements

Important:

- For dynamic NAT to work, you must configure at least one **translation** pool.
 - For static rules with twice-nat options, you must configure a **twice-nat** pool.
 - Interface-based pools automatically include main (first) IP address assigned to the specified interface.
-

Pool Selection Priority

When multiple pools are configured, VyOS uses the following selection priority:

1. **Static mappings:** Always use the specific external address defined in the rule.
2. **Dynamic NAT:** Use available addresses from translation pools in the order they were configured.
3. **Twice-NAT:** Use addresses from twice-nat pools for secondary translation.

Note: As soon as you have configured interfaces and pool, the NAT44 is operational.

Static Rules Configuration

Static NAT rules provide predictable and consistent mappings between private and public IP addresses. They are essential for:

- **Destination NAT (DNAT):** Allowing external hosts to access services in the private network.
- **Server publishing:** Making internal services available from the Internet.
- **Consistent mappings:** Ensuring the same private IP always maps to the same public IP.

Unlike dynamic NAT that uses a pool of addresses, static rules create one-to-one mappings that persist until explicitly removed.

Basic Static Rule Configuration

To create a static NAT rule, you need to define the local (internal) and external (public) address mappings:

```
set vpp nat nat44 static rule <rule-number> local address <internal-ip>
set vpp nat nat44 static rule <rule-number> external address <external-ip>
```

Where:

- <rule-number> is a unique identifier for the rule
- <internal-ip> is the private IP address in your local network
- <external-ip> is the public IP address that external hosts will use

This basic configuration creates a static one-to-one mapping. Traffic from outside to the external IP will be translated to the internal IP, and vice versa.

Port-based Static Rules

For more granular control, you can create port-specific static rules. This is useful when you want to publish specific services:

```
set vpp nat nat44 static rule <rule-number> local address <internal-ip>
set vpp nat nat44 static rule <rule-number> local port <internal-port>
set vpp nat nat44 static rule <rule-number> external address <external-ip>
set vpp nat nat44 static rule <rule-number> external port <external-port>
set vpp nat nat44 static rule <rule-number> protocol <protocol>
```

Where:

- <internal-port> and <external-port> are the port numbers used by the connection.
- <protocol> specifies the protocol (tcp, udp, icmp).

Important: If you do not specify ports and protocol, the rule will apply to *all* traffic between the specified internal and external addresses.

Rules must contain either both ports and protocol, or neither.

Advanced Static Rule Options

VyOS NAT44 supports several advanced options for static rules:

Twice-NAT

Twice-NAT performs both source and destination NAT. When an external host accesses an internal service, the source IP of such a connection is translated to an address from the twice-NAT address pool.

This is practical in scenarios where internal services cannot connect to public networks, so they see such traffic as internal.

The twice-NAT option can be enabled with the following command:

```
set vpp nat nat44 static rule <rule-number> options twice-nat
```

Self Twice-NAT

Self Twice-NAT is used when a local host needs to access itself via the external address:

```
set vpp nat nat44 static rule <rule-number> options self-twice-nat
```

This option rewrites source IP addresses on packets sent only from a local address to an external address configured in a rule.

Important:

- Using `self-twice-nat` option requires you to set the interface connected to the local network as both inside and outside, because both source and destination NAT need to be applied.
 - External IP address used in static rules must belong to one of the configured translation pools.
-

Out-to-In Only

Restricts the rule to only apply to traffic from outside to inside interfaces:

```
set vpp nat nat44 static rule <rule-number> options out-to-in-only
```

This prevents the creation of sessions from the inside interface, making it a purely DNAT rule.

Force Twice-NAT Address

When using twice-nat, you can force the use of a specific IP address from the twice-nat address pool:

```
set vpp nat nat44 static rule <rule-number> options twice-nat-address <ip-address>
```

Rule Description

To document your rules, you can add a description:

```
set vpp nat nat44 static rule <rule-number> description <description>
```

Static Rules Configuration Examples

Full one-to-one NAT mapping:

```
set vpp nat nat44 static rule 100 local address 192.168.1.10
set vpp nat nat44 static rule 100 external address 203.0.113.10
set vpp nat nat44 static rule 100 description "One-to-one mapping"
```

Port-specific SSH access:

```
set vpp nat nat44 static rule 200 local address 192.168.1.20
set vpp nat nat44 static rule 200 local port 22
set vpp nat nat44 static rule 200 external address 203.0.113.10
set vpp nat nat44 static rule 200 external port 2222
set vpp nat nat44 static rule 200 protocol tcp
set vpp nat nat44 static rule 200 description "SSH access to server"
```

Twice-NAT for local service access:

```
set vpp nat nat44 static rule 300 local address 192.168.1.30
set vpp nat nat44 static rule 300 local port 80
set vpp nat nat44 static rule 300 external address 203.0.113.10
set vpp nat nat44 static rule 300 external port 80
set vpp nat nat44 static rule 300 protocol tcp
set vpp nat nat44 static rule 300 options twice-nat
set vpp nat nat44 static rule 300 description "Web service with twice-nat"
```

Note: When using twice-nat or self-twice-nat options, ensure you have configured a twice-nat address pool using:

```
set vpp nat nat44 address-pool twice-nat address <twice-nat-ip-range>
```


Exclude Rules Configuration

Exclude rules allow you to prevent specific traffic from undergoing NAT translation. This is particularly useful for:

- **Router management:** Allowing SSH access to the router itself from external networks.
- **Service bypass:** Excluding specific services from NAT processing
- **Traffic forwarding:** Allowing forwarded traffic to bypass NAT with 1-to-1 mapping.

Exclude rules take precedence over both dynamic and static NAT rules, ensuring that matching traffic bypasses NAT processing. For forwarded traffic, exclude rules create invisible 1-to-1 mappings that allow packets to pass through without NAT modifications.

Basic Exclude Rule Configuration

To create an exclude rule, you need to specify the traffic characteristics that should bypass NAT. You can configure exclude rules in two ways:

Option 1: Using local address

```
set vpp nat nat44 exclude rule <rule-number> local-address <internal-ip>
```

Option 2: Using external interface

```
set vpp nat nat44 exclude rule <rule-number> external-interface <interface-name>
```

Where:

- **<rule-number>** is a unique identifier for the exclude rule.
- **<internal-ip>** is the local IP address that should be excluded from NAT.
- **<interface-name>** is the external interface where the traffic originates.

Important: You must use either `local-address` OR `external-interface` in an exclude rule, but not both simultaneously. These options are mutually exclusive.

Port-specific Exclude Rules

For more granular control, you can exclude only specific ports and protocols. You can combine port and protocol specifications with either `local-address` or `external-interface`:

With local address:

```
set vpp nat nat44 exclude rule <rule-number> local-address <internal-ip>
```

```
set vpp nat nat44 exclude rule <rule-number> local-port <port-number>
```

```
set vpp nat nat44 exclude rule <rule-number> protocol <protocol>
```

With external interface:

```
set vpp nat nat44 exclude rule <rule-number> external-interface <interface-name>
```

```
set vpp nat nat44 exclude rule <rule-number> local-port <port-number>
```

```
set vpp nat nat44 exclude rule <rule-number> protocol <protocol>
```

Where:

- <port-number> is the specific port to exclude (1-65535)
- <protocol> can be tcp, udp, icmp, or all (default)

Rule Documentation

Add descriptions to your exclude rules for better management:

```
set vpp nat nat44 exclude rule <rule-number> description <description>
```

Exclude Rules Configuration Examples

Exclude SSH access to router:

```
# Allow external SSH access to router without NAT
set vpp nat nat44 exclude rule 10 local-address 192.168.1.1
set vpp nat nat44 exclude rule 10 local-port 22
set vpp nat nat44 exclude rule 10 protocol tcp
set vpp nat nat44 exclude rule 10 description "SSH access to router"
```

Exclude SNMP monitoring:

```
# Allow SNMP monitoring without NAT translation
set vpp nat nat44 exclude rule 20 local-port 161
set vpp nat nat44 exclude rule 20 protocol udp
set vpp nat nat44 exclude rule 20 external-interface eth1
set vpp nat nat44 exclude rule 20 description "SNMP monitoring"
```

Exclude all traffic to router management interface:

```
# Exclude all traffic to router's management IP
set vpp nat nat44 exclude rule 30 local-address 192.168.100.1
set vpp nat nat44 exclude rule 30 description "Management interface bypass"
```

Exclude all traffic from external interface:

```
# Exclude all traffic from external interface (alternative approach)
set vpp nat nat44 exclude rule 31 external-interface eth1
set vpp nat nat44 exclude rule 31 description "External interface bypass"
```

Exclude forwarded traffic for specific service:

```
# Allow external access to internal server without NAT translation
set vpp nat nat44 exclude rule 40 local-address 192.168.1.50
set vpp nat nat44 exclude rule 40 local-port 8080
set vpp nat nat44 exclude rule 40 protocol tcp
set vpp nat nat44 exclude rule 40 description "Direct access to internal service"
```

Common Use Cases

Router Administration:

Exclude rules are essential when you need to manage the router from external networks. Without exclude rules, NAT would attempt to translate the router's own traffic, potentially breaking management connections.

Service Monitoring:

Network monitoring systems often need direct access to router services. Exclude rules ensure that monitoring traffic bypasses NAT translation.

Routing Protocols:

Some routing protocols or network services may require direct communication without NAT interference.

Traffic Forwarding:

Exclude rules also work for forwarded traffic between networks. Without exclude rules, traffic from external to local networks must either match a static rule or be dropped. With exclude rules, traffic can bypass NAT processing with invisible 1-to-1 mappings.

Important: Exclude rules affect both traffic destined for the router itself and forwarded traffic flowing through the router. For forwarded traffic, exclude rules create transparent 1-to-1 mappings that allow packets to pass without NAT modifications, while from the outside perspective, the traffic appears to bypass NAT entirely.

Advanced NAT44 Settings

VyOS provides additional NAT44 settings for fine-tuning performance and behavior.

Session Timeouts

NAT44 maintains translation sessions with configurable timeout values for different protocols:

```
set vpp nat nat44 timeout icmp <seconds>
```

Set the timeout for ICMP sessions (Default: 60 seconds).

```
set vpp nat nat44 timeout tcp-established <seconds>
```

Set the timeout for established TCP connections (Default: 7440 seconds or 2 hours 4 minutes).

```
set vpp nat nat44 timeout tcp-transitory <seconds>
```

Set the timeout for transitory TCP connections (setup/teardown) (Default: 240 seconds or 4 minutes).

```
set vpp nat nat44 timeout udp <seconds>
```

Set the timeout for UDP sessions (Default: 300 seconds or 5 minutes).

Example:

```
# Customize timeouts for high-traffic environment
set vpp nat nat44 timeout tcp-established 3600
set vpp nat nat44 timeout udp 600
set vpp nat nat44 timeout icmp 30
```

Session Limits

Control the maximum number of concurrent NAT sessions:

set vpp nat nat44 session-limit <number>

Set the maximum number of NAT sessions per worker thread (Default: 64512).

This setting helps prevent memory exhaustion and ensures predictable performance under high load.

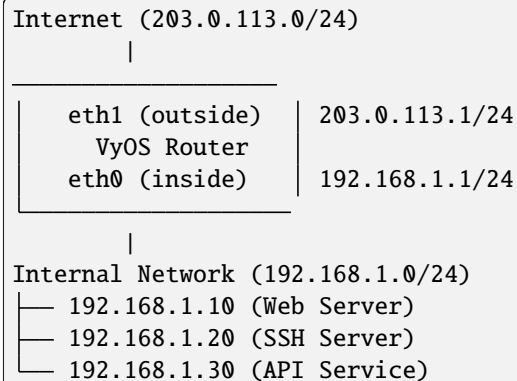
Example:

```
# Increase session limit for high-capacity deployment
set vpp nat nat44 session-limit 100000
```

Complete Configuration Example

Here's a complete example showing how to configure VyOS NAT44 for a typical network setup:

Network Topology:



Configuration:

```
# Configure interfaces
set vpp nat nat44 interface inside eth0
set vpp nat nat44 interface outside eth1

# Configure address pools
set vpp nat nat44 address-pool translation address 203.0.113.10-203.0.113.50
set vpp nat nat44 address-pool twice-nat address 203.0.113.100-203.0.113.110

# Exclude rules for router management
set vpp nat nat44 exclude rule 10 local-address 203.0.113.1
set vpp nat nat44 exclude rule 10 local-port 22
set vpp nat nat44 exclude rule 10 protocol tcp
set vpp nat nat44 exclude rule 10 description "SSH access to router"

set vpp nat nat44 exclude rule 11 local-address 203.0.113.1
set vpp nat nat44 exclude rule 11 local-port 443
set vpp nat nat44 exclude rule 11 protocol tcp
set vpp nat nat44 exclude rule 11 description "HTTPS access to router web interface"
```

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```
# Static rule for web server (HTTP)
set vpp nat nat44 static rule 100 local address 192.168.1.10
set vpp nat nat44 static rule 100 local port 80
set vpp nat nat44 static rule 100 external address 203.0.113.10
set vpp nat nat44 static rule 100 external port 80
set vpp nat nat44 static rule 100 protocol tcp
set vpp nat nat44 static rule 100 description "Public web server"

# Static rule for web server (HTTPS)
set vpp nat nat44 static rule 101 local address 192.168.1.10
set vpp nat nat44 static rule 101 local port 443
set vpp nat nat44 static rule 101 external address 203.0.113.10
set vpp nat nat44 static rule 101 external port 443
set vpp nat nat44 static rule 101 protocol tcp
set vpp nat nat44 static rule 101 description "Public web server HTTPS"

# Static rule for SSH server with custom port
set vpp nat nat44 static rule 200 local address 192.168.1.20
set vpp nat nat44 static rule 200 local port 22
set vpp nat nat44 static rule 200 external address 203.0.113.11
set vpp nat nat44 static rule 200 external port 2222
set vpp nat nat44 static rule 200 protocol tcp
set vpp nat nat44 static rule 200 description "SSH access"

# Static rule for API service (out-to-in only for security)
set vpp nat nat44 static rule 300 local address 192.168.1.30
set vpp nat nat44 static rule 300 local port 8080
set vpp nat nat44 static rule 300 external address 203.0.113.12
set vpp nat nat44 static rule 300 external port 8080
set vpp nat nat44 static rule 300 protocol tcp
set vpp nat nat44 static rule 300 options out-to-in-only
set vpp nat nat44 static rule 300 description "API service (No Internet access for it)"
```

Best Practices and Troubleshooting

Recommendations

- **Use exclude rules** for router management services like SSH
- **Use out-to-in-only for services that do not need access to external networks.**
- **Limit port ranges** in static rules to only necessary ports.
- **Document all rules** using descriptions for easier management.
- **Use non-standard ports for publishing SSH and other administrative services.**
- **Configure appropriate pool sizes based on expected concurrent connections** in your network.

Common Configuration Issues

Static rules not working:

1. Verify that the external IP address is included in an address pool
2. Check that interfaces are correctly configured as inside or outside
3. Ensure firewall rules allow the traffic

Twice-NAT not functioning:

1. Confirm twice-nat pool is configured
2. Verify static rules have the correct twice-nat option
3. Check that both translation and twice-nat pools are properly defined

Router management access issues:

1. Verify exclude rules are configured for management services
2. Check that local-address matches the router's interface IP
3. Ensure external-interface is correctly specified

Forwarded traffic from external networks not bypassing NAT:

1. Verify exclude rules are configured for the specific traffic flow
2. Check that local-address matches the destination IP in the internal network
3. Ensure protocol and port specifications match the traffic requirements

Operational Commands

Monitor NAT44 status and active connections using VyOS operational commands:

show vpp nat nat44 addresses

Display configured NAT44 address pools.

show vpp nat nat44 interfaces

Show which interfaces are configured as inside or outside for NAT44.

show vpp nat nat44 sessions

Display active NAT44 translation sessions.

show vpp nat nat44 static

Show all configured static NAT mappings.

show vpp nat nat44 summary

Display a summary of NAT44 and statistics.

VPP Dataplane in VyOS supports two types of NAT:

NAT44

This type is a classic NAT implementation where you can configure static and dynamic NAT rules. It supports both source and destination NAT. While the configuration may look a bit unusual compared to traditional NAT implementations, it provides flexibility in network configurations.

CGNAT

CGNAT is a special type of NAT44, which is highly useful when you have multiple local customers and a limited number of public IP addresses. It shares the public IP address space fairly between customers by using a combination of IP address and port number to distinguish between them.

ISPs often use this NAT type to provide internet access to customers.

It supports only source NAT.

CGNAT also supports exclude rules (identity mappings) to bypass translation for selected local addresses or protocol/port tuples.

12.4.7 VPP sFlow Configuration

VPP Dataplane in VyOS supports sFlow for traffic monitoring and analysis.

The VPP Dataplane integration works hand-in-hand with normal kernel sFlow agent, which is responsible for collecting and exporting sFlow samples. VPP itself is responsible for generating the samples.

To enable sFlow in VPP, you first need to configure the service using the same steps as for normal kernel sFlow agent, as described in [sFlow](#). Then you can enable sFlow on VPP interfaces.

Then, you need to enable sFlow on the VPP interfaces you want to monitor. This is done using the following commands:

```
set vpp sflow interface <interface-name>
```

This will enable sFlow on the specified interface. You can repeat this command for each interface you want to monitor.

Note: sFlow collects statistics only for traffic *received* on the interface. If you want to monitor traffic *sent* on the interface, you need to enable sFlow on the corresponding interface in the opposite direction.

Optionally, you can specify the number of bytes from each packet that should be included in the sFlow sample using the following command:

```
set vpp sflow header-bytes <bytes>
```

This defines the size of the packet header (in bytes) captured for each sFlow sample.

The sampling rate is configured globally under the `system sflow` section and automatically applied to VPP sFlow. This ensures consistent sampling behavior between the system and VPP, and prevents configuration conflicts.

Finally, you need to enable integration between VPP and the kernel sFlow agent using the following command:

```
set system sflow vpp
```

After this, collecting and exporting sFlow samples will be handled by the kernel sFlow agent, while VPP will generate the samples.

12.4.8 VPP Initialization

When VPP Dataplane is configured and the configuration is committed, VyOS will attempt to start VPP and initialize all interfaces assigned to it. During this process the following steps occur:

1. VyOS checks that the system meets all requirements for VPP operation. If any requirement is not met, VPP will not start and an error message will be displayed.
2. VPP is started and its initial configuration is applied.
3. All interfaces assigned to VPP are initialized and brought up.
4. A special virtual interfaces are reinstalled to the kernel with the same names as interfaces that were attached to VPP to maintain compatibility with the configuration.
5. VyOS configuration initializes those virtual interfaces, so that features that exist only in kernel dataplane continue to operate.

12.5 VPP Dataplane Troubleshooting

This page shows you how to collect diagnostic information to troubleshoot VPP dataplane issues. These techniques help you resolve problems yourself and provide support teams with the information they need.

Collecting the right diagnostic data is crucial for effective troubleshooting.

12.5.1 Packet Capture (PCAP)

Packet capture is a valuable debugging tool for analyzing network traffic and identifying issues with packet processing, routing, and filtering.

`pcap trace` in VPP captures packets at different states: received (rx), transmitted (tx), and dropped (drop).

Starting Packet Capture

Command syntax:

```
sudo vppctl pcap trace [rx] [tx] [drop] [max <n>] [intfc <interface-name|any>] [file <name>] [max-bytes-per-pkt <n>]
```

Parameters:

- `rx` - Capture received packets
- `tx` - Capture transmitted packets
- `drop` - Capture dropped packets
- `max <n>` - Depth of the local buffer. After `n` packets arrive, the buffer flushes to file. When the next `n` packets arrive, the file overwrites with new data. (default: 100)
- `intfc <interface-name|any>` - Specify an interface or use `any` for all interfaces (default: `any`)
- `file <name>` - Output filename. The PCAP file is stored in the `/tmp/` directory.
- `max-bytes-per-pkt <n>` - Maximum bytes to capture per packet (must be `>= 32`, `<= 9000`)

Examples:


```
# Start capturing tx packets with specific parameters
sudo vppctl pcap trace tx max 35 intfc eth1 file vpp_eth1.pcap

# Capture all packet types from any interface
sudo vppctl pcap trace rx tx drop max 1000 intfc any file vpp_capture.pcap max-bytes-per-
↳ pkt 128
```

Monitoring Capture Status

To check the capture status:

```
sudo vppctl pcap trace status
```

This command displays:

- Whether capture is active
- Capture parameters
- Number of packets captured
- Output file location

Stopping Packet Capture

Warning: VPP does not automatically stop packet captures. If left running, captures consume resources indefinitely. Always stop captures when you're done with them.

To stop the active packet capture:

```
sudo vppctl pcap trace off
```

Example output when stopping:

```
Write 35 packets to /tmp/vpp_eth1.pcap, and stop capture...
```

Notes:

- PCAP files are stored in the /tmp/ directory.
- Existing files are overwritten.
- If you don't specify a filename, default names are used: /tmp/rx.pcap, /tmp/tx.pcap, and /tmp/rxandtx.pcap.
- Large captures consume significant disk space—monitor available space.
- Stop captures promptly to avoid filling storage.

12.5.2 Packet Tracing

VPP packet tracing shows how packets flow through the VPP processing graph, including which nodes process each packet and what transformations occur.

Warning: Tracing generates large amounts of data, especially on high-traffic systems. Limit the number of traced packets to avoid overwhelming the system.

Basic Packet Tracing Commands

Start tracing

To start tracing packets at a specific graph node:

```
sudo vppctl trace add <input-graph-node> <pkts> [verbose]
```

- **<input-graph-node>** - Graph node name where tracing starts (for example, `dppdk-input`, `ethernet-input`, or `ip4-input`).
- **<pkts>** - Number of packets to trace (for example, 100).
- **[verbose]** - Optional flag to include detailed buffer information in the trace output.

Common node names for tracing:

- `dppdk-input`: Packets received from DPDK interfaces
- `ethernet-input`: Ethernet frame processing
- `ip4-input`: IPv4 packet processing
- `ip6-input`: IPv6 packet processing
- `ip4-lookup`: IPv4 routing table lookup
- `ip6-lookup`: IPv6 routing table lookup

View traces

After packets are traced, view the results:

```
sudo vppctl show trace [max COUNT]
```

- **[max COUNT]** - Optional limit on number of packets to display (default: all)

Clear traces

After reviewing traces, clear them to free up resources:

```
sudo vppctl clear trace
```

Example Workflow

```
# Add traces for 100 packets on dpdk-input node
sudo vppctl trace add dpdk-input 100

# Send some traffic, then view results
sudo vppctl show trace

# Clear traces for next test
sudo vppctl clear trace
```

Understanding Trace Output

Trace output shows how packets flow through VPP processing nodes:

Packet 1

```
01:00:09:508438: dpdk-input
  eth2 rx queue 0
  buffer 0x8533: current data 0, length 98, buffer-pool 0, ref-count 1, trace handle_
  ↪ 0x1000000
    ext-hdr-valid
  PKT MBUF: port 1, nb_segs 1, pkt_len 98
    buf_len 1828, data_len 98, ol_flags 0x0, data_off 128, phys_addr 0x78814d40
    packet_type 0x0 l2_len 0 l3_len 0 outer_l2_len 0 outer_l3_len 0
    rss 0x0 fdir.hi 0x0 fdir.lo 0x0
  IP4: 0c:87:6c:4e:00:01 -> 0c:de:0d:e2:00:02
  ICMP: 192.168.102.2 -> 192.168.99.3
    tos 0x00, ttl 64, length 84, checksum 0xb88d dscp CS0 ecn NON_ECN
    fragment id 0x37c5, flags DONT_FRAGMENT
  ICMP echo_request checksum 0x64e id 3024
01:00:09:508449: ethernet-input
  frame: flags 0x1, hw-if-index 2, sw-if-index 2
  IP4: 0c:87:6c:4e:00:01 -> 0c:de:0d:e2:00:02
01:00:09:508455: ip4-input
  ICMP: 192.168.102.2 -> 192.168.99.3
    tos 0x00, ttl 64, length 84, checksum 0xb88d dscp CS0 ecn NON_ECN
    fragment id 0x37c5, flags DONT_FRAGMENT
  ICMP echo_request checksum 0x64e id 3024
01:00:09:508458: ip4-sv-reassembly-feature
  [not-fragmented]
01:00:09:508460: nat-pre-in2out
  in2out next_index 2 arc_next_index 10
01:00:09:508462: nat44-ed-in2out
  NAT44_IN2OUT_ED_FAST_PATH: sw_if_index 2, next index 10, session 0, translation result
  ↪ 'success' via i2of
  i2of match: saddr 192.168.102.2 sport 3024 daddr 192.168.99.3 dport 3024 proto ICMP_
  ↪ fib_idx 0 rewrite: saddr 192.168.99.1 daddr 192.168.99.3 icmp-id 3024 txfib 0
  o2if match: saddr 192.168.99.3 sport 3024 daddr 192.168.99.1 dport 3024 proto ICMP fib_
  ↪ idx 0 rewrite: saddr 192.168.99.3 daddr 192.168.102.2 icmp-id 3024 txfib 0
  search key local 192.168.102.2:3024 remote 192.168.99.3:3024 proto ICMP fib 0 thread-
  ↪ index 0 session-index 0
```

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```

01:00:09:508469: ip4-lookup
  fib 0 dpo-idx 10 flow hash: 0x00000000
  ICMP: 192.168.99.1 -> 192.168.99.3
    tos 0x00, ttl 64, length 84, checksum 0xbb8e dscp CS0 ecn NON_ECN
    fragment id 0x37c5, flags DONT_FRAGMENT
  ICMP echo_request checksum 0x64e id 3024
01:00:09:508472: ip4-rewrite
  tx_sw_if_index 1 dpo-idx 10 : ipv4 via 192.168.99.3 eth1: mtu:1500 next:5 flags:[]
  ↳ 0ccea70400010cde0de200010800 flow hash: 0x00000000
    00000000: 0ccea70400010cde0de2000108004500005437c540003f01bc8ec0a86301c0a8
    00000020: 63030800064e0bd00d9a52c2d26800000000f4490000000000001011
01:00:09:508474: eth1-output
  eth1 flags 0x0038000d
  IP4: 0c:de:0d:e2:00:01 -> 0c:ce:a7:04:00:01
  ICMP: 192.168.99.1 -> 192.168.99.3
    tos 0x00, ttl 63, length 84, checksum 0xbc8e dscp CS0 ecn NON_ECN
    fragment id 0x37c5, flags DONT_FRAGMENT
  ICMP echo_request checksum 0x64e id 3024
01:00:09:508477: eth1-tx
  eth1 tx queue 0
  buffer 0x8533: current data 0, length 98, buffer-pool 0, ref-count 1, trace handle
  ↳ 0x1000000
    ext-hdr-valid
    natted l2-hdr-offset 0 l3-hdr-offset 14
  PKT MBUF: port 1, nb_segs 1, pkt_len 98
    buf_len 1828, data_len 98, ol_flags 0x0, data_off 128, phys_addr 0x78814d40
    packet_type 0x0 l2_len 0 l3_len 0 outer_l2_len 0 outer_l3_len 0
    rss 0x0 fdir.hi 0x0 fdir.lo 0x0
  IP4: 0c:de:0d:e2:00:01 -> 0c:ce:a7:04:00:01
  ICMP: 192.168.99.1 -> 192.168.99.3
    tos 0x00, ttl 63, length 84, checksum 0xbc8e dscp CS0 ecn NON_ECN
    fragment id 0x37c5, flags DONT_FRAGMENT
  ICMP echo_request checksum 0x64e id 3024

```

In this example, the trace shows:

- The packet is received on eth2 interface at the dpdk-input node.
- It flows through ethernet-input and ip4-input nodes.
- NAT translation occurs at the nat44-ed-in2out node, changing the source IP.
- The packet is routed through ip4-lookup and ip4-rewrite nodes.
- It transmits out of eth1 interface at the eth1-tx node.

12.5.3 Additional Diagnostic Information

When reporting issues to support teams or performing advanced troubleshooting, collect additional diagnostic information.

Before/After Traffic Analysis

Before you send traffic:

```
sudo vppctl clear hardware-interfaces
sudo vppctl clear interfaces
sudo vppctl clear error
sudo vppctl clear runtime
```

After you send traffic:

```
sudo vppctl show version verbose
sudo vppctl show hardware-interfaces
sudo vppctl show interface address
sudo vppctl show interface
sudo vppctl show runtime
sudo vppctl show error
```

Core System Information

Memory and buffer information:

```
sudo vppctl show memory api-segment stats-segment numa-heaps main-heap map verbose
sudo vppctl show buffers
sudo vppctl show physmem detail
sudo vppctl show physmem map
```

Runtime and performance data:

```
sudo vppctl show cpu
sudo vppctl show threads
sudo vppctl show runtime
sudo vppctl show node counters
```

Protocol-Specific Information

Layer 2 data (if configured):

```
sudo vppctl show l2fib
sudo vppctl show bridge-domain
```

IPv4 data (if configured):

```
sudo vppctl show ip fib
sudo vppctl show ip neighbors
```

IPv6 data (if configured):

```
sudo vppctl show ip6 fib
sudo vppctl show ip6 neighbors
```

MPLS data (if configured):

```
sudo vppctl show mpls fib
sudo vppctl show mpls tunnel
```

12.5.4 Creating Support Packages

Use the automated diagnostic collection script to gather comprehensive VPP troubleshooting information when contacting support or reporting issues.

VPP Diagnostic Collection Script

Create the diagnostic collection script:

```
#!/usr/bin/env python3
"""VyOS VPP Diagnostic Collection Script"""

import datetime
import shutil
import subprocess
import tarfile
from pathlib import Path

def run_cmd(cmd, output_file, diag_dir):
    """Run command and save output to file."""
    try:
        result = subprocess.run(
            cmd, shell=True, capture_output=True, text=True, timeout=30
        )
        content = f"Command: {cmd}\nExit code: {result.returncode}\nTimestamp: {datetime.
↪ datetime.now()}\n{'-' * 50}\n"
        if result.stdout:
            content += f"\nSTDOUT:\n{result.stdout}"
        if result.stderr:
            content += f"\nSTDERR:\n{result.stderr}"
        (diag_dir / output_file).write_text(content)
    except Exception as e:
        (diag_dir / output_file).write_text(f"Command: {cmd}\nERROR: {e}")

def collect_diagnostics():
    """Collect all VPP diagnostics and create archive."""
    timestamp = datetime.datetime.now().strftime("%Y%m%d-%H%M%S")
    diag_dir = Path.home() / f"vpp-diagnostics-{timestamp}"

    # VPP commands to collect
    commands = [
```

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```

("sudo vppctl show version verbose cmdline", "vpp-version.txt"),
("sudo vppctl show hardware-interfaces", "hardware-interfaces.txt"),
("sudo vppctl show interface address", "interface-addresses.txt"),
("sudo vppctl show interface", "interfaces.txt"),
("sudo vppctl show errors", "errors.txt"),
("sudo vppctl show runtime", "runtime.txt"),
(
    "sudo vppctl show memory api-segment stats-segment numa-heaps main-heap map_
↪verbose",
    "memory.txt",
),
("sudo vppctl show buffers", "buffers.txt"),
("sudo vppctl show physmem detail", "physmem.txt"),
("sudo vppctl show physmem map", "physmem-map.txt"),
("sudo vppctl show cpu", "cpu.txt"),
("sudo vppctl show threads", "threads.txt"),
("sudo vppctl show node counters", "node-counters.txt"),
("sudo vppctl show l2fib", "l2fib.txt"),
("sudo vppctl show bridge-domain", "bridge-domains.txt"),
("sudo vppctl show ip fib", "ip4-fib.txt"),
("sudo vppctl show ip neighbors", "ip4-neighbors.txt"),
("sudo vppctl show ip6 fib", "ip6-fib.txt"),
("sudo vppctl show ip6 neighbors", "ip6-neighbors.txt"),
("sudo vppctl show mpls fib", "mpls-fib.txt"),
("sudo vppctl show mpls tunnel", "mpls-tunnels.txt"),
("sudo vppctl show trace", "packet-traces.txt"),
]

try:
    # Create diagnostics directory
    diag_dir.mkdir(parents=True, exist_ok=True)

    # Collect VPP data
    for cmd, output_file in commands:
        run_cmd(cmd, output_file, diag_dir)

    # Collect PCAP files
    pcap_files = list(Path("/tmp").glob("*.pcap"))
    if pcap_files:
        pcap_dir = diag_dir / "pcap-files"
        pcap_dir.mkdir(exist_ok=True)
        for pcap_file in pcap_files:
            try:
                shutil.copy2(pcap_file, pcap_dir)
            except (PermissionError, OSError):
                pass

    # Create archive
    archive_name = f"vpp-diagnostics-{timestamp}.tar.gz"
    archive_path = Path.home() / archive_name

    with tarfile.open(archive_path, "w:gz") as tar:

```

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```

        tar.add(diag_dir, arcname=diag_dir.name)

    # Cleanup
    shutil.rmtree(diag_dir)

    print(f"VPP diagnostics collected: {archive_path}")
    return archive_path

except Exception as e:
    if diag_dir.exists():
        shutil.rmtree(diag_dir)
    print(f"Collection failed: {e}")
    return None

def main():
    """Main function."""
    collect_diagnostics()

if __name__ == "__main__":
    main()

```

Save this script as `/config/scripts/vpp-collect-diagnostics`

Installation and Usage

1. Make the script executable

```
sudo chmod +x /config/scripts/vpp-collect-diagnostics
```

2. Run VPP diagnostic collection

The script automatically collects all diagnostics and stores them in your home directory.

```
/config/scripts/vpp-collect-diagnostics
```

3. Generate VyOS tech-support archive separately You can also generate a tech-support archive with system-wide diagnostics:

```
generate tech-support archive
```

What the Script Collects

- **System information:** Version details, build information, command-line parameters.
- **Interface data:** Hardware interfaces, interface addresses, statistics, and configurations.
- **Performance metrics:** Runtime statistics, error counters, node counters, CPU, and thread information.
- **Memory analysis:** Memory usage (API segment, stats segment, NUMA heaps, main heap), buffers, and physical memory.
- **Layer 2 data:** L2 forwarding table (L2FIB) and bridge domain configurations.
- **IPv4 data:** IPv4 forwarding table (FIB) and IPv4 neighbor table.

- **IPv6 data:** IPv6 forwarding table (FIB) and IPv6 neighbor table.
- **MPLS data:** MPLS forwarding table (FIB) and MPLS tunnel information.
- **Packet traces:** Captured packet traces (if available).
- **Packet captures:** PCAP files from the /tmp directory (if available).

CONTRIBUTING

13.1 Build VyOS

13.1.1 Prerequisites

There are different ways you can build VyOS. Building using a *Docker* container is the easiest way because all dependencies are managed for you. Alternatively, you can set up your own build machine and run a *Native Build* build.

Note: Starting with VyOS 1.4, only source code and Debian package repositories of the rolling release (the *rolling* branch) are publicly available.

The source code and pre-built Debian package repositories of LTS releases are only available to subscription holders (customers and active community members with contributors subscriptions).

The following includes the build process for VyOS rolling release.

This will guide you through the process of building a VyOS ISO using *Docker*. This process has been tested on clean installs of Debian Bookworm.

Native Build

To build VyOS natively, you need a properly configured build host with Debian Bookworm installed.

To get started, clone the repository to your local machine:

```
$ sudo make clean
$ sudo ./build-vyos-image --architecture amd64 --build-by "j.randomhacker@vyos.io"
↳ generic
```

For required packages, refer to the *docker/Dockerfile* file in the *repository*. The *./build-vyos-image* script will also warn you if any dependencies are missing.

Docker

Installing [Docker](#) and prerequisites:

Hint: Docker versions are updated frequently. The following examples may become outdated.

```
# Add Docker's official GPG key:
sudo apt-get update
sudo apt-get install ca-certificates curl gnupg
sudo install -m 0755 -d /etc/apt/keyrings
curl -fsSL https://download.docker.com/linux/debian/gpg | sudo gpg --dearmor -o /etc/apt/
↳keyrings/docker.gpg
sudo chmod a+r /etc/apt/keyrings/docker.gpg

# Add the repository to Apt sources:
echo \
  "deb [arch=$(dpkg --print-architecture) signed-by=/etc/apt/keyrings/docker.gpg] https:/
↳/download.docker.com/linux/debian \
  $(. /etc/os-release && echo "$VERSION_CODENAME") stable" | \
  sudo tee /etc/apt/sources.list.d/docker.list > /dev/null

sudo apt-get update
sudo apt-get install docker-ce docker-ce-cli containerd.io docker-buildx-plugin docker-
↳compose-plugin
```

To use Docker without `sudo`, add your current non-root user to the `docker` group: `sudo usermod -aG docker yourusername`.

Hint: Adding a user to the `docker` group grants privileges equivalent to `root`. It is recommended to remove the non-root user from the `docker` group after building the VyOS ISO. See also [Docker as non-root](#).

Note: The build process must run on a local file system. Building on SMB or NFS shares will cause the container to fail. VirtualBox shared folders are also not supported because block device operations are not implemented.

Build Container

The container can be built by hand or by fetching the pre-built one from DockerHub. It is recommended to use the pre-built containers from the [VyOS DockerHub organisation](#). The container is built from Docker packages automatically after every commit to the `vyos-build` repository (this process may take 2-3 hours).

Note: If you use the pre-built container, it will be automatically downloaded from DockerHub if it is not found on your local machine when you build the ISO.

Dockerhub

To manually download the container from DockerHub, run:

```
$ docker pull vyos/vyos-build:rolling # For VyOS rolling release
```

Build from source

The container can also be built directly from source:

```
$ git clone -b rolling --single-branch https://github.com/vyos/vyos-build
$ cd vyos-build
$ docker build -t vyos/vyos-build:rolling docker
```

Note: VyOS switched to Debian Bookworm (12) in the rolling branch. Due to software version updates, it is recommended to use the official Docker Hub image to build VyOS ISO.

Tips and Tricks

You can create Bash aliases to easily launch the latest container per release train (rolling). Add the following to your `.bash_aliases` file:

```
alias vybld='docker pull vyos/vyos-build:rolling && docker run --rm -it \
  -v "$(pwd)":/vyos \
  -v "$HOME/.gitconfig":/etc/gitconfig \
  -v "$HOME/.bash_aliases":/home/vyos_bld/.bash_aliases \
  -v "$HOME/.bashrc":/home/vyos_bld/.bashrc \
  -w /vyos --privileged --sysctl net.ipv6.conf.lo.disable_ipv6=0 \
  -e GOSU_UID=$(id -u) -e GOSU_GID=$(id -g) \
  vyos/vyos-build:rolling bash'
```

Now you have a new alias `vybld` that launches development containers in your current working directory.

Note: Some VyOS packages (namely `vyos-1x`) come with build-time tests which verify some of the internal library calls that they work as expected. Those tests are carried out through the Python Unittest module. If you want to build the `vyos-1x` package (which is our main development package) you need to start your Docker container using the following argument: `--sysctl net.ipv6.conf.lo.disable_ipv6=0`, otherwise those tests will fail.

13.1.2 Build ISO

Now that you understand the prerequisites, you can build a VyOS ISO from source. First, fetch the latest source code from GitHub:

```
$ git clone -b rolling --single-branch https://github.com/vyos/vyos-build
```

Now you can begin a fresh VyOS ISO build. Change to the vyos-build directory and run:

```
$ cd vyos-build
$ docker run --rm -it --privileged -v $(pwd):/vyos -w /vyos vyos/vyos-build:rolling bash
```

Start the build:

```
vyos_bld@8153428c7e1f:/vyos$ sudo make clean
vyos_bld@8153428c7e1f:/vyos$ sudo ./build-vyos-image --architecture amd64 --build-by "j.
randomhacker@vyos.io" generic
```

When the build is successful, find the resulting ISO in the build directory as `live-image-[architecture].hybrid.iso`.

Customize

You can customize the ISO with the following configure options. Generate the full and current list with `./build-vyos-image --help`:

```
$ vyos_bld@8153428c7e1f:/vyos$ sudo ./build-vyos-image --help
I: Checking if packages required for VyOS image build are installed
usage: build-vyos-image [-h] [--architecture ARCHITECTURE]
[--build-by BUILD_BY] [--debian-mirror DEBIAN_MIRROR]
[--debian-security-mirror DEBIAN_SECURITY_MIRROR]
[--pbuilder-debian-mirror PBUILDER_DEBIAN_MIRROR]
[--vyos-mirror VYOS_MIRROR] [--build-type BUILD_TYPE]
[--version VERSION] [--build-comment BUILD_COMMENT] [--debug] [--dry-run]
[--custom-apt-entry CUSTOM_APT_ENTRY] [--custom-apt-key CUSTOM_APT_KEY]
[--custom-package CUSTOM_PACKAGE]
    [build_flavor]

positional arguments:
build_flavor            Build flavor

optional arguments:
-h, --help              show this help message and exit
--architecture ARCHITECTURE
                        Image target architecture (amd64 or arm64)
--build-by BUILD_BY     Builder identifier (e.g. jrandomhacker@example.net)
--debian-mirror DEBIAN_MIRROR
                        Debian repository mirror
--debian-security-mirror DEBIAN_SECURITY_MIRROR
                        Debian security updates mirror
--pbuilder-debian-mirror PBUILDER_DEBIAN_MIRROR
                        Debian repository mirror for pbuilder env bootstrap
--vyos-mirror VYOS_MIRROR
```

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```

VyOS package mirror
--build-type BUILD_TYPE
    Build type, release or development
--version VERSION      Version number (release builds only)
--build-comment BUILD_COMMENT
    Optional build comment
--debug                Enable debug output
--dry-run              Check build configuration and exit
--custom-apt-entry CUSTOM_APT_ENTRY
    Custom APT entry
--custom-apt-key CUSTOM_APT_KEY
    Custom APT key file
--custom-package CUSTOM_PACKAGE
    Custom package to install from repositories

```

ISO Build Issues

There are (rare) situations where building an ISO image is not possible at all due to a broken package feed in the background. APT is not very good at reporting the root cause of the issue. Your ISO build will likely fail with a more or less similar looking error message:

```

The following packages have unmet dependencies:
 vyos-1x : Depends: accel-ppp but it is not installable
E: Unable to correct problems, you have held broken packages.
...
make: *** [Makefile:30: iso] Error 1

```

To debug the build process and gain additional information of what could be the root cause, you need to use `chroot` to change into the build directory. This is explained in the following step by step procedure:

```
vyos_bld ece068908a5b:/vyos [rolling] # sudo chroot build/chroot /bin/bash
```

We now need to mount some required, volatile filesystems

```

(live)root@ece068908a5b:/# mount -t proc none /proc
(live)root@ece068908a5b:/# mount -t sysfs none /sys
(live)root@ece068908a5b:/# mount -t devtmpfs none /dev

```

We now are free to run any command we would like to use for debugging, e.g. re-installing the failed package after updating the repository.

```

(live)root@ece068908a5b:/# apt-get update; apt-get install vyos-1x
...
The following packages have unmet dependencies:
 vyos-1x : Depends: accel-ppp but it is not installable
E: Unable to correct problems, you have held broken packages.

```

Now it's time to fix the package mirror and rerun the last step until the package installation succeeds again!

Linux Kernel

The Linux kernel used by VyOS is heavily tied to the ISO build process. The file `data/defaults.json` hosts a JSON definition of the kernel version used `kernel_version` and the `kernel_flavor` of the kernel which represents the kernel's `LOCAL_VERSION`. Both together form the kernel version variable in the system:

```
vyos@vyos:~$ uname -r
6.1.52-amd64-vyos
```

- Accel-PPP
- Intel NIC drivers
- Intel QAT

Each of those modules holds a dependency on the kernel version and if you are lucky enough to receive an ISO build error which sounds like:

```
I: Create initramfs if it does not exist.
Extra argument '6.1.52-amd64-vyos'
...
E: config/hooks/live/17-gen_initramfs.chroot failed (exit non-zero). You should check ↵
↵ for errors.
```

The most obvious reasons could be:

- `vyos-build` repo is outdated, please `git pull` to update to the latest release kernel version from us.
- You have your own custom kernel `*.deb` packages in the `packages` folder but neglected to create all required out-of tree modules like Accel-PPP, Intel QAT or Intel NIC drivers

Building The Kernel

The kernel build is quite easy, most of the required steps can be found in the `vyos-build/packages/linux-kernel/Jenkinsfile` but we will walk you through it.

Clone the kernel source to `vyos-build/packages/linux-kernel/`:

```
$ cd vyos-build/packages/linux-kernel/
$ git clone https://git.kernel.org/pub/scm/linux/kernel/git/stable/linux.git
```

Check out the required kernel version - see `vyos-build/data/defaults.json` file (example uses kernel 4.19.146):

```
$ cd vyos-build/packages/linux-kernel/linux
$ git checkout v4.19.146
```

Now you can use the helper script `build-kernel.sh`, which completes all the necessary steps: applying required patches from the `vyos-build/packages/linux-kernel/patches` folder, copying the kernel configuration `x86_64_vyos_defconfig` to the correct location, and building the Debian packages.

Note: Building the kernel will take some time depending on the speed and quantity of your CPU/cores and disk speed. Expect 20 minutes (or even longer) on lower end hardware.

```
(18:59) vyos_bld 412374ca36b8:/vyos/vyos-build/packages/linux-kernel [rolling] # ./build-
↵kernel.sh
```

When complete, you will have kernel binary packages to use in your custom ISO build. Place all *.deb files in the vyos-build/packages folder, where the build process will use them automatically.

Firmware

If you upgrade your kernel or include new drivers you may need new firmware. This builds a new vyos-linux-firmware package using the included helper scripts.

```
$ cd vyos-build/packages/linux-kernel
$ git clone https://git.kernel.org/pub/scm/linux/kernel/git/firmware/linux-firmware.git
$ ./build-linux-firmware.sh
$ cp vyos-linux-firmware_*.deb ../
```

The script automatically detects which firmware blobs are needed based on the built drivers. If detection fails, you can manually add files to vyos-build/packages/linux-kernel/build-linux-firmware.sh:

```
ADD_FW_FILES="iwlwifi* ath11k/QCA6390/*/*.bin"
```

Building Out-Of-Tree Modules

Building the kernel is one step. You must also build required out-of-tree modules so the ABIs match. Refer to vyos-build/packages/linux-kernel/Jenkinsfile for all required modules and their versions. We show you how to build the currently required modules.

Accel-PPP

First, clone the source code and check out the appropriate version:

```
$ cd vyos-build/packages/linux-kernel
$ git clone https://github.com/accel-ppp/accel-ppp.git
```

Use the helper script and patches to build the package. Run the following command:

```
$ ./build-accel-ppp.sh
```

After compiling the packages you will find yourself the newly generated *.deb binaries in vyos-build/packages/linux-kernel from which you can copy them to the vyos-build/packages folder for inclusion during the ISO build.

Intel NIC

The Intel NIC drivers do not come from a Git repository. VyOS fetches the tarballs from a mirror and compiles them. Use the following wrapper script to build all driver modules:

```
./build-intel-drivers.sh
```

After compilation, find the generated *.deb binaries in vyos-build/packages/linux-kernel. Copy them to the vyos-build/packages folder for inclusion in the ISO build.

Intel QAT

The Intel QAT (Quick Assist Technology) drivers do not come from a Git repository. VyOS fetches the tarballs from 01.org, Intel's open-source website. Use the following wrapper script to build all driver modules:

```
$ ./build-intel-qat.sh
```

After compiling the packages you will find yourself the newly generated *.deb binaries in vyos-build/packages/linux-kernel from which you can copy them to the vyos-build/packages folder for inclusion during the ISO build.

Packages

If you are brave enough to build your own ISO image with any modified package from VyOS's GitHub organisation, this is the place for you.

Any modified package may be an altered version (e.g., vyos-1x) that you want to test before filing a pull request on GitHub.

Building an ISO with a customized package is the same as building a regular ISO image. Place your modified *.deb package inside the packages folder within vyos-build. The build process will automatically use your custom package during the ISO build.

Troubleshooting

Debian APT does not provide verbose error messages. If your ISO build fails and you suspect an APT dependencies or installation issue, you can apply this patch to increase APT verbosity during the ISO build.

```
diff --git i/scripts/live-build-config w/scripts/live-build-config
index 1b3b454..3696e4e 100755
--- i/scripts/live-build-config
+++ w/scripts/live-build-config
@@ -57,7 +57,8 @@ lb config noauto \
     --firmware-binary false \
     --updates true \
     --security true \
-    --apt-options "--yes -oAcquire::Check-Valid-Until=false" \
+    --apt-options "--yes -oAcquire::Check-Valid-Until=false -oDebug::BuildDeps=true -
+oDebug::pkgDepCache::AutoInstall=true \
+    -oDebug::pkgDepCache::Marker=true -
+oDebug::pkgProblemResolver=true -oDebug::Acquire::gpgv=true" \
     --apt-indices false
     "${@}"

```

13.1.3 Packages

VyOS comes with specific packages that cannot be found in any Debian mirror. These packages are located in the [VyOS GitHub project](#) in source format and can easily be compiled into custom Debian (*.deb) packages.

The easiest way to compile your package is with the *Docker* container mentioned earlier, as it includes all required dependencies for all VyOS related packages.

Assuming you want to build the vyos-1x package and modify it for your needs, first clone the repository from GitHub:

```
$ git clone --recurse-submodules https://github.com/vyos/vyos-1x
```

Build

Launch the Docker container and build the package:

```
# For VyOS 1.3 (equuleus, rolling)
$ docker run --rm -it --privileged -v $(pwd):/vyos -w /vyos vyos/vyos-build:rolling bash

# Change to source directory
$ cd vyos-1x

# Build DEB
$ dpkg-buildpackage -uc -us -tc -b
```

After a minute or two, the generated DEB packages are located next to the vyos-1x source directory:

```
# ls -al ../vyos-1x*.deb
-rw-r--r-- 1 vyos_bld vyos_bld 567420 Aug  3 12:01 ../vyos-1x_1.3dev0-1847-gb6dcb0a8_all.
↳ deb
-rw-r--r-- 1 vyos_bld vyos_bld  3808 Aug  3 12:01 ../vyos-1x-vmware_1.3dev0-1847-
↳ gb6dcb0a8_amd64.deb
```

Install

To test your newly created package, you can SCP it to a running VyOS instance and install the new *.deb package to replace the current one.

Install the package using the following commands:

```
vyos@vyos:~$ dpkg --install /tmp/vyos-1x_<version>_all.deb
```

You can also place the generated *.deb in your ISO build environment to include it in a custom ISO. See [Linux Kernel](#) for more information.

Warning: Any packages in the packages directory will be added to the ISO during the build, replacing upstream packages. Delete both the source directories and built DEB packages if you want to build an ISO from purely upstream packages.

13.2 Development

Learn how to contribute to VyOS.

13.2.1 Architecture overview

VyOS source code is hosted on GitHub in the VyOS organization: <https://github.com/vyos>

VyOS is composed of multiple modules spread across different repositories. Some modules contain forks of upstream packages and are periodically synced. VyOS consolidates most packages into the `vyos-1x` repository while maintaining a consistent structure. The base code is being rewritten from Perl and Bash to Python using an XML-based CLI interface definition.

VyOS ISO build scripts are hosted in the `vyos-build` repository. See the `vyos-build` repository [README.md](#) file for more information on building VyOS ISO images.

13.2.2 Contributing code

Warning: You must sign the [Contributor License Agreement](#) for your contributions to be accepted.

VyOS is open-source and welcomes patches. All submissions must adhere to these guidelines:

- Each commit addresses a single issue or feature.
- Each commit message references a [Phabricator](#) task ID (for example, T1234).
- Each commit is associated with a username and email address to identify the author (see [Configure your Git identity](#)).
- Only submit bugfixes in packages other than <https://github.com/vyos/vyos-1x>.
- Commits follow the [coding guidelines](#) outlined below.

Determining package ownership

To determine which VyOS package contains a file you want to modify, use Debian's `dpkg -S` command on your running VyOS installation.

Submitting your code

Fork the repository and submit a GitHub pull request. This is the preferred way to contribute changes to VyOS.

To fork a VyOS repository:

1. Append `/fork` to the repository URL on GitHub. For example, to fork `vyos-1x`, use: <https://github.com/vyos/vyos-1x/fork>
2. Clone your fork or add it as a remote to your local repository:
 - Clone: `git clone https://github.com/<user>/vyos-1x.git`
 - Add remote: `git remote add myfork https://github.com/<user>/vyos-1x.git`
3. Configure your Git identity:

```
git config --global user.name "J. Random Hacker"
git config --global user.email "jrhacker@example.net"
```

4. Make your changes and add files to the Git index:
 - Single file: `git add myfile`
 - Directory: `git add somedir/*`
5. Commit your changes with a meaningful headline and [Phabricator](#) reference:


```
git commit
```
6. Push to your fork and create a GitHub pull request:


```
git push
```

Alternatively, you can export commits as patches and send them to maintainers@vyos.net or attach them directly to the [Phabricator](#) task:

- Export last commit: `git format-patch`
- Export last two commits: `git format-patch -2`

13.2.3 Commit messages

For guidance on writing commit messages, review the file history with `git log path/to/file.txt`.

Every change must be associated with a task number (prefixed with **T**) and a component. If no bug report or feature request exists for your changes, create a [Phabricator](#) task first. Reference the task ID in your commit message:

- `ddclient: T1030: auto create runtime directories`
- `Jenkins: add current Git commit ID to build description`

If your pull request lacks a [Phabricator](#) reference, maintainers will request that you amend the commit message.

Writing good commit messages

Follow the format described in the [Git documentation](#) and [Chris Beams' guide](#).

Commit message format:

1. **Summary line** (50 characters recommended, 80 maximum): Include the component prefix and [Phabricator](#) reference (for example, `snmp: T1111:` or `ethernet: T2222:`). Concatenate multiple components with colons (for example, `snmp: ethernet: T3333`).
2. **Blank line**: Separate the summary from the body. This blank line is critical.
- 4) **Message body** with details:
 - Describe what changed, why, and how. This helps with `git bisect`.
 - Wrap text at 72 characters for readability with `git log` on an 80x25 terminal.
 - Reference previous commits when applicable: After commit `abcd12ef` (`"snmp: this is a headline"`) a Python import statement is missing, throwing the following exception: `ABCDEF`
- 5) **Cherry-pick option**: Always use the `-x` option when back-porting or forward-porting commits:


```
git cherry-pick -x <commit>
```

This appends (cherry picked from commit <ID>) to the commit message, making bisecting easier.

- 6) **Single responsibility:** Each commit must be self-contained. Do not fix multiple bugs in a single commit. Use `git add --patch` to stage only the parts related to one issue.

Constraints:

- Bugfixes are only accepted for packages other than <https://github.com/vyos/vyos-1x>. New functionality must use the new XML/Python interface, not old-style templates (node.def files and Perl/Bash code).

13.2.4 Coding guidelines

VyOS maintains consistent coding standards to help contributors navigate the codebase and understand its logic.

Formatting

- **Python:** Use 4 spaces per indentation level. Tabs **must not** be used.
- **XML:** Use 2 spaces per indentation level. Tabs **must not** be used.

Use tools like VIM extensions (xmllint) to enforce correct indentation. Add this to your .vimrc file:

```
au FileType xml setlocal equalprg=xmllint\ --format\ --recover\ -\ 2>/dev/null
```

Then use `gg=G` in command mode to run the linter.

Text generation

Use a template processor for generating config files:

- **Jinja2** is the default template processor for VyOS code.
- Built-in string formatting **may** be used for simple line-oriented formats (for example, iptables rules) where every line is self-contained.
- Template processors **must** be used for structured, multi-line formats (for example, ISC DHCPd configuration).

Python code

Configuration scripts and operation mode scripts written in Python3 should follow these guidelines:

- Wrap lines at 80 characters. This improves readability when browsing GitHub on mobile devices and reads well in side-by-side diffs.

Structure your scripts with these functions:

```
#!/usr/bin/env python3
#
# Copyright (C) 2020 VyOS maintainers and contributors
#
# This program is free software; you can redistribute it and/or modify
# it under the terms of the GNU General Public License version 2 or later as
# published by the Free Software Foundation.
#
# This program is distributed in the hope that it will be useful,
# but WITHOUT ANY WARRANTY; without even the implied warranty of
```

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```

# MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE. See the
# GNU General Public License for more details.
#
# You should have received a copy of the GNU General Public License
# along with this program. If not, see <http://www.gnu.org/licenses/>.

import sys

from vyos.config import Config
from vyos import ConfigError

def get_config(config=None):
    if config:
        conf = config
    else:
        conf = Config()

    # Base path to CLI nodes
    base = ['...', '...']
    # Convert the VyOS config to an abstract internal representation
    config_data = conf.get_config_dict(base, key_mangling('-', '_'), get_first_key=True)
    return config_data

def verify(config):
    # Verify that configuration is valid
    if invalid:
        raise ConfigError("Descriptive message")

def generate(config):
    # Generate daemon configs
    pass

def apply(config):
    # Apply the generated configs to the live system
    pass

try:
    c = get_config()
    verify(c)
    generate(c)
    apply(c)
except ConfigError as e:
    print(e)
    sys.exit(1)

```

`get_config()`: This function converts a VyOS config object to an abstract internal representation. No other function may call the `vyos.config.Config` object directly. Limiting config reads to one function makes it easier to modify the config syntax in the future. Additionally, this design improves testability since you can construct an internal representation by hand rather than mocking the entire config subsystem.

`verify()`: This function validates the internal representation. It must raise `ConfigError` with a descriptive message if the config is invalid. It **must not** make any changes to the system. This design enables future features like commit dry-run (“commit test” as in JunOS) where the system can abort a commit before making changes.

`generate()`: This function generates config files for system components.

`apply()`: This function applies the generated configuration to the live system. Prefer non-disruptive reload when possible. Disruptive operations like daemon restarts are acceptable only when:

- The component does not support non-disruptive reload, or
- The expected service degradation is minimal (for example, auxiliary services like LLDPd)

For high-impact services (VPN daemons, routing protocols), make effort to determine if changes can be applied non-disruptively before resorting to restarts.

Never modify active configuration directly unless absolutely necessary. Instead, generate configuration files and apply them with a single command like `service reload` through `systemd`. For example, save iptables rules to a file and load them with `iptables-restore` rather than executing iptables commands one by one.

The `apply()` and `generate()` functions may raise `ConfigError` if the daemon fails to start with the updated config. However, this is not a substitute for proper config validation in the `verify()` function. Make reasonable effort to verify that generated configuration is valid and will be accepted by the daemon, including cross-checks with other VyOS configuration subtrees when necessary.

Exceptions like `VyOSError` (raised by `vyos.config.Config` on improper operations) should not be silenced or caught. While this may produce less polished error output for users, it generates better bug reports and helps maintainers debug issues.

For reference implementations, see `ntp.py` or `interfaces-bonding.py` (for tag nodes) in the `vyos-1x` repository.

Other considerations: `vyos-configd`

All scripts now run under the config daemon and must conform to these requirements:

1. The signature and first four lines of `get_config(...)` **must** be as specified above.
2. Each of `get_config`, `verify`, `apply`, and `generate` **must** appear with the correct signatures, even if they are a no-op.
3. Config objects other than those in `get_config` **must not** appear.
4. The legacy function `my_set` **must not** appear. Modifications to active config **should not** appear in new code (alternative mechanisms may be used if absolutely necessary).

13.2.5 XML for CLI definitions

XML interface definitions define the VyOS CLI structure. Before VyOS 1.2 (crux), these files were created manually. After a redesign, new-style templates are automatically generated from XML input files.

VyOS interface definitions come with a RelaxNG schema located in the `vyos-1x` repository. This schema is a modified version from VyConf (VyOS 2.0). VyOS 1.2.x interface definitions are reusable in future VyOS versions with minimal changes.

Schemas provide two benefits:

- Complete grammar verification
- Automatic validation against the schema

The `build-command-templates` script converts XML definitions to old-style templates and verifies them against the schema. A bad definition causes the package build to fail. While the XML format is verbose, no other format provides this level of verification. Specialized XML editors can help manage verbosity.

Example XML interface definition:

```

<?xml version="1.0"?>
<!-- Cron configuration -->
<interfaceDefinition>
  <node name="system">
    <children>
      <node name="task-scheduler">
        <properties>
          <help>Task scheduler settings</help>
        </properties>
        <children>
          <tagNode name="task" owner="{vyos_conf_scripts_dir}/task_scheduler.py">
            <properties>
              <help>Scheduled task</help>
              <valueHelp>
                <format>&lt;string&gt;</format>
                <description>Task name</description>
              </valueHelp>
              <priority>999</priority>
            </properties>
            <children>
              <leafNode name="crontab-spec">
                <properties>
                  <help>UNIX crontab time specification string</help>
                </properties>
              </leafNode>
              <leafNode name="interval">
                <properties>
                  <help>Execution interval</help>
                  <valueHelp>
                    <format>&lt;minutes&gt;</format>
                    <description>Execution interval in minutes</description>
                  </valueHelp>
                  <valueHelp>
                    <format>&lt;minutes&gt;m</format>
                    <description>Execution interval in minutes</description>
                  </valueHelp>
                  <valueHelp>
                    <format>&lt;hours&gt;h</format>
                    <description>Execution interval in hours</description>
                  </valueHelp>
                  <valueHelp>
                    <format>&lt;days&gt;d</format>
                    <description>Execution interval in days</description>
                  </valueHelp>
                  <constraint>
                    <regex>[1-9]([0-9]*)([mhd]{0,1})</regex>
                  </constraint>
                </properties>
              </leafNode>
              <node name="executable">
                <properties>
                  <help>Executable path and arguments</help>
                </properties>
            </children>
          </tagNode>
        </children>
      </node>
    </children>
  </node>
</interfaceDefinition>

```

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```

    <children>
      <leafNode name="path">
        <properties>
          <help>Path to executable</help>
        </properties>
      </leafNode>
      <leafNode name="arguments">
        <properties>
          <help>Arguments passed to the executable</help>
        </properties>
      </leafNode>
    </children>
  </node>
</children>
</tagNode>
</children>
</node>
</children>
</node>
</interfaceDefinition>

```

XML definitions are purely declarative and contain no logic. All logic for generating config files, restarting services, and related tasks is implemented in configuration scripts.

Template Processors

XML interface definition files use the `.xml.in` file extension (implemented in [T1843](#)). These files use the GCC preprocessor to reduce code duplication in common areas:

- VIF (including VIF-S and VIF-C)
- Address configuration
- Description
- Enabled/Disabled state

Instead of repeating XML nodes, use include files with predefined features:

- [IPv4](#), [IPv6](#), and [DHCP\(v6\)](#) address assignment.
- [IPv4](#) and [IPv6](#) address assignment.
- [VLAN \(VIF\)](#) definition.
- [MAC](#) address assignment.

The `.in` files are preprocessed and stored in the `interface-definitions` folder. The `scripts/build-command-templates` script then operates on this folder to generate all required CLI nodes.

Example preprocessor output:

```

$ make interface_definitions
install -d -m 0755 build/interface-definitions
install -d -m 0755 build/op-mode-definitions
Generating build/interface-definitions/intel_qat.xml from interface-definitions/intel_
↳ qat.xml.in

```

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```
Generating build/interface-definitions/interfaces-bonding.xml from interface-definitions/  
↪ interfaces-bonding.xml.in  
Generating build/interface-definitions/cron.xml from interface-definitions/cron.xml.in  
Generating build/interface-definitions/pppoe-server.xml from interface-definitions/pppoe-  
↪ server.xml.in  
Generating build/interface-definitions/mdns-repeater.xml from interface-definitions/mdns-  
↪ repeater.xml.in  
Generating build/interface-definitions/tftp-server.xml from interface-definitions/tftp-  
↪ server.xml.in  
[...]
```

Command Definition Guidelines

Use of Numbers

Avoid using numbers in command names unless the number is part of a protocol name or similar. For example, protocols `ospfv3` is appropriate, but `server-1` is questionable.

Help Strings

Follow these guidelines for consistent, readable help strings:

Capitalization and Punctuation

- Capitalize the first word of every help string.
- Do not use a period at the end of help strings.

This standard mirrors network device CLIs and improves aesthetics.

Examples:

- Good: “Frobnication algorithm”
- Bad: “frobnication algorithm”
- Bad: “Frobnication algorithm.”
- Incorrect: “frobnication algorithm.”

Abbreviations and Acronyms

- Capitalize all abbreviations and acronyms.

Examples:

- Good: “TCP connection timeout”
- Bad: “tcp connection timeout”
- Bad: “Tcp connection timeout”
- Capitalize acronyms to distinguish them from normal words.

Examples:

- Good: RADIUS (remote authentication for dial-in user services)
- Bad: radius (unless referring to circular distance)
- Follow accepted spelling conventions for mixed-case abbreviations. If it contains “over” or “version”, use lowercase. Follow RFC or standard spellings when they exist.

Examples:

- Good: PPPoE, IPsec
- Bad: PPPOE, IPSEC
- Bad: pppoe, ipsec

Verbs

- Avoid verbs. If a verb can be omitted, omit it.

Examples:

- Good: “TCP connection timeout”
- Bad: “Set TCP connection timeout”
- When a verb is essential, use it. For example: “Disable IPv6 forwarding on all interfaces” for `set system ipv6 disable-forwarding`.
- Use infinitive form for necessary verbs.

Examples:

- Good: “Disable IPv6 forwarding”
- Bad: “Disables IPv6 forwarding”

13.2.6 C++ Backend Code

The VyOS CLI parser combines bash, bash-completion helpers, and the C++ backend library `vyatta-cfg`. This section references common CLI commands and their C/C++ entry points:

`set`:

- <https://github.com/vyos/vyatta-cfg/blob/0f42786a0b3/src/cstore/cstore.cpp#L352>
- <https://github.com/vyos/vyatta-cfg/blob/0f42786a0b3/src/cstore/cstore.cpp#L2549>

`commit`:

- <https://github.com/vyos/vyatta-cfg/blob/0f42786a0b3/src/commit/commit-algorithm.cpp#L1252>

13.3 Contributor License Agreement

Before we can accept your contributions to VyOS, you must sign a **Contributor License Agreement (CLA)**.

This is a standard open-source practice that protects both you and the project.

The process is straightforward and fully automated:

1. **Review the CLA document**

Find the CLA text in our [GitHub repository](#).

2. **Submit a pull request**

When you open a pull request, a CLA bot automatically checks whether all commit authors have signed the CLA.

3. **Follow the bot's instructions**

If the CLA has not been signed, the bot leaves a comment with instructions. Reply to that comment with the suggested text to sign the CLA.

4. **Wait for confirmation**

The CLA bot verifies your response and updates the pull request status. Once all commit authors have signed, the bot confirms that the CLA requirement is met and unlocks the pull request for merging.

Note: Each commit author must sign the CLA.

If your pull request includes commits from multiple contributors, each one must sign the CLA before the pull request can be accepted.

Once you sign the CLA, it remains valid for all your past and future contributions to VyOS under the same GitHub identity.

13.4 Issues/Feature requests

13.4.1 Bug Report/Issue

Issues and bugs occur in every software project, and VyOS is no exception.

I found a bug, what should I do?

When you find a potential bug, first:

- Consult the [documentation](#) to ensure you configured your system correctly.
- Check if the VyOS community has identified a workaround for the bug through [Slack](#) or the [VyOS Forum](#).

Ensure the bug is reproducible

Include the following information when reporting a bug:

- A sequence of configuration commands or a complete configuration file needed to recreate the bug. Avoid partial configurations: a sequence of commands is easy to paste and a complete configuration is easy to load, but a partial config is hard to reconstruct.
- Describe the expected behavior and how it differs from what you observe. Include command outputs or traffic dumps. Explain briefly why these outputs are incorrect and what the correct behavior should be.
- A sequence of actions that trigger the bug. While not always possible, this helps developers and community members confirm the issue and verify fixes.
- If the bug is a regression, specify the VyOS version where the feature worked correctly (any working version is acceptable). Identify the exact version that the feature stopped working, if possible.

If you are uncertain whether the behavior is a bug or what the correct behavior is, or if you lack a reliable reproducing procedure, post on the forum or ask in chat first. If you have a subscription, create a support ticket. The team and community can help identify the issue, work around it, and create an actionable bug report.

Report a Bug

To open a bug report or feature request, create an account on vyos.dev, the public issue tracker for VyOS.

When creating a new issue, select the appropriate project and:

- Provide as much information as you can.
- Specify which VyOS version you are using: `run show version`.
- Explain how to reproduce the bug.

13.4.2 Feature Requests

Have an idea to improve VyOS or need a feature that would benefit all users? Before submitting a feature request, search the public issue tracker vyos.dev to check if a request already exists. You can also enhance an existing request by providing additional information.

Create a task before starting work on a feature, even if it is a trivial feature. The task tracker generates release notes, so all work must be reflected in the tracker.

Include at least the following information:

- Provide a detailed description of the feature: what it is, how it works, and how you would use it. Maintainers may not have experience with every feature, protocol, and tool in VyOS. Detailed information helps VyOS contributors and maintainers test new features they are unfamiliar with.
- Include proposed CLI syntax if the feature requires new commands. Provide both configuration and operational mode commands if both are needed.

Consider including the following information:

- Is the feature already supported by the underlying component (FreeRangeRouting, nftables, Kea, etc.)?
- How would you configure the feature manually within that component?
- Are there any limitations to using the feature (hardware support, resource usage)?
- Are there any adverse or non-obvious interactions with other features? Should the feature be mutually exclusive?
- Any relevant documentation or references about the feature.

You do not need to provide all this information, but if you can, it simplifies developers' work considerably. Research these questions when possible.

13.4.3 Task auto-closing

A special task status exists for when all work by maintainers and contributors is complete: **Needs reporter action**.

VyOS assigns this status to:

- Feature requests that do not include required information and need clarification.
- Bug reports that lack reproducing procedures.
- Tasks that are implemented and tested by the implementation author, but require testing in the real-world environment that only the reporter can replicate (for example, hardware VyOS does not support or specific network conditions).

When a task is set to **Needs reporter action**:

- If the reporter does not respond within two weeks, the task bot adds a comment ("Any news?") to remind the reporter.
- If there is still no response after another two weeks, the task is closed automatically.

We do not auto-close tasks with any other status and do not close tasks due to lack of maintainer activity.

13.5 Upstream Packages

Many base system packages are pulled straight from Debian's `main` and `contrib` repositories, but there are exceptions. If you only want to build a fresh ISO image, you can skip this section. This information may be useful for a deeper dive into VyOS.

System packages that are not directly pulled from Debian are built through a separate build system, `build.py` in the `vyos-build` repository.

13.5.1 Overview

Previously, VyOS used Jenkins for building upstream packages. With the move away from Jenkins, the build system was replaced with a Python-based solution using `build.py` and `package.toml` configuration files.

Each package directory contains:

- A `package.toml` configuration file that defines how the package is built.
- A symlink to the common `build.py` script in the build system.

13.5.2 Building Packages

To build a package, navigate to the package directory and execute the build script:

```
cd package-build/<package-name>
./build.py
```

The script will:

1. Check out the source code from the configured repository.

2. Apply any patches defined in the configuration.
3. Execute pre-build hooks (if configured).
4. Build the package using the specified build command.
5. Generate both binary (.deb) packages and source tarballs.

13.5.3 Package Configuration (package.toml)

Each package directory contains a `package.toml` file that defines the build parameters. The key configuration fields are:

name

The package name (e.g., `frr`)

commit_id

The specific commit, tag, or branch to check out from the source repository (e.g., `stable/10.5`)

scm_url

The Git URL of the upstream source repository (e.g., `https://github.com/FRRouting/frr.git`)

build_cmd

The command to execute for building the package. This replaces what was previously defined in the Jenkins Jenkinsfile.

Default if not specified: `dpkg-buildpackage -uc -us -tc -F --source-option=--tar-ignore=.git --source-option=--tar-ignore=.github`

Example with custom build command:

```
build_cmd = "sudo dpkg -i ../*.deb; dpkg-buildpackage -us -uc -tc -b -Ppkg.frr.
↳rtrlib, pkg.frr.lua"
```

pre_build_hook (Optional)

A shell command or script that executes after the repository is checked out and before the build process begins. This allows you to perform preparatory tasks such as:

- Creating directories
- Copying files
- Running custom setup scripts
- Installing dependencies

Single command example:

```
pre_build_hook = "echo 'Preparing build environment'"
```

Multi-line commands example:

```
pre_build_hook = """
mkdir -p ../hello/vyos
mkdir -p ../vyos
cp example.txt ../vyos
"""
```

Combined commands and scripts:

```
pre_build_hook = "ls -l; ./script.sh"
```

apply_patches (Optional)

Boolean flag to control whether patches should be applied. Defaults to True.

```
apply_patches = false
```

prepare_package (Optional)

Boolean flag to enable package preparation. When set to True, the `install_data` configuration is used.

install_data (Optional)

Data used for package preparation when `prepare_package` is enabled.

13.5.4 Example package.toml file

Here's an example configuration for the FRRouting (FRR) package:

```
name = "frr"
commit_id = "stable/10.5"
scm_url = "https://github.com/FRRouting/frr.git"
build_cmd = "sudo dpkg -i ../*.deb; dpkg-buildpackage -us -uc -tc -b -Ppkg.frr.rtrlib,
↳ pkg.frr.lua"
```

13.5.5 Build Output

After running `./build.py`, the following artifacts are generated in the package directory:

- `.deb` files - Binary Debian packages ready for installation
- `.tar.gz` files - Source tarballs of the checked-out repositories
- Additional build artifacts as produced by the Debian build system

The build script also creates build dependency packages (`*build-deps*.deb`), which are automatically cleaned up after the build completes.

13.6 Debugging

Two flags are available to help debug configuration scripts. Configuration loading issues manifest during boot, so these flags are passed as kernel boot parameters.

13.6.1 ISO image build

If you have trouble compiling your own ISO image or debugging Jenkins issues, follow the steps at *ISO Build Issues*.

13.6.2 System Startup

Debug system startup by examining the configuration file loading from `/config/config.boot`. Extend the kernel command-line in the bootloader to enable this.

Kernel

- **vyos-debug** - Add this parameter to the Linux boot line to produce timing results for script execution during commit. If you see an unexpected delay during manual or boot commit, this parameter helps identify bottlenecks. The internal flag is `VYOS_DEBUG`, found in `vyatta-cfg`. Output is directed to `/var/log/vyatta/cfg-stdout.log`.
- **vyos-config-debug** - During development, coding errors can cause commit failures on boot, potentially preventing CLI initialization. This kernel boot parameter ensures access to the system as user `vyos` and logs a Python stack trace to `/tmp/boot-config-trace`. The file is created only if the configuration load fails.

13.6.3 Live System

Several flags can be set to change VyOS behavior at runtime. Toggle these flags using environment variables or by creating files.

For each feature, create a file called `vyos.feature.debug` to enable it. If a parameter is required, place it as the first line inside the file.

Place the file in `/tmp` for one-time debugging (the file is removed on reboot) or in `/config` to persist permanently.

For example, `/tmp/vyos.ifconfig.debug` can be created to enable interface debugging.

You can also enable debugging using environment variables. The environment variable name follows the convention `VYOS_FEATURE_DEBUG`.

For example, `export VYOS_IFCONFIG_DEBUG=""` in your `vbash` has the same effect as `touch /tmp/vyos.ifconfig.debug`.

- **ifconfig** - Display all commands and their responses from the OS on screen for inspection.
- **command** - Display all commands and their responses from the OS on screen for inspection.
- **developer** - When a command fails, start a PDB post-mortem session instead of showing a standard error message. This allows developers to debug issues interactively. Because the debugger waits for input, it can prevent the router from booting, so only enable this permanently on production systems if you are ready for potential boot failures.
- **log** - Send all commands used by VyOS to a log file for inspection. This is useful in rare cases when you need to see what the OS is doing, including during boot. The default file is `/tmp/full-log`, but you can change it.

Note: To retrieve debug output on the command line, disable `vyos-configd` in addition. You can do this one-time with `sudo systemctl stop vyos-configd` or permanently with `sudo systemctl disable vyos-configd`.

FRR

Recent versions use the `vyos.frr` framework. The Python class is located in `vyos-1x:python/vyos/frr.py`. It includes an embedded debugger similar to the one in `vyos.ifconfig`.

Enable debugging by running: `touch /tmp/vyos.frr.debug`

Debug Python code with PDB

Sometimes it is useful to debug Python code interactively on the live system rather than in an IDE. You can do this using `pdb`.

Assuming you want to debug a Python script called by an op-mode command, find the script by looking up the op-mode definitions, then edit it on the live system using `vi /usr/libexec/vyos/op_mode/show_xyz.py`

Insert the following statement right before the section where you want to investigate a problem (for example, a statement you see in a backtrace): `import pdb; pdb.set_trace()`

Optionally, surround this statement with an `if` condition that triggers only for the conditions you are interested in.

When you run `show xyz` and your condition triggers, you enter the Python debugger:

```
> /usr/libexec/vyos/op_mode/show_nat_translations.py(109)process()
-> rule_type = rule.get('type', '')
(Pdb)
```

You can type `help` to get an overview of the available commands, and `help` command to get more information on each command.

Common useful commands include:

- examine variables using `pp(var)`
- continue execution using `cont`
- get a backtrace using `bt`

Config Migration Scripts

Starting with VyOS 1.5, a new mechanism is used for config migration that improves migration performance. New migrators use only the new format with a `migration()` function.

```
from vyos.configtree import ConfigTree
base = ['vpn', 'ipsec']
def migrate(config: ConfigTree) -> None:
    if not config.exists(base):
        # Nothing to do
        return
    # do your stuff here
```

New-style migration scripts can no longer run on their own. However, the new migration subsystem handler includes a test kit:

```
vyos@vyos:~$ /usr/libexec/vyos/run-config-migration.py --help
usage: run-config-migration.py [-h] [--test-script TEST_SCRIPT] [--output-file OUTPUT_
↵FILE] [--force] config_file
```

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```
positional arguments:
  config_file          configuration file to migrate

options:
  -h, --help            show this help message and exit
  --test-script TEST_SCRIPT
                        test named script
  --output-file OUTPUT_FILE
                        write to named output file instead of config file
  --force                force run of all migration scripts
```

To test your migration, run:

```
vyos@vyos:~$ /usr/libexec/vyos/run-config-migration.py --test-script /opt/vyatta/etc/
↪config-migrate/migrate/quagga/11-to-12 --output-file /tmp/foo /tmp/static-route-basic
vyos@vyos:~$ cat /tmp/foo
```

The file `/tmp/foo` contains the migrated configuration.

Configuration Error on System Boot

Running the latest rolling releases sometimes exposes bugs due to edge cases missed in design. File these bugs via [Phabricator](#), but you can help narrow down the issue by following these steps:

1. Log in to your VyOS system.
2. Enter configuration mode: `configure`
3. Reload your boot configuration: `load`

You should see a Python backtrace that helps identify the issue. Attach it to the [Phabricator](#) task.

Boot Timing

During the migration and rewrite of functionality from Perl to Python, system boot time increased significantly. You can analyze and graph boot time to see detailed call sequences during startup.

This uses the `systemd-bootchart` package, which is installed by default on VyOS 1.3 (equuleus) and later. Configuration is versioned for comparable results. Refer to [bootchart.conf](#) for the configuration file.

To enable boot time graphing, add the following to the kernel command line: `init=/usr/lib/systemd/systemd-bootchart`

You can also make this permanent by editing `/boot/grub/grub.cfg`.

13.6.4 Priorities

VyOS CLI depends heavily on priorities. Every CLI node has a corresponding `node.def` file and possibly an attached script. Nodes can have priorities, and on system boot or any `commit` to the configuration, scripts execute from lowest to highest priority. This provides deterministic behavior.

To debug priority issues or see script execution order, use the `/opt/vyatta/sbin/priority.pl` script, which lists the execution order of scripts.

13.7 Testing

One of the major features introduced in VyOS 1.3 is an automated test framework. When you assemble an ISO image, several things can go wrong. VyOS uses this framework to detect issues before they cause downstream problems.

This section describes how the automated testing process at VyOS works.

13.7.1 Smoketests

Smoketests execute predefined VyOS CLI commands and check if the desired daemon or service configuration is rendered.

When an ISO image is assembled by the [VyOS CI](#), the `BUILD_SMOKETEST` parameter is enabled by default. This extends the ISO configuration line with the following packages:

```
def CUSTOM_PACKAGES = ''
if (params.BUILD_SMOKETESTS)
    CUSTOM_PACKAGES = '--custom-package vyos-1x-smoketest'
```

If you plan to build your own custom ISO image and want to use VyOS's smoketests, ensure that you have the `vyos-1x-smoketest` package installed.

The `make test` command from the `vyos-build` repository launches a new QEMU instance, and the ISO image is first installed to the virtual hard disk.

After the first boot into the newly installed system, the main Smoketest script is executed. It can be found at `/usr/bin/vyos-smoketest`.

The script searches for executable test cases under `/usr/libexec/vyos/tests/smoke/cli/` and executes them one by one.

Note: Smoketests will alter the system configuration. If you are logged in remotely, you may lose your connection to the system.

Note: To enable smoketest debugging (print the CLI set commands used), run: `touch /tmp/vyos.smoketest.debug`.

Manual Smoketest Run

Each test is contained in its own file, so you can execute a single Smoketest manually by running the Python test script.

Example:

```
vyos@vyos:~$ /usr/libexec/vyos/tests/smoke/cli/test_protocols_bgp.py
test_bgp_01_simple (__main__.TestProtocolsBGP) ... ok
test_bgp_02_neighbors (__main__.TestProtocolsBGP) ... ok
test_bgp_03_peer_groups (__main__.TestProtocolsBGP) ... ok
test_bgp_04_afip4 (__main__.TestProtocolsBGP) ... ok
test_bgp_05_afip6 (__main__.TestProtocolsBGP) ... ok
test_bgp_06_listen_range (__main__.TestProtocolsBGP) ... ok
test_bgp_07_l2vpn_evpn (__main__.TestProtocolsBGP) ... ok
test_bgp_08_zebra_route_map (__main__.TestProtocolsBGP) ... ok
test_bgp_09_distance_and_flowspec (__main__.TestProtocolsBGP) ... ok
test_bgp_10_vrf_simple (__main__.TestProtocolsBGP) ... ok
test_bgp_11_confederation (__main__.TestProtocolsBGP) ... ok
test_bgp_12_v6_link_local (__main__.TestProtocolsBGP) ... ok
test_bgp_13_solo (__main__.TestProtocolsBGP) ... ok

-----
Ran 13 tests in 348.191s

OK
```

Interface-based tests

Our smoketests not only test daemons and services, but also check if interface configuration works as expected. There is a common base class named `base_interfaces_test.py` that holds all the common code for interface tests.

These common tests consist of:

- Add one or more IP addresses
- DHCP client and DHCPv6 prefix delegation
- MTU size
- IP and IPv6 options
- Port description
- Port disable
- VLANs (QinQ and regular 802.1q)
- ...

Note: When you are working on interface configuration and want to test if the Smoketests pass, you would normally lose the remote SSH connection to your DUT (Device Under Test). To handle this, some interface-based tests can be called with an environment variable beforehand to limit the number of interfaces used in the test. By default, all interfaces (e.g., all Ethernet interfaces) are used.

```

vyos@vyos:~$ TEST_ETH="eth1 eth2" /usr/libexec/vyos/tests/smoke/cli/test_interfaces_
↳ bonding.py
test_add_multiple_ip_addresses (__main__.BondingInterfaceTest) ... ok
test_add_single_ip_address (__main__.BondingInterfaceTest) ... ok
test_bonding_hash_policy (__main__.BondingInterfaceTest) ... ok
test_bonding_lacp_rate (__main__.BondingInterfaceTest) ... ok
test_bonding_min_links (__main__.BondingInterfaceTest) ... ok
test_bonding_remove_member (__main__.BondingInterfaceTest) ... ok
test_dhcpv6_client_options (__main__.BondingInterfaceTest) ... ok
test_dhcpv6pd_auto_sla_id (__main__.BondingInterfaceTest) ... ok
test_dhcpv6pd_manual_sla_id (__main__.BondingInterfaceTest) ... ok
test_interface_description (__main__.BondingInterfaceTest) ... ok
test_interface_disable (__main__.BondingInterfaceTest) ... ok
test_interface_ip_options (__main__.BondingInterfaceTest) ... ok
test_interface_ipv6_options (__main__.BondingInterfaceTest) ... ok
test_interface_mtu (__main__.BondingInterfaceTest) ... ok
test_ipv6_link_local_address (__main__.BondingInterfaceTest) ... ok
test_mtu_1200_no_ipv6_interface (__main__.BondingInterfaceTest) ... ok
test_span_mirror (__main__.BondingInterfaceTest) ... ok
test_vif_8021q_interfaces (__main__.BondingInterfaceTest) ... ok
test_vif_8021q_lower_up_down (__main__.BondingInterfaceTest) ... ok
test_vif_8021q_mtu_limits (__main__.BondingInterfaceTest) ... ok
test_vif_8021q_qos_change (__main__.BondingInterfaceTest) ... ok
test_vif_s_8021ad_vlan_interfaces (__main__.BondingInterfaceTest) ... ok
test_vif_s_protocol_change (__main__.BondingInterfaceTest) ... ok

-----
Ran 23 tests in 244.694s

OK

```

This will limit the bond interface test to use only eth1 and eth2 as member ports.

13.7.2 Config Load Tests

The other part of our tests are called “config load tests.” Config load tests sequentially load arbitrary configuration files to verify that configuration migration scripts work as designed and that a given set of functionality can still be loaded with a fresh VyOS ISO image.

The configurations are all derived from production systems and can act as test cases or as references for enabling certain features. The configurations can be found here: <https://github.com/vyos/vyos-1x/tree/current/smoketest/configs>

The entire test is controlled by the main wrapper script `/usr/bin/vyos-configtest`. It behaves in the same way as the main smoketest script. It scans the folder for potential configuration files and issues a load command for each file.

Manual config load test

You do not have to load all configurations sequentially; you can also load individual test configurations manually.

```
vyos@vyos:~$ configure
load[edit]

vyos@vyos# load /usr/libexec/vyos/tests/config/ospf-small
Loading configuration from '/usr/libexec/vyos/tests/config/ospf-small'
Load complete. Use 'commit' to make changes effective.
[edit]
vyos@vyos# compare
[edit interfaces ethernet eth0]
-hw-id 00:50:56:bf:c5:6d
[edit interfaces ethernet eth1]
+duplex auto
-hw-id 00:50:56:b3:38:c5
+speed auto
[edit interfaces]
-ethernet eth2 {
-  hw-id 00:50:56:b3:9c:1d
-}
-vti vti1 {
-  address 192.0.2.1/30
-}
...

vyos@vyos# commit
vyos@vyos#
```

Note: Some configurations have preconditions that must be met. These most likely include generation of cryptographic keys before the config can be applied; otherwise, you will get a commit error. If you are interested in how those preconditions are fulfilled, check the [vyos-build](#) repository and the `scripts/check-qemu-install` file.

WRITE DOCUMENTATION

We encourage every VyOS user to help us improve our documentation as we have a deficit like most software projects. This not only helps you when reading but also everyone else.

Warning: Please read and sign the *Contributor License Agreement* before submitting any documentation updates.

If you are willing to contribute to our documentation this is the definite guide how to do so.

Note: In contrast to submitting code patches, there is no requirement that you open up a [Phabricator](#) task prior to submitting a Pull-Request to the documentation.

VyOS documentation is written in reStructuredText and generated to Read the Docs pages with Sphinx, as per the Python tradition. We welcome all sorts of contributions to the documentation. Not just new additions but also corrections to existing documentation.

The documentation source is kept in the Git repository at <https://github.com/vyos/vyos-documentation> and you can follow the instructions in the [README.md](#) to build and test your changes.

You can either install Sphinx and build the documentation locally, or use the [Dockerfile](#) to build it in a container.

14.1 Guidelines

There are a few things to keep in mind when contributing to the documentation, for the sake of consistency and readability.

The following is a quick summary of the rules:

- Use American English at all times. It's always a good idea to run your text through a grammar and spell checker, such as [Grammarly](#).
- Don't forget to update `index.rst` when adding a new node.
- Try not to exceed 80 characters per line, but don't break URLs over this.
- Properly quote commands, filenames and brief code snippets with double backticks.
- Use literal blocks for longer snippets.
- Leave a newline before and after a header.
- Indent with two spaces.
- When in doubt, follow the style of existing documentation.

And finally, remember that the reStructuredText files aren't exclusively for generating HTML and PDF. They should be human-readable and easily perused from a console.

14.2 Page content

All RST files must follow the same TOC Level syntax and have to start with

```
#####  
Title  
#####
```

The configuration mode folder and the articles cover the specific level of the commands. The exact level depends on the command. This should provide stability for URLs used in the forum or blogpost.

For example:

- `set firewall zone` is written in `firewall/zone.rst`
- `set interfaces ethernet` is written in `interfaces/ethernet.rst`

In the configuration part of the page, all possible configuration options should be documented. Use `.. cfgcmd::` described above.

Related operation command must be documented in the next part of the article. Use `::opcmd..` for these commands.

Each page must contain the following parts:

14.2.1 1. Theoretical information

Theoretical information required for users to understand the next document sections:

- a simple explanation of what is this page about, why or when it is required to be used
- references to standards, RFCs

14.2.2 2. Configuration description

Describe CLI items related to the service or use case. Each config line or section must be explained, using information provided in the 1st part of the page.

14.2.3 3. Configuration examples

Practical examples of the service or use case configuration. They must contain topology maps (if applicable) and short descriptions.

14.2.4 4. Known issues

This section must contain a list of:

- known issues or potential problems for the service or use case
- workarounds for known issues (if any exist)

14.2.5 5. Debugging

Described procedures for debugging a service:

- how to collect logs or other debugging information (like `show` commands output)
- how to read and what to search for in logs and collected information
- what are indicators of good and bad states in debugging outputs

14.3 Style Guide

14.3.1 Formatting and Sphinxmarkup

TOC Level

We use the following syntax for Headlines.

```
#####
Title
#####

*****
Chapters
*****

Sections
=====

Subsections
-----

Subsubsections
^^^^^^^^^^^^^^^^^^

Paragraphs
^^^^^^^^^^^^
```

Cross-References

A plugin will be used to generate a reference label for each headline. To reference a page or a section in the documentation use the `{ref}` command.

For example, you want to reference the headline **VLAN** in the **ethernet.rst** page. The plugin generates the label based on the headline and the file path.

```
{ref}`configuration/interfaces/ethernet:vlan
```

to use an alternative hyperlink use it this way:

```
{ref}`Check out VLAN<configuration/interfaces/ethernet:vlan>
```

handle build errors

The plugin will warn on build if a headline has a duplicate name in the same document. To prevent this warning, you have to put a custom link on top of the headline.

```
Section A
=====

Lorem ipsum dolor sit amet, consetetur sadipscing elitr

Example
-----

Lorem ipsum dolor sit amet, consetetur sadipscing elitr

Section B
=====

Lorem ipsum dolor sit amet, consetetur sadipscing elitr

.. _section B example:

Example
-----

Lorem ipsum dolor sit amet, consetetur sadipscing elitr
```

Address space

Note the following RFCs ([RFC 5737](#), [RFC 3849](#), [RFC 5389](#) and [RFC 7042](#)), which describe the reserved public IP addresses and autonomous system numbers for the documentation:

- 192.0.2.0/24
- 198.51.100.0/24
- 203.0.113.0/24
- 2001:db8::/32
- 16bit ASN: 64496 - 64511
- 32bit ASN: 65536 - 65551

- Unicast MAC Addresses: 00-53-00 to 00-53-FF
- Multicast MAC-Addresses: 90-10-00 to 90-10-FF

Please do not use other public address space.

Line length

Limit all lines to a maximum of 80 characters.

Except in `.. code-block::` because it uses the html tag `<pre>` and renders the same line format from the source rst file.

Autolinter

Each GitHub pull request is automatically linted to check the address space and line length.

Sometimes it is necessary to provide real IP addresses like in the *Configuration Blueprints*. For this, please use the sphinx comment syntax `.. stop_vyoslint` to stop the linter and `.. start_vyoslint` to start.

Custom Sphinx-doc Markup

Custom commands have been developed for writing the documentation. Please make yourself comfortable with those commands as this eases the way we render the documentation.

cfgcmd

When documenting CLI commands, use the `.. cfgcmd::` directive for all configuration mode commands. An explanation of the described command should be added below this statement. Replace all variable contents with `<value>` or something similar.

With those custom commands, it will be possible to render them in a more descriptive way in the resulting HTML/PDF manual.

```
.. cfgcmd:: protocols static arp <ipaddress> hwaddr <macaddress>
```

This will configure a static ARP entry, always resolving ``192.0.2.100`` to ``00:53:27:de:23:aa``.

For an inline configuration level command, use `:cfgcmd:`

```
:cfgcmd:`set interface ethernet eth0`
```

To extract a defaultvalue from the XML definitions add a `:defaultvalue:` to `.. cfgcmd::` directive. To have this feature locally, the `vyos-lx` submodule must be initialized before. Please be aware to not update the submodule in your PR.

```
.. cfgcmd:: set system conntrack table-size <1-500000000>
:defaultvalue:
```

The connection tracking table contains one entry for each connection being tracked by the system.

opcmd

When documenting operational level commands, use the `.. opcmd::` directive. An explanation of the described command should be added below this statement.

With those custom commands, it is possible to render them in a more descriptive way in the resulting HTML/PDF manual.

```
.. opcmd:: show protocols static arp
```

Display all known ARP table entries spanning across all interfaces

For an inline operational level command, use `:opcmd:`

```
:opcmd:`add system image`
```

cmdinclude

To minimize redundancy, there is a special include directive. It includes a txt file and replace the `{{ var0 }}` - `{{ var9 }}` with the correct value.

```
.. cmdinclude:: ../include/interface-address.txt
:var0: ethernet
:var1: eth1
```

the content of interface-address.txt looks like this

```
.. cfgcmd:: set interfaces {{ var0 }} <interface> address <address | dhcp |
dhcpv6>
```

Configure interface ``<interface>`` with one or more interface addresses.

* ****address**** can be specified multiple times as IPv4 and/or IPv6 address, e.g. 192.0.2.1/24 and/or 2001:db8::1/64
 * ****dhcp**** interface address is received by DHCP from a DHCP server on this segment.
 * ****dhcpv6**** interface address is received by DHCPv6 from a DHCPv6 server on this segment.

Example:

```
.. code-block:: none
```

```
set interfaces {{ var0 }} {{ var1 }} address 192.0.2.1/24
set interfaces {{ var0 }} {{ var1 }} address 192.0.2.2/24
set interfaces {{ var0 }} {{ var1 }} address 2001:db8::ffff/64
set interfaces {{ var0 }} {{ var1 }} address 2001:db8:100::ffff/64
```

vytask

When referencing to VyOS Phabricator Tasks, there is a custom Sphinx Markup command called `vytask` that automatically renders to a proper Phabricator URL. This is heavily used in the release-notes section.

```
* {vytask}`T1605` Fixed regression in L2TP/IPsec server
* {vytask}`T1613` Netflow/sFlow captures IPv6 traffic correctly
```

14.4 Forking Workflow

The Forking Workflow is fundamentally different from other popular Git workflows. Instead of using a single server-side repository to act as the “central” codebase, it gives every developer their own server-side repository. This means that each contributor has not one, but two Git repositories: a private local one and a public server-side one.

The main advantage of the Forking Workflow is that contributions can be integrated without the need for everybody to push to a single central repository. Developers push to their own server-side repositories, and only the project maintainer can push to the official repository. This allows the maintainer to accept commits from any developer without giving them write access to the official codebase.

Note: Updates to our documentation should be delivered by a GitHub pull-request. This requires you already have a GitHub account.

- Fork this project on GitHub <https://github.com/vyos/vyos-documentation/fork>
- Clone fork to local machine, then change to that directory `$ cd vyos-documentation`
- Install the requirements `$ pip install -r requirements.txt` (or something similar)
- Create a new branch for your work, use a descriptive name of your work: `$ git checkout -b <branch-name>`
- Make all your changes - please keep our commit rules in mind (`prepare_commit`). This mainly applies to proper commit messages describing your change (how and why). Please check out the documentation of [Sphinx-doc](#) or [reStructuredText](#) if you are not familiar with it. This is used for writing our docs. Additional directives how to write in RST can be obtained from [reStructuredTextDirectives](#).
- Check your changes by locally building the documentation `$ make livehtml`. Sphinx will build the html files in the `docs/_build` folder. We provide you with a Docker container for an easy-to-use user experience. Check the [README.md](#) file of this repository.
- View modified files by calling `$ git status`. You will get an overview of all files modified by you. You can add individual files to the Git Index in the next step.
- Add modified files to Git index `$ git add path/to/filename` or add all unstaged files `$ git add ..` All files added to the Git index will be part of you following Git commit.
- Commit your changes with the message, `$ git commit -m "<commit message>"` or use `$ git commit -v` to have your configured editor launched. You can type in a commit message. Again please make yourself comfortable without rules (`prepare_commit`).
- Push commits to your GitHub project: `$ git push -u origin <branch-name>`
- Submit pull-request. In GitHub visit the main repository and you should see a banner suggesting to make a pull request. Fill out the form and describe what you do.

- Once pull requests have been approved, you may want to locally update your forked repository too. First you'll have to add a second remote called `upstream` which points to our main repository. `$ git remote add upstream https://github.com/vyos/vyos-documentation.git`

Check your configured remote repositories:

```
$ git remote -v
origin    https://github.com/<username>/vyos-documentation.git (fetch)
origin    https://github.com/<username>/vyos-documentation.git (push)
upstream  https://github.com/vyos/vyos-documentation.git (fetch)
upstream  https://github.com/vyos/vyos-documentation.git (push)
```

Your remote repo on Github is called `origin`, while the original repo you have forked is called `upstream`. Now you can locally update your forked repo.

```
$ git fetch upstream
$ git checkout rolling
$ git merge upstream/rolling
```

- If you also want to update your fork on GitHub, use the following: `$ git push origin rolling`

COVERAGE

Overview over all commands that are documented in the `.. cfgcmd::` or `.. opcmd::` directives.

The build process takes all XML definition files from the `vyos-1x` repository and a periodical export of all VyOS commands and extracts each leaf command, or executable command. The script compares only the fixed part of a command. All variables and values are removed and the string of the commands are compared.

For example, take the following two commands:

- documentation: `interfaces ethernet <interface> address <address | dhcp | dhcpv6>`
- xml: `interfaces ethernet <ethernet> address <address>`
- VyOS: `interfaces ethernet <text> address <value>`

There are 3 kinds of issues with commands in the following tables:

Not documented yet

- An XML command is not found in `.. cfgcmd::` or `.. opcmd::` directives.
- The command should be documented.

Nothing found in XML Definitions

- `.. cfgcmd::` or `.. opcmd::` The command is not found in the XML definition.
- The command location may have changed in the XML definition, the feature is no longer supported in VyOS, or there is a typo in the command.

Nothing found in VyOS

- `.. cfgcmd::` or `.. opcmd::` The command is not found in VyOS documentation or the XML definition.
- The command location may have changed in the XML definition, the feature is no longer supported in VyOS, or there is a typo in the command.

The final list of commands are shown in the following two tables:

15.1 Configuration Commands

Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set</code>	Nothing found in XML Definitions	Nothing found in VyOS

continues on next page

Table 1 – continued from previous page

Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	Nothing found in XML Definitions	cluster dead-interval <value>
×	not yet documented	Nothing found in XML Definitions	cluster group <text> auto-failback <value>
×	not yet documented	Nothing found in XML Definitions	cluster group <text> monitor <value>
×	not yet documented	Nothing found in XML Definitions	cluster group <text> primary <value>
×	not yet documented	Nothing found in XML Definitions	cluster group <text> secondary <value>
×	not yet documented	Nothing found in XML Definitions	cluster group <text> service <value>
×	not yet documented	Nothing found in XML Definitions	cluster interface <value>
×	not yet documented	Nothing found in XML Definitions	cluster keepalive-interval <value>
×	not yet documented	Nothing found in XML Definitions	cluster mcast-group <value>
×	not yet documented	Nothing found in XML Definitions	cluster monitor-dead-interval <value>
×	not yet documented	Nothing found in XML Definitions	cluster pre-shared-secret <value>
×	<code>comment <config node> "comment text"</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>commit</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>commit-confirm <minutes></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>compare <saved / N> <M></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set container name <name> allow-host-networks</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set container name <name> allow-host-pid</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set container name <name> arguments <arguments></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set container name <name> capability <text></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set container name <name> command <command></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set container name <name> cpu-quota <num></code>	Nothing found in XML Definitions	Nothing found in VyOS

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Table 1 – continued from previous page

Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set container name <name> description <text></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set container name <name> device <devicename> destination <path></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set container name <name> device <devicename> source <path></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set container name <name> disable</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set container name <name> entrypoint <entrypoint></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set container name <name> environment <key> value <value></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set container name <name> gid <number></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set container name <name> health-check</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set container name <name> health-check command <command></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set container name <name> health-check interval <interval></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set container name <name> health-check retries <retries></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set container name <name> health-check timeout <timeout></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set container name <name> host-name <hostname></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set container name <name> image</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set container name <name> label <label> value <value></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set container name <name> log-driver [k8s-file journald none]</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set container name <name> memory <MB></code>	Nothing found in XML Definitions	Nothing found in VyOS

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Table 1 – continued from previous page

Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set container name <name> name-server <address></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set container name <name> network <networkname></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set container name <name> network <networkname> address <address></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set container name <name> port <portname> destination <portnumber></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set container name <name> port <portname> protocol <tcp udp></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set container name <name> port <portname> source <portnumber></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set container name <name> restart [no on-failure always]</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set container name <name> sysctl parameter <parameter> value <value></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set container name <name> tmpfs <tmpfsname> destination <path></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set container name <name> tmpfs <tmpfsname> size <MB></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set container name <name> uid <number></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set container name <name> volume <volumename> destination <path></code>	Nothing found in XML Definitions	Nothing found in VyOS

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Table 1 – continued from previous page

Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set container name <name> volume <volumename> mode <ro / rw></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set container name <name> volume <volumename> source <path></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	container.xml.in: container name <name> allow-host-networks	container name <text> allow-host-networks
×	not yet documented	container.xml.in: container name <name> arguments	container name <text> arguments <value>
×	not yet documented	Nothing found in XML Definitions	container name <text> cap-add <value>
×	not yet documented	container.xml.in: container name <name> capability	Nothing found in VyOS
×	not yet documented	container.xml.in: container name <name> command	container name <text> command <value>
×	not yet documented	container.xml.in: container name <name> description	container name <text> description <value>
×	not yet documented	container.xml.in: container name <name> device <device> destination	container name <text> device <text> destination <value>
×	not yet documented	container.xml.in: container name <name> device <device> source	container name <text> device <text> source <value>
×	not yet documented	container.xml.in: container name <name> disable	container name <text> disable
×	not yet documented	container.xml.in: container name <name> entrypoint	container name <text> entrypoint <value>
×	not yet documented	container.xml.in: container name <name> environment <environment> value	container name <text> environment <text> value <value>
×	not yet documented	container.xml.in: container name <name> gid	Nothing found in VyOS
×	not yet documented	container.xml.in: container name <name> host-name	container name <text> host-name <value>

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Table 1 – continued from previous page

Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	container.xml.in: container name <name> image	container name <text> im- age <value>
×	not yet documented	container.xml.in: container name <name> label <label> value	Nothing found in VyOS
×	not yet documented	container.xml.in: container name <name> memory	container name <text> memory <value>
×	not yet documented	container.xml.in: container name <name> network <network> address	container name <text> network <text> address <value>
×	not yet documented	container.xml.in: container name <name> port <port> destination	container name <text> port <text> destination <value>
×	not yet documented	container.xml.in: container name <name> port <port> listen-address	container name <text> port <text> listen-address <value>
×	not yet documented	container.xml.in: container name <name> port <port> protocol	container name <text> port <text> protocol <value>
×	not yet documented	container.xml.in: container name <name> port <port> source	container name <text> port <text> source <value>
×	not yet documented	container.xml.in: container name <name> restart	container name <text> restart <value>
×	not yet documented	container.xml.in: container name <name> shared-memory	container name <text> shared-memory <value>
×	not yet documented	container.xml.in: container name <name> uid	Nothing found in VyOS
×	not yet documented	container.xml.in: container name <name> volume <volume> destina- tion	container name <text> volume <text> destination <value>
×	not yet documented	container.xml.in: container name <name> volume <volume> mode	container name <text> volume <text> mode <value>
×	not yet documented	container.xml.in: container name <name> volume <volume> propa- gation	container name <text> volume <text> propaga- tion <value>
×	not yet documented	container.xml.in: container name <name> volume <volume> source	container name <text> volume <text> source <value>

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Table 1 – continued from previous page

Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set container network <name></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set container network <name> description</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set container network <name> mtu <number></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set container network <name> no-name-server</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set container network <name> prefix <ipv4/ipv6></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set container network <name> vrf <name></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	container.xml.in: container network <network> description	container network <text> description <value>
×	not yet documented	container.xml.in: container network <network> prefix	container network <text> prefix <value>
×	not yet documented	container.xml.in: container network <network> vrf	container network <text> vrf <value>
×	<code>set container registry <name></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set container registry <name> authentication password</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set container registry <name> authentication username</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set container registry <name> disable</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set container registry <name> insecure</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set container registry <name> mirror address <address></code>	Nothing found in XML Definitions	Nothing found in VyOS

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Table 1 – continued from previous page

Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set container registry <name> mirror host-name <host-name></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set container registry <name> mirror path <path></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set container registry <name> mirror port <port></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	container.xml.in: container registry <registry> authentication password	container registry <text> authentication password <value>
×	not yet documented	container.xml.in: container registry <registry> authentication username	container registry <text> authentication username <value>
×	not yet documented	container.xml.in: container registry <registry> disable	container registry <text> disable
×	<code>copy</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>delete</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>delete protocols static route 0.0.0.0/0</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>exit [discard]</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall bridge ...</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall bridge <options></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall bridge [forward output] filter rule <1-999999> set connection-mark <0-2147483647></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall bridge [prerouting forward output] filter rule <1-999999> set dscp <0-63></code>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set firewall bridge [prerouting / forward / output] filter rule <1-999999> set hop-limit <0-255></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall bridge [prerouting / forward / output] filter rule <1-999999> set mark <1-2147483647></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall bridge [prerouting / forward / output] filter rule <1-999999> set tcp-mss <500-1460></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall bridge [prerouting / forward / output] filter rule <1-999999> set ttl <0-255></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall bridge forward filter default-action [accept drop]</code>	firewall.xml.in: firewall bridge forward fil- ter default-action	Nothing found in VyOS
×	<code>set firewall bridge forward filter default-log</code>	firewall.xml.in: firewall bridge forward fil- ter default-log	Nothing found in VyOS
×	not yet documented	firewall.xml.in: firewall bridge forward fil- ter description	Nothing found in VyOS
×	<code>set firewall bridge forward filter rule <1-999999> action [accept continue / drop jump queue return]</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall bridge forward filter rule <1-999999> description <text></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall bridge forward filter rule <1-999999> disable</code>	Nothing found in XML Definitions	Nothing found in VyOS

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Table 1 – continued from previous page

Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set firewall bridge forward filter rule <1-999999> ethernet-type [802.1q 802.1ad arp ipv4 ipv6]</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall bridge forward filter rule <1-999999> jump-target <text></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall bridge forward filter rule <1-999999> log</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall bridge forward filter rule <1-999999> log-options group <0-65535></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall bridge forward filter rule <1-999999> log-options level [emerg alert crit err warn notice info debug]</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall bridge forward filter rule <1-999999> log-options queue-threshold <0-65535></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall bridge forward filter rule <1-999999> log-options snapshot-length <0-9000></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall bridge forward filter rule <1-999999> queue-options bypass</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall bridge forward filter rule <1-999999> queue-options fanout</code>	Nothing found in XML Definitions	Nothing found in VyOS

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Table 1 – continued from previous page

Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set firewall bridge forward filter rule <1-999999> queue <0-65535></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall bridge forward filter rule <1-999999> vlan ethernet-type [802.1q 802.1ad arp ipv4 ipv6]</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall bridge forward filter rule <1-999999> vlan id <0-4096></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall bridge forward filter rule <1-999999> vlan priority <0-7></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	firewall.xml.in: firewall bridge forward filter rule <rule> action	Nothing found in VyOS
×	not yet documented	firewall.xml.in: firewall bridge forward filter rule <rule> destination mac-address	Nothing found in VyOS
×	not yet documented	firewall.xml.in: firewall bridge forward filter rule <rule> disable	Nothing found in VyOS
×	not yet documented	firewall.xml.in: firewall bridge forward filter rule <rule> inbound-interface group	Nothing found in VyOS
×	not yet documented	firewall.xml.in: firewall bridge forward filter rule <rule> inbound-interface name	Nothing found in VyOS
×	not yet documented	firewall.xml.in: firewall bridge forward filter rule <rule> jump-target	Nothing found in VyOS
×	not yet documented	firewall.xml.in: firewall bridge forward filter rule <rule> log	Nothing found in VyOS
×	not yet documented	firewall.xml.in: firewall bridge forward filter rule <rule> log-options group	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	firewall.xml.in: firewall bridge forward filter rule <rule> log-options level	Nothing found in VyOS
×	not yet documented	firewall.xml.in: firewall bridge forward filter rule <rule> log-options queue-threshold	Nothing found in VyOS
×	not yet documented	firewall.xml.in: firewall bridge forward filter rule <rule> log-options snapshot-length	Nothing found in VyOS
×	not yet documented	firewall.xml.in: firewall bridge forward filter rule <rule> outbound-interface group	Nothing found in VyOS
×	not yet documented	firewall.xml.in: firewall bridge forward filter rule <rule> outbound-interface name	Nothing found in VyOS
×	not yet documented	firewall.xml.in: firewall bridge forward filter rule <rule> queue	Nothing found in VyOS
×	not yet documented	firewall.xml.in: firewall bridge forward filter rule <rule> queue-options	Nothing found in VyOS
×	not yet documented	firewall.xml.in: firewall bridge forward filter rule <rule> source mac-address	Nothing found in VyOS
×	not yet documented	firewall.xml.in: firewall bridge forward filter rule <rule> vlan id	Nothing found in VyOS
×	not yet documented	firewall.xml.in: firewall bridge forward filter rule <rule> vlan priority	Nothing found in VyOS
×	<i>set firewall bridge input filter default-action [accept drop]</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set firewall bridge input filter default-log</i>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set firewall bridge input filter rule <1-999999> action [accept continue drop jump queue return]</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall bridge input filter rule <1-999999> description <text></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall bridge input filter rule <1-999999> disable</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall bridge input filter rule <1-999999> ethernet-type [802. 1q 802.1ad arp ipv4 ipv6]</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall bridge input filter rule <1-999999> jump-target <text></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall bridge input filter rule <1-999999> log</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall bridge input filter rule <1-999999> log-options group <0-65535></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall bridge input filter rule <1-999999> log-options level [emerg alert crit err warn notice info debug]</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall bridge input filter rule <1-999999> log-options queue-threshold <0-65535></code>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set firewall bridge input filter rule <1-999999> log-options snapshot-length <0-9000></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall bridge input filter rule <1-999999> queue-options bypass</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall bridge input filter rule <1-999999> queue-options fanout</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall bridge input filter rule <1-999999> queue <0-65535></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall bridge input filter rule <1-999999> vlan ethernet-type [802. 1q 802.1ad arp ipv4 ipv6]</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall bridge input filter rule <1-999999> vlan id <0-4096></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall bridge input filter rule <1-999999> vlan priority <0-7></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall bridge name <name> default-action [accept continue drop jump reject return]</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall bridge name <name> default-jump-target <text></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall bridge name <name> default-log</code>	Nothing found in XML Definitions	Nothing found in VyOS

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Table 1 – continued from previous page

Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set firewall bridge name <name> description <text></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall bridge name <name> rule <1-999999> action [accept continue drop jump queue return]</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall bridge name <name> rule <1-999999> description <text></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall bridge name <name> rule <1-999999> disable</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall bridge name <name> rule <1-999999> ethernet-type [802. 1q 802.1ad arp ipv4 ipv6]</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall bridge name <name> rule <1-999999> jump-target <text></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall bridge name <name> rule <1-999999> log</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall bridge name <name> rule <1-999999> log-options group <0-65535></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall bridge name <name> rule <1-999999> log-options level [emerg alert crit err warn notice info debug]</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall bridge name <name> rule <1-999999> log-options queue-threshold <0-65535></code>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set firewall bridge name <name> rule <1-999999> log-options snapshot-length <0-9000></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall bridge name <name> rule <1-999999> queue-options bypass</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall bridge name <name> rule <1-999999> queue-options fanout</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall bridge name <name> rule <1-999999> queue <0-65535></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall bridge name <name> rule <1-999999> vlan ethernet-type [802. 1q 802.1ad arp ipv4 ipv6]</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall bridge name <name> rule <1-999999> vlan id <0-4096></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall bridge name <name> rule <1-999999> vlan priority <0-7></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	firewall.xml.in: firewall bridge name <name> default-action	Nothing found in VyOS
×	not yet documented	firewall.xml.in: firewall bridge name <name> default-jump- target	Nothing found in VyOS
×	not yet documented	firewall.xml.in: firewall bridge name <name> default-log	Nothing found in VyOS
×	not yet documented	firewall.xml.in: firewall bridge name <name> description	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	firewall.xml.in: firewall bridge name <name> rule <rule> action	Nothing found in VyOS
×	not yet documented	firewall.xml.in: firewall bridge name <name> rule <rule> destination mac-address	Nothing found in VyOS
×	not yet documented	firewall.xml.in: firewall bridge name <name> rule <rule> disable	Nothing found in VyOS
×	not yet documented	firewall.xml.in: firewall bridge name <name> rule <rule> inbound-interface group	Nothing found in VyOS
×	not yet documented	firewall.xml.in: firewall bridge name <name> rule <rule> inbound-interface name	Nothing found in VyOS
×	not yet documented	firewall.xml.in: firewall bridge name <name> rule <rule> jump-target	Nothing found in VyOS
×	not yet documented	firewall.xml.in: firewall bridge name <name> rule <rule> log	Nothing found in VyOS
×	not yet documented	firewall.xml.in: firewall bridge name <name> rule <rule> log-options group	Nothing found in VyOS
×	not yet documented	firewall.xml.in: firewall bridge name <name> rule <rule> log-options level	Nothing found in VyOS
×	not yet documented	firewall.xml.in: firewall bridge name <name> rule <rule> log- options queue-threshold	Nothing found in VyOS
×	not yet documented	firewall.xml.in: firewall bridge name <name> rule <rule> log- options snapshot-length	Nothing found in VyOS
×	not yet documented	firewall.xml.in: firewall bridge name <name> rule <rule> outbound-interface group	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	firewall.xml.in: firewall bridge name <name> rule <rule> outbound-interface name	Nothing found in VyOS
×	not yet documented	firewall.xml.in: firewall bridge name <name> rule <rule> queue	Nothing found in VyOS
×	not yet documented	firewall.xml.in: firewall bridge name <name> rule <rule> queue-options	Nothing found in VyOS
×	not yet documented	firewall.xml.in: firewall bridge name <name> rule <rule> source mac-address	Nothing found in VyOS
×	not yet documented	firewall.xml.in: firewall bridge name <name> rule <rule> vlan id	Nothing found in VyOS
×	not yet documented	firewall.xml.in: firewall bridge name <name> rule <rule> vlan priority	Nothing found in VyOS
×	<i>set firewall bridge output filter default-action [accept drop]</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set firewall bridge output filter default-log</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set firewall bridge output filter rule <1-999999> action [accept continue drop jump queue return]</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set firewall bridge output filter rule <1-999999> description <text></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set firewall bridge output filter rule <1-999999> disable</i>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set firewall bridge output filter rule <1-999999> ethernet-type [802. 1q 802.1ad arp ipv4 ipv6]</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall bridge output filter rule <1-999999> jump-target <text></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall bridge output filter rule <1-999999> log</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall bridge output filter rule <1-999999> log-options group <0-65535></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall bridge output filter rule <1-999999> log-options level [emerg alert crit err warn notice info debug]</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall bridge output filter rule <1-999999> log-options queue-threshold <0-65535></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall bridge output filter rule <1-999999> log-options snapshot-length <0-9000></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall bridge output filter rule <1-999999> queue-options bypass</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall bridge output filter rule <1-999999> queue-options fanout</code>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set firewall bridge output filter rule <1-999999> queue <0-65535></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall bridge output filter rule <1-999999> vlan ethernet-type [802.1q 802.1ad arp ipv4 ipv6]</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall bridge output filter rule <1-999999> vlan id <0-4096></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall bridge output filter rule <1-999999> vlan priority <0-7></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall bridge prerouting filter default-action [accept drop]</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall bridge prerouting filter default-log</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall bridge prerouting filter rule <1-999999> action [accept continue drop jump notrack queue return]</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall bridge prerouting filter rule <1-999999> description <text></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall bridge prerouting filter rule <1-999999> disable</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall bridge prerouting filter rule <1-999999> ethernet-type [802.1q 802.1ad arp ipv4 ipv6]</code>	Nothing found in XML Definitions	Nothing found in VyOS

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Table 1 – continued from previous page

Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set firewall bridge prerouting filter rule <1-999999> jump-target <text></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall bridge prerouting filter rule <1-999999> log</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall bridge prerouting filter rule <1-999999> log-options group <0-65535></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall bridge prerouting filter rule <1-999999> log-options level [emerg alert crit err warn notice info debug]</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall bridge prerouting filter rule <1-999999> log-options queue-threshold <0-65535></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall bridge prerouting filter rule <1-999999> log-options snapshot-length <0-9000></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall bridge prerouting filter rule <1-999999> queue-options bypass</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall bridge prerouting filter rule <1-999999> queue-options fanout</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall bridge prerouting filter rule <1-999999> queue <0-65535></code>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set firewall bridge prerouting filter rule <1-999999> vlan ethernet-type [802.1q 802.1ad arp ipv4 ipv6]</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall bridge prerouting filter rule <1-999999> vlan id <0-4096></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall bridge prerouting filter rule <1-999999> vlan priority <0-7></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall flowtable ...</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall flowtable <flow_table_name> description <text></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall flowtable <flow_table_name> interface <iface></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall flowtable <flow_table_name> offload <hardware software></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	firewall.xml.in: firewall flowtable <flowtable> descrip- tion	Nothing found in VyOS
×	not yet documented	firewall.xml.in: firewall flowtable <flowtable> interface	Nothing found in VyOS
×	not yet documented	firewall.xml.in: firewall flowtable <flowtable> offload	Nothing found in VyOS
×	<code>set firewall [ipv4 ipv6] forward filter rule <1-999999> action offload</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall [ipv4 ipv6] forward filter rule <1-999999> offload-target <flowtable></code>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set firewall global-options ..</code>	Nothing found in XML Definitions	Nothing found in VyOS
✓	<code>set firewall global-options all-ping [enable disable]</code>	firewall.xml.in: firewall global-options all-ping	firewall global-options all-ping <value>
×	<code>set firewall global-options apply-to-bridged-traf [ipv4 ipv6]</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall global-options apply-to-bridged-traf ipv4</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall global-options apply-to-bridged-traf ipv6</code>	Nothing found in XML Definitions	Nothing found in VyOS
✓	<code>set firewall global-options broadcast-ping [enable disable]</code>	firewall.xml.in: firewall global-options broadcast-ping	firewall global-options broadcast-ping <value>
×	not yet documented	firewall.xml.in: firewall global-options directed-broadcast	Nothing found in VyOS
✓	<code>set firewall global-options ip-src-route [enable disable]</code>	firewall.xml.in: firewall global-options ip-src-route	firewall global-options ip-src-route <value>
✓	<code>set firewall global-options ipv6-receive-redirect [enable disable]</code>	firewall.xml.in: firewall global-options ipv6-receive-redirects	firewall global-options ipv6-receive-redirects <value>
×	not yet documented	firewall.xml.in: firewall global-options ipv6-source-validation	Nothing found in VyOS
✓	<code>set firewall global-options ipv6-src-route [enable disable]</code>	firewall.xml.in: firewall global-options ipv6-src-route	firewall global-options ipv6-src-route <value>
✓	<code>set firewall global-options log-martians [enable disable]</code>	firewall.xml.in: firewall global-options log-martians	firewall global-options log-martians <value>
✓	<code>set firewall global-options receive-redirects [enable disable]</code>	firewall.xml.in: firewall global-options receive-redirects	firewall global-options receive-redirects <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	firewall.xml.in: firewall global-options resolver-cache	firewall global-options resolver-cache
×	not yet documented	firewall.xml.in: firewall global-options resolver-interval	firewall global-options resolver-interval <value>
✓	<i>set firewall global-options send-redirects [enable disable]</i>	firewall.xml.in: firewall global-options send-redirects	firewall global-options send-redirects <value>
✓	<i>set firewall global-options source-validation [strict loose disable]</i>	firewall.xml.in: firewall global-options source-validation	firewall global-options source-validation <value>
×	<i>set firewall global-options state-policy established action [accept drop reject]</i>	firewall.xml.in: firewall global-options state-policy established action	Nothing found in VyOS
×	<i>set firewall global-options state-policy established log</i>	firewall.xml.in: firewall global-options state-policy established log	Nothing found in VyOS
×	<i>set firewall global-options state-policy established log-level [emerg alert crit err warn notice info debug]</i>	firewall.xml.in: firewall global-options state-policy established log-level	Nothing found in VyOS
×	<i>set firewall global-options state-policy invalid action [accept drop reject]</i>	firewall.xml.in: firewall global-options state-policy invalid action	Nothing found in VyOS
×	<i>set firewall global-options state-policy invalid log</i>	firewall.xml.in: firewall global-options state-policy invalid log	Nothing found in VyOS
×	<i>set firewall global-options state-policy invalid log-level [emerg alert crit err warn notice info debug]</i>	firewall.xml.in: firewall global-options state-policy invalid log- level	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set firewall global-options state-policy related action [accept drop reject]</code>	firewall.xml.in: firewall global-options state-policy related action	Nothing found in VyOS
×	<code>set firewall global-options state-policy related log</code>	firewall.xml.in: firewall global-options state-policy related log	Nothing found in VyOS
×	<code>set firewall global-options state-policy related log-level [emerg alert crit err warn notice info debug]</code>	firewall.xml.in: firewall global-options state-policy related log- level	Nothing found in VyOS
✓	<code>set firewall global-options syn-cookies [enable disable]</code>	firewall.xml.in: firewall global-options syn-cookies	firewall global-options syn-cookies <value>
×	<code>set firewall global-options timeout icmp <1-21474836></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall global-options timeout other <1-21474836></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall global-options timeout tcp close <1-21474836></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall global-options timeout tcp close-wait <1-21474836></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall global-options timeout tcp established <1-21474836></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall global-options timeout tcp fin-wait <1-21474836></code>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set firewall global-options timeout tcp last-ack <1-21474836></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall global-options timeout tcp syn-recv <1-21474836></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall global-options timeout tcp syn-sent <1-21474836></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall global-options timeout tcp time-wait <1-21474836></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall global-options timeout udp other <1-21474836></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall global-options timeout udp stream <1-21474836></code>	Nothing found in XML Definitions	Nothing found in VyOS
✓	<code>set firewall global-options twa-hazards-protection [enable disable]</code>	firewall.xml.in: firewall global-options twa-hazards-protection	firewall global-options twa-hazards-protection <value>
×	<code>set firewall group ...</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall group address-group <name> address [address address range]</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall group address-group <name> description <text></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	firewall.xml.in: firewall group address-group <address-group> address	firewall group address-group <text> address <value>
×	not yet documented	firewall.xml.in: firewall group address-group <address-group> description	firewall group address-group <text> description <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	firewall.xml.in: firewall group address-group <address-group> include	firewall group address-group <text> include <value>
×	<i>set firewall group domain-group <name> address <domain></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set firewall group domain-group <name> description <text></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	firewall.xml.in: firewall group domain-group <domain-group> address	firewall group domain-group <text> address <value>
×	not yet documented	firewall.xml.in: firewall group domain-group <domain-group> description	firewall group domain-group <text> description <value>
×	<i>set firewall group dynamic-group address-group <name></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set firewall group dynamic-group address-group <name> description <text></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	firewall.xml.in: firewall group dynamic-group address-group <address-group> description	Nothing found in VyOS
×	<i>set firewall group dynamic-group ipv6-address-group <name></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set firewall group dynamic-group ipv6-address-group <name> description <text></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	firewall.xml.in: firewall group dynamic-group ipv6-address-group <ipv6-address-group> description	Nothing found in VyOS
×	<i>set firewall group interface-group <name> description <text></i>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<i>set firewall group interface-group</i> <i><name> interface</i> <i><text></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	firewall.xml.in: firewall group interface-group <interface-group> description	firewall group interface-group <text> description <value>
×	not yet documented	firewall.xml.in: firewall group interface-group <interface-group> include	firewall group interface-group <text> include <value>
×	not yet documented	firewall.xml.in: firewall group interface-group <interface-group> interface	firewall group interface-group <text> interface <value>
×	<i>set firewall group ipv6-address-group</i> <i><name> address</i> <i><address></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set firewall group ipv6-address-group</i> <i><name> description</i> <i><text></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	firewall.xml.in: firewall group ipv6-address-group <ipv6-address-group> address	firewall group ipv6-address-group <text> address <value>
×	not yet documented	firewall.xml.in: firewall group ipv6-address-group <ipv6-address-group> description	firewall group ipv6-address-group <text> description <value>
×	not yet documented	firewall.xml.in: firewall group ipv6-address-group <ipv6-address-group> include	firewall group ipv6-address-group <text> include <value>
×	<i>set firewall group ipv6-network-group</i> <i><name> description</i> <i><text></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set firewall group ipv6-network-group</i> <i><name> network</i> <i><CIDR></i>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	firewall.xml.in: firewall group ipv6- network-group <ipv6- network-group> descrip- tion	firewall group ipv6- network-group <text> description <value>
×	not yet documented	firewall.xml.in: firewall group ipv6- network-group <ipv6- network-group> include	firewall group ipv6- network-group <text> include <value>
×	not yet documented	firewall.xml.in: firewall group ipv6- network-group <ipv6- network-group> network	firewall group ipv6- network-group <text> network <value>
×	<i>set firewall group mac-group <name> description <text></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set firewall group mac-group <name> mac-address <mac-address></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	firewall.xml.in: firewall group mac-group <mac-group> description <value>	firewall group mac- group <text> description <value>
×	not yet documented	firewall.xml.in: firewall group mac-group <mac-group> include	firewall group mac-group <text> include <value>
×	not yet documented	firewall.xml.in: firewall group mac- group <mac-group> mac-address	firewall group mac-group <text> mac-address <value>
×	<i>set firewall group network-group <name> description <text></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set firewall group network-group <name> network <CIDR></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	firewall.xml.in: firewall group network- group <network-group> description	firewall group network- group <text> description <value>
×	not yet documented	firewall.xml.in: firewall group network- group <network-group> include	firewall group network- group <text> include <value>
×	not yet documented	firewall.xml.in: firewall group network- group <network-group> network	firewall group network- group <text> network <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set firewall group port-group <name> description <text></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall group port-group <name> port [portname portnumber startport-endport]</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	firewall.xml.in: firewall group port-group <port-group> description	firewall group port-group <text> description <value>
×	not yet documented	firewall.xml.in: firewall group port-group <port-group> include	firewall group port-group <text> include <value>
×	not yet documented	firewall.xml.in: firewall group port-group <port-group> port	firewall group port-group <text> port <value>
×	<code>set firewall group remote-group <name> description <text></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall group remote-group <name> url <http(s) url></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 ...</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 [input forward] filter rule <1-999999> action synproxy</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 [forward input output] filter rule <1-999999> add-address-to-group destination-address address-group <name></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 [forward input output] filter rule <1-999999> add-address-to-group source-address address-group <name></code>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set firewall ipv4 [forward input output] filter rule <1-999999> add-address-to-group [destination-address source-address] timeout <timeout></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 [input forward] filter rule <1-999999> protocol tcp</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 [input forward] filter rule <1-999999> synproxy tcp mss <501-65535></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 [input forward] filter rule <1-999999> synproxy tcp window-scale <1-14></code>	Nothing found in XML Definitions	Nothing found in VyOS
✓	<code>set firewall ipv4 forward filter default-action [accept drop]</code>	firewall.xml.in: firewall ipv4 forward filter default-action	firewall ipv4 forward filter default-action <value>
×	<code>set firewall ipv4 forward filter default-log</code>	firewall.xml.in: firewall ipv4 forward filter default-log	Nothing found in VyOS
×	not yet documented	firewall.xml.in: firewall ipv4 forward filter description	firewall ipv4 forward filter description <value>
×	<code>set firewall ipv4 forward filter rule <1-999999> action [accept continue drop jump queue reject return synproxy]</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 forward filter rule <1-999999> connection-mark <1-2147483647></code>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set firewall ipv4 forward filter rule <1-999999> connection-status nat [destination / source]</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 forward filter rule <1-999999> conntrack-helper <module></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 forward filter rule <1-999999> description <text></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 forward filter rule <1-999999> destination address [address / addressrange / CIDR]</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 forward filter rule <1-999999> destination address-mask [address]</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 forward filter rule <1-999999> destination fqdn <fqdn></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 forward filter rule <1-999999> destination geoip asn <1-4294967294></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 forward filter rule <1-999999> destination geoip country-code <country></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 forward filter rule <1-999999> destination geoip inverse-match</code>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set firewall ipv4 forward filter rule <1-999999> destination group address-group <name !name></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 forward filter rule <1-999999> destination group domain-group <name !name></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 forward filter rule <1-999999> destination group dynamic-address-group <name !name></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 forward filter rule <1-999999> destination group mac-group <name !name></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 forward filter rule <1-999999> destination group network-group <name !name></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 forward filter rule <1-999999> destination group port-group <name !name></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 forward filter rule <1-999999> destination port [1-65535 portname start-end]</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 forward filter rule <1-999999> disable</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 forward filter rule <1-999999> dscp [0-63 start-end]</code>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set firewall ipv4 forward filter rule <1-999999> dscp-exclude [0-63 / start-end]</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 forward filter rule <1-999999> fragment [match-frag / match-non-frag]</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 forward filter rule <1-999999> icmp [code type] <0-255></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 forward filter rule <1-999999> icmp type-name <text></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 forward filter rule <1-999999> inbound-interface group <iface_group></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 forward filter rule <1-999999> inbound-interface name <iface></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 forward filter rule <1-999999> ipsec [match-ipsec-in / match-ipsec-out / match-none-in / match-none-out]</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 forward filter rule <1-999999> jump-target <text></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 forward filter rule <1-999999> limit burst <0-4294967295></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 forward filter rule <1-999999> limit rate <text></code>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set firewall ipv4 forward filter rule <1-999999> log</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 forward filter rule <1-999999> log-options group <0-65535></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 forward filter rule <1-999999> log-options level [emerg alert crit err warn notice info debug]</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 forward filter rule <1-999999> log-options queue-threshold <0-65535></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 forward filter rule <1-999999> log-options snapshot-length <0-9000></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 forward filter rule <1-999999> outbound-interface group <iface_group></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 forward filter rule <1-999999> outbound-interface name <iface></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 forward filter rule <1-999999> packet-length-exclude <text></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 forward filter rule <1-999999> packet-length <text></code>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set firewall ipv4 forward filter rule <1-999999> packet-type [broadcast host multicast other]</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 forward filter rule <1-999999> protocol [<text> <0-255> all tcp_udp]</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 forward filter rule <1-999999> queue-options bypass</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 forward filter rule <1-999999> queue-options fanout</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 forward filter rule <1-999999> queue <0-65535></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 forward filter rule <1-999999> recent count <1-255></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 forward filter rule <1-999999> recent time [second minute hour]</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 forward filter rule <1-999999> set connection-mark <0-2147483647></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 forward filter rule <1-999999> set dscp <0-63></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 forward filter rule <1-999999> set mark <1-2147483647></code>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set firewall ipv4 forward filter rule <1-999999> set tcp-mss <500-1460></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 forward filter rule <1-999999> set ttl <0-255></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 forward filter rule <1-999999> source address [address addressrange CIDR]</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 forward filter rule <1-999999> source address-mask [address]</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 forward filter rule <1-999999> source fqdn <fqdn></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 forward filter rule <1-999999> source geoip asn <1-4294967294></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 forward filter rule <1-999999> source geoip country-code <country></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 forward filter rule <1-999999> source geoip inverse-match</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 forward filter rule <1-999999> source group address-group <name !name></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 forward filter rule <1-999999> source group domain-group <name !name></code>	Nothing found in XML Definitions	Nothing found in VyOS

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Table 1 – continued from previous page

Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set firewall ipv4 forward filter rule <1-999999> source group dynamic-address-group <name !name></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 forward filter rule <1-999999> source group mac-group <name !name></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 forward filter rule <1-999999> source group network-group <name !name></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 forward filter rule <1-999999> source group port-group <name !name></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 forward filter rule <1-999999> source mac-address <mac-address></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 forward filter rule <1-999999> source port [1-65535 portname start-end]</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 forward filter rule <1-999999> state [established invalid new related]</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 forward filter rule <1-999999> tcp flags [not] <text></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 forward filter rule <1-999999> time startdate <text></code>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set firewall ipv4 forward filter rule <1-999999> time starttime <text></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 forward filter rule <1-999999> time stopdate <text></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 forward filter rule <1-999999> time stoptime <text></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 forward filter rule <1-999999> time weekdays <text></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 forward filter rule <1-999999> ttl <eq / gt / lt> <0-255></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	firewall.xml.in: firewall ipv4 forward filter rule <rule> action	firewall ipv4 forward filter rule <text> action <value>
×	not yet documented	firewall.xml.in: firewall ipv4 forward filter rule <rule> add-address-to-group destination-address address-group	Nothing found in VyOS
×	not yet documented	firewall.xml.in: firewall ipv4 forward filter rule <rule> add-address-to-group destination-address timeout	Nothing found in VyOS
×	not yet documented	firewall.xml.in: firewall ipv4 forward filter rule <rule> add-address-to-group source-address address-group	Nothing found in VyOS
×	not yet documented	firewall.xml.in: firewall ipv4 forward filter rule <rule> add-address-to-group source-address timeout	Nothing found in VyOS
×	not yet documented	firewall.xml.in: firewall ipv4 forward filter rule <rule> connection-mark	firewall ipv4 forward filter rule <text> connection-mark <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	firewall.xml.in: firewall ipv4 forward filter rule <rule> connection-status nat	firewall ipv4 forward filter rule <text> connection-status nat <value>
×	not yet documented	firewall.xml.in: firewall ipv4 forward filter rule <rule> conntrack-helper	Nothing found in VyOS
×	not yet documented	firewall.xml.in: firewall ipv4 forward filter rule <rule> description	firewall ipv4 forward filter rule <text> description <value>
×	not yet documented	firewall.xml.in: firewall ipv4 forward filter rule <rule> destination address	firewall ipv4 forward filter rule <text> destination address <value>
×	not yet documented	firewall.xml.in: firewall ipv4 forward filter rule <rule> destination address-mask	firewall ipv4 forward filter rule <text> destination address-mask <value>
×	not yet documented	firewall.xml.in: firewall ipv4 forward filter rule <rule> destination fqdn	firewall ipv4 forward filter rule <text> destination fqdn <value>
×	not yet documented	firewall.xml.in: firewall ipv4 forward filter rule <rule> destination geoip country-code	firewall ipv4 forward filter rule <text> destination geoip country-code <value>
×	not yet documented	firewall.xml.in: firewall ipv4 forward filter rule <rule> destination geoip inverse-match	firewall ipv4 forward filter rule <text> destination geoip inverse-match
×	not yet documented	firewall.xml.in: firewall ipv4 forward filter rule <rule> destination group address-group	firewall ipv4 forward filter rule <text> destination group address-group <value>
×	not yet documented	firewall.xml.in: firewall ipv4 forward filter rule <rule> destination group domain-group	firewall ipv4 forward filter rule <text> destination group domain-group <value>
×	not yet documented	firewall.xml.in: firewall ipv4 forward filter rule <rule> destination group dynamic-address-group	Nothing found in VyOS
×	not yet documented	firewall.xml.in: firewall ipv4 forward filter rule <rule> destination group mac-group	firewall ipv4 forward filter rule <text> destination group mac-group <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	firewall.xml.in: firewall ipv4 forward filter rule <rule> destination group network-group	firewall ipv4 forward filter rule <text> destination group network-group <value>
×	not yet documented	firewall.xml.in: firewall ipv4 forward filter rule <rule> destination group port-group	firewall ipv4 forward filter rule <text> destination group port-group <value>
×	not yet documented	firewall.xml.in: firewall ipv4 forward filter rule <rule> destination mac-address	firewall ipv4 forward filter rule <text> destination mac-address <value>
×	not yet documented	firewall.xml.in: firewall ipv4 forward filter rule <rule> destination port	firewall ipv4 forward filter rule <text> destination port <value>
×	not yet documented	firewall.xml.in: firewall ipv4 forward filter rule <rule> disable	firewall ipv4 forward filter rule <text> disable
×	not yet documented	firewall.xml.in: firewall ipv4 forward filter rule <rule> dscp	firewall ipv4 forward filter rule <text> dscp <value>
×	not yet documented	firewall.xml.in: firewall ipv4 forward filter rule <rule> dscp-exclude	firewall ipv4 forward filter rule <text> dscp-exclude <value>
×	not yet documented	firewall.xml.in: firewall ipv4 forward filter rule <rule> fragment match-frag	firewall ipv4 forward filter rule <text> fragment match-frag
×	not yet documented	firewall.xml.in: firewall ipv4 forward filter rule <rule> fragment match-non-frag	firewall ipv4 forward filter rule <text> fragment match-non-frag
×	not yet documented	firewall.xml.in: firewall ipv4 forward filter rule <rule> icmp code	firewall ipv4 forward filter rule <text> icmp code <value>
×	not yet documented	firewall.xml.in: firewall ipv4 forward filter rule <rule> icmp type	firewall ipv4 forward filter rule <text> icmp type <value>
×	not yet documented	firewall.xml.in: firewall ipv4 forward filter rule <rule> icmp type-name	firewall ipv4 forward filter rule <text> icmp type-name <value>
×	not yet documented	firewall.xml.in: firewall ipv4 forward filter rule <rule> inbound-interface group	Nothing found in VyOS

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Table 1 – continued from previous page

Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	Nothing found in XML Definitions	firewall ipv4 forward filter rule <text> inbound-interface interface-group <value>
×	not yet documented	Nothing found in XML Definitions	firewall ipv4 forward filter rule <text> inbound-interface interface-name <value>
×	not yet documented	firewall.xml.in: firewall ipv4 forward filter rule <rule> inbound-interface name	Nothing found in VyOS
×	not yet documented	firewall.xml.in: firewall ipv4 forward filter rule <rule> ipsec match-ipsec	firewall ipv4 forward filter rule <text> ipsec match-ipsec
×	not yet documented	firewall.xml.in: firewall ipv4 forward filter rule <rule> ipsec match-none	firewall ipv4 forward filter rule <text> ipsec match-none
×	not yet documented	firewall.xml.in: firewall ipv4 forward filter rule <rule> jump-target	firewall ipv4 forward filter rule <text> jump-target <value>
×	not yet documented	firewall.xml.in: firewall ipv4 forward filter rule <rule> limit burst	firewall ipv4 forward filter rule <text> limit burst <value>
×	not yet documented	firewall.xml.in: firewall ipv4 forward filter rule <rule> limit rate	firewall ipv4 forward filter rule <text> limit rate <value>
×	not yet documented	firewall.xml.in: firewall ipv4 forward filter rule <rule> log	firewall ipv4 forward filter rule <text> log <value>
×	not yet documented	firewall.xml.in: firewall ipv4 forward filter rule <rule> log-options group	firewall ipv4 forward filter rule <text> log-options group <value>
×	not yet documented	firewall.xml.in: firewall ipv4 forward filter rule <rule> log-options level	firewall ipv4 forward filter rule <text> log-options level <value>
×	not yet documented	firewall.xml.in: firewall ipv4 forward filter rule <rule> log-options queue-threshold	firewall ipv4 forward filter rule <text> log-options queue-threshold <value>
×	not yet documented	firewall.xml.in: firewall ipv4 forward filter rule <rule> log-options snapshot-length	firewall ipv4 forward filter rule <text> log-options snapshot-length <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	firewall.xml.in: firewall ipv4 forward filter rule <rule> mark	Nothing found in VyOS
×	not yet documented	firewall.xml.in: firewall ipv4 forward filter rule <rule> offload-target	Nothing found in VyOS
×	not yet documented	firewall.xml.in: firewall ipv4 forward filter rule <rule> outbound-interface group	Nothing found in VyOS
×	not yet documented	Nothing found in XML Definitions	firewall ipv4 forward filter rule <text> outbound-interface interface-group <value>
×	not yet documented	Nothing found in XML Definitions	firewall ipv4 forward filter rule <text> outbound-interface interface-name <value>
×	not yet documented	firewall.xml.in: firewall ipv4 forward filter rule <rule> outbound-interface name	Nothing found in VyOS
×	not yet documented	firewall.xml.in: firewall ipv4 forward filter rule <rule> packet-length	firewall ipv4 forward filter rule <text> packet-length <value>
×	not yet documented	firewall.xml.in: firewall ipv4 forward filter rule <rule> packet-length-exclude	firewall ipv4 forward filter rule <text> packet-length-exclude <value>
×	not yet documented	firewall.xml.in: firewall ipv4 forward filter rule <rule> packet-type	firewall ipv4 forward filter rule <text> packet-type <value>
×	not yet documented	firewall.xml.in: firewall ipv4 forward filter rule <rule> protocol	firewall ipv4 forward filter rule <text> protocol <value>
×	not yet documented	firewall.xml.in: firewall ipv4 forward filter rule <rule> queue	firewall ipv4 forward filter rule <text> queue <value>
×	not yet documented	firewall.xml.in: firewall ipv4 forward filter rule <rule> queue-options	firewall ipv4 forward filter rule <text> queue-options <value>
×	not yet documented	firewall.xml.in: firewall ipv4 forward filter rule <rule> recent count	firewall ipv4 forward filter rule <text> recent count <value>
×	not yet documented	firewall.xml.in: firewall ipv4 forward filter rule <rule> recent time	firewall ipv4 forward filter rule <text> recent time <value>

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Table 1 – continued from previous page

Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	firewall.xml.in: firewall ipv4 forward filter rule <rule> source address	firewall ipv4 forward filter rule <text> source address <value>
×	not yet documented	firewall.xml.in: firewall ipv4 forward filter rule <rule> source address-mask	firewall ipv4 forward filter rule <text> source address-mask <value>
×	not yet documented	firewall.xml.in: firewall ipv4 forward filter rule <rule> source fqdn	firewall ipv4 forward filter rule <text> source fqdn <value>
×	not yet documented	firewall.xml.in: firewall ipv4 forward filter rule <rule> source geoip country-code	firewall ipv4 forward filter rule <text> source geoip country-code <value>
×	not yet documented	firewall.xml.in: firewall ipv4 forward filter rule <rule> source geoip inverse-match	firewall ipv4 forward filter rule <text> source geoip inverse-match
×	not yet documented	firewall.xml.in: firewall ipv4 forward filter rule <rule> source group address-group	firewall ipv4 forward filter rule <text> source group address-group <value>
×	not yet documented	firewall.xml.in: firewall ipv4 forward filter rule <rule> source group domain-group	firewall ipv4 forward filter rule <text> source group domain-group <value>
×	not yet documented	firewall.xml.in: firewall ipv4 forward filter rule <rule> source group dynamic-address-group	Nothing found in VyOS
×	not yet documented	firewall.xml.in: firewall ipv4 forward filter rule <rule> source group mac-group	firewall ipv4 forward filter rule <text> source group mac-group <value>
×	not yet documented	firewall.xml.in: firewall ipv4 forward filter rule <rule> source group network-group	firewall ipv4 forward filter rule <text> source group network-group <value>
×	not yet documented	firewall.xml.in: firewall ipv4 forward filter rule <rule> source group port-group	firewall ipv4 forward filter rule <text> source group port-group <value>
×	not yet documented	firewall.xml.in: firewall ipv4 forward filter rule <rule> source mac-address	firewall ipv4 forward filter rule <text> source mac-address <value>
×	not yet documented	firewall.xml.in: firewall ipv4 forward filter rule <rule> source port	firewall ipv4 forward filter rule <text> source port <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	firewall.xml.in: firewall ipv4 forward filter rule <rule> state	Nothing found in VyOS
×	not yet documented	Nothing found in XML Definitions	firewall ipv4 forward filter rule <text> state established <value>
×	not yet documented	Nothing found in XML Definitions	firewall ipv4 forward filter rule <text> state invalid <value>
×	not yet documented	Nothing found in XML Definitions	firewall ipv4 forward filter rule <text> state new <value>
×	not yet documented	Nothing found in XML Definitions	firewall ipv4 forward filter rule <text> state related <value>
×	not yet documented	firewall.xml.in: firewall ipv4 forward filter rule <rule> synproxy tcp mss	Nothing found in VyOS
×	not yet documented	firewall.xml.in: firewall ipv4 forward filter rule <rule> synproxy tcp window-scale	Nothing found in VyOS
×	not yet documented	firewall.xml.in: firewall ipv4 forward filter rule <rule> tcp flags ack	firewall ipv4 forward filter rule <text> tcp flags ack
×	not yet documented	firewall.xml.in: firewall ipv4 forward filter rule <rule> tcp flags cwr	firewall ipv4 forward filter rule <text> tcp flags cwr
×	not yet documented	firewall.xml.in: firewall ipv4 forward filter rule <rule> tcp flags ecn	firewall ipv4 forward filter rule <text> tcp flags ecn
×	not yet documented	firewall.xml.in: firewall ipv4 forward filter rule <rule> tcp flags fin	firewall ipv4 forward filter rule <text> tcp flags fin
×	not yet documented	firewall.xml.in: firewall ipv4 forward filter rule <rule> tcp flags not ack	firewall ipv4 forward filter rule <text> tcp flags not ack
×	not yet documented	firewall.xml.in: firewall ipv4 forward filter rule <rule> tcp flags not cwr	firewall ipv4 forward filter rule <text> tcp flags not cwr
×	not yet documented	firewall.xml.in: firewall ipv4 forward filter rule <rule> tcp flags not ecn	firewall ipv4 forward filter rule <text> tcp flags not ecn

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	firewall.xml.in: firewall ipv4 forward filter rule <rule> tcp flags not fin	firewall ipv4 forward filter rule <text> tcp flags not fin
×	not yet documented	firewall.xml.in: firewall ipv4 forward filter rule <rule> tcp flags not psh	firewall ipv4 forward fil- ter rule <text> tcp flags not psh
×	not yet documented	firewall.xml.in: firewall ipv4 forward filter rule <rule> tcp flags not rst	firewall ipv4 forward filter rule <text> tcp flags not rst
×	not yet documented	firewall.xml.in: firewall ipv4 forward filter rule <rule> tcp flags not syn	firewall ipv4 forward fil- ter rule <text> tcp flags not syn
×	not yet documented	firewall.xml.in: firewall ipv4 forward filter rule <rule> tcp flags not urg	firewall ipv4 forward fil- ter rule <text> tcp flags not urg
×	not yet documented	firewall.xml.in: firewall ipv4 forward filter rule <rule> tcp flags psh	firewall ipv4 forward filter rule <text> tcp flags psh
×	not yet documented	firewall.xml.in: firewall ipv4 forward filter rule <rule> tcp flags rst	firewall ipv4 forward filter rule <text> tcp flags rst
×	not yet documented	firewall.xml.in: firewall ipv4 forward filter rule <rule> tcp flags syn	firewall ipv4 forward filter rule <text> tcp flags syn
×	not yet documented	firewall.xml.in: firewall ipv4 forward filter rule <rule> tcp flags urg	firewall ipv4 forward filter rule <text> tcp flags urg
×	not yet documented	firewall.xml.in: firewall ipv4 forward filter rule <rule> tcp mss	firewall ipv4 forward fil- ter rule <text> tcp mss <value>
×	not yet documented	firewall.xml.in: firewall ipv4 forward filter rule <rule> time startdate	firewall ipv4 forward filter rule <text> time startdate <value>
×	not yet documented	firewall.xml.in: firewall ipv4 forward filter rule <rule> time starttime	firewall ipv4 forward filter rule <text> time starttime <value>
×	not yet documented	firewall.xml.in: firewall ipv4 forward filter rule <rule> time stopdate	firewall ipv4 forward filter rule <text> time stopdate <value>
×	not yet documented	firewall.xml.in: firewall ipv4 forward filter rule <rule> time stoptime	firewall ipv4 forward filter rule <text> time stoptime <value>
×	not yet documented	firewall.xml.in: firewall ipv4 forward filter rule <rule> time weekdays	firewall ipv4 forward filter rule <text> time weekdays <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	firewall.xml.in: firewall ipv4 forward filter rule <rule> ttl eq	firewall ipv4 forward filter rule <text> ttl eq <value>
×	not yet documented	firewall.xml.in: firewall ipv4 forward filter rule <rule> ttl gt	firewall ipv4 forward filter rule <text> ttl gt <value>
×	not yet documented	firewall.xml.in: firewall ipv4 forward filter rule <rule> ttl lt	firewall ipv4 forward filter rule <text> ttl lt <value>
✓	<code>set firewall ipv4 input filter default-action [accept drop]</code>	firewall.xml.in: firewall ipv4 input filter default-action	firewall ipv4 input filter default-action <value>
×	<code>set firewall ipv4 input filter default-log</code>	firewall.xml.in: firewall ipv4 input filter default-log	Nothing found in VyOS
×	not yet documented	firewall.xml.in: firewall ipv4 input filter description	firewall ipv4 input filter description <value>
×	<code>set firewall ipv4 input filter rule <1-999999> action [accept continue drop jump queue reject return synproxy]</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 input filter rule <1-999999> connection-mark <1-2147483647></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 input filter rule <1-999999> connection-status nat [destination source]</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 input filter rule <1-999999> conntrack-helper <module></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 input filter rule <1-999999> description <text></code>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set firewall ipv4 input filter rule <1-999999> destination address [address addressrange CIDR]</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 input filter rule <1-999999> destination address-mask [address]</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 input filter rule <1-999999> destination fqdn <fqdn></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 input filter rule <1-999999> destination geoup asn <1-4294967294></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 input filter rule <1-999999> destination geoup country-code <country></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 input filter rule <1-999999> destination geoup inverse-match</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 input filter rule <1-999999> destination group address-group <name !name></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 input filter rule <1-999999> destination group domain-group <name / !name></code>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set firewall ipv4 input filter rule <1-999999> destination group dynamic-address-group <name !name></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 input filter rule <1-999999> destination group mac-group <name !name></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 input filter rule <1-999999> destination group network-group <name !name></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 input filter rule <1-999999> destination group port-group <name !name></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 input filter rule <1-999999> destination port [1-65535 portname start-end]</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 input filter rule <1-999999> disable</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 input filter rule <1-999999> dscp [0-63 start-end]</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 input filter rule <1-999999> dscp-exclude [0-63 start-end]</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 input filter rule <1-999999> fragment [match-frag match-non-frag]</code>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set firewall ipv4 input filter rule <1-999999> icmp [code type] <0-255></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 input filter rule <1-999999> icmp type-name <text></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 input filter rule <1-999999> inbound-interface group <iface_group></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 input filter rule <1-999999> inbound-interface name <iface></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 input filter rule <1-999999> ipsec [match-ipsec-in match-none-in]</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 input filter rule <1-999999> jump-target <text></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 input filter rule <1-999999> limit burst <0-4294967295></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 input filter rule <1-999999> limit rate <text></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 input filter rule <1-999999> log</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 input filter rule <1-999999> log-options group <0-65535></code>	Nothing found in XML Definitions	Nothing found in VyOS

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Table 1 – continued from previous page

Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set firewall ipv4 input filter rule <1-999999> log-options level [emerg alert crit err warn notice info debug]</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 input filter rule <1-999999> log-options queue-threshold <0-65535></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 input filter rule <1-999999> log-options snapshot-length <0-9000></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 input filter rule <1-999999> packet-length-exclude <text></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 input filter rule <1-999999> packet-length <text></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 input filter rule <1-999999> packet-type [broadcast host multicast other]</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 input filter rule <1-999999> protocol [<text> <0-255> all tcp_udp]</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 input filter rule <1-999999> queue-options bypass</code>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set firewall ipv4 input filter rule <1-999999> queue-options fanout</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 input filter rule <1-999999> queue <0-65535></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 input filter rule <1-999999> recent count <1-255></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 input filter rule <1-999999> recent time [second minute hour]</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 input filter rule <1-999999> source address [address addressrange CIDR]</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 input filter rule <1-999999> source address-mask [address]</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 input filter rule <1-999999> source fqdn <fqdn></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 input filter rule <1-999999> source geoip asn <1-4294967294></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 input filter rule <1-999999> source geoip country-code <country></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 input filter rule <1-999999> source geoip inverse-match</code>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set firewall ipv4 input filter rule <1-999999> source group address-group <name !name></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 input filter rule <1-999999> source group domain-group <name !name></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 input filter rule <1-999999> source group dynamic-address-group <name !name></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 input filter rule <1-999999> source group mac-group <name !name></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 input filter rule <1-999999> source group network-group <name !name></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 input filter rule <1-999999> source group port-group <name !name></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 input filter rule <1-999999> source mac-address <mac-address></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 input filter rule <1-999999> source port [1-65535 portname start-end]</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 input filter rule <1-999999> state [established invalid new related]</code>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set firewall ipv4 input filter rule <1-999999> tcp flags [not] <text></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 input filter rule <1-999999> time startdate <text></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 input filter rule <1-999999> time starttime <text></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 input filter rule <1-999999> time stopdate <text></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 input filter rule <1-999999> time stoptime <text></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 input filter rule <1-999999> time weekdays <text></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 input filter rule <1-999999> ttl <eq / gt / lt> <0-255></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	firewall.xml.in: firewall ipv4 input filter rule <rule> action	firewall ipv4 input filter rule <text> action <value>
×	not yet documented	firewall.xml.in: firewall ipv4 input filter rule <rule> add-address-to-group destination-address address-group	Nothing found in VyOS
×	not yet documented	firewall.xml.in: firewall ipv4 input filter rule <rule> add-address-to-group destination-address timeout	Nothing found in VyOS
×	not yet documented	firewall.xml.in: firewall ipv4 input filter rule <rule> add-address-to-group source-address address-group	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	firewall.xml.in: firewall ipv4 input filter rule <rule> add-address- to-group source-address timeout	Nothing found in VyOS
×	not yet documented	firewall.xml.in: firewall ipv4 input filter rule <rule> connection- mark	firewall ipv4 input filter rule <text> connection- mark <value>
×	not yet documented	firewall.xml.in: firewall ipv4 input filter rule <rule> connection- status nat	firewall ipv4 input filter rule <text> connection- status nat <value>
×	not yet documented	firewall.xml.in: firewall ipv4 input filter rule <rule> conntrack- helper	Nothing found in VyOS
×	not yet documented	firewall.xml.in: firewall ipv4 input filter rule <rule> description	firewall ipv4 input filter rule <text> description <value>
×	not yet documented	firewall.xml.in: firewall ipv4 input filter rule <rule> destination ad- dress	firewall ipv4 input filter rule <text> destination ad- dress <value>
×	not yet documented	firewall.xml.in: firewall ipv4 input filter rule <rule> destination address-mask	firewall ipv4 input filter rule <text> destination address-mask <value>
×	not yet documented	firewall.xml.in: firewall ipv4 input filter rule <rule> destination fqdn	firewall ipv4 input filter rule <text> destination fqdn <value>
×	not yet documented	firewall.xml.in: firewall ipv4 input filter rule <rule> destination geoip country-code	firewall ipv4 input filter rule <text> destination geoip country-code <value>
×	not yet documented	firewall.xml.in: firewall ipv4 input filter rule <rule> destination geoip inverse-match	firewall ipv4 input filter rule <text> destination geoip inverse-match
×	not yet documented	firewall.xml.in: firewall ipv4 input filter rule <rule> destination group address-group	firewall ipv4 input filter rule <text> destination group address-group <value>
×	not yet documented	firewall.xml.in: firewall ipv4 input filter rule <rule> destination group domain-group	firewall ipv4 input filter rule <text> destination group domain-group <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	firewall.xml.in: firewall ipv4 input filter rule <rule> destination group dynamic-address- group	Nothing found in VyOS
×	not yet documented	firewall.xml.in: firewall ipv4 input filter rule <rule> destination group mac-group <value> group mac-group	firewall ipv4 input filter rule <text> destination group mac-group <value>
×	not yet documented	firewall.xml.in: firewall ipv4 input filter rule <rule> destination group network-group group network-group	firewall ipv4 input filter rule <text> destination group network-group <value>
×	not yet documented	firewall.xml.in: firewall ipv4 input filter rule <rule> destination group port-group <value> group port-group	firewall ipv4 input filter rule <text> destination group port-group <value>
×	not yet documented	firewall.xml.in: firewall ipv4 input filter rule <rule> destination mac-address <value> mac-address	firewall ipv4 input filter rule <text> destination mac-address <value>
×	not yet documented	firewall.xml.in: firewall ipv4 input filter rule <rule> destination port <value> port	firewall ipv4 input filter rule <text> destination port <value>
×	not yet documented	firewall.xml.in: firewall ipv4 input filter rule <rule> disable rule <rule> disable	firewall ipv4 input filter rule <text> disable
×	not yet documented	firewall.xml.in: firewall ipv4 input filter rule <rule> dscp rule <rule> dscp	firewall ipv4 input filter rule <text> dscp <value>
×	not yet documented	firewall.xml.in: firewall ipv4 input filter rule <rule> dscp-exclude rule <rule> dscp-exclude	firewall ipv4 input filter rule <text> dscp-exclude <value>
×	not yet documented	firewall.xml.in: firewall ipv4 input filter rule <rule> fragment match-frag match-frag	firewall ipv4 input filter rule <text> fragment match-frag
×	not yet documented	firewall.xml.in: firewall ipv4 input filter rule <rule> fragment match-non-frag match-non-frag	firewall ipv4 input filter rule <text> fragment match-non-frag
×	not yet documented	firewall.xml.in: firewall ipv4 input filter rule <rule> icmp code rule <rule> icmp code	firewall ipv4 input filter rule <text> icmp code <value>
×	not yet documented	firewall.xml.in: firewall ipv4 input filter rule <rule> icmp type rule <rule> icmp type	firewall ipv4 input filter rule <text> icmp type <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	firewall.xml.in: firewall ipv4 input filter rule <rule> icmp type- name	firewall ipv4 input filter rule <text> icmp type- name <value>
×	not yet documented	firewall.xml.in: firewall ipv4 input filter rule <rule> inbound- interface group	Nothing found in VyOS
×	not yet documented	Nothing found in XML Definitions	firewall ipv4 input filter rule <text> inbound- interface interface-group <value>
×	not yet documented	Nothing found in XML Definitions	firewall ipv4 input filter rule <text> inbound- interface interface-name <value>
×	not yet documented	firewall.xml.in: firewall ipv4 input filter rule <rule> inbound- interface name	Nothing found in VyOS
×	not yet documented	firewall.xml.in: firewall ipv4 input filter rule <rule> ipsec match- ipsec	firewall ipv4 input filter rule <text> ipsec match- ipsec
×	not yet documented	firewall.xml.in: firewall ipv4 input filter rule <rule> ipsec match- none	firewall ipv4 input filter rule <text> ipsec match- none
×	not yet documented	firewall.xml.in: firewall ipv4 input filter rule <rule> jump-target	firewall ipv4 input filter rule <text> jump-target <value>
×	not yet documented	firewall.xml.in: firewall ipv4 input filter rule <rule> limit burst	firewall ipv4 input filter rule <text> limit burst <value>
×	not yet documented	firewall.xml.in: firewall ipv4 input filter rule <rule> limit rate	firewall ipv4 input filter rule <text> limit rate <value>
×	not yet documented	firewall.xml.in: firewall ipv4 input filter rule <rule> log	firewall ipv4 input filter rule <text> log <value>
×	not yet documented	firewall.xml.in: firewall ipv4 input filter rule <rule> log-options group	firewall ipv4 input filter rule <text> log-options group <value>
×	not yet documented	firewall.xml.in: firewall ipv4 input filter rule <rule> log-options level	firewall ipv4 input filter rule <text> log-options level <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	firewall.xml.in: firewall ipv4 input filter rule <rule> log-options queue-threshold	firewall ipv4 input filter rule <text> log-options queue-threshold <value>
×	not yet documented	firewall.xml.in: firewall ipv4 input filter rule <rule> log-options snapshot-length	firewall ipv4 input filter rule <text> log-options snapshot-length <value>
×	not yet documented	firewall.xml.in: firewall ipv4 input filter rule <rule> mark	Nothing found in VyOS
×	not yet documented	firewall.xml.in: firewall ipv4 input filter rule <rule> packet-length	firewall ipv4 input filter rule <text> packet-length <value>
×	not yet documented	firewall.xml.in: firewall ipv4 input filter rule <rule> packet-length- exclude	firewall ipv4 input filter rule <text> packet-length- exclude <value>
×	not yet documented	firewall.xml.in: firewall ipv4 input filter rule <rule> packet-type	firewall ipv4 input filter rule <text> packet-type <value>
×	not yet documented	firewall.xml.in: firewall ipv4 input filter rule <rule> protocol	firewall ipv4 input fil- ter rule <text> protocol <value>
×	not yet documented	firewall.xml.in: firewall ipv4 input filter rule <rule> queue	firewall ipv4 input filter rule <text> queue <value>
×	not yet documented	firewall.xml.in: firewall ipv4 input filter rule <rule> queue-options	firewall ipv4 input filter rule <text> queue-options <value>
×	not yet documented	firewall.xml.in: firewall ipv4 input filter rule <rule> recent count	firewall ipv4 input filter rule <text> recent count <value>
×	not yet documented	firewall.xml.in: firewall ipv4 input filter rule <rule> recent time	firewall ipv4 input filter rule <text> recent time <value>
×	not yet documented	firewall.xml.in: firewall ipv4 input filter rule <rule> source address	firewall ipv4 input filter rule <text> source address <value>
×	not yet documented	firewall.xml.in: firewall ipv4 input fil- ter rule <rule> source address-mask	firewall ipv4 input fil- ter rule <text> source address-mask <value>
×	not yet documented	firewall.xml.in: firewall ipv4 input filter rule <rule> source fqdn	firewall ipv4 input filter rule <text> source fqdn <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	firewall.xml.in: firewall ipv4 input filter rule <rule> source geoip country-code <value>	firewall ipv4 input filter rule <text> source geoip country-code <value>
×	not yet documented	firewall.xml.in: firewall ipv4 input filter rule <rule> source geoip inverse-match	firewall ipv4 input filter rule <text> source geoip inverse-match
×	not yet documented	firewall.xml.in: firewall ipv4 input filter rule <rule> source group address-group	firewall ipv4 input filter rule <text> source group address-group <value>
×	not yet documented	firewall.xml.in: firewall ipv4 input filter rule <rule> source group domain-group	firewall ipv4 input filter rule <text> source group domain-group <value>
×	not yet documented	firewall.xml.in: firewall ipv4 input filter rule <rule> source group dynamic-address-group	Nothing found in VyOS
×	not yet documented	firewall.xml.in: firewall ipv4 input filter rule <rule> source group mac-group	firewall ipv4 input filter rule <text> source group mac-group <value>
×	not yet documented	firewall.xml.in: firewall ipv4 input filter rule <rule> source group network-group	firewall ipv4 input filter rule <text> source group network-group <value>
×	not yet documented	firewall.xml.in: firewall ipv4 input filter rule <rule> source group port-group	firewall ipv4 input filter rule <text> source group port-group <value>
×	not yet documented	firewall.xml.in: firewall ipv4 input filter rule <rule> source mac- address	firewall ipv4 input filter rule <text> source mac- address <value>
×	not yet documented	firewall.xml.in: firewall ipv4 input filter rule <rule> source port	firewall ipv4 input filter rule <text> source port <value>
×	not yet documented	firewall.xml.in: firewall ipv4 input filter rule <rule> state	Nothing found in VyOS
×	not yet documented	Nothing found in XML Definitions	firewall ipv4 input filter rule <text> state estab- lished <value>
×	not yet documented	Nothing found in XML Definitions	firewall ipv4 input filter rule <text> state invalid <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	Nothing found in XML Definitions	firewall ipv4 input filter rule <text> state new <value>
×	not yet documented	Nothing found in XML Definitions	firewall ipv4 input filter rule <text> state related <value>
×	not yet documented	firewall.xml.in: firewall ipv4 input filter rule <rule> synproxy tcp mss	Nothing found in VyOS
×	not yet documented	firewall.xml.in: firewall ipv4 input filter rule <rule> synproxy tcp window-scale	Nothing found in VyOS
×	not yet documented	firewall.xml.in: firewall ipv4 input filter rule <rule> tcp flags ack	firewall ipv4 input filter rule <text> tcp flags ack
×	not yet documented	firewall.xml.in: firewall ipv4 input filter rule <rule> tcp flags cwr	firewall ipv4 input filter rule <text> tcp flags cwr
×	not yet documented	firewall.xml.in: firewall ipv4 input filter rule <rule> tcp flags ecn	firewall ipv4 input filter rule <text> tcp flags ecn
×	not yet documented	firewall.xml.in: firewall ipv4 input filter rule <rule> tcp flags fin	firewall ipv4 input filter rule <text> tcp flags fin
×	not yet documented	firewall.xml.in: firewall ipv4 input filter rule <rule> tcp flags not ack	firewall ipv4 input filter rule <text> tcp flags not ack
×	not yet documented	firewall.xml.in: firewall ipv4 input filter rule <rule> tcp flags not cwr	firewall ipv4 input filter rule <text> tcp flags not cwr
×	not yet documented	firewall.xml.in: firewall ipv4 input filter rule <rule> tcp flags not ecn	firewall ipv4 input filter rule <text> tcp flags not ecn
×	not yet documented	firewall.xml.in: firewall ipv4 input filter rule <rule> tcp flags not fin	firewall ipv4 input filter rule <text> tcp flags not fin
×	not yet documented	firewall.xml.in: firewall ipv4 input filter rule <rule> tcp flags not psh	firewall ipv4 input filter rule <text> tcp flags not psh
×	not yet documented	firewall.xml.in: firewall ipv4 input filter rule <rule> tcp flags not rst	firewall ipv4 input filter rule <text> tcp flags not rst

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	firewall.xml.in: firewall ipv4 input filter rule <rule> tcp flags not syn	firewall ipv4 input filter rule <text> tcp flags not syn
×	not yet documented	firewall.xml.in: firewall ipv4 input filter rule <rule> tcp flags not urg	firewall ipv4 input filter rule <text> tcp flags not urg
×	not yet documented	firewall.xml.in: firewall ipv4 input filter rule <rule> tcp flags psh	firewall ipv4 input filter rule <text> tcp flags psh
×	not yet documented	firewall.xml.in: firewall ipv4 input filter rule <rule> tcp flags rst	firewall ipv4 input filter rule <text> tcp flags rst
×	not yet documented	firewall.xml.in: firewall ipv4 input filter rule <rule> tcp flags syn	firewall ipv4 input filter rule <text> tcp flags syn
×	not yet documented	firewall.xml.in: firewall ipv4 input filter rule <rule> tcp flags urg	firewall ipv4 input filter rule <text> tcp flags urg
×	not yet documented	firewall.xml.in: firewall ipv4 input filter rule <rule> tcp mss <value>	firewall ipv4 input fil- ter rule <text> tcp mss <value>
×	not yet documented	firewall.xml.in: firewall ipv4 input filter rule <rule> time startdate <value>	firewall ipv4 input filter rule <text> time startdate <value>
×	not yet documented	firewall.xml.in: firewall ipv4 input filter rule <rule> time starttime <value>	firewall ipv4 input filter rule <text> time starttime <value>
×	not yet documented	firewall.xml.in: firewall ipv4 input filter rule <rule> time stopdate <value>	firewall ipv4 input filter rule <text> time stopdate <value>
×	not yet documented	firewall.xml.in: firewall ipv4 input filter rule <rule> time stoptime <value>	firewall ipv4 input filter rule <text> time stoptime <value>
×	not yet documented	firewall.xml.in: firewall ipv4 input filter rule <rule> time weekdays <value>	firewall ipv4 input filter rule <text> time weekdays <value>
×	not yet documented	firewall.xml.in: firewall ipv4 input filter rule <rule> ttl eq <value>	firewall ipv4 input filter rule <text> ttl eq <value>
×	not yet documented	firewall.xml.in: firewall ipv4 input filter rule <rule> ttl gt <value>	firewall ipv4 input filter rule <text> ttl gt <value>
×	not yet documented	firewall.xml.in: firewall ipv4 input filter rule <rule> ttl lt <value>	firewall ipv4 input filter rule <text> ttl lt <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set firewall ipv4 name <name> default-action [accept drop jump queue reject return]</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 name <name> default-jump-target <text></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 name <name> default-log</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 name <name> description <text></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 name <name> rule <1-999999> action [accept continue drop jump queue reject return]</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 name <name> rule <1-999999> add-address-to-group destination-address address-group <name></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 name <name> rule <1-999999> add-address-to-group source-address address-group <name></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 name <name> rule <1-999999> add-address-to-group [destination-address source-address] timeout <timeout></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 name <name> rule <1-999999> connection-mark <1-2147483647></code>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set firewall ipv4 name <name> rule <1-999999> connection-status nat [destination / source]</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 name <name> rule <1-999999> conntrack-helper <module></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 name <name> rule <1-999999> description <text></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 name <name> rule <1-999999> destination address [address / addressrange / CIDR]</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 name <name> rule <1-999999> destination address-mask [address]</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 name <name> rule <1-999999> destination fqdn <fqdn></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 name <name> rule <1-999999> destination geoip asn <1-4294967294></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 name <name> rule <1-999999> destination geoip country-code <country></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 name <name> rule <1-999999> destination geoip inverse-match</code>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set firewall ipv4 name <name> rule <1-999999> destination group address-group <name !name></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 name <name> rule <1-999999> destination group domain-group <name !name></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 name <name> rule <1-999999> destination group dynamic-address-group <name !name></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 name <name> rule <1-999999> destination group mac-group <name !name></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 name <name> rule <1-999999> destination group network-group <name !name></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 name <name> rule <1-999999> destination group port-group <name !name></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 name <name> rule <1-999999> destination port [1-65535 portname start-end]</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 name <name> rule <1-999999> disable</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 name <name> rule <1-999999> dscp [0-63 start-end]</code>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set firewall ipv4 name <name> rule <1-999999> dscp-exclude [0-63 / start-end]</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 name <name> rule <1-999999> fragment [match-frag / match-non-frag]</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 name <name> rule <1-999999> icmp [code type] <0-255></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 name <name> rule <1-999999> icmp type-name <text></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 name <name> rule <1-999999> inbound-interface group <iface_group></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 name <name> rule <1-999999> inbound-interface name <iface></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 name <name> rule <1-999999> ipsec [match-ipsec-in / match-ipsec-out / match-none-in / match-none-out]</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 name <name> rule <1-999999> jump-target <text></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 name <name> rule <1-999999> limit burst <0-4294967295></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 name <name> rule <1-999999> limit rate <text></code>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set firewall ipv4 name <name> rule <1-999999> log</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 name <name> rule <1-999999> log-options group <0-65535></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 name <name> rule <1-999999> log-options level [emerg alert crit err warn notice info debug]</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 name <name> rule <1-999999> log-options queue-threshold <0-65535></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 name <name> rule <1-999999> log-options snapshot-length <0-9000></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 name <name> rule <1-999999> outbound-interface group <iface_group></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 name <name> rule <1-999999> outbound-interface name <iface></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 name <name> rule <1-999999> packet-length-exclude <text></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 name <name> rule <1-999999> packet-length <text></code>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set firewall ipv4 name <name> rule <1-999999> packet-type [broadcast host multicast other]</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 name <name> rule <1-999999> protocol [<text> <0-255> all tcp_udp]</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 name <name> rule <1-999999> queue-options bypass</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 name <name> rule <1-999999> queue-options fanout</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 name <name> rule <1-999999> queue <0-65535></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 name <name> rule <1-999999> recent count <1-255></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 name <name> rule <1-999999> recent time [second minute hour]</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 name <name> rule <1-999999> source address [address addressrange CIDR]</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 name <name> rule <1-999999> source address-mask [address]</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 name <name> rule <1-999999> source fqdn <fqdn></code>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set firewall ipv4 name <name> rule <1-999999> source geoip asn <1-4294967294></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 name <name> rule <1-999999> source geoip country-code <country></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 name <name> rule <1-999999> source geoip inverse-match</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 name <name> rule <1-999999> source group address-group <name !name></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 name <name> rule <1-999999> source group domain-group <name !name></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 name <name> rule <1-999999> source group dynamic-address-group <name !name></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 name <name> rule <1-999999> source group mac-group <name !name></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 name <name> rule <1-999999> source group network-group <name !name></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 name <name> rule <1-999999> source group port-group <name !name></code>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set firewall ipv4 name <name> rule <1-999999> source mac-address <mac-address></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 name <name> rule <1-999999> source port [1-65535 portname start-end]</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 name <name> rule <1-999999> state [established invalid new related]</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 name <name> rule <1-999999> tcp flags [not] <text></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 name <name> rule <1-999999> time startdate <text></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 name <name> rule <1-999999> time starttime <text></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 name <name> rule <1-999999> time stopdate <text></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 name <name> rule <1-999999> time stoptime <text></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 name <name> rule <1-999999> time weekdays <text></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 name <name> rule <1-999999> ttl <eq gt lt> <0-255></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	firewall.xml.in: firewall ipv4 name <name> default-action	firewall ipv4 name <text> default-action <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	firewall.xml.in: firewall ipv4 name <name> default-jump- target	firewall ipv4 name <text> default-jump-target <value>
×	not yet documented	firewall.xml.in: firewall ipv4 name <name> default-log	Nothing found in VyOS
×	not yet documented	firewall.xml.in: firewall ipv4 name <name> description	firewall ipv4 name <text> description <value>
×	not yet documented	Nothing found in XML Definitions	firewall ipv4 name <text> enable-default-log
×	not yet documented	firewall.xml.in: firewall ipv4 name <name> rule <rule> action	firewall ipv4 name <text> rule <text> action <value>
×	not yet documented	firewall.xml.in: firewall ipv4 name <name> rule <rule> add-address-to-group destination-address address-group	Nothing found in VyOS
×	not yet documented	firewall.xml.in: firewall ipv4 name <name> rule <rule> add-address-to-group destination-address time- out	Nothing found in VyOS
×	not yet documented	firewall.xml.in: firewall ipv4 name <name> rule <rule> add-address-to-group source-address address- group	Nothing found in VyOS
×	not yet documented	firewall.xml.in: firewall ipv4 name <name> rule <rule> add-address-to-group source-address timeout	Nothing found in VyOS
×	not yet documented	firewall.xml.in: firewall ipv4 name <name> rule <rule> connection-mark	firewall ipv4 name <text> rule <text> connection- mark <value>
×	not yet documented	firewall.xml.in: firewall ipv4 name <name> rule <rule> connection-status nat	firewall ipv4 name <text> rule <text> connection- status nat <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	firewall.xml.in: firewall ipv4 name <name> rule <rule> conntrack-helper	Nothing found in VyOS
×	not yet documented	firewall.xml.in: firewall ipv4 name <name> rule <rule> description	firewall ipv4 name <text> rule <text> description <value>
×	not yet documented	firewall.xml.in: firewall ipv4 name <name> rule <rule> destination address	firewall ipv4 name <text> rule <text> destination ad- dress <value>
×	not yet documented	firewall.xml.in: firewall ipv4 name <name> rule <rule> destination address-mask	firewall ipv4 name <text> rule <text> destination address-mask <value>
×	not yet documented	firewall.xml.in: firewall ipv4 name <name> rule <rule> destination fqdn	firewall ipv4 name <text> rule <text> destination fqdn <value>
×	not yet documented	firewall.xml.in: firewall ipv4 name <name> rule <rule> desti- nation geoip country-code	firewall ipv4 name <text> rule <text> destination geoip country-code <value>
×	not yet documented	firewall.xml.in: firewall ipv4 name <name> rule <rule> destination geoip inverse- match	firewall ipv4 name <text> rule <text> destination geoip inverse-match
×	not yet documented	firewall.xml.in: firewall ipv4 name <name> rule <rule> destination group address- group	firewall ipv4 name <text> rule <text> destination group address-group <value>
×	not yet documented	firewall.xml.in: firewall ipv4 name <name> rule <rule> destination group domain- group	firewall ipv4 name <text> rule <text> destination group domain-group <value>
×	not yet documented	firewall.xml.in: firewall ipv4 name <name> rule <rule> des- tination group dynamic- address-group	Nothing found in VyOS
×	not yet documented	firewall.xml.in: firewall ipv4 name <name> rule <rule> desti- nation group mac-group	firewall ipv4 name <text> rule <text> destination group mac-group <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	firewall.xml.in: firewall ipv4 name <name> rule <rule> des- tination group network- group	firewall ipv4 name <text> rule <text> destination group network-group <value>
×	not yet documented	firewall.xml.in: firewall ipv4 name <name> rule <rule> desti- nation group port-group	firewall ipv4 name <text> rule <text> destination group port-group <value>
×	not yet documented	firewall.xml.in: firewall ipv4 name <name> rule <rule> destination mac-address	firewall ipv4 name <text> rule <text> destination mac-address <value>
×	not yet documented	firewall.xml.in: firewall ipv4 name <name> rule <rule> destination port	firewall ipv4 name <text> rule <text> destination port <value>
×	not yet documented	firewall.xml.in: firewall ipv4 name <name> rule <rule> disable	firewall ipv4 name <text> rule <text> disable
×	not yet documented	firewall.xml.in: firewall ipv4 name <name> rule <rule> dscp	firewall ipv4 name <text> rule <text> dscp <value>
×	not yet documented	firewall.xml.in: firewall ipv4 name <name> rule <rule> dscp-exclude	firewall ipv4 name <text> rule <text> dscp-exclude <value>
×	not yet documented	firewall.xml.in: firewall ipv4 name <name> rule <rule> fragment match-frag	firewall ipv4 name <text> rule <text> fragment match-frag
×	not yet documented	firewall.xml.in: firewall ipv4 name <name> rule <rule> fragment match-non-frag	firewall ipv4 name <text> rule <text> fragment match-non-frag
×	not yet documented	firewall.xml.in: firewall ipv4 name <name> rule <rule> icmp code	firewall ipv4 name <text> rule <text> icmp code <value>
×	not yet documented	firewall.xml.in: firewall ipv4 name <name> rule <rule> icmp type	firewall ipv4 name <text> rule <text> icmp type <value>
×	not yet documented	firewall.xml.in: firewall ipv4 name <name> rule <rule> icmp type-name	firewall ipv4 name <text> rule <text> icmp type- name <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	firewall.xml.in: firewall ipv4 name <name> rule <rule> inbound-interface group	Nothing found in VyOS
×	not yet documented	Nothing found in XML Definitions	firewall ipv4 name <text> rule <text> inbound- interface interface-group <value>
×	not yet documented	Nothing found in XML Definitions	firewall ipv4 name <text> rule <text> inbound- interface interface-name <value>
×	not yet documented	firewall.xml.in: firewall ipv4 name <name> rule <rule> inbound-interface name	Nothing found in VyOS
×	not yet documented	firewall.xml.in: firewall ipv4 name <name> rule <rule> ipsec match-ipsec	firewall ipv4 name <text> rule <text> ipsec match- ipsec
×	not yet documented	firewall.xml.in: firewall ipv4 name <name> rule <rule> ipsec match-none	firewall ipv4 name <text> rule <text> ipsec match- none
×	not yet documented	firewall.xml.in: firewall ipv4 name <name> rule <rule> jump-target	firewall ipv4 name <text> rule <text> jump-target <value>
×	not yet documented	firewall.xml.in: firewall ipv4 name <name> rule <rule> limit burst	firewall ipv4 name <text> rule <text> limit burst <value>
×	not yet documented	firewall.xml.in: firewall ipv4 name <name> rule <rule> limit rate	firewall ipv4 name <text> rule <text> limit rate <value>
×	not yet documented	firewall.xml.in: firewall ipv4 name <name> rule <rule> log	firewall ipv4 name <text> rule <text> log <value>
×	not yet documented	firewall.xml.in: firewall ipv4 name <name> rule <rule> log-options group	firewall ipv4 name <text> rule <text> log-options group <value>
×	not yet documented	firewall.xml.in: firewall ipv4 name <name> rule <rule> log-options level	firewall ipv4 name <text> rule <text> log-options level <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	firewall.xml.in: firewall ipv4 name <name> rule <rule> log- options queue-threshold	firewall ipv4 name <text> rule <text> log-options queue-threshold <value>
×	not yet documented	firewall.xml.in: firewall ipv4 name <name> rule <rule> log- options snapshot-length	firewall ipv4 name <text> rule <text> log-options snapshot-length <value>
×	not yet documented	firewall.xml.in: firewall ipv4 name <name> rule <rule> mark	Nothing found in VyOS
×	not yet documented	firewall.xml.in: firewall ipv4 name <name> rule <rule> offload-target	Nothing found in VyOS
×	not yet documented	firewall.xml.in: firewall ipv4 name <name> rule <rule> outbound-interface group	Nothing found in VyOS
×	not yet documented	Nothing found in XML Definitions	firewall ipv4 name <text> rule <text> outbound- interface interface-group <value>
×	not yet documented	Nothing found in XML Definitions	firewall ipv4 name <text> rule <text> outbound- interface interface-name <value>
×	not yet documented	firewall.xml.in: firewall ipv4 name <name> rule <rule> outbound-interface name	Nothing found in VyOS
×	not yet documented	firewall.xml.in: firewall ipv4 name <name> rule <rule> packet-length	firewall ipv4 name <text> rule <text> packet-length <value>
×	not yet documented	firewall.xml.in: firewall ipv4 name <name> rule <rule> packet-length-exclude	firewall ipv4 name <text> rule <text> packet-length- exclude <value>
×	not yet documented	firewall.xml.in: firewall ipv4 name <name> rule <rule> packet-type	firewall ipv4 name <text> rule <text> packet-type <value>
×	not yet documented	firewall.xml.in: firewall ipv4 name <name> rule <rule> protocol	firewall ipv4 name <text> rule <text> protocol <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	firewall.xml.in: firewall ipv4 name <name> rule <rule> queue	firewall ipv4 name <text> rule <text> queue <value>
×	not yet documented	firewall.xml.in: firewall ipv4 name <name> rule <rule> queue-options	firewall ipv4 name <text> rule <text> queue-options <value>
×	not yet documented	firewall.xml.in: firewall ipv4 name <name> rule <rule> recent count	firewall ipv4 name <text> rule <text> recent count <value>
×	not yet documented	firewall.xml.in: firewall ipv4 name <name> rule <rule> recent time	firewall ipv4 name <text> rule <text> recent time <value>
×	not yet documented	firewall.xml.in: firewall ipv4 name <name> rule <rule> source address	firewall ipv4 name <text> rule <text> source address <value>
×	not yet documented	firewall.xml.in: firewall ipv4 name <name> rule <rule> source address-mask	firewall ipv4 name <text> rule <text> source address-mask <value>
×	not yet documented	firewall.xml.in: firewall ipv4 name <name> rule <rule> source fqdn	firewall ipv4 name <text> rule <text> source fqdn <value>
×	not yet documented	firewall.xml.in: firewall ipv4 name <name> rule <rule> source geoip country-code	firewall ipv4 name <text> rule <text> source geoip country-code <value>
×	not yet documented	firewall.xml.in: firewall ipv4 name <name> rule <rule> source geoip inverse-match	firewall ipv4 name <text> rule <text> source geoip inverse-match
×	not yet documented	firewall.xml.in: firewall ipv4 name <name> rule <rule> source group address-group	firewall ipv4 name <text> rule <text> source group address-group <value>
×	not yet documented	firewall.xml.in: firewall ipv4 name <name> rule <rule> source group domain-group	firewall ipv4 name <text> rule <text> source group domain-group <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	firewall.xml.in: firewall ipv4 name <name> rule <rule> source group dynamic- address-group	Nothing found in VyOS
×	not yet documented	firewall.xml.in: firewall ipv4 name <name> rule <rule> source group mac-group	firewall ipv4 name <text> rule <text> source group mac-group <value>
×	not yet documented	firewall.xml.in: firewall ipv4 name <name> rule <rule> source group network- group	firewall ipv4 name <text> rule <text> source group network-group <value>
×	not yet documented	firewall.xml.in: firewall ipv4 name <name> rule <rule> source group port-group	firewall ipv4 name <text> rule <text> source group port-group <value>
×	not yet documented	firewall.xml.in: firewall ipv4 name <name> rule <rule> source mac-address	firewall ipv4 name <text> rule <text> source mac- address <value>
×	not yet documented	firewall.xml.in: firewall ipv4 name <name> rule <rule> source port	firewall ipv4 name <text> rule <text> source port <value>
×	not yet documented	firewall.xml.in: firewall ipv4 name <name> rule <rule> state	Nothing found in VyOS
×	not yet documented	Nothing found in XML Definitions	firewall ipv4 name <text> rule <text> state estab- lished <value>
×	not yet documented	Nothing found in XML Definitions	firewall ipv4 name <text> rule <text> state invalid <value>
×	not yet documented	Nothing found in XML Definitions	firewall ipv4 name <text> rule <text> state new <value>
×	not yet documented	Nothing found in XML Definitions	firewall ipv4 name <text> rule <text> state related <value>
×	not yet documented	firewall.xml.in: firewall ipv4 name <name> rule <rule> synproxy tcp mss	Nothing found in VyOS
×	not yet documented	firewall.xml.in: firewall ipv4 name <name> rule <rule> syn- proxy tcp window-scale	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	firewall.xml.in: firewall ipv4 name <name> rule <rule> tcp flags ack	firewall ipv4 name <text> rule <text> tcp flags ack
×	not yet documented	firewall.xml.in: firewall ipv4 name <name> rule <rule> tcp flags cwr	firewall ipv4 name <text> rule <text> tcp flags cwr
×	not yet documented	firewall.xml.in: firewall ipv4 name <name> rule <rule> tcp flags ecn	firewall ipv4 name <text> rule <text> tcp flags ecn
×	not yet documented	firewall.xml.in: firewall ipv4 name <name> rule <rule> tcp flags fin	firewall ipv4 name <text> rule <text> tcp flags fin
×	not yet documented	firewall.xml.in: firewall ipv4 name <name> rule <rule> tcp flags not ack	firewall ipv4 name <text> rule <text> tcp flags not ack
×	not yet documented	firewall.xml.in: firewall ipv4 name <name> rule <rule> tcp flags not cwr	firewall ipv4 name <text> rule <text> tcp flags not cwr
×	not yet documented	firewall.xml.in: firewall ipv4 name <name> rule <rule> tcp flags not ecn	firewall ipv4 name <text> rule <text> tcp flags not ecn
×	not yet documented	firewall.xml.in: firewall ipv4 name <name> rule <rule> tcp flags not fin	firewall ipv4 name <text> rule <text> tcp flags not fin
×	not yet documented	firewall.xml.in: firewall ipv4 name <name> rule <rule> tcp flags not psh	firewall ipv4 name <text> rule <text> tcp flags not psh
×	not yet documented	firewall.xml.in: firewall ipv4 name <name> rule <rule> tcp flags not rst	firewall ipv4 name <text> rule <text> tcp flags not rst
×	not yet documented	firewall.xml.in: firewall ipv4 name <name> rule <rule> tcp flags not syn	firewall ipv4 name <text> rule <text> tcp flags not syn
×	not yet documented	firewall.xml.in: firewall ipv4 name <name> rule <rule> tcp flags not urg	firewall ipv4 name <text> rule <text> tcp flags not urg

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	firewall.xml.in: firewall ipv4 name <name> rule <rule> tcp flags psh	firewall ipv4 name <text> rule <text> tcp flags psh
×	not yet documented	firewall.xml.in: firewall ipv4 name <name> rule <rule> tcp flags rst	firewall ipv4 name <text> rule <text> tcp flags rst
×	not yet documented	firewall.xml.in: firewall ipv4 name <name> rule <rule> tcp flags syn	firewall ipv4 name <text> rule <text> tcp flags syn
×	not yet documented	firewall.xml.in: firewall ipv4 name <name> rule <rule> tcp flags urg	firewall ipv4 name <text> rule <text> tcp flags urg
×	not yet documented	firewall.xml.in: firewall ipv4 name <name> rule <rule> tcp mss	firewall ipv4 name <text> rule <text> tcp mss <value>
×	not yet documented	firewall.xml.in: firewall ipv4 name <name> rule <rule> time startdate	firewall ipv4 name <text> rule <text> time startdate <value>
×	not yet documented	firewall.xml.in: firewall ipv4 name <name> rule <rule> time starttime	firewall ipv4 name <text> rule <text> time starttime <value>
×	not yet documented	firewall.xml.in: firewall ipv4 name <name> rule <rule> time stopdate	firewall ipv4 name <text> rule <text> time stopdate <value>
×	not yet documented	firewall.xml.in: firewall ipv4 name <name> rule <rule> time stoptime	firewall ipv4 name <text> rule <text> time stoptime <value>
×	not yet documented	firewall.xml.in: firewall ipv4 name <name> rule <rule> time weekdays	firewall ipv4 name <text> rule <text> time weekdays <value>
×	not yet documented	firewall.xml.in: firewall ipv4 name <name> rule <rule> ttl eq	firewall ipv4 name <text> rule <text> ttl eq <value>
×	not yet documented	firewall.xml.in: firewall ipv4 name <name> rule <rule> ttl gt	firewall ipv4 name <text> rule <text> ttl gt <value>
×	not yet documented	firewall.xml.in: firewall ipv4 name <name> rule <rule> ttl lt	firewall ipv4 name <text> rule <text> ttl lt <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
✓	<code>set firewall ipv4 output filter default-action [accept drop]</code>	firewall.xml.in: firewall ipv4 output filter default-action	firewall ipv4 output filter default-action <value>
×	<code>set firewall ipv4 output filter default-log</code>	firewall.xml.in: firewall ipv4 output filter default-log	Nothing found in VyOS
×	not yet documented	firewall.xml.in: firewall ipv4 output filter description	firewall ipv4 output filter description <value>
×	<code>set firewall ipv4 output filter rule <1-999999> action [accept continue drop jump queue reject return]</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 output filter rule <1-999999> connection-mark <1-2147483647></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 output filter rule <1-999999> connection-status nat [destination source]</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 output filter rule <1-999999> conntrack-helper <module></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 output filter rule <1-999999> description <text></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 output filter rule <1-999999> destination address [address addressrange CIDR]</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 output filter rule <1-999999> destination address-mask [address]</code>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set firewall ipv4 output filter rule <1-999999> destination fqdn <fqdn></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 output filter rule <1-999999> destination geoip asn <1-4294967294></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 output filter rule <1-999999> destination geoip country-code <country></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 output filter rule <1-999999> destination geoip inverse-match</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 output filter rule <1-999999> destination group address-group <name !name></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 output filter rule <1-999999> destination group domain-group <name / !name></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 output filter rule <1-999999> destination group dynamic-address-group <name !name></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 output filter rule <1-999999> destination group mac-group <name / !name></code>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set firewall ipv4 output filter rule <1-999999> destination group network-group <name !name></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 output filter rule <1-999999> destination group port-group <name !name></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 output filter rule <1-999999> destination port [1-65535 portname start-end]</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 output filter rule <1-999999> disable</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 output filter rule <1-999999> dscp [0-63 start-end]</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 output filter rule <1-999999> dscp-exclude [0-63 start-end]</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 output filter rule <1-999999> fragment [match-frag match-non-frag]</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 output filter rule <1-999999> icmp [code type] <0-255></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 output filter rule <1-999999> icmp type-name <text></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 output filter rule <1-999999> ipsec [match-ipsec-out match-none-out]</code>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set firewall ipv4 output filter rule <1-999999> jump-target <text></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 output filter rule <1-999999> limit burst <0-4294967295></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 output filter rule <1-999999> limit rate <text></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 output filter rule <1-999999> log</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 output filter rule <1-999999> log-options group <0-65535></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 output filter rule <1-999999> log-options level [emerg alert crit err warn notice info debug]</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 output filter rule <1-999999> log-options queue-threshold <0-65535></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 output filter rule <1-999999> log-options snapshot-length <0-9000></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 output filter rule <1-999999> outbound-interface group <iface_group></code>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set firewall ipv4 output filter rule <1-999999> outbound-interface name <iface></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 output filter rule <1-999999> packet-length-exclude <text></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 output filter rule <1-999999> packet-length <text></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 output filter rule <1-999999> packet-type [broadcast host multicast other]</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 output filter rule <1-999999> protocol [<text> <0-255> all tcp_udp]</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 output filter rule <1-999999> queue-options bypass</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 output filter rule <1-999999> queue-options fanout</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 output filter rule <1-999999> queue <0-65535></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 output filter rule <1-999999> recent count <1-255></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 output filter rule <1-999999> recent time [second minute hour]</code>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set firewall ipv4 output filter rule <1-999999> source address [address addressrange CIDR]</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 output filter rule <1-999999> source address-mask [address]</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 output filter rule <1-999999> source fqdn <fqdn></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 output filter rule <1-999999> source geoip asn <1-4294967294></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 output filter rule <1-999999> source geoip country-code <country></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 output filter rule <1-999999> source geoip inverse-match</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 output filter rule <1-999999> source group address-group <name !name></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 output filter rule <1-999999> source group domain-group <name !name></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 output filter rule <1-999999> source group dynamic-address-group <name !name></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 output filter rule <1-999999> source group mac-group <name !name></code>	Nothing found in XML Definitions	Nothing found in VyOS

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Table 1 – continued from previous page

Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set firewall ipv4 output filter rule <1-999999> source group network-group <name !name></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 output filter rule <1-999999> source group port-group <name !name></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 output filter rule <1-999999> source mac-address <mac-address></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 output filter rule <1-999999> source port [1-65535 portname start-end]</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 output filter rule <1-999999> state [established invalid new related]</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 output filter rule <1-999999> tcp flags [not] <text></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 output filter rule <1-999999> time startdate <text></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 output filter rule <1-999999> time starttime <text></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 output filter rule <1-999999> time stopdate <text></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 output filter rule <1-999999> time stoptime <text></code>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set firewall ipv4 output filter rule <1-999999> time weekdays <text></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 output filter rule <1-999999> ttl <eq / gt / lt> <0-255></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	firewall.xml.in: firewall ipv4 output filter rule <rule> action	firewall ipv4 output filter rule <text> action <value>
×	not yet documented	firewall.xml.in: firewall ipv4 output filter rule <rule> add-address-to-group destination-address address-group	Nothing found in VyOS
×	not yet documented	firewall.xml.in: firewall ipv4 output filter rule <rule> add-address-to-group destination-address timeout	Nothing found in VyOS
×	not yet documented	firewall.xml.in: firewall ipv4 output filter rule <rule> add-address-to-group source-address address-group	Nothing found in VyOS
×	not yet documented	firewall.xml.in: firewall ipv4 output filter rule <rule> add-address-to-group source-address timeout	Nothing found in VyOS
×	not yet documented	firewall.xml.in: firewall ipv4 output filter rule <rule> connection-mark	firewall ipv4 output filter rule <text> connection-mark <value>
×	not yet documented	firewall.xml.in: firewall ipv4 output filter rule <rule> connection-status nat	firewall ipv4 output filter rule <text> connection-status nat <value>
×	not yet documented	firewall.xml.in: firewall ipv4 output filter rule <rule> conntrack-helper	Nothing found in VyOS
×	not yet documented	firewall.xml.in: firewall ipv4 output filter rule <rule> description	firewall ipv4 output filter rule <text> description <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	firewall.xml.in: firewall ipv4 output filter rule <rule> destination address	firewall ipv4 output filter rule <text> destination address <value>
×	not yet documented	firewall.xml.in: firewall ipv4 output filter rule <rule> destination address-mask	firewall ipv4 output filter rule <text> destination address-mask <value>
×	not yet documented	firewall.xml.in: firewall ipv4 output filter rule <rule> destination fqdn	firewall ipv4 output filter rule <text> destination fqdn <value>
×	not yet documented	firewall.xml.in: firewall ipv4 output filter rule <rule> destination geoip country-code	firewall ipv4 output filter rule <text> destination geoip country-code <value>
×	not yet documented	firewall.xml.in: firewall ipv4 output filter rule <rule> destination geoip inverse-match	firewall ipv4 output filter rule <text> destination geoip inverse-match
×	not yet documented	firewall.xml.in: firewall ipv4 output filter rule <rule> destination group address-group	firewall ipv4 output filter rule <text> destination group address-group <value>
×	not yet documented	firewall.xml.in: firewall ipv4 output filter rule <rule> destination group domain-group	firewall ipv4 output filter rule <text> destination group domain-group <value>
×	not yet documented	firewall.xml.in: firewall ipv4 output filter rule <rule> destination group dynamic-address-group	Nothing found in VyOS
×	not yet documented	firewall.xml.in: firewall ipv4 output filter rule <rule> destination group mac-group	firewall ipv4 output filter rule <text> destination group mac-group <value>
×	not yet documented	firewall.xml.in: firewall ipv4 output filter rule <rule> destination group network-group	firewall ipv4 output filter rule <text> destination group network-group <value>
×	not yet documented	firewall.xml.in: firewall ipv4 output filter rule <rule> destination group port-group	firewall ipv4 output filter rule <text> destination group port-group <value>
×	not yet documented	firewall.xml.in: firewall ipv4 output filter rule <rule> destination mac-address	firewall ipv4 output filter rule <text> destination mac-address <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	firewall.xml.in: firewall ipv4 output filter rule <rule> destination port	firewall ipv4 output filter rule <text> destination port <value>
×	not yet documented	firewall.xml.in: firewall ipv4 output filter rule <rule> disable	firewall ipv4 output filter rule <text> disable
×	not yet documented	firewall.xml.in: firewall ipv4 output filter rule <rule> dscp	firewall ipv4 output filter rule <text> dscp <value>
×	not yet documented	firewall.xml.in: firewall ipv4 output filter rule <rule> dscp-exclude	firewall ipv4 output filter rule <text> dscp-exclude <value>
×	not yet documented	firewall.xml.in: firewall ipv4 output filter rule <rule> fragment match-frag	firewall ipv4 output filter rule <text> fragment match-frag
×	not yet documented	firewall.xml.in: firewall ipv4 output filter rule <rule> fragment match-non-frag	firewall ipv4 output filter rule <text> fragment match-non-frag
×	not yet documented	firewall.xml.in: firewall ipv4 output filter rule <rule> icmp code	firewall ipv4 output filter rule <text> icmp code <value>
×	not yet documented	firewall.xml.in: firewall ipv4 output filter rule <rule> icmp type	firewall ipv4 output filter rule <text> icmp type <value>
×	not yet documented	firewall.xml.in: firewall ipv4 output filter rule <rule> icmp type-name	firewall ipv4 output filter rule <text> icmp type-name <value>
×	not yet documented	Nothing found in XML Definitions	firewall ipv4 output filter rule <text> ipsec match-ipsec
×	not yet documented	Nothing found in XML Definitions	firewall ipv4 output filter rule <text> ipsec match-none
×	not yet documented	firewall.xml.in: firewall ipv4 output filter rule <rule> jump-target	firewall ipv4 output filter rule <text> jump-target <value>
×	not yet documented	firewall.xml.in: firewall ipv4 output filter rule <rule> limit burst	firewall ipv4 output filter rule <text> limit burst <value>
×	not yet documented	firewall.xml.in: firewall ipv4 output filter rule <rule> limit rate	firewall ipv4 output filter rule <text> limit rate <value>
×	not yet documented	firewall.xml.in: firewall ipv4 output filter rule <rule> log	firewall ipv4 output filter rule <text> log <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	firewall.xml.in: firewall ipv4 output filter rule <rule> log-options group	firewall ipv4 output filter rule <text> log-options group <value>
×	not yet documented	firewall.xml.in: firewall ipv4 output filter rule <rule> log-options level	firewall ipv4 output filter rule <text> log-options level <value>
×	not yet documented	firewall.xml.in: firewall ipv4 output filter rule <rule> log-options queue-threshold	firewall ipv4 output filter rule <text> log-options queue-threshold <value>
×	not yet documented	firewall.xml.in: firewall ipv4 output filter rule <rule> log-options snapshot-length	firewall ipv4 output filter rule <text> log-options snapshot-length <value>
×	not yet documented	firewall.xml.in: firewall ipv4 output filter rule <rule> mark	Nothing found in VyOS
×	not yet documented	firewall.xml.in: firewall ipv4 output filter rule <rule> outbound-interface group	Nothing found in VyOS
×	not yet documented	Nothing found in XML Definitions	firewall ipv4 output filter rule <text> outbound-interface interface-group <value>
×	not yet documented	Nothing found in XML Definitions	firewall ipv4 output filter rule <text> outbound-interface interface-name <value>
×	not yet documented	firewall.xml.in: firewall ipv4 output filter rule <rule> outbound-interface name	Nothing found in VyOS
×	not yet documented	firewall.xml.in: firewall ipv4 output filter rule <rule> packet-length	firewall ipv4 output filter rule <text> packet-length <value>
×	not yet documented	firewall.xml.in: firewall ipv4 output filter rule <rule> packet-length-exclude	firewall ipv4 output filter rule <text> packet-length-exclude <value>
×	not yet documented	firewall.xml.in: firewall ipv4 output filter rule <rule> packet-type	firewall ipv4 output filter rule <text> packet-type <value>
×	not yet documented	firewall.xml.in: firewall ipv4 output filter rule <rule> protocol	firewall ipv4 output filter rule <text> protocol <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	firewall.xml.in: firewall ipv4 output filter rule <rule> queue	firewall ipv4 output filter rule <text> queue <value>
×	not yet documented	firewall.xml.in: firewall ipv4 output filter rule <rule> queue-options	firewall ipv4 output filter rule <text> queue-options <value>
×	not yet documented	firewall.xml.in: firewall ipv4 output filter rule <rule> recent count	firewall ipv4 output filter rule <text> recent count <value>
×	not yet documented	firewall.xml.in: firewall ipv4 output filter rule <rule> recent time	firewall ipv4 output fil- ter rule <text> recent time <value>
×	not yet documented	firewall.xml.in: firewall ipv4 output filter rule <rule> source address	firewall ipv4 output filter rule <text> source address <value>
×	not yet documented	firewall.xml.in: firewall ipv4 output fil- ter rule <rule> source address-mask	firewall ipv4 output fil- ter rule <text> source address-mask <value>
×	not yet documented	firewall.xml.in: firewall ipv4 output filter rule <rule> source fqdn	firewall ipv4 output filter rule <text> source fqdn <value>
×	not yet documented	firewall.xml.in: firewall ipv4 output filter rule <rule> source geoip country-code	firewall ipv4 output filter rule <text> source geoip country-code <value>
×	not yet documented	firewall.xml.in: firewall ipv4 output filter rule <rule> source geoip inverse-match	firewall ipv4 output filter rule <text> source geoip inverse-match
×	not yet documented	firewall.xml.in: firewall ipv4 output filter rule <rule> source group address-group	firewall ipv4 output filter rule <text> source group address-group <value>
×	not yet documented	firewall.xml.in: firewall ipv4 output filter rule <rule> source group domain-group	firewall ipv4 output filter rule <text> source group domain-group <value>
×	not yet documented	firewall.xml.in: firewall ipv4 output filter rule <rule> source group dynamic-address-group	Nothing found in VyOS
×	not yet documented	firewall.xml.in: firewall ipv4 output filter rule <rule> source group mac-group	firewall ipv4 output filter rule <text> source group mac-group <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	firewall.xml.in: firewall ipv4 output filter rule <rule> source group network-group <value>	firewall ipv4 output filter rule <text> source group network-group <value>
×	not yet documented	firewall.xml.in: firewall ipv4 output filter rule <rule> source group port-group	firewall ipv4 output filter rule <text> source group port-group <value>
×	not yet documented	firewall.xml.in: firewall ipv4 output filter rule <rule> source mac-address	firewall ipv4 output filter rule <text> source mac-address <value>
×	not yet documented	firewall.xml.in: firewall ipv4 output filter rule <rule> source port	firewall ipv4 output filter rule <text> source port <value>
×	not yet documented	firewall.xml.in: firewall ipv4 output filter rule <rule> state	Nothing found in VyOS
×	not yet documented	Nothing found in XML Definitions	firewall ipv4 output filter rule <text> state established <value>
×	not yet documented	Nothing found in XML Definitions	firewall ipv4 output filter rule <text> state invalid <value>
×	not yet documented	Nothing found in XML Definitions	firewall ipv4 output filter rule <text> state new <value>
×	not yet documented	Nothing found in XML Definitions	firewall ipv4 output filter rule <text> state related <value>
×	not yet documented	firewall.xml.in: firewall ipv4 output filter rule <rule> synproxy tcp mss	Nothing found in VyOS
×	not yet documented	firewall.xml.in: firewall ipv4 output filter rule <rule> synproxy tcp window-scale	Nothing found in VyOS
×	not yet documented	firewall.xml.in: firewall ipv4 output filter rule <rule> tcp flags ack	firewall ipv4 output filter rule <text> tcp flags ack
×	not yet documented	firewall.xml.in: firewall ipv4 output filter rule <rule> tcp flags cwr	firewall ipv4 output filter rule <text> tcp flags cwr
×	not yet documented	firewall.xml.in: firewall ipv4 output filter rule <rule> tcp flags ecn	firewall ipv4 output filter rule <text> tcp flags ecn

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	firewall.xml.in: firewall ipv4 output filter rule <rule> tcp flags fin	firewall ipv4 output filter rule <text> tcp flags fin
×	not yet documented	firewall.xml.in: firewall ipv4 output filter rule <rule> tcp flags not ack	firewall ipv4 output filter rule <text> tcp flags not ack
×	not yet documented	firewall.xml.in: firewall ipv4 output filter rule <rule> tcp flags not cwr	firewall ipv4 output filter rule <text> tcp flags not cwr
×	not yet documented	firewall.xml.in: firewall ipv4 output filter rule <rule> tcp flags not ecn	firewall ipv4 output filter rule <text> tcp flags not ecn
×	not yet documented	firewall.xml.in: firewall ipv4 output filter rule <rule> tcp flags not fin	firewall ipv4 output filter rule <text> tcp flags not fin
×	not yet documented	firewall.xml.in: firewall ipv4 output filter rule <rule> tcp flags not psh	firewall ipv4 output filter rule <text> tcp flags not psh
×	not yet documented	firewall.xml.in: firewall ipv4 output filter rule <rule> tcp flags not rst	firewall ipv4 output filter rule <text> tcp flags not rst
×	not yet documented	firewall.xml.in: firewall ipv4 output filter rule <rule> tcp flags not syn	firewall ipv4 output filter rule <text> tcp flags not syn
×	not yet documented	firewall.xml.in: firewall ipv4 output filter rule <rule> tcp flags not urg	firewall ipv4 output filter rule <text> tcp flags not urg
×	not yet documented	firewall.xml.in: firewall ipv4 output filter rule <rule> tcp flags psh	firewall ipv4 output filter rule <text> tcp flags psh
×	not yet documented	firewall.xml.in: firewall ipv4 output filter rule <rule> tcp flags rst	firewall ipv4 output filter rule <text> tcp flags rst
×	not yet documented	firewall.xml.in: firewall ipv4 output filter rule <rule> tcp flags syn	firewall ipv4 output filter rule <text> tcp flags syn
×	not yet documented	firewall.xml.in: firewall ipv4 output filter rule <rule> tcp flags urg	firewall ipv4 output filter rule <text> tcp flags urg
×	not yet documented	firewall.xml.in: firewall ipv4 output filter rule <rule> tcp mss	firewall ipv4 output fil- ter rule <text> tcp mss <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	firewall.xml.in: firewall ipv4 output filter rule <rule> time startdate	firewall ipv4 output filter rule <text> time startdate <value>
×	not yet documented	firewall.xml.in: firewall ipv4 output filter rule <rule> time starttime	firewall ipv4 output filter rule <text> time starttime <value>
×	not yet documented	firewall.xml.in: firewall ipv4 output filter rule <rule> time stopdate	firewall ipv4 output filter rule <text> time stopdate <value>
×	not yet documented	firewall.xml.in: firewall ipv4 output filter rule <rule> time stoptime	firewall ipv4 output filter rule <text> time stoptime <value>
×	not yet documented	firewall.xml.in: firewall ipv4 output filter rule <rule> time weekdays	firewall ipv4 output filter rule <text> time weekdays <value>
×	not yet documented	firewall.xml.in: firewall ipv4 output filter rule <rule> ttl eq	firewall ipv4 output filter rule <text> ttl eq <value>
×	not yet documented	firewall.xml.in: firewall ipv4 output filter rule <rule> ttl gt	firewall ipv4 output filter rule <text> ttl gt <value>
×	not yet documented	firewall.xml.in: firewall ipv4 output filter rule <rule> ttl lt	firewall ipv4 output filter rule <text> ttl lt <value>
×	<code>set firewall ipv4 output [filter raw] rule <1-999999> set connection-mark <0-2147483647></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 output [filter raw] rule <1-999999> set dscp <0-63></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 output [filter raw] rule <1-999999> set mark <1-2147483647></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 output [filter raw] rule <1-999999> set tcp-mss <500-1460></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 output [filter raw] rule <1-999999> set ttl <0-255></code>	Nothing found in XML Definitions	Nothing found in VyOS

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Table 1 – continued from previous page

Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set firewall ipv4 prerouting raw rule <1-999999> set dscp <0-63></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 prerouting raw rule <1-999999> set mark <1-2147483647></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 prerouting raw rule <1-999999> set tcp-mss <500-1460></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv4 prerouting raw rule <1-999999> set ttl <0-255></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 ...</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 [input forward] filter rule <1-999999> action synproxy</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 [forward input output] filter rule <1-999999> add-address-to-group destination-address address-group <name></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 [forward input output] filter rule <1-999999> add-address-to-group source-address address-group <name></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 [forward input output] filter rule <1-999999> add-address-to-group [destination-address source-address] timeout <timeout></code>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set firewall ipv6 [input / forward] filter rule <1-999999> protocol tcp</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 [input / forward] filter rule <1-999999> synproxy tcp mss <501-65535></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 [input / forward] filter rule <1-999999> synproxy tcp window-scale <1-14></code>	Nothing found in XML Definitions	Nothing found in VyOS
✓	<code>set firewall ipv6 forward filter default-action [accept drop]</code>	firewall.xml.in: firewall ipv6 forward filter default-action	firewall ipv6 forward filter default-action <value>
×	<code>set firewall ipv6 forward filter default-log</code>	firewall.xml.in: firewall ipv6 forward filter default-log	Nothing found in VyOS
×	not yet documented	firewall.xml.in: firewall ipv6 forward filter description	firewall ipv6 forward filter description <value>
×	<code>set firewall ipv6 forward filter rule <1-999999> action [accept continue drop jump queue reject return synproxy]</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 forward filter rule <1-999999> connection-mark <1-2147483647></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 forward filter rule <1-999999> connection-status nat [destination source]</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 forward filter rule <1-999999> description <text></code>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set firewall ipv6 forward filter rule <1-999999> destination address [address addressrange CIDR]</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 forward filter rule <1-999999> destination address-mask [address]</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 forward filter rule <1-999999> destination fqdn <fqdn></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 forward filter rule <1-999999> destination geoip asn <1-4294967294></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 forward filter rule <1-999999> destination geoip country-code <country></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 forward filter rule <1-999999> destination geoip inverse-match</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 forward filter rule <1-999999> destination group address-group <name !name></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 forward filter rule <1-999999> destination group domain-group <name !name></code>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set firewall ipv6 forward filter rule <1-999999> destination group dynamic-address-group <name !name></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 forward filter rule <1-999999> destination group mac-group <name !name></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 forward filter rule <1-999999> destination group network-group <name !name></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 forward filter rule <1-999999> destination group port-group <name !name></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 forward filter rule <1-999999> destination port [1-65535 portname start-end]</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 forward filter rule <1-999999> disable</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 forward filter rule <1-999999> dscp [0-63 start-end]</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 forward filter rule <1-999999> dscp-exclude [0-63 start-end]</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 forward filter rule <1-999999> fragment [match-frag match-non-frag]</code>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set firewall ipv6 forward filter rule <1-999999> hop-limit <eq gt lt> <0-255></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 forward filter rule <1-999999> icmpv6 [code type] <0-255></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 forward filter rule <1-999999> icmpv6 type-name <text></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 forward filter rule <1-999999> inbound-interface group <iface_group></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 forward filter rule <1-999999> inbound-interface name <iface></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 forward filter rule <1-999999> ipsec [match-ipsec-in match-ipsec-out match-none-in match-none-out]</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 forward filter rule <1-999999> jump-target <text></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 forward filter rule <1-999999> limit burst <0-4294967295></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 forward filter rule <1-999999> limit rate <text></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 forward filter rule <1-999999> log</code>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set firewall ipv6 forward filter rule <1-999999> log-options group <0-65535></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 forward filter rule <1-999999> log-options level [emerg alert crit err warn notice info debug]</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 forward filter rule <1-999999> log-options queue-threshold <0-65535></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 forward filter rule <1-999999> log-options snapshot-length <0-9000></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 forward filter rule <1-999999> outbound-interface group <iface_group></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 forward filter rule <1-999999> outbound-interface name <iface></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 forward filter rule <1-999999> packet-length-exclude <text></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 forward filter rule <1-999999> packet-length <text></code>	Nothing found in XML Definitions	Nothing found in VyOS

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Table 1 – continued from previous page

Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set firewall ipv6 forward filter rule <1-999999> packet-type [broadcast host multicast other]</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 forward filter rule <1-999999> protocol [<text> <0-255> all tcp_udp]</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 forward filter rule <1-999999> queue-options bypass</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 forward filter rule <1-999999> queue-options fanout</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 forward filter rule <1-999999> queue <0-65535></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 forward filter rule <1-999999> recent count <1-255></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 forward filter rule <1-999999> recent time [second minute hour]</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 forward filter rule <1-999999> recent time <second minute hour></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 forward filter rule <1-999999> set connection-mark <0-2147483647></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 forward filter rule <1-999999> set dscp <0-63></code>	Nothing found in XML Definitions	Nothing found in VyOS

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Table 1 – continued from previous page

Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set firewall ipv6 forward filter rule <1-999999> set hop-limit <0-255></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 forward filter rule <1-999999> set mark <1-2147483647></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 forward filter rule <1-999999> set tcp-mss <500-1460></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 forward filter rule <1-999999> source address [address addressrange CIDR]</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 forward filter rule <1-999999> source address-mask [address]</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 forward filter rule <1-999999> source fqdn <fqdn></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 forward filter rule <1-999999> source geoip asn <1-4294967294></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 forward filter rule <1-999999> source geoip country-code <country></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 forward filter rule <1-999999> source geoip inverse-match</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 forward filter rule <1-999999> source group address-group <name !name></code>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set firewall ipv6 forward filter rule <1-999999> source group domain-group <name !name></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 forward filter rule <1-999999> source group dynamic-address-group <name !name></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 forward filter rule <1-999999> source group mac-group <name !name></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 forward filter rule <1-999999> source group network-group <name !name></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 forward filter rule <1-999999> source group port-group <name !name></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 forward filter rule <1-999999> source mac-address <mac-address></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 forward filter rule <1-999999> source port [1-65535 portname start-end]</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 forward filter rule <1-999999> state [established invalid new related]</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 forward filter rule <1-999999> tcp flags [not] <text></code>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set firewall ipv6 forward filter rule <1-999999> time startdate <text></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 forward filter rule <1-999999> time starttime <text></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 forward filter rule <1-999999> time stopdate <text></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 forward filter rule <1-999999> time stoptime <text></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 forward filter rule <1-999999> time weekdays <text></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	firewall.xml.in: firewall ipv6 forward filter rule <rule> action	firewall ipv6 forward filter rule <text> action <value>
×	not yet documented	firewall.xml.in: firewall ipv6 forward filter rule <rule> add-address-to-group destination-address address-group	Nothing found in VyOS
×	not yet documented	firewall.xml.in: firewall ipv6 forward filter rule <rule> add-address-to-group destination-address timeout	Nothing found in VyOS
×	not yet documented	firewall.xml.in: firewall ipv6 forward filter rule <rule> add-address-to-group source-address address-group	Nothing found in VyOS
×	not yet documented	firewall.xml.in: firewall ipv6 forward filter rule <rule> add-address-to-group source-address timeout	Nothing found in VyOS
×	not yet documented	firewall.xml.in: firewall ipv6 forward filter rule <rule> connection-mark	firewall ipv6 forward filter rule <text> connection-mark <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	firewall.xml.in: firewall ipv6 forward filter rule <rule> connection-status nat	firewall ipv6 forward filter rule <text> connection-status nat <value>
×	not yet documented	firewall.xml.in: firewall ipv6 forward filter rule <rule> conntrack-helper	Nothing found in VyOS
×	not yet documented	firewall.xml.in: firewall ipv6 forward filter rule <rule> description	firewall ipv6 forward filter rule <text> description <value>
×	not yet documented	firewall.xml.in: firewall ipv6 forward filter rule <rule> destination address	firewall ipv6 forward filter rule <text> destination address <value>
×	not yet documented	firewall.xml.in: firewall ipv6 forward filter rule <rule> destination address-mask	firewall ipv6 forward filter rule <text> destination address-mask <value>
×	not yet documented	firewall.xml.in: firewall ipv6 forward filter rule <rule> destination fqdn	firewall ipv6 forward filter rule <text> destination fqdn <value>
×	not yet documented	firewall.xml.in: firewall ipv6 forward filter rule <rule> destination geoip country-code	firewall ipv6 forward filter rule <text> destination geoip country-code <value>
×	not yet documented	firewall.xml.in: firewall ipv6 forward filter rule <rule> destination geoip inverse-match	firewall ipv6 forward filter rule <text> destination geoip inverse-match
×	not yet documented	firewall.xml.in: firewall ipv6 forward filter rule <rule> destination group address-group	firewall ipv6 forward filter rule <text> destination group address-group <value>
×	not yet documented	firewall.xml.in: firewall ipv6 forward filter rule <rule> destination group domain-group	firewall ipv6 forward filter rule <text> destination group domain-group <value>
×	not yet documented	firewall.xml.in: firewall ipv6 forward filter rule <rule> destination group dynamic-address-group	Nothing found in VyOS
×	not yet documented	firewall.xml.in: firewall ipv6 forward filter rule <rule> destination group mac-group	firewall ipv6 forward filter rule <text> destination group mac-group <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	firewall.xml.in: firewall ipv6 forward filter rule <rule> destination group network-group	firewall ipv6 forward filter rule <text> destination group network-group <value>
×	not yet documented	firewall.xml.in: firewall ipv6 forward filter rule <rule> destination group port-group	firewall ipv6 forward filter rule <text> destination group port-group <value>
×	not yet documented	firewall.xml.in: firewall ipv6 forward filter rule <rule> destination mac-address	firewall ipv6 forward filter rule <text> destination mac-address <value>
×	not yet documented	firewall.xml.in: firewall ipv6 forward filter rule <rule> destination port	firewall ipv6 forward filter rule <text> destination port <value>
×	not yet documented	firewall.xml.in: firewall ipv6 forward filter rule <rule> disable	firewall ipv6 forward filter rule <text> disable
×	not yet documented	firewall.xml.in: firewall ipv6 forward filter rule <rule> dscp	firewall ipv6 forward filter rule <text> dscp <value>
×	not yet documented	firewall.xml.in: firewall ipv6 forward filter rule <rule> dscp-exclude	firewall ipv6 forward filter rule <text> dscp-exclude <value>
×	not yet documented	firewall.xml.in: firewall ipv6 forward filter rule <rule> fragment match-frag	firewall ipv6 forward filter rule <text> fragment match-frag
×	not yet documented	firewall.xml.in: firewall ipv6 forward filter rule <rule> fragment match-non-frag	firewall ipv6 forward filter rule <text> fragment match-non-frag
×	not yet documented	firewall.xml.in: firewall ipv6 forward filter rule <rule> hop-limit eq	firewall ipv6 forward filter rule <text> hop-limit eq <value>
×	not yet documented	firewall.xml.in: firewall ipv6 forward filter rule <rule> hop-limit gt	firewall ipv6 forward filter rule <text> hop-limit gt <value>
×	not yet documented	firewall.xml.in: firewall ipv6 forward filter rule <rule> hop-limit lt	firewall ipv6 forward filter rule <text> hop-limit lt <value>
×	not yet documented	firewall.xml.in: firewall ipv6 forward filter rule <rule> icmpv6 code	firewall ipv6 forward filter rule <text> icmpv6 code <value>
×	not yet documented	firewall.xml.in: firewall ipv6 forward filter rule <rule> icmpv6 type	firewall ipv6 forward filter rule <text> icmpv6 type <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	firewall.xml.in: firewall ipv6 forward filter rule <rule> icmpv6 type-name	firewall ipv6 forward filter rule <text> icmpv6 type-name <value>
×	not yet documented	firewall.xml.in: firewall ipv6 forward filter rule <rule> inbound-interface group	Nothing found in VyOS
×	not yet documented	Nothing found in XML Definitions	firewall ipv6 forward filter rule <text> inbound-interface interface-group <value>
×	not yet documented	Nothing found in XML Definitions	firewall ipv6 forward filter rule <text> inbound-interface interface-name <value>
×	not yet documented	firewall.xml.in: firewall ipv6 forward filter rule <rule> inbound-interface name	Nothing found in VyOS
×	not yet documented	firewall.xml.in: firewall ipv6 forward filter rule <rule> ipsec match-ipsec	firewall ipv6 forward filter rule <text> ipsec match-ipsec
×	not yet documented	firewall.xml.in: firewall ipv6 forward filter rule <rule> ipsec match-none	firewall ipv6 forward filter rule <text> ipsec match-none
×	not yet documented	firewall.xml.in: firewall ipv6 forward filter rule <rule> jump-target	firewall ipv6 forward filter rule <text> jump-target <value>
×	not yet documented	firewall.xml.in: firewall ipv6 forward filter rule <rule> limit burst	firewall ipv6 forward filter rule <text> limit burst <value>
×	not yet documented	firewall.xml.in: firewall ipv6 forward filter rule <rule> limit rate	firewall ipv6 forward filter rule <text> limit rate <value>
×	not yet documented	firewall.xml.in: firewall ipv6 forward filter rule <rule> log	firewall ipv6 forward filter rule <text> log <value>
×	not yet documented	firewall.xml.in: firewall ipv6 forward filter rule <rule> log-options group	firewall ipv6 forward filter rule <text> log-options group <value>
×	not yet documented	firewall.xml.in: firewall ipv6 forward filter rule <rule> log-options level	firewall ipv6 forward filter rule <text> log-options level <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	firewall.xml.in: firewall ipv6 forward filter rule <rule> log-options queue-threshold	firewall ipv6 forward filter rule <text> log-options queue-threshold <value>
×	not yet documented	firewall.xml.in: firewall ipv6 forward filter rule <rule> log-options snapshot-length	firewall ipv6 forward filter rule <text> log-options snapshot-length <value>
×	not yet documented	firewall.xml.in: firewall ipv6 forward filter rule <rule> mark	Nothing found in VyOS
×	not yet documented	firewall.xml.in: firewall ipv6 forward filter rule <rule> offload-target	Nothing found in VyOS
×	not yet documented	firewall.xml.in: firewall ipv6 forward filter rule <rule> outbound-interface group	Nothing found in VyOS
×	not yet documented	Nothing found in XML Definitions	firewall ipv6 forward filter rule <text> outbound-interface interface-group <value>
×	not yet documented	Nothing found in XML Definitions	firewall ipv6 forward filter rule <text> outbound-interface interface-name <value>
×	not yet documented	firewall.xml.in: firewall ipv6 forward filter rule <rule> outbound-interface name	Nothing found in VyOS
×	not yet documented	firewall.xml.in: firewall ipv6 forward filter rule <rule> packet-length	firewall ipv6 forward filter rule <text> packet-length <value>
×	not yet documented	firewall.xml.in: firewall ipv6 forward filter rule <rule> packet-length-exclude	firewall ipv6 forward filter rule <text> packet-length-exclude <value>
×	not yet documented	firewall.xml.in: firewall ipv6 forward filter rule <rule> packet-type	firewall ipv6 forward filter rule <text> packet-type <value>
×	not yet documented	firewall.xml.in: firewall ipv6 forward filter rule <rule> protocol	firewall ipv6 forward filter rule <text> protocol <value>
×	not yet documented	firewall.xml.in: firewall ipv6 forward filter rule <rule> queue	firewall ipv6 forward filter rule <text> queue <value>
×	not yet documented	firewall.xml.in: firewall ipv6 forward filter rule <rule> queue-options	firewall ipv6 forward filter rule <text> queue-options <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	firewall.xml.in: firewall ipv6 forward filter rule <rule> recent count	firewall ipv6 forward filter rule <text> recent count <value>
×	not yet documented	firewall.xml.in: firewall ipv6 forward filter rule <rule> recent time	firewall ipv6 forward filter rule <text> recent time <value>
×	not yet documented	firewall.xml.in: firewall ipv6 forward filter rule <rule> source address	firewall ipv6 forward filter rule <text> source address <value>
×	not yet documented	firewall.xml.in: firewall ipv6 forward filter rule <rule> source address-mask	firewall ipv6 forward filter rule <text> source address-mask <value>
×	not yet documented	firewall.xml.in: firewall ipv6 forward filter rule <rule> source fqdn	firewall ipv6 forward filter rule <text> source fqdn <value>
×	not yet documented	firewall.xml.in: firewall ipv6 forward filter rule <rule> source geoip country-code	firewall ipv6 forward filter rule <text> source geoip country-code <value>
×	not yet documented	firewall.xml.in: firewall ipv6 forward filter rule <rule> source geoip inverse-match	firewall ipv6 forward filter rule <text> source geoip inverse-match
×	not yet documented	firewall.xml.in: firewall ipv6 forward filter rule <rule> source group address-group	firewall ipv6 forward filter rule <text> source group address-group <value>
×	not yet documented	firewall.xml.in: firewall ipv6 forward filter rule <rule> source group domain-group	firewall ipv6 forward filter rule <text> source group domain-group <value>
×	not yet documented	firewall.xml.in: firewall ipv6 forward filter rule <rule> source group dynamic-address-group	Nothing found in VyOS
×	not yet documented	firewall.xml.in: firewall ipv6 forward filter rule <rule> source group mac-group	firewall ipv6 forward filter rule <text> source group mac-group <value>
×	not yet documented	firewall.xml.in: firewall ipv6 forward filter rule <rule> source group network-group	firewall ipv6 forward filter rule <text> source group network-group <value>
×	not yet documented	firewall.xml.in: firewall ipv6 forward filter rule <rule> source group port-group	firewall ipv6 forward filter rule <text> source group port-group <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	firewall.xml.in: firewall ipv6 forward filter rule <rule> source mac-address	firewall ipv6 forward filter rule <text> source mac-address <value>
×	not yet documented	firewall.xml.in: firewall ipv6 forward filter rule <rule> source port	firewall ipv6 forward filter rule <text> source port <value>
×	not yet documented	firewall.xml.in: firewall ipv6 forward filter rule <rule> state	Nothing found in VyOS
×	not yet documented	Nothing found in XML Definitions	firewall ipv6 forward filter rule <text> state established <value>
×	not yet documented	Nothing found in XML Definitions	firewall ipv6 forward filter rule <text> state invalid <value>
×	not yet documented	Nothing found in XML Definitions	firewall ipv6 forward filter rule <text> state new <value>
×	not yet documented	Nothing found in XML Definitions	firewall ipv6 forward filter rule <text> state related <value>
×	not yet documented	firewall.xml.in: firewall ipv6 forward filter rule <rule> synproxy tcp mss	Nothing found in VyOS
×	not yet documented	firewall.xml.in: firewall ipv6 forward filter rule <rule> synproxy tcp window-scale	Nothing found in VyOS
×	not yet documented	firewall.xml.in: firewall ipv6 forward filter rule <rule> tcp flags ack	firewall ipv6 forward filter rule <text> tcp flags ack
×	not yet documented	firewall.xml.in: firewall ipv6 forward filter rule <rule> tcp flags cwr	firewall ipv6 forward filter rule <text> tcp flags cwr
×	not yet documented	firewall.xml.in: firewall ipv6 forward filter rule <rule> tcp flags ecn	firewall ipv6 forward filter rule <text> tcp flags ecn
×	not yet documented	firewall.xml.in: firewall ipv6 forward filter rule <rule> tcp flags fin	firewall ipv6 forward filter rule <text> tcp flags fin
×	not yet documented	firewall.xml.in: firewall ipv6 forward filter rule <rule> tcp flags not ack	firewall ipv6 forward filter rule <text> tcp flags not ack

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	firewall.xml.in: firewall ipv6 forward filter rule <rule> tcp flags not cwr	firewall ipv6 forward fil- ter rule <text> tcp flags not cwr
×	not yet documented	firewall.xml.in: firewall ipv6 forward filter rule <rule> tcp flags not ecn	firewall ipv6 forward fil- ter rule <text> tcp flags not ecn
×	not yet documented	firewall.xml.in: firewall ipv6 forward filter rule <rule> tcp flags not fin	firewall ipv6 forward filter rule <text> tcp flags not fin
×	not yet documented	firewall.xml.in: firewall ipv6 forward filter rule <rule> tcp flags not psh	firewall ipv6 forward fil- ter rule <text> tcp flags not psh
×	not yet documented	firewall.xml.in: firewall ipv6 forward filter rule <rule> tcp flags not rst	firewall ipv6 forward filter rule <text> tcp flags not rst
×	not yet documented	firewall.xml.in: firewall ipv6 forward filter rule <rule> tcp flags not syn	firewall ipv6 forward fil- ter rule <text> tcp flags not syn
×	not yet documented	firewall.xml.in: firewall ipv6 forward filter rule <rule> tcp flags not urg	firewall ipv6 forward fil- ter rule <text> tcp flags not urg
×	not yet documented	firewall.xml.in: firewall ipv6 forward filter rule <rule> tcp flags psh	firewall ipv6 forward filter rule <text> tcp flags psh
×	not yet documented	firewall.xml.in: firewall ipv6 forward filter rule <rule> tcp flags rst	firewall ipv6 forward filter rule <text> tcp flags rst
×	not yet documented	firewall.xml.in: firewall ipv6 forward filter rule <rule> tcp flags syn	firewall ipv6 forward filter rule <text> tcp flags syn
×	not yet documented	firewall.xml.in: firewall ipv6 forward filter rule <rule> tcp flags urg	firewall ipv6 forward filter rule <text> tcp flags urg
×	not yet documented	firewall.xml.in: firewall ipv6 forward filter rule <rule> tcp mss <value>	firewall ipv6 forward fil- ter rule <text> tcp mss <value>
×	not yet documented	firewall.xml.in: firewall ipv6 forward filter rule <rule> time startdate <value>	firewall ipv6 forward filter rule <text> time startdate <value>
×	not yet documented	firewall.xml.in: firewall ipv6 forward filter rule <rule> time starttime <value>	firewall ipv6 forward filter rule <text> time starttime <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	firewall.xml.in: firewall ipv6 forward filter rule <rule> time stopdate	firewall ipv6 forward filter rule <text> time stopdate <value>
×	not yet documented	firewall.xml.in: firewall ipv6 forward filter rule <rule> time stoptime	firewall ipv6 forward filter rule <text> time stoptime <value>
×	not yet documented	firewall.xml.in: firewall ipv6 forward filter rule <rule> time weekdays	firewall ipv6 forward filter rule <text> time weekdays <value>
✓	<code>set firewall ipv6 input filter default-action [accept drop]</code>	firewall.xml.in: firewall ipv6 input filter default-action	firewall ipv6 input filter default-action <value>
×	<code>set firewall ipv6 input filter default-log</code>	firewall.xml.in: firewall ipv6 input filter default-log	Nothing found in VyOS
×	not yet documented	firewall.xml.in: firewall ipv6 input filter description	firewall ipv6 input filter description <value>
×	<code>set firewall ipv6 input filter rule <1-999999> action [accept continue drop jump queue reject return synproxy]</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 input filter rule <1-999999> connection-mark <1-2147483647></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 input filter rule <1-999999> connection-status nat [destination source]</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 input filter rule <1-999999> description <text></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 input filter rule <1-999999> destination address [address addressrange CIDR]</code>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set firewall ipv6 input filter rule <1-999999> destination address-mask [address]</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 input filter rule <1-999999> destination fqdn <fqdn></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 input filter rule <1-999999> destination geoip asn <1-4294967294></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 input filter rule <1-999999> destination geoip country-code <country></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 input filter rule <1-999999> destination geoip inverse-match</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 input filter rule <1-999999> destination group address-group <name !name></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 input filter rule <1-999999> destination group domain-group <name / !name></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 input filter rule <1-999999> destination group dynamic-address-group <name !name></code>	Nothing found in XML Definitions	Nothing found in VyOS

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Table 1 – continued from previous page

Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set firewall ipv6 input filter rule <1-999999> destination group mac-group <name !name></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 input filter rule <1-999999> destination group network-group <name !name></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 input filter rule <1-999999> destination group port-group <name !name></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 input filter rule <1-999999> destination port [1-65535 portname start-end]</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 input filter rule <1-999999> disable</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 input filter rule <1-999999> dscp [0-63 start-end]</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 input filter rule <1-999999> dscp-exclude [0-63 start-end]</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 input filter rule <1-999999> fragment [match-frag match-non-frag]</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 input filter rule <1-999999> hop-limit <eq gt lt> <0-255></code>	Nothing found in XML Definitions	Nothing found in VyOS

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Table 1 – continued from previous page

Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set firewall ipv6 input filter rule <1-999999> icmpv6 [code type] <0-255></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 input filter rule <1-999999> icmpv6 type-name <text></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 input filter rule <1-999999> inbound-interface group <iface_group></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 input filter rule <1-999999> inbound-interface name <iface></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 input filter rule <1-999999> ipsec [match-ipsec-in match-none-in]</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 input filter rule <1-999999> jump-target <text></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 input filter rule <1-999999> limit burst <0-4294967295></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 input filter rule <1-999999> limit rate <text></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 input filter rule <1-999999> log</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 input filter rule <1-999999> log-options group <0-65535></code>	Nothing found in XML Definitions	Nothing found in VyOS

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Table 1 – continued from previous page

Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set firewall ipv6 input filter rule <1-999999> log-options level [emerg alert crit err warn notice info debug]</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 input filter rule <1-999999> log-options queue-threshold <0-65535></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 input filter rule <1-999999> log-options snapshot-length <0-9000></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 input filter rule <1-999999> packet-length-exclude <text></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 input filter rule <1-999999> packet-length <text></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 input filter rule <1-999999> packet-type [broadcast host multicast other]</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 input filter rule <1-999999> protocol [<text> <0-255> all tcp_udp]</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 input filter rule <1-999999> queue-options bypass</code>	Nothing found in XML Definitions	Nothing found in VyOS

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Table 1 – continued from previous page

Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set firewall ipv6 input filter rule <1-999999> queue-options fanout</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 input filter rule <1-999999> queue <0-65535></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 input filter rule <1-999999> recent count <1-255></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 input filter rule <1-999999> recent time [second minute hour]</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 input filter rule <1-999999> recent time <second minute hour></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 input filter rule <1-999999> source address [address addressrange CIDR]</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 input filter rule <1-999999> source address-mask [address]</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 input filter rule <1-999999> source fqdn <fqdn></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 input filter rule <1-999999> source geoip asn <1-4294967294></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 input filter rule <1-999999> source geoip country-code <country></code>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set firewall ipv6 input filter rule <1-999999> source geoip inverse-match</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 input filter rule <1-999999> source group address-group <name !name></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 input filter rule <1-999999> source group domain-group <name !name></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 input filter rule <1-999999> source group dynamic-address-group <name !name></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 input filter rule <1-999999> source group mac-group <name !name></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 input filter rule <1-999999> source group network-group <name !name></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 input filter rule <1-999999> source group port-group <name !name></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 input filter rule <1-999999> source mac-address <mac-address></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 input filter rule <1-999999> source port [1-65535 portname start-end]</code>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set firewall ipv6 input filter rule <1-999999> state [established invalid new related]</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 input filter rule <1-999999> tcp flags [not] <text></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 input filter rule <1-999999> time startdate <text></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 input filter rule <1-999999> time starttime <text></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 input filter rule <1-999999> time stopdate <text></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 input filter rule <1-999999> time stoptime <text></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 input filter rule <1-999999> time weekdays <text></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	firewall.xml.in: firewall ipv6 input filter rule <rule> action	firewall ipv6 input filter rule <text> action <value>
×	not yet documented	firewall.xml.in: firewall ipv6 input filter rule <rule> add-address- to-group destination- address address-group	Nothing found in VyOS
×	not yet documented	firewall.xml.in: firewall ipv6 input filter rule <rule> add-address- to-group destination- address timeout	Nothing found in VyOS
×	not yet documented	firewall.xml.in: firewall ipv6 input filter rule <rule> add-address- to-group source-address address-group	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	firewall.xml.in: firewall ipv6 input filter rule <rule> add-address- to-group source-address timeout	Nothing found in VyOS
×	not yet documented	firewall.xml.in: firewall ipv6 input filter rule <rule> connection- mark	firewall ipv6 input filter rule <text> connection- mark <value>
×	not yet documented	firewall.xml.in: firewall ipv6 input filter rule <rule> connection- status nat	firewall ipv6 input filter rule <text> connection- status nat <value>
×	not yet documented	firewall.xml.in: firewall ipv6 input filter rule <rule> conntrack- helper	Nothing found in VyOS
×	not yet documented	firewall.xml.in: firewall ipv6 input filter rule <rule> description	firewall ipv6 input filter rule <text> description <value>
×	not yet documented	firewall.xml.in: firewall ipv6 input filter rule <rule> destination ad- dress	firewall ipv6 input filter rule <text> destination ad- dress <value>
×	not yet documented	firewall.xml.in: firewall ipv6 input filter rule <rule> destination address-mask	firewall ipv6 input filter rule <text> destination address-mask <value>
×	not yet documented	firewall.xml.in: firewall ipv6 input filter rule <rule> destination fqdn	firewall ipv6 input filter rule <text> destination fqdn <value>
×	not yet documented	firewall.xml.in: firewall ipv6 input filter rule <rule> destination geoip country-code	firewall ipv6 input filter rule <text> destination geoip country-code <value>
×	not yet documented	firewall.xml.in: firewall ipv6 input filter rule <rule> destination geoip inverse-match	firewall ipv6 input filter rule <text> destination geoip inverse-match
×	not yet documented	firewall.xml.in: firewall ipv6 input filter rule <rule> destination group address-group	firewall ipv6 input filter rule <text> destination group address-group <value>
×	not yet documented	firewall.xml.in: firewall ipv6 input filter rule <rule> destination group domain-group	firewall ipv6 input filter rule <text> destination group domain-group <value>

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Table 1 – continued from previous page

Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	firewall.xml.in: firewall ipv6 input filter rule <rule> destination group dynamic-address- group	Nothing found in VyOS
×	not yet documented	firewall.xml.in: firewall ipv6 input filter rule <rule> destination group mac-group <value>	firewall ipv6 input filter rule <text> destination group mac-group <value>
×	not yet documented	firewall.xml.in: firewall ipv6 input filter rule <rule> destination group network-group <value>	firewall ipv6 input filter rule <text> destination group network-group <value>
×	not yet documented	firewall.xml.in: firewall ipv6 input filter rule <rule> destination group port-group <value>	firewall ipv6 input filter rule <text> destination group port-group <value>
×	not yet documented	firewall.xml.in: firewall ipv6 input filter rule <rule> destination mac-address <value>	firewall ipv6 input filter rule <text> destination mac-address <value>
×	not yet documented	firewall.xml.in: firewall ipv6 input filter rule <rule> destination port <value>	firewall ipv6 input filter rule <text> destination port <value>
×	not yet documented	firewall.xml.in: firewall ipv6 input filter rule <rule> disable	firewall ipv6 input filter rule <text> disable
×	not yet documented	firewall.xml.in: firewall ipv6 input filter rule <rule> dscp	firewall ipv6 input filter rule <text> dscp <value>
×	not yet documented	firewall.xml.in: firewall ipv6 input filter rule <rule> dscp-exclude <value>	firewall ipv6 input filter rule <text> dscp-exclude <value>
×	not yet documented	firewall.xml.in: firewall ipv6 input filter rule <rule> fragment match-frag	firewall ipv6 input filter rule <text> fragment match-frag
×	not yet documented	firewall.xml.in: firewall ipv6 input filter rule <rule> fragment match-non-frag	firewall ipv6 input filter rule <text> fragment match-non-frag
×	not yet documented	firewall.xml.in: firewall ipv6 input filter rule <rule> hop-limit eq <value>	firewall ipv6 input filter rule <text> hop-limit eq <value>
×	not yet documented	firewall.xml.in: firewall ipv6 input filter rule <rule> hop-limit gt <value>	firewall ipv6 input filter rule <text> hop-limit gt <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	firewall.xml.in: firewall ipv6 input filter rule <rule> hop-limit lt	firewall ipv6 input filter rule <text> hop-limit lt <value>
×	not yet documented	firewall.xml.in: firewall ipv6 input filter rule <rule> icmpv6 code	firewall ipv6 input filter rule <text> icmpv6 code <value>
×	not yet documented	firewall.xml.in: firewall ipv6 input filter rule <rule> icmpv6 type	firewall ipv6 input filter rule <text> icmpv6 type <value>
×	not yet documented	firewall.xml.in: firewall ipv6 input filter rule <rule> icmpv6 type- name	firewall ipv6 input filter rule <text> icmpv6 type- name <value>
×	not yet documented	firewall.xml.in: firewall ipv6 input filter rule <rule> inbound- interface group	Nothing found in VyOS
×	not yet documented	Nothing found in XML Definitions	firewall ipv6 input filter rule <text> inbound- interface interface-group <value>
×	not yet documented	Nothing found in XML Definitions	firewall ipv6 input filter rule <text> inbound- interface interface-name <value>
×	not yet documented	firewall.xml.in: firewall ipv6 input filter rule <rule> inbound- interface name	Nothing found in VyOS
×	not yet documented	firewall.xml.in: firewall ipv6 input filter rule <rule> ipsec match- ipsec	firewall ipv6 input filter rule <text> ipsec match- ipsec
×	not yet documented	firewall.xml.in: firewall ipv6 input filter rule <rule> ipsec match- none	firewall ipv6 input filter rule <text> ipsec match- none
×	not yet documented	firewall.xml.in: firewall ipv6 input filter rule <rule> jump-target	firewall ipv6 input filter rule <text> jump-target <value>
×	not yet documented	firewall.xml.in: firewall ipv6 input filter rule <rule> limit burst	firewall ipv6 input filter rule <text> limit burst <value>
×	not yet documented	firewall.xml.in: firewall ipv6 input filter rule <rule> limit rate	firewall ipv6 input filter rule <text> limit rate <value>
×	not yet documented	firewall.xml.in: firewall ipv6 input filter rule <rule> log	firewall ipv6 input filter rule <text> log <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	firewall.xml.in: firewall ipv6 input filter rule <rule> log-options group	firewall ipv6 input filter rule <text> log-options group <value>
×	not yet documented	firewall.xml.in: firewall ipv6 input filter rule <rule> log-options level	firewall ipv6 input filter rule <text> log-options level <value>
×	not yet documented	firewall.xml.in: firewall ipv6 input filter rule <rule> log-options queue-threshold	firewall ipv6 input filter rule <text> log-options queue-threshold <value>
×	not yet documented	firewall.xml.in: firewall ipv6 input filter rule <rule> log-options snapshot-length	firewall ipv6 input filter rule <text> log-options snapshot-length <value>
×	not yet documented	firewall.xml.in: firewall ipv6 input filter rule <rule> mark	Nothing found in VyOS
×	not yet documented	firewall.xml.in: firewall ipv6 input filter rule <rule> packet-length	firewall ipv6 input filter rule <text> packet-length <value>
×	not yet documented	firewall.xml.in: firewall ipv6 input filter rule <rule> packet-length- exclude	firewall ipv6 input filter rule <text> packet-length- exclude <value>
×	not yet documented	firewall.xml.in: firewall ipv6 input filter rule <rule> packet-type	firewall ipv6 input filter rule <text> packet-type <value>
×	not yet documented	firewall.xml.in: firewall ipv6 input filter rule <rule> protocol	firewall ipv6 input fil- ter rule <text> protocol <value>
×	not yet documented	firewall.xml.in: firewall ipv6 input filter rule <rule> queue	firewall ipv6 input filter rule <text> queue <value>
×	not yet documented	firewall.xml.in: firewall ipv6 input filter rule <rule> queue-options	firewall ipv6 input filter rule <text> queue-options <value>
×	not yet documented	firewall.xml.in: firewall ipv6 input filter rule <rule> recent count	firewall ipv6 input filter rule <text> recent count <value>
×	not yet documented	firewall.xml.in: firewall ipv6 input filter rule <rule> recent time	firewall ipv6 input filter rule <text> recent time <value>
×	not yet documented	firewall.xml.in: firewall ipv6 input filter rule <rule> source address	firewall ipv6 input filter rule <text> source address <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	firewall.xml.in: firewall ipv6 input filter rule <rule> source address-mask	firewall ipv6 input filter rule <text> source address-mask <value>
×	not yet documented	firewall.xml.in: firewall ipv6 input filter rule <rule> source fqdn	firewall ipv6 input filter rule <text> source fqdn <value>
×	not yet documented	firewall.xml.in: firewall ipv6 input filter rule <rule> source geoip country-code	firewall ipv6 input filter rule <text> source geoip country-code <value>
×	not yet documented	firewall.xml.in: firewall ipv6 input filter rule <rule> source geoip inverse-match	firewall ipv6 input filter rule <text> source geoip inverse-match
×	not yet documented	firewall.xml.in: firewall ipv6 input filter rule <rule> source group address-group	firewall ipv6 input filter rule <text> source group address-group <value>
×	not yet documented	firewall.xml.in: firewall ipv6 input filter rule <rule> source group domain-group	firewall ipv6 input filter rule <text> source group domain-group <value>
×	not yet documented	firewall.xml.in: firewall ipv6 input filter rule <rule> source group dynamic-address-group	Nothing found in VyOS
×	not yet documented	firewall.xml.in: firewall ipv6 input filter rule <rule> source group mac-group	firewall ipv6 input filter rule <text> source group mac-group <value>
×	not yet documented	firewall.xml.in: firewall ipv6 input filter rule <rule> source group network-group	firewall ipv6 input filter rule <text> source group network-group <value>
×	not yet documented	firewall.xml.in: firewall ipv6 input filter rule <rule> source group port-group	firewall ipv6 input filter rule <text> source group port-group <value>
×	not yet documented	firewall.xml.in: firewall ipv6 input filter rule <rule> source mac-address	firewall ipv6 input filter rule <text> source mac-address <value>
×	not yet documented	firewall.xml.in: firewall ipv6 input filter rule <rule> source port	firewall ipv6 input filter rule <text> source port <value>
×	not yet documented	firewall.xml.in: firewall ipv6 input filter rule <rule> state	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	Nothing found in XML Definitions	firewall ipv6 input filter rule <text> state established <value>
×	not yet documented	Nothing found in XML Definitions	firewall ipv6 input filter rule <text> state invalid <value>
×	not yet documented	Nothing found in XML Definitions	firewall ipv6 input filter rule <text> state new <value>
×	not yet documented	Nothing found in XML Definitions	firewall ipv6 input filter rule <text> state related <value>
×	not yet documented	firewall.xml.in: firewall ipv6 input filter rule <rule> synproxy tcp mss	Nothing found in VyOS
×	not yet documented	firewall.xml.in: firewall ipv6 input filter rule <rule> synproxy tcp window-scale	Nothing found in VyOS
×	not yet documented	firewall.xml.in: firewall ipv6 input filter rule <rule> tcp flags ack	firewall ipv6 input filter rule <text> tcp flags ack
×	not yet documented	firewall.xml.in: firewall ipv6 input filter rule <rule> tcp flags cwr	firewall ipv6 input filter rule <text> tcp flags cwr
×	not yet documented	firewall.xml.in: firewall ipv6 input filter rule <rule> tcp flags ecn	firewall ipv6 input filter rule <text> tcp flags ecn
×	not yet documented	firewall.xml.in: firewall ipv6 input filter rule <rule> tcp flags fin	firewall ipv6 input filter rule <text> tcp flags fin
×	not yet documented	firewall.xml.in: firewall ipv6 input filter rule <rule> tcp flags not ack	firewall ipv6 input filter rule <text> tcp flags not ack
×	not yet documented	firewall.xml.in: firewall ipv6 input filter rule <rule> tcp flags not cwr	firewall ipv6 input filter rule <text> tcp flags not cwr
×	not yet documented	firewall.xml.in: firewall ipv6 input filter rule <rule> tcp flags not ecn	firewall ipv6 input filter rule <text> tcp flags not ecn
×	not yet documented	firewall.xml.in: firewall ipv6 input filter rule <rule> tcp flags not fin	firewall ipv6 input filter rule <text> tcp flags not fin

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	firewall.xml.in: firewall ipv6 input filter rule <rule> tcp flags not psh	firewall ipv6 input filter rule <text> tcp flags not psh
×	not yet documented	firewall.xml.in: firewall ipv6 input filter rule <rule> tcp flags not rst	firewall ipv6 input filter rule <text> tcp flags not rst
×	not yet documented	firewall.xml.in: firewall ipv6 input filter rule <rule> tcp flags not syn	firewall ipv6 input filter rule <text> tcp flags not syn
×	not yet documented	firewall.xml.in: firewall ipv6 input filter rule <rule> tcp flags not urg	firewall ipv6 input filter rule <text> tcp flags not urg
×	not yet documented	firewall.xml.in: firewall ipv6 input filter rule <rule> tcp flags psh	firewall ipv6 input filter rule <text> tcp flags psh
×	not yet documented	firewall.xml.in: firewall ipv6 input filter rule <rule> tcp flags rst	firewall ipv6 input filter rule <text> tcp flags rst
×	not yet documented	firewall.xml.in: firewall ipv6 input filter rule <rule> tcp flags syn	firewall ipv6 input filter rule <text> tcp flags syn
×	not yet documented	firewall.xml.in: firewall ipv6 input filter rule <rule> tcp flags urg	firewall ipv6 input filter rule <text> tcp flags urg
×	not yet documented	firewall.xml.in: firewall ipv6 input filter rule <rule> tcp mss	firewall ipv6 input fil- ter rule <text> tcp mss <value>
×	not yet documented	firewall.xml.in: firewall ipv6 input filter rule <rule> time startdate	firewall ipv6 input filter rule <text> time startdate <value>
×	not yet documented	firewall.xml.in: firewall ipv6 input filter rule <rule> time starttime	firewall ipv6 input filter rule <text> time starttime <value>
×	not yet documented	firewall.xml.in: firewall ipv6 input filter rule <rule> time stopdate	firewall ipv6 input filter rule <text> time stopdate <value>
×	not yet documented	firewall.xml.in: firewall ipv6 input filter rule <rule> time stoptime	firewall ipv6 input filter rule <text> time stoptime <value>
×	not yet documented	firewall.xml.in: firewall ipv6 input filter rule <rule> time weekdays	firewall ipv6 input filter rule <text> time weekdays <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set firewall ipv6 name <name> default-action [accept drop jump queue reject return]</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 name <name> default-jump-target <text></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 name <name> default-log</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 name <name> description <text></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 name <name> rule <1-999999> action [accept continue drop jump queue reject return]</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 name <name> rule <1-999999> add-address-to-group destination-address address-group <name></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 name <name> rule <1-999999> add-address-to-group source-address address-group <name></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 name <name> rule <1-999999> add-address-to-group [destination-address source-address] timeout <timeout></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 name <name> rule <1-999999> connection-mark <1-2147483647></code>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set firewall ipv6 name <name> rule <1-999999> connection-status nat [destination / source]</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 name <name> rule <1-999999> description <text></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 name <name> rule <1-999999> destination address [address / addressrange / CIDR]</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 name <name> rule <1-999999> destination address-mask [address]</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 name <name> rule <1-999999> destination fqdn <fqdn></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 name <name> rule <1-999999> destination geoip asn <1-4294967294></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 name <name> rule <1-999999> destination geoip country-code <country></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 name <name> rule <1-999999> destination geoip inverse-match</code>	Nothing found in XML Definitions	Nothing found in VyOS

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Table 1 – continued from previous page

Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set firewall ipv6 name <name> rule <1-999999> destination group address-group <name !name></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 name <name> rule <1-999999> destination group domain-group <name !name></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 name <name> rule <1-999999> destination group dynamic-address-group <name !name></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 name <name> rule <1-999999> destination group mac-group <name !name></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 name <name> rule <1-999999> destination group network-group <name !name></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 name <name> rule <1-999999> destination group port-group <name !name></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 name <name> rule <1-999999> destination port [1-65535 portname start-end]</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 name <name> rule <1-999999> disable</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 name <name> rule <1-999999> dscp [0-63 start-end]</code>	Nothing found in XML Definitions	Nothing found in VyOS

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Table 1 – continued from previous page

Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set firewall ipv6 name <name> rule <1-999999> dscp-exclude [0-63 / start-end]</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 name <name> rule <1-999999> fragment [match-frag / match-non-frag]</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 name <name> rule <1-999999> hop-limit <eq gt lt> <0-255></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 name <name> rule <1-999999> icmpv6 [code type] <0-255></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 name <name> rule <1-999999> icmpv6 type-name <text></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 name <name> rule <1-999999> inbound-interface group <iface_group></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 name <name> rule <1-999999> inbound-interface name <iface></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 name <name> rule <1-999999> ipsec [match-ipsec-in / match-ipsec-out / match-none-in / match-none-out]</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 name <name> rule <1-999999> jump-target <text></code>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set firewall ipv6 name <name> rule <1-999999> limit burst <0-4294967295></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 name <name> rule <1-999999> limit rate <text></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 name <name> rule <1-999999> log</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 name <name> rule <1-999999> log-options group <0-65535></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 name <name> rule <1-999999> log-options level [emerg alert crit err warn notice info debug]</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 name <name> rule <1-999999> log-options queue-threshold <0-65535></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 name <name> rule <1-999999> log-options snapshot-length <0-9000></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 name <name> rule <1-999999> outbound-interface group <iface_group></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 name <name> rule <1-999999> outbound-interface name <iface></code>	Nothing found in XML Definitions	Nothing found in VyOS

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Table 1 – continued from previous page

Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set firewall ipv6 name <name> rule <1-999999> packet-length-exclude <text></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 name <name> rule <1-999999> packet-length <text></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 name <name> rule <1-999999> packet-type [broadcast host multicast other]</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 name <name> rule <1-999999> protocol [<text> <0-255> all tcp_udp]</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 name <name> rule <1-999999> queue-options bypass</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 name <name> rule <1-999999> queue-options fanout</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 name <name> rule <1-999999> queue <0-65535></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 name <name> rule <1-999999> recent count <1-255></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 name <name> rule <1-999999> recent time [second minute hour]</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 name <name> rule <1-999999> recent time <second minute hour></code>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set firewall ipv6 name <name> rule <1-999999> source address [address addressrange CIDR]</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 name <name> rule <1-999999> source address-mask [address]</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 name <name> rule <1-999999> source fqdn <fqdn></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 name <name> rule <1-999999> source geoip asn <1-4294967294></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 name <name> rule <1-999999> source geoip country-code <country></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 name <name> rule <1-999999> source geoip inverse-match</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 name <name> rule <1-999999> source group address-group <name !name></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 name <name> rule <1-999999> source group domain-group <name !name></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 name <name> rule <1-999999> source group dynamic-address-group <name !name></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 name <name> rule <1-999999> source group mac-group <name !name></code>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set firewall ipv6 name <name> rule <1-999999> source group network-group <name !name></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 name <name> rule <1-999999> source group port-group <name !name></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 name <name> rule <1-999999> source mac-address <mac-address></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 name <name> rule <1-999999> source port [1-65535 portname start-end]</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 name <name> rule <1-999999> state [established invalid new related]</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 name <name> rule <1-999999> tcp flags [not] <text></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 name <name> rule <1-999999> time startdate <text></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 name <name> rule <1-999999> time starttime <text></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 name <name> rule <1-999999> time stopdate <text></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 name <name> rule <1-999999> time stoptime <text></code>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set firewall ipv6 name <name> rule <1-999999> time weekdays <text></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	firewall.xml.in: firewall ipv6 name <name> default-action	firewall ipv6 name <text> default-action <value>
×	not yet documented	firewall.xml.in: firewall ipv6 name <name> default-jump-target	firewall ipv6 name <text> default-jump-target <value>
×	not yet documented	firewall.xml.in: firewall ipv6 name <name> default-log	Nothing found in VyOS
×	not yet documented	firewall.xml.in: firewall ipv6 name <name> description	firewall ipv6 name <text> description <value>
×	not yet documented	Nothing found in XML Definitions	firewall ipv6 name <text> enable-default-log
×	not yet documented	firewall.xml.in: firewall ipv6 name <name> rule <rule> action	firewall ipv6 name <text> rule <text> action <value>
×	not yet documented	firewall.xml.in: firewall ipv6 name <name> rule <rule> add-address-to-group destination-address address-group	Nothing found in VyOS
×	not yet documented	firewall.xml.in: firewall ipv6 name <name> rule <rule> add-address-to-group destination-address time-out	Nothing found in VyOS
×	not yet documented	firewall.xml.in: firewall ipv6 name <name> rule <rule> add-address-to-group source-address address-group	Nothing found in VyOS
×	not yet documented	firewall.xml.in: firewall ipv6 name <name> rule <rule> add-address-to-group source-address timeout	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	firewall.xml.in: firewall ipv6 name <name> rule <rule> connection-mark	firewall ipv6 name <text> rule <text> connection- mark <value>
×	not yet documented	firewall.xml.in: firewall ipv6 name <name> rule <rule> connection-status nat	firewall ipv6 name <text> rule <text> connection- status nat <value>
×	not yet documented	firewall.xml.in: firewall ipv6 name <name> rule <rule> conntrack-helper	Nothing found in VyOS
×	not yet documented	firewall.xml.in: firewall ipv6 name <name> rule <rule> description	firewall ipv6 name <text> rule <text> description <value>
×	not yet documented	firewall.xml.in: firewall ipv6 name <name> rule <rule> destination address	firewall ipv6 name <text> rule <text> destination ad- dress <value>
×	not yet documented	firewall.xml.in: firewall ipv6 name <name> rule <rule> destination address-mask	firewall ipv6 name <text> rule <text> destination address-mask <value>
×	not yet documented	firewall.xml.in: firewall ipv6 name <name> rule <rule> destination fqdn	firewall ipv6 name <text> rule <text> destination fqdn <value>
×	not yet documented	firewall.xml.in: firewall ipv6 name <name> rule <rule> desti- nation geoip country-code	firewall ipv6 name <text> rule <text> destination geoip country-code <value>
×	not yet documented	firewall.xml.in: firewall ipv6 name <name> rule <rule> destination geoip inverse- match	firewall ipv6 name <text> rule <text> destination geoip inverse-match
×	not yet documented	firewall.xml.in: firewall ipv6 name <name> rule <rule> destination group address- group	firewall ipv6 name <text> rule <text> destination group address-group <value>
×	not yet documented	firewall.xml.in: firewall ipv6 name <name> rule <rule> destination group domain- group	firewall ipv6 name <text> rule <text> destination group domain-group <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	firewall.xml.in: firewall ipv6 name <name> rule <rule> destination group dynamic-address-group	Nothing found in VyOS
×	not yet documented	firewall.xml.in: firewall ipv6 name <name> rule <rule> destination group mac-group	firewall ipv6 name <text> rule <text> destination group mac-group <value>
×	not yet documented	firewall.xml.in: firewall ipv6 name <name> rule <rule> destination group network-group	firewall ipv6 name <text> rule <text> destination group network-group <value>
×	not yet documented	firewall.xml.in: firewall ipv6 name <name> rule <rule> destination group port-group	firewall ipv6 name <text> rule <text> destination group port-group <value>
×	not yet documented	firewall.xml.in: firewall ipv6 name <name> rule <rule> destination mac-address	firewall ipv6 name <text> rule <text> destination mac-address <value>
×	not yet documented	firewall.xml.in: firewall ipv6 name <name> rule <rule> destination port	firewall ipv6 name <text> rule <text> destination port <value>
×	not yet documented	firewall.xml.in: firewall ipv6 name <name> rule <rule> disable	firewall ipv6 name <text> rule <text> disable
×	not yet documented	firewall.xml.in: firewall ipv6 name <name> rule <rule> dscp	firewall ipv6 name <text> rule <text> dscp <value>
×	not yet documented	firewall.xml.in: firewall ipv6 name <name> rule <rule> dscp-exclude	firewall ipv6 name <text> rule <text> dscp-exclude <value>
×	not yet documented	firewall.xml.in: firewall ipv6 name <name> rule <rule> fragment match-frag	firewall ipv6 name <text> rule <text> fragment match-frag
×	not yet documented	firewall.xml.in: firewall ipv6 name <name> rule <rule> fragment match-non-frag	firewall ipv6 name <text> rule <text> fragment match-non-frag
×	not yet documented	firewall.xml.in: firewall ipv6 name <name> rule <rule> hop-limit eq	firewall ipv6 name <text> rule <text> hop-limit eq <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	firewall.xml.in: firewall ipv6 name <name> rule <rule> hop-limit gt	firewall ipv6 name <text> rule <text> hop-limit gt <value>
×	not yet documented	firewall.xml.in: firewall ipv6 name <name> rule <rule> hop-limit lt	firewall ipv6 name <text> rule <text> hop-limit lt <value>
×	not yet documented	firewall.xml.in: firewall ipv6 name <name> rule <rule> icmpv6 code	firewall ipv6 name <text> rule <text> icmpv6 code <value>
×	not yet documented	firewall.xml.in: firewall ipv6 name <name> rule <rule> icmpv6 type	firewall ipv6 name <text> rule <text> icmpv6 type <value>
×	not yet documented	firewall.xml.in: firewall ipv6 name <name> rule <rule> icmpv6 type-name	firewall ipv6 name <text> rule <text> icmpv6 type- name <value>
×	not yet documented	firewall.xml.in: firewall ipv6 name <name> rule <rule> inbound-interface group	Nothing found in VyOS
×	not yet documented	Nothing found in XML Definitions	firewall ipv6 name <text> rule <text> inbound- interface interface-group <value>
×	not yet documented	Nothing found in XML Definitions	firewall ipv6 name <text> rule <text> inbound- interface interface-name <value>
×	not yet documented	firewall.xml.in: firewall ipv6 name <name> rule <rule> inbound-interface name	Nothing found in VyOS
×	not yet documented	firewall.xml.in: firewall ipv6 name <name> rule <rule> ipsec match-ipsec	firewall ipv6 name <text> rule <text> ipsec match- ipsec
×	not yet documented	firewall.xml.in: firewall ipv6 name <name> rule <rule> ipsec match-none	firewall ipv6 name <text> rule <text> ipsec match- none
×	not yet documented	firewall.xml.in: firewall ipv6 name <name> rule <rule> jump-target	firewall ipv6 name <text> rule <text> jump-target <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	firewall.xml.in: firewall ipv6 name <name> rule <rule> limit burst	firewall ipv6 name <text> rule <text> limit burst <value>
×	not yet documented	firewall.xml.in: firewall ipv6 name <name> rule <rule> limit rate	firewall ipv6 name <text> rule <text> limit rate <value>
×	not yet documented	firewall.xml.in: firewall ipv6 name <name> rule <rule> log	firewall ipv6 name <text> rule <text> log <value>
×	not yet documented	firewall.xml.in: firewall ipv6 name <name> rule <rule> log-options group	firewall ipv6 name <text> rule <text> log-options group <value>
×	not yet documented	firewall.xml.in: firewall ipv6 name <name> rule <rule> log-options level	firewall ipv6 name <text> rule <text> log-options level <value>
×	not yet documented	firewall.xml.in: firewall ipv6 name <name> rule <rule> log- options queue-threshold	firewall ipv6 name <text> rule <text> log-options queue-threshold <value>
×	not yet documented	firewall.xml.in: firewall ipv6 name <name> rule <rule> log- options snapshot-length	firewall ipv6 name <text> rule <text> log-options snapshot-length <value>
×	not yet documented	firewall.xml.in: firewall ipv6 name <name> rule <rule> mark	Nothing found in VyOS
×	not yet documented	firewall.xml.in: firewall ipv6 name <name> rule <rule> offload-target	Nothing found in VyOS
×	not yet documented	firewall.xml.in: firewall ipv6 name <name> rule <rule> outbound-interface group	Nothing found in VyOS
×	not yet documented	Nothing found in XML Definitions	firewall ipv6 name <text> rule <text> outbound- interface interface-group <value>
×	not yet documented	Nothing found in XML Definitions	firewall ipv6 name <text> rule <text> outbound- interface interface-name <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	firewall.xml.in: firewall ipv6 name <name> rule <rule> outbound-interface name	Nothing found in VyOS
×	not yet documented	firewall.xml.in: firewall ipv6 name <name> rule <rule> packet-length	firewall ipv6 name <text> rule <text> packet-length <value>
×	not yet documented	firewall.xml.in: firewall ipv6 name <name> rule <rule> packet-length-exclude	firewall ipv6 name <text> rule <text> packet-length- exclude <value>
×	not yet documented	firewall.xml.in: firewall ipv6 name <name> rule <rule> packet-type	firewall ipv6 name <text> rule <text> packet-type <value>
×	not yet documented	firewall.xml.in: firewall ipv6 name <name> rule <rule> protocol	firewall ipv6 name <text> rule <text> protocol <value>
×	not yet documented	firewall.xml.in: firewall ipv6 name <name> rule <rule> queue	firewall ipv6 name <text> rule <text> queue <value>
×	not yet documented	firewall.xml.in: firewall ipv6 name <name> rule <rule> queue-options	firewall ipv6 name <text> rule <text> queue-options <value>
×	not yet documented	firewall.xml.in: firewall ipv6 name <name> rule <rule> recent count	firewall ipv6 name <text> rule <text> recent count <value>
×	not yet documented	firewall.xml.in: firewall ipv6 name <name> rule <rule> recent time	firewall ipv6 name <text> rule <text> recent time <value>
×	not yet documented	firewall.xml.in: firewall ipv6 name <name> rule <rule> source address	firewall ipv6 name <text> rule <text> source address <value>
×	not yet documented	firewall.xml.in: firewall ipv6 name <name> rule <rule> source address-mask	firewall ipv6 name <text> rule <text> source address-mask <value>
×	not yet documented	firewall.xml.in: firewall ipv6 name <name> rule <rule> source fqdn	firewall ipv6 name <text> rule <text> source fqdn <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	firewall.xml.in: firewall ipv6 name <name> rule <rule> source geoip country-code	firewall ipv6 name <text> rule <text> source geoip country-code <value>
×	not yet documented	firewall.xml.in: firewall ipv6 name <name> rule <rule> source geoip inverse- match	firewall ipv6 name <text> rule <text> source geoip inverse-match
×	not yet documented	firewall.xml.in: firewall ipv6 name <name> rule <rule> source group address- group	firewall ipv6 name <text> rule <text> source group address-group <value>
×	not yet documented	firewall.xml.in: firewall ipv6 name <name> rule <rule> source group domain- group	firewall ipv6 name <text> rule <text> source group domain-group <value>
×	not yet documented	firewall.xml.in: firewall ipv6 name <name> rule <rule> source group dynamic- address-group	Nothing found in VyOS
×	not yet documented	firewall.xml.in: firewall ipv6 name <name> rule <rule> source group mac-group	firewall ipv6 name <text> rule <text> source group mac-group <value>
×	not yet documented	firewall.xml.in: firewall ipv6 name <name> rule <rule> source group network- group	firewall ipv6 name <text> rule <text> source group network-group <value>
×	not yet documented	firewall.xml.in: firewall ipv6 name <name> rule <rule> source group port-group	firewall ipv6 name <text> rule <text> source group port-group <value>
×	not yet documented	firewall.xml.in: firewall ipv6 name <name> rule <rule> source mac-address	firewall ipv6 name <text> rule <text> source mac- address <value>
×	not yet documented	firewall.xml.in: firewall ipv6 name <name> rule <rule> source port	firewall ipv6 name <text> rule <text> source port <value>
×	not yet documented	firewall.xml.in: firewall ipv6 name <name> rule <rule> state	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	Nothing found in XML Definitions	firewall ipv6 name <text> rule <text> state established <value>
×	not yet documented	Nothing found in XML Definitions	firewall ipv6 name <text> rule <text> state invalid <value>
×	not yet documented	Nothing found in XML Definitions	firewall ipv6 name <text> rule <text> state new <value>
×	not yet documented	Nothing found in XML Definitions	firewall ipv6 name <text> rule <text> state related <value>
×	not yet documented	firewall.xml.in: firewall ipv6 name <name> rule <rule> synproxy tcp mss	Nothing found in VyOS
×	not yet documented	firewall.xml.in: firewall ipv6 name <name> rule <rule> syn- proxy tcp window-scale	Nothing found in VyOS
×	not yet documented	firewall.xml.in: firewall ipv6 name <name> rule <rule> tcp flags ack	firewall ipv6 name <text> rule <text> tcp flags ack
×	not yet documented	firewall.xml.in: firewall ipv6 name <name> rule <rule> tcp flags cwr	firewall ipv6 name <text> rule <text> tcp flags cwr
×	not yet documented	firewall.xml.in: firewall ipv6 name <name> rule <rule> tcp flags ecn	firewall ipv6 name <text> rule <text> tcp flags ecn
×	not yet documented	firewall.xml.in: firewall ipv6 name <name> rule <rule> tcp flags fin	firewall ipv6 name <text> rule <text> tcp flags fin
×	not yet documented	firewall.xml.in: firewall ipv6 name <name> rule <rule> tcp flags not ack	firewall ipv6 name <text> rule <text> tcp flags not ack
×	not yet documented	firewall.xml.in: firewall ipv6 name <name> rule <rule> tcp flags not cwr	firewall ipv6 name <text> rule <text> tcp flags not cwr
×	not yet documented	firewall.xml.in: firewall ipv6 name <name> rule <rule> tcp flags not ecn	firewall ipv6 name <text> rule <text> tcp flags not ecn

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	firewall.xml.in: firewall ipv6 name <name> rule <rule> tcp flags not fin	firewall ipv6 name <text> rule <text> tcp flags not fin
×	not yet documented	firewall.xml.in: firewall ipv6 name <name> rule <rule> tcp flags not psh	firewall ipv6 name <text> rule <text> tcp flags not psh
×	not yet documented	firewall.xml.in: firewall ipv6 name <name> rule <rule> tcp flags not rst	firewall ipv6 name <text> rule <text> tcp flags not rst
×	not yet documented	firewall.xml.in: firewall ipv6 name <name> rule <rule> tcp flags not syn	firewall ipv6 name <text> rule <text> tcp flags not syn
×	not yet documented	firewall.xml.in: firewall ipv6 name <name> rule <rule> tcp flags not urg	firewall ipv6 name <text> rule <text> tcp flags not urg
×	not yet documented	firewall.xml.in: firewall ipv6 name <name> rule <rule> tcp flags psh	firewall ipv6 name <text> rule <text> tcp flags psh
×	not yet documented	firewall.xml.in: firewall ipv6 name <name> rule <rule> tcp flags rst	firewall ipv6 name <text> rule <text> tcp flags rst
×	not yet documented	firewall.xml.in: firewall ipv6 name <name> rule <rule> tcp flags syn	firewall ipv6 name <text> rule <text> tcp flags syn
×	not yet documented	firewall.xml.in: firewall ipv6 name <name> rule <rule> tcp flags urg	firewall ipv6 name <text> rule <text> tcp flags urg
×	not yet documented	firewall.xml.in: firewall ipv6 name <name> rule <rule> tcp mss	firewall ipv6 name <text> rule <text> tcp mss <value>
×	not yet documented	firewall.xml.in: firewall ipv6 name <name> rule <rule> time startdate	firewall ipv6 name <text> rule <text> time startdate <value>
×	not yet documented	firewall.xml.in: firewall ipv6 name <name> rule <rule> time starttime	firewall ipv6 name <text> rule <text> time starttime <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	firewall.xml.in: firewall ipv6 name <name> rule <rule> time stopdate	firewall ipv6 name <text> rule <text> time stopdate <value>
×	not yet documented	firewall.xml.in: firewall ipv6 name <name> rule <rule> time stoptime	firewall ipv6 name <text> rule <text> time stoptime <value>
×	not yet documented	firewall.xml.in: firewall ipv6 name <name> rule <rule> time weekdays	firewall ipv6 name <text> rule <text> time weekdays <value>
✓	<i>set firewall ipv6 output filter default-action [accept drop]</i>	firewall.xml.in: firewall ipv6 output filter default-action	firewall ipv6 output filter default-action <value>
×	<i>set firewall ipv6 output filter default-log</i>	firewall.xml.in: firewall ipv6 output filter default-log	Nothing found in VyOS
×	not yet documented	firewall.xml.in: firewall ipv6 output filter description	firewall ipv6 output filter description <value>
×	<i>set firewall ipv6 output filter rule <1-999999> action [accept continue drop jump queue reject return]</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set firewall ipv6 output filter rule <1-999999> connection-mark <1-2147483647></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set firewall ipv6 output filter rule <1-999999> connection-status nat [destination source]</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set firewall ipv6 output filter rule <1-999999> description <text></i>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set firewall ipv6 output filter rule <1-999999> destination address [address addressrange CIDR]</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 output filter rule <1-999999> destination address-mask [address]</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 output filter rule <1-999999> destination fqdn <fqdn></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 output filter rule <1-999999> destination geoup asn <1-4294967294></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 output filter rule <1-999999> destination geoup country-code <country></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 output filter rule <1-999999> destination geoup inverse-match</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 output filter rule <1-999999> destination group address-group <name !name></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 output filter rule <1-999999> destination group domain-group <name / !name></code>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set firewall ipv6 output filter rule <1-999999> destination group dynamic-address-group <name !name></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 output filter rule <1-999999> destination group mac-group <name !name></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 output filter rule <1-999999> destination group network-group <name !name></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 output filter rule <1-999999> destination group port-group <name !name></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 output filter rule <1-999999> destination port [1-65535 portname start-end]</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 output filter rule <1-999999> disable</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 output filter rule <1-999999> dscp [0-63 start-end]</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 output filter rule <1-999999> dscp-exclude [0-63 start-end]</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 output filter rule <1-999999> fragment [match-frag match-non-frag]</code>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set firewall ipv6 output filter rule <1-999999> hop-limit <eq gt lt> <0-255></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 output filter rule <1-999999> icmpv6 [code type] <0-255></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 output filter rule <1-999999> icmpv6 type-name <text></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 output filter rule <1-999999> ipsec [match-ipsec-out match-none-out]</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 output filter rule <1-999999> jump-target <text></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 output filter rule <1-999999> limit burst <0-4294967295></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 output filter rule <1-999999> limit rate <text></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 output filter rule <1-999999> log</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 output filter rule <1-999999> log-options group <0-65535></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 output filter rule <1-999999> log-options level [emerg alert crit err warn notice info debug]</code>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set firewall ipv6 output filter rule <1-999999> log-options queue-threshold <0-65535></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 output filter rule <1-999999> log-options snapshot-length <0-9000></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 output filter rule <1-999999> outbound-interface group <iface_group></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 output filter rule <1-999999> outbound-interface name <iface></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 output filter rule <1-999999> packet-length-exclude <text></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 output filter rule <1-999999> packet-length <text></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 output filter rule <1-999999> packet-type [broadcast host multicast other]</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 output filter rule <1-999999> protocol [<text> <0-255> all tcp_udp]</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 output filter rule <1-999999> queue-options bypass</code>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set firewall ipv6 output filter rule <1-999999> queue-options fanout</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 output filter rule <1-999999> queue <0-65535></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 output filter rule <1-999999> recent count <1-255></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 output filter rule <1-999999> recent time [second minute hour]</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 output filter rule <1-999999> recent time <second minute hour></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 output filter rule <1-999999> source address [address addressrange CIDR]</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 output filter rule <1-999999> source address-mask [address]</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 output filter rule <1-999999> source fqdn <fqdn></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 output filter rule <1-999999> source geoip asn <1-4294967294></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 output filter rule <1-999999> source geoip country-code <country></code>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set firewall ipv6 output filter rule <1-999999> source geoip inverse-match</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 output filter rule <1-999999> source group address-group <name !name></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 output filter rule <1-999999> source group domain-group <name !name></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 output filter rule <1-999999> source group dynamic-address-group <name !name></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 output filter rule <1-999999> source group mac-group <name !name></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 output filter rule <1-999999> source group network-group <name !name></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 output filter rule <1-999999> source group port-group <name !name></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 output filter rule <1-999999> source mac-address <mac-address></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 output filter rule <1-999999> source port [1-65535 portname start-end]</code>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set firewall ipv6 output filter rule <1-999999> state [established invalid new related]</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 output filter rule <1-999999> tcp flags [not] <text></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 output filter rule <1-999999> time startdate <text></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 output filter rule <1-999999> time starttime <text></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 output filter rule <1-999999> time stopdate <text></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 output filter rule <1-999999> time stoptime <text></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 output filter rule <1-999999> time weekdays <text></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	firewall.xml.in: firewall ipv6 output filter rule <rule> action	firewall ipv6 output filter rule <text> action <value>
×	not yet documented	firewall.xml.in: firewall ipv6 output filter rule <rule> add-address-to-group destination-address address-group	Nothing found in VyOS
×	not yet documented	firewall.xml.in: firewall ipv6 output filter rule <rule> add-address-to-group destination-address timeout	Nothing found in VyOS
×	not yet documented	firewall.xml.in: firewall ipv6 output filter rule <rule> add-address-to-group source-address address-group	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	firewall.xml.in: firewall ipv6 output filter rule <rule> add-address- to-group source-address timeout	Nothing found in VyOS
×	not yet documented	firewall.xml.in: firewall ipv6 output filter rule <rule> connection- mark	firewall ipv6 output fil- ter rule <text> connection- mark <value>
×	not yet documented	firewall.xml.in: firewall ipv6 output filter rule <rule> connection- status nat	firewall ipv6 output fil- ter rule <text> connection- status nat <value>
×	not yet documented	firewall.xml.in: firewall ipv6 output fil- ter rule <rule> conntrack- helper	Nothing found in VyOS
×	not yet documented	firewall.xml.in: firewall ipv6 output filter rule <rule> description	firewall ipv6 output fil- ter rule <text> description <value>
×	not yet documented	firewall.xml.in: firewall ipv6 output filter rule <rule> destination ad- dress	firewall ipv6 output filter rule <text> destination ad- dress <value>
×	not yet documented	firewall.xml.in: firewall ipv6 output fil- ter rule <rule> destination address-mask	firewall ipv6 output fil- ter rule <text> destination address-mask <value>
×	not yet documented	firewall.xml.in: firewall ipv6 output fil- ter rule <rule> destination fqdn	firewall ipv6 output fil- ter rule <text> destination fqdn <value>
×	not yet documented	firewall.xml.in: firewall ipv6 output fil- ter rule <rule> destination geoip country-code	firewall ipv6 output fil- ter rule <text> destina- tion geoip country-code <value>
×	not yet documented	firewall.xml.in: firewall ipv6 output fil- ter rule <rule> destination geoip inverse-match	firewall ipv6 output fil- ter rule <text> destination geoip inverse-match
×	not yet documented	firewall.xml.in: firewall ipv6 output fil- ter rule <rule> destination group address-group	firewall ipv6 output fil- ter rule <text> destina- tion group address-group <value>
×	not yet documented	firewall.xml.in: firewall ipv6 output fil- ter rule <rule> destination group domain-group	firewall ipv6 output fil- ter rule <text> destina- tion group domain-group <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	firewall.xml.in: firewall ipv6 output filter rule <rule> destination group dynamic-address- group	Nothing found in VyOS
×	not yet documented	firewall.xml.in: firewall ipv6 output fil- ter rule <rule> destination group mac-group	firewall ipv6 output fil- ter rule <text> destination group mac-group <value>
×	not yet documented	firewall.xml.in: firewall ipv6 output fil- ter rule <rule> destination group network-group	firewall ipv6 output fil- ter rule <text> destina- tion group network-group <value>
×	not yet documented	firewall.xml.in: firewall ipv6 output fil- ter rule <rule> destination group port-group	firewall ipv6 output fil- ter rule <text> destination group port-group <value>
×	not yet documented	firewall.xml.in: firewall ipv6 output fil- ter rule <rule> destination mac-address	firewall ipv6 output fil- ter rule <text> destination mac-address <value>
×	not yet documented	firewall.xml.in: firewall ipv6 output fil- ter rule <rule> destination port	firewall ipv6 output fil- ter rule <text> destination port <value>
×	not yet documented	firewall.xml.in: firewall ipv6 output filter rule <rule> disable	firewall ipv6 output filter rule <text> disable
×	not yet documented	firewall.xml.in: firewall ipv6 output filter rule <rule> dscp	firewall ipv6 output filter rule <text> dscp <value>
×	not yet documented	firewall.xml.in: firewall ipv6 output filter rule <rule> dscp-exclude	firewall ipv6 output filter rule <text> dscp-exclude <value>
×	not yet documented	firewall.xml.in: firewall ipv6 output fil- ter rule <rule> fragment match-frag	firewall ipv6 output fil- ter rule <text> fragment match-frag
×	not yet documented	firewall.xml.in: firewall ipv6 output fil- ter rule <rule> fragment match-non-frag	firewall ipv6 output fil- ter rule <text> fragment match-non-frag
×	not yet documented	firewall.xml.in: firewall ipv6 output filter rule <rule> hop-limit eq	firewall ipv6 output filter rule <text> hop-limit eq <value>
×	not yet documented	firewall.xml.in: firewall ipv6 output filter rule <rule> hop-limit gt	firewall ipv6 output filter rule <text> hop-limit gt <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	firewall.xml.in: firewall ipv6 output filter rule <rule> hop-limit lt	firewall ipv6 output filter rule <text> hop-limit lt <value>
×	not yet documented	firewall.xml.in: firewall ipv6 output filter rule <rule> icmpv6 code	firewall ipv6 output filter rule <text> icmpv6 code <value>
×	not yet documented	firewall.xml.in: firewall ipv6 output filter rule <rule> icmpv6 type	firewall ipv6 output filter rule <text> icmpv6 type <value>
×	not yet documented	firewall.xml.in: firewall ipv6 output filter rule <rule> icmpv6 type-name	firewall ipv6 output filter rule <text> icmpv6 type-name <value>
×	not yet documented	Nothing found in XML Definitions	firewall ipv6 output filter rule <text> ipsec match-ipsec
×	not yet documented	Nothing found in XML Definitions	firewall ipv6 output filter rule <text> ipsec match-none
×	not yet documented	firewall.xml.in: firewall ipv6 output filter rule <rule> jump-target	firewall ipv6 output filter rule <text> jump-target <value>
×	not yet documented	firewall.xml.in: firewall ipv6 output filter rule <rule> limit burst	firewall ipv6 output filter rule <text> limit burst <value>
×	not yet documented	firewall.xml.in: firewall ipv6 output filter rule <rule> limit rate	firewall ipv6 output filter rule <text> limit rate <value>
×	not yet documented	firewall.xml.in: firewall ipv6 output filter rule <rule> log	firewall ipv6 output filter rule <text> log <value>
×	not yet documented	firewall.xml.in: firewall ipv6 output filter rule <rule> log-options group	firewall ipv6 output filter rule <text> log-options group <value>
×	not yet documented	firewall.xml.in: firewall ipv6 output filter rule <rule> log-options level	firewall ipv6 output filter rule <text> log-options level <value>
×	not yet documented	firewall.xml.in: firewall ipv6 output filter rule <rule> log-options queue-threshold	firewall ipv6 output filter rule <text> log-options queue-threshold <value>
×	not yet documented	firewall.xml.in: firewall ipv6 output filter rule <rule> log-options snapshot-length	firewall ipv6 output filter rule <text> log-options snapshot-length <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	firewall.xml.in: firewall ipv6 output filter rule <rule> mark	Nothing found in VyOS
×	not yet documented	firewall.xml.in: firewall ipv6 output filter rule <rule> outbound-interface group	Nothing found in VyOS
×	not yet documented	Nothing found in XML Definitions	firewall ipv6 output filter rule <text> outbound-interface interface-group <value>
×	not yet documented	Nothing found in XML Definitions	firewall ipv6 output filter rule <text> outbound-interface interface-name <value>
×	not yet documented	firewall.xml.in: firewall ipv6 output filter rule <rule> outbound-interface name	Nothing found in VyOS
×	not yet documented	firewall.xml.in: firewall ipv6 output filter rule <rule> packet-length	firewall ipv6 output filter rule <text> packet-length <value>
×	not yet documented	firewall.xml.in: firewall ipv6 output filter rule <rule> packet-length-exclude	firewall ipv6 output filter rule <text> packet-length-exclude <value>
×	not yet documented	firewall.xml.in: firewall ipv6 output filter rule <rule> packet-type	firewall ipv6 output filter rule <text> packet-type <value>
×	not yet documented	firewall.xml.in: firewall ipv6 output filter rule <rule> protocol	firewall ipv6 output filter rule <text> protocol <value>
×	not yet documented	firewall.xml.in: firewall ipv6 output filter rule <rule> queue	firewall ipv6 output filter rule <text> queue <value>
×	not yet documented	firewall.xml.in: firewall ipv6 output filter rule <rule> queue-options	firewall ipv6 output filter rule <text> queue-options <value>
×	not yet documented	firewall.xml.in: firewall ipv6 output filter rule <rule> recent count	firewall ipv6 output filter rule <text> recent count <value>
×	not yet documented	firewall.xml.in: firewall ipv6 output filter rule <rule> recent time	firewall ipv6 output filter rule <text> recent time <value>
×	not yet documented	firewall.xml.in: firewall ipv6 output filter rule <rule> source address	firewall ipv6 output filter rule <text> source address <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	firewall.xml.in: firewall ipv6 output filter rule <rule> source address-mask	firewall ipv6 output filter rule <text> source address-mask <value>
×	not yet documented	firewall.xml.in: firewall ipv6 output filter rule <rule> source fqdn	firewall ipv6 output filter rule <text> source fqdn <value>
×	not yet documented	firewall.xml.in: firewall ipv6 output filter rule <rule> source geoip country-code	firewall ipv6 output filter rule <text> source geoip country-code <value>
×	not yet documented	firewall.xml.in: firewall ipv6 output filter rule <rule> source geoip inverse-match	firewall ipv6 output filter rule <text> source geoip inverse-match
×	not yet documented	firewall.xml.in: firewall ipv6 output filter rule <rule> source group address-group	firewall ipv6 output filter rule <text> source group address-group <value>
×	not yet documented	firewall.xml.in: firewall ipv6 output filter rule <rule> source group domain-group	firewall ipv6 output filter rule <text> source group domain-group <value>
×	not yet documented	firewall.xml.in: firewall ipv6 output filter rule <rule> source group dynamic-address-group	Nothing found in VyOS
×	not yet documented	firewall.xml.in: firewall ipv6 output filter rule <rule> source group mac-group	firewall ipv6 output filter rule <text> source group mac-group <value>
×	not yet documented	firewall.xml.in: firewall ipv6 output filter rule <rule> source group network-group	firewall ipv6 output filter rule <text> source group network-group <value>
×	not yet documented	firewall.xml.in: firewall ipv6 output filter rule <rule> source group port-group	firewall ipv6 output filter rule <text> source group port-group <value>
×	not yet documented	firewall.xml.in: firewall ipv6 output filter rule <rule> source mac-address	firewall ipv6 output filter rule <text> source mac-address <value>
×	not yet documented	firewall.xml.in: firewall ipv6 output filter rule <rule> source port	firewall ipv6 output filter rule <text> source port <value>
×	not yet documented	firewall.xml.in: firewall ipv6 output filter rule <rule> state	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	Nothing found in XML Definitions	firewall ipv6 output filter rule <text> state established <value>
×	not yet documented	Nothing found in XML Definitions	firewall ipv6 output filter rule <text> state invalid <value>
×	not yet documented	Nothing found in XML Definitions	firewall ipv6 output filter rule <text> state new <value>
×	not yet documented	Nothing found in XML Definitions	firewall ipv6 output filter rule <text> state related <value>
×	not yet documented	firewall.xml.in: firewall ipv6 output filter rule <rule> synproxy tcp mss	Nothing found in VyOS
×	not yet documented	firewall.xml.in: firewall ipv6 output filter rule <rule> synproxy tcp window-scale	Nothing found in VyOS
×	not yet documented	firewall.xml.in: firewall ipv6 output filter rule <rule> tcp flags ack	firewall ipv6 output filter rule <text> tcp flags ack
×	not yet documented	firewall.xml.in: firewall ipv6 output filter rule <rule> tcp flags cwr	firewall ipv6 output filter rule <text> tcp flags cwr
×	not yet documented	firewall.xml.in: firewall ipv6 output filter rule <rule> tcp flags ecn	firewall ipv6 output filter rule <text> tcp flags ecn
×	not yet documented	firewall.xml.in: firewall ipv6 output filter rule <rule> tcp flags fin	firewall ipv6 output filter rule <text> tcp flags fin
×	not yet documented	firewall.xml.in: firewall ipv6 output filter rule <rule> tcp flags not ack	firewall ipv6 output filter rule <text> tcp flags not ack
×	not yet documented	firewall.xml.in: firewall ipv6 output filter rule <rule> tcp flags not cwr	firewall ipv6 output filter rule <text> tcp flags not cwr
×	not yet documented	firewall.xml.in: firewall ipv6 output filter rule <rule> tcp flags not ecn	firewall ipv6 output filter rule <text> tcp flags not ecn
×	not yet documented	firewall.xml.in: firewall ipv6 output filter rule <rule> tcp flags not fin	firewall ipv6 output filter rule <text> tcp flags not fin

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	firewall.xml.in: firewall ipv6 output filter rule <rule> tcp flags not psh	firewall ipv6 output filter rule <text> tcp flags not psh
×	not yet documented	firewall.xml.in: firewall ipv6 output filter rule <rule> tcp flags not rst	firewall ipv6 output filter rule <text> tcp flags not rst
×	not yet documented	firewall.xml.in: firewall ipv6 output filter rule <rule> tcp flags not syn	firewall ipv6 output filter rule <text> tcp flags not syn
×	not yet documented	firewall.xml.in: firewall ipv6 output filter rule <rule> tcp flags not urg	firewall ipv6 output filter rule <text> tcp flags not urg
×	not yet documented	firewall.xml.in: firewall ipv6 output filter rule <rule> tcp flags psh	firewall ipv6 output filter rule <text> tcp flags psh
×	not yet documented	firewall.xml.in: firewall ipv6 output filter rule <rule> tcp flags rst	firewall ipv6 output filter rule <text> tcp flags rst
×	not yet documented	firewall.xml.in: firewall ipv6 output filter rule <rule> tcp flags syn	firewall ipv6 output filter rule <text> tcp flags syn
×	not yet documented	firewall.xml.in: firewall ipv6 output filter rule <rule> tcp flags urg	firewall ipv6 output filter rule <text> tcp flags urg
×	not yet documented	firewall.xml.in: firewall ipv6 output filter rule <rule> tcp mss	firewall ipv6 output fil- ter rule <text> tcp mss <value>
×	not yet documented	firewall.xml.in: firewall ipv6 output filter rule <rule> time startdate	firewall ipv6 output filter rule <text> time startdate <value>
×	not yet documented	firewall.xml.in: firewall ipv6 output filter rule <rule> time starttime	firewall ipv6 output filter rule <text> time starttime <value>
×	not yet documented	firewall.xml.in: firewall ipv6 output filter rule <rule> time stopdate	firewall ipv6 output filter rule <text> time stopdate <value>
×	not yet documented	firewall.xml.in: firewall ipv6 output filter rule <rule> time stoptime	firewall ipv6 output filter rule <text> time stoptime <value>
×	not yet documented	firewall.xml.in: firewall ipv6 output filter rule <rule> time weekdays	firewall ipv6 output filter rule <text> time weekdays <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set firewall ipv6 output [filter raw] rule <1-999999> set connection-mark <0-2147483647></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 output [filter raw] rule <1-999999> set dscp <0-63></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 output [filter raw] rule <1-999999> set hop-limit <0-255></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 output [filter raw] rule <1-999999> set mark <1-2147483647></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 output [filter raw] rule <1-999999> set tcp-mss <500-1460></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 prerouting raw rule <1-999999> set dscp <0-63></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 prerouting raw rule <1-999999> set hop-limit <0-255></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 prerouting raw rule <1-999999> set mark <1-2147483647></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall ipv6 prerouting raw rule <1-999999> set tcp-mss <500-1460></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall zone ...</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall zone <name> default-action [drop reject]</code>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set firewall zone <Destination Zone> default-firewall ipv6-name <ipv6-rule-set-name></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall zone <Destination Zone> default-firewall name <ipv4-rule-set-name></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall zone <name> default-log</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall zone <name> description</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall zone <Destination Zone> from <Source Zone> firewall ipv6-name <ipv6-rule-set-name></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall zone <Destination Zone> from <Source Zone> firewall name <ipv4-rule-set-name></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall zone <name> interface <interface></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set firewall zone <name> local-zone</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	firewall.xml.in: firewall zone <zone> default-action	Nothing found in VyOS
×	not yet documented	firewall.xml.in: firewall zone <zone> default-log	Nothing found in VyOS
×	not yet documented	firewall.xml.in: firewall zone <zone> de- scription	Nothing found in VyOS
×	not yet documented	firewall.xml.in: firewall zone <zone> from <from> firewall ipv6-name	Nothing found in VyOS
×	not yet documented	firewall.xml.in: firewall zone <zone> from <from> firewall name	Nothing found in VyOS
×	not yet documented	firewall.xml.in: firewall zone <zone> inter- face	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	firewall.xml.in: firewall zone <zone> intra-zone-filtering action	Nothing found in VyOS
×	not yet documented	firewall.xml.in: firewall zone <zone> intra-zone-filtering fire- wall ipv6-name	Nothing found in VyOS
×	not yet documented	firewall.xml.in: firewall zone <zone> intra-zone-filtering fire- wall name	Nothing found in VyOS
×	not yet documented	firewall.xml.in: firewall zone <zone> local-zone	Nothing found in VyOS
×	<i>generate system login username <username> otp-key hotp-time rate-limit <1-10> rate-time <15-600> window-size <1-21></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	high-availability.xml.in: high-availability disable	high-availability disable
×	not yet documented	high-availability.xml.in: high-availability virtual- server <virtual-server> address	high-availability virtual- server <text> address <value>
×	not yet documented	high-availability.xml.in: high-availability virtual- server <virtual-server> algorithm	high-availability virtual- server <text> algorithm <value>
×	not yet documented	high-availability.xml.in: high-availability virtual- server <virtual-server> delay-loop	high-availability virtual- server <text> delay-loop <value>
×	not yet documented	high-availability.xml.in: high-availability virtual- server <virtual-server> forward-method	high-availability virtual- server <text> forward- method <value>
×	not yet documented	high-availability.xml.in: high-availability virtual- server <virtual-server> fwmark	high-availability virtual- server <text> fwmark <value>
×	not yet documented	high-availability.xml.in: high-availability virtual- server <virtual-server> persistence-timeout	high-availability virtual- server <text> persistence- timeout <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	high-availability.xml.in: high-availability virtual-server <virtual-server> port	high-availability virtual-server <text> port <value>
×	not yet documented	high-availability.xml.in: high-availability virtual-server <virtual-server> protocol	high-availability virtual-server <text> protocol <value>
×	not yet documented	high-availability.xml.in: high-availability virtual-server <virtual-server> real-server <real-server> connection-timeout	high-availability virtual-server <text> real-server <text> connection-timeout <value>
×	not yet documented	high-availability.xml.in: high-availability virtual-server <virtual-server> real-server <real-server> health-check script	high-availability virtual-server <text> real-server <text> health-check script <value>
×	not yet documented	high-availability.xml.in: high-availability virtual-server <virtual-server> real-server <real-server> port	high-availability virtual-server <text> real-server <text> port <value>
✓	set high-availability vrrp global-parameters garp interval <0.000-1000>	high-availability.xml.in: high-availability vrrp global-parameters garp interval	high-availability vrrp global-parameters garp interval <value>
✓	set high-availability vrrp global-parameters garp master-delay <1-255>	high-availability.xml.in: high-availability vrrp global-parameters garp master-delay	high-availability vrrp global-parameters garp master-delay <value>
✓	set high-availability vrrp global-parameters garp master-refresh <1-600>	high-availability.xml.in: high-availability vrrp global-parameters garp master-refresh	high-availability vrrp global-parameters garp master-refresh <value>
✓	set high-availability vrrp global-parameters garp master-refresh-repeat <1-600>	high-availability.xml.in: high-availability vrrp global-parameters garp master-refresh-repeat	high-availability vrrp global-parameters garp master-refresh-repeat <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
✓	<i>set high-availability vrrp global-parameters garp master-repeat <1-600></i>	high-availability.xml.in: high-availability vrrp global-parameters garp master-repeat	high-availability vrrp global-parameters garp master-repeat <value>
×	not yet documented	high-availability.xml.in: high-availability vrrp global-parameters startup- delay	high-availability vrrp global-parameters startup- delay <value>
×	<i>set high-availability vrrp global-parameters startup_delay <1-600></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	high-availability.xml.in: high-availability vrrp global-parameters version	high-availability vrrp global-parameters version <value>
×	<i>set high-availability vrrp global-parameters version 2/3</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	high-availability.xml.in: high-availability vrrp group <group> address <address> interface	high-availability vrrp group <text> address <text> interface <value>
×	not yet documented	high-availability.xml.in: high-availability vrrp group <group> advertise- interval	high-availability vrrp group <text> advertise- interval <value>
×	not yet documented	high-availability.xml.in: high-availability vrrp group <group> authenti- cation password	high-availability vrrp group <text> authentica- tion password <value>
×	not yet documented	high-availability.xml.in: high-availability vrrp group <group> authenti- cation type	high-availability vrrp group <text> authentica- tion type <value>
×	not yet documented	high-availability.xml.in: high-availability vrrp group <group> descrip- tion	high-availability vrrp group <text> description <value>
×	not yet documented	high-availability.xml.in: high-availability vrrp group <group> disable	high-availability vrrp group <text> disable

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	Nothing found in XML Definitions	high-availability vrrp group <text> excluded-address <value>
×	not yet documented	high-availability.xml.in: high-availability vrrp group <group> excluded-address <excluded-address> interface	Nothing found in VyOS
✓	<code>set high-availability vrrp group <name> garp interval <0.000-1000></code>	high-availability.xml.in: high-availability vrrp group <group> garp interval	high-availability vrrp group <text> garp interval <value>
✓	<code>set high-availability vrrp group <name> garp master-delay <1-255></code>	high-availability.xml.in: high-availability vrrp group <group> garp master-delay	high-availability vrrp group <text> garp master-delay <value>
✓	<code>set high-availability vrrp group <name> garp master-refresh <1-600></code>	high-availability.xml.in: high-availability vrrp group <group> garp master-refresh	high-availability vrrp group <text> garp master-refresh <value>
✓	<code>set high-availability vrrp group <name> garp master-refresh-repeat <1-600></code>	high-availability.xml.in: high-availability vrrp group <group> garp master-refresh-repeat	high-availability vrrp group <text> garp master-refresh-repeat <value>
✓	<code>set high-availability vrrp group <name> garp master-repeat <1-600></code>	high-availability.xml.in: high-availability vrrp group <group> garp master-repeat	high-availability vrrp group <text> garp master-repeat <value>
×	not yet documented	high-availability.xml.in: high-availability vrrp group <group> health-check failure-count	high-availability vrrp group <text> health-check failure-count <value>
×	not yet documented	high-availability.xml.in: high-availability vrrp group <group> health-check interval	high-availability vrrp group <text> health-check interval <value>
×	not yet documented	high-availability.xml.in: high-availability vrrp group <group> health-check ping	high-availability vrrp group <text> health-check ping <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	high-availability.xml.in: high-availability vrrp group <group> health- check script	high-availability vrrp group <text> health-check script <value>
×	not yet documented	high-availability.xml.in: high-availability vrrp group <group> hello- source-address	high-availability vrrp group <text> hello- source-address <value>
×	not yet documented	high-availability.xml.in: high-availability vrrp group <group> interface	high-availability vrrp group <text> interface <value>
×	not yet documented	high-availability.xml.in: high-availability vrrp group <group> no- preempt	high-availability vrrp group <text> no-preempt
×	not yet documented	high-availability.xml.in: high-availability vrrp group <group> peer- address	high-availability vrrp group <text> peer-address <value>
×	not yet documented	high-availability.xml.in: high-availability vrrp group <group> preempt- delay	high-availability vrrp group <text> preempt- delay <value>
×	not yet documented	high-availability.xml.in: high-availability vrrp group <group> priority	high-availability vrrp group <text> priority <value>
×	not yet documented	high-availability.xml.in: high-availability vrrp group <group> rfc3768- compatibility	high-availability vrrp group <text> rfc3768- compatibility
×	not yet documented	high-availability.xml.in: high-availability vrrp group <group> track exclude-vrrp-interface	high-availability vrrp group <text> track exclude-vrrp-interface
×	not yet documented	high-availability.xml.in: high-availability vrrp group <group> track interface	high-availability vrrp group <text> track inter- face <value>
×	not yet documented	high-availability.xml.in: high-availability vrrp group <group> transition- script backup	high-availability vrrp group <text> transition- script backup <value>
×	not yet documented	high-availability.xml.in: high-availability vrrp group <group> transition- script fault	high-availability vrrp group <text> transition- script fault <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	high-availability.xml.in: high-availability vrrp group <group> transition- script master	high-availability vrrp group <text> transition- script master <value>
×	not yet documented	high-availability.xml.in: high-availability vrrp group <group> transition- script stop	high-availability vrrp group <text> transition- script stop <value>
×	not yet documented	high-availability.xml.in: high-availability vrrp group <group> vrid	high-availability vrrp group <text> vrid <value>
×	not yet documented	high-availability.xml.in: high-availability vrrp snmp	Nothing found in VyOS
×	not yet documented	high-availability.xml.in: high-availability vrrp sync-group <sync-group> health-check failure-count	Nothing found in VyOS
×	not yet documented	high-availability.xml.in: high-availability vrrp sync-group <sync-group> health-check interval	Nothing found in VyOS
×	not yet documented	high-availability.xml.in: high-availability vrrp sync-group <sync-group> health-check ping	Nothing found in VyOS
×	not yet documented	high-availability.xml.in: high-availability vrrp sync-group <sync-group> health-check script	Nothing found in VyOS
×	not yet documented	high-availability.xml.in: high-availability vrrp sync-group <sync-group> member	high-availability vrrp sync-group <text> mem- ber <value>
×	not yet documented	high-availability.xml.in: high-availability vrrp sync-group <sync-group> transition-script backup	high-availability vrrp sync-group <text> transition-script backup <value>
×	not yet documented	high-availability.xml.in: high-availability vrrp sync-group <sync-group> transition-script fault	high-availability vrrp sync-group <text> transition-script fault <value>
×	not yet documented	high-availability.xml.in: high-availability vrrp sync-group <sync-group> transition-script master	high-availability vrrp sync-group <text> transition-script mas- ter <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	high-availability.xml.in: high-availability vrrp sync-group <sync-group> transition-script stop <value>	high-availability vrrp sync-group <text> transition-script stop <value>
×	<code>set interfaces <dummy ethernet bonding bridge pppoe> <interface> vrf <name></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set interfaces bonding <interface> arp-monitor interval <time></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set interfaces bonding <interface> arp-monitor target <address></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set interfaces bonding <interface> evpn es-df-pref <1-65535></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set interfaces bonding <interface> evpn es-id <1-16777215 10-byte ID></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set interfaces bonding <interface> evpn es-sys-mac <xx:xx:xx:xx:xx:xx></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set interfaces bonding <interface> hash-policy <policy></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set interfaces bonding <interface> lacp-rate <slow/fast></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set interfaces bonding <interface> member interface <member></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set interfaces bonding <interface> min-links <0-16></code>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set interfaces bonding <interface> mode <802.3ad / active-backup / broadcast / round-robin / transmit-load-balance / adaptive-load-balance / xor-hash></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set interfaces bonding <interface> primary <interface></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set interfaces bonding <interface> system-mac <mac address></code>	Nothing found in XML Definitions	Nothing found in VyOS
✓	<code>set interfaces bonding <interface> address <address / dhcp / dhcpv6></code>	inter-faces_bonding.xml.in: interfaces bonding <bonding> address	interfaces bonding <text> address <value>
×	not yet documented	inter-faces_bonding.xml.in: interfaces bonding <bonding> arp-monitor interval	interfaces bonding <text> arp-monitor interval <value>
×	not yet documented	inter-faces_bonding.xml.in: interfaces bonding <bonding> arp-monitor target	interfaces bonding <text> arp-monitor target <value>
✓	<code>set interfaces bonding <interface> description <description></code>	inter-faces_bonding.xml.in: interfaces bonding <bonding> description	interfaces bonding <text> description <value>
✓	<code>set interfaces bonding <interface> dhcp-options client-id <description></code>	inter-faces_bonding.xml.in: interfaces bonding <bonding> dhcp-options client-id	interfaces bonding <text> dhcp-options client-id <value>
✓	<code>set interfaces bonding <interface> dhcp-options default-route-distance <distance></code>	inter-faces_bonding.xml.in: interfaces bonding <bonding> dhcp-options default-route-distance	interfaces bonding <text> dhcp-options default-route-distance <value>
✓	<code>set interfaces bonding <interface> dhcp-options host-name <hostname></code>	inter-faces_bonding.xml.in: interfaces bonding <bonding> dhcp-options host-name	interfaces bonding <text> dhcp-options host-name <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	inter-faces_bonding.xml.in: interfaces bonding <bonding> dhcp-options mtu	interfaces bonding <text> dhcp-options mtu
✓	<i>set interfaces bonding <interface> dhcp-options no-default-route</i>	inter-faces_bonding.xml.in: interfaces bonding <bonding> dhcp-options no-default-route	interfaces bonding <text> dhcp-options no-default-route
✓	<i>set interfaces bonding <interface> dhcp-options reject <address></i>	inter-faces_bonding.xml.in: interfaces bonding <bonding> dhcp-options reject	interfaces bonding <text> dhcp-options reject <value>
×	<i>set interfaces bonding <interface> dhcp-options user-class <string></i>	inter-faces_bonding.xml.in: interfaces bonding <bonding> dhcp-options user-class	Nothing found in VyOS
✓	<i>set interfaces bonding <interface> dhcp-options vendor-class-id <vendor-id></i>	inter-faces_bonding.xml.in: interfaces bonding <bonding> dhcp-options vendor-class-id	interfaces bonding <text> dhcp-options vendor-class-id <value>
✓	<i>set interfaces bonding <interface> dhcpv6-options duid <duid></i>	inter-faces_bonding.xml.in: interfaces bonding <bonding> dhcpv6-options duid	interfaces bonding <text> dhcpv6-options duid <value>
✓	<i>set interfaces bonding <interface> dhcpv6-options no-release</i>	inter-faces_bonding.xml.in: interfaces bonding <bonding> dhcpv6-options no-release	interfaces bonding <text> dhcpv6-options no-release
✓	<i>set interfaces bonding <interface> dhcpv6-options parameters-only</i>	inter-faces_bonding.xml.in: interfaces bonding <bonding> dhcpv6-options parameters-only	interfaces bonding <text> dhcpv6-options parameters-only
✓	<i>set interfaces bonding <interface> dhcpv6-options pd <id> interface <delegatee> address <address></i>	inter-faces_bonding.xml.in: interfaces bonding <bonding> dhcpv6-options pd <pd> interface <interface> address	interfaces bonding <text> dhcpv6-options pd <text> interface <text> address <value>
✓	<i>set interfaces bonding <interface> dhcpv6-options pd <id> interface <delegatee> sla-id <id></i>	inter-faces_bonding.xml.in: interfaces bonding <bonding> dhcpv6-options pd <pd> interface <interface> sla-id	interfaces bonding <text> dhcpv6-options pd <text> interface <text> sla-id <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
✓	<i>set interfaces bonding <interface> dhcpv6-options pd <id> length <length></i>	inter-faces_bonding.xml.in: interfaces bonding <bonding> dhcpv6-options pd <pd> length	interfaces bonding <text> dhcpv6-options pd <text> length <value>
✓	<i>set interfaces bonding <interface> dhcpv6-options rapid-commit</i>	inter-faces_bonding.xml.in: interfaces bonding <bonding> dhcpv6-options rapid-commit	interfaces bonding <text> dhcpv6-options rapid-commit
✓	<i>set interfaces bonding <interface> dhcpv6-options temporary</i>	inter-faces_bonding.xml.in: interfaces bonding <bonding> dhcpv6-options temporary	interfaces bonding <text> dhcpv6-options temporary
✓	<i>set interfaces bonding <interface> disable</i>	inter-faces_bonding.xml.in: interfaces bonding <bonding> disable	interfaces bonding <text> disable
×	<i>set interfaces bonding <interface> disable-flow-control</i>	Nothing found in XML Definitions	Nothing found in VyOS
✓	<i>set interfaces bonding <interface> disable-link-detect</i>	inter-faces_bonding.xml.in: interfaces bonding <bonding> disable-link-detect	interfaces bonding <text> disable-link-detect
×	not yet documented	inter-faces_bonding.xml.in: interfaces bonding <bonding> evpn es-df-pref	Nothing found in VyOS
×	not yet documented	inter-faces_bonding.xml.in: interfaces bonding <bonding> evpn es-id	Nothing found in VyOS
×	not yet documented	inter-faces_bonding.xml.in: interfaces bonding <bonding> evpn es-sys-mac	Nothing found in VyOS
×	<i>set interfaces bonding <interface> evpn uplink</i>	inter-faces_bonding.xml.in: interfaces bonding <bonding> evpn uplink	Nothing found in VyOS
×	not yet documented	inter-faces_bonding.xml.in: interfaces bonding <bonding> hash-policy	interfaces bonding <text> hash-policy <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
✓	<i>set interfaces bonding <interface> ip adjust-mss <mss / clamp-mss-to-pmtu></i>	inter-faces_bonding.xml.in: interfaces bonding <bonding> ip adjust-mss	interfaces bonding <text> ip adjust-mss <value>
✓	<i>set interfaces bonding <interface> ip arp-cache-timeout</i>	inter-faces_bonding.xml.in: interfaces bonding <bonding> ip arp-cache-timeout	interfaces bonding <text> ip arp-cache-timeout <value>
✓	<i>set interfaces bonding <interface> ip disable-arp-filter</i>	inter-faces_bonding.xml.in: interfaces bonding <bonding> ip disable-arp-filter	interfaces bonding <text> ip disable-arp-filter
✓	<i>set interfaces bonding <interface> ip disable-forwarding</i>	inter-faces_bonding.xml.in: interfaces bonding <bonding> ip disable-forwarding	interfaces bonding <text> ip disable-forwarding
✓	<i>set interfaces bonding <interface> ip enable-arp-accept</i>	inter-faces_bonding.xml.in: interfaces bonding <bonding> ip enable-arp-accept	interfaces bonding <text> ip enable-arp-accept
✓	<i>set interfaces bonding <interface> ip enable-arp-announce</i>	inter-faces_bonding.xml.in: interfaces bonding <bonding> ip enable-arp-announce	interfaces bonding <text> ip enable-arp-announce
✓	<i>set interfaces bonding <interface> ip enable-arp-ignore</i>	inter-faces_bonding.xml.in: interfaces bonding <bonding> ip enable-arp-ignore	interfaces bonding <text> ip enable-arp-ignore
✓	<i>set interfaces bonding <interface> ip enable-directed-broadcast</i>	inter-faces_bonding.xml.in: interfaces bonding <bonding> ip enable-directed-broadcast	interfaces bonding <text> ip enable-directed-broadcast
✓	<i>set interfaces bonding <interface> ip enable-proxy-arp</i>	inter-faces_bonding.xml.in: interfaces bonding <bonding> ip enable-proxy-arp	interfaces bonding <text> ip enable-proxy-arp
✓	<i>set interfaces bonding <interface> ip proxy-arp-pvlan</i>	inter-faces_bonding.xml.in: interfaces bonding <bonding> ip proxy-arp-pvlan	interfaces bonding <text> ip proxy-arp-pvlan
✓	<i>set interfaces bonding <interface> ip source-validation <strict loose disable></i>	inter-faces_bonding.xml.in: interfaces bonding <bonding> ip source-validation	interfaces bonding <text> ip source-validation <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
✓	<i>set interfaces bonding <interface> ipv6 accept-dad <0 / 1 / 2></i>	inter-faces_bonding.xml.in: interfaces bonding <bonding> ipv6 accept-dad	interfaces bonding <text> ipv6 accept-dad <value>
✓	<i>set interfaces bonding <interface> ipv6 address autoconf</i>	inter-faces_bonding.xml.in: interfaces bonding <bonding> ipv6 address autoconf	interfaces bonding <text> ipv6 address autoconf
✓	<i>set interfaces bonding <interface> ipv6 address eui64 <prefix></i>	inter-faces_bonding.xml.in: interfaces bonding <bonding> ipv6 address eui64	interfaces bonding <text> ipv6 address eui64 <value>
✓	<i>set interfaces bonding <interface> ipv6 address no-default-link-local</i>	inter-faces_bonding.xml.in: interfaces bonding <bonding> ipv6 address no-default-link-local	interfaces bonding <text> ipv6 address no-default-link-local
✓	<i>set interfaces bonding <interface> ipv6 adjust-mss <mss / clamp-mss-to-pmtu></i>	inter-faces_bonding.xml.in: interfaces bonding <bonding> ipv6 adjust-mss	interfaces bonding <text> ipv6 adjust-mss <value>
×	not yet documented	inter-faces_bonding.xml.in: interfaces bonding <bonding> ipv6 base-reachable-time	Nothing found in VyOS
✓	<i>set interfaces bonding <interface> ipv6 disable-forwarding</i>	inter-faces_bonding.xml.in: interfaces bonding <bonding> ipv6 disable-forwarding	interfaces bonding <text> ipv6 disable-forwarding
✓	<i>set interfaces bonding <interface> ipv6 dup-addr-detect-transmits <n></i>	inter-faces_bonding.xml.in: interfaces bonding <bonding> ipv6 dup-addr-detect-transmits	interfaces bonding <text> ipv6 dup-addr-detect-transmits <value>
×	not yet documented	inter-faces_bonding.xml.in: interfaces bonding <bonding> ipv6 source-validation	Nothing found in VyOS
×	not yet documented	inter-faces_bonding.xml.in: interfaces bonding <bonding> lacp-rate	interfaces bonding <text> lacp-rate <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
✓	<i>set interfaces bonding <interface> mac <xx:xx:xx:xx:xx:xx></i>	inter-faces_bonding.xml.in: interfaces bonding <bonding> mac	interfaces bonding <text> mac <value>
×	not yet documented	inter-faces_bonding.xml.in: interfaces bonding <bonding> member interface	interfaces bonding <text> member interface <value>
×	not yet documented	inter-faces_bonding.xml.in: interfaces bonding <bonding> mii-mon-interval	interfaces bonding <text> mii-mon-interval <value>
×	not yet documented	inter-faces_bonding.xml.in: interfaces bonding <bonding> min-links	interfaces bonding <text> min-links <value>
✓	<i>set interfaces bonding <interface> mirror egress <monitor-interface></i>	inter-faces_bonding.xml.in: interfaces bonding <bonding> mirror egress	interfaces bonding <text> mirror egress <value>
✓	<i>set interfaces bonding <interface> mirror ingress <monitor-interface></i>	inter-faces_bonding.xml.in: interfaces bonding <bonding> mirror ingress	interfaces bonding <text> mirror ingress <value>
×	not yet documented	inter-faces_bonding.xml.in: interfaces bonding <bonding> mode	interfaces bonding <text> mode <value>
✓	<i>set interfaces bonding <interface> mtu <mtu></i>	inter-faces_bonding.xml.in: interfaces bonding <bonding> mtu	interfaces bonding <text> mtu <value>
×	not yet documented	inter-faces_bonding.xml.in: interfaces bonding <bonding> primary	interfaces bonding <text> primary <value>
×	not yet documented	inter-faces_bonding.xml.in: interfaces bonding <bonding> redirect	interfaces bonding <text> redirect <value>
×	<i>set interfaces bonding <interface> vif <vlan-id></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	inter-faces_bonding.xml.in: interfaces bonding <bonding> vif-s <vif-s> address	interfaces bonding <text> vif-s <text> address <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	inter-faces_bonding.xml.in: interfaces bonding <bonding> vif-s <vif-s> description	interfaces bonding <text> vif-s <text> description <value>
×	not yet documented	inter-faces_bonding.xml.in: interfaces bonding <bonding> vif-s <vif-s> dhcp-options client-id	interfaces bonding <text> vif-s <text> dhcp-options client-id <value>
×	not yet documented	inter-faces_bonding.xml.in: interfaces bonding <bonding> vif-s <vif-s> dhcp-options default-route-distance	interfaces bonding <text> vif-s <text> dhcp-options default-route-distance <value>
×	not yet documented	inter-faces_bonding.xml.in: interfaces bonding <bonding> vif-s <vif-s> dhcp-options host-name	interfaces bonding <text> vif-s <text> dhcp-options host-name <value>
×	not yet documented	inter-faces_bonding.xml.in: interfaces bonding <bonding> vif-s <vif-s> dhcp-options mtu	interfaces bonding <text> vif-s <text> dhcp-options mtu
×	not yet documented	inter-faces_bonding.xml.in: interfaces bonding <bonding> vif-s <vif-s> dhcp-options no-default-route	interfaces bonding <text> vif-s <text> dhcp-options no-default-route
×	not yet documented	inter-faces_bonding.xml.in: interfaces bonding <bonding> vif-s <vif-s> dhcp-options reject	interfaces bonding <text> vif-s <text> dhcp-options reject <value>
×	not yet documented	inter-faces_bonding.xml.in: interfaces bonding <bonding> vif-s <vif-s> dhcp-options user-class	Nothing found in VyOS
×	not yet documented	inter-faces_bonding.xml.in: interfaces bonding <bonding> vif-s <vif-s> dhcp-options vendor-class-id	interfaces bonding <text> vif-s <text> dhcp-options vendor-class-id <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	inter-faces_bonding.xml.in: interfaces bonding <bonding> vif-s <vif-s> dhcpv6-options duid	interfaces bonding <text> vif-s <text> dhcpv6-options duid <value>
×	not yet documented	inter-faces_bonding.xml.in: interfaces bonding <bonding> vif-s <vif-s> dhcpv6-options no-release	interfaces bonding <text> vif-s <text> dhcpv6-options no-release
×	not yet documented	inter-faces_bonding.xml.in: interfaces bonding <bonding> vif-s <vif-s> dhcpv6-options parameters-only	interfaces bonding <text> vif-s <text> dhcpv6-options parameters-only
×	not yet documented	inter-faces_bonding.xml.in: interfaces bonding <bonding> vif-s <vif-s> dhcpv6-options pd <pd> interface <interface> address	interfaces bonding <text> vif-s <text> dhcpv6-options pd <text> interface <text> address <value>
×	not yet documented	inter-faces_bonding.xml.in: interfaces bonding <bonding> vif-s <vif-s> dhcpv6-options pd <pd> interface <interface> sla-id	interfaces bonding <text> vif-s <text> dhcpv6-options pd <text> interface <text> sla-id <value>
×	not yet documented	inter-faces_bonding.xml.in: interfaces bonding <bonding> vif-s <vif-s> dhcpv6-options pd <pd> length	interfaces bonding <text> vif-s <text> dhcpv6-options pd <text> length <value>
×	not yet documented	inter-faces_bonding.xml.in: interfaces bonding <bonding> vif-s <vif-s> dhcpv6-options rapid-commit	interfaces bonding <text> vif-s <text> dhcpv6-options rapid-commit
×	not yet documented	inter-faces_bonding.xml.in: interfaces bonding <bonding> vif-s <vif-s> dhcpv6-options temporary	interfaces bonding <text> vif-s <text> dhcpv6-options temporary
×	not yet documented	inter-faces_bonding.xml.in: interfaces bonding <bonding> vif-s <vif-s> disable	interfaces bonding <text> vif-s <text> disable

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	inter-faces_bonding.xml.in: interfaces bonding <bonding> vif-s <vif-s> disable-link-detect	interfaces bonding <text> vif-s <text> disable-link-detect
×	not yet documented	inter-faces_bonding.xml.in: interfaces bonding <bonding> vif-s <vif-s> ip adjust-mss	interfaces bonding <text> vif-s <text> ip adjust-mss <value>
×	not yet documented	inter-faces_bonding.xml.in: interfaces bonding <bonding> vif-s <vif-s> ip arp-cache-timeout	interfaces bonding <text> vif-s <text> ip arp-cache-timeout <value>
×	not yet documented	inter-faces_bonding.xml.in: interfaces bonding <bonding> vif-s <vif-s> ip disable-arp-filter	interfaces bonding <text> vif-s <text> ip disable-arp-filter
×	not yet documented	inter-faces_bonding.xml.in: interfaces bonding <bonding> vif-s <vif-s> ip disable-forwarding	interfaces bonding <text> vif-s <text> ip disable-forwarding
×	not yet documented	inter-faces_bonding.xml.in: interfaces bonding <bonding> vif-s <vif-s> ip enable-arp-accept	interfaces bonding <text> vif-s <text> ip enable-arp-accept
×	not yet documented	inter-faces_bonding.xml.in: interfaces bonding <bonding> vif-s <vif-s> ip enable-arp-announce	interfaces bonding <text> vif-s <text> ip enable-arp-announce
×	not yet documented	inter-faces_bonding.xml.in: interfaces bonding <bonding> vif-s <vif-s> ip enable-arp-ignore	interfaces bonding <text> vif-s <text> ip enable-arp-ignore
×	not yet documented	inter-faces_bonding.xml.in: interfaces bonding <bonding> vif-s <vif-s> ip enable-directed-broadcast	interfaces bonding <text> vif-s <text> ip enable-directed-broadcast

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	inter-faces_bonding.xml.in: interfaces bonding <bonding> vif-s <vif-s> ip enable-proxy-arp	interfaces bonding <text> vif-s <text> ip enable-proxy-arp
×	not yet documented	inter-faces_bonding.xml.in: interfaces bonding <bonding> vif-s <vif-s> ip proxy-arp-pvlan	interfaces bonding <text> vif-s <text> ip proxy-arp-pvlan
×	not yet documented	inter-faces_bonding.xml.in: interfaces bonding <bonding> vif-s <vif-s> ip source-validation	interfaces bonding <text> vif-s <text> ip source-validation <value>
×	not yet documented	inter-faces_bonding.xml.in: interfaces bonding <bonding> vif-s <vif-s> ipv6 accept-dad	interfaces bonding <text> vif-s <text> ipv6 accept-dad <value>
×	not yet documented	inter-faces_bonding.xml.in: interfaces bonding <bonding> vif-s <vif-s> ipv6 address autoconf	interfaces bonding <text> vif-s <text> ipv6 address autoconf
×	not yet documented	inter-faces_bonding.xml.in: interfaces bonding <bonding> vif-s <vif-s> ipv6 address eui64	interfaces bonding <text> vif-s <text> ipv6 address eui64 <value>
×	not yet documented	inter-faces_bonding.xml.in: interfaces bonding <bonding> vif-s <vif-s> ipv6 address no-default-link-local	interfaces bonding <text> vif-s <text> ipv6 address no-default-link-local
×	not yet documented	inter-faces_bonding.xml.in: interfaces bonding <bonding> vif-s <vif-s> ipv6 adjust-mss	interfaces bonding <text> vif-s <text> ipv6 adjust-mss <value>
×	not yet documented	inter-faces_bonding.xml.in: interfaces bonding <bonding> vif-s <vif-s> ipv6 base-reachable-time	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	inter-faces_bonding.xml.in: interfaces bonding <bonding> vif-s <vif-s> ipv6 disable-forwarding	interfaces bonding <text> vif-s <text> ipv6 disable-forwarding
×	not yet documented	inter-faces_bonding.xml.in: interfaces bonding <bonding> vif-s <vif-s> ipv6 dup-addr-detect-transmits	interfaces bonding <text> vif-s <text> ipv6 dup-addr-detect-transmits <value>
×	not yet documented	inter-faces_bonding.xml.in: interfaces bonding <bonding> vif-s <vif-s> ipv6 source-validation	Nothing found in VyOS
×	not yet documented	inter-faces_bonding.xml.in: interfaces bonding <bonding> vif-s <vif-s> mac	interfaces bonding <text> vif-s <text> mac <value>
×	not yet documented	inter-faces_bonding.xml.in: interfaces bonding <bonding> vif-s <vif-s> mirror egress	interfaces bonding <text> vif-s <text> mirror egress <value>
×	not yet documented	inter-faces_bonding.xml.in: interfaces bonding <bonding> vif-s <vif-s> mirror ingress	interfaces bonding <text> vif-s <text> mirror ingress <value>
×	not yet documented	inter-faces_bonding.xml.in: interfaces bonding <bonding> vif-s <vif-s> mtu	interfaces bonding <text> vif-s <text> mtu <value>
×	not yet documented	inter-faces_bonding.xml.in: interfaces bonding <bonding> vif-s <vif-s> protocol	interfaces bonding <text> vif-s <text> protocol <value>
×	not yet documented	inter-faces_bonding.xml.in: interfaces bonding <bonding> vif-s <vif-s> redirect	interfaces bonding <text> vif-s <text> redirect <value>
×	not yet documented	inter-faces_bonding.xml.in: interfaces bonding <bonding> vif-s <vif-s> vif-c <vif-c> address	interfaces bonding <text> vif-s <text> vif-c <text> address <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	inter-faces_bonding.xml.in: interfaces bonding <bonding> vif-s <vif-s> vif-c <vif-c> description	interfaces bonding <text> vif-s <text> vif-c <text> description <value>
×	not yet documented	inter-faces_bonding.xml.in: interfaces bonding <bonding> vif-s <vif-s> vif-c <vif-c> dhcp-options client-id	interfaces bonding <text> vif-s <text> vif-c <text> dhcp-options client-id <value>
×	not yet documented	inter-faces_bonding.xml.in: interfaces bonding <bonding> vif-s <vif-s> vif-c <vif-c> dhcp-options default-route-distance	interfaces bonding <text> vif-s <text> vif-c <text> dhcp-options default-route-distance <value>
×	not yet documented	inter-faces_bonding.xml.in: interfaces bonding <bonding> vif-s <vif-s> vif-c <vif-c> dhcp-options host-name	interfaces bonding <text> vif-s <text> vif-c <text> dhcp-options host-name <value>
×	not yet documented	inter-faces_bonding.xml.in: interfaces bonding <bonding> vif-s <vif-s> vif-c <vif-c> dhcp-options mtu	interfaces bonding <text> vif-s <text> vif-c <text> dhcp-options mtu
×	not yet documented	inter-faces_bonding.xml.in: interfaces bonding <bonding> vif-s <vif-s> vif-c <vif-c> dhcp-options no-default-route	interfaces bonding <text> vif-s <text> vif-c <text> dhcp-options no-default-route
×	not yet documented	inter-faces_bonding.xml.in: interfaces bonding <bonding> vif-s <vif-s> vif-c <vif-c> dhcp-options reject	interfaces bonding <text> vif-s <text> vif-c <text> dhcp-options reject <value>
×	not yet documented	inter-faces_bonding.xml.in: interfaces bonding <bonding> vif-s <vif-s> vif-c <vif-c> dhcp-options user-class	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	inter-faces_bonding.xml.in: interfaces bonding <bonding> vif-s <vif-s> vif-c <vif-c> dhcp-options vendor-class-id	interfaces bonding <text> vif-s <text> vif-c <text> dhcp-options vendor-class-id <value>
×	not yet documented	inter-faces_bonding.xml.in: interfaces bonding <bonding> vif-s <vif-s> vif-c <vif-c> dhcpv6-options duid	interfaces bonding <text> vif-s <text> vif-c <text> dhcpv6-options duid <value>
×	not yet documented	inter-faces_bonding.xml.in: interfaces bonding <bonding> vif-s <vif-s> vif-c <vif-c> dhcpv6-options no-release	interfaces bonding <text> vif-s <text> vif-c <text> dhcpv6-options no-release
×	not yet documented	inter-faces_bonding.xml.in: interfaces bonding <bonding> vif-s <vif-s> vif-c <vif-c> dhcpv6-options parameters-only	interfaces bonding <text> vif-s <text> vif-c <text> dhcpv6-options parameters-only
×	not yet documented	inter-faces_bonding.xml.in: interfaces bonding <bonding> vif-s <vif-s> vif-c <vif-c> dhcpv6-options pd <pd> interface <interface> address	interfaces bonding <text> vif-s <text> vif-c <text> dhcpv6-options pd <text> interface <text> address <value>
×	not yet documented	inter-faces_bonding.xml.in: interfaces bonding <bonding> vif-s <vif-s> vif-c <vif-c> dhcpv6-options pd <pd> interface <interface> sla-id	interfaces bonding <text> vif-s <text> vif-c <text> dhcpv6-options pd <text> interface <text> sla-id <value>
×	not yet documented	inter-faces_bonding.xml.in: interfaces bonding <bonding> vif-s <vif-s> vif-c <vif-c> dhcpv6-options pd <pd> length	interfaces bonding <text> vif-s <text> vif-c <text> dhcpv6-options pd <text> length <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	inter-faces_bonding.xml.in: interfaces bonding <bonding> vif-s <vif-s> vif-c <vif-c> dhcpv6-options rapid-commit	interfaces bonding <text> vif-s <text> vif-c <text> dhcpv6-options rapid-commit
×	not yet documented	inter-faces_bonding.xml.in: interfaces bonding <bonding> vif-s <vif-s> vif-c <vif-c> dhcpv6-options temporary	interfaces bonding <text> vif-s <text> vif-c <text> dhcpv6-options temporary
×	not yet documented	inter-faces_bonding.xml.in: interfaces bonding <bonding> vif-s <vif-s> vif-c <vif-c> disable	interfaces bonding <text> vif-s <text> vif-c <text> disable
×	not yet documented	inter-faces_bonding.xml.in: interfaces bonding <bonding> vif-s <vif-s> vif-c <vif-c> disable-link-detect	interfaces bonding <text> vif-s <text> vif-c <text> disable-link-detect
×	not yet documented	inter-faces_bonding.xml.in: interfaces bonding <bonding> vif-s <vif-s> vif-c <vif-c> ip adjust-mss	interfaces bonding <text> vif-s <text> vif-c <text> ip adjust-mss <value>
×	not yet documented	inter-faces_bonding.xml.in: interfaces bonding <bonding> vif-s <vif-s> vif-c <vif-c> ip arp-cache-timeout	interfaces bonding <text> vif-s <text> vif-c <text> ip arp-cache-timeout <value>
×	not yet documented	inter-faces_bonding.xml.in: interfaces bonding <bonding> vif-s <vif-s> vif-c <vif-c> ip disable-arp-filter	interfaces bonding <text> vif-s <text> vif-c <text> ip disable-arp-filter
×	not yet documented	inter-faces_bonding.xml.in: interfaces bonding <bonding> vif-s <vif-s> vif-c <vif-c> ip disable-forwarding	interfaces bonding <text> vif-s <text> vif-c <text> ip disable-forwarding

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	inter- faces_bonding.xml.in: interfaces bonding <bond- ing> vif-s <vif-s> vif-c <vif-c> ip enable-arp- accept	interfaces bonding <text> vif-s <text> vif-c <text> ip enable-arp-accept
×	not yet documented	inter- faces_bonding.xml.in: interfaces bonding <bond- ing> vif-s <vif-s> vif-c <vif-c> ip enable-arp- announce	interfaces bonding <text> vif-s <text> vif-c <text> ip enable-arp-announce
×	not yet documented	inter- faces_bonding.xml.in: interfaces bonding <bond- ing> vif-s <vif-s> vif-c <vif-c> ip enable-arp- ignore	interfaces bonding <text> vif-s <text> vif-c <text> ip enable-arp-ignore
×	not yet documented	inter- faces_bonding.xml.in: interfaces bonding <bonding> vif-s <vif- s> vif-c <vif-c> ip enable-directed-broadcast	interfaces bonding <text> vif-s <text> vif-c <text> ip enable-directed-broadcast
×	not yet documented	inter- faces_bonding.xml.in: interfaces bonding <bond- ing> vif-s <vif-s> vif-c <vif-c> ip enable-proxy- arp	interfaces bonding <text> vif-s <text> vif-c <text> ip enable-proxy-arp
×	not yet documented	inter- faces_bonding.xml.in: interfaces bonding <bond- ing> vif-s <vif-s> vif-c <vif-c> ip proxy-arp- pvlan	interfaces bonding <text> vif-s <text> vif-c <text> ip proxy-arp-pvlan
×	not yet documented	inter- faces_bonding.xml.in: interfaces bonding <bonding> vif-s <vif- s> vif-c <vif-c> ip source-validation	interfaces bonding <text> vif-s <text> vif-c <text> ip source-validation <value>
×	not yet documented	inter- faces_bonding.xml.in: interfaces bonding <bond- ing> vif-s <vif-s> vif-c <vif-c> ipv6 accept-dad	interfaces bonding <text> vif-s <text> vif-c <text> ipv6 accept-dad <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	inter-faces_bonding.xml.in: interfaces bonding <bonding> vif-s <vif-s> vif-c <vif-c> ipv6 address autoconf	interfaces bonding <text> vif-s <text> vif-c <text> ipv6 address autoconf
×	not yet documented	inter-faces_bonding.xml.in: interfaces bonding <bonding> vif-s <vif-s> vif-c <vif-c> ipv6 address eui64	interfaces bonding <text> vif-s <text> vif-c <text> ipv6 address eui64 <value>
×	not yet documented	inter-faces_bonding.xml.in: interfaces bonding <bonding> vif-s <vif-s> vif-c <vif-c> ipv6 address no-default-link-local	interfaces bonding <text> vif-s <text> vif-c <text> ipv6 address no-default-link-local
×	not yet documented	inter-faces_bonding.xml.in: interfaces bonding <bonding> vif-s <vif-s> vif-c <vif-c> ipv6 adjust-mss	interfaces bonding <text> vif-s <text> vif-c <text> ipv6 adjust-mss <value>
×	not yet documented	inter-faces_bonding.xml.in: interfaces bonding <bonding> vif-s <vif-s> vif-c <vif-c> ipv6 base-reachable-time	Nothing found in VyOS
×	not yet documented	inter-faces_bonding.xml.in: interfaces bonding <bonding> vif-s <vif-s> vif-c <vif-c> ipv6 disable-forwarding	interfaces bonding <text> vif-s <text> vif-c <text> ipv6 disable-forwarding
×	not yet documented	inter-faces_bonding.xml.in: interfaces bonding <bonding> vif-s <vif-s> vif-c <vif-c> ipv6 dup-addr-detect-transmits	interfaces bonding <text> vif-s <text> vif-c <text> ipv6 dup-addr-detect-transmits <value>
×	not yet documented	inter-faces_bonding.xml.in: interfaces bonding <bonding> vif-s <vif-s> vif-c <vif-c> ipv6 source-validation	Nothing found in VyOS

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Table 1 – continued from previous page

Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	inter- faces_bonding.xml.in: interfaces bonding <bond- ing> vif-s <vif-s> vif-c <vif-c> mac	interfaces bonding <text> vif-s <text> vif-c <text> mac <value>
×	not yet documented	inter- faces_bonding.xml.in: interfaces bonding <bond- ing> vif-s <vif-s> vif-c <vif-c> mirror egress	interfaces bonding <text> vif-s <text> vif-c <text> mirror egress <value>
×	not yet documented	inter- faces_bonding.xml.in: interfaces bonding <bond- ing> vif-s <vif-s> vif-c <vif-c> mirror ingress	interfaces bonding <text> vif-s <text> vif-c <text> mirror ingress <value>
×	not yet documented	inter- faces_bonding.xml.in: interfaces bonding <bond- ing> vif-s <vif-s> vif-c <vif-c> mtu	interfaces bonding <text> vif-s <text> vif-c <text> mtu <value>
×	not yet documented	inter- faces_bonding.xml.in: interfaces bonding <bond- ing> vif-s <vif-s> vif-c <vif-c> redirect	interfaces bonding <text> vif-s <text> vif-c <text> redirect <value>
×	not yet documented	inter- faces_bonding.xml.in: interfaces bonding <bond- ing> vif-s <vif-s> vif-c <vif-c> vrf	interfaces bonding <text> vif-s <text> vif-c <text> vrf <value>
×	not yet documented	inter- faces_bonding.xml.in: interfaces bonding <bond- ing> vif-s <vif-s> vrf	interfaces bonding <text> vif-s <text> vrf <value>
✓	set interfaces bonding <interface> vif <vlan-id> address <address / dhcp / dhcpv6>	inter- faces_bonding.xml.in: interfaces bonding <bond- ing> vif <vif> address	interfaces bonding <text> vif <text> address <value>
✓	set interfaces bonding <interface> vif <vlan-id> description <description>	inter- faces_bonding.xml.in: interfaces bonding <bond- ing> vif <vif> description	interfaces bonding <text> vif <text> description <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
✓	<code>set interfaces bonding <interface> vif <vlan-id> dhcp-options client-id <description></code>	inter-faces_bonding.xml.in: interfaces bonding <bonding> vif <vif> dhcp-options client-id	interfaces bonding <text> vif <text> dhcp-options client-id <value>
✓	<code>set interfaces bonding <interface> vif <vlan-id> dhcp-options default-route-distance <distance></code>	inter-faces_bonding.xml.in: interfaces bonding <bonding> vif <vif> dhcp-options default-route-distance	interfaces bonding <text> vif <text> dhcp-options default-route-distance <value>
✓	<code>set interfaces bonding <interface> vif <vlan-id> dhcp-options host-name <hostname></code>	inter-faces_bonding.xml.in: interfaces bonding <bonding> vif <vif> dhcp-options host-name	interfaces bonding <text> vif <text> dhcp-options host-name <value>
×	not yet documented	inter-faces_bonding.xml.in: interfaces bonding <bonding> vif <vif> dhcp-options mtu	interfaces bonding <text> vif <text> dhcp-options mtu
✓	<code>set interfaces bonding <interface> vif <vlan-id> dhcp-options no-default-route</code>	inter-faces_bonding.xml.in: interfaces bonding <bonding> vif <vif> dhcp-options no-default-route	interfaces bonding <text> vif <text> dhcp-options no-default-route
✓	<code>set interfaces bonding <interface> vif <vlan-id> dhcp-options reject <address></code>	inter-faces_bonding.xml.in: interfaces bonding <bonding> vif <vif> dhcp-options reject	interfaces bonding <text> vif <text> dhcp-options reject <value>
×	<code>set interfaces bonding <interface> vif <vlan-id> dhcp-options user-class <string></code>	inter-faces_bonding.xml.in: interfaces bonding <bonding> vif <vif> dhcp-options user-class	Nothing found in VyOS
✓	<code>set interfaces bonding <interface> vif <vlan-id> dhcp-options vendor-class-id <vendor-id></code>	inter-faces_bonding.xml.in: interfaces bonding <bonding> vif <vif> dhcp-options vendor-class-id	interfaces bonding <text> vif <text> dhcp-options vendor-class-id <value>
✓	<code>set interfaces bonding <interface> vif <vlan-id> dhcpv6-options duid <duid></code>	inter-faces_bonding.xml.in: interfaces bonding <bonding> vif <vif> dhcpv6-options duid	interfaces bonding <text> vif <text> dhcpv6-options duid <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
✓	<code>set interfaces bonding <interface> vif <vlan-id> dhcpv6-options no-release</code>	inter-faces_bonding.xml.in: interfaces bonding <bonding> vif <vif> dhcpv6-options no-release	interfaces bonding <text> vif <text> dhcpv6-options no-release
✓	<code>set interfaces bonding <interface> vif <vlan-id> dhcpv6-options parameters-only</code>	inter-faces_bonding.xml.in: interfaces bonding <bonding> vif <vif> dhcpv6-options parameters-only	interfaces bonding <text> vif <text> dhcpv6-options parameters-only
✓	<code>set interfaces bonding <interface> vif <vlan-id> dhcpv6-options pd <id> interface <delegatee> address <address></code>	inter-faces_bonding.xml.in: interfaces bonding <bonding> vif <vif> dhcpv6-options pd <pd> interface <interface> address	interfaces bonding <text> vif <text> dhcpv6-options pd <text> interface <text> address <value>
✓	<code>set interfaces bonding <interface> vif <vlan-id> dhcpv6-options pd <id> interface <delegatee> sla-id <id></code>	inter-faces_bonding.xml.in: interfaces bonding <bonding> vif <vif> dhcpv6-options pd <pd> interface <interface> sla-id	interfaces bonding <text> vif <text> dhcpv6-options pd <text> interface <text> sla-id <value>
✓	<code>set interfaces bonding <interface> vif <vlan-id> dhcpv6-options pd <id> length <length></code>	inter-faces_bonding.xml.in: interfaces bonding <bonding> vif <vif> dhcpv6-options pd <pd> length	interfaces bonding <text> vif <text> dhcpv6-options pd <text> length <value>
✓	<code>set interfaces bonding <interface> vif <vlan-id> dhcpv6-options rapid-commit</code>	inter-faces_bonding.xml.in: interfaces bonding <bonding> vif <vif> dhcpv6-options rapid-commit	interfaces bonding <text> vif <text> dhcpv6-options rapid-commit
✓	<code>set interfaces bonding <interface> vif <vlan-id> dhcpv6-options temporary</code>	inter-faces_bonding.xml.in: interfaces bonding <bonding> vif <vif> dhcpv6-options temporary	interfaces bonding <text> vif <text> dhcpv6-options temporary
✓	<code>set interfaces bonding <interface> vif <vlan-id> disable</code>	inter-faces_bonding.xml.in: interfaces bonding <bonding> vif <vif> disable	interfaces bonding <text> vif <text> disable
✓	<code>set interfaces bonding <interface> vif <vlan-id> disable-link-detect</code>	inter-faces_bonding.xml.in: interfaces bonding <bonding> vif <vif> disable-link-detect	interfaces bonding <text> vif <text> disable-link-detect

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	inter-faces_bonding.xml.in: interfaces bonding <bonding> vif <vif> egress-qos	interfaces bonding <text> vif <text> egress-qos <value>
×	not yet documented	inter-faces_bonding.xml.in: interfaces bonding <bonding> vif <vif> ingress-qos	interfaces bonding <text> vif <text> ingress-qos <value>
✓	<i>set interfaces bonding <interface> vif <vlan-id> ip adjust-mss <mss / clamp-mss-to-pmtu></i>	inter-faces_bonding.xml.in: interfaces bonding <bonding> vif <vif> ip adjust-mss	interfaces bonding <text> vif <text> ip adjust-mss <value>
✓	<i>set interfaces bonding <interface> vif <vlan-id> ip arp-cache-timeout</i>	inter-faces_bonding.xml.in: interfaces bonding <bonding> vif <vif> ip arp-cache-timeout	interfaces bonding <text> vif <text> ip arp-cache-timeout <value>
✓	<i>set interfaces bonding <interface> vif <vlan-id> ip disable-arp-filter</i>	inter-faces_bonding.xml.in: interfaces bonding <bonding> vif <vif> ip disable-arp-filter	interfaces bonding <text> vif <text> ip disable-arp-filter
✓	<i>set interfaces bonding <interface> vif <vlan-id> ip disable-forwarding</i>	inter-faces_bonding.xml.in: interfaces bonding <bonding> vif <vif> ip disable-forwarding	interfaces bonding <text> vif <text> ip disable-forwarding
✓	<i>set interfaces bonding <interface> vif <vlan-id> ip enable-arp-accept</i>	inter-faces_bonding.xml.in: interfaces bonding <bonding> vif <vif> ip enable-arp-accept	interfaces bonding <text> vif <text> ip enable-arp-accept
✓	<i>set interfaces bonding <interface> vif <vlan-id> ip enable-arp-announce</i>	inter-faces_bonding.xml.in: interfaces bonding <bonding> vif <vif> ip enable-arp-announce	interfaces bonding <text> vif <text> ip enable-arp-announce
✓	<i>set interfaces bonding <interface> vif <vlan-id> ip enable-arp-ignore</i>	inter-faces_bonding.xml.in: interfaces bonding <bonding> vif <vif> ip enable-arp-ignore	interfaces bonding <text> vif <text> ip enable-arp-ignore
✓	<i>set interfaces bonding <interface> vif <vlan-id> ip enable-directed-broadcast</i>	inter-faces_bonding.xml.in: interfaces bonding <bonding> vif <vif> ip enable-directed-broadcast	interfaces bonding <text> vif <text> ip enable-directed-broadcast

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Table 1 – continued from previous page

Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
✓	<code>set interfaces bonding <interface> vif <vlan-id> ip enable-proxy-arp</code>	inter-faces_bonding.xml.in: interfaces bonding <bonding> vif <vif> ip enable-proxy-arp	interfaces bonding <text> vif <text> ip enable-proxy-arp
✓	<code>set interfaces bonding <interface> vif <vlan-id> ip proxy-arp-pvlan</code>	inter-faces_bonding.xml.in: interfaces bonding <bonding> vif <vif> ip proxy-arp-pvlan	interfaces bonding <text> vif <text> ip proxy-arp-pvlan
✓	<code>set interfaces bonding <interface> vif <vlan-id> ip source-validation <strict loose disable></code>	inter-faces_bonding.xml.in: interfaces bonding <bonding> vif <vif> ip source-validation	interfaces bonding <text> vif <text> ip source-validation <value>
✓	<code>set interfaces bonding <interface> vif <vlan-id> ipv6 accept-dad <0 1 2></code>	inter-faces_bonding.xml.in: interfaces bonding <bonding> vif <vif> ipv6 accept-dad	interfaces bonding <text> vif <text> ipv6 accept-dad <value>
✓	<code>set interfaces bonding <interface> vif <vlan-id> ipv6 address autoconf</code>	inter-faces_bonding.xml.in: interfaces bonding <bonding> vif <vif> ipv6 address autoconf	interfaces bonding <text> vif <text> ipv6 address autoconf
✓	<code>set interfaces bonding <interface> vif <vlan-id> ipv6 address eui64 <prefix></code>	inter-faces_bonding.xml.in: interfaces bonding <bonding> vif <vif> ipv6 address eui64	interfaces bonding <text> vif <text> ipv6 address eui64 <value>
✓	<code>set interfaces bonding <interface> vif <vlan-id> ipv6 address no-default-link-local</code>	inter-faces_bonding.xml.in: interfaces bonding <bonding> vif <vif> ipv6 address no-default-link-local	interfaces bonding <text> vif <text> ipv6 address no-default-link-local
✓	<code>set interfaces bonding <interface> vif <vlan-id> ipv6 adjust-mss <mss clamp-mss-to-pmtu></code>	inter-faces_bonding.xml.in: interfaces bonding <bonding> vif <vif> ipv6 adjust-mss	interfaces bonding <text> vif <text> ipv6 adjust-mss <value>
×	not yet documented	inter-faces_bonding.xml.in: interfaces bonding <bonding> vif <vif> ipv6 base-reachable-time	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
✓	<i>set interfaces bonding <interface> vif <vlan-id> ipv6 disable-forwarding</i>	inter-faces_bonding.xml.in: interfaces bonding <bonding> vif <vif> ipv6 disable-forwarding	interfaces bonding <text> vif <text> ipv6 disable-forwarding
✓	<i>set interfaces bonding <interface> vif <vlan-id> ipv6 dup-addr-detect-transmits <n></i>	inter-faces_bonding.xml.in: interfaces bonding <bonding> vif <vif> ipv6 dup-addr-detect-transmits	interfaces bonding <text> vif <text> ipv6 dup-addr-detect-transmits <value>
×	not yet documented	inter-faces_bonding.xml.in: interfaces bonding <bonding> vif <vif> ipv6 source-validation	Nothing found in VyOS
✓	<i>set interfaces bonding <interface> vif <vlan-id> mac <xx:xx:xx:xx:xx:xx></i>	inter-faces_bonding.xml.in: interfaces bonding <bonding> vif <vif> mac	interfaces bonding <text> vif <text> mac <value>
×	not yet documented	inter-faces_bonding.xml.in: interfaces bonding <bonding> vif <vif> mirror egress	interfaces bonding <text> vif <text> mirror egress <value>
×	not yet documented	inter-faces_bonding.xml.in: interfaces bonding <bonding> vif <vif> mirror ingress	interfaces bonding <text> vif <text> mirror ingress <value>
✓	<i>set interfaces bonding <interface> vif <vlan-id> mtu <mtu></i>	inter-faces_bonding.xml.in: interfaces bonding <bonding> vif <vif> mtu	interfaces bonding <text> vif <text> mtu <value>
×	not yet documented	inter-faces_bonding.xml.in: interfaces bonding <bonding> vif <vif> redirect	interfaces bonding <text> vif <text> redirect <value>
✓	<i>set interfaces bonding <interface> vif <vlan-id> vrf <vrf></i>	inter-faces_bonding.xml.in: interfaces bonding <bonding> vif <vif> vrf	interfaces bonding <text> vif <text> vrf <value>
✓	<i>set interfaces bonding <interface> vrf <vrf></i>	inter-faces_bonding.xml.in: interfaces bonding <bonding> vrf	interfaces bonding <text> vrf <value>
×	<i>set interfaces bridge <interface> aging <time></i>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set interfaces bridge <interface> enable-vlan</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set interfaces bridge <interface> forwarding-delay <delay></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set interfaces bridge <interface> hello-time <interval></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set interfaces bridge <interface> igmp querier</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set interfaces bridge <interface> igmp snooping</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set interfaces bridge <interface> max-age <time></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set interfaces bridge <interface> member interface <member></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set interfaces bridge <interface> member interface <member> allowed-vlan <vlan-id></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set interfaces bridge <interface> member interface <member> cost <cost></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set interfaces bridge <interface> member interface <member> disable-learning</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set interfaces bridge <interface> member interface <member> native-vlan <vlan-id></code>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set interfaces bridge <interface> member interface <member> priority <priority></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set interfaces bridge <interface> protocol <802.1ad / 802.1q></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set interfaces bridge <interface> stp</code>	Nothing found in XML Definitions	Nothing found in VyOS
✓	<code>set interfaces bridge <interface> address <address / dhcp / dhcpv6></code>	interfaces_bridge.xml.in: interfaces bridge <bridge> address	interfaces bridge <text> address <value>
×	not yet documented	interfaces_bridge.xml.in: interfaces bridge <bridge> aging	interfaces bridge <text> aging <value>
✓	<code>set interfaces bridge <interface> description <description></code>	interfaces_bridge.xml.in: interfaces bridge <bridge> description	interfaces bridge <text> description <value>
✓	<code>set interfaces bridge <interface> dhcp-options client-id <description></code>	interfaces_bridge.xml.in: interfaces bridge <bridge> dhcp-options client-id	interfaces bridge <text> dhcp-options client-id <value>
✓	<code>set interfaces bridge <interface> dhcp-options default-route-distance <distance></code>	interfaces_bridge.xml.in: interfaces bridge <bridge> dhcp-options default- route-distance	interfaces bridge <text> dhcp-options default- route-distance <value>
✓	<code>set interfaces bridge <interface> dhcp-options host-name <hostname></code>	interfaces_bridge.xml.in: interfaces bridge <bridge> dhcp-options host-name	interfaces bridge <text> dhcp-options host-name <value>
×	not yet documented	interfaces_bridge.xml.in: interfaces bridge <bridge> dhcp-options mtu	interfaces bridge <text> dhcp-options mtu
✓	<code>set interfaces bridge <interface> dhcp-options no-default-route</code>	interfaces_bridge.xml.in: interfaces bridge <bridge> dhcp-options no-default- route	interfaces bridge <text> dhcp-options no-default- route
✓	<code>set interfaces bridge <interface> dhcp-options reject <address></code>	interfaces_bridge.xml.in: interfaces bridge <bridge> dhcp-options reject	interfaces bridge <text> dhcp-options reject <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<i>set interfaces bridge <interface> dhcp-options user-class <string></i>	interfaces_bridge.xml.in: interfaces bridge <bridge> dhcp-options user-class	Nothing found in VyOS
✓	<i>set interfaces bridge <interface> dhcp-options vendor-class-id <vendor-id></i>	interfaces_bridge.xml.in: interfaces bridge <bridge> dhcp-options vendor- class-id	interfaces bridge <text> dhcp-options vendor- class-id <value>
✓	<i>set interfaces bridge <interface> dhcpv6-options duid <duid></i>	interfaces_bridge.xml.in: interfaces bridge <bridge> dhcpv6-options duid	interfaces bridge <text> dhcpv6-options duid <value>
✓	<i>set interfaces bridge <interface> dhcpv6-options no-release</i>	interfaces_bridge.xml.in: interfaces bridge <bridge> dhcpv6-options no-release	interfaces bridge <text> dhcpv6-options no-release
✓	<i>set interfaces bridge <interface> dhcpv6-options parameters-only</i>	interfaces_bridge.xml.in: interfaces bridge <bridge> dhcpv6-options parameters-only	interfaces bridge <text> dhcpv6-options parameters-only
✓	<i>set interfaces bridge <interface> dhcpv6-options pd <id> interface <delegatee> address <address></i>	interfaces_bridge.xml.in: interfaces bridge <bridge> dhcpv6-options pd <pd> interface <interface> ad- dress	interfaces bridge <text> dhcpv6-options pd <text> interface <text> address <value>
✓	<i>set interfaces bridge <interface> dhcpv6-options pd <id> interface <delegatee> sla-id <id></i>	interfaces_bridge.xml.in: interfaces bridge <bridge> dhcpv6-options pd <pd> interface <interface> sla- id	interfaces bridge <text> dhcpv6-options pd <text> interface <text> sla-id <value>
✓	<i>set interfaces bridge <interface> dhcpv6-options pd <id> length <length></i>	interfaces_bridge.xml.in: interfaces bridge <bridge> dhcpv6-options pd <pd> length	interfaces bridge <text> dhcpv6-options pd <text> length <value>
✓	<i>set interfaces bridge <interface> dhcpv6-options rapid-commit</i>	interfaces_bridge.xml.in: interfaces bridge <bridge> dhcpv6-options rapid- commit	interfaces bridge <text> dhcpv6-options rapid- commit
✓	<i>set interfaces bridge <interface> dhcpv6-options temporary</i>	interfaces_bridge.xml.in: interfaces bridge <bridge> dhcpv6-options temporary	interfaces bridge <text> dhcpv6-options temporary
✓	<i>set interfaces bridge <interface> disable</i>	interfaces_bridge.xml.in: interfaces bridge <bridge> disable	interfaces bridge <text> disable

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<i>set interfaces bridge <interface> disable-flow-control</i>	Nothing found in XML Definitions	Nothing found in VyOS
✓	<i>set interfaces bridge <interface> disable-link-detect</i>	interfaces_bridge.xml.in: interfaces bridge <bridge> disable-link-detect	interfaces bridge <text> disable-link-detect
×	not yet documented	interfaces_bridge.xml.in: interfaces bridge <bridge> enable-vlan	interfaces bridge <text> enable-vlan
×	not yet documented	interfaces_bridge.xml.in: interfaces bridge <bridge> forwarding-delay	interfaces bridge <text> forwarding-delay <value>
×	not yet documented	interfaces_bridge.xml.in: interfaces bridge <bridge> hello-time	interfaces bridge <text> hello-time <value>
×	not yet documented	interfaces_bridge.xml.in: interfaces bridge <bridge> igmp querier	interfaces bridge <text> igmp querier
×	not yet documented	interfaces_bridge.xml.in: interfaces bridge <bridge> igmp snooping	interfaces bridge <text> igmp snooping
✓	<i>set interfaces bridge <interface> ip adjust-mss <mss / clamp-mss-to-pmtu></i>	interfaces_bridge.xml.in: interfaces bridge <bridge> ip adjust-mss	interfaces bridge <text> ip adjust-mss <value>
✓	<i>set interfaces bridge <interface> ip arp-cache-timeout</i>	interfaces_bridge.xml.in: interfaces bridge <bridge> ip arp-cache-timeout	interfaces bridge <text> ip arp-cache-timeout <value>
✓	<i>set interfaces bridge <interface> ip disable-arp-filter</i>	interfaces_bridge.xml.in: interfaces bridge <bridge> ip disable-arp-filter	interfaces bridge <text> ip disable-arp-filter
✓	<i>set interfaces bridge <interface> ip disable-forwarding</i>	interfaces_bridge.xml.in: interfaces bridge <bridge> ip disable-forwarding	interfaces bridge <text> ip disable-forwarding
✓	<i>set interfaces bridge <interface> ip enable-arp-accept</i>	interfaces_bridge.xml.in: interfaces bridge <bridge> ip enable-arp-accept	interfaces bridge <text> ip enable-arp-accept
✓	<i>set interfaces bridge <interface> ip enable-arp-announce</i>	interfaces_bridge.xml.in: interfaces bridge <bridge> ip enable-arp-announce	interfaces bridge <text> ip enable-arp-announce
✓	<i>set interfaces bridge <interface> ip enable-arp-ignore</i>	interfaces_bridge.xml.in: interfaces bridge <bridge> ip enable-arp-ignore	interfaces bridge <text> ip enable-arp-ignore

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
✓	<code>set interfaces bridge <interface> ip enable-directed-broadcast</code>	interfaces_bridge.xml.in: interfaces bridge <bridge> ip enable-directed-broadcast	interfaces bridge <text> ip enable-directed-broadcast
✓	<code>set interfaces bridge <interface> ip enable-proxy-arp</code>	interfaces_bridge.xml.in: interfaces bridge <bridge> ip enable-proxy-arp	interfaces bridge <text> ip enable-proxy-arp
✓	<code>set interfaces bridge <interface> ip proxy-arp-pvlan</code>	interfaces_bridge.xml.in: interfaces bridge <bridge> ip proxy-arp-pvlan	interfaces bridge <text> ip proxy-arp-pvlan
✓	<code>set interfaces bridge <interface> ip source-validation <strict loose disable></code>	interfaces_bridge.xml.in: interfaces bridge <bridge> ip source-validation	interfaces bridge <text> ip source-validation <value>
✓	<code>set interfaces bridge <interface> ipv6 accept-dad <0 1 2></code>	interfaces_bridge.xml.in: interfaces bridge <bridge> ipv6 accept-dad	interfaces bridge <text> ipv6 accept-dad <value>
✓	<code>set interfaces bridge <interface> ipv6 address autoconf</code>	interfaces_bridge.xml.in: interfaces bridge <bridge> ipv6 address autoconf	interfaces bridge <text> ipv6 address autoconf
✓	<code>set interfaces bridge <interface> ipv6 address eui64 <prefix></code>	interfaces_bridge.xml.in: interfaces bridge <bridge> ipv6 address eui64	interfaces bridge <text> ipv6 address eui64 <value>
✓	<code>set interfaces bridge <interface> ipv6 address no-default-link-local</code>	interfaces_bridge.xml.in: interfaces bridge <bridge> ipv6 address no-default-link-local	interfaces bridge <text> ipv6 address no-default-link-local
✓	<code>set interfaces bridge <interface> ipv6 adjust-mss <mss clamp-mss-to-pmtu></code>	interfaces_bridge.xml.in: interfaces bridge <bridge> ipv6 adjust-mss	interfaces bridge <text> ipv6 adjust-mss <value>
×	not yet documented	interfaces_bridge.xml.in: interfaces bridge <bridge> ipv6 base-reachable-time	Nothing found in VyOS
✓	<code>set interfaces bridge <interface> ipv6 disable-forwarding</code>	interfaces_bridge.xml.in: interfaces bridge <bridge> ipv6 disable-forwarding	interfaces bridge <text> ipv6 disable-forwarding
✓	<code>set interfaces bridge <interface> ipv6 dup-addr-detect-transmits <n></code>	interfaces_bridge.xml.in: interfaces bridge <bridge> ipv6 dup-addr-detect-transmits	interfaces bridge <text> ipv6 dup-addr-detect-transmits <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	interfaces_bridge.xml.in: interfaces bridge <bridge> ipv6 source-validation	Nothing found in VyOS
✓	set interfaces bridge <interface> mac <xx:xx:xx:xx:xx:xx>	interfaces_bridge.xml.in: interfaces bridge <bridge> mac	interfaces bridge <text> mac <value>
×	not yet documented	interfaces_bridge.xml.in: interfaces bridge <bridge> max-age	interfaces bridge <text> max-age <value>
×	not yet documented	interfaces_bridge.xml.in: interfaces bridge <bridge> member interface <inter- face> allowed-vlan	interfaces bridge <text> member interface <text> allowed-vlan <value>
×	not yet documented	interfaces_bridge.xml.in: interfaces bridge <bridge> member interface <inter- face> cost	interfaces bridge <text> member interface <text> cost <value>
×	not yet documented	interfaces_bridge.xml.in: interfaces bridge <bridge> member interface <inter- face> isolated	interfaces bridge <text> member interface <text> isolated
×	not yet documented	interfaces_bridge.xml.in: interfaces bridge <bridge> member interface <inter- face> native-vlan	interfaces bridge <text> member interface <text> native-vlan <value>
×	not yet documented	interfaces_bridge.xml.in: interfaces bridge <bridge> member interface <inter- face> priority	interfaces bridge <text> member interface <text> priority <value>
✓	set interfaces bridge <interface> mirror egress <monitor-interface>	interfaces_bridge.xml.in: interfaces bridge <bridge> mirror egress	interfaces bridge <text> mirror egress <value>
✓	set interfaces bridge <interface> mirror ingress <monitor-interface>	interfaces_bridge.xml.in: interfaces bridge <bridge> mirror ingress	interfaces bridge <text> mirror ingress <value>
✓	set interfaces bridge <interface> mtu <mtu>	interfaces_bridge.xml.in: interfaces bridge <bridge> mtu	interfaces bridge <text> mtu <value>
×	not yet documented	interfaces_bridge.xml.in: interfaces bridge <bridge> priority	interfaces bridge <text> priority <value>
×	not yet documented	interfaces_bridge.xml.in: interfaces bridge <bridge> protocol	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	interfaces_bridge.xml.in: interfaces bridge <bridge> redirect	interfaces bridge <text> redirect <value>
×	not yet documented	interfaces_bridge.xml.in: interfaces bridge <bridge> stp	interfaces bridge <text> stp
×	<i>set interfaces bridge <interface> vif <vlan-id></i>	Nothing found in XML Definitions	Nothing found in VyOS
✓	<i>set interfaces bridge <interface> vif <vlan-id> address <address dhcp dhcpv6></i>	interfaces_bridge.xml.in: interfaces bridge <bridge> vif <vif> address	interfaces bridge <text> vif <text> address <value>
✓	<i>set interfaces bridge <interface> vif <vlan-id> description <description></i>	interfaces_bridge.xml.in: interfaces bridge <bridge> vif <vif> description	interfaces bridge <text> vif <text> description <value>
✓	<i>set interfaces bridge <interface> vif <vlan-id> dhcp-options client-id <description></i>	interfaces_bridge.xml.in: interfaces bridge <bridge> vif <vif> dhcp-options client-id	interfaces bridge <text> vif <text> dhcp-options client-id <value>
✓	<i>set interfaces bridge <interface> vif <vlan-id> dhcp-options default-route-distance <distance></i>	interfaces_bridge.xml.in: interfaces bridge <bridge> vif <vif> dhcp-options default-route-distance	interfaces bridge <text> vif <text> dhcp-options default-route-distance <value>
✓	<i>set interfaces bridge <interface> vif <vlan-id> dhcp-options host-name <hostname></i>	interfaces_bridge.xml.in: interfaces bridge <bridge> vif <vif> dhcp-options host-name	interfaces bridge <text> vif <text> dhcp-options host-name <value>
×	not yet documented	interfaces_bridge.xml.in: interfaces bridge <bridge> vif <vif> dhcp-options mtu	interfaces bridge <text> vif <text> dhcp-options mtu
✓	<i>set interfaces bridge <interface> vif <vlan-id> dhcp-options no-default-route</i>	interfaces_bridge.xml.in: interfaces bridge <bridge> vif <vif> dhcp-options no- default-route	interfaces bridge <text> vif <text> dhcp-options no-default-route

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
✓	<code>set interfaces bridge <interface> vif <vlan-id> dhcp-options reject <address></code>	interfaces_bridge.xml.in: interfaces bridge <bridge> vif <vif> dhcp-options re- ject	interfaces bridge <text> vif <text> dhcp-options re- ject <value>
×	<code>set interfaces bridge <interface> vif <vlan-id> dhcp-options user-class <string></code>	interfaces_bridge.xml.in: interfaces bridge <bridge> vif <vif> dhcp-options user-class	Nothing found in VyOS
✓	<code>set interfaces bridge <interface> vif <vlan-id> dhcp-options vendor-class-id <vendor-id></code>	interfaces_bridge.xml.in: interfaces bridge <bridge> vif <vif> dhcp-options vendor-class-id	interfaces bridge <text> vif <text> dhcp-options vendor-class-id <value>
✓	<code>set interfaces bridge <interface> vif <vlan-id> dhcpv6-options duid <duid></code>	interfaces_bridge.xml.in: interfaces bridge <bridge> vif <vif> dhcpv6-options duid	interfaces bridge <text> vif <text> dhcpv6-options duid <value>
✓	<code>set interfaces bridge <interface> vif <vlan-id> dhcpv6-options no-release</code>	interfaces_bridge.xml.in: interfaces bridge <bridge> vif <vif> dhcpv6-options no-release	interfaces bridge <text> vif <text> dhcpv6-options no-release
✓	<code>set interfaces bridge <interface> vif <vlan-id> dhcpv6-options parameters-only</code>	interfaces_bridge.xml.in: interfaces bridge <bridge> vif <vif> dhcpv6-options parameters-only	interfaces bridge <text> vif <text> dhcpv6-options parameters-only
✓	<code>set interfaces bridge <interface> vif <vlan-id> dhcpv6-options pd <id> interface <delegatee> address <address></code>	interfaces_bridge.xml.in: interfaces bridge <bridge> vif <vif> dhcpv6-options pd <pd> interface <inter- face> address	interfaces bridge <text> vif <text> dhcpv6-options pd <text> interface <text> address <value>
✓	<code>set interfaces bridge <interface> vif <vlan-id> dhcpv6-options pd <id> interface <delegatee> sla-id <id></code>	interfaces_bridge.xml.in: interfaces bridge <bridge> vif <vif> dhcpv6-options pd <pd> interface <inter- face> sla-id	interfaces bridge <text> vif <text> dhcpv6-options pd <text> interface <text> sla-id <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
✓	<code>set interfaces bridge <interface> vif <vlan-id> dhcpv6-options pd <id> length <length></code>	interfaces_bridge.xml.in: interfaces bridge <bridge> vif <vif> dhcpv6-options pd <pd> length	interfaces bridge <text> vif <text> dhcpv6-options pd <text> length <value>
✓	<code>set interfaces bridge <interface> vif <vlan-id> dhcpv6-options rapid-commit</code>	interfaces_bridge.xml.in: interfaces bridge <bridge> vif <vif> dhcpv6-options rapid-commit	interfaces bridge <text> vif <text> dhcpv6-options rapid-commit
✓	<code>set interfaces bridge <interface> vif <vlan-id> dhcpv6-options temporary</code>	interfaces_bridge.xml.in: interfaces bridge <bridge> vif <vif> dhcpv6-options temporary	interfaces bridge <text> vif <text> dhcpv6-options temporary
✓	<code>set interfaces bridge <interface> vif <vlan-id> disable</code>	interfaces_bridge.xml.in: interfaces bridge <bridge> vif <vif> disable	interfaces bridge <text> vif <text> disable
✓	<code>set interfaces bridge <interface> vif <vlan-id> disable-link-detect</code>	interfaces_bridge.xml.in: interfaces bridge <bridge> vif <vif> disable-link- detect	interfaces bridge <text> vif <text> disable-link- detect
×	not yet documented	interfaces_bridge.xml.in: interfaces bridge <bridge> vif <vif> egress-qos	interfaces bridge <text> vif <text> egress-qos <value>
×	not yet documented	interfaces_bridge.xml.in: interfaces bridge <bridge> vif <vif> ingress-qos	interfaces bridge <text> vif <text> ingress-qos <value>
✓	<code>set interfaces bridge <interface> vif <vlan-id> ip adjust-mss <mss / clamp-mss-to-pmtu></code>	interfaces_bridge.xml.in: interfaces bridge <bridge> vif <vif> ip adjust-mss	interfaces bridge <text> vif <text> ip adjust-mss <value>
✓	<code>set interfaces bridge <interface> vif <vlan-id> ip arp-cache-timeout</code>	interfaces_bridge.xml.in: interfaces bridge <bridge> vif <vif> ip arp-cache- timeout	interfaces bridge <text> vif <text> ip arp-cache- timeout <value>
✓	<code>set interfaces bridge <interface> vif <vlan-id> ip disable-arp-filter</code>	interfaces_bridge.xml.in: interfaces bridge <bridge> vif <vif> ip disable-arp- filter	interfaces bridge <text> vif <text> ip disable-arp- filter
✓	<code>set interfaces bridge <interface> vif <vlan-id> ip disable-forwarding</code>	interfaces_bridge.xml.in: interfaces bridge <bridge> vif <vif> ip disable- forwarding	interfaces bridge <text> vif <text> ip disable- forwarding

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
✓	<code>set interfaces bridge <interface> vif <vlan-id> ip enable-arp-accept</code>	interfaces_bridge.xml.in: interfaces bridge <bridge> vif <vif> ip enable-arp-accept	interfaces bridge <text> vif <text> ip enable-arp-accept
✓	<code>set interfaces bridge <interface> vif <vlan-id> ip enable-arp-announce</code>	interfaces_bridge.xml.in: interfaces bridge <bridge> vif <vif> ip enable-arp-announce	interfaces bridge <text> vif <text> ip enable-arp-announce
✓	<code>set interfaces bridge <interface> vif <vlan-id> ip enable-arp-ignore</code>	interfaces_bridge.xml.in: interfaces bridge <bridge> vif <vif> ip enable-arp-ignore	interfaces bridge <text> vif <text> ip enable-arp-ignore
✓	<code>set interfaces bridge <interface> vif <vlan-id> ip enable-directed-broad</code>	interfaces_bridge.xml.in: interfaces bridge <bridge> vif <vif> ip enable-directed-broadcast	interfaces bridge <text> vif <text> ip enable-directed-broadcast
✓	<code>set interfaces bridge <interface> vif <vlan-id> ip enable-proxy-arp</code>	interfaces_bridge.xml.in: interfaces bridge <bridge> vif <vif> ip enable-proxy-arp	interfaces bridge <text> vif <text> ip enable-proxy-arp
✓	<code>set interfaces bridge <interface> vif <vlan-id> ip proxy-arp-pvlan</code>	interfaces_bridge.xml.in: interfaces bridge <bridge> vif <vif> ip proxy-arp-pvlan	interfaces bridge <text> vif <text> ip proxy-arp-pvlan
✓	<code>set interfaces bridge <interface> vif <vlan-id> ip source-validation <strict loose disable></code>	interfaces_bridge.xml.in: interfaces bridge <bridge> vif <vif> ip source-validation	interfaces bridge <text> vif <text> ip source-validation <value>
✓	<code>set interfaces bridge <interface> vif <vlan-id> ipv6 accept-dad <0 1 2></code>	interfaces_bridge.xml.in: interfaces bridge <bridge> vif <vif> ipv6 accept-dad	interfaces bridge <text> vif <text> ipv6 accept-dad <value>
✓	<code>set interfaces bridge <interface> vif <vlan-id> ipv6 address autoconf</code>	interfaces_bridge.xml.in: interfaces bridge <bridge> vif <vif> ipv6 address autoconf	interfaces bridge <text> vif <text> ipv6 address autoconf
✓	<code>set interfaces bridge <interface> vif <vlan-id> ipv6 address eui64 <prefix></code>	interfaces_bridge.xml.in: interfaces bridge <bridge> vif <vif> ipv6 address eui64	interfaces bridge <text> vif <text> ipv6 address eui64 <value>
✓	<code>set interfaces bridge <interface> vif <vlan-id> ipv6 address no-default-link-local</code>	interfaces_bridge.xml.in: interfaces bridge <bridge> vif <vif> ipv6 address no-default-link-local	interfaces bridge <text> vif <text> ipv6 address no-default-link-local

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
✓	<code>set interfaces bridge <interface> vif <vlan-id> ipv6 adjust-mss <mss / clamp-mss-to-pmtu></code>	interfaces_bridge.xml.in: interfaces bridge <bridge> vif <vif> ipv6 adjust-mss	interfaces bridge <text> vif <text> ipv6 adjust-mss <value>
×	not yet documented	interfaces_bridge.xml.in: interfaces bridge <bridge> vif <vif> ipv6 base- reachable-time	Nothing found in VyOS
✓	<code>set interfaces bridge <interface> vif <vlan-id> ipv6 disable-forwarding</code>	interfaces_bridge.xml.in: interfaces bridge <bridge> vif <vif> ipv6 disable- forwarding	interfaces bridge <text> vif <text> ipv6 disable- forwarding
✓	<code>set interfaces bridge <interface> vif <vlan-id> ipv6 dup-addr-detect-trans <n></code>	interfaces_bridge.xml.in: interfaces bridge <bridge> vif <vif> ipv6 dup-addr- detect-transmits	interfaces bridge <text> vif <text> ipv6 dup-addr- detect-transmits <value>
×	not yet documented	interfaces_bridge.xml.in: interfaces bridge <bridge> vif <vif> ipv6 source- validation	Nothing found in VyOS
✓	<code>set interfaces bridge <interface> vif <vlan-id> mac <xx:xx:xx:xx:xx:xx></code>	interfaces_bridge.xml.in: interfaces bridge <bridge> vif <vif> mac	interfaces bridge <text> vif <text> mac <value>
×	not yet documented	interfaces_bridge.xml.in: interfaces bridge <bridge> vif <vif> mirror egress	interfaces bridge <text> vif <text> mirror egress <value>
×	not yet documented	interfaces_bridge.xml.in: interfaces bridge <bridge> vif <vif> mirror ingress	interfaces bridge <text> vif <text> mirror ingress <value>
✓	<code>set interfaces bridge <interface> vif <vlan-id> mtu <mtu></code>	interfaces_bridge.xml.in: interfaces bridge <bridge> vif <vif> mtu	interfaces bridge <text> vif <text> mtu <value>
×	not yet documented	interfaces_bridge.xml.in: interfaces bridge <bridge> vif <vif> redirect	interfaces bridge <text> vif <text> redirect <value>
✓	<code>set interfaces bridge <interface> vif <vlan-id> vrf <vrf></code>	interfaces_bridge.xml.in: interfaces bridge <bridge> vif <vif> vrf	interfaces bridge <text> vif <text> vrf <value>
✓	<code>set interfaces bridge <interface> vrf <vrf></code>	interfaces_bridge.xml.in: interfaces bridge <bridge> vrf	interfaces bridge <text> vrf <value>
✓	<code>set interfaces dummy <interface> address <address></code>	interfaces_dummy.xml.in: interfaces dummy <dummy> address	interfaces dummy <text> address <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
✓	<code>set interfaces dummy <interface> description <description></code>	interfaces_dummy.xml.in: interfaces dummy <dummy> description	interfaces dummy <text> description <value>
✓	<code>set interfaces dummy <interface> disable</code>	interfaces_dummy.xml.in: interfaces dummy <dummy> disable	interfaces dummy <text> disable
×	not yet documented	interfaces_dummy.xml.in: interfaces dummy <dummy> ip disable-forwarding	interfaces dummy <text> ip disable-forwarding
×	not yet documented	interfaces_dummy.xml.in: interfaces dummy <dummy> ip source-validation	interfaces dummy <text> ip source-validation <value>
×	not yet documented	interfaces_dummy.xml.in: interfaces dummy <dummy> ipv6 address eui64	interfaces dummy <text> ipv6 address eui64 <value>
×	not yet documented	interfaces_dummy.xml.in: interfaces dummy <dummy> ipv6 address no-default-link-local	interfaces dummy <text> ipv6 address no-default-link-local
×	not yet documented	interfaces_dummy.xml.in: interfaces dummy <dummy> ipv6 disable-forwarding	interfaces dummy <text> ipv6 disable-forwarding
×	not yet documented	interfaces_dummy.xml.in: interfaces dummy <dummy> mirror egress	interfaces dummy <text> mirror egress <value>
×	not yet documented	interfaces_dummy.xml.in: interfaces dummy <dummy> mirror ingress	interfaces dummy <text> mirror ingress <value>
×	not yet documented	interfaces_dummy.xml.in: interfaces dummy <dummy> mtu	interfaces dummy <text> mtu <value>
×	not yet documented	interfaces_dummy.xml.in: interfaces dummy <dummy> netns	interfaces dummy <text> netns <value>
×	not yet documented	interfaces_dummy.xml.in: interfaces dummy <dummy> redirect	interfaces dummy <text> redirect <value>
✓	<code>set interfaces dummy <interface> vrf <vrf></code>	interfaces_dummy.xml.in: interfaces dummy <dummy> vrf	interfaces dummy <text> vrf <value>
×	<code>set interfaces ethernet <interface> duplex <auto full half></code>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<i>set interfaces ethernet <interface> interrupt-coalescing adaptive-rx</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set interfaces ethernet <interface> interrupt-coalescing adaptive-tx</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set interfaces ethernet <interface> interrupt-coalescing cqe-mode-rx</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set interfaces ethernet <interface> interrupt-coalescing cqe-mode-tx</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set interfaces ethernet <interface> interrupt-coalescing pkt-rate-high <number></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set interfaces ethernet <interface> interrupt-coalescing pkt-rate-low <number></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set interfaces ethernet <interface> interrupt-coalescing rx-frame-high <number></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set interfaces ethernet <interface> interrupt-coalescing rx-frame-low <number></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set interfaces ethernet <interface> interrupt-coalescing rx-frames-irq <number></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set interfaces ethernet <interface> interrupt-coalescing rx-frames <number></i>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<i>set interfaces ethernet <interface> interrupt-coalescing rx-usecs-high <number></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set interfaces ethernet <interface> interrupt-coalescing rx-usecs-irq <number></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set interfaces ethernet <interface> interrupt-coalescing rx-usecs-low <number></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set interfaces ethernet <interface> interrupt-coalescing rx-usecs <0-16384></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set interfaces ethernet <interface> interrupt-coalescing sample-interval <number></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set interfaces ethernet <interface> interrupt-coalescing stats-block-usecs <number></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set interfaces ethernet <interface> interrupt-coalescing tx-aggr-max-bytes <number></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set interfaces ethernet <interface> interrupt-coalescing tx-aggr-max-frames <number></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set interfaces ethernet <interface> interrupt-coalescing tx-aggr-time-usecs <number></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set interfaces ethernet <interface> interrupt-coalescing tx-frame-high <number></i>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set interfaces ethernet <interface> interrupt-coalescing tx-frame-low <number></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set interfaces ethernet <interface> interrupt-coalescing tx-frames-irq <number></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set interfaces ethernet <interface> interrupt-coalescing tx-frames <number></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set interfaces ethernet <interface> interrupt-coalescing tx-usecs-high <number></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set interfaces ethernet <interface> interrupt-coalescing tx-usecs-irq <number></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set interfaces ethernet <interface> interrupt-coalescing tx-usecs-low <number></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set interfaces ethernet <interface> interrupt-coalescing tx-usecs <0-16384></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set interfaces ethernet <interface> offload <lro tso gso gro rps sg></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set interfaces ethernet <interface> ring-buffer rx <value></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set interfaces ethernet <interface> ring-buffer tx <value></code>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set interfaces ethernet <interface> speed <auto 10 100 1000 2500 5000 10000 25000 40000 50000 100000></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set interfaces ethernet <interface> switchdev</code>	Nothing found in XML Definitions	Nothing found in VyOS
✓	<code>set interfaces ethernet <interface> address <address dhcp dhcpv6></code>	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> address	interfaces ethernet <text> address <value>
✓	<code>set interfaces ethernet <interface> description <description></code>	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> description	interfaces ethernet <text> description <value>
✓	<code>set interfaces ethernet <interface> dhcp-options client-id <description></code>	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> dhcp-options client-id	interfaces ethernet <text> dhcp-options client-id <value>
✓	<code>set interfaces ethernet <interface> dhcp-options default-route-distance <distance></code>	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> dhcp-options default-route-distance	interfaces ethernet <text> dhcp-options default-route-distance <value>
✓	<code>set interfaces ethernet <interface> dhcp-options host-name <hostname></code>	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> dhcp-options host-name	interfaces ethernet <text> dhcp-options host-name <value>
×	not yet documented	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> dhcp-options mtu	interfaces ethernet <text> dhcp-options mtu
✓	<code>set interfaces ethernet <interface> dhcp-options no-default-route</code>	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> dhcp-options no-default-route	interfaces ethernet <text> dhcp-options no-default-route
✓	<code>set interfaces ethernet <interface> dhcp-options reject <address></code>	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> dhcp-options reject	interfaces ethernet <text> dhcp-options reject <value>

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Table 1 – continued from previous page

Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<i>set interfaces ethernet <interface> dhcp-options user-class <string></i>	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> dhcp-options user-class	Nothing found in VyOS
✓	<i>set interfaces ethernet <interface> dhcp-options vendor-class-id <vendor-id></i>	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> dhcp-options vendor-class-id	interfaces ethernet <text> dhcp-options vendor-class-id <value>
✓	<i>set interfaces ethernet <interface> dhcpv6-options duid <duid></i>	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> dhcpv6-options duid	interfaces ethernet <text> dhcpv6-options duid <value>
✓	<i>set interfaces ethernet <interface> dhcpv6-options no-release</i>	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> dhcpv6-options no-release	interfaces ethernet <text> dhcpv6-options no-release
✓	<i>set interfaces ethernet <interface> dhcpv6-options parameters-only</i>	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> dhcpv6-options parameters-only	interfaces ethernet <text> dhcpv6-options parameters-only
✓	<i>set interfaces ethernet <interface> dhcpv6-options pd <id> interface <delegatee> address <address></i>	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> dhcpv6-options pd <pd> interface <interface> address	interfaces ethernet <text> dhcpv6-options pd <text> interface <text> address <value>
✓	<i>set interfaces ethernet <interface> dhcpv6-options pd <id> interface <delegatee> sla-id <id></i>	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> dhcpv6-options pd <pd> interface <interface> sla-id	interfaces ethernet <text> dhcpv6-options pd <text> interface <text> sla-id <value>
✓	<i>set interfaces ethernet <interface> dhcpv6-options pd <id> length <length></i>	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> dhcpv6-options pd <pd> length	interfaces ethernet <text> dhcpv6-options pd <text> length <value>
✓	<i>set interfaces ethernet <interface> dhcpv6-options rapid-commit</i>	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> dhcpv6-options rapid-commit	interfaces ethernet <text> dhcpv6-options rapid-commit

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Table 1 – continued from previous page

Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
✓	<i>set interfaces ethernet <interface> dhcpv6-options temporary</i>	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> dhcpv6-options temporary	interfaces ethernet <text> dhcpv6-options temporary
✓	<i>set interfaces ethernet <interface> disable</i>	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> disable	interfaces ethernet <text> disable
✓	<i>set interfaces ethernet <interface> disable-flow-control</i>	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> disable-flow-control	interfaces ethernet <text> disable-flow-control
✓	<i>set interfaces ethernet <interface> disable-link-detect</i>	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> disable-link-detect	interfaces ethernet <text> disable-link-detect
×	not yet documented	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> duplex	interfaces ethernet <text> duplex <value>
✓	<i>set interfaces ethernet <interface> eapol ca-certificate <name></i>	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> eapol ca-certificate	interfaces ethernet <text> eapol ca-certificate <value>
✓	<i>set interfaces ethernet <interface> eapol certificate <name></i>	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> eapol certificate	interfaces ethernet <text> eapol certificate <value>
×	not yet documented	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> eapol passphrase	interfaces ethernet <text> eapol passphrase <value>
×	<i>set interfaces ethernet <interface> evpn uplink</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> hw-id	interfaces ethernet <text> hw-id <value>
✓	<i>set interfaces ethernet <interface> ip adjust-mss <mss / clamp-mss-to-pmtu></i>	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> ip adjust-mss	interfaces ethernet <text> ip adjust-mss <value>
✓	<i>set interfaces ethernet <interface> ip arp-cache-timeout</i>	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> ip arp-cache-timeout	interfaces ethernet <text> ip arp-cache-timeout <value>

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Table 1 – continued from previous page

Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
✓	<i>set interfaces ethernet <interface> ip disable-arp-filter</i>	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> ip disable-arp-filter	interfaces ethernet <text> ip disable-arp-filter
✓	<i>set interfaces ethernet <interface> ip disable-forwarding</i>	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> ip disable-forwarding	interfaces ethernet <text> ip disable-forwarding
✓	<i>set interfaces ethernet <interface> ip enable-arp-accept</i>	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> ip enable-arp-accept	interfaces ethernet <text> ip enable-arp-accept
✓	<i>set interfaces ethernet <interface> ip enable-arp-announce</i>	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> ip enable-arp-announce	interfaces ethernet <text> ip enable-arp-announce
✓	<i>set interfaces ethernet <interface> ip enable-arp-ignore</i>	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> ip enable-arp-ignore	interfaces ethernet <text> ip enable-arp-ignore
✓	<i>set interfaces ethernet <interface> ip enable-directed-broadcast</i>	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> ip enable-directed-broadcast	interfaces ethernet <text> ip enable-directed-broadcast
✓	<i>set interfaces ethernet <interface> ip enable-proxy-arp</i>	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> ip enable-proxy-arp	interfaces ethernet <text> ip enable-proxy-arp
✓	<i>set interfaces ethernet <interface> ip proxy-arp-pvlan</i>	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> ip proxy-arp-pvlan	interfaces ethernet <text> ip proxy-arp-pvlan
✓	<i>set interfaces ethernet <interface> ip source-validation <strict loose disable></i>	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> ip source-validation	interfaces ethernet <text> ip source-validation <value>
✓	<i>set interfaces ethernet <interface> ipv6 accept-dad <0 1 2></i>	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> ipv6 accept-dad	interfaces ethernet <text> ipv6 accept-dad <value>
✓	<i>set interfaces ethernet <interface> ipv6 address autoconf</i>	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> ipv6 address autoconf	interfaces ethernet <text> ipv6 address autoconf

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Table 1 – continued from previous page

Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
✓	<i>set interfaces ethernet <interface> ipv6 address eui64 <prefix></i>	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> ipv6 address eui64	interfaces ethernet <text> ipv6 address eui64 <value>
✓	<i>set interfaces ethernet <interface> ipv6 address no-default-link-local</i>	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> ipv6 address no-default-link-local	interfaces ethernet <text> ipv6 address no-default-link-local
✓	<i>set interfaces ethernet <interface> ipv6 adjust-mss <mss / clamp-mss-to-pmtu></i>	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> ipv6 adjust-mss	interfaces ethernet <text> ipv6 adjust-mss <value>
×	not yet documented	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> ipv6 base-reachable-time	Nothing found in VyOS
✓	<i>set interfaces ethernet <interface> ipv6 disable-forwarding</i>	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> ipv6 disable-forwarding	interfaces ethernet <text> ipv6 disable-forwarding
✓	<i>set interfaces ethernet <interface> ipv6 dup-addr-detect-trans.<n></i>	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> ipv6 dup-addr-detect-transmits	interfaces ethernet <text> ipv6 dup-addr-detect-transmits <value>
×	not yet documented	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> ipv6 source-validation	Nothing found in VyOS
✓	<i>set interfaces ethernet <interface> mac <xx:xx:xx:xx:xx:xx></i>	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> mac	interfaces ethernet <text> mac <value>
✓	<i>set interfaces ethernet <interface> mirror egress <monitor-interface></i>	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> mirror egress	interfaces ethernet <text> mirror egress <value>
✓	<i>set interfaces ethernet <interface> mirror ingress <monitor-interface></i>	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> mirror ingress	interfaces ethernet <text> mirror ingress <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
✓	<code>set interfaces ethernet <interface> mtu <mtu></code>	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> mtu	interfaces ethernet <text> mtu <value>
×	not yet documented	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> offload gro	interfaces ethernet <text> offload gro
×	not yet documented	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> offload gso	interfaces ethernet <text> offload gso
×	not yet documented	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> offload hw-tc-offload	Nothing found in VyOS
×	not yet documented	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> offload lro	interfaces ethernet <text> offload lro
×	not yet documented	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> offload rfs	interfaces ethernet <text> offload rfs
×	not yet documented	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> offload rps	interfaces ethernet <text> offload rps
×	not yet documented	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> offload sg	interfaces ethernet <text> offload sg
×	not yet documented	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> offload tso	interfaces ethernet <text> offload tso
×	not yet documented	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> redirect	interfaces ethernet <text> redirect <value>
×	not yet documented	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> ring-buffer rx	interfaces ethernet <text> ring-buffer rx <value>
×	not yet documented	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> ring-buffer tx	interfaces ethernet <text> ring-buffer tx <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	inter- faces_ethernet.xml.in: interfaces ethernet <ether- net> speed	interfaces ethernet <text> speed <value>
×	<i>set interfaces ethernet <interface> vif <vlan-id></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	inter- faces_ethernet.xml.in: interfaces ethernet <ether- net> vif-s <vif-s> address	interfaces ethernet <text> vif-s <text> address <value>
×	not yet documented	inter- faces_ethernet.xml.in: interfaces ethernet <ether- net> vif-s <vif-s> descrip- tion	interfaces ethernet <text> vif-s <text> description <value>
×	not yet documented	inter- faces_ethernet.xml.in: interfaces ethernet <ether- net> vif-s <vif-s> dhcp- options client-id	interfaces ethernet <text> vif-s <text> dhcp-options client-id <value>
×	not yet documented	inter- faces_ethernet.xml.in: interfaces ethernet <eth- ernet> vif-s <vif-s> dhcp-options default- route-distance	interfaces ethernet <text> vif-s <text> dhcp-options default-route-distance <value>
×	not yet documented	inter- faces_ethernet.xml.in: interfaces ethernet <ether- net> vif-s <vif-s> dhcp- options host-name	interfaces ethernet <text> vif-s <text> dhcp-options host-name <value>
×	not yet documented	inter- faces_ethernet.xml.in: interfaces ethernet <ether- net> vif-s <vif-s> dhcp- options mtu	interfaces ethernet <text> vif-s <text> dhcp-options mtu
×	not yet documented	inter- faces_ethernet.xml.in: interfaces ethernet <ether- net> vif-s <vif-s> dhcp- options no-default-route	interfaces ethernet <text> vif-s <text> dhcp-options no-default-route
×	not yet documented	inter- faces_ethernet.xml.in: interfaces ethernet <ether- net> vif-s <vif-s> dhcp- options reject	interfaces ethernet <text> vif-s <text> dhcp-options reject <value>

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Table 1 – continued from previous page

Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> vif-s <vif-s> dhcp-options user-class	Nothing found in VyOS
×	not yet documented	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> vif-s <vif-s> dhcp-options vendor-class-id	interfaces ethernet <text> vif-s <text> dhcp-options vendor-class-id <value>
×	not yet documented	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> vif-s <vif-s> dhcpv6-options duid	interfaces ethernet <text> vif-s <text> dhcpv6-options duid <value>
×	not yet documented	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> vif-s <vif-s> dhcpv6-options no-release	interfaces ethernet <text> vif-s <text> dhcpv6-options no-release
×	not yet documented	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> vif-s <vif-s> dhcpv6-options parameters-only	interfaces ethernet <text> vif-s <text> dhcpv6-options parameters-only
×	not yet documented	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> vif-s <vif-s> dhcpv6-options pd <pd> interface <interface> address	interfaces ethernet <text> vif-s <text> dhcpv6-options pd <text> interface <text> address <value>
×	not yet documented	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> vif-s <vif-s> dhcpv6-options pd <pd> interface <interface> sla-id	interfaces ethernet <text> vif-s <text> dhcpv6-options pd <text> interface <text> sla-id <value>
×	not yet documented	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> vif-s <vif-s> dhcpv6-options pd <pd> length	interfaces ethernet <text> vif-s <text> dhcpv6-options pd <text> length <value>
×	not yet documented	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> vif-s <vif-s> dhcpv6-options rapid-commit	interfaces ethernet <text> vif-s <text> dhcpv6-options rapid-commit

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> vif-s <vif-s> dhcpv6-options temporary	interfaces ethernet <text> vif-s <text> dhcpv6-options temporary
×	not yet documented	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> vif-s <vif-s> disable	interfaces ethernet <text> vif-s <text> disable
×	not yet documented	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> vif-s <vif-s> disable-link-detect	interfaces ethernet <text> vif-s <text> disable-link-detect
×	not yet documented	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> vif-s <vif-s> ip adjust-mss	interfaces ethernet <text> vif-s <text> ip adjust-mss <value>
×	not yet documented	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> vif-s <vif-s> ip arp-cache-timeout	interfaces ethernet <text> vif-s <text> ip arp-cache-timeout <value>
×	not yet documented	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> vif-s <vif-s> ip disable-arp-filter	interfaces ethernet <text> vif-s <text> ip disable-arp-filter
×	not yet documented	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> vif-s <vif-s> ip disable-forwarding	interfaces ethernet <text> vif-s <text> ip disable-forwarding
×	not yet documented	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> vif-s <vif-s> ip enable-arp-accept	interfaces ethernet <text> vif-s <text> ip enable-arp-accept
×	not yet documented	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> vif-s <vif-s> ip enable-arp-announce	interfaces ethernet <text> vif-s <text> ip enable-arp-announce
×	not yet documented	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> vif-s <vif-s> ip enable-arp-ignore	interfaces ethernet <text> vif-s <text> ip enable-arp-ignore

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Table 1 – continued from previous page

Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> vif-s <vif-s> ip enable-directed-broadcast	interfaces ethernet <text> vif-s <text> ip enable-directed-broadcast
×	not yet documented	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> vif-s <vif-s> ip enable-proxy-arp	interfaces ethernet <text> vif-s <text> ip enable-proxy-arp
×	not yet documented	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> vif-s <vif-s> ip proxy-arp-pvlan	interfaces ethernet <text> vif-s <text> ip proxy-arp-pvlan
×	not yet documented	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> vif-s <vif-s> ip source-validation	interfaces ethernet <text> vif-s <text> ip source-validation <value>
×	not yet documented	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> vif-s <vif-s> ipv6 accept-dad	interfaces ethernet <text> vif-s <text> ipv6 accept-dad <value>
×	not yet documented	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> vif-s <vif-s> ipv6 address autoconf	interfaces ethernet <text> vif-s <text> ipv6 address autoconf
×	not yet documented	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> vif-s <vif-s> ipv6 address eui64	interfaces ethernet <text> vif-s <text> ipv6 address eui64 <value>
×	not yet documented	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> vif-s <vif-s> ipv6 address no-default-link-local	interfaces ethernet <text> vif-s <text> ipv6 address no-default-link-local
×	not yet documented	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> vif-s <vif-s> ipv6 adjust-mss	interfaces ethernet <text> vif-s <text> ipv6 adjust-mss <value>

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Table 1 – continued from previous page

Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> vif-s <vif-s> ipv6 base-reachable-time	Nothing found in VyOS
×	not yet documented	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> vif-s <vif-s> ipv6 disable-forwarding	interfaces ethernet <text> vif-s <text> ipv6 disable-forwarding
×	not yet documented	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> vif-s <vif-s> ipv6 dup-addr-detect-transmits	interfaces ethernet <text> vif-s <text> ipv6 dup-addr-detect-transmits <value>
×	not yet documented	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> vif-s <vif-s> ipv6 source-validation	Nothing found in VyOS
×	not yet documented	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> vif-s <vif-s> mac	interfaces ethernet <text> vif-s <text> mac <value>
×	not yet documented	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> vif-s <vif-s> mirror egress	interfaces ethernet <text> vif-s <text> mirror egress <value>
×	not yet documented	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> vif-s <vif-s> mirror ingress	interfaces ethernet <text> vif-s <text> mirror ingress <value>
×	not yet documented	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> vif-s <vif-s> mtu	interfaces ethernet <text> vif-s <text> mtu <value>
×	not yet documented	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> vif-s <vif-s> protocol	interfaces ethernet <text> vif-s <text> protocol <value>
×	not yet documented	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> vif-s <vif-s> redirect	interfaces ethernet <text> vif-s <text> redirect <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
✓	<code>set interfaces ethernet <interface> vif-s <vlan-id> vif-c <vlan-id> address <address / dhcp / dhcpv6></code>	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> vif-s <vif-s> vif-c <vif-c> address	interfaces ethernet <text> vif-s <text> vif-c <text> address <value>
✓	<code>set interfaces ethernet <interface> vif-s <vlan-id> vif-c <vlan-id> description <description></code>	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> vif-s <vif-s> vif-c <vif-c> description	interfaces ethernet <text> vif-s <text> vif-c <text> description <value>
✓	<code>set interfaces ethernet <interface> vif-s <vlan-id> vif-c <vlan-id> dhcp-options client-id <description></code>	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> vif-s <vif-s> vif-c <vif-c> dhcp-options client-id	interfaces ethernet <text> vif-s <text> vif-c <text> dhcp-options client-id <value>
✓	<code>set interfaces ethernet <interface> vif-s <vlan-id> vif-c <vlan-id> dhcp-options default-route-distance <distance></code>	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> vif-s <vif-s> vif-c <vif-c> dhcp-options default-route-distance	interfaces ethernet <text> vif-s <text> vif-c <text> dhcp-options default-route-distance <value>
✓	<code>set interfaces ethernet <interface> vif-s <vlan-id> vif-c <vlan-id> dhcp-options host-name <hostname></code>	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> vif-s <vif-s> vif-c <vif-c> dhcp-options host-name	interfaces ethernet <text> vif-s <text> vif-c <text> dhcp-options host-name <value>
×	not yet documented	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> vif-s <vif-s> vif-c <vif-c> dhcp-options mtu	interfaces ethernet <text> vif-s <text> vif-c <text> dhcp-options mtu
✓	<code>set interfaces ethernet <interface> vif-s <vlan-id> vif-c <vlan-id> dhcp-options no-default-route</code>	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> vif-s <vif-s> vif-c <vif-c> dhcp-options no-default-route	interfaces ethernet <text> vif-s <text> vif-c <text> dhcp-options no-default-route

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
✓	<code>set interfaces ethernet <interface> vif-s <vlan-id> vif-c <vlan-id> dhcp-options reject <address></code>	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> vif-s <vif-s> vif-c <vif-c> dhcp-options reject	interfaces ethernet <text> vif-s <text> vif-c <text> dhcp-options reject <value>
×	<code>set interfaces ethernet <interface> vif-s <vlan-id> vif-c <vlan-id> dhcp-options user-class <string></code>	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> vif-s <vif-s> vif-c <vif-c> dhcp-options user-class	Nothing found in VyOS
✓	<code>set interfaces ethernet <interface> vif-s <vlan-id> vif-c <vlan-id> dhcp-options vendor-class-id <vendor-id></code>	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> vif-s <vif-s> vif-c <vif-c> dhcp-options vendor-class-id	interfaces ethernet <text> vif-s <text> vif-c <text> dhcp-options vendor-class-id <value>
✓	<code>set interfaces ethernet <interface> vif-s <vlan-id> vif-c <vlan-id> dhcpv6-options duid <duid></code>	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> vif-s <vif-s> vif-c <vif-c> dhcpv6-options duid	interfaces ethernet <text> vif-s <text> vif-c <text> dhcpv6-options duid <value>
✓	<code>set interfaces ethernet <interface> vif-s <vlan-id> vif-c <vlan-id> dhcpv6-options no-release</code>	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> vif-s <vif-s> vif-c <vif-c> dhcpv6-options no-release	interfaces ethernet <text> vif-s <text> vif-c <text> dhcpv6-options no-release
✓	<code>set interfaces ethernet <interface> vif-s <vlan-id> vif-c <vlan-id> dhcpv6-options parameters-only</code>	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> vif-s <vif-s> vif-c <vif-c> dhcpv6-options parameters-only	interfaces ethernet <text> vif-s <text> vif-c <text> dhcpv6-options parameters-only
✓	<code>set interfaces ethernet <interface> vif-s <vlan-id> vif-c <vlan-id> dhcpv6-options pd <id> interface <delegatee> address <address></code>	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> vif-s <vif-s> vif-c <vif-c> dhcpv6-options pd <pd> interface <interface> address	interfaces ethernet <text> vif-s <text> vif-c <text> dhcpv6-options pd <text> interface <text> address <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
✓	<code>set interfaces ethernet <interface> vif-s <vlan-id> vif-c <vlan-id> dhcpv6-options pd <id> interface <delegatee> sla-id <id></code>	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> vif-s <vif-s> vif-c <vif-c> dhcpv6-options pd <pd> interface <interface> sla-id	interfaces ethernet <text> vif-s <text> vif-c <text> dhcpv6-options pd <text> interface <text> sla-id <value>
✓	<code>set interfaces ethernet <interface> vif-s <vlan-id> vif-c <vlan-id> dhcpv6-options pd <id> length <length></code>	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> vif-s <vif-s> vif-c <vif-c> dhcpv6-options pd <pd> length	interfaces ethernet <text> vif-s <text> vif-c <text> dhcpv6-options pd <text> length <value>
✓	<code>set interfaces ethernet <interface> vif-s <vlan-id> vif-c <vlan-id> dhcpv6-options rapid-commit</code>	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> vif-s <vif-s> vif-c <vif-c> dhcpv6-options rapid-commit	interfaces ethernet <text> vif-s <text> vif-c <text> dhcpv6-options rapid-commit
✓	<code>set interfaces ethernet <interface> vif-s <vlan-id> vif-c <vlan-id> dhcpv6-options temporary</code>	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> vif-s <vif-s> vif-c <vif-c> dhcpv6-options temporary	interfaces ethernet <text> vif-s <text> vif-c <text> dhcpv6-options temporary
✓	<code>set interfaces ethernet <interface> vif-s <vlan-id> vif-c <vlan-id> disable</code>	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> vif-s <vif-s> vif-c <vif-c> disable	interfaces ethernet <text> vif-s <text> vif-c <text> disable
✓	<code>set interfaces ethernet <interface> vif-s <vlan-id> vif-c <vlan-id> disable-link-detect</code>	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> vif-s <vif-s> vif-c <vif-c> disable-link-detect	interfaces ethernet <text> vif-s <text> vif-c <text> disable-link-detect
✓	<code>set interfaces ethernet <interface> vif-s <vlan-id> vif-c <vlan-id> ip adjust-mss <mss clamp-mss-to-pmtu></code>	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> vif-s <vif-s> vif-c <vif-c> ip adjust-mss	interfaces ethernet <text> vif-s <text> vif-c <text> ip adjust-mss <value>

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Table 1 – continued from previous page

Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
✓	<i>set interfaces ethernet <interface> vif-s <vlan-id> vif-c <vlan-id> ip arp-cache-timeout</i>	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> vif-s <vif-s> vif-c <vif-c> ip arp-cache-timeout	interfaces ethernet <text> vif-s <text> vif-c <text> ip arp-cache-timeout <value>
✓	<i>set interfaces ethernet <interface> vif-s <vlan-id> vif-c <vlan-id> ip disable-arp-filter</i>	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> vif-s <vif-s> vif-c <vif-c> ip disable-arp-filter	interfaces ethernet <text> vif-s <text> vif-c <text> ip disable-arp-filter
✓	<i>set interfaces ethernet <interface> vif-s <vlan-id> vif-c <vlan-id> ip disable-forwarding</i>	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> vif-s <vif-s> vif-c <vif-c> ip disable-forwarding	interfaces ethernet <text> vif-s <text> vif-c <text> ip disable-forwarding
✓	<i>set interfaces ethernet <interface> vif-s <vlan-id> vif-c <vlan-id> ip enable-arp-accept</i>	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> vif-s <vif-s> vif-c <vif-c> ip enable-arp-accept	interfaces ethernet <text> vif-s <text> vif-c <text> ip enable-arp-accept
✓	<i>set interfaces ethernet <interface> vif-s <vlan-id> vif-c <vlan-id> ip enable-arp-announce</i>	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> vif-s <vif-s> vif-c <vif-c> ip enable-arp-announce	interfaces ethernet <text> vif-s <text> vif-c <text> ip enable-arp-announce
✓	<i>set interfaces ethernet <interface> vif-s <vlan-id> vif-c <vlan-id> ip enable-arp-ignore</i>	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> vif-s <vif-s> vif-c <vif-c> ip enable-arp-ignore	interfaces ethernet <text> vif-s <text> vif-c <text> ip enable-arp-ignore
✓	<i>set interfaces ethernet <interface> vif-s <vlan-id> vif-c <vlan-id> ip enable-directed-broadcast</i>	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> vif-s <vif-s> vif-c <vif-c> ip enable-directed-broadcast	interfaces ethernet <text> vif-s <text> vif-c <text> ip enable-directed-broadcast
✓	<i>set interfaces ethernet <interface> vif-s <vlan-id> vif-c <vlan-id> ip enable-proxy-arp</i>	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> vif-s <vif-s> vif-c <vif-c> ip enable-proxy-arp	interfaces ethernet <text> vif-s <text> vif-c <text> ip enable-proxy-arp

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
✓	<code>set interfaces ethernet <interface> vif-s <vlan-id> vif-c <vlan-id> ip proxy-arp-pvlan</code>	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> vif-s <vif-s> vif-c <vif-c> ip proxy-arp-pvlan	interfaces ethernet <text> vif-s <text> vif-c <text> ip proxy-arp-pvlan
✓	<code>set interfaces ethernet <interface> vif-s <vlan-id> vif-c <vlan-id> ip source-validation <strict loose disable></code>	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> vif-s <vif-s> vif-c <vif-c> ip source-validation	interfaces ethernet <text> vif-s <text> vif-c <text> ip source-validation <value>
✓	<code>set interfaces ethernet <interface> vif-s <vlan-id> vif-c <vlan-id> ipv6 accept-dad <0 1 2></code>	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> vif-s <vif-s> vif-c <vif-c> ipv6 accept-dad	interfaces ethernet <text> vif-s <text> vif-c <text> ipv6 accept-dad <value>
✓	<code>set interfaces ethernet <interface> vif-s <vlan-id> vif-c <vlan-id> ipv6 address autoconf</code>	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> vif-s <vif-s> vif-c <vif-c> ipv6 address autoconf	interfaces ethernet <text> vif-s <text> vif-c <text> ipv6 address autoconf
✓	<code>set interfaces ethernet <interface> vif-s <vlan-id> vif-c <vlan-id> ipv6 address eui64 <prefix></code>	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> vif-s <vif-s> vif-c <vif-c> ipv6 address eui64	interfaces ethernet <text> vif-s <text> vif-c <text> ipv6 address eui64 <value>
✓	<code>set interfaces ethernet <interface> vif-s <vlan-id> vif-c <vlan-id> ipv6 address no-default-link-local</code>	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> vif-s <vif-s> vif-c <vif-c> ipv6 address no-default-link-local	interfaces ethernet <text> vif-s <text> vif-c <text> ipv6 address no-default-link-local
✓	<code>set interfaces ethernet <interface> vif-s <vlan-id> vif-c <vlan-id> ipv6 adjust-mss <mss clamp-mss-to-pmtu></code>	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> vif-s <vif-s> vif-c <vif-c> ipv6 adjust-mss	interfaces ethernet <text> vif-s <text> vif-c <text> ipv6 adjust-mss <value>
×	not yet documented	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> vif-s <vif-s> vif-c <vif-c> ipv6 base-reachable-time	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
✓	<i>set interfaces ethernet <interface> vif-s <vlan-id> vif-c <vlan-id> ipv6 disable-forwarding</i>	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> vif-s <vif-s> vif-c <vif-c> ipv6 disable-forwarding	interfaces ethernet <text> vif-s <text> vif-c <text> ipv6 disable-forwarding
✓	<i>set interfaces ethernet <interface> vif-s <vlan-id> vif-c <vlan-id> ipv6 dup-addr-detect-transmits <n></i>	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> vif-s <vif-s> vif-c <vif-c> ipv6 dup-addr-detect-transmits	interfaces ethernet <text> vif-s <text> vif-c <text> ipv6 dup-addr-detect-transmits <value>
×	not yet documented	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> vif-s <vif-s> vif-c <vif-c> ipv6 source-validation	Nothing found in VyOS
✓	<i>set interfaces ethernet <interface> vif-s <vlan-id> vif-c <vlan-id> mac <xx:xx:xx:xx:xx:xx></i>	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> vif-s <vif-s> vif-c <vif-c> mac	interfaces ethernet <text> vif-s <text> vif-c <text> mac <value>
×	not yet documented	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> vif-s <vif-s> vif-c <vif-c> mirror egress	interfaces ethernet <text> vif-s <text> vif-c <text> mirror egress <value>
×	not yet documented	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> vif-s <vif-s> vif-c <vif-c> mirror ingress	interfaces ethernet <text> vif-s <text> vif-c <text> mirror ingress <value>
✓	<i>set interfaces ethernet <interface> vif-s <vlan-id> vif-c <vlan-id> mtu <mtu></i>	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> vif-s <vif-s> vif-c <vif-c> mtu	interfaces ethernet <text> vif-s <text> vif-c <text> mtu <value>
×	not yet documented	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> vif-s <vif-s> vif-c <vif-c> redirect	interfaces ethernet <text> vif-s <text> vif-c <text> redirect <value>
✓	<i>set interfaces ethernet <interface> vif-s <vlan-id> vif-c <vlan-id> vrf <vrf></i>	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> vif-s <vif-s> vif-c <vif-c> vrf	interfaces ethernet <text> vif-s <text> vif-c <text> vrf <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> vif-s <vif-s> vrf	interfaces ethernet <text> vif-s <text> vrf <value>
✓	<code>set interfaces ethernet <interface> vif <vlan-id> address <address> dhcp dhcpv6></code>	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> vif <vif> address	interfaces ethernet <text> vif <text> address <value>
✓	<code>set interfaces ethernet <interface> vif <vlan-id> description <description></code>	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> vif <vif> description	interfaces ethernet <text> vif <text> description <value>
✓	<code>set interfaces ethernet <interface> vif <vlan-id> dhcp-options client-id <description></code>	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> vif <vif> dhcp-options client-id	interfaces ethernet <text> vif <text> dhcp-options client-id <value>
✓	<code>set interfaces ethernet <interface> vif <vlan-id> dhcp-options default-route-distance <distance></code>	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> vif <vif> dhcp-options default-route-distance	interfaces ethernet <text> vif <text> dhcp-options default-route-distance <value>
✓	<code>set interfaces ethernet <interface> vif <vlan-id> dhcp-options host-name <hostname></code>	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> vif <vif> dhcp-options host-name	interfaces ethernet <text> vif <text> dhcp-options host-name <value>
×	not yet documented	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> vif <vif> dhcp-options mtu	interfaces ethernet <text> vif <text> dhcp-options mtu
✓	<code>set interfaces ethernet <interface> vif <vlan-id> dhcp-options no-default-route</code>	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> vif <vif> dhcp-options no-default-route	interfaces ethernet <text> vif <text> dhcp-options no-default-route
✓	<code>set interfaces ethernet <interface> vif <vlan-id> dhcp-options reject <address></code>	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> vif <vif> dhcp-options reject	interfaces ethernet <text> vif <text> dhcp-options reject <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set interfaces ethernet <interface> vif <vlan-id> dhcp-options user-class <string></code>	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> vif <vif> dhcp-options user-class	Nothing found in VyOS
✓	<code>set interfaces ethernet <interface> vif <vlan-id> dhcp-options vendor-class-id <vendor-id></code>	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> vif <vif> dhcp-options vendor-class-id	interfaces ethernet <text> vif <text> dhcp-options vendor-class-id <value>
✓	<code>set interfaces ethernet <interface> vif <vlan-id> dhcpv6-options duid <duid></code>	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> vif <vif> dhcpv6-options duid	interfaces ethernet <text> vif <text> dhcpv6-options duid <value>
✓	<code>set interfaces ethernet <interface> vif <vlan-id> dhcpv6-options no-release</code>	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> vif <vif> dhcpv6-options no-release	interfaces ethernet <text> vif <text> dhcpv6-options no-release
✓	<code>set interfaces ethernet <interface> vif <vlan-id> dhcpv6-options parameters-only</code>	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> vif <vif> dhcpv6-options parameters-only	interfaces ethernet <text> vif <text> dhcpv6-options parameters-only
✓	<code>set interfaces ethernet <interface> vif <vlan-id> dhcpv6-options pd <id> interface <delegatee> address <address></code>	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> vif <vif> dhcpv6-options pd <pd> interface <interface> address	interfaces ethernet <text> vif <text> dhcpv6-options pd <text> interface <text> address <value>
✓	<code>set interfaces ethernet <interface> vif <vlan-id> dhcpv6-options pd <id> interface <delegatee> sla-id <id></code>	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> vif <vif> dhcpv6-options pd <pd> interface <interface> sla-id	interfaces ethernet <text> vif <text> dhcpv6-options pd <text> interface <text> sla-id <value>
✓	<code>set interfaces ethernet <interface> vif <vlan-id> dhcpv6-options pd <id> length <length></code>	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> vif <vif> dhcpv6-options pd <pd> length	interfaces ethernet <text> vif <text> dhcpv6-options pd <text> length <value>

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Table 1 – continued from previous page

Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
✓	<i>set interfaces ethernet <interface> vif <vlan-id> dhcpv6-options rapid-commit</i>	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> vif <vif> dhcpv6-options rapid-commit	interfaces ethernet <text> vif <text> dhcpv6-options rapid-commit
✓	<i>set interfaces ethernet <interface> vif <vlan-id> dhcpv6-options temporary</i>	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> vif <vif> dhcpv6-options temporary	interfaces ethernet <text> vif <text> dhcpv6-options temporary
✓	<i>set interfaces ethernet <interface> vif <vlan-id> disable</i>	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> vif <vif> disable	interfaces ethernet <text> vif <text> disable
✓	<i>set interfaces ethernet <interface> vif <vlan-id> disable-link-detect</i>	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> vif <vif> disable-link-detect	interfaces ethernet <text> vif <text> disable-link-detect
×	not yet documented	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> vif <vif> egress-qos	interfaces ethernet <text> vif <text> egress-qos <value>
×	not yet documented	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> vif <vif> ingress-qos	interfaces ethernet <text> vif <text> ingress-qos <value>
✓	<i>set interfaces ethernet <interface> vif <vlan-id> ip adjust-mss <mss / clamp-mss-to-pmtu></i>	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> vif <vif> ip adjust-mss	interfaces ethernet <text> vif <text> ip adjust-mss <value>
✓	<i>set interfaces ethernet <interface> vif <vlan-id> ip arp-cache-timeout</i>	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> vif <vif> ip arp-cache-timeout	interfaces ethernet <text> vif <text> ip arp-cache-timeout <value>
✓	<i>set interfaces ethernet <interface> vif <vlan-id> ip disable-arp-filter</i>	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> vif <vif> ip disable-arp-filter	interfaces ethernet <text> vif <text> ip disable-arp-filter
✓	<i>set interfaces ethernet <interface> vif <vlan-id> ip disable-forwarding</i>	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> vif <vif> ip disable-forwarding	interfaces ethernet <text> vif <text> ip disable-forwarding

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Table 1 – continued from previous page

Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
✓	<code>set interfaces ethernet <interface> vif <vlan-id> ip enable-arp-accept</code>	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> vif <vif> ip enable-arp-accept	interfaces ethernet <text> vif <text> ip enable-arp-accept
✓	<code>set interfaces ethernet <interface> vif <vlan-id> ip enable-arp-announce</code>	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> vif <vif> ip enable-arp-announce	interfaces ethernet <text> vif <text> ip enable-arp-announce
✓	<code>set interfaces ethernet <interface> vif <vlan-id> ip enable-arp-ignore</code>	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> vif <vif> ip enable-arp-ignore	interfaces ethernet <text> vif <text> ip enable-arp-ignore
✓	<code>set interfaces ethernet <interface> vif <vlan-id> ip enable-directed-broadcast</code>	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> vif <vif> ip enable-directed-broadcast	interfaces ethernet <text> vif <text> ip enable-directed-broadcast
✓	<code>set interfaces ethernet <interface> vif <vlan-id> ip enable-proxy-arp</code>	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> vif <vif> ip enable-proxy-arp	interfaces ethernet <text> vif <text> ip enable-proxy-arp
✓	<code>set interfaces ethernet <interface> vif <vlan-id> ip proxy-arp-pvlan</code>	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> vif <vif> ip proxy-arp-pvlan	interfaces ethernet <text> vif <text> ip proxy-arp-pvlan
✓	<code>set interfaces ethernet <interface> vif <vlan-id> ip source-validation <strict loose disable></code>	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> vif <vif> ip source-validation	interfaces ethernet <text> vif <text> ip source-validation <value>
✓	<code>set interfaces ethernet <interface> vif <vlan-id> ipv6 accept-dad <0 1 2></code>	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> vif <vif> ipv6 accept-dad	interfaces ethernet <text> vif <text> ipv6 accept-dad <value>
✓	<code>set interfaces ethernet <interface> vif <vlan-id> ipv6 address autoconf</code>	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> vif <vif> ipv6 address autoconf	interfaces ethernet <text> vif <text> ipv6 address autoconf

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
✓	<i>set interfaces ethernet <interface> vif <vlan-id> ipv6 address eui64 <prefix></i>	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> vif <vif> ipv6 address eui64	interfaces ethernet <text> vif <text> ipv6 address eui64 <value>
✓	<i>set interfaces ethernet <interface> vif <vlan-id> ipv6 address no-default-link-local</i>	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> vif <vif> ipv6 address no-default-link-local	interfaces ethernet <text> vif <text> ipv6 address no-default-link-local
✓	<i>set interfaces ethernet <interface> vif <vlan-id> ipv6 adjust-mss <mss / clamp-mss-to-pmtu></i>	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> vif <vif> ipv6 adjust-mss	interfaces ethernet <text> vif <text> ipv6 adjust-mss <value>
×	not yet documented	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> vif <vif> ipv6 base-reachable-time	Nothing found in VyOS
✓	<i>set interfaces ethernet <interface> vif <vlan-id> ipv6 disable-forwarding</i>	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> vif <vif> ipv6 disable-forwarding	interfaces ethernet <text> vif <text> ipv6 disable-forwarding
✓	<i>set interfaces ethernet <interface> vif <vlan-id> ipv6 dup-addr-detect-trans <n></i>	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> vif <vif> ipv6 dup-addr-detect-transmits	interfaces ethernet <text> vif <text> ipv6 dup-addr-detect-transmits <value>
×	not yet documented	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> vif <vif> ipv6 source-validation	Nothing found in VyOS
✓	<i>set interfaces ethernet <interface> vif <vlan-id> mac <xx:xx:xx:xx:xx:xx></i>	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> vif <vif> mac	interfaces ethernet <text> vif <text> mac <value>
×	not yet documented	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> vif <vif> mirror egress	interfaces ethernet <text> vif <text> mirror egress <value>
×	not yet documented	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> vif <vif> mirror ingress	interfaces ethernet <text> vif <text> mirror ingress <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
✓	<i>set interfaces ethernet <interface> vif <vlan-id> mtu <mtu></i>	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> vif <vif> mtu	interfaces ethernet <text> vif <text> mtu <value>
×	not yet documented	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> vif <vif> redirect	interfaces ethernet <text> vif <text> redirect <value>
✓	<i>set interfaces ethernet <interface> vif <vlan-id> vrf <vrf></i>	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> vif <vif> vrf	interfaces ethernet <text> vif <text> vrf <value>
✓	<i>set interfaces ethernet <interface> vrf <vrf></i>	inter-faces_ethernet.xml.in: interfaces ethernet <ethernet> vrf	interfaces ethernet <text> vrf <value>
✓	<i>set interfaces geneve <interface> address <address></i>	interfaces_geneve.xml.in: interfaces geneve <geneve> address	interfaces geneve <text> address <value>
✓	<i>set interfaces geneve <interface> description <description></i>	interfaces_geneve.xml.in: interfaces geneve <geneve> description	interfaces geneve <text> description <value>
✓	<i>set interfaces geneve <interface> disable</i>	interfaces_geneve.xml.in: interfaces geneve <geneve> disable	interfaces geneve <text> disable
×	<i>set interfaces geneve gnv0 remote <address></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set interfaces geneve gnv0 vni <vni></i>	Nothing found in XML Definitions	Nothing found in VyOS
✓	<i>set interfaces geneve <interface> ip adjust-mss <mss / clamp-mss-to-pmtu></i>	interfaces_geneve.xml.in: interfaces geneve <geneve> ip adjust-mss	interfaces geneve <text> ip adjust-mss <value>
✓	<i>set interfaces geneve <interface> ip arp-cache-timeout</i>	interfaces_geneve.xml.in: interfaces geneve <geneve> ip arp-cache-timeout	interfaces geneve <text> ip arp-cache-timeout <value>
✓	<i>set interfaces geneve <interface> ip disable-arp-filter</i>	interfaces_geneve.xml.in: interfaces geneve <geneve> ip disable-arp-filter	interfaces geneve <text> ip disable-arp-filter
✓	<i>set interfaces geneve <interface> ip disable-forwarding</i>	interfaces_geneve.xml.in: interfaces geneve <geneve> ip disable-forwarding	interfaces geneve <text> ip disable-forwarding

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
✓	<code>set interfaces geneve <interface> ip enable-arp-accept</code>	interfaces_geneve.xml.in: interfaces geneve <geneve> ip enable-arp-accept	interfaces geneve <text> ip enable-arp-accept
✓	<code>set interfaces geneve <interface> ip enable-arp-announce</code>	interfaces_geneve.xml.in: interfaces geneve <geneve> ip enable-arp-announce	interfaces geneve <text> ip enable-arp-announce
✓	<code>set interfaces geneve <interface> ip enable-arp-ignore</code>	interfaces_geneve.xml.in: interfaces geneve <geneve> ip enable-arp-ignore	interfaces geneve <text> ip enable-arp-ignore
✓	<code>set interfaces geneve <interface> ip enable-directed-broadcast</code>	interfaces_geneve.xml.in: interfaces geneve <geneve> ip enable-directed-broadcast	interfaces geneve <text> ip enable-directed-broadcast
✓	<code>set interfaces geneve <interface> ip enable-proxy-arp</code>	interfaces_geneve.xml.in: interfaces geneve <geneve> ip enable-proxy-arp	interfaces geneve <text> ip enable-proxy-arp
✓	<code>set interfaces geneve <interface> ip proxy-arp-pvlan</code>	interfaces_geneve.xml.in: interfaces geneve <geneve> ip proxy-arp-pvlan	interfaces geneve <text> ip proxy-arp-pvlan
✓	<code>set interfaces geneve <interface> ip source-validation <strict loose disable></code>	interfaces_geneve.xml.in: interfaces geneve <geneve> ip source-validation	interfaces geneve <text> ip source-validation <value>
✓	<code>set interfaces geneve <interface> ipv6 accept-dad <0 1 2></code>	interfaces_geneve.xml.in: interfaces geneve <geneve> ipv6 accept-dad	interfaces geneve <text> ipv6 accept-dad <value>
✓	<code>set interfaces geneve <interface> ipv6 address autoconf</code>	interfaces_geneve.xml.in: interfaces geneve <geneve> ipv6 address autoconf	interfaces geneve <text> ipv6 address autoconf
✓	<code>set interfaces geneve <interface> ipv6 address eui64 <prefix></code>	interfaces_geneve.xml.in: interfaces geneve <geneve> ipv6 address eui64	interfaces geneve <text> ipv6 address eui64 <value>
✓	<code>set interfaces geneve <interface> ipv6 address no-default-link-local</code>	interfaces_geneve.xml.in: interfaces geneve <geneve> ipv6 address no-default-link-local	interfaces geneve <text> ipv6 address no-default-link-local
✓	<code>set interfaces geneve <interface> ipv6 adjust-mss <mss clamp-mss-to-pmtu></code>	interfaces_geneve.xml.in: interfaces geneve <geneve> ipv6 adjust-mss	interfaces geneve <text> ipv6 adjust-mss <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	interfaces_geneve.xml.in: interfaces geneve <geneve> ipv6 base-reachable-time	Nothing found in VyOS
✓	<code>set interfaces geneve <interface> ipv6 disable-forwarding</code>	interfaces_geneve.xml.in: interfaces geneve <geneve> ipv6 disable-forwarding	interfaces geneve <text> ipv6 disable-forwarding
✓	<code>set interfaces geneve <interface> ipv6 dup-addr-detect-trans <n></code>	interfaces_geneve.xml.in: interfaces geneve <geneve> ipv6 dup-addr-detect-transmits	interfaces geneve <text> ipv6 dup-addr-detect-transmits <value>
×	not yet documented	interfaces_geneve.xml.in: interfaces geneve <geneve> ipv6 source-validation	Nothing found in VyOS
✓	<code>set interfaces geneve <interface> mac <xx:xx:xx:xx:xx:xx></code>	interfaces_geneve.xml.in: interfaces geneve <geneve> mac	interfaces geneve <text> mac <value>
×	not yet documented	interfaces_geneve.xml.in: interfaces geneve <geneve> mirror egress	interfaces geneve <text> mirror egress <value>
×	not yet documented	interfaces_geneve.xml.in: interfaces geneve <geneve> mirror ingress	interfaces geneve <text> mirror ingress <value>
✓	<code>set interfaces geneve <interface> mtu <mtu></code>	interfaces_geneve.xml.in: interfaces geneve <geneve> mtu	interfaces geneve <text> mtu <value>
×	not yet documented	interfaces_geneve.xml.in: interfaces geneve <geneve> parameters ip df	interfaces geneve <text> parameters ip df <value>
×	not yet documented	interfaces_geneve.xml.in: interfaces geneve <geneve> parameters ip innerproto	interfaces geneve <text> parameters ip innerproto
×	not yet documented	interfaces_geneve.xml.in: interfaces geneve <geneve> parameters ip tos	interfaces geneve <text> parameters ip tos <value>
×	not yet documented	interfaces_geneve.xml.in: interfaces geneve <geneve> parameters ip ttl	interfaces geneve <text> parameters ip ttl <value>
×	not yet documented	interfaces_geneve.xml.in: interfaces geneve <geneve> parameters ipv6 flowlabel	interfaces geneve <text> parameters ipv6 flowlabel <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	interfaces_geneve.xml.in: interfaces geneve <geneve> redirect	interfaces geneve <text> redirect <value>
×	not yet documented	interfaces_geneve.xml.in: interfaces geneve <geneve> remote	interfaces geneve <text> remote <value>
×	not yet documented	interfaces_geneve.xml.in: interfaces geneve <geneve> vni	interfaces geneve <text> vni <value>
×	<i>set interfaces gnv0 <interface> port <port></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	interfaces_input.xml.in: interfaces input <input> description	interfaces input <text> de- scription <value>
×	not yet documented	interfaces_input.xml.in: interfaces input <input> disable	interfaces input <text> dis- able
×	not yet documented	interfaces_input.xml.in: interfaces input <input> redirect	interfaces input <text> redirect <value>
×	<i>set interfaces l2tpv3 <interface> encapsulation <udp / ip></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set interfaces l2tpv3 <interface> peer-session-id <id></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set interfaces l2tpv3 <interface> peer-tunnel-id <id></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set interfaces l2tpv3 <interface> remote <address></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set interfaces l2tpv3 <interface> session-id <id></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set interfaces l2tpv3 <interface> source-address <address></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set interfaces l2tpv3 <interface> tunnel-id <id></i>	Nothing found in XML Definitions	Nothing found in VyOS
✓	<i>set interfaces l2tpv3 <interface> address <address></i>	interfaces_l2tpv3.xml.in: interfaces l2tpv3 <l2tpv3> address	interfaces l2tpv3 <text> address <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
✓	<i>set interfaces l2tpv3 <interface> description <description></i>	interfaces_l2tpv3.xml.in: interfaces l2tpv3 <l2tpv3> description	interfaces l2tpv3 <text> description <value>
×	not yet documented	interfaces_l2tpv3.xml.in: interfaces l2tpv3 <l2tpv3> destination-port	interfaces l2tpv3 <text> destination-port <value>
✓	<i>set interfaces l2tpv3 <interface> disable</i>	interfaces_l2tpv3.xml.in: interfaces l2tpv3 <l2tpv3> disable	interfaces l2tpv3 <text> disable
×	<i>set interfaces l2tpv3 <interface> disable-flow-control</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set interfaces l2tpv3 <interface> disable-link-detect</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	interfaces_l2tpv3.xml.in: interfaces l2tpv3 <l2tpv3> encapsulation	interfaces l2tpv3 <text> encapsulation <value>
✓	<i>set interfaces l2tpv3 <interface> ip adjust-mss <mss / clamp-mss-to-pmtu></i>	interfaces_l2tpv3.xml.in: interfaces l2tpv3 <l2tpv3> ip adjust-mss	interfaces l2tpv3 <text> ip adjust-mss <value>
✓	<i>set interfaces l2tpv3 <interface> ip arp-cache-timeout</i>	interfaces_l2tpv3.xml.in: interfaces l2tpv3 <l2tpv3> ip arp-cache-timeout	interfaces l2tpv3 <text> ip arp-cache-timeout <value>
✓	<i>set interfaces l2tpv3 <interface> ip disable-arp-filter</i>	interfaces_l2tpv3.xml.in: interfaces l2tpv3 <l2tpv3> ip disable-arp-filter	interfaces l2tpv3 <text> ip disable-arp-filter
✓	<i>set interfaces l2tpv3 <interface> ip disable-forwarding</i>	interfaces_l2tpv3.xml.in: interfaces l2tpv3 <l2tpv3> ip disable-forwarding	interfaces l2tpv3 <text> ip disable-forwarding
✓	<i>set interfaces l2tpv3 <interface> ip enable-arp-accept</i>	interfaces_l2tpv3.xml.in: interfaces l2tpv3 <l2tpv3> ip enable-arp-accept	interfaces l2tpv3 <text> ip enable-arp-accept
✓	<i>set interfaces l2tpv3 <interface> ip enable-arp-announce</i>	interfaces_l2tpv3.xml.in: interfaces l2tpv3 <l2tpv3> ip enable-arp-announce	interfaces l2tpv3 <text> ip enable-arp-announce
✓	<i>set interfaces l2tpv3 <interface> ip enable-arp-ignore</i>	interfaces_l2tpv3.xml.in: interfaces l2tpv3 <l2tpv3> ip enable-arp-ignore	interfaces l2tpv3 <text> ip enable-arp-ignore

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
✓	<code>set interfaces l2tpv3 <interface> ip enable-directed-broad</code>	<code>interfaces_l2tpv3.xml.in: interfaces l2tpv3 <l2tpv3> ip enable-directed-broadcast</code>	<code>interfaces l2tpv3 <text> ip enable-directed-broadcast</code>
✓	<code>set interfaces l2tpv3 <interface> ip enable-proxy-arp</code>	<code>interfaces_l2tpv3.xml.in: interfaces l2tpv3 <l2tpv3> ip enable-proxy-arp</code>	<code>interfaces l2tpv3 <text> ip enable-proxy-arp</code>
✓	<code>set interfaces l2tpv3 <interface> ip proxy-arp-pvlan</code>	<code>interfaces_l2tpv3.xml.in: interfaces l2tpv3 <l2tpv3> ip proxy-arp-pvlan</code>	<code>interfaces l2tpv3 <text> ip proxy-arp-pvlan</code>
✓	<code>set interfaces l2tpv3 <interface> ip source-validation <strict loose disable></code>	<code>interfaces_l2tpv3.xml.in: interfaces l2tpv3 <l2tpv3> ip source-validation</code>	<code>interfaces l2tpv3 <text> ip source-validation <value></code>
✓	<code>set interfaces l2tpv3 <interface> ipv6 accept-dad <0 1 2></code>	<code>interfaces_l2tpv3.xml.in: interfaces l2tpv3 <l2tpv3> ipv6 accept-dad</code>	<code>interfaces l2tpv3 <text> ipv6 accept-dad <value></code>
✓	<code>set interfaces l2tpv3 <interface> ipv6 address autoconf</code>	<code>interfaces_l2tpv3.xml.in: interfaces l2tpv3 <l2tpv3> ipv6 address autoconf</code>	<code>interfaces l2tpv3 <text> ipv6 address autoconf</code>
✓	<code>set interfaces l2tpv3 <interface> ipv6 address eui64 <prefix></code>	<code>interfaces_l2tpv3.xml.in: interfaces l2tpv3 <l2tpv3> ipv6 address eui64</code>	<code>interfaces l2tpv3 <text> ipv6 address eui64 <value></code>
✓	<code>set interfaces l2tpv3 <interface> ipv6 address no-default-link-local</code>	<code>interfaces_l2tpv3.xml.in: interfaces l2tpv3 <l2tpv3> ipv6 address no-default-link-local</code>	<code>interfaces l2tpv3 <text> ipv6 address no-default-link-local</code>
✓	<code>set interfaces l2tpv3 <interface> ipv6 adjust-mss <mss clamp-mss-to-pmtu></code>	<code>interfaces_l2tpv3.xml.in: interfaces l2tpv3 <l2tpv3> ipv6 adjust-mss</code>	<code>interfaces l2tpv3 <text> ipv6 adjust-mss <value></code>
×	not yet documented	<code>interfaces_l2tpv3.xml.in: interfaces l2tpv3 <l2tpv3> ipv6 base-reachable-time</code>	Nothing found in VyOS
✓	<code>set interfaces l2tpv3 <interface> ipv6 disable-forwarding</code>	<code>interfaces_l2tpv3.xml.in: interfaces l2tpv3 <l2tpv3> ipv6 disable-forwarding</code>	<code>interfaces l2tpv3 <text> ipv6 disable-forwarding</code>
✓	<code>set interfaces l2tpv3 <interface> ipv6 dup-addr-detect-transmits <n></code>	<code>interfaces_l2tpv3.xml.in: interfaces l2tpv3 <l2tpv3> ipv6 dup-addr-detect-transmits</code>	<code>interfaces l2tpv3 <text> ipv6 dup-addr-detect-transmits <value></code>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	interfaces_l2tpv3.xml.in: interfaces l2tpv3 <l2tpv3> ipv6 source-validation	Nothing found in VyOS
×	<code>set interfaces l2tpv3 <interface> mac <xx:xx:xx:xx:xx:xx></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	interfaces_l2tpv3.xml.in: interfaces l2tpv3 <l2tpv3> mirror egress	interfaces l2tpv3 <text> mirror egress <value>
×	not yet documented	interfaces_l2tpv3.xml.in: interfaces l2tpv3 <l2tpv3> mirror ingress	interfaces l2tpv3 <text> mirror ingress <value>
✓	<code>set interfaces l2tpv3 <interface> mtu <mtu></code>	interfaces_l2tpv3.xml.in: interfaces l2tpv3 <l2tpv3> mtu	interfaces l2tpv3 <text> mtu <value>
×	not yet documented	interfaces_l2tpv3.xml.in: interfaces l2tpv3 <l2tpv3> peer-session-id	interfaces l2tpv3 <text> peer-session-id <value>
×	not yet documented	interfaces_l2tpv3.xml.in: interfaces l2tpv3 <l2tpv3> peer-tunnel-id	interfaces l2tpv3 <text> peer-tunnel-id <value>
×	not yet documented	interfaces_l2tpv3.xml.in: interfaces l2tpv3 <l2tpv3> remote	interfaces l2tpv3 <text> remote <value>
×	not yet documented	interfaces_l2tpv3.xml.in: interfaces l2tpv3 <l2tpv3> session-id	interfaces l2tpv3 <text> session-id <value>
×	not yet documented	interfaces_l2tpv3.xml.in: interfaces l2tpv3 <l2tpv3> source-address	interfaces l2tpv3 <text> source-address <value>
×	not yet documented	interfaces_l2tpv3.xml.in: interfaces l2tpv3 <l2tpv3> source-port	interfaces l2tpv3 <text> source-port <value>
×	not yet documented	interfaces_l2tpv3.xml.in: interfaces l2tpv3 <l2tpv3> tunnel-id	interfaces l2tpv3 <text> tunnel-id <value>
✓	<code>set interfaces l2tpv3 <interface> vrf <vrf></code>	interfaces_l2tpv3.xml.in: interfaces l2tpv3 <l2tpv3> vrf	interfaces l2tpv3 <text> vrf <value>
✓	<code>set interfaces loopback <interface> address <address></code>	inter- faces_loopback.xml.in: interfaces loopback <loop- back> address	interfaces loopback <text> address <value>
✓	<code>set interfaces loopback <interface> description <description></code>	inter- faces_loopback.xml.in: interfaces loopback <loop- back> description	interfaces loopback <text> description <value>

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Table 1 – continued from previous page

Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	inter- faces_loopback.xml.in: interfaces loopback <loop- back> ip source-validation	interfaces loopback <text> ip source-validation <value>
×	not yet documented	inter- faces_loopback.xml.in: interfaces loopback <loop- back> mirror egress	interfaces loopback <text> mirror egress <value>
×	not yet documented	inter- faces_loopback.xml.in: interfaces loopback <loop- back> mirror ingress	interfaces loopback <text> mirror ingress <value>
×	not yet documented	inter- faces_loopback.xml.in: interfaces loopback <loop- back> redirect	interfaces loopback <text> redirect <value>
×	<i>set interfaces macsec <interface> security cipher <gcm-aes-128/gcm-aes-</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set interfaces macsec <interface> security encrypt</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set interfaces macsec <interface> security mka cak <key></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set interfaces macsec <interface> security mka ckn <key></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set interfaces macsec <interface> security mka priority <priority></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set interfaces macsec <interface> security replay-window <window></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set interfaces macsec <interface> security static key <key></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set interfaces macsec <interface> security static peer <peer> disable</i>	Nothing found in XML Definitions	Nothing found in VyOS

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Table 1 – continued from previous page

Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set interfaces macsec <interface> security static peer <peer> key <key></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set interfaces macsec <interface> security static peer <peer> mac <mac address></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set interfaces macsec <interface> source-interface <physical-source></code>	Nothing found in XML Definitions	Nothing found in VyOS
✓	<code>set interfaces macsec <interface> address <address / dhcp / dhcpv6></code>	interfaces_macsec.xml.in: interfaces macsec <mac- sec> address	interfaces macsec <text> address <value>
✓	<code>set interfaces macsec <interface> description <description></code>	interfaces_macsec.xml.in: interfaces macsec <mac- sec> description	interfaces macsec <text> description <value>
✓	<code>set interfaces macsec <interface> dhcp-options client-id <description></code>	interfaces_macsec.xml.in: interfaces macsec <mac- sec> dhcp-options client- id	interfaces macsec <text> dhcp-options client-id <value>
✓	<code>set interfaces macsec <interface> dhcp-options default-route-distanc <distance></code>	interfaces_macsec.xml.in: interfaces macsec <mac- sec> dhcp-options default- route-distance	interfaces macsec <text> dhcp-options default- route-distance <value>
✓	<code>set interfaces macsec <interface> dhcp-options host-name <hostname></code>	interfaces_macsec.xml.in: interfaces macsec <mac- sec> dhcp-options host- name	interfaces macsec <text> dhcp-options host-name <value>
×	not yet documented	interfaces_macsec.xml.in: interfaces macsec <mac- sec> dhcp-options mtu	interfaces macsec <text> dhcp-options mtu
✓	<code>set interfaces macsec <interface> dhcp-options no-default-route</code>	interfaces_macsec.xml.in: interfaces macsec <mac- sec> dhcp-options no- default-route	interfaces macsec <text> dhcp-options no-default- route
✓	<code>set interfaces macsec <interface> dhcp-options reject <address></code>	interfaces_macsec.xml.in: interfaces macsec <mac- sec> dhcp-options reject	interfaces macsec <text> dhcp-options reject <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<i>set interfaces macsec <interface> dhcp-options user-class <string></i>	interfaces_macsec.xml.in: interfaces macsec <mac- sec> dhcp-options user- class	Nothing found in VyOS
✓	<i>set interfaces macsec <interface> dhcp-options vendor-class-id <vendor-id></i>	interfaces_macsec.xml.in: interfaces macsec <mac- sec> dhcp-options vendor- class-id	interfaces macsec <text> dhcp-options vendor- class-id <value>
✓	<i>set interfaces macsec <interface> dhcpv6-options duid <duid></i>	interfaces_macsec.xml.in: interfaces macsec <mac- sec> dhcpv6-options duid	interfaces macsec <text> dhcpv6-options duid <value>
✓	<i>set interfaces macsec <interface> dhcpv6-options no-release</i>	interfaces_macsec.xml.in: interfaces macsec <mac- sec> dhcpv6-options no- release	interfaces macsec <text> dhcpv6-options no-release
✓	<i>set interfaces macsec <interface> dhcpv6-options parameters-only</i>	interfaces_macsec.xml.in: interfaces macsec <mac- sec> dhcpv6-options parameters-only	interfaces macsec <text> dhcpv6-options parameters-only
✓	<i>set interfaces macsec <interface> dhcpv6-options pd <id> interface <delegatee> address <address></i>	interfaces_macsec.xml.in: interfaces macsec <mac- sec> dhcpv6-options pd <pd> interface <inter- face> address	interfaces macsec <text> dhcpv6-options pd <text> interface <text> address <value>
✓	<i>set interfaces macsec <interface> dhcpv6-options pd <id> interface <delegatee> sla-id <id></i>	interfaces_macsec.xml.in: interfaces macsec <mac- sec> dhcpv6-options pd <pd> interface <inter- face> sla-id	interfaces macsec <text> dhcpv6-options pd <text> interface <text> sla-id <value>
✓	<i>set interfaces macsec <interface> dhcpv6-options pd <id> length <length></i>	interfaces_macsec.xml.in: interfaces macsec <mac- sec> dhcpv6-options pd <pd> length	interfaces macsec <text> dhcpv6-options pd <text> length <value>
✓	<i>set interfaces macsec <interface> dhcpv6-options rapid-commit</i>	interfaces_macsec.xml.in: interfaces macsec <mac- sec> dhcpv6-options rapid-commit	interfaces macsec <text> dhcpv6-options rapid- commit
✓	<i>set interfaces macsec <interface> dhcpv6-options temporary</i>	interfaces_macsec.xml.in: interfaces macsec <mac- sec> dhcpv6-options tem- porary	interfaces macsec <text> dhcpv6-options temporary
✓	<i>set interfaces macsec <interface> disable</i>	interfaces_macsec.xml.in: interfaces macsec <mac- sec> disable	interfaces macsec <text> disable

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<i>set interfaces macsec <interface> disable-flow-control</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set interfaces macsec <interface> disable-link-detect</i>	Nothing found in XML Definitions	Nothing found in VyOS
✓	<i>set interfaces macsec <interface> ip adjust-mss <mss / clamp-mss-to-pmtu></i>	interfaces_macsec.xml.in: interfaces macsec <mac- sec> ip adjust-mss	interfaces macsec <text> ip adjust-mss <value>
✓	<i>set interfaces macsec <interface> ip arp-cache-timeout</i>	interfaces_macsec.xml.in: interfaces macsec <mac- sec> ip arp-cache-timeout	interfaces macsec <text> ip arp-cache-timeout <value>
✓	<i>set interfaces macsec <interface> ip disable-arp-filter</i>	interfaces_macsec.xml.in: interfaces macsec <mac- sec> ip disable-arp-filter	interfaces macsec <text> ip disable-arp-filter
✓	<i>set interfaces macsec <interface> ip disable-forwarding</i>	interfaces_macsec.xml.in: interfaces macsec <mac- sec> ip disable-forwarding	interfaces macsec <text> ip disable-forwarding
✓	<i>set interfaces macsec <interface> ip enable-arp-accept</i>	interfaces_macsec.xml.in: interfaces macsec <mac- sec> ip enable-arp-accept	interfaces macsec <text> ip enable-arp-accept
✓	<i>set interfaces macsec <interface> ip enable-arp-announce</i>	interfaces_macsec.xml.in: interfaces macsec <mac- sec> ip enable-arp- announce	interfaces macsec <text> ip enable-arp-announce
✓	<i>set interfaces macsec <interface> ip enable-arp-ignore</i>	interfaces_macsec.xml.in: interfaces macsec <mac- sec> ip enable-arp-ignore	interfaces macsec <text> ip enable-arp-ignore
✓	<i>set interfaces macsec <interface> ip enable-directed-broad</i>	interfaces_macsec.xml.in: interfaces macsec <mac- sec> ip enable-directed- broadcast	interfaces macsec <text> ip enable-directed- broadcast
✓	<i>set interfaces macsec <interface> ip enable-proxy-arp</i>	interfaces_macsec.xml.in: interfaces macsec <mac- sec> ip enable-proxy-arp	interfaces macsec <text> ip enable-proxy-arp
✓	<i>set interfaces macsec <interface> ip proxy-arp-pvlan</i>	interfaces_macsec.xml.in: interfaces macsec <mac- sec> ip proxy-arp-pvlan	interfaces macsec <text> ip proxy-arp-pvlan
✓	<i>set interfaces macsec <interface> ip source-validation <strict loose disable></i>	interfaces_macsec.xml.in: interfaces macsec <mac- sec> ip source-validation	interfaces macsec <text> ip source-validation <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
✓	<code>set interfaces macsec <interface> ipv6 accept-dad <0 / 1 2></code>	interfaces_macsec.xml.in: interfaces macsec <mac- sec> ipv6 accept-dad	interfaces macsec <text> ipv6 accept-dad <value>
✓	<code>set interfaces macsec <interface> ipv6 address autoconf</code>	interfaces_macsec.xml.in: interfaces macsec <mac- sec> ipv6 address auto- conf	interfaces macsec <text> ipv6 address autoconf
✓	<code>set interfaces macsec <interface> ipv6 address eui64 <prefix></code>	interfaces_macsec.xml.in: interfaces macsec <mac- sec> ipv6 address eui64	interfaces macsec <text> ipv6 address eui64 <value>
✓	<code>set interfaces macsec <interface> ipv6 address no-default-link-local</code>	interfaces_macsec.xml.in: interfaces macsec <mac- sec> ipv6 address no- default-link-local	interfaces macsec <text> ipv6 address no-default- link-local
✓	<code>set interfaces macsec <interface> ipv6 adjust-mss <mss / clamp-mss-to-pmtu></code>	interfaces_macsec.xml.in: interfaces macsec <mac- sec> ipv6 adjust-mss	interfaces macsec <text> ipv6 adjust-mss <value>
×	not yet documented	interfaces_macsec.xml.in: interfaces macsec <mac- sec> ipv6 base-reachable- time	Nothing found in VyOS
✓	<code>set interfaces macsec <interface> ipv6 disable-forwarding</code>	interfaces_macsec.xml.in: interfaces macsec <macsec> ipv6 disable- forwarding	interfaces macsec <text> ipv6 disable-forwarding
✓	<code>set interfaces macsec <interface> ipv6 dup-addr-detect-trans <n></code>	interfaces_macsec.xml.in: interfaces macsec <mac- sec> ipv6 dup-addr- detect-transmits	interfaces macsec <text> ipv6 dup-addr-detect- transmits <value>
×	not yet documented	interfaces_macsec.xml.in: interfaces macsec <macsec> ipv6 source- validation	Nothing found in VyOS
×	<code>set interfaces macsec <interface> mac <xx:xx:xx:xx:xx:xx></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	interfaces_macsec.xml.in: interfaces macsec <mac- sec> mirror egress	interfaces macsec <text> mirror egress <value>
×	not yet documented	interfaces_macsec.xml.in: interfaces macsec <mac- sec> mirror ingress	interfaces macsec <text> mirror ingress <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
✓	<code>set interfaces macsec <interface> mtu <mtu></code>	<code>interfaces_macsec.xml.in: interfaces macsec <macsec> mtu</code>	<code>interfaces macsec <text> mtu <value></code>
×	not yet documented	<code>interfaces_macsec.xml.in: interfaces macsec <macsec> redirect</code>	<code>interfaces macsec <text> redirect <value></code>
×	not yet documented	<code>interfaces_macsec.xml.in: interfaces macsec <macsec> security cipher</code>	<code>interfaces macsec <text> security cipher <value></code>
×	not yet documented	<code>interfaces_macsec.xml.in: interfaces macsec <macsec> security encrypt</code>	<code>interfaces macsec <text> security encrypt</code>
×	not yet documented	<code>interfaces_macsec.xml.in: interfaces macsec <macsec> security mka cak</code>	<code>interfaces macsec <text> security mka cak <value></code>
×	not yet documented	<code>interfaces_macsec.xml.in: interfaces macsec <macsec> security mka ckn</code>	<code>interfaces macsec <text> security mka ckn <value></code>
×	not yet documented	<code>interfaces_macsec.xml.in: interfaces macsec <macsec> security mka priority</code>	<code>interfaces macsec <text> security mka priority <value></code>
×	not yet documented	<code>interfaces_macsec.xml.in: interfaces macsec <macsec> security replay-window</code>	<code>interfaces macsec <text> security replay-window <value></code>
×	not yet documented	<code>interfaces_macsec.xml.in: interfaces macsec <macsec> security static key</code>	<code>interfaces macsec <text> security static key <value></code>
×	not yet documented	<code>interfaces_macsec.xml.in: interfaces macsec <macsec> security static peer <peer> disable</code>	<code>interfaces macsec <text> security static peer <text> disable</code>
×	not yet documented	<code>interfaces_macsec.xml.in: interfaces macsec <macsec> security static peer <peer> key</code>	<code>interfaces macsec <text> security static peer <text> key <value></code>
×	not yet documented	<code>interfaces_macsec.xml.in: interfaces macsec <macsec> security static peer <peer> mac</code>	<code>interfaces macsec <text> security static peer <text> mac <value></code>
×	not yet documented	<code>interfaces_macsec.xml.in: interfaces macsec <macsec> source-interface</code>	<code>interfaces macsec <text> source-interface <value></code>
✓	<code>set interfaces macsec <interface> vrf <vrf></code>	<code>interfaces_macsec.xml.in: interfaces macsec <macsec> vrf</code>	<code>interfaces macsec <text> vrf <value></code>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set interfaces openvpn <interface> authentication password <text></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set interfaces openvpn <interface> authentication username <text></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set interfaces openvpn <interface> description <description></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set interfaces openvpn <interface> device-type <tap tun></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set interfaces openvpn <interface> disable</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set interfaces openvpn <interface> encryption cipher < 3des aes128 aes128gcm aes192 aes192gcm aes256 aes256gcm none ></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set interfaces openvpn <interface> encryption data-ciphers-fallback < 3des aes128 aes128gcm aes192 aes192gcm aes256 aes256gcm none ></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set interfaces openvpn <interface> encryption data-ciphers < 3des aes128 aes128gcm aes192 aes192gcm aes256 aes256gcm none ></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set interfaces openvpn <interface> hash <md5 sha1 sha256 ...></code>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set interfaces openvpn <interface> keep-alive failure-count <value></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set interfaces openvpn <interface> keep-alive interval <value></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set interfaces openvpn <interface> local-address <address></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set interfaces openvpn <interface> local-host <address></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set interfaces openvpn <interface> local-port <port></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set interfaces openvpn <interface> mirror egress <monitor-interface></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set interfaces openvpn <interface> mirror ingress <monitor-interface></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set interfaces openvpn <interface> mode <site-to-site / server / client></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set interfaces openvpn <interface> offload dco</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set interfaces openvpn <interface> openvpn-option <text></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set interfaces openvpn <interface> persistent-tunnel</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set interfaces openvpn <interface> protocol <udp / tcp-passive / tcp-active ></code>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set interfaces openvpn <interface> redirect <interface></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set interfaces openvpn <interface> remote-address <address></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set interfaces openvpn <interface> remote-host <address / host></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set interfaces openvpn <interface> remote-port <port></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set interfaces openvpn <interface> replace-default-route</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set interfaces openvpn <interface> server bridge disable</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set interfaces openvpn <interface> server bridge gateway <ipv4 address></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set interfaces openvpn <interface> server bridge start <ipv4 address></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set interfaces openvpn <interface> server bridge stop <ipv4 address></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set interfaces openvpn <interface> server bridge subnet-mask <ipv4 subnet mask></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set interfaces openvpn <interface> server client-ip-pool start <address></code>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set interfaces openvpn <interface> server client-ip-pool stop <address></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set interfaces openvpn <interface> server client-ip-pool subnet <netmask></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set interfaces openvpn <interface> server client-ipv6-pool base <ipv6addr/ bits></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set interfaces openvpn <interface> server client <name></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set interfaces openvpn <interface> server client <name> disable</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set interfaces openvpn <interface> server client <name> ip <address></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set interfaces openvpn <interface> server client <name> push-route <subnet></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set interfaces openvpn <interface> server client <name> subnet <subnet></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set interfaces openvpn <interface> server domain-name <name></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set interfaces openvpn <interface> server max-connections <1-4096></code>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set interfaces openvpn <interface> server mfa totp challenge <enable / disable></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set interfaces openvpn <interface> server mfa totp digits <1-65535></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set interfaces openvpn <interface> server mfa totp drift <1-65535></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set interfaces openvpn <interface> server mfa totp slop <1-65535></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set interfaces openvpn <interface> server mfa totp step <1-65535></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set interfaces openvpn <interface> server name-server <address></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set interfaces openvpn <interface> server push-route <subnet></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set interfaces openvpn <interface> server reject-unconfigured-c</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set interfaces openvpn <interface> server subnet <subnet></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set interfaces openvpn <interface> server topology < net30 / point-to-point / subnet></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set interfaces openvpn <interface> shared-secret-key <key></code>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set interfaces openvpn <interface> tls auth-key <key></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set interfaces openvpn <interface> tls ca-certificate <name></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set interfaces openvpn <interface> tls certificate <name></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set interfaces openvpn <interface> tls crypt-key</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set interfaces openvpn <interface> tls dh-params</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set interfaces openvpn <interface> tls peer-fingerprint <text></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set interfaces openvpn <interface> tls role <active passive></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set interfaces openvpn <interface> tls tls-version-min <1.0 1.1 1.2 1.3 ></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set interfaces openvpn <interface> use-lzo-compression</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set interfaces openvpn <interface> vrf <name></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	inter- faces_openvpn.xml.in: interfaces openvpn <open- vpn> authentication pass- word	interfaces openvpn <text> authentication password <value>
×	not yet documented	inter- faces_openvpn.xml.in: interfaces openvpn <open- vpn> authentication user- name	interfaces openvpn <text> authentication username <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	inter-faces_openvpn.xml.in: interfaces openvpn <open-vpn> description	interfaces openvpn <text> description <value>
×	not yet documented	inter-faces_openvpn.xml.in: interfaces openvpn <open-vpn> device-type	interfaces openvpn <text> device-type <value>
×	not yet documented	inter-faces_openvpn.xml.in: interfaces openvpn <open-vpn> disable	interfaces openvpn <text> disable
×	not yet documented	inter-faces_openvpn.xml.in: interfaces openvpn <open-vpn> encryption cipher	interfaces openvpn <text> encryption cipher <value>
×	not yet documented	inter-faces_openvpn.xml.in: interfaces openvpn <open-vpn> encryption ncp-ciphers	interfaces openvpn <text> encryption ncp-ciphers <value>
×	not yet documented	inter-faces_openvpn.xml.in: interfaces openvpn <open-vpn> hash	interfaces openvpn <text> hash <value>
✓	<i>set interfaces openvpn <interface> ip adjust-mss <mss / clamp-mss-to-pmtu></i>	inter-faces_openvpn.xml.in: interfaces openvpn <open-vpn> ip adjust-mss	interfaces openvpn <text> ip adjust-mss <value>
✓	<i>set interfaces openvpn <interface> ip arp-cache-timeout</i>	inter-faces_openvpn.xml.in: interfaces openvpn <open-vpn> ip arp-cache-timeout	interfaces openvpn <text> ip arp-cache-timeout <value>
✓	<i>set interfaces openvpn <interface> ip disable-arp-filter</i>	inter-faces_openvpn.xml.in: interfaces openvpn <open-vpn> ip disable-arp-filter	interfaces openvpn <text> ip disable-arp-filter
✓	<i>set interfaces openvpn <interface> ip disable-forwarding</i>	inter-faces_openvpn.xml.in: interfaces openvpn <open-vpn> ip disable-forwarding	interfaces openvpn <text> ip disable-forwarding
✓	<i>set interfaces openvpn <interface> ip enable-arp-accept</i>	inter-faces_openvpn.xml.in: interfaces openvpn <open-vpn> ip enable-arp-accept	interfaces openvpn <text> ip enable-arp-accept

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
✓	<code>set interfaces openvpn <interface> ip enable-arp-announce</code>	inter- faces_openvpn.xml.in: interfaces openvpn <openvpn> ip enable-arp- announce	interfaces openvpn <text> ip enable-arp-announce
✓	<code>set interfaces openvpn <interface> ip enable-arp-ignore</code>	inter- faces_openvpn.xml.in: interfaces openvpn <open- vpn> ip enable-arp-ignore	interfaces openvpn <text> ip enable-arp-ignore
✓	<code>set interfaces openvpn <interface> ip enable-directed-broad</code>	inter- faces_openvpn.xml.in: interfaces openvpn <open- vpn> ip enable-directed- broadcast	interfaces openvpn <text> ip enable-directed- broadcast
✓	<code>set interfaces openvpn <interface> ip enable-proxy-arp</code>	inter- faces_openvpn.xml.in: interfaces openvpn <open- vpn> ip enable-proxy-arp	interfaces openvpn <text> ip enable-proxy-arp
✓	<code>set interfaces openvpn <interface> ip proxy-arp-pvlan</code>	inter- faces_openvpn.xml.in: interfaces openvpn <open- vpn> ip proxy-arp-pvlan	interfaces openvpn <text> ip proxy-arp-pvlan
✓	<code>set interfaces openvpn <interface> ip source-validation <strict loose disable></code>	inter- faces_openvpn.xml.in: interfaces openvpn <open- vpn> ip source-validation	interfaces openvpn <text> ip source-validation <value>
✓	<code>set interfaces openvpn <interface> ipv6 accept-dad <0 1 2></code>	inter- faces_openvpn.xml.in: interfaces openvpn <open- vpn> ipv6 accept-dad	interfaces openvpn <text> ipv6 accept-dad <value>
✓	<code>set interfaces openvpn <interface> ipv6 address autoconf</code>	inter- faces_openvpn.xml.in: interfaces openvpn <open- vpn> ipv6 address auto- conf	interfaces openvpn <text> ipv6 address autoconf
✓	<code>set interfaces openvpn <interface> ipv6 address eui64 <prefix></code>	inter- faces_openvpn.xml.in: interfaces openvpn <open- vpn> ipv6 address eui64	interfaces openvpn <text> ipv6 address eui64 <value>
✓	<code>set interfaces openvpn <interface> ipv6 address no-default-link-local</code>	inter- faces_openvpn.xml.in: interfaces openvpn <open- vpn> ipv6 address no- default-link-local	interfaces openvpn <text> ipv6 address no-default- link-local

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
✓	<i>set interfaces openvpn <interface> ipv6 adjust-mss <mss / clamp-mss-to-pmtu></i>	inter-faces_openvpn.xml.in: interfaces openvpn <open-vpn> ipv6 adjust-mss	interfaces openvpn <text> ipv6 adjust-mss <value>
×	not yet documented	inter-faces_openvpn.xml.in: interfaces openvpn <open-vpn> ipv6 base-reachable-time	Nothing found in VyOS
✓	<i>set interfaces openvpn <interface> ipv6 disable-forwarding</i>	inter-faces_openvpn.xml.in: interfaces openvpn <openvpn> ipv6 disable-forwarding	interfaces openvpn <text> ipv6 disable-forwarding
✓	<i>set interfaces openvpn <interface> ipv6 dup-addr-detect-transmits <n></i>	inter-faces_openvpn.xml.in: interfaces openvpn <openvpn> ipv6 dup-addr-detect-transmits	interfaces openvpn <text> ipv6 dup-addr-detect-transmits <value>
×	not yet documented	inter-faces_openvpn.xml.in: interfaces openvpn <openvpn> ipv6 source-validation	Nothing found in VyOS
×	not yet documented	inter-faces_openvpn.xml.in: interfaces openvpn <open-vpn> keep-alive failure-count	interfaces openvpn <text> keep-alive failure-count <value>
×	not yet documented	inter-faces_openvpn.xml.in: interfaces openvpn <open-vpn> keep-alive interval	interfaces openvpn <text> keep-alive interval <value>
×	not yet documented	inter-faces_openvpn.xml.in: interfaces openvpn <open-vpn> local-address <local-address> subnet-mask	interfaces openvpn <text> local-address <text> subnet-mask <value>
×	not yet documented	inter-faces_openvpn.xml.in: interfaces openvpn <open-vpn> local-host	interfaces openvpn <text> local-host <value>
×	not yet documented	inter-faces_openvpn.xml.in: interfaces openvpn <open-vpn> local-port	interfaces openvpn <text> local-port <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	inter-faces_openvpn.xml.in: interfaces openvpn <open-vpn> mirror egress	interfaces openvpn <text> mirror egress <value>
×	not yet documented	inter-faces_openvpn.xml.in: interfaces openvpn <open-vpn> mirror ingress	interfaces openvpn <text> mirror ingress <value>
×	not yet documented	inter-faces_openvpn.xml.in: interfaces openvpn <open-vpn> mode	interfaces openvpn <text> mode <value>
×	not yet documented	inter-faces_openvpn.xml.in: interfaces openvpn <open-vpn> offload dco	interfaces openvpn <text> offload dco
×	not yet documented	inter-faces_openvpn.xml.in: interfaces openvpn <open-vpn> openvpn-option	interfaces openvpn <text> openvpn-option <value>
×	not yet documented	inter-faces_openvpn.xml.in: interfaces openvpn <open-vpn> persistent-tunnel	interfaces openvpn <text> persistent-tunnel
×	not yet documented	inter-faces_openvpn.xml.in: interfaces openvpn <open-vpn> protocol	interfaces openvpn <text> protocol <value>
×	not yet documented	inter-faces_openvpn.xml.in: interfaces openvpn <open-vpn> redirect	interfaces openvpn <text> redirect <value>
×	not yet documented	inter-faces_openvpn.xml.in: interfaces openvpn <open-vpn> remote-address	interfaces openvpn <text> remote-address <value>
×	not yet documented	inter-faces_openvpn.xml.in: interfaces openvpn <open-vpn> remote-host	interfaces openvpn <text> remote-host <value>
×	not yet documented	inter-faces_openvpn.xml.in: interfaces openvpn <open-vpn> remote-port	interfaces openvpn <text> remote-port <value>
×	not yet documented	inter-faces_openvpn.xml.in: interfaces openvpn <open-vpn> replace-default-route local	interfaces openvpn <text> replace-default-route local <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	inter-faces_openvpn.xml.in: interfaces openvpn <open-vpn> server client-ip-pool disable	interfaces openvpn <text> server client-ip-pool disable
×	not yet documented	inter-faces_openvpn.xml.in: interfaces openvpn <open-vpn> server client-ip-pool start	interfaces openvpn <text> server client-ip-pool start <value>
×	not yet documented	inter-faces_openvpn.xml.in: interfaces openvpn <open-vpn> server client-ip-pool stop	interfaces openvpn <text> server client-ip-pool stop <value>
×	not yet documented	inter-faces_openvpn.xml.in: interfaces openvpn <open-vpn> server client-ip-pool subnet-mask	interfaces openvpn <text> server client-ip-pool subnet-mask <value>
×	not yet documented	inter-faces_openvpn.xml.in: interfaces openvpn <open-vpn> server client-ipv6-pool base	interfaces openvpn <text> server client-ipv6-pool base <value>
×	not yet documented	inter-faces_openvpn.xml.in: interfaces openvpn <open-vpn> server client-ipv6-pool disable	interfaces openvpn <text> server client-ipv6-pool disable
×	not yet documented	inter-faces_openvpn.xml.in: interfaces openvpn <openvpn> server client <client> disable	interfaces openvpn <text> server client <text> disable
×	not yet documented	inter-faces_openvpn.xml.in: interfaces openvpn <openvpn> server client <client> ip	interfaces openvpn <text> server client <text> ip <value>
×	not yet documented	inter-faces_openvpn.xml.in: interfaces openvpn <openvpn> server client <client> push-route	interfaces openvpn <text> server client <text> push-route <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	inter-faces_openvpn.xml.in: interfaces openvpn <openvpn> server client <client> subnet	interfaces openvpn <text> server client <text> subnet <value>
×	not yet documented	inter-faces_openvpn.xml.in: interfaces openvpn <open- vpn> server domain-name	interfaces openvpn <text> server domain-name <value>
×	not yet documented	inter-faces_openvpn.xml.in: interfaces openvpn <openvpn> server max- connections	interfaces openvpn <text> server max-connections <value>
×	not yet documented	inter-faces_openvpn.xml.in: interfaces openvpn <open- vpn> server mfa totp chal- lenge	interfaces openvpn <text> server mfa totp challenge <value>
×	not yet documented	inter-faces_openvpn.xml.in: interfaces openvpn <open- vpn> server mfa totp digits	interfaces openvpn <text> server mfa totp digits <value>
×	not yet documented	inter-faces_openvpn.xml.in: interfaces openvpn <open- vpn> server mfa totp drift	interfaces openvpn <text> server mfa totp drift <value>
×	not yet documented	inter-faces_openvpn.xml.in: interfaces openvpn <open- vpn> server mfa totp slop	interfaces openvpn <text> server mfa totp slop <value>
×	not yet documented	inter-faces_openvpn.xml.in: interfaces openvpn <open- vpn> server mfa totp step	interfaces openvpn <text> server mfa totp step <value>
×	not yet documented	inter-faces_openvpn.xml.in: interfaces openvpn <open- vpn> server name-server	interfaces openvpn <text> server name-server <value>
×	not yet documented	inter-faces_openvpn.xml.in: interfaces openvpn <open- vpn> server push-route <push-route> metric	interfaces openvpn <text> server push-route <text> metric <value>
×	not yet documented	inter-faces_openvpn.xml.in: interfaces openvpn <openvpn> server reject- unconfigured-clients	interfaces openvpn <text> server reject- unconfigured-clients

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	inter-faces_openvpn.xml.in: interfaces openvpn <open-vpn> server subnet	interfaces openvpn <text> server subnet <value>
×	not yet documented	inter-faces_openvpn.xml.in: interfaces openvpn <open-vpn> server topology	interfaces openvpn <text> server topology <value>
×	not yet documented	inter-faces_openvpn.xml.in: interfaces openvpn <open-vpn> shared-secret-key	interfaces openvpn <text> shared-secret-key <value>
×	not yet documented	inter-faces_openvpn.xml.in: interfaces openvpn <open-vpn> tls auth-key	interfaces openvpn <text> tls auth-key <value>
×	not yet documented	inter-faces_openvpn.xml.in: interfaces openvpn <open-vpn> tls ca-certificate	interfaces openvpn <text> tls ca-certificate <value>
×	not yet documented	inter-faces_openvpn.xml.in: interfaces openvpn <open-vpn> tls certificate	interfaces openvpn <text> tls certificate <value>
×	not yet documented	inter-faces_openvpn.xml.in: interfaces openvpn <open-vpn> tls crypt-key	interfaces openvpn <text> tls crypt-key <value>
×	not yet documented	inter-faces_openvpn.xml.in: interfaces openvpn <open-vpn> tls dh-params	interfaces openvpn <text> tls dh-params <value>
×	not yet documented	inter-faces_openvpn.xml.in: interfaces openvpn <open-vpn> tls peer-fingerprint	interfaces openvpn <text> tls peer-fingerprint <value>
×	not yet documented	inter-faces_openvpn.xml.in: interfaces openvpn <open-vpn> tls role	interfaces openvpn <text> tls role <value>
×	not yet documented	inter-faces_openvpn.xml.in: interfaces openvpn <open-vpn> tls tls-version-min	interfaces openvpn <text> tls tls-version-min <value>
×	not yet documented	inter-faces_openvpn.xml.in: interfaces openvpn <open-vpn> use-lzo-compression	interfaces openvpn <text> use-lzo-compression

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	inter-faces_openvpn.xml.in: interfaces openvpn <open-vpn> vrf	interfaces openvpn <text> vrf <value>
×	<i>set interfaces pppoe <interface> access-concentrator <name></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set interfaces pppoe <interface> authentication password <password></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set interfaces pppoe <interface> authentication username <username></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set interfaces pppoe <interface> connect-on-demand</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set interfaces pppoe <interface> default-route-distance <distance></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set interfaces pppoe <interface> holdoff <time></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set interfaces pppoe <interface> idle-timeout <time></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set interfaces pppoe <interface> ip adjust-mss <mss / clamp-mss-to-pmtu></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set interfaces pppoe <interface> ip disable-forwarding</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set interfaces pppoe <interface> ip source-validation <strict loose disable></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set interfaces pppoe <interface> ipv6 address autoconf</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set interfaces pppoe <interface> ipv6 adjust-mss <mss / clamp-mss-to-pmtu></i>	Nothing found in XML Definitions	Nothing found in VyOS

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Table 1 – continued from previous page

Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<i>set interfaces pppoe <interface> ipv6 disable-forwarding</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set interfaces pppoe <interface> local-address <address></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set interfaces pppoe <interface> mru <mru></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set interfaces pppoe <interface> no-default-route</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set interfaces pppoe <interface> no-peer-dns</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set interfaces pppoe <interface> remote-address <address></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set interfaces pppoe <interface> service-name <name></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set interfaces pppoe <interface> source-interface <source-interface></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	interfaces_pppoe.xml.in: interfaces pppoe <pppoe> access-concentrator	interfaces pppoe <text> access-concentrator <value>
×	not yet documented	interfaces_pppoe.xml.in: interfaces pppoe <pppoe> authentication password	interfaces pppoe <text> authentication password <value>
×	not yet documented	interfaces_pppoe.xml.in: interfaces pppoe <pppoe> authentication username	interfaces pppoe <text> authentication username <value>
×	not yet documented	interfaces_pppoe.xml.in: interfaces pppoe <pppoe> connect-on-demand	interfaces pppoe <text> connect-on-demand
×	not yet documented	interfaces_pppoe.xml.in: interfaces pppoe <pppoe> default-route-distance	interfaces pppoe <text> default-route-distance <value>
✓	<i>set interfaces pppoe <interface> description <description></i>	interfaces_pppoe.xml.in: interfaces pppoe <pppoe> description	interfaces pppoe <text> description <value>
×	not yet documented	interfaces_pppoe.xml.in: interfaces pppoe <pppoe> dhcpv6-options duid	interfaces pppoe <text> dhcpv6-options duid <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	interfaces_pppoe.xml.in: interfaces pppoe <pppoe> dhcpv6-options no-release	interfaces pppoe <text> dhcpv6-options no-release
×	not yet documented	interfaces_pppoe.xml.in: interfaces pppoe <pppoe> dhcpv6-options parameters-only	interfaces pppoe <text> dhcpv6-options parameters-only
✓	<i>set interfaces pppoe <interface> dhcpv6-options pd <id> interface <delegatee> address <address></i>	interfaces_pppoe.xml.in: interfaces pppoe <pppoe> dhcpv6-options pd <pd> interface <interface> address	interfaces pppoe <text> dhcpv6-options pd <text> interface <text> address <value>
✓	<i>set interfaces pppoe <interface> dhcpv6-options pd <id> interface <delegatee> sla-id <id></i>	interfaces_pppoe.xml.in: interfaces pppoe <pppoe> dhcpv6-options pd <pd> interface <interface> sla-id	interfaces pppoe <text> dhcpv6-options pd <text> interface <text> sla-id <value>
✓	<i>set interfaces pppoe <interface> dhcpv6-options pd <id> length <length></i>	interfaces_pppoe.xml.in: interfaces pppoe <pppoe> dhcpv6-options pd <pd> length	interfaces pppoe <text> dhcpv6-options pd <text> length <value>
×	not yet documented	interfaces_pppoe.xml.in: interfaces pppoe <pppoe> dhcpv6-options rapid-commit	interfaces pppoe <text> dhcpv6-options rapid-commit
×	not yet documented	interfaces_pppoe.xml.in: interfaces pppoe <pppoe> dhcpv6-options temporary	interfaces pppoe <text> dhcpv6-options temporary
✓	<i>set interfaces pppoe <interface> disable</i>	interfaces_pppoe.xml.in: interfaces pppoe <pppoe> disable	interfaces pppoe <text> disable
×	not yet documented	interfaces_pppoe.xml.in: interfaces pppoe <pppoe> holdoff	interfaces pppoe <text> holdoff <value>
×	not yet documented	interfaces_pppoe.xml.in: interfaces pppoe <pppoe> host-uniq	interfaces pppoe <text> host-uniq <value>
×	not yet documented	interfaces_pppoe.xml.in: interfaces pppoe <pppoe> idle-timeout	interfaces pppoe <text> idle-timeout <value>
×	not yet documented	interfaces_pppoe.xml.in: interfaces pppoe <pppoe> ip adjust-mss	interfaces pppoe <text> ip adjust-mss <value>
×	not yet documented	interfaces_pppoe.xml.in: interfaces pppoe <pppoe> ip disable-forwarding	interfaces pppoe <text> ip disable-forwarding

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	interfaces_pppoe.xml.in: interfaces pppoe <pppoe> ip source-validation	interfaces pppoe <text> ip source-validation <value>
×	not yet documented	interfaces_pppoe.xml.in: interfaces pppoe <pppoe> ipv6 address autoconf	interfaces pppoe <text> ipv6 address autoconf
×	not yet documented	interfaces_pppoe.xml.in: interfaces pppoe <pppoe> ipv6 adjust-mss	interfaces pppoe <text> ipv6 adjust-mss <value>
×	not yet documented	interfaces_pppoe.xml.in: interfaces pppoe <pppoe> ipv6 disable-forwarding	interfaces pppoe <text> ipv6 disable-forwarding
×	not yet documented	interfaces_pppoe.xml.in: interfaces pppoe <pppoe> local-address	interfaces pppoe <text> local-address <value>
×	not yet documented	interfaces_pppoe.xml.in: interfaces pppoe <pppoe> mirror egress	interfaces pppoe <text> mirror egress <value>
×	not yet documented	interfaces_pppoe.xml.in: interfaces pppoe <pppoe> mirror ingress	interfaces pppoe <text> mirror ingress <value>
×	not yet documented	interfaces_pppoe.xml.in: interfaces pppoe <pppoe> mru	Nothing found in VyOS
✓	<i>set interfaces pppoe <interface> mtu <mtu></i>	interfaces_pppoe.xml.in: interfaces pppoe <pppoe> mtu	interfaces pppoe <text> mtu <value>
×	not yet documented	interfaces_pppoe.xml.in: interfaces pppoe <pppoe> no-default-route	interfaces pppoe <text> no-default-route
×	not yet documented	interfaces_pppoe.xml.in: interfaces pppoe <pppoe> no-peer-dns	interfaces pppoe <text> no-peer-dns
×	not yet documented	interfaces_pppoe.xml.in: interfaces pppoe <pppoe> redirect	interfaces pppoe <text> redirect <value>
×	not yet documented	interfaces_pppoe.xml.in: interfaces pppoe <pppoe> remote-address	interfaces pppoe <text> remote-address <value>
×	not yet documented	interfaces_pppoe.xml.in: interfaces pppoe <pppoe> service-name	interfaces pppoe <text> service-name <value>
×	not yet documented	interfaces_pppoe.xml.in: interfaces pppoe <pppoe> source-interface	interfaces pppoe <text> source-interface <value>
✓	<i>set interfaces pppoe <interface> vrf <vrf></i>	interfaces_pppoe.xml.in: interfaces pppoe <pppoe> vrf	interfaces pppoe <text> vrf <value>

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Table 1 – continued from previous page

Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set interfaces pseudo-ethernet <interface> source-interface <ethX></code>	Nothing found in XML Definitions	Nothing found in VyOS
✓	<code>set interfaces pseudo-ethernet <interface> address <address dhcp dhcpv6></code>	interfaces_pseudo-ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> address	interfaces pseudo-ethernet <text> address <value>
×	<code>set interfaces pseudo-ethernet <interface> anycast-gateway</code>	Nothing found in XML Definitions	Nothing found in VyOS
✓	<code>set interfaces pseudo-ethernet <interface> description <description></code>	interfaces_pseudo-ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> description	interfaces pseudo-ethernet <text> description <value>
✓	<code>set interfaces pseudo-ethernet <interface> dhcp-options client-id <description></code>	interfaces_pseudo-ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> dhcp-options client-id	interfaces pseudo-ethernet <text> dhcp-options client-id <value>
✓	<code>set interfaces pseudo-ethernet <interface> dhcp-options default-route-distance <distance></code>	interfaces_pseudo-ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> dhcp-options default-route-distance	interfaces pseudo-ethernet <text> dhcp-options default-route-distance <value>
✓	<code>set interfaces pseudo-ethernet <interface> dhcp-options host-name <hostname></code>	interfaces_pseudo-ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> dhcp-options host-name	interfaces pseudo-ethernet <text> dhcp-options host-name <value>
×	not yet documented	interfaces_pseudo-ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> dhcp-options mtu	interfaces pseudo-ethernet <text> dhcp-options mtu
✓	<code>set interfaces pseudo-ethernet <interface> dhcp-options no-default-route</code>	interfaces_pseudo-ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> dhcp-options no-default-route	interfaces pseudo-ethernet <text> dhcp-options no-default-route

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
✓	<code>set interfaces pseudo-ethernet <interface> dhcp-options reject <address></code>	interfaces_pseudo- ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> dhcp- options reject	interfaces pseudo-ethernet <text> dhcp-options reject <value>
×	<code>set interfaces pseudo-ethernet <interface> dhcp-options user-class <string></code>	interfaces_pseudo- ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> dhcp- options user-class	Nothing found in VyOS
✓	<code>set interfaces pseudo-ethernet <interface> dhcp-options vendor-class-id <vendor-id></code>	interfaces_pseudo- ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> dhcp- options vendor-class-id	interfaces pseudo-ethernet <text> dhcp-options vendor-class-id <value>
✓	<code>set interfaces pseudo-ethernet <interface> dhcpv6-options duid <duid></code>	interfaces_pseudo- ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> dhcpv6-options duid	interfaces pseudo-ethernet <text> dhcpv6-options duid <value>
✓	<code>set interfaces pseudo-ethernet <interface> dhcpv6-options no-release</code>	interfaces_pseudo- ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> dhcpv6-options no-release	interfaces pseudo-ethernet <text> dhcpv6-options no- release
✓	<code>set interfaces pseudo-ethernet <interface> dhcpv6-options parameters-only</code>	interfaces_pseudo- ethernet.xml.in: interfaces pseudo- ethernet <pseudo- ethernet> dhcpv6-options parameters-only	interfaces pseudo-ethernet <text> dhcpv6-options parameters-only
✓	<code>set interfaces pseudo-ethernet <interface> dhcpv6-options pd <id> interface <delegatee> address <address></code>	interfaces_pseudo- ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> dhcpv6-options pd <pd> interface <interface> address	interfaces pseudo-ethernet <text> dhcpv6-options pd <text> interface <text> ad- dress <value>
✓	<code>set interfaces pseudo-ethernet <interface> dhcpv6-options pd <id> interface <delegatee> sla-id <id></code>	interfaces_pseudo- ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> dhcpv6-options pd <pd> interface <interface> sla-id	interfaces pseudo-ethernet <text> dhcpv6-options pd <text> interface <text> sla-id <value>

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Table 1 – continued from previous page

Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
✓	<i>set interfaces pseudo-ethernet <interface> dhcpv6-options pd <id> length <length></i>	interfaces_pseudo-ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> dhcpv6-options pd <pd> length	interfaces pseudo-ethernet <text> dhcpv6-options pd <text> length <value>
✓	<i>set interfaces pseudo-ethernet <interface> dhcpv6-options rapid-commit</i>	interfaces_pseudo-ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> dhcpv6-options rapid-commit	interfaces pseudo-ethernet <text> dhcpv6-options rapid-commit
✓	<i>set interfaces pseudo-ethernet <interface> dhcpv6-options temporary</i>	interfaces_pseudo-ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> dhcpv6-options temporary	interfaces pseudo-ethernet <text> dhcpv6-options temporary
✓	<i>set interfaces pseudo-ethernet <interface> disable</i>	interfaces_pseudo-ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> disable	interfaces pseudo-ethernet <text> disable
×	<i>set interfaces pseudo-ethernet <interface> disable-flow-control</i>	Nothing found in XML Definitions	Nothing found in VyOS
✓	<i>set interfaces pseudo-ethernet <interface> disable-link-detect</i>	interfaces_pseudo-ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> disable-link-detect	interfaces pseudo-ethernet <text> disable-link-detect
✓	<i>set interfaces pseudo-ethernet <interface> ip adjust-mss <mss / clamp-mss-to-pmtu></i>	interfaces_pseudo-ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> ip adjust-mss	interfaces pseudo-ethernet <text> ip adjust-mss <value>
✓	<i>set interfaces pseudo-ethernet <interface> ip arp-cache-timeout</i>	interfaces_pseudo-ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> ip arp-cache-timeout	interfaces pseudo-ethernet <text> ip arp-cache-timeout <value>
✓	<i>set interfaces pseudo-ethernet <interface> ip disable-arp-filter</i>	interfaces_pseudo-ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> ip disable-arp-filter	interfaces pseudo-ethernet <text> ip disable-arp-filter

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
✓	<i>set interfaces pseudo-ethernet <interface> ip disable-forwarding</i>	interfaces_pseudo-ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> ip disable-forwarding	interfaces pseudo-ethernet <text> ip disable-forwarding
✓	<i>set interfaces pseudo-ethernet <interface> ip enable-arp-accept</i>	interfaces_pseudo-ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> ip enable-arp-accept	interfaces pseudo-ethernet <text> ip enable-arp-accept
✓	<i>set interfaces pseudo-ethernet <interface> ip enable-arp-announce</i>	interfaces_pseudo-ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> ip enable-arp-announce	interfaces pseudo-ethernet <text> ip enable-arp-announce
✓	<i>set interfaces pseudo-ethernet <interface> ip enable-arp-ignore</i>	interfaces_pseudo-ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> ip enable-arp-ignore	interfaces pseudo-ethernet <text> ip enable-arp-ignore
✓	<i>set interfaces pseudo-ethernet <interface> ip enable-directed-broadcast</i>	interfaces_pseudo-ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> ip enable-directed-broadcast	interfaces pseudo-ethernet <text> ip enable-directed-broadcast
✓	<i>set interfaces pseudo-ethernet <interface> ip enable-proxy-arp</i>	interfaces_pseudo-ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> ip enable-proxy-arp	interfaces pseudo-ethernet <text> ip enable-proxy-arp
✓	<i>set interfaces pseudo-ethernet <interface> ip proxy-arp-pvlan</i>	interfaces_pseudo-ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> ip proxy-arp-pvlan	interfaces pseudo-ethernet <text> ip proxy-arp-pvlan
✓	<i>set interfaces pseudo-ethernet <interface> ip source-validation <strict loose disable></i>	interfaces_pseudo-ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> ip source-validation	interfaces pseudo-ethernet <text> ip source-validation <value>
✓	<i>set interfaces pseudo-ethernet <interface> ipv6 accept-dad <0 1 2></i>	interfaces_pseudo-ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> ipv6 accept-dad	interfaces pseudo-ethernet <text> ipv6 accept-dad <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
✓	<i>set interfaces pseudo-ethernet <interface> ipv6 address autoconf</i>	interfaces_pseudo-ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> ipv6 address autoconf	interfaces pseudo-ethernet <text> ipv6 address auto-conf
✓	<i>set interfaces pseudo-ethernet <interface> ipv6 address eui64 <prefix></i>	interfaces_pseudo-ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> ipv6 address eui64	interfaces pseudo-ethernet <text> ipv6 address eui64 <value>
✓	<i>set interfaces pseudo-ethernet <interface> ipv6 address no-default-link-local</i>	interfaces_pseudo-ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> ipv6 address no-default-link-local	interfaces pseudo-ethernet <text> ipv6 address no-default-link-local
✓	<i>set interfaces pseudo-ethernet <interface> ipv6 adjust-mss <mss / clamp-mss-to-pmtu></i>	interfaces_pseudo-ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> ipv6 adjust-mss	interfaces pseudo-ethernet <text> ipv6 adjust-mss <value>
×	not yet documented	interfaces_pseudo-ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> ipv6 base-reachable-time	Nothing found in VyOS
✓	<i>set interfaces pseudo-ethernet <interface> ipv6 disable-forwarding</i>	interfaces_pseudo-ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> ipv6 disable-forwarding	interfaces pseudo-ethernet <text> ipv6 disable-forwarding
✓	<i>set interfaces pseudo-ethernet <interface> ipv6 dup-addr-detect-transmits <n></i>	interfaces_pseudo-ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> ipv6 dup-addr-detect-transmits	interfaces pseudo-ethernet <text> ipv6 dup-addr-detect-transmits <value>
×	not yet documented	interfaces_pseudo-ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> ipv6 source-validation	Nothing found in VyOS
✓	<i>set interfaces pseudo-ethernet <interface> mac <xx:xx:xx:xx:xx:xx></i>	interfaces_pseudo-ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> mac	interfaces pseudo-ethernet <text> mac <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	interfaces_pseudo-ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> mirror egress	interfaces pseudo-ethernet <text> mirror egress <value>
×	not yet documented	interfaces_pseudo-ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> mirror ingress	interfaces pseudo-ethernet <text> mirror ingress <value>
×	not yet documented	interfaces_pseudo-ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> mode	interfaces pseudo-ethernet <text> mode <value>
✓	<i>set interfaces pseudo-ethernet <interface> mtu <mtu></i>	interfaces_pseudo-ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> mtu	interfaces pseudo-ethernet <text> mtu <value>
×	not yet documented	interfaces_pseudo-ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> redirect	interfaces pseudo-ethernet <text> redirect <value>
×	not yet documented	interfaces_pseudo-ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> source-interface	interfaces pseudo-ethernet <text> source-interface <value>
×	<i>set interfaces pseudo-ethernet <interface> vif <vlan-id></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	interfaces_pseudo-ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> vif-s <vif-s> address	interfaces pseudo-ethernet <text> vif-s <text> address <value>
×	not yet documented	interfaces_pseudo-ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> vif-s <vif-s> description	interfaces pseudo-ethernet <text> vif-s <text> description <value>
×	not yet documented	interfaces_pseudo-ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> vif-s <vif-s> dhcp-options client-id	interfaces pseudo-ethernet <text> vif-s <text> dhcp-options client-id <value>

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Table 1 – continued from previous page

Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	interfaces_pseudo-ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> vif-s <vif-s> dhcp-options default-route-distance	interfaces pseudo-ethernet <text> vif-s <text> dhcp-options default- route-distance <value>
×	not yet documented	interfaces_pseudo-ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> vif-s <vif-s> dhcp-options host-name	interfaces pseudo-ethernet <text> vif-s <text> dhcp-options host-name <value>
×	not yet documented	interfaces_pseudo-ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> vif-s <vif-s> dhcp-options mtu	interfaces pseudo-ethernet <text> vif-s <text> dhcp- options mtu
×	not yet documented	interfaces_pseudo-ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> vif-s <vif-s> dhcp-options no-default-route	interfaces pseudo-ethernet <text> vif-s <text> dhcp- options no-default-route
×	not yet documented	interfaces_pseudo-ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> vif-s <vif-s> dhcp-options reject	interfaces pseudo-ethernet <text> vif-s <text> dhcp- options reject <value>
×	not yet documented	interfaces_pseudo-ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> vif-s <vif-s> dhcp-options user-class	Nothing found in VyOS
×	not yet documented	interfaces_pseudo-ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> vif-s <vif-s> dhcp-options vendor-class-id	interfaces pseudo-ethernet <text> vif-s <text> dhcp- options vendor-class-id <value>
×	not yet documented	interfaces_pseudo-ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> vif-s <vif-s> dhcpv6-options duid	interfaces pseudo-ethernet <text> vif-s <text> dhcpv6-options duid <value>

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Table 1 – continued from previous page

Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	interfaces_pseudo-ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> vif-s <vif-s> dhcpv6-options no-release	interfaces pseudo-ethernet <text> vif-s <text> dhcpv6-options no-release
×	not yet documented	interfaces_pseudo-ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> vif-s <vif-s> dhcpv6-options parameters-only	interfaces pseudo-ethernet <text> vif-s <text> dhcpv6-options parameters-only
×	not yet documented	interfaces_pseudo-ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> vif-s <vif-s> dhcpv6-options pd <pd> interface <inter- face> address	interfaces pseudo-ethernet <text> vif-s <text> dhcpv6-options pd <text> interface <text> address <value>
×	not yet documented	interfaces_pseudo-ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> vif-s <vif-s> dhcpv6-options pd <pd> interface <inter- face> sla-id	interfaces pseudo-ethernet <text> vif-s <text> dhcpv6-options pd <text> interface <text> sla-id <value>
×	not yet documented	interfaces_pseudo-ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> vif-s <vif-s> dhcpv6-options pd <pd> length	interfaces pseudo-ethernet <text> vif-s <text> dhcpv6-options pd <text> length <value>
×	not yet documented	interfaces_pseudo-ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> vif-s <vif-s> dhcpv6-options rapid-commit	interfaces pseudo-ethernet <text> vif-s <text> dhcpv6-options rapid- commit
×	not yet documented	interfaces_pseudo-ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> vif-s <vif-s> dhcpv6-options temporary	interfaces pseudo-ethernet <text> vif-s <text> dhcpv6-options tempo- rary
×	not yet documented	interfaces_pseudo-ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> vif-s <vif-s> disable	interfaces pseudo-ethernet <text> vif-s <text> disable

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	interfaces_pseudo-ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> vif-s <vif-s> disable-link-detect	interfaces pseudo-ethernet <text> vif-s <text> disable-link-detect
×	not yet documented	interfaces_pseudo-ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> vif-s <vif-s> ip adjust-mss	interfaces pseudo-ethernet <text> vif-s <text> ip adjust-mss <value>
×	not yet documented	interfaces_pseudo-ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> vif-s <vif-s> ip arp-cache-timeout	interfaces pseudo-ethernet <text> vif-s <text> ip arp- cache-timeout <value>
×	not yet documented	interfaces_pseudo-ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> vif-s <vif-s> ip disable-arp-filter	interfaces pseudo-ethernet <text> vif-s <text> ip disable-arp-filter
×	not yet documented	interfaces_pseudo-ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> vif-s <vif-s> ip disable-forwarding	interfaces pseudo-ethernet <text> vif-s <text> ip disable-forwarding
×	not yet documented	interfaces_pseudo-ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> vif-s <vif-s> ip enable-arp-accept	interfaces pseudo-ethernet <text> vif-s <text> ip enable-arp-accept
×	not yet documented	interfaces_pseudo-ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> vif-s <vif-s> ip enable-arp-announce	interfaces pseudo-ethernet <text> vif-s <text> ip enable-arp-announce
×	not yet documented	interfaces_pseudo-ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> vif-s <vif-s> ip enable-arp-ignore	interfaces pseudo-ethernet <text> vif-s <text> ip enable-arp-ignore

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	interfaces_pseudo-ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> vif-s <vif-s> ip enable-directed-broadcast	interfaces pseudo-ethernet <text> vif-s <text> ip enable-directed-broadcast
×	not yet documented	interfaces_pseudo-ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> vif-s <vif-s> ip enable-proxy-arp	interfaces pseudo-ethernet <text> vif-s <text> ip enable-proxy-arp
×	not yet documented	interfaces_pseudo-ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> vif-s <vif-s> ip proxy-arp-pvlan	interfaces pseudo-ethernet <text> vif-s <text> ip proxy-arp-pvlan
×	not yet documented	interfaces_pseudo-ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> vif-s <vif-s> ip source-validation	interfaces pseudo-ethernet <text> vif-s <text> ip source-validation <value>
×	not yet documented	interfaces_pseudo-ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> vif-s <vif-s> ipv6 accept-dad	interfaces pseudo-ethernet <text> vif-s <text> ipv6 accept-dad <value>
×	not yet documented	interfaces_pseudo-ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> vif-s <vif-s> ipv6 address autoconf	interfaces pseudo-ethernet <text> vif-s <text> ipv6 address autoconf
×	not yet documented	interfaces_pseudo-ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> vif-s <vif-s> ipv6 address eui64	interfaces pseudo-ethernet <text> vif-s <text> ipv6 address eui64 <value>
×	not yet documented	interfaces_pseudo-ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> vif-s <vif-s> ipv6 address no-default-link-local	interfaces pseudo-ethernet <text> vif-s <text> ipv6 address no-default-link-local

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Table 1 – continued from previous page

Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	interfaces_pseudo-ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> vif-s <vif-s> ipv6 adjust-mss	interfaces pseudo-ethernet <text> vif-s <text> ipv6 adjust-mss <value>
×	not yet documented	interfaces_pseudo-ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> vif-s <vif-s> ipv6 base-reachable-time	Nothing found in VyOS
×	not yet documented	interfaces_pseudo-ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> vif-s <vif-s> ipv6 disable-forwarding	interfaces pseudo-ethernet <text> vif-s <text> ipv6 disable-forwarding
×	not yet documented	interfaces_pseudo-ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> vif-s <vif-s> ipv6 dup-addr-detect-transmits	interfaces pseudo-ethernet <text> vif-s <text> ipv6 dup-addr-detect-transmits <value>
×	not yet documented	interfaces_pseudo-ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> vif-s <vif-s> ipv6 source-validation	Nothing found in VyOS
×	not yet documented	interfaces_pseudo-ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> vif-s <vif-s> mac	interfaces pseudo-ethernet <text> vif-s <text> mac <value>
×	not yet documented	interfaces_pseudo-ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> vif-s <vif-s> mirror egress	interfaces pseudo-ethernet <text> vif-s <text> mirror egress <value>
×	not yet documented	interfaces_pseudo-ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> vif-s <vif-s> mirror ingress	interfaces pseudo-ethernet <text> vif-s <text> mirror ingress <value>
×	not yet documented	interfaces_pseudo-ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> vif-s <vif-s> mtu	interfaces pseudo-ethernet <text> vif-s <text> mtu <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	interfaces_pseudo-ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> vif-s <vif-s> protocol	interfaces pseudo-ethernet <text> vif-s <text> proto- col <value>
×	not yet documented	interfaces_pseudo-ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> vif-s <vif-s> redirect	interfaces pseudo-ethernet <text> vif-s <text> redi- rect <value>
×	not yet documented	interfaces_pseudo-ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> vif-s <vif-s> vif-c <vif-c> address	interfaces pseudo-ethernet <text> vif-s <text> vif-c <text> address <value>
×	not yet documented	interfaces_pseudo-ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> vif-s <vif-s> vif-c <vif-c> description	interfaces pseudo-ethernet <text> vif-s <text> vif- c <text> description <value>
×	not yet documented	interfaces_pseudo-ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> vif-s <vif-s> vif-c <vif-c> dhcp-options client-id	interfaces pseudo-ethernet <text> vif-s <text> vif- c <text> dhcp-options client-id <value>
×	not yet documented	interfaces_pseudo-ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> vif-s <vif-s> vif-c <vif-c> dhcp-options default- route-distance	interfaces pseudo-ethernet <text> vif-s <text> vif- c <text> dhcp-options default-route-distance <value>
×	not yet documented	interfaces_pseudo-ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> vif-s <vif-s> vif-c <vif-c> dhcp-options host-name	interfaces pseudo-ethernet <text> vif-s <text> vif-c <text> dhcp-options host- name <value>
×	not yet documented	interfaces_pseudo-ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> vif-s <vif-s> vif-c <vif-c> dhcp-options mtu	interfaces pseudo-ethernet <text> vif-s <text> vif-c <text> dhcp-options mtu

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	interfaces_pseudo-ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> vif-s <vif-s> vif-c <vif-c> dhcp-options no-default-route	interfaces pseudo-ethernet <text> vif-s <text> vif-c <text> dhcp-options no-default-route
×	not yet documented	interfaces_pseudo-ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> vif-s <vif-s> vif-c <vif-c> dhcp-options reject	interfaces pseudo-ethernet <text> vif-s <text> vif-c <text> dhcp-options reject <value>
×	not yet documented	interfaces_pseudo-ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> vif-s <vif-s> vif-c <vif-c> dhcp-options user-class	Nothing found in VyOS
×	not yet documented	interfaces_pseudo-ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> vif-s <vif-s> vif-c <vif-c> dhcp-options vendor-class-id	interfaces pseudo-ethernet <text> vif-s <text> vif-c <text> dhcp-options vendor-class-id <value>
×	not yet documented	interfaces_pseudo-ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> vif-s <vif-s> vif-c <vif-c> dhcpv6-options duid	interfaces pseudo-ethernet <text> vif-s <text> vif-c <text> dhcpv6-options duid <value>
×	not yet documented	interfaces_pseudo-ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> vif-s <vif-s> vif-c <vif-c> dhcpv6-options no-release	interfaces pseudo-ethernet <text> vif-s <text> vif-c <text> dhcpv6-options no-release
×	not yet documented	interfaces_pseudo-ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> vif-s <vif-s> vif-c <vif-c> dhcpv6-options parameters-only	interfaces pseudo-ethernet <text> vif-s <text> vif-c <text> dhcpv6-options parameters-only

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	interfaces_pseudo-ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> vif-s <vif-s> vif-c <vif-c> dhcpv6-options pd <pd> interface <interface> address	interfaces pseudo-ethernet <text> vif-s <text> vif-c <text> dhcpv6-options pd <text> interface <text> address <value>
×	not yet documented	interfaces_pseudo-ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> vif-s <vif-s> vif-c <vif-c> dhcpv6-options pd <pd> interface <interface> sla-id	interfaces pseudo-ethernet <text> vif-s <text> vif-c <text> dhcpv6-options pd <text> interface <text> sla-id <value>
×	not yet documented	interfaces_pseudo-ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> vif-s <vif-s> vif-c <vif-c> dhcpv6-options pd <pd> length	interfaces pseudo-ethernet <text> vif-s <text> vif-c <text> dhcpv6-options pd <text> length <value>
×	not yet documented	interfaces_pseudo-ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> vif-s <vif-s> vif-c <vif-c> dhcpv6-options rapid-commit	interfaces pseudo-ethernet <text> vif-s <text> vif-c <text> dhcpv6-options rapid-commit
×	not yet documented	interfaces_pseudo-ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> vif-s <vif-s> vif-c <vif-c> dhcpv6-options temporary	interfaces pseudo-ethernet <text> vif-s <text> vif-c <text> dhcpv6-options temporary
×	not yet documented	interfaces_pseudo-ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> vif-s <vif-s> vif-c <vif-c> disable	interfaces pseudo-ethernet <text> vif-s <text> vif-c <text> disable
×	not yet documented	interfaces_pseudo-ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> vif-s <vif-s> vif-c <vif-c> disable-link-detect	interfaces pseudo-ethernet <text> vif-s <text> vif-c <text> disable-link-detect

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	interfaces_pseudo-ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> vif-s <vif-s> vif-c <vif-c> ip adjust-mss	interfaces pseudo-ethernet <text> vif-s <text> vif-c <text> ip adjust-mss <value>
×	not yet documented	interfaces_pseudo-ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> vif-s <vif-s> vif-c <vif-c> ip arp-cache-timeout	interfaces pseudo-ethernet <text> vif-s <text> vif-c <text> ip arp-cache- timeout <value>
×	not yet documented	interfaces_pseudo-ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> vif-s <vif-s> vif-c <vif-c> ip disable-arp-filter	interfaces pseudo-ethernet <text> vif-s <text> vif-c <text> ip disable-arp-filter
×	not yet documented	interfaces_pseudo-ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> vif-s <vif-s> vif-c <vif-c> ip disable-forwarding	interfaces pseudo-ethernet <text> vif-s <text> vif-c <text> ip disable- forwarding
×	not yet documented	interfaces_pseudo-ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> vif-s <vif-s> vif-c <vif-c> ip enable-arp-accept	interfaces pseudo-ethernet <text> vif-s <text> vif- c <text> ip enable-arp- accept
×	not yet documented	interfaces_pseudo-ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> vif-s <vif-s> vif-c <vif-c> ip enable-arp-announce	interfaces pseudo-ethernet <text> vif-s <text> vif- c <text> ip enable-arp- announce
×	not yet documented	interfaces_pseudo-ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> vif-s <vif-s> vif-c <vif-c> ip enable-arp-ignore	interfaces pseudo-ethernet <text> vif-s <text> vif- c <text> ip enable-arp- ignore
×	not yet documented	interfaces_pseudo-ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> vif-s <vif-s> vif-c <vif-c> ip enable-directed-broadcast	interfaces pseudo-ethernet <text> vif-s <text> vif-c <text> ip enable-directed- broadcast

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Table 1 – continued from previous page

Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	interfaces_pseudo-ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> vif-s <vif-s> vif-c <vif-c> ip enable-proxy-arp	interfaces pseudo-ethernet <text> vif-s <text> vif- c <text> ip enable-proxy- arp
×	not yet documented	interfaces_pseudo-ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> vif-s <vif-s> vif-c <vif-c> ip proxy-arp-pvlan	interfaces pseudo-ethernet <text> vif-s <text> vif-c <text> ip proxy-arp-pvlan
×	not yet documented	interfaces_pseudo-ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> vif-s <vif-s> vif-c <vif-c> ip source-validation	interfaces pseudo-ethernet <text> vif-s <text> vif-c <text> ip source- validation <value>
×	not yet documented	interfaces_pseudo-ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> vif-s <vif-s> vif-c <vif-c> ipv6 accept-dad	interfaces pseudo-ethernet <text> vif-s <text> vif- c <text> ipv6 accept-dad <value>
×	not yet documented	interfaces_pseudo-ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> vif-s <vif-s> vif-c <vif-c> ipv6 address autoconf	interfaces pseudo-ethernet <text> vif-s <text> vif-c <text> ipv6 address auto- conf
×	not yet documented	interfaces_pseudo-ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> vif-s <vif-s> vif-c <vif-c> ipv6 address eui64	interfaces pseudo-ethernet <text> vif-s <text> vif-c <text> ipv6 address eui64 <value>
×	not yet documented	interfaces_pseudo-ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> vif-s <vif-s> vif-c <vif-c> ipv6 address no-default-link- local	interfaces pseudo-ethernet <text> vif-s <text> vif-c <text> ipv6 address no- default-link-local
×	not yet documented	interfaces_pseudo-ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> vif-s <vif-s> vif-c <vif-c> ipv6 adjust-mss	interfaces pseudo-ethernet <text> vif-s <text> vif- c <text> ipv6 adjust-mss <value>

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Table 1 – continued from previous page

Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	interfaces_pseudo-ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> vif-s <vif-s> vif-c <vif-c> ipv6 base-reachable-time	Nothing found in VyOS
×	not yet documented	interfaces_pseudo-ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> vif-s <vif-s> vif-c <vif-c> ipv6 disable-forwarding	interfaces pseudo-ethernet <text> vif-s <text> vif-c <text> ipv6 disable-forwarding
×	not yet documented	interfaces_pseudo-ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> vif-s <vif-s> vif-c <vif-c> ipv6 dup-addr-detect-transmits	interfaces pseudo-ethernet <text> vif-s <text> vif-c <text> ipv6 dup-addr-detect-transmits <value>
×	not yet documented	interfaces_pseudo-ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> vif-s <vif-s> vif-c <vif-c> ipv6 source-validation	Nothing found in VyOS
×	not yet documented	interfaces_pseudo-ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> vif-s <vif-s> vif-c <vif-c> mac	interfaces pseudo-ethernet <text> vif-s <text> vif-c <text> mac <value>
×	not yet documented	interfaces_pseudo-ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> vif-s <vif-s> vif-c <vif-c> mirror egress	interfaces pseudo-ethernet <text> vif-s <text> vif-c <text> mirror egress <value>
×	not yet documented	interfaces_pseudo-ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> vif-s <vif-s> vif-c <vif-c> mirror ingress	interfaces pseudo-ethernet <text> vif-s <text> vif-c <text> mirror ingress <value>
×	not yet documented	interfaces_pseudo-ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> vif-s <vif-s> vif-c <vif-c> mtu	interfaces pseudo-ethernet <text> vif-s <text> vif-c <text> mtu <value>

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Table 1 – continued from previous page

Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	interfaces_pseudo-ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> vif-s <vif-s> vif-c <vif-c> redirect	interfaces pseudo-ethernet <text> vif-s <text> vif-c <text> redirect <value>
×	not yet documented	interfaces_pseudo-ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> vif-s <vif-s> vif-c <vif-c> vrf	interfaces pseudo-ethernet <text> vif-s <text> vif-c <text> vrf <value>
×	not yet documented	interfaces_pseudo-ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> vif-s <vif-s> vrf	interfaces pseudo-ethernet <text> vif-s <text> vrf <value>
✓	<i>set interfaces pseudo-ethernet <interface> vif <vlan-id> address <address dhcp dhcpv6></i>	interfaces_pseudo-ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> vif <vif> address	interfaces pseudo-ethernet <text> vif <text> address <value>
✓	<i>set interfaces pseudo-ethernet <interface> vif <vlan-id> description <description></i>	interfaces_pseudo-ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> vif <vif> description	interfaces pseudo-ethernet <text> vif <text> descrip- tion <value>
✓	<i>set interfaces pseudo-ethernet <interface> vif <vlan-id> dhcp-options client-id <description></i>	interfaces_pseudo-ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> vif <vif> dhcp-options client-id	interfaces pseudo-ethernet <text> vif <text> dhcp- options client-id <value>
✓	<i>set interfaces pseudo-ethernet <interface> vif <vlan-id> dhcp-options default-route-distanc <distance></i>	interfaces_pseudo-ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> vif <vif> dhcp-options default-route-distance	interfaces pseudo-ethernet <text> vif <text> dhcp- options default-route- distance <value>
✓	<i>set interfaces pseudo-ethernet <interface> vif <vlan-id> dhcp-options host-name <hostname></i>	interfaces_pseudo-ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> vif <vif> dhcp-options host-name	interfaces pseudo-ethernet <text> vif <text> dhcp- options host-name <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	interfaces_pseudo-ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> vif <vif> dhcp-options mtu	interfaces pseudo-ethernet <text> vif <text> dhcp-options mtu
✓	<i>set interfaces pseudo-ethernet <interface> vif <vlan-id> dhcp-options no-default-route</i>	interfaces_pseudo-ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> vif <vif> dhcp-options no-default-route	interfaces pseudo-ethernet <text> vif <text> dhcp-options no-default-route
✓	<i>set interfaces pseudo-ethernet <interface> vif <vlan-id> dhcp-options reject <address></i>	interfaces_pseudo-ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> vif <vif> dhcp-options reject	interfaces pseudo-ethernet <text> vif <text> dhcp-options reject <value>
×	<i>set interfaces pseudo-ethernet <interface> vif <vlan-id> dhcp-options user-class <string></i>	interfaces_pseudo-ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> vif <vif> dhcp-options user-class	Nothing found in VyOS
✓	<i>set interfaces pseudo-ethernet <interface> vif <vlan-id> dhcp-options vendor-class-id <vendor-id></i>	interfaces_pseudo-ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> vif <vif> dhcp-options vendor-class-id	interfaces pseudo-ethernet <text> vif <text> dhcp-options vendor-class-id <value>
✓	<i>set interfaces pseudo-ethernet <interface> vif <vlan-id> dhcpv6-options duid <duid></i>	interfaces_pseudo-ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> vif <vif> dhcpv6-options duid	interfaces pseudo-ethernet <text> vif <text> dhcpv6-options duid <value>
✓	<i>set interfaces pseudo-ethernet <interface> vif <vlan-id> dhcpv6-options no-release</i>	interfaces_pseudo-ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> vif <vif> dhcpv6-options no-release	interfaces pseudo-ethernet <text> vif <text> dhcpv6-options no-release
✓	<i>set interfaces pseudo-ethernet <interface> vif <vlan-id> dhcpv6-options parameters-only</i>	interfaces_pseudo-ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> vif <vif> dhcpv6-options parameters-only	interfaces pseudo-ethernet <text> vif <text> dhcpv6-options parameters-only

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
✓	<i>set interfaces pseudo-ethernet <interface> vif <vlan-id> dhcpv6-options pd <id> interface <delegatee> address <address></i>	interfaces_pseudo-ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> vif <vif> dhcpv6-options pd <pd> interface <interface> address	interfaces pseudo-ethernet <text> vif <text> dhcpv6-options pd <text> interface <text> address <value>
✓	<i>set interfaces pseudo-ethernet <interface> vif <vlan-id> dhcpv6-options pd <id> interface <delegatee> sla-id <id></i>	interfaces_pseudo-ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> vif <vif> dhcpv6-options pd <pd> interface <interface> sla-id	interfaces pseudo-ethernet <text> vif <text> dhcpv6-options pd <text> interface <text> sla-id <value>
✓	<i>set interfaces pseudo-ethernet <interface> vif <vlan-id> dhcpv6-options pd <id> length <length></i>	interfaces_pseudo-ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> vif <vif> dhcpv6-options pd <pd> length	interfaces pseudo-ethernet <text> vif <text> dhcpv6-options pd <text> length <value>
✓	<i>set interfaces pseudo-ethernet <interface> vif <vlan-id> dhcpv6-options rapid-commit</i>	interfaces_pseudo-ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> vif <vif> dhcpv6-options rapid-commit	interfaces pseudo-ethernet <text> vif <text> dhcpv6-options rapid-commit
✓	<i>set interfaces pseudo-ethernet <interface> vif <vlan-id> dhcpv6-options temporary</i>	interfaces_pseudo-ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> vif <vif> dhcpv6-options temporary	interfaces pseudo-ethernet <text> vif <text> dhcpv6-options temporary
✓	<i>set interfaces pseudo-ethernet <interface> vif <vlan-id> disable</i>	interfaces_pseudo-ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> vif <vif> disable	interfaces pseudo-ethernet <text> vif <text> disable
✓	<i>set interfaces pseudo-ethernet <interface> vif <vlan-id> disable-link-detect</i>	interfaces_pseudo-ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> vif <vif> disable-link-detect	interfaces pseudo-ethernet <text> vif <text> disable-link-detect

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	interfaces_pseudo-ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> vif <vif> egress-qos	interfaces pseudo-ethernet <text> vif <text> egress-qos <value>
×	not yet documented	interfaces_pseudo-ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> vif <vif> ingress-qos	interfaces pseudo-ethernet <text> vif <text> ingress-qos <value>
✓	<i>set interfaces pseudo-ethernet <interface> vif <vlan-id> ip adjust-mss <mss / clamp-mss-to-pmtu></i>	interfaces_pseudo-ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> vif <vif> ip adjust-mss	interfaces pseudo-ethernet <text> vif <text> ip adjust-mss <value>
✓	<i>set interfaces pseudo-ethernet <interface> vif <vlan-id> ip arp-cache-timeout</i>	interfaces_pseudo-ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> vif <vif> ip arp-cache-timeout	interfaces pseudo-ethernet <text> vif <text> ip arp-cache-timeout <value>
✓	<i>set interfaces pseudo-ethernet <interface> vif <vlan-id> ip disable-arp-filter</i>	interfaces_pseudo-ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> vif <vif> ip disable-arp-filter	interfaces pseudo-ethernet <text> vif <text> ip disable-arp-filter
✓	<i>set interfaces pseudo-ethernet <interface> vif <vlan-id> ip disable-forwarding</i>	interfaces_pseudo-ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> vif <vif> ip disable-forwarding	interfaces pseudo-ethernet <text> vif <text> ip disable-forwarding
✓	<i>set interfaces pseudo-ethernet <interface> vif <vlan-id> ip enable-arp-accept</i>	interfaces_pseudo-ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> vif <vif> ip enable-arp-accept	interfaces pseudo-ethernet <text> vif <text> ip enable-arp-accept
✓	<i>set interfaces pseudo-ethernet <interface> vif <vlan-id> ip enable-arp-announce</i>	interfaces_pseudo-ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> vif <vif> ip enable-arp-announce	interfaces pseudo-ethernet <text> vif <text> ip enable-arp-announce

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
✓	<code>set interfaces pseudo-ethernet <interface> vif <vlan-id> ip enable-arp-ignore</code>	interfaces_pseudo- ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> vif <vif> ip enable-arp- ignore	interfaces pseudo-ethernet <text> vif <text> ip enable-arp-ignore
✓	<code>set interfaces pseudo-ethernet <interface> vif <vlan-id> ip enable-directed-broad</code>	interfaces_pseudo- ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> vif <vif> ip enable-directed- broadcast	interfaces pseudo-ethernet <text> vif <text> ip enable-directed-broadcast
✓	<code>set interfaces pseudo-ethernet <interface> vif <vlan-id> ip enable-proxy-arp</code>	interfaces_pseudo- ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> vif <vif> ip enable-proxy-arp	interfaces pseudo-ethernet <text> vif <text> ip enable-proxy-arp
✓	<code>set interfaces pseudo-ethernet <interface> vif <vlan-id> ip proxy-arp-pvlan</code>	interfaces_pseudo- ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> vif <vif> ip proxy-arp-pvlan	interfaces pseudo-ethernet <text> vif <text> ip proxy- arp-pvlan
✓	<code>set interfaces pseudo-ethernet <interface> vif <vlan-id> ip source-validation <strict loose disable></code>	interfaces_pseudo- ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> vif <vif> ip source-validation	interfaces pseudo-ethernet <text> vif <text> ip source-validation <value>
✓	<code>set interfaces pseudo-ethernet <interface> vif <vlan-id> ipv6 accept-dad <0 1 2></code>	interfaces_pseudo- ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> vif <vif> ipv6 accept-dad	interfaces pseudo-ethernet <text> vif <text> ipv6 accept-dad <value>
✓	<code>set interfaces pseudo-ethernet <interface> vif <vlan-id> ipv6 address autoconf</code>	interfaces_pseudo- ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> vif <vif> ipv6 address auto- conf	interfaces pseudo-ethernet <text> vif <text> ipv6 ad- dress autoconf
✓	<code>set interfaces pseudo-ethernet <interface> vif <vlan-id> ipv6 address eui64 <prefix></code>	interfaces_pseudo- ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> vif <vif> ipv6 address eui64	interfaces pseudo-ethernet <text> vif <text> ipv6 ad- dress eui64 <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
✓	<i>set interfaces pseudo-ethernet <interface> vif <vlan-id> ipv6 address no-default-link-local</i>	interfaces_pseudo-ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> vif <vif> ipv6 address no-default-link-local	interfaces pseudo-ethernet <text> vif <text> ipv6 address no-default-link-local
✓	<i>set interfaces pseudo-ethernet <interface> vif <vlan-id> ipv6 adjust-mss <mss / clamp-mss-to-pmtu></i>	interfaces_pseudo-ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> vif <vif> ipv6 adjust-mss	interfaces pseudo-ethernet <text> vif <text> ipv6 adjust-mss <value>
×	not yet documented	interfaces_pseudo-ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> vif <vif> ipv6 base-reachable-time	Nothing found in VyOS
✓	<i>set interfaces pseudo-ethernet <interface> vif <vlan-id> ipv6 disable-forwarding</i>	interfaces_pseudo-ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> vif <vif> ipv6 disable-forwarding	interfaces pseudo-ethernet <text> vif <text> ipv6 disable-forwarding
✓	<i>set interfaces pseudo-ethernet <interface> vif <vlan-id> ipv6 dup-addr-detect-transmits <n></i>	interfaces_pseudo-ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> vif <vif> ipv6 dup-addr-detect-transmits	interfaces pseudo-ethernet <text> vif <text> ipv6 dup-addr-detect-transmits <value>
×	not yet documented	interfaces_pseudo-ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> vif <vif> ipv6 source-validation	Nothing found in VyOS
✓	<i>set interfaces pseudo-ethernet <interface> vif <vlan-id> mac <xx:xx:xx:xx:xx:xx></i>	interfaces_pseudo-ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> vif <vif> mac	interfaces pseudo-ethernet <text> vif <text> mac <value>
×	not yet documented	interfaces_pseudo-ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> vif <vif> mirror egress	interfaces pseudo-ethernet <text> vif <text> mirror egress <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	interfaces_pseudo-ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> vif <vif> mirror ingress	interfaces pseudo-ethernet <text> vif <text> mirror ingress <value>
✓	<i>set interfaces pseudo-ethernet <interface> vif <vlan-id> mtu <mtu></i>	interfaces_pseudo-ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> vif <vif> mtu	interfaces pseudo-ethernet <text> vif <text> mtu <value>
×	not yet documented	interfaces_pseudo-ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> vif <vif> redirect	interfaces pseudo-ethernet <text> vif <text> redirect <value>
✓	<i>set interfaces pseudo-ethernet <interface> vif <vlan-id> vrf <vrf></i>	interfaces_pseudo-ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> vif <vif> vrf	interfaces pseudo-ethernet <text> vif <text> vrf <value>
✓	<i>set interfaces pseudo-ethernet <interface> vrf <vrf></i>	interfaces_pseudo-ethernet.xml.in: interfaces pseudo-ethernet <pseudo-ethernet> vrf	interfaces pseudo-ethernet <text> vrf <value>
×	<i>set interfaces sstpc <interface> default-route-distance <distance></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set interfaces sstpc <interface> ip adjust-mss <mss / clamp-mss-to-pmtu></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set interfaces sstpc <interface> ip disable-forwarding</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set interfaces sstpc <interface> ip source-validation <strict loose disable></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set interfaces sstpc <interface> no-default-route</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set interfaces sstpc <interface> no-peer-dns</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set interfaces sstpc <interface> server <address></i>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	interfaces_sstpc.xml.in: interfaces sstpc <sstpc> authentication password <value>	interfaces sstpc <text> authentication password <value>
×	not yet documented	interfaces_sstpc.xml.in: interfaces sstpc <sstpc> authentication username <value>	interfaces sstpc <text> authentication username <value>
×	not yet documented	interfaces_sstpc.xml.in: interfaces sstpc <sstpc> default-route-distance <value>	interfaces sstpc <text> default-route-distance <value>
✓	set interfaces sstpc <interface> description <description>	interfaces_sstpc.xml.in: interfaces sstpc <sstpc> description	interfaces sstpc <text> de- scription <value>
✓	set interfaces sstpc <interface> disable	interfaces_sstpc.xml.in: interfaces sstpc <sstpc> disable	interfaces sstpc <text> dis- able
✓	set interfaces sstpc <interface> mtu <mtu>	interfaces_sstpc.xml.in: interfaces sstpc <sstpc> mtu	interfaces sstpc <text> mtu <value>
×	not yet documented	interfaces_sstpc.xml.in: interfaces sstpc <sstpc> no-default-route	interfaces sstpc <text> no- default-route
×	not yet documented	interfaces_sstpc.xml.in: interfaces sstpc <sstpc> no-peer-dns	interfaces sstpc <text> no- peer-dns
×	not yet documented	interfaces_sstpc.xml.in: interfaces sstpc <sstpc> port	interfaces sstpc <text> port <value>
×	not yet documented	interfaces_sstpc.xml.in: interfaces sstpc <sstpc> server	interfaces sstpc <text> server <value>
×	not yet documented	interfaces_sstpc.xml.in: interfaces sstpc <sstpc> ssl ca-certificate	interfaces sstpc <text> ssl ca-certificate <value>
✓	set interfaces sstpc <interface> vrf <vrf>	interfaces_sstpc.xml.in: interfaces sstpc <sstpc> vrf	interfaces sstpc <text> vrf <value>
×	not yet documented	interfaces_tunnel.xml.in: interfaces tunnel <tunnel> 6rd-prefix	interfaces tunnel <text> 6rd-prefix <value>
×	not yet documented	interfaces_tunnel.xml.in: interfaces tunnel <tunnel> 6rd-relay-prefix	interfaces tunnel <text> 6rd-relay-prefix <value>
✓	set interfaces tunnel <interface> address <address>	interfaces_tunnel.xml.in: interfaces tunnel <tunnel> address	interfaces tunnel <text> address <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
✓	<i>set interfaces tunnel <interface> description <description></i>	interfaces_tunnel.xml.in: interfaces tunnel <tunnel> description	interfaces tunnel <text> description <value>
✓	<i>set interfaces tunnel <interface> disable</i>	interfaces_tunnel.xml.in: interfaces tunnel <tunnel> disable	interfaces tunnel <text> disable
×	<i>set interfaces tunnel <interface> disable-flow-control</i>	Nothing found in XML Definitions	Nothing found in VyOS
✓	<i>set interfaces tunnel <interface> disable-link-detect</i>	interfaces_tunnel.xml.in: interfaces tunnel <tunnel> disable-link-detect	interfaces tunnel <text> disable-link-detect
×	not yet documented	interfaces_tunnel.xml.in: interfaces tunnel <tunnel> enable-multicast	interfaces tunnel <text> enable-multicast
×	not yet documented	interfaces_tunnel.xml.in: interfaces tunnel <tunnel> encapsulation	interfaces tunnel <text> encapsulation <value>
✓	<i>set interfaces tunnel <interface> ip adjust-mss <mss / clamp-mss-to-pmtu></i>	interfaces_tunnel.xml.in: interfaces tunnel <tunnel> ip adjust-mss	interfaces tunnel <text> ip adjust-mss <value>
✓	<i>set interfaces tunnel <interface> ip arp-cache-timeout</i>	interfaces_tunnel.xml.in: interfaces tunnel <tunnel> ip arp-cache-timeout	interfaces tunnel <text> ip arp-cache-timeout <value>
✓	<i>set interfaces tunnel <interface> ip disable-arp-filter</i>	interfaces_tunnel.xml.in: interfaces tunnel <tunnel> ip disable-arp-filter	interfaces tunnel <text> ip disable-arp-filter
✓	<i>set interfaces tunnel <interface> ip disable-forwarding</i>	interfaces_tunnel.xml.in: interfaces tunnel <tunnel> ip disable-forwarding	interfaces tunnel <text> ip disable-forwarding
✓	<i>set interfaces tunnel <interface> ip enable-arp-accept</i>	interfaces_tunnel.xml.in: interfaces tunnel <tunnel> ip enable-arp-accept	interfaces tunnel <text> ip enable-arp-accept
✓	<i>set interfaces tunnel <interface> ip enable-arp-announce</i>	interfaces_tunnel.xml.in: interfaces tunnel <tunnel> ip enable-arp-announce	interfaces tunnel <text> ip enable-arp-announce
✓	<i>set interfaces tunnel <interface> ip enable-arp-ignore</i>	interfaces_tunnel.xml.in: interfaces tunnel <tunnel> ip enable-arp-ignore	interfaces tunnel <text> ip enable-arp-ignore

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Table 1 – continued from previous page

Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
✓	<code>set interfaces tunnel <interface> ip enable-directed-broadcast</code>	interfaces_tunnel.xml.in: interfaces tunnel <tunnel> ip enable-directed-broadcast	interfaces tunnel <text> ip enable-directed-broadcast
✓	<code>set interfaces tunnel <interface> ip enable-proxy-arp</code>	interfaces_tunnel.xml.in: interfaces tunnel <tunnel> ip enable-proxy-arp	interfaces tunnel <text> ip enable-proxy-arp
✓	<code>set interfaces tunnel <interface> ip proxy-arp-pvlan</code>	interfaces_tunnel.xml.in: interfaces tunnel <tunnel> ip proxy-arp-pvlan	interfaces tunnel <text> ip proxy-arp-pvlan
✓	<code>set interfaces tunnel <interface> ip source-validation <strict loose disable></code>	interfaces_tunnel.xml.in: interfaces tunnel <tunnel> ip source-validation	interfaces tunnel <text> ip source-validation <value>
✓	<code>set interfaces tunnel <interface> ipv6 accept-dad <0 1 2></code>	interfaces_tunnel.xml.in: interfaces tunnel <tunnel> ipv6 accept-dad	interfaces tunnel <text> ipv6 accept-dad <value>
✓	<code>set interfaces tunnel <interface> ipv6 address autoconf</code>	interfaces_tunnel.xml.in: interfaces tunnel <tunnel> ipv6 address autoconf	interfaces tunnel <text> ipv6 address autoconf
✓	<code>set interfaces tunnel <interface> ipv6 address eui64 <prefix></code>	interfaces_tunnel.xml.in: interfaces tunnel <tunnel> ipv6 address eui64	interfaces tunnel <text> ipv6 address eui64 <value>
✓	<code>set interfaces tunnel <interface> ipv6 address no-default-link-local</code>	interfaces_tunnel.xml.in: interfaces tunnel <tunnel> ipv6 address no-default-link-local	interfaces tunnel <text> ipv6 address no-default-link-local
✓	<code>set interfaces tunnel <interface> ipv6 adjust-mss <mss clamp-mss-to-pmtu></code>	interfaces_tunnel.xml.in: interfaces tunnel <tunnel> ipv6 adjust-mss	interfaces tunnel <text> ipv6 adjust-mss <value>
×	not yet documented	interfaces_tunnel.xml.in: interfaces tunnel <tunnel> ipv6 base-reachable-time	Nothing found in VyOS
✓	<code>set interfaces tunnel <interface> ipv6 disable-forwarding</code>	interfaces_tunnel.xml.in: interfaces tunnel <tunnel> ipv6 disable-forwarding	interfaces tunnel <text> ipv6 disable-forwarding
✓	<code>set interfaces tunnel <interface> ipv6 dup-addr-detect-transmits <n></code>	interfaces_tunnel.xml.in: interfaces tunnel <tunnel> ipv6 dup-addr-detect-transmits	interfaces tunnel <text> ipv6 dup-addr-detect-transmits <value>

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Table 1 – continued from previous page

Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	interfaces_tunnel.xml.in: interfaces tunnel <tunnel> ipv6 source-validation	Nothing found in VyOS
×	not yet documented	interfaces_tunnel.xml.in: interfaces tunnel <tunnel> mirror egress	interfaces tunnel <text> mirror egress <value>
×	not yet documented	interfaces_tunnel.xml.in: interfaces tunnel <tunnel> mirror ingress	interfaces tunnel <text> mirror ingress <value>
✓	set interfaces tunnel <interface> mtu <mtu>	interfaces_tunnel.xml.in: interfaces tunnel <tunnel> mtu	interfaces tunnel <text> mtu <value>
×	not yet documented	interfaces_tunnel.xml.in: interfaces tunnel <tunnel> parameters erspan direc- tion	interfaces tunnel <text> parameters erspan direc- tion <value>
×	not yet documented	interfaces_tunnel.xml.in: interfaces tunnel <tunnel> parameters erspan hw-id	interfaces tunnel <text> parameters erspan hw-id <value>
×	not yet documented	interfaces_tunnel.xml.in: interfaces tunnel <tunnel> parameters erspan index	interfaces tunnel <text> parameters erspan index <value>
×	not yet documented	interfaces_tunnel.xml.in: interfaces tunnel <tunnel> parameters erspan version	interfaces tunnel <text> parameters erspan version <value>
×	not yet documented	interfaces_tunnel.xml.in: interfaces tunnel <tunnel> parameters ip ignore-df	interfaces tunnel <text> parameters ip ignore-df
×	not yet documented	interfaces_tunnel.xml.in: interfaces tunnel <tunnel> parameters ip key	interfaces tunnel <text> parameters ip key <value>
×	not yet documented	interfaces_tunnel.xml.in: interfaces tunnel <tunnel> parameters ip no-pmtu- discovery	interfaces tunnel <text> parameters ip no-pmtu- discovery
×	not yet documented	interfaces_tunnel.xml.in: interfaces tunnel <tunnel> parameters ip tos	interfaces tunnel <text> parameters ip tos <value>
×	not yet documented	interfaces_tunnel.xml.in: interfaces tunnel <tunnel> parameters ip ttl	interfaces tunnel <text> parameters ip ttl <value>
×	not yet documented	interfaces_tunnel.xml.in: interfaces tunnel <tun- nel> parameters ipv6 en- caplimit	interfaces tunnel <text> parameters ipv6 en- caplimit <value>
×	not yet documented	interfaces_tunnel.xml.in: interfaces tunnel <tunnel> parameters ipv6 flowlabel	interfaces tunnel <text> parameters ipv6 flowlabel <value>

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Table 1 – continued from previous page

Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	interfaces_tunnel.xml.in: interfaces tunnel <tunnel> parameters ipv6 hoplimit	interfaces tunnel <text> parameters ipv6 hoplimit <value>
×	not yet documented	interfaces_tunnel.xml.in: interfaces tunnel <tunnel> parameters ipv6 tclass	interfaces tunnel <text> parameters ipv6 tclass <value>
×	not yet documented	interfaces_tunnel.xml.in: interfaces tunnel <tunnel> redirect	interfaces tunnel <text> redirect <value>
×	not yet documented	interfaces_tunnel.xml.in: interfaces tunnel <tunnel> remote	interfaces tunnel <text> remote <value>
×	not yet documented	interfaces_tunnel.xml.in: interfaces tunnel <tunnel> source-address	interfaces tunnel <text> source-address <value>
×	not yet documented	interfaces_tunnel.xml.in: interfaces tunnel <tunnel> source-interface	interfaces tunnel <text> source-interface <value>
✓	<i>set interfaces tunnel <interface> vrf <vrf></i>	interfaces_tunnel.xml.in: interfaces tunnel <tunnel> vrf	interfaces tunnel <text> vrf <value>
✓	<i>set interfaces virtual-ethernet <interface> address <address dhcp dhcpv6></i>	interfaces_virtual- ethernet.xml.in: interfaces virtual-ethernet <virtual-ethernet> address	interfaces virtual-ethernet <text> address <value>
✓	<i>set interfaces virtual-ethernet <interface> description <description></i>	interfaces_virtual- ethernet.xml.in: interfaces virtual-ethernet <virtual-ethernet> de- scription	interfaces virtual-ethernet <text> description <value>
×	not yet documented	interfaces_virtual- ethernet.xml.in: interfaces virtual-ethernet <virtual-ethernet> dhcp- options client-id	interfaces virtual-ethernet <text> dhcp-options client-id <value>
×	not yet documented	interfaces_virtual- ethernet.xml.in: interfaces virtual-ethernet <virtual-ethernet> dhcp- options default-route- distance	interfaces virtual-ethernet <text> dhcp-options default-route-distance <value>
×	not yet documented	interfaces_virtual- ethernet.xml.in: interfaces virtual-ethernet <virtual-ethernet> dhcp- options host-name	interfaces virtual-ethernet <text> dhcp-options host- name <value>

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Table 1 – continued from previous page

Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	interfaces_virtual-ethernet.xml.in: interfaces virtual-ethernet <virtual-ethernet> dhcp-options mtu	interfaces virtual-ethernet <text> dhcp-options mtu
×	not yet documented	interfaces_virtual-ethernet.xml.in: interfaces virtual-ethernet <virtual-ethernet> dhcp-options no-default-route	interfaces virtual-ethernet <text> dhcp-options no-default-route
×	not yet documented	interfaces_virtual-ethernet.xml.in: interfaces virtual-ethernet <virtual-ethernet> dhcp-options reject	interfaces virtual-ethernet <text> dhcp-options reject <value>
×	not yet documented	interfaces_virtual-ethernet.xml.in: interfaces virtual-ethernet <virtual-ethernet> dhcp-options user-class	Nothing found in VyOS
×	not yet documented	interfaces_virtual-ethernet.xml.in: interfaces virtual-ethernet <virtual-ethernet> dhcp-options vendor-class-id	interfaces virtual-ethernet <text> dhcp-options vendor-class-id <value>
×	not yet documented	interfaces_virtual-ethernet.xml.in: interfaces virtual-ethernet <virtual-ethernet> dhcpv6-options duid	interfaces virtual-ethernet <text> dhcpv6-options duid <value>
×	not yet documented	interfaces_virtual-ethernet.xml.in: interfaces virtual-ethernet <virtual-ethernet> dhcpv6-options no-release	interfaces virtual-ethernet <text> dhcpv6-options no-release
×	not yet documented	interfaces_virtual-ethernet.xml.in: interfaces virtual-ethernet <virtual-ethernet> dhcpv6-options parameters-only	interfaces virtual-ethernet <text> dhcpv6-options parameters-only
×	not yet documented	interfaces_virtual-ethernet.xml.in: interfaces virtual-ethernet <virtual-ethernet> dhcpv6-options pd <pd> interface <interface> address	interfaces virtual-ethernet <text> dhcpv6-options pd <text> interface <text> address <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	interfaces_virtual-ethernet.xml.in: interfaces virtual-ethernet <virtual-ethernet> dhcpv6-options pd <pd> interface <interface> sla-id	interfaces virtual-ethernet <text> dhcpv6-options pd <text> interface <text> sla-id <value>
×	not yet documented	interfaces_virtual-ethernet.xml.in: interfaces virtual-ethernet <virtual-ethernet> dhcpv6-options pd <pd> length	interfaces virtual-ethernet <text> dhcpv6-options pd <text> length <value>
×	not yet documented	interfaces_virtual-ethernet.xml.in: interfaces virtual-ethernet <virtual-ethernet> dhcpv6-options rapid-commit	interfaces virtual-ethernet <text> dhcpv6-options rapid-commit
×	not yet documented	interfaces_virtual-ethernet.xml.in: interfaces virtual-ethernet <virtual-ethernet> dhcpv6-options temporary	interfaces virtual-ethernet <text> dhcpv6-options temporary
✓	<i>set interfaces virtual-ethernet <interface> disable</i>	interfaces_virtual-ethernet.xml.in: interfaces virtual-ethernet <virtual-ethernet> disable	interfaces virtual-ethernet <text> disable
×	not yet documented	interfaces_virtual-ethernet.xml.in: interfaces virtual-ethernet <virtual-ethernet> peer-name	interfaces virtual-ethernet <text> peer-name <value>
×	<i>set interfaces virtual-ethernet <interface> vif <vlan-id></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	interfaces_virtual-ethernet.xml.in: interfaces virtual-ethernet <virtual-ethernet> vif-s <vif-s> address	interfaces virtual-ethernet <text> vif-s <text> address <value>
×	not yet documented	interfaces_virtual-ethernet.xml.in: interfaces virtual-ethernet <virtual-ethernet> vif-s <vif-s> description	interfaces virtual-ethernet <text> vif-s <text> description <value>

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Table 1 – continued from previous page

Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	interfaces_virtual-ethernet.xml.in: interfaces virtual-ethernet <virtual-ethernet> vif-s <vif-s> dhcp-options client-id	interfaces virtual-ethernet <text> vif-s <text> dhcp- options client-id <value>
×	not yet documented	interfaces_virtual-ethernet.xml.in: interfaces virtual-ethernet <virtual-ethernet> vif-s <vif-s> dhcp-options default-route-distance	interfaces virtual-ethernet <text> vif-s <text> dhcp-options default- route-distance <value>
×	not yet documented	interfaces_virtual-ethernet.xml.in: interfaces virtual-ethernet <virtual-ethernet> vif-s <vif-s> dhcp-options host-name	interfaces virtual-ethernet <text> vif-s <text> dhcp-options host-name <value>
×	not yet documented	interfaces_virtual-ethernet.xml.in: interfaces virtual-ethernet <virtual-ethernet> vif-s <vif-s> dhcp-options mtu	interfaces virtual-ethernet <text> vif-s <text> dhcp- options mtu
×	not yet documented	interfaces_virtual-ethernet.xml.in: interfaces virtual-ethernet <virtual-ethernet> vif-s <vif-s> dhcp-options no-default-route	interfaces virtual-ethernet <text> vif-s <text> dhcp- options no-default-route
×	not yet documented	interfaces_virtual-ethernet.xml.in: interfaces virtual-ethernet <virtual-ethernet> vif-s <vif-s> dhcp-options reject	interfaces virtual-ethernet <text> vif-s <text> dhcp- options reject <value>
×	not yet documented	interfaces_virtual-ethernet.xml.in: interfaces virtual-ethernet <virtual-ethernet> vif-s <vif-s> dhcp-options user-class	Nothing found in VyOS
×	not yet documented	interfaces_virtual-ethernet.xml.in: interfaces virtual-ethernet <virtual-ethernet> vif-s <vif-s> dhcp-options vendor-class-id	interfaces virtual-ethernet <text> vif-s <text> dhcp- options vendor-class-id <value>

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Table 1 – continued from previous page

Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	interfaces_virtual-ethernet.xml.in: interfaces virtual-ethernet <virtual-ethernet> vif-s <vif-s> dhcpv6-options duid	interfaces virtual-ethernet <text> vif-s <text> dhcpv6-options duid <value>
×	not yet documented	interfaces_virtual-ethernet.xml.in: interfaces virtual-ethernet <virtual-ethernet> vif-s <vif-s> dhcpv6-options no-release	interfaces virtual-ethernet <text> vif-s <text> dhcpv6-options no-release
×	not yet documented	interfaces_virtual-ethernet.xml.in: interfaces virtual-ethernet <virtual-ethernet> vif-s <vif-s> dhcpv6-options parameters-only	interfaces virtual-ethernet <text> vif-s <text> dhcpv6-options parameters-only
×	not yet documented	interfaces_virtual-ethernet.xml.in: interfaces virtual-ethernet <virtual-ethernet> vif-s <vif-s> dhcpv6-options pd <pd> interface <inter- face> address	interfaces virtual-ethernet <text> vif-s <text> dhcpv6-options pd <text> interface <text> address <value>
×	not yet documented	interfaces_virtual-ethernet.xml.in: interfaces virtual-ethernet <virtual-ethernet> vif-s <vif-s> dhcpv6-options pd <pd> interface <inter- face> sla-id	interfaces virtual-ethernet <text> vif-s <text> dhcpv6-options pd <text> interface <text> sla-id <value>
×	not yet documented	interfaces_virtual-ethernet.xml.in: interfaces virtual-ethernet <virtual-ethernet> vif-s <vif-s> dhcpv6-options pd <pd> length	interfaces virtual-ethernet <text> vif-s <text> dhcpv6-options pd <text> length <value>
×	not yet documented	interfaces_virtual-ethernet.xml.in: interfaces virtual-ethernet <virtual-ethernet> vif-s <vif-s> dhcpv6-options rapid-commit	interfaces virtual-ethernet <text> vif-s <text> dhcpv6-options rapid- commit

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	interfaces_virtual-ethernet.xml.in: interfaces virtual-ethernet <virtual-ethernet> vif-s <vif-s> dhcpv6-options temporary	interfaces virtual-ethernet <text> vif-s <text> dhcpv6-options tempo- rary
×	not yet documented	interfaces_virtual-ethernet.xml.in: interfaces virtual-ethernet <virtual-ethernet> vif-s <vif-s> disable	interfaces virtual-ethernet <text> vif-s <text> disable
×	not yet documented	interfaces_virtual-ethernet.xml.in: interfaces virtual-ethernet <virtual-ethernet> vif-s <vif-s> disable-link-detect	interfaces virtual-ethernet <text> vif-s <text> disable-link-detect
×	not yet documented	interfaces_virtual-ethernet.xml.in: interfaces virtual-ethernet <virtual-ethernet> vif-s <vif-s> ip adjust-mss	interfaces virtual-ethernet <text> vif-s <text> ip adjust-mss <value>
×	not yet documented	interfaces_virtual-ethernet.xml.in: interfaces virtual-ethernet <virtual-ethernet> vif-s <vif-s> ip arp-cache- timeout	interfaces virtual-ethernet <text> vif-s <text> ip arp- cache-timeout <value>
×	not yet documented	interfaces_virtual-ethernet.xml.in: interfaces virtual-ethernet <virtual-ethernet> vif-s <vif-s> ip disable-arp- filter	interfaces virtual-ethernet <text> vif-s <text> ip disable-arp-filter
×	not yet documented	interfaces_virtual-ethernet.xml.in: interfaces virtual-ethernet <virtual-ethernet> vif-s <vif-s> ip disable- forwarding	interfaces virtual-ethernet <text> vif-s <text> ip disable-forwarding
×	not yet documented	interfaces_virtual-ethernet.xml.in: interfaces virtual-ethernet <virtual-ethernet> vif-s <vif-s> ip enable-arp- accept	interfaces virtual-ethernet <text> vif-s <text> ip enable-arp-accept

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Table 1 – continued from previous page

Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	interfaces_virtual-ethernet.xml.in: interfaces virtual-ethernet <virtual-ethernet> vif-s <vif-s> ip enable-arp-announce	interfaces virtual-ethernet <text> vif-s <text> ip enable-arp-announce
×	not yet documented	interfaces_virtual-ethernet.xml.in: interfaces virtual-ethernet <virtual-ethernet> vif-s <vif-s> ip enable-arp-ignore	interfaces virtual-ethernet <text> vif-s <text> ip enable-arp-ignore
×	not yet documented	interfaces_virtual-ethernet.xml.in: interfaces virtual-ethernet <virtual-ethernet> vif-s <vif-s> ip enable-directed-broadcast	interfaces virtual-ethernet <text> vif-s <text> ip enable-directed-broadcast
×	not yet documented	interfaces_virtual-ethernet.xml.in: interfaces virtual-ethernet <virtual-ethernet> vif-s <vif-s> ip enable-proxy-arp	interfaces virtual-ethernet <text> vif-s <text> ip enable-proxy-arp
×	not yet documented	interfaces_virtual-ethernet.xml.in: interfaces virtual-ethernet <virtual-ethernet> vif-s <vif-s> ip proxy-arp-pvlan	interfaces virtual-ethernet <text> vif-s <text> ip proxy-arp-pvlan
×	not yet documented	interfaces_virtual-ethernet.xml.in: interfaces virtual-ethernet <virtual-ethernet> vif-s <vif-s> ip source-validation	interfaces virtual-ethernet <text> vif-s <text> ip source-validation <value>
×	not yet documented	interfaces_virtual-ethernet.xml.in: interfaces virtual-ethernet <virtual-ethernet> vif-s <vif-s> ipv6 accept-dad	interfaces virtual-ethernet <text> vif-s <text> ipv6 accept-dad <value>
×	not yet documented	interfaces_virtual-ethernet.xml.in: interfaces virtual-ethernet <virtual-ethernet> vif-s <vif-s> ipv6 address autoconf	interfaces virtual-ethernet <text> vif-s <text> ipv6 address autoconf

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Table 1 – continued from previous page

Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	interfaces_virtual-ethernet.xml.in: interfaces virtual-ethernet <virtual-ethernet> vif-s <vif-s> ipv6 address eui64	interfaces virtual-ethernet <text> vif-s <text> ipv6 address eui64 <value>
×	not yet documented	interfaces_virtual-ethernet.xml.in: interfaces virtual-ethernet <virtual-ethernet> vif-s <vif-s> ipv6 address no-default-link-local	interfaces virtual-ethernet <text> vif-s <text> ipv6 address no-default-link- local
×	not yet documented	interfaces_virtual-ethernet.xml.in: interfaces virtual-ethernet <virtual-ethernet> vif-s <vif-s> ipv6 adjust-mss	interfaces virtual-ethernet <text> vif-s <text> ipv6 adjust-mss <value>
×	not yet documented	interfaces_virtual-ethernet.xml.in: interfaces virtual-ethernet <virtual-ethernet> vif-s <vif-s> ipv6 base- reachable-time	Nothing found in VyOS
×	not yet documented	interfaces_virtual-ethernet.xml.in: interfaces virtual-ethernet <virtual-ethernet> vif-s <vif-s> ipv6 disable- forwarding	interfaces virtual-ethernet <text> vif-s <text> ipv6 disable-forwarding
×	not yet documented	interfaces_virtual-ethernet.xml.in: interfaces virtual-ethernet <virtual-ethernet> vif-s <vif-s> ipv6 dup-addr- detect-transmits	interfaces virtual-ethernet <text> vif-s <text> ipv6 dup-addr-detect-transmits <value>
×	not yet documented	interfaces_virtual-ethernet.xml.in: interfaces virtual-ethernet <virtual-ethernet> vif-s <vif-s> ipv6 source- validation	Nothing found in VyOS
×	not yet documented	interfaces_virtual-ethernet.xml.in: interfaces virtual-ethernet <virtual-ethernet> vif-s <vif-s> mac	interfaces virtual-ethernet <text> vif-s <text> mac <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	interfaces_virtual-ethernet.xml.in: interfaces virtual-ethernet <virtual-ethernet> vif-s <vif-s> mirror egress	interfaces virtual-ethernet <text> vif-s <text> mirror egress <value>
×	not yet documented	interfaces_virtual-ethernet.xml.in: interfaces virtual-ethernet <virtual-ethernet> vif-s <vif-s> mirror ingress	interfaces virtual-ethernet <text> vif-s <text> mirror ingress <value>
×	not yet documented	interfaces_virtual-ethernet.xml.in: interfaces virtual-ethernet <virtual-ethernet> vif-s <vif-s> mtu	interfaces virtual-ethernet <text> vif-s <text> mtu <value>
×	not yet documented	interfaces_virtual-ethernet.xml.in: interfaces virtual-ethernet <virtual-ethernet> vif-s <vif-s> protocol	interfaces virtual-ethernet <text> vif-s <text> proto- col <value>
×	not yet documented	interfaces_virtual-ethernet.xml.in: interfaces virtual-ethernet <virtual-ethernet> vif-s <vif-s> redirect	interfaces virtual-ethernet <text> vif-s <text> redi- rect <value>
✓	set interfaces virtual-ethernet <interface> vif-s <vlan-id> vif-c <vlan-id> address <address dhcp dhcpv6>	interfaces_virtual-ethernet.xml.in: interfaces virtual-ethernet <virtual-ethernet> vif-s <vif-s> vif-c <vif-c> address	interfaces virtual-ethernet <text> vif-s <text> vif-c <text> address <value>
✓	set interfaces virtual-ethernet <interface> vif-s <vlan-id> vif-c <vlan-id> description <description>	interfaces_virtual-ethernet.xml.in: interfaces virtual-ethernet <virtual-ethernet> vif-s <vif-s> vif-c <vif-c> description	interfaces virtual-ethernet <text> vif-s <text> vif- c <text> description <value>
✓	set interfaces virtual-ethernet <interface> vif-s <vlan-id> vif-c <vlan-id> dhcp-options client-id <description>	interfaces_virtual-ethernet.xml.in: interfaces virtual-ethernet <virtual-ethernet> vif-s <vif-s> vif-c <vif-c> dhcp-options client-id	interfaces virtual-ethernet <text> vif-s <text> vif- c <text> dhcp-options client-id <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
✓	<i>set interfaces virtual-ethernet <interface> vif-s <vlan-id> vif-c <vlan-id> dhcp-options default-route-distance <distance></i>	interfaces_virtual-ethernet.xml.in: interfaces virtual-ethernet <virtual-ethernet> vif-s <vif-s> vif-c <vif-c> dhcp-options default-route-distance	interfaces virtual-ethernet <text> vif-s <text> vif-c <text> dhcp-options default-route-distance <value>
✓	<i>set interfaces virtual-ethernet <interface> vif-s <vlan-id> vif-c <vlan-id> dhcp-options host-name <hostname></i>	interfaces_virtual-ethernet.xml.in: interfaces virtual-ethernet <virtual-ethernet> vif-s <vif-s> vif-c <vif-c> dhcp-options host-name	interfaces virtual-ethernet <text> vif-s <text> vif-c <text> dhcp-options host-name <value>
×	not yet documented	interfaces_virtual-ethernet.xml.in: interfaces virtual-ethernet <virtual-ethernet> vif-s <vif-s> vif-c <vif-c> dhcp-options mtu	interfaces virtual-ethernet <text> vif-s <text> vif-c <text> dhcp-options mtu
✓	<i>set interfaces virtual-ethernet <interface> vif-s <vlan-id> vif-c <vlan-id> dhcp-options no-default-route</i>	interfaces_virtual-ethernet.xml.in: interfaces virtual-ethernet <virtual-ethernet> vif-s <vif-s> vif-c <vif-c> dhcp-options no-default-route	interfaces virtual-ethernet <text> vif-s <text> vif-c <text> dhcp-options no-default-route
✓	<i>set interfaces virtual-ethernet <interface> vif-s <vlan-id> vif-c <vlan-id> dhcp-options reject <address></i>	interfaces_virtual-ethernet.xml.in: interfaces virtual-ethernet <virtual-ethernet> vif-s <vif-s> vif-c <vif-c> dhcp-options reject	interfaces virtual-ethernet <text> vif-s <text> vif-c <text> dhcp-options reject <value>
×	<i>set interfaces virtual-ethernet <interface> vif-s <vlan-id> vif-c <vlan-id> dhcp-options user-class <string></i>	interfaces_virtual-ethernet.xml.in: interfaces virtual-ethernet <virtual-ethernet> vif-s <vif-s> vif-c <vif-c> dhcp-options user-class	Nothing found in VyOS

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Table 1 – continued from previous page

Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
✓	<code>set interfaces virtual-ethernet <interface> vif-s <vlan-id> vif-c <vlan-id> dhcp-options vendor-class-id <vendor-id></code>	interfaces_virtual- ethernet.xml.in: interfaces virtual-ethernet <virtual-ethernet> vif-s <vif-s> vif-c <vif-c> dhcp-options vendor- class-id	interfaces virtual-ethernet <text> vif-s <text> vif- c <text> dhcp-options vendor-class-id <value>
✓	<code>set interfaces virtual-ethernet <interface> vif-s <vlan-id> vif-c <vlan-id> dhcpv6-options duid <duid></code>	interfaces_virtual- ethernet.xml.in: interfaces virtual-ethernet <virtual-ethernet> vif-s <vif-s> vif-c <vif-c> dhcpv6-options duid	interfaces virtual-ethernet <text> vif-s <text> vif- c <text> dhcpv6-options duid <value>
✓	<code>set interfaces virtual-ethernet <interface> vif-s <vlan-id> vif-c <vlan-id> dhcpv6-options no-release</code>	interfaces_virtual- ethernet.xml.in: interfaces virtual-ethernet <virtual-ethernet> vif-s <vif-s> vif-c <vif-c> dhcpv6-options no-release	interfaces virtual-ethernet <text> vif-s <text> vif-c <text> dhcpv6-options no- release
✓	<code>set interfaces virtual-ethernet <interface> vif-s <vlan-id> vif-c <vlan-id> dhcpv6-options parameters-only</code>	interfaces_virtual- ethernet.xml.in: interfaces virtual-ethernet <virtual-ethernet> vif-s <vif-s> vif-c <vif- c> dhcpv6-options parameters-only	interfaces virtual-ethernet <text> vif-s <text> vif- c <text> dhcpv6-options parameters-only
✓	<code>set interfaces virtual-ethernet <interface> vif-s <vlan-id> vif-c <vlan-id> dhcpv6-options pd <id> interface <delegatee> address <address></code>	interfaces_virtual- ethernet.xml.in: interfaces virtual-ethernet <virtual-ethernet> vif-s <vif-s> vif-c <vif-c> dhcpv6-options pd <pd> interface <interface> address	interfaces virtual-ethernet <text> vif-s <text> vif-c <text> dhcpv6-options pd <text> interface <text> ad- dress <value>
✓	<code>set interfaces virtual-ethernet <interface> vif-s <vlan-id> vif-c <vlan-id> dhcpv6-options pd <id> interface <delegatee> sla-id <id></code>	interfaces_virtual- ethernet.xml.in: interfaces virtual-ethernet <virtual-ethernet> vif-s <vif-s> vif-c <vif-c> dhcpv6-options pd <pd> interface <interface> sla-id	interfaces virtual-ethernet <text> vif-s <text> vif- c <text> dhcpv6-options pd <text> interface <text> sla-id <value>

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Table 1 – continued from previous page

Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
✓	<code>set interfaces virtual-ethernet <interface> vif-s <vlan-id> vif-c <vlan-id> dhcpv6-options pd <id> length <length></code>	interfaces_virtual- ethernet.xml.in: interfaces virtual-ethernet <virtual-ethernet> vif-s <vif-s> vif-c <vif-c> dhcpv6-options pd <pd> length	interfaces virtual-ethernet <text> vif-s <text> vif-c <text> dhcpv6-options pd <text> length <value>
✓	<code>set interfaces virtual-ethernet <interface> vif-s <vlan-id> vif-c <vlan-id> dhcpv6-options rapid-commit</code>	interfaces_virtual- ethernet.xml.in: interfaces virtual-ethernet <virtual-ethernet> vif-s <vif-s> vif-c <vif-c> dhcpv6-options rapid- commit	interfaces virtual-ethernet <text> vif-s <text> vif- c <text> dhcpv6-options rapid-commit
✓	<code>set interfaces virtual-ethernet <interface> vif-s <vlan-id> vif-c <vlan-id> dhcpv6-options temporary</code>	interfaces_virtual- ethernet.xml.in: interfaces virtual-ethernet <virtual-ethernet> vif- s <vif-s> vif-c <vif-c> dhcpv6-options temporary	interfaces virtual-ethernet <text> vif-s <text> vif- c <text> dhcpv6-options temporary
✓	<code>set interfaces virtual-ethernet <interface> vif-s <vlan-id> vif-c <vlan-id> disable</code>	interfaces_virtual- ethernet.xml.in: interfaces virtual-ethernet <virtual-ethernet> vif-s <vif-s> vif-c <vif-c> disable	interfaces virtual-ethernet <text> vif-s <text> vif-c <text> disable
✓	<code>set interfaces virtual-ethernet <interface> vif-s <vlan-id> vif-c <vlan-id> disable-link-detect</code>	interfaces_virtual- ethernet.xml.in: interfaces virtual-ethernet <virtual-ethernet> vif-s <vif-s> vif-c <vif-c> disable-link-detect	interfaces virtual-ethernet <text> vif-s <text> vif-c <text> disable-link-detect
✓	<code>set interfaces virtual-ethernet <interface> vif-s <vlan-id> vif-c <vlan-id> ip adjust-mss <mss / clamp-mss-to-pmtu></code>	interfaces_virtual- ethernet.xml.in: interfaces virtual-ethernet <virtual-ethernet> vif-s <vif-s> vif-c <vif-c> ip adjust-mss	interfaces virtual-ethernet <text> vif-s <text> vif-c <text> ip adjust-mss <value>
✓	<code>set interfaces virtual-ethernet <interface> vif-s <vlan-id> vif-c <vlan-id> ip arp-cache-timeout</code>	interfaces_virtual- ethernet.xml.in: interfaces virtual-ethernet <virtual-ethernet> vif-s <vif-s> vif-c <vif-c> ip arp-cache-timeout	interfaces virtual-ethernet <text> vif-s <text> vif-c <text> ip arp-cache- timeout <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
✓	<code>set interfaces virtual-ethernet <interface> vif-s <vlan-id> vif-c <vlan-id> ip disable-arp-filter</code>	interfaces_virtual- ethernet.xml.in: interfaces virtual-ethernet <virtual-ethernet> vif-s <vif-s> vif-c <vif-c> ip disable-arp-filter	interfaces virtual-ethernet <text> vif-s <text> vif-c <text> ip disable-arp-filter
✓	<code>set interfaces virtual-ethernet <interface> vif-s <vlan-id> vif-c <vlan-id> ip disable-forwarding</code>	interfaces_virtual- ethernet.xml.in: interfaces virtual-ethernet <virtual-ethernet> vif-s <vif-s> vif-c <vif-c> ip disable-forwarding	interfaces virtual-ethernet <text> vif-s <text> vif-c <text> ip disable- forwarding
✓	<code>set interfaces virtual-ethernet <interface> vif-s <vlan-id> vif-c <vlan-id> ip enable-arp-accept</code>	interfaces_virtual- ethernet.xml.in: interfaces virtual-ethernet <virtual-ethernet> vif-s <vif-s> vif-c <vif-c> ip enable-arp-accept	interfaces virtual-ethernet <text> vif-s <text> vif- c <text> ip enable-arp- accept
✓	<code>set interfaces virtual-ethernet <interface> vif-s <vlan-id> vif-c <vlan-id> ip enable-arp-announce</code>	interfaces_virtual- ethernet.xml.in: interfaces virtual-ethernet <virtual-ethernet> vif-s <vif-s> vif-c <vif-c> ip enable-arp-announce	interfaces virtual-ethernet <text> vif-s <text> vif- c <text> ip enable-arp- announce
✓	<code>set interfaces virtual-ethernet <interface> vif-s <vlan-id> vif-c <vlan-id> ip enable-arp-ignore</code>	interfaces_virtual- ethernet.xml.in: interfaces virtual-ethernet <virtual-ethernet> vif-s <vif-s> vif-c <vif-c> ip enable-arp-ignore	interfaces virtual-ethernet <text> vif-s <text> vif- c <text> ip enable-arp- ignore
✓	<code>set interfaces virtual-ethernet <interface> vif-s <vlan-id> vif-c <vlan-id> ip enable-directed-broad</code>	interfaces_virtual- ethernet.xml.in: interfaces virtual-ethernet <virtual-ethernet> vif-s <vif-s> vif-c <vif-c> ip enable-directed-broadcast	interfaces virtual-ethernet <text> vif-s <text> vif-c <text> ip enable-directed- broadcast
✓	<code>set interfaces virtual-ethernet <interface> vif-s <vlan-id> vif-c <vlan-id> ip enable-proxy-arp</code>	interfaces_virtual- ethernet.xml.in: interfaces virtual-ethernet <virtual-ethernet> vif-s <vif-s> vif-c <vif-c> ip enable-proxy-arp	interfaces virtual-ethernet <text> vif-s <text> vif- c <text> ip enable-proxy- arp
✓	<code>set interfaces virtual-ethernet <interface> vif-s <vlan-id> vif-c <vlan-id> ip proxy-arp-pvlan</code>	interfaces_virtual- ethernet.xml.in: interfaces virtual-ethernet <virtual-ethernet> vif-s <vif-s> vif-c <vif-c> ip proxy-arp-pvlan	interfaces virtual-ethernet <text> vif-s <text> vif-c <text> ip proxy-arp-pvlan

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
✓	<code>set interfaces virtual-ethernet <interface> vif-s <vlan-id> vif-c <vlan-id> ip source-validation <strict loose disable></code>	interfaces_virtual- ethernet.xml.in: interfaces virtual-ethernet <virtual-ethernet> vif-s <vif-s> vif-c <vif-c> ip source-validation	interfaces virtual-ethernet <text> vif-s <text> vif-c <text> ip source- validation <value>
✓	<code>set interfaces virtual-ethernet <interface> vif-s <vlan-id> vif-c <vlan-id> ipv6 accept-dad <0 1 2></code>	interfaces_virtual- ethernet.xml.in: interfaces virtual-ethernet <virtual-ethernet> vif-s <vif-s> vif-c <vif-c> ipv6 accept-dad	interfaces virtual-ethernet <text> vif-s <text> vif- c <text> ipv6 accept-dad <value>
✓	<code>set interfaces virtual-ethernet <interface> vif-s <vlan-id> vif-c <vlan-id> ipv6 address autoconf</code>	interfaces_virtual- ethernet.xml.in: interfaces virtual-ethernet <virtual-ethernet> vif-s <vif-s> vif-c <vif-c> ipv6 address autoconf	interfaces virtual-ethernet <text> vif-s <text> vif-c <text> ipv6 address auto- conf
✓	<code>set interfaces virtual-ethernet <interface> vif-s <vlan-id> vif-c <vlan-id> ipv6 address eui64 <prefix></code>	interfaces_virtual- ethernet.xml.in: interfaces virtual-ethernet <virtual-ethernet> vif-s <vif-s> vif-c <vif-c> ipv6 address eui64	interfaces virtual-ethernet <text> vif-s <text> vif-c <text> ipv6 address eui64 <value>
✓	<code>set interfaces virtual-ethernet <interface> vif-s <vlan-id> vif-c <vlan-id> ipv6 address no-default-link-local</code>	interfaces_virtual- ethernet.xml.in: interfaces virtual-ethernet <virtual-ethernet> vif-s <vif-s> vif-c <vif-c> ipv6 address no-default-link- local	interfaces virtual-ethernet <text> vif-s <text> vif-c <text> ipv6 address no- default-link-local
✓	<code>set interfaces virtual-ethernet <interface> vif-s <vlan-id> vif-c <vlan-id> ipv6 adjust-mss <mss clamp-mss-to-pmtu></code>	interfaces_virtual- ethernet.xml.in: interfaces virtual-ethernet <virtual-ethernet> vif-s <vif-s> vif-c <vif-c> ipv6 adjust-mss	interfaces virtual-ethernet <text> vif-s <text> vif- c <text> ipv6 adjust-mss <value>
×	not yet documented	interfaces_virtual- ethernet.xml.in: interfaces virtual-ethernet <virtual-ethernet> vif-s <vif-s> vif-c <vif-c> ipv6 base-reachable-time	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
✓	<code>set interfaces virtual-ethernet <interface> vif-s <vlan-id> vif-c <vlan-id> ipv6 disable-forwarding</code>	interfaces_virtual-ethernet.xml.in: interfaces virtual-ethernet <virtual-ethernet> vif-s <vif-s> vif-c <vif-c> ipv6 disable-forwarding	interfaces virtual-ethernet <text> vif-s <text> vif-c <text> ipv6 disable-forwarding
✓	<code>set interfaces virtual-ethernet <interface> vif-s <vlan-id> vif-c <vlan-id> ipv6 dup-addr-detect-transmits <n></code>	interfaces_virtual-ethernet.xml.in: interfaces virtual-ethernet <virtual-ethernet> vif-s <vif-s> vif-c <vif-c> ipv6 dup-addr-detect-transmits	interfaces virtual-ethernet <text> vif-s <text> vif-c <text> ipv6 dup-addr-detect-transmits <value>
×	not yet documented	interfaces_virtual-ethernet.xml.in: interfaces virtual-ethernet <virtual-ethernet> vif-s <vif-s> vif-c <vif-c> ipv6 source-validation	Nothing found in VyOS
✓	<code>set interfaces virtual-ethernet <interface> vif-s <vlan-id> vif-c <vlan-id> mac <xx:xx:xx:xx:xx:xx></code>	interfaces_virtual-ethernet.xml.in: interfaces virtual-ethernet <virtual-ethernet> vif-s <vif-s> vif-c <vif-c> mac	interfaces virtual-ethernet <text> vif-s <text> vif-c <text> mac <value>
×	not yet documented	interfaces_virtual-ethernet.xml.in: interfaces virtual-ethernet <virtual-ethernet> vif-s <vif-s> vif-c <vif-c> mirror egress	interfaces virtual-ethernet <text> vif-s <text> vif-c <text> mirror egress <value>
×	not yet documented	interfaces_virtual-ethernet.xml.in: interfaces virtual-ethernet <virtual-ethernet> vif-s <vif-s> vif-c <vif-c> mirror ingress	interfaces virtual-ethernet <text> vif-s <text> vif-c <text> mirror ingress <value>
✓	<code>set interfaces virtual-ethernet <interface> vif-s <vlan-id> vif-c <vlan-id> mtu <mtu></code>	interfaces_virtual-ethernet.xml.in: interfaces virtual-ethernet <virtual-ethernet> vif-s <vif-s> vif-c <vif-c> mtu	interfaces virtual-ethernet <text> vif-s <text> vif-c <text> mtu <value>
×	not yet documented	interfaces_virtual-ethernet.xml.in: interfaces virtual-ethernet <virtual-ethernet> vif-s <vif-s> vif-c <vif-c> redirect	interfaces virtual-ethernet <text> vif-s <text> vif-c <text> redirect <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
✓	<code>set interfaces virtual-ethernet <interface> vif-s <vlan-id> vif-c <vlan-id> vrf <vrf></code>	interfaces_virtual-ethernet.xml.in: interfaces virtual-ethernet <virtual-ethernet> vif-s <vif-s> vif-c <vif-c> vrf	interfaces virtual-ethernet <text> vif-s <text> vif-c <text> vrf <value>
×	not yet documented	interfaces_virtual-ethernet.xml.in: interfaces virtual-ethernet <virtual-ethernet> vif-s <vif-s> vrf	interfaces virtual-ethernet <text> vif-s <text> vrf <value>
✓	<code>set interfaces virtual-ethernet <interface> vif <vlan-id> address <address dhcp dhcpv6></code>	interfaces_virtual-ethernet.xml.in: interfaces virtual-ethernet <virtual-ethernet> vif <vif> address	interfaces virtual-ethernet <text> vif <text> address <value>
✓	<code>set interfaces virtual-ethernet <interface> vif <vlan-id> description <description></code>	interfaces_virtual-ethernet.xml.in: interfaces virtual-ethernet <virtual-ethernet> vif <vif> description	interfaces virtual-ethernet <text> vif <text> description <value>
✓	<code>set interfaces virtual-ethernet <interface> vif <vlan-id> dhcp-options client-id <description></code>	interfaces_virtual-ethernet.xml.in: interfaces virtual-ethernet <virtual-ethernet> vif <vif> dhcp-options client-id	interfaces virtual-ethernet <text> vif <text> dhcp-options client-id <value>
✓	<code>set interfaces virtual-ethernet <interface> vif <vlan-id> dhcp-options default-route-distance <distance></code>	interfaces_virtual-ethernet.xml.in: interfaces virtual-ethernet <virtual-ethernet> vif <vif> dhcp-options default-route-distance	interfaces virtual-ethernet <text> vif <text> dhcp-options default-route-distance <value>
✓	<code>set interfaces virtual-ethernet <interface> vif <vlan-id> dhcp-options host-name <hostname></code>	interfaces_virtual-ethernet.xml.in: interfaces virtual-ethernet <virtual-ethernet> vif <vif> dhcp-options host-name	interfaces virtual-ethernet <text> vif <text> dhcp-options host-name <value>
×	not yet documented	interfaces_virtual-ethernet.xml.in: interfaces virtual-ethernet <virtual-ethernet> vif <vif> dhcp-options mtu	interfaces virtual-ethernet <text> vif <text> dhcp-options mtu

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
✓	<i>set interfaces virtual-ethernet <interface> vif <vlan-id> dhcp-options no-default-route</i>	interfaces_virtual-ethernet.xml.in: interfaces virtual-ethernet <virtual-ethernet> vif <vif> dhcp-options no-default-route	interfaces virtual-ethernet <text> vif <text> dhcp-options no-default-route
✓	<i>set interfaces virtual-ethernet <interface> vif <vlan-id> dhcp-options reject <address></i>	interfaces_virtual-ethernet.xml.in: interfaces virtual-ethernet <virtual-ethernet> vif <vif> dhcp-options reject	interfaces virtual-ethernet <text> vif <text> dhcp-options reject <value>
×	<i>set interfaces virtual-ethernet <interface> vif <vlan-id> dhcp-options user-class <string></i>	interfaces_virtual-ethernet.xml.in: interfaces virtual-ethernet <virtual-ethernet> vif <vif> dhcp-options user-class	Nothing found in VyOS
✓	<i>set interfaces virtual-ethernet <interface> vif <vlan-id> dhcp-options vendor-class-id <vendor-id></i>	interfaces_virtual-ethernet.xml.in: interfaces virtual-ethernet <virtual-ethernet> vif <vif> dhcp-options vendor-class-id	interfaces virtual-ethernet <text> vif <text> dhcp-options vendor-class-id <value>
✓	<i>set interfaces virtual-ethernet <interface> vif <vlan-id> dhcpv6-options duid <duid></i>	interfaces_virtual-ethernet.xml.in: interfaces virtual-ethernet <virtual-ethernet> vif <vif> dhcpv6-options duid	interfaces virtual-ethernet <text> vif <text> dhcpv6-options duid <value>
✓	<i>set interfaces virtual-ethernet <interface> vif <vlan-id> dhcpv6-options no-release</i>	interfaces_virtual-ethernet.xml.in: interfaces virtual-ethernet <virtual-ethernet> vif <vif> dhcpv6-options no-release	interfaces virtual-ethernet <text> vif <text> dhcpv6-options no-release
✓	<i>set interfaces virtual-ethernet <interface> vif <vlan-id> dhcpv6-options parameters-only</i>	interfaces_virtual-ethernet.xml.in: interfaces virtual-ethernet <virtual-ethernet> vif <vif> dhcpv6-options parameters-only	interfaces virtual-ethernet <text> vif <text> dhcpv6-options parameters-only

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
✓	<code>set interfaces virtual-ethernet <interface> vif <vlan-id> dhcpv6-options pd <id> interface <delegatee> address <address></code>	interfaces_virtual- ethernet.xml.in: interfaces virtual-ethernet <virtual-ethernet> vif <vif> dhcpv6-options pd <pd> interface <inter- face> address	interfaces virtual-ethernet <text> vif <text> dhcpv6- options pd <text> interface <text> address <value>
✓	<code>set interfaces virtual-ethernet <interface> vif <vlan-id> dhcpv6-options pd <id> interface <delegatee> sla-id <id></code>	interfaces_virtual- ethernet.xml.in: interfaces virtual-ethernet <virtual-ethernet> vif <vif> dhcpv6-options pd <pd> interface <inter- face> sla-id	interfaces virtual-ethernet <text> vif <text> dhcpv6- options pd <text> interface <text> sla-id <value>
✓	<code>set interfaces virtual-ethernet <interface> vif <vlan-id> dhcpv6-options pd <id> length <length></code>	interfaces_virtual- ethernet.xml.in: interfaces virtual-ethernet <virtual-ethernet> vif <vif> dhcpv6-options pd <pd> length	interfaces virtual-ethernet <text> vif <text> dhcpv6- options pd <text> length <value>
✓	<code>set interfaces virtual-ethernet <interface> vif <vlan-id> dhcpv6-options rapid-commit</code>	interfaces_virtual- ethernet.xml.in: interfaces virtual-ethernet <virtual-ethernet> vif <vif> dhcpv6-options rapid-commit	interfaces virtual-ethernet <text> vif <text> dhcpv6- options rapid-commit
✓	<code>set interfaces virtual-ethernet <interface> vif <vlan-id> dhcpv6-options temporary</code>	interfaces_virtual- ethernet.xml.in: interfaces virtual-ethernet <virtual-ethernet> vif <vif> dhcpv6-options temporary	interfaces virtual-ethernet <text> vif <text> dhcpv6- options temporary
✓	<code>set interfaces virtual-ethernet <interface> vif <vlan-id> disable</code>	interfaces_virtual- ethernet.xml.in: interfaces virtual-ethernet <virtual-ethernet> vif <vif> disable	interfaces virtual-ethernet <text> vif <text> disable
✓	<code>set interfaces virtual-ethernet <interface> vif <vlan-id> disable-link-detect</code>	interfaces_virtual- ethernet.xml.in: interfaces virtual-ethernet <virtual-ethernet> vif <vif> disable-link-detect	interfaces virtual-ethernet <text> vif <text> disable- link-detect

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	interfaces_virtual-ethernet.xml.in: interfaces_virtual-ethernet <virtual-ethernet> vif <vif> egress-qos	interfaces_virtual-ethernet <text> vif <text> egress-qos <value>
×	not yet documented	interfaces_virtual-ethernet.xml.in: interfaces_virtual-ethernet <virtual-ethernet> vif <vif> ingress-qos	interfaces_virtual-ethernet <text> vif <text> ingress-qos <value>
✓	<i>set interfaces virtual-ethernet <interface> vif <vlan-id> ip adjust-mss <mss / clamp-mss-to-pmtu></i>	interfaces_virtual-ethernet.xml.in: interfaces_virtual-ethernet <virtual-ethernet> vif <vif> ip adjust-mss	interfaces_virtual-ethernet <text> vif <text> ip adjust-mss <value>
✓	<i>set interfaces virtual-ethernet <interface> vif <vlan-id> ip arp-cache-timeout</i>	interfaces_virtual-ethernet.xml.in: interfaces_virtual-ethernet <virtual-ethernet> vif <vif> ip arp-cache-timeout	interfaces_virtual-ethernet <text> vif <text> ip arp-cache-timeout <value>
✓	<i>set interfaces virtual-ethernet <interface> vif <vlan-id> ip disable-arp-filter</i>	interfaces_virtual-ethernet.xml.in: interfaces_virtual-ethernet <virtual-ethernet> vif <vif> ip disable-arp-filter	interfaces_virtual-ethernet <text> vif <text> ip disable-arp-filter
✓	<i>set interfaces virtual-ethernet <interface> vif <vlan-id> ip disable-forwarding</i>	interfaces_virtual-ethernet.xml.in: interfaces_virtual-ethernet <virtual-ethernet> vif <vif> ip disable-forwarding	interfaces_virtual-ethernet <text> vif <text> ip disable-forwarding
✓	<i>set interfaces virtual-ethernet <interface> vif <vlan-id> ip enable-arp-accept</i>	interfaces_virtual-ethernet.xml.in: interfaces_virtual-ethernet <virtual-ethernet> vif <vif> ip enable-arp-accept	interfaces_virtual-ethernet <text> vif <text> ip enable-arp-accept
✓	<i>set interfaces virtual-ethernet <interface> vif <vlan-id> ip enable-arp-announce</i>	interfaces_virtual-ethernet.xml.in: interfaces_virtual-ethernet <virtual-ethernet> vif <vif> ip enable-arp-announce	interfaces_virtual-ethernet <text> vif <text> ip enable-arp-announce

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
✓	<code>set interfaces virtual-ethernet <interface> vif <vlan-id> ip enable-arp-ignore</code>	interfaces_virtual- ethernet.xml.in: interfaces virtual-ethernet <virtual-ethernet> vif <vif> ip enable-arp- ignore	interfaces virtual-ethernet <text> vif <text> ip enable-arp-ignore
✓	<code>set interfaces virtual-ethernet <interface> vif <vlan-id> ip enable-directed-broad</code>	interfaces_virtual- ethernet.xml.in: interfaces virtual-ethernet <virtual-ethernet> vif <vif> ip enable-directed- broadcast	interfaces virtual-ethernet <text> vif <text> ip enable-directed-broadcast
✓	<code>set interfaces virtual-ethernet <interface> vif <vlan-id> ip enable-proxy-arp</code>	interfaces_virtual- ethernet.xml.in: interfaces virtual-ethernet <virtual-ethernet> vif <vif> ip enable-proxy-arp	interfaces virtual-ethernet <text> vif <text> ip enable-proxy-arp
✓	<code>set interfaces virtual-ethernet <interface> vif <vlan-id> ip proxy-arp-pvlan</code>	interfaces_virtual- ethernet.xml.in: interfaces virtual-ethernet <virtual-ethernet> vif <vif> ip proxy-arp-pvlan	interfaces virtual-ethernet <text> vif <text> ip proxy- arp-pvlan
✓	<code>set interfaces virtual-ethernet <interface> vif <vlan-id> ip source-validation <strict loose disable></code>	interfaces_virtual- ethernet.xml.in: interfaces virtual-ethernet <virtual-ethernet> vif <vif> ip source-validation	interfaces virtual-ethernet <text> vif <text> ip source-validation <value>
✓	<code>set interfaces virtual-ethernet <interface> vif <vlan-id> ipv6 accept-dad <0 1 2></code>	interfaces_virtual- ethernet.xml.in: interfaces virtual-ethernet <virtual-ethernet> vif <vif> ipv6 accept-dad	interfaces virtual-ethernet <text> vif <text> ipv6 accept-dad <value>
✓	<code>set interfaces virtual-ethernet <interface> vif <vlan-id> ipv6 address autoconf</code>	interfaces_virtual- ethernet.xml.in: interfaces virtual-ethernet <virtual-ethernet> vif <vif> ipv6 address auto- conf	interfaces virtual-ethernet <text> vif <text> ipv6 ad- dress autoconf
✓	<code>set interfaces virtual-ethernet <interface> vif <vlan-id> ipv6 address eui64 <prefix></code>	interfaces_virtual- ethernet.xml.in: interfaces virtual-ethernet <virtual-ethernet> vif <vif> ipv6 address eui64	interfaces virtual-ethernet <text> vif <text> ipv6 ad- dress eui64 <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
✓	<i>set interfaces virtual-ethernet <interface> vif <vlan-id> ipv6 address no-default-link-local</i>	interfaces_virtual-ethernet.xml.in: interfaces virtual-ethernet <virtual-ethernet> vif <vif> ipv6 address no-default-link-local	interfaces virtual-ethernet <text> vif <text> ipv6 address no-default-link-local
✓	<i>set interfaces virtual-ethernet <interface> vif <vlan-id> ipv6 adjust-mss <mss / clamp-mss-to-pmtu></i>	interfaces_virtual-ethernet.xml.in: interfaces virtual-ethernet <virtual-ethernet> vif <vif> ipv6 adjust-mss	interfaces virtual-ethernet <text> vif <text> ipv6 adjust-mss <value>
×	not yet documented	interfaces_virtual-ethernet.xml.in: interfaces virtual-ethernet <virtual-ethernet> vif <vif> ipv6 base-reachable-time	Nothing found in VyOS
✓	<i>set interfaces virtual-ethernet <interface> vif <vlan-id> ipv6 disable-forwarding</i>	interfaces_virtual-ethernet.xml.in: interfaces virtual-ethernet <virtual-ethernet> vif <vif> ipv6 disable-forwarding	interfaces virtual-ethernet <text> vif <text> ipv6 disable-forwarding
✓	<i>set interfaces virtual-ethernet <interface> vif <vlan-id> ipv6 dup-addr-detect-transmits <n></i>	interfaces_virtual-ethernet.xml.in: interfaces virtual-ethernet <virtual-ethernet> vif <vif> ipv6 dup-addr-detect-transmits	interfaces virtual-ethernet <text> vif <text> ipv6 dup-addr-detect-transmits <value>
×	not yet documented	interfaces_virtual-ethernet.xml.in: interfaces virtual-ethernet <virtual-ethernet> vif <vif> ipv6 source-validation	Nothing found in VyOS
✓	<i>set interfaces virtual-ethernet <interface> vif <vlan-id> mac <xx:xx:xx:xx:xx:xx></i>	interfaces_virtual-ethernet.xml.in: interfaces virtual-ethernet <virtual-ethernet> vif <vif> mac	interfaces virtual-ethernet <text> vif <text> mac <value>
×	not yet documented	interfaces_virtual-ethernet.xml.in: interfaces virtual-ethernet <virtual-ethernet> vif <vif> mirror egress	interfaces virtual-ethernet <text> vif <text> mirror egress <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	interfaces_virtual-ethernet.xml.in: interfaces_virtual-ethernet <virtual-ethernet> vif <vif> mirror ingress	interfaces_virtual-ethernet <text> vif <text> mirror ingress <value>
✓	<i>set interfaces virtual-ethernet <interface> vif <vlan-id> mtu <mtu></i>	interfaces_virtual-ethernet.xml.in: interfaces_virtual-ethernet <virtual-ethernet> vif <vif> mtu	interfaces_virtual-ethernet <text> vif <text> mtu <value>
×	not yet documented	interfaces_virtual-ethernet.xml.in: interfaces_virtual-ethernet <virtual-ethernet> vif <vif> redirect	interfaces_virtual-ethernet <text> vif <text> redirect <value>
✓	<i>set interfaces virtual-ethernet <interface> vif <vlan-id> vrf <vrf></i>	interfaces_virtual-ethernet.xml.in: interfaces_virtual-ethernet <virtual-ethernet> vif <vif> vrf	interfaces_virtual-ethernet <text> vif <text> vrf <value>
✓	<i>set interfaces virtual-ethernet <interface> vrf <vrf></i>	interfaces_virtual-ethernet.xml.in: interfaces_virtual-ethernet <virtual-ethernet> vrf	interfaces_virtual-ethernet <text> vrf <value>
×	<i>set interfaces vpp bonding <vppbondN></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set interfaces vpp bonding <vppbondN> address <ip-address/prefix></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set interfaces vpp bonding <vppbondN> description <description></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set interfaces vpp bonding <vppbondN> disable</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set interfaces vpp bonding <vppbondN> hash-policy <policy></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set interfaces vpp bonding <vppbondN> mac <mac-address></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set interfaces vpp bonding <vppbondN> member interface <interface-name></i>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set interfaces vpp bonding <vppbondN> mode <mode></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set interfaces vpp bonding <vppbondN> mtu <size></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set interfaces vpp bonding <vppbondN> vif <vlan-id></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set interfaces vpp bonding <vppbondN> vif <vlan-id> address <ip-address/prefix></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set interfaces vpp bonding <vppbondN> vif <vlan-id> description <description></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set interfaces vpp bonding <vppbondN> vif <vlan-id> disable</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set interfaces vpp bonding <vppbondN> vif <vlan-id> mtu <size></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set interfaces vpp bridge <vppbrN></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set interfaces vpp bridge <vppbrN> description <description></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set interfaces vpp bridge <vppbrN> member interface <interface-name></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set interfaces vpp bridge <vppbrN> member interface <loopback-interface> bvi</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set interfaces vpp gre <vppgreN></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set interfaces vpp gre <vppgreN> address <ip-address/prefix></code>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set interfaces vpp gre <vppgreN> description <description></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set interfaces vpp gre <vppgreN> disable</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set interfaces vpp gre <vppgreN> mtu <size></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set interfaces vpp gre <vppgreN> remote <address></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set interfaces vpp gre <vppgreN> source-address <address></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set interfaces vpp gre <vppgreN> tunnel-type <type></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set interfaces vpp ipip <vppipipN></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set interfaces vpp ipip <vppipipN> address <ip-address/prefix></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set interfaces vpp ipip <vppipipN> description <description></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set interfaces vpp ipip <vppipipN> disable</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set interfaces vpp ipip <vppipipN> mtu <size></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set interfaces vpp ipip <vppipipN> remote <address></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set interfaces vpp ipip <vppipipN> source-address <address></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set interfaces vpp loopback <vpploN></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set interfaces vpp loopback <vpploN> address <ip-address/prefix></code>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set interfaces vpp loopback <vpplon> description <description></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set interfaces vpp loopback <vpplon> disable</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set interfaces vpp loopback <vpplon> mtu <size></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set interfaces vpp loopback <vpplon> vif <vlan-id></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set interfaces vpp loopback <vpplon> vif <vlan-id> address <ip-address/prefix></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set interfaces vpp loopback <vpplon> vif <vlan-id> description <description></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set interfaces vpp loopback <vpplon> vif <vlan-id> disable</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set interfaces vpp loopback <vpplon> vif <vlan-id> mtu <size></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set interfaces vpp vxlan <vppvxlann></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set interfaces vpp vxlan <vppvxlann> address <ip-address/prefix></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set interfaces vpp vxlan <vppvxlann> description <description></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set interfaces vpp vxlan <vppvxlann> disable</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set interfaces vpp vxlan <vppvxlann> mtu <size></code>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set interfaces vpp vxlan <vppvxlanN> remote <address></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set interfaces vpp vxlan <vppvxlanN> source-address <address></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set interfaces vpp vxlan <vppvxlanN> vni <vni></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set interfaces vpp xconnect <vppxconN></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set interfaces vpp xconnect <vppxconN> description <description></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set interfaces vpp xconnect <vppxconN> member interface <interface-name></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set interfaces vti <interface> mirror egress <monitor-interface></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set interfaces vti <interface> mirror ingress <monitor-interface></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set interfaces vti <interface> redirect <interface></code>	Nothing found in XML Definitions	Nothing found in VyOS
✓	<code>set interfaces vti <interface> address <address></code>	interfaces_vti.xml.in: interfaces vti <vti> address	interfaces vti <text> address <value>
✓	<code>set interfaces vti <interface> description <description></code>	interfaces_vti.xml.in: interfaces vti <vti> description	interfaces vti <text> description <value>
✓	<code>set interfaces vti <interface> disable</code>	interfaces_vti.xml.in: interfaces vti <vti> disable	interfaces vti <text> disable
✓	<code>set interfaces vti <interface> ip adjust-mss <mss / clamp-mss-to-pmtu></code>	interfaces_vti.xml.in: interfaces vti <vti> ip adjust-mss	interfaces vti <text> ip adjust-mss <value>
✓	<code>set interfaces vti <interface> ip arp-cache-timeout</code>	interfaces_vti.xml.in: interfaces vti <vti> ip arp-cache-timeout	interfaces vti <text> ip arp-cache-timeout <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
✓	<code>set interfaces vti <interface> ip disable-arp-filter</code>	interfaces_vti.xml.in: interfaces vti <vti> ip disable-arp-filter	interfaces vti <text> ip disable-arp-filter
✓	<code>set interfaces vti <interface> ip disable-forwarding</code>	interfaces_vti.xml.in: interfaces vti <vti> ip disable-forwarding	interfaces vti <text> ip disable-forwarding
✓	<code>set interfaces vti <interface> ip enable-arp-accept</code>	interfaces_vti.xml.in: interfaces vti <vti> ip enable-arp-accept	interfaces vti <text> ip enable-arp-accept
✓	<code>set interfaces vti <interface> ip enable-arp-announce</code>	interfaces_vti.xml.in: interfaces vti <vti> ip enable-arp-announce	interfaces vti <text> ip enable-arp-announce
✓	<code>set interfaces vti <interface> ip enable-arp-ignore</code>	interfaces_vti.xml.in: interfaces vti <vti> ip enable-arp-ignore	interfaces vti <text> ip enable-arp-ignore
✓	<code>set interfaces vti <interface> ip enable-directed-broadcast</code>	interfaces_vti.xml.in: interfaces vti <vti> ip enable-directed-broadcast	interfaces vti <text> ip enable-directed-broadcast
✓	<code>set interfaces vti <interface> ip enable-proxy-arp</code>	interfaces_vti.xml.in: interfaces vti <vti> ip enable-proxy-arp	interfaces vti <text> ip enable-proxy-arp
✓	<code>set interfaces vti <interface> ip proxy-arp-pvlan</code>	interfaces_vti.xml.in: interfaces vti <vti> ip proxy-arp-pvlan	interfaces vti <text> ip proxy-arp-pvlan
✓	<code>set interfaces vti <interface> ip source-validation <strict loose disable></code>	interfaces_vti.xml.in: interfaces vti <vti> ip source-validation	interfaces vti <text> ip source-validation <value>
✓	<code>set interfaces vti <interface> ipv6 accept-dad <0 1 2></code>	interfaces_vti.xml.in: interfaces vti <vti> ipv6 accept-dad	interfaces vti <text> ipv6 accept-dad <value>
✓	<code>set interfaces vti <interface> ipv6 address autoconf</code>	interfaces_vti.xml.in: interfaces vti <vti> ipv6 address autoconf	interfaces vti <text> ipv6 address autoconf
✓	<code>set interfaces vti <interface> ipv6 address eui64 <prefix></code>	interfaces_vti.xml.in: interfaces vti <vti> ipv6 address eui64	interfaces vti <text> ipv6 address eui64 <value>
✓	<code>set interfaces vti <interface> ipv6 address no-default-link-local</code>	interfaces_vti.xml.in: interfaces vti <vti> ipv6 address no-default-link- local	interfaces vti <text> ipv6 address no-default-link- local
✓	<code>set interfaces vti <interface> ipv6 adjust-mss <mss clamp-mss-to-pmtu></code>	interfaces_vti.xml.in: interfaces vti <vti> ipv6 adjust-mss	interfaces vti <text> ipv6 adjust-mss <value>

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Table 1 – continued from previous page

Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	interfaces_vti.xml.in: interfaces vti <vti> ipv6 base-reachable-time	Nothing found in VyOS
✓	<i>set interfaces vti <interface> ipv6 disable-forwarding</i>	interfaces_vti.xml.in: interfaces vti <vti> ipv6 disable-forwarding	interfaces vti <text> ipv6 disable-forwarding
✓	<i>set interfaces vti <interface> ipv6 dup-addr-detect-trans <n></i>	interfaces_vti.xml.in: interfaces vti <vti> ipv6 dup-addr-detect-transmits	interfaces vti <text> ipv6 dup-addr-detect-transmits <value>
×	not yet documented	interfaces_vti.xml.in: interfaces vti <vti> ipv6 source-validation	Nothing found in VyOS
×	not yet documented	interfaces_vti.xml.in: interfaces vti <vti> mirror egress	interfaces vti <text> mir- ror egress <value>
×	not yet documented	interfaces_vti.xml.in: interfaces vti <vti> mirror ingress	interfaces vti <text> mir- ror ingress <value>
✓	<i>set interfaces vti <interface> mtu <mtu></i>	interfaces_vti.xml.in: interfaces vti <vti> mtu	interfaces vti <text> mtu <value>
×	not yet documented	interfaces_vti.xml.in: interfaces vti <vti> redi- rect	interfaces vti <text> redi- rect <value>
✓	<i>set interfaces vti <interface> vrf <vrf></i>	interfaces_vti.xml.in: interfaces vti <vti> vrf	interfaces vti <text> vrf <value>
×	<i>set interfaces vxlan <interface> gpe</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set interfaces vxlan <interface> group <address></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set interfaces vxlan <interface> parameters external</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set interfaces vxlan <interface> parameters neighbor-suppress</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set interfaces vxlan <interface> parameters nolearning</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set interfaces vxlan <interface> parameters vni-filter</i>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set interfaces vxlan <interface> port <port></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set interfaces vxlan <interface> remote <address></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set interfaces vxlan <interface> source-address <address></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set interfaces vxlan <interface> source-interface <interface></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set interfaces vxlan <interface> vlan-to-vni <vlan> vni <vni></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set interfaces vxlan <interface> vni <number></code>	Nothing found in XML Definitions	Nothing found in VyOS
✓	<code>set interfaces vxlan <interface> address <address></code>	interfaces_vxlan.xml.in: interfaces vxlan <vxlan> address	interfaces vxlan <text> address <value>
✓	<code>set interfaces vxlan <interface> description <description></code>	interfaces_vxlan.xml.in: interfaces vxlan <vxlan> description	interfaces vxlan <text> description <value>
✓	<code>set interfaces vxlan <interface> disable</code>	interfaces_vxlan.xml.in: interfaces vxlan <vxlan> disable	interfaces vxlan <text> disable
×	<code>set interfaces vxlan <interface> disable-flow-control</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set interfaces vxlan <interface> disable-link-detect</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	Nothing found in XML Definitions	interfaces vxlan <text> external
×	not yet documented	interfaces_vxlan.xml.in: interfaces vxlan <vxlan> gpe	interfaces vxlan <text> gpe
×	not yet documented	interfaces_vxlan.xml.in: interfaces vxlan <vxlan> group	interfaces vxlan <text> group <value>
✓	<code>set interfaces vxlan <interface> ip adjust-mss <mss / clamp-mss-to-pmtu></code>	interfaces_vxlan.xml.in: interfaces vxlan <vxlan> ip adjust-mss	interfaces vxlan <text> ip adjust-mss <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
✓	<code>set interfaces vxlan <interface> ip arp-cache-timeout</code>	interfaces_vxlan.xml.in: interfaces vxlan <vxlan> ip arp-cache-timeout	interfaces vxlan <text> ip arp-cache-timeout <value>
✓	<code>set interfaces vxlan <interface> ip disable-arp-filter</code>	interfaces_vxlan.xml.in: interfaces vxlan <vxlan> ip disable-arp-filter	interfaces vxlan <text> ip disable-arp-filter
✓	<code>set interfaces vxlan <interface> ip disable-forwarding</code>	interfaces_vxlan.xml.in: interfaces vxlan <vxlan> ip disable-forwarding	interfaces vxlan <text> ip disable-forwarding
✓	<code>set interfaces vxlan <interface> ip enable-arp-accept</code>	interfaces_vxlan.xml.in: interfaces vxlan <vxlan> ip enable-arp-accept	interfaces vxlan <text> ip enable-arp-accept
✓	<code>set interfaces vxlan <interface> ip enable-arp-announce</code>	interfaces_vxlan.xml.in: interfaces vxlan <vxlan> ip enable-arp-announce	interfaces vxlan <text> ip enable-arp-announce
✓	<code>set interfaces vxlan <interface> ip enable-arp-ignore</code>	interfaces_vxlan.xml.in: interfaces vxlan <vxlan> ip enable-arp-ignore	interfaces vxlan <text> ip enable-arp-ignore
✓	<code>set interfaces vxlan <interface> ip enable-directed-broadcast</code>	interfaces_vxlan.xml.in: interfaces vxlan <vxlan> ip enable-directed-broadcast	interfaces vxlan <text> ip enable-directed-broadcast
✓	<code>set interfaces vxlan <interface> ip enable-proxy-arp</code>	interfaces_vxlan.xml.in: interfaces vxlan <vxlan> ip enable-proxy-arp	interfaces vxlan <text> ip enable-proxy-arp
✓	<code>set interfaces vxlan <interface> ip proxy-arp-pvlan</code>	interfaces_vxlan.xml.in: interfaces vxlan <vxlan> ip proxy-arp-pvlan	interfaces vxlan <text> ip proxy-arp-pvlan
✓	<code>set interfaces vxlan <interface> ip source-validation <strict loose disable></code>	interfaces_vxlan.xml.in: interfaces vxlan <vxlan> ip source-validation	interfaces vxlan <text> ip source-validation <value>
✓	<code>set interfaces vxlan <interface> ipv6 accept-dad <0 1 2></code>	interfaces_vxlan.xml.in: interfaces vxlan <vxlan> ipv6 accept-dad	interfaces vxlan <text> ipv6 accept-dad <value>
✓	<code>set interfaces vxlan <interface> ipv6 address autoconf</code>	interfaces_vxlan.xml.in: interfaces vxlan <vxlan> ipv6 address autoconf	interfaces vxlan <text> ipv6 address autoconf
✓	<code>set interfaces vxlan <interface> ipv6 address eui64 <prefix></code>	interfaces_vxlan.xml.in: interfaces vxlan <vxlan> ipv6 address eui64	interfaces vxlan <text> ipv6 address eui64 <value>
✓	<code>set interfaces vxlan <interface> ipv6 address no-default-link-local</code>	interfaces_vxlan.xml.in: interfaces vxlan <vxlan> ipv6 address no-default-link-local	interfaces vxlan <text> ipv6 address no-default-link-local

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
✓	<code>set interfaces vxlan</code> <code><interface> ipv6</code> <code>adjust-mss <mss /</code> <code>clamp-mss-to-pmtu></code>	interfaces_vxlan.xml.in: interfaces vxlan <vxlan> ipv6 adjust-mss	interfaces vxlan <text> ipv6 adjust-mss <value>
×	not yet documented	interfaces_vxlan.xml.in: interfaces vxlan <vxlan> ipv6 base-reachable-time	Nothing found in VyOS
✓	<code>set interfaces vxlan</code> <code><interface> ipv6</code> <code>disable-forwarding</code>	interfaces_vxlan.xml.in: interfaces vxlan <vxlan> ipv6 disable-forwarding	interfaces vxlan <text> ipv6 disable-forwarding
✓	<code>set interfaces vxlan</code> <code><interface> ipv6</code> <code>dup-addr-detect-trans</code> <code><n></code>	interfaces_vxlan.xml.in: interfaces vxlan <vxlan> ipv6 dup-addr-detect-transmits	interfaces vxlan <text> ipv6 dup-addr-detect-transmits <value>
×	not yet documented	interfaces_vxlan.xml.in: interfaces vxlan <vxlan> ipv6 source-validation	Nothing found in VyOS
✓	<code>set interfaces vxlan</code> <code><interface> mac</code> <code><xx:xx:xx:xx:xx:xx></code>	interfaces_vxlan.xml.in: interfaces vxlan <vxlan> mac	interfaces vxlan <text> mac <value>
×	not yet documented	interfaces_vxlan.xml.in: interfaces vxlan <vxlan> mirror egress	interfaces vxlan <text> mirror egress <value>
×	not yet documented	interfaces_vxlan.xml.in: interfaces vxlan <vxlan> mirror ingress	interfaces vxlan <text> mirror ingress <value>
✓	<code>set interfaces vxlan</code> <code><interface> mtu</code> <code><mtu></code>	interfaces_vxlan.xml.in: interfaces vxlan <vxlan> mtu	interfaces vxlan <text> mtu <value>
×	not yet documented	interfaces_vxlan.xml.in: interfaces vxlan <vxlan> parameters external	Nothing found in VyOS
×	not yet documented	interfaces_vxlan.xml.in: interfaces vxlan <vxlan> parameters ip df	interfaces vxlan <text> pa- rameters ip df <value>
×	not yet documented	interfaces_vxlan.xml.in: interfaces vxlan <vxlan> parameters ip tos	interfaces vxlan <text> pa- rameters ip tos <value>
×	not yet documented	interfaces_vxlan.xml.in: interfaces vxlan <vxlan> parameters ip ttl	interfaces vxlan <text> pa- rameters ip ttl <value>
×	not yet documented	interfaces_vxlan.xml.in: interfaces vxlan <vxlan> parameters ipv6 flowlabel	interfaces vxlan <text> parameters ipv6 flowlabel <value>
×	not yet documented	interfaces_vxlan.xml.in: interfaces vxlan <vxlan> parameters neighbor-suppress	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	interfaces_vxlan.xml.in: interfaces vxlan <vxlan> parameters nolearning	interfaces vxlan <text> pa- rameters nolearning
×	not yet documented	interfaces_vxlan.xml.in: interfaces vxlan <vxlan> parameters vni-filter	Nothing found in VyOS
×	not yet documented	interfaces_vxlan.xml.in: interfaces vxlan <vxlan> port	interfaces vxlan <text> port <value>
×	not yet documented	interfaces_vxlan.xml.in: interfaces vxlan <vxlan> redirect	interfaces vxlan <text> redirect <value>
×	not yet documented	interfaces_vxlan.xml.in: interfaces vxlan <vxlan> remote	interfaces vxlan <text> re- mote <value>
×	not yet documented	interfaces_vxlan.xml.in: interfaces vxlan <vxlan> source-address	interfaces vxlan <text> source-address <value>
×	not yet documented	interfaces_vxlan.xml.in: interfaces vxlan <vxlan> source-interface	interfaces vxlan <text> source-interface <value>
×	not yet documented	interfaces_vxlan.xml.in: interfaces vxlan <vxlan> vlan-to-vni <vlan-to-vni> vni	Nothing found in VyOS
×	not yet documented	interfaces_vxlan.xml.in: interfaces vxlan <vxlan> vni	interfaces vxlan <text> vni <value>
✓	set interfaces vxlan <interface> vrf <vrf>	interfaces_vxlan.xml.in: interfaces vxlan <vxlan> vrf	interfaces vxlan <text> vrf <value>
×	set interfaces wireguard <interface> private-key <private-key>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	inter- faces_wireguard.xml.in: interfaces wireguard <wireguard> address	interfaces wireguard <text> address <value>
×	not yet documented	inter- faces_wireguard.xml.in: interfaces wireguard <wireguard> description	interfaces wireguard <text> description <value>
×	not yet documented	inter- faces_wireguard.xml.in: interfaces wireguard <wireguard> disable	interfaces wireguard <text> disable

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Table 1 – continued from previous page

Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	inter- faces_wireguard.xml.in: interfaces wireguard <wireguard> fwmark	interfaces wireguard <text> fwmark <value>
×	not yet documented	inter- faces_wireguard.xml.in: interfaces wireguard <wireguard> ip adjust- mss	interfaces wireguard <text> ip adjust-mss <value>
×	not yet documented	inter- faces_wireguard.xml.in: interfaces wireguard <wireguard> ip arp- cache-timeout	interfaces wireguard <text> ip arp-cache- timeout <value>
×	not yet documented	inter- faces_wireguard.xml.in: interfaces wireguard <wireguard> ip disable- arp-filter	interfaces wireguard <text> ip disable-arp-filter
×	not yet documented	inter- faces_wireguard.xml.in: interfaces wireguard <wireguard> ip disable- forwarding	interfaces wireguard <text> ip disable- forwarding
×	not yet documented	inter- faces_wireguard.xml.in: interfaces wireguard <wireguard> ip enable- arp-accept	interfaces wireguard <text> ip enable-arp- accept
×	not yet documented	inter- faces_wireguard.xml.in: interfaces wireguard <wireguard> ip enable- arp-announce	interfaces wireguard <text> ip enable-arp- announce
×	not yet documented	inter- faces_wireguard.xml.in: interfaces wireguard <wireguard> ip enable- arp-ignore	interfaces wireguard <text> ip enable-arp- ignore
×	not yet documented	inter- faces_wireguard.xml.in: interfaces wireguard <wireguard> ip enable- directed-broadcast	interfaces wireguard <text> ip enable-directed- broadcast
×	not yet documented	inter- faces_wireguard.xml.in: interfaces wireguard <wireguard> ip enable- proxy-arp	interfaces wireguard <text> ip enable-proxy- arp

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	inter-faces_wireguard.xml.in: interfaces wireguard <wireguard> ip proxy-arp-pvlan	interfaces wireguard <text> ip proxy-arp-pvlan
×	not yet documented	inter-faces_wireguard.xml.in: interfaces wireguard <wireguard> ip source-validation	interfaces wireguard <text> ip source-validation <value>
×	not yet documented	inter-faces_wireguard.xml.in: interfaces wireguard <wireguard> ipv6 accept-dad	interfaces wireguard <text> ipv6 accept-dad <value>
×	not yet documented	inter-faces_wireguard.xml.in: interfaces wireguard <wireguard> ipv6 address autoconf	interfaces wireguard <text> ipv6 address autoconf
×	not yet documented	inter-faces_wireguard.xml.in: interfaces wireguard <wireguard> ipv6 address eui64	interfaces wireguard <text> ipv6 address eui64 <value>
×	not yet documented	inter-faces_wireguard.xml.in: interfaces wireguard <wireguard> ipv6 address no-default-link-local	interfaces wireguard <text> ipv6 address no-default-link-local
×	not yet documented	inter-faces_wireguard.xml.in: interfaces wireguard <wireguard> ipv6 adjust-mss	interfaces wireguard <text> ipv6 adjust-mss <value>
×	not yet documented	inter-faces_wireguard.xml.in: interfaces wireguard <wireguard> ipv6 base-reachable-time	Nothing found in VyOS
×	not yet documented	inter-faces_wireguard.xml.in: interfaces wireguard <wireguard> ipv6 disable-forwarding	interfaces wireguard <text> ipv6 disable-forwarding

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	inter-faces_wireguard.xml.in: interfaces wireguard <wireguard> ipv6 dup-addr-detect-transmits	interfaces wireguard <text> ipv6 dup-addr-detect-transmits <value>
×	not yet documented	inter-faces_wireguard.xml.in: interfaces wireguard <wireguard> ipv6 source-validation	Nothing found in VyOS
×	not yet documented	inter-faces_wireguard.xml.in: interfaces wireguard <wireguard> mirror egress	interfaces wireguard <text> mirror egress <value>
×	not yet documented	inter-faces_wireguard.xml.in: interfaces wireguard <wireguard> mirror ingress	interfaces wireguard <text> mirror ingress <value>
×	not yet documented	inter-faces_wireguard.xml.in: interfaces wireguard <wireguard> mtu	interfaces wireguard <text> mtu <value>
×	not yet documented	inter-faces_wireguard.xml.in: interfaces wireguard <wireguard> peer <peer> address	interfaces wireguard <text> peer <text> address <value>
×	not yet documented	inter-faces_wireguard.xml.in: interfaces wireguard <wireguard> peer <peer> allowed-ips	interfaces wireguard <text> peer <text> allowed-ips <value>
×	not yet documented	inter-faces_wireguard.xml.in: interfaces wireguard <wireguard> peer <peer> description	interfaces wireguard <text> peer <text> description <value>
×	not yet documented	inter-faces_wireguard.xml.in: interfaces wireguard <wireguard> peer <peer> disable	interfaces wireguard <text> peer <text> disable
×	not yet documented	inter-faces_wireguard.xml.in: interfaces wireguard <wireguard> peer <peer> persistent-keepalive	interfaces wireguard <text> peer <text> persistent-keepalive <value>

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Table 1 – continued from previous page

Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	inter-faces_wireguard.xml.in: interfaces wireguard <wireguard> peer <peer> port	interfaces wireguard <text> peer <text> port <value>
×	not yet documented	inter-faces_wireguard.xml.in: interfaces wireguard <wireguard> peer <peer> preshared-key	interfaces wireguard <text> peer <text> preshared-key <value>
×	not yet documented	inter-faces_wireguard.xml.in: interfaces wireguard <wireguard> peer <peer> public-key	interfaces wireguard <text> peer <text> public- key <value>
✓	<i>set interfaces wireguard <interface> per-client-thread</i>	inter-faces_wireguard.xml.in: interfaces wireguard <wireguard> per-client- thread	interfaces wireguard <text> per-client-thread
×	not yet documented	inter-faces_wireguard.xml.in: interfaces wireguard <wireguard> port	interfaces wireguard <text> port <value>
×	not yet documented	inter-faces_wireguard.xml.in: interfaces wireguard <wireguard> private-key	interfaces wireguard <text> private-key <value>
×	not yet documented	inter-faces_wireguard.xml.in: interfaces wireguard <wireguard> redirect	interfaces wireguard <text> redirect <value>
×	not yet documented	inter-faces_wireguard.xml.in: interfaces wireguard <wireguard> vrf	interfaces wireguard <text> vrf <value>
×	<i>set interfaces wireless <interface> capabilities he antenna-pattern-fixed</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set interfaces wireless <interface> capabilities he beamform <single-user-beamform / single-user-beamforme / multi-user-beamformer.</i>	Nothing found in XML Definitions	Nothing found in VyOS

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Table 1 – continued from previous page

Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set interfaces wireless <interface> capabilities he bss-color <number></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set interfaces wireless <interface> capabilities he center-channel-freq <freq-1 freq-2> <number></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set interfaces wireless <interface> capabilities he channel-set-width <number></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set interfaces wireless <interface> capabilities he coding-scheme <number></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set interfaces wireless <interface> capabilities ht 40mhz-incapable</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set interfaces wireless <interface> capabilities ht auto-powersave</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set interfaces wireless <interface> capabilities ht channel-set-width <ht20 ht40+ ht40-></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set interfaces wireless <interface> capabilities ht delayed-block-ack</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set interfaces wireless <interface> capabilities ht dsss-cck-40</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set interfaces wireless <interface> capabilities ht greenfield</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set interfaces wireless <interface> capabilities ht ldpc</code>	Nothing found in XML Definitions	Nothing found in VyOS

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Table 1 – continued from previous page

Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set interfaces wireless <interface> capabilities ht lsig-protection</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set interfaces wireless <interface> capabilities ht max-amsdu <3839 7935></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set interfaces wireless <interface> capabilities ht short-gi <20 40></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set interfaces wireless <interface> capabilities ht smps <static dynamic></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set interfaces wireless <interface> capabilities ht stbc rx <num></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set interfaces wireless <interface> capabilities ht stbc tx</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set interfaces wireless <interface> capabilities require-he</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set interfaces wireless <interface> capabilities require-ht</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set interfaces wireless <interface> capabilities require-vht</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set interfaces wireless <interface> capabilities vht antenna-count <count></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set interfaces wireless <interface> capabilities vht antenna-pattern-fixed</code>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set interfaces wireless <interface> capabilities vht beamform <single-user-beamform / single-user-beamforme / multi-user-beamformer / multi-user-beamformee</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set interfaces wireless <interface> capabilities vht center-channel-freq <freq-1 freq-2> <number></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set interfaces wireless <interface> capabilities vht channel-set-width <0 1 2 3></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set interfaces wireless <interface> capabilities vht ldpc</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set interfaces wireless <interface> capabilities vht link-adaptation</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set interfaces wireless <interface> capabilities vht max-mpdu-exp <value></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set interfaces wireless <interface> capabilities vht max-mpdu <value></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set interfaces wireless <interface> capabilities vht short-gi <80 160></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set interfaces wireless <interface> capabilities vht stbc rx <num></code>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<i>set interfaces wireless <interface> capabilities vht stbc tx</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set interfaces wireless <interface> capabilities vht tx-powersave</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set interfaces wireless <interface> capabilities vht vht-cf</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set interfaces wireless <interface> channel <number></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set interfaces wireless <interface> disable-broadcast-ssi</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set interfaces wireless <interface> enable-bf-protection</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set interfaces wireless <interface> expunge-failing-stations</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set interfaces wireless <interface> isolate-stations</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set interfaces wireless <interface> max-stations <count></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set interfaces wireless <interface> mgmt-frame-protection</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set interfaces wireless <interface> mode <a b g n ac ax></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set interfaces wireless <interface> physical-device <device></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set interfaces wireless <interface> reduce-transmit-power <number></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set interfaces wireless <interface> ssid <ssid></i>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<i>set interfaces wireless <interface> type <access-point / station / monitor></i>	Nothing found in XML Definitions	Nothing found in VyOS
✓	<i>set interfaces wireless <interface> address <address / dhcp / dhcpv6></i>	inter-faces_wireless.xml.in: interfaces wireless <wireless> address	interfaces wireless <text> address <value>
×	not yet documented	inter-faces_wireless.xml.in: interfaces wireless <wireless> capabilities ht 40mhz-incapable	interfaces wireless <text> capabilities ht 40mhz-incapable
×	not yet documented	inter-faces_wireless.xml.in: interfaces wireless <wireless> capabilities ht auto-powersave	interfaces wireless <text> capabilities ht auto-powersave
×	not yet documented	inter-faces_wireless.xml.in: interfaces wireless <wireless> capabilities ht channel-set-width	interfaces wireless <text> capabilities ht channel-set-width <value>
×	not yet documented	inter-faces_wireless.xml.in: interfaces wireless <wireless> capabilities ht delayed-block-ack	interfaces wireless <text> capabilities ht delayed-block-ack
×	not yet documented	inter-faces_wireless.xml.in: interfaces wireless <wireless> capabilities ht dsss-cck-40	interfaces wireless <text> capabilities ht dsss-cck-40
×	not yet documented	inter-faces_wireless.xml.in: interfaces wireless <wireless> capabilities ht greenfield	interfaces wireless <text> capabilities ht greenfield
×	not yet documented	inter-faces_wireless.xml.in: interfaces wireless <wireless> capabilities ht ldpc	interfaces wireless <text> capabilities ht ldpc
×	not yet documented	inter-faces_wireless.xml.in: interfaces wireless <wireless> capabilities ht lsig-protection	interfaces wireless <text> capabilities ht lsig-protection

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Table 1 – continued from previous page

Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	inter- faces_wireless.xml.in: interfaces wireless <wire- less> capabilities ht max-amsdu	interfaces wireless <text> capabilities ht max-amsdu <value>
×	not yet documented	inter- faces_wireless.xml.in: interfaces wireless <wire- less> capabilities ht short-gi	interfaces wireless <text> capabilities ht short-gi <value>
×	not yet documented	inter- faces_wireless.xml.in: interfaces wireless <wire- less> capabilities ht smps	interfaces wireless <text> capabilities ht smps <value>
×	not yet documented	inter- faces_wireless.xml.in: interfaces wireless <wire- less> capabilities ht stbc rx	interfaces wireless <text> capabilities ht stbc rx <value>
×	not yet documented	inter- faces_wireless.xml.in: interfaces wireless <wire- less> capabilities ht stbc tx	interfaces wireless <text> capabilities ht stbc tx
×	not yet documented	inter- faces_wireless.xml.in: interfaces wireless <wire- less> capabilities require-ht	interfaces wireless <text> capabilities require-ht
×	not yet documented	inter- faces_wireless.xml.in: interfaces wireless <wire- less> capabilities require-vht	interfaces wireless <text> capabilities require-vht
×	not yet documented	inter- faces_wireless.xml.in: interfaces wireless <wire- less> capabilities vht antenna-count	interfaces wireless <text> capabilities vht antenna-count <value>
×	not yet documented	inter- faces_wireless.xml.in: interfaces wireless <wire- less> capabilities vht antenna-pattern-fixed	interfaces wireless <text> capabilities vht antenna-pattern-fixed
×	not yet documented	inter- faces_wireless.xml.in: interfaces wireless <wire- less> capabilities vht beamform	interfaces wireless <text> capabilities vht beamform <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	inter- faces_wireless.xml.in: interfaces wireless <wire- less> capabilities vht center-channel-freq freq-1	interfaces wireless <text> capabilities vht center- channel-freq freq-1 <value>
×	not yet documented	inter- faces_wireless.xml.in: interfaces wireless <wire- less> capabilities vht center-channel-freq freq-2	interfaces wireless <text> capabilities vht center- channel-freq freq-2 <value>
×	not yet documented	inter- faces_wireless.xml.in: interfaces wireless <wire- less> capabilities vht channel-set-width	interfaces wireless <text> capabilities vht channel- set-width <value>
×	not yet documented	inter- faces_wireless.xml.in: interfaces wireless <wire- less> capabilities vht ldpc	interfaces wireless <text> capabilities vht ldpc
×	not yet documented	inter- faces_wireless.xml.in: interfaces wireless <wire- less> capabilities vht link- adaptation	interfaces wireless <text> capabilities vht link- adaptation <value>
×	not yet documented	inter- faces_wireless.xml.in: interfaces wireless <wire- less> capabilities vht max- mpdu	interfaces wireless <text> capabilities vht max-mpdu <value>
×	not yet documented	inter- faces_wireless.xml.in: interfaces wireless <wire- less> capabilities vht max- mpdu-exp	interfaces wireless <text> capabilities vht max- mpdu-exp <value>
×	not yet documented	inter- faces_wireless.xml.in: interfaces wireless <wire- less> capabilities vht short-gi	interfaces wireless <text> capabilities vht short-gi <value>
×	not yet documented	inter- faces_wireless.xml.in: interfaces wireless <wire- less> capabilities vht stbc rx	interfaces wireless <text> capabilities vht stbc rx <value>
×	not yet documented	inter- faces_wireless.xml.in: interfaces wireless <wire- less> capabilities vht stbc tx	interfaces wireless <text> capabilities vht stbc tx

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Table 1 – continued from previous page

Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	inter-faces_wireless.xml.in: interfaces wireless <wireless> capabilities vht tx-powersave	interfaces wireless <text> capabilities vht tx-powersave
×	not yet documented	inter-faces_wireless.xml.in: interfaces wireless <wireless> capabilities vht vht-cf	interfaces wireless <text> capabilities vht vht-cf
×	not yet documented	inter-faces_wireless.xml.in: interfaces wireless <wireless> channel	interfaces wireless <text> channel <value>
×	not yet documented	inter-faces_wireless.xml.in: interfaces wireless <wireless> country-code	interfaces wireless <text> country-code <value>
✓	<i>set interfaces wireless <interface> description <description></i>	inter-faces_wireless.xml.in: interfaces wireless <wireless> description	interfaces wireless <text> description <value>
✓	<i>set interfaces wireless <interface> dhcp-options client-id <description></i>	inter-faces_wireless.xml.in: interfaces wireless <wireless> dhcp-options client-id	interfaces wireless <text> dhcp-options client-id <value>
✓	<i>set interfaces wireless <interface> dhcp-options default-route-distance <distance></i>	inter-faces_wireless.xml.in: interfaces wireless <wireless> dhcp-options default-route-distance	interfaces wireless <text> dhcp-options default-route-distance <value>
✓	<i>set interfaces wireless <interface> dhcp-options host-name <hostname></i>	inter-faces_wireless.xml.in: interfaces wireless <wireless> dhcp-options host-name	interfaces wireless <text> dhcp-options host-name <value>
×	not yet documented	inter-faces_wireless.xml.in: interfaces wireless <wireless> dhcp-options mtu	interfaces wireless <text> dhcp-options mtu
✓	<i>set interfaces wireless <interface> dhcp-options no-default-route</i>	inter-faces_wireless.xml.in: interfaces wireless <wireless> dhcp-options no-default-route	interfaces wireless <text> dhcp-options no-default-route

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Table 1 – continued from previous page

Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
✓	<i>set interfaces wireless <interface> dhcp-options reject <address></i>	inter-faces_wireless.xml.in: interfaces wireless <wireless> dhcp-options reject	interfaces wireless <text> dhcp-options reject <value>
×	<i>set interfaces wireless <interface> dhcp-options user-class <string></i>	inter-faces_wireless.xml.in: interfaces wireless <wireless> dhcp-options user-class	Nothing found in VyOS
✓	<i>set interfaces wireless <interface> dhcp-options vendor-class-id <vendor-id></i>	inter-faces_wireless.xml.in: interfaces wireless <wireless> dhcp-options vendor-class-id	interfaces wireless <text> dhcp-options vendor-class-id <value>
✓	<i>set interfaces wireless <interface> dhcpv6-options duid <duid></i>	inter-faces_wireless.xml.in: interfaces wireless <wireless> dhcpv6-options duid	interfaces wireless <text> dhcpv6-options duid <value>
✓	<i>set interfaces wireless <interface> dhcpv6-options no-release</i>	inter-faces_wireless.xml.in: interfaces wireless <wireless> dhcpv6-options no-release	interfaces wireless <text> dhcpv6-options no-release
✓	<i>set interfaces wireless <interface> dhcpv6-options parameters-only</i>	inter-faces_wireless.xml.in: interfaces wireless <wireless> dhcpv6-options parameters-only	interfaces wireless <text> dhcpv6-options parameters-only
✓	<i>set interfaces wireless <interface> dhcpv6-options pd <id> interface <delegatee> address <address></i>	inter-faces_wireless.xml.in: interfaces wireless <wireless> dhcpv6-options pd <pd> interface <interface> address	interfaces wireless <text> dhcpv6-options pd <text> interface <text> address <value>
✓	<i>set interfaces wireless <interface> dhcpv6-options pd <id> interface <delegatee> sla-id <id></i>	inter-faces_wireless.xml.in: interfaces wireless <wireless> dhcpv6-options pd <pd> interface <interface> sla-id	interfaces wireless <text> dhcpv6-options pd <text> interface <text> sla-id <value>
✓	<i>set interfaces wireless <interface> dhcpv6-options pd <id> length <length></i>	inter-faces_wireless.xml.in: interfaces wireless <wireless> dhcpv6-options pd <pd> length	interfaces wireless <text> dhcpv6-options pd <text> length <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
✓	<i>set interfaces wireless <interface> dhcpv6-options rapid-commit</i>	inter-faces_wireless.xml.in: interfaces wireless <wireless> dhcpv6-options rapid-commit	interfaces wireless <text> dhcpv6-options rapid-commit
✓	<i>set interfaces wireless <interface> dhcpv6-options temporary</i>	inter-faces_wireless.xml.in: interfaces wireless <wireless> dhcpv6-options temporary	interfaces wireless <text> dhcpv6-options temporary
✓	<i>set interfaces wireless <interface> disable</i>	inter-faces_wireless.xml.in: interfaces wireless <wireless> disable	interfaces wireless <text> disable
×	not yet documented	inter-faces_wireless.xml.in: interfaces wireless <wireless> disable-broadcast-ssid	interfaces wireless <text> disable-broadcast-ssid
×	<i>set interfaces wireless <interface> disable-flow-control</i>	Nothing found in XML Definitions	Nothing found in VyOS
✓	<i>set interfaces wireless <interface> disable-link-detect</i>	inter-faces_wireless.xml.in: interfaces wireless <wireless> disable-link-detect	interfaces wireless <text> disable-link-detect
×	not yet documented	inter-faces_wireless.xml.in: interfaces wireless <wireless> expunge-failing-stations	interfaces wireless <text> expunge-failing-stations
×	not yet documented	inter-faces_wireless.xml.in: interfaces wireless <wireless> hw-id	interfaces wireless <text> hw-id <value>
✓	<i>set interfaces wireless <interface> ip adjust-mss <mss / clamp-mss-to-pmtu></i>	inter-faces_wireless.xml.in: interfaces wireless <wireless> ip adjust-mss	interfaces wireless <text> ip adjust-mss <value>
✓	<i>set interfaces wireless <interface> ip arp-cache-timeout</i>	inter-faces_wireless.xml.in: interfaces wireless <wireless> ip arp-cache-timeout	interfaces wireless <text> ip arp-cache-timeout <value>
✓	<i>set interfaces wireless <interface> ip disable-arp-filter</i>	inter-faces_wireless.xml.in: interfaces wireless <wireless> ip disable-arp-filter	interfaces wireless <text> ip disable-arp-filter

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Table 1 – continued from previous page

Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
✓	<code>set interfaces wireless <interface> ip disable-forwarding</code>	inter-faces_wireless.xml.in: interfaces wireless <wireless> ip disable-forwarding	interfaces wireless <text> ip disable-forwarding
✓	<code>set interfaces wireless <interface> ip enable-arp-accept</code>	inter-faces_wireless.xml.in: interfaces wireless <wireless> ip enable-arp-accept	interfaces wireless <text> ip enable-arp-accept
✓	<code>set interfaces wireless <interface> ip enable-arp-announce</code>	inter-faces_wireless.xml.in: interfaces wireless <wireless> ip enable-arp-announce	interfaces wireless <text> ip enable-arp-announce
✓	<code>set interfaces wireless <interface> ip enable-arp-ignore</code>	inter-faces_wireless.xml.in: interfaces wireless <wireless> ip enable-arp-ignore	interfaces wireless <text> ip enable-arp-ignore
✓	<code>set interfaces wireless <interface> ip enable-directed-broadcast</code>	inter-faces_wireless.xml.in: interfaces wireless <wireless> ip enable-directed-broadcast	interfaces wireless <text> ip enable-directed-broadcast
✓	<code>set interfaces wireless <interface> ip enable-proxy-arp</code>	inter-faces_wireless.xml.in: interfaces wireless <wireless> ip enable-proxy-arp	interfaces wireless <text> ip enable-proxy-arp
✓	<code>set interfaces wireless <interface> ip proxy-arp-pvlan</code>	inter-faces_wireless.xml.in: interfaces wireless <wireless> ip proxy-arp-pvlan	interfaces wireless <text> ip proxy-arp-pvlan
✓	<code>set interfaces wireless <interface> ip source-validation <strict loose disable></code>	inter-faces_wireless.xml.in: interfaces wireless <wireless> ip source-validation	interfaces wireless <text> ip source-validation <value>
✓	<code>set interfaces wireless <interface> ipv6 accept-dad <0 1 2></code>	inter-faces_wireless.xml.in: interfaces wireless <wireless> ipv6 accept-dad	interfaces wireless <text> ipv6 accept-dad <value>
✓	<code>set interfaces wireless <interface> ipv6 address autoconf</code>	inter-faces_wireless.xml.in: interfaces wireless <wireless> ipv6 address autoconf	interfaces wireless <text> ipv6 address autoconf
✓	<code>set interfaces wireless <interface> ipv6 address eui64 <prefix></code>	inter-faces_wireless.xml.in: interfaces wireless <wireless> ipv6 address eui64	interfaces wireless <text> ipv6 address eui64 <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
✓	<i>set interfaces wireless <interface> ipv6 address no-default-link-local</i>	inter-faces_wireless.xml.in: interfaces wireless <wireless> ipv6 address no-default-link-local	interfaces wireless <text> ipv6 address no-default-link-local
✓	<i>set interfaces wireless <interface> ipv6 adjust-mss <mss / clamp-mss-to-pmtu></i>	inter-faces_wireless.xml.in: interfaces wireless <wireless> ipv6 adjust-mss	interfaces wireless <text> ipv6 adjust-mss <value>
×	not yet documented	inter-faces_wireless.xml.in: interfaces wireless <wireless> ipv6 base-reachable-time	Nothing found in VyOS
✓	<i>set interfaces wireless <interface> ipv6 disable-forwarding</i>	inter-faces_wireless.xml.in: interfaces wireless <wireless> ipv6 disable-forwarding	interfaces wireless <text> ipv6 disable-forwarding
✓	<i>set interfaces wireless <interface> ipv6 dup-addr-detect-trans<n></i>	inter-faces_wireless.xml.in: interfaces wireless <wireless> ipv6 dup-addr-detect-transmits	interfaces wireless <text> ipv6 dup-addr-detect-transmits <value>
×	not yet documented	inter-faces_wireless.xml.in: interfaces wireless <wireless> ipv6 source-validation	Nothing found in VyOS
×	not yet documented	inter-faces_wireless.xml.in: interfaces wireless <wireless> isolate-stations	interfaces wireless <text> isolate-stations
✓	<i>set interfaces wireless <interface> mac <xx:xx:xx:xx:xx:xx></i>	inter-faces_wireless.xml.in: interfaces wireless <wireless> mac	interfaces wireless <text> mac <value>
×	not yet documented	inter-faces_wireless.xml.in: interfaces wireless <wireless> max-stations	interfaces wireless <text> max-stations <value>
×	not yet documented	inter-faces_wireless.xml.in: interfaces wireless <wireless> mgmt-frame-protection	interfaces wireless <text> mgmt-frame-protection <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	inter-faces_wireless.xml.in: interfaces wireless <wireless> mirror egress	interfaces wireless <text> mirror egress <value>
×	not yet documented	inter-faces_wireless.xml.in: interfaces wireless <wireless> mirror ingress	interfaces wireless <text> mirror ingress <value>
×	not yet documented	inter-faces_wireless.xml.in: interfaces wireless <wireless> mode	interfaces wireless <text> mode <value>
×	<i>set interfaces wireless <interface> mtu <mtu></i>	Nothing found in XML Definitions	Nothing found in VyOS
✓	<i>set interfaces wireless <interface> per-client-thread</i>	inter-faces_wireless.xml.in: interfaces wireless <wireless> per-client-thread	interfaces wireless <text> per-client-thread
×	not yet documented	inter-faces_wireless.xml.in: interfaces wireless <wireless> physical-device	interfaces wireless <text> physical-device <value>
×	not yet documented	inter-faces_wireless.xml.in: interfaces wireless <wireless> redirect	interfaces wireless <text> redirect <value>
×	not yet documented	inter-faces_wireless.xml.in: interfaces wireless <wireless> reduce-transmit-power	interfaces wireless <text> reduce-transmit-power <value>
×	not yet documented	inter-faces_wireless.xml.in: interfaces wireless <wireless> security station-address accept mac	interfaces wireless <text> security station-address accept mac <value>
×	not yet documented	inter-faces_wireless.xml.in: interfaces wireless <wireless> security station-address deny mac	interfaces wireless <text> security station-address deny mac <value>
×	not yet documented	inter-faces_wireless.xml.in: interfaces wireless <wireless> security station-address mode	interfaces wireless <text> security station-address mode <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	inter- faces_wireless.xml.in: interfaces wireless <wire- less> security wep key	interfaces wireless <text> security wep key <value>
×	not yet documented	inter- faces_wireless.xml.in: interfaces wireless <wire- less> security wpa cipher	interfaces wireless <text> security wpa cipher <value>
×	not yet documented	inter- faces_wireless.xml.in: interfaces wireless <wire- less> security wpa group- cipher	interfaces wireless <text> security wpa group-cipher <value>
×	not yet documented	inter- faces_wireless.xml.in: interfaces wireless <wire- less> security wpa mode	interfaces wireless <text> security wpa mode <value>
×	not yet documented	inter- faces_wireless.xml.in: interfaces wireless <wire- less> security wpa passphrase	interfaces wireless <text> security wpa passphrase <value>
×	not yet documented	inter- faces_wireless.xml.in: interfaces wireless <wire- less> security wpa radius server <server> account- ing	interfaces wireless <text> security wpa radius server <text> accounting
×	not yet documented	inter- faces_wireless.xml.in: interfaces wireless <wire- less> security wpa radius server <server> disable	interfaces wireless <text> security wpa radius server <text> disable
×	not yet documented	inter- faces_wireless.xml.in: interfaces wireless <wire- less> security wpa radius server <server> key	interfaces wireless <text> security wpa radius server <text> key <value>
×	not yet documented	inter- faces_wireless.xml.in: interfaces wireless <wire- less> security wpa radius server <server> port	interfaces wireless <text> security wpa radius server <text> port <value>
×	not yet documented	inter- faces_wireless.xml.in: interfaces wireless <wire- less> security wpa radius source-address	interfaces wireless <text> security wpa radius source-address <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	inter- faces_wireless.xml.in: interfaces wireless <wire- less> ssid	interfaces wireless <text> ssid <value>
×	not yet documented	inter- faces_wireless.xml.in: interfaces wireless <wire- less> type	interfaces wireless <text> type <value>
×	<i>set interfaces wireless <interface> vif <vlan-id></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	inter- faces_wireless.xml.in: interfaces wireless <wire- less> vif-s <vif-s> address	interfaces wireless <text> vif-s <text> address <value>
×	not yet documented	inter- faces_wireless.xml.in: interfaces wireless <wire- less> vif-s <vif-s> description	interfaces wireless <text> vif-s <text> description <value>
×	not yet documented	inter- faces_wireless.xml.in: interfaces wireless <wire- less> vif-s <vif-s> dhcp- options client-id	interfaces wireless <text> vif-s <text> dhcp-options client-id <value>
×	not yet documented	inter- faces_wireless.xml.in: interfaces wireless <wire- less> vif-s <vif-s> dhcp-options default- route-distance	interfaces wireless <text> vif-s <text> dhcp-options default-route-distance <value>
×	not yet documented	inter- faces_wireless.xml.in: interfaces wireless <wire- less> vif-s <vif-s> dhcp- options host-name	interfaces wireless <text> vif-s <text> dhcp-options host-name <value>
×	not yet documented	inter- faces_wireless.xml.in: interfaces wireless <wire- less> vif-s <vif-s> dhcp- options mtu	interfaces wireless <text> vif-s <text> dhcp-options mtu
×	not yet documented	inter- faces_wireless.xml.in: interfaces wireless <wire- less> vif-s <vif-s> dhcp- options no-default-route	interfaces wireless <text> vif-s <text> dhcp-options no-default-route

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	inter- faces_wireless.xml.in: interfaces wireless <wire- less> vif-s <vif-s> dhcp- options reject	interfaces wireless <text> vif-s <text> dhcp-options reject <value>
×	not yet documented	inter- faces_wireless.xml.in: interfaces wireless <wire- less> vif-s <vif-s> dhcp- options user-class	Nothing found in VyOS
×	not yet documented	inter- faces_wireless.xml.in: interfaces wireless <wire- less> vif-s <vif-s> dhcp- options vendor-class-id	interfaces wireless <text> vif-s <text> dhcp-options vendor-class-id <value>
×	not yet documented	inter- faces_wireless.xml.in: interfaces wireless <wire- less> vif-s <vif-s> dhcpv6-options duid	interfaces wireless <text> vif-s <text> dhcpv6- options duid <value>
×	not yet documented	inter- faces_wireless.xml.in: interfaces wireless <wire- less> vif-s <vif-s> dhcpv6-options no-release	interfaces wireless <text> vif-s <text> dhcpv6- options no-release
×	not yet documented	inter- faces_wireless.xml.in: interfaces wireless <wireless> vif-s <vif- s> dhcpv6-options parameters-only	interfaces wireless <text> vif-s <text> dhcpv6- options parameters-only
×	not yet documented	inter- faces_wireless.xml.in: interfaces wireless <wire- less> vif-s <vif-s> dhcpv6-options pd <pd> interface <interface> address	interfaces wireless <text> vif-s <text> dhcpv6- options pd <text> in- terface <text> address <value>
×	not yet documented	inter- faces_wireless.xml.in: interfaces wireless <wire- less> vif-s <vif-s> dhcpv6-options pd <pd> interface <interface> sla-id	interfaces wireless <text> vif-s <text> dhcpv6- options pd <text> in- terface <text> sla-id <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	inter- faces_wireless.xml.in: interfaces wireless <wire- less> vif-s <vif-s> dhcpv6-options pd <pd> length	interfaces wireless <text> vif-s <text> dhcpv6- options pd <text> length <value>
×	not yet documented	inter- faces_wireless.xml.in: interfaces wireless <wire- less> vif-s <vif-s> dhcpv6-options rapid- commit	interfaces wireless <text> vif-s <text> dhcpv6- options rapid-commit
×	not yet documented	inter- faces_wireless.xml.in: interfaces wireless <wire- less> vif-s <vif-s> dhcpv6-options tempo- rary	interfaces wireless <text> vif-s <text> dhcpv6- options temporary
×	not yet documented	inter- faces_wireless.xml.in: interfaces wireless <wire- less> vif-s <vif-s> disable	interfaces wireless <text> vif-s <text> disable
×	not yet documented	inter- faces_wireless.xml.in: interfaces wireless <wire- less> vif-s <vif-s> disable- link-detect	interfaces wireless <text> vif-s <text> disable-link- detect
×	not yet documented	inter- faces_wireless.xml.in: interfaces wireless <wire- less> vif-s <vif-s> ip adjust-mss	interfaces wireless <text> vif-s <text> ip adjust-mss <value>
×	not yet documented	inter- faces_wireless.xml.in: interfaces wireless <wire- less> vif-s <vif-s> ip arp- cache-timeout	interfaces wireless <text> vif-s <text> ip arp-cache- timeout <value>
×	not yet documented	inter- faces_wireless.xml.in: interfaces wireless <wire- less> vif-s <vif-s> ip disable-arp-filter	interfaces wireless <text> vif-s <text> ip disable-arp- filter
×	not yet documented	inter- faces_wireless.xml.in: interfaces wireless <wire- less> vif-s <vif-s> ip disable-forwarding	interfaces wireless <text> vif-s <text> ip disable- forwarding

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	inter- faces_wireless.xml.in: interfaces wireless <wire- less> vif-s <vif-s> ip enable-arp-accept	interfaces wireless <text> vif-s <text> ip enable-arp- accept
×	not yet documented	inter- faces_wireless.xml.in: interfaces wireless <wire- less> vif-s <vif-s> ip enable-arp-announce	interfaces wireless <text> vif-s <text> ip enable-arp- announce
×	not yet documented	inter- faces_wireless.xml.in: interfaces wireless <wire- less> vif-s <vif-s> ip enable-arp-ignore	interfaces wireless <text> vif-s <text> ip enable-arp- ignore
×	not yet documented	inter- faces_wireless.xml.in: interfaces wireless <wire- less> vif-s <vif-s> ip enable-directed-broadcast	interfaces wireless <text> vif-s <text> ip enable- directed-broadcast
×	not yet documented	inter- faces_wireless.xml.in: interfaces wireless <wire- less> vif-s <vif-s> ip enable-proxy-arp	interfaces wireless <text> vif-s <text> ip enable- proxy-arp
×	not yet documented	inter- faces_wireless.xml.in: interfaces wireless <wire- less> vif-s <vif-s> ip proxy-arp-pvlan	interfaces wireless <text> vif-s <text> ip proxy-arp- pvlan
×	not yet documented	inter- faces_wireless.xml.in: interfaces wireless <wire- less> vif-s <vif-s> ip source-validation	interfaces wireless <text> vif-s <text> ip source- validation <value>
×	not yet documented	inter- faces_wireless.xml.in: interfaces wireless <wire- less> vif-s <vif-s> ipv6 accept-dad	interfaces wireless <text> vif-s <text> ipv6 accept- dad <value>
×	not yet documented	inter- faces_wireless.xml.in: interfaces wireless <wire- less> vif-s <vif-s> ipv6 address autoconf	interfaces wireless <text> vif-s <text> ipv6 address autoconf

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	inter-faces_wireless.xml.in: interfaces wireless <wireless> vif-s <vif-s> ipv6 address eui64	interfaces wireless <text> vif-s <text> ipv6 address eui64 <value>
×	not yet documented	inter-faces_wireless.xml.in: interfaces wireless <wireless> vif-s <vif-s> ipv6 address no-default-link-local	interfaces wireless <text> vif-s <text> ipv6 address no-default-link-local
×	not yet documented	inter-faces_wireless.xml.in: interfaces wireless <wireless> vif-s <vif-s> ipv6 adjust-mss	interfaces wireless <text> vif-s <text> ipv6 adjust-mss <value>
×	not yet documented	inter-faces_wireless.xml.in: interfaces wireless <wireless> vif-s <vif-s> ipv6 base-reachable-time	Nothing found in VyOS
×	not yet documented	inter-faces_wireless.xml.in: interfaces wireless <wireless> vif-s <vif-s> ipv6 disable-forwarding	interfaces wireless <text> vif-s <text> ipv6 disable-forwarding
×	not yet documented	inter-faces_wireless.xml.in: interfaces wireless <wireless> vif-s <vif-s> ipv6 dup-addr-detect-transmits	interfaces wireless <text> vif-s <text> ipv6 dup-addr-detect-transmits <value>
×	not yet documented	inter-faces_wireless.xml.in: interfaces wireless <wireless> vif-s <vif-s> ipv6 source-validation	Nothing found in VyOS
×	not yet documented	inter-faces_wireless.xml.in: interfaces wireless <wireless> vif-s <vif-s> mac	interfaces wireless <text> vif-s <text> mac <value>
×	not yet documented	inter-faces_wireless.xml.in: interfaces wireless <wireless> vif-s <vif-s> mirror egress	interfaces wireless <text> vif-s <text> mirror egress <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	inter- faces_wireless.xml.in: interfaces wireless <wire- less> vif-s <vif-s> mirror ingress	interfaces wireless <text> vif-s <text> mirror ingress <value>
×	not yet documented	inter- faces_wireless.xml.in: interfaces wireless <wire- less> vif-s <vif-s> mtu	interfaces wireless <text> vif-s <text> mtu <value>
×	not yet documented	inter- faces_wireless.xml.in: interfaces wireless <wire- less> vif-s <vif-s> proto- col	interfaces wireless <text> vif-s <text> protocol <value>
×	not yet documented	inter- faces_wireless.xml.in: interfaces wireless <wire- less> vif-s <vif-s> redirect	interfaces wireless <text> vif-s <text> redirect <value>
✓	<i>set interfaces wireless <interface> vif-s <vlan-id> vif-c <vlan-id> address <address / dhcp / dhcpv6></i>	inter- faces_wireless.xml.in: interfaces wireless <wire- less> vif-s <vif-s> vif-c <vif-c> address	interfaces wireless <text> vif-s <text> vif-c <text> address <value>
✓	<i>set interfaces wireless <interface> vif-s <vlan-id> vif-c <vlan-id> description <description></i>	inter- faces_wireless.xml.in: interfaces wireless <wire- less> vif-s <vif-s> vif-c <vif-c> description	interfaces wireless <text> vif-s <text> vif-c <text> description <value>
✓	<i>set interfaces wireless <interface> vif-s <vlan-id> vif-c <vlan-id> dhcp-options client-id <description></i>	inter- faces_wireless.xml.in: interfaces wireless <wire- less> vif-s <vif-s> vif-c <vif-c> dhcp-options client-id	interfaces wireless <text> vif-s <text> vif-c <text> dhcp-options client-id <value>
✓	<i>set interfaces wireless <interface> vif-s <vlan-id> vif-c <vlan-id> dhcp-options default-route-distanc <distance></i>	inter- faces_wireless.xml.in: interfaces wireless <wire- less> vif-s <vif-s> vif-c <vif-c> dhcp-options default-route-distance	interfaces wireless <text> vif-s <text> vif-c <text> dhcp-options default- route-distance <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
✓	<i>set interfaces wireless <interface> vif-s <vlan-id> vif-c <vlan-id> dhcp-options host-name <hostname></i>	inter-faces_wireless.xml.in: interfaces wireless <wireless> vif-s <vif-s> vif-c <vif-c> dhcp-options host-name	interfaces wireless <text> vif-s <text> vif-c <text> dhcp-options host-name <value>
×	not yet documented	inter-faces_wireless.xml.in: interfaces wireless <wireless> vif-s <vif-s> vif-c <vif-c> dhcp-options mtu	interfaces wireless <text> vif-s <text> vif-c <text> dhcp-options mtu
✓	<i>set interfaces wireless <interface> vif-s <vlan-id> vif-c <vlan-id> dhcp-options no-default-route</i>	inter-faces_wireless.xml.in: interfaces wireless <wireless> vif-s <vif-s> vif-c <vif-c> dhcp-options no-default-route	interfaces wireless <text> vif-s <text> vif-c <text> dhcp-options no-default-route
✓	<i>set interfaces wireless <interface> vif-s <vlan-id> vif-c <vlan-id> dhcp-options reject <address></i>	inter-faces_wireless.xml.in: interfaces wireless <wireless> vif-s <vif-s> vif-c <vif-c> dhcp-options reject	interfaces wireless <text> vif-s <text> vif-c <text> dhcp-options reject <value>
×	<i>set interfaces wireless <interface> vif-s <vlan-id> vif-c <vlan-id> dhcp-options user-class <string></i>	inter-faces_wireless.xml.in: interfaces wireless <wireless> vif-s <vif-s> vif-c <vif-c> dhcp-options user-class	Nothing found in VyOS
✓	<i>set interfaces wireless <interface> vif-s <vlan-id> vif-c <vlan-id> dhcp-options vendor-class-id <vendor-id></i>	inter-faces_wireless.xml.in: interfaces wireless <wireless> vif-s <vif-s> vif-c <vif-c> dhcp-options vendor-class-id	interfaces wireless <text> vif-s <text> vif-c <text> dhcp-options vendor-class-id <value>
✓	<i>set interfaces wireless <interface> vif-s <vlan-id> vif-c <vlan-id> dhcpv6-options duid <duid></i>	inter-faces_wireless.xml.in: interfaces wireless <wireless> vif-s <vif-s> vif-c <vif-c> dhcpv6-options duid	interfaces wireless <text> vif-s <text> vif-c <text> dhcpv6-options duid <value>
✓	<i>set interfaces wireless <interface> vif-s <vlan-id> vif-c <vlan-id> dhcpv6-options no-release</i>	inter-faces_wireless.xml.in: interfaces wireless <wireless> vif-s <vif-s> vif-c <vif-c> dhcpv6-options no-release	interfaces wireless <text> vif-s <text> vif-c <text> dhcpv6-options no-release

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
✓	<code>set interfaces wireless <interface> vif-s <vlan-id> vif-c <vlan-id> dhcpv6-options parameters-only</code>	inter-faces_wireless.xml.in: interfaces wireless <wireless> vif-s <vif-s> vif-c <vif-c> dhcpv6-options parameters-only	interfaces wireless <text> vif-s <text> vif-c <text> dhcpv6-options parameters-only
✓	<code>set interfaces wireless <interface> vif-s <vlan-id> vif-c <vlan-id> dhcpv6-options pd <id> interface <delegatee> address <address></code>	inter-faces_wireless.xml.in: interfaces wireless <wireless> vif-s <vif-s> vif-c <vif-c> dhcpv6-options pd <pd> interface <interface> address	interfaces wireless <text> vif-s <text> vif-c <text> dhcpv6-options pd <text> interface <text> address <value>
✓	<code>set interfaces wireless <interface> vif-s <vlan-id> vif-c <vlan-id> dhcpv6-options pd <id> interface <delegatee> sla-id <id></code>	inter-faces_wireless.xml.in: interfaces wireless <wireless> vif-s <vif-s> vif-c <vif-c> dhcpv6-options pd <pd> interface <interface> sla-id	interfaces wireless <text> vif-s <text> vif-c <text> dhcpv6-options pd <text> interface <text> sla-id <value>
✓	<code>set interfaces wireless <interface> vif-s <vlan-id> vif-c <vlan-id> dhcpv6-options pd <id> length <length></code>	inter-faces_wireless.xml.in: interfaces wireless <wireless> vif-s <vif-s> vif-c <vif-c> dhcpv6-options pd <pd> length	interfaces wireless <text> vif-s <text> vif-c <text> dhcpv6-options pd <text> length <value>
✓	<code>set interfaces wireless <interface> vif-s <vlan-id> vif-c <vlan-id> dhcpv6-options rapid-commit</code>	inter-faces_wireless.xml.in: interfaces wireless <wireless> vif-s <vif-s> vif-c <vif-c> dhcpv6-options rapid-commit	interfaces wireless <text> vif-s <text> vif-c <text> dhcpv6-options rapid-commit
✓	<code>set interfaces wireless <interface> vif-s <vlan-id> vif-c <vlan-id> dhcpv6-options temporary</code>	inter-faces_wireless.xml.in: interfaces wireless <wireless> vif-s <vif-s> vif-c <vif-c> dhcpv6-options temporary	interfaces wireless <text> vif-s <text> vif-c <text> dhcpv6-options temporary
✓	<code>set interfaces wireless <interface> vif-s <vlan-id> vif-c <vlan-id> disable</code>	inter-faces_wireless.xml.in: interfaces wireless <wireless> vif-s <vif-s> vif-c <vif-c> disable	interfaces wireless <text> vif-s <text> vif-c <text> disable

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
✓	<code>set interfaces wireless <interface> vif-s <vlan-id> vif-c <vlan-id> disable-link-detect</code>	inter- faces_wireless.xml.in: interfaces wireless <wire- less> vif-s <vif-s> vif-c <vif-c> disable-link- detect	interfaces wireless <text> vif-s <text> vif-c <text> disable-link-detect
✓	<code>set interfaces wireless <interface> vif-s <vlan-id> vif-c <vlan-id> ip adjust-mss <mss clamp-mss-to-pmtu></code>	inter- faces_wireless.xml.in: interfaces wireless <wire- less> vif-s <vif-s> vif-c <vif-c> ip adjust-mss	interfaces wireless <text> vif-s <text> vif-c <text> ip adjust-mss <value>
✓	<code>set interfaces wireless <interface> vif-s <vlan-id> vif-c <vlan-id> ip arp-cache-timeout</code>	inter- faces_wireless.xml.in: interfaces wireless <wire- less> vif-s <vif-s> vif- c <vif-c> ip arp-cache- timeout	interfaces wireless <text> vif-s <text> vif-c <text> ip arp-cache-timeout <value>
✓	<code>set interfaces wireless <interface> vif-s <vlan-id> vif-c <vlan-id> ip disable-arp-filter</code>	inter- faces_wireless.xml.in: interfaces wireless <wire- less> vif-s <vif-s> vif- c <vif-c> ip disable-arp- filter	interfaces wireless <text> vif-s <text> vif-c <text> ip disable-arp-filter
✓	<code>set interfaces wireless <interface> vif-s <vlan-id> vif-c <vlan-id> ip disable-forwarding</code>	inter- faces_wireless.xml.in: interfaces wireless <wire- less> vif-s <vif-s> vif-c <vif-c> ip disable- forwarding	interfaces wireless <text> vif-s <text> vif-c <text> ip disable-forwarding
✓	<code>set interfaces wireless <interface> vif-s <vlan-id> vif-c <vlan-id> ip enable-arp-accept</code>	inter- faces_wireless.xml.in: interfaces wireless <wire- less> vif-s <vif-s> vif- c <vif-c> ip enable-arp- accept	interfaces wireless <text> vif-s <text> vif-c <text> ip enable-arp-accept
✓	<code>set interfaces wireless <interface> vif-s <vlan-id> vif-c <vlan-id> ip enable-arp-announce</code>	inter- faces_wireless.xml.in: interfaces wireless <wire- less> vif-s <vif-s> vif- c <vif-c> ip enable-arp- announce	interfaces wireless <text> vif-s <text> vif-c <text> ip enable-arp-announce
✓	<code>set interfaces wireless <interface> vif-s <vlan-id> vif-c <vlan-id> ip enable-arp-ignore</code>	inter- faces_wireless.xml.in: interfaces wireless <wire- less> vif-s <vif-s> vif- c <vif-c> ip enable-arp- ignore	interfaces wireless <text> vif-s <text> vif-c <text> ip enable-arp-ignore

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
✓	<i>set interfaces wireless <interface> vif-s <vlan-id> vif-c <vlan-id> ip enable-directed-broad</i>	inter-faces_wireless.xml.in: interfaces wireless <wireless> vif-s <vif-s> vif-c <vif-c> ip enable-directed-broadcast	interfaces wireless <text> vif-s <text> vif-c <text> ip enable-directed-broadcast
✓	<i>set interfaces wireless <interface> vif-s <vlan-id> vif-c <vlan-id> ip enable-proxy-arp</i>	inter-faces_wireless.xml.in: interfaces wireless <wireless> vif-s <vif-s> vif-c <vif-c> ip enable-proxy-arp	interfaces wireless <text> vif-s <text> vif-c <text> ip enable-proxy-arp
✓	<i>set interfaces wireless <interface> vif-s <vlan-id> vif-c <vlan-id> ip proxy-arp-pvlan</i>	inter-faces_wireless.xml.in: interfaces wireless <wireless> vif-s <vif-s> vif-c <vif-c> ip proxy-arp-pvlan	interfaces wireless <text> vif-s <text> vif-c <text> ip proxy-arp-pvlan
✓	<i>set interfaces wireless <interface> vif-s <vlan-id> vif-c <vlan-id> ip source-validation <strict loose disable></i>	inter-faces_wireless.xml.in: interfaces wireless <wireless> vif-s <vif-s> vif-c <vif-c> ip source-validation	interfaces wireless <text> vif-s <text> vif-c <text> ip source-validation <value>
✓	<i>set interfaces wireless <interface> vif-s <vlan-id> vif-c <vlan-id> ipv6 accept-dad <0 1 2></i>	inter-faces_wireless.xml.in: interfaces wireless <wireless> vif-s <vif-s> vif-c <vif-c> ipv6 accept-dad	interfaces wireless <text> vif-s <text> vif-c <text> ipv6 accept-dad <value>
✓	<i>set interfaces wireless <interface> vif-s <vlan-id> vif-c <vlan-id> ipv6 address autoconf</i>	inter-faces_wireless.xml.in: interfaces wireless <wireless> vif-s <vif-s> vif-c <vif-c> ipv6 address autoconf	interfaces wireless <text> vif-s <text> vif-c <text> ipv6 address autoconf
✓	<i>set interfaces wireless <interface> vif-s <vlan-id> vif-c <vlan-id> ipv6 address eui64 <prefix></i>	inter-faces_wireless.xml.in: interfaces wireless <wireless> vif-s <vif-s> vif-c <vif-c> ipv6 address eui64	interfaces wireless <text> vif-s <text> vif-c <text> ipv6 address eui64 <value>
✓	<i>set interfaces wireless <interface> vif-s <vlan-id> vif-c <vlan-id> ipv6 address no-default-link-local</i>	inter-faces_wireless.xml.in: interfaces wireless <wireless> vif-s <vif-s> vif-c <vif-c> ipv6 address no-default-link-local	interfaces wireless <text> vif-s <text> vif-c <text> ipv6 address no-default-link-local

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
✓	<i>set interfaces wireless <interface> vif-s <vlan-id> vif-c <vlan-id> ipv6 adjust-mss <mss / clamp-mss-to-pmtu></i>	inter-faces_wireless.xml.in: interfaces wireless <wireless> vif-s <vif-s> vif-c <vif-c> ipv6 adjust-mss	interfaces wireless <text> vif-s <text> vif-c <text> ipv6 adjust-mss <value>
×	not yet documented	inter-faces_wireless.xml.in: interfaces wireless <wireless> vif-s <vif-s> vif-c <vif-c> ipv6 base-reachable-time	Nothing found in VyOS
✓	<i>set interfaces wireless <interface> vif-s <vlan-id> vif-c <vlan-id> ipv6 disable-forwarding</i>	inter-faces_wireless.xml.in: interfaces wireless <wireless> vif-s <vif-s> vif-c <vif-c> ipv6 disable-forwarding	interfaces wireless <text> vif-s <text> vif-c <text> ipv6 disable-forwarding
✓	<i>set interfaces wireless <interface> vif-s <vlan-id> vif-c <vlan-id> ipv6 dup-addr-detect-trans<n></i>	inter-faces_wireless.xml.in: interfaces wireless <wireless> vif-s <vif-s> vif-c <vif-c> ipv6 dup-addr-detect-transmits	interfaces wireless <text> vif-s <text> vif-c <text> ipv6 dup-addr-detect-transmits <value>
×	not yet documented	inter-faces_wireless.xml.in: interfaces wireless <wireless> vif-s <vif-s> vif-c <vif-c> ipv6 source-validation	Nothing found in VyOS
✓	<i>set interfaces wireless <interface> vif-s <vlan-id> vif-c <vlan-id> mac <xx:xx:xx:xx:xx:xx></i>	inter-faces_wireless.xml.in: interfaces wireless <wireless> vif-s <vif-s> vif-c <vif-c> mac	interfaces wireless <text> vif-s <text> vif-c <text> mac <value>
×	not yet documented	inter-faces_wireless.xml.in: interfaces wireless <wireless> vif-s <vif-s> vif-c <vif-c> mirror egress	interfaces wireless <text> vif-s <text> vif-c <text> mirror egress <value>
×	not yet documented	inter-faces_wireless.xml.in: interfaces wireless <wireless> vif-s <vif-s> vif-c <vif-c> mirror ingress	interfaces wireless <text> vif-s <text> vif-c <text> mirror ingress <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
✓	<code>set interfaces wireless <interface> vif-s <vlan-id> vif-c <vlan-id> mtu <mtu></code>	inter-faces_wireless.xml.in: interfaces wireless <wireless> vif-s <vif-s> vif-c <vif-c> mtu	interfaces wireless <text> vif-s <text> vif-c <text> mtu <value>
×	not yet documented	inter-faces_wireless.xml.in: interfaces wireless <wireless> vif-s <vif-s> vif-c <vif-c> redirect	interfaces wireless <text> vif-s <text> vif-c <text> redirect <value>
✓	<code>set interfaces wireless <interface> vif-s <vlan-id> vif-c <vlan-id> vrf <vrf></code>	inter-faces_wireless.xml.in: interfaces wireless <wireless> vif-s <vif-s> vif-c <vif-c> vrf	interfaces wireless <text> vif-s <text> vif-c <text> vrf <value>
×	not yet documented	inter-faces_wireless.xml.in: interfaces wireless <wireless> vif-s <vif-s> vrf	interfaces wireless <text> vif-s <text> vrf <value>
✓	<code>set interfaces wireless <interface> vif <vlan-id> address <address> dhcp dhcpv6></code>	inter-faces_wireless.xml.in: interfaces wireless <wireless> vif <vif> address	interfaces wireless <text> vif <text> address <value>
✓	<code>set interfaces wireless <interface> vif <vlan-id> description <description></code>	inter-faces_wireless.xml.in: interfaces wireless <wireless> vif <vif> description	interfaces wireless <text> vif <text> description <value>
✓	<code>set interfaces wireless <interface> vif <vlan-id> dhcp-options client-id <description></code>	inter-faces_wireless.xml.in: interfaces wireless <wireless> vif <vif> dhcp-options client-id	interfaces wireless <text> vif <text> dhcp-options client-id <value>
✓	<code>set interfaces wireless <interface> vif <vlan-id> dhcp-options default-route-distance <distance></code>	inter-faces_wireless.xml.in: interfaces wireless <wireless> vif <vif> dhcp-options default-route-distance	interfaces wireless <text> vif <text> dhcp-options default-route-distance <value>
✓	<code>set interfaces wireless <interface> vif <vlan-id> dhcp-options host-name <hostname></code>	inter-faces_wireless.xml.in: interfaces wireless <wireless> vif <vif> dhcp-options host-name	interfaces wireless <text> vif <text> dhcp-options host-name <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	inter-faces_wireless.xml.in: interfaces wireless <wireless> vif <vif> dhcp-options mtu	interfaces wireless <text> vif <text> dhcp-options mtu
✓	<i>set interfaces wireless <interface> vif <vlan-id> dhcp-options no-default-route</i>	inter-faces_wireless.xml.in: interfaces wireless <wireless> vif <vif> dhcp-options no-default-route	interfaces wireless <text> vif <text> dhcp-options no-default-route
✓	<i>set interfaces wireless <interface> vif <vlan-id> dhcp-options reject <address></i>	inter-faces_wireless.xml.in: interfaces wireless <wireless> vif <vif> dhcp-options reject	interfaces wireless <text> vif <text> dhcp-options reject <value>
×	<i>set interfaces wireless <interface> vif <vlan-id> dhcp-options user-class <string></i>	inter-faces_wireless.xml.in: interfaces wireless <wireless> vif <vif> dhcp-options user-class	Nothing found in VyOS
✓	<i>set interfaces wireless <interface> vif <vlan-id> dhcp-options vendor-class-id <vendor-id></i>	inter-faces_wireless.xml.in: interfaces wireless <wireless> vif <vif> dhcp-options vendor-class-id	interfaces wireless <text> vif <text> dhcp-options vendor-class-id <value>
✓	<i>set interfaces wireless <interface> vif <vlan-id> dhcpv6-options duid <duid></i>	inter-faces_wireless.xml.in: interfaces wireless <wireless> vif <vif> dhcpv6-options duid	interfaces wireless <text> vif <text> dhcpv6-options duid <value>
✓	<i>set interfaces wireless <interface> vif <vlan-id> dhcpv6-options no-release</i>	inter-faces_wireless.xml.in: interfaces wireless <wireless> vif <vif> dhcpv6-options no-release	interfaces wireless <text> vif <text> dhcpv6-options no-release
✓	<i>set interfaces wireless <interface> vif <vlan-id> dhcpv6-options parameters-only</i>	inter-faces_wireless.xml.in: interfaces wireless <wireless> vif <vif> dhcpv6-options parameters-only	interfaces wireless <text> vif <text> dhcpv6-options parameters-only
✓	<i>set interfaces wireless <interface> vif <vlan-id> dhcpv6-options pd <id> interface <delegatee> address <address></i>	inter-faces_wireless.xml.in: interfaces wireless <wireless> vif <vif> dhcpv6-options pd <pd> interface <interface> address	interfaces wireless <text> vif <text> dhcpv6-options pd <text> interface <text> address <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
✓	<i>set interfaces wireless <interface> vif <vlan-id> dhcpv6-options pd <id> interface <delegatee> sla-id <id></i>	inter-faces_wireless.xml.in: interfaces wireless <wireless> vif <vif> dhcpv6-options pd <pd> interface <interface> sla-id	interfaces wireless <text> vif <text> dhcpv6-options pd <text> interface <text> sla-id <value>
✓	<i>set interfaces wireless <interface> vif <vlan-id> dhcpv6-options pd <id> length <length></i>	inter-faces_wireless.xml.in: interfaces wireless <wireless> vif <vif> dhcpv6-options pd <pd> length	interfaces wireless <text> vif <text> dhcpv6-options pd <text> length <value>
✓	<i>set interfaces wireless <interface> vif <vlan-id> dhcpv6-options rapid-commit</i>	inter-faces_wireless.xml.in: interfaces wireless <wireless> vif <vif> dhcpv6-options rapid-commit	interfaces wireless <text> vif <text> dhcpv6-options rapid-commit
✓	<i>set interfaces wireless <interface> vif <vlan-id> dhcpv6-options temporary</i>	inter-faces_wireless.xml.in: interfaces wireless <wireless> vif <vif> dhcpv6-options temporary	interfaces wireless <text> vif <text> dhcpv6-options temporary
✓	<i>set interfaces wireless <interface> vif <vlan-id> disable</i>	inter-faces_wireless.xml.in: interfaces wireless <wireless> vif <vif> disable	interfaces wireless <text> vif <text> disable
✓	<i>set interfaces wireless <interface> vif <vlan-id> disable-link-detect</i>	inter-faces_wireless.xml.in: interfaces wireless <wireless> vif <vif> disable-link-detect	interfaces wireless <text> vif <text> disable-link-detect
×	not yet documented	inter-faces_wireless.xml.in: interfaces wireless <wireless> vif <vif> egress-qos	interfaces wireless <text> vif <text> egress-qos <value>
×	not yet documented	inter-faces_wireless.xml.in: interfaces wireless <wireless> vif <vif> ingress-qos	interfaces wireless <text> vif <text> ingress-qos <value>
✓	<i>set interfaces wireless <interface> vif <vlan-id> ip adjust-mss <mss / clamp-mss-to-pmtu></i>	inter-faces_wireless.xml.in: interfaces wireless <wireless> vif <vif> ip adjust-mss	interfaces wireless <text> vif <text> ip adjust-mss <value>

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Table 1 – continued from previous page

Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
✓	<i>set interfaces wireless <interface> vif <vlan-id> ip arp-cache-timeout</i>	inter-faces_wireless.xml.in: interfaces wireless <wireless> vif <vif> ip arp-cache-timeout	interfaces wireless <text> vif <text> ip arp-cache-timeout <value>
✓	<i>set interfaces wireless <interface> vif <vlan-id> ip disable-arp-filter</i>	inter-faces_wireless.xml.in: interfaces wireless <wireless> vif <vif> ip disable-arp-filter	interfaces wireless <text> vif <text> ip disable-arp-filter
✓	<i>set interfaces wireless <interface> vif <vlan-id> ip disable-forwarding</i>	inter-faces_wireless.xml.in: interfaces wireless <wireless> vif <vif> ip disable-forwarding	interfaces wireless <text> vif <text> ip disable-forwarding
✓	<i>set interfaces wireless <interface> vif <vlan-id> ip enable-arp-accept</i>	inter-faces_wireless.xml.in: interfaces wireless <wireless> vif <vif> ip enable-arp-accept	interfaces wireless <text> vif <text> ip enable-arp-accept
✓	<i>set interfaces wireless <interface> vif <vlan-id> ip enable-arp-announce</i>	inter-faces_wireless.xml.in: interfaces wireless <wireless> vif <vif> ip enable-arp-announce	interfaces wireless <text> vif <text> ip enable-arp-announce
✓	<i>set interfaces wireless <interface> vif <vlan-id> ip enable-arp-ignore</i>	inter-faces_wireless.xml.in: interfaces wireless <wireless> vif <vif> ip enable-arp-ignore	interfaces wireless <text> vif <text> ip enable-arp-ignore
✓	<i>set interfaces wireless <interface> vif <vlan-id> ip enable-directed-broadcast</i>	inter-faces_wireless.xml.in: interfaces wireless <wireless> vif <vif> ip enable-directed-broadcast	interfaces wireless <text> vif <text> ip enable-directed-broadcast
✓	<i>set interfaces wireless <interface> vif <vlan-id> ip enable-proxy-arp</i>	inter-faces_wireless.xml.in: interfaces wireless <wireless> vif <vif> ip enable-proxy-arp	interfaces wireless <text> vif <text> ip enable-proxy-arp
✓	<i>set interfaces wireless <interface> vif <vlan-id> ip proxy-arp-pvlan</i>	inter-faces_wireless.xml.in: interfaces wireless <wireless> vif <vif> ip proxy-arp-pvlan	interfaces wireless <text> vif <text> ip proxy-arp-pvlan

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Table 1 – continued from previous page

Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
✓	<code>set interfaces wireless <interface> vif <vlan-id> ip source-validation <strict loose disable></code>	inter-faces_wireless.xml.in: interfaces wireless <wireless> vif <vif> ip source-validation	interfaces wireless <text> vif <text> ip source-validation <value>
✓	<code>set interfaces wireless <interface> vif <vlan-id> ipv6 accept-dad <0 1 2></code>	inter-faces_wireless.xml.in: interfaces wireless <wireless> vif <vif> ipv6 accept-dad	interfaces wireless <text> vif <text> ipv6 accept-dad <value>
✓	<code>set interfaces wireless <interface> vif <vlan-id> ipv6 address autoconf</code>	inter-faces_wireless.xml.in: interfaces wireless <wireless> vif <vif> ipv6 address autoconf	interfaces wireless <text> vif <text> ipv6 address autoconf
✓	<code>set interfaces wireless <interface> vif <vlan-id> ipv6 address eui64 <prefix></code>	inter-faces_wireless.xml.in: interfaces wireless <wireless> vif <vif> ipv6 address eui64	interfaces wireless <text> vif <text> ipv6 address eui64 <value>
✓	<code>set interfaces wireless <interface> vif <vlan-id> ipv6 address no-default-link-local</code>	inter-faces_wireless.xml.in: interfaces wireless <wireless> vif <vif> ipv6 address no-default-link-local	interfaces wireless <text> vif <text> ipv6 address no-default-link-local
✓	<code>set interfaces wireless <interface> vif <vlan-id> ipv6 adjust-mss <mss clamp-mss-to-pmtu></code>	inter-faces_wireless.xml.in: interfaces wireless <wireless> vif <vif> ipv6 adjust-mss	interfaces wireless <text> vif <text> ipv6 adjust-mss <value>
×	not yet documented	inter-faces_wireless.xml.in: interfaces wireless <wireless> vif <vif> ipv6 base-reachable-time	Nothing found in VyOS
✓	<code>set interfaces wireless <interface> vif <vlan-id> ipv6 disable-forwarding</code>	inter-faces_wireless.xml.in: interfaces wireless <wireless> vif <vif> ipv6 disable-forwarding	interfaces wireless <text> vif <text> ipv6 disable-forwarding
✓	<code>set interfaces wireless <interface> vif <vlan-id> ipv6 dup-addr-detect-transmits <n></code>	inter-faces_wireless.xml.in: interfaces wireless <wireless> vif <vif> ipv6 dup-addr-detect-transmits	interfaces wireless <text> vif <text> ipv6 dup-addr-detect-transmits <value>

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Table 1 – continued from previous page

Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	inter- faces_wireless.xml.in: interfaces wireless <wire- less> vif <vif> ipv6 source-validation	Nothing found in VyOS
✓	set interfaces wireless <interface> vif <vlan-id> mac <xx:xx:xx:xx:xx:xx>	inter- faces_wireless.xml.in: interfaces wireless <wire- less> vif <vif> mac	interfaces wireless <text> vif <text> mac <value>
×	not yet documented	inter- faces_wireless.xml.in: interfaces wireless <wire- less> vif <vif> mirror egress	interfaces wireless <text> vif <text> mirror egress <value>
×	not yet documented	inter- faces_wireless.xml.in: interfaces wireless <wire- less> vif <vif> mirror ingress	interfaces wireless <text> vif <text> mirror ingress <value>
✓	set interfaces wireless <interface> vif <vlan-id> mtu <mtu>	inter- faces_wireless.xml.in: interfaces wireless <wire- less> vif <vif> mtu	interfaces wireless <text> vif <text> mtu <value>
×	not yet documented	inter- faces_wireless.xml.in: interfaces wireless <wire- less> vif <vif> redirect	interfaces wireless <text> vif <text> redirect <value>
✓	set interfaces wireless <interface> vif <vlan-id> vrf <vrf>	inter- faces_wireless.xml.in: interfaces wireless <wire- less> vif <vif> vrf	interfaces wireless <text> vif <text> vrf <value>
✓	set interfaces wireless <interface> vrf <vrf>	inter- faces_wireless.xml.in: interfaces wireless <wire- less> vrf	interfaces wireless <text> vrf <value>
×	set interfaces wwan <interface> apn <apn>	Nothing found in XML Definitions	Nothing found in VyOS
✓	set interfaces wwan <interface> address <address dhcp dhcpv6>	interfaces_wwan.xml.in: interfaces wwan <wwan> address	interfaces wwan <text> address <value>
×	not yet documented	interfaces_wwan.xml.in: interfaces wwan <wwan> apn	interfaces wwan <text> apn <value>
×	not yet documented	interfaces_wwan.xml.in: interfaces wwan <wwan> authentication password	interfaces wwan <text> authentication password <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	interfaces_wwan.xml.in: interfaces wwan <wwan> authentication username <value>	interfaces wwan <text> authentication username <value>
×	not yet documented	interfaces_wwan.xml.in: interfaces wwan <wwan> connect-on-demand	interfaces wwan <text> connect-on-demand
✓	<i>set interfaces wwan <interface> description <description></i>	interfaces_wwan.xml.in: interfaces wwan <wwan> description	interfaces wwan <text> description <value>
✓	<i>set interfaces wwan <interface> dhcp-options client-id <description></i>	interfaces_wwan.xml.in: interfaces wwan <wwan> dhcp-options client-id	interfaces wwan <text> dhcp-options client-id <value>
✓	<i>set interfaces wwan <interface> dhcp-options default-route-distance <distance></i>	interfaces_wwan.xml.in: interfaces wwan <wwan> dhcp-options default- route-distance	interfaces wwan <text> dhcp-options default- route-distance <value>
✓	<i>set interfaces wwan <interface> dhcp-options host-name <hostname></i>	interfaces_wwan.xml.in: interfaces wwan <wwan> dhcp-options host-name	interfaces wwan <text> dhcp-options host-name <value>
×	not yet documented	interfaces_wwan.xml.in: interfaces wwan <wwan> dhcp-options mtu	interfaces wwan <text> dhcp-options mtu
✓	<i>set interfaces wwan <interface> dhcp-options no-default-route</i>	interfaces_wwan.xml.in: interfaces wwan <wwan> dhcp-options no-default- route	interfaces wwan <text> dhcp-options no-default- route
✓	<i>set interfaces wwan <interface> dhcp-options reject <address></i>	interfaces_wwan.xml.in: interfaces wwan <wwan> dhcp-options reject	interfaces wwan <text> dhcp-options reject <value>
×	<i>set interfaces wwan <interface> dhcp-options user-class <string></i>	interfaces_wwan.xml.in: interfaces wwan <wwan> dhcp-options user-class	Nothing found in VyOS
✓	<i>set interfaces wwan <interface> dhcp-options vendor-class-id <vendor-id></i>	interfaces_wwan.xml.in: interfaces wwan <wwan> dhcp-options vendor- class-id	interfaces wwan <text> dhcp-options vendor- class-id <value>
✓	<i>set interfaces wwan <interface> dhcpv6-options duid <duid></i>	interfaces_wwan.xml.in: interfaces wwan <wwan> dhcpv6-options duid	interfaces wwan <text> dhcpv6-options duid <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
✓	<code>set interfaces wwan <interface> dhcpv6-options no-release</code>	interfaces_wwan.xml.in: interfaces wwan <wwan> dhcpv6-options no-release	interfaces wwan <text> dhcpv6-options no-release
✓	<code>set interfaces wwan <interface> dhcpv6-options parameters-only</code>	interfaces_wwan.xml.in: interfaces wwan <wwan> dhcpv6-options parameters-only	interfaces wwan <text> dhcpv6-options parameters-only
✓	<code>set interfaces wwan <interface> dhcpv6-options pd <id> interface <delegatee> address <address></code>	interfaces_wwan.xml.in: interfaces wwan <wwan> dhcpv6-options pd <pd> interface <interface> ad- dress	interfaces wwan <text> dhcpv6-options pd <text> interface <text> address <value>
✓	<code>set interfaces wwan <interface> dhcpv6-options pd <id> interface <delegatee> sla-id <id></code>	interfaces_wwan.xml.in: interfaces wwan <wwan> dhcpv6-options pd <pd> interface <interface> sla- id	interfaces wwan <text> dhcpv6-options pd <text> interface <text> sla-id <value>
✓	<code>set interfaces wwan <interface> dhcpv6-options pd <id> length <length></code>	interfaces_wwan.xml.in: interfaces wwan <wwan> dhcpv6-options pd <pd> length	interfaces wwan <text> dhcpv6-options pd <text> length <value>
✓	<code>set interfaces wwan <interface> dhcpv6-options rapid-commit</code>	interfaces_wwan.xml.in: interfaces wwan <wwan> dhcpv6-options rapid- commit	interfaces wwan <text> dhcpv6-options rapid- commit
✓	<code>set interfaces wwan <interface> dhcpv6-options temporary</code>	interfaces_wwan.xml.in: interfaces wwan <wwan> dhcpv6-options temporary	interfaces wwan <text> dhcpv6-options temporary
✓	<code>set interfaces wwan <interface> disable</code>	interfaces_wwan.xml.in: interfaces wwan <wwan> disable	interfaces wwan <text> disable
✓	<code>set interfaces wwan <interface> disable-link-detect</code>	interfaces_wwan.xml.in: interfaces wwan <wwan> disable-link-detect	interfaces wwan <text> disable-link-detect
✓	<code>set interfaces wwan <interface> ip adjust-mss <mss / clamp-mss-to-pmtu></code>	interfaces_wwan.xml.in: interfaces wwan <wwan> ip adjust-mss	interfaces wwan <text> ip adjust-mss <value>
✓	<code>set interfaces wwan <interface> ip arp-cache-timeout</code>	interfaces_wwan.xml.in: interfaces wwan <wwan> ip arp-cache-timeout	interfaces wwan <text> ip arp-cache-timeout <value>
✓	<code>set interfaces wwan <interface> ip disable-arp-filter</code>	interfaces_wwan.xml.in: interfaces wwan <wwan> ip disable-arp-filter	interfaces wwan <text> ip disable-arp-filter

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
✓	<code>set interfaces wwan <interface> ip disable-forwarding</code>	interfaces_wwan.xml.in: interfaces wwan <wwan> ip disable-forwarding	interfaces wwan <text> ip disable-forwarding
✓	<code>set interfaces wwan <interface> ip enable-arp-accept</code>	interfaces_wwan.xml.in: interfaces wwan <wwan> ip enable-arp-accept	interfaces wwan <text> ip enable-arp-accept
✓	<code>set interfaces wwan <interface> ip enable-arp-announce</code>	interfaces_wwan.xml.in: interfaces wwan <wwan> ip enable-arp-announce	interfaces wwan <text> ip enable-arp-announce
✓	<code>set interfaces wwan <interface> ip enable-arp-ignore</code>	interfaces_wwan.xml.in: interfaces wwan <wwan> ip enable-arp-ignore	interfaces wwan <text> ip enable-arp-ignore
✓	<code>set interfaces wwan <interface> ip enable-directed-broadcast</code>	interfaces_wwan.xml.in: interfaces wwan <wwan> ip enable-directed-broadcast	interfaces wwan <text> ip enable-directed-broadcast
✓	<code>set interfaces wwan <interface> ip enable-proxy-arp</code>	interfaces_wwan.xml.in: interfaces wwan <wwan> ip enable-proxy-arp	interfaces wwan <text> ip enable-proxy-arp
✓	<code>set interfaces wwan <interface> ip proxy-arp-pvlan</code>	interfaces_wwan.xml.in: interfaces wwan <wwan> ip proxy-arp-pvlan	interfaces wwan <text> ip proxy-arp-pvlan
✓	<code>set interfaces wwan <interface> ip source-validation <strict loose disable></code>	interfaces_wwan.xml.in: interfaces wwan <wwan> ip source-validation	interfaces wwan <text> ip source-validation <value>
✓	<code>set interfaces wwan <interface> ipv6 accept-dad <0 1 2></code>	interfaces_wwan.xml.in: interfaces wwan <wwan> ipv6 accept-dad	interfaces wwan <text> ipv6 accept-dad <value>
✓	<code>set interfaces wwan <interface> ipv6 address autoconf</code>	interfaces_wwan.xml.in: interfaces wwan <wwan> ipv6 address autoconf	interfaces wwan <text> ipv6 address autoconf
✓	<code>set interfaces wwan <interface> ipv6 address eui64 <prefix></code>	interfaces_wwan.xml.in: interfaces wwan <wwan> ipv6 address eui64	interfaces wwan <text> ipv6 address eui64 <value>
✓	<code>set interfaces wwan <interface> ipv6 address no-default-link-local</code>	interfaces_wwan.xml.in: interfaces wwan <wwan> ipv6 address no-default-link-local	interfaces wwan <text> ipv6 address no-default-link-local
✓	<code>set interfaces wwan <interface> ipv6 adjust-mss <mss clamp-mss-to-pmtu></code>	interfaces_wwan.xml.in: interfaces wwan <wwan> ipv6 adjust-mss	interfaces wwan <text> ipv6 adjust-mss <value>
×	not yet documented	interfaces_wwan.xml.in: interfaces wwan <wwan> ipv6 base-reachable-time	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
✓	<code>set interfaces wwan</code> <code><interface> ipv6</code> <code>disable-forwarding</code>	interfaces_wwan.xml.in: interfaces wwan <wwan> ipv6 disable-forwarding	interfaces wwan <text> ipv6 disable-forwarding
✓	<code>set interfaces wwan</code> <code><interface> ipv6</code> <code>dup-addr-detect-trans</code> <code><n></code>	interfaces_wwan.xml.in: interfaces wwan <wwan> ipv6 dup-addr-detect- transmits	interfaces wwan <text> ipv6 dup-addr-detect- transmits <value>
×	not yet documented	interfaces_wwan.xml.in: interfaces wwan <wwan> ipv6 source-validation	Nothing found in VyOS
×	not yet documented	interfaces_wwan.xml.in: interfaces wwan <wwan> mirror egress	interfaces wwan <text> mirror egress <value>
×	not yet documented	interfaces_wwan.xml.in: interfaces wwan <wwan> mirror ingress	interfaces wwan <text> mirror ingress <value>
✓	<code>set interfaces wwan</code> <code><interface> mtu</code> <code><mtu></code>	interfaces_wwan.xml.in: interfaces wwan <wwan> mtu	interfaces wwan <text> mtu <value>
×	not yet documented	interfaces_wwan.xml.in: interfaces wwan <wwan> redirect	interfaces wwan <text> redirect <value>
✓	<code>set interfaces wwan</code> <code><interface> vrf</code> <code><vrf></code>	interfaces_wwan.xml.in: interfaces wwan <wwan> vrf	interfaces wwan <text> vrf <value>
×	<code>set load-balancing</code> <code>haproxy backend</code> <code><name> balance</code> <code><balance></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set load-balancing</code> <code>haproxy backend</code> <code><name> health-check</code> <code><protocol></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set load-balancing</code> <code>haproxy backend</code> <code><name> http-check</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set load-balancing</code> <code>haproxy backend</code> <code><name> http-check</code> <code>expect <condition></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set load-balancing</code> <code>haproxy backend</code> <code><name> http-check</code> <code>method <method></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set load-balancing</code> <code>haproxy backend</code> <code><name> http-check</code> <code>uri <path></code>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set load-balancing haproxy backend <name> http-response-headers <header-name> value <header-value></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set load-balancing haproxy backend <name> logging facility <facility> level <level></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set load-balancing haproxy backend <name> mode <mode></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set load-balancing haproxy backend <name> server <name> address <x.x.x.x></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set load-balancing haproxy backend <name> server <name> check</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set load-balancing haproxy backend <name> server <name> check port <port></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set load-balancing haproxy backend <name> server <name> port <port></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set load-balancing haproxy backend <name> server <name> send-proxy</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set load-balancing haproxy backend <name> server <name> send-proxy-v2</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set load-balancing haproxy backend <name> ssl ca-certificate <ca-certificate></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set load-balancing haproxy backend <name> ssl no-verify</code>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set load-balancing haproxy backend <name> timeout check <seconds></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set load-balancing haproxy backend <name> timeout connect <seconds></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set load-balancing haproxy backend <name> timeout server <seconds></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set load-balancing haproxy global-parameters logging facility <facility> level <level></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set load-balancing haproxy global-parameters max-connections <num></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set load-balancing haproxy global-parameters ssl-bind-ciphers <ciphers></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set load-balancing haproxy global-parameters tls-version-min <version></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set load-balancing haproxy service <name> backend <name></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set load-balancing haproxy service <name> http-compression algorithm <gzip deflate identity raw-deflate></code>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set load-balancing haproxy service <name> http-compression mime-type <mime-type></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set load-balancing haproxy service <name> http-response-headers <header-name> value <header-value></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set load-balancing haproxy service <name> listen-address <address></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set load-balancing haproxy service <name> logging facility <facility> level <level></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set load-balancing haproxy service <name> mode <tcp/http></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set load-balancing haproxy service <name> port <port></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set load-balancing haproxy service <name> rule <rule> domain-name <name></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set load-balancing haproxy service <name> rule <rule> redirect-location <url></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set load-balancing haproxy service <name> rule <rule> set backend <name></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set load-balancing haproxy service <name> rule <rule> ssl <sni></code>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set load-balancing haproxy service <name> rule <rule> url-path <match> <url></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set load-balancing haproxy service <name> ssl certificate <name></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set load-balancing haproxy service <name> timeout client <seconds></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set load-balancing haproxy timeout check <seconds></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set load-balancing haproxy timeout client <seconds></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set load-balancing haproxy timeout connect <seconds></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set load-balancing haproxy timeout server <seconds></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	load-balancing_reverse-proxy.xml.in: load-balancing reverse-proxy backend <backend> balance	load-balancing reverse-proxy backend <text> balance <value>
×	not yet documented	load-balancing_reverse-proxy.xml.in: load-balancing reverse-proxy backend <backend> description	load-balancing reverse-proxy backend <text> description <value>
×	not yet documented	load-balancing_reverse-proxy.xml.in: load-balancing reverse-proxy backend <backend> http-check expect status	Nothing found in VyOS
×	not yet documented	load-balancing_reverse-proxy.xml.in: load-balancing reverse-proxy backend <backend> http-check expect string	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	load-balancing_reverse-proxy.xml.in: load-balancing reverse-proxy backend <backend> http-check method	Nothing found in VyOS
×	not yet documented	load-balancing_reverse-proxy.xml.in: load-balancing reverse-proxy backend <backend> http-check uri	Nothing found in VyOS
×	not yet documented	load-balancing_reverse-proxy.xml.in: load-balancing reverse-proxy backend <backend> mode	load-balancing reverse-proxy backend <text> mode <value>
×	not yet documented	load-balancing_reverse-proxy.xml.in: load-balancing reverse-proxy backend <backend> parameters http-check	load-balancing reverse-proxy backend <text> parameters http-check
×	not yet documented	load-balancing_reverse-proxy.xml.in: load-balancing reverse-proxy backend <backend> rule <rule> domain-name	load-balancing reverse-proxy backend <text> rule <text> domain-name <value>
×	not yet documented	load-balancing_reverse-proxy.xml.in: load-balancing reverse-proxy backend <backend> rule <rule> set redirect-location	load-balancing reverse-proxy backend <text> rule <text> set redirect-location <value>
×	not yet documented	load-balancing_reverse-proxy.xml.in: load-balancing reverse-proxy backend <backend> rule <rule> set server	load-balancing reverse-proxy backend <text> rule <text> set server <value>
×	not yet documented	load-balancing_reverse-proxy.xml.in: load-balancing reverse-proxy backend <backend> rule <rule> ssl	load-balancing reverse-proxy backend <text> rule <text> ssl <value>
×	not yet documented	load-balancing_reverse-proxy.xml.in: load-balancing reverse-proxy backend <backend> rule <rule> url-path begin	load-balancing reverse-proxy backend <text> rule <text> url-path begin <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	load-balancing_reverse-proxy.xml.in: load-balancing reverse-proxy backend <backend> rule <rule> url-path end	load-balancing reverse-proxy backend <text> rule <text> url-path end <value>
×	not yet documented	load-balancing_reverse-proxy.xml.in: load-balancing reverse-proxy backend <backend> rule <rule> url-path exact	load-balancing reverse-proxy backend <text> rule <text> url-path exact <value>
×	not yet documented	load-balancing_reverse-proxy.xml.in: load-balancing reverse-proxy backend <backend> server <server> address	load-balancing reverse-proxy backend <text> server <text> address <value>
×	not yet documented	load-balancing_reverse-proxy.xml.in: load-balancing reverse-proxy backend <backend> server <server> backup	Nothing found in VyOS
×	not yet documented	load-balancing_reverse-proxy.xml.in: load-balancing reverse-proxy backend <backend> server <server> check	load-balancing reverse-proxy backend <text> server <text> check
×	not yet documented	load-balancing_reverse-proxy.xml.in: load-balancing reverse-proxy backend <backend> server <server> port	load-balancing reverse-proxy backend <text> server <text> port <value>
×	not yet documented	load-balancing_reverse-proxy.xml.in: load-balancing reverse-proxy backend <backend> server <server> send-proxy	load-balancing reverse-proxy backend <text> server <text> send-proxy
×	not yet documented	load-balancing_reverse-proxy.xml.in: load-balancing reverse-proxy backend <backend> server <server> send-proxy-v2	load-balancing reverse-proxy backend <text> server <text> send-proxy-v2
×	not yet documented	load-balancing_reverse-proxy.xml.in: load-balancing reverse-proxy backend <backend> ssl ca-certificate	load-balancing reverse-proxy backend <text> ssl ca-certificate <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	load-balancing_reverse-proxy.xml.in: load-balancing reverse-proxy backend <backend> ssl no-verify	Nothing found in VyOS
×	not yet documented	load-balancing_reverse-proxy.xml.in: load-balancing reverse-proxy backend <backend> timeout check	load-balancing reverse-proxy backend <text> timeout check <value>
×	not yet documented	load-balancing_reverse-proxy.xml.in: load-balancing reverse-proxy backend <backend> timeout connect	load-balancing reverse-proxy backend <text> timeout connect <value>
×	not yet documented	load-balancing_reverse-proxy.xml.in: load-balancing reverse-proxy backend <backend> timeout server	load-balancing reverse-proxy backend <text> timeout server <value>
×	not yet documented	load-balancing_reverse-proxy.xml.in: load-balancing reverse-proxy global-parameters max-connections	load-balancing reverse-proxy global-parameters max-connections <value>
×	not yet documented	load-balancing_reverse-proxy.xml.in: load-balancing reverse-proxy global-parameters ssl-bind-ciphers	load-balancing reverse-proxy global-parameters ssl-bind-ciphers <value>
×	not yet documented	load-balancing_reverse-proxy.xml.in: load-balancing reverse-proxy global-parameters tls-version-min	load-balancing reverse-proxy global-parameters tls-version-min <value>
×	not yet documented	load-balancing_reverse-proxy.xml.in: load-balancing reverse-proxy service <service> backend	load-balancing reverse-proxy service <text> backend <value>
×	not yet documented	load-balancing_reverse-proxy.xml.in: load-balancing reverse-proxy service <service> description	load-balancing reverse-proxy service <text> description <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	load-balancing_reverse-proxy.xml.in: load-balancing reverse-proxy service <service> listen-address	load-balancing reverse-proxy service <text> listen-address <value>
×	not yet documented	load-balancing_reverse-proxy.xml.in: load-balancing reverse-proxy service <service> mode	load-balancing reverse-proxy service <text> mode <value>
×	not yet documented	load-balancing_reverse-proxy.xml.in: load-balancing reverse-proxy service <service> port	load-balancing reverse-proxy service <text> port <value>
×	not yet documented	load-balancing_reverse-proxy.xml.in: load-balancing reverse-proxy service <service> redirect-http-to-https	load-balancing reverse-proxy service <text> redirect-http-to-https
×	not yet documented	load-balancing_reverse-proxy.xml.in: load-balancing reverse-proxy service <service> rule <rule> domain-name	load-balancing reverse-proxy service <text> rule <text> domain-name <value>
×	not yet documented	load-balancing_reverse-proxy.xml.in: load-balancing reverse-proxy service <service> rule <rule> set backend	load-balancing reverse-proxy service <text> rule <text> set backend <value>
×	not yet documented	load-balancing_reverse-proxy.xml.in: load-balancing reverse-proxy service <service> rule <rule> set redirect-location	load-balancing reverse-proxy service <text> rule <text> set redirect-location <value>
×	not yet documented	load-balancing_reverse-proxy.xml.in: load-balancing reverse-proxy service <service> rule <rule> ssl	load-balancing reverse-proxy service <text> rule <text> ssl <value>
×	not yet documented	load-balancing_reverse-proxy.xml.in: load-balancing reverse-proxy service <service> rule <rule> url-path begin	load-balancing reverse-proxy service <text> rule <text> url-path begin <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	load-balancing_reverse-proxy.xml.in: load-balancing reverse-proxy service <service> rule <rule> url-path end	load-balancing reverse-proxy service <text> rule <text> url-path end <value>
×	not yet documented	load-balancing_reverse-proxy.xml.in: load-balancing reverse-proxy service <service> rule <rule> url-path exact	load-balancing reverse-proxy service <text> rule <text> url-path exact <value>
×	not yet documented	load-balancing_reverse-proxy.xml.in: load-balancing reverse-proxy service <service> ssl certificate	load-balancing reverse-proxy service <text> ssl certificate <value>
×	not yet documented	load-balancing_reverse-proxy.xml.in: load-balancing reverse-proxy service <service> tcp-request inspect-delay	Nothing found in VyOS
×	not yet documented	load-balancing_reverse-proxy.xml.in: load-balancing reverse-proxy vrf	load-balancing reverse-proxy vrf <value>
×	not yet documented	load-balancing_wan.xml.in: load-balancing wan disable-source-nat	load-balancing wan disable-source-nat
×	not yet documented	load-balancing_wan.xml.in: load-balancing wan enable-local-traffic	load-balancing wan enable-local-traffic
×	not yet documented	load-balancing_wan.xml.in: load-balancing wan flush-connections	load-balancing wan flush-connections
×	not yet documented	load-balancing_wan.xml.in: load-balancing wan hook	load-balancing wan hook <value>
×	not yet documented	load-balancing_wan.xml.in: load-balancing wan interface-health <interface-health> failure-count	load-balancing wan interface-health <text> failure-count <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	load-balancing_wan.xml.in: load-balancing wan interface-health <interface-health> nex- thop	load-balancing wan interface-health <text> nexthop <value>
×	not yet documented	load-balancing_wan.xml.in: load-balancing wan interface-health <interface-health> success-count	load-balancing wan interface-health <text> success-count <value>
×	not yet documented	load-balancing_wan.xml.in: load-balancing wan interface-health <interface-health> test <test> resp-time	load-balancing wan interface-health <text> test <text> resp-time <value>
×	not yet documented	load-balancing_wan.xml.in: load-balancing wan interface-health <interface-health> test <test> target	load-balancing wan interface-health <text> test <text> target <value>
×	not yet documented	load-balancing_wan.xml.in: load-balancing wan interface-health <interface-health> test <test> test-script	load-balancing wan interface-health <text> test <text> test-script <value>
×	not yet documented	load-balancing_wan.xml.in: load-balancing wan interface-health <interface-health> test <test> ttl-limit	load-balancing wan interface-health <text> test <text> ttl-limit <value>
×	not yet documented	load-balancing_wan.xml.in: load-balancing wan interface-health <interface-health> test <test> type	load-balancing wan interface-health <text> test <text> type <value>
×	not yet documented	load-balancing_wan.xml.in: load-balancing wan rule <rule> description	load-balancing wan rule <text> description <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	load-balancing_wan.xml.in: load-balancing wan rule <rule> destination address	load-balancing wan rule <text> destination address <value>
×	not yet documented	load-balancing_wan.xml.in: load-balancing wan rule <rule> destination port	load-balancing wan rule <text> destination port <value>
×	not yet documented	load-balancing_wan.xml.in: load-balancing wan rule <rule> exclude	load-balancing wan rule <text> exclude
×	not yet documented	load-balancing_wan.xml.in: load-balancing wan rule <rule> failover	load-balancing wan rule <text> failover
×	not yet documented	load-balancing_wan.xml.in: load-balancing wan rule <rule> inbound-interface	load-balancing wan rule <text> inbound-interface <value>
×	not yet documented	load-balancing_wan.xml.in: load-balancing wan rule <rule> interface <interface> weight	load-balancing wan rule <text> interface <text> weight <value>
×	not yet documented	load-balancing_wan.xml.in: load-balancing wan rule <rule> limit burst	load-balancing wan rule <text> limit burst <value>
×	not yet documented	load-balancing_wan.xml.in: load-balancing wan rule <rule> limit period	load-balancing wan rule <text> limit period <value>
×	not yet documented	load-balancing_wan.xml.in: load-balancing wan rule <rule> limit rate	load-balancing wan rule <text> limit rate <value>
×	not yet documented	load-balancing_wan.xml.in: load-balancing wan rule <rule> limit threshold	load-balancing wan rule <text> limit threshold <value>
×	not yet documented	load-balancing_wan.xml.in: load-balancing wan rule <rule> per-packet-balancing	load-balancing wan rule <text> per-packet-balancing

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	load-balancing_wan.xml.in: load-balancing wan rule <rule> protocol	load-balancing wan rule <text> protocol <value>
×	not yet documented	load-balancing_wan.xml.in: load-balancing wan rule <rule> source address	load-balancing wan rule <text> source address <value>
×	not yet documented	load-balancing_wan.xml.in: load-balancing wan rule <rule> source port	load-balancing wan rule <text> source port <value>
×	not yet documented	load-balancing_wan.xml.in: load-balancing wan sticky-connections in-bound	load-balancing wan sticky-connections in-bound
×	<i>load <URI></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	nat64.xml.in: nat64 source rule <rule> description	Nothing found in VyOS
×	not yet documented	nat64.xml.in: nat64 source rule <rule> disable	Nothing found in VyOS
×	not yet documented	nat64.xml.in: nat64 source rule <rule> match mark	Nothing found in VyOS
×	not yet documented	nat64.xml.in: nat64 source rule <rule> source prefix	Nothing found in VyOS
×	not yet documented	nat64.xml.in: nat64 source rule <rule> translation pool <pool> address	Nothing found in VyOS
×	not yet documented	nat64.xml.in: nat64 source rule <rule> translation pool <pool> description	Nothing found in VyOS
×	not yet documented	nat64.xml.in: nat64 source rule <rule> translation pool <pool> disable	Nothing found in VyOS
×	not yet documented	nat64.xml.in: nat64 source rule <rule> translation pool <pool> port	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	nat64.xml.in: nat64 source rule <rule> translation pool <pool> protocol icmp	Nothing found in VyOS
×	not yet documented	nat64.xml.in: nat64 source rule <rule> translation pool <pool> protocol tcp	Nothing found in VyOS
×	not yet documented	nat64.xml.in: nat64 source rule <rule> translation pool <pool> protocol udp	Nothing found in VyOS
×	not yet documented	nat66.xml.in: nat66 destination rule <rule> description	nat66 destination rule <text> description <value>
×	not yet documented	nat66.xml.in: nat66 destination rule <rule> destination address	nat66 destination rule <text> destination address <value>
×	not yet documented	nat66.xml.in: nat66 destination rule <rule> destination port	nat66 destination rule <text> destination port <value>
×	not yet documented	nat66.xml.in: nat66 destination rule <rule> disable	nat66 destination rule <text> disable
×	not yet documented	nat66.xml.in: nat66 destination rule <rule> exclude	nat66 destination rule <text> exclude
×	not yet documented	Nothing found in XML Definitions	nat66 destination rule <text> inbound-interface <value>
×	not yet documented	nat66.xml.in: nat66 destination rule <rule> inbound-interface name	Nothing found in VyOS
×	not yet documented	nat66.xml.in: nat66 destination rule <rule> log	nat66 destination rule <text> log
×	not yet documented	nat66.xml.in: nat66 destination rule <rule> protocol	nat66 destination rule <text> protocol <value>
×	not yet documented	nat66.xml.in: nat66 destination rule <rule> source address	nat66 destination rule <text> source address <value>
×	not yet documented	nat66.xml.in: nat66 destination rule <rule> source port	nat66 destination rule <text> source port <value>
×	not yet documented	nat66.xml.in: nat66 destination rule <rule> translation address	nat66 destination rule <text> translation address <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	nat66.xml.in: nat66 destination rule <rule> translation port	nat66 destination rule <text> translation port <value>
×	not yet documented	nat66.xml.in: nat66 source rule <rule> description	nat66 source rule <text> description <value>
×	not yet documented	nat66.xml.in: nat66 source rule <rule> destination port	nat66 source rule <text> destination port <value>
×	not yet documented	nat66.xml.in: nat66 source rule <rule> destination prefix	nat66 source rule <text> destination prefix <value>
×	not yet documented	nat66.xml.in: nat66 source rule <rule> disable	nat66 source rule <text> disable
×	not yet documented	nat66.xml.in: nat66 source rule <rule> exclude	nat66 source rule <text> exclude
×	not yet documented	nat66.xml.in: nat66 source rule <rule> log	nat66 source rule <text> log
×	not yet documented	Nothing found in XML Definitions	nat66 source rule <text> outbound-interface <value>
×	not yet documented	nat66.xml.in: nat66 source rule <rule> outbound-interface name	Nothing found in VyOS
×	not yet documented	nat66.xml.in: nat66 source rule <rule> protocol	nat66 source rule <text> protocol <value>
×	not yet documented	nat66.xml.in: nat66 source rule <rule> source port	nat66 source rule <text> source port <value>
×	not yet documented	nat66.xml.in: nat66 source rule <rule> source prefix	nat66 source rule <text> source prefix <value>
×	not yet documented	nat66.xml.in: nat66 source rule <rule> translation address	nat66 source rule <text> translation address <value>
×	not yet documented	nat66.xml.in: nat66 source rule <rule> translation port	nat66 source rule <text> translation port <value>
×	<i>set nat cgnat log-allocation</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set nat cgnat pool external <pool-name> external-port-range <port-range></i>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set nat cgnat pool external <pool-name> per-user-limit port <num></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set nat cgnat pool external <pool-name> range [address / address range / network] [seq]</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set nat cgnat pool internal <pool-name> range [address range / network]</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set nat cgnat rule <num> source pool <internal-pool-name></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set nat cgnat rule <num> translation pool <external-pool-name></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	nat.xml.in: nat destination rule <rule> description	nat destination rule <text> description <value>
×	not yet documented	nat.xml.in: nat destination rule <rule> destination address	nat destination rule <text> destination address <value>
×	not yet documented	nat.xml.in: nat destination rule <rule> destination group address-group	nat destination rule <text> destination group address-group <value>
×	not yet documented	nat.xml.in: nat destination rule <rule> destination group domain-group	nat destination rule <text> destination group domain-group <value>
×	not yet documented	nat.xml.in: nat destination rule <rule> destination group mac-group	nat destination rule <text> destination group mac-group <value>
×	not yet documented	nat.xml.in: nat destination rule <rule> destination group network-group	nat destination rule <text> destination group network-group <value>
×	not yet documented	nat.xml.in: nat destination rule <rule> destination group port-group	nat destination rule <text> destination group port-group <value>
×	not yet documented	nat.xml.in: nat destination rule <rule> destination port	nat destination rule <text> destination port <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	nat.xml.in: nat destination rule <rule> disable	nat destination rule <text> disable
×	not yet documented	nat.xml.in: nat destination rule <rule> exclude	nat destination rule <text> exclude
×	not yet documented	Nothing found in XML Definitions	nat destination rule <text> inbound-interface <value>
×	not yet documented	nat.xml.in: nat destination rule <rule> inbound-interface group	Nothing found in VyOS
×	not yet documented	nat.xml.in: nat destination rule <rule> inbound-interface name	Nothing found in VyOS
×	not yet documented	nat.xml.in: nat destination rule <rule> load-balance backend <backend> weight	nat destination rule <text> load-balance backend <text> weight <value>
×	not yet documented	nat.xml.in: nat destination rule <rule> load-balance hash	nat destination rule <text> load-balance hash <value>
×	not yet documented	nat.xml.in: nat destination rule <rule> log	nat destination rule <text> log
×	not yet documented	nat.xml.in: nat destination rule <rule> packet-type	nat destination rule <text> packet-type <value>
×	not yet documented	nat.xml.in: nat destination rule <rule> protocol	nat destination rule <text> protocol <value>
×	not yet documented	nat.xml.in: nat destination rule <rule> source address	nat destination rule <text> source address <value>
×	not yet documented	nat.xml.in: nat destination rule <rule> source group address- group	nat destination rule <text> source group address- group <value>
×	not yet documented	nat.xml.in: nat destination rule <rule> source group domain- group	nat destination rule <text> source group domain- group <value>
×	not yet documented	nat.xml.in: nat destination rule <rule> source group mac-group	nat destination rule <text> source group mac-group <value>
×	not yet documented	nat.xml.in: nat destination rule <rule> source group network- group	nat destination rule <text> source group network- group <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	nat.xml.in: nat destination rule <rule> source group port-group	nat destination rule <text> source group port-group <value>
×	not yet documented	nat.xml.in: nat destination rule <rule> source port	nat destination rule <text> source port <value>
×	not yet documented	nat.xml.in: nat destination rule <rule> translation address	nat destination rule <text> translation address <value>
×	not yet documented	nat.xml.in: nat destination rule <rule> translation options address-mapping	nat destination rule <text> translation options address-mapping <value>
×	not yet documented	nat.xml.in: nat destination rule <rule> translation options port- mapping	nat destination rule <text> translation options port- mapping <value>
×	not yet documented	nat.xml.in: nat destination rule <rule> translation port	nat destination rule <text> translation port <value>
×	not yet documented	nat.xml.in: nat destination rule <rule> translation redirect port	nat destination rule <text> translation redirect port <value>
×	<i>set nat [source / destination] rule <rule> load-balance backend <x.x.x.x> weight <1-100></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set nat [source / destination] rule <rule> load-balance hash [source-address / destination-address / source-port / destination-port / random]</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	nat.xml.in: nat source rule <rule> de- scription	nat source rule <text> de- scription <value>
×	not yet documented	nat.xml.in: nat source rule <rule> des- tination address	nat source rule <text> des- tination address <value>
×	not yet documented	nat.xml.in: nat source rule <rule> destination group address- group	nat source rule <text> destination group address- group <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	nat.xml.in: nat source rule <rule> destination group domain-group	nat source rule <text> destination group domain-group <value>
×	not yet documented	nat.xml.in: nat source rule <rule> destination group mac-group	nat source rule <text> destination group mac-group <value>
×	not yet documented	nat.xml.in: nat source rule <rule> destination group network-group	nat source rule <text> destination group network-group <value>
×	not yet documented	nat.xml.in: nat source rule <rule> destination group port-group	nat source rule <text> destination group port-group <value>
×	not yet documented	nat.xml.in: nat source rule <rule> destination port	nat source rule <text> destination port <value>
×	not yet documented	nat.xml.in: nat source rule <rule> disable	nat source rule <text> disable
×	not yet documented	nat.xml.in: nat source rule <rule> exclude	nat source rule <text> exclude
×	not yet documented	nat.xml.in: nat source rule <rule> load-balance backend <backend> weight	nat source rule <text> load-balance backend <text> weight <value>
×	not yet documented	nat.xml.in: nat source rule <rule> load-balance hash	nat source rule <text> load-balance hash <value>
×	not yet documented	nat.xml.in: nat source rule <rule> log	nat source rule <text> log
×	not yet documented	Nothing found in XML Definitions	nat source rule <text> outbound-interface <value>
×	not yet documented	nat.xml.in: nat source rule <rule> outbound-interface group	Nothing found in VyOS
×	not yet documented	nat.xml.in: nat source rule <rule> outbound-interface name	Nothing found in VyOS
×	not yet documented	nat.xml.in: nat source rule <rule> packet-type	nat source rule <text> packet-type <value>
×	not yet documented	nat.xml.in: nat source rule <rule> protocol	nat source rule <text> protocol <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	nat.xml.in: nat source rule <rule> source address	nat source rule <text> source address <value>
×	not yet documented	nat.xml.in: nat source rule <rule> source group address-group	nat source rule <text> source group address-group <value>
×	not yet documented	nat.xml.in: nat source rule <rule> source group domain-group	nat source rule <text> source group domain-group <value>
×	not yet documented	nat.xml.in: nat source rule <rule> source group mac-group	nat source rule <text> source group mac-group <value>
×	not yet documented	nat.xml.in: nat source rule <rule> source group network-group	nat source rule <text> source group network-group <value>
×	not yet documented	nat.xml.in: nat source rule <rule> source group port-group	nat source rule <text> source group port-group <value>
×	not yet documented	nat.xml.in: nat source rule <rule> source port	nat source rule <text> source port <value>
×	not yet documented	nat.xml.in: nat source rule <rule> translation address	nat source rule <text> translation address <value>
×	not yet documented	nat.xml.in: nat source rule <rule> translation options address-mapping	nat source rule <text> translation options address-mapping <value>
×	not yet documented	nat.xml.in: nat source rule <rule> translation options port-mapping	nat source rule <text> translation options port-mapping <value>
×	not yet documented	nat.xml.in: nat source rule <rule> translation port	nat source rule <text> translation port <value>
×	not yet documented	nat.xml.in: nat static rule <rule> description	nat static rule <text> description <value>
×	not yet documented	nat.xml.in: nat static rule <rule> destination address	nat static rule <text> destination address <value>
×	not yet documented	nat.xml.in: nat static rule <rule> inbound-interface	nat static rule <text> inbound-interface <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	nat.xml.in: nat static rule <rule> translation address	nat static rule <text> translation address <value>
×	not yet documented	netns.xml.in: netns name <name> description	netns name <text> description <value>
×	<i>set pki ca <name> certificate</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set pki ca <name> crt</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set pki ca <name> description</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set pki ca <name> private key</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set pki ca <name> private password-protected</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	pki.xml.in: pki ca <ca> certificate	pki ca <text> certificate <value>
×	not yet documented	pki.xml.in: pki ca <ca> crt	pki ca <text> crt <value>
×	not yet documented	pki.xml.in: pki ca <ca> description	pki ca <text> description <value>
×	not yet documented	pki.xml.in: pki ca <ca> private key	pki ca <text> private key <value>
×	not yet documented	pki.xml.in: pki ca <ca> private password-protected	pki ca <text> private password-protected
×	not yet documented	pki.xml.in: pki ca <ca> revoke	pki ca <text> revoke
×	<i>set pki certificate <name> acme domain-name <name></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set pki certificate <name> acme email <address></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set pki certificate <name> acme listen-address <address></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set pki certificate <name> acme rsa-key-size <2048 3072 4096></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set pki certificate <name> acme url <url></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set pki certificate <name> certificate</i>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<i>set pki certificate <name> description</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set pki certificate <name> private key</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set pki certificate <name> private password-protected</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set pki certificate <name> revoke</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	pki.xml.in: pki certificate <certificate> acme domain-name	Nothing found in VyOS
×	not yet documented	pki.xml.in: pki certificate <certificate> acme email	Nothing found in VyOS
×	not yet documented	pki.xml.in: pki certificate <certificate> acme listen-address	Nothing found in VyOS
×	not yet documented	pki.xml.in: pki certificate <certificate> acme rsa-key-size	Nothing found in VyOS
×	not yet documented	pki.xml.in: pki certificate <certificate> acme url	Nothing found in VyOS
×	not yet documented	pki.xml.in: pki certificate <certificate> certificate	pki certificate <text> certificate <value>
×	not yet documented	pki.xml.in: pki certificate <certificate> description	pki certificate <text> description <value>
×	not yet documented	pki.xml.in: pki certificate <certificate> private key	pki certificate <text> private key <value>
×	not yet documented	pki.xml.in: pki certificate <certificate> private password-protected	pki certificate <text> private password-protected
×	not yet documented	pki.xml.in: pki certificate <certificate> revoke	pki certificate <text> revoke
×	not yet documented	pki.xml.in: pki dh <dh> parameters	pki dh <text> parameters <value>
×	not yet documented	pki.xml.in: pki key-pair <key-pair> private key	pki key-pair <text> private key <value>
×	not yet documented	pki.xml.in: pki key-pair <key-pair> private password-protected	pki key-pair <text> private password-protected

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	pki.xml.in: pki key-pair <key-pair> public key	pki key-pair <text> public key <value>
×	not yet documented	pki.xml.in: pki openssh <openssh> private key	Nothing found in VyOS
×	not yet documented	pki.xml.in: pki openssh <openssh> private password- protected	Nothing found in VyOS
×	not yet documented	pki.xml.in: pki openssh <openssh> public key	Nothing found in VyOS
×	not yet documented	pki.xml.in: pki openssh <openssh> public type	Nothing found in VyOS
×	not yet documented	pki.xml.in: pki openvpn shared-secret <shared-secret> key	pki openvpn shared-secret <text> key <value>
×	not yet documented	pki.xml.in: pki openvpn shared-secret <shared-secret> version	pki openvpn shared-secret <text> version <value>
×	not yet documented	pki.xml.in: pki x509 default country	pki x509 default country <value>
×	not yet documented	pki.xml.in: pki x509 default locality	pki x509 default locality <value>
×	not yet documented	pki.xml.in: pki x509 default organiza- tion	pki x509 default organiza- tion <value>
×	not yet documented	pki.xml.in: pki x509 default state	pki x509 default state <value>
×	<i>set policy access-list6 <text></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set policy access-list6 <text> description <text></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set policy access-list6 <text> rule <1-65535> action <permit/deny></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set policy access-list6 <text> rule <1-65535> source <any/exact-match/netw</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	policy.xml.in: policy access-list6 <access-list6> description	policy access-list6 <text> description <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	policy.xml.in: policy access-list6 <access-list6> rule <rule> action	policy access-list6 <text> rule <text> action <value>
×	not yet documented	policy.xml.in: policy access-list6 <access-list6> rule <rule> description	policy access-list6 <text> rule <text> description <value>
×	not yet documented	policy.xml.in: policy access-list6 <access-list6> rule <rule> source any	policy access-list6 <text> rule <text> source any
×	not yet documented	policy.xml.in: policy access-list6 <access-list6> rule <rule> source exact-match	policy access-list6 <text> rule <text> source exact- match
×	not yet documented	policy.xml.in: policy access-list6 <access-list6> rule <rule> source network	policy access-list6 <text> rule <text> source network <value>
×	<i>set policy access-list <acl_number></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set policy access-list <acl_number> description <text></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set policy access-list <acl_number> rule <1-65535> <destination/source> <any/host/inverse-mas.</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set policy access-list <acl_number> rule <1-65535> action <permit/deny></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	policy.xml.in: policy access-list <access- list> description	policy access-list <text> description <value>
×	not yet documented	policy.xml.in: policy access-list <access- list> rule <rule> action	policy access-list <text> rule <text> action <value>
×	not yet documented	policy.xml.in: policy access-list <access- list> rule <rule> descrip- tion	policy access-list <text> rule <text> description <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	policy.xml.in: policy access-list <access-list> rule <rule> destination any	policy access-list <text> rule <text> destination any
×	not yet documented	policy.xml.in: policy access-list <access-list> rule <rule> destination host	policy access-list <text> rule <text> destination host <value>
×	not yet documented	policy.xml.in: policy access-list <access-list> rule <rule> destination inverse-mask	policy access-list <text> rule <text> destination inverse-mask <value>
×	not yet documented	policy.xml.in: policy access-list <access-list> rule <rule> destination network	policy access-list <text> rule <text> destination network <value>
×	not yet documented	policy.xml.in: policy access-list <access-list> rule <rule> source any	policy access-list <text> rule <text> source any
×	not yet documented	policy.xml.in: policy access-list <access-list> rule <rule> source host	policy access-list <text> rule <text> source host <value>
×	not yet documented	policy.xml.in: policy access-list <access-list> rule <rule> source inverse-mask	policy access-list <text> rule <text> source inverse-mask <value>
×	not yet documented	policy.xml.in: policy access-list <access-list> rule <rule> source network	policy access-list <text> rule <text> source network <value>
×	<i>set policy as-path-list <text></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set policy as-path-list <text> description <text></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set policy as-path-list <text> rule <1-65535> action <permit/deny></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set policy as-path-list <text> rule <1-65535> description <text></i>	Nothing found in XML Definitions	Nothing found in VyOS

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Table 1 – continued from previous page

Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set policy as-path-list <text> rule <1-65535> regex <text></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	policy.xml.in: policy as-path-list <as-path-list> description	policy as-path-list <text> description <value>
×	not yet documented	policy.xml.in: policy as-path-list <as-path-list> rule <rule> action	policy as-path-list <text> rule <text> action <value>
×	not yet documented	policy.xml.in: policy as-path-list <as-path-list> rule <rule> description	policy as-path-list <text> rule <text> description <value>
×	not yet documented	policy.xml.in: policy as-path-list <as-path-list> rule <rule> regex	policy as-path-list <text> rule <text> regex <value>
×	<code>set policy community-list <text></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set policy community-list <text> description <text></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set policy community-list <text> rule <1-65535> action <permit/deny></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set policy community-list <text> rule <1-65535> description <text></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set policy community-list <text> rule <1-65535> regex <aa:nn/local-AS/no-ad</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	policy.xml.in: policy community-list <community-list> de- scription	policy community-list <text> description <value>
×	not yet documented	policy.xml.in: policy community-list <community-list> rule <rule> action	policy community-list <text> rule <text> action <value>

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Table 1 – continued from previous page

Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	policy.xml.in: policy community-list <community-list> rule <rule> description	policy community-list <text> rule <text> de- scription <value>
×	not yet documented	policy.xml.in: policy community-list <community-list> rule <rule> regex	policy community-list <text> rule <text> regex <value>
×	<code>set policy extcommunity-list <text></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set policy extcommunity-list <text> description <text></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set policy extcommunity-list <text> rule <1-65535> action <permit/deny></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set policy extcommunity-list <text> rule <1-65535> description <text></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set policy extcommunity-list <text> rule <1-65535> regex <text></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	policy.xml.in: policy extcommunity-list <extcommunity-list> description	policy extcommunity- list <text> description <value>
×	not yet documented	policy.xml.in: policy extcommunity-list <extcommunity-list> rule <rule> action	policy extcommunity-list <text> rule <text> action <value>
×	not yet documented	policy.xml.in: policy extcommunity-list <extcommunity-list> rule <rule> description	policy extcommunity-list <text> rule <text> descrip- tion <value>
×	not yet documented	policy.xml.in: policy extcommunity-list <extcommunity-list> rule <rule> regex	policy extcommunity-list <text> rule <text> regex <value>
×	<code>set policy large-community-list <text></code>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set policy large-community-list <text> description <text></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set policy large-community-list <text> rule <1-65535> action <permit/deny></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set policy large-community-list <text> rule <1-65535> description <text></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set policy large-community-list <text> rule <1-65535> regex <aa:nn:nn></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	policy.xml.in: policy large-community-list <large-community-list> description	policy large-community-list <text> description <value>
×	not yet documented	policy.xml.in: policy large-community-list <large-community-list> rule <rule> action	policy large-community-list <text> rule <text> action <value>
×	not yet documented	policy.xml.in: policy large-community-list <large-community-list> rule <rule> description	policy large-community-list <text> rule <text> description <value>
×	not yet documented	policy.xml.in: policy large-community-list <large-community-list> rule <rule> regex	policy large-community-list <text> rule <text> regex <value>
×	<code>set policy local-route6 rule <1-32765> destination address <h:h:h:h:h:h:h/h:h:h></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set policy local-route6 rule <1-32765> destination port <1-65535></code>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set policy local-route6 rule <1-32765> fwmark <1-2147483647></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set policy local-route6 rule <1-32765> inbound-interface <interface></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set policy local-route6 rule <1-32765> protocol <protocol></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set policy local-route6 rule <1-32765> set table <1-200/main></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set policy local-route6 rule <1-32765> set vrf <vrf/default></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set policy local-route6 rule <1-32765> source address <h:h:h:h:h:h:h:h/h:h:x></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set policy local-route6 rule <1-32765> source port <1-65535></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	Nothing found in XML Definitions	policy local-route6 rule <text> destination <value>
×	not yet documented	policy_local-route.xml.in: policy local-route6 rule <rule> destination address	Nothing found in VyOS
×	not yet documented	policy_local-route.xml.in: policy local-route6 rule <rule> destination port	Nothing found in VyOS
×	not yet documented	policy_local-route.xml.in: policy local-route6 rule <rule> fwmark	policy local-route6 rule <text> fwmark <value>
×	not yet documented	policy_local-route.xml.in: policy local-route6 rule <rule> inbound-interface	policy local-route6 rule <text> inbound-interface <value>
×	not yet documented	policy_local-route.xml.in: policy local-route6 rule <rule> protocol	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	policy_local-route.xml.in: policy local-route6 rule <rule> set table	policy local-route6 rule <text> set table <value>
×	not yet documented	Nothing found in XML Definitions	policy local-route6 rule <text> source <value>
×	not yet documented	policy_local-route.xml.in: policy local-route6 rule <rule> source address	Nothing found in VyOS
×	not yet documented	policy_local-route.xml.in: policy local-route6 rule <rule> source port	Nothing found in VyOS
×	<i>set policy local-route rule <1-32765> destination address <x.x.x.x/x.x.x.x/x></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set policy local-route rule <1-32765> destination port <1-65535></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set policy local-route rule <1-32765> fwmark <1-2147483647></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set policy local-route rule <1-32765> inbound-interface <interface></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set policy local-route rule <1-32765> protocol <protocol></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set policy local-route rule <1-32765> set table <1-200/main></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set policy local-route rule <1-32765> set vrf <vrf/default></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set policy local-route rule <1-32765> source address <x.x.x.x/x. x.x.x/x></i>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<i>set policy local-route rule <1-32765> source port <1-65535></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	Nothing found in XML Definitions	policy local-route rule <text> destination <value>
×	not yet documented	policy_local-route.xml.in: policy local-route rule <rule> destination address	Nothing found in VyOS
×	not yet documented	policy_local-route.xml.in: policy local-route rule <rule> destination port	Nothing found in VyOS
×	not yet documented	policy_local-route.xml.in: policy local-route rule <rule> fwmark	policy local-route rule <text> fwmark <value>
×	not yet documented	policy_local-route.xml.in: policy local-route rule <rule> inbound-interface	policy local-route rule <text> inbound-interface <value>
×	not yet documented	policy_local-route.xml.in: policy local-route rule <rule> protocol	Nothing found in VyOS
×	not yet documented	policy_local-route.xml.in: policy local-route rule <rule> set table	policy local-route rule <text> set table <value>
×	not yet documented	Nothing found in XML Definitions	policy local-route rule <text> source <value>
×	not yet documented	policy_local-route.xml.in: policy local-route rule <rule> source address	Nothing found in VyOS
×	not yet documented	policy_local-route.xml.in: policy local-route rule <rule> source port	Nothing found in VyOS
×	<i>set policy prefix-list6 PL6-EXAMPLE-NAME rule 10 action 'permit'</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set policy prefix-list6 PL6-EXAMPLE-NAME rule 10 le '128'</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set policy prefix-list6 PL6-EXAMPLE-NAME rule 10 prefix '2001:db8:0:0::/64'</i>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set policy prefix-list6 PL6-EXAMPLE-NAME rule 20 action 'permit'</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set policy prefix-list6 PL6-EXAMPLE-NAME rule 20 le '128'</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set policy prefix-list6 PL6-EXAMPLE-NAME rule 20 prefix '2001:db8:0:1::/ 64'</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set policy prefix-list6 PL6-EXAMPLE-NAME rule 30 action 'permit'</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set policy prefix-list6 PL6-EXAMPLE-NAME rule 30 le '128'</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set policy prefix-list6 PL6-EXAMPLE-NAME rule 30 prefix '2001:db8:0:2::/ 64'</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set policy prefix-list6 <text></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set policy prefix-list6 <text> description <text></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set policy prefix-list6 <text> rule <1-65535> action <permit/deny></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set policy prefix-list6 <text> rule <1-65535> description <text></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set policy prefix-list6 <text> rule <1-65535> ge <0-128></code>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set policy prefix-list6 <text> rule <1-65535> le <0-128></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set policy prefix-list6 <text> rule <1-65535> prefix <h:h:h:h:h:h:h:h/x></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	policy.xml.in: policy prefix-list6 <prefix- list6> description	policy prefix-list6 <text> description <value>
×	not yet documented	policy.xml.in: policy prefix-list6 <prefix- list6> rule <rule> action	policy prefix-list6 <text> rule <text> action <value>
×	not yet documented	policy.xml.in: policy prefix-list6 <prefix- list6> rule <rule> descrip- tion	policy prefix-list6 <text> rule <text> description <value>
×	not yet documented	policy.xml.in: policy prefix-list6 <prefix- list6> rule <rule> ge	policy prefix-list6 <text> rule <text> ge <value>
×	not yet documented	policy.xml.in: policy prefix-list6 <prefix- list6> rule <rule> le	policy prefix-list6 <text> rule <text> le <value>
×	not yet documented	policy.xml.in: policy prefix-list6 <prefix- list6> rule <rule> prefix	policy prefix-list6 <text> rule <text> prefix <value>
×	<code>set policy prefix-list PL4-EXAMPLE-NAME rule 10 action 'permit'</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set policy prefix-list PL4-EXAMPLE-NAME rule 10 le '32'</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set policy prefix-list PL4-EXAMPLE-NAME rule 10 prefix '192. 0.2.0/24'</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set policy prefix-list PL4-EXAMPLE-NAME rule 20 action 'permit'</code>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set policy prefix-list PL4-EXAMPLE-NAME rule 20 le '32'</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set policy prefix-list PL4-EXAMPLE-NAME rule 20 prefix '198. 51.100.0/24'</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set policy prefix-list PL4-EXAMPLE-NAME rule 30 action 'permit'</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set policy prefix-list PL4-EXAMPLE-NAME rule 30 le '32'</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set policy prefix-list PL4-EXAMPLE-NAME rule 30 prefix '203. 0.113.0/24'</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set policy prefix-list <text></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set policy prefix-list <text> description <text></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set policy prefix-list <text> rule <1-65535> action <permit/deny></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set policy prefix-list <text> rule <1-65535> description <text></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set policy prefix-list <text> rule <1-65535> ge <0-32></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set policy prefix-list <text> rule <1-65535> le <0-32></code>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set policy prefix-list <text> rule <1-65535> prefix <x.x.x.x/ x></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	policy.xml.in: policy prefix-list <prefix-list> description	policy prefix-list <text> description <value>
×	not yet documented	policy.xml.in: policy prefix-list <prefix-list> rule <rule> action	policy prefix-list <text> rule <text> action <value>
×	not yet documented	policy.xml.in: policy prefix-list <prefix-list> rule <rule> description	policy prefix-list <text> rule <text> description <value>
×	not yet documented	policy.xml.in: policy prefix-list <prefix-list> rule <rule> ge	policy prefix-list <text> rule <text> ge <value>
×	not yet documented	policy.xml.in: policy prefix-list <prefix-list> rule <rule> le	policy prefix-list <text> rule <text> le <value>
×	not yet documented	policy.xml.in: policy prefix-list <prefix-list> rule <rule> prefix	policy prefix-list <text> rule <text> prefix <value>
×	<code>set policy route-map <text></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set policy route-map <text> description <text></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set policy route-map <text> rule <1-65535> action <permit/deny></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set policy route-map <text> rule <1-65535> call <text></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set policy route-map <text> rule <1-65535> continue <1-65535></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set policy route-map <text> rule <1-65535> description <text></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set policy route-map <text> rule <1-65535> match as-path <text></code>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set policy route-map <text> rule <1-65535> match community community-list <text></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set policy route-map <text> rule <1-65535> match community exact-match</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set policy route-map <text> rule <1-65535> match extcommunity <text></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set policy route-map <text> rule <1-65535> match interface <text></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set policy route-map <text> rule <1-65535> match ip address access-list <1-2699></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set policy route-map <text> rule <1-65535> match ip address prefix-len <0-32></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set policy route-map <text> rule <1-65535> match ip address prefix-list <text></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set policy route-map <text> rule <1-65535> match ip nexthop access-list <1-2699></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set policy route-map <text> rule <1-65535> match ip nexthop address <x.x.x.x></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set policy route-map <text> rule <1-65535> match ip nexthop prefix-len <0-32></code>	Nothing found in XML Definitions	Nothing found in VyOS

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Table 1 – continued from previous page

Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set policy route-map</code> <text> rule <1-65535> match ip nexthop prefix-list <text>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set policy route-map</code> <text> rule <1-65535> match ip nexthop type <blackhole>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set policy route-map</code> <text> rule <1-65535> match ip route-source access-list <1-2699>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set policy route-map</code> <text> rule <1-65535> match ip route-source prefix-list <text>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set policy route-map</code> <text> rule <1-65535> match ipv6 address access-list <text>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set policy route-map</code> <text> rule <1-65535> match ipv6 address prefix-len <0-128>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set policy route-map</code> <text> rule <1-65535> match ipv6 address prefix-list <text>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set policy route-map</code> <text> rule <1-65535> match ipv6 nexthop <h:h:h:h:h:h:h>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set policy route-map</code> <text> rule <1-65535> match large-community large-community-list <text>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set policy route-map</code> <code><text> rule</code> <code><1-65535> match</code> <code>local-preference</code> <code><0-4294967295></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set policy route-map</code> <code><text> rule</code> <code><1-65535> match</code> <code>metric <1-65535></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set policy</code> <code>route-map <text></code> <code>rule <1-65535></code> <code>match origin</code> <code><egp/igp/incomplete></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set policy route-map</code> <code><text> rule</code> <code><1-65535> match</code> <code>peer <x.x.x.x></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set policy route-map</code> <code><text> rule</code> <code><1-65535> match</code> <code>protocol <protocol></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set policy route-map</code> <code><text> rule</code> <code><1-65535> match rpki</code> <code><invalid/notfound/val</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set policy route-map</code> <code><text> rule</code> <code><1-65535> match</code> <code>source-vrf <text></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set policy route-map</code> <code><text> rule</code> <code><1-65535> match</code> <code>tag <1-65535></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set policy route-map</code> <code><text> rule</code> <code><1-65535> on-match</code> <code>goto <1-65535></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set policy route-map</code> <code><text> rule</code> <code><1-65535> on-match</code> <code>next</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set policy route-map</code> <code><text> rule</code> <code><1-65535> set</code> <code>aggregator <as/ip></code> <code><1-4294967295/x.x.x.x></code>	Nothing found in XML Definitions	Nothing found in VyOS

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Table 1 – continued from previous page

Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set policy route-map</code> <code><text> rule</code> <code><1-65535> set</code> <code>as-path exclude</code> <code><1-4294967295 </code> <code>all></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set policy</code> <code>route-map <text></code> <code>rule <1-65535></code> <code>set as-path</code> <code>prepend-last-as</code> <code><n></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set policy route-map</code> <code><text> rule</code> <code><1-65535> set</code> <code>as-path prepend</code> <code><1-4294967295></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set policy route-map</code> <code><text> rule</code> <code><1-65535> set</code> <code>atomic-aggregate</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set policy</code> <code>route-map <text></code> <code>rule <1-65535></code> <code>set community</code> <code><add/replace></code> <code><community></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set policy route-map</code> <code><text> rule</code> <code><1-65535> set</code> <code>community delete</code> <code><text></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set policy route-map</code> <code><text> rule</code> <code><1-65535> set</code> <code>community none</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set policy route-map</code> <code><text> rule</code> <code><1-65535> set</code> <code>distance <0-255></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set policy route-map</code> <code><text> rule</code> <code><1-65535> set</code> <code>extcommunity</code> <code>bandwidth-non-transit</code>	Nothing found in XML Definitions	Nothing found in VyOS

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Table 1 – continued from previous page

Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set policy route-map</code> <code><text> rule</code> <code><1-65535> set</code> <code>extcommunity</code> <code>bandwidth</code> <code><1-25600/cumulative/n</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set policy route-map</code> <code><text> rule</code> <code><1-65535> set</code> <code>extcommunity none</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set policy route-map</code> <code><text> rule</code> <code><1-65535> set</code> <code>extcommunity rt</code> <code><text></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set policy route-map</code> <code><text> rule</code> <code><1-65535> set</code> <code>extcommunity soo</code> <code><text></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set policy route-map</code> <code><text> rule</code> <code><1-65535> set</code> <code>ip-next-hop <x.x.x.x></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set policy</code> <code>route-map <text></code> <code>rule <1-65535></code> <code>set ip-next-hop</code> <code>peer-address</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set policy</code> <code>route-map <text></code> <code>rule <1-65535></code> <code>set ip-next-hop</code> <code>unchanged</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set policy route-map</code> <code><text> rule</code> <code><1-65535> set</code> <code>ipv6-next-hop</code> <code><global/local></code> <code><h:h:h:h:h:h:h></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set policy route-map</code> <code><text> rule</code> <code><1-65535> set</code> <code>ipv6-next-hop</code> <code>peer-address</code>	Nothing found in XML Definitions	Nothing found in VyOS

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Table 1 – continued from previous page

Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set policy route-map</code> <code><text> rule</code> <code><1-65535> set</code> <code>ipv6-next-hop</code> <code>prefer-global</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set policy route-map</code> <code><text> rule</code> <code><1-65535> set</code> <code>large-community</code> <code><add/replace></code> <code><GA:LDP1:LDP2></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set policy route-map</code> <code><text> rule</code> <code><1-65535> set</code> <code>large-community</code> <code>delete <text></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set policy route-map</code> <code><text> rule</code> <code><1-65535> set</code> <code>large-community</code> <code>none</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set policy route-map</code> <code><text> rule</code> <code><1-65535> set</code> <code>local-preference</code> <code><0-4294967295></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set policy</code> <code>route-map <text></code> <code>rule <1-65535></code> <code>set metric-type</code> <code><type-1/type-2></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set policy</code> <code>route-map <text></code> <code>rule <1-65535></code> <code>set metric <+/-</code> <code>-metric/0-4294967295/</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set policy route-map</code> <code><text> rule</code> <code><1-65535> set origin</code> <code><igp/egp/incomplete></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set policy route-map</code> <code><text> rule</code> <code><1-65535> set</code> <code>originator-id <x.</code> <code>x.x.x></code>	Nothing found in XML Definitions	Nothing found in VyOS

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Table 1 – continued from previous page

Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set policy route-map <text> rule <1-65535> set src <x.x.x.x/h:h:h:h:h:h:h:h></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set policy route-map <text> rule <1-65535> set table <1-200></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set policy route-map <text> rule <1-65535> set tag <1-65535></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set policy route-map <text> rule <1-65535> set weight <0-4294967295></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	policy.xml.in: policy route-map <route-map> description	policy route-map <text> description <value>
×	not yet documented	policy.xml.in: policy route-map <route-map> rule <rule> action	policy route-map <text> rule <text> action <value>
×	not yet documented	policy.xml.in: policy route-map <route-map> rule <rule> call	policy route-map <text> rule <text> call <value>
×	not yet documented	policy.xml.in: policy route-map <route-map> rule <rule> continue	policy route-map <text> rule <text> continue <value>
×	not yet documented	policy.xml.in: policy route-map <route-map> rule <rule> description	policy route-map <text> rule <text> description <value>
×	not yet documented	policy.xml.in: policy route-map <route-map> rule <rule> match as-path	policy route-map <text> rule <text> match as-path <value>
×	not yet documented	policy.xml.in: policy route-map <route-map> rule <rule> match community community-list	policy route-map <text> rule <text> match community community-list <value>
×	not yet documented	policy.xml.in: policy route-map <route-map> rule <rule> match community exact-match	policy route-map <text> rule <text> match community exact-match

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	policy.xml.in: policy route-map <route-map> rule <rule> match evpn default-route	policy route-map <text> rule <text> match evpn default-route
×	not yet documented	policy.xml.in: policy route-map <route-map> rule <rule> match evpn rd	policy route-map <text> rule <text> match evpn rd <value>
×	not yet documented	policy.xml.in: policy route-map <route-map> rule <rule> match evpn route-type	policy route-map <text> rule <text> match evpn route-type <value>
×	not yet documented	policy.xml.in: policy route-map <route-map> rule <rule> match evpn vni	policy route-map <text> rule <text> match evpn vni <value>
×	not yet documented	policy.xml.in: policy route-map <route-map> rule <rule> match extcommunity	policy route-map <text> rule <text> match extcommunity <value>
×	not yet documented	policy.xml.in: policy route-map <route-map> rule <rule> match interface	policy route-map <text> rule <text> match interface <value>
×	not yet documented	policy.xml.in: policy route-map <route-map> rule <rule> match ip address access-list	policy route-map <text> rule <text> match ip address access-list <value>
×	not yet documented	policy.xml.in: policy route-map <route-map> rule <rule> match ip address prefix-len	policy route-map <text> rule <text> match ip address prefix-len <value>
×	not yet documented	policy.xml.in: policy route-map <route-map> rule <rule> match ip address prefix-list	policy route-map <text> rule <text> match ip address prefix-list <value>
×	not yet documented	policy.xml.in: policy route-map <route-map> rule <rule> match ip nexthop access-list	policy route-map <text> rule <text> match ip nexthop access-list <value>
×	not yet documented	policy.xml.in: policy route-map <route-map> rule <rule> match ip nexthop address	policy route-map <text> rule <text> match ip nexthop address <value>
×	not yet documented	policy.xml.in: policy route-map <route-map> rule <rule> match ip nexthop prefix-len	policy route-map <text> rule <text> match ip nexthop prefix-len <value>

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Table 1 – continued from previous page

Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	policy.xml.in: policy route-map <route-map> rule <rule> match ip nexthop prefix-list	policy route-map <text> rule <text> match ip nexthop prefix-list <value>
×	not yet documented	policy.xml.in: policy route-map <route-map> rule <rule> match ip nexthop type	policy route-map <text> rule <text> match ip nexthop type <value>
×	not yet documented	policy.xml.in: policy route-map <route-map> rule <rule> match ip route-source access-list	policy route-map <text> rule <text> match ip route-source access-list <value>
×	not yet documented	policy.xml.in: policy route-map <route-map> rule <rule> match ip route-source prefix-list	policy route-map <text> rule <text> match ip route-source prefix-list <value>
×	not yet documented	policy.xml.in: policy route-map <route-map> rule <rule> match ipv6 address access-list	policy route-map <text> rule <text> match ipv6 address access-list <value>
×	not yet documented	policy.xml.in: policy route-map <route-map> rule <rule> match ipv6 address prefix-len	policy route-map <text> rule <text> match ipv6 address prefix-len <value>
×	not yet documented	policy.xml.in: policy route-map <route-map> rule <rule> match ipv6 address prefix-list	policy route-map <text> rule <text> match ipv6 address prefix-list <value>
×	not yet documented	policy.xml.in: policy route-map <route-map> rule <rule> match ipv6 nexthop access-list	policy route-map <text> rule <text> match ipv6 nexthop access-list <value>
×	not yet documented	policy.xml.in: policy route-map <route-map> rule <rule> match ipv6 nexthop address	policy route-map <text> rule <text> match ipv6 nexthop address <value>
×	not yet documented	policy.xml.in: policy route-map <route-map> rule <rule> match ipv6 nexthop prefix-list	policy route-map <text> rule <text> match ipv6 nexthop prefix-list <value>
×	not yet documented	policy.xml.in: policy route-map <route-map> rule <rule> match ipv6 nexthop type	policy route-map <text> rule <text> match ipv6 nexthop type <value>
×	not yet documented	policy.xml.in: policy route-map <route-map> rule <rule> match large-community large-community-list	policy route-map <text> rule <text> match large-community large-community-list <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	policy.xml.in: policy route-map <route-map> rule <rule> match local-preference	policy route-map <text> rule <text> match local-preference <value>
×	not yet documented	policy.xml.in: policy route-map <route-map> rule <rule> match metric	policy route-map <text> rule <text> match metric <value>
×	not yet documented	policy.xml.in: policy route-map <route-map> rule <rule> match origin	policy route-map <text> rule <text> match origin <value>
×	not yet documented	policy.xml.in: policy route-map <route-map> rule <rule> match peer	policy route-map <text> rule <text> match peer <value>
×	not yet documented	policy.xml.in: policy route-map <route-map> rule <rule> match protocol	policy route-map <text> rule <text> match protocol <value>
×	not yet documented	policy.xml.in: policy route-map <route-map> rule <rule> match rpki	policy route-map <text> rule <text> match rpki <value>
×	not yet documented	policy.xml.in: policy route-map <route-map> rule <rule> match tag	policy route-map <text> rule <text> match tag <value>
×	not yet documented	policy.xml.in: policy route-map <route-map> rule <rule> on-match goto	policy route-map <text> rule <text> on-match goto <value>
×	not yet documented	policy.xml.in: policy route-map <route-map> rule <rule> on-match next	policy route-map <text> rule <text> on-match next
×	not yet documented	policy.xml.in: policy route-map <route-map> rule <rule> set aggregator as	policy route-map <text> rule <text> set aggregator as <value>
×	not yet documented	policy.xml.in: policy route-map <route-map> rule <rule> set aggregator ip	policy route-map <text> rule <text> set aggregator ip <value>
×	not yet documented	policy.xml.in: policy route-map <route-map> rule <rule> set as-path exclude	policy route-map <text> rule <text> set as-path exclude <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	policy.xml.in: policy route-map <route-map> rule <rule> set as-path prepend	policy route-map <text> rule <text> set as-path prepend <value>
×	not yet documented	policy.xml.in: policy route-map <route-map> rule <rule> set as-path prepend-last-as	policy route-map <text> rule <text> set as-path prepend-last-as <value>
×	not yet documented	policy.xml.in: policy route-map <route-map> rule <rule> set atomic-aggregate	policy route-map <text> rule <text> set atomic-aggregate
×	not yet documented	policy.xml.in: policy route-map <route-map> rule <rule> set community add	policy route-map <text> rule <text> set community add <value>
×	not yet documented	policy.xml.in: policy route-map <route-map> rule <rule> set community delete	policy route-map <text> rule <text> set community delete <value>
×	not yet documented	policy.xml.in: policy route-map <route-map> rule <rule> set community none	policy route-map <text> rule <text> set community none
×	not yet documented	policy.xml.in: policy route-map <route-map> rule <rule> set community replace	policy route-map <text> rule <text> set community replace <value>
×	not yet documented	policy.xml.in: policy route-map <route-map> rule <rule> set distance	policy route-map <text> rule <text> set distance <value>
×	not yet documented	policy.xml.in: policy route-map <route-map> rule <rule> set evpn gateway ipv4	policy route-map <text> rule <text> set evpn gateway ipv4 <value>
×	not yet documented	policy.xml.in: policy route-map <route-map> rule <rule> set evpn gateway ipv6	policy route-map <text> rule <text> set evpn gateway ipv6 <value>
×	not yet documented	policy.xml.in: policy route-map <route-map> rule <rule> set extcommunity bandwidth	policy route-map <text> rule <text> set extcommunity bandwidth <value>
×	not yet documented	policy.xml.in: policy route-map <route-map> rule <rule> set extcommunity bandwidth-non-transitive	policy route-map <text> rule <text> set extcommunity bandwidth-non-transitive

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Table 1 – continued from previous page

Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	policy.xml.in: policy route-map <route-map> rule <rule> set extcommunity none	policy route-map <text> rule <text> set extcommunity none
×	not yet documented	policy.xml.in: policy route-map <route-map> rule <rule> set extcommunity rt	policy route-map <text> rule <text> set extcommunity rt <value>
×	not yet documented	policy.xml.in: policy route-map <route-map> rule <rule> set extcommunity soo	policy route-map <text> rule <text> set extcommunity soo <value>
×	not yet documented	policy.xml.in: policy route-map <route-map> rule <rule> set ip-next-hop	policy route-map <text> rule <text> set ip-next-hop <value>
×	not yet documented	policy.xml.in: policy route-map <route-map> rule <rule> set ipv6-next-hop global	policy route-map <text> rule <text> set ipv6-next-hop global <value>
×	not yet documented	policy.xml.in: policy route-map <route-map> rule <rule> set ipv6-next-hop local	policy route-map <text> rule <text> set ipv6-next-hop local <value>
×	not yet documented	policy.xml.in: policy route-map <route-map> rule <rule> set ipv6-next-hop peer-address	policy route-map <text> rule <text> set ipv6-next-hop peer-address
×	not yet documented	policy.xml.in: policy route-map <route-map> rule <rule> set ipv6-next-hop prefer-global	policy route-map <text> rule <text> set ipv6-next-hop prefer-global
×	not yet documented	policy.xml.in: policy route-map <route-map> rule <rule> set l3vpn-nexthop encapsulation gre	policy route-map <text> rule <text> set l3vpn-nexthop encapsulation gre
×	not yet documented	policy.xml.in: policy route-map <route-map> rule <rule> set large-community add	policy route-map <text> rule <text> set large-community add <value>
×	not yet documented	policy.xml.in: policy route-map <route-map> rule <rule> set large-community delete	policy route-map <text> rule <text> set large-community delete <value>
×	not yet documented	policy.xml.in: policy route-map <route-map> rule <rule> set large-community none	policy route-map <text> rule <text> set large-community none

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Table 1 – continued from previous page

Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	policy.xml.in: policy route-map <route-map> rule <rule> set large-community replace	policy route-map <text> rule <text> set large-community replace <value>
×	not yet documented	policy.xml.in: policy route-map <route-map> rule <rule> set local-preference	policy route-map <text> rule <text> set local-preference <value>
×	not yet documented	policy.xml.in: policy route-map <route-map> rule <rule> set metric	policy route-map <text> rule <text> set metric <value>
×	not yet documented	policy.xml.in: policy route-map <route-map> rule <rule> set metric-type	policy route-map <text> rule <text> set metric-type <value>
×	not yet documented	policy.xml.in: policy route-map <route-map> rule <rule> set origin	policy route-map <text> rule <text> set origin <value>
×	not yet documented	policy.xml.in: policy route-map <route-map> rule <rule> set originator-id	policy route-map <text> rule <text> set originator-id <value>
×	not yet documented	policy.xml.in: policy route-map <route-map> rule <rule> set src	policy route-map <text> rule <text> set src <value>
×	not yet documented	policy.xml.in: policy route-map <route-map> rule <rule> set table	policy route-map <text> rule <text> set table <value>
×	not yet documented	policy.xml.in: policy route-map <route-map> rule <rule> set tag	policy route-map <text> rule <text> set tag <value>
×	not yet documented	policy.xml.in: policy route-map <route-map> rule <rule> set weight	policy route-map <text> rule <text> set weight <value>
×	<i>set policy route6 <name> default-log</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set policy route6 <name> description <text></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set policy route6 <name> interface <interface></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set policy route6 <name> rule <n> action drop</i>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set policy route6 <name> rule <n> connection-mark <1-2147483647></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set policy route6 <name> rule <n> description <text></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set policy route6 <name> rule <n> destination address <match_criteria></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set policy route6 <name> rule <n> destination group <address-group/domain <text></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set policy route6 <name> rule <n> destination port <match_criteria></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set policy route6 <name> rule <n> disable</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set policy route6 <name> rule <n> dscp-exclude <text></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set policy route6 <name> rule <n> dscp <text></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set policy route6 <name> rule <n> fragment <match-grag/match-non</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set policy route6 <name> rule <n> icmpv6 <code type></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set policy route6 <name> rule <n> icmpv6 type-name <text></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set policy route6 <name> rule <n> ipsec <match-ipsec/match-no</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set policy route6 <name> rule <n> limit burst <0-4294967295></code>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set policy route6 <name> rule <n> limit rate <text></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set policy route6 <name> rule <n> log <enable/disable></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set policy route6 <name> rule <n> mark <match_criteria></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set policy route6 <name> rule <n> packet-length-exclude <text></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set policy route6 <name> rule <n> packet-length <text></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set policy route6 <name> rule <n> packet-type [broadcast host multicast other]</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set policy route6 <name> rule <n> protocol <text 0-255 tcp_udp all ></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set policy route6 <name> rule <n> recent count <1-255></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set policy route6 <name> rule <n> recent time <1-4294967295></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set policy route6 <name> rule <n> set connection-mark <1-2147483647></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set policy route6 <name> rule <n> set dscp <0-63></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set policy route6 <name> rule <n> set mark <1-2147483647></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set policy route6 <name> rule <n> set table <main 1-200></code>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set policy route6 <name> rule <n> set tcp-mss <500-1460></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set policy route6 <name> rule <n> set vrf <default text ></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set policy route6 <name> rule <n> source address <match_criteria></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set policy route6 <name> rule <n> source group <address-group/domain <text></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set policy route6 <name> rule <n> state <established invalid new related></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set policy route6 <name> rule <n> tcp flags <text></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set policy route6 <name> rule <n> time monthdays <text></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set policy route6 <name> rule <n> time startdate <text></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set policy route6 <name> rule <n> time starttime <text></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set policy route6 <name> rule <n> time stopdate <text></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set policy route6 <name> rule <n> time stoptime <text></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set policy route6 <name> rule <n> time utc</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set policy route6 <name> rule <n> time weekdays <text></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	policy_route.xml.in: policy route6 <route6> default-log	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	policy_route.xml.in: policy route6 <route6> de- scription	policy route6 <text> de- scription <value>
×	not yet documented	Nothing found in XML Definitions	policy route6 <text> enable-default-log
×	not yet documented	policy_route.xml.in: policy route6 <route6> in- terface	policy route6 <text> inter- face <value>
×	<i>set policy route6 rule <n> hop-limit <eq / gt / lt> <0-255></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	policy_route.xml.in: policy route6 <route6> rule <rule> action	policy route6 <text> rule <text> action <value>
×	not yet documented	policy_route.xml.in: policy route6 <route6> rule <rule> connection- mark	policy route6 <text> rule <text> connection-mark <value>
×	not yet documented	policy_route.xml.in: policy route6 <route6> rule <rule> description	policy route6 <text> rule <text> description <value>
×	not yet documented	policy_route.xml.in: policy route6 <route6> rule <rule> destination address	policy route6 <text> rule <text> destination address <value>
×	not yet documented	policy_route.xml.in: policy route6 <route6> rule <rule> destination group address-group	policy route6 <text> rule <text> destination group address-group <value>
×	not yet documented	policy_route.xml.in: policy route6 <route6> rule <rule> destination group domain-group	policy route6 <text> rule <text> destination group domain-group <value>
×	not yet documented	policy_route.xml.in: policy route6 <route6> rule <rule> destination group mac-group	policy route6 <text> rule <text> destination group mac-group <value>
×	not yet documented	policy_route.xml.in: policy route6 <route6> rule <rule> destination group network-group	policy route6 <text> rule <text> destination group network-group <value>
×	not yet documented	policy_route.xml.in: policy route6 <route6> rule <rule> destination group port-group	policy route6 <text> rule <text> destination group port-group <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	policy_route.xml.in: policy route6 <route6> rule <rule> destination port	policy route6 <text> rule <text> destination port <value>
×	not yet documented	policy_route.xml.in: policy route6 <route6> rule <rule> disable	policy route6 <text> rule <text> disable
×	not yet documented	policy_route.xml.in: policy route6 <route6> rule <rule> dscp	policy route6 <text> rule <text> dscp <value>
×	not yet documented	policy_route.xml.in: policy route6 <route6> rule <rule> dscp-exclude	policy route6 <text> rule <text> dscp-exclude <value>
×	not yet documented	policy_route.xml.in: policy route6 <route6> rule <rule> fragment match-frag	policy route6 <text> rule <text> fragment match- frag
×	not yet documented	policy_route.xml.in: policy route6 <route6> rule <rule> fragment match-non-frag	policy route6 <text> rule <text> fragment match- non-frag
×	not yet documented	policy_route.xml.in: policy route6 <route6> rule <rule> hop-limit eq	policy route6 <text> rule <text> hop-limit eq <value>
×	not yet documented	policy_route.xml.in: policy route6 <route6> rule <rule> hop-limit gt	policy route6 <text> rule <text> hop-limit gt <value>
×	not yet documented	policy_route.xml.in: policy route6 <route6> rule <rule> hop-limit lt	policy route6 <text> rule <text> hop-limit lt <value>
×	not yet documented	policy_route.xml.in: policy route6 <route6> rule <rule> icmpv6 type	policy route6 <text> rule <text> icmpv6 type <value>
×	not yet documented	policy_route.xml.in: policy route6 <route6> rule <rule> ipsec match- ipsec	policy route6 <text> rule <text> ipsec match-ipsec
×	not yet documented	policy_route.xml.in: policy route6 <route6> rule <rule> ipsec match- none	policy route6 <text> rule <text> ipsec match-none
×	not yet documented	policy_route.xml.in: policy route6 <route6> rule <rule> limit burst	policy route6 <text> rule <text> limit burst <value>
×	not yet documented	policy_route.xml.in: policy route6 <route6> rule <rule> limit rate	policy route6 <text> rule <text> limit rate <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	policy_route.xml.in: policy route6 <route6> rule <rule> log	policy route6 <text> rule <text> log <value>
×	not yet documented	policy_route.xml.in: policy route6 <route6> rule <rule> mark	Nothing found in VyOS
×	not yet documented	policy_route.xml.in: policy route6 <route6> rule <rule> packet-length	policy route6 <text> rule <text> packet-length <value>
×	not yet documented	policy_route.xml.in: policy route6 <route6> rule <rule> packet-length-exclude	policy route6 <text> rule <text> packet-length-exclude <value>
×	not yet documented	policy_route.xml.in: policy route6 <route6> rule <rule> packet-type	policy route6 <text> rule <text> packet-type <value>
×	not yet documented	policy_route.xml.in: policy route6 <route6> rule <rule> protocol	policy route6 <text> rule <text> protocol <value>
×	not yet documented	policy_route.xml.in: policy route6 <route6> rule <rule> recent count	policy route6 <text> rule <text> recent count <value>
×	not yet documented	policy_route.xml.in: policy route6 <route6> rule <rule> recent time	policy route6 <text> rule <text> recent time <value>
×	not yet documented	policy_route.xml.in: policy route6 <route6> rule <rule> set connection-mark	policy route6 <text> rule <text> set connection-mark <value>
×	not yet documented	policy_route.xml.in: policy route6 <route6> rule <rule> set dscp	policy route6 <text> rule <text> set dscp <value>
×	not yet documented	policy_route.xml.in: policy route6 <route6> rule <rule> set mark	policy route6 <text> rule <text> set mark <value>
×	not yet documented	policy_route.xml.in: policy route6 <route6> rule <rule> set table	policy route6 <text> rule <text> set table <value>
×	not yet documented	policy_route.xml.in: policy route6 <route6> rule <rule> set tcp-mss	policy route6 <text> rule <text> set tcp-mss <value>
×	not yet documented	policy_route.xml.in: policy route6 <route6> rule <rule> source address	policy route6 <text> rule <text> source address <value>
×	not yet documented	policy_route.xml.in: policy route6 <route6> rule <rule> source group address-group	policy route6 <text> rule <text> source group address-group <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	policy_route.xml.in: policy route6 <route6> rule <rule> source group domain-group	policy route6 <text> rule <text> source group domain-group <value>
×	not yet documented	policy_route.xml.in: policy route6 <route6> rule <rule> source group mac-group	policy route6 <text> rule <text> source group mac- group <value>
×	not yet documented	policy_route.xml.in: policy route6 <route6> rule <rule> source group network-group	policy route6 <text> rule <text> source group network-group <value>
×	not yet documented	policy_route.xml.in: policy route6 <route6> rule <rule> source group port-group	policy route6 <text> rule <text> source group port- group <value>
×	not yet documented	policy_route.xml.in: policy route6 <route6> rule <rule> source mac- address	policy route6 <text> rule <text> source mac-address <value>
×	not yet documented	policy_route.xml.in: policy route6 <route6> rule <rule> source port	policy route6 <text> rule <text> source port <value>
×	not yet documented	policy_route.xml.in: policy route6 <route6> rule <rule> state	Nothing found in VyOS
×	not yet documented	Nothing found in XML Definitions	policy route6 <text> rule <text> state established <value>
×	not yet documented	Nothing found in XML Definitions	policy route6 <text> rule <text> state invalid <value>
×	not yet documented	Nothing found in XML Definitions	policy route6 <text> rule <text> state new <value>
×	not yet documented	Nothing found in XML Definitions	policy route6 <text> rule <text> state related <value>
×	not yet documented	policy_route.xml.in: policy route6 <route6> rule <rule> tcp flags ack	policy route6 <text> rule <text> tcp flags ack
×	not yet documented	policy_route.xml.in: policy route6 <route6> rule <rule> tcp flags cwr	policy route6 <text> rule <text> tcp flags cwr
×	not yet documented	policy_route.xml.in: policy route6 <route6> rule <rule> tcp flags ecn	policy route6 <text> rule <text> tcp flags ecn
×	not yet documented	policy_route.xml.in: policy route6 <route6> rule <rule> tcp flags fin	policy route6 <text> rule <text> tcp flags fin

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	policy_route.xml.in: policy route6 <route6> rule <rule> tcp flags not ack	policy route6 <text> rule <text> tcp flags not ack
×	not yet documented	policy_route.xml.in: policy route6 <route6> rule <rule> tcp flags not cwr	policy route6 <text> rule <text> tcp flags not cwr
×	not yet documented	policy_route.xml.in: policy route6 <route6> rule <rule> tcp flags not ecn	policy route6 <text> rule <text> tcp flags not ecn
×	not yet documented	policy_route.xml.in: policy route6 <route6> rule <rule> tcp flags not fin	policy route6 <text> rule <text> tcp flags not fin
×	not yet documented	policy_route.xml.in: policy route6 <route6> rule <rule> tcp flags not psh	policy route6 <text> rule <text> tcp flags not psh
×	not yet documented	policy_route.xml.in: policy route6 <route6> rule <rule> tcp flags not rst	policy route6 <text> rule <text> tcp flags not rst
×	not yet documented	policy_route.xml.in: policy route6 <route6> rule <rule> tcp flags not syn	policy route6 <text> rule <text> tcp flags not syn
×	not yet documented	policy_route.xml.in: policy route6 <route6> rule <rule> tcp flags not urg	policy route6 <text> rule <text> tcp flags not urg
×	not yet documented	policy_route.xml.in: policy route6 <route6> rule <rule> tcp flags psh	policy route6 <text> rule <text> tcp flags psh
×	not yet documented	policy_route.xml.in: policy route6 <route6> rule <rule> tcp flags rst	policy route6 <text> rule <text> tcp flags rst
×	not yet documented	policy_route.xml.in: policy route6 <route6> rule <rule> tcp flags syn	policy route6 <text> rule <text> tcp flags syn
×	not yet documented	policy_route.xml.in: policy route6 <route6> rule <rule> tcp flags urg	policy route6 <text> rule <text> tcp flags urg
×	not yet documented	Nothing found in XML Definitions	policy route6 <text> rule <text> tcp mss <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	policy_route.xml.in: policy route6 <route6> rule <rule> time month- days	policy route6 <text> rule <text> time monthdays <value>
×	not yet documented	policy_route.xml.in: policy route6 <route6> rule <rule> time startdate	policy route6 <text> rule <text> time startdate <value>
×	not yet documented	policy_route.xml.in: policy route6 <route6> rule <rule> time starttime	policy route6 <text> rule <text> time starttime <value>
×	not yet documented	policy_route.xml.in: policy route6 <route6> rule <rule> time stopdate	policy route6 <text> rule <text> time stopdate <value>
×	not yet documented	policy_route.xml.in: policy route6 <route6> rule <rule> time stoptime	policy route6 <text> rule <text> time stoptime <value>
×	not yet documented	policy_route.xml.in: policy route6 <route6> rule <rule> time utc	policy route6 <text> rule <text> time utc
×	not yet documented	policy_route.xml.in: policy route6 <route6> rule <rule> time weekdays	policy route6 <text> rule <text> time weekdays <value>
×	<i>set policy route</i> <i><name> default-log</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set policy route</i> <i><name> description</i> <i><text></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set policy route</i> <i><name> interface</i> <i><interface></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set policy route</i> <i><name> rule <n></i> <i>action drop</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set policy route</i> <i><name> rule <n></i> <i>connection-mark</i> <i><1-2147483647></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set policy route</i> <i><name> rule <n></i> <i>description <text></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set policy route</i> <i><name> rule <n></i> <i>destination address</i> <i><match_criteria></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set policy route</i> <i><name> rule <n></i> <i>destination group</i> <i><address-group/domain</i> <i><text></i>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set policy route</code> <code><name> rule <n></code> <code>destination port</code> <code><match_criteria></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set policy route</code> <code><name> rule <n></code> <code>disable</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set policy route</code> <code><name> rule <n></code> <code>dscp-exclude <text></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set policy route</code> <code><name> rule <n></code> <code>dscp</code> <code><text></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set policy route</code> <code><name> rule</code> <code><n> fragment</code> <code><match-grag/match-non</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set policy route</code> <code><name> rule <n></code> <code>icmp</code> <code><code type></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set policy route</code> <code><name> rule <n></code> <code>icmp</code> <code>type-name <text></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set policy</code> <code>route <name></code> <code>rule <n> ipsec</code> <code><match-ipsec/match-no</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set policy route</code> <code><name> rule</code> <code><n> limit burst</code> <code><0-4294967295></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set policy route</code> <code><name> rule <n></code> <code>limit rate <text></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set policy route</code> <code><name> rule <n></code> <code>log</code> <code><enable/disable></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set policy route</code> <code><name> rule <n></code> <code>mark</code> <code><match_criteria></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set policy route</code> <code><name> rule <n></code> <code>packet-length-exclude</code> <code><text></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set policy route</code> <code><name> rule <n></code> <code>packet-length</code> <code><text></code>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set policy route <name> rule <n> packet-type [broadcast host multicast other]</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set policy route <name> rule <n> protocol <text 0-255 tcp_udp all ></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set policy route <name> rule <n> recent count <1-255></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set policy route <name> rule <n> recent time <1-4294967295></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set policy route <name> rule <n> set connection-mark <1-2147483647></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set policy route <name> rule <n> set dscp <0-63></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set policy route <name> rule <n> set mark <1-2147483647></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set policy route <name> rule <n> set table <main 1-200></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set policy route <name> rule <n> set tcp-mss <500-1460></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set policy route <name> rule <n> set vrf <default text ></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set policy route <name> rule <n> source address <match_criteria></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set policy route <name> rule <n> source group <address-group/domain <text></code>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set policy route <name> rule <n> state <established invalid new related></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set policy route <name> rule <n> tcp flags <text></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set policy route <name> rule <n> time monthdays <text></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set policy route <name> rule <n> time startdate <text></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set policy route <name> rule <n> time starttime <text></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set policy route <name> rule <n> time stopdate <text></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set policy route <name> rule <n> time stoptime <text></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set policy route <name> rule <n> time utc</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set policy route <name> rule <n> time weekdays <text></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	policy_route.xml.in: policy route <route> default-log	Nothing found in VyOS
×	not yet documented	policy_route.xml.in: policy route <route> de- scription	policy route <text> de- scription <value>
×	not yet documented	Nothing found in XML Definitions	policy route <text> enable-default-log
×	not yet documented	policy_route.xml.in: policy route <route> inter- face	policy route <text> inter- face <value>
×	<code>set policy route rule <n> ttl <eq gt lt> <0-255></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	policy_route.xml.in: policy route <route> rule <rule> action	policy route <text> rule <text> action <value>
×	not yet documented	policy_route.xml.in: policy route <route> rule <rule> connection-mark	policy route <text> rule <text> connection-mark <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	policy_route.xml.in: policy route <route> rule <rule> description	policy route <text> rule <text> description <value>
×	not yet documented	policy_route.xml.in: policy route <route> rule <rule> destination address	policy route <text> rule <text> destination address <value>
×	not yet documented	policy_route.xml.in: policy route <route> rule <rule> destination group address-group	policy route <text> rule <text> destination group address-group <value>
×	not yet documented	policy_route.xml.in: policy route <route> rule <rule> destination group domain-group	policy route <text> rule <text> destination group domain-group <value>
×	not yet documented	policy_route.xml.in: policy route <route> rule <rule> destination group mac-group	policy route <text> rule <text> destination group mac-group <value>
×	not yet documented	policy_route.xml.in: policy route <route> rule <rule> destination group network-group	policy route <text> rule <text> destination group network-group <value>
×	not yet documented	policy_route.xml.in: policy route <route> rule <rule> destination group port-group	policy route <text> rule <text> destination group port-group <value>
×	not yet documented	policy_route.xml.in: policy route <route> rule <rule> destination port	policy route <text> rule <text> destination port <value>
×	not yet documented	policy_route.xml.in: policy route <route> rule <rule> disable	policy route <text> rule <text> disable
×	not yet documented	policy_route.xml.in: policy route <route> rule <rule> dscp	policy route <text> rule <text> dscp <value>
×	not yet documented	policy_route.xml.in: policy route <route> rule <rule> dscp-exclude	policy route <text> rule <text> dscp-exclude <value>
×	not yet documented	policy_route.xml.in: policy route <route> rule <rule> fragment match- frag	policy route <text> rule <text> fragment match- frag
×	not yet documented	policy_route.xml.in: policy route <route> rule <rule> fragment match- non-frag	policy route <text> rule <text> fragment match- non-frag
×	not yet documented	policy_route.xml.in: policy route <route> rule <rule> icmp code	policy route <text> rule <text> icmp code <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	policy_route.xml.in: policy route <route> rule <rule> icmp type	policy route <text> rule <text> icmp type <value>
×	not yet documented	policy_route.xml.in: policy route <route> rule <rule> icmp type-name	policy route <text> rule <text> icmp type-name <value>
×	not yet documented	policy_route.xml.in: policy route <route> rule <rule> ipsec match-ipsec	policy route <text> rule <text> ipsec match-ipsec
×	not yet documented	policy_route.xml.in: policy route <route> rule <rule> ipsec match-none	policy route <text> rule <text> ipsec match-none
×	not yet documented	policy_route.xml.in: policy route <route> rule <rule> limit burst	policy route <text> rule <text> limit burst <value>
×	not yet documented	policy_route.xml.in: policy route <route> rule <rule> limit rate	policy route <text> rule <text> limit rate <value>
×	not yet documented	policy_route.xml.in: policy route <route> rule <rule> log	policy route <text> rule <text> log <value>
×	not yet documented	policy_route.xml.in: policy route <route> rule <rule> mark	Nothing found in VyOS
×	not yet documented	policy_route.xml.in: policy route <route> rule <rule> packet-length	policy route <text> rule <text> packet-length <value>
×	not yet documented	policy_route.xml.in: policy route <route> rule <rule> packet-length-exclude	policy route <text> rule <text> packet-length-exclude <value>
×	not yet documented	policy_route.xml.in: policy route <route> rule <rule> packet-type	policy route <text> rule rule <text> packet-type <value>
×	not yet documented	policy_route.xml.in: policy route <route> rule <rule> protocol	policy route <text> rule <text> protocol <value>
×	not yet documented	policy_route.xml.in: policy route <route> rule <rule> recent count	policy route <text> rule <text> recent count <value>
×	not yet documented	policy_route.xml.in: policy route <route> rule <rule> recent time	policy route <text> rule rule <text> recent time <value>
×	not yet documented	policy_route.xml.in: policy route <route> rule <rule> set connection-mark	policy route <text> rule <text> set connection-mark <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	policy_route.xml.in: policy route <route> rule <rule> set dscp	policy route <text> rule <text> set dscp <value>
×	not yet documented	policy_route.xml.in: policy route <route> rule <rule> set mark	policy route <text> rule <text> set mark <value>
×	not yet documented	policy_route.xml.in: policy route <route> rule <rule> set table	policy route <text> rule <text> set table <value>
×	not yet documented	policy_route.xml.in: policy route <route> rule <rule> set tcp-mss	policy route <text> rule rule <text> set tcp-mss <value>
×	not yet documented	policy_route.xml.in: policy route <route> rule <rule> source address	policy route <text> rule <text> source address <value>
×	not yet documented	policy_route.xml.in: policy route <route> rule <rule> source group address-group	policy route <text> rule <text> source group address-group <value>
×	not yet documented	policy_route.xml.in: policy route <route> rule <rule> source group domain-group	policy route <text> rule <text> source group domain-group <value>
×	not yet documented	policy_route.xml.in: policy route <route> rule <rule> source group mac- group	policy route <text> rule <text> source group mac- group <value>
×	not yet documented	policy_route.xml.in: policy route <route> rule <rule> source group network-group	policy route <text> rule <text> source group network-group <value>
×	not yet documented	policy_route.xml.in: policy route <route> rule <rule> source group port- group	policy route <text> rule <text> source group port- group <value>
×	not yet documented	policy_route.xml.in: policy route <route> rule <rule> source mac- address	policy route <text> rule <text> source mac-address <value>
×	not yet documented	policy_route.xml.in: policy route <route> rule <rule> source port	policy route <text> rule rule <text> source port <value>
×	not yet documented	policy_route.xml.in: policy route <route> rule <rule> state	Nothing found in VyOS
×	not yet documented	Nothing found in XML Definitions	policy route <text> rule <text> state established <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	Nothing found in XML Definitions	policy route <text> rule <text> state invalid <value>
×	not yet documented	Nothing found in XML Definitions	policy route <text> rule <text> state new <value>
×	not yet documented	Nothing found in XML Definitions	policy route <text> rule <text> state related <value>
×	not yet documented	policy_route.xml.in: policy route <route> rule <rule> tcp flags ack	policy route <text> rule <text> tcp flags ack
×	not yet documented	policy_route.xml.in: policy route <route> rule <rule> tcp flags cwr	policy route <text> rule <text> tcp flags cwr
×	not yet documented	policy_route.xml.in: policy route <route> rule <rule> tcp flags ecn	policy route <text> rule <text> tcp flags ecn
×	not yet documented	policy_route.xml.in: policy route <route> rule <rule> tcp flags fin	policy route <text> rule <text> tcp flags fin
×	not yet documented	policy_route.xml.in: policy route <route> rule <rule> tcp flags not ack	policy route <text> rule <text> tcp flags not ack
×	not yet documented	policy_route.xml.in: policy route <route> rule <rule> tcp flags not cwr	policy route <text> rule <text> tcp flags not cwr
×	not yet documented	policy_route.xml.in: policy route <route> rule <rule> tcp flags not ecn	policy route <text> rule <text> tcp flags not ecn
×	not yet documented	policy_route.xml.in: policy route <route> rule <rule> tcp flags not fin	policy route <text> rule <text> tcp flags not fin
×	not yet documented	policy_route.xml.in: policy route <route> rule <rule> tcp flags not psh	policy route <text> rule <text> tcp flags not psh
×	not yet documented	policy_route.xml.in: policy route <route> rule <rule> tcp flags not rst	policy route <text> rule <text> tcp flags not rst
×	not yet documented	policy_route.xml.in: policy route <route> rule <rule> tcp flags not syn	policy route <text> rule <text> tcp flags not syn
×	not yet documented	policy_route.xml.in: policy route <route> rule <rule> tcp flags not urg	policy route <text> rule <text> tcp flags not urg
×	not yet documented	policy_route.xml.in: policy route <route> rule <rule> tcp flags psh	policy route <text> rule <text> tcp flags psh

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	policy_route.xml.in: policy route <route> rule <rule> tcp flags rst	policy route <text> rule <text> tcp flags rst
×	not yet documented	policy_route.xml.in: policy route <route> rule <rule> tcp flags syn	policy route <text> rule <text> tcp flags syn
×	not yet documented	policy_route.xml.in: policy route <route> rule <rule> tcp flags urg	policy route <text> rule <text> tcp flags urg
×	not yet documented	Nothing found in XML Definitions	policy route <text> rule <text> tcp mss <value>
×	not yet documented	policy_route.xml.in: policy route <route> rule <rule> time monthdays	policy route <text> rule <text> time monthdays <value>
×	not yet documented	policy_route.xml.in: policy route <route> rule <rule> time startdate	policy route <text> rule <text> time startdate <value>
×	not yet documented	policy_route.xml.in: policy route <route> rule <rule> time starttime	policy route <text> rule <text> time starttime <value>
×	not yet documented	policy_route.xml.in: policy route <route> rule <rule> time stopdate	policy route <text> rule <text> time stopdate <value>
×	not yet documented	policy_route.xml.in: policy route <route> rule <rule> time stoptime	policy route <text> rule <text> time stoptime <value>
×	not yet documented	policy_route.xml.in: policy route <route> rule <rule> time utc	policy route <text> rule <text> time utc
×	not yet documented	policy_route.xml.in: policy route <route> rule <rule> time weekdays	policy route <text> rule <text> time weekdays <value>
×	not yet documented	policy_route.xml.in: policy route <route> rule <rule> ttl eq	policy route <text> rule <text> ttl eq <value>
×	not yet documented	policy_route.xml.in: policy route <route> rule <rule> ttl gt	policy route <text> rule <text> ttl gt <value>
×	not yet documented	policy_route.xml.in: policy route <route> rule <rule> ttl lt	policy route <text> rule <text> ttl lt <value>
×	<i>set protocols babel distribute-list <ipv4/ipv6> access-list <in/out> <number></i>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set protocols babel distribute-list <ipv4/ipv6> interface <interface> access-list <in/out> <number></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set protocols babel distribute-list <ipv4/ipv6> interface <interface> prefix-list <in/out> <name></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set protocols babel distribute-list <ipv4/ipv6> prefix-list <in/out> <name></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	protocols_babel.xml.in: protocols babel distribute- list ipv4 access-list in	protocols babel distribute- list ipv4 access-list in <value>
×	not yet documented	protocols_babel.xml.in: protocols babel distribute- list ipv4 access-list out	protocols babel distribute- list ipv4 access-list out <value>
×	not yet documented	protocols_babel.xml.in: protocols babel distribute- list ipv4 interface <inter- face> access-list in	protocols babel distribute- list ipv4 interface <text> access-list in <value>
×	not yet documented	protocols_babel.xml.in: protocols babel distribute- list ipv4 interface <inter- face> access-list out	protocols babel distribute- list ipv4 interface <text> access-list out <value>
×	not yet documented	protocols_babel.xml.in: protocols babel distribute- list ipv4 interface <inter- face> prefix-list in	protocols babel distribute- list ipv4 interface <text> prefix-list in <value>
×	not yet documented	protocols_babel.xml.in: protocols babel distribute- list ipv4 interface <inter- face> prefix-list out	protocols babel distribute- list ipv4 interface <text> prefix-list out <value>
×	not yet documented	protocols_babel.xml.in: protocols babel distribute- list ipv4 prefix-list in	protocols babel distribute- list ipv4 prefix-list in <value>
×	not yet documented	protocols_babel.xml.in: protocols babel distribute- list ipv4 prefix-list out	protocols babel distribute- list ipv4 prefix-list out <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	protocols_babel.xml.in: protocols babel distribute- list ipv6 access-list in	protocols babel distribute- list ipv6 access-list in <value>
×	not yet documented	protocols_babel.xml.in: protocols babel distribute- list ipv6 access-list out	protocols babel distribute- list ipv6 access-list out <value>
×	not yet documented	protocols_babel.xml.in: protocols babel distribute- list ipv6 interface <inter- face> access-list in	protocols babel distribute- list ipv6 interface <text> access-list in <value>
×	not yet documented	protocols_babel.xml.in: protocols babel distribute- list ipv6 interface <inter- face> access-list out	protocols babel distribute- list ipv6 interface <text> access-list out <value>
×	not yet documented	protocols_babel.xml.in: protocols babel distribute- list ipv6 interface <inter- face> prefix-list in	protocols babel distribute- list ipv6 interface <text> prefix-list in <value>
×	not yet documented	protocols_babel.xml.in: protocols babel distribute- list ipv6 interface <inter- face> prefix-list out	protocols babel distribute- list ipv6 interface <text> prefix-list out <value>
×	not yet documented	protocols_babel.xml.in: protocols babel distribute- list ipv6 prefix-list in	protocols babel distribute- list ipv6 prefix-list in <value>
×	not yet documented	protocols_babel.xml.in: protocols babel distribute- list ipv6 prefix-list out	protocols babel distribute- list ipv6 prefix-list out <value>
×	<i>set protocols babel interface <interface></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set protocols babel interface <interface> channel <1-254/interfering/no</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set protocols babel interface <interface> enable-timestamps</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set protocols babel interface <interface> hello-interval <20-655340></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set protocols babel interface <interface> max-rtt-penalty <0-65535></i>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set protocols babel interface <interface> rtt-decay <1-256></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set protocols babel interface <interface> rtt-max <1-65535></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set protocols babel interface <interface> rtt-min <1-65535></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set protocols babel interface <interface> rxcost <1-65534></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set protocols babel interface <interface> split-horizon <default/disable/enab</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set protocols babel interface <interface> type <auto/wired/wireless></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set protocols babel interface <interface> update-interval <20-655340></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	protocols_babel.xml.in: protocols babel interface <interface> channel	protocols babel interface <text> channel <value>
×	not yet documented	protocols_babel.xml.in: protocols babel interface <interface> enable- timestamps	protocols babel interface <text> enable-timestamps
×	not yet documented	protocols_babel.xml.in: protocols babel interface <interface> hello-interval	protocols babel interface <text> hello-interval <value>
×	not yet documented	protocols_babel.xml.in: protocols babel interface <interface> max-rtt- penalty	protocols babel interface <text> max-rtt-penalty <value>
×	not yet documented	protocols_babel.xml.in: protocols babel interface <interface> rtt-decay	protocols babel interface <text> rtt-decay <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	protocols_babel.xml.in: protocols babel interface <interface> rtt-max	protocols babel interface <text> rtt-max <value>
×	not yet documented	protocols_babel.xml.in: protocols babel interface <interface> rtt-min	protocols babel interface <text> rtt-min <value>
×	not yet documented	protocols_babel.xml.in: protocols babel interface <interface> rxcost	protocols babel interface <text> rxcost <value>
×	not yet documented	protocols_babel.xml.in: protocols babel interface <interface> split-horizon	protocols babel inter- face <text> split-horizon <value>
×	not yet documented	protocols_babel.xml.in: protocols babel interface <interface> type	protocols babel interface <text> type <value>
×	not yet documented	protocols_babel.xml.in: protocols babel interface <interface> update- interval	protocols babel interface <text> update-interval <value>
✓	<i>set protocols babel parameters diversity</i>	protocols_babel.xml.in: protocols babel param- eters diversity	protocols babel param- eters diversity
×	not yet documented	protocols_babel.xml.in: protocols babel param- eters diversity-factor	protocols babel param- eters diversity-factor <value>
×	<i>set protocols babel parameters diversity-factor <1-256></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	protocols_babel.xml.in: protocols babel param- eters resend-delay	protocols babel param- eters resend-delay <value>
×	<i>set protocols babel parameters resend-delay <20-655340></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	protocols_babel.xml.in: protocols babel param- eters smoothing-half-life	protocols babel param- eters smoothing-half-life <value>
×	<i>set protocols babel parameters smoothing-half-life <0-65534></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set protocols babel redistribute <ipv4/ipv6> <route source></i>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	protocols_babel.xml.in: protocols babel redis- tribute ipv4 bgp	protocols babel redis- tribute ipv4 bgp
×	not yet documented	protocols_babel.xml.in: protocols babel redis- tribute ipv4 connected	protocols babel redis- tribute ipv4 connected
×	not yet documented	protocols_babel.xml.in: protocols babel redis- tribute ipv4 eigrp	protocols babel redis- tribute ipv4 eigrp
×	not yet documented	protocols_babel.xml.in: protocols babel redis- tribute ipv4 isis	protocols babel redis- tribute ipv4 isis
×	not yet documented	protocols_babel.xml.in: protocols babel redis- tribute ipv4 kernel	protocols babel redis- tribute ipv4 kernel
×	not yet documented	protocols_babel.xml.in: protocols babel redis- tribute ipv4 nhrp	protocols babel redis- tribute ipv4 nhrp
×	not yet documented	protocols_babel.xml.in: protocols babel redis- tribute ipv4 ospf	protocols babel redis- tribute ipv4 ospf
×	not yet documented	protocols_babel.xml.in: protocols babel redis- tribute ipv4 rip	protocols babel redis- tribute ipv4 rip
×	not yet documented	protocols_babel.xml.in: protocols babel redis- tribute ipv4 static	protocols babel redis- tribute ipv4 static
×	not yet documented	protocols_babel.xml.in: protocols babel redis- tribute ipv6 bgp	protocols babel redis- tribute ipv6 bgp
×	not yet documented	protocols_babel.xml.in: protocols babel redis- tribute ipv6 connected	protocols babel redis- tribute ipv6 connected
×	not yet documented	protocols_babel.xml.in: protocols babel redis- tribute ipv6 isis	protocols babel redis- tribute ipv6 isis
×	not yet documented	protocols_babel.xml.in: protocols babel redis- tribute ipv6 kernel	protocols babel redis- tribute ipv6 kernel
×	not yet documented	protocols_babel.xml.in: protocols babel redis- tribute ipv6 nhrp	protocols babel redis- tribute ipv6 nhrp
×	not yet documented	protocols_babel.xml.in: protocols babel redis- tribute ipv6 ospfv3	protocols babel redis- tribute ipv6 ospfv3
×	not yet documented	protocols_babel.xml.in: protocols babel redis- tribute ipv6 ripng	protocols babel redis- tribute ipv6 ripng

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	protocols_babel.xml.in: protocols babel redis- tribute ipv6 static	protocols babel redis- tribute ipv6 static
×	<i>set protocols bfd peer <address></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set protocols bfd peer <address> echo-mode</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set protocols bfd peer <address> interval echo-interval <10-60000></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set protocols bfd peer <address> interval multiplier <2-255></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set protocols bfd peer <address> interval receive <10-60000></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set protocols bfd peer <address> interval transmit <10-60000></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set protocols bfd peer <address> minimum-ttl <1-254></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set protocols bfd peer <address> multihop</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set protocols bfd peer <address> passive</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set protocols bfd peer <address> profile <profile></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set protocols bfd peer <address> shutdown</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set protocols bfd peer <address> source address <address></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set protocols bfd peer <address> source interface <interface></i>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<i>set protocols bfd peer <address> vrf <vrf></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	protocols_bfd.xml.in: protocols bfd peer <peer> echo-mode	protocols bfd peer <text> echo-mode
×	not yet documented	protocols_bfd.xml.in: protocols bfd peer <peer> interval echo-interval	protocols bfd peer <text> interval echo-interval <value>
×	not yet documented	protocols_bfd.xml.in: protocols bfd peer <peer> interval multiplier	protocols bfd peer <text> interval multiplier <value>
×	not yet documented	protocols_bfd.xml.in: protocols bfd peer <peer> interval receive	protocols bfd peer <text> interval receive <value>
×	not yet documented	protocols_bfd.xml.in: protocols bfd peer <peer> interval transmit	protocols bfd peer <text> interval transmit <value>
×	not yet documented	protocols_bfd.xml.in: protocols bfd peer <peer> minimum-ttl	Nothing found in VyOS
×	not yet documented	protocols_bfd.xml.in: protocols bfd peer <peer> multihop	protocols bfd peer <text> multihop
×	not yet documented	protocols_bfd.xml.in: protocols bfd peer <peer> passive	protocols bfd peer <text> passive
×	not yet documented	protocols_bfd.xml.in: protocols bfd peer <peer> profile	protocols bfd peer <text> profile <value>
×	not yet documented	protocols_bfd.xml.in: protocols bfd peer <peer> shutdown	protocols bfd peer <text> shutdown
×	not yet documented	protocols_bfd.xml.in: protocols bfd peer <peer> source address	protocols bfd peer <text> source address <value>
×	not yet documented	protocols_bfd.xml.in: protocols bfd peer <peer> source interface	protocols bfd peer <text> source interface <value>
×	not yet documented	protocols_bfd.xml.in: protocols bfd peer <peer> vrf	protocols bfd peer <text> vrf <value>
×	not yet documented	protocols_bfd.xml.in: protocols bfd profile <profile> echo-mode	protocols bfd profile <text> echo-mode
×	not yet documented	protocols_bfd.xml.in: protocols bfd profile <profile> interval echo-interval	protocols bfd profile <text> interval echo-interval <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	protocols_bfd.xml.in: protocols bfd profile <profile> interval multiplier	protocols bfd profile <text> interval multiplier <value>
×	not yet documented	protocols_bfd.xml.in: protocols bfd profile <profile> interval receive	protocols bfd profile <text> interval receive <value>
×	not yet documented	protocols_bfd.xml.in: protocols bfd profile <profile> interval transmit	protocols bfd profile <text> interval transmit <value>
×	not yet documented	protocols_bfd.xml.in: protocols bfd profile <profile> minimum-ttl	Nothing found in VyOS
×	not yet documented	protocols_bfd.xml.in: protocols bfd profile <profile> passive	protocols bfd profile <text> passive
×	not yet documented	protocols_bfd.xml.in: protocols bfd profile <profile> shutdown	protocols bfd profile <text> shutdown
×	<i>set protocols bgp address-family <ipv4-unicast/ipv6-un aggregate-address <prefix></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set protocols bgp address-family <ipv4-unicast/ipv6-un aggregate-address <prefix> as-set</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set protocols bgp address-family <ipv4-unicast/ipv6-un aggregate-address <prefix> summary-only</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set protocols bgp address-family <ipv4-unicast/ipv6-un maximum-paths <ebgp/ibgp> <number></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set protocols bgp address-family <ipv4-unicast/ipv6-un network <prefix></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set protocols bgp address-family <ipv4-unicast/ipv6-un redistribute</i>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<i>set protocols bgp address-family <ipv4-unicast/ipv6-un redistribute metric <number></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set protocols bgp address-family <ipv4-unicast/ipv6-un redistribute route-map <name></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	protocols_bgp.xml.in: protocols bgp address-family ipv4-flowspec local-install interface	protocols bgp address-family ipv4-flowspec local-install interface <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp address-family ipv4-labeled-unicast aggregate-address <aggregate-address> as-set	protocols bgp address-family ipv4-labeled-unicast aggregate-address <text> as-set
×	not yet documented	protocols_bgp.xml.in: protocols bgp address-family ipv4-labeled-unicast aggregate-address <aggregate-address> route-map	protocols bgp address-family ipv4-labeled-unicast aggregate-address <text> route-map <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp address-family ipv4-labeled-unicast aggregate-address <aggregate-address> summary-only	protocols bgp address-family ipv4-labeled-unicast aggregate-address <text> summary-only
×	not yet documented	protocols_bgp.xml.in: protocols bgp address-family ipv4-labeled-unicast maximum-paths ebgp	Nothing found in VyOS
×	not yet documented	protocols_bgp.xml.in: protocols bgp address-family ipv4-labeled-unicast maximum-paths ibgp	Nothing found in VyOS
×	not yet documented	protocols_bgp.xml.in: protocols bgp address-family ipv4-labeled-unicast network <network> backdoor	protocols bgp address-family ipv4-labeled-unicast network <text> backdoor

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	protocols_bgp.xml.in: protocols bgp address-family ipv4-labeled-unicast network <network> route-map	protocols bgp address-family ipv4-labeled-unicast network <text> route-map <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp address-family ipv4-multicast aggregate-address <aggregate-address> as-set	protocols bgp address-family ipv4-multicast aggregate-address <text> as-set
×	not yet documented	protocols_bgp.xml.in: protocols bgp address-family ipv4-multicast aggregate-address <aggregate-address> route-map	protocols bgp address-family ipv4-multicast aggregate-address <text> route-map <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp address-family ipv4-multicast aggregate-address <aggregate-address> summary-only	protocols bgp address-family ipv4-multicast aggregate-address <text> summary-only
×	not yet documented	protocols_bgp.xml.in: protocols bgp address-family ipv4-multicast distance external	protocols bgp address-family ipv4-multicast distance external <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp address-family ipv4-multicast distance internal	protocols bgp address-family ipv4-multicast distance internal <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp address-family ipv4-multicast distance local	protocols bgp address-family ipv4-multicast distance local <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp address-family ipv4-multicast distance prefix <prefix> distance	protocols bgp address-family ipv4-multicast distance prefix <text> distance <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp address-family ipv4-multicast network <network> backdoor	protocols bgp address-family ipv4-multicast network <text> backdoor

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	protocols_bgp.xml.in: protocols bgp address-family ipv4-multicast network <network> route-map	protocols bgp address-family ipv4-multicast network <text> route-map <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp address-family ipv4-unicast aggregate-address <aggregate-address> as-set	protocols bgp address-family ipv4-unicast aggregate-address <text> as-set
×	not yet documented	protocols_bgp.xml.in: protocols bgp address-family ipv4-unicast aggregate-address <aggregate-address> route-map	protocols bgp address-family ipv4-unicast aggregate-address <text> route-map <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp address-family ipv4-unicast aggregate-address <aggregate-address> summary-only	protocols bgp address-family ipv4-unicast aggregate-address <text> summary-only
×	not yet documented	protocols_bgp.xml.in: protocols bgp address-family ipv4-unicast distance external	protocols bgp address-family ipv4-unicast distance external <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp address-family ipv4-unicast distance internal	protocols bgp address-family ipv4-unicast distance internal <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp address-family ipv4-unicast distance local	protocols bgp address-family ipv4-unicast distance local <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp address-family ipv4-unicast distance prefix <prefix> distance	protocols bgp address-family ipv4-unicast distance prefix <text> distance <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp address-family ipv4-unicast export vpn	protocols bgp address-family ipv4-unicast export vpn
×	not yet documented	protocols_bgp.xml.in: protocols bgp address-family ipv4-unicast import vpn	protocols bgp address-family ipv4-unicast import vpn

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	protocols_bgp.xml.in: protocols bgp address-family ipv4-unicast import vrf	protocols bgp address-family ipv4-unicast import vrf <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp address-family ipv4-unicast label vpn allocation-mode per-nexthop	protocols bgp address-family ipv4-unicast label vpn allocation-mode per-nexthop
×	not yet documented	protocols_bgp.xml.in: protocols bgp address-family ipv4-unicast label vpn export	protocols bgp address-family ipv4-unicast label vpn export <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp address-family ipv4-unicast maximum-paths ebgp	protocols bgp address-family ipv4-unicast maximum-paths ebgp <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp address-family ipv4-unicast maximum-paths ibgp	protocols bgp address-family ipv4-unicast maximum-paths ibgp <value>
×	<i>set protocols bgp address-family ipv4-unicast network <prefix> backdoor</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	protocols_bgp.xml.in: protocols bgp address-family ipv4-unicast network <network> backdoor	protocols bgp address-family ipv4-unicast network <text> backdoor
×	not yet documented	protocols_bgp.xml.in: protocols bgp address-family ipv4-unicast network <network> route-map	protocols bgp address-family ipv4-unicast network <text> route-map <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp address-family ipv4-unicast nexthop vpn export	Nothing found in VyOS
×	not yet documented	protocols_bgp.xml.in: protocols bgp address-family ipv4-unicast rd vpn export	protocols bgp address-family ipv4-unicast rd vpn export <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp address-family ipv4-unicast redistribute babel metric	protocols bgp address-family ipv4-unicast redistribute babel metric <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	protocols_bgp.xml.in: protocols bgp address-family ipv4-unicast redistribute babel route-map	protocols bgp address-family ipv4-unicast redistribute babel route-map <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp address-family ipv4-unicast redistribute connected metric	protocols bgp address-family ipv4-unicast redistribute connected metric <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp address-family ipv4-unicast redistribute connected route-map	protocols bgp address-family ipv4-unicast redistribute connected route-map <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp address-family ipv4-unicast redistribute isis metric	protocols bgp address-family ipv4-unicast redistribute isis metric <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp address-family ipv4-unicast redistribute isis route-map	protocols bgp address-family ipv4-unicast redistribute isis route-map <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp address-family ipv4-unicast redistribute kernel metric	protocols bgp address-family ipv4-unicast redistribute kernel metric <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp address-family ipv4-unicast redistribute kernel route-map	protocols bgp address-family ipv4-unicast redistribute kernel route-map <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp address-family ipv4-unicast redistribute ospf metric	protocols bgp address-family ipv4-unicast redistribute ospf metric <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp address-family ipv4-unicast redistribute ospf route-map	protocols bgp address-family ipv4-unicast redistribute ospf route-map <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp address-family ipv4-unicast redistribute rip metric	protocols bgp address-family ipv4-unicast redistribute rip metric <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp address-family ipv4-unicast redistribute rip route-map	protocols bgp address-family ipv4-unicast redistribute rip route-map <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	protocols_bgp.xml.in: protocols bgp address-family ipv4-unicast redistribute static metric	protocols bgp address-family ipv4-unicast redistribute static metric <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp address-family ipv4-unicast redistribute static route-map	protocols bgp address-family ipv4-unicast redistribute static route-map <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp address-family ipv4-unicast redistribute table	protocols bgp address-family ipv4-unicast redistribute table <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp address-family ipv4-unicast route-map vpn export	protocols bgp address-family ipv4-unicast route-map vpn export <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp address-family ipv4-unicast route-map vpn import	protocols bgp address-family ipv4-unicast route-map vpn import <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp address-family ipv4-unicast route-target vpn both	protocols bgp address-family ipv4-unicast route-target vpn both <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp address-family ipv4-unicast route-target vpn export	protocols bgp address-family ipv4-unicast route-target vpn export <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp address-family ipv4-unicast route-target vpn import	protocols bgp address-family ipv4-unicast route-target vpn import <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp address-family ipv4-unicast sid vpn export	Nothing found in VyOS
×	not yet documented	protocols_bgp.xml.in: protocols bgp address-family ipv4-vpn network <network> label	protocols bgp address-family ipv4-vpn network <text> label <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp address-family ipv4-vpn network <network> rd	protocols bgp address-family ipv4-vpn network <text> rd <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp address-family ipv6-flowspec local-install interface	protocols bgp address-family ipv6-flowspec local-install interface <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	protocols_bgp.xml.in: protocols bgp address-family ipv6-labeled-unicast aggregate-address <aggregate-address> as-set	protocols bgp address-family ipv6-labeled-unicast aggregate-address <text> as-set
×	not yet documented	protocols_bgp.xml.in: protocols bgp address-family ipv6-labeled-unicast aggregate-address <aggregate-address> route-map	protocols bgp address-family ipv6-labeled-unicast aggregate-address <text> route-map <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp address-family ipv6-labeled-unicast aggregate-address <aggregate-address> summary-only	protocols bgp address-family ipv6-labeled-unicast aggregate-address <text> summary-only
×	not yet documented	protocols_bgp.xml.in: protocols bgp address-family ipv6-labeled-unicast network <network> backdoor	protocols bgp address-family ipv6-labeled-unicast network <text> backdoor
×	not yet documented	protocols_bgp.xml.in: protocols bgp address-family ipv6-labeled-unicast network <network> route-map	protocols bgp address-family ipv6-labeled-unicast network <text> route-map <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp address-family ipv6-multicast aggregate-address <aggregate-address> as-set	protocols bgp address-family ipv6-multicast aggregate-address <text> as-set
×	not yet documented	protocols_bgp.xml.in: protocols bgp address-family ipv6-multicast aggregate-address <aggregate-address> route-map	protocols bgp address-family ipv6-multicast aggregate-address <text> route-map <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp address-family ipv6-multicast aggregate-address <aggregate-address> summary-only	protocols bgp address-family ipv6-multicast aggregate-address <text> summary-only

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	protocols_bgp.xml.in: protocols bgp address-family ipv6-multicast distance external <value>	protocols bgp address-family ipv6-multicast distance external <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp address-family ipv6-multicast distance internal <value>	protocols bgp address-family ipv6-multicast distance internal <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp address-family ipv6-multicast distance local <value>	protocols bgp address-family ipv6-multicast distance local <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp address-family ipv6-multicast distance prefix <prefix> distance <value>	protocols bgp address-family ipv6-multicast distance prefix <text> distance <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp address-family ipv6-multicast network <network> path-limit <value>	protocols bgp address-family ipv6-multicast network <text> path-limit <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp address-family ipv6-multicast network <network> route-map <value>	protocols bgp address-family ipv6-multicast network <text> route-map <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp address-family ipv6-unicast aggregate-address <aggregate-address> as-set	protocols bgp address-family ipv6-unicast aggregate-address <text> as-set
×	not yet documented	protocols_bgp.xml.in: protocols bgp address-family ipv6-unicast aggregate-address <aggregate-address> route-map <value>	protocols bgp address-family ipv6-unicast aggregate-address <text> route-map <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp address-family ipv6-unicast aggregate-address <aggregate-address> summary-only	protocols bgp address-family ipv6-unicast aggregate-address <text> summary-only
×	not yet documented	protocols_bgp.xml.in: protocols bgp address-family ipv6-unicast distance external <value>	protocols bgp address-family ipv6-unicast distance external <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	protocols_bgp.xml.in: protocols bgp address-family ipv6-unicast distance internal	protocols bgp address-family ipv6-unicast distance internal <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp address-family ipv6-unicast distance local	protocols bgp address-family ipv6-unicast distance local <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp address-family ipv6-unicast distance prefix <prefix> distance	protocols bgp address-family ipv6-unicast distance prefix <text> distance <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp address-family ipv6-unicast export vpn	protocols bgp address-family ipv6-unicast export vpn
×	not yet documented	protocols_bgp.xml.in: protocols bgp address-family ipv6-unicast import vpn	protocols bgp address-family ipv6-unicast import vpn
×	not yet documented	protocols_bgp.xml.in: protocols bgp address-family ipv6-unicast import vrf	protocols bgp address-family ipv6-unicast import vrf <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp address-family ipv6-unicast label vpn allocation-mode per-nexthop	protocols bgp address-family ipv6-unicast label vpn allocation-mode per-nexthop
×	not yet documented	protocols_bgp.xml.in: protocols bgp address-family ipv6-unicast label vpn export	protocols bgp address-family ipv6-unicast label vpn export <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp address-family ipv6-unicast maximum-paths ebgp	protocols bgp address-family ipv6-unicast maximum-paths ebgp <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp address-family ipv6-unicast maximum-paths ibgp	protocols bgp address-family ipv6-unicast maximum-paths ibgp <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp address-family ipv6-unicast network <network> path-limit	protocols bgp address-family ipv6-unicast network <text> path-limit <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	protocols_bgp.xml.in: protocols bgp address-family ipv6-unicast network <network> route-map	protocols bgp address-family ipv6-unicast network <text> route-map <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp address-family ipv6-unicast nexthop vpn export	Nothing found in VyOS
×	not yet documented	protocols_bgp.xml.in: protocols bgp address-family ipv6-unicast rd vpn export	protocols bgp address-family ipv6-unicast rd vpn export <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp address-family ipv6-unicast redistribute babel metric	protocols bgp address-family ipv6-unicast redistribute babel metric <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp address-family ipv6-unicast redistribute babel route-map	protocols bgp address-family ipv6-unicast redistribute babel route-map <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp address-family ipv6-unicast redistribute connected metric	protocols bgp address-family ipv6-unicast redistribute connected metric <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp address-family ipv6-unicast redistribute connected route-map	protocols bgp address-family ipv6-unicast redistribute connected route-map <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp address-family ipv6-unicast redistribute kernel metric	protocols bgp address-family ipv6-unicast redistribute kernel metric <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp address-family ipv6-unicast redistribute kernel route-map	protocols bgp address-family ipv6-unicast redistribute kernel route-map <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp address-family ipv6-unicast redistribute ospfv3 metric	protocols bgp address-family ipv6-unicast redistribute ospfv3 metric <value>

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Table 1 – continued from previous page

Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	protocols_bgp.xml.in: protocols bgp address-family ipv6-unicast redistribute ospfv3 route-map	protocols bgp address-family ipv6-unicast redistribute ospfv3 route-map <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp address-family ipv6-unicast redistribute ripng metric	protocols bgp address-family ipv6-unicast redistribute ripng metric <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp address-family ipv6-unicast redistribute ripng route-map	protocols bgp address-family ipv6-unicast redistribute ripng route-map <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp address-family ipv6-unicast redistribute static metric	protocols bgp address-family ipv6-unicast redistribute static metric <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp address-family ipv6-unicast redistribute static route-map	protocols bgp address-family ipv6-unicast redistribute static route-map <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp address-family ipv6-unicast redistribute table	protocols bgp address-family ipv6-unicast redistribute table <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp address-family ipv6-unicast route-map vpn export	protocols bgp address-family ipv6-unicast route-map vpn export <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp address-family ipv6-unicast route-map vpn import	protocols bgp address-family ipv6-unicast route-map vpn import <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp address-family ipv6-unicast route-target vpn both	protocols bgp address-family ipv6-unicast route-target vpn both <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp address-family ipv6-unicast route-target vpn export	protocols bgp address-family ipv6-unicast route-target vpn export <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp address-family ipv6-unicast route-target vpn import	protocols bgp address-family ipv6-unicast route-target vpn import <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	protocols_bgp.xml.in: protocols bgp address-family ipv6-unicast sid vpn export	Nothing found in VyOS
×	not yet documented	protocols_bgp.xml.in: protocols bgp address-family ipv6-vpn network <network> label	protocols bgp address-family ipv6-vpn network <text> label <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp address-family ipv6-vpn network <network> rd	protocols bgp address-family ipv6-vpn network <text> rd <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp address-family l2vpn-evpn advertise-all-vni	protocols bgp address-family l2vpn-evpn advertise-all-vni
×	not yet documented	protocols_bgp.xml.in: protocols bgp address-family l2vpn-evpn advertise-default-gw	protocols bgp address-family l2vpn-evpn advertise-default-gw
×	not yet documented	protocols_bgp.xml.in: protocols bgp address-family l2vpn-evpn advertise-pip	protocols bgp address-family l2vpn-evpn advertise-pip <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp address-family l2vpn-evpn advertise-svi-ip	protocols bgp address-family l2vpn-evpn advertise-svi-ip
×	not yet documented	protocols_bgp.xml.in: protocols bgp address-family l2vpn-evpn advertise ipv4 unicast route-map	protocols bgp address-family l2vpn-evpn advertise ipv4 unicast route-map <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp address-family l2vpn-evpn advertise ipv6 unicast route-map	protocols bgp address-family l2vpn-evpn advertise ipv6 unicast route-map <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp address-family l2vpn-evpn default-originate ipv4	Nothing found in VyOS
×	not yet documented	protocols_bgp.xml.in: protocols bgp address-family l2vpn-evpn default-originate ipv6	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	protocols_bgp.xml.in: protocols bgp address-family l2vpn-evpn disable-ead-evi-rx	Nothing found in VyOS
×	not yet documented	protocols_bgp.xml.in: protocols bgp address-family l2vpn-evpn disable-ead-evi-tx	Nothing found in VyOS
×	not yet documented	protocols_bgp.xml.in: protocols bgp address-family l2vpn-evpn ead-es-frag evi-limit	Nothing found in VyOS
×	not yet documented	protocols_bgp.xml.in: protocols bgp address-family l2vpn-evpn ead-es-route-target export	Nothing found in VyOS
×	not yet documented	protocols_bgp.xml.in: protocols bgp address-family l2vpn-evpn flooding disable	protocols bgp address-family l2vpn-evpn flooding disable
×	not yet documented	protocols_bgp.xml.in: protocols bgp address-family l2vpn-evpn flooding head-end-replication	protocols bgp address-family l2vpn-evpn flooding head-end-replication
×	not yet documented	protocols_bgp.xml.in: protocols bgp address-family l2vpn-evpn mac-vrf soo	Nothing found in VyOS
×	not yet documented	protocols_bgp.xml.in: protocols bgp address-family l2vpn-evpn rd <value>	protocols bgp address-family l2vpn-evpn rd <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp address-family l2vpn-evpn route-target both	protocols bgp address-family l2vpn-evpn route-target both <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp address-family l2vpn-evpn route-target export	protocols bgp address-family l2vpn-evpn route-target export <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp address-family l2vpn-evpn route-target import	protocols bgp address-family l2vpn-evpn route-target import <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp address-family l2vpn-evpn rt-auto-derive	protocols bgp address-family l2vpn-evpn rt-auto-derive

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	protocols_bgp.xml.in: protocols bgp address-family l2vpn-evpn vni <vni> advertise-default-gw	protocols bgp address-family l2vpn-evpn vni <text> advertise-default-gw
×	not yet documented	protocols_bgp.xml.in: protocols bgp address-family l2vpn-evpn vni <vni> advertise-svi-ip	protocols bgp address-family l2vpn-evpn vni <text> advertise-svi-ip
×	not yet documented	protocols_bgp.xml.in: protocols bgp address-family l2vpn-evpn vni <vni> rd	protocols bgp address-family l2vpn-evpn vni <text> rd <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp address-family l2vpn-evpn vni <vni> route-target both	protocols bgp address-family l2vpn-evpn vni <text> route-target both <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp address-family l2vpn-evpn vni <vni> route-target export	protocols bgp address-family l2vpn-evpn vni <text> route-target export <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp address-family l2vpn-evpn vni <vni> route-target import	protocols bgp address-family l2vpn-evpn vni <text> route-target import <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp bmp mirror-buffer-limit	Nothing found in VyOS
×	not yet documented	protocols_bgp.xml.in: protocols bgp bmp target <target> address	Nothing found in VyOS
×	not yet documented	protocols_bgp.xml.in: protocols bgp bmp target <target> max-retry	Nothing found in VyOS
×	not yet documented	protocols_bgp.xml.in: protocols bgp bmp target <target> min-retry	Nothing found in VyOS
×	not yet documented	protocols_bgp.xml.in: protocols bgp bmp target <target> mirror	Nothing found in VyOS
×	not yet documented	protocols_bgp.xml.in: protocols bgp bmp target <target> monitor ipv4-unicast post-policy	Nothing found in VyOS
×	not yet documented	protocols_bgp.xml.in: protocols bgp bmp target <target> monitor ipv4-unicast pre-policy	Nothing found in VyOS

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Table 1 – continued from previous page

Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	protocols_bgp.xml.in: protocols bgp bmp target <target> monitor ipv6-unicast post-policy	Nothing found in VyOS
×	not yet documented	protocols_bgp.xml.in: protocols bgp bmp target <target> monitor ipv6-unicast pre-policy	Nothing found in VyOS
×	not yet documented	protocols_bgp.xml.in: protocols bgp bmp target <target> port	Nothing found in VyOS
×	not yet documented	protocols_bgp.xml.in: protocols bgp interface <interface> mpls forwarding	protocols bgp interface <text> mpls forwarding
×	not yet documented	protocols_bgp.xml.in: protocols bgp listen limit	protocols bgp listen limit <value>
×	<i>set protocols bgp listen limit <number></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set protocols bgp listen range <prefix> peer-group <name></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	protocols_bgp.xml.in: protocols bgp listen range <range> peer-group	protocols bgp listen range <text> peer-group <value>
×	<i>set protocols bgp neighbor <address/interface> address-family <ipv4-unicast/ipv6-unallowas-in number <number></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set protocols bgp neighbor <address/interface> address-family <ipv4-unicast/ipv6-unas-override</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set protocols bgp neighbor <address/interface> address-family <ipv4-unicast/ipv6-unattribute-unchanged <as-path/med/next-hop</i>	Nothing found in XML Definitions	Nothing found in VyOS

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Table 1 – continued from previous page

Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<i>set protocols bgp neighbor <address/interface> address-family <ipv4-unicast/ipv6-un capability orf <receive/send></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set protocols bgp neighbor <address/interface> address-family <ipv4-unicast/ipv6-un default-originate [route-map <name>]</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set protocols bgp neighbor <address/interface> address-family <ipv4-unicast/ipv6-un distribute-list <export/import> <number></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set protocols bgp neighbor <address/interface> address-family <ipv4-unicast/ipv6-un filter-list <export/import> <name></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set protocols bgp neighbor <address/interface> address-family <ipv4-unicast/ipv6-un maximum-prefix <number></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set protocols bgp neighbor <address/interface> address-family <ipv4-unicast/ipv6-un nexthop-self</i>	Nothing found in XML Definitions	Nothing found in VyOS

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Table 1 – continued from previous page

Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<i>set protocols bgp neighbor <address/interface> address-family <ipv4-unicast/ipv6-un prefix-list <export/import> <name></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set protocols bgp neighbor <address/interface> address-family <ipv4-unicast/ipv6-un remove-private-as</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set protocols bgp neighbor <address/interface> address-family <ipv4-unicast/ipv6-un route-map <export/import> <name></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set protocols bgp neighbor <address> address-family <ipv4-unicast/ipv6-un route-reflector-clien</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set protocols bgp neighbor <address/interface> address-family <ipv4-unicast/ipv6-un soft-reconfiguration inbound</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set protocols bgp neighbor <address/interface> address-family <ipv4-unicast/ipv6-un unsuppress-map <name></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set protocols bgp neighbor <address/interface> address-family <ipv4-unicast/ipv6-un weight <number></i>	Nothing found in XML Definitions	Nothing found in VyOS

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Table 1 – continued from previous page

Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set protocols bgp neighbor <address interface> advertisement-interva <seconds></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set protocols bgp neighbor <neighbor> bfd</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set protocols bgp neighbor <address interface> capability dynamic</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set protocols bgp neighbor <address interface> capability extended-nexthop</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set protocols bgp neighbor <address interface> description <text></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set protocols bgp neighbor <address interface> disable-capability-ne</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set protocols bgp neighbor <address interface> disable-connected-che</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set protocols bgp neighbor <address interface> disable-send-communit <extended standard></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set protocols bgp neighbor <address interface> ebgp-multihop <number></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set protocols bgp neighbor <address interface> local-as <asn> [no-prepend] [replace-as]</code>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<i>set protocols bgp neighbor <address interface> local-role <role> [strict]</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set protocols bgp neighbor <address interface> override-capability</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set protocols bgp neighbor <address interface> passive</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set protocols bgp neighbor <address interface> password <text></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set protocols bgp neighbor <address interface> peer-group <name></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set protocols bgp neighbor <address interface> remote-as <asn></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set protocols bgp neighbor <address interface> remote-as auto</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set protocols bgp neighbor <address interface> remote-as external</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set protocols bgp neighbor <address interface> remote-as internal</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set protocols bgp neighbor <address interface> shutdown</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set protocols bgp neighbor <address interface> solo</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set protocols bgp neighbor <address interface> strict-capability-mat</i>	Nothing found in XML Definitions	Nothing found in VyOS

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Table 1 – continued from previous page

Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<i>set protocols bgp neighbor <address/interface> ttl-security hops <number></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set protocols bgp neighbor <address/interface> update-source <address/interface></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv4-flowspec filter-list export	protocols bgp neighbor <text> address-family ipv4-flowspec filter-list export <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv4-flowspec filter-list import	protocols bgp neighbor <text> address-family ipv4-flowspec filter-list import <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv4-flowspec prefix-list export	protocols bgp neighbor <text> address-family ipv4-flowspec prefix-list export <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv4-flowspec prefix-list import	protocols bgp neighbor <text> address-family ipv4-flowspec prefix-list import <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv4-flowspec route-map export	protocols bgp neighbor <text> address-family ipv4-flowspec route-map export <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv4-flowspec route-map import	protocols bgp neighbor <text> address-family ipv4-flowspec route-map import <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv4-flowspec route-reflector-client	protocols bgp neighbor <text> address-family ipv4-flowspec route- reflector-client

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv4-flowspec route-server-client	protocols bgp neighbor <text> address-family ipv4-flowspec route- server-client
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv4-flowspec soft-reconfiguration inbound	protocols bgp neighbor <text> address-family ipv4-flowspec soft- reconfiguration inbound
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv4-labeled- unicast addpath-tx-all	protocols bgp neighbor <text> address-family ipv4-labeled-unicast addpath-tx-all
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv4-labeled- unicast addpath-tx-per-as	protocols bgp neighbor <text> address-family ipv4-labeled-unicast addpath-tx-per-as
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv4-labeled- unicast allowas-in number	protocols bgp neighbor <text> address-family ipv4-labeled-unicast allowas-in number <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv4-labeled- unicast as-override	protocols bgp neighbor <text> address-family ipv4-labeled-unicast as-override
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv4-labeled- unicast attribute- unchanged as-path	protocols bgp neighbor <text> address-family ipv4-labeled-unicast attribute-unchanged as-path
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv4-labeled- unicast attribute- unchanged med	protocols bgp neighbor <text> address-family ipv4-labeled-unicast attribute-unchanged med
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv4-labeled- unicast attribute- unchanged next-hop	protocols bgp neighbor <text> address-family ipv4-labeled-unicast attribute-unchanged next-hop

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Table 1 – continued from previous page

Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address-family family ipv4-labeled-unicast capability orf prefix-list receive	protocols bgp neighbor <text> address-family ipv4-labeled-unicast ca- pability orf prefix-list receive
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address-family family ipv4-labeled-unicast capability orf prefix-list send	protocols bgp neighbor <text> address-family ipv4-labeled-unicast ca- pability orf prefix-list send
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address-family family ipv4-labeled-unicast conditionally-advertise advertise-map <value>	protocols bgp neighbor <text> address-family ipv4-labeled-unicast conditionally-advertise advertise-map <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address-family family ipv4-labeled-unicast conditionally-advertise exist-map	protocols bgp neighbor <text> address-family ipv4-labeled-unicast conditionally-advertise exist-map <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address-family family ipv4-labeled-unicast conditionally-advertise non-exist-map	protocols bgp neighbor <text> address-family ipv4-labeled-unicast conditionally-advertise non-exist-map <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address-family family ipv4-labeled-unicast default-originate route-map	protocols bgp neighbor <text> address-family ipv4-labeled-unicast default-originate route- map <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address-family family ipv4-labeled-unicast disable-send-community extended	protocols bgp neighbor <text> address-family ipv4-labeled-unicast disable-send-community extended
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address-family family ipv4-labeled-unicast disable-send-community standard	protocols bgp neighbor <text> address-family ipv4-labeled-unicast disable-send-community standard

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address-family family ipv4-labeled-unicast unicast distribute-list export	protocols bgp neighbor <text> address-family ipv4-labeled-unicast distribute-list export <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address-family family ipv4-labeled-unicast unicast distribute-list import	protocols bgp neighbor <text> address-family ipv4-labeled-unicast distribute-list import <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address-family family ipv4-labeled-unicast unicast filter-list export	protocols bgp neighbor <text> address-family ipv4-labeled-unicast filter-list export <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address-family family ipv4-labeled-unicast unicast filter-list import	protocols bgp neighbor <text> address-family ipv4-labeled-unicast filter-list import <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address-family family ipv4-labeled-unicast unicast maximum-prefix	protocols bgp neighbor <text> address-family ipv4-labeled-unicast maximum-prefix <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address-family family ipv4-labeled-unicast unicast maximum-prefix-out	protocols bgp neighbor <text> address-family ipv4-labeled-unicast maximum-prefix-out <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address-family family ipv4-labeled-unicast unicast nexthop-self force	protocols bgp neighbor <text> address-family ipv4-labeled-unicast nexthop-self force
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address-family family ipv4-labeled-unicast unicast prefix-list export	protocols bgp neighbor <text> address-family ipv4-labeled-unicast prefix-list export <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address-family family ipv4-labeled-unicast unicast prefix-list import	protocols bgp neighbor <text> address-family ipv4-labeled-unicast prefix-list import <value>

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Table 1 – continued from previous page

Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	Nothing found in XML Definitions	protocols bgp neighbor <text> address-family ipv4-labeled-unicast remove-private-as
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address-family ipv4-labeled-unicast remove-private-as all	Nothing found in VyOS
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address-family ipv4-labeled-unicast route-map export	protocols bgp neighbor <text> address-family ipv4-labeled-unicast route-map export <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address-family ipv4-labeled-unicast route-map import	protocols bgp neighbor <text> address-family ipv4-labeled-unicast route-map import <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address-family ipv4-labeled-unicast route-reflector-client	protocols bgp neighbor <text> address-family ipv4-labeled-unicast route-reflector-client
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address-family ipv4-labeled-unicast route-server-client	protocols bgp neighbor <text> address-family ipv4-labeled-unicast route-server-client
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address-family ipv4-labeled-unicast soft-reconfiguration inbound	protocols bgp neighbor <text> address-family ipv4-labeled-unicast soft-reconfiguration inbound
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address-family ipv4-labeled-unicast unsuppress-map	protocols bgp neighbor <text> address-family ipv4-labeled-unicast unsuppress-map <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address-family ipv4-labeled-unicast weight	protocols bgp neighbor <text> address-family ipv4-labeled-unicast weight <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv4-multicast addpath-tx-all	protocols bgp neighbor <text> address-family ipv4-multicast addpath- tx-all
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv4-multicast addpath-tx-per-as	protocols bgp neighbor <text> address-family ipv4-multicast addpath- tx-per-as
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv4-multicast allowas-in number	protocols bgp neighbor <text> address-family ipv4-multicast allowas-in number <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv4-multicast as-override	protocols bgp neighbor <text> address-family ipv4-multicast as-override
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv4-multicast attribute-unchanged as-path	protocols bgp neighbor <text> address-family ipv4-multicast attribute- unchanged as-path
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv4-multicast attribute-unchanged med	protocols bgp neighbor <text> address-family ipv4-multicast attribute- unchanged med
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv4-multicast attribute-unchanged next-hop	protocols bgp neighbor <text> address-family ipv4-multicast attribute- unchanged next-hop
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv4-multicast capability orf prefix-list receive	protocols bgp neighbor <text> address-family ipv4-multicast capability orf prefix-list receive
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv4-multicast capability orf prefix-list send	protocols bgp neighbor <text> address-family ipv4-multicast capability orf prefix-list send

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv4-multicast conditionally-advertise advertise-map	protocols bgp neigh- bor <text> address- family ipv4-multicast conditionally-advertise advertise-map <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv4-multicast conditionally-advertise exist-map	protocols bgp neigh- bor <text> address- family ipv4-multicast conditionally-advertise exist-map <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv4-multicast conditionally-advertise non-exist-map	protocols bgp neigh- bor <text> address- family ipv4-multicast conditionally-advertise non-exist-map <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv4-multicast default-originate route- map	protocols bgp neighbor <text> address-family ipv4-multicast default- originate route-map <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv4-multicast disable-send-community extended	protocols bgp neighbor <text> address-family ipv4-multicast disable- send-community extended
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv4-multicast disable-send-community standard	protocols bgp neighbor <text> address-family ipv4-multicast disable- send-community standard
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv4-multicast distribute-list export	protocols bgp neighbor <text> address-family ipv4-multicast distribute- list export <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv4-multicast distribute-list import	protocols bgp neighbor <text> address-family ipv4-multicast distribute- list import <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv4-multicast filter-list export	protocols bgp neighbor <text> address-family ipv4-multicast filter-list export <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv4-multicast filter-list import	protocols bgp neighbor <text> address-family ipv4-multicast filter-list import <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv4-multicast maximum-prefix	protocols bgp neighbor <text> address-family ipv4-multicast maximum- prefix <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv4-multicast maximum-prefix-out	protocols bgp neighbor <text> address-family ipv4-multicast maximum- prefix-out <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv4-multicast nexthop-self force	protocols bgp neighbor <text> address-family ipv4-multicast nexthop- self force
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv4-multicast prefix-list export	protocols bgp neighbor <text> address-family ipv4-multicast prefix-list export <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv4-multicast prefix-list import	protocols bgp neighbor <text> address-family ipv4-multicast prefix-list import <value>
×	not yet documented	Nothing found in XML Definitions	protocols bgp neighbor <text> address-family ipv4-multicast remove- private-as
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv4-multicast remove-private-as all	Nothing found in VyOS
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv4-multicast route-map export	protocols bgp neighbor <text> address-family ipv4-multicast route-map export <value>

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Table 1 – continued from previous page

Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv4-multicast route-map import	protocols bgp neighbor <text> address-family ipv4-multicast route-map import <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv4-multicast route-reflector-client	protocols bgp neighbor <text> address-family ipv4-multicast route- reflector-client
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv4-multicast route-server-client	protocols bgp neighbor <text> address-family ipv4-multicast route- server-client
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv4-multicast soft-reconfiguration inbound	protocols bgp neighbor <text> address-family ipv4-multicast soft- reconfiguration inbound
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv4-multicast unsuppress-map	protocols bgp neigh- bor <text> address- family ipv4-multicast unsuppress-map <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv4-multicast weight	protocols bgp neighbor <text> address-family ipv4-multicast weight <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv4-unicast addpath-tx-all	protocols bgp neighbor <text> address-family ipv4-unicast addpath-tx- all
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv4-unicast addpath-tx-per-as	protocols bgp neighbor <text> address-family ipv4-unicast addpath-tx- per-as
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv4-unicast allowas-in number	protocols bgp neighbor <text> address-family ipv4-unicast allowas-in number <value>

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Table 1 – continued from previous page

Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv4-unicast as-override	protocols bgp neighbor <text> address-family ipv4-unicast as-override
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv4-unicast attribute-unchanged as-path	protocols bgp neighbor <text> address-family ipv4-unicast attribute- unchanged as-path
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv4-unicast attribute-unchanged med	protocols bgp neighbor <text> address-family ipv4-unicast attribute- unchanged med
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv4-unicast attribute-unchanged next-hop	protocols bgp neighbor <text> address-family ipv4-unicast attribute- unchanged next-hop
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv4-unicast capability orf prefix-list receive	protocols bgp neighbor <text> address-family ipv4-unicast capability orf prefix-list receive
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv4-unicast capability orf prefix-list send	protocols bgp neighbor <text> address-family ipv4-unicast capability orf prefix-list send
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv4-unicast conditionally-advertise advertise-map	protocols bgp neigh- bor <text> address- family ipv4-unicast conditionally-advertise advertise-map <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv4-unicast conditionally-advertise exist-map	protocols bgp neigh- bor <text> address- family ipv4-unicast conditionally-advertise exist-map <value>

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Table 1 – continued from previous page

Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv4-unicast conditionally-advertise non-exist-map <value>	protocols bgp neigh- bor <text> address- family ipv4-unicast conditionally-advertise non-exist-map <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv4-unicast default-originate route- map	protocols bgp neighbor <text> address-family ipv4-unicast default- originate route-map <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv4-unicast disable-send-community extended	protocols bgp neighbor <text> address-family ipv4-unicast disable-send- community extended
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv4-unicast disable-send-community standard	protocols bgp neighbor <text> address-family ipv4-unicast disable-send- community standard
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv4-unicast distribute-list export	protocols bgp neighbor <text> address-family ipv4-unicast distribute-list export <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv4-unicast distribute-list import	protocols bgp neighbor <text> address-family ipv4-unicast distribute-list import <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv4-unicast filter-list export	protocols bgp neighbor <text> address-family ipv4-unicast filter-list export <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv4-unicast filter-list import	protocols bgp neighbor <text> address-family ipv4-unicast filter-list import <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv4-unicast maximum-prefix	protocols bgp neighbor <text> address-family ipv4-unicast maximum- prefix <value>

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Table 1 – continued from previous page

Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv4-unicast maximum-prefix-out	protocols bgp neighbor <text> address-family ipv4-unicast maximum- prefix-out <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv4-unicast nexthop-self force	protocols bgp neighbor <text> address-family ipv4-unicast nexthop-self force
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv4-unicast prefix-list export	protocols bgp neighbor <text> address-family ipv4-unicast prefix-list export <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv4-unicast prefix-list import	protocols bgp neighbor <text> address-family ipv4-unicast prefix-list import <value>
×	not yet documented	Nothing found in XML Definitions	protocols bgp neighbor <text> address-family ipv4-unicast remove- private-as
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv4-unicast remove-private-as all	Nothing found in VyOS
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv4-unicast route-map export	protocols bgp neighbor <text> address-family ipv4-unicast route-map export <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv4-unicast route-map import	protocols bgp neighbor <text> address-family ipv4-unicast route-map import <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv4-unicast route-reflector-client	protocols bgp neighbor <text> address-family ipv4-unicast route- reflector-client
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv4-unicast route-server-client	protocols bgp neighbor <text> address-family ipv4-unicast route-server- client

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv4-unicast soft-reconfiguration inbound	protocols bgp neigh- bor <text> address- family ipv4-unicast soft-reconfiguration inbound
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv4-unicast unsuppress-map	protocols bgp neighbor <text> address-family ipv4-unicast unsuppress- map <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv4-unicast weight	protocols bgp neighbor <text> address-family ipv4-unicast weight <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv4-vpn addpath- tx-all	protocols bgp neighbor <text> address-family ipv4-vpn addpath-tx-all
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv4-vpn addpath- tx-per-as	protocols bgp neighbor <text> address-family ipv4-vpn addpath-tx-per- as
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv4-vpn allowas- in number	protocols bgp neighbor <text> address-family ipv4-vpn allowas-in number <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv4-vpn as- override	protocols bgp neighbor <text> address-family ipv4-vpn as-override
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv4-vpn attribute- unchanged as-path	protocols bgp neighbor <text> address-family ipv4-vpn attribute- unchanged as-path
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv4-vpn attribute- unchanged med	protocols bgp neighbor <text> address-family ipv4-vpn attribute- unchanged med

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv4-vpn attribute- unchanged next-hop	protocols bgp neighbor <text> address-family ipv4-vpn attribute- unchanged next-hop
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv4-vpn conditionally-advertise advertise-map	protocols bgp neighbor <text> address-family ipv4-vpn conditionally- advertise advertise-map <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv4-vpn conditionally-advertise exist-map	protocols bgp neighbor <text> address-family ipv4-vpn conditionally- advertise exist-map <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv4-vpn conditionally-advertise non-exist-map	protocols bgp neighbor <text> address-family ipv4-vpn conditionally- advertise non-exist-map <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv4-vpn disable- send-community extended	protocols bgp neighbor <text> address-family ipv4-vpn disable-send- community extended
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv4-vpn disable- send-community standard	protocols bgp neighbor <text> address-family ipv4-vpn disable-send- community standard
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv4-vpn distribute-list export	protocols bgp neighbor <text> address-family ipv4-vpn distribute-list export <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv4-vpn distribute-list import	protocols bgp neighbor <text> address-family ipv4-vpn distribute-list import <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv4-vpn filter-list export	protocols bgp neighbor <text> address-family ipv4-vpn filter-list export <value>

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Table 1 – continued from previous page

Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv4-vpn filter-list import	protocols bgp neighbor <text> address-family ipv4-vpn filter-list import <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv4-vpn maximum-prefix	protocols bgp neighbor <text> address-family ipv4-vpn maximum-prefix <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv4-vpn maximum-prefix-out	protocols bgp neighbor <text> address-family ipv4-vpn maximum- prefix-out <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv4-vpn nexthop- self force	protocols bgp neighbor <text> address-family ipv4-vpn nexthop-self force
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv4-vpn prefix-list export	protocols bgp neighbor <text> address-family ipv4-vpn prefix-list export <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv4-vpn prefix-list import	protocols bgp neighbor <text> address-family ipv4-vpn prefix-list import <value>
×	not yet documented	Nothing found in XML Definitions	protocols bgp neighbor <text> address-family ipv4-vpn remove-private- as
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv4-vpn remove- private-as all	Nothing found in VyOS
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv4-vpn route- map export	protocols bgp neighbor <text> address-family ipv4-vpn route-map export <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv4-vpn route- map import	protocols bgp neighbor <text> address-family ipv4-vpn route-map import <value>

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Table 1 – continued from previous page

Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv4-vpn route- reflector-client	protocols bgp neighbor <text> address-family ipv4-vpn route-reflector- client
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv4-vpn route- server-client	protocols bgp neighbor <text> address-family ipv4-vpn route-server- client
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv4-vpn soft- reconfiguration inbound	protocols bgp neigh- bor <text> address- family ipv4-vpn soft- reconfiguration inbound
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv4-vpn unsuppress-map	protocols bgp neighbor <text> address-family ipv4-vpn unsuppress-map <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv4-vpn weight	protocols bgp neighbor <text> address-family ipv4-vpn weight <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv6-flowspec filter-list export	protocols bgp neighbor <text> address-family ipv6-flowspec filter-list export <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv6-flowspec filter-list import	protocols bgp neighbor <text> address-family ipv6-flowspec filter-list import <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv6-flowspec prefix-list export	protocols bgp neighbor <text> address-family ipv6-flowspec prefix-list export <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv6-flowspec prefix-list import	protocols bgp neighbor <text> address-family ipv6-flowspec prefix-list import <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv6-flowspec route-map export	protocols bgp neighbor <text> address-family ipv6-flowspec route-map export <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv6-flowspec route-map import	protocols bgp neighbor <text> address-family ipv6-flowspec route-map import <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv6-flowspec route-reflector-client	protocols bgp neighbor <text> address-family ipv6-flowspec route- reflector-client
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv6-flowspec route-server-client	protocols bgp neighbor <text> address-family ipv6-flowspec route- server-client
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv6-flowspec soft-reconfiguration inbound	protocols bgp neighbor <text> address-family ipv6-flowspec soft- reconfiguration inbound
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv6-labeled- unicast addpath-tx-all	protocols bgp neighbor <text> address-family ipv6-labeled-unicast addpath-tx-all
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv6-labeled- unicast addpath-tx-per-as	protocols bgp neighbor <text> address-family ipv6-labeled-unicast addpath-tx-per-as
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv6-labeled- unicast allowas-in number	protocols bgp neighbor <text> address-family ipv6-labeled-unicast allowas-in number <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv6-labeled- unicast as-override	protocols bgp neighbor <text> address-family ipv6-labeled-unicast as-override
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv6-labeled- unicast attribute- unchanged as-path	protocols bgp neighbor <text> address-family ipv6-labeled-unicast attribute-unchanged as-path

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Table 1 – continued from previous page

Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv6-labeled- unicast attribute- unchanged med	protocols bgp neighbor <text> address-family ipv6-labeled-unicast attribute-unchanged med
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv6-labeled- unicast attribute- unchanged next-hop	protocols bgp neighbor <text> address-family ipv6-labeled-unicast attribute-unchanged next-hop
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv6-labeled- unicast capability orf prefix-list receive	protocols bgp neighbor <text> address-family ipv6-labeled-unicast ca- pability orf prefix-list receive
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv6-labeled- unicast capability orf prefix-list send	protocols bgp neighbor <text> address-family ipv6-labeled-unicast ca- pability orf prefix-list send
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv6-labeled- unicast conditionally- advertise advertise-map	protocols bgp neighbor <text> address-family ipv6-labeled-unicast conditionally-advertise advertise-map <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv6-labeled- unicast conditionally- advertise exist-map	protocols bgp neighbor <text> address-family ipv6-labeled-unicast conditionally-advertise exist-map <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv6-labeled- unicast conditionally- advertise non-exist-map	protocols bgp neighbor <text> address-family ipv6-labeled-unicast conditionally-advertise non-exist-map <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv6-labeled- unicast default-originate route-map	protocols bgp neighbor <text> address-family ipv6-labeled-unicast default-originate route- map <value>

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Table 1 – continued from previous page

Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address-family family ipv6-labeled-unicast disable-send-community community extended	protocols bgp neighbor <text> address-family ipv6-labeled-unicast disable-send-community extended
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address-family family ipv6-labeled-unicast disable-send-community community standard	protocols bgp neighbor <text> address-family ipv6-labeled-unicast disable-send-community standard
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address-family family ipv6-labeled-unicast distribute-list export	protocols bgp neighbor <text> address-family ipv6-labeled-unicast distribute-list export <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address-family family ipv6-labeled-unicast distribute-list import	protocols bgp neighbor <text> address-family ipv6-labeled-unicast distribute-list import <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address-family family ipv6-labeled-unicast filter-list export	protocols bgp neighbor <text> address-family ipv6-labeled-unicast filter-list export <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address-family family ipv6-labeled-unicast filter-list import	protocols bgp neighbor <text> address-family ipv6-labeled-unicast filter-list import <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address-family family ipv6-labeled-unicast maximum-prefix	protocols bgp neighbor <text> address-family ipv6-labeled-unicast maximum-prefix <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address-family family ipv6-labeled-unicast maximum-prefix-out	protocols bgp neighbor <text> address-family ipv6-labeled-unicast maximum-prefix-out <value>

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Table 1 – continued from previous page

Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address-family family ipv6-labeled-unicast nextthop-local unchanged	protocols bgp neighbor <text> address-family ipv6-labeled-unicast nextthop-local unchanged
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address-family family ipv6-labeled-unicast nextthop-self force	protocols bgp neighbor <text> address-family ipv6-labeled-unicast nextthop-self force
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address-family family ipv6-labeled-unicast prefix-list export	protocols bgp neighbor <text> address-family ipv6-labeled-unicast prefix-list export <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address-family family ipv6-labeled-unicast prefix-list import	protocols bgp neighbor <text> address-family ipv6-labeled-unicast prefix-list import <value>
×	not yet documented	Nothing found in XML Definitions	protocols bgp neighbor <text> address-family ipv6-labeled-unicast remove-private-as
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address-family family ipv6-labeled-unicast remove-private-as all	Nothing found in VyOS
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address-family family ipv6-labeled-unicast route-map export	protocols bgp neighbor <text> address-family ipv6-labeled-unicast route-map export <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address-family family ipv6-labeled-unicast route-map import	protocols bgp neighbor <text> address-family ipv6-labeled-unicast route-map import <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address-family family ipv6-labeled-unicast route-reflector-client	protocols bgp neighbor <text> address-family ipv6-labeled-unicast route-reflector-client

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Table 1 – continued from previous page

Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv6-labeled- unicast route-server-client	protocols bgp neighbor <text> address-family ipv6-labeled-unicast route-server-client
×	not yet documented	protocols_bgp.xml.in: protocols bgp neigh- bor <neighbor> address-family ipv6- labeled-unicast soft- reconfiguration inbound	protocols bgp neighbor <text> address-family ipv6-labeled-unicast soft- reconfiguration inbound
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv6-labeled- unicast unsuppress-map	protocols bgp neighbor <text> address-family ipv6-labeled-unicast unsuppress-map <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv6-labeled- unicast weight	protocols bgp neighbor <text> address-family ipv6-labeled-unicast weight <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv6-multicast addpath-tx-all	protocols bgp neighbor <text> address-family ipv6-multicast addpath- tx-all
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv6-multicast addpath-tx-per-as	protocols bgp neighbor <text> address-family ipv6-multicast addpath- tx-per-as
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv6-multicast allowas-in number	protocols bgp neighbor <text> address-family ipv6-multicast allowas-in number <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv6-multicast as-override	protocols bgp neighbor <text> address-family ipv6-multicast as-override
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv6-multicast attribute-unchanged as-path	protocols bgp neighbor <text> address-family ipv6-multicast attribute- unchanged as-path

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv6-multicast attribute-unchanged med	protocols bgp neighbor <text> address-family ipv6-multicast attribute- unchanged med
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv6-multicast attribute-unchanged next-hop	protocols bgp neighbor <text> address-family ipv6-multicast attribute- unchanged next-hop
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv6-multicast conditionally-advertise advertise-map	protocols bgp neigh- bor <text> address- family ipv6-multicast conditionally-advertise advertise-map <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv6-multicast conditionally-advertise exist-map	protocols bgp neigh- bor <text> address- family ipv6-multicast conditionally-advertise exist-map <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv6-multicast conditionally-advertise non-exist-map	protocols bgp neigh- bor <text> address- family ipv6-multicast conditionally-advertise non-exist-map <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv6-multicast default-originate route- map	protocols bgp neighbor <text> address-family ipv6-multicast default- originate route-map <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv6-multicast disable-send-community extended	protocols bgp neighbor <text> address-family ipv6-multicast disable- send-community extended
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv6-multicast disable-send-community standard	protocols bgp neighbor <text> address-family ipv6-multicast disable- send-community standard

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Table 1 – continued from previous page

Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address-family ipv6-multicast distribute-list export <value>	protocols bgp neighbor <text> address-family ipv6-multicast distribute-list export <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address-family ipv6-multicast distribute-list import <value>	protocols bgp neighbor <text> address-family ipv6-multicast distribute-list import <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address-family ipv6-multicast filter-list export <value>	protocols bgp neighbor <text> address-family ipv6-multicast filter-list export <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address-family ipv6-multicast filter-list import <value>	protocols bgp neighbor <text> address-family ipv6-multicast filter-list import <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address-family ipv6-multicast maximum-prefix <value>	protocols bgp neighbor <text> address-family ipv6-multicast maximum-prefix <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address-family ipv6-multicast maximum-prefix-out <value>	protocols bgp neighbor <text> address-family ipv6-multicast maximum-prefix-out <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address-family ipv6-multicast nexthop-local unchanged	protocols bgp neighbor <text> address-family ipv6-multicast nexthop-local unchanged
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address-family ipv6-multicast nexthop-self force	protocols bgp neighbor <text> address-family ipv6-multicast nexthop-self force
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address-family ipv6-multicast prefix-list export <value>	protocols bgp neighbor <text> address-family ipv6-multicast prefix-list export <value>

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Table 1 – continued from previous page

Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv6-multicast prefix-list import	protocols bgp neighbor <text> address-family ipv6-multicast prefix-list import <value>
×	not yet documented	Nothing found in XML Definitions	protocols bgp neighbor <text> address-family ipv6-multicast remove- private-as
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv6-multicast remove-private-as all	Nothing found in VyOS
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv6-multicast route-map export	protocols bgp neighbor <text> address-family ipv6-multicast route-map export <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv6-multicast route-map import	protocols bgp neighbor <text> address-family ipv6-multicast route-map import <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv6-multicast route-reflector-client	protocols bgp neighbor <text> address-family ipv6-multicast route- reflector-client
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv6-multicast route-server-client	protocols bgp neighbor <text> address-family ipv6-multicast route- server-client
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv6-multicast soft-reconfiguration inbound	protocols bgp neighbor <text> address-family ipv6-multicast soft- reconfiguration inbound
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv6-multicast unsuppress-map	protocols bgp neigh- bor <text> address- family ipv6-multicast unsuppress-map <value>

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Table 1 – continued from previous page

Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv6-multicast weight	protocols bgp neighbor <text> address-family ipv6-multicast weight <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv6-unicast addpath-tx-all	protocols bgp neighbor <text> address-family ipv6-unicast addpath-tx- all
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv6-unicast addpath-tx-per-as	protocols bgp neighbor <text> address-family ipv6-unicast addpath-tx- per-as
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv6-unicast allowas-in number	protocols bgp neighbor <text> address-family ipv6-unicast allowas-in number <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv6-unicast as-override	protocols bgp neighbor <text> address-family ipv6-unicast as-override
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv6-unicast attribute-unchanged as-path	protocols bgp neighbor <text> address-family ipv6-unicast attribute- unchanged as-path
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv6-unicast attribute-unchanged med	protocols bgp neighbor <text> address-family ipv6-unicast attribute- unchanged med
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv6-unicast attribute-unchanged next-hop	protocols bgp neighbor <text> address-family ipv6-unicast attribute- unchanged next-hop
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv6-unicast capability orf prefix-list receive	protocols bgp neighbor <text> address-family ipv6-unicast capability orf prefix-list receive

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv6-unicast capability orf prefix-list send	protocols bgp neighbor <text> address-family ipv6-unicast capability orf prefix-list send
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv6-unicast conditionally-advertise advertise-map <value>	protocols bgp neigh- bor <text> address- family ipv6-unicast conditionally-advertise advertise-map <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv6-unicast conditionally-advertise exist-map <value>	protocols bgp neigh- bor <text> address- family ipv6-unicast conditionally-advertise exist-map <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv6-unicast conditionally-advertise non-exist-map <value>	protocols bgp neigh- bor <text> address- family ipv6-unicast conditionally-advertise non-exist-map <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv6-unicast default-originate route- map	protocols bgp neighbor <text> address-family ipv6-unicast default- originate route-map <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv6-unicast disable-send-community extended	protocols bgp neighbor <text> address-family ipv6-unicast disable-send- community extended
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv6-unicast disable-send-community standard	protocols bgp neighbor <text> address-family ipv6-unicast disable-send- community standard
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv6-unicast distribute-list export	protocols bgp neighbor <text> address-family ipv6-unicast distribute-list export <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address-family family ipv6-unicast distribute-list import	protocols bgp neighbor <text> address-family ipv6-unicast distribute-list import <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address-family family ipv6-unicast filter-list export	protocols bgp neighbor <text> address-family ipv6-unicast filter-list export <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address-family family ipv6-unicast filter-list import	protocols bgp neighbor <text> address-family ipv6-unicast filter-list import <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address-family family ipv6-unicast maximum-prefix	protocols bgp neighbor <text> address-family ipv6-unicast maximum- prefix <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address-family family ipv6-unicast maximum-prefix-out	protocols bgp neighbor <text> address-family ipv6-unicast maximum- prefix-out <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address-family family ipv6-unicast nexthop-local unchanged	protocols bgp neighbor <text> address-family ipv6-unicast nexthop- local unchanged
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address-family family ipv6-unicast nexthop-self force	protocols bgp neighbor <text> address-family ipv6-unicast nexthop-self force
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address-family family ipv6-unicast prefix-list export	protocols bgp neighbor <text> address-family ipv6-unicast prefix-list export <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address-family family ipv6-unicast prefix-list import	protocols bgp neighbor <text> address-family ipv6-unicast prefix-list import <value>
×	not yet documented	Nothing found in XML Definitions	protocols bgp neighbor <text> address-family ipv6-unicast remove- private-as

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv6-unicast remove-private-as all	Nothing found in VyOS
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv6-unicast route-map export	protocols bgp neighbor <text> address-family ipv6-unicast route-map export <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv6-unicast route-map import	protocols bgp neighbor <text> address-family ipv6-unicast route-map import <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv6-unicast route-reflector-client	protocols bgp neighbor <text> address-family ipv6-unicast route- reflector-client
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv6-unicast route-server-client	protocols bgp neighbor <text> address-family ipv6-unicast route-server- client
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv6-unicast soft-reconfiguration inbound	protocols bgp neigh- bor <text> address- family ipv6-unicast soft-reconfiguration inbound
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv6-unicast unsuppress-map	protocols bgp neighbor <text> address-family ipv6-unicast unsuppress- map <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv6-unicast weight	protocols bgp neighbor <text> address-family ipv6-unicast weight <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv6-vpn addpath- tx-all	protocols bgp neighbor <text> address-family ipv6-vpn addpath-tx-all

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv6-vpn addpath- tx-per-as	protocols bgp neighbor <text> address-family ipv6-vpn addpath-tx-per- as
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv6-vpn allowas- in number	protocols bgp neighbor <text> address-family ipv6-vpn allowas-in number <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv6-vpn as- override	protocols bgp neighbor <text> address-family ipv6-vpn as-override
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv6-vpn attribute- unchanged as-path	protocols bgp neighbor <text> address-family ipv6-vpn attribute- unchanged as-path
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv6-vpn attribute- unchanged med	protocols bgp neighbor <text> address-family ipv6-vpn attribute- unchanged med
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv6-vpn attribute- unchanged next-hop	protocols bgp neighbor <text> address-family ipv6-vpn attribute- unchanged next-hop
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv6-vpn conditionally-advertise advertise-map	protocols bgp neighbor <text> address-family ipv6-vpn conditionally- advertise advertise-map <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv6-vpn conditionally-advertise exist-map	protocols bgp neighbor <text> address-family ipv6-vpn conditionally- advertise exist-map <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv6-vpn conditionally-advertise non-exist-map	protocols bgp neighbor <text> address-family ipv6-vpn conditionally- advertise non-exist-map <value>

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Table 1 – continued from previous page

Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv6-vpn disable- send-community extended	protocols bgp neighbor <text> address-family ipv6-vpn disable-send- community extended
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv6-vpn disable- send-community standard	protocols bgp neighbor <text> address-family ipv6-vpn disable-send- community standard
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv6-vpn distribute-list export	protocols bgp neighbor <text> address-family ipv6-vpn distribute-list export <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv6-vpn distribute-list import	protocols bgp neighbor <text> address-family ipv6-vpn distribute-list import <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv6-vpn filter-list export	protocols bgp neighbor <text> address-family ipv6-vpn filter-list export <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv6-vpn filter-list import	protocols bgp neighbor <text> address-family ipv6-vpn filter-list import <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv6-vpn maximum-prefix	protocols bgp neighbor <text> address-family ipv6-vpn maximum-prefix <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv6-vpn maximum-prefix-out	protocols bgp neighbor <text> address-family ipv6-vpn maximum- prefix-out <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv6-vpn nexthop- local unchanged	protocols bgp neighbor <text> address-family ipv6-vpn nexthop-local unchanged

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv6-vpn nexthop- self force	protocols bgp neighbor <text> address-family ipv6-vpn nexthop-self force
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv6-vpn prefix-list export	protocols bgp neighbor <text> address-family ipv6-vpn prefix-list export <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv6-vpn prefix-list import	protocols bgp neighbor <text> address-family ipv6-vpn prefix-list import <value>
×	not yet documented	Nothing found in XML Definitions	protocols bgp neighbor <text> address-family ipv6-vpn remove-private- as
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv6-vpn remove- private-as all	Nothing found in VyOS
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv6-vpn route- map export	protocols bgp neighbor <text> address-family ipv6-vpn route-map export <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv6-vpn route- map import	protocols bgp neighbor <text> address-family ipv6-vpn route-map import <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv6-vpn route- reflector-client	protocols bgp neighbor <text> address-family ipv6-vpn route-reflector- client
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv6-vpn route- server-client	protocols bgp neighbor <text> address-family ipv6-vpn route-server- client
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv6-vpn soft- reconfiguration inbound	protocols bgp neigh- bor <text> address- family ipv6-vpn soft- reconfiguration inbound

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv6-vpn unsuppress-map	protocols bgp neighbor <text> address-family ipv6-vpn unsuppress-map <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family ipv6-vpn weight	protocols bgp neighbor <text> address-family ipv6-vpn weight <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family l2vpn-evpn allowas-in number	protocols bgp neighbor <text> address-family l2vpn-evpn allowas-in number <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family l2vpn-evpn attribute-unchanged as-path	protocols bgp neighbor <text> address-family l2vpn-evpn attribute- unchanged as-path
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family l2vpn-evpn attribute-unchanged med	protocols bgp neighbor <text> address-family l2vpn-evpn attribute- unchanged med
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family l2vpn-evpn attribute-unchanged next-hop	protocols bgp neighbor <text> address-family l2vpn-evpn attribute- unchanged next-hop
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family l2vpn-evpn nexthop-self force	protocols bgp neighbor <text> address-family l2vpn-evpn nexthop-self force
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family l2vpn-evpn route- map export	protocols bgp neighbor <text> address-family l2vpn-evpn route-map export <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family l2vpn-evpn route- map import	protocols bgp neighbor <text> address-family l2vpn-evpn route-map import <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family l2vpn-evpn route- reflector-client	protocols bgp neighbor <text> address-family l2vpn-evpn route- reflector-client
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family l2vpn-evpn route- server-client	protocols bgp neighbor <text> address-family l2vpn-evpn route-server- client
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> address- family l2vpn-evpn soft- reconfiguration inbound	protocols bgp neigh- bor <text> address- family l2vpn-evpn soft- reconfiguration inbound
×	not yet documented	protocols_bgp.xml.in: protocols bgp neigh- bor <neighbor> advertisement-interval	protocols bgp neighbor <text> advertisement- interval <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> bfd check- control-plane-failure	protocols bgp neighbor <text> bfd check-control- plane-failure
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> bfd profile	protocols bgp neighbor <text> bfd profile <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> capability dynamic	protocols bgp neighbor <text> capability dynamic
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> capability extended-nexthop	protocols bgp neigh- bor <text> capability extended-nexthop
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> capability software-version	Nothing found in VyOS
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> description	protocols bgp neigh- bor <text> description <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> disable- capability-negotiation	protocols bgp neighbor <text> disable-capability- negotiation
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> disable- connected-check	protocols bgp neighbor <text> disable-connected- check

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> ebgp-multihop	protocols bgp neighbor <text> ebgp-multihop <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> enforce-first-as	protocols bgp neighbor <text> enforce-first-as
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> graceful-restart	protocols bgp neighbor <text> graceful-restart <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> interface peer-group	protocols bgp neighbor <text> interface peer-group <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> interface remote-as	protocols bgp neighbor <text> interface remote-as <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> interface source-interface	protocols bgp neighbor <text> interface source-interface <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> interface v6only peer-group	protocols bgp neighbor <text> interface v6only peer-group <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> interface v6only remote-as	protocols bgp neighbor <text> interface v6only remote-as <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> local-as <local-as> no-prepend replace-as	protocols bgp neighbor <text> local-as <text> no-prepend replace-as
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> local-role <local-role> strict	protocols bgp neighbor <text> local-role <text> strict
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> override-capability	protocols bgp neighbor <text> override-capability
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> passive	protocols bgp neighbor <text> passive

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> password	protocols bgp neighbor <text> password <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> path-attribute discard	protocols bgp neighbor <text> path-attribute discard <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> path-attribute treat-as-withdraw	Nothing found in VyOS
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> peer-group	protocols bgp neigh- bor <text> peer-group <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> port	protocols bgp neighbor <text> port <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> remote-as	protocols bgp neighbor <text> remote-as <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> shutdown	protocols bgp neighbor <text> shutdown
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> solo	protocols bgp neighbor <text> solo
×	not yet documented	protocols_bgp.xml.in: protocols bgp neigh- bor <neighbor> strict- capability-match	protocols bgp neighbor <text> strict-capability- match
×	not yet documented	protocols_bgp.xml.in: protocols bgp neigh- bor <neighbor> timers connect	protocols bgp neighbor <text> timers connect <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp neigh- bor <neighbor> timers holdtime	protocols bgp neighbor <text> timers holdtime <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp neigh- bor <neighbor> timers keepalive	protocols bgp neighbor <text> timers keepalive <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> ttl-security hops	protocols bgp neighbor <text> ttl-security hops <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp neighbor <neighbor> update-source	protocols bgp neighbor <text> update-source <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<i>set protocols bgp parameters allow-martian-nexthop</i>	protocols_bgp.xml.in: protocols bgp parameters allow-martian-nexthop	Nothing found in VyOS
✓	<i>set protocols bgp parameters always-compare-med</i>	protocols_bgp.xml.in: protocols bgp parameters always-compare-med	protocols bgp parameters always-compare-med
✓	<i>set protocols bgp parameters bestpath as-path confed</i>	protocols_bgp.xml.in: protocols bgp parameters bestpath as-path confed	protocols bgp parameters bestpath as-path confed
✓	<i>set protocols bgp parameters bestpath as-path ignore</i>	protocols_bgp.xml.in: protocols bgp parameters bestpath as-path ignore	protocols bgp parameters bestpath as-path ignore
✓	<i>set protocols bgp parameters bestpath as-path multipath-relax</i>	protocols_bgp.xml.in: protocols bgp param- eters bestpath as-path multipath-relax	protocols bgp param- eters bestpath as-path multipath-relax
×	not yet documented	protocols_bgp.xml.in: protocols bgp parameters bestpath bandwidth	protocols bgp param- eters bestpath bandwidth <value>
✓	<i>set protocols bgp parameters bestpath compare-routerid</i>	protocols_bgp.xml.in: protocols bgp parameters bestpath compare-routerid	protocols bgp parameters bestpath compare-routerid
×	not yet documented	protocols_bgp.xml.in: protocols bgp parameters bestpath med	protocols bgp parameters bestpath med <value>
×	<i>set protocols bgp parameters bestpath med confed</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set protocols bgp parameters bestpath med missing-as-worst</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	protocols_bgp.xml.in: protocols bgp param- eters bestpath peer-type multipath-relax	protocols bgp param- eters bestpath peer-type multipath-relax
×	not yet documented	protocols_bgp.xml.in: protocols bgp parameters cluster-id	protocols bgp parameters cluster-id <value>
×	<i>set protocols bgp parameters cluster-id <id></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	protocols_bgp.xml.in: protocols bgp parameters conditional-advertisement timer	protocols bgp parameters conditional-advertisement timer <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp parameters confederation identifier	protocols bgp parameters confederation identifier <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<i>set protocols bgp parameters confederation identifier <asn></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	protocols_bgp.xml.in: protocols bgp parameters confederation peers	protocols bgp parameters confederation peers <value>
×	<i>set protocols bgp parameters confederation peers <nsbasn></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	protocols_bgp.xml.in: protocols bgp parameters dampening half-life	protocols bgp parameters dampening half-life <value>
×	<i>set protocols bgp parameters dampening half-life <minutes></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	protocols_bgp.xml.in: protocols bgp parameters dampening max-suppress-time	protocols bgp parameters dampening max-suppress-time <value>
×	<i>set protocols bgp parameters dampening max-suppress-time <seconds></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	protocols_bgp.xml.in: protocols bgp parameters dampening re-use	protocols bgp parameters dampening re-use <value>
×	<i>set protocols bgp parameters dampening re-use <seconds></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	protocols_bgp.xml.in: protocols bgp parameters dampening start-suppress-time	protocols bgp parameters dampening start-suppress-time <value>
×	<i>set protocols bgp parameters dampening start-suppress-time <seconds></i>	Nothing found in XML Definitions	Nothing found in VyOS
✓	<i>set protocols bgp parameters default local-pref</i>	protocols_bgp.xml.in: protocols bgp parameters default local-pref	protocols bgp parameters default local-pref <value>
✓	<i>set protocols bgp parameters deterministic-med</i>	protocols_bgp.xml.in: protocols bgp parameters deterministic-med	protocols bgp parameters deterministic-med
×	not yet documented	protocols_bgp.xml.in: protocols bgp parameters disable-ebgp-connected-route-check	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<i>set protocols bgp parameters distance global <external/internal/lo <distance></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	protocols_bgp.xml.in: protocols bgp parameters distance global external	protocols bgp parameters distance global external <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp parameters distance global internal	protocols bgp parameters distance global internal <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp parameters distance global local	protocols bgp parameters distance global local <value>
×	<i>set protocols bgp parameters distance prefix <subnet> distance <distance></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	protocols_bgp.xml.in: protocols bgp parameters distance prefix <prefix> distance	protocols bgp parameters distance prefix <text> distance <value>
✓	<i>set protocols bgp parameters ebgp-requires-policy</i>	protocols_bgp.xml.in: protocols bgp parameters ebgp-requires-policy	protocols bgp parameters ebgp-requires-policy
×	not yet documented	protocols_bgp.xml.in: protocols bgp parameters fast-convergence	protocols bgp parameters fast-convergence
×	not yet documented	protocols_bgp.xml.in: protocols bgp parameters graceful-restart stalepath-time	protocols bgp parameters graceful-restart stalepath-time <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp parameters graceful-shutdown	protocols bgp parameters graceful-shutdown
×	<i>set protocols bgp parameters input-queue-limit <messages></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	protocols_bgp.xml.in: protocols bgp parameters labeled-unicast	Nothing found in VyOS
×	<i>set protocols bgp parameters labeled-unicast <explicit-null / ipv4-explicit-null / ipv6-explicit-null></i>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
✓	<i>set protocols bgp parameters log-neighbor-changes</i>	protocols_bgp.xml.in: protocols bgp parameters log-neighbor-changes	protocols bgp parameters log-neighbor-changes
×	not yet documented	protocols_bgp.xml.in: protocols bgp parameters minimum-holdtime	protocols bgp parameters minimum-holdtime <value>
✓	<i>set protocols bgp parameters network-import-check</i>	protocols_bgp.xml.in: protocols bgp parameters network-import-check	protocols bgp parameters network-import-check
✓	<i>set protocols bgp parameters no-client-to-client-r</i>	protocols_bgp.xml.in: protocols bgp parameters no-client-to-client- reflection	protocols bgp parameters no-client-to-client- reflection
✓	<i>set protocols bgp parameters no-fast-external-fail</i>	protocols_bgp.xml.in: protocols bgp parameters no-fast-external-failover	protocols bgp parameters no-fast-external-failover
×	<i>set protocols bgp parameters no-hard-administrativ</i>	protocols_bgp.xml.in: protocols bgp parameters no-hard-administrative- reset	Nothing found in VyOS
×	<i>set protocols bgp parameters no-ipv6-auto-ra</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	protocols_bgp.xml.in: protocols bgp parameters no-suppress-duplicates	protocols bgp parameters no-suppress-duplicates
×	<i>set protocols bgp parameters output-queue-limit <messages></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	protocols_bgp.xml.in: protocols bgp parameters reject-as-sets	protocols bgp parameters reject-as-sets
×	not yet documented	protocols_bgp.xml.in: protocols bgp parameters route-reflector-allow- route-reflector-allow- outbound-policy	protocols bgp parameters route-reflector-allow- outbound-policy
×	not yet documented	protocols_bgp.xml.in: protocols bgp parameters router-id	protocols bgp parameters router-id <value>
×	<i>set protocols bgp parameters router-id <id></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	protocols_bgp.xml.in: protocols bgp parameters shutdown	protocols bgp parameters shutdown
×	not yet documented	protocols_bgp.xml.in: protocols bgp parameters suppress-fib-pending	protocols bgp parameters suppress-fib-pending

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	protocols_bgp.xml.in: protocols bgp parameters tcp-keepalive idle	protocols bgp parameters tcp-keepalive idle <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp parameters tcp-keepalive interval	protocols bgp parameters tcp-keepalive interval <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp parameters tcp-keepalive probes	protocols bgp parameters tcp-keepalive probes <value>
×	<i>set protocols bgp parameters update-delay establish-wait <seconds></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set protocols bgp parameters update-delay max-delay <seconds></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set protocols bgp peer-group <name></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set protocols bgp peer-group <neighbor> bfd</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address- family ipv4-labeled- unicast addpath-tx-all	Nothing found in VyOS
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address- family ipv4-labeled- unicast addpath-tx-per-as	Nothing found in VyOS
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer- group <peer-group> address-family ipv4- labeled-unicast allowas-in number	Nothing found in VyOS
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address- family ipv4-labeled- unicast as-override	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family ipv4-labeled-unicast attribute-unchanged as-path	Nothing found in VyOS
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family ipv4-labeled-unicast attribute-unchanged med	Nothing found in VyOS
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family ipv4-labeled-unicast attribute-unchanged next-hop	Nothing found in VyOS
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family ipv4-labeled-unicast capability orf prefix-list receive	Nothing found in VyOS
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family ipv4-labeled-unicast capability orf prefix-list send	Nothing found in VyOS
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family ipv4-labeled-unicast conditionally-advertise advertise-map	Nothing found in VyOS
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family ipv4-labeled-unicast conditionally-advertise exist-map	Nothing found in VyOS
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family ipv4-labeled-unicast conditionally-advertise non-exist-map	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family ipv4-labeled-unicast default-originate route-map	Nothing found in VyOS
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family ipv4-labeled-unicast disable-send-community extended	Nothing found in VyOS
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family ipv4-labeled-unicast disable-send-community standard	Nothing found in VyOS
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family ipv4-labeled-unicast distribute-list export	Nothing found in VyOS
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family ipv4-labeled-unicast distribute-list import	Nothing found in VyOS
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family ipv4-labeled-unicast filter-list export	Nothing found in VyOS
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family ipv4-labeled-unicast filter-list import	Nothing found in VyOS
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family ipv4-labeled-unicast maximum-prefix	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family ipv4-labeled-unicast maximum-prefix-out	Nothing found in VyOS
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family ipv4-labeled-unicast nexthop-self force	Nothing found in VyOS
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family ipv4-labeled-unicast prefix-list export	Nothing found in VyOS
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family ipv4-labeled-unicast prefix-list import	Nothing found in VyOS
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family ipv4-labeled-unicast remove-private-as all	Nothing found in VyOS
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family ipv4-labeled-unicast route-map export	Nothing found in VyOS
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family ipv4-labeled-unicast route-map import	Nothing found in VyOS
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family ipv4-labeled-unicast route-reflector-client	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family ipv4-labeled-unicast route-server-client	Nothing found in VyOS
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family ipv4-labeled-unicast soft-reconfiguration inbound	Nothing found in VyOS
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family ipv4-labeled-unicast unsuppress-map	Nothing found in VyOS
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family ipv4-labeled-unicast weight	Nothing found in VyOS
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family ipv4-unicast addpath-tx-all	protocols bgp peer-group <text> address-family ipv4-unicast addpath-tx-all
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family ipv4-unicast addpath-tx-per-as	protocols bgp peer-group <text> address-family ipv4-unicast addpath-tx-per-as
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family ipv4-unicast allowas-in number	protocols bgp peer-group <text> address-family ipv4-unicast allowas-in number <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family ipv4-unicast as-override	protocols bgp peer-group <text> address-family ipv4-unicast as-override
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family ipv4-unicast attribute-unchanged as-path	protocols bgp peer-group <text> address-family ipv4-unicast attribute-unchanged as-path

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family ipv4-unicast attribute- unchanged med	protocols bgp peer-group <text> address-family ipv4-unicast attribute- unchanged med
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family ipv4-unicast attribute- unchanged next-hop	protocols bgp peer-group <text> address-family ipv4-unicast attribute- unchanged next-hop
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family ipv4-unicast capability orf prefix-list receive	protocols bgp peer-group <text> address-family ipv4-unicast capability orf prefix-list receive
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family ipv4-unicast capability orf prefix-list send	protocols bgp peer-group <text> address-family ipv4-unicast capability orf prefix-list send
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family ipv4-unicast conditionally-advertise advertise-map	protocols bgp peer-group <text> address-family ipv4-unicast conditionally-advertise advertise-map <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family ipv4-unicast conditionally-advertise exist-map	protocols bgp peer-group <text> address-family ipv4-unicast conditionally-advertise exist-map <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family ipv4-unicast conditionally-advertise non-exist-map	protocols bgp peer-group <text> address-family ipv4-unicast conditionally-advertise non-exist-map <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family ipv4-unicast default-originate route-map	protocols bgp peer-group <text> address-family ipv4-unicast default-originate route-map <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family family ipv4-unicast disable-send-community extended	protocols bgp peer-group <text> address-family ipv4-unicast disable-send-community extended
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family family ipv4-unicast disable-send-community standard	protocols bgp peer-group <text> address-family ipv4-unicast disable-send-community standard
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family family ipv4-unicast distribute-list export	protocols bgp peer-group <text> address-family ipv4-unicast distribute-list export <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family family ipv4-unicast distribute-list import	protocols bgp peer-group <text> address-family ipv4-unicast distribute-list import <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family family ipv4-unicast filter-list export	protocols bgp peer-group <text> address-family ipv4-unicast filter-list export <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family family ipv4-unicast filter-list import	protocols bgp peer-group <text> address-family ipv4-unicast filter-list import <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family family ipv4-unicast maximum-prefix	protocols bgp peer-group <text> address-family ipv4-unicast maximum-prefix <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family family ipv4-unicast maximum-prefix-out	protocols bgp peer-group <text> address-family ipv4-unicast maximum-prefix-out <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family family ipv4-unicast nexthop-self force	protocols bgp peer-group <text> address-family ipv4-unicast nexthop-self force

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family family ipv4-unicast prefix-list export	protocols bgp peer-group <text> address-family ipv4-unicast prefix-list export <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family family ipv4-unicast prefix-list import	protocols bgp peer-group <text> address-family ipv4-unicast prefix-list import <value>
×	not yet documented	Nothing found in XML Definitions	protocols bgp peer-group <text> address-family ipv4-unicast remove-private-as
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family family ipv4-unicast remove-private-as all	Nothing found in VyOS
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family family ipv4-unicast route-map export	protocols bgp peer-group <text> address-family ipv4-unicast route-map export <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family family ipv4-unicast route-map import	protocols bgp peer-group <text> address-family ipv4-unicast route-map import <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family family ipv4-unicast route-reflector-client	protocols bgp peer-group <text> address-family ipv4-unicast route-reflector-client
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family family ipv4-unicast route-server-client	protocols bgp peer-group <text> address-family ipv4-unicast route-server-client
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family family ipv4-unicast soft-reconfiguration inbound	protocols bgp peer-group <text> address-family family ipv4-unicast soft-reconfiguration inbound

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family ipv4-unicast unsuppress-map unsuppress-map	protocols bgp peer-group <text> address-family ipv4-unicast unsuppress-map <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family ipv4-unicast weight weight	protocols bgp peer-group <text> address-family ipv4-unicast weight <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family ipv4-vpn addpath-tx-all addpath-tx-all	protocols bgp peer-group <text> address-family ipv4-vpn addpath-tx-all
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family ipv4-vpn addpath-tx-per-as addpath-tx-per-as	protocols bgp peer-group <text> address-family ipv4-vpn addpath-tx-per-as
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family ipv4-vpn allowas-in allowas-in number	protocols bgp peer-group <text> address-family ipv4-vpn allowas-in number <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family ipv4-vpn as-override as-override	protocols bgp peer-group <text> address-family ipv4-vpn as-override
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family ipv4-vpn attribute-unchanged as-path	protocols bgp peer-group <text> address-family ipv4-vpn attribute-unchanged as-path
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family ipv4-vpn attribute-unchanged med	protocols bgp peer-group <text> address-family ipv4-vpn attribute-unchanged med
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family ipv4-vpn attribute-unchanged next-hop	protocols bgp peer-group <text> address-family ipv4-vpn attribute-unchanged next-hop

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family ipv4-vpn conditionally-advertise advertise-map	protocols bgp peer-group <text> address-family ipv4-vpn conditionally- advertise advertise-map <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family ipv4-vpn conditionally-advertise exist-map	protocols bgp peer-group <text> address-family ipv4-vpn conditionally- advertise exist-map <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family ipv4-vpn conditionally-advertise non-exist-map	protocols bgp peer-group <text> address-family ipv4-vpn conditionally- advertise non-exist-map <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family ipv4-vpn disable-send-community extended	protocols bgp peer-group <text> address-family ipv4-vpn disable-send- community extended
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family ipv4-vpn disable-send-community standard	protocols bgp peer-group <text> address-family ipv4-vpn disable-send- community standard
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family ipv4-vpn distribute-list export	protocols bgp peer-group <text> address-family ipv4-vpn distribute-list export <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family ipv4-vpn distribute-list import	protocols bgp peer-group <text> address-family ipv4-vpn distribute-list import <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address- family ipv4-vpn filter-list export	protocols bgp peer-group <text> address-family ipv4-vpn filter-list export <value>

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Table 1 – continued from previous page

Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family ipv4-vpn filter-list import <value>	protocols bgp peer-group <text> address-family ipv4-vpn filter-list import <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family ipv4-vpn maximum-prefix <value>	protocols bgp peer-group <text> address-family ipv4-vpn maximum-prefix <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family ipv4-vpn maximum-prefix-out <value>	protocols bgp peer-group <text> address-family ipv4-vpn maximum-prefix-out <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family ipv4-vpn nexthop-self force	protocols bgp peer-group <text> address-family ipv4-vpn nexthop-self force
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family ipv4-vpn prefix-list export <value>	protocols bgp peer-group <text> address-family ipv4-vpn prefix-list export <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family ipv4-vpn prefix-list import <value>	protocols bgp peer-group <text> address-family ipv4-vpn prefix-list import <value>
×	not yet documented	Nothing found in XML Definitions	protocols bgp peer-group <text> address-family ipv4-vpn remove-private-as
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family ipv4-vpn remove-private-as all	Nothing found in VyOS
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family ipv4-vpn route-map export <value>	protocols bgp peer-group <text> address-family ipv4-vpn route-map export <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family ipv4-vpn route-map import <value>	protocols bgp peer-group <text> address-family ipv4-vpn route-map import <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family ipv4-vpn route-reflector-client	protocols bgp peer-group <text> address-family ipv4-vpn route-reflector-client
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family ipv4-vpn route-server-client	protocols bgp peer-group <text> address-family ipv4-vpn route-server-client
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family ipv4-vpn soft-reconfiguration inbound	protocols bgp peer-group <text> address-family ipv4-vpn soft-reconfiguration inbound
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family ipv4-vpn unsuppress-map	protocols bgp peer-group <text> address-family ipv4-vpn unsuppress-map <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family ipv4-vpn weight	protocols bgp peer-group <text> address-family ipv4-vpn weight <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family ipv6-labeled-unicast addpath-tx-all	Nothing found in VyOS
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family ipv6-labeled-unicast addpath-tx-per-as	Nothing found in VyOS
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family ipv6-labeled-unicast allowas-in number	Nothing found in VyOS
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family ipv6-labeled-unicast as-override	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family ipv6-labeled-unicast attribute-unchanged as-path	Nothing found in VyOS
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family ipv6-labeled-unicast attribute-unchanged med	Nothing found in VyOS
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family ipv6-labeled-unicast attribute-unchanged next-hop	Nothing found in VyOS
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family ipv6-labeled-unicast capability orf prefix-list receive	Nothing found in VyOS
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family ipv6-labeled-unicast capability orf prefix-list send	Nothing found in VyOS
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family ipv6-labeled-unicast conditionally-advertise advertise-map	Nothing found in VyOS
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family ipv6-labeled-unicast conditionally-advertise exist-map	Nothing found in VyOS
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family ipv6-labeled-unicast conditionally-advertise non-exist-map	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family ipv6-labeled-unicast default-originate route-map	Nothing found in VyOS
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family ipv6-labeled-unicast disable-send-community extended	Nothing found in VyOS
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family ipv6-labeled-unicast disable-send-community standard	Nothing found in VyOS
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family ipv6-labeled-unicast distribute-list export	Nothing found in VyOS
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family ipv6-labeled-unicast distribute-list import	Nothing found in VyOS
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family ipv6-labeled-unicast filter-list export	Nothing found in VyOS
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family ipv6-labeled-unicast filter-list import	Nothing found in VyOS
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family ipv6-labeled-unicast maximum-prefix	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family ipv6-labeled-unicast maximum-prefix-out	Nothing found in VyOS
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family ipv6-labeled-unicast nexthop-local unchanged	Nothing found in VyOS
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family ipv6-labeled-unicast nexthop-self force	Nothing found in VyOS
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family ipv6-labeled-unicast prefix-list export	Nothing found in VyOS
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family ipv6-labeled-unicast prefix-list import	Nothing found in VyOS
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family ipv6-labeled-unicast remove-private-as all	Nothing found in VyOS
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family ipv6-labeled-unicast route-map export	Nothing found in VyOS
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family ipv6-labeled-unicast route-map import	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family ipv6-labeled-unicast route-reflector-client	Nothing found in VyOS
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family ipv6-labeled-unicast route-server-client	Nothing found in VyOS
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family ipv6-labeled-unicast soft-reconfiguration inbound	Nothing found in VyOS
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family ipv6-labeled-unicast unsuppress-map	Nothing found in VyOS
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family ipv6-labeled-unicast weight	Nothing found in VyOS
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family ipv6-unicast addpath-tx-all	protocols bgp peer-group <text> address-family ipv6-unicast addpath-tx-all
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family ipv6-unicast addpath-tx-per-as	protocols bgp peer-group <text> address-family ipv6-unicast addpath-tx-per-as
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family ipv6-unicast allowas-in number	protocols bgp peer-group <text> address-family ipv6-unicast allowas-in number <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family ipv6-unicast as-override	protocols bgp peer-group <text> address-family ipv6-unicast as-override

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family ipv6-unicast attribute- unchanged as-path	protocols bgp peer-group <text> address-family ipv6-unicast attribute- unchanged as-path
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family ipv6-unicast attribute- unchanged med	protocols bgp peer-group <text> address-family ipv6-unicast attribute- unchanged med
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family ipv6-unicast attribute- unchanged next-hop	protocols bgp peer-group <text> address-family ipv6-unicast attribute- unchanged next-hop
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family ipv6-unicast capability orf prefix-list receive	protocols bgp peer-group <text> address-family ipv6-unicast capability orf prefix-list receive
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family ipv6-unicast capability orf prefix-list send	protocols bgp peer-group <text> address-family ipv6-unicast capability orf prefix-list send
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family ipv6-unicast conditionally-advertise advertise-map	protocols bgp peer-group <text> address-family ipv6-unicast conditionally-advertise advertise-map <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family ipv6-unicast conditionally-advertise exist-map	protocols bgp peer-group <text> address-family ipv6-unicast conditionally-advertise exist-map <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family ipv6-unicast conditionally-advertise non-exist-map	protocols bgp peer-group <text> address-family ipv6-unicast conditionally-advertise non-exist-map <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family ipv6-unicast default-originate route-map <value>	protocols bgp peer-group <text> address-family ipv6-unicast default-originate route-map <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family ipv6-unicast disable-send-community extended	protocols bgp peer-group <text> address-family ipv6-unicast disable-send-community extended
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family ipv6-unicast disable-send-community standard	protocols bgp peer-group <text> address-family ipv6-unicast disable-send-community standard
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family ipv6-unicast distribute-list export <value>	protocols bgp peer-group <text> address-family ipv6-unicast distribute-list export <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family ipv6-unicast distribute-list import	protocols bgp peer-group <text> address-family ipv6-unicast distribute-list import <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family ipv6-unicast filter-list export	protocols bgp peer-group <text> address-family ipv6-unicast filter-list export <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family ipv6-unicast filter-list import	protocols bgp peer-group <text> address-family ipv6-unicast filter-list import <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family ipv6-unicast maximum-prefix	protocols bgp peer-group <text> address-family ipv6-unicast maximum-prefix <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family ipv6-unicast maximum-prefix-out	protocols bgp peer-group <text> address-family ipv6-unicast maximum-prefix-out <value>

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Table 1 – continued from previous page

Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family family ipv6-unicast nexthop-local unchanged	protocols bgp peer-group <text> address-family ipv6-unicast nexthop-local unchanged
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family family ipv6-unicast nexthop-self force	protocols bgp peer-group <text> address-family ipv6-unicast nexthop-self force
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family family ipv6-unicast prefix-list export	protocols bgp peer-group <text> address-family ipv6-unicast prefix-list export <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family family ipv6-unicast prefix-list import	protocols bgp peer-group <text> address-family ipv6-unicast prefix-list import <value>
×	not yet documented	Nothing found in XML Definitions	protocols bgp peer-group <text> address-family ipv6-unicast remove-private-as
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family family ipv6-unicast remove-private-as all	Nothing found in VyOS
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family family ipv6-unicast route-map export	protocols bgp peer-group <text> address-family ipv6-unicast route-map export <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family family ipv6-unicast route-map import	protocols bgp peer-group <text> address-family ipv6-unicast route-map import <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family family ipv6-unicast route-reflector-client	protocols bgp peer-group <text> address-family ipv6-unicast route-reflector-client
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family family ipv6-unicast route-server-client	protocols bgp peer-group <text> address-family ipv6-unicast route-server-client

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family ipv6-unicast soft-reconfiguration inbound	protocols bgp peer-group <text> address-family ipv6-unicast soft-reconfiguration inbound
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family ipv6-unicast unsuppress-map unsuppress-map	protocols bgp peer-group <text> address-family ipv6-unicast unsuppress-map <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family ipv6-unicast weight weight	protocols bgp peer-group <text> address-family ipv6-unicast weight <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family ipv6-vpn addpath-tx-all addpath-tx-all	protocols bgp peer-group <text> address-family ipv6-vpn addpath-tx-all
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family ipv6-vpn addpath-tx-per-as addpath-tx-per-as	protocols bgp peer-group <text> address-family ipv6-vpn addpath-tx-per-as
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family ipv6-vpn allowas-in number allowas-in number	protocols bgp peer-group <text> address-family ipv6-vpn allowas-in number <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family ipv6-vpn as-override as-override	protocols bgp peer-group <text> address-family ipv6-vpn as-override
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family ipv6-vpn attribute-unchanged as-path	protocols bgp peer-group <text> address-family ipv6-vpn attribute-unchanged as-path
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family ipv6-vpn attribute-unchanged med	protocols bgp peer-group <text> address-family ipv6-vpn attribute-unchanged med

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family ipv6-vpn attribute-unchanged next-hop	protocols bgp peer-group <text> address-family ipv6-vpn attribute-unchanged next-hop
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family ipv6-vpn conditionally-advertise advertise-map	protocols bgp peer-group <text> address-family ipv6-vpn conditionally-advertise advertise-map <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family ipv6-vpn conditionally-advertise exist-map	protocols bgp peer-group <text> address-family ipv6-vpn conditionally-advertise exist-map <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family ipv6-vpn conditionally-advertise non-exist-map	protocols bgp peer-group <text> address-family ipv6-vpn conditionally-advertise non-exist-map <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family ipv6-vpn disable-send-community extended	protocols bgp peer-group <text> address-family ipv6-vpn disable-send-community extended
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family ipv6-vpn disable-send-community standard	protocols bgp peer-group <text> address-family ipv6-vpn disable-send-community standard
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family ipv6-vpn distribute-list export	protocols bgp peer-group <text> address-family ipv6-vpn distribute-list export <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family ipv6-vpn distribute-list import	protocols bgp peer-group <text> address-family ipv6-vpn distribute-list import <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family ipv6-vpn filter-list export export	protocols bgp peer-group <text> address-family ipv6-vpn filter-list export <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family ipv6-vpn filter-list import import	protocols bgp peer-group <text> address-family ipv6-vpn filter-list import <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family ipv6-vpn maximum-prefix maximum-prefix	protocols bgp peer-group <text> address-family ipv6-vpn maximum-prefix <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family ipv6-vpn maximum-prefix-out maximum-prefix-out	protocols bgp peer-group <text> address-family ipv6-vpn maximum-prefix-out <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family ipv6-vpn nexthop-local nexthop-local unchanged	protocols bgp peer-group <text> address-family ipv6-vpn nexthop-local unchanged
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family ipv6-vpn nexthop-self nexthop-self force	protocols bgp peer-group <text> address-family ipv6-vpn nexthop-self force
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family ipv6-vpn prefix-list prefix-list export	protocols bgp peer-group <text> address-family ipv6-vpn prefix-list export <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family ipv6-vpn prefix-list import	protocols bgp peer-group <text> address-family ipv6-vpn prefix-list import <value>
×	not yet documented	Nothing found in XML Definitions	protocols bgp peer-group <text> address-family ipv6-vpn remove-private-as
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family ipv6-vpn remove-private-as remove-private-as all	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family ipv6-vpn route-map export	protocols bgp peer-group <text> address-family ipv6-vpn route-map export <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family ipv6-vpn route-map import	protocols bgp peer-group <text> address-family ipv6-vpn route-map import <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family ipv6-vpn route-reflector-client	protocols bgp peer-group <text> address-family ipv6-vpn route-reflector- client
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family ipv6-vpn route-server-client	protocols bgp peer-group <text> address-family ipv6-vpn route-server- client
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family ipv6- vpn soft-reconfiguration inbound	protocols bgp peer-group <text> address- family ipv6-vpn soft- reconfiguration inbound
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family ipv6-vpn unsuppress-map	protocols bgp peer-group <text> address-family ipv6-vpn unsuppress-map <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address- family ipv6-vpn weight	protocols bgp peer-group <text> address-family ipv6-vpn weight <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family l2vpn- evpn allowas-in number	protocols bgp peer-group <text> address-family l2vpn-evpn allowas-in number <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family l2vpn- evpn attribute-unchanged as-path	protocols bgp peer-group <text> address-family l2vpn-evpn attribute- unchanged as-path

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family l2vpn-evpn attribute-unchanged med	protocols bgp peer-group <text> address-family l2vpn-evpn attribute-unchanged med
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family l2vpn-evpn attribute-unchanged next-hop	protocols bgp peer-group <text> address-family l2vpn-evpn attribute-unchanged next-hop
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family l2vpn-evpn nexthop-self force	protocols bgp peer-group <text> address-family l2vpn-evpn nexthop-self force
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family l2vpn-evpn route-map export	protocols bgp peer-group <text> address-family l2vpn-evpn route-map export <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family l2vpn-evpn route-map import	protocols bgp peer-group <text> address-family l2vpn-evpn route-map import <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family l2vpn-evpn route-reflector-client	protocols bgp peer-group <text> address-family l2vpn-evpn route-reflector-client
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family l2vpn-evpn route-server-client	protocols bgp peer-group <text> address-family l2vpn-evpn route-server-client
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> address-family l2vpn-evpn soft-reconfiguration inbound	protocols bgp peer-group <text> address-family l2vpn-evpn soft-reconfiguration inbound
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> bfd check-control-plane-failure	protocols bgp peer-group <text> bfd check-control-plane-failure

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> bfd profile	protocols bgp peer-group <text> bfd profile <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> capability dynamic	protocols bgp peer-group <text> capability dynamic
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> capability extended-nexthop	protocols bgp peer-group <text> capability extended-nexthop
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> capability software-version	Nothing found in VyOS
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> description	protocols bgp peer-group <text> description <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> disable- capability-negotiation	protocols bgp peer-group <text> disable-capability- negotiation
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> disable- connected-check	protocols bgp peer-group <text> disable-connected- check
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> ebgp- multihop	protocols bgp peer-group <text> ebgp-multihop <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> graceful- restart	protocols bgp peer-group <text> graceful-restart <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> local-as <local-as> no-prepend replace-as	protocols bgp peer-group <text> local-as <text> no- prepend replace-as
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> local-role <local-role> strict	protocols bgp peer-group <text> local-role <text> strict
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> override- capability	protocols bgp peer-group <text> override-capability

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> passive	protocols bgp peer-group <text> passive
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> password	protocols bgp peer-group <text> password <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> path-attribute discard	protocols bgp peer-group <text> path-attribute discard <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> path-attribute treat-as-withdraw	Nothing found in VyOS
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> port	protocols bgp peer-group <text> port <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> remote-as	protocols bgp peer-group <text> remote-as <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> shutdown	protocols bgp peer-group <text> shutdown
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> ttl-security hops	protocols bgp peer-group <text> ttl-security hops <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp peer-group <peer-group> update-source	protocols bgp peer-group <text> update-source <value>
×	not yet documented	protocols_bgp.xml.in: protocols bgp sid vpn per-vrf export	Nothing found in VyOS
×	not yet documented	protocols_bgp.xml.in: protocols bgp srv6 locator	Nothing found in VyOS
×	not yet documented	protocols_bgp.xml.in: protocols bgp system-as	protocols bgp system-as <value>
×	<i>set protocols bgp system-as <asn></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	protocols_bgp.xml.in: protocols bgp timers hold-time	protocols bgp timers hold-time <value>
×	<i>set protocols bgp timers holdtime <seconds></i>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	protocols_bgp.xml.in: protocols bgp timers keepalive	protocols bgp timers keepalive <value>
×	<i>set protocols bgp timers keepalive <seconds></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	protocols_eigrp.xml.in: protocols eigrp local-as	Nothing found in VyOS
×	not yet documented	protocols_eigrp.xml.in: protocols eigrp maximum- paths	Nothing found in VyOS
×	not yet documented	protocols_eigrp.xml.in: protocols eigrp metric weights	Nothing found in VyOS
×	not yet documented	protocols_eigrp.xml.in: protocols eigrp network	Nothing found in VyOS
×	not yet documented	protocols_eigrp.xml.in: protocols eigrp passive- interface	Nothing found in VyOS
×	not yet documented	protocols_eigrp.xml.in: protocols eigrp redis- tribute	Nothing found in VyOS
×	not yet documented	protocols_eigrp.xml.in: protocols eigrp router-id	Nothing found in VyOS
×	not yet documented	protocols_eigrp.xml.in: protocols eigrp variance	Nothing found in VyOS
✓	<i>set protocols failover route <subnet> next-hop <address> check policy <policy></i>	protocols_failover.xml.in: protocols failover route <route> next-hop <next- hop> check policy	protocols failover route <text> next-hop <text> check policy <value>
✓	<i>set protocols failover route <subnet> next-hop <address> check port <port></i>	protocols_failover.xml.in: protocols failover route <route> next-hop <next- hop> check port	protocols failover route <text> next-hop <text> check port <value>
✓	<i>set protocols failover route <subnet> next-hop <address> check target <target-address></i>	protocols_failover.xml.in: protocols failover route <route> next-hop <next- hop> check target	protocols failover route <text> next-hop <text> check target <value>
✓	<i>set protocols failover route <subnet> next-hop <address> check timeout <timeout></i>	protocols_failover.xml.in: protocols failover route <route> next-hop <next- hop> check timeout	protocols failover route <text> next-hop <text> check timeout <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
✓	<code>set protocols failover route <subnet> next-hop <address> check type <protocol></code>	protocols_failover.xml.in: protocols failover route <route> next-hop <next-hop> check type	protocols failover route <text> next-hop <text> check type <value>
✓	<code>set protocols failover route <subnet> next-hop <address> interface <interface></code>	protocols_failover.xml.in: protocols failover route <route> next-hop <next-hop> interface	protocols failover route <text> next-hop <text> interface <value>
✓	<code>set protocols failover route <subnet> next-hop <address> metric <1-255></code>	protocols_failover.xml.in: protocols failover route <route> next-hop <next-hop> metric	protocols failover route <text> next-hop <text> metric <value>
×	<code>set protocols failover route <subnet> next-hop <address> onlink</code>	protocols_failover.xml.in: protocols failover route <route> next-hop <next-hop> onlink	Nothing found in VyOS
✓	<code>set protocols igmp-proxy disable</code>	protocols_igmp-proxy.xml.in: protocols igmp-proxy disable	protocols igmp-proxy disable
✓	<code>set protocols igmp-proxy disable-quickleave</code>	protocols_igmp-proxy.xml.in: protocols igmp-proxy disable-quickleave	protocols igmp-proxy disable-quickleave
×	<code>set protocols igmp-proxy interface <interface> alt-subnet <network></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set protocols igmp-proxy interface <interface> role <upstream / downstream></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set protocols igmp-proxy interface <interface> threshold <1-255></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set protocols igmp-proxy interface <interface> whitelist <network></code>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	protocols_igmp-proxy.xml.in: protocols igmp-proxy interface <interface> alt-subnet	protocols igmp-proxy interface <text> alt-subnet <value>
×	not yet documented	protocols_igmp-proxy.xml.in: protocols igmp-proxy interface <interface> role	protocols igmp-proxy interface <text> role <value>
×	not yet documented	protocols_igmp-proxy.xml.in: protocols igmp-proxy interface <interface> threshold	protocols igmp-proxy interface <text> threshold <value>
×	not yet documented	protocols_igmp-proxy.xml.in: protocols igmp-proxy interface <interface> whitelist	protocols igmp-proxy interface <text> whitelist <value>
×	not yet documented	Nothing found in XML Definitions	protocols igmp interface <text> join <text> source <value>
×	not yet documented	Nothing found in XML Definitions	protocols igmp interface <text> query-interval <value>
×	not yet documented	Nothing found in XML Definitions	protocols igmp interface <text> query-max-response-time <value>
×	not yet documented	Nothing found in XML Definitions	protocols igmp interface <text> version <value>
×	<i>set protocols isis advertise-high-metric</i>	protocols_isis.xml.in: protocols isis advertise-high-metrics	Nothing found in VyOS
×	<i>set protocols isis advertise-passive-only</i>	protocols_isis.xml.in: protocols isis advertise-passive-only	Nothing found in VyOS
×	not yet documented	protocols_isis.xml.in: protocols isis area-password md5	protocols isis area-password md5 <value>
×	<i>set protocols isis area-password md5 <text></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	protocols_isis.xml.in: protocols isis area-password plaintext-password	protocols isis area-password plaintext-password <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<i>set protocols isis area-password plaintext-password <text></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set protocols isis default-information originate ipv4 level-1</i>	Nothing found in XML Definitions	Nothing found in VyOS
✓	<i>set protocols isis default-information originate ipv4 level-1 always</i>	protocols_isis.xml.in: protocols isis default-information originate ipv4 level-1 always	protocols isis default-information originate ipv4 level-1 always
×	not yet documented	protocols_isis.xml.in: protocols isis default-information originate ipv4 level-1 metric	protocols isis default-information originate ipv4 level-1 metric <value>
×	<i>set protocols isis default-information originate ipv4 level-1 metric <0-16777215></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	protocols_isis.xml.in: protocols isis default-information originate ipv4 level-1 route-map	protocols isis default-information originate ipv4 level-1 route-map <value>
×	<i>set protocols isis default-information originate ipv4 level-1 route-map <name></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set protocols isis default-information originate ipv4 level-2</i>	Nothing found in XML Definitions	Nothing found in VyOS
✓	<i>set protocols isis default-information originate ipv4 level-2 always</i>	protocols_isis.xml.in: protocols isis default-information originate ipv4 level-2 always	protocols isis default-information originate ipv4 level-2 always
×	not yet documented	protocols_isis.xml.in: protocols isis default-information originate ipv4 level-2 metric	protocols isis default-information originate ipv4 level-2 metric <value>
×	<i>set protocols isis default-information originate ipv4 level-2 metric <0-16777215></i>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	protocols_isis.xml.in: protocols isis default-information originate ipv4 level-2 route-map	protocols isis default-information originate ipv4 level-2 route-map <value>
×	<i>set protocols isis default-information originate ipv4 level-2 route-map <name></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set protocols isis default-information originate ipv6 level-1</i>	Nothing found in XML Definitions	Nothing found in VyOS
✓	<i>set protocols isis default-information originate ipv6 level-1 always</i>	protocols_isis.xml.in: protocols isis default-information originate ipv6 level-1 always	protocols isis default-information originate ipv6 level-1 always
×	not yet documented	protocols_isis.xml.in: protocols isis default-information originate ipv6 level-1 metric	protocols isis default-information originate ipv6 level-1 metric <value>
×	<i>set protocols isis default-information originate ipv6 level-1 metric <0-16777215></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	protocols_isis.xml.in: protocols isis default-information originate ipv6 level-1 route-map	protocols isis default-information originate ipv6 level-1 route-map <value>
×	<i>set protocols isis default-information originate ipv6 level-1 route-map <name></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set protocols isis default-information originate ipv6 level-2</i>	Nothing found in XML Definitions	Nothing found in VyOS
✓	<i>set protocols isis default-information originate ipv6 level-2 always</i>	protocols_isis.xml.in: protocols isis default-information originate ipv6 level-2 always	protocols isis default-information originate ipv6 level-2 always
×	not yet documented	protocols_isis.xml.in: protocols isis default-information originate ipv6 level-2 metric	protocols isis default-information originate ipv6 level-2 metric <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<i>set protocols isis default-information originate ipv6 level-2 metric <0-16777215></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	protocols_isis.xml.in: protocols isis default-information originate ipv6 level-2 route-map	protocols isis default-information originate ipv6 level-2 route-map <value>
×	<i>set protocols isis default-information originate ipv6 level-2 route-map <name></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	protocols_isis.xml.in: protocols isis domain-password md5	protocols isis domain-password md5 <value>
×	<i>set protocols isis domain-password md5 <text></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	protocols_isis.xml.in: protocols isis domain-password plaintext-password	protocols isis domain-password plaintext-password <value>
×	<i>set protocols isis domain-password plaintext-password <text></i>	Nothing found in XML Definitions	Nothing found in VyOS
✓	<i>set protocols isis dynamic-hostname</i>	protocols_isis.xml.in: protocols isis dynamic-hostname	protocols isis dynamic-hostname
×	<i>set protocols isis fast-reroute lfa local load-sharing disable level-1</i>	protocols_isis.xml.in: protocols isis fast-reroute lfa local load-sharing disable level-1	Nothing found in VyOS
×	<i>set protocols isis fast-reroute lfa local load-sharing disable level-2</i>	protocols_isis.xml.in: protocols isis fast-reroute lfa local load-sharing disable level-2	Nothing found in VyOS
×	<i>set protocols isis fast-reroute lfa local priority-limit <critical high medium> level-1</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set protocols isis fast-reroute lfa local priority-limit <critical high medium> level-2</i>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	protocols_isis.xml.in: protocols isis fast-reroute lfa local priority-limit critical level-1	Nothing found in VyOS
×	not yet documented	protocols_isis.xml.in: protocols isis fast-reroute lfa local priority-limit critical level-2	Nothing found in VyOS
×	not yet documented	protocols_isis.xml.in: protocols isis fast-reroute lfa local priority-limit high level-1	Nothing found in VyOS
×	not yet documented	protocols_isis.xml.in: protocols isis fast-reroute lfa local priority-limit high level-2	Nothing found in VyOS
×	not yet documented	protocols_isis.xml.in: protocols isis fast-reroute lfa local priority-limit medium level-1	Nothing found in VyOS
×	not yet documented	protocols_isis.xml.in: protocols isis fast-reroute lfa local priority-limit medium level-2	Nothing found in VyOS
×	<i>set protocols isis fast-reroute lfa local tiebreaker <downstream / lowest-backup-metric / node-protecting> index <1-255> level-1</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set protocols isis fast-reroute lfa local tiebreaker <downstream / lowest-backup-metric / node-protecting> index <1-255> level-2</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	protocols_isis.xml.in: protocols isis fast-reroute lfa local tiebreaker downstream index <index> level-1	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	protocols_isis.xml.in: protocols isis fast-reroute lfa local tiebreaker down- stream index <index> level-2	Nothing found in VyOS
×	not yet documented	protocols_isis.xml.in: protocols isis fast-reroute lfa local tiebreaker lowest- backup-metric index <in- dex> level-1	Nothing found in VyOS
×	not yet documented	protocols_isis.xml.in: protocols isis fast-reroute lfa local tiebreaker lowest- backup-metric index <in- dex> level-2	Nothing found in VyOS
×	not yet documented	protocols_isis.xml.in: protocols isis fast-reroute lfa local tiebreaker node- protecting index <index> level-1	Nothing found in VyOS
×	not yet documented	protocols_isis.xml.in: protocols isis fast-reroute lfa local tiebreaker node- protecting index <index> level-2	Nothing found in VyOS
×	<i>set protocols isis fast-reroute lfa remote prefix-list <name> level-1</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set protocols isis fast-reroute lfa remote prefix-list <name> level-2</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	protocols_isis.xml.in: protocols isis fast-reroute lfa remote prefix-list <prefix-list> level-1	Nothing found in VyOS
×	not yet documented	protocols_isis.xml.in: protocols isis fast-reroute lfa remote prefix-list <prefix-list> level-2	Nothing found in VyOS
×	<i>set protocols isis interface <interface></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set protocols isis interface <interface> bfd</i>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<i>set protocols isis interface <interface> bfd profile <profile-name></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set protocols isis interface <interface> circuit-type <level-1 level-1-2 level-2-only></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set protocols isis interface <interface> fast-reroute lfa level-1 enable</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set protocols isis interface <interface-1> fast-reroute lfa level-1 exclude interface <interface-2></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set protocols isis interface <interface> fast-reroute lfa level-2 enable</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set protocols isis interface <interface-1> fast-reroute lfa level-2 exclude interface <interface-2></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set protocols isis interface <interface> fast-reroute remote-lfa level-1 maximum-metric <1-16777215></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set protocols isis interface <interface> fast-reroute remote-lfa level-1 tunnel mpls-ldp</i>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<i>set protocols isis interface <interface> fast-reroute remote-lfa level-2 maximum-metric <1-16777215></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set protocols isis interface <interface> fast-reroute remote-lfa level-2 tunnel mpls-ldp</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set protocols isis interface <interface> fast-reroute ti-lfa level-1</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set protocols isis interface <interface> fast-reroute ti-lfa level-1 node-protection</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set protocols isis interface <interface> fast-reroute ti-lfa level-1 node-protection link-fallback</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set protocols isis interface <interface> fast-reroute ti-lfa level-2</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set protocols isis interface <interface> fast-reroute ti-lfa level-2 node-protection</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set protocols isis interface <interface> fast-reroute ti-lfa level-2 node-protection link-fallback</i>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<i>set protocols isis interface <interface> hello-interval <1-600></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set protocols isis interface <interface> hello-multiplier <2-100></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set protocols isis interface <interface> hello-padding</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set protocols isis interface <interface> ldp-sync disable</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set protocols isis interface <interface> ldp-sync holddown <0-10000></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set protocols isis interface <interface> metric <0-16777215></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set protocols isis interface <interface> network point-to-point</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set protocols isis interface <interface> no-three-way-handshak</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set protocols isis interface <interface> passive</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set protocols isis interface <interface> password md5 <text></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set protocols isis interface <interface> password plaintext-password <text></i>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<i>set protocols isis interface <interface> priority <0-127></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set protocols isis interface <interface> psnp-interval <0-127></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	protocols_isis.xml.in: protocols isis interface <interface> bfd profile	protocols isis interface <text> bfd profile <value>
×	not yet documented	protocols_isis.xml.in: protocols isis interface <interface> circuit-type	protocols isis inter- face <text> circuit-type <value>
×	not yet documented	protocols_isis.xml.in: protocols isis interface <interface> hello-interval	protocols isis interface <text> hello-interval <value>
×	not yet documented	protocols_isis.xml.in: protocols isis inter- face <interface> hello- multiplier	protocols isis interface <text> hello-multiplier <value>
×	not yet documented	protocols_isis.xml.in: protocols isis interface <interface> hello-padding	protocols isis interface <text> hello-padding
×	not yet documented	protocols_isis.xml.in: protocols isis interface <interface> ldp-sync disable	protocols isis interface <text> ldp-sync disable
×	not yet documented	protocols_isis.xml.in: protocols isis interface <interface> ldp-sync holddown	protocols isis interface <text> ldp-sync holddown <value>
×	not yet documented	protocols_isis.xml.in: protocols isis interface <interface> metric	protocols isis interface <text> metric <value>
×	not yet documented	protocols_isis.xml.in: protocols isis interface <interface> network point-to-point	protocols isis interface <text> network point-to- point
×	not yet documented	protocols_isis.xml.in: protocols isis interface <interface> no-three-way- handshake	protocols isis interface <text> no-three-way- handshake
×	not yet documented	protocols_isis.xml.in: protocols isis interface <interface> passive	protocols isis interface <text> passive

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	protocols_isis.xml.in: protocols isis interface <interface> password md5	protocols isis interface <text> password md5 <value>
×	not yet documented	protocols_isis.xml.in: protocols isis interface <interface> password plaintext-password	protocols isis interface <text> password plaintext- password <value>
×	not yet documented	protocols_isis.xml.in: protocols isis interface <interface> priority	protocols isis interface <text> priority <value>
×	not yet documented	protocols_isis.xml.in: protocols isis interface <interface> psnp-interval	protocols isis interface <text> psnp-interval <value>
×	<i>set protocols isis ldp-sync</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	protocols_isis.xml.in: protocols isis ldp-sync holddown	protocols isis ldp-sync holddown <value>
×	<i>set protocols isis ldp-sync holddown <0-10000></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	protocols_isis.xml.in: protocols isis level	protocols isis level <value>
×	<i>set protocols isis level <level-1 / level-1-2 / level-2></i>	Nothing found in XML Definitions	Nothing found in VyOS
✓	<i>set protocols isis log-adjacency-changes</i>	protocols_isis.xml.in: protocols isis log- adjacency-changes	protocols isis log- adjacency-changes
×	not yet documented	protocols_isis.xml.in: protocols isis lsp-gen- interval	protocols isis lsp-gen- interval <value>
×	<i>set protocols isis lsp-gen-interval <1-120></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	protocols_isis.xml.in: protocols isis lsp-mtu	protocols isis lsp-mtu <value>
×	<i>set protocols isis lsp-mtu <128-4352></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	protocols_isis.xml.in: protocols isis lsp-refresh- interval	protocols isis lsp-refresh- interval <value>
×	<i>set protocols isis lsp-refresh-interval <2-65235></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	protocols_isis.xml.in: protocols isis max-lsp- lifetime	protocols isis max-lsp- lifetime <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set protocols isis max-lsp-lifetime <350-65535></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	protocols_isis.xml.in: protocols isis metric-style	protocols isis metric-style <value>
×	<code>set protocols isis metric-style <narrow transition wide></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	protocols_isis.xml.in: protocols isis net	protocols isis net <value>
×	<code>set protocols isis net <network-entity-title</code>	Nothing found in XML Definitions	Nothing found in VyOS
✓	<code>set protocols isis purge-originator</code>	protocols_isis.xml.in: protocols isis purge-originator	protocols isis purge-originator
×	<code>set protocols isis redistribute ipv4 <bgp connected kernel nhrp ospf rip babel static> level-1</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set protocols isis redistribute ipv4 <bgp connected kernel nhrp ospf rip babel static> level-1 metric <0-16777215></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set protocols isis redistribute ipv4 <bgp connected kernel nhrp ospf rip babel static> level-1 route-map <name></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set protocols isis redistribute ipv4 <bgp connected kernel nhrp ospf rip babel static> level-2</code>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set protocols isis redistribute ipv4 <bgp connected kernel nhrp ospf rip babel static> level-2 metric <0-16777215></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set protocols isis redistribute ipv4 <bgp connected kernel nhrp ospf rip babel static> level-2 route-map <name></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	protocols_isis.xml.in: protocols isis redistribute ipv4 babel level-1 metric	protocols isis redistribute ipv4 babel level-1 metric <value>
×	not yet documented	protocols_isis.xml.in: protocols isis redistribute ipv4 babel level-1 route-map	protocols isis redistribute ipv4 babel level-1 route-map <value>
×	not yet documented	protocols_isis.xml.in: protocols isis redistribute ipv4 babel level-2 metric	protocols isis redistribute ipv4 babel level-2 metric <value>
×	not yet documented	protocols_isis.xml.in: protocols isis redistribute ipv4 babel level-2 route-map	protocols isis redistribute ipv4 babel level-2 route-map <value>
×	not yet documented	protocols_isis.xml.in: protocols isis redistribute ipv4 bgp level-1 metric	protocols isis redistribute ipv4 bgp level-1 metric <value>
×	not yet documented	protocols_isis.xml.in: protocols isis redistribute ipv4 bgp level-1 route-map	protocols isis redistribute ipv4 bgp level-1 route-map <value>
×	not yet documented	protocols_isis.xml.in: protocols isis redistribute ipv4 bgp level-2 metric	protocols isis redistribute ipv4 bgp level-2 metric <value>
×	not yet documented	protocols_isis.xml.in: protocols isis redistribute ipv4 bgp level-2 route-map	protocols isis redistribute ipv4 bgp level-2 route-map <value>
×	not yet documented	protocols_isis.xml.in: protocols isis redistribute ipv4 connected level-1 metric	protocols isis redistribute ipv4 connected level-1 metric <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	protocols_isis.xml.in: protocols isis redistribute ipv4 connected level-1 route-map	protocols isis redistribute ipv4 connected level-1 route-map <value>
×	not yet documented	protocols_isis.xml.in: protocols isis redistribute ipv4 connected level-2 metric	protocols isis redistribute ipv4 connected level-2 metric <value>
×	not yet documented	protocols_isis.xml.in: protocols isis redistribute ipv4 connected level-2 route-map	protocols isis redistribute ipv4 connected level-2 route-map <value>
×	not yet documented	protocols_isis.xml.in: protocols isis redistribute ipv4 kernel level-1 metric	protocols isis redistribute ipv4 kernel level-1 metric <value>
×	not yet documented	protocols_isis.xml.in: protocols isis redistribute ipv4 kernel level-1 route- map	protocols isis redistribute ipv4 kernel level-1 route- map <value>
×	not yet documented	protocols_isis.xml.in: protocols isis redistribute ipv4 kernel level-2 metric	protocols isis redistribute ipv4 kernel level-2 metric <value>
×	not yet documented	protocols_isis.xml.in: protocols isis redistribute ipv4 kernel level-2 route- map	protocols isis redistribute ipv4 kernel level-2 route- map <value>
×	not yet documented	protocols_isis.xml.in: protocols isis redistribute ipv4 ospf level-1 metric	protocols isis redistribute ipv4 ospf level-1 metric <value>
×	not yet documented	protocols_isis.xml.in: protocols isis redistribute ipv4 ospf level-1 route- map	protocols isis redistribute ipv4 ospf level-1 route- map <value>
×	not yet documented	protocols_isis.xml.in: protocols isis redistribute ipv4 ospf level-2 metric	protocols isis redistribute ipv4 ospf level-2 metric <value>
×	not yet documented	protocols_isis.xml.in: protocols isis redistribute ipv4 ospf level-2 route- map	protocols isis redistribute ipv4 ospf level-2 route- map <value>
×	not yet documented	protocols_isis.xml.in: protocols isis redistribute ipv4 rip level-1 metric	protocols isis redistribute ipv4 rip level-1 metric <value>
×	not yet documented	protocols_isis.xml.in: protocols isis redistribute ipv4 rip level-1 route-map	protocols isis redistribute ipv4 rip level-1 route-map <value>
×	not yet documented	protocols_isis.xml.in: protocols isis redistribute ipv4 rip level-2 metric	protocols isis redistribute ipv4 rip level-2 metric <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	protocols_isis.xml.in: protocols isis redistribute ipv4 rip level-2 route-map	protocols isis redistribute ipv4 rip level-2 route-map <value>
×	not yet documented	protocols_isis.xml.in: protocols isis redistribute ipv4 static level-1 metric	protocols isis redistribute ipv4 static level-1 metric <value>
×	not yet documented	protocols_isis.xml.in: protocols isis redistribute ipv4 static level-1 route- map	protocols isis redistribute ipv4 static level-1 route- map <value>
×	not yet documented	protocols_isis.xml.in: protocols isis redistribute ipv4 static level-2 metric	protocols isis redistribute ipv4 static level-2 metric <value>
×	not yet documented	protocols_isis.xml.in: protocols isis redistribute ipv4 static level-2 route- map	protocols isis redistribute ipv4 static level-2 route- map <value>
×	<i>set protocols isis redistribute ipv6 <bgp connected kernel ospf6 ripng babel static> level-1</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set protocols isis redistribute ipv6 <bgp connected kernel ospf6 ripng babel static> level-1 metric <0-16777215></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set protocols isis redistribute ipv6 <bgp connected kernel ospf6 ripng babel static> level-1 route-map <name></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set protocols isis redistribute ipv6 <bgp connected kernel ospf6 ripng babel static> level-2</i>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set protocols isis redistribute ipv6 <bgp connected kernel ospf6 ripng babel static> level-2 metric <0-16777215></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set protocols isis redistribute ipv6 <bgp connected kernel ospf6 ripng babel static> level-2 route-map <name></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	protocols_isis.xml.in: protocols isis redistribute ipv6 babel level-1 metric	protocols isis redistribute ipv6 babel level-1 metric <value>
×	not yet documented	protocols_isis.xml.in: protocols isis redistribute ipv6 babel level-1 route-map	protocols isis redistribute ipv6 babel level-1 route-map <value>
×	not yet documented	protocols_isis.xml.in: protocols isis redistribute ipv6 babel level-2 metric	protocols isis redistribute ipv6 babel level-2 metric <value>
×	not yet documented	protocols_isis.xml.in: protocols isis redistribute ipv6 babel level-2 route-map	protocols isis redistribute ipv6 babel level-2 route-map <value>
×	not yet documented	protocols_isis.xml.in: protocols isis redistribute ipv6 bgp level-1 metric	protocols isis redistribute ipv6 bgp level-1 metric <value>
×	not yet documented	protocols_isis.xml.in: protocols isis redistribute ipv6 bgp level-1 route-map	protocols isis redistribute ipv6 bgp level-1 route-map <value>
×	not yet documented	protocols_isis.xml.in: protocols isis redistribute ipv6 bgp level-2 metric	protocols isis redistribute ipv6 bgp level-2 metric <value>
×	not yet documented	protocols_isis.xml.in: protocols isis redistribute ipv6 bgp level-2 route-map	protocols isis redistribute ipv6 bgp level-2 route-map <value>
×	not yet documented	protocols_isis.xml.in: protocols isis redistribute ipv6 connected level-1 metric	protocols isis redistribute ipv6 connected level-1 metric <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	protocols_isis.xml.in: protocols isis redistribute ipv6 connected level-1 route-map	protocols isis redistribute ipv6 connected level-1 route-map <value>
×	not yet documented	protocols_isis.xml.in: protocols isis redistribute ipv6 connected level-2 metric	protocols isis redistribute ipv6 connected level-2 metric <value>
×	not yet documented	protocols_isis.xml.in: protocols isis redistribute ipv6 connected level-2 route-map	protocols isis redistribute ipv6 connected level-2 route-map <value>
×	not yet documented	protocols_isis.xml.in: protocols isis redistribute ipv6 kernel level-1 metric	protocols isis redistribute ipv6 kernel level-1 metric <value>
×	not yet documented	protocols_isis.xml.in: protocols isis redistribute ipv6 kernel level-1 route- map	protocols isis redistribute ipv6 kernel level-1 route- map <value>
×	not yet documented	protocols_isis.xml.in: protocols isis redistribute ipv6 kernel level-2 metric	protocols isis redistribute ipv6 kernel level-2 metric <value>
×	not yet documented	protocols_isis.xml.in: protocols isis redistribute ipv6 kernel level-2 route- map	protocols isis redistribute ipv6 kernel level-2 route- map <value>
×	not yet documented	protocols_isis.xml.in: protocols isis redistribute ipv6 ospf6 level-1 metric	protocols isis redistribute ipv6 ospf6 level-1 metric <value>
×	not yet documented	protocols_isis.xml.in: protocols isis redistribute ipv6 ospf6 level-1 route- map	protocols isis redistribute ipv6 ospf6 level-1 route- map <value>
×	not yet documented	protocols_isis.xml.in: protocols isis redistribute ipv6 ospf6 level-2 metric	protocols isis redistribute ipv6 ospf6 level-2 metric <value>
×	not yet documented	protocols_isis.xml.in: protocols isis redistribute ipv6 ospf6 level-2 route- map	protocols isis redistribute ipv6 ospf6 level-2 route- map <value>
×	not yet documented	protocols_isis.xml.in: protocols isis redistribute ipv6 ripng level-1 metric	protocols isis redistribute ipv6 ripng level-1 metric <value>
×	not yet documented	protocols_isis.xml.in: protocols isis redistribute ipv6 ripng level-1 route- map	protocols isis redistribute ipv6 ripng level-1 route- map <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	protocols_isis.xml.in: protocols isis redistribute ipv6 ripng level-2 metric	protocols isis redistribute ipv6 ripng level-2 metric <value>
×	not yet documented	protocols_isis.xml.in: protocols isis redistribute ipv6 ripng level-2 route- map	protocols isis redistribute ipv6 ripng level-2 route- map <value>
×	not yet documented	protocols_isis.xml.in: protocols isis redistribute ipv6 static level-1 metric	protocols isis redistribute ipv6 static level-1 metric <value>
×	not yet documented	protocols_isis.xml.in: protocols isis redistribute ipv6 static level-1 route- map	protocols isis redistribute ipv6 static level-1 route- map <value>
×	not yet documented	protocols_isis.xml.in: protocols isis redistribute ipv6 static level-2 metric	protocols isis redistribute ipv6 static level-2 metric <value>
×	not yet documented	protocols_isis.xml.in: protocols isis redistribute ipv6 static level-2 route- map	protocols isis redistribute ipv6 static level-2 route- map <value>
×	not yet documented	protocols_isis.xml.in: protocols isis segment- routing global-block high-label-value	protocols isis segment- routing global-block high-label-value <value>
×	<i>set protocols isis segment-routing global-block high-label-value <16-1048575></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	protocols_isis.xml.in: protocols isis segment- routing global-block low-label-value	protocols isis segment- routing global-block low-label-value <value>
×	<i>set protocols isis segment-routing global-block low-label-value <16-1048575></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	protocols_isis.xml.in: protocols isis segment- routing local-block high-label-value	protocols isis segment- routing local-block high-label-value <value>
×	<i>set protocols isis segment-routing local-block high-label-value <16-1048575></i>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	protocols_isis.xml.in: protocols isis segment- routing local-block low-label-value	protocols isis segment- routing local-block low-label-value <value>
×	<i>set protocols isis segment-routing local-block low-label-value <16-1048575></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	protocols_isis.xml.in: protocols isis segment- routing maximum-label- depth	protocols isis segment- routing maximum-label- depth <value>
×	<i>set protocols isis segment-routing maximum-label-depth <1-16></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set protocols isis segment-routing prefix <address> absolute <no-php-flag / explicit-null></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set protocols isis segment-routing prefix <address> absolute value <16-1048575></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set protocols isis segment-routing prefix <address> index <no-php-flag / explicit-null></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set protocols isis segment-routing prefix <address> index value <0-65535></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	protocols_isis.xml.in: protocols isis segment- routing prefix <prefix> absolute explicit-null	protocols isis segment- routing prefix <text> absolute explicit-null
×	not yet documented	protocols_isis.xml.in: protocols isis segment- routing prefix <prefix> absolute no-php-flag	protocols isis segment- routing prefix <text> absolute no-php-flag

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	protocols_isis.xml.in: protocols isis segment- routing prefix <prefix> absolute value	protocols isis segment- routing prefix <text> absolute value <value>
×	not yet documented	protocols_isis.xml.in: protocols isis segment- routing prefix <prefix> index explicit-null	protocols isis segment- routing prefix <text> index explicit-null
×	not yet documented	protocols_isis.xml.in: protocols isis segment- routing prefix <prefix> index no-php-flag	protocols isis segment- routing prefix <text> index no-php-flag
×	not yet documented	protocols_isis.xml.in: protocols isis segment- routing prefix <prefix> index value	protocols isis segment- routing prefix <text> index value <value>
×	<i>set protocols isis segment-routing srv6 interface <interface></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set protocols isis segment-routing srv6 locator <locator></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set protocols isis segment-routing srv6 node-msd max-end-d <0-255></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set protocols isis segment-routing srv6 node-msd max-end-pop <0-255></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set protocols isis segment-routing srv6 node-msd max-h-encaps <0-255></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set protocols isis segment-routing srv6 node-msd max-segs-left <0-255></i>	Nothing found in XML Definitions	Nothing found in VyOS
✓	<i>set protocols isis set-attached-bit</i>	protocols_isis.xml.in: protocols isis set-attached- bit	protocols isis set-attached- bit
✓	<i>set protocols isis set-overload-bit</i>	protocols_isis.xml.in: protocols isis set- overload-bit	protocols isis set- overload-bit

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	protocols_isis.xml.in: protocols isis spf-delay-ietf holddown	protocols isis spf-delay-ietf holddown <value>
×	<i>set protocols isis spf-delay-ietf holddown <0-60000></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	protocols_isis.xml.in: protocols isis spf-delay-ietf init-delay	protocols isis spf-delay-ietf init-delay <value>
×	<i>set protocols isis spf-delay-ietf init-delay <0-60000></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	protocols_isis.xml.in: protocols isis spf-delay-ietf long-delay	protocols isis spf-delay-ietf long-delay <value>
×	<i>set protocols isis spf-delay-ietf long-delay <0-60000></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	protocols_isis.xml.in: protocols isis spf-delay-ietf short-delay	protocols isis spf-delay-ietf short-delay <value>
×	<i>set protocols isis spf-delay-ietf short-delay <0-60000></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	protocols_isis.xml.in: protocols isis spf-delay-ietf time-to-learn	protocols isis spf-delay-ietf time-to-learn <value>
×	<i>set protocols isis spf-delay-ietf time-to-learn <0-60000></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	protocols_isis.xml.in: protocols isis spf-interval	protocols isis spf-interval <value>
×	<i>set protocols isis spf-interval <1-120></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set protocols isis topology <ipv4-multicast / ipv4-mgmt / ipv6-unicast / ipv6-multicast / ipv6-mgmt / ipv6-dstsrc></i>	Nothing found in XML Definitions	Nothing found in VyOS

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Table 1 – continued from previous page

Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	protocols_isis.xml.in: protocols isis traffic- engineering address	protocols isis traffic- engineering address <value>
×	<i>set protocols isis traffic-engineering address <ipv4-address></i>	Nothing found in XML Definitions	Nothing found in VyOS
✓	<i>set protocols isis traffic-engineering enable</i>	protocols_isis.xml.in: protocols isis traffic- engineering enable	protocols isis traffic- engineering enable
×	<i>set protocols isis traffic-engineering export</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	protocols_mpls.xml.in: protocols mpls interface	protocols mpls interface <value>
×	<i>set protocols mpls interface <interface></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	protocols_mpls.xml.in: protocols mpls ldp alloca- tion ipv4 access-list	protocols mpls ldp al- location ipv4 access-list <value>
×	<i>set protocols mpls ldp allocation ipv4 access-list <1-2699></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	protocols_mpls.xml.in: protocols mpls ldp alloca- tion ipv6 access-list6	protocols mpls ldp allo- cation ipv6 access-list6 <value>
×	<i>set protocols mpls ldp allocation ipv6 access-list6 <1-2699></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	protocols_mpls.xml.in: protocols mpls ldp discov- ery hello-ipv4-holdtime	protocols mpls ldp discov- ery hello-ipv4-holdtime <value>
×	<i>set protocols mpls ldp discovery hello-ipv4-holdtime <1-65535></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	protocols_mpls.xml.in: protocols mpls ldp discov- ery hello-ipv4-interval	protocols mpls ldp discov- ery hello-ipv4-interval <value>
×	<i>set protocols mpls ldp discovery hello-ipv4-interval <1-65535></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	protocols_mpls.xml.in: protocols mpls ldp discov- ery hello-ipv6-holdtime	protocols mpls ldp discov- ery hello-ipv6-holdtime <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<i>set protocols mpls ldp discovery hello-ipv6-holdtime <1-65535></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	protocols_mpls.xml.in: protocols mpls ldp discovery hello-ipv6-interval	protocols mpls ldp discovery hello-ipv6-interval <value>
×	<i>set protocols mpls ldp discovery hello-ipv6-interval <1-65535></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	protocols_mpls.xml.in: protocols mpls ldp discovery session-ipv4-holdtime	protocols mpls ldp discovery session-ipv4-holdtime <value>
×	<i>set protocols mpls ldp discovery session-ipv4-holdtime <15-65535></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	protocols_mpls.xml.in: protocols mpls ldp discovery session-ipv6-holdtime	protocols mpls ldp discovery session-ipv6-holdtime <value>
×	<i>set protocols mpls ldp discovery session-ipv6-holdtime <15-65535></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	protocols_mpls.xml.in: protocols mpls ldp discovery transport-ipv4-address	protocols mpls ldp discovery transport-ipv4-address <value>
×	<i>set protocols mpls ldp discovery transport-ipv4-address <address></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	protocols_mpls.xml.in: protocols mpls ldp discovery transport-ipv6-address	protocols mpls ldp discovery transport-ipv6-address <value>
×	<i>set protocols mpls ldp discovery transport-ipv6-address <address></i>	Nothing found in XML Definitions	Nothing found in VyOS
✓	<i>set protocols mpls ldp export ipv4 explicit-null</i>	protocols_mpls.xml.in: protocols mpls ldp export ipv4 explicit-null	protocols mpls ldp export ipv4 explicit-null
×	not yet documented	protocols_mpls.xml.in: protocols mpls ldp export ipv4 export-filter filter-access-list	protocols mpls ldp export ipv4 export-filter filter-access-list <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<i>set protocols mpls ldp export ipv4 export-filter filter-access-list <1-2699></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	protocols_mpls.xml.in: protocols mpls ldp ex- port ipv4 export-filter neighbor-access-list	protocols mpls ldp ex- port ipv4 export-filter neighbor-access-list <value>
×	<i>set protocols mpls ldp export ipv4 export-filter neighbor-access-list <1-2699></i>	Nothing found in XML Definitions	Nothing found in VyOS
✓	<i>set protocols mpls ldp export ipv6 explicit-null</i>	protocols_mpls.xml.in: protocols mpls ldp export ipv6 explicit-null	protocols mpls ldp export ipv6 explicit-null
×	not yet documented	protocols_mpls.xml.in: protocols mpls ldp ex- port ipv6 export-filter filter-access-list6	protocols mpls ldp ex- port ipv6 export-filter filter-access-list6 <value>
×	<i>set protocols mpls ldp export ipv6 export-filter filter-access-list6 <1-2699></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	protocols_mpls.xml.in: protocols mpls ldp ex- port ipv6 export-filter neighbor-access-list6	protocols mpls ldp ex- port ipv6 export-filter neighbor-access-list6 <value>
×	<i>set protocols mpls ldp export ipv6 export-filter neighbor-access-list6 <1-2699></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	protocols_mpls.xml.in: protocols mpls ldp im- port ipv4 import-filter filter-access-list	protocols mpls ldp im- port ipv4 import-filter filter-access-list <value>
×	<i>set protocols mpls ldp import ipv4 import-filter filter-access-list <1-2699></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	protocols_mpls.xml.in: protocols mpls ldp im- port ipv4 import-filter neighbor-access-list	protocols mpls ldp im- port ipv4 import-filter neighbor-access-list <value>

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Table 1 – continued from previous page

Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<i>set protocols mpls ldp import ipv4 import-filter neighbor-access-list <1-2699></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	protocols_mpls.xml.in: protocols mpls ldp im- port ipv6 import-filter filter-access-list6	protocols mpls ldp im- port ipv6 import-filter filter-access-list6 <value>
×	<i>set protocols mpls ldp import ipv6 import-filter filter-access-list6 <1-2699></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	protocols_mpls.xml.in: protocols mpls ldp im- port ipv6 import-filter neighbor-access-list6	protocols mpls ldp im- port ipv6 import-filter neighbor-access-list6 <value>
×	<i>set protocols mpls ldp import ipv6 import-filter neighbor-access-list6 <1-2699></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	protocols_mpls.xml.in: protocols mpls ldp inter- face	protocols mpls ldp inter- face <value>
×	<i>set protocols mpls ldp interface <interface></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set protocols mpls ldp interface <interface> disable-establish-hel</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set protocols mpls ldp neighbor <ipv4-address> password <password></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set protocols mpls ldp neighbor <ipv4-address> session-holdtime <15-65535></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set protocols mpls ldp neighbor <ipv4-address> ttl-security <disable 1-254></i>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	protocols_mpls.xml.in: protocols mpls ldp neighbor <neighbor> password	protocols mpls ldp neighbor <text> password <value>
×	not yet documented	protocols_mpls.xml.in: protocols mpls ldp neighbor <neighbor> session-holdtime	protocols mpls ldp neighbor <text> session-holdtime <value>
×	not yet documented	protocols_mpls.xml.in: protocols mpls ldp neighbor <neighbor> ttl-security	protocols mpls ldp neighbor <text> ttl-security <value>
✓	<i>set protocols mpls ldp parameters cisco-interop-tlv</i>	protocols_mpls.xml.in: protocols mpls ldp parameters cisco-interop-tlv	protocols mpls ldp parameters cisco-interop-tlv
✓	<i>set protocols mpls ldp parameters ordered-control</i>	protocols_mpls.xml.in: protocols mpls ldp parameters ordered-control	protocols mpls ldp parameters ordered-control
✓	<i>set protocols mpls ldp parameters transport-prefer-ipv4</i>	protocols_mpls.xml.in: protocols mpls ldp parameters transport-prefer-ipv4	protocols mpls ldp parameters transport-prefer-ipv4
×	not yet documented	protocols_mpls.xml.in: protocols mpls ldp router-id	protocols mpls ldp router-id <value>
×	<i>set protocols mpls ldp router-id <address></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	protocols_mpls.xml.in: protocols mpls ldp targeted-neighbor ipv4 address	protocols mpls ldp targeted-neighbor ipv4 address <value>
×	<i>set protocols mpls ldp targeted-neighbor ipv4 address <address></i>	Nothing found in XML Definitions	Nothing found in VyOS
✓	<i>set protocols mpls ldp targeted-neighbor ipv4 enable</i>	protocols_mpls.xml.in: protocols mpls ldp targeted-neighbor ipv4 enable	protocols mpls ldp targeted-neighbor ipv4 enable
×	not yet documented	protocols_mpls.xml.in: protocols mpls ldp targeted-neighbor ipv4 hello-holdtime	protocols mpls ldp targeted-neighbor ipv4 hello-holdtime <value>
×	<i>set protocols mpls ldp targeted-neighbor ipv4 hello-holdtime <1-65535></i>	Nothing found in XML Definitions	Nothing found in VyOS

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Table 1 – continued from previous page

Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	protocols_mpls.xml.in: protocols mpls ldp targeted-neighbor ipv4 hello-interval	protocols mpls ldp targeted-neighbor ipv4 hello-interval <value>
×	<i>set protocols mpls ldp targeted-neighbor ipv4 hello-interval <1-65535></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	protocols_mpls.xml.in: protocols mpls ldp targeted-neighbor ipv6 address	protocols mpls ldp targeted-neighbor ipv6 address <value>
×	<i>set protocols mpls ldp targeted-neighbor ipv6 address <address></i>	Nothing found in XML Definitions	Nothing found in VyOS
✓	<i>set protocols mpls ldp targeted-neighbor ipv6 enable</i>	protocols_mpls.xml.in: protocols mpls ldp targeted-neighbor ipv6 enable	protocols mpls ldp targeted-neighbor ipv6 enable
×	not yet documented	protocols_mpls.xml.in: protocols mpls ldp targeted-neighbor ipv6 hello-holdtime	protocols mpls ldp targeted-neighbor ipv6 hello-holdtime <value>
×	<i>set protocols mpls ldp targeted-neighbor ipv6 hello-holdtime <1-65535></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	protocols_mpls.xml.in: protocols mpls ldp targeted-neighbor ipv6 hello-interval	protocols mpls ldp targeted-neighbor ipv6 hello-interval <value>
×	<i>set protocols mpls ldp targeted-neighbor ipv6 hello-interval <1-65535></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	protocols_mpls.xml.in: protocols mpls parameters maximum-ttl	protocols mpls parameters maximum-ttl <value>
×	<i>set protocols mpls parameters maximum-ttl <1-255></i>	Nothing found in XML Definitions	Nothing found in VyOS
✓	<i>set protocols mpls parameters no-propagate-ttl</i>	protocols_mpls.xml.in: protocols mpls parameters no-propagate-ttl	protocols mpls parameters no-propagate-ttl

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set protocols nhrp tunnel <tunnel> authentication <secret></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set protocols nhrp tunnel <tunnel> holdtime <timeout></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set protocols nhrp tunnel <tunnel> map tunnel-ip <tunnel-ip> nbma <nbma-ip></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set protocols nhrp tunnel <tunnel> mtu <mtu></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set protocols nhrp tunnel <tunnel> multicast <nbma-ip></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set protocols nhrp tunnel <tunnel> network-id <network-id></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set protocols nhrp tunnel <tunnel> nhs tunnel-ip <tunnel-ip> nbma <nbma-ip></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set protocols nhrp tunnel <tunnel> redirect</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set protocols nhrp tunnel <tunnel> registration-no-unique</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set protocols nhrp tunnel <tunnel> shortcut</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	protocols_nhrp.xml.in: protocols nhrp tunnel protocol <tunnel> cisco-authentication	protocols nhrp tunnel <text> cisco-authentication <value>
×	not yet documented	protocols_nhrp.xml.in: protocols nhrp tunnel <tunnel> dynamic-map <dynamic-map> nbma-domain-name	protocols nhrp tunnel <text> dynamic-map <text> nbma-domain-name <value>
×	not yet documented	protocols_nhrp.xml.in: protocols nhrp tunnel <tunnel> holding-time	protocols nhrp tunnel <text> holding-time <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	protocols_nhrp.xml.in: protocols nhrp tunnel <tunnel> map <map> cisco	protocols nhrp tunnel <text> map <text> cisco
×	not yet documented	protocols_nhrp.xml.in: protocols nhrp tunnel <tunnel> map <map> nbma-address	protocols nhrp tunnel <text> map <text> nbma- address <value>
×	not yet documented	protocols_nhrp.xml.in: protocols nhrp tunnel <tunnel> map <map> register	protocols nhrp tunnel <text> map <text> register
×	not yet documented	protocols_nhrp.xml.in: protocols nhrp tunnel <tunnel> multicast	protocols nhrp tunnel <text> multicast <value>
×	not yet documented	protocols_nhrp.xml.in: protocols nhrp tunnel <tunnel> non-caching	protocols nhrp tunnel <text> non-caching
×	not yet documented	protocols_nhrp.xml.in: protocols nhrp tunnel <tunnel> redirect	protocols nhrp tunnel <text> redirect
×	not yet documented	protocols_nhrp.xml.in: protocols nhrp tunnel <tunnel> shortcut	protocols nhrp tunnel <text> shortcut
×	not yet documented	protocols_nhrp.xml.in: protocols nhrp tunnel <tunnel> shortcut- destination	protocols nhrp tun- nel <text> shortcut- destination
×	not yet documented	protocols_nhrp.xml.in: protocols nhrp tunnel <tunnel> shortcut-target <shortcut-target> holding- time	protocols nhrp tunnel <text> shortcut-target <text> holding-time <value>
×	<i>set protocols openfabric domain <name> domain-password <plaintext-password/m <password></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set protocols openfabric domain <name> fabric-tier <0-14></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set protocols openfabric domain <name> interface <interface> address-family <ipv4/ipv6></i>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set protocols openfabric domain <name> interface <interface> csnp-interval <1-600></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set protocols openfabric domain <name> interface <interface> hello-interval <1-600></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set protocols openfabric domain <name> interface <interface> hello-multiplier <2-100></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set protocols openfabric domain <name> interface <interface> metric <0-16777215></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set protocols openfabric domain <name> interface <interface> passive</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set protocols openfabric domain <name> interface <interface> password <plaintext-password/m <password></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set protocols openfabric domain <name> interface <interface> psnp-interval <0-120></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set protocols openfabric domain <name> log-adjacency-changes</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set protocols openfabric domain <name> lsp-gen-interval <1-120></code>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<i>set protocols openfabric domain <name> lsp-refresh-interval <1-65235></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set protocols openfabric domain <name> max-lsp-lifetime <360-65535></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set protocols openfabric domain <name> purge-originator</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set protocols openfabric domain <name> set-overload-bit</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set protocols openfabric domain <name> spf-interval <1-120></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set protocols openfabric net <network-entity-title</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	protocols_ospf.xml.in: protocols ospf access-list <access-list> export	protocols ospf access-list <text> export <value>
×	not yet documented	protocols_ospf.xml.in: protocols ospf aggregation timer	protocols ospf aggregation timer <value>
×	<i>set protocols ospf aggregation timer <seconds></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set protocols ospf area <number> area-type nssa</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set protocols ospf area <number> area-type nssa default-cost <number></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set protocols ospf area <number> area-type nssa no-summary</i>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set protocols ospf area <number> area-type nssa translate <always/candidate/nev</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set protocols ospf area <number> area-type stub</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set protocols ospf area <number> area-type stub default-cost <number></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set protocols ospf area <number> area-type stub no-summary</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set protocols ospf area <number> authentication md5</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set protocols ospf area <number> authentication plaintext-password</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set protocols ospf area <number> export-list <acl_number></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set protocols ospf area <number> import-list <acl_number></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set protocols ospf area <number> network <A.B.C.D/ M></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set protocols ospf area <number> range <A.B.C.D/M> [cost <number>]</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set protocols ospf area <number> range <A.B.C.D/M> not-advertise</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set protocols ospf area <number> range <A.B.C.D/M> substitute <E.F.G.H/ M></code>	Nothing found in XML Definitions	Nothing found in VyOS

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Table 1 – continued from previous page

Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set protocols ospf area <number> shortcut <default/disable/enab</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set protocols ospf area <number> virtual-link <A.B.C.D></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	protocols_ospf.xml.in: protocols ospf area <area> area-type normal	protocols ospf area <text> area-type normal
×	not yet documented	protocols_ospf.xml.in: protocols ospf area <area> area-type nssa default-cost	protocols ospf area <text> area-type nssa default-cost <value>
×	not yet documented	protocols_ospf.xml.in: protocols ospf area <area> area-type nssa no-summary	protocols ospf area <text> area-type nssa no-summary
×	not yet documented	protocols_ospf.xml.in: protocols ospf area <area> area-type nssa translate	protocols ospf area <text> area-type nssa translate <value>
×	not yet documented	protocols_ospf.xml.in: protocols ospf area <area> area-type stub default-cost	protocols ospf area <text> area-type stub default-cost <value>
×	not yet documented	protocols_ospf.xml.in: protocols ospf area <area> area-type stub no-summary	protocols ospf area <text> area-type stub no-summary
×	not yet documented	protocols_ospf.xml.in: protocols ospf area <area> authentication	protocols ospf area <text> authentication <value>
×	not yet documented	protocols_ospf.xml.in: protocols ospf area <area> export-list	protocols ospf area <text> export-list <value>
×	not yet documented	protocols_ospf.xml.in: protocols ospf area <area> import-list	protocols ospf area <text> import-list <value>
×	not yet documented	protocols_ospf.xml.in: protocols ospf area <area> network	protocols ospf area <text> network <value>
×	not yet documented	protocols_ospf.xml.in: protocols ospf area <area> range <range> cost	protocols ospf area <text> range <text> cost <value>
×	not yet documented	protocols_ospf.xml.in: protocols ospf area <area> range <range> not-advertise	protocols ospf area <text> range <text> not-advertise

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	protocols_ospf.xml.in: protocols ospf area <area> range <range> substitute	protocols ospf area <text> range <text> substitute <value>
×	not yet documented	protocols_ospf.xml.in: protocols ospf area <area> shortcut	protocols ospf area <text> shortcut <value>
×	not yet documented	protocols_ospf.xml.in: protocols ospf area <area> virtual-link <virtual-link> authentication md5 key-id <key-id> md5-key	protocols ospf area <text> virtual-link <text> authen- tication md5 key-id <text> md5-key <value>
×	not yet documented	protocols_ospf.xml.in: protocols ospf area <area> virtual-link <virtual-link> authentication plaintext- password	protocols ospf area <text> virtual-link <text> authentication plaintext- password <value>
×	not yet documented	protocols_ospf.xml.in: protocols ospf area <area> virtual-link <virtual-link> dead-interval	protocols ospf area <text> virtual-link <text> dead- interval <value>
×	not yet documented	protocols_ospf.xml.in: protocols ospf area <area> virtual-link <virtual-link> hello-interval	protocols ospf area <text> virtual-link <text> hello- interval <value>
×	not yet documented	protocols_ospf.xml.in: protocols ospf area <area> virtual-link <virtual-link> retransmit-interval	protocols ospf area <text> virtual-link <text> retransmit-interval <value>
×	not yet documented	protocols_ospf.xml.in: protocols ospf area <area> virtual-link <virtual-link> transmit-delay	protocols ospf area <text> virtual-link <text> transmit-delay <value>
×	not yet documented	protocols_ospf.xml.in: protocols ospf auto-cost reference-bandwidth	protocols ospf auto-cost reference-bandwidth <value>
×	<i>set protocols ospf auto-cost reference-bandwidth <number></i>	Nothing found in XML Definitions	Nothing found in VyOS
✓	<i>set protocols ospf capability opaque</i>	protocols_ospf.xml.in: protocols ospf capability opaque	protocols ospf capability opaque
×	<i>set protocols ospf default-information originate [always] [metric <number>] [metric-type <1 2>] [route-map <name>]</i>	Nothing found in XML Definitions	Nothing found in VyOS

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Table 1 – continued from previous page

Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	protocols_ospf.xml.in: protocols ospf default- information originate always	protocols ospf default- information originate always
×	not yet documented	protocols_ospf.xml.in: protocols ospf default- information originate metric	protocols ospf default- information originate metric <value>
×	not yet documented	protocols_ospf.xml.in: protocols ospf default- information originate metric-type	protocols ospf default- information originate metric-type <value>
×	not yet documented	protocols_ospf.xml.in: protocols ospf default- information originate route-map	protocols ospf default- information originate route-map <value>
×	not yet documented	protocols_ospf.xml.in: protocols ospf default- metric	protocols ospf default- metric <value>
×	<i>set protocols ospf default-metric <number></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	protocols_ospf.xml.in: protocols ospf distance global	protocols ospf distance global <value>
×	<i>set protocols ospf distance global <distance></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set protocols ospf distance ospf <external/inter-area/ <distance></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	protocols_ospf.xml.in: protocols ospf distance ospf external	protocols ospf distance ospf external <value>
×	not yet documented	protocols_ospf.xml.in: protocols ospf distance ospf inter-area	protocols ospf distance ospf inter-area <value>
×	not yet documented	protocols_ospf.xml.in: protocols ospf distance ospf intra-area	protocols ospf distance ospf intra-area <value>
×	<i>set protocols ospf graceful-restart [grace-period (1-1800)]</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	protocols_ospf.xml.in: protocols ospf graceful- restart grace-period	protocols ospf graceful- restart grace-period <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<i>set protocols ospf graceful-restart helper enable [router-id A.B.C.D]</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	protocols_ospf.xml.in: protocols ospf graceful-restart helper enable router-id	protocols ospf graceful-restart helper enable router-id <value>
✓	<i>set protocols ospf graceful-restart helper no-strict-lsa-checkin</i>	protocols_ospf.xml.in: protocols ospf graceful-restart helper no-strict-lsa-checking	protocols ospf graceful-restart helper no-strict-lsa-checking
✓	<i>set protocols ospf graceful-restart helper planned-only</i>	protocols_ospf.xml.in: protocols ospf graceful-restart helper planned-only	protocols ospf graceful-restart helper planned-only
✓	<i>set protocols ospf graceful-restart helper supported-grace-time</i>	protocols_ospf.xml.in: protocols ospf graceful-restart helper supported-grace-time	protocols ospf graceful-restart helper supported-grace-time <value>
×	<i>set protocols ospf interface <interface> area <x.x.x.x x></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set protocols ospf interface <interface> authentication md5 key-id <id> md5-key <text></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set protocols ospf interface <interface> authentication plaintext-password <text></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set protocols ospf interface <interface> bandwidth <number></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set protocols ospf interface <interface> bfd</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set protocols ospf interface <interface> cost <number></i>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set protocols ospf interface <interface> dead-interval <number></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set protocols ospf interface <interface> hello-interval <number></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set protocols ospf interface <interface> hello-multiplier <number></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set protocols ospf interface <interface> ldp-sync disable</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set protocols ospf interface <interface> ldp-sync holddown <seconds></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set protocols ospf interface <interface> mtu-ignore</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set protocols ospf interface <interface> network <type></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set protocols ospf interface <interface> passive [disable]</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set protocols ospf interface <interface> priority <number></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set protocols ospf interface <interface> retransmit-interval <number></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set protocols ospf interface <interface> transmit-delay <number></code>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	protocols_ospf.xml.in: protocols ospf interface <interface> area	protocols ospf interface <text> area <value>
×	not yet documented	protocols_ospf.xml.in: protocols ospf interface <interface> authentication md5 key-id <key-id> md5-key	protocols ospf interface <text> authentication md5 key-id <text> md5-key <value>
×	not yet documented	protocols_ospf.xml.in: protocols ospf interface <interface> authentication plaintext-password	protocols ospf inter- face <text> authentica- tion plaintext-password <value>
×	not yet documented	protocols_ospf.xml.in: protocols ospf interface <interface> bandwidth	protocols ospf interface <text> bandwidth <value>
×	not yet documented	protocols_ospf.xml.in: protocols ospf interface <interface> bfd profile	protocols ospf interface <text> bfd profile <value>
×	not yet documented	protocols_ospf.xml.in: protocols ospf interface <interface> cost	protocols ospf interface <text> cost <value>
×	not yet documented	protocols_ospf.xml.in: protocols ospf interface <interface> dead-interval	protocols ospf interface <text> dead-interval <value>
×	not yet documented	protocols_ospf.xml.in: protocols ospf interface <interface> hello-interval	protocols ospf interface <text> hello-interval <value>
×	not yet documented	protocols_ospf.xml.in: protocols ospf inter- face <interface> hello- multiplier	protocols ospf interface <text> hello-multiplier <value>
×	not yet documented	protocols_ospf.xml.in: protocols ospf interface <interface> ldp-sync disable	protocols ospf interface <text> ldp-sync disable
×	not yet documented	protocols_ospf.xml.in: protocols ospf interface <interface> ldp-sync holddown	protocols ospf interface <text> ldp-sync holddown <value>
×	not yet documented	protocols_ospf.xml.in: protocols ospf interface <interface> mtu-ignore	protocols ospf interface <text> mtu-ignore
×	not yet documented	protocols_ospf.xml.in: protocols ospf interface <interface> network	protocols ospf interface <text> network <value>
×	not yet documented	protocols_ospf.xml.in: protocols ospf inter- face <interface> passive disable	protocols ospf interface <text> passive disable

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	protocols_ospf.xml.in: protocols ospf interface <interface> priority	protocols ospf interface <text> priority <value>
×	not yet documented	protocols_ospf.xml.in: protocols ospf interface <interface> retransmit- interval	protocols ospf interface <text> retransmit-interval <value>
×	not yet documented	protocols_ospf.xml.in: protocols ospf interface <interface> transmit-delay	protocols ospf interface <text> transmit-delay <value>
×	<i>set protocols ospf ldp-sync</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	protocols_ospf.xml.in: protocols ospf ldp-sync holddown	protocols ospf ldp-sync holddown <value>
×	<i>set protocols ospf ldp-sync holddown <seconds></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set protocols ospf log-adjacency-changes [detail]</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	protocols_ospf.xml.in: protocols ospf log- adjacency-changes detail	protocols ospf log- adjacency-changes detail
×	<i>set protocols ospf max-metric router-lsa <administrative/on-sh <seconds>/on-startup <seconds>></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	protocols_ospf.xml.in: protocols ospf max-metric router-lsa administrative	protocols ospf max-metric router-lsa administrative
×	not yet documented	protocols_ospf.xml.in: protocols ospf max-metric router-lsa on-shutdown	protocols ospf max-metric router-lsa on-shutdown <value>
×	not yet documented	protocols_ospf.xml.in: protocols ospf max-metric router-lsa on-startup	protocols ospf max-metric router-lsa on-startup <value>
×	not yet documented	protocols_ospf.xml.in: protocols ospf maximum- paths	protocols ospf maximum- paths <value>
×	<i>set protocols ospf maximum-paths <1-64></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	protocols_ospf.xml.in: protocols ospf mpls-te en- able	protocols ospf mpls-te en- able

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	protocols_ospf.xml.in: protocols ospf mpls-te router-address	protocols ospf mpls-te router-address <value>
×	<i>set protocols ospf neighbor <A.B.C.D></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set protocols ospf neighbor <A.B.C. D> poll-interval <seconds></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set protocols ospf neighbor <A.B.C.D> priority <number></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	protocols_ospf.xml.in: protocols ospf neighbor <neighbor> poll-interval	protocols ospf neigh- bor <text> poll-interval <value>
×	not yet documented	protocols_ospf.xml.in: protocols ospf neighbor <neighbor> priority	protocols ospf neighbor <text> priority <value>
×	not yet documented	protocols_ospf.xml.in: protocols ospf parameters abr-type	protocols ospf parameters abr-type <value>
×	<i>set protocols ospf parameters abr-type <cisco/ibm/shortcut/s</i>	Nothing found in XML Definitions	Nothing found in VyOS
✓	<i>set protocols ospf parameters opaque-lsa</i>	protocols_ospf.xml.in: protocols ospf parameters opaque-lsa	protocols ospf parameters opaque-lsa
✓	<i>set protocols ospf parameters rfc1583-compatibility</i>	protocols_ospf.xml.in: protocols ospf parameters rfc1583-compatibility	protocols ospf parameters rfc1583-compatibility
×	not yet documented	protocols_ospf.xml.in: protocols ospf parameters router-id	protocols ospf parameters router-id <value>
×	<i>set protocols ospf parameters router-id <rid></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	protocols_ospf.xml.in: protocols ospf passive- interface	protocols ospf passive- interface <value>
×	<i>set protocols ospf passive-interface default</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set protocols ospf redistribute <route source></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set protocols ospf redistribute <route source> metric-type <1/2></i>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<i>set protocols ospf redistribute <route source> metric <number></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set protocols ospf redistribute <route source> route-map <name></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	protocols_ospf.xml.in: protocols ospf redistribute babel metric	protocols ospf redistribute babel metric <value>
×	not yet documented	protocols_ospf.xml.in: protocols ospf redistribute babel metric-type	protocols ospf redistribute babel metric-type <value>
×	not yet documented	protocols_ospf.xml.in: protocols ospf redistribute babel route-map	protocols ospf redistribute babel route-map <value>
×	not yet documented	protocols_ospf.xml.in: protocols ospf redistribute bgp metric	protocols ospf redistribute bgp metric <value>
×	not yet documented	protocols_ospf.xml.in: protocols ospf redistribute bgp metric-type	protocols ospf redistribute bgp metric-type <value>
×	not yet documented	protocols_ospf.xml.in: protocols ospf redistribute bgp route-map	protocols ospf redistribute bgp route-map <value>
×	not yet documented	protocols_ospf.xml.in: protocols ospf redistribute connected metric	protocols ospf redistribute connected metric <value>
×	not yet documented	protocols_ospf.xml.in: protocols ospf redistribute connected metric-type	protocols ospf redistribute connected metric-type <value>
×	not yet documented	protocols_ospf.xml.in: protocols ospf redistribute connected route-map	protocols ospf redistribute connected route-map <value>
×	not yet documented	protocols_ospf.xml.in: protocols ospf redistribute isis metric	protocols ospf redistribute isis metric <value>
×	not yet documented	protocols_ospf.xml.in: protocols ospf redistribute isis metric-type	protocols ospf redistribute isis metric-type <value>
×	not yet documented	protocols_ospf.xml.in: protocols ospf redistribute isis route-map	protocols ospf redistribute isis route-map <value>
×	not yet documented	protocols_ospf.xml.in: protocols ospf redistribute kernel metric	protocols ospf redistribute kernel metric <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	protocols_ospf.xml.in: protocols ospf redistribute kernel metric-type	protocols ospf redis- tribute kernel metric-type <value>
×	not yet documented	protocols_ospf.xml.in: protocols ospf redistribute kernel route-map	protocols ospf redistribute kernel route-map <value>
×	not yet documented	protocols_ospf.xml.in: protocols ospf redistribute rip metric	protocols ospf redistribute rip metric <value>
×	not yet documented	protocols_ospf.xml.in: protocols ospf redistribute rip metric-type	protocols ospf redistribute rip metric-type <value>
×	not yet documented	protocols_ospf.xml.in: protocols ospf redistribute rip route-map	protocols ospf redistribute rip route-map <value>
×	not yet documented	protocols_ospf.xml.in: protocols ospf redistribute static metric	protocols ospf redistribute static metric <value>
×	not yet documented	protocols_ospf.xml.in: protocols ospf redistribute static metric-type	protocols ospf redistribute static metric-type <value>
×	not yet documented	protocols_ospf.xml.in: protocols ospf redistribute static route-map	protocols ospf redistribute static route-map <value>
×	not yet documented	protocols_ospf.xml.in: protocols ospf redistribute table <table> metric	protocols ospf redis- tribute table <text> metric <value>
×	not yet documented	protocols_ospf.xml.in: protocols ospf redistribute table <table> metric-type	protocols ospf redistribute table <text> metric-type <value>
×	not yet documented	protocols_ospf.xml.in: protocols ospf redistribute table <table> route-map	protocols ospf redistribute table <text> route-map <value>
×	not yet documented	protocols_ospf.xml.in: protocols ospf refresh timers	protocols ospf refresh timers <value>
×	<i>set protocols ospf refresh timers <seconds></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	protocols_ospf.xml.in: protocols ospf segment- routing global-block high- label-value	protocols ospf segment- routing global-block high- label-value <value>
×	<i>set protocols ospf segment-routing global-block high-label-value <16-1048575></i>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	protocols_ospf.xml.in: protocols ospf segment- routing global-block low- label-value	protocols ospf segment- routing global-block low- label-value <value>
×	<i>set protocols ospf segment-routing global-block low-label-value <16-1048575></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	protocols_ospf.xml.in: protocols ospf segment- routing local-block high- label-value	protocols ospf segment- routing local-block high- label-value <value>
×	<i>set protocols ospf segment-routing local-block high-label-value <16-1048575></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	protocols_ospf.xml.in: protocols ospf segment- routing local-block low- label-value	protocols ospf segment- routing local-block low- label-value <value>
×	<i>set protocols ospf segment-routing local-block low-label-value <16-1048575></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	protocols_ospf.xml.in: protocols ospf segment- routing maximum-label- depth	protocols ospf segment- routing maximum-label- depth <value>
×	<i>set protocols ospf segment-routing maximum-label-depth <1-16></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set protocols ospf segment-routing prefix <address> index <no-php-flag / explicit-null></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set protocols ospf segment-routing prefix <address> index value <0-65535></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	protocols_ospf.xml.in: protocols ospf segment- routing prefix <prefix> in- dex explicit-null	protocols ospf segment- routing prefix <text> index explicit-null

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	protocols_ospf.xml.in: protocols ospf segment- routing prefix <prefix> in- dex no-php-flag	protocols ospf segment- routing prefix <text> index no-php-flag
×	not yet documented	protocols_ospf.xml.in: protocols ospf segment- routing prefix <prefix> in- dex value	protocols ospf segment- routing prefix <text> index value <value>
×	not yet documented	protocols_ospf.xml.in: protocols ospf summary- address <summary- address> no-advertise	protocols ospf summary- address <text> no- advertise
×	not yet documented	protocols_ospf.xml.in: protocols ospf summary- address <summary- address> tag	protocols ospf summary- address <text> tag <value>
×	<code>set protocols ospf summary-address x.x.x.x/y [tag (1-4294967295)]</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set protocols ospf summary-address x.x. x.x/y no-advertise</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set protocols ospf timers throttle spf <delay/initial-holdti- <seconds></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	protocols_ospf.xml.in: protocols ospf timers throttle spf delay	protocols ospf timers throttle spf delay <value>
×	not yet documented	protocols_ospf.xml.in: protocols ospf timers throttle spf initial- holdtime	protocols ospf timers throttle spf initial- holdtime <value>
×	not yet documented	protocols_ospf.xml.in: protocols ospf timers throttle spf max-holdtime	protocols ospf timers throttle spf max-holdtime <value>
×	<code>set protocols ospfv3 area <number> range <prefix></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set protocols ospfv3 area <number> range <prefix> not-advertise</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	protocols_ospfv3.xml.in: protocols ospfv3 area <area> area-type nssa default-information- originate	protocols ospfv3 area <text> area-type nssa default-information- originate

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	protocols_ospfv3.xml.in: protocols ospfv3 area <area> area-type nssa no-summary	protocols ospfv3 area <text> area-type nssa no-summary
×	not yet documented	protocols_ospfv3.xml.in: protocols ospfv3 area <area> area-type stub no-summary	protocols ospfv3 area <text> area-type stub no-summary
×	not yet documented	protocols_ospfv3.xml.in: protocols ospfv3 area <area> export-list	protocols ospfv3 area <text> export-list <value>
×	not yet documented	protocols_ospfv3.xml.in: protocols ospfv3 area <area> import-list	protocols ospfv3 area <text> import-list <value>
×	not yet documented	protocols_ospfv3.xml.in: protocols ospfv3 area <area> range <range> advertise	protocols ospfv3 area <text> range <text> advertise
×	not yet documented	protocols_ospfv3.xml.in: protocols ospfv3 area <area> range <range> not-advertise	protocols ospfv3 area <text> range <text> not-advertise
×	not yet documented	protocols_ospfv3.xml.in: protocols ospfv3 auto-cost reference-bandwidth	protocols ospfv3 auto- cost reference-bandwidth <value>
×	not yet documented	protocols_ospfv3.xml.in: protocols ospfv3 default- information originate al- ways	protocols ospfv3 default- information originate al- ways
×	not yet documented	protocols_ospfv3.xml.in: protocols ospfv3 default- information originate met- ric	protocols ospfv3 default- information originate met- ric <value>
×	not yet documented	protocols_ospfv3.xml.in: protocols ospfv3 default- information originate metric-type	protocols ospfv3 default- information originate metric-type <value>
×	not yet documented	protocols_ospfv3.xml.in: protocols ospfv3 default- information originate route-map	protocols ospfv3 default- information originate route-map <value>
×	not yet documented	protocols_ospfv3.xml.in: protocols ospfv3 distance global	protocols ospfv3 distance global <value>
×	<i>set protocols ospfv3 distance global <distance></i>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<i>set protocols ospfv3 distance ospfv3 <external/inter-area/> <distance></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	protocols_ospfv3.xml.in: protocols ospfv3 distance ospfv3 external	protocols ospfv3 distance ospfv3 external <value>
×	not yet documented	protocols_ospfv3.xml.in: protocols ospfv3 distance ospfv3 inter-area	protocols ospfv3 distance ospfv3 inter-area <value>
×	not yet documented	protocols_ospfv3.xml.in: protocols ospfv3 distance ospfv3 intra-area	protocols ospfv3 distance ospfv3 intra-area <value>
×	<i>set protocols ospfv3 graceful-restart [grace-period (1-1800)]</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	protocols_ospfv3.xml.in: protocols ospfv3 graceful-restart grace-period	protocols ospfv3 graceful-restart grace-period <value>
×	<i>set protocols ospfv3 graceful-restart helper enable [router-id A.B.C.D]</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	protocols_ospfv3.xml.in: protocols ospfv3 graceful-restart helper enable router-id	protocols ospfv3 graceful-restart helper enable router-id <value>
✓	<i>set protocols ospfv3 graceful-restart helper lsa-check-disable</i>	protocols_ospfv3.xml.in: protocols ospfv3 graceful-restart helper lsa-check-disable	protocols ospfv3 graceful-restart helper lsa-check-disable
✓	<i>set protocols ospfv3 graceful-restart helper planned-only</i>	protocols_ospfv3.xml.in: protocols ospfv3 graceful-restart helper planned-only	protocols ospfv3 graceful-restart helper planned-only
✓	<i>set protocols ospfv3 graceful-restart helper supported-grace-time</i>	protocols_ospfv3.xml.in: protocols ospfv3 graceful-restart helper supported-grace-time	protocols ospfv3 graceful-restart helper supported-grace-time <value>
×	<i>set protocols ospfv3 interface <interface> area <number></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set protocols ospfv3 interface <interface> bfd</i>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set protocols ospfv3 interface <interface> dead-interval <number></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set protocols ospfv3 interface <interface> hello-interval <number></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set protocols ospfv3 interface <interface> ipv6 cost <number></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set protocols ospfv3 interface <interface> mtu-ignore</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set protocols ospfv3 interface <interface> network <type></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set protocols ospfv3 interface <interface> passive</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set protocols ospfv3 interface <interface> priority <number></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set protocols ospfv3 interface <interface> retransmit-interval <number></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set protocols ospfv3 interface <interface> transmit-delay <number></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	protocols_ospfv3.xml.in: protocols ospfv3 interface <interface> area	protocols ospfv3 interface <text> area <value>
×	not yet documented	protocols_ospfv3.xml.in: protocols ospfv3 interface <interface> bfd profile	protocols ospfv3 interface <text> bfd profile <value>
×	not yet documented	protocols_ospfv3.xml.in: protocols ospfv3 interface <interface> cost	protocols ospfv3 interface <text> cost <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	protocols_ospfv3.xml.in: protocols ospfv3 interface <interface> dead-interval	protocols ospfv3 interface <text> dead-interval <value>
×	not yet documented	protocols_ospfv3.xml.in: protocols ospfv3 interface <interface> hello-interval	protocols ospfv3 interface <text> hello-interval <value>
×	not yet documented	protocols_ospfv3.xml.in: protocols ospfv3 interface <interface> ifmtu	protocols ospfv3 interface <text> ifmtu <value>
×	not yet documented	protocols_ospfv3.xml.in: protocols ospfv3 interface <interface> instance-id	protocols ospfv3 interface <text> instance-id <value>
×	not yet documented	protocols_ospfv3.xml.in: protocols ospfv3 interface <interface> mtu-ignore	protocols ospfv3 interface <text> mtu-ignore
×	not yet documented	protocols_ospfv3.xml.in: protocols ospfv3 interface <interface> network	protocols ospfv3 interface <text> network <value>
×	not yet documented	protocols_ospfv3.xml.in: protocols ospfv3 interface <interface> passive	protocols ospfv3 interface <text> passive
×	not yet documented	protocols_ospfv3.xml.in: protocols ospfv3 interface <interface> priority	protocols ospfv3 interface <text> priority <value>
×	not yet documented	protocols_ospfv3.xml.in: protocols ospfv3 interface <interface> retransmit-interval	protocols ospfv3 interface <text> retransmit-interval <value>
×	not yet documented	protocols_ospfv3.xml.in: protocols ospfv3 interface <interface> transmit-delay	protocols ospfv3 interface <text> transmit-delay <value>
×	not yet documented	protocols_ospfv3.xml.in: protocols ospfv3 log-adjacency-changes detail	protocols ospfv3 log-adjacency-changes detail
×	not yet documented	protocols_ospfv3.xml.in: protocols ospfv3 parameters router-id	protocols ospfv3 parameters router-id <value>
×	<i>set protocols ospfv3 parameters router-id <rid></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set protocols ospfv3 redistribute <route source></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	protocols_ospfv3.xml.in: protocols ospfv3 redistribute babel metric	Nothing found in VyOS
×	not yet documented	protocols_ospfv3.xml.in: protocols ospfv3 redistribute babel metric-type	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	protocols_ospfv3.xml.in: protocols ospfv3 redis- tribute babel route-map	protocols ospfv3 redis- tribute babel route-map <value>
×	not yet documented	protocols_ospfv3.xml.in: protocols ospfv3 redis- tribute bgp metric	Nothing found in VyOS
×	not yet documented	protocols_ospfv3.xml.in: protocols ospfv3 redis- tribute bgp metric-type	Nothing found in VyOS
×	not yet documented	protocols_ospfv3.xml.in: protocols ospfv3 redis- tribute bgp route-map	protocols ospfv3 redis- tribute bgp route-map <value>
×	not yet documented	protocols_ospfv3.xml.in: protocols ospfv3 redis- tribute connected metric	Nothing found in VyOS
×	not yet documented	protocols_ospfv3.xml.in: protocols ospfv3 re- distribute connected metric-type	Nothing found in VyOS
×	not yet documented	protocols_ospfv3.xml.in: protocols ospfv3 re- distribute connected route-map	protocols ospfv3 re- distribute connected route-map <value>
×	not yet documented	protocols_ospfv3.xml.in: protocols ospfv3 redis- tribute isis metric	Nothing found in VyOS
×	not yet documented	protocols_ospfv3.xml.in: protocols ospfv3 redis- tribute isis metric-type	Nothing found in VyOS
×	not yet documented	protocols_ospfv3.xml.in: protocols ospfv3 redis- tribute isis route-map	Nothing found in VyOS
×	not yet documented	protocols_ospfv3.xml.in: protocols ospfv3 redis- tribute kernel metric	Nothing found in VyOS
×	not yet documented	protocols_ospfv3.xml.in: protocols ospfv3 redis- tribute kernel metric-type	Nothing found in VyOS
×	not yet documented	protocols_ospfv3.xml.in: protocols ospfv3 redis- tribute kernel route-map	protocols ospfv3 redis- tribute kernel route-map <value>
×	not yet documented	protocols_ospfv3.xml.in: protocols ospfv3 redis- tribute ripng metric	Nothing found in VyOS
×	not yet documented	protocols_ospfv3.xml.in: protocols ospfv3 redis- tribute ripng metric-type	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	protocols_ospfv3.xml.in: protocols ospfv3 redis- tribute ripng route-map	protocols ospfv3 redis- tribute ripng route-map <value>
×	not yet documented	protocols_ospfv3.xml.in: protocols ospfv3 redis- tribute static metric	Nothing found in VyOS
×	not yet documented	protocols_ospfv3.xml.in: protocols ospfv3 redis- tribute static metric-type	Nothing found in VyOS
×	not yet documented	protocols_ospfv3.xml.in: protocols ospfv3 redis- tribute static route-map	protocols ospfv3 redis- tribute static route-map <value>
×	<code>set protocols pim6 interface <interface-name></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set protocols pim6 interface <interface-name> mld disable</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set protocols pim6 interface <interface-name> mld interval <1-65535></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set protocols pim6 interface <interface-name> mld join <multicast-address></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set protocols pim6 interface <interface-name> mld join <multicast-address> source <source-address></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set protocols pim6 interface <interface-name> mld last-member-query-cou <1-255></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set protocols pim6 interface <interface-name> mld last-member-query-int <100-6553500></code>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set protocols pim6 interface <interface-name> mld max-response-time <100-6553500></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set protocols pim6 interface <interface-name> mld version <1-2></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	protocols_pim6.xml.in: protocols pim6 interface <interface> dr-priority	Nothing found in VyOS
×	not yet documented	protocols_pim6.xml.in: protocols pim6 interface <interface> hello	Nothing found in VyOS
×	not yet documented	protocols_pim6.xml.in: protocols pim6 interface <interface> mld disable	Nothing found in VyOS
×	not yet documented	protocols_pim6.xml.in: protocols pim6 interface <interface> mld interval	Nothing found in VyOS
×	not yet documented	protocols_pim6.xml.in: protocols pim6 interface <interface> mld join <join> source	Nothing found in VyOS
×	not yet documented	protocols_pim6.xml.in: protocols pim6 inter- face <interface> mld last-member-query-count	Nothing found in VyOS
×	not yet documented	protocols_pim6.xml.in: protocols pim6 interface <interface> mld last- member-query-interval	Nothing found in VyOS
×	not yet documented	protocols_pim6.xml.in: protocols pim6 inter- face <interface> mld max-response-time	Nothing found in VyOS
×	not yet documented	protocols_pim6.xml.in: protocols pim6 interface <interface> mld version	Nothing found in VyOS
×	not yet documented	protocols_pim6.xml.in: protocols pim6 interface <interface> no-bsm	Nothing found in VyOS
×	not yet documented	protocols_pim6.xml.in: protocols pim6 interface <interface> no-unicast- bsm	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	protocols_pim6.xml.in: protocols pim6 interface <interface> passive	Nothing found in VyOS
×	not yet documented	protocols_pim6.xml.in: protocols pim6 join-prune-interval	Nothing found in VyOS
×	not yet documented	protocols_pim6.xml.in: protocols pim6 keep-alive-timer	Nothing found in VyOS
×	not yet documented	protocols_pim6.xml.in: protocols pim6 packets	Nothing found in VyOS
×	not yet documented	protocols_pim6.xml.in: protocols pim6 register-suppress-time	Nothing found in VyOS
×	not yet documented	protocols_pim6.xml.in: protocols pim6 rp address <address> group	Nothing found in VyOS
×	not yet documented	protocols_pim6.xml.in: protocols pim6 rp address <address> prefix-list6	Nothing found in VyOS
×	not yet documented	protocols_pim6.xml.in: protocols pim6 rp keep-alive-timer	Nothing found in VyOS
×	<i>set protocols pim ecmp</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set protocols pim ecmp rebalance</i>	protocols_pim.xml.in: protocols pim ecmp rebalance	Nothing found in VyOS
×	not yet documented	protocols_pim.xml.in: protocols pim igmp watermark-warning	Nothing found in VyOS
×	<i>set protocols pim igmp watermark-warning <1-65535></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set protocols pim interface <interface> bfd</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set protocols pim interface <interface> bfd profile <name></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set protocols pim interface <interface> dr-priority <1-4294967295></i>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set protocols pim interface <interface> hello <1-180></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set protocols pim interface <interface> igmp</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set protocols pim interface <interface> igmp disable</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set protocols pim interface <interface> igmp join <multicast-address> source-address <IP-address></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set protocols pim interface <interface> igmp query-interval <1-1800></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set protocols pim interface <interface> igmp query-max-response-ti. <10-250></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set protocols pim interface <interface> igmp version <2-3></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set protocols pim interface <interface> no-bsm</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set protocols pim interface <interface> no-unicast-bsm</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set protocols pim interface <interface> passive</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set protocols pim interface <interface> source-address <ip-address></code>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	protocols_pim.xml.in: protocols pim interface <interface> bfd profile	Nothing found in VyOS
×	not yet documented	protocols_pim.xml.in: protocols pim interface <interface> dr-priority	protocols pim interface <text> dr-priority <value>
×	not yet documented	protocols_pim.xml.in: protocols pim interface <interface> hello	protocols pim interface <text> hello <value>
×	not yet documented	protocols_pim.xml.in: protocols pim interface <interface> igmp disable	Nothing found in VyOS
×	not yet documented	protocols_pim.xml.in: protocols pim interface <interface> igmp join <join> source-address	Nothing found in VyOS
×	not yet documented	protocols_pim.xml.in: protocols pim inter- face <interface> igmp query-interval	Nothing found in VyOS
×	not yet documented	protocols_pim.xml.in: protocols pim inter- face <interface> igmp query-max-response-time	Nothing found in VyOS
×	not yet documented	protocols_pim.xml.in: protocols pim interface <interface> igmp version	Nothing found in VyOS
×	not yet documented	protocols_pim.xml.in: protocols pim interface <interface> no-bsm	Nothing found in VyOS
×	not yet documented	protocols_pim.xml.in: protocols pim interface <interface> no-unicast- bsm	Nothing found in VyOS
×	not yet documented	protocols_pim.xml.in: protocols pim interface <interface> passive	Nothing found in VyOS
×	not yet documented	protocols_pim.xml.in: protocols pim interface <interface> source- address	Nothing found in VyOS
×	not yet documented	protocols_pim.xml.in: protocols pim join-prune- interval	Nothing found in VyOS
×	<i>set protocols pim join-prune-interval <1-65535></i>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	protocols_pim.xml.in: protocols pim keep-alive-timer	Nothing found in VyOS
×	<i>set protocols pim keep-alive-timer <1-65535></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set protocols pim no-v6-secondary</i>	protocols_pim.xml.in: protocols pim no-v6-secondary	Nothing found in VyOS
×	not yet documented	protocols_pim.xml.in: protocols pim packets	Nothing found in VyOS
×	<i>set protocols pim packets <1-255></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	protocols_pim.xml.in: protocols pim register-accept-list prefix-list	Nothing found in VyOS
×	<i>set protocols pim register-accept-list prefix-list <prefix-list></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	protocols_pim.xml.in: protocols pim register-suppress-time	Nothing found in VyOS
×	<i>set protocols pim register-suppress-tim <1-65535></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set protocols pim rp address <ip-address> group <prefix></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	protocols_pim.xml.in: protocols pim rp address <address> group	protocols pim rp address <text> group <value>
×	not yet documented	protocols_pim.xml.in: protocols pim rp keep-alive-timer	protocols pim rp keep-alive-timer <value>
×	<i>set protocols pim rp keep-alive-timer <1-65535></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	protocols_pim.xml.in: protocols pim spt-switchover infinity-and-beyond prefix-list	Nothing found in VyOS
×	<i>set protocols pim spt-switchover infinity-and-beyond prefix-list <list></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	protocols_pim.xml.in: protocols pim ssm prefix-list	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<i>set protocols pim ssm prefix-list <list></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	protocols_rip.xml.in: protocols rip default-distance	protocols rip default-distance <value>
×	<i>set protocols rip default-distance <1-255></i>	Nothing found in XML Definitions	Nothing found in VyOS
✓	<i>set protocols rip default-information originate</i>	protocols_rip.xml.in: protocols rip default-information originate	protocols rip default-information originate
×	not yet documented	protocols_rip.xml.in: protocols rip default-metric	protocols rip default-metric <value>
×	<i>set protocols rip default-metric <1-16></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set protocols rip distribute-list access-list <in/out> <0-4294967295></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	protocols_rip.xml.in: protocols rip distribute-list access-list in	protocols rip distribute-list access-list in <value>
×	not yet documented	protocols_rip.xml.in: protocols rip distribute-list access-list out	protocols rip distribute-list access-list out <value>
×	<i>set protocols rip distribute-list interface <interface> access-list <in/out> <0-4294967295></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set protocols rip distribute-list interface <interface> prefix-list <in/out> <name></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	protocols_rip.xml.in: protocols rip distribute-list interface <interface> access-list in	protocols rip distribute-list interface <text> access-list in <value>
×	not yet documented	protocols_rip.xml.in: protocols rip distribute-list interface <interface> access-list out	protocols rip distribute-list interface <text> access-list out <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	protocols_rip.xml.in: protocols rip distribute- list interface <interface> prefix-list in	protocols rip distribute-list interface <text> prefix-list in <value>
×	not yet documented	protocols_rip.xml.in: protocols rip distribute- list interface <interface> prefix-list out	protocols rip distribute-list interface <text> prefix-list out <value>
×	<i>set protocols rip distribute-list prefix-list <in/out> <name></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	protocols_rip.xml.in: protocols rip distribute-list prefix-list in	protocols rip distribute-list prefix-list in <value>
×	not yet documented	protocols_rip.xml.in: protocols rip distribute-list prefix-list out	protocols rip distribute-list prefix-list out <value>
×	<i>set protocols rip interface <interface></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set protocols rip interface <interface> authentication md5 <id> password <text></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set protocols rip interface <interface> authentication plaintext-password <text></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set protocols rip interface <interface> split-horizon disable</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set protocols rip interface <interface> split-horizon poison-reverse</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	protocols_rip.xml.in: protocols rip interface <interface> authentication md5 <md5> password	protocols rip interface <text> authentication md5 <text> password <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	protocols_rip.xml.in: protocols rip interface <interface> authentication plaintext-password <value>	protocols rip interface <text> authentication plaintext-password <value>
×	not yet documented	protocols_rip.xml.in: protocols rip interface <in- terface> receive version <value>	protocols rip interface <text> receive version <value>
×	not yet documented	protocols_rip.xml.in: protocols rip interface <in- terface> send version <value>	protocols rip interface <text> send version <value>
×	not yet documented	protocols_rip.xml.in: protocols rip interface <in- terface> split-horizon dis- able	protocols rip interface <text> split-horizon disable
×	not yet documented	protocols_rip.xml.in: protocols rip interface <interface> split-horizon poison-reverse	protocols rip interface <text> split-horizon poison-reverse
×	not yet documented	protocols_rip.xml.in: protocols rip neighbor <value>	protocols rip neighbor <value>
×	<i>set protocols rip neighbor <A.B.C.D></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	protocols_rip.xml.in: protocols rip network <value>	protocols rip network <value>
×	<i>set protocols rip network-distance <A. B.C.D/M> access-list <name></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set protocols rip network-distance <A.B.C.D/M> distance <1-255></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	protocols_rip.xml.in: protocols rip network- distance <network- distance> access-list <value>	protocols rip network- distance <text> access-list <value>
×	not yet documented	protocols_rip.xml.in: protocols rip network- distance <network- distance> distance <value>	protocols rip network- distance <text> distance <value>
×	<i>set protocols rip network <A.B.C.D/M></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	protocols_ripng.xml.in: protocols ripng aggregate- address <value>	protocols ripng aggregate- address <value>
×	not yet documented	protocols_ripng.xml.in: protocols ripng default- information originate	protocols ripng default- information originate

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	protocols_ripng.xml.in: protocols ripng default-metric	protocols ripng default-metric <value>
×	not yet documented	protocols_ripng.xml.in: protocols ripng distribute-list access-list in	protocols ripng distribute-list access-list in <value>
×	not yet documented	protocols_ripng.xml.in: protocols ripng distribute-list access-list out	protocols ripng distribute-list access-list out <value>
×	not yet documented	protocols_ripng.xml.in: protocols ripng distribute-list interface <interface> access-list in	protocols ripng distribute-list interface <text> access-list in <value>
×	not yet documented	protocols_ripng.xml.in: protocols ripng distribute-list interface <interface> access-list out	protocols ripng distribute-list interface <text> access-list out <value>
×	not yet documented	protocols_ripng.xml.in: protocols ripng distribute-list interface <interface> prefix-list in	protocols ripng distribute-list interface <text> prefix-list in <value>
×	not yet documented	protocols_ripng.xml.in: protocols ripng distribute-list interface <interface> prefix-list out	protocols ripng distribute-list interface <text> prefix-list out <value>
×	not yet documented	protocols_ripng.xml.in: protocols ripng distribute-list prefix-list in	protocols ripng distribute-list prefix-list in <value>
×	not yet documented	protocols_ripng.xml.in: protocols ripng distribute-list prefix-list out	protocols ripng distribute-list prefix-list out <value>
×	not yet documented	protocols_ripng.xml.in: protocols ripng interface <interface> split-horizon disable	protocols ripng interface <text> split-horizon disable
×	not yet documented	protocols_ripng.xml.in: protocols ripng interface <interface> split-horizon poison-reverse	protocols ripng interface <text> split-horizon poison-reverse
×	not yet documented	protocols_ripng.xml.in: protocols ripng network	protocols ripng network <value>
×	not yet documented	protocols_ripng.xml.in: protocols ripng passive-interface	protocols ripng passive-interface <value>
×	not yet documented	protocols_ripng.xml.in: protocols ripng redistribute babel metric	protocols ripng redistribute babel metric <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	protocols_ripng.xml.in: protocols ripng redis- tribute babel route-map	protocols ripng redis- tribute babel route-map <value>
×	not yet documented	protocols_ripng.xml.in: protocols ripng redis- tribute bgp metric	protocols ripng re- distribute bgp metric <value>
×	not yet documented	protocols_ripng.xml.in: protocols ripng redis- tribute bgp route-map	protocols ripng redis- tribute bgp route-map <value>
×	not yet documented	protocols_ripng.xml.in: protocols ripng redis- tribute connected metric	protocols ripng redis- tribute connected metric <value>
×	not yet documented	protocols_ripng.xml.in: protocols ripng re- distribute connected route-map	protocols ripng re- distribute connected route-map <value>
×	not yet documented	protocols_ripng.xml.in: protocols ripng redis- tribute kernel metric	protocols ripng redis- tribute kernel metric <value>
×	not yet documented	protocols_ripng.xml.in: protocols ripng redis- tribute kernel route-map	protocols ripng redis- tribute kernel route-map <value>
×	not yet documented	protocols_ripng.xml.in: protocols ripng redis- tribute ospfv3 metric	protocols ripng redis- tribute ospfv3 metric <value>
×	not yet documented	protocols_ripng.xml.in: protocols ripng redis- tribute ospfv3 route-map	protocols ripng redis- tribute ospfv3 route-map <value>
×	not yet documented	protocols_ripng.xml.in: protocols ripng redis- tribute static metric	protocols ripng redis- tribute static metric <value>
×	not yet documented	protocols_ripng.xml.in: protocols ripng redis- tribute static route-map	protocols ripng redis- tribute static route-map <value>
×	not yet documented	protocols_ripng.xml.in: protocols ripng route	protocols ripng route <value>
×	not yet documented	protocols_ripng.xml.in: protocols ripng route-map	protocols ripng route-map <value>
×	not yet documented	protocols_ripng.xml.in: protocols ripng timers garbage-collection	protocols ripng timers garbage-collection <value>
×	not yet documented	protocols_ripng.xml.in: protocols ripng timers timeout	protocols ripng timers timeout <value>
×	not yet documented	protocols_ripng.xml.in: protocols ripng timers up- date	protocols ripng timers up- date <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	protocols_rip.xml.in: protocols_rip_passive-interface	protocols_rip_passive-interface <value>
×	<i>set protocols rip passive-interface <interface></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set protocols rip passive-interface default</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set protocols rip redistribute <babel/bgp/connected/</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set protocols rip redistribute <babel/bgp/connected/ metric <1-16></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set protocols rip redistribute <babel/bgp/connected/ route-map <name></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	protocols_rip.xml.in: protocols_rip_redistribute_babel_metric	protocols_rip_redistribute_babel_metric <value>
×	not yet documented	protocols_rip.xml.in: protocols_rip_redistribute_babel_route-map	protocols_rip_redistribute_babel_route-map <value>
×	not yet documented	protocols_rip.xml.in: protocols_rip_redistribute_bgp_metric	protocols_rip_redistribute_bgp_metric <value>
×	not yet documented	protocols_rip.xml.in: protocols_rip_redistribute_bgp_route-map	protocols_rip_redistribute_bgp_route-map <value>
×	not yet documented	protocols_rip.xml.in: protocols_rip_redistribute_connected_metric	protocols_rip_redistribute_connected_metric <value>
×	not yet documented	protocols_rip.xml.in: protocols_rip_redistribute_connected_route-map	protocols_rip_redistribute_connected_route-map <value>
×	not yet documented	protocols_rip.xml.in: protocols_rip_redistribute_isis_metric	protocols_rip_redistribute_isis_metric <value>
×	not yet documented	protocols_rip.xml.in: protocols_rip_redistribute_isis_route-map	protocols_rip_redistribute_isis_route-map <value>
×	not yet documented	protocols_rip.xml.in: protocols_rip_redistribute_kernel_metric	protocols_rip_redistribute_kernel_metric <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	protocols_rip.xml.in: protocols rip redistribute kernel route-map	protocols rip redistribute kernel route-map <value>
×	not yet documented	protocols_rip.xml.in: protocols rip redistribute ospf metric	protocols rip redistribute ospf metric <value>
×	not yet documented	protocols_rip.xml.in: protocols rip redistribute ospf route-map	protocols rip redistribute ospf route-map <value>
×	not yet documented	protocols_rip.xml.in: protocols rip redistribute static metric	protocols rip redistribute static metric <value>
×	not yet documented	protocols_rip.xml.in: protocols rip redistribute static route-map	protocols rip redistribute static route-map <value>
×	not yet documented	protocols_rip.xml.in: protocols rip route	protocols rip route <value>
×	not yet documented	protocols_rip.xml.in: protocols rip route-map	protocols rip route-map <value>
×	<i>set protocols rip route <A.B.C.D/M></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	protocols_rip.xml.in: protocols rip timers garbage-collection	protocols rip timers garbage-collection <value>
×	<i>set protocols rip timers garbage-collection <5-2147483647></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	protocols_rip.xml.in: protocols rip timers time- out	protocols rip timers time- out <value>
×	<i>set protocols rip timers timeout <5-2147483647></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	protocols_rip.xml.in: protocols rip timers update	protocols rip timers update <value>
×	<i>set protocols rip timers update <5-2147483647></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	protocols_rip.xml.in: protocols rip version	protocols rip version <value>
×	<i>set protocols rpki cache <address> port <1-65535></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set protocols rpki cache <address> preference <1-255></i>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<i>set protocols rpki cache <address> ssh key <name></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set protocols rpki cache <address> ssh username <user></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	protocols_rpki.xml.in: protocols rpki cache <cache> port	protocols rpki cache <text> port <value>
×	not yet documented	protocols_rpki.xml.in: protocols rpki cache <cache> preference	protocols rpki cache <text> preference <value>
×	not yet documented	protocols_rpki.xml.in: protocols rpki cache <cache> ssh key	Nothing found in VyOS
×	not yet documented	Nothing found in XML Definitions	protocols rpki cache <text> ssh known-hosts-file <value>
×	not yet documented	Nothing found in XML Definitions	protocols rpki cache <text> ssh private-key-file <value>
×	not yet documented	Nothing found in XML Definitions	protocols rpki cache <text> ssh public-key-file <value>
×	not yet documented	protocols_rpki.xml.in: protocols rpki cache <cache> ssh username	protocols rpki cache <text> ssh username <value>
×	not yet documented	protocols_rpki.xml.in: protocols rpki expire-interval	Nothing found in VyOS
×	<i>set protocols rpki expire-interval <600-172800></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	protocols_rpki.xml.in: protocols rpki polling-period	protocols rpki polling-period <value>
×	<i>set protocols rpki polling-period <1-86400></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	protocols_rpki.xml.in: protocols rpki retry-interval	Nothing found in VyOS
×	<i>set protocols rpki retry-interval <1-7200></i>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	protocols_segment-routing.xml.in: protocols segment-routing interface <interface> srv6 hmac	Nothing found in VyOS
×	not yet documented	protocols_segment-routing.xml.in: protocols segment-routing srv6 locator <locator> behavior-usid	Nothing found in VyOS
×	not yet documented	protocols_segment-routing.xml.in: protocols segment-routing srv6 locator <locator> block-len	Nothing found in VyOS
×	not yet documented	protocols_segment-routing.xml.in: protocols segment-routing srv6 locator <locator> func-bits	Nothing found in VyOS
×	not yet documented	protocols_segment-routing.xml.in: protocols segment-routing srv6 locator <locator> node-len	Nothing found in VyOS
×	not yet documented	protocols_segment-routing.xml.in: protocols segment-routing srv6 locator <locator> prefix	Nothing found in VyOS
×	not yet documented	protocols_static_arp.xml.in: protocols static arp interface <interface> address <address> description	protocols static arp interface <text> address <text> description <value>
✓	<i>set protocols static arp interface <interface> address <host> mac <mac></i>	protocols_static_arp.xml.in: protocols static arp interface <interface> address <address> mac	protocols static arp interface <text> address <text> mac <value>
×	<i>set protocols static mroute <subnet> interface <interface></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set protocols static mroute <subnet> interface <interface> disable</i>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set protocols static mroute <subnet> interface <interface> distance <1-255></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set protocols static mroute <subnet> next-hop <address></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set protocols static mroute <subnet> next-hop <address> disable</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set protocols static mroute <subnet> next-hop <address> distance <1-255></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	proto-cols_static_multicast.xml.in protocols static multicast interface-route <interface-route> next-hop-interface <next-hop-interface> distance	protocols static multicast interface-route <text> next-hop-interface <text> distance <value>
×	not yet documented	proto-cols_static_multicast.xml.in protocols static multicast route <route> next-hop <next-hop> distance	protocols static multicast route <text> next-hop <text> distance <value>
×	not yet documented	protocols_static_neighbor-proxy.xml.in: protocols static neighbor-proxy arp <arp> interface	Nothing found in VyOS
×	not yet documented	protocols_static_neighbor-proxy.xml.in: protocols static neighbor-proxy nd <nd> interface	Nothing found in VyOS
×	not yet documented	protocols_static.xml.in: protocols static route-map	protocols static route-map <value>
×	<code>set protocols static route 0.0.0.0/0 next-hop <address></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set protocols static route6 <subnet> blackhole</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set protocols static route6 <subnet> blackhole distance <distance></code>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set protocols static route6 <subnet> blackhole tag <tag></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set protocols static route6 <subnet> interface <interface></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set protocols static route6 <subnet> interface <interface> disable</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set protocols static route6 <subnet> interface <interface> distance <distance></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set protocols static route6 <subnet> interface <interface> segments <segments></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set protocols static route6 <subnet> next-hop <address></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set protocols static route6 <subnet> next-hop <address> bfd</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set protocols static route6 <subnet> next-hop <address> bfd multi-hop source-address <source-address></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set protocols static route6 <subnet> next-hop <address> bfd profile <profile></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set protocols static route6 <subnet> next-hop <address> disable</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set protocols static route6 <subnet> next-hop <address> distance <distance></code>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set protocols static route6 <subnet> next-hop <address> segments <segments></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set protocols static route6 <subnet> reject</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set protocols static route6 <subnet> reject distance <distance></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set protocols static route6 <subnet> reject tag <tag></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	protocols_static.xml.in: protocols static route6 <route6> blackhole distance	protocols static route6 <text> blackhole distance <value>
×	not yet documented	protocols_static.xml.in: protocols static route6 <route6> blackhole tag	protocols static route6 <text> blackhole tag <value>
×	not yet documented	protocols_static.xml.in: protocols static route6 <route6> description	protocols static route6 <text> description <value>
×	not yet documented	protocols_static.xml.in: protocols static route6 <route6> interface <inter- face> disable	protocols static route6 <text> interface <text> disable
×	not yet documented	protocols_static.xml.in: protocols static route6 <route6> interface <inter- face> distance	protocols static route6 <text> interface <text> distance <value>
×	not yet documented	protocols_static.xml.in: protocols static route6 <route6> interface <inter- face> segments	Nothing found in VyOS
×	not yet documented	protocols_static.xml.in: protocols static route6 <route6> interface <inter- face> vrf	protocols static route6 <text> interface <text> vrf <value>
×	not yet documented	protocols_static.xml.in: protocols static route6 <route6> next-hop <next- hop> bfd multi-hop source <source> profile	protocols static route6 <text> next-hop <text> bfd multi-hop source <text> profile <value>
×	not yet documented	protocols_static.xml.in: protocols static route6 <route6> next-hop <next- hop> bfd profile	protocols static route6 <text> next-hop <text> bfd profile <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	protocols_static.xml.in: protocols static route6 <route6> next-hop <next-hop> disable	protocols static route6 <text> next-hop <text> disable
×	not yet documented	protocols_static.xml.in: protocols static route6 <route6> next-hop <next-hop> distance	protocols static route6 <text> next-hop <text> distance <value>
×	not yet documented	protocols_static.xml.in: protocols static route6 <route6> next-hop <next-hop> interface	protocols static route6 <text> next-hop <text> interface <value>
×	not yet documented	protocols_static.xml.in: protocols static route6 <route6> next-hop <next-hop> segments	Nothing found in VyOS
×	not yet documented	protocols_static.xml.in: protocols static route6 <route6> next-hop <next-hop> vrf	protocols static route6 <text> next-hop <text> vrf <value>
×	not yet documented	protocols_static.xml.in: protocols static route6 <route6> reject distance	protocols static route6 <text> reject distance <value>
×	not yet documented	protocols_static.xml.in: protocols static route6 <route6> reject tag	protocols static route6 <text> reject tag <value>
×	<i>set protocols static route <subnet> blackhole</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set protocols static route <subnet> blackhole distance <distance></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set protocols static route <subnet> blackhole tag <tag></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set protocols static route <subnet> dhcp-interface <interface></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set protocols static route <subnet> interface <interface></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set protocols static route <subnet> interface <interface> disable</i>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set protocols static route <subnet> interface <interface> distance <distance></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set protocols static route <subnet> next-hop <address></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set protocols static route <subnet> next-hop <address> bfd</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set protocols static route <subnet> next-hop <address> bfd multi-hop source-address <source-address></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set protocols static route <subnet> next-hop <address> bfd profile <profile></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set protocols static route <subnet> next-hop <address> disable</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set protocols static route <subnet> next-hop <address> distance <distance></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set protocols static route <subnet> reject</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set protocols static route <subnet> reject distance <distance></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set protocols static route <subnet> reject tag <tag></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	protocols_static.xml.in: protocols static route <route> blackhole dis- tance	protocols static route <text> blackhole distance <value>
×	not yet documented	protocols_static.xml.in: protocols static route <route> blackhole tag	protocols static route <text> blackhole tag <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	protocols_static.xml.in: protocols static route <route> description	protocols static route <text> description <value>
×	not yet documented	protocols_static.xml.in: protocols static route <route> dhcp-interface	protocols static route <text> dhcp-interface <value>
×	not yet documented	protocols_static.xml.in: protocols static route <route> interface <inter- face> disable	protocols static route <text> interface <text> disable
×	not yet documented	protocols_static.xml.in: protocols static route <route> interface <inter- face> distance	protocols static route <text> interface <text> distance <value>
×	not yet documented	protocols_static.xml.in: protocols static route <route> interface <inter- face> vrf	protocols static route <text> interface <text> vrf <value>
×	not yet documented	protocols_static.xml.in: protocols static route <route> next-hop <next- hop> bfd multi-hop source <source> profile	protocols static route <text> next-hop <text> bfd multi-hop source <text> profile <value>
×	not yet documented	protocols_static.xml.in: protocols static route <route> next-hop <next- hop> bfd profile	protocols static route <text> next-hop <text> bfd profile <value>
×	not yet documented	protocols_static.xml.in: protocols static route <route> next-hop <next- hop> disable	protocols static route <text> next-hop <text> disable
×	not yet documented	protocols_static.xml.in: protocols static route <route> next-hop <next- hop> distance	protocols static route <text> next-hop <text> distance <value>
×	not yet documented	protocols_static.xml.in: protocols static route <route> next-hop <next- hop> interface	protocols static route <text> next-hop <text> interface <value>
×	not yet documented	protocols_static.xml.in: protocols static route <route> next-hop <next- hop> vrf	protocols static route <text> next-hop <text> vrf <value>
×	not yet documented	protocols_static.xml.in: protocols static route <route> reject distance	protocols static route <text> reject distance <value>
×	not yet documented	protocols_static.xml.in: protocols static route <route> reject tag	protocols static route <text> reject tag <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	protocols_static.xml.in: protocols static table <table> description	protocols static table <text> description <value>
×	not yet documented	protocols_static.xml.in: protocols static table <table> route6 <route6> blackhole distance	protocols static table <text> route6 <text> blackhole distance <value>
×	not yet documented	protocols_static.xml.in: protocols static table <table> route6 <route6> blackhole tag	protocols static table <text> route6 <text> blackhole tag <value>
×	not yet documented	protocols_static.xml.in: protocols static table <table> route6 <route6> description	protocols static table <text> route6 <text> description <value>
×	not yet documented	protocols_static.xml.in: protocols static table <table> route6 <route6> interface <interface> disable	protocols static table <text> route6 <text> interface <text> disable
×	not yet documented	protocols_static.xml.in: protocols static table <table> route6 <route6> interface <interface> distance	protocols static table <text> route6 <text> interface <text> distance <value>
×	not yet documented	protocols_static.xml.in: protocols static table <table> route6 <route6> interface <interface> segments	Nothing found in VyOS
×	not yet documented	protocols_static.xml.in: protocols static table <table> route6 <route6> interface <interface> vrf	protocols static table <text> route6 <text> interface <text> vrf <value>
×	not yet documented	protocols_static.xml.in: protocols static table <table> route6 <route6> next-hop <next-hop> bfd multi-hop source <source> profile	protocols static table <text> route6 <text> next-hop <text> bfd multi-hop source <text> profile <value>
×	not yet documented	protocols_static.xml.in: protocols static table <table> route6 <route6> next-hop <next-hop> bfd profile	protocols static table <text> route6 <text> next-hop <text> bfd profile <value>
×	not yet documented	protocols_static.xml.in: protocols static table <table> route6 <route6> next-hop <next-hop> disable	protocols static table <text> route6 <text> next-hop <text> disable

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	protocols_static.xml.in: protocols static table <table> route6 <route6> next-hop <next-hop> distance	protocols static table <text> route6 <text> next-hop <text> distance <value>
×	not yet documented	protocols_static.xml.in: protocols static table <table> route6 <route6> next-hop <next-hop> interface	protocols static table <text> route6 <text> next-hop <text> interface <value>
×	not yet documented	protocols_static.xml.in: protocols static table <table> route6 <route6> next-hop <next-hop> segments	Nothing found in VyOS
×	not yet documented	protocols_static.xml.in: protocols static table <table> route6 <route6> next-hop <next-hop> vrf	protocols static table <text> route6 <text> next-hop <text> vrf <value>
×	not yet documented	protocols_static.xml.in: protocols static table <table> route6 <route6> reject distance	protocols static table <text> route6 <text> reject distance <value>
×	not yet documented	protocols_static.xml.in: protocols static table <table> route6 <route6> reject tag	protocols static table <text> route6 <text> reject tag <value>
×	not yet documented	protocols_static.xml.in: protocols static table <table> route <route> black-hole distance	protocols static table <text> route <text> black-hole distance <value>
×	not yet documented	protocols_static.xml.in: protocols static table <table> route <route> black-hole tag	protocols static table <text> route <text> blackhole tag <value>
×	not yet documented	protocols_static.xml.in: protocols static table <table> route <route> description	protocols static table <text> route <text> description <value>
×	not yet documented	protocols_static.xml.in: protocols static table <table> route <route> dhcp-interface	protocols static table <text> route <text> dhcp-interface <value>
×	not yet documented	protocols_static.xml.in: protocols static table <table> route <route> interface <interface> disable	protocols static table <text> route <text> interface <text> disable

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	protocols_static.xml.in: protocols static table <table> route <route> interface <interface> distance	protocols static table <text> route <text> interface <text> distance <value>
×	not yet documented	protocols_static.xml.in: protocols static table <table> route <route> interface <interface> vrf	protocols static table <text> route <text> interface <text> vrf <value>
×	not yet documented	protocols_static.xml.in: protocols static table <table> route <route> next-hop <next-hop> bfd multi-hop source <source> profile	protocols static table <text> route <text> next-hop <text> bfd multi-hop source <text> profile <value>
×	not yet documented	protocols_static.xml.in: protocols static table <table> route <route> next-hop <next-hop> bfd profile	protocols static table <text> route <text> next-hop <text> bfd profile <value>
×	not yet documented	protocols_static.xml.in: protocols static table <table> route <route> next-hop <next-hop> disable	protocols static table <text> route <text> next-hop <text> disable
×	not yet documented	protocols_static.xml.in: protocols static table <table> route <route> next-hop <next-hop> distance	protocols static table <text> route <text> next-hop <text> distance <value>
×	not yet documented	protocols_static.xml.in: protocols static table <table> route <route> next-hop <next-hop> interface	protocols static table <text> route <text> next-hop <text> interface <value>
×	not yet documented	protocols_static.xml.in: protocols static table <table> route <route> next-hop <next-hop> vrf	protocols static table <text> route <text> next-hop <text> vrf <value>
×	not yet documented	protocols_static.xml.in: protocols static table <table> route <route> reject distance	protocols static table <text> route <text> reject distance <value>
×	not yet documented	protocols_static.xml.in: protocols static table <table> route <route> reject tag	protocols static table <text> route <text> reject tag <value>
×	<i>set protocols traffic-engineering admin-group <admin-group-name> bit-position <0-31></i>	Nothing found in XML Definitions	Nothing found in VyOS

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Table 1 – continued from previous page

Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set protocols traffic-engineering interface <ifname> admin-group <admin-group-name></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set protocols traffic-engineering interface <ifname> max-bandwidth <1-4294967295></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set protocols traffic-engineering interface <ifname> max-reservable-bandwidth <1-4294967295></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set protocols traffic-engineering interface <ifname> metric <1-4294967295></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	qos.xml.in: qos interface <interface> egress	qos interface <text> egress <value>
×	not yet documented	qos.xml.in: qos interface <interface> ingress	qos interface <text> ingress <value>
×	<code>set qos policy cake <text> bandwidth <value></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set qos policy cake <text> description</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set qos policy cake <text> flow-isolation blind</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set qos policy cake <text> flow-isolation dst-host</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set qos policy cake <text> flow-isolation dual-dst-host</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set qos policy cake <text> flow-isolation dual-src-host</code>	Nothing found in XML Definitions	Nothing found in VyOS

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Table 1 – continued from previous page

Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<i>set qos policy cake <text> flow-isolation flow</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set qos policy cake <text> flow-isolation host</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set qos policy cake <text> flow-isolation nat</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set qos policy cake <text> flow-isolation src-host</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set qos policy cake <text> flow-isolation triple-isolate</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set qos policy cake <text> rtt</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	qos.xml.in: qos policy cake <cake> bandwidth	qos policy cake <text> bandwidth <value>
×	not yet documented	qos.xml.in: qos policy cake <cake> description	qos policy cake <text> de- scription <value>
×	not yet documented	qos.xml.in: qos policy cake <cake> flow-isolation blind	qos policy cake <text> flow-isolation blind
×	not yet documented	qos.xml.in: qos policy cake <cake> flow-isolation dst-host	qos policy cake <text> flow-isolation dst-host
×	not yet documented	qos.xml.in: qos policy cake <cake> flow-isolation dual-dst- host	qos policy cake <text> flow-isolation dual-dst- host
×	not yet documented	qos.xml.in: qos policy cake <cake> flow-isolation dual-src- host	qos policy cake <text> flow-isolation dual-src- host
×	not yet documented	qos.xml.in: qos policy cake <cake> flow-isolation flow	qos policy cake <text> flow-isolation flow
×	not yet documented	qos.xml.in: qos policy cake <cake> flow-isolation host	qos policy cake <text> flow-isolation host
×	not yet documented	qos.xml.in: qos policy cake <cake> flow-isolation nat	qos policy cake <text> flow-isolation nat

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Table 1 – continued from previous page

Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	qos.xml.in: qos policy cake <cake> flow-isolation src-host	qos policy cake <text> flow-isolation src-host
×	not yet documented	qos.xml.in: qos policy cake <cake> flow-isolation triple- isolate	Nothing found in VyOS
×	not yet documented	qos.xml.in: qos policy cake <cake> rtt	qos policy cake <text> rtt <value>
×	<i>set qos policy drop-tail <policy-name> queue-limit <number-of-packets></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	qos.xml.in: qos policy drop-tail <drop-tail> description	qos policy drop-tail <text> description <value>
×	not yet documented	qos.xml.in: qos policy drop-tail <drop-tail> queue-limit	qos policy drop-tail <text> queue-limit <value>
×	<i>set qos policy fair-queue <policy-name></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set qos policy fair-queue <policy-name> hash-interval <seconds></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set qos policy fair-queue <policy-name> queue-limit <limit></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	qos.xml.in: qos policy fair-queue <fair-queue> description	qos policy fair-queue <text> description <value>
×	not yet documented	qos.xml.in: qos policy fair-queue <fair-queue> hash- interval	qos policy fair-queue <text> hash-interval <value>
×	not yet documented	qos.xml.in: qos policy fair-queue <fair-queue> queue-limit	qos policy fair-queue <text> queue-limit <value>
×	<i>set qos policy fq-codel <policy name> codel-quantum <bytes></i>	Nothing found in XML Definitions	Nothing found in VyOS

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Table 1 – continued from previous page

Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set qos policy fq-codel <policy-name> flows <number-of-flows></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set qos policy fq-codel <policy-name> interval <milliseconds></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set qos policy fq-codel <policy-name> queue-limit <number-of-packets></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set qos policy fq-codel <policy-name> target <milliseconds></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	qos.xml.in: qos policy fq-codel <fq-codel> codel-quantum	qos policy fq-codel <text> codel-quantum <value>
×	not yet documented	qos.xml.in: qos policy fq-codel <fq-codel> description	qos policy fq-codel <text> description <value>
×	not yet documented	qos.xml.in: qos policy fq-codel <fq-codel> flows	qos policy fq-codel <text> flows <value>
×	not yet documented	qos.xml.in: qos policy fq-codel <fq-codel> interval	qos policy fq-codel <text> interval <value>
×	not yet documented	qos.xml.in: qos policy fq-codel <fq-codel> queue-limit	qos policy fq-codel <text> queue-limit <value>
×	not yet documented	qos.xml.in: qos policy fq-codel <fq-codel> target	qos policy fq-codel <text> target <value>
×	<code>set qos policy limiter <policy-name> class <class-ID> bandwidth <rate></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set qos policy limiter <policy-name> class <class-ID> burst <burst-size></code>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set qos policy limiter</code> <code><policy-name></code> <code>class <class ID></code> <code>match <match-name></code> <code>description</code> <code><description></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set qos policy limiter</code> <code><policy-name> class</code> <code><class ID> priority</code> <code><value></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set qos policy limiter</code> <code><policy-name></code> <code>default bandwidth</code> <code><rate></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set qos policy limiter</code> <code><policy-name></code> <code>default burst</code> <code><burst-size></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	qos.xml.in: qos policy limiter <lim-iter> class <class> band- width	qos policy limiter <text> class <text> bandwidth <value>
×	not yet documented	qos.xml.in: qos policy limiter <lim-iter> class <class> burst	qos policy limiter <text> class <text> burst <value>
×	not yet documented	qos.xml.in: qos policy limiter <lim-iter> class <class> de- scription	qos policy limiter <text> class <text> description <value>
×	not yet documented	qos.xml.in: qos policy limiter <lim-iter> class <class> exceed	qos policy limiter <text> class <text> exceed <value>
×	not yet documented	qos.xml.in: qos policy limiter <lim-iter> class <class> match <match> description	qos policy limiter <text> class <text> match <text> description <value>
×	not yet documented	qos.xml.in: qos policy limiter <lim-iter> class <class> match <match> ether destination	qos policy limiter <text> class <text> match <text> ether destination <value>
×	not yet documented	qos.xml.in: qos policy limiter <lim-iter> class <class> match <match> ether protocol	qos policy limiter <text> class <text> match <text> ether protocol <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	qos.xml.in: qos policy limiter <lim- iter> class <class> match <match> ether source	qos policy limiter <text> class <text> match <text> ether source <value>
×	not yet documented	qos.xml.in: qos policy limiter <lim- iter> class <class> match <match> interface	qos policy limiter <text> class <text> match <text> interface <value>
×	not yet documented	qos.xml.in: qos policy limiter <lim- iter> class <class> match <match> ip destination ad- dress	qos policy limiter <text> class <text> match <text> ip destination address <value>
×	not yet documented	qos.xml.in: qos policy limiter <lim- iter> class <class> match <match> ip destination port	qos policy limiter <text> class <text> match <text> ip destination port <value>
×	not yet documented	qos.xml.in: qos policy limiter <lim- iter> class <class> match <match> ip dscp	qos policy limiter <text> class <text> match <text> ip dscp <value>
×	not yet documented	qos.xml.in: qos policy limiter <lim- iter> class <class> match <match> ip max-length	qos policy limiter <text> class <text> match <text> ip max-length <value>
×	not yet documented	qos.xml.in: qos policy limiter <lim- iter> class <class> match <match> ip protocol	qos policy limiter <text> class <text> match <text> ip protocol <value>
×	not yet documented	qos.xml.in: qos policy limiter <lim- iter> class <class> match <match> ip source address	qos policy limiter <text> class <text> match <text> ip source address <value>
×	not yet documented	qos.xml.in: qos policy limiter <lim- iter> class <class> match <match> ip source port	qos policy limiter <text> class <text> match <text> ip source port <value>
×	not yet documented	qos.xml.in: qos policy limiter <lim- iter> class <class> match <match> ip tcp ack	qos policy limiter <text> class <text> match <text> ip tcp ack
×	not yet documented	qos.xml.in: qos policy limiter <lim- iter> class <class> match <match> ip tcp syn	qos policy limiter <text> class <text> match <text> ip tcp syn

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	qos.xml.in: qos policy limiter <lim-iter> class <class> match <match> ipv6 destination address	qos policy limiter <text> class <text> match <text> ipv6 destination address <value>
×	not yet documented	qos.xml.in: qos policy limiter <lim-iter> class <class> match <match> ipv6 destination port	qos policy limiter <text> class <text> match <text> ipv6 destination port <value>
×	not yet documented	qos.xml.in: qos policy limiter <lim-iter> class <class> match <match> ipv6 dscp	qos policy limiter <text> class <text> match <text> ipv6 dscp <value>
×	not yet documented	qos.xml.in: qos policy limiter <lim-iter> class <class> match <match> ipv6 max-length	qos policy limiter <text> class <text> match <text> ipv6 max-length <value>
×	not yet documented	qos.xml.in: qos policy limiter <lim-iter> class <class> match <match> ipv6 protocol	qos policy limiter <text> class <text> match <text> ipv6 protocol <value>
×	not yet documented	qos.xml.in: qos policy limiter <lim-iter> class <class> match <match> ipv6 source address	qos policy limiter <text> class <text> match <text> ipv6 source address <value>
×	not yet documented	qos.xml.in: qos policy limiter <lim-iter> class <class> match <match> ipv6 source port	qos policy limiter <text> class <text> match <text> ipv6 source port <value>
×	not yet documented	qos.xml.in: qos policy limiter <lim-iter> class <class> match <match> ipv6 tcp ack	qos policy limiter <text> class <text> match <text> ipv6 tcp ack
×	not yet documented	qos.xml.in: qos policy limiter <lim-iter> class <class> match <match> ipv6 tcp syn	qos policy limiter <text> class <text> match <text> ipv6 tcp syn
×	not yet documented	qos.xml.in: qos policy limiter <lim-iter> class <class> match <match> mark	qos policy limiter <text> class <text> match <text> mark <value>
×	not yet documented	qos.xml.in: qos policy limiter <lim-iter> class <class> match <match> vif	qos policy limiter <text> class <text> match <text> vif <value>

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Table 1 – continued from previous page

Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	qos.xml.in: qos policy limiter <lim-iter> class <class> mtu	Nothing found in VyOS
×	not yet documented	qos.xml.in: qos policy limiter <lim-iter> class <class> not-exceed	qos policy limiter <text> class <text> not-exceed <value>
×	not yet documented	qos.xml.in: qos policy limiter <lim-iter> class <class> priority	qos policy limiter <text> class <text> priority <value>
×	not yet documented	qos.xml.in: qos policy limiter <lim-iter> default bandwidth	qos policy limiter <text> default bandwidth <value>
×	not yet documented	qos.xml.in: qos policy limiter <lim-iter> default burst	qos policy limiter <text> default burst <value>
×	not yet documented	qos.xml.in: qos policy limiter <lim-iter> default exceed	qos policy limiter <text> default exceed <value>
×	not yet documented	qos.xml.in: qos policy limiter <lim-iter> default mtu	Nothing found in VyOS
×	not yet documented	qos.xml.in: qos policy limiter <lim-iter> default not-exceed	qos policy limiter <text> default not-exceed <value>
×	not yet documented	qos.xml.in: qos policy limiter <lim-iter> description	qos policy limiter <text> description <value>
×	<i>set qos policy network-emulator <policy-name> bandwidth <rate></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set qos policy network-emulator <policy-name> burst <burst-size></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set qos policy network-emulator <policy-name> corruption <percent></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set qos policy network-emulator <policy-name> delay <delay></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set qos policy network-emulator <policy-name> loss <percent></i>	Nothing found in XML Definitions	Nothing found in VyOS

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Table 1 – continued from previous page

Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	qos.xml.in: qos policy network-emulator <network-emulator> bandwidth	qos policy network-emulator <text> bandwidth <value>
×	not yet documented	qos.xml.in: qos policy network-emulator <network-emulator> corruption	qos policy network-emulator <text> corruption <value>
×	not yet documented	qos.xml.in: qos policy network-emulator <network-emulator> delay	qos policy network-emulator <text> delay <value>
×	not yet documented	qos.xml.in: qos policy network-emulator <network-emulator> description	qos policy network-emulator <text> description <value>
×	not yet documented	qos.xml.in: qos policy network-emulator <network-emulator> duplicate	qos policy network-emulator <text> duplicate <value>
×	not yet documented	qos.xml.in: qos policy network-emulator <network-emulator> loss	qos policy network-emulator <text> loss <value>
×	not yet documented	qos.xml.in: qos policy network-emulator <network-emulator> queue-limit	qos policy network-emulator <text> queue-limit <value>
×	not yet documented	qos.xml.in: qos policy network-emulator <network-emulator> reordering	qos policy network-emulator <text> reordering <value>
×	<i>set qos policy priority-queue <policy-name> class <class-ID> queue-limit <limit></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	qos.xml.in: qos policy priority-queue <priority-queue> class <class> codel-quantum	qos policy priority-queue <text> class <text> codel-quantum <value>
×	not yet documented	qos.xml.in: qos policy priority-queue <priority-queue> class <class> description	qos policy priority-queue <text> class <text> description <value>
×	not yet documented	qos.xml.in: qos policy priority-queue <priority-queue> class <class> flows	qos policy priority-queue <text> class <text> flows <value>

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Table 1 – continued from previous page

Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	qos.xml.in: qos policy priority-queue <priority-queue> class <class> interval	qos policy priority-queue <text> class <text> interval <value>
×	not yet documented	qos.xml.in: qos policy priority-queue <priority-queue> class <class> match <match> description	qos policy priority-queue <text> class <text> match <text> description <value>
×	not yet documented	qos.xml.in: qos policy priority-queue <priority-queue> class <class> match <match> ether destination	qos policy priority-queue <text> class <text> match <text> ether destination <value>
×	not yet documented	qos.xml.in: qos policy priority-queue <priority-queue> class <class> match <match> ether protocol	qos policy priority-queue <text> class <text> match <text> ether protocol <value>
×	not yet documented	qos.xml.in: qos policy priority-queue <priority-queue> class <class> match <match> ether source	qos policy priority-queue <text> class <text> match <text> ether source <value>
×	not yet documented	qos.xml.in: qos policy priority-queue <priority-queue> class <class> match <match> interface	qos policy priority-queue <text> class <text> match <text> interface <value>
×	not yet documented	qos.xml.in: qos policy priority-queue <priority-queue> class <class> match <match> ip destination address	qos policy priority-queue <text> class <text> match <text> ip destination address <value>
×	not yet documented	qos.xml.in: qos policy priority-queue <priority-queue> class <class> match <match> ip destination port	qos policy priority-queue <text> class <text> match <text> ip destination port <value>
×	not yet documented	qos.xml.in: qos policy priority-queue <priority-queue> class <class> match <match> ip dscp	qos policy priority-queue <text> class <text> match <text> ip dscp <value>
×	not yet documented	qos.xml.in: qos policy priority-queue <priority-queue> class <class> match <match> ip max-length	qos policy priority-queue <text> class <text> match <text> ip max-length <value>

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Table 1 – continued from previous page

Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	qos.xml.in: qos policy priority-queue <priority-queue> class <class> match <match> ip protocol	qos policy priority-queue <text> class <text> match <text> ip protocol <value>
×	not yet documented	qos.xml.in: qos policy priority-queue <priority-queue> class <class> match <match> ip source address	qos policy priority-queue <text> class <text> match <text> ip source address <value>
×	not yet documented	qos.xml.in: qos policy priority-queue <priority-queue> class <class> match <match> ip source port	qos policy priority-queue <text> class <text> match <text> ip source port <value>
×	not yet documented	qos.xml.in: qos policy priority-queue <priority-queue> class <class> match <match> ip tcp ack	qos policy priority-queue <text> class <text> match <text> ip tcp ack
×	not yet documented	qos.xml.in: qos policy priority-queue <priority-queue> class <class> match <match> ip tcp syn	qos policy priority-queue <text> class <text> match <text> ip tcp syn
×	not yet documented	qos.xml.in: qos policy priority-queue <priority-queue> class <class> match <match> ipv6 destination address	qos policy priority-queue <text> class <text> match <text> ipv6 destination ad- dress <value>
×	not yet documented	qos.xml.in: qos policy priority-queue <priority-queue> class <class> match <match> ipv6 destination port	qos policy priority-queue <text> class <text> match <text> ipv6 destination port <value>
×	not yet documented	qos.xml.in: qos policy priority-queue <priority-queue> class <class> match <match> ipv6 dscp	qos policy priority-queue <text> class <text> match <text> ipv6 dscp <value>
×	not yet documented	qos.xml.in: qos policy priority-queue <priority-queue> class <class> match <match> ipv6 max-length	qos policy priority-queue <text> class <text> match <text> ipv6 max-length <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	qos.xml.in: qos policy priority-queue <priority-queue> class <class> match <match> ipv6 protocol	qos policy priority-queue <text> class <text> match <text> ipv6 protocol <value>
×	not yet documented	qos.xml.in: qos policy priority-queue <priority-queue> class <class> match <match> ipv6 source address	qos policy priority-queue <text> class <text> match <text> ipv6 source address <value>
×	not yet documented	qos.xml.in: qos policy priority-queue <priority-queue> class <class> match <match> ipv6 source port	qos policy priority-queue <text> class <text> match <text> ipv6 source port <value>
×	not yet documented	qos.xml.in: qos policy priority-queue <priority-queue> class <class> match <match> ipv6 tcp ack	qos policy priority-queue <text> class <text> match <text> ipv6 tcp ack
×	not yet documented	qos.xml.in: qos policy priority-queue <priority-queue> class <class> match <match> ipv6 tcp syn	qos policy priority-queue <text> class <text> match <text> ipv6 tcp syn
×	not yet documented	qos.xml.in: qos policy priority-queue <priority-queue> class <class> match <match> mark	qos policy priority-queue <text> class <text> match <text> mark <value>
×	not yet documented	qos.xml.in: qos policy priority-queue <priority-queue> class <class> match <match> vif	qos policy priority-queue <text> class <text> match <text> vif <value>
×	not yet documented	qos.xml.in: qos policy priority-queue <priority-queue> class <class> queue-limit	qos policy priority-queue <text> class <text> queue- limit <value>
×	not yet documented	qos.xml.in: qos policy priority-queue <priority-queue> class <class> queue-type	qos policy priority-queue <text> class <text> queue- type <value>
×	not yet documented	qos.xml.in: qos policy priority-queue <priority-queue> class <class> target	qos policy priority-queue <text> class <text> target <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	qos.xml.in: qos policy priority-queue <priority-queue> default code1-quantum	qos policy priority-queue <text> default code1- quantum <value>
×	not yet documented	qos.xml.in: qos policy priority-queue <priority-queue> default flows	qos policy priority-queue <text> default flows <value>
×	not yet documented	qos.xml.in: qos policy priority-queue <priority-queue> default interval	qos policy priority-queue <text> default interval <value>
×	not yet documented	qos.xml.in: qos policy priority-queue <priority-queue> default queue-limit	qos policy priority-queue <text> default queue-limit <value>
×	not yet documented	qos.xml.in: qos policy priority-queue <priority-queue> default queue-type	qos policy priority-queue <text> default queue-type <value>
×	not yet documented	qos.xml.in: qos policy priority-queue <priority-queue> default target	qos policy priority-queue <text> default target <value>
×	not yet documented	qos.xml.in: qos policy priority-queue <priority-queue> descrip- tion	qos policy priority- queue <text> description <value>
×	<i>set qos policy random-detect <policy-name> bandwidth <bandwidth></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set qos policy random-detect <policy-name> precedence <IP-precedence-value> average-packet <bytes></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set qos policy random-detect <policy-name> precedence <IP-precedence-value> mark-probability <value></i>	Nothing found in XML Definitions	Nothing found in VyOS

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Table 1 – continued from previous page

Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<i>set qos policy random-detect <policy-name> precedence <IP-precedence-value> maximum-threshold <packets></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set qos policy random-detect <policy-name> precedence <IP-precedence-value> minimum-threshold <packets></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set qos policy random-detect <policy-name> precedence <IP-precedence-value> queue-limit <packets></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	qos.xml.in: qos policy random-detect <random-detect> bandwidth	qos policy random-detect <text> bandwidth <value>
×	not yet documented	qos.xml.in: qos policy random-detect <random-detect> description	qos policy random-detect <text> description <value>
×	not yet documented	qos.xml.in: qos policy random-detect <random-detect> precedence <precedence> average-packet	qos policy random-detect <text> precedence <text> average-packet <value>
×	not yet documented	qos.xml.in: qos policy random-detect <random-detect> precedence <precedence> mark-probability	qos policy random-detect <text> precedence <text> mark-probability <value>
×	not yet documented	qos.xml.in: qos policy random-detect <random-detect> precedence <precedence> maximum-threshold	qos policy random-detect <text> precedence <text> maximum-threshold <value>
×	not yet documented	qos.xml.in: qos policy random-detect <random-detect> precedence <precedence> minimum-threshold	qos policy random-detect <text> precedence <text> minimum-threshold <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	qos.xml.in: qos policy random-detect <random-detect> precedence precedence <precedence> queue-limit	qos policy random-detect <text> precedence <text> queue-limit <value>
×	<i>set qos policy rate-control <policy-name> bandwidth <rate></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set qos policy rate-control <policy-name> burst <burst-size></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set qos policy rate-control <policy-name> latency</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	qos.xml.in: qos policy rate-control <rate-control> bandwidth	qos policy rate-control <text> bandwidth <value>
×	not yet documented	qos.xml.in: qos policy rate-control <rate-control> burst	qos policy rate-control <text> burst <value>
×	not yet documented	qos.xml.in: qos policy rate-control <rate-control> description	qos policy rate-control <text> description <value>
×	not yet documented	qos.xml.in: qos policy rate-control <rate-control> latency	qos policy rate-control <text> latency <value>
×	<i>set qos policy round-robin <policy name> class <class-ID> quantum <packets></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set qos policy round-robin <policy name> class queue-limit <packets></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	qos.xml.in: qos policy round-robin <round-robin> class <class> codel-quantum	qos policy round-robin <text> class <text> codel-quantum <value>
×	not yet documented	qos.xml.in: qos policy round-robin <round-robin> class <class> description	qos policy round-robin <text> class <text> description <value>

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Table 1 – continued from previous page

Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	qos.xml.in: qos policy round-robin <round-robin> class <class> flows	qos policy round-robin <text> class <text> flows <value>
×	not yet documented	qos.xml.in: qos policy round-robin <round-robin> class <class> interval	qos policy round-robin <text> class <text> interval <value>
×	not yet documented	qos.xml.in: qos policy round-robin <round-robin> class <class> match <match> description	qos policy round-robin <text> class <text> match <text> description <value>
×	not yet documented	qos.xml.in: qos policy round-robin <round-robin> class <class> match <match> ether destination	qos policy round-robin <text> class <text> match <text> ether destination <value>
×	not yet documented	qos.xml.in: qos policy round-robin <round-robin> class <class> match <match> ether protocol	qos policy round-robin <text> class <text> match <text> ether protocol <value>
×	not yet documented	qos.xml.in: qos policy round-robin <round-robin> class <class> match <match> ether source	qos policy round-robin <text> class <text> match <text> ether source <value>
×	not yet documented	qos.xml.in: qos policy round-robin <round-robin> class <class> match <match> interface	qos policy round-robin <text> class <text> match <text> interface <value>
×	not yet documented	qos.xml.in: qos policy round-robin <round-robin> class <class> match <match> ip destination address	qos policy round-robin <text> class <text> match <text> ip destination address <value>
×	not yet documented	qos.xml.in: qos policy round-robin <round-robin> class <class> match <match> ip destination port	qos policy round-robin <text> class <text> match <text> ip destination port <value>
×	not yet documented	qos.xml.in: qos policy round-robin <round-robin> class <class> match <match> ip dscp	qos policy round-robin <text> class <text> match <text> ip dscp <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	qos.xml.in: qos policy round-robin <round-robin> class <class> match <match> ip max-length	qos policy round-robin <text> class <text> match <text> ip max-length <value>
×	not yet documented	qos.xml.in: qos policy round-robin <round-robin> class <class> match <match> ip protocol	qos policy round-robin <text> class <text> match <text> ip protocol <value>
×	not yet documented	qos.xml.in: qos policy round-robin <round-robin> class <class> match <match> ip source address	qos policy round-robin <text> class <text> match <text> ip source address <value>
×	not yet documented	qos.xml.in: qos policy round-robin <round-robin> class <class> match <match> ip source port	qos policy round-robin <text> class <text> match <text> ip source port <value>
×	not yet documented	qos.xml.in: qos policy round-robin <round-robin> class <class> match <match> ip tcp ack	qos policy round-robin <text> class <text> match <text> ip tcp ack
×	not yet documented	qos.xml.in: qos policy round-robin <round-robin> class <class> match <match> ip tcp syn	qos policy round-robin <text> class <text> match <text> ip tcp syn
×	not yet documented	qos.xml.in: qos policy round-robin <round-robin> class <class> match <match> ipv6 destination address	qos policy round-robin <text> class <text> match <text> ipv6 destination address <value>
×	not yet documented	qos.xml.in: qos policy round-robin <round-robin> class <class> match <match> ipv6 destination port	qos policy round-robin <text> class <text> match <text> ipv6 destination port <value>
×	not yet documented	qos.xml.in: qos policy round-robin <round-robin> class <class> match <match> ipv6 dscp	qos policy round-robin <text> class <text> match <text> ipv6 dscp <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	qos.xml.in: qos policy round-robin <round-robin> class <class> match <match> ipv6 max-length	qos policy round-robin <text> class <text> match <text> ipv6 max-length <value>
×	not yet documented	qos.xml.in: qos policy round-robin <round-robin> class <class> match <match> ipv6 protocol	qos policy round-robin <text> class <text> match <text> ipv6 protocol <value>
×	not yet documented	qos.xml.in: qos policy round-robin <round-robin> class <class> match <match> ipv6 source address	qos policy round-robin <text> class <text> match <text> ipv6 source address <value>
×	not yet documented	qos.xml.in: qos policy round-robin <round-robin> class <class> match <match> ipv6 source port	qos policy round-robin <text> class <text> match <text> ipv6 source port <value>
×	not yet documented	qos.xml.in: qos policy round-robin <round-robin> class <class> match <match> ipv6 tcp ack	qos policy round-robin <text> class <text> match <text> ipv6 tcp ack
×	not yet documented	qos.xml.in: qos policy round-robin <round-robin> class <class> match <match> ipv6 tcp syn	qos policy round-robin <text> class <text> match <text> ipv6 tcp syn
×	not yet documented	qos.xml.in: qos policy round-robin <round-robin> class <class> match <match> mark	qos policy round-robin <text> class <text> match <text> mark <value>
×	not yet documented	qos.xml.in: qos policy round-robin <round-robin> class <class> match <match> vif	qos policy round-robin <text> class <text> match <text> vif <value>
×	not yet documented	qos.xml.in: qos policy round-robin <round-robin> class <class> quantum	qos policy round-robin <text> class <text> quantum <value>
×	not yet documented	qos.xml.in: qos policy round-robin <round-robin> class <class> queue-limit	qos policy round-robin <text> class <text> queue-limit <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	qos.xml.in: qos policy round-robin <round-robin> class <class> queue-type	qos policy round-robin <text> class <text> queue-type <value>
×	not yet documented	qos.xml.in: qos policy round-robin <round-robin> class <class> target	qos policy round-robin <text> class <text> target <value>
×	not yet documented	qos.xml.in: qos policy round-robin <round-robin> default code1-quantum	qos policy round-robin <text> default code1- quantum <value>
×	not yet documented	qos.xml.in: qos policy round-robin <round-robin> default flows	qos policy round-robin <text> default flows <value>
×	not yet documented	qos.xml.in: qos policy round-robin <round-robin> default interval	qos policy round-robin <text> default interval <value>
×	not yet documented	qos.xml.in: qos policy round-robin <round-robin> default queue-limit	qos policy round-robin <text> default queue-limit <value>
×	not yet documented	qos.xml.in: qos policy round-robin <round-robin> default queue-type	qos policy round-robin <text> default queue-type <value>
×	not yet documented	qos.xml.in: qos policy round-robin <round-robin> default target	qos policy round-robin <text> default target <value>
×	not yet documented	qos.xml.in: qos policy round-robin <round-robin> description	qos policy round-robin <text> description <value>
×	not yet documented	qos.xml.in: qos policy shaper-hfsc <shaper-hfsc> bandwidth	qos policy shaper-hfsc <text> bandwidth <value>
×	not yet documented	qos.xml.in: qos policy shaper-hfsc <shaper-hfsc> class <class> description	qos policy shaper-hfsc <text> class <text> description <value>
×	not yet documented	qos.xml.in: qos policy shaper-hfsc <shaper-hfsc> class <class> linkshare d	qos policy shaper-hfsc <text> class <text> linkshare d <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	qos.xml.in: qos policy shaper-hfsc <shaper-hfsc> class <class> linkshare m1	qos policy shaper-hfsc <text> class <text> linkshare m1 <value>
×	not yet documented	qos.xml.in: qos policy shaper-hfsc <shaper-hfsc> class <class> linkshare m2	qos policy shaper-hfsc <text> class <text> linkshare m2 <value>
×	not yet documented	qos.xml.in: qos policy shaper-hfsc <shaper-hfsc> class <class> match <match> description	qos policy shaper-hfsc <text> class <text> match <text> description <value>
×	not yet documented	qos.xml.in: qos policy shaper-hfsc <shaper-hfsc> class <class> match <match> ether destination	qos policy shaper-hfsc <text> class <text> match <text> ether destination <value>
×	not yet documented	qos.xml.in: qos policy shaper-hfsc <shaper-hfsc> class <class> match <match> ether protocol	qos policy shaper-hfsc <text> class <text> match <text> ether protocol <value>
×	not yet documented	qos.xml.in: qos policy shaper-hfsc <shaper-hfsc> class <class> match <match> ether source	qos policy shaper-hfsc <text> class <text> match <text> ether source <value>
×	not yet documented	qos.xml.in: qos policy shaper-hfsc <shaper-hfsc> class <class> match <match> interface	qos policy shaper-hfsc <text> class <text> match <text> interface <value>
×	not yet documented	qos.xml.in: qos policy shaper-hfsc <shaper-hfsc> class <class> match <match> ip destination address	qos policy shaper-hfsc <text> class <text> match <text> ip destination address <value>
×	not yet documented	qos.xml.in: qos policy shaper-hfsc <shaper-hfsc> class <class> match <match> ip destination port	qos policy shaper-hfsc <text> class <text> match <text> ip destination port <value>
×	not yet documented	qos.xml.in: qos policy shaper-hfsc <shaper-hfsc> class <class> match <match> ip dscp	qos policy shaper-hfsc <text> class <text> match <text> ip dscp <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	qos.xml.in: qos policy shaper-hfsc <shaper-hfsc> class <class> match <match> ip max-length	qos policy shaper-hfsc <text> class <text> match <text> ip max-length <value>
×	not yet documented	qos.xml.in: qos policy shaper-hfsc <shaper-hfsc> class <class> match <match> ip protocol	qos policy shaper-hfsc <text> class <text> match <text> ip protocol <value>
×	not yet documented	qos.xml.in: qos policy shaper-hfsc <shaper-hfsc> class <class> match <match> ip source address	qos policy shaper-hfsc <text> class <text> match <text> ip source address <value>
×	not yet documented	qos.xml.in: qos policy shaper-hfsc <shaper-hfsc> class <class> match <match> ip source port	qos policy shaper-hfsc <text> class <text> match <text> ip source port <value>
×	not yet documented	qos.xml.in: qos policy shaper-hfsc <shaper-hfsc> class <class> match <match> ip tcp ack	qos policy shaper-hfsc <text> class <text> match <text> ip tcp ack
×	not yet documented	qos.xml.in: qos policy shaper-hfsc <shaper-hfsc> class <class> match <match> ip tcp syn	qos policy shaper-hfsc <text> class <text> match <text> ip tcp syn
×	not yet documented	qos.xml.in: qos policy shaper-hfsc <shaper-hfsc> class <class> match <match> ipv6 destination address	qos policy shaper-hfsc <text> class <text> match <text> ipv6 destination address <value>
×	not yet documented	qos.xml.in: qos policy shaper-hfsc <shaper-hfsc> class <class> match <match> ipv6 destination port	qos policy shaper-hfsc <text> class <text> match <text> ipv6 destination port <value>
×	not yet documented	qos.xml.in: qos policy shaper-hfsc <shaper-hfsc> class <class> match <match> ipv6 dscp	qos policy shaper-hfsc <text> class <text> match <text> ipv6 dscp <value>

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Table 1 – continued from previous page

Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	qos.xml.in: qos policy shaper-hfsc <shaper-hfsc> class <class> match <match> ipv6 max-length	qos policy shaper-hfsc <text> class <text> match <text> ipv6 max-length <value>
×	not yet documented	qos.xml.in: qos policy shaper-hfsc <shaper-hfsc> class <class> match <match> ipv6 protocol	qos policy shaper-hfsc <text> class <text> match <text> ipv6 protocol <value>
×	not yet documented	qos.xml.in: qos policy shaper-hfsc <shaper-hfsc> class <class> match <match> ipv6 source address	qos policy shaper-hfsc <text> class <text> match <text> ipv6 source address <value>
×	not yet documented	qos.xml.in: qos policy shaper-hfsc <shaper-hfsc> class <class> match <match> ipv6 source port	qos policy shaper-hfsc <text> class <text> match <text> ipv6 source port <value>
×	not yet documented	qos.xml.in: qos policy shaper-hfsc <shaper-hfsc> class <class> match <match> ipv6 tcp ack	qos policy shaper-hfsc <text> class <text> match <text> ipv6 tcp ack
×	not yet documented	qos.xml.in: qos policy shaper-hfsc <shaper-hfsc> class <class> match <match> ipv6 tcp syn	qos policy shaper-hfsc <text> class <text> match <text> ipv6 tcp syn
×	not yet documented	qos.xml.in: qos policy shaper-hfsc <shaper-hfsc> class <class> match <match> mark	qos policy shaper-hfsc <text> class <text> match <text> mark <value>
×	not yet documented	qos.xml.in: qos policy shaper-hfsc <shaper-hfsc> class <class> match <match> vif	qos policy shaper-hfsc <text> class <text> match <text> vif <value>
×	not yet documented	qos.xml.in: qos policy shaper-hfsc <shaper-hfsc> class <class> realtime d	qos policy shaper-hfsc <text> class <text> realtime d <value>
×	not yet documented	qos.xml.in: qos policy shaper-hfsc <shaper-hfsc> class <class> realtime m1	qos policy shaper-hfsc <text> class <text> realtime m1 <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	qos.xml.in: qos policy shaper-hfsc <shaper-hfsc> class <class> realtime m2	qos policy shaper-hfsc <text> class <text> realtime m2 <value>
×	not yet documented	qos.xml.in: qos policy shaper-hfsc <shaper-hfsc> class <class> upperlimit d	qos policy shaper-hfsc <text> class <text> upperlimit d <value>
×	not yet documented	qos.xml.in: qos policy shaper-hfsc <shaper-hfsc> class <class> upperlimit m1	qos policy shaper-hfsc <text> class <text> upperlimit m1 <value>
×	not yet documented	qos.xml.in: qos policy shaper-hfsc <shaper-hfsc> class <class> upperlimit m2	qos policy shaper-hfsc <text> class <text> upperlimit m2 <value>
×	not yet documented	qos.xml.in: qos policy shaper-hfsc <shaper-hfsc> default linkshare d	qos policy shaper-hfsc <text> default linkshare d <value>
×	not yet documented	qos.xml.in: qos policy shaper-hfsc <shaper-hfsc> default linkshare m1	qos policy shaper-hfsc <text> default linkshare m1 <value>
×	not yet documented	qos.xml.in: qos policy shaper-hfsc <shaper-hfsc> default linkshare m2	qos policy shaper-hfsc <text> default linkshare m2 <value>
×	not yet documented	qos.xml.in: qos policy shaper-hfsc <shaper-hfsc> default realtime d	qos policy shaper-hfsc <text> default realtime d <value>
×	not yet documented	qos.xml.in: qos policy shaper-hfsc <shaper-hfsc> default realtime m1	qos policy shaper-hfsc <text> default realtime m1 <value>
×	not yet documented	qos.xml.in: qos policy shaper-hfsc <shaper-hfsc> default realtime m2	qos policy shaper-hfsc <text> default realtime m2 <value>
×	not yet documented	qos.xml.in: qos policy shaper-hfsc <shaper-hfsc> default upperlimit d	qos policy shaper-hfsc <text> default upperlimit d <value>
×	not yet documented	qos.xml.in: qos policy shaper-hfsc <shaper-hfsc> default upperlimit m1	qos policy shaper-hfsc <text> default upperlimit m1 <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	qos.xml.in: qos policy shaper-hfsc <shaper-hfsc> default upperlimit m2	qos policy shaper-hfsc <text> default upperlimit m2 <value>
×	not yet documented	qos.xml.in: qos policy shaper-hfsc <shaper-hfsc> description	qos policy shaper-hfsc <text> description <value>
×	<i>set qos policy shaper <policy-name> bandwidth <rate></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set qos policy shaper <policy-name> class <class-ID> bandwidth <rate></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set qos policy shaper <policy-name> class <class-ID> burst <bytes></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set qos policy shaper <policy-name> class <class-ID> ceiling <bandwidth></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set qos policy shaper <policy-name> class <class-ID> priority <0-7></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	qos.xml.in: qos policy shaper <shaper> bandwidth	qos policy shaper <text> bandwidth <value>
×	not yet documented	qos.xml.in: qos policy shaper <shaper> class <class> average-packet	Nothing found in VyOS
×	not yet documented	qos.xml.in: qos policy shaper <shaper> class <class> bandwidth	qos policy shaper <text> class <text> bandwidth <value>
×	not yet documented	qos.xml.in: qos policy shaper <shaper> class <class> burst	qos policy shaper <text> class <text> burst <value>
×	not yet documented	qos.xml.in: qos policy shaper <shaper> class <class> ceiling	qos policy shaper <text> class <text> ceiling <value>
×	not yet documented	qos.xml.in: qos policy shaper <shaper> class <class> codel-quantum	qos policy shaper <text> class <text> codel- quantum <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	qos.xml.in: qos policy shaper <shaper> class <class> description	qos policy shaper <text> class <text> description <value>
×	not yet documented	qos.xml.in: qos policy shaper <shaper> class <class> flows	qos policy shaper <text> class <text> flows <value>
×	not yet documented	qos.xml.in: qos policy shaper <shaper> class <class> interval	qos policy shaper <text> class <text> interval <value>
×	not yet documented	qos.xml.in: qos policy shaper <shaper> class <class> mark-probability	Nothing found in VyOS
×	not yet documented	qos.xml.in: qos policy shaper <shaper> class <class> match <match> descrip- tion	qos policy shaper <text> class <text> match <text> description <value>
×	not yet documented	qos.xml.in: qos policy shaper <shaper> class <class> match <match> ether destination	qos policy shaper <text> class <text> match <text> ether destination <value>
×	not yet documented	qos.xml.in: qos policy shaper <shaper> class <class> match <match> ether protocol	qos policy shaper <text> class <text> match <text> ether protocol <value>
×	not yet documented	qos.xml.in: qos policy shaper <shaper> class <class> match <match> ether source	qos policy shaper <text> class <text> match <text> ether source <value>
×	not yet documented	qos.xml.in: qos policy shaper <shaper> class <class> match <match> interface	qos policy shaper <text> class <text> match <text> interface <value>
×	not yet documented	qos.xml.in: qos policy shaper <shaper> class <class> match <match> ip desti- nation address	qos policy shaper <text> class <text> match <text> ip destination address <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	qos.xml.in: qos policy shaper <shaper> class <class> match <match> ip destination port	qos policy shaper <text> class <text> match <text> ip destination port <value>
×	not yet documented	qos.xml.in: qos policy shaper <shaper> class <class> match <match> ip dscp	qos policy shaper <text> class <text> match <text> ip dscp <value>
×	not yet documented	qos.xml.in: qos policy shaper <shaper> class <class> match <match> ip max-length	qos policy shaper <text> class <text> match <text> ip max-length <value>
×	not yet documented	qos.xml.in: qos policy shaper <shaper> class <class> match <match> ip protocol	qos policy shaper <text> class <text> match <text> ip protocol <value>
×	not yet documented	qos.xml.in: qos policy shaper <shaper> class <class> match <match> ip source address	qos policy shaper <text> class <text> match <text> ip source address <value>
×	not yet documented	qos.xml.in: qos policy shaper <shaper> class <class> match <match> ip source port	qos policy shaper <text> class <text> match <text> ip source port <value>
×	not yet documented	qos.xml.in: qos policy shaper <shaper> class <class> match <match> ip tcp ack	qos policy shaper <text> class <text> match <text> ip tcp ack
×	not yet documented	qos.xml.in: qos policy shaper <shaper> class <class> match <match> ip tcp syn	qos policy shaper <text> class <text> match <text> ip tcp syn
×	not yet documented	qos.xml.in: qos policy shaper <shaper> class <class> match <match> ipv6 destination address	qos policy shaper <text> class <text> match <text> ipv6 destination address <value>
×	not yet documented	qos.xml.in: qos policy shaper <shaper> class <class> match <match> ipv6 destination port	qos policy shaper <text> class <text> match <text> ipv6 destination port <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	qos.xml.in: qos policy shaper <shaper> class <class> match <match> ipv6 dscp	qos policy shaper <text> class <text> match <text> ipv6 dscp <value>
×	not yet documented	qos.xml.in: qos policy shaper <shaper> class <class> match <match> ipv6 max-length	qos policy shaper <text> class <text> match <text> ipv6 max-length <value>
×	not yet documented	qos.xml.in: qos policy shaper <shaper> class <class> match <match> ipv6 protocol	qos policy shaper <text> class <text> match <text> ipv6 protocol <value>
×	not yet documented	qos.xml.in: qos policy shaper <shaper> class <class> match <match> ipv6 source address	qos policy shaper <text> class <text> match <text> ipv6 source address <value>
×	not yet documented	qos.xml.in: qos policy shaper <shaper> class <class> match <match> ipv6 source port	qos policy shaper <text> class <text> match <text> ipv6 source port <value>
×	not yet documented	qos.xml.in: qos policy shaper <shaper> class <class> match <match> ipv6 tcp ack	qos policy shaper <text> class <text> match <text> ipv6 tcp ack
×	not yet documented	qos.xml.in: qos policy shaper <shaper> class <class> match <match> ipv6 tcp syn	qos policy shaper <text> class <text> match <text> ipv6 tcp syn
×	not yet documented	qos.xml.in: qos policy shaper <shaper> class <class> match <match> mark	qos policy shaper <text> class <text> match <text> mark <value>
×	not yet documented	qos.xml.in: qos policy shaper <shaper> class <class> match <match> vif	qos policy shaper <text> class <text> match <text> vif <value>
×	not yet documented	qos.xml.in: qos policy shaper <shaper> class <class> maximum-threshold	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	qos.xml.in: qos policy shaper <shaper> class <class> minimum-threshold	Nothing found in VyOS
×	not yet documented	qos.xml.in: qos policy shaper <shaper> class <class> priority	qos policy shaper <text> class <text> priority <value>
×	not yet documented	qos.xml.in: qos policy shaper <shaper> class <class> queue-limit	qos policy shaper <text> class <text> queue-limit <value>
×	not yet documented	qos.xml.in: qos policy shaper <shaper> class <class> queue-type	qos policy shaper <text> class <text> queue-type <value>
×	not yet documented	qos.xml.in: qos policy shaper <shaper> class <class> set-dscp	qos policy shaper <text> class <text> set-dscp <value>
×	not yet documented	qos.xml.in: qos policy shaper <shaper> class <class> target	qos policy shaper <text> class <text> target <value>
×	not yet documented	qos.xml.in: qos policy shaper <shaper> default average- packet	Nothing found in VyOS
×	not yet documented	qos.xml.in: qos policy shaper <shaper> default band- width	qos policy shaper <text> default bandwidth <value>
×	not yet documented	qos.xml.in: qos policy shaper <shaper> default burst	qos policy shaper <text> default burst <value>
×	not yet documented	qos.xml.in: qos policy shaper <shaper> default ceil- ing	qos policy shaper <text> default ceiling <value>
×	not yet documented	qos.xml.in: qos policy shaper <shaper> default codel- quantum	qos policy shaper <text> default codel-quantum <value>
×	not yet documented	qos.xml.in: qos policy shaper <shaper> default flows	qos policy shaper <text> default flows <value>
×	not yet documented	qos.xml.in: qos policy shaper <shaper> default interval	qos policy shaper <text> default interval <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	qos.xml.in: qos policy shaper <shaper> default mark- probability	Nothing found in VyOS
×	not yet documented	qos.xml.in: qos policy shaper <shaper> default maximum-threshold	Nothing found in VyOS
×	not yet documented	qos.xml.in: qos policy shaper <shaper> default minimum-threshold	Nothing found in VyOS
×	not yet documented	qos.xml.in: qos policy shaper <shaper> default pri- ority	qos policy shaper <text> default priority <value>
×	not yet documented	qos.xml.in: qos policy shaper <shaper> default queue- limit	qos policy shaper <text> default queue-limit <value>
×	not yet documented	qos.xml.in: qos policy shaper <shaper> default queue- type	qos policy shaper <text> default queue-type <value>
×	not yet documented	qos.xml.in: qos policy shaper <shaper> default set- dscp	qos policy shaper <text> default set-dscp <value>
×	not yet documented	qos.xml.in: qos policy shaper <shaper> default tar- get	qos policy shaper <text> default target <value>
×	not yet documented	qos.xml.in: qos policy shaper <shaper> description	qos policy shaper <text> description <value>
×	rename	Nothing found in XML Definitions	Nothing found in VyOS
×	rollback <N>	Nothing found in XML Definitions	Nothing found in VyOS
×	run	Nothing found in XML Definitions	Nothing found in VyOS
×	save	Nothing found in XML Definitions	Nothing found in VyOS
✓	set service broadcast-relay disable	service_broadcast- relay.xml.in: service broadcast-relay disable	service broadcast-relay disable

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set service broadcast-relay id <1-99> address <ipv4-address></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set service broadcast-relay id <1-99> description <description></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set service broadcast-relay id <1-99> disable</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set service broadcast-relay id <1-99> interface <interface></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set service broadcast-relay id <1-99> port <1-65535></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	service_broadcast-relay.xml.in: service broadcast-relay id <id> address	service broadcast-relay id <text> address <value>
×	not yet documented	service_broadcast-relay.xml.in: service broadcast-relay id <id> description	service broadcast-relay id <text> description <value>
×	not yet documented	service_broadcast-relay.xml.in: service broadcast-relay id <id> disable	service broadcast-relay id <text> disable
×	not yet documented	service_broadcast-relay.xml.in: service broadcast-relay id <id> interface	service broadcast-relay id <text> interface <value>
×	not yet documented	service_broadcast-relay.xml.in: service broadcast-relay id <id> port	service broadcast-relay id <text> port <value>
×	not yet documented	service_config-sync.xml.in: service config-sync mode	service config-sync mode <value>
×	<code>set service config-sync mode <load set></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	service_config-sync.xml.in: service config-sync secondary address	service config-sync secondary address <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<i>set service config-sync secondary address <address></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	service_config-sync.xml.in: service config-sync secondary key	service config-sync secondary key <value>
×	<i>set service config-sync secondary key <key></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set service config-sync secondary port <1-65535></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	service_config-sync.xml.in: service config-sync secondary timeout	service config-sync secondary timeout <value>
×	<i>set service config-sync secondary timeout <1-3600></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	Nothing found in XML Definitions	service config-sync section <value>
×	<i>set service config-sync section <section></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	service_config-sync.xml.in: service config-sync section firewall	Nothing found in VyOS
×	not yet documented	service_config-sync.xml.in: service config-sync section interfaces bonding	Nothing found in VyOS
×	not yet documented	service_config-sync.xml.in: service config-sync section interfaces bridge	Nothing found in VyOS
×	not yet documented	service_config-sync.xml.in: service config-sync section interfaces dummy	Nothing found in VyOS
×	not yet documented	service_config-sync.xml.in: service config-sync section interfaces ethernet	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	service_config-sync.xml.in: service config-sync section interfaces geneve	Nothing found in VyOS
×	not yet documented	service_config-sync.xml.in: service config-sync section interfaces input	Nothing found in VyOS
×	not yet documented	service_config-sync.xml.in: service config-sync section interfaces l2tpv3	Nothing found in VyOS
×	not yet documented	service_config-sync.xml.in: service config-sync section interfaces loopback	Nothing found in VyOS
×	not yet documented	service_config-sync.xml.in: service config-sync section interfaces macsec	Nothing found in VyOS
×	not yet documented	service_config-sync.xml.in: service config-sync section interfaces openvpn	Nothing found in VyOS
×	not yet documented	service_config-sync.xml.in: service config-sync section interfaces pppoe	Nothing found in VyOS
×	not yet documented	service_config-sync.xml.in: service config-sync section interfaces pseudo-ethernet	Nothing found in VyOS
×	not yet documented	service_config-sync.xml.in: service config-sync section interfaces sstp	Nothing found in VyOS
×	not yet documented	service_config-sync.xml.in: service config-sync section interfaces tunnel	Nothing found in VyOS
×	not yet documented	service_config-sync.xml.in: service config-sync section interfaces virtual-ethernet	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	service_config-sync.xml.in: service config-sync section interfaces vti	Nothing found in VyOS
×	not yet documented	service_config-sync.xml.in: service config-sync section interfaces vxlan	Nothing found in VyOS
×	not yet documented	service_config-sync.xml.in: service config-sync section interfaces wireguard	Nothing found in VyOS
×	not yet documented	service_config-sync.xml.in: service config-sync section interfaces wireless	Nothing found in VyOS
×	not yet documented	service_config-sync.xml.in: service config-sync section interfaces wwan	Nothing found in VyOS
×	not yet documented	service_config-sync.xml.in: service config-sync section nat	Nothing found in VyOS
×	not yet documented	service_config-sync.xml.in: service config-sync section nat66	Nothing found in VyOS
×	not yet documented	service_config-sync.xml.in: service config-sync section pki	Nothing found in VyOS
×	not yet documented	service_config-sync.xml.in: service config-sync section policy	Nothing found in VyOS
×	not yet documented	service_config-sync.xml.in: service config-sync section protocols babel	Nothing found in VyOS
×	not yet documented	service_config-sync.xml.in: service config-sync section protocols bfd	Nothing found in VyOS
×	not yet documented	service_config-sync.xml.in: service config-sync section protocols bgp	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	service_config-sync.xml.in: service config-sync section protocols failover	Nothing found in VyOS
×	not yet documented	service_config-sync.xml.in: service config-sync section protocols igmp-proxy	Nothing found in VyOS
×	not yet documented	service_config-sync.xml.in: service config-sync section protocols isis	Nothing found in VyOS
×	not yet documented	service_config-sync.xml.in: service config-sync section protocols mpls	Nothing found in VyOS
×	not yet documented	service_config-sync.xml.in: service config-sync section protocols nhrp	Nothing found in VyOS
×	not yet documented	service_config-sync.xml.in: service config-sync section protocols ospf	Nothing found in VyOS
×	not yet documented	service_config-sync.xml.in: service config-sync section protocols ospfv3	Nothing found in VyOS
×	not yet documented	service_config-sync.xml.in: service config-sync section protocols pim	Nothing found in VyOS
×	not yet documented	service_config-sync.xml.in: service config-sync section protocols pim6	Nothing found in VyOS
×	not yet documented	service_config-sync.xml.in: service config-sync section protocols rip	Nothing found in VyOS
×	not yet documented	service_config-sync.xml.in: service config-sync section protocols ripng	Nothing found in VyOS
×	not yet documented	service_config-sync.xml.in: service config-sync section protocols rpki	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	service_config-sync.xml.in: service config-sync section protocols segment-routing	Nothing found in VyOS
×	not yet documented	service_config-sync.xml.in: service config-sync section protocols static	Nothing found in VyOS
×	not yet documented	service_config-sync.xml.in: service config-sync section qos interface	Nothing found in VyOS
×	not yet documented	service_config-sync.xml.in: service config-sync section qos policy	Nothing found in VyOS
×	not yet documented	service_config-sync.xml.in: service config-sync section service console-server	Nothing found in VyOS
×	not yet documented	service_config-sync.xml.in: service config-sync section service dhcp-relay	Nothing found in VyOS
×	not yet documented	service_config-sync.xml.in: service config-sync section service dhcp-server	Nothing found in VyOS
×	not yet documented	service_config-sync.xml.in: service config-sync section service dhcpv6-relay	Nothing found in VyOS
×	not yet documented	service_config-sync.xml.in: service config-sync section service dhcpv6-server	Nothing found in VyOS
×	not yet documented	service_config-sync.xml.in: service config-sync section service dns	Nothing found in VyOS
×	not yet documented	service_config-sync.xml.in: service config-sync section service lldp	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	service_config-sync.xml.in: service config-sync section service mdns	Nothing found in VyOS
×	not yet documented	service_config-sync.xml.in: service config-sync section service monitoring	Nothing found in VyOS
×	not yet documented	service_config-sync.xml.in: service config-sync section service ndp-proxy	Nothing found in VyOS
×	not yet documented	service_config-sync.xml.in: service config-sync section service ntp	Nothing found in VyOS
×	not yet documented	service_config-sync.xml.in: service config-sync section service snmp	Nothing found in VyOS
×	not yet documented	service_config-sync.xml.in: service config-sync section service tftp-server	Nothing found in VyOS
×	not yet documented	service_config-sync.xml.in: service config-sync section service webproxy	Nothing found in VyOS
×	not yet documented	service_config-sync.xml.in: service config-sync section system conntrack	Nothing found in VyOS
×	not yet documented	service config-sync section system flow-accounting	Nothing found in VyOS
×	not yet documented	service_config-sync.xml.in: service config-sync section system option	Nothing found in VyOS
×	not yet documented	service_config-sync.xml.in: service config-sync section system sflow	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	service_config-sync.xml.in: service config-sync section system static-host-mapping	Nothing found in VyOS
×	not yet documented	service_config-sync.xml.in: service config-sync section system sysctl	Nothing found in VyOS
×	not yet documented	service_config-sync.xml.in: service config-sync section vpn	Nothing found in VyOS
×	not yet documented	service_config-sync.xml.in: service config-sync section vrf	Nothing found in VyOS
✓	<i>set service conntrack-sync accept-protocol</i>	service_conntrack-sync.xml.in: service conntrack-sync accept-protocol	service conntrack-sync accept-protocol <value>
✓	<i>set service conntrack-sync disable-external-cache</i>	service_conntrack-sync.xml.in: service conntrack-sync disable-external-cache	service conntrack-sync disable-external-cache
×	<i>set service conntrack-sync disable-syslog</i>	service_conntrack-sync.xml.in: service conntrack-sync disable-syslog	Nothing found in VyOS
×	not yet documented	service_conntrack-sync.xml.in: service conntrack-sync event-listen-queue-size	service conntrack-sync event-listen-queue-size <value>
×	<i>set service conntrack-sync event-listen-queue-size <size></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	service_conntrack-sync.xml.in: service conntrack-sync expect-sync	service conntrack-sync expect-sync <value>
×	<i>set service conntrack-sync expect-sync <all ftp h323 nfs sip></i>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	service_conntrack-sync.xml.in: service conntrack-sync failover-mechanism vrrp sync-group	service conntrack-sync failover-mechanism vrrp sync-group <value>
×	<i>set service conntrack-sync failover-mechanism vrrp sync-group <group></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	service_conntrack-sync.xml.in: service conntrack-sync ignore-address	service conntrack-sync ignore-address <value>
×	<i>set service conntrack-sync ignore-address <x.x. x.x></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set service conntrack-sync interface <name></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set service conntrack-sync interface <name> peer <address></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set service conntrack-sync interface <name> port <port></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	service_conntrack-sync.xml.in: service conntrack-sync in- terface <interface> peer	service conntrack-sync interface <text> peer <value>
×	not yet documented	service_conntrack-sync.xml.in: service conntrack-sync in- terface <interface> port	service conntrack-sync interface <text> port <value>
×	not yet documented	service_conntrack-sync.xml.in: service conntrack-sync listen-address	service conntrack-sync listen-address <value>
×	<i>set service conntrack-sync listen-address <ipv4address></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	service_conntrack-sync.xml.in: service conntrack-sync mcast-group	service conntrack-sync mcast-group <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set service conntrack-sync mcast-group <x.x.x.x></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set service conntrack-sync purge-timeout <timeout></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set service conntrack-sync startup-resync</code>	service_conntrack-sync.xml.in: service conntrack-sync startup-resync	Nothing found in VyOS
×	not yet documented	service_conntrack-sync.xml.in: service conntrack-sync sync-queue-size	service conntrack-sync sync-queue-size <value>
×	<code>set service conntrack-sync sync-queue-size <size></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set service console-server device <device> alias <alias></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set service console-server device <device> data-bits <7 8></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set service console-server device <device> description <description></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set service console-server device <device> parity <even odd none></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set service console-server device <device> speed <300 1200 2400 4800 9600 19200 38400 57600 115200></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set service console-server device <device> ssh port <1-65535></code>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	set service console-server device <device> stop-bits <1 2>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	service_console-server.xml.in: service console-server device <device> alias	service console-server device <text> alias <value>
×	not yet documented	service_console-server.xml.in: service console-server device <device> data-bits	service console-server device <text> data-bits <value>
×	not yet documented	service_console-server.xml.in: service console-server device <device> description	service console-server device <text> description <value>
×	not yet documented	service_console-server.xml.in: service console-server device <device> parity	service console-server device <text> parity <value>
×	not yet documented	service_console-server.xml.in: service console-server device <device> speed	service console-server device <text> speed <value>
×	not yet documented	service_console-server.xml.in: service console-server device <device> ssh port	service console-server device <text> ssh port <value>
×	not yet documented	service_console-server.xml.in: service console-server device <device> stop-bits	service console-server device <text> stop-bits <value>
✓	set service dhcp-relay disable	service_dhcp-relay.xml.in: service dhcp-relay disable	service dhcp-relay disable
×	not yet documented	service_dhcp-relay.xml.in: service dhcp-relay interface	service dhcp-relay interface <value>
×	set service dhcp-relay interface <interface>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	service_dhcp-relay.xml.in: service dhcp-relay listen-interface	service dhcp-relay listen-interface <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<i>set service dhcp-relay listen-interface <interface></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	service_dhcp-relay.xml.in: service dhcp-relay relay-options hop-count	service dhcp-relay relay-options hop-count <value>
×	<i>set service dhcp-relay relay-options hop-count <count></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	service_dhcp-relay.xml.in: service dhcp-relay relay-options max-size	service dhcp-relay relay-options max-size <value>
×	<i>set service dhcp-relay relay-options max-size <size></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	service_dhcp-relay.xml.in: service dhcp-relay relay-options relay-agents-packets	service dhcp-relay relay-options relay-agents-packets <value>
×	<i>set service dhcp-relay relay-options relay-agents-packets <append discard forward replace></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set service dhcp-relay relay-options relay-agents-packets discard</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	service_dhcp-relay.xml.in: service dhcp-relay server	service dhcp-relay server <value>
×	<i>set service dhcp-relay server <server></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	service_dhcp-relay.xml.in: service dhcp-relay upstream-interface	service dhcp-relay upstream-interface <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set service dhcp-relay upstream-interface <interface></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set service dhcp-server client-class <name> relay-agent-informati circuit-id <value></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set service dhcp-server client-class <name> relay-agent-informati remote-id <value></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	service_dhcp-server.xml.in: service dhcp-server disable	service dhcp-server disable
✓	<code>set service dhcp-server dynamic-dns-update</code>	service_dhcp-server.xml.in: service dhcp-server dynamic-dns-update	service dhcp-server dynamic-dns-update
×	<code>set service dhcp-server dynamic-dns-update [forward reverse]-dom <domain-name> dns-server <number> address <ip-address></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set service dhcp-server dynamic-dns-update [forward reverse]-dom <domain-name> dns-server <number> port <port></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set service dhcp-server dynamic-dns-update [forward reverse]-dom <domain-name> key-name <tsig-key-name></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set service dhcp-server dynamic-dns-update conflict-resolution <enable disable></code>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set service dhcp-server dynamic-dns-update generated-prefix <prefix></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set service dhcp-server dynamic-dns-update hostname-char-replace <character string></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set service dhcp-server dynamic-dns-update hostname-char-set <character string></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set service dhcp-server dynamic-dns-update override-client-updat <enable disable></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set service dhcp-server dynamic-dns-update override-no-update <enable disable></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set service dhcp-server dynamic-dns-update qualifying-suffix <suffix></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set service dhcp-server dynamic-dns-update replace-client-name [never always when-present when-not-present]</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set service dhcp-server dynamic-dns-update send-updates <enable disable></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set service dhcp-server dynamic-dns-update tsig-key <key-name> algorithm <algorithm></code>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<i>set service dhcp-server dynamic-dns-update tsig-key <key-name> secret <key-secret></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set service dhcp-server dynamic-dns-update ttl-percent <0-100></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set service dhcp-server dynamic-dns-update update-on-renew <enable disable></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	Nothing found in XML Definitions	service dhcp-server failover name <value>
×	not yet documented	Nothing found in XML Definitions	service dhcp-server failover remote <value>
×	not yet documented	Nothing found in XML Definitions	service dhcp-server failover source-address <value>
×	not yet documented	Nothing found in XML Definitions	service dhcp-server failover status <value>
×	not yet documented	service_dhcp-server.xml.in: service dhcp-server global-parameters	service dhcp-server global-parameters <value>
×	<i>set service dhcp-server high-availability mode [active-active active-passive]</i>	service_dhcp-server.xml.in: service dhcp-server high-availability mode	Nothing found in VyOS
×	not yet documented	service_dhcp-server.xml.in: service dhcp-server high-availability name	Nothing found in VyOS
×	<i>set service dhcp-server high-availability name <name></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	service_dhcp-server.xml.in: service dhcp-server high-availability remote	Nothing found in VyOS
×	<i>set service dhcp-server high-availability remote <address></i>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	service_dhcp-server.xml.in: service dhcp-server high-availability source-address	Nothing found in VyOS
×	<i>set service dhcp-server high-availability source-address <address></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	service_dhcp-server.xml.in: service dhcp-server high-availability status	Nothing found in VyOS
×	<i>set service dhcp-server high-availability status <primary / secondary></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	service_dhcp-server.xml.in: service dhcp-server host-decl-name	service dhcp-server host-decl-name
✓	<i>set service dhcp-server hostfile-update</i>	service_dhcp-server.xml.in: service dhcp-server hostfile-update	service dhcp-server hostfile-update
×	not yet documented	service_dhcp-server.xml.in: service dhcp-server listen-address	service dhcp-server listen-address <value>
×	<i>set service dhcp-server listen-address <address></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set service dhcp-server log-level <fatal / error / warn / info / debug></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set service dhcp-server shared-network-name <name> authoritative</i>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set service dhcp-server shared-network-name <name> option bootfile-name <filename></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set service dhcp-server shared-network-name <name> option bootfile-server <address></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set service dhcp-server shared-network-name <name> option bootfile-size <size></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set service dhcp-server shared-network-name <name> option captive-portal <url></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set service dhcp-server shared-network-name <name> option capwap-controller <address></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set service dhcp-server shared-network-name <name> option client-prefix-length <prefix-length></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set service dhcp-server shared-network-name <name> option default-router <address></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set service dhcp-server shared-network-name <name> option domain-name <domain-name></code>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set service dhcp-server shared-network-name <name> option domain-search <domain-name></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set service dhcp-server shared-network-name <name> option interface-mtu <mtu></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set service dhcp-server shared-network-name <name> option ip-forwarding</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set service dhcp-server shared-network-name <name> option ipv6-only-preferred <seconds></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set service dhcp-server shared-network-name <name> option name-server <address></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set service dhcp-server shared-network-name <name> option ntp-server <address></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set service dhcp-server shared-network-name <name> option pop-server <address></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set service dhcp-server shared-network-name <name> option server-identifier <address></code>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set service dhcp-server shared-network-name <name> option smtp-server <address></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set service dhcp-server shared-network-name <name> option static-route <subnet> next-hop <address></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set service dhcp-server shared-network-name <name> option tftp-server-name <server-name></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set service dhcp-server shared-network-name <name> option time-offset <seconds></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set service dhcp-server shared-network-name <name> option time-server <address></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set service dhcp-server shared-network-name <name> option time-zone <timezone></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set service dhcp-server shared-network-name <name> option vendor-option <vendor> <option-name> <value></code>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set service dhcp-server shared-network-name <name> option wins-server <address></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set service dhcp-server shared-network-name <name> option wpad-url <url></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set service dhcp-server shared-network-name <subnet-name> subnet <CIDR> client-class <class-name></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set service dhcp-server shared-network-name <name> subnet <subnet> exclude <address></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set service dhcp-server shared-network-name <name> subnet <subnet> lease <time></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set service dhcp-server shared-network-name <name> subnet <subnet> option default-router <address></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set service dhcp-server shared-network-name <name> subnet <subnet> option domain-name <domain-name></code>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set service dhcp-server shared-network-name <name> subnet <subnet> option domain-search <domain-name></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set service dhcp-server shared-network-name <name> subnet <subnet> option name-server <address></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set service dhcp-server shared-network-name <name> subnet <subnet> option vendor-option <vendor> <option-name> <value></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set service dhcp-server shared-network-name <subnet-name> subnet <CIDR> range <range-name> client-class <class-name></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set service dhcp-server shared-network-name <name> subnet <subnet> range <n> start <address></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set service dhcp-server shared-network-name <name> subnet <subnet> range <n> stop <address></code>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set service dhcp-server shared-network-name <name> subnet <subnet> static-mapping <hostname> duid <identifier></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set service dhcp-server shared-network-name <name> subnet <subnet> static-mapping <hostname> ip-address <address></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set service dhcp-server shared-network-name <name> subnet <subnet> static-mapping <hostname> mac <address></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set service dhcp-server shared-network-name <name> subnet <subnet> subnet-id <id></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	service_dhcp-server.xml.in: service dhcp-server shared-network-name <shared-network-name> authoritative	service dhcp-server shared-network-name <text> authoritative
×	not yet documented	service_dhcp-server.xml.in: service dhcp-server shared-network-name <shared-network-name> description	service dhcp-server shared-network-name <text> description <value>
×	not yet documented	service_dhcp-server.xml.in: service dhcp-server shared-network-name <shared-network-name> disable	service dhcp-server shared-network-name <text> disable

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	service_dhcp-server.xml.in: service dhcp-server shared-network-name <shared-network-name> domain-name	service dhcp-server shared-network-name <text> domain-name <value>
×	not yet documented	service_dhcp-server.xml.in: service dhcp-server shared-network-name <shared-network-name> domain-search	service dhcp-server shared-network-name <text> domain-search <value>
×	not yet documented	service_dhcp-server.xml.in: service dhcp-server shared-network-name <shared-network-name> name-server	service dhcp-server shared-network-name <text> name-server <value>
×	not yet documented	service_dhcp-server.xml.in: service dhcp-server shared-network-name <shared-network-name> ntp-server	service dhcp-server shared-network-name <text> ntp-server <value>
×	not yet documented	service_dhcp-server.xml.in: service dhcp-server shared-network-name <shared-network-name> ping-check	service dhcp-server shared-network-name <text> ping-check
×	not yet documented	service_dhcp-server.xml.in: service dhcp-server shared-network-name <shared-network-name> shared-network-parameters	service dhcp-server shared-network-name <text> shared-network-parameters <value>
×	not yet documented	service_dhcp-server.xml.in: service dhcp-server shared-network-name <shared-network-name> subnet <subnet> bootfile-name	service dhcp-server shared-network-name <text> subnet <text> bootfile-name <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	service_dhcp-server.xml.in: service dhcp-server shared-network-name <shared-network-name> subnet <subnet> bootfile-server	service dhcp-server shared-network-name <text> subnet <text> bootfile-server <value>
×	not yet documented	service_dhcp-server.xml.in: service dhcp-server shared-network-name <shared-network-name> subnet <subnet> bootfile-size	service dhcp-server shared-network-name <text> subnet <text> bootfile-size <value>
×	not yet documented	service_dhcp-server.xml.in: service dhcp-server shared-network-name <shared-network-name> subnet <subnet> client-prefix-length	service dhcp-server shared-network-name <text> subnet <text> client-prefix-length <value>
×	not yet documented	service_dhcp-server.xml.in: service dhcp-server shared-network-name <shared-network-name> subnet <subnet> default-router	service dhcp-server shared-network-name <text> subnet <text> default-router <value>
×	not yet documented	service_dhcp-server.xml.in: service dhcp-server shared-network-name <shared-network-name> subnet <subnet> description	service dhcp-server shared-network-name <text> subnet <text> description <value>
×	not yet documented	service_dhcp-server.xml.in: service dhcp-server shared-network-name <shared-network-name> subnet <subnet> domain-name	service dhcp-server shared-network-name <text> subnet <text> domain-name <value>
×	not yet documented	service_dhcp-server.xml.in: service dhcp-server shared-network-name <shared-network-name> subnet <subnet> domain-search	service dhcp-server shared-network-name <text> subnet <text> domain-search <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	service_dhcp-server.xml.in: service dhcp-server shared-network-name <shared-network-name> subnet <subnet> enable-failover	service dhcp-server shared-network-name <text> subnet <text> enable-failover
×	not yet documented	service_dhcp-server.xml.in: service dhcp-server shared-network-name <shared-network-name> subnet <subnet> exclude	service dhcp-server shared-network-name <text> subnet <text> exclude <value>
×	not yet documented	service_dhcp-server.xml.in: service dhcp-server shared-network-name <shared-network-name> subnet <subnet> ip-forwarding	service dhcp-server shared-network-name <text> subnet <text> ip-forwarding
×	not yet documented	service_dhcp-server.xml.in: service dhcp-server shared-network-name <shared-network-name> subnet <subnet> ipv6-only-preferred	service dhcp-server shared-network-name <text> subnet <text> ipv6-only-preferred <value>
×	not yet documented	service_dhcp-server.xml.in: service dhcp-server shared-network-name <shared-network-name> subnet <subnet> lease	service dhcp-server shared-network-name <text> subnet <text> lease <value>
×	not yet documented	service_dhcp-server.xml.in: service dhcp-server shared-network-name <shared-network-name> subnet <subnet> name-server	service dhcp-server shared-network-name <text> subnet <text> name-server <value>
×	not yet documented	service_dhcp-server.xml.in: service dhcp-server shared-network-name <shared-network-name> subnet <subnet> ntp-server	service dhcp-server shared-network-name <text> subnet <text> ntp-server <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	service_dhcp-server.xml.in: service dhcp-server shared-network-name <shared-network-name> subnet <subnet> ping-check	service dhcp-server shared-network-name <text> subnet <text> ping-check
×	not yet documented	service_dhcp-server.xml.in: service dhcp-server shared-network-name <shared-network-name> subnet <subnet> pop-server	service dhcp-server shared-network-name <text> subnet <text> pop-server <value>
×	not yet documented	service_dhcp-server.xml.in: service dhcp-server shared-network-name <shared-network-name> subnet <subnet> range <range> start	service dhcp-server shared-network-name <text> subnet <text> range <text> start <value>
×	not yet documented	service_dhcp-server.xml.in: service dhcp-server shared-network-name <shared-network-name> subnet <subnet> range <range> stop	service dhcp-server shared-network-name <text> subnet <text> range <text> stop <value>
×	not yet documented	service_dhcp-server.xml.in: service dhcp-server shared-network-name <shared-network-name> subnet <subnet> server-identifier	service dhcp-server shared-network-name <text> subnet <text> server-identifier <value>
×	not yet documented	service_dhcp-server.xml.in: service dhcp-server shared-network-name <shared-network-name> subnet <subnet> smtp-server	service dhcp-server shared-network-name <text> subnet <text> smtp-server <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	service_dhcp-server.xml.in: service dhcp-server shared-network-name <shared-network-name> subnet <subnet> static-mapping <static-mapping> disable	service dhcp-server shared-network-name <text> subnet <text> static-mapping <text> disable
×	not yet documented	service_dhcp-server.xml.in: service dhcp-server shared-network-name <shared-network-name> subnet <subnet> static-mapping <static-mapping> ip-address	service dhcp-server shared-network-name <text> subnet <text> static-mapping <text> ip-address <value>
×	not yet documented	service_dhcp-server.xml.in: service dhcp-server shared-network-name <shared-network-name> subnet <subnet> static-mapping <static-mapping> mac-address	service dhcp-server shared-network-name <text> subnet <text> static-mapping <text> mac-address <value>
×	not yet documented	service_dhcp-server.xml.in: service dhcp-server shared-network-name <shared-network-name> subnet <subnet> static-mapping <static-mapping> static-mapping-parameters	service dhcp-server shared-network-name <text> subnet <text> static-mapping <text> static-mapping-parameters <value>
×	not yet documented	service_dhcp-server.xml.in: service dhcp-server shared-network-name <shared-network-name> subnet <subnet> static-route <static-route> next-hop	service dhcp-server shared-network-name <text> subnet <text> static-route <text> next-hop <value>
×	not yet documented	service_dhcp-server.xml.in: service dhcp-server shared-network-name <shared-network-name> subnet <subnet> subnet-parameters	service dhcp-server shared-network-name <text> subnet <text> subnet-parameters <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	service_dhcp-server.xml.in: service dhcp-server shared-network-name <shared-network-name> subnet <subnet> tftp-server-name	service dhcp-server shared-network-name <text> subnet <text> tftp-server-name <value>
×	not yet documented	service_dhcp-server.xml.in: service dhcp-server shared-network-name <shared-network-name> subnet <subnet> time-offset	service dhcp-server shared-network-name <text> subnet <text> time-offset <value>
×	not yet documented	service_dhcp-server.xml.in: service dhcp-server shared-network-name <shared-network-name> subnet <subnet> time-server	service dhcp-server shared-network-name <text> subnet <text> time-server <value>
×	not yet documented	service_dhcp-server.xml.in: service dhcp-server shared-network-name <shared-network-name> subnet <subnet> vendor-option ubiquiti unifi-controller	service dhcp-server shared-network-name <text> subnet <text> vendor-option ubiquiti unifi-controller <value>
×	not yet documented	service_dhcp-server.xml.in: service dhcp-server shared-network-name <shared-network-name> subnet <subnet> wins-server	service dhcp-server shared-network-name <text> subnet <text> wins-server <value>
×	not yet documented	service_dhcp-server.xml.in: service dhcp-server shared-network-name <shared-network-name> subnet <subnet> wpad-url	service dhcp-server shared-network-name <text> subnet <text> wpad-url <value>
✓	<i>set service dhcpv6-relay disable</i>	service_dhcpv6-relay.xml.in: service dhcpv6-relay disable	service dhcpv6-relay disable

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<i>set service dhcpv6-relay listen-interface <interface></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	service_dhcpv6-relay.xml.in: service dhcpv6-relay listen-interface <listen-interface> address	service dhcpv6-relay listen-interface <text> address <value>
×	not yet documented	service_dhcpv6-relay.xml.in: service dhcpv6-relay max-hop-count	service dhcpv6-relay max-hop-count <value>
×	<i>set service dhcpv6-relay max-hop-count <count></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set service dhcpv6-relay upstream-interface <interface> address <server></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	service_dhcpv6-relay.xml.in: service dhcpv6-relay upstream-interface <upstream-interface> address	service dhcpv6-relay upstream-interface <text> address <value>
✓	<i>set service dhcpv6-relay use-interface-id-option</i>	service_dhcpv6-relay.xml.in: service dhcpv6-relay use-interface-id-option	service dhcpv6-relay use-interface-id-option
×	not yet documented	service_dhcpv6-server.xml.in: service dhcpv6-server disable	service dhcpv6-server disable
×	not yet documented	service_dhcpv6-server.xml.in: service dhcpv6-server global-parameters name-server	service dhcpv6-server global-parameters name-server <value>
×	<i>set service dhcpv6-server log-level <fatal / error / warn / info / debug></i>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	service_dhcpv6-server.xml.in: service dhcpv6-server preference	service dhcpv6-server preference <value>
×	<code>set service dhcpv6-server preference <preference value></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set service dhcpv6-server shared-network-name <name> interface <interface></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set service dhcpv6-server shared-network-name <name> option captive-portal <url></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set service dhcpv6-server shared-network-name <name> option capwap-controller <address></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set service dhcpv6-server shared-network-name <name> option domain-search <domain-name></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set service dhcpv6-server shared-network-name <name> option info-refresh-time <seconds></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set service dhcpv6-server shared-network-name <name> option name-server <address></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set service dhcpv6-server shared-network-name <name> option nis-domain <domain-name></code>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set service dhcpv6-server shared-network-name <name> option nis-server <address></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set service dhcpv6-server shared-network-name <name> option nisplus-domain <domain-name></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set service dhcpv6-server shared-network-name <name> option nisplus-server <address></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set service dhcpv6-server shared-network-name <name> option sip-server <address></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set service dhcpv6-server shared-network-name <name> option snmp-server <address></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set service dhcpv6-server shared-network-name <name> option time-zone <timezone></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set service dhcpv6-server shared-network-name <name> option vendor-option <vendor> <option-name> <value></code>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set service dhcpv6-server shared-network-name <name> subnet <prefix> lease-time {default maximum minimum} <seconds></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set service dhcpv6-server shared-network-name <name> subnet <prefix> prefix-delegation prefix <pd-prefix> delegated-length <length></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set service dhcpv6-server shared-network-name <name> subnet <prefix> prefix-delegation prefix <pd-prefix> excluded-prefix-length <length></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set service dhcpv6-server shared-network-name <name> subnet <prefix> prefix-delegation prefix <pd-prefix> excluded-prefix <exclude-prefix></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set service dhcpv6-server shared-network-name <name> subnet <prefix> prefix-delegation prefix <pd-prefix> prefix-length <length></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set service dhcpv6-server shared-network-name <name> subnet <prefix> range <n> start <address></code>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set service dhcpv6-server shared-network-name <name> subnet <prefix> range <n> stop <address></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set service dhcpv6-server shared-network-name <name> subnet <prefix> static-mapping <hostname> duid <identifier></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set service dhcpv6-server shared-network-name <name> subnet <prefix> static-mapping <hostname> ipv6-address <address></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set service dhcpv6-server shared-network-name <name> subnet <prefix> static-mapping <hostname> ipv6-prefix <delegated-prefix></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set service dhcpv6-server shared-network-name <name> subnet <prefix> subnet-id <id></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	service_dhcpv6-server.xml.in: service dhcpv6-server shared-network-name <shared-network-name> common-options domain-search	service dhcpv6-server shared-network-name <text> common-options domain-search <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	service_dhcpv6-server.xml.in: service dhcpv6-server shared-network-name <shared-network-name> common-options info-refresh-time	service dhcpv6-server shared-network-name <text> common-options info-refresh-time <value>
×	not yet documented	service_dhcpv6-server.xml.in: service dhcpv6-server shared-network-name <shared-network-name> common-options name-server	service dhcpv6-server shared-network-name <text> common-options name-server <value>
×	not yet documented	service_dhcpv6-server.xml.in: service dhcpv6-server shared-network-name <shared-network-name> description	service dhcpv6-server shared-network-name <text> description <value>
×	not yet documented	service_dhcpv6-server.xml.in: service dhcpv6-server shared-network-name <shared-network-name> disable	service dhcpv6-server shared-network-name <text> disable
×	not yet documented	service_dhcpv6-server.xml.in: service dhcpv6-server shared-network-name <shared-network-name> subnet <subnet> address-range prefix <prefix> temporary	service dhcpv6-server shared-network-name <text> subnet <text> address-range prefix <text> temporary
×	not yet documented	service_dhcpv6-server.xml.in: service dhcpv6-server shared-network-name <shared-network-name> subnet <subnet> address-range start <start> stop	service dhcpv6-server shared-network-name <text> subnet <text> address-range start <text> stop <value>
×	not yet documented	service_dhcpv6-server.xml.in: service dhcpv6-server shared-network-name <shared-network-name> subnet <subnet> domain-search	service dhcpv6-server shared-network-name <text> subnet <text> domain-search <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	service_dhcpv6-server.xml.in: service dhcpv6-server shared-network-name <shared-network-name> subnet <subnet> lease-time default	service dhcpv6-server shared-network-name <text> subnet <text> lease-time default <value>
×	not yet documented	service_dhcpv6-server.xml.in: service dhcpv6-server shared-network-name <shared-network-name> subnet <subnet> lease-time maximum	service dhcpv6-server shared-network-name <text> subnet <text> lease-time maximum <value>
×	not yet documented	service_dhcpv6-server.xml.in: service dhcpv6-server shared-network-name <shared-network-name> subnet <subnet> lease-time minimum	service dhcpv6-server shared-network-name <text> subnet <text> lease-time minimum <value>
×	not yet documented	service_dhcpv6-server.xml.in: service dhcpv6-server shared-network-name <shared-network-name> subnet <subnet> name-server	service dhcpv6-server shared-network-name <text> subnet <text> name-server <value>
×	not yet documented	service_dhcpv6-server.xml.in: service dhcpv6-server shared-network-name <shared-network-name> subnet <subnet> nis-domain	service dhcpv6-server shared-network-name <text> subnet <text> nis-domain <value>
×	not yet documented	service_dhcpv6-server.xml.in: service dhcpv6-server shared-network-name <shared-network-name> subnet <subnet> nis-server	service dhcpv6-server shared-network-name <text> subnet <text> nis-server <value>
×	not yet documented	service_dhcpv6-server.xml.in: service dhcpv6-server shared-network-name <shared-network-name> subnet <subnet> nisplus-domain	service dhcpv6-server shared-network-name <text> subnet <text> nisplus-domain <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	service_dhcpv6-server.xml.in: service dhcpv6-server shared-network-name <shared-network-name> subnet <subnet> nisplus-server	service dhcpv6-server shared-network-name <text> subnet <text> nisplus-server <value>
×	not yet documented	service_dhcpv6-server.xml.in: service dhcpv6-server shared-network-name <shared-network-name> subnet <subnet> prefix-delegation start <start> prefix-length	service dhcpv6-server shared-network-name <text> subnet <text> prefix-delegation start <text> prefix-length <value>
×	not yet documented	service_dhcpv6-server.xml.in: service dhcpv6-server shared-network-name <shared-network-name> subnet <subnet> prefix-delegation start <start> stop	service dhcpv6-server shared-network-name <text> subnet <text> prefix-delegation start <text> stop <value>
×	not yet documented	service_dhcpv6-server.xml.in: service dhcpv6-server shared-network-name <shared-network-name> subnet <subnet> sip-server	service dhcpv6-server shared-network-name <text> subnet <text> sip-server <value>
×	not yet documented	service_dhcpv6-server.xml.in: service dhcpv6-server shared-network-name <shared-network-name> subnet <subnet> sntp-server	service dhcpv6-server shared-network-name <text> subnet <text> sntp-server <value>
×	not yet documented	service_dhcpv6-server.xml.in: service dhcpv6-server shared-network-name <shared-network-name> subnet <subnet> static-mapping <static-mapping> disable	service dhcpv6-server shared-network-name <text> subnet <text> static-mapping <text> disable

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	service_dhcpv6-server.xml.in: service dhcpv6-server shared-network-name <shared-network-name> subnet <subnet> static-mapping <static-mapping> identifier	service dhcpv6-server shared-network-name <text> subnet <text> static-mapping <text> identifier <value>
×	not yet documented	service_dhcpv6-server.xml.in: service dhcpv6-server shared-network-name <shared-network-name> subnet <subnet> static-mapping <static-mapping> ipv6-address	service dhcpv6-server shared-network-name <text> subnet <text> static-mapping <text> ipv6-address <value>
×	not yet documented	service_dhcpv6-server.xml.in: service dhcpv6-server shared-network-name <shared-network-name> subnet <subnet> static-mapping <static-mapping> ipv6-prefix	service dhcpv6-server shared-network-name <text> subnet <text> static-mapping <text> ipv6-prefix <value>
×	not yet documented	service_dhcpv6-server.xml.in: service dhcpv6-server shared-network-name <shared-network-name> subnet <subnet> vendor-option cisco tftp-server	service dhcpv6-server shared-network-name <text> subnet <text> vendor-option cisco tftp-server <value>
×	not yet documented	Nothing found in XML Definitions	service dns dynamic address <text> rfc2136 <text> description <value>
×	not yet documented	Nothing found in XML Definitions	service dns dynamic address <text> rfc2136 <text> host-name <value>
×	not yet documented	Nothing found in XML Definitions	service dns dynamic address <text> rfc2136 <text> key <value>
×	not yet documented	Nothing found in XML Definitions	service dns dynamic address <text> rfc2136 <text> server <value>
×	not yet documented	Nothing found in XML Definitions	service dns dynamic address <text> rfc2136 <text> ttl <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	Nothing found in XML Definitions	service dns dynamic address <text> rfc2136 <text> zone <value>
×	not yet documented	Nothing found in XML Definitions	service dns dynamic address <text> service <text> description <value>
×	not yet documented	Nothing found in XML Definitions	service dns dynamic address <text> service <text> host-name <value>
×	not yet documented	Nothing found in XML Definitions	service dns dynamic address <text> service <text> ip-version <value>
×	not yet documented	Nothing found in XML Definitions	service dns dynamic address <text> service <text> password <value>
×	not yet documented	Nothing found in XML Definitions	service dns dynamic address <text> service <text> protocol <value>
×	not yet documented	Nothing found in XML Definitions	service dns dynamic address <text> service <text> server <value>
×	not yet documented	Nothing found in XML Definitions	service dns dynamic address <text> service <text> username <value>
×	not yet documented	Nothing found in XML Definitions	service dns dynamic address <text> service <text> zone <value>
×	not yet documented	Nothing found in XML Definitions	service dns dynamic address <text> web-options skip <value>
×	not yet documented	Nothing found in XML Definitions	service dns dynamic address <text> web-options url <value>
×	not yet documented	service_dns_dynamic.xml.in: service dns dynamic interval	Nothing found in VyOS
×	<i>set service dns dynamic interval <60-3600></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set service dns dynamic name <service-name> address interface <interface></i>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set service dns dynamic name <service-name> address web <url></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set service dns dynamic name <service-name> address web skip <pattern></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set service dns dynamic name <service-name> description <text></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set service dns dynamic name <service-name> host-name <hostname></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set service dns dynamic name <service-name> ip-version 'ipv6'</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set service dns dynamic name <service-name> key <filename></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set service dns dynamic name <service-name> password <password></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set service dns dynamic name <service-name> protocol <protocol></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set service dns dynamic name <service-name> server <server></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set service dns dynamic name <service-name> ttl <ttl></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set service dns dynamic name <service-name> username <username></code>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<i>set service dns dynamic name <service-name> zone <zone></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	ser-vice_dns_dynamic.xml.in: service dns dynamic name <name> address interface	Nothing found in VyOS
×	not yet documented	ser-vice_dns_dynamic.xml.in: service dns dynamic name <name> address web skip	Nothing found in VyOS
×	not yet documented	ser-vice_dns_dynamic.xml.in: service dns dynamic name <name> address web url	Nothing found in VyOS
×	not yet documented	ser-vice_dns_dynamic.xml.in: service dns dynamic name <name> description	Nothing found in VyOS
×	not yet documented	ser-vice_dns_dynamic.xml.in: service dns dynamic name <name> expiry-time	Nothing found in VyOS
×	not yet documented	ser-vice_dns_dynamic.xml.in: service dns dynamic name <name> host-name	Nothing found in VyOS
×	not yet documented	ser-vice_dns_dynamic.xml.in: service dns dynamic name <name> ip-version	Nothing found in VyOS
×	not yet documented	ser-vice_dns_dynamic.xml.in: service dns dynamic name <name> key	Nothing found in VyOS
×	not yet documented	ser-vice_dns_dynamic.xml.in: service dns dynamic name <name> password	Nothing found in VyOS
×	not yet documented	ser-vice_dns_dynamic.xml.in: service dns dynamic name <name> protocol	Nothing found in VyOS
×	not yet documented	ser-vice_dns_dynamic.xml.in: service dns dynamic name <name> server	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	service_dns_dynamic.xml.in: service dns dynamic name <name> ttl	Nothing found in VyOS
×	not yet documented	service_dns_dynamic.xml.in: service dns dynamic name <name> username	Nothing found in VyOS
×	not yet documented	service_dns_dynamic.xml.in: service dns dynamic name <name> wait-time	Nothing found in VyOS
×	not yet documented	service_dns_dynamic.xml.in: service dns dynamic name <name> zone	Nothing found in VyOS
×	not yet documented	Nothing found in XML Definitions	service dns dynamic time-out <value>
×	not yet documented	service_dns_dynamic.xml.in: service dns dynamic vrf	Nothing found in VyOS
×	not yet documented	service_dns_forwarding.xml.in: service dns forwarding allow-from	service dns forwarding allow-from <value>
×	<i>set service dns forwarding allow-from <network></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set service dns forwarding authoritative-domain <domain-name> disable</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set service dns forwarding authoritative-domain <domain-name> records <type> <name> disable</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set service dns forwarding authoritative-domain <domain-name> records <type> <name> ttl <seconds></i>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set service dns forwarding authoritative-domain <domain-name> records a <name> address <x.x.x.x></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set service dns forwarding authoritative-domain <domain-name> records aaaa <name> address <h:h:h:h:h:h:h:h></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set service dns forwarding authoritative-domain <domain-name> records cname <name> target <target-domain-name></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set service dns forwarding authoritative-domain <domain-name> records naptr <name> rule <rule-number> <option> <value></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set service dns forwarding authoritative-domain <domain-name> records ns <name> target <target-name></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set service dns forwarding authoritative-domain <domain-name> records ptr <name> target <target-name></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set service dns forwarding authoritative-domain <domain-name> records spf <name> value <value></code>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set service dns forwarding authoritative-domain <domain-name> records srv <name> entry <entry-number> [hostname port priority weight] <value></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set service dns forwarding authoritative-domain <domain-name> records txt <name> value <value></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	ser- vice_dns_forwarding.xml.ir service dns forwarding authoritative-domain <authoritative-domain> disable	service dns forwarding authoritative-domain <text> disable
×	not yet documented	ser- vice_dns_forwarding.xml.ir service dns forwarding authoritative-domain <authoritative-domain> records aaaa <aaaa> address	service dns forwarding authoritative-domain <text> records aaaa <text> address <value>
×	not yet documented	ser- vice_dns_forwarding.xml.ir service dns forwarding authoritative-domain <authoritative-domain> records aaaa <aaaa> disable	service dns forwarding authoritative-domain <text> records aaaa <text> disable
×	not yet documented	ser- vice_dns_forwarding.xml.ir service dns forwarding authoritative-domain <authoritative-domain> records aaaa <aaaa> ttl	service dns forwarding authoritative-domain <text> records aaaa <text> ttl <value>
×	not yet documented	ser- vice_dns_forwarding.xml.ir service dns forwarding authoritative-domain <authoritative-domain> records a <a> address	service dns forwarding authoritative-domain <text> records a <text> address <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	ser- vice_dns_forwarding.xml.ir service dns forwarding authoritative-domain <authoritative-domain> records a <a> disable	service dns forwarding authoritative-domain <text> records a <text> disable
×	not yet documented	ser- vice_dns_forwarding.xml.ir service dns forwarding authoritative-domain <authoritative-domain> records a <a> ttl	service dns forwarding authoritative-domain <text> records a <text> ttl <value>
×	not yet documented	ser- vice_dns_forwarding.xml.ir service dns forwarding authoritative-domain <authoritative-domain> records cname <cname> disable	service dns forwarding authoritative-domain <text> records cname <text> disable
×	not yet documented	ser- vice_dns_forwarding.xml.ir service dns forwarding authoritative-domain <authoritative-domain> records cname <cname> target	service dns forwarding authoritative-domain <text> records cname <text> target <value>
×	not yet documented	ser- vice_dns_forwarding.xml.ir service dns forwarding authoritative-domain <authoritative-domain> records cname <cname> ttl	service dns forwarding authoritative-domain <text> records cname <text> ttl <value>
×	not yet documented	ser- vice_dns_forwarding.xml.ir service dns forwarding authoritative-domain <authoritative-domain> records mx <mx> disable	service dns forwarding authoritative-domain <text> records mx <text> disable
×	not yet documented	ser- vice_dns_forwarding.xml.ir service dns forwarding authoritative-domain <authoritative-domain> records mx <mx> server <server> priority	service dns forwarding authoritative-domain <text> records mx <text> server <text> priority <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	ser- vice_dns_forwarding.xml.ir service dns forwarding authoritative-domain <authoritative-domain> records mx <mx> ttl	service dns forwarding authoritative-domain <text> records mx <text> ttl <value>
×	not yet documented	ser- vice_dns_forwarding.xml.ir service dns forwarding authoritative-domain <authoritative-domain> records naptr <naptr> disable	service dns forwarding authoritative-domain <text> records naptr <text> disable
×	not yet documented	ser- vice_dns_forwarding.xml.ir service dns forwarding authoritative-domain <authoritative-domain> records naptr <naptr> rule <rule> lookup-a	service dns forwarding authoritative-domain <text> records naptr <text> rule <text> lookup- a
×	not yet documented	ser- vice_dns_forwarding.xml.ir service dns forwarding authoritative-domain <authoritative-domain> records naptr <naptr> rule <rule> lookup-srv	service dns forwarding authoritative-domain <text> records naptr <text> rule <text> lookup- srv
×	not yet documented	ser- vice_dns_forwarding.xml.ir service dns forwarding authoritative-domain <authoritative-domain> records naptr <naptr> rule <rule> order	service dns forwarding authoritative-domain <text> records naptr <text> rule <text> order <value>
×	not yet documented	ser- vice_dns_forwarding.xml.ir service dns forwarding authoritative-domain <authoritative-domain> records naptr <naptr> rule <rule> preference	service dns forwarding authoritative-domain <text> records naptr <text> rule <text> prefer- ence <value>
×	not yet documented	ser- vice_dns_forwarding.xml.ir service dns forwarding authoritative-domain <authoritative-domain> records naptr <naptr> rule <rule> protocol-specific	service dns forwarding authoritative-domain <text> records naptr <text> rule <text> protocol-specific

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	ser- vice_dns_forwarding.xml.ir service dns forwarding authoritative-domain <authoritative-domain> records naptr <naptr> rule <rule> regexp	service dns forwarding authoritative-domain <text> records naptr <text> rule <text> regexp <value>
×	not yet documented	ser- vice_dns_forwarding.xml.ir service dns forwarding authoritative-domain <authoritative-domain> records naptr <naptr> rule <rule> replacement	service dns forwarding authoritative-domain <text> records naptr <text> rule <text> re- placement <value>
×	not yet documented	ser- vice_dns_forwarding.xml.ir service dns forwarding authoritative-domain <authoritative-domain> records naptr <naptr> rule <rule> resolve-uri	service dns forwarding authoritative-domain <text> records naptr <text> rule <text> resolve- uri
×	not yet documented	ser- vice_dns_forwarding.xml.ir service dns forwarding authoritative-domain <authoritative-domain> records naptr <naptr> rule <rule> service	service dns forwarding authoritative-domain <text> records naptr <text> rule <text> service <value>
×	not yet documented	ser- vice_dns_forwarding.xml.ir service dns forwarding authoritative-domain <authoritative-domain> records naptr <naptr> ttl	service dns forwarding authoritative-domain <text> records naptr <text> ttl <value>
×	not yet documented	ser- vice_dns_forwarding.xml.ir service dns forwarding authoritative-domain <authoritative-domain> records ns <ns> disable	service dns forwarding authoritative-domain <text> records ns <text> disable
×	not yet documented	ser- vice_dns_forwarding.xml.ir service dns forwarding authoritative-domain <authoritative-domain> records ns <ns> target	service dns forwarding authoritative-domain <text> records ns <text> target <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	ser- vice_dns_forwarding.xml.ir service dns forwarding authoritative-domain <authoritative-domain> records ns <ns> ttl	service dns forwarding authoritative-domain <text> records ns <text> ttl <value>
×	not yet documented	ser- vice_dns_forwarding.xml.ir service dns forwarding authoritative-domain <authoritative-domain> records ptr <ptr> disable	service dns forwarding authoritative-domain <text> records ptr <text> disable
×	not yet documented	ser- vice_dns_forwarding.xml.ir service dns forwarding authoritative-domain <authoritative-domain> records ptr <ptr> target	service dns forwarding authoritative-domain <text> records ptr <text> target <value>
×	not yet documented	ser- vice_dns_forwarding.xml.ir service dns forwarding authoritative-domain <authoritative-domain> records ptr <ptr> ttl	service dns forwarding authoritative-domain <text> records ptr <text> ttl <value>
×	not yet documented	ser- vice_dns_forwarding.xml.ir service dns forwarding authoritative-domain <authoritative-domain> records spf <spf> disable	service dns forwarding authoritative-domain <text> records spf <text> disable
×	not yet documented	ser- vice_dns_forwarding.xml.ir service dns forwarding authoritative-domain <authoritative-domain> records spf <spf> ttl	service dns forwarding authoritative-domain <text> records spf <text> ttl <value>
×	not yet documented	ser- vice_dns_forwarding.xml.ir service dns forwarding authoritative-domain <authoritative-domain> records spf <spf> value	service dns forwarding authoritative-domain <text> records spf <text> value <value>
×	not yet documented	ser- vice_dns_forwarding.xml.ir service dns forwarding authoritative-domain <authoritative-domain> records srv <srv> disable	service dns forwarding authoritative-domain <text> records srv <text> disable

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	ser- vice_dns_forwarding.xml.ir service dns forwarding authoritative-domain <authoritative-domain> records srv <srv> entry <entry> hostname	service dns forwarding authoritative-domain <text> records srv <text> entry <text> hostname <value>
×	not yet documented	ser- vice_dns_forwarding.xml.ir service dns forwarding authoritative-domain <authoritative-domain> records srv <srv> entry <entry> port	service dns forwarding authoritative-domain <text> records srv <text> entry <text> port <value>
×	not yet documented	ser- vice_dns_forwarding.xml.ir service dns forwarding authoritative-domain <authoritative-domain> records srv <srv> entry <entry> priority	service dns forwarding authoritative-domain <text> records srv <text> entry <text> priority <value>
×	not yet documented	ser- vice_dns_forwarding.xml.ir service dns forwarding authoritative-domain <authoritative-domain> records srv <srv> entry <entry> weight	service dns forwarding authoritative-domain <text> records srv <text> entry <text> weight <value>
×	not yet documented	ser- vice_dns_forwarding.xml.ir service dns forwarding authoritative-domain <authoritative-domain> records srv <srv> ttl	service dns forwarding authoritative-domain <text> records srv <text> ttl <value>
×	not yet documented	ser- vice_dns_forwarding.xml.ir service dns forwarding authoritative-domain <authoritative-domain> records txt <txt> disable	service dns forwarding authoritative-domain <text> records txt <text> disable
×	not yet documented	ser- vice_dns_forwarding.xml.ir service dns forwarding authoritative-domain <authoritative-domain> records txt <txt> ttl	service dns forwarding authoritative-domain <text> records txt <text> ttl <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	ser-vice_dns_forwarding.xml.ir service dns forwarding authoritative-domain <text> records txt <text> value <value>	service dns forwarding authoritative-domain <text> records txt <text> value <value>
×	not yet documented	ser-vice_dns_forwarding.xml.ir service dns forwarding cache-size	service dns forwarding cache-size <value>
×	<i>set service dns forwarding cache-size <0-2147483647></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	ser-vice_dns_forwarding.xml.ir service dns forwarding dhcp	service dns forwarding dhcp <value>
×	<i>set service dns forwarding dhcp <interface></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	ser-vice_dns_forwarding.xml.ir service dns forwarding dns64-prefix	service dns forwarding dns64-prefix <value>
×	not yet documented	ser-vice_dns_forwarding.xml.ir service dns forwarding dnssec	service dns forwarding dnssec <value>
×	<i>set service dns forwarding dnssec <off process-no-validate process log-fail validate></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set service dns forwarding domain <domain-name> addnta</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set service dns forwarding domain <domain-name> name-server <address></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set service dns forwarding domain <domain-name> recursion-desired</i>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	ser- vice_dns_forwarding.xml.ir service dns forwarding do- main <domain> addnta	service dns forwarding do- main <text> addnta
×	not yet documented	ser- vice_dns_forwarding.xml.ir service dns forwarding domain <domain> name- server <name-server> port	service dns forwarding do- main <text> name-server <text> port <value>
×	not yet documented	ser- vice_dns_forwarding.xml.ir service dns forwarding do- main <domain> recursion- desired	service dns forwarding domain <text> recursion- desired
×	not yet documented	ser- vice_dns_forwarding.xml.ir service dns forwarding exclude-throttle-address	Nothing found in VyOS
✓	<i>set service dns forwarding ignore-hosts-file</i>	ser- vice_dns_forwarding.xml.ir service dns forwarding ignore-hosts-file	service dns forwarding ignore-hosts-file
×	not yet documented	ser- vice_dns_forwarding.xml.ir service dns forwarding listen-address	service dns forwarding listen-address <value>
×	<i>set service dns forwarding listen-address <address></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set service dns forwarding name-server <address> port <port></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	ser- vice_dns_forwarding.xml.ir service dns forwarding name-server <name- server> port	service dns forwarding name-server <text> port <value>
×	not yet documented	ser- vice_dns_forwarding.xml.ir service dns forwarding negative-ttl	service dns forwarding negative-ttl <value>
×	<i>set service dns forwarding negative-ttl <0-7200></i>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
✓	<i>set service dns forwarding no-serve-rfc1918</i>	ser-vice_dns_forwarding.xml.ir service dns forwarding no-serve-rfc1918	service dns forwarding no-serve-rfc1918
×	not yet documented	ser-vice_dns_forwarding.xml.ir service dns forwarding options ecs-add-for	Nothing found in VyOS
×	not yet documented	ser-vice_dns_forwarding.xml.ir service dns forwarding options ecs-ipv4-bits	Nothing found in VyOS
×	not yet documented	ser-vice_dns_forwarding.xml.ir service dns forwarding options edns-subnet-allow-list	Nothing found in VyOS
×	not yet documented	ser-vice_dns_forwarding.xml.ir service dns forwarding port	service dns forwarding port <value>
×	not yet documented	ser-vice_dns_forwarding.xml.ir service dns forwarding serve-stale-extension	Nothing found in VyOS
×	not yet documented	ser-vice_dns_forwarding.xml.ir service dns forwarding source-address	service dns forwarding source-address <value>
×	<i>set service dns forwarding source-address <address></i>	Nothing found in XML Definitions	Nothing found in VyOS
✓	<i>set service dns forwarding system</i>	ser-vice_dns_forwarding.xml.ir service dns forwarding system	service dns forwarding system
×	not yet documented	ser-vice_dns_forwarding.xml.ir service dns forwarding timeout	service dns forwarding timeout <value>
×	<i>set service dns forwarding timeout <10-60000></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set service event-handler event <event-handler name></i>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set service event-handler event <event-handler name> filter pattern <regex></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set service event-handler event <event-handler name> filter syslog-identifier <syslogid name></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set service event-handler event <event-handler name> script arguments <arguments></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set service event-handler event <event-handler name> script environment <env name> value <env value></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set service event-handler event <event-handler name> script path <path to script></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	service_event-handler.xml.in: service event-handler event <event> filter pattern	service event-handler event <text> filter pattern <value>
×	not yet documented	service_event-handler.xml.in: service event-handler event <event> filter syslog-identifier	service event-handler event <text> filter syslog-identifier <value>
×	not yet documented	service_event-handler.xml.in: service event-handler event <event> script arguments	service event-handler event <text> script arguments <value>
×	not yet documented	service_event-handler.xml.in: service event-handler event <event> script environment <environment> value	service event-handler event <text> script environment <text> value <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	service_event-handler.xml.in: service event-handler event <event> script path <value>	service event-handler event <text> script path <value>
×	not yet documented	service_https.xml.in: service https allow-client address	Nothing found in VyOS
×	<code>set service https allow-client address <address></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	Nothing found in XML Definitions	service https api-restrict virtual-host <value>
×	not yet documented	service_https.xml.in: service https api cors allow-origin	service https api cors allow-origin <value>
×	not yet documented	service_https.xml.in: service https api debug	service https api debug
✓	<code>set service https api graphql authentication expiration</code>	service_https.xml.in: service https api graphql authentication expiration	service https api graphql authentication expiration <value>
✓	<code>set service https api graphql authentication secret-length</code>	service_https.xml.in: service https api graphql authentication secret-length	service https api graphql authentication secret-length <value>
×	not yet documented	service_https.xml.in: service https api graphql authentication type	service https api graphql authentication type <value>
×	<code>set service https api graphql authentication type <key token></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set service https api graphql cors allow-origin <origin></code>	Nothing found in XML Definitions	Nothing found in VyOS
✓	<code>set service https api graphql introspection</code>	service_https.xml.in: service https api graphql introspection	service https api graphql introspection
×	<code>set service https api keys id <name> key <apikey></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	service_https.xml.in: service https api keys id <id> key	service https api keys id <text> key <value>
×	not yet documented	Nothing found in XML Definitions	service https api port <value>
×	<code>set service https api rest</code>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set service https api rest debug</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set service https api rest strict</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	Nothing found in XML Definitions	service https api socket
×	not yet documented	service_https.xml.in: service https api strict	service https api strict
×	not yet documented	service_https.xml.in: service https certificates ca-certificate	service https certificates ca-certificate <value>
×	<code>set service https certificates ca-certificate <name></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	Nothing found in XML Definitions	service https certificates certbot domain-name <value>
×	not yet documented	Nothing found in XML Definitions	service https certificates certbot email <value>
×	not yet documented	service_https.xml.in: service https certificates certificate	service https certificates certificate <value>
×	<code>set service https certificates certificate <name></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	service_https.xml.in: service https certificates dh-params	Nothing found in VyOS
×	<code>set service https certificates dh-params <name></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set service https enable-http-redirect</code>	service_https.xml.in: service https enable-http-redirect	Nothing found in VyOS
×	not yet documented	service_https.xml.in: service https listen-address	Nothing found in VyOS
×	<code>set service https listen-address <address></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	service_https.xml.in: service https port	Nothing found in VyOS
×	<code>set service https port <number></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	service_https.xml.in: service https request-body-size-limit	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set service https request-body-size-limit <size></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	service_https.xml.in: service https tls-version	Nothing found in VyOS
×	<code>set service https tls-version <1.2 / 1.3></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	Nothing found in XML Definitions	service https virtual-host <text> allow-client address <value>
×	not yet documented	Nothing found in XML Definitions	service https virtual-host <text> listen-address <value>
×	not yet documented	Nothing found in XML Definitions	service https virtual-host <text> listen-port <value>
×	not yet documented	Nothing found in XML Definitions	service https virtual-host <text> server-name <value>
×	not yet documented	service_https.xml.in: service https vrf	service https vrf <value>
×	<code>set service https vrf <name></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	service_ids_ddos-protection.xml.in: service ids ddos-protection alert-script	service ids ddos-protection alert-script <value>
×	not yet documented	service_ids_ddos-protection.xml.in: service ids ddos-protection ban-time	service ids ddos-protection ban-time <value>
×	not yet documented	service_ids_ddos-protection.xml.in: service ids ddos-protection direction	service ids ddos-protection direction <value>
×	not yet documented	service_ids_ddos-protection.xml.in: service ids ddos-protection excluded-network	service ids ddos-protection excluded-network <value>
×	not yet documented	service_ids_ddos-protection.xml.in: service ids ddos-protection listen-interface	service ids ddos-protection listen-interface <value>
×	not yet documented	service_ids_ddos-protection.xml.in: service ids ddos-protection mode	service ids ddos-protection mode <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	service_ids_ddos-protection.xml.in: service_ids ddos-protection network <value>	service_ids ddos-protection network <value>
×	not yet documented	service_ids_ddos-protection.xml.in: service_ids ddos-protection sflow listen-address <value>	service_ids ddos-protection sflow listen-address <value>
×	not yet documented	service_ids_ddos-protection.xml.in: service_ids ddos-protection sflow port <value>	service_ids ddos-protection sflow port <value>
×	not yet documented	service_ids_ddos-protection.xml.in: service_ids ddos-protection threshold general fps <value>	service_ids ddos-protection threshold general fps <value>
×	not yet documented	service_ids_ddos-protection.xml.in: service_ids ddos-protection threshold general mbps <value>	service_ids ddos-protection threshold general mbps <value>
×	not yet documented	service_ids_ddos-protection.xml.in: service_ids ddos-protection threshold general pps <value>	service_ids ddos-protection threshold general pps <value>
×	not yet documented	service_ids_ddos-protection.xml.in: service_ids ddos-protection threshold icmp fps <value>	service_ids ddos-protection threshold icmp fps <value>
×	not yet documented	service_ids_ddos-protection.xml.in: service_ids ddos-protection threshold icmp mbps <value>	service_ids ddos-protection threshold icmp mbps <value>
×	not yet documented	service_ids_ddos-protection.xml.in: service_ids ddos-protection threshold icmp pps <value>	service_ids ddos-protection threshold icmp pps <value>
×	not yet documented	service_ids_ddos-protection.xml.in: service_ids ddos-protection threshold tcp fps <value>	service_ids ddos-protection threshold tcp fps <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	service_ids_ddos-protection.xml.in: service_ids_ddos-protection threshold tcp mbps	service_ids_ddos-protection threshold tcp mbps <value>
×	not yet documented	service_ids_ddos-protection.xml.in: service_ids_ddos-protection threshold tcp pps	service_ids_ddos-protection threshold tcp pps <value>
×	not yet documented	service_ids_ddos-protection.xml.in: service_ids_ddos-protection threshold udp fps	service_ids_ddos-protection threshold udp fps <value>
×	not yet documented	service_ids_ddos-protection.xml.in: service_ids_ddos-protection threshold udp mbps	service_ids_ddos-protection threshold udp mbps <value>
×	not yet documented	service_ids_ddos-protection.xml.in: service_ids_ddos-protection threshold udp pps	service_ids_ddos-protection threshold udp pps <value>
×	<i>set service ipoe-server authentication interface <interface> mac <MAC></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set service ipoe-server authentication interface <interface> mac <MAC> rate-limit download <bandwidth></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set service ipoe-server authentication interface <interface> mac <MAC> rate-limit upload <bandwidth></i>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set service ipoe-server authentication interface <interface> mac <MAC> vlan <vlan-id></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	service_ipoe-server.xml.in: service ipoe-server authentication interface <interface> mac <mac> rate-limit download	service ipoe-server authentication interface <text> mac <text> rate-limit download <value>
×	not yet documented	service_ipoe-server.xml.in: service ipoe-server authentication interface <interface> mac <mac> rate-limit upload	service ipoe-server authentication interface <text> mac <text> rate-limit upload <value>
×	not yet documented	service_ipoe-server.xml.in: service ipoe-server authentication interface <interface> mac <mac> vlan	service ipoe-server authentication interface <text> mac <text> vlan <value>
×	not yet documented	service_ipoe-server.xml.in: service ipoe-server authentication mode	service ipoe-server authentication mode <value>
×	<code>set service ipoe-server authentication mode <local radius></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	service_ipoe-server.xml.in: service ipoe-server authentication radius accounting-interim-interval	service ipoe-server authentication radius accounting-interim-interval <value>
×	not yet documented	service_ipoe-server.xml.in: service ipoe-server authentication radius acct-interim-jitter	service ipoe-server authentication radius acct-interim-jitter <value>
×	not yet documented	service_ipoe-server.xml.in: service ipoe-server authentication radius acct-timeout	service ipoe-server authentication radius acct-timeout <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<i>set service ipoe-server authentication radius acct-timeout <timeout></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	service_ipoe-server.xml.in: service ipoe-server authentication radius dynamic-author key	service ipoe-server authentication radius dynamic-author key <value>
×	<i>set service ipoe-server authentication radius dynamic-author key <secret></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	service_ipoe-server.xml.in: service ipoe-server authentication radius dynamic-author port	service ipoe-server authentication radius dynamic-author port <value>
×	<i>set service ipoe-server authentication radius dynamic-author port <port></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	service_ipoe-server.xml.in: service ipoe-server authentication radius dynamic-author server	service ipoe-server authentication radius dynamic-author server <value>
×	<i>set service ipoe-server authentication radius dynamic-author server <address></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	service_ipoe-server.xml.in: service ipoe-server authentication radius max-try	service ipoe-server authentication radius max-try <value>
×	<i>set service ipoe-server authentication radius max-try <number></i>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	service_ipoe-server.xml.in: service ipoe-server authentication radius nas-identifier	service ipoe-server authentication radius nas-identifier <value>
×	<i>set service ipoe-server authentication radius nas-identifier <identifier></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	service_ipoe-server.xml.in: service ipoe-server authentication radius nas-ip-address	service ipoe-server authentication radius nas-ip-address <value>
×	<i>set service ipoe-server authentication radius nas-ip-address <address></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	service_ipoe-server.xml.in: service ipoe-server authentication radius preallocate-vif	service ipoe-server authentication radius preallocate-vif
×	not yet documented	service_ipoe-server.xml.in: service ipoe-server authentication radius rate-limit attribute	service ipoe-server authentication radius rate-limit attribute <value>
×	<i>set service ipoe-server authentication radius rate-limit attribute <attribute></i>	Nothing found in XML Definitions	Nothing found in VyOS
✓	<i>set service ipoe-server authentication radius rate-limit enable</i>	service_ipoe-server.xml.in: service ipoe-server authentication radius rate-limit enable	service ipoe-server authentication radius rate-limit enable
×	not yet documented	service_ipoe-server.xml.in: service ipoe-server authentication radius rate-limit multiplier	service ipoe-server authentication radius rate-limit multiplier <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
✓	<code>set service ipoe-server authentication radius rate-limit vendor</code>	service_ipoe-server.xml.in: service ipoe-server authentication radius rate-limit vendor	service ipoe-server authentication radius rate-limit vendor <value>
×	<code>set service ipoe-server authentication radius server <server> disable</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set service ipoe-server authentication radius server <server> fail-time <time></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set service ipoe-server authentication radius server <server> key <secret></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set service ipoe-server authentication radius server <server> port <port></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	service_ipoe-server.xml.in: service ipoe-server authentication radius server <server> acct-port	service ipoe-server authentication radius server <text> acct-port <value>
×	not yet documented	service_ipoe-server.xml.in: service ipoe-server authentication radius server <server> disable	service ipoe-server authentication radius server <text> disable
×	not yet documented	service_ipoe-server.xml.in: service ipoe-server authentication radius server <server> disable-accounting	service ipoe-server authentication radius server <text> disable-accounting
×	not yet documented	service_ipoe-server.xml.in: service ipoe-server authentication radius server <server> fail-time	service ipoe-server authentication radius server <text> fail-time <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	service_ipoe-server.xml.in: service ipoe-server authentication radius server <server> key	service ipoe-server authentication radius server <text> key <value>
×	not yet documented	service_ipoe-server.xml.in: service ipoe-server authentication radius server <server> port	service ipoe-server authentication radius server <text> port <value>
×	not yet documented	service_ipoe-server.xml.in: service ipoe-server authentication radius source-address	service ipoe-server authentication radius source-address <value>
×	set service ipoe-server authentication radius source-address <address>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	service_ipoe-server.xml.in: service ipoe-server authentication radius timeout	service ipoe-server authentication radius timeout <value>
×	set service ipoe-server authentication radius timeout <timeout>	Nothing found in XML Definitions	Nothing found in VyOS
×	set service ipoe-server client-ip-pool <POOL-NAME> next-pool <NEXT-POOL-NAME>	Nothing found in XML Definitions	Nothing found in VyOS
×	set service ipoe-server client-ip-pool <POOL-NAME> range <X.X.X.X-X.X.X.X / X.X.X.X/X>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	Nothing found in XML Definitions	service ipoe-server client-ip-pool name <text> gateway-address <value>
×	not yet documented	Nothing found in XML Definitions	service ipoe-server client-ip-pool name <text> next-pool <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	Nothing found in XML Definitions	service ipoe-server client-ip-pool name <text> subnet <value>
×	not yet documented	service_ipoe-server.xml.in: service ipoe-server client-ip-pool <client-ip-pool> next-pool	Nothing found in VyOS
×	not yet documented	service_ipoe-server.xml.in: service ipoe-server client-ip-pool <client-ip-pool> range	Nothing found in VyOS
×	<i>set service ipoe-server client-ipv6-pool <IPv6-POOL-NAME> delegate <address> delegation-prefix <number-of-bits></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set service ipoe-server client-ipv6-pool <IPv6-POOL-NAME> prefix <address> mask <number-of-bits></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	service_ipoe-server.xml.in: service ipoe-server client-ipv6-pool <client-ipv6-pool> delegate <delegate> delegation-prefix	service ipoe-server client-ipv6-pool delegate <text> delegation-prefix <value>
×	not yet documented	service_ipoe-server.xml.in: service ipoe-server client-ipv6-pool <client-ipv6-pool> prefix <prefix> mask	service ipoe-server client-ipv6-pool prefix <text> mask <value>
×	not yet documented	service_ipoe-server.xml.in: service ipoe-server default-ipv6-pool	Nothing found in VyOS
×	<i>set service ipoe-server default-ipv6-pool <IPv6-POOL-NAME></i>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	service_ipoe-server.xml.in: service ipoe-server default-pool	Nothing found in VyOS
×	<i>set service ipoe-server default-pool <POOL-NAME></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	service_ipoe-server.xml.in: service ipoe-server de- scription	Nothing found in VyOS
×	<i>set service ipoe-server description <description></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	service_ipoe-server.xml.in: service ipoe-server extended-scripts on- change	Nothing found in VyOS
×	<i>set service ipoe-server extended-scripts on-change <path_to_script></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	service_ipoe-server.xml.in: service ipoe-server extended-scripts on-down	Nothing found in VyOS
×	<i>set service ipoe-server extended-scripts on-down <path_to_script></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	service_ipoe-server.xml.in: service ipoe-server extended-scripts on-pre- up	Nothing found in VyOS
×	<i>set service ipoe-server extended-scripts on-pre-up <path_to_script></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	service_ipoe-server.xml.in: service ipoe-server extended-scripts on-up	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set service ipoe-server extended-scripts on-up <path_to_script></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	service_ipoe-server.xml.in: service ipoe-server gateway-address	Nothing found in VyOS
×	<code>set service ipoe-server gateway-address <x.x.x.x/x></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set service ipoe-server interface <interface> client-subnet <x.x.x.x/x></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set service ipoe-server interface <interface> external-dhcp dhcp-relay <x.x.x.x></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set service ipoe-server interface <interface> external-dhcp giaddr <x.x.x.x></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set service ipoe-server interface <interface> mode <12 13></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set service ipoe-server interface <interface> network <shared vlan></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	service_ipoe-server.xml.in: service ipoe-server interface <interface> client-subnet	service ipoe-server interface <text> client-subnet <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	service_ipoe-server.xml.in: service ipoe-server interface <interface> external-dhcp dhcp-relay	service ipoe-server interface <text> external-dhcp dhcp-relay <value>
×	not yet documented	service_ipoe-server.xml.in: service ipoe-server interface <interface> external-dhcp giaddr	service ipoe-server interface <text> external-dhcp giaddr <value>
×	not yet documented	service_ipoe-server.xml.in: service ipoe-server interface <interface> mode	service ipoe-server interface <text> mode <value>
×	not yet documented	service_ipoe-server.xml.in: service ipoe-server interface <interface> network	service ipoe-server interface <text> network <value>
×	not yet documented	service_ipoe-server.xml.in: service ipoe-server interface <interface> vlan	service ipoe-server interface <text> vlan <value>
×	not yet documented	service_ipoe-server.xml.in: service ipoe-server limits burst	Nothing found in VyOS
×	<i>set service ipoe-server limits burst <value></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	service_ipoe-server.xml.in: service ipoe-server limits connection-limit	Nothing found in VyOS
×	<i>set service ipoe-server limits connection-limit <value></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	service_ipoe-server.xml.in: service ipoe-server limits timeout	Nothing found in VyOS
×	<i>set service ipoe-server limits timeout <value></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set service ipoe-server max-concurrent-sessio</i>	service_ipoe-server.xml.in: service ipoe-server max-concurrent-sessions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	service_ipoe-server.xml.in: service ipoe-server name-server	service ipoe-server name-server <value>
×	<i>set service ipoe-server name-server <address></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	service_ipoe-server.xml.in: service ipoe-server shaper fwmark	Nothing found in VyOS
×	<i>set service ipoe-server shaper fwmark <1-2147483647></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set service ipoe-server snmp master-agent</i>	service_ipoe-server.xml.in: service ipoe-server snmp master-agent	Nothing found in VyOS
×	<i>set service lldp</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set service lldp interface <interface></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set service lldp interface <interface> mode [disable rx-tx rx tx]</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	service_lldp.xml.in: service lldp interface <interface> disable	service lldp interface <text> disable
×	not yet documented	service_lldp.xml.in: service lldp interface <interface> location coordinate-based altitude	service lldp interface <text> location coordinate-based altitude <value>
×	not yet documented	service_lldp.xml.in: service lldp interface <interface> location coordinate-based datum	service lldp interface <text> location coordinate-based datum <value>
×	not yet documented	service_lldp.xml.in: service lldp interface <interface> location coordinate-based latitude	service lldp interface <text> location coordinate-based latitude <value>
×	not yet documented	service_lldp.xml.in: service lldp interface <interface> location coordinate-based longitude	service lldp interface <text> location coordinate-based longitude <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	service_lldp.xml.in: service lldp interface <interface> location elin	service lldp interface <text> location elin <value>
×	<i>set service lldp legacy-protocols <cdp edp fdp sonmp></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	service_lldp.xml.in: service lldp legacy-protocols cdp	service lldp legacy-protocols cdp
×	not yet documented	service_lldp.xml.in: service lldp legacy-protocols edp	service lldp legacy-protocols edp
×	not yet documented	service_lldp.xml.in: service lldp legacy-protocols fdp	service lldp legacy-protocols fdp
×	not yet documented	service_lldp.xml.in: service lldp legacy-protocols sonmp	service lldp legacy-protocols sonmp
×	not yet documented	service_lldp.xml.in: service lldp management-address	service lldp management-address <value>
×	<i>set service lldp management-address <address></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set service lldp snmp</i>	service_lldp.xml.in: service lldp snmp	Nothing found in VyOS
×	not yet documented	Nothing found in XML Definitions	service lldp snmp enable
×	not yet documented	service_mdns_repeater.xml.in: service mdns repeater allow-service	service mdns repeater allow-service <value>
×	<i>set service mdns repeater allow-service <service></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	service_mdns_repeater.xml.in: service mdns repeater browse-domain	service mdns repeater browse-domain <value>
×	<i>set service mdns repeater browse-domain <domain></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set service mdns repeater cache-entries <entries></i>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
✓	<i>set service mdns repeater disable</i>	ser- vice_mdns_repeater.xml.in: service mdns repeater dis- able	service mdns repeater dis- able
×	not yet documented	ser- vice_mdns_repeater.xml.in: service mdns repeater in- terface	service mdns repeater in- terface <value>
×	<i>set service mdns repeater interface <interface></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	ser- vice_mdns_repeater.xml.in: service mdns repeater ip- version	Nothing found in VyOS
×	<i>set service mdns repeater ip-version <ipv4 ipv6 both></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	ser- vice_mdns_repeater.xml.in: service mdns repeater vrrp-disable	service mdns repeater vrrp-disable
×	<i>set service monitoring prometheus blackbox-exporter listen-address <address></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set service monitoring prometheus blackbox-exporter port <port></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set service monitoring prometheus blackbox-exporter vrf <name></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set service monitoring prometheus frr-exporter listen-address <address></i>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set service monitoring prometheus frr-exporter port <port></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set service monitoring prometheus frr-exporter vrf <name></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set service monitoring prometheus node-exporter collectors textfile</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set service monitoring prometheus node-exporter listen-address <address></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set service monitoring prometheus node-exporter port <port></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set service monitoring prometheus node-exporter vrf <name></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	service_monitoring_telegraf.x1 service_monitoring_telegraf azure-data-explorer authentication client-id <value>	service monitoring telegraf azure-data-explorer authentication client-id
×	<code>set service monitoring telegraf azure-data-explorer authentication client-id <client-id></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	service_monitoring_telegraf.x1 service_monitoring_telegraf azure-data-explorer authentication client-secret <secret>	service monitoring telegraf azure-data-explorer authentication client-secret <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set service monitoring telegraf azure-data-explorer authentication client-secret <client-secret></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	service_monitoring_telegraf.x1 service monitoring telegraf azure-data-explorer authentication tenant-id <value>	service monitoring telegraf azure-data-explorer authentication tenant-id <value>
×	<code>set service monitoring telegraf azure-data-explorer authentication tenant-id <tenant-id></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	service_monitoring_telegraf.x1 service monitoring telegraf azure-data-explorer database <value>	service monitoring telegraf azure-data-explorer database <value>
×	<code>set service monitoring telegraf azure-data-explorer database <name></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	service_monitoring_telegraf.x1 service monitoring telegraf azure-data-explorer group-metrics <value>	service monitoring telegraf azure-data-explorer group-metrics <value>
×	<code>set service monitoring telegraf azure-data-explorer group-metrics <single-table / table-per-metric></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	service_monitoring_telegraf.x1 service monitoring telegraf azure-data-explorer table <value>	service monitoring telegraf azure-data-explorer table <value>
×	<code>set service monitoring telegraf azure-data-explorer table <name></code>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	service_monitoring_telegraf.xl service_monitoring_telegraf azure-data-explorer url	service_monitoring_telegraf azure-data-explorer url <value>
×	<i>set service monitoring telegraf azure-data-explorer url <url></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set service monitoring telegraf bucket <bucket></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	service_monitoring_telegraf.xl service_monitoring_telegraf influxdb authentication organization	service_monitoring_telegraf influxdb authentication organization <value>
×	<i>set service monitoring telegraf influxdb authentication organization <organization></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	service_monitoring_telegraf.xl service_monitoring_telegraf influxdb authentication token	service_monitoring_telegraf influxdb authentication token <value>
×	<i>set service monitoring telegraf influxdb authentication token <token></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	service_monitoring_telegraf.xl service_monitoring_telegraf influxdb bucket	service_monitoring_telegraf influxdb bucket <value>
×	not yet documented	service_monitoring_telegraf.xl service_monitoring_telegraf influxdb port	service_monitoring_telegraf influxdb port <value>
×	<i>set service monitoring telegraf influxdb port <port></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	service_monitoring_telegraf.xl service_monitoring_telegraf influxdb url	service_monitoring_telegraf influxdb url <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set service monitoring telegraf influxdb url <url></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set service monitoring telegraf loki authentication password <password></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set service monitoring telegraf loki authentication username <username></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set service monitoring telegraf loki metric-name-label <label></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set service monitoring telegraf loki port <port></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set service monitoring telegraf loki url <url></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set service monitoring telegraf prometheus-client</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	ser- vice_monitoring_telegraf.x1 service monitoring tele- graf prometheus-client allow-from	service monitoring tele- graf prometheus-client allow-from <value>
×	<code>set service monitoring telegraf prometheus-client allow-from <prefix></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	ser- vice_monitoring_telegraf.x1 service monitoring tele- graf prometheus-client au- thentication password	service monitoring tele- graf prometheus-client authentication password <value>
×	<code>set service monitoring telegraf prometheus-client authentication password <password></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	ser- vice_monitoring_telegraf.x1 service monitoring tele- graf prometheus-client au- thentication username	service monitoring tele- graf prometheus-client authentication username <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<i>set service monitoring telegraf prometheus-client authentication username <username></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	service_monitoring_telegraf.listen-address	service monitoring telegraf prometheus-client listen-address <value>
×	<i>set service monitoring telegraf prometheus-client listen-address <address></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	service_monitoring_telegraf.metric-version	service monitoring telegraf prometheus-client metric-version <value>
×	<i>set service monitoring telegraf prometheus-client metric-version <1 2></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	service_monitoring_telegraf.port	service monitoring telegraf prometheus-client port <value>
×	<i>set service monitoring telegraf prometheus-client port <port></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	service_monitoring_telegraf.source	service monitoring telegraf source <value>
✓	<i>set service monitoring telegraf splunk authentication insecure</i>	service_monitoring_telegraf.splunk_authentication_insecure	service monitoring telegraf splunk authentication insecure
×	not yet documented	service_monitoring_telegraf.splunk_authentication_token	service monitoring telegraf splunk authentication token <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set service monitoring telegraf splunk authentication token <token></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set service monitoring telegraf splunk authentication url <url></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	service_monitoring_telegraf.xml: service monitoring telegraf splunk url	service monitoring telegraf splunk url <value>
×	not yet documented	service_monitoring_telegraf.xml: service monitoring telegraf vrf	service monitoring telegraf vrf <value>
×	not yet documented	service_monitoring_zabbix-agent.xml.in: service monitoring zabbix-agent directory	service monitoring zabbix-agent directory <value>
×	not yet documented	service_monitoring_zabbix-agent.xml.in: service monitoring zabbix-agent host-name	service monitoring zabbix-agent host-name <value>
×	not yet documented	service_monitoring_zabbix-agent.xml.in: service monitoring zabbix-agent limits buffer-flush-interval	service monitoring zabbix-agent limits buffer-flush-interval <value>
×	not yet documented	service_monitoring_zabbix-agent.xml.in: service monitoring zabbix-agent limits buffer-size	service monitoring zabbix-agent limits buffer-size <value>
×	not yet documented	service_monitoring_zabbix-agent.xml.in: service monitoring zabbix-agent listen-address	service monitoring zabbix-agent listen-address <value>
×	not yet documented	service_monitoring_zabbix-agent.xml.in: service monitoring zabbix-agent log debug-level	service monitoring zabbix-agent log debug-level <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	service_monitoring_zabbix-agent.xml.in: service monitoring zabbix-agent log remote-commands	service monitoring zabbix-agent log remote-commands
×	not yet documented	service_monitoring_zabbix-agent.xml.in: service monitoring zabbix-agent log size	service monitoring zabbix-agent log size <value>
×	not yet documented	service_monitoring_zabbix-agent.xml.in: service monitoring zabbix-agent port	service monitoring zabbix-agent port <value>
×	not yet documented	service_monitoring_zabbix-agent.xml.in: service monitoring zabbix-agent server	service monitoring zabbix-agent server <value>
×	not yet documented	service_monitoring_zabbix-agent.xml.in: service monitoring zabbix-agent server-active <server-active> port	service monitoring zabbix-agent server-active <text> port <value>
×	not yet documented	service_monitoring_zabbix-agent.xml.in: service monitoring zabbix-agent timeout	service monitoring zabbix-agent timeout <value>
×	not yet documented	service_ndp-proxy.xml.in: service ndp-proxy inter- face <interface> disable	Nothing found in VyOS
×	not yet documented	service_ndp-proxy.xml.in: service ndp-proxy inter- face <interface> enable- router-bit	Nothing found in VyOS
×	not yet documented	service_ndp-proxy.xml.in: service ndp-proxy inter- face <interface> prefix <prefix> disable	Nothing found in VyOS
×	not yet documented	service_ndp-proxy.xml.in: service ndp-proxy inter- face <interface> prefix <prefix> interface	Nothing found in VyOS
×	not yet documented	service_ndp-proxy.xml.in: service ndp-proxy inter- face <interface> prefix <prefix> mode	Nothing found in VyOS
×	not yet documented	service_ndp-proxy.xml.in: service ndp-proxy inter- face <interface> timeout	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	service_ndp-proxy.xml.in: service ndp-proxy inter- face <interface> ttl	Nothing found in VyOS
×	not yet documented	service_ndp-proxy.xml.in: service ndp-proxy route- refresh	Nothing found in VyOS
×	not yet documented	service_ntp.xml.in: service ntp allow-client address	service ntp allow-client address <value>
×	set service ntp allow-client address <address>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	service_ntp.xml.in: service ntp interface	service ntp interface <value>
×	set service ntp leap-second [ignore/smear/system/	service_ntp.xml.in: service ntp leap-second	Nothing found in VyOS
×	not yet documented	service_ntp.xml.in: service ntp listen-address	service ntp listen-address <value>
×	set service ntp listen-address <address>	Nothing found in XML Definitions	Nothing found in VyOS
×	set service ntp ptp	Nothing found in XML Definitions	Nothing found in VyOS
×	set service ntp ptp port <port>	Nothing found in XML Definitions	Nothing found in VyOS
×	set service ntp server <address>	Nothing found in XML Definitions	Nothing found in VyOS
×	set service ntp server <address> <noselect nts pool prefer ptp interleave>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	service_ntp.xml.in: service ntp server <server> noselect	service ntp server <text> noselect
×	not yet documented	service_ntp.xml.in: service ntp server <server> nts	service ntp server <text> nts
×	not yet documented	service_ntp.xml.in: service ntp server <server> pool	service ntp server <text> pool
×	not yet documented	service_ntp.xml.in: service ntp server <server> prefer	service ntp server <text> prefer
×	set service ntp timestamp interface <interface>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	service_ntp.xml.in: service ntp vrf	service ntp vrf <value>
×	<i>set service ntp vrf <name></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	service_pppoe- server.xml.in: service pppoe-server access-concentrator	service pppoe-server access-concentrator <value>
×	<i>set service pppoe-server access-concentrator <name></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set service pppoe-server authentication local-users username <user> disable</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set service pppoe-server authentication local-users username <name> password <password></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set service pppoe-server authentication local-users username <user> rate-limit download <bandwidth></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set service pppoe-server authentication local-users username <user> rate-limit upload <bandwidth></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set service pppoe-server authentication local-users username <user> static-ip <address></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	service_pppoe- server.xml.in: service pppoe-server authentication local-users username <username> disable	service pppoe-server authentication local-users username <text> disable

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	service_pppoe-server.xml.in: service pppoe-server authentication local-users username <username> password	service pppoe-server authentication local-users username <text> password <value>
×	not yet documented	service_pppoe-server.xml.in: service pppoe-server authentication local-users username <username> rate-limit download	service pppoe-server authentication local-users username <text> rate-limit download <value>
×	not yet documented	service_pppoe-server.xml.in: service pppoe-server authentication local-users username <username> rate-limit upload	service pppoe-server authentication local-users username <text> rate-limit upload <value>
×	not yet documented	service_pppoe-server.xml.in: service pppoe-server authentication local-users username <username> static-ip	service pppoe-server authentication local-users username <text> static-ip <value>
×	not yet documented	service_pppoe-server.xml.in: service pppoe-server authentication mode	service pppoe-server authentication mode <value>
×	<i>set service pppoe-server authentication mode <local radius></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	service_pppoe-server.xml.in: service pppoe-server authentication protocols	service pppoe-server authentication protocols <value>
×	<i>set service pppoe-server authentication protocols <pap chap mschap mschap-v2></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	service_pppoe-server.xml.in: service pppoe-server authentication radius accounting-interim-interval	service pppoe-server authentication radius accounting-interim-interval <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	service_pppoe-server.xml.in: service pppoe-server authentication radius acct-interim-jitter	service pppoe-server authentication radius acct-interim-jitter <value>
×	not yet documented	service_pppoe-server.xml.in: service pppoe-server authentication radius acct-timeout	service pppoe-server authentication radius acct-timeout <value>
×	<i>set service pppoe-server authentication radius acct-timeout <timeout></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	service_pppoe-server.xml.in: service pppoe-server authentication radius called-sid-format	service pppoe-server authentication radius called-sid-format <value>
×	not yet documented	service_pppoe-server.xml.in: service pppoe-server authentication radius dynamic-author key	service pppoe-server authentication radius dynamic-author key <value>
×	<i>set service pppoe-server authentication radius dynamic-author key <secret></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	service_pppoe-server.xml.in: service pppoe-server authentication radius dynamic-author port	service pppoe-server authentication radius dynamic-author port <value>
×	<i>set service pppoe-server authentication radius dynamic-author port <port></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	service_pppoe-server.xml.in: service pppoe-server authentication radius dynamic-author server	service pppoe-server authentication radius dynamic-author server <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set service pppoe-server authentication radius dynamic-author server <address></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	service_pppoe-server.xml.in: service pppoe-server authentication radius max-try	service pppoe-server authentication radius max-try <value>
×	<code>set service pppoe-server authentication radius max-try <number></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	service_pppoe-server.xml.in: service pppoe-server authentication radius nas-identifier	service pppoe-server authentication radius nas-identifier <value>
×	<code>set service pppoe-server authentication radius nas-identifier <identifier></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	service_pppoe-server.xml.in: service pppoe-server authentication radius nas-ip-address	service pppoe-server authentication radius nas-ip-address <value>
×	<code>set service pppoe-server authentication radius nas-ip-address <address></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	service_pppoe-server.xml.in: service pppoe-server authentication radius preallocate-vif	service pppoe-server authentication radius preallocate-vif
×	not yet documented	service_pppoe-server.xml.in: service pppoe-server authentication radius rate-limit attribute	service pppoe-server authentication radius rate-limit attribute <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set service pppoe-server authentication radius rate-limit attribute <attribute></code>	Nothing found in XML Definitions	Nothing found in VyOS
✓	<code>set service pppoe-server authentication radius rate-limit enable</code>	service_pppoe-server.xml.in: service pppoe-server authentication radius rate-limit enable	service pppoe-server authentication radius rate-limit enable
×	not yet documented	service_pppoe-server.xml.in: service pppoe-server authentication radius rate-limit multiplier	service pppoe-server authentication radius rate-limit multiplier <value>
✓	<code>set service pppoe-server authentication radius rate-limit vendor</code>	service_pppoe-server.xml.in: service pppoe-server authentication radius rate-limit vendor	service pppoe-server authentication radius rate-limit vendor <value>
×	<code>set service pppoe-server authentication radius server <server> disable</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set service pppoe-server authentication radius server <server> fail-time <time></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set service pppoe-server authentication radius server <server> key <secret></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set service pppoe-server authentication radius server <server> port <port></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	service_pppoe-server.xml.in: service pppoe-server authentication radius server <server> acct-port	service pppoe-server authentication radius server <text> acct-port <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	service_pppoe-server.xml.in: service pppoe-server authentication radius server <server> disable	service pppoe-server authentication radius server <text> disable
×	not yet documented	service_pppoe-server.xml.in: service pppoe-server authentication radius server <server> disable-accounting	service pppoe-server authentication radius server <text> disable-accounting
×	not yet documented	service_pppoe-server.xml.in: service pppoe-server authentication radius server <server> fail-time	service pppoe-server authentication radius server <text> fail-time <value>
×	not yet documented	service_pppoe-server.xml.in: service pppoe-server authentication radius server <server> key	service pppoe-server authentication radius server <text> key <value>
×	not yet documented	service_pppoe-server.xml.in: service pppoe-server authentication radius server <server> port	service pppoe-server authentication radius server <text> port <value>
×	not yet documented	service_pppoe-server.xml.in: service pppoe-server authentication radius source-address	service pppoe-server authentication radius source-address <value>
×	set service pppoe-server authentication radius source-address <address>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	service_pppoe-server.xml.in: service pppoe-server authentication radius timeout	service pppoe-server authentication radius timeout <value>
×	set service pppoe-server authentication radius timeout <timeout>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set service pppoe-server client-ip-pool <POOL-NAME> next-pool <NEXT-POOL-NAME></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set service pppoe-server client-ip-pool <POOL-NAME> range <x.x.x.x-x.x.x.x / x.x.x.x/x></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	Nothing found in XML Definitions	service pppoe-server client-ip-pool name <text> gateway-address <value>
×	not yet documented	Nothing found in XML Definitions	service pppoe-server client-ip-pool name <text> next-pool <value>
×	not yet documented	Nothing found in XML Definitions	service pppoe-server client-ip-pool name <text> subnet <value>
×	not yet documented	service_pppoe- server.xml.in: service pppoe-server client-ip-pool <client-ip- pool> next-pool	Nothing found in VyOS
×	not yet documented	service_pppoe- server.xml.in: service pppoe-server client-ip-pool <client-ip- pool> range	Nothing found in VyOS
×	not yet documented	Nothing found in XML Definitions	service pppoe-server client-ip-pool start <value>
×	not yet documented	Nothing found in XML Definitions	service pppoe-server client-ip-pool stop <value>
×	not yet documented	Nothing found in XML Definitions	service pppoe-server client-ip-pool subnet <value>
×	<code>set service pppoe-server client-ipv6-pool <IPv6-POOL-NAME> delegate <address> delegation-prefix <number-of-bits></code>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<i>set service pppoe-server client-ipv6-pool <IPv6-POOL-NAME> prefix <address> mask <number-of-bits></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	service_pppoe-server.xml.in: service pppoe-server client-ipv6-pool <client-ipv6-pool> delegate <delegate> delegation-prefix	service pppoe-server client-ipv6-pool delegate <text> delegation-prefix <value>
×	not yet documented	service_pppoe-server.xml.in: service pppoe-server client-ipv6-pool <client-ipv6-pool> prefix <prefix> mask	service pppoe-server client-ipv6-pool prefix <text> mask <value>
×	not yet documented	service_pppoe-server.xml.in: service pppoe-server default-ipv6-pool	Nothing found in VyOS
×	<i>set service pppoe-server default-ipv6-pool <IPv6-POOL-NAME></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	service_pppoe-server.xml.in: service pppoe-server default-pool	Nothing found in VyOS
×	<i>set service pppoe-server default-pool <POOL-NAME></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	service_pppoe-server.xml.in: service pppoe-server description	Nothing found in VyOS
×	<i>set service pppoe-server description <description></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	service_pppoe-server.xml.in: service pppoe-server extended-scripts on-change	service pppoe-server extended-scripts on-change <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<i>set service pppoe-server extended-scripts on-change <path_to_script></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	service_pppoe-server.xml.in: service pppoe-server extended-scripts on-down	service pppoe-server extended-scripts on-down <value>
×	<i>set service pppoe-server extended-scripts on-down <path_to_script></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	service_pppoe-server.xml.in: service pppoe-server extended-scripts on-pre-up	service pppoe-server extended-scripts on-pre-up <value>
×	<i>set service pppoe-server extended-scripts on-pre-up <path_to_script></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	service_pppoe-server.xml.in: service pppoe-server extended-scripts on-up	service pppoe-server extended-scripts on-up <value>
×	<i>set service pppoe-server extended-scripts on-up <path_to_script></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	service_pppoe-server.xml.in: service pppoe-server gateway-address	service pppoe-server gateway-address <value>
×	<i>set service pppoe-server gateway-address <address></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set service pppoe-server interface <interface></i>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<i>set service pppoe-server interface <interface> vlan <id range></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	service_pppoe-server.xml.in: service pppoe-server interface <interface> vlan	service pppoe-server interface <text> vlan <value>
×	not yet documented	service_pppoe-server.xml.in: service pppoe-server limits burst	service pppoe-server limits burst <value>
×	<i>set service pppoe-server limits burst <value></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	service_pppoe-server.xml.in: service pppoe-server limits connection-limit	service pppoe-server limits connection-limit <value>
×	<i>set service pppoe-server limits connection-limit <value></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	service_pppoe-server.xml.in: service pppoe-server limits timeout	service pppoe-server limits timeout <value>
×	<i>set service pppoe-server limits timeout <value></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set service pppoe-server max-concurrent-sessio</i>	service_pppoe-server.xml.in: service pppoe-server max-concurrent-sessions	Nothing found in VyOS
✓	<i>set service pppoe-server mtu</i>	service_pppoe-server.xml.in: service pppoe-server mtu	service pppoe-server mtu <value>
×	not yet documented	service_pppoe-server.xml.in: service pppoe-server name-server	service pppoe-server name-server <value>
×	<i>set service pppoe-server name-server <address></i>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<i>set service pppoe-server pado-delay <number-of-ms> sessions <number-of-sessions></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	service_pppoe-server.xml.in: service pppoe-server pado-delay <pado-delay> sessions	service pppoe-server pado-delay <text> sessions <value>
×	not yet documented	Nothing found in XML Definitions	service pppoe-server ppp-options ccp
×	<i>set service pppoe-server ppp-options disable-ccp</i>	service_pppoe-server.xml.in: service pppoe-server ppp-options disable-ccp	Nothing found in VyOS
×	not yet documented	service_pppoe-server.xml.in: service pppoe-server ppp-options interface-cache	service pppoe-server ppp-options interface-cache <value>
×	<i>set service pppoe-server ppp-options interface-cache <number></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	service_pppoe-server.xml.in: service pppoe-server ppp-options ipv4	service pppoe-server ppp-options ipv4 <value>
×	<i>set service pppoe-server ppp-options ipv4 <require prefer allow deny></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	service_pppoe-server.xml.in: service pppoe-server ppp-options ipv6	service pppoe-server ppp-options ipv6 <value>
×	<i>set service pppoe-server ppp-options ipv6-accept-peer-inte</i>	service_pppoe-server.xml.in: service pppoe-server ppp-options ipv6-accept-peer-interface-id	Nothing found in VyOS
×	not yet documented	Nothing found in XML Definitions	service pppoe-server ppp-options ipv6-accept-peer-intf-id

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	service_pppoe-server.xml.in: service pppoe-server ppp-options ipv6-interface-id	Nothing found in VyOS
×	<code>set service pppoe-server ppp-options ipv6-interface-id <random / x:x:x:x></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	Nothing found in XML Definitions	service pppoe-server ppp-options ipv6-intf-id <value>
×	not yet documented	service_pppoe-server.xml.in: service pppoe-server ppp-options ipv6-peer-interface-id	Nothing found in VyOS
×	not yet documented	Nothing found in XML Definitions	service pppoe-server ppp-options ipv6-peer-intf-id <value>
×	<code>set service pppoe-server ppp-options ipv6 <require prefer allow deny></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	service_pppoe-server.xml.in: service pppoe-server ppp-options lcp-echo-failure	service pppoe-server ppp-options lcp-echo-failure <value>
×	<code>set service pppoe-server ppp-options lcp-echo-failure <number></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	service_pppoe-server.xml.in: service pppoe-server ppp-options lcp-echo-interval	service pppoe-server ppp-options lcp-echo-interval <value>
×	<code>set service pppoe-server ppp-options lcp-echo-interval <interval></code>	Nothing found in XML Definitions	Nothing found in VyOS
✓	<code>set service pppoe-server ppp-options lcp-echo-timeout</code>	service_pppoe-server.xml.in: service pppoe-server ppp-options lcp-echo-timeout	service pppoe-server ppp-options lcp-echo-timeout <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	service_pppoe-server.xml.in: service pppoe-server ppp-options min-mtu	service pppoe-server ppp-options min-mtu <value>
×	set service pppoe-server ppp-options min-mtu <number>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	service_pppoe-server.xml.in: service pppoe-server ppp-options mppe	service pppoe-server ppp-options mppe <value>
×	set service pppoe-server ppp-options mppe <require prefer deny>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	service_pppoe-server.xml.in: service pppoe-server ppp-options mru	service pppoe-server ppp-options mru <value>
×	set service pppoe-server ppp-options mru <number>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	service_pppoe-server.xml.in: service pppoe-server service-name	service pppoe-server service-name <value>
×	set service pppoe-server service-name <names>	Nothing found in XML Definitions	Nothing found in VyOS
✓	set service pppoe-server session-control	service_pppoe-server.xml.in: service pppoe-server session-control	service pppoe-server session-control <value>
×	not yet documented	service_pppoe-server.xml.in: service pppoe-server shaper fwmark	service pppoe-server shaper fwmark <value>
×	set service pppoe-server shaper fwmark <1-2147483647>	Nothing found in XML Definitions	Nothing found in VyOS
✓	set service pppoe-server snmp master-agent	service_pppoe-server.xml.in: service pppoe-server snmp master-agent	service pppoe-server snmp master-agent

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	service_pppoe-server.xml.in: service pppoe-server wins-server	service pppoe-server wins-server <value>
×	<code>set service pppoe-server wins-server <address></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set service router-advert interface <interface> ...</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set service router-advert interface <interface> nat64prefix <prefix/mask></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set service router-advert interface <interface> no-send-advert</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set service router-advert interface <interface> no-send-interval</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set service router-advert interface <interface> prefix <prefix/mask></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	service_router-advert.xml.in: service router-advert interface <interface> default-lifetime	service router-advert interface <text> default-lifetime <value>
×	not yet documented	service_router-advert.xml.in: service router-advert interface <interface> default-preference	service router-advert interface <text> default-preference <value>
×	not yet documented	service_router-advert.xml.in: service router-advert interface <interface> dnssl	service router-advert interface <text> dnssl <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	service_router-advert.xml.in: service router-advert interface <interface> hop-limit	service router-advert interface <text> hop-limit <value>
×	not yet documented	service_router-advert.xml.in: service router-advert interface <interface> interval max	service router-advert interface <text> interval max <value>
×	not yet documented	service_router-advert.xml.in: service router-advert interface <interface> interval min	service router-advert interface <text> interval min <value>
×	not yet documented	service_router-advert.xml.in: service router-advert interface <interface> link-mtu	service router-advert interface <text> link-mtu <value>
×	not yet documented	service_router-advert.xml.in: service router-advert interface <interface> managed-flag	service router-advert interface <text> managed-flag
×	not yet documented	service_router-advert.xml.in: service router-advert interface <interface> name-server	service router-advert interface <text> name-server <value>
×	not yet documented	service_router-advert.xml.in: service router-advert interface <interface> name-server-lifetime	service router-advert interface <text> name-server-lifetime <value>
×	not yet documented	service_router-advert.xml.in: service router-advert interface <interface> nat64prefix <nat64prefix> valid-lifetime	Nothing found in VyOS
×	not yet documented	service_router-advert.xml.in: service router-advert interface <interface> no-send-advert	service router-advert interface <text> no-send-advert
×	not yet documented	service_router-advert.xml.in: service router-advert interface <interface> other-config-flag	service router-advert interface <text> other-config-flag

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	service_router-advert.xml.in: service router-advert interface <interface> prefix <prefix> decrement-lifetime	service router-advert interface <text> prefix <text> decrement-lifetime
×	not yet documented	service_router-advert.xml.in: service router-advert interface <interface> prefix <prefix> deprecate-prefix	service router-advert interface <text> prefix <text> deprecate-prefix
×	not yet documented	service_router-advert.xml.in: service router-advert interface <interface> prefix <prefix> no-autonomous-flag	service router-advert interface <text> prefix <text> no-autonomous-flag
×	not yet documented	service_router-advert.xml.in: service router-advert interface <interface> prefix <prefix> no-on-link-flag	service router-advert interface <text> prefix <text> no-on-link-flag
×	not yet documented	service_router-advert.xml.in: service router-advert interface <interface> prefix <prefix> preferred-lifetime	service router-advert interface <text> prefix <text> preferred-lifetime <value>
×	not yet documented	service_router-advert.xml.in: service router-advert interface <interface> prefix <prefix> valid-lifetime	service router-advert interface <text> prefix <text> valid-lifetime <value>
×	not yet documented	service_router-advert.xml.in: service router-advert interface <interface> reachable-time	service router-advert interface <text> reachable-time <value>
×	not yet documented	service_router-advert.xml.in: service router-advert interface <interface> retrans-timer	service router-advert interface <text> retrans-timer <value>
×	not yet documented	service_router-advert.xml.in: service router-advert interface <interface> route <route> no-remove-route	service router-advert interface <text> route <text> no-remove-route

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	service_router-advert.xml.in: service router-advert interface <interface> route <route> route-preference	service router-advert interface <text> route <text> route-preference <value>
×	not yet documented	service_router-advert.xml.in: service router-advert interface <interface> route <route> valid-lifetime	service router-advert interface <text> route <text> valid-lifetime <value>
×	not yet documented	service_router-advert.xml.in: service router-advert interface <interface> source-address	service router-advert interface <text> source-address <value>
×	not yet documented	service_salt-minion.xml.in: service salt-minion hash	service salt-minion hash <value>
×	not yet documented	service_salt-minion.xml.in: service salt-minion id	service salt-minion id <value>
×	not yet documented	service_salt-minion.xml.in: service salt-minion interval	service salt-minion interval <value>
×	not yet documented	service_salt-minion.xml.in: service salt-minion master	service salt-minion master <value>
×	not yet documented	service_salt-minion.xml.in: service salt-minion master-key	service salt-minion master-key <value>
×	not yet documented	service_salt-minion.xml.in: service salt-minion source-interface	service salt-minion source-interface <value>
×	not yet documented	service_sla.xml.in: service sla owamp-server port	service sla owamp-server port <value>
×	not yet documented	service_sla.xml.in: service sla twamp-server port	service sla twamp-server port <value>
×	not yet documented	service_snmp.xml.in: service snmp community <community> authorization	service snmp community <text> authorization <value>
×	not yet documented	service_snmp.xml.in: service snmp community <community> client	service snmp community <text> client <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	service_snmp.xml.in: service snmp community <community> network	service snmp community <text> network <value>
×	not yet documented	service_snmp.xml.in: service snmp contact	service snmp contact <value>
×	not yet documented	service_snmp.xml.in: service snmp description	service snmp description <value>
×	not yet documented	service_snmp.xml.in: service snmp listen- address <listen-address> port	service snmp listen- address <text> port <value>
×	not yet documented	service_snmp.xml.in: service snmp location	service snmp location <value>
×	not yet documented	service_snmp.xml.in: service snmp mib inter- face	Nothing found in VyOS
×	not yet documented	service_snmp.xml.in: service snmp mib interface-max	Nothing found in VyOS
×	not yet documented	service_snmp.xml.in: service snmp oid-enable	service snmp oid-enable <value>
×	not yet documented	service_snmp.xml.in: service snmp protocol	service snmp protocol <value>
×	not yet documented	service_snmp.xml.in: service snmp script- extensions extension- name <extension-name> script	service snmp script- extensions extension- name <text> script <value>
×	not yet documented	service_snmp.xml.in: service snmp smux-peer	service snmp smux-peer <value>
×	not yet documented	service_snmp.xml.in: service snmp trap-source	service snmp trap-source <value>
×	not yet documented	service_snmp.xml.in: service snmp trap-target <trap-target> community	service snmp trap- target <text> community <value>
×	not yet documented	service_snmp.xml.in: service snmp trap-target <trap-target> port	service snmp trap-target <text> port <value>
×	not yet documented	service_snmp.xml.in: service snmp v3 engineid	service snmp v3 engineid <value>
×	not yet documented	service_snmp.xml.in: service snmp v3 group <group> mode	service snmp v3 group <text> mode <value>
×	not yet documented	service_snmp.xml.in: service snmp v3 group <group> seclevel	service snmp v3 group <text> seclevel <value>
×	not yet documented	service_snmp.xml.in: service snmp v3 group <group> view	service snmp v3 group <text> view <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	service_snmp.xml.in: service snmp v3 trap- target <trap-target> auth encrypted-password <value>	service snmp v3 trap- target <text> auth encrypted-password <value>
×	not yet documented	service_snmp.xml.in: service snmp v3 trap- target <trap-target> auth plaintext-password <value>	service snmp v3 trap- target <text> auth plaintext-password <value>
×	not yet documented	service_snmp.xml.in: service snmp v3 trap- target <trap-target> auth type <value>	service snmp v3 trap- target <text> auth type <value>
×	not yet documented	service_snmp.xml.in: service snmp v3 trap- target <trap-target> port <value>	service snmp v3 trap- target <text> port <value>
×	not yet documented	service_snmp.xml.in: service snmp v3 trap- target <trap-target> pri- vacy encrypted-password <value>	service snmp v3 trap- target <text> privacy encrypted-password <value>
×	not yet documented	service_snmp.xml.in: service snmp v3 trap- target <trap-target> pri- vacy plaintext-password <value>	service snmp v3 trap- target <text> privacy plaintext-password <value>
×	not yet documented	service_snmp.xml.in: service snmp v3 trap- target <trap-target> privacy type <value>	service snmp v3 trap- target <text> privacy type <value>
×	not yet documented	service_snmp.xml.in: service snmp v3 trap- target <trap-target> protocol <value>	service snmp v3 trap- target <text> protocol <value>
×	not yet documented	service_snmp.xml.in: service snmp v3 trap- target <trap-target> type <value>	service snmp v3 trap- target <text> type <value>
×	not yet documented	service_snmp.xml.in: service snmp v3 trap- target <trap-target> user <value>	service snmp v3 trap- target <text> user <value>
×	not yet documented	service_snmp.xml.in: service snmp v3 user <user> auth encrypted- password <value>	service snmp v3 user <text> auth encrypted- password <value>
×	not yet documented	service_snmp.xml.in: service snmp v3 user <user> auth plaintext- password <value>	service snmp v3 user <text> auth plaintext- password <value>
×	not yet documented	service_snmp.xml.in: service snmp v3 user <user> auth type <value>	service snmp v3 user <text> auth type <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	service_snmp.xml.in: service snmp v3 user <user> group	service snmp v3 user <text> group <value>
×	not yet documented	service_snmp.xml.in: service snmp v3 user <user> mode	service snmp v3 user <text> mode <value>
×	not yet documented	service_snmp.xml.in: service snmp v3 user <user> privacy encrypted- password	service snmp v3 user <text> privacy encrypted- password <value>
×	not yet documented	service_snmp.xml.in: service snmp v3 user <user> privacy plaintext- password	service snmp v3 user <text> privacy plaintext- password <value>
×	not yet documented	service_snmp.xml.in: service snmp v3 user <user> privacy type	service snmp v3 user <text> privacy type <value>
×	not yet documented	service_snmp.xml.in: service snmp v3 view <view> oid <oid> exclude	service snmp v3 view <text> oid <text> exclude <value>
×	not yet documented	service_snmp.xml.in: service snmp v3 view <view> oid <oid> mask	service snmp v3 view <text> oid <text> mask <value>
×	not yet documented	service_snmp.xml.in: service snmp vrf	service snmp vrf <value>
×	set service ssh access-control <allow deny> <group user> <name>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	service_ssh.xml.in: service ssh access-control allow group	service ssh access-control allow group <value>
×	not yet documented	service_ssh.xml.in: service ssh access-control allow user	service ssh access-control allow user <value>
×	not yet documented	service_ssh.xml.in: service ssh access-control deny group	service ssh access-control deny group <value>
×	not yet documented	service_ssh.xml.in: service ssh access-control deny user	service ssh access-control deny user <value>
×	set service ssh cipher <cipher>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	service_ssh.xml.in: service ssh ciphers	service ssh ciphers <value>
×	not yet documented	service_ssh.xml.in: service ssh client- keepalive-interval	service ssh client- keepalive-interval <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set service ssh client-keepalive-interval <interval></code>	Nothing found in XML Definitions	Nothing found in VyOS
✓	<code>set service ssh disable-host-validation</code>	service_ssh.xml.in: service ssh disable-host-validation	service ssh disable-host-validation
✓	<code>set service ssh disable-password-auth</code>	service_ssh.xml.in: service ssh disable-password-authentication	service ssh disable-password-authentication
×	<code>set service ssh dynamic-protection</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	service_ssh.xml.in: service ssh dynamic-protection allow-from <value>	service ssh dynamic-protection allow-from <value>
×	<code>set service ssh dynamic-protection allow-from <address / prefix></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	service_ssh.xml.in: service ssh dynamic-protection block-time <value>	service ssh dynamic-protection block-time <value>
×	<code>set service ssh dynamic-protection block-time <sec></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	service_ssh.xml.in: service ssh dynamic-protection detect-time <value>	service ssh dynamic-protection detect-time <value>
×	<code>set service ssh dynamic-protection detect-time <sec></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	service_ssh.xml.in: service ssh dynamic-protection threshold <value>	service ssh dynamic-protection threshold <value>
×	<code>set service ssh dynamic-protection threshold <sec></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set service ssh fido pin-required</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set service ssh fido touch-required</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	service_ssh.xml.in: service ssh hostkey-algorithm <value>	service ssh hostkey-algorithm <value>
×	not yet documented	service_ssh.xml.in: service ssh key-exchange <value>	service ssh key-exchange <value>
×	<code>set service ssh key-exchange <kex></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	service_ssh.xml.in: service ssh listen-address <value>	service ssh listen-address <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set service ssh listen-address <address></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	service_ssh.xml.in: service ssh loglevel	service ssh loglevel <value>
×	<code>set service ssh loglevel <quiet / fatal / error / info / verbose></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	service_ssh.xml.in: service ssh mac	service ssh mac <value>
×	<code>set service ssh mac <mac></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	service_ssh.xml.in: service ssh port	service ssh port <value>
×	<code>set service ssh port <port></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set service ssh pubkey-accepted-algor <name></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	service_ssh.xml.in: service ssh rekey data	service ssh rekey data <value>
×	not yet documented	service_ssh.xml.in: service ssh rekey time	service ssh rekey time <value>
×	<code>set service ssh trusted-user-ca <name></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	service_ssh.xml.in: service ssh vrf	service ssh vrf <value>
×	<code>set service ssh vrf <name></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set service suricata address-group <name> address <address></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set service suricata address-group <name> group <name></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set service suricata interface <interface></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set service suricata log eve filename <filename></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set service suricata log eve filetype <regular / syslog></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set service suricata log eve type <type></code>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set service suricata port-group <name> group <name></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set service suricata port-group <name> port <port></code>	Nothing found in XML Definitions	Nothing found in VyOS
✓	<code>set service tftp-server allow-upload</code>	service_tftp-server.xml.in: service tftp-server allow-upload	service tftp-server allow-upload
×	not yet documented	service_tftp-server.xml.in: service tftp-server directory	service tftp-server directory <value>
×	<code>set service tftp-server directory <directory></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set service tftp-server listen-address <address></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set service tftp-server listen-address <address> vrf <name></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	service_tftp-server.xml.in: service tftp-server listen-address <listen-address> vrf	service tftp-server listen-address <text> vrf <value>
×	not yet documented	service_tftp-server.xml.in: service tftp-server port	service tftp-server port <value>
×	<code>set service tftp-server port <1-65535></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	Nothing found in XML Definitions	service upnp friendly-name <value>
×	not yet documented	Nothing found in XML Definitions	service upnp listen <value>
×	not yet documented	Nothing found in XML Definitions	service upnp nat-pmp
×	not yet documented	Nothing found in XML Definitions	service upnp pcp-lifetime max <value>
×	not yet documented	Nothing found in XML Definitions	service upnp pcp-lifetime min <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	Nothing found in XML Definitions	service upnp presentation-url <value>
×	not yet documented	Nothing found in XML Definitions	service upnp rule <text> action <value>
×	not yet documented	Nothing found in XML Definitions	service upnp rule <text> disable
×	not yet documented	Nothing found in XML Definitions	service upnp rule <text> external-port-range <value>
×	not yet documented	Nothing found in XML Definitions	service upnp rule <text> internal-port-range <value>
×	not yet documented	Nothing found in XML Definitions	service upnp rule <text> ip <value>
×	not yet documented	Nothing found in XML Definitions	service upnp secure-mode
×	not yet documented	Nothing found in XML Definitions	service upnp stun host <value>
×	not yet documented	Nothing found in XML Definitions	service upnp stun port <value>
×	not yet documented	Nothing found in XML Definitions	service upnp wan-interface <value>
×	not yet documented	Nothing found in XML Definitions	service upnp wan-ip <value>
×	not yet documented	service_webproxy.xml.in: service webproxy append-domain	service webproxy append-domain <value>
×	<i>set service webproxy append-domain <domain></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	service_webproxy.xml.in: service webproxy authentication children	service webproxy authentication children <value>
×	<i>set service webproxy authentication children <number></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	service_webproxy.xml.in: service webproxy authentication credentials-ttl	service webproxy authentication credentials-ttl <value>
×	<i>set service webproxy authentication credentials-ttl <time></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	service_webproxy.xml.in: service webproxy authentication ldap base-dn	service webproxy authentication ldap base-dn <value>
×	<i>set service webproxy authentication ldap base-dn <base-dn></i>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	service_webproxy.xml.in: service webproxy authentication ldap bind-dn	service webproxy authentication ldap bind-dn <value>
×	<i>set service webproxy authentication ldap bind-dn <bind-dn></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	service_webproxy.xml.in: service webproxy authentication ldap filter-expression	service webproxy authentication ldap filter-expression <value>
×	<i>set service webproxy authentication ldap filter-expression <expr></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	service_webproxy.xml.in: service webproxy authentication ldap password	service webproxy authentication ldap password <value>
×	<i>set service webproxy authentication ldap password <password></i>	Nothing found in XML Definitions	Nothing found in VyOS
✓	<i>set service webproxy authentication ldap persistent-connection</i>	service_webproxy.xml.in: service webproxy authentication ldap persistent-connection	service webproxy authentication ldap persistent-connection
×	not yet documented	service_webproxy.xml.in: service webproxy authentication ldap port	service webproxy authentication ldap port <value>
×	<i>set service webproxy authentication ldap port <port></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	service_webproxy.xml.in: service webproxy authentication ldap server	service webproxy authentication ldap server <value>
×	<i>set service webproxy authentication ldap server <server></i>	Nothing found in XML Definitions	Nothing found in VyOS
✓	<i>set service webproxy authentication ldap use-ssl</i>	service_webproxy.xml.in: service webproxy authentication ldap use-ssl	service webproxy authentication ldap use-ssl
×	not yet documented	service_webproxy.xml.in: service webproxy authentication ldap username-attribute	service webproxy authentication ldap username-attribute <value>
×	<i>set service webproxy authentication ldap username-attribute <attr></i>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	service_webproxy.xml.in: service webproxy authentication ldap version	service webproxy authentication ldap version <value>
×	<i>set service webproxy authentication ldap version <2 3></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	service_webproxy.xml.in: service webproxy authentication method	service webproxy authentication method <value>
×	<i>set service webproxy authentication method <ldap></i>	Nothing found in XML Definitions	Nothing found in VyOS
✓	<i>set service webproxy authentication realm</i>	service_webproxy.xml.in: service webproxy authentication realm	service webproxy authentication realm <value>
×	not yet documented	service_webproxy.xml.in: service webproxy cache-peer <cache-peer> address	service webproxy cache-peer <text> address <value>
×	not yet documented	service_webproxy.xml.in: service webproxy cache-peer <cache-peer> http-port	service webproxy cache-peer <text> http-port <value>
×	not yet documented	service_webproxy.xml.in: service webproxy cache-peer <cache-peer> icp-port	service webproxy cache-peer <text> icp-port <value>
×	not yet documented	service_webproxy.xml.in: service webproxy cache-peer <cache-peer> options	service webproxy cache-peer <text> options <value>
×	not yet documented	service_webproxy.xml.in: service webproxy cache-peer <cache-peer> type	service webproxy cache-peer <text> type <value>
×	not yet documented	service_webproxy.xml.in: service webproxy cache-size	service webproxy cache-size <value>
×	<i>set service webproxy cache-size <size></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	service_webproxy.xml.in: service webproxy default-port	service webproxy default-port <value>
×	<i>set service webproxy default-port <port></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	service_webproxy.xml.in: service webproxy disable-access-log	service webproxy disable-access-log
×	not yet documented	service_webproxy.xml.in: service webproxy domain-block	service webproxy domain-block <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set service webproxy domain-block <domain></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	service_webproxy.xml.in: service webproxy domain-noncache	service webproxy domain-noncache <value>
×	<code>set service webproxy domain-noncache <domain></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set service webproxy listen-address <address></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set service webproxy listen-address <address> disable-transparent</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set service webproxy listen-address <address> port <port></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	service_webproxy.xml.in: service webproxy listen-address <listen-address> disable-transparent	service webproxy listen-address <text> disable-transparent
×	not yet documented	service_webproxy.xml.in: service webproxy listen-address <listen-address> port	service webproxy listen-address <text> port <value>
×	not yet documented	service_webproxy.xml.in: service webproxy maximum-object-size	service webproxy maximum-object-size <value>
×	not yet documented	service_webproxy.xml.in: service webproxy mem-cache-size	service webproxy mem-cache-size <value>
×	not yet documented	service_webproxy.xml.in: service webproxy minimum-object-size	service webproxy minimum-object-size <value>
×	not yet documented	service_webproxy.xml.in: service webproxy outgoing-address	service webproxy outgoing-address <value>
×	not yet documented	service_webproxy.xml.in: service webproxy reply-block-mime	service webproxy reply-block-mime <value>
×	<code>set service webproxy reply-block-mime <mime></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	service_webproxy.xml.in: service webproxy reply-body-max-size	service webproxy reply-body-max-size <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<i>set service webproxy reply-body-max-size <size></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	service_webproxy.xml.in: service webproxy safe-ports	service webproxy safe-ports <value>
×	<i>set service webproxy safe-ports <port></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	service_webproxy.xml.in: service webproxy ssl-safe-ports	service webproxy ssl-safe-ports <value>
×	<i>set service webproxy ssl-safe-ports <port></i>	Nothing found in XML Definitions	Nothing found in VyOS
✓	<i>set service webproxy url-filtering disable</i>	service_webproxy.xml.in: service webproxy url-filtering disable	service webproxy url-filtering disable
×	not yet documented	service_webproxy.xml.in: service webproxy url-filtering squidguard allow-category	service webproxy url-filtering squidguard allow-category <value>
×	not yet documented	service_webproxy.xml.in: service webproxy url-filtering squidguard allow-ipaddr-url	service webproxy url-filtering squidguard allow-ipaddr-url
×	not yet documented	service_webproxy.xml.in: service webproxy url-filtering squidguard auto-update update-hour	service webproxy url-filtering squidguard auto-update update-hour <value>
×	not yet documented	service_webproxy.xml.in: service webproxy url-filtering squidguard block-category	service webproxy url-filtering squidguard block-category <value>
×	not yet documented	service_webproxy.xml.in: service webproxy url-filtering squidguard default-action	service webproxy url-filtering squidguard default-action <value>
×	not yet documented	service_webproxy.xml.in: service webproxy url-filtering squidguard enable-safe-search	service webproxy url-filtering squidguard enable-safe-search
×	not yet documented	service_webproxy.xml.in: service webproxy url-filtering squidguard local-block	service webproxy url-filtering squidguard local-block <value>
×	not yet documented	service_webproxy.xml.in: service webproxy url-filtering squidguard local-block-keyword	service webproxy url-filtering squidguard local-block-keyword <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	service_webproxy.xml.in: service webproxy url- filtering squidguard local-block-url	service webproxy url- filtering squidguard local-block-url <value>
×	not yet documented	service_webproxy.xml.in: service webproxy url- filtering squidguard local-ok	service webproxy url- filtering squidguard local-ok <value>
×	not yet documented	service_webproxy.xml.in: service webproxy url- filtering squidguard local-ok-url	service webproxy url- filtering squidguard local-ok-url <value>
×	not yet documented	service_webproxy.xml.in: service webproxy url- filtering squidguard log	service webproxy url- filtering squidguard log <value>
×	not yet documented	service_webproxy.xml.in: service webproxy url- filtering squidguard redirect-url	service webproxy url- filtering squidguard redirect-url <value>
×	not yet documented	service_webproxy.xml.in: service webproxy url- filtering squidguard rule <rule> allow-category	service webproxy url- filtering squidguard rule <text> allow-category <value>
×	not yet documented	service_webproxy.xml.in: service webproxy url- filtering squidguard rule <rule> allow-ipaddr-url	service webproxy url- filtering squidguard rule <text> allow-ipaddr-url
×	not yet documented	service_webproxy.xml.in: service webproxy url- filtering squidguard rule <rule> block-category	service webproxy url- filtering squidguard rule <text> block-category <value>
×	not yet documented	service_webproxy.xml.in: service webproxy url- filtering squidguard rule <rule> default-action	service webproxy url- filtering squidguard rule <text> default-action <value>
×	not yet documented	service_webproxy.xml.in: service webproxy url- filtering squidguard rule <rule> enable-safe-search	service webproxy url- filtering squidguard rule <text> enable-safe-search
×	not yet documented	service_webproxy.xml.in: service webproxy url- filtering squidguard rule <rule> local-block	service webproxy url- filtering squidguard rule <text> local-block <value>
×	not yet documented	service_webproxy.xml.in: service webproxy url- filtering squidguard rule <rule> local-block- keyword	service webproxy url- filtering squidguard rule <text> local-block- keyword <value>

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Table 1 – continued from previous page

Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	service_webproxy.xml.in: service webproxy url- filtering squidguard rule <rule> local-block-url	service webproxy url- filtering squidguard rule <text> local-block-url <value>
×	not yet documented	service_webproxy.xml.in: service webproxy url- filtering squidguard rule <rule> local-ok	service webproxy url- filtering squidguard rule <text> local-ok <value>
×	not yet documented	service_webproxy.xml.in: service webproxy url- filtering squidguard rule <rule> local-ok-url	service webproxy url- filtering squidguard rule <text> local-ok-url <value>
×	not yet documented	service_webproxy.xml.in: service webproxy url- filtering squidguard rule <rule> log	service webproxy url- filtering squidguard rule <text> log <value>
×	not yet documented	service_webproxy.xml.in: service webproxy url- filtering squidguard rule <rule> redirect-url	service webproxy url- filtering squidguard rule <text> redirect-url <value>
×	not yet documented	service_webproxy.xml.in: service webproxy url- filtering squidguard rule <rule> source-group	service webproxy url- filtering squidguard rule <text> source-group <value>
×	not yet documented	service_webproxy.xml.in: service webproxy url- filtering squidguard rule <rule> time-period	service webproxy url- filtering squidguard rule <text> time-period <value>
×	not yet documented	service_webproxy.xml.in: service webproxy url- filtering squidguard source-group <source- group> address	service webproxy url- filtering squidguard source-group <text> address <value>
×	not yet documented	service_webproxy.xml.in: service webproxy url- filtering squidguard source-group <source- group> description	service webproxy url- filtering squidguard source-group <text> description <value>
×	not yet documented	service_webproxy.xml.in: service webproxy url- filtering squidguard source-group <source- group> domain	service webproxy url- filtering squidguard source-group <text> domain <value>
×	not yet documented	service_webproxy.xml.in: service webproxy url- filtering squidguard source-group <source- group> ldap-ip-search	service webproxy url- filtering squidguard source-group <text> ldap-ip-search <value>

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Table 1 – continued from previous page

Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	service_webproxy.xml.in: service webproxy url- filtering squidguard source-group <source- group> ldap-user-search	service webproxy url- filtering squidguard source-group <text> ldap-user-search <value>
×	not yet documented	service_webproxy.xml.in: service webproxy url- filtering squidguard source-group <source- group> user	service webproxy url- filtering squidguard source-group <text> user <value>
×	not yet documented	service_webproxy.xml.in: service webproxy url- filtering squidguard time- period <time-period> days <days> time	service webproxy url- filtering squidguard time-period <text> days <text> time <value>
×	not yet documented	service_webproxy.xml.in: service webproxy url- filtering squidguard time- period <time-period> description	service webproxy url- filtering squidguard time- period <text> description <value>
×	<i>show</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>sh system login authentication user <username> otp <full key-b32 qrcode uri></i>	Nothing found in XML Definitions	Nothing found in VyOS
✓	<i>set system acceleration qat</i>	sys- tem_acceleration.xml.in: system acceleration qat	system acceleration qat
×	not yet documented	system_config- management.xml.in: system config- management commit- archive location	system config- management commit- archive location <value>
×	<i>set system config-management commit-archive location <URI></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	system_config- management.xml.in: system config- management commit- archive source-address	system config- management commit- archive source-address <value>
×	<i>set system config-management commit-archive vrf <name></i>	Nothing found in XML Definitions	Nothing found in VyOS

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Table 1 – continued from previous page

Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	system_config-management.xml.in: system config-management commit-revisions	system config-management commit-revisions <value>
×	<i>set system config-management commit-revisions <N></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	system_conntrack.xml.in: system conntrack expect-table-size	system conntrack expect-table-size <value>
×	<i>set system conntrack expect-table-size <1-500000000></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	system_conntrack.xml.in: system conntrack flow-accounting	Nothing found in VyOS
×	not yet documented	system_conntrack.xml.in: system conntrack hash-size	system conntrack hash-size <value>
×	<i>set system conntrack hash-size <1-500000000></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	system_conntrack.xml.in: system conntrack ignore ipv4 rule <rule> description	Nothing found in VyOS
×	not yet documented	system_conntrack.xml.in: system conntrack ignore ipv4 rule <rule> destination address	Nothing found in VyOS
×	not yet documented	system_conntrack.xml.in: system conntrack ignore ipv4 rule <rule> destination group address-group	Nothing found in VyOS
×	not yet documented	system_conntrack.xml.in: system conntrack ignore ipv4 rule <rule> destination group domain-group	Nothing found in VyOS
×	not yet documented	system_conntrack.xml.in: system conntrack ignore ipv4 rule <rule> destination group network-group	Nothing found in VyOS
×	not yet documented	system_conntrack.xml.in: system conntrack ignore ipv4 rule <rule> destination group port-group	Nothing found in VyOS

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Table 1 – continued from previous page

Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	system_conntrack.xml.in: system conntrack ignore ipv4 rule <rule> destina- tion port	Nothing found in VyOS
×	not yet documented	system_conntrack.xml.in: system conntrack ignore ipv4 rule <rule> inbound- interface	Nothing found in VyOS
×	not yet documented	system_conntrack.xml.in: system conntrack ignore ipv4 rule <rule> protocol	Nothing found in VyOS
×	not yet documented	system_conntrack.xml.in: system conntrack ignore ipv4 rule <rule> source address	Nothing found in VyOS
×	not yet documented	system_conntrack.xml.in: system conntrack ignore ipv4 rule <rule> source group address-group	Nothing found in VyOS
×	not yet documented	system_conntrack.xml.in: system conntrack ignore ipv4 rule <rule> source group domain-group	Nothing found in VyOS
×	not yet documented	system_conntrack.xml.in: system conntrack ignore ipv4 rule <rule> source group network-group	Nothing found in VyOS
×	not yet documented	system_conntrack.xml.in: system conntrack ignore ipv4 rule <rule> source group port-group	Nothing found in VyOS
×	not yet documented	system_conntrack.xml.in: system conntrack ignore ipv4 rule <rule> source port	Nothing found in VyOS
×	not yet documented	system_conntrack.xml.in: system conntrack ignore ipv4 rule <rule> tcp flags ack	Nothing found in VyOS
×	not yet documented	system_conntrack.xml.in: system conntrack ignore ipv4 rule <rule> tcp flags cwr	Nothing found in VyOS
×	not yet documented	system_conntrack.xml.in: system conntrack ignore ipv4 rule <rule> tcp flags ecn	Nothing found in VyOS

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Table 1 – continued from previous page

Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	system_conntrack.xml.in: system conntrack ignore ipv4 rule <rule> tcp flags fin	Nothing found in VyOS
×	not yet documented	system_conntrack.xml.in: system conntrack ignore ipv4 rule <rule> tcp flags not ack	Nothing found in VyOS
×	not yet documented	system_conntrack.xml.in: system conntrack ignore ipv4 rule <rule> tcp flags not cwr	Nothing found in VyOS
×	not yet documented	system_conntrack.xml.in: system conntrack ignore ipv4 rule <rule> tcp flags not ecn	Nothing found in VyOS
×	not yet documented	system_conntrack.xml.in: system conntrack ignore ipv4 rule <rule> tcp flags not fin	Nothing found in VyOS
×	not yet documented	system_conntrack.xml.in: system conntrack ignore ipv4 rule <rule> tcp flags not psh	Nothing found in VyOS
×	not yet documented	system_conntrack.xml.in: system conntrack ignore ipv4 rule <rule> tcp flags not rst	Nothing found in VyOS
×	not yet documented	system_conntrack.xml.in: system conntrack ignore ipv4 rule <rule> tcp flags not syn	Nothing found in VyOS
×	not yet documented	system_conntrack.xml.in: system conntrack ignore ipv4 rule <rule> tcp flags not urg	Nothing found in VyOS
×	not yet documented	system_conntrack.xml.in: system conntrack ignore ipv4 rule <rule> tcp flags psh	Nothing found in VyOS
×	not yet documented	system_conntrack.xml.in: system conntrack ignore ipv4 rule <rule> tcp flags rst	Nothing found in VyOS
×	not yet documented	system_conntrack.xml.in: system conntrack ignore ipv4 rule <rule> tcp flags syn	Nothing found in VyOS

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Table 1 – continued from previous page

Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	system_conntrack.xml.in: system conntrack ignore ipv4 rule <rule> tcp flags urg	Nothing found in VyOS
×	not yet documented	system_conntrack.xml.in: system conntrack ignore ipv6 rule <rule> descrip- tion	Nothing found in VyOS
×	not yet documented	system_conntrack.xml.in: system conntrack ignore ipv6 rule <rule> destina- tion address	Nothing found in VyOS
×	not yet documented	system_conntrack.xml.in: system conntrack ignore ipv6 rule <rule> destina- tion group address-group	Nothing found in VyOS
×	not yet documented	system_conntrack.xml.in: system conntrack ignore ipv6 rule <rule> destina- tion group domain-group	Nothing found in VyOS
×	not yet documented	system_conntrack.xml.in: system conntrack ignore ipv6 rule <rule> destina- tion group mac-group	Nothing found in VyOS
×	not yet documented	system_conntrack.xml.in: system conntrack ignore ipv6 rule <rule> destina- tion group network-group	Nothing found in VyOS
×	not yet documented	system_conntrack.xml.in: system conntrack ignore ipv6 rule <rule> destina- tion group port-group	Nothing found in VyOS
×	not yet documented	system_conntrack.xml.in: system conntrack ignore ipv6 rule <rule> destina- tion port	Nothing found in VyOS
×	not yet documented	system_conntrack.xml.in: system conntrack ignore ipv6 rule <rule> inbound- interface	Nothing found in VyOS
×	not yet documented	system_conntrack.xml.in: system conntrack ignore ipv6 rule <rule> protocol	Nothing found in VyOS
×	not yet documented	system_conntrack.xml.in: system conntrack ignore ipv6 rule <rule> source address	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	system_conntrack.xml.in: system conntrack ignore ipv6 rule <rule> source group address-group	Nothing found in VyOS
×	not yet documented	system_conntrack.xml.in: system conntrack ignore ipv6 rule <rule> source group domain-group	Nothing found in VyOS
×	not yet documented	system_conntrack.xml.in: system conntrack ignore ipv6 rule <rule> source group mac-group	Nothing found in VyOS
×	not yet documented	system_conntrack.xml.in: system conntrack ignore ipv6 rule <rule> source group network-group	Nothing found in VyOS
×	not yet documented	system_conntrack.xml.in: system conntrack ignore ipv6 rule <rule> source group port-group	Nothing found in VyOS
×	not yet documented	system_conntrack.xml.in: system conntrack ignore ipv6 rule <rule> source port	Nothing found in VyOS
×	not yet documented	system_conntrack.xml.in: system conntrack ignore ipv6 rule <rule> tcp flags ack	Nothing found in VyOS
×	not yet documented	system_conntrack.xml.in: system conntrack ignore ipv6 rule <rule> tcp flags cwr	Nothing found in VyOS
×	not yet documented	system_conntrack.xml.in: system conntrack ignore ipv6 rule <rule> tcp flags ecn	Nothing found in VyOS
×	not yet documented	system_conntrack.xml.in: system conntrack ignore ipv6 rule <rule> tcp flags fin	Nothing found in VyOS
×	not yet documented	system_conntrack.xml.in: system conntrack ignore ipv6 rule <rule> tcp flags not ack	Nothing found in VyOS
×	not yet documented	system_conntrack.xml.in: system conntrack ignore ipv6 rule <rule> tcp flags not cwr	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	system_conntrack.xml.in: system conntrack ignore ipv6 rule <rule> tcp flags not ecn	Nothing found in VyOS
×	not yet documented	system_conntrack.xml.in: system conntrack ignore ipv6 rule <rule> tcp flags not fin	Nothing found in VyOS
×	not yet documented	system_conntrack.xml.in: system conntrack ignore ipv6 rule <rule> tcp flags not psh	Nothing found in VyOS
×	not yet documented	system_conntrack.xml.in: system conntrack ignore ipv6 rule <rule> tcp flags not rst	Nothing found in VyOS
×	not yet documented	system_conntrack.xml.in: system conntrack ignore ipv6 rule <rule> tcp flags not syn	Nothing found in VyOS
×	not yet documented	system_conntrack.xml.in: system conntrack ignore ipv6 rule <rule> tcp flags not urg	Nothing found in VyOS
×	not yet documented	system_conntrack.xml.in: system conntrack ignore ipv6 rule <rule> tcp flags psh	Nothing found in VyOS
×	not yet documented	system_conntrack.xml.in: system conntrack ignore ipv6 rule <rule> tcp flags rst	Nothing found in VyOS
×	not yet documented	system_conntrack.xml.in: system conntrack ignore ipv6 rule <rule> tcp flags syn	Nothing found in VyOS
×	not yet documented	system_conntrack.xml.in: system conntrack ignore ipv6 rule <rule> tcp flags urg	Nothing found in VyOS
×	<i>set system conntrack ignore [ipv4 ipv6] rule <1-999999> description <text></i>	Nothing found in XML Definitions	system conntrack ignore rule <text> description <value>
×	<i>set system conntrack ignore [ipv4 ipv6] rule <1-999999> destination address <ip-address></i>	Nothing found in XML Definitions	system conntrack ignore rule <text> destination ad- dress <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set system conntrack ignore [ipv4 ipv6] rule <1-999999> destination port <port></code>	Nothing found in XML Definitions	system conntrack ignore rule <text> destination port <value>
×	<code>set system conntrack ignore [ipv4 ipv6] rule <1-999999> inbound-interface <interface></code>	Nothing found in XML Definitions	system conntrack ignore rule <text> inbound-interface <value>
×	<code>set system conntrack ignore [ipv4 ipv6] rule <1-999999> protocol <protocol></code>	Nothing found in XML Definitions	system conntrack ignore rule <text> protocol <value>
×	<code>set system conntrack ignore [ipv4 ipv6] rule <1-999999> source address <ip-address></code>	Nothing found in XML Definitions	system conntrack ignore rule <text> source address <value>
×	<code>set system conntrack ignore [ipv4 ipv6] rule <1-999999> source port <port></code>	Nothing found in XML Definitions	system conntrack ignore rule <text> source port <value>
×	<code>set system conntrack ignore [ipv4 ipv6] rule <1-999999> tcp flags [not] <text></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set system conntrack log event destroy</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set system conntrack log event destroy icmp</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set system conntrack log event destroy other</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set system conntrack log event destroy tcp</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set system conntrack log event destroy udp</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set system conntrack log event new</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set system conntrack log event new icmp</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set system conntrack log event new other</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set system conntrack log event new tcp</code>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<i>set system conntrack log event new udp</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set system conntrack log event update</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set system conntrack log event update icmp</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set system conntrack log event update other</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set system conntrack log event update tcp</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set system conntrack log event update udp</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	system_conntrack.xml.in: system_conntrack_log_icmp_destroy	system_conntrack_log_icmp_destroy
×	not yet documented	system_conntrack.xml.in: system_conntrack_log_icmp_new	system_conntrack_log_icmp_new
×	not yet documented	system_conntrack.xml.in: system_conntrack_log_icmp_update	system_conntrack_log_icmp_update
×	<i>set system conntrack log log-level <info / debug></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	system_conntrack.xml.in: system_conntrack_log_other_destroy	system_conntrack_log_other_destroy
×	not yet documented	system_conntrack.xml.in: system_conntrack_log_other_new	system_conntrack_log_other_new
×	not yet documented	system_conntrack.xml.in: system_conntrack_log_other_update	system_conntrack_log_other_update
×	<i>set system conntrack log queue-size <100-999999></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	system_conntrack.xml.in: system_conntrack_log_tcp_destroy	system_conntrack_log_tcp_destroy
×	not yet documented	system_conntrack.xml.in: system_conntrack_log_tcp_new	system_conntrack_log_tcp_new
×	not yet documented	system_conntrack.xml.in: system_conntrack_log_tcp_update	system_conntrack_log_tcp_update

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<i>set system conntrack log timestamp</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	system_conntrack.xml.in: system conntrack log udp destroy	system conntrack log udp destroy
×	not yet documented	system_conntrack.xml.in: system conntrack log udp new	system conntrack log udp new
×	not yet documented	system_conntrack.xml.in: system conntrack log udp update	system conntrack log udp update
✓	<i>set system conntrack modules ftp</i>	system_conntrack.xml.in: system conntrack modules ftp	system conntrack modules ftp
✓	<i>set system conntrack modules h323</i>	system_conntrack.xml.in: system conntrack modules h323	system conntrack modules h323
✓	<i>set system conntrack modules nfs</i>	system_conntrack.xml.in: system conntrack modules nfs	system conntrack modules nfs
✓	<i>set system conntrack modules pptp</i>	system_conntrack.xml.in: system conntrack modules pptp	system conntrack modules pptp
✓	<i>set system conntrack modules sip</i>	system_conntrack.xml.in: system conntrack modules sip	system conntrack modules sip
✓	<i>set system conntrack modules sqlnet</i>	system_conntrack.xml.in: system conntrack modules sqlnet	system conntrack modules sqlnet
✓	<i>set system conntrack modules tftp</i>	system_conntrack.xml.in: system conntrack modules tftp	system conntrack modules tftp
×	not yet documented	system_conntrack.xml.in: system conntrack table- size	system conntrack table- size <value>
×	<i>set system conntrack table-size <1-500000000></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	system_conntrack.xml.in: system conntrack tcp half- open-connections	system conntrack tcp half- open-connections <value>
×	<i>set system conntrack tcp half-open-connections <1-21474836></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	system_conntrack.xml.in: system conntrack tcp loose	system conntrack tcp loose <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<i>set system conntrack tcp loose <enable / disable></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	system_conntrack.xml.in: system conntrack tcp max-retrans	system conntrack tcp max-retrans <value>
×	<i>set system conntrack tcp max-retrans <1-2147483647></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	system_conntrack.xml.in: system conntrack timeout custom ipv4 rule <rule> description	Nothing found in VyOS
×	not yet documented	system_conntrack.xml.in: system conntrack timeout custom ipv4 rule <rule> destination address	Nothing found in VyOS
×	not yet documented	system_conntrack.xml.in: system conntrack timeout custom ipv4 rule <rule> destination port	Nothing found in VyOS
×	not yet documented	system_conntrack.xml.in: system conntrack timeout custom ipv4 rule <rule> inbound-interface	Nothing found in VyOS
×	not yet documented	system_conntrack.xml.in: system conntrack timeout custom ipv4 rule <rule> protocol tcp close	Nothing found in VyOS
×	not yet documented	system_conntrack.xml.in: system conntrack timeout custom ipv4 rule <rule> protocol tcp close-wait	Nothing found in VyOS
×	not yet documented	system_conntrack.xml.in: system conntrack timeout custom ipv4 rule <rule> protocol tcp established	Nothing found in VyOS
×	not yet documented	system_conntrack.xml.in: system conntrack timeout custom ipv4 rule <rule> protocol tcp fin-wait	Nothing found in VyOS
×	not yet documented	system_conntrack.xml.in: system conntrack timeout custom ipv4 rule <rule> protocol tcp last-ack	Nothing found in VyOS
×	not yet documented	system_conntrack.xml.in: system conntrack timeout custom ipv4 rule <rule> protocol tcp syn-recv	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	system_conntrack.xml.in: system conntrack timeout custom ipv4 rule <rule> protocol tcp syn-sent	Nothing found in VyOS
×	not yet documented	system_conntrack.xml.in: system conntrack timeout custom ipv4 rule <rule> protocol tcp time-wait	Nothing found in VyOS
×	not yet documented	system_conntrack.xml.in: system conntrack timeout custom ipv4 rule <rule> protocol udp replied	Nothing found in VyOS
×	not yet documented	system_conntrack.xml.in: system conntrack timeout custom ipv4 rule <rule> protocol udp unreplied	Nothing found in VyOS
×	not yet documented	system_conntrack.xml.in: system conntrack timeout custom ipv4 rule <rule> source address	Nothing found in VyOS
×	not yet documented	system_conntrack.xml.in: system conntrack timeout custom ipv4 rule <rule> source port	Nothing found in VyOS
×	not yet documented	system_conntrack.xml.in: system conntrack timeout custom ipv6 rule <rule> description	Nothing found in VyOS
×	not yet documented	system_conntrack.xml.in: system conntrack timeout custom ipv6 rule <rule> destination address	Nothing found in VyOS
×	not yet documented	system_conntrack.xml.in: system conntrack timeout custom ipv6 rule <rule> destination port	Nothing found in VyOS
×	not yet documented	system_conntrack.xml.in: system conntrack timeout custom ipv6 rule <rule> inbound-interface	Nothing found in VyOS
×	not yet documented	system_conntrack.xml.in: system conntrack timeout custom ipv6 rule <rule> protocol tcp close	Nothing found in VyOS
×	not yet documented	system_conntrack.xml.in: system conntrack timeout custom ipv6 rule <rule> protocol tcp close-wait	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	system_conntrack.xml.in: system conntrack timeout custom ipv6 rule <rule> protocol tcp established	Nothing found in VyOS
×	not yet documented	system_conntrack.xml.in: system conntrack timeout custom ipv6 rule <rule> protocol tcp fin-wait	Nothing found in VyOS
×	not yet documented	system_conntrack.xml.in: system conntrack timeout custom ipv6 rule <rule> protocol tcp last-ack	Nothing found in VyOS
×	not yet documented	system_conntrack.xml.in: system conntrack timeout custom ipv6 rule <rule> protocol tcp syn-recv	Nothing found in VyOS
×	not yet documented	system_conntrack.xml.in: system conntrack timeout custom ipv6 rule <rule> protocol tcp syn-sent	Nothing found in VyOS
×	not yet documented	system_conntrack.xml.in: system conntrack timeout custom ipv6 rule <rule> protocol tcp time-wait	Nothing found in VyOS
×	not yet documented	system_conntrack.xml.in: system conntrack timeout custom ipv6 rule <rule> protocol udp replied	Nothing found in VyOS
×	not yet documented	system_conntrack.xml.in: system conntrack timeout custom ipv6 rule <rule> protocol udp unreplied	Nothing found in VyOS
×	not yet documented	system_conntrack.xml.in: system conntrack timeout custom ipv6 rule <rule> source address	Nothing found in VyOS
×	not yet documented	system_conntrack.xml.in: system conntrack timeout custom ipv6 rule <rule> source port	Nothing found in VyOS
×	<i>set system conntrack timeout custom [ipv4 ipv6] rule <1-999999> description <test></i>	Nothing found in XML Definitions	system conntrack timeout custom rule <text> de- scription <value>

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Table 1 – continued from previous page

Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set system conntrack timeout custom [ipv4 ipv6] rule <1-999999> destination address <ip-address></code>	Nothing found in XML Definitions	system conntrack timeout custom rule <text> destination address <value>
×	<code>set system conntrack timeout custom [ipv4 ipv6] rule <1-999999> destination port <value></code>	Nothing found in XML Definitions	system conntrack timeout custom rule <text> destination port <value>
×	not yet documented	Nothing found in XML Definitions	system conntrack timeout custom rule <text> inbound-interface <value>
×	not yet documented	Nothing found in XML Definitions	system conntrack timeout custom rule <text> protocol icmp <value>
×	not yet documented	Nothing found in XML Definitions	system conntrack timeout custom rule <text> protocol other <value>
×	<code>set system conntrack timeout custom [ipv4 ipv6] rule <1-999999> protocol tcp close <1-21474836></code>	Nothing found in XML Definitions	system conntrack timeout custom rule <text> protocol tcp close <value>
×	<code>set system conntrack timeout custom [ipv4 ipv6] rule <1-999999> protocol tcp close-wait <1-21474836></code>	Nothing found in XML Definitions	system conntrack timeout custom rule <text> protocol tcp close-wait <value>
×	<code>set system conntrack timeout custom [ipv4 ipv6] rule <1-999999> protocol tcp established <1-21474836></code>	Nothing found in XML Definitions	system conntrack timeout custom rule <text> protocol tcp established <value>
×	<code>set system conntrack timeout custom [ipv4 ipv6] rule <1-999999> protocol tcp fin-wait <1-21474836></code>	Nothing found in XML Definitions	system conntrack timeout custom rule <text> protocol tcp fin-wait <value>

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Table 1 – continued from previous page

Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set system conntrack timeout custom [ipv4 ipv6] rule <1-999999> protocol tcp last-ack <1-21474836></code>	Nothing found in XML Definitions	system conntrack timeout custom rule <text> protocol tcp last-ack <value>
×	<code>set system conntrack timeout custom [ipv4 ipv6] rule <1-999999> protocol tcp syn-recv <1-21474836></code>	Nothing found in XML Definitions	system conntrack timeout custom rule <text> protocol tcp syn-recv <value>
×	<code>set system conntrack timeout custom [ipv4 ipv6] rule <1-999999> protocol tcp syn-sent <1-21474836></code>	Nothing found in XML Definitions	system conntrack timeout custom rule <text> protocol tcp syn-sent <value>
×	<code>set system conntrack timeout custom [ipv4 ipv6] rule <1-999999> protocol tcp time-wait <1-21474836></code>	Nothing found in XML Definitions	system conntrack timeout custom rule <text> protocol tcp time-wait <value>
×	not yet documented	Nothing found in XML Definitions	system conntrack timeout custom rule <text> protocol udp other <value>
×	<code>set system conntrack timeout custom [ipv4 ipv6] rule <1-999999> protocol udp replied <1-21474836></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	Nothing found in XML Definitions	system conntrack timeout custom rule <text> protocol udp stream <value>
×	<code>set system conntrack timeout custom [ipv4 ipv6] rule <1-999999> protocol udp unreplied <1-21474836></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set system conntrack timeout custom [ipv4 ipv6] rule <1-999999> source address <ip-address></code>	Nothing found in XML Definitions	system conntrack timeout custom rule <text> source address <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<i>set system conntrack timeout custom [ipv4 ipv6] rule <1-999999> source port <value></i>	Nothing found in XML Definitions	system conntrack timeout custom rule <text> source port <value>
×	not yet documented	system_conntrack.xml.in: system conntrack timeout icmp	system conntrack timeout icmp <value>
×	not yet documented	system_conntrack.xml.in: system conntrack timeout other	system conntrack timeout other <value>
×	not yet documented	system_conntrack.xml.in: system conntrack timeout tcp close	system conntrack timeout tcp close <value>
×	not yet documented	system_conntrack.xml.in: system conntrack timeout tcp close-wait	system conntrack timeout tcp close-wait <value>
×	not yet documented	system_conntrack.xml.in: system conntrack timeout tcp established	system conntrack timeout tcp established <value>
×	not yet documented	system_conntrack.xml.in: system conntrack timeout tcp fin-wait	system conntrack timeout tcp fin-wait <value>
×	not yet documented	system_conntrack.xml.in: system conntrack timeout tcp last-ack	system conntrack timeout tcp last-ack <value>
×	not yet documented	system_conntrack.xml.in: system conntrack timeout tcp syn-recv	system conntrack timeout tcp syn-recv <value>
×	not yet documented	system_conntrack.xml.in: system conntrack timeout tcp syn-sent	system conntrack timeout tcp syn-sent <value>
×	not yet documented	system_conntrack.xml.in: system conntrack timeout tcp time-wait	system conntrack timeout tcp time-wait <value>
×	not yet documented	system_conntrack.xml.in: system conntrack timeout udp other	system conntrack timeout udp other <value>
×	not yet documented	system_conntrack.xml.in: system conntrack timeout udp stream	system conntrack timeout udp stream <value>
×	<i>set system console device <device></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set system console device <device> kernel</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set system console device <device> speed <speed></i>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	system_console.xml.in: system console device <device> speed	system console device <text> speed <value>
×	not yet documented	system_console.xml.in: system console powersave	system console powersave
×	not yet documented	system_domain- name.xml.in: system domain-name	system domain-name <value>
×	<i>set system domain-name <domain></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	system_domain- search.xml.in: system domain-search	Nothing found in VyOS
×	<i>set system domain-search <domain></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	Nothing found in XML Definitions	system domain-search do- main <value>
×	not yet documented	system_flow- accounting.xml.in: system flow-accounting buffer-size	system flow-accounting buffer-size <value>
×	<i>set system flow-accounting buffer-size <buffer size></i>	Nothing found in XML Definitions	Nothing found in VyOS
✓	<i>set system flow-accounting disable-imt</i>	system_flow- accounting.xml.in: system flow-accounting disable-imt	system flow-accounting disable-imt
✓	<i>set system flow-accounting enable-egress</i>	system_flow- accounting.xml.in: system flow-accounting enable-egress	system flow-accounting enable-egress
×	not yet documented	system_flow- accounting.xml.in: system flow-accounting interface	system flow-accounting interface <value>
×	<i>set system flow-accounting interface <interface></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	system_flow- accounting.xml.in: system flow-accounting netflow engine-id	system flow-accounting netflow engine-id <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<i>set system flow-accounting netflow engine-id <id></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	system_flow-accounting.xml.in: system flow-accounting netflow max-flows	system flow-accounting netflow max-flows <value>
×	<i>set system flow-accounting netflow max-flows <n></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	system_flow-accounting.xml.in: system flow-accounting netflow sampling-rate	system flow-accounting netflow sampling-rate <value>
×	<i>set system flow-accounting netflow sampling-rate <rate></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set system flow-accounting netflow server <address></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	system_flow-accounting.xml.in: system flow-accounting netflow server <server> port	system flow-accounting netflow server <text> port <value>
×	not yet documented	system_flow-accounting.xml.in: system flow-accounting netflow source-address	system flow-accounting netflow source-address <value>
×	<i>set system flow-accounting netflow source-ip <address></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	system_flow-accounting.xml.in: system flow-accounting netflow timeout expiry-interval	system flow-accounting netflow timeout expiry-interval <value>
×	<i>set system flow-accounting netflow timeout expiry-interval <interval></i>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	system_flow-accounting.xml.in: system flow-accounting netflow timeout flow-generic	system flow-accounting netflow timeout flow-generic <value>
×	not yet documented	system_flow-accounting.xml.in: system flow-accounting netflow timeout icmp	system flow-accounting netflow timeout icmp <value>
×	not yet documented	system_flow-accounting.xml.in: system flow-accounting netflow timeout max-active-life	system flow-accounting netflow timeout max-active-life <value>
×	not yet documented	system_flow-accounting.xml.in: system flow-accounting netflow timeout tcp-fin	system flow-accounting netflow timeout tcp-fin <value>
×	not yet documented	system_flow-accounting.xml.in: system flow-accounting netflow timeout tcp-generic	system flow-accounting netflow timeout tcp-generic <value>
×	not yet documented	system_flow-accounting.xml.in: system flow-accounting netflow timeout tcp-rst	system flow-accounting netflow timeout tcp-rst <value>
×	not yet documented	system_flow-accounting.xml.in: system flow-accounting netflow timeout udp	system flow-accounting netflow timeout udp <value>
×	not yet documented	system_flow-accounting.xml.in: system flow-accounting netflow version	system flow-accounting netflow version <value>
×	<i>set system flow-accounting netflow version <version></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	system_flow-accounting.xml.in: system flow-accounting packet-length	system flow-accounting packet-length <value>
×	not yet documented	system_flow-accounting.xml.in: system flow-accounting sflow agent-address	system flow-accounting sflow agent-address <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	system_flow-accounting.xml.in: system flow-accounting sflow sampling-rate <value>	system flow-accounting sflow sampling-rate <value>
×	not yet documented	system_flow-accounting.xml.in: system flow-accounting sflow server <server> port <value>	system flow-accounting sflow server <text> port <value>
×	not yet documented	system_flow-accounting.xml.in: system flow-accounting sflow source-address <value>	system flow-accounting sflow source-address <value>
×	not yet documented	system_flow-accounting.xml.in: system flow-accounting syslog-facility <value>	system flow-accounting syslog-facility <value>
×	<i>set system flow-accounting syslog-facility <facility></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	system_flow-accounting.xml.in: system flow-accounting vrf <value>	system flow-accounting vrf <value>
✓	<i>set system frr bmp</i>	system_frr.xml.in: system frr bmp	system frr bmp
×	not yet documented	system_frr.xml.in: system frr descriptors	Nothing found in VyOS
×	<i>set system frr descriptors <number></i>	Nothing found in XML Definitions	Nothing found in VyOS
✓	<i>set system frr irdp</i>	system_frr.xml.in: system frr irdp	system frr irdp
×	<i>set system frr profile <traditional / datacenter></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set system frr snmp <daemon></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	system_frr.xml.in: system frr snmp bgpd	system frr snmp bgpd
×	not yet documented	system_frr.xml.in: system frr snmp isisd	system frr snmp isisd
×	not yet documented	system_frr.xml.in: system frr snmp ldpd	system frr snmp ldpd
×	not yet documented	system_frr.xml.in: system frr snmp ospf6d	system frr snmp ospf6d
×	not yet documented	system_frr.xml.in: system frr snmp ospfd	system frr snmp ospfd

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	system_frr.xml.in: system frr snmp ripd	system frr snmp ripd
×	not yet documented	system_frr.xml.in: system frr snmp zebra	system frr snmp zebra
×	not yet documented	system_host-name.xml.in: system host-name	system host-name <value>
×	<i>set system host-name <hostname></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	system_ip.xml.in: system ip arp table-size	system ip arp table-size <value>
×	<i>set system ip arp table-size <number></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set system ip disable-directed-broa</i>	Nothing found in XML Definitions	system ip disable-directed-broadcast
✓	<i>set system ip disable-forwarding</i>	system_ip.xml.in: system ip disable-forwarding	system ip disable-forwarding
×	<i>set system ip import-table <table-id></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set system ip import-table <table-id> distance <distance></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set system ip import-table <table-id> route-map <route-map></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	system_ip.xml.in: system ip multipath ignore-unreachable-nexthops	system ip multipath ignore-unreachable-nexthops
✓	<i>set system ip multipath layer4-hashing</i>	system_ip.xml.in: system ip multipath layer4-hashing	system ip multipath layer4-hashing
×	<i>set system ip nht no-resolve-via-default</i>	system_ip.xml.in: system ip nht no-resolve-via-default	Nothing found in VyOS
×	<i>set system ip protocol <protocol> route-map <route-map></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	system_ip.xml.in: system ip protocol <protocol> route-map	system ip protocol <text> route-map <value>
×	not yet documented	system_ip.xml.in: system ip tcp mss base	system ip tcp mss base <value>
×	not yet documented	system_ip.xml.in: system ip tcp mss floor	system ip tcp mss floor <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	system_ip.xml.in: system ip tcp mss probing	system ip tcp mss probing <value>
✓	<i>set system ipv6 disable-forwarding</i>	system_ipv6.xml.in: system ipv6 disable- forwarding	system ipv6 disable- forwarding
✓	<i>set system ipv6 multipath layer4-hashing</i>	system_ipv6.xml.in: system ipv6 multipath layer4-hashing	system ipv6 multipath layer4-hashing
×	not yet documented	system_ipv6.xml.in: system ipv6 neighbor table-size	system ipv6 neighbor table-size <value>
×	<i>set system ipv6 neighbor table-size <number></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set system ipv6 nht no-resolve-via-default</i>	system_ipv6.xml.in: system ipv6 nht no- resolve-via-default	Nothing found in VyOS
×	<i>set system ipv6 protocol <protocol> route-map <route-map></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	system_ipv6.xml.in: system ipv6 protocol <protocol> route-map	system ipv6 protocol <text> route-map <value>
✓	<i>set system ipv6 strict-dad</i>	system_ipv6.xml.in: system ipv6 strict-dad	system ipv6 strict-dad
×	not yet documented	system_lcd.xml.in: system lcd device	system lcd device <value>
×	<i>set system lcd device <device></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	system_lcd.xml.in: system lcd model	system lcd model <value>
×	<i>set system lcd model <model></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	sys- tem_login_banner.xml.in: system login banner post- login	system login banner post- login <value>
×	<i>set system login banner post-login <message></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	sys- tem_login_banner.xml.in: system login banner pre- login	system login banner pre- login <value>
×	<i>set system login banner pre-login <message></i>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	system_login.xml.in: system login max-login-session	system login max-login-session <value>
×	<i>set system login max-login-session <number></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	system_login.xml.in: system login radius security-mode	Nothing found in VyOS
×	<i>set system login radius server <address> disable</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set system login radius server <address> key <secret></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set system login radius server <address> port <port></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set system login radius server <address> timeout <timeout></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	system_login.xml.in: system login radius server <server> disable	system login radius server <text> disable
×	not yet documented	system_login.xml.in: system login radius server <server> key	system login radius server <text> key <value>
×	not yet documented	system_login.xml.in: system login radius server <server> port	system login radius server <text> port <value>
×	not yet documented	system_login.xml.in: system login radius server <server> priority	system login radius server <text> priority <value>
×	not yet documented	system_login.xml.in: system login radius server <server> timeout	system login radius server <text> timeout <value>
×	not yet documented	system_login.xml.in: system login radius source-address	system login radius source-address <value>
×	<i>set system login radius source-address <address></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	system_login.xml.in: system login radius vrf	system login radius vrf <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set system login radius vrf <name></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	system_login.xml.in: system login tacacs security-mode	Nothing found in VyOS
×	<code>set system login tacacs server <address> disable</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set system login tacacs server <address> key <secret></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set system login tacacs server <address> port <port></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set system login tacacs server <address> timeout <timeout></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	system_login.xml.in: system login tacacs server <server> disable	system login tacacs server <text> disable
×	not yet documented	system_login.xml.in: system login tacacs server <server> key	system login tacacs server <text> key <value>
×	not yet documented	system_login.xml.in: system login tacacs server <server> port	system login tacacs server <text> port <value>
×	not yet documented	system_login.xml.in: system login tacacs source-address	system login tacacs source-address <value>
×	<code>set system login tacacs source-address <address></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	system_login.xml.in: system login tacacs timeout	system login tacacs timeout <value>
×	not yet documented	system_login.xml.in: system login tacacs vrf	system login tacacs vrf <value>
×	<code>set system login tacacs vrf <name></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	system_login.xml.in: system login timeout	system login timeout <value>
×	<code>set system login timeout <timeout></code>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set system login user <name> authentication encrypted-password <password></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set system login user <username> authentication otp key <key></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set system login user <username> authentication otp rate-limit <limit></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set system login user <username> authentication otp rate-time <seconds></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set system login user <username> authentication otp window-size <size></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set system login user <name> authentication plaintext-password <password></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set system login user <name> authentication principal <principal></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set system login user <username> authentication public-keys <identifier> key <key></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set system login user <username> authentication public-keys <identifier> options <options></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set system login user <username> authentication public-keys <identifier> type <type></code>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set system login user <name> disable</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set system login user <name> full-name "<string>"</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	system_login.xml.in: system login user <user> authentication encrypted-password	system login user <text> authentication encrypted-password <value>
×	not yet documented	system_login.xml.in: system login user <user> authentication otp key	system login user <text> authentication otp key <value>
×	not yet documented	system_login.xml.in: system login user <user> authentication otp rate-limit	system login user <text> authentication otp rate-limit <value>
×	not yet documented	system_login.xml.in: system login user <user> authentication otp rate-time	system login user <text> authentication otp rate-time <value>
×	not yet documented	system_login.xml.in: system login user <user> authentication otp window-size	system login user <text> authentication otp window-size <value>
×	not yet documented	system_login.xml.in: system login user <user> authentication plaintext-password	system login user <text> authentication plaintext-password <value>
×	not yet documented	system_login.xml.in: system login user <user> authentication public-keys <public-keys> key	system login user <text> authentication public-keys <text> key <value>
×	not yet documented	system_login.xml.in: system login user <user> authentication public-keys <public-keys> options	system login user <text> authentication public-keys <text> options <value>
×	not yet documented	system_login.xml.in: system login user <user> authentication public-keys <public-keys> type	system login user <text> authentication public-keys <text> type <value>
×	not yet documented	system_login.xml.in: system login user <user> disable	Nothing found in VyOS
×	not yet documented	system_login.xml.in: system login user <user> full-name	system login user <text> full-name <value>
×	not yet documented	system_login.xml.in: system login user <user> home-directory	system login user <text> home-directory <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	system_logs.xml.in: system logs logrotate atop max-size	system logs logrotate atop max-size <value>
×	not yet documented	system_logs.xml.in: system logs logrotate atop rotate	system logs logrotate atop rotate <value>
×	not yet documented	system_logs.xml.in: system logs logrotate mes- sages max-size	system logs logrotate mes- sages max-size <value>
×	not yet documented	system_logs.xml.in: system logs logrotate mes- sages rotate	system logs logrotate mes- sages rotate <value>
×	not yet documented	system_name- server.xml.in: system name-server	system name-server <value>
×	set system name-server <address>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	system_option.xml.in: system option ctrl-alt- delete	system option ctrl-alt- delete <value>
×	set system option ctrl-alt-delete <ignore reboot poweroff>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	system_option.xml.in: system option http-client source-address	system option http-client source-address <value>
×	set system option http-client source-address <address>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	system_option.xml.in: system option http-client source-interface	system option http-client source-interface <value>
×	set system option http-client source-interface <interface>	Nothing found in XML Definitions	Nothing found in VyOS
×	set system option kernel amd-pstate-driver <mode>	Nothing found in XML Definitions	Nothing found in VyOS
×	set system option kernel cpu disable-nmi-watchdog	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set system option kernel cpu isolate-cpus <cpu-range></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set system option kernel cpu nohz-full <cpu-range></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set system option kernel cpu rcu-no-cbs <cpu-range></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set system option kernel disable-hpet</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set system option kernel disable-mce</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set system option kernel disable-mitigations</code>	system_option.xml.in: system option kernel disable-mitigations	Nothing found in VyOS
×	<code>set system option kernel disable-power-saving</code>	system_option.xml.in: system option kernel disable-power-saving	Nothing found in VyOS
×	<code>set system option kernel disable-softlockup</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set system option kernel memory hugepage-size <size> hugepage-count '<count>'</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set system option kernel quiet</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	system_option.xml.in: system option keyboard-layout	system option keyboard-layout <value>
×	<code>set system option keyboard-layout <us fr de fi no dk></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	system_option.xml.in: system option performance	system option performance <value>
×	<code>set system option performance <throughput latency></code>	Nothing found in XML Definitions	Nothing found in VyOS
✓	<code>set system option reboot-on-panic</code>	system_option.xml.in: system option reboot-on-panic	system option reboot-on-panic

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Table 1 – continued from previous page

Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set system option reboot-on-upgrade-fail [timeout <min>]</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set system option reboot-on-upgrade-fail <timeout></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set system option resource-limits max-map-count <value></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set system option resource-limits shmmax <value></code>	Nothing found in XML Definitions	Nothing found in VyOS
✓	<code>set system option root-partition-auto-resize</code>	system_option.xml.in: system option root-partition-auto-resize	system option root-partition-auto-resize
×	not yet documented	system_option.xml.in: system option ssh-client source-address	system option ssh-client source-address <value>
×	<code>set system option ssh-client source-address <address></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	system_option.xml.in: system option ssh-client source-interface	system option ssh-client source-interface <value>
×	<code>set system option ssh-client source-interface <interface></code>	Nothing found in XML Definitions	Nothing found in VyOS
✓	<code>set system option startup-beep</code>	system_option.xml.in: system option startup-beep	system option startup-beep
×	not yet documented	system_option.xml.in: system option time-format	system option time-format <value>
×	not yet documented	system_proxy.xml.in: system proxy password	system proxy password <value>
×	<code>set system proxy password <password></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	system_proxy.xml.in: system proxy port	system proxy port <value>
×	<code>set system proxy port <port></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	system_proxy.xml.in: system proxy url	system proxy url <value>
×	<code>set system proxy url <url></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	system_proxy.xml.in: system proxy username	system proxy username <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<i>set system proxy username <username></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	system_sflow.xml.in: system sflow agent-address	system sflow agent-address <value>
×	<i>set system sflow agent-address <address></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	system_sflow.xml.in: system sflow agent-interface	system sflow agent-interface <value>
×	<i>set system sflow agent-interface <interface></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	system_sflow.xml.in: system sflow drop-monitor-limit	system sflow drop-monitor-limit <value>
×	<i>set system sflow drop-monitor-limit <limit></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set system sflow enable-egress</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	system_sflow.xml.in: system sflow interface	system sflow interface <value>
×	<i>set system sflow interface <interface></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	system_sflow.xml.in: system sflow polling	system sflow polling <value>
×	<i>set system sflow polling <sec></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	system_sflow.xml.in: system sflow sampling-rate	system sflow sampling-rate <value>
×	<i>set system sflow sampling-rate <rate></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set system sflow server <address> port <port></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	system_sflow.xml.in: system sflow server <server> port	system sflow server <text> port <value>
×	<i>set system sflow vpp</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	system_sflow.xml.in: system sflow vrf	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set system static-host-mapping host-name <hostname> alias <alias></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set system static-host-mapping host-name <hostname> inet <address></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	system_static-host-mapping.xml.in: system static-host-mapping host-name <host-name> alias	system static-host-mapping host-name <text> alias <value>
×	not yet documented	system_static-host-mapping.xml.in: system static-host-mapping host-name <host-name> inet	system static-host-mapping host-name <text> inet <value>
×	<code>set system sysctl parameter <parameter> value <value></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	system_sysctl.xml.in: system sysctl parameter <parameter> value	system sysctl parameter <text> value <value>
×	<code>set system syslog console facility <keyword> level <keyword></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	system_syslog.xml.in: system syslog console facility <facility> level	system syslog console facility <text> level <value>
×	not yet documented	system_syslog.xml.in: system syslog file <file> archive file	system syslog file <text> archive file <value>
×	not yet documented	system_syslog.xml.in: system syslog file <file> archive size	system syslog file <text> archive size <value>
×	not yet documented	system_syslog.xml.in: system syslog file <file> facility <facility> level	system syslog file <text> facility <text> level <value>
×	not yet documented	system_syslog.xml.in: system syslog global facility <facility> level	system syslog global facility <text> level <value>
×	not yet documented	system_syslog.xml.in: system syslog global marker interval	system syslog global marker interval <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	system_syslog.xml.in: system syslog global preserve-fqdn	system syslog global preserve-fqdn
×	not yet documented	system_syslog.xml.in: system syslog host <host> facility <facility> level <value>	system syslog host <text> facility <text> level <value>
×	not yet documented	system_syslog.xml.in: system syslog host <host> format octet-counted	system syslog host <text> format octet-counted
×	not yet documented	system_syslog.xml.in: system syslog host <host> port	system syslog host <text> port <value>
×	not yet documented	system_syslog.xml.in: system syslog host <host> protocol	system syslog host <text> protocol <value>
×	<i>set system syslog local <filename> facility <keyword> level <keyword></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set system syslog marker disable</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set system syslog marker interval <number></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set system syslog preserve-fqdn</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set system syslog remote <address> facility <keyword> level <keyword></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set system syslog remote <address> format include-timezone</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set system syslog remote <address> format octet-counted</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set system syslog remote <address> port <port></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set system syslog remote <address> protocol <udp tcp></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set system syslog remote <address> source-address <address></i>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set system syslog remote <address> tls</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set system syslog remote <address> tls auth-mode <anon fingerprint certvalid name></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set system syslog remote <address> tls ca-certificate <ca_name></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set system syslog remote <address> tls certificate <cert_name></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set system syslog remote <address> tls permitted-peer <peer></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set system syslog remote <address> vrf <name></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	system_syslog.xml.in: system syslog user <user> facility <facility> level	system syslog user <text> facility <text> level <value>
×	not yet documented	system_syslog.xml.in: system syslog vrf	system syslog vrf <value>
×	<code>set system task-scheduler task <task> crontab-spec <spec></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set system task-scheduler task <task> executable arguments <args></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set system task-scheduler task <task> executable path <path></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set system task-scheduler task <task> interval <interval></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	system_task-scheduler.xml.in: system task-scheduler task <task> crontab-spec	system task-scheduler task <text> crontab-spec <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	system_task-scheduler.xml.in: system task-scheduler task <task> executable arguments	system task-scheduler task <text> executable arguments <value>
×	not yet documented	system_task-scheduler.xml.in: system task-scheduler task <task> executable path	system task-scheduler task <text> executable path <value>
×	not yet documented	system_task-scheduler.xml.in: system task-scheduler task <task> interval	system task-scheduler task <text> interval <value>
×	not yet documented	system_time-zone.xml.in: system time-zone	system time-zone <value>
×	set system time-zone <timezone>	Nothing found in XML Definitions	Nothing found in VyOS
✓	set system update-check auto-check	system_update-check.xml.in: system update-check auto-check	system update-check auto-check
×	not yet documented	system_update-check.xml.in: system update-check url	system update-check url <value>
×	set system update-check url <url>	Nothing found in XML Definitions	Nothing found in VyOS
×	set system watchdog	Nothing found in XML Definitions	Nothing found in VyOS
×	set system watchdog module <module-name>	Nothing found in XML Definitions	Nothing found in VyOS
×	set system watchdog reboot-timeout <seconds>	Nothing found in XML Definitions	Nothing found in VyOS
×	set system watchdog shutdown-timeout <seconds>	Nothing found in XML Definitions	Nothing found in VyOS
×	set system watchdog timeout <seconds>	Nothing found in XML Definitions	Nothing found in VyOS
×	set system wireless country-code <cc>	Nothing found in XML Definitions	Nothing found in VyOS
×	set traffic-policy network-emulator <policy-name> queue-limit <limit>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<i>set traffic-policy network-emulator <policy-name> reordering <percent></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set vpn ipsec authentication psk <name> dhcp-interface</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	vpn_ipsec.xml.in: vpn ipsec authentication psk <psk> dhcp-interface	vpn ipsec authentication psk <text> dhcp-interface <value>
×	not yet documented	vpn_ipsec.xml.in: vpn ipsec authentication psk <psk> id	vpn ipsec authentication psk <text> id <value>
×	<i>set vpn ipsec authentication psk id <id></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	vpn_ipsec.xml.in: vpn ipsec authentication psk <psk> secret	vpn ipsec authentication psk <text> secret <value>
×	<i>set vpn ipsec authentication psk secret-type <type></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set vpn ipsec authentication psk secret <secret></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set vpn ipsec authentication x509 passphrase <passphrase></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	vpn_ipsec.xml.in: vpn ipsec disable-uniqreqids	vpn ipsec disable-uniqreqids
✓	<i>set vpn ipsec esp-group <name> compression</i>	vpn_ipsec.xml.in: vpn ipsec esp-group <esp-group> compression	vpn ipsec esp-group <text> compression
×	<i>set vpn ipsec esp-group <name> disable-rekey</i>	Nothing found in XML Definitions	Nothing found in VyOS
✓	<i>set vpn ipsec esp-group <name> life-bytes <bytes></i>	vpn_ipsec.xml.in: vpn ipsec esp-group <esp-group> life-bytes	vpn ipsec esp-group <text> life-bytes <value>
✓	<i>set vpn ipsec esp-group <name> life-packets <packets></i>	vpn_ipsec.xml.in: vpn ipsec esp-group <esp-group> life-packets	vpn ipsec esp-group <text> life-packets <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
✓	<code>set vpn ipsec esp-group <name> lifetime <timeout></code>	vpn_ipsec.xml.in: vpn ipsec esp-group <esp-group> lifetime	vpn ipsec esp-group <text> lifetime <value>
✓	<code>set vpn ipsec esp-group <name> mode <mode></code>	vpn_ipsec.xml.in: vpn ipsec esp-group <esp-group> mode	vpn ipsec esp-group <text> mode <value>
✓	<code>set vpn ipsec esp-group <name> pfs <dh-group></code>	vpn_ipsec.xml.in: vpn ipsec esp-group <esp-group> pfs	vpn ipsec esp-group <text> pfs <value>
✓	<code>set vpn ipsec esp-group <name> proposal <number> encryption <encryption></code>	vpn_ipsec.xml.in: vpn ipsec esp-group <esp-group> proposal <proposal> encryption	vpn ipsec esp-group <text> proposal <text> encryption <value>
✓	<code>set vpn ipsec esp-group <name> proposal <number> hash <hash></code>	vpn_ipsec.xml.in: vpn ipsec esp-group <esp-group> proposal <proposal> hash	vpn ipsec esp-group <text> proposal <text> hash <value>
✓	<code>set vpn ipsec ike-group <name> close-action <action></code>	vpn_ipsec.xml.in: vpn ipsec ike-group <ike-group> close-action	vpn ipsec ike-group <text> close-action <value>
✓	<code>set vpn ipsec ike-group <name> dead-peer-detection action <action></code>	vpn_ipsec.xml.in: vpn ipsec ike-group <ike-group> dead-peer- detection action	vpn ipsec ike-group <text> dead-peer-detection action <value>
✓	<code>set vpn ipsec ike-group <name> dead-peer-detection interval <interval></code>	vpn_ipsec.xml.in: vpn ipsec ike-group <ike-group> dead-peer- detection interval	vpn ipsec ike-group <text> dead-peer-detection inter- val <value>
✓	<code>set vpn ipsec ike-group <name> dead-peer-detection timeout <timeout></code>	vpn_ipsec.xml.in: vpn ipsec ike-group <ike-group> dead-peer- detection timeout	vpn ipsec ike-group <text> dead-peer-detection time- out <value>
×	not yet documented	vpn_ipsec.xml.in: vpn ipsec ike-group <ike-group> disable-mobike	vpn ipsec ike-group <text> disable-mobike
✓	<code>set vpn ipsec ike-group <name> ikev2-reauth</code>	vpn_ipsec.xml.in: vpn ipsec ike-group <ike-group> ikev2-reauth	vpn ipsec ike-group <text> ikev2-reauth
✓	<code>set vpn ipsec ike-group <name> key-exchange</code>	vpn_ipsec.xml.in: vpn ipsec ike-group <ike-group> key-exchange	vpn ipsec ike-group <text> key-exchange <value>
✓	<code>set vpn ipsec ike-group <name> lifetime</code>	vpn_ipsec.xml.in: vpn ipsec ike-group <ike-group> lifetime	vpn ipsec ike-group <text> lifetime <value>
✓	<code>set vpn ipsec ike-group <name> mode</code>	vpn_ipsec.xml.in: vpn ipsec ike-group <ike-group> mode	vpn ipsec ike-group <text> mode <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
✓	<code>set vpn ipsec ike-group <name> proposal <number> dh-group <dh-group number></code>	vpn_ipsec.xml.in: vpn ipsec ike-group <ike-group> proposal <proposal> dh-group	vpn ipsec ike-group <text> proposal <text> dh-group <value>
✓	<code>set vpn ipsec ike-group <name> proposal <number> encryption <encryption></code>	vpn_ipsec.xml.in: vpn ipsec ike-group <ike-group> proposal <proposal> encryption	vpn ipsec ike-group <text> proposal <text> encryp- tion <value>
✓	<code>set vpn ipsec ike-group <name> proposal <number> hash <hash></code>	vpn_ipsec.xml.in: vpn ipsec ike-group <ike-group> proposal <proposal> hash	vpn ipsec ike-group <text> proposal <text> hash <value>
✓	<code>set vpn ipsec ike-group <name> proposal <number> prf <prf></code>	vpn_ipsec.xml.in: vpn ipsec ike-group <ike-group> proposal <proposal> prf	vpn ipsec ike-group <text> proposal <text> prf <value>
✓	<code>set vpn ipsec interface <name></code>	vpn_ipsec.xml.in: vpn ipsec interface	vpn ipsec interface <value>
×	<code>set vpn ipsec interface <INTERFACE></code>	Nothing found in XML Definitions	Nothing found in VyOS
✓	<code>set vpn ipsec log level <number></code>	vpn_ipsec.xml.in: vpn ipsec log level	vpn ipsec log level <value>
✓	<code>set vpn ipsec log subsystem <name></code>	vpn_ipsec.xml.in: vpn ipsec log subsystem	vpn ipsec log subsystem <value>
✓	<code>set vpn ipsec options disable-route-autoins</code>	vpn_ipsec.xml.in: vpn ipsec options disable- route-autoinstall	vpn ipsec options disable- route-autoinstall
✓	<code>set vpn ipsec options flexvpn</code>	vpn_ipsec.xml.in: vpn ipsec options flexvpn	vpn ipsec options flexvpn
✓	<code>set vpn ipsec options interface <name></code>	vpn_ipsec.xml.in: vpn ipsec options interface	vpn ipsec options interface <value>
×	<code>set vpn ipsec options retransmission attempts</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set vpn ipsec options retransmission base</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set vpn ipsec options retransmission timeout</code>	Nothing found in XML Definitions	Nothing found in VyOS
✓	<code>set vpn ipsec options virtual-ip</code>	vpn_ipsec.xml.in: vpn ipsec options virtual- ip	vpn ipsec options virtual- ip

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set vpn ipsec profile <profile-name> authentication mode pre-shared-secret</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set vpn ipsec profile <profile-name> authentication pre-shared-secret <secret></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set vpn ipsec profile <profile-name> bind tunnel <tunnel name></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set vpn ipsec profile <profile-name> esp-group 'ESP-HUB'</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set vpn ipsec profile <profile-name> ike-group 'IKE-HUB'</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	vpn_ipsec.xml.in: vpn ipsec profile <profile> authentication mode	vpn ipsec profile <text> authentication mode <value>
×	not yet documented	vpn_ipsec.xml.in: vpn ipsec profile <profile> authentication pre-shared-secret	vpn ipsec profile <text> authentication pre-shared-secret <value>
×	not yet documented	vpn_ipsec.xml.in: vpn ipsec profile <profile> bind tunnel	vpn ipsec profile <text> bind tunnel <value>
×	not yet documented	vpn_ipsec.xml.in: vpn ipsec profile <profile> disable	vpn ipsec profile <text> disable
×	not yet documented	vpn_ipsec.xml.in: vpn ipsec profile <profile> esp-group	vpn ipsec profile <text> esp-group <value>
×	not yet documented	vpn_ipsec.xml.in: vpn ipsec profile <profile> ike-group	vpn ipsec profile <text> ike-group <value>
×	not yet documented	vpn_ipsec.xml.in: vpn ipsec remote-access connection <connection> authentication client-mode	vpn ipsec remote-access connection <text> au- thentication client-mode <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	vpn_ipsec.xml.in: vpn ipsec remote-access connection <connection> authentication eap-id	Nothing found in VyOS
×	not yet documented	vpn_ipsec.xml.in: vpn ipsec remote-access connection <connection> authentication local-id	vpn ipsec remote-access connection <text> authentication local-id <value>
×	not yet documented	vpn_ipsec.xml.in: vpn ipsec remote-access connection <connection> authentication local-users username <username> disable	vpn ipsec remote-access connection <text> authentication local-users username <text> disable
×	not yet documented	vpn_ipsec.xml.in: vpn ipsec remote-access connection <connection> authentication local-users username <username> password	vpn ipsec remote-access connection <text> authentication local-users username <text> password <value>
×	not yet documented	vpn_ipsec.xml.in: vpn ipsec remote-access connection <connection> authentication pre-shared-secret	vpn ipsec remote-access connection <text> authentication pre-shared-secret <value>
×	not yet documented	vpn_ipsec.xml.in: vpn ipsec remote-access connection <connection> authentication server-mode	vpn ipsec remote-access connection <text> authentication server-mode <value>
×	not yet documented	vpn_ipsec.xml.in: vpn ipsec remote-access connection <connection> authentication x509 ca-certificate	vpn ipsec remote-access connection <text> authentication x509 ca-certificate <value>
×	not yet documented	vpn_ipsec.xml.in: vpn ipsec remote-access connection <connection> authentication x509 certificate	vpn ipsec remote-access connection <text> authentication x509 certificate <value>
×	not yet documented	vpn_ipsec.xml.in: vpn ipsec remote-access connection <connection> authentication x509 passphrase	vpn ipsec remote-access connection <text> authentication x509 passphrase <value>
×	not yet documented	vpn_ipsec.xml.in: vpn ipsec remote-access connection <connection> description	vpn ipsec remote-access connection <text> description <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	vpn_ipsec.xml.in: vpn ipsec remote-access connection <connection> dhcp-interface	Nothing found in VyOS
×	not yet documented	vpn_ipsec.xml.in: vpn ipsec remote-access connection <connection> disable	vpn ipsec remote-access connection <text> disable
×	not yet documented	vpn_ipsec.xml.in: vpn ipsec remote-access connection <connection> esp-group	vpn ipsec remote-access connection <text> esp- group <value>
×	not yet documented	vpn_ipsec.xml.in: vpn ipsec remote-access connection <connection> ike-group	vpn ipsec remote-access connection <text> ike- group <value>
×	not yet documented	vpn_ipsec.xml.in: vpn ipsec remote-access connection <connection> local-address	vpn ipsec remote-access connection <text> local- address <value>
×	not yet documented	vpn_ipsec.xml.in: vpn ipsec remote-access connection <connection> local port	vpn ipsec remote-access connection <text> local port <value>
×	not yet documented	vpn_ipsec.xml.in: vpn ipsec remote-access connection <connection> local prefix	vpn ipsec remote-access connection <text> local prefix <value>
×	not yet documented	vpn_ipsec.xml.in: vpn ipsec remote-access connection <connection> pool	vpn ipsec remote-access connection <text> pool <value>
×	not yet documented	vpn_ipsec.xml.in: vpn ipsec remote-access connection <connection> replay-window	Nothing found in VyOS
×	not yet documented	vpn_ipsec.xml.in: vpn ipsec remote-access connection <connection> timeout	vpn ipsec remote-access connection <text> timeout <value>
×	not yet documented	vpn_ipsec.xml.in: vpn ipsec remote-access connection <connection> unique	vpn ipsec remote-access connection <text> unique <value>
×	not yet documented	vpn_ipsec.xml.in: vpn ipsec remote-access dhcp interface	vpn ipsec remote-access dhcp interface <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	vpn_ipsec.xml.in: vpn ipsec remote-access dhcp server	vpn ipsec remote-access dhcp server <value>
×	not yet documented	vpn_ipsec.xml.in: vpn ipsec remote-access pool <pool> exclude	vpn ipsec remote-access pool <text> exclude <value>
×	not yet documented	vpn_ipsec.xml.in: vpn ipsec remote-access pool <pool> name-server	vpn ipsec remote-access pool <text> name-server <value>
×	not yet documented	vpn_ipsec.xml.in: vpn ipsec remote-access pool <pool> prefix	vpn ipsec remote-access pool <text> prefix <value>
×	not yet documented	vpn_ipsec.xml.in: vpn ipsec remote-access radius nas-identifier	vpn ipsec remote-access radius nas-identifier <value>
×	not yet documented	vpn_ipsec.xml.in: vpn ipsec remote-access radius server <server> dis- able	vpn ipsec remote-access radius server <text> dis- able
×	not yet documented	vpn_ipsec.xml.in: vpn ipsec remote-access radius server <server> disable-accounting	vpn ipsec remote-access radius server <text> disable-accounting
×	not yet documented	vpn_ipsec.xml.in: vpn ipsec remote-access radius server <server> key	vpn ipsec remote-access radius server <text> key <value>
×	not yet documented	vpn_ipsec.xml.in: vpn ipsec remote-access radius server <server> port	vpn ipsec remote-access radius server <text> port <value>
×	not yet documented	vpn_ipsec.xml.in: vpn ipsec remote-access radius source-address	Nothing found in VyOS
×	not yet documented	vpn_ipsec.xml.in: vpn ipsec remote-access radius timeout	vpn ipsec remote-access radius timeout <value>
×	<i>set vpn ipsec site-to-site peer <name> authentication local-id <id></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set vpn ipsec site-to-site peer <name> authentication mode <mode></i>	Nothing found in XML Definitions	Nothing found in VyOS

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Table 1 – continued from previous page

Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set vpn ipsec site-to-site peer <name> authentication remote-id <id></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set vpn ipsec site-to-site peer <name> authentication rsa local-key <key></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set vpn ipsec site-to-site peer <name> authentication rsa passphrase <passphrase></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set vpn ipsec site-to-site peer <name> authentication rsa remote-key <key></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set vpn ipsec site-to-site peer <name> authentication use-x509-id <id></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set vpn ipsec site-to-site peer <name> authentication x509 ca-certificate <name></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set vpn ipsec site-to-site peer <name> authentication x509 certificate <name></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set vpn ipsec site-to-site peer <name> connection-type <type></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set vpn ipsec site-to-site peer <name> default-esp-group <name></code>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set vpn ipsec site-to-site peer <name> description <description></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set vpn ipsec site-to-site peer <name> dhcp-interface <interface></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set vpn ipsec site-to-site peer <peer-name> force-udp-encapsulati</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set vpn ipsec site-to-site peer <name> ike-group <name></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set vpn ipsec site-to-site peer <name> ikev2-reauth <yes/no/inherit></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set vpn ipsec site-to-site peer <name> local-address <address></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set vpn ipsec site-to-site peer <name> remote-address <address></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set vpn ipsec site-to-site peer <name> replay-window <size></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set vpn ipsec site-to-site peer <name> tunnel <number> disable</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set vpn ipsec site-to-site peer <name> tunnel <number> esp-group <name></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set vpn ipsec site-to-site peer <name> tunnel <number> local port <number></code>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set vpn ipsec site-to-site peer <name> tunnel <number> local prefix <network></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set vpn ipsec site-to-site peer <name> tunnel <number> priority <number></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set vpn ipsec site-to-site peer <name> tunnel <number> protocol <name></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set vpn ipsec site-to-site peer <name> tunnel <number> remote port <number></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set vpn ipsec site-to-site peer <name> tunnel <number> remote prefix <network></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set vpn ipsec site-to-site peer <name> virtual-address <address></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set vpn ipsec site-to-site peer <name> vti bind <interface></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set vpn ipsec site-to-site peer <name> vti esp-group <name></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set vpn ipsec site-to-site peer <name> vti traffic-selector local prefix <network></code>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<i>set vpn ipsec site-to-site peer <name> vti traffic-selector remote prefix <network></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	vpn_ipsec.xml.in: vpn ipsec site-to-site peer <peer> authentication local-id	vpn ipsec site-to-site peer <text> authentication local-id <value>
×	not yet documented	vpn_ipsec.xml.in: vpn ipsec site-to-site peer <peer> authentication mode	vpn ipsec site-to-site peer <text> authentication mode <value>
×	not yet documented	vpn_ipsec.xml.in: vpn ipsec site-to-site peer <peer> authentication remote-id	vpn ipsec site-to-site peer <text> authentication remote-id <value>
×	not yet documented	vpn_ipsec.xml.in: vpn ipsec site-to-site peer <peer> authentication rsa local-key	vpn ipsec site-to-site peer <text> authentication rsa local-key <value>
×	not yet documented	vpn_ipsec.xml.in: vpn ipsec site-to-site peer <peer> authentication rsa passphrase	vpn ipsec site-to-site peer <text> authentication rsa passphrase <value>
×	not yet documented	vpn_ipsec.xml.in: vpn ipsec site-to-site peer <peer> authentication rsa remote-key	vpn ipsec site-to-site peer <text> authentication rsa remote-key <value>
×	not yet documented	vpn_ipsec.xml.in: vpn ipsec site-to-site peer <peer> authentication use- x509-id	vpn ipsec site-to-site peer <text> authentication use- x509-id
×	not yet documented	vpn_ipsec.xml.in: vpn ipsec site-to-site peer <peer> authentication x509 ca-certificate	vpn ipsec site-to-site peer <text> authentication x509 ca-certificate <value>
×	not yet documented	vpn_ipsec.xml.in: vpn ipsec site-to-site peer <peer> authentication x509 certificate	vpn ipsec site-to-site peer <text> authentication x509 certificate <value>
×	not yet documented	vpn_ipsec.xml.in: vpn ipsec site-to-site peer <peer> authentication x509 passphrase	vpn ipsec site-to-site peer <text> authentication x509 passphrase <value>
×	not yet documented	vpn_ipsec.xml.in: vpn ipsec site-to-site peer <peer> connection-type	vpn ipsec site-to-site peer <text> connection-type <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	vpn_ipsec.xml.in: vpn ipsec site-to-site peer <peer> default-esp-group	vpn ipsec site-to-site peer <text> default-esp-group <value>
×	not yet documented	vpn_ipsec.xml.in: vpn ipsec site-to-site peer <peer> description	vpn ipsec site-to-site peer <text> description <value>
×	not yet documented	vpn_ipsec.xml.in: vpn ipsec site-to-site peer <peer> dhcp-interface	vpn ipsec site-to-site peer <text> dhcp-interface <value>
×	not yet documented	vpn_ipsec.xml.in: vpn ipsec site-to-site peer <peer> disable	vpn ipsec site-to-site peer <text> disable
×	not yet documented	vpn_ipsec.xml.in: vpn ipsec site-to-site peer <peer> force-udp-encapsulation	vpn ipsec site-to-site peer <text> force-udp-encapsulation
×	not yet documented	vpn_ipsec.xml.in: vpn ipsec site-to-site peer <peer> ike-group	vpn ipsec site-to-site peer <text> ike-group <value>
×	not yet documented	vpn_ipsec.xml.in: vpn ipsec site-to-site peer <peer> ikev2-reauth	vpn ipsec site-to-site peer <text> ikev2-reauth <value>
×	not yet documented	vpn_ipsec.xml.in: vpn ipsec site-to-site peer <peer> local-address	vpn ipsec site-to-site peer <text> local-address <value>
×	not yet documented	vpn_ipsec.xml.in: vpn ipsec site-to-site peer <peer> remote-address	vpn ipsec site-to-site peer <text> remote-address <value>
×	not yet documented	vpn_ipsec.xml.in: vpn ipsec site-to-site peer <peer> replay-window	Nothing found in VyOS
×	not yet documented	vpn_ipsec.xml.in: vpn ipsec site-to-site peer <peer> tunnel <tunnel> disable	vpn ipsec site-to-site peer <text> tunnel <text> disable
×	not yet documented	vpn_ipsec.xml.in: vpn ipsec site-to-site peer <peer> tunnel <tunnel> esp-group	vpn ipsec site-to-site peer <text> tunnel <text> esp-group <value>
×	not yet documented	vpn_ipsec.xml.in: vpn ipsec site-to-site peer <peer> tunnel <tunnel> local port	vpn ipsec site-to-site peer <text> tunnel <text> local port <value>
×	not yet documented	vpn_ipsec.xml.in: vpn ipsec site-to-site peer <peer> tunnel <tunnel> local prefix	vpn ipsec site-to-site peer <text> tunnel <text> local prefix <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	vpn_ipsec.xml.in: vpn ipsec site-to-site peer <peer> tunnel <tunnel> priority	vpn ipsec site-to-site peer <text> tunnel <text> prior- ity <value>
×	not yet documented	vpn_ipsec.xml.in: vpn ipsec site-to-site peer <peer> tunnel <tunnel> protocol	vpn ipsec site-to-site peer <text> tunnel <text> pro- tocol <value>
×	not yet documented	vpn_ipsec.xml.in: vpn ipsec site-to-site peer <peer> tunnel <tunnel> remote port	vpn ipsec site-to-site peer <text> tunnel <text> re- mote port <value>
×	not yet documented	vpn_ipsec.xml.in: vpn ipsec site-to-site peer <peer> tunnel <tunnel> remote prefix	vpn ipsec site-to-site peer <text> tunnel <text> re- mote prefix <value>
×	not yet documented	vpn_ipsec.xml.in: vpn ipsec site-to-site peer <peer> virtual-address	vpn ipsec site-to-site peer <text> virtual-address <value>
×	not yet documented	vpn_ipsec.xml.in: vpn ipsec site-to-site peer <peer> vti bind	vpn ipsec site-to-site peer <text> vti bind <value>
×	not yet documented	vpn_ipsec.xml.in: vpn ipsec site-to-site peer <peer> vti esp-group	vpn ipsec site-to-site peer <text> vti esp-group <value>
×	<i>set vpn l2tp remote-access authentication local-users username <user> disable</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set vpn l2tp remote-access authentication local-users username <user> password <pass></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set vpn l2tp remote-access authentication local-users username <user> rate-limit download <bandwidth></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set vpn l2tp remote-access authentication local-users username <user> rate-limit upload <bandwidth></i>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<i>set vpn l2tp remote-access authentication local-users username <user> static-ip <address></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	vpn_l2tp.xml.in: vpn l2tp remote-access authentication local-users username <username> disable	vpn l2tp remote-access authentication local-users username <text> disable
×	not yet documented	vpn_l2tp.xml.in: vpn l2tp remote-access authentication local-users username <username> password	vpn l2tp remote-access authentication local-users username <text> password <value>
×	not yet documented	vpn_l2tp.xml.in: vpn l2tp remote-access authentication local-users username <username> rate-limit download	vpn l2tp remote-access authentication local-users username <text> rate-limit download <value>
×	not yet documented	vpn_l2tp.xml.in: vpn l2tp remote-access authentication local-users username <username> rate-limit upload	vpn l2tp remote-access authentication local-users username <text> rate-limit upload <value>
×	not yet documented	vpn_l2tp.xml.in: vpn l2tp remote-access authentication local-users username <username> static-ip	vpn l2tp remote-access authentication local-users username <text> static-ip <value>
×	not yet documented	vpn_l2tp.xml.in: vpn l2tp remote-access authentication mode	vpn l2tp remote-access authentication mode <value>
×	<i>set vpn l2tp remote-access authentication mode <local radius></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	Nothing found in XML Definitions	vpn l2tp remote-access authentication mppe <value>
×	not yet documented	vpn_l2tp.xml.in: vpn l2tp remote-access authentication protocols	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set vpn l2tp remote-access authentication protocols <pap / chap / mschap / mschap-v2></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	vpn_l2tp.xml.in: vpn l2tp remote-access authentication radius accounting-interim-interval	vpn l2tp remote-access authentication radius accounting-interim-interval <value>
×	not yet documented	vpn_l2tp.xml.in: vpn l2tp remote-access authentication radius acct-interim-jitter	Nothing found in VyOS
×	not yet documented	vpn_l2tp.xml.in: vpn l2tp remote-access authentication radius acct-timeout	vpn l2tp remote-access authentication radius acct-timeout <value>
×	<code>set vpn l2tp remote-access authentication radius acct-timeout <timeout></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	Nothing found in XML Definitions	vpn l2tp remote-access authentication radius dae-server ip-address <value>
×	not yet documented	Nothing found in XML Definitions	vpn l2tp remote-access authentication radius dae-server port <value>
×	not yet documented	Nothing found in XML Definitions	vpn l2tp remote-access authentication radius dae-server secret <value>
×	not yet documented	vpn_l2tp.xml.in: vpn l2tp remote-access authentication radius dynamic-author key	Nothing found in VyOS
×	<code>set vpn l2tp remote-access authentication radius dynamic-author key <secret></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	vpn_l2tp.xml.in: vpn l2tp remote-access authentication radius dynamic-author port	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<i>set vpn l2tp remote-access authentication radius dynamic-author port <port></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	vpn_l2tp.xml.in: vpn l2tp remote-access authentication radius dynamic-author server	Nothing found in VyOS
×	<i>set vpn l2tp remote-access authentication radius dynamic-author server <address></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	vpn_l2tp.xml.in: vpn l2tp remote-access authentication radius max-try	vpn l2tp remote-access authentication radius max-try <value>
×	<i>set vpn l2tp remote-access authentication radius max-try <number></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	vpn_l2tp.xml.in: vpn l2tp remote-access authentication radius nas-identifier	vpn l2tp remote-access authentication radius nas-identifier <value>
×	<i>set vpn l2tp remote-access authentication radius nas-identifier <identifier></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	vpn_l2tp.xml.in: vpn l2tp remote-access authentication radius nas-ip-address	vpn l2tp remote-access authentication radius nas-ip-address <value>
×	<i>set vpn l2tp remote-access authentication radius nas-ip-address <address></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	vpn_l2tp.xml.in: vpn l2tp remote-access authentication radius preallocate-vif	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	vpn_l2tp.xml.in: vpn l2tp remote-access authentication radius rate-limit attribute	vpn l2tp remote-access authentication radius rate-limit attribute <value>
×	<i>set vpn l2tp remote-access authentication radius rate-limit attribute <attribute></i>	Nothing found in XML Definitions	Nothing found in VyOS
✓	<i>set vpn l2tp remote-access authentication radius rate-limit enable</i>	vpn_l2tp.xml.in: vpn l2tp remote-access authentication radius rate-limit enable	vpn l2tp remote-access authentication radius rate-limit enable
×	not yet documented	vpn_l2tp.xml.in: vpn l2tp remote-access authentication radius rate-limit multiplier	vpn l2tp remote-access authentication radius rate-limit multiplier <value>
✓	<i>set vpn l2tp remote-access authentication radius rate-limit vendor</i>	vpn_l2tp.xml.in: vpn l2tp remote-access authentication radius rate-limit vendor	vpn l2tp remote-access authentication radius rate-limit vendor <value>
×	<i>set vpn l2tp remote-access authentication radius server <server> disable</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set vpn l2tp remote-access authentication radius server <server> fail-time <time></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set vpn l2tp remote-access authentication radius server <server> key <secret></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set vpn l2tp remote-access authentication radius server <server> port <port></i>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	vpn_l2tp.xml.in: vpn l2tp remote-access authentication radius server <server> acct-port	Nothing found in VyOS
×	not yet documented	vpn_l2tp.xml.in: vpn l2tp remote-access authentication radius server <server> disable	vpn l2tp remote-access authentication radius server <text> disable
×	not yet documented	vpn_l2tp.xml.in: vpn l2tp remote-access authentication radius server <server> disable-accounting	vpn l2tp remote-access authentication radius server <text> disable-accounting
×	not yet documented	vpn_l2tp.xml.in: vpn l2tp remote-access authentication radius server <server> fail-time	vpn l2tp remote-access authentication radius server <text> fail-time <value>
×	not yet documented	vpn_l2tp.xml.in: vpn l2tp remote-access authentication radius server <server> key	vpn l2tp remote-access authentication radius server <text> key <value>
×	not yet documented	vpn_l2tp.xml.in: vpn l2tp remote-access authentication radius server <server> port	vpn l2tp remote-access authentication radius server <text> port <value>
×	not yet documented	vpn_l2tp.xml.in: vpn l2tp remote-access authentication radius source-address	vpn l2tp remote-access authentication radius source-address <value>
×	<i>set vpn l2tp remote-access authentication radius source-address <address></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	vpn_l2tp.xml.in: vpn l2tp remote-access authentication radius timeout	vpn l2tp remote-access authentication radius timeout <value>
×	<i>set vpn l2tp remote-access authentication radius timeout <timeout></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	Nothing found in XML Definitions	vpn l2tp remote-access authentication require <value>
×	not yet documented	Nothing found in XML Definitions	vpn l2tp remote-access ccp-disable

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set vpn l2tp remote-access client-ip-pool <POOL-NAME> next-pool <NEXT-POOL-NAME></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set vpn l2tp remote-access client-ip-pool <POOL-NAME> range <x.x.x.x-x.x.x.x / x.x.x.x/x></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	vpn_l2tp.xml.in: vpn l2tp remote-access client-ip-pool <client-ip- pool> next-pool	Nothing found in VyOS
×	not yet documented	vpn_l2tp.xml.in: vpn l2tp remote-access client-ip-pool <client-ip- pool> range	Nothing found in VyOS
×	not yet documented	Nothing found in XML Definitions	vpn l2tp remote-access client-ip-pool start <value>
×	not yet documented	Nothing found in XML Definitions	vpn l2tp remote-access client-ip-pool stop <value>
×	not yet documented	Nothing found in XML Definitions	vpn l2tp remote-access client-ip-pool subnet <value>
×	<code>set vpn l2tp remote-access client-ipv6-pool <IPv6-POOL-NAME> delegate <address> delegation-prefix <number-of-bits></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set vpn l2tp remote-access client-ipv6-pool <IPv6-POOL-NAME> prefix <address> mask <number-of-bits></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	vpn_l2tp.xml.in: vpn l2tp remote-access client-ipv6-pool <client- ipv6-pool> delegate <delegate> delegation- prefix	vpn l2tp remote-access client-ipv6-pool delegate <text> delegation-prefix <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	vpn_l2tp.xml.in: vpn l2tp remote-access client-ipv6-pool <client- ipv6-pool> prefix <pre- fix> mask	vpn l2tp remote-access client-ipv6-pool prefix <text> mask <value>
×	not yet documented	vpn_l2tp.xml.in: vpn l2tp remote-access default-ipv6-pool	Nothing found in VyOS
×	<i>set vpn l2tp remote-access default-ipv6-pool <IPv6-POOL-NAME></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	vpn_l2tp.xml.in: vpn l2tp remote-access default-pool	Nothing found in VyOS
×	<i>set vpn l2tp remote-access default-pool <POOL-NAME></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	vpn_l2tp.xml.in: vpn l2tp remote-access de- scription	vpn l2tp remote-access de- scription <value>
×	<i>set vpn l2tp remote-access description <description></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	Nothing found in XML Definitions	vpn l2tp remote-access dhcp-interface <value>
×	not yet documented	vpn_l2tp.xml.in: vpn l2tp remote-access extended-scripts on- change	Nothing found in VyOS
×	<i>set vpn l2tp remote-access extended-scripts on-change <path_to_script></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	vpn_l2tp.xml.in: vpn l2tp remote-access extended-scripts on-down	Nothing found in VyOS
×	<i>set vpn l2tp remote-access extended-scripts on-down <path_to_script></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	vpn_l2tp.xml.in: vpn l2tp remote-access extended-scripts on-pre- up	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<i>set vpn l2tp remote-access extended-scripts on-pre-up <path_to_script></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	vpn_l2tp.xml.in: vpn l2tp remote-access extended-scripts on-up	Nothing found in VyOS
×	<i>set vpn l2tp remote-access extended-scripts on-up <path_to_script></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	vpn_l2tp.xml.in: vpn l2tp remote-access gateway-address	vpn l2tp remote-access gateway-address <value>
×	<i>set vpn l2tp remote-access gateway-address <gateway></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	Nothing found in XML Definitions	vpn l2tp remote-access idle <value>
×	not yet documented	vpn_l2tp.xml.in: vpn l2tp remote-access ipsec-settings authentication mode	vpn l2tp remote-access ipsec-settings authentication mode <value>
×	<i>set vpn l2tp remote-access ipsec-settings authentication mode <pre-shared-secret / x509></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	vpn_l2tp.xml.in: vpn l2tp remote-access ipsec-settings authentication pre-shared-secret	vpn l2tp remote-access ipsec-settings authentication pre-shared-secret <value>
×	<i>set vpn l2tp remote-access ipsec-settings authentication pre-shared-secret <secret></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	vpn_l2tp.xml.in: vpn l2tp remote-access ipsec-settings authentication x509 ca-certificate	vpn l2tp remote-access ipsec-settings authentication x509 ca-certificate <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	vpn_l2tp.xml.in: vpn l2tp remote-access ipsec-settings authentication x509 certificate	vpn l2tp remote-access ipsec-settings authentication x509 certificate <value>
×	not yet documented	vpn_l2tp.xml.in: vpn l2tp remote-access ipsec-settings authentication x509 passphrase	vpn l2tp remote-access ipsec-settings authentication x509 passphrase <value>
×	not yet documented	vpn_l2tp.xml.in: vpn l2tp remote-access ipsec-settings esp-group	vpn l2tp remote-access ipsec-settings esp-group <value>
×	not yet documented	vpn_l2tp.xml.in: vpn l2tp remote-access ipsec-settings ike-group	vpn l2tp remote-access ipsec-settings ike-group <value>
×	not yet documented	vpn_l2tp.xml.in: vpn l2tp remote-access ipsec-settings ike-lifetime	vpn l2tp remote-access ipsec-settings ike-lifetime <value>
×	not yet documented	vpn_l2tp.xml.in: vpn l2tp remote-access ipsec-settings lifetime	vpn l2tp remote-access ipsec-settings lifetime <value>
×	not yet documented	vpn_l2tp.xml.in: vpn l2tp remote-access limits burst	Nothing found in VyOS
×	<i>set vpn l2tp remote-access limits burst <value></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	vpn_l2tp.xml.in: vpn l2tp remote-access limits connection-limit	Nothing found in VyOS
×	<i>set vpn l2tp remote-access limits connection-limit <value></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	vpn_l2tp.xml.in: vpn l2tp remote-access limits timeout	Nothing found in VyOS
×	<i>set vpn l2tp remote-access limits timeout <value></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	vpn_l2tp.xml.in: vpn l2tp remote-access lns host-name	vpn l2tp remote-access lns host-name <value>
×	<i>set vpn l2tp remote-access lns host-name <hostname></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	vpn_l2tp.xml.in: vpn l2tp remote-access lns shared-secret	vpn l2tp remote-access lns shared-secret <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set vpn l2tp remote-access lns shared-secret <secret></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set vpn l2tp remote-access max-concurrent-sessio</code>	vpn_l2tp.xml.in: vpn l2tp remote-access max-concurrent-sessions	Nothing found in VyOS
✓	<code>set vpn l2tp remote-access mtu</code>	vpn_l2tp.xml.in: vpn l2tp remote-access mtu	vpn l2tp remote-access mtu <value>
×	not yet documented	vpn_l2tp.xml.in: vpn l2tp remote-access name-server	vpn l2tp remote-access name-server <value>
×	<code>set vpn l2tp remote-access name-server <address></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	vpn_l2tp.xml.in: vpn l2tp remote-access outside-address	vpn l2tp remote-access outside-address <value>
×	<code>set vpn l2tp remote-access ppp-options disable-ccp</code>	vpn_l2tp.xml.in: vpn l2tp remote-access ppp-options disable-ccp	Nothing found in VyOS
×	not yet documented	vpn_l2tp.xml.in: vpn l2tp remote-access ppp-options interface-cache	Nothing found in VyOS
×	<code>set vpn l2tp remote-access ppp-options interface-cache <number></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	vpn_l2tp.xml.in: vpn l2tp remote-access ppp-options ipv4	Nothing found in VyOS
×	<code>set vpn l2tp remote-access ppp-options ipv4 <require prefer allow deny></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	vpn_l2tp.xml.in: vpn l2tp remote-access ppp-options ipv6	vpn l2tp remote-access ppp-options ipv6 <value>
×	<code>set vpn l2tp remote-access ppp-options ipv6-accept-peer-inte</code>	vpn_l2tp.xml.in: vpn l2tp remote-access ppp-options ipv6-accept-peer-interface-id	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	Nothing found in XML Definitions	vpn l2tp remote-access ppp-options ipv6-accept-peer-intf-id
×	not yet documented	vpn_l2tp.xml.in: vpn l2tp remote-access ppp-options ipv6-interface-id	Nothing found in VyOS
×	<i>set vpn l2tp remote-access ppp-options ipv6-interface-id <random / x:x:x:x></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	Nothing found in XML Definitions	vpn l2tp remote-access ppp-options ipv6-intf-id <value>
×	not yet documented	vpn_l2tp.xml.in: vpn l2tp remote-access ppp-options ipv6-peer-interface-id	Nothing found in VyOS
×	not yet documented	Nothing found in XML Definitions	vpn l2tp remote-access ppp-options ipv6-peer-intf-id <value>
×	<i>set vpn l2tp remote-access ppp-options ipv6 <require / prefer / allow / deny></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	vpn_l2tp.xml.in: vpn l2tp remote-access ppp-options lcp-echo-failure	vpn l2tp remote-access ppp-options lcp-echo-failure <value>
×	<i>set vpn l2tp remote-access ppp-options lcp-echo-failure <number></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	vpn_l2tp.xml.in: vpn l2tp remote-access ppp-options lcp-echo-interval	vpn l2tp remote-access ppp-options lcp-echo-interval <value>
×	<i>set vpn l2tp remote-access ppp-options lcp-echo-interval <interval></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set vpn l2tp remote-access ppp-options lcp-echo-timeout</i>	vpn_l2tp.xml.in: vpn l2tp remote-access ppp-options lcp-echo-timeout	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	vpn_l2tp.xml.in: vpn l2tp remote-access ppp-options min-mtu	Nothing found in VyOS
×	<i>set vpn l2tp remote-access ppp-options min-mtu <number></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	vpn_l2tp.xml.in: vpn l2tp remote-access ppp-options mppe	Nothing found in VyOS
×	<i>set vpn l2tp remote-access ppp-options mppe <require prefer deny></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	vpn_l2tp.xml.in: vpn l2tp remote-access ppp-options mru	Nothing found in VyOS
×	<i>set vpn l2tp remote-access ppp-options mru <number></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	vpn_l2tp.xml.in: vpn l2tp remote-access shaper fwmark	Nothing found in VyOS
×	<i>set vpn l2tp remote-access shaper fwmark <1-2147483647></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set vpn l2tp remote-access snmp master-agent</i>	vpn_l2tp.xml.in: vpn l2tp remote-access snmp master-agent	Nothing found in VyOS
×	not yet documented	vpn_l2tp.xml.in: vpn l2tp remote-access wins-server	vpn l2tp remote-access wins-server <value>
×	<i>set vpn l2tp remote-access wins-server <address></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	vpn_openconnect.xml.in: vpn openconnect account- ing mode radius	vpn openconnect account- ing mode radius
×	not yet documented	vpn_openconnect.xml.in: vpn openconnect account- ing radius server <server> disable	vpn openconnect account- ing radius server <text> disable

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	vpn_openconnect.xml.in: vpn openconnect account- ing radius server <server> key	vpn openconnect account- ing radius server <text> key <value>
×	not yet documented	vpn_openconnect.xml.in: vpn openconnect account- ing radius server <server> port	vpn openconnect account- ing radius server <text> port <value>
×	not yet documented	vpn_openconnect.xml.in: vpn openconnect authenti- cation group	vpn openconnect authenti- cation group <value>
×	not yet documented	vpn_openconnect.xml.in: vpn openconnect authen- tication identity-based- config default-config	vpn openconnect authen- tication identity-based- config default-config <value>
×	not yet documented	vpn_openconnect.xml.in: vpn openconnect authen- tication identity-based- config directory	vpn openconnect au- thentication identity- based-config directory <value>
×	not yet documented	vpn_openconnect.xml.in: vpn openconnect authen- tication identity-based- config disable	vpn openconnect authen- tication identity-based- config disable
×	not yet documented	vpn_openconnect.xml.in: vpn openconnect authen- tication identity-based- config mode	vpn openconnect authen- tication identity-based- config mode <value>
×	not yet documented	vpn_openconnect.xml.in: vpn openconnect authen- tication local-users user- name <username> disable	vpn openconnect authen- tication local-users user- name <text> disable
×	not yet documented	vpn_openconnect.xml.in: vpn openconnect authen- tication local-users user- name <username> otp in- terval	vpn openconnect authen- tication local-users user- name <text> otp interval <value>
×	not yet documented	vpn_openconnect.xml.in: vpn openconnect authen- tication local-users user- name <username> otp key	vpn openconnect au- thentication local-users username <text> otp key <value>
×	not yet documented	vpn_openconnect.xml.in: vpn openconnect authen- tication local-users user- name <username> otp otp- length	vpn openconnect au- thentication local-users username <text> otp otp-length <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	vpn_openconnect.xml.in: vpn openconnect authentication local-users username <username> otp token-type	vpn openconnect authentication local-users username <text> otp token-type <value>
×	not yet documented	vpn_openconnect.xml.in: vpn openconnect authentication local-users username <username> password	vpn openconnect authentication local-users username <text> password <value>
×	not yet documented	vpn_openconnect.xml.in: vpn openconnect authentication mode local	vpn openconnect authentication mode local <value>
×	not yet documented	vpn_openconnect.xml.in: vpn openconnect authentication mode radius	vpn openconnect authentication mode radius
×	not yet documented	vpn_openconnect.xml.in: vpn openconnect authentication radius groupconfig	vpn openconnect authentication radius groupconfig <value>
×	not yet documented	vpn_openconnect.xml.in: vpn openconnect authentication radius server <server> disable	vpn openconnect authentication radius server <text> disable
×	not yet documented	vpn_openconnect.xml.in: vpn openconnect authentication radius server <server> key	vpn openconnect authentication radius server <text> key <value>
×	not yet documented	vpn_openconnect.xml.in: vpn openconnect authentication radius server <server> port	vpn openconnect authentication radius server <text> port <value>
×	not yet documented	vpn_openconnect.xml.in: vpn openconnect authentication radius source-address	vpn openconnect authentication radius source-address <value>
×	not yet documented	vpn_openconnect.xml.in: vpn openconnect authentication radius timeout	vpn openconnect authentication radius timeout <value>
×	not yet documented	vpn_openconnect.xml.in: vpn openconnect http-security-headers	Nothing found in VyOS
×	not yet documented	vpn_openconnect.xml.in: vpn openconnect listen-address	vpn openconnect listen-address <value>
×	not yet documented	vpn_openconnect.xml.in: vpn openconnect listen-ports tcp	vpn openconnect listen-ports tcp <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	vpn_openconnect.xml.in: vpn openconnect listen-ports udp	vpn openconnect listen-ports udp <value>
×	not yet documented	vpn_openconnect.xml.in: vpn openconnect network-settings client-ip-settings subnet	vpn openconnect network-settings client-ip-settings subnet <value>
×	not yet documented	vpn_openconnect.xml.in: vpn openconnect network-settings client-ipv6-pool mask	vpn openconnect network-settings client-ipv6-pool mask <value>
×	not yet documented	vpn_openconnect.xml.in: vpn openconnect network-settings client-ipv6-pool prefix	vpn openconnect network-settings client-ipv6-pool prefix <value>
×	not yet documented	vpn_openconnect.xml.in: vpn openconnect network-settings name-server	vpn openconnect network-settings name-server <value>
×	not yet documented	vpn_openconnect.xml.in: vpn openconnect network-settings push-route	vpn openconnect network-settings push-route <value>
×	not yet documented	vpn_openconnect.xml.in: vpn openconnect network-settings split-dns	vpn openconnect network-settings split-dns <value>
×	not yet documented	vpn_openconnect.xml.in: vpn openconnect network-settings tunnel-all-dns	vpn openconnect network-settings tunnel-all-dns <value>
×	not yet documented	vpn_openconnect.xml.in: vpn openconnect ssl ca-certificate	vpn openconnect ssl ca-certificate <value>
×	not yet documented	vpn_openconnect.xml.in: vpn openconnect ssl certificate	vpn openconnect ssl certificate <value>
×	not yet documented	vpn_openconnect.xml.in: vpn openconnect ssl passphrase	vpn openconnect ssl passphrase <value>
×	<i>set vpn pptp remote-access authentication local-users username <user> disable</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set vpn pptp remote-access authentication local-users username <user> password <pass></i>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set vpn pptp remote-access authentication local-users username <user> rate-limit download <bandwidth></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set vpn pptp remote-access authentication local-users username <user> rate-limit upload <bandwidth></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set vpn pptp remote-access authentication local-users username <user> static-ip <address></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	vpn_pptp.xml.in: vpn pptp remote-access authentication local-users username <username> disable	vpn pptp remote-access authentication local-users username <text> disable
×	not yet documented	vpn_pptp.xml.in: vpn pptp remote-access authentication local-users username <username> password	vpn pptp remote-access authentication local-users username <text> password <value>
×	not yet documented	vpn_pptp.xml.in: vpn pptp remote-access authentication local-users username <username> rate-limit download	Nothing found in VyOS
×	not yet documented	vpn_pptp.xml.in: vpn pptp remote-access authentication local-users username <username> rate-limit upload	Nothing found in VyOS
×	not yet documented	vpn_pptp.xml.in: vpn pptp remote-access authentication local-users username <username> static-ip	vpn pptp remote-access authentication local-users username <text> static-ip <value>
×	not yet documented	vpn_pptp.xml.in: vpn pptp remote-access authentication mode	vpn pptp remote-access authentication mode <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set vpn pptp remote-access authentication mode <local radius></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	Nothing found in XML Definitions	vpn pptp remote-access authentication mppe <value>
×	not yet documented	vpn_pptp.xml.in: vpn pptp remote-access authentication protocols	Nothing found in VyOS
×	<code>set vpn pptp remote-access authentication protocols <pap chap mschap mschap-v2></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	vpn_pptp.xml.in: vpn pptp remote-access authentication radius accounting-interim-interval	vpn pptp remote-access authentication radius accounting-interim-interval <value>
×	not yet documented	vpn_pptp.xml.in: vpn pptp remote-access authentication radius acct-interim-jitter	vpn pptp remote-access authentication radius acct-interim-jitter <value>
×	not yet documented	vpn_pptp.xml.in: vpn pptp remote-access authentication radius acct-timeout	vpn pptp remote-access authentication radius acct-timeout <value>
×	<code>set vpn pptp remote-access authentication radius acct-timeout <timeout></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	vpn_pptp.xml.in: vpn pptp remote-access authentication radius dynamic-author key	vpn pptp remote-access authentication radius dynamic-author key <value>
×	<code>set vpn pptp remote-access authentication radius dynamic-author key <secret></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	vpn_pptp.xml.in: vpn pptp remote-access authentication radius dynamic-author port	vpn pptp remote-access authentication radius dynamic-author port <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<i>set vpn pptp remote-access authentication radius dynamic-author port <port></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	vpn_pptp.xml.in: vpn pptp remote-access authentication radius dynamic-author server	vpn pptp remote-access authentication radius dynamic-author server <value>
×	<i>set vpn pptp remote-access authentication radius dynamic-author server <address></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	vpn_pptp.xml.in: vpn pptp remote-access authentication radius max-try	vpn pptp remote-access authentication radius max-try <value>
×	<i>set vpn pptp remote-access authentication radius max-try <number></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	vpn_pptp.xml.in: vpn pptp remote-access authentication radius nas-identifier	vpn pptp remote-access authentication radius nas-identifier <value>
×	<i>set vpn pptp remote-access authentication radius nas-identifier <identifier></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	vpn_pptp.xml.in: vpn pptp remote-access authentication radius nas-ip-address	vpn pptp remote-access authentication radius nas-ip-address <value>
×	<i>set vpn pptp remote-access authentication radius nas-ip-address <address></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	vpn_pptp.xml.in: vpn pptp remote-access authentication radius preallocate-vif	vpn pptp remote-access authentication radius preallocate-vif

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	vpn_pptp.xml.in: vpn pptp remote-access authentication radius rate-limit attribute	vpn pptp remote-access authentication radius rate- limit attribute <value>
×	<i>set vpn pptp remote-access authentication radius rate-limit attribute <attribute></i>	Nothing found in XML Definitions	Nothing found in VyOS
✓	<i>set vpn pptp remote-access authentication radius rate-limit enable</i>	vpn_pptp.xml.in: vpn pptp remote-access authentication radius rate-limit enable	vpn pptp remote-access authentication radius rate-limit enable
×	not yet documented	vpn_pptp.xml.in: vpn pptp remote-access authentication radius rate-limit multiplier	vpn pptp remote-access authentication radius rate-limit multiplier <value>
✓	<i>set vpn pptp remote-access authentication radius rate-limit vendor</i>	vpn_pptp.xml.in: vpn pptp remote-access authentication radius rate-limit vendor	vpn pptp remote-access authentication radius rate-limit vendor <value>
×	<i>set vpn pptp remote-access authentication radius server <server> disable</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set vpn pptp remote-access authentication radius server <server> fail-time <time></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set vpn pptp remote-access authentication radius server <server> key <secret></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set vpn pptp remote-access authentication radius server <server> port <port></i>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	vpn_pptp.xml.in: vpn pptp remote-access authentication radius server <server> acct-port	vpn pptp remote-access authentication radius server <text> acct-port <value>
×	not yet documented	vpn_pptp.xml.in: vpn pptp remote-access authentication radius server <server> disable	vpn pptp remote-access authentication radius server <text> disable
×	not yet documented	vpn_pptp.xml.in: vpn pptp remote-access authentication radius server <server> disable- accounting	vpn pptp remote-access authentication radius server <text> disable- accounting
×	not yet documented	vpn_pptp.xml.in: vpn pptp remote-access authentication radius server <server> fail-time	vpn pptp remote-access authentication radius server <text> fail-time <value>
×	not yet documented	vpn_pptp.xml.in: vpn pptp remote-access authentication radius server <server> key	vpn pptp remote-access authentication radius server <text> key <value>
×	not yet documented	vpn_pptp.xml.in: vpn pptp remote-access authentication radius server <server> port	vpn pptp remote-access authentication radius server <text> port <value>
×	not yet documented	vpn_pptp.xml.in: vpn pptp remote-access authentication radius source-address	vpn pptp remote-access authentication radius source-address <value>
×	<i>set vpn pptp remote-access authentication radius source-address <address></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	vpn_pptp.xml.in: vpn pptp remote-access authentication radius timeout	vpn pptp remote-access authentication radius timeout <value>
×	<i>set vpn pptp remote-access authentication radius timeout <timeout></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	Nothing found in XML Definitions	vpn pptp remote-access authentication require <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<i>set vpn pptp remote-access client-ip-pool <POOL-NAME> next-pool <NEXT-POOL-NAME></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set vpn pptp remote-access client-ip-pool <POOL-NAME> range <x.x.x.x-x.x.x.x / x.x.x.x/x></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	vpn_pptp.xml.in: vpn pptp remote-access client-ip-pool <client-ip-pool> next-pool	Nothing found in VyOS
×	not yet documented	vpn_pptp.xml.in: vpn pptp remote-access client-ip-pool <client-ip-pool> range	Nothing found in VyOS
×	not yet documented	Nothing found in XML Definitions	vpn pptp remote-access client-ip-pool start <value>
×	not yet documented	Nothing found in XML Definitions	vpn pptp remote-access client-ip-pool stop <value>
×	<i>set vpn pptp remote-access client-ipv6-pool <IPv6-POOL-NAME> delegate <address> delegation-prefix <number-of-bits></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set vpn pptp remote-access client-ipv6-pool <IPv6-POOL-NAME> prefix <address> mask <number-of-bits></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	vpn_pptp.xml.in: vpn pptp remote-access client-ipv6-pool <client-ipv6-pool> delegate <delegate> delegation-prefix	Nothing found in VyOS

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Table 1 – continued from previous page

Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	vpn_pptp.xml.in: vpn pptp remote-access client-ipv6-pool <client- ipv6-pool> prefix <pre- fix> mask	Nothing found in VyOS
×	not yet documented	vpn_pptp.xml.in: vpn pptp remote-access default-ipv6-pool	Nothing found in VyOS
×	<i>set vpn pptp remote-access default-ipv6-pool <IPv6-POOL-NAME></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	vpn_pptp.xml.in: vpn pptp remote-access default-pool	Nothing found in VyOS
×	<i>set vpn pptp remote-access default-pool <POOL-NAME></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	vpn_pptp.xml.in: vpn pptp remote-access description	Nothing found in VyOS
×	<i>set vpn pptp remote-access description <description></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	vpn_pptp.xml.in: vpn pptp remote-access extended-scripts on- change	Nothing found in VyOS
×	<i>set vpn pptp remote-access extended-scripts on-change <path_to_script></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	vpn_pptp.xml.in: vpn pptp remote-access extended-scripts on-down	Nothing found in VyOS
×	<i>set vpn pptp remote-access extended-scripts on-down <path_to_script></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	vpn_pptp.xml.in: vpn pptp remote-access extended-scripts on-pre- up	Nothing found in VyOS

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Table 1 – continued from previous page

Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<i>set vpn pptp remote-access extended-scripts on-pre-up <path_to_script></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	vpn_pptp.xml.in: vpn pptp remote-access extended-scripts on-up	Nothing found in VyOS
×	<i>set vpn pptp remote-access extended-scripts on-up <path_to_script></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	vpn_pptp.xml.in: vpn pptp remote-access gateway-address	vpn pptp remote-access gateway-address <value>
×	<i>set vpn pptp remote-access gateway-address <gateway></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	vpn_pptp.xml.in: vpn pptp remote-access limits burst	Nothing found in VyOS
×	<i>set vpn pptp remote-access limits burst <value></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	vpn_pptp.xml.in: vpn pptp remote-access limits connection-limit	Nothing found in VyOS
×	<i>set vpn pptp remote-access limits connection-limit <value></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	vpn_pptp.xml.in: vpn pptp remote-access limits timeout	Nothing found in VyOS
×	<i>set vpn pptp remote-access limits timeout <value></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set vpn pptp remote-access max-concurrent-sessio</i>	vpn_pptp.xml.in: vpn pptp remote-access max-concurrent-sessions	Nothing found in VyOS
✓	<i>set vpn pptp remote-access mtu</i>	vpn_pptp.xml.in: vpn pptp remote-access mtu	vpn pptp remote-access mtu <value>
×	not yet documented	vpn_pptp.xml.in: vpn pptp remote-access name-server	vpn pptp remote-access name-server <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<i>set vpn ptp remote-access name-server <address></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	vpn_pttp.xml.in: vpn ptp remote-access outside-address	vpn ptp remote-access outside-address <value>
×	<i>set vpn ptp remote-access ppp-options disable-ccp</i>	vpn_pttp.xml.in: vpn ptp remote-access ppp-options disable-ccp	Nothing found in VyOS
×	not yet documented	vpn_pttp.xml.in: vpn ptp remote-access ppp-options interface-cache	Nothing found in VyOS
×	<i>set vpn ptp remote-access ppp-options interface-cache <number></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	vpn_pttp.xml.in: vpn ptp remote-access ppp-options ipv4	Nothing found in VyOS
×	<i>set vpn ptp remote-access ppp-options ipv4 <require prefer allow deny></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	vpn_pttp.xml.in: vpn ptp remote-access ppp-options ipv6	Nothing found in VyOS
×	<i>set vpn ptp remote-access ppp-options ipv6-accept-peer-interface-id</i>	vpn_pttp.xml.in: vpn ptp remote-access ppp-options ipv6-accept-peer-interface-id	Nothing found in VyOS
×	not yet documented	vpn_pttp.xml.in: vpn ptp remote-access ppp-options ipv6-interface-id	Nothing found in VyOS
×	<i>set vpn ptp remote-access ppp-options ipv6-interface-id <random x:x:x:x></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	vpn_pttp.xml.in: vpn ptp remote-access ppp-options ipv6-peer-interface-id	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<i>set vpn pptp remote-access ppp-options ipv6 <require prefer allow deny></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	vpn_pptp.xml.in: vpn pptp remote-access ppp-options lcp-echo-failure	Nothing found in VyOS
×	<i>set vpn pptp remote-access ppp-options lcp-echo-failure <number></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	vpn_pptp.xml.in: vpn pptp remote-access ppp-options lcp-echo-interval	Nothing found in VyOS
×	<i>set vpn pptp remote-access ppp-options lcp-echo-interval <interval></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set vpn pptp remote-access ppp-options lcp-echo-timeout</i>	vpn_pptp.xml.in: vpn pptp remote-access ppp-options lcp-echo-timeout	Nothing found in VyOS
×	not yet documented	vpn_pptp.xml.in: vpn pptp remote-access ppp-options min-mtu	Nothing found in VyOS
×	<i>set vpn pptp remote-access ppp-options min-mtu <number></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	vpn_pptp.xml.in: vpn pptp remote-access ppp-options mppe	Nothing found in VyOS
×	<i>set vpn pptp remote-access ppp-options mppe <require prefer deny></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	vpn_pptp.xml.in: vpn pptp remote-access ppp-options mru	Nothing found in VyOS
×	<i>set vpn pptp remote-access ppp-options mru <number></i>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	vpn_pptp.xml.in: vpn pptp remote-access shaper fwmark	Nothing found in VyOS
×	<i>set vpn pptp remote-access shaper fwmark <1-2147483647></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set vpn pptp remote-access snmp master-agent</i>	vpn_pptp.xml.in: vpn pptp remote-access snmp master-agent	Nothing found in VyOS
×	not yet documented	vpn_pptp.xml.in: vpn pptp remote-access wins-server	vpn pptp remote-access wins-server <value>
×	<i>set vpn pptp remote-access wins-server <address></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set vpn sstp authentication local-users username <user> disable</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set vpn sstp authentication local-users username <user> password <pass></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set vpn sstp authentication local-users username <user> rate-limit download <bandwidth></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set vpn sstp authentication local-users username <user> rate-limit upload <bandwidth></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set vpn sstp authentication local-users username <user> static-ip <address></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	vpn_sstp.xml.in: vpn sstp authentication local-users username <username> disable	vpn sstp authentication local-users username <text> disable

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	vpn_sstp.xml.in: vpn sstp authentication local-users username <username> password	vpn sstp authentication local-users username <text> password <value>
×	not yet documented	vpn_sstp.xml.in: vpn sstp authentication local-users username <username> rate-limit download	vpn sstp authentication local-users username <text> rate-limit down- load <value>
×	not yet documented	vpn_sstp.xml.in: vpn sstp authentication local-users username <username> rate-limit upload	vpn sstp authentication local-users username <text> rate-limit upload <value>
×	not yet documented	vpn_sstp.xml.in: vpn sstp authentication local-users username <username> static-ip	vpn sstp authentication local-users username <text> static-ip <value>
×	not yet documented	vpn_sstp.xml.in: vpn sstp authentication mode	vpn sstp authentication mode <value>
×	<i>set vpn sstp authentication mode <local radius></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	vpn_sstp.xml.in: vpn sstp authentication protocols	vpn sstp authentication protocols <value>
×	<i>set vpn sstp authentication protocols <pap chap mschap mschap-v2></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	vpn_sstp.xml.in: vpn sstp authentication ra- dius accounting-interim- interval	vpn sstp authentication ra- dius accounting-interim- interval <value>
×	not yet documented	vpn_sstp.xml.in: vpn sstp authentication ra- dius acct-interim-jitter	vpn sstp authentication radius acct-interim-jitter <value>
×	not yet documented	vpn_sstp.xml.in: vpn sstp authentication ra- dius acct-timeout	vpn sstp authentication ra- dius acct-timeout <value>
×	<i>set vpn sstp authentication radius acct-timeout <timeout></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	vpn_sstp.xml.in: vpn sstp authentication ra- dius dynamic-author key	vpn sstp authentication ra- dius dynamic-author key <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<i>set vpn sstp authentication radius dynamic-author key <secret></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	vpn_sstp.xml.in: vpn sstp authentication radius dynamic-author port	vpn sstp authentication radius dynamic-author port <value>
×	<i>set vpn sstp authentication radius dynamic-author port <port></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	vpn_sstp.xml.in: vpn sstp authentication radius dynamic-author server	vpn sstp authentication radius dynamic-author server <value>
×	<i>set vpn sstp authentication radius dynamic-author server <address></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	vpn_sstp.xml.in: vpn sstp authentication radius max-try	vpn sstp authentication radius max-try <value>
×	<i>set vpn sstp authentication radius max-try <number></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	vpn_sstp.xml.in: vpn sstp authentication radius nas-identifier	vpn sstp authentication radius nas-identifier <value>
×	<i>set vpn sstp authentication radius nas-identifier <identifier></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	vpn_sstp.xml.in: vpn sstp authentication radius nas-ip-address	vpn sstp authentication radius nas-ip-address <value>
×	<i>set vpn sstp authentication radius nas-ip-address <address></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	vpn_sstp.xml.in: vpn sstp authentication radius preallocate-vif	vpn sstp authentication radius preallocate-vif

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	vpn_sstp.xml.in: vpn sstp authentication radius rate-limit attribute	vpn sstp authentication radius rate-limit attribute <value>
×	<i>set vpn sstp authentication radius rate-limit attribute <attribute></i>	Nothing found in XML Definitions	Nothing found in VyOS
✓	<i>set vpn sstp authentication radius rate-limit enable</i>	vpn_sstp.xml.in: vpn sstp authentication radius rate-limit enable	vpn sstp authentication radius rate-limit enable
×	not yet documented	vpn_sstp.xml.in: vpn sstp authentication radius rate-limit multiplier	vpn sstp authentication radius rate-limit multiplier <value>
✓	<i>set vpn sstp authentication radius rate-limit vendor</i>	vpn_sstp.xml.in: vpn sstp authentication radius rate-limit vendor	vpn sstp authentication radius rate-limit vendor <value>
×	<i>set vpn sstp authentication radius server <server> disable</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set vpn sstp authentication radius server <server> fail-time <time></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set vpn sstp authentication radius server <server> key <secret></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set vpn sstp authentication radius server <server> port <port></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	vpn_sstp.xml.in: vpn sstp authentication radius server <server> acct-port	vpn sstp authentication radius server <text> acct-port <value>
×	not yet documented	vpn_sstp.xml.in: vpn sstp authentication radius server <server> disable	vpn sstp authentication radius server <text> disable

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	vpn_sstp.xml.in: vpn sstp authentication radius server <server> disable-accounting	vpn sstp authentication ra- dius server <text> disable- accounting
×	not yet documented	vpn_sstp.xml.in: vpn sstp authentication ra- dius server <server> fail- time	vpn sstp authentication ra- dius server <text> fail- time <value>
×	not yet documented	vpn_sstp.xml.in: vpn sstp authentication ra- dius server <server> key	vpn sstp authentication radius server <text> key <value>
×	not yet documented	vpn_sstp.xml.in: vpn sstp authentication ra- dius server <server> port	vpn sstp authentication ra- dius server <text> port <value>
×	not yet documented	vpn_sstp.xml.in: vpn sstp authentication ra- dius source-address	vpn sstp authentication radius source-address <value>
×	<i>set vpn sstp authentication radius source-address <address></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	vpn_sstp.xml.in: vpn sstp authentication ra- dius timeout	vpn sstp authentication ra- dius timeout <value>
×	<i>set vpn sstp authentication radius timeout <timeout></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set vpn sstp client-ip-pool <POOL-NAME> next-pool <NEXT-POOL-NAME></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set vpn sstp client-ip-pool <POOL-NAME> range <x.x.x.x-x.x.x.x / x.x.x.x/x></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	vpn_sstp.xml.in: vpn sstp client-ip-pool <client-ip-pool> next-pool	Nothing found in VyOS
×	not yet documented	vpn_sstp.xml.in: vpn sstp client-ip-pool <client-ip-pool> range	Nothing found in VyOS
×	not yet documented	Nothing found in XML Definitions	vpn sstp client-ip-pool subnet <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set vpn sstp client-ipv6-pool <IPv6-POOL-NAME> delegate <address> delegation-prefix <number-of-bits></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set vpn sstp client-ipv6-pool <IPv6-POOL-NAME> prefix <address> mask <number-of-bits></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	vpn_sstp.xml.in: vpn sstp client-ipv6- pool <client-ipv6-pool> delegate <delegate> delegation-prefix	vpn sstp client-ipv6- pool delegate <text> delegation-prefix <value>
×	not yet documented	vpn_sstp.xml.in: vpn sstp client-ipv6-pool <client-ipv6-pool> prefix <prefix> mask	vpn sstp client-ipv6- pool prefix <text> mask <value>
×	not yet documented	vpn_sstp.xml.in: vpn sstp default-ipv6-pool	Nothing found in VyOS
×	<code>set vpn sstp default-ipv6-pool <IPv6-POOL-NAME></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	vpn_sstp.xml.in: vpn sstp default-pool	Nothing found in VyOS
×	<code>set vpn sstp default-pool <POOL-NAME></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	vpn_sstp.xml.in: vpn sstp description	Nothing found in VyOS
×	<code>set vpn sstp description <description></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	vpn_sstp.xml.in: vpn sstp extended-scripts on-change	Nothing found in VyOS
×	<code>set vpn sstp extended-scripts on-change <path_to_script></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	vpn_sstp.xml.in: vpn sstp extended-scripts on-down	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<i>set vpn sstp extended-scripts on-down <path_to_script></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	vpn_sstp.xml.in: vpn sstp extended-scripts on-pre-up	Nothing found in VyOS
×	<i>set vpn sstp extended-scripts on-pre-up <path_to_script></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	vpn_sstp.xml.in: vpn sstp extended-scripts on-up	Nothing found in VyOS
×	<i>set vpn sstp extended-scripts on-up <path_to_script></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	vpn_sstp.xml.in: vpn sstp gateway-address	vpn sstp gateway-address <value>
×	<i>set vpn sstp gateway-address <gateway></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set vpn sstp host-name <hostname></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	vpn_sstp.xml.in: vpn sstp limits burst	Nothing found in VyOS
×	<i>set vpn sstp limits burst <value></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	vpn_sstp.xml.in: vpn sstp limits connection-limit	Nothing found in VyOS
×	<i>set vpn sstp limits connection-limit <value></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	vpn_sstp.xml.in: vpn sstp limits timeout	Nothing found in VyOS
×	<i>set vpn sstp limits timeout <value></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set vpn sstp max-concurrent-sessio</i>	vpn_sstp.xml.in: vpn sstp max-concurrent-sessions	Nothing found in VyOS
✓	<i>set vpn sstp mtu</i>	vpn_sstp.xml.in: vpn sstp mtu	vpn sstp mtu <value>
×	not yet documented	vpn_sstp.xml.in: vpn sstp name-server	vpn sstp name-server <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set vpn sstp name-server <address></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	vpn_sstp.xml.in: vpn sstp port	vpn sstp port <value>
×	<code>set vpn sstp ppp-options disable-ccp</code>	vpn_sstp.xml.in: vpn sstp ppp-options disable-ccp	Nothing found in VyOS
×	not yet documented	vpn_sstp.xml.in: vpn sstp ppp-options interface-cache	Nothing found in VyOS
×	<code>set vpn sstp ppp-options interface-cache <number></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	vpn_sstp.xml.in: vpn sstp ppp-options ipv4	vpn sstp ppp-options ipv4 <value>
×	<code>set vpn sstp ppp-options ipv4 <require prefer allow deny></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	vpn_sstp.xml.in: vpn sstp ppp-options ipv6	vpn sstp ppp-options ipv6 <value>
×	<code>set vpn sstp ppp-options ipv6-accept-peer-interface-id</code>	vpn_sstp.xml.in: vpn sstp ppp-options ipv6-accept-peer-interface-id	Nothing found in VyOS
×	not yet documented	vpn_sstp.xml.in: vpn sstp ppp-options ipv6-interface-id	Nothing found in VyOS
×	<code>set vpn sstp ppp-options ipv6-interface-id <random x:x:x:x></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	vpn_sstp.xml.in: vpn sstp ppp-options ipv6-peer-interface-id	Nothing found in VyOS
×	<code>set vpn sstp ppp-options ipv6 <require prefer allow deny></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	vpn_sstp.xml.in: vpn sstp ppp-options lcp-echo-failure	vpn sstp ppp-options lcp-echo-failure <value>
×	<code>set vpn sstp ppp-options lcp-echo-failure <number></code>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	vpn_sstp.xml.in: vpn sstp ppp-options lcp-echo-interval	vpn sstp ppp-options lcp-echo-interval <value>
×	<i>set vpn sstp ppp-options lcp-echo-interval <interval></i>	Nothing found in XML Definitions	Nothing found in VyOS
✓	<i>set vpn sstp ppp-options lcp-echo-timeout</i>	vpn_sstp.xml.in: vpn sstp ppp-options lcp-echo-timeout	vpn sstp ppp-options lcp-echo-timeout <value>
×	not yet documented	vpn_sstp.xml.in: vpn sstp ppp-options min-mtu	Nothing found in VyOS
×	<i>set vpn sstp ppp-options min-mtu <number></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	vpn_sstp.xml.in: vpn sstp ppp-options mppe	vpn sstp ppp-options mppe <value>
×	<i>set vpn sstp ppp-options mppe <require prefer deny></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	vpn_sstp.xml.in: vpn sstp ppp-options mru	Nothing found in VyOS
×	<i>set vpn sstp ppp-options mru <number></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	vpn_sstp.xml.in: vpn sstp shaper fwmark	Nothing found in VyOS
×	<i>set vpn sstp shaper fwmark <1-2147483647></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set vpn sstp snmp master-agent</i>	vpn_sstp.xml.in: vpn sstp snmp master-agent	Nothing found in VyOS
×	not yet documented	vpn_sstp.xml.in: vpn sstp ssl ca-certificate	vpn sstp ssl ca-certificate <value>
×	<i>set vpn sstp ssl ca-certificate <file></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	vpn_sstp.xml.in: vpn sstp ssl certificate	vpn sstp ssl certificate <value>
×	<i>set vpn sstp ssl certificate <file></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	vpn_sstp.xml.in: vpn sstp wins-server	Nothing found in VyOS
×	<i>set vpn sstp wins-server <address></i>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set vpp nat cgnat exclude rule <rule-number> description <description></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set vpp nat cgnat exclude rule <rule-number> local-address <local-address></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set vpp nat cgnat exclude rule <rule-number> local-port <1-65535></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set vpp nat cgnat exclude rule <rule-number> protocol <tcp/udp/icmp/all></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set vpp nat cgnat interface inside <inside-interface></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set vpp nat cgnat interface outside <outside-interface></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set vpp nat cgnat rule <rule-number> description <description></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set vpp nat cgnat rule <rule-number> inside-prefix <inside-prefix></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set vpp nat cgnat rule <rule-number> outside-prefix <outside-prefix></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set vpp nat nat44 address-pool translation address <ip-address / ip-address-range></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set vpp nat nat44 address-pool translation interface <interface-name></code>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set vpp nat nat44 address-pool twice-nat address <ip-address / ip-address-range></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set vpp nat nat44 address-pool twice-nat interface <interface-name></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set vpp nat nat44 exclude rule <rule-number> description <description></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set vpp nat nat44 exclude rule <rule-number> external-interface <interface-name></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set vpp nat nat44 exclude rule <rule-number> local-address <internal-ip></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set vpp nat nat44 exclude rule <rule-number> local-port <port-number></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set vpp nat nat44 exclude rule <rule-number> protocol <protocol></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set vpp nat nat44 interface inside <inside-interface></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set vpp nat nat44 interface outside <outside-interface></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set vpp nat nat44 session-limit <number></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set vpp nat nat44 static rule <rule-number> description <description></code>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set vpp nat nat44 static rule <rule-number> external address <external-ip></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set vpp nat nat44 static rule <rule-number> external port <external-port></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set vpp nat nat44 static rule <rule-number> local address <internal-ip></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set vpp nat nat44 static rule <rule-number> local port <internal-port></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set vpp nat nat44 static rule <rule-number> options out-to-in-only</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set vpp nat nat44 static rule <rule-number> options self-twice-nat</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set vpp nat nat44 static rule <rule-number> options twice-nat</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set vpp nat nat44 static rule <rule-number> options twice-nat-address <ip-address></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set vpp nat nat44 static rule <rule-number> protocol <protocol></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set vpp nat nat44 timeout icmp <seconds></code>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set vpp nat nat44 timeout tcp-established <seconds></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set vpp nat nat44 timeout tcp-transitory <seconds></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set vpp nat nat44 timeout udp <seconds></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set vpp settings allow-unsupported-nic</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set vpp settings ignore-kernel-routes</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set vpp settings interface-rx-mode <mode></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set vpp settings interface <interface-name></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set vpp settings interface <interface-name> num-rx-desc <value></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set vpp settings interface <interface-name> num-rx-queues <value></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set vpp settings interface <interface-name> num-tx-desc <value></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set vpp settings interface <interface-name> num-tx-queues <value></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set vpp settings ipsec-acceleration</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set vpp settings lcp netlink batch-delay-ms <milliseconds></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set vpp settings lcp netlink batch-size <number></code>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<i>set vpp settings lcp netlink rx-buffer-size <number></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set vpp settings logging default-level <level></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set vpp settings poll-sleep-usec <delay></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set vpp settings resource-allocation buffers buffers-per-numa <value></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set vpp settings resource-allocation buffers data-size <value></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set vpp settings resource-allocation buffers page-size <value></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set vpp settings resource-allocation cpu-cores <core-number></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set vpp settings resource-allocation ipv6 hash-buckets <value></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set vpp settings resource-allocation ipv6 heap-size <value></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set vpp settings resource-allocation mac-limit <value></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set vpp settings resource-allocation memory main-heap-page-size <size></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>set vpp settings resource-allocation memory main-heap-size <size></i>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set vpp settings resource-allocation memory physmem-max-size <size></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set vpp settings resource-allocation memory stats page-size <size></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set vpp settings resource-allocation memory stats size <size></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set vpp sflow header-bytes <bytes></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set vpp sflow interface <interface-name></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set vrf <name> ip protocol <protocol> route-map <route-map></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set vrf <name> ipv6 protocol <protocol> route-map <route-map></code>	Nothing found in XML Definitions	Nothing found in VyOS
✓	<code>set vrf bind-to-all</code>	vrf.xml.in: vrf bind-to-all	vrf bind-to-all
×	<code>set vrf name <name> ip nht no-resolve-via-default</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set vrf name <name> ipv6 nht no-resolve-via-default</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set vrf name <name> protocols bgp address-family <ipv4-unicast/ipv6-un <import/export> vpn</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set vrf name <name> protocols bgp address-family <ipv4-unicast/ipv6-un import vrf <name></code>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>set vrf name <name> protocols bgp address-family <ipv4-unicast/ipv6-un label vpn allocation-mode per-nexthop</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set vrf name <name> protocols bgp address-family <ipv4-unicast/ipv6-un label vpn export <0-1048575/auto></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set vrf name <name> protocols bgp address-family <ipv4-unicast/ipv6-un rd vpn export <asn:nn/address:nn></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set vrf name <name> protocols bgp address-family <ipv4-unicast/ipv6-un route-map vpn <import/export> [route-map <name>]</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set vrf name <name> protocols bgp address-family <ipv4-unicast/ipv6-un route-map vrf import [route-map <name>]</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set vrf name <name> protocols bgp address-family <ipv4-unicast/ipv6-un route-target vpn <import/export/both> [RTLIST]</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set vrf name <name> protocols bgp interface <interface> mpls forwarding</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set vrf name <name> table <id></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	vrf.xml.in: vrf name <name> descrip- tion	vrf name <text> descrip- tion <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	vrf.xml.in: vrf name <name> disable	vrf name <text> disable
×	not yet documented	vrf.xml.in: vrf name <name> ip disable-forwarding	vrf name <text> ip disable-forwarding
×	not yet documented	vrf.xml.in: vrf name <name> ip nht no-resolve-via-default	Nothing found in VyOS
×	not yet documented	vrf.xml.in: vrf name <name> ip proto- col <protocol> route-map	vrf name <text> ip pro- tocol <text> route-map <value>
×	not yet documented	vrf.xml.in: vrf name <name> ipv6 disable-forwarding	vrf name <text> ipv6 disable-forwarding
×	not yet documented	vrf.xml.in: vrf name <name> ipv6 nht no-resolve-via-default	Nothing found in VyOS
×	not yet documented	vrf.xml.in: vrf name <name> ipv6 protocol <protocol> route- map	vrf name <text> ipv6 protocol <text> route-map <value>
×	not yet documented	vrf.xml.in: vrf name <name> proto- cols bgp address-family ipv4-flowspec local-install interface	vrf name <text> protocols bgp address-family ipv4- flowspec local-install in- terface <value>
×	not yet documented	vrf.xml.in: vrf name <name> proto- cols bgp address-family ipv4-labeled-unicast aggregate-address <aggregate-address> as-set	vrf name <text> protocols bgp address-family ipv4- labeled-unicast aggregate- address <text> as-set
×	not yet documented	vrf.xml.in: vrf name <name> proto- cols bgp address-family ipv4-labeled-unicast aggregate-address <aggregate-address> route-map	vrf name <text> protocols bgp address-family ipv4- labeled-unicast aggregate- address <text> route-map <value>
×	not yet documented	vrf.xml.in: vrf name <name> proto- cols bgp address-family ipv4-labeled-unicast aggregate-address <aggregate-address> summary-only	vrf name <text> protocols bgp address-family ipv4- labeled-unicast aggregate- address <text> summary- only

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp address-family ipv4-labeled-unicast maximum-paths ebgp	Nothing found in VyOS
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp address-family ipv4-labeled-unicast maximum-paths ibgp	Nothing found in VyOS
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp address-family ipv4-labeled-unicast network <network> backdoor	vrf name <text> protocols bgp address-family ipv4-labeled-unicast network <text> backdoor
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp address-family ipv4-labeled-unicast network <network> route-map	vrf name <text> protocols bgp address-family ipv4-labeled-unicast network <text> route-map <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp address-family ipv4-multicast aggregate-address <aggregate-address> as-set	vrf name <text> protocols bgp address-family ipv4-multicast aggregate-address <text> as-set
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp address-family ipv4-multicast aggregate-address <aggregate-address> route-map	vrf name <text> protocols bgp address-family ipv4-multicast aggregate-address <text> route-map <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp address-family ipv4-multicast aggregate-address <aggregate-address> summary-only	vrf name <text> protocols bgp address-family ipv4-multicast aggregate-address <text> summary-only
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp address-family ipv4-multicast distance external	vrf name <text> protocols bgp address-family ipv4-multicast distance external <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp address-family ipv4-multicast distance internal	vrf name <text> protocols bgp address-family ipv4-multicast distance internal <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp address-family ipv4-multicast distance local	vrf name <text> protocols bgp address-family ipv4-multicast distance local <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp address-family ipv4-multicast distance prefix <prefix> distance	vrf name <text> protocols bgp address-family ipv4-multicast distance prefix <text> distance <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp address-family ipv4-multicast network <network> backdoor	vrf name <text> protocols bgp address-family ipv4-multicast network <text> backdoor
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp address-family ipv4-multicast network <network> route-map	vrf name <text> protocols bgp address-family ipv4-multicast network <text> route-map <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp address-family ipv4-unicast aggregate-address <aggregate-address> as-set	vrf name <text> protocols bgp address-family ipv4-unicast aggregate-address <text> as-set
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp address-family ipv4-unicast aggregate-address <aggregate-address> route-map	vrf name <text> protocols bgp address-family ipv4-unicast aggregate-address <text> route-map <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp address-family ipv4-unicast aggregate-address <aggregate-address> summary-only	vrf name <text> protocols bgp address-family ipv4-unicast aggregate-address <text> summary-only
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp address-family ipv4-unicast distance external	vrf name <text> protocols bgp address-family ipv4-unicast distance external <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp address-family ipv4-unicast distance internal	vrf name <text> protocols bgp address-family ipv4-unicast distance internal <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp address-family ipv4-unicast distance local	vrf name <text> protocols bgp address-family ipv4-unicast distance local <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp address-family ipv4-unicast distance prefix <prefix> distance	vrf name <text> protocols bgp address-family ipv4-unicast distance prefix <text> distance <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp address-family ipv4-unicast export vpn	vrf name <text> protocols bgp address-family ipv4-unicast export vpn
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp address-family ipv4-unicast import vpn	vrf name <text> protocols bgp address-family ipv4-unicast import vpn
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp address-family ipv4-unicast import vrf	vrf name <text> protocols bgp address-family ipv4-unicast import vrf <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp address-family ipv4-unicast label vpn allocation-mode per-nexthop	vrf name <text> protocols bgp address-family ipv4-unicast label vpn allocation-mode per-nexthop
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp address-family ipv4-unicast label vpn export	vrf name <text> protocols bgp address-family ipv4-unicast label vpn export <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp address-family ipv4-unicast maximum-paths ebgp	vrf name <text> protocols bgp address-family ipv4-unicast maximum-paths ebgp <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp address-family ipv4-unicast maximum-paths ibgp	vrf name <text> protocols bgp address-family ipv4-unicast maximum-paths ibgp <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp address-family ipv4-unicast network <network> backdoor	vrf name <text> protocols bgp address-family ipv4-unicast network <text> backdoor

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp address-family ipv4-unicast network <text> <network> route-map	vrf name <text> protocols bgp address-family ipv4-unicast network <text> route-map <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp address-family ipv4-unicast nexthop vpn export	Nothing found in VyOS
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp address-family ipv4-unicast rd vpn export	vrf name <text> protocols bgp address-family ipv4-unicast rd vpn export <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp address-family ipv4-unicast redistribute babel metric	vrf name <text> protocols bgp address-family ipv4-unicast redistribute babel metric <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp address-family ipv4-unicast redistribute babel route-map	vrf name <text> protocols bgp address-family ipv4-unicast redistribute babel route-map <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp address-family ipv4-unicast redistribute connected metric	vrf name <text> protocols bgp address-family ipv4-unicast redistribute connected metric <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp address-family ipv4-unicast redistribute connected route-map	vrf name <text> protocols bgp address-family ipv4-unicast redistribute connected route-map <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp address-family ipv4-unicast redistribute isis metric	vrf name <text> protocols bgp address-family ipv4-unicast redistribute isis metric <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp address-family ipv4-unicast redistribute isis route-map	vrf name <text> protocols bgp address-family ipv4-unicast redistribute isis route-map <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp address-family ipv4-unicast redistribute kernel metric	vrf name <text> protocols bgp address-family ipv4-unicast redistribute kernel metric <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp address-family ipv4-unicast redistribute kernel route-map	vrf name <text> protocols bgp address-family ipv4-unicast redistribute kernel route-map <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp address-family ipv4-unicast redistribute ospf metric	vrf name <text> protocols bgp address-family ipv4-unicast redistribute ospf metric <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp address-family ipv4-unicast redistribute ospf route-map	vrf name <text> protocols bgp address-family ipv4-unicast redistribute ospf route-map <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp address-family ipv4-unicast redistribute rip metric	vrf name <text> protocols bgp address-family ipv4-unicast redistribute rip metric <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp address-family ipv4-unicast redistribute rip route-map	vrf name <text> protocols bgp address-family ipv4-unicast redistribute rip route-map <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp address-family ipv4-unicast redistribute static metric	vrf name <text> protocols bgp address-family ipv4-unicast redistribute static metric <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp address-family ipv4-unicast redistribute static route-map	vrf name <text> protocols bgp address-family ipv4-unicast redistribute static route-map <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp address-family ipv4-unicast redistribute table	vrf name <text> protocols bgp address-family ipv4-unicast redistribute table <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp address-family ipv4-unicast route-map vpn export	vrf name <text> protocols bgp address-family ipv4-unicast route-map vpn export <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp address-family ipv4-unicast route-map vpn import	vrf name <text> protocols bgp address-family ipv4-unicast route-map vpn import <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp address-family ipv4-unicast route-target vpn both	vrf name <text> protocols bgp address-family ipv4-unicast route-target vpn both <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp address-family ipv4-unicast route-target vpn export	vrf name <text> protocols bgp address-family ipv4-unicast route-target vpn export <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp address-family ipv4-unicast route-target vpn import	vrf name <text> protocols bgp address-family ipv4-unicast route-target vpn import <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp address-family ipv4-unicast sid vpn export	Nothing found in VyOS
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp address-family ipv4-vpn network <network> label	vrf name <text> protocols bgp address-family ipv4-vpn network <text> label <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp address-family ipv4-vpn network <network> rd	vrf name <text> protocols bgp address-family ipv4-vpn network <text> rd <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp address-family ipv6-flowspec local-install interface	vrf name <text> protocols bgp address-family ipv6-flowspec local-install interface <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp address-family ipv6-labeled-unicast aggregate-address <aggregate-address> as-set	vrf name <text> protocols bgp address-family ipv6-labeled-unicast aggregate-address <text> as-set

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp address-family ipv6-labeled-unicast aggregate-address <aggregate-address> route-map	vrf name <text> protocols bgp address-family ipv6-labeled-unicast aggregate-address <text> route-map <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp address-family ipv6-labeled-unicast aggregate-address <aggregate-address> summary-only	vrf name <text> protocols bgp address-family ipv6-labeled-unicast aggregate-address <text> summary-only
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp address-family ipv6-labeled-unicast network <network> backdoor	vrf name <text> protocols bgp address-family ipv6-labeled-unicast network <text> backdoor
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp address-family ipv6-labeled-unicast network <network> route-map	vrf name <text> protocols bgp address-family ipv6-labeled-unicast network <text> route-map <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp address-family ipv6-multicast aggregate-address <aggregate-address> as-set	vrf name <text> protocols bgp address-family ipv6-multicast aggregate-address <text> as-set
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp address-family ipv6-multicast aggregate-address <aggregate-address> route-map	vrf name <text> protocols bgp address-family ipv6-multicast aggregate-address <text> route-map <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp address-family ipv6-multicast aggregate-address <aggregate-address> summary-only	vrf name <text> protocols bgp address-family ipv6-multicast aggregate-address <text> summary-only
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp address-family ipv6-multicast distance external	vrf name <text> protocols bgp address-family ipv6-multicast distance external <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp address-family ipv6-multicast distance internal	vrf name <text> protocols bgp address-family ipv6-multicast distance internal <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp address-family ipv6-multicast distance local	vrf name <text> protocols bgp address-family ipv6-multicast distance local <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp address-family ipv6-multicast distance prefix <prefix> distance	vrf name <text> protocols bgp address-family ipv6-multicast distance prefix <text> distance <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp address-family ipv6-multicast network <network> path-limit	vrf name <text> protocols bgp address-family ipv6-multicast network <text> path-limit <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp address-family ipv6-multicast network <network> route-map	vrf name <text> protocols bgp address-family ipv6-multicast network <text> route-map <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp address-family ipv6-unicast aggregate-address <aggregate-address> as-set	vrf name <text> protocols bgp address-family ipv6-unicast aggregate-address <text> as-set
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp address-family ipv6-unicast aggregate-address <aggregate-address> route-map	vrf name <text> protocols bgp address-family ipv6-unicast aggregate-address <text> route-map <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp address-family ipv6-unicast aggregate-address <aggregate-address> summary-only	vrf name <text> protocols bgp address-family ipv6-unicast aggregate-address <text> summary-only
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp address-family ipv6-unicast distance external	vrf name <text> protocols bgp address-family ipv6-unicast distance external <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp address-family ipv6-unicast distance internal	vrf name <text> protocols bgp address-family ipv6-unicast distance internal <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp address-family ipv6-unicast distance local	vrf name <text> protocols bgp address-family ipv6-unicast distance local <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp address-family ipv6-unicast distance prefix <prefix> distance	vrf name <text> protocols bgp address-family ipv6-unicast distance prefix <text> distance <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp address-family ipv6-unicast export vpn	vrf name <text> protocols bgp address-family ipv6-unicast export vpn
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp address-family ipv6-unicast import vpn	vrf name <text> protocols bgp address-family ipv6-unicast import vpn
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp address-family ipv6-unicast import vrf	vrf name <text> protocols bgp address-family ipv6-unicast import vrf <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp address-family ipv6-unicast label vpn allocation-mode per-nexthop	vrf name <text> protocols bgp address-family ipv6-unicast label vpn allocation-mode per-nexthop
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp address-family ipv6-unicast label vpn export	vrf name <text> protocols bgp address-family ipv6-unicast label vpn export <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp address-family ipv6-unicast maximum-paths ebgp	vrf name <text> protocols bgp address-family ipv6-unicast maximum-paths ebgp <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp address-family ipv6-unicast maximum-paths ibgp	vrf name <text> protocols bgp address-family ipv6-unicast maximum-paths ibgp <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp address-family ipv6-unicast network <network> path-limit	vrf name <text> protocols bgp address-family ipv6-unicast network <text> path-limit <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp address-family ipv6-unicast network <network> route-map	vrf name <text> protocols bgp address-family ipv6-unicast network <text> route-map <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp address-family ipv6-unicast nexthop vpn export	Nothing found in VyOS
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp address-family ipv6-unicast rd vpn export	vrf name <text> protocols bgp address-family ipv6-unicast rd vpn export <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp address-family ipv6-unicast redistribute babel metric	vrf name <text> protocols bgp address-family ipv6-unicast redistribute babel metric <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp address-family ipv6-unicast redistribute babel route-map	vrf name <text> protocols bgp address-family ipv6-unicast redistribute babel route-map <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp address-family ipv6-unicast redistribute connected metric	vrf name <text> protocols bgp address-family ipv6-unicast redistribute connected metric <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp address-family ipv6-unicast redistribute connected route-map	vrf name <text> protocols bgp address-family ipv6-unicast redistribute connected route-map <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp address-family ipv6-unicast redistribute kernel metric	vrf name <text> protocols bgp address-family ipv6-unicast redistribute kernel metric <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp address-family ipv6-unicast redistribute kernel route-map	vrf name <text> protocols bgp address-family ipv6-unicast redistribute kernel route-map <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp address-family ipv6-unicast redistribute ospfv3 metric	vrf name <text> protocols bgp address-family ipv6-unicast redistribute ospfv3 metric <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp address-family ipv6-unicast redistribute ospfv3 route-map	vrf name <text> protocols bgp address-family ipv6-unicast redistribute ospfv3 route-map <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp address-family ipv6-unicast redistribute ripng metric	vrf name <text> protocols bgp address-family ipv6-unicast redistribute ripng metric <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp address-family ipv6-unicast redistribute ripng route-map	vrf name <text> protocols bgp address-family ipv6-unicast redistribute ripng route-map <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp address-family ipv6-unicast redistribute static metric	vrf name <text> protocols bgp address-family ipv6-unicast redistribute static metric <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp address-family ipv6-unicast redistribute static route-map	vrf name <text> protocols bgp address-family ipv6-unicast redistribute static route-map <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp address-family ipv6-unicast redistribute table	vrf name <text> protocols bgp address-family ipv6-unicast redistribute table <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp address-family ipv6-unicast route-map vpn export	vrf name <text> protocols bgp address-family ipv6-unicast route-map vpn export <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp address-family ipv6-unicast route-map vpn import	vrf name <text> protocols bgp address-family ipv6-unicast route-map vpn import <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp address-family ipv6-unicast route-target vpn both	vrf name <text> protocols bgp address-family ipv6-unicast route-target vpn both <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp address-family ipv6-unicast route-target vpn export	vrf name <text> protocols bgp address-family ipv6-unicast route-target vpn export <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp address-family ipv6-unicast route-target vpn import	vrf name <text> protocols bgp address-family ipv6-unicast route-target vpn import <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp address-family ipv6-unicast sid vpn export	Nothing found in VyOS
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp address-family ipv6-vpn network <network> label	vrf name <text> protocols bgp address-family ipv6-vpn network <text> label <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp address-family ipv6-vpn network <network> rd	vrf name <text> protocols bgp address-family ipv6-vpn network <text> rd <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp address-family l2vpn-evpn advertise-all-vni	vrf name <text> protocols bgp address-family l2vpn-evpn advertise-all-vni
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp address-family l2vpn-evpn advertise-default-gw	vrf name <text> protocols bgp address-family l2vpn-evpn advertise-default-gw
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp address-family l2vpn-evpn advertise-pip	vrf name <text> protocols bgp address-family l2vpn-evpn advertise-pip <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp address-family l2vpn-evpn advertise-svi-ip	vrf name <text> protocols bgp address-family l2vpn-evpn advertise-svi-ip

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp address-family l2vpn-evpn advertise ipv4 unicast route-map	vrf name <text> protocols bgp address-family l2vpn-evpn advertise ipv4 unicast route-map <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp address-family l2vpn-evpn advertise ipv6 unicast route-map	vrf name <text> protocols bgp address-family l2vpn-evpn advertise ipv6 unicast route-map <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp address-family l2vpn-evpn default-originate ipv4	Nothing found in VyOS
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp address-family l2vpn-evpn default-originate ipv6	Nothing found in VyOS
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp address-family l2vpn-evpn disable-ead-evi-rx	Nothing found in VyOS
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp address-family l2vpn-evpn disable-ead-evi-tx	Nothing found in VyOS
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp address-family l2vpn-evpn ead-es-frag evi-limit	Nothing found in VyOS
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp address-family l2vpn-evpn ead-es-route-target export	Nothing found in VyOS
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp address-family l2vpn-evpn flooding disable	vrf name <text> protocols bgp address-family l2vpn-evpn flooding disable

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp address-family l2vpn-evpn flooding head-end-replication	vrf name <text> protocols bgp address-family l2vpn-evpn flooding head-end-replication
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp address-family l2vpn-evpn mac-vrf soo	Nothing found in VyOS
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp address-family l2vpn-evpn rd	vrf name <text> protocols bgp address-family l2vpn-evpn rd <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp address-family l2vpn-evpn route-target both	vrf name <text> protocols bgp address-family l2vpn-evpn route-target both <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp address-family l2vpn-evpn route-target export	vrf name <text> protocols bgp address-family l2vpn-evpn route-target export <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp address-family l2vpn-evpn route-target import	vrf name <text> protocols bgp address-family l2vpn-evpn route-target import <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp address-family l2vpn-evpn rt-auto-derive	vrf name <text> protocols bgp address-family l2vpn-evpn rt-auto-derive
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp address-family l2vpn-evpn vni <vni> advertise-default-gw	vrf name <text> protocols bgp address-family l2vpn-evpn vni <text> advertise-default-gw
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp address-family l2vpn-evpn vni <vni> advertise-svi-ip	vrf name <text> protocols bgp address-family l2vpn-evpn vni <text> advertise-svi-ip
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp address-family l2vpn-evpn vni <vni> rd	vrf name <text> protocols bgp address-family l2vpn-evpn vni <text> rd <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp address-family l2vpn-evpn vni <vni> route-target both	vrf name <text> protocols bgp address-family l2vpn-evpn vni <text> route-target both <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp address-family l2vpn-evpn vni <vni> route-target export	vrf name <text> protocols bgp address-family l2vpn-evpn vni <text> route-target export <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp address-family l2vpn-evpn vni <vni> route-target import	vrf name <text> protocols bgp address-family l2vpn-evpn vni <text> route-target import <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp bmp mirror-buffer-limit	Nothing found in VyOS
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp bmp target <target> address	Nothing found in VyOS
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp bmp target <target> max-retry	Nothing found in VyOS
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp bmp target <target> min-retry	Nothing found in VyOS
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp bmp target <target> mirror	Nothing found in VyOS
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp bmp target <target> monitor ipv4-unicast post-policy	Nothing found in VyOS
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp bmp target <target> monitor ipv4-unicast pre-policy	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp bmp target <target> monitor ipv6-unicast post-policy	Nothing found in VyOS
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp bmp target <target> monitor ipv6-unicast pre-policy	Nothing found in VyOS
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp bmp target <target> port	Nothing found in VyOS
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp interface <interface> mpls forwarding	vrf name <text> protocols bgp interface <text> mpls forwarding
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp listen limit	vrf name <text> protocols bgp listen limit <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp listen range <range> peer-group	vrf name <text> protocols bgp listen range <text> peer-group <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv4-flowspec filter-list export	vrf name <text> protocols bgp neighbor <text> address-family ipv4-flowspec filter-list export <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv4-flowspec filter-list import	vrf name <text> protocols bgp neighbor <text> address-family ipv4-flowspec filter-list import <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv4-flowspec prefix-list export	vrf name <text> protocols bgp neighbor <text> address-family ipv4-flowspec prefix-list export <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv4-flowspec prefix-list import	vrf name <text> protocols bgp neighbor <text> address-family ipv4-flowspec prefix-list import <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv4-flowspec route-map export	vrf name <text> protocols bgp neighbor <text> address-family ipv4-flowspec route-map export <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv4-flowspec route-map import	vrf name <text> protocols bgp neighbor <text> address-family ipv4-flowspec route-map import <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv4-flowspec route-reflector-client	vrf name <text> protocols bgp neighbor <text> address-family ipv4-flowspec route-reflector-client
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv4-flowspec route-server-client	vrf name <text> protocols bgp neighbor <text> address-family ipv4-flowspec route-server-client
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv4-flowspec soft-reconfiguration inbound	vrf name <text> protocols bgp neighbor <text> address-family ipv4-flowspec soft-reconfiguration inbound
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv4-labeled-unicast addpath-tx-all	vrf name <text> protocols bgp neighbor <text> address-family ipv4-labeled-unicast addpath-tx-all
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv4-labeled-unicast addpath-tx-per-as	vrf name <text> protocols bgp neighbor <text> address-family ipv4-labeled-unicast addpath-tx-per-as
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv4-labeled-unicast allowas-in number	vrf name <text> protocols bgp neighbor <text> address-family ipv4-labeled-unicast allowas-in number <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv4-labeled-unicast as-override	vrf name <text> protocols bgp neighbor <text> address-family ipv4-labeled-unicast as-override
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv4-labeled-unicast attribute-unchanged as-path	vrf name <text> protocols bgp neighbor <text> address-family ipv4-labeled-unicast attribute-unchanged as-path
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv4-labeled-unicast attribute-unchanged med	vrf name <text> protocols bgp neighbor <text> address-family ipv4-labeled-unicast attribute-unchanged med
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv4-labeled-unicast attribute-unchanged next-hop	vrf name <text> protocols bgp neighbor <text> address-family ipv4-labeled-unicast attribute-unchanged next-hop
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv4-labeled-unicast capability orf prefix-list receive	vrf name <text> protocols bgp neighbor <text> address-family ipv4-labeled-unicast capability orf prefix-list receive
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv4-labeled-unicast capability orf prefix-list send	vrf name <text> protocols bgp neighbor <text> address-family ipv4-labeled-unicast capability orf prefix-list send
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv4-labeled-unicast conditionally-advertise advertise-map	vrf name <text> protocols bgp neighbor <text> address-family ipv4-labeled-unicast conditionally-advertise advertise-map <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv4-labeled-unicast conditionally-advertise exist-map	vrf name <text> protocols bgp neighbor <text> address-family ipv4-labeled-unicast conditionally-advertise exist-map <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv4-labeled-unicast conditionally-advertise non-exist-map	vrf name <text> protocols bgp neighbor <text> address-family ipv4-labeled-unicast conditionally-advertise non-exist-map <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv4-labeled-unicast default-originate route-map	vrf name <text> protocols bgp neighbor <text> address-family ipv4-labeled-unicast default-originate route-map <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv4-labeled-unicast disable-send-community extended	vrf name <text> protocols bgp neighbor <text> address-family ipv4-labeled-unicast disable-send-community extended
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv4-labeled-unicast disable-send-community standard	vrf name <text> protocols bgp neighbor <text> address-family ipv4-labeled-unicast disable-send-community standard
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv4-labeled-unicast distribute-list export	vrf name <text> protocols bgp neighbor <text> address-family ipv4-labeled-unicast distribute-list export <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv4-labeled-unicast distribute-list import	vrf name <text> protocols bgp neighbor <text> address-family ipv4-labeled-unicast distribute-list import <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv4-labeled-unicast filter-list export	vrf name <text> protocols bgp neighbor <text> address-family ipv4-labeled-unicast filter-list export <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv4-labeled-unicast filter-list import	vrf name <text> protocols bgp neighbor <text> address-family ipv4-labeled-unicast filter-list import <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv4-labeled-unicast maximum-prefix	vrf name <text> protocols bgp neighbor <text> address-family ipv4-labeled-unicast maximum-prefix <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv4-labeled-unicast maximum-prefix-out	vrf name <text> protocols bgp neighbor <text> address-family ipv4-labeled-unicast maximum-prefix-out <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv4-labeled-unicast nexthop-self force	vrf name <text> protocols bgp neighbor <text> address-family ipv4-labeled-unicast nexthop-self force
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv4-labeled-unicast prefix-list export	vrf name <text> protocols bgp neighbor <text> address-family ipv4-labeled-unicast prefix-list export <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv4-labeled-unicast prefix-list import	vrf name <text> protocols bgp neighbor <text> address-family ipv4-labeled-unicast prefix-list import <value>
×	not yet documented	Nothing found in XML Definitions	vrf name <text> protocols bgp neighbor <text> address-family ipv4-labeled-unicast remove-private-as

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv4-labeled-unicast remove-private-as all	Nothing found in VyOS
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv4-labeled-unicast route-map export	vrf name <text> protocols bgp neighbor <text> address-family ipv4-labeled-unicast route-map export <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv4-labeled-unicast route-map import	vrf name <text> protocols bgp neighbor <text> address-family ipv4-labeled-unicast route-map import <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv4-labeled-unicast route-reflector-client	vrf name <text> protocols bgp neighbor <text> address-family ipv4-labeled-unicast route-reflector-client
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv4-labeled-unicast route-server-client	vrf name <text> protocols bgp neighbor <text> address-family ipv4-labeled-unicast route-server-client
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv4-labeled-unicast soft-reconfiguration inbound	vrf name <text> protocols bgp neighbor <text> address-family ipv4-labeled-unicast soft-reconfiguration inbound
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv4-labeled-unicast unsuppress-map	vrf name <text> protocols bgp neighbor <text> address-family ipv4-labeled-unicast unsuppress-map <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv4-labeled-unicast weight	vrf name <text> protocols bgp neighbor <text> address-family ipv4-labeled-unicast weight <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv4-multicast addpath-tx-all	vrf name <text> protocols bgp neighbor <text> address-family ipv4-multicast addpath-tx-all
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv4-multicast addpath-tx-per-as	vrf name <text> protocols bgp neighbor <text> address-family ipv4-multicast addpath-tx-per-as
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv4-multicast allowas-in number	vrf name <text> protocols bgp neighbor <text> address-family ipv4-multicast allowas-in number <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv4-multicast as-override	vrf name <text> protocols bgp neighbor <text> address-family ipv4-multicast as-override
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv4-multicast attribute-unchanged as-path	vrf name <text> protocols bgp neighbor <text> address-family ipv4-multicast attribute-unchanged as-path
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv4-multicast attribute-unchanged med	vrf name <text> protocols bgp neighbor <text> address-family ipv4-multicast attribute-unchanged med
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv4-multicast attribute-unchanged next-hop	vrf name <text> protocols bgp neighbor <text> address-family ipv4-multicast attribute-unchanged next-hop
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv4-multicast capability orf prefix-list receive	vrf name <text> protocols bgp neighbor <text> address-family ipv4-multicast capability orf prefix-list receive

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv4-multicast capability orf prefix-list send	vrf name <text> protocols bgp neighbor <text> address-family ipv4-multicast capability orf prefix-list send
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv4-multicast conditionally-advertise advertise-map	vrf name <text> protocols bgp neighbor <text> address-family ipv4-multicast conditionally-advertise advertise-map <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv4-multicast conditionally-advertise exist-map	vrf name <text> protocols bgp neighbor <text> address-family ipv4-multicast conditionally-advertise exist-map <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv4-multicast conditionally-advertise non-exist-map	vrf name <text> protocols bgp neighbor <text> address-family ipv4-multicast conditionally-advertise non-exist-map <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv4-multicast default-originate route-map	vrf name <text> protocols bgp neighbor <text> address-family ipv4-multicast default-originate route-map <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv4-multicast disable-send-community extended	vrf name <text> protocols bgp neighbor <text> address-family ipv4-multicast disable-send-community extended
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv4-multicast disable-send-community standard	vrf name <text> protocols bgp neighbor <text> address-family ipv4-multicast disable-send-community standard

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv4-multicast distribute-list export <value>	vrf name <text> protocols bgp neighbor <text> address-family ipv4-multicast distribute-list export <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv4-multicast distribute-list import	vrf name <text> protocols bgp neighbor <text> address-family ipv4-multicast distribute-list import <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv4-multicast filter-list export <value>	vrf name <text> protocols bgp neighbor <text> address-family ipv4-multicast filter-list export <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv4-multicast filter-list import <value>	vrf name <text> protocols bgp neighbor <text> address-family ipv4-multicast filter-list import <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv4-multicast maximum-prefix	vrf name <text> protocols bgp neighbor <text> address-family ipv4-multicast maximum-prefix <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv4-multicast maximum-prefix-out	vrf name <text> protocols bgp neighbor <text> address-family ipv4-multicast maximum-prefix-out <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv4-multicast nexthop-self force	vrf name <text> protocols bgp neighbor <text> address-family ipv4-multicast nexthop-self force
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv4-multicast prefix-list export <value>	vrf name <text> protocols bgp neighbor <text> address-family ipv4-multicast prefix-list export <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv4-multicast prefix-list import	vrf name <text> protocols bgp neighbor <text> address-family ipv4-multicast prefix-list import <value>
×	not yet documented	Nothing found in XML Definitions	vrf name <text> protocols bgp neighbor <text> address-family ipv4-multicast remove-private-as
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv4-multicast remove-private-as all	Nothing found in VyOS
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv4-multicast route-map export	vrf name <text> protocols bgp neighbor <text> address-family ipv4-multicast route-map export <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv4-multicast route-map import	vrf name <text> protocols bgp neighbor <text> address-family ipv4-multicast route-map import <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv4-multicast route-reflector-client	vrf name <text> protocols bgp neighbor <text> address-family ipv4-multicast route-reflector-client
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv4-multicast route-server-client	vrf name <text> protocols bgp neighbor <text> address-family ipv4-multicast route-server-client
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv4-multicast soft-reconfiguration inbound	vrf name <text> protocols bgp neighbor <text> address-family ipv4-multicast soft-reconfiguration inbound

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv4-multicast unsuppress-map	vrf name <text> protocols bgp neighbor <text> address-family ipv4-multicast unsuppress-map <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv4-multicast weight	vrf name <text> protocols bgp neighbor <text> address-family ipv4-multicast weight <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv4-unicast addpath-tx-all	vrf name <text> protocols bgp neighbor <text> address-family ipv4-unicast addpath-tx-all
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv4-unicast addpath-tx-per-as	vrf name <text> protocols bgp neighbor <text> address-family ipv4-unicast addpath-tx-per-as
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv4-unicast allowas-in number	vrf name <text> protocols bgp neighbor <text> address-family ipv4-unicast allowas-in number <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv4-unicast as-override	vrf name <text> protocols bgp neighbor <text> address-family ipv4-unicast as-override
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv4-unicast attribute-unchanged as-path	vrf name <text> protocols bgp neighbor <text> address-family ipv4-unicast attribute-unchanged as-path
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv4-unicast attribute-unchanged med	vrf name <text> protocols bgp neighbor <text> address-family ipv4-unicast attribute-unchanged med

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv4-unicast attribute-unchanged next-hop	vrf name <text> protocols bgp neighbor <text> address-family ipv4-unicast attribute-unchanged next-hop
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv4-unicast capability orf prefix-list receive	vrf name <text> protocols bgp neighbor <text> address-family ipv4-unicast capability orf prefix-list receive
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv4-unicast capability orf prefix-list send	vrf name <text> protocols bgp neighbor <text> address-family ipv4-unicast capability orf prefix-list send
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv4-unicast conditionally-advertise advertise-map	vrf name <text> protocols bgp neighbor <text> address-family ipv4-unicast conditionally-advertise advertise-map <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv4-unicast conditionally-advertise exist-map	vrf name <text> protocols bgp neighbor <text> address-family ipv4-unicast conditionally-advertise exist-map <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv4-unicast conditionally-advertise non-exist-map	vrf name <text> protocols bgp neighbor <text> address-family ipv4-unicast conditionally-advertise non-exist-map <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv4-unicast default-originate route-map	vrf name <text> protocols bgp neighbor <text> address-family ipv4-unicast default-originate route-map <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv4-unicast disable-send-community extended	vrf name <text> protocols bgp neighbor <text> address-family ipv4-unicast disable-send-community extended
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv4-unicast disable-send-community standard	vrf name <text> protocols bgp neighbor <text> address-family ipv4-unicast disable-send-community standard
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv4-unicast distribute-list export	vrf name <text> protocols bgp neighbor <text> address-family ipv4-unicast distribute-list export <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv4-unicast distribute-list import	vrf name <text> protocols bgp neighbor <text> address-family ipv4-unicast distribute-list import <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv4-unicast filter-list export	vrf name <text> protocols bgp neighbor <text> address-family ipv4-unicast filter-list export <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv4-unicast filter-list import	vrf name <text> protocols bgp neighbor <text> address-family ipv4-unicast filter-list import <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv4-unicast maximum-prefix	vrf name <text> protocols bgp neighbor <text> address-family ipv4-unicast maximum-prefix <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv4-unicast maximum-prefix-out	vrf name <text> protocols bgp neighbor <text> address-family ipv4-unicast maximum-prefix-out <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv4-unicast nexthop-self force	vrf name <text> protocols bgp neighbor <text> address-family ipv4-unicast nexthop-self force
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv4-unicast prefix-list export	vrf name <text> protocols bgp neighbor <text> address-family ipv4-unicast prefix-list export <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv4-unicast prefix-list import	vrf name <text> protocols bgp neighbor <text> address-family ipv4-unicast prefix-list import <value>
×	not yet documented	Nothing found in XML Definitions	vrf name <text> protocols bgp neighbor <text> address-family ipv4-unicast remove-private-as
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv4-unicast remove-private-as all	Nothing found in VyOS
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv4-unicast route-map export	vrf name <text> protocols bgp neighbor <text> address-family ipv4-unicast route-map export <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv4-unicast route-map import	vrf name <text> protocols bgp neighbor <text> address-family ipv4-unicast route-map import <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv4-unicast route-reflector-client	vrf name <text> protocols bgp neighbor <text> address-family ipv4-unicast route-reflector-client
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv4-unicast route-server-client	vrf name <text> protocols bgp neighbor <text> address-family ipv4-unicast route-server-client

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv4-unicast soft-reconfiguration inbound	vrf name <text> protocols bgp neighbor <text> address-family ipv4-unicast soft-reconfiguration inbound
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv4-unicast unsuppress-map	vrf name <text> protocols bgp neighbor <text> address-family ipv4-unicast unsuppress-map <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv4-unicast weight	vrf name <text> protocols bgp neighbor <text> address-family ipv4-unicast weight <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv4-vpn addpath-tx-all	vrf name <text> protocols bgp neighbor <text> address-family ipv4-vpn addpath-tx-all
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv4-vpn addpath-tx-per-as	vrf name <text> protocols bgp neighbor <text> address-family ipv4-vpn addpath-tx-per-as
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv4-vpn allowas-in number	vrf name <text> protocols bgp neighbor <text> address-family ipv4-vpn allowas-in number <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv4-vpn as-override	vrf name <text> protocols bgp neighbor <text> address-family ipv4-vpn as-override
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv4-vpn attribute-unchanged as-path	vrf name <text> protocols bgp neighbor <text> address-family ipv4-vpn attribute-unchanged as-path
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv4-vpn attribute-unchanged med	vrf name <text> protocols bgp neighbor <text> address-family ipv4-vpn attribute-unchanged med

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv4-vpn attribute-unchanged next-hop	vrf name <text> protocols bgp neighbor <text> address-family ipv4-vpn attribute-unchanged next-hop
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv4-vpn conditionally-advertise advertise-map	vrf name <text> protocols bgp neighbor <text> address-family ipv4-vpn conditionally-advertise advertise-map <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv4-vpn conditionally-advertise exist-map	vrf name <text> protocols bgp neighbor <text> address-family ipv4-vpn conditionally-advertise exist-map <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv4-vpn conditionally-advertise non-exist-map	vrf name <text> protocols bgp neighbor <text> address-family ipv4-vpn conditionally-advertise non-exist-map <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv4-vpn disable-send-community extended	vrf name <text> protocols bgp neighbor <text> address-family ipv4-vpn disable-send-community extended
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv4-vpn disable-send-community standard	vrf name <text> protocols bgp neighbor <text> address-family ipv4-vpn disable-send-community standard
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv4-vpn distribute-list export	vrf name <text> protocols bgp neighbor <text> address-family ipv4-vpn distribute-list export <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv4-vpn distribute-list import	vrf name <text> protocols bgp neighbor <text> address-family ipv4-vpn distribute-list import <value>

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Table 1 – continued from previous page

Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv4-vpn filter-list export <value>	vrf name <text> protocols bgp neighbor <text> address-family ipv4-vpn filter-list export <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv4-vpn filter-list import <value>	vrf name <text> protocols bgp neighbor <text> address-family ipv4-vpn filter-list import <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv4-vpn maximum-prefix <value>	vrf name <text> protocols bgp neighbor <text> address-family ipv4-vpn maximum-prefix <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv4-vpn maximum-prefix-out <value>	vrf name <text> protocols bgp neighbor <text> address-family ipv4-vpn maximum-prefix-out <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv4-vpn nexthop-self force	vrf name <text> protocols bgp neighbor <text> address-family ipv4-vpn nexthop-self force
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv4-vpn prefix-list export <value>	vrf name <text> protocols bgp neighbor <text> address-family ipv4-vpn prefix-list export <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv4-vpn prefix-list import <value>	vrf name <text> protocols bgp neighbor <text> address-family ipv4-vpn prefix-list import <value>
×	not yet documented	Nothing found in XML Definitions	vrf name <text> protocols bgp neighbor <text> address-family ipv4-vpn remove-private-as
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv4-vpn remove-private-as all	Nothing found in VyOS
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv4-vpn route-map export <value>	vrf name <text> protocols bgp neighbor <text> address-family ipv4-vpn route-map export <value>

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Table 1 – continued from previous page

Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv4-vpn route-map import <value>	vrf name <text> protocols bgp neighbor <text> address-family ipv4-vpn route-map import <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv4-vpn route-reflector-client	vrf name <text> protocols bgp neighbor <text> address-family ipv4-vpn route-reflector-client
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv4-vpn route-server-client	vrf name <text> protocols bgp neighbor <text> address-family ipv4-vpn route-server-client
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv4-vpn soft-reconfiguration inbound	vrf name <text> protocols bgp neighbor <text> address-family ipv4-vpn soft-reconfiguration inbound
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv4-vpn unsuppress-map	vrf name <text> protocols bgp neighbor <text> address-family ipv4-vpn unsuppress-map <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv4-vpn weight	vrf name <text> protocols bgp neighbor <text> address-family ipv4-vpn weight <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv6-flowspec filter-list export	vrf name <text> protocols bgp neighbor <text> address-family ipv6-flowspec filter-list export <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv6-flowspec filter-list import	vrf name <text> protocols bgp neighbor <text> address-family ipv6-flowspec filter-list import <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv6-flowspec prefix-list export	vrf name <text> protocols bgp neighbor <text> address-family ipv6-flowspec prefix-list export <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv6-flowspec prefix-list import	vrf name <text> protocols bgp neighbor <text> address-family ipv6-flowspec prefix-list import <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv6-flowspec route-map export	vrf name <text> protocols bgp neighbor <text> address-family ipv6-flowspec route-map export <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv6-flowspec route-map import	vrf name <text> protocols bgp neighbor <text> address-family ipv6-flowspec route-map import <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv6-flowspec route-reflector-client	vrf name <text> protocols bgp neighbor <text> address-family ipv6-flowspec route-reflector-client
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv6-flowspec route-server-client	vrf name <text> protocols bgp neighbor <text> address-family ipv6-flowspec route-server-client
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv6-flowspec soft-reconfiguration inbound	vrf name <text> protocols bgp neighbor <text> address-family ipv6-flowspec soft-reconfiguration inbound
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv6-labeled-unicast addpath-tx-all	vrf name <text> protocols bgp neighbor <text> address-family ipv6-labeled-unicast addpath-tx-all
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv6-labeled-unicast addpath-tx-per-as	vrf name <text> protocols bgp neighbor <text> address-family ipv6-labeled-unicast addpath-tx-per-as

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv6-labeled-unicast allowas-in number	vrf name <text> protocols bgp neighbor <text> address-family ipv6-labeled-unicast allowas-in number <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv6-labeled-unicast as-override	vrf name <text> protocols bgp neighbor <text> address-family ipv6-labeled-unicast as-override
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv6-labeled-unicast attribute-unchanged as-path	vrf name <text> protocols bgp neighbor <text> address-family ipv6-labeled-unicast attribute-unchanged as-path
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv6-labeled-unicast attribute-unchanged med	vrf name <text> protocols bgp neighbor <text> address-family ipv6-labeled-unicast attribute-unchanged med
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv6-labeled-unicast attribute-unchanged next-hop	vrf name <text> protocols bgp neighbor <text> address-family ipv6-labeled-unicast attribute-unchanged next-hop
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv6-labeled-unicast capability orf prefix-list receive	vrf name <text> protocols bgp neighbor <text> address-family ipv6-labeled-unicast capability orf prefix-list receive
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv6-labeled-unicast capability orf prefix-list send	vrf name <text> protocols bgp neighbor <text> address-family ipv6-labeled-unicast capability orf prefix-list send

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv6-labeled-unicast conditionally-advertise advertise-map	vrf name <text> protocols bgp neighbor <text> address-family ipv6-labeled-unicast conditionally-advertise advertise-map <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv6-labeled-unicast conditionally-advertise exist-map	vrf name <text> protocols bgp neighbor <text> address-family ipv6-labeled-unicast conditionally-advertise exist-map <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv6-labeled-unicast conditionally-advertise non-exist-map	vrf name <text> protocols bgp neighbor <text> address-family ipv6-labeled-unicast conditionally-advertise non-exist-map <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv6-labeled-unicast default-originate route-map	vrf name <text> protocols bgp neighbor <text> address-family ipv6-labeled-unicast default-originate route-map <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv6-labeled-unicast disable-send-community extended	vrf name <text> protocols bgp neighbor <text> address-family ipv6-labeled-unicast disable-send-community extended
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv6-labeled-unicast disable-send-community standard	vrf name <text> protocols bgp neighbor <text> address-family ipv6-labeled-unicast disable-send-community standard
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv6-labeled-unicast distribute-list export	vrf name <text> protocols bgp neighbor <text> address-family ipv6-labeled-unicast distribute-list export <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv6-labeled-unicast distribute-list import	vrf name <text> protocols bgp neighbor <text> address-family ipv6-labeled-unicast distribute-list import <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv6-labeled-unicast filter-list export	vrf name <text> protocols bgp neighbor <text> address-family ipv6-labeled-unicast filter-list export <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv6-labeled-unicast filter-list import	vrf name <text> protocols bgp neighbor <text> address-family ipv6-labeled-unicast filter-list import <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv6-labeled-unicast maximum-prefix	vrf name <text> protocols bgp neighbor <text> address-family ipv6-labeled-unicast maximum-prefix <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv6-labeled-unicast maximum-prefix-out	vrf name <text> protocols bgp neighbor <text> address-family ipv6-labeled-unicast maximum-prefix-out <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv6-labeled-unicast nexthop-local unchanged	vrf name <text> protocols bgp neighbor <text> address-family ipv6-labeled-unicast nexthop-local unchanged
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv6-labeled-unicast nexthop-self force	vrf name <text> protocols bgp neighbor <text> address-family ipv6-labeled-unicast nexthop-self force
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv6-labeled-unicast prefix-list export	vrf name <text> protocols bgp neighbor <text> address-family ipv6-labeled-unicast prefix-list export <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv6-labeled-unicast prefix-list import	vrf name <text> protocols bgp neighbor <text> address-family ipv6-labeled-unicast prefix-list import <value>
×	not yet documented	Nothing found in XML Definitions	vrf name <text> protocols bgp neighbor <text> address-family ipv6-labeled-unicast remove-private-as
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv6-labeled-unicast remove-private-as all	Nothing found in VyOS
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv6-labeled-unicast route-map export	vrf name <text> protocols bgp neighbor <text> address-family ipv6-labeled-unicast route-map export <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv6-labeled-unicast route-map import	vrf name <text> protocols bgp neighbor <text> address-family ipv6-labeled-unicast route-map import <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv6-labeled-unicast route-reflector-client	vrf name <text> protocols bgp neighbor <text> address-family ipv6-labeled-unicast route-reflector-client
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv6-labeled-unicast route-server-client	vrf name <text> protocols bgp neighbor <text> address-family ipv6-labeled-unicast route-server-client
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv6-labeled-unicast soft-reconfiguration inbound	vrf name <text> protocols bgp neighbor <text> address-family ipv6-labeled-unicast soft-reconfiguration inbound

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv6-labeled-unicast unsuppress-map	vrf name <text> protocols bgp neighbor <text> address-family ipv6-labeled-unicast unsuppress-map <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv6-labeled-unicast weight	vrf name <text> protocols bgp neighbor <text> address-family ipv6-labeled-unicast weight <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv6-multicast addpath-tx-all	vrf name <text> protocols bgp neighbor <text> address-family ipv6-multicast addpath-tx-all
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv6-multicast addpath-tx-per-as	vrf name <text> protocols bgp neighbor <text> address-family ipv6-multicast addpath-tx-per-as
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv6-multicast allowas-in number	vrf name <text> protocols bgp neighbor <text> address-family ipv6-multicast allowas-in number <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv6-multicast as-override	vrf name <text> protocols bgp neighbor <text> address-family ipv6-multicast as-override
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv6-multicast attribute-unchanged as-path	vrf name <text> protocols bgp neighbor <text> address-family ipv6-multicast attribute-unchanged as-path
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv6-multicast attribute-unchanged med	vrf name <text> protocols bgp neighbor <text> address-family ipv6-multicast attribute-unchanged med

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv6-multicast attribute-unchanged next-hop	vrf name <text> protocols bgp neighbor <text> address-family ipv6-multicast attribute-unchanged next-hop
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv6-multicast conditionally-advertise advertise-map	vrf name <text> protocols bgp neighbor <text> address-family ipv6-multicast conditionally-advertise advertise-map <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv6-multicast conditionally-advertise exist-map	vrf name <text> protocols bgp neighbor <text> address-family ipv6-multicast conditionally-advertise exist-map <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv6-multicast conditionally-advertise non-exist-map	vrf name <text> protocols bgp neighbor <text> address-family ipv6-multicast conditionally-advertise non-exist-map <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv6-multicast default-originate route-map	vrf name <text> protocols bgp neighbor <text> address-family ipv6-multicast default-originate route-map <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv6-multicast disable-send-community extended	vrf name <text> protocols bgp neighbor <text> address-family ipv6-multicast disable-send-community extended
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv6-multicast disable-send-community standard	vrf name <text> protocols bgp neighbor <text> address-family ipv6-multicast disable-send-community standard

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv6-multicast distribute-list export <value>	vrf name <text> protocols bgp neighbor <text> address-family ipv6-multicast distribute-list export <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv6-multicast distribute-list import <value>	vrf name <text> protocols bgp neighbor <text> address-family ipv6-multicast distribute-list import <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv6-multicast filter-list export <value>	vrf name <text> protocols bgp neighbor <text> address-family ipv6-multicast filter-list export <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv6-multicast filter-list import <value>	vrf name <text> protocols bgp neighbor <text> address-family ipv6-multicast filter-list import <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv6-multicast maximum-prefix <value>	vrf name <text> protocols bgp neighbor <text> address-family ipv6-multicast maximum-prefix <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv6-multicast maximum-prefix-out <value>	vrf name <text> protocols bgp neighbor <text> address-family ipv6-multicast maximum-prefix-out <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv6-multicast nexthop-local unchanged	vrf name <text> protocols bgp neighbor <text> address-family ipv6-multicast nexthop-local unchanged
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv6-multicast nexthop-self force	vrf name <text> protocols bgp neighbor <text> address-family ipv6-multicast nexthop-self force

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv6-multicast prefix-list export	vrf name <text> protocols bgp neighbor <text> address-family ipv6-multicast prefix-list export <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv6-multicast prefix-list import	vrf name <text> protocols bgp neighbor <text> address-family ipv6-multicast prefix-list import <value>
×	not yet documented	Nothing found in XML Definitions	vrf name <text> protocols bgp neighbor <text> address-family ipv6-multicast remove-private-as
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv6-multicast remove-private-as all	Nothing found in VyOS
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv6-multicast route-map export	vrf name <text> protocols bgp neighbor <text> address-family ipv6-multicast route-map export <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv6-multicast route-map import	vrf name <text> protocols bgp neighbor <text> address-family ipv6-multicast route-map import <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv6-multicast route-reflector-client	vrf name <text> protocols bgp neighbor <text> address-family ipv6-multicast route-reflector-client
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv6-multicast route-server-client	vrf name <text> protocols bgp neighbor <text> address-family ipv6-multicast route-server-client

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv6-multicast soft-reconfiguration inbound	vrf name <text> protocols bgp neighbor <text> address-family ipv6-multicast soft-reconfiguration inbound
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv6-multicast unsuppress-map	vrf name <text> protocols bgp neighbor <text> address-family ipv6-multicast unsuppress-map <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv6-multicast weight	vrf name <text> protocols bgp neighbor <text> address-family ipv6-multicast weight <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv6-unicast addpath-tx-all	vrf name <text> protocols bgp neighbor <text> address-family ipv6-unicast addpath-tx-all
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv6-unicast addpath-tx-per-as	vrf name <text> protocols bgp neighbor <text> address-family ipv6-unicast addpath-tx-per-as
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv6-unicast allowas-in number	vrf name <text> protocols bgp neighbor <text> address-family ipv6-unicast allowas-in number <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv6-unicast as-override	vrf name <text> protocols bgp neighbor <text> address-family ipv6-unicast as-override
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv6-unicast attribute-unchanged as-path	vrf name <text> protocols bgp neighbor <text> address-family ipv6-unicast attribute-unchanged as-path

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv6-unicast attribute-unchanged med	vrf name <text> protocols bgp neighbor <text> address-family ipv6-unicast attribute-unchanged med
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv6-unicast attribute-unchanged next-hop	vrf name <text> protocols bgp neighbor <text> address-family ipv6-unicast attribute-unchanged next-hop
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv6-unicast capability orf prefix-list receive	vrf name <text> protocols bgp neighbor <text> address-family ipv6-unicast capability orf prefix-list receive
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv6-unicast capability orf prefix-list send	vrf name <text> protocols bgp neighbor <text> address-family ipv6-unicast capability orf prefix-list send
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv6-unicast conditionally-advertise advertise-map	vrf name <text> protocols bgp neighbor <text> address-family ipv6-unicast conditionally-advertise advertise-map <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv6-unicast conditionally-advertise exist-map	vrf name <text> protocols bgp neighbor <text> address-family ipv6-unicast conditionally-advertise exist-map <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv6-unicast conditionally-advertise non-exist-map	vrf name <text> protocols bgp neighbor <text> address-family ipv6-unicast conditionally-advertise non-exist-map <value>

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Table 1 – continued from previous page

Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv6-unicast default-originate route-map	vrf name <text> protocols bgp neighbor <text> address-family ipv6-unicast default-originate route-map <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv6-unicast disable-send-community extended	vrf name <text> protocols bgp neighbor <text> address-family ipv6-unicast disable-send-community extended
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv6-unicast disable-send-community standard	vrf name <text> protocols bgp neighbor <text> address-family ipv6-unicast disable-send-community standard
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv6-unicast distribute-list export	vrf name <text> protocols bgp neighbor <text> address-family ipv6-unicast distribute-list export <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv6-unicast distribute-list import	vrf name <text> protocols bgp neighbor <text> address-family ipv6-unicast distribute-list import <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv6-unicast filter-list export	vrf name <text> protocols bgp neighbor <text> address-family ipv6-unicast filter-list export <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv6-unicast filter-list import	vrf name <text> protocols bgp neighbor <text> address-family ipv6-unicast filter-list import <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv6-unicast maximum-prefix	vrf name <text> protocols bgp neighbor <text> address-family ipv6-unicast maximum-prefix <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv6-unicast maximum-prefix-out	vrf name <text> protocols bgp neighbor <text> address-family ipv6-unicast maximum-prefix-out <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv6-unicast nexthop-local unchanged	vrf name <text> protocols bgp neighbor <text> address-family ipv6-unicast nexthop-local unchanged
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv6-unicast nexthop-self force	vrf name <text> protocols bgp neighbor <text> address-family ipv6-unicast nexthop-self force
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv6-unicast prefix-list export	vrf name <text> protocols bgp neighbor <text> address-family ipv6-unicast prefix-list export <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv6-unicast prefix-list import	vrf name <text> protocols bgp neighbor <text> address-family ipv6-unicast prefix-list import <value>
×	not yet documented	Nothing found in XML Definitions	vrf name <text> protocols bgp neighbor <text> address-family ipv6-unicast remove-private-as
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv6-unicast remove-private-as all	Nothing found in VyOS
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv6-unicast route-map export	vrf name <text> protocols bgp neighbor <text> address-family ipv6-unicast route-map export <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv6-unicast route-map import	vrf name <text> protocols bgp neighbor <text> address-family ipv6-unicast route-map import <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv6-unicast route-reflector-client	vrf name <text> protocols bgp neighbor <text> address-family ipv6-unicast route-reflector-client
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv6-unicast route-server-client	vrf name <text> protocols bgp neighbor <text> address-family ipv6-unicast route-server-client
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv6-unicast soft-reconfiguration inbound	vrf name <text> protocols bgp neighbor <text> address-family ipv6-unicast soft-reconfiguration inbound
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv6-unicast unsuppress-map	vrf name <text> protocols bgp neighbor <text> address-family ipv6-unicast unsuppress-map <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv6-unicast weight	vrf name <text> protocols bgp neighbor <text> address-family ipv6-unicast weight <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv6-vpn addpath-tx-all	vrf name <text> protocols bgp neighbor <text> address-family ipv6-vpn addpath-tx-all
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv6-vpn addpath-tx-per-as	vrf name <text> protocols bgp neighbor <text> address-family ipv6-vpn addpath-tx-per-as
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv6-vpn allowas-in number	vrf name <text> protocols bgp neighbor <text> address-family ipv6-vpn allowas-in number <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv6-vpn as-override	vrf name <text> protocols bgp neighbor <text> address-family ipv6-vpn as-override

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv6-vpn attribute-unchanged as-path	vrf name <text> protocols bgp neighbor <text> address-family ipv6-vpn attribute-unchanged as-path
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv6-vpn attribute-unchanged med	vrf name <text> protocols bgp neighbor <text> address-family ipv6-vpn attribute-unchanged med
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv6-vpn attribute-unchanged next-hop	vrf name <text> protocols bgp neighbor <text> address-family ipv6-vpn attribute-unchanged next-hop
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv6-vpn conditionally-advertise advertise-map	vrf name <text> protocols bgp neighbor <text> address-family ipv6-vpn conditionally-advertise advertise-map <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv6-vpn conditionally-advertise exist-map	vrf name <text> protocols bgp neighbor <text> address-family ipv6-vpn conditionally-advertise exist-map <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv6-vpn conditionally-advertise non-exist-map	vrf name <text> protocols bgp neighbor <text> address-family ipv6-vpn conditionally-advertise non-exist-map <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv6-vpn disable-send-community extended	vrf name <text> protocols bgp neighbor <text> address-family ipv6-vpn disable-send-community extended
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv6-vpn disable-send-community standard	vrf name <text> protocols bgp neighbor <text> address-family ipv6-vpn disable-send-community standard

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv6-vpn distribute-list export	vrf name <text> protocols bgp neighbor <text> address-family ipv6-vpn distribute-list export <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv6-vpn distribute-list import	vrf name <text> protocols bgp neighbor <text> address-family ipv6-vpn distribute-list import <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv6-vpn filter-list export	vrf name <text> protocols bgp neighbor <text> address-family ipv6-vpn filter-list export <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv6-vpn filter-list import	vrf name <text> protocols bgp neighbor <text> address-family ipv6-vpn filter-list import <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv6-vpn maximum-prefix	vrf name <text> protocols bgp neighbor <text> address-family ipv6-vpn maximum-prefix <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv6-vpn maximum-prefix-out	vrf name <text> protocols bgp neighbor <text> address-family ipv6-vpn maximum-prefix-out <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv6-vpn nexthop-local unchanged	vrf name <text> protocols bgp neighbor <text> address-family ipv6-vpn nexthop-local unchanged
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv6-vpn nexthop-self force	vrf name <text> protocols bgp neighbor <text> address-family ipv6-vpn nexthop-self force
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv6-vpn prefix-list export	vrf name <text> protocols bgp neighbor <text> address-family ipv6-vpn prefix-list export <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv6-vpn prefix-list import <value>	vrf name <text> protocols bgp neighbor <text> address-family ipv6-vpn prefix-list import <value>
×	not yet documented	Nothing found in XML Definitions	vrf name <text> protocols bgp neighbor <text> address-family ipv6-vpn remove-private-as
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv6-vpn remove-private-as all	Nothing found in VyOS
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv6-vpn route-map export	vrf name <text> protocols bgp neighbor <text> address-family ipv6-vpn route-map export <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv6-vpn route-map import	vrf name <text> protocols bgp neighbor <text> address-family ipv6-vpn route-map import <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv6-vpn route-reflector-client	vrf name <text> protocols bgp neighbor <text> address-family ipv6-vpn route-reflector-client
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv6-vpn route-server-client	vrf name <text> protocols bgp neighbor <text> address-family ipv6-vpn route-server-client
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv6-vpn soft-reconfiguration inbound	vrf name <text> protocols bgp neighbor <text> address-family ipv6-vpn soft-reconfiguration inbound
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv6-vpn unsuppress-map	vrf name <text> protocols bgp neighbor <text> address-family ipv6-vpn unsuppress-map <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family ipv6-vpn weight <value>	vrf name <text> protocols bgp neighbor <text> address-family ipv6-vpn weight <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family l2vpn-evpn allowas-in number	vrf name <text> protocols bgp neighbor <text> address-family l2vpn-evpn allowas-in number <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family l2vpn-evpn attribute-unchanged as-path	vrf name <text> protocols bgp neighbor <text> address-family l2vpn-evpn attribute-unchanged as-path
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family l2vpn-evpn attribute-unchanged med	vrf name <text> protocols bgp neighbor <text> address-family l2vpn-evpn attribute-unchanged med
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family l2vpn-evpn attribute-unchanged next-hop	vrf name <text> protocols bgp neighbor <text> address-family l2vpn-evpn attribute-unchanged next-hop
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family l2vpn-evpn nexthop-self force	vrf name <text> protocols bgp neighbor <text> address-family l2vpn-evpn nexthop-self force
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family l2vpn-evpn route-map export	vrf name <text> protocols bgp neighbor <text> address-family l2vpn-evpn route-map export <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family l2vpn-evpn route-map import	vrf name <text> protocols bgp neighbor <text> address-family l2vpn-evpn route-map import <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family l2vpn-evpn route-reflector-client	vrf name <text> protocols bgp neighbor <text> address-family l2vpn-evpn route-reflector-client
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family l2vpn-evpn route-server-client	vrf name <text> protocols bgp neighbor <text> address-family l2vpn-evpn route-server-client
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> address-family l2vpn-evpn soft-reconfiguration inbound	vrf name <text> protocols bgp neighbor <text> address-family l2vpn-evpn soft-reconfiguration inbound
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> advertisement-interval	vrf name <text> protocols bgp neighbor <text> advertisement-interval <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> bfd check-control-plane-failure	vrf name <text> protocols bgp neighbor <text> bfd check-control-plane-failure
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> bfd profile	vrf name <text> protocols bgp neighbor <text> bfd profile <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> capability dynamic	vrf name <text> protocols bgp neighbor <text> capability dynamic
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> capability extended-nexthop	vrf name <text> protocols bgp neighbor <text> capability extended-nexthop
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> capability software-version	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> description	vrf name <text> protocols bgp neighbor <text> description <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> disable-capability-negotiation	vrf name <text> protocols bgp neighbor <text> disable-capability-negotiation
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> disable-connected-check	vrf name <text> protocols bgp neighbor <text> disable-connected-check
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> ebgp-multihop	vrf name <text> protocols bgp neighbor <text> ebgp-multihop <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> enforce-first-as	vrf name <text> protocols bgp neighbor <text> enforce-first-as
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> graceful-restart	vrf name <text> protocols bgp neighbor <text> graceful-restart <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> interface peer-group	vrf name <text> protocols bgp neighbor <text> interface peer-group <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> interface remote-as	vrf name <text> protocols bgp neighbor <text> interface remote-as <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> interface source-interface	vrf name <text> protocols bgp neighbor <text> interface source-interface <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> interface v6only peer-group	vrf name <text> protocols bgp neighbor <text> interface v6only peer-group <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> interface v6only remote-as	vrf name <text> protocols bgp neighbor <text> interface v6only remote-as <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> local-as <local-as> no-prepend replace-as	vrf name <text> protocols bgp neighbor <text> local-as <text> no-prepend replace-as
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> local-role <local-role> strict	vrf name <text> protocols bgp neighbor <text> local-role <text> strict
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> override-capability	vrf name <text> protocols bgp neighbor <text> override-capability
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> passive	vrf name <text> protocols bgp neighbor <text> passive
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> password	vrf name <text> protocols bgp neighbor <text> password <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> path-attribute discard	vrf name <text> protocols bgp neighbor <text> path-attribute discard <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> path-attribute treat-as-withdraw	Nothing found in VyOS
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> peer-group	vrf name <text> protocols bgp neighbor <text> peer-group <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> port	vrf name <text> protocols bgp neighbor <text> port <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> remote-as	vrf name <text> protocols bgp neighbor <text> remote-as <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> shutdown	vrf name <text> protocols bgp neighbor <text> shutdown

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> solo	vrf name <text> protocols bgp neighbor <text> solo
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> strict-capability-match	vrf name <text> protocols bgp neighbor <text> strict-capability-match
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> timers connect	vrf name <text> protocols bgp neighbor <text> timers connect <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> timers holdtime	vrf name <text> protocols bgp neighbor <text> timers holdtime <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> timers keepalive	vrf name <text> protocols bgp neighbor <text> timers keepalive <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> ttl-security hops	vrf name <text> protocols bgp neighbor <text> ttl-security hops <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp neighbor <neighbor> update-source	vrf name <text> protocols bgp neighbor <text> update-source <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp parameters allow-martian-nexthop	Nothing found in VyOS
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp parameters always-compare-med	vrf name <text> protocols bgp parameters always-compare-med
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp parameters best-path as-path confed	vrf name <text> protocols bgp parameters best-path as-path confed
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp parameters best-path as-path ignore	vrf name <text> protocols bgp parameters best-path as-path ignore

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp parameters bestpath as-path multipath-relax	vrf name <text> protocols bgp parameters bestpath as-path multipath-relax
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp parameters bestpath bandwidth	vrf name <text> protocols bgp parameters bestpath bandwidth <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp parameters bestpath compare-routerid	vrf name <text> protocols bgp parameters bestpath compare-routerid
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp parameters bestpath med	vrf name <text> protocols bgp parameters bestpath med <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp parameters bestpath peer-type multipath-relax	vrf name <text> protocols bgp parameters bestpath peer-type multipath-relax
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp parameters cluster-id	vrf name <text> protocols bgp parameters cluster-id <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp parameters conditional-advertisement timer	vrf name <text> protocols bgp parameters conditional-advertisement timer <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp parameters confederation identifier	vrf name <text> protocols bgp parameters confederation identifier <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp parameters confederation peers	vrf name <text> protocols bgp parameters confederation peers <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp parameters dampening half-life	vrf name <text> protocols bgp parameters dampening half-life <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp parameters dampening max-suppress-time	vrf name <text> protocols bgp parameters dampening max-suppress-time <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp parameters dampening re-use	vrf name <text> protocols bgp parameters dampening re-use <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp parameters dampening start-suppress-time	vrf name <text> protocols bgp parameters dampening start-suppress-time <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp parameters default local-pref	vrf name <text> protocols bgp parameters default local-pref <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp parameters deterministic-med	vrf name <text> protocols bgp parameters deterministic-med
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp parameters disable-ebgp-connected-route-check	Nothing found in VyOS
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp parameters distance global external	vrf name <text> protocols bgp parameters distance global external <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp parameters distance global internal	vrf name <text> protocols bgp parameters distance global internal <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp parameters distance global local	vrf name <text> protocols bgp parameters distance global local <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp parameters distance prefix <prefix> distance	vrf name <text> protocols bgp parameters distance prefix <text> distance <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp parameters ebgp-requires-policy	vrf name <text> protocols bgp parameters ebgp-requires-policy
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp parameters fast-convergence	vrf name <text> protocols bgp parameters fast-convergence

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp parameters graceful-restart stalepath-time	vrf name <text> protocols bgp parameters graceful-restart stalepath-time <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp parameters graceful-shutdown	vrf name <text> protocols bgp parameters graceful-shutdown
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp parameters labeled-unicast	Nothing found in VyOS
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp parameters log-neighbor-changes	vrf name <text> protocols bgp parameters log-neighbor-changes
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp parameters minimum-holdtime	vrf name <text> protocols bgp parameters minimum-holdtime <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp parameters network-import-check	vrf name <text> protocols bgp parameters network-import-check
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp parameters no-client-to-client-reflection	vrf name <text> protocols bgp parameters no-client-to-client-reflection
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp parameters no-fast-external-failover	vrf name <text> protocols bgp parameters no-fast-external-failover
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp parameters no-hard-administrative-reset	Nothing found in VyOS
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp parameters no-suppress-duplicates	vrf name <text> protocols bgp parameters no-suppress-duplicates
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp parameters reject-as-sets	vrf name <text> protocols bgp parameters reject-as-sets

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp parameters route-reflector-allow-outbound-policy	vrf name <text> protocols bgp parameters route-reflector-allow-outbound-policy
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp parameters router-id	vrf name <text> protocols bgp parameters router-id <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp parameters shutdown	vrf name <text> protocols bgp parameters shutdown
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp parameters suppress-fib-pending	vrf name <text> protocols bgp parameters suppress-fib-pending
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp parameters tcp-keepalive idle	vrf name <text> protocols bgp parameters tcp-keepalive idle <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp parameters tcp-keepalive interval	vrf name <text> protocols bgp parameters tcp-keepalive interval <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp parameters tcp-keepalive probes	vrf name <text> protocols bgp parameters tcp-keepalive probes <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv4-labeled-unicast addpath-tx-all	Nothing found in VyOS
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv4-labeled-unicast addpath-tx-per-as	Nothing found in VyOS
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv4-labeled-unicast allowas-in number	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv4-labeled-unicast as-override	Nothing found in VyOS
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv4-labeled-unicast attribute-unchanged as-path	Nothing found in VyOS
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv4-labeled-unicast attribute-unchanged med	Nothing found in VyOS
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv4-labeled-unicast attribute-unchanged next-hop	Nothing found in VyOS
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv4-labeled-unicast capability orf prefix-list receive	Nothing found in VyOS
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv4-labeled-unicast capability orf prefix-list send	Nothing found in VyOS
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv4-labeled-unicast conditionally-advertise advertise-map	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv4-labeled-unicast conditionally-advertise exist-map	Nothing found in VyOS
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv4-labeled-unicast conditionally-advertise non-exist-map	Nothing found in VyOS
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv4-labeled-unicast default-originate route-map	Nothing found in VyOS
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv4-labeled-unicast disable-send-community extended	Nothing found in VyOS
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv4-labeled-unicast disable-send-community standard	Nothing found in VyOS
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv4-labeled-unicast distribute-list export	Nothing found in VyOS
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv4-labeled-unicast distribute-list import	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv4-labeled-unicast filter-list export	Nothing found in VyOS
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv4-labeled-unicast filter-list import	Nothing found in VyOS
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv4-labeled-unicast maximum-prefix	Nothing found in VyOS
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv4-labeled-unicast maximum-prefix-out	Nothing found in VyOS
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv4-labeled-unicast nexthop-self force	Nothing found in VyOS
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv4-labeled-unicast prefix-list export	Nothing found in VyOS
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv4-labeled-unicast prefix-list import	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv4-labeled-unicast remove-private-as all	Nothing found in VyOS
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv4-labeled-unicast route-map export	Nothing found in VyOS
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv4-labeled-unicast route-map import	Nothing found in VyOS
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv4-labeled-unicast route-reflector-client	Nothing found in VyOS
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv4-labeled-unicast route-server-client	Nothing found in VyOS
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv4-labeled-unicast soft-reconfiguration inbound	Nothing found in VyOS
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv4-labeled-unicast unsuppress-map	Nothing found in VyOS
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv4-labeled-unicast weight	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv4-unicast addpath-tx-all	vrf name <text> protocols bgp peer-group <text> address-family ipv4-unicast addpath-tx-all
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv4-unicast addpath-tx-per-as	vrf name <text> protocols bgp peer-group <text> address-family ipv4-unicast addpath-tx-per-as
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv4-unicast allowas-in number	vrf name <text> protocols bgp peer-group <text> address-family ipv4-unicast allowas-in number <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv4-unicast as-override	vrf name <text> protocols bgp peer-group <text> address-family ipv4-unicast as-override
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv4-unicast attribute-unchanged as-path	vrf name <text> protocols bgp peer-group <text> address-family ipv4-unicast attribute-unchanged as-path
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv4-unicast attribute-unchanged med	vrf name <text> protocols bgp peer-group <text> address-family ipv4-unicast attribute-unchanged med
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv4-unicast attribute-unchanged next-hop	vrf name <text> protocols bgp peer-group <text> address-family ipv4-unicast attribute-unchanged next-hop

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv4-unicast capability orf prefix-list receive	vrf name <text> protocols bgp peer-group <text> address-family ipv4-unicast capability orf prefix-list receive
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv4-unicast capability orf prefix-list send	vrf name <text> protocols bgp peer-group <text> address-family ipv4-unicast capability orf prefix-list send
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv4-unicast conditionally-advertise advertise-map	vrf name <text> protocols bgp peer-group <text> address-family ipv4-unicast conditionally-advertise advertise-map <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv4-unicast conditionally-advertise exist-map	vrf name <text> protocols bgp peer-group <text> address-family ipv4-unicast conditionally-advertise exist-map <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv4-unicast conditionally-advertise non-exist-map	vrf name <text> protocols bgp peer-group <text> address-family ipv4-unicast conditionally-advertise non-exist-map <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv4-unicast default-originate route-map	vrf name <text> protocols bgp peer-group <text> address-family ipv4-unicast default-originate route-map <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv4-unicast disable-send-community extended	vrf name <text> protocols bgp peer-group <text> address-family ipv4-unicast disable-send-community extended

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv4-unicast disable-send-community standard	vrf name <text> protocols bgp peer-group <text> address-family ipv4-unicast disable-send-community standard
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv4-unicast distribute-list export	vrf name <text> protocols bgp peer-group <text> address-family ipv4-unicast distribute-list export <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv4-unicast distribute-list import	vrf name <text> protocols bgp peer-group <text> address-family ipv4-unicast distribute-list import <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv4-unicast filter-list export	vrf name <text> protocols bgp peer-group <text> address-family ipv4-unicast filter-list export <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv4-unicast filter-list import	vrf name <text> protocols bgp peer-group <text> address-family ipv4-unicast filter-list import <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv4-unicast maximum-prefix	vrf name <text> protocols bgp peer-group <text> address-family ipv4-unicast maximum-prefix <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv4-unicast maximum-prefix-out	vrf name <text> protocols bgp peer-group <text> address-family ipv4-unicast maximum-prefix-out <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv4-unicast nexthop-self force	vrf name <text> protocols bgp peer-group <text> address-family ipv4-unicast nexthop-self force

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Table 1 – continued from previous page

Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv4-unicast prefix-list export	vrf name <text> protocols bgp peer-group <text> address-family ipv4-unicast prefix-list export <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv4-unicast prefix-list import	vrf name <text> protocols bgp peer-group <text> address-family ipv4-unicast prefix-list import <value>
×	not yet documented	Nothing found in XML Definitions	vrf name <text> protocols bgp peer-group <text> address-family ipv4-unicast remove-private-as
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv4-unicast remove-private-as all	Nothing found in VyOS
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv4-unicast route-map export	vrf name <text> protocols bgp peer-group <text> address-family ipv4-unicast route-map export <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv4-unicast route-map import	vrf name <text> protocols bgp peer-group <text> address-family ipv4-unicast route-map import <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv4-unicast route-reflector-client	vrf name <text> protocols bgp peer-group <text> address-family ipv4-unicast route-reflector-client
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv4-unicast route-server-client	vrf name <text> protocols bgp peer-group <text> address-family ipv4-unicast route-server-client

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Table 1 – continued from previous page

Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv4-unicast soft-reconfiguration inbound	vrf name <text> protocols bgp peer-group <text> address-family ipv4-unicast soft-reconfiguration inbound
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv4-unicast unsuppress-map	vrf name <text> protocols bgp peer-group <text> address-family ipv4-unicast unsuppress-map <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv4-unicast weight	vrf name <text> protocols bgp peer-group <text> address-family ipv4-unicast weight <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv4-vpn addpath-tx-all	vrf name <text> protocols bgp peer-group <text> address-family ipv4-vpn addpath-tx-all
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv4-vpn addpath-tx-per-as	vrf name <text> protocols bgp peer-group <text> address-family ipv4-vpn addpath-tx-per-as
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv4-vpn allowas-in number	vrf name <text> protocols bgp peer-group <text> address-family ipv4-vpn allowas-in number <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv4-vpn as-override	vrf name <text> protocols bgp peer-group <text> address-family ipv4-vpn as-override
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv4-vpn attribute-unchanged as-path	vrf name <text> protocols bgp peer-group <text> address-family ipv4-vpn attribute-unchanged as-path

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv4-vpn attribute-unchanged med	vrf name <text> protocols bgp peer-group <text> address-family ipv4-vpn attribute-unchanged med
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv4-vpn attribute-unchanged next-hop	vrf name <text> protocols bgp peer-group <text> address-family ipv4-vpn attribute-unchanged next-hop
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv4-vpn conditionally-advertise advertise-map <value>	vrf name <text> protocols bgp peer-group <text> address-family ipv4-vpn conditionally-advertise advertise-map <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv4-vpn conditionally-advertise exist-map	vrf name <text> protocols bgp peer-group <text> address-family ipv4-vpn conditionally-advertise exist-map <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv4-vpn conditionally-advertise non-exist-map	vrf name <text> protocols bgp peer-group <text> address-family ipv4-vpn conditionally-advertise non-exist-map <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv4-vpn disable-send-community extended	vrf name <text> protocols bgp peer-group <text> address-family ipv4-vpn disable-send-community extended
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv4-vpn disable-send-community standard	vrf name <text> protocols bgp peer-group <text> address-family ipv4-vpn disable-send-community standard
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv4-vpn distribute-list export	vrf name <text> protocols bgp peer-group <text> address-family ipv4-vpn distribute-list export <value>

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Table 1 – continued from previous page

Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv4-vpn distribute-list import	vrf name <text> protocols bgp peer-group <text> address-family ipv4-vpn distribute-list import <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv4-vpn filter-list export	vrf name <text> protocols bgp peer-group <text> address-family ipv4-vpn filter-list export <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv4-vpn filter-list import	vrf name <text> protocols bgp peer-group <text> address-family ipv4-vpn filter-list import <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv4-vpn maximum-prefix	vrf name <text> protocols bgp peer-group <text> address-family ipv4-vpn maximum-prefix <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv4-vpn maximum-prefix-out	vrf name <text> protocols bgp peer-group <text> address-family ipv4-vpn maximum-prefix-out <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv4-vpn nexthop-self force	vrf name <text> protocols bgp peer-group <text> address-family ipv4-vpn nexthop-self force
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv4-vpn prefix-list export	vrf name <text> protocols bgp peer-group <text> address-family ipv4-vpn prefix-list export <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv4-vpn prefix-list import	vrf name <text> protocols bgp peer-group <text> address-family ipv4-vpn prefix-list import <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	Nothing found in XML Definitions	vrf name <text> protocols bgp peer-group <text> address-family ipv4-vpn remove-private-as
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv4-vpn remove-private-as all	Nothing found in VyOS
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv4-vpn route-map export	vrf name <text> protocols bgp peer-group <text> address-family ipv4-vpn route-map export <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv4-vpn route-map import	vrf name <text> protocols bgp peer-group <text> address-family ipv4-vpn route-map import <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv4-vpn route-reflector-client	vrf name <text> protocols bgp peer-group <text> address-family ipv4-vpn route-reflector-client
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv4-vpn route-server-client	vrf name <text> protocols bgp peer-group <text> address-family ipv4-vpn route-server-client
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv4-vpn soft-reconfiguration inbound	vrf name <text> protocols bgp peer-group <text> address-family ipv4-vpn soft-reconfiguration inbound
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv4-vpn unsuppress-map	vrf name <text> protocols bgp peer-group <text> address-family ipv4-vpn unsuppress-map <value>

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Table 1 – continued from previous page

Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv4-vpn weight	vrf name <text> protocols bgp peer-group <text> address-family ipv4-vpn weight <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv6-labeled-unicast addpath-tx-all	Nothing found in VyOS
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv6-labeled-unicast addpath-tx-per-as	Nothing found in VyOS
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv6-labeled-unicast allowas-in number	Nothing found in VyOS
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv6-labeled-unicast as-override	Nothing found in VyOS
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv6-labeled-unicast attribute-unchanged as-path	Nothing found in VyOS
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv6-labeled-unicast attribute-unchanged med	Nothing found in VyOS
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv6-labeled-unicast attribute-unchanged next-hop	Nothing found in VyOS

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Table 1 – continued from previous page

Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv6-labeled-unicast capability orf prefix-list receive	Nothing found in VyOS
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv6-labeled-unicast capability orf prefix-list send	Nothing found in VyOS
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv6-labeled-unicast conditionally-advertise advertise-map	Nothing found in VyOS
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv6-labeled-unicast conditionally-advertise exist-map	Nothing found in VyOS
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv6-labeled-unicast conditionally-advertise non-exist-map	Nothing found in VyOS
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv6-labeled-unicast default-originate route-map	Nothing found in VyOS
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv6-labeled-unicast disable-send-community extended	Nothing found in VyOS

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Table 1 – continued from previous page

Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv6-labeled-unicast disable-send-community standard	Nothing found in VyOS
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv6-labeled-unicast distribute-list export	Nothing found in VyOS
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv6-labeled-unicast distribute-list import	Nothing found in VyOS
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv6-labeled-unicast filter-list export	Nothing found in VyOS
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv6-labeled-unicast filter-list import	Nothing found in VyOS
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv6-labeled-unicast maximum-prefix	Nothing found in VyOS
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv6-labeled-unicast maximum-prefix-out	Nothing found in VyOS

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Table 1 – continued from previous page

Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv6-labeled-unicast nexthop-local unchanged	Nothing found in VyOS
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv6-labeled-unicast nexthop-self force	Nothing found in VyOS
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv6-labeled-unicast prefix-list export	Nothing found in VyOS
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv6-labeled-unicast prefix-list import	Nothing found in VyOS
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv6-labeled-unicast remove-private-as all	Nothing found in VyOS
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv6-labeled-unicast route-map export	Nothing found in VyOS
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv6-labeled-unicast route-map import	Nothing found in VyOS

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Table 1 – continued from previous page

Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv6-labeled-unicast route-reflector-client	Nothing found in VyOS
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv6-labeled-unicast route-server-client	Nothing found in VyOS
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv6-labeled-unicast soft-reconfiguration inbound	Nothing found in VyOS
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv6-labeled-unicast unsuppress-map	Nothing found in VyOS
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv6-labeled-unicast weight	Nothing found in VyOS
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv6-unicast addpath-tx-all	vrf name <text> protocols bgp peer-group <text> address-family ipv6-unicast addpath-tx-all
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv6-unicast addpath-tx-per-as	vrf name <text> protocols bgp peer-group <text> address-family ipv6-unicast addpath-tx-per-as
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv6-unicast allowas-in number	vrf name <text> protocols bgp peer-group <text> address-family ipv6-unicast allowas-in number <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv6-unicast as-override	vrf name <text> protocols bgp peer-group <text> address-family ipv6-unicast as-override
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv6-unicast attribute-unchanged as-path	vrf name <text> protocols bgp peer-group <text> address-family ipv6-unicast attribute-unchanged as-path
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv6-unicast attribute-unchanged med	vrf name <text> protocols bgp peer-group <text> address-family ipv6-unicast attribute-unchanged med
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv6-unicast attribute-unchanged next-hop	vrf name <text> protocols bgp peer-group <text> address-family ipv6-unicast attribute-unchanged next-hop
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv6-unicast capability orf prefix-list receive	vrf name <text> protocols bgp peer-group <text> address-family ipv6-unicast capability orf prefix-list receive
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv6-unicast capability orf prefix-list send	vrf name <text> protocols bgp peer-group <text> address-family ipv6-unicast capability orf prefix-list send
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv6-unicast conditionally-advertise advertise-map	vrf name <text> protocols bgp peer-group <text> address-family ipv6-unicast conditionally-advertise advertise-map <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv6-unicast conditionally-advertise exist-map	vrf name <text> protocols bgp peer-group <text> address-family ipv6-unicast conditionally-advertise exist-map <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv6-unicast conditionally-advertise non-exist-map	vrf name <text> protocols bgp peer-group <text> address-family ipv6-unicast conditionally-advertise non-exist-map <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv6-unicast default-originate route-map	vrf name <text> protocols bgp peer-group <text> address-family ipv6-unicast default-originate route-map <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv6-unicast disable-send-community extended	vrf name <text> protocols bgp peer-group <text> address-family ipv6-unicast disable-send-community extended
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv6-unicast disable-send-community standard	vrf name <text> protocols bgp peer-group <text> address-family ipv6-unicast disable-send-community standard
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv6-unicast distribute-list export	vrf name <text> protocols bgp peer-group <text> address-family ipv6-unicast distribute-list export <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv6-unicast distribute-list import	vrf name <text> protocols bgp peer-group <text> address-family ipv6-unicast distribute-list import <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv6-unicast filter-list export	vrf name <text> protocols bgp peer-group <text> address-family ipv6-unicast filter-list export <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv6-unicast filter-list import	vrf name <text> protocols bgp peer-group <text> address-family ipv6-unicast filter-list import <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv6-unicast maximum-prefix	vrf name <text> protocols bgp peer-group <text> address-family ipv6-unicast maximum-prefix <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv6-unicast maximum-prefix-out	vrf name <text> protocols bgp peer-group <text> address-family ipv6-unicast maximum-prefix-out <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv6-unicast nexthop-local unchanged	vrf name <text> protocols bgp peer-group <text> address-family ipv6-unicast nexthop-local unchanged
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv6-unicast nexthop-self force	vrf name <text> protocols bgp peer-group <text> address-family ipv6-unicast nexthop-self force
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv6-unicast prefix-list export	vrf name <text> protocols bgp peer-group <text> address-family ipv6-unicast prefix-list export <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv6-unicast prefix-list import	vrf name <text> protocols bgp peer-group <text> address-family ipv6-unicast prefix-list import <value>

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Table 1 – continued from previous page

Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	Nothing found in XML Definitions	vrf name <text> protocols bgp peer-group <text> address-family ipv6-unicast remove-private-as
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv6-unicast remove-private-as all	Nothing found in VyOS
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv6-unicast route-map export	vrf name <text> protocols bgp peer-group <text> address-family ipv6-unicast route-map export <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv6-unicast route-map import	vrf name <text> protocols bgp peer-group <text> address-family ipv6-unicast route-map import <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv6-unicast route-reflector-client	vrf name <text> protocols bgp peer-group <text> address-family ipv6-unicast route-reflector-client
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv6-unicast route-server-client	vrf name <text> protocols bgp peer-group <text> address-family ipv6-unicast route-server-client
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv6-unicast soft-reconfiguration inbound	vrf name <text> protocols bgp peer-group <text> address-family ipv6-unicast soft-reconfiguration inbound
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv6-unicast unsuppress-map	vrf name <text> protocols bgp peer-group <text> address-family ipv6-unicast unsuppress-map <value>

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Table 1 – continued from previous page

Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv6-unicast weight	vrf name <text> protocols bgp peer-group <text> address-family ipv6-unicast weight <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv6-vpn addpath-tx-all	vrf name <text> protocols bgp peer-group <text> address-family ipv6-vpn addpath-tx-all
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv6-vpn addpath-tx-per-as	vrf name <text> protocols bgp peer-group <text> address-family ipv6-vpn addpath-tx-per-as
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv6-vpn allowas-in number	vrf name <text> protocols bgp peer-group <text> address-family ipv6-vpn allowas-in number <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv6-vpn as-override	vrf name <text> protocols bgp peer-group <text> address-family ipv6-vpn as-override
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv6-vpn attribute-unchanged as-path	vrf name <text> protocols bgp peer-group <text> address-family ipv6-vpn attribute-unchanged as-path
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv6-vpn attribute-unchanged med	vrf name <text> protocols bgp peer-group <text> address-family ipv6-vpn attribute-unchanged med
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv6-vpn attribute-unchanged next-hop	vrf name <text> protocols bgp peer-group <text> address-family ipv6-vpn attribute-unchanged next-hop

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Table 1 – continued from previous page

Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv6-vpn conditionally-advertise advertise-map <value>	vrf name <text> protocols bgp peer-group <text> address-family ipv6-vpn conditionally-advertise advertise-map <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv6-vpn conditionally-advertise exist-map	vrf name <text> protocols bgp peer-group <text> address-family ipv6-vpn conditionally-advertise exist-map <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv6-vpn conditionally-advertise non-exist-map	vrf name <text> protocols bgp peer-group <text> address-family ipv6-vpn conditionally-advertise non-exist-map <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv6-vpn disable-send-community extended	vrf name <text> protocols bgp peer-group <text> address-family ipv6-vpn disable-send-community extended
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv6-vpn disable-send-community standard	vrf name <text> protocols bgp peer-group <text> address-family ipv6-vpn disable-send-community standard
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv6-vpn distribute-list export	vrf name <text> protocols bgp peer-group <text> address-family ipv6-vpn distribute-list export <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv6-vpn distribute-list import	vrf name <text> protocols bgp peer-group <text> address-family ipv6-vpn distribute-list import <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv6-vpn filter-list export	vrf name <text> protocols bgp peer-group <text> address-family ipv6-vpn filter-list export <value>

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Table 1 – continued from previous page

Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv6-vpn filter-list import	vrf name <text> protocols bgp peer-group <text> address-family ipv6-vpn filter-list import <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv6-vpn maximum-prefix	vrf name <text> protocols bgp peer-group <text> address-family ipv6-vpn maximum-prefix <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv6-vpn maximum-prefix-out	vrf name <text> protocols bgp peer-group <text> address-family ipv6-vpn maximum-prefix-out <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv6-vpn nexthop-local unchanged	vrf name <text> protocols bgp peer-group <text> address-family ipv6-vpn nexthop-local unchanged
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv6-vpn nexthop-self force	vrf name <text> protocols bgp peer-group <text> address-family ipv6-vpn nexthop-self force
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv6-vpn prefix-list export	vrf name <text> protocols bgp peer-group <text> address-family ipv6-vpn prefix-list export <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv6-vpn prefix-list import	vrf name <text> protocols bgp peer-group <text> address-family ipv6-vpn prefix-list import <value>
×	not yet documented	Nothing found in XML Definitions	vrf name <text> protocols bgp peer-group <text> address-family ipv6-vpn remove-private-as

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv6-vpn remove-private-as all	Nothing found in VyOS
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv6-vpn route-map export	vrf name <text> protocols bgp peer-group <text> address-family ipv6-vpn route-map export <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv6-vpn route-map import	vrf name <text> protocols bgp peer-group <text> address-family ipv6-vpn route-map import <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv6-vpn route-reflector-client	vrf name <text> protocols bgp peer-group <text> address-family ipv6-vpn route-reflector-client
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv6-vpn route-server-client	vrf name <text> protocols bgp peer-group <text> address-family ipv6-vpn route-server-client
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv6-vpn soft-reconfiguration inbound	vrf name <text> protocols bgp peer-group <text> address-family ipv6-vpn soft-reconfiguration inbound
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv6-vpn unsuppress-map	vrf name <text> protocols bgp peer-group <text> address-family ipv6-vpn unsuppress-map <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family ipv6-vpn weight	vrf name <text> protocols bgp peer-group <text> address-family ipv6-vpn weight <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family l2vpn-evpn allowas-in number	vrf name <text> protocols bgp peer-group <text> address-family l2vpn-evpn allowas-in number <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family l2vpn-evpn attribute-unchanged as-path	vrf name <text> protocols bgp peer-group <text> address-family l2vpn-evpn attribute-unchanged as-path
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family l2vpn-evpn attribute-unchanged med	vrf name <text> protocols bgp peer-group <text> address-family l2vpn-evpn attribute-unchanged med
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family l2vpn-evpn attribute-unchanged next-hop	vrf name <text> protocols bgp peer-group <text> address-family l2vpn-evpn attribute-unchanged next-hop
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family l2vpn-evpn nexthop-self force	vrf name <text> protocols bgp peer-group <text> address-family l2vpn-evpn nexthop-self force
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family l2vpn-evpn route-map export	vrf name <text> protocols bgp peer-group <text> address-family l2vpn-evpn route-map export <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family l2vpn-evpn route-map import	vrf name <text> protocols bgp peer-group <text> address-family l2vpn-evpn route-map import <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family l2vpn-evpn route-reflector-client	vrf name <text> protocols bgp peer-group <text> address-family l2vpn-evpn route-reflector-client

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family l2vpn-evpn route-server-client	vrf name <text> protocols bgp peer-group <text> address-family l2vpn-evpn route-server-client
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> address-family l2vpn-evpn soft-reconfiguration inbound	vrf name <text> protocols bgp peer-group <text> address-family l2vpn-evpn soft-reconfiguration inbound
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> bfd check-control-plane-failure	vrf name <text> protocols bgp peer-group <text> bfd check-control-plane-failure
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> bfd profile	vrf name <text> protocols bgp peer-group <text> bfd profile <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> capability dynamic	vrf name <text> protocols bgp peer-group <text> capability dynamic
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> capability extended-nexthop	vrf name <text> protocols bgp peer-group <text> capability extended-nexthop
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> capability software-version	Nothing found in VyOS
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> description	vrf name <text> protocols bgp peer-group <text> description <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> disable-capability-negotiation	vrf name <text> protocols bgp peer-group <text> disable-capability-negotiation

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> disable-connected-check	vrf name <text> protocols bgp peer-group <text> disable-connected-check
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> ebgp-multihop	vrf name <text> protocols bgp peer-group <text> ebgp-multihop <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> graceful-restart	vrf name <text> protocols bgp peer-group <text> graceful-restart <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> local-as <local-as> no-prepend replace-as	vrf name <text> protocols bgp peer-group <text> local-as <text> no-prepend replace-as
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> local-role <local-role> strict	vrf name <text> protocols bgp peer-group <text> local-role <text> strict
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> override-capability	vrf name <text> protocols bgp peer-group <text> override-capability
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> passive	vrf name <text> protocols bgp peer-group <text> passive
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> password	vrf name <text> protocols bgp peer-group <text> password <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> path-attribute discard	vrf name <text> protocols bgp peer-group <text> path-attribute discard <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> path-attribute treat-as-withdraw	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> port	vrf name <text> protocols bgp peer-group <text> port <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> remote-as	vrf name <text> protocols bgp peer-group <text> remote-as <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> shutdown	vrf name <text> protocols bgp peer-group <text> shutdown
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> ttl-security hops	vrf name <text> protocols bgp peer-group <text> ttl-security hops <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp peer-group <peer-group> update-source	vrf name <text> protocols bgp peer-group <text> update-source <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp sid vpn per-vrf export	Nothing found in VyOS
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp srv6 locator	Nothing found in VyOS
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp system-as	vrf name <text> protocols bgp system-as <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp timers holdtime	vrf name <text> protocols bgp timers holdtime <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols bgp timers keepalive	vrf name <text> protocols bgp timers keepalive <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols eigrp local-as	Nothing found in VyOS
×	not yet documented	vrf.xml.in: vrf name <name> protocols eigrp maximum-paths	Nothing found in VyOS
×	not yet documented	vrf.xml.in: vrf name <name> protocols eigrp metric weights	Nothing found in VyOS
×	not yet documented	vrf.xml.in: vrf name <name> protocols eigrp network	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	vrf.xml.in: vrf name <name> protocols eigrp passive-interface	Nothing found in VyOS
×	not yet documented	vrf.xml.in: vrf name <name> protocols eigrp redistribute	Nothing found in VyOS
×	not yet documented	vrf.xml.in: vrf name <name> protocols eigrp router-id	Nothing found in VyOS
×	not yet documented	vrf.xml.in: vrf name <name> protocols eigrp variance	Nothing found in VyOS
×	not yet documented	vrf.xml.in: vrf name <name> protocols isis advertise-high-metrics	Nothing found in VyOS
×	not yet documented	vrf.xml.in: vrf name <name> protocols isis advertise-passive-only	Nothing found in VyOS
×	not yet documented	vrf.xml.in: vrf name <name> protocols isis area-password md5 md5	vrf name <text> protocols isis area-password md5 <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols isis area-password plaintext-password	vrf name <text> protocols isis area-password plaintext-password <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols isis default-information originate ipv4 level-1 always	vrf name <text> protocols isis default-information originate ipv4 level-1 always
×	not yet documented	vrf.xml.in: vrf name <name> protocols isis default-information originate ipv4 level-1 metric	vrf name <text> protocols isis default-information originate ipv4 level-1 metric <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols isis default-information originate ipv4 level-1 route-map	vrf name <text> protocols isis default-information originate ipv4 level-1 route-map <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols isis default-information originate ipv4 level-2 always	vrf name <text> protocols isis default-information originate ipv4 level-2 always

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	vrf.xml.in: vrf name <name> protocols isis default-information originate ipv4 level-2 metric	vrf name <text> protocols isis default-information originate ipv4 level-2 metric <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols isis default-information originate ipv4 level-2 route-map	vrf name <text> protocols isis default-information originate ipv4 level-2 route-map <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols isis default-information originate ipv6 level-1 always	vrf name <text> protocols isis default-information originate ipv6 level-1 always
×	not yet documented	vrf.xml.in: vrf name <name> protocols isis default-information originate ipv6 level-1 metric	vrf name <text> protocols isis default-information originate ipv6 level-1 metric <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols isis default-information originate ipv6 level-1 route-map	vrf name <text> protocols isis default-information originate ipv6 level-1 route-map <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols isis default-information originate ipv6 level-2 always	vrf name <text> protocols isis default-information originate ipv6 level-2 always
×	not yet documented	vrf.xml.in: vrf name <name> protocols isis default-information originate ipv6 level-2 metric	vrf name <text> protocols isis default-information originate ipv6 level-2 metric <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols isis default-information originate ipv6 level-2 route-map	vrf name <text> protocols isis default-information originate ipv6 level-2 route-map <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols isis domain-password md5	vrf name <text> protocols isis domain-password md5 <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols isis domain-password plaintext-password	vrf name <text> protocols isis domain-password plaintext-password <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	vrf.xml.in: vrf name <name> protocols isis dynamic-hostname	vrf name <text> protocols isis dynamic-hostname
×	not yet documented	vrf.xml.in: vrf name <name> protocols isis fast-reroute lfa local load-sharing disable level-1	Nothing found in VyOS
×	not yet documented	vrf.xml.in: vrf name <name> protocols isis fast-reroute lfa local load-sharing disable level-2	Nothing found in VyOS
×	not yet documented	vrf.xml.in: vrf name <name> protocols isis fast-reroute lfa local priority-limit critical level-1	Nothing found in VyOS
×	not yet documented	vrf.xml.in: vrf name <name> protocols isis fast-reroute lfa local priority-limit critical level-2	Nothing found in VyOS
×	not yet documented	vrf.xml.in: vrf name <name> protocols isis fast-reroute lfa local priority-limit high level-1	Nothing found in VyOS
×	not yet documented	vrf.xml.in: vrf name <name> protocols isis fast-reroute lfa local priority-limit high level-2	Nothing found in VyOS
×	not yet documented	vrf.xml.in: vrf name <name> protocols isis fast-reroute lfa local priority-limit medium level-1	Nothing found in VyOS
×	not yet documented	vrf.xml.in: vrf name <name> protocols isis fast-reroute lfa local priority-limit medium level-2	Nothing found in VyOS
×	not yet documented	vrf.xml.in: vrf name <name> protocols isis fast-reroute lfa local tiebreaker downstream index <index> level-1	Nothing found in VyOS

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	vrf.xml.in: vrf name <name> protocols isis fast-reroute lfa local tiebreaker downstream index <index> level-2	Nothing found in VyOS
×	not yet documented	vrf.xml.in: vrf name <name> protocols isis fast-reroute lfa local tiebreaker lowest-backup-metric index <index> level-1	Nothing found in VyOS
×	not yet documented	vrf.xml.in: vrf name <name> protocols isis fast-reroute lfa local tiebreaker lowest-backup-metric index <index> level-2	Nothing found in VyOS
×	not yet documented	vrf.xml.in: vrf name <name> protocols isis fast-reroute lfa local tiebreaker node-protecting index <index> level-1	Nothing found in VyOS
×	not yet documented	vrf.xml.in: vrf name <name> protocols isis fast-reroute lfa local tiebreaker node-protecting index <index> level-2	Nothing found in VyOS
×	not yet documented	vrf.xml.in: vrf name <name> protocols isis fast-reroute lfa remote prefix-list <prefix-list> level-1	Nothing found in VyOS
×	not yet documented	vrf.xml.in: vrf name <name> protocols isis fast-reroute lfa remote prefix-list <prefix-list> level-2	Nothing found in VyOS
×	not yet documented	vrf.xml.in: vrf name <name> protocols isis interface <interface> bfd profile	vrf name <text> protocols isis interface <text> bfd profile <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols isis interface <interface> circuit-type	vrf name <text> protocols isis interface <text> circuit-type <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	vrf.xml.in: vrf name <name> protocols isis interface <interface> hello-interval	vrf name <text> protocols isis interface <text> hello-interval <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols isis interface <interface> hello-multiplier	vrf name <text> protocols isis interface <text> hello-multiplier <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols isis interface <interface> hello-padding	vrf name <text> protocols isis interface <text> hello-padding
×	not yet documented	vrf.xml.in: vrf name <name> protocols isis interface <interface> ldp-sync disable	vrf name <text> protocols isis interface <text> ldp-sync disable
×	not yet documented	vrf.xml.in: vrf name <name> protocols isis interface <interface> ldp-sync holddown	vrf name <text> protocols isis interface <text> ldp-sync holddown <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols isis interface <interface> metric	vrf name <text> protocols isis interface <text> metric <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols isis interface <interface> network point-to-point	vrf name <text> protocols isis interface <text> network point-to-point
×	not yet documented	vrf.xml.in: vrf name <name> protocols isis interface <interface> no-three-way-handshake	vrf name <text> protocols isis interface <text> no-three-way-handshake
×	not yet documented	vrf.xml.in: vrf name <name> protocols isis interface <interface> passive	vrf name <text> protocols isis interface <text> passive
×	not yet documented	vrf.xml.in: vrf name <name> protocols isis interface <interface> password md5	vrf name <text> protocols isis interface <text> password md5 <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols isis interface <interface> password plaintext-password	vrf name <text> protocols isis interface <text> password plaintext-password <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	vrf.xml.in: vrf name <name> protocols isis interface <interface> priority	vrf name <text> protocols isis interface <text> priority <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols isis interface <interface> psnp-interval	vrf name <text> protocols isis interface <text> psnp-interval <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols isis ldp-sync hold-down	vrf name <text> protocols isis ldp-sync hold-down <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols isis level	vrf name <text> protocols isis level <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols isis log-adjacency-changes	vrf name <text> protocols isis log-adjacency-changes
×	not yet documented	vrf.xml.in: vrf name <name> protocols isis lsp-gen-interval	vrf name <text> protocols isis lsp-gen-interval <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols isis lsp-mtu	vrf name <text> protocols isis lsp-mtu <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols isis lsp-refresh-interval	vrf name <text> protocols isis lsp-refresh-interval <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols isis max-lsp-lifetime	vrf name <text> protocols isis max-lsp-lifetime <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols isis metric-style	vrf name <text> protocols isis metric-style <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols isis net	vrf name <text> protocols isis net <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols isis purge-originator	vrf name <text> protocols isis purge-originator
×	not yet documented	vrf.xml.in: vrf name <name> protocols isis redistribute ipv4 babel level-1 metric	vrf name <text> protocols isis redistribute ipv4 babel level-1 metric <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols isis redistribute ipv4 babel level-1 route-map	vrf name <text> protocols isis redistribute ipv4 babel level-1 route-map <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	vrf.xml.in: vrf name <name> protocols isis redistribute ipv4 babel level-2 metric	vrf name <text> protocols isis redistribute ipv4 babel level-2 metric <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols isis redistribute ipv4 babel level-2 route-map	vrf name <text> protocols isis redistribute ipv4 babel level-2 route-map <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols isis redistribute ipv4 bgp level-1 metric	vrf name <text> protocols isis redistribute ipv4 bgp level-1 metric <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols isis redistribute ipv4 bgp level-1 route-map	vrf name <text> protocols isis redistribute ipv4 bgp level-1 route-map <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols isis redistribute ipv4 bgp level-2 metric	vrf name <text> protocols isis redistribute ipv4 bgp level-2 metric <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols isis redistribute ipv4 bgp level-2 route-map	vrf name <text> protocols isis redistribute ipv4 bgp level-2 route-map <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols isis redistribute ipv4 connected level-1 metric	vrf name <text> protocols isis redistribute ipv4 connected level-1 metric <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols isis redistribute ipv4 connected level-1 route-map	vrf name <text> protocols isis redistribute ipv4 connected level-1 route-map <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols isis redistribute ipv4 connected level-2 metric	vrf name <text> protocols isis redistribute ipv4 connected level-2 metric <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols isis redistribute ipv4 connected level-2 route-map	vrf name <text> protocols isis redistribute ipv4 connected level-2 route-map <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols isis redistribute ipv4 kernel level-1 metric	vrf name <text> protocols isis redistribute ipv4 kernel level-1 metric <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	vrf.xml.in: vrf name <name> protocols isis redistribute ipv4 kernel level-1 route-map	vrf name <text> protocols isis redistribute ipv4 kernel level-1 route-map <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols isis redistribute ipv4 kernel level-2 metric	vrf name <text> protocols isis redistribute ipv4 kernel level-2 metric <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols isis redistribute ipv4 kernel level-2 route-map	vrf name <text> protocols isis redistribute ipv4 kernel level-2 route-map <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols isis redistribute ipv4 ospf level-1 metric	vrf name <text> protocols isis redistribute ipv4 ospf level-1 metric <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols isis redistribute ipv4 ospf level-1 route-map	vrf name <text> protocols isis redistribute ipv4 ospf level-1 route-map <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols isis redistribute ipv4 ospf level-2 metric	vrf name <text> protocols isis redistribute ipv4 ospf level-2 metric <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols isis redistribute ipv4 ospf level-2 route-map	vrf name <text> protocols isis redistribute ipv4 ospf level-2 route-map <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols isis redistribute ipv4 rip level-1 metric	vrf name <text> protocols isis redistribute ipv4 rip level-1 metric <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols isis redistribute ipv4 rip level-1 route-map	vrf name <text> protocols isis redistribute ipv4 rip level-1 route-map <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols isis redistribute ipv4 rip level-2 metric	vrf name <text> protocols isis redistribute ipv4 rip level-2 metric <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols isis redistribute ipv4 rip level-2 route-map	vrf name <text> protocols isis redistribute ipv4 rip level-2 route-map <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols isis redistribute ipv4 static level-1 metric	vrf name <text> protocols isis redistribute ipv4 static level-1 metric <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	vrf.xml.in: vrf name <name> protocols isis redistribute ipv4 static level-1 route-map	vrf name <text> protocols isis redistribute ipv4 static level-1 route-map <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols isis redistribute ipv4 static level-2 metric	vrf name <text> protocols isis redistribute ipv4 static level-2 metric <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols isis redistribute ipv4 static level-2 route-map	vrf name <text> protocols isis redistribute ipv4 static level-2 route-map <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols isis redistribute ipv6 babel level-1 metric	vrf name <text> protocols isis redistribute ipv6 babel level-1 metric <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols isis redistribute ipv6 babel level-1 route-map	vrf name <text> protocols isis redistribute ipv6 babel level-1 route-map <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols isis redistribute ipv6 babel level-2 metric	vrf name <text> protocols isis redistribute ipv6 babel level-2 metric <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols isis redistribute ipv6 babel level-2 route-map	vrf name <text> protocols isis redistribute ipv6 babel level-2 route-map <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols isis redistribute ipv6 bgp level-1 metric	vrf name <text> protocols isis redistribute ipv6 bgp level-1 metric <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols isis redistribute ipv6 bgp level-1 route-map	vrf name <text> protocols isis redistribute ipv6 bgp level-1 route-map <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols isis redistribute ipv6 bgp level-2 metric	vrf name <text> protocols isis redistribute ipv6 bgp level-2 metric <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols isis redistribute ipv6 bgp level-2 route-map	vrf name <text> protocols isis redistribute ipv6 bgp level-2 route-map <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols isis redistribute ipv6 connected level-1 metric	vrf name <text> protocols isis redistribute ipv6 connected level-1 metric <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	vrf.xml.in: vrf name <name> protocols isis redistribute ipv6 connected level-1 route-map	vrf name <text> protocols isis redistribute ipv6 connected level-1 route-map <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols isis redistribute ipv6 connected level-2 metric	vrf name <text> protocols isis redistribute ipv6 connected level-2 metric <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols isis redistribute ipv6 connected level-2 route-map	vrf name <text> protocols isis redistribute ipv6 connected level-2 route-map <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols isis redistribute ipv6 kernel level-1 metric	vrf name <text> protocols isis redistribute ipv6 kernel level-1 metric <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols isis redistribute ipv6 kernel level-1 route-map	vrf name <text> protocols isis redistribute ipv6 kernel level-1 route-map <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols isis redistribute ipv6 kernel level-2 metric	vrf name <text> protocols isis redistribute ipv6 kernel level-2 metric <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols isis redistribute ipv6 kernel level-2 route-map	vrf name <text> protocols isis redistribute ipv6 kernel level-2 route-map <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols isis redistribute ipv6 ospf6 level-1 metric	vrf name <text> protocols isis redistribute ipv6 ospf6 level-1 metric <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols isis redistribute ipv6 ospf6 level-1 route-map	vrf name <text> protocols isis redistribute ipv6 ospf6 level-1 route-map <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols isis redistribute ipv6 ospf6 level-2 metric	vrf name <text> protocols isis redistribute ipv6 ospf6 level-2 metric <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols isis redistribute ipv6 ospf6 level-2 route-map	vrf name <text> protocols isis redistribute ipv6 ospf6 level-2 route-map <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	vrf.xml.in: vrf name <name> protocols isis redistribute ipv6 ripng level-1 metric	vrf name <text> protocols isis redistribute ipv6 ripng level-1 metric <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols isis redistribute ipv6 ripng level-1 route-map	vrf name <text> protocols isis redistribute ipv6 ripng level-1 route-map <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols isis redistribute ipv6 ripng level-2 metric	vrf name <text> protocols isis redistribute ipv6 ripng level-2 metric <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols isis redistribute ipv6 ripng level-2 route-map	vrf name <text> protocols isis redistribute ipv6 ripng level-2 route-map <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols isis redistribute ipv6 static level-1 metric	vrf name <text> protocols isis redistribute ipv6 static level-1 metric <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols isis redistribute ipv6 static level-1 route-map	vrf name <text> protocols isis redistribute ipv6 static level-1 route-map <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols isis redistribute ipv6 static level-2 metric	vrf name <text> protocols isis redistribute ipv6 static level-2 metric <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols isis redistribute ipv6 static level-2 route-map	vrf name <text> protocols isis redistribute ipv6 static level-2 route-map <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols isis segment-routing global-block high-label-value	vrf name <text> protocols isis segment-routing global-block high-label-value <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols isis segment-routing global-block low-label-value	vrf name <text> protocols isis segment-routing global-block low-label-value <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols isis segment-routing local-block high-label-value	vrf name <text> protocols isis segment-routing local-block high-label-value <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	vrf.xml.in: vrf name <name> protocols isis segment-routing local-block low-label-value	vrf name <text> protocols isis segment-routing local-block low-label-value <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols isis segment-routing maximum-label-depth	vrf name <text> protocols isis segment-routing maximum-label-depth <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols isis segment-routing prefix <prefix> absolute explicit-null	vrf name <text> protocols isis segment-routing prefix <text> absolute explicit-null
×	not yet documented	vrf.xml.in: vrf name <name> protocols isis segment-routing prefix <prefix> absolute no-php-flag	vrf name <text> protocols isis segment-routing prefix <text> absolute no-php-flag
×	not yet documented	vrf.xml.in: vrf name <name> protocols isis segment-routing prefix <prefix> absolute value	vrf name <text> protocols isis segment-routing prefix <text> absolute value <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols isis segment-routing prefix <prefix> index explicit-null	vrf name <text> protocols isis segment-routing prefix <text> index explicit-null
×	not yet documented	vrf.xml.in: vrf name <name> protocols isis segment-routing prefix <prefix> index no-php-flag	vrf name <text> protocols isis segment-routing prefix <text> index no-php-flag
×	not yet documented	vrf.xml.in: vrf name <name> protocols isis segment-routing prefix <prefix> index value	vrf name <text> protocols isis segment-routing prefix <text> index value <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols isis set-attached-bit	vrf name <text> protocols isis set-attached-bit
×	not yet documented	vrf.xml.in: vrf name <name> protocols isis set-overload-bit	vrf name <text> protocols isis set-overload-bit
×	not yet documented	vrf.xml.in: vrf name <name> protocols isis spf-delay-ietf holddown	vrf name <text> protocols isis spf-delay-ietf holddown <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	vrf.xml.in: vrf name <name> protocols isis spf-delay-ietf init-delay	vrf name <text> protocols isis spf-delay-ietf init-delay <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols isis spf-delay-ietf long-delay	vrf name <text> protocols isis spf-delay-ietf long-delay <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols isis spf-delay-ietf short-delay	vrf name <text> protocols isis spf-delay-ietf short-delay <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols isis spf-delay-ietf time-to-learn	vrf name <text> protocols isis spf-delay-ietf time-to-learn <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols isis spf-interval	vrf name <text> protocols isis spf-interval <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols isis traffic-engineering address	vrf name <text> protocols isis traffic-engineering address <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols isis traffic-engineering enable	vrf name <text> protocols isis traffic-engineering enable
×	not yet documented	vrf.xml.in: vrf name <name> protocols ospf access-list <access-list> export	vrf name <text> protocols ospf access-list <text> export <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols ospf aggregation timer	vrf name <text> protocols ospf aggregation timer <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols ospf area <area> area-type normal	vrf name <text> protocols ospf area <text> area-type normal
×	not yet documented	vrf.xml.in: vrf name <name> protocols ospf area <area> area-type nssa default-cost	vrf name <text> protocols ospf area <text> area-type nssa default-cost <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols ospf area <area> area-type nssa no-summary	vrf name <text> protocols ospf area <text> area-type nssa no-summary

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	vrf.xml.in: vrf name <name> protocols ospf area <area> area-type nssa translate	vrf name <text> protocols ospf area <text> area-type nssa translate <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols ospf area <area> area-type stub default-cost	vrf name <text> protocols ospf area <text> area-type stub default-cost <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols ospf area <area> area-type stub no-summary	vrf name <text> protocols ospf area <text> area-type stub no-summary
×	not yet documented	vrf.xml.in: vrf name <name> protocols ospf area <area> authentication	vrf name <text> protocols ospf area <text> authentication <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols ospf area <area> export-list	vrf name <text> protocols ospf area <text> export-list <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols ospf area <area> import-list	vrf name <text> protocols ospf area <text> import-list <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols ospf area <area> network	vrf name <text> protocols ospf area <text> network <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols ospf area <area> range <range> cost	vrf name <text> protocols ospf area <text> range <text> cost <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols ospf area <area> range <range> not-advertise	vrf name <text> protocols ospf area <text> range <text> not-advertise
×	not yet documented	vrf.xml.in: vrf name <name> protocols ospf area <area> range <range> substitute	vrf name <text> protocols ospf area <text> range <text> substitute <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols ospf area <area> shortcut	vrf name <text> protocols ospf area <text> shortcut <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	vrf.xml.in: vrf name <name> protocols ospf area <area> virtual-link <virtual-link> authentication md5 key-id <key-id> md5-key	vrf name <text> protocols ospf area <text> virtual-link <text> authentication md5 key-id <text> md5-key <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols ospf area <area> virtual-link <virtual-link> authentication plaintext-password	vrf name <text> protocols ospf area <text> virtual-link <text> authentication plaintext-password <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols ospf area <area> virtual-link <virtual-link> dead-interval	vrf name <text> protocols ospf area <text> virtual-link <text> dead-interval <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols ospf area <area> virtual-link <virtual-link> hello-interval	vrf name <text> protocols ospf area <text> virtual-link <text> hello-interval <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols ospf area <area> virtual-link <virtual-link> retransmit-interval	vrf name <text> protocols ospf area <text> virtual-link <text> retransmit-interval <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols ospf area <area> virtual-link <virtual-link> transmit-delay	vrf name <text> protocols ospf area <text> virtual-link <text> transmit-delay <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols ospf auto-cost reference-bandwidth	vrf name <text> protocols ospf auto-cost reference-bandwidth <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols ospf capability opaque	vrf name <text> protocols ospf capability opaque
×	not yet documented	vrf.xml.in: vrf name <name> protocols ospf default-information originate	vrf name <text> protocols ospf default-information originate always

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	vrf.xml.in: vrf name <name> protocols ospf default-information originate metric	vrf name <text> protocols ospf default-information originate metric <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols ospf default-information originate metric-type	vrf name <text> protocols ospf default-information originate metric-type <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols ospf default-information originate route-map	vrf name <text> protocols ospf default-information originate route-map <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols ospf default-metric	vrf name <text> protocols ospf default-metric <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols ospf distance global	vrf name <text> protocols ospf distance global <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols ospf distance ospf external	vrf name <text> protocols ospf distance ospf external <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols ospf distance ospf inter-area	vrf name <text> protocols ospf distance ospf inter-area <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols ospf distance ospf intra-area	vrf name <text> protocols ospf distance ospf intra-area <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols ospf graceful-restart grace-period	vrf name <text> protocols ospf graceful-restart grace-period <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols ospf graceful-restart helper enable router-id	vrf name <text> protocols ospf graceful-restart helper enable router-id <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols ospf graceful-restart helper no-strict-lsa-checking	vrf name <text> protocols ospf graceful-restart helper no-strict-lsa-checking

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	vrf.xml.in: vrf name <name> protocols ospf graceful-restart helper planned-only	vrf name <text> protocols ospf graceful-restart helper planned-only
×	not yet documented	vrf.xml.in: vrf name <name> protocols ospf graceful-restart helper supported-grace-time	vrf name <text> protocols ospf graceful-restart helper supported-grace-time <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols ospf interface <interface> area	vrf name <text> protocols ospf interface <text> area <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols ospf interface <interface> authentication md5 key-id <key-id> md5-key	vrf name <text> protocols ospf interface <text> authentication md5 key-id <text> md5-key <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols ospf interface <interface> authentication plaintext-password	vrf name <text> protocols ospf interface <text> authentication plaintext-password <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols ospf interface <interface> bandwidth	vrf name <text> protocols ospf interface <text> bandwidth <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols ospf interface <interface> bfd profile	vrf name <text> protocols ospf interface <text> bfd profile <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols ospf interface <interface> cost	vrf name <text> protocols ospf interface <text> cost <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols ospf interface <interface> dead-interval	vrf name <text> protocols ospf interface <text> dead-interval <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols ospf interface <interface> hello-interval	vrf name <text> protocols ospf interface <text> hello-interval <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols ospf interface <interface> hello-multiplier	vrf name <text> protocols ospf interface <text> hello-multiplier <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	vrf.xml.in: vrf name <name> protocols ospf interface <interface> ldp-sync disable	vrf name <text> protocols ospf interface <text> ldp-sync disable
×	not yet documented	vrf.xml.in: vrf name <name> protocols ospf interface <interface> ldp-sync holddown	vrf name <text> protocols ospf interface <text> ldp-sync holddown <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols ospf interface <interface> mtu-ignore	vrf name <text> protocols ospf interface <text> mtu-ignore
×	not yet documented	vrf.xml.in: vrf name <name> protocols ospf interface <interface> network	vrf name <text> protocols ospf interface <text> network <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols ospf interface <interface> passive disable	vrf name <text> protocols ospf interface <text> passive disable
×	not yet documented	vrf.xml.in: vrf name <name> protocols ospf interface <interface> priority	vrf name <text> protocols ospf interface <text> priority <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols ospf interface <interface> retransmit-interval	vrf name <text> protocols ospf interface <text> retransmit-interval <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols ospf interface <interface> transmit-delay	vrf name <text> protocols ospf interface <text> transmit-delay <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols ospf ldp-sync holddown	vrf name <text> protocols ospf ldp-sync holddown <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols ospf log-adjacency-changes detail	vrf name <text> protocols ospf log-adjacency-changes detail
×	not yet documented	vrf.xml.in: vrf name <name> protocols ospf max-metric router-lsa administrative	vrf name <text> protocols ospf max-metric router-lsa administrative
×	not yet documented	vrf.xml.in: vrf name <name> protocols ospf max-metric router-lsa on-shutdown	vrf name <text> protocols ospf max-metric router-lsa on-shutdown <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	vrf.xml.in: vrf name <name> protocols ospf max-metric router-lsa on-startup	vrf name <text> protocols ospf max-metric router-lsa on-startup <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols ospf maximum-paths	vrf name <text> protocols ospf maximum-paths <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols ospf mpls-te enable	vrf name <text> protocols ospf mpls-te enable
×	not yet documented	vrf.xml.in: vrf name <name> protocols ospf mpls-te router-address	vrf name <text> protocols ospf mpls-te router-address <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols ospf neighbor <neighbor> poll-interval	vrf name <text> protocols ospf neighbor <text> poll-interval <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols ospf neighbor <neighbor> priority	vrf name <text> protocols ospf neighbor <text> priority <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols ospf parameters abr-type	vrf name <text> protocols ospf parameters abr-type <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols ospf parameters opaque-lsa	vrf name <text> protocols ospf parameters opaque-lsa
×	not yet documented	vrf.xml.in: vrf name <name> protocols ospf parameters rfc1583-compatibility	vrf name <text> protocols ospf parameters rfc1583-compatibility
×	not yet documented	vrf.xml.in: vrf name <name> protocols ospf parameters router-id	vrf name <text> protocols ospf parameters router-id <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols ospf passive-interface	vrf name <text> protocols ospf passive-interface <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols ospf redistribute babel metric	vrf name <text> protocols ospf redistribute babel metric <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols ospf redistribute babel metric-type	vrf name <text> protocols ospf redistribute babel metric-type <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	vrf.xml.in: vrf name <name> protocols ospf redistribute babel route-map	vrf name <text> protocols ospf redistribute babel route-map <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols ospf redistribute bgp metric	vrf name <text> protocols ospf redistribute bgp metric <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols ospf redistribute bgp metric-type	vrf name <text> protocols ospf redistribute bgp metric-type <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols ospf redistribute bgp route-map	vrf name <text> protocols ospf redistribute bgp route-map <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols ospf redistribute connected metric	vrf name <text> protocols ospf redistribute connected metric <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols ospf redistribute connected metric-type	vrf name <text> protocols ospf redistribute connected metric-type <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols ospf redistribute connected route-map	vrf name <text> protocols ospf redistribute connected route-map <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols ospf redistribute isis metric	vrf name <text> protocols ospf redistribute isis metric <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols ospf redistribute isis metric-type	vrf name <text> protocols ospf redistribute isis metric-type <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols ospf redistribute isis route-map	vrf name <text> protocols ospf redistribute isis route-map <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols ospf redistribute kernel metric	vrf name <text> protocols ospf redistribute kernel metric <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols ospf redistribute kernel metric-type	vrf name <text> protocols ospf redistribute kernel metric-type <value>

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Table 1 – continued from previous page

Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	vrf.xml.in: vrf name <name> protocols ospf redistribute kernel route-map	vrf name <text> protocols ospf redistribute kernel route-map <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols ospf redistribute rip metric	vrf name <text> protocols ospf redistribute rip metric <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols ospf redistribute rip metric-type	vrf name <text> protocols ospf redistribute rip metric-type <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols ospf redistribute rip route-map	vrf name <text> protocols ospf redistribute rip route-map <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols ospf redistribute static metric	vrf name <text> protocols ospf redistribute static metric <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols ospf redistribute static metric-type	vrf name <text> protocols ospf redistribute static metric-type <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols ospf redistribute static route-map	vrf name <text> protocols ospf redistribute static route-map <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols ospf redistribute table <table> metric	vrf name <text> protocols ospf redistribute table <text> metric <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols ospf redistribute table <table> metric-type	vrf name <text> protocols ospf redistribute table <text> metric-type <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols ospf redistribute table <table> route-map	vrf name <text> protocols ospf redistribute table <text> route-map <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols ospf refresh timers	vrf name <text> protocols ospf refresh timers <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols ospf segment-routing global-block high-label-value	vrf name <text> protocols ospf segment-routing global-block high-label-value <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	vrf.xml.in: vrf name <name> protocols ospf segment-routing global-block low-label-value	vrf name <text> protocols ospf segment-routing global-block low-label-value <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols ospf segment-routing local-block high-label-value	vrf name <text> protocols ospf segment-routing local-block high-label-value <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols ospf segment-routing local-block low-label-value	vrf name <text> protocols ospf segment-routing local-block low-label-value <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols ospf segment-routing maximum-label-depth	vrf name <text> protocols ospf segment-routing maximum-label-depth <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols ospf segment-routing prefix <prefix> index explicit-null	vrf name <text> protocols ospf segment-routing prefix <text> index explicit-null
×	not yet documented	vrf.xml.in: vrf name <name> protocols ospf segment-routing prefix <prefix> index no-php-flag	vrf name <text> protocols ospf segment-routing prefix <text> index no-php-flag
×	not yet documented	vrf.xml.in: vrf name <name> protocols ospf segment-routing prefix <prefix> index value	vrf name <text> protocols ospf segment-routing prefix <text> index value <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols ospf summary-address <summary-address> no-advertise	vrf name <text> protocols ospf summary-address <text> no-advertise
×	not yet documented	vrf.xml.in: vrf name <name> protocols ospf summary-address <summary-address> tag	vrf name <text> protocols ospf summary-address <text> tag <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols ospf timers throttle spf delay	vrf name <text> protocols ospf timers throttle spf delay <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	vrf.xml.in: vrf name <name> protocols ospf timers throttle spf initial-holdtime	vrf name <text> protocols ospf timers throttle spf initial-holdtime <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols ospf timers throttle spf max-holdtime	vrf name <text> protocols ospf timers throttle spf max-holdtime <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols ospfv3 area <area> area-type nssa default-information-originate	vrf name <text> protocols ospfv3 area <text> area-type nssa default-information-originate
×	not yet documented	vrf.xml.in: vrf name <name> protocols ospfv3 area <area> area-type nssa no-summary	vrf name <text> protocols ospfv3 area <text> area-type nssa no-summary
×	not yet documented	vrf.xml.in: vrf name <name> protocols ospfv3 area <area> area-type stub no-summary	vrf name <text> protocols ospfv3 area <text> area-type stub no-summary
×	not yet documented	vrf.xml.in: vrf name <name> protocols ospfv3 area <area> export-list	vrf name <text> protocols ospfv3 area <text> export-list <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols ospfv3 area <area> import-list	vrf name <text> protocols ospfv3 area <text> import-list <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols ospfv3 area <area> range <range> advertise	vrf name <text> protocols ospfv3 area <text> range <text> advertise
×	not yet documented	vrf.xml.in: vrf name <name> protocols ospfv3 area <area> range <range> not-advertise	vrf name <text> protocols ospfv3 area <text> range <text> not-advertise
×	not yet documented	vrf.xml.in: vrf name <name> protocols ospfv3 auto-cost reference-bandwidth	vrf name <text> protocols ospfv3 auto-cost reference-bandwidth <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols ospfv3 default-information originate always	vrf name <text> protocols ospfv3 default-information originate always

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	vrf.xml.in: vrf name <name> protocols ospfv3 default-information originate metric	vrf name <text> protocols ospfv3 default-information originate metric <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols ospfv3 default-information originate metric-type	vrf name <text> protocols ospfv3 default-information originate metric-type <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols ospfv3 default-information originate route-map	vrf name <text> protocols ospfv3 default-information originate route-map <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols ospfv3 distance global	vrf name <text> protocols ospfv3 distance global <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols ospfv3 distance ospfv3 external	vrf name <text> protocols ospfv3 distance ospfv3 external <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols ospfv3 distance ospfv3 inter-area	vrf name <text> protocols ospfv3 distance ospfv3 inter-area <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols ospfv3 distance ospfv3 intra-area	vrf name <text> protocols ospfv3 distance ospfv3 intra-area <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols ospfv3 graceful-restart grace-period	vrf name <text> protocols ospfv3 graceful-restart grace-period <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols ospfv3 graceful-restart helper enable router-id	vrf name <text> protocols ospfv3 graceful-restart helper enable router-id <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols ospfv3 graceful-restart helper lsa-check-disable	vrf name <text> protocols ospfv3 graceful-restart helper lsa-check-disable

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	vrf.xml.in: vrf name <name> protocols ospfv3 graceful-restart helper planned-only	vrf name <text> protocols ospfv3 graceful-restart helper planned-only
×	not yet documented	vrf.xml.in: vrf name <name> protocols ospfv3 graceful-restart helper supported-grace-time	vrf name <text> protocols ospfv3 graceful-restart helper supported-grace-time <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols ospfv3 interface <interface> area	vrf name <text> protocols ospfv3 interface <text> area <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols ospfv3 interface <interface> bfd profile	vrf name <text> protocols ospfv3 interface <text> bfd profile <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols ospfv3 interface <interface> cost	vrf name <text> protocols ospfv3 interface <text> cost <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols ospfv3 interface <interface> dead-interval	vrf name <text> protocols ospfv3 interface <text> dead-interval <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols ospfv3 interface <interface> hello-interval	vrf name <text> protocols ospfv3 interface <text> hello-interval <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols ospfv3 interface <interface> ifmtu	vrf name <text> protocols ospfv3 interface <text> ifmtu <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols ospfv3 interface <interface> instance-id	vrf name <text> protocols ospfv3 interface <text> instance-id <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols ospfv3 interface <interface> mtu-ignore	vrf name <text> protocols ospfv3 interface <text> mtu-ignore
×	not yet documented	vrf.xml.in: vrf name <name> protocols ospfv3 interface <interface> network	vrf name <text> protocols ospfv3 interface <text> network <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	vrf.xml.in: vrf name <name> protocols ospfv3 interface <interface> passive	vrf name <text> protocols ospfv3 interface <text> passive
×	not yet documented	vrf.xml.in: vrf name <name> protocols ospfv3 interface <interface> priority	vrf name <text> protocols ospfv3 interface <text> priority <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols ospfv3 interface <interface> retransmit-interval	vrf name <text> protocols ospfv3 interface <text> retransmit-interval <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols ospfv3 interface <interface> transmit-delay	vrf name <text> protocols ospfv3 interface <text> transmit-delay <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols ospfv3 log-adjacency-changes detail	vrf name <text> protocols ospfv3 log-adjacency-changes detail
×	not yet documented	vrf.xml.in: vrf name <name> protocols ospfv3 parameters router-id	vrf name <text> protocols ospfv3 parameters router-id <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols ospfv3 redistribute babel metric	Nothing found in VyOS
×	not yet documented	vrf.xml.in: vrf name <name> protocols ospfv3 redistribute babel metric-type	Nothing found in VyOS
×	not yet documented	vrf.xml.in: vrf name <name> protocols ospfv3 redistribute babel route-map	vrf name <text> protocols ospfv3 redistribute babel route-map <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols ospfv3 redistribute bgp metric	Nothing found in VyOS
×	not yet documented	vrf.xml.in: vrf name <name> protocols ospfv3 redistribute bgp metric-type	Nothing found in VyOS
×	not yet documented	vrf.xml.in: vrf name <name> protocols ospfv3 redistribute bgp route-map	vrf name <text> protocols ospfv3 redistribute bgp route-map <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	vrf.xml.in: vrf name <name> protocols ospfv3 redistribute connected metric	Nothing found in VyOS
×	not yet documented	vrf.xml.in: vrf name <name> protocols ospfv3 redistribute connected metric-type	Nothing found in VyOS
×	not yet documented	vrf.xml.in: vrf name <name> protocols ospfv3 redistribute connected route-map	vrf name <text> protocols ospfv3 redistribute connected route-map <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols ospfv3 redistribute isis metric	Nothing found in VyOS
×	not yet documented	vrf.xml.in: vrf name <name> protocols ospfv3 redistribute isis metric-type	Nothing found in VyOS
×	not yet documented	vrf.xml.in: vrf name <name> protocols ospfv3 redistribute isis route-map	Nothing found in VyOS
×	not yet documented	vrf.xml.in: vrf name <name> protocols ospfv3 redistribute kernel metric	Nothing found in VyOS
×	not yet documented	vrf.xml.in: vrf name <name> protocols ospfv3 redistribute kernel metric-type	Nothing found in VyOS
×	not yet documented	vrf.xml.in: vrf name <name> protocols ospfv3 redistribute kernel route-map	vrf name <text> protocols ospfv3 redistribute kernel route-map <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols ospfv3 redistribute ripng metric	Nothing found in VyOS
×	not yet documented	vrf.xml.in: vrf name <name> protocols ospfv3 redistribute ripng metric-type	Nothing found in VyOS
×	not yet documented	vrf.xml.in: vrf name <name> protocols ospfv3 redistribute ripng route-map	vrf name <text> protocols ospfv3 redistribute ripng route-map <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	vrf.xml.in: vrf name <name> protocols ospfv3 redistribute static metric	Nothing found in VyOS
×	not yet documented	vrf.xml.in: vrf name <name> protocols ospfv3 redistribute static metric-type	Nothing found in VyOS
×	not yet documented	vrf.xml.in: vrf name <name> protocols ospfv3 redistribute static route-map	vrf name <text> protocols ospfv3 redistribute static route-map <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols static route6 <route6> blackhole distance	vrf name <text> protocols static route6 <text> blackhole distance <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols static route6 <route6> blackhole tag	vrf name <text> protocols static route6 <text> blackhole tag <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols static route6 <route6> description	vrf name <text> protocols static route6 <text> description <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols static route6 <route6> interface <interface> disable	vrf name <text> protocols static route6 <text> interface <text> disable
×	not yet documented	vrf.xml.in: vrf name <name> protocols static route6 <route6> interface <interface> distance	vrf name <text> protocols static route6 <text> interface <text> distance <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols static route6 <route6> interface <interface> segments	Nothing found in VyOS
×	not yet documented	vrf.xml.in: vrf name <name> protocols static route6 <route6> interface <interface> vrf	vrf name <text> protocols static route6 <text> interface <text> vrf <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols static route6 <route6> next-hop <next-hop> bfd multi-hop source <source> profile	vrf name <text> protocols static route6 <text> next-hop <text> bfd multi-hop source <text> profile <value>

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Table 1 – continued from previous page

Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	vrf.xml.in: vrf name <name> protocols static route6 <route6> next-hop <next-hop> bfd profile	vrf name <text> protocols static route6 <text> next-hop <text> bfd profile <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols static route6 <route6> next-hop <next-hop> disable	vrf name <text> protocols static route6 <text> next-hop <text> disable
×	not yet documented	vrf.xml.in: vrf name <name> protocols static route6 <route6> next-hop <next-hop> distance	vrf name <text> protocols static route6 <text> next-hop <text> distance <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols static route6 <route6> next-hop <next-hop> interface	vrf name <text> protocols static route6 <text> next-hop <text> interface <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols static route6 <route6> next-hop <next-hop> segments	Nothing found in VyOS
×	not yet documented	vrf.xml.in: vrf name <name> protocols static route6 <route6> next-hop <next-hop> vrf	vrf name <text> protocols static route6 <text> next-hop <text> vrf <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols static route6 <route6> reject distance	vrf name <text> protocols static route6 <text> reject distance <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols static route6 <route6> reject tag	vrf name <text> protocols static route6 <text> reject tag <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols static route <route> blackhole distance	vrf name <text> protocols static route <text> blackhole distance <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols static route <route> blackhole tag	vrf name <text> protocols static route <text> blackhole tag <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols static route <route> description	vrf name <text> protocols static route <text> description <value>

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Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	vrf.xml.in: vrf name <name> protocols static route <route> dhcp-interface	vrf name <text> protocols static route <text> dhcp-interface <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols static route <route> interface <interface> disable	vrf name <text> protocols static route <text> interface <text> disable
×	not yet documented	vrf.xml.in: vrf name <name> protocols static route <route> interface <interface> distance	vrf name <text> protocols static route <text> interface <text> distance <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols static route <route> interface <interface> vrf	vrf name <text> protocols static route <text> interface <text> vrf <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols static route <route> next-hop <next-hop> bfd multi-hop source <source> profile	vrf name <text> protocols static route <text> next-hop <text> bfd multi-hop source <text> profile <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols static route <route> next-hop <next-hop> bfd profile	vrf name <text> protocols static route <text> next-hop <text> bfd profile <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols static route <route> next-hop <next-hop> disable	vrf name <text> protocols static route <text> next-hop <text> disable
×	not yet documented	vrf.xml.in: vrf name <name> protocols static route <route> next-hop <next-hop> distance	vrf name <text> protocols static route <text> next-hop <text> distance <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols static route <route> next-hop <next-hop> interface	vrf name <text> protocols static route <text> next-hop <text> interface <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols static route <route> next-hop <next-hop> vrf	vrf name <text> protocols static route <text> next-hop <text> vrf <value>

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Table 1 – continued from previous page

Status 955/10086	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	vrf.xml.in: vrf name <name> protocols static route <route> reject distance	vrf name <text> protocols static route <text> reject distance <value>
×	not yet documented	vrf.xml.in: vrf name <name> protocols static route <route> reject tag	vrf name <text> protocols static route <text> reject tag <value>
×	not yet documented	vrf.xml.in: vrf name <name> table	vrf name <text> table <value>
×	not yet documented	vrf.xml.in: vrf name <name> vni	vrf name <text> vni <value>

15.2 Operational Commands

Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<i>/config/scripts/vpp-collect-diagnosti</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	Nothing found in XML Definitions	add container image <text>
×	<i>add container image <containername></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>add raid <RAID-1-device> member <disk-partition></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	Nothing found in XML Definitions	add raid <text> member <text>
×	not yet documented	Nothing found in XML Definitions	add system image <text>
×	<i>add system image <url path> [latest] [vrf name] [username user [password pass]]</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	Nothing found in XML Definitions	add system image <text> username <text> password <text>
×	not yet documented	Nothing found in XML Definitions	add system image <text> vrf <text>
×	not yet documented	Nothing found in XML Definitions	add system image <text> vrf <text> username <text> password <text>
×	not yet documented	terminal.xml.in: set builtin <builtin>	set builtin <text>

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Table 2 – continued from previous page

Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	Nothing found in XML Definitions	clear console
×	not yet documented	clear-dhcp-server-lease.xml.in: clear dhcp-server lease <lease>	clear dhcp-server lease <text>
×	not yet documented	Nothing found in XML Definitions	clear dns dynamic cache
×	not yet documented	flow-accounting-op.xml.in: clear flow-accounting counters	clear flow-accounting counters
×	not yet documented	counters.xml.in: clear interfaces bonding <bonding> counters	clear interfaces bonding <text> counters
×	not yet documented	counters.xml.in: clear interfaces bridge <bridge> counters	clear interfaces bridge <text> counters
×	not yet documented	Nothing found in XML Definitions	clear interfaces connection <text>
×	not yet documented	counters.xml.in: clear interfaces counters	clear interfaces counters
×	not yet documented	counters.xml.in: clear interfaces dummy <dummy> counters	clear interfaces dummy <text> counters
×	not yet documented	counters.xml.in: clear interfaces ethernet <ethernet> counters	clear interfaces ethernet <text> counters
×	not yet documented	counters.xml.in: clear interfaces geneve <geneve> counters	clear interfaces geneve <text> counters
×	not yet documented	counters.xml.in: clear interfaces input <input> counters	clear interfaces input <text> counters
×	not yet documented	counters.xml.in: clear interfaces l2tpv3 <l2tpv3> counters	clear interfaces l2tpv3 <text> counters
×	not yet documented	counters.xml.in: clear interfaces loopback <loopback> counters	clear interfaces loopback <text> counters
×	not yet documented	counters.xml.in: clear interfaces macsec <macsec> counters	clear interfaces macsec <text> counters
×	not yet documented	counters.xml.in: clear interfaces openvpn <openvpn> counters	clear interfaces openvpn <text> counters
×	not yet documented	counters.xml.in: clear interfaces pppoe <pppoe> counters	clear interfaces pppoe <text> counters

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Table 2 – continued from previous page

Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	counters.xml.in: clear interfaces pseudo- ethernet <pseudo- ethernet> counters	clear interfaces pseudo- ethernet <text> counters
×	not yet documented	counters.xml.in: clear interfaces sstp <sstp> counters	clear interfaces sstp <text> counters
×	not yet documented	counters.xml.in: clear interfaces tunnel <tunnel> counters	clear interfaces tunnel <text> counters
×	not yet documented	counters.xml.in: clear interfaces virtual- ethernet <virtual- ethernet> counters	clear interfaces virtual- ethernet <text> counters
×	not yet documented	counters.xml.in: clear interfaces vti <vti> counters	clear interfaces vti <text> counters
×	not yet documented	counters.xml.in: clear interfaces vxlan <vxlan> counters	clear interfaces vxlan <text> counters
×	not yet documented	counters.xml.in: clear interfaces wireguard <wireguard> counters	clear interfaces wireguard <text> counters
×	not yet documented	counters.xml.in: clear interfaces wireless <wireless> counters	clear interfaces wireless <text> counters
×	not yet documented	counters.xml.in: clear interfaces wwan <wwan> counters	clear interfaces wwan <text> counters
×	not yet documented	clear-ip.xml.in: clear ip prefix-list <prefix- list>	clear ip prefix-list <text> <text>
×	not yet documented	clear-ip.xml.in: clear ip prefix-list <prefix- list> node.tag	Nothing found in VyOS
×	not yet documented	clear-ipv6.xml.in: clear ipv6 prefix-list <prefix-list>	clear ipv6 prefix-list <text> <text>
×	not yet documented	clear-ipv6.xml.in: clear ipv6 prefix-list <prefix-list> node.tag	Nothing found in VyOS
×	not yet documented	clear-log.xml.in: clear log	clear log
×	not yet documented	clear-session.xml.in: clear session <session>	clear session <text>
×	not yet documented	Nothing found in XML Definitions	clone system config <text>
×	not yet documented	Nothing found in XML Definitions	clone system config <text> from <text>

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Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	configure.xml.in: configure	configure
×	not yet documented	connect.xml.in: connect console <con- sole>	connect console <text>
×	<i>connect console <device / alias></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	Nothing found in XML Definitions	connect container <text>
×	not yet documented	connect.xml.in: connect interface <inter- face>	connect interface <text>
×	<i>connect interface <interface></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	terminal.xml.in: set console keymap	set console keymap
×	not yet documented	Nothing found in XML Definitions	copy file <text> to <text>
×	not yet documented	date.xml.in: set date <date>	set date <text>
×	not yet documented	Nothing found in XML Definitions	delete container image <text>
×	<i>delete container image <image id/all> [force]</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	Nothing found in XML Definitions	delete file <text>
×	not yet documented	Nothing found in XML Definitions	delete log file <text>
×	<i>delete raid <RAID-1-device> member <disk-partition></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	raid.xml.in: delete raid <raid> by-id member <member>	Nothing found in VyOS
×	not yet documented	raid.xml.in: delete raid <raid> member <member>	delete raid <text> member <text>
×	<i>delete system image [image-name]</i>	Nothing found in XML Definitions	delete system image <text>
×	not yet documented	disconnect.xml.in: disconnect interface <in- terface>	disconnect interface <text>
×	<i>disconnect interface <interface></i>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	force-arp.xml.in: force arp duplicate interface <interface> address <address>	force arp duplicate interface <text> address <text>
×	not yet documented	force-arp.xml.in: force arp reply interface <interface> address <address>	force arp reply interface <text> address <text>
×	not yet documented	force-arp.xml.in: force arp reply interface <interface> address <address> count <count>	force arp reply interface <text> address <text> count <text>
×	not yet documented	force-arp.xml.in: force arp request interface <interface> address <address>	force arp request interface <text> address <text>
×	not yet documented	force-arp.xml.in: force arp request interface <interface> address <address> count <count>	force arp request interface <text> address <text> count <text>
×	not yet documented	Nothing found in XML Definitions	force cluster active
×	not yet documented	Nothing found in XML Definitions	force cluster standby
×	not yet documented	force-commit-archive.xml.in: force commit-archive	Nothing found in VyOS
×	<i>force ipv6-nd interface <interface> address <ipv6-address></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	force-ipv6-nd.xml.in: force ipv6-nd interface <interface> address <address>	force ipv6-nd interface <text> address <text>
×	not yet documented	force-ipv6-rd.xml.in: force ipv6-rd interface <interface>	force ipv6-rd interface <text>
×	<i>force ipv6-rd interface <interface> [address <ipv6-address>]</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	force-ipv6-rd.xml.in: force ipv6-rd interface <interface> address <address>	force ipv6-rd interface <text> address <text>
×	not yet documented	force-mtu-host.xml.in: force mtu host <host>	force mtu host <text>

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Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	force-mtu-host.xml.in: force mtu host <host> in- terface <interface>	force mtu host <text> in- terface <text>
×	not yet documented	force-netns.xml.in: force netns <netns>	force netns <text>
×	not yet documented	ntp.xml.in: force ntp synchronization	Nothing found in VyOS
×	not yet documented	ntp.xml.in: force ntp synchronization vrf <vrf>	Nothing found in VyOS
×	not yet documented	force-wamp.xml.in: force owping <owping>	force owping <text>
×	not yet documented	force-root-partition-auto- resize.xml.in: force root-partition-auto- resize	force root-partition-auto- resize
×	not yet documented	force-wamp.xml.in: force twping <twping>	force twping <text>
×	not yet documented	force-vrf.xml.in: force vrf <vrf>	force vrf <text>
×	<i>force vrf <name></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>format disk <disk-device1> like <disk-device2></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	Nothing found in XML Definitions	format disk <text> like <text>
×	not yet documented	Nothing found in XML Definitions	generate container image <text> path <text>
×	not yet documented	generate_firewall_rule- resequence.xml.in: generate firewall rule- resequence	Nothing found in VyOS
×	not yet documented	generate_firewall_rule- resequence.xml.in: generate firewall rule- resequence start <start>	Nothing found in VyOS
×	not yet documented	generate_firewall_rule- resequence.xml.in: generate firewall rule- resequence start <start> step <step>	Nothing found in VyOS
×	not yet documented	generate-interfaces- debug-archive.xml.in: generate interfaces debug- archive	generate interfaces debug- archive

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Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	generate-ipsec-debug-archive.xml.in: generate ipsec debug-archive	generate ipsec debug-archive
×	not yet documented	generate-ipsec-profile.xml.in: generate ipsec profile ios-remote-access <ios-remote-access> remote <remote>	generate ipsec profile ios-remote-access <text> remote <text>
×	not yet documented	generate-ipsec-profile.xml.in: generate ipsec profile ios-remote-access <ios-remote-access> remote <remote> name <name>	generate ipsec profile ios-remote-access <text> remote <text> name <text>
×	not yet documented	generate-ipsec-profile.xml.in: generate ipsec profile ios-remote-access <ios-remote-access> remote <remote> name <name> profile <profile>	generate ipsec profile ios-remote-access <text> remote <text> name <text> profile <text>
×	not yet documented	generate-ipsec-profile.xml.in: generate ipsec profile ios-remote-access <ios-remote-access> remote <remote> profile <profile>	generate ipsec profile ios-remote-access <text> remote <text> profile <text>
×	not yet documented	generate-ipsec-profile.xml.in: generate ipsec profile ios-remote-access <ios-remote-access> remote <remote> profile <profile> name <name>	generate ipsec profile ios-remote-access <text> remote <text> profile <text> name <text>
×	not yet documented	generate-ipsec-profile.xml.in: generate ipsec profile windows-remote-access <windows-remote-access> remote <remote>	generate ipsec profile windows-remote-access <text> remote <text>

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Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	generate-ipsec-profile.xml.in: generate ipsec profile windows-remote-access <windows-remote-access> remote <remote> name <name>	generate ipsec profile windows-remote-access <text> remote <text> name <text>
×	not yet documented	generate-macsec-key.xml.in: generate macsec mka cak gcm-aes-128	generate macsec mka cak gcm-aes-128
×	not yet documented	generate-macsec-key.xml.in: generate macsec mka cak gcm-aes-256	generate macsec mka cak gcm-aes-256
×	not yet documented	generate-macsec-key.xml.in: generate macsec mka ckn	generate macsec mka ckn
×	not yet documented	generate-openconnect-user-key.xml.in: generate openconnect username <username> otp-key hotp-time	generate openconnect username <text> otp-key hotp-time
×	not yet documented	generate-openconnect-user-key.xml.in: generate openconnect username <username> otp-key hotp-time digits <digits>	generate openconnect username <text> otp-key hotp-time digits <text>
×	not yet documented	generate-openconnect-user-key.xml.in: generate openconnect username <username> otp-key hotp-time digits <digits> interval <interval>	generate openconnect username <text> otp-key hotp-time digits <text> interval <text>
×	not yet documented	generate-openconnect-user-key.xml.in: generate openconnect username <username> otp-key hotp-time interval <interval>	generate openconnect username <text> otp-key hotp-time interval <text>
×	not yet documented	generate-openconnect-user-key.xml.in: generate openconnect username <username> otp-key hotp-time interval <interval> digits <digits>	generate openconnect username <text> otp-key hotp-time interval <text> digits <text>

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Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<i>generate openvpn client-config interface <interface> ca <name> certificate <name></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	generate-openvpn-config-client.xml.in: generate openvpn client-config interface <interface> ca <ca> certificate <certificate>	generate openvpn client-config interface <text> ca <text> certificate <text>
×	not yet documented	generate-openvpn-config-client.xml.in: generate openvpn client-config interface <interface> ca <ca> certificate <certificate> key <key>	generate openvpn client-config interface <text> ca <text> certificate <text> key <text>
×	<i>generate pki ca</i>	Nothing found in XML Definitions	generate pki ca
×	not yet documented	Nothing found in XML Definitions	generate pki ca file <text>
×	not yet documented	Nothing found in XML Definitions	generate pki ca install <text>
×	<i>generate pki ca install <name></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	Nothing found in XML Definitions	generate pki ca sign <text>
×	<i>generate pki ca sign <ca-name></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>generate pki ca sign <ca-name> install <name></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	Nothing found in XML Definitions	generate pki ca sign <text> file <text>
×	not yet documented	Nothing found in XML Definitions	generate pki ca sign <text> install <text>
×	<i>generate pki certificate</i>	Nothing found in XML Definitions	generate pki certificate
×	not yet documented	Nothing found in XML Definitions	generate pki certificate file <text>
×	not yet documented	Nothing found in XML Definitions	generate pki certificate install <text>
×	<i>generate pki certificate install <name></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>generate pki certificate self-signed</i>	Nothing found in XML Definitions	generate pki certificate self-signed

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Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	Nothing found in XML Definitions	generate pki certificate self-signed file <text>
×	not yet documented	Nothing found in XML Definitions	generate pki certificate self-signed install <text>
×	<i>generate pki certificate self-signed install <name></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	Nothing found in XML Definitions	generate pki certificate sign <text>
×	<i>generate pki certificate sign <ca-name></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>generate pki certificate sign <ca-name> install <name></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	Nothing found in XML Definitions	generate pki certificate sign <text> file <text>
×	not yet documented	Nothing found in XML Definitions	generate pki certificate sign <text> install <text>
×	not yet documented	Nothing found in XML Definitions	generate pki crl <text>
×	not yet documented	Nothing found in XML Definitions	generate pki crl <text> file <text>
×	not yet documented	Nothing found in XML Definitions	generate pki crl <text> install
×	<i>generate pki dh</i>	Nothing found in XML Definitions	generate pki dh
×	not yet documented	Nothing found in XML Definitions	generate pki dh file <text>
×	not yet documented	Nothing found in XML Definitions	generate pki dh install <text>
×	<i>generate pki dh install <name></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	Nothing found in XML Definitions	generate pki key-pair
×	not yet documented	Nothing found in XML Definitions	generate pki key-pair file <text>
×	not yet documented	Nothing found in XML Definitions	generate pki key-pair install <text>
×	<i>generate pki openvpn shared-secret</i>	Nothing found in XML Definitions	generate pki openvpn shared-secret
×	not yet documented	Nothing found in XML Definitions	generate pki openvpn shared-secret file <text>
×	not yet documented	Nothing found in XML Definitions	generate pki openvpn shared-secret install <text>

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×	<i>generate pki openvpn shared-secret install <name></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	Nothing found in XML Definitions	generate pki ssh-key
×	not yet documented	Nothing found in XML Definitions	generate pki ssh-key file <text>
×	not yet documented	Nothing found in XML Definitions	generate pki ssh-key install <text>
×	<i>generate pki wireguard key-pair</i>	Nothing found in XML Definitions	generate pki wireguard key-pair
×	<i>generate pki wireguard key-pair install <interface></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	Nothing found in XML Definitions	generate pki wireguard key-pair install interface <text>
×	<i>generate pki wireguard key-pair install interface <interface></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>generate pki wireguard preshared-key</i>	Nothing found in XML Definitions	generate pki wireguard preshared-key
×	<i>generate pki wireguard preshared-key install <peer></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>generate pki wireguard preshared-key install interface <interface> peer <peer></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	Nothing found in XML Definitions	generate pki wireguard preshared-key install interface <text> peer <text>
×	<i>generate public-key-command user <username> path <location></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	generate-public-key-command.xml.in: generate public-key-command user <user> path <path>	generate public-key-command user <text> path <text>

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Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	generate-ssh-server-key.xml.in: generate ssh client-key <client-key>	generate ssh client-key <text>
×	<i>generate ssh client-key /config/auth/id_rsa_rpki</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>generate ssh client-key /path/to/private_key</i>	Nothing found in XML Definitions	Nothing found in VyOS
✓	<i>generate ssh server-key</i>	generate-ssh-server-key.xml.in: generate ssh server-key	generate ssh server-key
×	not yet documented	generate-system-login-user.xml.in: generate system login username <username> otp-key hotp-time	generate system login username <text> otp-key hotp-time
×	not yet documented	generate-system-login-user.xml.in: generate system login username <username> otp-key hotp-time rate-limit <rate-limit>	generate system login username <text> otp-key hotp-time rate-limit <text>
×	not yet documented	generate-system-login-user.xml.in: generate system login username <username> otp-key hotp-time rate-limit <rate-limit> rate-time <rate-time>	generate system login username <text> otp-key hotp-time rate-limit <text> rate-time <text>
×	not yet documented	generate-system-login-user.xml.in: generate system login username <username> otp-key hotp-time rate-limit <rate-limit> rate-time <rate-time> window-size <window-size>	generate system login username <text> otp-key hotp-time rate-limit <text> rate-time <text> window-size <text>
×	not yet documented	generate-system-login-user.xml.in: generate system login username <username> otp-key hotp-time window-size <window-size>	generate system login username <text> otp-key hotp-time window-size <text>

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Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	generate-system-login-user.xml.in: generate system login username <username> otp-key hotp-time window-size <window-size> rate-limit <rate-limit>	generate system login username <text> otp-key hotp-time window-size <text> rate-limit <text>
×	not yet documented	generate-system-login-user.xml.in: generate system login username <username> otp-key hotp-time window-size <window-size> rate-limit <rate-limit> rate-time <rate-time>	generate system login username <text> otp-key hotp-time window-size <text> rate-limit <text> rate-time <text>
✓	<i>generate tech-support archive</i>	generate_tech-support_archive.xml.in: generate tech-support archive <archive>	generate tech-support archive <text>
×	<i>generate wireguard client-config</i> <i><name> interface</i> <i><interface> server</i> <i><ip/fqdn> address</i> <i><client-ip></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	generate-wireguard.xml.in: generate wireguard client-config <client-config> interface <interface> server <server>	generate wireguard client-config <text> interface <text> server <text>
×	not yet documented	generate-wireguard.xml.in: generate wireguard client-config <client-config> interface <interface> server <server> address <address>	generate wireguard client-config <text> interface <text> server <text> address <text>
×	not yet documented	generate-wireguard.xml.in: generate wireguard client-config <client-config> interface <interface> server <server> address <address> address <address>	generate wireguard client-config <text> interface <text> server <text> address <text> address <text>

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×	<i>import pki ca <name> file <Path to CA certificate file></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>import pki ca <name> key-file <Path to private key file></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	Nothing found in XML Definitions	import pki ca <text> file <text>
×	not yet documented	Nothing found in XML Definitions	import pki ca <text> key-file <text>
×	<i>import pki certificate <name> file <path to certificate></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>import pki certificate <name> key-file <path to private key></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	Nothing found in XML Definitions	import pki certificate <text> file <text>
×	not yet documented	Nothing found in XML Definitions	import pki certificate <text> key-file <text>
×	not yet documented	Nothing found in XML Definitions	import pki crl <text> file <text>
×	not yet documented	Nothing found in XML Definitions	import pki dh <text> file <text>
×	not yet documented	Nothing found in XML Definitions	import pki key-pair <text> private-file <text>
×	not yet documented	Nothing found in XML Definitions	import pki key-pair <text> public-file <text>
×	<i>import pki openvpn shared-secret <name> file <path to OpenVPN secret key></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	Nothing found in XML Definitions	import pki openvpn shared-secret <text> file <text>
×	not yet documented	Nothing found in XML Definitions	install image
×	not yet documented	monitor-bandwidth-test.xml.in: monitor bandwidth-test accept	monitor bandwidth-test accept
×	not yet documented	monitor-bandwidth-test.xml.in: monitor bandwidth-test accept tcp	monitor bandwidth-test accept tcp

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×	not yet documented	monitor-bandwidth-test.xml.in: monitor bandwidth-test accept udp	monitor bandwidth-test accept udp
×	not yet documented	monitor-bandwidth-test.xml.in: monitor bandwidth-test initiate tcp <tcp>	monitor bandwidth-test initiate tcp <text>
×	not yet documented	monitor-bandwidth-test.xml.in: monitor bandwidth-test initiate udp <udp>	monitor bandwidth-test initiate udp <text>
×	not yet documented	monitor-bandwidth.xml.in: monitor bandwidth interface <interface>	monitor bandwidth interface <text>
×	not yet documented	monitor-bridge.xml.in: monitor bridge	monitor bridge
×	not yet documented	monitor-bridge.xml.in: monitor bridge fdb	monitor bridge fdb
×	not yet documented	monitor-bridge.xml.in: monitor bridge link	monitor bridge link
×	not yet documented	monitor-bridge.xml.in: monitor bridge mdb	monitor bridge mdb
×	not yet documented	monitor-command.xml.in: monitor command <command>	Nothing found in VyOS
×	not yet documented	monitor-command.xml.in: monitor command diff <diff>	Nothing found in VyOS
×	not yet documented	monitor-log.xml.in: monitor log	monitor log
×	not yet documented	monitor-log.xml.in: monitor log certbot	Nothing found in VyOS
×	not yet documented	monitor-log.xml.in: monitor log color	monitor log color
×	not yet documented	monitor-log.xml.in: monitor log conntrack-sync	monitor log conntrack-sync
×	not yet documented	monitor-log.xml.in: monitor log console-server	monitor log console-server
×	not yet documented	Nothing found in XML Definitions	monitor log container <text>
×	not yet documented	monitor-log.xml.in: monitor log dhcp client	monitor log dhcp client
×	not yet documented	monitor-log.xml.in: monitor log dhcp client interface <interface>	monitor log dhcp client interface <text>

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×	not yet documented	monitor-log.xml.in: monitor log dhcp server	monitor log dhcp server
×	not yet documented	monitor-log.xml.in: monitor log dhcpv6 client	monitor log dhcpv6 client
×	not yet documented	monitor-log.xml.in: monitor log dhcpv6 client interface <text> interface <interface>	monitor log dhcpv6 client interface <text>
×	not yet documented	monitor-log.xml.in: monitor log dhcpv6 server	monitor log dhcpv6 server
×	not yet documented	Nothing found in XML Definitions	monitor log dns dynamic
×	not yet documented	Nothing found in XML Definitions	monitor log dns forward- ing
×	not yet documented	monitor-log.xml.in: monitor log flow- accounting	monitor log flow- accounting
×	not yet documented	monitor-log.xml.in: monitor log ids ddos- protection	monitor log ids ddos- protection
×	not yet documented	monitor-log.xml.in: monitor log ipoe-server	monitor log ipoe-server
×	not yet documented	monitor-log.xml.in: monitor log ipsec	monitor log ipsec
×	not yet documented	monitor-log.xml.in: monitor log kernel	monitor log kernel
×	not yet documented	monitor-log.xml.in: monitor log l2tp	monitor log l2tp
×	not yet documented	monitor-log.xml.in: monitor log macsec	monitor log macsec
×	not yet documented	monitor-log.xml.in: monitor log macsec inter- face <interface>	monitor log macsec inter- face <text>
×	<i>monitor log mdns repeater</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	monitor-log.xml.in: monitor log ndp-proxy	Nothing found in VyOS
×	not yet documented	monitor-log.xml.in: monitor log nhrp	monitor log nhrp
×	not yet documented	monitor-log.xml.in: monitor log ntp	monitor log ntp
×	not yet documented	monitor-log.xml.in: monitor log openconnect	monitor log openconnect
×	not yet documented	monitor-log.xml.in: monitor log openvpn	monitor log openvpn
×	not yet documented	monitor-log.xml.in: monitor log openvpn inter- face <interface>	monitor log openvpn inter- face <text>
×	not yet documented	monitor-log.xml.in: monitor log pppoe	monitor log pppoe

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×	not yet documented	monitor-log.xml.in: monitor log pppoe-server	monitor log pppoe-server
×	not yet documented	monitor-log.xml.in: monitor log pppoe interface <interface>	monitor log pppoe interface <text>
×	not yet documented	monitor-log.xml.in: monitor log pptp	monitor log pptp
×	not yet documented	monitor-log.xml.in: monitor log protocol bfd	monitor log protocol bfd
×	not yet documented	monitor-log.xml.in: monitor log protocol bgp	monitor log protocol bgp
×	not yet documented	monitor-log.xml.in: monitor log protocol isis	monitor log protocol isis
×	not yet documented	monitor-log.xml.in: monitor log protocol mpls	monitor log protocol mpls
×	not yet documented	monitor-log.xml.in: monitor log protocol multicast	monitor log protocol multicast
×	not yet documented	monitor-log.xml.in: monitor log protocol nhrp	monitor log protocol nhrp
×	not yet documented	monitor-log.xml.in: monitor log protocol ospf	monitor log protocol ospf
×	not yet documented	monitor-log.xml.in: monitor log protocol ospfv3	monitor log protocol ospfv3
×	not yet documented	monitor-log.xml.in: monitor log protocol rip	monitor log protocol rip
×	not yet documented	monitor-log.xml.in: monitor log protocol ripng	monitor log protocol ripng
×	not yet documented	monitor-log.xml.in: monitor log protocol static	monitor log protocol static
×	not yet documented	monitor-log.xml.in: monitor log router-advert	monitor log router-advert
×	not yet documented	monitor-log.xml.in: monitor log snmp	monitor log snmp
✓	<i>monitor log ssh</i>	monitor-log.xml.in: monitor log ssh	monitor log ssh
×	<i>monitor log ssh dynamic-protection</i>	monitor-log.xml.in: monitor log ssh dynamic-protection	Nothing found in VyOS
×	not yet documented	monitor-log.xml.in: monitor log sstp	monitor log sstp
×	not yet documented	monitor-log.xml.in: monitor log sstpc	monitor log sstpc
×	not yet documented	monitor-log.xml.in: monitor log sstpc interface <interface>	monitor log sstpc interface <text>
×	not yet documented	monitor-log.xml.in: monitor log vpn	monitor log vpn

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Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	Nothing found in XML Definitions	monitor log vpp
×	not yet documented	monitor-log.xml.in: monitor log vrrp	monitor log vrrp
×	not yet documented	Nothing found in XML Definitions	monitor log webproxy
×	not yet documented	Nothing found in XML Definitions	monitor log webproxy access-log
×	not yet documented	Nothing found in XML Definitions	monitor log webproxy cache-log
×	not yet documented	monitor-ndp.xml.in: monitor ndp	monitor ndp
×	not yet documented	monitor-ndp.xml.in: monitor ndp interface <interface>	monitor ndp interface <text>
×	not yet documented	monitor-ndp.xml.in: monitor ndp interface <interface> type <type>	monitor ndp interface <text> type <text>
×	not yet documented	monitor-ndp.xml.in: monitor ndp type <type>	monitor ndp type <text>
×	not yet documented	monitor-protocol.xml.in: monitor protocol bgp disable all	monitor protocol bgp disable all
×	not yet documented	monitor-protocol.xml.in: monitor protocol bgp disable allow-martians	monitor protocol bgp disable allow-martians
×	not yet documented	monitor-protocol.xml.in: monitor protocol bgp disable as4	monitor protocol bgp disable as4
×	not yet documented	monitor-protocol.xml.in: monitor protocol bgp disable bestpath <bestpath>	monitor protocol bgp disable bestpath <text>
×	not yet documented	monitor-protocol.xml.in: monitor protocol bgp disable flowspec	monitor protocol bgp disable flowspec
×	not yet documented	monitor-protocol.xml.in: monitor protocol bgp disable keepalives	monitor protocol bgp disable keepalives
×	not yet documented	monitor-protocol.xml.in: monitor protocol bgp disable labelpool	monitor protocol bgp disable labelpool
×	not yet documented	monitor-protocol.xml.in: monitor protocol bgp disable neighbor-events	monitor protocol bgp disable neighbor-events
×	not yet documented	monitor-protocol.xml.in: monitor protocol bgp disable nht	monitor protocol bgp disable nht

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Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	monitor-protocol.xml.in: monitor protocol bgp disable pbr	monitor protocol bgp disable pbr
×	not yet documented	monitor-protocol.xml.in: monitor protocol bgp disable rib	monitor protocol bgp disable rib
×	not yet documented	monitor-protocol.xml.in: monitor protocol bgp disable update-groups	monitor protocol bgp disable update-groups
×	not yet documented	monitor-protocol.xml.in: monitor protocol bgp disable updates	monitor protocol bgp disable updates
×	not yet documented	monitor-protocol.xml.in: monitor protocol bgp disable vnc	monitor protocol bgp disable vnc
×	not yet documented	monitor-protocol.xml.in: monitor protocol bgp disable vnc import-bi-attach	monitor protocol bgp disable vnc import-bi-attach
×	not yet documented	monitor-protocol.xml.in: monitor protocol bgp disable vnc import-del-remote	monitor protocol bgp disable vnc import-del-remote
×	not yet documented	monitor-protocol.xml.in: monitor protocol bgp disable vnc rfapi-query	monitor protocol bgp disable vnc rfapi-query
×	not yet documented	monitor-protocol.xml.in: monitor protocol bgp disable vnc verbose	monitor protocol bgp disable vnc verbose
×	not yet documented	monitor-protocol.xml.in: monitor protocol bgp enable allow-martians	monitor protocol bgp enable allow-martians
×	not yet documented	monitor-protocol.xml.in: monitor protocol bgp enable as4	monitor protocol bgp enable as4
×	not yet documented	monitor-protocol.xml.in: monitor protocol bgp enable bestpath <text>	monitor protocol bgp enable bestpath <text>
×	not yet documented	monitor-protocol.xml.in: monitor protocol bgp enable flowspec	monitor protocol bgp enable flowspec
×	not yet documented	monitor-protocol.xml.in: monitor protocol bgp enable keepalives	monitor protocol bgp enable keepalives
×	not yet documented	monitor-protocol.xml.in: monitor protocol bgp enable labelpool	monitor protocol bgp enable labelpool
×	not yet documented	monitor-protocol.xml.in: monitor protocol bgp enable neighbor-events	monitor protocol bgp enable neighbor-events

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Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	monitor-protocol.xml.in: monitor protocol bgp enable nht	monitor protocol bgp enable nht
×	not yet documented	monitor-protocol.xml.in: monitor protocol bgp enable pbr	monitor protocol bgp enable pbr
×	not yet documented	monitor-protocol.xml.in: monitor protocol bgp enable rib	monitor protocol bgp enable rib
×	not yet documented	monitor-protocol.xml.in: monitor protocol bgp enable update-groups	monitor protocol bgp enable update-groups
×	not yet documented	monitor-protocol.xml.in: monitor protocol bgp enable updates	monitor protocol bgp enable updates
×	not yet documented	monitor-protocol.xml.in: monitor protocol bgp enable vnc	monitor protocol bgp enable vnc
×	not yet documented	monitor-protocol.xml.in: monitor protocol bgp enable vnc import-bi-attach	monitor protocol bgp enable vnc import-bi-attach
×	not yet documented	monitor-protocol.xml.in: monitor protocol bgp enable vnc import-del-remote	monitor protocol bgp enable vnc import-del-remote
×	not yet documented	monitor-protocol.xml.in: monitor protocol bgp enable vnc rfapi-query	monitor protocol bgp enable vnc rfapi-query
×	not yet documented	monitor-protocol.xml.in: monitor protocol bgp enable vnc verbose	monitor protocol bgp enable vnc verbose
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospf disable event	monitor protocol ospf disable event
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospf disable ism	monitor protocol ospf disable ism
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospf disable ism events	monitor protocol ospf disable ism events
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospf disable ism status	monitor protocol ospf disable ism status
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospf disable ism timers	monitor protocol ospf disable ism timers
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospf disable lsa	monitor protocol ospf disable lsa

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Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospf disable lsa flooding	monitor protocol ospf disable lsa flooding
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospf disable lsa generate	monitor protocol ospf disable lsa generate
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospf disable lsa install	monitor protocol ospf disable lsa install
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospf disable lsa refresh	monitor protocol ospf disable lsa refresh
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospf disable nsm	monitor protocol ospf disable nsm
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospf disable nsm events	monitor protocol ospf disable nsm events
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospf disable nsm status	monitor protocol ospf disable nsm status
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospf disable nsm timers	monitor protocol ospf disable nsm timers
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospf disable nssa	monitor protocol ospf disable nssa
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospf disable packet	monitor protocol ospf disable packet
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospf disable packet all	monitor protocol ospf disable packet all
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospf disable packet all detail	monitor protocol ospf disable packet all detail
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospf disable packet all recv	monitor protocol ospf disable packet all recv
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospf disable packet all recv detail	monitor protocol ospf disable packet all recv detail
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospf disable packet all send	monitor protocol ospf disable packet all send
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospf disable packet all send detail	monitor protocol ospf disable packet all send detail

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Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospf disable packet dd	monitor protocol ospf disable packet dd
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospf disable packet dd detail	monitor protocol ospf disable packet dd detail
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospf disable packet dd recv	monitor protocol ospf disable packet dd recv
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospf disable packet dd recv detail	monitor protocol ospf disable packet dd recv detail
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospf disable packet dd send	monitor protocol ospf disable packet dd send
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospf disable packet dd send detail	monitor protocol ospf disable packet dd send detail
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospf disable packet hello	monitor protocol ospf disable packet hello
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospf disable packet hello detail	monitor protocol ospf disable packet hello detail
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospf disable packet hello recv	monitor protocol ospf disable packet hello recv
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospf disable packet hello recv detail	monitor protocol ospf disable packet hello recv detail
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospf disable packet hello send	monitor protocol ospf disable packet hello send
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospf disable packet hello send detail	monitor protocol ospf disable packet hello send detail
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospf disable packet ls-ack	monitor protocol ospf disable packet ls-ack
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospf disable packet ls-ack detail	monitor protocol ospf disable packet ls-ack detail
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospf disable packet ls-ack recv	monitor protocol ospf disable packet ls-ack recv

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Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospf disable packet ls-ack rcv detail	monitor protocol ospf disable packet ls-ack rcv detail
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospf disable packet ls-ack send	monitor protocol ospf disable packet ls-ack send
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospf disable packet ls-ack send detail	monitor protocol ospf disable packet ls-ack send detail
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospf disable packet ls-request	monitor protocol ospf disable packet ls-request
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospf disable packet ls-request detail	monitor protocol ospf disable packet ls-request detail
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospf disable packet ls-request rcv	monitor protocol ospf disable packet ls-request rcv
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospf disable packet ls-request rcv detail	monitor protocol ospf disable packet ls-request rcv detail
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospf disable packet ls-request send	monitor protocol ospf disable packet ls-request send
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospf disable packet ls-request send detail	monitor protocol ospf disable packet ls-request send detail
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospf disable packet ls-update	monitor protocol ospf disable packet ls-update
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospf disable packet ls-update detail	monitor protocol ospf disable packet ls-update detail
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospf disable packet ls-update rcv	monitor protocol ospf disable packet ls-update rcv
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospf disable packet ls-update rcv detail	monitor protocol ospf disable packet ls-update rcv detail
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospf disable packet ls-update send	monitor protocol ospf disable packet ls-update send

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Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospf disable packet ls-update send detail	monitor protocol ospf disable packet ls-update send detail
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospf disable rib	monitor protocol ospf disable rib
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospf disable rib interface	monitor protocol ospf disable rib interface
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospf disable rib redistribute	monitor protocol ospf disable rib redistribute
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospf enable event	monitor protocol ospf enable event
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospf enable ism	monitor protocol ospf enable ism
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospf enable ism events	monitor protocol ospf enable ism events
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospf enable ism status	monitor protocol ospf enable ism status
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospf enable ism timers	monitor protocol ospf enable ism timers
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospf enable lsa	monitor protocol ospf enable lsa
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospf enable lsa flooding	monitor protocol ospf enable lsa flooding
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospf enable lsa generate	monitor protocol ospf enable lsa generate
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospf enable lsa install	monitor protocol ospf enable lsa install
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospf enable lsa refresh	monitor protocol ospf enable lsa refresh
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospf enable nsm	monitor protocol ospf enable nsm
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospf enable nsm events	monitor protocol ospf enable nsm events

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Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospf enable nsm status	monitor protocol ospf enable nsm status
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospf enable nsm timers	monitor protocol ospf enable nsm timers
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospf enable nssa	monitor protocol ospf enable nssa
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospf enable packet	monitor protocol ospf enable packet
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospf enable packet all	monitor protocol ospf enable packet all
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospf enable packet all detail	monitor protocol ospf enable packet all detail
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospf enable packet all recv	monitor protocol ospf enable packet all recv
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospf enable packet all recv detail	monitor protocol ospf enable packet all recv detail
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospf enable packet all send	monitor protocol ospf enable packet all send
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospf enable packet all send detail	monitor protocol ospf enable packet all send detail
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospf enable packet dd	monitor protocol ospf enable packet dd
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospf enable packet dd detail	monitor protocol ospf enable packet dd detail
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospf enable packet dd recv	monitor protocol ospf enable packet dd recv
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospf enable packet dd recv detail	monitor protocol ospf enable packet dd recv detail
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospf enable packet dd send	monitor protocol ospf enable packet dd send
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospf enable packet dd send detail	monitor protocol ospf enable packet dd send detail

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Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospf enable packet hello	monitor protocol ospf enable packet hello
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospf enable packet hello detail	monitor protocol ospf enable packet hello detail
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospf enable packet hello recv	monitor protocol ospf enable packet hello recv
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospf enable packet hello recv detail	monitor protocol ospf enable packet hello recv detail
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospf enable packet hello send	monitor protocol ospf enable packet hello send
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospf enable packet hello send detail	monitor protocol ospf enable packet hello send detail
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospf enable packet ls-ack	monitor protocol ospf enable packet ls-ack
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospf enable packet ls-ack detail	monitor protocol ospf enable packet ls-ack detail
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospf enable packet ls-ack recv	monitor protocol ospf enable packet ls-ack recv
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospf enable packet ls-ack recv detail	monitor protocol ospf enable packet ls-ack recv detail
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospf enable packet ls-ack send	monitor protocol ospf enable packet ls-ack send
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospf enable packet ls-ack send detail	monitor protocol ospf enable packet ls-ack send detail
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospf enable packet ls-request	monitor protocol ospf enable packet ls-request
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospf enable packet ls-request detail	monitor protocol ospf enable packet ls-request detail

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Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospf enable packet ls-request rcv	monitor protocol ospf enable packet ls-request rcv
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospf enable packet ls-request rcv detail	monitor protocol ospf enable packet ls-request rcv detail
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospf enable packet ls-request send	monitor protocol ospf enable packet ls-request send
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospf enable packet ls-request send detail	monitor protocol ospf enable packet ls-request send detail
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospf enable packet ls-update	monitor protocol ospf enable packet ls-update
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospf enable packet ls-update detail	monitor protocol ospf enable packet ls-update detail
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospf enable packet ls-update rcv	monitor protocol ospf enable packet ls-update rcv
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospf enable packet ls-update rcv detail	monitor protocol ospf enable packet ls-update rcv detail
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospf enable packet ls-update send	monitor protocol ospf enable packet ls-update send
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospf enable packet ls-update send detail	monitor protocol ospf enable packet ls-update send detail
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospf enable rib	monitor protocol ospf enable rib
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospf enable rib interface	monitor protocol ospf enable rib interface
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospf enable rib redistribute	monitor protocol ospf enable rib redistribute
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospfv3 disable abr	monitor protocol ospfv3 disable abr

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Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospfv3 disable asbr	monitor protocol ospfv3 disable asbr
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospfv3 disable border-routers	monitor protocol ospfv3 disable border-routers
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospfv3 disable border-routers area-id	monitor protocol ospfv3 disable border-routers area-id
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospfv3 disable border-routers router-id	monitor protocol ospfv3 disable border-routers router-id
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospfv3 disable flooding	monitor protocol ospfv3 disable flooding
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospfv3 disable interface	monitor protocol ospfv3 disable interface
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospfv3 disable lsa	monitor protocol ospfv3 disable lsa
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospfv3 disable lsa as-external	monitor protocol ospfv3 disable lsa as-external
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospfv3 disable lsa inter-prefix	monitor protocol ospfv3 disable lsa inter-prefix
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospfv3 disable lsa inter-router	monitor protocol ospfv3 disable lsa inter-router
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospfv3 disable lsa intra-prefix	monitor protocol ospfv3 disable lsa intra-prefix
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospfv3 disable lsa link	monitor protocol ospfv3 disable lsa link
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospfv3 disable lsa network	monitor protocol ospfv3 disable lsa network
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospfv3 disable lsa router	monitor protocol ospfv3 disable lsa router
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospfv3 disable lsa unknown	monitor protocol ospfv3 disable lsa unknown

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Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospfv3 disable message	monitor protocol ospfv3 disable message
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospfv3 disable message all	monitor protocol ospfv3 disable message all
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospfv3 disable message dbdesc	monitor protocol ospfv3 disable message dbdesc
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospfv3 disable message hello	monitor protocol ospfv3 disable message hello
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospfv3 disable message lsack	monitor protocol ospfv3 disable message lsack
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospfv3 disable message lsreq	monitor protocol ospfv3 disable message lsreq
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospfv3 disable message lsupdate	monitor protocol ospfv3 disable message lsupdate
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospfv3 disable message unknown	monitor protocol ospfv3 disable message unknown
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospfv3 disable neighbor	monitor protocol ospfv3 disable neighbor
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospfv3 disable neighbor event	monitor protocol ospfv3 disable neighbor event
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospfv3 disable neighbor state	monitor protocol ospfv3 disable neighbor state
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospfv3 disable rib	monitor protocol ospfv3 disable rib
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospfv3 disable rib recv	monitor protocol ospfv3 disable rib recv
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospfv3 disable rib send	monitor protocol ospfv3 disable rib send
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospfv3 disable route	monitor protocol ospfv3 disable route
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospfv3 disable route inter-area	monitor protocol ospfv3 disable route inter-area

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Table 2 – continued from previous page

Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospfv3 disable route intra-area	monitor protocol ospfv3 disable route intra-area
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospfv3 disable route memory	monitor protocol ospfv3 disable route memory
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospfv3 disable route table	monitor protocol ospfv3 disable route table
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospfv3 disable spf	monitor protocol ospfv3 disable spf
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospfv3 disable spf database	monitor protocol ospfv3 disable spf database
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospfv3 disable spf process	monitor protocol ospfv3 disable spf process
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospfv3 disable spf time	monitor protocol ospfv3 disable spf time
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospfv3 enable abr	monitor protocol ospfv3 enable abr
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospfv3 enable asbr	monitor protocol ospfv3 enable asbr
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospfv3 enable border-routers	monitor protocol ospfv3 enable border-routers
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospfv3 enable border-routers area-id	monitor protocol ospfv3 enable border-routers area-id
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospfv3 enable border-routers router-id	monitor protocol ospfv3 enable border-routers router-id
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospfv3 enable flooding	monitor protocol ospfv3 enable flooding
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospfv3 enable interface	monitor protocol ospfv3 enable interface
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospfv3 enable lsa	monitor protocol ospfv3 enable lsa

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Table 2 – continued from previous page

Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospfv3 enable lsa as-external	monitor protocol ospfv3 enable lsa as-external
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospfv3 enable lsa inter-prefix	monitor protocol ospfv3 enable lsa inter-prefix
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospfv3 enable lsa inter-router	monitor protocol ospfv3 enable lsa inter-router
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospfv3 enable lsa intra-prefix	monitor protocol ospfv3 enable lsa intra-prefix
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospfv3 enable lsa link	monitor protocol ospfv3 enable lsa link
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospfv3 enable lsa network	monitor protocol ospfv3 enable lsa network
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospfv3 enable lsa router	monitor protocol ospfv3 enable lsa router
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospfv3 enable lsa unknown	monitor protocol ospfv3 enable lsa unknown
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospfv3 enable message	monitor protocol ospfv3 enable message
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospfv3 enable message all	monitor protocol ospfv3 enable message all
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospfv3 enable message dbdesc	monitor protocol ospfv3 enable message dbdesc
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospfv3 enable message hello	monitor protocol ospfv3 enable message hello
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospfv3 enable message lsack	monitor protocol ospfv3 enable message lsack
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospfv3 enable message lsreq	monitor protocol ospfv3 enable message lsreq
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospfv3 enable message lupdate	monitor protocol ospfv3 enable message lupdate
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospfv3 enable message unknown	monitor protocol ospfv3 enable message unknown

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Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospfv3 enable neighbor	monitor protocol ospfv3 enable neighbor
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospfv3 enable neighbor event	monitor protocol ospfv3 enable neighbor event
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospfv3 enable neighbor state	monitor protocol ospfv3 enable neighbor state
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospfv3 enable rib	monitor protocol ospfv3 enable rib
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospfv3 enable rib recv	monitor protocol ospfv3 enable rib recv
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospfv3 enable rib send	monitor protocol ospfv3 enable rib send
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospfv3 enable route	monitor protocol ospfv3 enable route
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospfv3 enable route inter-area	monitor protocol ospfv3 enable route inter-area
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospfv3 enable route intra-area	monitor protocol ospfv3 enable route intra-area
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospfv3 enable route memory	monitor protocol ospfv3 enable route memory
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospfv3 enable route table	monitor protocol ospfv3 enable route table
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospfv3 enable spf	monitor protocol ospfv3 enable spf
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospfv3 enable spf database	monitor protocol ospfv3 enable spf database
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospfv3 enable spf process	monitor protocol ospfv3 enable spf process
×	not yet documented	monitor-protocol.xml.in: monitor protocol ospfv3 enable spf time	monitor protocol ospfv3 enable spf time
×	not yet documented	monitor-protocol.xml.in: monitor protocol rib dis- able events	monitor protocol rib dis- able events

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Table 2 – continued from previous page

Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	monitor-protocol.xml.in: monitor protocol rib disable kernel	monitor protocol rib disable kernel
×	not yet documented	monitor-protocol.xml.in: monitor protocol rib disable mpls	monitor protocol rib disable mpls
×	not yet documented	monitor-protocol.xml.in: monitor protocol rib disable nexthop	monitor protocol rib disable nexthop
×	not yet documented	monitor-protocol.xml.in: monitor protocol rib disable packet	monitor protocol rib disable packet
×	not yet documented	monitor-protocol.xml.in: monitor protocol rib disable packet detail	monitor protocol rib disable packet detail
×	not yet documented	monitor-protocol.xml.in: monitor protocol rib disable packet recv	monitor protocol rib disable packet recv
×	not yet documented	monitor-protocol.xml.in: monitor protocol rib disable packet send	monitor protocol rib disable packet send
×	not yet documented	monitor-protocol.xml.in: monitor protocol rib disable rib	monitor protocol rib disable rib
×	not yet documented	monitor-protocol.xml.in: monitor protocol rib disable rib detailed	monitor protocol rib disable rib detailed
×	not yet documented	monitor-protocol.xml.in: monitor protocol rib enable events	monitor protocol rib enable events
×	not yet documented	monitor-protocol.xml.in: monitor protocol rib enable kernel	monitor protocol rib enable kernel
×	not yet documented	monitor-protocol.xml.in: monitor protocol rib enable mpls	monitor protocol rib enable mpls
×	not yet documented	monitor-protocol.xml.in: monitor protocol rib enable nexthop	monitor protocol rib enable nexthop
×	not yet documented	monitor-protocol.xml.in: monitor protocol rib enable packet	monitor protocol rib enable packet
×	not yet documented	monitor-protocol.xml.in: monitor protocol rib enable packet detail	monitor protocol rib enable packet detail
×	not yet documented	monitor-protocol.xml.in: monitor protocol rib enable packet recv	monitor protocol rib enable packet recv

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Table 2 – continued from previous page

Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	monitor-protocol.xml.in: monitor protocol rib enable packet send	monitor protocol rib enable packet send
×	not yet documented	monitor-protocol.xml.in: monitor protocol rib enable rib	monitor protocol rib enable rib
×	not yet documented	monitor-protocol.xml.in: monitor protocol rib enable rib detailed	monitor protocol rib enable rib detailed
×	not yet documented	monitor-protocol.xml.in: monitor protocol rip disable all	monitor protocol rip disable all
×	not yet documented	monitor-protocol.xml.in: monitor protocol rip disable events	monitor protocol rip disable events
×	not yet documented	monitor-protocol.xml.in: monitor protocol rip disable packet	monitor protocol rip disable packet
×	not yet documented	monitor-protocol.xml.in: monitor protocol rip disable packet recv	monitor protocol rip disable packet recv
×	not yet documented	monitor-protocol.xml.in: monitor protocol rip disable packet send	monitor protocol rip disable packet send
×	not yet documented	monitor-protocol.xml.in: monitor protocol rip disable rib	monitor protocol rip disable rib
×	not yet documented	monitor-protocol.xml.in: monitor protocol rip enable events	monitor protocol rip enable events
×	not yet documented	monitor-protocol.xml.in: monitor protocol rip enable packet	monitor protocol rip enable packet
×	not yet documented	monitor-protocol.xml.in: monitor protocol rip enable packet recv	monitor protocol rip enable packet recv
×	not yet documented	monitor-protocol.xml.in: monitor protocol rip enable packet send	monitor protocol rip enable packet send
×	not yet documented	monitor-protocol.xml.in: monitor protocol rip enable rib	monitor protocol rip enable rib
×	not yet documented	monitor-protocol.xml.in: monitor protocol ripng disable all	monitor protocol ripng disable all
×	not yet documented	monitor-protocol.xml.in: monitor protocol ripng disable events	monitor protocol ripng disable events

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Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	monitor-protocol.xml.in: monitor protocol ripng disable packet	monitor protocol ripng disable packet
×	not yet documented	monitor-protocol.xml.in: monitor protocol ripng disable packet recv	monitor protocol ripng disable packet recv
×	not yet documented	monitor-protocol.xml.in: monitor protocol ripng disable packet send	monitor protocol ripng disable packet send
×	not yet documented	monitor-protocol.xml.in: monitor protocol ripng disable rib	monitor protocol ripng disable rib
×	not yet documented	monitor-protocol.xml.in: monitor protocol ripng en- able events	monitor protocol ripng en- able events
×	not yet documented	monitor-protocol.xml.in: monitor protocol ripng en- able packet	monitor protocol ripng en- able packet
×	not yet documented	monitor-protocol.xml.in: monitor protocol ripng en- able packet recv	monitor protocol ripng en- able packet recv
×	not yet documented	monitor-protocol.xml.in: monitor protocol ripng en- able packet send	monitor protocol ripng en- able packet send
×	not yet documented	monitor-protocol.xml.in: monitor protocol ripng en- able rib	monitor protocol ripng en- able rib
×	<i>monitor traceroute</i> <i><destination></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	traffic-dump.xml.in: monitor traffic interface <interface>	monitor traffic interface <text>
×	not yet documented	traffic-dump.xml.in: monitor traffic interface <interface> filter <filter>	monitor traffic interface <text> filter <text>
×	not yet documented	traffic-dump.xml.in: monitor traffic interface <interface> save <save>	monitor traffic interface <text> save <text>
×	not yet documented	traffic-dump.xml.in: monitor traffic interface <interface> save <save> filter <filter>	monitor traffic interface <text> save <text> filter <text>
×	not yet documented	traffic-dump.xml.in: monitor traffic interface <interface> verbose	monitor traffic interface <text> verbose
×	not yet documented	traffic-dump.xml.in: monitor traffic interface <interface> verbose filter <filter>	monitor traffic interface <text> verbose filter <text>

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Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	traffic-dump.xml.in: monitor traffic interface <interface> verbose save <save>	monitor traffic interface <text> verbose save <text>
×	not yet documented	traffic-dump.xml.in: monitor traffic interface <interface> verbose save <save> filter <filter>	monitor traffic interface <text> verbose save <text> filter <text>
×	not yet documented	mtr.xml.in: mtr <mtr>	Nothing found in VyOS
×	not yet documented	mtr.xml.in: mtr <mtr> node.tag	Nothing found in VyOS
×	not yet documented	ping.xml.in: ping <ping>	ping <text> <text>
×	<i>ping <destination></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>ping <host> vrf <name></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	ping.xml.in: ping <ping> node.tag	Nothing found in VyOS
×	<i>ping tcp <host> port <port> [count <n>] [interface <ifname>] [vrf <name>] [source-address <address>]</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	poweroff.xml.in: poweroff	poweroff
×	not yet documented	poweroff.xml.in: poweroff at <at>	poweroff at <text>
×	not yet documented	poweroff.xml.in: poweroff at <at> date <date>	poweroff at <text> date <text>
×	not yet documented	poweroff.xml.in: poweroff cancel	poweroff cancel
×	not yet documented	poweroff.xml.in: poweroff in <in>	poweroff in <text>
×	not yet documented	poweroff.xml.in: poweroff now	poweroff now
×	not yet documented	pppoe-server.xml.in: set pppoe-server maintenance-mode cancel	set pppoe-server maintenance-mode cancel
×	not yet documented	pppoe-server.xml.in: set pppoe-server maintenance-mode enable	set pppoe-server maintenance-mode enable
×	not yet documented	reboot.xml.in: reboot	reboot
×	not yet documented	reboot.xml.in: reboot at <at>	reboot at <text>

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Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	reboot.xml.in: reboot at <at> date <date>	reboot at <text> date <text>
×	not yet documented	reboot.xml.in: reboot cancel	reboot cancel
×	not yet documented	reboot.xml.in: reboot in <in>	reboot in <text>
×	not yet documented	reboot.xml.in: reboot now	reboot now
×	not yet documented	Nothing found in XML Definitions	release dhcp interface <text>
×	not yet documented	Nothing found in XML Definitions	release dhcpv6 interface <text>
×	not yet documented	Nothing found in XML Definitions	rename system image <text> <text>
×	<i>renew certbot</i>	pki.xml.in: renew certbot	Nothing found in VyOS
×	not yet documented	dhcp.xml.in: renew dhcp interface <in- terface>	renew dhcp interface <text>
×	not yet documented	dhcp.xml.in: renew dhcpv6 interface <interface>	renew dhcpv6 interface <text>
×	not yet documented	reset-bgp.xml.in: reset bgp <bgp>	reset bgp <text>
×	not yet documented	reset-bgp.xml.in: reset bgp 1-4294967295 <1-4294967295>	reset bgp 1-4294967295 <text>
×	not yet documented	reset-bgp.xml.in: reset bgp 1-4294967295 <1-4294967295> in	reset bgp 1-4294967295 <text> in
×	not yet documented	reset-bgp.xml.in: reset bgp 1-4294967295 <1-4294967295> in prefix-filter	reset bgp 1-4294967295 <text> in prefix-filter
×	not yet documented	reset-bgp.xml.in: reset bgp 1-4294967295 <1-4294967295> message-stats	reset bgp 1-4294967295 <text> message-stats
×	not yet documented	reset-bgp.xml.in: reset bgp 1-4294967295 <1-4294967295> out	reset bgp 1-4294967295 <text> out
×	not yet documented	reset-bgp.xml.in: reset bgp 1-4294967295 <1-4294967295> soft	reset bgp 1-4294967295 <text> soft
×	not yet documented	reset-bgp.xml.in: reset bgp 1-4294967295 <1-4294967295> soft in	reset bgp 1-4294967295 <text> soft in

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Table 2 – continued from previous page

Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	reset-bgp.xml.in: reset bgp 1-4294967295 <1-4294967295> soft out	reset bgp 1-4294967295 <text> soft out
×	<i>reset bgp</i> <i><ipv4/ipv6></i> <i><address> [soft</i> <i>[in/out]]</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>reset bgp</i> <i><ipv4/ipv6></i> <i>external</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>reset bgp</i> <i><ipv4/ipv6></i> <i>peer-group <name></i> <i>[soft [in/out]]</i>	Nothing found in XML Definitions	Nothing found in VyOS
✓	<i>reset bgp all</i>	reset-bgp.xml.in: reset bgp all	reset bgp all
×	not yet documented	reset-bgp.xml.in: reset bgp external	reset bgp external
×	not yet documented	reset-bgp.xml.in: reset bgp external in	reset bgp external in
×	not yet documented	reset-bgp.xml.in: reset bgp external in prefix-filter	reset bgp external in prefix-filter
×	not yet documented	reset-bgp.xml.in: reset bgp external message-stats	reset bgp external message-stats
×	not yet documented	reset-bgp.xml.in: reset bgp external out	reset bgp external out
×	not yet documented	reset-bgp.xml.in: reset bgp external soft	reset bgp external soft
×	not yet documented	reset-bgp.xml.in: reset bgp external soft in	reset bgp external soft in
×	not yet documented	reset-bgp.xml.in: reset bgp external soft out	reset bgp external soft out
×	not yet documented	reset-bgp.xml.in: reset bgp <bgp> in	reset bgp <text> in
×	not yet documented	reset-bgp.xml.in: reset bgp <bgp> in prefix- filter	reset bgp <text> in prefix- filter
×	not yet documented	reset-bgp.xml.in: reset bgp ipv4 <ipv4>	reset bgp ipv4 <text>
×	not yet documented	reset-bgp.xml.in: reset bgp ipv4 1- 4294967295 <1- 4294967295>	reset bgp ipv4 1- 4294967295 <text>
×	not yet documented	reset-bgp.xml.in: reset bgp ipv4 1- 4294967295 <1- 4294967295> in	reset bgp ipv4 1- 4294967295 <text> in

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Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	reset-bgp.xml.in: reset bgp ipv4 1- 4294967295 <1- 4294967295> in prefix- filter	reset bgp ipv4 1- 4294967295 <text> in prefix-filter
×	not yet documented	reset-bgp.xml.in: reset bgp ipv4 1- 4294967295 <1- 4294967295> message- stats	reset bgp ipv4 1- 4294967295 <text> message-stats
×	not yet documented	reset-bgp.xml.in: reset bgp ipv4 1- 4294967295 <1- 4294967295> out	reset bgp ipv4 1- 4294967295 <text> out
×	not yet documented	reset-bgp.xml.in: reset bgp ipv4 1- 4294967295 <1- 4294967295> soft	reset bgp ipv4 1- 4294967295 <text> soft
×	not yet documented	reset-bgp.xml.in: reset bgp ipv4 1- 4294967295 <1- 4294967295> soft in	reset bgp ipv4 1- 4294967295 <text> soft in
×	not yet documented	reset-bgp.xml.in: reset bgp ipv4 1- 4294967295 <1- 4294967295> soft out	reset bgp ipv4 1- 4294967295 <text> soft out
×	not yet documented	reset-bgp.xml.in: reset bgp ipv4 all	reset bgp ipv4 all
×	not yet documented	reset-bgp.xml.in: reset bgp ipv4 external	reset bgp ipv4 external
×	not yet documented	reset-bgp.xml.in: reset bgp ipv4 external in	reset bgp ipv4 external in
×	not yet documented	reset-bgp.xml.in: reset bgp ipv4 external in prefix-filter	reset bgp ipv4 external in prefix-filter
×	not yet documented	reset-bgp.xml.in: reset bgp ipv4 external message-stats	reset bgp ipv4 external message-stats
×	not yet documented	reset-bgp.xml.in: reset bgp ipv4 external out	reset bgp ipv4 external out
×	not yet documented	reset-bgp.xml.in: reset bgp ipv4 external soft	reset bgp ipv4 external soft
×	not yet documented	reset-bgp.xml.in: reset bgp ipv4 external soft in	reset bgp ipv4 external soft in
×	not yet documented	reset-bgp.xml.in: reset bgp ipv4 external soft out	reset bgp ipv4 external soft out

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Table 2 – continued from previous page

Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	reset-bgp.xml.in: reset bgp ipv4 <ipv4> in	reset bgp ipv4 <text> in
×	not yet documented	reset-bgp.xml.in: reset bgp ipv4 <ipv4> in prefix-filter	reset bgp ipv4 <text> in prefix-filter
×	not yet documented	reset-bgp.xml.in: reset bgp ipv4 <ipv4> message-stats	reset bgp ipv4 <text> message-stats
×	not yet documented	reset-bgp.xml.in: reset bgp ipv4 <ipv4> out	reset bgp ipv4 <text> out
×	not yet documented	reset-bgp.xml.in: reset bgp ipv4 peer-group <peer-group>	reset bgp ipv4 peer-group <text>
×	not yet documented	reset-bgp.xml.in: reset bgp ipv4 peer-group <peer-group> in	reset bgp ipv4 peer-group <text> in
×	not yet documented	reset-bgp.xml.in: reset bgp ipv4 peer- group <peer-group> in prefix-filter	reset bgp ipv4 peer-group <text> in prefix-filter
×	not yet documented	reset-bgp.xml.in: reset bgp ipv4 peer-group <peer-group> message- stats	reset bgp ipv4 peer-group <text> message-stats
×	not yet documented	reset-bgp.xml.in: reset bgp ipv4 peer-group <peer-group> out	reset bgp ipv4 peer-group <text> out
×	not yet documented	reset-bgp.xml.in: reset bgp ipv4 peer-group <peer-group> soft	reset bgp ipv4 peer-group <text> soft
×	not yet documented	reset-bgp.xml.in: reset bgp ipv4 peer-group <peer-group> soft in	reset bgp ipv4 peer-group <text> soft in
×	not yet documented	reset-bgp.xml.in: reset bgp ipv4 peer-group <peer-group> soft out	reset bgp ipv4 peer-group <text> soft out
×	not yet documented	reset-bgp.xml.in: reset bgp ipv4 <ipv4> soft	reset bgp ipv4 <text> soft
×	not yet documented	reset-bgp.xml.in: reset bgp ipv4 <ipv4> soft in	reset bgp ipv4 <text> soft in
×	not yet documented	reset-bgp.xml.in: reset bgp ipv4 <ipv4> soft out	reset bgp ipv4 <text> soft out
×	not yet documented	reset-bgp.xml.in: reset bgp ipv6 <ipv6>	reset bgp ipv6 <text>

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Table 2 – continued from previous page

Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	reset-bgp.xml.in: reset bgp ipv6 1- 4294967295 <1- 4294967295>	reset bgp ipv6 1- 4294967295 <text>
×	not yet documented	reset-bgp.xml.in: reset bgp ipv6 1- 4294967295 <1- 4294967295> in	reset bgp ipv6 1- 4294967295 <text> in
×	not yet documented	reset-bgp.xml.in: reset bgp ipv6 1- 4294967295 <1- 4294967295> in prefix- filter	reset bgp ipv6 1- 4294967295 <text> in prefix-filter
×	not yet documented	reset-bgp.xml.in: reset bgp ipv6 1- 4294967295 <1- 4294967295> message- stats	reset bgp ipv6 1- 4294967295 <text> message-stats
×	not yet documented	reset-bgp.xml.in: reset bgp ipv6 1- 4294967295 <1- 4294967295> out	reset bgp ipv6 1- 4294967295 <text> out
×	not yet documented	reset-bgp.xml.in: reset bgp ipv6 1- 4294967295 <1- 4294967295> soft	reset bgp ipv6 1- 4294967295 <text> soft
×	not yet documented	reset-bgp.xml.in: reset bgp ipv6 1- 4294967295 <1- 4294967295> soft in	reset bgp ipv6 1- 4294967295 <text> soft in
×	not yet documented	reset-bgp.xml.in: reset bgp ipv6 1- 4294967295 <1- 4294967295> soft out	reset bgp ipv6 1- 4294967295 <text> soft out
×	<i>reset bgp ipv6</i> <i><address></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	reset-bgp.xml.in: reset bgp ipv6 all	reset bgp ipv6 all
×	not yet documented	reset-bgp.xml.in: reset bgp ipv6 external	reset bgp ipv6 external
×	not yet documented	reset-bgp.xml.in: reset bgp ipv6 external in	reset bgp ipv6 external in
×	not yet documented	reset-bgp.xml.in: reset bgp ipv6 external in prefix-filter	reset bgp ipv6 external in prefix-filter
×	not yet documented	reset-bgp.xml.in: reset bgp ipv6 external message-stats	reset bgp ipv6 external message-stats

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Table 2 – continued from previous page

Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	reset-bgp.xml.in: reset bgp ipv6 external out	reset bgp ipv6 external out
×	not yet documented	reset-bgp.xml.in: reset bgp ipv6 external soft	reset bgp ipv6 external soft
×	not yet documented	reset-bgp.xml.in: reset bgp ipv6 external soft in	reset bgp ipv6 external soft in
×	not yet documented	reset-bgp.xml.in: reset bgp ipv6 external soft out	reset bgp ipv6 external soft out
×	not yet documented	reset-bgp.xml.in: reset bgp ipv6 <ipv6> in	reset bgp ipv6 <text> in
×	not yet documented	reset-bgp.xml.in: reset bgp ipv6 <ipv6> in prefix-filter	reset bgp ipv6 <text> in prefix-filter
×	not yet documented	reset-bgp.xml.in: reset bgp ipv6 <ipv6> message-stats	reset bgp ipv6 <text> message-stats
×	not yet documented	reset-bgp.xml.in: reset bgp ipv6 <ipv6> out	reset bgp ipv6 <text> out
×	not yet documented	reset-bgp.xml.in: reset bgp ipv6 peer-group <peer-group>	reset bgp ipv6 peer-group <text>
×	not yet documented	reset-bgp.xml.in: reset bgp ipv6 peer-group <peer-group> in	reset bgp ipv6 peer-group <text> in
×	not yet documented	reset-bgp.xml.in: reset bgp ipv6 peer-group <peer-group> in prefix-filter	reset bgp ipv6 peer-group <text> in prefix-filter
×	not yet documented	reset-bgp.xml.in: reset bgp ipv6 peer-group <peer-group> message-stats	reset bgp ipv6 peer-group <text> message-stats
×	not yet documented	reset-bgp.xml.in: reset bgp ipv6 peer-group <peer-group> out	reset bgp ipv6 peer-group <text> out
×	not yet documented	reset-bgp.xml.in: reset bgp ipv6 peer-group <peer-group> soft	reset bgp ipv6 peer-group <text> soft
×	not yet documented	reset-bgp.xml.in: reset bgp ipv6 peer-group <peer-group> soft in	reset bgp ipv6 peer-group <text> soft in
×	not yet documented	reset-bgp.xml.in: reset bgp ipv6 peer-group <peer-group> soft out	reset bgp ipv6 peer-group <text> soft out
×	not yet documented	reset-bgp.xml.in: reset bgp ipv6 <ipv6> soft	reset bgp ipv6 <text> soft

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Table 2 – continued from previous page

Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	reset-bgp.xml.in: reset bgp ipv6 <ipv6> soft in	reset bgp ipv6 <text> soft in
×	not yet documented	reset-bgp.xml.in: reset bgp ipv6 <ipv6> soft out	reset bgp ipv6 <text> soft out
×	not yet documented	reset-bgp.xml.in: reset bgp l2vpn evpn <evpn>	reset bgp l2vpn evpn <text>
×	not yet documented	reset-bgp.xml.in: reset bgp l2vpn evpn 1-4294967295 <1- 4294967295>	reset bgp l2vpn evpn 1- 4294967295 <text>
×	not yet documented	reset-bgp.xml.in: reset bgp l2vpn evpn 1-4294967295 <1- 4294967295> in	reset bgp l2vpn evpn 1- 4294967295 <text> in
×	not yet documented	reset-bgp.xml.in: reset bgp l2vpn evpn 1-4294967295 <1- 4294967295> in prefix- filter	reset bgp l2vpn evpn 1-4294967295 <text> in prefix-filter
×	not yet documented	reset-bgp.xml.in: reset bgp l2vpn evpn 1-4294967295 <1- 4294967295> message- stats	reset bgp l2vpn evpn 1-4294967295 <text> message-stats
×	not yet documented	reset-bgp.xml.in: reset bgp l2vpn evpn 1-4294967295 <1- 4294967295> out	reset bgp l2vpn evpn 1- 4294967295 <text> out
×	not yet documented	reset-bgp.xml.in: reset bgp l2vpn evpn 1-4294967295 <1- 4294967295> soft	reset bgp l2vpn evpn 1- 4294967295 <text> soft
×	not yet documented	reset-bgp.xml.in: reset bgp l2vpn evpn 1-4294967295 <1- 4294967295> soft in	reset bgp l2vpn evpn 1- 4294967295 <text> soft in
×	not yet documented	reset-bgp.xml.in: reset bgp l2vpn evpn 1-4294967295 <1- 4294967295> soft out	reset bgp l2vpn evpn 1- 4294967295 <text> soft out
×	not yet documented	reset-bgp.xml.in: reset bgp l2vpn evpn all	reset bgp l2vpn evpn all
×	not yet documented	reset-bgp.xml.in: reset bgp l2vpn evpn exter- nal	reset bgp l2vpn evpn exter- nal

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Table 2 – continued from previous page

Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	reset-bgp.xml.in: reset bgp l2vpn evpn external in	reset bgp l2vpn evpn external in
×	not yet documented	reset-bgp.xml.in: reset bgp l2vpn evpn external in prefix-filter	reset bgp l2vpn evpn external in prefix-filter
×	not yet documented	reset-bgp.xml.in: reset bgp l2vpn evpn external message-stats	reset bgp l2vpn evpn external message-stats
×	not yet documented	reset-bgp.xml.in: reset bgp l2vpn evpn external out	reset bgp l2vpn evpn external out
×	not yet documented	reset-bgp.xml.in: reset bgp l2vpn evpn external soft	reset bgp l2vpn evpn external soft
×	not yet documented	reset-bgp.xml.in: reset bgp l2vpn evpn external soft in	reset bgp l2vpn evpn external soft in
×	not yet documented	reset-bgp.xml.in: reset bgp l2vpn evpn external soft out	reset bgp l2vpn evpn external soft out
×	not yet documented	reset-bgp.xml.in: reset bgp l2vpn evpn <evpn> in	reset bgp l2vpn evpn <text> in
×	not yet documented	reset-bgp.xml.in: reset bgp l2vpn evpn <evpn> in prefix-filter	reset bgp l2vpn evpn <text> in prefix-filter
×	not yet documented	reset-bgp.xml.in: reset bgp l2vpn evpn <evpn> message-stats	reset bgp l2vpn evpn <text> message-stats
×	not yet documented	reset-bgp.xml.in: reset bgp l2vpn evpn <evpn> out	reset bgp l2vpn evpn <text> out
×	not yet documented	reset-bgp.xml.in: reset bgp l2vpn evpn peer-group <peer-group>	reset bgp l2vpn evpn peer-group <text>
×	not yet documented	reset-bgp.xml.in: reset bgp l2vpn evpn peer-group <peer-group> in	reset bgp l2vpn evpn peer-group <text> in
×	not yet documented	reset-bgp.xml.in: reset bgp l2vpn evpn peer-group <peer-group> in prefix-filter	reset bgp l2vpn evpn peer-group <text> in prefix-filter
×	not yet documented	reset-bgp.xml.in: reset bgp l2vpn evpn peer-group <peer-group> message-stats	reset bgp l2vpn evpn peer-group <text> message-stats

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Table 2 – continued from previous page

Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	reset-bgp.xml.in: reset bgp l2vpn evpn peer-group <peer-group> out	reset bgp l2vpn evpn peer-group <text> out
×	not yet documented	reset-bgp.xml.in: reset bgp l2vpn evpn peer-group <peer-group> soft	reset bgp l2vpn evpn peer-group <text> soft
×	not yet documented	reset-bgp.xml.in: reset bgp l2vpn evpn peer-group <peer-group> soft in	reset bgp l2vpn evpn peer-group <text> soft in
×	not yet documented	reset-bgp.xml.in: reset bgp l2vpn evpn peer-group <peer-group> soft out	reset bgp l2vpn evpn peer-group <text> soft out
×	not yet documented	reset-bgp.xml.in: reset bgp l2vpn evpn <evpn> soft	reset bgp l2vpn evpn <text> soft
×	not yet documented	reset-bgp.xml.in: reset bgp l2vpn evpn <evpn> soft in	reset bgp l2vpn evpn <text> soft in
×	not yet documented	reset-bgp.xml.in: reset bgp l2vpn evpn <evpn> soft out	reset bgp l2vpn evpn <text> soft out
×	not yet documented	reset-bgp.xml.in: reset bgp <bgp> message-stats	reset bgp <text> message-stats
×	not yet documented	reset-bgp.xml.in: reset bgp <bgp> out	reset bgp <text> out
×	not yet documented	reset-bgp.xml.in: reset bgp peer-group <peer-group>	reset bgp peer-group <text>
×	not yet documented	reset-bgp.xml.in: reset bgp peer-group <peer-group> in	reset bgp peer-group <text> in
×	not yet documented	reset-bgp.xml.in: reset bgp peer-group <peer-group> in prefix-filter	reset bgp peer-group <text> in prefix-filter
×	not yet documented	reset-bgp.xml.in: reset bgp peer-group <peer-group> message-stats	reset bgp peer-group <text> message-stats
×	not yet documented	reset-bgp.xml.in: reset bgp peer-group <peer-group> out	reset bgp peer-group <text> out
×	not yet documented	reset-bgp.xml.in: reset bgp peer-group <peer-group> soft	reset bgp peer-group <text> soft

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Table 2 – continued from previous page

Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	reset-bgp.xml.in: reset bgp peer-group <peer-group> soft in	reset bgp peer-group <text> soft in
×	not yet documented	reset-bgp.xml.in: reset bgp peer-group <peer-group> soft out	reset bgp peer-group <text> soft out
×	not yet documented	reset-bgp.xml.in: reset bgp prefix <prefix>	reset bgp prefix <text>
×	not yet documented	reset-bgp.xml.in: reset bgp <bgp> soft	reset bgp <text> soft
×	not yet documented	reset-bgp.xml.in: reset bgp <bgp> soft in	reset bgp <text> soft in
×	not yet documented	reset-bgp.xml.in: reset bgp <bgp> soft out	reset bgp <text> soft out
×	not yet documented	Nothing found in XML Definitions	reset bgp vrf <text> <text>
×	not yet documented	reset-bgp.xml.in: reset bgp vrf <vrf> 1-4294967295 <1- 4294967295>	reset bgp vrf <text> 1- 4294967295 <text>
×	not yet documented	reset-bgp.xml.in: reset bgp vrf <vrf> 1-4294967295 <1- 4294967295> in	reset bgp vrf <text> 1- 4294967295 <text> in
×	not yet documented	reset-bgp.xml.in: reset bgp vrf <vrf> 1-4294967295 <1- 4294967295> in prefix- filter	reset bgp vrf <text> 1- 4294967295 <text> in prefix-filter
×	not yet documented	reset-bgp.xml.in: reset bgp vrf <vrf> 1-4294967295 <1- 4294967295> message- stats	reset bgp vrf <text> 1-4294967295 <text> message-stats
×	not yet documented	reset-bgp.xml.in: reset bgp vrf <vrf> 1-4294967295 <1- 4294967295> out	reset bgp vrf <text> 1- 4294967295 <text> out
×	not yet documented	reset-bgp.xml.in: reset bgp vrf <vrf> 1-4294967295 <1- 4294967295> soft	reset bgp vrf <text> 1- 4294967295 <text> soft
×	not yet documented	reset-bgp.xml.in: reset bgp vrf <vrf> 1-4294967295 <1- 4294967295> soft in	reset bgp vrf <text> 1- 4294967295 <text> soft in

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Table 2 – continued from previous page

Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	reset-bgp.xml.in: reset bgp vrf <vrf> 1-4294967295 <1- 4294967295> soft out	reset bgp vrf <text> 1- 4294967295 <text> soft out
×	not yet documented	reset-bgp.xml.in: reset bgp vrf <vrf> all	reset bgp vrf <text> all
×	not yet documented	reset-bgp.xml.in: reset bgp vrf <vrf> exter- nal	reset bgp vrf <text> exter- nal
×	not yet documented	reset-bgp.xml.in: reset bgp vrf <vrf> exter- nal in	reset bgp vrf <text> exter- nal in
×	not yet documented	reset-bgp.xml.in: reset bgp vrf <vrf> exter- nal in prefix-filter	reset bgp vrf <text> exter- nal in prefix-filter
×	not yet documented	reset-bgp.xml.in: reset bgp vrf <vrf> exter- nal message-stats	reset bgp vrf <text> exter- nal message-stats
×	not yet documented	reset-bgp.xml.in: reset bgp vrf <vrf> exter- nal out	reset bgp vrf <text> exter- nal out
×	not yet documented	reset-bgp.xml.in: reset bgp vrf <vrf> exter- nal soft	reset bgp vrf <text> exter- nal soft
×	not yet documented	reset-bgp.xml.in: reset bgp vrf <vrf> exter- nal soft in	reset bgp vrf <text> exter- nal soft in
×	not yet documented	reset-bgp.xml.in: reset bgp vrf <vrf> exter- nal soft out	reset bgp vrf <text> exter- nal soft out
×	not yet documented	Nothing found in XML Definitions	reset bgp vrf <text> <text> in
×	not yet documented	Nothing found in XML Definitions	reset bgp vrf <text> <text> in prefix-filter
×	not yet documented	reset-bgp.xml.in: reset bgp vrf <vrf> ipv4 <ipv4>	reset bgp vrf <text> ipv4 <text>
×	not yet documented	reset-bgp.xml.in: reset bgp vrf <vrf> ipv4 1-4294967295 <1-4294967295>	reset bgp vrf <text> ipv4 1-4294967295 <text>
×	not yet documented	reset-bgp.xml.in: reset bgp vrf <vrf> ipv4 1-4294967295 <1-4294967295> in	reset bgp vrf <text> ipv4 1-4294967295 <text> in

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Table 2 – continued from previous page

Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	reset-bgp.xml.in: reset bgp vrf <vrf> ipv4 1-4294967295 <1-4294967295> in prefix-filter	reset bgp vrf <text> ipv4 1-4294967295 <text> in prefix-filter
×	not yet documented	reset-bgp.xml.in: reset bgp vrf <vrf> ipv4 1-4294967295 <1- 4294967295> message- stats	reset bgp vrf <text> ipv4 1-4294967295 <text> message-stats
×	not yet documented	reset-bgp.xml.in: reset bgp vrf <vrf> ipv4 1-4294967295 <1-4294967295> out	reset bgp vrf <text> ipv4 1-4294967295 <text> out
×	not yet documented	reset-bgp.xml.in: reset bgp vrf <vrf> ipv4 1-4294967295 <1-4294967295> soft	reset bgp vrf <text> ipv4 1-4294967295 <text> soft
×	not yet documented	reset-bgp.xml.in: reset bgp vrf <vrf> ipv4 1-4294967295 <1-4294967295> soft in	reset bgp vrf <text> ipv4 1-4294967295 <text> soft in
×	not yet documented	reset-bgp.xml.in: reset bgp vrf <vrf> ipv4 1-4294967295 <1-4294967295> soft out	reset bgp vrf <text> ipv4 1-4294967295 <text> soft out
×	not yet documented	reset-bgp.xml.in: reset bgp vrf <vrf> ipv4 all	reset bgp vrf <text> ipv4 all
×	not yet documented	reset-bgp.xml.in: reset bgp vrf <vrf> ipv4 external	reset bgp vrf <text> ipv4 external
×	not yet documented	reset-bgp.xml.in: reset bgp vrf <vrf> ipv4 external in	reset bgp vrf <text> ipv4 external in
×	not yet documented	reset-bgp.xml.in: reset bgp vrf <vrf> ipv4 external in prefix-filter	reset bgp vrf <text> ipv4 external in prefix-filter
×	not yet documented	reset-bgp.xml.in: reset bgp vrf <vrf> ipv4 external message-stats	reset bgp vrf <text> ipv4 external message-stats
×	not yet documented	reset-bgp.xml.in: reset bgp vrf <vrf> ipv4 external out	reset bgp vrf <text> ipv4 external out
×	not yet documented	reset-bgp.xml.in: reset bgp vrf <vrf> ipv4 external soft	reset bgp vrf <text> ipv4 external soft
×	not yet documented	reset-bgp.xml.in: reset bgp vrf <vrf> ipv4 external soft in	reset bgp vrf <text> ipv4 external soft in

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Table 2 – continued from previous page

Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	reset-bgp.xml.in: reset bgp vrf <vrf> ipv4 external soft out	reset bgp vrf <text> ipv4 external soft out
×	not yet documented	reset-bgp.xml.in: reset bgp vrf <vrf> ipv4 <ipv4> in	reset bgp vrf <text> ipv4 <text> in
×	not yet documented	reset-bgp.xml.in: reset bgp vrf <vrf> ipv4 <ipv4> in prefix-filter	reset bgp vrf <text> ipv4 <text> in prefix-filter
×	not yet documented	reset-bgp.xml.in: reset bgp vrf <vrf> ipv4 <ipv4> message-stats	reset bgp vrf <text> ipv4 <text> message-stats
×	not yet documented	reset-bgp.xml.in: reset bgp vrf <vrf> ipv4 <ipv4> out	reset bgp vrf <text> ipv4 <text> out
×	not yet documented	reset-bgp.xml.in: reset bgp vrf <vrf> ipv4 peer-group <peer-group>	reset bgp vrf <text> ipv4 peer-group <text>
×	not yet documented	reset-bgp.xml.in: reset bgp vrf <vrf> ipv4 peer-group <peer-group> in	reset bgp vrf <text> ipv4 peer-group <text> in
×	not yet documented	reset-bgp.xml.in: reset bgp vrf <vrf> ipv4 peer-group <peer-group> in prefix-filter	reset bgp vrf <text> ipv4 peer-group <text> in prefix-filter
×	not yet documented	reset-bgp.xml.in: reset bgp vrf <vrf> ipv4 peer-group <peer-group> message-stats	reset bgp vrf <text> ipv4 peer-group <text> message-stats
×	not yet documented	reset-bgp.xml.in: reset bgp vrf <vrf> ipv4 peer-group <peer-group> out	reset bgp vrf <text> ipv4 peer-group <text> out
×	not yet documented	reset-bgp.xml.in: reset bgp vrf <vrf> ipv4 peer-group <peer-group> soft	reset bgp vrf <text> ipv4 peer-group <text> soft
×	not yet documented	reset-bgp.xml.in: reset bgp vrf <vrf> ipv4 peer-group <peer-group> soft in	reset bgp vrf <text> ipv4 peer-group <text> soft in
×	not yet documented	reset-bgp.xml.in: reset bgp vrf <vrf> ipv4 peer-group <peer-group> soft out	reset bgp vrf <text> ipv4 peer-group <text> soft out
×	not yet documented	reset-bgp.xml.in: reset bgp vrf <vrf> ipv4 <ipv4> soft	reset bgp vrf <text> ipv4 <text> soft

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Table 2 – continued from previous page

Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	reset-bgp.xml.in: reset bgp vrf <vrf> ipv4 <ipv4> soft in	reset bgp vrf <text> ipv4 <text> soft in
×	not yet documented	reset-bgp.xml.in: reset bgp vrf <vrf> ipv4 <ipv4> soft out	reset bgp vrf <text> ipv4 <text> soft out
×	not yet documented	reset-bgp.xml.in: reset bgp vrf <vrf> ipv6 <ipv6>	reset bgp vrf <text> ipv6 <text>
×	not yet documented	reset-bgp.xml.in: reset bgp vrf <vrf> ipv6 1-4294967295 <1-4294967295>	reset bgp vrf <text> ipv6 1-4294967295 <text>
×	not yet documented	reset-bgp.xml.in: reset bgp vrf <vrf> ipv6 1-4294967295 <1-4294967295> in	reset bgp vrf <text> ipv6 1-4294967295 <text> in
×	not yet documented	reset-bgp.xml.in: reset bgp vrf <vrf> ipv6 1-4294967295 <1-4294967295> in prefix-filter	reset bgp vrf <text> ipv6 1-4294967295 <text> in prefix-filter
×	not yet documented	reset-bgp.xml.in: reset bgp vrf <vrf> ipv6 1-4294967295 <1- 4294967295> message- stats	reset bgp vrf <text> ipv6 1-4294967295 <text> message-stats
×	not yet documented	reset-bgp.xml.in: reset bgp vrf <vrf> ipv6 1-4294967295 <1-4294967295> out	reset bgp vrf <text> ipv6 1-4294967295 <text> out
×	not yet documented	reset-bgp.xml.in: reset bgp vrf <vrf> ipv6 1-4294967295 <1-4294967295> soft	reset bgp vrf <text> ipv6 1-4294967295 <text> soft
×	not yet documented	reset-bgp.xml.in: reset bgp vrf <vrf> ipv6 1-4294967295 <1-4294967295> soft in	reset bgp vrf <text> ipv6 1-4294967295 <text> soft in
×	not yet documented	reset-bgp.xml.in: reset bgp vrf <vrf> ipv6 1-4294967295 <1-4294967295> soft out	reset bgp vrf <text> ipv6 1-4294967295 <text> soft out
×	not yet documented	reset-bgp.xml.in: reset bgp vrf <vrf> ipv6 all	reset bgp vrf <text> ipv6 all
×	not yet documented	reset-bgp.xml.in: reset bgp vrf <vrf> ipv6 external	reset bgp vrf <text> ipv6 external

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Table 2 – continued from previous page

Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	reset-bgp.xml.in: reset bgp vrf <vrf> ipv6 external in	reset bgp vrf <text> ipv6 external in
×	not yet documented	reset-bgp.xml.in: reset bgp vrf <vrf> ipv6 external in prefix-filter	reset bgp vrf <text> ipv6 external in prefix-filter
×	not yet documented	reset-bgp.xml.in: reset bgp vrf <vrf> ipv6 external message-stats	reset bgp vrf <text> ipv6 external message-stats
×	not yet documented	reset-bgp.xml.in: reset bgp vrf <vrf> ipv6 external out	reset bgp vrf <text> ipv6 external out
×	not yet documented	reset-bgp.xml.in: reset bgp vrf <vrf> ipv6 external soft	reset bgp vrf <text> ipv6 external soft
×	not yet documented	reset-bgp.xml.in: reset bgp vrf <vrf> ipv6 external soft in	reset bgp vrf <text> ipv6 external soft in
×	not yet documented	reset-bgp.xml.in: reset bgp vrf <vrf> ipv6 external soft out	reset bgp vrf <text> ipv6 external soft out
×	not yet documented	reset-bgp.xml.in: reset bgp vrf <vrf> ipv6 <ipv6> in	reset bgp vrf <text> ipv6 <text> in
×	not yet documented	reset-bgp.xml.in: reset bgp vrf <vrf> ipv6 <ipv6> in prefix-filter	reset bgp vrf <text> ipv6 <text> in prefix-filter
×	not yet documented	reset-bgp.xml.in: reset bgp vrf <vrf> ipv6 <ipv6> message-stats	reset bgp vrf <text> ipv6 <text> message-stats
×	not yet documented	reset-bgp.xml.in: reset bgp vrf <vrf> ipv6 <ipv6> out	reset bgp vrf <text> ipv6 <text> out
×	not yet documented	reset-bgp.xml.in: reset bgp vrf <vrf> ipv6 peer-group <peer-group>	reset bgp vrf <text> ipv6 peer-group <text>
×	not yet documented	reset-bgp.xml.in: reset bgp vrf <vrf> ipv6 peer-group <peer-group> in	reset bgp vrf <text> ipv6 peer-group <text> in
×	not yet documented	reset-bgp.xml.in: reset bgp vrf <vrf> ipv6 peer-group <peer-group> in prefix-filter	reset bgp vrf <text> ipv6 peer-group <text> in prefix-filter
×	not yet documented	reset-bgp.xml.in: reset bgp vrf <vrf> ipv6 peer-group <peer-group> message-stats	reset bgp vrf <text> ipv6 peer-group <text> message-stats

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Table 2 – continued from previous page

Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	reset-bgp.xml.in: reset bgp vrf <vrf> ipv6 peer-group <peer-group> out	reset bgp vrf <text> ipv6 peer-group <text> out
×	not yet documented	reset-bgp.xml.in: reset bgp vrf <vrf> ipv6 peer-group <peer-group> soft	reset bgp vrf <text> ipv6 peer-group <text> soft
×	not yet documented	reset-bgp.xml.in: reset bgp vrf <vrf> ipv6 peer-group <peer-group> soft in	reset bgp vrf <text> ipv6 peer-group <text> soft in
×	not yet documented	reset-bgp.xml.in: reset bgp vrf <vrf> ipv6 peer-group <peer-group> soft out	reset bgp vrf <text> ipv6 peer-group <text> soft out
×	not yet documented	reset-bgp.xml.in: reset bgp vrf <vrf> ipv6 <ipv6> soft	reset bgp vrf <text> ipv6 <text> soft
×	not yet documented	reset-bgp.xml.in: reset bgp vrf <vrf> ipv6 <ipv6> soft in	reset bgp vrf <text> ipv6 <text> soft in
×	not yet documented	reset-bgp.xml.in: reset bgp vrf <vrf> ipv6 <ipv6> soft out	reset bgp vrf <text> ipv6 <text> soft out
×	not yet documented	reset-bgp.xml.in: reset bgp vrf <vrf> l2vpn evpn <evpn>	reset bgp vrf <text> l2vpn evpn <text>
×	not yet documented	reset-bgp.xml.in: reset bgp vrf <vrf> l2vpn evpn 1-4294967295 <1- 4294967295>	reset bgp vrf <text> l2vpn evpn 1-4294967295 <text>
×	not yet documented	reset-bgp.xml.in: reset bgp vrf <vrf> l2vpn evpn 1-4294967295 <1- 4294967295> in	reset bgp vrf <text> l2vpn evpn 1-4294967295 <text> in
×	not yet documented	reset-bgp.xml.in: reset bgp vrf <vrf> l2vpn evpn 1-4294967295 <1- 4294967295> in prefix- filter	reset bgp vrf <text> l2vpn evpn 1-4294967295 <text> in prefix-filter
×	not yet documented	reset-bgp.xml.in: reset bgp vrf <vrf> l2vpn evpn 1-4294967295 <1- 4294967295> message- stats	reset bgp vrf <text> l2vpn evpn 1-4294967295 <text> message-stats

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Table 2 – continued from previous page

Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	reset-bgp.xml.in: reset bgp vrf <vrf> l2vpn evpn 1-4294967295 <1- 4294967295> out	reset bgp vrf <text> l2vpn evpn 1-4294967295 <text> out
×	not yet documented	reset-bgp.xml.in: reset bgp vrf <vrf> l2vpn evpn 1-4294967295 <1- 4294967295> soft	reset bgp vrf <text> l2vpn evpn 1-4294967295 <text> soft
×	not yet documented	reset-bgp.xml.in: reset bgp vrf <vrf> l2vpn evpn 1-4294967295 <1- 4294967295> soft in	reset bgp vrf <text> l2vpn evpn 1-4294967295 <text> soft in
×	not yet documented	reset-bgp.xml.in: reset bgp vrf <vrf> l2vpn evpn 1-4294967295 <1- 4294967295> soft out	reset bgp vrf <text> l2vpn evpn 1-4294967295 <text> soft out
×	not yet documented	reset-bgp.xml.in: reset bgp vrf <vrf> l2vpn evpn all	reset bgp vrf <text> l2vpn evpn all
×	not yet documented	reset-bgp.xml.in: reset bgp vrf <vrf> l2vpn evpn external	reset bgp vrf <text> l2vpn evpn external
×	not yet documented	reset-bgp.xml.in: reset bgp vrf <vrf> l2vpn evpn external in	reset bgp vrf <text> l2vpn evpn external in
×	not yet documented	reset-bgp.xml.in: reset bgp vrf <vrf> l2vpn evpn external in prefix- filter	reset bgp vrf <text> l2vpn evpn external in prefix- filter
×	not yet documented	reset-bgp.xml.in: reset bgp vrf <vrf> l2vpn evpn external message- stats	reset bgp vrf <text> l2vpn evpn external message- stats
×	not yet documented	reset-bgp.xml.in: reset bgp vrf <vrf> l2vpn evpn external out	reset bgp vrf <text> l2vpn evpn external out
×	not yet documented	reset-bgp.xml.in: reset bgp vrf <vrf> l2vpn evpn external soft	reset bgp vrf <text> l2vpn evpn external soft
×	not yet documented	reset-bgp.xml.in: reset bgp vrf <vrf> l2vpn evpn external soft in	reset bgp vrf <text> l2vpn evpn external soft in
×	not yet documented	reset-bgp.xml.in: reset bgp vrf <vrf> l2vpn evpn external soft out	reset bgp vrf <text> l2vpn evpn external soft out
×	not yet documented	reset-bgp.xml.in: reset bgp vrf <vrf> l2vpn evpn <evpn> in	reset bgp vrf <text> l2vpn evpn <text> in

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Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	reset-bgp.xml.in: reset bgp vrf <vrf> l2vpn evpn <evpn> in prefix- filter	reset bgp vrf <text> l2vpn evpn <text> in prefix-filter
×	not yet documented	reset-bgp.xml.in: reset bgp vrf <vrf> l2vpn evpn <evpn> message- stats	reset bgp vrf <text> l2vpn evpn <text> message-stats
×	not yet documented	reset-bgp.xml.in: reset bgp vrf <vrf> l2vpn evpn <evpn> out	reset bgp vrf <text> l2vpn evpn <text> out
×	not yet documented	reset-bgp.xml.in: reset bgp vrf <vrf> l2vpn evpn peer-group <peer-group>	reset bgp vrf <text> l2vpn evpn peer-group <text>
×	not yet documented	reset-bgp.xml.in: reset bgp vrf <vrf> l2vpn evpn peer-group <peer-group> in	reset bgp vrf <text> l2vpn evpn peer-group <text> in
×	not yet documented	reset-bgp.xml.in: reset bgp vrf <vrf> l2vpn evpn peer-group <peer- group> in prefix-filter	reset bgp vrf <text> l2vpn evpn peer-group <text> in prefix-filter
×	not yet documented	reset-bgp.xml.in: reset bgp vrf <vrf> l2vpn evpn peer-group <peer- group> message-stats	reset bgp vrf <text> l2vpn evpn peer-group <text> message-stats
×	not yet documented	reset-bgp.xml.in: reset bgp vrf <vrf> l2vpn evpn peer-group <peer-group> out	reset bgp vrf <text> l2vpn evpn peer-group <text> out
×	not yet documented	reset-bgp.xml.in: reset bgp vrf <vrf> l2vpn evpn peer-group <peer-group> soft	reset bgp vrf <text> l2vpn evpn peer-group <text> soft
×	not yet documented	reset-bgp.xml.in: reset bgp vrf <vrf> l2vpn evpn peer-group <peer-group> soft in	reset bgp vrf <text> l2vpn evpn peer-group <text> soft in
×	not yet documented	reset-bgp.xml.in: reset bgp vrf <vrf> l2vpn evpn peer-group <peer-group> soft out	reset bgp vrf <text> l2vpn evpn peer-group <text> soft out
×	not yet documented	reset-bgp.xml.in: reset bgp vrf <vrf> l2vpn evpn <evpn> soft	reset bgp vrf <text> l2vpn evpn <text> soft
×	not yet documented	reset-bgp.xml.in: reset bgp vrf <vrf> l2vpn evpn <evpn> soft in	reset bgp vrf <text> l2vpn evpn <text> soft in

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Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	reset-bgp.xml.in: reset bgp vrf <vrf> l2vpn evpn <evpn> soft out	reset bgp vrf <text> l2vpn evpn <text> soft out
×	not yet documented	Nothing found in XML Definitions	reset bgp vrf <text> <text> message-stats
×	not yet documented	reset-bgp.xml.in: reset bgp vrf <vrf> node.tag	Nothing found in VyOS
×	not yet documented	reset-bgp.xml.in: reset bgp vrf <vrf> node.tag in	Nothing found in VyOS
×	not yet documented	reset-bgp.xml.in: reset bgp vrf <vrf> node.tag in prefix-filter	Nothing found in VyOS
×	not yet documented	reset-bgp.xml.in: reset bgp vrf <vrf> node.tag message-stats	Nothing found in VyOS
×	not yet documented	reset-bgp.xml.in: reset bgp vrf <vrf> node.tag out	Nothing found in VyOS
×	not yet documented	reset-bgp.xml.in: reset bgp vrf <vrf> node.tag soft	Nothing found in VyOS
×	not yet documented	reset-bgp.xml.in: reset bgp vrf <vrf> node.tag soft in	Nothing found in VyOS
×	not yet documented	reset-bgp.xml.in: reset bgp vrf <vrf> node.tag soft out	Nothing found in VyOS
×	not yet documented	Nothing found in XML Definitions	reset bgp vrf <text> <text> out
×	not yet documented	reset-bgp.xml.in: reset bgp vrf <vrf> peer- group <peer-group>	reset bgp vrf <text> peer- group <text>
×	not yet documented	reset-bgp.xml.in: reset bgp vrf <vrf> peer- group <peer-group> in	reset bgp vrf <text> peer- group <text> in
×	not yet documented	reset-bgp.xml.in: reset bgp vrf <vrf> peer- group <peer-group> in prefix-filter	reset bgp vrf <text> peer- group <text> in prefix- filter
×	not yet documented	reset-bgp.xml.in: reset bgp vrf <vrf> peer-group <peer-group> message-stats	reset bgp vrf <text> peer- group <text> message- stats
×	not yet documented	reset-bgp.xml.in: reset bgp vrf <vrf> peer- group <peer-group> out	reset bgp vrf <text> peer- group <text> out

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Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	reset-bgp.xml.in: reset bgp vrf <vrf> peer-group <peer-group> soft	reset bgp vrf <text> peer-group <text> soft
×	not yet documented	reset-bgp.xml.in: reset bgp vrf <vrf> peer-group <peer-group> soft in	reset bgp vrf <text> peer-group <text> soft in
×	not yet documented	reset-bgp.xml.in: reset bgp vrf <vrf> peer-group <peer-group> soft out	reset bgp vrf <text> peer-group <text> soft out
×	not yet documented	reset-bgp.xml.in: reset bgp vrf <vrf> prefix <prefix>	reset bgp vrf <text> prefix <text>
×	not yet documented	Nothing found in XML Definitions	reset bgp vrf <text> <text> soft
×	not yet documented	Nothing found in XML Definitions	reset bgp vrf <text> <text> soft in
×	not yet documented	Nothing found in XML Definitions	reset bgp vrf <text> <text> soft out
×	not yet documented	reset-conntrack.xml.in: reset conntrack	reset conntrack
×	not yet documented	Nothing found in XML Definitions	reset conntrack-sync external-cache
×	not yet documented	Nothing found in XML Definitions	reset conntrack-sync internal-cache
×	not yet documented	dns-dynamic.xml.in: reset dns dynamic	Nothing found in VyOS
×	<i>reset dns forwarding <all domain></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	dns-forwarding.xml.in: reset dns forwarding all	reset dns forwarding all
×	not yet documented	dns-forwarding.xml.in: reset dns forwarding domain <domain>	reset dns forwarding domain <text>
×	not yet documented	ipv4-route.xml.in: reset ip arp address <address>	reset ip arp address <text>
×	not yet documented	ipv4-route.xml.in: reset ip arp interface <interface>	reset ip arp interface <text>
×	not yet documented	ipv4-route.xml.in: reset ip arp table	reset ip arp table
×	not yet documented	reset-ip-bgp.xml.in: reset ip bgp <bgp>	reset ip bgp <text>
×	not yet documented	reset-ip-bgp.xml.in: reset ip bgp 1-4294967295 <1-4294967295>	reset ip bgp 1-4294967295 <text>

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Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	reset-ip-bgp.xml.in: reset ip bgp 1-4294967295 <1-4294967295> in	reset ip bgp 1-4294967295 <text> in
×	not yet documented	reset-ip-bgp.xml.in: reset ip bgp 1-4294967295 <1-4294967295> in prefix-filter	reset ip bgp 1-4294967295 <text> in prefix-filter
×	not yet documented	reset-ip-bgp.xml.in: reset ip bgp 1-4294967295 <1-4294967295> message-stats	reset ip bgp 1-4294967295 <text> message-stats
×	not yet documented	reset-ip-bgp.xml.in: reset ip bgp 1-4294967295 <1-4294967295> out	reset ip bgp 1-4294967295 <text> out
×	not yet documented	reset-ip-bgp.xml.in: reset ip bgp 1-4294967295 <1-4294967295> soft	reset ip bgp 1-4294967295 <text> soft
×	not yet documented	reset-ip-bgp.xml.in: reset ip bgp 1-4294967295 <1-4294967295> soft in	reset ip bgp 1-4294967295 <text> soft in
×	not yet documented	reset-ip-bgp.xml.in: reset ip bgp 1-4294967295 <1-4294967295> soft out	reset ip bgp 1-4294967295 <text> soft out
×	not yet documented	reset-ip-bgp.xml.in: reset ip bgp all	reset ip bgp all
×	not yet documented	reset-ip-bgp.xml.in: reset ip bgp dampening <dampening>	reset ip bgp dampening <text> <text>
×	not yet documented	reset-ip-bgp.xml.in: reset ip bgp dampening <dampening> node.tag	Nothing found in VyOS
×	not yet documented	reset-ip-bgp.xml.in: reset ip bgp external	reset ip bgp external
×	not yet documented	reset-ip-bgp.xml.in: reset ip bgp external in	reset ip bgp external in
×	not yet documented	reset-ip-bgp.xml.in: reset ip bgp external in prefix-filter	reset ip bgp external in prefix-filter
×	not yet documented	reset-ip-bgp.xml.in: reset ip bgp external message-stats	reset ip bgp external message-stats
×	not yet documented	reset-ip-bgp.xml.in: reset ip bgp external out	reset ip bgp external out
×	not yet documented	reset-ip-bgp.xml.in: reset ip bgp external soft	reset ip bgp external soft
×	not yet documented	reset-ip-bgp.xml.in: reset ip bgp external soft in	reset ip bgp external soft in

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Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	reset-ip-bgp.xml.in: reset ip bgp external soft out	reset ip bgp external soft out
×	not yet documented	reset-ip-bgp.xml.in: reset ip bgp <bgp> in	reset ip bgp <text> in
×	not yet documented	reset-ip-bgp.xml.in: reset ip bgp <bgp> in prefix-filter	reset ip bgp <text> in prefix-filter
×	not yet documented	reset-ip-bgp.xml.in: reset ip bgp <bgp> message-stats	reset ip bgp <text> message-stats
×	not yet documented	reset-ip-bgp.xml.in: reset ip bgp <bgp> out	reset ip bgp <text> out
×	not yet documented	reset-ip-bgp.xml.in: reset ip bgp peer-group <peer-group>	reset ip bgp peer-group <text>
×	not yet documented	reset-ip-bgp.xml.in: reset ip bgp peer-group <peer-group> in	reset ip bgp peer-group <text> in
×	not yet documented	reset-ip-bgp.xml.in: reset ip bgp peer-group <peer-group> in prefix-filter	reset ip bgp peer-group <text> in prefix-filter
×	not yet documented	reset-ip-bgp.xml.in: reset ip bgp peer-group <peer-group> message-stats	reset ip bgp peer-group <text> message-stats
×	not yet documented	reset-ip-bgp.xml.in: reset ip bgp peer-group <peer-group> out	reset ip bgp peer-group <text> out
×	not yet documented	reset-ip-bgp.xml.in: reset ip bgp peer-group <peer-group> soft	reset ip bgp peer-group <text> soft
×	not yet documented	reset-ip-bgp.xml.in: reset ip bgp peer-group <peer-group> soft in	reset ip bgp peer-group <text> soft in
×	not yet documented	reset-ip-bgp.xml.in: reset ip bgp peer-group <peer-group> soft out	reset ip bgp peer-group <text> soft out
×	not yet documented	reset-ip-bgp.xml.in: reset ip bgp <bgp> soft	reset ip bgp <text> soft
×	not yet documented	reset-ip-bgp.xml.in: reset ip bgp <bgp> soft in	reset ip bgp <text> soft in
×	not yet documented	reset-ip-bgp.xml.in: reset ip bgp <bgp> soft out	reset ip bgp <text> soft out
×	not yet documented	Nothing found in XML Definitions	reset ip bgp vrf <text> <text>
×	not yet documented	reset-ip-bgp.xml.in: reset ip bgp vrf <vrf> all	reset ip bgp vrf <text> all

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Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	reset-ip-bgp.xml.in: reset ip bgp vrf <vrf> node.tag	Nothing found in VyOS
×	not yet documented	reset-ip-igmp.xml.in: reset ip igmp interfaces	reset ip igmp interfaces
×	not yet documented	reset-ip-multicast.xml.in: reset ip multicast route	reset ip multicast route
×	not yet documented	Nothing found in XML Definitions	reset ipoe-server session interface <text>
×	not yet documented	Nothing found in XML Definitions	reset ipoe-server session sid <text>
×	not yet documented	Nothing found in XML Definitions	reset ipoe-server session username <text>
×	not yet documented	ipv4-route.xml.in: reset ip route cache <cache>	reset ip route cache <text>
×	<i>reset ipv6 neighbors <address / interface></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	ipv6-route.xml.in: reset ipv6 neighbors ad- dress <address>	reset ipv6 neighbors ad- dress <text>
×	not yet documented	ipv6-route.xml.in: reset ipv6 neighbors inter- face <interface>	reset ipv6 neighbors inter- face <text>
✓	<i>reset ipv6 route cache</i>	ipv6-route.xml.in: reset ipv6 route cache <cache>	reset ipv6 route cache <text>
×	not yet documented	reset-mpls.xml.in: reset mpls ldp neighbor <neighbor>	reset mpls ldp neighbor <text>
×	<i>reset mpls ldp neighbor <address></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	Nothing found in XML Definitions	reset nhrp flush
×	not yet documented	Nothing found in XML Definitions	reset nhrp flush tunnel <text>
×	not yet documented	Nothing found in XML Definitions	reset nhrp purge
×	not yet documented	Nothing found in XML Definitions	reset nhrp purge tunnel <text>
×	not yet documented	Nothing found in XML Definitions	reset openvpn client <text>
×	<i>reset openvpn client <text></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	Nothing found in XML Definitions	reset openvpn interface <text>

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Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<i>reset openvpn interface <interface></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	Nothing found in XML Definitions	reset pppoe-server all
×	not yet documented	Nothing found in XML Definitions	reset pppoe-server interface <text>
×	not yet documented	Nothing found in XML Definitions	reset pppoe-server username <text>
×	not yet documented	rpki.xml.in: reset rpki	reset rpki
×	not yet documented	Nothing found in XML Definitions	reset terminal
×	not yet documented	Nothing found in XML Definitions	reset vpn ipsec profile <text> tunnel <text>
×	not yet documented	Nothing found in XML Definitions	reset vpn ipsec profile <text> tunnel <text> remote-host <text>
×	not yet documented	Nothing found in XML Definitions	reset vpn ipsec remote-access all
×	not yet documented	Nothing found in XML Definitions	reset vpn ipsec remote-access user <text>
×	<i>reset vpn ipsec site-to-site all</i>	Nothing found in XML Definitions	reset vpn ipsec site-to-site all
×	not yet documented	Nothing found in XML Definitions	reset vpn ipsec site-to-site peer <text>
×	<i>reset vpn ipsec site-to-site peer <name></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>reset vpn ipsec site-to-site peer <name> tunnel <number></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>reset vpn ipsec site-to-site peer <name> vti <number></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	Nothing found in XML Definitions	reset vpn ipsec site-to-site peer <text> tunnel <text>
×	not yet documented	Nothing found in XML Definitions	reset vpn ipsec site-to-site peer <text> vti
×	not yet documented	reset-vpn.xml.in: reset vpn l2tp all	reset vpn l2tp all
×	not yet documented	reset-vpn.xml.in: reset vpn l2tp interface <interface>	reset vpn l2tp interface <text>
×	not yet documented	reset-vpn.xml.in: reset vpn l2tp user <user>	reset vpn l2tp user <text>
×	not yet documented	reset-vpn.xml.in: reset vpn pptp all	reset vpn pptp all

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Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	reset-vpn.xml.in: reset vpn pptp interface <interface>	reset vpn pptp interface <text>
×	not yet documented	reset-vpn.xml.in: reset vpn pptp user <user>	reset vpn pptp user <text>
×	not yet documented	reset-vpn.xml.in: reset vpn sstp all	reset vpn sstp all
×	not yet documented	reset-vpn.xml.in: reset vpn sstp interface <interface>	reset vpn sstp interface <text>
×	not yet documented	reset-vpn.xml.in: reset vpn sstp user <user>	reset vpn sstp user <text>
×	not yet documented	restart-frr.xml.in: restart all	restart all
×	not yet documented	restart-frr.xml.in: restart babel	restart babel
×	not yet documented	restart-frr.xml.in: restart bfd	restart bfd
×	not yet documented	restart-frr.xml.in: restart bgp	restart bgp
×	not yet documented	Nothing found in XML Definitions	restart cluster
×	not yet documented	Nothing found in XML Definitions	restart conntrack-sync
×	not yet documented	Nothing found in XML Definitions	restart container <text>
×	<i>restart container <containername></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>restart dhcp relay-agent</i>	Nothing found in XML Definitions	restart dhcp relay-agent
×	<i>restart dhcp server</i>	Nothing found in XML Definitions	restart dhcp server
×	<i>restart dhcpv6 relay-agent</i>	Nothing found in XML Definitions	restart dhcpv6 relay-agent
×	<i>restart dhcpv6 server</i>	Nothing found in XML Definitions	restart dhcpv6 server
×	not yet documented	Nothing found in XML Definitions	restart dns dynamic
×	<i>restart dns forwarding</i>	Nothing found in XML Definitions	restart dns forwarding
×	not yet documented	Nothing found in XML Definitions	restart flow-accounting
✓	<i>restart igmp-proxy</i>	igmp-proxy.xml.in: restart igmp-proxy	restart igmp-proxy
×	not yet documented	ipoe-server.xml.in: restart ipoe-server	restart ipoe-server
×	<i>restart ipsec</i>	Nothing found in XML Definitions	restart ipsec

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Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	restart-frr.xml.in: restart isis	restart isis
×	not yet documented	restart-frr.xml.in: restart ldp	restart ldp
×	<i>restart mdns repeater</i>	mdns-reflector.xml.in: restart mdns repeater	Nothing found in VyOS
×	not yet documented	openconnect.xml.in: restart openconnect-server	restart openconnect-server
×	not yet documented	restart-frr.xml.in: restart ospf	restart ospf
×	not yet documented	restart-frr.xml.in: restart ospfv3	restart ospfv3
×	not yet documented	Nothing found in XML Definitions	restart pppoe-server
×	not yet documented	restart-frr.xml.in: restart rip	restart rip
×	not yet documented	restart-frr.xml.in: restart ripng	restart ripng
×	not yet documented	restart-router- advert.xml.in: restart router-advert	Nothing found in VyOS
×	not yet documented	restart-snmp.xml.in: restart snmp	restart snmp
✓	<i>restart ssh</i>	restart-ssh.xml.in: restart ssh	restart ssh
×	not yet documented	restart-frr.xml.in: restart static	restart static
×	<i>restart suricata</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	vrrp.xml.in: restart vrrp	restart vrrp
×	not yet documented	Nothing found in XML Definitions	restart wan-load-balance
×	not yet documented	Nothing found in XML Definitions	restart webproxy
×	not yet documented	restart-frr.xml.in: restart zebra	restart zebra
×	<i>run generate macsec mka cak <gcm-aes-128/gcm-aes-</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>run generate macsec mka ckn</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	show-arp.xml.in: show arp	show arp
×	not yet documented	show-arp.xml.in: show arp interface <inter- face>	show arp interface <text>
×	not yet documented	show-babel.xml.in: show babel interface	show babel interface

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Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	show-babel.xml.in: show babel neighbor <neighbor>	show babel neighbor <text>
×	not yet documented	show-babel.xml.in: show babel route	show babel route
×	not yet documented	show-bfd.xml.in: show bfd peer <peer>	show bfd peer <text>
×	not yet documented	show-bfd.xml.in: show bfd peer <peer> counters	show bfd peer <text> counters
✓	<i>show bfd peers</i>	show-bfd.xml.in: show bfd peers	show bfd peers
×	not yet documented	show-bfd.xml.in: show bfd peers brief	show bfd peers brief
×	not yet documented	show-bfd.xml.in: show bfd peers counters	show bfd peers counters
×	not yet documented	show-bfd.xml.in: show bfd static routes	show bfd static routes
×	not yet documented	show-bgp.xml.in: show bgp	show bgp
×	<i>show bgp <ipv4/ipv6></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>show bgp <ipv4/ipv6> <address/prefix></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>show bgp <ipv4/ipv6> community-list <name></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>show bgp <ipv4/ipv6> community <value></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>show bgp <ipv4/ipv6> dampening dampened-paths</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>show bgp <ipv4/ipv6> dampening flap-statistics</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>show bgp <ipv4/ipv6> filter-list <name></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>show bgp <ipv4/ipv6> neighbors <address> advertised-routes</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>show bgp <ipv4/ipv6> neighbors <address> dampened-routes</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>show bgp <ipv4/ipv6> neighbors <address> received-routes</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>show bgp <ipv4/ipv6> neighbors <address> routes</i>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<i>show bgp <ipv4/ipv6> regexp <text></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>show bgp <ipv4/ipv6> summary</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>show bgp <ipv4/ipv6> vpn</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>show bgp <ipv4/ipv6> vpn summary</i>	Nothing found in XML Definitions	Nothing found in VyOS
✓	<i>show bgp cidr-only</i>	show-bgp.xml.in: show bgp cidr-only	show bgp cidr-only
×	not yet documented	show-bgp.xml.in: show bgp cidr-only wide	show bgp cidr-only wide
×	not yet documented	show-bgp.xml.in: show bgp community	show bgp community <text>
×	not yet documented	show-bgp.xml.in: show bgp community-list <community-list>	show bgp community-list <text>
×	not yet documented	show-bgp.xml.in: show bgp community-list <community-list> exact-match	show bgp community-list <text> exact-match
×	not yet documented	show-bgp.xml.in: show bgp community accept-own	show bgp community accept-own
×	not yet documented	show-bgp.xml.in: show bgp community accept-own-nexthop	show bgp community accept-own-nexthop
×	not yet documented	show-bgp.xml.in: show bgp community blackhole	show bgp community blackhole
×	not yet documented	show-bgp.xml.in: show bgp community exact-match	show bgp community <text> exact-match
×	not yet documented	show-bgp.xml.in: show bgp community graceful-shutdown	show bgp community graceful-shutdown
×	not yet documented	show-bgp.xml.in: show bgp community llgr-stale	show bgp community llgr-stale
×	not yet documented	show-bgp.xml.in: show bgp community local-AS	show bgp community local-AS
×	not yet documented	show-bgp.xml.in: show bgp community no-advertise	show bgp community no-advertise
×	not yet documented	show-bgp.xml.in: show bgp community no-export	show bgp community no-export

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Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	show-bgp.xml.in: show bgp community no-llgr	show bgp community no-llgr
×	not yet documented	show-bgp.xml.in: show bgp community no-peer	show bgp community no-peer
×	not yet documented	show-bgp.xml.in: show bgp community route-filter-translated-v4	show bgp community route-filter-translated-v4
×	not yet documented	show-bgp.xml.in: show bgp community route-filter-translated-v6	show bgp community route-filter-translated-v6
×	not yet documented	show-bgp.xml.in: show bgp community route-filter-v4	show bgp community route-filter-v4
×	not yet documented	show-bgp.xml.in: show bgp community route-filter-v6	show bgp community route-filter-v6
×	not yet documented	show-bgp.xml.in: show bgp dampening dampened-paths	show bgp dampening dampened-paths
×	not yet documented	show-bgp.xml.in: show bgp dampening flap-statistics	show bgp dampening flap-statistics
×	not yet documented	show-bgp.xml.in: show bgp dampening parameters	show bgp dampening parameters
×	not yet documented	show-bgp.xml.in: show bgp filter-list <filter-list>	Nothing found in VyOS
×	not yet documented	show-bgp.xml.in: show bgp ipv4	show bgp ipv4 <text>
×	not yet documented	show-bgp.xml.in: show bgp ipv4 <ipv4> bestpath	show bgp ipv4 <text> bestpath
×	not yet documented	show-bgp.xml.in: show bgp ipv4 community <text>	show bgp ipv4 community <text>
×	not yet documented	show-bgp.xml.in: show bgp ipv4 community-list <community-list>	show bgp ipv4 community-list <text>
×	not yet documented	show-bgp.xml.in: show bgp ipv4 community-list <community-list> exact-match	show bgp ipv4 community-list <text> exact-match
×	not yet documented	show-bgp.xml.in: show bgp ipv4 community accept-own	show bgp ipv4 community accept-own

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Table 2 – continued from previous page

Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	show-bgp.xml.in: show bgp ipv4 community accept-own-nexthop	show bgp ipv4 community accept-own-nexthop
×	not yet documented	show-bgp.xml.in: show bgp ipv4 community blackhole	show bgp ipv4 community blackhole
×	not yet documented	show-bgp.xml.in: show bgp ipv4 community exact-match	show bgp ipv4 community <text> exact-match
×	not yet documented	show-bgp.xml.in: show bgp ipv4 community graceful-shutdown	show bgp ipv4 community graceful-shutdown
×	not yet documented	show-bgp.xml.in: show bgp ipv4 community llgr-stale	show bgp ipv4 community llgr-stale
×	not yet documented	show-bgp.xml.in: show bgp ipv4 community local-AS	show bgp ipv4 community local-AS
×	not yet documented	show-bgp.xml.in: show bgp ipv4 community no-advertise	show bgp ipv4 community no-advertise
×	not yet documented	show-bgp.xml.in: show bgp ipv4 community no-export	show bgp ipv4 community no-export
×	not yet documented	show-bgp.xml.in: show bgp ipv4 community no-llgr	show bgp ipv4 community no-llgr
×	not yet documented	show-bgp.xml.in: show bgp ipv4 community no-peer	show bgp ipv4 community no-peer
×	not yet documented	show-bgp.xml.in: show bgp ipv4 community route-filter-translated-v4	show bgp ipv4 community route-filter-translated-v4
×	not yet documented	show-bgp.xml.in: show bgp ipv4 community route-filter-translated-v6	show bgp ipv4 community route-filter-translated-v6
×	not yet documented	show-bgp.xml.in: show bgp ipv4 community route-filter-v4	show bgp ipv4 community route-filter-v4
×	not yet documented	show-bgp.xml.in: show bgp ipv4 community route-filter-v6	show bgp ipv4 community route-filter-v6
×	not yet documented	show-bgp.xml.in: show bgp ipv4 dampening dampened-paths	show bgp ipv4 dampening dampened-paths
×	not yet documented	show-bgp.xml.in: show bgp ipv4 dampening flap-statistics	show bgp ipv4 dampening flap-statistics

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Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	show-bgp.xml.in: show bgp ipv4 dampening parameters	show bgp ipv4 dampening parameters
×	not yet documented	show-bgp.xml.in: show bgp ipv4 filter-list <filter-list>	Nothing found in VyOS
×	not yet documented	show-bgp.xml.in: show bgp ipv4 flowspec <text>	show bgp ipv4 flowspec <text>
×	not yet documented	show-bgp.xml.in: show bgp ipv4 flowspec <flowspec> bestpath	show bgp ipv4 flowspec <text> bestpath
×	not yet documented	show-bgp.xml.in: show bgp ipv4 flowspec community	show bgp ipv4 flowspec community <text>
×	not yet documented	show-bgp.xml.in: show bgp ipv4 flowspec community-list <community-list>	show bgp ipv4 flowspec community-list <text>
×	not yet documented	show-bgp.xml.in: show bgp ipv4 flowspec community-list <community-list> exact-match	show bgp ipv4 flowspec community-list <text> exact-match
×	not yet documented	show-bgp.xml.in: show bgp ipv4 flowspec community accept-own	show bgp ipv4 flowspec community accept-own
×	not yet documented	show-bgp.xml.in: show bgp ipv4 flowspec community accept-own-nexthop	show bgp ipv4 flowspec community accept-own-nexthop
×	not yet documented	show-bgp.xml.in: show bgp ipv4 flowspec community blackhole	show bgp ipv4 flowspec community blackhole
×	not yet documented	show-bgp.xml.in: show bgp ipv4 flowspec community exact-match	show bgp ipv4 flowspec community <text> exact-match
×	not yet documented	show-bgp.xml.in: show bgp ipv4 flowspec community graceful-shutdown	show bgp ipv4 flowspec community graceful-shutdown
×	not yet documented	show-bgp.xml.in: show bgp ipv4 flowspec community llgr-stale	show bgp ipv4 flowspec community llgr-stale
×	not yet documented	show-bgp.xml.in: show bgp ipv4 flowspec community local-AS	show bgp ipv4 flowspec community local-AS
×	not yet documented	show-bgp.xml.in: show bgp ipv4 flowspec community no-advertise	show bgp ipv4 flowspec community no-advertise

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Table 2 – continued from previous page

Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	show-bgp.xml.in: show bgp ipv4 flowspec community no-export	show bgp ipv4 flowspec community no-export
×	not yet documented	show-bgp.xml.in: show bgp ipv4 flowspec community no-llgr	show bgp ipv4 flowspec community no-llgr
×	not yet documented	show-bgp.xml.in: show bgp ipv4 flowspec community no-peer	show bgp ipv4 flowspec community no-peer
×	not yet documented	show-bgp.xml.in: show bgp ipv4 flowspec community route-filter- translated-v4	show bgp ipv4 flowspec community route-filter- translated-v4
×	not yet documented	show-bgp.xml.in: show bgp ipv4 flowspec community route-filter- translated-v6	show bgp ipv4 flowspec community route-filter- translated-v6
×	not yet documented	show-bgp.xml.in: show bgp ipv4 flowspec community route-filter-v4	show bgp ipv4 flowspec community route-filter-v4
×	not yet documented	show-bgp.xml.in: show bgp ipv4 flowspec community route-filter-v6	show bgp ipv4 flowspec community route-filter-v6
×	not yet documented	show-bgp.xml.in: show bgp ipv4 flowspec dampening dampened- paths	show bgp ipv4 flowspec dampening dampened- paths
×	not yet documented	show-bgp.xml.in: show bgp ipv4 flowspec dampening flap-statistics	show bgp ipv4 flowspec dampening flap-statistics
×	not yet documented	show-bgp.xml.in: show bgp ipv4 flowspec dampening parameters	show bgp ipv4 flowspec dampening parameters
×	not yet documented	show-bgp.xml.in: show bgp ipv4 flowspec detail	show bgp ipv4 flowspec detail
×	not yet documented	show-bgp.xml.in: show bgp ipv4 flowspec filter-list <filter-list>	Nothing found in VyOS
×	not yet documented	show-bgp.xml.in: show bgp ipv4 flowspec large-community	show bgp ipv4 flowspec large-community <text>
×	not yet documented	show-bgp.xml.in: show bgp ipv4 flowspec large-community-list <large-community-list>	show bgp ipv4 flowspec large-community-list <text>

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Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	show-bgp.xml.in: show bgp ipv4 flowspec large-community-list <large-community-list> exact-match	show bgp ipv4 flowspec large-community-list <text> exact-match
×	not yet documented	show-bgp.xml.in: show bgp ipv4 flowspec large-community <large- community> exact-match	show bgp ipv4 flowspec large-community <text> exact-match
×	not yet documented	show-bgp.xml.in: show bgp ipv4 flowspec <flowspec> longer- prefixes	show bgp ipv4 flowspec <text> longer-prefixes
×	not yet documented	show-bgp.xml.in: show bgp ipv4 flowspec <flowspec> multipath	show bgp ipv4 flowspec <text> multipath
×	not yet documented	show-bgp.xml.in: show bgp ipv4 flowspec neighbors <neighbors>	show bgp ipv4 flowspec neighbors <text>
×	not yet documented	show-bgp.xml.in: show bgp ipv4 flowspec neighbors <neighbors> advertised-routes	show bgp ipv4 flowspec neighbors <text> advertised-routes
×	not yet documented	show-bgp.xml.in: show bgp ipv4 flowspec neighbors <neighbors> dampened-routes	show bgp ipv4 flowspec neighbors <text> dampened-routes
×	not yet documented	show-bgp.xml.in: show bgp ipv4 flowspec neighbors <neighbors> filtered-routes	show bgp ipv4 flowspec neighbors <text> filtered- routes
×	not yet documented	show-bgp.xml.in: show bgp ipv4 flowspec neighbors <neighbors> flap-statistics	show bgp ipv4 flowspec neighbors <text> flap- statistics
×	not yet documented	show-bgp.xml.in: show bgp ipv4 flowspec neighbors <neighbors> prefix-counts	show bgp ipv4 flowspec neighbors <text> prefix- counts
×	not yet documented	show-bgp.xml.in: show bgp ipv4 flowspec neighbors <neighbors> received-routes	show bgp ipv4 flowspec neighbors <text> received-routes
×	not yet documented	show-bgp.xml.in: show bgp ipv4 flowspec neighbors <neighbors> re- ceived prefix-filter	show bgp ipv4 flowspec neighbors <text> received prefix-filter

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Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	show-bgp.xml.in: show bgp ipv4 flowspec neighbors <neighbors> routes	show bgp ipv4 flowspec neighbors <text> routes
×	not yet documented	show-bgp.xml.in: show bgp ipv4 flowspec prefix-list <prefix-list>	show bgp ipv4 flowspec prefix-list <text>
×	not yet documented	show-bgp.xml.in: show bgp ipv4 flowspec regexp <regexp>	show bgp ipv4 flowspec regexp <text>
×	not yet documented	show-bgp.xml.in: show bgp ipv4 flowspec route-map <route-map>	show bgp ipv4 flowspec route-map <text>
×	not yet documented	show-bgp.xml.in: show bgp ipv4 flowspec statistics	show bgp ipv4 flowspec statistics
×	not yet documented	show-bgp.xml.in: show bgp ipv4 flowspec summary	show bgp ipv4 flowspec summary
×	not yet documented	show-bgp.xml.in: show bgp ipv4 flowspec summary established	show bgp ipv4 flowspec summary established
×	not yet documented	show-bgp.xml.in: show bgp ipv4 flowspec summary failed	show bgp ipv4 flowspec summary failed
×	not yet documented	show-bgp.xml.in: show bgp ipv4 flowspec wide	show bgp ipv4 flowspec wide
×	not yet documented	show-bgp.xml.in: show bgp ipv4 large- community	show bgp ipv4 large- community <text>
×	not yet documented	show-bgp.xml.in: show bgp ipv4 large- community-list <large- community-list>	show bgp ipv4 large- community-list <text>
×	not yet documented	show-bgp.xml.in: show bgp ipv4 large- community-list <large- community-list> exact- match	show bgp ipv4 large- community-list <text> exact-match
×	not yet documented	show-bgp.xml.in: show bgp ipv4 large- community <large- community> exact-match	show bgp ipv4 large- community <text> exact- match
×	not yet documented	show-bgp.xml.in: show bgp ipv4 <ipv4> longer-prefixes	show bgp ipv4 <text> longer-prefixes

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Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	show-bgp.xml.in: show bgp ipv4 <ipv4> multipath	show bgp ipv4 <text> mul- tipath
×	not yet documented	show-bgp.xml.in: show bgp ipv4 neighbors <neighbors>	show bgp ipv4 neighbors <text>
×	not yet documented	show-bgp.xml.in: show bgp ipv4 neighbors <neighbors> advertised- routes	show bgp ipv4 neighbors <text> advertised-routes
×	not yet documented	show-bgp.xml.in: show bgp ipv4 neighbors <neighbors> dampened- routes	show bgp ipv4 neighbors <text> dampened-routes
×	not yet documented	show-bgp.xml.in: show bgp ipv4 neighbors <neighbors> filtered- routes	show bgp ipv4 neighbors <text> filtered-routes
×	not yet documented	show-bgp.xml.in: show bgp ipv4 neighbors <neighbors> flap-statistics	show bgp ipv4 neighbors <text> flap-statistics
×	not yet documented	show-bgp.xml.in: show bgp ipv4 neighbors <neighbors> prefix-counts	show bgp ipv4 neighbors <text> prefix-counts
×	not yet documented	show-bgp.xml.in: show bgp ipv4 neighbors <neighbors> received- routes	show bgp ipv4 neighbors <text> received-routes
×	not yet documented	show-bgp.xml.in: show bgp ipv4 neighbors <neighbors> received prefix-filter	show bgp ipv4 neigh- bors <text> received prefix-filter
×	not yet documented	show-bgp.xml.in: show bgp ipv4 neighbors <neighbors> routes	show bgp ipv4 neighbors <text> routes
×	not yet documented	show-bgp.xml.in: show bgp ipv4 prefix-list <prefix-list>	show bgp ipv4 prefix-list <text>
×	not yet documented	show-bgp.xml.in: show bgp ipv4 regexp <regexp>	show bgp ipv4 regexp <text>
×	not yet documented	show-bgp.xml.in: show bgp ipv4 route-map <route-map>	show bgp ipv4 route-map <text>
×	not yet documented	show-bgp.xml.in: show bgp ipv4 statistics	show bgp ipv4 statistics
×	not yet documented	show-bgp.xml.in: show bgp ipv4 summary	show bgp ipv4 summary

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Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	show-bgp.xml.in: show bgp ipv4 summary established	show bgp ipv4 summary established
×	not yet documented	show-bgp.xml.in: show bgp ipv4 summary failed	show bgp ipv4 summary failed
×	not yet documented	show-bgp.xml.in: show bgp ipv4 vpn	show bgp ipv4 vpn <text>
×	not yet documented	show-bgp.xml.in: show bgp ipv4 vpn <vpn> bestpath	show bgp ipv4 vpn <text> bestpath
×	not yet documented	show-bgp.xml.in: show bgp ipv4 vpn com- munity	show bgp ipv4 vpn com- munity <text>
×	not yet documented	show-bgp.xml.in: show bgp ipv4 vpn community-list <community-list>	show bgp ipv4 vpn community-list <text>
×	not yet documented	show-bgp.xml.in: show bgp ipv4 vpn community-list <community-list> exact- match	show bgp ipv4 vpn community-list <text> exact-match
×	not yet documented	show-bgp.xml.in: show bgp ipv4 vpn com- munity accept-own	show bgp ipv4 vpn com- munity accept-own
×	not yet documented	show-bgp.xml.in: show bgp ipv4 vpn community accept-own- nexthop	show bgp ipv4 vpn community accept-own- nexthop
×	not yet documented	show-bgp.xml.in: show bgp ipv4 vpn com- munity blackhole	show bgp ipv4 vpn com- munity blackhole
×	not yet documented	show-bgp.xml.in: show bgp ipv4 vpn com- munity exact-match	show bgp ipv4 vpn community <text> exact- match
×	not yet documented	show-bgp.xml.in: show bgp ipv4 vpn com- munity graceful-shutdown	show bgp ipv4 vpn com- munity graceful-shutdown
×	not yet documented	show-bgp.xml.in: show bgp ipv4 vpn com- munity llgr-stale	show bgp ipv4 vpn com- munity llgr-stale
×	not yet documented	show-bgp.xml.in: show bgp ipv4 vpn com- munity local-AS	show bgp ipv4 vpn com- munity local-AS
×	not yet documented	show-bgp.xml.in: show bgp ipv4 vpn com- munity no-advertise	show bgp ipv4 vpn com- munity no-advertise

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Table 2 – continued from previous page

Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	show-bgp.xml.in: show bgp ipv4 vpn community no-export	show bgp ipv4 vpn community no-export
×	not yet documented	show-bgp.xml.in: show bgp ipv4 vpn community no-llgr	show bgp ipv4 vpn community no-llgr
×	not yet documented	show-bgp.xml.in: show bgp ipv4 vpn community no-peer	show bgp ipv4 vpn community no-peer
×	not yet documented	show-bgp.xml.in: show bgp ipv4 vpn community route-filter-translated-v4	show bgp ipv4 vpn community route-filter-translated-v4
×	not yet documented	show-bgp.xml.in: show bgp ipv4 vpn community route-filter-translated-v6	show bgp ipv4 vpn community route-filter-translated-v6
×	not yet documented	show-bgp.xml.in: show bgp ipv4 vpn community route-filter-v4	show bgp ipv4 vpn community route-filter-v4
×	not yet documented	show-bgp.xml.in: show bgp ipv4 vpn community route-filter-v6	show bgp ipv4 vpn community route-filter-v6
×	not yet documented	show-bgp.xml.in: show bgp ipv4 vpn dampening dampened-paths	show bgp ipv4 vpn dampening dampened-paths
×	not yet documented	show-bgp.xml.in: show bgp ipv4 vpn dampening flap-statistics	show bgp ipv4 vpn dampening flap-statistics
×	not yet documented	show-bgp.xml.in: show bgp ipv4 vpn dampening parameters	show bgp ipv4 vpn dampening parameters
×	not yet documented	show-bgp.xml.in: show bgp ipv4 vpn filter-list <filter-list>	Nothing found in VyOS
×	not yet documented	show-bgp.xml.in: show bgp ipv4 vpn large-community	show bgp ipv4 vpn large-community <text>
×	not yet documented	show-bgp.xml.in: show bgp ipv4 vpn large-community-list <large-community-list>	show bgp ipv4 vpn large-community-list <text>
×	not yet documented	show-bgp.xml.in: show bgp ipv4 vpn large-community-list <large-community-list> exact-match	show bgp ipv4 vpn large-community-list <text> exact-match

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Table 2 – continued from previous page

Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	show-bgp.xml.in: show bgp ipv4 vpn large-community <large- community> exact-match	show bgp ipv4 vpn large- community <text> exact- match
×	not yet documented	show-bgp.xml.in: show bgp ipv4 vpn <vpn> longer-prefixes	show bgp ipv4 vpn <text> longer-prefixes
×	not yet documented	show-bgp.xml.in: show bgp ipv4 vpn <vpn> multipath	show bgp ipv4 vpn <text> multipath
×	not yet documented	show-bgp.xml.in: show bgp ipv4 vpn neigh- bors <neighbors>	show bgp ipv4 vpn neigh- bors <text>
×	not yet documented	show-bgp.xml.in: show bgp ipv4 vpn neighbors <neighbors> advertised-routes	show bgp ipv4 vpn neigh- bors <text> advertised- routes
×	not yet documented	show-bgp.xml.in: show bgp ipv4 vpn neighbors <neighbors> dampened-routes	show bgp ipv4 vpn neigh- bors <text> dampened- routes
×	not yet documented	show-bgp.xml.in: show bgp ipv4 vpn neigh- bors <neighbors> filtered- routes	show bgp ipv4 vpn neigh- bors <text> filtered-routes
×	not yet documented	show-bgp.xml.in: show bgp ipv4 vpn neighbors <neighbors> flap-statistics	show bgp ipv4 vpn neigh- bors <text> flap-statistics
×	not yet documented	show-bgp.xml.in: show bgp ipv4 vpn neigh- bors <neighbors> prefix- counts	show bgp ipv4 vpn neigh- bors <text> prefix-counts
×	not yet documented	show-bgp.xml.in: show bgp ipv4 vpn neighbors <neighbors> received-routes	show bgp ipv4 vpn neigh- bors <text> received- routes
×	not yet documented	show-bgp.xml.in: show bgp ipv4 vpn neigh- bors <neighbors> received prefix-filter	show bgp ipv4 vpn neigh- bors <text> received prefix-filter
×	not yet documented	show-bgp.xml.in: show bgp ipv4 vpn neigh- bors <neighbors> routes	show bgp ipv4 vpn neigh- bors <text> routes
×	not yet documented	show-bgp.xml.in: show bgp ipv4 vpn prefix- list <prefix-list>	show bgp ipv4 vpn prefix- list <text>

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Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	show-bgp.xml.in: show bgp ipv4 vpn regexp <regexp>	show bgp ipv4 vpn regexp <text>
×	not yet documented	show-bgp.xml.in: show bgp ipv4 vpn route- map <route-map>	show bgp ipv4 vpn route- map <text>
×	not yet documented	show-bgp.xml.in: show bgp ipv4 vpn statis- tics	show bgp ipv4 vpn statis- tics
×	not yet documented	show-bgp.xml.in: show bgp ipv4 vpn sum- mary	show bgp ipv4 vpn sum- mary
×	not yet documented	show-bgp.xml.in: show bgp ipv4 vpn sum- mary established	show bgp ipv4 vpn sum- mary established
×	not yet documented	show-bgp.xml.in: show bgp ipv4 vpn sum- mary failed	show bgp ipv4 vpn sum- mary failed
×	not yet documented	show-bgp.xml.in: show bgp ipv4 vpn wide	show bgp ipv4 vpn wide
×	not yet documented	show-bgp.xml.in: show bgp ipv4 wide	show bgp ipv4 wide
×	not yet documented	show-bgp.xml.in: show bgp ipv6	show bgp ipv6 <text>
×	not yet documented	show-bgp.xml.in: show bgp ipv6 <ipv6> bestpath	show bgp ipv6 <text> bestpath
×	not yet documented	show-bgp.xml.in: show bgp ipv6 community	show bgp ipv6 community <text>
×	not yet documented	show-bgp.xml.in: show bgp ipv6 community-list <community-list>	show bgp ipv6 community-list <text>
×	not yet documented	show-bgp.xml.in: show bgp ipv6 community-list <community-list> exact- match	show bgp ipv6 community-list <text> exact-match
×	not yet documented	show-bgp.xml.in: show bgp ipv6 community accept-own	show bgp ipv6 community accept-own
×	not yet documented	show-bgp.xml.in: show bgp ipv6 community accept-own-nexthop	show bgp ipv6 community accept-own-nexthop
×	not yet documented	show-bgp.xml.in: show bgp ipv6 community blackhole	show bgp ipv6 community blackhole

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Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	show-bgp.xml.in: show bgp ipv6 community exact-match	show bgp ipv6 community <text> exact-match
×	not yet documented	show-bgp.xml.in: show bgp ipv6 community graceful-shutdown	show bgp ipv6 community graceful-shutdown
×	not yet documented	show-bgp.xml.in: show bgp ipv6 community llgr-stale	show bgp ipv6 community llgr-stale
×	not yet documented	show-bgp.xml.in: show bgp ipv6 community local-AS	show bgp ipv6 community local-AS
×	not yet documented	show-bgp.xml.in: show bgp ipv6 community no-advertise	show bgp ipv6 community no-advertise
×	not yet documented	show-bgp.xml.in: show bgp ipv6 community no-export	show bgp ipv6 community no-export
×	not yet documented	show-bgp.xml.in: show bgp ipv6 community no-llgr	show bgp ipv6 community no-llgr
×	not yet documented	show-bgp.xml.in: show bgp ipv6 community no-peer	show bgp ipv6 community no-peer
×	not yet documented	show-bgp.xml.in: show bgp ipv6 community route-filter-translated-v4	show bgp ipv6 community route-filter-translated-v4
×	not yet documented	show-bgp.xml.in: show bgp ipv6 community route-filter-translated-v6	show bgp ipv6 community route-filter-translated-v6
×	not yet documented	show-bgp.xml.in: show bgp ipv6 community route-filter-v4	show bgp ipv6 community route-filter-v4
×	not yet documented	show-bgp.xml.in: show bgp ipv6 community route-filter-v6	show bgp ipv6 community route-filter-v6
×	not yet documented	show-bgp.xml.in: show bgp ipv6 dampening dampened-paths	show bgp ipv6 dampening dampened-paths
×	not yet documented	show-bgp.xml.in: show bgp ipv6 dampening flap-statistics	show bgp ipv6 dampening flap-statistics
×	not yet documented	show-bgp.xml.in: show bgp ipv6 dampening parameters	show bgp ipv6 dampening parameters
×	not yet documented	show-bgp.xml.in: show bgp ipv6 filter-list <filter-list>	Nothing found in VyOS

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Table 2 – continued from previous page

Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	show-bgp.xml.in: show bgp ipv6 large-community	show bgp ipv6 large-community <text>
×	not yet documented	show-bgp.xml.in: show bgp ipv6 large-community-list <large-community-list>	show bgp ipv6 large-community-list <text>
×	not yet documented	show-bgp.xml.in: show bgp ipv6 large-community-list <large-community-list> exact-match	show bgp ipv6 large-community-list <text> exact-match
×	not yet documented	show-bgp.xml.in: show bgp ipv6 large-community <large-community> exact-match	show bgp ipv6 large-community <text> exact-match
×	not yet documented	show-bgp.xml.in: show bgp ipv6 <ipv6> longer-prefixes	show bgp ipv6 <text> longer-prefixes
×	not yet documented	show-bgp.xml.in: show bgp ipv6 <ipv6> multipath	show bgp ipv6 <text> multipath
×	not yet documented	show-bgp.xml.in: show bgp ipv6 neighbors <neighbors>	show bgp ipv6 neighbors <text>
×	not yet documented	show-bgp.xml.in: show bgp ipv6 neighbors <neighbors> advertised-routes	show bgp ipv6 neighbors <text> advertised-routes
×	not yet documented	show-bgp.xml.in: show bgp ipv6 neighbors <neighbors> dampened-routes	show bgp ipv6 neighbors <text> dampened-routes
×	not yet documented	show-bgp.xml.in: show bgp ipv6 neighbors <neighbors> filtered-routes	show bgp ipv6 neighbors <text> filtered-routes
×	not yet documented	show-bgp.xml.in: show bgp ipv6 neighbors <neighbors> flap-statistics	show bgp ipv6 neighbors <text> flap-statistics
×	not yet documented	show-bgp.xml.in: show bgp ipv6 neighbors <neighbors> prefix-counts	show bgp ipv6 neighbors <text> prefix-counts
×	not yet documented	show-bgp.xml.in: show bgp ipv6 neighbors <neighbors> received-routes	show bgp ipv6 neighbors <text> received-routes

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Table 2 – continued from previous page

Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	show-bgp.xml.in: show bgp ipv6 neighbors <neighbors> received prefix-filter	show bgp ipv6 neighbors <text> received prefix-filter
×	not yet documented	show-bgp.xml.in: show bgp ipv6 neighbors <neighbors> routes	show bgp ipv6 neighbors <text> routes
×	not yet documented	show-bgp.xml.in: show bgp ipv6 prefix-list <prefix-list>	show bgp ipv6 prefix-list <text>
×	not yet documented	show-bgp.xml.in: show bgp ipv6 regexp <regexp>	show bgp ipv6 regexp <text>
×	not yet documented	show-bgp.xml.in: show bgp ipv6 route-map <route-map>	show bgp ipv6 route-map <text>
×	not yet documented	show-bgp.xml.in: show bgp ipv6 statistics	show bgp ipv6 statistics
×	not yet documented	show-bgp.xml.in: show bgp ipv6 summary	show bgp ipv6 summary
×	not yet documented	show-bgp.xml.in: show bgp ipv6 summary established	show bgp ipv6 summary established
×	not yet documented	show-bgp.xml.in: show bgp ipv6 summary failed	show bgp ipv6 summary failed
×	not yet documented	show-bgp.xml.in: show bgp ipv6 vpn	show bgp ipv6 vpn <text>
×	not yet documented	show-bgp.xml.in: show bgp ipv6 vpn <vpn> bestpath	show bgp ipv6 vpn <text> bestpath
×	not yet documented	show-bgp.xml.in: show bgp ipv6 vpn com- munity	show bgp ipv6 vpn com- munity <text>
×	not yet documented	show-bgp.xml.in: show bgp ipv6 vpn community-list <community-list>	show bgp ipv6 vpn community-list <text>
×	not yet documented	show-bgp.xml.in: show bgp ipv6 vpn community-list <community-list> exact- match	show bgp ipv6 vpn community-list <text> exact-match
×	not yet documented	show-bgp.xml.in: show bgp ipv6 vpn com- munity accept-own	show bgp ipv6 vpn com- munity accept-own

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Table 2 – continued from previous page

Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	show-bgp.xml.in: show bgp ipv6 vpn community accept-own- nexthop	show bgp ipv6 vpn community accept-own- nexthop
×	not yet documented	show-bgp.xml.in: show bgp ipv6 vpn com- munity blackhole	show bgp ipv6 vpn com- munity blackhole
×	not yet documented	show-bgp.xml.in: show bgp ipv6 vpn com- munity exact-match	show bgp ipv6 vpn community <text> exact- match
×	not yet documented	show-bgp.xml.in: show bgp ipv6 vpn com- munity graceful-shutdown	show bgp ipv6 vpn com- munity graceful-shutdown
×	not yet documented	show-bgp.xml.in: show bgp ipv6 vpn com- munity llgr-stale	show bgp ipv6 vpn com- munity llgr-stale
×	not yet documented	show-bgp.xml.in: show bgp ipv6 vpn com- munity local-AS	show bgp ipv6 vpn com- munity local-AS
×	not yet documented	show-bgp.xml.in: show bgp ipv6 vpn com- munity no-advertise	show bgp ipv6 vpn com- munity no-advertise
×	not yet documented	show-bgp.xml.in: show bgp ipv6 vpn com- munity no-export	show bgp ipv6 vpn com- munity no-export
×	not yet documented	show-bgp.xml.in: show bgp ipv6 vpn com- munity no-llgr	show bgp ipv6 vpn com- munity no-llgr
×	not yet documented	show-bgp.xml.in: show bgp ipv6 vpn com- munity no-peer	show bgp ipv6 vpn com- munity no-peer
×	not yet documented	show-bgp.xml.in: show bgp ipv6 vpn community route-filter- translated-v4	show bgp ipv6 vpn community route-filter- translated-v4
×	not yet documented	show-bgp.xml.in: show bgp ipv6 vpn community route-filter- translated-v6	show bgp ipv6 vpn community route-filter- translated-v6
×	not yet documented	show-bgp.xml.in: show bgp ipv6 vpn com- munity route-filter-v4	show bgp ipv6 vpn com- munity route-filter-v4
×	not yet documented	show-bgp.xml.in: show bgp ipv6 vpn com- munity route-filter-v6	show bgp ipv6 vpn com- munity route-filter-v6
×	not yet documented	show-bgp.xml.in: show bgp ipv6 vpn damp- ening dampened-paths	show bgp ipv6 vpn damp- ening dampened-paths

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Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	show-bgp.xml.in: show bgp ipv6 vpn damp- ening flap-statistics	show bgp ipv6 vpn damp- ening flap-statistics
×	not yet documented	show-bgp.xml.in: show bgp ipv6 vpn damp- ening parameters	show bgp ipv6 vpn damp- ening parameters
×	not yet documented	show-bgp.xml.in: show bgp ipv6 vpn filter- list <filter-list>	Nothing found in VyOS
×	not yet documented	show-bgp.xml.in: show bgp ipv6 vpn large- community	show bgp ipv6 vpn large- community <text>
×	not yet documented	show-bgp.xml.in: show bgp ipv6 vpn large-community-list <large-community-list>	show bgp ipv6 vpn large- community-list <text>
×	not yet documented	show-bgp.xml.in: show bgp ipv6 vpn large-community-list <large-community-list> exact-match	show bgp ipv6 vpn large- community-list <text> exact-match
×	not yet documented	show-bgp.xml.in: show bgp ipv6 vpn large-community <large- community> exact-match	show bgp ipv6 vpn large- community <text> exact- match
×	not yet documented	show-bgp.xml.in: show bgp ipv6 vpn <vpn> longer-prefixes	show bgp ipv6 vpn <text> longer-prefixes
×	not yet documented	show-bgp.xml.in: show bgp ipv6 vpn <vpn> multipath	show bgp ipv6 vpn <text> multipath
×	not yet documented	show-bgp.xml.in: show bgp ipv6 vpn neigh- bors <neighbors>	show bgp ipv6 vpn neigh- bors <text>
×	not yet documented	show-bgp.xml.in: show bgp ipv6 vpn neighbors <neighbors> advertised-routes	show bgp ipv6 vpn neigh- bors <text> advertised- routes
×	not yet documented	show-bgp.xml.in: show bgp ipv6 vpn neighbors <neighbors> dampened-routes	show bgp ipv6 vpn neigh- bors <text> dampened- routes
×	not yet documented	show-bgp.xml.in: show bgp ipv6 vpn neigh- bors <neighbors> filtered- routes	show bgp ipv6 vpn neigh- bors <text> filtered-routes

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Table 2 – continued from previous page

Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	show-bgp.xml.in: show bgp ipv6 vpn neighbors <neighbors> flap-statistics	show bgp ipv6 vpn neighbors <text> flap-statistics
×	not yet documented	show-bgp.xml.in: show bgp ipv6 vpn neighbors <neighbors> prefix-counts	show bgp ipv6 vpn neighbors <text> prefix-counts
×	not yet documented	show-bgp.xml.in: show bgp ipv6 vpn neighbors <neighbors> received-routes	show bgp ipv6 vpn neighbors <text> received-routes
×	not yet documented	show-bgp.xml.in: show bgp ipv6 vpn neighbors <neighbors> received prefix-filter	show bgp ipv6 vpn neighbors <text> received prefix-filter
×	not yet documented	show-bgp.xml.in: show bgp ipv6 vpn neighbors <neighbors> routes	show bgp ipv6 vpn neighbors <text> routes
×	not yet documented	show-bgp.xml.in: show bgp ipv6 vpn prefix-list <prefix-list>	show bgp ipv6 vpn prefix-list <text>
×	not yet documented	show-bgp.xml.in: show bgp ipv6 vpn regexp <regexp>	show bgp ipv6 vpn regexp <text>
×	not yet documented	show-bgp.xml.in: show bgp ipv6 vpn route-map <route-map>	show bgp ipv6 vpn route-map <text>
×	not yet documented	show-bgp.xml.in: show bgp ipv6 vpn statistics	show bgp ipv6 vpn statistics
×	not yet documented	show-bgp.xml.in: show bgp ipv6 vpn summary	show bgp ipv6 vpn summary
×	not yet documented	show-bgp.xml.in: show bgp ipv6 vpn summary established	show bgp ipv6 vpn summary established
×	not yet documented	show-bgp.xml.in: show bgp ipv6 vpn summary failed	show bgp ipv6 vpn summary failed
×	not yet documented	show-bgp.xml.in: show bgp ipv6 vpn wide	show bgp ipv6 vpn wide
×	not yet documented	show-bgp.xml.in: show bgp ipv6 wide	show bgp ipv6 wide
×	not yet documented	show-bgp.xml.in: show bgp l2vpn evpn	show bgp l2vpn evpn <text>
×	not yet documented	show-bgp.xml.in: show bgp l2vpn evpn all overlay	show bgp l2vpn evpn all overlay

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Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	show-bgp.xml.in: show bgp l2vpn evpn all tags	show bgp l2vpn evpn all tags
×	not yet documented	show-bgp.xml.in: show bgp l2vpn evpn community <community>	show bgp l2vpn evpn community <text>
×	not yet documented	show-bgp.xml.in: show bgp l2vpn evpn community <community> exact-match	show bgp l2vpn evpn community <text> exact-match
×	not yet documented	show-bgp.xml.in: show bgp l2vpn evpn es	show bgp l2vpn evpn es
×	not yet documented	show-bgp.xml.in: show bgp l2vpn evpn es-evi	show bgp l2vpn evpn es-evi
×	not yet documented	show-bgp.xml.in: show bgp l2vpn evpn es-evi detail	show bgp l2vpn evpn es-evi detail
×	not yet documented	show-bgp.xml.in: show bgp l2vpn evpn es-evi vni <vni>	show bgp l2vpn evpn es-evi vni <text>
×	not yet documented	show-bgp.xml.in: show bgp l2vpn evpn es-vrf	Nothing found in VyOS
×	not yet documented	show-bgp.xml.in: show bgp l2vpn evpn es detail	show bgp l2vpn evpn es detail
×	not yet documented	show-bgp.xml.in: show bgp l2vpn evpn import-rt	show bgp l2vpn evpn import-rt
×	not yet documented	show-bgp.xml.in: show bgp l2vpn evpn large-community <large-community>	show bgp l2vpn evpn large-community <text>
×	not yet documented	show-bgp.xml.in: show bgp l2vpn evpn large-community-list <large-community-list>	show bgp l2vpn evpn large-community-list <text>
×	not yet documented	show-bgp.xml.in: show bgp l2vpn evpn large-community-list <large-community-list> exact-match	show bgp l2vpn evpn large-community-list <text> exact-match
×	not yet documented	show-bgp.xml.in: show bgp l2vpn evpn large-community <large-community> exact-match	show bgp l2vpn evpn large-community <text> exact-match

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Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	show-bgp.xml.in: show bgp l2vpn evpn neighbors <neighbors>	show bgp l2vpn evpn neighbors <text>
×	not yet documented	show-bgp.xml.in: show bgp l2vpn evpn neighbors <neighbors> advertised-routes	show bgp l2vpn evpn neighbors <text> advertised-routes
×	not yet documented	show-bgp.xml.in: show bgp l2vpn evpn neighbors <neighbors> routes	show bgp l2vpn evpn neighbors <text> routes
×	not yet documented	show-bgp.xml.in: show bgp l2vpn evpn next- hops	Nothing found in VyOS
×	not yet documented	show-bgp.xml.in: show bgp l2vpn evpn rd <rd>	show bgp l2vpn evpn rd <text>
×	not yet documented	show-bgp.xml.in: show bgp l2vpn evpn rd <rd> overlay	show bgp l2vpn evpn rd <text> overlay
×	not yet documented	show-bgp.xml.in: show bgp l2vpn evpn rd <rd> tags	show bgp l2vpn evpn rd <text> tags
×	not yet documented	show-bgp.xml.in: show bgp l2vpn evpn route	show bgp l2vpn evpn route
×	not yet documented	show-bgp.xml.in: show bgp l2vpn evpn route detail	show bgp l2vpn evpn route detail
×	not yet documented	show-bgp.xml.in: show bgp l2vpn evpn route type 1	show bgp l2vpn evpn route type 1
×	not yet documented	show-bgp.xml.in: show bgp l2vpn evpn route type 2	show bgp l2vpn evpn route type 2
×	not yet documented	show-bgp.xml.in: show bgp l2vpn evpn route type 3	show bgp l2vpn evpn route type 3
×	not yet documented	show-bgp.xml.in: show bgp l2vpn evpn route type 4	show bgp l2vpn evpn route type 4
×	not yet documented	show-bgp.xml.in: show bgp l2vpn evpn route type 5	show bgp l2vpn evpn route type 5
×	not yet documented	show-bgp.xml.in: show bgp l2vpn evpn route type ead	show bgp l2vpn evpn route type ead
×	not yet documented	show-bgp.xml.in: show bgp l2vpn evpn route type es	show bgp l2vpn evpn route type es

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Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	show-bgp.xml.in: show bgp l2vpn evpn route type macip	show bgp l2vpn evpn route type macip
×	not yet documented	show-bgp.xml.in: show bgp l2vpn evpn route type multicast	show bgp l2vpn evpn route type multicast
×	not yet documented	show-bgp.xml.in: show bgp l2vpn evpn route type prefix	show bgp l2vpn evpn route type prefix
×	not yet documented	show-bgp.xml.in: show bgp l2vpn evpn route vni <vni>	show bgp l2vpn evpn route vni <text>
×	not yet documented	show-bgp.xml.in: show bgp l2vpn evpn statistics	show bgp l2vpn evpn statistics
×	not yet documented	show-bgp.xml.in: show bgp l2vpn evpn sum- mary	show bgp l2vpn evpn sum- mary
×	not yet documented	show-bgp.xml.in: show bgp l2vpn evpn sum- mary established	show bgp l2vpn evpn sum- mary established
×	not yet documented	show-bgp.xml.in: show bgp l2vpn evpn sum- mary failed	show bgp l2vpn evpn sum- mary failed
×	not yet documented	show-bgp.xml.in: show bgp l2vpn evpn vni	show bgp l2vpn evpn vni <text>
×	not yet documented	show-bgp.xml.in: show bgp large- community	show bgp large- community <text>
×	not yet documented	show-bgp.xml.in: show bgp large- community-list <large- community-list>	show bgp large- community-list <text>
×	not yet documented	show-bgp.xml.in: show bgp large- community-list <large- community-list> exact-match	show bgp large- community-list <text> exact-match
×	not yet documented	show-bgp.xml.in: show bgp large- community <large- community> exact-match	show bgp large- community <text> exact-match
×	not yet documented	show-bgp.xml.in: show bgp mac hash	show bgp mac hash
×	not yet documented	show-bgp.xml.in: show bgp martian next- hop	show bgp martian next- hop
×	not yet documented	show-bgp.xml.in: show bgp memory	show bgp memory

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Table 2 – continued from previous page

Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	show-bgp.xml.in: show bgp neighbors <neighbors>	show bgp neighbors <text>
×	not yet documented	show-bgp.xml.in: show bgp neighbors <neighbors> advertised- routes	show bgp neighbors <text> advertised-routes
×	not yet documented	show-bgp.xml.in: show bgp neighbors <neighbors> dampened- routes	show bgp neighbors <text> dampened-routes
×	not yet documented	show-bgp.xml.in: show bgp neighbors <neighbors> filtered- routes	show bgp neighbors <text> filtered-routes
×	not yet documented	show-bgp.xml.in: show bgp neighbors <neighbors> flap-statistics	show bgp neighbors <text> flap-statistics
×	not yet documented	show-bgp.xml.in: show bgp neighbors <neighbors> prefix-counts	show bgp neighbors <text> prefix-counts
×	not yet documented	show-bgp.xml.in: show bgp neighbors <neighbors> received- routes	show bgp neighbors <text> received-routes
×	not yet documented	show-bgp.xml.in: show bgp neighbors <neighbors> received prefix-filter	show bgp neighbors <text> received prefix- filter
×	not yet documented	show-bgp.xml.in: show bgp neighbors <neighbors> routes	show bgp neighbors <text> routes
×	not yet documented	show-bgp.xml.in: show bgp nexthop <nex- thop>	show bgp nexthop <text>
×	not yet documented	show-bgp.xml.in: show bgp nexthop <nex- thop> detail	show bgp nexthop <text> detail
×	not yet documented	show-bgp.xml.in: show bgp prefix-list <prefix-list>	show bgp prefix-list <text>
×	not yet documented	show-bgp.xml.in: show bgp regexp <regexp>	show bgp regexp <text>
×	not yet documented	show-bgp.xml.in: show bgp route-map <route-map>	show bgp route-map <text>
×	not yet documented	show-bgp.xml.in: show bgp segment-routing srv6	Nothing found in VyOS

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Table 2 – continued from previous page

Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	show-bgp.xml.in: show bgp statistics	show bgp statistics
×	not yet documented	show-bgp.xml.in: show bgp statistics-all	show bgp statistics-all
×	not yet documented	show-bgp.xml.in: show bgp summary	show bgp summary
×	not yet documented	show-bgp.xml.in: show bgp summary estab- lished	show bgp summary estab- lished
×	not yet documented	show-bgp.xml.in: show bgp summary failed	show bgp summary failed
×	not yet documented	show-bgp.xml.in: show bgp vni <vni>	show bgp vni <text>
×	not yet documented	show-bgp.xml.in: show bgp vni <vni> type 1	show bgp vni <text> type 1
×	not yet documented	show-bgp.xml.in: show bgp vni <vni> type 2	show bgp vni <text> type 2
×	not yet documented	show-bgp.xml.in: show bgp vni <vni> type 3	show bgp vni <text> type 3
×	not yet documented	show-bgp.xml.in: show bgp vni <vni> type ead	show bgp vni <text> type ead
×	not yet documented	show-bgp.xml.in: show bgp vni <vni> type macip	show bgp vni <text> type macip
×	not yet documented	show-bgp.xml.in: show bgp vni <vni> type multicast	show bgp vni <text> type multicast
×	not yet documented	show-bgp.xml.in: show bgp vni <vni> vtep <vtep>	show bgp vni <text> vtep <text>
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf>	show bgp vrf <text>
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> com- munity	show bgp vrf <text> com- munity <text>
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> community-list <community-list>	show bgp vrf <text> community-list <text>
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> community-list <community-list> exact- match	show bgp vrf <text> community-list <text> exact-match
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> com- munity accept-own	show bgp vrf <text> com- munity accept-own

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Table 2 – continued from previous page

Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> community accept-own- nexthop	show bgp vrf <text> community accept-own- nexthop
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> com- munity blackhole	show bgp vrf <text> com- munity blackhole
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> com- munity exact-match	show bgp vrf <text> community <text> exact- match
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> com- munity graceful-shutdown	show bgp vrf <text> com- munity graceful-shutdown
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> com- munity llgr-stale	show bgp vrf <text> com- munity llgr-stale
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> com- munity local-AS	show bgp vrf <text> com- munity local-AS
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> com- munity no-advertise	show bgp vrf <text> com- munity no-advertise
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> com- munity no-export	show bgp vrf <text> com- munity no-export
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> com- munity no-llgr	show bgp vrf <text> com- munity no-llgr
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> com- munity no-peer	show bgp vrf <text> com- munity no-peer
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> community route-filter- translated-v4	show bgp vrf <text> community route-filter- translated-v4
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> community route-filter- translated-v6	show bgp vrf <text> community route-filter- translated-v6
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> com- munity route-filter-v4	show bgp vrf <text> com- munity route-filter-v4
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> com- munity route-filter-v6	show bgp vrf <text> com- munity route-filter-v6
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> damp- ening dampened-paths	show bgp vrf <text> dampening dampened- paths

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Table 2 – continued from previous page

Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> damp- ening flap-statistics	show bgp vrf <text> dampening flap-statistics
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> damp- ening parameters	show bgp vrf <text> dampening parameters
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> filter- list <filter-list>	Nothing found in VyOS
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv4	show bgp vrf <text> ipv4 <text>
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv4 <ipv4> bestpath	show bgp vrf <text> ipv4 <text> bestpath
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv4 community	show bgp vrf <text> ipv4 community <text>
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv4 community-list <community-list>	show bgp vrf <text> ipv4 community-list <text>
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv4 community-list <community-list> exact- match	show bgp vrf <text> ipv4 community-list <text> exact-match
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv4 community accept-own	show bgp vrf <text> ipv4 community accept-own
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv4 community accept-own- nexthop	show bgp vrf <text> ipv4 community accept-own- nexthop
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv4 community blackhole	show bgp vrf <text> ipv4 community blackhole
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv4 community exact-match	show bgp vrf <text> ipv4 community <text> exact- match
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv4 community graceful- shutdown	show bgp vrf <text> ipv4 community graceful- shutdown
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv4 community llgr-stale	show bgp vrf <text> ipv4 community llgr-stale
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv4 community local-AS	show bgp vrf <text> ipv4 community local-AS

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Table 2 – continued from previous page

Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv4 community no-advertise	show bgp vrf <text> ipv4 community no-advertise
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv4 community no-export	show bgp vrf <text> ipv4 community no-export
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv4 community no-llgr	show bgp vrf <text> ipv4 community no-llgr
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv4 community no-peer	show bgp vrf <text> ipv4 community no-peer
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv4 community route-filter- translated-v4	show bgp vrf <text> ipv4 community route-filter- translated-v4
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv4 community route-filter- translated-v6	show bgp vrf <text> ipv4 community route-filter- translated-v6
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv4 community route-filter-v4	show bgp vrf <text> ipv4 community route-filter-v4
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv4 community route-filter-v6	show bgp vrf <text> ipv4 community route-filter-v6
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv4 dampening dampened- paths	show bgp vrf <text> ipv4 dampening dampened- paths
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv4 dampening flap-statistics	show bgp vrf <text> ipv4 dampening flap-statistics
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv4 dampening parameters	show bgp vrf <text> ipv4 dampening parameters
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv4 filter-list <filter-list>	Nothing found in VyOS
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv4 flowspec	show bgp vrf <text> ipv4 flowspec <text>
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv4 flowspec <flowspec> best- path	show bgp vrf <text> ipv4 flowspec <text> bestpath
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv4 flowspec community	show bgp vrf <text> ipv4 flowspec community <text>

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Table 2 – continued from previous page

Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv4 flowspec community-list <community-list>	show bgp vrf <text> ipv4 flowspec community-list <text>
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv4 flowspec community- list <community-list> exact-match	show bgp vrf <text> ipv4 flowspec community-list <text> exact-match
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv4 flowspec community accept-own	show bgp vrf <text> ipv4 flowspec community accept-own
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv4 flowspec community accept-own-nexthop	show bgp vrf <text> ipv4 flowspec community accept-own-nexthop
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv4 flowspec community blackhole	show bgp vrf <text> ipv4 flowspec community blackhole
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv4 flowspec community exact-match	show bgp vrf <text> ipv4 flowspec community <text> exact-match
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv4 flowspec community graceful-shutdown	show bgp vrf <text> ipv4 flowspec community graceful-shutdown
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv4 flowspec community llgr- stale	show bgp vrf <text> ipv4 flowspec community llgr- stale
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv4 flowspec community local-AS	show bgp vrf <text> ipv4 flowspec community local-AS
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv4 flowspec community no- advertise	show bgp vrf <text> ipv4 flowspec community no- advertise
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv4 flowspec community no- export	show bgp vrf <text> ipv4 flowspec community no- export
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv4 flowspec community no- llgr	show bgp vrf <text> ipv4 flowspec community no- llgr

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Table 2 – continued from previous page

Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv4 flowspec community no- peer	show bgp vrf <text> ipv4 flowspec community no- peer
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv4 flowspec community route-filter-translated-v4	show bgp vrf <text> ipv4 flowspec community route-filter-translated-v4
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv4 flowspec community route-filter-translated-v6	show bgp vrf <text> ipv4 flowspec community route-filter-translated-v6
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv4 flowspec community route-filter-v4	show bgp vrf <text> ipv4 flowspec community route-filter-v4
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv4 flowspec community route-filter-v6	show bgp vrf <text> ipv4 flowspec community route-filter-v6
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv4 flowspec dampening dampened-paths	show bgp vrf <text> ipv4 flowspec dampening dampened-paths
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv4 flowspec dampening flap- statistics	show bgp vrf <text> ipv4 flowspec dampening flap- statistics
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv4 flowspec dampening pa- rameters	show bgp vrf <text> ipv4 flowspec dampening pa- rameters
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv4 flowspec detail	show bgp vrf <text> ipv4 flowspec detail
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv4 flowspec filter-list <filter- list>	Nothing found in VyOS
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv4 flowspec large-community	show bgp vrf <text> ipv4 flowspec large-community <text>
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv4 flowspec large- community-list <large- community-list>	show bgp vrf <text> ipv4 flowspec large- community-list <text>

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Table 2 – continued from previous page

Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv4 flowspec large- community-list <large- community-list> exact- match	show bgp vrf <text> ipv4 flowspec large- community-list <text> exact-match
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv4 flowspec large-community <large-community> exact- match	show bgp vrf <text> ipv4 flowspec large-community <text> exact-match
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv4 flowspec <flowspec> longer-prefixes	show bgp vrf <text> ipv4 flowspec <text> longer-prefixes
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv4 flowspec <flowspec> mul- tipath	show bgp vrf <text> ipv4 flowspec <text> multipath
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv4 flowspec neighbors <neighbors>	show bgp vrf <text> ipv4 flowspec neighbors <text>
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv4 flowspec neighbors <neighbors> advertised- routes	show bgp vrf <text> ipv4 flowspec neighbors <text> advertised-routes
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv4 flowspec neighbors <neighbors> dampened- routes	show bgp vrf <text> ipv4 flowspec neighbors <text> dampened-routes
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv4 flowspec neighbors <neighbors> filtered- routes	show bgp vrf <text> ipv4 flowspec neighbors <text> filtered-routes
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv4 flowspec neighbors <neighbors> flap-statistics	show bgp vrf <text> ipv4 flowspec neighbors <text> flap-statistics
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv4 flowspec neighbors <neighbors> prefix-counts	show bgp vrf <text> ipv4 flowspec neighbors <text> prefix-counts

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Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv4 flowspec neighbors <neighbors> received- routes	show bgp vrf <text> ipv4 flowspec neighbors <text> received-routes
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv4 flowspec neighbors <neighbors> received prefix-filter	show bgp vrf <text> ipv4 flowspec neighbors <text> received prefix-filter
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv4 flowspec neighbors <neighbors> routes	show bgp vrf <text> ipv4 flowspec neighbors <text> routes
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv4 flowspec prefix-list <prefix-list>	show bgp vrf <text> ipv4 flowspec prefix-list <text>
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv4 flowspec regexp <regexp>	show bgp vrf <text> ipv4 flowspec regexp <text>
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv4 flowspec route-map <route-map>	show bgp vrf <text> ipv4 flowspec route-map <text>
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv4 flowspec statistics	show bgp vrf <text> ipv4 flowspec statistics
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv4 flowspec summary	show bgp vrf <text> ipv4 flowspec summary
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv4 flowspec summary estab- lished	show bgp vrf <text> ipv4 flowspec summary estab- lished
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv4 flowspec summary failed	show bgp vrf <text> ipv4 flowspec summary failed
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv4 flowspec wide	show bgp vrf <text> ipv4 flowspec wide
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv4 large-community	show bgp vrf <text> ipv4 large-community <text>
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv4 large-community-list <large-community-list>	show bgp vrf <text> ipv4 large-community-list <text>

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Table 2 – continued from previous page

Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv4 large-community-list <large-community-list> exact-match	show bgp vrf <text> ipv4 large-community-list <text> exact-match
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv4 large-community <large- community> exact-match	show bgp vrf <text> ipv4 large-community <text> exact-match
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv4 <ipv4> longer-prefixes	show bgp vrf <text> ipv4 <text> longer-prefixes
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv4 <ipv4> multipath	show bgp vrf <text> ipv4 <text> multipath
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv4 neighbors <neighbors>	show bgp vrf <text> ipv4 neighbors <text>
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv4 neighbors <neighbors> advertised-routes	show bgp vrf <text> ipv4 neighbors <text> advertised-routes
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv4 neighbors <neighbors> dampened-routes	show bgp vrf <text> ipv4 neighbors <text> dampened-routes
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv4 neighbors <neighbors> filtered-routes	show bgp vrf <text> ipv4 neighbors <text> filtered- routes
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv4 neighbors <neighbors> flap-statistics	show bgp vrf <text> ipv4 neighbors <text> flap-statistics
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv4 neighbors <neighbors> prefix-counts	show bgp vrf <text> ipv4 neighbors <text> prefix- counts
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv4 neighbors <neighbors> received-routes	show bgp vrf <text> ipv4 neighbors <text> received-routes
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv4 neighbors <neighbors> re- ceived prefix-filter	show bgp vrf <text> ipv4 neighbors <text> received prefix-filter

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Table 2 – continued from previous page

Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv4 neighbors <neighbors> routes	show bgp vrf <text> ipv4 neighbors <text> routes
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv4 prefix-list <prefix-list>	show bgp vrf <text> ipv4 prefix-list <text>
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv4 regex <regex>	show bgp vrf <text> ipv4 regex <text>
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv4 route-map <route-map>	show bgp vrf <text> ipv4 route-map <text>
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv4 statistics	show bgp vrf <text> ipv4 statistics
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv4 summary	show bgp vrf <text> ipv4 summary
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv4 summary established	show bgp vrf <text> ipv4 summary established
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv4 summary failed	show bgp vrf <text> ipv4 summary failed
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv4 vpn	show bgp vrf <text> ipv4 vpn <text>
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv4 vpn <vpn> bestpath	show bgp vrf <text> ipv4 vpn <text> bestpath
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv4 vpn community	show bgp vrf <text> ipv4 vpn community <text>
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv4 vpn community-list <community-list>	show bgp vrf <text> ipv4 vpn community-list <text>
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv4 vpn community- list <community-list> exact-match	show bgp vrf <text> ipv4 vpn community-list <text> exact-match
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv4 vpn community accept-own	show bgp vrf <text> ipv4 vpn community accept-own

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Table 2 – continued from previous page

Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv4 vpn community accept-own-nexthop	show bgp vrf <text> ipv4 vpn community accept-own-nexthop
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv4 vpn community blackhole	show bgp vrf <text> ipv4 vpn community blackhole
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv4 vpn community exact-match	show bgp vrf <text> ipv4 vpn community <text> exact-match
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv4 vpn community graceful- shutdown	show bgp vrf <text> ipv4 vpn community graceful- shutdown
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv4 vpn community llgr-stale	show bgp vrf <text> ipv4 vpn community llgr-stale
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv4 vpn community local-AS	show bgp vrf <text> ipv4 vpn community local-AS
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv4 vpn community no-advertise	show bgp vrf <text> ipv4 vpn community no-advertise
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv4 vpn community no-export	show bgp vrf <text> ipv4 vpn community no-export
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv4 vpn community no-llgr	show bgp vrf <text> ipv4 vpn community no-llgr
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv4 vpn community no-peer	show bgp vrf <text> ipv4 vpn community no-peer
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv4 vpn community route-filter-translated-v4	show bgp vrf <text> ipv4 vpn community route-filter-translated-v4
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv4 vpn community route-filter-translated-v6	show bgp vrf <text> ipv4 vpn community route-filter-translated-v6
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv4 vpn community route-filter-v4	show bgp vrf <text> ipv4 vpn community route-filter-v4

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Table 2 – continued from previous page

Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv4 vpn community route-filter-v6	show bgp vrf <text> ipv4 vpn community route-filter-v6
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv4 vpn dampening dampened-paths	show bgp vrf <text> ipv4 vpn dampening dampened-paths
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv4 vpn dampening flap-statistics	show bgp vrf <text> ipv4 vpn dampening flap-statistics
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv4 vpn dampening param- eters	show bgp vrf <text> ipv4 vpn dampening param- eters
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv4 vpn filter-list <filter-list>	Nothing found in VyOS
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv4 vpn large-community	show bgp vrf <text> ipv4 vpn large-community <text>
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv4 vpn large-community-list <large-community-list>	show bgp vrf <text> ipv4 vpn large-community-list <text>
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv4 vpn large-community-list <large-community-list> exact-match	show bgp vrf <text> ipv4 vpn large-community-list <text> exact-match
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv4 vpn large-community <large-community> exact-match	show bgp vrf <text> ipv4 vpn large-community <text> exact-match
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv4 vpn <vpn> longer-prefixes	show bgp vrf <text> ipv4 vpn <text> longer-prefixes
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv4 vpn <vpn> multipath	show bgp vrf <text> ipv4 vpn <text> multipath
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv4 vpn neighbors <neigh- bors>	show bgp vrf <text> ipv4 vpn neighbors <text>

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Table 2 – continued from previous page

Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv4 vpn neighbors <neigh- bors> advertised-routes	show bgp vrf <text> ipv4 vpn neighbors <text> advertised-routes
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv4 vpn neighbors <neigh- bors> dampened-routes	show bgp vrf <text> ipv4 vpn neighbors <text> dampened-routes
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv4 vpn neighbors <neigh- bors> filtered-routes	show bgp vrf <text> ipv4 vpn neighbors <text> filtered-routes
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv4 vpn neighbors <neigh- bors> flap-statistics	show bgp vrf <text> ipv4 vpn neighbors <text> flap- statistics
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv4 vpn neighbors <neigh- bors> prefix-counts	show bgp vrf <text> ipv4 vpn neighbors <text> prefix-counts
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv4 vpn neighbors <neigh- bors> received-routes	show bgp vrf <text> ipv4 vpn neighbors <text> received-routes
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv4 vpn neighbors <neighbors> received prefix-filter	show bgp vrf <text> ipv4 vpn neighbors <text> re- ceived prefix-filter
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv4 vpn neighbors <neigh- bors> routes	show bgp vrf <text> ipv4 vpn neighbors <text> routes
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv4 vpn prefix-list <prefix- list>	show bgp vrf <text> ipv4 vpn prefix-list <text>
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv4 vpn regexp <regexp>	show bgp vrf <text> ipv4 vpn regexp <text>
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv4 vpn route-map <route-map>	show bgp vrf <text> ipv4 vpn route-map <text>
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv4 vpn statistics	show bgp vrf <text> ipv4 vpn statistics

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Table 2 – continued from previous page

Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv4 vpn summary	show bgp vrf <text> ipv4 vpn summary
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv4 vpn summary established	show bgp vrf <text> ipv4 vpn summary established
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv4 vpn summary failed	show bgp vrf <text> ipv4 vpn summary failed
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv4 vpn wide	show bgp vrf <text> ipv4 vpn wide
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv4 wide	show bgp vrf <text> ipv4 wide
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv6 <text>	show bgp vrf <text> ipv6 <text>
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv6 <ipv6> bestpath	show bgp vrf <text> ipv6 <text> bestpath
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv6 community	show bgp vrf <text> ipv6 community <text>
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv6 community-list <community-list>	show bgp vrf <text> ipv6 community-list <text>
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv6 community-list <community-list> exact- match	show bgp vrf <text> ipv6 community-list <text> exact-match
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv6 community accept-own	show bgp vrf <text> ipv6 community accept-own
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv6 community accept-own- nexthop	show bgp vrf <text> ipv6 community accept-own- nexthop
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv6 community blackhole	show bgp vrf <text> ipv6 community blackhole
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv6 community exact-match	show bgp vrf <text> ipv6 community <text> exact- match
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv6 community graceful- shutdown	show bgp vrf <text> ipv6 community graceful- shutdown

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Table 2 – continued from previous page

Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv6 community llgr-stale	show bgp vrf <text> ipv6 community llgr-stale
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv6 community local-AS	show bgp vrf <text> ipv6 community local-AS
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv6 community no-advertise	show bgp vrf <text> ipv6 community no-advertise
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv6 community no-export	show bgp vrf <text> ipv6 community no-export
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv6 community no-llgr	show bgp vrf <text> ipv6 community no-llgr
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv6 community no-peer	show bgp vrf <text> ipv6 community no-peer
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv6 community route-filter- translated-v4	show bgp vrf <text> ipv6 community route-filter- translated-v4
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv6 community route-filter- translated-v6	show bgp vrf <text> ipv6 community route-filter- translated-v6
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv6 community route-filter-v4	show bgp vrf <text> ipv6 community route-filter-v4
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv6 community route-filter-v6	show bgp vrf <text> ipv6 community route-filter-v6
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv6 dampening dampened- paths	show bgp vrf <text> ipv6 dampening dampened- paths
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv6 dampening flap-statistics	show bgp vrf <text> ipv6 dampening flap-statistics
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv6 dampening parameters	show bgp vrf <text> ipv6 dampening parameters
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv6 filter-list <filter-list>	Nothing found in VyOS
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv6 large-community	show bgp vrf <text> ipv6 large-community <text>

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Table 2 – continued from previous page

Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv6 large-community-list <large-community-list>	show bgp vrf <text> ipv6 large-community-list <text>
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv6 large-community-list <large-community-list> exact-match	show bgp vrf <text> ipv6 large-community-list <text> exact-match
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv6 large-community <large- community> exact-match	show bgp vrf <text> ipv6 large-community <text> exact-match
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv6 <ipv6> longer-prefixes	show bgp vrf <text> ipv6 <text> longer-prefixes
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv6 <ipv6> multipath	show bgp vrf <text> ipv6 <text> multipath
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv6 neighbors <neighbors>	show bgp vrf <text> ipv6 neighbors <text>
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv6 neighbors <neighbors> advertised-routes	show bgp vrf <text> ipv6 neighbors <text> advertised-routes
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv6 neighbors <neighbors> dampened-routes	show bgp vrf <text> ipv6 neighbors <text> dampened-routes
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv6 neighbors <neighbors> filtered-routes	show bgp vrf <text> ipv6 neighbors <text> filtered- routes
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv6 neighbors <neighbors> flap-statistics	show bgp vrf <text> ipv6 neighbors <text> flap-statistics
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv6 neighbors <neighbors> prefix-counts	show bgp vrf <text> ipv6 neighbors <text> prefix- counts
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv6 neighbors <neighbors> received-routes	show bgp vrf <text> ipv6 neighbors <text> received-routes

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Table 2 – continued from previous page

Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv6 neighbors <neighbors> re- ceived prefix-filter	show bgp vrf <text> ipv6 neighbors <text> received prefix-filter
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv6 neighbors <neighbors> routes	show bgp vrf <text> ipv6 neighbors <text> routes
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv6 prefix-list <prefix-list>	show bgp vrf <text> ipv6 prefix-list <text>
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv6 regex <regex>	show bgp vrf <text> ipv6 regex <text>
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv6 route-map <route-map>	show bgp vrf <text> ipv6 route-map <text>
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv6 statistics	show bgp vrf <text> ipv6 statistics
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv6 summary	show bgp vrf <text> ipv6 summary
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv6 summary established	show bgp vrf <text> ipv6 summary established
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv6 summary failed	show bgp vrf <text> ipv6 summary failed
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv6 vpn	show bgp vrf <text> ipv6 vpn <text>
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv6 vpn <vpn> bestpath	show bgp vrf <text> ipv6 vpn <text> bestpath
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv6 vpn community	show bgp vrf <text> ipv6 vpn community <text>
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv6 vpn community-list <community-list>	show bgp vrf <text> ipv6 vpn community-list <text>
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv6 vpn community- list <community-list> exact-match	show bgp vrf <text> ipv6 vpn community-list <text> exact-match

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Table 2 – continued from previous page

Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv6 vpn community accept-own	show bgp vrf <text> ipv6 vpn community accept-own
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv6 vpn community accept-own-nexthop	show bgp vrf <text> ipv6 vpn community accept-own-nexthop
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv6 vpn community blackhole	show bgp vrf <text> ipv6 vpn community blackhole
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv6 vpn community exact-match	show bgp vrf <text> ipv6 vpn community <text> exact-match
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv6 vpn community graceful- shutdown	show bgp vrf <text> ipv6 vpn community graceful- shutdown
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv6 vpn community llgr-stale	show bgp vrf <text> ipv6 vpn community llgr-stale
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv6 vpn community local-AS	show bgp vrf <text> ipv6 vpn community local-AS
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv6 vpn community no-advertise	show bgp vrf <text> ipv6 vpn community no-advertise
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv6 vpn community no-export	show bgp vrf <text> ipv6 vpn community no-export
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv6 vpn community no-llgr	show bgp vrf <text> ipv6 vpn community no-llgr
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv6 vpn community no-peer	show bgp vrf <text> ipv6 vpn community no-peer
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv6 vpn community route-filter-translated-v4	show bgp vrf <text> ipv6 vpn community route-filter-translated-v4
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv6 vpn community route-filter-translated-v6	show bgp vrf <text> ipv6 vpn community route-filter-translated-v6

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Table 2 – continued from previous page

Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv6 vpn community route-filter-v4	show bgp vrf <text> ipv6 vpn community route-filter-v4
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv6 vpn community route-filter-v6	show bgp vrf <text> ipv6 vpn community route-filter-v6
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv6 vpn dampening dampened-paths	show bgp vrf <text> ipv6 vpn dampening dampened-paths
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv6 vpn dampening flap-statistics	show bgp vrf <text> ipv6 vpn dampening flap-statistics
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv6 vpn dampening param- eters	show bgp vrf <text> ipv6 vpn dampening param- eters
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv6 vpn filter-list <filter-list>	Nothing found in VyOS
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv6 vpn large-community	show bgp vrf <text> ipv6 vpn large-community <text>
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv6 vpn large-community-list <large-community-list>	show bgp vrf <text> ipv6 vpn large-community-list <text>
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv6 vpn large-community-list <large-community-list> exact-match	show bgp vrf <text> ipv6 vpn large-community-list <text> exact-match
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv6 vpn large-community <large-community> exact-match	show bgp vrf <text> ipv6 vpn large-community <text> exact-match
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv6 vpn <vpn> longer-prefixes	show bgp vrf <text> ipv6 vpn <text> longer-prefixes
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv6 vpn <vpn> multipath	show bgp vrf <text> ipv6 vpn <text> multipath

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Table 2 – continued from previous page

Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv6 vpn neighbors <neigh- bors>	show bgp vrf <text> ipv6 vpn neighbors <text>
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv6 vpn neighbors <neigh- bors> advertised-routes	show bgp vrf <text> ipv6 vpn neighbors <text> advertised-routes
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv6 vpn neighbors <neigh- bors> dampened-routes	show bgp vrf <text> ipv6 vpn neighbors <text> dampened-routes
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv6 vpn neighbors <neigh- bors> filtered-routes	show bgp vrf <text> ipv6 vpn neighbors <text> filtered-routes
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv6 vpn neighbors <neigh- bors> flap-statistics	show bgp vrf <text> ipv6 vpn neighbors <text> flap- statistics
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv6 vpn neighbors <neigh- bors> prefix-counts	show bgp vrf <text> ipv6 vpn neighbors <text> prefix-counts
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv6 vpn neighbors <neigh- bors> received-routes	show bgp vrf <text> ipv6 vpn neighbors <text> received-routes
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv6 vpn neighbors <neighbors> received prefix-filter	show bgp vrf <text> ipv6 vpn neighbors <text> re- ceived prefix-filter
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv6 vpn neighbors <neigh- bors> routes	show bgp vrf <text> ipv6 vpn neighbors <text> routes
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv6 vpn prefix-list <prefix- list>	show bgp vrf <text> ipv6 vpn prefix-list <text>
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv6 vpn regexp <regexp>	show bgp vrf <text> ipv6 vpn regexp <text>
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv6 vpn route-map <route-map>	show bgp vrf <text> ipv6 vpn route-map <text>

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Table 2 – continued from previous page

Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv6 vpn statistics	show bgp vrf <text> ipv6 vpn statistics
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv6 vpn summary	show bgp vrf <text> ipv6 vpn summary
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv6 vpn summary established	show bgp vrf <text> ipv6 vpn summary established
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv6 vpn summary failed	show bgp vrf <text> ipv6 vpn summary failed
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv6 vpn wide	show bgp vrf <text> ipv6 vpn wide
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> ipv6 wide	show bgp vrf <text> ipv6 wide
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> l2vpn evpn	show bgp vrf <text> l2vpn evpn <text>
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> l2vpn evpn all overlay	show bgp vrf <text> l2vpn evpn all overlay
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> l2vpn evpn all tags	show bgp vrf <text> l2vpn evpn all tags
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> l2vpn evpn community <com- munity>	show bgp vrf <text> l2vpn evpn community <text>
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> l2vpn evpn community <com- munity> exact-match	show bgp vrf <text> l2vpn evpn community <text> exact-match
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> l2vpn evpn es	show bgp vrf <text> l2vpn evpn es
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> l2vpn evpn es-evi	show bgp vrf <text> l2vpn evpn es-evi
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> l2vpn evpn es-evi detail	show bgp vrf <text> l2vpn evpn es-evi detail
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> l2vpn evpn es-evi vni <vni>	show bgp vrf <text> l2vpn evpn es-evi vni <text>

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Table 2 – continued from previous page

Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> l2vpn evpn es-vrf	Nothing found in VyOS
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> l2vpn evpn es detail	show bgp vrf <text> l2vpn evpn es detail
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> l2vpn evpn import-rt	show bgp vrf <text> l2vpn evpn import-rt
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> l2vpn evpn large-community <large-community>	show bgp vrf <text> l2vpn evpn large-community <text>
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> l2vpn evpn large-community-list <large-community-list>	show bgp vrf <text> l2vpn evpn large-community-list <text>
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> l2vpn evpn large-community-list <large-community-list> exact-match	show bgp vrf <text> l2vpn evpn large-community-list <text> exact-match
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> l2vpn evpn large-community <large-community> exact-match	show bgp vrf <text> l2vpn evpn large-community <text> exact-match
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> l2vpn evpn neighbors <neigh- bors>	show bgp vrf <text> l2vpn evpn neighbors <text>
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> l2vpn evpn neighbors <neigh- bors> advertised-routes	show bgp vrf <text> l2vpn evpn neighbors <text> advertised-routes
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> l2vpn evpn neighbors <neigh- bors> routes	show bgp vrf <text> l2vpn evpn neighbors <text> routes
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> l2vpn evpn next-hops	Nothing found in VyOS
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> l2vpn evpn rd <rd>	show bgp vrf <text> l2vpn evpn rd <text>
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> l2vpn evpn rd <rd> overlay	show bgp vrf <text> l2vpn evpn rd <text> overlay

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Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> l2vpn evpn rd <rd> tags	show bgp vrf <text> l2vpn evpn rd <text> tags
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> l2vpn evpn route	show bgp vrf <text> l2vpn evpn route
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> l2vpn evpn route detail	show bgp vrf <text> l2vpn evpn route detail
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> l2vpn evpn route type 1	show bgp vrf <text> l2vpn evpn route type 1
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> l2vpn evpn route type 2	show bgp vrf <text> l2vpn evpn route type 2
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> l2vpn evpn route type 3	show bgp vrf <text> l2vpn evpn route type 3
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> l2vpn evpn route type 4	show bgp vrf <text> l2vpn evpn route type 4
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> l2vpn evpn route type 5	show bgp vrf <text> l2vpn evpn route type 5
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> l2vpn evpn route type ead	show bgp vrf <text> l2vpn evpn route type ead
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> l2vpn evpn route type es	show bgp vrf <text> l2vpn evpn route type es
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> l2vpn evpn route type macip	show bgp vrf <text> l2vpn evpn route type macip
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> l2vpn evpn route type multicast	show bgp vrf <text> l2vpn evpn route type multicast
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> l2vpn evpn route type prefix	show bgp vrf <text> l2vpn evpn route type prefix
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> l2vpn evpn route vni <vni>	show bgp vrf <text> l2vpn evpn route vni <text>
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> l2vpn evpn statistics	show bgp vrf <text> l2vpn evpn statistics
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> l2vpn evpn summary	show bgp vrf <text> l2vpn evpn summary

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Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> l2vpn evpn summary established	show bgp vrf <text> l2vpn evpn summary established
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> l2vpn evpn summary failed	show bgp vrf <text> l2vpn evpn summary failed
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> l2vpn evpn vni	show bgp vrf <text> l2vpn evpn vni <text>
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> large- community	show bgp vrf <text> large- community <text>
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> large-community-list <large-community-list>	show bgp vrf <text> large- community-list <text>
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> large-community-list <large-community-list> exact-match	show bgp vrf <text> large- community-list <text> exact-match
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> large-community <large- community> exact-match	show bgp vrf <text> large- community <text> exact- match
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> mar- tian next-hop	show bgp vrf <text> mar- tian next-hop
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> neigh- bors <neighbors>	show bgp vrf <text> neighbors <text>
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> neighbors <neighbors> advertised-routes	show bgp vrf <text> neighbors <text> advertised-routes
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> neighbors <neighbors> dampened-routes	show bgp vrf <text> neighbors <text> dampened-routes
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> neigh- bors <neighbors> filtered- routes	show bgp vrf <text> neighbors <text> filtered- routes
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> neighbors <neighbors> flap-statistics	show bgp vrf <text> neighbors <text> flap- statistics

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Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> neighbors <neighbors> prefix-counts	show bgp vrf <text> neighbors <text> prefix-counts
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> neighbors <neighbors> received-routes	show bgp vrf <text> neighbors <text> received-routes
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> neighbors <neighbors> received prefix-filter	show bgp vrf <text> neighbors <text> received prefix-filter
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> neighbors <neighbors> routes	show bgp vrf <text> neighbors <text> routes
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> nexthop <nexthop>	show bgp vrf <text> nexthop <text>
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> nexthop <nexthop> detail	show bgp vrf <text> nexthop <text> detail
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> prefix-list <prefix-list>	show bgp vrf <text> prefix-list <text>
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> regexp <regexp>	show bgp vrf <text> regexp <text>
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> route-map <route-map>	show bgp vrf <text> route-map <text>
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> statistics	show bgp vrf <text> statistics
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> summary	show bgp vrf <text> summary
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> summary established	show bgp vrf <text> summary established
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> summary failed	show bgp vrf <text> summary failed
×	not yet documented	show-bgp.xml.in: show bgp vrf <vrf> wide	show bgp vrf <text> wide
×	not yet documented	show-bgp.xml.in: show bgp wide	show bgp wide
✓	<i>show bridge</i>	show-bridge.xml.in: show bridge <bridge>	show bridge <text>

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Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<i>show bridge <name> fdb</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>show bridge <name> macs</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>show bridge <name> mdb</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	show-bridge.xml.in: show bridge <bridge> detail	show bridge <text> detail
×	not yet documented	show-bridge.xml.in: show bridge <bridge> fdb	show bridge <text> fdb
×	not yet documented	show-bridge.xml.in: show bridge <bridge> mdb	show bridge <text> mdb
×	not yet documented	show-bridge.xml.in: show bridge <bridge> nexthop-group	show bridge <text> nexthop-group
×	not yet documented	show-bridge.xml.in: show bridge vlan	show bridge vlan
×	not yet documented	show-bridge.xml.in: show bridge vlan tunnel	show bridge vlan tunnel
×	not yet documented	show-bridge.xml.in: show bridge vni	Nothing found in VyOS
×	not yet documented	Nothing found in XML Definitions	show cluster status
✓	<i>show configuration</i>	show-configuration.xml.in: show configuration	show configuration
×	not yet documented	show-configuration.xml.in: show configuration all	show configuration all
✓	<i>show configuration commands</i>	show-configuration.xml.in: show configuration commands	show configuration commands
×	not yet documented	show-configuration.xml.in: show configuration files	show configuration files
✓	<i>show configuration json</i>	show-configuration.xml.in: show configuration json	show configuration json
✓	<i>show configuration json pretty</i>	show-configuration.xml.in: show configuration json pretty	show configuration json pretty

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Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<i>show configuration secondary sync [commands] [running candidate saved [<config-node-path>]]</i>	Nothing found in XML Definitions	Nothing found in VyOS
✓	<i>show conntrack-sync cache external</i>	conntrack-sync.xml.in: show conntrack-sync cache external	show conntrack-sync cache external
×	not yet documented	conntrack-sync.xml.in: show conntrack-sync cache external expect	show conntrack-sync cache external expect
×	not yet documented	conntrack-sync.xml.in: show conntrack-sync cache external main	show conntrack-sync cache external main
✓	<i>show conntrack-sync cache internal</i>	conntrack-sync.xml.in: show conntrack-sync cache internal	show conntrack-sync cache internal
×	not yet documented	conntrack-sync.xml.in: show conntrack-sync cache internal expect	show conntrack-sync cache internal expect
×	not yet documented	conntrack-sync.xml.in: show conntrack-sync cache internal main	show conntrack-sync cache internal main
✓	<i>show conntrack-sync statistics</i>	conntrack-sync.xml.in: show conntrack-sync statistics	show conntrack-sync statistics
✓	<i>show conntrack-sync status</i>	conntrack-sync.xml.in: show conntrack-sync status	show conntrack-sync status
×	not yet documented	show-conntrack.xml.in: show conntrack statistics	show conntrack statistics
×	<i>show conntrack table <ipv4/ipv6></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>show conntrack table <ipv4/ipv6> vrf <vrf-name></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	show-conntrack.xml.in: show conntrack table ipv4	show conntrack table ipv4
×	not yet documented	show-conntrack.xml.in: show conntrack table ipv6	show conntrack table ipv6
✓	<i>show console-server ports</i>	show-console-server.xml.in: show console-server ports	show console-server ports
✓	<i>show console-server user</i>	show-console-server.xml.in: show console-server user	show console-server user
×	<i>show container</i>	Nothing found in XML Definitions	show container

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Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<i>show container image</i>	Nothing found in XML Definitions	show container image
×	not yet documented	Nothing found in XML Definitions	show container log <text>
×	<i>show container log <containername></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>show container network</i>	Nothing found in XML Definitions	show container network
×	not yet documented	Nothing found in XML Definitions	show date
×	not yet documented	Nothing found in XML Definitions	show date utc
×	not yet documented	Nothing found in XML Definitions	show date utc maya
×	not yet documented	Nothing found in XML Definitions	show dhcp client leases
×	not yet documented	Nothing found in XML Definitions	show dhcp client leases interface <text>
×	<i>show dhcp server leases</i>	Nothing found in XML Definitions	show dhcp server leases
×	<i>show dhcp server leases origin [local / remote]</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	Nothing found in XML Definitions	show dhcp server leases pool <text>
×	<i>show dhcp server leases pool <pool></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	Nothing found in XML Definitions	show dhcp server leases sort <text>
×	<i>show dhcp server leases sort <key></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	Nothing found in XML Definitions	show dhcp server leases state <text>
×	<i>show dhcp server leases state <state></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>show dhcp server statistics</i>	Nothing found in XML Definitions	show dhcp server statistics
×	not yet documented	Nothing found in XML Definitions	show dhcp server statistics pool <text>
×	<i>show dhcp server statistics pool <pool></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>show dhcpv6 server leases</i>	Nothing found in XML Definitions	show dhcpv6 server leases
×	not yet documented	Nothing found in XML Definitions	show dhcpv6 server leases pool <text>
×	<i>show dhcpv6 server leases pool <pool></i>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	Nothing found in XML Definitions	show dhcpv6 server leases sort <text>
×	<i>show dhcpv6 server leases sort <key></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	Nothing found in XML Definitions	show dhcpv6 server leases state <text>
×	<i>show dhcpv6 server leases state <state></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	disks.xml.in: show disk <disk> format	show disk <text> format
×	<i>show disk sda format</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	Nothing found in XML Definitions	show dns dynamic status
×	not yet documented	Nothing found in XML Definitions	show dns forwarding statistics
×	not yet documented	show-environment.xml.in: show environment sensors	show environment sensors
×	not yet documented	show-evpn.xml.in: show evpn access-vlan	show evpn access-vlan <text> <text>
×	not yet documented	show-evpn.xml.in: show evpn access-vlan detail	show evpn access-vlan detail
×	not yet documented	show-evpn.xml.in: show evpn access-vlan <access-vlan> node.tag	Nothing found in VyOS
×	not yet documented	show-evpn.xml.in: show evpn arp-cache vni <vni>	show evpn arp-cache vni <text>
×	not yet documented	show-evpn.xml.in: show evpn mac vni <vni>	show evpn mac vni <text>
×	not yet documented	show-evpn.xml.in: show evpn next-hops vni <vni>	show evpn next-hops vni <text>
×	not yet documented	show-evpn.xml.in: show evpn rmac vni <vni>	show evpn rmac vni <text>
×	not yet documented	Nothing found in XML Definitions	show file <text>
✓	<i>show firewall</i>	firewall.xml.in: show firewall	show firewall
×	<i>show firewall bridge</i>	firewall.xml.in: show firewall bridge	Nothing found in VyOS
×	<i>show firewall bridge forward filter</i>	firewall.xml.in: show firewall bridge forward filter	Nothing found in VyOS
×	not yet documented	firewall.xml.in: show firewall bridge forward filter detail	Nothing found in VyOS

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Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	firewall.xml.in: show firewall bridge forward filter rule <rule>	Nothing found in VyOS
×	<i>show firewall bridge forward filter rule <rule></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	firewall.xml.in: show firewall bridge forward filter rule <rule> detail	Nothing found in VyOS
×	not yet documented	firewall.xml.in: show firewall bridge name <name>	Nothing found in VyOS
×	<i>show firewall bridge name <name></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>show firewall bridge name <name> rule <rule></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	firewall.xml.in: show firewall bridge name <name> detail	Nothing found in VyOS
×	not yet documented	firewall.xml.in: show firewall bridge name <name> rule <rule>	Nothing found in VyOS
×	not yet documented	firewall.xml.in: show firewall bridge name <name> rule <rule> detail	Nothing found in VyOS
✓	<i>show firewall group</i>	firewall.xml.in: show firewall group	show firewall group <text>
×	<i>show firewall group <name></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	firewall.xml.in: show firewall group detail	Nothing found in VyOS
×	not yet documented	firewall.xml.in: show firewall ipv4	show firewall ipv4
×	<i>show firewall ipv4 [forward input output] filter</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>show firewall ipv4 [forward input output] filter rule <1-999999></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	firewall.xml.in: show firewall ipv4 forward filter	show firewall ipv4 forward filter
×	not yet documented	firewall.xml.in: show firewall ipv4 forward filter detail	Nothing found in VyOS

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Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	firewall.xml.in: show firewall ipv4 forward filter rule <rule>	show firewall ipv4 forward filter rule <text>
×	not yet documented	firewall.xml.in: show firewall ipv4 forward filter rule <rule> detail	Nothing found in VyOS
×	not yet documented	firewall.xml.in: show firewall ipv4 input filter	show firewall ipv4 input filter
×	not yet documented	firewall.xml.in: show firewall ipv4 input filter detail	Nothing found in VyOS
×	not yet documented	firewall.xml.in: show firewall ipv4 input filter rule <rule>	show firewall ipv4 input filter rule <text>
×	not yet documented	firewall.xml.in: show firewall ipv4 input filter rule <rule> detail	Nothing found in VyOS
×	not yet documented	firewall.xml.in: show firewall ipv4 name <name>	show firewall ipv4 name <text>
×	<i>show firewall ipv4 name <name></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>show firewall ipv4 name <name> rule <1-999999></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	firewall.xml.in: show firewall ipv4 name <name> detail	Nothing found in VyOS
×	not yet documented	firewall.xml.in: show firewall ipv4 name <name> rule <rule>	show firewall ipv4 name <text> rule <text>
×	not yet documented	firewall.xml.in: show firewall ipv4 name <name> rule <rule> detail	Nothing found in VyOS
×	not yet documented	firewall.xml.in: show firewall ipv4 output filter	show firewall ipv4 output filter
×	not yet documented	firewall.xml.in: show firewall ipv4 output filter detail	Nothing found in VyOS
×	not yet documented	firewall.xml.in: show firewall ipv4 output filter rule <rule>	show firewall ipv4 output filter rule <text>
×	not yet documented	firewall.xml.in: show firewall ipv4 output filter rule <rule> detail	Nothing found in VyOS
×	not yet documented	firewall.xml.in: show firewall ipv6	show firewall ipv6

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Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<code>show firewall ipv6 [forward input output] filter</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>show firewall ipv6 [forward input output] filter rule <1-999999></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	firewall.xml.in: show firewall ipv6 forward filter	show firewall ipv6 forward filter
×	not yet documented	firewall.xml.in: show firewall ipv6 forward filter detail	Nothing found in VyOS
×	not yet documented	firewall.xml.in: show firewall ipv6 forward filter rule <text> filter rule <rule>	show firewall ipv6 forward filter rule <text>
×	not yet documented	firewall.xml.in: show firewall ipv6 forward filter rule <rule> detail	Nothing found in VyOS
×	not yet documented	firewall.xml.in: show firewall ipv6 input filter	show firewall ipv6 input filter
×	not yet documented	firewall.xml.in: show firewall ipv6 input filter detail	Nothing found in VyOS
×	not yet documented	firewall.xml.in: show firewall ipv6 input filter rule <text> filter rule <rule>	show firewall ipv6 input filter rule <text>
×	not yet documented	firewall.xml.in: show firewall ipv6 input filter rule <rule> detail	Nothing found in VyOS
×	not yet documented	Nothing found in XML Definitions	show firewall ipv6 ipv6-name <text>
×	<code>show firewall ipv6 ipv6-name <name></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>show firewall ipv6 ipv6-name <name> rule <1-999999></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	Nothing found in XML Definitions	show firewall ipv6 ipv6-name <text> rule <text>
×	not yet documented	firewall.xml.in: show firewall ipv6 name <name>	Nothing found in VyOS
×	<code>show firewall ipv6 name <name> rule <1-999999></code>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	firewall.xml.in: show firewall ipv6 name <name> detail	Nothing found in VyOS

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Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	firewall.xml.in: show firewall ipv6 name <name> rule <rule>	Nothing found in VyOS
×	not yet documented	firewall.xml.in: show firewall ipv6 name <name> rule <rule> detail	Nothing found in VyOS
×	not yet documented	firewall.xml.in: show firewall ipv6 output filter	show firewall ipv6 output filter
×	not yet documented	firewall.xml.in: show firewall ipv6 output filter detail	Nothing found in VyOS
×	not yet documented	firewall.xml.in: show firewall ipv6 output filter rule <rule>	show firewall ipv6 output filter rule <text>
×	not yet documented	firewall.xml.in: show firewall ipv6 output filter rule <rule> detail	Nothing found in VyOS
✓	<i>show firewall statistics</i>	firewall.xml.in: show firewall statistics	show firewall statistics
×	not yet documented	firewall.xml.in: show firewall statistics de- tail	Nothing found in VyOS
✓	<i>show firewall summary</i>	firewall.xml.in: show firewall summary	show firewall summary
×	<i>show firewall zone-policy</i>	firewall.xml.in: show firewall zone-policy	Nothing found in VyOS
×	not yet documented	firewall.xml.in: show firewall zone-policy zone <zone>	Nothing found in VyOS
×	<i>show firewall zone-policy zone <zone></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	Nothing found in XML Definitions	show flow-accounting
×	not yet documented	Nothing found in XML Definitions	show flow-accounting in- terface <text>
×	<i>show flow-accounting interface <interface></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>show flow-accounting interface <interface> host <address></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	Nothing found in XML Definitions	show flow-accounting in- terface <text> host <text>
×	not yet documented	Nothing found in XML Definitions	show flow-accounting in- terface <text> port <text>

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Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	Nothing found in XML Definitions	show flow-accounting interface <text> top <text>
×	not yet documented	show-hardware.xml.in: show hardware cpu	show hardware cpu
×	not yet documented	show-hardware.xml.in: show hardware cpu detail	show hardware cpu detail
×	not yet documented	show-hardware.xml.in: show hardware cpu summary	show hardware cpu summary
×	not yet documented	show-hardware.xml.in: show hardware dmi	show hardware dmi
×	not yet documented	show-hardware.xml.in: show hardware mem	show hardware mem
×	not yet documented	show-hardware.xml.in: show hardware pci	show hardware pci
×	not yet documented	show-hardware.xml.in: show hardware pci detail	show hardware pci detail
×	not yet documented	show-hardware.xml.in: show hardware storage nvme	show hardware storage nvme
×	not yet documented	show-hardware.xml.in: show hardware storage scsi	show hardware storage scsi
×	not yet documented	show-hardware.xml.in: show hardware storage scsi detail	show hardware storage scsi detail
×	not yet documented	show-hardware.xml.in: show hardware storage smart <smart>	show hardware storage smart <text>
✓	<i>show hardware usb</i>	show-hardware.xml.in: show hardware usb	show hardware usb
×	not yet documented	show-hardware.xml.in: show hardware usb detail	show hardware usb detail
✓	<i>show hardware usb serial</i>	show-hardware.xml.in: show hardware usb serial	show hardware usb serial
×	not yet documented	show-history.xml.in: show history <history>	show history <text>
×	not yet documented	show-history.xml.in: show history brief	show history brief
×	not yet documented	show-host.xml.in: show host date	show host date
×	not yet documented	show-host.xml.in: show host domain	show host domain
×	not yet documented	show-host.xml.in: show host lookup <lookup>	show host lookup <text>
×	not yet documented	show-host.xml.in: show host name	show host name

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Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	show-host.xml.in: show host os	show host os
×	not yet documented	show-interfaces.xml.in: show interfaces	show interfaces
✓	<i>show interfaces bonding</i>	show-interfaces-bonding.xml.in: show interfaces bonding	show interfaces bonding <text>
×	<i>show interfaces bonding <interface></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>show interfaces bonding <interface> detail</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	show-interfaces-bonding.xml.in: show interfaces bonding <bonding> brief	show interfaces bonding <text> brief
×	not yet documented	show-interfaces-bonding.xml.in: show interfaces bonding detail	show interfaces bonding <text> detail
×	not yet documented	show-interfaces-bonding.xml.in: show interfaces bonding slaves	show interfaces bonding slaves
×	not yet documented	show-interfaces-bonding.xml.in: show interfaces bonding <bonding> vif <vif>	show interfaces bonding <text> vif <text>
×	not yet documented	show-interfaces-bonding.xml.in: show interfaces bonding <bonding> vif <vif> brief	show interfaces bonding <text> vif <text> brief
×	not yet documented	show-interfaces-bridge.xml.in: show interfaces bridge	show interfaces bridge <text>
×	not yet documented	show-interfaces-bridge.xml.in: show interfaces bridge <bridge> brief	show interfaces bridge <text> brief
×	not yet documented	show-interfaces-bridge.xml.in: show interfaces bridge de- tail	show interfaces bridge de- tail
×	not yet documented	show-interfaces.xml.in: show interfaces counters	show interfaces counters
×	not yet documented	show-interfaces.xml.in: show interfaces detail	show interfaces detail

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Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
✓	<i>show interfaces dummy</i>	show-interfaces-dummy.xml.in: show interfaces dummy	show interfaces dummy <text>
×	<i>show interfaces dummy <interface></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	show-interfaces-dummy.xml.in: show interfaces dummy <dummy> brief	show interfaces dummy <text> brief
×	not yet documented	show-interfaces-dummy.xml.in: show interfaces dummy detail	show interfaces dummy detail
✓	<i>show interfaces ethernet</i>	show-interfaces-ethernet.xml.in: show interfaces ethernet	show interfaces ethernet <text>
×	<i>show interfaces ethernet <interface></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>show interfaces ethernet <interface> physical</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>show interfaces ethernet <interface> physical offload</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>show interfaces ethernet <interface> transceiver</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	show-interfaces-ethernet.xml.in: show interfaces ethernet <ethernet> brief	show interfaces ethernet <text> brief
×	not yet documented	show-interfaces-ethernet.xml.in: show interfaces ethernet detail	show interfaces ethernet detail
×	not yet documented	show-interfaces-ethernet.xml.in: show interfaces ethernet <ethernet> identify	show interfaces ethernet <text> identify
×	not yet documented	show-interfaces-ethernet.xml.in: show interfaces ethernet <ethernet> physical	show interfaces ethernet <text> physical
×	not yet documented	show-interfaces-ethernet.xml.in: show interfaces ethernet <ethernet> physical offload	show interfaces ethernet <text> physical offload

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Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	show-interfaces-ethernet.xml.in: show interfaces ethernet <ethernet> statistics	show interfaces ethernet <text> statistics
×	not yet documented	show-interfaces-ethernet.xml.in: show interfaces ethernet <ethernet> transceiver	show interfaces ethernet <text> transceiver
×	not yet documented	show-interfaces-ethernet.xml.in: show interfaces ethernet <ethernet> vif <vif>	show interfaces ethernet <text> vif <text>
×	not yet documented	show-interfaces-ethernet.xml.in: show interfaces ethernet <ethernet> vif <vif> brief	show interfaces ethernet <text> vif <text> brief
×	not yet documented	show-interfaces-geneve.xml.in: show interfaces geneve	show interfaces geneve <text>
×	not yet documented	show-interfaces-geneve.xml.in: show interfaces geneve <geneve> brief	show interfaces geneve <text> brief
×	not yet documented	show-interfaces-geneve.xml.in: show interfaces geneve de- tail	show interfaces geneve de- tail
×	not yet documented	show-interfaces-input.xml.in: show interfaces input	show interfaces input <text>
×	not yet documented	show-interfaces-input.xml.in: show interfaces input <in- put> brief	show interfaces input <text> brief
×	not yet documented	show-interfaces-input.xml.in: show interfaces input de- tail	show interfaces input de- tail
×	not yet documented	show-interfaces-l2tpv3.xml.in: show interfaces l2tpv3	show interfaces l2tpv3 <text>
×	not yet documented	show-interfaces-l2tpv3.xml.in: show interfaces l2tpv3 <l2tpv3> brief	show interfaces l2tpv3 <text> brief
×	not yet documented	show-interfaces-l2tpv3.xml.in: show interfaces l2tpv3 de- tail	show interfaces l2tpv3 de- tail

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Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
✓	<i>show interfaces loopback</i>	show-interfaces-loopback.xml.in: show interfaces loopback	show interfaces loopback <text>
×	not yet documented	show-interfaces-loopback.xml.in: show interfaces loopback <loopback> brief	show interfaces loopback <text> brief
×	not yet documented	show-interfaces-loopback.xml.in: show interfaces loopback detail	show interfaces loopback detail
×	<i>show interfaces loopback lo</i>	Nothing found in XML Definitions	Nothing found in VyOS
✓	<i>show interfaces macsec</i>	show-interfaces-macsec.xml.in: show interfaces macsec <macsec>	show interfaces macsec <text>
×	<i>show interfaces macsec <interface></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	openvpn.xml.in: show interfaces openvpn <openvpn>	show interfaces openvpn <text>
×	not yet documented	openvpn.xml.in: show interfaces openvpn <openvpn> brief	show interfaces openvpn <text> brief
×	not yet documented	openvpn.xml.in: show interfaces openvpn detail	show interfaces openvpn detail
×	not yet documented	openvpn.xml.in: show interfaces openvpn <openvpn> user <user> mfa qr code	show interfaces openvpn <text> user <text> mfa qr- code
×	not yet documented	openvpn.xml.in: show interfaces openvpn <openvpn> user <user> mfa secret	show interfaces openvpn <text> user <text> mfa se- cret
×	not yet documented	openvpn.xml.in: show interfaces openvpn <openvpn> user <user> mfa uri	show interfaces openvpn <text> user <text> mfa uri
×	not yet documented	show-interfaces-pppoe.xml.in: show interfaces pppoe	show interfaces pppoe <text>
×	<i>show interfaces pppoe <interface></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>show interfaces pppoe <interface> queue</i>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	show-interfaces-pppoe.xml.in: show interfaces pppoe detail	show interfaces pppoe detail
×	not yet documented	show-interfaces-pppoe.xml.in: show interfaces pppoe <pppoe> log	show interfaces pppoe <text> log
×	not yet documented	show-interfaces-pppoe.xml.in: show interfaces pppoe <pppoe> statistics	show interfaces pppoe <text> statistics
×	not yet documented	show-interfaces-pseudo-ethernet.xml.in: show interfaces pseudo-ethernet	show interfaces pseudo-ethernet <text>
×	not yet documented	show-interfaces-pseudo-ethernet.xml.in: show interfaces pseudo-ethernet <pseudo-ethernet> brief	show interfaces pseudo-ethernet <text> brief
×	not yet documented	show-interfaces-pseudo-ethernet.xml.in: show interfaces pseudo-ethernet detail	show interfaces pseudo-ethernet detail
×	not yet documented	show-interfaces-sstp.xml.in: show interfaces sstp	show interfaces sstp <text>
×	<i>show interfaces sstp <interface></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	show-interfaces-sstp.xml.in: show interfaces sstp detail	show interfaces sstp detail
×	not yet documented	show-interfaces-sstp.xml.in: show interfaces sstp <sstp> log	show interfaces sstp <text> log
×	not yet documented	show-interfaces-sstp.xml.in: show interfaces sstp <sstp> statistics	show interfaces sstp <text> statistics
×	not yet documented	show-interfaces.xml.in: show interfaces summary	Nothing found in VyOS
×	not yet documented	Nothing found in XML Definitions	show interfaces system
×	not yet documented	Nothing found in XML Definitions	show interfaces system enabled

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Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	show-interfaces-tunnel.xml.in: show interfaces tunnel	show interfaces tunnel <text>
×	not yet documented	show-interfaces-tunnel.xml.in: show interfaces tunnel <tunnel> brief	show interfaces tunnel <text> brief
×	not yet documented	show-interfaces-tunnel.xml.in: show interfaces tunnel detail	show interfaces tunnel detail
✓	<i>show interfaces virtual-ethernet</i>	show-interfaces-virtual-ethernet.xml.in: show interfaces virtual-ethernet	show interfaces virtual-ethernet <text>
×	<i>show interfaces virtual-ethernet <interface></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	show-interfaces-virtual-ethernet.xml.in: show interfaces virtual-ethernet <virtual-ethernet> brief	show interfaces virtual-ethernet <text> brief
×	not yet documented	show-interfaces-virtual-ethernet.xml.in: show interfaces virtual-ethernet detail	show interfaces virtual-ethernet detail
×	not yet documented	Nothing found in XML Definitions	show interfaces vrrp <text>
×	not yet documented	Nothing found in XML Definitions	show interfaces vrrp <text> brief
×	not yet documented	Nothing found in XML Definitions	show interfaces vrrp detail
×	not yet documented	show-interfaces-vti.xml.in: show interfaces vti	show interfaces vti <text>
×	<i>show interfaces vti <vtiX></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>show interfaces vti <vtiX> brief</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	show-interfaces-vti.xml.in: show interfaces vti <vti> brief	show interfaces vti <text> brief
×	not yet documented	show-interfaces-vti.xml.in: show interfaces vti detail	show interfaces vti detail

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Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	show-interfaces-vxlan.xml.in: show interfaces vxlan	show interfaces vxlan <text>
×	not yet documented	show-interfaces-vxlan.xml.in: show interfaces vxlan <vxlan> brief	show interfaces vxlan <text> brief
×	not yet documented	show-interfaces-vxlan.xml.in: show interfaces vxlan de- tail	show interfaces vxlan de- tail
✓	<i>show interfaces wireguard</i>	show-interfaces-wireguard.xml.in: show interfaces wireguard	show interfaces wireguard <text>
×	<i>show interfaces wireguard <interface></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>show interfaces wireguard <interface> public-key</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	show-interfaces-wireguard.xml.in: show interfaces wireguard <wireguard> allowed-ips	show interfaces wireguard <text> allowed-ips
×	not yet documented	show-interfaces-wireguard.xml.in: show interfaces wireguard detail	show interfaces wireguard detail
×	not yet documented	show-interfaces-wireguard.xml.in: show interfaces wireguard <wireguard> endpoints	show interfaces wireguard <text> endpoints
×	not yet documented	show-interfaces-wireguard.xml.in: show interfaces wireguard <wireguard> peers	show interfaces wireguard <text> peers
×	not yet documented	show-interfaces-wireguard.xml.in: show interfaces wireguard <wireguard> public-key	show interfaces wireguard <text> public-key
×	not yet documented	show-interfaces-wireguard.xml.in: show interfaces wireguard <wireguard> summary	show interfaces wireguard <text> summary
×	<i>show interfaces wireguard wg01 summary</i>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	show-interfaces-wireless.xml.in: show interfaces wireless <wireless>	show interfaces wireless <text>
×	<i>show interfaces wireless <wlanX></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>show interfaces wireless <wlanX> brief</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>show interfaces wireless <wlanX> queue</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>show interfaces wireless <wlanX> scan</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	show-interfaces-wireless.xml.in: show interfaces wireless <wireless> brief	show interfaces wireless <text> brief
✓	<i>show interfaces wireless detail</i>	show-interfaces-wireless.xml.in: show interfaces wireless detail	show interfaces wireless detail
✓	<i>show interfaces wireless info</i>	show-interfaces-wireless.xml.in: show interfaces wireless info	show interfaces wireless info
×	not yet documented	show-interfaces-wireless.xml.in: show interfaces wireless <wireless> scan	show interfaces wireless <text> scan
×	not yet documented	show-interfaces-wireless.xml.in: show interfaces wireless <wireless> scan detail	show interfaces wireless <text> scan detail
×	not yet documented	show-interfaces-wireless.xml.in: show interfaces wireless <wireless> stations	show interfaces wireless <text> stations
×	not yet documented	show-interfaces-wireless.xml.in: show interfaces wireless <wireless> vif <vif>	show interfaces wireless <text> vif <text>
×	not yet documented	show-interfaces-wireless.xml.in: show interfaces wireless <wireless> vif <vif> brief	show interfaces wireless <text> vif <text> brief

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Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	show-interfaces- wwan.xml.in: show interfaces wwan	show interfaces wwan <text>
×	<i>show interfaces wwan</i> <interface>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>show interfaces</i> <i>wwan</i> <interface> <i>capabilities</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>show interfaces wwan</i> <interface> <i>firmware</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>show interfaces wwan</i> <interface> <i>imei</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>show interfaces wwan</i> <interface> <i>imsi</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>show interfaces wwan</i> <interface> <i>model</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>show interfaces wwan</i> <interface> <i>msisdn</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>show interfaces</i> <i>wwan</i> <interface> <i>revision</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>show interfaces wwan</i> <interface> <i>signal</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>show interfaces wwan</i> <interface> <i>sim</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>show interfaces wwan</i> <interface> <i>summary</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	show-interfaces- wwan.xml.in: show interfaces wwan <wwan> capabilities	show interfaces wwan <text> capabilities
×	not yet documented	show-interfaces- wwan.xml.in: show interfaces wwan de- tail	show interfaces wwan <text> detail
×	not yet documented	show-interfaces- wwan.xml.in: show interfaces wwan <wwan> firmware	show interfaces wwan <text> firmware
×	not yet documented	show-interfaces- wwan.xml.in: show interfaces wwan <wwan> imei	show interfaces wwan <text> imei
×	not yet documented	show-interfaces- wwan.xml.in: show interfaces wwan <wwan> imsi	show interfaces wwan <text> imsi

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Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	show-interfaces- wwan.xml.in: show interfaces wwan <wwan> log	show interfaces wwan <text> log
×	not yet documented	show-interfaces- wwan.xml.in: show interfaces wwan <wwan> model	show interfaces wwan <text> model
×	not yet documented	show-interfaces- wwan.xml.in: show interfaces wwan <wwan> msisdn	show interfaces wwan <text> msisdn
×	not yet documented	show-interfaces- wwan.xml.in: show interfaces wwan <wwan> revision	show interfaces wwan <text> revision
×	not yet documented	show-interfaces- wwan.xml.in: show interfaces wwan <wwan> signal	show interfaces wwan <text> signal
×	not yet documented	show-interfaces- wwan.xml.in: show interfaces wwan <wwan> sim	show interfaces wwan <text> sim
×	not yet documented	show-ip-access-paths- prefix-community- lists.xml.in: show ip access-list <access-list>	show ip access-list <text>
×	not yet documented	show-ip-access-paths- prefix-community- lists.xml.in: show ip as-path-access-list <as-path-access-list>	show ip as-path-access-list <text>
×	not yet documented	show-ip-bgp.xml.in: show ip bgp	show ip bgp
×	not yet documented	show-ip-bgp.xml.in: show ip bgp attribute-info	show ip bgp attribute-info
×	not yet documented	show-ip-bgp.xml.in: show ip bgp cidr-only	show ip bgp cidr-only
×	not yet documented	show-ip-bgp.xml.in: show ip bgp community	show ip bgp community <text>
×	not yet documented	show-ip-bgp.xml.in: show ip bgp community- info	show ip bgp community- info
×	not yet documented	show-ip-bgp.xml.in: show ip bgp community- list <community-list>	show ip bgp community- list <text>

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Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	show-ip-bgp.xml.in: show ip bgp community- list <community-list> exact-match	show ip bgp community- list <text> exact-match
×	not yet documented	show-ip-bgp.xml.in: show ip bgp community accept-own	show ip bgp community accept-own
×	not yet documented	show-ip-bgp.xml.in: show ip bgp community accept-own-nexthop	show ip bgp community accept-own-nexthop
×	not yet documented	show-ip-bgp.xml.in: show ip bgp community blackhole	show ip bgp community blackhole
×	not yet documented	show-ip-bgp.xml.in: show ip bgp community exact-match	show ip bgp community <text> exact-match
×	not yet documented	show-ip-bgp.xml.in: show ip bgp community graceful-shutdown	show ip bgp community graceful-shutdown
×	not yet documented	show-ip-bgp.xml.in: show ip bgp community llgr-stale	show ip bgp community llgr-stale
×	not yet documented	show-ip-bgp.xml.in: show ip bgp community local-AS	show ip bgp community local-AS
×	not yet documented	show-ip-bgp.xml.in: show ip bgp community no-advertise	show ip bgp community no-advertise
×	not yet documented	show-ip-bgp.xml.in: show ip bgp community no-export	show ip bgp community no-export
×	not yet documented	show-ip-bgp.xml.in: show ip bgp community no-llgr	show ip bgp community no-llgr
×	not yet documented	show-ip-bgp.xml.in: show ip bgp community no-peer	show ip bgp community no-peer
×	not yet documented	show-ip-bgp.xml.in: show ip bgp community route-filter-translated-v4	show ip bgp community route-filter-translated-v4
×	not yet documented	show-ip-bgp.xml.in: show ip bgp community route-filter-translated-v6	show ip bgp community route-filter-translated-v6
×	not yet documented	show-ip-bgp.xml.in: show ip bgp community route-filter-v4	show ip bgp community route-filter-v4
×	not yet documented	show-ip-bgp.xml.in: show ip bgp community route-filter-v6	show ip bgp community route-filter-v6

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Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	show-ip-bgp.xml.in: show ip bgp dampening dampened-paths	show ip bgp dampening dampened-paths
×	not yet documented	show-ip-bgp.xml.in: show ip bgp dampening flap-statistics	show ip bgp dampening flap-statistics
×	not yet documented	show-ip-bgp.xml.in: show ip bgp dampening parameters	show ip bgp dampening parameters
×	not yet documented	show-ip-bgp.xml.in: show ip bgp filter-list <filter-list>	Nothing found in VyOS
×	not yet documented	show-ip-bgp.xml.in: show ip bgp ipv4 unicast <unicast>	show ip bgp ipv4 unicast <text>
×	not yet documented	show-ip-bgp.xml.in: show ip bgp ipv4 unicast cidr-only	show ip bgp ipv4 unicast cidr-only
×	not yet documented	show-ip-bgp.xml.in: show ip bgp ipv4 unicast community <community>	show ip bgp ipv4 unicast community <text>
×	not yet documented	show-ip-bgp.xml.in: show ip bgp ipv4 unicast community-list <community-list>	show ip bgp ipv4 unicast community-list <text>
×	not yet documented	show-ip-bgp.xml.in: show ip bgp ipv4 unicast community-list <community-list> exact-match	show ip bgp ipv4 unicast community-list <text> exact-match
×	not yet documented	show-ip-bgp.xml.in: show ip bgp ipv4 unicast neighbors <neighbors>	show ip bgp ipv4 unicast neighbors <text>
×	not yet documented	show-ip-bgp.xml.in: show ip bgp ipv4 unicast neighbors <neighbors> advertised-routes	show ip bgp ipv4 unicast neighbors <text> advertised-routes
×	not yet documented	show-ip-bgp.xml.in: show ip bgp ipv4 unicast neighbors <neighbors> filtered-routes	show ip bgp ipv4 unicast neighbors <text> filtered-routes
×	not yet documented	show-ip-bgp.xml.in: show ip bgp ipv4 unicast neighbors <neighbors> prefix-counts	show ip bgp ipv4 unicast neighbors <text> prefix-counts
×	not yet documented	show-ip-bgp.xml.in: show ip bgp ipv4 unicast neighbors <neighbors> received-routes	show ip bgp ipv4 unicast neighbors <text> received-routes

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Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	show-ip-bgp.xml.in: show ip bgp ipv4 unicast neighbors <neighbors> routes	show ip bgp ipv4 unicast neighbors <text> routes
×	not yet documented	show-ip-bgp.xml.in: show ip bgp ipv4 unicast paths	show ip bgp ipv4 unicast paths
×	not yet documented	show-ip-bgp.xml.in: show ip bgp ipv4 unicast prefix-list <prefix-list>	show ip bgp ipv4 unicast prefix-list <text>
×	not yet documented	show-ip-bgp.xml.in: show ip bgp ipv4 unicast regexp <regexp>	show ip bgp ipv4 unicast regexp <text>
×	not yet documented	show-ip-bgp.xml.in: show ip bgp ipv4 unicast route-map <route-map>	show ip bgp ipv4 unicast route-map <text>
×	not yet documented	show-ip-bgp.xml.in: show ip bgp ipv4 unicast summary	show ip bgp ipv4 unicast summary
×	not yet documented	show-ip-bgp.xml.in: show ip bgp large- community	show ip bgp large- community <text>
×	not yet documented	show-ip-bgp.xml.in: show ip bgp large- community-info	show ip bgp large- community-info
×	not yet documented	show-ip-bgp.xml.in: show ip bgp large- community-list <large- community-list>	show ip bgp large- community-list <text>
×	not yet documented	show-ip-bgp.xml.in: show ip bgp large- community-list <large- community-list> exact- match	show ip bgp large- community-list <text> exact-match
×	not yet documented	show-ip-bgp.xml.in: show ip bgp large- community <large- community> exact-match	show ip bgp large- community <text> exact- match
×	not yet documented	show-ip-bgp.xml.in: show ip bgp memory	show ip bgp memory
×	not yet documented	show-ip-bgp.xml.in: show ip bgp neighbors <neighbors>	show ip bgp neighbors <text>
×	not yet documented	show-ip-bgp.xml.in: show ip bgp neighbors <neighbors> advertised- routes	show ip bgp neighbors <text> advertised-routes

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Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	show-ip-bgp.xml.in: show ip bgp neighbors <neighbors> dampened-routes	show ip bgp neighbors <text> dampened-routes
×	not yet documented	show-ip-bgp.xml.in: show ip bgp neighbors <neighbors> filtered-routes	show ip bgp neighbors <text> filtered-routes
×	not yet documented	show-ip-bgp.xml.in: show ip bgp neighbors <neighbors> flap-statistics	show ip bgp neighbors <text> flap-statistics
×	not yet documented	show-ip-bgp.xml.in: show ip bgp neighbors <neighbors> prefix-counts	show ip bgp neighbors <text> prefix-counts
×	not yet documented	show-ip-bgp.xml.in: show ip bgp neighbors <neighbors> received-routes	show ip bgp neighbors <text> received-routes
×	not yet documented	show-ip-bgp.xml.in: show ip bgp neighbors <neighbors> received prefix-filter	show ip bgp neighbors <text> received prefix-filter
×	not yet documented	show-ip-bgp.xml.in: show ip bgp neighbors <neighbors> routes	show ip bgp neighbors <text> routes
×	not yet documented	show-ip-bgp.xml.in: show ip bgp paths	show ip bgp paths
×	not yet documented	show-ip-bgp.xml.in: show ip bgp prefix-list <prefix-list>	show ip bgp prefix-list <text>
×	not yet documented	show-ip-bgp.xml.in: show ip bgp regexp <reg-exp>	show ip bgp regexp <text>
×	not yet documented	show-ip-bgp.xml.in: show ip bgp route-map <route-map>	show ip bgp route-map <text>
×	not yet documented	show-ip-bgp.xml.in: show ip bgp statistics	show ip bgp statistics
×	not yet documented	show-ip-bgp.xml.in: show ip bgp summary	show ip bgp summary
×	not yet documented	show-ip-bgp.xml.in: show ip bgp summary es-established	show ip bgp summary es-established
×	not yet documented	show-ip-bgp.xml.in: show ip bgp summary failed	show ip bgp summary failed
×	not yet documented	show-ip-bgp.xml.in: show ip bgp vrf <vrf>	show ip bgp vrf <text>

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Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	show-ip-bgp.xml.in: show ip bgp vrf <vrf> attribute-info	show ip bgp vrf <text> attribute-info
×	not yet documented	show-ip-bgp.xml.in: show ip bgp vrf <vrf> cidr- only	show ip bgp vrf <text> cidr-only
×	not yet documented	show-ip-bgp.xml.in: show ip bgp vrf <vrf> community	show ip bgp vrf <text> community <text>
×	not yet documented	show-ip-bgp.xml.in: show ip bgp vrf <vrf> community-info	show ip bgp vrf <text> community-info
×	not yet documented	show-ip-bgp.xml.in: show ip bgp vrf <vrf> community-list <community-list>	show ip bgp vrf <text> community-list <text>
×	not yet documented	show-ip-bgp.xml.in: show ip bgp vrf <vrf> community-list <community-list> exact- match	show ip bgp vrf <text> community-list <text> exact-match
×	not yet documented	show-ip-bgp.xml.in: show ip bgp vrf <vrf> community accept-own	show ip bgp vrf <text> community accept-own
×	not yet documented	show-ip-bgp.xml.in: show ip bgp vrf <vrf> community accept-own- nexthop	show ip bgp vrf <text> community accept-own- nexthop
×	not yet documented	show-ip-bgp.xml.in: show ip bgp vrf <vrf> community blackhole	show ip bgp vrf <text> community blackhole
×	not yet documented	show-ip-bgp.xml.in: show ip bgp vrf <vrf> community exact-match	show ip bgp vrf <text> community <text> exact- match
×	not yet documented	show-ip-bgp.xml.in: show ip bgp vrf <vrf> community graceful- shutdown	show ip bgp vrf <text> community graceful- shutdown
×	not yet documented	show-ip-bgp.xml.in: show ip bgp vrf <vrf> community llgr-stale	show ip bgp vrf <text> community llgr-stale
×	not yet documented	show-ip-bgp.xml.in: show ip bgp vrf <vrf> community local-AS	show ip bgp vrf <text> community local-AS
×	not yet documented	show-ip-bgp.xml.in: show ip bgp vrf <vrf> community no-advertise	show ip bgp vrf <text> community no-advertise

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Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	show-ip-bgp.xml.in: show ip bgp vrf <vrf> community no-export	show ip bgp vrf <text> community no-export
×	not yet documented	show-ip-bgp.xml.in: show ip bgp vrf <vrf> community no-llgr	show ip bgp vrf <text> community no-llgr
×	not yet documented	show-ip-bgp.xml.in: show ip bgp vrf <vrf> community no-peer	show ip bgp vrf <text> community no-peer
×	not yet documented	show-ip-bgp.xml.in: show ip bgp vrf <vrf> community route-filter- translated-v4	show ip bgp vrf <text> community route-filter- translated-v4
×	not yet documented	show-ip-bgp.xml.in: show ip bgp vrf <vrf> community route-filter- translated-v6	show ip bgp vrf <text> community route-filter- translated-v6
×	not yet documented	show-ip-bgp.xml.in: show ip bgp vrf <vrf> community route-filter-v4	show ip bgp vrf <text> community route-filter-v4
×	not yet documented	show-ip-bgp.xml.in: show ip bgp vrf <vrf> community route-filter-v6	show ip bgp vrf <text> community route-filter-v6
×	not yet documented	show-ip-bgp.xml.in: show ip bgp vrf <vrf> dampening dampened- paths	show ip bgp vrf <text> dampening dampened- paths
×	not yet documented	show-ip-bgp.xml.in: show ip bgp vrf <vrf> dampening flap-statistics	show ip bgp vrf <text> dampening flap-statistics
×	not yet documented	show-ip-bgp.xml.in: show ip bgp vrf <vrf> dampening parameters	show ip bgp vrf <text> dampening parameters
×	not yet documented	show-ip-bgp.xml.in: show ip bgp vrf <vrf> filter-list <filter-list>	Nothing found in VyOS
×	not yet documented	show-ip-bgp.xml.in: show ip bgp vrf <vrf> ipv4 unicast <unicast>	show ip bgp vrf <text> ipv4 unicast <text>
×	not yet documented	show-ip-bgp.xml.in: show ip bgp vrf <vrf> ipv4 unicast cidr-only	show ip bgp vrf <text> ipv4 unicast cidr-only
×	not yet documented	show-ip-bgp.xml.in: show ip bgp vrf <vrf> ipv4 unicast community <com- munity>	show ip bgp vrf <text> ipv4 unicast community <text>

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Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	show-ip-bgp.xml.in: show ip bgp vrf <vrf> ipv4 unicast community- list <community-list>	show ip bgp vrf <text> ipv4 unicast community- list <text>
×	not yet documented	show-ip-bgp.xml.in: show ip bgp vrf <vrf> ipv4 unicast community- list <community-list> exact-match	show ip bgp vrf <text> ipv4 unicast community- list <text> exact-match
×	not yet documented	show-ip-bgp.xml.in: show ip bgp vrf <vrf> ipv4 unicast neighbors <neigh- bors>	show ip bgp vrf <text> ipv4 unicast neighbors <text>
×	not yet documented	show-ip-bgp.xml.in: show ip bgp vrf <vrf> ipv4 unicast neighbors <neigh- bors> advertised-routes	show ip bgp vrf <text> ipv4 unicast neighbors <text> advertised-routes
×	not yet documented	show-ip-bgp.xml.in: show ip bgp vrf <vrf> ipv4 unicast neighbors <neigh- bors> filtered-routes	show ip bgp vrf <text> ipv4 unicast neighbors <text> filtered-routes
×	not yet documented	show-ip-bgp.xml.in: show ip bgp vrf <vrf> ipv4 unicast neighbors <neigh- bors> prefix-counts	show ip bgp vrf <text> ipv4 unicast neighbors <text> prefix-counts
×	not yet documented	show-ip-bgp.xml.in: show ip bgp vrf <vrf> ipv4 unicast neighbors <neigh- bors> received-routes	show ip bgp vrf <text> ipv4 unicast neighbors <text> received-routes
×	not yet documented	show-ip-bgp.xml.in: show ip bgp vrf <vrf> ipv4 unicast neighbors <neigh- bors> routes	show ip bgp vrf <text> ipv4 unicast neighbors <text> routes
×	not yet documented	show-ip-bgp.xml.in: show ip bgp vrf <vrf> ipv4 unicast paths	show ip bgp vrf <text> ipv4 unicast paths
×	not yet documented	show-ip-bgp.xml.in: show ip bgp vrf <vrf> ipv4 unicast prefix-list <prefix- list>	show ip bgp vrf <text> ipv4 unicast prefix-list <text>
×	not yet documented	show-ip-bgp.xml.in: show ip bgp vrf <vrf> ipv4 unicast regexp <regexp>	show ip bgp vrf <text> ipv4 unicast regexp <text>
×	not yet documented	show-ip-bgp.xml.in: show ip bgp vrf <vrf> ipv4 unicast route-map <route- map>	show ip bgp vrf <text> ipv4 unicast route-map <text>

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Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	show-ip-bgp.xml.in: show ip bgp vrf <vrf> ipv4 unicast summary	show ip bgp vrf <text> ipv4 unicast summary
×	not yet documented	show-ip-bgp.xml.in: show ip bgp vrf <vrf> large-community	show ip bgp vrf <text> large-community <text>
×	not yet documented	show-ip-bgp.xml.in: show ip bgp vrf <vrf> large-community-info	show ip bgp vrf <text> large-community-info
×	not yet documented	show-ip-bgp.xml.in: show ip bgp vrf <vrf> large-community-list <large-community-list>	show ip bgp vrf <text> large-community-list <text>
×	not yet documented	show-ip-bgp.xml.in: show ip bgp vrf <vrf> large-community-list <large-community-list> exact-match	show ip bgp vrf <text> large-community-list <text> exact-match
×	not yet documented	show-ip-bgp.xml.in: show ip bgp vrf <vrf> large-community <large- community> exact-match	show ip bgp vrf <text> large-community <text> exact-match
×	not yet documented	show-ip-bgp.xml.in: show ip bgp vrf <vrf> memory	show ip bgp vrf <text> memory
×	not yet documented	show-ip-bgp.xml.in: show ip bgp vrf <vrf> neighbors <neighbors>	show ip bgp vrf <text> neighbors <text>
×	not yet documented	show-ip-bgp.xml.in: show ip bgp vrf <vrf> neighbors <neighbors> advertised-routes	show ip bgp vrf <text> neighbors <text> advertised-routes
×	not yet documented	show-ip-bgp.xml.in: show ip bgp vrf <vrf> neighbors <neighbors> dampened-routes	show ip bgp vrf <text> neighbors <text> dampened-routes
×	not yet documented	show-ip-bgp.xml.in: show ip bgp vrf <vrf> neighbors <neighbors> filtered-routes	show ip bgp vrf <text> neighbors <text> filtered- routes
×	not yet documented	show-ip-bgp.xml.in: show ip bgp vrf <vrf> neighbors <neighbors> flap-statistics	show ip bgp vrf <text> neighbors <text> flap- statistics
×	not yet documented	show-ip-bgp.xml.in: show ip bgp vrf <vrf> neighbors <neighbors> prefix-counts	show ip bgp vrf <text> neighbors <text> prefix- counts

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Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	show-ip-bgp.xml.in: show ip bgp vrf <vrf> neighbors <neighbors> received-routes	show ip bgp vrf <text> neighbors <text> received-routes
×	not yet documented	show-ip-bgp.xml.in: show ip bgp vrf <vrf> neighbors <neighbors> re- ceived prefix-filter	show ip bgp vrf <text> neighbors <text> received prefix-filter
×	not yet documented	show-ip-bgp.xml.in: show ip bgp vrf <vrf> neighbors <neighbors> routes	show ip bgp vrf <text> neighbors <text> routes
×	not yet documented	show-ip-bgp.xml.in: show ip bgp vrf <vrf> paths	show ip bgp vrf <text> paths
×	not yet documented	show-ip-bgp.xml.in: show ip bgp vrf <vrf> prefix-list <prefix-list>	show ip bgp vrf <text> prefix-list <text>
×	not yet documented	show-ip-bgp.xml.in: show ip bgp vrf <vrf> reg- exp <regex>	show ip bgp vrf <text> regex <text>
×	not yet documented	show-ip-bgp.xml.in: show ip bgp vrf <vrf> route-map <route-map>	show ip bgp vrf <text> route-map <text>
×	not yet documented	show-ip-bgp.xml.in: show ip bgp vrf <vrf> statistics	show ip bgp vrf <text> statistics
×	not yet documented	show-ip-bgp.xml.in: show ip bgp vrf <vrf> summary	show ip bgp vrf <text> summary
×	not yet documented	show-ip-bgp.xml.in: show ip bgp vrf <vrf> summary established	show ip bgp vrf <text> summary established
×	not yet documented	show-ip-bgp.xml.in: show ip bgp vrf <vrf> summary failed	show ip bgp vrf <text> summary failed
×	not yet documented	show-ip-bgp.xml.in: show ip bgp vrf <vrf> wide	show ip bgp vrf <text> wide
×	not yet documented	show-ip-bgp.xml.in: show ip bgp wide	show ip bgp wide
×	not yet documented	show-ip-access-paths- prefix-community- lists.xml.in: show ip community-list <community-list>	show ip community-list <text>

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Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	show-ip-access-paths-prefix-community-lists.xml.in: show ip extcommunity-list <extcommunity-list>	show ip extcommunity-list <text>
×	not yet documented	show-ip-access-paths-prefix-community-lists.xml.in: show ip forwarding	show ip forwarding
×	not yet documented	Nothing found in XML Definitions	show ip groups
×	not yet documented	show-ip-igmp.xml.in: show ip igmp groups	show ip igmp groups
×	not yet documented	show-ip-igmp.xml.in: show ip igmp interface	Nothing found in VyOS
×	not yet documented	Nothing found in XML Definitions	show ip igmp interfaces
×	not yet documented	show-ip-igmp.xml.in: show ip igmp join	show ip igmp join
×	not yet documented	show-ip-igmp.xml.in: show ip igmp sources	show ip igmp sources
×	not yet documented	show-ip-igmp.xml.in: show ip igmp statistics	show ip igmp statistics
×	not yet documented	show-ip-access-paths-prefix-community-lists.xml.in: show ip large-community-list <large-community-list>	show ip large-community-list <text>
×	not yet documented	multicast-group.xml.in: show ip multicast group	Nothing found in VyOS
×	not yet documented	multicast-group.xml.in: show ip multicast group interface <interface>	Nothing found in VyOS
×	not yet documented	show-ip-multicast.xml.in: show ip multicast interface	show ip multicast interface
×	not yet documented	show-ip-multicast.xml.in: show ip multicast route	show ip multicast route
×	not yet documented	show-ip-multicast.xml.in: show ip multicast summary	show ip multicast summary
×	not yet documented	show-ip.xml.in: show ip neighbors	show ip neighbors
×	not yet documented	show-ip.xml.in: show ip neighbors interface <interface>	show ip neighbors interface <text>
×	not yet documented	show-ip.xml.in: show ip neighbors state <state>	show ip neighbors state <text>

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Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	<i>show ip nhrp cache</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>show ip nhrp nhs</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>show ip nhrp shortcut</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	show-ip.xml.in: show ip nht	Nothing found in VyOS
×	not yet documented	show-ip.xml.in: show ip nht vrf <vrf>	Nothing found in VyOS
×	<i>show ipoe-server sessions</i>	Nothing found in XML Definitions	show ipoe-server sessions
×	not yet documented	Nothing found in XML Definitions	show ipoe-server statistics
×	not yet documented	show-ip-ospf.xml.in: show ip ospf	show ip ospf
✓	<i>show ip ospf border-routers</i>	show-ip-ospf.xml.in: show ip ospf border-routers	show ip ospf border-routers
✓	<i>show ip ospf database</i>	show-ip-ospf.xml.in: show ip ospf database	show ip ospf database
×	<i>show ip ospf database <type> [A.B.C.D] [adv-router <A.B.C.D> self-originate]</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	show-ip-ospf.xml.in: show ip ospf database asbr-summary <asbr-summary>	show ip ospf database asbr-summary <text>
×	not yet documented	show-ip-ospf.xml.in: show ip ospf database asbr-summary <asbr-summary> adv-router <adv-router>	show ip ospf database asbr-summary <text> adv-router <text>
×	not yet documented	show-ip-ospf.xml.in: show ip ospf database asbr-summary <asbr-summary> self-originate	show ip ospf database asbr-summary <text> self-originate
×	not yet documented	show-ip-ospf.xml.in: show ip ospf database external <external>	show ip ospf database external <text>
×	not yet documented	show-ip-ospf.xml.in: show ip ospf database external <external> adv-router <adv-router>	show ip ospf database external <text> adv-router <text>

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Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	show-ip-ospf.xml.in: show ip ospf database external <external> self-originate	show ip ospf database external <text> self- originate
✓	<i>show ip ospf database max-age</i>	show-ip-ospf.xml.in: show ip ospf database max-age	show ip ospf database max-age
×	not yet documented	show-ip-ospf.xml.in: show ip ospf database net- work <network>	show ip ospf database net- work <text>
×	not yet documented	show-ip-ospf.xml.in: show ip ospf database network <network> adv-router <adv-router>	show ip ospf database network <text> adv-router <text>
×	not yet documented	show-ip-ospf.xml.in: show ip ospf database network <network> self-originate	show ip ospf database net- work <text> self-originate
×	not yet documented	show-ip-ospf.xml.in: show ip ospf database nssa-external <nssa- external>	show ip ospf database nssa-external <text>
×	not yet documented	show-ip-ospf.xml.in: show ip ospf database nssa-external <nssa- external> adv-router <adv-router>	show ip ospf database nssa-external <text> adv-router <text>
×	not yet documented	show-ip-ospf.xml.in: show ip ospf database nssa-external <nssa- external> self-originate	show ip ospf database nssa-external <text> self-originate
×	not yet documented	show-ip-ospf.xml.in: show ip ospf database opaque-area <opaque- area>	show ip ospf database opaque-area <text>
×	not yet documented	show-ip-ospf.xml.in: show ip ospf database opaque-area <opaque- area> adv-router <adv- router>	show ip ospf database opaque-area <text> adv-router <text>
×	not yet documented	show-ip-ospf.xml.in: show ip ospf database opaque-area <opaque- area> self-originate	show ip ospf database opaque-area <text> self-originate
×	not yet documented	show-ip-ospf.xml.in: show ip ospf database opaque-as <opaque-as>	show ip ospf database opaque-as <text>

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Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	show-ip-ospf.xml.in: show ip ospf database opaque-as <opaque-as> adv-router <adv-router>	show ip ospf database opaque-as <text> adv- router <text>
×	not yet documented	show-ip-ospf.xml.in: show ip ospf database opaque-as <opaque-as> self-originate	show ip ospf database opaque-as <text> self- originate
×	not yet documented	show-ip-ospf.xml.in: show ip ospf database opaque-link <opaque- link>	show ip ospf database opaque-link <text>
×	not yet documented	show-ip-ospf.xml.in: show ip ospf database opaque-link <opaque- link> adv-router <adv- router>	show ip ospf database opaque-link <text> adv- router <text>
×	not yet documented	show-ip-ospf.xml.in: show ip ospf database opaque-link <opaque- link> self-originate	show ip ospf database opaque-link <text> self- originate
×	not yet documented	show-ip-ospf.xml.in: show ip ospf database router <router>	show ip ospf database router <text>
×	not yet documented	show-ip-ospf.xml.in: show ip ospf database router <router> adv-router <adv-router>	show ip ospf database router <text> adv-router <text>
×	not yet documented	show-ip-ospf.xml.in: show ip ospf database router <router> self- originate	show ip ospf database router <text> self- originate
×	not yet documented	show-ip-ospf.xml.in: show ip ospf database self- originate	show ip ospf database self- originate
×	not yet documented	show-ip-ospf.xml.in: show ip ospf database summary <summary>	show ip ospf database summary <text>
×	not yet documented	show-ip-ospf.xml.in: show ip ospf database summary <summary> adv-router <adv-router>	show ip ospf database summary <text> adv- router <text>
×	not yet documented	show-ip-ospf.xml.in: show ip ospf database summary <summary> self-originate	show ip ospf database summary <text> self- originate
×	not yet documented	show-ip-ospf.xml.in: show ip ospf graceful- restart helper	show ip ospf graceful- restart helper

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Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
✓	<i>show ip ospf interface [<interface>]</i>	show-ip-ospf.xml.in: show ip ospf interface <interface>	show ip ospf interface <text>
×	not yet documented	show-ip-ospf.xml.in: show ip ospf mpls ldp-sync	show ip ospf mpls ldp-sync
×	not yet documented	show-ip-ospf.xml.in: show ip ospf mpls ldp-sync interface <interface>	show ip ospf mpls ldp-sync interface <text>
✓	<i>show ip ospf neighbor</i>	show-ip-ospf.xml.in: show ip ospf neighbor <neighbor>	show ip ospf neighbor <text>
×	<i>show ip ospf neighbor <interface></i>	Nothing found in XML Definitions	Nothing found in VyOS
✓	<i>show ip ospf neighbor detail</i>	show-ip-ospf.xml.in: show ip ospf neighbor detail	show ip ospf neighbor detail
✓	<i>show ip ospf route [detail]</i>	show-ip-ospf.xml.in: show ip ospf route	show ip ospf route
×	not yet documented	show-ip-ospf.xml.in: show ip ospf route detail	show ip ospf route detail
×	not yet documented	show-ip-ospf.xml.in: show ip ospf vrf <vrf>	show ip ospf vrf <text>
×	not yet documented	show-ip-ospf.xml.in: show ip ospf vrf <vrf> border-routers	show ip ospf vrf <text> border-routers
×	not yet documented	show-ip-ospf.xml.in: show ip ospf vrf <vrf> database	show ip ospf vrf <text> database
×	not yet documented	show-ip-ospf.xml.in: show ip ospf vrf <vrf> database asbr-summary <asbr-summary>	show ip ospf vrf <text> database asbr-summary <text>
×	not yet documented	show-ip-ospf.xml.in: show ip ospf vrf <vrf> database asbr-summary <asbr-summary> adv-router <adv-router>	show ip ospf vrf <text> database asbr-summary <text> adv-router <text>
×	not yet documented	show-ip-ospf.xml.in: show ip ospf vrf <vrf> database asbr-summary <asbr-summary> self-originate	show ip ospf vrf <text> database asbr-summary <text> self-originate
×	not yet documented	show-ip-ospf.xml.in: show ip ospf vrf <vrf> database external <external>	show ip ospf vrf <text> database external <text>

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Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	show-ip-ospf.xml.in: show ip ospf vrf <vrf> database external <external> adv-router <adv-router>	show ip ospf vrf <text> database external <text> adv-router <text>
×	not yet documented	show-ip-ospf.xml.in: show ip ospf vrf <vrf> database external <external> self-originate	show ip ospf vrf <text> database external <text> self-originate
×	not yet documented	show-ip-ospf.xml.in: show ip ospf vrf <vrf> database max-age	show ip ospf vrf <text> database max-age
×	not yet documented	show-ip-ospf.xml.in: show ip ospf vrf <vrf> database network <network>	show ip ospf vrf <text> database network <text>
×	not yet documented	show-ip-ospf.xml.in: show ip ospf vrf <vrf> database network <network> adv-router <adv-router>	show ip ospf vrf <text> database network <text> adv-router <text>
×	not yet documented	show-ip-ospf.xml.in: show ip ospf vrf <vrf> database network <network> self-originate	show ip ospf vrf <text> database network <text> self-originate
×	not yet documented	show-ip-ospf.xml.in: show ip ospf vrf <vrf> database nssa-external <nssa-external>	show ip ospf vrf <text> database nssa-external <text>
×	not yet documented	show-ip-ospf.xml.in: show ip ospf vrf <vrf> database nssa-external <nssa-external> adv-router <adv-router>	show ip ospf vrf <text> database nssa-external <text> adv-router <text>
×	not yet documented	show-ip-ospf.xml.in: show ip ospf vrf <vrf> database nssa-external <nssa-external> self-originate	show ip ospf vrf <text> database nssa-external <text> self-originate
×	not yet documented	show-ip-ospf.xml.in: show ip ospf vrf <vrf> database opaque-area <opaque-area>	show ip ospf vrf <text> database opaque-area <text>
×	not yet documented	show-ip-ospf.xml.in: show ip ospf vrf <vrf> database opaque-area <opaque-area> adv-router <adv-router>	show ip ospf vrf <text> database opaque-area <text> adv-router <text>

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Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	show-ip-ospf.xml.in: show ip ospf vrf <vrf> database opaque-area <opaque-area> self-originate	show ip ospf vrf <text> database opaque-area <text> self-originate
×	not yet documented	show-ip-ospf.xml.in: show ip ospf vrf <vrf> database opaque-as <opaque-as>	show ip ospf vrf <text> database opaque-as <text>
×	not yet documented	show-ip-ospf.xml.in: show ip ospf vrf <vrf> database opaque-as <opaque-as> adv-router <adv-router>	show ip ospf vrf <text> database opaque-as <text> adv-router <text>
×	not yet documented	show-ip-ospf.xml.in: show ip ospf vrf <vrf> database opaque-as <opaque-as> self-originate	show ip ospf vrf <text> database opaque-as <text> self-originate
×	not yet documented	show-ip-ospf.xml.in: show ip ospf vrf <vrf> database opaque-link <opaque-link>	show ip ospf vrf <text> database opaque-link <text>
×	not yet documented	show-ip-ospf.xml.in: show ip ospf vrf <vrf> database opaque-link <opaque-link> adv-router <adv-router>	show ip ospf vrf <text> database opaque-link <text> adv-router <text>
×	not yet documented	show-ip-ospf.xml.in: show ip ospf vrf <vrf> database opaque-link <opaque-link> self-originate	show ip ospf vrf <text> database opaque-link <text> self-originate
×	not yet documented	show-ip-ospf.xml.in: show ip ospf vrf <vrf> database router <router>	show ip ospf vrf <text> database router <text>
×	not yet documented	show-ip-ospf.xml.in: show ip ospf vrf <vrf> database router <router> adv-router <adv-router>	show ip ospf vrf <text> database router <text> adv-router <text>
×	not yet documented	show-ip-ospf.xml.in: show ip ospf vrf <vrf> database router <router> self-originate	show ip ospf vrf <text> database router <text> self-originate
×	not yet documented	show-ip-ospf.xml.in: show ip ospf vrf <vrf> database self-originate	show ip ospf vrf <text> database self-originate

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Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	show-ip-ospf.xml.in: show ip ospf vrf <vrf> database summary <summary>	show ip ospf vrf <text> database summary <text>
×	not yet documented	show-ip-ospf.xml.in: show ip ospf vrf <vrf> database summary <summary> adv-router <adv-router>	show ip ospf vrf <text> database summary <text> adv-router <text>
×	not yet documented	show-ip-ospf.xml.in: show ip ospf vrf <vrf> database summary <summary> self-originate	show ip ospf vrf <text> database summary <text> self-originate
×	not yet documented	show-ip-ospf.xml.in: show ip ospf vrf <vrf> graceful-restart helper	show ip ospf vrf <text> graceful-restart helper
×	not yet documented	show-ip-ospf.xml.in: show ip ospf vrf <vrf> interface <interface>	show ip ospf vrf <text> interface <text>
×	not yet documented	show-ip-ospf.xml.in: show ip ospf vrf <vrf> mpls ldp-sync	show ip ospf vrf <text> mpls ldp-sync
×	not yet documented	show-ip-ospf.xml.in: show ip ospf vrf <vrf> mpls ldp-sync interface <interface>	show ip ospf vrf <text> mpls ldp-sync interface <text>
×	not yet documented	show-ip-ospf.xml.in: show ip ospf vrf <vrf> neighbor <neighbor>	show ip ospf vrf <text> neighbor <text>
×	not yet documented	show-ip-ospf.xml.in: show ip ospf vrf <vrf> neighbor detail	show ip ospf vrf <text> neighbor detail
×	not yet documented	show-ip-ospf.xml.in: show ip ospf vrf <vrf> route	show ip ospf vrf <text> route
×	not yet documented	show-ip-ospf.xml.in: show ip ospf vrf <vrf> route detail	show ip ospf vrf <text> route detail
×	not yet documented	show-ip-pim.xml.in: show ip pim assert	Nothing found in VyOS
×	not yet documented	show-ip-pim.xml.in: show ip pim assert-internal	Nothing found in VyOS
×	not yet documented	show-ip-pim.xml.in: show ip pim assert-metric	Nothing found in VyOS
×	not yet documented	show-ip-pim.xml.in: show ip pim assert-winner-metric	Nothing found in VyOS

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Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	show-ip-pim.xml.in: show ip pim bsm-database	Nothing found in VyOS
×	not yet documented	show-ip-pim.xml.in: show ip pim bsr	Nothing found in VyOS
×	not yet documented	show-ip-pim.xml.in: show ip pim bsrp-info	Nothing found in VyOS
×	not yet documented	show-ip-pim.xml.in: show ip pim channel	Nothing found in VyOS
×	not yet documented	show-ip-pim.xml.in: show ip pim group-type	Nothing found in VyOS
×	not yet documented	show-ip-pim.xml.in: show ip pim interface	Nothing found in VyOS
×	not yet documented	Nothing found in XML Definitions	show ip pim interfaces
×	not yet documented	show-ip-pim.xml.in: show ip pim join	show ip pim join
×	not yet documented	show-ip-pim.xml.in: show ip pim local- membership	Nothing found in VyOS
×	not yet documented	show-ip-pim.xml.in: show ip pim neighbor	show ip pim neighbor
×	not yet documented	show-ip-pim.xml.in: show ip pim nexthop	show ip pim nexthop
×	not yet documented	Nothing found in XML Definitions	show ip pim rp
×	not yet documented	show-ip-pim.xml.in: show ip pim rp-info	Nothing found in VyOS
×	not yet documented	show-ip-pim.xml.in: show ip pim rpf	show ip pim rpf
×	not yet documented	show-ip-pim.xml.in: show ip pim secondary	Nothing found in VyOS
×	not yet documented	show-ip-pim.xml.in: show ip pim state	show ip pim state
×	not yet documented	show-ip-pim.xml.in: show ip pim statistics	show ip pim statistics
×	not yet documented	show-ip-pim.xml.in: show ip pim upstream	show ip pim upstream
×	not yet documented	show-ip-pim.xml.in: show ip pim upstream- join-desired	Nothing found in VyOS
×	not yet documented	show-ip-pim.xml.in: show ip pim upstream-rpf	Nothing found in VyOS
×	not yet documented	show-ip-pim.xml.in: show ip pim vxlan-groups	Nothing found in VyOS
×	not yet documented	show-ip-ports.xml.in: show ip ports	show ip ports

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Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	show-ip-access-paths-prefix-community-lists.xml.in: show ip prefix-list <prefix-list>	show ip prefix-list <text>
×	not yet documented	show-ip-access-paths-prefix-community-lists.xml.in: show ip protocol	show ip protocol
✓	<i>show ip rip</i>	show-ip-rip.xml.in: show ip rip	show ip rip
✓	<i>show ip rip status</i>	show-ip-rip.xml.in: show ip rip status	show ip rip status
×	not yet documented	show-ip-route.xml.in: show ip route <route>	show ip route <text>
×	<i>show ip route 0.0.0.0</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	show-ip-route.xml.in: show ip route bgp	show ip route bgp
×	not yet documented	show-ip-route.xml.in: show ip route cache <cache>	show ip route cache <text>
×	not yet documented	show-ip-route.xml.in: show ip route connected	show ip route connected
×	not yet documented	show-ip-route.xml.in: show ip route forward <forward>	show ip route forward <text>
×	not yet documented	show-ip-route.xml.in: show ip route isis	show ip route isis
×	not yet documented	show-ip-route.xml.in: show ip route kernel	show ip route kernel
×	not yet documented	show-ip-route.xml.in: show ip route <route> longer-prefixes	show ip route <text> longer-prefixes
×	not yet documented	show-ip-route.xml.in: show ip route ospf	show ip route ospf
×	not yet documented	show-ip-route.xml.in: show ip route rip	show ip route rip
×	not yet documented	show-ip-route.xml.in: show ip route static	show ip route static
×	not yet documented	show-ip-route.xml.in: show ip route summary	show ip route summary
×	not yet documented	show-ip-route.xml.in: show ip route summary table <table>	show ip route summary table <text>
×	not yet documented	show-ip-route.xml.in: show ip route supernets-only	show ip route supernets-only

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Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	show-ip-route.xml.in: show ip route table <table>	show ip route table <text>
×	not yet documented	show-ip-route.xml.in: show ip route tag <tag>	show ip route tag <text>
×	not yet documented	show-ip-route.xml.in: show ip route vrf <vrf>	show ip route vrf <text> <text>
×	<i>show ip route vrf <name></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	show-ip-route.xml.in: show ip route vrf <vrf> bgp	show ip route vrf <text> bgp
×	not yet documented	show-ip-route.xml.in: show ip route vrf <vrf> connected	show ip route vrf <text> connected
×	not yet documented	show-ip-route.xml.in: show ip route vrf <vrf> isis	show ip route vrf <text> isis
×	not yet documented	show-ip-route.xml.in: show ip route vrf <vrf> kernel	show ip route vrf <text> kernel
×	not yet documented	Nothing found in XML Definitions	show ip route vrf <text> <text> longer-prefixes
×	not yet documented	show-ip-route.xml.in: show ip route vrf <vrf> node.tag	Nothing found in VyOS
×	not yet documented	show-ip-route.xml.in: show ip route vrf <vrf> node.tag longer-prefixes	Nothing found in VyOS
×	not yet documented	show-ip-route.xml.in: show ip route vrf <vrf> ospf	show ip route vrf <text> ospf
×	not yet documented	show-ip-route.xml.in: show ip route vrf <vrf> rip	show ip route vrf <text> rip
×	not yet documented	show-ip-route.xml.in: show ip route vrf <vrf> static	show ip route vrf <text> static
×	not yet documented	show-ip-route.xml.in: show ip route vrf <vrf> summary	show ip route vrf <text> summary
×	not yet documented	show-ip-route.xml.in: show ip route vrf <vrf> supernets-only	show ip route vrf <text> supernets-only
×	not yet documented	show-ip-route.xml.in: show ip route vrf <vrf> tag <tag>	show ip route vrf <text> tag <text>
✓	<i>show ipv6 access-list</i>	show-ipv6.xml.in: show ipv6 access-list <access-list>	show ipv6 access-list <text>

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Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
✓	<i>show ipv6 forwarding</i>	show-ipv6.xml.in: show ipv6 forwarding	show ipv6 forwarding
×	<i>show ipv6 groups</i>	Nothing found in XML Definitions	show ipv6 groups
×	not yet documented	show-ipv6-mld.xml.in: show ipv6 mld groups	Nothing found in VyOS
×	not yet documented	show-ipv6-mld.xml.in: show ipv6 mld interface	Nothing found in VyOS
×	not yet documented	show-ipv6-mld.xml.in: show ipv6 mld joins	Nothing found in VyOS
×	not yet documented	show-ipv6-mld.xml.in: show ipv6 mld statistics	Nothing found in VyOS
×	not yet documented	multicast-group.xml.in: show ipv6 multicast group	Nothing found in VyOS
×	not yet documented	multicast-group.xml.in: show ipv6 multicast group interface <interface>	Nothing found in VyOS
×	<i>show ipv6 neighbors</i>	Nothing found in XML Definitions	show ipv6 neighbors
×	not yet documented	Nothing found in XML Definitions	show ipv6 neighbors interface <text>
×	not yet documented	Nothing found in XML Definitions	show ipv6 neighbors state <text>
×	not yet documented	show-ipv6.xml.in: show ipv6 nht	Nothing found in VyOS
×	not yet documented	show-ipv6.xml.in: show ipv6 nht vrf <vrf>	Nothing found in VyOS
✓	<i>show ipv6 ospfv3</i>	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3	show ipv6 ospfv3
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 area <area>	show ipv6 ospfv3 area <text>
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 area <area> router <router>	show ipv6 ospfv3 area <text> router <text>
✓	<i>show ipv6 ospfv3 border-routers</i>	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 border-routers <border-routers>	show ipv6 ospfv3 border-routers <text>
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 border-routers detail	show ipv6 ospfv3 border-routers detail
✓	<i>show ipv6 ospfv3 database</i>	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database	show ipv6 ospfv3 database <text> <text>
×	<i>show ipv6 ospfv3 database <type> [A.B.C.D] [adv-router <A.B.C.D> self-originate]</i>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	Nothing found in XML Definitions	show ipv6 ospfv3 database <text> adv-router <text> detail
×	not yet documented	Nothing found in XML Definitions	show ipv6 ospfv3 database <text> adv-router <text> dump
×	not yet documented	Nothing found in XML Definitions	show ipv6 ospfv3 database <text> adv-router <text> internal
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database adv-router <adv-router> linkstate-id <linkstate-id> detail	show ipv6 ospfv3 database <text> adv-router <text> linkstate-id <text> detail
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database adv-router <adv-router> linkstate-id <linkstate-id> dump	show ipv6 ospfv3 database <text> adv-router <text> linkstate-id <text> dump
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database adv-router <adv-router> linkstate-id <linkstate-id> internal	show ipv6 ospfv3 database <text> adv-router <text> linkstate-id <text> internal
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database any <any>	show ipv6 ospfv3 database any <text> <text>
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database any any <any> detail	show ipv6 ospfv3 database any any <text> detail
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database any any <any> dump	show ipv6 ospfv3 database any any <text> dump
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database any any <any> internal	show ipv6 ospfv3 database any any <text> internal
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database any <any> detail	show ipv6 ospfv3 database any <text> <text> detail
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database any <any> dump	show ipv6 ospfv3 database any <text> <text> dump
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database any <any> internal	show ipv6 ospfv3 database any <text> <text> internal
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database any <any> node.tag	Nothing found in VyOS

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Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database any <any> node.tag detail	Nothing found in VyOS
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database any <any> node.tag dump	Nothing found in VyOS
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database any <any> node.tag internal	Nothing found in VyOS
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database as-external	show ipv6 ospfv3 database as-external <text> <text>
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database as-external adv-router <adv-router> detail	show ipv6 ospfv3 database as-external adv-router <text> detail
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database as-external adv-router <adv-router> dump	show ipv6 ospfv3 database as-external adv-router <text> dump
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database as-external adv-router <adv-router> internal	show ipv6 ospfv3 database as-external adv-router <text> internal
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database as-external adv-router <adv-router> linkstate-id <linkstate-id> detail	show ipv6 ospfv3 database as-external adv-router <text> linkstate-id <text> detail
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database as-external adv-router <adv-router> linkstate-id <linkstate-id> dump	show ipv6 ospfv3 database as-external adv-router <text> linkstate-id <text> dump
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database as-external adv-router <adv-router> linkstate-id <linkstate-id> internal	show ipv6 ospfv3 database as-external adv-router <text> linkstate-id <text> internal
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database as-external any <any>	show ipv6 ospfv3 database as-external any <text>
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database as-external any <any> de- tail	show ipv6 ospfv3 database as-external any <text> de- tail

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Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database as-external any <any> dump	show ipv6 ospfv3 database as-external any <text> dump
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database as-external any <any> in- ternal	show ipv6 ospfv3 database as-external any <text> in- ternal
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database as-external <as-external> detail	show ipv6 ospfv3 database as-external <text> <text> detail
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database as-external <as-external> dump	show ipv6 ospfv3 database as-external <text> <text> dump
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database as-external <as-external> internal	show ipv6 ospfv3 database as-external <text> <text> internal
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database as-external linkstate-id <linkstate-id> detail	show ipv6 ospfv3 database as-external linkstate-id <text> detail
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database as-external linkstate-id <linkstate-id> dump	show ipv6 ospfv3 database as-external linkstate-id <text> dump
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database as-external linkstate-id <linkstate-id> internal	show ipv6 ospfv3 database as-external linkstate-id <text> internal
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database as-external <as-external> node.tag	Nothing found in VyOS
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database as-external <as-external> node.tag detail	Nothing found in VyOS
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database as-external <as-external> node.tag dump	Nothing found in VyOS
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database as-external <as-external> node.tag internal	Nothing found in VyOS

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Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database as-external <as-external> self-originated	show ipv6 ospfv3 database as-external self-originated
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database as-external <as-external> self-originated detail	show ipv6 ospfv3 database as-external self-originated detail
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database as-external <as-external> self-originated dump	show ipv6 ospfv3 database as-external self-originated dump
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database as-external <as-external> self-originated internal	show ipv6 ospfv3 database as-external self-originated internal
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database detail	show ipv6 ospfv3 database <text> <text> detail
×	not yet documented	Nothing found in XML Definitions	show ipv6 ospfv3 database <text> <text> dump
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database group-membership	show ipv6 ospfv3 database group-membership <text>
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database group-membership adv- router <adv-router> detail	show ipv6 ospfv3 database group-membership adv- router <text> detail
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database group-membership adv- router <adv-router> dump	show ipv6 ospfv3 database group-membership adv- router <text> dump
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database group-membership adv- router <adv-router> internal	show ipv6 ospfv3 database group-membership adv- router <text> internal
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database group-membership adv- router <adv-router> linkstate-id <linkstate-id> detail	show ipv6 ospfv3 database group-membership adv- router <text> linkstate-id <text> detail

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Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database group-membership adv- router <adv-router> linkstate-id <linkstate-id> dump	show ipv6 ospfv3 database group-membership adv- router <text> linkstate-id <text> dump
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database group-membership adv- router <adv-router> linkstate-id <linkstate-id> internal	show ipv6 ospfv3 database group-membership adv- router <text> linkstate-id <text> internal
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database group-membership detail	show ipv6 ospfv3 database group-membership <text> detail
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database group-membership dump	show ipv6 ospfv3 database group-membership <text> dump
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database group-membership internal	show ipv6 ospfv3 database group-membership <text> internal
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database group-membership linkstate-id <linkstate-id> detail	show ipv6 ospfv3 database group-membership linkstate-id <text> detail
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database group-membership linkstate-id <linkstate-id> dump	show ipv6 ospfv3 database group-membership linkstate-id <text> dump
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database group-membership linkstate-id <linkstate-id> internal	show ipv6 ospfv3 database group-membership linkstate-id <text> internal
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database group-membership node.tag	Nothing found in VyOS
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database group-membership node.tag detail	Nothing found in VyOS
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database group-membership node.tag dump	Nothing found in VyOS

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Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database group-membership node.tag internal	Nothing found in VyOS
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database group-membership node.tag self-originated	Nothing found in VyOS
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database group-membership node.tag self-originated detail	Nothing found in VyOS
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database group-membership node.tag self-originated dump	Nothing found in VyOS
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database group-membership node.tag self-originated internal	Nothing found in VyOS
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database group-membership self- originated	show ipv6 ospfv3 database group-membership self- originated
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database group-membership self- originated detail	show ipv6 ospfv3 database group-membership self- originated detail
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database group-membership self- originated dump	show ipv6 ospfv3 database group-membership self- originated dump
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database group-membership self- originated internal	show ipv6 ospfv3 database group-membership self- originated internal
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database inter-prefix	show ipv6 ospfv3 database inter-prefix <text>
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database inter-prefix adv-router <adv-router> detail	show ipv6 ospfv3 database inter-prefix adv-router <text> detail

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Table 2 – continued from previous page

Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database inter-prefix adv-router <adv-router> dump	show ipv6 ospfv3 database inter-prefix adv-router <text> dump
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database inter-prefix adv-router <adv-router> internal	show ipv6 ospfv3 database inter-prefix adv-router <text> internal
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database inter-prefix adv-router <adv-router> linkstate-id <linkstate-id> detail	show ipv6 ospfv3 database inter-prefix adv-router <text> linkstate-id <text> detail
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database inter-prefix adv-router <adv-router> linkstate-id <linkstate-id> dump	show ipv6 ospfv3 database inter-prefix adv-router <text> linkstate-id <text> dump
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database inter-prefix adv-router <adv-router> linkstate-id <linkstate-id> internal	show ipv6 ospfv3 database inter-prefix adv-router <text> linkstate-id <text> internal
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database inter-prefix detail	show ipv6 ospfv3 database inter-prefix <text> detail
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database inter-prefix dump	show ipv6 ospfv3 database inter-prefix <text> dump
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database inter-prefix internal	show ipv6 ospfv3 database inter-prefix <text> internal
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database inter-prefix linkstate-id <linkstate-id> detail	show ipv6 ospfv3 database inter-prefix linkstate-id <text> detail
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database inter-prefix linkstate-id <linkstate-id> dump	show ipv6 ospfv3 database inter-prefix linkstate-id <text> dump
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database inter-prefix linkstate-id <linkstate-id> internal	show ipv6 ospfv3 database inter-prefix linkstate-id <text> internal
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database inter-prefix node.tag	Nothing found in VyOS

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Table 2 – continued from previous page

Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database inter-prefix node.tag detail	Nothing found in VyOS
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database inter-prefix node.tag dump	Nothing found in VyOS
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database inter-prefix node.tag internal	Nothing found in VyOS
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database inter-prefix node.tag self-originated	Nothing found in VyOS
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database inter-prefix node.tag self-originated detail	Nothing found in VyOS
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database inter-prefix node.tag self-originated dump	Nothing found in VyOS
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database inter-prefix node.tag self-originated internal	Nothing found in VyOS
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database inter-prefix self-originated	show ipv6 ospfv3 database inter-prefix self-originated
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database inter-prefix self-originated detail	show ipv6 ospfv3 database inter-prefix self-originated detail
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database inter-prefix self-originated dump	show ipv6 ospfv3 database inter-prefix self-originated dump
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database inter-prefix self-originated internal	show ipv6 ospfv3 database inter-prefix self-originated internal
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database inter-router	show ipv6 ospfv3 database inter-router <text>
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database inter-router adv-router <adv-router> detail	show ipv6 ospfv3 database inter-router adv-router <text> detail

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Table 2 – continued from previous page

Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database inter-router adv-router <adv-router> dump	show ipv6 ospfv3 database inter-router adv-router <text> dump
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database inter-router adv-router <adv-router> internal	show ipv6 ospfv3 database inter-router adv-router <text> internal
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database inter-router adv-router <adv-router> linkstate-id <linkstate-id> detail	show ipv6 ospfv3 database inter-router adv-router <text> linkstate-id <text> detail
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database inter-router adv-router <adv-router> linkstate-id <linkstate-id> dump	show ipv6 ospfv3 database inter-router adv-router <text> linkstate-id <text> dump
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database inter-router adv-router <adv-router> linkstate-id <linkstate-id> internal	show ipv6 ospfv3 database inter-router adv-router <text> linkstate-id <text> internal
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database inter-router detail	show ipv6 ospfv3 database inter-router <text> detail
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database inter-router dump	show ipv6 ospfv3 database inter-router <text> dump
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database inter-router internal	show ipv6 ospfv3 database inter-router <text> internal
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database inter-router linkstate-id <linkstate-id> detail	show ipv6 ospfv3 database inter-router linkstate-id <text> detail
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database inter-router linkstate-id <linkstate-id> dump	show ipv6 ospfv3 database inter-router linkstate-id <text> dump
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database inter-router linkstate-id <linkstate-id> internal	show ipv6 ospfv3 database inter-router linkstate-id <text> internal
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database inter-router node.tag	Nothing found in VyOS

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Table 2 – continued from previous page

Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database inter-router node.tag detail	Nothing found in VyOS
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database inter-router node.tag dump	Nothing found in VyOS
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database inter-router node.tag internal	Nothing found in VyOS
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database inter-router node.tag self- originated	Nothing found in VyOS
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database inter-router node.tag self- originated detail	Nothing found in VyOS
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database inter-router node.tag self- originated dump	Nothing found in VyOS
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database inter-router node.tag self- originated internal	Nothing found in VyOS
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database inter-router self-originated	show ipv6 ospfv3 database inter-router self-originated
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database inter-router self-originated detail	show ipv6 ospfv3 database inter-router self-originated detail
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database inter-router self-originated dump	show ipv6 ospfv3 database inter-router self-originated dump
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database inter-router self-originated internal	show ipv6 ospfv3 database inter-router self-originated internal
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database internal	show ipv6 ospfv3 database <text> <text> internal
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database intra-prefix	show ipv6 ospfv3 database intra-prefix <text>

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Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database intra-prefix adv-router <adv-router> detail	show ipv6 ospfv3 database intra-prefix adv-router <text> detail
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database intra-prefix adv-router <adv-router> dump	show ipv6 ospfv3 database intra-prefix adv-router <text> dump
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database intra-prefix adv-router <adv-router> internal	show ipv6 ospfv3 database intra-prefix adv-router <text> internal
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database intra-prefix adv-router <adv-router> linkstate-id <linkstate-id> detail	show ipv6 ospfv3 database intra-prefix adv-router <text> linkstate-id <text> detail
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database intra-prefix adv-router <adv-router> linkstate-id <linkstate-id> dump	show ipv6 ospfv3 database intra-prefix adv-router <text> linkstate-id <text> dump
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database intra-prefix adv-router <adv-router> linkstate-id <linkstate-id> internal	show ipv6 ospfv3 database intra-prefix adv-router <text> linkstate-id <text> internal
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database intra-prefix detail	show ipv6 ospfv3 database intra-prefix <text> detail
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database intra-prefix dump	show ipv6 ospfv3 database intra-prefix <text> dump
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database intra-prefix internal	show ipv6 ospfv3 database intra-prefix <text> internal
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database intra-prefix linkstate-id <linkstate-id> detail	show ipv6 ospfv3 database intra-prefix linkstate-id <text> detail
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database intra-prefix linkstate-id <linkstate-id> dump	show ipv6 ospfv3 database intra-prefix linkstate-id <text> dump
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database intra-prefix linkstate-id <linkstate-id> internal	show ipv6 ospfv3 database intra-prefix linkstate-id <text> internal

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Table 2 – continued from previous page

Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database intra-prefix node.tag	Nothing found in VyOS
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database intra-prefix node.tag detail	Nothing found in VyOS
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database intra-prefix node.tag dump	Nothing found in VyOS
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database intra-prefix node.tag internal	Nothing found in VyOS
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database intra-prefix node.tag self- originated	Nothing found in VyOS
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database intra-prefix node.tag self- originated detail	Nothing found in VyOS
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database intra-prefix node.tag self- originated dump	Nothing found in VyOS
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database intra-prefix node.tag self- originated internal	Nothing found in VyOS
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database intra-prefix self-originated	show ipv6 ospfv3 database intra-prefix self-originated
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database intra-prefix self-originated detail	show ipv6 ospfv3 database intra-prefix self-originated detail
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database intra-prefix self-originated dump	show ipv6 ospfv3 database intra-prefix self-originated dump
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database intra-prefix self-originated internal	show ipv6 ospfv3 database intra-prefix self-originated internal
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database link	show ipv6 ospfv3 database link <text>

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Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database link adv-router <adv- router> detail	show ipv6 ospfv3 database link adv-router <text> de- tail
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database link adv-router <adv- router> dump	show ipv6 ospfv3 database link adv-router <text> dump
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database link adv-router <adv- router> internal	show ipv6 ospfv3 database link adv-router <text> in- ternal
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database link adv-router <adv- router> linkstate-id <linkstate-id> detail	show ipv6 ospfv3 database link adv-router <text> linkstate-id <text> detail
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database link adv-router <adv- router> linkstate-id <linkstate-id> dump	show ipv6 ospfv3 database link adv-router <text> linkstate-id <text> dump
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database link adv-router <adv- router> linkstate-id <linkstate-id> internal	show ipv6 ospfv3 database link adv-router <text> linkstate-id <text> inter- nal
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database link detail	show ipv6 ospfv3 database link <text> detail
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database link dump	show ipv6 ospfv3 database link <text> dump
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database link internal	show ipv6 ospfv3 database link <text> internal
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database link linkstate-id <linkstate-id> detail	show ipv6 ospfv3 database link linkstate-id <text> de- tail
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database link linkstate-id <linkstate-id> dump	show ipv6 ospfv3 database link linkstate-id <text> dump
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database link linkstate-id <linkstate-id> internal	show ipv6 ospfv3 database link linkstate-id <text> in- ternal

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Table 2 – continued from previous page

Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database link node.tag	Nothing found in VyOS
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database link node.tag detail	Nothing found in VyOS
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database link node.tag dump	Nothing found in VyOS
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database link node.tag internal	Nothing found in VyOS
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database link node.tag self-originated	Nothing found in VyOS
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database link node.tag self-originated detail	Nothing found in VyOS
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database link node.tag self-originated dump	Nothing found in VyOS
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database link node.tag self-originated internal	Nothing found in VyOS
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database link self-originated	show ipv6 ospfv3 database link self-originated
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database link self-originated detail	show ipv6 ospfv3 database link self-originated detail
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database link self-originated dump	show ipv6 ospfv3 database link self-originated dump
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database link self-originated internal	show ipv6 ospfv3 database link self-originated internal
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database linkstate-id <linkstate-id> detail	show ipv6 ospfv3 database <text> linkstate-id <text> detail
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database linkstate-id <linkstate-id> dump	show ipv6 ospfv3 database <text> linkstate-id <text> dump

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Table 2 – continued from previous page

Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database linkstate-id <linkstate-id> internal	show ipv6 ospfv3 database <text> linkstate-id <text> internal
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database network	show ipv6 ospfv3 database network <text>
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database network adv-router <adv- router> detail	show ipv6 ospfv3 database network adv-router <text> detail
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database network adv-router <adv- router> dump	show ipv6 ospfv3 database network adv-router <text> dump
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database network adv-router <adv- router> internal	show ipv6 ospfv3 database network adv-router <text> internal
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database network adv-router <adv- router> linkstate-id <linkstate-id> detail	show ipv6 ospfv3 database network adv-router <text> linkstate-id <text> detail
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database network adv-router <adv- router> linkstate-id <linkstate-id> dump	show ipv6 ospfv3 database network adv-router <text> linkstate-id <text> dump
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database network adv-router <adv- router> linkstate-id <linkstate-id> internal	show ipv6 ospfv3 database network adv-router <text> linkstate-id <text> internal
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database network detail	show ipv6 ospfv3 database network <text> detail
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database network dump	show ipv6 ospfv3 database network <text> dump
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database network internal	show ipv6 ospfv3 database network <text> internal
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database network linkstate-id <linkstate-id> detail	show ipv6 ospfv3 database network linkstate-id <text> detail

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Table 2 – continued from previous page

Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database network linkstate-id <linkstate-id> dump	show ipv6 ospfv3 database network linkstate-id <text> dump
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database network linkstate-id <linkstate-id> internal	show ipv6 ospfv3 database network linkstate-id <text> internal
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database network node.tag	Nothing found in VyOS
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database network node.tag detail	Nothing found in VyOS
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database network node.tag dump	Nothing found in VyOS
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database network node.tag internal	Nothing found in VyOS
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database network node.tag self- originated	Nothing found in VyOS
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database network node.tag self- originated detail	Nothing found in VyOS
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database network node.tag self- originated dump	Nothing found in VyOS
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database network node.tag self- originated internal	Nothing found in VyOS
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database network self-originated	show ipv6 ospfv3 database network self-originated
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database network self-originated detail	show ipv6 ospfv3 database network self-originated detail
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database network self-originated dump	show ipv6 ospfv3 database network self-originated dump

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Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database network self-originated in- ternal	show ipv6 ospfv3 database network self-originated in- ternal
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database node.tag	Nothing found in VyOS
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database node.tag adv-router <adv- router> detail	Nothing found in VyOS
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database node.tag adv-router <adv- router> dump	Nothing found in VyOS
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database node.tag adv-router <adv- router> internal	Nothing found in VyOS
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database node.tag adv-router <adv- router> linkstate-id <linkstate-id> detail	Nothing found in VyOS
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database node.tag adv-router <adv- router> linkstate-id <linkstate-id> dump	Nothing found in VyOS
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database node.tag adv-router <adv- router> linkstate-id <linkstate-id> internal	Nothing found in VyOS
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database node.tag detail	Nothing found in VyOS
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database node.tag dump	Nothing found in VyOS
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database node.tag internal	Nothing found in VyOS
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database node.tag linkstate-id <linkstate-id> detail	Nothing found in VyOS

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Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database node.tag linkstate-id <linkstate-id> dump	Nothing found in VyOS
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database node.tag linkstate-id <linkstate-id> internal	Nothing found in VyOS
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database node.tag node.tag	Nothing found in VyOS
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database node.tag node.tag detail	Nothing found in VyOS
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database node.tag node.tag dump	Nothing found in VyOS
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database node.tag node.tag internal	Nothing found in VyOS
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database node.tag node.tag self- originated	Nothing found in VyOS
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database node.tag node.tag self- originated detail	Nothing found in VyOS
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database node.tag node.tag self- originated dump	Nothing found in VyOS
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database node.tag node.tag self- originated internal	Nothing found in VyOS
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database node.tag self-originated	Nothing found in VyOS
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database node.tag self-originated detail	Nothing found in VyOS
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database node.tag self-originated dump	Nothing found in VyOS

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Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database node.tag self-originated internal	Nothing found in VyOS
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database router	show ipv6 ospfv3 database router <text>
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database router adv-router <adv- router> detail	show ipv6 ospfv3 database router adv-router <text> detail
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database router adv-router <adv- router> dump	show ipv6 ospfv3 database router adv-router <text> dump
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database router adv-router <adv- router> internal	show ipv6 ospfv3 database router adv-router <text> internal
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database router adv-router <adv- router> linkstate-id <linkstate-id> detail	show ipv6 ospfv3 database router adv-router <text> linkstate-id <text> detail
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database router adv-router <adv- router> linkstate-id <linkstate-id> dump	show ipv6 ospfv3 database router adv-router <text> linkstate-id <text> dump
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database router adv-router <adv- router> linkstate-id <linkstate-id> internal	show ipv6 ospfv3 database router adv-router <text> linkstate-id <text> internal
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database router detail	show ipv6 ospfv3 database router <text> detail
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database router dump	show ipv6 ospfv3 database router <text> dump
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database router internal	show ipv6 ospfv3 database router <text> internal
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database router linkstate- id <linkstate-id> detail	show ipv6 ospfv3 database router linkstate-id <text> detail

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Table 2 – continued from previous page

Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database router linkstate-id <linkstate-id> dump	show ipv6 ospfv3 database router linkstate-id <text> dump
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database router linkstate-id <linkstate-id> internal	show ipv6 ospfv3 database router linkstate-id <text> internal
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database router node.tag	Nothing found in VyOS
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database router node.tag detail	Nothing found in VyOS
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database router node.tag dump	Nothing found in VyOS
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database router node.tag internal	Nothing found in VyOS
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database router node.tag self-originated	Nothing found in VyOS
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database router node.tag self-originated detail	Nothing found in VyOS
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database router node.tag self-originated dump	Nothing found in VyOS
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database router node.tag self-originated internal	Nothing found in VyOS
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database router self-originated	show ipv6 ospfv3 database router self-originated
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database router self-originated detail	show ipv6 ospfv3 database router self-originated detail
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database router self-originated dump	show ipv6 ospfv3 database router self-originated dump

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Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database router self-originated in- ternal	show ipv6 ospfv3 database router self-originated in- ternal
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database self-originated	show ipv6 ospfv3 database self-originated
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database self-originated detail	show ipv6 ospfv3 database self-originated detail
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database self-originated dump	show ipv6 ospfv3 database self-originated dump
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database self-originated internal	show ipv6 ospfv3 database self-originated internal
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database type-7	show ipv6 ospfv3 database type-7 <text>
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database type-7 adv-router <adv- router> detail	show ipv6 ospfv3 database type-7 adv-router <text> detail
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database type-7 adv-router <adv- router> dump	show ipv6 ospfv3 database type-7 adv-router <text> dump
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database type-7 adv-router <adv- router> internal	show ipv6 ospfv3 database type-7 adv-router <text> internal
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database type-7 adv-router <adv- router> linkstate-id <linkstate-id> detail	show ipv6 ospfv3 database type-7 adv-router <text> linkstate-id <text> detail
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database type-7 adv-router <adv- router> linkstate-id <linkstate-id> dump	show ipv6 ospfv3 database type-7 adv-router <text> linkstate-id <text> dump
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database type-7 adv-router <adv- router> linkstate-id <linkstate-id> internal	show ipv6 ospfv3 database type-7 adv-router <text> linkstate-id <text> internal
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database type-7 detail	show ipv6 ospfv3 database type-7 <text> detail

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Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database type-7 dump	show ipv6 ospfv3 database type-7 <text> dump
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database type-7 internal	show ipv6 ospfv3 database type-7 <text> internal
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database type-7 linkstate-id <linkstate-id> detail	show ipv6 ospfv3 database type-7 linkstate-id <text> detail
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database type-7 linkstate-id <linkstate-id> dump	show ipv6 ospfv3 database type-7 linkstate-id <text> dump
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database type-7 linkstate-id <linkstate-id> internal	show ipv6 ospfv3 database type-7 linkstate-id <text> internal
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database type-7 node.tag	Nothing found in VyOS
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database type-7 node.tag detail	Nothing found in VyOS
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database type-7 node.tag dump	Nothing found in VyOS
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database type-7 node.tag internal	Nothing found in VyOS
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database type-7 node.tag self- originated	Nothing found in VyOS
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database type-7 node.tag self- originated detail	Nothing found in VyOS
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database type-7 node.tag self- originated dump	Nothing found in VyOS
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database type-7 node.tag self- originated internal	Nothing found in VyOS
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database type-7 self-originated	show ipv6 ospfv3 database type-7 self-originated

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Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database type-7 self-originated de- tail	show ipv6 ospfv3 database type-7 self-originated de- tail
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database type-7 self-originated dump	show ipv6 ospfv3 database type-7 self-originated dump
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 database type-7 self-originated in- ternal	show ipv6 ospfv3 database type-7 self-originated in- ternal
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 graceful-restart helper	show ipv6 ospfv3 graceful-restart helper
✓	<i>show ipv6 ospfv3 interface [prefix] [<interface> [prefix]]</i>	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 inter- face <interface>	show ipv6 ospfv3 inter- face <text>
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 inter- face <interface> prefix <prefix>	show ipv6 ospfv3 inter- face prefix <text>
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 inter- face <interface> prefix <prefix> detail	show ipv6 ospfv3 inter- face prefix <text> detail
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 inter- face <interface> prefix <prefix> match	show ipv6 ospfv3 inter- face prefix <text> match
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 linkstate detail	show ipv6 ospfv3 linkstate detail
×	not yet documented	Nothing found in XML Definitions	show ipv6 ospfv3 linkstate network <text> <text>
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 linkstate network <network> node.tag	Nothing found in VyOS
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 linkstate router <router>	show ipv6 ospfv3 linkstate router <text>
✓	<i>show ipv6 ospfv3 neighbor</i>	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 neigh- bor	show ipv6 ospfv3 neigh- bor
✓	<i>show ipv6 ospfv3 neighbor detail</i>	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 neigh- bor detail	show ipv6 ospfv3 neigh- bor detail

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Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
✓	<i>show ipv6 ospfv3 neighbor drchoice</i>	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 neighbor drchoice	show ipv6 ospfv3 neighbor drchoice
✓	<i>show ipv6 ospfv3 redistribute</i>	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 redistribute	show ipv6 ospfv3 redistribute
✓	<i>show ipv6 ospfv3 route</i>	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 route <route>	show ipv6 ospfv3 route <text>
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 route detail	show ipv6 ospfv3 route detail
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 route external-1	show ipv6 ospfv3 route external-1
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 route external-1 detail	show ipv6 ospfv3 route external-1 detail
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 route external-2	show ipv6 ospfv3 route external-2
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 route external-2 detail	show ipv6 ospfv3 route external-2 detail
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 route inter-area	show ipv6 ospfv3 route inter-area
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 route inter-area detail	show ipv6 ospfv3 route inter-area detail
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 route intra-area	show ipv6 ospfv3 route intra-area
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 route intra-area detail	show ipv6 ospfv3 route intra-area detail
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 route <text> longer <route> longer	show ipv6 ospfv3 route <text> longer
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 route <text> match <route> match	show ipv6 ospfv3 route <text> match
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 route <text> match detail <route> match detail	show ipv6 ospfv3 route <text> match detail
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 route summary	show ipv6 ospfv3 route summary

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Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf>	show ipv6 ospfv3 vrf <text>
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> area <area>	show ipv6 ospfv3 vrf <text> area <text>
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> area <area> router <router>	show ipv6 ospfv3 vrf <text> area <text> router <text>
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> border-routers <border-routers>	show ipv6 ospfv3 vrf <text> border-routers <text>
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> border-routers detail	show ipv6 ospfv3 vrf <text> border-routers detail
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database	show ipv6 ospfv3 vrf <text> database <text> <text>
×	not yet documented	Nothing found in XML Definitions	show ipv6 ospfv3 vrf <text> database <text> adv-router <text> detail
×	not yet documented	Nothing found in XML Definitions	show ipv6 ospfv3 vrf <text> database <text> adv-router <text> dump
×	not yet documented	Nothing found in XML Definitions	show ipv6 ospfv3 vrf <text> database <text> adv-router <text> internal
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database adv-router <adv-router> linkstate-id <linkstate-id> detail	show ipv6 ospfv3 vrf <text> database <text> adv-router <text> linkstate-id <text> detail
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database adv-router <adv-router> linkstate-id <linkstate-id> dump	show ipv6 ospfv3 vrf <text> database <text> adv-router <text> linkstate-id <text> dump
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database adv-router <adv-router> linkstate-id <linkstate-id> internal	show ipv6 ospfv3 vrf <text> database <text> adv-router <text> linkstate-id <text> internal
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database any <any>	show ipv6 ospfv3 vrf <text> database any <text> <text>

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Table 2 – continued from previous page

Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database any any <any> detail	show ipv6 ospfv3 vrf <text> database any any <text> detail
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database any any <any> dump	show ipv6 ospfv3 vrf <text> database any any <text> dump
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database any any <any> internal	show ipv6 ospfv3 vrf <text> database any any <text> internal
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database any <any> detail	show ipv6 ospfv3 vrf <text> database any <text> <text> detail
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database any <any> dump	show ipv6 ospfv3 vrf <text> database any <text> <text> dump
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database any <any> internal	show ipv6 ospfv3 vrf <text> database any <text> <text> internal
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database any <any> node.tag	Nothing found in VyOS
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database any <any> node.tag detail	Nothing found in VyOS
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database any <any> node.tag dump	Nothing found in VyOS
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database any <any> node.tag internal	Nothing found in VyOS
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database as-external <adv-router> <adv-router> detail	show ipv6 ospfv3 vrf <text> database as- external <text> <text> adv-router <text> detail

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Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database as-external adv-router <adv-router> dump	show ipv6 ospfv3 vrf <text> database as- external adv-router <text> dump
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database as-external adv-router <adv-router> internal	show ipv6 ospfv3 vrf <text> database as- external adv-router <text> internal
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database as-external adv-router <adv-router> linkstate-id <linkstate-id> detail	show ipv6 ospfv3 vrf <text> database as- external adv-router <text> linkstate-id <text> detail
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database as-external adv-router <adv-router> linkstate-id <linkstate-id> dump	show ipv6 ospfv3 vrf <text> database as- external adv-router <text> linkstate-id <text> dump
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database as-external adv-router <adv-router> linkstate-id <linkstate-id> internal	show ipv6 ospfv3 vrf <text> database as- external adv-router <text> linkstate-id <text> internal
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database as-external any <any>	show ipv6 ospfv3 vrf <text> database as- external any <text>
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database as-external any <any> detail	show ipv6 ospfv3 vrf <text> database as- external any <text> detail
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database as-external any <any> dump	show ipv6 ospfv3 vrf <text> database as- external any <text> dump
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database as-external any <any> internal	show ipv6 ospfv3 vrf <text> database as- external any <text> internal
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database as-external <as-external> detail	show ipv6 ospfv3 vrf <text> database as- external <text> <text> detail

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Table 2 – continued from previous page

Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database as-external <as-external> dump	show ipv6 ospfv3 vrf <text> database as-external <text> <text> dump
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database as-external <as-external> internal	show ipv6 ospfv3 vrf <text> database as-external <text> <text> internal
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database as-external linkstate-id <linkstate-id> detail	show ipv6 ospfv3 vrf <text> database as-external linkstate-id <text> detail
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database as-external linkstate-id <linkstate-id> dump	show ipv6 ospfv3 vrf <text> database as-external linkstate-id <text> dump
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database as-external linkstate-id <linkstate-id> internal	show ipv6 ospfv3 vrf <text> database as-external linkstate-id <text> internal
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database as-external <as-external> node.tag	Nothing found in VyOS
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database as-external <as-external> node.tag detail	Nothing found in VyOS
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database as-external <as-external> node.tag dump	Nothing found in VyOS
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database as-external <as-external> node.tag internal	Nothing found in VyOS
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database as-external <as-external> self-originated	show ipv6 ospfv3 vrf <text> database as-external self-originated

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Table 2 – continued from previous page

Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database as-external <as-external> self-originated detail	show ipv6 ospfv3 vrf <text> database as-external self-originated detail
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database as-external <as-external> self-originated dump	show ipv6 ospfv3 vrf <text> database as-external self-originated dump
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database as-external <as-external> self-originated internal	show ipv6 ospfv3 vrf <text> database as-external self-originated internal
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database detail	show ipv6 ospfv3 vrf <text> database <text> <text> detail
×	not yet documented	Nothing found in XML Definitions	show ipv6 ospfv3 vrf <text> database <text> <text> dump
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database group-membership	show ipv6 ospfv3 vrf <text> database group-membership <text>
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database group-membership adv-router <adv-router> detail	show ipv6 ospfv3 vrf <text> database group-membership adv-router <text> detail
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database group-membership adv-router <adv-router> dump	show ipv6 ospfv3 vrf <text> database group-membership adv-router <text> dump
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database group-membership adv-router <adv-router> internal	show ipv6 ospfv3 vrf <text> database group-membership adv-router <text> internal
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database group-membership adv-router <adv-router> linkstate-id <linkstate-id> detail	show ipv6 ospfv3 vrf <text> database group-membership adv-router <text> linkstate-id <text> detail

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Table 2 – continued from previous page

Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database group- membership adv-router <adv-router> linkstate-id <linkstate-id> dump	show ipv6 ospfv3 vrf <text> database group- membership adv-router <text> linkstate-id <text> dump
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database group- membership adv-router <adv-router> linkstate-id <linkstate-id> internal	show ipv6 ospfv3 vrf <text> database group- membership adv-router <text> linkstate-id <text> internal
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database group- membership detail	show ipv6 ospfv3 vrf <text> database group- membership <text> detail
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database group- membership dump	show ipv6 ospfv3 vrf <text> database group- membership <text> dump
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database group- membership internal	show ipv6 ospfv3 vrf <text> database group- membership <text> internal
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database group- membership linkstate-id <linkstate-id> detail	show ipv6 ospfv3 vrf <text> database group- membership linkstate-id <text> detail
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database group- membership linkstate-id <linkstate-id> dump	show ipv6 ospfv3 vrf <text> database group- membership linkstate-id <text> dump
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database group- membership linkstate-id <linkstate-id> internal	show ipv6 ospfv3 vrf <text> database group- membership linkstate-id <text> internal
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database group- membership node.tag	Nothing found in VyOS
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database group- membership node.tag detail	Nothing found in VyOS

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Table 2 – continued from previous page

Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database group- membership node.tag dump	Nothing found in VyOS
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database group- membership node.tag internal	Nothing found in VyOS
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database group- membership node.tag self-originated	Nothing found in VyOS
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database group- membership node.tag self-originated detail	Nothing found in VyOS
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database group- membership node.tag self-originated dump	Nothing found in VyOS
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database group- membership node.tag self-originated internal	Nothing found in VyOS
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database group-membership self- originated	show ipv6 ospfv3 vrf <text> database group-membership self- originated
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database group-membership self- originated detail	show ipv6 ospfv3 vrf <text> database group-membership self- originated detail
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database group-membership self- originated dump	show ipv6 ospfv3 vrf <text> database group-membership self- originated dump

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Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database group-membership self- originated internal	show ipv6 ospfv3 vrf <text> database group-membership self- originated internal
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database inter- prefix	show ipv6 ospfv3 vrf <text> database inter- prefix <text>
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database inter- prefix adv-router <adv- router> detail	show ipv6 ospfv3 vrf <text> database inter- prefix adv-router <text> detail
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database inter- prefix adv-router <adv- router> dump	show ipv6 ospfv3 vrf <text> database inter- prefix adv-router <text> dump
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database inter- prefix adv-router <adv- router> internal	show ipv6 ospfv3 vrf <text> database inter- prefix adv-router <text> internal
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database inter- prefix adv-router <adv- router> linkstate-id <linkstate-id> detail	show ipv6 ospfv3 vrf <text> database inter- prefix adv-router <text> linkstate-id <text> detail
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database inter- prefix adv-router <adv- router> linkstate-id <linkstate-id> dump	show ipv6 ospfv3 vrf <text> database inter- prefix adv-router <text> linkstate-id <text> dump
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database inter- prefix adv-router <adv- router> linkstate-id <linkstate-id> internal	show ipv6 ospfv3 vrf <text> database inter- prefix adv-router <text> linkstate-id <text> internal
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database inter- prefix <text> detail	show ipv6 ospfv3 vrf <text> database inter- prefix <text> detail

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Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database inter- prefix dump	show ipv6 ospfv3 vrf <text> database inter- prefix <text> dump
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database inter- prefix internal	show ipv6 ospfv3 vrf <text> database inter- prefix <text> internal
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database inter-prefix linkstate-id <linkstate-id> detail	show ipv6 ospfv3 vrf <text> database inter- prefix linkstate-id <text> detail
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database inter-prefix linkstate-id <linkstate-id> dump	show ipv6 ospfv3 vrf <text> database inter- prefix linkstate-id <text> dump
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database inter-prefix linkstate-id <linkstate-id> internal	show ipv6 ospfv3 vrf <text> database inter- prefix linkstate-id <text> internal
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database inter- prefix node.tag	Nothing found in VyOS
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database inter- prefix node.tag detail	Nothing found in VyOS
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database inter- prefix node.tag dump	Nothing found in VyOS
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database inter- prefix node.tag internal	Nothing found in VyOS
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database inter- prefix node.tag self- originated	Nothing found in VyOS
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database inter- prefix node.tag self- originated detail	Nothing found in VyOS

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Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database inter- prefix node.tag self- originated dump	Nothing found in VyOS
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database inter- prefix node.tag self- originated internal	Nothing found in VyOS
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database inter- prefix self-originated	show ipv6 ospfv3 vrf <text> database inter- prefix self-originated
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database inter- prefix self-originated detail	show ipv6 ospfv3 vrf <text> database inter- prefix self-originated detail
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database inter- prefix self-originated dump	show ipv6 ospfv3 vrf <text> database inter- prefix self-originated dump
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database inter- prefix self-originated internal	show ipv6 ospfv3 vrf <text> database inter- prefix self-originated internal
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database inter- router	show ipv6 ospfv3 vrf <text> database inter- router <text>
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database inter- router adv-router <adv- router> detail	show ipv6 ospfv3 vrf <text> database inter- router adv-router <text> detail
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database inter- router adv-router <adv- router> dump	show ipv6 ospfv3 vrf <text> database inter- router adv-router <text> dump
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database inter- router adv-router <adv- router> internal	show ipv6 ospfv3 vrf <text> database inter- router adv-router <text> internal

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Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database inter- router adv-router <adv- router> linkstate-id <linkstate-id> detail	show ipv6 ospfv3 vrf <text> database inter- router adv-router <text> linkstate-id <text> detail
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database inter- router adv-router <adv- router> linkstate-id <linkstate-id> dump	show ipv6 ospfv3 vrf <text> database inter- router adv-router <text> linkstate-id <text> dump
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database inter- router adv-router <adv- router> linkstate-id <linkstate-id> internal	show ipv6 ospfv3 vrf <text> database inter- router adv-router <text> linkstate-id <text> internal
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database inter- router detail	show ipv6 ospfv3 vrf <text> database inter- router <text> detail
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database inter- router dump	show ipv6 ospfv3 vrf <text> database inter- router <text> dump
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database inter- router internal	show ipv6 ospfv3 vrf <text> database inter- router <text> internal
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database inter- router linkstate-id <linkstate-id> detail	show ipv6 ospfv3 vrf <text> database inter- router linkstate-id <text> detail
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database inter- router linkstate-id <linkstate-id> dump	show ipv6 ospfv3 vrf <text> database inter- router linkstate-id <text> dump
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database inter- router linkstate-id <linkstate-id> internal	show ipv6 ospfv3 vrf <text> database inter- router linkstate-id <text> internal
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database inter- router node.tag	Nothing found in VyOS

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Table 2 – continued from previous page

Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database inter- router node.tag detail	Nothing found in VyOS
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database inter- router node.tag dump	Nothing found in VyOS
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database inter- router node.tag internal	Nothing found in VyOS
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database inter- router node.tag self- originated	Nothing found in VyOS
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database inter- router node.tag self- originated detail	Nothing found in VyOS
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database inter- router node.tag self- originated dump	Nothing found in VyOS
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database inter- router node.tag self- originated internal	Nothing found in VyOS
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database inter- router self-originated	show ipv6 ospfv3 vrf <text> database inter- router self-originated
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database inter- router self-originated detail	show ipv6 ospfv3 vrf <text> database inter- router self-originated detail
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database inter- router self-originated dump	show ipv6 ospfv3 vrf <text> database inter- router self-originated dump

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Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database inter- router self-originated internal	show ipv6 ospfv3 vrf <text> database inter- router self-originated internal
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database internal	show ipv6 ospfv3 vrf <text> database <text> <text> internal
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database intra- prefix	show ipv6 ospfv3 vrf <text> database intra- prefix <text>
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database intra- prefix adv-router <adv- router> detail	show ipv6 ospfv3 vrf <text> database intra- prefix adv-router <text> detail
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database intra- prefix adv-router <adv- router> dump	show ipv6 ospfv3 vrf <text> database intra- prefix adv-router <text> dump
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database intra- prefix adv-router <adv- router> internal	show ipv6 ospfv3 vrf <text> database intra- prefix adv-router <text> internal
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database intra- prefix adv-router <adv- router> linkstate-id <linkstate-id> detail	show ipv6 ospfv3 vrf <text> database intra- prefix adv-router <text> linkstate-id <text> detail
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database intra- prefix adv-router <adv- router> linkstate-id <linkstate-id> dump	show ipv6 ospfv3 vrf <text> database intra- prefix adv-router <text> linkstate-id <text> dump
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database intra- prefix adv-router <adv- router> linkstate-id <linkstate-id> internal	show ipv6 ospfv3 vrf <text> database intra- prefix adv-router <text> linkstate-id <text> internal
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database intra- prefix	show ipv6 ospfv3 vrf <text> database intra- prefix <text> detail

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Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database intra- prefix dump	show ipv6 ospfv3 vrf <text> database intra- prefix <text> dump
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database intra- prefix internal	show ipv6 ospfv3 vrf <text> database intra- prefix <text> internal
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database intra-prefix linkstate-id <linkstate-id> detail	show ipv6 ospfv3 vrf <text> database intra- prefix linkstate-id <text> detail
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database intra-prefix linkstate-id <linkstate-id> dump	show ipv6 ospfv3 vrf <text> database intra- prefix linkstate-id <text> dump
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database intra-prefix linkstate-id <linkstate-id> internal	show ipv6 ospfv3 vrf <text> database intra- prefix linkstate-id <text> internal
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database intra- prefix node.tag	Nothing found in VyOS
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database intra- prefix node.tag detail	Nothing found in VyOS
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database intra- prefix node.tag dump	Nothing found in VyOS
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database intra- prefix node.tag internal	Nothing found in VyOS
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database intra- prefix node.tag self- originated	Nothing found in VyOS
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database intra- prefix node.tag self- originated detail	Nothing found in VyOS

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Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database intra- prefix node.tag self- originated dump	Nothing found in VyOS
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database intra- prefix node.tag self- originated internal	Nothing found in VyOS
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database intra- prefix self-originated	show ipv6 ospfv3 vrf <text> database intra- prefix self-originated
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database intra- prefix self-originated detail	show ipv6 ospfv3 vrf <text> database intra- prefix self-originated detail
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database intra- prefix self-originated dump	show ipv6 ospfv3 vrf <text> database intra- prefix self-originated dump
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database intra- prefix self-originated internal	show ipv6 ospfv3 vrf <text> database intra- prefix self-originated internal
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database link	show ipv6 ospfv3 vrf <text> database link <text>
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database link adv-router <adv-router> detail	show ipv6 ospfv3 vrf <text> database link adv-router <text> detail
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database link adv-router <adv-router> dump	show ipv6 ospfv3 vrf <text> database link adv-router <text> dump
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database link adv-router <adv-router> internal	show ipv6 ospfv3 vrf <text> database link adv-router <text> internal

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Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database link adv-router <adv-router> linkstate-id <linkstate-id> detail	show ipv6 ospfv3 vrf <text> database link adv- router <text> linkstate-id <text> detail
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database link adv-router <adv-router> linkstate-id <linkstate-id> dump	show ipv6 ospfv3 vrf <text> database link adv- router <text> linkstate-id <text> dump
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database link adv-router <adv-router> linkstate-id <linkstate-id> internal	show ipv6 ospfv3 vrf <text> database link adv- router <text> linkstate-id <text> internal
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database link detail	show ipv6 ospfv3 vrf <text> database link <text> detail
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database link dump	show ipv6 ospfv3 vrf <text> database link <text> dump
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database link internal	show ipv6 ospfv3 vrf <text> database link <text> internal
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database link linkstate-id <linkstate-id> detail	show ipv6 ospfv3 vrf <text> database link linkstate-id <text> detail
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database link linkstate-id <linkstate-id> dump	show ipv6 ospfv3 vrf <text> database link linkstate-id <text> dump
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database link linkstate-id <linkstate-id> internal	show ipv6 ospfv3 vrf <text> database link linkstate-id <text> internal
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database link node.tag	Nothing found in VyOS

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Table 2 – continued from previous page

Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database link node.tag detail	Nothing found in VyOS
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database link node.tag dump	Nothing found in VyOS
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database link node.tag internal	Nothing found in VyOS
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database link node.tag self-originated	Nothing found in VyOS
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database link node.tag self-originated detail	Nothing found in VyOS
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database link node.tag self-originated dump	Nothing found in VyOS
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database link node.tag self-originated internal	Nothing found in VyOS
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database link self-originated	show ipv6 ospfv3 vrf <text> database link self-originated
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database link self-originated detail	show ipv6 ospfv3 vrf <text> database link self-originated detail
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database link self-originated dump	show ipv6 ospfv3 vrf <text> database link self-originated dump
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database link self-originated internal	show ipv6 ospfv3 vrf <text> database link self-originated internal

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Table 2 – continued from previous page

Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database linkstate- id <linkstate-id> detail	show ipv6 ospfv3 vrf <text> database <text> linkstate-id <text> detail
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database linkstate- id <linkstate-id> dump	show ipv6 ospfv3 vrf <text> database <text> linkstate-id <text> dump
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database linkstate- id <linkstate-id> internal	show ipv6 ospfv3 vrf <text> database <text> linkstate-id <text> internal
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database network <text>	show ipv6 ospfv3 vrf <text> database network <text>
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database network adv-router <adv-router> detail	show ipv6 ospfv3 vrf <text> database network adv-router <text> detail
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database network adv-router <adv-router> dump	show ipv6 ospfv3 vrf <text> database network adv-router <text> dump
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database network adv-router <adv-router> internal	show ipv6 ospfv3 vrf <text> database network adv-router <text> internal
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database network adv-router <adv-router> linkstate-id <linkstate-id> detail	show ipv6 ospfv3 vrf <text> database net- work adv-router <text> linkstate-id <text> detail
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database network adv-router <adv-router> linkstate-id <linkstate-id> dump	show ipv6 ospfv3 vrf <text> database net- work adv-router <text> linkstate-id <text> dump
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database network adv-router <adv-router> linkstate-id <linkstate-id> internal	show ipv6 ospfv3 vrf <text> database net- work adv-router <text> linkstate-id <text> internal

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Table 2 – continued from previous page

Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database network detail	show ipv6 ospfv3 vrf <text> database network <text> detail
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database network dump	show ipv6 ospfv3 vrf <text> database network <text> dump
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database network internal	show ipv6 ospfv3 vrf <text> database network <text> internal
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database network linkstate-id <linkstate-id> detail	show ipv6 ospfv3 vrf <text> database network linkstate-id <text> detail
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database network linkstate-id <linkstate-id> dump	show ipv6 ospfv3 vrf <text> database network linkstate-id <text> dump
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database network linkstate-id <linkstate-id> internal	show ipv6 ospfv3 vrf <text> database network linkstate-id <text> internal
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database network node.tag	Nothing found in VyOS
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database network node.tag detail	Nothing found in VyOS
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database network node.tag dump	Nothing found in VyOS
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database network node.tag internal	Nothing found in VyOS
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database network node.tag self-originated	Nothing found in VyOS

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Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database network node.tag self-originated detail	Nothing found in VyOS
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database network node.tag self-originated dump	Nothing found in VyOS
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database network node.tag self-originated internal	Nothing found in VyOS
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database network self-originated	show ipv6 ospfv3 vrf <text> database network self-originated
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database network self-originated detail	show ipv6 ospfv3 vrf <text> database network self-originated detail
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database network self-originated dump	show ipv6 ospfv3 vrf <text> database network self-originated dump
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database network self-originated internal	show ipv6 ospfv3 vrf <text> database network self-originated internal
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database node.tag	Nothing found in VyOS
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database node.tag adv-router <adv-router> detail	Nothing found in VyOS
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database node.tag adv-router <adv-router> dump	Nothing found in VyOS
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database node.tag adv-router <adv-router> internal	Nothing found in VyOS

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Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database node.tag adv-router <adv-router> linkstate-id <linkstate-id> detail	Nothing found in VyOS
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database node.tag adv-router <adv-router> linkstate-id <linkstate-id> dump	Nothing found in VyOS
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database node.tag adv-router <adv-router> linkstate-id <linkstate-id> internal	Nothing found in VyOS
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database node.tag detail	Nothing found in VyOS
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database node.tag dump	Nothing found in VyOS
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database node.tag internal	Nothing found in VyOS
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database node.tag linkstate-id <linkstate-id> detail	Nothing found in VyOS
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database node.tag linkstate-id <linkstate-id> dump	Nothing found in VyOS
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database node.tag linkstate-id <linkstate-id> internal	Nothing found in VyOS
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database node.tag node.tag	Nothing found in VyOS

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Table 2 – continued from previous page

Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database node.tag node.tag detail	Nothing found in VyOS
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database node.tag node.tag dump	Nothing found in VyOS
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database node.tag node.tag internal	Nothing found in VyOS
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database node.tag node.tag self-originated	Nothing found in VyOS
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database node.tag node.tag self-originated detail	Nothing found in VyOS
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database node.tag node.tag self-originated dump	Nothing found in VyOS
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database node.tag node.tag self-originated internal	Nothing found in VyOS
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database node.tag self-originated	Nothing found in VyOS
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database node.tag self-originated detail	Nothing found in VyOS
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database node.tag self-originated dump	Nothing found in VyOS
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database node.tag self-originated internal	Nothing found in VyOS

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Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database router	show ipv6 ospfv3 vrf <text> database router <text>
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database router adv-router <adv-router> detail	show ipv6 ospfv3 vrf <text> database router adv-router <text> detail
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database router adv-router <adv-router> dump	show ipv6 ospfv3 vrf <text> database router adv-router <text> dump
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database router adv-router <adv-router> internal	show ipv6 ospfv3 vrf <text> database router adv-router <text> internal
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database router adv-router <adv-router> linkstate-id <linkstate-id> detail	show ipv6 ospfv3 vrf <text> database router adv-router <text> linkstate-id <text> detail
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database router adv-router <adv-router> linkstate-id <linkstate-id> dump	show ipv6 ospfv3 vrf <text> database router adv-router <text> linkstate-id <text> dump
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database router adv-router <adv-router> linkstate-id <linkstate-id> internal	show ipv6 ospfv3 vrf <text> database router adv-router <text> linkstate-id <text> internal
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database router detail	show ipv6 ospfv3 vrf <text> database router <text> detail
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database router dump	show ipv6 ospfv3 vrf <text> database router <text> dump
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database router internal	show ipv6 ospfv3 vrf <text> database router <text> internal

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Table 2 – continued from previous page

Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database router linkstate-id <linkstate-id> detail	show ipv6 ospfv3 vrf <text> database router linkstate-id <text> detail
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database router linkstate-id <linkstate-id> dump	show ipv6 ospfv3 vrf <text> database router linkstate-id <text> dump
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database router linkstate-id <linkstate-id> internal	show ipv6 ospfv3 vrf <text> database router linkstate-id <text> internal
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database router node.tag	Nothing found in VyOS
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database router node.tag detail	Nothing found in VyOS
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database router node.tag dump	Nothing found in VyOS
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database router node.tag internal	Nothing found in VyOS
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database router node.tag self-originated	Nothing found in VyOS
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database router node.tag self-originated detail	Nothing found in VyOS
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database router node.tag self-originated dump	Nothing found in VyOS

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Table 2 – continued from previous page

Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database router node.tag self-originated internal	Nothing found in VyOS
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database router self-originated	show ipv6 ospfv3 vrf <text> database router self-originated
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database router self-originated detail	show ipv6 ospfv3 vrf <text> database router self-originated detail
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database router self-originated dump	show ipv6 ospfv3 vrf <text> database router self-originated dump
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database router self-originated internal	show ipv6 ospfv3 vrf <text> database router self-originated internal
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database self- originated	show ipv6 ospfv3 vrf <text> database self- originated
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database self- originated detail	show ipv6 ospfv3 vrf <text> database self- originated detail
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database self- originated dump	show ipv6 ospfv3 vrf <text> database self- originated dump
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database self- originated internal	show ipv6 ospfv3 vrf <text> database self- originated internal
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database type-7 <text>	show ipv6 ospfv3 vrf <text> database type-7 <text>
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database type-7 adv-router <adv-router> detail	show ipv6 ospfv3 vrf <text> database type-7 adv-router <text> detail

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Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database type-7 adv-router <adv-router> dump	show ipv6 ospfv3 vrf <text> database type-7 adv-router <text> dump
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database type-7 adv-router <adv-router> internal	show ipv6 ospfv3 vrf <text> database type-7 adv-router <text> internal
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database type-7 adv-router <adv-router> linkstate-id <linkstate-id> detail	show ipv6 ospfv3 vrf <text> database type-7 adv-router <text> linkstate-id <text> detail
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database type-7 adv-router <adv-router> linkstate-id <linkstate-id> dump	show ipv6 ospfv3 vrf <text> database type-7 adv-router <text> linkstate-id <text> dump
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database type-7 adv-router <adv-router> linkstate-id <linkstate-id> internal	show ipv6 ospfv3 vrf <text> database type-7 adv-router <text> linkstate-id <text> internal
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database type-7 adv-router <adv-router> linkstate-id <linkstate-id> detail	show ipv6 ospfv3 vrf <text> database type-7 <text> detail
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database type-7 adv-router <adv-router> linkstate-id <linkstate-id> dump	show ipv6 ospfv3 vrf <text> database type-7 <text> dump
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database type-7 adv-router <adv-router> linkstate-id <linkstate-id> internal	show ipv6 ospfv3 vrf <text> database type-7 <text> internal
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database type-7 adv-router <adv-router> linkstate-id <linkstate-id> detail	show ipv6 ospfv3 vrf <text> database type-7 linkstate-id <text> detail

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Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database type-7 linkstate-id <linkstate-id> dump	show ipv6 ospfv3 vrf <text> database type-7 linkstate-id <text> dump
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database type-7 linkstate-id <linkstate-id> internal	show ipv6 ospfv3 vrf <text> database type-7 linkstate-id <text> internal
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database type-7 node.tag	Nothing found in VyOS
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database type-7 node.tag detail	Nothing found in VyOS
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database type-7 node.tag dump	Nothing found in VyOS
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database type-7 node.tag internal	Nothing found in VyOS
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database type-7 node.tag self-originated	Nothing found in VyOS
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database type-7 node.tag self-originated detail	Nothing found in VyOS
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database type-7 node.tag self-originated dump	Nothing found in VyOS
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database type-7 node.tag self-originated internal	Nothing found in VyOS
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database type-7 self-originated	show ipv6 ospfv3 vrf <text> database type-7 self-originated

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Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database type-7 self-originated detail	show ipv6 ospfv3 vrf <text> database type-7 self-originated detail
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database type-7 self-originated dump	show ipv6 ospfv3 vrf <text> database type-7 self-originated dump
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> database type-7 self-originated internal	show ipv6 ospfv3 vrf <text> database type-7 self-originated internal
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> graceful-restart helper	show ipv6 ospfv3 vrf <text> graceful-restart helper
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> interface <inter- face>	show ipv6 ospfv3 vrf <text> interface <text>
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> interface <inter- face> prefix <prefix>	show ipv6 ospfv3 vrf <text> interface prefix <text>
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> interface <inter- face> prefix <prefix> detail	show ipv6 ospfv3 vrf <text> interface prefix <text> detail
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> interface <inter- face> prefix <prefix> match	show ipv6 ospfv3 vrf <text> interface prefix <text> match
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> linkstate detail	show ipv6 ospfv3 vrf <text> linkstate detail
×	not yet documented	Nothing found in XML Definitions	show ipv6 ospfv3 vrf <text> linkstate network <text> <text>
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> linkstate network <network> node.tag	Nothing found in VyOS
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> linkstate router <router>	show ipv6 ospfv3 vrf <text> linkstate router <text>

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Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> neighbor	show ipv6 ospfv3 vrf <text> neighbor
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> neighbor detail	show ipv6 ospfv3 vrf <text> neighbor detail
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> neighbor drchoice	show ipv6 ospfv3 vrf <text> neighbor drchoice
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> redistribute	show ipv6 ospfv3 vrf <text> redistribute
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> route <route>	show ipv6 ospfv3 vrf <text> route <text>
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> route detail	show ipv6 ospfv3 vrf <text> route detail
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> route external-1	show ipv6 ospfv3 vrf <text> route external-1
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> route external-1 detail	show ipv6 ospfv3 vrf <text> route external-1 detail
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> route external-2	show ipv6 ospfv3 vrf <text> route external-2
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> route external-2 detail	show ipv6 ospfv3 vrf <text> route external-2 detail
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> route inter-area	show ipv6 ospfv3 vrf <text> route inter-area
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> route inter-area detail	show ipv6 ospfv3 vrf <text> route inter-area detail
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> route intra-area	show ipv6 ospfv3 vrf <text> route intra-area
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> route intra-area detail	show ipv6 ospfv3 vrf <text> route intra-area detail

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Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> route <route> longer	show ipv6 ospfv3 vrf <text> route <text> longer
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> route <route> match	show ipv6 ospfv3 vrf <text> route <text> match
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> route <route> match detail	show ipv6 ospfv3 vrf <text> route <text> match detail
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrf <vrf> route summary	show ipv6 ospfv3 vrf <text> route summary
×	not yet documented	show-ipv6-ospfv3.xml.in: show ipv6 ospfv3 vrfs	show ipv6 ospfv3 vrfs
×	not yet documented	show-ipv6-pim.xml.in: show ipv6 pim bsm- database	Nothing found in VyOS
×	not yet documented	show-ipv6-pim.xml.in: show ipv6 pim bsr	Nothing found in VyOS
×	not yet documented	show-ipv6-pim.xml.in: show ipv6 pim bsrp-info	Nothing found in VyOS
×	not yet documented	show-ipv6-pim.xml.in: show ipv6 pim channel	Nothing found in VyOS
×	not yet documented	show-ipv6-pim.xml.in: show ipv6 pim interface	Nothing found in VyOS
×	not yet documented	show-ipv6-pim.xml.in: show ipv6 pim join	Nothing found in VyOS
×	not yet documented	show-ipv6-pim.xml.in: show ipv6 pim local- membership	Nothing found in VyOS
×	not yet documented	show-ipv6-pim.xml.in: show ipv6 pim neighbor	Nothing found in VyOS
×	not yet documented	show-ipv6-pim.xml.in: show ipv6 pim nexthop	Nothing found in VyOS
×	not yet documented	show-ipv6-pim.xml.in: show ipv6 pim rp-info	Nothing found in VyOS
×	not yet documented	show-ipv6-pim.xml.in: show ipv6 pim rpf	Nothing found in VyOS
×	not yet documented	show-ipv6-pim.xml.in: show ipv6 pim secondary	Nothing found in VyOS
×	not yet documented	show-ipv6-pim.xml.in: show ipv6 pim state	Nothing found in VyOS
×	not yet documented	show-ipv6-pim.xml.in: show ipv6 pim statistics	Nothing found in VyOS
×	not yet documented	show-ipv6-pim.xml.in: show ipv6 pim upstream	Nothing found in VyOS

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Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	show-ipv6-pim.xml.in: show ipv6 pim upstream-join-desired	Nothing found in VyOS
×	not yet documented	show-ipv6-pim.xml.in: show ipv6 pim upstream-rpf	Nothing found in VyOS
✓	<i>show ipv6 prefix-list</i>	show-ipv6-prefix-list.xml.in: show ipv6 prefix-list <prefix-list>	show ipv6 prefix-list <text> <text>
×	not yet documented	show-ipv6-prefix-list.xml.in: show ipv6 prefix-list detail <detail>	show ipv6 prefix-list detail <text>
×	not yet documented	Nothing found in XML Definitions	show ipv6 prefix-list <text> <text> first-match
×	not yet documented	Nothing found in XML Definitions	show ipv6 prefix-list <text> <text> longer
×	not yet documented	show-ipv6-prefix-list.xml.in: show ipv6 prefix-list <prefix-list> node.tag	Nothing found in VyOS
×	not yet documented	show-ipv6-prefix-list.xml.in: show ipv6 prefix-list <prefix-list> node.tag first-match	Nothing found in VyOS
×	not yet documented	show-ipv6-prefix-list.xml.in: show ipv6 prefix-list <prefix-list> node.tag longer	Nothing found in VyOS
×	not yet documented	show-ipv6-prefix-list.xml.in: show ipv6 prefix-list <prefix-list> seq <seq>	show ipv6 prefix-list <text> seq <text>
×	not yet documented	show-ipv6-prefix-list.xml.in: show ipv6 prefix-list summary <summary>	show ipv6 prefix-list summary <text>
✓	<i>show ipv6 ripng</i>	show-ipv6.xml.in: show ipv6 ripng	show ipv6 ripng
✓	<i>show ipv6 ripng status</i>	show-ipv6.xml.in: show ipv6 ripng status	show ipv6 ripng status
✓	<i>show ipv6 route</i>	show-ipv6-route.xml.in: show ipv6 route <route>	show ipv6 route <text>
×	not yet documented	show-ipv6-route.xml.in: show ipv6 route bgp	show ipv6 route bgp

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Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	show-ipv6-route.xml.in: show ipv6 route cache <cache>	show ipv6 route cache <text>
×	not yet documented	show-ipv6-route.xml.in: show ipv6 route connected	show ipv6 route connected
×	not yet documented	show-ipv6-route.xml.in: show ipv6 route forward <forward>	show ipv6 route forward <text>
×	not yet documented	show-ipv6-route.xml.in: show ipv6 route isis	show ipv6 route isis
×	not yet documented	show-ipv6-route.xml.in: show ipv6 route kernel	show ipv6 route kernel
×	not yet documented	show-ipv6-route.xml.in: show ipv6 route <text> longer-prefixes	show ipv6 route <text> longer-prefixes
×	not yet documented	show-ipv6-route.xml.in: show ipv6 route ospfv3	show ipv6 route ospfv3
×	not yet documented	show-ipv6-route.xml.in: show ipv6 route ripng	show ipv6 route ripng
×	not yet documented	show-ipv6-route.xml.in: show ipv6 route static	show ipv6 route static
×	not yet documented	show-ipv6-route.xml.in: show ipv6 route summary	show ipv6 route summary
×	not yet documented	show-ipv6-route.xml.in: show ipv6 route summary table <table>	show ipv6 route summary table <text>
×	not yet documented	show-ipv6-route.xml.in: show ipv6 route table <table>	show ipv6 route table <text>
×	not yet documented	show-ipv6-route.xml.in: show ipv6 route tag <tag>	show ipv6 route tag <text>
×	not yet documented	show-ipv6-route.xml.in: show ipv6 route vrf <vrf>	show ipv6 route vrf <text>
×	<i>show ipv6 route vrf <name></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	show-ipv6-route.xml.in: show ipv6 route vrf <vrf> bgp	show ipv6 route vrf <text> bgp
×	not yet documented	show-ipv6-route.xml.in: show ipv6 route vrf <vrf> connected	show ipv6 route vrf <text> connected
×	not yet documented	show-ipv6-route.xml.in: show ipv6 route vrf <vrf> isis	show ipv6 route vrf <text> isis
×	not yet documented	show-ipv6-route.xml.in: show ipv6 route vrf <vrf> kernel	show ipv6 route vrf <text> kernel

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Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	show-ipv6-route.xml.in: show ipv6 route vrf <vrf> node.tag	Nothing found in VyOS
×	not yet documented	show-ipv6-route.xml.in: show ipv6 route vrf <vrf> node.tag longer-prefixes	Nothing found in VyOS
×	not yet documented	show-ipv6-route.xml.in: show ipv6 route vrf <vrf> ospfv3	show ipv6 route vrf <text> ospfv3
×	not yet documented	show-ipv6-route.xml.in: show ipv6 route vrf <vrf> ripng	show ipv6 route vrf <text> ripng
×	not yet documented	show-ipv6-route.xml.in: show ipv6 route vrf <vrf> static	show ipv6 route vrf <text> static
×	not yet documented	show-ipv6-route.xml.in: show ipv6 route vrf <vrf> summary	show ipv6 route vrf <text> summary
×	not yet documented	show-ipv6-route.xml.in: show ipv6 route vrf <vrf> supernets-only	show ipv6 route vrf <text> supernets-only
×	not yet documented	show-ipv6-route.xml.in: show ipv6 route vrf <vrf> table <table>	show ipv6 route vrf <text> table <text>
×	not yet documented	show-ipv6-route.xml.in: show ipv6 route vrf <vrf> tag <tag>	show ipv6 route vrf <text> tag <text>
×	not yet documented	show-isis.xml.in: show isis database <database>	show isis database <text>
×	not yet documented	show-isis.xml.in: show isis database detail	show isis database detail
×	not yet documented	show-isis.xml.in: show isis fast-reroute sum- mary	Nothing found in VyOS
×	not yet documented	show-isis.xml.in: show isis fast-reroute sum- mary level-1	Nothing found in VyOS
×	not yet documented	show-isis.xml.in: show isis fast-reroute sum- mary level-2	Nothing found in VyOS
×	not yet documented	show-isis.xml.in: show isis hostname	show isis hostname
×	not yet documented	show-isis.xml.in: show isis interface <inter- face>	show isis interface <text>
×	not yet documented	show-isis.xml.in: show isis interface detail	show isis interface detail

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Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	show-isis.xml.in: show isis mpls-te interface <interface>	show isis mpls-te interface <text>
×	not yet documented	show-isis.xml.in: show isis mpls-te router	show isis mpls-te router
×	not yet documented	show-isis.xml.in: show isis mpls ldp-sync	show isis mpls ldp-sync
×	not yet documented	show-isis.xml.in: show isis mpls ldp-sync interface <interface>	show isis mpls ldp-sync interface <text>
×	not yet documented	show-isis.xml.in: show isis neighbor <neighbor>	show isis neighbor <text>
×	not yet documented	show-isis.xml.in: show isis neighbor detail	show isis neighbor detail
×	not yet documented	show-isis.xml.in: show isis route	show isis route
×	not yet documented	show-isis.xml.in: show isis route level-1	show isis route level-1
×	not yet documented	show-isis.xml.in: show isis route level-2	show isis route level-2
✓	<i>show isis route prefix-sid</i>	show-isis.xml.in: show isis route prefix-sid	show isis route prefix-sid
✓	<i>show isis segment-routing node</i>	show-isis.xml.in: show isis segment-routing node	show isis segment-routing node
×	not yet documented	show-isis.xml.in: show isis spf-delay-ietf	show isis spf-delay-ietf
×	not yet documented	show-isis.xml.in: show isis summary	show isis summary
×	not yet documented	show-isis.xml.in: show isis topology	show isis topology
×	not yet documented	show-isis.xml.in: show isis topology level-1	show isis topology level-1
×	not yet documented	show-isis.xml.in: show isis topology level-2	show isis topology level-2
×	not yet documented	show-isis.xml.in: show isis vrf <vrf> database <database>	show isis vrf <text> database <text>
×	not yet documented	show-isis.xml.in: show isis vrf <vrf> database detail	show isis vrf <text> database detail
×	not yet documented	show-isis.xml.in: show isis vrf <vrf> fast- reroute summary	Nothing found in VyOS
×	not yet documented	show-isis.xml.in: show isis vrf <vrf> fast- reroute summary level-1	Nothing found in VyOS

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Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	show-isis.xml.in: show isis vrf <vrf> fast-reroute summary level-2	Nothing found in VyOS
×	not yet documented	show-isis.xml.in: show isis vrf <vrf> host-name	show isis vrf <text> host-name
×	not yet documented	show-isis.xml.in: show isis vrf <vrf> interface <interface>	show isis vrf <text> interface <text>
×	not yet documented	show-isis.xml.in: show isis vrf <vrf> interface detail	show isis vrf <text> interface detail
×	not yet documented	show-isis.xml.in: show isis vrf <vrf> mpls-te interface <interface>	show isis vrf <text> mpls-te interface <text>
×	not yet documented	show-isis.xml.in: show isis vrf <vrf> mpls-te router	show isis vrf <text> mpls-te router
×	not yet documented	show-isis.xml.in: show isis vrf <vrf> mpls ldp-sync	show isis vrf <text> mpls ldp-sync
×	not yet documented	show-isis.xml.in: show isis vrf <vrf> mpls ldp-sync interface <interface>	show isis vrf <text> mpls ldp-sync interface <text>
×	not yet documented	show-isis.xml.in: show isis vrf <vrf> neighbor <neighbor>	show isis vrf <text> neighbor <text>
×	not yet documented	show-isis.xml.in: show isis vrf <vrf> neighbor detail	show isis vrf <text> neighbor detail
×	not yet documented	show-isis.xml.in: show isis vrf <vrf> route	show isis vrf <text> route
×	not yet documented	show-isis.xml.in: show isis vrf <vrf> route level-1	show isis vrf <text> route level-1
×	not yet documented	show-isis.xml.in: show isis vrf <vrf> route level-2	show isis vrf <text> route level-2
×	not yet documented	show-isis.xml.in: show isis vrf <vrf> route prefix-sid	show isis vrf <text> route prefix-sid
×	not yet documented	show-isis.xml.in: show isis vrf <vrf> segment-routing node	show isis vrf <text> segment-routing node
×	not yet documented	show-isis.xml.in: show isis vrf <vrf> spf-delay-ietf	show isis vrf <text> spf-delay-ietf

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Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	show-isis.xml.in: show isis vrf <vrf> summary	show isis vrf <text> summary
×	not yet documented	show-isis.xml.in: show isis vrf <vrf> topology	show isis vrf <text> topology
×	not yet documented	show-isis.xml.in: show isis vrf <vrf> topology level-1	show isis vrf <text> topology level-1
×	not yet documented	show-isis.xml.in: show isis vrf <vrf> topology level-2	show isis vrf <text> topology level-2
×	not yet documented	l2tp-server.xml.in: show l2tp-server sessions	show l2tp-server sessions
×	not yet documented	l2tp-server.xml.in: show l2tp-server statistics	show l2tp-server statistics
×	not yet documented	show-license.xml.in: show license	show license
✓	<i>show lldp neighbors</i>	lldp.xml.in: show lldp neighbors	show lldp neighbors
×	<i>show lldp neighbors detail</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	lldp.xml.in: show lldp neighbors interface <interface>	show lldp neighbors interface <text>
×	<i>show lldp neighbors interface <interface></i>	Nothing found in XML Definitions	Nothing found in VyOS
✓	<i>show log [all authorization cluster conntrack-sync ...]</i>	show-log.xml.in: show log	show log <text>
×	not yet documented	show-log.xml.in: show log all	show log all
×	not yet documented	show-log.xml.in: show log audit	show log audit
×	not yet documented	show-log.xml.in: show log authorization	show log authorization
×	not yet documented	show-log.xml.in: show log certbot	Nothing found in VyOS
×	not yet documented	show-log.xml.in: show log cluster	show log cluster
×	not yet documented	show-log.xml.in: show log conntrack-sync	show log conntrack-sync
✓	<i>show log console-server</i>	show-log.xml.in: show log console-server	show log console-server
×	not yet documented	Nothing found in XML Definitions	show log container <text>

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Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
✓	<i>show log dhcp client</i>	show-log.xml.in: show log dhcp client	show log dhcp client
×	not yet documented	show-log.xml.in: show log dhcp client interface <interface>	show log dhcp client interface <text>
×	<i>show log dhcp client interface <interface></i>	Nothing found in XML Definitions	Nothing found in VyOS
✓	<i>show log dhcp server</i>	show-log.xml.in: show log dhcp server	show log dhcp server
✓	<i>show log dhcpv6 client</i>	show-log.xml.in: show log dhcpv6 client	show log dhcpv6 client
×	not yet documented	show-log.xml.in: show log dhcpv6 client interface <interface>	show log dhcpv6 client interface <text>
×	<i>show log dhcpv6 client interface <interface></i>	Nothing found in XML Definitions	Nothing found in VyOS
✓	<i>show log dhcpv6 server</i>	show-log.xml.in: show log dhcpv6 server	show log dhcpv6 server
×	not yet documented	Nothing found in XML Definitions	show log dns dynamic
×	not yet documented	Nothing found in XML Definitions	show log dns forwarding
×	<i>show log firewall</i>	show-log.xml.in: show log firewall	Nothing found in VyOS
×	<i>show log firewall bridge</i>	show-log.xml.in: show log firewall bridge	Nothing found in VyOS
×	<i>show log firewall bridge forward</i>	show-log.xml.in: show log firewall bridge forward	Nothing found in VyOS
×	<i>show log firewall bridge forward filter</i>	show-log.xml.in: show log firewall bridge forward filter	Nothing found in VyOS
×	not yet documented	show-log.xml.in: show log firewall bridge forward filter rule <rule>	Nothing found in VyOS
×	<i>show log firewall bridge forward filter rule <rule></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	show-log.xml.in: show log firewall bridge name <name>	Nothing found in VyOS
×	<i>show log firewall bridge name <name></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>show log firewall bridge name <name> rule <rule></i>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	show-log.xml.in: show log firewall bridge name <name> rule <rule>	Nothing found in VyOS
×	<i>show log firewall ipv4 [forward input output name]</i>	show-log.xml.in: show log firewall ipv4	Nothing found in VyOS
×	<i>show log firewall ipv4 [forward input output] filter</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>show log firewall ipv4 [forward input output] filter rule <rule></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	show-log.xml.in: show log firewall ipv4 for- ward	Nothing found in VyOS
×	not yet documented	show-log.xml.in: show log firewall ipv4 for- ward filter	Nothing found in VyOS
×	not yet documented	show-log.xml.in: show log firewall ipv4 for- ward filter rule <rule>	Nothing found in VyOS
×	not yet documented	show-log.xml.in: show log firewall ipv4 in- put	Nothing found in VyOS
×	not yet documented	show-log.xml.in: show log firewall ipv4 in- put filter	Nothing found in VyOS
×	not yet documented	show-log.xml.in: show log firewall ipv4 in- put filter rule <rule>	Nothing found in VyOS
×	not yet documented	show-log.xml.in: show log firewall ipv4 name <name>	Nothing found in VyOS
×	<i>show log firewall ipv4 name <name></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>show log firewall ipv4 name <name> rule <rule></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	show-log.xml.in: show log firewall ipv4 name <name> rule <rule>	Nothing found in VyOS
×	not yet documented	show-log.xml.in: show log firewall ipv4 out- put	Nothing found in VyOS

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Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	show-log.xml.in: show log firewall ipv4 out- put filter	Nothing found in VyOS
×	not yet documented	show-log.xml.in: show log firewall ipv4 out- put filter rule <rule>	Nothing found in VyOS
×	<i>show log firewall ipv6 [forward input output name]</i>	show-log.xml.in: show log firewall ipv6	Nothing found in VyOS
×	not yet documented	Nothing found in XML Definitions	show log firewall ipv6- name <text>
×	not yet documented	Nothing found in XML Definitions	show log firewall ipv6- name <text> rule <text>
×	<i>show log firewall ipv6 [forward input output] filter</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>show log firewall ipv6 [forward input output] filter rule <rule></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	show-log.xml.in: show log firewall ipv6 for- ward	Nothing found in VyOS
×	not yet documented	show-log.xml.in: show log firewall ipv6 for- ward filter	Nothing found in VyOS
×	not yet documented	show-log.xml.in: show log firewall ipv6 for- ward filter rule <rule>	Nothing found in VyOS
×	not yet documented	show-log.xml.in: show log firewall ipv6 in- put	Nothing found in VyOS
×	not yet documented	show-log.xml.in: show log firewall ipv6 in- put filter	Nothing found in VyOS
×	not yet documented	show-log.xml.in: show log firewall ipv6 in- put filter rule <rule>	Nothing found in VyOS
×	not yet documented	show-log.xml.in: show log firewall ipv6 name <name>	Nothing found in VyOS
×	<i>show log firewall ipv6 name <name></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>show log firewall ipv6 name <name> rule <rule></i>	Nothing found in XML Definitions	Nothing found in VyOS

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Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	show-log.xml.in: show log firewall ipv6 name <name> rule <rule>	Nothing found in VyOS
×	not yet documented	show-log.xml.in: show log firewall ipv6 out- put	Nothing found in VyOS
×	not yet documented	show-log.xml.in: show log firewall ipv6 out- put filter	Nothing found in VyOS
×	not yet documented	show-log.xml.in: show log firewall ipv6 out- put filter rule <rule>	Nothing found in VyOS
×	not yet documented	Nothing found in XML Definitions	show log firewall name <text>
×	not yet documented	Nothing found in XML Definitions	show log firewall name <text> rule <text>
×	not yet documented	show-log.xml.in: show log flow-accounting	show log flow-accounting
×	not yet documented	show-log.xml.in: show log https	show log https
×	not yet documented	show-log.xml.in: show log ids ddos- protection	show log ids ddos- protection
×	not yet documented	show-log.xml.in: show log image <image>	show log image <text>
×	<i>show log image <name> [all authorization directory file <file name> tail <lines>]</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	show-log.xml.in: show log image <image> all	show log image <text> all
×	not yet documented	show-log.xml.in: show log image <image> authorization	show log image <text> au- thorization
×	not yet documented	show-log.xml.in: show log image <image> tail <tail>	show log image <text> tail <text>
×	not yet documented	show-login.xml.in: show login	show login
×	not yet documented	show-login.xml.in: show login groups	show login groups
×	not yet documented	show-login.xml.in: show login level	show login level
×	not yet documented	show-login.xml.in: show login user	show login user

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×	not yet documented	show-log.xml.in: show log ipoe-server	show log ipoe-server
✓	<i>show log ipsec</i>	show-log.xml.in: show log ipsec	show log ipsec
×	not yet documented	show-log.xml.in: show log kernel	show log kernel
×	not yet documented	show-log.xml.in: show log l2tp	show log l2tp
✓	<i>show log lldp</i>	show-log.xml.in: show log lldp	show log lldp
×	not yet documented	show-log.xml.in: show log macsec	show log macsec
×	not yet documented	show-log.xml.in: show log macsec interface <interface>	show log macsec interface <text>
×	<i>show log mdns repeater</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	show-log.xml.in: show log nat	show log nat
×	not yet documented	show-log.xml.in: show log ndp-proxy	Nothing found in VyOS
×	not yet documented	show-log.xml.in: show log nhrp	show log nhrp
×	not yet documented	show-log.xml.in: show log ntp	show log ntp
×	not yet documented	show-log.xml.in: show log openconnect	show log openconnect
✓	<i>show log openvpn</i>	show-log.xml.in: show log openvpn	show log openvpn
×	not yet documented	show-log.xml.in: show log openvpn inter- face <interface>	show log openvpn inter- face <text>
×	<i>show log openvpn interface <interface></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	show-log.xml.in: show log pppoe	show log pppoe
×	not yet documented	show-log.xml.in: show log pppoe-server	show log pppoe-server
×	not yet documented	show-log.xml.in: show log pppoe interface <interface>	show log pppoe interface <text>
×	not yet documented	show-log.xml.in: show log pptp	show log pptp
×	not yet documented	show-log.xml.in: show log protocol bfd	show log protocol bfd
×	not yet documented	show-log.xml.in: show log protocol bgp	show log protocol bgp

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×	not yet documented	show-log.xml.in: show log protocol isis	show log protocol isis
×	not yet documented	show-log.xml.in: show log protocol mpls	show log protocol mpls
×	not yet documented	show-log.xml.in: show log protocol multi-cast	show log protocol multi-cast
×	not yet documented	show-log.xml.in: show log protocol nhrp	show log protocol nhrp
×	not yet documented	show-log.xml.in: show log protocol ospf	show log protocol ospf
×	not yet documented	show-log.xml.in: show log protocol ospfv3	show log protocol ospfv3
×	not yet documented	show-log.xml.in: show log protocol rip	show log protocol rip
×	not yet documented	show-log.xml.in: show log protocol ripng	show log protocol ripng
×	not yet documented	show-log.xml.in: show log protocol static	show log protocol static
×	not yet documented	show-log.xml.in: show log router-advert	show log router-advert
×	not yet documented	show-log.xml.in: show log snmp	show log snmp
✓	<i>show log ssh</i>	show-log.xml.in: show log ssh	show log ssh
×	<i>show log ssh dynamic-protection</i>	show-log.xml.in: show log ssh dynamic-protection	Nothing found in VyOS
×	not yet documented	show-log.xml.in: show log sstp	show log sstp
×	not yet documented	show-log.xml.in: show log sstpc	show log sstpc
×	not yet documented	show-log.xml.in: show log sstpc interface <interface>	show log sstpc interface <text>
×	not yet documented	show-log.xml.in: show log tail	show log tail <text>
×	not yet documented	show-log.xml.in: show log vpn	show log vpn
×	not yet documented	Nothing found in XML Definitions	show log vpp
×	not yet documented	show-log.xml.in: show log vrrp	show log vrrp
×	not yet documented	show-log.xml.in: show log webproxy	show log webproxy
×	not yet documented	show-monitoring.xml.in: show monitoring	show monitoring

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Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
✓	<i>show mpls ldp binding</i>	show-mpls.xml.in: show mpls ldp binding <binding>	show mpls ldp binding <text>
×	<i>show mpls ldp binding <prefix></i>	Nothing found in XML Definitions	Nothing found in VyOS
✓	<i>show mpls ldp binding detail</i>	show-mpls.xml.in: show mpls ldp binding <binding> detail	show mpls ldp binding <text> detail
×	not yet documented	show-mpls.xml.in: show mpls ldp binding local-label <local-label>	show mpls ldp binding local-label <text>
×	<i>show mpls ldp binding local-label <label></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	show-mpls.xml.in: show mpls ldp binding local-label <local-label> detail	show mpls ldp binding local-label <text> detail
×	not yet documented	show-mpls.xml.in: show mpls ldp binding local-label <local-label> neighbor <neighbor>	show mpls ldp binding local-label <text> neigh- bor <text>
×	not yet documented	show-mpls.xml.in: show mpls ldp binding local-label <local-label> remote-label <remote-label>	show mpls ldp binding local-label <text> remote- label <text>
×	not yet documented	show-mpls.xml.in: show mpls ldp binding neighbor <neighbor>	show mpls ldp binding neighbor <text>
×	<i>show mpls ldp binding neighbor <address></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	show-mpls.xml.in: show mpls ldp binding neighbor <neighbor> de- tail	show mpls ldp binding neighbor <text> detail
×	not yet documented	show-mpls.xml.in: show mpls ldp binding neighbor <neighbor> local-label <local-label>	show mpls ldp binding neighbor <text> local- label <text>
×	not yet documented	show-mpls.xml.in: show mpls ldp binding neighbor <neighbor> remote-label <remote-label>	show mpls ldp binding neighbor <text> remote- label <text>

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Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	show-mpls.xml.in: show mpls ldp binding remote-label <remote-label>	show mpls ldp binding remote-label <text>
×	<i>show mpls ldp binding remote-label <label></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	show-mpls.xml.in: show mpls ldp binding remote-label <remote-label> detail	show mpls ldp binding remote-label <text> detail
×	not yet documented	show-mpls.xml.in: show mpls ldp binding remote-label <remote-label> local-label <local-label>	show mpls ldp binding remote-label <text> local-label <text>
×	not yet documented	show-mpls.xml.in: show mpls ldp binding remote-label <remote-label> neighbor <neighbor>	show mpls ldp binding remote-label <text> neighbor <text>
✓	<i>show mpls ldp discovery</i>	show-mpls.xml.in: show mpls ldp discovery	show mpls ldp discovery
✓	<i>show mpls ldp discovery detail</i>	show-mpls.xml.in: show mpls ldp discovery detail	show mpls ldp discovery detail
✓	<i>show mpls ldp interface</i>	show-mpls.xml.in: show mpls ldp interface	show mpls ldp interface
✓	<i>show mpls ldp neighbor</i>	show-mpls.xml.in: show mpls ldp neighbor <neighbor>	show mpls ldp neighbor <text>
×	<i>show mpls ldp neighbor <address></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>show mpls ldp neighbor <address> capabilities</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>show mpls ldp neighbor <address> detail</i>	Nothing found in XML Definitions	Nothing found in VyOS
✓	<i>show mpls ldp neighbor capabilities</i>	show-mpls.xml.in: show mpls ldp neighbor <neighbor> capabilities	show mpls ldp neighbor <text> capabilities
✓	<i>show mpls ldp neighbor detail</i>	show-mpls.xml.in: show mpls ldp neighbor <neighbor> detail	show mpls ldp neighbor <text> detail
✓	<i>show mpls pseudowire</i>	show-mpls.xml.in: show mpls pseudowire	show mpls pseudowire
✓	<i>show mpls table</i>	show-mpls.xml.in: show mpls table	show mpls table

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Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	nat66.xml.in: show nat66 destination rules	show nat66 destination rules
×	not yet documented	nat66.xml.in: show nat66 destination statistics	show nat66 destination statistics
×	not yet documented	nat66.xml.in: show nat66 destination translations	show nat66 destination translations
×	not yet documented	nat66.xml.in: show nat66 destination translations address <address>	show nat66 destination translations address <text>
×	not yet documented	nat66.xml.in: show nat66 source rules	show nat66 source rules
×	not yet documented	nat66.xml.in: show nat66 source statistics	show nat66 source statistics
×	not yet documented	nat66.xml.in: show nat66 source translations	show nat66 source translations
×	not yet documented	nat66.xml.in: show nat66 source translations address <address>	show nat66 source translations address <text>
×	<i>show nat cgnat allocation</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>show nat cgnat allocation external-address <address></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>show nat cgnat allocation internal-address <address></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	nat.xml.in: show nat destination rules	show nat destination rules
×	not yet documented	nat.xml.in: show nat destination statistics	show nat destination statistics
×	not yet documented	nat.xml.in: show nat destination translations	show nat destination translations
×	not yet documented	nat.xml.in: show nat destination translations address <address>	show nat destination translations address <text>
×	not yet documented	nat.xml.in: show nat source rules	show nat source rules
×	not yet documented	nat.xml.in: show nat source statistics	show nat source statistics

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Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	nat.xml.in: show nat source translations	show nat source translations
×	not yet documented	nat.xml.in: show nat source translations address <text> show nat source translations address <address>	show nat source translations address <text>
×	not yet documented	show-netns.xml.in: show netns	show netns
×	not yet documented	nhrp.xml.in: show nhrp interface	show nhrp interface
×	not yet documented	nhrp.xml.in: show nhrp tunnel	show nhrp tunnel
×	not yet documented	Nothing found in XML Definitions	show ntp
×	not yet documented	Nothing found in XML Definitions	show ntp system
×	not yet documented	Nothing found in XML Definitions	show openconnect-server sessions
×	not yet documented	Nothing found in XML Definitions	show openconnect-server user <text> otp full
×	not yet documented	Nothing found in XML Definitions	show openconnect-server user <text> otp key-b32
×	not yet documented	Nothing found in XML Definitions	show openconnect-server user <text> otp key-hex
×	not yet documented	Nothing found in XML Definitions	show openconnect-server user <text> otp qrcode
×	not yet documented	Nothing found in XML Definitions	show openconnect-server user <text> otp uri
✓	<i>show openvpn client</i>	openvpn.xml.in: show openvpn client	show openvpn client
✓	<i>show openvpn server</i>	openvpn.xml.in: show openvpn server	show openvpn server
✓	<i>show openvpn site-to-site</i>	openvpn.xml.in: show openvpn site-to-site	show openvpn site-to-site
×	not yet documented	Nothing found in XML Definitions	show pki
×	<i>show pki ca</i>	Nothing found in XML Definitions	show pki ca <text>
×	<i>show pki ca <name></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	Nothing found in XML Definitions	show pki ca <text> pem
×	<i>show pki certificate</i>	Nothing found in XML Definitions	show pki certificate <text>
×	<i>show pki certificate <name></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	Nothing found in XML Definitions	show pki certificate <text> fingerprint <text>

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×	not yet documented	Nothing found in XML Definitions	show pki certificate <text> pem
×	<i>show pki crl</i>	Nothing found in XML Definitions	show pki crl <text>
×	not yet documented	Nothing found in XML Definitions	show pki crl <text> pem
×	not yet documented	policy-route.xml.in: show policy route <route>	show policy route <text>
×	not yet documented	policy-route.xml.in: show policy route6 <route6>	show policy route6 <text>
×	not yet documented	policy-route.xml.in: show policy route6 <route6> rule <rule>	show policy route6 <text> rule <text>
×	not yet documented	policy-route.xml.in: show policy route <route> rule <rule>	show policy route <text> rule <text>
×	not yet documented	show-poweroff.xml.in: show poweroff	show poweroff
×	not yet documented	Nothing found in XML Definitions	show pppoe-server interfaces
×	<i>show pppoe-server sessions</i>	Nothing found in XML Definitions	show pppoe-server sessions
×	not yet documented	Nothing found in XML Definitions	show pppoe-server statistics
✓	<i>show pptp-server sessions</i>	pptp-server.xml.in: show pptp-server sessions	show pptp-server sessions
×	not yet documented	pptp-server.xml.in: show pptp-server statistics	show pptp-server statistics
✓	<i>show protocols static arp</i>	show-protocols.xml.in: show protocols static arp	show protocols static arp
✓	<i>show protocols static arp interface <interface></i>	show-protocols.xml.in: show protocols static arp interface <interface>	show protocols static arp interface <text>
×	not yet documented	show-raid.xml.in: show raid <raid>	show raid <text>
×	<i>show raid <RAID-1-device></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	show-reboot.xml.in: show reboot	show reboot
×	not yet documented	Nothing found in XML Definitions	show remote-config openvpn <text> remote-platform <text>
×	not yet documented	show-reverse-proxy.xml.in: show reverse-proxy	show reverse-proxy
×	not yet documented	show-route-map.xml.in: show route-map <route-map>	show route-map <text>

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×	not yet documented	Nothing found in XML Definitions	show rpki cache-connection
×	not yet documented	Nothing found in XML Definitions	show rpki cache-server
×	not yet documented	Nothing found in XML Definitions	show rpki prefix-table
×	not yet documented	show-segment-routing.xml.in: show segment-routing srv6 locator	Nothing found in VyOS
×	not yet documented	sflow.xml.in: show sflow	show sflow
×	not yet documented	snmp.xml.in: show snmp community <community>	show snmp community <text>
×	not yet documented	snmp.xml.in: show snmp community <community> host <host>	show snmp community <text> host <text>
×	not yet documented	snmp.xml.in: show snmp mib ifmib	show snmp mib ifmib
×	not yet documented	snmp.xml.in: show snmp mib ifmib ifAlias <ifAlias>	show snmp mib ifmib ifAlias <text>
×	not yet documented	snmp.xml.in: show snmp mib ifmib ifDescr <ifDescr>	show snmp mib ifmib ifDescr <text>
×	not yet documented	snmp.xml.in: show snmp mib ifmib ifIndex <ifIndex>	show snmp mib ifmib ifIndex <text>
×	not yet documented	snmp.xml.in: show snmp v3	show snmp v3
×	not yet documented	snmp.xml.in: show snmp v3 certificates	show snmp v3 certificates
×	not yet documented	snmp.xml.in: show snmp v3 group	show snmp v3 group
×	not yet documented	snmp.xml.in: show snmp v3 trap-target	show snmp v3 trap-target
×	not yet documented	snmp.xml.in: show snmp v3 user	show snmp v3 user
×	not yet documented	snmp.xml.in: show snmp v3 view	show snmp v3 view
×	<i>show ssh dynamic-protection</i>	show-ssh.xml.in: show ssh dynamic- protection	Nothing found in VyOS
×	<i>show ssh fingerprints</i>	show-ssh.xml.in: show ssh fingerprints	Nothing found in VyOS
×	<i>show ssh fingerprints ascii</i>	show-ssh.xml.in: show ssh fingerprints ascii	Nothing found in VyOS

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✓	<i>show sstp-server sessions</i>	sstp-server.xml.in: show sstp-server sessions	show sstp-server sessions
×	not yet documented	sstp-server.xml.in: show sstp-server statistics	show sstp-server statistics
✓	<i>show system acceleration qat</i>	show-acceleration.xml.in: show system acceleration qat	show system acceleration qat
×	<i>show system acceleration qat device <device> config</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>show system acceleration qat device <device> flows</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	show-acceleration.xml.in: show system acceleration qat device <device> config	show system acceleration qat device <text> config
×	not yet documented	show-acceleration.xml.in: show system acceleration qat device <device> flows	show system acceleration qat device <text> flows
✓	<i>show system acceleration qat interrupts</i>	show-acceleration.xml.in: show system acceleration qat interrupts	show system acceleration qat interrupts
✓	<i>show system acceleration qat status</i>	show-acceleration.xml.in: show system acceleration qat status	show system acceleration qat status
✓	<i>show system commit</i>	show-system.xml.in: show system commit	show system commit
×	not yet documented	show-system.xml.in: show system commit diff <diff>	show system commit diff <text>
×	<i>show system commit diff <number></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	show-system.xml.in: show system commit file <file>	show system commit file <text>
×	not yet documented	show-system.xml.in: show system commit file <file> compare <compare>	show system commit file <text> compare <text>
×	not yet documented	show-system.xml.in: show system commit file <file> compare <compare> commands	show system commit file <text> compare <text> commands
×	not yet documented	show-system.xml.in: show system connections	show system connections

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Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	show-system.xml.in: show system connections tcp	show system connections tcp
×	not yet documented	show-system.xml.in: show system connections tcp all	show system connections tcp all
×	not yet documented	show-system.xml.in: show system connections tcp numeric	show system connections tcp numeric
×	not yet documented	show-system.xml.in: show system connections udp	show system connections udp
×	not yet documented	show-system.xml.in: show system connections udp numeric	show system connections udp numeric
×	not yet documented	show-system.xml.in: show system cpu	show system cpu
✓	<i>show system image</i>	system-image.xml.in: show system image	show system image
×	not yet documented	system-image.xml.in: show system image details	Nothing found in VyOS
×	not yet documented	Nothing found in XML Definitions	show system image storage
×	not yet documented	Nothing found in XML Definitions	show system image version
×	not yet documented	show-system.xml.in: show system kernel-messages	show system kernel-messages
×	not yet documented	show-system.xml.in: show system login authentication user <user> otp	Nothing found in VyOS
×	not yet documented	show-system.xml.in: show system login authentication user <user> otp full	Nothing found in VyOS
×	not yet documented	show-system.xml.in: show system login authentication user <user> otp key-b32	Nothing found in VyOS
×	not yet documented	show-system.xml.in: show system login authentication user <user> otp qr-code	Nothing found in VyOS
×	not yet documented	show-system.xml.in: show system login authentication user <user> otp uri	Nothing found in VyOS
×	not yet documented	show-system.xml.in: show system login users	show system login users

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Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	show-system.xml.in: show system login users all	show system login users all
×	not yet documented	show-system.xml.in: show system login users locked	show system login users locked
×	not yet documented	show-system.xml.in: show system login users other	show system login users other
×	not yet documented	show-system.xml.in: show system login users vyos	show system login users vyos
×	not yet documented	show-system.xml.in: show system memory	show system memory
×	not yet documented	show-system.xml.in: show system memory cache	show system memory cache
×	not yet documented	show-system.xml.in: show system memory detail	show system memory detail
×	not yet documented	show-system.xml.in: show system memory routing-daemons	show system memory routing-daemons
×	not yet documented	show-system.xml.in: show system processes	show system processes
×	not yet documented	show-system.xml.in: show system processes extensive	show system processes extensive
×	not yet documented	show-system.xml.in: show system processes summary	show system processes summary
×	not yet documented	show-system.xml.in: show system processes tree	show system processes tree
×	not yet documented	show-system.xml.in: show system routing-daemons	show system routing-daemons
×	not yet documented	show-system.xml.in: show system storage	show system storage
×	not yet documented	show-system.xml.in: show system updates	show system updates
×	not yet documented	show-system.xml.in: show system uptime	show system uptime
×	not yet documented	show-table.xml.in: show table	show table
×	not yet documented	Nothing found in XML Definitions	show tech-support
×	not yet documented	Nothing found in XML Definitions	show tech-support brief

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Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	Nothing found in XML Definitions	show tech-support brief save <text>
×	not yet documented	Nothing found in XML Definitions	show tech-support brief save-uncompressed <text>
×	not yet documented	Nothing found in XML Definitions	show tech-support private
×	not yet documented	Nothing found in XML Definitions	show tech-support private brief
×	not yet documented	Nothing found in XML Definitions	show tech-support private brief save <text>
×	not yet documented	Nothing found in XML Definitions	show tech-support private brief save-uncompressed <text>
×	not yet documented	Nothing found in XML Definitions	show tech-support private save <text>
×	not yet documented	Nothing found in XML Definitions	show tech-support private save-uncompressed <text>
×	not yet documented	show-techsupport_report.xml.in: show tech-support report	show tech-support report
×	not yet documented	Nothing found in XML Definitions	show tech-support save <text>
×	not yet documented	Nothing found in XML Definitions	show tech-support save-uncompressed <text>
×	not yet documented	show-users.xml.in: show users	show users
×	not yet documented	show-users.xml.in: show users recent <recent>	show users recent <text>
✓	<i>show version</i>	show-version.xml.in: show version	show version
×	not yet documented	show-version.xml.in: show version all	show version all
✓	<i>show version frr</i>	show-version.xml.in: show version frr	show version frr
×	not yet documented	show-version.xml.in: show version funny	show version funny
✓	<i>show version kernel</i>	show-version.xml.in: show version kernel	show version kernel
×	not yet documented	show-virtual-server.xml.in: show virtual-server	show virtual-server
×	not yet documented	vpn-ipsec.xml.in: show vpn debug	show vpn debug
×	not yet documented	vpn-ipsec.xml.in: show vpn debug peer <peer>	show vpn debug peer <text>

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Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	vpn-ipsec.xml.in: show vpn debug peer <peer> tunnel <tunnel>	show vpn debug peer <text> tunnel <text>
✓	<i>show vpn ike sa</i>	vpn-ipsec.xml.in: show vpn ike sa	show vpn ike sa
×	not yet documented	vpn-ipsec.xml.in: show vpn ike sa nat- traversal	show vpn ike sa nat- traversal
×	not yet documented	vpn-ipsec.xml.in: show vpn ike sa peer <peer>	show vpn ike sa peer <text>
✓	<i>show vpn ike secrets</i>	vpn-ipsec.xml.in: show vpn ike secrets	show vpn ike secrets
✓	<i>show vpn ike status</i>	vpn-ipsec.xml.in: show vpn ike status	show vpn ike status
✓	<i>show vpn ipsec connections</i>	vpn-ipsec.xml.in: show vpn ipsec connec- tions	show vpn ipsec connec- tions
✓	<i>show vpn ipsec policy</i>	vpn-ipsec.xml.in: show vpn ipsec policy	show vpn ipsec policy
×	not yet documented	vpn-ipsec.xml.in: show vpn ipsec remote- access connection-id <connection-id>	show vpn ipsec remote- access connection-id <text>
×	not yet documented	vpn-ipsec.xml.in: show vpn ipsec remote- access detail	show vpn ipsec remote- access detail
×	not yet documented	vpn-ipsec.xml.in: show vpn ipsec remote- access summary	show vpn ipsec remote- access summary
×	not yet documented	vpn-ipsec.xml.in: show vpn ipsec remote- access username <user- name>	show vpn ipsec remote- access username <text>
✓	<i>show vpn ipsec sa [detail]</i>	vpn-ipsec.xml.in: show vpn ipsec sa	show vpn ipsec sa
×	not yet documented	vpn-ipsec.xml.in: show vpn ipsec sa detail	show vpn ipsec sa detail
✓	<i>show vpn ipsec state</i>	vpn-ipsec.xml.in: show vpn ipsec state	show vpn ipsec state
✓	<i>show vpn ipsec status</i>	vpn-ipsec.xml.in: show vpn ipsec status	show vpn ipsec status
×	<i>show vpp acl</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>show vpp acl ip</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>show vpp acl ip interface</i>	Nothing found in XML Definitions	Nothing found in VyOS

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×	<i>show vpp acl ip tag-name <tag-name></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>show vpp acl mac</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>show vpp acl mac interface</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>show vpp acl mac tag-name <tag-name></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>show vpp nat cgnat exclude-rules</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>show vpp nat cgnat interfaces</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>show vpp nat cgnat mappings</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>show vpp nat cgnat sessions</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>show vpp nat nat44 addresses</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>show vpp nat nat44 interfaces</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>show vpp nat nat44 sessions</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>show vpp nat nat44 static</i>	Nothing found in XML Definitions	Nothing found in VyOS
×	<i>show vpp nat nat44 summary</i>	Nothing found in XML Definitions	Nothing found in VyOS
✓	<i>show vrf</i>	show-vrf.xml.in: show vrf <vrf>	show vrf <text>
×	<i>show vrf <name></i>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	show-vrf.xml.in: show vrf <vrf> processes	show vrf <text> processes
×	not yet documented	show-vrf.xml.in: show vrf <vrf> vni	Nothing found in VyOS
×	not yet documented	Nothing found in XML Definitions	show vrrp
×	not yet documented	Nothing found in XML Definitions	show vrrp detail
×	not yet documented	Nothing found in XML Definitions	show vrrp statistics
×	not yet documented	Nothing found in XML Definitions	show wan-load-balance
×	not yet documented	Nothing found in XML Definitions	show wan-load-balance connection
×	not yet documented	Nothing found in XML Definitions	show wan-load-balance status
×	not yet documented	Nothing found in XML Definitions	show wan-load-balance status with-dns

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Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	Nothing found in XML Definitions	show webproxy blacklist categories
×	not yet documented	Nothing found in XML Definitions	show webproxy log
×	not yet documented	Nothing found in XML Definitions	show webproxy update-log
×	not yet documented	show-zebra.xml.in: show zebra	show zebra
×	not yet documented	show-zebra.xml.in: show zebra client summary	show zebra client summary
×	not yet documented	show-zebra.xml.in: show zebra dplane	show zebra dplane
×	not yet documented	show-zebra.xml.in: show zebra router table summary	show zebra router table summary
×	not yet documented	Nothing found in XML Definitions	show zone-policy
×	not yet documented	Nothing found in XML Definitions	show zone-policy zone <text>
×	<code>sudo chmod +x /config/scripts/vpp-collect-diagnostics</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>sudo vppctl clear trace</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>sudo vppctl pcap trace [rx] [tx] [drop] [max <n>] [intfc <interface-name/any>] [file <name>] [max-bytes-per-pkt <n>]</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>sudo vppctl pcap trace off</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>sudo vppctl pcap trace status</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>sudo vppctl show trace [max COUNT]</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>sudo vppctl trace add <input-graph-node> <pkts> [verbose]</code>	Nothing found in XML Definitions	Nothing found in VyOS
×	<code>set system image default-boot [image-name]</code>	Nothing found in XML Definitions	set system image default-boot <text>
×	not yet documented	telnet.xml.in: telnet to <to>	telnet to <text>

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Status 108/3404	Documentation	XML	in VyOS 1.4-rolling-202308240020
×	not yet documented	telnet.xml.in: telnet to <to> port <port>	telnet to <text> port <text>
×	not yet documented	terminal.xml.in: set terminal key query-help <query-help>	set terminal key query-help <text>
×	not yet documented	terminal.xml.in: set terminal length <length>	set terminal length <text>
×	not yet documented	terminal.xml.in: set terminal pager <pager>	set terminal pager <text>
×	not yet documented	terminal.xml.in: set terminal width <width>	set terminal width <text>
×	not yet documented	traceroute.xml.in: traceroute <traceroute>	traceroute <text> <text>
×	traceroute <destination>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	traceroute.xml.in: traceroute <traceroute> node.tag	Nothing found in VyOS
×	traceroute vrf <name> [ipv4 ipv6] <host>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	container.xml.in: update container image <image>	update container image <text>
×	update container image <containername>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	Nothing found in XML Definitions	update dns dynamic
✓	update geoip	geoip.xml.in: update geoip	update geoip
×	update suricata	Nothing found in XML Definitions	Nothing found in VyOS
✓	update webproxy blacklists	webproxy.xml.in: update webproxy blacklists	update webproxy blacklists
×	update webproxy blacklists category <category>	Nothing found in XML Definitions	Nothing found in VyOS
×	not yet documented	webproxy.xml.in: update webproxy blacklists vrf <vrf>	update webproxy blacklists vrf <text>
×	not yet documented	wake-on-lan.xml.in: wake-on-lan interface <interface> host <host>	wake-on-lan interface <text> host <text>

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